IBM Fusion HCI

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What's new

Each release offers new functions and improvements. IBM is constantly updating the infrastructure, security, and stability of IBM Fusion HCI to improve your experience. Review this information for a high-level summary of the new features and changes in each release.

AI enhancements

Content-Aware Storage (CAS) service

Simplified CAS deployment with native RBD support
: In CAS v1.1.5, CAS backend leverages the native RADOS Block Device (RBD) support introduced in IBM Storage Scale CNSA v6.0.1 to provision local cache filesystems by using RBD disks from Data Foundation. Enhanced automated deployment script eliminates additional setup steps and provides a more streamlined deployment experience. For details on setting up local cache filesystem, see Installing CAS without a remotely mounted IBM Storage Scale filesystem.

Network File System (NFS) data source support with CAS
: CAS introduces an NFS data source type, enabling you to bring enterprise data stored on NFS exports into your RAG applications. As with all other supported data source types, CAS leverages IBM Docling Multimodal services and NVIDIA NeMo Retriever Extraction microservices for optimized content processing. By utilizing IBM Storage Scale as a cache layer, CAS eliminates the need to maintain duplicate copies of data. This reduces storage costs and operational complexity while accelerating access to enterprise content for AI workloads. For more information, see Creating a data source.

Domain-level file access control
: CAS now supports domain-level configuration for file permission checking, allowing users to enable or disable file-level access control on a per-domain basis. This enhancement removes the need for separate scripts by providing built-in configuration through the user interface. With this capability, administrators can apply file-level controls selectively across domains while continuing to use an integrated identity provider with IBM Storage Scale for enforcing access policies. For more information, see Modifying file-level access control for a domain.

Zero-trust network policy enforcement
: Content-Aware Storage (CAS) now automatically enforces a zero-trust network communication model across its internal services by deploying and managing Kubernetes NetworkPolicies. This behavior is configurable through the CasInstall custom resource, allowing administrators to control policy enforcement as part of the deployment configuration. For more information, see Configuring network policies.

Installation and image mirroring precheck utilities

Call Home support during installation
: Call Home can now be configured as part of the IBM Fusion HCI installation. By providing support and connectivity details upfront, Call Home becomes fully functional when IBM Fusion and its services are installed. No post-installation setup is required.

A new, simplified proxy configuration experience is available during setup, with support for multiple connectivity options:

- Direct connection
- Open proxy
- Proxy with basic authentication
- Proxy with certificate-based authentication

This ensures seamless connectivity in both restricted and unrestricted network environments from day one. For more information, see Installing IBM Fusion HCI.

Preinstalled images for simplified IBM Fusion installation
: IBM Fusion now includes Red Hat® and IBM Fusion images preinstalled within the installation package, eliminating the need to mirror images before deployment. This enhancement simplifies installation in offline or air-gapped environments, reduces setup time, and streamlines the overall deployment process. For more information, see Installing IBM Fusion HCI.

Improved image mirroring efficiency
: IBM Fusion now simplifies the image mirroring process by removing the dependency on the **`ibm-pak`** tool. You can now complete mirroring by using only the **`oc-mirror`** utility. In addition, the mirroring process is more efficient, as it copies only the required release images, resulting in reduced image size and lower storage requirements.

Automated Stage 2 installation
: IBM Fusion HCI now supports automated Stage 2 installation through CLI-based automation, enabling a fully “as-code” deployment experience. This removes the need for UI-based execution and allows users to initiate and complete installations using configuration files and automation scripts, improving consistency and simplifying multi-rack deployments.

Upgrade enhancements

Simplified upgrade path for IBM Fusion HCI
: The upgrade process for IBM Fusion HCI is streamlined. Users currently on 2.11.x can now directly upgrade to 2.13.0 on specific configurations, bypassing the need for intermediate 2.12.x releases. For details on supported configurations, see Enhanced upgrade experience.

Fully automated service node upgrades
: IBM Fusion now supports fully automated service node upgrades through a dedicated upgrade agent that orchestrates the entire upgrade process without manual intervention. This enhancement streamlines the service node upgrade workflow and reduces operational effort. When a service node upgrade becomes available, you can trigger and monitor the upgrade directly from the IBM Fusion UI. For more information, see Upgrading the service node.

Service management

Automatic storage class selection

IBM Fusion now automatically configures storage classes for its data services based on best-practice settings, reducing the need for manual setup and simplifying deployment. Manual configuration is required only when IBM Fusion storage is not used, providing flexibility for custom storage environments.

Security

Automated hardware credential rotation

IBM Fusion HCI now automates hardware credential rotation, removing the need for administrators to manually manage and track credential updates as part of operational best practices. The platform also supports integration with an external Vault instance, enabling customers to rotate secrets in OpenShift® based on their own security policies. IBM Fusion automatically synchronizes these updates with the underlying hardware, ensuring consistent and compliant credential management across the environment. For more information, see Password rotation.

Integrated secret management

IBM Fusion HCI now includes integrated Vault-based secret management, enabling secure storage and centralized management of credentials, secrets, and cryptographic keys. It enhances platform security through strong access controls, audit logging, and encryption of data at rest and in transit, including multi-key unseal mechanisms and TLS-secured communication. It supports key infrastructure use cases such as managing node credentials, network device access, and SSH keys. For more information, see Using Vault with IBM Fusion HCI.

Hardware enhancements

Enhanced node addition experience in the IBM Fusion HCI user interface

IBM Fusion HCI introduces an enhanced user experience when adding nodes to OpenShift Container Platform. The new functionality provides real-time feedback and visibility into the node addition process.

Precheck enhancements

- Nodes in the discovered tab now undergo a series of prechecks to verify their readiness for addition to OpenShift Container Platform, helping to identify and prevent common configuration issues such as misconfigured DNS.
- Nodes are labeled as **Ready to add** or **Not ready** based on precheck results.
- Clicking a node reveals detailed information about failed prechecks, enabling swift troubleshooting.

Real-time node addition progress

- After prechecks pass, the node addition progress is displayed in IBM Fusion HCI user interface.
- You can monitor the status of node addition by clicking the node, providing transparency into the process.

These enhancements improve the overall user experience, reduce potential errors, and increase productivity when managing OpenShift Container Platform nodes through IBM Fusion HCI user interface. For more information, see Configuring nodes for management.

Service node upsize for existing IBM Fusion HCI deployments

IBM Fusion HCI now supports adding a Service Node to an existing running deployment, enabling customers to seamlessly upsize their infrastructure without requiring a new installation or system redeployment. This enhancement is supported on both Gen1 and Gen2 racks.

By introducing a Service Node into an operational IBM Fusion HCI environment, customers can unlock the full set of Service Node capabilities, including enhanced lifecycle management, monitoring, supportability, and platform services. The feature provides a non-disruptive path to expand system functionality and improve operational efficiency while preserving the existing cluster deployment.

Key Benefits:

- Add a Service Node to an already deployed IBM Fusion HCI system.
- Supported on both Gen1 and Gen2 rack architectures.
- No need to redeploy or rebuild the existing cluster.
- Enables access to all Service Node features and services after the upsize.

Optimized node restart for Lenovo firmware upgrades

The Lenovo node firmware upgrade process is optimized to reduce the number of nodes restarts required, resulting in faster and more efficient upgrades compared to previous releases. For more information, see Node firmware.

Automated rack reset

IBM Fusion HCI now introduces an automated rack reset capability that returns the rack to a clean, factory-ready state. This feature simplifies reinstallation and recovery by eliminating manual reset steps and ensuring a consistent baseline configuration across all components. For more information, see Rack reset.

Serviceability enhancements

Enhanced log collection and automatic sanitization

IBM Fusion now supports targeted log collection for individual nodes and switches, reducing collection time and generating smaller, more relevant log packages for troubleshooting.

In addition, IBM Fusion includes automatic log package sanitization, which helps protect sensitive data when you collect diagnostic logs for troubleshooting and support. The system automatically omits or masks sensitive information from log packages, for example:

- MAC addresses
- IP addresses
- Subdomains
- Domains
- Gateways
- DNS and DHCP information

Automatic log sanitization is enabled by default, ensuring that diagnostic logs are safe to share with IBM Support without requiring manual cleanup. This capability reduces security risk while simplifying support interactions. For more information, see Automatic cleanup of log packages.

Collection of OS logs from the service node using IBM Fusion UI

IBM Fusion now supports collection of operating system (OS) logs directly from the service node to help troubleshooting and system diagnostics. For more information, see Nodes.

Reduced event noise and improved maintenance visibility
IBM Fusion now reduces event noise by automatically resolving compute, installation, and network-related critical issues once they are fixed, while enhanced deduplication minimizes duplicate alerts. During planned maintenance operations such as OpenShift upgrades, node maintenance initiated from the IBM Fusion UI, or switch upgrades, the system suppresses unnecessary critical alerts and degraded health indicators. This results in fewer, more relevant events and improved operational clarity.

IBM Lakehouse-Table service [Technology preview]

The Lakehouse-Table service is a Kubernetes-native solution that simplifies the creation and management of lakehouse infrastructure on IBM Fusion. This service provides a streamlined experience for administrators to provision and manage Data Lakehouse capabilities, including object storage integration and Iceberg catalog management, and integrates seamlessly with watsonx.data, enabling a comprehensive data lakehouse experience. For more information, see Lakehouse-Table [Technology preview].

Backup & Restore enhancements

Enhanced error messaging
Error messages now clearly indicate the specific component that triggered the error, enabling you to quickly identify and troubleshoot issues. This improvement reduces resolution time and allows you to potentially resolve problems independently or provide support with precise information to expedite the resolution process.

Optimized Backup & Restore pod resources
The Pod Resource Recommendation Tool is now available, enabling you to optimize the resources that are allocated to Backup & Restore pods. This tool analyzes historical data from a specified reference period to recommend optimal resource levels, ensuring that cluster resources are used efficiently. For more information, see Pod Resource Recommendation and Patch Tool.

Fusion Data Foundation support

IBM Fusion now includes support for Fusion Data Foundation 4.21. Key highlights in Fusion Data Foundation 4.21 include:

- Container Native Storage Access (CNSA) integration
- Erasure coding support
- Multi-volume consistency groups for RBD and CephFS volumes
- CephFS volumes support using non-default StorageClasses
- Enhanced protected applications list view
- Automatic resource cleanup for discovered VMs during DR operations
- Minimal IAM support for Multicloud Object Gateway
- Enhanced Multicloud Object Gateway database backup
- Developer access to object browser
- Internal mode RGW support with object browser
- HCP clusters support enhancements

For more information, see New features and Enhancements topics in *Fusion Data Foundation 4.21 release notes*. For known issues and limitations, see Known issues in 4.21 topic.

IBM Data Cataloging enhancements

Enhanced data classification

AI-powered content-based tagging
IBM Data Cataloging introduces AI-powered content-based tagging to automatically classify unstructured data based on actual file content, rather than relying solely on metadata. By leveraging large language models (LLMs) and external GPU acceleration, the solution analyzes data at scale and suggests relevant business tags with minimal manual effort. This capability standardizes content organization across multiple data sources, improving data discovery and governance. For more information, see Content-based AI-enabled autotagging.

Improved data governance
IBM Data Cataloging enhances Content-Aware Storage (CAS) by propagating tags that are assigned to cataloged data. Tags, whether created manually, through auto-tagging policies, or through AI, are consolidated and propagated to the underlying data in CAS.

This enrichment enables CAS enrich content ingestion with tags attributes, resulting in more accurate semantic search and improved governance across domains. For more information, see Content-Aware Storage (CAS) APIs.

Simplified Integration with Content-Aware Storage

One click installation and enablement
Announcing new capability to ease IBM Data Cataloging and Content-Aware Storage integration. This capability simplifies the integration between IBM Data Cataloging and Content-Aware Storage with a one-click installation and enablement experience. Users can deploy and activate IBM Data Cataloging directly from the CAS interface in just two steps: Install and Enable.

This streamlined setup enhances operational efficiency and allows CAS to leverage advanced scanning and filtering capabilities for more granular data ingestion control. For more information, see One-click installation and enablement integration experience.

Data insights dashboards
Announcing new IBM Data Cataloging and CAS introduce data insights dashboards that provide immediate visibility into data once sources are connected and scanned. This experience is enabled when both IBM Data Cataloging and CAS are integrated. The provided insights surface key indicators of data composition on each established data source. For more information, see IBM Data Cataloging and Content-Aware Storage insights.

Security fixes and updates

Enhance IBM Data Cataloging service stability and protect against known vulnerabilities.

UI enhancements

Enhanced Service Node UI for installation and lifecycle management

IBM Fusion HCI introduces a new Service Node dashboard, which serves as the default landing page when no specific page is specified. This dashboard provides guided access to Stage 1 and Stage 2 installation workflows, along with a high-level overview of the OpenShift cluster, including quick access to the OpenShift UI and a summary of node status and network health.

After cluster installation, the dashboard also enables administrators to reset the IBM Fusion HCI rack to factory settings, supporting streamlined reinstallation and lifecycle management from a single interface.

New currency support

- Content-Aware Storage 1.1.5
- OpenShift Container Platform 4.21
- IBM Storage Scale Erasure Code Edition (ECE) or Global Data Platform service 5.2.3.7
- IBM Storage Scale Data Management Edition or Global Data Platform service 6.0.1.0 and 6.0.0.2
- Backup & Restore 2.13.0
- IBM Data Cataloging 2.5.3
- For supported IBM Cloud Paks versions, see IBM Cloud Paks support for IBM Fusion.
- For supported IBM Maximo® versions, see IBM Maximo® Applications support for IBM Fusion.

For a complete list of currency support, see Support matrix for IBM Fusion versions.

Product overview

IBM Fusion is a container-native hybrid cloud data platform that offers simplified deployment and data management for Kubernetes applications on Red Hat® OpenShift® Container Platform. IBM Fusion is designed to meet the storage requirements of modern, stateful Kubernetes applications and to make it easy to deploy and manage container-native applications and their data on Red Hat OpenShift Container Platform.

IBM Fusion is an advanced storage and backup solution that is designed to simplify data accessibility and availability across hybrid clouds. Companies can expand data availability across complex hybrid clouds for greater business performance and resilience. With the IBM Fusion solutions, organizations manage only a single copy of data. They need not create duplicate data when moving application workloads across the enterprise, easing management functions when you streamline analytics and AI.

IBM Fusion provides a streamlined way for organizations to discover, secure, protect, and manage data from the edge, to the core data center, to the public cloud.

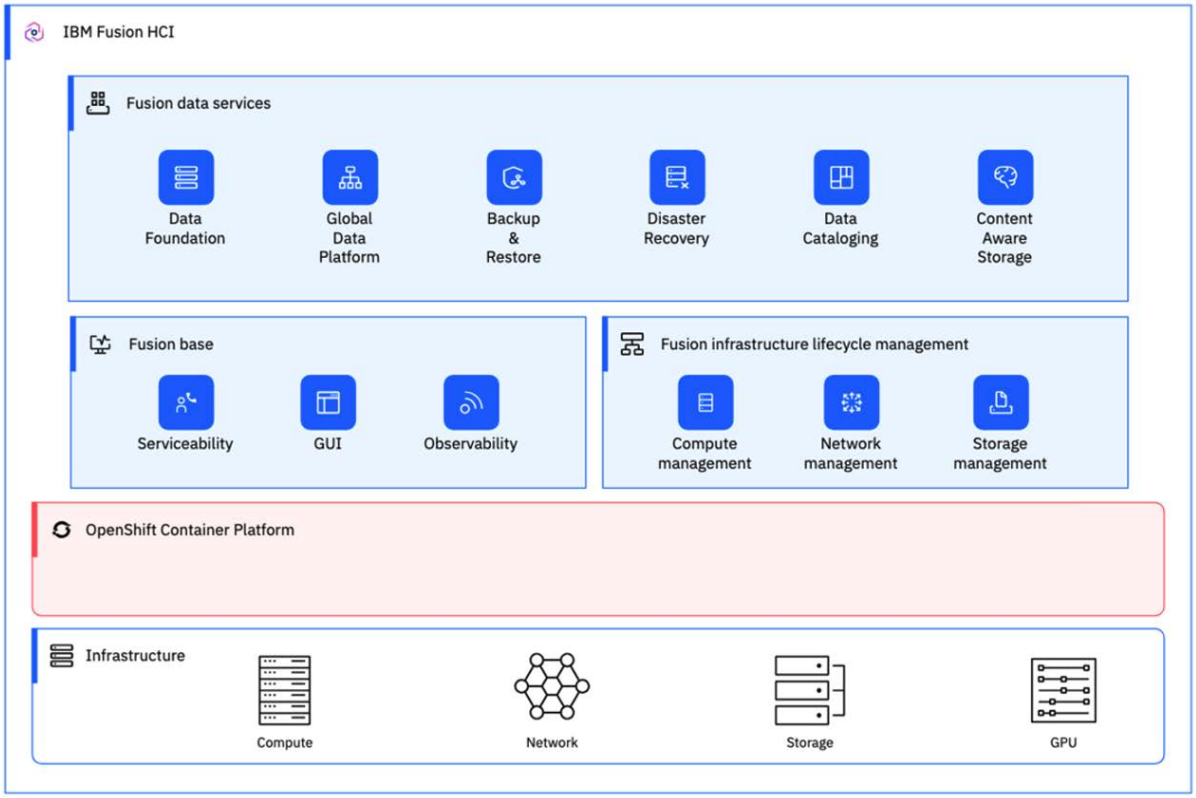
IBM Fusion is available as two deployments, namely IBM Fusion HCI and IBM Fusion.

IBM Fusion HCI

IBM Fusion HCI is a purpose-built, hyper-converged architecture that is designed to deploy bare metal Red Hat OpenShift container management and deployment software alongside IBM Fusion software. For consistent and rapid deployment and management, it features an appliance form-factor, hyper-converged infrastructure along with integrated software-defined storage to meet the storage requirements of modern, stateful Kubernetes applications. Built with a storage platform that includes the essential elements necessary for mission-critical containers and hybrid cloud, the IBM Fusion provides a comprehensive infrastructure with compute, networking, and storage resources, including a data platform and global data services for Red Hat OpenShift.
The following *Figure 1. High-level conceptual overview of IBM Fusion HCI* provides a high-level conceptual overview of IBM Fusion HCI architecture:

- The IBM Fusion HCI provides a fully integrated infrastructure, which combines compute, networking, and storage resources optimized for containerized workloads.
- Designed to run on Bare Metal Red Hat OpenShift Container Platform, it supports the building and running of cloud-native applications.
- The Fusion Base offers essential services such as serviceability for issue diagnosis and resolution, a user interface for interaction with IBM Fusion HCI, and observability for monitoring system performance and health.
- The Fusion infrastructure management layer, built on top of OpenShift, simplifies lifecycle management of compute nodes, networking, and storage, ensuring operational efficiency.
- The Fusion services layer delivers a suite of solutions to secure and manage data across hybrid cloud environments.

Figure 1. High-level conceptual overview of IBM Fusion HCI

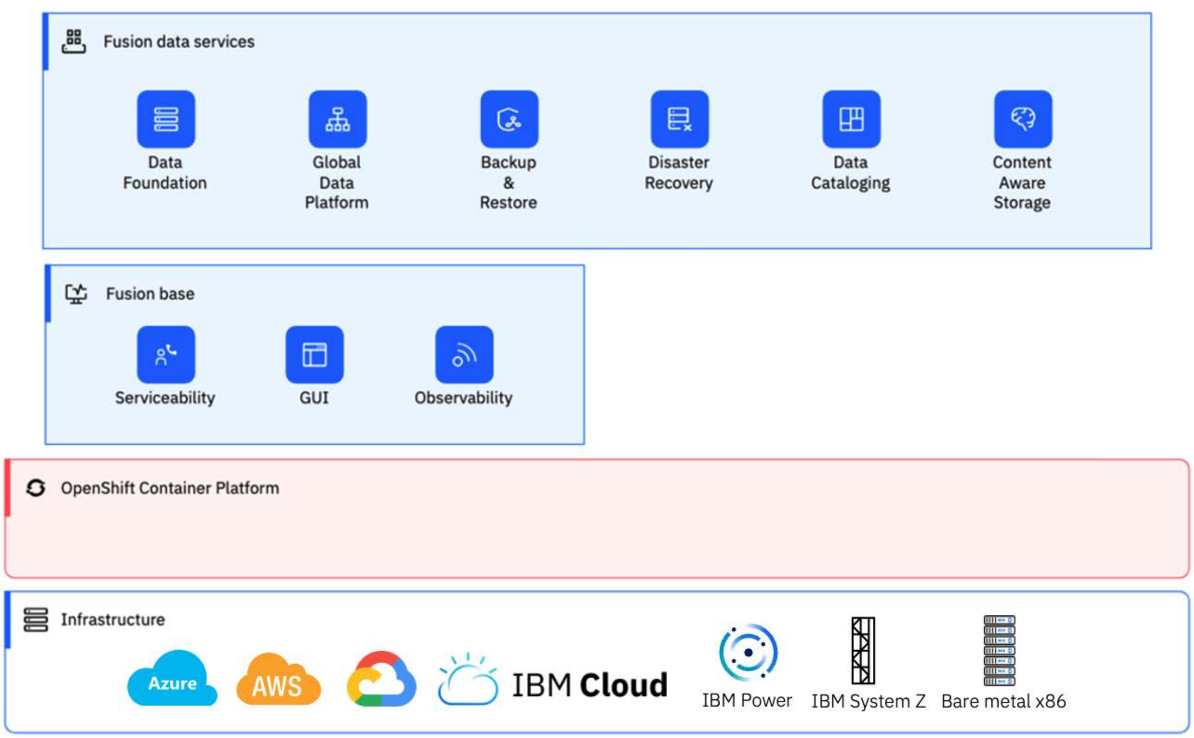


IBM Fusion

IBM Fusion is software-defined storage with protection, backup, and caching elements and can be run on existing hardware resources.
The following *Figure 2. High-level conceptual overview of IBM Fusion* provides a high-level conceptual architecture overview:

- IBM Fusion is designed to run on OpenShift Container Platform across your infrastructure and various supported platforms, enabling the building and running of cloud-native applications.
- Built atop OpenShift, Fusion Base provides foundational services such as serviceability for efficient issue diagnosis and resolution, a user interface for seamless interaction with IBM Fusion, and observability for system performance insights.
- Layered above Fusion Base, Fusion services deliver a suite of solutions to secure and manage data across hybrid cloud environments.

Figure 2. High-level conceptual overview of IBM Fusion



For more information about IBM Fusion, see [IBM Fusion documentation](#).

- **AI-enhanced integration**
  IBM Fusion HCI provides AI-driven data management capabilities through Model Context Protocol (MCP) integrations and content-based automation features.
- **Hardware overview**
  Review the system layout and hardware summary of the IBM Fusion HCI. The IBM Fusion HCI mainly contains the Lenovo and Dell hardware components.
- **Deployment configurations**
  The stand-alone rack, multiple rack, and disaster recovery deployment configurations are available in IBM Fusion HCI. The multi rack topology offers these variants: high-availability multi-rack, 3 zone high-availability multi-rack, two availability zones with an arbiter node in a third zone, and expansion rack.
- **IBM Storage Expert Care**
  IBM Storage Expert Care is a simplified method of selecting services and support for systems at the time of the purchase. IBM Storage Expert Care is designed to simplify and standardize the support approach for IBM Fusion HCIwith simple straightforward pricing and selection of services.
- **Security in IBM Fusion**
  IBM Fusion is a secure platform to deploy your applications.
- **Verifying image signatures**
  Digital signatures provide a way for consumers of content to ensure that what they download is both authentic (it originated from the expected source) and has integrity (it is what we expect it to be). All images for IBM Fusion are signed. This page describes how to verify the signatures on those images.

AI-enhanced integration

IBM Fusion HCI provides AI-driven data management capabilities through Model Context Protocol (MCP) integrations and content-based automation features.

AI-driven capabilities

- Automatically gather observability data, system events, and inventory information for your clusters with smart agents. For more information, see AI-enabled observability with Red Hat OpenShift Lightspeed.
- Search and interact with vector stores by using natural language queries through AI chat applications. For more information, see Integrating Model Context Protocol (MCP) with CAS.
- Simplify tool discovery, execution, and lifecycle management of a unified set of cataloging and data management tools without requiring direct LLM-to-tool integrations. For more information, see IBM Data Cataloging MCP server.
- Analyze document content to automatically generate metadata tags through content-based autotagging. For more information, see Content-based AI-enabled autotagging.

Hardware overview

Review the system layout and hardware summary of the IBM Fusion HCI. The IBM Fusion HCI mainly contains the Lenovo and Dell hardware components.

The IBM Fusion HCI is a hyper-converged infrastructure solution that integrates compute, storage, and networking resources into a single, scalable platform. The IBM Fusion HCI is built using with two reliable hardware options for your business needs.

- Dell hardware
- Lenovo hardware

You can choose from either "Lenovo" or "Dell" hardware option to build a robust IBM Fusion HCI management software. Here is an overview of the key hardware components:

Compute servers
: The compute servers that are used in IBM Fusion HCI compute servers are based on industry-standard x86 architecture, offering high-performance, reliability, and flexibility. They are available on Lenovo (9155-C10 and 9155-C14) and Dell (9155-D10 and 9155-D14) hardware. For more information, see Compute servers.

GPU servers
: The IBM Fusion is a comprehensive management software that uses GPU servers to accelerate various workloads, including artificial intelligence, high-performance computing, and data analytics. They are available on Lenovo (9155-G03) and Dell (9155-DG1, DG2, DG3, and DG4) hardware. For more information, see GPU servers.

Network switches
: The IBM Fusion HCI management software uses management and high-speed switches to facilitate communication between nodes, manage traffic, and ensure high availability. They are available on Lenovo (9155-S01, S02, S04, S05) and Dell (9155-S04 and 9155-S05) hardware. For more information, see Network switches.

Storage servers
: IBM Fusion HCI management software uses a range of storage servers to provide a scalable and efficient storage solution. They are available on Lenovo (9155-C10 and 9155-C14), Dell (9155-D10 and 9155-D14). For more information, see Storage servers.

Power Distribution Units (PDUs)
: The PDUs are designed to provide reliable and efficient power distribution to the appliances components. It enables remote power on/off, restart, and power cycle control of connected devices, reducing the need for physical intervention. For more information, see Power Distribution Units (PDUs).

Service node
: The service node is a critical component of a distributed network, responsible for providing a specific set of capabilities to the network. For more information, see Service node.

Hardware configuration

The IBM Fusion HCI management software offers you with two options for obtaining a rack: either you can buy a rack directly from the IBM, or you can arrange for a rack yourself. For more information about rack options, see IBM Fusion HCI rack ordering options.

Rack configuration
: The rack configuration includes the physical arrangement of servers, storage systems, and network devices.

Rack configuration	Description
Rack size	42U
Networking	• 2x Ethernet high-speed switches • 2x Ethernet management switches

Compute only or compute-storage servers	• 3x servers with 2-10 NVMe drives per server
	• KVM connected to the service node
	• Servers in RU2, RU3, and RU4 become the OpenShift® control plane servers
	• You can choose any one of the following server configuration options:
	  ◦ 3x 32 core servers
	  ◦ 3x 64 core server

Storage	2x 7.68 TB NVMe PCIe drives or 3.84 TB NVMe drives Important: No mixing of 7.68 TB and 3.84 TB drives is allowed within a rack.

Scalable options

To customize your rack configuration, the following options are available:

Rack configuration	Description
Additional compute-storage servers	Add up to 10 additional compute-storage servers, for a maximum total of 16 servers
Additional storage	Add more storage to compute-storage servers using: • 7.68 TB NVMe PCIe drives (up to 10 drives per server) • 3.84 TB NVMe PCIe drives (up to 10 drives per server) Important: No mixing of 7.68 TB and 3.84 TB drives is allowed within a rack.
Additional compute power	Add compute-only servers to increase compute power

Drives and usable storage capacities

To determine the usable storage capacities for drives, utilize the IBM Sales Configurator tool. This tool provides an accurate calculation of the available storage space, taking into account various factors that affect storage capacity for the IBM Fusion HCI solution.

Weight of miscellaneous rack parts

The following table lists the weight of the rack configurations that include miscellaneous parts beyond individual components like cables, Constellation rack and so on:

- **Dell hardware**:

Table 1. Weight of the Dell hardware miscellaneous rack parts

Component	Weight (lbs)	Weight (kg)	Model
SN3700 200GbE switch	30.8	14.0	S05
SN2201 1GbE switch	16.3	7.4	S04
SN3700C switch	27.5	12.5	S01
7316-TF5 console	12.0	5.5	TF5
R660XL 32-core 256GB server, 0 drives	47.8	21.7	D10
R660XL 32-core 256GB server, 2 drives	52.2	23.7	D10
R660XL 32-core 256GB server, 10 drives	69.8	31.7	D10
R660XL 64-core 1024GB server, 0 drives	47.8	21.7	D14
R660XL 64-core 1024GB server, 2 drives	52.2	23.7	D14
R660XL 64-core 1024GB server, 10 drives	69.8	31.7	D14
R660 8-core service node	44.8	44.8	DB1
Constellation rack	398.2	181.0	R42
Intelligent switched PDU+	9.5	4.3	NA
All cables, rails, and so on.	395.5	179.8	NA

- **Lenovo hardware**:

Table 2. Weight of the Lenovo hardware miscellaneous rack parts

Component	Weight (lbs)	Weight (kg)	Model
SN3700 200GbE switch	30.8	14.0	S05
SN2201 1GbE switch	16.3	7.4	S04
SN3700C switch	27.5	12.5	S01
7316-TF5 console	12.0	5.5	TF5
AS4610 switch	11.8	5.4	S02
SR665 GPU server	85.5	38.9	G01
SR645-0 32-core 256GB server	39.1	17.8	C00
SR645-2 32-core 256GB server	40.2	18.3	C01 with 2 drives
SR645-10 32-core 256GB server	44.6	20.3	C01 with 10 drives
SR645-0 64-core 1024GB server	39.1	17.8	C04
SR645-2 64-core 1024GB server	40.2	18.3	C05 with 2 drives
SR645-10 64-core 1024GB server	40.2	18.3	C05 with 10 drives
SR630 V3 32-core	45.9	20.8	C10

SR630 V3 64-core	45.9	20.8	C14
SR630 V3 8-core	40.6	18.5	B01
Constellation rack The **9155-R42** is also known as the **7965-S42** used by Power. For hardware specifications, see Model 7965-S42 rack specifications.	398.2	181.0	R42
Intelligent switched PDU+	9.5	4.3	NA
All cables, rails, and so on.	395.5	179.8	NA
SR675 V3 GPU server	72.1 lbs	32.8 kg	G03 with 1 GPUs
SR675 V3 GPU server	87.5 lbs	39.8 kg	G03 with 8 GPUs

- **Compute servers**
  Use this section to get a detailed understanding and configuration details of compute servers that are used in the IBM Fusion HCI appliance.
- **GPU servers**
  Use this section to get a detailed understanding and configuration details of GPU servers that are used in the IBM Fusion HCI appliance.
- **Network switches**
  The IBM Fusion HCI is configured with a dual-network architecture, designed to optimize performance, management, resiliency, and redundancy.
- **Compute storage servers**
  Use this section to get a detailed understanding and configuration details of compute storage servers that are used in the IBM Fusion HCI appliance.
- **Service node**
  Use this section to get a detailed understanding and configuration details of service nodes that are used in the IBM Fusion HCI appliance.
- **Power Distribution Units (PDUs)**
  Use this section as a guidance for PDU positions in the IBM Fusion HCI.
- **Supported PDU power cords**
  Find out which power distribution unit (PDU) power cords are supported for your system.
- **Power consumption specification for client-provided PDUs**
  The following power consumption information for IBM Fusion HCI components is provided as a planning aid for you to connect these components to PDUs that you provide.
- **Power consumption by configuration**
  Before you up size the nodes, consider the power consumption data that is depicted in the following *Power consumption* tables.

Compute servers

Use this section to get a detailed understanding and configuration details of compute servers that are used in the IBM Fusion HCI appliance.

Compute-only nodes are designed for applications that require high processing power and can use external storage solutions. These nodes are ideal for workloads that depend on shared storage or have low storage requirements and provide the necessary resources for workloads to run efficiently. Also, the compute servers are used a control nodes and worker nodes in the OpenShift® cluster.

The available compute servers are:

- Dell hardware - 9155-D10 and 9155-D14. For configuration details, see Dell hardware.
- Lenovo hardware - 9155-C10 and 9155-C14. For configuration details, see Lenovo hardware.

You can have either three 9155-C10/D10 and 9155-C14/D14 servers with storage as a minimum configuration, and also you can expand your IBM Fusion HCI beyond the minimum three servers by adding combinations of models C10/D10 and D10/D14. Each rack can support up to 16 servers. Each compute server can have a minimum of two storage drives that can be increased up to a maximum of ten storage drives.
Important: You cannot mix and match different drive capacities within the same rack. Choose one capacity and use it consistently throughout.

Key capabilities

The following are the key capabilities of the compute server:

- Compute servers offer customizable CPU and memory allocation, allowing you to tailor resources to meet specific workload requirements.
- You can adjust the amount of CPU and memory in each server to optimize performance and cost.
- Compute servers can be configured as storage-rich servers, providing additional storage capacity for workloads that require it.

Advantages

The following are the key benefits of the compute servers:

Improved resource utilization
: By customizing CPU and memory allocation, you can ensure that workloads have the necessary resources to run efficiently, reducing waste and optimizing utilization.

Enhanced flexibility
: Compute servers can be easily scaled up or down to accommodate changing workload demands, making it easier to adapt to evolving business needs.

Cost effective
: With the ability to customize resources, you can optimize costs by allocating only the necessary resources for each workload, reducing unnecessary expenses.

Hardware configuration

Dell

9155-D10 (32 core)

- Dell R660XL
- 2x 16-core Intel Xeon 4th Gen Gold 6426Y 2.5GHz 185W processors
- 256GB RAM (16x 16GB DDR5 DIMMs)
- Optional memory upgrade to 512GB (32x 16GB DIMMs)
- 0-10x 7.68TB or 3.84TB PCIe NVMe drives
- 1x NVIDIA ConnectX-6 DX dual-port 100GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- 2x 960GB M.2 NVMe OS drives (RAID 1)

9155-D14 (64 core)

- Dell R660XL
- 2x 32-core Intel Xeon 4th Gen Gold 6438N 2.0GHz 205W processors
- 1024GB RAM (16x 64GB DDR5 DIMMs)
- Optional memory upgrade to 2048 GB (32x 64GB DIMMs)
- 0-10x 7.68TB or 3.84TB PCIe Gen 5 NVMe drives
- 1x NVIDIA ConnectX-6 DX dual-port 100GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- 2x 960GB M.2 NVMe OS drives (RAID 1)

Lenovo

9155-C10 (32 core)

- Lenovo SR630 V3
- 2x 16-core Intel Gold 6426Y 2.5GHz 185W processors (“Sapphire Rapids”)
- 256GB RAM (16x 16GB DDR5 DIMMs), expandable to 512GB RAM
- 0-10x 7.68TB or 3.84TB PCIe generation 5 NVMe drives
- 1x NVIDIA ConnectX-6 DX dual-port 100GbE PCIe NIC
- 1x NVIDIA ConnectX-6 LX dual-port 25GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- 2x 960GB M.2 NVMe OS drives (RAID 1)

9155-C14 (64 core)

- Lenovo SR630 V3
- 2x 32-core Intel Gold 6438 2.0GHz 205W processors (“Sapphire Rapids”)
- 1024GB RAM (16x 64GB DDR5 DIMMs), expandable to 2048GB RAM
- 0-10x 7.68TB or 3.84TB PCIe generation 5 NVMe drives
- 1x NVIDIA ConnectX-6 DX dual-port 100GbE PCIe NIC
- 1x NVIDIA ConnectX-6 LX dual-port 25GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- 2x 960GB M.2 NVMe OS drives (RAID 1)

GPU servers

Use this section to get a detailed understanding and configuration details of GPU servers that are used in the IBM Fusion HCI appliance.

The GPU is a specialized server node in the IBM Fusion HCI appliance that is equipped with one or more graphics processing units. These nodes are designed to handle compute-intensive workloads that require massive parallel processing, such as AI and high-performance computing applications. The available GPU servers are:

- Dell hardware - 9155-DG1, DG2, DG3, and DG4. For configuration details, see Dell hardware.
- Lenovo hardware - 9155-G03 and 9155-G04. For configuration details, see Lenovo hardware.

Key capabilities

The following are the key capabilities of the GPU servers:

- Provides massive parallel processing capabilities, making them ideal for workloads that require high-performance computing, such as scientific simulations, data analytics, and AI model training.
- Improves the performance of compute-intensive workloads, reducing processing times and increasing overall system efficiency.
- GPU nodes can be virtualized, allowing multiple virtual machines to share the same GPU resources, improving resource utilization and reducing costs.

Advantages

The following are the key benefits of the GPU servers:

Improved performance and efficiency
: GPU servers can significantly improve the performance and efficiency of compute-intensive workloads.

Increased agility
: GPU servers can be quickly provisioned and deployed, allowing organizations to rapidly respond to changing business needs.

Reduced costs
: GPU servers can help reduce costs by improving resource utilization and reducing the need for specialized hardware.

Simplified management
: GPU servers are fully integrated with the IBM Fusion HCI platform, providing a single, easy-to-manage interface for administrators.

Hardware configuration

Dell
Hardware configuration of 9155-DG1 GPU server:

- Dell XE7745 4U 2-socket server
- 2x 64-core AMD EPYC 5 9575 3.3 GHz 400 W processors
- 2304 GB DDR5 6400 MT/s RAM (24x 64 GB DIMMs)
- 4x NVIDIA L40S 48 GB GPU PCIe Gen 4
- 2x 960 GB M.2 SED NVMe hot-swappable OS drives (RAID 1)
- 4x NVIDIA ConnectX-6 DX dual-port 100 GbE PCIe NICs
- Broadcom Quad Port 1 GbE PCIe Ethernet Adapter
- 8x 3200 W Titanium power supplies

Important:
The 9155-DG2, DG3, and DG4 GPU servers contain the same configuration similar to 9155-DG1, but they come up with different GPU cards configuration:

- 9155-DG2 - 8x NVIDIA PCIe Gen 4 L40S instead of 4x NVIDIA PCIe Gen 4 L40S.
- 9155-DG3 - 4x NVIDIA H200 NVL 141 GB GPU PCIe Gen 5 instead of 4x NVIDIA PCIe Gen 4 L40S.
- 9155-DG4 - 8x NVIDIA H200 NVL 141 GB GPU PCIe Gen 5 instead of 4x NVIDIA PCIe Gen 4 L40S.

Lenovo
Hardware configuration of 9155-G03 GPU server:

- SR675 V3 8DW PCIe GPU Base
- 2x AMD EPYC 9254 24C 200W 2.9GHz
- (1-8)x NVIDIA L40S 48GB GPU Gen4 PCIe adapter cards
- NVIDIA H100 NVL 94GB PCIe Gen5 Passive GPU
  Two H100 NVL can be inter-connected with NVLink to efficiently support models up to 188 GB.
- AMD MI210 PCIe GPU Adapter Card
- 768GB RAM (24x 32GB TruDDR5 4800MHz (2Rx8) RDIMM-A)
- Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- ConnectX-6 Dx dual port 100GbE network interface card
- ConnectX-6 Lx dual port 25GbE network interface card
- 2x M.2 960GB NVMe drives for the operating system and the drives are in a RAID 1 configuration
- 3U height

Important: Each G03 server supports between 1 and 8 GPU PCIe cards, with options including NVIDIA L40S, NVIDIA H100 NVL, and AMD MI210. However, you cannot mix different GPU models within the same server.
Hardware configuration of 9155-G04 GPU server:

- SR675 V3 PCIe GPU Base
- 2x AMD EPYC 9555 64-Core CPU
- (1-8)x NVIDIA H200 NVL 141GB GPU
- (1-8)x NVIDIA RTX PRO 6000 Blackwell Server Edition 96GB GPU
- 1152GB RAM (12x 96GB) and expandable to 2304 GB RAM (24x 96GB)
- Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- ConnectX-7 Dx dual port 200GbE network interface card
- ConnectX-6 Lx dual port 25GbE network interface card
- 2x M.2 960GB NVMe drives for the operating system and the drives are in a RAID 1 configuration
- 2x 3.84TB NVMe SSD
- 3U height

Important: Each G04 server supports between 1 and 8 GPU PCIe cards, with options including NVIDIA H200 NVL, and NVIDIA RTXPRO6000. However, you cannot mix different GPU models within the same server.

Network switches

The IBM Fusion HCI is configured with a dual-network architecture, designed to optimize performance, management, resiliency, and redundancy.

The IBM Fusion HCI appliance features a robust networking infrastructure, designed to provide high-speed connectivity and redundancy for optimal performance and reliability. To ensure high availability, each rack is equipped with two high-speed and management switches.

The IBM Fusion HCI appliance comes with two types of networks:

High-speed network
Used for the OpenShift® cluster, this network can be connected to the data center network. This network is dedicated to handling high-speed data transfer between the storage cluster and applications, ensuring optimal performance and low latency.

Management network
Used by IBM Fusion HCI to manage the compute, storage, and switches in the appliance. This network is used for managing the servers, monitoring their health, and performing administrative tasks, ensuring the overall system's stability and security.

Key capabilities

The following are the key capabilities of the network switches:

- High-speed connectivity enables fast data transfer and communication between nodes in the OpenShift cluster.

- Ensures that the appliance remains operational even in the event of a switch failure.
- Allows Fusion to coordinate capabilities like firmware upgrades and power operations.

Advantages

The following are the key benefits of the network switches:

Improved performance
: High-speed connectivity and redundancy ensure optimal performance and minimize downtime.

Simplified management
: Centralized management of the appliance's compute, storage, and switches streamlines operations.

Scalability
: Supports multi-rack deployments with spine switches, enabling easy expansion of the appliance.

Connectivity

100GbE high-speed network switches (Model S01 and S05)
: The high-speed network is built around a pair of 32-port, 200Gb Ethernet switches. The switches are configured together using MLAG to create a redundant pair. All of the compute-storage servers and the GPU servers have a 2-port, 100Gb Ethernet adapter. One port on the adapter is connected to the first high-speed switch and the second port is connected to the second high-speed switch. This 100GbE connections are reserved for use by the storage cluster. All of the compute-storage servers and the GPU servers also have a 2-port, 25Gb Ethernet adapter. Using breakout cables that split the 100GbE ports on the switch into four 25GbE ports, one port on the server's 25 GbE network adapter is connected to the first high-speed switch and the second port is connected to the second high-speed switch. The OpenShift network and the workload runs through this switch.

200GbE high-speed network switches (Model S05)
: The high-speed network is built around a pair of 32-port, 200Gb Ethernet switches. The switches are configured together using MLAG to create a redundant pair. All of the compute-storage servers have a 2-port, 100Gb Ethernet adapter. One port on the adapter is connected to the first high-speed switch and the second port is connected to the second high-speed switch.

1GbE management network switches (Model S04)
: The management network is built around a pair of 48-port, 1Gb Ethernet switches. The BMC of each server and the management of switches run through this 1Gb management switch.

Compute storage servers

Use this section to get a detailed understanding and configuration details of compute storage servers that are used in the IBM Fusion HCI appliance.

Compute-storage servers combine processing power with integrated storage, providing a self-contained solution for applications that require both compute and storage resources. This configuration is suitable for workloads that require low-latency access to data or have high storage demands.

The available compute servers are:

- Dell hardware - 9155-D10 and 9155-D14. For configuration details, see Dell hardware.
- Lenovo hardware - 9155-C10 and 9155-C14. For configuration details, see Lenovo hardware.

You can have either three 9155-C10/D10 and 9155-C14/D14 servers with storage as a minimum configuration, and also you can expand your IBM Fusion HCI beyond the minimum three servers by adding combinations of models C10/D10 and D10/D14. Each rack can support up to 16 servers. Each compute server can have a minimum of two storage drives that can be increased up to a maximum of ten storage drives.
Important: You cannot mix and match different drive capacities within the same rack. Choose one capacity and use it consistently throughout.

Key capabilities

The following are the key capabilities of the compute storage server:

- You can easily scaled up or down to meet changing workload demands. This scalability is achieved through the addition or removal of nodes, allowing organizations to quickly adapt to changing business needs.
- Compute storage server provides a single, unified storage pool for all workloads. This integrated storage is optimized for virtualized workloads and provides features such as deduplication, compression, and thin provisioning.
- Provides robust data protection and security features, including encryption, replication, and snapshots.
- Provides high-performance computing capabilities, with support for multiple CPU cores, high-speed memory, and low-latency storage.

Advantages

The following are the key benefits of the compute storage servers:

Reduced complexity
: Compute storage servers can simplify infrastructure management by integrating compute, storage, and networking resources into a single platform.

Improved scalability
: Easy to adapt to changing workload demands, reducing the need for costly hardware upgrades.

Increased efficiently
: The integrated storage and high-performance computing capabilities improve overall system efficiency, reducing power consumption and cooling costs.

Cost efficiency

The simplified management and scalability of compute storage servers reduce the total cost of ownership, making it an attractive solution for organizations of all sizes.

Hardware configuration

Dell

9155-D10 (32 core)

- Dell R660XL
- 2x 16-core Intel Xeon 4th Gen Gold 6426Y 2.5GHz 185W processors
- 256GB RAM (16x 16GB DDR5 DIMMs)
- Optional memory upgrade to 512GB (32x 16GB DIMMs)
- 0-10x 7.68TB or 3.84TB PCIe NVMe drives
- 1x NVIDIA ConnectX-6 DX dual-port 100GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- 2x 960GB M.2 NVMe OS drives (RAID 1)

9155-D14 (64 core)

- Dell R660XL
- 2x 32-core Intel Xeon 4th Gen Gold 6438N 2.0GHz 205W processors
- 1024GB RAM (16x 64GB DDR5 DIMMs)
- Optional memory upgrade to 2048 GB (32x 64GB DIMMs)
- 0-10x 7.68TB or 3.84TB PCIe Gen 5 NVMe drives
- 1x NVIDIA ConnectX-6 DX dual-port 100GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- 2x 960GB M.2 NVMe OS drives (RAID 1)

Lenovo

9155-C10 (32 core)

- Lenovo SR630 V3
- 2x 16-core Intel Gold 6426Y 2.5GHz 185W processors (“Sapphire Rapids”)
- 256GB RAM (16x 16GB DDR5 DIMMs), expandable to 512GB RAM
- 0-10x 7.68TB or 3.84TB PCIe generation 5 NVMe drives
- 1x NVIDIA ConnectX-6 DX dual-port 100GbE PCIe NIC
- 1x NVIDIA ConnectX-6 LX dual-port 25GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- 2x 960GB M.2 NVMe OS drives (RAID 1)

9155-C14 (64 core)

- Lenovo SR630 V3
- 2x 32-core Intel Gold 6438 2.0GHz 205W processors (“Sapphire Rapids”)
- 1024GB RAM (16x 64GB DDR5 DIMMs), expandable to 2048GB RAM
- 0-10x 7.68TB or 3.84TB PCIe generation 5 NVMe drives
- 1x NVIDIA ConnectX-6 DX dual-port 100GbE PCIe NIC
- 1x NVIDIA ConnectX-6 LX dual-port 25GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
- 2x 960GB M.2 NVMe OS drives (RAID 1)

Service node

Use this section to get a detailed understanding and configuration details of service nodes that are used in the IBM Fusion HCI appliance.

A service node is a critical component of a distributed network, responsible for providing a specific set of capabilities to the network. It acts as a bastion node for IBM Fusion HCI. The service node is pre-wired and integrated as a part of the IBM Fusion HCI with default configuration and it acts as a jump server, provides serviceability functions, and enhances overall infrastructure management. The available compute servers are:

- Dell hardware - 9155-DB1. For configuration details, see Dell hardware.
- Lenovo hardware - 9155-B01. For configuration details, see Lenovo hardware.

Key capabilities

The following are the key capabilities of the service node:

- Act as a jump server for secure remote access.
- Acts as the central control point for managing clusters, virtual machines, storage, and networking within the IBM Fusion HCI.
- Provides a secure entry point for administrators and support teams.
- Ensures system health monitoring, logging, and alerting for smooth operations.

Advantages

The following are the key benefits of the service node:

Centralized administration
: Administer nodes and switches from the service node, even when the OpenShift® Container Platform cluster or the IBM Fusion user interface is down.

Simplified troubleshooting

- Enable remote support sessions to eliminate the need for live debugging with the service node.
- Initiate remote debugging of IBM Fusion HCI problems by accessing nodes and switches through the service node, even when the IBM Fusion HCI user interface is inaccessible.

Easy log collection
Collect logs from switches and nodes using the service node, even when the IBM Fusion user interface is down.

Simplified recovery
If you need to reset IBM Fusion HCI to its original configuration, you can redeploy the OpenShift cluster from the service node.

Hardware configuration

Dell
9155-DB1

- Dell R660
- 1x 8-core Intel Xeon 4th Gen processor
- 128GB RAM (8x16GB DDR5 DIMMs)
- 2x 960GB M.2 NVMe OS drives (RAID 1)
- 1x NVIDIA ConnectX-6 LX dual-port 25GbE PCIe NIC
- 1x Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter

Lenovo
9155-B01

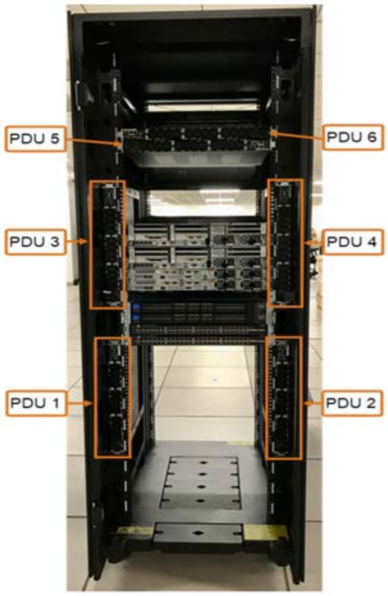
- Lenovo SR630 V3
- Intel Xeon Bronze 3408U 8C 125W 1.8 GHz processor
- 128GB RAM (8x ThinkSystem 16GB TruDDR5 4800MHz (1Rx8) RDIMM)
- ThinkSystem Intel I350 1 GbE RJ45 4-port OCP Ethernet Adapter
- ThinkSystem Mellanox ConnectX-6 Lx 10/25 GbE SFP28 2-Port PCIe Ethernet Adapter
- M.2 7450 PRO 960 GB Read Intensive NVMe PCIe 0.4 x 4 NHS SSD
- M.2 NVMe 2-Bay Raid enable kit
- 750 W 230V/115V Platinum hot-swap Gen2 power supply v3
- V3 1U x16/x16 BF PCIe Gen5 Riser 1

Power Distribution Units (PDUs)

Use this section as a guidance for PDU positions in the IBM Fusion HCI.

Additional PDUs may be added depending upon the configuration of the system. These additional PDUs get horizontally mounted in the space above PDU 6.

Figure 1. IBM Fusion HCI PDU positions



Important: There may be more than 6 PDUs for racks that have one or more GPU servers. Those servers are added in pairs, horizontally mounted, and placed on top of PDUs 5 and 6.
All the PDUs that are installed in the rack are needed and all of them must be connected to power.

Note: For independent or redundant power feeds, the two power sources must be split between the left and lower (odd numbered) PDUs and the right and upper (even numbered) PDUs.
There are two possible models of the PDU and the one you get depends on your power connection:

- For all single-phase power and for wye-wired three-phase power, there is one PDU feature code. See *Supported PDU power cords are ECJN with Souriau inlet* table in Supported PDU power cords.
- For delta-wired three-phase power (typically used only in N America), there is a different PDU feature code. See *Supported PDU power cords for PDU feature codes ECJQ with Amphenol inlet* table in Supported PDU power cords.

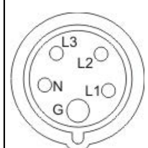
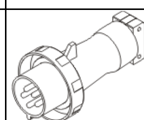
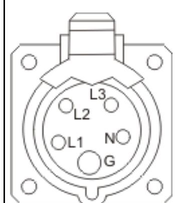
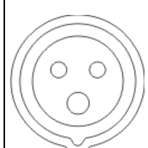
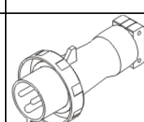
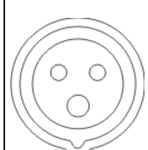
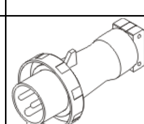
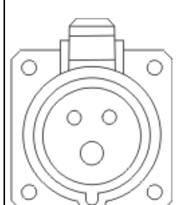
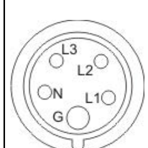
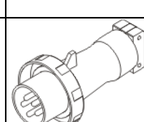
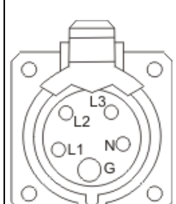
For more information about the line cords for power connections, see Supported PDU power cords. For more information about power prerequisites for IBM Fusion, see General power information.
For more information about power distribution in IBM Fusion HCI MTM (Gen 1), see Hardware overview of a stand-alone IBM Fusion HCI.

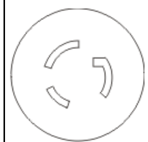
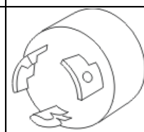
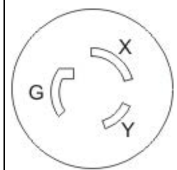
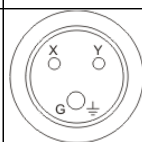
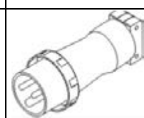
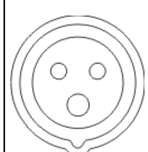
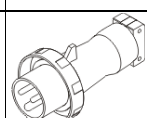
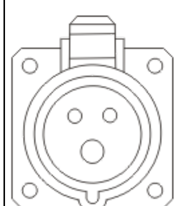
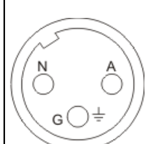
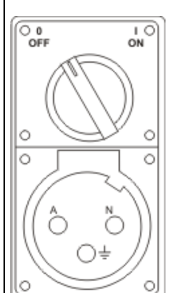
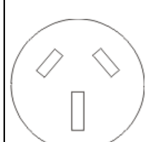
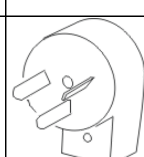
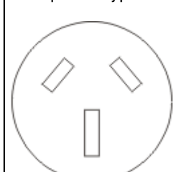
Supported PDU power cords

Find out which power distribution unit (PDU) power cords are supported for your system.

Use the following table to determine the appropriate PDU power cord to use with your system in your country.

Table 1. Supported PDU power cords for PDU feature code ECJN with Souriau inlet and ECJJ PDUs

Feature code (FC)	Description • Voltage • Amperage • Phase • Length • Wall plug	IBM® shipped plug	View of plug	Matched female connector (on cord)	Matched female wall receptacle (on wall)	IBM part number	Countries
6489	Power cord, PDU to wall • 230 V ac output • 32 A • Three-phase wye • 4.3 m (14 ft) • IEC 309, 3P+N+G	Plug type 532P6W 		Connector type 532C6W	Receptacle type 532R6W 	39M5413	Europe, Middle East, Africa (EMEA), China, Asia Pacific
6491	Power cord, PDU to wall • 230 V ac • 63 A • Single phase[1] • 4.3 m (14 ft) • IEC 309, P+N+G	Plug type 363P6W 		Connector type 363C6W	Receptacle type 363P6W	39M5415	Europe, Middle East, Africa (EMEA), China, Asia Pacific
6492	Power cord, PDU to wall • 200 - 208 V ac or 240 V ac • 60 A plug (48 A derated) • Single phase[1] • 4.3 m (14 ft) • IEC 309, 2P+G	Plug type 360P6W 		Connector type 360C6W	Receptacle type 360P6W 	39M5417	United States, Canada, Latin America, Japan, and Taiwan
6653	Power cord, PDU to wall • 230 V ac output • 16 A • 3-phase wye • 4.3 m (14 ft) • IEC 309, 3P+N+G	Plug type 516P6W 		Connector type 516C6W	Receptacle type 516R6W 	39M5412	Switzerland

Feature code (FC)	**Description** • Voltage • Amperage • Phase • Length • Wall plug	IBM® shipped plug	View of plug	Matched female connector (on cord)	Matched female wall receptacle (on wall)	IBM part number	Countries
6654	Power cord, PDU to wall • 200 - 208 V ac or 240 V ac • 30 A plug (24 A derated) • Single phase[1] • 4.3 m (14 ft) • NEMA L6-30	Plug type NEMA L6-30P 			Receptacle type NEMA L6-30R 	39M5416	United States, Canada, Latin America, Japan, and Taiwan
6655	Power cord, PDU to wall • 200 - 208 V ac or 240 V ac • 30 A plug (24 A derated) • Single phase[1] • 4.3 m (14 ft) • RS 3750DP (Watertight)					39M5418	United States, Canada, Latin America, Japan, and Taiwan
6656	Power cord, PDU to wall • 230 V ac • 32 A • Single phase[1] • 4.3 m (14 ft) • IEC 309, P+N+G	Plug type 60309 		Connector type 60309	Receptacle type 60309 	39M5414	Europe, Middle East, Africa (EMEA), China, Asia Pacific
6657	Power cord, PDU to wall • 230 - 240 V ac • 32 A • Single phase[1] • 4.3 m (14 ft) • PDL	Plug type 56P332 		Connector type 56P332	Receptacle type 56CV332 	39M5419	Australia and New Zealand
6658	Power cord, PDU to wall • 220 V ac • 30 A plug (24 A derated) • Single phase[1] • 4.3 m (14 ft) • Korean plug SJ-P3302	Plug type KP 32A 		Connector type KP	Receptacle type KP 	39M5420	South Korea

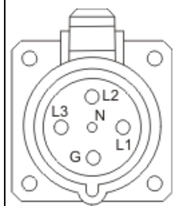
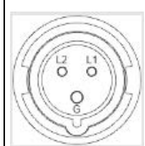
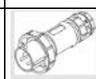
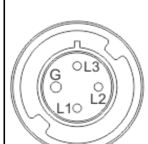
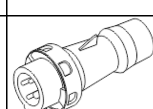
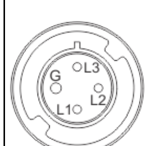
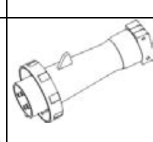
Feature code (FC)	Description • Voltage • Amperage • Phase • Length • Wall plug	IBM® shipped plug	View of plug	Matched female connector (on cord)	Matched female wall receptacle (on wall)	IBM part number	Countries
6667	Power cord, PDU to wall • 230 - 240 V ac output • 32 A • 3-phase wye • 4.3 m (14 ft) • PDL 56P532	Plug type 56P532 		Connector type 56P532	Receptacle type 56P532 	69Y1619	Australia and New Zealand
ELC1	Power cord, PDU to wall • 250 V ac output • 30 - 32 A • Single phase • 4.3 m (14 ft) • IEC 309, P+N+G	Plug type 330P6W 		Connector type 330C6W	Receptacle type 330R6W	03JJ281	United States, Canada, Mexico, and Japan
ELC2	Power cord, PDU to wall • 415 V ac output • 30 - 32 A • 3-phase wye • 4.3 m (14 ft)	Plug type 530P6W 		Connector type 530C6W	Receptacle type 530R6W	03FM499	United States, Canada, Mexico, and Japan
Note: The single phase wiring is line-to-line and the expected voltage input range is 200 - 240 V AC.							

Table 2. Supported PDU power cords for PDU feature code ECJQ with Amphenol inlet and ECJL PDUs

Feature code (FC)	Description • Voltage • Amperag e • Phase • Length • Wall plug	IBM shipped plug	View of plug	Matched female connector (on cord)	Matched female wall receptacle (on wall)	IBM part number	Countries
ECJ5	• 200 - 240 V ac • 24 A • Three phase delta • 4.3 m (14 ft) • IEC 309, 3P+N+G	Plug type 430P9W 		Connector type 430C9W	Receptacle type 430R9W	02WN660	United States, Canada, Latin America, Japan, and Taiwan
ECJ7	• 200 - 240 V ac • 48 A • Three phase delta • 4.3 m (14 ft) • IEC 309, 3P+G	Plug type 460P9W 		Connector type 460C9W	Receptacle type 460R9W	02WN658	United States, Canada, Latin America, Japan, and Taiwan

Power consumption specification for client-provided PDUs

The following power consumption information for IBM Fusion HCI components is provided as a planning aid for you to connect these components to PDUs that you provide.

Dell hardware

The following lists the power consumption information for Dell hardware components:

Product	kVA	Amps	Power Inlets	Inlet	Watts
9155-DB1 service node	0.259	1.29	2	C14	251
9155-D10 with 256GB RAM, no NVMe drives	0.555	2.77	2	C14	538
9155-D10 with 256GB RAM, 10x NVMe drives	0.709	3.55	2	C14	688
9155-D10 with 512GB server, no NVMe drives	0.575	2.88	2	C14	558
9155-D10 with 512GB server, 10x NVMe drives	0.730	3.65	2	C14	708
9155-D14 with 1024GB RAM, no NVMe drives	0.651	3.25	2	C14	631
9155-C14 with 1024GB RAM, 10x NVMe drives	0.805	4.03	2	C14	781
9155-C14 with 2048GB RAM, no NVMe drives	0.690	3.45	2	C14	669
9155-C14 with 2048GB RAM, 10x NVMe drives	0.844	4.22	2	C14	819
9155-DG1 with 4x L40S	4.867	24.34	8	C20	4721
9155-DG2 with 8x L40S	6.320	31.60	8	C20	6130
9155-DG3 with 4x H200	5.481	27.41	8	C20	5317
9155-DG4 with 8x H200	8.639	43.20	8	C20	8380
9155-TF5 pull out console	0.051	0.26	1	C14	36
9155-S04 1 GbE management switch	0.105	0.53	2	C14	100
9155-S05 200 GbE high-speed switch	0.263	1.32	2	C14	250

Lenovo hardware

The following lists the power consumption information for Lenovo hardware components:

Product	kVA	Amps	Power Inlets	Inlet	Watts
9155-C10 with 256GB RAM, no NVMe drives	0.725	3.62	2	C14	703
9155-C10 with 256GB RAM, 10x NVMe drives	0.776	3.88	2	C14	753
9155-C10 with 512GB server, no NVMe drives	0.818	4.09	2	C14	793
9155-C10 with 512GB server, 10x NVMe drives	0.869	4.35	2	C14	843
9155-C14 with 1024GB RAM, no NVMe drives	0.794	3.97	2	C14	770
9155-C14 with 1024GB RAM, 10x NVMe drives	0.845	4.23	2	C14	820
9155-C14 with 2048GB RAM, no NVMe drives	0.913	4.57	2	C14	886
9155-C14 with 2048GB RAM, 10x NVMe drives	0.965	4.82	2	C14	936
9155-TF5	0.051	0.26	1	C14	36
9155-C00	0.628	3.14	2	C14	609
9155-C00 with 512GB RAM	0.703	3.52	2	C14	682
9155-C00 with 1024GB RAM	0.752	3.76	2	C14	729
9155-C01	0.679	3.40	2	C14	659
9155-C01 with 512GB RAM	0.755	3.77	2	C14	732
9155-C01 with 1024GB RAM	0.803	4.02	2	C14	779
9155-C04	0.849	4.25	2	C14	824
9155-C04 with 2048GB RAM	0.869	4.35	2	C14	843
9155-C05	0.897	4.48	2	C14	870
9155-C05 with 2048GB RAM	0.921	4.60	2	C14	893
9155-F01	0.539	2.70	2	C14	534
9155-G01	1.487	7.43	2	C14	1442
9155-G02 Standard	1.771	8.86	2	C14	1,718
9155-G03	3.392	16.96	4	C20	3290
9155-G03 with 2 H100 NVL GPUs	1.742	8.71	4	C20	1690
9155-G03 with 4 H100 NVL GPUs	2.402	12.01	4	C20	2330
9155-G03 with 8 H100 NVL GPUs	3.722	18.61	4	C20	3610
9155-G03 with 2x L40S GPUs	1.66	8.3	4	C20	1610
9155-G03 with 4x L40S GPUs	2.237	11.19	4	C20	2170
9155-G03 with 8x L40S GPUs	3.392	16.96	4	C20	3290
9155-G04 with 2x H200 NVL GPU servers with 1152 GB	2.488	12.44	4	C20	2413
9155-G04 with 2x H200 NVL GPU servers with 2304 GB	2.605	13.03	4	C20	2527
9155-G04 with 4x H200 NVL GPU servers with 1152 GB	3.686	18.43	4	C20	3575
9155-G04 with 4x H200 NVL GPU servers with 2304 GB	3.799	18.99	4	C20	3685
9155-G04 with 8x H200 NVL GPU servers with 1152 GB	6.064	30.32	4	C20	5882
9155-G04 with 8x H200 NVL GPU servers with 2304 GB	6.179	30.90	4	C20	5994
9155-G04 with 2x RTX Pro 6000 GPU servers with 1152GB	2.336	11.68	4	C20	2266
9155-G04 with 2x RTX Pro 6000 GPU servers with 2304GB	2.452	12.26	4	C20	2378

Product	kVA	Amps	Power Inlets	Inlet	Watts
9155-G04 with 4x RTX Pro 6000 GPU servers with 1152GB	3.384	16.92	4	C20	3282
9155-G04 with 4x RTX Pro 6000 GPU servers with 2304GB	3.501	17.51	4	C20	3396
9155-G04 with 8x RTX Pro 6000 GPU servers with 1152GB	5.451	27.25	4	C20	5287
9155-G04 with 8x RTX Pro 6000 GPU servers with 2304GB	5.561	27.80	4	C20	5394
9155-S01	0.255	1.27	2	C14	242
9155-S02	0.097	0.48	2	C14	92
9155-S04	0.105	0.53	2	C14	100
9155-S05	0.263	1.32	2	C14	250

Power consumption by configuration

Before you up size the nodes, consider the power consumption data that is depicted in the following *Power consumption* tables.

Table 1. 32 Core Servers with 256GB RAM, 10 NVMe Drives per Gen 2 Compute/Storage Node

Number of Compute or Storage nodes	6	8	10	12	14	16
Power consumption (kW)	5.25	6.76	8.27	9.77	11.28	12.78
Power Consumption with 1xG03 H100 NVL GPU (kW)	8.86	10.37	11.88	13.38	14.89	n/a
Power Consumption with 2xG03 H100 NVL GPU (kW)	12.47	13.98	15.49	16.99	18.50	n/a
Power Consumption with 3xG03 H100 NVL GPU (kW)	16.08	17.59	19.10	20.60	n/a	n/a
Power Consumption with 4xG03 H100 NVL GPU (kW)	19.69	21.20	22.71	24.21	n/a	n/a
Power Consumption with 1xG04 H200 NVL GPU (kW)	11.14	12.64	14.15	15.65	17.16	NA
Power Consumption with 2xG04 H200 NVL GPU (kW)	17.02	18.52	20.03	21.54	23.04	n/a
Power Consumption with 3xG04 H200 NVL GPU (kW)	22.90	24.41	25.91	27.42	n/a	n/a
Power Consumption with 4xG04 H200 NVL GPU (kW)	28.78	30.29	31.79	33.3.0	n/a	n/a
Power Consumption with 1xG04 RTX Pro 6000 GPU (kW)	10.54	12.05	13.55	15.06	16.57	n/a
Power Consumption with 2xG04 RTX Pro 6000 GPU (kW)	15.83	17.33	18.84	20.35	21.85	n/a
Power Consumption with 3xG04 RTX Pro 6000 GPU (kW)	21.12	22.62	24.13	25.63	n/a	n/a
Power Consumption with 4xG04 RTX Pro 6000 GPU (kW)	26.40	27.91	29.41	30.92	n/a	n/a

Note: These values assume 8x L40S GPUs in each G03 server. Reduce by 0.28 kW for each L40S that is less than the maximum.

Table 2. 64 Core Servers with 1024 GB RAM, 10 NVMe Drives per Gen 2 Compute/Storage Node and 32 core control plane servers

Number of Compute or Storage nodes	6	8	10	12	14	16
Power consumption (kW)	5.46	7.10	8.74	10.38	12.02	13.66
Power Consumption with 1xG03 GPU (kW)	9.07	10.71	12.35	13.99	15.63	n/a
Power Consumption with 2xG03 GPU (kW)	12.68	14.32	15.96	21.21	19.24	n/a
Power Consumption with 3xG03 GPU (kW)	16.29	17.93	19.57	20.21	n/a	n/a
Power Consumption with 4xG03 GPU (kW)	19.90	21.54	23.18	24.82	n/a	n/a
Power Consumption with 1xG04 H200 NVL GPU (kW)	11.34	12.98	14.62	16.26	17.90	n/a
Power Consumption with 2xG04 H200 NVL GPU (kW)	17.22	18.86	20.50	22.14	23.78	n/a
Power Consumption with 3xG04 H200 NVL GPU (kW)	23.10	24.74	26.38	28.02	NA	n/a
Power Consumption with 4xG04 H200 NVL GPU (kW)	28.98	30.62	32.26	33.90	n/a	n/a
Power Consumption with 1xG04 RTX Pro 6000 GPU (kW)	10.74	12.38	14.02	15.66	17.30	n/a
Power Consumption with 2xG04 RTX Pro 6000 GPU (kW)	16.03	17.67	19.31	20.95	22.59	n/a
Power Consumption with 3xG04 RTX Pro 6000 GPU (kW)	21.32	22.96	24.60	26.24	n/a	n/a
Power Consumption with 4xG04 RTX Pro 6000 GPU (kW)	26.60	28.24	29.88	31.52	n/a	n/a

Note: These values assume 8x L40S GPUs in each G03 server. Reduce by 0.28 kW for each L40S that is less than the maximum.

Deployment configurations

The stand-alone rack, multiple rack, and disaster recovery deployment configurations are available in IBM Fusion HCI. The multi rack topology offers these variants: high-availability multi-rack, 3 zone high-availability multi-rack, two availability zones with an arbiter node in a third zone, and expansion rack.

Stand-alone rack topology

The stand-alone rack of IBM Fusion HCI consists of management switches, high speed switches, PDUs, storage nodes, compute nodes, and GPU nodes to form a single OpenShift® cluster. For a single rack setup, you can simplify the deployment and begin with three nodes.

You can upsize the node to scale up the capacity of a stand-alone rack. It does not have fault tolerance for a complete rack failure.

For more information about each of the components of a Stand-alone rack, see [Hardware overview](#).

High-availability multi-rack and 3 zone HA rack topology

IBM Fusion HCI comes with the bootstrapping software that is installed from the factory for installing Fusion HCI in data centre. For high available multi rack, the IBM support representative completes the initial verification and physically connects a minimum of three racks to network and power. Then, they conduct the network setup of the first two racks which are called auxiliary racks. Network setup of 3rd, also known as last rack, is done after first two racks. This network setup validates hardware and wiring of rack and connects the rack to the data centre network. This procedure configures all the default nodes. During network setup of last rack, software collects network configuration details from first two racks automatically and shows consolidated appliance view. After the network setup is complete for all three racks, the control plane is created across the three racks. In this setup, the cluster remains operational even when a rack fails, which ensures resilience and continuity. The IBM SSR provides stage 2 URL to customer to continue next phase of the installation.

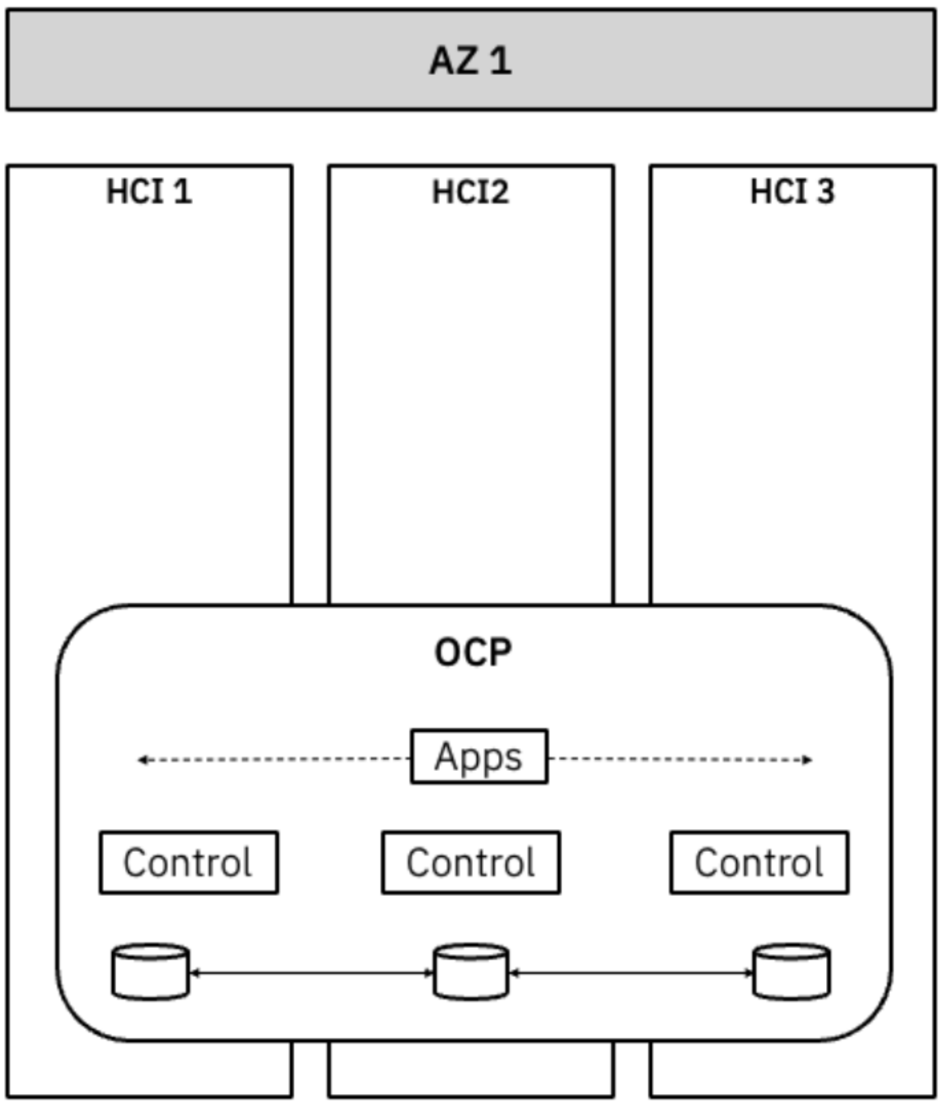
In high-availability multi-rack, at installation time, you can connect three IBM Fusion HCI racks to create a single large OpenShift Container Platform cluster. The control nodes are distributed across multiple racks for fault tolerance. This topology also spreads Fusion Data Foundation replicas across the racks to withstand the loss an entire rack of infrastructure without impacting the availability of the platform.
Note: Order three similar racks from IBM factory with a minimum of six storage nodes per rack.
IBM Fusion HCI provides flexible rack configuration within availability zones, allowing precise control over failure domains. You have the option of setting up high-availability multi-rack in two different ways. When you select an availability zone, you can choose to deploy all the racks within a single physical zone (three virtual zones) or distribute them across three physical availability zones. Spreading deployments across different physical locations enhances resilience by ensuring physical isolation, with separate power and networking infrastructure. This guarantees that Fusion remains operational even if one zone experiences an outage.

Single Zone HA

The OpenShift control plane is distributed across three racks. The Fusion Data Foundation replicas are distributed across three racks containing storage nodes.

The following diagram shows a single physical availability zone. In this setup, IBM Fusion HCI provides failure domain as a rack.

Figure 1. Three failure domains within a single physical availability zone



3-Zone HA

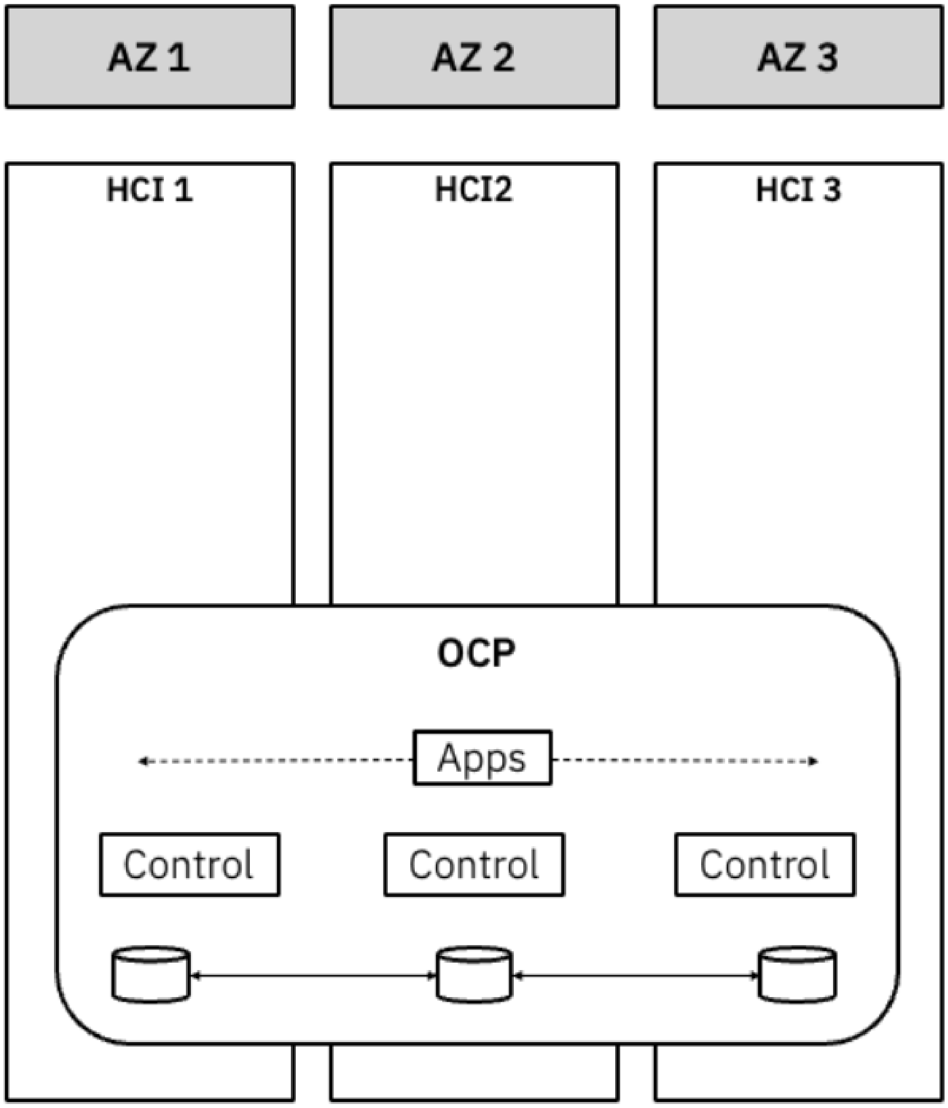
IBM Fusion HCI supports 3 zone HA where the racks are distributed across distinct locations within a region that are designed to be isolated from failures occurring in other zones. For 3 zone HA, each rack must be connected to the customer switches in such a way that they are in the same broadcast domain.

In a high-availability multi-rack or three Availability Zone (AZ) deployment, each node is automatically assigned a distinct zone label. This ensures that all nodes within a single rack share the same zone label, effectively creating three distinct zones for load distribution.

Deploy racks in each of the two or three availability zones to achieve a three AZ deployment of IBM Fusion HCI. The OpenShift control plane is distributed across racks in each of the three AZs. Fusion Data Foundation replicas are distributed across storage nodes in each AZ. Storage nodes are placed in the same rack as the control nodes. Additional racks of compute or GPU only servers provide additional worker nodes, and are placed in each AZ.

The following diagram shows how the OpenShift control plane and storage replicas are stretched across racks in separate availability zones.

Figure 2. OpenShift control plane and storage replicas are stretched across racks in separate availability zones



For both high-availability multi-rack and 3-Zone HA, Fusion Data Foundation is the only supported storage type. The Fusion Data Foundation storage spreads its resources in these three racks so that you do not lose data when a rack goes down either because of maintenance or power failure. The three side-by-side (adjacent) appliances act as a single unit and hosts a single Fusion Data Foundation storage and OpenShift Container Platform cluster. To achieve high availability, 1 control node (Rack Unit 2) is available in each of these three racks so even when one rack goes down, the cluster stays available. The components in each of the three racks are the same as a single rack.

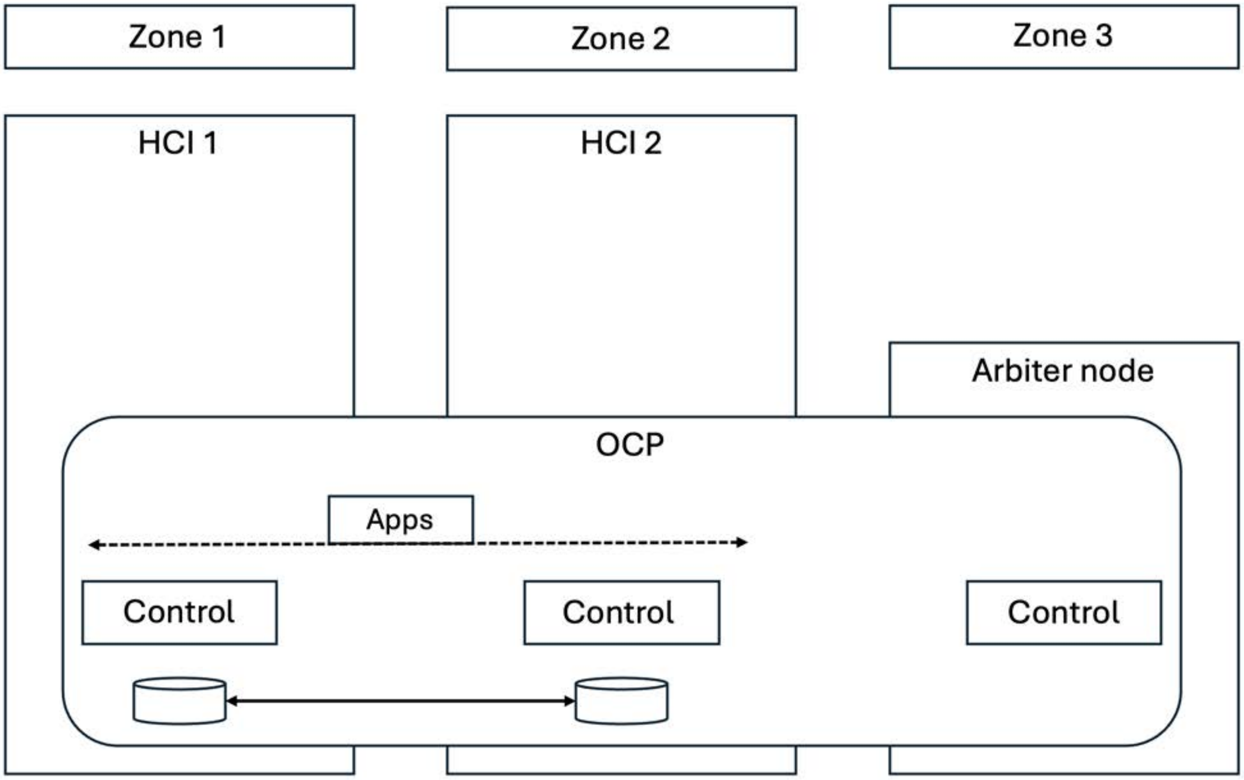
The service node can be made available in any one of the racks. The rack to which the service node is connected is designated and used as the base rack.

For more information about component details and hardware configuration, see Hardware overview. For more information about the storage configuration for multi-rack, see Fusion Data Foundation storage compatibility across rack configurations. For more information about the known issues and limitations in multi-rack, see Multi-rack issues.
Important: The high-availability multi-rack is only supported with Fusion Data Foundation storage 4.16 and higher.

2AZ and arbiter node topology

IBM Fusion HCI supports a two-availability-zone and arbiter node (2AZ + arbiter node) configuration, in which two racks are deployed across separate availability zones within a region, while an arbiter node (or control node) is deployed in a third zone. This configuration ensures that one zone or rack can remain isolated if a failure occurs in another zone. The OpenShift Container Platform control plane is formed using the control node of each rack and the arbiter node, ensuring high availability for the OpenShift Container Platform cluster.

Figure 3. 2AZ + arbiter node topology



Fusion Data Foundation is the only supported storage type for the 2AZ + arbiter node configuration. Fusion Data Foundation distributes its resources across the two racks to ensure that no data is lost if a rack goes down due to maintenance or a power failure. For more information, see [2AZ and arbiter node configuration with Fusion Data Foundation].

The 2AZ + arbiter node topology can be deployed in a metropolitan-spanned data centers where latency between servers stays within specific limits. For more information on network requirements and planning, see [Network for two-availability-zone and arbiter node (2AZ + arbiter node) configuration].

Expansion rack topology

The expansion rack extends the capacity of an existing running cluster, which can be stand-alone system, high-availability multi-rack setup, or a previously expanded configuration. Expansion allows you to start with just one node. For Fusion Data Foundation based topology, all the networks are connected to the customer switch.

Disaster recovery topology

IBM Fusion HCI supports disaster recovery (DR) through two primary configurations: Metro-DR and Regional-DR.

The Metro-DR topology provides synchronous data replication between two IBM Fusion HCI clusters located within metropolitan distances (with latency under 40 milliseconds). Global Data Platform is the only supported storage type for the Metro-DR topology. For information on Metro-DR topology configuration, see Metro-DR configuration for Global Data Platform.

The Regional-DR topology uses asynchronous data replication between clusters located at long distances or two geographically separated clusters. While it introduces a slight delay in data synchronization, it is ideal for scenarios where geographical separation is necessary for disaster recovery. Fusion Data Foundation is the only supported storage type for the Regional-DR topology. For information on Regional-DR topology configuration, see Regional-DR configuration for Fusion Data Foundation.

IBM Storage Expert Care

IBM Storage Expert Care is a simplified method of selecting services and support for systems at the time of the purchase. IBM Storage Expert Care is designed to simplify and standardize the support approach for IBM Fusion HCIwith simple straightforward pricing and selection of services.

For more information about IBM Storage Expert Care for IBM Fusion, see Support reference guide.

Security in IBM Fusion

IBM Fusion is a secure platform to deploy your applications.

The following security measures are followed for IBM Fusion product:

- Penetration testing to check for exploitable vulnerabilities
- Threat modelling and threat assessment to ensure that threats are assessed and resolved
- Static scan to maintain high code quality and to ensure that no security vulnerabilities are left in the code
- Dynamic scan to detect and remove runtime vulnerabilities

- Open Source Scan to detect and remove open source vulnerabilities in open source libraries that are used by IBM Fusion
- Security and Privacy by Design (SPbD) compliance certified by IBM BISO to ensure security and privacy compliance
- Container Software certifications (IBM Certification and Red Hat Certifications) to ensure adherence to container security standards.
- Automated password rotation in IBM Fusion HCI ensures hardware (nodes) credentials are regularly updated to maintain security compliance, reduce unauthorized access risk, and eliminate manual intervention. For more information about enabling password rotation, see Enabling password rotation for IBM Fusion HCI system.

Verifying image signatures

Digital signatures provide a way for consumers of content to ensure that what they download is both authentic (it originated from the expected source) and has integrity (it is what we expect it to be). All images for IBM Fusion are signed. This page describes how to verify the signatures on those images.

Before you begin

Your machine must have these command line tools installed (they can usually be installed on Linux using the package manager):

- GNU Privacy Guard v2
- OpenSSL
- skopeo

The IBM Fusion public key must exist on the same machine. Copy the following into a text editor, and save it in a file named **storage-fusion.pub.asc**:

```
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQINBGPOH0IBEADA253RAyBkEFFO7d9gfu+hdG58qQUe2H+4m5MuLc0zy/ulhkGg
YGBWFBCutkMp9F/jvtTrhf8r0oSlrIIzyJHFpM6qnlLftkbzQlpaj8yT+aPpNgVm
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C2KBy/YWmpWaMAeqNLRRYcHCUb9HZCfWchPlywXWIIggjL3D8vrPePWITpblC4wF
Tl6utwKHKM1CUqjn20j6q2Mh/7Z7g5XwaqmtoDm8cpx1bRY2ChHHxPXzbQARAQAB
tCpJQk0gU3BlY3RydW0gRnVzaW9uIEhDSSA8cHNpcnRAdXMuaWJtLmNvbT6JAjoE
EwEIACQFAmPOH0ICGw8FCwkIBwIGFQoJCAsCBBYCAwECHgEFCQPCZv8ACgkQHS4H
jVltgQNUmQ/+J9xUtUs1PwzAB/LpoqHGOmyZqz6EmQbaRiPzD4VSxcTYbpSKVb07
5H1sVR0QgDR9OUt5cbc4d9NBtFGiY8UeiCSL+MPjy5ksjFmbsxLRKEYZjB/qiOHv
c8viMH4y0boquomjeqGL1qajKyUvG1h99S5osGJVmPa2tmxMhTai57+aDX5SAKS4
RO1fd52K4nZw5+iqy95fXoEkPNwdw6Xm2fZ10F+kZ2dS8y+gu45hEaa/h+3P2myJ
37mp+336cfrqmqIv0pXD64et8zsQRNYqr2QEUSNsJ7ok4RkRqlpku5RYTV7T7TqH
g88rLAY1vxw4RzkQMve+t1IqgLANG9A+/fbQUQ/yshjCmvqmzB54VrOjOIwMWVtL
J+u1jbrdX1kFPogei6c0v3CBNe1khO6+1amBPTazKONbuMhBvaseX16x4Mk5r0FS
4FLNDEjktk4qUQAkEZk/Sugozp6oQpMYmKxPIt/7snLL9e6naF98TosKrpUCkme3
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Wn3d9bGePGCMwPNwn3Auo8w/qMvHNAHykwTq1T1r/1vrmOnx+EEar2J2ZXqoA82P
4hcXWG4auoSA3y9tjAVLWChZdawFcMqcHZlHxER9/GOnY3hP+WSmv8o=
=YUDL
-----END PGP PUBLIC KEY BLOCK-----
```

You must have a list of images to verify. To get a list of container images used, see the procedure in Downloading container images.

In this procedure, the following example image is used: **icr.io/cpopen/isf-operator-catalog:2.13.0-linux.amd64**.

Procedure

1. Run the following command to delete all previous IBM Fusion public keys:

   ```
   sudo gpg2 --delete-key "Fusion"
   ```

   To confirm the key deletion, enter 'Y'. If you have previous IBM Fusion public keys, repeat this step until gpg2 reports:

   ```
   gpg: key "Fusion" not found: Not found
   gpg: Fusion: delete key failed: Not found
   ```

2. Import the IBM Fusion public key on the machine that you prepared according to the *Before you begin* section:

   ```
   sudo gpg2 --import storage-fusion.pub.asc
   ```

   Note: This step must be done only once on each machine you use for signature verification.
3. Calculate the fingerprint:

   ```
   fingerprint=$(sudo gpg2 --fingerprint --with-colons "Fusion" | grep fpr | tr -d 'fpr:')
   ```

   This command stores the key's fingerprint in an environment variable that is called **fingerprint**, which is needed for the command to verify the signature. When you exit your shell session, the variable gets deleted. The next time that you log in to your machine, you can set it again by rerunning the command.

4. Create a directory for the image and use **skopeo** to pull it into local storage:

   ```
   mkdir images
   FUSION_VERSION=<version of Fusion being installed/verified>
   skopeo copy docker://icr.io/cpopen/isf-operator-catalog:${FUSION_VERSION}-linux.amd64 dir:./images
   ```

For example, to check the signatures on 2.13.0:

```
mkdir images
FUSION_VERSION=2.13.0
skopeo copy docker://icr.io/cpopen/isf-operator-catalog:${FUSION_VERSION}-linux.amd64 dir:./images
```

This command downloads the image as a set of files and places them in the `images` directory (or another directory that you choose).
Note: There is a manifest file that is named `images/manifest.json`, and a signature file named `images/signature-1`. You reference both these files in the next step (in the command to verify the signature).
Log in to the `icr.io` entitlement registry to pull the image. Alternatively, you can pass your credentials directly to `skopeo` with `--src-creds iamapikey:<YOUR ENTITLEMENT KEY>`.

5. Verify the signature:

```
FUSION_VERSION=<version of Fusion being installed/verified>
sudo skopeo standalone-verify ./images/manifest.json icr.io/cpopen/isf-operator-catalog:${FUSION_VERSION}-linux.amd64 
${fingerprint} ./images/signature-1
```

For example, to check the signatures on 2.13.0:

```
# FUSION_VERSION=2.13.0
sudo skopeo standalone-verify ./images/manifest.json icr.io/cpopen/isf-operator-catalog:${FUSION_VERSION}-linux.amd64 
${fingerprint} ./images/signature-1
```

Installing

This topic describes the IBM Fusion HCI planning and installation. It also includes prerequisites that you need for a successful setup and installation of IBM Fusion HCI.

The IBM Fusion HCI comes with a bootstrapping software from the factory. The IBM Service Support Representatives (SSR) complete the initial verification and physically connect the system to network and power. Then, they setup the network of the appliance, which connects the appliance to the data centre network. This procedure configures network for all the appliance switches and nodes (three controllers). If you ordered more nodes, then they get installed as well. After the network setup is complete, you receive a link to install IBM Fusion HCI. The installation process includes setting up a three-node Red Hat® OpenShift® cluster, followed by installing IBM Fusion software.

The IBM Fusion HCI installation wizard provides a seamless and accelerated experience to get the system up and running with a few clicks. It progressively gathers the information in the following stages of IBM Fusion HCI installation:

1. In the **Network Setup** wizard stage (stage 1), the SSR, in collaboration with your network administrator, configures switches, VLANs, aggregating links, and validates DHCP and DNS configuration. Your NTP server is also configured on each node as a hardware clock.
2. In the **Software installation** wizard stage (stage 2), you can continue with the installation from a remote host by using the details that are provided at the end of the previous stage (Network setup). In this stage, you accept the license agreement contracts, provide the image registry details, optionally customize the Red Hat OpenShift and storage network details, and optionally configure a custom certificate for OpenShift Ingress. At the end of this stage, IBM Fusion HCI completes the installation of three nodes OpenShift cluster and management software.

Post the installation of stages, you can do the following next steps:

1. Proceed to add more nodes to the cluster.
2. Configure Fusion Data Foundation or Global Data Platform storage on the cluster.
3. Install IBM Fusion services of your choice.

If you want to bring your own rack, contact IBM Support.

- **Planning and prerequisites**
  Thorough planning ensures that you have everything to meet all the prerequisites for a successful setup and installation of the system.
- **Installing IBM Fusion HCI**
  Use this procedure to install IBM Fusion HCI stage 2.
- **Quay on IBM Fusion HCI**
  How to setup Quay and S3 for Quay.
- **Validating IBM Fusion HCI installation**
  After the IBM Fusion HCI is installed, validate the success of the installation.
- **Installation issues**
  Use the troubleshooting tips and tricks in IBM Fusion installation.
- **Collecting log files of final installation**
  Three operators get involved in the final installation. The `Installer` operator adds compute nodes to the OpenShift Container Platform cluster, and orchestrates the overall final installation. The `Storage` operator installs the IBM Storage Scale cluster.
- **Installing services**
  In the service-based IBM Fusion HCI , choose the features that you want to deploy.
- **Uninstalling services**
  You can uninstall IBM Fusion services by using the command line.

Planning and prerequisites

Thorough planning ensures that you have everything to meet all the prerequisites for a successful setup and installation of the system.

Planning minimizes errors during installation and enables a quicker upgrade or installation. This planning information helps you integrate the system into your data center, plan power and environmental needs, and prepare for any unique configurations you need to handle.

To ensure that the planning is completed successfully, assign a planning project manager to provide a documented plan that includes the following items:

- A timeline for the activities

- Each major activity phase and the intended outcome
- A list of responsibilities and the person assigned
- A current system diagram and configuration listing. Include all hardware content, software content, cabling, and other pertinent configuration items, particularly if this is a modification to an existing system
- A key contacts list, including off-hours contact information for all key task or activities participants
- A plan for communicating appropriate elements of your plan with key personnel.

- **IBM Fusion HCI rack ordering options**
  The IBM Fusion HCI offers two hardware configuration options that you can order based on your requirements: a rack-based system and a client provided rack.
- **Blueprints and deployment guidance**
  IBM Fusion is an application platform designed for running containers, virtual machines, and AI workloads. It has a set of blueprints that provide guidance on how to deploy different types of workloads. The blueprints provide guidance on topics such as resiliency, performance, and data services. They provide t-shirt sizes and guidance on how to scale out.
- **Site-readiness**
  The selection of a site for information technology equipment is the first consideration in planning and preparing for the installation.
- **System requirements**
  System requirements provide the details of the resources allocation and management of the IBM Fusion HCI and its services to help ensure efficient and effective operation.
- **Choosing your storage type**
  Guidance to choose between Fusion Data Foundation and IBM Storage Scale storage.
- **Networking**
  This topic provides a high-level overview of the network architecture of the IBM Fusion HCI.
- **General Metro-DR prerequisites**
  Prerequisites for Metro-DR between two OpenShift® Container Platform clusters.
- **Converting a rack from online proxy to offline using external Quay**
  Use the following instructions to convert a rack from online proxy to offline using external quay on the IBM Fusion HCI cluster.
- **Enterprise registry for IBM Fusion HCI installation**
  During the installation process, IBM Fusion HCI installs Red Hat® OpenShift Container Platform and IBM Fusion software using images that are hosted in the Red Hat and IBM entitled registries. If you want to use your enterprise registry, you can install both OpenShift and IBM Fusion HCI software from images that are maintained in a container registry that you manage.
- **Converting a rack from internal registry to offline using enterprise registry**
  Use the following instructions to convert a rack from internal registry to offline using enterprise registry on the IBM Fusion HCI cluster.
- **Activating Red Hat OpenShift Container Platform subscriptions purchased from IBM**
  If you purchased your Red Hat OpenShift Container Platform subscription from IBM, then you must activate your OpenShift subscription with Red Hat and activate OpenShift support with IBM.
- **Activating IBM Fusion HCI Software to be downloaded**
  You need IBM Fusion HCI software entitlement to access IBM Fusion HCI images that are used to install your IBM Fusion HCI appliance.
- **How to Activate NVIDIA AI Enterprise Subscriptions**
  NVIDIA H100 NVL and H200 NVL GPUs are bundled with a five-year subscription to NVIDIA AI Enterprise.
- **Installation prerequisites**
  Go through these general prerequisites and ensure that they are all met before you go ahead with your installation.

IBM Fusion HCI rack ordering options

The IBM Fusion HCI offers two hardware configuration options that you can order based on your requirements: a rack-based system and a client provided rack.

Rack-Based System

The rack based IBM Fusion HCI includes an R42 unit rack, with nodes, switches, and Power Distribution Units (PDUs) pre-stacked and pre-wired at the factory. To begin the installation, you only need to connect your data center network to the high-speed switch. This configuration ensures minimal setup effort at your premises and streamlines deployment.

Client provided rack

In this client provided rack option, you must provide the rack and PDUs. However, you can also order the PDUs from IBM. A technical assessment is essential to ensure compatibility of your rack and PDUs for seamless integration. IBM provides the nodes, switches, and network cables to connect the nodes and switches. The IBM Service Support Representatives (SSR) does the assembly at your premises.

Blueprints and deployment guidance

IBM Fusion is an application platform designed for running containers, virtual machines, and AI workloads. It has a set of blueprints that provide guidance on how to deploy different types of workloads. The blueprints provide guidance on topics such as resiliency, performance, and data services. They provide t-shirt sizes and guidance on how to scale out.

To view the blueprints documents, see the following links:

- AI Applications with IBM Fusion
- OpenShift Virtualization with IBM Fusion Blueprints
- Containerized workloads with IBM Fusion

Site-readiness

The selection of a site for information technology equipment is the first consideration in planning and preparing for the installation.

Determine whether a new site is to be constructed or alterations are to be performed on an existing site. This section provides specific information on building location, structure, and space requirements for present and future needs.

Power and communication facilities must be available in the quantities that are required for operation. If these are inadequate, contact the utility company to determine whether additional services can be made available.

Pollution, flooding, radio or radar interference, and hazards that are caused by nearby industries can cause problems to information technology equipment and recorded media. Any indication of exposure in these areas should be recognized and included in the planning of the installation.

Ensure that IBM Fusion appliance is installed in a restricted access location, such that the area is accessible only to skilled and instructed persons with proper authorization.

- **Static electricity and floor resistance**
  Use these guidelines to minimize static electricity buildup in the data center.
- **Space requirements**
  The floor area that is required for the equipment is determined by the specific systems to be installed, the location of columns, the floor loading capacity, and the provisions for future expansion.
- **Floor construction and floor loading**
  Use the following formulas to calculate the floor loads for the system.
- **Computer room location**
  Computer room location is affected by several factors, such as considerations for safety and fire prevention.
- **Vibration and shock**
  Use this information to plan for possible vibration and shock in the data center.
- **Acoustics**
  Acoustic noise emission data enables you to assess noise levels for your data processing equipment.
- **Electromagnetic compatibility**
  Use this information to plan for system installation in an environment that has a high electromagnetic-radiated field.
- **Material and data storage protection**
  Special safety considerations are required when storing data or other material.
- **General power information**
  Reliable electrical power is required for the proper functioning of your data processing equipment.
- **Air conditioning determination**
  The air conditioning system must provide year-round temperature and humidity control as a result of the heat dissipated during equipment operation.

Static electricity and floor resistance

Use these guidelines to minimize static electricity buildup in the data center.

Floor covering material can contribute to the buildup of high static electrical charges as a result of the motion of people, carts, and furniture in contact with the floor material. Abrupt discharge of the static charges causes discomfort to personnel and might cause malfunction of electronic equipment.

The following list explains how to minimize static buildup and discharge:

- Maintain the relative humidity of the room within the system operating limits. Choose a control point that normally keeps the humidity between 35 percent and 60 percent. See Air conditioning determination for more information.
- Provide a conductive path to the ground from a metallic raised floor structure including the metal panels.
- Ground the raised floor metallic support structure (stringer, pedestals) to building steel at several places within the room. The number of ground points is based on the size of the room. The larger the room, the more ground points are required.
- Ensure that the maximum resistance for the flooring system is 2 x 10^{10} ohms, measured between the floor surface and the building (or an applicable ground reference). Flooring material with a lower resistance will further decrease static buildup and discharge. For safety, the floor covering and flooring system should provide a resistance of no less than 150 kilohms when measured between any two points on the floor space 1 m (3 ft) apart.
- Maintenance of antistatic floor coverings (carpet and tile) should be in agreement with the individual supplier's recommendations. Carpeted floor coverings must meet electrical conductivity requirements. Use only antistatic materials with low-propensity ratings.
- Use ESD-resistant furniture with conductive casters to prevent static buildup.

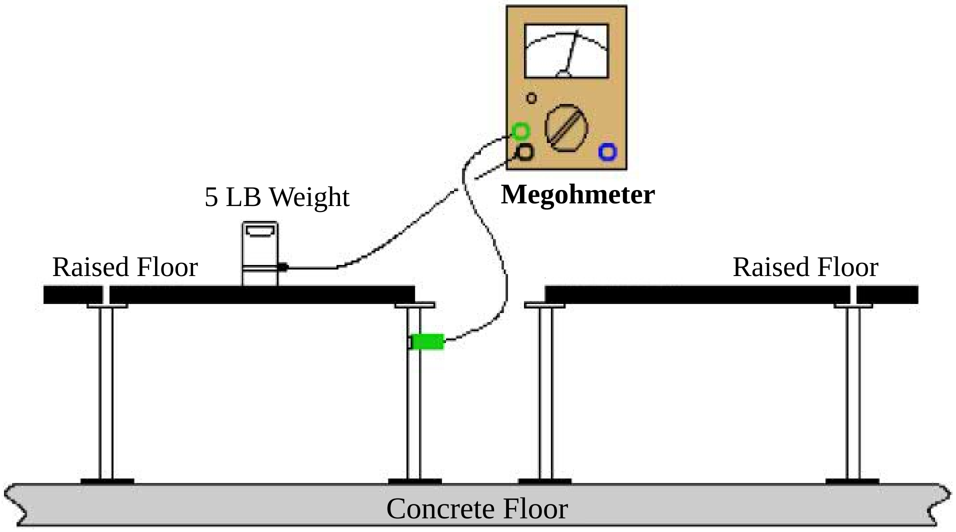
Measuring floor resistance

The following equipment is required for measuring floor resistance:

- A test instrument similar to an AEMC-1000 megohmmeter is required for measuring floor conductivity.

The following figure shows the typical test connection to measure floor conductivity.

Figure 1. Typical test connection to measure floor conductivity



Space requirements

The floor area that is required for the equipment is determined by the specific systems to be installed, the location of columns, the floor loading capacity, and the provisions for future expansion.

When the amount of space is determined, allow more space for furniture, carts, and storage cabinets. Additional space, not necessarily in the computer area, is required for air conditioning, electrical, security systems, fire protection equipment, and storage of tapes, forms, and other supplies. Plan to store all combustible materials in properly designed and protected storage areas.

Additional space might be needed to access the system, for example, rack-door-opening clearance.

For information about floor loading and weight distribution for your system, see *Floor construction and floor loading* Floor construction and floor loading.

For more information about the depth, height, and width measurements of a IBM Fusion HCI, see Planning for the rack.

Floor construction and floor loading

Use the following formulas to calculate the floor loads for the system.

A floor loading assessment is the evaluation of the concrete subfloor, not the raised floor. The weight of the raised floor is considered in the floor loading formula.

The building floor must support the weight of the equipment to be installed. Although older devices might impose 345 kg/m^2 (75 lb/ft^2) on the building floor, a typical system design imposes a load of no more than 340 kg/m^2 (70 lb/ft^2). The following pounds-per-square-foot (lb/ft^2) formula is used to calculate floor loading. For assistance with floor load evaluation, contact a structural engineer.

Floor Loading is: (machine weight + (15 lb/ft2 x 0.5 svc clear) + (10 lb/ft2 x total area))/ total area

- The floor loading should not exceed 240 kg/m^2 (50 lb/ft^2) with a partition allowance of 100 kg/m^2 (20 lb/ft^2) for a total floor load rating of 340 kg/m^2 (70 lb/ft^2).
- The raised-floor weight plus the cable weight adds 50 kg/m^2 (10 lb/ft^2) uniformly across the total area that is used in calculations and is included in the 340 kg/m^2 (70 lb/ft^2) floor loading. The total area is defined as: machine area + 0.5 service clearance.
- When the service clearance area is also used to distribute machine weight (weight distribution/service clearance), 75 kg/m^2 (15 lb/ft^2) is considered for personnel and equipment traffic. The distribution weight is applied over 0.5 of the clearance up to a maximum of 760 mm (30 in.) as measured from the machine frame.

Computer room location

Computer room location is affected by several factors, such as considerations for safety and fire prevention.

Before selecting a location for the computer, give attention to these guidelines:

- The computer room must be in a noncombustible or fire-resistant building or room.
- The computer room must not be above, below, or next to areas where hazardous materials or gases are stored, manufactured, or processed. If the computer must be located near such an area, take extra precautions to safeguard the area.

- If the computer room is below ground level, provide adequate drainage.

Safety consideration and fire prevention

Safety is a vital factor when planning computer installation. This consideration is reflected in the choice of the computer location, building materials used, fire prevention equipment, air conditioning and electrical systems, and personnel training.

If an inconsistency occurs between IBM recommendations and any local or national regulation, the more stringent of the recommendations or regulations must take precedence. The National Fire Protection Association standard, NFPA 75, provides guidelines for protection of information technology equipment. The customer is responsible for adherence to governmental regulations.

- Computer room walls must have a minimum of a 1-hour-fire-resistance rating and extend from the structural floor to the structural ceiling (slab-to-slab).
- In rooms used for critical operations, it is preferable to install processors in 1-hour-fire-rated rooms separate from the main computer room.
- If the computer room has one or more outside walls that are adjacent to a building that is susceptible to fire, consider taking the following precautionary actions:
    - Installing shatterproof windows in the computer room to improve the safety of personnel and equipment from flying debris and water damage. Usually, windows in the computer room are undesirable because of security concerns, and the negative effect they have on temperature control. They can cause excessive heating in the summer, and excessive cooling in the winter.
    - Installing sprinklers outside the windows to protect them with a blanket of water if a fire occurs in the adjacent area.
    - Sealing the windows with masonry.
- Where a false (or hung) ceiling or insulating material is to be added, ensure that it is a noncombustible or fire-resistant material. All duct work must be noncombustible. If combustible material is used in the space between the structural ceiling and the false ceiling, appropriate protection must be provided.
- A raised floor that is installed over the structural floor must be constructed of noncombustible or fire-retardant materials. If the structural floor is of combustible material, it must be protected by water sprinklers on the ceiling of the room below.
  Note: Before the equipment is installed, the space between the raised and the structural floors must be cleared of debris. This space must also be checked periodically after installation to keep it free of accumulated dust, possible debris, and unused cables.
- The roof, ceiling, and floor above the computer room and the storage area for recorded media must be watertight. Liquid piping, roof drains, and other potential sources of liquid damage must be rerouted around the area.
- The space under the raised floor in the computer room must be provided with drainage to protect against flooding or trapped water.
- Waste material containers must be constructed of metal with a frame-suppressant lid.

Fire prevention equipment in a computer room

Fire prevention equipment in the computer room must be installed as an added safety measure. A fire suppression system is the responsibility of the customer. Your insurance underwriter, local fire marshall, and local building inspector are all parties that must be consulted in selecting a fire suppression system that provides the correct level of coverage and protection. IBM® designs and manufactures equipment to internal and external standards that require certain environments for reliable operation. Because IBM does not test any equipment for compatibility with fire suppression systems, IBM does not make compatibility claims of any kind nor does IBM provide recommendations on fire suppression systems.

- An early-warning fire detection system must be installed to protect the computer room and storage areas for recorded media. This system must activate both an audible and a visual alarm in the rooms and at a monitored central station.
- Portable carbon dioxide fire extinguishers, of suitable size and number, must be provided in the computer room for use on electrical equipment. Dry chemical extinguishers must not be used in the data center.
- Portable, pressurized-water extinguishers must be provided for combustible material such as paper.
- Extinguishers must be readily accessible to individuals in the area, and extinguisher locations must be marked so they are visible.
- Automatic sprinkler systems and gaseous total flooding systems are acceptable forms of fixed protection. For information on environmentally friendly gases for total flooding systems, consult NFPA 2001 titled Standard on Clean Agent Fire Extinguishing Systems.
- Special consideration must be used if you prefer a gaseous total flooding system. If a gaseous total flooding system is installed, include a time delay feature that allows investigation and evacuation from the covered area of the gaseous total flooding system. A cross-zoned detection system is suggested.
- The protected area must be evacuated whenever the gaseous total flooding system or its controls are being serviced. Additionally, a master Disarm switch, available for use by the system service personnel, is required. With the switch set in the off position, the detonators that are used to release the gaseous total flooding system must be made inoperative, even if the circuit fails elsewhere in the system. This switch must be placed in the off (manual) position before servicing begins to prevent possible accidental discharge of the gaseous total flooding system.
- Alternatives to ordinary wet pipe sprinkler systems might include dry pipe systems or preaction systems. Water flows into preaction systems only if triggered by smoke or heat detectors. The detection systems must be independent of gaseous total flooding system detection systems. The On-Off type of sprinkler head is not suggested because it is more prone to leakage.

To determine the proper fire protection required for the computer room, consult with your insurance underwriter and your local code authority.

Vibration and shock

Use this information to plan for possible vibration and shock in the data center.

It might be necessary to install the information technology equipment in an area subject to minor vibration. The following information supplies vibration and shock limits for your equipment and some basic definitions concerning vibration. The vibration levels normally present in computer-room and industrial installations are well within the indicated levels.

However, mounting the equipment in racks, stackers, or similar equipment might increase the risk of vibration-related problems. It is important to consult the manufacturer of such equipment to ensure that vibration factors will not exceed the specifications that are provided in the following tables.

Some useful definitions of vibration follow:

Acceleration
: Normally measured in g multiples of the acceleration because of the force of gravity. If the frequency is also known for a sine wave, acceleration can be calculated from displacement. (g: The unit of acceleration caused by the force of gravity.)

Continuous
: The presence of vibration over an extended period and cause a sustained resonant response in the equipment.

Displacement

Magnitude of the wave shape; normally given in peak-to-peak displacement in English or metric units:

- Normally used to measure floor vibration at low frequencies
- If the frequency is also known, it can be converted to displacement g for a sine wave.
  Note: Many measuring instruments can convert displacement to go for either sinusoidal or complex wave shapes.

Peak
: The maximum value of a sinusoidal or random vibration. This can be expressed as peak-to-peak in cases of sinusoidal vibration displacement.

Random
: A complex vibration wave form varying in amplitude and frequency content.

rms (root mean square)
: The long-term average of the acceleration or amplitude values. Normally used as a measure of overall vibration for random vibration.

Shock
: Intermittent inputs that occur and then decay to zero prior to a recurrence of the event. Typical examples are foot traffic, fork lifts in aisles, and external events such as railroad, highway traffic, or construction activities (including blasting).

Sinusoidal
: Vibration with the characteristic shape of the classical sine wave (for example, 60-Hz AC power).

Transient
: Vibrations that are intermittent and do not cause a sustained resonant response in the equipment.

If you need to make any calculations or require information regarding the previous definitions, consult a mechanical engineer, a vibration consulting engineer, or your seller.

The three classes of a vibration environment are shown in the following table.

Table 1. Vibration environment

Class	Vibration environment
V1	Floor-mounted machines in an office environment
V2	Table-top and wall-mounted machines
V3	Heavy industrial and mobile equipment

A summary of the vibration limits for each of the three classes is shown in the following table. A legend follows the table.

Note: Vibration levels at any discrete frequency should not exceed a level of 1/2 the g rms values for the class that is listed in the following table.

Table 2. Operational vibration and shock limits

Class	g rms	g peak	Mils	Shock
V1 L	0.10	0.30	3.4	3 g at 3 ms
V1 H	0.05	0.15	1.7	3 g at 3 ms
V2	0.10	0.30	3.4	3 g at 3 ms
V3	0.27	0.80	9.4	application dependent

L
: Light with weight less than 600 kg

H
: Heavy with weight equal to or greater than 600 kg

g rms
: Overall average g level over the 5 to 500 Hz frequency range

g peak
: Maximum real-time instantaneous peak value of the vibration time history wave form (excluding events that are defined as shocks).

Mils
: Peak-to-peak displacement of a discrete frequency in the 5 to 17 Hz range. One mil equals 0.001 inch

Shock
: Amplitude and pulse width of a classical 1/2 sine shock pulse

The values that are given in the Operational vibration and shock limits table are based on worst-case field data measured at customer installations for current and previously released products. The vibration and shock environment will not exceed these values except for abnormal cases involving earthquakes or direct impacts. Your seller can contact the IBM® Standards Authority for Vibration and Shock in case of specific technical questions.

Earthquakes

Special frame-strengthening features might be required in earthquake prone areas. Local codes might require the information technology equipment to be tied down to the concrete floor. If sufficient information on equipment tie down is not provided in the product's physical planning documentation, consult with your seller.

Acoustics

Acoustic noise emission data enables you to assess noise levels for your data processing equipment.

Acoustic noise emission data on IBM® products is provided for the benefit of installation planners and consultants to help predict acoustical noise levels in data centers and other installations of information technology and telecommunications equipment. Such noise declarations also enable you to compare noise levels of one product to another and to compare the levels to any applicable specifications. The format of the data that is provided conforms to ISO 9296: Acoustics - Declared Noise Emission Values of Computer and Business Equipment. The measurement procedures used to acquire the data conform to International Standard ISO 7779 and its American National Standard equivalent ANSI S12.10.

The following terms are used to present acoustical data.

- $\mathbf{L_{WAd}}$ is the declared (upper limit) A-weighted sound power level for a random sample of machines.

- **L_{pAm}** is the mean value of the A-weighted sound pressure levels either at the operator position or at the bystander (1-meter) positions for a random sample of machines.
- **$\langle L_{pA} \rangle_m$** is the mean value of the space-averaged sound-pressure-emission levels at the 1-meter positions for a random sample of machines.

Acoustical treatment of data centers or other rooms, in which the equipment is installed, is suggested to achieve lower noise levels. Lower noise levels tend to enhance employee productivity and avoid mental fatigue, improve communications, reduce employee complaints, and generally improve employee comfort. Proper room design, including the use of acoustical treatment, might require the services of a specialist in acoustics.

The total noise level of an installation with information technology and telecommunications equipment is an accumulation of all the noise sources in the room. This level is affected by the physical arrangement of the products on the floor, the sound reflective (or absorptive) characteristics of the room surfaces, and the noise from other data center support equipment such as air conditioning units and backup power equipment. Noise levels might be reduced with proper spacing and orientation of the various noise-emitting equipment. Provide sufficient space around such machines. The farther apart they can be placed, the lower the overall room noise will be.

In smaller installations, such as small offices and general business areas, pay additional attention to the location of equipment relative to the work areas of the employees. At work areas, consider locating computers next to the desk rather than on top of it. Small systems must be located as far away from personnel as possible. Locate nearby work areas away from the exhaust of computer equipment.

The use of absorptive materials can reduce the overall noise level in most installations. Effective and economical sound reduction can be achieved by using a sound-absorptive ceiling. The use of acoustically absorbing free-standing barriers can reduce the direct noise, increase room absorption and provide privacy. The use of absorptive material, such as carpeting on the floor, results in further reduction of the sound level in the room. Any carpeting that is used in a computer room must meet the electrical continuity requirements that are stated in *Static electricity and floor resistance*. To prevent computer room noise from reaching adjacent office areas, walls must be constructed from the structural floor to the structural ceiling. Also, ensure that doors and walls are properly sealed. Acoustical treatment of overhead ducts might further reduce noise that is transmitted to or from other rooms.

Many IBM large system products are offered with optional acoustical front and rear doors to help attenuate the noise of the product itself. If noise exposure is a concern for the installation planners or employees, inquiries must be made to IBM on the availability of such product options.

Electromagnetic compatibility

Use this information to plan for system installation in an environment that has a high electromagnetic-radiated field.

Information technology equipment installation might occasionally be planned in an area that has a high electromagnetic-radiated field environment. This condition results when the information technology equipment is near a radio frequency source such as a radio-transmitting antenna (AM, FM, TV, or two-core radio), civilian and military radar, and certain industrial machines (rf induction heaters, rf arc welders, and insulation testers). If any of these sources are near the proposed site, a planning review might be appropriate to assess the environment and determine whether any special installation or product considerations are advisable to reduce interference. Consult your seller. Workstations that are located near devices like transformers or buried electrical conduits can experience jitter on the workstation display in the presence of strong magnetic fields.

Most products can tolerate low-frequency to very-high-frequency rf levels of 3 volts per meter. Field strengths greater than 3 volts per meter might cause operational or serviceability problems. Products have different tolerance levels to electromagnetic-radiated fields in different frequency ranges. Radar (frequency of 1300 MHz, and 2800 MHz) signals with field strengths of a maximum of 5 volts per meter are acceptable. If problems occur, reorientation of the system or selective shielding might be required.

Two-core radio or cellular telephone usage must be properly controlled in the computer room. To reduce the likelihood of a problem, the following suggestions must be considered when operating such equipment:

- Keep hand-held transmitters (for example, walkie-talkies, radio paging, and cellular telephones) a minimum of 1.5 m (5 ft) from information technology equipment.
- Use only an operator-controlled transmitting device (no automatic transmissions). Develop specific rules, such as do not transmit within 1.5 m (5 ft) of a fully covered operating system. If covers are open, do not transmit.
- Choose the minimum output power that accomplishes your communication needs.

Extremely low frequency (ELF) fields

Except for some video display cathode ray tubes (CRT), most information technology equipment is tolerant of extremely low frequency (ELF) electromagnetic fields. The video displays that use cathode ray tubes are more sensitive because they use electromagnetic fields to position the electron beam in normal operation. The extremely low frequency range covers frequencies between 0 - 300 Hz. It is also referred to as electrical power frequency because most world electrical power is generated at either 50 or 60 Hz.

Many products tolerate ELF electromagnetic fields in the following ranges:

- Cathode ray tube video display: 15 - 20 milligauss
- Liquid crystal display (LCD): 10 Gausses
- Magnetic tape equipment: 20 Gausses
- Disk drive equipment: 20 Gausses
- Processors or systems: 20 Gausses

Typical information technology centers exhibit an ambient electromagnetic field of 3 - 8 milligauss. Some equipment within a center might, under normal operation, produce fields in excess of 100 milligauss. Examples of equipment that produces large magnetic fields include: power distribution units, electric motors, electrical transformers, laser printers, and uninterruptible power systems. However, magnetic field density decreases rapidly with distance. If a CRT display is located near equipment that produces large electromagnetic fields, the display might exhibit distortion such as poor focus, change in image shape or slight motion in static display images. Moving the CRT away from the equipment might remedy the problem.

Material and data storage protection

Special safety considerations are required when storing data or other material.

Consider the following factors:

- Any data or material that is stored in the computer room, whether in the form of magnetic tapes, paper tapes, cards, or paper forms, must be limited to the minimum needed for safe, efficient operation and must be enclosed in metal cabinets or fire-resistant containers when not in use.
- For security purposes, and protection against fire, a separate room for material storage is strongly suggested. This room must be constructed of fire-resistant material (minimum 2-hour-fire-resistance rating). An approved fixed extinguishing system is suggested. Fixed extinguishing systems include automatic sprinklers and approved total flooding gaseous systems.

If continuity of operation is critical, plan a remote storage location for vital records if a disaster occurs. Key considerations in the choice of an off-site location for data storage are that the area is:

- Not subject to the same risk that might occur in the computer room.
- Suitable for long-term storage of hardcopy records and magnetic media files.

Air conditioning systems

In most installations, the computer area is controlled by a separate air conditioning system. Therefore, emergency power-off switches for the equipment and air conditioning must be placed in convenient locations, preferably near the console operator and next to the main exit doors. For more information, see the National Fire Protection Association standard, NFPA 70 article 645.

- When the regular building air conditioning system is used, with supplemental units in the computer area, the supplemental units would then be handled as previously stated. The regular building air conditioning system must have an audible alarm to alert maintenance personnel of an emergency.
- Fire dampers must be located in all air ducts at fire walls.
- The air filters in the air conditioning system must contain noncombustible or self-extinguishing material.

Electrical systems

Provide a mainline disconnect control for the computer equipment at a remote location. The remote controls must be in a convenient location, preferably near the console operator and next to the main exit doors. They must be next to the power-off switch for the air conditioning system and must be properly marked. A light must be installed to indicate when power is on. The National Electric Code (NFPA 70) article 645 states that a single disconnecting means to control both the electronic equipment and the HVAC system is permitted.

- If continuity of operation is essential, a standby power source must be installed.
- It is advisable to install an automatic battery-operated lighting unit to illuminate an area if a power or lighting circuit failure occurs. This unit is wired to and controlled by the lighting circuit.
- Watertight connectors are suggested under raised floors because of the moisture exposures (water pipe leaks, high humidity levels) under raised floors.

General power information

Reliable electrical power is required for the proper functioning of your data processing equipment.

IBM® information technology equipment requires a reliable electrical power source that is free from interference or disturbance. Electrical power companies generally supply power of sufficient quality. The Power quality, Voltage and frequency limits, Power load, and Power source topics provide the guidance and specifications that are needed to meet the requirements of the equipment. Qualified personnel must ensure that the electrical power distribution system is safe and meets local and national codes. They must also ensure that the voltage that is measured at the power receptacle is within the specified tolerance for the equipment. In addition, a separate power feeder is required for items such as lighting and air conditioning. A properly installed electrical power system helps to provide for reliable operation of your IBM equipment.

Other factors to consider when planning and installing the electrical system include a means of providing a low impedance conducting path to ground (path to earth) and lightning protection. Depending on the geographical location, special considerations may be required for lightning protection. Your electrical contractor should meet all local and national electrical code requirements. Building electrical power is normally derived from a three-phase power distribution system. General office areas are normally provided with single-phase power outlets, and data processing rooms are provided with three-phase power.

Some IBM IT equipment and devices may require standard three-phase power; others may require single-phase power. The power requirements for each device are specified in the individual system specifications for that system. Nominal voltage, plugs, receptacles, and in some cases, conduit and back boxes are listed in the specific system specifications. Refer to the respective system specifications to determine the power requirements. Ensure that existing branch circuit outlets are the correct type and are properly grounded.

- **Power quality**
  The quality of electrical power significantly impacts the performance of sensitive electronic equipment. These guidelines ensure that quality electrical power is provided to your data center.
- **Voltage and frequency limits**
  Voltage and frequency limits must be maintained to ensure proper functioning of your system.
- **Power load**
  A preliminary sizing for total power load can be obtained by adding the total power requirements for all devices to be connected.
- **Power source**
  These guidelines help to ensure that your data center has a quality power source.
- **Dual-power installation configurations**
  These dual-power installation configurations allow you to leverage the fully-redundant power features of your system.

Power quality

The quality of electrical power significantly impacts the performance of sensitive electronic equipment. These guidelines ensure that quality electrical power is provided to your data center.

IBM® equipment can tolerate some power disturbances or transients. However, large disturbances can cause equipment power failures or errors. Transients can come into the site on the power utility company lines but are often caused by electrical equipment that is installed in the building. For example, transients can be produced by welders, cranes, motors, induction heaters, elevators, copy machines, and other office equipment. The best way to prevent problems that are caused by power disturbances is to have transient-producing equipment on a separate power service than the one that supplies power to your information technology equipment.

Ground or earth

When used in reference to electrical power systems, Ground is a conducting connection between an electrical circuit and the earth or some conducting body that serves in place of the earth. The term ground is the most common name that is used, however it is also referred to as earth or terra in several international geographies. In this topic, these terms and other local language equivalents are interchangeable.

Ground is a critical component of an electrical power distribution system. A properly installed ground system allows for safe operation of equipment that is connected to the electrical power source under normal and electrical or equipment fault conditions. The life safety function of ground and grounding methods is addressed by the appropriate local and national electrical wiring codes. In the United States, this code is known as the National Electric Code or publication 70 of the National Fire Protection Association. Many countries have adopted the National Electric Code or have developed an equivalent code.

The National Electric Code and its equivalents have a primary objective to provide safe operation of electrical power distribution systems and electrical equipment installations. Compliance with these codes does not guarantee efficient operation of equipment connected to the power distribution systems. When sensitive electronic equipment is connected, there are often times when additional ground connections may be required. Typically, additional ground connections are recommended when there is a concern for high frequency or radio frequency (RF) interference, which may impact electronic circuits. These additional ground requirements will be found with the installation documentation for specific equipment. Additional ground requirements may also be recommendations from engineering or data center evaluations, reviews or surveys. Local or national codes allow for these additional grounds to be installed.

Grounding

IBM equipment, unless double insulated, has power cords containing an insulated grounding conductor (color-coded green or green with yellow stripe) that connects the frame of the equipment to the ground terminal at the power receptacle. The power receptacles for IBM equipment are identified in the equipment documentation and should match the equipment power plug. In some cases, there may be options for different manufacturer equivalent receptacles. IBM equipment plugs should not be changed or altered to match existing connectors or receptacles. To do so may create a safety hazard and void product warranty. The connectors or receptacles for IBM equipment should be installed to a branch circuit with an equipment grounding conductor, connected to the grounding bus bar in the branch-circuit distribution panel. The grounding bus bar in the panel should then be connected back to the service entrance or suitable building ground by an equipment grounding conductor.

Information technology equipment must be properly grounded. It is recommended that an insulated green wire ground, the same size as the phase wire, be installed between the branch circuit panel and the receptacle.

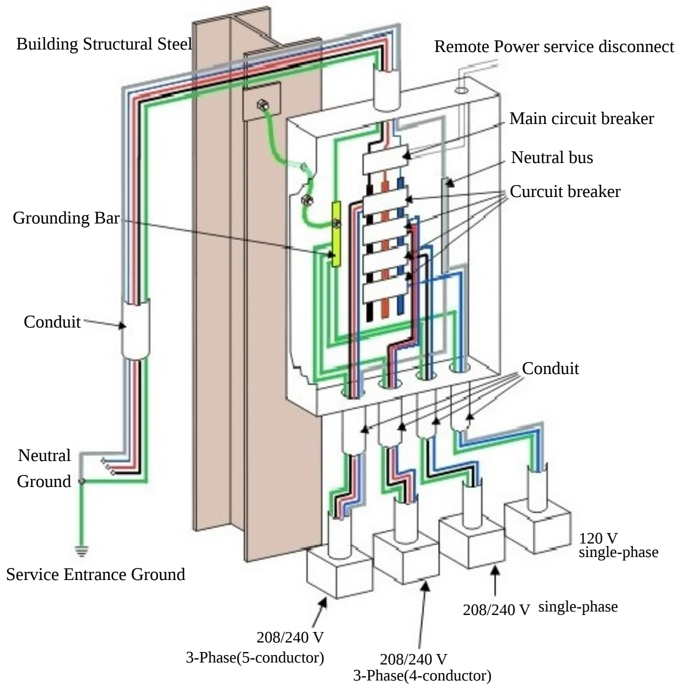
For personnel safety, the ground must have a sufficiently low impedance to limit the voltage to the ground and to facilitate the operation of protective devices in the circuit. For example, the ground path shall not exceed 1 ohm for 120-volt, 20-ampere branch circuit devices.

The ground path impedance limit is 0.5 ohms for 120 volt branch circuits that are protected by 30 ampere circuit breakers. The limit is 0.1 ohms for 120 volt 60 to 100 ampere circuits.

All grounds entering the room should be interconnected somewhere within the building to provide a common ground potential. This includes any separate power sources, lighting and convenience outlets, and other grounded objects, such as building steel, plumbing, and duct work.

The equipment grounding conductor must be electrically attached to both the enclosure of the computer power center and the connector grounding terminal. Conduit must not be used as the only grounding means, and it must be connected in parallel with any grounding conductors that it contains.

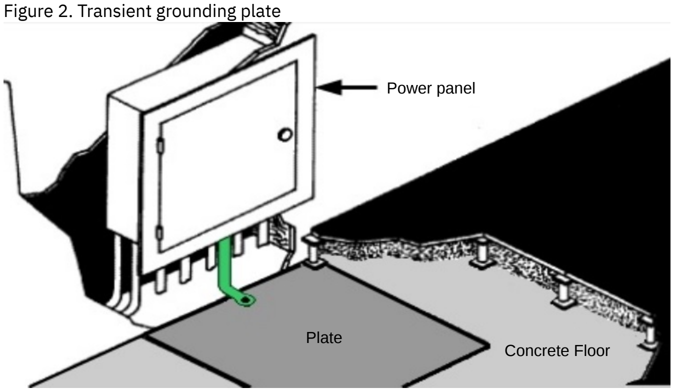
Figure 1. Transient grounding plate



Transient grounding

To minimize the effects of high-frequency electrical noise, the branch circuit power panel servicing the equipment should be mounted in contact with bare building steel or connected to it by a short length of cable. If this is not possible, a metal area of at least 1 m^2 (10 ft^2) in contact with masonry can be used. The plate should be connected to the green-conductor common.

Figure 2. Transient grounding plate



The preferred connection is with a braided strap. If a braided strap is not available, the connection should consist of no. 12 AWG (3.3 mm or 0.0051 in.) or larger conductor and should not be more than 1.5 m (5 ft) long. To minimize this length, the preferred connection of this braided strap or conductor is to the nearest portion of the enclosure on the panel, if the enclosure is electrically continuous from the green-conductor common point to this point of connection.

The raised-floor-supporting substructure can be used as a substitute for the transient plate if the structure has a consistently low-impedance path. If the raised floor has stringers or other subframing that makes electrical connection between the pedestals, the floor itself can be used for the signal reference plane. Some raised floors are stringerless and the floor tiles lock into isolated pedestals by gravity alone. If there is no reliable electrical connection between the pedestals, a signal reference grid can be constructed by connecting the pedestals together with conductors. A minimal grid would interconnect every other pedestal in the immediate area of the power panel and extend at least 3 m (10 ft) in all directions.

Figure 3. Transient grounding using the raised floor support structure

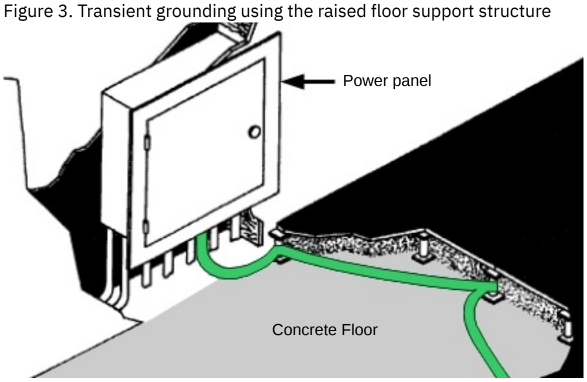
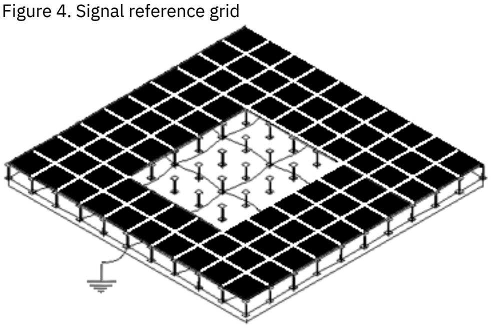


Figure 4. Signal reference grid



Stranded bare or insulated conductor of at least no. 8 AWG (8 mm or 0.0124 in.) copper is required. This conductor provides a low-impedance path and is strong enough to make physical damage unlikely. Any connection method is acceptable as long as it provides a reliable electrical and mechanical connection.

A customer's self-contained, separately derived power system (computer power centers, transformers, motor generators), installed on a raised floor, has the same requirements.

Voltage and frequency limits

Voltage and frequency limits must be maintained to ensure proper functioning of your system.

The phase-to-phase steady-state voltage must be maintained within plus six percent to minus 10 percent of the normal rated voltage, which is measured at the receptacle when the system is operating. A voltage surge or sag condition must not exceed plus 15 percent or minus 18 percent of the nominal voltage and must return to within a steady-state tolerance of plus 6 percent or minus 10 percent of normal rated voltage within 0.5 second.

Some systems might require special considerations and might have more or less restrictive specifications. See the individual system specifications for actual requirements. Because of the possibility of brownouts (planned voltage reduction by the utility company) or other marginal voltage conditions, installing a voltage monitor might be advisable.

The phase frequency must be maintained at 50 or 60 Hz + 0.5 Hz.

The value of any of the three phase-to-phase equipment voltages in the three-phase system must not differ by more than 2.5 percent from the arithmetic average of the three voltages. All three line-to-line voltages must be within the limits previously specified.

The maximum total harmonic content of the power system voltage waveforms on the equipment feeder must not exceed 5 percent with the equipment operating.

Power load

A preliminary sizing for total power load can be obtained by adding the total power requirements for all devices to be connected.

For a more precise analysis of power distribution system requirements, you can request an IBM® System Power Profile Program printout from your seller. The System Power Profile Program, controlled and operated by the service office installation planning representative, provides a vector analysis rather than an arithmetic summation of total power. The vector analysis takes into consideration power factor and phase relationships. In addition, it considers waveform distortions caused by the load and inrush requirements. Additional capacity should be planned for future expansion. Contact your service office installation planning representative for information on how to obtain a System Power Profile.

Primary power problem areas

Your system is designed to operate on the normal power supplied by most electrical utility companies. However, possible computer malfunctions can be caused by outside (radiated or conducted) transient electrical noise signals being superimposed on the power line to the computer. To guard against this interference, power distribution design should comply with the specifications discussed in this topic.

Failures caused by the power source are basically of three types:

- Power line disturbances, such as, short duration dips in voltage as well as prolonged outages. If the frequency of such power failures is not acceptable for your operation, installing standby or buffered power might be necessary.
- Transient electrical noise superimposed on power lines might be caused by a variety of industrial, medical, communication, or other equipment:
  - Within the computing facilities
  - Adjacent to the computing facilities
  - In the vicinity of the power company's distribution lines

  Switching large electrical loads can cause problems, even though the source is on a different branch circuit. If you suspect such a condition, it might be advisable to provide a separate, dedicated feeder or transformer for your system directly from your power source.

If the transient-producing devices have been eliminated from the feeder and the computer room power panel and power line disturbances are still present, it might be necessary for you to install isolation equipment (for example, transformers, motor generators, or other power conditioning equipment).

Lightning protection

Installing lightning protection devices is recommended on the computer power source when:

- The primary power is supplied by an overhead power service.
- The utility company installs lightning protectors on the primary power source.
- The area is subject to electrical storms or an equivalent type of power surge.

Lightning protection for communication wiring

Be sure to install lightning protection devices to protect communication wiring and equipment from surges and transients induced into the communication wiring. In any area subject to lightning, surge suppressors should be installed at each end of every outdoor cable installation, whether installed above the ground (aerial) or buried below the ground.

Information about lightning surge suppressors for communication wiring systems and recommended installation methods for outdoor communication cables can be found in the manuals for the specific type of data processing system that is being considered.

Power source

These guidelines help to ensure that your data center has a quality power source.

The primary power source is normally a wye-type or delta-type, three-phase service coming from a service entrance or a separately derived source with appropriate overcurrent protection and suitable ground (service entrance or building ground). A three-phase, five-wire power distribution system should be provided for flexibility in your data processing installation. However, depending on the type of equipment that is installed , a single-phase distribution system might be sufficient. The five wire system enables you to provide power for three-phase line-to-line, single phase line-to-line, and single phase line-to-neutral. The five wires consist of three phase conductors, one neutral conductor, and one insulated equipment grounding conductor (green, or green with yellow trace).

Conduit must not be used as the only grounding means.

Power panel feeders

Ensure that the feeder wires to the branch-circuit distribution panel are large enough to handle the total system power load. For optimal results, ensure that these feeders service no other loads. See Power quality for more information.

Branch circuits

The computer branch circuit panel should be in an unobstructed, well-lighted area in the computer room.

The individual branch circuits on the panel should be protected by suitable circuit breakers properly rated according to manufacturer specifications and applicable codes. Each circuit breaker should be labeled to identify the branch circuit it is controlling. The receptacle should also be labeled.

Where a branch circuit and receptacle are installed to service your system, it is recommended that the grounding conductor of the branch circuit is insulated and equal in size to the phase conductors. The grounding conductor is an insulated, dedicated-equipment-grounding conductor, not the neutral.

Branch circuit receptacles that are installed under a raised floor should be within 0.9 m (3 ft) of the system that they supply power to. If the branch circuits are contained in a metallic conduit, either rigid or nonrigid, the conduit system should be grounded. This is accomplished by bonding the conduit to the power distribution panel, which in turn, is tied to the building or transformer ground.

Power cords are supplied in 4.3 m (14 ft) lengths unless otherwise noted in the system specifications. The length is measured from the exit symbol on the plan views. Some power plugs furnished by your seller are watertight, and should be located under the computer room raised floor.

Phase rotation

The three-phase power receptacles for some equipment, such as printers, must be wired for correct phase rotation. When looking at the face of the receptacle and counting clockwise from the ground pin, the sequence is phase 1, phase 2, and phase 3.

Emergency power control

A disconnecting means should be provided to disconnect the power from all electronic equipment in the computer room. This disconnecting means should be controlled from locations readily accessible to the operator at the principal exit doors. A similar disconnecting means to disconnect the air conditioning system serving this area should be available. Consult the local and national codes to determine the requirements for your installation. National Electric Code (NFPA 70) article 645 provides the requirements for this room EPO.

Convenience outlets

A suitable number of convenience outlets should be installed in the computer room and the Service Representative area for use by building maintenance personnel and service representatives. Convenience outlets should be on the lighting or other building circuits, not on the computer power panel or feeder. Under no circumstances are the service conveniences outlets on your systems to be used for any purpose other than normal servicing.

Dual-power installation configurations

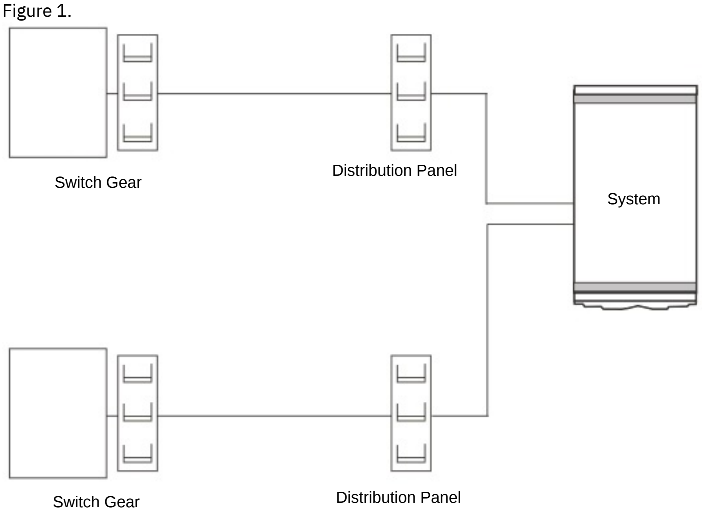
These dual-power installation configurations allow you to leverage the fully-redundant power features of your system.

Some models are designed with a fully redundant power system. The possible power installation configurations are listed as follows.

Dual-power installation: Redundant distribution panel and switch

This configuration requires that the system receives power from two separate power distribution panels. Each distribution panel receives power from a separate piece of building switch gear. This level of redundancy is not available in most facilities.

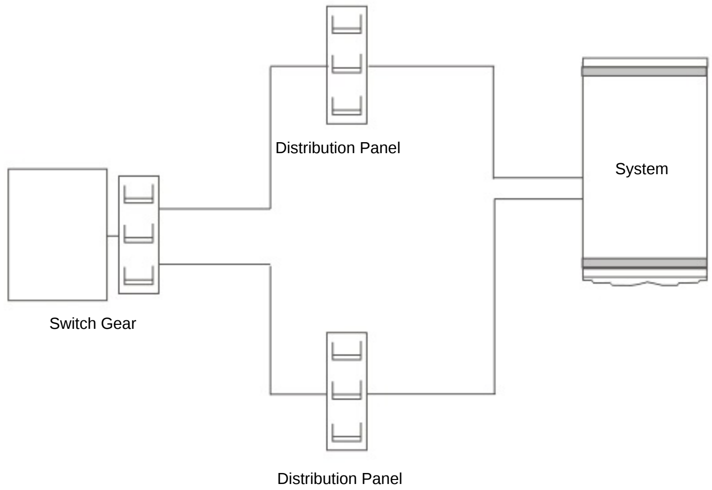
Figure 1.



Dual-power installation: Redundant distribution panel

This configuration requires that the system receives power from two separate power distribution panels. The two distribution panels receive power from the same piece of building switch gear. Most facilities should be able to achieve this level of redundancy.

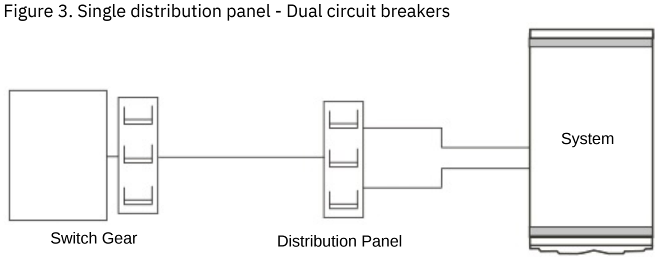
Figure 2. Dual power installation - Redundant distribution panel



Single distribution panel: Dual circuit breakers

This configuration requires that the system receives power from two separate circuit breakers in a single power panel. This configuration does not make full use of the redundancy that is provided by the processor. However, it is acceptable if a second power distribution panel is not available.

Figure 3. Single distribution panel - Dual circuit breakers



Air conditioning determination

The air conditioning system must provide year-round temperature and humidity control as a result of the heat dissipated during equipment operation.

Heat dissipation ratings are given in the server specifications for each server. Air conditioning units must not be powered from the computer power panel because of the high starting current drawn by their compressor units. The feeder line for the air conditioning system and the computer room power must not be in the same conduit.

Consider the following factors when you determine the air conditioning capacity necessary for installation:

- Information technology equipment heat dissipation
- Number of personnel
- Lighting requirements
- Amount of fresh air introduced
- Possible reheating of circulated air
- Heat conduction through outer walls and windows
- Ceiling height
- Area of floors
- Number and placement of door openings
- Number and height of partitions

Most servers are air-cooled by internal blowers. A separate air conditioning system is suggested for data processing installation. A separate system might be required for small systems or individual servers that are intended for operation when the building air conditioning system is not adequate or is not operational. Server heat dissipation loads are given on the server specifications for each server. See the environmental requirements in the server specifications for your server.

System requirements

System requirements provide the details of the resources allocation and management of the IBM Fusion HCI and its services to help ensure efficient and effective operation.

System requirements outline the minimum and recommended specifications CPU, memory, and storage for a system to function correctly, including:

- Request - minimum requirements required for basic functions
- Limit - maximum requirements that are required for smooth and efficient operation

To ensure efficient and effective operation, the following resource allocation and management details are recommended for IBM Fusion HCI management software and its services, based on a cluster configuration with three control nodes and three compute nodes. Also, the resource sizing depends on the number of nodes for components that are marked as "Per node".

Important: Fusion Data Foundation can be deployed on Infra or worker nodes, whereas all other IBM Fusion services including Base must be deployed on worker nodes.

IBM Fusion management software

IBM Fusion	vCPU				Memory (GiB)				Storage (GiB)	
	Control node		Worker node		Control node		Worker node			
	Request	Limit	Request	Limit	Request	Limit	Request	Limit	Request	Limit
`ibm-spectrum-fusion-ns` namespace	5.5	12	0.5	3	5	16.5	0.6	2.5	500	NA
`openshift-monitoring` namespace	0.008	NA	NA	NA	0.2	NA	NA	NA	500	NA

Each additional multi-rack or expansion rack added to the system introduces four new switches, so increasing the overall capacity by four extra pods.

IBM Fusion	vCPU				Memory (GiB)				Storage (GiB)	
	Control node		Worker node		Control node		Worker node			
	Request	Limit	Request	Limit	Request	Limit	Request	Limit	Request	Limit
`openshift-monitoring` namespace	0.002 x 4	NA	NA	NA	0.1 x 4	NA	NA	NA	NA	NA

Fusion Data Foundation

The following table lists out the system requirements that must be met to install the Fusion Data Foundation service.

Component	vCPU			Memory (GiB)			Storage (GiB)
	Profile	Request	Limit	Profile	Request	Limit	
Fusion Data Foundation	Balanced	13 (number of storage drives x 2)	NA	Balanced	26	NA	50.5
MCG standalone	NA	3	NA	NA	15.5	NA	50.5

Important: Resource sizing might vary depending on the number of nodes for components that are marked as "Per node". By default, IBM Fusion HCI servers come with two disks. If you want to add more disks to your IBM Fusion HCI, the storage requirement changes based on the total disk count.

The following table lists out the system requirements that must be met for an additional worker node in the Fusion Data Foundation service.

Component	vCPU			Memory (GiB)			Storage (GiB)
	Profile	Request	Limit	Profile	Request	Limit	
Fusion Data Foundation	Balanced	13 + (number of storage drives x 2)	NA	Balanced	26 + (number of storage drives x 5)	NA	50.5
MCG standalone	NA	3	NA	NA	15.5	NA	50.5

Global Data Platform

The following table lists out the system requirements that must be met to install the Global Data Platform service.

To install the Global Data Platform using control nodes, the following requirements must be met:

Component	vCPU		Memory (GiB)		Storage (GiB)
	Request	Limit	Request	Limit	
Global Data Platform	10	NA	71	NA	25 per node
Global Data Platform remote mount	4	NA	11	NA	25 per node

To install the Global Data Platform using worker nodes, the following requirements must be met:

Component	vCPU			Memory (GiB)			Storage (GiB)
	Role	Request	Limit	Role	Request	Limit	
Global Data Platform	Storage	8	NA	Storage	64	NA	25
	Client	2	NA	Client	16	NA	
	AFM	16	NA	AFM	162	NA	
Global Data Platform remote mount	NA	2	NA	NA	5	NA	25

Backup & Restore

The following table lists out the system requirements that must be met to install the Backup & Restore service.

Component	vCPU				Memory (GiB)				Storage (GiB)	
	Control node		Worker node		Control node		Worker node			
	Request	Limit	Request	Limit	Request	Limit	Request	Limit	Request	Limit
Hub	NA	NA	4.91	31.1	NA	NA	17.91	42.76	Ephemeral: 3.35 PVC: 200	Ephemeral: 8.07 PVC: NA
Spoke	NA	NA	1.39	14	NA	NA	2.1	13.87	Ephemeral: 0.88 PVC: 10 minimum	Ephemeral: 3.15 PVC: NA

Important: Resource sizing might vary depending on the number of nodes for components that are marked as "Per node".

The following table lists out the system requirements that must be met for an additional job in the Backup & Restore service.

Component	vCPU		Memory (GiB)		Storage (GiB)	
	Request	Limit	Request	Limit	Request	Limit
Each additional node for Backup & Restore (Beyond the first three) Applicable for both Hub and Spoke	0.05	4	0.24	2	0.02	0.5
Per Job Additional Requirements Data Mover (1 per PVC, max 5 per node)	2	4	4	16	4.89	4.89

Per Job Additional Requirements Legacy Restic Data Mover (1 per PVC, max 3 per node)	0.1	2	3.9	14.65	0.4	1.95
Per Job Additional Requirements Maintenance (after every job and once per day per application/policy combination)	0.5	4	0.5	4	4.89	4.89

IBM Data Cataloging

The following table lists out the system requirements that must be met to install the IBM Data Cataloging service.

Component	vCPU				Memory (GiB)				Storage (GiB)	
	Control node		Worker node		Control node		Worker node			
	Request	Limit	Request	Limit	Request	Limit	Request	Limit	Request	Limit
IBM Data Cataloging	NA	NA	18	72	NA	NA	26	115	120	NA

Important: For IBM Data Cataloging service, no changes are required as the footprint on IBM Data Cataloging does not change for additional nodes.

Content-Aware Storage (CAS)

The following table lists out the system requirements that must be met to install the CAS service.

Note: For Content-Aware Storage with more than 12 TB of ingested data, contact IBM Sales representatives for your intended workloads.

Component	vCPUs	Memory	Storage
Content-Aware Storage with less than 12 TB of ingested data	Request: 9 Limit: 25	Request: 22 GiB Limit: 24 GiB	The minimum required filesystem size is 550 GB*
`nv-ingest` Requirements	Request: 30 Limit: 60	Request: 90 GiB Limit: 290 GiB	100 GB
`docling_multimodal` - Additional requirements	Request: 4.5 Limit: 21	Request: 13 GiB Limit: 44 GiB	100 GB
Minimum requirements for each additional Collection	Request: 1 Limit: 2	Request: 10 GiB Limit: 10 GiB	1 GB

* For more information about IBM Spectrum Scale Container Native Storage Access, see Hardware requirements for IBM Storage Scale Container Native Storage Access.

GPU resources

Configuration	GPU worker nodes	Non-GPU worker nodes	Production with Multi-Instance GPU (MIG)* enabled	Production without Multi-Instance GPU (MIG) enabled
Content-Aware Storage with less than 12 TB of ingested data	1	2	2 or more MIG Capable NVIDIA GPUs (A100 80 GB, H100, H200, RTX PRO 6000)	1 NVIDIA GPU for each NVIDIA NIM service** configured (L40S, A100 H100, H200, RTX PRO 6000)

* For more information about MIG, see Configuring a Multi-Instance GPU (MIG) with CAS.

** For more information about the NVIDIA NIM services, see Select NVIDIA Inference Microservices (NIMs).

Choosing your storage type

Guidance to choose between Fusion Data Foundation and IBM Storage Scale storage.

Fusion Data Foundation is designed to provide persistent storage and cluster data management. It supports containerized environments and integrates seamlessly with Red Hat® OpenShift®. It supports file, block, and object storage, and provides a consistent storage experience across hybrid cloud environments.

The IBM Storage Scale storage is designed for building a global data platform for AI, high-performance computing (HPC), advanced analytics, and other demanding workloads. It supports file and object storage for unstructured data.

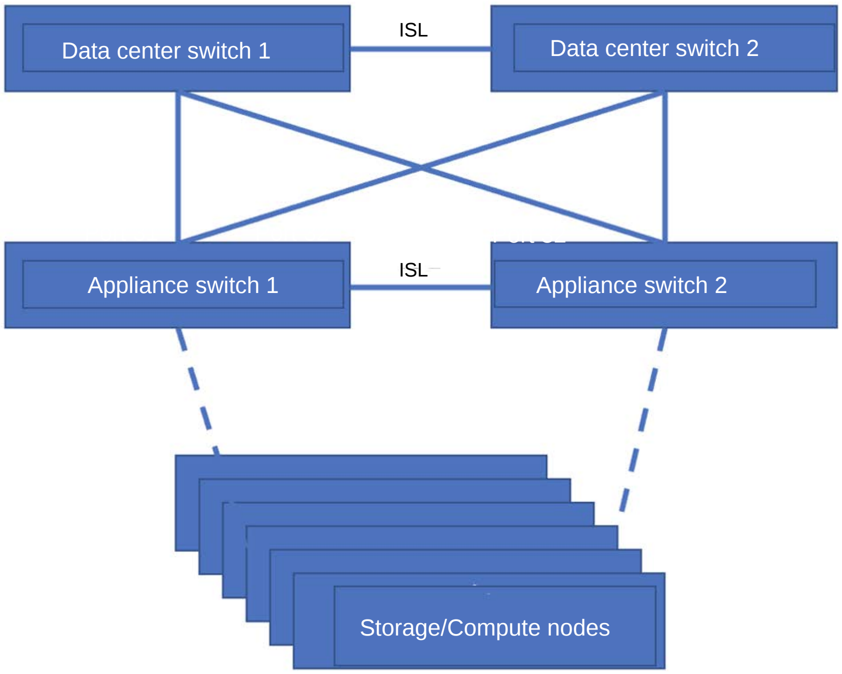
Fusion Data Foundation is tailored for containerized environments, multi cluster, virtualization, and OpenShift deployments, while IBM Storage Scale is more towards AI, HPC, and analytics workloads.

After installing the IBM Fusion HCI, you have the flexibility to install either of these storage types (IBM Storage Scale or Fusion Data Foundation) as a service through the user interface.

Networking

This topic provides a high-level overview of the network architecture of the IBM Fusion HCI.

The IBM Fusion HCI system contains two high-speed switches that connect to the data center network. The storage and compute nodes within the appliance are connected to the data center network indirectly through these high-speed switches, forming a switch-to-switch connection. This setup treats the appliance switches as leaf switches, recommending a connection to core switches for integration with the larger network.



Planning a Connection between IBM Fusion HCI and Your Network

IBM Fusion HCI is a hyperconverged infrastructure solution that combines compute, storage, and networking resources into a single, easy-to-manage platform. To integrate IBM Fusion HCI with your existing network, you need to plan a connection that ensures seamless communication and data exchange between the two environments.

Before establishing a connection, ensure that you have:

- A clear understanding of your network architecture and topology
- The necessary network infrastructure, including switches, routers, and firewalls
- IBM Fusion HCI hardware and software requirements met
- A plan for network segmentation and security

You can connect IBM Fusion HCI to your network as follows:

- Connect IBM Fusion HCI directly to your network by using a dedicated Ethernet cable.
- Configure the network settings on IBM Fusion HCI to match your network configuration.
  After establishing a connection, you configure DHCP, DNS, and NTP server connections. For more information, see Planning and prerequisites.

Network connectivity for different storage options

The IBM Fusion HCI offers two storage options: Global Data Platform and Fusion Data Foundation. Understanding the network differences between these two options is crucial for effective planning.

Global Data Platform

- The uplink carries only the OpenShift® Bare Metal VLAN.
- In a multi-rack setup, internal storage and management traffic also use the spine switches.
  Important: Global Data Platform supports expansion multi-rack configurations only.

Fusion Data Foundation

- The uplink carries both the OpenShift VLAN and the management or provisioning VLAN 4091.
- No spine switches are present in a Fusion Data Foundation rack.
- Fusion Data Foundation supports multiple rack configurations, including:
  - High-Availability (HA) multi-rack
  - Expansion multi-rack
  - 3 zone multi-rack
  - 2 zone + arbiter

For more information on Fusion Data Foundation network architecture, see Deployment configurations. For high-availability multi-rack and expansion racks with storage type FDF prerequisites (also applicable to multi AZ racks), refer to Network prerequisites for high-availability racks.

IBM Fusion HCI network configuration for SSR setup

After receiving the IBM Fusion HCI appliance, an IBM Systems Service Representative (SSR) will visit your data center to set it up. To complete the setup, the SSR needs information about your internal network. Fill out the following Planning Worksheet and share it with your IBM representative.

- **Planning and prerequisites**
  To ensure a smooth installation of the IBM Fusion HCI in your data center, review the following network requirements and configuration options. This information is intended for your network administrator.
- **Firewall requirements for IBM Fusion HCI**
  IBM Fusion HCI requires outbound access to external sites for accessing image registries and sending telemetry data to IBM and Red Hat®. This outbound access can be optionally routed through a proxy server. Configure your data center's firewall rules or proxy access control list (ACL) to meet the requirements for IBM Fusion HCI.
- **Proxy configuration**
  Proxy configuration is needed only in a connected install through proxy.
- **Configuring Scale daemon network IP parameters**
  This is an optional configuration and is only needed when the default scale daemon IPs conflicts with some of your workloads. An additional IPVLAN Container Network Interface (CNI) network named as daemon-network is created for scale core pods. By default, IP addresses are assigned for Scale daemon network. You can override the default IP addresses before you begin the Final installation of IBM Fusion HCI. If the default IPs are being overridden ensure that they are configured in a private subnet.
- **Network cable and transceiver options**
  This topic has general network cable requirements.

Planning and prerequisites

To ensure a smooth installation of the IBM Fusion HCI in your data center, review the following network requirements and configuration options. This information is intended for your network administrator.

Initial setup and network connection
: The IBM Fusion HCI connects to your data center's network through two high-speed switches. These switches are linked to your core network through a single port channel, which serves as the gateway between the appliance and your network. This connection enables:

  - Administration of the appliance and OpenShift®
  - Network traffic in and out of the cluster

Pre-installation network preparation
: Before the IBM Fusion HCI appliance installation, your network team must prepare your network. An IBM Systems Service Representative (SSR) performs the initial configuration of the appliance and connects it to your pre-configured network using the information you provide.
  Important: Complete the planning worksheets to provide IBM Service Support Representative (SSR) with the IBM Fusion HCI network setup plan.
  To download the worksheets, see IBM Storage Fusion HCI Installation worksheets.

  When you fill the worksheet, check with your network team about whether the CIDR ranges that you plan to use are free on your network.

Pre-installation validation utilities
: IBM Fusion HCI pre-installation validation utilities provide automated checks to verify environment readiness for IBM Fusion HCI deployments. These scripts validate critical software dependencies, network configuration, connectivity, credentials, and external access requirements, helping identify and resolve issues before installation.

  IBM Fusion HCI Management software precheck and validation
  : The openshift-fusion-mgmt-software-prechecks.py script is an interactive pre-installation validation utility designed to assess readiness for IBM Fusion HCI management cluster deployment. The script guides users through environment-specific inputs and validates required software, registry access, credentials, certificates, and external connectivity for both connected and air-gapped installations.

    The script validates the following areas to ensure installation readiness:

    - Container registry validation
    - Credential and configuration validation
    - X.509 Certificate validation
    - Network and endpoint connectivity
    - Installation mode validation
    - Validation result tracking and summarization
    - Log generation for audit and troubleshooting

  IBM Fusion HCI network precheck and validation
  : The fusion-hci-network-prechecks.py script is a network validation utility that verifies connectivity, configuration, and infrastructure readiness for IBM Fusion HCI deployments. The script supports both interactive and CLI execution modes and performs comprehensive network-level checks across addressing, name resolution, connectivity, and time synchronization.

    The script includes the following network validation capabilities:

    - IP configuration validation
    - DNS validation
    - Network connectivity validation
    - NTP configuration and synchronization checks
    - Network interface validation
    - Parallel execution of validation tasks
    - Input validation and error handling
    - Logging and output reporting

Configuring DNS and DHCP

To set up an OpenShift cluster, you need to configure the Domain Name System (DNS) and Dynamic Host Configuration Protocol (DHCP) for each node in the appliance. It involves creating DHCP entries for each MAC address and setting up forward and reverse DNS entries for each IP address.

Before configuring DNS and DHCP, ensure that you have:

- Each node must have a pre-configured MAC address and a MAC address for the bootstrap VM.
- IBM provides a list of all MAC addresses for configuration purposes.

Follow the instructions to create DNS and DHCP entries:

Create DHCP entries
: Create a DHCP entry for each MAC address provided by IBM. It ensures that each node and service node receives an IP address.

Create DNS entries
: Create forward and reverse DNS entries for each IP address that is assigned to the nodes and service nodes.

The following are some key considerations while you create DNS and DHCP entries:

- Ensure that your DHCP server can provide infinite leases. It is required to set a permanent IP address for each node.
- Set the DHCP expiration time to 4294967295 seconds (as specified by RFC 2131) to enable infinite leases.
- In RHEL 8, `dhcpd` does not provide infinite leases. As an alternative, use `dnsmasq` to serve dynamic IP addresses with infinite lease times.

To serve dynamic IP addresses with infinite lease times using the provisioner node, it is recommended using `dnsmasq` instead of `dhcpd`.
For a detailed guide on setting up DNS and DHCP for IBM Fusion HCI, refer to Setting up DHCP for IBM Fusion HCI.

Configuring static IP

To configure static IP, you need to obtain the following IP addresses:

- **Gateway IP**: The IP address of the gateway.
- **DNS IP**: The IP address of the DNS server.
- **Service node IP**: The IP address of the service node.
  For a detailed guide on setting up static IP for IBM Fusion HCI, refer to Setting up Static IP.

These IP addresses are used during the cluster configuration process.

Configuring NTP Server

Ensure that the network administrator has configured an NTP server that is connected to your network.

NTP server requirements

- The IBM Fusion HCI requires a connection to an NTP server to coordinate time across the nodes in the OpenShift cluster.
- Ensure that the NTP server is accessible on the network where the IBM Fusion HCI is connected.

Providing NTP server information

- Provide the IP address of the NTP server in the planning worksheet. The IBM SSR requires this information to complete the initial configuration of the appliance.
- You can provide multiple NTP servers as comma-separated values.

Note: According to Red Hat®, to maintain optimal synchronization in an OpenShift Container Platform cluster, use a minimum of four NTP servers. If four NTPs are not available, then use one or three NTPs servers but never two because it can introduce synchronization issues.

LACP topology

To ensure high availability and redundancy, it is recommended to configure the IBM Fusion HCI high-speed switches to connect to the data center switches using link aggregation.

Configuration recommendations

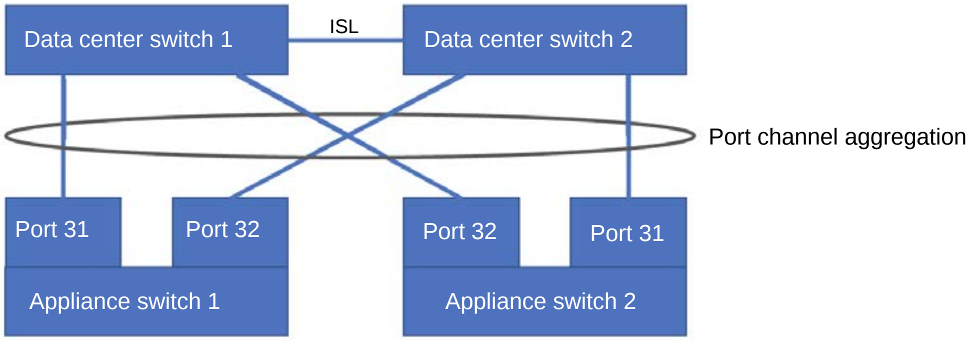
- Use both ports 31 and 32 on each high-speed switch to connect to the data center switches.
- This configuration creates a total of four links, which are aggregated into a single port channel with four times the bandwidth.

Benefits

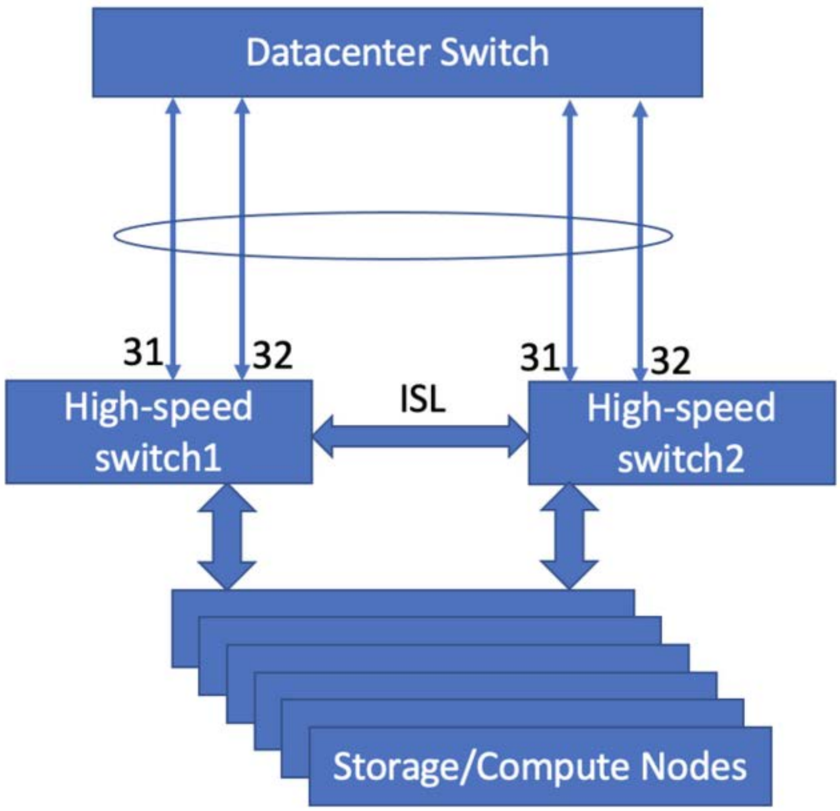
- Provides redundancy and high availability: if any link or switch fails, traffic is automatically balanced between the remaining links.
- Increases bandwidth: the aggregated port channel has four times the bandwidth of a single link.

Switch configuration options

- **Recommended Configuration**: Use two high-speed switches, each connected to the data center switches with two ports (31 and 32).

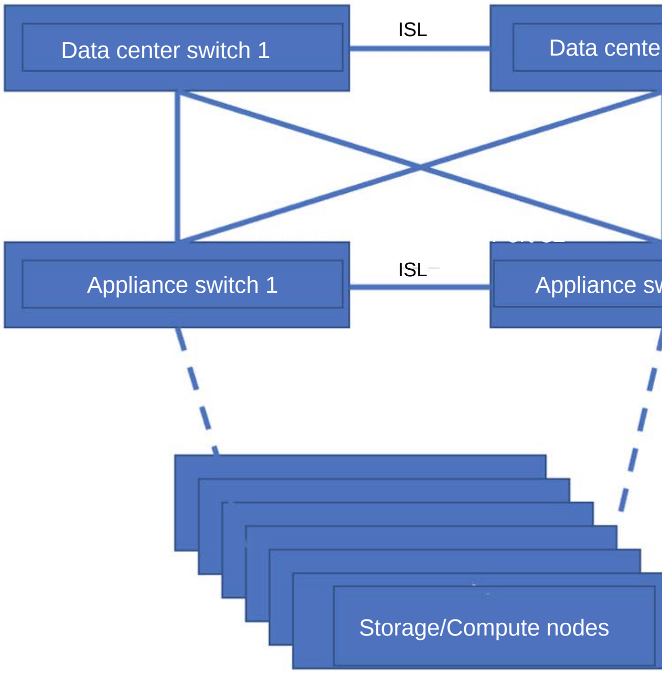


- **Alternative Configuration**: If you prefer to use only one switch, combine data center switch 1 and data center switch 2 with four ports.



By following these configuration recommendations, you can ensure a reliable and high-bandwidth connection between the IBM Fusion HCI and the data center switches.

It is recommended to use an LACP topology to connect the IBM Fusion HCI switches to the data center switches.
Note: LACP is the preferred choice. Support is not available for non-LACP.
Link Aggregation Control Protocol (LACP) is an IEEE standard that is defined in IEEE 802.3ad to dynamically build an Etherchannel. Here, ISL refers to Inter Switch Link.



For IBM Fusion HCI, the recommended LACP settings are as follows:

- Rate setting is Fast
- Mode is Active-Active setting

For more information about network cable requirements to connect the appliance switches to the data center switches, see Network cable and transceiver options.

VLAN planning

The IBM Fusion HCI System requires two VLANs for communication between the data center switches and the appliance high-speed switches.

Supported VLAN range
The IBM Fusion HCI System supports VLANs from 1 to 4094.

The following are the required VLANs:

OpenShift customer VLAN

- A default VLAN used to access OpenShift.
- Requires a name and ID, which must be recorded in the planning worksheet for initial configuration.
  For example, **`OpenShift-Customer`** (name) and **`100`** (ID).

Native VLAN

- Handles discard traffic.
- Typically uses VLAN ID 1, but verify your data centers default ID.
  For example: **`Native`** (name) and **`1`** (ID).

Important: The storage VLAN is applicable only for the Global Data Platform. Data Foundation does not contain a storage VLAN.

Storage VLAN

- Used by the internal 100G storage network.
- The default value is 3201, but can be changed during initial configuration.
  For example: **`Storage`** (name) and **`3201`** (ID).

Reserved VLAN IDs
The following VLAN IDs are reserved and must not be used:

- VLAN 4091 (internal Provisioning network)
- VLANs 3725-3999 (internal switches).

VLAN naming conventions

- Link names and VLAN IDs must not contain special characters.
- Interface names can be up to 15 characters long.
- The first character cannot be a number, and dashes (-) are not allowed in the name.

Important considerations

- All internal VLANs used within the IBM Fusion HCI must have unique VLAN IDs.
- The system does not provide routing between VLANs; routing functions must be provided at the data center core network.

- It is recommended to configure the uplink in trunk mode rather than access mode as it allows addition of VLANs to the uplink.

Network for service node

Network overview for service node

The node supports three types of networks, namely provisioning network, Bare Metal network, and Customer DC network. Both provisioning network and Bare Metal network are mandatory and the Customer DC network is optional.

To connect the DC network directly to the service node, connect RJ45 network cable to service node slot 4 port 4. The Port 4 of the quad-port OpenShift Container Platform 1Gb NIC is used to connect the Service Node to the client network. IA Cat5e (or better) Ethernet cable with RJ45 connectors is needed to make the connection from the Service Node to the client switch. IBM does not provide this cable so you must arrange cat5e cable that is commonly available in any data center.

- **Bare Metal network**
  Bare Metal node has two 100G ports (with Data Foundation rack), which internally connects to the high-speed switch of the rack. This network is used to reach the OpenShift cluster on the rack. This network is configured by the IBM Fusion HCI installation process.
- **Provisioning network**
  It has two ports 1G Nic to connect to the provisioning network of the OpenShift. The device comes pre-wired and pre-configured right from the factory.
- **Customer DC network**
  This network is used to access service node directly for maintenance and out of band management purposes. It is recommended that you provide an IP, gateway, subnetmask to the SSR at the time of initial setup. Also, check whether you filled in network planing sheet with the right details.

  It has 1 GbE Nic that is used to connect to the customer data centre for out of band access.

For more information about the hardware configuration of the service node, see Service node.

Prerequisites for service node

Ensure that you meet the following prerequisites for the service node:

- A separate VLAN must be available for the Customer DC connectivity to the service node.
- Ensure to provide the DHCP-assigned IP address to the bare metal network interface of the service node.
  Note: The DHCP and DNS lookup and reverse lookup records are required for service node similar to other OpenShift nodes.
- Ensure that the following firewall ports are allowed for the service node:
  - Port 22 for SSH access.
  - Port 443, 3000, and 3900 for stage 2 user interface (the 443 and 3900 are allocated for IMM console access through port forwarding on service node if needed).

Network for multi-rack HA

Planning and prerequisites for high-availability multi-rack, 3-zone high-availability, and expansion racks:

- The provisioning network (VLAN 4091) and OpenShift Bare Metal network must be extended between the racks and must be on the same layer2 domain across all three racks or sites.
- The recommended uplink bandwidth requirement is 25G and higher.
- The latency must be in the range of less than 10 ms RTT. For more information, see Guidance for OCP Deployments Spanning Multiple Sites.
- For Data Foundation multi-rack setup, ensure that you meet the following mandatory requirements from lab switches setup:
  1. The provisioner VLAN 4091 must be extended on all the intermediate switches between the racks.
  2. The OpenShift Container Platform VLAN must be extended on all the intermediate switches between the racks.
- Each rack must have the same number of storage nodes. Ensure that the number of NVMe drives are uniform across storage nodes in the multi-rack solution.

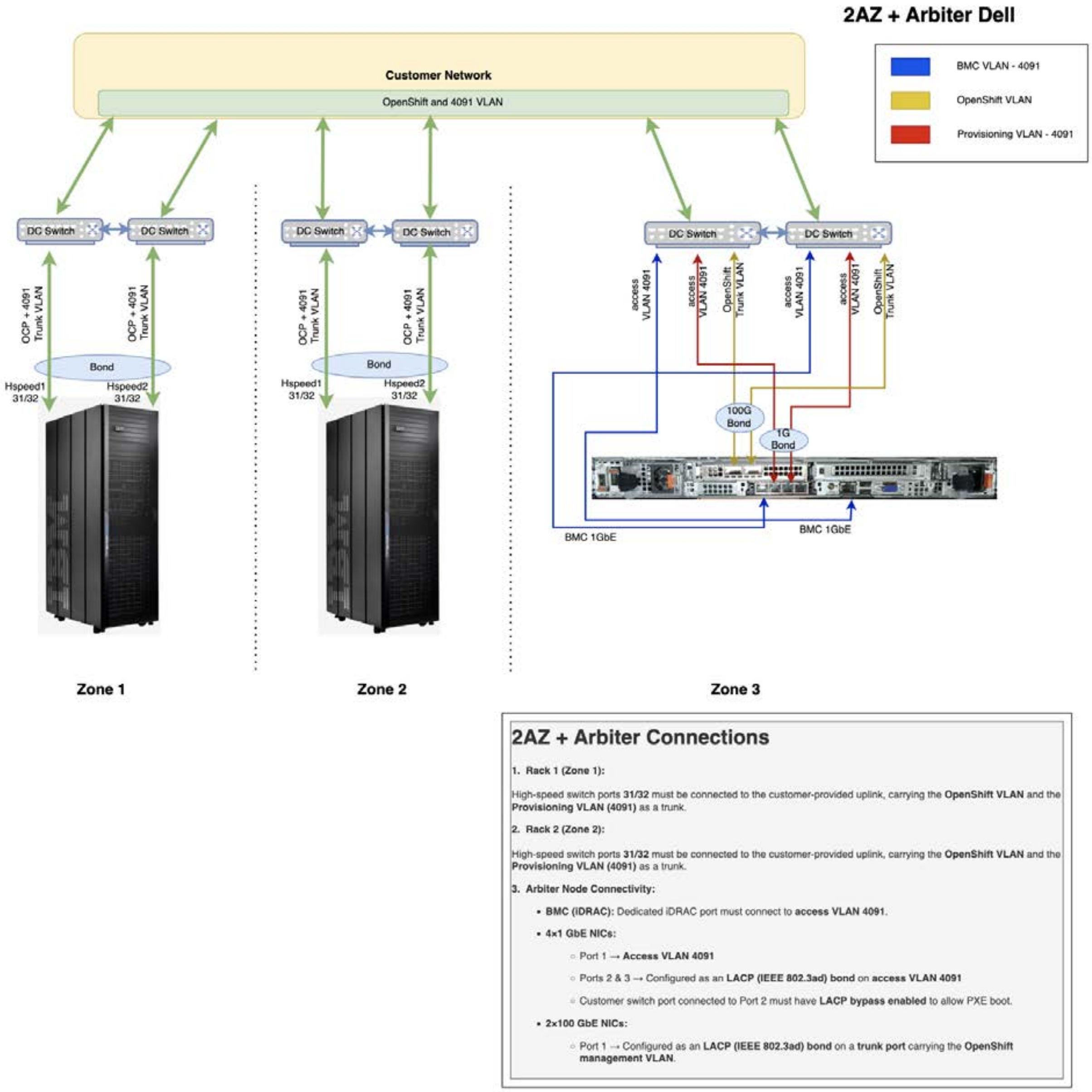
Network for two-availability-zone and arbiter node (2AZ + arbiter node) configuration

Ensure that the following network planning and prerequisites are met:

- Two IBM Fusion HCI are placed in two different zones, and the arbiter node is placed in a third zone.
- The latency between servers or zones must be less than 10 ms round-trip time (RTT). For more information, see Guidance for OCP Deployments Spanning Multiple Sites.
  Note: If you are planning a 2AZ + arbiter node deployment with latency higher than 10 ms RTT, contact IBM Support.
- All nodes in both racks and the arbiter node must be of the same type. For example, if nodes in both racks are Dell type, the arbiter node must also be Dell type.
- The arbiter node is connected to the switch using the following ports:
  - 1G provisioning port
  - 1G IMM port
- OpenShift Container Platform VLANs and Provisioning VLANs (4091) are extended to all three zones for mutual discovery and communication.
- If you use DHCP, ensure that DHCP and DNS are configured for both racks and the arbiter node by following the instructions that are mentioned in Setting up DHCP for IBM Fusion HCI and Setting up DNS.
- If you use static IP, ensure that static IP and DNS are configured for both racks and the arbiter node by following the instructions that are mentioned in Setting up Static IP and Setting up DNS.
- All nodes of both racks and the arbiter node are powered on.
- The Stretch cluster (2 AZ) tile is selected, and a unique value in the Availability zone field are added for both racks during installation using the IBM Fusion HCI web console (stage 1).
  Note: The keyword `arbiter` cannot be used as a Availability zone field value.
- Stage 1 deployment can start with any rack, which is considered as the first rack. After Stage 1 on the first rack completes, start Stage 1 on the last rack. During this process, the Arbiter node and the first rack are discovered. After Stage 1 completes on the last rack, all discovered zones appear on the final page. The order of the intermediate racks does not matter. Any rack can be treated as the last rack.
- The same number of SSD or NVMe disks are available across storage nodes.
- The firewall on the provisioner or service node is disabled. Otherwise, the racks and arbiter node cannot discover each other.

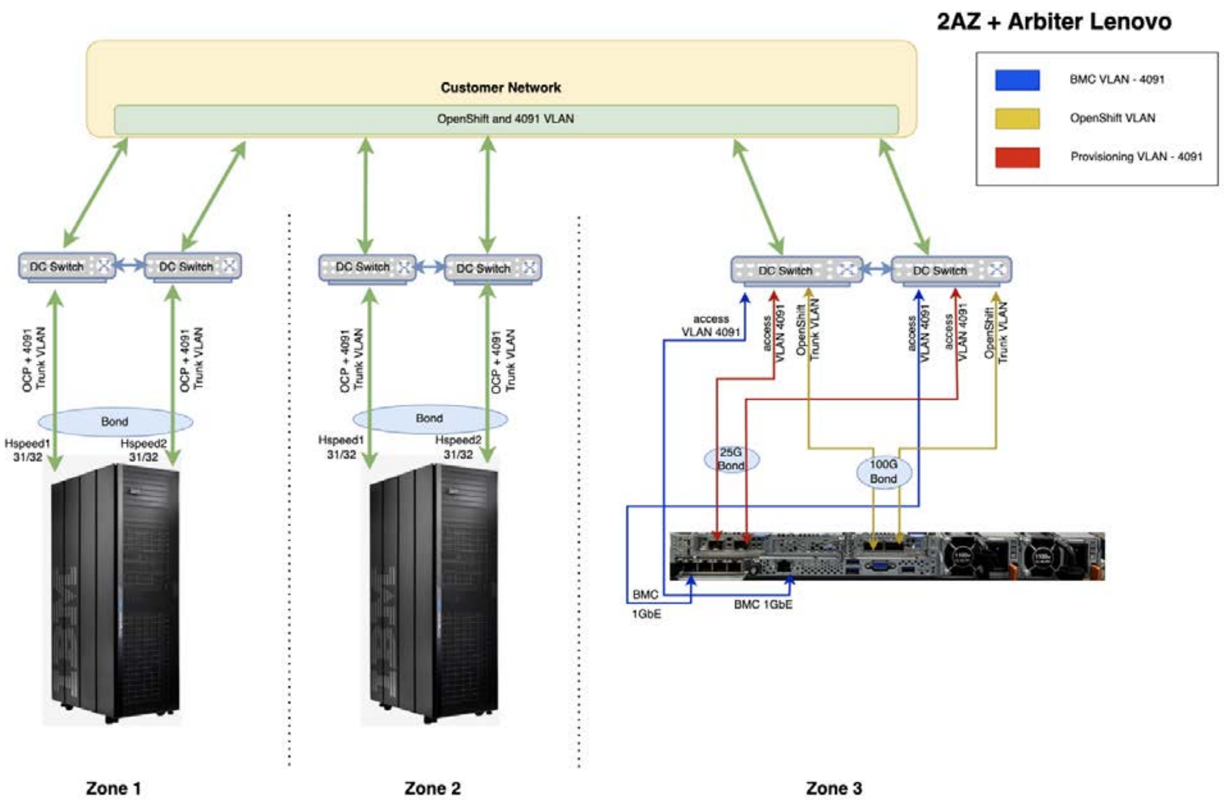
Network configurations for 2AZ + arbiter node are as follows:

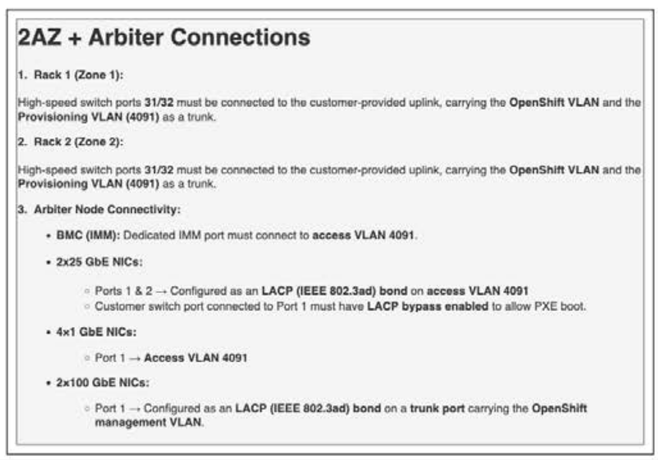
Figure 1. Network configurations for Dell



Important: As shown in Figure 1, four 1 GbE Ethernet cables with RJ45 connectors and two 100 GbE AOC cables with QSFP28 transceivers are provided to connect the arbiter node to the network. No additional cables are required from the client.

Figure 2. Network configurations for Lenovo





2AZ + Arbiter Connections

1. **Rack 1 (Zone 1):**

High-speed switch ports **31/32** must be connected to the customer-provided uplink, carrying the **OpenShift VLAN** and the **Provisioning VLAN (4091)** as a trunk.

2. **Rack 2 (Zone 2):**

High-speed switch ports **31/32** must be connected to the customer-provided uplink, carrying the **OpenShift VLAN** and the **Provisioning VLAN (4091)** as a trunk.

3. **Arbiter Node Connectivity:**

- **BMC (IMM):** Dedicated IMM port must connect to **access VLAN 4091**.
- **2x25 GbE NICs:**
  - Ports 1 & 2 → Configured as an **LACP (IEEE 802.3ad) bond** on **access VLAN 4091**
  - Customer switch port connected to Port 1 must have **LACP bypass enabled** to allow PXE boot.
- **4x1 GbE NICs:**
  - Port 1 → **Access VLAN 4091**
- **2x100 GbE NICs:**
  - Port 1 → Configured as an **LACP (IEEE 802.3ad) bond** on a **trunk port** carrying the **OpenShift management VLAN.**

Important: As shown in Figure 2, two 1 GbE Ethernet cables with RJ45 connectors and two 100 GbE AOC cables with QSFP28 transceivers are provided to connect the arbiter node to the network. Two additional 1 GbE Ethernet cables are included as spares. The client must provide two 25 GbE cables with SFP28 connectors.

Network planning and prerequisites for remote support connection

Remote support is a service that enables IBM Support representative to remotely access your rack and accomplish essential tasks, such as installation, maintenance, and troubleshooting. To create a remote support connection, ensure that the additional configuration is required with a customers firewall or proxy, the following table describes required IP and port configuration.

Table 1. Ports, Relay Hosts, and IP Information

Hostname / GEO	IP	Ports
aos.us.ihost.com	72.15.208.234	443
Americas Broker (4.0 Sessions)	150.238.213.135	443
Americas Broker (4.0 Sessions)	72.15.223.60	443
Americas Broker (4.0 Sessions)	72.15.223.62	443

To a remote support connection to work, the customer must allow encrypted TLS outbound traffic for the server `72.15.223.60` on port 443, `150.238.213.135` on port 443 or `72.15.223.62` on port 443.

To use geographically specific relay servers and realize improved throughput, the customer must also allow encrypted non-SSL (TLS encryption) outbound traffic to one of the geographically specific relay servers:

- 72.15.208.234 aos.us.ihost.com (hosted in North America, best for most geographies)
- 150.238.213.135 aosback.us.ihost.com (hosted in North America, best for most geographies)
- 72.15.223.60 aosrelay1.us.ihost.com (hosted in North America, best for most geographies)
- 72.15.223.62 aoshats.us.ihost.com (hosted in North America, best for most geographies)

Remote support connection automatically chooses the broker, which provides the best end-to-end performance. All broker servers are available from all geographies, with performance typically better from the server closest to the customer system.

For more information about remote support, see Remote support.

MTU requirement for Backup & Restore Hub and Spoke configuration

This prerequisite is applicable when you create a relationship between two IBM Fusion HCI OpenShift Container Platform clusters. If you plan to make one IBM Fusion HCI as Backup & Restore Hub and another as Spoke, then make sure that the following things are met before you install IBM Fusion HCI:

1. The same MTU (maximum transmission unit) values are set on both clusters.
2. Confirm with network admins that planned MTU value is supported by your data center infrastructure as well.
3. All the devices and routers between these two clusters are configured with the same MTU.

Planning for Hosted Control Plane

In the same CIDR range (number of clusters), you must have a set of free available IPs (no DHCP or DNS is required). It is based on the number of Hosted Control Plane clusters that you plan to create in the rack.

Planning and prerequisites for remote mount support

Ensure that the following prerequisites are met for remote mount support:

For IBM Storage Scale Erasure Code Edition (ECE)

- The storage VLAN (default 3201) must be available and configured on the customer network. If the default storage VLAN 3201 is not available on the customer network, contact IBM Support to change the default VLAN on IBM Fusion HCI.
- The default gateway from the storage VLAN must be configured and reachable from IBM Fusion HCI.
- The routing must be in place on the customer network from the storage gateway to the external IBM Storage Scale System
- The IBM Storage Scale System must also be configured to support MTU 9000 along with other devices en route router, switches. The default MTU of IBM Fusion HCI is 9000.

For Red Hat OpenShift Data Foundation

- The routing must be in place on the customer network from the storage gateway to the external IBM Storage Scale System
- The IBM Storage Scale System must also be configured to support MTU 9000 along with other devices en route router, switches. The default MTU of IBM Fusion HCI is 9000.

Configuring Spanning Tree Protocol (STP)

Primary recommendation: Disable STP on IBM Fusion HCI switches
: For the highest stability and performance, disable STP on the IBM Fusion HCI leaf switches.

  - Reason: The IBM Fusion HCI switches act as edge or leaf nodes in a controlled Multi-Chassis Link Aggregation (MLAG) configuration.
  - Loop prevention: LACP (802.3ad) and the MLAG peer link manages redundancy and loop prevention.
  - Safety: Since these leaf switches with no downstream switches are connected, the risk of a loop is localized to the server ports.

Alternative configuration for enabling STP
: If your network policy requires STP to be enabled for all infrastructure, or if there is a risk of accidental loops from downstream patching, follow these strict configuration guidelines to ensure compatibility with the IBM Fusion HCI cluster.

  Match the mode: RSTP (802.1w)
  : The IBM Fusion HCI switches strictly use RSTP. Your uplink switches must match with this configuration or you can use MSTP.

    Note: Do not use Cisco's PVRST or PVST+ on the uplink, as the IBM Fusion HCI switches do not process per-VLAN BPDUs correctly, leading to blocked ports or 30 second listening or learning delays.

  Root bridge placement
  : Configure the customer uplink or core switches as the STP root bridge. IBM Fusion HCI switches use a fixed priority of `61440` and must not become the root of the network topology.

  Uplink port configuration
  : The ports on the customer switch connecting to the IBM Fusion HCI bond must be configured as normal (network) ports.

    Ensure the following settings:

    - BPDU Guard: Must be disabled on the uplink bond.
    - BPDU Filter: Must be disabled. Enabling this setting prevents the network from detecting loops.
    - PortFast or Edge: Must be disabled. These ports connect to other switches (the IBM Fusion HCI pair) and must perform the standard RSTP handshake.

Network definitions

This section provides a comprehensive overview of network definitions, including key terms and concepts.
Important: While you go through these definitions, different networking teams use different terminologies. For example, Mellanox or Cisco.

Use LACP aggregation
: Whether the recommended LACP topology is being used.
  Note: LACP is the preferred choice. Use the `no aggregation` topology only if LACP is not possible as it does not provide the redundancy most clients require.

Link name
: The name to assign to the aggregated link created by the recommended LAG topology.

Lag ID
: The default LAG ID is 250, and you only need to customize the ID if it is already in use on your network. A scenario where this might occur is if you have multiple IBM Fusion HCI appliances on the same network.

Spanning tree enabled
: If you are not using the recommended LAG topology, specify whether spanning tree is enabled on the network or not.

OpenShift VLAN name
: The name to assign to the OpenShift Customer VLAN.

OpenShift VLAN ID
: The VLAN ID to assign to the OpenShift Customer VLAN.

VLAN
: A virtual LAN (VLAN) is an isolated broadcast domain that is created within a switch. Each VLAN created within a switch is isolated from other VLANs. The network traffic can pass from one VLAN to another by adding a routing device. The routing functions must be provided at the data center core network.

Native VLAN ID
: The VLAN ID to assign to the Native VLAN if you aren't using the default of 1.

Storage VLAN ID
: The VLAN ID to assign to the 100G storage network if you aren't using the default of 3201. You might only normally customize this if you are configuring a Metro Disaster Recovery configuration between two IBM Fusion HCI clusters.

MLAG
: A Multi-Chassis Link Aggregation Group (MLAG) allows for multi-system link aggregation and facilitates active-active uplinks of access layer switches. An MLAG with no Spanning Tree configured avoids the wasted bandwidth that is associated with links that are blocked by the spanning tree.
To ensure high availability for the system, IBM Fusion requires MLAG and link aggregation for the switches.

Switch LAG ID
: This Switch LAG ID is used by high-speed switches inside the IBM Fusion rack, which is not related to the LACP ID on (external) switch. It is unique for each IBM Fusion rack and can be less than 250.
The high-speed switch (MLAG PAIR) that gets connected to customer switch must have a unique system MAC address (MLAG ID). This MLAG ID is used as source MAC address for traffic that is sourced from MLAG PAIR, such as STP BPDUs. The MLAG ID is internally derived from the Switch LAG ID. The specific Switch LAG ID is used as the last octet for "**`44:38:39:ff:00:xx`**".

Layer 2 connections
: The connections from all the IBM Fusion high-speed switches to the data center core network and customer management network are all layer 2 connections. IBM Fusion supports the Link Aggregate Control Protocol (LACP) type of layer 2 aggregation.
The following characteristics describe layer 2 connections:

- Layer 2 is considered switching and is done at the hardware layer.
- Layer 2 is in the same broadcast domain or local network.
- Layer 2 finds adjacent partners by MAC address.

Layer 3 connections
: IBM Fusion does not participate in any layer 3 routing or firewall functions. These functions are done in the data center core network.
The following characteristics describe layer 3 connections:

- Layer 3 is considered routing and is done at the software layer.
- Layer 3 knows how to traverse multiple networks (hops).
- Layer 3 finds adjacent partners by IP address.

Spanning tree
: In the networking stages of the installation for IBM Fusion, an option to enable or disable the Spanning Tree is available. When you enable it, the spanning tree is enabled on the high-speed switches. By default, the option is disabled on these switches.The high-speed switches in the IBM Fusion rack support only rapid spanning-tree (RSTP) modes. It is also compatible with PVST and PVST+. Customer data networks that are running older spanning-tree methods (MST) are not supported.
Note: IBM Fusion sets the spanning-tree bridge priority to `32768` as the default value. To avoid IBM Fusion high-speed switches from becoming the spanning-tree root, the customer switch must have priority less than `32768`. For more information, see Spanning Tree and Rapid Spanning Tree - STP.

Port type
: Specify whether you are using VLAN trunking or access ports.

Access
: An access port provides access to a single VLAN only. Typically, the packets on an access port are raw Ethernet frames (untagged packets).

Trunk
: A trunk connection is used to pass traffic from multiple VLANs between two switches. All trunks use the IEEE 802.1Q standard. This link can be a single wire or one of the aggregations methods.

Transceiver
: Specify the type of cable that is being used to connect the HCI high-speed switches with your data center switches. See Hardware Compatibility List.

NTP server
: The IP address of the NTP server that is used by IBM Fusion HCI.

Aggregate links
: Link aggregation is to combine multiple network connections in parallel.
The standard-based negotiation protocol, which is known as IEEE 802.1ax Link Aggregation Control Protocol (LACP), is a way to dynamically build an Etherchannel.

- **Static or DHCP IP address assignment methods**
  IP address assignment is a critical settings on your network devices. You must properly assign IP addresses for your hosts so that it participates on your network. IBM Fusion offers you the option to choose between DHCP reservation-based IP management and Static IP (recommended) for your needs.

Static or DHCP IP address assignment methods

IP address assignment is a critical settings on your network devices. You must properly assign IP addresses for your hosts so that it participates on your network. IBM Fusion offers you the option to choose between DHCP reservation-based IP management and Static IP (recommended) for your needs.

Static
: These addresses do not change that makes it easy to find on the network. The static IP option eliminates the need to create Mac based reservations in DHCP server of your data center.

Dynamic
: Automatically assigned to network devices from a Dynamic Host Configuration Protocol (DHCP) server. You must create Mac based reservations in DHCP server of your data center.

Advantages of Static IP address assignment method

Before you select your IP address assignment method, consider the following advantages of the Static IP method:

- The Static IP addresses method requires less configuration effort, as you only need to set up DNS. Unlike the Dynamic IP addresses method, you do not have to configure the DHCP.
- When you scale out by adding new nodes, the effort is minimal as you only need to set up DNS. Unlike the Dynamic IP addresses method, there is no need to configure the DHCP for each new node.
- In the event of a NIC replacement in the field, a new MAC address is assigned, which requires an update to the DHCP configuration. However, with Static IP addresses, this step is eliminated, as the IP address remains unchanged.
- If your network does not have a dedicated DHCP server or it is mounted as an add-on service, then the Static IP addresses option eliminates the need for DHCP deployment.

- **Configuring sub-domains**
  A network administrator must configure a subdomain or sub-zone where the canonical name extension is the cluster name.
- **Setting up DHCP for IBM Fusion HCI**
  As a prerequisite to IBM Fusion HCI installation, update Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) with the information that is detailed in the planning worksheet for your system.
- **Setting up Static IP**
  Steps to set up the static IP for IBM Fusion HCI appliance.
- **Setting up DNS**
  Steps to set up the DNS server for IBM Fusion HCI appliance. It is applicable for both static and DHCP.

Configuring sub-domains

A network administrator must configure a subdomain or sub-zone where the canonical name extension is the cluster name.

```
<cluster_name>.<base_domain>
```

For example:

```
test-cluster.example.com
```

Red Hat® OpenShift® Container Platform includes functionality that uses cluster membership information to generate A/AAAA records. This resolves the node names to their IP addresses. After the nodes are registered with the API, the cluster can disperse node information without using `CoreDNS-mDNS`. This eliminates the network traffic that is associated with multicast DNS.

Setting up DHCP for IBM Fusion HCI

As a prerequisite to IBM Fusion HCI installation, update Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) with the information that is detailed in the planning worksheet for your system.

Before you begin

Download the installation preparation worksheet and review the network preparation section. To download the worksheets, see IBM Fusion HCI Installation worksheets.

- Ensure that you have the MAC addresses from your IBM Technical Seller with whom you engaged with. If you do not have your IBM Technical Seller contact information, see the planning worksheets for contact details.
  To download the worksheets, see IBM Fusion HCI Installation worksheets.

- The sample files that are used in this section are based on standard DNS/DHCP services available on Red Hat® Enterprise Linux 8.

About this task

Important: If you are using static IP, then you do not have to do a DHCP setup.
You can access IBM Fusion OpenShift® Container Platform cluster nodes on your network. As such, each cluster node requires an IP address that is accessible on this network. During installation, each cluster node obtains its IP address from an external DHCP server based on the MAC address of the cluster node. As a prerequisite for IBM Fusion HCI installation, you must reserve few IP addresses for OpenShift:
Note: All these IPs must be in the same range.

- One IP address for the OpenShift API server endpoint
  For example:
  - API in DNS lookup file.

    ```
    api                    IN   A       10.44.100.143
    ```

  - API in DNS reverse lookup file.

    ```
    143      IN       PTR      api.isf.mycompany.com.
    ```

- One IP address for the wildcard ingress endpoint (to access your workloads)
  For example, ingress in DNS.

  ```
  ; Ingress LB (apps)
  *.apps            IN      A       10.44.100.144
  ```

- One IP address for each cluster node (control, compute, AFM, and GPU nodes)
- One IP address for the bootstrap node (it is temporary and is only required during the OpenShift Container Platform installation)
  For example:
  - Bootstrap in DHCP.

    ```
    host bootstrap.isf.mycompany.com { option host-name "bootstrap.isf.mycompany.com"; hardware ethernet
    00:16:3e:e0:31:65; fixed-address 10.44.100. 151; }
    ```

  - Bootstrap in DNS lookup.

    ```
    bootstrap    IN   A    10.44.100.151
    ```

  - Bootstrap in DNS reverse lookup.

    ```
    151      IN       PTR      bootstrap.isf.mycompany.com.
    ```

Note:

- For each cluster node and the bootstrap node, you must add DHCP reservations to your DHCP configuration.

A single sample DHCP reservation record for a cluster node looks similar to the following example:

```
host control-1-ru2.isf.mycompany.com { option host-name "control-1-ru2.isf.mycompany.com"; hardware ethernet 04:3f:72:f7:2f:76;
fixed-address 10.44.100.145;}
```

IBM provides the node's hostname (for example, control-1-rux) and hardware Ethernet (MAC) address and you cannot change them. The fixed-address that you provide is the IP address that gets assigned to the cluster node.

A fully functional DNS environment is a requirement for IBM Fusion HCI and OpenShift Enterprise to work correctly. Adding entries into the /etc/hosts file is not enough because that file is not copied into containers that are running on the platform. For more information about DHCP and DNS, see OpenShift documentation.

Mapping of rack unit with OpenShift host name in a single rack with 9155-C10, 9155-C14, 9155-G03 servers:

For rack that supports 9155-C00, 9155-C01, 9155-C04, 9155-C05, 9155-G02, 9155-G03, and 9155-F01 servers, see IBM Fusion HCI 2.8.x documentation for host name conventions.

For rack that supports 9155-C010, 9155-C014, G03, and Service node, use the following table as reference for host name conventions:

Rack unit	Host name	Node type
RU2	control-1-ru2.<domainname>	Storage or compute-only
RU3	control-1-ru3.<domainname>	Storage or compute-only
RU4	control-1-ru4.<domainname>	Storage or compute-only
RU5	compute-1-ru5.<domainname>	Storage or compute-only
RU6	compute-1-ru6.<domainname>	Storage or compute-only
RU7	compute-1-ru7.<domainname>	Storage or compute-only
RU23	servicenode-1.<domainname>	Service node

After RU7 any type of nodes can be installed on the next available RU. The naming convention is based on the RU number in the format **compute-1-ru<x>.<domainname>**.

Mapping of rack unit with OpenShift host name for a high availability multi rack cluster:

Rack unit	Host name			Node type
	Rack 1	**Rack 2**	**Rack 3**	
RU2	control-1-ru2.<domainname>	control-2-ru2.<domainname>	control-3-ru2.<domainname>	Storage or compute-only
RU3	compute-1-ru3.<domainname>	compute-2-ru3.<domainname>	compute-3-ru3.<domainname>	Storage or compute-only
RU4	compute-1-ru4.<domainname>	compute-2-ru4.<domainname>	compute-3-ru4.<domainname>	Storage or compute-only
RU5	compute-1-ru5.<domainname>	compute-2-ru5.<domainname>	compute-3-ru5.<domainname>	Storage or compute-only
RU6	compute-1-ru6.<domainname>	compute-2-ru6.<domainname>	compute-3-ru6.<domainname>	Storage or compute-only
RU7	compute-1-ru7.<domainname>	compute-2-ru7.<domainname>	compute-3-ru7.<domainname>	Storage or compute-only
RU23	servicenode-1.<domainname>	compute-2-ru23.<domainname>	compute-3-ru23.<domainname>	Compute-only and Service node in base rack

After RU7 any type of nodes can be installed on the next available RU. The naming convention is based on the RU number in the format **compute-1-ru<x>.<domainname>**. In a multi-rack setup, the service node can be added to any one of the racks.

Important: Ensure that in a two-availability-zone and arbiter node (2AZ + arbiter node) configuration, the arbiter node hostname is configured as **arbiter-999.<domain.com>**.

Procedure

1. Place your DHCP servers in the same VLAN as IBM Fusion HCI unless the DHCP relay agent is used.

2. Verify whether the DNS server is accessible from outside the IBM Fusion HCI by using either the `ping <server-ipaddress>` or the `nslookup <server-ipaddress>` command.
   Note: The DNS server can be anywhere but must be reachable, that is, the installation and cluster must have access to your network.
3. After the verification is complete, set up DHCP reservation for each node. For more information about setting up the DHCP, see Setting up DHCP.
4. Set up the DNS reservation for each node. For more information about setting up the DNS, see Setting up DNS.
   Here, ensure that you use the same `domain-name-servers` value and option host name that you used in DHCP configuration as they are interconnected. Ensure that the fixed addresses that you configure here is same in DHCP as well.
   Note: Ensure that the subdomain consists of lowercase alphanumeric characters, '-' or '.', and starts and ends with an alphanumeric character. For example, `example.com`. The regex used for validation is 'a-z0-9?(\.a-z0-9?)*')",

- **Setting up DHCP**
  Steps to set up the DHCP server for IBM Fusion HCI appliance.

Setting up DHCP

Steps to set up the DHCP server for IBM Fusion HCI appliance.

About this task

Important: If you are using static IP, then you do not have to do a DHCP setup.
All storage and GPU nodes in the IBM Fusion appliance must be in the same network segment (subnet). The nodes must have access to a DHCP server to provide IP addresses and other network parameters, for example, default gateway and DNS address.
Note: The samples in this procedure are based on Red Hat® Enterprise Linux. If you are on Microsoft® Linux®, then use equivalent commands.
Use the DHCP server to provide all the nodes of the OpenShift® cluster that you are about to install, with IP information that corresponds to the mac address of the node. It is the reason for you to use fixed-address directive for DHCP to statically map all the mac addresses of the nodes to the required IP addresses. Also, the DHCP server must provide these DHCP options in its DHCP responses for the nodes to be configured correctly:

```
option routers
option broadcast-address
option subnet-mask
option domain-name-servers
option domain-name
option domain-search
option host-name
```

Note: For the host-name option, it must be full FQDN value.

Procedure

1. Edit `/etc/dhcp/dhcpd.conf` file to update values for routers, broadcast-address, subnet-mask, domain-name-servers.
   Note: The following sample `dhcpd.conf` (DHCP configuration file) provides bootstrap IP addresses for the control nodes and computes nodes in the `10.44.100.0/24` subnet. In this example, the OpenShift cluster name is `isf` and the DNS domain is `mycompany.com`. Together, the cluster name and domain name form the cluster subdomain. The MAC addresses are provided by IBM.

```
authoritative;
ddns-update-style interim;
default-lease-time 489776228;
max-lease-time 489776228;


       option domain-name-servers      10.44.100.142;

       subnet 10.44.100.128 netmask 255.255.255.128 {
        interface bond0;
        option routers                  10.44.100.129;
        option broadcast-address        10.44.100.255;
        option subnet-mask              255.255.255.128;
        group {

                option domain-name              "isf.mycompany.com";
                option domain-search            "isf.mycompany.com";
                host control-1-ru2.isf.mycompany.com { option host-name "control-1-ru2.isf.mycompany.com"; hardware 
ethernet 04:3f:72:f7:2f:76; fixed-address 10.44.100.145; }
                host control-1-ru3.isf.mycompany.com { option host-name "control-1-ru3.isf.mycompany.com"; hardware 
ethernet 04:3f:72:f7:31:2e; fixed-address 10.44.100.146; }
                host control-1-ru4.isf.mycompany.com { option host-name "control-1-ru4.isf.mycompany.com"; hardware 
ethernet 04:3f:72:f5:a7:2a; fixed-address 10.44.100.147; }

                host compute-1-ru5.isf.mycompany.com { option host-name "compute-1-ru5.isf.mycompany.com"; hardware 
ethernet 04:3f:72:f5:a7:26; fixed-address 10.44.100.148; }
                host compute-1-ru6.isf.mycompany.com { option host-name "compute-1-ru6.isf.mycompany.com"; hardware 
ethernet 04:3f:72:f5:a7:16; fixed-address 10.44.100.149; }
                host compute-1-ru7.isf.mycompany.com { option host-name "compute-1-ru7.isf.mycompany.com"; hardware 
ethernet b8:59:9f:e4:46:6e; fixed-address 10.44.100.150; }

              # The compute-1-ru23.isf.mycompany.com is for service node
        host servicenode-1.gen2003.mycompany.com { option host-name "servicenode-1.gen2003.mycompany.com"; hardware 
ethernet a0:88:c2:e5:fc:74; fixed-address 10.0.20.222; 

}
        deny unknown-clients;
```

Note: Here, the same `domain-name-servers` value and option host name must be used when you configure in DNS as they are interconnected. Ensure that the fixed addresses that you configure here must be the same in DNS as well.

This example is for a single rack. If you want to configure DHCP for a high availability multi rack cluster or expansion rack, then add rest of the nodes in the cluster according to your host name. For more information about the hostname, see Setting up DHCP for IBM Fusion HCI.

If you want to configure using dnsmasq:

```
user=dnsmasq
group=dnsmasq
interface=ens19
domain=isf.mycompany.com
dhcp-range=10.10.9.10,10.10.9.127,9.9.9.0,infinite
conf-dir=/etc/dnsmasq.d,.rpmnew,.rpmsave,.rpmorig



dhcp-host=08:c0:eb:ff:40:66,control-0,10.10.9.12,infinite,set:control-0
dhcp-option=tag:control-0,12,control-0.isf.mycompany.com

dhcp-host=08:c0:eb:ff:43:36,control-1,10.10.9.13,infinite,set:control-1
dhcp-option=tag:control-1,12,control-1.isf.mycompany.com

dhcp-host=08:c0:eb:ff:40:86,control-2,10.10.9.14,infinite,set:control-2
dhcp-option=tag:control-2,12,control-2.isf.mycompany.com

dhcp-host=08:c0:eb:ff:2f:86,compute-0,10.10.9.15,infinite,set:compute-0
dhcp-option=tag:compute-0,12,compute-0.isf.mycompany.com

dhcp-host=08:c0:eb:ff:3e:92,compute-1,10.10.9.16,infinite,set:compute-1
dhcp-option=tag:compute-1,12,compute-1.isf.mycompany.com

dhcp-host=08:c0:eb:ff:30:42,compute-2,10.10.9.17,infinite,set:compute-2
dhcp-option=tag:compute-2,12,compute-2.isf.mycompany.com

dhcp-host=08:c0:eb:cb:09:2e,compute-13,10.10.9.18,infinite,set:compute-13
dhcp-option=tag:compute-13,12,compute-13.isf.mycompany.com

dhcp-host=08:c0:eb:cb:08:8e,compute-14,10.10.9.19,infinite,set:compute-14
dhcp-option=tag:compute-14,12,compute-14.isf.mycompany.com

dhcp-host=b8:ce:f6:74:ef:d6,compute-15,10.10.9.20,infinite,set:compute-15
dhcp-option=tag:compute-15,12,compute-15.isf.mycompany.com

dhcp-host=08:c0:eb:ff:31:66,compute-16,10.10.9.21,infinite,set:compute-16
dhcp-option=tag:compute-16,12,compute-16.isf.mycompany.com

address=/api.isf.mycompany.com/10.10.9.6
address=/api-int.isf.mycompany.com/10.10.9.6
address=/apps.isf.mycompany.com/10.10.9.7
```

2. After you update the configuration file, restart the service for the changes to take effect.
   For example, on Linux®, run the following command:

   ```
   sudo /etc/init.d/dnsmasq restart
   ```

   or

   ```
   sudo systemctl restart network-manager
   ```

   Note: The first command works on RHEL 7 or lower. For RHEL 8 or higher, use **sudo systemctl restart network-manager**/

Setting up Static IP

Steps to set up the static IP for IBM Fusion HCI appliance.

Before you begin

- Download the installation preparation worksheet and review the network preparation section. To download the worksheets, see IBM Fusion HCI Installation worksheets.
- If you plan to use Static IP instead of DHCP, you must have selected the Static IP assignment method in Stage 1 of the installation.
- A fully functional DNS environment is a requirement for IBM Fusion HCI and OpenShift® Enterprise to work correctly. Adding entries into the /etc/hosts file is not enough because that file is not copied into containers that are running on the platform.

About this task

Important: If you are using static IP, then you do not have to do a DHCP setup.

Procedure

1. Place your static servers in the same VLAN as IBM Fusion HCI.
2. Verify whether the DNS server is accessible from outside the IBM Fusion HCI by using either the `ping <server-ipaddress>` or the `nslookup <server-ipaddress>` command.
   Note: The DNS server can be anywhere but must be reachable, that is, the installation and cluster must have access to your network.
3. Set up the DNS reservation for each node. For more information about setting up the DNS, see Setting up DNS.

Here, ensure that you use the same `domain-name-servers` value and option host name that you used in Static IP as they are interconnected. Ensure that the fixed addresses that you configure here is same in Static IP as well.

Setting up DNS

Steps to set up the DNS server for IBM Fusion HCI appliance. It is applicable for both static and DHCP.

Before you begin

- Ensure that the VIP address for Ingress and API are in the same subnet as machine CIDR.
- For IBM Fusion HCI with static IP, configure DNS server to disallow `nslookup` on IP range `169.253.0.0/24`. This IP range belongs to the provisioning network and DNS must not try to resolve it. On Linux named server do the following configuration changes:
  1. Edit `/etc/named.conf` to add the following content:

     ```
     zone "253.169.in-addr.arpa" {
         type master;
         file "/var/named/db.blocked-253.169";
     };
     ```

  2. Create zone config file `/var/named/db.blocked-253.169` and add the following configuration:

     ```
     $TTL 86400
     @   IN  SOA localhost. root.localhost. (
             1        ; Serial
             3600     ; Refresh
             1800     ; Retry
             604800   ; Expire
             86400 )  ; Minimum

         IN  NS  localhost.
     ```

  3. Run the following command to restart the named server:

     ```
     sudo systemctl restart named
     ```

  4. Test whether the `nslookup` on the IP range returns no response:

     ```
     nslookup 169.253.2.53 <DNS server IP>
     ```

     Output:

     ```
     ** server can't find 53.2.253.169.in-addr.arpa: NXDOMAIN
     ```

About this task

Forward and reverse lookup must match to ensure that no network spoofing exists. It is a security issue where the hostname points to a wrong IP address. For MAC address, IBM Fusion HCI installation needs static IPs for its nodes and each node has its own specific hostname. To facilitate this requirement, DHCP is setup to map MAC address with an IP and hostname. In DNS service, similar mapping is configured between IP and hostname and vice-verse.
Note: The samples in this procedure are based on Red Hat® Enterprise Linux. If you are on Microsoft® Linux®, then use equivalent commands.

Procedure

1. Run the following command to suspend updates to all dynamic zones:

   ```
   rndc freeze
   ```

2. Edit your forward lookup (zonefile.db) at /var/named/ folder:
   Note: Forward look up returns the IP address of a hostname.
   Samples forward lookup file:

   ```
   $TTL 1W
   @       IN      SOA     ns1.isf.mycompany.com.  root (
                           2021052608      ; serial
                           3H              ; refresh (3 hours)
                           30M             ; retry (30 minutes)
                           2W              ; expiry (2 weeks)
                           1W )            ; minimum (1 week)
           IN      NS      ns1.isf.mycompany.com.
           IN      MX 10   smtp.isf.mycompany.com.
   ;
   ns1         IN  A       10.9.20.17
   smtp    IN      A       10.9.20.17
   ;
   api                 IN  A       10.9.20.127
   ;
   ; Ingress LB (apps)
   *.apps          IN      A       10.9.20.128
   ;
   bootstrap   IN  A   10.0.20.135
   ;
   ; Create entries for the master nodes
   control-1-ru2   IN      A       10.9.20.129
   control-1-ru3   IN      A       10.9.20.130
   control-1-ru4   IN      A       10.9.20.131
   ```

```
;
; Create entries for the worker nodes
compute-1-ru5    IN      A       10.9.20.132
compute-1-ru6    IN      A       10.9.20.133
compute-1-ru7    IN      A       10.9.20.134
servicenode-1    IN      A       10.9.20.222
;
;
;EOF
```

Note: The servicenode-<rackid> is used for servicenode hostname and for single rack it is always 1. For multi rack, it depends on the rackid used for the base rack.

3. Edit the reverse lookup file (reverse.db) at /var/named/reverse.db folder.
   Reverse look up returns the hostname of an IP address.

   Sample reverse lookup file:

```
$TTL 1W
@       IN      SOA     ns1.isf.mycompany.com.  root (
                        2021052608      ; serial
                        3H              ; refresh (3 hours)
                        30M             ; retry (30 minutes)
                        2W              ; expiry (2 weeks)
                        1W )     ; minimum (1 week)
        IN      NS      ns1.isf.mycompany.com.
; gen2003 rack
127     IN      PTR     api.isf.mycompany.com.
135     IN      PTR     bootstrap.isf.mycompany.com.
;
; syntax is "last octet" and the host must have fqdn with trailing dot
129     IN      PTR     control-1-ru2.isf.mycompany.com.
130     IN      PTR     control-1-ru3.isf.mycompany.com.
131     IN      PTR     control-1-ru4.isf.mycompany.com.
132     IN      PTR     compute-1-ru5.isf.mycompany.com.
133     IN      PTR     compute-1-ru6.isf.mycompany.com.
134     IN      PTR     compute-1-ru7.isf.mycompany.com.
222     IN      PTR     servicenode-1.isf.mycompany.com.
;
```

   In this example, the IP addresses 10.9.20.*127-222* points to the corresponding fully qualified domain name.

   Note: Ensure that the subdomain consists of lowercase alphanumeric characters, '-' or '.', and starts and ends with an alphanumeric character. For example, `example.com`. The regex used for validation is 'a-z0-9?(\.a-z0-9?)*')",

4. Ensure that your named.conf file in the /etc folder contains details of your forward and reverse lookup files.
   Sample named.conf file:

```
[root@provisioner named]# cat /etc/named.conf

options {
        listen-on port 53 { any; };
        listen-on-v6 port 53 { ::1; };
        directory       "/var/named";
        dump-file       "/var/named/data/cache_dump.db";
        statistics-file "/var/named/data/named_stats.txt";
        memstatistics-file "/var/named/data/named_mem_stats.txt";
        allow-query     { any; };
    .......
    .......
###### Add what's between these comments ###########
zone "isf.mycompany.com" IN {
        type    master;
        file    "zonefile.db";
};
zone "100.44.10.in-addr.arpa" IN {
        type    master;
        file    "reverse.db";
};
########################################################
include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
```

5. On the DNS server, run the following commands to open the firewall port for DNS:

```
firewall-cmd --zone=public --add-service=dns --permanent
firewall-cmd --reload
```

6. Restart the DNS service.

```
sudo systemctl restart named.service
```

Firewall requirements for IBM Fusion HCI

IBM Fusion HCI requires outbound access to external sites for accessing image registries and sending telemetry data to IBM and Red Hat®. This outbound access can be optionally routed through a proxy server. Configure your data center's firewall rules or proxy access control list (ACL) to meet the requirements for IBM Fusion HCI.

Inbound access for OpenShift cluster

The following lists inbound access for the OpenShift® cluster.

URL	Port	Function
API server and apps	**TCP**: 443 and 6443	Required for communication with OpenShift API server and apps URL

Outbound access for Red Hat OpenShift and IBM Fusion HCI image registries

The following lists outbound access for Red Hat OpenShift and IBM Fusion HCI image registries.

URL	Port	Function
`icr.io`	443	IBM Entitled Registry and IBM Cloud Paks foundational services catalog source
`cp.icr.io`	443	IBM Entitled Registry and IBM Cloud Paks foundational services catalog source
`gcr.io`	443	Images from Google Cloud's Container Registry
`registry.redhat.io`	443	Provides core container images
`*.quay.io` Note: If wildcard is not allowed, add `cdn01.quay.io` `cdn02.quay.io` `cdn03.quay.io`	443	Provides core container images
`*.openshiftapps.com` Note: If your firewall does not accept wildcard, then use the complete URL.	443	Provides Red Hat Enterprise Linux CoreOS (RHCOS) images
`cert-api.access.redhat.com`	443	Required for Telemetry
`access.redhat.com`	443	Required for Telemetry
`api.access.redhat.com`	443	Required for Telemetry
`infogw.api.openshift.com`	443	Required for Telemetry
`console.redhat.com/api/ingress`	443	Required for Telemetry and for `insights-operator`
`cloud.redhat.com/api/ingress`	443	Required for Telemetry and for `insights-operator`
`mirror.openshift.com`	443	Required to access mirrored installation content and images. This site is also a source of release image signatures, although the Cluster Version Operator needs only a single functioning source.
`storage.googleapis.com/openshift-release`	443	A source of release image signatures, although the Cluster Version Operator needs only a single functioning source.
`.apps.<cluster_name>.<base_domain>`	443	Required to access the default cluster routes unless you set an ingress wildcard during installation.
`quayio-production-s3.s3.amazonaws.com`	443	Required to access Quay image content in AWS.
`api.openshift.com`	443	Required both for your cluster token and to check whether updates are available for the cluster.
`art-rhcos-ci.s3.amazonaws.com`	443	Required to download Red Hat Enterprise Linux CoreOS (RHCOS) images.
`console.redhat.com/openshift`	443	Required for your cluster token.
`cloud.redhat.com/openshift`	443	Required for your cluster token.
`registry.access.redhat.com`	443	Required for `odo` CLI.
`sso.redhat.com`	443	The `https://console.redhat.com/openshift` site uses authentication from `sso.redhat.com`

Note: You can use the wildcard `*.quay.io` and `*.openshiftapps.com` instead of `cdn0[1-3].quay.io` and `rhcos-redirector.apps.art.xq1c.p1.openshiftapps.com` in your allowlist.
The entries from row 4 till the end are from Red Hat OpenShift documentation. For the actual list from Red Hat OpenShift, see https://docs.redhat.com/en/documentation/openshift_container_platform/4.21/html/installation_configuration/configuring-firewall.
Note: If you are using a private image registry to mirror Red Hat OpenShift and IBM Fusion HCI images, then set the host and port of your registry (`<your registry host><your registry port>`).
Important: After you go online, do not remove the firewall settings or switch between online and offline; otherwise, your cluster might go down. An uninterrupted connection to the repository is needed for both online and offline.

Service node and remote support

For service node, ensure that firewall port 22 is allocated for SSH access.

The remote support works seamlessly with most firewalls. Usually, remote support connections are possible without any firewall reconfiguration. It requires access to outbound ports at both ends of a connection, so there is normally no need to open holes in firewalls. Remote support also supports configuring a proxy for communication.

Call Home

Configure your data center's firewall rules or proxy access control list (ACL) to allow the manual and automated Service and Support request feature to communicate with the servers. To configure firewall, consider the following points for planning and prerequisites:

- Source IP for Call Home outbound connections is the OpenShift control node.
- The Host name, IP, and port details of Call Home servers:

Table 1. IP addresses of the IBM call home servers

Host name	IP address	Ports	Protocol
esupport.ibm.com	192.148.6.11	443	https
ecurep.ibm.com	192.109.81.21	443	https

Important: Though the table gives IP address guidance, ensure that you check the following pages for latest information: IP addresses and https://www.ibm.com/support/pages/node/6853429.

Metro-DR

The following table lists outbound access from a IBM Fusion HCI cluster for Metro sync DR:

IP address	TCP or UDP or IP protocol	Additional detail
<TieBreaker IP address>	**TCP**: 1191 and 12345	Required for IBM Fusion HCI storage to talk to tiebreaker storage
<Other site storage network>	**TCP**: 1191 and 12345	Required for stretch cluster deployment across the sites.

IP address	TCP or UDP or IP protocol	Additional detail
<Other site Red Hat OpenShift network>	**UDP**: 4500	Required for IPsec encapsulation traffic.
<Other site Red Hat OpenShift network>	**ESP IP protocol**: 50	Required for IPsec tunnel.
<API and apps IP of the other site>	**TCP**: 443 and 6443	Required for OpenShift and IBM Fusion management software stack to stay in sync.
<Other site storage network>	ICMP (ICMP ECHO (ping))	Required for stretch cluster deployment across the sites.

The following UDP port is required to be allowed only when there is a NAT device between the two site.

IP address	TCP or UDP or IP protocol	Additional detail
<Other site master nodes>	**UDP**: 4490	Required for NAT traversal discovery.

The following table lists outbound access from tiebreaker VM for Metro sync DR:

IP address	TCP	Additional detail
<IBM Fusion storage network of both sites	**TCP**: 1191 and 12345	Required for tiebreaker storage to talk to IBM Fusion HCI storage.

For more information firewall details, see General Metro-DR prerequisites.

Scale (GDP) ephemeral ports

Range	Protocol	Service name	Components involved in communication
60000 - 61000	TCP	GPFS ephemeral port range	Intra-cluster

The ephemeral port range is automatically set to 60000-61000. Firewall ports must be opened according to the defined ephemeral port range. If commands such as **mmlsmgr** and **mmcrfs** hangs, it indicates that the ephemeral port range is not configured correctly.
As mentioned in this topic, pods use port 12345 for ssh. If it is for a different source or destination, then mention accordingly.
For more information about Scale port number recommendation, see Scale documentation.

Proxy configuration

Proxy configuration is needed only in a connected install through proxy.

- Note: As a prerequisite, check the following points for proxy limitations:
  - Ensure that there are no restrictions on the maximum file size for download through proxy or that it must be adjusted adequately.
  - Ensure that there is enough bandwidth available, or otherwise installation might run into errors due to download timeouts
  - Ensure that there are no quotas or download limits set or adjusted appropriately, or else pods might get into a crash loop due to download limitations.
  - Do not use special characters in proxy credentials.
- If you plan to setup Metro-DR and both sites use proxy, then site1 API must be in the allowed list of site2 proxy and site2 API must be in the allowed list of site1 proxy.
- If you are going to install through your enterprise registry and that is also through your proxy, then add `http://<enterprise registry host name>`
- If you plan to use a proxy server for internet access with IBM Fusion HCI for configuration, add the following list to access control list (ACL) on proxy server:
  - `registry.redhat.io`
  - `redhat.com`
  - `registry.connect.redhat.com`
  - `console.redhat.com`
  - `cloud.redhat.com`
  - `access.redhat.com`
  - `registry.access.redhat.com`
  - `api.access.redhat.com`
  - `quay.io`
  - `cdn.quay.io`
  - `cdn01.quay.io`
  - `cdn02.quay.io`
  - `cdn03.quay.io`
  - `cdn04.quay.io`
  - `cdn05.quay.io`
  - `cdn06.quay.io`
  - `sso.redhat.com`
  - `oauth-openshift.apps.<cluster_name>.<base_domain>`
  - `canary-openshift-ingress-canary.apps.<cluster_name>.<base_domain>`
  - `console-openshift-console.apps.<cluster_name>.<base_domain>`
  - `api.openshift.com`
  - `infogw.api.openshift.com`
  - `mirror.openshift.com`
  - `docker.com`
  - `docker.io`
  - `dseasb33srnrn.cloudfront.net`
  - `rhc4tp-prod-z8cxf-image-registry-us-east-1-evenkyleffocxqvofrk.s3.dualstack.us-east-1.amazonaws.com`
  - `storage.googleapis.com`
  - `pkg-containers.githubusercontent.com`
  - `ghcr.io`
  - `www.okd.io`
- If your IBM Fusion appliance contains GPU nodes, you must configure the following sites:
  - `cloud.openshift.com`
  - `.nvcr.io`
  - `.d3c2pjnrr68kpx.cloudfront.net`
  - `containers.nvcr.io`
  - `authn.nvcr.io`
  - `api.ngc.nvidia.com`
  - `catalog.ngc.nvidia.com`
- The following firewall sites must be configured for IBM Cloud Container registry:

- `icr.io`
    - `cp.icr.io`
    - `dd0.icr.io`
    - `dd2.icr.io`

  If you're located in China, you must also allow the following hosts:
    - `dd1-icr.ibm-zh.com`
    - `dd3-icr.ibm-zh.com`
- The following firewall site must be configured for Me tor-DR set up:
    - `gcr.io`
- The following firewall sites must be configured to interact with IBM or Red Hat® services:
    - `http://www.ibm.com`
    - `http://www.redhat.com`
- The following firewall sites must be configured for Call Home:
    - `www.ecurep.ibm.com`
    - `esupport.ibm.com`
- The following endpoints must be configured for Remote Support:
    - aosrelay1.us.ihost.com
    - aosback.us.ihost.com
    - aoshats.us.ihost.com

Configuring Scale daemon network IP parameters

This is an optional configuration and is only needed when the default scale daemon IPs conflicts with some of your workloads. An additional IPVLAN Container Network Interface (CNI) network named as daemon-network is created for scale core pods. By default, IP addresses are assigned for Scale daemon network. You can override the default IP addresses before you begin the Final installation of IBM Fusion HCI. If the default IPs are being overridden ensure that they are configured in a private subnet.

About this task

Scale core pods have two networks, namely admin and daemon network. Admin network is the default pod network and pod service domain, whereas the daemon network is used for communication between the Scale core pods.

Compute nodes parts of the Scale cluster are annotated to provide the daemon network name, IPv4 address, gateway, and optionally the machine address.

By default, for Scale daemon network, the private IPv4 addresses is in the range of 192.168.128.6 to 192.168.191.254. The default gateway for this network is 192.168.128.1. This default allocated IP range can be overridden if necessary.

Network cable and transceiver options

This topic has general network cable requirements.

Note: IBM Fusion HCI does not include any cables or transceivers for connecting the system to external switches. If you need cables and transceivers for external connections, then explicitly add them to an order.

Network cable requirements

Connect port 31 or both ports 31 and 32 of both IBM Fusion HCI high-speed switches to the data center switches. The supported speeds are 1, 10, 25, 40, and 100 GbE. The ports on the high-speed switches are 100 GbE QSFP28 ports. You can use a splitter (breakout or fan-out) cable or an adapter to connect the high-speed switches of IBM Fusion HCI to the data center switches.

Purchase the needed DAC/AOC/QSFP and fibre cables from IBM or anther supplier depending on your needs. Discuss with your network administrators or consult an IBM Support representative before you purchase the cables (minimum of 2). If you plan to use both 31 and 32 ports, then you must purchase four cables.

The following table lists cable and transceiver options for the 9155-S01 high-speed switch:

Feature Code (FC) or Request for Price Quotation (RPQ)	Description	Purpose	Country of Origin (COO)	Part Number (PN)
FC EB49	A connector that supports plugging a SFP28 transceiver into a QSFP28 port.	To adapt a QSFP28 100GbE port so that a SFP28 25GbE transceiver can be plugged into it.	ID	01FT845
FC EB57	A 40GbE QSFP+ base-SR4 optical transceiver with MPO connector.	To connect a IBM Fusion HCI 100 GbE switch port to a 40GbE client network port (cable must be supplied separately).	TH	78P6916
FC EB59	A 100GbE QSFP28 MTP/MPO SR4 850nm optical transceiver.	To connect a IBM Fusion HCI 100 GbE switch port to a 100GbE client network port (cable must be supplied separately).	CN	78P4988
FC AKBP	A 200GbE SR4 QSFP56 850nm optical transceiver with MPO-12 connector.	To connect a IBM Fusion HCI 200 GbE switch port to a 200GbE client network port (cable must be supplied separately).	CN	78P7682
FC EB5T	A 10 meter QSFP28 Ethernet active optical cable with integrated transceivers.	To connect a IBM Fusion HCI 100 GbE port to your 100 GbE switch port. These are AOC cables and come with integrated 100GbE transceivers at both ends of the cable.	TH	01FT724
FC EB5W	A 30 meter QSFP28 Ethernet active optical cable with integrated transceivers.	To connect a IBM Fusion HCI 100 GbE port to your 100 GbE switch port. These are AOC cables and come with integrated 100GbE transceivers at both ends of the cable.	TH	01FT726

Feature Code (FC) or Request for Price Quotation (RPQ)	Description	Purpose	Country of Origin (COO)	Part Number (PN)
RPQ 8S1916	A 10GbE SFP+ SR optical transceiver.	Transceiver to connect a IBM Fusion HCI 100 GbE port to your 10 GbE network. The 10GbE cable must be supplied separately. To plug this transceiver into a IBM Fusion HCI 100GbE port, you must use the adapter FC EB49.	MY	78P6987
RPQ 8S1929	A 1 Gb optical transceiver for cables with RJ-45 connectors.	Transceiver to connect a IBM Fusion HCI 100 GbE port to your 1 GbE network. The 1 GbE cable must be supplied separately. To plug this transceiver into a IBM Fusion HCI 100GbE port, you must use the adapter FC EB49.	MY	03FP283
FC EB47	A 25GbE SFP28 optical transceiver with LC connector.	Transceiver to connect a IBM Fusion HCI 100 GbE port to your 25 GbE network. The 25 GbE cable must be supplied separately. To plug this transceiver into a IBM Fusion HCI 100GbE port, you must use the adapter FC EB49.	ID	78P5153
FC ECBD	A 5 meter LC-LC Fiber OM3 MMF cable.	This is only an optical cable and has no transceivers. Use this with transceiver that has an LC-type connector.	TW	45D4774
FC ECBE	A 25 meter LC-LC Fiber OM3 MMF cable.	This is only an optical cable and has no transceivers. Use it with transceiver that has an LC-type connector.	TW	15R8848

General Metro-DR prerequisites

Prerequisites for Metro-DR between two OpenShift® Container Platform clusters.

Meet the following requirements:
Note: Metro-DR solution is only supported on Global Data Platform storage.

Setting up the tiebreaker

- Install IBM Fusion HCI in two separate data centers within a metropolitan distance from each other. It is recommended to have the third site to host the tiebreaker, and a VM is needed to host the tie breaker. For the actual procedure to setup, see Preparing the tiebreaker.

Setting up a network

- Network connectivity must be available between all three sites for both the Red Hat® OpenShift network and the IBM Fusion HCI storage network. The IBM Fusion HCI storage network is responsible for the synchronous data replication between both sites, and a high-bandwidth L3 connection must be used that has less than 40 milliseconds of latency and within 150 km. It is recommended that all switches used to connect the two clusters support jumbo frames as it improves performance.
- The IBM Storage Scale core pod that runs on every IBM Fusion HCI node at both sites is assigned an IP address on the storage network. As the number of nodes in a cluster can grow over time, a default IP address range is automatically generated for each cluster. This default IP address range is displayed during the installation GUI for each cluster. If needed, you can provide a different range of IP addresses for the storage network. The IP address range must be different for each site and must be reachable across sites.
- By default, IBM Fusion HCI reserves the 3201 VLAN ID for the storage network. If you need to customize the VLAN ID, record the customization in the planning worksheet. IBM SSR needs the completed worksheet to do the initial configuration of the appliance.
- For more information about how to connect both the sites, see Metro-DR configuration for Global Data Platform.
- **Jumbo frames**
  - Configure all the network devices through which the traffic flows from one site to another with Maximum Transmission Unit (MTU) size 9216 or higher. If your network does not support jumbo frames all the way from site 1 to site 2, then do the following steps:
    - Do not enable Jumbo frames during the installation of site 2 and apply the same preference to site 1.

    For steps to enable or disable, see Configure GDP storage topic.
  - If you enable jumbo frames on the site 1, then you must enable it on the site 2 and on all network devices in between.
  - Both the client and IBM Fusion HCI switches must have matching packet sizes.
  - For optimal performance, IBM recommends that you enable the jumbo frames.

Setting up communication between the clusters

The Metro-DR solution involves network traffic between two IBM Fusion HCI clusters and the tie breaker. Update the firewall for the three sites to allow traffic between each of the three endpoints in the solution.

- Site one and site two must allow communication between the IP addresses of the two IBM Fusion HCI clusters on both Red Hat OpenShift network and the IBM Fusion HCI storage network. On the Red Hat OpenShift, the network communications across sites take place over UDP port 4500. On the storage network, communication across sites takes place over TCP ports 1191 and 12345.
- Site one and the tie breaker site must allow communication between the IP address on the storage network of site one cluster and the IP address of the tie breaker virtual machine. This communication takes place over 1191 and 12345.
- Site two and the tie breaker site must allow communication between the IP address on the storage network of the site two clusters and the IP address of the tie breaker virtual machine. This communication takes place over 1191 and 12345.
- For Metro-DR allowed ports, see Firewall requirements for IBM Fusion HCI.

Storage network

- A default IP range is provided out of the box for the storage network.
- If needed, you can provide a different range of IP addresses for the storage network; however, it must be different across sites.
- Storage VLANs and storage networks are customizable for a single rack and must use only a private network for the storage network.

Admin network

Each cluster has a pod network that needs to communicate with each other in a Metro-DR setup. To achieve this communication, make sure that the pod network is different across the sites. By default, different values are available for site 1 and site 2 during installation, but you can override them.

- **Preparing the tiebreaker**
  Planning and preparation to setup the tiebreaker.

Preparing the tiebreaker

Planning and preparation to setup the tiebreaker.

Important:

- Ensure that you install the **ncat** package for the tiebreaker.
- Ensure that you have entry for the **gpfs-tiebreaker** in the DNS server along with the IP that you choose for the tiebreaker.

Connect with your network administrator for the tiebreaker setup. You must connect the tiebreaker to the daemon network of the two OpenShift® Container Platform clusters.

- The Metro-DR configuration must be complete between the two sites.
- When you want to upgrade from IBM Fusion 2.12.x Metro-DR setup to IBM Fusion 2.13.0 Metro-DR setup, skip this procedure and upgrade your tiebreaker to IBM Storage Scale 5.2.3.7. For the procedure to upgrade, see Upgrading tiebreaker from IBM Storage Scale 5.2.3.x to 5.2.3.7.
- **Secure Boot enabled system:**
  For RHEL, ensure either of the following for the IBM Storage Scale to work properly in a Secure Boot enabled system:
  - Disable Secure Boot in BIOS.
  - Sign into IBM Storage Scale kernel module manually.

  For more information about how to check Secure Boot enablement, see https://access.redhat.com/articles/5337691.
- Hardware requirements are CPU 2 cores, Memory 4G, a raw disk with not less than 20 GB.
- Add entry in the Domain Name System (DNS) for the tie-breaker: **gpfs-tiebreaker**.
- For software requirements, see Software requirements for IBM Storage Scale.
  - For RHEL, ensure either of the following for IBM Storage Scale to work properly in a Secure Boot enabled system:
    - Disable Secure Boot in BIOS.
    - Sign into IBM Storage Scale kernel module manually.
- For tiebreaker allowed ports, see Firewall requirements for IBM Fusion HCI.
- Software preparation:
  1. SSH to the tiebreaker VM.
  2. Run the following commands to install Python 3:

     ```
     yum install python3
     ```

  3. Download the IBM Storage Scale Data Management 5.2.3.7 from IBM Entitled System Support:
     a. Log in to IBM Entitled System Support - https://www.ibm.com/servers/eserver/ess/landing/index.html.
     b. Go to My entitled software > Software Downloads.
     c. In the Product, search for 5771-PP7 (IBM Fusion HCI) and click Add Product.
        - The **IBM Storage Scale Data Mgmt 5.2.3.7 x86 64 Linux install** is available for Storage Scale Data Management.
        - Select cd776379.tar.gz to download.
  4. Extract the IBM Storage Scale package:

     ```
     chmod +x cd776379.tar.gz
     ./cd776379.tar.gz
     ```

  5. Go to /usr/lpp/mmfs/5.2.3.7/ansible-toolkit directory.
  6. Run the following setup command to install Ansible.

     ```
     ./spectrumscale setup -s <Tiebreaker IP>
     ```

  7. Create /usr/lpp/mmfs/5.2.3.7/ansible-toolkit/ansible/vars/tiebreaker_nodedefinition.json file.
  8. Update the content of **tiebreaker_nodedefinition.json** as follows:

     ```
     {
     "scale_cluster": {
      "scale_version": "5.2.3.7",
      "ssh_authorizedkeys": "<ssh_authorizedkeys>",
      "ssh_privatekey": "<ssh_privatekey>",
      "ssh_publickey": "<ssh_publickey>"
     },
     "tiebreaker_node": [
      {
        "fqdn": "<Tiebreaker IP>"
      }
     ]
     }
     ```

     <ssh_authorizedkeys>
     Run the following command to get the **ssh_authorizedkeys** value:

     ```
     oc get secret ibm-spectrum-scale-core-ssh-key-secret -n ibm-spectrum-scale -ojsonpath="{.data.ssh-authorizedkeys}"
     ```

     <ssh_privatekey>
     Run the following command to get the **ssh-privatekey** value:

     ```
     oc get secret ibm-spectrum-scale-core-ssh-key-secret -n ibm-spectrum-scale -ojsonpath="{.data.ssh-privatekey}"
     ```

     <ssh_publickey>
     Run the following command to get the **ssh_publickey** value:

     ```
     oc get secret ibm-spectrum-scale-core-ssh-key-secret -n ibm-spectrum-scale -ojsonpath="{.data.ssh-publickey}"
     ```

     <Tiebreaker IP>
     Enter your tiebreaker IP address.

  9. Copy **playbook_tiebreakernode_install.yml** and **set_json_variables_tb.yml** from the sample directory to the Ansible directory:

```
cp /usr/lpp/mmfs/5.2.3.7/ansible-toolkit/ansible/sample/playbook_tiebreakernode_install.yml 
/usr/lpp/mmfs/5.2.3.7/ansible-toolkit/ansible/sample/set_json_variables_tb.yml  /usr/lpp/mmfs/5.2.3.7/ansible-
toolkit/ansible/
```

10. Run the following command to configure the SSH passwordless:

```
sed -i 's/scale_private_public_key_config: false/scale_private_public_key_config: true/g' 
/usr/lpp/mmfs/5.2.3.7/ansible-toolkit/ansible/playbook_tiebreakernode_install.yml
```

11. Go to directory `/usr/lpp/mmfs/5.2.3.7/ansible-toolkit/ansible/` and run the playbook to install the tiebreaker node:

```
ansible-playbook playbook_tiebreakernode_install.yml
```

Important:
    - Change the port from default 22 to 12345. Otherwise, you cannot log in to the tiebreaker VM.
    - If tiebreaker VM goes down, then you need to run the playbook again to install the tiebreaker node. After restart, the node port is automatically set to 22 because that is the default.
12. Run `lsblk` to get the device name of the raw disk attached to the tiebreaker VM. The device name gets used later to add the tiebreaker into the IBM Storage Scale cluster.
13. Run the following **echo** command to encode the disk name string:

```
echo "<Disk name eg: /dev/vda>" | base64
```

As a prerequisite to run base64, you must install base64 or jq.
14. From either the site 1 or site 2, run the following command to patch the Metro-DR config secret to include the tiebreaker disk details:

```
oc patch secret isf-metrodr-config-secret -n ibm-spectrum-fusion-ns -p '{"data":{"TieBreakerDevice":"Encoded disk 
name from previous command"}}'
```

Note: Enter double quotes for the tiebreaker device name while you patch the Metro-DR config secrets.
15. Set up the tiebreaker. For the actual steps to set up, see Upgrade and upsize considerations in Metro-DR section.

Converting a rack from online proxy to offline using external Quay

Use the following instructions to convert a rack from online proxy to offline using external quay on the IBM Fusion HCI cluster.

Procedure

1. Mirror the images of the cluster to an external repo. For more information, see Enterprise registry for IBM Fusion HCI installation.
   Note: Mirroring of all operators must be of the same version, as it is in the online cluster.
2. Remove cluster-wise proxy details:

   Remove proxy details from `userconfig_secret`
      a. Log in to the OpenShift web console and go to Workloads > Secrets.
      b. Click userconfig-secret.
      c. From the Actions dropdown, select Edit Secret.
      d. Update the following values:
         i. Set `isProxy` to `false`. The default value is `true` since the cluster is installed as proxy.
         ii. Clear the values for `proxy_host`, `proxy_password`, `proxy_port`, and `proxy_username` (for example: `proxy_password=''`).

   Remove proxy details from the proxy custom resource (CR):
      a. In the OpenShift web console, go to Administration > CustomResourceDefinitions, and search for `Proxy`.
      b. Click the `Proxy` named CR, and then click Instances.
      c. Click cluster and edit the YAML.
      d. Remove the proxy details under the `spec` section (`httpProxy`, `httpsProxy`, and `noProxy`), and save the changes.
         This step triggers a Machine Config Pool (MCP) roll out. Allow the rollout to complete.

3. If the certificate is self-sign certificate or not a well-known authority signed certificate, add the certificate to the cluster manually:

```
oc create configmap registry-cas -n openshift-config --from-file=[repo]..[port]=[certificate.crt]
```

```
oc patch image.config.openshift.io/cluster --type=merge -p '{"spec":{"additionalTrustedCA":{"name":"registry-cas"}}}'
```

   To verify that `registry-cas` is added to the nodes, follow these steps:
   a. Log in to the node:

```
oc debug node/[nodename]
```

   b. Run the following command on the node to switch to the host environment:

```
chroot /host
```

   c. Go to the certificates directory:

```
cd /etc/docker/certs.d
```

   d. List the contents of the directory:

```
ls
```

   The certificate must be present in this directory.

4. Disable call home and AOS: In the offline system, call home is not supported. For more information, see Disabling Call Home.
   a. In the IBM Fusion user interface, go to Settings > Support Toggle to disable remote support.
   b. Click Edit for call home and disable the call home support.

Note:
- For offline system, call home can be enabled by configuring proxy for call home. For more information, see Configuring Call Home with proxy.
- For offline system, AOS can be enabled by configuring proxy for AOS. For more information, see Enabling remote support through user interface.

5. Convert the rack from online to offline:
   a. Add IDMS, ITMS, and catalog source for the offline repository, for IBM Fusion, Red Hat and services.
   b. Add a new registry pull secret.
      This action triggers a MCP rollout. Allow the rollout to complete.
   c. Remove online pull-secrets from the cluster.
      This action triggers a MCP rollout. Allow the rollout to complete.
   d. Run the following command to disable the default catalog:

      ```
      oc patch OperatorHub cluster --type=json -p='[{"op":"add","path":"/spec/disableAllDefaultSources","value":true}]'
      ```
6. Update `CatalogSource` `fusion-catalog` image to offline repo path for `isf-operator-catalog`.

   ```
   spec:
     displayName: IBM Storage Fusion Catalog
     image:
   ```

   Verify no issues are observed after updating `CatalogSource`.

Results

MCP is triggered and the cluster is converted to offline.

Enterprise registry for IBM Fusion HCI installation

During the installation process, IBM Fusion HCI installs Red Hat® OpenShift® Container Platform and IBM Fusion software using images that are hosted in the Red Hat and IBM entitled registries. If you want to use your enterprise registry, you can install both OpenShift and IBM Fusion HCI software from images that are maintained in a container registry that you manage.

Though the installation experience of using your enterprise registry and IBM entitled and Red Hat registry are the same, you are responsible for setting up and populating your private enterprise registry. IBM Fusion supports both JFrog Artifactory and Quay enterprise registries.

These mirroring procedures guide you to download the images of IBM Fusion HCI and all its services at once or individually depending on your specific needs. Before you initiate the mirroring process, verify that all prerequisites are met to ensure its success, regardless of whether you choose the end-to-end procedure or selective procedure.

- **Mirroring prerequisites**
  Complete the following perquisites before you mirror images of IBM Fusion and its services.
- **End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry**
  If you are planing a disconnected or offline installation, then you must mirror images to your enterprise registry. You can use this procedure to complete an integrated mirroring of IBM Fusion HCI and its services **Backup & Restore**, **IBM Data Cataloging**, **IBM Software Hub**, and **Content-Aware Storage**. Otherwise, you can mirror them individually per your requirement.
- **Mirroring OpenShift Container Platform release images to enterprise registry**
  Mirror the OpenShift Container Platform image repository to your registry to use during installation or upgrade.
- **Mirroring Red Hat operator images to enterprise registry**
  Mirror the Red Hat operator images to your enterprise registry.
- **Mirroring OpenShift Container Platform release images to enterprise registry for Hosted Control Plane**
  Mirror the OpenShift Container Platform image repository to your registry to use during installation or upgrade of the Hosted Control Plane.
- **Mirroring individual images**
  There may be scenarios wherein you may want to mirror individual images separately.
- **Troubleshooting and known issues in offline mirroring**
  Issues can occur during the offline mirroring of images. This topic outlines known issues and offers troubleshooting guidance.

Mirroring prerequisites

Complete the following perquisites before you mirror images of IBM Fusion and its services.

Prepare your mirroring host

- The mirror host must have access to the internet and enterprise registry.
- Ensure that you install the following tools on your system from where you can connect to Red Hat® registry and enterprise registry:
  - Install Docker or Podman to create, run, and maintain images.
    If you are using self-signed certificate registry, then ensure that you have updated the container tools configuration with insecure registries to skip certificate validation. For more information about how to update, see Update container tools configuration.
  - Install skopeo for Global Data Platform storage image copy operation. For more information to install skopeo, see https://github.com/containers/skopeo.
  - Install OC command tool from any of the Red Hat OpenShift® Container Platform cluster:
    1. Log in to the system from where you want to run commands.
    2. Log in to OpenShift Container Platform.
    3. Click bell icon and select Command line tools.
    4. Click the appropriate oc download option based on your operating system.

    Alternatively, you can also download the platform based `oc` client.

Note: For example, if the latest version is 4.18.10, then use the following link to download `oc` client for Linux platform: https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/4.18.10/openshift-client-linux-4.18.10.tar.gz.

- Install the `oc-mirror` CLI plug-in. For more information about `oc-mirror` CLI plug-in installation, see Installing the oc-mirror OpenShift CLI plugin.
  Note: Ensure that you install the `oc-mirror` CLI plug-in of **4.20.6** version due to RH bug.
- Download pull-secret.txt.
  Note: After downloading the file, ensure it is saved with a `.json` file extension.
  To download the pull-secret, see https://console.redhat.com/openshift/install/pull-secret and follow the instructions.

  You can either use podman command to add private registry credentials to downloaded pull-secret.json or manually edit the downloaded `pull-secret` with registry credentials. For example:

```
podman login registryhost.com:443 --authfile="pull-secret.json"
```

  Add a new section of key-value pair under `auths`. For example,

```
"<Your enterprise registry>:<port>": {
"auth": "<base64 encoded 'user_name:password'>",
email: "<your email>" }
```

  As a prerequisite to run base64, you must install base64 or jq.
  See the following sample values:

```
{
"auths": {
   "cloud.openshift.com": {
      ....
   "registryhost.com:443": {
       "auth": "dXNlcl9uYW1lOnBhc3N3b3Jk",
        "email": "user_name@ibm.com"
    }
  }
}
```

  Here, `user_name` and `password` are credentials to connect to enterprise registry.
  Note: If you want to use multiple repositories, add an auth section for both repositories.
  To authenticate to quay.io by using the username and password from the pull-secret, do the following steps:

  1. Look for quay.io in the json and copy `auth`. Example:

```
  "auths": {
    "quay.io": {
      "auth": "b3BlbnNoaWZ0LXJTczOGY2YzNjNDM2ZWI0JRSDdUQU45RFBWVUNXM0VZQUlVVDBOQTU3VFM2RE1JMg==",
      "email": <email-id>
    }
  }
}
```

     It is a base64 encoded value of `username:password`.
  2. Get the username/password.

```
echo <auth-content> | base64 -decode
```

     Example echo command:

```
echo "b3BlbnNoaWZ0LXJTczOGY2YzNjNDM2ZWI0JRSDdUQU45RFBWVUNXM0VZQUlVVDBOQTU3VFM2RE1JMg==" | base64 -decode
```

     Example output:

```
openshift-release+ocmaccess0ab5738f6c3c42:CXKR2TSBQH7TAN9
```

     Here, `openshift-release+ocmaccess0ab5738f6c3c42` is the user name and `CXKR2TSBQH7TAN9` is the password.

  Command line to get the username/password :echo <auth-content> | base64 –decode

- Ensure that you have entitlement key to access IBM Fusion HCI appliance images. For more information about entitlement key, see Activating IBM Fusion HCI Software to be downloaded.

General prerequisites

- Before you proceed with mirroring, be aware of key considerations, known issues, and troubleshooting information. See Troubleshooting and known issues in offline mirroring.
- Provide the port number as follows:

  For installation
    If you want to use the default port (443), then make sure you provide the port number.
    For example, `<Your enterprise registry>:9443`.

    If you want to use custom port, then provide the custom details.

    To verify, log in to `your registry:port` by using docker or podman. Use the following **podman** command:

```
"podman login registryhost:registryport"
```

- The registry must have at least one directory path specified.
  For example:

```
https://<enterprise registry host>:<enterprise registry port>/<mandatory root path>
```

- Ensure that your secure enterprise registry is already setup and ready for use. Support is available for Quay and registries with self signed certificate.

Important: Always mirror the most recent images at any point in time to your registry before installation.
- Run a pull command in your network to test the network speed. If that time is more than 2 to 5 minutes, there may be an overall reduction in network speed that can cause installation failure. Example command:

```
time podman pull cp.icr.io/cp/isf/isf-compute-
operator@sha256:414275e851972db2b7172642f7f7827c42da14914061bd5d82ea0584e6ce7fc8
```

Single or multiple repositories

You can either host your images in a single repository or host your OpenShift and IBM Fusion images separately. Choose the latter option if you want to manage their OpenShift images separately from other software.

Single repository
: If you want to use Single repository, mirror all the images to IBM Fusion images repository, so your **$LOCAL_OCP_REGISTRY** remains same as **$LOCAL_ISF_REGISTRY**.

Multiple repositories
: If you want to use Multiple repositories, mirror the OpenShift images to the OpenShift images repository (**$LOCAL_OCP_REGISTRY**) and all other images to the IBM Fusion images repository (**$LOCAL_ISF_REGISTRY**).

Considerations for your enterprise registry

- There must not be a huge latency between cluster nodes and your enterprise registry. Slow image pull can cause OpenShift Container Platform installation to fail.
- You must have a container image registry that supports Docker v2-2 in the location that hosts the Red Hat OpenShift Container Platform cluster. For more information about image manifest, see opm CLI reference .
  Important: Artifactory is the recommended registry because it supports the following must have capabilities:
  - Import and export
  - Untagged images

  JFrog Artifactory version 7.55.8 is tested on IBM Fusion HCI. The port that is specified in the URL is used to login and pull the images from the enterprise repository. For a secured enterprise registry, specify 443. If you do not provide the port value, then no default port is considered in its absence. The API key is the only supported authentication method.
-
- For disconnected installation, you need a firewall type proxy server and for an air break installation, the registry must have import and export capabilities.
- If you are using self-signed certificate registry, then ensure that you have updated the container tools configuration with insecure registries to skip certificate validation.
  - For Podman, see https://docs.podman.io/en/stable/markdown/podman-login.1.html.
  - For Docker, follow the steps to skip certificate validation:
    - To trust the certificate in the Docker daemon:
      For Linux, copy the **domain.crt** file to **/etc/docker/certs.d/myregistrydomain.com:5000/ca.crt** on your mirroring host, Where **domain.crt** is a self-signed generated certificate and **myregistrydomain.com** is your Docker registry and 5000 is port for registry.
    - For configuring an insecure registry:
      Note: It is insecure and not recommended.
      Edit the **daemon.json** file that present in the default location of **/etc/docker/daemon.json** on the Linux Server.

      If the **daemon.json** file does not exist, then create it. Ensure it contains the following contents:

      ```
      {
        "insecure-registries" : ["myregistrydomain.com:5000"]
      }
      ```
- Offline upgrade cons

  The offline mirroring process includes applying **ImageDigestMirrorSet** or **ImageContentSourcePolicy** files to your cluster. Any changes to the IDMS or ICSP on clusters triggers a **MachineConfigPool** rollout, which can cause nodes to reboot. This includes adding, updating, or deleting IDMS or ICSP files.
  There are two options for offline upgrade mirroring:

  Mirroring images to the same location as the previous offline installation
  : This gives the IDMS or ICSP files that are identical to your previous installation, and you can choose to not apply the IDMS or ICSP files or **oc apply** the file and OpenShift recognizes there is no change in content and exit gracefully.

  Mirror images to a different location as the previous offline installation
  : Because the generated IDMS or ICSP is tailored specifically to the target offline registry, mirroring to a new location every time gives a different IDMS or ICSP every time. Applying a new IDMS or ICSP triggers the MCP rollout and contains the potential to reboot nodes, which can be disruptive to the health of the OpenShift cluster.
  Additionally, if you have done multiple offline upgrades using this method, then your cluster likely have multiple ICSP or IDMS files, some of which are old and unused. Again, deleting a IDMS or ICSP can trigger a disruptive MCP rollout. It is recommended to not delete any IDMS or ICSP files from the cluster until you have completed your upgrade to your target version of IBM Fusion HCI.
  However, it is recommended you to edit the generated IDMS and change the **metadata.name** to unique name to avoid overwriting an existing IDMS on your cluster.

End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry

If you are planing a disconnected or offline installation, then you must mirror images to your enterprise registry. You can use this procedure to complete an integrated mirroring of IBM Fusion HCI and its services **Backup & Restore**, **IBM Data Cataloging**, **IBM Software Hub**, and **Content-Aware Storage**. Otherwise, you can mirror them individually per your requirement.

Before you begin

- Before you mirror, go through the prerequisites. See Mirroring prerequisites.
- Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute **podman** for **docker** in the commands.
- Configure Common Environment Variables:
  Important: Ensure that you review the offline upgrade considerations before proceeding with upgrade mirroring.
  Define the following environment variables for your target container registry. These values are used in each of the subsequent steps to mirror all the related components.

  ```
  export LOCAL_ISF_REGISTRY="<Your container registry host>:<port>"
  export LOCAL_ISF_REPOSITORY="<Your image path>"
  export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
  ```

  LOCAL_ISF_REGISTRY is your target container registry.

  LOCAL_ISF_REPOSITORY is the image path in which you want to mirror the images. You can choose your own repository paths. For example, **hci-images/isf** or **hci-images**. See the following sample values:

  ```
  export LOCAL_ISF_REGISTRY="registryhost.com:443"
  export LOCAL_ISF_REPOSITORY="mirror-fusion-services-images"
  export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
  ```

About this task

Points to note about this task:

- Run all commands as root user.
- In the commands, replace **<your enterprise registry>** with your enterprise registry.
- Common tasks to complete this procedure.
  1. Install **oc** CLI plug-in. For steps to install, see Mirroring prerequisites.
  2. Set common variables used by all subsequent mirroring steps, such as the target registry and target registry location.
  3. IBM Fusion complete installation:
     Mirror and install IBM Fusion and its related services (Backup & Restore, IBM Data Cataloging, IBM Software Hub, and Content-Aware Storage). For steps, see Procedure.
- IBM Fusion custom installation:
  To perform a custom installation of IBM Fusion and its components, determine which components you want to install, and then see the following topics for installation instructions.
  - Mirror Red Hat® OpenShift® Container Platform release images. For more information, see Mirroring OpenShift Container Platform release images to enterprise registry.
  - Mirror images for Red Hat operator. For more information, see Mirroring Red Hat operator images to enterprise registry.
  - IBM Fusion HCI images. For more information, see Mirroring IBM Fusion HCI images
  - Mirror Global Data Platform images. For more information, see Mirroring IBM Storage Scale images.
  - Mirror Fusion Data Foundation images. For more information, see Fusion Data Foundation images.
  - Mirror Backup & Restore images. For more information, see Mirroring Backup & Restore images.
  - Mirror IBM Data Cataloging images. For more information, see Mirroring IBM Data Cataloging images.
  - Mirror IBM Software Hub images. For more information, see Mirroring IBM Software Hub, watsonx.ai, and watsonx Orchestrate images.
  - CAS images. For more information, see Mirroring Content-Aware Storage (CAS) images.
- For Hosted Control Plane upgrades, follow the step described in the following procedure.

Procedure

Mirror IBM Fusion and related Backup & Restore, IBM Data Cataloging, IBM Software Hub, and CAS services.

1. Run the following command to login to the Docker registry with your Red Hat enterprise credentials:

   ```
   docker login registry.redhat.io -u <Red Hat enterprise registry username> -p <Red Hat enterprise registry password>
   ```

   Log in to the IBM Entitled Container Registry using the IBM entitlement key:

   ```
   docker login cp.icr.io -u cp -p <your entitlement key>
   ```

   Important:
   - Ensure that your entitlement key for IBM Fusion HCI contains the correct entitlement.
   - Ensure you review the offline upgrade considerations before proceeding with upgrade mirroring.

   Set the following environment variables:

   ```
   export LOCAL_ISF_REGISTRY="<Your container registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<Your image path>"
   export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
   ```

   Note: Port is a non-mandatory value when setting the **LOCAL_ISF_REGISTRY** variable. You can ignore this if your enterprise registry is accessible and has a secure connection.
   Sample value for without port:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com"
   ```

   See the following sample values:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com:443"
   export LOCAL_ISF_REPOSITORY="fusion-mirror"
   ```

`LOCAL_ISF_REGISTRY` is your entitlement registry.

`LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images.

2. Run the command to login to the Docker registry with your enterprise registry credentials.

```
docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
```

3. Ensure that the `redhat-oadp-operator` and `amq-streams` operator packages are present in your cluster.
   Note: If you did not mirror `redhat-oadp-operator` and `amq-streams` from the Red Hat packages previously, follow the steps that are provided in the Mirroring Red Hat operator images to enterprise registry.
   Note:
   - Run the following command to get list of mirrored Red Hat packages available on your cluster.

     ```
     oc get packagemanifests | grep -i "Red Hat Operators"
     ```

   - If you do not get `redhat-oadp-operator` and `amq-streams` operator packages from the previous step, then you must follow the step 1.
   - Ensure that you also add existing packages along with new one in ImageSetConfiguration file. Otherwise, old packages can be lost from the Red Hat operator index image.
4. Create the image set configuration for IBM Fusion HCI.
   `imageset-config-fusion.yaml`

```
cat << EOF > imageset-config-fusion.yaml
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v2alpha1
mirror:
  operators:
  - catalog: icr.io/cpopen/ibm-fsh-operator-
catalog@sha256:61812c3fe5f4c58e18177398e39f62523cac3dc9d08e4e00e7af8486e9ac146c
    packages:
    - name: ibm-fsh-operator
      channels:
      - name: v1.0
    full: true
  - catalog: icr.io/cpopen/ibm-isf-cas-operator-
catalog@sha256:361a92013a2faf3e58a98fee85a8149caa80eb9b216f84b5384944dff6a7187f
    packages:
    - name: ibm-isf-cas-operator
      channels:
      - name: v1.0
      - name: v1.1
    full: true
  - catalog: icr.io/cpopen/ibm-usage-metering-operator-
catalog@sha256:a97cef95c73ac6dacea010818757b4ed9b7440f4c0ec4154a091f248c68d1c9e
    packages:
    - name: ibm-usage-metering-operator
      channels:
      - name: v1.0
    full: true
  - catalog: icr.io/cpopen/idp-server-operator-
catalog@sha256:1b93488e0bf5c421dace9bae65fd992695c1e63fa12e7b82f03cf852d3fa02ad
    packages:
    - name: guardian-dm-operator
      channels:
      - name: v2.0
    - name: guardian-dp-operator
      channels:
      - name: v2.0
    - name: guardian-mongo-operator
      channels:
      - name: v2.0
    - name: ibm-dataprotectionagent
      channels:
      - name: v2.0
    - name: ibm-dataprotectionserver
      channels:
      - name: v2.0
    - name: redis-operator
      channels:
      - name: v2.0
    full: true
  - catalog: icr.io/cpopen/isf-operator-catalog@sha256:7e580c4cd056dbffd6a80dd117597e52530bde0b5cb7c9d31321f33cd01adb3e
    packages:
    - name: isf-operator
      channels:
      - name: v2.0
      minVersion: 2.13.0
      maxVersion: 2.13.0
  additionalImages:
  - name: cp.icr.io/cp/dcs/auth@sha256:6cb6e582d02fb017dbbc1c736f3409a08823fef56381747f13952bd8a613cabb
  - name: cp.icr.io/cp/dcs/connmgr-scheduler@sha256:cca24250b9bbfdfe8c6c85c613c5fa1a56d2b0fbbb0e48a4c56fc3f50ea338a2
  - name: cp.icr.io/cp/dcs/connmgr@sha256:cbefb44b2d448b36f90e4fca7973ab66bc1fbfd799eaa1c13b5185bf116f6f09
  - name: cp.icr.io/cp/dcs/db-schema@sha256:534c98417f909a248698281bc244ef1441b6a01dccf5a3055242c13956576c9a
  - name: cp.icr.io/cp/dcs/db2whrest@sha256:3233681ab2ec32515653f5659d6c02f8e12fbfddaa9d65747dfe07a62de1bfd2
  - name: cp.icr.io/cp/dcs/dcs-ai-mcp-server@sha256:d8af262ff80ee897797d322bbd8f8382cf4b823033901aea37d5531b27565365
  - name: cp.icr.io/cp/dcs/import-service@sha256:1003a8694eff91c163a72b88c19baeb5cd47a97d152ab64ac910b8b8b693f096
  - name: cp.icr.io/cp/dcs/import-tags-app@sha256:c37128ef0ad7f7add93c72c42c8c1902f75a2907e683af5d7af99d04ff4ea381
  - name: cp.icr.io/cp/dcs/isd-proxy@sha256:5056d6a265b2a37123806976c56722b86c5c20dad6d2f5933838c8d076b1e120
  - name: cp.icr.io/cp/dcs/keystone@sha256:b3128687e35f78bc25cefbe7254f625b5f01f344ae190d161cdbf79057109aa6
  - name: cp.icr.io/cp/dcs/llm-proxy@sha256:fce3aebc677eba45434f16039d3c678ccf4c1c71d709ab3b13b5aebd79c928b6
  - name: cp.icr.io/cp/dcs/metadata-extender@sha256:9a988643df596a1bbfb1094874273c3c1ca879e8a645677090409b4de1f37ac9
  - name: cp.icr.io/cp/dcs/mo-agent-extract@sha256:2f4d7b0d5ada12d39932f687cea031d88beb9cdac791b48cdca29ed23a9860ca
  - name: cp.icr.io/cp/dcs/mo-ubi-init@sha256:927bb277f576f2a772faf8dc9978a6592d5d08bfad3cdce81da180a9095a34c2
```

```
  - name: cp.icr.io/cp/dcs/moconsumer@sha256:2f1d3fdec099951cfc5f6788d4a4d944d66ec0aa8e39c9688239f1ecad38729f
  - name: cp.icr.io/cp/dcs/moproducer@sha256:c984c05bc47939e22cec6f4c8c0a9f365b71b7582c3711a36a59b96cc546da78
  - name: cp.icr.io/cp/dcs/must-gather@sha256:3f655e31efb04862c52333f113322a9169ea54d88f627177707453fd48acec50
  - name: cp.icr.io/cp/dcs/policyengine@sha256:ffe2fac7ee8bd49be48a142818ca760c6482018d9a79941296ef8d3424c1949c
  - name: cp.icr.io/cp/dcs/remote-content-search@sha256:7c0a39bcf04c1063fffc866ec8513066beb398dfd28cdb436b21cf7c4754f7c8
  - name: cp.icr.io/cp/dcs/reports@sha256:a95d112e43f1ecf7f962f04354c64be6052319eb0c5cabf94c9af325648d9f46
  - name: cp.icr.io/cp/dcs/scaleafmdatamover@sha256:19aa20ce2b493f420e03758313e02f0351cf72836c15905f3c4b7285603e29c0
  - name: cp.icr.io/cp/dcs/scaleilmdatamover@sha256:725614999df502d09ba0621c4a2c317ea18818cb288ad38d0c46f98c78648ba9
  - name: cp.icr.io/cp/dcs/sdmonitor@sha256:2d853b11e533a9eee9e922034478289ed82d4641a16e9c797fafbaaff8fb1fdf
  - name: cp.icr.io/cp/dcs/tikaserver@sha256:11a8175a621e6a5fb093ff41d71eb20f8dca8a4186a24a843cc2713f634b382d
  - name: cp.icr.io/cp/dcs/uibackend@sha256:1bfd41169c4450852872b0f8d190d1b24b12a46040eb6a167437a2f8086c798f
  - name: cp.icr.io/cp/dcs/uifrontend@sha256:58979c1c53c116e5f90e814166a129a1fa8587831f617147328b5b0a5843d3f1
  - name: cp.icr.io/cp/dcs/wkc-connector@sha256:3a68bf843669cce6f000ca14ecd5d42a6727fd8055de5ac157b9ecf360fd5c59
  - name: cp.icr.io/cp/fusion-hci/isf-validate-
entitlement@sha256:1a0dbf7c537f02dc0091e3abebae0ccac83da6aa147529f5de49af0f23cd9e8e
  - name: icr.io/cpopen/db2u-day2-operator@sha256:67e656ec5ad43ef62550852f42b346ee68b4f2e97867980e822819687aa740ba
  - name: icr.io/cpopen/db2u-day2-operator@sha256:6e87d78d96101ec3a8dfca7464e4196bd34b1bc700b7ef1b1de5add00a591e55
  - name: icr.io/cpopen/db2u-day2-operator@sha256:af106b46b923745688ef2952da39add7d38b46768e18bf2e2147b4b05e755f00
  - name: icr.io/cpopen/db2u-operator@sha256:9062cf50e0d1df382878bf3ad4d5c5cfdcdc801666e419cd3c6a62838445978c
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  - name: icr.io/db2u/db2u.veleroplugin@sha256:6905ab5396f6b86c0eb70513b9b5f871c4d3832efee7694e8187451b4d8115a4
  - name: icr.io/db2u/db2u.veleroplugin@sha256:6e3141301930950e3a0671dd8d74a7148a7ef790284e607fd3d6870edcbd823a
  - name: icr.io/db2u/db2u.veleroplugin@sha256:73d8c40d160da82b2218b40ac5e1b065339db8d7bf6914ac694ebba071a858ad
  - name: icr.io/db2u/db2u.veleroplugin@sha256:7a890053b98b6e510f9697175071bc5bae00cdd1ea671c2949dfec9a61830403
  - name: icr.io/db2u/db2u.veleroplugin@sha256:ce73ee327f940e12bc22e20324d72acd7266f4adf730469d4fbbec482481f8e3
  - name: icr.io/db2u/db2u.veleroplugin@sha256:d327a193a266c982c30bc009a8def0c84d75dd4f28bf3806bee82e8b173946a0
  - name: icr.io/db2u/db2u.veleroplugin@sha256:f4f2e9e3afdcaa7a9e4e522bcdb2aaa6b79db97361968bdf1428961b1ada878d
  - name: icr.io/db2u/db2u.watsonquery@sha256:33d6ee3007e62284cf7852cfd337393a0bfb86432df599fd971c31836abaa28b
  - name: icr.io/db2u/db2u.watsonquery@sha256:48cce86af50ba003dbed9ce54b0240b9eec59a2702ab93bae58a649d7fe777ce
  - name: icr.io/db2u/db2u@sha256:0ecd06074f6ce0f090f93b2b5ed51e41679154d638dc2031d64498aab2c0de68
  - name: icr.io/db2u/db2u@sha256:323e043a92970faa921733dbd4d474bab3e80dc105fc6fe4b9c5e0d2160e1856
  - name: icr.io/db2u/db2u@sha256:64d1fc881fa7c4f9203a1e43b498464333768623cc1494d42e2d8aac21b4af3c
  - name: icr.io/db2u/db2u@sha256:9678623a716ceb4ed9367812b842cd56227ed15858edae08f7bcd5961dfa5e2d
  - name: icr.io/db2u/db2u@sha256:b8701f8d719345ee156c2490d8a0fc9ee08a2cfca7c4537d728c729337da9b77
  - name: icr.io/db2u/db2u@sha256:d282a83ba1d016035d9e7c8f489b3599cc3e39826890e927856a5232b6f6013a
  - name: icr.io/db2u/db2u@sha256:d7c92b368064d15ddf6bcc3d96dc19f876603f5a9fb453185724607e2b966567
  - name: icr.io/db2u/db2u@sha256:ef563ca49bf4ad388e8af360a5a1d87fd756d0e9329c6a181f3a37c29429b9c6
  - name: icr.io/db2u/db2u@sha256:f95b4ad43756f1f256df6f9b8cc79757db430de06a0fea22b78261490a73649a
  - name: icr.io/db2u/etcd@sha256:3c5d8ff1c77f146b6db7224bcfc92d6d64a832332cc087a162b2cda81ea99eef
  - name: icr.io/db2u/etcd@sha256:84dda9076ae58f2f02abba6102ab8325e4c301984b2588cb751e46245c636940
  - name: icr.io/db2u/etcd@sha256:9fa13ffeab820c5a62bc5bdb41347ef5cfd5244584117f0f3bd28cfe8690d358
  - name: icr.io/db2u/etcd@sha256:a5123f1a68e08bbda547eb653b724b917c9ec4d4cb6f447b066c4b71c6828909
  - name: icr.io/db2u/etcd@sha256:af1d052b60f599452ecbb44a060173922286e296cbf52d5a551a485cc28900e6
  - name: icr.io/db2u/etcd@sha256:b211cdc9c094ecad0f8bb3d0b714331937bb26582e450b3babec158f9857976a
  - name: icr.io/db2u/etcd@sha256:b617959f55b413b22a5bb72a80c19c7ff966a695dd4793903c9044165296aa3c
  - name: icr.io/db2u/etcd@sha256:cc1f51f48d7d95bd25a1658490adbdc0b7d8d0429e23ff621b3b8b457d51e271
  - name: icr.io/db2u/etcd@sha256:e386ff2ca65b99c63c6166de9a649875891f493091754aa4dc0e83e92dae7cb2
  blockedImages:
  - name: icr.io/cpopen/db2u-day2-operator@sha256:91a7ebbbf0667cf18e6686145dcb9acf2df2d616ee61073309100247e3b7e6c3
  - name: icr.io/cpopen/db2u-operator@sha256:4a938eb64e44e5bf155293e1939c4c8f2bcee9603647167994a94dbd79ced316
  - name: icr.io/cpopen/ibm-db2uoperator-bundle@sha256:8fc35a4bdb7db8231f0258986a3dcad014b80059681a55ce4cd2a87659c52849
  - name: icr.io/cpopen/ibm-db2uoperator-catalog:2.13.0-
22871@sha256:3325c4e4ec118c186f80b8336e5a3a5d8e2c2d7a1f06d97c9fee1cc4c90d68dc
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:12713664b4753551ae3ac303c25c4df938a4394e241e1f9bbc526e74a8ad562e
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:4b890828cc50851a42e5fec6bb929d1f71578aeb143794cbfaa1207eb0d7d048
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:890d04affc2beb381a483162aa3e998599b308ca1615ecd63e66ba04d94c224a
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:b711c164a62ccd4db955ea67afb0e225601429b6406d98fbad31c5fea13a86ea
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:d35b06c1d03707642c6641aff7d481be128064b3aabb384e5aac43715da45d04
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:e57279502a5d605931494771e8de0b382d3ddb1a5d744820f70977bb777bdaf9
```

```
  - name: icr.io/db2u/db2u.db2wh@sha256:1594796ebbe52b158c12eeb8e5d4faa791f6c9a28f96e33844a05f0282b4fc1f
  - name: icr.io/db2u/db2u.db2wh@sha256:394a2a0c640a884c1f63332bce1a8d5ad73baba0d3b3a05b4fa876e35bf9d82e
  - name: icr.io/db2u/db2u.dv.api@sha256:f8ed8a89397c5aa2d6b61ac2ed9d55e3b6351c3c0396355c17d7e44a48038def
  - name: icr.io/db2u/db2u.dv.caching@sha256:d5fe8bbd87c5d4e4424f5bd71ae3bf35769ea744322b152128ab563412eb68b0
  - name: icr.io/db2u/db2u.dv.utils@sha256:83ec4f59d30685a3ab43c74ee20336a16635935f339b78c4744058504e110f5e
  - name: icr.io/db2u/db2u.hurricane@sha256:7e7d7b1df0122120aeb33864b00cd5b9d8a2beab0ee6e5bbe273b0b7ea4f0e14
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:22bb255ac34cff1d99255a52a94b6560880f5f51937fdfcedba1ccd668688d9c
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:61c8a176b449e0aa7638a524297a80ece4ec52700d99299ae855db5b54609438
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:7906196d7910eada7f94bb375f4d6d7e5174992a6a0b3aa3308ce12283df45a8
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:7ea95db7c6249ad9b115c5934b4849f68b633024db937bb7835ec3cb4ab6a897
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:b67acd5ee75833ec1e10705a0c4e7ba0f4be85ab03ed37527b56250ee3448cb9
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:d5e452c1f3f5dcacb747713e56dcd006be6ba291bbf305352cd411a9bae57ee6
  - name: icr.io/db2u/db2u.instdb@sha256:4e8688e4964cbbeb4a5329294a0d995a0a45a9bec237e3f0a18491f0bce97784
  - name: icr.io/db2u/db2u.instdb@sha256:5ad3fc61b531781b212275eed3c3632ba0b7cd7d5cb48cf321b3f2308b66c421
  - name: icr.io/db2u/db2u.instdb@sha256:829a95722e17914869a4e224f35ace529607e71a1d2235f6e6fe7f33378d1cb5
  - name: icr.io/db2u/db2u.instdb@sha256:a6c008e42a208fe1407cf8e9a9c5e4d97eb923aee7269821fad9bd037b407357
  - name: icr.io/db2u/db2u.instdb@sha256:cc300234190d53a01cbe7349b628df0518553b4cd6795bebfbbacd66ca5ae29b
  - name: icr.io/db2u/db2u.instdb@sha256:d1f0a4cd8591bc509208ab01d0268881d4a8529d53bae147e570de5a0ead3bb1
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:344840dc5a825a755f9701792bfc5f8e7766b776665c88e83a04bec8a1cd85ac
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:7313133d3b83dbf4a214c73e809b21b431bf81f4ff3c027a4ebe0608c254c9fc
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:7e7ac8e3f66e45c99e4fcf8673e09b18e2c0f7a0d6f72c3f86eb2eb157fabaf8
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:88574364d46b83ca98f20569e7ba0533fd38dd6a3cdff23bc374a629de1379e3
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:b8288381e8b2538464097db4328641648a44c5e9671bbc502af2e997b39abb41
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:f84b460effa534c060592012f01d85f16d340557931e3532f5f6e0b4b892b266
  - name: icr.io/db2u/db2u.mustgather@sha256:51707eef34bad0eeeecfb09c5746782851a0b4d6af562b626dc667ec5e50c723
  - name: icr.io/db2u/db2u.mustgather@sha256:517ae4be20211a68c11d5848f6330ec156bff90c3c64057d273b52d4bb98dae9
  - name: icr.io/db2u/db2u.mustgather@sha256:536bf34b07defd3bc624a7a3973e6d5e701d5fdb949978cfcb1dbd546968c475
  - name: icr.io/db2u/db2u.mustgather@sha256:628ac4c7d5e9b6f527cf451f397da5a8723956a283fa9a3dfe1db2dfc91bc858
  - name: icr.io/db2u/db2u.mustgather@sha256:6c91db906f8dd87bef288595ef2393d910f124c8d31440731af8b260fd2fa55d
  - name: icr.io/db2u/db2u.mustgather@sha256:f0a3a3dd13c7d133d39b25234977a150993334ae45049cb51a7fffc15943cd9a
  - name: icr.io/db2u/db2u.qrep@sha256:3c731b912f43a68b8ab8fef2bd61a4fe5287fac5c251046a160a049730f60f5f
  - name: icr.io/db2u/db2u.qrep@sha256:44724bda2d5336f17bec950e40195f4cb6199291a22a3c6d3d9b47407e0155f1
  - name: icr.io/db2u/db2u.qrep@sha256:9512f41becbcfc6ebf8afcaf35521478fd87e2b2881d856d443b5bb882cb8d79
  - name: icr.io/db2u/db2u.qrep@sha256:d6bcb17bd3ff2776ee3588ddebd946cade42a76f38666ccfc425635b09a6eb12
  - name: icr.io/db2u/db2u.rest@sha256:6e1f90125ad8a4740bf388d16f44e73737379f4dd9b881bc2559aea2d60d9976
  - name: icr.io/db2u/db2u.rest@sha256:778a3bd330aa7c9aa7cf405551856e92d3d8b40d75a509054d8c115e73e7b795
  - name: icr.io/db2u/db2u.rest@sha256:a12723d1671e2321286c22cfad30352c7860cd95f8099e45b4847036ccc0830e
  - name: icr.io/db2u/db2u.rest@sha256:a8158630c0ad0a21592a9e08891035d0d02d7b406c15f4740db77fec125b03ce
  - name: icr.io/db2u/db2u.rest@sha256:bf6576eecd916419d2d521289970094e0c73ded5de5154030678d6543c60a2de
  - name: icr.io/db2u/db2u.rest@sha256:d000f87783bf8963423064de28d069e6877aed8ec94d572c6b22a27313fec8ff
  - name: icr.io/db2u/db2u.restricted@sha256:27e1777bf955d26ce713cb8d6ffc6ee1d8d76c79dcbe94c07d1ec154b12519bd
  - name: icr.io/db2u/db2u.restricted@sha256:33a70ec4786221efe5e5fd397951111fc90ea3b75842ef9558c889034564e91b
  - name: icr.io/db2u/db2u.restricted@sha256:4e4397c45c10f09f723ff11b3b51bb7adb492898d652501abd85511cc660ccc6
  - name: icr.io/db2u/db2u.restricted@sha256:6b204a6026e176bc7c7404fd41ede9399439cf5347698c95b838a78454ace6d9
  - name: icr.io/db2u/db2u.restricted@sha256:9b388eab7b17d543bf9c731a2583bd2606ba6199873ab914a3cdd2065e231385
  - name: icr.io/db2u/db2u.restricted@sha256:c75c667f44a6ab63447d0e72ad2743beb6f34f29f0cdee9b83d83c72ebb2457e
  - name: icr.io/db2u/db2u.tools@sha256:02f1771f4a00db39a5ef741dc39eb84caedfd7393092607994500d9b481b9a54
  - name: icr.io/db2u/db2u.tools@sha256:35ae6c6618b443c49a17ff32f5ba0c098e46fc542482c02e94857eb378ace419
  - name: icr.io/db2u/db2u.tools@sha256:83c657db8954d116a97daae90957ed8b31a89055913de08c92bbe5d30d3c71ec
  - name: icr.io/db2u/db2u.tools@sha256:adeffc101962a8181b4ceba587d046808c6440ca4a516cabdb04f973b5e299e9
  - name: icr.io/db2u/db2u.tools@sha256:e89e56844720d8512e0e0ff65a0331719bc4c203cabf2eb1da329a6dd3d00c70
  - name: icr.io/db2u/db2u.tools@sha256:f22e5bbc28263a51aa45c4a42dfaabee793322594c32a58a13914b977aca300f
  - name: icr.io/db2u/db2u.update.image@sha256:318c6f072f745c191feecb34576bb68681dd9db3644cc32191fba34274332852
  - name: icr.io/db2u/db2u.update.image@sha256:56ec63418fba021fb1eafe54bc6d62106dfa9819a3ce88707fabcadb98b239aa
  - name: icr.io/db2u/db2u.update.image@sha256:5af408ae4cb77592ee6726abe3a281b9a7fac500231d05b45b0a2b05ed88293d
  - name: icr.io/db2u/db2u.update.image@sha256:836a33719793c32744c0f830890633cd1648f282585ae672d3067bd146b1d467
  - name: icr.io/db2u/db2u.update.image@sha256:9a5875da2221ed1e2725a085ec7abd2e0418379686e337987f9496367e7c7a61
  - name: icr.io/db2u/db2u.update.image@sha256:d327bb0e5cc2a453d2fad23b629ba4862b5af9f794fc7f10ecf9d46e5d107f19
  - name: icr.io/db2u/db2u.veleroplugin@sha256:1a2d4657bf14fa60ffa1b0dbca50b696f1ff4ffd28f4c470db67b0ecca5d6fb4
  - name: icr.io/db2u/db2u.veleroplugin@sha256:51114e7afaa01a9002289017a22a2f5c9c0eceeb8f2df857c8148dbb5651e46b
  - name: icr.io/db2u/db2u.veleroplugin@sha256:796518c555504dd1e5b77504ca07397c114ac6753adbf8c8de41dcd2e1b57065
  - name: icr.io/db2u/db2u.veleroplugin@sha256:a2b0b6ff986b618fe880acec97d4af1673f4bffda385e05b6e14301c1b51d472
  - name: icr.io/db2u/db2u.veleroplugin@sha256:b3ecbc09923c288fa8d21d9e36a23a61cbec71a0c43bdb1e8213688bab571626
  - name: icr.io/db2u/db2u.veleroplugin@sha256:f87644162951b7210e450f567dacf718d3480a2f3468b36131b7a43286a6c44f
  - name: icr.io/db2u/db2u.watsonquery@sha256:897237498f224723b5818c50cd37d218032c4a05ebe2bda15bebe99d13ef0521
  - name: icr.io/db2u/db2u@sha256:01bdc550ea7d727fa5715f96c61f7ffe763ee38fc1875132187c40e09d08a6aa
  - name: icr.io/db2u/db2u@sha256:427657ba472b22f66b68d2a4f2ea7564572853083508ae2a8dbf770d1a181b85
  - name: icr.io/db2u/db2u@sha256:4319988ff6a8dff6a97368f253fed5cb268eb0c20de4db9dedd77cc47214f806
  - name: icr.io/db2u/db2u@sha256:6dda7e511b92ffc83c4b92e7b5ed99b5b619a5986af4a86d75c5f4425917dd91
  - name: icr.io/db2u/db2u@sha256:94226fa2a0fa69f8438265f481ffc05e830a75b9bb4a04f2a2b1f8d239758b09
  - name: icr.io/db2u/db2u@sha256:adc37d1fdfde105a516b8b891c494627703d90e9104181b2959a98e6630a4e94
  - name: icr.io/db2u/etcd@sha256:0c79ec0ad02a7e810ce4bc094fe85ad869ac973494161e85dbf32e394fcccf6c
  - name: icr.io/db2u/etcd@sha256:1a3362799bd9402aba7f99eed179c2747a520b37c994e63ad9e2b5fdeb4d5bd9
  - name: icr.io/db2u/etcd@sha256:1befb2b817dc6f7bf3f2fe3f5f2862b638d791d1284facddf789f245775e575c
  - name: icr.io/db2u/etcd@sha256:234c073c178f60ff8e8e29fdfda25b1e908ea0655ff61f3a4e29a51eefdea8eb
  - name: icr.io/db2u/etcd@sha256:6e0c097d8e6d9cdacc71f80312dd96b8a0d50a07ad48a333fce0bfc190f58553
  - name: icr.io/db2u/etcd@sha256:bb5c436b88c57715ffc991d0ab5a05979a4c092c65ad98296af2b1349926c45c
EOF
```

5. Run the following `oc-mirror` command to mirror the images.

```
oc mirror --v2 --config imageset-config-fusion.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-verify=false
```

6. Apply the IDMS file to your cluster:
   Note: Replace the variable `$TARGET_PATH` with your registry details where images are mirrored.

```
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: ibm-spectrum-fusion
spec:
  imageDigestMirrors:
  - mirrors:
    - $TARGET_PATH/cp
    source: cp.icr.io/cp
```

```
  - mirrors:
    - $TARGET_PATH/cpopen
    source: icr.io/cpopen
  - mirrors:
    - $TARGET_PATH/db2u
    source: icr.io/db2u
  - mirrors:
    - $TARGET_PATH/minio
    source: quay.io/minio
  - mirrors:
    - $TARGET_PATH/openshift4
    source: registry.redhat.io/openshift4
```

7. Create a **CatalogSource** (required only for upgrade).
   Note: Replace the variable **$TARGET_PATH** with your registry details where images are mirrored.

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: ibm-spectrum-fusion-catalog
  namespace: openshift-marketplace
spec:
  displayName: ibm-spectrum-fusion-2.13.0-linux-amd64
  publisher: IBM
  image: ${TARGET_PATH}/cpopen/isf-operator-
catalog@sha256:7e580c4cd056dbffd6a80dd117597e52530bde0b5cb7c9d31321f33cd01adb3e
  sourceType: grpc
  grpcPodConfig:
    nodeSelector:
      kubernetes.io/arch: amd64
```

8. If you planned to install the IBM Data Cataloging service, then apply the below catalog source.

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: ibm-spectrum-discover-catalog
  namespace: openshift-marketplace
spec:
  displayName: ibm-spectrum-discover-253.0.0
  publisher: IBM
  image: ${TARGET_PATH}/cpopen/ibm-spectrum-discover-operator-
catalog@sha256:c8b42eafa039542123c4ba29ef650c4a41996a28fd4504fb7c4e2e62fedab5b1
  sourceType: grpc
```

9. If you planned to install the Content-Aware Storage service, then apply the below catalog source.

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: ibm-cas-catalog
  namespace: openshift-marketplace
spec:
  displayName: ibm-cas-1.1.5-linux-amd64
  publisher: IBM
  image: ${TARGET_PATH}/cpopen/ibm-isf-cas-operator-
catalog@sha256:182acd92bd262539c5cc13470465719cf56398188bc6dfa230ccec1b34d8ecf7
  sourceType: grpc
  grpcPodConfig:
    nodeSelector:
      kubernetes.io/arch: amd64
```

10. Hosted Control Plane upgrade:
    a. To upgrade Hosted Control Planes, mirror the IBM Fusion image of the upgradeable version. For more information, see Mirroring IBM Fusion images.
    b. After mirroring, apply the generated IBM Fusion catalog source in IBM Fusion HCI base cluster.
       Note: Verify whether mirror of IBM Fusion images is added in the **HostedCluster** custom resource (CR).

What to do next

After you complete the end-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry, proceed with Mirroring OpenShift Container Platform release images to enterprise registry.

Mirroring OpenShift Container Platform release images to enterprise registry

Mirror the OpenShift® Container Platform image repository to your registry to use during installation or upgrade.

Before you begin

- Before you mirror, go through the prerequisites. See Mirroring prerequisites.
- Ensure that you go through the *Before you begin* section and *About the task* section of End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.
- Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute **podman** for **docker** in the commands.

About this task

For more information about Red Hat® OpenShift Container Platform, see Red Hat OpenShift Documentation.

Procedure

1. Set the following environment variables:

```
export OCP_RELEASE=<your OCP version>
export PRODUCT_REPO='openshift-release-dev'
export RELEASE_NAME="ocp-release"
export ARCHITECTURE=x86_64
export LOCAL_SECRET_JSON='<relative path to pull-secret.json>'
export LOCAL_OCP_REGISTRY='<Your enterprise registry host>:<port>'
export LOCAL_OCP_REPOSITORY='<Your image path>'
```

   For 2.13.0 fresh installation, the OCP_RELEASE values are 4.18.22 or 4.20.24.
   See the following sample values:

```
export LOCAL_SECRET_JSON='/home/mirror/pull-secret.json'
export LOCAL_OCP_REGISTRY='registryhost.com:443'
export LOCAL_OCP_REPOSITORY='mirror-ocp'
```

   `LOCAL_SECRET_JSON` is relative path for your `pull-secret.json` file.

   `LOCAL_OCP_REGISTRY` is your entitlement registry.

   `LOCAL_OCP_REPOSITORY` is the image path, in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images or hci-images/isf.

   Note: Change the values of `LOCAL_SECRET_JSON`, `LOCAL_OCP_REGISTRY`, and `LOCAL_OCP_REPOSITORY`. Keep the other values as is.
2. Run the command to login to the Docker registry with your enterprise registry credentials.

```
docker login $LOCAL_OCP_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
```

3. Run the command to create catalog mirror images:

```
oc adm release mirror -a ${LOCAL_SECRET_JSON} --
from=quay.io/${PRODUCT_REPO}/${RELEASE_NAME}:${OCP_RELEASE}-${ARCHITECTURE} --
to=${LOCAL_OCP_REGISTRY}/${LOCAL_OCP_REPOSITORY} --to-release-
image=${LOCAL_OCP_REGISTRY}/${LOCAL_OCP_REPOSITORY}:${OCP_RELEASE}-${ARCHITECTURE} --print-mirror-instructions=idms --
insecure=true
```

   After successful mirroring, a confirmation message gets displayed. For example:

```
Success
Update image:  <Your enterprise registry>:<port>/<Your image path>:4.18.22-x86_64
Mirror prefix: <Your enterprise registry>:<port>/<Your image path>
Mirror prefix: <Your enterprise registry>:<port>/<Your image path>:4.18.22-x86_64
```

   Note: If your enterprise registry is configured using self-signed certificate or if you get an error `x509: certificate signed by unknown authority`, then use `--insecure=true` to successfully complete the mirroring for OpenShift release image.
   Sample command:

```
oc adm release mirror -a /root/pull-secret.json --from=quay.io/openshift-release-dev/ocp-release:4.18.22-x86_64 --
to=registryhost.com:443/mirror-ocp --to-release-image=registryhost.com:443/mirror-ocp-x86_64 --print-mirror-
instructions=idms --insecure=true
```

4. To verify whether the images were successfully mirrored, check that the command output contains the following information:

```
imageContentSources:
- mirrors:
  - <Your enterprise registry>:<port>/ocp4/openshift4
  source: quay.io/openshift-release-dev/ocp-release
- mirrors:
  - <Your enterprise registry>:<port>/ocp4/openshift4
  source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
```

   See the following sample values:

```
imageContentSources:
- mirrors:
  - registryhost.com:443/mirror-ocp
  source: quay.io/openshift-release-dev/ocp-release
- mirrors:
  - registryhost.com:443/mirror-ocp
  source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
```

What to do next

After you complete with mirroring OpenShift Container Platform release images to enterprise registry, proceed with Mirroring Red Hat operator images to enterprise registry.

Mirroring Red Hat operator images to enterprise registry

Mirror the Red Hat® operator images to your enterprise registry.

Before you begin

- Before you mirror, go through the prerequisites. See Mirroring prerequisites.
- Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute `podman` for `docker` in the commands.
- Ensure that you install the `oc-mirror OpenShift` CLI plug-in. For more information about `oc-mirror OpenShift` CLI plug-in installation, see Installing the oc-mirror OpenShift CLI plugin.
- Ensure that you must use the recent enterprise registry images for mirroring.

Procedure

1. Log in to quay.io and run the following command to login to the Docker registry with your Red Hat enterprise credentials:

   ```
   podman login registry.redhat.io -u <Red Hat enterprise registry username> -p <Red Hat enterprise registry password>
   ```

   Set the following environment variables:

   ```
   export LOCAL_ISF_REGISTRY="<Your enterprise registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<Your image path>"
   export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
   export OCP_VERSION="<your selected 4.x version>"
   echo "$TARGET_PATH"
   ```

   Note:
   - The Red Hat OpenShift® Container Platform version variable value must be same as the version you used during the OpenShift mirroring. For example, if you mirror Red Hat OpenShift Container Platform 4.20 images, then set the OCP_VERSION value as 4.20.
   - Port is a non-mandatory value when you set the `LOCAL_ISF_REGISTRY` variable. You can ignore this if your enterprise registry is accessible and has a secure connection.

   Sample value for without port:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com"
   ```

   See the following sample values:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com:443"
   export LOCAL_ISF_REPOSITORY="fusion-mirror"
   ```

   `LOCAL_ISF_REGISTRY` is your entitlement registry.

   `LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images

2. Run the command to login to the Docker registry with your enterprise registry credentials.

   ```
   podman login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

3. Create an image set configuration file for Red Hat packages that are required for IBM Fusion installation.
   For example:

   ```
   cat << EOF > imageset-config-redhatoperator.yaml
   kind: ImageSetConfiguration
   apiVersion: mirror.openshift.io/v2alpha1
   mirror:
     operators:
     - catalog: registry.redhat.io/redhat/certified-operator-index:v$OCP_VERSION
       packages:
       - name: cloudnative-pg
     - catalog: registry.redhat.io/redhat/redhat-operator-index:v$OCP_VERSION
       packages:
       - name: kubernetes-nmstate-operator
       - name: redhat-oadp-operator
         channels:
         - name: stable-1.4
           minVersion: "1.4.2-0"
         - name: stable-1.5
           minVersion: "1.5.0-0"
       - name: amq-streams
         channels:
         - name: stable
           minVersion: "3.0.0-0"
       - name: nfd
       - name: node-maintenance-operator
       - name: rhods-operator
       - name: openshift-cert-manager-operator
       - name: kubevirt-hyperconverged
       - name: metallb-operator
       - name: multicluster-engine
       - name: local-storage-operator
       - name: lvms-operator
       - name: kernel-module-management
     additionalImages:
     - name: registry.redhat.io/ubi9/ubi:latest
     - name: registry.redhat.io/ubi9/ubi-minimal:latest
     - name: registry.redhat.io/ubi9/ubi-micro:latest
   EOF
   ```

   Note:
   - The self-node-remediation operator is used to enable automatic workload failover in case of unplanned node unavailability.

- The redhat-oadp-operator and amq-streams packages are required to install the Red Hat OADP operator and AMQ Streams operator. It is a prerequisite to deploy the IBM Backup & Restore service. If you plan to use the IBM Backup & Restore service, retain it in the commands, or else you can skip it.
- The local-storage-operator and lvms-operator packages are required to install the IBM Fusion Data Foundation service. If you plan to use the IBM Fusion Data Foundation service, retain it in the commands, or else you can skip it.
- The amq-streams package is required to install the IBM Data Cataloging service. If you plan to use the IBM IBM Data Cataloging service, retain it in the commands, or else you can skip it.
- The **`kubevirt-hyperconverged`** is required to install the OpenShift virtualization operator.
- The **`cloudnative-pg`** package is required to install the CAS service. If you plan to use the CAS service, add it in the commands, or else you can skip it.
- The **`kubevirt-hyperconverged`**, **`metallb-operator`**, and **`multicluster-engine`** operators are required to deploy Hosted Control Plane clusters.
- The **`node-maintenance-operator`** is required for node maintenance operations on both the Hub and Hosted Control Plane clusters.
- The **`kubernetes-nmstate-operator`** package is required for Stage2, that manages node network configuration
- The **`node-feature-discovery(nfd)`** package is required when the cluster contains GPU nodes or specialized hardware that needs to be discovered and labeled. Essential for GPU enabled workloads and hardware accelerated services.
- The **`rhods-operator`** package is required for Software Hub and IBM watsonx services.
- The **`openshift-cert-manager-operator`** package is required for Software Hub and IBM watsonx services to manage TLS certificates and secure communications between service components.

4. Run the following `oc` command to mirror the images from the specified image set configuration to a specified registry:

   ```
   oc mirror --v2 --config imageset-config-redhatoperator.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-verify=false
   ```

   This can take 1-2 hours to complete.

   Example output:

   ```
   Rendering catalog image "<TARGET_PATH>/redhat/redhat-operator-index:v4.16" with file-based catalog
   Writing image mapping to oc-mirror-workspace/results-1693810862/mapping.txt
   Writing CatalogSource manifests to oc-mirror-workspace/results-1693810862
   Writing ICSP manifests to oc-mirror-workspace/results-1693810862
   ```

5. Important: You can run steps 5, 6, and 7 only when the cluster is up for a freshly installed rack.
   Apply the generated **`ImageDigestMirrorSet`** to the cluster:

   ```
   oc apply -f working-dir/cluster-resources/idms-oc-mirror.yaml
   ```

   Example **`ImageDigestMirrorSet`**:

   ```
   apiVersion: config.openshift.io/v1
   kind: ImageDigestMirrorSet
   metadata:
     annotations:
       createdAt: Tuesday, 07-Apr-26 14:48:07 UTC createdBy: oc-mirror v2 oc-mirror_version: 4.20.0-202511252120.p2.g5b23c4a.asse
     name: idms-operator-0
   spec:
     imageDigestMirrors:
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/migration-toolkit-virtualization
       source: registry.redhat.io/migration-toolkit-virtualization
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/openshift4
       source: registry.redhat.io/openshift4
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/amq-streams
       source: registry.redhat.io/amq-streams
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/rhel9
       source: registry.redhat.io/rhel9
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/workload-availability
       source: registry.redhat.io/workload-availability
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/cert-manager
       source: registry.redhat.io/cert-manager
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/rhoai
       source: registry.redhat.io/rhoai
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/multicluster-engine
       source: registry.redhat.io/multicluster-engine
     - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/oadp
       source: registry.redhat.io/oadp
      - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/rhel7
       source: registry.redhat.io/rhel7
      - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new
       source: registry.redhat.io
      - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/container-native-virtualization
       source: registry.redhat.io/container-native-virtualization
      - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/rhaii-early-access
       source: registry.redhat.io/rhaii-early-access
      - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/lvms4
       source: registry.redhat.io/lvms4
      - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/rhaiis
       source: registry.redhat.io/rhaiis
      - mirrors:
       - r11-ru22-w-l.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/rhelai1
   ```

```
    source: registry.redhat.io/rhelai1
  - mirrors:
    - r11-ru22-w-1.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/ubi9
    source: registry.redhat.io/ubi9
  - mirrors:
    - r11-ru22-w-1.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/ubi8
    source: registry.redhat.io/ubi8
  - mirrors:
    - r11-ru22-w-1.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/cluster-observability-operator
    source: registry.redhat.io/cluster-observability-operator
  - mirrors:
    - r11-ru22-w-1.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/openshift-service-mesh
    source: registry.redhat.io/openshift-service-mesh
  - mirrors:
    - r11-ru22-w-1.quay-service.fusion.tadn.ibm.com:8443/temptest/test-rh-new/rhel8
    source: registry.redhat.io/rhel8
```

6. Apply the generated `CatalogSource` to the cluster:

```
oc apply -f catalogSource-cs-redhat-operator-index.yaml
oc apply -f catalogSource-cs-certified-operator-index.yaml
```

For example:

```
oc apply -f working-dir/cluster-resources/cs-redhat-operator-index-v4-20.yaml
```

7. For offline environments, update the `redhat-operators` CatalogSource image with the following mirrored image:

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: redhat-operators
  namespace: openshift-marketplace
spec:
  displayName: Red Hat Operators
  image: $TARGET_PATH/redhat/redhat-operator-index:v$OCP_VERSION
  publisher: Red Hat
  sourceType: grpc
```

Mirroring OpenShift Container Platform release images to enterprise registry for Hosted Control Plane

Mirror the OpenShift® Container Platform image repository to your registry to use during installation or upgrade of the Hosted Control Plane.

Before you begin

1. Before you mirror, go through the prerequisites. See Mirroring prerequisites.
2. Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute `podman` for `docker` in the commands.

Procedure

1. Set the following environment variables:

```
export LOCAL_ISF_REGISTRY="<Your enterprise registry host>:<port>"
export LOCAL_ISF_REPOSITORY="<Your image path>"
export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
export OCP_VERSION="<your selected OCP version>
export OCP_MIN_VERSION="<your selected  4.x.y-build>"
export OCP_MAX_VERSION="<your selected 4.x.y-build>"
echo "$TARGET_PATH"
```

2. Run the command to login to the Docker registry with your enterprise registry credentials.

```
podman login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
```

3. Create an image set configuration file for the OpenShift Container Platform.

```
cat << EOF > hcp-ocp-images-imageset.yaml
apiVersion: mirror.openshift.io/v2alpha1
kind: ImageSetConfiguration
mirror:
  platform:
    channels:
      - name: stable-$OCP_VERSION
        minVersion: $OCP_MIN_VERSION
        maxVersion: $OCP_MAX_VERSION
        type: ocp
    kubeVirtContainer: true
    graph: true
EOF
```

For example:

```
apiVersion: mirror.openshift.io/v2alpha1
kind: ImageSetConfiguration
mirror:
```

```
  platform:
    channels:
      - name: stable-4.18
        minVersion: 4.18.22
        maxVersion: 4.18.22
        type: ocp
    kubeVirtContainer: true
    graph: true
```

Note: In case of Bare Metal Hosted Control Plane deployment, remove `kubeVirtContainer: true` flag.

4. Run the command to create catalog mirror images:

```
oc-mirror --v2 --config hcp-ocp-images-imageset.yaml --workspace file://mirror-file docker://$TARGET_PATH
```

It might take 10 -15 minutes to complete.
Example output:

```
[INFO]    : === Results ===
[INFO]    :   release images mirrored successfully
[INFO]    : mirror-file/working-dir/cluster-resources/idms-oc-mirror.yaml file created
#…
```

5. Upon successful execution, apply the generated IDMS configuration to the cluster.
Example IDMS:

```
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: idms-release-0
spec:
  imageDigestMirrors:
  - mirrors:
    - <target path>/openshift/release
    source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
  - mirrors:
    - <target path>/openshift/release-images
    source: quay.io/openshift-release-dev/ocp-release
```

Mirroring individual images

There may be scenarios wherein you may want to mirror individual images separately.

You may choose to mirror individual images to suit customized requirements, manage resource usage, or greater control over the mirroring.

- **Mirroring IBM Fusion HCI images**
  Mirror the IBM Fusion HCI images to your enterprise registry.
- **Mirroring IBM Storage Scale images**
  Mirror the IBM Storage Scale images to your enterprise registry.
- **Mirroring Fusion Data Foundation images**
  Mirror the Fusion Data Foundation images deployed on Red Hat OpenShift Container Platform to your enterprise registry by using `ImageDigestMirrorSet`.
- **Mirroring Backup & Restore images**
  Mirror the Backup & Restore images to your enterprise registry.
- **Mirroring IBM Data Cataloging images**
  Mirror the IBM Data Cataloging images to your enterprise registry.
- **Mirroring Content-Aware Storage (CAS) images**
  IBM Fusion HCI supports a range of CAS versions, and it can upgrade independently of IBM Fusion HCI within a certain range. In 2.13.0, the supported CAS version range is any version < v1.1.0. You can run this procedure multiple times to upgrade to each newer version of CAS within the supported range.
- **Mirroring IBM Software Hub, watsonx.ai, and watsonx Orchestrate images**
  Mirror the IBM Software Hub, watsonx.ai, watsonx Orchestrate, and Red Hat OpenShift AI images to your enterprise registry.

Mirroring IBM Fusion HCI images

Mirror the IBM Fusion HCI images to your enterprise registry.

Before you begin

1. Before you mirror, go through the prerequisites. See Mirroring prerequisites.
2. Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute `podman` for `docker` in the commands.
3. For more information about the version support, see IBM Fusion HCI support matrix.

About this task

Use this procedure only necessary if you plan to mirror specific sub-components of IBM Fusion HCI instead of following the full end-to-end mirroring instructions. For more information about the end-to-end mirroring of and all its services, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.

High level tasks for mirroring:

- Log in to the relevant Docker registries.
- Define some convenience variables that are used to configure and mirror the Fusion base component.

- Run the oc-mirror CLI tool that mirrors the Fusion base components.
- Run the mirroring steps for the additional components you want to include in your installation.

Note: If you choose to mirror specific sub-components, it is recommended that you use the same `TARGET_PATH` variable so that all the components mirror in the same location, and the generated `ImageDigestMirrorSet` is same across all components.

Procedure

1. Run the following command to login to the Docker registry with your Red Hat® enterprise credentials:

   ```
   docker login registry.redhat.io -u <Red Hat enterprise registry username> -p <Red Hat enterprise registry password>
   ```

   Log in to the IBM Entitled Container Registry using the IBM entitlement key:

   ```
   docker login cp.icr.io -u cp -p <your entitlement key>
   ```

   Important:
   - Ensure that your entitlement key for IBM Fusion HCI contains the correct entitlement.
   - Ensure you review the offline upgrade considerations before proceeding with upgrade mirroring.

   Set the following environment variables:

   ```
   export LOCAL_ISF_REGISTRY="<Your container registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<Your image path>"
   export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
   ```

   Note: Port is a non-mandatory value when setting the `LOCAL_ISF_REGISTRY` variable. You can ignore this if your enterprise registry is accessible and has a secure connection.
   Sample value for without port:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com"
   ```

   See the following sample values:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com:443"
   export LOCAL_ISF_REPOSITORY="fusion-mirror"
   ```

   `LOCAL_ISF_REGISTRY` is your entitlement registry.

   `LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images.

2. Run the command to login to the Docker registry with your enterprise registry credentials.

   ```
   docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

3. Create the image set configuration for IBM Fusion HCI.
   `imageset-config-fusion.yaml`

   ```
   cat << EOF > imageset-config-fusion.yaml
   kind: ImageSetConfiguration
   apiVersion: mirror.openshift.io/v2alpha1
   mirror:
     operators:
     - catalog: icr.io/cpopen/isf-operator-catalog@sha256:7e580c4cd056dbffd6a80dd117597e52530bde0b5cb7c9d31321f33cd01adb3e
       packages:
       - name: isf-operator
         channels:
         - name: v2.0
         minVersion: 2.13.0
         maxVersion: 2.13.0
     additionalImages:
     - name: cp.icr.io/cp/fusion-hci/isf-validate-
   entitlement@sha256:1a0dbf7c537f02dc0091e3abebae0ccac83da6aa147529f5de49af0f23cd9e8e
     - name: icr.io/cpopen/isf-operator-catalog:2.13.0-
   20329@sha256:7e580c4cd056dbffd6a80dd117597e52530bde0b5cb7c9d31321f33cd01adb3e
   EOF
   ```

4. Run the following oc-mirror command to mirror the images.

   ```
   oc mirror --v2 --config imageset-config-fusion.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-verify=false
   ```

5. Important: You can run steps 5 and 6 only when the cluster is up for a freshly installed rack.
   Apply the IDMS file to your cluster:
   Note: Replace the variable `$TARGET_PATH` with your registry details where images are mirrored.

   ```
   apiVersion: config.openshift.io/v1
   kind: ImageDigestMirrorSet
   metadata:
     name: ibm-spectrum-fusion
   spec:
     imageDigestMirrors:
     - mirrors:
       - $TARGET_PATH/cp
       source: cp.icr.io/cp
     - mirrors:
       - $TARGET_PATH/cpopen
       source: icr.io/cpopen
     - mirrors:
       - $TARGET_PATH/openshift4
       source: registry.redhat.io/openshift4
   ```

6. Create a **CatalogSource** (required only for upgrade).
   Note: Replace the variable **$TARGET_PATH** with your registry details where images are mirrored.

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: ibm-spectrum-fusion-catalog
  namespace: openshift-marketplace
spec:
  displayName: ibm-spectrum-fusion-2.13.0-linux-amd64
  publisher: IBM
  image: ${TARGET_PATH}/cpopen/isf-operator-
catalog@sha256:7e580c4cd056dbffd6a80dd117597e52530bde0b5cb7c9d31321f33cd01adb3e
  sourceType: grpc
  grpcPodConfig:
    nodeSelector:
      kubernetes.io/arch: amd64
```

7. Follow the steps to mirror the Usage Metering Service (UMS):
   Important: This step is required for mirroring IBM Fusion HCI images. Do not skip.
   a. Create the image set configuration for UMS.

   ```
   cat << EOF > imageset-config-ums.yaml
   kind: ImageSetConfiguration
   apiVersion: mirror.openshift.io/v2alpha1
   mirror:
    operators:
     - catalog: icr.io/cpopen/ibm-usage-metering-operator-
   catalog@sha256:a97cef95c73ac6dacea010818757b4ed9b7440f4c0ec4154a091f248c68d1c9e
      full: true
      packages:
       - name: ibm-usage-metering-operator
        channels:
         - name: v1.0
    additionalImages:
    - name: icr.io/cpopen/ibm-usage-metering-operator-
   catalog@sha256:a97cef95c73ac6dacea010818757b4ed9b7440f4c0ec4154a091f248c68d1c9e
   EOF
   ```

   b. Run the following oc command to mirror the images.

   ```
   oc mirror --v2 --config imageset-config-ums.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-
   verify=false
   ```

   c. Important: You can run this step only when the cluster is up for a freshly installed rack.
   Apply the **ImageDigestMirrorSet** file to your cluster:
   Note: Replace the variable **$TARGET_PATH** with your registry details where images are mirrored.

   ```
   apiVersion: config.openshift.io/v1
   kind: ImageDigestMirrorSet
   metadata:
     name: ibm-usage-metering
   spec:
     imageDigestMirrors:
     - mirrors:
       - ${TARGET_PATH}/cpopen
       source: icr.io/cpopen
   ```

   In case of any failure, refer Troubleshooting and known issues in offline mirroring to get the resolution steps.

8. Important: The Node Feature Discovery (NFD) and NVIDIA GPU operator images need to be mirrored only if GPU nodes are present or if you are planning to add GPU nodes.
   Follow the steps to mirroring the NFD catalog source:
   Note:
   - If the cluster already contains an existing **redhat-operators** catalog source, perform the mirroring using the same path specified in that catalog source.
   - Verify the **<ocp-version>** installed on the cluster.

   a. Run the following command to check the package name of the NFD operator:

   ```
   oc-mirror list operators --catalog=registry.redhat.io/redhat/redhat-operator-index:v<ocp-version> --version=<ocp-
   version>
   ```

   Here **<ocp-version>** refers to OpenShift® version installed on the cluster.

   For example:

   ```
   oc-mirror list operators --catalog=registry.redhat.io/redhat/redhat-operator-index:v4.12 --version=4.12
   ```

   b. Run the following commands to create the **imageset-config-nfd.yaml**.

   ```
   oc mirror init --registry ${TARGET_PATH}/oc-mirror-nfd-metadata > imageset-config-nfd.yaml
   ```

   c. Edit the **imageset-config-nfd.yaml** file as follows:
   - Set **skipTLS** to true.
   - Set catalog to
     registry.redhat.io/redhat/redhat-operator-index:v<ocp-version>. Here **<ocp-version>** refers to OpenShift version installed on the cluster.
     For example:

     ```
     registry.redhat.io/redhat/redhat-operator-index:v4.12 (assuming 4.12 is installed on the cluster)
     ```

   - Set packages name to nfd.

For example:

```
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v1alpha2
storageConfig:
  registry:
    imageURL: ${TARGET_PATH}/oc-mirror-nfd-metadata
    skipTLS: true
mirror:
operators:
- catalog: registry.redhat.io/redhat/redhat-operator-index:v<OCP_VERSION>
packages:
- name: nfd
additionalImages:
- name: registry.redhat.io/ubi8/ubi:latest
helm: {}
```

Note: If Red Hat operator catalog is running on your cluster, prune and push all earlier packages, including the NFD . Otherwise, old packages get lost from the Red Hat operator index image. For more information about Red Hat operator images, see Mirroring Red Hat operator images to enterprise registry.

d. Run the following command to mirror the image set to the private mirror registry:

```
oc-mirror --dest-skip-tls=true --config=./imageset-config-nfd.yaml docker://${TARGET_PATH}
```

This command generates an output directory, for example:

```
oc-mirror-workspace/results-1670391296/
```

The directory contains **mapping.txt**, **CatalogSource manifests**, and **ImageContentSourcePolicy (ICSP) manifests**. Example output:

```
Rendering catalog image
"ec2-3-16-218-255.us-east-2.compute.amazonaws.com:5000/redhat/redhat-operator-index:v4.11"
with file-based catalog

Writing image mapping to oc-mirror-workspace/results-1670391296/mapping.txt
Writing CatalogSource manifests to oc-mirror-workspace/results-1670391296
Writing ICSP manifests to oc-mirror-workspace/results-1670391296
```

e. Important:
   - You can run steps 8.e, 8.f, and 8.g only when the cluster is up for a freshly installed rack.
   - If the **redhat-operators** catalog source already exists in the cluster, then skip the following step.

Rename the generated catalog source name to **redhat-operators** from **cs-redhat-operator-index**.

```
kind: CatalogSource
metadata:
  name: cs-redhat-operator-index   --> change to redhat-operators
  namespace: openshift-marketplace
```

Apply this edited catalogsource yaml file using following command:

```
oc apply oc-mirror-workspace/results-1759765853/catalogSource-cs-redhat-operator-index.yaml
```

f. Convert the generated ICSPs (generic-0, operator-0, and release-0) to IDMS equivalents. Sample IDMS.yaml:

```
---
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: generic-0
spec:
  imageDigestMirrors:
  - mirrors:
    - ${TARGET_PATH}/ubi8
    source: registry.redhat.io/ubi8
---
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  labels:
    operators.openshift.org/catalog: "true"
  name: operator-0
spec:
  imageDigestMirrors:
  - mirrors:
    - ${TARGET_PATH}/openshift4
    source: registry.redhat.io/openshift4
---
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: release-0
spec:
  imageDigestMirrors:
  - mirrors:
    - ${TARGET_PATH}/openshift/release
    source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
  - mirrors:
    - ${TARGET_PATH}/openshift/release-images
    source: quay.io/openshift-release-dev/ocp-release
```

Apply these three IDMSs on the cluster using following command:

```
oc apply -f imageDigestMirrorSet.yaml
```

g. Create and apply ITMS equivalent for the generated ICSP operator-0.
For example:

```
apiVersion: config.openshift.io/v1
kind: ImageTagMirrorSet
metadata:
  name: operator-0
spec:
  imageTagMirrors:
    - mirrors:
        - '${TARGET_PATH}/openshift4'
      source: registry.redhat.io/openshift4
```

Apply this ITMS using following command:

```
oc apply -f imageTagMirrorSet.yaml
```

h. Search for the `ose-cluster-nfd-rhel9-operator` image in the mapping.txt file:
For example:

```
registry.redhat.io/openshift4/ose-cluster-nfd-rhel9-
operator@sha256:ce25bd4baaf2c29e3212c69e243c943e4eb507dad4a85a7b040dafb427920523=${TARGET_PATH}/openshift4/ose-
cluster-nfd-rhel9-operator:54e0411b
```

Mirror the image for `ose-cluster-nfd-rhel9-operator` in digest format into the private registry.

```
skopeo copy --all --dest-tls-verify=false \docker://registry.redhat.io/openshift4/ose-cluster-nfd-rhel9-
operator@sha256:ce25bd4baaf2c29e3212c69e243c943e4eb507dad4a85a7b040dafb427920523 \
  docker://${TARGET_PATH}/openshift4/ose-cluster-nfd-rhel9-
operator@sha256:ce25bd4baaf2c29e3212c69e243c943e4eb507dad4a85a7b040dafb427920523
```

i. Mirror the images required by the NFD custom resource. For procedure, follow Steps 1 and 2 from the Red Hat documentation.

9. Follow the steps to mirroring the GPU catalog source to install the NVIDIA GPU operator:

a. Run the following command to check for the correct package name for the GPU operator.

```
oc-mirror list operators --catalog=registry.redhat.io/redhat/certified-operator-index:v<ocp-version> --version=<ocp-
version>
oc-mirror list operators --catalog=registry.redhat.io/redhat/certified-operator-index:v<ocp-version> --package=gpu-
operator-certified
```

Example output:

```
NAME                    DISPLAY NAME         DEFAULT CHANNEL
gpu-operator-certified  NVIDIA GPU Operator  v25.3

PACKAGE                 CHANNEL  HEAD
gpu-operator-certified  stable   gpu-operator-certified.v25.3.4
gpu-operator-certified  v1.10    gpu-operator-certified.v1.10.1
gpu-operator-certified  v1.11    gpu-operator-certified.v1.11.1
gpu-operator-certified  v25.3    gpu-operator-certified.v25.3.4
```

b. Run the following command to create the `imageset-config-gpu.yaml` file.

```
oc-mirror init --registry ${TARGET_PATH}/oc-mirror-gpu-metadata > imageset-config-gpu.yaml
```

c. Edit the `imageset-config-gpu.yaml` file as follows:

- Set `skipTLS` to true.
- Set catalog to `registry.redhat.io/redhat/certified-operator-index:v<ocp-version>`.
- Set packages name to `gpu-operator-certified`.
  For example:

```
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v1alpha2
storageConfig:
  registry:
    imageURL: ${TARGET_PATH}/oc-mirror-gpu-metadata
    skipTLS: true
mirror:
  platform:
    channels:
    - name: stable-<ocp-version>
      type: ocp
  operators:
  - catalog: registry.redhat.io/redhat/certified-operator-index:v<ocp-version>
    packages:
    - name: gpu-operator-certified
  additionalImages:
  - name: registry.redhat.io/ubi8/ubi:latest
  helm: {}
```

To mirror only v25.3.4, make the same edits as the preceding example and also specify the minversion value:

```
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v1alpha2
storageConfig:
  registry:
    imageURL: ${TARGET_PATH}/oc-mirror-gpu-metadata
    skipTLS: true
mirror:
  platform:
    channels:
    - name: stable-<ocp-version>
```

```
      type: ocp
  operators:
  - catalog: registry.redhat.io/redhat/certified-operator-index:v<ocp-version>
    packages:
    - name: gpu-operator-certified
      minversion: "25.3.4"
  additionalImages:
  - name: registry.redhat.io/ubi8/ubi:latest
  helm: {}
```

d. Run the following command to mirror the GPU operator catalog source.

```
oc-mirror --dest-skip-tls=true --skip-missing --continue-on-error --config=./imageset-config-gpu.yaml
docker://${TARGET_PATH}
```

The `oc-mirror` tool generates a folder similar to `oc-mirror-workspace/results-<ID>`.

e. Important: You can run steps 9.e, 9.f, and 9.g only when the cluster is up for a freshly installed rack.
Run the following command to create the catalog source for the GPU operator.
Note: Before applying the YAML, edit the name of the catalog source with certified-operator.

```
oc apply oc-mirror-workspace/results-<ID>/catalogSource-cs-certified-operator-index.yaml
```

f. Create and apply the equivalent IDMS for all the generated ICSPs in the path `oc-mirror-workspace/results-1759765853/imageContentSourcePolicy.yaml`.

g. Create and apply the equivalent ITMS for the generated operator-0 ICSP.
Note: Update the ICSP names to reflect GPU-specific configurations. For example, rename operator-0 to operator-0-gpu.

```
# cat oc-mirror-workspace/results-<ID>/imageContentSourcePolicy.yaml
---
apiVersion: operator.openshift.io/v1alpha1
kind: ImageContentSourcePolicy
metadata:
  name: generic-0
spec:
  repositoryDigestMirrors:
  - mirrors:
    - ${TARGET_PATH}/ubi8
    source: registry.redhat.io/ubi8
---
apiVersion: operator.openshift.io/v1alpha1
kind: ImageContentSourcePolicy
metadata:
  labels:
    operators.openshift.org/catalog: "true"
  name: operator-0
spec:
  repositoryDigestMirrors:
  - mirrors:
    - ${TARGET_PATH}/nvidia
    source: nvcr.io/nvidia
  - mirrors:
    - ${TARGET_PATH}/nvidia
    source: registry.connect.redhat.com/nvidia
---
apiVersion: operator.openshift.io/v1alpha1
kind: ImageContentSourcePolicy
metadata:
  name: release-0
spec:
  repositoryDigestMirrors:
  - mirrors:
    - ${TARGET_PATH}/openshift/release
    source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
  - mirrors:
    - ${TARGET_PATH}/openshift/release-images
    source: quay.io/openshift-release-dev/ocp-release
```

h. Use the `skopeo` CLI tool to copy the NVIDIA driver image from `nvcr.io` to your mirror registry, by running the following command:

```
skopeo copy \
docker://nvcr.io/nvidia/driver:<GPU Driver versions>-rhcos<ocp_version> \
docker://${TARGET_PATH}/nvidia/driver:<GPU Driver versions>-rhcos<ocp_version>
```

For example:

```
skopeo copy \
docker://nvcr.io/nvidia/driver:580.82.07-rhcos<ocp_version> \
docker://${TARGET_PATH}/nvidia/driver:580.82.07-rhcos<ocp_version>
```

Mirroring IBM Storage Scale images

Mirror the IBM Storage Scale images to your enterprise registry.

Before you begin

- Use this procedure to mirror IBM Storage Scale images to enable the Global Data Platform or Global Data Platform Remote Mount service in IBM Fusion HCI on Red Hat® OpenShift® Container Platform 4.20 or earlier.

- For Red Hat OpenShift Container Platform 4.21:

  IBM Storage Scale Container Native Storage Access (CNSA) is integrated with Fusion Data Foundation through the remote mount feature.

  To enable the Global Data Platform Remote Mount service in IBM Fusion HCI:
  1. Mirror the Fusion Data Foundation images as described in [Mirroring Fusion Data Foundation images](#).
  2. Mirror the IBM Storage Scale 6.0.1.0 images as described in [Mirroring IBM Storage Scale 6.0.1.0 images](#).
- Before you mirror, go through the prerequisites. See [Mirroring prerequisites](#).
- Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute **`podman`** for **`docker`** in the commands.
- For more information about the version support, see [IBM Fusion HCI support matrix](#).
- For more information on offline setup for network restricted Red Hat OpenShift Container Platform clusters, see [Air-gaped installs](#).

Procedure

1. Log in to the IBM Entitled Container Registry by using the IBM entitlement key:

   ```
   docker login cp.icr.io -u cp -p <your entitlement key>
   ```

   Note: Your entitlement key for IBM Fusion HCI must contain the correct entitlement.
   Set the following environment variables:

   ```
   export LOCAL_ISF_REGISTRY="<Your enterprise registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<Your image path>"
   IFS='/' read -r NAMESPACE PREFIX <<< "$LOCAL_ISF_REPOSITORY"
   if [[ "$PREFIX" != "" ]]; then export TARGET_PATH="$LOCAL_ISF_REGISTRY/$NAMESPACE/$PREFIX"; export REPO_PREFIX=$(echo 
   "$PREFIX"| sed -r 's/\//-/g')-; export NAMESPACE="$NAMESPACE"; else export TARGET_PATH="$LOCAL_ISF_REGISTRY/$NAMESPACE"; 
   export REPO_PREFIX=""; fi
   #verify both variables set correctly
   echo "$TARGET_PATH"
   echo "$NAMESPACE"
   echo "$REPO_PREFIX"
   ```

   Note: Port is a non-mandatory value when you set the **`LOCAL_ISF_REGISTRY`** variable. If your enterprise registry is accessible and has a secure connection, ignore it.
   Sample value for without port:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com"
   ```

   See the following sample values:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com:443"
   export LOCAL_ISF_REPOSITORY="fusion-mirror"
   ```

   `LOCAL_ISF_REGISTRY` is your entitlement registry.

   `LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images.

2. Run the command to login to the Docker registry with your enterprise registry credentials.

   ```
   docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password> --tls-
   verify=false
   ```

3. From the mirroring host, run the following commands to copy the IBM Storage Scale images to the artifactory and quay registry:
   a. Global Data Platform 5.2.3.7
      - Artifactory:

        ```
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/erasure-code/ibm-spectrum-scale-
        daemon@sha256:c3a1a1a2632ff80a4f7a5064fabb8ad6b7d0bd24f8dc4195595a96556feab7a0 docker://$TARGET_PATH/erasure-
        code/ibm-spectrum-scale-daemon@sha256:c3a1a1a2632ff80a4f7a5064fabb8ad6b7d0bd24f8dc4195595a96556feab7a0
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/data-management/ibm-spectrum-scale-
        daemon@sha256:d81a8887d2a6b7fa450bc6074a053a8f29ee7e4406c8baaba5c45526135d6763 docker://$TARGET_PATH/data-
        management/ibm-spectrum-scale-daemon@sha256:d81a8887d2a6b7fa450bc6074a053a8f29ee7e4406c8baaba5c45526135d6763
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-core-
        init@sha256:dd3899c7e2b50041d81b37aa3b1b2c028dc12691d465f3f8acd5c3ce22714b16 docker://$TARGET_PATH/ibm-spectrum-
        scale-core-init@sha256:dd3899c7e2b50041d81b37aa3b1b2c028dc12691d465f3f8acd5c3ce22714b16
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
        gui@sha256:324e700c049ee9eb5239c6baada0eeb1b9389877fcb0f3366e596d294d5c82e9 docker://$TARGET_PATH/ibm-spectrum-
        scale-gui@sha256:324e700c049ee9eb5239c6baada0eeb1b9389877fcb0f3366e596d294d5c82e9
        skopeo copy --all --preserve-digests 
        docker://cp.icr.io/cp/gpfs/postgres@sha256:ef257d85f76e48da1c64832459b59fcaba1a4dac97bf5d7450c77753542eee94 
        docker://$TARGET_PATH/postgres@sha256:ef257d85f76e48da1c64832459b59fcaba1a4dac97bf5d7450c77753542eee94
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
        logs@sha256:31a5c265df2a3183885e601369435f986baa6c8375ed7d6b827422b464015389 docker://$TARGET_PATH/ibm-spectrum-
        scale-logs@sha256:31a5c265df2a3183885e601369435f986baa6c8375ed7d6b827422b464015389
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
        pmcollector@sha256:2398d15d59a0a66603a510d9196d5a30517787c89fe347b2507b06d343f1c2cd docker://$TARGET_PATH/ibm-
        spectrum-scale-pmcollector@sha256:2398d15d59a0a66603a510d9196d5a30517787c89fe347b2507b06d343f1c2cd
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
        monitor@sha256:e32160671466a3794364c099712cf69761530d3728b548a5b951f6f6a9d3b541 docker://$TARGET_PATH/ibm-
        spectrum-scale-monitor@sha256:e32160671466a3794364c099712cf69761530d3728b548a5b951f6f6a9d3b541
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-grafana-
        bridge@sha256:b45261f1b673e4f933d512ce7ac31b72da1b4fa1229809181bf6f089cadbd6ed docker://$TARGET_PATH/ibm-
        spectrum-scale-grafana-bridge@sha256:b45261f1b673e4f933d512ce7ac31b72da1b4fa1229809181bf6f089cadbd6ed
        skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
        coredns@sha256:18058dc1e5ee3febb327a13d846fa80533de365788e3e03ed00e358ef45581ff docker://$TARGET_PATH/ibm-
        spectrum-scale-coredns@sha256:18058dc1e5ee3febb327a13d846fa80533de365788e3e03ed00e358ef45581ff
        skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-must-
        gather@sha256:c9b091bd73585a1361689bbb11a78f76883185f7a76cc872bb77c0a4c9ab2b8c docker://$TARGET_PATH/ibm-
        ```

```
spectrum-scale-must-gather@sha256:c9b091bd73585a1361689bbb11a78f76883185f7a76cc872bb77c0a4c9ab2b8c
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-
snapshotter@sha256:5f4bb469fec51147ce157329dab598c758da1b018bad6dad26f0ff469326d769
docker://$TARGET_PATH/csi/csi-
snapshotter@sha256:5f4bb469fec51147ce157329dab598c758da1b018bad6dad26f0ff469326d769
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-
attacher@sha256:69888dba58159c8bc0d7c092b9fb97900c9ca8710d088b0b7ea7bd9052df86f6 docker://$TARGET_PATH/csi/csi-
attacher@sha256:69888dba58159c8bc0d7c092b9fb97900c9ca8710d088b0b7ea7bd9052df86f6
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-
provisioner@sha256:d5e46da8aff7d73d6f00c761dae94472bcda6e78f4f17b3802dc89d44de0111b
docker://$TARGET_PATH/csi/csi-
provisioner@sha256:d5e46da8aff7d73d6f00c761dae94472bcda6e78f4f17b3802dc89d44de0111b
skopeo copy --all --preserve-digests
docker://cp.icr.io/cp/gpfs/csi/livenessprobe@sha256:2c5f9dc4ea5ac5509d93c664ae7982d4ecdec40ca7b0638c24e5b16243b8
360f
docker://$TARGET_PATH/csi/livenessprobe@sha256:2c5f9dc4ea5ac5509d93c664ae7982d4ecdec40ca7b0638c24e5b16243b8360f
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-node-driver-
registrar@sha256:d7138bcc3aa5f267403d45ad4292c95397e421ea17a0035888850f424c7de25d docker://$TARGET_PATH/csi/csi-
node-driver-registrar@sha256:d7138bcc3aa5f267403d45ad4292c95397e421ea17a0035888850f424c7de25d
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-
resizer@sha256:8ddd178ba5d08973f1607f9b84619b58320948de494b31c9d7cd5375b316d6d4 docker://$TARGET_PATH/csi/csi-
resizer@sha256:8ddd178ba5d08973f1607f9b84619b58320948de494b31c9d7cd5375b316d6d4
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/ibm-spectrum-scale-csi-
driver@sha256:35ffde86b9805b83862a07acff838d086006cd3277424c8630de4697218a31e6 docker://$TARGET_PATH/csi/ibm-
spectrum-scale-csi-driver@sha256:35ffde86b9805b83862a07acff838d086006cd3277424c8630de4697218a31e6
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-csi-
operator@sha256:a247e6bde9a851d80b671b01150dbe7267129b4650cf50cfef046ac28b2d9c8d docker://$TARGET_PATH/ibm-
spectrum-scale-csi-operator@sha256:a247e6bde9a851d80b671b01150dbe7267129b4650cf50cfef046ac28b2d9c8d
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-
operator@sha256:ec39d48e72c086961ad7b65384f09af137977cbb4b8b218aa7d572c3c7501046 docker://$TARGET_PATH/ibm-
spectrum-scale-operator@sha256:ec39d48e72c086961ad7b65384f09af137977cbb4b8b218aa7d572c3c7501046
```

- Quay registry:

```
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/erasure-code/ibm-spectrum-scale-
daemon@sha256:c3a1a1a2632ff80a4f7a5064fabb8ad6b7d0bd24f8dc4195595a96556feab7a0 docker://$TARGET_PATH/erasure-
code/ibm-spectrum-scale-daemon:c3a1a1a2632ff80a4f7a5064fabb8ad6b7d0bd24f8dc4195595a96556feab7a0 --tls-
verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/data-management/ibm-spectrum-scale-
daemon@sha256:d81a8887d2a6b7fa450bc6074a053a8f29ee7e4406c8baaba5c45526135d6763 docker://$TARGET_PATH/data-
management/ibm-spectrum-scale-daemon:d81a8887d2a6b7fa450bc6074a053a8f29ee7e4406c8baaba5c45526135d6763 --tls-
verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-core-
init@sha256:dd3899c7e2b50041d81b37aa3b1b2c028dc12691d465f3f8acd5c3ce22714b16 docker://$TARGET_PATH/ibm-spectrum-
scale-core-init:dd3899c7e2b50041d81b37aa3b1b2c028dc12691d465f3f8acd5c3ce22714b16 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
gui@sha256:324e700c049ee9eb5239c6baada0eeb1b9389877fcb0f3366e596d294d5c82e9 docker://$TARGET_PATH/ibm-spectrum-
scale-gui:324e700c049ee9eb5239c6baada0eeb1b9389877fcb0f3366e596d294d5c82e9 --tls-verify=false
skopeo copy --all --preserve-digests
docker://cp.icr.io/cp/gpfs/postgres@sha256:ef257d85f76e48da1c64832459b59fcaba1a4dac97bf5d7450c77753542eee94
docker://$TARGET_PATH/postgres:ef257d85f76e48da1c64832459b59fcaba1a4dac97bf5d7450c77753542eee94 --tls-
verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
logs@sha256:31a5c265df2a3183885e601369435f986baa6c8375ed7d6b827422b464015389 docker://$TARGET_PATH/ibm-spectrum-
scale-logs:31a5c265df2a3183885e601369435f986baa6c8375ed7d6b827422b464015389 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
pmcollector@sha256:2398d15d59a0a66603a510d9196d5a30517787c89fe347b2507b06d343f1c2cd docker://$TARGET_PATH/ibm-
spectrum-scale-pmcollector:2398d15d59a0a66603a510d9196d5a30517787c89fe347b2507b06d343f1c2cd --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
monitor@sha256:e32160671466a3794364c099712cf69761530d3728b548a5b951f6f6a9d3b541 docker://$TARGET_PATH/ibm-
spectrum-scale-monitor:e32160671466a3794364c099712cf69761530d3728b548a5b951f6f6a9d3b541 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-grafana-
bridge@sha256:b45261f1b673e4f933d512ce7ac31b72da1b4fa1229809181bf6f089cadbd6ed docker://$TARGET_PATH/ibm-
spectrum-scale-grafana-bridge:b45261f1b673e4f933d512ce7ac31b72da1b4fa1229809181bf6f089cadbd6ed --tls-
verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-
coredns@sha256:18058dc1e5ee3febb327a13d846fa80533de365788e3e03ed00e358ef45581ff docker://$TARGET_PATH/ibm-
spectrum-scale-coredns:18058dc1e5ee3febb327a13d846fa80533de365788e3e03ed00e358ef45581ff --tls-verify=false
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-must-
gather@sha256:c9b091bd73585a1361689bbb11a78f76883185f7a76cc872bb77c0a4c9ab2b8c docker://$TARGET_PATH/ibm-
spectrum-scale-must-gather:c9b091bd73585a1361689bbb11a78f76883185f7a76cc872bb77c0a4c9ab2b8c --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-
snapshotter@sha256:5f4bb469fec51147ce157329dab598c758da1b018bad6dad26f0ff469326d769
docker://$TARGET_PATH/csi/csi-snapshotter:5f4bb469fec51147ce157329dab598c758da1b018bad6dad26f0ff469326d769 --
tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-
attacher@sha256:69888dba58159c8bc0d7c092b9fb97900c9ca8710d088b0b7ea7bd9052df86f6 docker://$TARGET_PATH/csi/csi-
attacher:69888dba58159c8bc0d7c092b9fb97900c9ca8710d088b0b7ea7bd9052df86f6 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-
provisioner@sha256:d5e46da8aff7d73d6f00c761dae94472bcda6e78f4f17b3802dc89d44de0111b
docker://$TARGET_PATH/csi/csi-provisioner:d5e46da8aff7d73d6f00c761dae94472bcda6e78f4f17b3802dc89d44de0111b --
tls-verify=false
skopeo copy --all --preserve-digests
docker://cp.icr.io/cp/gpfs/csi/livenessprobe@sha256:2c5f9dc4ea5ac5509d93c664ae7982d4ecdec40ca7b0638c24e5b16243b8
360f docker://$TARGET_PATH/csi/livenessprobe:2c5f9dc4ea5ac5509d93c664ae7982d4ecdec40ca7b0638c24e5b16243b8360f --
tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-node-driver-
registrar@sha256:d7138bcc3aa5f267403d45ad4292c95397e421ea17a0035888850f424c7de25d docker://$TARGET_PATH/csi/csi-
node-driver-registrar:d7138bcc3aa5f267403d45ad4292c95397e421ea17a0035888850f424c7de25d --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-
resizer@sha256:8ddd178ba5d08973f1607f9b84619b58320948de494b31c9d7cd5375b316d6d4 docker://$TARGET_PATH/csi/csi-
resizer:8ddd178ba5d08973f1607f9b84619b58320948de494b31c9d7cd5375b316d6d4 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/ibm-spectrum-scale-csi-
driver@sha256:35ffde86b9805b83862a07acff838d086006cd3277424c8630de4697218a31e6 docker://$TARGET_PATH/csi/ibm-
spectrum-scale-csi-driver:35ffde86b9805b83862a07acff838d086006cd3277424c8630de4697218a31e6 --tls-verify=false
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-csi-
```

```
operator@sha256:a247e6bde9a851d80b671b01150dbe7267129b4650cf50cfef046ac28b2d9c8d docker://$TARGET_PATH/ibm-spectrum-scale-csi-operator:a247e6bde9a851d80b671b01150dbe7267129b4650cf50cfef046ac28b2d9c8d --tls-verify=false
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-operator@sha256:ec39d48e72c086961ad7b65384f09af137977cbb4b8b218aa7d572c3c7501046 docker://$TARGET_PATH/ibm-spectrum-scale-operator:ec39d48e72c086961ad7b65384f09af137977cbb4b8b218aa7d572c3c7501046 --tls-verify=false
```

If you are using Metro-DR, also run the following commands:

Artifactory:

```
skopeo copy --insecure-policy --preserve-digests --all docker://registry.redhat.io/openshift4/ose-kube-rbac-proxy:v4.15 docker://$TARGET_PATH/openshift4/ose-kube-rbac-proxy:v4.15
skopeo copy --insecure-policy --preserve-digests --all docker://registry.redhat.io/openshift4/ose-kube-rbac-proxy:v4.15 docker://$LOCAL_ISF_REGISTRY/$NAMESPACE/$(echo $REPO_PREFIX)openshift4-ose-kube-rbac-proxy:v4.15
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/submariner-operator:0.22.1 docker://$TARGET_PATH/submariner/submariner-operator:0.22.1
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/submariner-gateway:0.22.1 docker://$TARGET_PATH/submariner/submariner-gateway:0.22.1
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/lighthouse-agent:0.22.1 docker://$TARGET_PATH/submariner/lighthouse-agent:0.22.1
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/lighthouse-coredns:0.22.1 docker://$TARGET_PATH/submariner/lighthouse-coredns:0.22.1
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/submariner-route-agent:0.22.1 docker://$TARGET_PATH/submariner/submariner-route-agent:0.22.1
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/nettest:0.22.1 docker://$TARGET_PATH/submariner/nettest:0.22.1
```

Quay registry:

```
skopeo copy --insecure-policy --preserve-digests --all docker://registry.redhat.io/openshift4/ose-kube-rbac-proxy:v4.15 docker://$TARGET_PATH/openshift4/ose-kube-rbac-proxy:v4.15 --tls-verify=false
skopeo copy --insecure-policy --preserve-digests --all docker://registry.redhat.io/openshift4/ose-kube-rbac-proxy:v4.15 docker://$LOCAL_ISF_REGISTRY/$NAMESPACE/$(echo $REPO_PREFIX)openshift4-ose-kube-rbac-proxy:v4.15 --tls-verify=false
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/submariner-operator:0.22.1 docker://$TARGET_PATH/submariner/submariner-operator:0.22.1 --tls-verify=false
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/submariner-gateway:0.22.1 docker://$TARGET_PATH/submariner/submariner-gateway:0.22.1 --tls-verify=false
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/lighthouse-agent:0.22.1 docker://$TARGET_PATH/submariner/lighthouse-agent:0.22.1 --tls-verify=false
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/lighthouse-coredns:0.22.1 docker://$TARGET_PATH/submariner/lighthouse-coredns:0.22.1 --tls-verify=false
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/submariner-route-agent:0.22.1 docker://$TARGET_PATH/submariner/submariner-route-agent:0.22.1 --tls-verify=false
skopeo copy --insecure-policy --preserve-digests --all docker://quay.io/submariner/nettest:0.22.1 docker://$TARGET_PATH/submariner/nettest:0.22.1 --tls-verify=false
```

b. Global Data Platform Remote Mount 6.0.0.2

- Artifactory:

```
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/data-management/ibm-spectrum-scale-daemon@sha256:feac3019cd5384ac327b7cf775177e146da024ecac2315d4312c8bb12bc3e417 docker://$TARGET_PATH/data-management/ibm-spectrum-scale-daemon@sha256:feac3019cd5384ac327b7cf775177e146da024ecac2315d4312c8bb12bc3e417
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-core-init@sha256:a362af4bdbbf8a1f867df22646439cbd791ef1038310c4a81ed81128f0cef70d docker://$TARGET_PATH/ibm-spectrum-scale-core-init@sha256:a362af4bdbbf8a1f867df22646439cbd791ef1038310c4a81ed81128f0cef70d
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-gui@sha256:05a1bb16e7ca717b279935e9678a92eaec97b530878f307418e95171d9934d2b docker://$TARGET_PATH/ibm-spectrum-scale-gui@sha256:05a1bb16e7ca717b279935e9678a92eaec97b530878f307418e95171d9934d2b
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/postgres@sha256:ef257d85f76e48da1c64832459b59fcaba1a4dac97bf5d7450c77753542eee94 docker://$TARGET_PATH/postgres@sha256:ef257d85f76e48da1c64832459b59fcaba1a4dac97bf5d7450c77753542eee94
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-logs@sha256:c0d8156a277041e4c8eb7206c428a8c8a1728b5117dd3c91d8bfe6672349b42b docker://$TARGET_PATH/ibm-spectrum-scale-logs@sha256:c0d8156a277041e4c8eb7206c428a8c8a1728b5117dd3c91d8bfe6672349b42b
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-pmcollector@sha256:61163d62fc93fef53bfbead12d0bbc32eebb612fa0dc3afe8e938bca318aac25 docker://$TARGET_PATH/ibm-spectrum-scale-pmcollector@sha256:61163d62fc93fef53bfbead12d0bbc32eebb612fa0dc3afe8e938bca318aac25
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-monitor@sha256:856b7b4978382ecacfa8235cc80bc87e721f768ef6999c23f3e4a0db66c589f2 docker://$TARGET_PATH/ibm-spectrum-scale-monitor@sha256:856b7b4978382ecacfa8235cc80bc87e721f768ef6999c23f3e4a0db66c589f2
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-grafana-bridge@sha256:d423ca26549177b434efddd1fb21932f31110c288cfd77a2390d3bdf8af8876d docker://$TARGET_PATH/ibm-spectrum-scale-grafana-bridge@sha256:d423ca26549177b434efddd1fb21932f31110c288cfd77a2390d3bdf8af8876d
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-coredns@sha256:d5f4690898d9468a22d92f6da530cd28966dffc4e8519de89236954a3815dd02 docker://$TARGET_PATH/ibm-spectrum-scale-coredns@sha256:d5f4690898d9468a22d92f6da530cd28966dffc4e8519de89236954a3815dd02
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-must-gather@sha256:a9c18c3f77a8ae89647b238b6504422fabc5bdc0aa59f45a7e179e2472086ec4 docker://$TARGET_PATH/ibm-spectrum-scale-must-gather@sha256:a9c18c3f77a8ae89647b238b6504422fabc5bdc0aa59f45a7e179e2472086ec4
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-snapshotter@sha256:bc7be893ecc3ad524194aa6573b2f5c06cd469bdf21a500ab6c99c2ba1c4d64d docker://$TARGET_PATH/csi/csi-snapshotter@sha256:bc7be893ecc3ad524194aa6573b2f5c06cd469bdf21a500ab6c99c2ba1c4d64d
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-attacher@sha256:5aaefc24f315b182233c8b6146077f8c32e274d864cb03c632206e78bd0302da docker://$TARGET_PATH/csi/csi-attacher@sha256:5aaefc24f315b182233c8b6146077f8c32e274d864cb03c632206e78bd0302da
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-provisioner@sha256:bb057f866177d5f4139a1527e594499cbe0feeb67b63aaca8679dfdf0a6016f9 docker://$TARGET_PATH/csi/csi-provisioner@sha256:bb057f866177d5f4139a1527e594499cbe0feeb67b63aaca8679dfdf0a6016f9
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/livenessprobe@sha256:88092d100909918ae0a768956cf78c88bc59cd7232720f7cdbdfb5d2e235001e docker://$TARGET_PATH/csi/livenessprobe@sha256:88092d100909918ae0a768956cf78c88bc59cd7232720f7cdbdfb5d2e235001e
```

```
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-node-driver-registrar@sha256:5244abbe87e01b35adeb8bb13882a74785df0c0619f8325c9e950395c3f72a97 docker://$TARGET_PATH/csi/csi-node-driver-registrar@sha256:5244abbe87e01b35adeb8bb13882a74785df0c0619f8325c9e950395c3f72a97
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-resizer@sha256:5e7cbb63fd497fa913caa21fee1a69f727c220c6fa83c5f8bb0995e2ad73a474 docker://$TARGET_PATH/csi/csi-resizer@sha256:5e7cbb63fd497fa913caa21fee1a69f727c220c6fa83c5f8bb0995e2ad73a474
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/ibm-spectrum-scale-csi-driver@sha256:dd589aa2abd408d2e9c8f6a0b001db27802f1c3175a95c54f7c2d4993f6ae8d4 docker://$TARGET_PATH/csi/ibm-spectrum-scale-csi-driver@sha256:dd589aa2abd408d2e9c8f6a0b001db27802f1c3175a95c54f7c2d4993f6ae8d4
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-operator@sha256:138ecf352bb886d6e2d8c81c6d94bad4b4b16ad36be472f5d858f060c845907a docker://$TARGET_PATH/ibm-spectrum-scale-operator@sha256:138ecf352bb886d6e2d8c81c6d94bad4b4b16ad36be472f5d858f060c845907a
```

- Quay registry:

```
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/data-management/ibm-spectrum-scale-daemon@sha256:feac3019cd5384ac327b7cf775177e146da024ecac2315d4312c8bb12bc3e417 docker://$TARGET_PATH/data-management/ibm-spectrum-scale-daemon:feac3019cd5384ac327b7cf775177e146da024ecac2315d4312c8bb12bc3e417 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-core-init@sha256:a362af4bdbbf8a1f867df22646439cbd791ef1038310c4a81ed81128f0cef70d docker://$TARGET_PATH/ibm-spectrum-scale-core-init:a362af4bdbbf8a1f867df22646439cbd791ef1038310c4a81ed81128f0cef70d --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-gui@sha256:05a1bb16e7ca717b279935e9678a92eaec97b530878f307418e95171d9934d2b docker://$TARGET_PATH/ibm-spectrum-scale-gui:05a1bb16e7ca717b279935e9678a92eaec97b530878f307418e95171d9934d2b --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/postgres@sha256:ef257d85f76e48da1c64832459b59fcaba1a4dac97bf5d7450c77753542eee94 docker://$TARGET_PATH/postgres:ef257d85f76e48da1c64832459b59fcaba1a4dac97bf5d7450c77753542eee94 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-logs@sha256:c0d8156a277041e4c8eb7206c428a8c8a1728b5117dd3c91d8bfe6672349b42b docker://$TARGET_PATH/ibm-spectrum-scale-logs:c0d8156a277041e4c8eb7206c428a8c8a1728b5117dd3c91d8bfe6672349b42b --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-pmcollector@sha256:61163d62fc93fef53bfbead12d0bbc32eebb612fa0dc3afe8e938bca318aac25 docker://$TARGET_PATH/ibm-spectrum-scale-pmcollector:61163d62fc93fef53bfbead12d0bbc32eebb612fa0dc3afe8e938bca318aac25 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-monitor@sha256:856b7b4978382ecacfa8235cc80bc87e721f768ef6999c23f3e4a0db66c589f2 docker://$TARGET_PATH/ibm-spectrum-scale-monitor:856b7b4978382ecacfa8235cc80bc87e721f768ef6999c23f3e4a0db66c589f2 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-grafana-bridge@sha256:d423ca26549177b434efddd1fb21932f31110c288cfd77a2390d3bdf8af8876d docker://$TARGET_PATH/ibm-spectrum-scale-grafana-bridge:d423ca26549177b434efddd1fb21932f31110c288cfd77a2390d3bdf8af8876d --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/ibm-spectrum-scale-coredns@sha256:d5f4690898d9468a22d92f6da530cd28966dffc4e8519de89236954a3815dd02 docker://$TARGET_PATH/ibm-spectrum-scale-coredns:d5f4690898d9468a22d92f6da530cd28966dffc4e8519de89236954a3815dd02 --tls-verify=false
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-must-gather@sha256:a9c18c3f77a8ae89647b238b6504422fabc5bdc0aa59f45a7e179e2472086ec4 docker://$TARGET_PATH/ibm-spectrum-scale-must-gather:a9c18c3f77a8ae89647b238b6504422fabc5bdc0aa59f45a7e179e2472086ec4 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-snapshotter@sha256:bc7be893ecc3ad524194aa6573b2f5c06cd469bdf21a500ab6c99c2ba1c4d64d docker://$TARGET_PATH/csi/csi-snapshotter:bc7be893ecc3ad524194aa6573b2f5c06cd469bdf21a500ab6c99c2ba1c4d64d --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-attacher@sha256:5aaefc24f315b182233c8b6146077f8c32e274d864cb03c632206e78bd0302da docker://$TARGET_PATH/csi/csi-attacher:5aaefc24f315b182233c8b6146077f8c32e274d864cb03c632206e78bd0302da --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-provisioner@sha256:bb057f866177d5f4139a1527e594499cbe0feeb67b63aaca8679dfdf0a6016f9 docker://$TARGET_PATH/csi/csi-provisioner:bb057f866177d5f4139a1527e594499cbe0feeb67b63aaca8679dfdf0a6016f9 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/livenessprobe@sha256:88092d100909918ae0a768956cf78c88bc59cd7232720f7cdbdfb5d2e235001e docker://$TARGET_PATH/csi/livenessprobe:88092d100909918ae0a768956cf78c88bc59cd7232720f7cdbdfb5d2e235001e --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-node-driver-registrar@sha256:5244abbe87e01b35adeb8bb13882a74785df0c0619f8325c9e950395c3f72a97 docker://$TARGET_PATH/csi/csi-node-driver-registrar:5244abbe87e01b35adeb8bb13882a74785df0c0619f8325c9e950395c3f72a97 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/csi-resizer@sha256:5e7cbb63fd497fa913caa21fee1a69f727c220c6fa83c5f8bb0995e2ad73a474 docker://$TARGET_PATH/csi/csi-resizer:5e7cbb63fd497fa913caa21fee1a69f727c220c6fa83c5f8bb0995e2ad73a474 --tls-verify=false
skopeo copy --all --preserve-digests docker://cp.icr.io/cp/gpfs/csi/ibm-spectrum-scale-csi-driver@sha256:dd589aa2abd408d2e9c8f6a0b001db27802f1c3175a95c54f7c2d4993f6ae8d4 docker://$TARGET_PATH/csi/ibm-spectrum-scale-csi-driver:dd589aa2abd408d2e9c8f6a0b001db27802f1c3175a95c54f7c2d4993f6ae8d4 --tls-verify=false
skopeo copy --all --preserve-digests docker://icr.io/cpopen/ibm-spectrum-scale-operator@sha256:138ecf352bb886d6e2d8c81c6d94bad4b4b16ad36be472f5d858f060c845907a docker://$TARGET_PATH/ibm-spectrum-scale-operator:138ecf352bb886d6e2d8c81c6d94bad4b4b16ad36be472f5d858f060c845907a --tls-verify=false
```

Note: All commands must be successful.

4. In the OpenShift Container Platform cluster, add **ImageDigestMirrorSet**.

   Note:

   - Replace the variable **$TARGET_PATH** with your registry details where images are mirrored.
   - Do not add white spaces while you update the mirror paths. If you want to do so, add values within double quotation marks. For example, if the new path where the images are mirrored is **registryhost.com:443/isf-new-path**, then encapsulate within double quotation marks ("") while you edit to avoid white spaces in the line.

```
- mirrors:
    - "registryhost.com:443/isf-new-path"
    - registryhost.com:443/old-path
    source: cp.icr.io/cp/spectrum/scale
```

```
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: isf-scale-idms
```

```
spec:
  imageDigestMirrors:
    - mirrors:
        - $TARGET_PATH
      source: cp.icr.io/cp/spectrum/scale
    - mirrors:
        - $TARGET_PATH
      source: cp.icr.io/cp/gpfs
    - mirrors:
        - $TARGET_PATH
      source: icr.io/cpopen
```

Note: Ensure that you use `ImageTagMirrorSet` only when you upgrade the IBM Storage Scale.

```
apiVersion: config.openshift.io/v1
kind: ImageTagMirrorSet
metadata:
  name: isf-scale-itms
spec:
  imageTagMirrors:
    - mirrors:
        - $TARGET_PATH/submariner
      source: quay.io/submariner
    - mirrors:
        - $TARGET_PATH/kubebuilder/kube-rbac-proxy
      source: gcr.io/kubebuilder/kube-rbac-proxy
```

- **Mirroring IBM Storage Scale 6.0.1.0 images**
  Mirror IBM Storage Scale 6.0.1.0 images to your enterprise registry for installation in a disconnected environment by using the `oc-mirror` plug-in.

Mirroring IBM Storage Scale 6.0.1.0 images

Mirror IBM Storage Scale 6.0.1.0 images to your enterprise registry for installation in a disconnected environment by using the `oc-mirror` plug-in.

Before you begin

Before you mirror images, review the prerequisites in Mirroring prerequisites.

- Ensure that Red Hat® OpenShift® Container Platform version 4.21 is installed.
- Ensure that the Red Hat operator images are mirrored. For more information, see Mirroring Red Hat operator images to enterprise registry.
- Ensure that the Fusion Data Foundation images are mirrored. For more information, see Mirroring Fusion Data Foundation images.
- Use either the `podman` or `docker` commands to log in to the registries. The examples in this procedure use `docker`. If you use `podman`, replace `docker` with `podman` in the commands.
- For more information about offline setup for network-restricted Red Hat OpenShift Container Platform clusters, see Air-gaped installs.

Procedure

1. Log in to the IBM Entitled Container Registry by using your IBM entitlement key:
   Command example:

   ```
   docker login cp.icr.io -u cp -p <your entitlement key>
   ```

2. Set the following environment variables:
   Command example:

   ```
   export LOCAL_ISF_REGISTRY="<your enterprise registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<your image path>"
   export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
   ```

3. Log in to your enterprise registry:
   Command example:

   ```
   docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

4. Create the image set configuration for Fusion Data Foundation.
   Command example where Fusion Data Foundation version is 4.21.7:

   ```
   cat << EOF > image-set-config.yaml
   kind: ImageSetConfiguration
   apiVersion: mirror.openshift.io/v1alpha2
   mirror:
     additionalImages:
     - name: cp.icr.io/cp/gpfs/csi/csi-
   attacher:v4.11.0@sha256:b74b05b39501565022883fc128002b4cb857a7bb6c858606bcb3fdedba0b0b80
     - name: cp.icr.io/cp/gpfs/csi/csi-node-driver-
   registrar:v2.16.0@sha256:ab482308a4921e28a6df09a16ab99a457e9af9641ff44fb1be1a690d07ce8b70
     - name: cp.icr.io/cp/gpfs/csi/csi-
   provisioner:v6.2.0@sha256:6be9f63ca4caa6c46aae55aa372500949d8a21473d72f819da1f746076b32d4e
     - name: cp.icr.io/cp/gpfs/csi/csi-resizer:v2.1.0@sha256:589e525cddef6d768e68da1f0bc9ffd0a24bf3add3dd010648eb7189976fde79
     - name: cp.icr.io/cp/gpfs/csi/csi-
   snapshotter:v8.5.0@sha256:da081c27e8a6d91f36042c1942362d0515ced8d06e18c11b8f893e58c4d6d797
     - name: cp.icr.io/cp/gpfs/csi/ibm-spectrum-scale-csi-
   driver:v3.1.0@sha256:35c2c45c0a8f6504cf50dda57fd0c827244e822e02febf449a37630fc6d01b9d
     - name: 
   cp.icr.io/cp/gpfs/csi/livenessprobe:v2.18.0@sha256:c4cc074199c045dd73ab85f28897e2a32f4d6f38ffdba4f3b13b8007ccbd3570
   ```

```
  - name: cp.icr.io/cp/gpfs/data-management/ibm-spectrum-scale-daemon:v6.0.1.0@sha256:3651a79cfd42e67416995af3442b667beef09c9c1417e406b7be20cb63497ddf
  - name: cp.icr.io/cp/gpfs/ibm-spectrum-scale-core-init:v6.0.1.0@sha256:1775f4d2c51ae9bef6d0d129c79efcfefb4d6d4008445d57b852d4c4397b2fe2
  - name: cp.icr.io/cp/gpfs/ibm-spectrum-scale-coredns:v6.0.1.0@sha256:781db6ee6019ae2468b57f48abc17033ccfcdb88beb555b2797ee4fe32a2731b
  - name: cp.icr.io/cp/gpfs/ibm-spectrum-scale-grafana-bridge:v6.0.1.0@sha256:3843a5db15d214355d7c80751b7ab771cbaef9be025eea04745386d9210914e5
  - name: cp.icr.io/cp/gpfs/ibm-spectrum-scale-gui:v6.0.1.0@sha256:fd85ae58cfdf4ef48b300f542967a0b2e626ff4015af86a5f31e648b76d11160
  - name: cp.icr.io/cp/gpfs/ibm-spectrum-scale-logs:v6.0.1.0@sha256:16754a6b3e1eaac73a3df6d6ded01d31e0f5d0cc75dc6f30d1cb8043d2dc0685
  - name: cp.icr.io/cp/gpfs/ibm-spectrum-scale-monitor:v6.0.1.0@sha256:a15797ec26b5e6b54de4396ed4fdb9303169908c04337694ff2d2f2e9372df87
  - name: cp.icr.io/cp/gpfs/ibm-spectrum-scale-pmcollector:v6.0.1.0@sha256:10dd10e7ab9c32d36b31924c266671573c48fe1fd4c43cb1a6d89d7644482a4c
  - name: cp.icr.io/cp/gpfs/postgres:18.1-alpine3.23@sha256:aa6eb304ddb6dd26df23d05db4e5cb05af8951cda3e0dc57731b771e0ef4ab29
  - name: icr.io/cpopen/ibm-spectrum-scale-must-gather:v6.0.1.0@sha256:39d5a03dcc657ce704299767e09408e52be040edf11d8a2cf4b72864178e4535
  - name: icr.io/cpopen/ibm-spectrum-scale-operator-bundle:v6.0.1.0@sha256:b29e753b855f26561dc320c90f70940d96400d79d04f6870a8aef3ca72b6e5f0
  - name: icr.io/cpopen/ibm-spectrum-scale-operator:v6.0.1.0@sha256:5e2095f878f45b561e17fc00725d6ca69653d12f7693d295b0a53a40046f1a45
EOF
```

5. Run the following command to mirror the images:
   Command example:

   ```
   oc mirror --v2 --config image-set-config.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-verify=false
   ```

6. Create an `ImageDigestMirrorSet` custom resource named `isf-fdf-idms.yaml`.
   Important: Replace `$TARGET_PATH` with the registry path where the images are mirrored.
   Command example:

   ```
   apiVersion: config.openshift.io/v1
   kind: ImageDigestMirrorSet
   metadata:
     name: ibm-spectrum-scale-dme-bundle
   spec:
     imageDigestMirrors:
     - mirrors:
       - $TARGET_PATH/cp
       source: cp.icr.io/cp
     - mirrors:
       - $TARGET_PATH/cpopen
       source: icr.io/cpopen
   ```

7. Apply the `ImageDigestMirrorSet` configuration.
   Command example:

   ```
   oc apply -f isf-fdf-idms.yaml
   ```

8. Verify that the images are pulled from your internal registry and that the operator pod is in the Running state.
   Command example:

   ```
   kubectl get pods -lapp.kubernetes.io/name=operator -n ibm-spectrum-scale
   ```

   Example output:

   ```
   NAME                                                   READY   STATUS    RESTARTS    AGE
   ibm-spectrum-scale-controller-manager-64d548f576-znrgc  1/1     Running   0           37m
   ```

Mirroring Fusion Data Foundation images

Mirror the Fusion Data Foundation images deployed on Red Hat® OpenShift® Container Platform to your enterprise registry by using `ImageDigestMirrorSet`.

Before you begin

- Use this procedure to mirror Fusion Data Foundation images to enable the Fusion Data Foundation service in IBM Fusion HCI.
- To enable the Global Data Platform Remote Mount service in IBM Fusion HCI on Red Hat OpenShift Container Platform 4.21:
  1. Use this procedure to mirror the Fusion Data Foundation images.
  2. Mirror the IBM Storage Scale 6.0.1.0 images as described in Mirroring IBM Storage Scale 6.0.1.0 images.
- Before you mirror, go through the prerequisites. See Mirroring prerequisites.
- Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute `podman` for `docker` in the commands.
- For more information about the version support, see IBM Fusion HCI support matrix.

About this task

The Fusion Data Foundation 4.16 with Red Hat OpenShift Container Platform 4.16 is used in the procedure as an example. Replace it with your appropriate supported version.

Procedure

1. Run the following command to login to Docker registry with your Red Hat® enterprise credentials:

```
docker login registry.redhat.io -u <Red Hat enterprise registry username> -p <Red Hat enterprise registry password>
```

2. Log in to the IBM Entitled Container Registry by using the IBM entitlement key:

```
docker login cp.icr.io -u cp -p <your entitlement key>
```

Note: Ensure that your entitlement key for IBM Fusion HCI contains the correct entitlement.
Set the following environment variables:

```
export LOCAL_ISF_REGISTRY="<Your enterprise registry host>:<port>"
export LOCAL_ISF_REPOSITORY="<Your image path>"
export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
export OCP_VERSION="<Your ocp version, eg 4.16>"
```

Note: Port is a non-mandatory value when you set the `LOCAL_ISF_REGISTRY` variable. If your enterprise registry is accessible and has a secure connection, then you can ignore it.
Sample value for without port:

```
export LOCAL_ISF_REGISTRY="registryhost.com"
```

See the following sample values:

```
export LOCAL_ISF_REGISTRY="registryhost.com:443"
export LOCAL_ISF_REPOSITORY="fusion-mirror"
```

`LOCAL_ISF_REGISTRY` is your entitlement registry.

`LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, in 2.13.0, the name can be hci-images/isfimages or hci-images.

3. Run the command to login to the Docker registry with your enterprise registry credentials:

```
docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
```

4. Create the image set configuration for Fusion Data Foundation.
If you are using Fusion Data Foundation 4.21, then use the following image set configuration:

```
imageset-config-fdf.yaml

cat << EOF > imageset-config-fdf.yaml
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v2alpha1
mirror:
 operators:
 - catalog: icr.io/cpopen/isf-data-foundation-catalog:v$OCP_VERSION
   packages:
     - name: "cephcsi-operator"
     - name: "mcg-operator"
     - name: "ocs-operator"
     - name: "odf-csi-addons-operator"
     - name: "odf-multicluster-orchestrator"
     - name: "odf-operator"
     - name: "odr-cluster-operator"
     - name: "odr-hub-operator"
     - name: "ocs-client-operator"
     - name: "odf-prometheus-operator"
     - name: "recipe"
     - name: "rook-ceph-operator"
     - name: "odf-dependencies"
     - name: "odf-external-snapshotter-operator"
     - name: "ibm-spectrum-scale-operator"
     - name: "cnsa-dependencies"
EOF
```

If you are using Fusion Data Foundation 4.20, then use the following image set configuration:

```
imageset-config-fdf.yaml

cat << EOF > imageset-config-fdf.yaml
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v2alpha1
mirror:
 operators:
 - catalog: icr.io/cpopen/isf-data-foundation-catalog:v$OCP_VERSION
   packages:
     - name: "cephcsi-operator"
     - name: "mcg-operator"
     - name: "ocs-operator"
     - name: "odf-csi-addons-operator"
     - name: "odf-multicluster-orchestrator"
     - name: "odf-operator"
     - name: "odr-cluster-operator"
     - name: "odr-hub-operator"
     - name: "ocs-client-operator"
     - name: "odf-prometheus-operator"
     - name: "recipe"
     - name: "rook-ceph-operator"
     - name: "odf-dependencies"
     - name: "odf-external-snapshotter-operator"
EOF
```

If you are using Fusion Data Foundation 4.18 and 4.19, then use the following image set configuration:

`imageset-config-fdf.yaml`

```
cat << EOF > imageset-config-fdf.yaml
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v2alpha1
mirror:
 operators:
 - catalog: icr.io/cpopen/isf-data-foundation-catalog:v$OCP_VERSION
   packages:
     - name: "cephcsi-operator"
     - name: "mcg-operator"
     - name: "ocs-operator"
     - name: "odf-csi-addons-operator"
     - name: "odf-multicluster-orchestrator"
     - name: "odf-operator"
     - name: "odr-cluster-operator"
     - name: "odr-hub-operator"
     - name: "ocs-client-operator"
     - name: "odf-prometheus-operator"
     - name: "recipe"
     - name: "rook-ceph-operator"
     - name: "odf-dependencies"
EOF
```

5. Run the following oc command to mirror the images.

   ```
   oc mirror --v2 --config imageset-config-fdf.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-verify=false
   ```

6. Run the following command to check whether the image got successfully mirrored to the registry:
   Note: If you are using a self signed registry, then export the `SSL_CERT_DIR=/root/quay-install/quay-rootCA` before running this command.
   The `/root/quay-install/quay-rootCA` path is where certificates are placed. For more information, see Red Hat solution.

   ```
   oc mirror list operators --catalog $TARGET_PATH/cpopen/isf-data-foundation-catalog:v$OCP_VERSION
   ```

   Sample output:

   ```
   WARN[0033] DEPRECATION NOTICE:

   Sqlite-based catalogs and their related subcommands are deprecated. Support for them will be removed in a future release. Plea
   to the new file-based catalog format.

   NAME                            DISPLAY NAME               DEFAULT              CHANNEL

   mcg-operator                    NooBaa                     Operator             stable-4.16

   ocs-client-operator             OpenShift Data Foundation  Client               stable-4.16

   ocs-operator                    Container                  Storage              stable-4.16

   odf-csi-addons-operator         CSI                        Addons               stable-4.16

   odf-multicluster-orchestrator   Multicluster               Orchestrator         stable-4.16

   odf-operator                    IBM Storage Fusion         Data Foundation      stable-4.16

   odr-cluster-operator            DR Cluster                 Operator             stable-4.16

   odr-hub-operator                DR Hub                     Operator              stable-4.16
   ```

7. Ensure that the local-storage-operator and lvms-operator operator packages are present in your cluster.
   Note: If you have not mirrored local-storage-operator and lvms-operator from the Red Hat packages previously, then follow the steps that are provided in the Mirroring Red Hat operator images to enterprise registry.
   Note:
   - For the regional disaster recovery (RDR) setup, create the following image set configuration for the local storage operator:

     ```
     cat << EOF > imageset-config-lso.yaml
     kind: ImageSetConfiguration
     apiVersion: mirror.openshift.io/v2alpha1
     mirror:
       operators:
        - catalog: registry.redhat.io/redhat/redhat-operator-index:v$OCP_VERSION
          packages:
              - name: "multicluster-engine"
              - name: "local-storage-operator"
              - name: "lvms-operator"
              - name: "redhat-oadp-operator"
              - name: "openshift-gitops-operator"
              - name: "submariner"
              - name: "advanced-cluster-management"
     EOF
     ```

   - Run the following **OC** command to mirror the images.

     ```
     oc mirror --config imageset-config-lso.yaml docker://${TARGET_PATH} --dest-skip-tls --ignore-history
     ```

8. Note: You can run steps 8, 9, and 10 only when the cluster is up for a freshly installed rack.
   Create an `ImageDigestMirrorSet` custom resource for Fusion Data Foundation. Apply the configuration by running the command:

   ```
   oc apply -f isf-fdf-idms.yaml
   ```

   See the following `ImageDigestMirrorSet`:
   Note: Replace the variable `$TARGET_PATH` with your registry details where images are mirrored.

```
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  labels:
    operators.openshift.org/catalog: "true"
  name: isf-fdf-idms
spec:
  imageDigestMirrors:
  - mirrors:
    - $TARGET_PATH/openshift4
    source: registry.redhat.io/openshift4
  - mirrors:
    - $TARGET_PATH/redhat
    source: registry.redhat.io/redhat
  - mirrors:
    - $TARGET_PATH/rhel9
    source: registry.redhat.io/rhel9
  - mirrors:
    - $TARGET_PATH/rhel8
    source: registry.redhat.io/rhel8
  - mirrors:
    - $TARGET_PATH/cp/df
    source: cp.icr.io/cp/df
  - mirrors:
    - $TARGET_PATH/cpopen
    source: cp.icr.io/cpopen
  - mirrors:
    - $TARGET_PATH/cpopen
    source: icr.io/cpopen
  - mirrors:
    - $TARGET_PATH/cp/df/ceph
    source: cp.icr.io/cp/df/ceph
  - mirrors:
    - $TARGET_PATH/odf4
    source: registry.redhat.io/odf4
  - mirrors:
    - $TARGET_PATH/lvms4
    source: registry.redhat.io/lvms4
  - mirrors:
    - $TARGET_PATH/rhceph
    source: registry.redhat.io/rhceph
```

9. Create a `CatalogSource` named `redhat-operators`. Apply the `CatalogSource` by running the command:

   ```
   oc apply -f redhat-operators-catalogsource.yaml
   ```

   Note: Do this step when the Fusion Data Foundation uses the local disk.

   ```
   apiVersion: operators.coreos.com/v1alpha1
   kind: CatalogSource
   metadata:
     name: redhat-operators
     namespace: openshift-marketplace
   spec:
     displayName: Red Hat Operators
     image: $TARGET_PATH/redhat/redhat-operator-index:v$OCP_VERSION
     publisher: Red Hat
     sourceType: grpc
   ```

10. Create an `ImageTagMirrorSet` for Fusion Data Foundation. Apply the configuration by running the command:

    ```
    oc apply -f isf-fdf-imagetagmirrorset.yaml
    ```

    See the following `ImageDigestMirrorSet`:

    ```
    apiVersion: config.openshift.io/v1
    kind: ImageTagMirrorSet
    metadata:
      name: isf-fdf
    spec:
      imageTagMirrors:
        - mirrors:
            - $TARGET_PATH/cpopen/isf-data-foundation-catalog
          source: icr.io/cpopen/isf-data-foundation-catalog
          mirrorSourcePolicy: AllowContactingSource
    ```

What to do next

- To enable the Global Data Platform Remote Mount service in IBM Fusion HCI on Red Hat OpenShift Container Platform 4.21, mirror the IBM Storage Scale 6.0.1.0 images as described in Mirroring IBM Storage Scale 6.0.1.0 images.
- For the RDR setup, if you want to install the multicluster engine operator in an offline system or a disconnected environment, see Installing in a disconnected environment in *Red Hat Advanced Cluster Management for Kubernetes* documentation.

Mirroring Backup & Restore images

Mirror the Backup & Restore images to your enterprise registry.

Before you begin

1. Before you mirror, go through the prerequisites. See Mirroring prerequisites.
2. Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute `podman` for `docker` in the commands.
3. For more information about the version support, see IBM Fusion HCI support matrix.

About this task

High level steps to mirror Backup & Restore service:

1. Log in to the relevant Docker registries
2. Define variables to configure and mirror the Backup & Restore component of IBM Fusion HCI.
3. Run the `oc-mirror` CLI tool to mirror the Backup & Restore components.

Note:

- Use this procedure only necessary if you plan to mirror specific sub-components of IBM Fusion instead of following the full end-to-end mirroring instructions. For more information about the end-to-end mirroring of IBM Fusion and all its services, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.
- If you choose to mirror specific sub-components, use the same TARGET_PATH variable so that all the components mirror into the same location, and the generated `ImageDigestMirrorSet` remains the same across all components.
- For more information about Air gap setup for network restricted Red Hat® OpenShift® Container Platform clusters, see Offline setup for network restricted Red Hat OpenShift Container Platform clusters.

Procedure

1. Log in to your Docker registry:
   a. Run the following command to login to the Docker registry with your Red Hat enterprise credentials:

   ```
   docker login registry.redhat.io -u <Red Hat enterprise registry username> -p <Red Hat enterprise registry password>
   ```

   b. To authenticate to quay.io by using the username and password from the pull-secret, do the following steps:
      i. Download the pull-secret, see https://console.redhat.com/openshift/install/pull-secret and follow the instructions.
      See the following sample values:

      ```
        "auths": {
          "quay.io": {
            "auth": "b3BlbnNoaWZ0LXJTczOGY2YzNjNDM2ZWI0JRSDdUQU45RFBWVUNXM0VZQUlVVDBOQTU3VFM2RE1JMg==",
            "email": <email-id>
          }
        }
      }
      ```

      ii. Get the username/password:

      ```
      echo <auth-content> | base64 -decode
      ```

      Example echo command:

      ```
      echo "b3BlbnNoaWZ0LXJTczOGY2YzNjNDM2ZWI0JRSDdUQU45RFBWVUNXM0VZQUlVVDBOQTU3VFM2RE1JMg==" | base64 -decode
      ```

      Example output:

      ```
      openshift-release+ocmaccess0ab5738f6c3c42:CXKR2TSBQH7TAN9
      ```

      Here, `openshift-release+ocmaccess0ab5738f6c3c42` is the user name and `CXKR2TSBQH7TAN9` is the password.
   c. Log in to the IBM Entitled Container Registry using the IBM entitlement key.

   ```
   docker login cp.icr.io -u cp -p <your entitlement key>
   ```

   Note: Ensure that your entitlement key for IBM Fusion HCI contains the correct entitlement.
   d. Set the following environment variables:
   Important: Ensure you review the offline upgrade considerations before proceeding with upgrade mirroring.

   ```
   export LOCAL_ISF_REGISTRY="<Your enterprise registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<Your image path>"
   export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
   echo "$TARGET_PATH"
   ```

   Note: Port is a non-mandatory value when setting the `LOCAL_ISF_REGISTRY` variable. You can ignore this if your enterprise registry is accessible and has a secure connection.
   Sample value without port:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com"
   ```

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com:443"
   export LOCAL_ISF_REPOSITORY="fusion-mirror"
   ```

   e. Run the command to login to the Docker registry with your enterprise registry credentials.

   ```
   docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

   `LOCAL_ISF_REGISTRY` is your entitlement registry.

   `LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images.

2. Ensure that the **redhat-oadp-operator** and **amq-streams** operator packages are present in your cluster.
Note: If you have not mirrored **redhat-oadp-operator** and **amq-streams** from the Red Hat packages previously, then follow the steps that are provided in the [Mirroring Red Hat operator images to enterprise registry](#).
Note:
    - Run the following command to get list of mirrored Red Hat packages available on your cluster.

      ```
      oc get packagemanifests | grep -i "Red Hat Operators"
      ```

    - If you do not get **redhat-oadp-operator** and **amq-streams** operator packages from the previous step, then you must follow the step [2](#).
    - Ensure that you also add existing packages along with new one in ImageSetConfiguration file. Otherwise, old packages can be lost from the Red Hat operator index image.
3. Run the command to login to the Docker registry with your enterprise registry credentials.
For docker users:

   ```
   docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

   For podman users:

   ```
   podman login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

4. Create the image set configuration for Backup & Restore.

   ```
   cat << EOF > imageset-config-bnr.yaml
   kind: ImageSetConfiguration
   apiVersion: mirror.openshift.io/v2alpha1
   mirror:
     operators:
     - catalog: icr.io/cpopen/idp-server-operator-
   catalog@sha256:1b93488e0bf5c421dace9bae65fd992695c1e63fa12e7b82f03cf852d3fa02ad
       packages:
       - name: guardian-dm-operator
         channels:
         - name: v2.0
       - name: guardian-dp-operator
         channels:
         - name: v2.0
       - name: guardian-mongo-operator
         channels:
         - name: v2.0
       - name: ibm-dataprotectionagent
         channels:
         - name: v2.0
       - name: ibm-dataprotectionserver
         channels:
         - name: v2.0
       - name: redis-operator
         channels:
         - name: v2.0
       full: true
     additionalImages:
     - name: icr.io/cpopen/idp-server-operator-catalog:2.13.0-
   26195@sha256:1b93488e0bf5c421dace9bae65fd992695c1e63fa12e7b82f03cf852d3fa02ad
   EOF
   ```

5. Run the following **oc-mirror** command to mirror the images.

   ```
   oc mirror --v2 --config imageset-config-bnr.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-verify=false
   ```

6. Important: You can run step 5 only when the cluster is up for a freshly installed rack.
Apply the **ImageDigestMirrorSet** file to your cluster:
Note: Replace the variable **$TARGET_PATH** with your registry details where images are mirrored.

   ```
   apiVersion: config.openshift.io/v1
   kind: ImageDigestMirrorSet
   metadata:
     name: ibm-fusion-bnr
   spec:
     imageDigestMirrors:
     - mirrors:
       - ${TARGET_PATH}/cp
       source: cp.icr.io/cp
     - mirrors:
       - ${TARGET_PATH}/cpopen
       source: icr.io/cpopen
     - mirrors:
       - ${TARGET_PATH}/minio
       source: quay.io/minio
     - mirrors:
       - ${TARGET_PATH}/openshift4
       source: registry.redhat.io/openshift4
   ```

   In case of any failure, refer [Troubleshooting and known issues](#) in offline mirroring to get the resolution steps.

Mirroring IBM Data Cataloging images

Mirror the IBM Data Cataloging images to your enterprise registry.

Before you begin

- Before you mirror, go through the prerequisites. See Mirroring prerequisites.
- Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute **podman** for **docker** in the commands.
- For more information about the version support, see IBM Fusion HCI support matrix.
- Ensure that you install the `oc-mirror OpenShift` CLI plug-in. For more information about `oc-mirror OpenShift` CLI plug-in installation, see Installing the oc-mirror OpenShift CLI plugin.

About this task

For more information about Air gap setup for network restricted Red Hat® OpenShift® Container Platform clusters, see Offline setup for network restricted Red Hat OpenShift Container Platform clusters.
Note:

- This procedure is only necessary if you plan to enable the IBM Data Cataloging IBM Fusion HCI service.
- The destination registry where all IBM Data Cataloging images are going to be copied need to support both OCI (`application/vnd.oci.image.manifest.v1+json`) and Docker V2 (`application/vnd.docker.distribution.manifest.v2+json`) manifest types.

Use this procedure only necessary if you plan to mirror specific sub-components of IBM Fusion HCI instead of following the full end-to-end mirroring instructions. For more information about the end-to-end mirroring of and all its services, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.

High level tasks of this procedure:

1. Log in to the relevant Docker registries.
2. Define some convenience variables used to configure and mirror the Data Cataloging component of Fusion.
3. Run the `oc-mirror` CLI tool mirrors the Data Cataloging components.

Note: If you choose to mirror specific sub-components, it is recommended that you use the same `TARGET_PATH` variable so that all the components mirror in the same location, and the generated `ImageDigestMirrorSet` is same across all components.

Procedure

1. Ensure that the `amq-streams` operator package is present in your cluster.
   Note: If you have not mirrored `amq-streams` from the Red Hat packages previously, then follow the steps that are provided in the Mirroring Red Hat operator images to enterprise registry.
   Note:
   - Run the following command to get list of mirrored Red Hat packages available on your cluster.

     ```
     oc get packagemanifests | grep -i "Red Hat Operators"
     ```

   - If you do not get `redhat-oadp-operator` operator package from the previous step, then you must follow the step 1.
   - Ensure that you also add existing packages along with new one in ImageSetConfiguration file. Otherwise, old packages can be lost from the Red Hat operator index image.
2. Run the following command to login the Red Hat enterprise registry and the IBM Entitled Container Registry using your credentials.
   Note: Ensure that your entitlement key for IBM Fusion contains the correct entitlement.
   For docker users:

   ```
   docker login registry.redhat.io -u <Red Hat enterprise registry username> -p <Red Hat enterprise registry password>
   ```

   ```
   docker login cp.icr.io -u cp -p <your entitlement key>
   ```

   For podman users:

   ```
   podman login registry.redhat.io -u <Red Hat enterprise registry username> -p <Red Hat enterprise registry password>
   ```

   ```
   podman login cp.icr.io -u cp -p <Your entitlement key>
   ```

3. Set the following environment variable.
   Important: Ensure you review the offline upgrade considerations before proceeding with upgrade mirroring.

   ```
   export LOCAL_ISF_REGISTRY="<Your enterprise registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<Your image path>"
   export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
   ```

   `LOCAL_ISF_REGISTRY` is your entitlement registry.

   `LOCAL_ISF_REGISTRY` is the registry image path in which you want to mirror the images. You can choose your own repository paths. For example, `hci-images/isf` or `hci-images`.

4. Run the command to login to the Docker registry with your enterprise registry credentials.
   For docker users:

   ```
   docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

   For podman users:

   ```
   podman login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

5. Create the image set configuration for IBM Data Cataloging.

   ```
   cat << EOF > imageset-config-dcs.yaml
   kind: ImageSetConfiguration
   apiVersion: mirror.openshift.io/v2alpha1
   mirror:
     additionalImages:
     - name: cp.icr.io/cp/dcs/auth@sha256:6cb6e582d02fb017dbbc1c736f3409a08823fef56381747f13952bd8a613cabb
     - name: cp.icr.io/cp/dcs/connmgr-scheduler@sha256:cca24250b9bbfdfe8c6c85c613c5fa1a56d2b0fbbb0e48a4c56fc3f50ea338a2
     - name: cp.icr.io/cp/dcs/connmgr@sha256:cbefb44b2d448b36f90e4fca7973ab66bc1fbfd799eaa1c13b5185bf116f6f09
   ```

```
  - name: cp.icr.io/cp/dcs/db-schema@sha256:534c98417f909a248698281bc244ef1441b6a01dccf5a3055242c13956576c9a
  - name: cp.icr.io/cp/dcs/db2whrest@sha256:3233681ab2ec32515653f5659d6c02f8e12fbfddaa9d65747dfe07a62de1bfd2
  - name: cp.icr.io/cp/dcs/dcs-ai-mcp-server@sha256:d8af262ff80ee897797d322bbd8f8382cf4b823033901aea37d5531b27565365
  - name: cp.icr.io/cp/dcs/import-service@sha256:1003a8694eff91c163a72b88c19baeb5cd47a97d152ab64ac910b8b8b693f096
  - name: cp.icr.io/cp/dcs/import-tags-app@sha256:c37128ef0ad7f7add93c72c42c8c1902f75a2907e683af5d7af99d04ff4ea381
  - name: cp.icr.io/cp/dcs/isd-proxy@sha256:5056d6a265b2a37123806976c56722b86c5c20dad6d2f5933838c8d076b1e120
  - name: cp.icr.io/cp/dcs/keystone@sha256:b3128687e35f78bc25cefbe7254f625b5f01f344ae190d161cdbf79057109aa6
  - name: cp.icr.io/cp/dcs/llm-proxy@sha256:fce3aebc677eba45434f16039d3c678ccf4c1c71d709ab3b13b5aebd79c928b6
  - name: cp.icr.io/cp/dcs/metadata-extender@sha256:9a988643df596a1bbfb1094874273c3c1ca879e8a645677090409b4de1f37ac9
  - name: cp.icr.io/cp/dcs/mo-agent-extract@sha256:2f4d7b0d5ada12d39932f687cea031d88beb9cdac791b48cdca29ed23a9860ca
  - name: cp.icr.io/cp/dcs/mo-ubi-init@sha256:927bb277f576f2a772faf8dc9978a6592d5d08bfad3cdce81da180a9095a34c2
  - name: cp.icr.io/cp/dcs/moconsumer@sha256:2f1d3fdec099951cfc5f6788d4a4d944d66ec0aa8e39c9688239f1ecad38729f
  - name: cp.icr.io/cp/dcs/moproducer@sha256:c984c05bc47939e22cec6f4c8c0a9f365b71b7582c3711a36a59b96cc546da78
  - name: cp.icr.io/cp/dcs/must-gather@sha256:3f655e31efb04862c52333f113322a9169ea54d88f627177707453fd48acec50
  - name: cp.icr.io/cp/dcs/policyengine@sha256:ffe2fac7ee8bd49be48a142818ca760c6482018d9a79941296ef8d3424c1949c
  - name: cp.icr.io/cp/dcs/remote-content-search@sha256:7c0a39bcf04c1063fffc866ec8513066beb398dfd28cdb436b21cf7c4754f7c8
  - name: cp.icr.io/cp/dcs/reports@sha256:a95d112e43f1ecf7f962f04354c64be6052319eb0c5cabf94c9af325648d9f46
  - name: cp.icr.io/cp/dcs/scaleafmdatamover@sha256:19aa20ce2b493f420e03758313e02f0351cf72836c15905f3c4b7285603e29c0
  - name: cp.icr.io/cp/dcs/scaleilmdatamover@sha256:725614999df502d09ba0621c4a2c317ea18818cb288ad38d0c46f98c78648ba9
  - name: cp.icr.io/cp/dcs/sdmonitor@sha256:2d853b11e533a9eee9e922034478289ed82d4641a16e9c797fafbaaff8fb1fdf
  - name: cp.icr.io/cp/dcs/tikaserver@sha256:11a8175a621e6a5fb093ff41d71eb20f8dca8a4186a24a843cc2713f634b382d
  - name: cp.icr.io/cp/dcs/uibackend@sha256:1bfd41169c4450852872b0f8d190d1b24b12a46040eb6a167437a2f8086c798f
  - name: cp.icr.io/cp/dcs/uifrontend@sha256:58979c1c53c116e5f90e814166a129a1fa8587831f617147328b5b0a5843d3f1
  - name: cp.icr.io/cp/dcs/wkc-connector@sha256:3a68bf843669cce6f000ca14ecd5d42a6727fd8055de5ac157b9ecf360fd5c59
  - name: icr.io/cpopen/db2u-day2-operator@sha256:67e656ec5ad43ef62550852f42b346ee68b4f2e97867980e822819687aa740ba
  - name: icr.io/cpopen/db2u-day2-operator@sha256:6e87d78d96101ec3a8dfca7464e4196bd34b1bc700b7ef1b1de5add00a591e55
  - name: icr.io/cpopen/db2u-day2-operator@sha256:af106b46b923745688ef2952da39add7d38b46768e18bf2e2147b4b05e755f00
  - name: icr.io/cpopen/db2u-operator@sha256:9062cf50e0d1df382878bf3ad4d5c5cfdcdc801666e419cd3c6a62838445978c
  - name: icr.io/cpopen/db2u-operator@sha256:b838f643c872fee3552f889c7977d593438e6b9583b03798555f44737d601e82
  - name: icr.io/cpopen/db2u-operator@sha256:df5b2ee589a381253f8d0797885c8f798351914edf9e506cc1ff261ad3156f09
  - name: icr.io/cpopen/ibm-db2uoperator-bundle@sha256:0f013b2188b1e44fd450b4753b66e38f0a3624cf7f0b1bdab01ab36105a5d08b
  - name: icr.io/cpopen/ibm-db2uoperator-bundle@sha256:10e960e1e26d3b64ceda847618b757e1d7e162a3378668d1f85a14d1b38ec55a
  - name: icr.io/cpopen/ibm-db2uoperator-bundle@sha256:9c3f363473b182a4291e2bf07c630bdc8907abb26b5ede6c36f55f3e07de6ac0
  - name: icr.io/cpopen/ibm-db2uoperator-catalog:253.0.0-26025-
hack@sha256:3d8a379ecc97328eee2e4d5c83f380c87f1b5a32c4097112116f24f287226830
  - name: icr.io/cpopen/ibm-db2uoperator-catalog:253.0.0-26025-
hack@sha256:d6b6fcd8be9359bc8854fb314a6a73b6b5018d3dda2a005a1d16f8b15b18d1a8
  - name: icr.io/cpopen/ibm-spectrum-discover-operator-
bundle@sha256:cced71d877089db30e7badc95a2113ae09136347777cae0ec49000a71beaac81
  - name: icr.io/cpopen/ibm-spectrum-discover-operator-catalog:253.0.0-26025-
hack@sha256:c8b42eafa039542123c4ba29ef650c4a41996a28fd4504fb7c4e2e62fedab5b1
  - name: icr.io/cpopen/ibm-spectrum-discover-
operator@sha256:0b97164be106bf8733de8e432c17afc9e597251c64d0a2c37f2da5edab70f9f1
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:1b0dab1db92c2a6c36ad24ac9d5aa52acde882dcbcff4e6f4ad8af4fa9b857b7
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:39baed72d1a73aa5a2fbf245d4fe20faa6762908acb6efe9659e6cc6bd3a5c6f
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:46e9621ee884f45907330610be5f5a4e4f5f978da2d35d4e0d77e70abd767386
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:4a34187ba2196012acfd2eab8fb06b07c679bc6c0c13fcc988676043afc1c968
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:67fb65c796d162a58648f2ede6a053e1b9fd76aab01c91aad9acf39c5afbb814
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:6a123d208d9936f9f0148588e944edc348da066570f11ba6627659acca79794a
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:9a9fcad406d1500f3b86f6820ca80b0fcf4ddc6b8e2b6796b60eb53e0c616a76
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:c93051ba7ba210c4ad1a4c904be3752355c5937daf10298a14fb02c098a24c71
  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:fffb214cf777b845fa4a4617ff59a5b14d4ef94ca7e430b684bcbc3db6c75939
  - name: icr.io/db2u/db2u.db2wh.restricted.v2@sha256:6855ca6e06372a9072e248ccb1c5770b65400b1226859402411a9c18e0426789
  - name: icr.io/db2u/db2u.db2wh.restricted.v2@sha256:c6c98d491a01e10bf244e39b86d0073b3c3da4ccefaa15c15e9db562f7da0da7
  - name: icr.io/db2u/db2u.db2wh@sha256:0aa85bc1e6c8fe891fc4b3f2137d2afef611a52e11134055a76ce3d5602b6200
  - name: icr.io/db2u/db2u.db2wh@sha256:3cc0814e4850663815b34a21ba3d480bebe4d312031315c192ad33d1ef0d6b09
  - name: icr.io/db2u/db2u.db2wh@sha256:a21f1de1cf2accdd0c0e94f045cda34f1d2b6a0b292bce33e66fb0eb93fbf45b
  - name: icr.io/db2u/db2u.db2wh@sha256:da3dcced05f576ff140647b8fcf73bc2e8aad9d329cfb34d042350c487798caf
  - name: icr.io/db2u/db2u.dv.api@sha256:c775f04c1cf735e4eb9b654f8307e69a294d33a182310b2068c631365ace03e2
  - name: icr.io/db2u/db2u.dv.api@sha256:fe5d3b18469795b2bf9635a0d85b1b84542dc1506aaa5f3f5568620c1b11bc39
  - name: icr.io/db2u/db2u.dv.caching@sha256:16b671ca11da90cb2032f52e0adfdb11763137255b2f8abd75d0907c3c36ee6c
  - name: icr.io/db2u/db2u.dv.caching@sha256:52ab0a6385636334cd3f405b62aa5719c4b600e59ac1f2636977300c3d798297
  - name: icr.io/db2u/db2u.dv.utils@sha256:8327012b2fbe2789aee4855a54125dfd114572790d08054b205ae9f24d679587
  - name: icr.io/db2u/db2u.dv.utils@sha256:ed9da17c3676c6928572927d820b953be7b49074047ff94e61962e4fbd596b62
  - name: icr.io/db2u/db2u.hurricane@sha256:2e8c303ad6878d2de0b80c72f47140b93b25abdfe320aaf3e0de0380edf3ddb5
  - name: icr.io/db2u/db2u.hurricane@sha256:eb88a59e26bf023af2cf97c9fe162be817dcb2afc6a8fab053fc06989a243875
  - name: icr.io/db2u/db2u.instdb.restricted.v2@sha256:8f49d7dab7152d689d057297894174bd3e5ce604b5357d776223bc243fccfa84
  - name: icr.io/db2u/db2u.instdb.restricted.v2@sha256:b7112c671588766b31a889582ceef49557117c72a73ce0c94e53bd7739879e3c
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:1036e55662f721284a535bc63a46f1d8c61fa17ce2eb4c4b90f147f69ec35290
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:28982d3ea268ab42a865f96f6f93ad9e36602c4f8dc0a0cf87d29e0570bd7d40
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:56a9a4bde204ac715f6762704f263c6f384412d370a1bcfd629f91a06995eed6
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:751cdc1796099c74ced03bda277dd1950166255d4aa0410a7f21e94276c68139
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:76ab047ae55cc5f1d34c46cbab64e33c725d38f93a9d74c62eb5426521718a99
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  - name: icr.io/db2u/db2u.auxiliary.auth@sha256:e57279502a5d605931494771e8de0b382d3ddb1a5d744820f70977bb777bdaf9
  - name: icr.io/db2u/db2u.db2wh@sha256:1594796ebbe52b158c12eeb8e5d4faa791f6c9a28f96e33844a05f0282b4fc1f
  - name: icr.io/db2u/db2u.db2wh@sha256:394a2a0c640a884c1f63332bce1a8d5ad73baba0d3b3a05b4fa876e35bf9d82e
  - name: icr.io/db2u/db2u.dv.api@sha256:f8ed8a89397c5aa2d6b61ac2ed9d55e3b6351c3c0396355c17d7e44a48038def
  - name: icr.io/db2u/db2u.dv.caching@sha256:d5fe8bbd87c5d4e4424f5bd71ae3bf35769ea744322b152128ab563412eb68b0
  - name: icr.io/db2u/db2u.dv.utils@sha256:83ec4f59d30685a3ab43c74ee20336a16635935f339b78c4744058504e110f5e
  - name: icr.io/db2u/db2u.hurricane@sha256:7e7d7b1df0122120aeb33864b00cd5b9d8a2beab0ee6e5bbe273b0b7ea4f0e14
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:22bb255ac34cff1d99255a52a94b6560880f5f51937fdfcedba1ccd668688d9c
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:61c8a176b449e0aa7638a524297a80ece4ec52700d99299ae855db5b54609438
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:7906196d7910eada7f94bb375f4d6d7e5174992a6a0b3aa3308ce12283df45a8
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:7ea95db7c6249ad9b115c5934b4849f68b633024db937bb7835ec3cb4ab6a897
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:b67acd5ee75833ec1e10705a0c4e7ba0f4be85ab03ed37527b56250ee3448cb9
  - name: icr.io/db2u/db2u.instdb.restricted@sha256:d5e452c1f3f5dcacb747713e56dcd006be6ba291bbf305352cd411a9bae57ee6
  - name: icr.io/db2u/db2u.instdb@sha256:4e8688e4964cbbeb4a5329294a0d995a0a45a9bec237e3f0a18491f0bce97784
  - name: icr.io/db2u/db2u.instdb@sha256:5ad3fc61b531781b212275eed3c3632ba0b7cd7d5cb48cf321b3f2308b66c421
```

```
  - name: icr.io/db2u/db2u.instdb@sha256:829a95722e17914869a4e224f35ace529607e71a1d2235f6e6fe7f33378d1cb5
  - name: icr.io/db2u/db2u.instdb@sha256:a6c008e42a208fe1407cf8e9a9c5e4d97eb923aee7269821fad9bd037b407357
  - name: icr.io/db2u/db2u.instdb@sha256:cc300234190d53a01cbe7349b628df0518553b4cd6795bebfbbacd66ca5ae29b
  - name: icr.io/db2u/db2u.instdb@sha256:d1f0a4cd8591bc509208ab01d0268881d4a8529d53bae147e570de5a0ead3bb1
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:344840dc5a825a755f9701792bfc5f8e7766b776665c88e83a04bec8a1cd85ac
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:7313133d3b83dbf4a214c73e809b21b431bf81f4ff3c027a4ebe0608c254c9fc
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:7e7ac8e3f66e45c99e4fcf8673e09b18e2c0f7a0d6f72c3f86eb2eb157fabaf8
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:88574364d46b83ca98f20569e7ba0533fd38dd6a3cdff23bc374a629de1379e3
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:b8288381e8b2538464097db4328641648a44c5e9671bbc502af2e997b39abb41
  - name: icr.io/db2u/db2u.logstreaming.fluentd@sha256:f84b460effa534c060592012f01d85f16d340557931e3532f5f6e0b4b892b266
  - name: icr.io/db2u/db2u.mustgather@sha256:51707eef34bad0eeeecfb09c5746782851a0b4d6af562b626dc667ec5e50c723
  - name: icr.io/db2u/db2u.mustgather@sha256:517ae4be20211a68c11d5848f6330ec156bff90c3c64057d273b52d4bb98dae9
  - name: icr.io/db2u/db2u.mustgather@sha256:536bf34b07defd3bc624a7a3973e6d5e701d5fdb949978cfcb1dbd546968c475
  - name: icr.io/db2u/db2u.mustgather@sha256:628ac4c7d5e9b6f527cf451f397da5a8723956a283fa9a3dfe1db2dfc91bc858
  - name: icr.io/db2u/db2u.mustgather@sha256:6c91db906f8dd87bef288595ef2393d910f124c8d31440731af8b260fd2fa55d
  - name: icr.io/db2u/db2u.mustgather@sha256:f0a3a3dd13c7d133d39b25234977a150993334ae45049cb51a7fffc15943cd9a
  - name: icr.io/db2u/db2u.qrep@sha256:3c731b912f43a68b8ab8fef2bd61a4fe5287fac5c251046a160a049730f60f5f
  - name: icr.io/db2u/db2u.qrep@sha256:44724bda2d5336f17bec950e40195f4cb6199291a22a3c6d3d9b47407e0155f1
  - name: icr.io/db2u/db2u.qrep@sha256:9512f41becbcfc6ebf8afcaf35521478fd87e2b2881d856d443b5bb882cb8d79
  - name: icr.io/db2u/db2u.qrep@sha256:d6bcb17bd3ff2776ee3588ddebd946cade42a76f38666ccfc425635b09a6eb12
  - name: icr.io/db2u/db2u.rest@sha256:6e1f90125ad8a4740bf388d16f44e73737379f4dd9b881bc2559aea2d60d9976
  - name: icr.io/db2u/db2u.rest@sha256:778a3bd330aa7c9aa7cf405551856e92d3d8b40d75a509054d8c115e73e7b795
  - name: icr.io/db2u/db2u.rest@sha256:a12723d1671e2321286c22cfad30352c7860cd95f8099e45b4847036ccc0830e
  - name: icr.io/db2u/db2u.rest@sha256:a8158630c0ad0a21592a9e08891035d0d02d7b406c15f4740db77fec125b03ce
  - name: icr.io/db2u/db2u.rest@sha256:bf6576eecd916419d2d521289970094e0c73ded5de5154030678d6543c60a2de
  - name: icr.io/db2u/db2u.rest@sha256:d000f87783bf8963423064de28d069e6877aed8ec94d572c6b22a27313fec8ff
  - name: icr.io/db2u/db2u.restricted@sha256:27e1777bf955d26ce713cb8d6ffc6ee1d8d76c79dcbe94c07d1ec154b12519bd
  - name: icr.io/db2u/db2u.restricted@sha256:33a70ec4786221efe5e5fd397951111fc90ea3b75842ef9558c889034564e91b
  - name: icr.io/db2u/db2u.restricted@sha256:4e4397c45c10f09f723ff11b3b51bb7adb492898d652501abd85511cc660ccc6
  - name: icr.io/db2u/db2u.restricted@sha256:6b204a6026e176bc7c7404fd41ede9399439cf5347698c95b838a78454ace6d9
  - name: icr.io/db2u/db2u.restricted@sha256:9b388eab7b17d543bf9c731a2583bd2606ba6199873ab914a3cdd2065e231385
  - name: icr.io/db2u/db2u.restricted@sha256:c75c667f44a6ab63447d0e72ad2743beb6f34f29f0cdee9b83d83c72ebb2457e
  - name: icr.io/db2u/db2u.tools@sha256:02f1771f4a00db39a5ef741dc39eb84caedfd7393092607994500d9b481b9a54
  - name: icr.io/db2u/db2u.tools@sha256:35ae6c6618b443c49a17ff32f5ba0c098e46fc542482c02e94857eb378ace419
  - name: icr.io/db2u/db2u.tools@sha256:83c657db8954d116a97daae90957ed8b31a89055913de08c92bbe5d30d3c71ec
  - name: icr.io/db2u/db2u.tools@sha256:adeffc101962a8181b4ceba587d046808c6440ca4a516cabdb04f973b5e299e9
  - name: icr.io/db2u/db2u.tools@sha256:e89e56844720d8512e0e0ff65a0331719bc4c203cabf2eb1da329a6dd3d00c70
  - name: icr.io/db2u/db2u.tools@sha256:f22e5bbc28263a51aa45c4a42dfaabee793322594c32a58a13914b977aca300f
  - name: icr.io/db2u/db2u.update.image@sha256:318c6f072f745c191feecb34576bb68681dd9db3644cc32191fba34274332852
  - name: icr.io/db2u/db2u.update.image@sha256:56ec63418fba021fb1eafe54bc6d62106dfa9819a3ce88707fabcadb98b239aa
  - name: icr.io/db2u/db2u.update.image@sha256:5af408ae4cb77592ee6726abe3a281b9a7fac500231d05b45b0a2b05ed88293d
  - name: icr.io/db2u/db2u.update.image@sha256:836a33719793c32744c0f830890633cd1648f282585ae672d3067bd146b1d467
  - name: icr.io/db2u/db2u.update.image@sha256:9a5875da2221ed1e2725a085ec7abd2e0418379686e337987f9496367e7c7a61
  - name: icr.io/db2u/db2u.update.image@sha256:d327bb0e5cc2a453d2fad23b629ba4862b5af9f794fc7f10ecf9d46e5d107f19
  - name: icr.io/db2u/db2u.veleroplugin@sha256:1a2d4657bf14fa60ffa1b0dbca50b696f1ff4ffd28f4c470db67b0ecca5d6fb4
  - name: icr.io/db2u/db2u.veleroplugin@sha256:51114e7afaa01a9002289017a22a2f5c9c0eceeb8f2df857c8148dbb5651e46b
  - name: icr.io/db2u/db2u.veleroplugin@sha256:796518c555504dd1e5b77504ca07397c114ac6753adbf8c8de41dcd2e1b57065
  - name: icr.io/db2u/db2u.veleroplugin@sha256:a2b0b6ff986b618fe880acec97d4af1673f4bffda385e05b6e14301c1b51d472
  - name: icr.io/db2u/db2u.veleroplugin@sha256:b3ecbc09923c288fa8d21d9e36a23a61cbec71a0c43bdb1e8213688bab571626
  - name: icr.io/db2u/db2u.veleroplugin@sha256:f87644162951b7210e450f567dacf718d3480a2f3468b36131b7a43286a6c44f
  - name: icr.io/db2u/db2u.watsonquery@sha256:897237498f224723b5818c50cd37d218032c4a05ebe2bda15bebe99d13ef0521
  - name: icr.io/db2u/db2u@sha256:01bdc550ea7d727fa5715f96c61f7ffe763ee38fc1875132187c40e09d08a6aa
  - name: icr.io/db2u/db2u@sha256:427657ba472b22f66b68d2a4f2ea7564572853083508ae2a8dbf770d1a181b85
  - name: icr.io/db2u/db2u@sha256:4319988ff6a8dff6a97368f253fed5cb268eb0c20de4db9dedd77cc47214f806
  - name: icr.io/db2u/db2u@sha256:6dda7e511b92ffc83c4b92e7b5ed99b5b619a5986af4a86d75c5f4425917dd91
  - name: icr.io/db2u/db2u@sha256:94226fa2a0fa69f8438265f481ffc05e830a75b9bb4a04f2a2b1f8d239758b09
  - name: icr.io/db2u/db2u@sha256:adc37d1fdfde105a516b8b891c494627703d90e9104181b2959a98e6630a4e94
  - name: icr.io/db2u/etcd@sha256:0c79ec0ad02a7e810ce4bc094fe85ad869ac973494161e85dbf32e394fcccf6c
  - name: icr.io/db2u/etcd@sha256:1a3362799bd9402aba7f99eed179c2747a520b37c994e63ad9e2b5fdeb4d5bd9
  - name: icr.io/db2u/etcd@sha256:1befb2b817dc6f7bf3f2fe3f5f2862b638d791d1284facddf789f245775e575c
  - name: icr.io/db2u/etcd@sha256:234c073c178f60ff8e8e29fdfda25b1e908ea0655ff61f3a4e29a51eefdea8eb
  - name: icr.io/db2u/etcd@sha256:6e0c097d8e6d9cdacc71f80312dd96b8a0d50a07ad48a333fce0bfc190f58553
  - name: icr.io/db2u/etcd@sha256:bb5c436b88c57715ffc991d0ab5a05979a4c092c65ad98296af2b1349926c45c
EOF
```

6. Run the following `oc-mirror` command to mirror the images.

```
oc mirror --v2 --config imageset-config-dcs.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-verify=false
```

7. Generate `oc-mirror` config files using V2:

```
oc ibm-pak generate mirror-manifests --oc-mirror-plugin=v2 --version "${CASE_VERSION}" "${CASE_NAME}" "${LOCAL_REGISTRY}"
```

8. Important: You can run steps 8 and 9 only when the cluster is up for a freshly installed rack.
   Apply the `ImageDigestMirrorSet` file to your cluster:
   Note: Replace the variable `$TARGET_PATH` with your registry details where images are mirrored.

```
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: ibm-spectrum-discover
spec:
  imageDigestMirrors:
  - mirrors:
    - ${TARGET_PATH}/cp
    source: cp.icr.io/cp
  - mirrors:
    - ${TARGET_PATH}/cpopen
    source: icr.io/cpopen
  - mirrors:
    - ${TARGET_PATH}/db2u
    source: icr.io/db2u
```

In case of any failure, refer Troubleshooting and known issues in offline mirroring to get the resolution steps.

9. Apply the following `catalogsource` to the cluster:

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: ibm-spectrum-discover-catalog
  namespace: openshift-marketplace
spec:
  displayName: ibm-spectrum-discover-253.0.0
  publisher: IBM
  image: ${TARGET_PATH}/cpopen/ibm-spectrum-discover-operator-
catalog@sha256:c8b42eafa039542123c4ba29ef650c4a41996a28fd4504fb7c4e2e62fedab5b1
  sourceType: grpc
```

Mirroring Content-Aware Storage (CAS) images

IBM Fusion HCI supports a range of CAS versions, and it can upgrade independently of IBM Fusion HCI within a certain range. In 2.13.0, the supported CAS version range is any version < v1.1.0. You can run this procedure multiple times to upgrade to each newer version of CAS within the supported range.

Before you begin

- Before you mirror, go through the prerequisites. See Mirroring prerequisites.
- Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute `podman` for `docker` in the commands.
- For more information about the version support, see IBM Fusion HCI support matrix.

About this task

High level steps to mirror CAS service:

1. Log in to the relevant Docker registries
2. Define variables to configure and mirror the CAS component of IBM Fusion HCI.
3. Run the `oc-mirror` CLI tool to mirror the CAS components.

Note:

- Use this procedure when you want to upgrade your offline CAS installation to a newer version, without having to upgrade the IBM Fusion installation.
- If you choose to mirror specific sub-components, use the same TARGET_PATH variable so that all the components mirror into the same location, and the generated `ImageDigestMirrorSet` remains the same across all components.
- For more information about Air gap setup for network restricted Red Hat® OpenShift® Container Platform clusters, see Offline setup for network restricted Red Hat OpenShift Container Platform clusters.

Procedure

1. Log in to your Docker registry:
   a. Run the following command to login to the Docker registry with your Red Hat enterprise credentials:

   ```
   docker login registry.redhat.io -u <Red Hat enterprise registry username> -p <Red Hat enterprise registry password>
   ```

   b. Log in to the IBM Entitled Container Registry using the IBM entitlement key.

   ```
   docker login cp.icr.io -u cp -p <your entitlement key>
   ```

   Note: Ensure that your entitlement key for IBM Fusion HCI contains the correct entitlement.

   c. Set the following environment variables:

   ```
   export LOCAL_ISF_REGISTRY="<Your enterprise registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<Your image path>"
   export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
   echo "$TARGET_PATH"
   ```

   Note: Port is a non-mandatory value when setting the `LOCAL_ISF_REGISTRY` variable. You can ignore this if your enterprise registry is accessible and has a secure connection.
   Sample value without port:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com"
   ```

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com:443"
   export LOCAL_ISF_REPOSITORY="fusion-mirror"
   ```

   d. Run the command to login to the Docker registry with your enterprise registry credentials.

   ```
   docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

   `LOCAL_ISF_REGISTRY` is your entitlement registry.

   `LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images.

2. Create the image set configuration for CAS.

```
cat << EOF > imageset-config-cas.yaml
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v2alpha1
```

```
mirror:
  operators:
  - catalog: icr.io/cpopen/ibm-isf-cas-operator-
catalog@sha256:182acd92bd262539c5cc13470465719cf56398188bc6dfa230ccec1b34d8ecf7
    packages:
    - name: ibm-isf-cas-operator
      channels:
      - name: v1.0
      - name: v1.1
    full: true
  additionalImages:
  - name: icr.io/cpopen/ibm-isf-cas-operator-catalog:1.1.5-
08971@sha256:182acd92bd262539c5cc13470465719cf56398188bc6dfa230ccec1b34d8ecf7
EOF
```

3. Run the following `oc-mirror` command to mirror the images.

```
oc mirror --v2 --config imageset-config-cas.yaml --workspace file://./ docker://${TARGET_PATH} --dest-tls-verify=false
```

4. Apply the `ImageDigestMirrorSet` file to your cluster:
   Note: Replace the variable `$TARGET_PATH` with your registry details where images are mirrored.

```
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: isf-cas
spec:
  imageDigestMirrors:
  - mirrors:
    - ${TARGET_PATH}/cp
    source: cp.icr.io/cp
  - mirrors:
    - ${TARGET_PATH}/cpopen
    source: icr.io/cpopen
  - mirrors:
    - ${TARGET_PATH}/openshift4
    source: registry.redhat.io/openshift4
```

   In case of any failure, refer Troubleshooting and known issues in offline mirroring to get the resolution steps.

   Note: After applying the catalog source, it gets deleted automatically and it recreates once CAS is deployed.
   There is no need to apply the generated catalog-source.yaml file in this directory as Fusion applies the CatalogSource for CAS.
5. Apply the following catalog source to the cluster.

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: ibm-cas-catalog
  namespace: openshift-marketplace
spec:
  displayName: ibm-cas-1.1.5-linux-amd64
  publisher: IBM
  image: ${TARGET_PATH}/cpopen/ibm-isf-cas-operator-
catalog@sha256:182acd92bd262539c5cc13470465719cf56398188bc6dfa230ccec1b34d8ecf7
  sourceType: grpc
  grpcPodConfig:
    nodeSelector:
      kubernetes.io/arch: amd64
```

Mirroring IBM Software Hub, watsonx.ai, and watsonx Orchestrate images

Mirror the IBM Software Hub, watsonx.ai, watsonx Orchestrate, and Red Hat® OpenShift® AI images to your enterprise registry.

Before you begin

- Before you mirror, go through the prerequisites. See Mirroring prerequisites.
- Use either podman or docker commands to login to the registries. The examples in the procedure used docker; if you are using podman, substitute `podman` for `docker` in the commands.
- For more information about the version support, see IBM Fusion HCI support matrix.
- Ensure that you install the `oc-mirror OpenShift` CLI plug-in. For more information about `oc-mirror OpenShift` CLI plug-in installation, see Installing the oc-mirror OpenShift CLI plugin.

About this task

For more information about Air gap setup for network restricted Red Hat OpenShift Container Platform clusters, see Offline setup for network restricted Red Hat OpenShift Container Platform clusters.
Note:

- This procedure is only necessary if you plan to enable the IBM Software Hub, watsonx.ai, watsonx Orchestrate, and Red Hat OpenShift AI services of IBM Fusion HCI.
- If you choose to mirror specific sub-components, it is recommended that you use the same `TARGET_PATH` variable so that all the components mirror in the same location, and the generated `ImageDigestMirrorSet` is same across all components.

Procedure

1. Log in to the IBM Entitled Container Registry using the IBM entitlement key:

```
docker login cp.icr.io -u cp -p <your entitlement key>
```

Set the following environment variables:

```
export LOCAL_ISF_REGISTRY="<Your container registry host>:<port>"
export LOCAL_ISF_REPOSITORY="<Your image path>"
export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
```

See the following sample values:

```
export LOCAL_ISF_REGISTRY="registryhost.com:443"
export LOCAL_ISF_REPOSITORY="fusion-mirror"
```

`LOCAL_ISF_REGISTRY` is your entitlement registry.

`LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images.

2. Run the following command to login to the docker registry with your enterprise registry credentials:

```
docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
```

3. Mirror the following mandatory IBM Software Hub components to your private registry:

IBM Software Hub
- `ibm-licensing`
- `cpfs`
- `cpd_platform`
- `ibm-cert-manager`

watsonx.ai
- `watsonx_ai`

watsonx Orchestrate
- `watsonx_orchestrate`

For detailed mirroring procedures and prerequisites of the services, see Preparing to run IBM Software Hub installs from a private container registry.

4. Download and install the following tools on your machine:
   a. `casectl` from Cloud Pak Open Content.
   b. `cpd-cli` from Cloud Pak for Data command line interface.
   Clone this Github repository to your machine.

5. Run the following commands to download the required cases for AI services:

```
export COMPONENTS=cpfs,cpd_platform,ibm-licensing,ibm-cert-manager,watsonx_ai,watsonx_orchestrate

cpd-cli manage case-download \
--components=${COMPONENTS} \
--release=5.2.2
```

After executing the case-download command, you should see list of cases required for `ai-services` in the path `/cpd-cli-workspace/olm-utils-workspace/work/offline/5.2.2/.ibm-pak/data/cases`.

You can also verify by running the following command:

```
`ls cpd-cli-workspace/olm-utils-workspace/work/offline/5.2.2/.ibm-pak/data/cases/`
```

```
ibm-ccs                        ibm-cp-datacore                ibm-etcd-operator              ibm-redis-cp
ibm-watsonx-ai-ifm             ibm-wsl
ibm-cert-manager               ibm-data-governor              ibm-licensing                  ibm-watson-assistant
ibm-watsonx-data               ibm-wsl-runtimes
ibm-cloud-native-postgresql    ibm-datarefinery               ibm-opensearch-operator        ibm-watson-gateway-
operator        ibm-watsonx-orchestrate        ibm-zen
ibm-cp-common-services         ibm-elasticsearch-operator     ibm-rabbitmq-operator          ibm-watsonx-ai
ibm-wml-cpd
```

To know the versions of all of these cases, run the following command:

```
for d in */; do
  echo "===== $d ====="
  ls "$d"
  echo
done
```

6. After the CASE files are downloaded, they must be mirrored to the private container registry with the appropriate versions.
   a. Create the CASE repository only once using the first command.

   ```
   ./casectl repo create --repoPath /<some-path>/temp
   ```

   b. Run these commands for each CASE along with its version.

   ```
   ./casectl repo update --repoPath /<some-path>/temp --addCase  ~/.ibm-
   pak/data/cases/<casename>/<caseversion>/<casename>-<caseversion>.tgz
   ./casectl repo updateoci --repoPath /<some-path>/temp  --addCase <casename> --addCaseVersion <caseversion> --
   artifactRegistry <private-artifactory-path>:<port>
   ```

   Here `<casename>` refers to the name of each case, and `<caseversion>` is the version of that particular case. You should have obtained a list of versions for each case from the previous step 5.

`<some-path>` is some random path to use based on the requirements.

For example:

```
./casectl repo create --repoPath /Documents/temp
```

```
./casectl repo update --repoPath /Documents/temp --addCase  ~/.ibm-pak/data/cases/ibm-cert-manager/4.2.18/ibm-cert-manager-4.2.18.tgz
```

```
./casectl repo updateoci --repoPath /Documents/temp  --addCase ibm-cert-manager --addCaseVersion 4.2.18 --
artifactRegistry <private-registry-path>
```

7. The `ibm-cp-common-services` CASE directory contains additional dependent CASE packages that must also be mirrored. To access these packages, navigate to the /cpd-cli-workspace/olm-utils-workspace/work/offline/5.2.2/.ibm-pak/data/cases/ibm-cp-common-services/<ibm-cp-common-services-version> path, where `<ibm-cp-common-services-version>` represents the CASE version of `ibm-cp-common-services`.
For an example:

```
% cd ibm-cp-common-services
ibm-cp-common-services % ls
4.15.0
ibm-cp-common-services % cd 4.15.0
4.15.0 % ls
caseDependencyMapping.csv                                                   ibm-cp-common-services-
4.15.0+20251023.100000-images.csv                       ibm-cs-install-4.15.0.tgz
charts                                                                      ibm-cp-common-services-
4.15.0+20251023.100000.tgz                              ibm-events-operator-5.2.1-airgap-metadata.yaml
component-set-config.yaml                                                   ibm-cs-iam-4.14.0-airgap-metadata.yaml
ibm-events-operator-5.2.1-images.csv
ibm-cloud-native-postgresql-5.22.0+20251001.142254.2660-airgap-metadata.yaml    ibm-cs-iam-4.14.0-images.csv
ibm-events-operator-5.2.1.tgz
ibm-cloud-native-postgresql-5.22.0+20251001.142254.2660-images.csv          ibm-cs-iam-4.14.0.tgz
resourceIndexes
ibm-cloud-native-postgresql-5.22.0+20251001.142254.2660.tgz                 ibm-cs-install-4.15.0-airgap-metadata.yaml
ibm-cp-common-services-4.15.0+20251023.100000-airgap-metadata.yaml          ibm-cs-install-4.15.0-images.csv
```

From the output, consider only the `.tgz` files and exclude the main `ibm-cp-common-services` CASE archive. The following are the additional CASE archives that must be mirrored:

```
ibm-cs-install-4.15.0.tgz
ibm-events-operator-5.2.1.tgz
ibm-cs-iam-4.14.0.tgz
ibm-cloud-native-postgresql-5.22.0+20251001.142254.2660.tgz
```

For each of the `.tgz` files found in the directory, run the following commands:

```
./casectl repo update --repoPath /Documents/temp --addCase  ~/.ibm-pak/data/cases/ibm-cp-common-services/<filename>.tgz
```

```
./casectl repo updateoci --repoPath /Documents/temp  --addCase <casename> --addCaseVersion <caseversion> --
artifactRegistry <private-registry-path>
```

Here, `<casename>` represents the name of the case contained within the `.tgz` files, and `<caseversion>` represents the version of these supplementary cases.
Derive the `<casename>` and `<caseversion>` from the `.tgz` file in the following manner:

File name	Case name	Case version
`ibm-cs-install-4.15.0.tgz`	`ibm-cs-install`	4.15.0
`ibm-cloud-native-postgresql-5.22.0+20251001.142254.2660.tgz`	`ibm-cloud-native-postgresql`	5.22.0

8. Mirror images for Red Hat OpenShift AI as follows:
   - Mirror images for Red Hat OpenShift AI version 2.2, which is required for installing IBM watsonx.ai and watsonx Orchestrate version 5.2.2. For detailed mirroring procedures and prerequisites, see Mirroring images to a private registry for a disconnected installation.
9. Log in to your Docker registry as follows:
   a. Log in to the IBM Entitled Container Registry by using the IBM entitlement key:

   ```
   docker login cp.icr.io -u cp -p <your entitlement key>
   ```

   Note: Your entitlement key for IBM Fusion HCI must contain the correct entitlement.
   Set the following environment variables:

   ```
   export LOCAL_ISF_REGISTRY="<Your enterprise registry host>:<port>"
   export LOCAL_ISF_REPOSITORY="<Your image path>"
   export TARGET_PATH="$LOCAL_ISF_REGISTRY/$LOCAL_ISF_REPOSITORY"
   echo "$TARGET_PATH"
   export SRC_REGISTRY="cp.icr.io"
   export SRC_NAMESPACE="fsh-dev"
   export TAG="v1.0.3"
   ```

   Note: Port is a non-mandatory value when you set the `LOCAL_ISF_REGISTRY` variable. If your enterprise registry is accessible and has a secure connection, ignore it.
   Sample value for without port:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com"
   ```

   See the following sample values:

   ```
   export LOCAL_ISF_REGISTRY="registryhost.com:443"
   export LOCAL_ISF_REPOSITORY="fusion-mirror"
   ```

   b. Run the command to login to the Docker registry with your enterprise registry credentials.

   ```
   docker login $LOCAL_ISF_REGISTRY -u <your enterprise registry username> -p <your enterprise registry password>
   ```

   `LOCAL_ISF_REGISTRY` is your entitlement registry.

`LOCAL_ISF_REPOSITORY` is the image path in which you want to mirror the images. You can choose your own repository paths. For example, hci-images/isf or hci-images.

c. Mirror operator catalog, image, and bundle.

   i. Copy operator catalog from source registry:

```
skopeo copy --all
"docker://${SRC_REGISTRY}/${SRC_NAMESPACE}/cpopen/ibm-fsh-operator-catalog:${TAG}"
"docker://${TARGET_PATH}/ibm-fsh-operator-catalog:${TAG}"
--tls-verify=false
```

   ii. Copy operator image from source registry:

```
skopeo copy --all
"docker://${SRC_REGISTRY}/${SRC_NAMESPACE}/cp/ibm-fsh-operator:${TAG}"
"docker://${TARGET_PATH}/ibm-fsh-operator:${TAG}"
--tls-verify=false
```

   iii. Copy operator bundle from source registry:

```
skopeo copy --all
"docker://${SRC_REGISTRY}/${SRC_NAMESPACE}/cpopen/ibm-fsh-operator-bundle:${TAG}"
"docker://${TARGET_PATH}/ibm-fsh-operator-bundle:${TAG}"
--tls-verify=false
```

d. Apply the IDMS and ITMS objects to your cluster:

```
---
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: ibm-fsh-idms
spec:
  imageDigestMirrors:
    - mirrors:
        - <Your enterprise registry host>:<port>/<Your mirror path>
      source: cp.icr.io/cpopen
    - mirrors:
        - <Your enterprise registry host>:<port>/<Your mirror path>
      source: icr.io/cpopen
```

```
---
apiVersion: config.openshift.io/v1
kind: ImageTagMirrorSet
metadata:
  name: ibm-fsh-itms
spec:
  imageTagMirrors:
    - mirrors:
        - <Your enterprise registry host>:<port>/<Your mirror path>
      source: cp.icr.io/cp
    - mirrors:
        - <Your enterprise registry host>:<port>/<Your mirror path>
      source: icr.io/cpopen
```

10. To access the private registry where the images are mirrored, log in using the `cpd-login` command. To do this, open a terminal in the `fsh-ai-operator-xxxx` pod within the `ibm-software-hub-operator` namespace and run the following command:

```
cpd-cli manage login-private-registry <private-registry-path>:<port> <username> <password>
```

11. To mirror the required images, run the following `skopeo` commands for `ibm-events-operator-catalog`, `ibm-events-operator-bundle`, and `kserve-controller`.

```
skopeo copy --all docker://icr.io/cpopen/ibm-events-operator-catalog:5.2.0 docker://$TARGET_PATH/ai-images/cpopen/ibm-
events-operator-catalog:5.2.0 --tls-verify=false
skopeo copy --all docker://icr.io/cpopen/ibm-events-operator-bundle:5.2.0   docker://$TARGET_PATH/ai-images/cpopen/ibm-
events-operator-bundle:5.2.0 --tls-verify=false
skopeo copy --all docker://quay.io/modh/kserve-
controller@sha256:67672b25f5e4c2ba5457d29f0c5ae5b7e0ec3b635bf58a79892306ddda30ac2a docker://$TARGET_PATH/ai-
images/rhoai/kserve-controller:sha256-67672b25f5e4c2ba5457d29f0c5ae5b7e0ec3b635bf58a79892306ddda30ac2a
skopeo copy --all docker://cp.icr.io/cp/cpd/ibm-slate-30m-
english@sha256:29bc84a89164a00215cc8565e2e70d7cf07b67ae099916946972533f1a221773  docker://$TARGET_PATH/ai-
images/cp/cpd/ibm-slate-30m-english@sha256:29bc84a89164a00215cc8565e2e70d7cf07b67ae099916946972533f1a221773
```

12. In the OpenShift Container Platform cluster, add `ImageDigestMirrorSet` and `ImageTagMirrorSet`:

`ImageDigestMirrorSet` (IDMS)

```
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: ai-services-idms
spec:
  imageDigestMirrors:
    - mirrors:
        - <private-registry-path>:port/ai-images/cp
      source: cp.icr.io/cp
    - mirrors:
        - <private-registry-path>:port/ai-images/cp/cpd
      source: cp.icr.io/cp/cpd
    - mirrors:
        - <private-registry-path>:port/ai-images/cpopen
      source: icr.io/cpopen
    - mirrors:
        - <private-registry-path>:port/ai-images/db2u
```

```
      source: icr.io/db2u
    - mirrors:
        - <private-registry-path>:port/ai-images/cpopen/cpfs
      source: icr.io/cpopen/cpfs
    - mirrors:
        - <private-registry-path>:port/ai-images/cpopen
      source: cp.icr.io/cpopen
    - mirrors:
        - <private-registry-path>:port/ai-images/rhoai
      source: quay.io/modh
```

`ImageTagMirrorSet` (ITMS)

```
apiVersion: config.openshift.io/v1
kind: ImageTagMirrorSet
metadata:
  name: ai-service-itms
spec:
  imageTagMirrors:
    - mirrors:
        - <private-registry-path>:port/ai-images/cp
      source: cp.icr.io/cp
    - mirrors:
        - <private-registry-path>:port/ai-images/cp/cpd
      source: cp.icr.io/cp/cpd
    - mirrors:
        - <private-registry-path>:port/ai-images/cpopen
      source: icr.io/cpopen
    - mirrors:
        - <private-registry-path>:port/ai-images/db2u
      source: icr.io/db2u
    - mirrors:
        - <private-registry-path>:port/ai-images/cpopen/cpfs
      source: icr.io/cpopen/cpfs
    - mirrors:
        - <private-registry-path>:port/ai-images/cpopen
      source: cp.icr.io/cpopen
    - mirrors:
        - <private-registry-path>:port/ai-images/rhoai
      source: quay.io/modh
```

Troubleshooting and known issues in offline mirroring

Issues can occur during the offline mirroring of images. This topic outlines known issues and offers troubleshooting guidance.

`oc-mirror` command fails with unauthorized error

Problem statement

The `oc-mirror` command fails with an `unauthorized: access to the requested resource is not authorized` error.

Cause

The error occurs due to one of the following reasons:

- Insufficient authentication credentials for the registries mentioned in the command.
- Incorrect or missing secrets in the docker configuration file.

Resolution

Follow the steps to resolve the issue:

1. Ensure you are logged in to the registries mentioned in the `oc-mirror` command.
2. Verify that the `~/.docker/config.json` file contains the correct secrets for the registries.
   - If the file does not exist, create it manually.
   - Add the required registry credentials to the file.
3. After completing the above steps, retry the `oc-mirror` command. If the issue persists, review the registry credentials and docker configuration file for accuracy.

`oc-mirror` v2 fails with an error `stat .: no such file or directory`

Problem statement

The `oc-mirror` v2 fails with an error `stat .: no such file or directory`.

Resolution

As a resolution, install the `oc-mirror` CLI plug-in of **4.20.6** version.

Use the latest versions of the oc and oc-mirror tools

Problem statement

Issues may occur when you do not use the latest version of the **oc** CLI or **oc-mirror** plug in.

Diagnosis

Red Hat® has identified several library compatibility issues in the older versions of these tools, which are resolved in the latest releases. When you run older versions of **oc** or **oc-mirror** commands, you may encounter messages similar to the following examples.

```
oc: /lib/x86_64-linux-gnu/libc.so.6: version `GLIBC_2.33' not found (required by oc)
oc: /lib/x86_64-linux-gnu/libc.so.6: version `GLIBC_2.34' not found (required by oc)
oc: /lib/x86_64-linux-gnu/libc.so.6: version `GLIBC_2.32' not found (required by oc)
```

Resolution

Download the latest available **oc** CLI tool, even if it is a newer than your Red Hat OpenShift® version. The **oc** CLI is backward compatible with older OpenShift versions. For example, your **oc** CLI can be on version 4.17 while your OpenShift cluster remains on 4.14.
In addition, ensure you obtain the latest version of the **oc-mirror** plugin. The **oc-mirror** plugin need not match the **oc** CLI version and can even be newer. For example, you can use an **oc-mirror** plugin 4.17 with an **oc** CLI 4.16, even if your cluster is on 4.14.

Login to both source and target registries before mirroring images

Problem statement

You may encounter issues where some or all images fail to copy, resulting in errors related to either authorization or authentication requirements.

Diagnosis

If you encounter messages similar to the following examples, then it indicates that you must login to the specified registries:

- ```
  error: unable to retrieve source image cp.icr.io/cp/isf/isf-prereq-operator manifest
  sha256:1d062784dd6818297bd8275d139d0dd7ac6cd2ae6a4188bb3258bcdad8ccedf4:
  Get "https://cp.icr.io/v2/cp/isf/isf-prereq-operator/manifests/sha256:1d062784dd6818297bd8275d139d0dd7ac6cd2ae6a4188bb325
  unauthorized: Authorization required. See https://cloud.ibm.com/docs/Registry?topic=Registry-troubleshoot-auth-req
  ```
- ```
  Checking push permissions for docker-na-public.artifactory.swg-devops.com
  error: error checking push permissions for docker-na-public.artifactory.swg-devops.com:
  creating push check transport for docker-na-public.artifactory.swg-devops.com failed:
  GET https://docker-na-public.artifactory.swg-devops.com/artifactory/api/docker/null/v2/token?scope=repository%3Aoffline%2
  Authentication is required
  ```

Resolution

As a resolution, log in to all the registries:

- registry.redhat.io
- icr.io
- quay.io
- Your target registry host

Also, create a `.docker/config.json` file in MGen with

- registry.redhat.io
- icr.io
- quay.io
- Your target registry host

If authorization errors persist even after logging, it indicates that your user does not have the right permissions to pull from the source registries or push to target registries. In such a case, contact your registry administrator regarding your user permissions

Check available disk space

Problem statement

If repeated retries of **oc-mirror** continue to fail, it is possible that the target registry has run out of disk space. Additionally, if you mirror to a local directory for offline install media, your local disk may be full.

Resolution

Check disk space usage and free up disk space if necessary.

Error with HTTP status: 504 Gateway Timeout or no space left on device

Problem statement

The **oc-mirror** operation may fail with an error message similar to the following example:

```
Rendering catalog image
"our.offline-registry.net/fusion-hci-2.9.0/cpopen/isf-operator-catalog:f34ee5" with file-based catalog
error: error rebuilding catalog images from file-based catalogs:
error regenerating the cache for our.offline-registry.net /fusion-hci-2.9.0/cpopen/isf-operator-catalog:f34ee5: exit status 1
```

Resolution

This is a known issue in the **oc-mirror** tool. For more information about how to resolve the issue, see the Red Hat article.

Error with HTTP status: 502 Bad Gateway

Problem statement

The **oc-mirror** operation may fail with an error message similar to the following example:

```
error: unable to retrieve source image registry.redhat.io/amq-streams/kafka-34-rhel8 manifest sha256:e4b2073b3001502b3bd02275e
error: unable to retrieve source image registry.access.redhat.com/ubi8/openjdk-11 manifest sha256:d9739b266415be7c915daf3c61f8
error: an error occurred during planning
```

Resolution

As a resolution, retry the **oc-mirror** command.

Error with HTTP status: 504 Gateway Timeout or no space left on the device

Problem statement

The **oc-mirror** operation may fail with an error message similar to the following example:

```
error: unable to retrieve source image registry.redhat.io/amq-streams/kafka-35-rhel8 manifest #3 from manifest list: received 
error: unable to retrieve source image registry.redhat.io/amq-streams/kafka-35-rhel8 manifest #1 from manifest list: received 
error: unable to retrieve source image registry.redhat.io/amq-streams/kafka-35-rhel8 manifest #1 from manifest list: received 
error: an error occurred during planning
```

Resolution

1. Run the following command:

```
rm -rf /root/.oc-mirror
```

2. Retry the **oc-mirror** command.

Converting a rack from internal registry to offline using enterprise registry

Use the following instructions to convert a rack from internal registry to offline using enterprise registry on the IBM Fusion HCI cluster.

Procedure

1. Mirror the images of the cluster to an external registry. For more information, see Enterprise registry for IBM Fusion HCI installation.
   Note: All images must be mirrored by using the same versions as those in the internal registry cluster.
2. Convert the rack from internal registry to offline:
   a. If the certificate is self-sign certificate or not a well-known authority signed certificate, add the certificate to the cluster manually:
      Note: Replace the following variables with appropriate values:
      - Replace `repo` with the enterprise registry where the images are mirrored.
      - Replace `port` with the port number of the enterprise registry.
      - Use `certificate.crt` as the file that contains the enterprise registry certificates.
      - If the registry does not require a port, use `--from-file=[repo]=[certificate.crt]` instead.

      ```
      oc create configmap registry-cas -n openshift-config --from-file=[repo]..[port]=[certificate.crt] --dry-run=client -o
      yaml | oc apply -f -
      ```

   b. Update the required IDMS, ITMS, or `catalogsources` that are generated during step 1 in the cluster. This action triggers an MachineConfigPool (MCP) rollout. Wait for it to complete.
   c. Update the pull-secret in the cluster:
      i. Go to OpenShift UI and search for secret `pull-secret` in the `openshift-config` namespace.
      ii. Click the Actions dropdown and select Edit Secret.
      iii. Click Add credentials and update the enterprise registry details.
      iv. Click Remove credentials to delete the previous entry of the internal registry.
      v. Click Save.
         This action triggers an MCP rollout. Wait for it to complete.

Results

The rack is converted from internal registry to offline using enterprise registry.

Activating Red Hat OpenShift Container Platform subscriptions purchased from IBM

If you purchased your Red Hat® OpenShift® Container Platform subscription from IBM, then you must activate your OpenShift subscription with Red Hat and activate OpenShift support with IBM.

About this task

IBM Fusion HCI base cluster can be used with any of the four distributions of OpenShift: Red Hat OpenShift Virtualization Engine (OVE), OpenShift Kubernetes Engine (OKE), OpenShift Container Platform, or OpenShift Platform Plus (OPP). For more information about the subscriptions, see Red Hat OpenShift distributions.

Procedure

1. Activate your OpenShift subscription with Red Hat.
   Open an account in the Red Hat Customer Portal: https://access.redhat.com/. If you already have an account, you do not need to open a new account.
   Contact Red Hat at openshift-orders@redhat.com, requesting OpenShift subscription activation. Provide your Red Hat Customer Portal id and IBM Software Order number for your OpenShift purchase in the email.

   You can find the Order Number on your purchase invoice. The Order Number can be a 6-character number, and it can be similar to the following sample formats:
   - LCxxxx
   - LDxxxx
   - A3xxxx
   - A4xxxx
2. Activate OpenShift support with IBM.

Complete the following steps to accept the IBM support agreement:

a. Access the Agreements for IBM Software Support Services website - www.ibm.biz/tss-agreements. On the website, select the Agreement for IBM Support for Commercial Open Source for the country or region in which the software is installed (Agreement).
b. Print and read the Agreement. The Agreement terms govern the Service. By your ordering, making payment for, or using the Service, you accept the terms and Agreement without modification. Service becomes subject to this Agreement when IBM or your IBM Business Partner accepts your order or payment.

- **Red Hat OpenShift distributions**
  IBM Fusion HCI base cluster can be used with any of the four distributions of OpenShift: Red Hat OpenShift Virtualization Engine (OVE), OpenShift Kubernetes Engine (OKE), OpenShift Container Platform, or OpenShift Platform Plus (OPP).

Red Hat OpenShift distributions

IBM Fusion HCI base cluster can be used with any of the four distributions of OpenShift®: Red Hat OpenShift Virtualization Engine (OVE), OpenShift Kubernetes Engine (OKE), OpenShift Container Platform, or OpenShift Platform Plus (OPP).

Important: You must follow the terms and conditions of Red Hat® to use these different distributions. For more information about these terms and conditions, see Red Hat Subscription Guide.

OpenShift Virtualization Engine (OVE)

For organizations that want to run and manage Virtual Machines (VMs) on the same platform as containers without a separate virtualization stack. For example, VMware.

IBM Fusion services permitted for use with OVE in support of virtual machines include:

- Fusion Data Foundation including MCG, Scale CNSA, Backup & Restore Spoke, Fusion Control Tower are permitted to run on infrastructure nodes.
- IBM Data Cataloging, Backup & Restore Hub and Fusion Base can be deployed as containers on OVE but must run on normal worker nodes.

OpenShift Kubernetes Engine (OKE)

For organizations who want a lightweight, Kubernetes-only platform with enterprise support.
Warning: IBM Fusion enables monitoring of user-defined objects at the time of installation. User-defined monitoring is not supported for users with OKE license. To remain compliant with your license agreement, disable user-defined monitoring. Continuing to use this feature can result in a violation of your contract. For more information on how to disable user-defined monitoring, see the Red Hat documentation.

OpenShift Container Platform

To provide a full-featured enterprise container platform with DevOps tools, monitoring, networking, and security. Also, for enterprises with complex, multi-cluster, or hybrid cloud environments that need unified governance and security.

OpenShift Platform Plus

For enterprises with complex, multi-cluster, or hybrid cloud environments that need unified governance and security.

Activating IBM Fusion HCI Software to be downloaded

You need IBM Fusion HCI software entitlement to access IBM Fusion HCI images that are used to install your IBM Fusion HCI appliance.

Before you begin

1. Sign up for an IBMid. If you already have an IBMid, skip this step. You can sign up for an IBMid here: https://www.ibm.com/account/reg/us-en/signup?formid=urx-19776A. It is a best practice to use your company email address for your IBMid.
2. Ensure that you have the following information:
   This information can be retrieved from the IBM Invoice of your IBM Fusion HCI purchase. You can request the invoice from your internal buyer, IBM Sales Representative, or your Business Partner. If you are a Business Partner, you can request the invoice from your Distributor.

   - IBM Customer Number (ICN)
     Clients can have multiple IBM Customer Numbers. You need the ICN that was used for your purchase of IBM Fusion HCI. Your seller or business partner can provide this number.

   - Order number
     It is also called an Internal Order Number. An Order number can be a 6-character number that might look like the following example formats:
     - LCxxxx
     - LDxxxx
     - A3xxxx
     - A4xxxx
   - Serial number of a IBM Fusion HCI component:
     Examples for 7-character serial numbers:
     - 85xxxxx
     - 78xxxxx
     - RMxxxxx
     - 66xxxxx
     - 68xxxxx

Method 1
: Serial number of IBM Fusion HCI Software, 5771-PP7. You can find multiple 5771-PP7 serial numbers in the invoice, but you need only one of them to complete this process.

Method 2
: Serial number of a IBM Fusion HCI hardware component. This method might be useful if you have your ICN but you do not have access to your IBM Invoice. IBM Fusion HCI hardware serial numbers can be found on the documents that are shipped with your IBM Fusion HCI. IBM Fusion HCI hardware components have machine type 9155. You find multiple serial numbers for 9155 components. You only need one serial number from one of the following components to proceed: 9155-C00, 9155-C01, 9155-C04, or 9155-C05.

Generate entitlement key

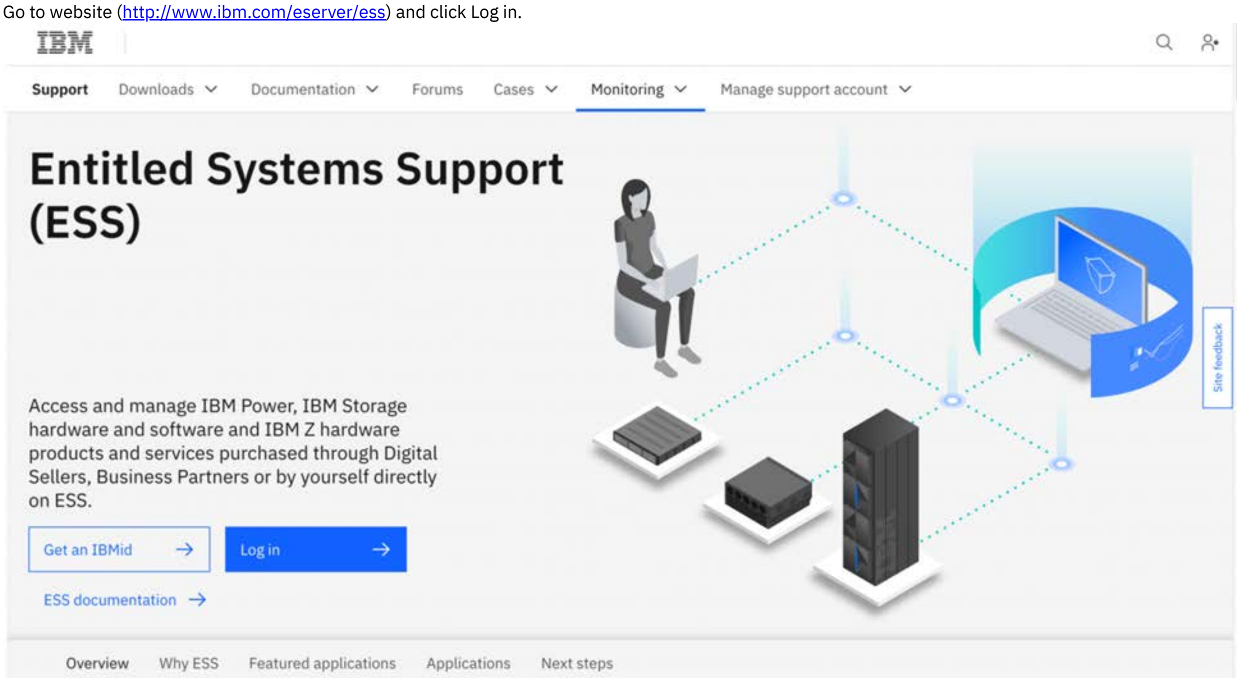
Entitled Systems Support (ESS)

Your purchase of IBM Fusion HCI Software, 5771-PP7, and IBM Fusion HCI hardware is recorded in the IBM Entitled Systems Support (ESS) portal (https://www.ibm.com/servers/eserver/ess/landing/index.html). For more information about the registration process, see https://www.ibm.com/docs/en/entitled-systems-support.

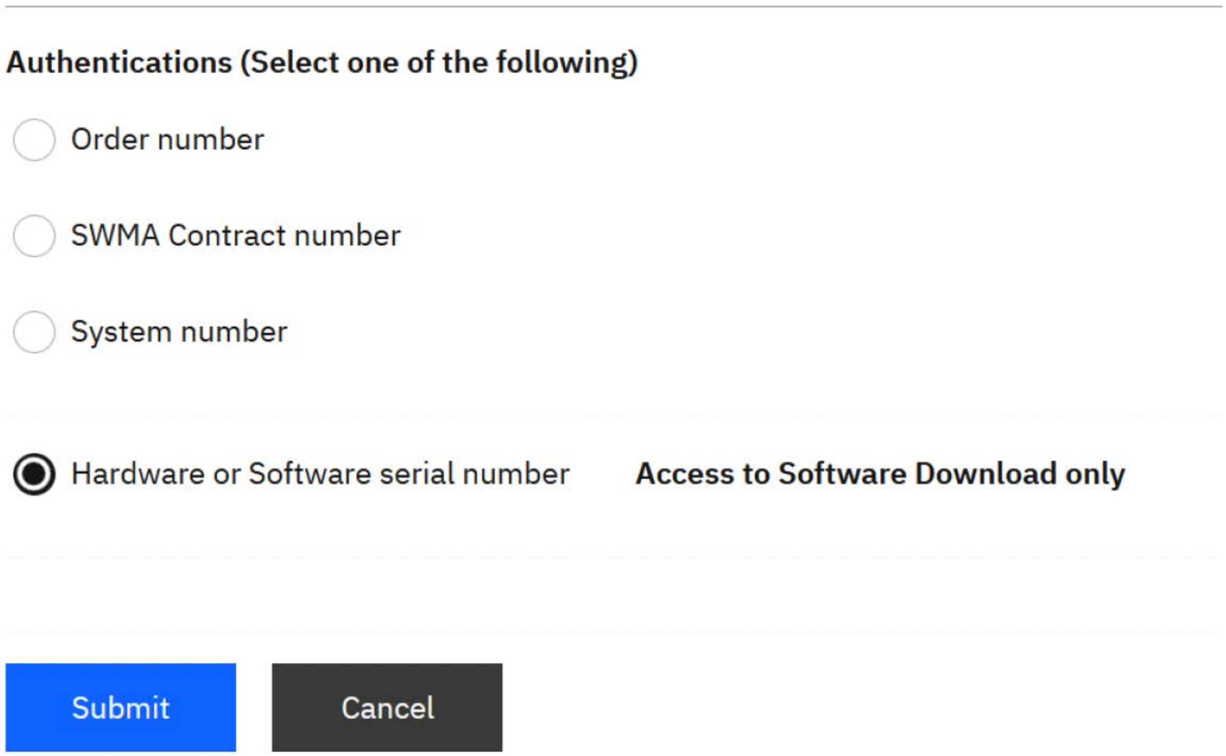
IBM Fusion HCI software is made available for installation into your OpenShift cluster by using an entitlement key that is generated with your IBMid in My IBM (https://www.ibm.com/account/us-en/). The entitlement key grants access to IBM Fusion HCI software that resides in the IBM Container Registry (ICR).

To enable your IBMid to generate an entitlement key with permission to access IBM Fusion HCI software in the ICR, you must follow the procedure to register your IBM Fusion HCI purchase in the IBM Entitled Systems Support portal:

1. Go to website (http://www.ibm.com/eserver/ess) and click Log in.



2. When the login is complete, click My Profile avatar in the title bar and select Register Customer Number.
3. Select the country and enter your IBM Customer Number (ICN).If you receive a message that the ICN is already registered, no further action is required.
4. Register an IBM Fusion HCI serial number.Use one of the serial numbers from either Method 1 or Method 2 that is mentioned in the *Before you begin* section.
5. In Authentications, select Hardware or Software serial number and then enter the IBM Fusion HCI serial number.



6. Click Submit.
7. Log in to IBM product services dashboard with your IBMid.
8. Go to Entitlement keys.
9. Click Add new key to generate a key.
   If you have an existing key, then click Copy to get the key and use.
10. Run the following commands to validate the key:

```
docker login cp.icr.io -u cp -p ${key}
```

```
docker pull  cp.icr.io/cp/isf/isf-validate-
entitlement@sha256:1a0dbf7c537f02dc0091e3abebae0ccac83da6aa147529f5de49af0f23cd9e8e
```

Note: After you complete the registration, start to pull the images even if the entitlement does not show up in the software container library. For installation procedure, see Installing IBM Fusion HCI.

Passport Advantage (PPA)

To generate the entitlement key with Passport Advantage (PPA), follow these steps:

1. Log in to the IBM container software library with an IBM ID and a password that is associated with the entitled software.
2. Go to Container software library > Passport Advantage Online.
3. Check whether IBM Fusion subscription is present. If present, go to Entitlement keys and click Add new key to generate a key.
4. Click Copy to copy the generated entitlement key.
5. Save the key to a secure location for future use.

How to Activate NVIDIA AI Enterprise Subscriptions

NVIDIA H100 NVL and H200 NVL GPUs are bundled with a five-year subscription to NVIDIA AI Enterprise.

You can activate your subscription by following the instructions here: Activate your NVIDIA AI Enterprise License.

To activate your NVIDIA AI Enterprise license, you require the serial number of your NVL GPU. To find the serial number, log into the OpenShift® Cluster as an administrator and run the following command:

```
oc get cmo monitoring-compute-1-ru25 -o jsonpath='{.status.nodes[0].processorInfos}'
```

If you have L40S GPU, then you can use the 90-day evaluation licenses available directly from NVIDIA NGC™. See NVIDIA AI Enterprise Trial.

Important: IBM does not provide support for NVIDIA AI Enterprise. You must work directly with NVIDIA to get support for your subscriptions.

Installation prerequisites

Go through these general prerequisites and ensure that they are all met before you go ahead with your installation.

- A connection to the internet is needed to install the software that operates the IBM Fusion HCI appliance. If you plan to do an offline installation of IBM Fusion, mirror external-operators and OpenShift® Container Platform image repositories to your registry. For the actual steps, see Enterprise registry for IBM Fusion HCI installation.
- IBM's delivery service removes the rack from the pallet, unpacks, and moves the rack to its final location on the data center floor. If you instruct to leave the pallet or rack packed in the staging area, then it is your responsibility to remove, unpack, and move the rack to its final location. Follow the printed manual to unbox the rack and locate it in your data center.
  If client-provided rack is used, then the following system components are delivered in two boxes that are shipped on one or more pallets:
  - First box has four switches (two High-speed and two management) and all nodes
  - Second box have rails, cables, and manuals
- Ensure that IBM Fusion HCI is installed in a restricted access location, such that the area is accessible only to skilled and instructed persons with proper authorization.
- If you plan to use a custom certificate, follow the OpenShift Container Platform guidelines to create a certificate. Consider the following points for the chain of certificates:
  - Wildcard certificate must be the first certificate in the file.
  - Any intermediate certificates must follow the wildcard certificate.
  - End of the file must have a root CA certificate.

  Note: Ensure that the certificate is well-known and not a self-signed certificate.
- Go through the Site-readiness topics and confirm whether your premise satisfies the requirements.
- The following are the high-level steps to set up and configure the hardware appliance:
  1. You must unpack the appliance and locate it in your data center. For planning and prerequisites, see Planning and prerequisites.
     If you purchased the system without an IBM rack, the IBM SSR unpacks the components from their boxes and installs them into the client-provided rack.
  2. The Service Support Representative (SSR) assists you in network cabling and power connections by using the details that are provided by your network team.
  3. The SSR assists you with the Network setup stage of the installation. The network setup includes the configuration of your network, such as high-speed switch and IP configuration. For more information, see Planning and prerequisites.
- If you plan to use a proxy server for internet access, then contact the IBM Support team. For more information about proxy configuration, see Proxy configuration.
- The Container Network Interface (CNI) network (daemon network) is created for IBM Spectrum® Storage Scale Erasure Code Edition (ECE) core pods. By default, IP addresses are assigned for scale daemon network. You can override the default IP address before you begin the Final installation. For the procedure, see Configuring Scale daemon network IP parameters.
  Note: You can override the default IP addresses only before you start the Final Installation wizard.
- If you plan to install site 2 in a Metro-DR setup, ensure that the following prerequisites are met:
  - IBM Storage Scale on the site 1 is healthy and all IBM Storage Scale core pods are up and running.
  - Ensure that the disk count is same on site 1 and site 2.
  - Ensure that site 2 has the same supported OpenShift Container Platform version as site 1.

  For general Metro-DR prerequisites, see General Metro-DR prerequisites.
- If you installed IBM Fusion HCI in the previous version using offline or online installation mode, then ensure that you do not change the mode during the upgrade to the new version. To change the installation mode, reinstall IBM Fusion HCI.
- Run the command to pull the image and verify that you have access to the IBM Fusion HCI images.
  Note: You can use either Podman or Docker to verify the access.

  ```
  podman login cp.icr.io -u cp -p "CLIENT ENTITLEMENT KEY"
  podman pull cp.icr.io/cp/isf/isf-validate-
  entitlement@sha256:1a0dbf7c537f02dc0091e3abebae0ccac83da6aa147529f5de49af0f23cd9e8e
  ```

  If the pull is successful, then you have a valid entitlement for IBM Fusion HCI images.
- For a high availability cluster, you must have received three racks from IBM with a minimum of six nodes per rack. Also, you must buy a pair of spine switches from IBM. Cables of required length to attach the spine switches can be IBM provided or client that is provided.

Installing IBM Fusion HCI

Use this procedure to install IBM Fusion HCI stage 2.

Before you begin

- IBM Fusion HCI requires Red Hat® OpenShift® Container Platform. If you purchased OpenShift subscriptions from IBM, see Activating Red Hat OpenShift Container Platform subscriptions purchased from IBM. For the supported version of Red Hat OpenShift Container Platform, see Support matrix. For other compatible OpenShift distributions, see Red Hat OpenShift distributions.
- Before the installation of IBM Fusion HCI, enable IBM Fusion HCI Software to be downloaded. For steps to enable, see Activating IBM Fusion HCI Software to be downloaded.
- Go through the installation prerequisites and ensure that they are all met before you go ahead with your installation. See Installation prerequisites.
- For two-availability-zone and arbiter node (2AZ + arbiter node) configuration, complete the network planning and meet all prerequisites mentioned in Network for two-availability-zone and arbiter node (2AZ + arbiter node) configuration.
- For the installation sequence in a disaster recovery setup, see Installing IBM Fusion HCI in a Metro-DR setup.
- To enable remote support so that the IBM Support team can assist with the installation, follow these steps:
  1. Go to the URL provided by your IBM Systems Support Representative (SSR).
     The format of the URL is `https://<host IP address>:3000/isfsetup`.
  2. In Login, enter the Username and Password and click Log in.
     The default credentials are Username `service-node-client` and Password `passw0rd`.
  3. In the License agreement window, select I accept the license agreement.
  4. In the title bar, click (?) (Help) and select Support.
  5. In the Support settings pane, enter the contact information, optional proxy server details, and enable the remote support option.
     After you enable the remote support, you can start or stop remote sessions. For the actual procedure, see Enabling remote support.

  If you have proxy connection, you can use this remote support feature for offline installation also.
- The rack discovery fails for HA MR with static IP. For more information about the workaround steps, see Workaround for HA MR static IP.

About this task

IBM Fusion HCI comes with the bootstrapping software that is installed from the factory. The IBM support representative completes the initial verification and physically connects the system to network and power. Then, they conduct the network setup of the appliance (stage 1), which connects the appliance to the data center network. This procedure configures all the default nodes (three controllers and three compute nodes). If you ordered more nodes, then they get installed as well. For high available multi rack, the IBM support representative completes the initial verification and physically connects the three racks to network and power. Then, they conduct the network setup of the first two racks which are called auxiliary racks. Network setup of the third, also known as last rack, is done after the first two racks. The network setup validates hardware and wiring of rack and connects the rack to the data centre network. This procedure configures all the default nodes. During network setup of last rack, software collects network configuration details from first two racks automatically and shows consolidated appliance view. After the network setup (stage 1) is completed for all three racks, IBM SSR provides stage 2 URL so you can continue next phase of install that is explained in the following procedure.

Note: For basic guidance on stage 2 of installation in a disaster recovery setup, see Installing IBM Fusion HCI in a Metro-DR setup.

Procedure

1. Go to the URL provided by your IBM Systems Support Representative (SSR).
   The format of the URL is **`https://<host IP address>:3000/isfsetup`**.
   Note: If you close your browser anytime before the completion of stage 2, then open the URL again to go back to its last state.
   Important: Do not add, change, or remove any files at any time.
2. In Login, enter the Username and Password and click Log in.
   The default credentials are Username `service-node-client` and Password `passw0rd`.
3. In the License agreement window, select I accept the license agreement.
4. In Create login, enter a new Username and Password for the service node.
   In the Create login page, you are prompted to create a different username and password. You need to create new credentials because the default credentials are deleted as soon as the stage 2 installation begins. The password must be at least 8 characters long and include uppercase letters, lowercase letters, numbers, and special characters. After you change the password, use the new username for subsequent logins. If you enter an incorrect password for the same username more than six times within one minute, access to that username is locked out for the next 10 minutes.

5. Click Next.
6. Configure support assistance.
   a. Enter contact information:
      In the Contact information section, provide the primary support contact details:
      i. Enter Company name.
      ii. Enter Customer number.
      iii. Enter Primary contact name.
      iv. Enter Primary email address.
      v. Enter Primary phone number:
         1. Select the country code.
         2. Enter the phone number.
   b. Provide the data center location:
      In the Data center section, enter the physical location details:
      i. Enter Address line 1.
      ii. Enter Address line 2 (optional).
      iii. Enter City.
      iv. Enter State/Province/Region.
      v. Enter Zip/Postal code.
      vi. Select the appropriate Country from the drop-down list.
   c. Configure Call Home settings:
      The Call Home feature enables automatic upload of telemetry and logs to IBM Support for faster issue resolution.
      Select a Connection type:
      - Direct connection: Uses a direct internet connection to IBM Support.
      - Open proxy: Requires proxy host and port.
      - Basic authentication: Requires a proxy with username and password.
      - Certificate authentication: Requires certificate-based proxy authentication.

      For more information, see Configuring Call Home with proxy.

   d. Configure remote access:
      The remote access feature enables secure, on-demand access for IBM Support to troubleshoot issues when required. Every session is initiated with explicit user approval.
      i. Toggle Remote access to Enabled to allow support connectivity.
      ii. Select a Connection type:
         - Direct connection: Uses a direct internet connection to IBM Support.
         - Open proxy: Requires proxy host and port.
         - Basic authentication: Requires a proxy with username and password.
7. Configure network:
   The Network precheck page runs an automatic network check against all nodes in the appliance. It checks whether each node has an assigned IP address and hostname. If it is high-availability multi-rack, then you can see the nodes of all the three racks.
   a. Check whether you can see all your nodes in Connected state.
      In a high-availability cluster, it displays all the nodes present in all three racks. If you observe all your nodes in connected state, click Next and go to step 8. Any nodes that do not pass the network check shows in Disconnected status.
   b. If one or more nodes are in a Disconnected state, do the following steps:

      DHCP
      : DHCP, this error means that either DHCP or DNS is not configured for the node. For more information about the prerequisite, see Setting up DHCP for IBM Fusion HCI and Setting up DNS.

      Static
      : The DNS configuration of the nodes may have issues. For more information about the prerequisite, see Setting up Static IP and Setting up DNS.

   c. Work with your network team and confirm that the DHCP or Static and DNS are configured for all nodes in the appliance.
   d. After you make all the changes, click Retry pre-check connection to initiate a new network check.

e. Click Next.

8. In the OpenShift version page, select the version of the OpenShift and click Next.
   Important:
   - For Dell hardware, the OpenShift 4.18.22 version is supported with only Data Foundation as a storage.
   - For Lenovo hardware, the OpenShift 4.18.22 and 4.20.24 versions are supported with both Global Data Platform and Data Foundation as a storage.
9. Set up your image registry:
   If you want to use your private image registry, you can install both Red Hat OpenShift and IBM Fusion HCI software from images that are maintained in a container registry that you manage. If you do not want to use your enterprise registry, then skip this step to install Red Hat OpenShift Container Platform and IBM Fusion software by using either the images that are hosted in the Red Hat and IBM entitled registries, or the internal registry hosted within IBM Fusion HCI.
   Important: Support is available for self signed, custom certificates, and certificates signed by a well-known CA authority. Also, the self-signed and ingress certificate must be RSA certificates.
   a. Choose whether to use the Public image registry, Private image registry, or Internal registry option.

<table>
<tr><th>Choice</th><th>Procedure</th></tr>
<tr><td>Public image registry</td><td>To use the public image registry, you need a pull secret and an entitlement key.
i. Enter the OpenShift Pull secret. It is an authorization token that stores Docker credentials that you can use to access a registry. Your cluster needs this secret to access and pull OpenShift images from the <code>quay.io</code> container registry. If you do not have a pull secret, click Get Pull secret. It takes you to https://cloud.redhat.com/openshift/install/pull-secret.
ii. Enter the IBM entitlement key. It is a product code that is used to pull images from IBM Entitlement Registry. Your cluster needs this key to gain access to IBM Fusion images in the IBM Entitlement Registry. If you do not have a key, click Get Entitlement key. It takes you to IBM Container Library. For steps to obtain the key, see the Activating IBM Fusion HCI Software to be downloaded.</td></tr>
<tr><td>Private image registry</td><td>If you select the Private image registry, then first mirror the Red Hat and IBM Fusion images to your private registry. For more information about mirroring, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.
You can choose to host the Red Hat and IBM Fusion images in separate repositories, or use the same repository.
• Single repository
Enter the following details for the enterprise registry.
• Enter the URL of the private registry in the Repository path.
For example,
<code>https://<enterprise registry>:<custom port>/<mirrorpath></code>
If you want to use custom port, then provide the custom port details.
• Enter the Username for the private registry.
• Enter the API key/ Password for the private registry.
• For Image repository > Registry certificate (Optional) section, provide a CA certificate to trust your private registry. If you select the option, then enter the following details:
• Select whether the file upload method is File upload or Text input.
• If you select File upload, drag and drop your file in the specified box. The maximum size of the file must be 1 MB and the supported file type is crt.
• If you select Text input, enter the Custom certificate.
• Multiple repositories
Enter the following details for both OpenShift images repository and IBM Fusion images repository:
• Enter the URL of the respective private image registry OpenShift images repository path or IBM Fusion images repository path in the Repository path field.
For example, URLs for OpenShift and IBM Fusion images repository paths:
<code>https://<enterprise registry for IBM Spectrum Fusion>:<custom port>/<mirrorpath>
or
https://<enterprise registry for Red Hat OpenShift>:<custom port>/<mirrorpath></code>
See the following sample values:
<code>https://registryhost.com:443/fusion-mirror
or
https://registryhost.com:443/mirror-ocp</code>
If private registry is configured with a custom port, then provide the port details in URL.
• Enter the Username for the private registry. Ensure that this user has access to the private registry.
• Enter the API key/ Password for the private registry.
• Enter the Proxy server URL (Optional). In both the OpenShift and IBM Fusion image repository sections, this option allows you to specify an alternative registry URL where images are not mirrored to a location but accessible through a proxy URL.
• For any of the Image repository > Registry certificate (Optional) section, provide a CA certificate to trust your private registry. If you select the option, then enter the following details:
• Select whether the file upload method is File upload or Text input.
• If you select File upload, drag and drop your file in the specified box. The maximum size of the file must be 1 MB and the supported file type is crt.
• If you select Text input, enter the Custom certificate.
Note:
• It is applicable for both single and multiple repo sections.
• The CA certificate is mandatory only for self-signed certificate registry. This certificate is used to trust your private registry.
• Ensure the uploaded CA certificate has external constraint set to true.
Run the following openssl command to verify:
<code>openssl x509 -in ssl.cert -noout -ext basicConstraints</code>
Expected output:
<code>`X509v3 Basic Constraints:<br>CA:TRUE`</code></td></tr>
</table>

Choice	Procedure
Internal registry	If you select the Internal registry option, no images need to be mirrored initially. Select the Internal registry tile and click Next to proceed with installation. The images and internal registry is validated and can take around 30-60 seconds before you proceed to the next page. Note: • If you select the Internal registry for installation, you must mirror the images within three months from the start of installation. • Follow the instructions in Converting a rack from internal registry to offline using enterprise registry to mirror the images and update the pull secret in your cluster. • Only Stage2, Fusion Data Foundation, and Global Data Platform Remote Mount service can be installed by using the Internal registry option. To use other services, you must mirror the required images to the enterprise registry by following the steps mentioned in Converting a rack from internal registry to offline using enterprise registry.

b. Select Connect through a proxy option in Proxy Settings to provide proxy server details to access to your network and image registries.
Enter the IP and port for the proxy server in the Host address and Port fields. If your proxy requires authentication, then enter a Username and Password.
c. Click Next.
If there are any missing images, then an error message is displayed. Click the View details link to know the details of the missing images. Correct the missing image error and click Next. This error can also occur when the IBM Fusion or the OpenShift server is temporarily down.

10. Configure network - OpenShift network and storage network.
The Network settings page shows the OpenShift network to setup OpenShift, as well as the storage configuration for the internal storage network of IBM Fusion. To enter values and customize network, see Network configuration.
11. Optional: Ingress certificate:
On the Ingress certificate page, you can optionally configure a custom certificate for OpenShift.
    a. Click the Configure now toggle to off to configure OpenShift with a self-signed certificate.
    By default, the Configure now toggle is set to on. However, it is recommended that you upload a certificate that is provided by Certificate Authority (CA). Applying a custom certificate during the installation ensures that the certificate is used immediately by OpenShift. If you do not apply custom certificate during installation, then you can do it later from OpenShift. For more information about how to apply custom certificate from OpenShift, see Ingress Operator in OpenShift.
    Important: Support is available for self signed, custom certificates, and certificates signed by a well known CA authority. Also, the self-signed and ingress certificate must be RSA certificates.
    b. Drag and drop to upload a .crt file of a size that does not exceed 1 MB or enter the details as text input.
    c. Enter the Private key and click Next. The OpenShift initialization page gets displayed.
    d. Click Finish.
12. Cluster initialization and next steps.
The final step of this phase of installation is to create a three node Red Hat OpenShift cluster and install IBM Fusion software. This minimal cluster is used in the next phase of the installation to orchestrate building out the cluster and configuring storage for IBM Fusion HCI.
    a. Monitor the progress of the OpenShift cluster.
    In case of failures, collect logs to analyze the errors. Click Learn more link to see details about the error and steps to troubleshoot the issue. After you fix the issue, click Retry. If you need to change any information entered in previous install steps, click Change install settings.
    After the OpenShift cluster gets successfully created, you can view the credentials for the OpenShift cluster.
    b. In the Next steps > Download and save credentials section, click Download Password and CoreOS key to download and save the kubeadmin password and CoreOS key.
    Important: If the Download Credentials step fails during stage 2 installation, see Contacting IBM Support.
    Note down the credentials before the installation proceeds with the next phase as you cannot access the OpenShift cluster without these credentials. It is also essential for debugging, system recovery, and reimaging or resetting. In the Downloading Password and CoreOS key window, confirm that you have saved the Password and CoreOS key, and click Continue.
    Note: For security reasons, after you download the password and CoreOS key, all the files that have credentials get deleted, such as kickstart, inventory. If there is a failure, then a error message is displayed with details. Fix the error and click Retry link. If retry does not solve the problem, see Delete files from provisioned node.
    A window displays the progress of the deletion of all the sensitive service node files. After completion, the window closes automatically and returns to the final page of the installation.
    c. To retrieve the password from the downloaded file in the future, run the following commands:
        i. Go to the Downloads folder:
        ```
        cd ~/Downloads
        ```
        ii. List the files in the folder:
        ```
        ls -ltr
        ```
        iii. Extract the contents of ocpkeys compressed file.
        iv. Go to the auth folder:
        ```
        cd clusterconfigs/auth
        ```
        v. Open kubeadmin-password in edit mode and copy the password:
        ```
        vi kubeadmin-password
        ```
        vi. Go to the extracted folder /install, save the CoreOS Key:
        ```
        id_rsa
        ```
        vii. In the installation folder, id_rsa is a CoreOS key that can be used to connect to CoreOS nodes. The folder also includes isfconfig sub folder where you can get the configuration data of the rack. Do not make updates to these files and keep it in safe location for future reference.
    d. In the Login section, copy the Username and Password and save it. Use it to log in to IBM Fusion HCI and Red Hat OpenShift. These credentials are configured as single sign-on between Red Hat OpenShift and IBM Fusion.
    Note: After you save the password and download the ocpkey.zip file, the URL points you to the OCP address. If your URL does not automatically point to the OCP address, then check the *Network Preparation* tab in your TDA installation worksheet to ensure that the DNS wildcard domain name is added to the DNS server. Test your connectivity with https://DNSentryingressendpointipaddress.
    e. Click IBM Fusion to go to the IBM Fusion HCI user interface.
    The login page of the IBM Fusion HCI console displays in a new browser tab.
    f. Enter the credentials that you noted down and click Log in to resume with the installation of other Fusion services.

Note: If there are any issues to access the console, then verify that your DNS server has a wildcard DNS A/AAAA or CNAME record that refers to OpenShift ingress. Test your connectivity <cluster_name>.<base_domain> with https://console-ibm-storage-fusion-ns.apps.<cluster_name>.<base_domain> .
If you encounter errors in the OpenShift installation wizard, see Installation issues. If you encounter errors in the Provisioning and software installation wizard, check the logs. For more information about accessing these logs, see Collecting log files of final installation.

Important: If there arises a need to redeploy IBM Fusion HCI stage 2 installation in the future, contact Contact IBM Support.

What to do next

- Set up a non-kubeadmin cluster-admin user and disable the temporary kubeadmin user. Set up an identity provider for the OpenShift Container Platform cluster. For more information about the management, see User management.
- To verify the installation, see Validating IBM Fusion HCI installation.
- To add more nodes and configure storage, see Scaling the cluster.
- For high-availability cluster, go to IBM Fusion HCI user interface and view rack details. For the procedure to view rack details, see Adding expansion racks.
- For the 2AZ + arbiter node configuration, go to Compute > Nodes in the OpenShift Container Platform web console, and verify that nodes of the cluster have the following names and roles:

  Table 1. Nodes of 2AZ + arbiter node deployment

Names	Roles
arbiter-999.<domain.com>	control-plane, master, worker
control-1-ru2.<domain.com>	control-plane, master, worker
control-2-ru2.<domain.com>	control-plane, master, worker

  After verifying the nodes, configure Fusion Data Foundation storage by following the instructions mentioned in 2AZ and arbiter node configuration with Fusion Data Foundation.

- Configure Global Data Platform service or Fusion Data Foundation service before you install additional services. IBM Fusion HCI does not support both Global Data Platform and Fusion Data Foundation services to coexist.
  Important:
  - The high-availability setup of IBM Fusion HCI does not support Fusion Data Foundation storage option.
  - If either Global Data Platform or Fusion Data Foundation are not configured or are configured but are not in a healthy state, then a few pods, like logcollector, might remain in a pending state. The pods come into a running state automatically after the storage is configured and is in a healthy state.

  For the procedure to install and work with services, see Fusion services.

- If your cluster is using Fusion Data Foundation for storage, ensure you specify the appropriate storage class as mentioned in the following snippet:

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: image-registry-storage
spec:
  storageClassName: ocs-storagecluster-cephfs
  accessModes:
    - ReadWriteMany
  resources:
    requests:
      storage: <desired-storage-size>
```

  Replace **desired-storage-size** with the amount of storage you want to allocate. For example, 500Gi, 1Ti.
  For Global Data Platform, you can proceed with the default storage class that is available on your cluster.

- Optionally, after the storage is available, configure the OpenShift Container Platform internal image registry. For the procedure to configure, see *Changing the image registry's management state* section and *Configuring registry storage for bare metal and other manual installations* section of OpenShift documentation.
  Run the following command to make this registry accessible outside the cluster.

```
oc patch configs.imageregistry.operator.openshift.io/cluster --patch '{"spec":{"defaultRoute":true}}' --type=merge
```

- If you want to install Noobaa service, then install Red Hat OpenShift Data Foundation and deploy it as Multicloud Object Gateway (MCG) only mode to provide object service. For the procedure, see Installing Data Foundation for MCG only.
- In a Metro-DR setup, run the following command on both site1 and site2 when stage 2 on site 2 is complete:

  Site 1

```
oc -n ibm-spectrum-scale-csi create configmap ibm-spectrum-scale-csi-config --from-
literal=VAR_DRIVER_DISCOVER_CG_FILESET=DISABLED
```

  Site 2

```
create namespace oc new-project ibm-spectrum-scale-csi
```

```
oc -n ibm-spectrum-scale-csi create configmap ibm-spectrum-scale-csi-config --from-
literal=VAR_DRIVER_DISCOVER_CG_FILESET=DISABLED
```

- **Network configuration**
  Steps to configure the Red Hat OpenShift network and storage network. Use the defaults as is and customize only for any special CIDRs network requirements.
- **Installing IBM Fusion HCI in a Metro-DR setup**
  Use the sequence mentioned in this procedure to install IBM Fusion HCI with Metro-DR setup.
- **Rack reset**
  Rack reset automates the process of resetting a rack to a clean, factory-ready state. Use this feature when you need to reinstall the rack or recover from configuration issues.

Network configuration

Steps to configure the Red Hat® OpenShift® network and storage network. Use the defaults as is and customize only for any special CIDRs network requirements.

About this task

The Network settings page shows the OpenShift network to setup OpenShift and the storage configuration for the internal storage network of IBM Fusion. You can use the default values in this page and customize values only whenever you have specific network requirements. In the OpenShift network section, you can override the default network CIDRs, which might collide with the data center networks. Also, it provides the flexibility to choose your own range during network planning.
Important:

- If you choose to edit the default values, ensure you remember the following points:
  - The CIDRs defined for the OpenShift network and Storage network do not overlap.
  - Each CIDR is large enough to handle nodes and workloads.
  - All IPs are unused by other systems on the network.
- For high-availability multi rack is only supported with Data Foundation storage and not with Global Data Platform.

Procedure

1. For Metro-DR setup, select whether it is the First cluster or Second cluster.
   Note: Metro-DR solution is only supported on Global Data Platform storage.
   If it is the second cluster, enter the value for Network configuration snippet:
   a. Log in to the user interface of the first cluster.
   b. Go to Disaster Reovery > Overview.
   c. In the Get started section, copy the network snippet.
   d. Go back to the installation page of the second cluster and paste it in Network configuration.
   During the Metro-DR stretch cluster installation on a secondary site, enable ping on ports 1191 and 12345. For more information about ports that must be enabled with pings, see IBM Storage Scale port usage.
2. In the OpenShift network section, enter the following network configuration for Red Hat OpenShift.
   Important:
   - 100.64.0.0/16 is reserved for `v4InternalSubnet` by OVN and must not be used for CIDR in OpenShift Container Platform installation.
   - Use unique network values for Metro-DR clusters. The following values are pre-populated with unique values. If you want to change these values, ensure that the network setting used between the two Metro-DR sites are not duplicated.

   a. Enter Pod network CIDR. It is the IP address pools from which pod IP addresses are allocated. The default value is 10.128.0.0/14.
   Note: The Pod network CIDR must be different between site 1 and site 2 of Metro-DR.
   b. Enter Pod network Host Prefix. It is the subnet prefix length to assign to each individual node.
   For example, if `hostPrefix` is set to 23, then each node is assigned a /23 subnet out of the given CIDR. A `hostPrefix` value of 23 provides 510 (2^(32 - 23) - 2) pod IP addresses. The default value is 23.

   c. Enter Service network CIDR. It is the IP address pool for the services. The default value is 172.30.0.0/16.
   Note: The Service network CIDR must be different between site 1 and site 2 of Metro-DR.
   d. In the Network MTU section, select either of the following values:
   - 9000 MTU is recommended option to maximize performance of features like Hosted Control Plane and Regional Disaster Recovery.
   - 1500 MTU is the default option for older switches to connect IBM Fusion HCI to external networks. Check with you network administrator to understand whether this option is supported.

   Important: Ensure that the MTU setting is the same for both clusters in a Metro-DR relationship. The chosen MTU must be same on the complete network infrastructure where the traffic from or to is expected in the cluster.
3. In the Storage network section, enter the following network configuration details for the internal storage network:
   Note: The CIDR address, Gateway address, and IP address range must be unique across site 1 and site 2. This is only available in case of Global Data Platform.
   a. Enter CIDR address. It is the network subnet that is used for the multus additional storage network of the scale core pods. The default value is 192.168.128.0/18.
   The list of reserved CIDRs are 169.254.169.0/29 and 100.88.0.0/16.

   b. For list of all reserved ranges, see Red Hat Documentation.
   c. Enter the Gateway address. It is the gateway address for the scale core Pods on the multus additional storage network that connects to the other site in a Metro-DR.
   Note: In Metro-DR, the gateway must be pre-configured in the customer data center.
   d. Enter the IP address range. It is the IP address pools from which scale core pod IP addresses are allocated for the multus additional storage network. The default range is 192.168.128.11 - 192.168.191.254.
   e. Click Next to go to the Custom certificate wizard page.

Installing IBM Fusion HCI in a Metro-DR setup

Use the sequence mentioned in this procedure to install IBM Fusion HCI with Metro-DR setup.

Before you begin

- For general Metro-DR prerequisites, see General Metro-DR prerequisites.
- For tiebreaker prerequisites, see Preparing the tiebreaker.

Important: Contact IBM support before you begin the installation of Metro-DR deployment.

About this task

During the installation planning, you can enable disaster recovery for your rack in either of these methods:

- If you already set up your first rack as Site 1. During the installation of Site 2, connect it to Site 1 in a Metro-DR relationship.
- You already got a standalone rack set up but not designated it as Site 1. During the installation of Site 2, use the connection snippet of Site 1 to establish a Metro-DR relationship between the two racks.

Procedure

1. Note: Perform Day 2 operations, such as expanding capacity by upsizing or adding nodes, only after the Metro-DR is successfully established and stable.
   Install site 1.
   For the procedure to install, see the single rack installation procedure in Installing IBM Fusion HCI.
2. Install site 2.
   For the procedure to install, see the single rack installation procedure in Installing IBM Fusion HCI.
3. For Metro-DR, set up the Tiebreaker. For the tiebreaker procedure, see Preparing the tiebreaker.

What to do next

Metro-DR
  Set up Metro-DR, Enable applications for Metro-DR, Failover applications from site 1 to site 2, and Failback applications from the mirrored site to the original site. For the procedure, see Metro-DR configuration for Global Data Platform.

Rack reset

Rack reset automates the process of resetting a rack to a clean, factory-ready state. Use this feature when you need to reinstall the rack or recover from configuration issues.

Overview

Rack reset prepares your IBM Fusion HCI rack for a fresh installation by:

- Discovering all hardware devices (compute nodes, switches, BMCs)
- Resetting devices to factory defaults (credentials, configurations)
- Applying baseline switch configurations
- Generating stage 1 installation files (kickstart.json, rackInfo.json)

After rack reset completes successfully, the rack is ready for stage 1 and stage 2 OpenShift® installation.

Key benefits

- Automated workflow: Eliminates manual configuration steps
- Consistent results: Resets all hardware to a known baseline state
- Efficient: Completes rack preparation in approximately 60-90 minutes
- Recovery capability: Automatically resumes from failure points without restarting

Use cases

Use rack reset in the following scenarios:

Table 1. Primary use cases

Scenario	Description	When to use
Fresh reinstallation	Complete reinstallation of IBM Fusion HCI.	When you need to start from a clean state.
Configuration recovery	Fix misconfigured hardware.	When hardware settings are corrupted.
Hardware replacement	Reset after replacing failed components.	After hardware maintenance.
Upgrade preparation	Clean state before major upgrades.	When recommended by IBM Support.

Warning: Rack reset is a destructive operation. All existing configurations are removed. All data on compute nodes is erased, and the OpenShift cluster is deleted. This operation cannot be undone. Always backup critical data before proceeding.

Prerequisites

Before performing rack reset, ensure that the following prerequisites are met.

Service node requirements

- The service node is installed and operational in the rack.
- It is typically located at RU23 (Rack Unit 23) for Day 0 service node topology and RU33 for Day 2 service node topology.
- It is connected to all rack components.
- Rack setup is completed up to Stage 2.
- The Vault must be active and unsealed. If the Vault is sealed, follow the instructions in Troubleshooting Vault to unseal it.
  If secret rotation is enabled on the cluster, ensure that it is completed successfully. For Vault-specific issues related to secret updation, see Common issues and resolution.
- Out of band IP is configured for the service node.

Network requirements

- All devices are connected to a management network.
- Forward and reverse DNS entries are configured for all devices.
- DHCP reservations are configured (if using DHCP).
- The `httpd` service must be running. If the service is stopped, restart by using the following commands:

```
systemctl start httpd
systemctl enable httpd
```

- Required firewall ports are open on the service node:
  - Port 22 (SSH)
  - Port 443 (HTTPS)
  - Port 3000 (Stage 2 UI)
  - Port 3900 (IMM console access)

Hardware requirements

- All compute nodes and switches are powered on.
- The cabling is complete and correct.
- All node Baseboard Management Controllers (BMCs) are reachable.
- Switch management interfaces are accessible.

Access and credentials

- Valid service node UI credentials are available.
- Re-enter your password to authorize the rack reset.
- Rack reset specific Vault credential secrets including `CNIMMSecret`, `SwitchSecret`, `KickStartSecret`, and `awxadmin` must be present in the HashiCorp Vault (managed automatically).

Time requirements

- Allocate 60-90 minutes for complete rack reset.
- Schedule an appropriate maintenance window for downtime.
- Ensure stable power and network connectivity during the operation.

Before you begin

- Backup all critical data. Contact IBM Support for backup procedures.
- Verify that all prerequisites are met.
  Use the following checklist to verify all prerequisites:
  - The service node is powered on and accessible through OOB network IP
  - All BMCs are reachable
  - All switches are powered on
  - Network connectivity is verified
  - DNS/DHCP is configured correctly
  - You have service node UI credentials
  - Maintenance window is scheduled
  - Critical data backed up
  - Stakeholders are notified
- For Day 2 service node topology and service node upgrade scenarios, back up the `vault-secret` and `vault-login-secret` from OpenShift Container Platform.

Preparing an upgraded service node (versions prior to 2.13) for rack reset

1. Access the service node.
2. Create 2.13 specific directory by running the following command:

   ```
   mkdir -p /home/kni/isfconfig/firmware/isf-2.13.0
   ```

3. Download `onecli` and `rhel min.iso` from the Enterprise Storage Server (ESS).
4. Copy `onecli` to path: /home/kni/isfconfig/firmware/isf-2.13.0/onecli
5. Copy `rhel min.iso` to path: /home/kni/isfconfig/firmware/isf-2.13.0/rhel
6. Copy privateRepoImages.json to path: /home/kni/isf-ui/conf/privateRepoImages/privateRepoImages.json
7. Retrieve the AWX ingress hostname:

   ```
   kubectl config use-context default --kubeconfig=/etc/rancher/k3s/k3s.yaml
   kubectl get ingress -n awx
   ```

8. Update `AWXDomain` with hostname obtained from earlier command in the file: /home/kni/isf-ui/conf/app.conf
9. Restart the UI service:

   ```
   sudo systemctl restart isf-ui.service
   ```

   For upgraded service nodes, a fully populated Kickstart file is not available on the node. As a result, the hardware inventory information displayed in the Rack reset UI might not be complete.

Performing rack reset

Use IBM Fusion HCI installer UI running on the service node to access the rack reset feature.

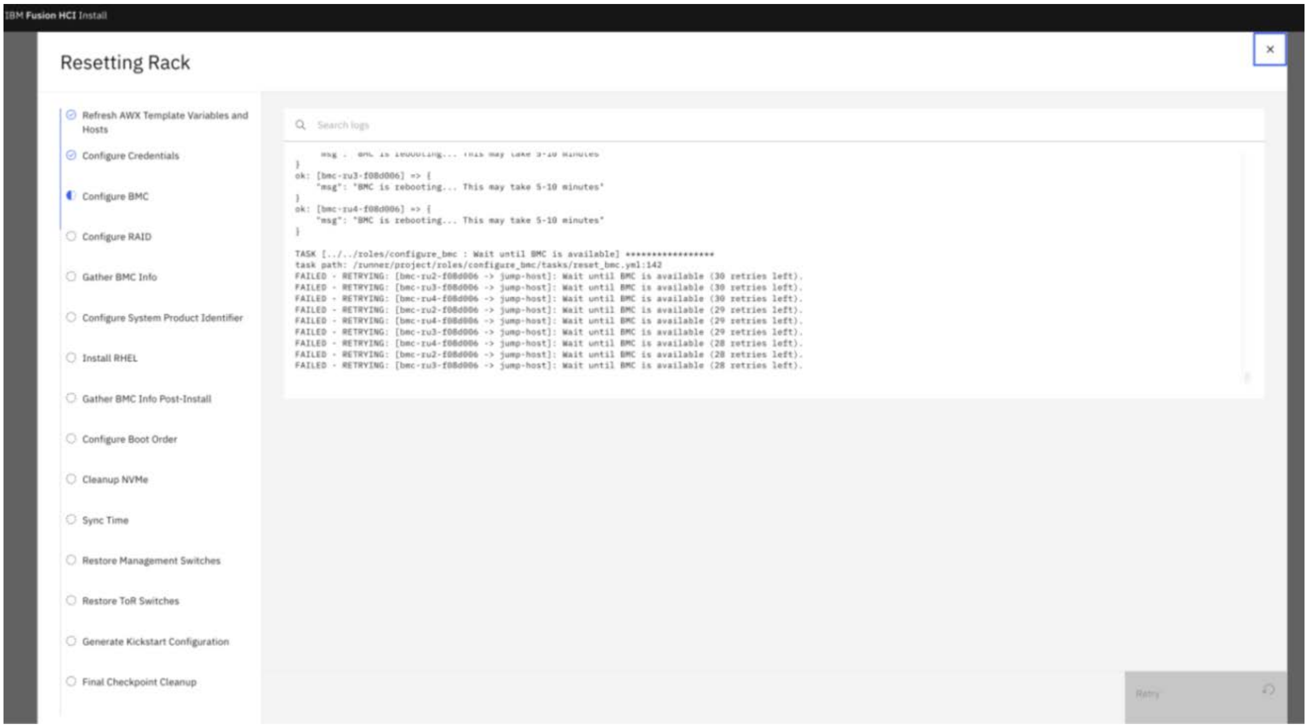
To perform rack reset, follow these steps:

1. Open a web browser (Google Chrome, Firefox, or Microsoft Edge recommended).
2. Go to the service node UI by using the out of band IP:

   ```
   https://<service-node-ip>:3000/dashboard
   ```

3. Accept the security certificate warning (self-signed certificate):
   - Click Advanced or Show Details.
   - Click Proceed to site or Accept Risk and Continue.

4. Log in to the service node UI:
   - Username: Your service node username
   - Password: Your service node password
5. Click Factory Reset.
6. In the Factory reset dialog, re-enter your password to confirm the operation.
   Note: This step is a security measure for destructive operations. Use the same password that you used to log in.
7. Click Reset.
   The rack reset workflow launches. An AWX workflow job is created on the service node, and the Progress monitoring page appears.



Rack reset workflow phases

The rack reset workflow executes the following phases automatically:

Phase 1: Device inventory creation (5-10 minutes)
    Discovers all hardware devices in the rack.

Phase 2: Server configuration (50 minutes)

- Factory reset of BMCs
- Configuration of BMCs
- RAID configuration
- System product identifier configuration
- RHEL installation
- Configuration of boot order of CoreOS
- Cleanup of NVMe disks
- BMC and OS time synchronization with NTP server

Phase 3: Switch configuration (15-20 minutes)

- Restore switches to IBM Fusion defaults

Phase 4: Generation of Stage 1 deliverables (5-10 minutes)

- Generate kickstart.json
- Generate rhel.json
- Generate <rack serial>_adaperMacAddress.json
- Generate rackInfo.json

Total estimated time: 60-90 minutes

Recovery procedures

Rack reset includes recovery mechanisms to handle failures and resume operations.

Automatic recovery (checkpoint-based)
    Rack reset includes automatic checkpoint-based recovery. If a workflow fails, you can re-launch it.

1. Review error logs to understand what failed.
2. Fix the underlying issue (for example, power on a device or fix network).
3. Re-launch rack reset from the UI.

The workflow automatically resumes from the last successful checkpoint.

Example:

```
Previous Run:
✅ Discovery complete
✅ Normalization complete
✅ Switch firmware complete
✖ Node firmware failed (node03 unreachable)


After fixing node03 and re-launching:
. Discovery skipped (already complete)
  Normalization skipped (already complete)
  Switch firmware skipped (already complete)
▶ Node firmware resumed (retry node03)
```

Manual recovery
If automatic recovery does not work, you can perform manual recovery.

Forced complete restart
To start the Rack reset workflow from the beginning, follow these steps:

1. Log in to the service node by using SSH:

   ```
   ssh kni@<service-node-ip>
   ```

2. Remove the rack reset checkpoint file:

   ```
   sudo rm /var/lib/awx/checkpoints/rack_reset/{rack_id}.json
   ```

3. Re-launch the Rack reset workflow from the service node UI.

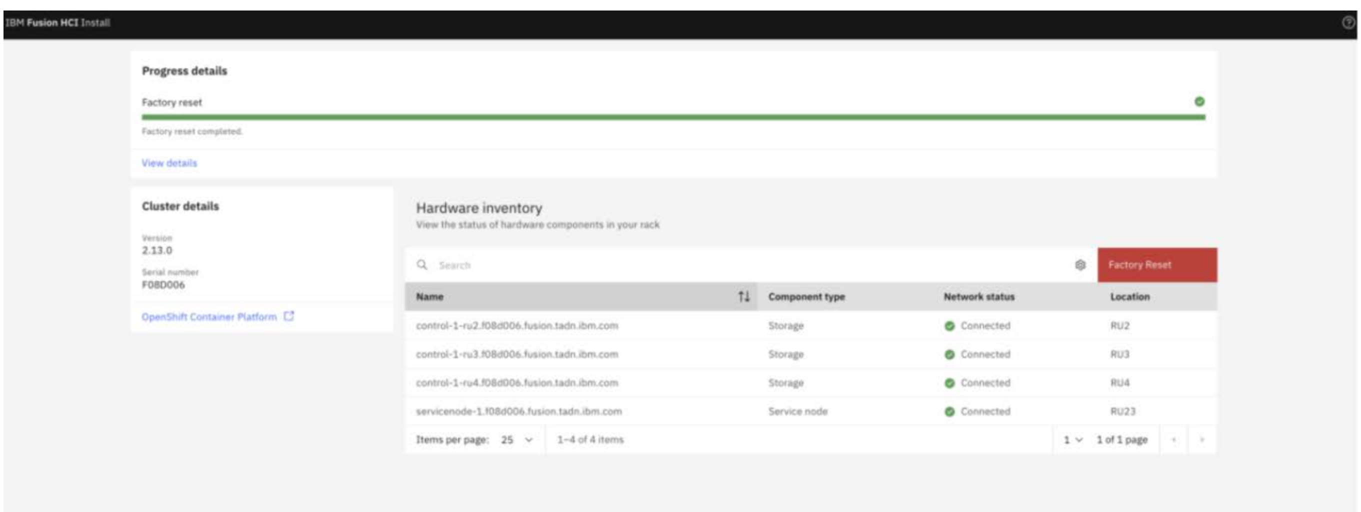
- **Monitoring rack reset**
  The service node UI provides real-time monitoring of the rack reset workflow, enabling you to track progress and respond to any issues that occur during the reset process.
- **Post-rack reset**
  After rack reset completes successfully, you must proceed with stage 1 and stage 2 installations to deploy OpenShift.
- **Troubleshooting and known issues for rack reset**
  Troubleshooting steps and known issues for rack reset.

Monitoring rack reset

The service node UI provides real-time monitoring of the rack reset workflow, enabling you to track progress and respond to any issues that occur during the reset process.

Real-time progress monitoring

The service node UI provides real-time monitoring of the rack reset workflow through a dedicated progress dashboard:



Status indicators

The workflow uses visual indicators to communicate the current state of the rack reset process:

Table 1. Status indicators

Icon	Status	Description
▶	In Progress	Workflow currently executing
✅	Completed	Workflow finished successfully

Icon	Status	Description
×	Failed	Workflow encountered an error

Handling workflow failures

If the workflow fails during execution, complete the following steps:

1. Review error messages in the logs.
2. Fix the underlying issue based on the error message guidance.
3. Re-launch rack reset from the service node UI.
4. The workflow automatically resumes from the last successful checkpoint.

Tip: The rack reset workflow includes automatic checkpoint recovery. When you re-launch after a failure, the workflow skips completed phases and resumes from where it stopped, saving time and avoiding redundant operations.

Post-rack reset

After rack reset completes successfully, you must proceed with stage 1 and stage 2 installations to deploy OpenShift®.

Stage 1 installation

Stage 1 installation prepares the infrastructure for OpenShift deployment.

Prerequisites

- Rack reset must be completed successfully.
- The service node is accessible.

Proceed with the stage 1 installation. For more information, contact IBM Support.

Preparing for stage 2 installation

Before starting stage 2 installation, complete the following preparation steps.

1. Generate an SSH key pair by running the following command:

   ```
   ssh-keygen -t rsa -b 4096 -f /home/kni/.ssh/id_rsa -N ""
   cat > /home/kni/.ssh/config << 'EOF'
   Host *
       StrictHostKeyChecking no
       UserKnownHostsFile=/dev/null
       IdentityFile /home/kni/.ssh/id_rsa
   EOF
   ```

2. Configure Vault based on your service node topology:
   - For Day 0 service node topology: Retrieve the Vault keys from the earlier downloaded keysAndPassword.zip file after successful stage 2 completion.
   - For Day 2 service node topology and service node upgrade scenario: Retrieve the Vault keys from the backup of `vault-secret` and `vault-login-secret` taken before executing rack reset.
     Note: The vault.config file must contain encoded content for the Vault root token and unseal keys.
3. Create the following files in the /var/vault directory path by using the vault.config file and `vaultLogin` content:
   - vault.config
   - unsealKey1
   - unsealKey2
   - unsealKey3
   - unsealKey4
   - unsealKey5
   - rootToken
   - vaultLogin

   Important: The content of all these files must be Base64-encoded.
   Example of vault.config file:

   ```
   Unseal Key 1: 20vcme9qFfZoJJ2Dl40lTkHMudcZ9yoSPG4O5rrPkpsa
   Unseal Key 2: KvsaHf3pocBmM/F8CEd+E9WknXnl8cp+Emdhjo7noV2L
   Unseal Key 3: /B5aCjleJwIKMloB7vF5f8kr7n3FiRgGvRlJMAt+kr04
   Unseal Key 4: zbgQhs8v1PnHrXeAR24z7ELFdjnlgpvqfME0MfWQbGCs
   Unseal Key 5: 3KFywLCGnRy0ry/Q279HB4aRR8H5aXWfQ8TkrSulaPoj

   Initial Root Token: hvs.etJ4YvVk5O26fqXS91jP0XFR

   Vault initialized with 5 key shares and a key threshold of 3. Please securely
   distribute the key shares printed above. When the Vault is re-sealed,
   restarted, or stopped, you must supply at least 3 of these keys to unseal it
   before it can start servicing requests.

   Vault does not store the generated root key. Without at least 3 keys to
   reconstruct the root key, Vault will remain permanently sealed!

   It is possible to generate new unseal keys, provided you have a quorum of
   existing unseal keys shares.
   ```

Stage 2 installation (OpenShift installation and IBM Fusion HCI setup)

Stage 2 installation deploys OpenShift and configures IBM Fusion HCI.

Prerequisites

- The stage 1 installation is completed successfully.
- The service node UI is accessible.

Proceed with the stage 2 installation. For more information, see Installing IBM Fusion HCI.

Troubleshooting and known issues for rack reset

Troubleshooting steps and known issues for rack reset.

Cannot access service node UI

Symptoms

- The browser cannot connect to `https://<service-node-ip>:3000`.
- The connection is timed out or refused.

Possible causes

- The service node powered off.
- Network connectivity issue.
- Firewall is blocking port `3000`.
- The service node is not running.

Resolution

1. Verify that the service node is powered on:

   ```
   ping <service-node-ip>
   ```

2. Check that the firewall allows port `3000`:

   ```
   # Run the following command on the service node
   sudo firewall-cmd --list-ports

   # It must show: 3000/tcp
   ```

3. Verify that the service is running:

   ```
   # SSH to service node
   ssh kni@<service-node-ip>

   # Check service status
   sudo systemctl status isf-ui
   ```

4. Restart the service if needed:

   ```
   sudo systemctl restart isf-ui
   ```

Failed to retrieve AWX credentials

Symptom
: When you click Factory reset, the Rack reset UI throws an error.

Possible causes

- The Vault is not in an active state.
- The Vault service is not running.
- The Vault is sealed.
- The `awxadmin` secret is not present in the Vault.

Resolution

1. Verify that the vault service is running:

   ```
   sudo systemctl status vault
   ```

2. Start the vault service, if not running:

   ```
   sudo systemctl start vault
   ```

3. Unseal the Vault, if Vault is sealed:

   ```
   sudo vault operator unseal <Vault unseal keys obtained from Openshift cluster or downloaded zip>
   ```

4. Check for `awxadmin` secret:

   ```
   sudo vault kv list secret/node
   ```

5. Create `awxadmin` secret if it does not exist in Vault:

```
export KUBECONFIG=/etc/rancher/k3s/k3s.yaml
kubectl get secret awx-admin-password -n awx -o jsonpath='{.data.password}' | base64 -d
sudo vault kv put secret/node/awxadmin username=admin password=<password obtained from previous step>
```

Failed to initialize AWX client

Symptom
: When you click Factory reset, the Rack reset UI throws an error.

Possible causes

- The AWX service is not running.
- AWX pods are not in a running state.
- Firewall issues

Resolution

1. Verify AWX service:

```
curl -k "https://localhost"
```

2. Verify whether the AWX pods are running:

```
kubectl get pods -n awx
```

3. Retrieve AWX logs and contact IBM Support:

```
kubectl logs -n awx deployment/awx-operator-controller-manager
kubectl logs -n awx <pod> --previous
```

Configure Boot Order workflow fails

Symptom
: The Configure Boot Order workflow fails for a few baseboard management controllers (BMCs).

Possible causes

- The BMC is not reachable.
- The BMC is in an inconsistent state.

Resolution

1. Check BMC reachability from the service node:

```
ping <BMC ipv6>
```

2. If BMC is reachable, run `resetsp`:
   a. Retrieve failed BMC credentials from `KickStartSecret` in Vault by using the following command:

   ```
   sudo vault kv get secret/KickStartSecret
   ```

   b. Log in to BMC by using the obtained IP and credentials.
   c. Run the `resetsp` command.
   d. Retry the Factory Reset workflow.

Known issues

- The Cluster details section from the Rack reset UI displays only the base version that is used during the initial cluster installation. However, after a rack reset, the rack is updated to the latest IBM Fusion version (2.13.0). For service nodes upgraded from earlier versions (2.11 or 2.12), the Rack reset UI might display the version with which the rack was initially configured.

Quay on IBM Fusion HCI

How to setup Quay and S3 for Quay.

To setup Quay and S3 for Quay, see https://access.redhat.com/products/red-hat-quay/.

Also, when the local storage is Global Data Platform service, see Data Foundation for MCG only.

When the local storage is Data Foundation service, you can use the Data Foundation service provided object storage. .

Validating IBM Fusion HCI installation

After the IBM Fusion HCI is installed, validate the success of the installation.

Verify the IBM Fusion HCI installation

Validate IBM Fusion HCI installation from the IBM Fusion user interface

1. Log in to IBM Fusion user interface by using the default user credentials: `Kubadmin/<password generated during the installation of IBM Fusion>`.
   Note: The default user has cluster administrator privileges. A single sign-on exists between the IBM Fusion user interface and the OpenShift® management console.
2. Go to Infrastructure > Overview and check the health of the nodes and switches. For more information about how to monitor the status, see Monitoring hardware from IBM Fusion HCI user interface.

Validate IBM Fusion operator from the OpenShift management console

1. Log in to the OpenShift web console.
2. Go to Overview page and in the dashboard check the status of your Cluster in the Status section. A green check mark indicates that the cluster is healthy.
3. From the menu navigation, click Operators > Installed Operators.
4. From the Installed Operators list, click IBM Fusion operator. The Details tab is open by default.
5. In the Details tab, check whether the Status is set to Succeeded.
6. From the menu navigation, click Compute > Nodes.
7. Check whether the default 6 nodes are available and are in Ready status. If you ordered more than six nodes, also verify their availability and statuses.
   Note: If the default 6 nodes are not in the Ready state, configure them. For more information, see Configuring nodes for management.

Installation issues

Use the troubleshooting tips and tricks in IBM Fusion installation.

Note: To log in to the provisioner node, as a kni user, SSH to service node. It is the same host that the IBM Service Support Representative (SSR) provided you before the stage 2 of the installation: `https://<host IP address>:3000/isfsetup`. For the actual step, see Installation procedure.

HA MR with static IP

Problem statement
: The rack discovery fails for HA MR with static IP.

Resolution
: For HA MR static IP setup, do the following workaround:

1. After you finish the stage 1 installation of rack 1, log in to the provisioning node RHEL.
2. Run the following command to change the directory:

   ```
   cd /home/kni/isfconfig
   ```

3. Run the following command to copy the JSON file:

   ```
   cp computeNodesDNSResult.json nodesDHCPResult.json
   ```

4. Complete the stage 1 installation of rack 2.
5. Log in to the provisioning node RHEL.
6. Run the following command to change the directory:

   ```
   cd /home/kni/isfconfig
   ```

7. Run the following command to copy the JSON file:

   ```
   cp computeNodesDNSResult.json nodesDHCPResult.json
   ```

8. In stage1 of rack3 (last rack of HA MR), go till the stage where all nodes are discovered and before you click Next to start the discovery of the neighboring rack.
9. Log in to provisioning node RHEL
10. Run the following command to change the directory:

    ```
    cd /home/kni/isfconfig
    ```

11. Run the following command to copy the JSON file:

    ```
    cp computeNodesDNSResult.json nodesDHCPResult.json
    ```

12. Go back to the stage 1 installation of rack 3 and click Next to start the rack discovery.

Red Hat OpenShift Container Platform cluster degradation

Problem statement
: You might observe unpredictable Red Hat® OpenShift® Container Platform cluster degradation issues that include but not limited to severe performance and operator degradation issues.

Resolution
: To avoid these issues, go through the Red Hat recommendations. For example, the following list has some of the high-level considerations (not all inclusive):

- Your cluster must have sufficient CPU and RAM resources that are allocated to etcd and compute nodes.
- The etcd disks must be high-performing with low read or write latency and high IOPs. You can use FIO test to check IOPS for etcd disks.
- There must not be high network latency among etcd members.

For more information and a complete list of considerations, see Red Hat Documentation.

DeadlineExceeded error

If you notice that the operator installation or upgrade fails with **`DeadlineExceeded error`**, see Operator installation or upgrade fails with DeadlineExceeded error.

Duplicate host entries in Stage2 Network pre-check page

Problem statement
When you retry stage 2, installation can fail due to multiple entries found for nodes in the list.
For example:

```
For node at RU5 the lookup of IP address from hostname compute-1-ru5.testdomain.com failed with error DNS lookup of hostname c
```

Resolution

1. Run the following command to verify duplicate entries:

   ```
   sudo vim /etc/hosts
   ```

2. Remove all the duplicate entries from file and save it.
3. In the Network validation page, click Retry pre-check connection.

- **Troubleshooting installation issues**
  Troubleshooting IBM Fusion HCI installation issues.
- **Operator installation or upgrade fails with DeadlineExceeded error**
  IBM Cloud Paks foundational services operator ClusterServiceVersion (CSV) status shows Failed and its InstallPlan status shows Failed after the subscription gets created.

Troubleshooting installation issues

Troubleshooting IBM Fusion HCI installation issues.

Failed to pull SCA certs error on OpenShift Container Platform console

Problem statement
The following error may occur in **oc get co** command or on OpenShift® Container Platform console:

```
Failed to pull SCA certs from https://api.openshift.com/api/accounts_mgmt/v1/certificates: OCM API https://api.openshift.com/a
```

Resolution
For the resolution, see Red Hat customer portal.

Empty node list in Network precheck wizard

Problem statement
The Network validation wizard page (Network setup stage 1) of the IBM Fusion HCI installation can have an empty node list with the Finish button in enabled state. The InlineNotification status may also show a `Connection complete!` status with a green checkmark that suggests you can proceed to the next step.
Similarly, Network precheck wizard page (Red Hat OpenShift installation stage 2) of IBM Fusion HCI may have an empty node list with the Next button enabled.

Resolution

1. If you run into this scenario, confirm that the nodes are connected before you proceed:
   Check the response of the endpoint **`<https://<host IP address>:3000/api/v1/verifyDHCP`**
2. Find the failed nodes from the response.
3. Manually verify the configuration of the node in DHCP and DNS.
   The IP addresses of the nodes can either be incorrect or not reachable.
4. Fix the issue and restart Network setup (stage 1) installation or Red Hat OpenShift installation (stage 2).

Nodes added as local host or local domain to Red Hat OpenShift Container Platform cluster

Resolution
If the Red Hat® OpenShift installation fails, then retry OpenShift installation wizard. If problem persists, contact IBM Support.

`ISF node exporter` pod in `container creating` state

Problem statement
You might encounter **`ISF node exporter`** pod in **`container creating`** state at the end of stage 2 installation. An example pod in **`container creating`** state:

```
2022-05-16T12:06:07.538Z ERROR controller-runtime.manager.controller.node Reconciler error {"reconciler group": "storage.isf.i
```

Resolution
Ignore this message and proceed to work with next steps as the error gets resolved by itself.

ImagePull failure during an installation

Problem statement
: The `ImagePull` failure can occur due to intermittent network or registry issues.

Resolution
: If an `ImagePull` failure occurs due to intermittent network or registry issue during IBM Fusion HCI installation, then restart the pod and retry. If the issue persists, contact IBM support.

Pull a container image from the `registry.connect.redhat.com`

Problem statement
: If you pull a container image from the `registry.connect.redhat.com`, it redirects to AWS S3. It is a known issue in Red Hat.

Diagnostic steps
: To verify the error message, do the following steps:

1. Log in to Red Hat OpenShift Container Platform control node.
2. Use the following commands to manually pull an image from `registry.connect.redhat.com`:

```
$ podman login registry.connect.redhat.com
$ podman pull registry.connect.redhat.com/seldonio/seldon-core-operator-bundle:latest
```

Note: An example image is used for illustration.

Resolution
: A portion of the content is hosted on `registry.connect.redhat.com` by using the following AWS S3 bucket: `rhc4tp-prod-z8cxf-image-registry-us-east-1-evenkyleffocxqvofrk.s3.dualstack.us-east-1.amazonaws.com`. Allow this domain so that OpenShift Container Platform can access it in your firewall.

NVIDIA GPU operator validator pod failure on OpenShift 4.18.22 and 4.18.23

Problem statement
: The NVIDIA GPU operator validator pod fails to initialize on OpenShift Container Platform versions 4.18.22 and 4.18.23 when using the `crun` container runtime. The pod fails with an `Init:CreateContainerError` message.
Note:

- This issue affects all versions of the NVIDIA GPU operator when deployed on the OpenShift Container Platform 4.18.22 and 4.18.23 versions.
- Clusters running OpenShift Container Platform versions before 4.18.22 or after 4.18.23 are not affected. Additionally, clusters that use `runc` as their default container runtime are also unaffected.

Diagnosis

1. Run the following command to verify the OpenShift Container Platform version.

```
oc version
```

2. Run the following command to check the container runtime used on the nodes, and check the `default_runtime` value in `/etc/crio/crio.conf`.

```
oc debug node/<node_name>
```

Example output:

```
$ oc debug node/nvd-srv-29.nvidia.example.com
Starting pod/nvd-srv-29nvidiaexamplecom-debug-fq54m ...
To use host binaries, run `chroot /host`
Pod IP: 10.0.0.1
If you don't see a command prompt, try pressing enter.
sh-5.1# chroot /host
sh-5.1# cat /etc/crio/crio.conf | grep default_runtime
# default_runtime is the _name_ of the OCI runtime to be used as the default.
# default_runtime = "crun"
# If no runtime handler is provided, the "default_runtime" will be used.
# inherit_default_runtime = false
# - inherit_default_runtime (optional, bool): when true the runtime_path,
# inherit_default_runtime = false
```

If you see default_runtime = "`crun`" (or if it is commented and `crun` is default), then your node is using `crun`.

Clusters that use `runc` are not impacted

3. Run the following command to verify the status of the NVIDIA operator validator pod.

```
oc get pods -n nvidia-gpu-operator | grep validator
```

Example output:

```
nvidia-operator-validator-xxxxx   0/1   Init:CreateContainerError   0   2m
```

4. Run the following command to verify the pod events or logs for the error message.

```
oc describe pod <validator_pod_name> -n nvidia-gpu-operator
```

Example output:

```
error executing hook `/usr/local/nvidia/toolkit/nvidia-container-runtime-hook` (exit code: 1)
```

Resolution
To resolve this issue, upgrade the OpenShift cluster to version 4.18.24 or later, or to version 4.19.

NVIDIA `CUDA` validator pod stuck in `Init:CrashLoopBackOff` on OpenShift 4

Problem statement
After installing NVIDIA GPU operator version 25.10.0 or 25.3.4 on an OpenShift cluster, certain pods in the **nvidia-gpu-operator** namespace may fail to initialize, resulting in an unhealthy GPU operator.

1. The **nvidia-operator-validator** pod remains stuck in the **Init:2/4** state.
2. The **nvidia-cuda-validator** pod enters a state of **Init:CrashLoopBackOff**.
3. Although the **nvidia-driver-daemonset** pod is running, the CUDA validation phases fail.

Diagnosis

1. Run the following command to get an overview of the pod states in the **nvidia-gpu-operator** namespace:

   ```
   oc get pods -n nvidia-gpu-operator -o wide
   ```

   Check for the following symptoms:
   a. The **nvidia-operator-validator** pod remains stuck in the **Init:2/4** state.
   b. The **nvidia-cuda-validator** pod enters a state of **Init:CrashLoopBackOff**.
2. Run the following command inside the driver **DaemonSet**:

   ```
   oc exec <nvidia-driver-daemonset-pod> -n nvidia-gpu-operator -c nvidia-driver-ctr -- nvidia-smi
   ```

   If **nvidia-smi** fails or cannot initialize the GPU, the kernel module may not be loading successfully.
3. Run the following command to check the validator pod logs:

   ```
   oc logs <nvidia-cuda-validator-pod> -n nvidia-gpu-operator -c cuda-validation
   ```

   Example failure message:

   ```
   Failed to allocate device vector A (error code initialization error)!
   [Vector addition of 50000 elements]
   ```

Resolution
Follow the steps to resolve the issue:

1. Create a **MachineConfig** that adds **nokaslr** to the kernel arguments for the worker nodes.
   For example:

   ```
   apiVersion: machineconfiguration.openshift.io/v1
   kind: MachineConfig
   metadata:
     labels:
       machineconfiguration.openshift.io/role: worker
     name: 99-disable-kaslr
   spec:
     kernelArguments:
       - nokaslr
   ```

   Run the following command to apply it:

   ```
   oc apply -f 99-disable-kaslr.yaml
   ```

2. Monitor the machine configuration pool state using the following command:

   ```
   oc get mcp
   ```

   The worker or GPU nodes automatically reboot to apply the new kernel argument. Wait until the MCP returns to **Updated=True** and **Degraded=False**.
3. After the nodes are back to **Ready** state, run the following command to verify that GPU components have recovered:

   ```
   oc get pods -n nvidia-gpu-operator
   ```

   Confirm the following:
   a. The **nvidia-operator-validator** pod is not stuck in the **Init:2/4** state.
   b. The **nvidia-cuda-validator** pod is not in a **CrashLoopBackOff** state.
   c. The driver pods are in a **Running** state.
4. After disabling KASLR and allowing nodes to reboot, all NVIDIA validation containers should initialize successfully, resolving the issue with validator pods being stuck.

Known issues

- During Red Hat OpenShift cluster creation, you cannot download logs but can monitor them on the user interface and download them after installation.
- If you observe an error **Configmap fusion platform not found in fusion namespace** in the prereq operator logs, then the error does not have any impact and can be ignored from the **isf-prereq-operator-controller-manager-xxxx** pod logs.
- If the Download Credentials step fails during stage 2 installation, contact contact IBM Support.

Operator installation or upgrade fails with DeadlineExceeded error

IBM Cloud Paks foundational services operator ClusterServiceVersion (CSV) status shows Failed and its InstallPlan status shows Failed after the subscription gets created.

Symptom

The foundational services operator CSV status shows as **Failed**. On the Red Hat® OpenShift® Container Platform cluster console, you see an error message similar to the following message:

```
Bundle unpacking failed. Reason: DeadlineExceeded, and Message: Job was active longer than specified deadline".
```

The InstallPlan status also shows as **Failed**.

1. Get InstallPlans that have the **Failed** status.

   ```
   oc get subscription <failed-operator-subscription> -n <operator-namespace> -o jsonpath='{.status.installPlanRef}'
   ```

   See the following sample output:

   ```
   {"apiVersion":"operators.coreos.com/v1alpha1","kind":"InstallPlan","name":"install-cpqk2","namespace":"cloudpak-control","reso
   ```

2. Confirm the status of the InstallPlan that you got in the previous command output.

   ```
   oc get installplan install-cpqk2 -n cloudpak-control -o jsonpath='{.status.installPlanRef}'
   ```

   Following is a sample output:

   ```
   Failed
   ```

3. Inspect the InstallPlan.

   ```
   oc get installplan install-cpqk2 -n cloudpak-control -o yaml | grep "unpack job not completed"
   ```

   See the following sample output:

   ```
   bundle unpacking failed. Reason: DeadlineExceeded, and Message: Job was active longer than specified deadline
   ```

Cause

Operator Lifecycle Manager (OLM) fails to unpack the operator bundle because the extract job failed (probably due to an issue of remote image access, or any other reason) and corrupted the configmap. Therefore, the operator manifest also most likely gets corrupted. When it is corrupted, any repeated installation attempts by using the same job and configmap fail.

In an air-gapped environment, in most cases the issue happens when the bundle image is unavailable in the private registry.

Resolution

For more information about how to resolve the issue, see Operator installation or upgrade fails with DeadlineExceeded in Red Hat OpenShift Container Platform. Complete the following steps to resolve the issue:

1. Find the corresponding job and configmap in the namespace where the CatalogSource is deployed. Narrow down your search by using the operator name or any other keyword.

   ```
   oc get job -n <catalog-namespace> -o json | jq -r '.items[] |
   select(.spec.template.spec.containers[].env[].value|contains ("<failed-operator-name>")) | .metadata.name'
   ```

2. Delete the job and the corresponding configmap that you got in the previous command. In most cases, the job and configmap have the same name.

   ```
   oc delete job <job-name> -n <catalog-namespace>
   ```

   ```
   oc delete configmap <job-name> -n <catalog-namespace>
   ```

3. Delete the **Failed** InstallPlan.

   ```
   oc delete installplan <operator-installplan-name> -n <operator-namespace>
   ```

4. Delete the subscription and CSV of the **Failed** operator.

   ```
   oc delete subscription <name-of-the-operator-subscription> -n <operator-namespace>
   ```

   ```
   oc delete csv <name-of-the-corresponding-CSV> -n <operator-namespace>
   ```

5. Delete the Operand Deployment Lifecycle Manager pod.

   ```
   oc delete pod -l name=operand-deployment-lifecycle-manager -n <your-foundational-services-namespace>
   ```

Collecting log files of final installation

Three operators get involved in the final installation. The **Installer** operator adds compute nodes to the OpenShift® Container Platform cluster, and orchestrates the overall final installation. The **Storage** operator installs the IBM Storage Scale cluster.

See these steps to get the IBM Fusion log file of all these three operators.

1. Log in to the cluster with the correct permissions to access the **ibm-spectrum-fusion-ns** namespace resource.
2. Run the following command to go to the **ibm-spectrum-fusion-ns** namespace:

   ```
   oc project ibm-spectrum-fusion-ns
   ```

   The following message is displayed as an output:

```
Now using project "ibm-spectrum-fusion-ns" on server "https://api.racke.mydomain.com:6443".
```

3. Run the following command to get the `Installer` operator Pod's name:

   ```
   oc get po -l control-plane=isf-installer-operator
   ```

   A sample result of this command is as follows.

   ```
   NAME                                                 READY   STATUS    RESTARTS   AGE
   isf-installer-operator-controller-manager-58546db557-qrcrn   2/2     Running   0          6h29m
   ```

   Run the following command to get the `Installer` operator log information:

   ```
   oc logs <Installer operator Pod's name> manager > ./installer-operator.log
   ```

   A sample command is as follows:

   ```
   oc logs isf-installer-operator-controller-manager-58546db557-qrcrn manager > ./installer-operator.log
   ```

   `isf-installer-operator-controller-manager-58546db557-qrcrn` is the `Installer` operator Pod's name. Replace this value with your `Installer` operator Pod's name. The value of the container name is always `manager`.
4. Run the following command to get the `Storage` operator Pod's name:

   ```
   oc get po -l control-plane=isf-storage-operator
   ```

   A sample result of this command is as follows.

   ```
   NAME                                                READY   STATUS              RESTARTS    AGE
   isf-storage-operator-controller-manager-7c6f945c97-bv7mp   2/2     Running             0           6h29m
   ```

   Run the following command to get the `Storage` operator log information:

   ```
   oc logs <Storage operator Pod's name> manager > ./storage-operator.log
   ```

   A sample command is as follows.

   ```
   oc logs isf-storage-operator-controller-manager-7c6f945c97-bv7mp manager > ./storage-operator.log
   ```

   `isf-storage-operator-controller-manager-7c6f945c97-bv7mp` is the `Storage` operator Pod's name. Replace this value with your `Storage` operator Pod's name. The value of the container name is always `manager`.
5. Run the following command to get the `BackupRestore` operator Pod's name:

   ```
   oc get po -l control-plane=isf-bkprstr-operator
   ```

   A sample result of this command is as follows.

   ```
   NAME                                                READY   STATUS              RESTARTS    AGE
   isf-bkprstr-operator-controller-manager-55b646456f-kcxcf   2/2     Running             0           6h29m
   ```

   Run the following command to get the `BackupRestore` operator log information:

   ```
   oc logs <BackupRestore operator Pod's name> manager > ./bkprstr-operator.log
   ```

   A sample command is as follows:

   ```
   oc logs isf-bkprstr-operator-controller-manager-55b646456f-kcxcf manager > ./bkprstr-operator.log
   ```

   `isf-bkprstr-operator-controller-manager-55b646456f-kcxcf` is the `BackupRestore` operator Pod's name. Replace this value with your `BackupRestore` operator Pod's name. The value of the container name is always `manager`.

Installing services

In the service-based IBM Fusion HCI , choose the features that you want to deploy.

For an overview on each of these services, see Fusion services.

As an administrator, you can install or enable the required services from the user interface, but make sure you meet the resource requirements of individual services.

In the Available section of the Services page, first install and configure Data Foundation service or Global Data Platform service as your storage. You can install Additional services only after you configure storage.

IBM Fusion supports preferred storage class selection during service installation by allowing each service to define an ordered list of storage classes in its FSD. The IBM Fusion UI reads this list, automatically selects the preferred storage class that exists in the cluster, and proceeds with installation without user input. If none of the preferred storage classes are available, the UI enables manual selection and records the user’s choice in the service FSD.

You can also manage the upgrade of these IBM Fusion services from the user interface. For steps to upgrade in IBM Fusion HCI, see Upgrading IBM Fusion HCI services.
Note: All IBM Fusion services are supported on the Bare Metal platform of IBM Fusion HCI.

Fusion Data Foundation
: Installing Fusion Data Foundation

Global Data Platform
: Installing Global Data Platform

Backup & Restore
: Install Backup & Restore

IBM Data Cataloging

Installing IBM Data Cataloging

CAS
Installing Content-Aware Storage (CAS)

Uninstalling services

You can uninstall IBM Fusion services by using the command line.

As a prerequisite to uninstallation of services, disable connections that are setup. For the actual procedure to disable, see Disabling the connection.

Important: If you want to do a complete uninstallation, then uninstall all your installed services in the sequence mentioned in this topic and then uninstall IBM Fusion.

- **Disabling the connection**
  Disable the connection between Hub and Spoke.
- **Uninstalling IBM Data Cataloging**
  Steps to uninstall IBM Data Cataloging by using command line interface.
- **Uninstalling Backup & Restore**
  Steps to uninstall IBM Fusion Backup & Restore by using command-line or command prompt interface.
- **Uninstalling Fusion Data Foundation**
  Use the following steps to uninstall Fusion Data Foundation by using command line interface.

Disabling the connection

Disable the connection between Hub and Spoke.

About this task

In Hub and Spoke, there is a connection CR in each cluster that represents the connection to the another(remote) cluster. Delete connection CR in the Spoke or Hub. If the network is down, then use either method 1 or method 2 to manually delete the connection CR in another cluster also.

Important: Disabling a spoke connection results in the unassignment of policies from spoke applications, and no new backups are initiated subsequently. Custom Resources (CRs) of applications without backups are deleted, and these applications no longer appear in the user interface.

Method 1: Using command line or command prompt

Procedure

1. In the Spoke, use the following command to get all the connection CR:

   ```
   oc get connection -n <fusion-namespace> -o=custom-
   columns='NAME:.metadata.name,ClusterName:.metadata.annotations.connection\.isf\.ibm\.com/cluster-name'
   ```

   Example command:

   ```
   oc get connection -n ibm-spectrum-fusion-ns -o=custom-
   columns='NAME:.metadata.name,ClusterName:.metadata.annotations.connection\.isf\.ibm\.com/cluster-name'
   ```

   Example output:

   ```
   NAME                     ClusterName
   connection-44ae8529aa    cp4d-vpc-98b7318c91b01bd72490e80cc2328915-0000.us-east.containers.appdomain.cloud
   ```

2. Check the `ClusterName` in connection CR, and delete the connection CR which has the same ClusterName with Hub:

   ```
   oc delete connection <connection-name>  -n <fusion-namespace>
   ```

   Example command:

   ```
   oc delete connection connection-44ae8529aa -n ibm-spectrum-fusion-ns
   ```

Method 2: By using OpenShift Container Platform console

Procedure

1. Log in to OpenShift Container Platform console.
2. Go to Administration > CustomResourceDefinitions > connections.application.isf.ibm.com > Instances.
3. Get the connection instance whose cluster name in the annotation is the same as the hub cluster name.
4. Delete this connection instance.

Uninstalling IBM Data Cataloging

Steps to uninstall IBM Data Cataloging by using command line interface.

Procedure

1. Run the following command and export IBM Fusion namespace as an environmental variable.

```
export FUSION_NS="<Fusion namespace>"
```

Important: Replace `<Fusion namespace>` with your namespace.
2. Run the following command to prevent catalog source creation.

```
oc -n ${FUSION_NS} patch fusionservicedefinition data-cataloging-service-definition --type='json' -p='[{"op": "replace",
"path": "/spec/onboarding/serviceOperatorSubscription/triggerCatSrcCreate", "value":false}]'
```

3. Run the following command to scale down prerequisite operator.

```
oc -n ${FUSION_NS} scale --replicas=0 deployment/isf-prereq-operator-controller-manager
```

4. Run the following commands to delete all the IBM Data Cataloging workload, networking and storage resources.

```
oc project default
export DCS_CR=$(oc get SpectrumDiscover -n ibm-data-cataloging -o jsonpath="{.items[*].metadata.name}")
oc delete SpectrumDiscover ${DCS_CR} -n ibm-data-cataloging --timeout=2m --wait=true
oc delete project ibm-data-cataloging --timeout=1m --wait=true
```

5. Run the following command to delete the IBM Fusion IBM Data Cataloging service instance object.

```
oc -n ${FUSION_NS} delete fusionserviceinstance data-cataloging-service-instance
```

6. Run the following command to delete the `SecurityContextConstraints` objects.
Important: The SCC objects are based on their namespace, and the namespace must be included in the name of the SCC object. The default namespace is `ibm-data-cataloging`, but you must change it if you use a different one.

```
oc delete securitycontextconstraints/isd-scc-ibm-data-cataloging
oc delete securitycontextconstraints/ibm-data-cataloging-c-isd-scc
```

7. Run the following command to delete the IBM Data Cataloging `ConsoleLink` object.

```
oc delete consolelink/data-cataloging
```

8. Run the following command to delete IBM Data Cataloging `CustomResourceDefinitions` from the cluster.

```
oc delete customresourcedefinition/spectrumdiscovers.spectrum-discover.ibm.com
oc delete customresourcedefinition/spectrumdiscoverapplications.spectrum-discover.ibm.com
```

9. Run the following command to scale up prerequisite operator.

```
oc -n ${FUSION_NS} scale --replicas=1 deployment/isf-prereq-operator-controller-manager
```

Uninstalling Backup & Restore

Steps to uninstall IBM Fusion Backup & Restore by using command-line or command prompt interface.

Before you begin

- Ensure you have the following installed:
  - Bash 5.0 or higher
  - OpenShift® Container Platform command line 4.14 or higher
  - `jq` 1.6 or higher version.

About this task

The uninstall-backup-restore.sh script has two optional arguments `-u` and `-d`. Use the `-u` argument to uninstall agent when the hub still exists.

Procedure

1. Log in to your cluster.
2. From your browser, go to this URL- https://github.com/IBM/storage-fusion/blob/master/backup-restore/uninstall/uninstall-backup-restore.sh.
3. Download uninstall-backup-restore.sh and run the script.

What to do next

Note: On the Topology page, it can take 10 minutes for the Connection Status of the spoke cluster to change from `Connected` to Unknown. Though in the Backedup applications page, you can see the previous application backups from the spoke cluster, no new backups are possible.
After uninstall of agent or server, you may want to delete the connection from the system. You can remove the connection for a spoke from the hub user interface, but the agent user interface does not have this capability. When you try to remove a connection from a hub, check your firewall. For more information, see Remove connections.

Uninstalling Fusion Data Foundation

Use the following steps to uninstall Fusion Data Foundation by using command line interface.

Before you begin

- Ensure that all workloads are stopped. Delete the associated persistent volume claims (PVCs) and clean up the namespace linked to it. Also, remove any other IBM Fusion services that use the Fusion Data Foundation service.
- If you are uninstalling Fusion Data Foundation 4.21 or later on Red Hat® OpenShift® Container Platform 4.21 or later, uninstall the Global Data Platform remote mount service if it is installed. For instructions, see Uninstalling Global Data Platform remote mount.
- If your installed version of Fusion Data Foundation is a provider cluster and has hosted clusters, delete the hosted clusters before proceeding with the uninstallation. For the procedure to clean up, see Cleaning up of a Hosted Control Plane cluster.

Procedure

1. Export the IBM Fusion namespace as an environmental variable.

   ```
   export FUSION_NS="<Storage Fusion namespace>"
   ```

   Important: Replace `<Storage Fusion namespace>` with your namespace.
2. Scale the IBM Fusion storage deployment to 0 replica.

   ```
   oc scale deployment --replicas=0 isf-cns-operator-controller-manager -n "$FUSION_NS"
   ```

3. Delete the Fusion Data Foundation CR.

   ```
   oc delete odfmanager odfmanager
   oc delete odfcluster odfcluster -n "$FUSION_NS"
   ```

4. Remove monitoring stack from Fusion Data Foundation.
   The procedure is same as that of Red Hat Fusion Data Foundation. For the procedure, see Removing monitoring stack Red Hat documentation.
5. Delete the internally used IBM Fusion PVCs of `isf-bkprstr-claim` and `logcollector`.

   ```
   oc delete pvc isf-bkprstr-claim logcollector -n "$FUSION_NS"
   ```

6. Uninstall steps for Fusion Data Foundation Service Consumer mode.
   Note: This step is used to uninstall Fusion Data Foundation client mode. If the current cluster has one or more Hosted Control Plane clusters, make sure to clean all the Hosted Control Plane clusters. Use the **oc** login to the different Hosted Control Plane clusters.
   a. Delete the `storageclassclaims`.

      ```
      oc delete storageclassclaims --all
      ```

   b. Delete the `storageclient` CR.

      ```
      oc delete storageclient storageclient -n openshift-storage-client
      ```

   c. Delete the finalizers from `CSIAddonsNode`.

      ```
      for name in $(oc get CSIAddonsNode -n openshift-storage-client -o name);
      do oc patch $name -p '{"metadata":{"finalizers":null}}' -n openshift-storage-client --type=merge;
      done
      ```

   d. Delete the `openshift-storage-client` namespace.

      ```
      oc delete ns openshift-storage-client
      ```

7. Uninstall the Fusion Data Foundation. For uninstallation steps, see Uninstalling Data foundation in Internal mode deployment on the IBM Support page.
   Important: Ensure that the `openshift-storage` namespace and all associated resources are deleted before you proceed to the next step.
8. Delete the IBM Fusion storageclass for its internal usage.

   ```
   oc delete sc ibm-spectrum-fusion-mgmt-sc
   ```

9. Delete the Fusion Data Foundation Service Instance CR.

   ```
   oc delete fusionserviceinstance odfmanager -n "$FUSION_NS"
   ```

10. Delete the Fusion Data Foundation Catalog Source.

    ```
    oc delete catalogsource isf-data-foundation-catalog -n openshift-marketplace
    ```

11. Recreate the PVCs for `isf-bkprstr-claim` and `logcollector`.

    ```
    oc apply -f - <<EOF
    kind: PersistentVolumeClaim
    apiVersion: v1
    metadata:
        name: isf-bkprstr-claim
        namespace: ibm-spectrum-fusion-ns
    spec:
        accessModes:
             - ReadWriteMany
        resources:
            requests:
               storage: 25Gi
        storageClassName: ibm-spectrum-fusion-mgmt-sc
        volumeMode: Filesystem
    ---
    kind: PersistentVolumeClaim
    apiVersion: v1
    metadata:
      name: logcollector
      namespace: ibm-spectrum-fusion-ns
    ```

```
spec:
  accessModes:
    - ReadWriteMany
  resources:
    requests:
      storage: 25Gi
  storageClassName: ibm-spectrum-fusion-mgmt-sc
  volumeMode: Filesystem
EOF
```

12. Scale the IBM Fusion storage deployment to 1 replica.

```
oc scale deployment --replicas=1 isf-cns-operator-controller-manager -n "$FUSION_NS"
```

13. Scale the `isf-bkprstr` deployment to 1 replica.

```
oc scale deployment --replicas=1 isf-bkprstr-operator-controller-manager -n ibm-spectrum-fusion-ns
```

14. Scale the `logcollector` deployment to 2 replica.

```
oc scale deployment --replicas=2 logcollector -n ibm-spectrum-fusion-ns
```

- **Uninstalling Global Data Platform remote mount**
  Uninstall Global Data Platform remote mount resources from Fusion Data Foundation.

Uninstalling Global Data Platform remote mount

Uninstall Global Data Platform remote mount resources from Fusion Data Foundation.

Before you begin

- Remove all filesystems from the IBM Fusion HCI user interface.
- Run the following commands to define namespace environmental variables:

```
export FUSION_NS=ibm-spectrum-fusion-ns
export SCALE_NS=ibm-spectrum-scale
```

- Ensure that all applications that access the storage filesystem through the IBM Storage Scale Container Storage Interface Driver (CSI) are stopped.
- Do the following steps to remove all applications:
  1. Stop all applications that access storage by using the IBM Storage Scale CSI driver.
  2. Delete all Persistent Volume Claims (PVCs) and Persistent Volumes (PVs) created by the IBM Storage Scale CSI driver.

About this task

This procedure applies to uninstalling Global Data Platform remote mount resources from Fusion Data Foundation 4.21 or later on Red Hat® OpenShift® Container Platform 4.21 or later.
Important: Do not perform this procedure if Global Data Platform is installed as local storage.

Procedure

1. Complete the following steps to delete the IBM Fusion storage custom resources:
   a. If encryption is configured, delete the encryption custom resources.
   Command example:

```
oc -n ${FUSION_NS} get EncryptionClient | grep -v NAME | awk '{print $1}' | xargs -I{} oc -n ${FUSION_NS} delete 
EncryptionClient {}
oc -n ${FUSION_NS} get EncryptionServer | grep -v NAME | awk '{print $1}' | xargs -I{} oc -n ${FUSION_NS} delete 
EncryptionServer {}
```

   b. Delete all filesystem custom resources.
   Command example:

```
oc -n ${FUSION_NS} get ScaleCluster | grep -v NAME | awk '{print $1}' | xargs -I{} oc -n ${FUSION_NS} delete 
ScaleCluster {}
```

   c. Verify that all `EncryptionServer` and `EncryptionClient` custom resources are deleted from IBM Fusion namespace.
   Command example:

```
oc -n ${FUSION_NS} get EncryptionServer
oc -n ${FUSION_NS} get EncryptionClient
```

   d. Verify that all the `ScaleCluster` custom resources are deleted from IBM Fusion namespace.
   Command example:

```
oc -n ${FUSION_NS} get ScaleCluster
```

   e. Delete sample PersistentVolume templates.
   Command example:

```
oc get pv -o name | grep 'ibm-spectrum-scale-fs-.*-template' | xargs -I{} oc delete {}
```

   f. Delete the `VolumeSnapshotClass`.
   Command example:

```
oc get volumesnapshotclass | grep spectrumscale.csi.ibm.com | awk '{print $1}' | xargs -I{} oc delete 
volumesnapshotclass {}
```

2. Complete the following steps to delete IBM Storage Scale Container Native resources:
    a. Record the IBM Storage Scale cluster name.
       Command example:

       ```
       oc -n ${SCALE_NS} get daemon -o yaml | grep clusterName: | awk -F " " '{print $2}'
       ```

       Example cluster name:

       ```
       ibm-storage-fusion-xx.xxx.example.com
       ```

       Note: Save the cluster name for later use.
    b. Delete the IBM Storage Scale Container Native cluster.
       Command example:

       ```
       oc delete cluster.scale.spectrum.ibm.com ibm-spectrum-scale
       ```

    c. Delete sample storage classes, PVCs, and PVs.
       Command example:

       ```
       oc -n ${SCALE_NS} get pvc -o name | grep pmcollector | xargs -I{} oc -n ${SCALE_NS} delete {}
       oc delete sc -lapp.kubernetes.io/instance=ibm-spectrum-scale,app.kubernetes.io/name=pmcollector
       oc delete pv -lapp.kubernetes.io/instance=ibm-spectrum-scale,app.kubernetes.io/name=pmcollector
       ```

3. Complete the following steps to delete IBM Storage Scale Container Native directories from nodes on which IBM Storage Scale is deployed:
    a. Identify the nodes on which IBM Storage Scale is deployed.
       Command example:

       ```
       kubectl get nodes -o jsonpath="{range .items[*]}{.metadata.name}{'\n'}"
       ```

       Tip: Use the `nodeSelector` to limit the set of nodes to clean up, or run cleanup on all nodes.
    b. Create a debug pod to remove the kernel modules and host path volume mounted directories used by IBM Storage Scale Container Native.
       Run the following command on each node:

       Command example:

       ```
       oc debug node/<openshift_node> -T -- chroot /host sh -c "rm -rf /var/mmfs; rm -rf /var/adm/ras; rmmod tracedev mmfs26
       mmfslinux;"
       ```

    c. Verify that no artifacts remain by running the following validation command:
       Command example:

       ```
       oc debug node/<openshift_node> -T -- chroot /host sh -c "ls /var/mmfs; ls /var/adm/ras; rmmod tracedev mmfs26 
       mmfslinux;"
       ```

    d. Delete IBM Storage Scale node labels created by IBM Storage Scale Container Native operator.
       Command example:

       ```
       # clean up all 'scale.spectrum.ibm.com' labels on the nodes
       kubectl get nodes -ojson | jq -r '.items[].metadata.labels' | \
       grep scale.spectrum.ibm.com | sort | cut -d: -f1 | \
       tr -d "\""| uniq | xargs -I{} kubectl label nodes --all {}\-

       # clean up 'scale' label on the nodes
       oc label node --all scale.spectrum.ibm.com/role-
       oc label node --all scale.spectrum.ibm.com/designation-
       oc label node --all scale-
       ```

4. Complete the following steps on the remote IBM Storage Scale cluster to delete access permission:
   Note: If your storage cluster is running on AWS ROSA, skip this step and revoke filesystem access directly from the storage cluster. For the procedure to revoke, see Cleanup on AWS ROSA.
    a. Run the following command to query the name of the containerized client cluster:
       Command example:

       ```
       mmauth show all | grep <scale_cluster_name>
       ```

       Replace `<scale_cluster_name>` with the cluster name recorded in step 2.a.

    b. Run the following command to remove the client cluster authorization:
       Command example:

       ```
       mmauth delete <scale_cluster_name>
       ```

5. Complete the following steps to disable Global Data Platform.
    a. Delete the `ScaleManager` custom resource.
       Command example:

       ```
       oc -n ${FUSION_NS} delete scalemanager scalemanager
       ```

    b. Delete the `ScaleManager` Fusion service instance.
       Command example:

       ```
       oc delete fusionserviceinstance scalemanager -n ibm-spectrum-fusion-ns
       ```

    c. Disable the Global Data Platform service.
       Command example:

       ```
       oc -n $SCALE_NS get csv | grep ibm-spectrum-scale-operator
       oc -n $SCALE_NS get csv -o name | grep ibm-spectrum-scale-operator | xargs -r -I{} oc -n $SCALE_NS patch {} --
       type=json -p='[{"op":"replace","path":"/spec/install/spec/deployments/0/spec/replicas","value":0}]'
       ```

Results

Global Data Platform remote mount resources are uninstalled from Fusion Data Foundation.

What to do next

If a complete cleanup is required, remove the Global Data Platform namespace and related custom resource definitions (CRDs) after the Fusion Data Foundation cleanup is complete. For the Fusion Data Foundation cleanup procedure, see Uninstalling Fusion Data Foundation.

To uninstall the IBM Storage Scale Container Native operator and remove associated Kubernetes objects and namespaces, run the following commands:

```
oc delete project ibm-spectrum-scale --wait=true --timeout=5m
oc delete crd approvalrequests.scale.spectrum.ibm.com cachevolumeoperations.scale.spectrum.ibm.com
cachevolumes.scale.spectrum.ibm.com callhomes.scale.spectrum.ibm.com clusterinterconnects.scale.spectrum.ibm.com
clusters.scale.spectrum.ibm.com compressionjobs.scale.spectrum.ibm.com consistencygroups.scale.spectrum.ibm.com
daemons.scale.spectrum.ibm.com diskjobs.scale.spectrum.ibm.com dnsconfigs.scale.spectrum.ibm.com dnss.scale.spectrum.ibm.com
encryptionconfigs.scale.spectrum.ibm.com filesystemrecoverydata.scale.spectrum.ibm.com filesystems.scale.spectrum.ibm.com
grafanabridges.scale.spectrum.ibm.com guis.scale.spectrum.ibm.com localdisks.scale.spectrum.ibm.com
pmcollectors.scale.spectrum.ibm.com recoveryconfigstores.scale.spectrum.ibm.com recoverygroups.scale.spectrum.ibm.com
regionaldrexports.scale.spectrum.ibm.com regionaldrs.scale.spectrum.ibm.com remoteclusters.scale.spectrum.ibm.com
restripefsjobs.scale.spectrum.ibm.com stretchclusterinitnodes.scale.spectrum.ibm.com stretchclusters.scale.spectrum.ibm.com
stretchclustertiebreakers.scale.spectrum.ibm.com upgradeapprovals.scale.spectrum.ibm.com volumes.scale.spectrum.ibm.com
```

Storage

IBM Fusion offers a variety of storage options to meet the demands of high-performance workloads and fulfill the storage needs of virtual machines.

Fusion storage services

Depending on your workload requirements, choose between either of the following services:

- Fusion Data Foundation storage is optimized for container workload management, ensuring efficient orchestration and resource allocation.
- Global Data Platform, on the other hand, is designed for complex AI processing, providing enhanced computational power and scalability.

Fusion Data Foundation

Fusion Data Foundation is designed to provide persistent storage and cluster data management. It supports containerized environments and integrates seamlessly with Red Hat® OpenShift®. It supports file, block, and object storage, and provides a consistent storage experience across hybrid cloud environments. For more information about Fusion Data Foundation service, see Fusion Data Foundation.

Object Storage
: Multi-Cloud Gateway (MCG) is a highly customizable data gateway for object storage. It defines resources by using Custom Resource Definitions (CRDs), and provides data services such as caching, tiering, mirroring, deduplication, encryption, and compression over many storage resources including private and public cloud object storage, file systems, and container PV/PVC. IBM Fusion supports MCG through Fusion Data Foundation. You can install MCG from the Services page of IBM Fusion. For more information about how to manage and work with MCG, see Managing hybrid and multicloud resource.

Global Data Platform

Global Data Platform is a clustered file system that provides concurrent access to a single file system or set of file systems from multiple nodes. The nodes can be SAN-attached, network-attached, a mixture of SAN-attached and network-attached, or in a shared-nothing cluster configuration. This enables high-performance access to this common set of data to support a scale-out solution or to provide a high availability platform. The IBM Storage Scale storage is designed for building a global data platform for AI, high-performance computing (HPC), advanced analytics, and other demanding workloads. It supports file and object storage for unstructured data. For more information about Global Data Platform service, see Global Data Platform. For more information about IBM Storage Scale service, see IBM Storage Scale documentation.

Object Storage
: Multi-Cloud Gateway (MCG) in Global Data Platform is an on-prem deployment. If the Global Data Platform ECE is installed, then you can see the MCG tile for it.

Global Data Platform Remote Mount

The Global Data Platform remote mount enables you to connect to a remote IBM Storage Scale cluster as if it were part of your local cluster. This is particularly useful for distributed workloads and hybrid cloud environments. For more information about, see Remote IBM Storage Scale storage cluster in IBM Fusion.

On Red Hat OpenShift Container Platform 4.20 or earlier and Fusion Data Foundation 4.20 or earlier, Fusion Data Foundation and the Global Data Platform remote mount service can be installed independently.

Starting with Red Hat OpenShift Container Platform 4.21 and Fusion Data Foundation 4.21 or later, install Fusion Data Foundation first. After installation, the Global Data Platform remote mount tile becomes available for configuration.

If you install Fusion Data Foundation as local storage, you must explicitly install and configure the remote mount service by using the Global Data Platform remote mount service tile. If you install Global Data Platform as local storage, the remote mount capability is available by default.

IBM Storage Ceph

When you install Fusion Data Foundation, you can connect or configure the Fusion Data Foundation with an external IBM Storage Ceph cluster to use it as a storage provider. As you connect and manage additional storage resources, you can scale your storage infrastructure as needed.

IBM FlashSystem

When you install Fusion Data Foundation, you can connect or configure the Fusion Data Foundation with an external IBM FlashSystem cluster to use it as a storage provider. As you connect and manage additional storage resources, you can scale your storage infrastructure as needed.

Combining these storage options

The different storage installation combinations that you can do using these options, see Storage deployment combinations from the user interface.

- **Storage deployment combinations from the user interface**
  The supported storage deployment options are Global Data Platform local, Global Data Platform remote mount, Fusion Data Foundation provider mode, and Global Data Platform with MCG only.
- **Deploying your workloads with Persistent volumes**
  This section shows you how you can provision Persistent volumes for application data persistence on IBM Fusion HCI.

Storage deployment combinations from the user interface

The supported storage deployment options are Global Data Platform local, Global Data Platform remote mount, Fusion Data Foundation provider mode, and Global Data Platform with MCG only.

In the Services page of the IBM Fusion user interface, you can see only the services that can be deployed on the cluster. It is based on your current storage configuration. The different storage installation combinations are as follows:

- Fusion Data Foundation service on IBM Fusion
-
- Fusion Data Foundation service with Global Data Platform remote mount
- Global Data Platform service
- Global Data Platform service with MCG only

Note: The Global Data Platform remote mount capability is available only for the following deployment modes:

- Standalone rack with Global Data Platform storage (newly installed with IBM Fusion 2.10.0 or later)
- Standalone rack with FDF (upgraded to IBM Fusion 2.10.0 or later / newly installed with IBM Fusion 2.10.0 or later)

Fusion Data Foundation service on IBM Fusion

Install Fusion Data Foundation on provider mode. For the installation procedure, see Installing Fusion Data Foundation.

Fusion Data Foundation service with Global Data Platform remote mount

1. Install Fusion Data Foundation on provider mode.
2. From the Available section of the Services page, install Global Data Platform remote mount. To know more about the remote mount, see Connecting to remote IBM Storage Scale file systems. The Global Data Platform remote mount is available only for IBM Fusion HCI with Fusion Data Foundation. Ensure the previous step is complete before you start this step.
3. After the installation of Global Data Platform completes successfully and the service is in healthy state, go to Storage > Remote file systems and add remote file systems. For more information about the procedure to add file systems, see Remote IBM Storage Scale storage cluster in IBM Fusion.

Global Data Platform service

1. Install Global Data Platform local. For the procedure to install, see Installing Global Data Platform.
2. Configure Global Data Platform. For the procedure to configure, see Configuring Global Data Platform local storage.

Global Data Platform service with MCG only

1. Install Global Data Platform local. For the procedure to install, see Installing Global Data Platform.
   Multi-Cloud Gateway (MCG) in Global Data Platform is an on-prem deployment. If the Global Data Platform ECE is installed, then you can see the MCG tile for it.
2. Configure Global Data Platform. For the procedure to configure, see Configuring Global Data Platform local storage.
3. Go to the Services page and install Data Foundation with MCG only from the Available section. For the procedure to install, see Installing Data Foundation for MCG only.

Deploying your workloads with Persistent volumes

This section shows you how you can provision Persistent volumes for application data persistence on IBM Fusion HCI.

A PersistentVolume (PV) is a representation of storage volume in the IBM Fusion cluster that is provisioned by an administrator, or dynamically provisioned by Kubernetes, to fulfill a request made in a PersistentVolumeClaim (PVC).

PersistentVolume and PersistentVolumeClaim are independent of pod life cycles and preserve data through restarting, rescheduling, and even deleting Pods. For example, PersistentVolumes lets you store your WordPress platform data outside the containers. This way, even if the containers are deleted, their data persists.

With PersistentVolumeClaims, you can consume abstract storage resources. It is common that users need PersistentVolumes with varying properties, such as performance, for different problems. Cluster administrators need to be able to offer a variety of PersistentVolumes that differ in more ways than size and access modes, without exposing users to the details of how those volumes are implemented. For these needs, there is the *StorageClass* resource.

A PVC is a request for storage of a certain storage class by a user that can be fulfilled by a PV. There are two ways PVs may be provisioned: statically or dynamically.

This section explains how to deploy an application that uses PVs and PVCs to store data. A PersistentVolume (PV) is a piece of storage in the cluster that has been manually provisioned by an administrator, or dynamically provisioned by Kubernetes using a StorageClass. For example, WordPress with Persistent Volumes to access and work with the storage of your workloads.

Static
: A cluster administrator creates a number of PVs. They carry the details of the real storage, which is available for use by cluster users. For more information on creating Static provisioning, see Static provisioning of Persistent Volume in filesystem.

Dynamic
: Dynamic volume provisioning allows storage volumes to be created on-demand. In dynamic provisioning when a PersistentVolumeClaim is created, a PersistentVolume is dynamically provisioned based on the StorageClass configuration. For more information on creating dynamic provisioning, see Dynamic provisioning of Persistent Volume in filesystem.

Storage classes
: The storage class options for and IBM Fusion HCI by using Container Storage Interface driver:
For more information about IBM Storage Scale Container Native , see https://www.ibm.com/docs/en/scalecontainernative?topic=overview-introduction.

  For more information on how to create storage classes in IBM Spectrum® Storage Scale Erasure Code Edition (ECE) by using Container Network Interface (CNI), see IBM Storage Scale Container Storage Interface Driver storage class .

  The IBM Storage Scale Container Native Storage Access (CNSA) allows the deployment of the cluster file system in a Red Hat® OpenShift® cluster by using a remote mount attached file system.

- **Static provisioning of Persistent Volume in filesystem**
  Persistent Volume storage that is statically provisioned by an administrator.
- **Dynamic provisioning of Persistent Volume in filesystem**
  You can create and manage storage classes for dynamic provisioning of workloads.

Static provisioning of Persistent Volume in filesystem

Persistent Volume storage that is statically provisioned by an administrator.

1. Configure Persistent Volume manifest file with a `volumeHandle` as described in this example:

```
kind: PersistentVolume
apiVersion: v1
metadata:
  name: ibm-spectrum-fusion-pv-images
  labels:
    cns.isf.ibm.com/creator-fusion: ''
spec:
  capacity:
    storage: 10Gi
  csi:
    driver: spectrumscale.csi.ibm.com
    volumeHandle: 14544310586575975773;096E4651:61DFAFB3;path=/mnt/ibm-spectrum-scale-fs-mcgpfs3-
14544310586575975773/images
  accessModes:
    - ReadWriteMany
  persistentVolumeReclaimPolicy: Retain
  volumeMode: Filesystem
status:
  phase: Available
```

   Here, the name is `ibm-spectrum-fusion-pv-images2`. The storage capacity is 10Gi. The `volumeHandle` value is `14544310586575975773;096E4651:61DFAFB3;path=/mnt/ibm-spectrum-scale-fs-mcgpfs3-14544310586575975773/images`.

2. Run the following apply command to create a Persistent Volume:

```
kubectl apply -f pv.yaml
```

Note:

- IBM Fusion creates static Persistent Volume templates for each filesystem.
- For IBM Fusion, static Persistent Volume for fileset supports backup but does not support backup for normal file or directory. If backup must be taken for normal file or directory, use dynamic Persistent Volume.

It is not applicable for IBM Fusion HCI.

Dynamic provisioning of Persistent Volume in filesystem

You can create and manage storage classes for dynamic provisioning of workloads.

Storage classes in IBM Storage Scale

There are two versions of storage classes that you can use with IBM Storage Scale. The default version works with all versions of IBM Storage Scale. For storage class examples, see Storage class (version 1) and Storage class (version 2). For the storage class to use with IBM Cloud Paks, IBM Storage Scale version (All), see Storage class to use with Cloud Paks.

Note: For IBM Storage Scale Container Storage Interface Driver volume expansion function, include **allowVolumeExpansion=true** in your Storage Classes. For information about how to set it in a storage class, see the example storage classes in this topic. For more information about volume expansion in CSI documentation, see Volume expansion.
The IBM Fusion detects the version of IBM Storage Scale, and creates storage class for each filesystem.

Procedure to create and apply a storage class YAML:

1. Create the following YAML and store it in the location of your choice. Retain this YAML until you create the storage class.
2. Run the following command to apply it:

```
oc apply -f <filename>
```

Storage class(version 1)
The default version that works with all versions of IBM Storage Scale. Example storage class (version 1) for IBM Storage Scale version (All):

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: ibm-spectrum-scale-csi-fileset-dependent-new
provisioner: spectrumscale.csi.ibm.com
parameters:
  volBackendFs: "ibm-spectrum-scale-fs-mcgpfs3-14544310586575975773"
  filesetType: "dependent"
  parentFileset: "primary-fileset-ibm-spectrum-scale-fs-mcgpfs3-14544310586575975773-2822920183822285462"
reclaimPolicy: Delete
allowVolumeExpansion: true
```

The file set path pattern is **primary-fileset-FS_CR_NAME-LocalClusterID**. If the fileset does not exist, then IBM Fusion creates it automatically from remote storage class.

For more information about the parameters, see Storage class.

Storage class(version 2)
This second storage class version requires IBM Storage Scale on Storage Cluster. Example storage class(version 2) for IBM Storage Scale version >= 5.1.3:

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: ibm-spectrum-scale-csi-storageclass-version2
provisioner: spectrumscale.csi.ibm.com
parameters:
  version: "2"
  volBackendFs: "mygpfs"
reclaimPolicy: Delete
allowVolumeExpansion: true
```

The optional parameter list depends on the storage class type.

For more information about the additional parameters, see Volume expansion.

Storage classes in Fusion Data Foundation

There are two versions of storage classes that you can use with Fusion Data Foundation:

- Filesystem storage class **ibm-spectrum-fusion-mgmt-sc** and **ocs-storagecluster-cephfs**
- Block storage class **ocs-storagecluster-ceph-rbd**

Storage class to use with Cloud Paks

- Storage class to use with Cloud Paks for IBM Storage Scale
- Storage class to use with Cloud Paks for Fusion Data Foundation

IBM Storage Scale
The **ibm-storage-fusion-cp-sc** storage class is the default for use with IBM Cloud Paks, IBM Storage Scale (All). For IBM Cloud Paks, the permissions parameter is set to **shared: true** on the storage class.

The **ibm-storage-fusion-cp-sc** storage class is available for IBM Fusion:

```
NAME                        PROVISIONER                  RECLAIMPOLICY   VOLUMEBINDINGMODE    ALLOWVOLUMEEXPANSION   A
ibm-storage-fusion-cp-sc    spectrumscale.csi.ibm.com    Delete          Immediate            true                   2
```

```
oc get sc ibm-storage-fusion-cp-sc -o yaml
```

```
allowVolumeExpansion: true
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  creationTimestamp: "2023-08-06T13:07:55Z"
  name: ibm-storage-fusion-cp-sc
  resourceVersion: "4107984"
  uid: c8be0a2f-e379-4498-9051-ce8d4cde8d93
parameters:
  shared: "true"
  version: "2"
```

```
  volBackendFs: ibmspectrum-fs
provisioner: spectrumscale.csi.ibm.com
reclaimPolicy: Delete
volumeBindingMode: Immediate
```

Data Foundation

Fusion Data Foundation defines storage classes for ReadWriteAfter (RWO) access and ReadWriteMany (RWX) access separately.
During the installation of IBM Cloud Paks, the `--block_storage_class` option must point to block storage class that supports ReadWriteAfter (RWO) access, which is `ocs-storagecluster-ceph-rbd`. The `--file_storage_class` option must point to a file storage class that supports ReadWriteMany (RWX) access, which is `ocs-storagecluster-cephfs`.

Note: For supported IBM Cloud Paks versions, see IBM Cloud Paks support for IBM Fusion.

Fusion services and capabilities

IBM Fusion Service provides scalable, integrated storage and data management solutions optimized for hybrid cloud and AI workloads on Red Hat® OpenShift®.

- **Fusion services**
  IBM Fusion Service provides scalable, integrated storage and data management solutions optimized for hybrid cloud and AI workloads on Red Hat OpenShift.
- **Disaster recovery**
  Disaster recovery offerings of IBM Fusion HCI.
- **High availability**
  High availability ensures that system services and data remain accessible with minimal downtime by distributing compute, control, and storage components across independent failure domains.
- **Multi-cluster IBM Fusion HCI using Hosted Control Plane**
  Create a Hosted Control Plane with Red Hat OpenShift Virtualization clusters or Bare Metal clusters (both internal and external) on IBM Fusion HCI.
- **Red Hat OpenShift Virtualization**
  IBM Fusion enables the coexistence of containers and VMs, with OpenShift Virtualization allowing VMs to be managed as Kubernetes resources. Beyond VM hosting, Fusion and OpenShift Virtualization offer virtual machine mobility, backup and restore, and additional capabilities to ensure high availability and disaster recovery.
- **Advanced workloads**
  IBM Fusion is built to provide enterprise ready deployments for IBM Cloud Paks, IBM watsonx, IBM Maximo®, and Db2 Warehouse.

Fusion services

IBM Fusion Service provides scalable, integrated storage and data management solutions optimized for hybrid cloud and AI workloads on Red Hat® OpenShift®.

IBM Fusion supports preferred storage class selection during service installation by allowing each service to define an ordered list of storage classes in its FSD. The IBM Fusion UI reads this list, automatically selects the preferred storage class that exists in the cluster, and proceeds with installation without user input. If none of the preferred storage classes are available, the UI enables manual selection and records the user’s choice in the service FSD.

The available services in IBM Fusion are as follows:

Global Data Platform

The Global Data Platform in IBM Fusion is a unified data management solution that simplifies access, sharing, and management of data across diverse environments. It offers scalable storage optimized for hybrid cloud and AI workloads, ensuring seamless data availability and mobility across multiple locations. It is a software-defined file and object storage that enables organizations to build a global data platform for artificial intelligence (AI), high-performance computing (HPC), advanced analytics, and other demanding workloads.

The Global Data Platform storage type provides the following features:

- File storage
- High availability through capacity-efficient erasure coding
- Metro and regional disaster recovery
- CSI snapshot support with built-in application consistency
- Encryption at rest
- Ability to mount file systems hosted by remote IBM Storage Scale clusters.

Fusion Data Foundation

You can use Fusion Data Foundation as storage service in IBM Fusion HCI. It enables you to manage Fusion Data Foundation services, and to deploy and scale storage clusters. The Fusion Data Foundation storage type provides the following features:

- Block, file, and object storage
- High availability through automatic data replication
- Metro and regional disaster recovery
- CSI snapshot support
- Encryption at rest

For more information about Fusion Data Foundation service, see Introduction to Fusion Data Foundation.

Backup & Restore

The Backup & Restore service provides the following features:

- Policy-driven backup of applications that run on Red Hat OpenShift
- Change block detection for Ceph RBD block volumes. It greatly reduces backup times for applications using block mode volumes, including OpenShift Virtualization VMs based on RBD block volumes.
- Orchestration of application consistent online backups through recipes
- Multi-cluster Backup & Restore by using a hub and spoke topology

The Backup & Restore provides application-centric backup and data recovery. For more information about the service and its architecture, see Backup & Restore overview.

IBM Data Cataloging

The IBM Data Cataloging provides data insight for exabyte-scale heterogeneous file, object, backup, and archive storage on premises and in the cloud. The software easily connects to these data sources to rapidly ingest, consolidate, and index metadata for billions of files and objects. For more information about the IBM Data Cataloging service and its architecture, see Data Cataloging.

Content-Aware Storage (CAS)

It leverages natural language processing (NLP) and AI technologies to extract semantic meaning from unstructured data, such as text, charts, graphs, and images. It allows for more efficient data retrieval and integration with AI workflows, particularly in retrieval-augmented generation (RAG) processes. For more information about the CAS service, see Content-Aware Storage (CAS).

- **Fusion Data Foundation**
  Welcome to the official documentation for IBM Fusion Data Foundation 4.21.
- **Global Data Platform**
  Learn how to install and configure Global Data Platform service.
- **Backup & Restore**
  The Backup & Restore service protects your workloads with tailored backups by using a hub and spoke model. You can restore your data rapidly from local snapshots without having to rely on external storage. Backups can be transferred to external object storage solutions like IBM Cloud, Microsoft Azure, Amazon Web Services, and any S3-compliant object storage. It allows the protection of the IBM Fusion Backup & Restore service itself to a S3 bucket.
- **Data Cataloging**
  Companies need the ability to use unstructured data to meet their business priorities.
- **Content-Aware Storage (CAS)**
  The IBM Fusion Content-Aware Storage service is engineered to enhance the value of customers' AI applications, such as retrieval augmented generation (RAG) for faster time to insights, reduced cost, improved performance, enhanced security, and streamlined operations.
- **Software Hub**
  The Software Hub is a comprehensive platform that simplifies the deployment and management of IBM software solutions, including IBM Software Hub, watsonx.ai and watsonx Orchestrate, on IBM Fusion HCI. It provides a streamlined and automated experience, reducing the complexity and manual effort associated with traditional deployment processes.
- **Lakehouse-Table [Technology preview]**
  The Lakehouse-Table service is a Kubernetes-native solution for building and managing lakehouse infrastructure on IBM Fusion. It provides a simplified experience for administrators to provision and manage Data Lakehouse capabilities.

Fusion Data Foundation

Welcome to the official documentation for IBM Fusion Data Foundation 4.21.

This documentation provides information on Fusion Data Foundation, including:

- Planning considerations required before deployment
- Step-by-step installation and deployment guidance
- Detailed configuration instructions for new features and enhancements
- Recommendations and procedures for managing the deployment
- Guidance on monitoring cluster health, metrics, and alerts
- Upgrade and migration procedures from previous versions
- Troubleshooting tips and best practices

The Fusion Data Foundation 4.21 release introduces a set of new features, enhancements, and bug fixes designed to improve performance, usability, and integration capabilities across supported environments.

Remember: Each IBM Fusion release documentation includes content only for the latest supported version of Fusion Data Foundation. To access documentation for earlier Fusion Data Foundation versions, refer to the corresponding IBM Fusion release documentation. For version compatibility details, see the Support matrix. It is recommended to use the latest Fusion Data Foundation version to ensure full support and access to the latest features and fixes.

- **Release notes**
  IBM Fusion Data Foundation 4.21 is now available. New features, enhancements, and known issues that pertain to Fusion Data Foundation 4.21 are included in this section.
- **Introduction to Fusion Data Foundation**
  IBM Fusion Data Foundation is a highly integrated collection of cloud storage and data services for Red Hat OpenShift Container Platform. It is available as a IBM Fusion storage service to facilitate simple deployment and management.
- **Installing Fusion Data Foundation**
  The Fusion Data Foundation service provides a foundational data layer for applications to function and interact with data in a simplified, consistent, and scalable manner. The provider mode is similar to an external Data Foundation configuration. It acts as the provider and base storage on the host cluster. You can provision new clusters within a managed rack and consume storage from the central provider cluster.
- **Installing Data Foundation for MCG only**
  The IBM Fusion HCI supports MCG only mode.
- **Configuring Fusion Data Foundation**
  Configure Fusion Data Foundation to set up storage services that support scalable, resilient, and flexible data management for OpenShift applications.

- **Preparing to connect to an external KMS server in Fusion Data Foundation**
  Procedure to prepare for the connection to an external KMS from Fusion Data Foundation.
- **Managing hybrid and multicloud resource**
  Use this information to learn how to manage hybrid and multicloud resources.
- **Replacing nodes and devices**
  Explains how to safely replace a node and storage devices in a Fusion Data Foundation cluster.
- **Fusion Data Foundation service install and upgrade issues**
  Use this troubleshooting information to resolve install and upgrade problems that are related to Fusion Data Foundation service.
- **Fusion Data Foundation service error scenarios**
  Use these troubleshooting information to know the problem and workaround when install or configure Fusion Data Foundation service.

Release notes

IBM Fusion Data Foundation 4.21 is now available. New features, enhancements, and known issues that pertain to Fusion Data Foundation 4.21 are included in this section.

- **New features**
  Each release offers new functionalities and improvements. Read this information for a high-level summary of the new features introduced in the Fusion Data Foundation 4.21 release.
- **Enhancements**
  Each release offers new functionalities and improvements. Read this information for a high-level summary of major enhancements introduced in the Fusion Data Foundation 4.21 release.
- **Known issues**
  This section describes the known issues in Fusion Data Foundation 4.21.

New features

Each release offers new functionalities and improvements. Read this information for a high-level summary of the new features introduced in the Fusion Data Foundation 4.21 release.

- CNSA integration with Fusion Data Foundation
- Erasure coding support in Fusion Data Foundation
- Disaster Recovery

CNSA integration with Fusion Data Foundation

IBM Fusion 2.13 introduces integration between Fusion Data Foundation and Container Native Storage Access (CNSA) by using the remote mount feature. This feature enables Fusion Data Foundation to deploy and manage the required IBM Storage Scale or Global Data Platform resources during installation. It also allows administrators to configure and attach external IBM Storage Scale clusters directly from the Red Hat® OpenShift® Container Platform 4.21 user interface.

The integration simplifies the installation, upgrade, and lifecycle management of remote mount functionality across IBM Fusion deployments, supports seamless upgrades and migrations from earlier releases, and ensures continuity for existing workloads while enabling a unified, UI-driven storage management experience.

Erasure coding support in Fusion Data Foundation

IBM Fusion 2.13 adds erasure coding support to Fusion Data Foundation through the Storage efficiency configuration option. This feature improves usable storage capacity while maintaining strong data protection, providing an alternative to replica-3 storage configurations.

With this capability, administrators can select the Storage efficiency option during storage configuration to apply erasure coding. Erasure coding splits data into data and parity chunks that are distributed across storage nodes, delivering fault tolerance with significantly higher usable capacity compared to performance-optimized configurations. Use this option for workloads such as archival storage, backups, and other capacity-driven use cases.

Erasure coding requires a minimum number of storage nodes, based on the selected data and parity layout. The minimum supported configuration requires six storage nodes. For example, a 4+2 configuration requires at least six nodes, while an 8+2 configuration requires at least ten nodes.

Important:

- Erasure coding is supported starting with Red Hat OpenShift Container Platform 4.21 and later and Fusion Data Foundation 4.21 and later.
- This feature is supported only for Block storage (RBD).
- Clusters upgraded from earlier releases do not support erasure coding.

Disaster Recovery

Multi-volume consistency for disaster recovery
: Multi-volume consistency group support is now fully supported again for both, RBD and CephFS volumes. Consistency groups are used by default for all newly protected applications. Already protected non-CG applications on an upgraded cluster applications should be unprotected and protected again to migrate to consistency groups.

Support for CephFS volumes using non-default StorageClasses
: Fusion Data Foundation DR solution supports data replication for CephFS volumes that use multiple or non-default StorageClasses, such as replica-2 file storage. This enhancement enables disaster recovery for workloads backed by customized CephFS configurations, including cases where different pools or secrets require separate StorageClass, VolumeSnapshotClass, and VolumeReplicationClass resources.

Enhanced disaster recovery capabilities in the RHACM KubeVirt UI
: **Disaster recovery integration for virtual machines**

The Red Hat Advanced Cluster Management for Kubernetes (RHACM) KubeVirt UI now integrates Fusion Data Foundation disaster recovery, allowing users to view VM protection status and initiate DR protection directly from the VM view. This capability is available for both GitOps-managed and discovered VMs, with disabled actions providing guidance when prerequisites are not met.

Improved visibility for failover and relocate operations

The RHACM UI now surfaces detailed, step-by-step progression for disaster recovery failover and relocate workflows using progression data from Ramen. Enhanced modals show each operation phase, status indicators, and any associated errors, improving transparency, troubleshooting speed, and overall observability of DR processes.

Enhancements

Each release offers new functionalities and improvements. Read this information for a high-level summary of major enhancements introduced in the Fusion Data Foundation 4.21 release.

- HCP clusters
- Disaster Recovery
- Multicloud Object Gateway

HCP clusters

Fusion Data Foundation support for HCP clusters
: Fusion Data Foundation is supported on hosted (HCP) clusters running OpenShift® Container Platform versions N, N-1, N-2, N-3, and N-4, where N represents the OpenShift Container Platform version deployed on the hub cluster.

Disaster Recovery

Enhanced protected applications list view for disaster recovery
: The protected Applications page shows managed (ApplicationSet and Subscription) and discovered applications together, providing a complete view of all workloads protected by disaster recovery. Multi-selection is enabled in the list view, laying the groundwork for grouped failover, relocate, and disable-DR operations for managed applications.

Automatic resource cleanup for discovered VMs during disaster recovery operations
: Disaster recovery workflows now automatically remove resources associated with DR-protected virtual machines during failover or relocate operations. This automation applies only to the discovered virtual machines enrolled for protection through Red Hat® Advanced Cluster Management for Kubernetes Fleet Virtualization and helps reduce manual steps while improving transition efficiency.

  For more information, see Virtual machine garbage collection automation.

Multicloud Object Gateway

Minimal IAM support for Multicloud Object Gateway
: Minimal IAM support is now available in the Multicloud Object Gateway to provide improved access control for multi-tenant environments. This enhancement enables administrators to define and manage user-level permissions more effectively, supporting scenarios where data scientists and other user groups require isolated and governed access to object storage resources. The feature lays the groundwork for more secure, flexible multi-tenant deployments by ensuring that users interact only with the buckets and operations they are authorized to access.

  For more information, see Creating and managing IAM user.

Enhanced Multicloud Object Gateway database backup
: Multicloud Object Gateway database backup is enhanced to provide more reliable protection against data loss and to complement the high-availability capabilities. The update adds support for automated scheduled backups with configurable frequency (daily, weekly, or monthly) and retention settings along with on-demand backup options through the MCG CLI or CRD. The UI now displays the time of the most recent backup and offers advanced configuration for Multicloud Object Gateway backup management. The system supports internal, external, and standalone Multicloud Object Gateway deployments by automatically detecting or allowing selection of snapshot StorageClasses.

  To configure Multicloud Object Gateway database backup, refer to the deployment procedures for your specific platform.

Developer access to the Fusion Data Foundation object browser
: Developer-level users can now be granted access to the Fusion Data Foundation object browser, enabling them to view and manage only their own object storage resources with restricted permissions. This enhancement supports flexible, role-appropriate access control and allows development teams to work more independently while preserving administrative boundaries.

  For more information, see Accessing the object browser.

Support for internal mode RGW with object browser
: The object browser now supports browsing, uploading, downloading, and managing RGW object. Operations include listing and creating buckets, navigating objects in a folder-like view, uploading files or directories, downloading objects, generating presigned URLs, viewing metadata, filtering and sorting results, previewing content, and creating prefixes. Bucket-level capabilities such as configuring expiration and access policies are also available.

  For more information, see Creating and managing buckets using the object browser.

Known issues

This section describes the known issues in Fusion Data Foundation 4.21.

- Disaster recovery
- Multicloud Object Gateway
- Ceph
- CSI driver
- IBM Storage Scale operator
- Fusion Data Foundation console

Disaster recovery

CIDR range does not persist in **`csiaddonsnode`** object when the respective node is down

When a node is down, the Classless Inter-Domain Routing (CIDR) information disappears from the **`csiaddonsnode`** object. This impacts the fencing mechanism when it is required to fence the impacted nodes.

Workaround:

Collect the CIDR information immediately after the **`NetworkFenceClass`** object is created.

(DFBUGS-2948)

DRPCs protect all persistent volume claims created on the same namespace

The namespaces that host multiple disaster recovery (DR) protected workloads protect all the persistent volume claims (PVCs) within the namespace for each DRPlacementControl resource in the same namespace on the hub cluster that does not specify and isolate PVCs based on the workload using its **`spec.pvcSelector`** field.

This results in PVCs that match the DRPlacementControl **`spec.pvcSelector`** across multiple workloads. Or, if the selector is missing across all workloads, replication management to potentially manage each PVC multiple times and cause data corruption or invalid operations based on individual DRPlacementControl actions.

Workaround:

Label PVCs that belong to a workload uniquely, and use the selected label as the DRPlacementControl **`spec.pvcSelector`** to disambiguate which DRPlacementControl protects and manages which subset of PVCs within a namespace. It is not possible to specify the **`spec.pvcSelector`** field for the DRPlacementControl using the user interface, hence the DRPlacementControl for such applications must be deleted and created using the command line.

Result:

PVCs are no longer managed by multiple DRPlacementControl resources and do not cause any operation and data inconsistencies.

(DFBUGS-1749)

Disabled **`PeerReady`** flag prevents changing the action to failover

The DR controller executes full reconciliation as and when needed. When a cluster becomes inaccessible, the DR controller performs a sanity check. If the workload is already relocated, this sanity check causes the **`PeerReady`** flag associated with the workload to be disabled, and the sanity check does not complete due to the cluster being offline. As a result, the disabled **`PeerReady`** flag prevents you from changing the action to failover.

Workaround:

Use the command-line interface to change the DR action to failover despite the disabled **`PeerReady`** flag.

(DFBUGS-665)

Information about **`lastGroupSyncTime`** is lost after hub recovery for the workloads which are primary on the unavailable managed cluster

Applications that are previously failed over to a managed cluster do not report a **`lastGroupSyncTime`**, thereby causing the trigger of the alert **`VolumeSynchronizationDelay`**. This is because when the ACM hub and a managed cluster that are part of the DRPolicy are unavailable, a new ACM hub cluster is reconstructed from the backup.

Workaround:

If the managed cluster to which the workload was failed over is unavailable, you can still failover to a surviving managed cluster.

(DFBUGS-376)

MCO operator reconciles the **`veleroNamespaceSecretKeyRef`** and **`CACertificates`** fields

When the Fusion Data Foundation operator is upgraded, the **`CACertificates`** and **`veleroNamespaceSecretKeyRef`** fields under **`s3StoreProfiles`** in the Ramen config are lost.

Workaround:

If the Ramen config has the custom values for the **`CACertificates`** and **`veleroNamespaceSecretKeyRef`** fields, then set those custom values after the upgrade is performed.

(DFBUGS-440)

For discovered apps with CephFS, sync stop after failover

For CephFS-based workloads, synchronization of discovered applications may stop at some point after a failover or relocation. This can occur with a **`Permission Denied`** error reported in the **`ReplicationSource`** status.

Workaround:

- **For Non-Discovered Applications**
  1. Delete the VolumeSnapshot.

     ```
     oc delete volumesnapshot -n <vrg-namespace> <volumesnapshot-name>
     ```

     The snapshot name usually starts with the PVC name followed by a timestamp.

2. Delete the VolSync Job.

```
oc delete job -n <vrg-namespace> <pvc-name>
```

The job name matches the PVC name.

- **For Discovered Applications**

Use the same steps as above, except `<namespace>` refers to the **application workload namespace**, not the VRG namespace.

- **For Workloads using Consistency Groups**
  - Delete the ReplicationGroupSource:

```
oc delete replicationgroupsource -n
<namespace> <name>
```

  - Delete all VolSync jobs in that namespace:

```
oc delete jobs --all -n
<namespace>
```

In this case, `<namespace>` refers to the namespace of the workload (either discovered or not), and `<name>` refers to the name of the ReplicationGroupSource resource.

(DFBUGS-2883)

Remove DR option is not available for discovered apps on the Virtual machines page

The Remove DR option is not available for discovered applications listed on the Virtual machines page.

Workaround:

1. Add the missing label to the DRPlacementControl.

```
{{oc label drplacementcontrol <drpcname> \
odf.console.selector/resourcetype=virtualmachine \
-n openshift-dr-ops}}
```

2. Add the `PROTECTED_VMS` recipe parameter with the virtual machine name as its value.

```
{{oc patch drplacementcontrol <drpcname> \
-n openshift-dr-ops \
--type='merge' \
-p '{"spec":{"kubeObjectProtection":{"recipeParameters":{"PROTECTED_VMS":["<vm-name>"]}}}}'}}
```

(DFBUGS-2823)

DR Status is not displayed for discovered apps on the Virtual machines page

DR Status is not displayed for discovered applications listed on the Virtual machines page.

Workaround:

1. Add the missing label to the DRPlacementControl.

```
{{oc label drplacementcontrol <drpcname> \
odf.console.selector/resourcetype=virtualmachine \
-n openshift-dr-ops}}
```

2. Add the `PROTECTED_VMS` recipe parameter with the virtual machine name as its value.

```
{{oc patch drplacementcontrol <drpcname> \
-n openshift-dr-ops \
--type='merge' \
-p '{"spec":{"kubeObjectProtection":{"recipeParameters":{"PROTECTED_VMS":["<vm-name>"]}}}}'}}
```

(DFBUGS-2822)

Secondary PVCs aren’t removed when DR protection is removed for discovered apps

On the secondary cluster, CephFS PVCs linked to a workload are usually managed by the VolumeReplicationGroup (VRG). However, when a workload is discovered using the Discovered Applications feature, the associated CephFS PVCs are not marked as VRG-owned. As a result, when the workload is disabled, these PVCs are not automatically cleaned up and become orphaned.

Workaround:

To clean up the orphaned CephFS PVCs after disabling DR protection for a discovered workload, manually delete them using the following command:

```
oc delete pvc <pvc-name> -n <pvc-namespace>
```

(DFBUGS-2827)

Multicloud Object Gateway

Unable to create new OBCs using Multicloud Object Gateway

When provisioning an NSFS bucket by using ObjectBucketClaim (OBC), the default filesystem path is expected to use the bucket name. However, if path is set in `OBC.Spec.AdditionalConfig`, it should take precedence. This behavior is currently inconsistent, resulting in failures when creating new OBCs.

(DFBUGS-3817)

Ceph

Poor CephFS performance on stretch clusters

Workloads with many small metadata operations might exhibit poor performance because of the arbitrary placement of metadata server pods (MDS) on multi-site Fusion Data Foundation clusters.

(DFBUGS-1753)

OSD pods restart during add capacity

OSD pods restart after performing cluster expansion by adding capacity to the cluster. However, no impact to the cluster is observed apart from pod restarting.

(DFBUGS-1426)

Ceph becomes inaccessible and IO is paused when connection is lost between the two data centers in stretch cluster

When two data centers lose connection with each other but are still connected to the Arbiter node, there is a flaw in the election logic that causes an infinite election among Ceph Monitors. As a result, the Monitors are unable to elect a leader and the Ceph cluster becomes unavailable. Also, IO is paused during the connection loss.

Workaround:

Shutdown the monitors of any one data zone by bringing down the zone nodes. Additionally, you can reset the connection scores of surviving Monitor pods.

As a result, Monitors can form a quorum and Ceph becomes available again and IOs resumes.

(DFBUGS-425)

SELinux relabelling issue with a very high number of files

When attaching volumes to pods in Red Hat® OpenShift® Container Platform, the pods sometimes do not start or take an excessive amount of time to start. This behavior is generic and it is tied to how SELinux relabelling is handled by Kubelet. This issue is observed with any filesystem based volumes having very high file counts. In Fusion Data Foundation, the issue is seen when using CephFS based volumes with a very high number of files. There are multiple ways to work around this issue. Depending on your business needs you can choose one of the workarounds from the knowledgebase solution https://access.redhat.com/solutions/6221251.

(RFE-3327)

CSI driver

Sync stops after PVC deselection

When a PersistentVolumeClaim (PVC) is added to or removed from a group by modifying its label to match or unmatch the group criteria, sync operations may unexpectedly stop. This occurs due to stale protected PVC entries remaining in the VolumeReplicationGroup (VRG) status.

Workaround:

Manually edit the VRG's status field to remove the stale protected PVC:

```
oc edit vrg <vrg-name> -n <vrg-namespace> --subresource=status
```

(DFBUGS-4012)

IBM Storage Scale operator

Scale Dashboard reports "Healthy" connection despite total storage I/O failure on worker nodes

The IBM Storage Scale (CNSA) operator fails to report a connection failure when the storage data path is severed between OpenShift worker nodes and the remote Scale cluster. While the data plane is functionally dead, the RemoteCluster custom resource (CR) and the Scale Dashboard UI continue to report a status of "Healthy/Available." This occurs because the health check relies solely on the reachability of the remote Scale REST API from the operator pod, ignoring the actual mount state and heartbeat status of the compute (daemon) pods on the worker nodes.

Workaround:

Perform manual verification by running `mmlscluster` from the CLI or run an `ls` command directly on the mount point to confirm the actual storage health and path status.

(DFBUGS-6498)

Fusion Data Foundation console

UI shows `WaitOnUserCleanUp` even when automatic cleanup is enabled

The UI incorrectly displays the `WaitOnUserCleanUp` status even when automatic cleanup is enabled for VMs. This occurs because the UI relies only on the `phase` and `progression` fields of the `DRPlacementControl` to determine cleanup behavior and does not evaluate the more granular `AutoCleanup` condition that explicitly indicates automatic cleanup.

Workaround:

There is no manual workaround required. This state is transient and clears automatically once the `progression` field advances to `Completed`. Manual cleanup should be avoided unless the `AutoCleanup` condition and its corresponding `reason` in the `DRPlacementControl` or VRG status indicate otherwise.

During automatic cleanup, the UI may briefly present a misleading status, which can cause temporary confusion until the cleanup completes.

(DFBUGS-5824)

DRPlacementControl shows `ProtectionError` even after successful relocation

When a relocation completes, the `DRPlacementControl` may continue to display a `ProtectionError` status. This occurs because the `Protected` condition in the DRPlacementControl status incorrectly reports an `Error` state, even though the relocation has finished (`phase: Relocated`, `progression: Completed`).

Workaround:

No direct workaround is available. Wait until retrying the `NoClusterDataConflict` condition is met.

The DR status in the UI remains in the `ProtectionError` state until the data conflict is resolved.

(DFBUGS-5823)

UI shows "Unauthorized" error and blank screen with loading temporarily during Fusion Data Foundation operator installation
During Fusion Data Foundation operator installation, sometimes the `InstallPlan` transiently goes missing which causes the page to show unknown status. This does not happen regularly. As a result, the messages and title go missing for a few seconds.

(DFBUGS-3574)

Introduction to Fusion Data Foundation

IBM Fusion Data Foundation is a highly integrated collection of cloud storage and data services for Red Hat OpenShift Container Platform. It is available as a IBM Fusion storage service to facilitate simple deployment and management.

Fusion Data Foundation services are primarily made available to applications in the form of storage classes that represent the following components:

- Block storage devices, catering primarily to database workloads, for example, Red Hat OpenShift Container Platform logging and monitoring, and PostgreSQL.
  Important: Block storage should be used for any workload only when it does not require sharing the data across multiple containers.
- Shared and distributed file system, catering primarily to software development, messaging, and data aggregation workloads, for example, Jenkins build sources and artifacts, Wordpress uploaded content, Red Hat OpenShift Container Platform registry, and messaging using JBoss AMQ.
- Multicloud object storage, featuring a lightweight S3 API endpoint that can abstract the storage and retrieval of data from multiple cloud object stores.
- On-premises object storage, featuring a robust S3 API endpoint that scales to tens of petabytes and billions of objects, primarily targeting data intensive applications, for example, the storage and access of row, columnar, and semi-structured data with applications like Spark, Presto, Red Hat AMQ Streams (Kafka), and even machine learning frameworks like TensorFlow and Pytorch.

Note:
Running PostgresSQL workload on CephFS persistent volume is not supported and it is recommended to use RADOS Block Device (RBD) volume.

Fusion Data Foundation version 4.x integrates a collection of software projects, including:

- Ceph, providing block storage, a shared and distributed file system, and on-premises object storage.
- Ceph CSI, to manage provisioning and lifecycle of persistent volumes and claims.
- NooBaa, providing a Multicloud Object Gateway (MCG).
- Fusion Data Foundation, rook-ceph, and NooBaa operators to initialize and manage Fusion Data Foundation services.

Installing Fusion Data Foundation

The Fusion Data Foundation service provides a foundational data layer for applications to function and interact with data in a simplified, consistent, and scalable manner. The provider mode is similar to an external Data Foundation configuration. It acts as the provider and base storage on the host cluster. You can provision new clusters within a managed rack and consume storage from the central provider cluster.

Before you begin

- It is recommended to install the latest version of Fusion Data Foundation and OpenShift® Container Platform. For detailed storage comparability matrix, see IBM Fusion support matrix.
- For Fusion Data Foundation 4.16, perform a fresh installation. Do not upgrade from versions 4.14 or 4.15.
- If you install Data Foundation, the Global Data Platform remote mount tile becomes available, but you must explicitly install and configure it.
  Important: If you install Fusion Data Foundation 4.21 or later and the Global Data Platform remote mount service is installed and configured, the service is automatically migrated to version 6.0.1.0. Review the supported migration scenarios before installation.
- If you install Data Foundation directly from the OpenShift Container Platform, then IBM Fusion HCI does not support its discovery.
- When LVM drives are available, IBM Fusion HCI automatically detects the presence of the control plane drives and configures them for LVM.

Procedure

1. Go to Services page in IBM Fusion user interface.
2. In the Available section, click the Data Foundation tile.
3. In the Storage providers section, click the Data Foundation tile to install and configure storage service.
   Note: You cannot install other dependent services until you install and configure a storage service.
4. In the Data Foundation page, go through the details about the service and click Install.
   The IBM Fusion HCI supports both provider mode and MCG only mode.
5. Choose the following options and click Install:
   - Automatic updates
     Note: If you select Automatic updates, then IBM Fusion auto upgrades Fusion Data Foundation version.
     This upgrades Fusion Data Foundation service within the subscription channel.

     For example, 4.16.x to 4.16.y. For storage support matrix, see IBM Fusion support matrix.

   The Fusion Data Foundation operator starts to deploy and you can see Data Foundation in the Installed section of the Services page. Initially, the status shows as Installing and progress of the installation is mentioned in percentage. After successful completion of the installation, the status changes to Running.
6. To validate the installation, do the following steps in OpenShift Container Platform web console:
   Go to Installed Operators to check whether the Fusion Data Foundation operator is listed and Status shows as Succeeded.

7. Verify the Fusion Data Foundation service installation.
   a. To verify the installation of Fusion Data Foundation on IBM Fusion, perform the checks specified in Verify the installation.
   b. Additionally, run the command to verify that Fusion Data Foundation service is installed successfully:

```
oc describe odfmanager/odfmanager -n <Fusionns>
```

Update `<Fusions>` with your namespace value.

Table 1. Install states Fusion Data Foundation service

Install State	Description
`Installing`	Fusion Data Foundation installation is ongoing, and there are no errors yet.
`Failing`	There is an installation error but IBM Fusion retries to install Fusion Data Foundation. If the problem is solved, this status changes to `Installing` or `Completed`.
`Completed`	Fusion Data Foundation installation completed successfully.

Table 2. Upgrade states Fusion Data Foundation service

Upgrade State	Description
`Not started`	Fusion Data Foundation upgrade has not started yet.
`Upgrading`	Fusion Data Foundation upgrade is ongoing, and there are no errors yet.
`Failing`	There is an installation error but IBM Fusion retries to upgrade Fusion Data Foundation. If the problem is solved, this status changes to `Upgrading` or `Completed`.
`Completed`	Fusion Data Foundation upgrade completed successfully.

Table 3. Health states Fusion Data Foundation service

State	Description
`Installing`	Fusion Data Foundation installation is ongoing.
`Upgrading`	Fusion Data Foundation upgrade is ongoing.
`Healthy`	Fusion Data Foundation installation completed successfully and the service is in normal state.
`Degraded`	Fusion Data Foundation installation completed successfully but the service is not in normal state.

8. Validate the installation.

Verify the Fusion Data Foundation installation was completed successfully from the OpenShift Container Platform console:

a. Go to Installed Operators to check whether the Fusion Data Foundation operator is listed and status is Succeeded.
b. Verify pods from the Red Hat®OpenShift Container Platform web console:
   i. Go to Workloads > Pods.
   ii. Select the `openshift-storage` namespace, where the Fusion Data Foundation is installed.
   iii. Check whether the following pods are running successfully:
      - `csi-addons-controller-manager`
      - `ocs-metrics-exporter`
      - `ocs-operator`
      - `odf-console`
      - `odf-operator-controller-manager`
      - `rook-ceph-operator`
      - `ceph-csi-controller-manager`
c. Alternatively, you can verify the installation by running the OC commands:

```
oc get pod -n openshift-storage
```

A sample result of the OC command is as follows:

```
[root@fu40 ~]# oc get pod -n openshift-storage
NAME                                              READY   STATUS              RESTARTS    AGE
csi-addons-controller-manager-bdb965b47-jvfhs     2/2     Running             0           119s
ocs-metrics-exporter-69cfc56fd9-jk49s             1/1     Running             0           2m45s
ocs-operator-6879c74556-pppj9                     1/1     Running             0           2m46s
odf-console-849f64fdd7-rqd97                      1/1     Running             0           3m3s
odf-operator-controller-manager-5bd4c85d6b-q8g2f  2/2     Running             0           3m3s
rook-ceph-operator-76699f976c-2zpms               1/1     Running             0           2m46s
```

d. Ensure that all the pods are up and running.
e. Verify the Fusion Data Foundation status as follows:
   i. Click Operators > Install operators. Select the namespace in which IBM Fusion was installed from the Project tab.
   ii. Select the IBM Fusion.
   iii. Go to Fusion Service Instance tab and click odfmanager from the list.
   iv. Open **YAML** tab and ensure that `.status.health` is Healthy.
f. Alternatively, you can verify the status by running the OC commands:

```
oc get fusionserviceinstances.service.isf.ibm.com -n ibm-spectrum-fusion-ns odfmanager -o 
jsonpath='{.status.health}{"\n"}'
```

g. Verify the local storage operator pods:
   i. Click Operators > Install operators. Select the `ibm-spectrum-fusion-ns` namespace from the Project tab.
   ii. Select the IBM Fusion.
   iii. Go to Fusion Service Instance tab and click odfmanager from the list.
   iv. Click YAML tab and check the `backingStorageType`.

```
- name: backingStorageType
  provided: true
  value: Provider
```

   v. Alternatively, you can verify the backing storage type by running the oc commands.

```
oc get fusionserviceinstances.service.isf.ibm.com -n ibm-spectrum-fusion-ns odfmanager -o 
jsonpath='{.spec.parameters}{"\n"}'
```

   vi. Verify pods from the Red HatOpenShift Container Platform web console:
      1. Go to Workloads > Pods.

2. Select the `openshift-local-storage` namespace, where the local storage operator is installed.
3. Check whether the following pods are running successfully: `local-storage-operator-xxxxx`
   For example: `local-storage-operator-75c55cf57d-pfjcp`

vii. Ensure that all pods are running.

viii. Alternatively, you can verify the installation by running the oc commands:

```
oc get pod -n openshift-local-storage
```

A sample result of the oc command is as follows:

```
[root@fu40 ~]# oc get pod -n openshift-local-storage
NAME                                     READY   STATUS    RESTARTS   AGE
local-storage-operator-75c55cf57d-pfjcp  1/1     Running   0          109s
```

What to do next

Configure local storage. For the procedure to configure, see Configuring Fusion Data Foundation.

- **Fusion Data Foundation storage compatibility across rack configurations**
  Describes the supported Fusion Data Foundation storage configurations across single-rack, multi-rack, and expansion-rack deployments in IBM Fusion HCI systems.
- **Installing a Fusion Data Foundation client**
  Install a Fusion Data Foundation client on Red Hat OpenShift Container Platform to enable the cluster to access storage from a Fusion Data Foundation provider cluster without managing local storage.

Fusion Data Foundation storage compatibility across rack configurations

Describes the supported Fusion Data Foundation storage configurations across single-rack, multi-rack, and expansion-rack deployments in IBM Fusion HCI systems.

Table 1. Fusion Data Foundation storage compatibility across IBM Fusion HCI rack configurations

Deployment topology	Fusion Data Foundation provider mode	Fusion Data Foundation Provider mode failure domain tolerance	Disk upsize	Node Upsize	Node configuration
Standalone rack	Supported	Node	Supported	Supported	• A minimum of 3 storage nodes is required. • Disk size and type must be the same.
HA rack and 3 zone HA rack	Supported	Rack	Supported	Supported	• A minimum of 3 storage nodes is required in each rack. • All racks must maintain consistency in the number of nodes, disk sizes, and disk types. This uniformity requirement extends to all node or disk upsizing operations.
2 availability zone (AZ) + Arbiter node rack	Supported	Rack	Supported	Supported	• A minimum of 3 storage nodes is required in each rack. • Both racks must maintain consistency in the number of nodes, disk sizes, and disk types. This uniformity requirement extends to all node or disk upsizing operations.
Expansion rack	Supported	Node	Supported	Supported	• A minimum of 3 storage nodes is required in the base rack. • Disk size and type must be the same.

For more information about multi-rack configuration, see Deployment configurations.

Installing a Fusion Data Foundation client

Install a Fusion Data Foundation client on Red Hat® OpenShift® Container Platform to enable the cluster to access storage from a Fusion Data Foundation provider cluster without managing local storage.

Before you begin

Before you begin, ensure that the following conditions are met:

- Provider and client clusters are running on Red Hat OpenShift Container Platform.
- The provider and client clusters are in the same network and can communicate with each other.
- The round-trip network latency between the two sites does not exceed 10 ms (RTT).
- Fusion Data Foundation is installed on the provider cluster.
- You have cluster administrator permissions.
- The OpenShift CLI (`oc`) is installed and configured.
- A mix of OpenShift Data Foundation and Fusion Data Foundation client-provider configurations is not supported.

About this task

Fusion Data Foundation supports a provider–consumer architecture in which storage resources are managed on a provider cluster and consumed by one or more client clusters. In this configuration, the client cluster does not provision or manage local storage. Instead, it consumes storage exposed by the provider cluster.

In this model:

- The provider cluster hosts and exposes storage services.
- The client cluster connects to the provider cluster and consumes its storage.

This approach is useful when you want to:

- Centralize storage management in a provider cluster
- Enable multiple clusters to share centralized storage, isolated for each connected client
- Reduce infrastructure overhead on client clusters

Follow this procedure when:

- An OpenShift Container Platform cluster act as a Fusion Data Foundation client
- Storage is centralized and managed on a separate provider cluster
- The client cluster must consume storage without deploying or managing local storage

This task includes configuration steps on both the provider and the client clusters.

Procedure

1. Configure the provider cluster for external client access.
   On the provider cluster, set up MetalLB, expose the provider endpoint, and generate the onboarding token. For instructions, see Configuring the provider cluster for external client access.

2. Configure the client cluster to access provider storage.
   Install the Fusion Data Foundation client and configure access to the provider storage. For instructions, see Configuring the client cluster to access provider storage.

Results

- The client cluster is successfully connected to the provider cluster.
- Fusion Data Foundation storage classes are available to the client cluster.
- The provider cluster lists the client cluster as a registered storage consumer.

- **Configuring the provider cluster for external client access**
  Configure the provider cluster to expose storage services and generate an onboarding token for client access.
- **Configuring the client cluster to access provider storage**
  Install and configure the Fusion Data Foundation client on the client cluster to access storage from the provider cluster.

Configuring the provider cluster for external client access

Configure the provider cluster to expose storage services and generate an onboarding token for client access.

Procedure

1. Install the MetalLB operator.
   To install the MetalLB operator, complete the following steps:
   a. Log in to the Red Hat® OpenShift® Container Platform web console.
   b. Go to the Ecosystem > Software Catalog page.
   c. Scroll or type MetalLB into the Filter by keyword box.
   d. Select MetalLB Operator from the results.
   e. Click Install.
   f. On the Install Operator page, complete the following configuration:
      i. Select stable from the Update channel list.
      ii. Select the required version from the Version list.
      iii. For Installation mode, select All namespaces on the cluster (default).
      iv. For Installed Namespace, select Operator recommended Namespace.
      v. For Update approval, select either Automatic (recommended) or Manual.
   g. Click Install.
   h. Go to the Installed Operators page and confirm that:
      - The MetalLB Operator appears in the list
      - Status is Succeeded
2. Create a MetalLB instance.
   Apply the following configuration:

```
apiVersion: metallb.io/v1beta1
kind: MetalLB
metadata:
  name: metallb
  namespace: metallb-system
```

3. Configure an IP address pool.
   Define a range of IP addresses to be assigned by MetalLB:

```
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: metallb
  namespace: metallb-system
spec:
  addresses:
  - 10.0.0.138-10.0.0.142
```

4. Create an L2 advertisement.
   Apply the following configuration:

```
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: l2advertisement
  namespace: metallb-system
spec:
  ipAddressPools:
  - metallb
```

5. Create a LoadBalancer service.
   Expose the provider API using a LoadBalancer:

```
apiVersion: v1
kind: Service
metadata:
  name: ocs-provider-server-load-balancer
  namespace: openshift-storage
  annotations:
    metallb.universe.tf/ip-allocated-from-pool: metallb
spec:
  type: LoadBalancer
  selector:
    app: ocsProviderApiServer
  ports:
    - name: provider
      protocol: TCP
      port: 50051
      targetPort: ocs-provider
      nodePort: 30756
```

   Important:
   Record the LoadBalancer IP address and port. For example: `storageProviderEndpoint: 10.0.0.138:50051`. You need this information when onboarding the client cluster.

6. Create a storage consumer instance.
   To create a storage consumer instance, complete the following steps:
   a. On the Red Hat OpenShift Container Platform web console, navigate to the Storage > Data Foundation > Storage Consumers page.
   b. Click Create Storage Consumer.
   c. Provide the required details, such as name and quota.
   d. Click Create.
7. Generate the onboarding token.
   To generate the onboarding token of the created storage consumer instance, complete the following steps:
   a. Click the overflow menu ⋮ icon for the new storage consumer instance.
   b. Select Generate client onboarding token.
   c. Copy the onboarding token for use on the client cluster.

Results

The provider cluster is configured, and the onboarding token and LoadBalancer endpoint are available for onboarding the client cluster.

Configuring the client cluster to access provider storage

Install and configure the Fusion Data Foundation client on the client cluster to access storage from the provider cluster.

Procedure

1. Install IBM Fusion on the client cluster.
   To install IBM Fusion on the client cluster, use the Red Hat® OpenShift® Container Platform web console. For example steps, see Deploying IBM Fusion in the *IBM Fusion* documentation. Make sure to use the documentation version that aligns with your setup.
2. Create the Fusion Data Foundation catalog if it is not already available.
   Apply the following configuration:

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: isf-data-foundation-catalog
  namespace: openshift-marketplace
spec:
  displayName: Data Foundation Catalog
  image: icr.io/cpopen/isf-data-foundation-catalog:v4.21
  publisher: IBM
  sourceType: grpc
```

```
  registryPoll:
    interval: 60m
```

Important:
The version **v4.21** in the **image** field is an example. Replace it with the appropriate version according to your deployment requirements.

The Fusion Data Foundation versions on the provider and client clusters must match the same minor version for onboarding to succeed. For example, a provider running version **4.21.z** supports onboarding only for clients running version **4.21.*** (any patch version). These are Fusion Data Foundation versions, not Red Hat OpenShift Container Platform versions.

3. Install the Fusion Data Foundation client operator.
   To install the Fusion Data Foundation client operator, complete the following steps:
   a. On the Red Hat OpenShift Container Platform web console, navigate to the Ecosystem > Software Catalog page.
   b. Scroll or type **Fusion Data Foundation** into the Filter by keyword box.
   c. Select **IBM FusionData Foundation Client** from the results.
   d. Select stable-X.XX from the Channel list.
   e. Select the required version from the Version list.
   f. Click Install.
   g. Go to the Installed Operators page and confirm that:
      - The **IBM FusionData Foundation Client** appears in the list
      - Status is Succeeded
4. Verify that all pods are running in the `openshift-storage` namespace.
   Command example:

   ```
   oc get pods -n openshift-storage
   ```

5. Create a storage client.
   To create a storage client, complete the following steps:
   a. Go to the Ecosystem > Installed Operators page.
   b. Select the installed Fusion Data Foundation client operator.
   c. Go to the Storage Client tab and click Create StorageClient.
   d. Provide the following values:
      - Name: Specify a name.
      - Data Foundation endpoint: Enter the LoadBalancer endpoint from the provider cluster.
      - Onbarding token: Paste the token generated on the provider cluster.
   e. Click Create.
6. Verify the following:
   a. On the Storage Client tab, storage client status is Connected.
   b. Go to Storage > StorageClasses, and verify that storage classes from the provider cluster are available.
   c. On the provider cluster, navigate to Storage > Data Foundation > Storage Consumers, and verify that the cluster ID, last heartbeat, and other fields are populated for the consumer.

Results

The client cluster is connected to the provider cluster and can use the provider storage.

Installing Data Foundation for MCG only

The IBM Fusion HCI supports MCG only mode.

Before you begin

Label at least two nodes:

1. Validate whether the nodes have necessary `cluster.ocs.openshift.io/openshift-storage` label:

   ```
   oc get node -l cluster.ocs.openshift.io/openshift-storage=
   ```

2. If `No resources found` gets displayed, then it means that no node exists with the label. Run the following command to label at least two worker nodes for ensuring High Availability:

   ```
   oc label node <worker-node-1> <worker-node-2> cluster.ocs.openshift.io/openshift-storage=''
   ```

3. Validate label is applied to the nodes:

   ```
   oc get node -l cluster.ocs.openshift.io/openshift-storage=
   ```

   Example output:

   ```
   NAME             STATUS   ROLES    AGE   VERSION
   <worker-node-1>  Ready    worker   28d   v1.28.12+396c881
   <worker-node-2>  Ready    worker   28d   v1.28.12+396c881
   ```

About this task

You must have a local Global Data Platform storage installed and in a running state. Otherwise, you cannot see the Data Foundation MCG Only tile for installation.

Multi-Cloud Gateway (MCG) in Global Data Platform is an on-prem deployment. If the Global Data Platform ECE is installed, then you can see the MCG tile for it.

Procedure

1. Go to Services page in IBM Fusion user interface.
2. In the Available section, click the Data Foundation MCG Only tile to install and configure storage service.
3. In the Data Foundation page, go through the details about the service and click Install.
   The IBM Fusion HCI supports both provider and MCG only modes.
   The Fusion Data Foundation operator starts to deploy and you can see Data Foundation MCG Only in the Installed section of the Services page. Initially, the status shows as Installing and the progress of the installation is mentioned in percentage. After installation completes successfully, the status changes to Running.
4. To verify, view Multi Cloud gateway in OpenShift®.
   For more information about Multi Cloud gateway in OpenShift, see [Managing hybrid and multicloud resource](#).

Configuring Fusion Data Foundation

Configure Fusion Data Foundation to set up storage services that support scalable, resilient, and flexible data management for OpenShift® applications.

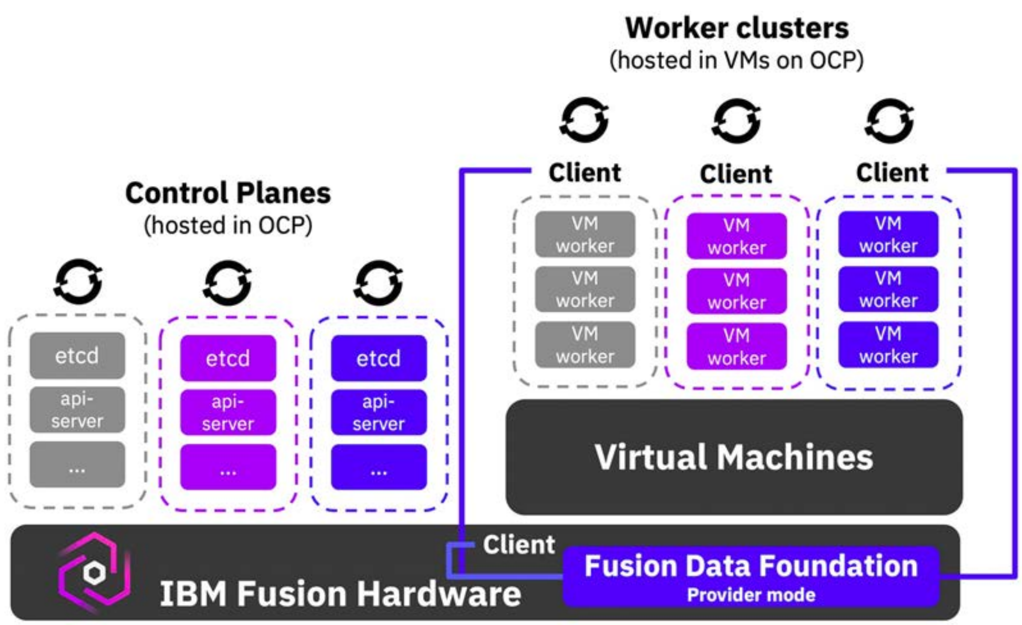
After the Fusion Data Foundation installation is complete, configure the storage system based on the device type selected during Fusion Data Foundation installation.

Fusion Data Foundation supports the following configuration mode:

Provider
: In provider mode, you can run a local Fusion Data Foundation storage cluster that serves as the central storage provider for multiple consumer clusters. It functions similarly to an external Fusion Data Foundation configuration, where the provider cluster acts as the provider and base storage on the host cluster. You can provision new clusters within a managed rack and have them consume storage from the central provider cluster. This mode is available from IBM Fusion HCI 2.8.1 onward with Fusion Data Foundation 4.16. For instructions, see [Configuring Fusion Data Foundation in provider mode](#).

In this mode, Fusion Data Foundation is layered on Red Hat OpenShift Service on bare metal and includes the Fusion Data Foundation provider. IBM Fusion HCI with Fusion Data Foundation supplies the base storage for all Hosted Control Plane-based clusters, as shown in the following figure:

Figure 1. Fusion Data Foundation provider mode



Fusion Data Foundation provider mode is a deployment configuration in which the OpenShift cluster running on bare metal serves local storage for OpenShift virtualization, standard applications, and hosted clusters that are running on top of OpenShift virtualization. Provider mode consists of two key components:

- **Provider** - The provider is a deployment mode for Fusion Data Foundation storage services, built on a minimum of three worker nodes using bare metal volumes. This mode allows external clusters to connect through the client operator. Fusion Data Foundation replicates data across these nodes, ensuring resilience and tolerance to node failures.
- **Client** - The client is a managed Container Storage Interface (CSI) deployment that connects to the provider to deliver a local storage class for hosted clusters.

For instructions on Fusion Data Foundation installation, see [Installing Fusion Data Foundation](#).

After completing the configuration, you can monitor capacity utilization and storage nodes, add nodes, and scale capacity as needed.

- **[Configuring Fusion Data Foundation in provider mode](#)**
  Configure Fusion Data Foundation storage in IBM Fusion HCI for provider mode.

Configuring Fusion Data Foundation in provider mode

Configure Fusion Data Foundation storage in IBM Fusion HCI for provider mode.

About this task

The provider mode is similar to an external Data Foundation configuration. It acts as the provider and base storage on host cluster. You can provision new clusters within a managed rack and consume storage from the central provider cluster. Hosting multiple OpenShift® clusters within a single HCI rack improves cluster density and brings down cost.
The consumer can be deployed on the IBM Fusion HCI host cluster.

Important: If Fusion Data Foundation is not configured or is configured but are not in a healthy state, then a few pods, like logcollector, might remain in a pending state. The pods come into a running state automatically once storage is configured and in a healthy state.

Procedure

1. Go to the Storage > Local storage page.
2. Click Configure storage.
3. In the Configure storage page, do the following configuration:
    a. Under Stoarge optimization section, select one of the following options based on your workload requirements:
        - Performance: Optimizes for high-speed operations and fast recovery for performance-critical workloads.

          Tip: This option is designed for workloads that require maximum performance and fast rebuild times. It provides higher read and write performance and faster rebuild and recovery times, but results in lower usable capacity. This option is ideal for databases, virtual machines, and applications that require low latency.
        - Storage efficiency: Optimizes for strong data protection with maximum usable capacity.

          Tip: This option uses erasure coding to significantly increase usable storage capacity compared to replication while maintaining strong data protection. Depending on the erasure coding profile, it provides approximately 67% usable capacity for 4+2 profile and 80% usable capacity for 8+2 profile, compared to approximately 33% usable capacity with triple replication. Rebuild times can be longer than the Performance option. This option is ideal for archival storage, backups, and workloads where capacity efficiency is paramount.
          The Storage efficiency option is based on erasure coding, which improves usable capacity while maintaining strong data protection. Erasure coding splits data into k data chunks and m parity chunks, which are distributed across storage nodes. To support this configuration, the environment must have at least k + m nodes.

          Examples:
            - **4 data chunks + 2 parity chunks (4+2) profile:**
                - Minimum nodes: 6
                - Usable capacity: ~67% (2/3 of raw capacity)
                - Overhead: 1.5×
            - **8 data chunks + 2 parity chunks (8+2) profile:**
                - Minimum nodes: 10
                - Usable capacity: ~80% of raw capacity)
                - Overhead: 1.25×

          Important:
            - Erasure coding is supported starting with Red Hat® OpenShift Container Platform 4.21 or later and Fusion Data Foundation 4.21 or later.
            - This feature is supported only for Block storage (RBD).
            - Clusters upgraded from earlier releases do not support erasure coding.

    b. In the Nodes section, select a minimum of six nodes from the node table based on the recommendation.
       By default, all available nodes are selected. If required, deselect nodes that you want to exclude from storage by clearing the corresponding check boxes.

       The listed nodes are candidate nodes for selection as Fusion Data Foundation storage nodes. Both compute and control plane nodes that have available SSD or NVMe disks are displayed as candidate nodes in the table.

       The node table includes the following details:
        - Name
        - Disks
        - Disk size (TiB)
        - vCPUs
        - Memory (GB)

       For High-Availability Multi-Rack configurations, the table also includes a Rack column.

       When you configure storage for a High-Availability Multi-Rack configuration, select the same number of nodes from each rack. This configuration ensures that the Fusion Data Foundation cluster continues to operate even if one rack becomes unavailable.

       For a two-availability-zone and arbiter node (2AZ + arbiter node) configuration:
        - Select the same number of nodes from each zone.
        - Select a minimum of two storage nodes from each zone.

       Note: The Next button remains disabled if an uneven number of nodes are selected across zones or if fewer than two nodes are selected from each zone.
       Based on your selection, the Summary section displays the usable capacity. The usable capacity includes raw capacity, number of nodes, CPUs, and memory.
       For example, the recommendation might be to select a minimum of six nodes with a usable capacity of 48 TiB.
    c. Click Next.
    d. In the Encryption settings section, select the location for the encryption key:
        - Store the encryption key in a secret in the cluster: Stores the encryption key locally within the cluster as a Kubernetes secret.
        - Store the encryption key in an external KMS: Stores the encryption key in an external Key Management System for enhanced security.
        - None: Disables encryption for the storage configuration.
4. Click Configure.
   The Data Foundation page displays Usable capacity, Health, and Storage nodes. After the configuration is complete, the Health section displays the health of the Storage cluster and Data resiliency.
   Note: Setting up local storage may take around 45-50 minutes.
5. To validate the configuration, do the following steps:
    a. Log in to OpenShift Container Platform console.

b. In the menu, go to Storage > StorageClasses.
c. In the StorageClasses page, check whether you see the following storage classes:
   - `ocs-storagecluster-cephfs` for Fusion Data Foundation storage file system
   - `ocs-storagecluster-ceph-rbd` for Ceph's RADOS Block Devices (RBD) storage class. It is the default storage class.
   - `openshift-storage.noobaa.io` storage class is for object storage. You can use it for MCG so that it gets generated either for the installation of Fusion Data Foundation or for Global Data Platform with Fusion Data Foundation MCG only mode.

What to do next

1. Create a `NetworkFenceClass` CR to fence and taint nodes, preventing volume corruption during non-graceful node shutdowns. For instructions, see Creating a NetworkFenceClass CR.
2. After the configuration is complete, you can view capacity utilization and storage nodes. You can add nodes to your storage configuration and scale its capacities. For instructions, see Scaling the cluster.

- **Creating a NetworkFenceClass CR**
  Create a `NetworkFenceClass` CR to fence and taint nodes during non-graceful node shutdowns to prevent volume corruption.

Creating a NetworkFenceClass CR

Create a `NetworkFenceClass` CR to fence and taint nodes during non-graceful node shutdowns to prevent volume corruption.

About this task

Note: Create one or more `NetworkFenceClass` CRs, depending on the number of storage clusters connected to the OpenShift® Container Platform cluster.

Procedure

1. Define a YAML file for the `NetworkFenceClass` CR and save it.
   For example:

```
apiVersion: csiaddons.openshift.io/v1alpha1
kind: NetworkFenceClass
metadata:
  name: odf-networkfenceclass
spec:
  provisioner: openshift-storage.rbd.csi.ceph.com
  parameters:
    clusterID: openshift-storage
    csiaddons.openshift.io/networkfence-secret-name: rook-csi-rbd-node
    csiaddons.openshift.io/networkfence-secret-namespace: openshift-storage
```

   Where,
   - `provisioner`: Specifies the name of the storage provisioner.
   - `parameters`: specifies storage provider-specific parameters.
   - `csiaddons.openshift.io/networkfence-secret-name`: specifies the name of the secret that is required for network fencing. Fetch it from `parameters.csi.storage.k8s.io/node-stage-secret-name` field in the `ocs-storagecluster-ceph-rbd` StorageClass.
   - `csiaddons.openshift.io/networkfence-secret-namespace`: specifies the namespace in which the secret is located. Fetch it from `parameters.csi.storage.k8s.io/node-stage-secret-namespace` field in the `ocs-storagecluster-ceph-rbd` StorageClass.
2. Verify the `csiaddonsnode` object details.
   The `csiaddonsnode` object belongs to the daemonset pod (RBD) that will have the IP address that needs to be fenced.
   Command example:

```
oc get csiaddonsnode -n openshift-storage
```

Preparing to connect to an external KMS server in Fusion Data Foundation

Procedure to prepare for the connection to an external KMS from Fusion Data Foundation.

Before you begin

- For external Key Management System (KMS), choose either HashiCorp Vault or Thales CipherTrust Manager.
- You must install Fusion Data Foundation from the Services page of the user interface and ensure that it is in running state.
  For the installation options and their procedures, see Installing Fusion Data Foundation.
- For HashiCorp Vault, select an unique path name as the backend path that follows the naming convention. If you change this path name later, then the data becomes inaccessible.
- For Thales CipherTrust Manager, enable the Key Management Interoperability Protocol.
- Ensure that you are using signed certificates on your KMS servers.

About this task

Fusion Data Foundation supports cluster-wide encryption (encryption-at-rest) for all the disks and Multicloud Object Gateway operations in the storage cluster. The keys are stored using a Kubernetes secret or an external KMS. When you store the keys by using a Kubernetes secret, no need for you to do manual steps. You can enable cluster-wide encryption when you deploy Fusion Data Foundation.

This procedure provides steps (manual part) to initialize encryption configuration with KMS before you enable encryption. For HashiCorp Vault, you can choose either the token authentication method or the Kubernetes authentication method.

For additional reference, see Red Hat Data Foundation Data encryption options.

- **Enabling encryption with the token authentication using HashiCorp Vault(manual part)**
  Configure token settings in vault server.
- **Enabling encryption with the Kubernetes authentication using HashiCorp Vault (manual part)**
  Configure Kubernetes settings in vault server.
- **Enabling encryption using Thales CipherTrust Manager (manual part)**
  Configure the Key Management Interoperability Protocol (KMIP) settings in Thales CipherTrust Manager server.

Enabling encryption with the token authentication using HashiCorp Vault(manual part)

Configure token settings in vault server.

Procedure

1. Enable the Key/Value (KV) backend path in Vault. Run the following command for Vault KV secret engine API, version 2, as it supports only version 2.

   ```
   vault secrets enable -path=odf kv-v2
   ```

   This is a one time settings. You can get the token from vault administrator directly, or you can login vault server and create a token with following steps. Use an unique path name as the backend path that follows the naming convention. You cannot change it later.
   Note: The example uses backend path **odf**.
2. Create a policy with certain permission on the secret using the following commands.

   ```
   echo '
   path "odf/*" {
     capabilities = ["create", "read", "update", "delete", "list"]
   }
   path "sys/mounts" {
   capabilities = ["read"]
   }'| vault policy write odf -
   ```

3. Create a token matching the policy.

   ```
   vault token create -policy=odf -format json
   ```

   Example output:

   Take a note for the `auth.client_token`

   ```
   # vault token create -policy=odf -format json
   {
   "request_id": "e7103f23-c67c-2412-77b0-b92a728fd5fd",
   "lease_id": "",
   "lease_duration": 0,
   "renewable": false,
   "data": null,
   "warnings": null,
   "auth": {
       "client_token": "hvs.CAESILkFSEnBgMgE0JxdbnBf_VUPxlvhCPTBBdx30jlbMfiMGh4KHGh2cy5QeFZVWnhOWGMxdlhvSU5YU0tNRjBBVGo",
       "accessor": "IFNbUOwWAzg5cyiUZGVlvf5o",
       "policies": [
       "default",
       "odf"
       ],
       "token_policies": [
       "default",
       "odf"
       ],
       "identity_policies": null,
       "metadata": null,
       "orphan": false,
       "entity_id": "",
       "lease_duration": 2764800,
       "renewable": true,
       "mfa_requirement": null
   }
   }
   ```

Enabling encryption with the Kubernetes authentication using HashiCorp Vault (manual part)

Configure Kubernetes settings in vault server.

Procedure

1. Get the `VAULT_SA_SECRET_NAME` in OpenShift® Container Platform:
   a. For OpenShift Container Platform 4.14, identify the secret name associated with the `serviceaccount` (SA) created previously.

   ```
   VAULT_SA_SECRET_NAME=$(oc -n openshift-storage get sa odf-vault-auth -o jsonpath="{.secrets[*]['name']}" | grep -o "[^[:space:]]*-token-[^[:space:]]*")
   ```

   Example output:

   ```
   [root@fu40 ~]# echo $VAULT_SA_SECRET_NAME
   odf-vault-auth-token-8kb2r
   ```

   b. For OpenShift Container Platform 4.15, assign a default value for `VAULT_SA_SECRET_NAME`"

   ```
   VAULT_SA_SECRET_NAME=odf-vault-auth-token
   ```

2. Get the parameters from OpenShift Container Platform cluster.
   a. Get `SA_JWT_TOKEN` and `SA_CA_CRT` from the secret.

   ```
   SA_JWT_TOKEN=$(oc -n openshift-storage get secret "$VAULT_SA_SECRET_NAME" -o jsonpath="{.data.token}" | base64 --decode; echo)
   ```

   b. Get `SA_CA_CRT` from the secret.

   ```
   SA_CA_CRT=$(oc -n openshift-storage get secret "$VAULT_SA_SECRET_NAME" -o jsonpath="{.data['ca\.crt']}" | base64 --decode; echo)
   ```

   c. Retrieve the OpenShift Container Platform cluster endpoint `OCP_HOST`.

   ```
   OCP_HOST=$(oc config view --minify --flatten -o jsonpath="{.clusters[0].cluster.server}")
   ```

   d. Fetch the service account issuer:

   ```
   oc proxy &
   proxy_pid=$!
   issuer="$( curl --silent http://127.0.0.1:8001/.well-known/openid-configuration | jq -r .issuer)"
   kill $proxy_pid
   ```

   Example output:

   ```
   # oc proxy &
   # proxy_pid=$!
   # issuer="$( curl --silent http://127.0.0.1:8001/.well-known/openid-configuration | jq -r .issuer)"
   # echo $issuer
   https://kubernetes.default.svc
   #     kill $proxy_pid
   ```

3. Apply the settings in the vault server.
   a. Use the information collected in the previous section to set up the Kubernetes authentication method in the vault server.

   ```
   vault auth enable kubernetes

   vault write auth/kubernetes/config \
             token_reviewer_jwt="$SA_JWT_TOKEN" \
             kubernetes_host="$OCP_HOST" \
             kubernetes_ca_cert="$SA_CA_CRT" \
             issuer="$issuer"
   ```

   Example output:

   ```
   # vault write auth/kubernetes/config \
   >           token_reviewer_jwt="$SA_JWT_TOKEN" \
   >           kubernetes_host="$OCP_HOST" \
   >           kubernetes_ca_cert="$SA_CA_CRT" \
   >           issuer="$issuer"
   Success! Data written to: auth/kubernetes/config
   ```

   b. Enable the Key/Value (KV) backend path in Vault. For the vault KV secret engine API, see version 2. Use a unique path name as the backend path. The name should not be changed, and it should be consistent with the value of Backend Path input that present in the user interface.

   ```
   vault secrets enable -path=<repalce-with-the-backend-path> kv-v2
   ```

   The example sample will use backend path odf.

   ```
   vault secrets enable -path=odf kv-v2
   ```

   c. Create a policy to restrict users to perform a write or delete operation on the secret using the following commands.

   ```
   echo '
   path "odf/*" {
     capabilities = ["create", "read", "update", "delete", "list"]
   }
   path "sys/mounts" {
   capabilities = ["read"]
   }'| vault policy write odf -
   ```

   d. Generate the `odf-rook-ceph-op` role using the following commands.

   ```
   vault write auth/kubernetes/role/odf-rook-ceph-op \
           bound_service_account_names=rook-ceph-system,rook-ceph-osd,noobaa \
           bound_service_account_namespaces=openshift-storage \
           policies=odf \
           ttl=1440h
   ```

   Example output:

```
#     vault write auth/kubernetes/role/odf-rook-ceph-op \
>             bound_service_account_names=rook-ceph-system,rook-ceph-osd,noobaa \
>             bound_service_account_namespaces=openshift-storage \
>             policies=odf \
>             ttl=1440h
Success! Data written to: auth/kubernetes/role/odf-rook-ceph-op
```

The role `odf-rook-ceph-op` is previously input in the role field used when you configure the KMS connection details in the IBM Fusion user interface.

e. Generate the `odf-rook-ceph-osd` role using the following commands.

```
vault write auth/kubernetes/role/odf-rook-ceph-osd \
        bound_service_account_names=rook-ceph-osd \
        bound_service_account_namespaces=openshift-storage \
        policies=odf \
        ttl=1440h
```

Example output:

```
#  vault write auth/kubernetes/role/odf-rook-ceph-osd \
>         bound_service_account_names=rook-ceph-osd \
>         bound_service_account_namespaces=openshift-storage \
>         policies=odf \
>         ttl=1440h
Success! Data written to: auth/kubernetes/role/odf-rook-ceph-osd
```

- **Enabling key rotation for Kubernetes authentication**
  Enabling key rotation for Kubernetes authentication helps to achieve the security common practices that require periodic encryption key rotation.
- **Disabling key rotation for Kubernetes authentication**
  Key rotation for Kubernetes persistent volume claims (PVCs) can be disabled at the storage class level or for a specific PVC by updating the relevant annotations and patch settings.

Enabling key rotation for Kubernetes authentication

Enabling key rotation for Kubernetes authentication helps to achieve the security common practices that require periodic encryption key rotation.

To enable key rotation, add the annotation *keyrotation.csiaddons.openshift.io/schedule: <value>* to either Namespace, StorageClass, or PersistentVolumeClaim in order of precedence.

Set the *<value>* either *@hourly*, *@daily*, *@weekly*, *@monthly*, or *@yearly* based on your requirement. If the *<value>* is empty, *@weekly* is set by default.

Important: Only RADOS Block Device (RBD)-related volumes support key rotation.
Examples of enabling key rotation with the value *@weekly* are as follows:

Namespace

For example, run the following command to get the *default* namespace details:

```
oc get namespace default
```

Example output:

```
NAME      STATUS   AGE
default   Active   5d2h
```

Run the following command to enable the key rotation to the default namespace:

```
$ oc annotate namespace default "keyrotation.csiaddons.openshift.io/schedule=@weekly"
```

Example output:

```
namespace/default annotated
```

StorageClass

For example, run the following command to get the *rbd-sc* StorageClass details:

```
oc get storageclass rbd-sc
```

Example output:

```
NAME       PROVISIONER         RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
rbd-sc     rbd.csi.ceph.com    Delete          Immediate           true                   5d2h
```

Run the following command to enable the key rotation to the *rbd-sc* StorageClass:

```
oc annotate storageclass rbd-sc "keyrotation.csiaddons.openshift.io/schedule=@weekly"
```

Example output:

```
storageclass.storage.k8s.io/rbd-sc annotated
```

PersistentVolumeClaim

For example, run the following command to get the *data-pvc* PersistentVolumeClaim details:

```
oc get pvc data-pvc
```

Example output:

```
NAME       STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS      AGE
data-pvc   Bound    pvc-f37b8582-4b04-4676-88dd-e1b95c6abf74   1Gi        RWO            default           20h
```

Run the following command to enable the key rotation to the *data-pvc* PersistentVolumeClaim:

```
oc annotate pvc data-pvc "keyrotation.csiaddons.openshift.io/schedule=@weekly"
```

Example output:

```
persistentvolumeclaim/data-pvc annotated
```

Replacing key rotation schedule

To replace the current key rotation schedule with a new one, follow these steps:

1. Run the following command to get the existing schedule details:

   ```
   oc get encryptionkeyrotationcronjobs.csiaddons.openshift.io
   ```

   Example output:

   ```
   NAME                     SCHEDULE    SUSPEND   ACTIVE   LASTSCHEDULE   AGE
   data-pvc-1642663516      @weekly                                       3s
   ```

2. Run the following command to overwrite the existing schedule with a new one:

   ```
   oc annotate pvc data-pvc "keyrotation.csiaddons.openshift.io/schedule=*/1 * * * *" --overwrite=true$
   ```

   Example output:

   ```
   persistentvolumeclaim/data-pvc annotated
   ```

3. Run the following command to check whether the new schedule is set:

   ```
   oc get encryptionkeyrotationcronjobs.csiaddons.openshift.io
   ```

   Example output:

   ```
   NAME                    SCHEDULE    SUSPEND   ACTIVE   LASTSCHEDULE   AGE
   data-pvc-1642664617     */1 * * * *                                  3s
   ```

Disabling key rotation for Kubernetes authentication

Key rotation for Kubernetes persistent volume claims (PVCs) can be disabled at the storage class level or for a specific PVC by updating the relevant annotations and patch settings.

You can disable key rotation for the following:

- All the persistent volume claims (PVCs) of storage class
- A specific PVC

Disabling key rotation for all PVCs of a storage class

To disable key rotation for all PVCs, update the annotation of the storage class:

```
oc get storageclass rbd-sc
NAME       PROVISIONER        RECLAIMPOLICY   VOLUMEBINDINGMODE   ALLOWVOLUMEEXPANSION   AGE
rbd-sc     rbd.csi.ceph.com   Delete          Immediate           true                   5d2h
```

```
oc annotate storageclass rbd-sc "keyrotation.csiaddons.openshift.io/enable: false"
storageclass.storage.k8s.io/rbd-sc annotated
```

Disabling key rotation for a specific persistent volume claim

1. Identify the `EncryptionKeyRotationCronJob` CR for the PVC you want to disable key rotation on:

   ```
   oc get encryptionkeyrotationcronjob -o jsonpath='{range .items[?(@.spec.jobTemplate.spec.target.persistentVolumeClaim=="
   <PVC_NAME>")]}{.metadata.name}{"\n"}{end}'
   ```

   Where `<PVC_NAME>` is the name of the PVC that you want to disable.
2. Apply the following to the `EncryptionKeyRotationCronJob`CR from the previous step to disable the key rotation:
   a. Update the `csiaddons.openshift.io/state` annotation from `managed` to `unmanaged`:

      ```
      oc annotate encryptionkeyrotationcronjob <encryptionkeyrotationcronjob_name> "csiaddons.openshift.io/state=unmanaged"
      --overwrite=true
      ```

      Where <encryptionkeyrotationcronjob_name> is the name of the `EncryptionKeyRotationCronJob` CR.
   b. Add `suspend: true` under the `spec` field:

      ```
      oc patch encryptionkeyrotationcronjob <encryptionkeyrotationcronjob_name> -p '{"spec": {"suspend": true}}' --
      type=merge.
      ```

3. Save and exit. The key rotation will be disabled for the PVC.

Enabling encryption using Thales CipherTrust Manager (manual part)

Configure the Key Management Interoperability Protocol (KMIP) settings in Thales CipherTrust Manager server.

About this task

IBM Fusion HCI with IBM Spectrum® Storage Scale Erasure Code Edition (ECE) does not support the Thales or Vormetric CipherTrust Manager, and that the current encryption CR implementation in CNSA or IBM Fusion HCI works only with IBM Security GKLM (with simplified setup).

Procedure

1. If KMIP client does not exist, then create it.
    a. From the Thales CipherTrust Manager user interface, select Products > KMIP > Client Profile > Add Profile.
    b. Add the username location to the Common Name (CN) field during profile creation.
2. Create a token.
    a. Go to KMIP > Registration Token > New Registration Token.
    b. Copy the token for the next step.
3. Register the client.
    a. Go to KMIP > Registered Clients > Add Client.
    b. Specify the name.
    c. Paste the Registration Token from the previous step.
    d. Click Save.
4. To download the Private Key and Client Certificate, click Save Private Key and Save Certificate respectively.
5. Create a KMIP interface.
    a. Go to Admin Settings > Interfaces > Add Interface.
    b. Select KMIP Key Management Interoperability Protocol and click Next.
    c. Select an available Port.
    d. Select Network Interface as `all`.
    e. Select `Interface Mode` for TLS. Verify client certificate and the username that is taken from the client certificate. The auth request is optional.
    f. Select the CA to be used, and click Save.
6. To get the server CA certificate, click ellipsis overflow menu of the newly created interface, and click Download Certificate.
   Whenever you configure encryption from the IBM Fusion user interface, use these downloaded files.

Managing hybrid and multicloud resource

Use this information to learn how to manage hybrid and multicloud resources.

The Multicloud Object Gateway (MCG) is a lightweight object storage service for OpenShift, allowing users to start small and then scale as needed on-premise, in multiple clusters, and with cloud-native storage.

- **Accessing the Multicloud Object Gateway with your applications**
  You can access the object service with any application targeting AWS S3 or code that uses AWS S3 Software Development Kit (SDK). Applications need to specify the Multicloud Object Gateway (MCG) endpoint, an access key, and a secret access key. You can use your terminal or the MCG command-line interface (CLI) to retrieve this information.
- **Adding storage resources for hybrid or Multicloud**
  Understand how to add and edit storage resources for hybrid or Multicloud.
- **Creating and managing buckets using the object browser**
  The object browser in the OpenShift Web Console enables you to create, view, and manage buckets and objects across different storage endpoints, including Multicloud Object Gateway (MCG) and internal mode RADOS Gateway (RGW) object storage.
- **Managing namespace buckets**
  Namespace buckets let you connect data repositories on different providers together, so you can interact with all of your data through a single unified view. Add the object bucket associated with each provider to the namespace bucket, and access your data through the namespace bucket to see all of your object buckets at once. This lets you write to your preferred storage provider while reading from multiple other storage providers, greatly reducing the cost of migrating to a new storage provider. Use this information to add namespace buckets using command-line interface, YAML, and user interface and for information about sharing and accessing legacy data.
- **Securing Multicloud Object Gateway**
  Ensure that the Multicloud Object Gateway is secure.
- **Mirroring data for hybrid and Multicloud buckets**
  Create bucket classes to mirror data for hybrid and Multicloud buckets. Mirroring data can be setup by using the OpenShift UI, YAML, or MCG command-line interface.
- **Bucket policies in the Multicloud Object Gateway**
  Fusion Data Foundation supports AWS S3 bucket policies. Bucket policies allow you to grant users access permissions for buckets and the objects in them. Use the information in this section to understand how to use bucket policies in Multicloud Object Gateway.
- **Bucket notification in Multicloud Object Gateway**
  With bucket notification, you can send a notification to an external server whenever there is an event in the bucket.
- **Multicloud Object Gateway bucket replication**
  Data replication from one Multicloud Object Gateway (MCG) bucket to another MCG bucket provides higher resiliency and better collaboration options. These buckets can be either data buckets or namespace buckets backed by any supported storage solution (S3, Azure, and so on). Replicate a bucket to another bucket and set replication policies using command-line interface and YAML.
- **Object Bucket Claim**
  An Object Bucket Claim can be used to request an S3 compatible bucket backend for your workloads. An object bucket claim creates a new bucket and an application account in NooBaa with permissions to the bucket, including a new access key and secret access key. The application account is allowed to access only a single bucket and can't create new buckets by default. Use this information to create, view, and delete object bucket claims.

- **Caching policy for object buckets**
  A cache bucket is a namespace bucket with a hub target and a cache target. The hub target is an S3 compatible large object storage bucket. The cache bucket is the local Multicloud Object Gateway bucket. You can create a cache bucket that caches an AWS bucket or an IBM COS bucket.
- **Lifecycle bucket configuration in Multicloud Object Gateway**
  Multicloud Object Gateway (MCG) lifecycle provides a way to reduce storage costs due to accumulated data objects.
- **Scaling Multicloud Object Gateway performance**
  The Multicloud Object Gateway (MCG) performance may vary from one environment to another. In some cases, specific applications require faster performance which can be easily addressed by scaling S3 endpoints.
- **Accessing the RADOS Object Gateway S3 endpoint**
  Users can access the RADOS Object Gateway (RGW) endpoint directly. The RGW route is created by default and is named `rook-ceph-rgw-ocs-storagecluster-cephobjectstore`.
- **Using TLS certificates for applications accessing RGW**
  Fetch TLS certificates stored as Kubernetes secrets and configure applications to use them when accessing the RADOS Object Gateway (RGW) over a secure connection.
- **Using the security token service of Multicloud Object Gateway to assume the role of another user**
  Multicloud Object Gateway (MCG) supports for a security token service (STS) similar to the one provided by Amazon Web Services. This STS of the MCG authorizes a user to assume the role of another user.

Accessing the Multicloud Object Gateway with your applications

You can access the object service with any application targeting AWS S3 or code that uses AWS S3 Software Development Kit (SDK). Applications need to specify the Multicloud Object Gateway (MCG) endpoint, an access key, and a secret access key. You can use your terminal or the MCG command-line interface (CLI) to retrieve this information.

Accessing the MCG buckets using the virtual-hosted style

If the client application tries to access `https://<bucket-name>.s3-openshift-storage.apps.mycluster-cluster.qe.rh-ocs.com`, a DNS entry is needed for `mcg-test-bucket.s3-openshift-storage.apps.mycluster-cluster.qe.rh-ocs.com` to point to the S3 Service.

<bucket-name> is the name of the MCG bucket, for example: `https://mcg-test-bucket.s3-openshift-storage.apps.mycluster-cluster.qe.rh-ocs.com`.

Important: Ensure that you have a DNS entry in order to point the client application to the MCG buckets using the virtual-hosted style.

- **Accessing the Multicloud Object Gateway from the terminal**
  Use this information to access the Multicloud Object Gateway from the terminal.
- **Accessing the Multicloud Object Gateway from the MCG command-line interface**
  Use this information to access the Multicloud Object Gateway from the MCG command-line interface.
- **Support of Multicloud Object Gateway data bucket APIs**
  The following table lists the Multicloud Object Gateway (MCG) data bucket APIs and their support levels.

Accessing the Multicloud Object Gateway from the terminal

Use this information to access the Multicloud Object Gateway from the terminal.

Before you begin

- A running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.
  - For IBM Power, use the following command:

    ```
    subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
    ```

  - For IBM Z infrastructure, use the following command:

    ```
    subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
    ```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.
  Note: Choose the correct Product Variant according to your architecture.

Procedure

Run the **describe** command to view information about the Multicloud Object Gateway (MCG) endpoint, including its access key (AWS_ACCESS_KEY_ID value) and secret access key (AWS_SECRET_ACCESS_KEY value).

```
oc describe noobaa -n openshift-storage
```

Example output, where **(1)** is the access key (AWS_ACCESS_KEY_ID value), **(2)** is the secret access key (AWS_SECRET_ACCESS_KEY value), and **(3)** is the MCG endpoint

```
Name:         noobaa
Namespace:    openshift-storage
Labels:       <none>
```

```
Annotations:  <none>
API Version:  noobaa.io/v1alpha1
Kind:         NooBaa
Metadata:
  Creation Timestamp:  2019-07-29T16:22:06Z
  Generation:          1
  Resource Version:    6718822
  Self Link:           /apis/noobaa.io/v1alpha1/namespaces/openshift-storage/noobaas/noobaa
  UID:                 019cfb4a-b21d-11e9-9a02-06c8de012f9e
Spec:
Status:
  Accounts:
    Admin:
      Secret Ref:
        Name:           noobaa-admin
        Namespace:      openshift-storage
  Actual Image:         noobaa/noobaa-core:4.0
  Observed Generation:  1
  Phase:                Ready
  Readme:

  Welcome to NooBaa!
  -----------------

  Welcome to NooBaa!
    -----------------
    NooBaa Core Version:
    NooBaa Operator Version:

    Lets get started:

    1. Connect to Management console:

      Read your mgmt console login information (email & password) from secret: "noobaa-admin".

        kubectl get secret noobaa-admin -n openshift-storage -o json | jq '.data|map_values(@base64d)'

      Open the management console service - take External IP/DNS or Node Port or use port forwarding:

        kubectl port-forward -n openshift-storage service/noobaa-mgmt 11443:443 &
        open https://localhost:11443

    2. Test S3 client:

      kubectl port-forward -n openshift-storage service/s3 10443:443 &how
(1)
      NOOBAA_ACCESS_KEY=$(kubectl get secret noobaa-admin -n openshift-storage -o json | jq -r
'.data.AWS_ACCESS_KEY_ID|@base64d')
(2)
      NOOBAA_SECRET_KEY=$(kubectl get secret noobaa-admin -n openshift-storage -o json | jq -r
'.data.AWS_SECRET_ACCESS_KEY|@base64d')
      alias s3='AWS_ACCESS_KEY_ID=$NOOBAA_ACCESS_KEY AWS_SECRET_ACCESS_KEY=$NOOBAA_SECRET_KEY aws --endpoint
https://localhost:10443 --no-verify-ssl s3'
      s3 ls


    Services:
      Service Mgmt:
        External DNS:
          https://noobaa-mgmt-openshift-storage.apps.mycluster-cluster.qe.rh-ocs.com
          https://a3406079515be11eaa3b70683061451e-1194613580.us-east-2.elb.amazonaws.com:443
        Internal DNS:
          https://noobaa-mgmt.openshift-storage.svc:443
        Internal IP:
          https://172.30.235.12:443
        Node Ports:
          https://10.0.142.103:31385
        Pod Ports:
          https://10.131.0.19:8443
      serviceS3:
        External DNS: (3)
          https://s3-openshift-storage.apps.mycluster-cluster.qe.rh-ocs.com
          https://a340f4e1315be11eaa3b70683061451e-943168195.us-east-2.elb.amazonaws.com:443
        Internal DNS:
          https://s3.openshift-storage.svc:443
        Internal IP:
          https://172.30.86.41:443
        Node Ports:
          https://10.0.142.103:31011
        Pod Ports:
          https://10.131.0.19:6443
```

Note: The output from the **oc describe noobaa** command lists the internal and external DNS names that are available. When using the internal DNS, the traffic is free. The external DNS uses Load Balancing to process the traffic, and therefore has a cost per hour.

Accessing the Multicloud Object Gateway from the MCG command-line interface

Use this information to access the Multicloud Object Gateway from the MCG command-line interface.

Before you begin

- A running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.
  - For IBM Power, use the following command:

    ```
    subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
    ```

  - For IBM Z infrastructure, use the following command:

    ```
    subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
    ```

Procedure

Run the **status** command to access the endpoint, access key, and secret access key.

```
odf noobaa status -n openshift-storage
```

Example output, where **(1)** is the endpoint, **(2)** is the access key, and **(3)** is secret access key.

```
INFO[0000] Namespace: openshift-storage
INFO[0000]
INFO[0000] CRD Status:
INFO[0003] ✅ Exists: CustomResourceDefinition "noobaas.noobaa.io"
INFO[0003] ✅ Exists: CustomResourceDefinition "backingstores.noobaa.io"
INFO[0003] ✅ Exists: CustomResourceDefinition "bucketclasses.noobaa.io"
INFO[0004] ✅ Exists: CustomResourceDefinition "objectbucketclaims.objectbucket.io"
INFO[0004] ✅ Exists: CustomResourceDefinition "objectbuckets.objectbucket.io"
INFO[0004]
INFO[0004] Operator Status:
INFO[0004] ✅ Exists: Namespace "openshift-storage"
INFO[0004] ✅ Exists: ServiceAccount "noobaa"
INFO[0005] ✅ Exists: Role "ocs-operator.v0.0.271-6g45f"
INFO[0005] ✅ Exists: RoleBinding "ocs-operator.v0.0.271-6g45f-noobaa-f9vpj"
INFO[0006] ✅ Exists: ClusterRole "ocs-operator.v0.0.271-fjhgh"
INFO[0006] ✅ Exists: ClusterRoleBinding "ocs-operator.v0.0.271-fjhgh-noobaa-pdxn5"
INFO[0006] ✅ Exists: Deployment "noobaa-operator"
INFO[0006]
INFO[0006] System Status:
INFO[0007] ✅ Exists: NooBaa "noobaa"
INFO[0007] ✅ Exists: StatefulSet "noobaa-core"
INFO[0007] ✅ Exists: Service "noobaa-mgmt"
INFO[0008] ✅ Exists: Service "s3"
INFO[0008] ✅ Exists: Secret "noobaa-server"
INFO[0008] ✅ Exists: Secret "noobaa-operator"
INFO[0008] ✅ Exists: Secret "noobaa-admin"
INFO[0009] ✅ Exists: StorageClass "openshift-storage.noobaa.io"
INFO[0009] ✅ Exists: BucketClass "noobaa-default-bucket-class"
INFO[0009] ✅ (Optional) Exists: BackingStore "noobaa-default-backing-store"
INFO[0010] ✅ (Optional) Exists: CredentialsRequest "noobaa-cloud-creds"
INFO[0010] ✅ (Optional) Exists: PrometheusRule "noobaa-prometheus-rules"
INFO[0010] ✅ (Optional) Exists: ServiceMonitor "noobaa-service-monitor"
INFO[0011] ✅ (Optional) Exists: Route "noobaa-mgmt"
INFO[0011] ✅ (Optional) Exists: Route "s3"
INFO[0011] ✅ Exists: PersistentVolumeClaim "db-noobaa-core-0"
INFO[0011] ✅ System Phase is "Ready"
INFO[0011] ✅ Exists:  "noobaa-admin"

#------------------#
#- Mgmt Addresses -#
#------------------#

ExternalDNS : [https://noobaa-mgmt-openshift-storage.apps.mycluster-cluster.qe.rh-ocs.com
https://a3406079515be11eaa3b70683061451e-1194613580.us-east-2.elb.amazonaws.com:443]
ExternalIP  : []
NodePorts   : [https://10.0.142.103:31385]
InternalDNS : [https://noobaa-mgmt.openshift-storage.svc:443]
InternalIP  : [https://172.30.235.12:443]
PodPorts    : [https://10.131.0.19:8443]

#--------------------#
#- Mgmt Credentials -#
#--------------------#

email    : admin@noobaa.io
password : HKLbH1rSuVU0I/souIkSiA==

#----------------#
#- S3 Addresses -#
#----------------#

(1)
ExternalDNS : [https://s3-openshift-storage.apps.mycluster-cluster.qe.rh-ocs.com https://a340f4e1315be11eaa3b70683061451e-
943168195.us-east-2.elb.amazonaws.com:443]
ExternalIP  : []
NodePorts   : [https://10.0.142.103:31011]
InternalDNS : [https://s3.openshift-storage.svc:443]
InternalIP  : [https://172.30.86.41:443]
PodPorts    : [https://10.131.0.19:6443]
```

```
#-----------------#
#- S3 Credentials -#
#-----------------#

(2)
AWS_ACCESS_KEY_ID     : jVmAsu9FsvRHYmfjTiHV
(3)
AWS_SECRET_ACCESS_KEY : E//420VNedJfATvVSmDz6FMtsSAzuBv6z180PT5c

#-----------------#
#- Backing Stores -#
#-----------------#

NAME                          TYPE     TARGET-BUCKET                                          PHASE    AGE
noobaa-default-backing-store  aws-s3   noobaa-backing-store-15dc896d-7fe0-4bed-9349-5942211b93c9  Ready   141h35m32s

#-----------------#
#- Bucket Classes -#
#-----------------#

NAME                          PLACEMENT                                                        PHASE    AGE
noobaa-default-bucket-class   {Tiers:[{Placement: BackingStores:[noobaa-default-backing-store]}]}  Ready   141h35m33s

#----------------#
#- Bucket Claims -#
#----------------#

No OBC's found.
```

Results

You now have the relevant endpoint, access key, and secret access key in order to connect to your applications.

For example:

If AWS S3 CLI is the application, the following command will list the buckets in Fusion Data Foundation:

```
AWS_ACCESS_KEY_ID=<AWS_ACCESS_KEY_ID>
AWS_SECRET_ACCESS_KEY=<AWS_SECRET_ACCESS_KEY>
aws --endpoint <ENDPOINT> --no-verify-ssl s3 ls
```

Support of Multicloud Object Gateway data bucket APIs

The following table lists the Multicloud Object Gateway (MCG) data bucket APIs and their support levels.

Data buckets	**Support**	
List buckets	Supported	
Delete bucket	Supported	Replication configuration is part of MCG bucket class configuration
Create bucket	Supported	A different set of canned ACLs
Post bucket	Not supported	
Put bucket	Partially supported	Replication configuration is part of MCG bucket class configuration
Bucket lifecycle	Partially supported	Object expiration only
Policy (Buckets, Objects)	Partially supported	Bucket policies are supported
Bucket Website	Supported	
Bucket ACLs (Get, Put)	Supported	A different set of canned ACLs
Bucket Location	Partialy	Returns a default value only
Bucket Notification	Not supported	
Bucket Object Versions	Supoorted	
Get Bucket Info (HEAD)	Supported	
Bucket Request Payment	Partially supported	Returns the bucket owner
Put Object	Supported	
Delete Object	Supported	
Get Object	Supported	
Object ACLs (Get, Put)	Supported	
Get Object Info (HEAD)	Supported	
POST Object	Supported	
Copy Object	Supported	
Multipart Uploads	Supported	
Object Tagging	Supported	
Storage Class	Not supported	

Adding storage resources for hybrid or Multicloud

Understand how to add and edit storage resources for hybrid or Multicloud.

- **Creating a new backing store**
  Create a new backing store to add storage resources.
- **Overriding the default backing store**
  Follow this procedure to override the default NooBaa backing store.
- **Adding storage resources for hybrid or Multicloud using the MCG command line interface**
  The Multicloud Object Gateway (MCG) simplifies the process of spanning data across the cloud provider and clusters. dd a backing storage that can be used by the MCG.
- **Creating an s3 compatible Multicloud Object Gateway backingstore**
  Create an S3 compatible MCG backing store for IBM Storage Ceph's RGW. When the RGW is deployed, Fusion Data Foundation operator creates an S3 compatible backingstore for MCG automatically.
- **Creating a new bucket class**
  Bucket class is a CRD representing a class of buckets that defines tiering policies and data placements for an Object Bucket Class (OBC). Use this procedure to create a bucket class in Fusion Data Foundation.
- **Editing a bucket class**
  Use this information to edit an existing bucket class.
- **Editing backing stores for bucket class**
  Use this procedure to edit an existing Multicloud Object Gateway (MCG) bucket class to change the underlying backing stores used in a bucket class.

Creating a new backing store

Create a new backing store to add storage resources.

Before you begin

Ensure you have Administrator access to Fusion Data Foundation.

Procedure

1. From the OpenShift Web Console, go to Storage > Data Foundation.
2. From the Backing Store tab, click Create Backing Store.
3. From the Create New Backing Store page:
   a. Enter a Backing Store Name.
   b. Select a Provider.
   c. Select a Region.
   d. Optional: Enter an Endpoint..
   e. Select a Secret from the drop-down list, or create your own secret. Optionally, you can Switch to Credentials view which lets you fill in the required secrets. For more information on creating an OCP secret, see Security > Authentication and authorization > Configuring identity providers > Creating the secret in Red Hat OpenShift Container Platform product documentation.

      Each backingstore requires a different secret. For more information on creating the secret for a particular backingstore, see Adding storage resources for hybrid or Multicloud using the MCG command line interface and follow the steps to add storage resources using a YAML.

      Note: This menu is relevant for all providers except Google Cloud and local PVC.
   f. Enter the Target bucket.
      The target bucket is a storage that is hosted on the remote cloud service. The target bucket enables you to create a connection that tells the MCG that it can use this bucket for the system.
4. Click Create Backing Store.

What to do next

Optionally, verify the creation of the backing store:

1. From the OpenShift Web Console, go to Storage > Data Foundation.
2. From the Backing Store tab to view all the backing stores.

Overriding the default backing store

Follow this procedure to override the default NooBaa backing store.

Before you begin

Download the Multicloud Object Gateway (MCG) command-line interface.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
 yum install mcg
```

Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.

- For IBM Power, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
  ```

- For IBM Z infrastructure, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
  ```

About this task

You can use the `manualDefaultBackingStore`flag to override the default NooBaa backing store and remove it if you do not want to use the default backing store configuration. This provides flexibility to customize your backing store configuration and tailor it to your specific needs. By leveraging this feature, you can further optimize your system and enhance its performance.

Procedure

1. Check if `noobaa-default-backing-store` is present:

   ```
   oc get backingstore
   NAME TYPE PHASE AGE
   noobaa-default-backing-store pv-pool Creating 102s
   ```

2. Patch the NooBaa CR to enable `manualDefaultBackingStore`:

   ```
   oc patch noobaa/noobaa --type json --patch='[{"op":"add","path":"/spec/manualDefaultBackingStore","value":true}]'
   ```

   Important: Use the Multicloud Object Gateway CLI to create a new backing store and update accounts.
3. Create a new default backing store to override the default backing store.
   For example:

   ```
   odf noobaa backingstore create pv-pool _NEW-DEFAULT-BACKING-STORE_ --num-volumes 1 --pv-size-gb 16
   ```

   a. Replace `NEW-DEFAULT-BACKING-STORE` with the name you want for your new default backing store.
4. Update the admin account to use the new default backing store as its default resource:

   ```
   odf noobaa account update admin@noobaa.io --new_default_resource=_NEW-DEFAULT-BACKING-STORE_
   ```

   a. Replace `NEW-DEFAULT-BACKING-STORE` with the name of the backing store from the previous step.
   Updating the default resource for admin accounts ensures that the new configuration is used throughout your system.

5. Configure the default-bucketclass to use the new default backingstore:

   ```
   oc patch Bucketclass noobaa-default-bucket-class -n openshift-storage --type=json --patch='[{"op": "replace", "path":
   "/spec/placementPolicy/tiers/0/backingStores/0", "value": "NEW-DEFAULT-BACKING-STORE"}]'
   ```

6. Optional: Delete the `noobaa-default-backing-store`.
   a. Delete all instances of and buckets associated with `noobaa-default-backing-store` and update the accounts using it as resource.
   b. Delete the `noobaa-default-backing-store`:

      ```
      oc delete backingstore noobaa-default-backing-store -n openshift-storage | oc patch -n openshift-storage
      backingstore/noobaa-default-backing-store --type json --patch='[ { "op": "remove", "path": "/metadata/finalizers" }
      ]'
      ```

      You must enable the `manualDefaultBackingStore` flag before proceeding. Additionally, it is crucial to update all accounts that use the default resource and delete all instances of and buckets associated with the default backing store to ensure a smooth transition.

Adding storage resources for hybrid or Multicloud using the MCG command line interface

The Multicloud Object Gateway (MCG) simplifies the process of spanning data across the cloud provider and clusters. dd a backing storage that can be used by the MCG.

Depending on your deployment type, follow the appropriate procedure to add storage resources by creating a backing storage.

For VMware deployments, skip this section and go to Creating an s3 compatible Multicloud Object Gateway backingstore.

- **Creating an AWS-backed backingstore**
  Create an AWS-backed backing store using the MCG command-line interface or a YAML file to add storage resources.
- **Creating an AWS-STS-backed backingstore**
  Creating an AWS STS-backed backingstore involves creating an AWS role with temporary credentials, installing the operator in an AWS STS cluster, and creating the backingstore using short-lived credentials.
- **Creating an IBM COS-backed backingstore**
  Create an IBM COS-backed backing store using the MCG command-line interface or a YAML file to add storage resources.
- **Creating an Azure-backed backingstore**
  Create an Azure-backed backing store using the MCG command-line interface or a YAML file to add storage resources.
- **Creating a GCP-backed backingstore**
  Create a GCP-backed backing store using the MCG command-line interface or a YAML file to add storage resources.
- **Creating a local Persistent Volume-backed backingstore**
  Create a local Persistent Volume-backed backing store using the MCG command-line interface or a YAML file to add storage resources.

Creating an AWS-backed backingstore

Create an AWS-backed backing store using the MCG command-line interface or a YAML file to add storage resources.

Before you begin, ensure the following:

- Fusion Data Foundation Platform is running.
- Download the MCG command-line interface for easier management.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
yum install mcg
```

Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

Create an AWS-backed backingstore using MCG command-line interface

From the MCG command-line interface, run the following command:

```
odf noobaa backingstore create aws-s3 <backingstore_name> --access-key=<AWS ACCESS KEY> --secret-key=<AWS SECRET ACCESS KEY> --target-bucket <bucket-name> -n openshift-storage
```

backingstore_name
: The name of the backingstore.

AWS ACCESS KEY and *AWS SECRET ACCESS KEY*
: The AWS access key ID and secret access key you created for this purpose.

bucket-name
: The existing AWS bucket name. This argument indicates to the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.

Example output:

```
INFO[0001] ✅ Exists: NooBaa "noobaa"
INFO[0002] ✅ Created: BackingStore "aws-resource"
INFO[0002] ✅ Created: Secret "backing-store-secret-aws-resource"
```

Create an AWS-backed backingstore using a YAML file

1. Create a secret with the credentials:

   ```
   apiVersion: v1
   kind: Secret
   metadata:
     name: <backingstore-secret-name>
     namespace: openshift-storage
   type: Opaque
   data:
     AWS_ACCESS_KEY_ID: <AWS ACCESS KEY ID ENCODED IN BASE64>
     AWS_SECRET_ACCESS_KEY: <AWS SECRET ACCESS KEY ENCODED IN BASE64>
   ```

   AWS ACCESS KEY and *AWS SECRET ACCESS KEY*
   : Supply and encode your own AWS access key ID and secret access key using Base64, and use the results for ***AWS ACCESS KEY ID ENCODED IN BASE64*** and ***AWS SECRET ACCESS KEY ENCODED IN BASE64***.

   backingstore-secret-name
   : The name of the backingstore secret created in Creating a new backing store.

2. Apply the following YAML for a specific backing store:

   ```
   apiVersion: noobaa.io/v1alpha1
   kind: BackingStore
   metadata:
     finalizers:
     - noobaa.io/finalizer
     labels:
       app: noobaa
     name: bs
     namespace: openshift-storage
   spec:
     awsS3:
       secret:
         name: <backingstore-secret-name>
         namespace: openshift-storage
       targetBucket: <bucket-name>
     type: aws-s3
   ```

   bucket-name
   : The existing AWS bucket name.

   backingstore-secret-name
   : The name of the backingstore secret created in the previous step.

Creating an AWS-STS-backed backingstore

Creating an AWS STS-backed backingstore involves creating an AWS role with temporary credentials, installing the operator in an AWS STS cluster, and creating the backingstore using short-lived credentials.

Amazon Web Services Security Token Service (AWS STS) is an AWS feature and it is a way to authenticate using short-lived credentials. Creating an AWS-STS-backed backingstore involves the following:

- Creating an AWS role using a script, which helps to get the temporary security credentials for the role session
- Installing {product-name-short} operator in AWS STS OpenShift cluster
- Creating backingstore in AWS STS OpenShift cluster

- **Creating an AWS role using a script**
  You need to create a role and pass the role Amazon resource name (ARN) while installing the {product-name-short} operator.
- **Installing OpenShift Data Foundation operator in AWS STS OpenShift cluster**
  Install the OpenShift Data Foundation operator in an AWS STS OpenShift cluster by annotating the subscription with the ARN and applying the required OIDC-based credentials.
- **Creating a new AWS STS backingstore**
  Create an AWS STS backingstore for the Multicloud Object Gateway using short-lived credentials obtained through OpenID Connect (OIDC) authentication.

Creating an AWS role using a script

You need to create a role and pass the role Amazon resource name (ARN) while installing the {product-name-short} operator.

Before you begin

Configure Red Hat OpenShift Container Platform cluster with AWS STS. For more information, see Configuring an AWS cluster to use short-term credentials.

Procedure

Create an AWS role using a script that matches OpenID Connect (OIDC) configuration for Multicloud Object Gateway (MCG) on OpenShift Data Foundation. The following example shows the details that are required to create the role:

```
{
    “Version”: “2012-10-17",
    “Statement”: [
        {
            “Effect”: “Allow”,
            “Principal”: {
                “Federated”: “arn:aws:iam::123456789123:oidc-provider/mybucket-oidc.s3.us-east-2.amazonaws.com”
            },
            “Action”: “sts:AssumeRoleWithWebIdentity”,
            “Condition”: {
                “StringEquals”: {
                    “mybucket-oidc.s3.us-east-2.amazonaws.com:sub”: [
                        “system:serviceaccount:openshift-storage:noobaa”,
                        “system:serviceaccount:openshift-storage:noobaa-endpoint”
                    ]
                }
            }
        }
    ]
}
```

where

`123456789123`
Is the AWS account ID

`mybucket`
Is the bucket name (using public bucket configuration)

`us-east-2`
Is the AWS region

`openshift-storage`
Is the namespace name

Sample script

```
#!/bin/bash
set -x

# This is a sample script to help you deploy MCG on AWS STS cluster.
# This script shows how to create role-policy and then create the role in AWS.
# For more information see: https://docs.openshift.com/rosa/authentication/assuming-an-aws-iam-role-for-a-service-account.html

# WARNING: This is a sample script. You need to adjust the variables based on your requirement.

# Variables :
# user variables - REPLACE these variables with your values:
ROLE_NAME="<role-name>" # role name that you pick in your AWS account
NAMESPACE="<namespace>" # namespace name where MCG is running. For OpenShift Data Foundation, it is openshift-storage.

# MCG variables
SERVICE_ACCOUNT_NAME_1="<service-account-name-1>" # The service account name of statefulset core and deployment operator (MCG
SERVICE_ACCOUNT_NAME_2="<service-account-name-2>" # The service account name of deployment endpoint (MCG endpoint)

# AWS variables
```

```
# Ensure these values are not empty (AWS_ACCOUNT_ID, OIDC_PROVIDER)
# AWS_ACCOUNT_ID is your AWS account number
AWS_ACCOUNT_ID=$(aws sts get-caller-identity --query "Account" --output text)
# If you want to create the role before using the cluster, replace this field too.
# The OIDC provider is in the structure:
# 1) <OIDC-bucket>.s3.<aws-region>.amazonaws.com. for OIDC bucket configurations are in an S3 public bucket
# 2) `<characters>.cloudfront.net` for OIDC bucket configurations in an S3 private bucket with a public CloudFront distributio
OIDC_PROVIDER=$(oc get authentication cluster -ojson | jq -r .spec.serviceAccountIssuer | sed -e "s/^https:\/\///")
# the permission (S3 full access)
POLICY_ARN_STRINGS="arn:aws:iam::aws:policy/AmazonS3FullAccess"

# Creating the role (with AWS command line interface)

read -r -d '' TRUST_RELATIONSHIP <<EOF
{
 "Version": "2012-10-17",
 "Statement": [
   {
        "Effect": "Allow",
        "Principal": {
        "Federated": "arn:aws:iam::${AWS_ACCOUNT_ID}:oidc-provider/${OIDC_PROVIDER}"
        },
        "Action": "sts:AssumeRoleWithWebIdentity",
        "Condition": {
        "StringEquals": {
        "${OIDC_PROVIDER}:sub": [
        "system:serviceaccount:${NAMESPACE}:${SERVICE_ACCOUNT_NAME_1}",
        "system:serviceaccount:${NAMESPACE}:${SERVICE_ACCOUNT_NAME_2}"
        ]
        }
        }
   }
 ]
}
EOF

echo "${TRUST_RELATIONSHIP}" > trust.json

aws iam create-role --role-name "$ROLE_NAME" --assume-role-policy-document file://trust.json --description "role for demo"

while IFS= read -r POLICY_ARN; do
   echo -n "Attaching $POLICY_ARN ... "
   aws iam attach-role-policy \
        --role-name "$ROLE_NAME" \
        --policy-arn "${POLICY_ARN}"
   echo "ok."
done <<< "$POLICY_ARN_STRINGS"
```

Installing OpenShift Data Foundation operator in AWS STS OpenShift cluster

Install the OpenShift Data Foundation operator in an AWS STS OpenShift cluster by annotating the subscription with the ARN and applying the required OIDC-based credentials.

Before you begin

- Configure OpenShift Container Platform cluster with AWS STS. For more information, see Configuring an AWS cluster to use short-term credentials.
- Create an AWS role using a script that matches OpenID Connect (OIDC) configuration. For more information, see Creating an AWS role using a script.

Procedure

Install Fusion Data Foundation Operator from the Operator Hub.

- During the installation add the role ARN in the **ARN Details** field.
- Ensure that the **Update approval** field is set to **Manual**.

Creating a new AWS STS backingstore

Create an AWS STS backingstore for the Multicloud Object Gateway using short-lived credentials obtained through OpenID Connect (OIDC) authentication.

Before you begin

- Configure Red Hat OpenShift Container Platform cluster with AWS STS. For more information, see Configuring an AWS cluster to use short-term credentials.
- Create an AWS role using a script that matches OpenID Connect (OIDC) configuration. For more information, see Creating an AWS role using a script.

Procedure

1. Install Multicloud Object Gateway (MCG).
   It is installed with the default backingstore by using the short-lived credentials.
2. After the MCG system is ready, you can create more backingstores of the type `aws-sts-s3` using the following MCG command line interface command:

```
odf noobaa backingstore create aws-sts-s3 <backingstore-name> --aws-sts-arn=<aws-sts-role-arn> --region=<region> --target-bucket=<target-bucket>
```

where

backingstore-name
: Name of the backingstore

aws-sts-role-arn
: The AWS STS role ARN which will assume role

region
: The AWS bucket region

target-bucket
: The target bucket name on the cloud

Creating an IBM COS-backed backingstore

Create an IBM COS-backed backing store using the MCG command-line interface or a YAML file to add storage resources.

Before you begin, ensure the following:

- Fusion Data Foundation Platform is running.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.
  - For IBM Power, use the following command:

    ```
    subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
    ```

  - For IBM Z infrastructure, use the following command:

    ```
    subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
    ```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

Create an IBM COS-backed backingstore using MCG command-line interface

From the MCG command-line interface, run the following command:

```
odf noobaa backingstore create ibm-cos <backingstore_name> --access-key=<IBM ACCESS KEY> --secret-key=<IBM SECRET ACCESS KEY> --endpoint=<IBM COS ENDPOINT> --target-bucket <bucket-name> -n openshift-storage
```

backingstore_name
: The name of the backingstore.

IBM ACCESS KEY and *IBM SECRET ACCESS KEY*
: An IBM access key ID, secret access key and the appropriate regional endpoint that corresponds to the location of the existing IBM bucket.
To generate the above keys on IBM cloud, you must include HMAC credentials while creating the service credentials for your target bucket.

bucket-name
: The existing IBM bucket name. This argument indicates to the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.
Example output:

```
INFO[0001] ✅ Exists: NooBaa "noobaa"
INFO[0002] ✅ Created: BackingStore "ibm-resource"
INFO[0002] ✅ Created: Secret "backing-store-secret-ibm-resource"
```

Create an IBM COS-backed backingstore using a YAML file

1. Create a secret with the credentials:

   ```
   apiVersion: v1
   kind: Secret
   metadata:
     name: <backingstore-secret-name>
     namespace: openshift-storage
   type: Opaque
   data:
     IBM_COS_ACCESS_KEY_ID: <IBM COS ACCESS KEY ID ENCODED IN BASE64>
     IBM_COS_SECRET_ACCESS_KEY: <IBM COS SECRET ACCESS KEY ENCODED IN BASE64>
   ```

   IBM ACCESS KEY and *IBM SECRET ACCESS KEY*
   : Provide and encode your own IBM COS access key ID and secret access key using Base64, and use the results in place of these attributes respectively.

   backingstore-secret-name
   : The name of the backingstore secret created in Creating a new backing store.

2. Apply the following YAML for a specific backing store:

```
apiVersion: noobaa.io/v1alpha1
kind: BackingStore
metadata:
  finalizers:
  - noobaa.io/finalizer
  labels:
    app: noobaa
  name: bs
  namespace: openshift-storage
spec:
  ibmCos:
    endpoint: <endpoint>
    secret:
      name: <backingstore-secret-name>
      namespace: openshift-storage
    targetBucket: <bucket-name>
  type: ibm-cos
```

bucket-name
An existing IBM COS bucket name.

endpoint
A regional endpoint that corresponds to the location of the existing IBM bucket name. This argument indicates to MCG about the endpoint to use for its backingstore, and subsequently, data storage and administration.

backingstore-secret-name
The name of the backingstore secret created in the previous step.

Creating an Azure-backed backingstore

Create an Azure-backed backing store using the MCG command-line interface or a YAML file to add storage resources.

Before you begin, ensure the following:

- Fusion Data Foundation Platform is running.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
  ```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

Create an Azure-backed backingstore using MCG command-line interface

From the MCG command-line interface, run the following command:

```
odf noobaa backingstore create azure-blob <backingstore_name> --account-key=<AZURE ACCOUNT KEY> --account-name=<AZURE ACCOUNT NAME> --target-blob-container <blob container name> -n openshift-storage
```

backingstore_name
The name of the backingstore.

AZURE ACCOUNT KEY and *AZURE ACCOUNT NAME*
An AZURE account key and account name you created for this purpose.

blob container name
An existing Azure blob container name. This argument indicates to MCG about the bucket to use as a target bucket for its backingstore, and subsequently, data storage and administration.

Example output:

```
INFO[0001] ✅ Exists: NooBaa "noobaa"
INFO[0002] ✅ Created: BackingStore "azure-resource"
INFO[0002] ✅ Created: Secret "backing-store-secret-azure-resource"
```

Create an Azure-backed backingstore using a YAML file

1. Create a secret with the credentials:

```
apiVersion: v1
kind: Secret
metadata:
  name: <backingstore-secret-name>
type: Opaque
data:
  AccountName: <AZURE ACCOUNT NAME ENCODED IN BASE64>
  AccountKey: <AZURE ACCOUNT KEY ENCODED IN BASE64>
```

AZURE ACCOUNT NAME ENCODED IN BASE64 and *AZURE ACCOUNT KEY ENCODED IN BASE64*
: Supply and encode your own Azure Account Name and Account Key using Base64, and use the results in place of these attributes respectively.

backingstore-secret-name
: A unique name of backingstore secret.

2. Apply the following YAML for a specific backingstore:

```
apiVersion: noobaa.io/v1alpha1
kind: BackingStore
metadata:
  finalizers:
  - noobaa.io/finalizer
  labels:
    app: noobaa
  name: bs
  namespace: openshift-storage
spec:
  azureBlob:
    secret:
      name: <backingstore-secret-name>
      namespace: openshift-storage
    targetBlobContainer: <blob-container-name>
  type: azure-blob
```

blob-container-name
: An existing Azure blob container name. This argument indicates to the MCG about the bucket to use as a target bucket for its backingstore, and subsequently, data storage and administration.

backingstore-secret-name
: The name of the backingstore secret created in Creating a new backing store.

Creating a GCP-backed backingstore

Create a GCP-backed backing store using the MCG command-line interface or a YAML file to add storage resources.

Before you begin, ensure the following:

- Fusion Data Foundation Platform is running.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
  ```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

Create a GCP-backed backingstore using MCG command-line interface

From the MCG command-line interface, run the following command:

```
odf noobaa backingstore create google-cloud-storage <backingstore_name> --private-key-json-file=<PATH TO GCP PRIVATE KEY JSON FILE> --target-bucket <GCP bucket name> -n openshift-storage
```

backingstore_name
: Name of the backingstore.

<PATH TO GCP PRIVATE KEY JSON FILE>
: A path to your GCP private key created for this purpose.

GCP bucket name
: An existing GCP object storage bucket name. This argument tells the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.
The output will be similar to the following:

```
INFO[0001] ✅ Exists: NooBaa "noobaa"
INFO[0002] ✅ Created: BackingStore "google-gcp"
INFO[0002] ✅ Created: Secret "backing-store-google-cloud-storage-gcp"
```

Create a GCP-backed backingstore using a YAML file

1. Create a secret with the credentials:

```
apiVersion: v1
kind: Secret
metadata:
  name: <backingstore-secret-name>
type: Opaque
```

```
data:
  GoogleServiceAccountPrivateKeyJson: <GCP PRIVATE KEY ENCODED IN BASE64>
```

GCP PRIVATE KEY ENCODED IN BASE64
: Provide and encode your own GCP service account private key using Base64, and use the results for this attribute.

backingstore-secret-name
: A unique name of the backingstore secret.

2. Apply the following YAML for a specific backing store:

```
apiVersion: noobaa.io/v1alpha1
kind: BackingStore
metadata:
  finalizers:
  - noobaa.io/finalizer
  labels:
    app: noobaa
  name: bs
  namespace: openshift-storage
spec:
  googleCloudStorage:
    secret:
      name: <backingstore-secret-name>
      namespace: openshift-storage
    targetBucket: <target bucket>
  type: google-cloud-storage
```

target bucket
: An existing Google storage bucket. This argument indicates to the MCG about the bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.

backingstore-secret-name
: The name of the backingstore secret created in Creating a new backing store.

Creating a local Persistent Volume-backed backingstore

Create a local Persistent Volume-backed backing store using the MCG command-line interface or a YAML file to add storage resources.

Before you begin, ensure the following:

- Fusion Data Foundation Platform is running.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.
  - For IBM Power, use the following command:

    ```
    subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
    ```

  - For IBM Z infrastructure, use the following command:

    ```
    subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
    ```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

Create a local Persistent Volume-backed backingstore using MCG command-line interface

From the MCG command-line interface, run the following command:
Note: This command must be run from within the `openshift-storage` namespace.

```
odf noobaa -n openshift-storage backingstore create pv-pool <backingstore_name> --num-volumes <NUMBER OF VOLUMES>  --pv-size-gb
<VOLUME SIZE> --request-cpu <CPU REQUEST> --request-memory <MEMORY REQUEST> --limit-cpu <CPU LIMIT> --limit-memory <MEMORY
LIMIT> --storage-class <LOCAL STORAGE CLASS>
```

backingstore_name
: The name of the backingstore.

NUMBER OF VOLUMES
: The number of volumes you would like to create. Increasing the number of volumes scales up the storage.

VOLUME SIZE
: Required size in GB of each volume.

CPU REQUEST
: Guaranteed amount of CPU requested in CPU unit `m`.

MEMORY REQUEST
: Guaranteed amount of memory requested.

CPU LIMIT
: Maximum amount of CPU that can be consumed in CPU unit `m`.

MEMORY LIMIT
: Maximum amount of memory that can be consumed.

LOCAL STORAGE CLASS
: The local storage class name, recommended to use `ocs-storagecluster-ceph-rbd`.

Create a local Persistent Volume-backed backingstore using a YAML file

Apply the following YAML for a specific backing store:

```
apiVersion: noobaa.io/v1alpha1
kind: BackingStore
metadata:
  finalizers:
  - noobaa.io/finalizer
  labels:
    app: noobaa
  name: <backingstore_name>
  namespace: openshift-storage
spec:
   pvPool:
    numVolumes: <NUMBER OF VOLUMES>
    resources:
      requests:
        storage: <VOLUME SIZE>
        cpu: <CPU REQUEST>
        memory: <MEMORY REQUEST>
      limits:
        cpu: <CPU LIMIT>
        memory: <MEMORY LIMIT>
    storageClass: <LOCAL STORAGE CLASS>
  type: pv-pool
```

backingstore_name
: The name of the backingstore.

NUMBER OF VOLUMES
: The number of volumes you would like to create. increasing the number of volumes scales up the storage.

VOLUME SIZE
: Required size in GB of each volume.

CPU REQUEST
: Guaranteed amount of CPU requested in CPU unit `m`.

MEMORY REQUEST
: Guaranteed amount of memory requested.

CPU LIMIT
: Maximum amount of CPU that can be consumed in CPU unit `m`.

MEMORY LIMIT
: Maximum amount of memory that can be consumed.

LOCAL STORAGE CLASS
: The local storage class name, recommended to use `ocs-storagecluster-ceph-rbd`.

Example output:

```
INFO[0001] ✅ Exists: NooBaa "noobaa"
INFO[0002] ✅ Exists: BackingStore "local-mcg-storage"
```

Creating an s3 compatible Multicloud Object Gateway backingstore

Create an S3 compatible MCG backing store for IBM Storage Ceph's RGW. When the RGW is deployed, Fusion Data Foundation operator creates an S3 compatible backingstore for MCG automatically.

The Multicloud Object Gateway (MCG) can use any S3 compatible object storage as a backing store. For example, IBM Storage Ceph's RADOS Object Gateway (RGW).

Creating an s3 compatible Multicloud Object Gateway backingstore using the MCG command-line interface

- Run the following command:
  Note: This command must be run from within the `openshift-storage` namespace.

  ```
  odf noobaa backingstore create s3-compatible rgw-resource --access-key=<RGW ACCESS KEY> --secret-key=<RGW SECRET KEY> --
  target-bucket=<bucket-name> --endpoint=<RGW endpoint> -n openshift-storage
  ```

  Example output:

  ```
  INFO[0001] ✅ Exists: NooBaa "noobaa"
  INFO[0002] ✅ Created: BackingStore "rgw-resource"
  INFO[0002] ✅ Created: Secret "backing-store-secret-rgw-resource"
  ```

- To get the *RGW ACCESS KEY* and *RGW SECRET KEY*, run the following command using your RGW user secret name:

  ```
  oc get secret <RGW USER SECRET NAME> -o yaml -n openshift-storage
  ```

  Decode the access key ID and the access key from Base64 and keep them.

  RGW ACCESS KEY
  RGW SECRET KEY
  : Decoded data

  bucket-name
  : An existing RGW bucket name. This argument tells the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.

  RGW endpoint
  : To get the *RGW endpoint*, see Accessing the RADOS Object Gateway S3 endpoint.

Creating an s3 compatible Multicloud Object Gateway backingstore using a YAML file

1. Create a `CephObjectStore` user. This also creates a secret containing the RGW credentials:

   ```
   apiVersion: ceph.rook.io/v1
   kind: CephObjectStoreUser
   metadata:
     name: <RGW-Username>
     namespace: openshift-storage
   spec:
     store: ocs-storagecluster-cephobjectstore
     displayName: "<Display-name>"
   ```

   RGW-Username
   : A unique RGW user name.

   Display-name
   : A display name

2. Apply the following YAML for an S3-Compatible backing store:

   ```
   apiVersion: noobaa.io/v1alpha1
   kind: BackingStore
   metadata:
     finalizers:
     - noobaa.io/finalizer
     labels:
       app: noobaa
     name: <backingstore-name>
     namespace: openshift-storage
   spec:
     s3Compatible:
       endpoint: <RGW endpoint>
       secret:
         name: <backingstore-secret-name>
         namespace: openshift-storage
       signatureVersion: v4
       targetBucket: <RGW-bucket-name>

     type: s3-compatible
   ```

   backingstore-secret-name
   : Name of the secret that was created with `CephObjectStore` in the previous step.

   bucket-name
   : An existing RGW bucket name. This argument tells the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.

   RGW endpoint
   : To get the *RGW endpoint*, see Accessing the RADOS Object Gateway S3 endpoint.

Creating a new bucket class

Bucket class is a CRD representing a class of buckets that defines tiering policies and data placements for an Object Bucket Class (OBC). Use this procedure to create a bucket class in Fusion Data Foundation.

Procedure

1. From the OpenShift Web Console, go to Storage > Data Foundation.
2. From the Bucket Class tab click Create Bucket Class.
3. On the Create new Bucket Class page, perform the following:
   a. Select the bucket class type by choosing one of the following options:
      - Standard: data will be consumed by a Multicloud Object Gateway (MCG), deduped, compressed and encrypted.
      - Namespace: data is stored on the NamespaceStores without performing de-duplication, compression or encryption.
        By default, Standard is selected.

   b. Enter a Bucket Class Name.
   c. Click Next.
   d. In Placement Policy, select Tier 1 - Policy Type and click Next.
      You can choose one of the options as per your requirements.

- Spread allows spreading of the data across the chosen resources.
    - Mirror allows full duplication of the data across the chosen resources.

e. To add another policy tier, click Add Tier.

f. Select at least one Backing Store resource from the available list if you have selected Tier 1 - Policy Type as Spread and click Next.
Alternatively, you can also Creating a new backing store.
Note: You need to select at least two backing stores when you select Policy Type as Mirror in step 3.d.

What to do next

To verify the creation of the new bucket class:

1. In the OpenShift Web Console, click **Storage** → **Data Foundation**.
2. Click the **Bucket Class** tab and search the new Bucket Class.

Editing a bucket class

Use this information to edit an existing bucket class.

Before you begin

Ensure you have Administrator access to OpenShift Web Console.

About this task

Use the following procedure to edit the bucket class components through the YAML file by clicking the edit button on the OpenShift web console.

Procedure

1. From the OpenShift Web Console, go to Storage > Data Foundation.
2. From the Bucket Class tab, next to the Bucket class you want to edit, click Action Menu > Edit Bucket Class.
   You are redirected to the YAML file.
3. Make the required changes in this file and click Save.

Editing backing stores for bucket class

Use this procedure to edit an existing Multicloud Object Gateway (MCG) bucket class to change the underlying backing stores used in a bucket class.

Before you begin

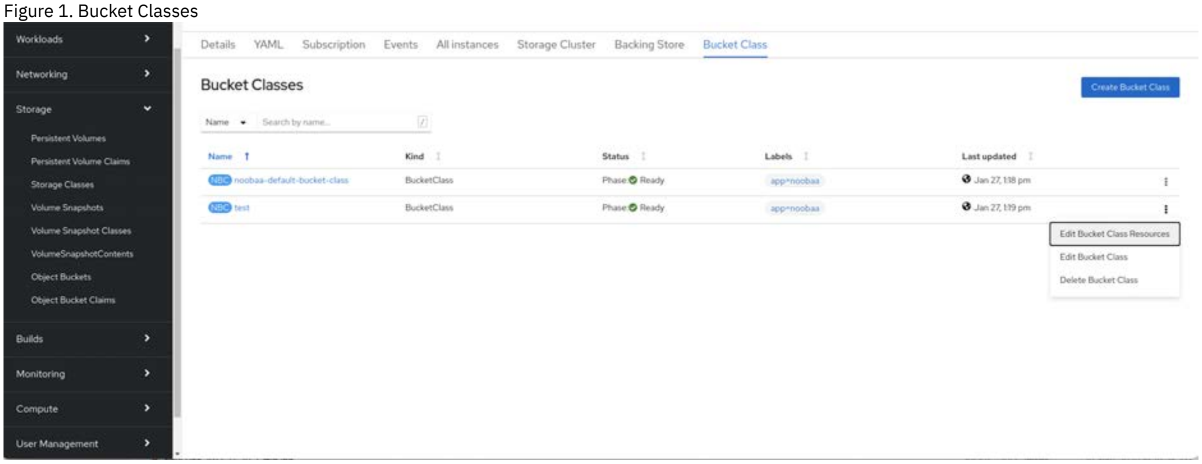
Ensure you have the following:

- Administrator access to OpenShift Web Console
- A bucket class
- Backing stores

Procedure

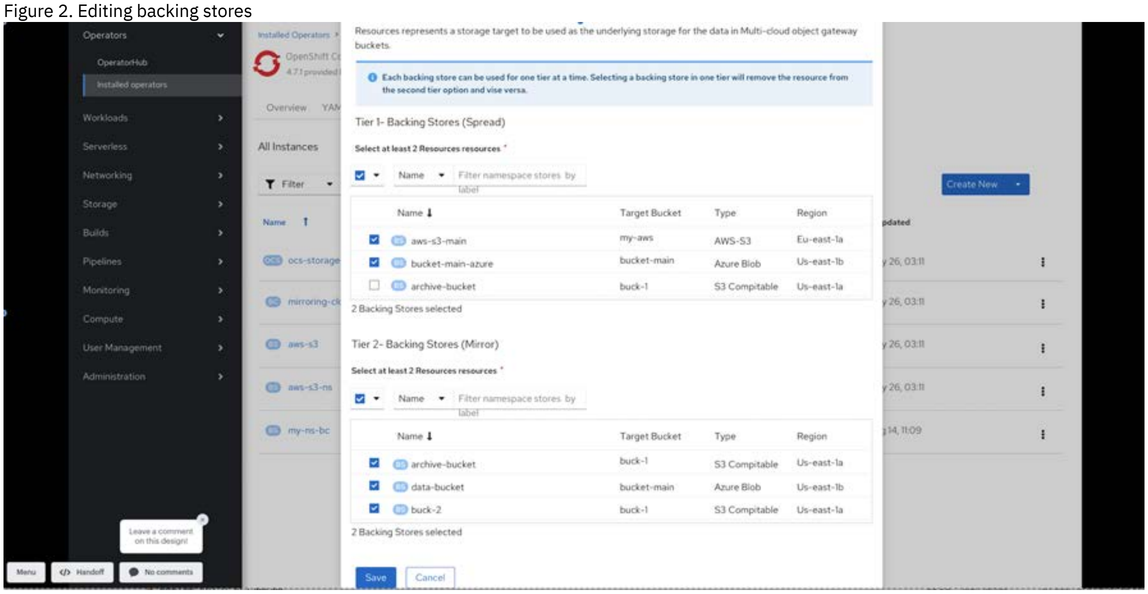
1. From the OpenShift Web Console, go to Storage > Data Foundation.
2. From the Bucket Class tab, next to the Bucket class you want to edit, click Action Menu > Edit Bucket Class Resources.

Figure 1. Bucket Classes



3. On the Edit Bucket Class Resources page, edit the bucket class resources.
   Edit the bucket class resources either by adding a backing store to the bucket class or by removing a backing store from the bucket class. You can also edit bucket class resources created with one or two tiers and different placement policies.
   - To add a backing store to the bucket class, select the name of the backing store.
   - To remove a backing store from the bucket class, clear the name of the backing store.

   Figure 2. Editing backing stores

4. Click Save.

Creating and managing buckets using the object browser

The object browser in the OpenShift Web Console enables you to create, view, and manage buckets and objects across different storage endpoints, including Multicloud Object Gateway (MCG) and internal mode RADOS Gateway (RGW) object storage.

A storage endpoint selector is available under Storage > Object Storage > Buckets, allowing you to switch between the MCG and RGW. After selecting an endpoint, the object browser presents a unified interface for creating and managing buckets, uploading and viewing objects, configuring policies, and performing related operations.

- **Accessing the object browser**
  Learn how to access the object browser based on your user role: administrators or developers.
- **Creating and managing buckets**
  Create, view, and manage buckets for MCG and RGW storage endpoints using the object browser in the OpenShift Web Console.
- **Managing objects**
  Managing objects in the Multicloud Object Gateway includes uploading, downloading, and organizing S3-compatible objects stored across backing stores.
- **Creating and managing lifecycle rules**
  S3 bucket lifecycle rules automate object expiration and transition actions in the Multicloud Object Gateway to manage storage costs and data retention.
- **Creating and managing IAM user**
  Use this section to create and manage IAM users for object storage access.

Accessing the object browser

Learn how to access the object browser based on your user role: administrators or developers.

The access flow depends on your user role:

- **Administrator users**

  Administrators have direct access to the object browser without additional login steps.

- **Developer users**

  Developers must sign in before accessing the object browser as illustrated in Signing in to the object browser as a developer user.

- **Signing in to the object browser as a developer user**
  Sign in to the object browser with Kubernetes Secret credentials.

Signing in to the object browser as a developer user

Sign in to the object browser with Kubernetes Secret credentials.

Before you begin

- Kubernetes Secret containing S3 credentials
- Namespace and Secret name details

Procedure

1. Open the object browser in the OpenShift Web Console.
2. Select the storage endpoint:
    - MCG
    - RGW

   Developers must log in separately for each endpoint.
3. Select the Secret namespace and name where the S3 credentials are stored.
4. Optional: Select Stay signed in to remain logged in across browser tabs or sessions. Click Sign in.
   Note:
    - The UI does not store credentials in local or session storage but only a reference to the Kubernetes Secret is stored.
    - Access level depends on the credentials provided.

Creating and managing buckets

Create, view, and manage buckets for MCG and RGW storage endpoints using the object browser in the OpenShift Web Console.

- **Creating new buckets using the object browser**
  Create new buckets using the object browser through Object Bucket Claims or the S3 API.
- **Listing bucket details using the object browser**
  View bucket details in the object browser for MCG or RGW storage endpoints.
- **Deleting or emptying bucket using object browser**
  Delete or empty buckets by using the object browser in the OpenShift Web Console.

Creating new buckets using the object browser

Create new buckets using the object browser through Object Bucket Claims or the S3 API.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then click the Buckets tab.
3. Choose one of the following options to create the buckets:
    - **Create via Object Bucket Claim**
      Note: For RGW object bucket claims (OBCs), you must use the credentials associated with the OBC to access the newly created bucket. RGW administrator credentials cannot be used to access these buckets.
        - Select a namespace.
        - Enter the ObjectBucketClaim name.
        - Select the storage class.
        - Select the bucket class.
        - Select the Enable replication to obtain higher resiliency of the objects stored in the buckets.
        - Click Create using ObjectBucketClaim.
    - **Create via S3 API**
        - Enter a name for the bucket.
        - Add tags with different key and value criteria to tag your bucket.
        - Click Create.

Listing bucket details using the object browser

View bucket details in the object browser for MCG or RGW storage endpoints.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then click the Buckets tab.
3. Select the required bucket and then click the Details tab.

Deleting or emptying bucket using object browser

Delete or empty buckets by using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW and click the Buckets tab.
3. From the bucket's options menu, select Empty bucket.
4. Enter the bucket name to confirm.
5. Click Empty bucket.
6. After the bucket is empty, open the options menu again.
7. Select Delete bucket.
8. Enter the bucket name to confirm and click Delete bucket.

Managing objects

Managing objects in the Multicloud Object Gateway includes uploading, downloading, and organizing S3-compatible objects stored across backing stores.

- **Listing objects and object details using MCG object browser**
  View objects and object details using the object browser in the OpenShift Web Console.
- **Downloading or previewing objects using the object browser**
  Download or preview objects using the object browser in the OpenShift Web Console.
- **Generating the presigned URL to share the object**
  Generate the presigned URL to share an object using the object browser in the OpenShift Web Console.
- **Uploading objects using the object browser**
  Upload objects using the object browser in the OpenShift Web Console.
- **Creating folder using the object browser**
  Create a folder using the object browser in the OpenShift Web Console.
- **Deleting objects using the object browser**
  Delete objects using the object browser in the OpenShift Web Console.
- **Enabling or suspending object versioning in a bucket using the object browser**
  Enable or suspend objects versioning in a bucket using the object browser in the OpenShift Web Console.
- **Listing object versions using the object browser**
  View object versions using the object browser in the OpenShift Web Console.

Listing objects and object details using MCG object browser

View objects and object details using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, click Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW and click the Buckets tab.
3. Select the required bucket and click the Objects tab.
4. Select the object to open the details in the sidebar.

Downloading or previewing objects using the object browser

Download or preview objects using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, click Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the required bucket.
3. Click the Objects tab.
4. From the object's options menu, select Download or Preview.
   Note: Preview behavior depends on browser support. Unsupported file types that cannot be displayed in a separate tab are downloaded automatically.

Generating the presigned URL to share the object

Generate the presigned URL to share an object using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, click Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. Click the Objects tab.
4. From the object's options menu, select Share with presigned URL.
5. Select the expiration time to set the validity period of the presigned URL.
6. Click Create.
7. Copy the generated URL.

Uploading objects using the object browser

Upload objects using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. Click the Objects tab.
4. In the Add objects section, do one of the following:
   - Click Upload to add a folder.
   - Drag and drop individual files to upload specific files.
5. Click the uploaded object to view its Overview and Versions tab.

Creating folder using the object browser

Create a folder using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, click Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. Click the Objects tab.
4. Click Create folder.
5. Enter a folder name.
6. Click Create.
   Note: The folder persists only after at least one object is uploaded inside it.

Deleting objects using the object browser

Delete objects using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, click Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. Click the Objects tab.
4. To delete objects, do one of the following:
   - **Multiple objects**: Select multiple objects using the checkbox and then from the Actions menu, select Delete.
   - **Single object**: Use the object's Action menu and select Delete.

   Note: Folders cannot be deleted directly. Empty folders disappear automatically.

Enabling or suspending object versioning in a bucket using the object browser

Enable or suspend objects versioning in a bucket using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. Click the Properties tab.
4. Toggle Versioning to Enabled or Disabled.
5. Click Enable or Suspend to confirm depending on your versioning selection.
   Remember: Update lifecycle rules after enabling versioning.

Listing object versions using the object browser

View object versions using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. Click the Objects tab.
4. Toggle List of versions.
   - When List all versions is OFF:

- The current version of the object excluding the delete markers is displayed.
- Any Action menu options apply to the current version.
- To view all versions of one object:
    - Click object and open the Versions tab from the sidebar.
- When List all versions is ON:
    - All object versions appear including deleted objects that have delete markers.
    - Deleting a version is permanent.
    - Removing a delete marker restores the most recent version. If no previous version exists, the object is permanently deleted.

Creating and managing lifecycle rules

S3 bucket lifecycle rules automate object expiration and transition actions in the Multicloud Object Gateway to manage storage costs and data retention.

- **Creating new lifecycle rules by using the object browser**
  Create lifecycle rules using the object browser in the OpenShift Web Console.
- **Editing lifecycle rules by using the object browser**
  Edit existing lifecycle rules using the object browser in the OpenShift Web Console.
- **Creating CORS rule**
  Configure cross-origin resource sharing (CORS) rules using the object browser in the OpenShift Web Console.
- **Editing CORS rule**
  Edit cross-origin resource sharing (CORS) rules using the object browser in the OpenShift Web Console.
- **Applying bucket policy to grant public or restricted access to objects**
  Apply or edit bucket policies using the object browser to control public or restricted access to objects.
- **Setting up public access limit to S3 resources**
  Configure public access restrictions for S3-compatible buckets by using the object browser.

Creating new lifecycle rules by using the object browser

Create lifecycle rules using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
   You can also create a new bucket using the steps from the Creating new buckets using the object browser.
3. Click the Management tab.
4. Click Create lifecycle rule.
5. Enter a name for the lifecycle rule.
6. Select a Rule scope:
    - Targeted (filters: prefix, tags, size)
    - Global (Bucket-wide)
7. Add one or more actions:
    - Delete an object after a specified time
      For a versioned bucket, the current version of the object is retained as a noncurrent version. For a nonversioned object, the object is deleted permanently.
    - Noncurrent versions of the objects
      This rule applies to only the versioned objects and deletes the older versions of the objects after they become noncurrent.
    - Incomplete multipart uploads
      This rule cleans up the abandoned uploads that were initiated but never completed after a specified time to prevent unnecessary storage costs. The object tags and size filter that were specified for the object with the targeted rule scope are not applicable for this rule and it deletes all multipart uploads from the bucket regardless of any filter.
    - Expired object delete markers
      This rule removes any unnecessary delete markers in versioned buckets that do not have any associated object versions, which might clutter bucket listings. The object tags and size filter that were specified for the object with the targeted rule scope are not applicable for this rule and it deletes all multipart uploads from the bucket regardless of any filter.

      Note:
        - The object tags and object size filters are not applicable for this rule. The expired object delete markers are removed from the bucket regardless of any filters that are configured.
        - This rule is not enabled when deleting an object after a specified time rule is selected.
8. Click Save.

Editing lifecycle rules by using the object browser

Edit existing lifecycle rules using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. In the bucket view, click the Management tab.
4. Locate the rule, then from the Action menu select Edit lifecycle rule.
5. Edit the required fields.
6. Click Save.

Creating CORS rule

Configure cross-origin resource sharing (CORS) rules using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. In the bucket view, click the Permission tab and then click the CORS tab.
4. Click Create CORS rule.
5. Provide a name for the rule.
6. Configure allowed origins*.
7. Select Allowed methods.
8. Configure allowed headers.
9. Enter the maximum age of preflight request in seconds.
10. Click Create.

Editing CORS rule

Edit cross-origin resource sharing (CORS) rules using the object browser in the OpenShift Web Console.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. In the bucket view, click the Permission tab and then click the CORS tab.
4. Select the CORS rule that you want to edit.
5. From the Actions menu, click Edit configuration.
6. Modify the required fields.
7. Click Save.

Applying bucket policy to grant public or restricted access to objects

Apply or edit bucket policies using the object browser to control public or restricted access to objects.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.

- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. In the bucket view, click the Permission tab.
4. In the Bucket policy tab, you can do one of the following:
   - When no policy exists, you can do one of the following:
     - Drag a file from another location or upload a file by using the Browse button.
     - Start from scratch or use a predefined policy configuration.

       Select Use a predefined policy configuration.

       - Create a new policy or choose a policy from Available policies.
       - Update the policy as required.
       - Click Apply policy.
       - Click Confirm to update the policy after reviewing the changes.
   - When a policy exists, use a predefined policy configuration by selecting the Edit bucket policy from the Action menu.
     - Select Use a predefined policy configuration.
     - Choose a policy from Available policies.
     - Update the policy as required.
     - Click Apply policy.
     - Click Confirm to update the policy after reviewing the changes.

   Note: After you apply the bucket policy configuration, it can take up to 120 seconds to take effect.

Setting up public access limit to S3 resources

Configure public access restrictions for S3-compatible buckets by using the object browser.

Before you begin

- **Administrators**: Administrator access to Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW, then open the bucket.
3. Select a bucket for which you need to set the public access limitation.
4. In the bucket view, click the Permission tab.
5. Click the Block public access tab.
6. Click Manage public access settings to set the limitations.
   Select one of the following options based on your requirement:
   - Block all public access.

     This blocks all forms of public access to the bucket and its objects.
   - Block public access to buckets and objects that are granted through new public bucket policies.

     This prevents new policies from making your data public.
   - Block public and cross-account access to buckets and objects through any public bucket policies.

     This blocks all public and cross-account access that is granted by any existing policies.
7. Click Save changes.

Creating and managing IAM user

Use this section to create and manage IAM users for object storage access.

Note the following considerations when creating and managing an IAM user:

- **Bucket policy principals and ARNs:** Bucket policies can reference either the account ID/ARN (for administrator or NooBaa CLI-created accounts) or the IAM user ARN. On upgraded clusters, bucket-policy principals previously defined using the email-based account format (account@noobaa.io) are automatically converted to the ARN format (arn:aws:iam::account_id:root).
- **S3 access requires a policy:** An IAM user's S3 requests return **AccessDenied** if no IAM policy is attached.
- **OBC account limitations:** Object Bucket Claim (OBC) accounts cannot create IAM users.
- **Default policy review:** IAM users created through the IAM console are automatically assigned a default policy that grants full access.

- **Creating an IAM user using the object browser**
  Learn how to create IAM users for object storage access using the object browser.

- **Generating additional access key for an IAM user**
  Learn how to generate an additional access key for an IAM user. You can generate one additional access key for the IAM user. IAM users can generate additional access key for themselves.
- **Activating or deactivating access key**
  You can generate one additional access key.
- **Deleting access key for an IAM user**
  Delete an access key for an IAM user.
- **Deleting an IAM user**
  Delete an IAM user from the object browser.

Creating an IAM user using the object browser

Learn how to create IAM users for object storage access using the object browser.

Before you begin

- **Administrators**: Administrator access to IBM Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW.
3. Click the IAM tab.
4. Click Create User.
5. Enter the required details in the Create User form.
   After the user is created, a window appears prompting you to download the secret key.
6. Click Download secret key, or copy the key and store it securely.
   Note:
   - When a new IAM user is created, an inline policy permitting S3 operations is generated automatically.
   - Access keys are also created automatically as part of the user creation workflow. No manual configuration is required.

Generating additional access key for an IAM user

Learn how to generate an additional access key for an IAM user. You can generate one additional access key for the IAM user. IAM users can generate additional access key for themselves.

Before you begin

- **Administrators**: Administrator access to IBM Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW.
3. Click the IAM tab.
4. Select the IAM user for whom you want to generate access key.
5. On the user details page, click the Access keys tab.
   The existing access key cards are displayed.
6. Click Generate another access key.
7. Optional: Enter a description for the new access key.
8. Click Generate.
   The newly generated access key card is displayed.

Results

The additional access key is generated and displayed in the Access keys tab.

Activating or deactivating access key

You can generate one additional access key.

Before you begin

- **Administrators**: Administrator access to IBM Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.

Note: Object Bucket Claim (OBC) based accounts do not have permissions to manage IAM users.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW.
3. Click the IAM tab.
4. Select the IAM user for whom you want to activate or deactivate access keys.
5. On the user details page, click the Access keys tab.
   The existing access key cards are displayed.
6. Select the Actions menu corresponding to the required Access key and select Activate/Deactivate Access key.
7. Click Activate or Deactivate depending on the existing status.

Deleting access key for an IAM user

Delete an access key for an IAM user.

Before you begin

- **Administrators**: Administrator access to IBM Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.
- An existing IAM user with at least one access key.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW.
3. Click the IAM tab.
4. Select the IAM user for whom you want to delete the access key.
5. On the user details page, click the Access keys tab.
   The list of access key cards are displayed.
6. Locate the access key you want to delete and open the Action menu on its card.
7. Select Delete access key.
8. You must deactivate the access key before deleting it. Click Deactivate.
9. To confirm deletion, type `delete` in the confirmation field.
10. Click Delete to permanently remove the access key.

Deleting an IAM user

Delete an IAM user from the object browser.

Before you begin

- **Administrators**: Administrator access to IBM Fusion Data Foundation.
- **Developers**: Logged in to the object browser using secret namespace and secret name. For more information see Accessing the object browser.
- Ensure that the selected storage endpoint (MCG or RGW) is deployed and its pods are healthy.
- All access keys and inline user policies associated with the user must be removed before deletion.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage.
2. From the Storage endpoint selector, choose MCG or RGW.
3. Click the IAM tab.
4. Select the IAM user you want to delete.
5. Click the Actions menu and select Delete user.
6. You must delete any existing access keys before deleting the user.
7. Type `delete` to confirm the user deletion.

Managing namespace buckets

Namespace buckets let you connect data repositories on different providers together, so you can interact with all of your data through a single unified view. Add the object bucket associated with each provider to the namespace bucket, and access your data through the namespace bucket to see all of your object buckets at once. This lets you write to your preferred storage provider while reading from multiple other storage providers, greatly reducing the cost of migrating to a new storage provider. Use this information to add namespace buckets using command-line interface, YAML, and user interface and for information about sharing and accessing legacy data.

Note: A namespace bucket can only be used if its write target is available and functional.

- **Amazon S3 API endpoints for objects in namespace buckets**
  Amazon S3 API endpoints supported for objects in namespace buckets list the operations that IBM Fusion Data Foundation supports when interacting with namespace bucket data.
- **Adding a namespace bucket using the Multicloud Object Gateway CLI and YAML**
  A namespace bucket can be added using either Multicloud Object Gateway command-line interface or YAML files.
- **Adding a namespace bucket using the OpenShift Container Platform user interface**
  You can add namespace buckets using the OpenShift Container Platform user interface.
- **Sharing legacy application data with cloud native application using S3 protocol**
  Many legacy applications use file systems to share data sets. You can access and share the legacy data in the file system by using the S3 operations.
- **Configuring or modifying the PublicAccessBlock configuration for S3 bucket**
  Learn how to restrict public access to S3 buckets, enhancing data security by configuring or modifying the PublicAccessBlock policy through `put-bucket-policy` S3 API.

Amazon S3 API endpoints for objects in namespace buckets

Amazon S3 API endpoints supported for objects in namespace buckets list the operations that IBM Fusion Data Foundation supports when interacting with namespace bucket data.

You can interact with objects in the namespace buckets using the Amazon Simple Storage Service (S3) API.

IBM Fusion Data Foundation supports the following namespace bucket operations:

- ListObjectVersions
- ListObjects
- PutObject
- CopyObject
- ListParts
- CreateMultipartUpload
- CompleteMultipartUpload
- UploadPart
- UploadPartCopy
- AbortMultipartUpload
- GetObjectAcl
- GetObject
- HeadObject
- DeleteObject
- DeleteObjects

See the Amazon S3 API reference documentation for the most up-to-date information about these operations and how to use them.

Related information

- Amazon S3 REST API Reference
- Amazon S3 CLI Reference

Adding a namespace bucket using the Multicloud Object Gateway CLI and YAML

A namespace bucket can be added using either Multicloud Object Gateway command-line interface or YAML files.

Depending on the type of your deployment and whether you want to use YAML or the Multicloud Object Gateway CLI, choose one of the following procedures to add a namespace bucket:

For more information about namespace buckets, see Managing namespace buckets.

- **Adding an AWS S3 namespace bucket using YAML**
  Add an AWS S3 namespace bucket using a YAML file.
- **Adding an IBM COS namespace bucket using YAML**
  Add an IBM COS namespace bucket using a YAML file.
- **Adding an AWS S3 namespace bucket using the Multicloud Object Gateway CLI**
  Add an AWS S3 namespace bucket using the Multicloud Object Gateway CLI.
- **Adding an IBM COS namespace bucket using the Multicloud Object Gateway CLI**
  Add an IBM COS namespace bucket using the Multicloud Object Gateway CLI.

Adding an AWS S3 namespace bucket using YAML

Add an AWS S3 namespace bucket using a YAML file.

Before you begin

- Install OpenShift Container Platform with Fusion Data Foundation operator.
- Ensure you have access to the Multicloud Object Gateway (MCG), see Accessing the Multicloud Object Gateway with your applications.

Procedure

1. Create a secret with the credentials:

```
apiVersion: v1
kind: Secret
metadata:
  name: <namespacestore-secret-name>
  type: Opaque
data:
  AWS_ACCESS_KEY_ID: <AWS ACCESS KEY ID ENCODED IN BASE64>
  AWS_SECRET_ACCESS_KEY: <AWS SECRET ACCESS KEY ENCODED IN BASE64>
```

namespacestore-secret-name
    Is a unique NamespaceStore name.
AWS ACCESS KEY ID ENCODED IN BASE64
AWS SECRET ACCESS KEY ENCODED IN BASE64
    You must provide and encode your own AWS access key ID and secret access key using **Base64**, and use the results in place of *AWS ACCESS KEY ID ENCODED IN BASE64* and *AWS SECRET ACCESS KEY ENCODED IN BASE64*.

2. Create a NamespaceStore resource using OpenShift custom resource definitions (CRDs).
A NamespaceStore represents underlying storage to be used as a **read** or **write** target for the data in the MCG namespace buckets.

To create a NamespaceStore resource, apply the following YAML:

```
apiVersion: noobaa.io/v1alpha1
kind: NamespaceStore
metadata:
  finalizers:
  - noobaa.io/finalizer
  labels:
    app: noobaa
  name: <resource-name>
  namespace: openshift-storage
spec:
  awsS3:
    secret:
      name: <namespacestore-secret-name>
      namespace: <namespace-secret>
    targetBucket: <target-bucket>
  type: aws-s3
```

resource-name
    The name you want to give to the resource.

namespacestore-secret-name
    The secret created in the previous step.

namespace-secret
    The namespace where the secret can be found.

target-bucket
    The target bucket you created for the NamespaceStore.

3. Create a namespace bucket class that defines a namespace policy for the namespace buckets.
The namespace policy requires a type of either single or multi.
    - A namespace policy of type single requires the following configuration:

```
apiVersion: noobaa.io/v1alpha1
kind: BucketClass
metadata:
  labels:
    app: noobaa
  name: <my-bucket-class>
  namespace: openshift-storage
spec:
  namespacePolicy:
    type:
    single:
      resource: <resource>
```

    my-bucket-class
        The unique namespace bucket class name.

    resource
        The name of a single NamespaceStore that defines the read and write target of the namespace bucket.

    - A namespace policy of type multi requires the following configuration:

```
apiVersion: noobaa.io/v1alpha1

kind: BucketClass
metadata:
  labels:
```

```
    app: noobaa
  name: my-bucket-class
  namespace: openshift-storage
spec:
  namespacePolicy:
    type: Multi
    multi:
      writeResource: <write-resource>
      readResources:
      - <read-resources>
      - <read-resources>
```

my-bucket-class
A unique bucket class name.

write-resource
The name of a single NamespaceStore that defines the **write** target of the namespace bucket.

read-resources
A list of the names of the NamespaceStores that defines the **read** targets of the namespace bucket.

4. Create a bucket using an Object Bucket Class (OBC) resource.
Use the bucket class defined in the earlier step using the following YAML:

```
apiVersion: objectbucket.io/v1alpha1
kind: ObjectBucketClaim
metadata:
  name: <resource-name>
  namespace: openshift-storage
spec:
  generateBucketName: <my-bucket>
  storageClassName: openshift-storage.noobaa.io
  additionalConfig:
    bucketclass: <my-bucket-class>
```

resource-name
The name you want to give to the resource.

my-bucket
The name you want to give to the bucket.

my-bucket-class
The bucket class created in the previous step.

After the OBC is provisioned by the operator, a bucket is created in the MCG, and the operator creates a **Secret** and **ConfigMap** with the same name and in the same namespace as that of the OBC.

Adding an IBM COS namespace bucket using YAML

Add an IBM COS namespace bucket using a YAML file.

Before you begin

- Install OpenShift Container Platform with Fusion Data Foundation operator.
- Access to the Multicloud Object Gateway (MCG), see Accessing the Multicloud Object Gateway with your applications.

Procedure

1. Create a secret with the credentials:

```
apiVersion: v1
kind: Secret
metadata:
  name: <namespacestore-secret-name>
  type: Opaque
data:
  IBM_COS_ACCESS_KEY_ID: <IBM COS ACCESS KEY ID ENCODED IN BASE64>
  IBM_COS_SECRET_ACCESS_KEY: <IBM COS SECRET ACCESS KEY ENCODED IN BASE64>
```

namespacestore-secret-name
A unique NamespaceStore name.

IBM COS ACCESS KEY ID ENCODED IN BASE64
IBM COS SECRET ACCESS KEY ENCODED IN BASE64
You must provide and encode your own IBM COS access key ID and secret access key using **Base64**, and use the results in place of *IBM COS ACCESS KEY ID ENCODED IN BASE64* and *IBM COS SECRET ACCESS KEY ENCODED IN BASE64*

2. Create a NamespaceStore resource using OpenShift custom resource definitions (CRDs).
A NamespaceStore represents underlying storage to be used as a **read** or **write** target for the data in the MCG namespace buckets.
To create a NamespaceStore resource, apply the following YAML:

```
apiVersion: noobaa.io/v1alpha1
kind: NamespaceStore
metadata:
  finalizers:
```

```
  - noobaa.io/finalizer
  labels:
    app: noobaa
  name: bs
  namespace: openshift-storage
spec:
  s3Compatible:
    endpoint: <IBM COS ENDPOINT>
    secret:
      name: <namespacestore-secret-name>
      namespace: <namespace-secret>
    signatureVersion: v2
    targetBucket: <target-bucket>
  type: ibm-cos
```

IBM COS ENDPOINT
: The appropriate IBM COS endpoint.

namespacestore-secret-name
: The secret created in step 1.

namespace-secret
: The namespace where the secret can be found.

target-bucket
: The target bucket you created for the NamespaceStore.

3. Create a namespace bucket class that defines a namespace policy for the namespace buckets.
   The namespace policy requires a type of either single or multi.
   - A namespace policy of type single requires the following configuration:

     ```
     apiVersion: noobaa.io/v1alpha1
     kind: BucketClass
     metadata:
       labels:
         app: noobaa
       name: <my-bucket-class>
       namespace: openshift-storage
     spec:
       namespacePolicy:
         type:
         single:
           resource: <resource>
     ```

     my-bucket-class
     : The unique namespace bucket class name.

     resource
     : The name of a single NamespaceStore that defines the read and write target of the namespace bucket.

   - A namespace policy of type multi requires the following configuration:

     ```
     apiVersion: noobaa.io/v1alpha1

     kind: BucketClass
     metadata:
       labels:
         app: noobaa
       name: my-bucket-class
       namespace: openshift-storage
     spec:
       namespacePolicy:
         type: Multi
         multi:
           writeResource: <write-resource>
           readResources:
           - <read-resources>
           - <read-resources>
     ```

     my-bucket-class
     : A unique bucket class name.

     write-resource
     : The name of a single NamespaceStore that defines the `write` target of the namespace bucket.

     read-resources
     : A list of the names of the NamespaceStores that defines the `read` targets of the namespace bucket.

4. Create a bucket using an Object Bucket Class (OBC) resource.
   Use the bucket class defined in the earlier step using the following YAML:

```
apiVersion: objectbucket.io/v1alpha1
kind: ObjectBucketClaim
metadata:
  name: <resource-name>
  namespace: openshift-storage
spec:
  generateBucketName: <my-bucket>
  storageClassName: openshift-storage.noobaa.io
  additionalConfig:
    bucketclass: <my-bucket-class>
```

resource-name
: The name you want to give to the resource.

my-bucket
: The name you want to give to the bucket.

my-bucket-class
: The bucket class created in the previous step.

After the OBC is provisioned by the operator, a bucket is created in the MCG, and the operator creates a `Secret` and `ConfigMap` with the same name and in the same namespace as that of the OBC.

Adding an AWS S3 namespace bucket using the Multicloud Object Gateway CLI

Add an AWS S3 namespace bucket using the Multicloud Object Gateway CLI.

Before you begin

- Install OpenShift Container Platform with Fusion Data Foundation operator.
- Ensure you have access to the Multicloud Object Gateway (MCG), see Accessing the Multicloud Object Gateway with your applications.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
  ```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

Procedure

1. In the MCG command-line interface, create a NamespaceStore resource.
   A NamespaceStore represents an underlying storage to be used as a `read` or `write` target for the data in MCG namespace buckets.

   ```
   odf noobaa namespacestore create aws-s3 <namespacestore> --access-key <AWS ACCESS KEY> --secret-key <AWS SECRET ACCESS KEY> --target-bucket <bucket-name> -n openshift-storage
   ```

   namespacestore
   : The name of the NamespaceStore.

   AWS ACCESS KEY and *AWS SECRET ACCESS KEY*
   : The AWS access key ID and secret access key you created for this purpose.

   bucket-name
   : The existing AWS bucket name. This argument tells the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.

2. Create a namespace bucket class that defines a namespace policy for the namespace buckets. The namespace policy can be either single or multi.
   - To create a namespace bucket class with a namespace policy of type single:

     ```
     odf noobaa bucketclass create namespace-bucketclass single <my-bucket-class> --resource <resource> -n openshift-storage
     ```

     resource-name
     : The name you want to give the resource.

     my-bucket-class
     : A unique bucket class name.

     resource
     : A single namespace-store that defines the `read` and `write` target of the namespace bucket.

   - To create a namespace bucket class with a namespace policy of type multi:

     ```
     odf noobaa bucketclass create namespace-bucketclass multi <my-bucket-class> --write-resource <write-resource> --read-resources <read-resources> -n openshift-storage
     ```

     resource-name
     : The name you want to give the resource.

     my-bucket-class
     : A unique bucket class name.

     write-resource
     : A single namespace-store that defines the `write` target of the namespace bucket.

     read-resources
     : A list of namespace-stores separated by commas that defines the `read` targets of the namespace bucket.

3. Create a bucket using an Object Bucket Class (OBC) resource that uses the bucket class defined in the previous step.

```
odf noobaa obc create my-bucket-claim -n openshift-storage --app-namespace my-app --bucketclass <custom-bucket-class>
```

bucket-name
A bucket name of your choice.

custom-bucket-class
The name of the bucket class created in the previous step.

After the OBC is provisioned by the operator, a bucket is created in the MCG, and the operator creates a **Secret** and a **ConfigMap** with the same name and in the same namespace as that of the OBC.

Adding an IBM COS namespace bucket using the Multicloud Object Gateway CLI

Add an IBM COS namespace bucket using the Multicloud Object Gateway CLI.

Before you begin

- Ensure you have a running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
yum install mcg
```

Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.
  - For IBM Power, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
```

  - For IBM Z infrastructure, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

Note: Choose the correct Product Variant according to your architecture.

Procedure

1. In the MCG command-line interface, create a NamespaceStore resource.
A NamespaceStore represents an underlying storage to be used as a **read** or **write** target for the data in the MCG namespace buckets.

```
odf noobaa namespacestore create ibm-cos <namespacestore> --endpoint <IBM COS ENDPOINT> --access-key <IBM ACCESS KEY> --secret-key <IBM SECRET ACCESS KEY> --target-bucket <bucket-name> -n openshift-storage
```

namespacestore
The name of the NamespaceStore.

IBM ACCESS KEY, *IBM SECRET ACCESS KEY*, **IBM COS ENDPOINT**
An IBM access key ID, secret access key, and the appropriate regional endpoint that corresponds to the location of the existing IBM bucket.

bucket-name
An existing IBM bucket name. This argument tells the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.

2. Create a namespace bucket class that defines a namespace policy for the namespace buckets. The namespace policy can be either single or multi.
   - To create a namespace bucket class with a namespace policy of type single:

```
odf noobaa bucketclass create namespace-bucketclass single <my-bucket-class> --resource <resource> -n openshift-storage
```

resource-name
The name you want to give the resource.
my-bucket-class
A unique bucket class name.
resource
A single namespace-store that defines the **read** and **write** target of the namespace bucket.

   - To create a namespace bucket class with a namespace policy of type multi:

```
odf noobaa bucketclass create namespace-bucketclass multi <my-bucket-class> --write-resource <write-resource> --read-resources <read-resources> -n openshift-storage
```

resource-name
The name you want to give the resource.

my-bucket-class
A unique bucket class name.

write-resource
A single namespace-store that defines the **write** target of the namespace bucket.

read-resources
: A list of namespace-stores separated by commas that defines the `read` targets of the namespace bucket.

3. Create a bucket using an Object Bucket Class (OBC) resource that uses the bucket class defined in the previous step.

```
odf noobaa obc create my-bucket-claim -n openshift-storage --app-namespace my-app --bucketclass <custom-bucket-class>
```

bucket-name
: A bucket name of your choice.

custom-bucket-class
: The name of the bucket class created in the previous step.

After the OBC is provisioned by the operator, a bucket is created in the MCG, and the operator creates a `Secret` and a `ConfigMap` with the same name and in the same namespace as that of the OBC.

Adding a namespace bucket using the OpenShift Container Platform user interface

You can add namespace buckets using the OpenShift Container Platform user interface.

Before you begin

- Install OpenShift Container Platform with Fusion Data Foundation operator.
- Access to the Multicloud Object Gateway (MCG).

About this task

For information about namespace buckets, see Managing namespace buckets.

Procedure

1. Log into the OpenShift Web Console and go to Storage > Object Storage > Namespace Store.
2. Create a `namespacestore` resource to be used in the namespace bucket.
   a. Click Create namespace store.
   b. Enter a namespacestore name.
   c. Choose a provider.
   d. Choose a region.
   e. Either select an existing secret, or click Switch to credentials to create a secret by entering a secret key and secret access key.
   f. Choose a target bucket.
   g. Click Create.
   h. Verify that the namespacestore is in the *Ready* state.
   i. Repeat these steps until you have the desired amount of resources.
3. From the Bucket Class tab, click Create a new Bucket Class.
   a. Select the Namespace.
   b. Enter a Bucket Class name.
   c. Optional: Add description.
   d. Click Next.
4. Choose a namespace policy type for your namespace bucket, and then click Next.
5. Select the target resources.
   - If your namespace policy type is Single, choose a read resource.
   - If your namespace policy type is Multi, choose read resources and a write resource.
   - If your namespace policy type is Cache, choose a Hub namespace store that defines the read and write target of the namespace bucket.
6. Click Next.
7. Review your new bucket class, and then click Create Bucketclass.
8. On the BucketClass page, verify that your newly created resource is in the *Created* phase.
9. Go to **Storage** → **Data Foundation**.
10. In the Status card, click Storage System and click the storage system link from the pop up that appears.
11. In the Object tab, go to Multicloud Object Gateway > Buckets > Namespace Buckets tab.
12. Click Create Namespace Bucket.
    a. From the Choose Name tab, specify a name for the namespace bucket and click Next.
    b. From the Set Placement tab:
       i. For Read Policy, select the check box for each namespace resource created in the earlier step that the namespace bucket should read data from.
       ii. If the namespace policy type you are using is Multi, then specify which namespace resource the namespace bucket should write data to within the Write Policy field.
    c. Click Next.
    d. Click Create.

What to do next

Verify that the namespace bucket is listed with a green check mark in the State column, the expected number of read resources, and the expected write resource name.

Sharing legacy application data with cloud native application using S3 protocol

Many legacy applications use file systems to share data sets. You can access and share the legacy data in the file system by using the S3 operations.

To share data:

- Export the pre-existing file system datasets, that is, RWX volume such as Ceph FileSystem (CephFS) or create a new file system datasets using the S3 protocol.
- Access file system datasets from both file system and S3 protocol.
- Configure S3 accounts and map them to the existing or a new file system unique identifiers (UIDs) and group identifiers (GIDs).

- **Creating a NamespaceStore to use a file system**
  Create a NamespaceStore to use a file system.
- **Creating accounts with NamespaceStore file system configuration**
  Create a Multicloud Object Gateway account with NamespaceStore file system configuration to enable file-system-backed object storage access by using S3 credentials.
- **Accessing legacy application data from the openshift-storage namespace**
  When using the Multicloud Object Gateway (MCG) NamespaceStore filesystem (NSFS) feature, you need to have the Persistent Volume Claim (PVC) where the data resides in the `openshift-storage` namespace. In almost all cases, the data you need to access is not in the `openshift-storage` namespace, but in the namespace that the legacy application uses. A PVC is used in order to access the data.

Creating a NamespaceStore to use a file system

Create a NamespaceStore to use a file system.

Before you begin

Ensure you have the following:

- OpenShift Container Platform with Fusion Data Foundation operator installed.
- Access to the Multicloud Object Gateway (MCG).

Procedure

1. Log into the OpenShift Web Console and go to Storage > Object Storage.
2. Go to the NamespaceStore tab to create NamespaceStore resources to be used in the namespace bucket.
3. Click Create namespacestore.
4. Enter a name for the NamespaceStore.
5. Choose Filesystem as the provider.
6. Choose the Persistent volume claim.
7. Enter a folder name.
   If the folder name exists, then that folder is used to create the NamespaceStore or else a folder with that name is created.
8. Click Create.
9. Verify the NamespaceStore is in the *Ready* state.

Creating accounts with NamespaceStore file system configuration

Create a Multicloud Object Gateway account with NamespaceStore file system configuration to enable file-system-backed object storage access by using S3 credentials.

Before you begin

Download the Multicloud Object Gateway (MCG) command-line interface:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
yum install mcg
```

About this task

You can either create a new account with `NamespaceStore` file system configuration or convert an existing normal account into a `NamespaceStore` file system account by editing the YAML.
Note: You cannot remove a `NamespaceStore` file system configuration from an account.

Procedure

Create a new account with `NamespaceStore` file system configuration by using the MCG command-line interface.

```
odf noobaa account create <noobaa-account-name> [flags]
```

For example:

```
odf noobaa account create testaccount --nsfs_account_config --gid 10001 --uid 10001 –default_resource fs_namespacestore
```

`allow_bucket_create`	Indicates whether the account is allowed to create new buckets. Supported values are `true` or `false`. Default value is `true`.
`default_resource`	The NamespaceStore resource on which the new buckets will be created when using the S3 CreateBucket operation. The NamespaceStore must be backed by a rwx (ReadWriteMany) persistent volume claim (PVC).

force_md5_etag	Enables md5 etag calculation for the user account.
new_buckets_path	The file system path where directories corresponding to new buckets will be created. The path is inside the file system of NamespaceStore file system PVCs where new directories are created to act as the file system mapping of newly created object bucket classes.
nsfs_account_config	A mandatory field that indicates if the account is used for NamespaceStore file system.
nsfs_only	Indicates whether the account is used only for NamespaceStore file system or not. Supported values are true or **false**. Default value is **false**. If it is set to 'true', it limits you from accessing other types of buckets.
uid	The user ID of the file system to which the MCG account will be mapped and it is used to access and manage data on the file system.
gid	The group ID of the file system to which the MCG account will be mapped and it is used to access and manage data on the file system.

The MCG system sends a response with the account configuration and its S3 credentials:

```
NooBaaAccount spec:
allow_bucket_creation: true
Allowed_buckets:
  full_permission: true
  permission_list: []
default_resource: noobaa-default-namespace-store
Nsfs_account_config:
  gid: 10001
  new_buckets_path: /
  nsfs_only: true
  uid: 10001
INFO[0006] ✅ Exists: Secret "noobaa-account-testaccount"
Connection info:
  AWS_ACCESS_KEY_ID      : <aws-access-key-id>
  AWS_SECRET_ACCESS_KEY  : <aws-secret-access-key>
```

What to do next

You can list all the custom resource definition (CRD) based accounts by using the following command:

```
odf noobaa account list

NAME          ALLOWED_BUCKETS   DEFAULT_RESOURCE               PHASE   AGE
testaccount   [*]               noobaa-default-backing-store   Ready   1m17s
```

If you are interested in a particular account, you can read its custom resource definition (CRD) directly by the account name:

```
oc get noobaaaccount/testaccount -o yaml
spec:
  allow_bucket_creation: true
  allowed_buckets:
    full_permission: true
    permission_list: []
  default_resource: noobaa-default-namespace-store
  nsfs_account_config:
    gid: 10001
    new_buckets_path: /
    nsfs_only: true
    uid: 10001
```

Accessing legacy application data from the openshift-storage namespace

When using the Multicloud Object Gateway (MCG) NamespaceStore filesystem (NSFS) feature, you need to have the Persistent Volume Claim (PVC) where the data resides in the **openshift-storage** namespace. In almost all cases, the data you need to access is not in the **openshift-storage** namespace, but in the namespace that the legacy application uses. A PVC is used in order to access the data.

About this task

To access data stored in another namespace, you need to create a PVC in the **openshift-storage** namespace that points to the same CephFS volume that the legacy application uses.

Procedure

1. Display the application namespace with **scc**, where *<application_namespace>* is the name of the application namespace.

   ```
   oc get ns <application_namespace> -o yaml | grep scc
   ```

   For example:

   ```
   oc get ns testnamespace -o yaml | grep scc

   openshift.io/sa.scc.mcs: s0:c26,c5
   openshift.io/sa.scc.supplemental-groups: 1000660000/10000
   openshift.io/sa.scc.uid-range: 1000660000/10000
   ```

2. Navigate into the application namespace:.

   ```
   oc project <application_namespace>
   ```

   For example:

   ```
   oc project testnamespace
   ```

3. Ensure that a ReadWriteMany (RWX) PVC is mounted on the pod that you want to consume from the noobaa S3 endpoint using the MCG NSFS feature:.

```
oc get pvc
```

```
NAME                                               STATUS VOLUME
CAPACITY ACCESS MODES STORAGECLASS                AGE
cephfs-write-workload-generator-no-cache-pv-claim  Bound  pvc-aa58fb91-c3d2-475b-bbee-68452a613e1a
10Gi     RWX          ocs-storagecluster-cephfs 12s
```

```
oc get pod
```

```
NAME                                                READY   STATUS              RESTARTS   AGE
cephfs-write-workload-generator-no-cache-1-cv892    1/1     Running             0          11s
```

4. Check the mount point of the Persistent Volume (PV) inside your pod.
   a. Get the volume name of the PV from the pod, where *<pod_name>* is the name of the pod.

   ```
   oc get pods <pod_name> -o jsonpath='{.spec.volumes[]}'
   ```

   For example:

   ```
   oc get pods cephfs-write-workload-generator-no-cache-1-cv892 -o jsonpath='{.spec.volumes[]}'
   ```

   ```
   {"name":"app-persistent-storage","persistentVolumeClaim":{"claimName":"cephfs-write-workload-generator-no-cache-pv-
   claim"}}
   ```

   In this example, the name of the volume for the PVC is `cephfs-write-workload-generator-no-cache-pv-claim`.

   b. List all the mounts in the pod, and check for the mount point of the volume that were identified in step 4.a.

   ```
   oc get pods <pod_name> -o jsonpath='{.spec.containers[].volumeMounts}'
   ```

   For example:

   ```
   oc get pods cephfs-write-workload-generator-no-cache-1-cv892 -o jsonpath='{.spec.containers[].volumeMounts}'
   ```

   ```
   [{"mountPath":"/mnt/pv","name":"app-persistent-storage"},
   {"mountPath":"/var/run/secrets/kubernetes.io/serviceaccount","name":"kube-api-access-8tnc5","readOnly":true}]
   ```

5. Confirm the mount point of the RWX PV in your pod, where *<mount_path>* is the path to the mount point that was identified in step 4.b.

```
oc exec -it <pod_name> -- df <mount_path>
```

For example:

```
oc exec -it cephfs-write-workload-generator-no-cache-1-cv892 -- df /mnt/pv
```

```
main
Filesystem
1K-blocks Used Available  Use%  Mounted on
172.30.202.87:6789,172.30.120.254:6789,172.30.77.247:6789:/volumes/csi/csi-vol-cc416d9e-dbf3-11ec-b286-
0a580a810213/edcfe4d5-bdcb-4b8e-8824-8a03ad94d67c
10485760  0    10485760   0%    /mnt/pv
```

6. Ensure that the UID and SELinux labels are the same as the ones that the legacy namespace uses:.

```
oc exec -it <pod_name> -- ls -latrZ <mount_path>
```

For example:

```
oc exec -it cephfs-write-workload-generator-no-cache-1-cv892 -- ls -latrZ /mnt/pv/
```

```
total 567
drwxrwxrwx. 3 root       root system_u:object_r:container_file_t:s0:c26,c5      2 May 25 06:35 .
-rw-r--r--. 1 1000660000 root system_u:object_r:container_file_t:s0:c26,c5 580138 May 25 06:35 fs_write_cephfs-write-
workload-generator-no-cache-1-cv892-data.log
drwxrwxrwx. 3 root       root system_u:object_r:container_file_t:s0:c26,c5     30 May 25 06:35 ..
```

7. Get the information of the legacy application RWX PV that you want to make accessible from the `openshift-storage` namespace, where *<pv_name>* is the name of the PV.

```
oc get pv | grep <pv_name>
```

For example:

```
oc get pv | grep pvc-aa58fb91-c3d2-475b-bbee-68452a613e1a
```

```
pvc-aa58fb91-c3d2-475b-bbee-68452a613e1a   10Gi       RWX            Delete            Bound    testnamespace/cephfs-write-
workload-generator-no-cache-pv-claim   ocs-storagecluster-cephfs               47s
```

8. Ensure that the PVC from the legacy application is accessible from the `openshift-storage` namespace so that one or more noobaa-endpoint pods can access the PVC.
   a. Find the values of the `subvolumePath` and `volumeHandle` from the `volumeAttributes`.
   You can get these values from the YAML description of the legacy application PV.

   ```
   oc get pv <pv_name> -o yaml
   ```

   For example:

   ```
   oc get pv pvc-aa58fb91-c3d2-475b-bbee-68452a613e1a -o yaml
   ```

   ```
   apiVersion: v1
   kind: PersistentVolume
   ```

```
metadata:
  annotations:
    pv.kubernetes.io/provisioned-by: openshift-storage.cephfs.csi.ceph.com
  creationTimestamp: "2022-05-25T06:27:49Z"
  finalizers:
  - kubernetes.io/pv-protection
  name: pvc-aa58fb91-c3d2-475b-bbee-68452a613e1a
  resourceVersion: "177458"
  uid: 683fa87b-5192-4ccf-af2f-68c6bcf8f500
spec:
  accessModes:
  - ReadWriteMany
  capacity:
    storage: 10Gi
  claimRef:
    apiVersion: v1
    kind: PersistentVolumeClaim
    name: cephfs-write-workload-generator-no-cache-pv-claim
    namespace: testnamespace
    resourceVersion: "177453"
    uid: aa58fb91-c3d2-475b-bbee-68452a613e1a
  csi:
    controllerExpandSecretRef:
      name: rook-csi-cephfs-provisioner
      namespace: openshift-storage
    driver: openshift-storage.cephfs.csi.ceph.com
    nodeStageSecretRef:
      name: rook-csi-cephfs-node
      namespace: openshift-storage
    volumeAttributes:
      clusterID: openshift-storage
      fsName: ocs-storagecluster-cephfilesystem
      storage.kubernetes.io/csiProvisionerIdentity: 1653458225664-8081-openshift-storage.cephfs.csi.ceph.com
      subvolumeName: csi-vol-cc416d9e-dbf3-11ec-b286-0a580a810213
      subvolumePath: /volumes/csi/csi-vol-cc416d9e-dbf3-11ec-b286-0a580a810213/edcfe4d5-bdcb-4b8e-8824-8a03ad94d67c
                volumeHandle: 0001-0011-openshift-storage-0000000000000001-cc416d9e-dbf3-11ec-b286-0a580a810213
  persistentVolumeReclaimPolicy: Delete
  storageClassName: ocs-storagecluster-cephfs
  volumeMode: Filesystem
status:
  phase: Bound
```

b. Use the **subvolumePath** and **volumeHandle** values that you identified in step 8.a to create a new PV and PVC object in the **openshift-storage** namespace that points to the same CephFS volume as the legacy application PV.
Example YAML file:

```
cat << EOF >> pv-openshift-storage.yaml
apiVersion: v1
kind: PersistentVolume
metadata:
  name: cephfs-pv-legacy-openshift-storage
spec:
  storageClassName: ""
  accessModes:
  - ReadWriteMany
  capacity:
    storage: 10Gi
                (1)
  csi:
    driver: openshift-storage.cephfs.csi.ceph.com
    nodeStageSecretRef:
      name: rook-csi-cephfs-node
      namespace: openshift-storage
    volumeAttributes:
    # Volume Attributes can be copied from the Source testnamespace PV
      "clusterID": "openshift-storage"
      "fsName": "ocs-storagecluster-cephfilesystem"
      "staticVolume": "true"
    # rootpath is the subvolumePath: you copied from the Source testnamespace PV
      "rootPath": /volumes/csi/csi-vol-cc416d9e-dbf3-11ec-b286-0a580a810213/edcfe4d5-bdcb-4b8e-8824-8a03ad94d67c
    volumeHandle: 0001-0011-openshift-storage-0000000000000001-cc416d9e-dbf3-11ec-b286-0a580a810213-clone
                (2)
  persistentVolumeReclaimPolicy: Retain
  volumeMode: Filesystem
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: cephfs-pvc-legacy
  namespace: openshift-storage
spec:
  storageClassName: ""
  accessModes:
  - ReadWriteMany
  resources:
    requests:
      storage: 10Gi
                (3)
  volumeMode: Filesystem
  # volumeName should be same as PV name
  volumeName: cephfs-pv-legacy-openshift-storage
EOF
```

(1)
: The storage capacity of the PV that you are creating in the **openshift-storage** namespace must be the same as the original PV.

(2)
: The volume handle for the target PV that you create in **openshift-storage** needs to have a different handle than the original application PV, for example, add **-clone** at the end of the volume handle.

(3)
: The storage capacity of the PVC that you are creating in the **openshift-storage** namespace must be the same as the original PVC.

c. Create the PV and PVC in the **openshift-storage** namespace using the YAML file specified in step 8.b, where *YAML_file* is the name of the YAML file.

```
oc create -f <YAML_file>
```

For example:

```
oc create -f pv-openshift-storage.yaml
```

```
persistentvolume/cephfs-pv-legacy-openshift-storage created
persistentvolumeclaim/cephfs-pvc-legacy created
```

d. Ensure that the PVC is available in the **openshift-storage** namespace:

```
oc get pvc -n openshift-storage
```

```
NAME                                      STATUS   VOLUME                               CAPACITY   ACCESS MODES
STORAGECLASS                       AGE
cephfs-pvc-legacy                         Bound    cephfs-pv-legacy-openshift-storage   10Gi       RWX
14s
```

e. Navigate into the **openshift-storage** project:

```
oc project openshift-storage
```

```
Now using project "openshift-storage" on server "https://api.cluster-5f6ng.5f6ng.sandbox65.opentlc.com:6443".
```

f. Create the NSFS namespacestore:

```
odf noobaa namespacestore create nsfs <nsfs_namespacestore> --pvc-name='<cephfs_pvc_name>' --fs-backend='CEPH_FS'
```

nsfs_namespacestore
: A name of the NSFS namespacestore.

cephfs_pvc_name
: A name of the CephFS PVC in the **openshift-storage** namespace.

For example:

```
odf noobaa namespacestore create nsfs legacy-namespace --pvc-name='cephfs-pvc-legacy' --fs-backend='CEPH_FS'
```

g. Ensure that the noobaa-endpoint pod restarts and that it successfully mounts the PVC at the NSFS namespacestore, for example, **/nsfs/legacy-namespace** mountpoint, where *noobaa_endpoint_pod_name* is the name of the **noobaa-endpoint** pod.

```
oc exec -it <noobaa_endpoint_pod_name> -- df -h /nsfs/<nsfs_namespacestore>
```

For example:

```
oc exec -it noobaa-endpoint-5875f467f5-546c6 -- df -h /nsfs/legacy-namespace
```

```
Filesystem
Size  Used Avail Use% Mounted on
172.30.202.87:6789,172.30.120.254:6789,172.30.77.247:6789:/volumes/csi/csi-vol-cc416d9e-dbf3-11ec-b286-
0a580a810213/edcfe4d5-bdcb-4b8e-8824-8a03ad94d67c   10G     0   10G   0% /nsfs/legacy-namespace
```

h. Create an MCG user account:

```
odf noobaa account create <user_account> --allow_bucket_create=true --new_buckets_path='/' --nsfs_only=true --
nsfs_account_config=true --gid <gid_number> --uid <uid_number> --default_resource='legacy-namespace'
```

user_account
: Specify the name of the MCG user account.

gid_number
: Specify the GID number.

uid_number
: Specify the UID number.

: Important: Use the same **UID** and **GID** as that of the legacy application. You can find it from the previous output.

For example:

```
odf noobaa account create leguser --allow_bucket_create=true --new_buckets_path='/' --nsfs_only=true --
nsfs_account_config=true --gid 0 --uid 1000660000 --default_resource='legacy-namespace'
```

i. Create an MCG bucket.

  i. Create a dedicated folder for S3 inside the NSFS share on the CephFS PV and PVC of the legacy application pod:

  ```
  oc exec -it <pod_name> -- mkdir <mount_path>/nsfs
  ```

  For example:

  ```
  oc exec -it cephfs-write-workload-generator-no-cache-1-cv892 -- mkdir /mnt/pv/nsfs
  ```

  ii. Create the MCG bucket using the nsfs/ path:

```
odf noobaa api bucket_api create_bucket '{
  "name": "<bucket_name>",
  "namespace":{
    "write_resource": { "resource": "<nsfs_namespacestore>", "path": "nsfs/" },
    "read_resources": [ { "resource": "<nsfs_namespacestore>", "path": "nsfs/" }]
  }
}'
```

For example:

```
odf noobaa api bucket_api create_bucket '{
  "name": "legacy-bucket",
  "namespace":{
    "write_resource": { "resource": "legacy-namespace", "path": "nsfs/" },
    "read_resources": [ { "resource": "legacy-namespace", "path": "nsfs/" }]
  }
}'
```

j. Check the SELinux labels of the folders residing in the PVCs in the legacy application and `openshift-storage` namespaces:

```
oc exec -it <noobaa_endpoint_pod_name> -n openshift-storage -- ls -ltraZ /nsfs/<nsfs_namespacstore>
```

For example:

```
oc exec -it noobaa-endpoint-5875f467f5-546c6 -n openshift-storage -- ls -ltraZ /nsfs/legacy-namespace


total 567
drwxrwxrwx. 3 root       root system_u:object_r:container_file_t:s0:c0,c26      2 May 25 06:35 .
-rw-r--r--. 1 1000660000 root system_u:object_r:container_file_t:s0:c0,c26 580138 May 25 06:35 fs_write_cephfs-write-workload-generator-no-cache-1-cv892-data.log
drwxrwxrwx. 3 root       root system_u:object_r:container_file_t:s0:c0,c26     30 May 25 06:35 ..
```

```
oc exec -it <pod_name> -- ls -latrZ <mount_path>
```

For example:

```
oc exec -it cephfs-write-workload-generator-no-cache-1-cv892 -- ls -latrZ /mnt/pv/
```

```
total 567
drwxrwxrwx. 3 root       root system_u:object_r:container_file_t:s0:c26,c5      2 May 25 06:35 .
-rw-r--r--. 1 1000660000 root system_u:object_r:container_file_t:s0:c26,c5 580138 May 25 06:35 fs_write_cephfs-write-workload-generator-no-cache-1-cv892-data.log
drwxrwxrwx. 3 root       root system_u:object_r:container_file_t:s0:c26,c5     30 May 25 06:35 ..
```

In these examples, you can see that the SELinux labels are not the same which results in permission denied or access issues.

9. Ensure that the legacy application and `openshift-storage` pods use the same SELinux labels on the files.
You can do this in one of the following ways:
   - Changing the default SELinux label on the legacy application project to match the one in the openshift-storage project
   - Modifying the SELinux label only for the deployment config that has the pod which mounts the legacy application PVC
10. Delete the NSFS namespacestore.
   a. Delete the MCG bucket:

   ```
   odf noobaa bucket delete <bucket_name>
   ```

   For example:

   ```
   odf noobaa bucket delete legacy-bucket
   ```

   b. Delete the MCG user account.

   ```
   odf noobaa account delete <user_account>
   ```

   For example:

   ```
   odf noobaa account delete leguser
   ```

   c. Delete the NSFS namespacestore.

   ```
   odf noobaa namespacestore delete <nsfs_namespacestore>
   ```

   For example:

   ```
   odf noobaa namespacestore delete legacy-namespace
   ```

11. Delete the PV and PVC.
Important: Before you delete the PV and PVC, ensure that the PV has a retain policy configured.

```
oc delete pv <cephfs_pv_name>
```

```
oc delete pvc <cephfs_pvc_name>
```

cephfs_pv_name
Specify the CephFS PV name of the legacy application.

cephfs_pvc_name
Specify the CephFS PVC name of the legacy application.

For example:

```
oc delete pv cephfs-pv-legacy-openshift-storage
```

For example:

```
oc delete pvc cephfs-pvc-legacy
```

- **Changing the default SELinux label on the legacy application project to match the one in the openshift-storage project**
  Ensure that the legacy application and `openshift-storage` pods use the same SELinux labels on the files, by changing the default SELinux label on the legacy application project.
- **Modifying the SELinux label only for the deployment config that has the pod which mounts the legacy application PVC**
  Ensure that the legacy application and `openshift-storage` pods use the same SELinux labels on the files, by modifying the SELinux label on the deployment config that has the pod which mounts the legacy application.

Changing the default SELinux label on the legacy application project to match the one in the openshift-storage project

Ensure that the legacy application and `openshift-storage` pods use the same SELinux labels on the files, by changing the default SELinux label on the legacy application project.

Procedure

1. Display the current `openshift-storage` namespace with `sa.scc.mcs`:

   ```
   oc get ns openshift-storage -o yaml | grep sa.scc.mcs
   openshift.io/sa.scc.mcs: s0:c26,c0
   ```

2. Edit the legacy application namespace, and modify the `sa.scc.mcs`.
   Use the value from the `sa.scc.mcs` of the `openshift-storage` namespace:

   ```
   oc edit ns <appplication_namespace>
   ```

   For example:

   ```
   oc edit ns testnamespace
   ```

   ```
   oc get ns <application_namespace> -o yaml | grep sa.scc.mcs
   ```

   For example:

   ```
   oc get ns testnamespace -o yaml | grep sa.scc.mcs
   openshift.io/sa.scc.mcs: s0:c26,c0
   ```

3. Restart the legacy application pod.
   A relabel of all the files take place and now the SELinux labels match with the `openshift-storage` deployment.

Modifying the SELinux label only for the deployment config that has the pod which mounts the legacy application PVC

Ensure that the legacy application and `openshift-storage` pods use the same SELinux labels on the files, by modifying the SELinux label on the deployment config that has the pod which mounts the legacy application.

Procedure

1. Create a new `scc` with the `MustRunAs` and `seLinuxOptions` options, with the Multi Category Security (MCS) that the `openshift-storage` project uses.
   Example YAML file:

   ```
   cat << EOF >> scc.yaml
   ```

   ```
   allowHostDirVolumePlugin: false
   allowHostIPC: false
   allowHostNetwork: false
   allowHostPID: false
   allowHostPorts: false
   allowPrivilegeEscalation: true
   allowPrivilegedContainer: false
   allowedCapabilities: null
   apiVersion: security.openshift.io/v1
   defaultAddCapabilities: null
   fsGroup:
     type: MustRunAs
   groups:
   - system:authenticated
   kind: SecurityContextConstraints
   metadata:
     annotations:
     name: restricted-pvselinux
   priority: null
   readOnlyRootFilesystem: false
   requiredDropCapabilities:
   - KILL
   - MKNOD
   - SETUID
   ```

```
- SETGID
runAsUser:
  type: MustRunAsRange
seLinuxContext:
  seLinuxOptions:
    level: s0:c26,c0
  type: MustRunAs
supplementalGroups:
  type: RunAsAny
users: []
volumes:
- configMap
- downwardAPI
- emptyDir
- persistentVolumeClaim
- projected
- secret
EOF
```

```
oc create -f scc.yaml
```

2. Create a service account for the deployment and add it to the newly created **scc**.
   a. Create a service account, where *<service_account_name>* is the name of the service account.

      ```
      oc create serviceaccount <service_account_name>
      ```

      For example:

      ```
      oc create serviceaccount testnamespacesa
      ```

   b. Add the service account to the newly created **scc**:

      ```
      oc adm policy add-scc-to-user restricted-pvselinux -z <service_account_name>
      ```

      For example:

      ```
      oc adm policy add-scc-to-user restricted-pvselinux -z testname
      ```

3. Patch the legacy application deployment so that it uses the newly created service account.
   With this change, you can specify the SELinux label in the deployment.

   ```
   oc patch dc/<pod_name> '{"spec":{"template":{"spec":{"serviceAccountName": "<service_account_name>"}}}}'
   ```

   For example:

   ```
   oc patch dc/cephfs-write-workload-generator-no-cache --patch '{"spec":{"template":{"spec":{"serviceAccountName": 
   "testnamespacesa"}}}}'
   ```

4. Edit the deployment to specify the security context to use at the SELinux label in the deployment configuration:

   ```
   oc edit dc <pod_name> -n <application_namespace>
   ```

   Add the following lines:

   ```
   spec:
    template:
       metadata:
         securityContext:
           seLinuxOptions:
             Level: <security_context_value>
   ```

   security_context_value
   : You can find this value when you run the command to create a dedicated folder for S3 inside the NSFS share on the CephFS PV and PVC of the legacy application pod.

   For example:

   ```
   oc edit dc cephfs-write-workload-generator-no-cache -n testnamespace
   ```

   ```
   spec:
    template:
       metadata:
         securityContext:
           seLinuxOptions:
             level: s0:c26,c0
   ```

5. Ensure that the security context to be used at the SELinux label in the deployment configuration is specified correctly.

   ```
   oc get dc <pod_name> -n <application_namespace> -o yaml | grep -A 2 securityContext
   ```

   For example:

   ```
   oc get dc cephfs-write-workload-generator-no-cache -n testnamespace -o yaml | grep -A 2 securityContext
   ```

   ```
   securityContext:
           seLinuxOptions:
             level: s0:c26,c0
   ```

   The legacy application is restarted and begins using the same SELinux labels as the **openshift-storage** namespace.

Configuring or modifying the PublicAccessBlock configuration for S3 bucket

Learn how to restrict public access to S3 buckets, enhancing data security by configuring or modifying the PublicAccessBlock policy through **put-bucket-policy** S3 API.

About this task

To apply PublicAccessBlock policy to the bucket, first apply a policy that grants public access, and then apply the PublicAccessBlock policy. The policy is applied through **put-bucket-policy** S3 API, which is similar to Amazon S3. For more information, see PutPublicAccessBlock in *Amazon S3 API Reference* Guide.

Important: The Multicloud Object Gateway (MCG) checks the PublicAccessBlock policy configuration for both the bucket that contains the object and the bucket owner's account during evaluating the PublicAccessBlock policy for a bucket or an object.
Restriction: Only bucket-wide policy is supported and not account-wide policy. Also, access control lists (ACLs) are not supported within the policy.

Procedure

1. Allow a general access policy.
   Command example:

```
{
  "Version": "2025-06-25",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:*",
      "Resource": [
        "arn:aws:s3:::<bucket_name>/*",
        "arn:aws:s3:::<bucket_name>"
      ]
    }
  ]
}
```

2. Set the PublicAccessBlock policy.
   The default PublicAccessBlock policy configuration is as follows:

```
{
    {
        "BlockPublicPolicy": false,
        "RestrictPublicBuckets": false
    }
}
```

   To block public policies and restrict public buckets, set the value to **true** as follows:

```
{
    {
        "BlockPublicPolicy": true,
        "RestrictPublicBuckets": true
    }
}
```

   The following configuration is not supported:

```
{
    {
        "BlockPublicAcls": false, <-- Not allowed even as false
        "IgnorePublicAcls": false, <-- Not allowed even as false
        "BlockPublicPolicy": true,
        "RestrictPublicBuckets": true
    }
}
```

3. Apply the PublicAccessBlock policy by using the **put-public-access-block** command.
   Command example:

```
alias s3api=AWS_ACCESS_KEY_ID=<access_key> AWS_SECRET_ACCESS_KEY=<secret_key> aws s3api --no-verify-ssl --endpoint 
<endpoint>  --no-verify put-public-access-block --bucket <bucket_name> --public-access-block-configuration 
file://<configuration_file_path>. * *
```

Securing Multicloud Object Gateway

Ensure that the Multicloud Object Gateway is secure.

- **Changing the default account credentials to ensure better security in the Multicloud Object Gateway**
  Change and rotate your Multicloud Object Gateway (MCG) account credentials using the command-line interface to prevent issues with applications and to ensure better account security.
- **Enabling secured mode deployment for Multicloud Object Gateway**
  You can specify a range of IP addresses that should be allowed to reach the Multicloud Object Gateway (MCG) load balancer services to enable secure mode deployment. This helps to control the IP addresses that can access the MCG services.
- **Disabling routes that enable external access to Multicloud Object Gateway**
  Disable **S3**, **STS**, and **noobaa-mgmt** routes by updating the storagecluster CR.

Changing the default account credentials to ensure better security in the Multicloud Object Gateway

Change and rotate your Multicloud Object Gateway (MCG) account credentials using the command-line interface to prevent issues with applications and to ensure better account security.

Before changing the default account credentials, ensure you have the following:

- A running Fusion Data Foundation cluster.
- Download the Multicloud Object Gateway (MCG) command-line interface for easier management. For instructions, see Accessing the Multicloud Object Gateway with your applications.

- **Resetting the noobaa account password**
  Change and rotate your Multicloud Object Gateway (MCG) account credentials by resetting the noobaa account password.
- **Setting Multicloud Object Gateway account credentials using CLI command**
  Use this procedure to set the Multicloud Object Gateway (MCG) account credentials using the command-line interface (CLI) command. You can update and verify the MCG account credentials manually by using the MCG CLI command.
- **Regenerating the S3 credentials for the accounts**
  Change and rotate your Multicloud Object Gateway (MCG) account credentials by regenerating the S3 credentials for the accounts.
- **Regenerating the S3 credentials for the OBC**
  Change and rotate your Multicloud Object Gateway (MCG) account credentials by regenerating the S3 credentials for the OBC.

Resetting the noobaa account password

Change and rotate your Multicloud Object Gateway (MCG) account credentials by resetting the noobaa account password.

Procedure

To reset the noobaa account password, run the following command:

```
odf noobaa account passwd <noobaa_account_name> [options]
```

```
odf noobaa account passwd
```

```
FATA[0000] ✗ Missing expected arguments: <noobaa_account_name>
```

```
Options:
    --new-password='': New Password for authentication - the best practice is to omit this flag, in that
    case the CLI will prompt to prompt and read it securely from the terminal to avoid leaking secrets in t
     he shell history
    --old-password='': Old Password for authentication - the best practice is to omit this flag, in that
    case the CLI will prompt to prompt and read it securely from the terminal to avoid leaking secrets in
    the shell history
    --retype-new-password='': Retype new Password for authentication - the best practice is to omit
    this flag, in that case the CLI will prompt to prompt and read it securely from the terminal to avoid
    leaking secrets in the shell history


Usage:
    odf noobaa account passwd <noobaa-account-name> [flags] [options]
```

```
Use "odf noobaa options" for a list of global command-line options (applies to all commands).
```

Example:

```
odf noobaa account passwd admin@noobaa.io
```

Example output:

```
Enter old-password: [got 24 characters]
Enter new-password: [got 7 characters]
Enter retype-new-password: [got 7 characters]
INFO[0017] ✅ Exists: Secret "noobaa-admin"
INFO[0017] ✅ Exists: NooBaa "noobaa"
INFO[0017] ✅ Exists: Service "noobaa-mgmt"
INFO[0017] ✅ Exists: Secret "noobaa-operator"
INFO[0017] ✅ Exists: Secret "noobaa-admin"
INFO[0017] ✈  RPC: account.reset_password() Request: {Email:admin@noobaa.io VerificationPassword:* Password:*}
WARN[0017] RPC: GetConnection creating connection to wss://localhost:58460/rpc/ 0xc000402ae0
INFO[0017] RPC: Connecting websocket (0xc000402ae0) &{RPC:0xc000501a40 Address:wss://localhost:58460/rpc/ State:init WS:<nil>
PendingRequests:map[] NextRequestID:0
Lock:{state:1 sema:0} ReconnectDelay:0s cancelPings:<nil>}
INFO[0017] RPC: Connected websocket (0xc000402ae0) &{RPC:0xc000501a40 Address:wss://localhost:58460/rpc/ State:init WS:<nil>
PendingRequests:map[] NextRequestID:0
Lock:{state:1 sema:0} ReconnectDelay:0s cancelPings:<nil>}
INFO[0020] ✅ RPC: account.reset_password() Response OK: took 2907.1ms
INFO[0020] ✅ Updated:  "noobaa-admin"
INFO[0020] ✅ Successfully reset the password for the account "admin@noobaa.io"
```

Important:
To access the admin account credentials run the **noobaa status** command from the terminal:

```
-------------------- Mgmt Credentials --------------------

email    : admin@noobaa.io
password : ***
```

Setting Multicloud Object Gateway account credentials using CLI command

Use this procedure to set the Multicloud Object Gateway (MCG) account credentials using the command-line interface (CLI) command. You can update and verify the MCG account credentials manually by using the MCG CLI command.

Before you begin

Ensure that the following prerequisites are met:

- A running Fusion Data Foundation cluster.
- Download the MCG CLI binary from the customer portal and make it executable.

Note:

- Choose the correct product variant according to your architecture.
- Available platforms are Linux (x86_64), Windows, and Mac OS.

Procedure

1. Run the following command to update the MCG account credentials:

   ```
   odf noobaa account credentials <noobaa-account-name> [options]
   ```

   Example:

   ```
   odf noobaa account credentials admin@noobaa.io
   ```

   Example output:

   ```
   odf noobaa account credentials admin@noobaa.io
   Enter access-key: [got 20 characters]
   Enter secret-key: [got 40 characters]
   INFO[0026] ✖ Not Found: NooBaaAccount "admin@noobaa.io"
   INFO[0026] ✅ Exists: NooBaa "noobaa"
   INFO[0026] ✅ Exists: Service "noobaa-mgmt"
   INFO[0026] ✅ Exists: Secret "noobaa-operator"
   INFO[0026] ✅ Exists: Secret "noobaa-admin"
   INFO[0026] ✈  RPC: account.update_account_keys() Request: {Email:admin@noobaa.io AccessKeys:{AccessKey:* SecretKey:}}
   WARN[0026] RPC: GetConnection creating connection to wss://localhost:33495/rpc/ 0xc000cd9980 INFO[0026] RPC: Connecting
   websocket (0xc000cd9980) &{RPC:0xc0001655e0 Address:wss://localhost:33495/rpc/ State:init WS:<nil> PendingRequests:map[]
   NextRequestID:0 Lock:{state:1 sema:0} ReconnectDelay:0s cancelPings:<nil>} INFO[0026] RPC: Connected websocket
   (0xc000cd9980) &{RPC:0xc0001655e0 Address:wss://localhost:33495/rpc/ State:init WS:<nil> PendingRequests:map[]
   NextRequestID:0 Lock:{state:1 sema:0} ReconnectDelay:0s cancelPings:<nil>} INFO[0026] ✅ RPC:
   account.update_account_keys() Response OK: took 42.7ms INFO[0026] ✈ RPC: account.read_account() Request:
   {Email:admin@noobaa.io} INFO[0026] ✅ RPC: account.read_account() Response OK: took 2.0ms INFO[0026] ✅ Updated: "noobaa-
   admin" INFO[0026] ✅ Successfully updated s3 credentials for the account "admin@noobaa.io" INFO[0026] ✅ Exists: Secret
   "noobaa-admin" Connection info: AWS_ACCESS_KEY_ID :  AWS_SECRET_ACCESS_KEY : *
   ```

   Credential complexity requirements:

   Access key

   The account access key must be 20 characters in length, and it must contain only alphanumeric characters.

   Secret key

   The secret key must be 40 characters in length, and it must contain alphanumeric characters and "+" and "/".

   Example:

   ```
   odf noobaa account credentials my-account --access-key=ABCDEF1234567890ABCD --secret-
   key=ABCDE12345+FGHIJ67890/KLMNOPQRSTUV123456
   ```

2. Run the following command to verify the credentials:

   ```
   odf noobaa account status <noobaa-account-name> --show-secrets
   ```

   Remember: Each user must have a unique access key and secret key.

Regenerating the S3 credentials for the accounts

Change and rotate your Multicloud Object Gateway (MCG) account credentials by regenerating the S3 credentials for the accounts.

Procedure

1. Get the account name.

- For listing the accounts, run the following command:

```
odf noobaa account list
```

Example output:

```
NAME           ALLOWED_BUCKETS    DEFAULT_RESOURCE               PHASE    AGE
account-test   [*]                noobaa-default-backing-store   Ready    14m17s
test2          [first.bucket]     noobaa-default-backing-store   Ready    3m12s
```

- Alternatively, run the **oc get noobaaaccount** command from the terminal:

```
oc get noobaaaccount
```

Example output:

```
NAME           PHASE    AGE
account-test   Ready    15m
test2          Ready    3m59s
```

2. To regenerate the noobaa account S3 credentials, run the following command:

```
odf noobaa account regenerate <noobaa_account_name> [options]
```

```
odf noobaa account regenerate
```

```
FATA[0000] ✗ Missing expected arguments: <noobaa-account-name>

Usage:
    odf noobaa account regenerate <noobaa-account-name> [flags] [options]

Use "odf noobaa options" for a list of global command-line options (applies to all commands).
```

3. After you run the **odf noobaa account regenerate** command it prompts the `This will invalidate all connections between S3 clients and NooBaa which are connected using the current credentials.` warning and asks for confirmation.
Example:

```
odf noobaa account regenerate account-test
```

Example output:

```
INFO[0000] You are about to regenerate an account's security credentials.
INFO[0000] This will invalidate all connections between S3 clients and NooBaa which are connected using the current
credentials.
INFO[0000] are you sure? y/n
```

After approving, the credentials are regenerated and eventually printed.

```
INFO[0015] ✅ Exists: Secret "noobaa-account-account-test"
Connection info:
AWS_ACCESS_KEY_ID      : ***
AWS_SECRET_ACCESS_KEY  : ***
```

Regenerating the S3 credentials for the OBC

Change and rotate your Multicloud Object Gateway (MCG) account credentials by regenerating the S3 credentials for the OBC.

Procedure

1. Get the OBC name.
   - Run the **odf noobaa obc list** command

```
odf noobaa obc list
```

Example output:

```
NAMESPACE   NAME       BUCKET-NAME                                   STORAGE-CLASS       BUCKET-CLASS
PHASE
default     obc-test   obc-test-35800e50-8978-461f-b7e0-7793080e26ba   default.noobaa.io   noobaa-default-bucket-
class   Bound
```

   - Run the **oc get obc** command from the terminal.

```
oc get obc
```

Example output:

```
NAME       STORAGE-CLASS       PHASE   AGE
obc-test   default.noobaa.io   Bound   38s
```

2. To regenerate the noobaa OBC S3 credentials, run the following command:

```
odf noobaa obc regenerate <bucket_claim_name> [options]
```

```
odf noobaa obc regenerate
FATA[0000] ✗ Missing expected arguments: <bucket-claim-name>

Usage:
   odf noobaa obc regenerate <bucket-claim-name> [flags] [options]
```

```
Use "odf noobaa options" for a list of global command-line options (applies to all commands).
```

3. After you run the `odf noobaa obc regenerate` command it prompts the `This will invalidate all connections between S3 clients and NooBaa which are connected using the current credentials.` warning and asks for confirmation.
   Example:

```
odf noobaa obc regenerate obc-test
```

Example output:

```
INFO[0000] You are about to regenerate an OBC's security credentials.
INFO[0000] This will invalidate all connections between S3 clients and NooBaa which are connected using the current
credentials.
INFO[0000] are you sure? y/n
```

After approving, the credentials are regenerated and eventually printed.

```
INFO[0022] ✅ RPC: bucket.read_bucket() Response OK: took 95.4ms

ObjectBucketClaim info:
  Phase                  : Bound
  ObjectBucketClaim      : kubectl get -n default objectbucketclaim obc-test
  ConfigMap              : kubectl get -n default configmap obc-test
  Secret                 : kubectl get -n default secret obc-test
  ObjectBucket           : kubectl get objectbucket obc-default-obc-test
  StorageClass           : kubectl get storageclass default.noobaa.io
  BucketClass            : kubectl get -n default bucketclass noobaa-default-bucket-class

Connection info:
 BUCKET_HOST            : s3.default.svc
 BUCKET_NAME            : obc-test-35800e50-8978-461f-b7e0-7793080e26ba
    BUCKET_PORT            : 443
    AWS_ACCESS_KEY_ID      : ***
    AWS_SECRET_ACCESS_KEY  : ***

Shell commands:
  AWS S3 Alias           : alias s3='AWS_ACCESS_KEY_ID=***
AWS_SECRET_ACCESS_KEY=*** aws s3 --no-verify-ssl --endpoint-url ***'

Bucket status:
  Name                   : obc-test-35800e50-8978-461f-b7e0-7793080e26ba
  Type                   : REGULAR
  Mode                   : OPTIMAL
  ResiliencyStatus       : OPTIMAL
  QuotaStatus            : QUOTA_NOT_SET
  Num Objects            : 0
  Data Size              : 0.000 B
  Data Size Reduced      : 0.000 B
  Data Space Avail       : 13.261 GB
  Num Objects Avail      : 9007199254740991
```

Enabling secured mode deployment for Multicloud Object Gateway

You can specify a range of IP addresses that should be allowed to reach the Multicloud Object Gateway (MCG) load balancer services to enable secure mode deployment. This helps to control the IP addresses that can access the MCG services.

Before you begin

- A running Fusion Data Foundation cluster.
- In case of a bare metal deployment, ensure that the load balancer controller supports setting the loadBalancerSourceRanges attribute in the Kubernetes services.

Procedure

Edit the NooBaa custom resource (CR) to specify the range of IP addresses that can access the MCG services after deploying Fusion Data Foundation.

```
oc edit noobaa -n openshift-storage noobaa
```

noobaa
: The NooBaa CR type that controls the NooBaa system deployment.

noobaa
: The name of the NooBaa CR.

For example:

```
...
spec:
  ...
  loadBalancerSourceSubnets:
    s3: ["10.0.0.0/16", "192.168.10.0/32"]
    sts:
      - "10.0.0.0/16"
      - "192.168.10.0/32"
...
```

loadBalancerSourceSubnets
A new field that can be added under **spec** in the NooBaa CR to specify the IP addresses that should have access to the NooBaa services.

In this example, all the IP addresses that are in the subnet 10.0.0.0/16 or 192.168.10.0/32 will be able to access MCG S3 and security token service (STS) while the other IP addresses are not allowed to access.

What to do next

To verify if the specified IP addresses are set, from the OpenShift Web Console, run the following command and check if the output matches with the IP addresses provided to MCG:

```
oc get svc -n openshift-storage <s3 | sts> -o=go-template='{{ .spec.loadBalancerSourceRanges }}'
```

Disabling routes that enable external access to Multicloud Object Gateway

Disable **S3**, **STS**, and **noobaa-mgmt** routes by updating the storagecluster CR.

About this task

You can disable **S3**, **STS**, and **noobaa-mgmt** routes by setting the following option in the storagecluster CR:

```
spec:
  multiCloudGateway:
    disableRoutes: true
```

This stops reconciling of all the three routes so that the routes are not recreated when they are deleted.

Procedure

1. Edit the storagecluster CR in one of the following ways:
   - Using the **oc patch** command:

     ```
     oc patch storagecluster ocs-storage -n openshift-storage --type=merge -p '{"spec":{"multiCloudGateway":
     {"disableRoutes":true}}}'
     ```

   - Using the interactive method:

     ```
     oc edit storagecluster ocs-storage -n openshift-storage
     ```

     - Add the following under **.spec**:

       ```
       multiCloudGateway:
         disableRoutes: true
       ```

2. To verify, delete the existing routes and check if they are listed:

   ```
   oc delete routes --all -n openshift-storage
   oc get routes -n openshift-storage
   ```

   The **S3**, **STS**, and **noobaa-mgmt** routes are no longer reconciled by the operator.

Mirroring data for hybrid and Multicloud buckets

Create bucket classes to mirror data for hybrid and Multicloud buckets. Mirroring data can be setup by using the OpenShift UI, YAML, or MCG command-line interface.

You can use the simplified process of the Multicloud Object Gateway (MCG) to span data across cloud providers and clusters. Before you create a bucket class that reflects the data management policy and mirroring, you must add a backing storage that can be used by the MCG. For information, see Adding storage resources for hybrid or Multicloud.

- **Creating bucket classes to mirror data using the MCG command-line-interface**
  Create bucket classes to mirror data for hybrid and Multicloud buckets using the MCG command-line interface.
- **Creating bucket classes to mirror data using a YAML**
  Create bucket classes to mirror data for hybrid and Multicloud buckets using a YAML file.

Creating bucket classes to mirror data using the MCG command-line-interface

Create bucket classes to mirror data for hybrid and Multicloud buckets using the MCG command-line interface.

Before you begin

Ensure to download Multicloud Object Gateway (MCG) command-line interface.

Procedure

1. From the Multicloud Object Gateway (MCG) command-line interface, run the following command to create a bucket class with a mirroring policy:

```
odf noobaa bucketclass create placement-bucketclass mirror-to-aws --backingstores=azure-resource,aws-resource --placement
Mirror
```

2. Set the newly created bucket class to a new bucket claim to generate a new bucket that will be mirrored between two locations:

```
odf noobaa obc create  mirrored-bucket --bucketclass=mirror-to-aws
```

Creating bucket classes to mirror data using a YAML

Create bucket classes to mirror data for hybrid and Multicloud buckets using a YAML file.

Procedure

1. Apply the following YAML.
   This YAML is a hybrid example that mirrors data between local Ceph storage and AWS.

```
apiVersion: noobaa.io/v1alpha1
kind: BucketClass
metadata:
  labels:
    app: noobaa
  name: <bucket-class-name>
  namespace: openshift-storage
spec:
  placementPolicy:
    tiers:
    - backingStores:
      - <backing-store-1>
      - <backing-store-2>
      placement: Mirror
```

2. Add the following lines to your standard Object Bucket Claim (OBC):

```
additionalConfig:
  bucketclass: mirror-to-aws
```

   For more information about OBCs, see Object Bucket Claim.

Bucket policies in the Multicloud Object Gateway

Fusion Data Foundation supports AWS S3 bucket policies. Bucket policies allow you to grant users access permissions for buckets and the objects in them. Use the information in this section to understand how to use bucket policies in Multicloud Object Gateway.

- **Introduction to bucket policies**
  Bucket policies are an access policy option available for you to grant permission to your AWS S3 buckets and objects. Bucket policies use JSON-based access policy language.
- **Using bucket policies in Multicloud Object Gateway**
  Use these instructions to use bucket policies in Multicloud Object Gateway.
- **Creating a user in the Multicloud Object Gateway**
  Create a user account in the Multicloud Object Gateway using the MCG command-line interface, optionally with a NamespaceStore file system configuration.

Introduction to bucket policies

Bucket policies are an access policy option available for you to grant permission to your AWS S3 buckets and objects. Bucket policies use JSON-based access policy language.

For more information about access policy language, see AWS Access Policy Language Overview.

Using bucket policies in Multicloud Object Gateway

Use these instructions to use bucket policies in Multicloud Object Gateway.

Before you begin

Ensure you have the following:

- A running Fusion Data Foundation Platform.
- Access to the Multicloud Object Gateway (MCG), see Accessing the Multicloud Object Gateway with your applications.

Procedure

1. Create the bucket policy in JSON format.
   For example:

```
{
    "Version": "NewVersion",
    "Statement": [
        {
            "Sid": "Example",
            "Effect": "Allow",
            "Principal": [
                    "john.doe@example.com"
            ],
            "Action": [
                "s3:GetObject"
            ],
            "Resource": [
                "arn:aws:s3:::john_bucket"
            ]
        }
    ]
}
```

2. Using AWS S3 client, use the **put-bucket-policy** command to apply the bucket policy to your S3 bucket:

```
aws --endpoint ENDPOINT --no-verify-ssl s3api put-bucket-policy --bucket MyBucket --policy BucketPolicy
```

   ENDPOINT
      An S3 endpoint.
   MyBucket
      A name of the bucket to set the policy on.
   BucketPolicy
      A bucket policy JSON file.
   --no-verify-ssl
      Add this if you are using the default self signed certificates.

   For example:

```
aws --endpoint https://s3-openshift-storage.apps.gogo44.noobaa.org --no-verify-ssl s3api put-bucket-policy -bucket
MyBucket --policy file://BucketPolicy
```

   For more information on the **put-bucket-policy** command, see AWS CLI Command Reference for put-bucket-policy.

   Note: The principal element specifies the user that is allowed or denied access to a resource, such as a bucket. Currently, only NooBaa accounts can be used as principals. In the case of object bucket claims, NooBaa automatically creates an account .

```
obc-account.<generated bucket name>@noobaa.io
```

   Note: Bucket policy conditions are not supported.

What to do next

There are many available elements for bucket policies with regard to access permissions.

- For details on these elements and examples of how they can be used to control the access permissions, see AWS Access Policy Language Overview.
- For more examples of bucket policies, see AWS Bucket Policy Examples.
- Fusion Data Foundation version 4.16 or higher introduces the bucket policy elements `NotPrincipal`, `NotAction`, and `NotResource`. For more information on these elements, see IAM JSON policy elements reference.

Creating a user in the Multicloud Object Gateway

Create a user account in the Multicloud Object Gateway using the MCG command-line interface, optionally with a NamespaceStore file system configuration.

Before you begin

- A running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
yum install mcg
```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.
  - For IBM Power, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
```

  - For IBM Z infrastructure, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

Procedure

Run the following command to create an MCG user account:

```
odf noobaa account create <noobaa-account-name> [--allow_bucket_create=true] [--default_resource=''] [--force_md5_etag=false]
[--gid=-1] [--new_buckets_path='/'] [--nsfs_account_config=false] [--nsfs_only=true] [--uid=-1]
```

<allow_bucket_create>
: Specify if this account be allowed to create new buckets.

<default_resource>
: Specify the default resource on which new buckets are created.

<force_md5_etag>
: Specify if `md5 etag` calculation be enabled for the account.

<gid_number>
: Specify the GID number.

<new_buckets_path>
: Specify the path where the new buckets are created.

<nsfs_account_config>
: Specify if NSFS account needs to be created.

<nsfs_only>
: Set if this account is used only for NSFS.

<uid_number>
: Specify the UID number.

Bucket notification in Multicloud Object Gateway

With bucket notification, you can send a notification to an external server whenever there is an event in the bucket.

Multicloud Object Gateway (MCG) supports several bucket event notification types. You can receive bucket notifications for activities such as flow creation, triggering data digestion, and so on. MCG supports the following event types that can be specified in the notification configuration:

- s3:TestEvent
- s3:ObjectCreated:*
- s3:ObjectCreated:Put
- s3:ObjectCreated:Post
- s3:ObjectCreated:Copy
- s3:ObjectCreated:CompleteMultipartUpload
- s3:ObjectRemoved:*
- s3:ObjectRemoved:Delete
- s3:ObjectRemoved:DeleteMarkerCreated
- s3:LifecycleExpiration:*
- s3:LifecycleExpiration:Delete
- s3:LifecycleExpiration:DeleteMarkerCreated
- s3:ObjectRestore:*
- s3:ObjectRestore:Post
- s3:ObjectRestore:Completed
- s3:ObjectRestore:Delete
- s3:ObjectTagging:*
- s3:ObjectTagging:Put
- s3:ObjectTagging:Delete

- **Configuring bucket notification in Multicloud Object Gateway**
  Learn how to configure bucket notifications in Multicloud Object Gateway (MCG).

Configuring bucket notification in Multicloud Object Gateway

Learn how to configure bucket notifications in Multicloud Object Gateway (MCG).

Before you begin

- Ensure to have one of the following before configuring:
  - Kafka cluster is deployed.

    For example, you can deploy Kafka by using `AMQ/strimzi` by referring to Getting Started with AMQ Streams on OpenShift.
  - An HTTP or HTTPs server is connected.

    For example, you can set up an HTTP server to log incoming HTTP requests so that you can observe them using the `oc logs` command as follows:

    ```
    cat http_logging_server.yaml
    apiVersion: v1
    ```

```
kind: List
metadata: {}
items:
- apiVersion: apps/v1
  kind: Deployment
  metadata:
    name: http-logger
  spec:
    replicas: 1
    selector:
      matchLabels:
        app: http-logger
    template:
      metadata:
        labels:
          app: http-logger
      spec:
        containers:
        - name: http-logger
          image: registry.redhat.io/ubi9/python-39:latest
          command:
            - /bin/sh
            - -c
            - |
              set -e  # Fail on any error
              mkdir -p /tmp/app
              pip install flask
              cat <<EOF > /tmp/app/server.py
              from flask import Flask, request
              app = Flask(__name__)
              @app.route("/", methods=["POST", "GET"])
              def log_request():
                  body = request.get_data(as_text=True)
                  print(body)  # Simple one-line logging per request
                  return "", 200
              if __name__ == "__main__":
                  app.run(host="0.0.0.0", port=8676, debug=True)
              EOF
              exec python /tmp/app/server.py
          ports:
          - containerPort: 8676
            protocol: TCP
          securityContext:
            runAsNonRoot: true
            allowPrivilegeEscalation: false
- apiVersion: v1
  kind: Service
  metadata:
    name: http-logger
    labels:
      app: http-logger
  spec:
    selector:
      app: http-logger
    ports:
      - port: 8676
        targetPort: 8676
        protocol: TCP
        name: http
```

```
oc create -f http_logging_server.yaml -n <http-server-namespace>
```

- Ensure that server is accessible from the MCG core pod from which the notifications are sent.
- Ensure to fetch MCG's credentials:

```
NOOBAA_ACCESS_KEY=$(oc extract secret/noobaa-admin -n openshift-storage --keys=AWS_ACCESS_KEY_ID --to=- 2>/dev/null); \
NOOBAA_SECRET_KEY=$(oc extract secret/noobaa-admin -n openshift-storage --keys=AWS_SECRET_ACCESS_KEY --to=- 2>/dev/null);
\
S3_ENDPOINT=https://$(oc get route s3 -n openshift-storage -o json | jq -r ".spec.host")
alias aws_alias='AWS_ACCESS_KEY_ID=$NOOBAA_ACCESS_KEY AWS_SECRET_ACCESS_KEY=$NOOBAA_SECRET_KEY aws --endpoint $S3_ENDPOINT
--no-verify-
```

Procedure

1. Describe connection using a json file, for example, connection.json on your local machine.
   - **An example for Kafka**:

     ```
     {
       "name": "kafka_notif_conn_file" <-- any string
       "notification_protocol": "kafka",
       "metadata.broker.list": "<kafka-service-name>.<project>.svc.cluster.local:<kafka-service-port>",
       "topic": "my-topic",<--  refer to an existing KafkaTopic resource
     }
     ```

     The structure under `metadata.broker.list` must be `<service-name>.<namespace>.svc.cluster.local:9092`, topic must refer to the name of an existing `kafkatopic` resource in the namespace.

   - **An example for HTTP or HTTPs**:

     ```
     {
       "name": "http_notif_connection_config",
       "notification_protocol": "http", <-- or "https"
       "agent_request_object": {
     ```

```
    "host": "<http-service>.<http-server-namespace>.svc.cluster.local",
    "port": <http-server-port>
   }
}
```

Additional options:

request_options_object
: Value is a JSON that is passed to nodejs' http(s) request (optional).
  Any field supported by nodejs' http(s) request option can be used, for example:

path'
: Used to specify the url path.

'auth'
: Used for http simple auth. Syntax for the value of 'auth' is: <name>:<password>.

2. Create a secret from the file in the `openshift-storage` namespace.
   For example:

```
oc create secret generic <connection-secret> --from-file=connect.json -n openshift-storage
```

3. Update the `NooBaa` CR in the `openshift-storage` namespace with the connection secret and an optional CephFS RWX PVC.
   For example:

```
oc patch noobaa noobaa --type='merge' -n openshift-storage -p '{
  "spec": {
    "bucketNotifications": {
      "connections": [
        {
          "name": <connection-secret>,
          "namespace": "openshift-storage"
        }
      ],
      "enabled": true,
      "pvc": "bn-pvc" <-- optional
    }
  }
}'
```

   Note: If PVC is not provided, MCG creates one automatically as needed.
   For example:

```
oc get pvc bn-pvc -n openshift-storage
NAME     STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS
VOLUMEATTRIBUTESCLASS   AGE
bn-pvc   Bound    pvc-24683f8d-48e4-4c6b-b108-d507ca2f4fd1   25Gi       RWX            ocs-storagecluster-cephfs   <unset>
25h
```

   Important: Names of the connection secret must be unique in the list even though the namespaces are different.
   Wait for the `noobaa-core` and `noobaa-endpoint` pods to restart before proceeding with the next step.

4. Use `S3:PutBucketNotification` on an MCG bucket by using the `noobaa-admin` credentials and S3 endpoint under an S3 alias.
   For example:

```
aws_alias s3api put-bucket-notification --bucket first.bucket --notification-configuration '{
  "TopicConfiguration": {
    "Id": "<notif_event_kafka>", <-- Unique string
    "Events": ["s3:ObjectCreated:*"], <-- a filter for events
    "Topic": "<connection-secret/connect.json>"
  }
}'
```

5. Verify that the bucket notification configuration is set on the bucket.
   For example:

```
aws_alias s3api get-bucket-notification-configuration --bucket first.bucket
{
    "TopicConfigurations": [
        {
            "Id": "notif_event_kafka",
            "TopicArn": "kafka-connection-secret/connect.json",
            "Events": [
                "s3:ObjectCreated:*"
            ]
        }
    ]
}
```

6. Add some objects on the bucket.

```
echo 'a' | aws_alias s3 cp - s3://first.bucket/a
```

   Wait for a while and query the topic messages to verify the expected notification has been sent and received.

   An example for Kafka:

```
oc -n your-kafka-project rsh my-cluster-kafka-0 bin/kafka-console-consumer.sh \
  --bootstrap-server my-cluster-kafka-bootstrap.myproject.svc.cluster.local:9092 \
  --topic my-topic \
  --from-beginning --timeout-ms 10000 | grep '^{.*}' | jq -c '.' | jq
```

   An example for HTTP or HTTPs:

```
oc logs deployment/http-logger -n <http-server-namespace> | grep '^{.*}' | jq
```

Example output:


```
{
  "Records": [
    {
      "eventVersion": "2.3",
      "eventSource": "noobaa:s3",
      "eventTime": "2024-11-27T12:44:21.987Z",
      "s3": {
        "s3SchemaVersion": "1.0",
        "object": {
          "sequencer": 10,
          "key": "a",
          "eTag": "60b725f10c9c85c70d97880dfe8191b3"
        },
        "bucket": {
          "name": "second.bucket",
          "ownerIdentity": {
            "principalId": "admin@noobaa.io"
          },
          "arn": "arn:aws:s3:::first.bucket"
        }
      },
      "eventName": "ObjectCreated:Put",
      "userIdentity": {
        "principalId": "noobaa"
      },
      "requestParameters": {
        "sourceIPAddress": "100.64.0.3"
      },
      "responseElements": {
        "x-amz-request-id": "m3zvo0cm-5239xd-bhj",
        "x-amz-id-2": "m3zvo0cm-5239xd-bhj"
      }
    }
  ]
}
```


Multicloud Object Gateway bucket replication

Data replication from one Multicloud Object Gateway (MCG) bucket to another MCG bucket provides higher resiliency and better collaboration options. These buckets can be either data buckets or namespace buckets backed by any supported storage solution (S3, Azure, and so on). Replicate a bucket to another bucket and set replication policies using command-line interface and YAML.

A replication policy is composed of a list of replication rules. Each rule defines the destination bucket, and can specify a filter based on an object key prefix. Configuring a complementing replication policy on the second bucket results in bidirectional replication.

- **Replicating a bucket to another bucket**
  This section describes how to replicate one bucket to another using CLI and YAML.
- **Setting a bucket class replication policy**
  Set a bucket class replication policy to automatically replicate all buckets created under a bucket class using the MCG command-line interface or a YAML file.
- **Enabling log based bucket replication**
  This section provides information about the log based bucket replication.
- **Bucket logging for Multicloud Object Gateway**
  Bucket logging helps in recording the S3 operations that are performed against the Multicloud Object Gateway (MCG) bucket for compliance, auditing, and optimization purposes.
- **Synchronizing versions in Multicloud Object Gateway bucket replication**
  Learn how to synchronize versions in Multicloud Object Gateway (MCG) bucket replication.
- **Obtaining metrics to reflect bucket replication state**
  Learn how to obtain bucket-level replication metrics to evaluate replication progress.

Replicating a bucket to another bucket

This section describes how to replicate one bucket to another using CLI and YAML.

You can set the bucket replication policy in two ways:

- Replicating a bucket to another bucket using the MCG command-line interface.
- Replicating a bucket to another bucket using a YAML.

- **Replicating a bucket to another bucket using the MCG command-line interface**
  Provide higher resiliency and better collaboration options by replicating a bucket to another bucket using the MCG command-line interface.
- **Replicating a bucket to another bucket using a YAML**
  Provide higher resiliency and better collaboration options by replicating a bucket to another bucket using the a YAML file.

Replicating a bucket to another bucket using the MCG command-line interface

Provide higher resiliency and better collaboration options by replicating a bucket to another bucket using the MCG command-line interface.

Before you begin

- Ensure you have a running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
yum install mcg
```

Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

Note: Choose the correct Product Variant according to your architecture.

About this task

Applications that require a Multicloud Object Gateway (MCG) bucket to have a specific replication policy can create an Object Bucket Claim (OBC) and define the replication policy parameter in a JSON file.

Procedure

From the MCG command-line interface, run the following command to create an OBC with a specific replication policy:

```
odf noobaa obc create <bucket-claim-name> -n openshift-storage --replication-policy /path/to/json-file.json
```

bucket-claim-name
: Specify the name of the bucket claim.

/path/to/json-file.json
: Is the path to a JSON file which defines the replication policy.

Example JSON file:

```
[{ "rule_id": "rule-1", "destination_bucket": "first.bucket", "filter": {"prefix": "repl"}}]
```

`"prefix"`
: (Optional:) It is the prefix of the object keys that should be replicated, and you can even leave it empty, for example, `{"prefix": ""}`.
For example:

```
odf noobaa obc create my-bucket-claim -n openshift-storage --replication-policy /path/to/json-file.json
```

Replicating a bucket to another bucket using a YAML

Provide higher resiliency and better collaboration options by replicating a bucket to another bucket using the a YAML file.

Before you begin

- Ensure you have a running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
yum install mcg
```

Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

Note: Choose the correct Product Variant according to your architecture.

About this task

Applications that require a Multicloud Object Gateway (MCG) data bucket to have a specific replication policy can create an Object Bucket Claim (OBC) and add the spec.additionalConfig.replication-policy parameter to the OBC. For more information about OBCs, see Object Bucket Claim.

Procedure

Apply the following YAML:

```
apiVersion: objectbucket.io/v1alpha1
kind: ObjectBucketClaim
metadata:
  name: <desired-bucket-claim>
  namespace: <desired-namespace>
```

```
spec:
  generateBucketName: <desired-bucket-name>
  storageClassName: openshift-storage.noobaa.io
  additionalConfig:
    replication-policy: [{ "rule_id": "<rule id>", "destination_bucket": "first.bucket", "filter": {"prefix": "<object name prefix>"}}]
```

desired-bucket-claim
: Specify the name of the bucket claim.

desired-namespace
: Specify the namespace.

desired-bucket-name>
: Specify the prefix of the bucket name.

rule_id
: Specify the ID number of the rule, for example, `{"rule_id": "rule-1"}`.

destination_bucket
: Specify the name of the destination bucket, for example, `{"destination_bucket": "first.bucket"}`.

object name prefix
: (Optional:) It is the prefix of the object keys that should be replicated, and you can even leave it empty, for example, `{"prefix": ""}`.

Setting a bucket class replication policy

Set a bucket class replication policy to automatically replicate all buckets created under a bucket class using the MCG command-line interface or a YAML file.

It is possible to set up a replication policy that automatically applies to all the buckets created under a certain bucket class. You can do this in two ways:

- Setting a bucket class replication policy using the MCG command-line interface.
- Setting a bucket class replication policy using a YAML.

- **Setting a bucket class replication policy using the MCG command-line interface**
  Provide higher resiliency and better collaboration options by setting a bucket class replication policy using the MCG command-line interface.
- **Setting a bucket class replication policy using a YAML**
  Provide higher resiliency and better collaboration options by setting a bucket class replication policy using a YAML file.

Setting a bucket class replication policy using the MCG command-line interface

Provide higher resiliency and better collaboration options by setting a bucket class replication policy using the MCG command-line interface.

Before you begin

- Ensure you have a running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
  ```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

About this task

Applications that require a Multicloud Object Gateway (MCG) bucket class to have a specific replication policy can create a `bucketclass` and define the replication-policy parameter in a JSON file.

It is possible to set a bucket class replication policy for two types of bucket classes:

- Placement
- Namespace

Procedure

From the MCG command-line interface, run the following command:

```
odf noobaa -n openshift-storage bucketclass create placement-bucketclass <bucketclass-name> --backingstores <backingstores> --replication-policy=/path/to/json-file.json
```

bucketclass-name
    Specify the name of the bucket class.

backingstores
    Specify the name of a backingstore. It is possible to pass several backingstores separated by commas.

/path/to/json-file.json
    Is the path to a JSON file which defines the replication policy.

Example JSON file:

```
[{ "rule_id": "rule-1", "destination_bucket": "first.bucket", "filter": {"prefix": "repl"}}]
```

"prefix"
    (Optional:) It is the prefix of the object keys that should be replicated, and you can even leave it empty, for example, `{"prefix": ""}`.
    For example:

```
noobaa -n openshift-storage bucketclass create placement-bucketclass bc --backingstores azure-blob-ns --replication-polic
```

This example creates a placement bucket class with a specific replication policy defined in the JSON file.

Setting a bucket class replication policy using a YAML

Provide higher resiliency and better collaboration options by setting a bucket class replication policy using a YAML file.

Before you begin

- Ensure you have a running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
yum install mcg
```

Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
```

- Alternatively, you can install the MCG package from the RPMs found at [Download Red Hat OpenShift Data Foundation page](#).

Note: Choose the correct Product Variant according to your architecture.

About this task

Applications that require a Multicloud Object Gateway (MCG) bucket class to have a specific replication policy can create a bucket class using the spec.replicationPolicy field.

Procedure

Apply the following YAML:

```
apiVersion: noobaa.io/v1alpha1
kind: BucketClass
metadata:
  labels:
    app: <desired-app-label>
  name: <desired-bucketclass-name>
  namespace: <desired-namespace>
spec:
  placementPolicy:
    tiers:
    - backingstores:
      - <backingstore>
      placement: Spread
  replicationPolicy: [{ "rule_id": "<rule id>", "destination_bucket": "first.bucket", "filter": {"prefix": "<object name
prefix>"}}]
```

This YAML is an example that creates a placement bucket class. Each Object bucket claim (OBC) object that is uploaded to the bucket is filtered based on the prefix and is replicated to `first.bucket`.

desired-app-label
    Specify a label for the app.

desired-bucketclass-name
    Specify the bucket class name.

desired-namespace
    Specify the namespace in which the bucket class gets created.

backingstore
    Specify the name of a backingstore. It is possible to pass several backingstores.

rule_id
: Specify the ID number of the rule, for example, `{"rule_id": "rule-1"}`.

destination_bucket
: Specify the name of the destination bucket, for example, `{"destination_bucket": "first.bucket"}`.

object name prefix
: (Optional:) It is the prefix of the object keys that should be replicated, and you can even leave it empty, for example, `{"prefix": ""}`.

Enabling log based bucket replication

This section provides information about the log based bucket replication.

When creating a bucket replication policy, you can use logs so that recent data is replicated more quickly, while the default scan-based replication works on replicating the rest of the data.

Important:
This feature requires setting up bucket logs on AWS or Azure. For more information about setting up AWS logs, see Enabling Amazon S3 server access logging. The AWS logs bucket needs to be created in the same region as the source NamespaceStore AWS bucket.

Note:
This feature is only supported in buckets that are backed by a NamespaceStore. Buckets backed by BackingStores cannot utilize log-based replication.

- **Enabling log based bucket replication for new namespace buckets using OpenShift Web Console in AWS environment**
  You can optimize replication by using the event logs of the Amazon Web Service(AWS) cloud environment. You enable log based bucket replication for new namespace buckets using the web console during the creation of namespace buckets.
- **Enabling log based bucket replication for existing namespace buckets using YAML**
  Enable log-based bucket replication for existing namespace buckets by editing the OBC YAML to add a replication policy with a destination bucket and a log bucket location.
- **Enabling log based bucket replication in Microsoft Azure**
  Enable log-based bucket replication to a Microsoft Azure namespace store by registering an Azure application, creating the required NamespaceStore, and applying the replication policy.
- **Enabling log-based bucket replication deletion**
  Enable deletion synchronization for log-based bucket replication so that objects deleted in the source bucket are also removed from the destination bucket.

Enabling log based bucket replication for new namespace buckets using OpenShift Web Console in AWS environment

You can optimize replication by using the event logs of the Amazon Web Service(AWS) cloud environment. You enable log based bucket replication for new namespace buckets using the web console during the creation of namespace buckets.

Before you begin

- Ensure that object logging is enabled in AWS. For more information, see the “Using the S3 console” section in Enabling Amazon S3 server access logging.
- Administrator access to OpenShift Web Console.

About this task

Procedure

1. In the OpenShift Web Console, navigate to **Storage** → **Object Storage** → **Object Bucket Claims**.
2. Click **Create ObjectBucketClaim**.
3. Enter the name of ObjectBucketName and select StorageClass and BucketClass.
4. Select the **Enable replication** check box to enable replication.
5. In the **Replication policy** section, select the **Optimize replication using event logs** check box.
6. Enter the name of the bucket that will contain the logs under **Event log Bucket**.
   If the logs are not stored in the root of the bucket, provide the full path without `s3://`
7. Enter a prefix to replicate only the objects whose name begins with the given prefix.

Enabling log based bucket replication for existing namespace buckets using YAML

Enable log-based bucket replication for existing namespace buckets by editing the OBC YAML to add a replication policy with a destination bucket and a log bucket location.

About this task

You can enable log based bucket replication for the existing buckets that are created using the command line interface or by applying an YAML, and not the buckets that are created using AWS S3 commands.

Procedure

Edit the YAML of the bucket's OBC to enable log based bucket replication. Add the following under `spec`:

```
replicationPolicy: '{"rules":[{"rule_id":"<RULE ID>", "destination_bucket":"<DEST>", "filter": {"prefix": "<PREFIX>"}}],
"log_replication_info": {"logs_location": {"logs_bucket": "<LOGS_BUCKET>"}}}'
```

`rule_id`
: Specify an ID of your choice for identifying the rule

`destination_bucket`
: Specify the name of the target MCG bucket that the objects are copied to

(optional) `{"filter": {"prefix": <>}}`
: Specify a prefix string that you can set to filter the objects that are replicated

`log_replication_info`
: Specify an object that contains data related to log-based replication optimization. `{"logs_location": {"logs_bucket": <>}}` is set to the location of the AWS S3 server access logs.

Enabling log based bucket replication in Microsoft Azure

Enable log-based bucket replication to a Microsoft Azure namespace store by registering an Azure application, creating the required NamespaceStore, and applying the replication policy.

Before you begin

- Refer to Microsoft Azure documentation and ensure that you have completed the following tasks in the Microsoft Azure portal:
  1. Ensure that have created a new application and noted down the name, application (client) ID, and directory (tenant) ID.

     For information, see Register an application.
  2. Ensure that a new client secret is created and the application secret is noted down.
  3. Ensure that a new Log Analytics workspace is created and its name and workspace ID is noted down.

     For information, see Create a Log Analytics workspace.
  4. Ensure that the `Reader` role is assigned under **Access control** and members are selected and the name of the application that you registered in the previous step is provided.

     For more information, see Assign Azure roles using the Azure portal.
  5. Ensure that a new storage account is created and the Access keys are noted down.
  6. In the **Monitoring** section of the storage account created, select a blob and in the **Diagnostic settings** screen, select only `StorageWrite` and `StorageDelete`, and in the destination details add the Log Analytics workspace that you created earlier. Ensure that a blob is selected in the **Diagnostic settings** screen of the **Monitoring** section of the storage account created. Also, ensure that only `StorageWrite` and `StorageDelete` is selected and in the destination details, the Log Analytics workspace that you created earlier is added.

     For more information, see Diagnostic settings in Azure Monitor.
  7. Ensure that two new containers for object source and object destination are created.
- Administrator access to OpenShift Web Console.

Procedure

1. Create a secret with credentials to be used by the `namespacestores`.

   ```
   apiVersion: v1
   kind: Secret
   metadata:
     name: <namespacestore-secret-name>
   type: Opaque
   data:
       TenantID: <AZURE TENANT ID ENCODED IN BASE64>
       ApplicationID: <AZURE APPLICATIOM ID ENCODED IN BASE64>
       ApplicationSecret: <AZURE APPLICATION SECRET ENCODED IN BASE64>
       LogsAnalyticsWorkspaceID: <AZURE LOG ANALYTICS WORKSPACE ID ENCODED IN BASE64>
       AccountName: <AZURE ACCOUNT NAME ENCODED IN BASE64>
       AccountKey: <AZURE ACCOUNT KEY ENCODED IN BASE64>
   ```
2. Create a `NamespaceStore` backed by a container created in Azure.
   For more information, see Adding a namespace bucket using the OpenShift Container Platform user interface.
3. Create a new Namespace-Bucketclass and OBC that utilizes it.
4. Check the object bucket name by looking in the YAML of target OBC, or by listing all S3 buckets, for example, `- s3 ls`.
5. Use the following template to apply an Azure replication policy on your source OBC by adding the following in its YAML, under `.spec`:

   ```
   replicationPolicy:'{"rules":[ {"rule_id":"ID goes here", "sync_deletions": "<true or false>"", "destination_bucket":object
   bucket name"}
    ], "log_replication_info":{"endpoint_type":"AZURE"}}'
   ```

   `sync_deletion`

Specify a boolean value, `true` or `false`.

`destination_bucket`
Ensure to use the name of the object bucket, and not the claim. The name can be retrieved using the `s3 ls` command, or by looking for the value in an OBC's YAML.

What to do next

- **Verification steps:**
  1. Write objects to the source bucket.
  2. Wait until MCG replicates them.
  3. Delete the objects from the source bucket.
  4. Verify the objects were removed from the target bucket.

Enabling log-based bucket replication deletion

Enable deletion synchronization for log-based bucket replication so that objects deleted in the source bucket are also removed from the destination bucket.

Before you begin

- Administrator access to OpenShift Web Console.
- AWS Server Access Logging configured for the desired bucket.

Procedure

1. In the OpenShift Web Console, navigate to Storage > Object Storage > Object Bucket Claims.
2. Click Create new Object bucket claim.
3. Optional: In the Replication rules section, select the Sync deletion checkbox for each rule separately.
4. Enter the name of the bucket that will contain the logs under Event log Bucket.
   If the logs are not stored in the root of the bucket, provide the full path without `s3://`.
5. Enter a prefix to replicate only the objects whose name begins with the given prefix.

Bucket logging for Multicloud Object Gateway

Bucket logging helps in recording the S3 operations that are performed against the Multicloud Object Gateway (MCG) bucket for compliance, auditing, and optimization purposes.

Bucket logging supports the following two options:

- Best-Effort: Bucket logging is recorded by using the Unified Data Platform (UDP) on a best-effort basis.
- Guaranteed: Bucket logging with this option creates a PersistentVolumeClaim (PVC) attached to the MCG pods and saves the logs to this PVC on a guaranteed basis, and then from the PVC to the log buckets. When this option is used, logging occurs twice for each S3 operation, as follows:
  - At the start of processing the request.
  - At the end, with the result of the S3 operation.

- **Enabling bucket logging by using the Best-Effort option**
  Use the NooBaa API or the S3 API to enable bucket logging for the best-effort option.
- **Enabling bucket logging by using the Guaranteed option**
  Use the default CephFS storage class or ReadWriteMany PersistentVolumeClaim (RWX PVC) to enable bucket logging for the guaranteed option.

Enabling bucket logging by using the Best-Effort option

Use the NooBaa API or the S3 API to enable bucket logging for the best-effort option.

Before you begin

- Ensure that you installed the Openshift Container Platform with the Fusion Data Foundation operator.
- Ensure that you have access to MCG.

  For more information, see Accessing the Multicloud Object Gateway with your applications.

Procedure

1. Create a data bucket where you can upload the objects.

   ```
   odf nb bucket create data.bucket
   ```

2. Create a log bucket where you want to store the logs for bucket operations.

   ```
   odf nb bucket create log.bucket
   ```

3. Configure bucket logging on data bucket with log bucket.

You can configure bucket logging by using the NooBaa API or the S3 API.

- By using the NooBaa API:

```
odf nb api bucket_api put_bucket_logging '{
   "name": "data.bucket",
   "log_bucket": "log.bucket",
   "log_prefix": "data-bucket-logs"
}'
```

- By using the S3 API:

  - Define an alias for the S3 API command.

    ```
    alias s3api_alias='AWS_ACCESS_KEY_ID=$NOOBAA_ACCESS_KEY AWS_SECRET_ACCESS_KEY=$NOOBAA_SECRET_KEY aws --endpoint
    https://localhost:10443 --no-verify-ssl s3api'
    ```

  - Create a setlogging.json file in the following format:

    ```
    {
      "LoggingEnabled": {
         "TargetBucket": "<log-bucket-name>",
         "TargetPrefix": "<prefix/empty-string>"
      }
    }
    ```

  - Configure bucket logging based on settings defined in the setlogging.json file.

    ```
    s3api_alias put-bucket-logging --endpoint <ep> --bucket <source-bucket> --bucket-logging-status
    file://setlogging.json --no-verify-ssl
    ```

4. Verify whether the bucket logging is set for the data bucket in one of the following ways:
   - By using the NooBaa API:

     ```
     odf nb api bucket_api get_bucket_logging '{
        "name": "data.bucket"
     }'
     ```

   - By using the S3 API:

     ```
     s3api_alias get-bucket-logging --no-verify-ssl --endpoint <ep> --bucket <source-bucket>
     ```

     The S3 operations can take up to 24 hours to get recorded in the logs bucket. The following example shows the recorded logs and how to download them:

     ```
     s3_alias cp s3://logs.bucket/data-bucket-logs/logs.bucket.bucket_data-bucket-logs_1719230150.log - | tail -n 2

     Jun 24 14:00:02 10-XXX-X-XXX.sts.openshift-storage.svc.cluster.local
     {"noobaa_bucket_logging":"true","op":"GET","bucket_owner":"operator@noobaa.io","source_bucket":"data.bucket","object_
     key":"/data.bucket?list-type=2&prefix=data-bucket-logs&delimiter=%2F&encoding-
     type=url","log_bucket":"logs.bucket","remote_ip":"100.XX.X.X","request_uri":"/data.bucket?list-type=2&prefix=data-
     bucket-logs&delimiter=%2F&encoding-type=url","request_id":"luv2XXXX-ctyg2k-12gs"} Jun 24 14:00:06 10-XXX-X-
     XXX.s3.openshift-storage.svc.cluster.local
     {"noobaa_bucket_logging":"true","op":"PUT","bucket_owner":"operator@noobaa.io","source_bucket":"data.bucket","object_
     key":"/data.bucket/B69EC83F-0177-44D8-A8D1-
     4A10C5A5AB0F.file","log_bucket":"logs.bucket","remote_ip":"100.XX.X.X","request_uri":"/data.bucket/B69EC83F-0177-
     44D8-A8D1-4A10C5A5AB0F.file","request_id":"luv2XXXX-9syea5-x5z"}
     ```

5. Optional: To disable bucket logging, run the following command:

   ```
   odf nb api bucket_api delete_bucket_logging '{
      "name": "data.bucket"
   }'
   ```

Enabling bucket logging by using the Guaranteed option

Use the default CephFS storage class or ReadWriteMany PersistentVolumeClaim (RWX PVC) to enable bucket logging for the guaranteed option.

Enable guaranteed bucket logging by using the NooBaa Custom Resource (CR) in one of the following ways:

- By using the default Cepheus storage class:

  ```
  bucketLogging:
  {
  loggingType: guaranteed
  }
  ```

- By using the RWX PVC that you created:

  Important: Ensure that the PVC supports RWX.

  ```
  bucketLogging:
  {
  loggingType: guaranteed
  bucketLoggingPVC: <pvc-name>
  }
  ```

Synchronizing versions in Multicloud Object Gateway bucket replication

Learn how to synchronize versions in Multicloud Object Gateway (MCG) bucket replication.

Before you begin

- A Multicloud Object Gateway (MCG) source bucket, which is created from an object bucket claim (OBC) and any MCG target bucket. For example, you can create the two buckets using OBCs using the MCG command line interface (CLI):

  Create a source bucket using an OBC:

  ```
  odf noobaa obc create source-bucket --exact
  odf noobaa obc create target-bucket --exact
  ```

  Where, `--exact` is optional.

- Ensure that S3 client aliases with MCG credentials and endpoint are set up.

  ```
  NOOBAA_ACCESS_KEY=$(oc extract secret/noobaa-admin -n openshift-storage --keys=AWS_ACCESS_KEY_ID --to=- 2>/dev/null); \
  NOOBAA_SECRET_KEY=$(oc extract secret/noobaa-admin -n openshift-storage --keys=AWS_SECRET_ACCESS_KEY --to=- 2>/dev/null); 
  \
  S3_ENDPOINT=https://$(oc get route s3 -n openshift-storage -o json | jq -r ".spec.host")
  ```

  ```
  alias common_s3='AWS_ACCESS_KEY_ID=$NOOBAA_ACCESS_KEY AWS_SECRET_ACCESS_KEY=$NOOBAA_SECRET_KEY aws --endpoint $S3_ENDPOINT 
  --no-verify-ssl'; \
  alias s3_alias='common_s3 s3'; \
  alias s3api_alias='common_s3 s3api'
  ```

- Ensure to enable versioning on both the source and target bucket by using the `put-bucket-versioning` command in the AWS S3 client:

  ```
  s3api_alias put-bucket-versioning --bucket source-bucket --versioning-configuration Status=Enabled
  ```

  ```
  s3api_alias put-bucket-versioning --bucket target-bucket --versioning-configuration Status=Enabled
  ```

Procedure

1. Patch or edit a replication policy with `sync_versions: true` to the source bucket's OBC:

   ```
   oc patch obc source-bucket -n openshift-storage --type=merge -p '{"spec": {"additionalConfig": {"replicationPolicy": "
   {\"rules\":[{\"rule_id\":\"replication-rule-a\",\"destination_bucket\":\"target-bucket\", \"sync_versions\": true}]}"}}}'
   ```

   Normal bucket replication replicates only the latest version, and the use of `sync_versions` add the functionality of replicating even the older versions, and in their original order.

   Note:
   - Delete markers are replicated if configured and using log base replication.
   - Delete version IDs are not replicated to avoid human errors.
2. Verify the replication to the target bucket by adding a few versions under an object key and waiting for them to get replicated to the target bucket:

   ```
   echo 'version_a' | s3_alias cp - s3://source-bucket/versioned_obj.txt
   ```

   ```
   echo 'version_b' | s3_alias cp - s3://source-bucket/versioned_obj.txt
   ```

   ```
   echo 'version_c' | s3_alias cp - s3://source-bucket/versioned_obj.txt
   ```

   After a few minutes, compare the versions of the object on both the buckets:

   ```
   s3api_alias list-object-versions --bucket source-bucket --prefix versioned_obj.txt | jq -r ".Versions[].ETag"
   "aaabf266d38a8e995cef03c13ee9a7f1"
   "181d8a23f59939c0cddec1692e05cdf3"
   "ebc538cc6ffa04a39263f4b1be2f832f"
   ```

   ```
   s3api_alias list-object-versions --bucket target-bucket --prefix versioned_obj.txt | jq -r ".Versions[].ETag"
   "aaabf266d38a8e995cef03c13ee9a7f1"
   "181d8a23f59939c0cddec1692e05cdf3"
   "ebc538cc6ffa04a39263f4b1be2f832f"
   ```

Obtaining metrics to reflect bucket replication state

Learn how to obtain bucket-level replication metrics to evaluate replication progress.

About this task

To determine if the data is safe or available on the secondary site, the replication progress per bucket is provided using the following metrics:

- Total number of objects scanned in the last replication cycle per bucket - `bucket_last_cycle_total_objects_num{bucket="bucket-name"}`
- Number of objects successfully replicated in the last cycle per bucket - `bucket_last_cycle_replicated_objects_num{bucket="bucket-name"}`
- Number of objects that failed replication in the last cycle per bucket - `bucket_last_cycle_error_objects_num{bucket="bucket-name"}`

Procedure

1. Use the following commands to obtain the metrics:

   ```
   JWT_TOKEN=$(oc get secret noobaa-metrics-auth-secret -n openshift-storage -o jsonpath=" {.data.metrics_token}
   " | base64 -d)
   ```

```
oc -n openshift-storage exec noobaa-core-0 – curl -k -H "Authorization: Bearer ${JWT_TOKEN}"
localhost:8080/metrics/bg_workers
```

For example:

```
noobaa-operator$ oc -n openshift-storage exec -it noobaa-core-0 -- curl -k -H "Authorization: Bearer ${JWT_TOKEN}"
http://localhost:8080/metrics/bg_workers | grep -E "bucket_name"| grep -v "NooBaa_replication_status"
Defaulted container "core" out of: core, noobaa-log-processor

NooBaa_bucket_last_cycle_total_objects_num{bucket_name="test-buck1"} 15
NooBaa_bucket_last_cycle_total_objects_num{bucket_name="test-buck2"} 25
NooBaa_bucket_last_cycle_replicated_objects_num{bucket_name="test-buck1"} 15
NooBaa_bucket_last_cycle_replicated_objects_num{bucket_name="test-buck2"} 25
NooBaa_bucket_last_cycle_error_objects_num{bucket_name="test-buck1"} 0
NooBaa_bucket_last_cycle_error_objects_num{bucket_name="test-buck2"} 0
```

2. Use the following syntax to view the metrics in the metrics section:

```
metric_name {bucket_name='bucket_name'}
```

For example:

For `NooBaa_bucket_last_cycle_total_objects_num`

```
NooBaa_bucket_last_cycle_total_objects_num {bucket_name='bubck1'}
```

For `NooBaa_bucket_last_cycle_replicated_objects_num`

```
NooBaa_bucket_last_cycle_replicated_objects_num {bucket_name='bubck1'}
```

For `NooBaa_bucket_last_cycle_error_objects_num`

```
NooBaa_bucket_last_cycle_error_objects_num {bucket_name='bubck1'}
```

Object Bucket Claim

An Object Bucket Claim can be used to request an S3 compatible bucket backend for your workloads. An object bucket claim creates a new bucket and an application account in NooBaa with permissions to the bucket, including a new access key and secret access key. The application account is allowed to access only a single bucket and can't create new buckets by default. Use this information to create, view, and delete object bucket claims.

Object Bucket Claim can be created in three ways: dynamically, using the command line interface, and using OpenShift Web Console.

- **Dynamic Object Bucket Claim**
  Similar to Persistent Volumes, you can add the details of the Object Bucket claim (OBC) to your application's YAML, and get the object service endpoint, access key, and secret access key available in a configuration map and secret. It is easy to read this information dynamically into environment variables of your application. Use this information to add the details of the Object Bucket claim (OBC) to your application's YAML.
- **Creating an Object Bucket Claim using the command line interface**
  Use this information to create an Object Bucket Claim using the command line interface.
- **Creating an Object Bucket Claim using the OpenShift Web Console**
  Use this information to create an Object Bucket Claim using the OpenShift Web Console.
- **Attaching an Object Bucket Claim to a deployment**
  Use this information to attach an Object Bucket Claim (OBC) to a deployment. Attaching to specific deployments can only be done after the OBCs are created.
- **Viewing object buckets using the OpenShift Web Console**
  You can view the details of object buckets created for Object Bucket Claims (OBCs) using the OpenShift Web Console.
- **Deleting Object Bucket Claims**
  When you no longer need the Object Bucket Claims, you can delete it from the namespace using Open Shift Web Console.

Dynamic Object Bucket Claim

Similar to Persistent Volumes, you can add the details of the Object Bucket claim (OBC) to your application's YAML, and get the object service endpoint, access key, and secret access key available in a configuration map and secret. It is easy to read this information dynamically into environment variables of your application. Use this information to add the details of the Object Bucket claim (OBC) to your application's YAML.

About this task

Note: The Multicloud Object Gateway endpoints uses self-signed certificates only if OpenShift uses self-signed certificates. Using signed certificates in OpenShift automatically replaces the Multicloud Object Gateway endpoints certificates with signed certificates. Get the certificate currently used by Multicloud Object Gateway by accessing the endpoint through the browser. For more information, see Accessing the Multicloud Object Gateway with your applications.

Procedure

1. Add the following line to your application YAML:
   These lines are the OBC itself.

```
apiVersion: objectbucket.io/v1alpha1
kind: ObjectBucketClaim
metadata:
  name: <obc-name>
spec:
```

```
  generateBucketName: <obc-bucket-name>
  storageClassName: openshift-storage.noobaa.io
```

<obc-name>
    Use a unique OBC name.

<obc-bucket-name>
    Use a unique bucket name for your OBC.

2. To automate the use of OBC, add more lines to the YAML file.
   For example:

```
apiVersion: batch/v1
kind: Job
metadata:
  name: testjob
spec:
  template:
    spec:
      restartPolicy: OnFailure
      containers:
        - image: <your application image>
          name: test
          env:
            - name: BUCKET_NAME
              valueFrom:
                configMapKeyRef:
                  name: <obc-name>
                  key: BUCKET_NAME
            - name: BUCKET_HOST
              valueFrom:
                configMapKeyRef:
                  name: <obc-name>
                  key: BUCKET_HOST
            - name: BUCKET_PORT
              valueFrom:
                configMapKeyRef:
                  name: <obc-name>
                  key: BUCKET_PORT
            - name: AWS_ACCESS_KEY_ID
              valueFrom:
                secretKeyRef:
                  name: <obc-name>
                  key: AWS_ACCESS_KEY_ID
            - name: AWS_SECRET_ACCESS_KEY
              valueFrom:
                secretKeyRef:
                  name: <obc-name>
                  key: AWS_SECRET_ACCESS_KEY
```

   The example is the mapping between the bucket claim result, which is a configuration map with data and a secret with the credentials. This specific job claims the Object Bucket from NooBaa, which creates a bucket and an account.

   <obc-name>
       Use your OBC name.

   <your application image>
       Use your application image.

3. Apply the updated YAML file, where *<yaml.file>* is the name of your YAML file.

```
oc apply -f <yaml.file>
```

4. To view the new configuration map, run the following command, where *<obc-name>* is the name of your OBC.

```
oc get cm <obc-name> -o yaml
```

   Expect the following environment variables in the output:

   BUCKET_HOST
       Endpoint to use in the application.

   BUCKET_HOST
       The port is related to the *BUCKET_HOST*. For example, if the `BUCKET_HOST` is `https://my.example.com`, and the `BUCKET_PORT` is 443, the endpoint for the object service would be `https://my.exaple.com:443`.

   BUCKET_NAME
       Requested or generated bucket name.

   AWS_ACCESS_KEY_ID
       Access key that is part of the credentials.

   AWS_SECRET_ACCESS_KEY
       Secret access key that is part of the credentials.

   Important: Retrieve the *AWS_ACCESS_KEY_ID* and *AWS_SECRET_ACCESS_KEY*. The names are used so that it is compatible with the AWS S3 API. You need to specify the keys while performing S3 operations, especially when you read, write or list from the Multicloud Object Gateway (MCG) bucket. The keys are encoded in Base64.
   Decode the keys before using them, using the following command, where *<obc_name>* specifies the name of the object bucket claim:

```
oc get secret <obc_name> -o yaml
```

Creating an Object Bucket Claim using the command line interface

Use this information to create an Object Bucket Claim using the command line interface.

Before you begin

Download the Multicloud Object Gateway (MCG) command-line interface.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
 yum install mcg
```

Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.

- For IBM Power, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
  ```

- For IBM Z infrastructure, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
  ```

About this task

When creating an Object Bucket Claim (OBC) using the command-line interface, you get a configuration map and a Secret that together contain all the information your application needs to use the object storage service.

Procedure

1. Use the command-line interface to generate the details of a new bucket and credentials.

   ```
   odf noobaa obc create <obc-name> -n openshift-storage
   ```

   Replace *<obc-name>* with a unique OBC name, for example, **myappobc**.

   Additionally, you can use the **--app-namespace** option to specify the namespace where the OBC configuration map and secret will be created, for example, **myapp-namespace**.

   For example:

   ```
   INFO[0001] ✅ Created: ObjectBucketClaim "test21obc"
   ```

2. Run the following command to view the OBC.

   ```
   oc get obc -n openshift-storage
   ```

   Example output:

   ```
   NAME        STORAGE-CLASS                 PHASE   AGE
   test21obc   openshift-storage.noobaa.io   Bound   38s
   ```

3. Run the following command to view the YAML file for the new OBC.

   ```
   oc get obc test21obc -o yaml -n openshift-storage
   ```

   Example YAML output:


   ```
   apiVersion: objectbucket.io/v1alpha1
   kind: ObjectBucketClaim
   metadata:
     creationTimestamp: "2019-10-24T13:30:07Z"
     finalizers:
     - objectbucket.io/finalizer
     generation: 2
     labels:
       app: noobaa
       bucket-provisioner: openshift-storage.noobaa.io-obc
       noobaa-domain: openshift-storage.noobaa.io
     name: test21obc
     namespace: openshift-storage
     resourceVersion: "40756"
     selfLink: /apis/objectbucket.io/v1alpha1/namespaces/openshift-storage/objectbucketclaims/test21obc
     uid: 64f04cba-f662-11e9-bc3c-0295250841af
   spec:
     ObjectBucketName: obc-openshift-storage-test21obc
     bucketName: test21obc-933348a6-e267-4f82-82f1-e59bf4fe3bb4
     generateBucketName: test21obc
     storageClassName: openshift-storage.noobaa.io
   status:
     phase: Bound
   ```


4. Inside of your **openshift-storage** namespace, you can find the configuration map and the secret to use this OBC.
   The CM and the secret have the same name as the OBC.

   ```
   oc get -n openshift-storage secret test21obc -o yaml
   ```

   For example:


   ```
   apiVersion: v1
   data:
     AWS_ACCESS_KEY_ID: c0M0R2xVanF3ODR3bHBkVW94cmY=
     AWS_SECRET_ACCESS_KEY: Wi9kcFluSWxHRzlWaFlzNk1hc0xma2JXcjM1MVhqa051S1BleXpmOQ==
   kind: Secret
   ```

```
metadata:
  creationTimestamp: "2019-10-24T13:30:07Z"
  finalizers:
  - objectbucket.io/finalizer
  labels:
    app: noobaa
    bucket-provisioner: openshift-storage.noobaa.io-obc
    noobaa-domain: openshift-storage.noobaa.io
  name: test21obc
  namespace: openshift-storage
  ownerReferences:
  - apiVersion: objectbucket.io/v1alpha1
    blockOwnerDeletion: true
    controller: true
    kind: ObjectBucketClaim
    name: test21obc
    uid: 64f04cba-f662-11e9-bc3c-0295250841af
  resourceVersion: "40751"
  selfLink: /api/v1/namespaces/openshift-storage/secrets/test21obc
  uid: 65117c1c-f662-11e9-9094-0a5305de57bb
type: Opaque
```

The secret gives you S3 access credentials.

5. Run the following command to see the configuration map:

```
oc get -n openshift-storage cm test21obc -o yaml
```

For example:

```
apiVersion: v1
data:
  BUCKET_HOST: 10.0.171.35
  BUCKET_NAME: test21obc-933348a6-e267-4f82-82f1-e59bf4fe3bb4
  BUCKET_PORT: "31242"
  BUCKET_REGION: ""
  BUCKET_SUBREGION: ""
kind: ConfigMap
metadata:
  creationTimestamp: "2019-10-24T13:30:07Z"
  finalizers:
  - objectbucket.io/finalizer
  labels:
    app: noobaa
    bucket-provisioner: openshift-storage.noobaa.io-obc
    noobaa-domain: openshift-storage.noobaa.io
  name: test21obc
  namespace: openshift-storage
  ownerReferences:
  - apiVersion: objectbucket.io/v1alpha1
    blockOwnerDeletion: true
    controller: true
    kind: ObjectBucketClaim
    name: test21obc
    uid: 64f04cba-f662-11e9-bc3c-0295250841af
  resourceVersion: "40752"
  selfLink: /api/v1/namespaces/openshift-storage/configmaps/test21obc
  uid: 651c6501-f662-11e9-9094-0a5305de57bb
```

The configuration map contains the S3 endpoint information for your application.

Creating an Object Bucket Claim using the OpenShift Web Console

Use this information to create an Object Bucket Claim using the OpenShift Web Console.

Before you begin

Ensure you have the following:

- Administrative access to the OpenShift Web Console.
- In order for your applications to communicate with the OBC, you need to use the configmap and secret. For more information, see Dynamic Object Bucket Claim.

Procedure

From the OpenShift Web Console, go to Storage > Object Bucket Claims > Create Object Bucket Claim.

1. Enter a name for your object bucket claim.
2. Select the appropriate storage class based on your deployment, either internal or external, from the drop-down menu.
   - **Internal mode**
     The following storage classes, which were created after deployment, are available for use:

     `ocs-storagecluster-ceph-rgw`
     : Uses the Ceph Object Gateway (RGW).

     `openshift-storage.noobaa.io`
     : Uses the Multicloud Object Gateway (MCG).

   - **External mode**
     The following storage classes, which were created after deployment, are available for use:

`ocs-external-storagecluster-ceph-rgw`
: uses the RGW.

`openshift-storage.noobaa.io`
: Uses the MCG.

Note: The RGW OBC storage class is only available with fresh installations of Fusion Data Foundation. It does not apply to clusters upgraded from previous OpenShift Data Foundation releases.

3. Click Create.
   After you create the OBC, you are redirected to its detail page.

Attaching an Object Bucket Claim to a deployment

Use this information to attach an Object Bucket Claim (OBC) to a deployment. Attaching to specific deployments can only be done after the OBCs are created.

Before you begin

Ensure you have Administrative access to the OpenShift Web Console.

Procedure

1. Go to Storage > Object Bucket Claim.
2. Click the Action menu next to Object Bucket Claim (OBC) you want to delete.
   a. From the drop-down menu, select Attach to deployment.
   b. Select the desired deployment from the Deployment Name list, then click Attach.

Viewing object buckets using the OpenShift Web Console

You can view the details of object buckets created for Object Bucket Claims (OBCs) using the OpenShift Web Console.

Before you begin

Ensure you have Administrative access to the OpenShift Web Console.

Procedure

1. Log into the Open Shift Web Console and go to Storage > Object Buckets.
   Alternatively, you can also navigate to the details page of a specific OBC, and click the Resource link to view the object buckets for that OBC.
2. Select the object bucket of which you want to see the details.
   After selected you are navigated to the Object Bucket Details page.

Deleting Object Bucket Claims

When you no longer need the Object Bucket Claims, you can delete it from the namespace using Open Shift Web Console.

Before you begin

Ensure you have Administrative access to the OpenShift Web Console.

Procedure

1. Go to Storage > Object Bucket Claims.
2. From next to the Object Bucket Claim (OBC) you want to delete, click Action Menu > Delete Object Bucket Claim
3. Click Delete.

Caching policy for object buckets

A cache bucket is a namespace bucket with a hub target and a cache target. The hub target is an S3 compatible large object storage bucket. The cache bucket is the local Multicloud Object Gateway bucket. You can create a cache bucket that caches an AWS bucket or an IBM COS bucket.

- **Creating an AWS cache bucket**
  Create an AWS cache bucket.
- **Creating an IBM COS cache bucket**
  Create an IBM COS cache bucket.

Creating an AWS cache bucket

Create an AWS cache bucket.

Before you begin

- A running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
  ```

  ```
  yum install mcg
  ```

  Note: Specify the appropriate architecture for enabling the repositories using the subscription manager. In case of IBM Z infrastructure, use the following command:

  ```
  subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
  ```

- Alternatively, you can install the MCG package from the RPMs found at Download Red Hat OpenShift Data Foundation page.

  Note: Choose the correct Product Variant according to your architecture.

About this task

This procedure describes creating a NamespaceStore. A NamespaceStore represents an underlying storage to be used as a read or write target for the data in the MCG namespace buckets.

Procedure

1. Create NamespaceStore in one of the following ways:
   - From the MCG command-line interface, run the following command:

     ```
     odf noobaa namespacestore create aws-s3 <namespacestore> --access-key <AWS ACCESS KEY> --secret-key <AWS SECRET ACCESS KEY> --target-bucket <bucket-name>
     ```

     namespacestore
       Name of the namespacestore

     AWS ACCESS KEY and *AWS SECRET ACCESS KEY*
       AWS access key ID and secret access key you created for this purpose.

     bucket-name
       An existing AWS bucket name. This argument tells the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.

   - Add storage resources by applying a YAML.
     - Create a secret with credentials, where *<namespacestore-secret-name>* is a unique name for the NamespaceStore secret.

       ```
       apiVersion: v1
       kind: Secret
       metadata:
         name: <namespacestore-secret-name>
       type: Opaque
       data:
         AWS_ACCESS_KEY_ID: <AWS ACCESS KEY ID ENCODED IN BASE64>
         AWS_SECRET_ACCESS_KEY: <AWS SECRET ACCESS KEY ENCODED IN BASE64>
       ```

       You must supply and encode your own AWS access key ID and secret access key using Base64, and use the results in place of *<AWS ACCESS KEY ID ENCODED IN BASE64>* and *<AWS SECRET ACCESS KEY ENCODED IN BASE64>*.

     - Apply the following YAML:

       ```
       apiVersion: noobaa.io/v1alpha1
       kind: NamespaceStore
       metadata:
         finalizers:
         - noobaa.io/finalizer
         labels:
           app: noobaa
         name: <namespacestore>
         namespace: openshift-storage
       spec:
         awsS3:
           secret:
             name: <namespacestore-secret-name>
             namespace: <namespace-secret>
           targetBucket: <target-bucket>
         type: aws-s3
       ```

       namespacestore
         A unique name for the NamespaceStore secret.

       namespacestore-secret-name
         A name for the NamespaceStore secret created in the previous step.

       namespace-secret
         Namespace used to create the secret in the previous step.

target-bucket
: the AWS S3 bucket you created for the NamespaceStore.

2. Run the following command to create a bucket class:

```
odf noobaa bucketclass create namespace-bucketclass cache <my-cache-bucket-class> --backingstores <backing-store> --hub-resource <namespacestore>
```

my-cache-bucket-class
: A unique bucket class name.

backing-store
: Name of the relevant backing store. You can also list more than one backing stores separated by commas.

namespacestore
: Name of the NamespaceStore created in the previous step.

3. Run the following command to create a bucket using an Object Bucket Claim (OBC) resource that uses the bucket class defined in the previous step.

```
odf noobaa obc create <my-bucket-claim> my-app --bucketclass <custom-bucket-class>
```

my-bucket-claim
: A unique object bucket claim name.

custom-bucket-class
: Name of the bucket class created in the previous step.

Creating an IBM COS cache bucket

Create an IBM COS cache bucket.

Before you begin

- A running Fusion Data Foundation Platform.
- Download the MCG command-line interface for easier management.

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-x86_64-rpms
```

```
yum install mcg
```

Note: Specify the appropriate architecture for enabling the repositories using the subscription manager.

  - For IBM Power, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-ppc64le-rpms
```

  - For IBM Z infrastructure, use the following command:

```
subscription-manager repos --enable=rh-odf-4-for-rhel-8-s390x-rpms
```

- Alternatively, you can install the MCG package from the RPMs found at [Download Red Hat OpenShift Data Foundation page](#).

Note: Choose the correct Product Variant according to your architecture.

About this task

This procedure describes creating a NamespaceStore. A NamespaceStore represents an underlying storage to be used as a read or write target for the data in the MCG namespace buckets.

Procedure

1. Create NamespaceStore in one of the following ways:
   - From the MCG command-line interface, run the following command:

```
odf noobaa namespacestore create ibm-cos <namespacestore> --endpoint <IBM COS ENDPOINT> --access-key <IBM ACCESS KEY> --
```

namespacestore
: Name of the NamespaceStore

IBM COS ENDPOINT
: An appropriate regional endpoint that corresponds to the location of the existing IBM bucket.

IBM ACCESS KEY and *IBM SECRET ACCESS KEY*
: IBM access key ID and secret access key you created for this purpose.

bucket-name
: An existing IBM bucket name. This argument tells the MCG which bucket to use as a target bucket for its backing store, and subsequently, data storage and administration.

   - Add storage resources by applying a YAML.
     - Create a secret with credentials, where *<namespacestore-secret-name>* is a unique name for the NamespaceStore secret.

```
apiVersion: v1
kind: Secret
metadata:
  name: <namespacestore-secret-name>
type: Opaque
data:
```

```
  IBM_COS_ACCESS_KEY_ID: <IBM COS ACCESS KEY ID ENCODED IN BASE64>
  IBM_COS_SECRET_ACCESS_KEY: <IBM COS SECRET ACCESS KEY ENCODED IN BASE64>
```

You must supply and encode your own IBM COS access key ID and secret access key using Base64, and use the results in place of *<IBM COS ACCESS KEY ID ENCODED IN BASE64>* and *<IBM COS SECRET ACCESS KEY ENCODED IN BASE64>*.

- Apply the following YAML:

```
apiVersion: noobaa.io/v1alpha1
kind: NamespaceStore
metadata:
  finalizers:
  - noobaa.io/finalizer
  labels:
    app: noobaa
  name: <namespacestore>
  namespace: openshift-storage
spec:
  s3Compatible:
    endpoint: <IBM COS ENDPOINT>
    secret:
      name: <backingstore-secret-name>
      namespace: <namespace-secret>
    signatureVersion: v2
    targetBucket: <target-bucket>
  type: ibm-cos
```

namespacestore
: A unique name for the NamespaceStore secret.

IBM COS ENDPOINT
: An appropriate regional endpoint that corresponds to the location of the existing IBM bucket.

backingstore-secret-name
: A name for the backing store secret created in the previous step.

namespace-secret
: Namespace used to create the secret in the previous step.

target-bucket
: the IBM COS bucket you created for the NamespaceStore.

2. Run the following command to create a bucket class:

```
odf noobaa bucketclass create namespace-bucketclass cache <my-bucket-class> --backingstores <backing-store> --hub-resource
<namespacestore>
```

my-bucket-class
: A unique bucket class name.

backing-store
: Name of the relevant backing store. You can also list more than one backing stores separated by commas.

namespacestore
: Name of the NamespaceStore created in the previous step.

3. Run the following command to create a bucket using an Object Bucket Claim (OBC) resource that uses the bucket class defined in the previous step.

```
odf noobaa obc create <my-bucket-claim> my-app --bucketclass <custom-bucket-class>
```

my-bucket-claim
: A unique object bucket claim name.

custom-bucket-class
: Name of the bucket class created in the previous step.

Lifecycle bucket configuration in Multicloud Object Gateway

Multicloud Object Gateway (MCG) lifecycle provides a way to reduce storage costs due to accumulated data objects.

Deletion of expired objects is a simplified way that enables handling of unused data. Data expiration is a part of Amazon Web Services (AWS) lifecycle management and sets an expiration date for automatic deletion. The minimal time resolution of the lifecycle expiration is one day. For more information, see Expiring objects.

AWS S3 API is used to configure lifecycle bucket in MCG. For information about the data bucket APIs and their support level, see Support of Multicloud Object Gateway data bucket APIs.

There are a few limitations with the expiratation rule API for MCG in comparison with AWS:

- `ExpiredObjectDeleteMarker` is accepted but it is not processed.
- No option to define specific non-current version’s expiration conditions

The expiration rule API for MCG doesn’t support transition rules like AWS does for both current and noncurrent objects.

You can configure the bucket lifecycle by using the MCG object browser. For more information, see Creating new lifecycle rules by using the object browser.

Scaling Multicloud Object Gateway performance

The Multicloud Object Gateway (MCG) performance may vary from one environment to another. In some cases, specific applications require faster performance which can be easily addressed by scaling S3 endpoints.

The MCG resource pool is a group of NooBaa daemon containers that provide two types of services enabled by default:

- Storage service
- S3 endpoint service

S3 endpoint service

The S3 endpoint is a service that every Multicloud Object Gateway (MCG) provides by default that handles the heavy lifting data digestion in the MCG. The endpoint service handles the inline data chunking, deduplication, compression, and encryption, and it accepts data placement instructions from the MCG.

- **Automatic scaling of MultiCloud Object Gateway endpoints**
  The number of MultiCloud Object Gateway (MCG) endpoints scale automatically when the load on the MCG S3 service increases or decreases.
- **Autoscaling RGW in Fusion Data Foundation using HPA and KEDA**
  Enable autoscaling for RGW (RADOS Gateway) in Fusion Data Foundation by integrating with OpenShift Horizontal Pod Autoscaler (HPA) with KEDA (Kubernetes Event-Driven Autoscaling).
- **Scaling the Multicloud Object Gateway with storage nodes**
  A storage node in the MCG is a NooBaa daemon container attached to one or more Persistent Volumes (PVs) and used for local object service data storage. NooBaa daemons can be deployed on Kubernetes nodes. This can be done by creating a Kubernetes pool consisting of `StatefulSet` pods.
- **Increasing CPU and memory for PV pool resources**
  Multicloud Object Gateway (MCG) default configuration supports low resource consumption. However, when you need to increase CPU and memory to accommodate specific workloads and to increase MCG performance for the workloads, it is possible to configure the required values for CPU and memory in the OpenShift Web Console.

Automatic scaling of MultiCloud Object Gateway endpoints

The number of MultiCloud Object Gateway (MCG) endpoints scale automatically when the load on the MCG S3 service increases or decreases.

Fusion Data Foundation clusters are deployed with one active MCG endpoint. Each MCG endpoint pod is configured by default with 1 CPU and 2Gi memory request, with limits matching the request. When the CPU load on the endpoint crosses over an 80% usage threshold for a consistent period of time, a second endpoint is deployed lowering the load on the first endpoint. When the average CPU load on both endpoints falls below the 80% threshold for a consistent period of time, one of the endpoints is deleted. This feature improves performance and serviceability of the MCG.

Autoscaling RGW in Fusion Data Foundation using HPA and KEDA

Enable autoscaling for RGW (RADOS Gateway) in Fusion Data Foundation by integrating with OpenShift Horizontal Pod Autoscaler (HPA) with KEDA (Kubernetes Event-Driven Autoscaling).

Before you begin

- All resources described in this procedure (Roles, RoleBindings, TriggerAuthentication, ScaledObject, and so on) must be created in the `openshift-storage` namespace.
- Custom Metrics Autoscaler Operator version must be 2.6 or later.

Procedure

1. Install the Custom Metrics Autoscaler Operator.
   a. Log in to the Red Hat® OpenShift® Container Platform web console.
   b. Click Ecosystem > Software Catalog.
   c. Scroll or type Custom Metrics Autoscaler into the Filter by keyword box to find the Custom Metrics Autoscaler Operator.
   d. Click Install.
2. Create a Custom Metrics Autoscaler controller and verify whether the following pods are running in the `openshift-keda` namespace:

```
oc get pods -n openshift-keda
NAME                                                  READY   STATUS    RESTARTS   AGE
custom-metrics-autoscaler-operator-6cb698bb4d-nnpmj   1/1     Running   0          103m
keda-admission-c6d879546-twprp                        1/1     Running   0          103m
keda-metrics-apiserver-68788fbd9b-h715n               1/1     Running   0          103m
keda-operator-664dc57859-mb49f                        1/1     Running   0          20m
```

3. Configure Custom Metrics Autoscaler with the Thanos Query Service.
   Prometheus scaler that is supported by Custom Metrics Autoscaler collects metrics through the Thanos Query service provided by OpenShift Monitoring. Because Thanos Query requires authentication, a ServiceAccount token must be used for bearer authentication.

   a. Create a service account.
      Command example:

```
oc create serviceaccount thanos -n openshift-storage
```

   b. Create a secret to generate the token.

```
apiVersion: v1
kind: Secret
metadata:
  name: thanos-token-secret
  namespace: openshift-storage
  annotations:
```

```
    kubernetes.io/service-account.name: "thanos"
type: kubernetes.io/service-account-token
```

Command example:

```
oc apply -f secret.yaml
```

c. Add the `cluster-monitoring-view` cluster role to the created Service Account in the `openshift-storage` namespace.
Command example:

```
oc adm policy add-cluster-role-to-user cluster-monitoring-view -z thanos
```

d. Create a `triggerAuthentication` CRD for the bearer authentication using the secret name.

```
apiVersion: keda.sh/v1alpha1
kind: TriggerAuthentication
metadata:
  name: keda-trigger-auth-prometheus
  namespace: openshift-storage
spec:
  secretTargetRef:
  - parameter: bearerToken
    name: thanos-token-secret
    key: token
  - parameter: ca
    name: thanos-token-secret
    key: ca.crt
```

4. Enable RGW autoscaling.
Command example:

```
oc annotate storagecluster ocs-storagecluster ocs.openshift.io/enable-rgw-autoscale="true" -n openshift-storage
```

This behavior takes precedence over manual scaling. You can revert this change by performing the following command:

```
oc annotate storagecluster ocs-storagecluster ocs.openshift.io/enable-rgw-autoscale- -n openshift-storage
```

5. Create ScaledObject for autoscaling RGW.
Autoscaling using the HPA feature of OpenShift triggers scaling based on custom metrics. The prometheus metrics related to RGW exported by the Ceph manager is used. The following example uses `ceph_rgw_put_obj_ops` metrics and scale up to five pods.

```
apiVersion: keda.sh/v1alpha1
kind: ScaledObject
metadata:
  name: rgw-scale-namespace
  namespace: openshift-storage
spec:
  scaleTargetRef:
    apiVersion: ceph.rook.io/v1
    kind: CephObjectStore
    name: ocs-storagecluster-cephobjectstore
    namespace: openshift-storage
  minReplicaCount: 1
  maxReplicaCount: 5
  triggers:
    - type: prometheus
      metadata:
        serverAddress: https://thanos-querier.openshift-monitoring.svc.cluster.local:9091
        metricName: ceph_rgw_put_obj_ops
        query: |
          sum(rate(ceph_rgw_op_put_obj_ops[2m]))
        threshold: "1"
        authModes: "bearer"
        namespace: openshift-storage
      authenticationRef:
        name: keda-trigger-auth-prometheus
```

Scaling the Multicloud Object Gateway with storage nodes

A storage node in the MCG is a NooBaa daemon container attached to one or more Persistent Volumes (PVs) and used for local object service data storage. NooBaa daemons can be deployed on Kubernetes nodes. This can be done by creating a Kubernetes pool consisting of `StatefulSet` pods.

Before you begin

Ensure you have a running Fusion Data Foundation cluster on OpenShift Container Platform with access to the Multicloud Object Gateway (MCG).

Procedure

1. Log in to the OpenShift Web Console.
2. From the MCG user interface, go to Overview > Add Storage Resources.
3. Click **Deploy Kubernetes Pool**.
4. In the Create Pool step, create the target pool for the future installed nodes.
5. In the Configure step, configure the number of requested pods and the size of each PV.
For each new pod, one PV is to be created
6. In the Review step, you can find the details of the new pool and select the deployment method you wish to use: local or external deployment.

If local deployment is selected, the Kubernetes nodes will deploy within the cluster. If external deployment is selected, you will be provided with a YAML file to run externally.

What to do next

All nodes will be assigned to the pool you chose in the first step. To verify, go to Resources > Storage resources > Resource name.

Increasing CPU and memory for PV pool resources

Multicloud Object Gateway (MCG) default configuration supports low resource consumption. However, when you need to increase CPU and memory to accommodate specific workloads and to increase MCG performance for the workloads, it is possible to configure the required values for CPU and memory in the OpenShift Web Console.

Procedure

1. From the OpenShift Web Console, go to Storage > Object storage > Backingstore tab.
2. Select the new backingstore.
3. Click Edit PV pool resources.
4. From the edit window, edit the following values, based on your requirements: based on the requirement.
   Mem, CPU, and Vol size
5. Click **Save**.

What to do next

To verify, check the resource values of the PV pool pods.

Accessing the RADOS Object Gateway S3 endpoint

Users can access the RADOS Object Gateway (RGW) endpoint directly. The RGW route is created by default and is named **`rook-ceph-rgw-ocs-storagecluster-cephobjectstore`**.

Using TLS certificates for applications accessing RGW

Fetch TLS certificates stored as Kubernetes secrets and configure applications to use them when accessing the RADOS Object Gateway (RGW) over a secure connection.

Before you begin

A running Fusion Data Foundation cluster.

About this task

Most of the S3 applications require TLS certificate in the forms such as an option included in the Deployment configuration file, passed as a file in the request, or stored in **`/etc/pki`** paths.

TLS certificates for RADOS Object Gateway (RGW) are stored as Kubernetes secret and you need to fetch the details from the secret.

Procedure

- For internal RGW server
  Get the TLS certificate and key from the kubernetes secret:

  ```
  oc get secrets/<secret_name> --template={{.data.tls.crt}} | base64 -d
  ```

  ```
  oc get secrets/<secret_name> --template={{.data.tls.key}} | base64 -d
  ```

  <secret_name>

  The default kubernetes secret name is **`<objectstore_name>-cos-ceph-rgw-tls-cert`**. Specify the name of the object store.

- For external RGW server
  Get the TLS certificate from the kubernetes secret:

  ```
  oc get secrets/<secret_name> --template={{.data.cert}} | base64 -d
  ```

  <secret_name>

  The default kubernetes secret name is **`ceph-rgw-tls-cert`** and it is an opaque type of secret. The key value for storing the TLS certificates is **`cert`**.

- **Accessing external RGW server**
  This section describes the two ways of accessing external RGW server

Accessing external RGW server

This section describes the two ways of accessing external RGW server

The S3 credentials such as `AccessKey` or `Secret Key` is stored in the secret generated by the Object Bucket Claim (OBC) creation and you can fetch the same by using the following commands:

Accessing External RGW server using Object Bucket Claims

```
oc get secret <object bucket claim name> -o jsonpath='{.data.AWS_SECRET_ACCESS_KEY}' | base64 --decode
oc get secret <object bucket claim name> -o jsonpath='{.data.AWS_ACCESS_KEY_ID}' | base64 --decode
```

Similarly, you can fetch the endpoint details from the `configmap` of OBC:

```
oc get cm <object bucket claim name> -o jsonpath='{.data.BUCKET_HOST}'
oc get cm <object bucket claim name> -o jsonpath='{.data.BUCKET_PORT}'
oc get cm <object bucket claim name> -o jsonpath='{.data.BUCKET_NAME}'
```

Accessing External RGW server using the `Ceph Object Store User` CR

You can fetch the S3 Credentials and endpoint details from the secret generated as part of the `Ceph Object Store User` CR:

```
oc get secret rook-ceph-object-user-<object-store-cr-name>-<object-user-cr-name> -o jsonpath='{.data.AccessKey}' | base64 --
decode
oc get secret rook-ceph-object-user-<object-store-cr-name>-<object-user-cr-name> -o jsonpath='{.data.SecretKey}' | base64 --
decode
oc get secret rook-ceph-object-user-<object-store-cr-name>-<object-user-cr-name> -o jsonpath='{.data.Endpoint}' | base64 --
decode
```

Important:
For both the access mechanisms, you can either request for new certificates from the administrator or reuse the certificates from the Kubernetes secret, `ceph-rgw-tls-cert`.

Using the security token service of Multicloud Object Gateway to assume the role of another user

Multicloud Object Gateway (MCG) supports for a security token service (STS) similar to the one provided by Amazon Web Services. This STS of the MCG authorizes a user to assume the role of another user.

About this task

To authorize other users to assume the role of a certain user, you need to assign a role configuration to the user. You can manage the configuration of roles that use the MCG command-line interface (CLI) tool.

The following example shows the role configuration that allows two MCG users (*assumer@mcg.test* and *assumer2@mcg.test*) to assume a certain user's role:

```
'{"role_name": "AllowTwoAssumers", "assume_role_policy": {"version": "2012-10-17", "statement": [ {"action":
["sts:AssumeRole"], "effect": "allow", "principal": ["assumer@mcg.test", "assumer2@mcg.test"]}]}}'
```

Procedure

1. Assign the role configuration by using the MCG CLI tool.

   ```
   mcg sts assign-role --email <assumed user's username> --role_config '{"role_name": "AllowTwoAssumers",
   "assume_role_policy": {"version": "2012-10-17", "statement": [ {"action": ["sts:AssumeRole"], "effect": "allow",
   "principal": ["assumer@mcg.test", "assumer2@mcg.test"]}]}}'
   ```

2. Collect the following information:
   - The access key ID and secret access key of the assumer (the user who assumes the role).
   - The MCG STS endpoint, which can be retrieved by using the command:

     ```
     oc -n openshift-storage get route
     ```

   - The access key ID of the assumed user.
   - The value of the *role_name* in your role configuration.
   - A name of your choice for the role session.
3. Assign the configuration role to the appropriate user.

   ```
   AWS_ACCESS_KEY_ID=<aws-access-key-id> AWS_SECRET_ACCESS_KEY=<aws-secret-access-key1> aws --endpoint-url <mcg-sts-endpoint>
   sts assume-role --role-arn arn:aws:sts::<assumed-user-access-key-id>:role/<role-name> --role-session-name <role-session-
   name>
   ```

   Note: Adding --no-verify-ssl might be necessary depending on the configuration of the cluster.
   The output contains the access key ID, secret access key, and session token that can be used for running actions while assuming the other user's role.

   You can use the credentials generated after the assume role steps, as shown in the following example:

   ```
   AWS_ACCESS_KEY_ID=<aws-access-key-id> AWS_SECRET_ACCESS_KEY=<aws-secret-access-key1> AWS_SESSION_TOKEN=<session token> aws
   --endpoint-url <mcg-s3-endpoint> s3 ls
   ```

Replacing nodes and devices

Explains how to safely replace a node and storage devices in a Fusion Data Foundation cluster.

Fusion Data Foundation node replacement can be performed proactively for an operational node and reactively for a failed node. See Fusion Data Foundation storage node failure to replace storage nodes on Fusion Data Foundation using local storage devices.

See Disk replacement and OpenShift dedicated OSD failure recovery to replace storage devices on Fusion Data Foundation using local storage devices.

Fusion Data Foundation service install and upgrade issues

Use this troubleshooting information to resolve install and upgrade problems that are related to Fusion Data Foundation service.

- ceph-csi-controller-manager pod in Crashloopbackoff state
- Data Foundation operator upgrade is not initiated
- Data Foundation ocs-client-operator-console-xxx pod is stuck while upgrade

Known issues in upgrade

- You cannot upgrade Fusion Data Foundation from 4.14 or 4.15 to 4.16 instead do a fresh installation of Fusion Data Foundation 4.16.

ceph-csi-controller-manager pod in Crashloopbackoff state

Problem statement
: After the upgrade of OpenShift® Container Platform 4.16 and Fusion Data Foundation 4.16 to 4.17, the Fusion Data Foundation goes into a degraded state.

Cause
: It error occurs because of the `Crashloopbackoff` state of the `ceph-csi-controller-manager` pod.

Resolution
: Do the following steps to resolve and validate this error:

1. Run the following example command to remove the limit from CSV:

```
oc patch csv cephcsi-operator.v4.17.1 --type json -p '[ {"op": "remove", "path":
"/spec/install/spec/deployments/0/spec/template/spec/containers/1/resources/limits"} ]'
clusterserviceversion.operators.coreos.com/cephcsi-operator.v4.17.1 patched
```

2. Run the following command to verify the pods status:

```
oc get pods
```

Example output:

```
NAME                                              READY   STATUS    RESTARTS      AGE
ceph-csi-controller-manager-675dc67f7b-f64pb      2/2     Running   0             3h39m
ceph-csi-controller-manager-8854c7f5-xdrvx        1/2     Running   0             6s
```

Data Foundation operator upgrade is not initiated

Problem statement
: Data Foundation operator upgrade is not initiated even though the service is installed successfully. If auto-upgrade is not enabled for Data Foundation during the upgrade to version 4.17 or later, some components might not upgrade automatically.

Resolution
: To ensure all components are upgraded successfully, verify that all `InstallPlans` in the `openshift-storage` namespace are approved. Follow the steps to verify and manually approve any pending `InstallPlans`:

1. Run the following command to retrieve the list of `InstallPlans` in the `openshift-storage` namespace.

```
oc get installplan -n openshift-storage
```

2. Check the `APPROVED` column in the output and make sure that all listed `InstallPlans` status set to `true`.
3. If any `InstallPlans` in the `APPROVED` column is set to `false`, then manually approve it by running the following command.

```
for ip in $(oc get installplan --no-headers -n openshift-storage | grep false | awk '{print $1}')
do
  oc patch installplan $ip -n openshift-storage --type merge --patch '{"spec":{"approved":true}}'
done
```

4. After approving all pending `InstallPlans`, repeat the steps 1 and 2 to confirm that all `InstallPlans` status set to true.

Data Foundation `ocs-client-operator-console-xxx` pod is stuck while upgrade

Problem statement
: During the Data Foundation service upgrade from version 4.18.2 to 4.18.3, an error occurred: the `odf-operator` CSV `ocs-operator.v4.18.3` is unavailable due to `ocs-client-operator-console-xxx` pod is stuck in `ContainerCreating` state.

Resolution

Delete the pod to resolve the issue using the following command.

```
oc delete pod <name> -nopenshift-storage
```

Fusion Data Foundation service error scenarios

Use these troubleshooting information to know the problem and workaround when install or configure Fusion Data Foundation service.

- Fusion Data Foundation storage node failure
- Disk replacement and OpenShift dedicated OSD failure recovery
- Storage client status reporter pod is stuck in terminating state
- Admission webhook warning
- Local storage operator unable to find candidate storage nodes
- Fusion Data Foundation capacity cannot be loaded
- Fusion Data Foundation cluster fails due to pending StorageClusterPreparing stage
- Deletion of StorageCluster doesn't delete connected StorageClient
- The StorageClient pod is in ContainerStatusUnknown and CrashLoopBackOff state

Admission webhook warning

After you create Object Bucket Claim, the following error occurs:

```
Admission webhook warning :- ObjectBucketClaim my-obj-bucket violates policy 299 - "unknown field "spec.ss||"' after object bucket 
```

For more information about this known issue, see Red Hat Known issues.

Local storage operator unable to find candidate storage nodes

Problem statement
: When you configure a Fusion Data Foundation cluster, you do not find any candidate storage nodes.

Cause
: When you configure Fusion Data Foundation cluster, only compute nodes with available disks (SSD/NVMe or HDD) get displayed in the Data Foundation page of IBM Fusion user interface. The following nodes get filtered out and do not display on the screen:

- Nodes have SSD/NVMe or HDD disks but they are not in available state
- The selected disk properties are not present in current node. For example, disk size or disk type.
- The total disk count (with same disk size, disk type) is less than 3.

Steps to verify whether you have the correct storage node candidates

1. In Red Hat OpenShift Container Platform console, go to Operators > Installed Operators.
2. Verify whether the `LocalStorage` operator is installed successfully.
3. Run the following command to get all the worker nodes:

   ```
   oc get node -l node-role.kubernetes.io/worker=
   ```

4. Run the following command to check if discovery results are created for all worker nodes.

   ```
   oc get localvolumediscoveryresult -n openshift-local-storage
   ```

5. Run the following command to confirm that none of the nodes have a Fusion Data Foundation storage label:

   ```
   oc get node -l cluster.ocs.openshift.io/openshift-storage=
   ```

Note: In Linux on IBM Z platform, disks might be formatted and partitioned first. For more information about this behavior, see Red Hat OpenShift Data Foundation on IBM Z and IBM LinuxONE - Reference Architecture section *4.1.1 and 4.1.2*.

If all the above checks pass, but the node still could not be seen in the IBM Fusion user interface, then contact IBM support.

Fusion Data Foundation capacity cannot be loaded

If you encounter this issue in the Data foundation page of IBM Fusion user interface, then contact IBM support.

Fusion Data Foundation cluster fails due to pending StorageClusterPreparing stage

Problem statement
: Here, the PVC is not created and the `odfcluster` status shows as follows:

```
conditions:
  - lastTransitionTime: "2022-12-01T15:09:47Z"
    message: storagecluster is not ready,install pending
    reason: StorageClusterPreparing
    status: "False"
    type: Ready
  phase: InProgress
  replica: 1
```

Diagnosis and resolution
: To diagnose and fix the problem, do the following steps:

1. Run the following command to open the `storagecluster` CR:

```
oc get storageclusters.ocs.openshift.io -n openshift-storage ocs-storagecluster -o yaml
```

2. Check whether the output of the command shows the following error message in the status:

```
ConfigMap "ocs-kms-connection-details" not found'
```

Output example:

```
status:
conditions:
- lastHeartbeatTime: "2023-03-29T08:01:10Z"
lastTransitionTime: "2023-03-29T07:49:47Z"
message: 'Error while reconciling: some StorageClasses were skipped while waiting
for pre-requisites to be met: [ocs-storagecluster-cephfs,ocs-storagecluster-ceph-rbd]'
reason: ReconcileFailed
status: "False"
type: ReconcileComplete
```

3. If you notice the error message, check the root-operator logs with the following command:

```
oc logs -n openshift-storage $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name)
```

Example output:

```
2023-03-29 07:55:41.297073 E | ceph-cluster-controller: failed  to reconcile CephCluster  "openshift-storage/ocs-storagec
cluster "ocs-storagecluster-cephcluster": failed to configure local ceph  cluster: failed to perform validation before cl
request.
URL: GET https://9.9.9.75:8200/v1/sys/mounts
Code: 403. Errors: * permission denied
```

Deletion of `StorageCluster` doesn't delete connected `StorageClient`

Workaround

Remove the finalizer from the `StorageClient` and ensure that the `StorageClient` instance is not present on the cleaned-up cluster.

Note: This issue occurs on a reinstalled or reprovisioned cluster.

The `StorageClient` pod is in `ContainerStatusUnknown` and `CrashLoopBackOff` state

Diagnosis

When Fusion Data Foundation is reinstalled and reconfigured, the `StorageClient` continues pointing to the old instance. As a result, no or few storage classes are created, and NooBaa DB PVCs remain in the pending state.

Workaround

Check the metadata of the `StorageClient` CR. It contains details from the last connection. Force delete the old StorageClient instance and wait for it to be recreated.
Note: This issue occurs on a reinstalled or reprovisioned cluster.

- **Fusion Data Foundation storage node failure**
  You can do a node replacement proactively for an operational node and reactively for a failed node. For a failed node backed by local storage devices, you must replace the Fusion Data Foundation storage node.
- **Disk replacement and OpenShift dedicated OSD failure recovery**
  Learn how to replace a faulty disk in a Red Hat OpenShift dedicated Object Storage Device (OSD) cluster.
- **Storage client status reporter pod is stuck in terminating state**
  Learn how to diagnose and fix issues causing the storage client status reporter pod to get stuck during termination.

Fusion Data Foundation storage node failure

You can do a node replacement proactively for an operational node and reactively for a failed node. For a failed node backed by local storage devices, you must replace the Fusion Data Foundation storage node.

Before you begin

IBM recommends that replacement nodes be configured with similar infrastructure, resources, and disks as the node planned for replacement.
Note: Contact IBM Support before you proceed with any of these fixes.

Procedure

Do the following steps to check for the occurrence of Fusion Data Foundation storage node failure and identify the failed node:

1. Set the Fusion Data Foundation cluster to maintenance mode:

```
oc label odfclusters.odf.isf.ibm.com -n ibm-spectrum-fusion-ns odfcluster "odf.isf.ibm.com/maintenanceMode=true"
```

Example output:

```
[root@fu40 ~]# oc label odfclusters.odf.isf.ibm.com -n ibm-spectrum-fusion-ns odfcluster
"odf.isf.ibm.com/maintenanceMode=true"
odfcluster.odf.isf.ibm.com/odfcluster labeled
```

2. Identify the failed node:
   a. Log in to IBM Fusion user interface.
   b. Go to Data foundation page and check for warning in the Health section for storage cluster.

Alternatively, you can use the **oc** command to identify the node:

```
oc get node -l cluster.ocs.openshift.io/openshift-storage=
```

Sample output:

```
[root@fu71-f09-vm3 ~]# oc get node -l cluster.ocs.openshift.io/openshift-storage=
NAME                                 STATUS     ROLES    AGE   VERSION
f09-prc4m-worker-cluster-b-9chb5   NotReady   worker   27d   v1.24.0+4f0dd4d
f09-prc4m-worker-cluster-c-mfb77   Ready      worker   31d   v1.24.0+4f0dd4d
f09-prc4m-worker-cluster-d-r5bxx   Ready      worker   27d   v1.24.0+4f0dd4d
```

3. Identify the failed mon (if any) and Red Hat OpenShift Dedicated pods that are running in the node, which is planned for replacement:
   In an operational storage node environment:

```
oc get pods -n openshift-storage -o wide | grep -i <node_name>
```

4. If the storage node failed in a failed storage node, there is no `node_name` for the failed pods. Filter the `pending` pods instead.

```
oc get pods -n openshift-storage -o wide | grep -i pending
```

Example output: The mon deployment is `rook-ceph-mon-d`, and the Red Hat OpenShift Dedicated deployment is `ook-ceph-osd-0`.

```
[root@fu71-f09-vm3 ~]# oc get pods -n openshift-storage -o wide | grep -i pending
 rook-ceph-mon-d-67686857d7-zv62c                                  0/2     Pending     0          8m50s   <none>
<none>                             <none>            <none>
 rook-ceph-osd-0-75b954c9bf-62xm4                                  0/2     Pending     0          8m50s   <none>
<none>                             <none>            <none>
```

5. Remove the failed objects.

Remove the failed node from `odfcluster` CR

```
spec:
  autoScaleUp: false
  creator: CreatedByFusion
  deviceSets:
  - capacity: "0"
    count: 3
    name: ocs-deviceset-ibm-spectrum-fusion-local
    storageClass: ibm-spectrum-fusion-local
  encryption:
    keyManagementService: {}
  localVolumeSetSpec:
    deviceTypes:
    - disk
    - part
    size: 2Ti
  storageNodes:
  - f09-prc4m-worker-cluster-d-r5bxx
  - f09-prc4m-worker-cluster-c-mfb77
  - f09-prc4m-worker-cluster-b-9chb5 <<-- this one, remove this line.
```

Remove the mon and Red Hat OpenShift Dedicated pods
   Scale down the deployments of the identified pods. The mon deployment is `rook-ceph-mon-d` and the Red Hat OpenShift Dedicated deployment is `rook-ceph-osd-0`.

```
oc scale deployment rook-ceph-mon-d --replicas=0 -n openshift-storage
oc scale deployment rook-ceph-osd-0 --replicas=0 -n openshift-storage
```

   Ensure that you confirm the values of `mon_id` and `osd_id`.

Remove the crashcollector pods
   Remove the crashcollector pods (if any). You must put scale replica to 0.

```
oc scale deployment --selector=app=rook-ceph-crashcollector,node_name=<node_name>  --replicas=0 -n openshift-
storage
```

Mark the failed node as unschedulable
   Mark the node as `SchedulingDisabled`.

```
oc adm cordon <node_name>
```

   Example command and output:

```
oc adm cordon f09-prc4m-worker-cluster-b-9chb5
node/f09-prc4m-worker-cluster-b-9chb5 cordoned
```

```
oc get node -l cluster.ocs.openshift.io/openshift-storage=
```

```
NAME                               STATUS                        ROLES    AGE   VERSION
f09-prc4m-worker-cluster-b-9chb5   NotReady,SchedulingDisabled   worker   28d   v1.24.0+4f0dd4d
f09-prc4m-worker-cluster-c-mfb77   Ready                         worker   31d   v1.24.0+4f0dd4d
f09-prc4m-worker-cluster-d-r5bxx   Ready                         worker   27d   v1.24.0+4f0dd4d
```

Remove the pods which are in Terminating state
   This step is for the failed storage node. You can ignore this step if your are removing an operational node.

```
oc get pods -A -o wide | grep -i <node_name> |  awk '{if ($4 == "Terminating") system ("oc -n " $1 " delete pods
" $2  " --grace-period=0 " " --force ")}'
```

   Example command and output:

```
oc get pods -A -o wide | grep -i f09-prc4m-worker-cluster-b-9chb5 |  awk '{if ($4 == "Terminating") system ("oc
-n " $1 " delete pods " $2  " --grace-period=0 " " --force ")}'
```

```
warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resour
pod "isf-data-protection-operator-controller-manager-5c7cf574d5ms4xx" force deleted
```

Drain the node

```
oc adm drain <node_name> --force --delete-emptydir-data=true --ignore-daemonsets
```

Example command and output:

```
oc adm drain f09-prc4m-worker-cluster-b-9chb5 --force --delete-emptydir-data=true --ignore-daemonsets
node/f09-prc4m-worker-cluster-b-9chb5 already cordoned
```

```
WARNING: ignoring DaemonSet-managed Pods: openshift-cluster-csi-drivers/vmware-vsphere-csi-driver-node-7696f, opensh
node/f09-prc4m-worker-cluster-b-9chb5 drained
```

Delete the node

Delete the failed node:

```
oc delete node <node_name>
```

If you do not want to destroy this node for test purpose, you can remove the storage label **cluster.ocs.openshift.io/openshift-storage=**.

```
oc label nodes/f09-prc4m-worker-cluster-b-9chb5 cluster.ocs.openshift.io/openshift-storage-
```

6. Add new OpenShift® compute nodes.
   a. Create a new compute node and ensure that the new node is in ready state.
   b. Update new node info in `odfcluster` CR.

   Edit the odfcluster cr and add new node name in **StorageNodes**

   ```
   oc edit odfclusters.odf.isf.ibm.com -n ibm-spectrum-fusion-ns odfcluster
   ```

   ```
   storageNodes:
       - f09-prc4m-worker-cluster-b-djplp  <<--- new node
       - f09-prc4m-worker-cluster-d-r5bxx
       - f09-prc4m-worker-cluster-c-mfb77
   ```

   Verify whether the new nodes are labeled successfully as storage node.

   ```
   oc get node -l cluster.ocs.openshift.io/openshift-storage
   ```

   ```
   NAME                               STATUS   ROLES    AGE   VERSION
   f09-prc4m-worker-cluster-b-djplp   Ready    worker   27d   v1.24.0+4f0dd4d <<-- this node
   f09-prc4m-worker-cluster-c-mfb77   Ready    worker   31d   v1.24.0+4f0dd4d
   f09-prc4m-worker-cluster-d-r5bxx   Ready    worker   27d   v1.24.0+4f0dd4d
   ```

   Verify whether the new PVs are created automatically. The local PV gets created automatically in a short time.

   ```
   oc get pv | grep Available
   ```

   ```
   local-pv-e97b23d7                          2Ti        RWO            Delete           Available
   ```

   c. Replace the failed Red Hat OpenShift Dedicated disks.
   Remove the failed Red Hat OpenShift Dedicated from the cluster. You can also specify multiple failed ODs. Use the correct **failed_osd_id**.

   The **failed_osd_id** is the integer in the pod name immediately after the **rook-ceph-osd** prefix. You can add comma separated Red Hat OpenShift Dedicated IDs in the command to remove more than one Red Hat OpenShift Dedicated, for example, **FAILED_OSD_IDS=0,1,2**.

   Remove the failed Red Hat OpenShift Dedicated:

   ```
   oc process -n openshift-storage ocs-osd-removal \
   -p FAILED_OSD_IDS=<failed_osd_id> FORCE_OSD_REMOVAL=true | oc create -n openshift-storage -f -
   ```

   Example output:

   ```
   [root@fu71-f09-vm3 ~]# oc process -n openshift-storage ocs-osd-removal -p FAILED_OSD_IDS=0
   FORCE_OSD_REMOVAL=true | oc create -n openshift-storage -f -
   Warning: would violate PodSecurity "restricted:latest": allowPrivilegeEscalation != false (container "operator"
   must set securityContext.allowPrivilegeEscalation=false), unrestricted capabilities (container "operator" must
   set securityContext.capabilities.drop=["ALL"]), runAsNonRoot != true (pod or container "operator" must set
   securityContext.runAsNonRoot=true), seccompProfile (pod or container "operator" must set
   securityContext.seccompProfile.type to "RuntimeDefault" or "Localhost")
   job.batch/ocs-osd-removal-job created
   ```

   Check the status of the **ocs-osd-removal-job** pod to verify whether the Red Hat OpenShift Dedicated got removed successfully. A status of Completed confirms that the Red Hat OpenShift Dedicated removal job succeeded.

   ```
   oc get pod -l job-name=ocs-osd-removal-job -n openshift-storage
   ```

   Example output:

   ```
   [root@fu71-f09-vm3 ~]# oc get pod -n openshift-storage  | grep ocs-osd-removal
   ocs-osd-removal-job-ls65l                                  0/1     Completed   0          23s
   ```

   Ensure that the Red Hat OpenShift Dedicated removal is completed:

   ```
   oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1 | egrep -i 'completed removal'
   ```

   Example output:

   ```
   [root@fu71-f09-vm3 ~]# oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1 | egrep -i
   'completed removal'
   ```

```
2022-11-24 15:19:28.750910 I | cephosd: completed removal of OSD 0
```

Delete the `ocs-osd-removal-job`:

```
oc delete -n openshift-storage job ocs-osd-removal-job
```

Delete the `Released PV` which is attached to the previous node.

```
oc get pv | grep -i released
```

```
local-pv-e4a12175                              2Ti        RWO            Delete            Released    openshift-storage/ocs-deviceset-ibm-spectrum-fusion-local-0-data-1m6gnz   ibm-spectrum-fusion-local                3h34m
[root@fu71-f09-vm3 ~]# oc delete pv local-pv-e4a12175
persistentvolume "local-pv-e4a12175" deleted
```

7. Recover the failed objects.
   a. Restart the mon deployment/pod:
      i. Update the nodeSelector in deployment with new node.

      ```
      oc edit deployment -n openshift-storage rook-ceph-mon-d
      ```

      ```
      nodeSelector:
              kubernetes.io/hostname: f09-prc4m-worker-cluster-b-djplp <<--new node
      ```

      ii. Scale the replica to 1 and wait till the mon pods are in running state.

      ```
      oc scale deployment rook-ceph-mon-d --replicas=1 -n openshift-storage
      ```

      Example:

      ```
      oc scale deployment rook-ceph-mon-d --replicas=1 -n openshift-storage
          deployment.apps/rook-ceph-mon-d scaled
      ```

      ```
      [root@fu71-f09-vm3 ~]# oc get pod -n openshift-storage | grep mon
          rook-ceph-mon-a-5bbb9dd98b-z54fx                          2/2     Running     0           4m45s
          rook-ceph-mon-b-7fdd8f958b-1k9g2                          2/2     Running     0           5m18s
          rook-ceph-mon-d-6945fbbfc5-nhhw8                          2/2     Running     0           5m41s
      ```

   b. Verify the Red Hat OpenShift Dedicated pods.
      Wait till all the pods are in running state.

      ```
      oc get pods -o wide -n openshift-storage| grep osd
      ```

      ```
      [root@fu71-f09-vm3 ~]# oc get pods -o wide -n openshift-storage| grep osd
      rook-ceph-osd-0-d559cc4fb-xspr8                                2/2     Running     0           4m58s
      10.129.6.49       f09-prc4m-worker-cluster-b-djplp   <none>           <none>
      rook-ceph-osd-1-6df7f9c669-n94md                               2/2     Running     0           5m20s
      10.128.6.95       f09-prc4m-worker-cluster-d-r5bxx   <none>           <none>
      rook-ceph-osd-2-5c5d48ff7c-sdd7l                               2/2     Running     0           5m17s
      10.129.4.125      f09-prc4m-worker-cluster-c-mfb77   <none>           <none>
      rook-ceph-osd-prepare-1bf7dd3d71fe899383e625dd0c27ea37-x9vtk   0/1     Completed   0           4h8m
      10.128.6.79       f09-prc4m-worker-cluster-d-r5bxx   <none>           <none>
      rook-ceph-osd-prepare-24272b5641dc95baffc7932d78894e3c-zhz8m   0/1     Completed   0           5m24s
      10.129.6.48       f09-prc4m-worker-cluster-b-djplp   <none>           <none>
      rook-ceph-osd-prepare-6f3c3b4626ec9888b6fbe5597afd55ea-zh7cp   0/1     Completed   0           4h8m
      10.129.4.113      f09-prc4m-worker-cluster-c-mfb77   <none>           <none>
      ```

   c. Verify the Red Hat OpenShift Dedicated encryption settings. If cluster wide encryption is enabled, make sure the “crypt” keyword beside the ocs-deviceset name(s)

      ```
      oc debug node/<new-node-name> -- chroot /host dmsetup ls
      ```

      Example output:

      ```
      # oc debug node/fu47 -- chroot /host dmsetup ls
      Starting pod/fu47-debug ...
      To use host binaries, run `chroot /host`
      ocs-deviceset-sc-lvs-0-data-0clwxf-block-dmcrypt        (253:0)
      ```

      If verification fails, contact IBM support.

8. Exit maintenance mode after all steps are completed.

   ```
   oc label odfclusters.odf.isf.ibm.com -n ibm-spectrum-fusion-ns odfcluster "odf.isf.ibm.com/maintenanceMode-"
   ```

   Example output:

   ```
   [root@fu40 ~]# oc label odfclusters.odf.isf.ibm.com -n ibm-spectrum-fusion-ns odfcluster
   "odf.isf.ibm.com/maintenanceMode-"
   odfcluster.odf.isf.ibm.com/odfcluster unlabeled
   ```

9. Go to Data foundation page in IBM Fusion user interface and check the health of the Storage cluster in the Health section.

Disk replacement and OpenShift dedicated OSD failure recovery

Learn how to replace a faulty disk in a Red Hat® OpenShift® dedicated Object Storage Device (OSD) cluster.

Procedure

1. Identify the failed OSD.
   Command example:

   ```
   oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
   ```

   An example output where osd-19 is failed and the status is `CrashLoopBackOff`:

   ```
   NAME                              READY   STATUS             RESTARTS        AGE     IP              NODE
   rook-ceph-osd-0-85fcb5fd9-5cvws    2/2     Running            0               9d      9.42.107.150    compute-1-ru7.isf-racka.
   rook-ceph-osd-1-5dd8bc8d9d-cbz7g   2/2     Running            0               37d     9.42.107.149    compute-1-ru6.isf-racka.
   rook-ceph-osd-10-7cc49487b5-bdw6w  2/2     Running            0               9d      9.42.107.150    compute-1-ru7.isf-racka.
   rook-ceph-osd-11-84cd6fb7d7-xpn22  2/2     Running            0               37d     9.42.107.148    compute-1-ru5.isf-racka.
   rook-ceph-osd-12-7bb579498c-25tzh  2/2     Running            0               9d      9.42.107.150    compute-1-ru7.isf-racka.
   rook-ceph-osd-13-866dbc7f57-kqqpn  2/2     Running            0               37d     9.42.107.149    compute-1-ru6.isf-racka.
   rook-ceph-osd-14-6f7f6dd89b-7skg8  2/2     Running            0               37d     9.42.107.148    compute-1-ru5.isf-racka.
   rook-ceph-osd-15-697c5b8577-9plff  2/2     Running            0               6h58m   9.42.107.146    control-1-ru3.isf-racka.
   rook-ceph-osd-16-8d78df98c-khbh7   2/2     Running            0               37d     9.42.107.145    control-1-ru2.isf-racka.
   rook-ceph-osd-17-c8ffbb5bf-q4xqb   2/2     Running            0               37d     9.42.107.145    control-1-ru2.isf-racka.
   rook-ceph-osd-18-65f847c8d4-zpz2f  2/2     Running            0               37d     9.42.107.145    control-1-ru2.isf-racka.
   rook-ceph-osd-19-85d475d68d-5kvmq  1/2     CrashLoopBackOff   6 (2m10s ago)   37d     9.42.107.145    control-1-ru2.isf-racka.
   .
   .
   .
    <none>           <none>
   rook-ceph-osd-7-7f79d4b76-jxz8x    2/2     Running            0               37d     9.42.107.149    compute-1-ru6.isf-racka.
   rook-ceph-osd-8-75497d6999-wz296   2/2     Running            0               37d     9.42.107.148    compute-1-ru5.isf-racka.
   rook-ceph-osd-9-56d5cc7c59-w29n2   2/2     Running            0               37d     9.42.107.149    compute-1-ru6.isf-racka.
   ```

2. Scale down `ocs-operator` first, then `rook-ceph-operator`.
   a. Scale down `ocs-operator`.
      i. Log in to the Red Hat OpenShift Container Platform web console and select your project.
      ii. Go to Workloads > Pods page and verify that `ocs-operator` is in the `Running` status in the Pods list.
      iii. Go to Workloads > Deployments page.
      iv. Click ocs-operator and check the Deployment details tab.

         For example, the Deployment details tab displays that `ocs-operator` has 1 Pod.

      v. Scale down the Pod number to 0.
      vi. Go back to the Pods tab and check for the `ocs-operator`.

         The `ocs-operator` does not appear or is no longer available in the Pods list.

   b. Scale down `rook-ceph-operator`.
      i. Go to Workloads > Pods page and verify that `rook-ceph-operator` is in the `Running` status in the Pods list.
      ii. Go to Workloads > Deployments page.
      iii. Click rook-ceph-operator and check the Deployment details tab.

         For example, the Deployment details tab displays that `rook-ceph-operator` has 1 Pod.

      iv. Scale down the Pod number to 0.
      v. Go back to the Pods tab and check for the `rook-ceph-operator`.

         The `rook-ceph-operator` does not appear or is no longer available in the Pods list.

3. Optional: Verify whether Pods for `ocs-operator` and `rook-ceph-operator` are removed as expected.
   Command example:

   ```
   oc get pods |grep operator
   ```

   Example output:

   ```
   noobaa-operator-79446cc789-gj66l                            1/1     Running           0               38d
   ocs-client-operator-console-7d85dc6bf9-vdbhn                1/1     Running           0               9d
   ocs-client-operator-controller-manager-85bbcc7bfd-znvx4     2/2     Running           1 (6d1h ago)    37d
   odf-operator-controller-manager-59465654c-jmj5z             2/2     Running           0               38d
   ```

4. Clean up failed OSD.
   a. Scale down the `rook-ceph-osd` deployment.
      Command example:

      ```
      osd_id_to_remove=<replace-it-with-osd-id>
      oc scale -n openshift-storage deployment rook-ceph-osd-${osd_id_to_remove} --replicas=0
      ```

      Example output:

      ```
      deployment.apps/rook-ceph-osd-19 scaled
      ```

   b. Verify that the `rook-ceph-osd` Pod for the faulty OSD is deleted.
      Command example:

      ```
      oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
      ```

      Example output:

      ```
      NAME                              READY   STATUS    RESTARTS   AGE    IP              NODE
      rook-ceph-osd-0-85fcb5fd9-5cvws    2/2     Running   0          9d     9.42.107.150    compute-1-ru7.isf-racka.rtp.ralei
      rook-ceph-osd-1-5dd8bc8d9d-cbz7g   2/2     Running   0          37d    9.42.107.149    compute-1-ru6.isf-racka.rtp.ralei
      rook-ceph-osd-10-7cc49487b5-bdw6w  2/2     Running   0          9d     9.42.107.150    compute-1-ru7.isf-racka.rtp.ralei
      rook-ceph-osd-11-84cd6fb7d7-xpn22  2/2     Running   0          37d    9.42.107.148    compute-1-ru5.isf-racka.rtp.ralei
      rook-ceph-osd-12-7bb579498c-25tzh  2/2     Running   0          9d     9.42.107.150    compute-1-ru7.isf-racka.rtp.ralei
      rook-ceph-osd-13-866dbc7f57-kqqpn  2/2     Running   0          37d    9.42.107.149    compute-1-ru6.isf-racka.rtp.ralei
      rook-ceph-osd-14-6f7f6dd89b-7skg8  2/2     Running   0          37d    9.42.107.148    compute-1-ru5.isf-racka.rtp.ralei
      ```

```
rook-ceph-osd-15-697c5b8577-9plff   2/2     Running   0          7h16m   9.42.107.146   control-1-ru3.isf-racka.rtp.ralei
rook-ceph-osd-16-8d78df98c-khbh7    2/2     Running   0          37d     9.42.107.145   control-1-ru2.isf-racka.rtp.ralei
rook-ceph-osd-17-c8ffbb5bf-q4xqb    2/2     Running   0          37d     9.42.107.145   control-1-ru2.isf-racka.rtp.ralei
rook-ceph-osd-18-65f847c8d4-zpz2f   2/2     Running   0          37d     9.42.107.145   control-1-ru2.isf-racka.rtp.ralei
rook-ceph-osd-2-5fd5548c56-gclrv    2/2     Running   0          37d     9.42.107.148   compute-1-ru5.isf-racka.rtp.ralei
rook-ceph-osd-20-b88868db5-h55h9    2/2     Running   0          37d     9.42.107.147   control-1-ru4.isf-racka.rtp.ralei
.
.
.
.
rook-ceph-osd-8-75497d6999-wz296    2/2     Running   0          37d     9.42.107.148   compute-1-ru5.isf-racka.rtp.ralei
rook-ceph-osd-9-56d5cc7c59-w29n2    2/2     Running   0          37d     9.42.107.149   compute-1-ru6.isf-racka.rtp.ralei
```

5. Remove the old OSD from the cluster.
   a. Delete any existing `ocs-osd-removal` jobs, if present.
      Command example:

      ```
      oc delete -n openshift-storage job ocs-osd-removal-job
      ```

      Example output:

      ```
      job.batch "ocs-osd-removal-job" deleted
      ```

   b. Remove the old OSD from the cluster
      Command example:

      ```
      oc process -n openshift-storage ocs-osd-removal -p FAILED_OSD_IDS=${osd_id_to_remove} FORCE_OSD_REMOVAL=true |oc
      create -n openshift-storage -f -
      ```

      Important: Ensure that you set the correct `osd_id_to_remove`.
      Example output:

      ```
      job.batch/ocs-osd-removal-job created
      ```

   c. Verify that the `ocs-osd-removal-job` has completed.
      Command example:

      ```
      oc get pod -l job-name=ocs-osd-removal-job -n openshift-storage
      ```

      Example output:

      ```
      NAME                          READY   STATUS      RESTARTS   AGE
      ocs-osd-removal-job-rh72h     0/1     Completed   0          39s
      ```

      After the `ocs-osd-removal-job` completes, the associated Persistent Volumes (PVs) are released, and the Persistent Volume Claims (PVCs) are either deleted or move to the `Pending` state.

   d. Check the PV state.
      Command example:

      ```
      oc get pv |grep Released
      ```

      Example output:

      ```
      local-pv-141605d2                        7153Gi     RWO            Delete           Released   openshift-storage/ocs-de
      ```

   e. If encryption is enabled, remove the associated configuration.
      Do the following steps:

      i. Retrieve PVC and node details.

         Command example:

         ```
         oc describe pv local-pv-141605d2
         ```

         Example output:

         ```
         Name:              local-pv-141605d2
         Labels:            kubernetes.io/hostname=control-1-ru2.isf-racka.rtp.raleigh.ibm.com
                            storage.openshift.com/owner-kind=LocalVolumeSet
                            storage.openshift.com/owner-name=ibm-spectrum-fusion-local
                            storage.openshift.com/owner-namespace=openshift-local-storage
         Annotations:       pv.kubernetes.io/bound-by-controller: yes
                            pv.kubernetes.io/provisioned-by: local-volume-provisioner-control-1-ru2.isf-racka.rtp.raleigh.ibm
                            storage.openshift.com/device-id: nvme-MZWLJ7T6HALA-000V5_S5LLNE0R400096
                            storage.openshift.com/device-name: nvme1n1
         Finalizers:        [kubernetes.io/pv-protection]
         StorageClass:      ibm-spectrum-fusion-local
         Status:            Released
         Claim:             openshift-storage/ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465
         Reclaim Policy:    Delete
         Access Modes:      RWO
         VolumeMode:        Block
         Capacity:          7153Gi
         Node Affinity:
           Required Terms:
             Term 0:        kubernetes.io/hostname in [control-1-ru2.isf-racka.rtp.raleigh.ibm.com]
         Message:
         Source:
             Type:  LocalVolume (a persistent volume backed by local storage on a node)
             Path:  /mnt/local-storage/ibm-spectrum-fusion-local/nvme-MZWLJ7T6HALA-000V5_S5LLNE0R400096
         Events:
           Type     Reason              Age                   From     Message
         ```

```
----     ------             ----                  ----     -------
Warning  VolumeFailedDelete  15s (x15 over 3m20s)  deleter  Error cleaning PV "local-pv-141605d2": failed to get v
```

ii. Remove the **dm-crypt**–managed device-mapper mapping from the OSD devices.

Command example:

```
oc debug node/<node name
chroot /host
dmsetup ls| grep <pvc name>
```

Example output:

```
ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt    (253:3)
```

iii. Remove the mapped device.

Command example:

```
cryptsetup luksClose --debug --verbose ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt
```

Example output:

```
# cryptsetup 2.6.0 processing "cryptsetup luksClose --debug --verbose ocs-deviceset-ibm-spectrum-fusion-local-0-data
# Verifying parameters for command close.
# Running command close.
# Installing SIGINT/SIGTERM handler.
# Unblocking interruption on signal.
# Allocating crypt device context by device ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt.
# Initialising device-mapper backend library.
# dm version   [ opencount flush ]   [16384] (*1)
# dm versions   [ opencount flush ]   [16384] (*1)
# Detected dm-ioctl version 4.48.0.
# Detected dm-crypt version 1.24.0.
# Device-mapper backend running with UDEV support enabled.
# dm status ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt  [ opencount noflush ]   [16384] (*
# Releasing device-mapper backend.
# Allocating context for crypt device (none).
# Initialising device-mapper backend library.
Underlying device for crypt device ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt disappeared.
# dm versions   [ opencount flush ]   [16384] (*1)
# dm table ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt  [ opencount flush securedata ]   [1
# dm versions   [ opencount flush ]   [16384] (*1)
# dm deps ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt  [ opencount flush ]   [16384] (*1)
# LUKS device header not available.
# Deactivating volume ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt.
# dm versions   [ opencount flush ]   [16384] (*1)
# dm status ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt  [ opencount noflush ]   [16384] (*
# dm versions   [ opencount flush ]   [16384] (*1)
# dm table ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt  [ opencount flush securedata ]   [1
# dm versions   [ opencount flush ]   [16384] (*1)
# dm deps ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt  [ opencount flush ]   [16384] (*1)
# dm versions   [ opencount flush ]   [16384] (*1)
# dm table ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt  [ opencount flush securedata ]   [1
# dm versions   [ opencount flush ]   [16384] (*1)
# Udev cookie 0xd4d2da5 (semid 0) created
# Udev cookie 0xd4d2da5 (semid 0) incremented to 1
# Udev cookie 0xd4d2da5 (semid 0) incremented to 2
# Udev cookie 0xd4d2da5 (semid 0) assigned to REMOVE task(2) with flags DISABLE_LIBRARY_FALLBACK         (0x20)
# dm remove ocs-deviceset-ibm-spectrum-fusion-local-0-data-32b6465-block-dmcrypt  [ opencount flush retryremove ]
# Udev cookie 0xd4d2da5 (semid 0) decremented to 0
# Udev cookie 0xd4d2da5 (semid 0) waiting for zero
# Udev cookie 0xd4d2da5 (semid 0) destroyed
# Releasing crypt device empty context.
# Releasing device-mapper backend.
Command successful.
```

f. Identify and delete the failed PV.
Do the following steps:

i. Identify the failed PV.

Command example:

```
oc get pv -l kubernetes.io/hostname=control-1-ru2.isf-racka.rtp.raleigh.ibm.com
```

The status of the failed PV shows as **Released**.

Example output:

```
NAME                CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS     CLAIM
local-pv-141605d2   7153Gi     RWO            Delete           Released   openshift-storage/ocs-deviceset-ibm-spectr
local-pv-4ce100b9   7153Gi     RWO            Delete           Bound      openshift-storage/ocs-deviceset-ibm-spectr
local-pv-5b320962   7153Gi     RWO            Delete           Bound      openshift-storage/ocs-deviceset-ibm-spectr
local-pv-6ba1664e   7153Gi     RWO            Delete           Bound      openshift-storage/ocs-deviceset-ibm-spectr
local-pv-89429641   7153Gi     RWO            Delete           Bound      openshift-storage/ocs-deviceset-ibm-spectr
local-pv-a3bb3040   7153Gi     RWO            Delete           Bound      openshift-storage/ocs-deviceset-ibm-spectr
local-pv-e9ed407b   7153Gi     RWO            Delete           Bound      openshift-storage/ocs-deviceset-ibm-spectr
```

ii. Delete the failed PV.

Command example:

```
oc delete pv local-pv-141605d2
```

Example output:

```
persistentvolume "local-pv-141605d2" deleted
```

iii. Verify that the failed PV is deleted.

Command example:

```
oc get pv -l kubernetes.io/hostname=control-1-ru2.isf-racka.rtp.raleigh.ibm.com
```

Example output:

```
NAME               CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAIM
local-pv-4ce100b9   7153Gi     RWO            Delete           Bound    openshift-storage/ocs-deviceset-ibm-spectrum
local-pv-5b320962   7153Gi     RWO            Delete           Bound    openshift-storage/ocs-deviceset-ibm-spectrum
local-pv-6ba1664e   7153Gi     RWO            Delete           Bound    openshift-storage/ocs-deviceset-ibm-spectrum
local-pv-89429641   7153Gi     RWO            Delete           Bound    openshift-storage/ocs-deviceset-ibm-spectrum
local-pv-a3bb3040   7153Gi     RWO            Delete           Bound    openshift-storage/ocs-deviceset-ibm-spectrum
local-pv-e9ed407b   7153Gi     RWO            Delete           Bound    openshift-storage/ocs-deviceset-ibm-spectrum
```

6. Verify that the stale OSD entry is removed.
   Command example:

```
oc get pods
```

Example output:

```
NAME                                                              READY   STATUS      RESTARTS       AGE
csi-addons-controller-manager-585444c85d-vlklj                    2/2     Running     0              38d
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-5rc9r            2/2     Running     0              37d
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-6zvlj            2/2     Running     0              37d
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-8rpcc            2/2     Running     0              37d
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-9xtqk            2/2     Running     4 (46h ago)    37d
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-k6jf9            2/2     Running     0              37d
openshift-storage.cephfs.csi.ceph.com-nodeplugin-796d9bb95        5/5     Running     0              37d
openshift-storage.cephfs.csi.ceph.com-nodeplugin-796d9bc86        5/5     Running     3 (9d ago)     9d
openshift-storage.cephfs.csi.ceph.com-ctrlplugin-srk79            2/2     Running     0              37d
openshift-storage.rbd.csi.ceph.com-ctrlplugin-dzfln               3/3     Running     0              37d
openshift-storage.rbd.csi.ceph.com-ctrlplugin-j5jk5               3/3     Running     5 (46h ago)    37d
openshift-storage.rbd.csi.ceph.com-ctrlplugin-lw5lt               3/3     Running     0              37d
openshift-storage.rbd.csi.ceph.com-ctrlplugin-mf6pp               3/3     Running     0              37d
openshift-storage.rbd.csi.ceph.com-ctrlplugin-nc9g9               3/3     Running     0              37d
openshift-storage.rbd.csi.ceph.com-nodeplugin-7cf4b47dd9-89j27    6/6     Running     0              37d
openshift-storage.rbd.csi.ceph.com-nodeplugin-7cf4b47dd9-9w4      6/6     Running     0              37d
openshift-storage.rbd.csi.ceph.com-ctrlplugin-tb2wf               3/3     Running     0              37d
noobaa-core-0                                                     2/2     Running     0              37d
noobaa-db-pg-0                                                    1/1     Running     0              37d
noobaa-endpoint-68c99f7ddf-vlfbs                                  1/1     Running     0              37d
noobaa-operator-79446cc789-gj66l                                  1/1     Running     0              38d
ocs-client-operator-console-7d85dc6bf9-vdbhn                      1/1     Running     0              9d
ocs-client-operator-controller-manager-85bbcc7bfd-znvx4           2/2     Running     1 (6d1h ago)   37d
ocs-metrics-exporter-858847c98f-m6x4l                             1/1     Running     0              37d
ocs-osd-removal-job-rh72h                                         0/1     Completed   0              22m
ocs-provider-server-6d49cffc97-q4vfm                              1/1     Running     0              37d
odf-console-69bdbf8bc4-xt7jh                                      1/1     Running     0              38d
odf-operator-controller-manager-59465654c-jmj5z                   2/2     Running     0              38d
rook-ceph-crashcollector-18f4004ce1313cc06e57f6eb215a1321-lhpn6   1/1     Running     0              37d
rook-ceph-crashcollector-3f4d91760a68a6dbc7579b3e8b5915dd-m84jq   1/1     Running     0              8d
rook-ceph-crashcollector-6ff2efb9f9fe11fb194447b2fbf5c1e4-85ztt   1/1     Running     0              9d
rook-ceph-crashcollector-74af1983e44d54775407e88bf7382e9e-vchft   1/1     Running     0              37d
rook-ceph-crashcollector-9178b690870ea615c9dc0b5706acedea-84kcs   1/1     Running     0              37d
rook-ceph-crashcollector-e67f06af5747db0daafd2ddc1be00528-wlkn8   1/1     Running     0              46h
rook-ceph-exporter-compute-1-ru5.isf-racka.rtp.raleigh.ibm7p7df   1/1     Running     0              37d
rook-ceph-exporter-compute-1-ru6.isf-racka.rtp.raleigh.ibmz7qk8   1/1     Running     0              37d
rook-ceph-exporter-compute-1-ru7.isf-racka.rtp.raleigh.ibm4v9ct   1/1     Running     0              8d
rook-ceph-exporter-control-1-ru2.isf-racka.rtp.raleigh.ibmrd6f9   1/1     Running     0              9d
rook-ceph-exporter-control-1-ru3.isf-racka.rtp.raleigh.ibms5wxb   1/1     Running     0              46h
rook-ceph-exporter-control-1-ru4.isf-racka.rtp.raleigh.ibmtq9dr   1/1     Running     0              37d
rook-ceph-mds-ocs-storagecluster-cephfilesystem-a-6b557764fgdv7   2/2     Running     10 (46h ago)   37d
rook-ceph-mds-ocs-storagecluster-cephfilesystem-b-6d8c97d72fjts   2/2     Running     10 (46h ago)   9d
rook-ceph-mgr-a-864fb44895-pl2n8                                  3/3     Running     0              46h
rook-ceph-mgr-b-7cf6b49bcd-q5k49                                  3/3     Running     0              37d
rook-ceph-mon-a-79897dd546-hbstb                                  2/2     Running     0              37d
rook-ceph-mon-c-bf487b7c7-flcpf                                   2/2     Running     0              37d
rook-ceph-mon-d-76f96d4fd4-h4h9t                                  2/2     Running     0              46h
rook-ceph-osd-0-85fcb5fd9-5cvws                                   2/2     Running     0              9d
rook-ceph-osd-1-5dd8bc8d9d-cbz7g                                  2/2     Running     0              37d
rook-ceph-osd-10-7cc49487b5-bdw6w                                 2/2     Running     0              9d
rook-ceph-osd-11-84cd6fb7d7-xpn22                                 2/2     Running     0              37d
rook-ceph-osd-12-7bb579498c-25tzh                                 2/2     Running     0              9d
rook-ceph-osd-13-866dbc7f57-kqqpn                                 2/2     Running     0              37d
rook-ceph-osd-14-6f7f6dd89b-7skg8                                 2/2     Running     0              37d
rook-ceph-osd-15-697c5b8577-9plff                                 2/2     Running     0              7h41m
rook-ceph-osd-16-8d78df98c-khbh7                                  2/2     Running     0              37d
rook-ceph-osd-17-c8ffbb5bf-q4xqb                                  2/2     Running     0              37d
rook-ceph-osd-18-65f847c8d4-zpz2f                                 2/2     Running     0              37d
rook-ceph-osd-2-5fd5548c56-gclrv                                  2/2     Running     0              37d
rook-ceph-osd-20-b88868db5-h55h9                                  2/2     Running     0              37d
rook-ceph-osd-21-556947cb6f-k29mm                                 2/2     Running     0              37d
rook-ceph-osd-22-5d79578d54-b9qlb                                 2/2     Running     0              37d
rook-ceph-osd-23-84bcb674c5-6jzxx                                 2/2     Running     0              37d
rook-ceph-osd-24-6df4949c8c-qvczd                                 2/2     Running     0              37d
rook-ceph-osd-26-5d7b7b6f46-vqgz4                                 2/2     Running     0              46h
rook-ceph-osd-27-59b8c85b78-wz4pb                                 2/2     Running     0              46h
rook-ceph-osd-28-54dc8db898-ml8r5                                 2/2     Running     0              46h
```

```
rook-ceph-osd-29-5666b4975c-qrpm5                               2/2     Running     0             46h
rook-ceph-osd-3-7cfc7899dc-px8xm                                2/2     Running     0             9d
rook-ceph-osd-30-7cc88bfcfc-xsnvc                               2/2     Running     0             37d
rook-ceph-osd-31-78cc567d99-8jhkh                               2/2     Running     0             37d
rook-ceph-osd-32-6c5fcdfb45-4jz2q                               2/2     Running     0             46h
rook-ceph-osd-33-694f48d4f8-4sflf                               2/2     Running     0             22d
rook-ceph-osd-4-6f4fb99d7-w7vrb                                 2/2     Running     0             37d
rook-ceph-osd-5-7b7ff5cfb5-blpj7                                2/2     Running     0             37d
rook-ceph-osd-6-5d9856b54f-bjpmm                                2/2     Running     0             9d
rook-ceph-osd-7-7f79d4b76-jxz8x                                 2/2     Running     0             37d
rook-ceph-osd-8-75497d6999-wz296                                2/2     Running     0             37d
rook-ceph-osd-9-56d5cc7c59-w29n2                                2/2     Running     0             37d
rook-ceph-rgw-ocs-storagecluster-cephobjectstore-a-668df44tbvtn 2/2     Running     0             37d
rook-ceph-tools-68d744d548-jwj5l                                1/1     Running     0             23d
storageclient-737342087af10580-status-reporter-29061474-tv52t   0/1     Completed   0             51s
ux-backend-server-7796f96896-kwtsf                              2/2     Running     0             38d
```

7. Add a new disk to the node.
   Important:
   - If a fresh or unused disk is inserted, a new PV is created in the **Available** state.
   - If a used disk is inserted as a replacement, it must be formatted for the corresponding PV to become available.

   After adding the new disk, verify the status of the new PV.

   Command example:

   ```
   oc get pv |grep local
   ```

   Example output:

   ```
   local-pv-159e1b8e                         7153Gi     RWO           Delete           Bound       openshift-storage/ocs-device
   local-pv-2ac8a85b                         7153Gi     RWO           Delete           Bound       openshift-storage/ocs-device
   local-pv-406f1af8                         7153Gi     RWO           Delete           Bound       openshift-storage/ocs-device
   local-pv-451f6341                         7153Gi     RWO           Delete           Bound       openshift-storage/ocs-device
   local-pv-4ce100b9                         7153Gi     RWO           Delete           Bound       openshift-storage/ocs-device
   local-pv-4da6a449                         7153Gi     RWO           Delete           Available
   local-pv-521ee04f                         7153Gi     RWO           Delete           Bound       openshift-storage/ocs-device
   .
   .
   .
   .
   local-pv-b7c4932c                         7153Gi     RWO           Delete           Bound       openshift-storage/ocs-device
   ```

8. Scale up **ocs-operator** first, then **rook-ceph-operator**.
   a. Scale up **ocs-operator**.
      i. Go to Workloads > Deployments page.
      ii. Click ocs-operator and check the Deployment details tab.

      For example, the Deployment details tab displays that **ocs-operator** has 0 Pod.

      iii. Scale up the Pod number to 1.
      iv. Go to Workloads > Pods page and verify that **ocs-operator** is in the **Running** status in the Pods list.
   b. Scale up **rook-ceph-operator**.
      i. Go to Workloads > Deployments page.
      ii. Click rook-ceph-operator and check the Deployment details tab.

      For example, the Deployment details tab displays that **rook-ceph-operator** has 0 Pod.

      iii. Scale up the Pod number to 1.
      iv. Go to Workloads > Pods page and verify that **rook-ceph-operator** is in the **Running** status in the Pods list.
9. Verify that the new OSD Pod is running.
   Command example:

   ```
   oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
   ```

   Example output:

   ```
   NAME                                READY   STATUS    RESTARTS   AGE    IP              NODE
   rook-ceph-osd-0-85fcb5fd9-5cvws     2/2     Running   0          9d     9.42.107.150    compute-1-ru7.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-1-5dd8bc8d9d-cbz7g    2/2     Running   0          37d    9.42.107.149    compute-1-ru6.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-10-7cc49487b5-bdw6w   2/2     Running   0          9d     9.42.107.150    compute-1-ru7.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-11-84cd6fb7d7-xpn22   2/2     Running   0          37d    9.42.107.148    compute-1-ru5.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-12-7bb579498c-25tzh   2/2     Running   0          9d     9.42.107.150    compute-1-ru7.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-13-866dbc7f57-kqqpn   2/2     Running   0          37d    9.42.107.149    compute-1-ru6.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-14-6f7f6dd89b-7skg8   2/2     Running   0          37d    9.42.107.148    compute-1-ru5.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-15-697c5b8577-9plff   2/2     Running   0          8h     9.42.107.146    control-1-ru3.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-16-8d78df98c-khbh7    2/2     Running   0          37d    9.42.107.145    control-1-ru2.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-17-c8ffbb5bf-q4xqb    2/2     Running   0          37d    9.42.107.145    control-1-ru2.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-18-65f847c8d4-zpz2f   2/2     Running   0          37d    9.42.107.145    control-1-ru2.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-19-74cc9c8bc6-cc8p4   2/2     Running   0          111s   9.42.107.145    control-1-ru2.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-2-5fd5548c56-gclrv    2/2     Running   0          37d    9.42.107.148    compute-1-ru5.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-20-b88868db5-h55h9    2/2     Running   0          37d    9.42.107.147    control-1-ru4.isf-racka.rtp.raleigh.ibm
   .
   .
   .
   rook-ceph-osd-8-75497d6999-wz296    2/2     Running   0          37d    9.42.107.148    compute-1-ru5.isf-racka.rtp.raleigh.ibm
   rook-ceph-osd-9-56d5cc7c59-w29n2    2/2     Running   0          37d    9.42.107.149    compute-1-ru6.isf-racka.rtp.raleigh.ibm
   ```

10. Verify that the PV and PVC are successfully created.
    Command example for PVC:

    ```
    oc get pvc -n openshift-storage |grep local-pv-4da6a449
    ```

Command example for PV:

```
oc get pv -n openshift-storage |grep local-pv-4da6a449
```

Example output for PVC:

```
ocs-deviceset-ibm-spectrum-fusion-local-0-data-35tw7jf   Bound    local-pv-4da6a449                                 7153Gi     RWO
```

Example output for PV:

```
local-pv-4da6a449                          7153Gi     RWO            Delete           Bound    openshift-storage/ocs-deviceset
```

Results

Faulty disk or failed OSD replacement is complete.

What to do next

Wait for the storage cluster to reconcile its health status. If it remains in the **Warning** state despite all Fusion Data Foundation Pods running, check for and resolve any previous crashes.

To verify the health status, do as follows:

- Log in to the IBM Fusion web console and go to Storage > Local storage to check the health of the local storage.
- In the Red Hat OpenShift Container Platform web console, go to Storage > Data Foundation and check the health of the storage.

Storage client status reporter pod is stuck in terminating state

Learn how to diagnose and fix issues causing the storage client status reporter pod to get stuck during termination.

If the storageclient-status-reporter pod gets stuck during termination, the following options don't work:

- Force deletion of the pod.
- Patching of the finalizer.

To fix this issue, remove the finalizer by using the following command:

```
for pod in $(oc get pod --field-selector=status.phase=Failed -oname -nopenshift-storage | grep storageclient)
do
    oc patch $pod -nopenshift-storage --type merge -p '{"status":{"qosClass":"BestEffort"}}' --subresource status
    oc patch $pod -nopenshift-storage --type merge -p '{"metadata":{"finalizers": []}}'
done
```

Global Data Platform

Learn how to install and configure Global Data Platform service.

- **Installing Global Data Platform**
  From the Services page, enable the IBM Fusion Global data platform service.
- **Configuring Global Data Platform local storage**
  Configure Global Data Platform as your storage, then configure storage nodes, and finally set its encryption.
- **Setting up encryption for storage**
  Setting encryption for the different storage types in IBM Fusion HCI .
- **Configuring storage**
  You can configure Global Data Platform as your storage.
- **Sharing data**
  Active file management (AFM) is a scalable, high-performance, file system caching layer, and enables sharing of data across clusters, even if the networks are unreliable or have high latency.
- **Global Data Platform service install and upgrade issues**
  Use this troubleshooting information to resolve install and upgrade problems that are related to Global Data Platform service.
- **Global Data Platform service issues**
  This section lists the troubleshooting tips and tricks when you use IBM Fusion storage.

Installing Global Data Platform

From the Services page, enable the IBM Fusion Global data platform service.

About this task

Ensure that you are aware of the following points before you install Global Data Platform:

- For more information about the compatibility of services on the different supported versions of OpenShift® Container Platform, see Support matrix.
- The IBM Fusion supports Global Data Platform version 5.2.3.4 and Global Data Platform Remote Mount version 6.0.0.1.
- Do NOT delete IBM Storage Scale pods. In many circumstances, the deletion of Scale pods has implications on the availability and data integrity.

- If Global Data Platform service is installed on a OpenShift Container Platform cluster, check whether the nodes are running with secure boot:

  `mokutil --sb-state`

  As a root user, disable secure boot with the following command:

  `mokutil --disable-validation`

Procedure

1. Go to the Services page in the IBM Fusion user interface.
2. In the Available section, click Global data platform tile to install and configure storage service.
   Note: You cannot install other dependent services until you install and configure a storage service.
3. In the Global Data Platform page, go through the details and click Install.
   If you have less than six storage nodes, then the Install is disabled. Go to the Nodes page > Discovered tab and add nodes to the cluster so that it has enough storage for Global Data Platform. After you add a minimum of six nodes, the Install is enabled and you continue with the installation.
   After you enable the Global Data Platform, you can view the service version and health status. In the Services page, you can see Global data platform tile under Installed section.

   Table 1. Health states Global Data Platform service

State	Description
`Healthy`	The `Running` and `Ready` states for the following items indicate that the service is healthy: • All the pods in `ibm-spectrum-scale` namespace are running. • Scale operator pod in `ibm-spectrum-scale-operator` namespace. • `ScaleManager` CR in the Fusion namespace. • All the pods in `ibm-spectrum-scale-csi` namespace are running. • GPFS Scale service is active on all the pods. • The filesystem is mounted on all the nodes.
`Degraded`	If any of the following conditions are not met, then the service goes to a degraded state. • Any one of the scale core pod statuses is not running per Recovery Group (RG) • GPFS service is not active on any one of the pods per RG. • If one of the PM collector pods is not running or all containers are not in the ready state. • If one of the GUI pods is not running or all containers are not in the ready state.
`Critical`	• 2 scale core pods are not running per RG with erasure code parity 4+2P • 2 or 3 scale core pods are not running per RG with erasure code parity 4+3P • 2 or 3 scale core pods are not running per RG with erasure code parity 8+3P • GPFS service is not active on 2 scale core pods per RG (4+2P per RG) • GPFS service is not active on 2 or 3 scale core pods per RG (4+3P per RG) • GPFS service is not active on 2 or 3 scale core pods per RG (8+3P per RG) • If both the PM collector pods are not running or all containers are not in the ready state. • If any of the CSI pods are not running or all containers are not in the ready state. • If both the GUI pods are not running or all containers are not in the ready state. • Add Filesystem is not mounted on more than the tolerable number of nodes.
Down	• More than 2 scale core pods are not running (4+2P per RG) • More than 3 scale core pods are not running (4+3P Per RG) • More than 3 scale core pods are not running (8+3P Per RG) • GPFS service not Active on more than 2 scale core pods per RG (4+2P per RG) • GPFS service not Active on more than 3 scale core pods per RG (4+3P Per RG) • GPFS service not Active on more than 3 scale core pods per RG (8+3P Per RG)

   From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification gets displayed. The notification disappears automatically after some time.
   For more information about Global data platform service issues during IBM Fusion HCI installation, see Installation issues.

4. Verify Global Data Platform installation.

   Validate Global Data Platform operators, deployments, and pods from the OpenShift management console
   a. Log in to the OpenShift web console.
   b. From the menu navigation, click Operators > Installed Operators.
   c. From the Installed Operators list, click IBM Storage Fusion operator. The Details tab is open by default.
   d. Open Scale tab, and check whether the storagemanager is in the Scales list and is healthy.
      Note: For Global Data Platform Remote Mount version 6.0.0.1, the `storage-manager` is not created.
   e. From the menu navigation, click Workloads > Deployments.
   f. Check whether the following namespaces and their deployments are available:

      `ibm-spectrum-fusion-ns` project
      - `isf-storage-service-dep`
      - `isf-storage-operator-controller-manager`

      `ibm-spectrum-scale-operator` project
      `ibm-spectrum-scale-controller-manager`

      `ibm-spectrum-scale-csi` project
      `ibm-spectrum-scale-csi-operator`
      Note: For Global Data Platform Remote Mount version 6.0.0.1, the `ibm-spectrum-scale-csi-operator` pod is no longer available.

   g. From the menu navigation, click Workloads > Pods.
   h. Check whether you can see the following pods:

      `ibm-spectrum-fusion-ns` project
      `isf-storage-operator-controller-manager`
      `isf-storage-service-dep`
      `ibm-spectrum-scale-csi` project

ibm-spectrum-scale-csi-operator
ibm-spectrum-scale-csi-provisioner
ibm-spectrum-scale-csi-attacher
ibm-spectrum-scale-csi

Note: For Global Data Platform Remote Mount version 6.0.0.1, the `ibm-spectrum-scale-csi-operator` pod is no longer available.

`ibm-spectrum-scale` project

compute-1-ruX
ibm-spectrum-scale-gui-0
ibm-spectrum-scale-gui-1
ibm-spectrum-scale-pmcollector-0
ibm-spectrum-scale-pmcollector-1

`ibm-spectrum-scale-operator`

ibm-spectrum-scale-controller-manager

Validate other pods from the CLI

a. To list all DNS pods, run the OC command:

```
./oc get pods -n ibm-spectrum-scale-dns
```

For example, see the following result:

```

oc get po -n ibm-spectrum-scale-dns
NAME             READY   STATUS    RESTARTS    AGE
coredns-4gmrg    1/1     Running   3           7d16h
coredns-bjpv7    1/1     Running   5           7d16h
coredns-hnrph    1/1     Running   3           7d16h
coredns-j4dxq    1/1     Running   3           7d16h
coredns-ln6gj    1/1     Running   3           7d16h
coredns-rtfdg    1/1     Running   3           7d16h
```

Ensure all the pods are up and running.

Verify the storage class.

Note: Ensure that the following storage classes are created.

```
oc get sc
```

What to do next

1. Configure storage to work with other dependent services. For the procedure to configure, see Configuring Global Data Platform local storage.
2. In the Hosted Control Plane cluster, go to Remote file systems page of the user interface and connect to remote file systems. For the procedure to connect to remote file systems, see Connecting to remote IBM Storage Scale file systems.
3. In a Hosted Control Plane cluster, to consume the remote file system by using Global Data Platform, see Static provisioning of Persistent Volume in filesystem and Dynamic provisioning of Persistent Volume in filesystem.
4. You can now install other available IBM Fusion services.

- **Global Data Platform storage support for multi rack**
  Compatibility of Global Data Platform storage service with IBM Fusion HCI multi-rack and expansion rack.

Global Data Platform storage support for multi rack

Compatibility of Global Data Platform storage service with IBM Fusion HCI multi-rack and expansion rack.

Table 1. Global Data Platform support on IBM Fusion HCI multi rack deployment topologies

Deployment topology	Fresh installation	Node upsize	Disk upsize	Remote mount
Expansion rack	Require a minimum of 6 storage nodes on the base rack	Any number of Compute-only, Compute-Storage, and GPU nodes can be added	Even number of disks with the same capacity can only be up sized	Supported

Configuring Global Data Platform local storage

Configure Global Data Platform as your storage, then configure storage nodes, and finally set its encryption.

Before you begin

- Install a local storage provider from the Services page. For the procedure to install, see Installing Global Data Platform.
- Plan whether you want to configure storage and set it for disaster recovery. If you choose the disaster recovery option, select whether you want to do a Metro-DR configuration or a Regional DR configuration.

About this task

The configuration of the local storage can take up to several hours depending on your cluster size or the amount of nodes you have in your cluster.

Important:

- If Global Data Platform is not configured or is configured but are not in a healthy state, then a few pods, like logcollector, might remain in a pending state. The pods come into a running state automatically once storage is configured and in a healthy state.

- Both Global Data Platform and Fusion Data Foundation cannot coexist as your storage.

Procedure

1. Go to Storage > Local storage.
2. Wait for the discovery of compute nodes to complete.
   After the discovery process completes, the Configure storage button is enabled.
3. Click Configure.
4. Optional: In the Configure local storage page, you can decide to select Set up Disaster Recovery or not.
   A disaster recovery configuration provides data replication between two HCI clusters by connecting the storage networks used by the clusters. The default is always disabled.
   If you selected the Set up Disaster Recovery option in the previous step, then the configuration section is displayed with Metro and Regional options. If you did not select the Set up Disaster Recovery option, then skip step 5 and 6.
5. Select Metro or Regional tile.

Option	Description
Metro	Select whether you want this rack to be your First site or Second site of the disaster recovery pair. First site a. Select First site tile. b. Go to step 6. Second site a. Select Second site tile. b. In the Connect to the first site section, paste the connection snippet of the first site. For the procedure to get the snippet, see Upgrade and upsize considerations in Metro-DR. c. Select whether you want to enable Jumbo frames or not. Jumbo frames improve the performance of the storage network between the two IBM Fusion HCI clusters in a disaster recovery pair. To know more about Jumbo frames and its prerequisites, see the *Jumbo frames section of* General Metro-DR prerequisites topic. d. Click Configure. The storage configuration of site 2 is same as the site 1 so the wizard closes after you click Configure, and the configuration of Global Data Platform service starts for the second site. e. Go to step 10 to view the details of the configured Global Data Platform.
Regional	Proceed with step 6. For asynchronous disaster recovery set up, go to Disaster recovery page after you install the Global Data Platform service.

6. Click Next.
7. In the Storage option wizard page, click Stronger data resiliency tile or Better storage efficiency tile.
   The Storage option wizard page provides storage capabilities of IBM Fusion with two storage configurations, Stronger data resiliency and Better storage efficiency. If there are eleven or more storage nodes in the appliance, you can choose between Stronger data resiliency and Better storage efficiency building block configurations. If there are less than eleven storage nodes in the appliance, the Stronger data resiliency building block configuration is only available.

   Note: The storage setting is predetermined to 4+3P erasure coding for a high availability cluster.
   A Stronger data resiliency option is an optimal set of (4+2P) building blocks based on the number of nodes in the rack. If enough nodes exist for the two building blocks, then IBM Fusion creates a two-building block configuration. This configuration increases the number of simultaneous node failures, which the appliance can withstand. For example, if you set up a rack and choose a (4+2P) building block configuration, then for every six nodes, you get two nodes or disk failure tolerance. If the number of nodes are less for two building block configuration, a Strong data resiliency option appears with one (4+2P) building block configuration. For example, if you set up a rack and choose a (4+2P) building block configuration, then for every six nodes, you get one node or disk failure tolerance.

   The Better storage efficiency option uses a single building block configuration, consisting of a minimum of eleven storage nodes. The building block resiliency is achieved by using (8+3p) erasure coding. It means that the cluster can withstand a maximum of three storage node failures, with the data recovered from other nodes in the building block.

   Note:
   - The building block configuration that you choose applies to all storage nodes that you add in the future when you expand the rack. For example, if you start with a single (4+2p) Stronger data resiliency building block and later add six additional storage nodes, then those new nodes are used to create a second (4+2p) Stronger data resiliency building block.
   - You do not have to configure the Global data platform for site 2 in a Metro-DR setup because IBM Fusion HCI automatically applies the configuration of the primary rack or site 1 to site 2.

   Important: You cannot change the building block configuration post the installation.
8. Expand Advanced and select Block size.
   An advanced setting is available that you can use to customize the Block size for IBM Fusion HCI's Global data platform. As OpenShift clusters run a mix of workloads with different I/O characteristics, it is recommended that you use the default 4 MiB block size that is optimized for mixed workloads.

   If the applications that get deployed to the OpenShift cluster are specifically workloads that drive large or small IOs, you can customize the block size for better performance.

   For single rack, the available block sizes for Stronger data resiliency or (4+2p) erasure code are 1 MiB and 4 MiB.

   For high-availability cluster, the available block sizes for Stronger data resiliency or (4+3p) erasure code are 1 MiB and 4 MiB.

   The available block sizes for Better storage efficiency or (8+3p) erasure code are 1 MiB, 4 MiB, and 16 MiB.

   Note: Considerations before you choose a block size:
   a. For general workloads, it is recommended to choose 4 MiB block size.
   b. If the cluster primarily deals with many small files, it is recommended to choose a small block size such as 1 MiB.
   c. For large files or streaming workloads, it is recommended to choose 16 MiB block size.

   For more information about block size, see Block size.
9. Click Configure.

10. Go to the Global Data Platform page in the IBM Fusion user interface.
    The Local storage page now includes Usable capacity, Health, and Nodes sections. Here, the Usable capacity is configuring storage and the Health is initializing.
    Note: Sometimes, it can take up to five minutes to show the summary of Usable capacity and Health sections.

    Usable capacity
        The amount of capacity that is available for storing data on a system after the RAID or mirroring technology is applied. Usable capacity is represented in a line graph.

    Health
        It includes the health of the following:
        - Storage cluster
        - Global Data Platform service

        The status gets displayed only after the provisioning is complete.

    Nodes
        It includes a list of all nodes that you used in your local storage configuration. The node details listed in the table are Name, Storage status, Type, Disk count, Disk size (TiB), CPU core, and Memory (GiB).
        In the Nodes section, the initial Storage status is Installing storage software. After the initial status, it can have the following statuses:

        Normal
            The component or the service that is hosted on the node is working as expected, and no active error events exist.

        Critical
            The component or the service that is hosted on the node failed because of the failure of some of its critical components.

        Degraded
            The node or the service that is hosted on the node is not working as expected. A problem occurred with the component, but it is not a complete failure.

        Unknown
            The status of the component or the service that is hosted on the node is not known.

        You can use the search option to filter and search for nodes. You can add nodes to scale up.

11. To enable encryption, click Configure encryption.
    If the encryption is already set up, then the Connection settings is displayed.
    For the procedure to configure encryption, see Configuring encryption for Global Data Platform storage.

What to do next

You can now add disks.

Setting up encryption for storage

Setting encryption for the different storage types in IBM Fusion HCI .

Global Data Platform

You can connect to an encrypted remote IBM Storage Scale file system.

- The encryption server is configured after the configuration of encryption workflow. After the encryption settings are added, you cannot disable the local storage encryption.
- In Global Data Platform storage, you can connect to an encrypted remote IBM Storage Scale file system.
- The local storage can have non-encrypted data and encrypted data after you add the encryption settings on the Configure encryption page.
- The local storage gets automatically encrypted from the point a key management server is added to .
- Remote filesystem encryption is configured on remote Scale before you add it to the user interface.
- To secure or encrypt your etcd data, follow the steps that are mentioned in the OpenShift® Container Platform documentation. See Enabling etcd encryption.

- **Configuring encryption for Global Data Platform storage**
  You can set the encryption for the IBM Fusion local storage.

Configuring encryption for Global Data Platform storage

You can set the encryption for the IBM Fusion local storage.

Before you begin

Prepare IBM® Security Guardium® Key Lifecycle Manager (GKLM) server for IBM Fusion. To establish an encryption-enabled environment, see part 1 of Simplified setup: Using GKLM with a self-signed certificate.

Ensure that you go through the firewall recommendations for GKLM. See Firewall recommendations for IBM GKLM

About this task

The IBM Storage Scale admin encrypts the remote file system. As a IBM Fusion user, you must connect to the same key management server so that encrypted data can be accessed.

Procedure

1. Go to the Storage > Remote file systems page in IBM Fusion user interface to configure encryption for remote IBM Storage Scale filesystem.
   For Global Data Platform local storage, go to Storage > Local storage
2. Click Connect in the Encryption tile.
3. Enter the following connection details:

   Hostname
   : For local storage, enter the Security Key Lifecycle Manager host name to connect.

   Backup host name
   : Optionally, enter secondary GKLM server host name.

   Port number (optional)
   : The REST port number connects IBM Fusion to Security Key Lifecycle Manager REST admin interface. The default port number is 9443.
     Note: It can depend on Security Key Lifecycle Manager version. See Firewall recommendations for IBM GKLM.

   User name
   : The administrator user name for GKLM Server. The default value is `GKLMAdmin`.

   Password
   : The administrator password for the GKLM Server.

4. Enter the following Certificate details.
   Note: TLS/KMIP Certificates for secure communication on the KMIP port, only require when the key server is running with a certificate chain from a Certificate Authority (CA) rather than with a self-signed server certificate. The certificates must be formatted as PEM-encoded X.509 certificates.

   Root certificate
   : The root CA certificate from the Certificate Authority.

   Endpoint certificate
   : The server certificate that is signed by a CA.

   Intermediate certificate (optional)
   : The intermediate CA certificates are required only when the server certificate is signed by one of them. If you have more intermediate certificates, then click Add intermediate certificate to add them.

5. For the local storage, to encrypt data, select NIST SP 800-131A or NIST SP 800-131AFAST for the Encryption algorithm.
6. For the remote file systems, enter the following values for File system tenants.

   Encryption tenant ID
   : Represents the keyspace configured on the GKLM server. All IBM Fusion systems that want to share or use encryption keys must use the same tenant ID.

   Remote key management ID
   : It is the remote key management ID. All nodes in the IBM Fusion system must use the same RKMID, which describes a combination of keyserver, tenant, and client on the remote scale cluster.
     Important: The DNS name for the key server or tenant must start with a letter or number. Special characters at the beginning of the DNS name are not supported and can result in an invalid RKMID.

   You can add more such Encryption tenant ID and Remote key management ID pairs.

   Run the following commands on the remote scale cluster to retrieve the values:

   ```
   mmkeyserv client show
   ```

   This command gives the tenant name. If the tenant name is displayed as (none), then first register client using **mmkeyserv** client register command. For more information about this command, see mmkeyserv command.
   To get Remote key management ID, run the following command on the remote scale cluster:

   ```
   mmkeyserv tenant show
   ```

7. Click Configure.

Configuring storage

You can configure Global Data Platform as your storage.

Important:

- If either Global Data Platform or Fusion Data Foundation are not configured or are configured but are not in a healthy state, then a few pods, like logcollector, might remain in a pending state. The pods come into a running state automatically once storage is configured and in a healthy state.
- Both Global Data Platform and Fusion Data Foundation cannot coexist as your storage.

Sharing data

Active file management (AFM) is a scalable, high-performance, file system caching layer, and enables sharing of data across clusters, even if the networks are unreliable or have high latency.

Use IBM Fusion HCI as a cache cluster. For AFM, you also need a home cluster. To set up a home cluster, see Setting up the home cluster.

The subsections provide insights on how you can work with Active file management (AFM) volume management in IBM Fusion HCI.

Retrieving credentials

As a user, you must authenticate the REST APIs for Active file management (AFM).

To retrieve the credentials (username and password) and use it for authentication, run the following `oc` command in the *ibm-spectrum-scale* namespace:

```
oc get secret/ibm-spectrum-scale-gui-secret -n ibm-spectrum-scale -o go-template --template="{{.data.username|base64decode}}"
```

```
oc get secret/ibm-spectrum-scale-gui-secret -n ibm-spectrum-scale -o go-template --template="{{.data.password|base64decode}}"
```

Active file management (AFM) fileset creation

Create an Active file management (AFM) fileset according to this procedure.

1. Create Active file management (AFM) fileset with IBM Storage Scale server and NFS or with any other NFS server

Prerequisite

- An NFS export is available from the home cluster with `no_root_squash`. The NFS export must have read/write permissions. See the following snippet:

  ```
  /test *(rw,nohide,insecure,no_subtree_check,sync,no_root_squash,fsid=101)
  ```

- To retrieve the route, run the following `oc` command:

  ```
  oc get route -n ibm-spectrum-scale
  ```

  Example output:

  ```
  NAME                    HOST/PORT                                                   PATH
  ibm-spectrum-scale-gui  ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf.ibm.com          ibm-spectrum-scale
  ```

- To retrieve the authorization token, run the following curl command:

  ```
  curl -X POST --header 'Content-Type: application/json' --header 'Accept: application/json' -u '<userName>:
  <password>' -d '{"type":"USER", "clusterName":"<clusterName>", "requestingUser":"<userName>",
  "accessToken":"access-token-1","authData": { "userName":"<userName>", "roles": {"<userName>":
  "PENDING"},"status": "PENDING" } }' https://<Scale Route URL>/scalemgmt/v2/access -k
  ```

  Where, <"userName">:<"password"> or<"userName"> are the username and password that is retrieved from the *Retrieving credentials* topic.

  "clusterName" is the OpenShift® cluster name.

  Example command:

  ```
  curl -X POST --header 'Content-Type: application/json' --header 'Accept: application/json' -u
  'admin:passw0rd' -d '{"type": "USER","clusterName": "fus0.sdi.dmz","requestingUser": "admin","accessToken":
  "access-token-1","authData": { "userName": "admin", "roles": {"admin": "PENDING"},"status": "PENDING" } }'
  https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.fus0.sdi.dmz/scalemgmt/v2/access -k
  ```

  Example output:

  ```
  { "clusterName" : "ibm-spectrum-scale.fus0.sdi.dmz", "status" : { "code" : 200, "message" : "The request
  finished successfully." }, "token" : "0ccfb637-d46c-dead-beef-f8438c2842d5" }
  ```

- Retrieve the filesystem name (*filesystem_name*) that is mounted on the OpenShift Container Platform cluster. Example command:

  ```
  curl -k -H "Authorization:Token <token retreived from the previous command>" -X GET --header 'content-
  type:application/json' https://<spectrum_scale_gui_route>/scalemgmt/v2/filesystems
  ```

  Example output:

  ```
  { "filesystems" : [ { "name" : "ibmspectrum-fs" } ], "status" : { "code" : 200, "message" : "The request
  finished successfully." }
  ```

  For more information about the REST API, see Filesystems: GET.

Procedure

Run the REST API to create the Active file management (AFM) fileset in IBM Fusion as follows:

```
curl -k -H "Authorization:Token <token retreived from the previous command>" -X POST --header 'content-
type:application/json' --header 'accept:application/json' -d '{
"filesetName": "<afm_fileset_name>",
"path": "/mnt/<filesystem_name>/<afm_fileset_name>",
"afmTarget": "nfs://<nfs_home_ip>/<export_path_on_home>",
"afmMode": "independent-writer"
}' https://<spectrum_scale_gui_route>/scalemgmt/v2/filesystems/<filesystem_name>/filesets
```

An example is as follows:

```
curl -k -H "Authorization:Token <token retreived from the previous command>" -X POST --header 'content-type:application/json' --header 'accept:application/json' -d '{
> "filesetName": "afmfsetnfs",
> "path": "/mnt/ibmspectrum-fs/afmfsetnfs",
> "afmTarget": "nfs://9.9.9.9/mnt/fs1/fset1",
> "afmMode": "independent-writer"
> }' https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf.ibm.com/scalemgmt/v2/filesystems/ibmspectrum-fs/filesets
```

2. Create AFM fileset with Object store as home

Prerequisite

- A bucket is available on an S3 compliant object store.
- Access key and secret key are available to access the bucket.
- Configure the AFM to Cloud Object Storage by using the keys that are created or generated on the Cloud Object Storage. For more information, see Configuring AFM to cloud object storage.

Procedure

- Run the REST API to create Active file management (AFM) fileset in IBM Fusion as follows:

```
curl -k -H "Authorization:Token <token retreived from the previous command>" -X POST --header 'content-type:application/json' --header 'accept:application/json' -d '{
 "filesetName": "<afm_fileset_name",
 "endpoint": "<s3_endpoint>",
 "useObjectFs": true,
 "bucket": "<bucket_name>",
 "mode": "<iw/ro>",
 "accessKey": "<access_key>",
 "secretKey": "<secret_key>"
 }' 'https://<spectrum_scale_gui_route>/scalemgmt/v2/filesystems/<filesystem_name>/filesets/cos'
```

An example is as follows:

```
curl -k -H "Authorization:Token <token retreived from the previous command>" -X POST --header 'content-type:application/json' --header 'accept:application/json' -d '{
 "filesetName": "cosfset1",
 "endpoint": "http://us-east-2@s3.us-east-2.amazonaws.com",
 "useObjectFs": true,
 "bucket": "mybucket",
 "mode": "iw",
 "accessKey": "xxxxxxxx",
"secretKey": "xxxxxxxxxxxxx"
 }' 'https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.fusion.cp.fyre.ibm.com/scalemgmt/v2/filesystems/ibmspectrum-fs/filesets/cos'
```

- Make note of the job ID. See the following sample text:

```
{
  "jobs" : [ {
    "jobId" : 5000000000543,
    "status" : "RUNNING",
    "submitted" : "2021-11-08 05:57:02,375",
    "completed" : "N/A",
    "runtime" : 2,
    "request" : {
      "data" : {
        "afmMode" : "read-only",
        "afmTarget" : "nfs://<ip_address>/gpfs/fs1/tgt1",
        "filesetName" : "roCacheFileset",
        "path" : "/mnt/rackhscale/roCacheFileset"
      },
      "type" : "POST",
      "url" : "/scalemgmt/v2/filesystems/rackhscale/filesets"
    },
    "result" : { },
    "pids" : [ ]
  } ],
  "status" : {
    "code" : 202,
    "message" : "The request was accepted for processing."
  }
}
```

3. Verification of Active file management (AFM) fileset creation configuration

Procedure

a. Check that the fileset creation job is successful as follows:

```
curl -k -H "Authorization:Token <token retreived from the previous command>" -X GET --header 'content-type:application/json' https://<spectrum_scale_gui_route>/scalemgmt/v2/jobs/<job_id>
```

b. Verify that Active file management (AFM) cache fileset is created and cache is active as follows:

```
curl -k -H "Authorization:Token <token retreived from the previous command>" -X GET --header 'content-type:application/json'
https://<spectrum_scale_gui_route>/scalemgmt/v2/filesystems/<filesystem_name>/afm/state
```

An example is as follows:

```
curl -k -H "Authorization:Token <token retreived from the previous command>" -X GET --header 'content-type:application/json' https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.fusion.cp.fyre.ibm.com/scalemgmt/v2/filesystems/ibmspectrum-fs/afm/state
{
  "filesetAfmStateList" : [ {
```

```
    "cacheState" : "Active",
    "filesetName" : "afmfsetnfs",
    "filesystemName" : "ibmspectrum-fs"
  }, {
    "cacheState" : "Active",
    "filesetName" : "afmpall",
    "filesystemName" : "ibmspectrum-fs"
  }, {
    "cacheState" : "Active",
    "filesetName" : "afms",
    "filesystemName" : "ibmspectrum-fs"
  } ],
  "status" : {
    "code" : 200,
    "message" : "The request finished successfully."
  }
}
```


Note: For the container logs, go to /var/log/cnlog/mgtsrv. To view the last few lines of the container log file, run the following tail command:

```
tail -f mgtsrv-trace-log in ibm-spectrum-scale-gui-0 pod
```

If you face errors, then it can be in the container log file of any of the two GUI pods. Check the log of `ibm-spectrum-scale-gui-1` as well.

Create Persistent Volume and Persistent Volume Claim

Create a Persistent Volume and Persistent Volume Claim.

You must create a persistent volume and a persistent volume claim. For more information, see Generating static provisioning manifests.

Global Data Platform service install and upgrade issues

Use this troubleshooting information to resolve install and upgrade problems that are related to Global Data Platform service.

Warning: Do NOT delete IBM Storage Scale pods. Deletion of Scale pods in many circumstances has implications on availability and data integrity.

IBM Storage Scale stuck with error

Problem statement

If IBM Storage Scale installation stuck with the error `ERROR Failed to define vdiskset {"controller": "filesystem", "controllerGroup": "scale.spectrum.ibm.com", "controllerKind": "Filesystem", "Filesystem": {"name":"ibmspectrum-fs","namespace":"ibm-spectrum-scale"}, "namespace": "ibm-spectrum-scale", "name": "ibmspectrum-fs", "reconcileID": "5fcf21a4-b488-44db-8db5-0f3c1ab695cd", "cmd": "/usr/lpp/mmfs/bin/mmvdisk vs define --vs ibmspectrum-fs-0 --rg rg1 --code 4+2P --bs 4M -`

Resolution

As a resolution, run the following command to manually create a recovery group.

```
mmvdisk recoverygroup create --recovery-group <RG name> --node-class <Node class name>
```

Upgrade does not complete as node pod is in `ContainerStatusUnknown` state

Problem statement

Global data platform upgrade does not complete because a compute node pod is in `ContainerStatusUnknown` state.

Workaround

Delete the pod in `ContainerStatusUnknown` state and restart the compute node.

Pod drain issues in Global data platform upgrade

Diagnosis and resolution

If the upgrade of Global data platform gets struck, then do the following diagnose and resolve:

1. Go through the logs from the scale operator.
2. Check whether you observe the following error:
   `ERROR Drain error when evicting pods`
3. If it is a drain error, then check whether the virtual machine's PVC were created ReadWriteOnce.
   The PVCs created by using ReadWriteAfter are not shareable or moveable and can cause the draining of the node.

   For example:

   ```
   oc get pods
   ```

   ```
   NAME                                        READY        STATUS                  RESTARTS        AGE

   compute-0                                   2/2         Running                   0              20h
   compute-1                                   2/2         Running                   0              24h
   compute-13                                  2/2         Running                   0              21h
   compute-14                                  2/2         Running                   3              21h
   compute-2                                   2/2         Running                   0              21h
   compute-3                                   2/2         Running                   0              21h
   ```

```
control-0                       2/2     Running                 0       23h
control-1                       2/2     Running                 0       23h
control-1-1                     2/2     Running                 0       23h
control1-1-reboot               0/1     ContainerStatusUnknown  1       23h
control-2                       2/2     Running                 0       23h
ibm-spectrum-scale-gui-0        4/4     Running                 0       20h
ibm-spectrum-scale-gui-1        4/4     Running                 0       23h
ibm-spectrum-scale-pmcollector-0  2/2   Running                 0       23h
ibm-spectrum-scale-pmcollector-1  2/2   Running                 0       20h
```

4. If virtual machine's PVC got created ReadWriteOnce, then stop the virtual machine to continue the upgrade of scale pods.
5. If the upgrade fails further, then check whether the reason can be due to an orphaned **control-1-reboot** container left in the **ibm-spectrum-scale** namespace. Delete the container to resume and complete the installation.

IBM Storage Scale might get stuck during upgrade

Resolution

Do the following steps:

1. Shut down all applications that use storage.
2. Scale down the IBM Storage Scale operator. Make the replica as 0 for deployment **ibm-spectrum-scale-controller-manager**.
   a. Use **ibm-spectrum-scale-operator** project:

   ```
   oc project ibm-spectrum-scale-operator
   ```

   b. Scale down the IBM Storage Scale operator.

   ```
   oc scale --replicas=0 deployment ibm-spectrum-scale-controller-manager -n ibm-spectrum-scale-operator
   ```

   c. Run the following command to confirm resources are not found in **ibm-spectrum-scale-operator** namespace.

   ```
   oc get po
   ```

3. Log in to the one of the IBM Storage Scale core pods and shut down the server.
   a. Log in to the server:

   ```
   oc rsh compute-0
   ```

   b. Shut down the server:

   ```
   mmshutdown -a
   ```

   Example output:

   ```
   Thu May 19 07:46:47 UTC 2022: mmshutdown: Starting force unmount of GPFS file systems
   Thu May 19 07:46:52 UTC 2022: mmshutdown: Shutting down GPFS daemons
   compute-13.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Shutting down!
   compute-14.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Shutting down!
   compute-0.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Shutting down!
   control-1.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Shutting down!
   control-0.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Shutting down!
   compute-1.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Shutting down!
   compute-2.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Shutting down!
   control-2.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Shutting down!
   compute-13.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  'shutdown' command about to kill process 1
   compute-13.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading modules from /lib/modules/4.18.0
   compute-14.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  'shutdown' command about to kill process 1
   compute-14.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading modules from /lib/modules/4.18.0
   compute-14.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  mmfsenv: Module mmfslinux is still in use.
   compute-13.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  mmfsenv: Module mmfslinux is still in use.
   compute-14.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading module mmfs26
   compute-13.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading module mmfs26
   compute-14.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unmount all GPFS file syste
   ..
   compute-2.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading module mmfs26
   control-2.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading module mmfs26
   control-2.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading module mmfslinux
   compute-0.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading module mmfslinux
   compute-2.ibm-spectrum-scale-core.ibm-spectrum-scale.svc.cluster.local:  Unloading module mmfslinux
   Thu May 19 07:47:03 UTC 2022: mmshutdown: Finished
   ```

4. Check whether the states are down:

   ```
   mmgetstate -a
   ```

   Sample output:

   ```
   Node number  Node name          GPFS state
   --------------------------------------------
              1  control-0-daemon   down
              2  control-1-daemon   down
              3  control-2-daemon   down
              4  compute-14-daemon  down
              5  compute-13-daemon  down
              6  compute-0-daemon   down
              7  compute-1-daemon   down
              8  compute-2-daemon   down
   ```

5. Delete all the pods in **ibm-spectrum-scale** namespace:

   ```
   oc delete pods --all -n ibm-spectrum-scale
   ```

6. To verify, get the list of pods:

```
oc get po -n ibm-spectrum-scale
```

Sample output:

```
NAME                               READY  STATUS   RESTARTS  AGE
 ibm-spectrum-scale-gui-0           0/4    Pending  0          7s
 ibm-spectrum-scale-pmcollector-0  0/2    Pending  0          7s
```

Note: Only IBM Storage Scale pods get deleted and not GUI and `pmcollector`.

IBM Storage Scale upgrade gets stuck

Cause
: The IBM Storage Scale upgrade gets stuck because of `isf-storage-service`. It happens whenever the `scaleoutstatus` or `scaleupStatus` is **In Progress** in the Scale CR.
Important: Do not update the latest image for the `isf-storage-service` when a IBM Storage Scale operation is in progress.

Workaround
: Steps to upgrade the IBM Storage Scale on the rack:

1. Go to the `isf-storage-service` in `ibm-spectrum-fusion-ns` project and change the image to `cp.icr.io/cp/isf/isf-storage-services:2.2.0-latest` or run the following OC command.

   ```
   oc patch deploy isf-storage-service-dep -n ibm-spectrum-fusion-ns --patch='{"spec":{"template":{"spec":{"containers":
   [{"name": "app", "image":"cp.icr.io/cp/isf/isf-storage-services:2.2.0-latest"}]}}}}
   ```

2. After the `isf-storage-service` pod goes to running state and points to the `cp.icr.io/cp/isf/isf-storage-services:2.2.0-latest` version, run the following API endpoint command in the `isf-storage-service` pod's terminal:

   ```
   curl -k https://isf-scale-svc/api/v1/upgradeExcludingOperator
   ```

   You can observe the new logs for `isf-storage-service` as well.
3. Set the Edition to `erasure-code` in the `Daemon CR`:

   ```
   oc patch daemon ibm-spectrum-scale \
    --type='json' -n ibm-spectrum-scale \
    -p="[{'op': 'replace', 'path': '/spec/edition', 'value': "erasure-code"}]"
   ```

4. Run the following curl command to deploy the operator on the Red Hat® OpenShift® Container Platform cluster:

   ```
   curl -k https://isf-scale-svc/api/v1/upgradeWithOperator
   ```

   You can observe new logs for `isf-storage-service`.
5. Wait for sometime and check whether the status of IBM Storage Scale core pod is in restart or running state in `ibm-spectrum-scale`.

CSI Pods experience scheduling problems post node drain

Problem statement
: Whenever a compute node is drained, an eviction of CSI PODs or sidecar PODs occur. The remaining available set of compute nodes cannot host the CSI PODs or sidecar PODs because of resource constraints at that point in time.

Resolution
: Ensure that a functional system exists with available compute nodes, having sufficient resources to accommodate evicted CSI PODs or sidecar PODs.

Global Data Platform upgrade progress

Problem statement
: During Global Data Platform upgrade, the progress percentage can go beyond 100% in a few cases. You can ignore this issue, as the percentage comes down to 100 % after the successful completion of the upgrade.

Scale installation might get stuck with ECE pods in `init` state

Problem statement
: The Scale installation might get stuck with ECE pods in `init` state with the following error:

```
The node already belongs to a GPFS cluster.
```

Resolution

1. Check whether the daemon-network IP of the pod is pingable.
2. If it is not pingable, then restart all available TORs.
3. After you restart TORs, check whether daemon-network IP is pingable.
4. Run the following command to manually clean up the nodes:
   For the given worker node, run the following **oc** command:

   ```
   oc debug node/<openshift_worker_node> -T -- chroot /host sh -c "rm -rf /var/mmfs"
   ```

5. Kill the pod and wait for it to come up again. The Scale installation resumes after all the ECE pods goes to running state.

Expansion rack fails for 4+2p setup

Problem statement

Sometimes, in a high-availability cluster, the expansion rack fails for 4+2p setup.

Resolution

- Do the following workaround steps to resolve the issue:
  1. In the OpenShift Container Platform user interface, go to Administration > CustomResourceDefinitions > Scale > Instances > storagemanager > Yaml and check whether the **scaleoutStatus: IN-PROGRESS**
  2. If **scaleoutStatus: IN-PROGRESS**, go to Administration > CustomResourceDefinitions > RecoveryGroup > Instances and check for the created RecoveryGroups.
  3. If more than two recovery groups are created, do the following actions:
     a. Go to Administration > CustomResourceDefinitions > RecoveryGroup > Instances > ..
     b. Find **RecoveryGroup** instance with suffix rg2.
     c. Click the ellipsis menu for that instance and select the Delete RecoveryGroup option.
  4. Do the following checks to validate whether the node upsize completed successfully:
     - In the IBM Fusion user interface, go to the IBM Storage Scale user interface by using the app switcher.
     - Click Recovery groups on the IBM Spectrum Scale RAID tile.
     - Click the entry of Recovery group server nodes.
     - Check whether all the newly added nodes for expansion rack are listed and is in Healthy state.
     - Create a sample PVC to check whether the storage can be provisioned.
  5. Check for the labels on each recovery group.
     a. In the OpenShift Container Platform console, go to Administration > CustomResourceDefinitions > RecoveryGroup > Instances > Yaml.
     b. Check whether the YAML contains either of these two labels:
        - **scale.spectrum.ibm.com/scale: up**
        - **scale.spectrum.ibm.com/scale: out**
     c. If it is yes for the previous step, remove those labels from each recovery group and click Save.

Known issues in installation

- The display unit of filesystem block size in **storagesetupcr** is MiB and that of **ibmspectrum-fs** is M. As IBM Storage Scale also uses MiB format internally, you can ignore this inconsistency.

Known issues in upgrade

- During upgrade, the IBM Fusion rack user interface, IBM Storage Scale user interface, Grafana endpoint, and Applications are not reachable for sometime.
- During the upgrade, an intermittent error **-1 node updated successfully out of 5 nodes** shows on the IBM Fusion user interface in the upgrade details page.
- During the upgrade, an intermittent error shows on the IBM Fusion user interface that the progress percentage for the Global Data Platform decreased. It occurs, especially when the upgrade for one node is completed.
- During the upgrade, an intermittent error **Global Data Platform upgrade failed** shows on the IBM Fusion user interface in the upgrade details page. Ignore the error because it might recover during the next reconciliation.

Global Data Platform service issues

This section lists the troubleshooting tips and tricks when you use IBM Fusion storage.

Warning: Do NOT delete IBM Storage Scale pods. Deletion of Scale pods in many circumstances has implications on availability and data integrity.

Filesystem status from Remote file systems is stuck in Connecting state

Cause
: The secure boot is not supported by Scale and CNSA.

Resolution

1. Check whether the nodes are running with secure boot:

   mokutil --sb-state

   ```
   SecureBoot enabled
   ```

2. As a root password, you can disable the secure boot with the following command:

   mokutil --disable-validation

   ```
   password length: 8~16
   input password:
   input password again:
   ```

Increased CPU usage on Scale pods after upgrading to Cloud Pak for Integration (CP4I)

Problem statement
: Scale pod CPU usage has increased due to changes in MQ streams application. It is because of IBM MQ's liveness and readiness probes.

Resolution
: The file handle reservation can be turned off by setting the environment variable **AMQ_NO_RESERVE_FD=1**. It fixes the high CPU usage.

Scale operator failed to add quorum label

Problem statement
: During Storage configuration, the recovery group status can be "`Waiting on some daemons in the recovery group to be restarted`".

Resolution

1. Run the following command to check the count of the quorum node:

```
oc get nodes -l scale.spectrum.ibm.com/designation=quorum
```

2. Check the number of quorum nodes. Whenever the node count is less than 10, there must 3 quorum nodes. If not, apply the following quorum label for the missing nodes:

```
scale.spectrum.ibm.com/designation=quorum
```

Remote filesystem connection goes to a Disconnected state

Problem statement
: If even one scale core pod is not ready due to node restart or other issues, then a problem occurs on the filesystem mount of that node and the filesystem CR reports errors. As a result, the filesystem goes to a Disconnected and the same is displayed on the IBM Fusion user interface.

Resolution
: Wait for the scale core pod to go to Ready state, and the filesystem automatically changes to Connected state.

PVCs stuck in pending state

Resolution
: If PVCs are stuck in a pending state for a long time, restart Scale GUI pods.

Nodes with pods scheduled for storage results in a `crashloobackoff` or container creating errors

Problem statement
: When install, upgrade, or upsize operations are in progress and the storage configuration is not yet done, the nodes with pods scheduled for storage results in a `crashloobackoff` or container creating errors.

Resolution
: To resolve this issue, disable schedule on these nodes before you begin these operations, and schedule the pods on nodes that are configured for storage.

File system is not mounted on some nodes

Problem statement
: After IBM Storage Scale upgrade, the file system is not mounted on some nodes.

Resolution
: Do the following workaround steps:

1. Delete the problematic pod where the file system is not mounted. Delete one pod at a time.
2. Refresh the pod overview page until the pod appears. Run `watch oc get pods`.
3. When the problematic pod comes up, set the replicas to 0 in the `Spec` section in the `ibm-spectrum-scale-operator` namespace. You must stop the operator deployment immediately. Ensure that no operator pod is running in `ibm-spectrum-scale-operator` namespace. Command for setting the operator replica to 0:

```
oc patch deploy ibm-spectrum-scale-controller-manager \
 --type='json' -n ibm-spectrum-scale-operator \
 -p='[{"op": "replace", "path": "/spec/replicas", "value": 0}]'
```

```
oc get deployment -n ibm-spectrum-scale-operator
```

Sample output:

```
I0420 16:56:23.579340    4184 request.go:645] Throttling request took 1.192496025s, request: GET:https://api.isf-rackk.rtp.raleigh.ibm.com:6443/apis/nmstate.io/v1beta1?timeout=32s
NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
ibm-spectrum-scale-controller-manager    0/0     0            0           28h
```

4. Wait till the problematic pod is in `init` container state and run the following command:

```
oc exec podname -- mmsdrrestore -p <any working core ip/name[Specify pod IP where FS is mounted]>
```

5. After the previous step is successful, set the replicas to 1 again in the `ibm-spectrum-scale-operator` deployment. Ensure that the operator pod is running in `ibm-spectrum-scale-operator` namespace.

   Run the following command for setting the operator replica to 0:

```
oc patch deploy ibm-spectrum-scale-controller-manager \
 --type='json' -n ibm-spectrum-scale-operator \
 -p='[{"op": "replace", "path": "/spec/replicas", "value": 1}]'
```

```
oc get deployment -n ibm-spectrum-scale-operator
```

   Sample output:

```
I0420 16:56:23.579340    4184 request.go:645] Throttling request took 1.192496025s, request: GET:https://api.isf-
rackk.rtp.raleigh.ibm.com:6443/apis/nmstate.io/v1beta1?timeout=32s
NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
ibm-spectrum-scale-controller-manager    1/1     0            0           28h
```

6. Wait for some time for the problematic pod to come in running state and file system gets mounted on this problematic node.

IBM Storage Scale user interface does not load properly or IBM Storage Scale CSI pods are in `CrashLoopBack` state

Problem statement
: Sometimes, IBM Storage Scale user interface might not load properly or IBM Storage Scale CSI pods are in **`CrashLoopBack`** state.

Resolution
: As a workaround, do the following steps:

1. Restart the GUI pods and check whether it is working properly.
2. If GUI pod restart does not open the IBM Storage Scale console, then restart the **`ibm-spectrum-scale-controller-manager`** pod and then restart GUI pod.

Not able to log in to IBM Storage Scale user interface

Cause
: The secret that is mounted in GUI pods /etc/gui-oauth/oauth.properties file and secret present in the **`ibm-spectrum-scale-gui-oauthclient`** are different.

```
gui-1
sh-4.4$ cd /etc/gui-oauth/
sh-4.4$ cat oauth.properties
secret=tjCK6kE0pmCe2d8b7azw
oauthServer=https://oauth-openshift.apps.isf-racka.rtp.raleigh.ibm.com
redirectURI=https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racka.rtp.raleigh.ibm.com/auth/redirect
clientID=ibm-spectrum-scale-gui-oauthclient
k8sAPIServer=https://172.30.0.1:443
sh-4.4$
gui_0
secret=tjCK6kE0pmCe2d8b7azw
oauthServer=https://oauth-openshift.apps.isf-racka.rtp.raleigh.ibm.com
redirectURI=https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racka.rtp.raleigh.ibm.com/auth/redirect
clientID=ibm-spectrum-scale-gui-oauthclient
k8sAPIServer=https://172.30.0.1:443
Oauth client config (API Explorer -> OAuthClient -> ibm-spectrum-scale-gui-oauthclient)
kind: OAuthClient
apiVersion: oauth.openshift.io/v1
metadata:
  name: ibm-spectrum-scale-gui-oauthclient
  uid: a9eb9d85-33b9-4cd7-99c8-ae0213032380
  resourceVersion: '314932'
  creationTimestamp: '2022-03-29T17:09:38Z'
  labels:
    app.kubernetes.io/instance: ibm-spectrum-scale
    app.kubernetes.io/name: gui
  ownerReferences:
apiVersion: scale.spectrum.ibm.com/v1beta1
      kind: Gui
      name: ibm-spectrum-scale-gui
      uid: da23c696-5c71-405b-b9ac-c7d0ccb4dd43
      controller: true
      blockOwnerDeletion: true
  managedFields:
manager: ibm-spectrum-scale/ibm-spectrum-scale-gui
      operation: Apply
      apiVersion: oauth.openshift.io/v1
      time: '2022-03-29T17:09:38Z'
      fieldsType: FieldsV1
      fieldsV1:
        'f:grantMethod': {}
        'f:metadata':
          'f:labels':
            'f:app.kubernetes.io/instance': {}
            'f:app.kubernetes.io/name': {}
          'f:ownerReferences':
            'k:{"uid":"da23c696-5c71-405b-b9ac-c7d0ccb4dd43"}':
              .: {}
              'f:apiVersion': {}
              'f:blockOwnerDeletion': {}
              'f:controller': {}
              'f:kind': {}
              'f:name': {}
              'f:uid': {}
        'f:redirectURIs':
          'v:"https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racka.rtp.raleigh.ibm.com/auth/redirect"': {}
        'f:secret': {}
*secret: bk2_q1FqwhVhoNqOl7ob*
redirectURIs:
>-
    https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racka.rtp.raleigh.ibm.com/auth/redirect
grantMethod: auto
```

Resolution

As a resolution, do the following steps:

1. As a resolution, restart the GUI pods.
2. If the previous step (step 1) does not resolve the issue, then restart IBM Storage Scale Operator pod and then restart the GUI pods.

Download log files of all deployments

If you cannot download log files of all deployments, individually download the log files.

Issues in node drain and node restart

Resolution
: To resolve such node related issues, see Issues related to IBM Fusion HCI node drains.

IBM Storage Scale cluster failed to recover after powering off and on the compute nodes

Diagnosis
: Identify if the nodes joining the cluster are in an out-of-tolerance situation.

1. mmhealth reported UNKNOWN RG status on some nodes.
2. mmrepquota commands have been generated repeatedly, likely by the GUI, which has caused many long waits.
3. Due to the large number of long waits, the cluster manager node was not responding to any **ts cmd** other than **tsctl nqStatus**.
   As a resolution, follow the steps defined in the Recover storage cluster from an unplanned multi-node restart or failure.

MCO rollout is not progressing for more than 5 hours during the scale upgrade

Cause
: Whenever the nodes are over committed and no pod limit is available, you might see delays in the policy rollout, ICSP, pull secret, and MCO rollout changes.

Resolution
: As a resolution, do the following step:

- IBM Fusion HCI nodes allow the maximum number of pods on a node as defined by OpenShift® Container Platform default and ensure to keep that count within the limit. Otherwise, you might see this issue.

IBM Storage Scale cluster status is in a critical state after power on/off the rack

Cause
: IBM Storage Scale cluster status is in critical state due to `unmounted_fs_check` is present on the `filesystem` CR.

Resolution
: As a resolution, wait until this `filesystem` CR status gets cleared; after that, the IBM Storage Scale cluster status automatically returns to a healthy state. If the issue persists for a long time, then contact IBM support.

Disk upsize issues

- Manual steps to test the `/dev` mount:
  After you add a new NVMe disk to the server, the new disk cannot be discovered in the IBM Storage Scale pod. The disks do not show up with /dev/nvme* device names in the POD, IBM Storage Scale need the device name /dev/nvme* to access the disks. Without the /dev/nvme* device name, the new disks cannot be added into recovery group.

  Resolution
  : Do the following manual steps to mount disks to /dev in the pod.

  1. Log in to each IBM Storage Scale storage pods.

     ```
     oc exec -it <pod name> -c <container> /bin/sh
     ```

  2. Stop the GPFS daemons on one or more nodes:

     ```
     mmshutdown
     ```

  3. Mount `/dev` from the host running the following command:

     ```
     mount -t devtmpfs none /dev
     ```

  4. Validate that the content of /dev is from host and you can see sub directories like block, bus, char.

     ```
     mmstartup
     ```

- **Recover storage cluster from an unplanned multi-node restart or failure**
  The service of IBM Storage Scale file system might hang if the storage nodes are restarted.
- **IBM Storage Scale file system not showing newly added file system size**
  The IBM Storage Scale file system is not showing the newly added file system size after adding six nodes as upsize.

Recover storage cluster from an unplanned multi-node restart or failure

The service of IBM Storage Scale file system might hang if the storage nodes are restarted.

The storage nodes might get restarted unexpectedly and might cause the disk to miss in the storage cluster. If the missing disks exceed the fault tolerance RAID code used that is used in the storage cluster, the file system might stop to provide service. It cannot finish the log recovery process for the lost nodes. It might also block the restarting nodes to join the cluster again.

Follow these steps to recover IBM Spectrum® Storage Scale Erasure Code Edition (ECE) from the out of fault tolerance.
Note: Out of fault tolerance in the IBM Storage Scale Erasure Code Edition (ECE) environment can be caused due to different reasons.

1. Identify the problem.
    a. Run the **mmfsadm dump waiters** command on the running node of the application. It displays the long waiters of I/O threads.
    b. Run the **mmgetstate** command. It displays the restart nodes in arbitrating state and others in active state.
    c. Run the **tslsrecgroup rg_1 --server --v2** command several times. It displays that Log Groups jumps between nodes.
    d. Check the mmfs.log file on all storage nodes. The log message shows up in loop on different nodes. See the message as follows:

    ```
    [E] Beginning to resign log group LG002 in recovery group rg1 ....
    ```

    It means that the user log group continues to resign.
2. Start the recovery process.
    a. Run the **mmlscluster** command to find all the quorum nodes.
    b. Run the **mmshutdown** command on all quorum nodes. This action causes all nodes to unmount from the file system.
    c. Run the **mmstartup** command on all quorum nodes.
    d. Run the **mmlsrecoverygroup rg1 -L --pdisk** to check whether the recovery group is active again and that no missing `pdisk` exists.
    e. Mount the file system again on all the client nodes and restart the application.

IBM Storage Scale file system not showing newly added file system size

The IBM Storage Scale file system is not showing the newly added file system size after adding six nodes as upsize.

About this task

A system with a single recovery group (setup with 4+2p erasure code) needs to do node upsizing with six nodes all together.

Procedure

1. Follow the steps to upsize nodes:
    a. Follow the standard node upsize process. For more information, see Configuring nodes for management.
    b. IBM Fusion HCI user interface shows the Running state for the newly added nodes.
    c. In the IBM Fusion HCI user interface, use the app switcher to go to the IBM Storage Scale user interface.
    d. Click Recovery groups on the IBM Storage Scale RAID tile.
    The entry for the recovery group server nodes column shows up empty. If true, perform the following steps. Otherwise, contact the IBM support.
    e. Edit the labels of newly added nodes, change the label `scale.spectrum.ibm.com/rg: rg1` to `scale.spectrum.ibm.com/rg: rg2`.
    f. Repeat the following process for each newly added node.
        i. Open the OpenShift® Container Platform user interface.
        ii. Go to Compute > Nodes > Details > Actions > Edit labels.
        iii. Remove the label `scale.spectrum.ibm.com/rg: rg1` and add the label `scale.spectrum.ibm.com/rg: rg2`.
        iv. Click Save.
2. Follow the steps to validate the upsized nodes:
    a. In the IBM Fusion HCI user interface, use the app switcher to go to the IBM Storage Scale user interface.
    b. Click Recovery groups on the IBM Storage Scale RAID tile.
    c. Click the entry of the recovery group server nodes column for rg2.
    d. Check all the newly added nodes are listed and in healthy state.
    e. Create a sample PVC to check that storage can be provisioned.

Remote IBM Storage Scale storage cluster in IBM Fusion

IBM Storage Scale remote mount feature enables you to connect to a remote IBM Storage Scale cluster as though it is a part of your local cluster.

About this task

- The Global Data Platform remote mount capability is available only for the following deployment modes:
    - Standalone rack with Global Data Platform (newly installed IBM Fusion versions 2.10.0 or later, that is compliant with network requirements with OpenShift® Container Platform 4.16 or later. For more information about the network requirements, see Planning and prerequisites for remote mount support.
    - Standalone rack with Fusion Data Foundation (newly installed with IBM Fusion versions 2.9.0 or later).
    - Expansion multirack with Global Data Platform storage (newly installed with IBM Fusion versions 2.9.0 or later).

  Important:
  On Red Hat® OpenShift Container Platform 4.20 or earlier and Fusion Data Foundation 4.20 or earlier, Fusion Data Foundation and the Global Data Platform remote mount service can be installed independently.

  Starting with Red Hat OpenShift Container Platform 4.21 and Fusion Data Foundation 4.21 or later, install Fusion Data Foundation first. After installation, the Global Data Platform remote mount tile becomes available for configuration.

  If you install Fusion Data Foundation as local storage, you must explicitly install and configure the remote mount service by using the Global Data Platform remote mount service tile. If you install Global Data Platform as local storage, the remote mount capability is available by default.

- You can run a container workload against an existing data set that is stored on an IBM Storage Scale file system. For example, an IBM Storage Scale file system contains lot of images and you want to run a container workload that specializes in image processing.

- The remote file system mount must be configured. For more information, see Connecting to remote IBM Storage Scale file systems.
- When the remote file system mount is created, the IBM Fusion automatically creates a statically provisioned volume that points to the file system. You must know how to identify this Persistent Volume.
- Optionally, modify the file system path that is specified in the Persistent Volume so that it points to a specific directory in the file system, rather than the root.
- Now a Persistent Volume Claim (PVC) can be made against the Persistent Volume. It can be accomplished by matching the **accessMode** and storage request against the specified in the Persistent Volume.
- A workload that uses the Persistent Volume Claim can view the contents of the file system.
- Configure the UID/GID to match the one that is used to write the data set on the IBM Storage Scale file system. If you do not configure, then the container workload cannot access the files, because it does not have the permission.

Procedure

1. Click Storage > Remote file systems menu in the IBM Fusion user interface.
   It lists all existing connections to file systems with their details in a table format. The details include File System, FS status, Cluster, Cluster status, and Storage class. You can enter keywords in Search to search for a connection. You can also filter records based on the status.
2. Click ellipsis of a remote file system record to do the following actions:
   - Details- Click to view details about the remote file system connection.
   - Edit cluster- Click to change the cluster of a file system connection.
   - Remove- Click to delete a file system record.

Connecting to remote IBM Storage Scale file systems

In IBM Fusion, from a containerized workload, you can access and store data from existing IBM Storage Scale clusters.

Before you begin

On Red Hat® OpenShift® Container Platform 4.20 or earlier and Fusion Data Foundation 4.20 or earlier, Fusion Data Foundation and the Global Data Platform remote mount service can be installed independently.

Starting with Red Hat OpenShift Container Platform 4.21 and Fusion Data Foundation 4.21 or later, install Fusion Data Foundation first. After installation, the Global Data Platform remote mount tile becomes available for configuration.

If you install Fusion Data Foundation as local storage, you must explicitly install and configure the remote mount service by using the Global Data Platform remote mount service tile. If you install Global Data Platform as local storage, the remote mount capability is available by default.

Firewall requirements
: Important: For details of IBM Storage Scale container native storage access firewall requirements, see Network and firewall requirements.

Network planning and prerequisites
: For network planning and prerequisites for remote mount support, see Network planning.

- The IBM Fusion admin must exchange information with the IBM Storage Scale admin, and provide instructions on how the IBM Storage Scale admin must configure the file system sharing on the IBM Storage Scale side.
  - As a OpenShift Container Platform administrator, confirm whether you configured IBM Storage Scale remote file system mounts during the installation of IBM Fusion. You can check whether the IBM Fusion user interface displays a Remote file systems menu option in the left-navigation.
    To determine whether the appropriate remote mount service is installed, check the Services page.
  - **Important:** The IBM Storage Scale administrator must do a remote filesystem configuration on the IBM Storage Scale Container Storage Interface Driver storage cluster by following **steps 1** to **6** of Storage cluster configuration for Container Storage Interface (CSI) on non-AWS environment (like on-premises).
  - The IBM Storage Scale administrator in turn provides you the necessary information about cluster, file system, encryption algorithms. For instructions to share, click Instructions link in the Add IBM Storage Scale file system slide out pane. For the Instructions link, see Storage cluster.
    Note: The details include host name, port, Cluster ID, IBM Storage Scale certificate, user name, and password.
  - You can enter cluster access credentials information that is provided by the IBM Storage Scale admin into the IBM Fusion user interface, or place it into a CR by using YAML.
  - Run the following precheck commands on the IBM Storage Scale GUI node:
    - Check whether the user that does the remote mount configuration has correct access rights:

      ```
      /usr/lpp/mmfs/gui/cli/lsuser | grep ContainerOperator | grep CsiAdmin
      ```

      By default, the user passwords expire after 90 days. If the security policy of your organization permits it, use the `-e 1` option on the **mkuser** command to create a user with a password that does not expire. Also, create a unique GUI user for each OpenShift Container Platform cluster on the same remote Scale cluster.

      If not present, create the user with both ContainerOperator and CsiAdmin roles.

      ```
      /usr/lpp/mmfs/gui/cli/mkuser csi-storage-gui-user -p passw0rd -g CsiAdmin,ContainerOperator
      ```

    - Run the following command to ensure all the nodes of your cluster are active:

      ```
      /usr/lpp/mmfs/bin/mmgetstate -a
      ```

      Sample output:

      ```
      Node number  Node name  GPFS state

                1          host1     active

                2          host2     active
      ```

```
3           host3       active
```

If any of the node is down, analysis the issue and start the node.

- Run the following command to validate whether the gpfsgui service is running:

  ```
  systemctl status  gpfsgui
  ```

  If gui service has issues, stop and start the service:

  ```
  systemctl stop gpfsgui.service
  systemctl start gpfsgui.service
  ```

  Log in to the GUI of `scalecluster` by using the validated username and password.

- To get more information about the cluster, run **/usr/lpp/mmfs/bin/mmlscluster** command.

About this task

All the IBM Storage Scale commands are run by the IBM Storage Scale administrator on the remote IBM Storage Scale cluster.

```
/usr/lpp/mmfs/bin/xxx
```

Procedure

1. Click Storage > Remote file systems menu in the IBM Fusion user interface.
2. Click Add file system.
   The Add IBM Spectrum Scale file system slide out pane gets displayed.
3. Enter the following details to add one or more file systems.
   You can add one or more file systems of the existing cluster or choose to connect to a new cluster.
   Note: The OpenShift Container Platform admin needs the IBM Storage Scale admin to provide all of this information. The CLI commands referenced in this topic are run by the IBM Storage Scale admin on the remote IBM Storage Scale cluster.
   - Connect to a new cluster option:
     - Enter the following cluster access credentials for IBM Storage Scale:

       Host name
       : The host name is the GUI REST API endpoint. It needs to have a user with ContainerOperator and CsiAdmin roles. You can enter up to a maximum of three hosts.

       Port
       : The port is the GUI port, that is, the REST API endpoint port. By default, the port is 443. If the remote cluster uses other ports, you can provide it.

       Cluster ID
       : Enter the ID of the cluster. Contact your IBM Storage Scale administrator to get the cluster ID.

       IBM Spectrum Scale certificate
       : Enter the IBM Storage Scale root CA. If you provide the root CA value, the IBM Fusion verifies the certificate chain and hostname of the IBM Storage Scale cluster. If you do not provide the root CA value, the IBM Fusion skips the verification step.

       User name
       : Enter the GUI user name of the cluster.
         Note: Ensure that this user has both container operator permissions and IBM Storage Scale Container Storage Interface Driver administrator permissions.

       Password
       : Enter the password for the GUI user name.

     - Define up to three nodes on the IBM Storage Scale cluster to establish connection. These nodes are contact nodes, which is used to communicate with remote storage cluster. The projects and data get distributed between these nodes.
       Run the following command to get node details:

       ```
       /usr/lpp/mmfs/bin/mmlscluster
       ```

     - Enter the name and IP address for node 1, node 2, and node 3.
       Note: The name value must consist of lower case alphanumeric characters, '-' or '.', and must start and end with an alphanumeric character.
     - Enter a Subnet value. It is mandatory for remote file systems whenever Global Data Platform is the local storage. This field takes IP address in CIDR notation. For example, 100.8.9.10./32.
       Note: You cannot edit the Subnet value after you add.
     - Enter the following details to identify the file system or system:
       - File system name - The name helps to identify the file system or systems that can be accessed through IBM Fusion. Run the following command to get the file system name:

         ```
         /usr/lpp/mmfs/bin/mmlsconfig
         ```

         The file system name must be an existing file system on the IBM Storage Scale cluster that you want to remote mount.

       - Storage class name - Enter a unique storage class name to add to this file system. The Storage class must not exist in the Red Hat OpenShift cluster.

       Note: Click Add to add more than one file system from this cluster. You can click the remove icon to delete a file system.
   - If you choose Select an existing connect cluster, then select the host name from the Select a cluster drop-down list. The cluster details are displayed, such as host name and name of the connected file system. Enter the following details to identify the file system:

- File system name - The name helps to identify the file system or systems that can be accessed through IBM Fusion. Run the following command to get the file system name:

  ```
  /usr/lpp/mmfs/bin/mmlsconfig
  ```

  The file system name must be an existing file system on the IBM Storage Scale cluster that you want to remote mount.

- Storage class name - Enter a unique storage class name to add to this file system. The Storage class must not exist in the Red Hat OpenShift cluster.

Click Add to include more than one file system from this cluster. You can click the remove icon to delete a file system.

4. Click Add file system.
   An information message informs you that the file system is connecting. After the connection is complete, a success message appears to confirm that the file system is connected. The new connection gets added to the Remote file system list.

Backup & Restore

The Backup & Restore service protects your workloads with tailored backups by using a hub and spoke model. You can restore your data rapidly from local snapshots without having to rely on external storage. Backups can be transferred to external object storage solutions like IBM Cloud, Microsoft Azure, Amazon Web Services, and any S3-compliant object storage. It allows the protection of the IBM Fusion Backup & Restore service itself to a S3 bucket.

- **Backup & Restore overview**
  You can protect your container workloads and virtual machines by using the IBM Fusion Backup & Restore service.
- **Install Backup & Restore**
  Protect your data with application-centric backups. Use local snapshots for quick recovery or transfer backups to external object storage for safe keeping. The IBM Fusion Backup & Restore service installation include backup hub and backup spoke.
- **Uninstalling Backup & Restore**
  Steps to uninstall IBM Fusion Backup & Restore by using command-line or command prompt interface.
- **Getting started**
  To quick start the service, define the StorageClass and make sure the VolumeSnapshotClass matches the StorageClass provisioner. Next, connect your clusters and set up the backup locations. Create and assign backup policies to the relevant applications, and regularly back up for continuity. This section also introduces you to the protection of the Backup & Restore service itself from data corruption and loss.
- **Backup & Restore of your workloads**
  The Backup & Restore service is designed to protect a wide range of workloads, including virtual machines, Kubernetes applications, AI workloads, and various IBM Cloud Paks. Additionally, it can also back up the Backup & Restore data management service to an S3 bucket.
- **Managing backup for multi-cluster environment**
  You can connect and manage cluster connections in IBM Fusion. It includes steps to establish connections and manage existing connections. You can also reconnect an OpenShift Container Platform cluster either after a disaster recovery event or a temporary unavailability.
- **Backup consistencies**
- **Performance and scaling**
- **FIPS-140-2**
  The Federal Information Processing Standard Publication 140-2 (FIPS-140-2) is a standard that defines a set of security requirements for the use of cryptographic modules. Law mandates this standard for the US government agencies and contractors and is also referenced in other international and industry specific standards.
- **Backup & Restore service installation and upgrade issues**
  Use this troubleshooting information to resolve install and upgrade problems that are related to Backup & Restore service.
- **Troubleshooting Backup & Restore service issues**
  List of known Backup & Restore issues in IBM Fusion.

Backup & Restore overview

You can protect your container workloads and virtual machines by using the IBM Fusion Backup & Restore service.

In the event of data loss or corruption, such as accidental deletion or malicious ransomware attacks, a backup and restore system is needed to ensure data integrity and availability. By taking periodic backups of applications and storing them safely, you can restore your workloads during disasters, thereby minimizing downtime and data loss. Whenever disaster occurs, you can restore the workloads in place or to an alternate cluster.

Backup & Restore service protects both containerized and virtual machine (VM) workloads, going beyond data protection to include the application's state and essential etcd resources. This capability is not only for stateful applications but also for stateless ones, ensuring complete restoration of functionality and configuration.

Backup and restore works by first choosing a secure backup location and then defining backup polices tailored to your application needs with frequency and retention parameters. You can then associate the workloads with defined backup policies to automate and simply the backup process. Finally, you can recover the backed up workloads either in their original location or to an alternate cluster based on the requirements.

For more complex scenarios or whenever you need to run tailored and repeatable workflows, you can make use of the advanced recipes provided by Backup & Restore service.

Backup & Restore supports both single-cluster mode and hub and spoke model. In a single cluster mode, all backup and restore operations are managed within a single cluster. In case of hub and spoke model, the hub acts as a central management point and spokes are individual clusters that run backup and restore tasks locally. You can choose this mode to manage backups across multiple clusters, and it is useful for large-scale environments where multiple clusters need to be managed with resilience in case of failures.

The IBM Fusion Backup & Restore service protection involves the backup of the Backup & Restore data management service itself to a S3 bucket.

- **Backup and restore planning**
  Plan resource requirements for backups.
- **How Backup & Restore service backups up application data**
  Backup & Restore service includes support for block volume mode PVs along with the backup of applications with filesystem volume mode PVs. It also supports incremental backups for stateful data (PVC) using Fusion Backup & Restore.

- **Backup & Restore service architecture**
  This topic explains the high-level workings of the Backup & Restore service to replicate data. It provides an overall idea of the internal components and their roles in the backup and restore operation, resulting in the final transfer of data to designated storage.
- **Hub and spoke model overview**
  Hub and spoke model allows a single administrator to manage backups for multiple applications that exist on the different clusters.

Backup and restore planning

Plan resource requirements for backups.

The first backup is a full backup. All subsequent backups are incremental.

The capacity required for a backup repository on a remote S3 object store depends on the initial size of all application PVCs and resources, the application data change rate, the backup frequency, and the retention settings. Because the repository is remote, the network must support transferring data volumes equivalent to the S3 object store size.

For example, consider an application that uses 2 TB of data and generates 300 MB of changed data per day. With a daily backup policy and 7-day retention, the initial backup transfers and stores 2 TB of data. Each subsequent daily incremental backup transfers and stores 300 MB.

IBM Fusion HCI supports backup and recovery operations on Red Hat® OpenShift® Container Platform environment with any storage provider that implemented the Container Storage Interface driver and meets the following prerequisites:

- The Container Storage Interface driver must be v1 or higher (no support for *alpha* or *beta* versions)
- The Container Storage Interface driver must support Volume Snapshot and Restore.

Note: It is also recommended that the `VolumeSnapshotClass` deletion policy is configured to delete snapshots after the corresponding `VolumeSnapshotContent` is deleted (`deletionPolicy: Delete`).

How Backup & Restore service backups up application data

Backup & Restore service includes support for block volume mode PVs along with the backup of applications with filesystem volume mode PVs. It also supports incremental backups for stateful data (PVC) using Fusion Backup & Restore.

You can backup an application data, and it consists of PVC data and resources. For the PVC data, you can use online backups or snapshots. You mount the snapshot as a separate PVC for independent processing of the backup without impacting the application. The first back is a full backup and all subsequent backups are incremental. The backups are store in a S3 repository or locally on the system using snapshots (not recommended). The detection and processing of Block volume mode PVs is automatic with no user configuration or changes to the input.

Backup and restore of PV

The Backup & Restore service can backup data hosted on both `volumeMode: Block` and `Filesystem persistent volumes (PV)`. The detection and processing of the `volumeMode` is automatic and does not need any user configuration or changes.

PV with volumeMode: Filesystem
: The pod mounts it as a filesystem directory.

PV with volumeMode: Block
: The pod mounts it as a raw block device without any filesystem on it. This mode provides the best performance as it eliminates the filesystem layer between the pod and the PV. A common application for block PVs is database and virtual machines, that is, Red Hat® OpenShift® Virtualization.

Example Block volume mode PersistentVolume definition.

```
kind: PersistentVolume
apiVersion: v1
metadata:
  name: local-pv-76de1a29
  uid: 42643cc7-ac33-4536-b31b-6774524ee590
spec:
  capacity:
    storage: 3576Gi
  local:
    path: /mnt/local-storage/mylocalvs/nvme-Dell_Ent_NVMe_CM6_RI_3.84TB_Y2B0A01HTCE8
  accessModes:
    - ReadWriteOnce
  claimRef:
    kind: PersistentVolumeClaim
    namespace: openshift-storage
    name: ocs-deviceset-mylocalvs-0-data-4xh44d
    uid: 9abba1a3-6cd9-4d61-9f3e-95b29dab2094
    apiVersion: v1
    resourceVersion: '71161'
  persistentVolumeReclaimPolicy: Delete
  storageClassName: mylocalvs
  volumeMode: Block
```

Backup and restore of PVC data

Stateful data (PVC) backups with Fusion Backup & Restore are incremental, meaning that only new or changed data since the previous backup is sent to the backup storage location. After the initial backup, which copies all stateful data on the PVCs, subsequent backups only transfer new or modified data. After the first backup, the entire data is never copied again; instead, backups continue to send only new or changed data.

RADOS Block Device (RBD) storage class based volumes

In RBD storage class based volumes, change block detection capability speeds up processing time of the backup.

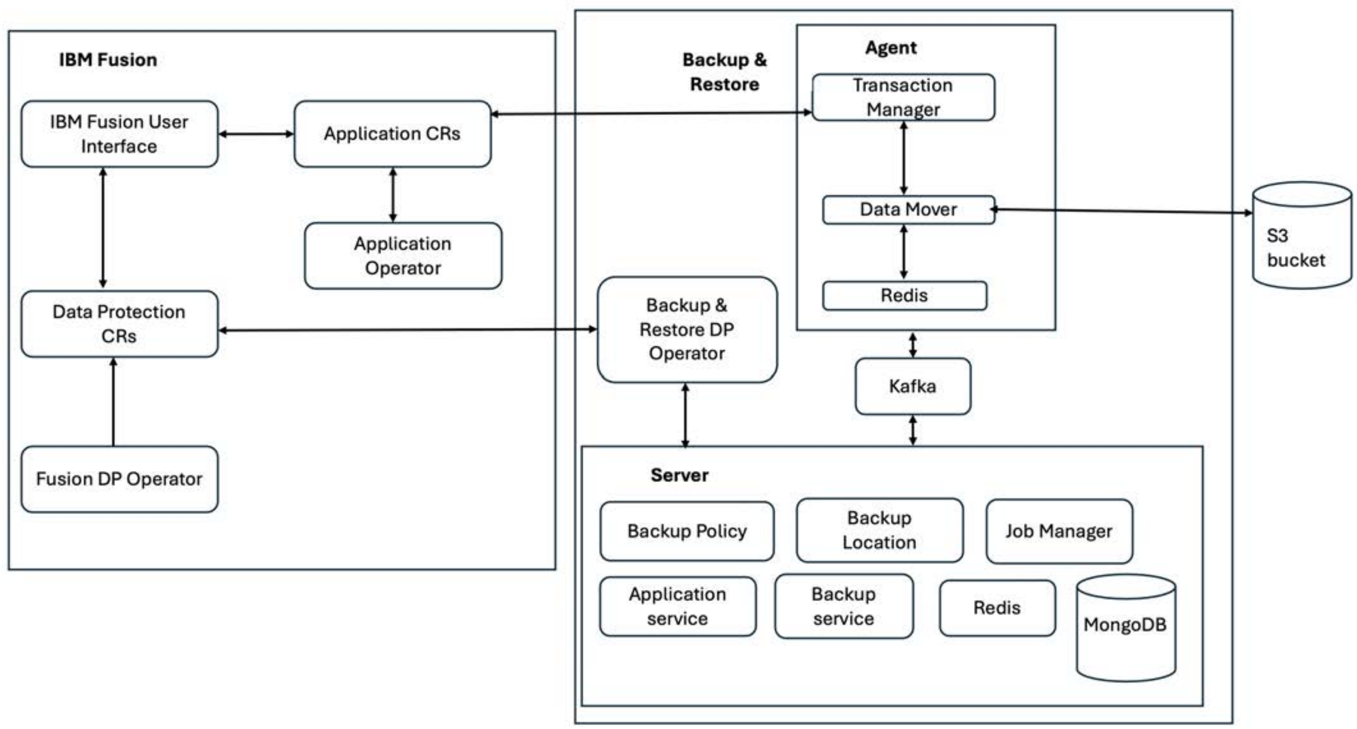
Resources

Kubernetes resource data is always copied in its entirety on each backup operation. Comparatively, the Kubernetes resource data represents a very small percentage of the total amount of data that gets transferred for the backup operation.

Backup & Restore service architecture

This topic explains the high-level workings of the Backup & Restore service to replicate data. It provides an overall idea of the internal components and their roles in the backup and restore operation, resulting in the final transfer of data to designated storage.

The following diagram depicts the workflow of a Backup & Restore hub and local agent.



The IBM Fusion CRs (Application and Data Protections CRs) and operators (Application and Fusion DP operator) interact with the Backup & Restore service. The Backup & Restore service includes Agent and Server components.

Agents handle specific tasks to perform up to ten backup or restore jobs at once:

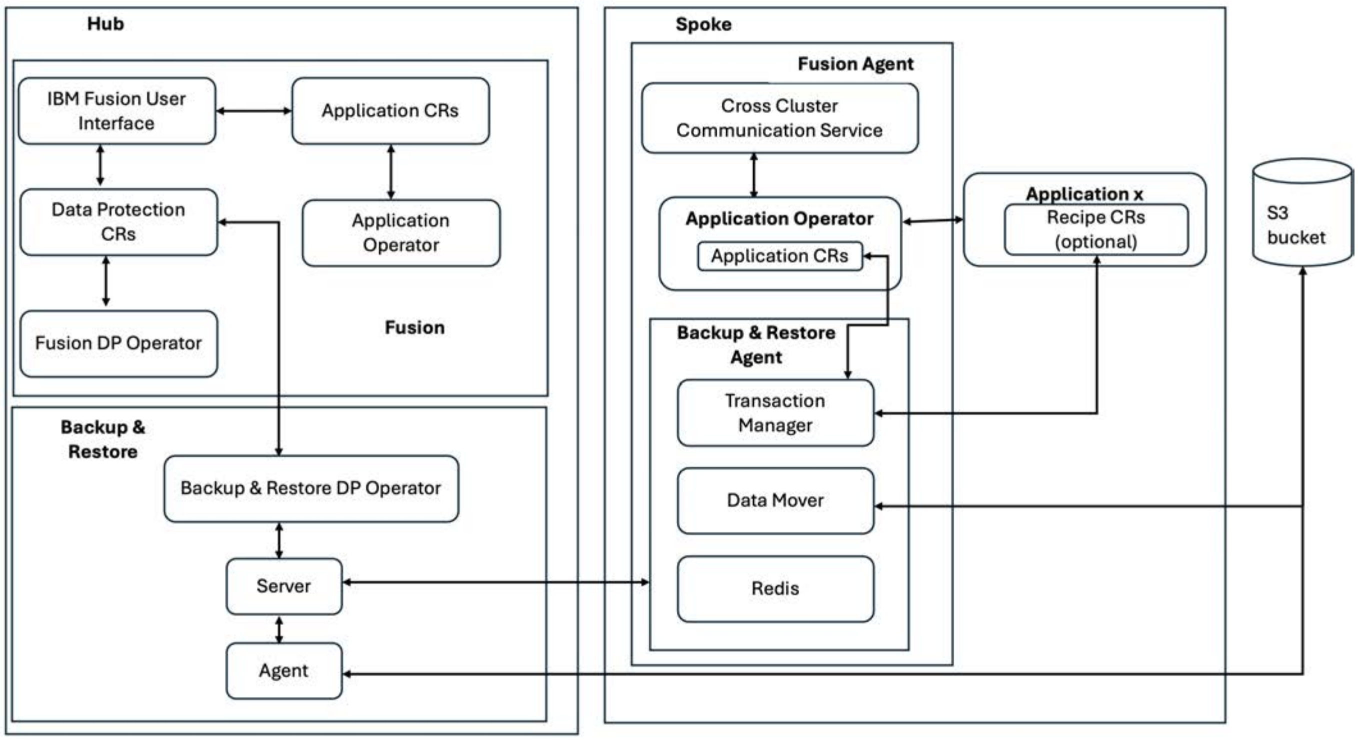
- The Transaction Manager facilitates the backup or restore job requests from the server. These actions include request for PVC snapshots, recipe processing, and status updates to the server.
- To process the backup, the Data Mover mounts the PVC snapshot to read the data, determines any deduplication, and then transfers the data to the designated S3 bucket.

The server includes components like Backup Policy, Backup Location, Job Manager, Application Service, Backup Service, Redis, and MongoDB.

- The Backup Policy manages scheduling and cleanup tasks through Policy CRs.
- the Backup Location validates BSL CRs without performing extensive validations as it directly communicates with the S3 bucket to verify connectivity and existence.
- The Job Manager distributes jobs to agents and monitors job progress to detect and cancel hung jobs. It also prevents duplicate jobs for the same application or policy combination from running at the same time.
- Applications Service maintains records of all existing applications across clusters.
- The Backup Service updates backup and restore Custom Resource with their respective statuses.
- Redis is employed to manage lock contentions; for instance, during scheduled backups involving replica instances, after one acquires a Redis lock, it initiates the task while others are dormant.
- MongoDB functions as the central repository for backup and metadata details.

Backup & Restore service uses Kafka as the primary communication medium among services, providing scalability and resilience. It also allows multiple replicas to process distinct messages concurrently.

The following diagram depicts the workflow of a Backup & Restore hub with multiple spokes.



In case of hub and spoke, it is similar but the agents are on different clusters. The Fusion UI, CRs, Backup & Restore server and local agent are the same except that it needs additional cross cluster communication.

When you attach a spoke cluster to the hub, install the Backup & Restore Agent service on the spoke. The installation of the Backup & Restore Agent service requires a temporary token provided by the Hub for initial setup of the secure cross-cluster communication between the hub and spoke.

Hub and spoke model overview

Hub and spoke model allows a single administrator to manage backups for multiple applications that exist on the different clusters.

The Hub is the single location from where the administrator can login and manage all the applications on all the spoke and hub clusters. The hub location hosts the server and facilitates the Backup & restore administrator to run backup and restore jobs across all clusters.

The hub includes all the Backup & restore server components and agent components for backing up components on its own cluster. The spoke has only the Backup & restore agent installed in it. The Backup & restore hub maintains all scheduling, retention, policy handling, location management, and everything else from a management point of view. The agent or spoke takes care of requests to facilitate Backup & restore jobs on that cluster.

Whenever you want to manage applications on a spoke cluster, you must install IBM Fusion and the Backup & restore Agent Service on that cluster. As part of the spoke agent installation process, a communication is established between the hub and spoke cluster. The spoke Agent installation requests for a connection snippet to be generated on the hub and you must provide it within an hour to the Agent installation. This temporary connection snippet generates a permanent communication between the hub and spoke by generating a certificate from the Certificate Authority (CA) of the hub. By default, this certificate expires after 90 days. IBM Fusion takes care of the needed steps to auto-renew a certificate before it expires.

IBM Fusion is also resilient to unreliable network connections and can recover with network disconnections between the hub and spoke clusters up to an hour .

In the Jobs page of the hub cluster, you can view all the backup and restore jobs across all clusters. You can filter jobs of a specific cluster. When a backup or restore job gets initiated, it gets remotely deployed to the cluster where the agent picks up the job to run it locally on that cluster.

The Service Protection feature provides the ability to restore applications from remote S3 backups (local snapshot only backups not supported) across all hub and spoke clusters. While the Service Protection backup is configured on the hub cluster, all backup information for spoke clusters is automatically included and no configuration on the spoke clusters are required.

In the event of a cluster crash, you can restore application backups on the replacement cluster.

Note: Service protection is just for backup restore on the hub cluster and not for other configurations that exist in IBM Fusion. For example, Red Hat® OpenShift® Container Platform cluster, disaster recovery, Red Hat OpenShift Data Foundation.
In the Topology page, you can see the hub cluster in the first row followed by all connected spoke clusters. The details of the connected clusters, such as health, status, success rate are available in this view.

The two available backup storage locations are the in place local snapshots and S3 remote object store locations. If you want to restore applications on another cluster, configure a S3 remote object store for the backup storage locations. The S3 remote object store is to store data on an external location. During instances wherein the cluster crashes, the backups that are stored in the local snapshots are lost along with it. Hence, it is important to offload your backups to an external storage location, and later restore the application on a replacement cluster.

You can share the locations and policies for applications that are backedup up across the cluster. From the hub cluster, you can apply policies to any number of applications across any number of clusters. All applications get backed up at the same time as they share policies and the schedules. However, it can cause resource contention in the network cluster. For example, all backups need network connection to a S3 location at the same time. Though it gives the advantage of maintaining limited number of policies, it might place a strain on the network. Similar to the policies, you can use one location to store every backup of all clusters. It can cause contention of network requests on the same backup location.

The Applications page shows only applications on that cluster and not any applications on the spoke cluster. The Backedup application page shows all applications that are protected across the cluster. To protect new applications in the cluster, click Protect apps in the Backedup application page, and select applications from a specific cluster.

Install Backup & Restore

Protect your data with application-centric backups. Use local snapshots for quick recovery or transfer backups to external object storage for safe keeping. The IBM Fusion Backup & Restore service installation include backup hub and backup spoke.

About this task

For more information about the hub and spoke architecture of Backup & Restore service, see Hub and spoke model overview.

- **Prerequisites**
  Ensure that you meet the prerequisites before you work with Backup & Restore service.
- **Backup & Restore hub**
  Protect your data with application-centric backups. Use local snapshots for quick recovery or transfer backups to external object storage for safe keeping.
- **Backup & Restore spoke**
  Protect your data with application-centric backups. Use local snapshots for quick recovery or transfer backups to external object storage for safe keeping.
- **Establishing connection between hub and spoke**
  Generate YAML from the hub cluster to establish mutual authentication between the hub and spoke clusters.

Prerequisites

Ensure that you meet the prerequisites before you work with Backup & Restore service.

1. Install either the Backup & Restore or Backup & Restore Agent service. For the procedure to install, see Install Backup & Restore.
2. Ensure that you define a StorageClass that is provisioned by a supported CSI driver. Check the `provisioner:` in the `Spec` to make sure that it is backed by CSI.
   Sample Storage Class for Fusion Data Foundation:

   ```
   apiVersion: storage.k8s.io/v1
   kind: StorageClass
   metadata:
     name: ocs-storagecluster-cephfs
   provisioner: openshift-storage.cephfs.csi.ceph.com
   reclaimPolicy: Delete
   allowVolumeExpansion: true
   volumeBindingMode: Immediate
   ```

   Note: The Storage Class must have the CSI Snapshot capability (`VolumeSnapshotClass`).
3. For backups and restores to run successfully, ensure that CSI compliant storage classes have `volumeBindingMode: Immediate` or `volumeBindingMode: WaitForFirstConsumer` and `allowVolumeExpansion: true`.
4. Ensure that the VolumeSnapshotClass has a `driver:` that matches the `provisioner:` in the StorageClass.
   Sample YAML for Red Hat® OpenShift® Data Foundation RBD storage:

   ```
   apiVersion: snapshot.storage.k8s.io/v1
   kind: VolumeSnapshotClass
   driver: openshift-storage.cephfs.csi.ceph.com
   metadata:
     name: ocs-storagecluster-cephfsplugin-snapclass
   deletionPolicy: Delete
   parameters:
     clusterID: openshift-storage
     csi.storage.k8s.io/snapshotter-secret-name: rook-csi-cephfs-provisioner
     csi.storage.k8s.io/snapshotter-secret-namespace: openshift-storage
   ```

Note: If you plan to create and use your own storage classes in IBM Fusion HCI, ensure that you have the VolumeSnapshotClass.

Backup & Restore hub

Protect your data with application-centric backups. Use local snapshots for quick recovery or transfer backups to external object storage for safe keeping.

Before you begin

Consider the following points before you begin installation:

- Firewall ports required for Hub and Spoke architecture:
  - Hub

    Must be able to make a TCP connection to the Spoke cluster API address
  - Spoke
    - Must be able to make a TCP connection to the Hub cluster API address

- Must be able to make a TCP connection address :443, where **<kafka-route>:443** can be found by running the following command on the hub:

  ```
  oc get route -n ibm-bnr -l app=kafka-bridge-rbac-proxy
  ```

  Example output:

  ```
  NAME                      HOST/PORT
  kafka-bridge              kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.containers.app
  kafka-bridge-rbac-proxy   kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.containers.app
  kafka-bridge-rbac-proxy1  kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.containers.app
  kafka-bridge-rbac-proxy2  kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.containers.app
  kafka-bridge-rbac-proxy3  kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.containers.app
  ```

- A route on the host that creates a DNS address exists for the Kubernetes API, which is enabled by default during the installation of Red Hat® OpenShift®. Check whether it is available and is resolvable from the spoke containers. Format of the URL is **api.<cluster-name>.<domain>** but is changeable. This is port 443 on all control plane nodes.
- A route to the Kafka Bridge creates a DNS address. Check whether it is available and is resolvable from the spoke containers. Uses port 443 on compute nodes. Run the following command and check the role in the output to know which nodes are compute nodes (needed for Kafka Bridge) and control plane nodes (for Kubernetes API connection):

  ```
  oc get nodes
  ```

  Example output:

  ```
  NAME STATUS ROLES AGE VERSION
  bootstrap.ocpfsn.pok.stglabs.ibm.com Ready worker 3d18h v1.25.14+20cda61
  master0.ocpfsn.pok.stglabs.ibm.com Ready control-plane,master 3d19h v1.25.14+20cda61
  master1.ocpfsn.pok.stglabs.ibm.com Ready control-plane,master 3d19h v1.25.14+20cda61
  master2.ocpfsn.pok.stglabs.ibm.com Ready control-plane,master 3d19h v1.25.14+20cda61
  worker0.ocpfsn.pok.stglabs.ibm.com Ready worker 3d18h v1.25.14+20cda61
  worker1.ocpfsn.pok.stglabs.ibm.com Ready worker 3d18h v1.25.14+20cda61
  worker3.ocpfsn.pok.stglabs.ibm.com Ready worker 2d2h v1.25.14+20cda61
  ```

  Alternatively, to check the roles from OpenShift console, do the following steps:
  1. Log in to the OpenShift console.
  2. Click Compute > Nodes menu.
  3. Check the role of the nodes.
- The following command can be used to get the cluster API address of a cluster:

  ```
  oc cluster-info
  ```

  For example:

  ```
  Kubernetes control plane is running at https://c109-e.us-east.containers.cloud.ibm.com:30363
  ```

Procedure

1. Go to Services page in IBM Fusion user interface.
2. In the Available section, click the Backup & Restore tile.
3. In the Backup & Restore page, go through the features and capabilities of the service and click Install.
   In the Install service window, the `ocs-storagecluster-ceph-rbd` class is set by default for the Backup & Restore service. The internal data catalog requires a minimum of 200 GB for ReadWriteAfter storage.
4. Click Install.
   The installation starts and a notification appears on the Services page. You can see the progress of the installation in the Services > Installed section. After the installation completes successfully, you can see the status as normal and a Get started link.
   After you enable the Backup & Restore, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification gets displayed. The notification disappears automatically after some time.

   The Backup & Restore menu in the spoke cluster includes the following sub-menus:
   - Topology
   - Backed up applications
   - Policies
   - Locations
   - Jobs
   - Service protection

   In the Backup & restore > Overview page, you have quick links to generate YAML, Connect locations, Create backup policies, and protect your applications.

   In case of other failures, go through the downloaded logs to understand the cause of the failure and fix the issue. For more information about service issues in IBM Fusion, see Backup & Restore service installation and upgrade issues.
5. Validate the installation:
   Note: Ensure that the Backup & Restore IBM Fusion service is enabled before validation.

   Verify that the Backup & Restore IBM Fusion HCI service operators from the OpenShift Container Platform web console:
   a. Go to Installed operators from OpenShift Container Platform web console.
   b. Select the project as **ibm-backup-restore**.
   c. Verify that the following operators show the status as **succeeded**.
      - Red Hat Integration - AMQ Streams
      - IBM Fusion Backup & Restore Hub
      - IBM Fusion Backup & Restore Spoke
      - OADP Operator

   d. Alternatively, you can verify the status of operators by running the following oc command:
      Note: More operators appear in the command line output than in the web console.

      ```
      oc get csv -n ibm-backup-restore
      ```

A sample result of the oc command is as follows:

```
NAME                              DISPLAY                                          VERSION     REPLACES                 PHASE
amqstreams.v2.3.0-1               Red Hat Integration - AMQ Streams                2.3.0-1   amqstreams.v2.3.0-0
Succeeded
guardian-dm-operator.v2.5.0       IBM Fusion Backup and Restore Data Mover         2.5.0
Succeeded
guardian-dp-operator.v2.5.0       IBM Fusion Backup and Restore Data Protection    2.5.0
Succeeded
guardian-mongo-operator.v2.5.0    IBM Fusion Backup and Restore Mongo              2.5.0
Succeeded
ibm-dataprotectionagent.v2.5.0    IBM Fusion Backup and Restore Agent              2.5.0
Succeeded
ibm-dataprotectionserver.v2.5.0   IBM Fusion Backup and Restore Server             2.5.0
Succeeded
oadp-operator.v1.1.2              OADP Operator                                    1.1.2     oadp-operator.v1.1.1
Succeeded
redis-operator.v2.5.0             IBM Fusion Backup and Restore Redis              2.5.0
Succeeded
```

.Ensure that the status shows **Succeeded**.

Verify the Backup & Restore pods from the OpenShift Container Platform console:

a. Go to Workloads > Pods.
b. Select the namespace, where you installed IBM Fusion Backup & Restore. In this case, select **ibm-backup-restore** namespace.
   It lists all the pods. Ensure that all pods are running.
c. Verify whether the following pods are running successfully:

```
amq-streams-cluster-operator
applicationsvc
backup-location-deployment
backup-service
backuppolicy-deployment
guardian-dm-controller-manager
guardian-dp-operator-controller-manager
guardian-kafka-cluster-entity-operator
guardian-kafka-cluster-kafka-0
guardian-kafka-cluster-kafka-1
guardian-kafka-cluster-kafka-2
guardian-kafka-cluster-zookeeper-0
guardian-kafka-cluster-zookeeper-1
guardian-kafka-cluster-zookeeper-2
guardian-minio-0
guardian-mongo-operator-controller-manager
ibm-dataprotectionagent-controller-manager
ibm-dataprotectionserver-catalog-ibm-backup-restore
ibm-dataprotectionserver-controller-manager
ibm-backup-restoreagent-controller-manager
ibm-backup-restoreserver-controller-manager
job-manager
mongodb-0
mongodb-1
mongodb-2
mongodb-ab-0
openshift-adp-controller-manager
redis-master-0
redis-operator-controller-manager
redis-replicas-0
redis-replicas-1
redis-replicas-2
transaction-manager
velero
```

d. Alternatively, you can verify the installation by running the following oc command:

```
oc get pods -n ibm-backup-restore
```

A sample result of the oc command is as follows:

```
NAME                                                                READY   STATUS      R
amq-streams-cluster-operator-v2.3.0-1-7d6fb79d84-jdkfh              1/1     Running     0
applicationsvc-55c9b4d6c9-6hdv7                                     1/1     Running     0
b0f64f9161e0882f278dde2eaa1ea9677f4a230a29180fcf21fc665761hvvxz     0/1     Completed   0
backup-location-deployment-6b565b856c-j4vjc                         1/1     Running     0
backup-service-54bf9988f6-47bpv                                     1/1     Running     0
backuppolicy-deployment-b997cc9bf-dfwnh                             1/1     Running     0
bc6176f08ef686ccde24395724b77ea07a586a0bd1fa27ebfd5d704d0dxv9pl     0/1     Completed   0
e349f7c16f02ad6c0c31e41ba2fb1750d5154b58537224d239fe47508872cj5     0/1     Completed   0
f3dd0cbe8cb98614cf163fc5372733148236ea2bdeb5efc6a5d5afe4c085qlk     0/1     Completed   0
ff53c6d827e0fd610a4392c4f9411beb0c785572d2fca1bda57e208650bwv2m     0/1     Completed   0
ffacc5cdaa1e0aa4f3b3c0021f3c87931aa7422f8415303d95457febc8nmzk7     0/1     Completed   0
guardian-dm-controller-manager-64d57bf9ff-28dqj                     2/2     Running     0
guardian-dp-operator-controller-manager-6f6d55f6f7-fhndb            2/2     Running     0
guardian-kafka-cluster-entity-operator-b59d699f7-5qxt8              3/3     Running     0
guardian-kafka-cluster-kafka-0                                      1/1     Running     0
guardian-kafka-cluster-kafka-1                                      1/1     Running     0
guardian-kafka-cluster-kafka-2                                      1/1     Running     0
guardian-kafka-cluster-zookeeper-0                                  1/1     Running     0
guardian-kafka-cluster-zookeeper-1                                  1/1     Running     0
guardian-kafka-cluster-zookeeper-2                                  1/1     Running     0
guardian-minio-0                                                    1/1     Running     0
guardian-mongo-operator-controller-manager-6f47776cb4-s6tkm         2/2     Running     0
ibm-dataprotectionagent-controller-manager-54d66f7975-lgdgh         2/2     Running     0
ibm-dataprotectionserver-catalog-ibm-backup-restore-k967t           1/1     Running     0
ibm-dataprotectionserver-controller-manager-749554d89f-q6gmx        2/2     Running     0
```

```
job-manager-859484bfc5-fzpt8                                  1/1     Running     0
mongodb-0                                                     2/2     Running     1
mongodb-1                                                     2/2     Running     1
mongodb-2                                                     2/2     Running     1
mongodb-ab-0                                                  1/1     Running     0
openshift-adp-controller-manager-59fb9f86f4-gdbhb             1/1     Running     0
redis-master-0                                                1/1     Running     0
redis-operator-controller-manager-647cf89ff7-wl8w2            2/2     Running     0
redis-replicas-0                                              1/1     Running     0
redis-replicas-1                                              1/1     Running     0
redis-replicas-2                                              1/1     Running     0
transaction-manager-5d9b59cf9f-hdzjm                          2/2     Running     0
velero-777d65c9b7-455fv                                       1/1     Running     0
```

Verify the Backup & Restore IBM Fusion service installation status from the OpenShift Container Platform web console:

a. Go to Installed operators from OpenShift Container Platform web console.
b. Select the namespace as `ibm-backup-restore`.
c. Select IBM Fusion Backup & Restore server.
d. Click the Data Protection server tab and select Data Protection server.
e. Select the YAML tab.
f. In the status section, makes sure the following:
   - `HealthStatuses` Shows the health status of all components listed. It may take 5 minutes or more for all components to show up as healthy.
   - Ensure the install status shows Complete and `progressPercentage` as 100.
g. Alternatively, you can verify the installation status by running the following oc command:

```
oc describe dataprotectionserver dataprotectionserver -n ibm-backup-restore
```

A sample result of the oc command is as follows:

```
Status:
  Health Statuses:
    Service Name:  applicationservice
    Status:        Healthy
    Service Name:  backuplocation
    Status:        Healthy
    Service Name:  backuppolicy
    Status:        Healthy
    Service Name:  backupservice
    Status:        Healthy
    Service Name:  jobmanager
    Status:        Healthy
    Service Name:  backupagent
    Status:        Healthy
    Service Name:  mongo
    Status:        Healthy
    Service Name:  redis
    Status:        Healthy
    Service Name:  kafka
    Status:        Healthy
  Install Status:
    Progress Percentage:  100
    Retry On Failure:     false
    Status:               Completed
  Installed Version:      2.5.0
  Upgrade In Progress:    false
  Upgrade Status:
    Retry On Failure:  false
```

h. Check the status section at the end of the CR and make sure the following:
   - The Install Status section shows the status of the install along with the percentage complete.
   - The install status shows the install status as Completed when the installation is successfully completed.
   - The health status section lists and shows components health status as healthy.
   - Note: Individual component health statuses may show `Unknown` or `Degraded` for up to five minutes, and show a healthy status when installation is complete.

What to do next

- Generate a connection snippet or YAML. For the procedure to generate, see Establishing connection between hub and spoke.
- You can now begin to protect your IBM Fusion applications.
  1. Add location to determine whether the network verification is needed. For the procedure to add a location, see Adding backup storage location.
  2. Creating backup policy.
  3. Assigning backup policy.

Backup & Restore spoke

Protect your data with application-centric backups. Use local snapshots for quick recovery or transfer backups to external object storage for safe keeping.

Before you begin

- Install backup hub. For the procedure to install, see Backup & Restore hub.
- Generate the YAML. This YAMLis required to establish mutual authentication between the two clusters. For the procedure to generate, see Establishing connection between hub and spoke.
- When you add a Spoke to a Hub, the version of the Spoke must be the same version as the Hub.
- Consider the following points before you begin installation:

- Firewall ports required for Hub and Spoke architecture:
  - Hub
    Must be able to make a TCP connection to the Spoke cluster API address
  - Spoke
    - Must be able to make a TCP connection to the Hub cluster API address
    - Must be able to make a TCP connection address :443, where `<kafka-route>:443` can be found by running the following command on the hub:

      ```
      oc get route -n ibm-bnr -l app=kafka-bridge-rbac-proxy
      ```

      Example output:

      ```
      NAME                       HOST/PORT
      kafka-bridge               kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.container
      kafka-bridge-rbac-proxy    kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.container
      kafka-bridge-rbac-proxy1   kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.container
      kafka-bridge-rbac-proxy2   kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.container
      kafka-bridge-rbac-proxy3   kafka-bridge-ibm-bnr.bnr-98b7318c91b01bd72490e80cc2328915-0000.ca-tor.container
      ```
  - A route on the host that creates a DNS address exists for the Kubernetes API, which is enabled by default during the installation of Red Hat® OpenShift®. Check whether it is available and is resolvable from the spoke containers. Format of the URL is `api.<cluster-name>.<domain>` but is changeable. This is port 443 on all control plane nodes.
  - A route to the Kafka Bridge creates a DNS address. Check whether it is available and is resolvable from the spoke containers. Uses port 443 on compute nodes. Run the following command and check the role in the output to know which nodes are compute nodes (needed for Kafka Bridge) and control plane nodes (for Kubernetes API connection):

    ```
    oc get nodes
    ```

    Example output:

    ```
    NAME STATUS ROLES AGE VERSION
    bootstrap.ocpfsn.pok.stglabs.ibm.com Ready worker 3d18h v1.25.14+20cda61
    master0.ocpfsn.pok.stglabs.ibm.com Ready control-plane,master 3d19h v1.25.14+20cda61
    master1.ocpfsn.pok.stglabs.ibm.com Ready control-plane,master 3d19h v1.25.14+20cda61
    master2.ocpfsn.pok.stglabs.ibm.com Ready control-plane,master 3d19h v1.25.14+20cda61
    worker0.ocpfsn.pok.stglabs.ibm.com Ready worker 3d18h v1.25.14+20cda61
    worker1.ocpfsn.pok.stglabs.ibm.com Ready worker 3d18h v1.25.14+20cda61
    worker3.ocpfsn.pok.stglabs.ibm.com Ready worker 2d2h v1.25.14+20cda61
    ```

    Alternatively, to check the roles from OpenShift console, do the following steps:
    1. Log in to the OpenShift console.
    2. Click Compute > Nodes menu.
    3. Check the role of the nodes.
  - The following command can be used to get the cluster API address of a cluster:

    ```
    oc cluster-info
    ```

    For example:

    ```
    Kubernetes control plane is running at https://c109-e.us-east.containers.cloud.ibm.com:30363
    ```

Procedure

1. Go to Services page in IBM Fusion user interface.
2. In the Available section, click the Backup & Restore Agent tile.
3. In the Backup & Restore page, go through the features and capabilities of the service and click Install.
   In the Install service window, the `ocs-storagecluster-ceph-rbd` class is set by default for the Backup & Restore service. The internal data catalog requires a minimum of 200 GB for ReadWriteAfter storage.
4. In the Install service window, select a Storage class that is used for the service.
   so select a storage class that supports this criteria.
5. Enter a connection snippet that is generated from the backup hub cluster.
   Important: When you install Spoke from the user interface, use the snippet. Use YAML option only when you do an automated deployment outside the IBM Fusion user interface.
   For more information about establishing connection between the Hub and Spoke, see Establishing connection between hub and spoke.
6. Click Install.
   A validation is done to check whether the connection is possible. If connection failed message appears, check the message and take corrective action. The installation starts and a notification appears on the Services page. You can see the progress of the installation in the Services > Installed section. After the installation completes successfully, you can see the status as normal and a Get started link.
   After you enable the Backup & Restore, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification gets displayed. The notification disappears automatically after some time.

   The Backup & Restore menu in the spoke cluster includes the following sub-menus:
   - Topology
   - Backed up applications

   In case of other failures, go through the downloaded logs to understand the cause of the failure and fix the issue. For more information about service issues in IBM Fusion, see Backup & Restore service installation and upgrade issues.
7. Validate the installation:
   Note: Ensure that the Backup & Restore IBM Fusion service is enabled before validation.

   Verify that the Backup & Restore IBM Fusion HCI service operators from the OpenShift Container Platform web console:
   a. Go to Installed operators from OpenShift Container Platform web console.
   b. Select the project as **ibm-backup-restore**.
   c. Verify that the following operators show the status as **succeeded**.

- Red Hat Integration - AMQ Streams
- IBM Fusion Backup & Restore Hub
- IBM Fusion Backup & Restore Spoke
- OADP Operator

d. Alternatively, you can verify the status of operators by running the following oc command:
Note: More operators appear in the command line output than in the web console.

```
oc get csv -n ibm-backup-restore
```

A sample result of the oc command is as follows:

```
NAME                               DISPLAY                                      VERSION    REPLACES                PHASE
amqstreams.v2.3.0-1                Red Hat Integration - AMQ Streams                   2.3.0-1   amqstreams.v2.3.0-0
Succeeded
guardian-dm-operator.v2.5.0        IBM Fusion Backup and Restore Data Mover            2.5.0
Succeeded
guardian-dp-operator.v2.5.0        IBM Fusion Backup and Restore Data Protection       2.5.0
Succeeded
guardian-mongo-operator.v2.5.0     IBM Fusion Backup and Restore Mongo                 2.5.0
Succeeded
ibm-dataprotectionagent.v2.5.0     IBM Fusion Backup and Restore Agent                 2.5.0
Succeeded
ibm-dataprotectionserver.v2.5.0    IBM Fusion Backup and Restore Server                2.5.0
Succeeded
oadp-operator.v1.1.2               OADP Operator                                       1.1.2     oadp-operator.v1.1.1
Succeeded
redis-operator.v2.5.0              IBM Fusion Backup and Restore Redis                 2.5.0
Succeeded
```

.Ensure that the status shows **Succeeded**.

Verify the Backup & Restore pods from the OpenShift Container Platform console:

a. Go to Workloads > Pods.
b. Select the namespace, where you installed IBM Fusion Backup & Restore. In this case, select **ibm-backup-restore** namespace.
It lists all the pods. Ensure that all pods are running.

c. Verify whether the following pods are running successfully:

```
amq-streams-cluster-operator
applicationsvc
backup-location-deployment
backup-service
backuppolicy-deployment
guardian-dm-controller-manager
guardian-dp-operator-controller-manager
guardian-kafka-cluster-entity-operator
guardian-kafka-cluster-kafka-0
guardian-kafka-cluster-kafka-1
guardian-kafka-cluster-kafka-2
guardian-kafka-cluster-zookeeper-0
guardian-kafka-cluster-zookeeper-1
guardian-kafka-cluster-zookeeper-2
guardian-minio-0
guardian-mongo-operator-controller-manager
ibm-dataprotectionagent-controller-manager
ibm-dataprotectionserver-catalog-ibm-backup-restore
ibm-dataprotectionserver-controller-manager
ibm-backup-restoreagent-controller-manager
ibm-backup-restoreserver-controller-manager
job-manager
mongodb-0
mongodb-1
mongodb-2
mongodb-ab-0
openshift-adp-controller-manager
redis-master-0
redis-operator-controller-manager
redis-replicas-0
redis-replicas-1
redis-replicas-2
transaction-manager
velero
```

d. Alternatively, you can verify the installation by running the following oc command:

```
oc get pods -n ibm-backup-restore
```

A sample result of the oc command is as follows:

```
NAME                                                              READY   STATUS      R
amq-streams-cluster-operator-v2.3.0-1-7d6fb79d84-jdkfh            1/1     Running     0
applicationsvc-55c9b4d6c9-6hdv7                                   1/1     Running     0
b0f64f9161e0882f278dde2eaa1ea9677f4a230a29180fcf21fc665761hvvxz   0/1     Completed   0
backup-location-deployment-6b565b856c-j4vjc                       1/1     Running     0
backup-service-54bf9988f6-47bpv                                   1/1     Running     0
backuppolicy-deployment-b997cc9bf-dfwnh                           1/1     Running     0
bc6176f08ef686ccde24395724b77ea07a586a0bd1fa27ebfd5d704d0dxv9pl   0/1     Completed   0
e349f7c16f02ad6c0c31e41ba2fb1750d5154b58537224d239fe47508872cj5   0/1     Completed   0
f3dd0cbe8cb98614cf163fc5372733148236ea2bdeb5efc6a5d5afe4c085qlk   0/1     Completed   0
ff53c6d827e0fd610a4392c4f9411beb0c785572d2fca1bda57e208650bwv2m   0/1     Completed   0
ffacc5cdaa1e0aa4f3b3c0021f3c87931aa7422f8415303d95457febc8nmzk7   0/1     Completed   0
guardian-dm-controller-manager-64d57bf9ff-28dqj                   2/2     Running     0
guardian-dp-operator-controller-manager-6f6d55f6f7-fhndb          2/2     Running     0
guardian-kafka-cluster-entity-operator-b59d699f7-5qxt8            3/3     Running     0
guardian-kafka-cluster-kafka-0                                    1/1     Running     0
```

```
guardian-kafka-cluster-kafka-1                                   1/1     Running     0
guardian-kafka-cluster-kafka-2                                   1/1     Running     0
guardian-kafka-cluster-zookeeper-0                               1/1     Running     0
guardian-kafka-cluster-zookeeper-1                               1/1     Running     0
guardian-kafka-cluster-zookeeper-2                               1/1     Running     0
guardian-minio-0                                                 1/1     Running     0
guardian-mongo-operator-controller-manager-6f47776cb4-s6tkm      2/2     Running     0
ibm-dataprotectionagent-controller-manager-54d66f7975-lgdgh      2/2     Running     0
ibm-dataprotectionserver-catalog-ibm-backup-restore-k967t        1/1     Running     0
ibm-dataprotectionserver-controller-manager-749554d89f-q6gmx     2/2     Running     0
job-manager-859484bfc5-fzpt8                                     1/1     Running     0
mongodb-0                                                        2/2     Running     1
mongodb-1                                                        2/2     Running     1
mongodb-2                                                        2/2     Running     1
mongodb-ab-0                                                     1/1     Running     0
openshift-adp-controller-manager-59fb9f86f4-gdbhb                1/1     Running     0
redis-master-0                                                   1/1     Running     0
redis-operator-controller-manager-647cf89ff7-wl8w2               2/2     Running     0
redis-replicas-0                                                 1/1     Running     0
redis-replicas-1                                                 1/1     Running     0
redis-replicas-2                                                 1/1     Running     0
transaction-manager-5d9b59cf9f-hdzjm                             2/2     Running     0
velero-777d65c9b7-455fv                                          1/1     Running     0
```

Verify the Backup & Restore IBM Fusion service installation status from the OpenShift Container Platform web console:

a. Go to Installed operators from OpenShift Container Platform web console.
b. Select the namespace as `ibm-backup-restore`.
c. Select IBM Fusion Backup & Restore server.
d. Click the Data Protection server tab and select Data Protection server.
e. Select the YAML tab.
f. In the status section, makes sure the following:
    - `HealthStatuses` Shows the health status of all components listed. It may take 5 minutes or more for all components to show up as healthy.
    - Ensure the install status shows Complete and `progressPercentage` as 100.
g. Alternatively, you can verify the installation status by running the following oc command:

```
oc describe dataprotectionserver dataprotectionserver -n ibm-backup-restore
```

A sample result of the oc command is as follows:

```
Status:
  Health Statuses:
    Service Name:  applicationservice
    Status:        Healthy
    Service Name:  backuplocation
    Status:        Healthy
    Service Name:  backuppolicy
    Status:        Healthy
    Service Name:  backupservice
    Status:        Healthy
    Service Name:  jobmanager
    Status:        Healthy
    Service Name:  backupagent
    Status:        Healthy
    Service Name:  mongo
    Status:        Healthy
    Service Name:  redis
    Status:        Healthy
    Service Name:  kafka
    Status:        Healthy
  Install Status:
    Progress Percentage:  100
    Retry On Failure:     false
    Status:               Completed
  Installed Version:      2.5.0
  Upgrade In Progress:    false
  Upgrade Status:
    Retry On Failure:  false
```

h. Check the status section at the end of the CR and make sure the following:
    - The Install Status section shows the status of the install along with the percentage complete.
    - The install status shows the install status as Completed when the installation is successfully completed.
    - The health status section lists and shows components health status as healthy.
    - Note: Individual component health statuses may show `Unknown` or `Degraded` for up to five minutes, and show a healthy status when installation is complete.

What to do next

- You can set the configuration parameters in ConfigMap guardian-configmap to change defaults for IBM Fusion Backup & restore agent. For more information about the parameters, see Backup & restore configuration parameters.
- Go to the Backup spoke cluster user interface > Overview page and click Launch Backup Hub to open the Backup hub.
  In the Backup & restore > Overview page, you have quick links to generate YAML, connect locations, create backup policies, and protect your applications.
- You can now begin to protect your IBM Fusion applications.
    1. Add location to determine whether the network verification is needed. For the procedure to add a location, see Adding backup storage location.
    2. Creating backup policy.
    3. Assigning backup policy.

Establishing connection between hub and spoke

Generate YAML from the hub cluster to establish mutual authentication between the hub and spoke clusters.

Procedure

1. Log in to the user interface of the hub cluster.
2. Click Backup & Restore > Overview.
3. Use any of the following methods to generate the YAML.

Option	Description
Use the Fusion UI	a. In the step 1: Connect your clusters (optional) section, click Connect clusters. b. From the drop-down list, select Use the Fusion UI. The connect your cluster > Use the Fusion UI window appears. The snippet starts to generate. c. After the snippet is generated successfully, click Copy snippet. d. Enter the copied connection snippet when prompted during the spoke installation. For the step in Spoke installation, see Enter connection snippet.
Automate deployment	a. In the step 1: Connect your clusters (optional) section, click Connect clusters. b. From the drop-down list, select Automate deployment. The connect your cluster > Automate deployment window appears. c. Select the storage class to use on the spoke where you want to run the YAML. d. Click Generate. The YAML starts to generate. e. In the Deploy using YAML section, click copy icon in Deployment YAML or click Download YAML. Alternatively, in the Or deploy using oc command section, click Copy command. Important: If you use a custom namespace, then do not use the Copy command because the encoded command is not editable and is of no value with a custom namespace. f. Apply the downloaded YAML or the copied **oc** command on the Spoke cluster.

4. To validate the connection, do the following steps:
   a. Click Backup & Restore > Topology.
   In the Topology page, you can see the hub cluster. In the Cluster table, you can see a list of spokes connected.
   b. In the Topology page > Cluster table, check whether there is an entry for the spoke and the Connection status column is Connected for it. The Service status is Normal.

Uninstalling Backup & Restore

Steps to uninstall IBM Fusion Backup & Restore by using command-line or command prompt interface.

Before you begin

- Ensure you have the following installed:
  - Bash 5.0 or higher
  - OpenShift® Container Platform command line 4.14 or higher
  - `jq` 1.6 or higher version.

About this task

The uninstall-backup-restore.sh script has two optional arguments `-u` and `-d`. Use the `-u` argument to uninstall agent when the hub still exists.

Procedure

1. Log in to your cluster.
2. From your browser, go to this URL- https://github.com/IBM/storage-fusion/blob/master/backup-restore/uninstall/uninstall-backup-restore.sh.
3. Download uninstall-backup-restore.sh and run the script.

What to do next

Note: On the Topology page, it can take 10 minutes for the Connection Status of the spoke cluster to change from `Connected` to Unknown. Though in the Backedup applications page, you can see the previous application backups from the spoke cluster, no new backups are possible.
After uninstall of agent or server, you may want to delete the connection from the system. You can remove the connection for a spoke from the hub user interface, but the agent user interface does not have this capability. When you try to remove a connection from a hub, check your firewall. For more information, see Remove connections.

Getting started

To quick start the service, define the StorageClass and make sure the VolumeSnapshotClass matches the StorageClass provisioner. Next, connect your clusters and set up the backup locations. Create and assign backup policies to the relevant applications, and regularly back up for continuity. This section also introduces you to the protection of the Backup & Restore service itself from data corruption and loss.

- **Quick start Backup & Restore service**
  The Overview page provides quick access to essential Backup & Restore service tasks, which enables you to quickly set up and use the service.

- **Service protection**
  The IBM Fusion Backup & Restore service protection involves the backup of the Backup & Restore data management service to a S3 bucket.

Quick start Backup & Restore service

The Overview page provides quick access to essential Backup & Restore service tasks, which enables you to quickly set up and use the service.

Get started

From the Get started pane, you can quickly add locations, create policies, and assign a policy to your applications.

Step 1: Connect your cluster
: Create connections to the clusters that you provide backup services for. If you are only backing up applications in this cluster, you can skip this step. Use any of the following methods to generate the YAML. For more information about the options, see Connect clusters.

Step 2: Connect backup locations
: Connect S3 compatible object storage locations that stores the backups.

Step 3: Create backup policies
: Define backup frequency, storage location, and how long they are retained.

Step 4: Protect your applications
: Assign backup policies to your applications.

Step 5: Protect your Backup & Restore service
: Protect the backup service by backing up the control plane to an S3 bucket. You can restore the backup service in another cluster in the event that this cluster fails.

Explore capabilities

Application resiliency
: Backup applications so they can be recovered in the event of cluster loss, data loss, or corruption. See Backup & Restore overview.

Application mobility
: Move applications between different clusters by utilizing Hub & Spoke cluster configurations. Application backups taken in one cluster can be restored to a separate cluster. See Restoring an application.

Isolate your systems
: Remote S3 backup location separates access between backups and applications. Your backups are not compromised even when a workload cluster is compromised. See Install Backup & Restore.

Orchestrate a backup or restore
: Use a custom, application-specific recipe to consistently automate and control the required backup or restore sequence of an application. See Orchestrate a backup or restore.

Service protection

The IBM Fusion Backup & Restore service protection involves the backup of the Backup & Restore data management service to a S3 bucket.

If the cluster or the storage containing metadata becomes unavailable, you can recover the Backup & Restore metadata required to restore application backups.

You can configure service backups by specifying locations, setting schedules, and managing credentials. Additionally, you can restore a service configuration from a backup, connect to various backup locations (such as Azure, IBM Cloud, and Amazon Web Services), enter credentials, and initiate the restore process.

For more information about how to configure and restore, see Configuring service backups and Recovering service backups.

Backup & Restore of your workloads

The Backup & Restore service is designed to protect a wide range of workloads, including virtual machines, Kubernetes applications, AI workloads, and various IBM Cloud Paks. Additionally, it can also back up the Backup & Restore data management service to an S3 bucket.

Backup of application and VMs: The workload includes the following data:

- PVs of the deployed workloads
- Namespace and cluster scope resources for deployed workloads
- Application VMs that use filesystem based PVCs are backed up and restored.
  Important: To backup application VMs, ensure that the following prerequisites are met:
  - Bare Metal cluster is available
  - OpenShift Virtualization feature is installed from the Operator Hub of OpenShift® Container Platform console.

  Support is available for VMs based on Filesystem PVCs (PersistentVolume has the `volumeMode: Filesystem` defined). The Block mode volumes (`volumeMode: Block`) support is also available.

  Note: The Backup & Restore uses Velero hooks that are present on an application or resource. This means that if your application has hooks, then they are automatically run by Velero at the time of backup or restore.

For more information about OpenShift Container Platform applications, see https://docs.openshift.com/.

- **Backup & Restore of your applications and virtual machines**
  To protect your applications, create storage location, storage policy, and then enroll your applications for backup and restore.
- **Service protection**
  The IBM Fusion Backup & Restore service protection involves the backup of the Backup & Restore data management service to a S3 bucket.
- **Self-service Backup & Restore**
  You can protect your namespace application with IBM Fusion Backup & Restore even as an application user without cluster rights or a user without IBM Fusion administration rights. Application owners can create IBM Fusion Backup & Restore Custom Resource (CRs) within the application's namespace to self-manage their application's backup and restore needs without the involvement of a cluster or IBM Fusion Backup & Restore administrator.

Backup & Restore of your applications and virtual machines

To protect your applications, create storage location, storage policy, and then enroll your applications for backup and restore.

You create backup storage locations and polices as a prerequisite to backup. Assign the policies to applications such that backups run based on the defined policy schedule. Only one backup per policy or application combination allowed at a time. Redundant backups are canceled. IBM Fusion Backup & Restore supports block level incremental backups, which detect change blocks and processes only the changes since the last snapshot. It is also applicable for backups of OpenShift® Container Platform virtualization VMs using block volumes.

- **Backup storage locations**
  You can create backups by assigning a backup policy to an application. The policy specifies how frequently to backup the application, and where to store the backup. The location where the backup is stored is called a backup location and it is a prerequisite to create a policy.
- **Backup policies**
  A backup policy is a set of rules and schedules that define how and when data is backed up. It outlines the frequency of backups, the type of data to be backed up, the retention period, and the storage location. The Backup & Restore service you can create and manage backup policies to ensure your data is protected according to your specific needs.
- **Running on demand backups**
  Backup policies are applied to applications to ensure scheduled backups occur at the specified frequency. However, there are instances where you may want to run a backup outside of the predefined schedule. The Backup & Restore provides you the option to initiate backups on demand.
- **Jobs**
  A job in the Backup & Restore service represents the process of data backup and recovery. You can monitor jobs, manage jobs, and troubleshooting any issues that may arise during the backup and recovery process.
- **Restoring an application**
  You can restore backed-up data to the same project, a different project, or even a different cluster, while specifying the desired target storage classes.
- **Backup and restore as code**
  Make use of the backup and restore services through custom resources (CRs) and integrate these functionalities with the application code. It manages application backups by using a declarative state and maintains your backup posture for an application alongside the application in Git.
- **Backup & restore service from OpenShift Container Platform**
  You can monitor and manage Backup & restore service related operations from the OpenShift Container Platform console.

Backup storage locations

You can create backups by assigning a backup policy to an application. The policy specifies how frequently to backup the application, and where to store the backup. The location where the backup is stored is called a backup location and it is a prerequisite to create a policy.

Local snapshots store the backup on the local cluster, but it is a good practice to specify an external object store as a backup location so that you do not lose backups upon local storage failures. IBM Fusion uses a secure connection to transfer the backups to the object storage provider of your choice.

Supports multiple backup storage locations:

You can backup your Red Hat® OpenShift® applications to local and object storage target locations, such as IBM Cloud®, Microsoft Azure, Amazon Web Services, and any S3 Compliant object storage.

For S3 compliant object storage, ensure that the ACL (access control list) is enabled, and that read permission is granted. Also, ensure that S3 is not in the same cluster or namespace as IBM Fusion Backup & Restore service. You can plan to use cluster services like MinIO or NooBaa but ensure to host them on different clusters. If your backups are in the same place as your primary data, then the data is not protected.
Important: Backup & Restore does not support OpenStack Swift of ANY version.
On the Locations page of the IBM Fusion user interface, you can configure at least one location to store your backed-up data. It supports a variety of object storage types and includes details such as name, status, type, used capacity, associated policies, and applications. You can perform actions like viewing details, editing credentials, and removing locations using the ellipsis menu.

To manage the backup location by using CRs, see Backup and restore as code.

- **Creating a secret**
  You must create a secret for S3 targets that use a self signed certificate before you create a backup storage location.
- **Adding a backup storage location**
  You can add new location from the Backup & restore page of the IBM Fusion or IBM Fusion HCI user interface.
- **Managing backup storage location details**
  You can view location details, update credentials (like access key and secret key for AWS, or account name and account key for Azure), access the YAML configuration, and remove the location record.

Creating a secret

You must create a secret for S3 targets that use a self signed certificate before you create a backup storage location.

Whenever you create a backup storage location for S3 compliant storage, you have the option to specify a certificate to authenticate the connection.

1. Extract the certificate from an S3 compliant service to a file. Use the **openssl** command to extract the certificate into the file **tls.crt**.
   Note: Ensure that the file name must be **tls.crt**.

   ```
   openssl s_client -connect <s3-service-name>-<s3-service-namespace>.apps.<fusion-hostname>.<domainname>:443 -showcerts \
    | sed -n '/BEGIN CERTIFICATE/,/END CERTIFICATE/p' > tls.crt
   ```

   For example, use the **openssl** command to extract the certificate from the **minio** service in the **minio-ns** namespace on a IBM Fusion cluster.

   ```
   echo | openssl s_client -connect minio-minio-ns.apps.myfusionhostname.mydomain:443 2>&1 | sed -ne '/-BEGIN
   CERTIFICATE-/,/-END CERTIFICATE-/p' > tls.crt
   ```

2. Run the **OC** command to create a generic secret in the IBM Fusion namespace using the **tls.crt** file.

   ```
   oc create secret generic <secret-name> --type=opaque --from-file=tls.crt -n <fusion-namespace>
   ```

   For example, in the default IBM Fusion namespace **ibm-spectrum-fusion-ns**.

   ```
   oc create secret generic minio-cert-secret --type=opaque --from-file=tls.crt -n ibm-spectrum-fusion-ns
   ```

   Note: Make a note of this secret name. You need it to create a backup storage location.

Adding a backup storage location

You can add new location from the Backup & restore page of the IBM Fusion or IBM Fusion HCI user interface.

Before you begin

If you plan to use a certificate for a S3 compliant backup storage location, then create a secret with the certificate as a prerequisite. For the procedure to create a secret, see Creating a secret.

About this task

Create the backup storage location in the specified sequence. High-level steps to dd a backup storage location:

1. Go to Backup & restore > Locations, and click Add location.
2. Enter Location name, type of object storage, and credentials for the storage.
3. If applicable, enter the certificate's secret name for S3 compliant storage locations.
4. Click Add.

Note: IBM does not support the creation of two backup storage locations that have both identical endpoint and bucket names.

Procedure

1. Log in to IBM Fusion user interface.
2. From the menu, click Backup & restore > Locations.
3. In the Locations page, click Add location.
   The Add backup location wizard page is displayed.
4. In the Add backup location, enter the Location name.
5. Select the type of object storage backup location. The different location types are Azure (Microsoft Object Storage), IBM Cloud (IBM Object Storage), AWS (Amazon Object Storage), MCG/NooBaa (Red Hat Object Storage), S3Compliant (Any Object Storage), and Storage Protect.
   For the procedure to add MCG/NooBaa location, see Using MCG or Noobaa as a backup storage location. For the procedure to add Storage Protect location, see Using IBM Storage Protect as a backup storage location.
   Note: S3 buckets must not enable expiration policies. For more information about this known issue, see S3 buckets must not enable expiration policies. Also, the bucket must not have an archive rule set.
6. Click Next.
7. In the Login credentials section, enter the following credentials to connect IBM Fusion to your backup location: Endpoint, Bucket, Access key, Secret key. If the location is Azure (Microsoft Object Storage), then enter the Account name and Account key instead of Access key and Secret key. If the location is Amazon AWS, then you must also enter the Region.
   Example for AWS endpoint:

   ```
   https://s3.us-west-1.amazonaws.com
   ```

8. In the Certificate settings (optional) section, enter the Secret name for the certificate.
   Note:
   - This setting is applicable only when you create an S3 compliant backup storage location type.
   - The endpoint URL must be an HTTPS protocol with a trusted connection. If the endpoint URL contains the HTTPS protocol, then you need to enter the name of the secret that contains the SSL certificate.
   - If you did not create a secret before you create the backup storage location, then you cannot complete further steps. Ensure that you cancel the operation and go back to the create secret step. For the procedure to create a secret, see Creating a secret.
   - If you plan to use a S3 location, check whether your permissions are valid with the cloud provider for that particular endpoint and bucket or equivalent.
     For certificates, run the following **openssl** command to check whether a Subject Alternative Name (SAN) exists:

     ```
     openssl x509 -in <filepath> -text
     ```

     The output displays a SAN field. This SAN field must match the endpoint host of the S3 bucket.

- Do not use "wildcard + self-signed certificate" as it is a major security risk. If installed system-wide, then on exposure, all encrypted communication can be decrypted.

Warning: For security reasons, HTTP connections to S3 backup storage locations are disabled by default. Enable HTTP access only in non-production environments. IBM strongly recommends avoiding this option whenever possible. To allow insecure connections, set the `allowInsecureS3` property to `true` in the `DataProtectionAgent` custom resource.

For example:

```
apiVersion: dataprotectionagent.idp.ibm.com/v1
kind: DataProtectionAgent
metadata:
  name: dpagent
  namespace: ibm-backup-restore
spec:
  transactionManager:
    allowInsecureS3: 'true'
```

9. Click Add.
   A success message gets displayed after adding the location.

- **Using MCG or Noobaa as a backup storage location**
  Create an MCG or Noobaa S3 target backup storage location.
- **Using IBM Storage Protect as a backup storage location**
  Backup OpenShift Container Platform applications to an IBM Storage Protect server and optionally make a copy of the data on tape.

Using MCG or Noobaa as a backup storage location

Create an MCG or Noobaa S3 target backup storage location.

Procedure

1. Log in to Red Hat® OpenShift® Container Platform.
2. Go to Storage > Object Bucket Claims.
3. In the Create ObjectBucketClaim page, create an Object Bucket claim with the following details: capture credential details.

   ObjectBucketClaim Name
   : Enter the name of the Object Bucket Claim. If you do not enter a value, a generic name gets generated.

   StorageClass
   : Select `openshift-storage.noobaa.io` class. It defines the object storage service and the bucket provision.

   BucketClass
   : Select `noobaa-default-bucket-class`.

4. Click Create.
5. Note down the Object Bucket Claim credentials like Endpoint, Bucket Name, Access Key, Secret key.
6. Extract the certificate from an S3 compliant service to a file. Use the `openssl` command to extract the certificate into the file `tls.crt`.
   Note: Ensure that the file name must be `tls.crt`.

   ```
   openssl s_client -connect <s3-service-name>-<s3-service-namespace>.apps.<fusion-hostname>.<domainname>:443 -showcerts \ |
   sed -n '/BEGIN CERTIFICATE/,/END CERTIFICATE/p' > tls.crt
   ```

   For example, use the `openssl`command to extract the certificate from the `MCG/Noobaa` service.

   ```
   export s3_url=$(oc get routes.route.openshift.io -n openshift-storage s3 -o jsonpath='{.spec.host}'):443
   echo $s3_url
   openssl s_client -connect $s3_url -showcerts \ | sed -n '/BEGIN CERTIFICATE/,/END CERTIFICATE/p' > tls.crt
   ```

7. Run the **OC** command to create a generic secret in the IBM Fusion namespace by using the `tls.crt` file.

   ```
   oc create secret generic <secret-name> --type=opaque --from-file=tls.crt -n <fusion-namespace>
   ```

   For example, in the default IBM Fusion namespace `ibm-spectrum-fusion-ns`:

   ```
   oc create secret generic bsl-cert --type=opaque --from-file=tls.crt -n ibm-spectrum-fusion-ns
   ```

8. Log in to IBM Fusion user interface.
9. From the menu, click Backup & restore > Locations.
10. In the Locations page, click Add location.
    The Add a backup location wizard page is displayed.
11. Enter the Login credentials and Certificate settings to create a backup storage location.
    In the Secret Name for Certificate field, enter the secret name created in step 7.
12. After the backup storage location is created, go to the Locations page and check whether the status is Connected.
13. If you want to create a S3 of NooBaa type, do the following steps:
    You can add the location as a separate NooBaa type instead of the S3, but you cannot do it from the UI.

    It requires additional steps to create the credentials secret and BSL YAML.
    a. Create a secret using the `AWS_ACCESS_KEY_ID` and `AWS_SECRET_ACCESS_KEY` from NooBaa.

    ```
    kind: Secret
    apiVersion: v1
    metadata:
      name: <noobaa bsl secret>
    ```

```
  namespace: <fusion namespace>
  labels:
    dp.isf.ibm.com/ownedBy: fbsl
    dp.isf.ibm.com/provider-name: isf-backup-restore
data:
  access-key-id: <base64-encoded AWS_ACCESS_KEY_ID>
  secret-access-key: <base64-encoded AWS_SECRET_ACCESS_KEY>
type: Opaque
```

If a YAML is used, encode the values using base64.

b. Copy the location of the S3 route to use as the endpoint for the BSL CR. Use noobaa CLI to get the list of buckets to choose from:

```
noobaa bucket list
```

c. Create the BSL CR with type `noobaa`:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: BackupStorageLocation
metadata:
  name: <noobaa bsl>
  namespace: <fusion namespace>
  labels:
    dp.isf.ibm.com/provider-name: isf-backup-restore
spec:
  credentialName: <noobaa bsl secret>
  params:
    bucket: <bucket name>
    endpoint: '<noobaa s3 route>'
  privateCertificateName: <optional noobaa certificate secret>
  provider: isf-backup-restore
  type: noobaa
```

Using IBM Storage Protect as a backup storage location

Backup OpenShift® Container Platform applications to an IBM Storage Protect server and optionally make a copy of the data on tape.

Before you begin

Ensure that you have IBM Storage Protect 8.1.18 server or higher.
Before you implement this solution, see IBM Storage Protect documentation to understand expected recovery time of data from tape based on the total amount of managed data on the tape.

About this task

If the IBM Storage Protect server already has an object agent defined, you must recreate the object agent to ensure that the public certificate is generated correctly. To recreate the object agent, you must do the following steps to remove the current object agent:

1. From the IBM Storage Protect server, issue the command `delete server <object_agent_name>`.
2. Uninstall the IBM Storage Protect object agent on the system by using the ibmspecified command from the `delete server` output.
3. Delete the object agent directory under the server instance directory on the machine that hosts the IBM Storage Protect installation.
4. Proceed with the steps in the procedure to define the IBM Storage Protect object agent.

Procedure

1. Configure the object agent service on the IBM Storage Protect server.
   a. Run the following command on IBM Storage Protect server:

   ```
   setopt objectagentsancertificate yes
   ```

   Note:
   - If you are using a self-signed certificate, then you need this step to allow the object agent to create a self-signed certificate with the Subject Alternate Name (SAN).
   - If you are using a CA-signed certificate, then this step is not required.

   b. Create the object agent service by using the `DEFINE SERVER` command.
   For details about the command, see IBM Storage Protect 8.1.18 documentation.
   Note: If you are using a self-signed certificate, then you must ibmspecify the HLAddress in dotted decimal format.
   c. Configure an object client as documented in IBM Storage Protect documentation.
   d. Create an object storage bucket. You can create a bucket either by using the S3 API or by using the following MinIO client command:
   `mc mb`
2. Configure a backup location from the IBM Fusion interface.
   a. Location name is any name of your choice.
   b. Location type is S3 Compliant.
   c. Endpoint is the URL using the dotted IP address of the object agent client, which was configured on the IBM Storage Protect server.
   d. Bucket is the bucket that was configured on the IBM Storage Protect server.
   e. Access key and Secret key are the keys that were generated when the object client was created on the IBM Storage Protect server.
   f. Secret name for certificate to create a secret to store the IBM Storage Protect public certificate as documented in the Adding a location or Adding a backup storage location.

   After the backup location is created, create a backup policy using the new backup location and assign one or more applications to the backup policy.
3. Copy data to and from tape:
   a. Optionally, copy IBM ibmspectrum Protect data to tape:

```
PROTECT STGPOOL
          Type=Local
```

You can schedule this command to run periodically.

b. To recover data that is no longer available in the primary storage pool, use the `REPAIR STGPOOL` command to recover the data from tape to the primary storage pool. For more information about the command, see IBM Storage Protect documentation.
c. Initiate the restore request from IBM Fusion.

Managing backup storage location details

You can view location details, update credentials (like access key and secret key for AWS, or account name and account key for Azure), access the YAML configuration, and remove the location record.

View location details

From the ellipsis overflow menu of the location record, click Details to open the <Location name> slide out pane.

- Expand the Policies and Applications to see a list of associated polices and applications. The number that is mentioned within parenthesis for Policies and Applications indicate the total number of associated polices and applications respectively.
- Click the edit icon to update Access key and Secret key. For Azure, the values are Account name and Account key.
- Click the launch YAML icon to view the YAML.

The details about the location vary based on the type. For example, Amazon AWS includes the following details: Type, Status, Used capacity, Endpoint, Bucket, Region, and associated Polices and Applications. The Region is only applicable for AWS locations.

Add location

Click Add Location to add a new location. For more information about the procedure, see Adding a backup storage location.

Edit location details

From the ellipsis overflow menu, click Edit to update the values of Access key and Secret key. For Azure, the values are Account name and Account key.

Alternatively, you can also select the location that you want to edit and click the edit icon in the side pane to modify the location details.
Note: You cannot update bucket details except credentials because two backup storage locations with different names can share the same credentials and endpoint.

Remove a location

From the ellipsis overflow menu of a location record, click Remove to delete the location record. Click Remove in the Confirm remove window. This action removes the location as an available backup location from IBM Fusion. However, you can still access this backup location through your storage service provider.

Backup policies

A backup policy is a set of rules and schedules that define how and when data is backed up. It outlines the frequency of backups, the type of data to be backed up, the retention period, and the storage location. The Backup & Restore service you can create and manage backup policies to ensure your data is protected according to your specific needs.

The Policies page lists all policies. You can search for policy records based on the backup location or other keywords, configure the table display, and perform actions like viewing, editing, or deleting policies from the ellipsis overflow menu.

- **Creating backup policy**
  You can add new backup policies from the Backup page of the user interface.
- **Managing backup policy**
  Click Manage policies to manage policies for an application and initiate backup now.

Creating backup policy

You can add new backup policies from the Backup page of the user interface.

Before you begin

As a prerequisite to create a backup policy with object storage, you must first add a backup storage location.

For more information about configuring the target backup storage Location, see Backup storage locations.

About this task

To manage the backup polices and backup location by using CRs, see Backup and restore as code.

High-level steps to create a policy:

1. In the Policies page, click Add policy to open the Create a backup policy panel.
2. Enter the policy name and select the desired frequency (Hourly, Daily, Weekly, Monthly, or Custom) with specific times and days.
3. Select between In place snapshot or Object storage for the backup location.
4. Enter the retention period.
5. Click Create policy.

Procedure

1. Log in to IBM Fusion user interface.
2. From the menu, click Backup & Restore > Policies.
3. Click Add policy.
   The Create a backup policy slide out pane displays on the screen.
4. In the Create a backup policy pane, enter the following details to create a new backup policy that meets your business needs:
   a. Enter the Policy name.
   b. Enter the Frequency details:
      i. Enter whether the frequency is Hourly, Daily, Weekly, Monthly, or Custom:

         Hourly
            Choose a time.

         Daily
            Choose a time.

         Weekly
            Select a day for Schedule and Choose a time.

         Monthly
            Choose a day from calendar and time.

         Custom
            Choose a specific day and time.

      ii. If you select frequency intervals of 1 day or more, the Time window section is visible on the screen for you to select a start time, end time, and time zone.
         Note: Consider the following points when you enter details in the Time Window:
         - The minimum time window is 4 hours from start to end time. An error message gets displayed whenever the time window is less than 4 hours.
         - When you change the start time, the end time is automatically adjusted to reflect the new start time plus the current time window duration.
         - When you change the end time, the duration between the start and end time is adjusted automatically, while the start time remains unchanged.
         - If the start and end times are identical, the duration is considered a 24-hour time window.
   c. Select either In place snapshot or Object storage from Backup Locations.

      In place snapshot
         For In place snapshot policies, the resource backup information is kept locally on the cluster in the backup and restore install namespace of the MinIO storage. The VolumeSnapshots and VolumeSnapshotContent objects are kept in the application namespace. The VolumeSnapshotContent objects that are not namespaced get stored at cluster scope. The restore job fails whenever you delete the VolumeSnapshots or VolumeSnapshotContent objects. At expiration time, the resource backup in MinIO, VolumeSnapshots, and VolumeSnapshotContents get deleted. The deletion of the used StorageClasses or VolumeSnapshotClass objects, which are used in backup and restore also causes the expiration to fail.
         Note: Capacity for in place snapshots is not available.

      Object storage
         Using the Object Storage option, you to choose from a list of available object storage. You can also use the Search option to filter the location records.

   d. Enter the retention period. It is the duration in which the backup copies exist in the storage location. Choose a value for its unit that can be days, weeks, months, or years.
      For example, 30 Days.
5. Click Create policy.
   A message gets displayed informing you about the successful creation of a new backup policy along with the policy name.

Managing backup policy

Click Manage policies to manage policies for an application and initiate backup now.

Procedure

1. Go to Backup & Restore > Backedup applications page.
2. Select one or more applications from the list.

Option	Description
Single application	a. Select an application. b. From the ellipsis overflow menu, select Manage policies. A Manage policies page is displayed with a list of all policies in the local cluster.

Multiple applications	a. Select multiple applications.
	b. Click Manage policies.
	A Manage policies page is displayed with a list of all policies in the local cluster. The following notification message is displayed:
	`Mixed policies`
	`the application selected do not share the same set of policies.`

3. To remove all policies, do the following steps:
   a. In the Manage policies page, click the Remove all policies toggle button.

    b. If you set the Remove all policies toggle button to on, then the policies table is disabled.
    c. In the Remove all assigned policies confirmation window, click OK to proceed.
4. Click Backup now to initiate an immediate backup of one or more applications.
5. Select one or more policies and click Save.

Setting up backup policies

After you create a policy, you can assign them to applications so that application data is backed up in the specified frequency and stored in the defined storage, such as object storage or in place snapshot. You can set up backups for applications of the local cluster from the Applications page. To setup backup for applications of both hub and spoke clusters, go to Backed up applications page.

Set up policies from the Applications page

Before you begin

For backup and restore to work, you must create volume snapshot class.

About this task

The high-level steps to set policies for applications are as follows:

1. Go to the Applications page.
2. Select one or more applications for which you want to assign a backup policy.
3. Assign one or more policies for the selected applications, and click Assign. You can also initiate an immediate backup.

Procedure

1. Go to Applications from the menu.
   For applications that do not have backup policy set, the Backup status columns of these records show No policy. If a policy is assigned, the name of the policy is displayed in the Policies column. If more than one backup policy is assigned to the application, the number of policies is displayed.
2. Select the check box from the Name column or click ellipsis menu and select Assign backup policy in the Policies column to assign a backup policy to a single application with no policy assigned.
   Note: If there are no policies assigned before, select the policy provider either Backup & Restore or Backup & Restore (legacy).
   To assign a policy to multiple applications, select the checkboxes in the Name column for the applications to which you want to assign the policy, and click Assign policy. The Assign policies window gets displayed and click Continue.
   The Assign backup policy slide out pane gets displayed.
3. In the Assign backup policy slide out pane, the Backup & Restore is selected by default as a policy provider and you can assign one or more backup policies to the selected application.
   The details of all available backup policies are displayed, such as location, locationtype, frequency, retention, and time.
4. Select Run backup now check box.
5. Click Assign to assign a backup policy.
   In the Applications page, you can observe that the Backup status of the selected application shows as Not backed up for applications that have not got backed up yet. After the backup set up is done, the Backup status of the selected application changes to Completed with a green check mark. Also, the Last backup time shows the backup time.
   Now you can check the status of your job from the Jobs tab.

Set up policies from Backed up applications page

Before you begin

As a prerequisite, you must have created policies.

- Ensure one or more policies exist.
- If you want the application VMs to be backed up and restored, then you need to ensure that the following prerequisites are met:
    - Bare Metal cluster is available
    - OpenShift Virtualization feature is installed from the Operator Hub of OpenShift® Container Platform console.

  Note: Support is available VMs based on Filesystem PVCs (PersistentVolume has the `volumeMode: Filesystem` defined). From 2.8 release, the Block mode volumes (`volumeMode: Block`) support is available as a technical preview.
  Recipes for VMs are not supported.

Ensure you are aware of the Backup & Restore service related issues before you protect your applications. For more information about issues and how to resolve them, see Troubleshooting Backup & Restore service issues.

Procedure

1. Go to Backup & Restore > Backed up applications.
2. Click Protect apps.
   The Protect applications wizard page gets displayed.
3. In the Select applications page, select a Hub or Spoke cluster on which you want to backup your applications.
   After you select the cluster, the number of unprotected applications on the cluster gets displayed. The applications table populates only unprotected applications and does not display applications that are already protected.
4. Select your application record(s). You can also select the entire list all together.
5. Click Next.
   The Assign policies wizard page gets displayed. The table includes Name, Location, Type, Frequency, Time, Retention, and Backup now column headings. The Type indicates the supported providers, such as Microsoft Azure, IBM Cloud®, S3 compliant object storage, Amazon Web Services, and Storage protect.
6. Select policies to assign to the applications. You can select a single to all policies for assignment.
7. Optionally, you can turn on the Backup now toggle for a record to immediately start the backup.

8. Click Assign.
   After the validation is successful, a success notification message is displayed. You can go to the Backed up applications page and search the applications table for the newly selected applications.

Managing your application workloads

Applications contain resources, such as microservices, databases, and other stateful data distributed across projects. IBM Fusion provides ways to do application-centric backup, which provides data loss protection for your applications.

Note: If IBM Storage Scale is used for storage, it must have a minimum of 5G PVC size to perform backup and restore operations.
Note: The Applications page shows only local cluster applications or remote DR applications. Remote Backup & Restore Spoke applications do not show up on the Hub cluster. To view Spoke applications, go to Backup & Restore > Backed up applications > Protect apps view.

Application list

The Applications page lists all your OpenShift® applications with their backup details, such as Name, Used, Capacity, DR status, Backup status, Last backup on, Success rate, and Policies.

Table 1. Application list

Application list	The Applications page lists all your OpenShift applications with their backup details, such as Name, Used, Capacity, Backup status, Last backup on, and Policies. Note: In the Applications page, if you continue to see applications that got deleted from OpenShift, check whether associated backups are available. If you no longer need to restore those backups, delete the Application CR instance of that application from the CRD.
Actions	View details Use the Search field to filter and find your record from the list. You can use the settings icon to customize the headings. To view the application details, click the Name link of the record or click View details in the ellipsis menu. Assign backup policy To assign backup policy, see Assign backup policy. Backup now To start the backup of an application right away, click Backup now in the ellipsis menu of the application record. Restore Restore the application backup. For the steps to restore an application resource, see Restoring an application.

Assign backup policy

Select applications with no policy assigned:

1. Select one or more application(s) and click Assign backup policy. Alternatively, for an application, click Assign backup policy from the ellipsis menu.
   The Assign a backup policy slide out pane gets displayed.
2. Select one or more backup policies. You can select Run backup now to initiate a backup. If you do not see any policy that suits your requirement, create a new policy. For the procedure to, create a new policy, see Managing backup policies for application workloads.
3. Click Assign .

Select multiple applications having a mix of policies:

1. Select one or more application(s) and click Assign backup policy. Alternatively, for an application, click Assign backup policy from the ellipsis menu.
   The Assign a backup policy slide out pane gets displayed.
2. In the Policy provider section, select which service you want to use as a policy provider.
3. Select one or more backup policies. You can select Run backup now to initiate a backup. If you do not see any policy that suits your requirement, create a new policy. For the procedure to, create a new policy, see Managing backup policies for application workloads.
4. Click Assign .
5. In the Confirm assignment confirmation page, go through the information about the number of applications that cannot be assigned as they use policies that are provided by another backup service.
6. Click OK.

Managing the application details

The application details are classified into Overview, Storage, Backups, and Resources tabs.

You can click the Assign a policy to open the Assign a backup policy page. For more information about Assign a backup policy page, see Setting up backup policies.

In addition, you can also click Assign policy to open the Assign a backup policy page. For more information about Assign a backup policy page, see Setting up backup policies.

Overview tab
The Overview tab includes the following details:

Table 1. Overview tab details

Element	Description

Element	Description
Storage	The Storage section provides a summary of how this application is utilizing storage resources. It represents overall capacity and usage of the PVCs in a line chart. The capacity usage of storage classes is represented as a pie chart. For example, 1200 GiB is the overall capacity and 300 GiB is the used capacity.
Events	The Events section includes critical and recent events that occurred on the application. Click View all to see all the events that are associated to this application.
Backups	The Backups section provides a summary of backup jobs. The percentage of successful backups, timestamp of the recent successful backup, and total number of available backups and backup policies. The storage capacity usage from a policy perspective is represented as a pie chart. For example, there are 3 policies and 600 GiB overall capacity. The "check_app" policy usage is 400 GiB, the "object_policy" usage is 150 GiB, and the "client26" sewage is 50 GiB.
Inventory	The Inventory section displays a clickable list of storage and resources. The number within parenthesis for Storage and Resources headings indicate the total number of available storage and resources respectively.

Storage tab

The Storage tab includes the following details:

Table 2. Storage tab details

Element	Description
Capacity usage by PVCs	Capacity usage by PVCs section displays the current application capacity usage by PVCs.
Capacity by storage class	Capacity by storage class section represents in a pie chart the total capacity of the storage class and the amount of used storage in GiB.
Storage classes	Storage classes section individually lists all the storage classes along with the total number of PVCs, total capacity of the PVC, and the total used capacity in GiB. The number that is specified within parentheses is the total number of storage classes. Note: A Persistent Volume (PV) storage is dynamically provisioned by using the storage classes. For more information about how to create storage classes, see Dynamic provisioning of Persistent Volume in filesystem.
Persistent Volume Claims	The Persistent Volume Claims section lists PVCs distributed across the associated storage classes along with their used and total capacities. Note: The PVC is a storage request and it uses the PV resources. It also does claim checks to these resources. The following details of the PVCs are available in the list: • The Name of the storage class. • The PVC status of the storage class. The values are `Bound` and `Unbound`. • The Used and Capacity indicates the used storage and the total capacity in GiB. • The Storage class and PV indicates the associated storage class and PV. Use the Search text box to filter the records in the list. You can filter based on PVC status and storage class parameters. You can further refine your search based on the different values of these parameters. For example, you can filter PVC status based on Bound, unbound, or All. You can also use the settings icon to choose the columns to display. Click Reset to default if you want the original system settings.

Backups tab

The Backups tab includes the following details:

Note: If a backup is in progress for the application, then a message is displayed with details about the percentage completion of the backup job.

Table 3. Backups tab details

Element	Description
Backups list	The Backups section lists all backup jobs. By default, the following details about a backup are available in the list: • The Time of backup. Click the Time link of a record to view details. You can sort records based on Time. The most recent • The Status of the backup. The various statuses are namely, Snapshot in progress, In progress, Completed, Failed, PartiallyFailed, Failed Validation. • The Policy used. • The Location where it is backed up. • The Size (GiB). Additionally, you can select following details available under Edit columns section by clicking the setting icon. • The Location. • The Duration. • The End time.
Search	Use the Search text box to filter the records in the list. You can also use the settings icon to customize the column display. Click Reset to default if you want the original system settings. For advanced search on the records, click the filter icon and select values for Policy, Status, or Location from the drop-down lists. After selecting the values, click Done. Click Clear filters to clear the selection made in these drop-down lists.
Usage section	The Usage section provides total number of available backups and used capacity.
Backup policies section	For individual associated policies, it provides Retention, Used Capacity, Last backup, and Location. You can use the ellipsis overflow menu to view details, edit, or remove a policy association from the application. If you want to assign policies from this section, click Assign. For the procedure to assign, see Setting up backup policies.

Element	Description
Actions from the ellipsis overflow menu of a backed up record	The ellipsis menu of your backed up record includes the following options: View details and Delete backup. View details The details slide out pane is displayed: • The Backup details section contains the time, status, policy, location, and name of the application. If the backup is available, then the status is Available.The Size and Expiration information is added to the Backup details section. Click (Launch YAML icon) to open and view the YAML. Click (Restore icon) to restore the backed up data. If the restore is in progress, then the Restore icon is disabled. • The Job details section contains the Job ID and Elapsed time. • The Inventory, Backup sequence, and Data transfer phases are displayed for monitoring. The backup's inline notification is displayed. If an error occurs in another of the statuses, then click View job details to know more information about the job. Delete backup Click Delete backup to delete the backup record.

Resources tab

The Resources tab lists all the included resources of the selected application in a table format with Name, Type, and Label headers.

Use the Search text box to filter the records in the list. For advanced search, click the filter icon and select a value from the Type drop down to list all resources that belong only to a selected type.

You can also use the settings icon to choose the columns to display. Click Reset to default if you want the original system settings.

Actions that you can perform on an application details page

Click Actions and select any of the following operations:

Table 4. Actions available in a application details page

Manage backup policies	Click Manage backup policies to select one or more backup policies for data protection. In the Assign policy slide out pane, click the Frequency and select run backup now to do an immediate backup. It overrides the frequency set in the policy and starts the on-demand backup job immediately. However, the scheduled backups continue to run as set in the associated policy. After you select, click Done to complete the operation. If the listed policies do not match your requirement, then create a new policy from the Backup Policies page. A message is displayed to inform the success of the policy assignment operation. Note: If you have not assigned a policy to this application, the Actions and Restore buttons are not visible. Instead Assign policy button is available.
Backup now	Click Backup now to backup the application data. In the Backup now slide out pane, verify the application, policy, backed up data location, and retention period. If there is more than one policy associated to the application, then select one or more policies and click Run <n> backup policies. Here, `n` indicates the number of policies that are selected to run backup. In the confirmation window, click Run backup policy to initiate the backup job.
Restore	Click Restore to restore the application backup. A Restore Application <application name> slide out pane gets displayed. For more information about how to work with the slide out pane, see Restoring an application.
Failover This action is available only in a Metro-DR setup.	Click Failover to register this application for disaster recovery. For more information about failover steps, see *Failover* section in Managing your application workloads. For more information about disaster recovery, see Upgrade and upsize considerations in Metro-DR. A message gets displayed confirming that the application is enabled for disaster recovery. Note: If the two sites of the Metro sync DR setup are in network disconnected state, then before you start the failover or failback operation on the surviving site, scale down the Ramen DR. To get the name of surviving site, run **mmlsmgr** command. After network issues are resolved, scale up the Ramen DR.
Remove disaster recovery This action is available only in a Metro-DR setup.	After an application is enabled for disaster recovery, the Actions shows Remove disaster recovery option. If you no longer need disaster recovery (fail back) for an application, click Remove disaster recovery to disable the protection for the application. In the confirmation message box, click Disable to proceed with the disable operation. In the Remove disaster recovery window, click Remove.

Note: The Application CR remains as long as there are PolicyAssignments or Backups associated with the application. The system checks for both, and the Application CR is retained until neither exists.To completely remove the application, you must delete all existing Backups and PolicyAssignments, and then delete the namespace.

Running on demand backups

Backup policies are applied to applications to ensure scheduled backups occur at the specified frequency. However, there are instances where you may want to run a backup outside of the predefined schedule. The Backup & Restore provides you the option to initiate backups on demand.

Before you begin

- When you restore an application from Hub to Spoke or Spoke to Hub cluster, ensure that the same StorageClass exists in both Hub and Spoke.
- In Backup & Restore, one hub cluster and more than one spoke clusters may be connected to easily manage application backups in multi-clusters. For various reasons, the OpenShift® Container Platform cluster may have problems and become unusable. If the cluster recovers after the expiry of the client cert, you must clean and setup to rejoin the recovered cluster to the Hub or Spoke. For the procedure to rejoin, see Reconnecting OpenShift Container Platform cluster.

About this task

- You can backup one policy or more policies.
- The label `velero.io/exclude-from-backup=true` is not supported by the Backup & Restore service.

Procedure

1. Go to Backup & Restore > Backedup applications page.
2. Select one or more applications from the list.
   Note: Applications using remote Global Data Platform storage are not supported for Backup & Restore.

Option	Description
Single application	a. Click the Backup now option from the ellipsis overflow menu of an application record. b. Click Backup in the Backup now confirmation window to start the backup operation of the application immediately. The `Initiating backup Starting backup on [app name]` notification is displayed.
Multiple applications	a. Select multiple applications and then click the global Backup now. When different policies are assigned for multiple records, then the Backup now window is displayed. b. Select the backup policies that are used to backup the selected applications. c. Click Backup. The Backup now confirmation window appears with a notification about the availability of multiple policies. d. In the Backup now confirmation window, you can optionally select Exclude applications with multiple policies. e. Click Backup in the Backup now confirmation window to start the backup operation of the application immediately. The `Initiating backup on [n] applications` notification is displayed. In the Backedup applications page, you can observe that the Backup status of the selected applications are in progress.

Jobs

A job in the Backup & Restore service represents the process of data backup and recovery. You can monitor jobs, manage jobs, and troubleshooting any issues that may arise during the backup and recovery process.

View backup and restore jobs

The Jobs page of the IBM Fusion user interface the backup and restore jobs are segregated in their respective tab pages.

In the Backups tab page, you can monitor the statuses of all backup jobs, which can be categorized as failed, pending, in-progress, canceled, or completed. You can sort these jobs based on the Started field. The Backups table provides a comprehensive overview of each job, including its name, associated cluster, application, status, time, policies, and recipes utilized. By clicking on the job name, you can access detailed information about each backup job. Similarly, you can view the list of all restore jobs in the Restore tab page.

In the job details, you can manage and track your backup and restore jobs, cancel ongoing jobs, rerun jobs, adjust job resources, and collect logs. It also enables you to quickly access detailed information about any specific job to investigate further or troubleshoot any issues.

In the Summary section, you can view the start time and duration of the job. You can toggle between the List view or Log view.

Within the Summary section, the backup jobs follow the Inventory, Backup sequence, and Data transfer order. The restore jobs follows the Inventory, Data transfer, and Restore sequence order.

Inventory

: In the Workloads section, you can see the number of deployments, deploymentConfigs, stateful sets, and pods.
In the Storage section, you can see the number of PVCs attached.

: In the Resources section, you can see the number of routes, secrets, services, and configMaps.

: You can also download the resource list to your local.

Backup sequence

: You can see the status of pre-hook, backing up PVCs, post-hook, and backing up etcd resources.

Data transfer

: You can view the backup location, backup size, upload speed, upload status, and the upload progress percentage.
Note: You can view the data transfer details only for the jobs such as backups to S3 object and restore from S3 object.

From the Actions menu, you can click View YAML to view the CR's YAML in the OpenShift® console.

Cancel jobs

In the Jobs page, click Cancel jobs to cancel an ongoing job or a job in the queue. Alternatively, you can cancel jobs by using the Backup or Restore CR. Add the following line to the CR specification to cancel jobs that are in queue or running state.

```
spec:
  jobControl: cancel
```

If you cancel during backup, the volume snapshots get removed as a part of cleanup.

If you issue a cancel during restore, the cleanup happens only when you choose a new or an existing namespace.
Note: When you cancel a restore job to the original namespace, cleanup does not happen and can result in a corrupt application after the cancellation completes.

Scale jobs

You can scale out the number of transaction manager pods to process more jobs at the same time. For example, run the following command to increase the number of agent pods to two:

```
oc scale deployments/transaction-manager --replicas=2 -n ibm-backup-restore
```

To the scale out from the OpenShift Container Platform console, do the following steps:

1. Change the project to the Backup & Restore install namespace. The default is **`ibm-backup-restore`**.
2. Search and select the **`transaction-manager`** deployment.
3. Click the scale up or down to increase or decrease the scale. It takes about 30 seconds for a replica to start in optimal conditions. Do not click multiple times instead wait for each pod to get to ready status.

Re-run jobs

From the Actions menu, select Run job again to restart a previously completed or failed job.

Download logs

Download the log package for the current cluster and optionally any spoke clusters. Select the Collect service logs option in the Actions menu and choose your spoke clusters and click Create. The logs provide a high-level overview of the status of the job, including start and end times, success or failure, and any error messages. They are useful for quickly checking the overall status of a job and identifying any immediate issues.

Restoring an application

You can restore backed-up data to the same project, a different project, or even a different cluster, while specifying the desired target storage classes.

Before you begin

- When you restore an application from the Hub cluster to a Spoke cluster, manually check whether the same storageclass that is used to define the PVCs on the Hub exists on the Spoke cluster.
- By default, if PVCs with the same name exist in the namespace, they get restored with a new name (original name + “-n” suffix with n=1,2,3). If you want to skip the restore of existing PVCs, manually create a Restore CR from the command line and set an optional **`spec`** field **`skipExistingPVC`** to true.
- You must have OpenShift® APIs for Data Protection (OADP) version 1.3.1.
- For the Datamover operator to restore CephFS and Global Data Platform 5.2.0+ snapshots by using the ReadOnlyMany access mode, ensure that the value of backingSnasphot is `true` in the storageclass.
- Manually modify the restore when the original application already exists on the cluster and creates conflicts with each other.

About this task

- You can use the following restore options for your applications:
  - From S3 to the same cluster and project
  - From S3 to the same cluster and existing project
  - From S3 to the same cluster and new project
  - From S3 to a different cluster
  - From S3 to IBM Software Hub.
  - Restore from snapshot
- Backups that are taken using an In place snapshot can be restored only to the same project, and not to any new or a different project that exists. If you try to restore a project that is deleted from Red Hat® OpenShift, then the restore to the same project with In place snapshot does not include PVCs.

- After you remove an Agent or Spoke connection from the Hub cluster, it does not show up in the Topology page. Also, you can restore the available (not expired) successful application backups with an object storage location from that cluster to a different cluster.
- When you do a restore operation, the restored files, directories, and mount points can have different permissions, group, and owner from the source that was backed up; it is an expected behavior of Red Hat OpenShift design. For more information, see OpenShift documentation and https://access.redhat.com/solutions/7007252. You can modify this behavior by using security context constraints (SCCs), parameters such as fsGroup, or a custom script or command such as **chmod** or **chown**.
- IBM Cloud Paks instance backups cannot be created when continuous backup is enabled on the MongoDB service.
- You can restore backup data to a different StorageClass provisioner on an alternate cluster, for example, restore data that was originally backed-up from a CephFS provisioned StorageClass to a Scale provisioned StorageClass.

Procedure

1. In the menu, click Applications.
2. In the Applications page, search and open your application that you want to restore.
3. In the application details page, go to the Backups tab and from the ellipsis overflow menu of the backup record, click Restore option.
   Alternatively, go to the Backed up applications page and select Restore option from the ellipsis overflow menu of an application record.
   A Restore Application <application name> slide out window gets displayed.
4. In the Restore Application <application name>, select a Cluster destination where you want the application to be restored.
5. Select a Project destination for the application data.

   Single namespace to the same or new cluster
      For a single namespace, you can choose from any of the following options:
      Note: For the single namespace to same or new cluster, the option Choose existing project is only available for same cluster restores.
      - If you select the Use same project option, then click Next and proceed.
      - If you select Choose existing project option, then select a Project from the list and click Next to proceed.
      - If you select Create a new project option, then enter the name of the project and click Next to proceed.

   Multi-namespace to the same or new cluster
      For multiple namespaces, you can choose from any of the following options:

    - If you select the Use same project option, then click Next and proceed.
    - If you select Create a new project option, then select Project suffix and click Next to proceed.
      The Project suffix is used to avoid name conflicts by associating it with a specific namespace. For example, if you select enter `123` for Project suffix, then in the preview you can see `<Project name>-one-123`. You can also choose different combinations of available suffixes in the Suffix preview.

   The details in the Summary section vary based on your selections.
6. If you have not selected the backups, then select the backups that you want to restore and click Next.
7. If you do not want to use the existing Target storage class in your cluster to restore the PVCs, then select Change target storage class and enter a storage class of the same type to change it.
   By default, restore occurs on a matching storage class name. If **`target storage class`** is specified, then restore happens on PVCs from this class. If the original storage class is non-existent or **`target storage class`** is not provided, the default storage class of the cluster is used.
   Example scenarios where there may be a need to change the Target storage class are as follows:
    - If there is a requirement for a filesystem PVC with RWX access mode that a particular storage class (such as RBD) does not support, consider using an alternative storage class that meets the requirement.
    - An application may require a storage class with specific attributes, such as a particular compression or encryption setting.

   The restore does not overwrite existing PVCs instead new PVCs are created with an annotated name. Modify your application deployment to use the new PVCs.
8. In the Missing etcd resources section, select Include missing etcd resources to restore non-PVC application resources that are included in this backup but do not exist in this Red Hat OpenShift project.
   Note: If you do not select the Missing etcd resources, the application-specific pods that need deployment resources or jobs may not start.
9. If you select Only restore a subset of selected PVCs, then select all PVCs or choose individual PVCs from the selected backup to restore.
10. Check the Summary and confirm the changes after restore.
11. Click Restore.
    A message appears informing that the restore of the application from its backup is in progress. The status of the backup record in the Backups tab is initially set to Restoring and then it changes to Completed.
12. Go to the application to confirm the success of the restore operation.

What to do next

Storage classes that are used by applications on the source cluster must exist or be created on the target cluster before you attempt to restore the applications that use those classes.

Backup and restore as code

Make use of the backup and restore services through custom resources (CRs) and integrate these functionalities with the application code. It manages application backups by using a declarative state and maintains your backup posture for an application alongside the application in Git.

You can use CRs to deploy backup storage locations, establish backup policies to each cluster, assign policies to applications, and perform backup and restore operations for applications and their resources.
Note: If you are not currently in the same namespace, then include **`-n <namespace>`** in all commands. If you are already in the IBM Fusion namespace (**`ibm-spectrum-fusion-ns`**), then **`-n <namespace>`** is not necessary.

Backup storage location

If you want to create a backup storage location, then define the type of the S3 endpoint, the name of the target S3 bucket, credentials to connect to the endpoint, and some optional parameters based on the type.

Create a backup storage location

```
oc create -f <storagelocation.yaml>
```

Sample storage location YAML file:

```
apiVersion: v1
data:
  access-key-id: AMIATNJ3JEMKR6GAUOLN
  secret-access-key: vgm9AJPztPkOygBFpBp2UzEErLBelTcp3JPdPn9c
kind: Secret
metadata:
  name: backup-storage-location-example-secret-0
  namespace: ibm-spectrum-fusion-ns
---
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: BackupStorageLocation
metadata:

  name: backup-storage-location-example
  namespace: ibm-spectrum-fusion-ns
spec:
  type: aws
  credentialName:  backup-storage-location-example-secret-0
  provider: isf-backup-restore
  params:
    region: us-west-2
    bucket: bucket-name
    endpoint: https://s3.us-west-2.amazonaws.com
```

In this sample, a secret and backup storage locations are created.

Note: The secret is created before the triple dashes, and the backup storage location is created after the triple dashes in the YAML file.
Run the following command to generate the cloud data:

```
printf "[default]\naws_access_key_id=minio\naws_secret_access_key=minio123\n" | base64 -w 0
```

Replace the value of **aws_access_key_id** and **aws_secret_access_key** with your value.

Modify backup storage location

```
oc edit fbsl backup-storage-location-example
```

You can modify the backup storage location details.

Modify secret

```
oc edit secret backup-storage-location-example-secret-0
```

You can modify the secret details.

Delete backup storage location

```
oc delete fbsl backup-storage-location-example
```

Note: You cannot delete a backup storage location if there are any existing backup policies or backups associated with it. For successful backups, they must surpass their retention period before IBM Fusion can automatically remove them. However, failed backups can be manually removed through the IBM Fusion web console.

Delete secret

```
oc delete secret backup-storage-location-example-secret-0
```

Get all backup storage locations

```
oc get fbsl
```

Backup policy

Create a backup policy

```
oc create -f <backuppolicy.yaml>
```

or

```
oc apply -f <backuppolicy.yaml>
```

Sample backup policy YAML files:

- Backup policy to take backups daily:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: BackupPolicy
metadata:
  name: daily-policy
  namespace: ibm-spectrum-fusion-ns
spec:
  provider: isf-backup-restore
  backupStorageLocation: backup-storage-location-example
  retention:
    number: 10
    unit: days
  schedule:
    cron: "30 10 * * *"
    timezone: America/Los_Angeles
```

- Backup policy to take backups weekly from Monday through Friday:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: BackupPolicy
metadata:
  name: backup-policy-weekdays
  namespace: ibm-spectrum-fusion-ns
spec:
  provider: isf-backup-restore
  backupStorageLocation: backup-storage-location-example
  retention:
    number: 4
    unit: weeks
  schedule:
    cron: "30 20 * * 1,2,3,4,5"
    timezone: America/Los_Angeles
```

Edit a backup policy

```
oc edit backuppolicy <backuppolicy name>
```

You cannot modify the policy name and CR name.

Delete a backup policy

```
oc delete backuppolicy <backuppolicy name>
```

Example:

```
oc delete backuppolicy daily-policy
```

If at least one reference exists a **backupObject** or **PolicyAssignment** Object, then you cannot delete the policy.

Get the basic information about a backup policy

```
oc get backuppolicy <backuppolicy name>
```

Get all the information about a backup policy

```
oc describe backuppolicy daily-policy
```

Alternatively, you can use the following sample OC command:

```
oc describe backuppolicy/daily-policy
```

Get all backup policies

```
oc get backuppolicies
```

Assign a policy to an application

BackupPolicyAssignment is a call to create backups by associating a backup policy with an application. You must assign a policy to an application to schedule backups.

```
oc create -f <policyassignment.yaml>
```

Note: The appCluster is the name of the cluster where this application resides. It is optional and only required for remote Spoke applications to designate which Spoke cluster to run the job against. If it is not provided, the assumption is that the jobs are for an application on the Hub cluster.
Sample assignment YAML file for daily backup:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: PolicyAssignment
metadata:
  name: backup-policy-assignment-example
  namespace: ibm-spectrum-fusion-ns
spec:
  application: application-sample
  backupPolicy: daily-policy
  appCluster: fusion-cluster
```

Run the following command to get the name:

```
oc get cluster
```

Example:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: PolicyAssignment
metadata:

  name: filebrowser-20220919-1834111-isf-auto-ibmcos-backuppolicy-20220919-183411
  namespace: ibm-spectrum-fusion-ns
spec:
  application: filebrowser-20220919-1834111
  backupPolicy: isf-auto-ibmcos-backuppolicy-20220919-183411
  appCluster: fusion-cluster
  runNow: true
```

The corresponding backup CRD file is as follows:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: BackupPolicy
metadata:

  name: isf-auto-ibmcos-backuppolicy-20220919-183411
  namespace: ibm-spectrum-fusion-ns
spec:
  backupStorageLocation: isf-ibm-location-20220919-183411
  provider: isf-backup-restore
  retention:
    number: 1
    unit: days
  schedule:
    cron: 51 18 1 * *
    timezone: America/Los_Angeles
```

Edit policy assignment

```
oc edit policyassignment <policyassignment-name>
```

Delete policy assignment

```
oc delete policyassignment <policyassignment-name>
```

Backup and restore

Create backup CR for on-demand backup

Here, one time application backup is taken by using an on-demand backup policy.

```
oc create -f <CR_Ondemand_backup.yaml>
```

Sample YAML file:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Backup
metadata:

  name: auto-fb-spoke1-20230702-1122581-azure-dpcos1-daily-202307102004
  namespace: ibm-spectrum-fusion-ns
spec:
  appCluster: apps.bnr-sno-sunshine.fusion-sno-ibm.com
  application: auto-fb-spoke1-20230702-1122581
  backupPolicy: azure-dpcos1-daily
```

Note: The appCluster is the name of the cluster where this application resides. It is optional and only required for remote Spoke applications to designate which Spoke cluster to run the job against. If it is not provided, the assumption is that the jobs are for an application on the Hub cluster.
Run the following command to get the name of the cluster where this application resides and add it as a value for appCluster:

```
oc get cluster
```

Delete backup request CR
A `DeleteBackupRequest` can be used to delete a backup CR and all the related resources (backup data, snapshot, and so on) of a backup CR.

```
oc create -f <delete_CR_Name.yaml>
```

Sample YAML file:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: DeleteBackupRequest
metadata:
  name: backup-example-delete-request
  namespace: ibm-spectrum-fusion-ns
spec:
  backup: backup-example
```

Restore an application

```
oc create -f <backup-restore.yaml>
```

Sample of YAML restore file:

The following sample restores the `backup-wordpress` backup.

For cross cluster restore, the `targetCluster` is provided. If `targetCluster` is not available, then default to the same cluster where the backup got originated from:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Restore
metadata:
  dp.isf.ibm.com/provider-name: isf-backup-restore
  name: restore-complete-wordpress-1689030046
  namespace: ibm-spectrum-fusion-ns
spec:
  backup: complete-wordpress-ibm-do-not-use-202307102253
  objectsToRestore:
    RESOURCES:
    - ALL
    v1/persistentvolumeclaim:
    - complete-wordpress/mysql-pv-claim
    - complete-wordpress/wp-pv-claim
  targetCluster: guard-vpc-dog-98b7318c91b01bd72490e80cc2328915-0000.us-south.containers.appdomain.cloud
```

Restore an application to alternate namespaces
You can restore application to alternate namespaces on same cluster or another cluster.
For example, application A comprising of three namespaces namely ns1, ns2, and ns3 are backed up, then you need to specify the `namespaceMapping` field in Restore CR to restore this application to alternate namespace.

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Restore
metadata:
  name: <custom-restore-name>
  namespace: ibm-spectrum-fusion-ns
spec:
  backup: <your-backup-job-name>
  namespaceMapping:
      ns1: ns1-new
      ns2: ns2-new
      ns3: ns3-new
```

Namespace variables automatically point to the correct namespaces during backup and restore. And it would be useful in scenarios, where you want to refer current namespace like exec hooks, mainly in restore cases.
Note: If a namespace or namespace field is missing in the `namespaceMapping`, then it restores to the original namespace.
An example of application that is backed up with three variables in Application CR.

```
apiVersion: application.isf.ibm.com/v1alpha1
kind: Application
...
spec:
  includedNamespaces:
    - ns1
    - ns2
    - ns3
  variables:
    - name: PARENT_NAMESPACE
      value: ns1
```

```
  - name: CHILD1_NAMESPACE
    value: ns2
  - name: CHILD2_NAMESPACE
    value: ns3
```

There are two methods to access these variables:

- The first method is using `${PARENT_NAMESPACE}`. It is valid only in backup scenario, and becomes invalid during restore as still referring to `ns1`.
- The second method is using new construct `${APP.${PARENT_NAMESPACE}.name}`, which substitutes namespace variables correctly. This refers to `ns1` in case of backup and `ns1-new` when restoring to alternate namespace.
  In a restore scenario within the same namespace, it refers to `ns1` only. This new construct can be utilized while specifying hook `namespace` field and exec `commands`.

List backups

```
oc get backups.data-protection.isf.ibm.com
```

List restores

```
oc get restores.data-protection.isf.ibm.com
```

List policy assignments

```
oc get policyassignments.data-protection.isf.ibm.com
```

Self-service Backup & Restore

Create a `RecoverBackup` CR
: To create a `RecoverBackup` CR, see Self-service Backup & Restore.

Service protection

Note: The service protection is supported only through the user interface and not from the command line.

Retrieve a backup logs and resources information

The following commands use jq to process the JSON output.

Get the Backup & Restore namespaces with the following commands using the former to retrieve the latter. Skip this step if you know the namespace used in the environment.
Note: You can run these commands only from the hub cluster, even when the Backup CR is created in a spoke cluster.

```
oc get spectrumfusion -A --no-headers | cut -d" " -f1
```

```
oc -n <fusion namespace> get fusionserviceinstance ibm-backup-restore-service-instance -o json | jq -rc
'[(.spec.parameters[]|"\n",.name,.value)]|@csv' | tr -d '"' | grep ",namespace," | cut -d"," -f3
```

Port forward to the backup service, then use curl or a web browser with either URL.

```
oc port-forward svc/backup-service 9443:9443 -n <backup and restore namespace> &
```

```
curl -k <logURL or resourcesURL> | jq -r '[.logs] | @sh'
```

Alternatively, the following steps may be used to run curl from the backup service pod:

1. Get the backup service's pod on the Backup & Restore namespace.

   ```
   oc get pod -n <backup and restore namespace> --field-selector=status.phase=Running --selector=app=backup-service | grep
   backup-service | head -n 1 | cut -d ' ' -f1
   ```

2. Run the curl command on the backup service pod.

   ```
   oc exec -it <backup-service-pod ID> -n <backup and restore namespace> -- curl -k <logURL or resourceURL> | jq -r '[.logs]
   | @sh'
   ```

Example:

```
: oc get spectrumfusion -A --no-headers | cut -d" " -f1
> ibm-spectrum-fusion-ns
```

```
: oc -n ibm-spectrum-fusion-ns get fusionserviceinstance ibm-backup-restore-service-instance -o json | jq -rc
'[(.spec.parameters[]|"\n",.name,.value)]|@csv' | tr -d '"' | grep ",namespace," | cut -d"," -f3
> ibm-backup-restore
```

```
: oc get pod -n ibm-backup-restore --field-selector=status.phase=Running --selector=app=backup-service | grep backup-service |
head -n 1 | cut -d ' ' -f1
> backup-service-cd597966b-btkgq
```

```
: oc exec -it backup-service-cd597966b-btkgq -n ibm-backup-restore -- curl -k https://localhost:9443/dataprotection/backup-
service/v1/logs/c2ed99be-6daa-4296-876e-6a027197ef0c/recipe | jq -r '[.logs] | @sh'
```

```
: oc exec -it backup-service-cd597966b-btkgq -n ibm-backup-restore -- curl -k https://localhost:9443/dataprotection/backup-
service/v1/logs/c2ed99be-6daa-4296-876e-6a027197ef0c/resources | jq -r '[.logs] | @sh'
```

Backup & restore service from OpenShift Container Platform

You can monitor and manage Backup & restore service related operations from the OpenShift® Container Platform console.

In the OpenShift Container Platform console, go to Storage > Backup & Restore. The Backup & Restore page includes the following tabs:

- Overview tab
  From this tab page, you can monitor storage consumption, backup storage activity, backup coverage, and backup and restore jobs.
- Policies tab
  From this tab page, you can do the following operations:
  - Create a backup policy
  - Edit an existing policy
  - Delete an existing policy
- Locations tab
  From this tab page, you can do the following operations:
  - Create a new backup storage location
  - Edit credentials of a backup location
  - Remove an existing backup location
- Applications tab
  From this tab page, you can do the following operations:
  - Assign backup policy to applications
  - Restore backups of applications
- Jobs tab
  From this tab page, you can do the following operations:
  - Delete backup jobs
  - Collect logs
  - View job summary

At any point in time, you can click Go to Fusion Backup & Restore to go to the Backup & restore page of the IBM Fusion user interface.

Prerequisites

- Enable Backup & restore service in IBM Fusion.
  For the steps to enable Backup & restore service from IBM Fusion user interface, see Install Backup & Restore.
- If for some reason, you can access OpenShift Container Platform console prior to or during ongoing IBM Fusion HCI installation, then wait for 10-20 seconds after installation to see the following notification in the OpenShift Container Platform console:

  ```
  Web console update is available
  ```

  Click Refresh web console link to refresh and view the Backup & restore service.
- The user interface works only for the supported OpenShift Container Platform 4.16. Do not try to access that user interface on any older OpenShift Container Platform versions.

Overview tab

The Overview tab page is a dashboard view to monitor Backup & restore service details.

Section	Description
Status	The Status section includes Statuses of Backup location and Applications. The Backup locations are either Active or inactive. The Active refers to the connected location status and inactive refers to not connected. The Applications subsection includes the following details: Backed up applications Applications with assigned policies Applications with recent backup Applications that are backed up and the recent or latest backup was successful (Completed or PartiallyFailed). Backup in progress Applications with recent or latest ongoing backup.
Backup & Restore jobs summary	The Backup & Restore jobs summary includes the following details: Queued jobs It includes the number of in progress and number of pending jobs. Success rate Note: The Success rate and other fields are dependent on the frequency drop down. The Success rate subsection displays the percentage of success rate and the number of jobs completed out of the total number of jobs. It also gives a break up of the number of completed, failed, and canceled jobs. The All, Backup, and Restore options display the completed, canceled and failed jobs accordingly. The Backup job count is also represented in a graph bar chart.

Section	Description

Backup summary	The Backup coverage and Most recent backup are represented in a circular chart. It also shows a legend of the color representation. The Backup coverage includes the following details:
	
	Backup up applications
	    Applications with assigned policies
	
	Not backed up applications
	    Applications without assigned policies
	
	The Most recent backup includes the following details:
	
	Backup completed
	    Applications that are backed up and its recent or latest backup was successful (Completed or PartiallyFailed)
	
	Backup failed
	    Applications that are backed up and recent or backup was not successful.

Storage consumption	The Object storage section includes the amount of GiB used.
Backup & Restore activity	The Backup & Restore activity lists ongoing and recent events related to application, backup, backup policy, backup storage location, policy assignment, and restore.

Policies tab

A backup policy defines a set of rules to backup application data. The following table describes the different sections and actions you can perform in the Policies tab page.

Section	Description or action
Backup policies table	The Backup policies table lists the policies with Name, Status, Location, Backup scheduled, Retention period, Application. You can search a backup policy record by its name from the list.
Backup policy record actions	From the ellipsis overflow menu of a policy record, you can do the following operations: • Edit backup policy • Edit YAML • Delete backup policy

Create backup policy	1. Click Create backup policy.
	The Create BackupPolicy page is displayed.
	
	2. Enter the following details of the policy:
	  • Enter the Policy name of the policy.
	  • In the Backup location section, select whether you want In place snapshot or Object storage. If you select Object storage, then select the Location from the drop down list.
	  • Select the Backup schedule for the backup. In Schedule, select the frequency. In At, select the time. In Time zone, select your zone.
	  • In the Retention period, select number for Schedule and Duration in `Days`, `Weeks`, `Months` or `Years`. It is the duration to retain the backup.
	3. Click Create.

Locations tab

A backup location is a prerequisite to create a policy. You must configure at least one location to store your backed-up data. IBM Fusion supports a variety of object storage types. The following table describes the different sections and actions you can perform in the Backup locations tab page.

Element or section	Description or action
Backup locations table	The Backup locations table includes Name, Status, Location type, Policies, Used capacity, and Protected applications.

Actions performed on a backup location record	From the ellipsis overflow menu of a backup location record, you can do the following operations:
	
	• Edit credentials
	  1. Update Endpoint, Access key, or Secret key values.
	  2. Click Save.
	• Edit YAML opens in YAML format. You can make changes and click Save.
	• Remove backup location
	In the confirmation window, click Remove to proceed with deletion.

Element or section	Description or action
Create a backup location	If you plan to use a certificate for a S3 compliant backup storage location, then as a prerequisite create a secret with the certificate as a prerequisite. For the procedure to create a secret, see Creating a secret. Note: IBM does not support the creation of two backup storage locations that have both identical endpoint and bucket names. 1. Click Create backup location. 2. Enter the details about the backup location: • Enter the Name of the backup location. • Select the Location Type of the backup location. The different location types are IBM Cloud (IBM Object Storage), Azure (Microsoft Object Storage), AWS (Amazon Object Storage), Spectrum Protect, and S3Compliant (Any Object Storage). Enter the following credentials to connect IBM Fusion to your backup location: Endpoint, Path/Bucket, Access key, and Secret key. If the location is Amazon Web Services, then you must also enter the Region. If the location is Microsoft Azure (Microsoft Object Storage), then enter the Account name and Account key instead of Access key and Secret key. • In the Certificate setting (optional), enter the Secret name for certificate.

Applications tab

Applications contain resources, such as microservices, databases, and other stateful data distributed across projects. IBM Fusion provides ways to do application-centric backup, which provides data loss protection for your applications. The following table describes the different sections and actions you can perform in the Applications tab page.

Element or section	Description or action
Applications table	The Applications table includes Name, Backup status, Backup policy, Location, Used/Total capacity, Last back up on, and Success rate. Here, the Backup status indicates the outcome of the most recent backup performed for an application. The Success rate indicates the number of successful backups performed for the application. It also indicates the availability of backup copies.
Actions performed on an application record	From the ellipsis overflow menu of an application record, you can do the following operations: • Assign backup policy • Restore It opens a Restore window. 1. In the Select a project destination section, select from the following options: Use the same project the application is already using Select the project in the current cluster where the application gets restored to. Restoring to an existing cluster delete the contents of the project and replaces it with the restored application. Use existing project Create new project If you select Create a new project, then enter a new project. Note: If you choose Existing project or `Create new project`, you can only specify a single namespace.You cannot restore multiple namespaces to a single alternate namespace. 2. Select a backup from the available list of backups. 3. Optionally, select Include missing etc resource to restore any etc resources that are not present in the existing OpenShift project. 4. Select the PVCs. By default, all PVCs in the back are restored. If you want to restore only certain volumes, then you can select individual PVCs. 5. Click Restore.
Assign backup policy	If you plan to use a certificate for a S3 compliant backup storage location, then as a prerequisite create a secret with the certificate as a prerequisite. For the procedure to create a secret, see Creating a secret. Note: IBM does not support the creation of two backup storage locations that have both identical endpoint and bucket names. 1. Click Assign backup policy. Alternatively, you can also click Assign backup policy from the ellipsis overflow menu of an application record. You can search for backup policy and filter records. 2. Select one or more polices. 3. If you want to initiate a backup right away, set the Run backup now toggle button to on. 4. Click Assign. If you set the Run backup now toggle button is set to on, the Backup status of the application in the Applications page is initially New. It then changes to Queued, Inventory in progress, Snapshot inprogress, and Complete. You can also check the Backup & Restore Jobs list in the Jobs tab page for similar status.

Jobs tab

The Jobs page displays the details of all the backup and restore jobs. The following table describes the different sections and actions you can perform in the Jobs tab page.

Element or section	Description or action

Element or section	Description or action
Backup & Restore jobs table	The Backup & Restore jobs table includes Name, Operation, Status, Start time, Completion time (Duration), Application, and Policy. You can filter the records based on Status and Operation. The Operation values are Backup and Restore. You can select a field and enter keywords in the Search to filter records. You can also choose to view the list of jobs in last 24 hours, last 7 Days, or last 30 days.
Actions to perform on a jobs record	The ellipsis overflow menu of a job record are as follows: • Delete backup job If you click Delete backup job, a confirmation window appears. If you are sure to delete the job, click Delete. • Collect logs Click Collect logs to go to the Jobs page of the IBM Fusion user interface. In the Jobs page, click Collect logs. • Job summary Click Jobs summary to go to the Jobs page of the IBM Fusion user interface. Here, you can view details like Summary (Inventory and Backup sequence) and Details (Duration, Created, Size, Policy, Application. Backup ID, Backup location).

Service protection

The IBM Fusion Backup & Restore service protection involves the backup of the Backup & Restore data management service to a S3 bucket.

In case of cluster failure, use the IBM Fusion Service Protection to recover all namespaces from the failed cluster into a new cluster. However, it assumes that the service protection backup got created and all application backups exist before the cluster failure. The high-level steps to backup and restore the service:

- Create a new instance of OpenShift® Container Platform cluster or use an exiting healthy cluster.
- Install Backup & Restore service.
- The Service Protection Recover service uses the backup to restore all resources used by the Backup & Restore service such as backup policy, location, policy assignment, Backup CRs.
- Restore all other applications from Backup CRs.

- **Configuring service backups**
  Backup a service configuration from the service backup.
- **Recovering service backups**
  Restore a service configuration from the service backup.

Configuring service backups

Backup a service configuration from the service backup.

About this task

In the Configure Service backups, select the location type and enter the credentials to access the location. After the discovery of existing service backups, the Service protection page is displayed with a list of service backups. If no service backups are available, define a schedule with frequency and retention period, and initiate a backup.
In Backup & Restore, one hub cluster and more than one spoke clusters can be connected to easily manage application backups in multi-clusters. For various reasons, the OpenShift® Container Platform cluster may have problems and become unusable. If the cluster recovers after the expiry of the client cert, you must clean and setup to rejoin the recovered cluster to the Hub or Spoke. For the procedure to rejoin, see Reconnecting OpenShift Container Platform cluster.

Note: Do not create virtual machines or PVCs on the IBM Fusion namespace, as it causes restoration failures of service protection backups.

Procedure

1. Go to Backup & Restore > Service protection.
   The Service protection page is displayed. Here, Configure Service backups and Recover service from backup tiles are available.
2. Select Configure Service backups.
   The Connect backup location wizard page is displayed.
3. In the Location type tab, select the location type and click Next.
   The available location types are Microsoft Azure, IBM Cloud®, S3 compliant object storage, Amazon Web Services, and Storage protect.
   The Location credentials tab page is displayed.
4. In the Location type tab, enter the credentials to connect to your location. Enter the endpoint and bucket to use for the service protection backups. For SSL secured object storage locations, enter the Secret name for certificate.
   There are two limitations that are not enforced:
   - While the user interface does not allow the created BSL to be assigned to any other policy, it is still possible to do so by using the CR. However, do not do this action.
   - The same endpoint or bucket must not be used for service protection configuration for more than one cluster deployments at a time.
   Note: After you set the location or bucket details, you cannot edit it later.
5. Click Add.
   Note: The discovery of existing service backups may take a minute or so. After the discovery completes, the BSL changes the status. For example, Connected.
   The Service protection page is displayed with a Service backups table.
6. If no service backups are available, click Define schedule.
   The Define schedule window is displayed.
7. In the Define schedule, enter the following details:

Frequency
: The frequency of backup. It can be Hourly, daily, Weekly, Monthly, Custom. For Custom, you can choose a value in minutes. Select the Timezone.

Retention period
: How long does the backup reside before automatic deletion. Select the number and its unit. For example, 30 Days.

Initiate Service backup now
: You can use this toggle to imitate the backup immediately.

8. Click Save.
   The Service protection page is displayed. If you had initiated service backup, then you can monitor the progress of the backup in the Service backups table. The Service backup table has Time, Status, Location, and Size. The Backup summary section is updated with details of Available backup, Used capacity, Last successful backup, and Backup policy.
9. If you had not initiated any backup, then click Backup now in the Service backup table.
   The Backup summary section is updated with details of Available backup, Used capacity, Last successful backup, and Backup policy.

Recovering service backups

Restore a service configuration from the service backup.

Before you begin

- Ensure that the IBM Fusion Backup & Restore (Legacy) service is not installed. This recovery replaces all existing objects with the objects in the service backup.
- If service protection requires rebuilding the hub cluster, backup your application data to a cloud-based storage. Also, backup any defined local S3 storage on your cluster to a cloud storage. The recovery procedure in this topic restores only the metadata for backups, which got saved to the cloud storage.
  For example, you have a MinIO storage service defined and configured with a backup storage location that uses the local storage. The MinIO application must have its cloud backup policy to restore the service first, and then any applications backed up to the local S3 storage.

  You can use the local storage for sensitive applications, but ensure that you save the database to the cloud or another S3 location within the firewall. Restore the local storage location from a cloud backup before you attempt to restore other applications backed up to that storage service.
- The IBM Fusion namespace must remain the same between the service backup and service restore. When you recover a service backup to a new IBM Fusion deployment, make sure that the deployment uses the same namespace as backup.

Procedure

1. Go to Backup & Restore > Service protection.
   The Service protection page is displayed. Here, Configure Service backups and Recover service from backup tiles are available.
2. Select Recover service from backup.
   The Add a backup location wizard page is displayed.
3. In the Location type tab, select the location type and click Next.
   The available location types are Microsoft Azure, IBM Cloud®, S3 compliant object storage, Amazon Web Services, and Storage protect.
   The Location credentials tab page is displayed.
4. In the Location credentials tab, enter the credentials to connect to your location. For SSL secured object storage locations, enter the Secret name for certificate. Provide the same S3 information or credentials as provided in the original Service Protection backup configuration.
5. Click Connect.
   The discovery process looks for existing service backups.

Option	Description
No service backups available	If no service backups are available, then `No service backups` message is displayed. Click Define schedule to configure. Note: It may take a while for the discovery of existing service backups to complete.
Service backups are available	Initiate restore by using either of the following methods: • Click Restore service. • From the ellipsis overflow menu of a service backup, select Restore. • Click Backup to display the Backup details slide out pane, and then click the Restore.

   The Restore service page is displayed.
6. Select a backup and click Restore.
   Note: This restore action reverts all Backup & Restore objects to the selected backup.
   When the Restore is in progress, the user interface disables Backup & Restore functions. However, you can still create CRs outside of the user interface.

7. Do the following steps based on the scenario existing on your cluster:

Option	Description
Empty cluster with no Backup & Restore	No action required
Empty cluster but Backup & Restore installed	a. Type `Recover`. b. Click Initiate service restore. Note: The recover action restores all backup policy, location, policy assignment, CRs including the ones used by Backup & Restore from service backup. However, any available Backup & Restore backups on the restore cluster get deleted. If you need them, copy the backup CRs.

   The Service protection page is displayed. A `Service restore is in progress` notification message is displayed. You can monitor the progress of restore in the Summary section. When the restore begins, the service protection page gets replaced by a job summary page for restore.

What to do next

1. Application backups are available to restore before the backup storage locations are connected. Wait for the backup storage location to be connected before you attempt to restore.
2. Storage classes that are used by applications on the original cluster must exist or be created on the restored cluster before you attempt to restore the applications that use those classes.
3. Scheduled backups resume after the service restore is complete. However, these backups fail until the applications are restored from a backup. You can ignore these failed backups. After the application is restored from a backup, subsequent backups will be successful.
4. When you do a service protection restore, the Backed up applications page does not display the applications on the Spoke clusters that are not reconnected to the new Hub cluster. After a Spoke is manually reconnected, it may take a while for the backed up applications on that Spoke to appear on the page.

Self-service Backup & Restore

You can protect your namespace application with IBM Fusion Backup & Restore even as an application user without cluster rights or a user without IBM Fusion administration rights. Application owners can create IBM Fusion Backup & Restore Custom Resource (CRs) within the application's namespace to self-manage their application's backup and restore needs without the involvement of a cluster or IBM Fusion Backup & Restore administrator.

Important: This feature is only supported for applications by using Backup & Restore Custom Resources (CRs) and not through IBM Fusion user interface.
All self-service Backup & Restore CRs, including scheduled Backup CRs exist in the namespace of the application.

To create Backup & Restore CRs in the application's namespace, you need to be an application namespace administrator, an application user, or service account with granted permission to the namespace. Application namespace administrators can directly create Backup & Restore CRs within their application namespace.

Non-admin application users and service accounts need the namespace administrator to first grant IBM Fusion Backup & Restore permissions through a standard OpenShift RoleBindings. The RoleBindings grant permissions for the user or service account in the application's namespace with the specific Backup & Restore Role.

Each of the Backup & Restore CRs have an 'admin', 'crdview', 'edit', and 'view' Roles available for them. They are named in the following format:

`<CR Name>.data-protection.isf.ibm.com-v1alpha1-<Role>`

The following Backup & Restore CRs have Role available for them:

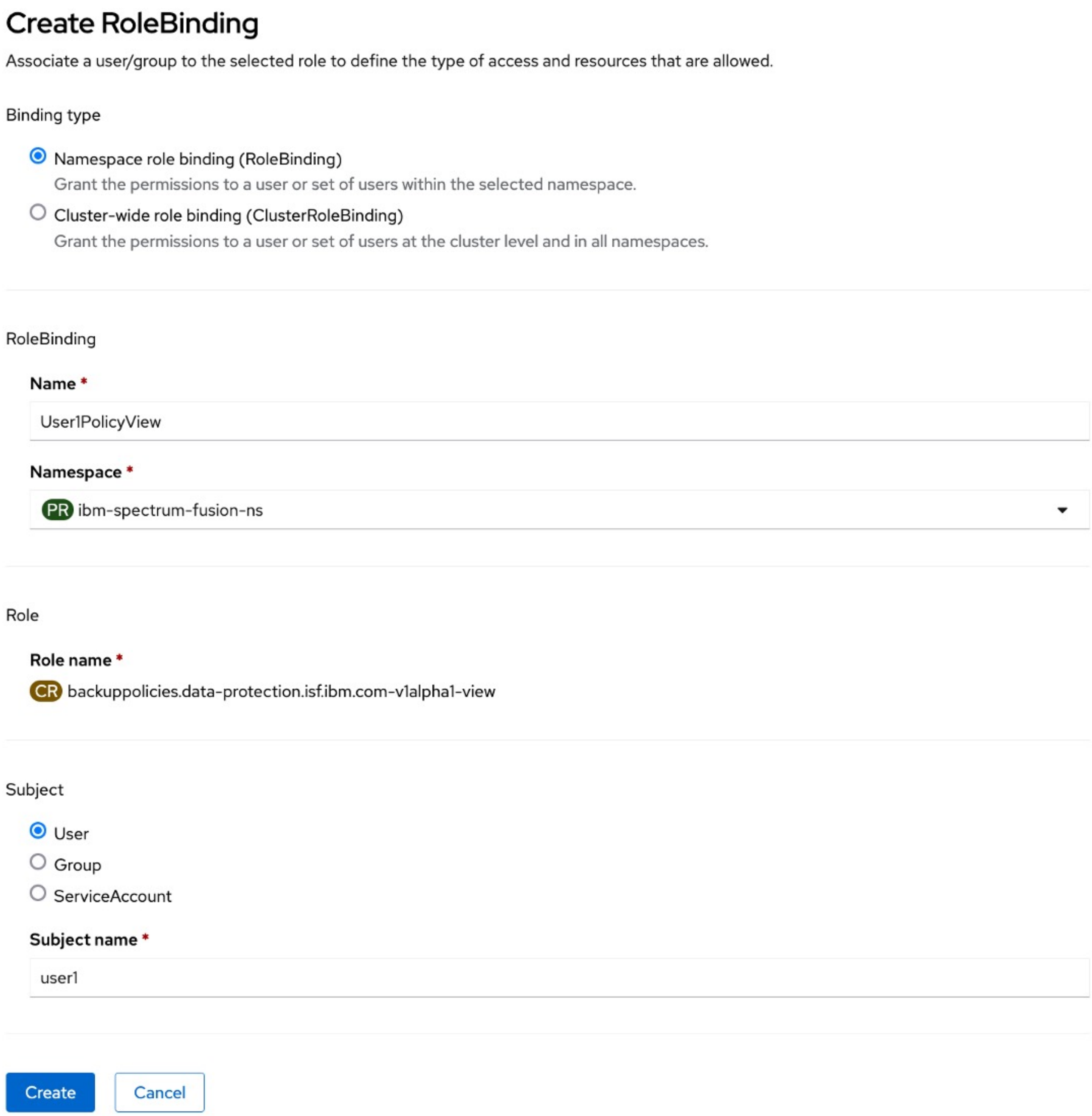
Role	CR
`backuppolicies.data-protection.isf.ibm.com`	Backup Policy
`backups.data-protection.isf.ibm.com`	Backup
`backupstoragelocations.data-protection.isf.ibm.com`	Backup Storage Location
`deletebackuprequets.data-protection.isf.ibm.com`	Delete Backup Request
`policyassignments.data-protection.isf.ibm.com`	Policy Assignment
`recipes.spp-data-protection.isf.ibm.com`	Recipe
`recoverbackupcrs.data-protection.isf.ibm.com`	Recover Backup CRs
`restores.data-protection.isf.ibm.com`	Restore

For example, the IBM Fusion Backup CR has the following Roles:

- `backups.data-protection.isf.ibm.com-v1alpha1-admin`
- `backups.data-protection.isf.ibm.com-v1alpha1-crdview`
- `backups.data-protection.isf.ibm.com-v1alpha1-edit`
- `backups.data-protection.isf.ibm.com-v1alpha1-view`

If you want to reference `backuppolicy` or `backupstoragelocation` that is in another namespace, then specify it in "namespace/BP" or "namespace/BSL" format. The user must have permissions for the namespace that is specific for the BSL or BP.
RoleBindings can be created with a mix of permissions to achieve the desired level of management. For example, the application administrator wants to restrict Restore actions while allowing another user to perform all other Backup actions. To achieve this, the application administrator must create RoleBindings in the application's namespace for that other user with all Backup & Restore administrative roles except the Restore CR. To create the RoleBindings, see OpenShift documentation.

As an example, consider the following screen shot of the OpenShift® RoleBinding creation page showing the process of creating a RoleBinding for the user1 user identity in the namespace1 namespace with the `view` Backup & Restore PolicyAssignment Role.



The same RoleBinding can be created from the **oc** command line:

```
oc create rolebinding User1PolicyView --role=backuppolicies.data-protection.isf.ibm.com-v1alpha1-view --user=user1 -n 
namespace1
```

Note: To retrieve logs and resources, see Backup and restore as code.

Service Accounts

It is a time-consuming process to create RoleBindings to multiple users individually. The Service Accounts provide another option for the self-service capability to a broad group of non-admin users. When creating a RoleBinding, you can select ServiceAccount option in the Subject section instead of an individual user. For more information about service accounts, see OpenShift documentation.

RecoverBackupCRs

If an application's namespace gets accidentally deleted or corrupted, the Backup & Restore CRs may prevent any potential recovery of the application from backups. It prevents the potential recovery of the application through an Backup CR.

Any successful `RecoverBackupCRs` action always recreates any prior Backup CRs. Recovery of other Backup & Restore CRs can be specified as options.

Example `RecoverBackupCRs` CR:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: RecoverBackupCRs
```

```
metadata:
   name: example
   namespace: <replacement namespace>
spec:
   application: <deleted application namespace>
   reassignPolicies: true
   recreateDeleteBackupRequestCRs: false
   recreateRestoreCRs: false
```

All spec fields are optional.

- `application`
  Name of the deleted application's namespace. If not specified, the Backup & Restore service makes use of the application name from the `metadata.namespace` field.
- `reassignPolicies`:
  Boolean to recreate any prior Policy Assignment CRs or not. Default is false.
- `recreateDeleteBackupRequestCRs`:
  Boolean to recreate any prior `DeleteBackupRequest` CRs or not. Default is false.
- `recreateRestoreCRs`
  Boolean to recreate any prior Restore CRs or not. Default is false.

To recover from an accidental deletion or corruption of the namespace or Backup & Restore CRs, do the following steps:

1. Create a replacement application namespace
2. Re-create the Backup & Restore RoleBindings that is same as the original namespace.
3. Create a `RecoverBackupCRs` CR in the replacement namespace that refers to the original namespace.

The Backup & Restore service detects the presence of this CR and performs verification checks to ensure that the administrative RoleBindings in the new namespace matches the deleted application. If it matches, then the Backup & Restore service recreates from its metadata all the relevant Backup & Restore. The desired Backup CR can then be used to restore the application from a prior backup.

Custom Backup & Restore Administrator

Another mode of operation for managing the Backup & Restore service is to grant a non-admin user the admin of the Backup & Restore service. To do this, create the RoleBindings of the user in the IBM Fusion namespace with the Backup & Restore admin Roles.

This user does not have IBM Fusion user interface access but can manage Backup & Restore activity of all applications on both hub and spoke clusters through Backup & Restore CRs within the IBM Fusion namespace.

Limitations

Limitations that are applicable for both Hub and Spoke:

- Only command line support is available
- Cross-cluster restore is not available
- Self-service jobs are not shown in the IBM Fusion UI
- Spoke self-service policies and backup storage locations are not shown in the IBM Fusion UI
- Applications with only self-service policies or completed self-service backup or restore jobs do not appear in the Backed up applications table.
- Applications configured with self-service policies do not appear in the Backed Up Applications > Protect Apps.
- On the Topology table, the backed up apps ratio and success rate may also be inaccurate as it includes self-service applications and self-service backup jobs.

Managing backup for multi-cluster environment

You can connect and manage cluster connections in IBM Fusion. It includes steps to establish connections and manage existing connections. You can also reconnect an OpenShift® Container Platform cluster either after a disaster recovery event or a temporary unavailability.

- **Connecting and managing cluster connections**
  You can connect clusters, manage actions on cluster records, and remove cluster configurations. You have options to connect clusters either through the Fusion UI or automated deployment.
- **Reconnecting OpenShift Container Platform cluster**
  The OpenShift Container Platform cluster that is connected in a disaster recovery relationship or a Backup & Restore service with Hub and Spoke can face disasters, leading to a temporary unusable state. After recovery, it can still display an `Unhealthy` or `UnManaged` status in remote clusters. In such instances, this cluster must be reconnected to its corresponding connections.
- **Disabling the connection**
  Disable the connection between Hub and Spoke.

Connecting and managing cluster connections

You can connect clusters, manage actions on cluster records, and remove cluster configurations. You have options to connect clusters either through the Fusion UI or automated deployment.

The Topology page displays the clusters' connections and you can perform the following activities:

- Connect clusters

- Actions on a cluster record
- Remove cluster configuration

Connect clusters

1. In the Backup & Restore menu, click Topology.
2. In the Topology page, click Connect clusters.
3. In the Connection method window, select Use the Fusion UI or Automate deployment.

Table 1. Cluster connection options

Option	Procedure
Use the Fusion UI	a. On your hub cluster, go to the > Topology. b. Click Connect clusters. c. Select Use the Fusion UI. The connect your cluster > Use the Fusion UI window appears. The YAML starts to generate. d. After the YAML is generated successfully, click Copy snippet. e. Go to the Spoke cluster and install IBM Fusion. f. Go to the Services page and install Backup & Restore Agent. g. Paste the copied snippet to proceed with the agent install. Note: If you use a custom namespace, then do not use the **Copy** command because the encoded command is not editable and is of no value with a custom namespace.
Automate deployment	To connect multiple Spoke clusters: a. On your hub cluster, go to the > Topology. b. Click Connect clusters. c. Select Automate deployment. The connect your cluster > Automate deployment window appears. d. Select the storage class to use on the spoke where you want to run the YAML. e. Click Generate. The YAML starts to generate. Here, you can either use the Deploy using YAML or Deploy using oc command option. f. In the Deploy using YAML option, copy or download the YAML. g. Apply the YAML on each of the Spoke clusters.

Actions on a cluster record

The Topology page lists the hub cluster and all spoke connections. Using the Actions, you can View details, View backed up apps, and remove connections.

In View details, a pane opens with the details of the cluster, namely Connection status, Service status, Job queue, any critical issues, and Backup success rate. You can also click View backed up apps in this pane to view applications that are local to a specific cluster.

In View backed up apps, view applications that are local to a specific cluster. The assumption here is that you assigned policies for these applications on the spoke.

You can also remove connections. See Remove connections.

Remove connections

You can search and remove connections.

From the Hub cluster, you can remove a Spoke or Agent cluster connection.
Note: The Backup & Restore service operations stops after the removal, however, all backups that are associated with this cluster remain available until they expire.
If the selected cluster is involved in a Regional DR, remove the Regional DR configuration before you remove the connection to the selected cluster.
If you remove the connection from the user interface, it removes the connection on both hub and spoke for that spoke. On the hub, there exists a connection for every spoke, so you must only remove the connection for the spoke to be uninstalled. There exists only one connection on the spoke or agent and hence you can remove it from the agent side safely. When you remove the connection on one cluster, the action removes it from both clusters.

Remove the connection first and then uninstall the agent.

1. Search for a connection.
2. From the ellipsis overflow menu of a connection record, click Remove connection.
   A Remove connection confirmation window is displayed.
3. Click Remove
   A notification appears to confirm the successful removal of the connection. The cluster is not available in the table after the successful removal. From the removed Agent user interface, the following banner notification is displayed:

   ```
   Unmanaged service The Backup & Restore Agent's connection to the Hub cluster has been removed.
   ```

Important: When a spoke cluster is removed, its application backups are deleted from storage when they expire. If the connection to the cluster is down but still configured, backups are retained until the connection is restored, and then deleted.
Alternatively, if you want to remove the connection from the hub side by using the **oc delete connection** command and more than one connection exists, then view the YAML and confirm the connected host that you want to remove.

If you want to remove the connection using **oc** commands, run the following commands:

```
oc get connection
```

```
NAME                     AGE
connection-7d9a76ce09    25h
```

```
oc delete connection connection-7d9a76ce09
```

```
connection.application.isf.ibm.com "connection-7d9a76ce09" deleted
```

Reconnecting OpenShift Container Platform cluster

The OpenShift® Container Platform cluster that is connected in a disaster recovery relationship or a Backup & Restore service with Hub and Spoke can face disasters, leading to a temporary unusable state. After recovery, it can still display an `Unhealthy` or `UnManaged` status in remote clusters. In such instances, this cluster must be reconnected to its corresponding connections.

About this task

If the cluster recovers before the expiration time, the cluster rejoins the connection automatically and no action is needed. However, if the cluster recovers after the expiration of the client cert, the connection must be cleaned and setup to rejoin the recovered cluster.

Procedure

1. Run the following example command to login to cluster-a and list the connection names.

   ```
   oc login to cluster-a
   ```

2. List the connection names.
   `oc get connections`
   It lists all connection CRs that are used for Backup & Restore and disaster recovery. The following example shows connections among clusters:
   - If cluster-a is in a DR relationship, a connection exists for the other cluster.
   - If cluster-a is a Backup & Restore hub, there exists a connection for each spoke.
   - If cluster-a is a Backup & Restore spoke, one connection to the hub exists.

   The output includes the API endpoint of cluster-b.
   Example output

   ```
   NAME                    API_ENDPOINT                                   PHASE       HEALTHY_TO   HEALTHY_FROM   AGE
   connection-8414b9c84d   https://api.cluster-b.mydomain:6443   InProgress   Unhealthy    Unknown         25d
   %
   ```

3. Save the name of connection CR in cluster-a (a cluster in the hub and spoke or DR clusters) in `connection-<id>` format.
4. Clean the connection.
   For the procedure to clean the connection, see Disabling the connection.
5. Setup the connection between clusters.
   a. Log in to the cluster-b and get the bootstrap token for cluster-b.

   ```
   oc login to cluster-b
   ```

   ```
   oc create token isf-application-operator-cluster-bootstrap -n <Fusion Namespace of cluster-b>
   ```

   Note: The version of **oc** command line must be greater or equal to 4.11.

   b. Log in to cluster-a.

   ```
   oc login to cluster-a
   ```

   c. Create init secret on cluster-a with the bootstrap token and API endpoint of cluster-b.
   For example:

   ```
   apiVersion: v1
   kind: Secret
   metadata:
      name: <Init Secret Name>
      namespace: <Fusion Namespace of cluster-a>
   stringData:
      apiserver: <cluster-b API Endpoint>
      bootstrapToken: <cluster-b Token Generated in Step 3.a>
   ```

   d. Create connection CR on cluster-a with this init secret in spec:

   ```
   apiVersion: application.isf.ibm.com/v1
   kind: Connection
   metadata:
     name: <Connection Name Saved in Step 1>
     namespace: <Fusion Namespace of cluster-a>
   spec:
     remoteCluster:
       apiEndpoint: <cluster-b API Endpoint>
       initSecretName: <Init Secret Name>
       connectionOperatorNamespace: <Fusion Namespace of cluster-b>
   ```

Disabling the connection

Disable the connection between Hub and Spoke.

About this task

In Hub and Spoke, there is a connection CR in each cluster that represents the connection to the another(remote) cluster. Delete connection CR in the Spoke or Hub. If the network is down, then use either method 1 or method 2 to manually delete the connection CR in another cluster also.

Important: Disabling a spoke connection results in the unassignment of policies from spoke applications, and no new backups are initiated subsequently. Custom Resources (CRs) of applications without backups are deleted, and these applications no longer appear in the user interface.

Method 1: Using command line or command prompt

Procedure

1. In the Spoke, use the following command to get all the connection CR:

   ```
   oc get connection -n <fusion-namespace> -o=custom-
   columns='NAME:.metadata.name,ClusterName:.metadata.annotations.connection\.isf\.ibm\.com/cluster-name'
   ```

   Example command:

   ```
   oc get connection -n ibm-spectrum-fusion-ns -o=custom-
   columns='NAME:.metadata.name,ClusterName:.metadata.annotations.connection\.isf\.ibm\.com/cluster-name'
   ```

   Example output:

   ```
   NAME                     ClusterName
   connection-44ae8529aa    cp4d-vpc-98b7318c91b01bd72490e80cc2328915-0000.us-east.containers.appdomain.cloud
   ```

2. Check the `ClusterName` in connection CR, and delete the connection CR which has the same ClusterName with Hub:

   ```
   oc delete connection <connection-name>  -n <fusion-namespace>
   ```

   Example command:

   ```
   oc delete connection connection-44ae8529aa -n ibm-spectrum-fusion-ns
   ```

Method 2: By using OpenShift Container Platform console

Procedure

1. Log in to OpenShift Container Platform console.
2. Go to Administration > CustomResourceDefinitions > connections.application.isf.ibm.com > Instances.
3. Get the connection instance whose cluster name in the annotation is the same as the hub cluster name.
4. Delete this connection instance.

Backup consistencies

- **Achieving data consistency for containers**
  IBM Fusion backs up application data consistently across all the Persistent Volumes (PVs). It ensures that the recovered data is error-free, reliable, and usable.
- **Achieving data consistency for virtual machine workloads**
  IBM Fusion backs up the data of the Red Hat® OpenShift® Virtualization virtual machines consistently.
- **Orchestrate a backup or restore**
  Use a custom, application-specific recipe to consistently automate and control the required backup or restore sequence of an application.

Achieving data consistency for containers

IBM Fusion backs up application data consistently across all the Persistent Volumes (PVs). It ensures that the recovered data is error-free, reliable, and usable.

The IBM Fusion supports application consistency and crash consistency approaches:

Application consistency
: In this approach, application I/O must be paused before a snapshot is taken by using recipes. Though it achieves the highest level of consistency, it comes at the cost of application downtime that can be disruptive. Depending on the application, the pause can be for a long time. To know more about how to create and use recipes, see Custom backup and restore workflows.

Crash consistency
: If IBM Storage Scale is used for storage (and not Red Hat® OpenShift® Data Foundation), then you can consider crash consistency. In this approach, application I/O is not paused at all and hence results in zero disruption. It is achieved by what is called a consistency group. All the PVs of an application are placed in a consistency group. During backup, a snapshot is taken for all the PVs in the group at the same instant in time to ensure that all the PVs are consistent with each other. But the disadvantage of this approach is that the application might have some data in memory that is not flushed to the PVs and not included in the snapshot.
  In IBM Fusion with IBM Storage Scale, all PVs in a namespace are placed in a consistency group automatically and you do not have to do anything.

Choose the type of data consistency that is appropriate for your application. If the application can restore itself without the data in memory with just what is there in the PVs, then backups are nondisruptive. Otherwise, you must pause I/O and disrupt the application.

Achieving data consistency for virtual machine workloads

IBM Fusion backs up the data of the Red Hat® OpenShift® Virtualization virtual machines consistently.

During virtual machine backups, consider the following options to provide data consistency within the virtual machine.

- **Crash consistency**- the backup includes data that is written to a disk. It is the equivalent to a virtual machine crash before a backup. To achieve this consistency, take a snapshot of the virtual machine PVCs without any additional orchestrations or hooks within the virtual machine.
- **File system consistency**- the file system freezes, writes, and flushes to the disk before a backup. The backup does not capture any data in the application buffers that is not yet written to the file system. Achieve this data consistency by using tools inside the virtual machine, which interacts with the file system to freeze write before a snapshot (backup) operation.
- **Application consistency**- data in memory buffers is written to disk before the application is backed up. It ensures that all application data is included in the backup. You can achieve this consistency by tools inside the virtual machine, which interacts with an application to take an application-specific action such as locking a database.

File system consistency

The following example diagram shows a virtual machine that runs within a namespace. Each virtual machine has a `virt-launcher` pod that runs the virtual machine instance. The `virt-launcher` pod runs an instance of `libvirt`, which is used to manage the virtual machine process.

When a namespace backup is requested, a virtual machine snapshot operation gets run. The `virt-launcher` pod includes Velero pre-snapshot (`pre.hook.backup`) and post-snapshot (`post.hoook.backup`) annotations to run the `kubevirt` tool `virt-freezer`. It communicates to a file system driver inside the Linux® or Microsoft® Windows virtual machine guest to freeze and unfreeze i/o operations around the virtual machine snapshot.

Configure and validate file system consistency for Linux virtual machines:

1. Ensure that `virt-launcher` pod has the appropriate Velero annotations:

   ```
   pre.hook.backup.velero.io/command:
                 '["/usr/bin/virt-freezer", "--freeze", "--name", "rhel9-lovely-louse",
                 "--namespace", "test-vm-ns"]'
   ```

   ```
   post.hook.backup.velero.io/command:
                 '["/usr/bin/virt-freezer", "--unfreeze", "--name", "rhel9-lovely-louse",
                 "--namespace", "test-vm-ns"]
   ```

2. Ensure that the QEMU guest agent (`qemu-guest-agent`) is installed in the Linux virtual machine. See [OpenShift documentation](#).
3. Add the default service account security context constraint `kubevirt-controller`:

   ```
   oc adm policy add-scc-to-user
                 kubevirt-controller -z default -n <namespace>
   ```

4. Disable SELinux inside the virtual machine.

Configure and validate file system consistency for Microsoft® Windows virtual machines:

1. Ensure that `virt-launcher` pod has the appropriate Velero annotations:

   ```
   pre.hook.backup.velero.io/command:
                 '["/usr/bin/virt-freezer", "--freeze", "--name", "rhel9-lovely-louse",
                 "--namespace", "test-vm-ns"]'
   ```

   ```
   post.hook.backup.velero.io/command:
                 '["/usr/bin/virt-freezer", "--unfreeze", "--name", "rhel9-lovely-louse",
                 "--namespace", "test-vm-ns"]
   ```

2. Ensure that the QEMU guest agent (`qemu-ga-x86_64`) is installed in the Microsoft Windows virtual machine and that the “QEMU Guest Agent” network service is running. See [OpenShift documentation](#).

3. Ensure that the `Volume Shadow Copy` service is running in the Microsoft Windows virtual machine.
4. Add the default service account security context `constraint` “`kubevirt-controller`“:

   ```
   oc adm policy
                 add-scc-to-user kubevirt-controller -z default -n
             <namespace>
   ```

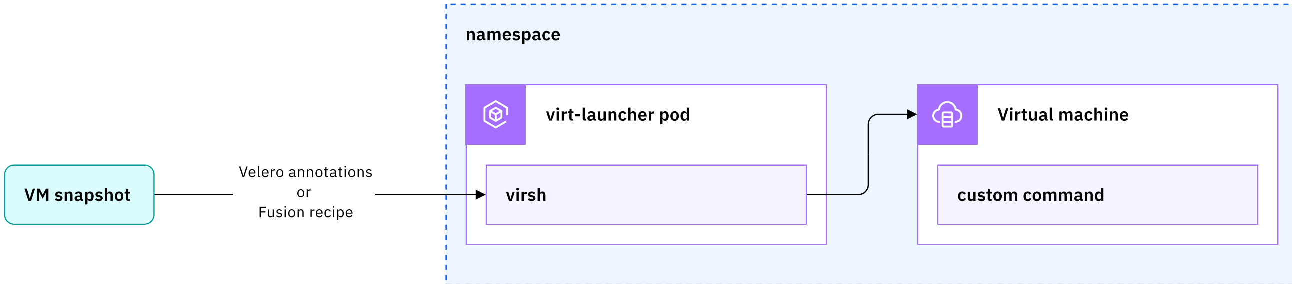
Crash consistency

You can achieve crash consistency by taking a snapshot of the virtual machine without engaging the QEMU guest agent to do a file system freeze. It an be done in one of following two ways:
Note: QEMU guest agent is a helper daemon. It is installed in the guest to exchange information between the host and guest, and to run command in the guest.

1. Remove the Velero pre-snapshot (`pre.hook.backup`) and post-snapshot (`post.hoook.backup`) annotations from the `virt-launcher` pod. This change does not persist across instantiations of the `virt-launcher` pod. The Velero annotations are automatically created along with a virtual machine and must be removed for each new instantiation of the `virt-launcher` pod.
2. Ensure that the QEMU guest agents are not installed or running in a virtual machine guest.

Application consistency

If you have a specific application consistency requirement that cannot be met by using the file system consistency, you can create a backup workflow that communicates directly with the application by using the virtual shell tool (`virsh`) instead of the `virt-freezer` tool. The following diagram depicts the scenario:



Consider the following alternative approaches run an application command by using `virsh`:

- Velero annotations to run an application command by using `virsh`:
  - You can directly modify the Velero annotations on the `virt-launcher` pod to run `virsh` (/usr/bin/virsh) instead of `virt-freezer` (/usr/bin/virt-freezer).
  - The modifications to the Velero annotations do not persist across instantiations of the `virt-launcher` pod. The Velero annotations are automatically created along with the virtual machine and must be modified for each new instantiation of the `virt-launcher` pod.
- Using IBM Fusion backup and restore recipes to run an application command by using `virsh`:
  Alternatively, you can create an IBM Fusion backup and restore recipe to run an application command by using `virsh`. Even though no differences exist in the outcome as compared to `virsh` by using the Velero annotations, consider the following points:
  - There might be instances wherein you must run more than one command to produce the wanted results. For example, issuing a command that locks a database for write and a second command to monitor or validate the lock. In these cases, a IBM Fusion recipe might be easier to work with.
  - If you decide to use an IBM Fusion recipe, the Velero annotations are also acceptable but it might produce undesirable results. For this reason, consider the removal of Velero annotations from the `virt-launcher` pod.
  - Use the recipe exec hooks to run `virsh` in the wanted sequence.

Orchestrate a backup or restore

Use a custom, application-specific recipe to consistently automate and control the required backup or restore sequence of an application.

For applications that support IBM Fusion Backup & Restore recipes, see Fusion Backup & Restore Recipes - Supported Applications.

Important: In Backup & Restore service, it is recommended to remove all Velero hooks from your applications before you use the recipe. Otherwise, the recipe and Velero hooks can contradict each other without your knowledge. If you want custom actions, then the recommendation is to use the recipe and not Velero hooks.

- **Custom backup and restore workflows**
  When you protect an application with Backup & Restore service, a default backup and restore workflow is used to protect an application. But while the backup and restore workflow is sufficient for some applications, there are some instances where you need to create a custom workflow for the backup and restore process.
- **Backup and restore of IBM Software Hub to the same or different cluster**
  IBM Software Hub is integrated with IBM Fusion HCI, which enables you to create online backups and to restore them to the same cluster or to a different cluster.
- **IBM Fusion repository for recipes**
  The IBM Fusion public GitHub repository provides helpful utilities for IBM Fusion and allows members of the broader IBM Fusion community to share and contribute tools and knowledge.

Custom backup and restore workflows

When you protect an application with Backup & Restore service, a default backup and restore workflow is used to protect an application. But while the backup and restore workflow is sufficient for some applications, there are some instances where you need to create a custom workflow for the backup and restore process.

- The default backup workflow that is used by the Backup & Restore service is as follows:
  1. Backup all namespace-scoped resources that are associated with the application and any dependent cluster-scoped resources.
  2. Run any `Velero pre-backup` hooks that are specified through pod annotations in an arbitrary order.
  3. Create snapshots of all PVCs associated with the application.
  4. Run any `Velero post-backup` hooks that are specified through pod annotations in an arbitrary order.
  5. If the backup policy specifies backup to an object storage, then copy the data to object storage.
- The default restore workflow that is used by the Backup & Restore service is as follows:
  1. Restore all PVCs.
  2. Restore all resources in an arbitrary order.
  3. Run `Velero post-restore` hooks that are specified through pod annotations in an arbitrary order.

You need a recipe to create a custom workflow for the backup and restore process. For more information, see Creating a Recipe.

- **Creating a Recipe**
  You need a recipe to create a custom workflow for the backup and restore process.
- **Assigning a Recipe**
  You can assign a recipe to your backup and restore operations.
- **Resource Transformation**
  Use the instructions to create a transform CR.

Creating a Recipe

You need a recipe to create a custom workflow for the backup and restore process.

The recipe custom resource has three basic specification elements:

1. Groups: A group defines a set of resources or PVCs that are processed together within a backup or restore. For example, a group can be a subset of specific resources that are specified with an exclude statement or an include statement.
2. Hooks: A hook can be used to start external scripts before and after snapshots, scale deployments up and down, or wait for certain conditions. For example, pods starting up and so on.
3. Workflows: A workflow defines the sequence of steps for a backup or restore operation, specifically the order in which the groups of resources and PVCs must be processed and any hooks that need to be applied.

For example:

```
apiVersion: spp-data-protection.isf.ibm.com/v1alpha1
kind: Recipe
metadata:
  name: wp-recipe
  namespace: wordpress
spec:
  appType: wordpress
  groups:
    - name: mysql_data
      type: volume
      labelSelector: app=wordpress
    - name: frontend
      labelSelector: tier in (frontend),app=wordpress
    - name: backend
      type: resource
      labelSelector: tier notin (frontend)
  hooks:
  - name: demoexechook
    type: exec
    namespace: ${GROUP.wp-app.namespace}
    nameSelector: wordpress-mysql.*
    ops:
    - name: pre
      command: >
        ["/bin/bash", "-c", "echo 'This is pretest' > /tmp/cfg_map_pre.txt"]
      container: mysql
    - name: post
      command: >
        ["/bin/bash", "-c", "echo 'This is posttest' > /tmp/cfg_map_post.txt"]
      container: mysql
  - name: demoscalehook
    type: scale
    namespace: ${GROUP.wp-app.namespace}
    selectResource: statefulset
    nameSelector: wordpress$
  workflows:
  - name: backup
    sequence:
    - hook: demoscalehook/down
    - group: frontend
    - group: backend
    - hook: demoscalehook/sync
    - hook: demoexechook/pre
    - group: mysql_data
    - hook: demoexechook/post
    - hook: demoscalehook/up
  - name: restore
    sequence:
    - group: mysql_data
    - group: backend
    - group: frontend
```

Specifying a group

Groups that are defined in the recipe refine the scope of the parent group, which is completely defined by the application CR. For more information about specifying groups in the application CR, see Assigning a Recipe CR with an application CR.
The following fields are mandatory for creating a group:

- `name`: name of the group that is specified in the workflow.
- `type`: type of group, either volume or resource.

The following fields are optional for creating a group:

- `LabelSelector`: Enables you to define multiple label selectors, and includes any resource or volume matching to the selectors in the backup and restore. For more on how to specify labels using or `LabelSelectors`, see Velero Backup API documentation .
- `backupRef`: Reference to a group used in a backup workflow. It is helpful for creating a subset of a backup group for a restore workflow.
- `includeResourceTypes`: list of resource type names to include.
  - It applies to resource groups only.
    For example, deployments.
  - If this field is not specified, then all resource types are included.
- `excludeResourceTypes`: list of resource type names to exclude.
  - It applies to resource groups only.
    For example, deployments.

- `excludeResourceTypes` has precedence over `includeResourceTypes`.
- `essential`: determines whether the group represents volumes or resources, which are essential to ensuring that the backup can produce a successful restore (true or false).
  - Groups specified as essential (default) need to be processed successfully. Otherwise, the overall backup operation fails.
  - If the backup operation fails, a rollback is initiated, and the workflow is not processed any further.
  - For non-essential groups, an unsuccessful operation of the group is tolerated; the processing of the workflow continues, and the overall backup is reported as successful while only this operation is marked as failed.
  - The flag is only considered whether the workflow is configured with `fail-on=essential-error`. See the workflow section for details.
- `labelSelector`: select volumes and resources by label. For more information about on specifying labels, see Labels and Selectors.
- `nameSelector`: select volume or associated workloads by object name.
  - This field applies to volume groups only.
  - When combined with `labelSelector` or logic applies.
- `selectResource`: select which resource types the fields `labelSelector` and `nameSelector` must apply when you are selecting PVCs directly or indirectly through the workloads that the PVCs are associated with.
  - This field applies to volume groups only.
- `includeClusterResources`: determines how cluster-scoped resources are processed (true, false, or not specified).
  - `not specified`: include only cluster-scoped resources that are associated with the namespace-scoped resources that are included in the groups.
  - `false`: do not include any cluster-scoped resources.
  - `true`: include all cluster-scoped resources (still considering other clauses and label selectors).
  - This field applies to only resource groups.
- `includedNamespaces`: specify list of namespaces that are included.
  Note: Use `includedNamespaces` and `excludedNamespaces` when you have applications that can stretch multiple namespaces. If the application contains only one namespace, then the namespace that is used in the recipe is obtained from the application (applications.application.isf.ibm.com) resource. In this case, make sure that your recipe is more portable by avoiding including the namespace declaration explicitly.
- `excludedNamespaces`: specify a list of namespaces that are excluded.
  - This parameter has precedence over `includedNamespaces`.
    Note: Use `includedNamespaces` and `excludedNamespaces` when you have applications that can stretch multiple namespaces. If the application contains only one namespace, then the namespace that is used in the recipe is obtained from the application (applications.application.isf.ibm.com) resource. In this case, make sure that your recipe is more portable by avoiding including the namespace declaration explicitly.
- `restoreOverwriteResources`: Specify whether to overwrite resources during restore. By default, the value is false. If this field is set, then you must specify `backupRef`.
- `restoreStatus`: This field helps to preserves the statuses of the resources.
  For example:

```
            - name: keycloak-resources
              type: resource
              includedNamespaces:
              - cp4i
             labelSelector: foundationservices.cloudpak.ibm.com=keycloak
             restoreStatus:
               includedResources:
               - integrationkeycloakclients.keycloak.integration.ibm.com
               - integrationkeycloakusers.keycloak.integration.ibm.com
```

  In this example, the statuses of the `integrationkeycloakclients.keycloak.integration.ibm.com` and `integrationkeycloakusers.keycloak.integration.ibm.com` are backed up and restored.

Specifying a hook

Hooks can be used to start external scripts before and after snapshots, scale deployments up and down, or wait for certain conditions (such as pods starting up). The following types of hooks are supported:

1. `exec`: Start arbitrary commands and scripts within target containers.
2. `scale`: scale workload resources up and down.
   Note: New resource types are support for both `exec` and `scale` hooks:
   - For `exec` hook: `daemonset`
   - For `scale` hook: `daemonset`, `job` and `cronjob`.
     Both Singular and fully qualified resource types are supported for `exec` and `scale` hooks.

     For example:

Singular	Plural
`pod`	`pods`
`deployment`	`apps/v1/deployments`
`satatefulset`	`apps/v1/satatefulsets`
`daemonset`	`apps/v1/daemonsets`
`job`	`batch/v1/jobs`
`cronjob`	`batch/v1/cronjobs`

3. `check`: check for conditions on workload resources (such as `readiness`, `replicaCount` and so on).
   Important: You can have multiple logical expressions in check hooks using AND (&&) and OR (||) operators. It is also possible to create complex groupings using parentheses.
   For example:

```
{$.spec.replicas} == {$.status.readyReplicas}

{$.spec.replicas} == {$.status.readyReplicas} || {$.metadata.generation} == {$.status.observedGeneration} &&
{$.spec.progressDeadlineSeconds} == {"500"}

({$.spec.replicas} == {$.status.readyReplicas}) || (({$.metadata.generation} == {$.status.observedGeneration}) &&
{$.spec.progressDeadlineSeconds} == {"500"})
```

4. The following are the additional hooks that you can create in the IBM Fusion recipe improve its ability to perform application backup and restore operations. For more information, see Additional hooks.
   - Job hooks
   - Label hooks

- Annotation hooks

The following fields are mandatory for creating a hook:

- `name`: the name of the hook that is specified in the workflow.
  - The name must be unique within the Recipe CR.
- `namespace`: the namespace to which the hook applies.
  - Namespace can be either specified directly (not recommended) or indirectly by the effective namespaces of a group `${GROUP..namespace}` or using a variable from the application CR `${VARIABLE_NAME`. In either case, you need to resolve to a single namespace.
- `type` : the type of hook, either exec, scale, or check.
- `selectResource`: workload resource type to that a hook applies. Specify the resource by using the fully qualified resource name. For example, `orchestrator.aiops.ibm.com/v1alpha1/installations`. The exception to this rule is that you can use the short names for commonly used resources pod, deployment, or `statefulset` (fully qualified resource name not required for these resources).
  - For exec hooks, defaults to pod.
  - This field is only required for hook types of scale or check.

  If you add a workload resource type that is specified by using the fully qualified resource name, add additional read permissions for the resource by updating the transaction manager `clusterroles` (`ibm-backup-restore` namespace) on the cluster (IBM Fusion hub or spoke) where the application is deployed. For example, if you plan to add the custom resource `orchestrator.aiops.ibm.com/v1alpha1/installations`, update the transaction manager `clusterroles` with the following command:

  ```
  oc edit clusterrole transaction-manager-ibm-backup-restore
  ```

  Add the following fields to the `clusterrole`:

  ```
  - verbs:
        - get
        - list
      apiGroups:
        - orchestrator.aiops.ibm.com
      resources:
        - installations
  ```

  Note: If you specify a resource of type pod, deployment, or `statefulset`, you do not have to update the transaction manager `clusterroles`.
- `labelSelector` or `orlabelSelector`: If specified, then the workload resource needs to match this label selector.
  Important:
  - The `labelSelector` includes `AND` condition for labels and both volume and resources of the recipe groups. It is comma separated (label) string that translates to AND operation later during resource selection.
  - The `orlabelSelector` includes `OR` condition for labels and both volume and resources of the recipe groups.
  - Either select `labelSelector` or `orlabelSelector`.
  - Either `labelSelector` or `nameSelector` is required. If both are specified, or logic applies.
- `nameSelector`: if specified, then the workload resource needs to match this expression.
  - Either `labelSelector` or `nameSelector` is required. If both are specified, or logic applies.

The following fields are optional for creating a hook:

- `singlePodOnly`: flag (true or false) that indicates whether to run a command on a single pod or on all pods that match the selector (when `selectResource=pod`). The hook is run on one of the matching pods only, but, which one is arbitrary.
  - For deployments and statefulsets (`selectResource=deployment or statefulset`), the option applies to each replicaset of the selected workload resources individually. For example, when two deployments are selected, the hook is run on one of the pods for each deployments replicaset, which pod is arbitrary.
  - This option applies only to exec hooks.
  - The default value is false.
- `essential`: flag (true or false) that indicates whether the hook is essential for a successful backup. Hooks that are defined as essential (which is the default) need to be processed successfully. Otherwise, the overall backup operation is considered as failed. In this case, a rollback is initiated, and the workflow is not processed any further.
  - For nonessential hooks, an unsuccessful execution is tolerated. The processing of the workflow continues, and the backup is reported as successful.
  - The flag is only considered whether the workflow is configured with `fail-on=essential-error`.
- `onError`: Determines the default behavior (fail or continue) if there is failures when an operation applies to multiple resources such as multiple pods. The option `fail` cancels the operation on the first occurrence of a failure, while `continue` continues processing of the remaining resources. In either case, the overall result of the operation is considered a failure.
- `timeout`: The default timeout (an integer value specified in seconds) applies to custom and built-in operations. If not specified in IBM Fusion 2.8.0, the default is 300 seconds. If not specified in IBM Fusion 2.8.1, the default is 10 days.

Ops specifications (optional)

- `ops`: set of operations that the hook can be started for.
  - Exec hooks provide you to specify any number of custom operations.
  - Other hook types have built-in operations.
- `ops/name`: name of the operation. This field needs to be unique within the hook.
- `ops/container`: the container where the command must be run.
  - The default is to pick an arbitrary container within the pod.
- `ops/command`: the command to run.
  - If you need multiple arguments, specify the command as a JSON array as single string, such as `["/usr/bin/uname", "-a"]`.
    Note: If you want to save embracing quotation marks so that you can use it within the command, use > to start a block scalar.
  - Variable substitution is applied, both intrinsic group namespace variables (`${GROUP...}`) and variables from application CR can be used.
  - For intrinsic group namespace variables, if there are multiple namespaces it yields a comma-separated list without spaces.
- `ops/onError`: specify fail or continue to provide the option to overwrite the identical option of the owning hook just for this operation. Defaults to the setting of the owning hook.
- `ops/timeout`: timeout (an integer value specified in seconds) applied to the specified operation. If specified, it overrides the timeout that is specified on the hook level. The hook is considered in error if the command exceeds the timeout.
  - Defaults to timeout set on hook level.
- `ops/inverseOp`: for rollback scenarios, it gives you the ability to rollback an operation by providing the name of another operation that reverses the effect of this operation.

For example, resume a database would be the revert operation for a failure on a suspend a database operation.

Chks specifications (optional)

- `chks`: set of checks that the hook can apply to the target workload. Check that hooks provide an option to specify any number of custom checks based on conditions that are formulated as `JsonPath` like expressions.
- `chks/name`: name of the check. The name must be unique within the hook.
- `chks/condition`: the condition that needs to be true to release the hook. Conditions are specified as `JsonPath` like expressions.
  - The expression is expected to be a Boolean.
- `chks/onError`: specify fail or continue to overwrite the identical option of the owning hook just for this operation. Defaults to the setting of the owning hook.
- `chks/timeout`: timeout (an integer value specified in seconds) applied to the specified operation. If specified, it overrides the timeout that is specified on the hook level. The hook is considered in error if the command exceeds the timeout.
  Examples of specifying conditions as `JsonPath` expressions:
  1. Check to see if the number of available replicas matches the specified number of replicas.
     ```
     condition: "{$.spec.replicas} == {$.status.readyReplicas}"
     ```
  2. Check if a replica is ready.
     ```
     condition: "{$.spec.replicas} ==
     {$.status.readyReplicas}"

     condition: "{$.status.containerStatuses[0].ready}
     == {True}"
     ```

Specifying a workflow
A workflow defines the sequence of steps for both backup and restore operations. At least one backup workflow (named "backup") and one restore workflow (named "restore") must be specified in the recipe CR. More or alternative workflows can be specified that can be referenced in on-demand backup or restore requests (overriding the default ones "backup" and "restore").

The following fields are mandatory for creating a workflow:

- `name`: name of workflow.
  - The names "backup" and "restore" are reserved and implicitly used by default for backup or restore.
- `sequence`: sequence of steps to be performed.
  - Sequence can refer to a series of groups, resources, and hooks in a specific order. The actions are processed sequentially in the specified order.

The following fields are optional for creating a workflow:

- `sequence/step/group`: refer to a group specified.
- `sequence/step/hook`: refer to a hook and one of its operations.
- `failOn`: determines the behavior if failures when processing groups or hooks. The following is one of these values:
  - `any-error`: If this value is specified, the operation fails and performs defined rollback operations if any step of the workflow fails.
  - `essential-error`: if this value is specified, the operation fails and performs defined rollback operations if any step of the workflow that is defined as essential fails.
  - `full-error`: if this value is specified, the entire workflow is attempted. The workflow is only considered a complete failure if all essential steps fail.

- **Additional hooks**
  Use this section to understand the additional hooks and guidelines on how to create additional hooks.
- **Dynamic recipe**
  Use this section to create a dynamic recipe and guidelines on how to create it.

Additional hooks

Use this section to understand the additional hooks and guidelines on how to create additional hooks.

The following newly added additional hooks in the IBM Fusion recipe improve its ability to perform application backup and restore operations.

- Job hook
- Label hook
- Annotation hook

Job hook

A job hook supports to run custom operations during specific events, such as backups, restores or other workflows. It provides a flexible way to run long-running commands, provide dedicated environments for tasks, or perform batch processing and one-time tasks.

Note: Ensure that you use job string format either JSON or YAML.

Sample JobHook: JobSpec as YAML string:

```
hooks:
    - jobs:
        - forceCreate: true
          name: my-job1
          onError: fail
          timeout: 60
          inverseOp: ""
      name: job-hook-test1
      namespace: workload
      onError: fail
      timeout: 60
      type: job
  jobs:
```

```
    - my-job1: |
        apiVersion: batch/v1
        kind: Job
        metadata:
          name: sample-job
          labels:
            label-key1: value1
          annotations:
            annotation-key1: value2
        spec:
          template:
            spec:
              containers:
              - command:
                - perl
                - --version
                image: perl:5.34.0
                name: perl-version
              restartPolicy: Never
```

Sample JobHook: JobSpec as JSON string:

```
hooks:
    - jobs:
        - forceCreate: true
          name: my-job1
          onError: fail
          timeout: 60
          inverseOp: ""
      name: job-hook-test1
      namespace: workload
      onError: fail
      timeout: 60
      type: job
  jobs:
    - my-job1: |
            {
              "apiVersion": "batch/v1",
              "kind": "Job",
              "metadata": {
                "name": "sample-job",
                "labels": {
                  "label-key1": "value1"
                },
                "annotations": {
                  "annotation-key1": "value2"
                }
              },
              "spec": {
                "template": {
                  "spec": {
                    "containers": [
                      {
                        "command": [
                          "perl",
                          "--version"
                        ],
                        "image": "perl:5.34.0",
                        "name": "perl-version"
                      }
                    ],
                    "restartPolicy": "Never"
                  }
                }
              }
            }
```

The default fields for creating a job hook:

- Job hook jobs **forceCreate**: defaults to **false**
- Job hook jobs **timeout**: if not present, defaults to job hook timeout
- Job hook jobs **onError**: defaults to **fail**
- Job hook **timeout**: defaults to **864000** seconds
- Job string **name**: defaults to job hook job name
- Job string **namespace**: defaults to job hook namespace
- Job string **apiVersion**: defaults to **batch/v1**
- Job string **kind**: defaults to **Job**
- Job string missing **backoffLimit** value: defaults to **0**

Label hook

A label hook modifies a resource by applying a new label or removing an existing one.

Sample LabelHook:

```
 hooks:
    - name: add-labels
      nameSelector: filebrowser*
      namespace: workload
      onError: fail
      timeout: 60
      ops:
```

```
        - labels: for-backup=true,myapp.io/app-name=filebrowser                # add labels
          name: filebrowser-resources-add-labels
          onError: fail
          overwrite: true
          timeout: 60
          inverseOp: filebrowser-resources-remove-labels
        - labels: for-backup-,myapp.io/app-name-                               # remove labels
          name: filebrowser-resources-remove-labels
          onError: fail
          overwrite: true
          timeout: 60
      selectResource: pod
      type: label
```

The default fields for creating a label hook:

- Ops `timeout`: defaults to label hook `timeout`
- Ops `onError`: defaults to `fail`
- Ops `overwrite`: defaults to `false`
- Label hook `timeout`: defaults to 864000 seconds
- Label hook `onError`: defaults to `fail`

Annotation hook

An annotation hook modifies a resource by applying a new annotation or removing an existing one.

Sample AnnotationHook:

```
hooks:
    - labelSelector: app=fb
      name: add-annotations
      namespace: workload
      onError: fail
      timeout: 60
      ops:
        - annotations: for-backup=true,myapp.io/app-name=filebrowser-workload         # add annotations
          name: filebrowser-resources-add-annotations
          onError: fail
          overwrite: true
          timeout: 60
          inverseOp: filebrowser-resources-remove-annotations
        - annotations: for-backup-,myapp.io/app-name-                                 # remove annotations
          name: filebrowser-resources-remove-annotations
          onError: fail
          overwrite: true
          timeout: 60
      selectResource: pod
      type: annotation
```

The default fields for creating an annotation hook:

- Ops `timeout`: defaults to annotation hook `timeout`
- Ops `onError`: defaults to `fail`
- Ops `overwrite`: defaults to `false`
- Annotation hook `timeout`: defaults to 864000 seconds
- Annotation hook `onError`: defaults to `fail`

Dynamic recipe

Use this section to create a dynamic recipe and guidelines on how to create it.

Dynamic recipe provides a capability to handle the backup and restore of applications with multiple sub-components dynamically. With a dynamic recipe, you can define a parent recipe that automatically discovers child recipes that provide extra workflow information, making your recipes more dynamic.

- Creating a parent recipe
- Creating a child recipe
- Adding new workflows with priorities
- Adding new workflows without a sequence
- Pre-recipe inventory workflow (Optional)
- Parallel backup hook execution
- Parallel restore hook execution

Some of the important considerations you should follow when you create a dynamic recipe:

- Ensure that child recipes correctly reference the parent recipe by using labels.
- You can define new workflows apart from reserved `backup` and `restore` workflows.
- If a child recipe uses the same workflow as the parent with a higher priority value, then the child recipe runs first.
- If a child recipe shares the same workflow and priority value as its parent, then the execution order is random but they run sequentially.
  The priority value can range from `-2,147,483,648 to 1,000,000,000`.

  For example:
    - A workflow with priority `-2,147,483,648` contain the lowest precedence.
    - A workflow with priority `1,000,000,000` contain the highest precedence.

Creating a parent recipe

In addition to the standard `backup` and `restore` `workflows`, you can define the new `workflows` in the dynamic recipe. These `workflows` enable dynamic behavior during the backup and restore processes.
Note: Parent and child can have same hook or group name but during recipe merge process child hook name is going to be prefixed automatically with child recipe name separated by period.
For example:

```
apiVersion: spp-data-protection.isf.ibm.com/v1alpha1
kind: Recipe
metadata:
  name: parent-recipe
  namespace: ibm-spectrum-fusion-ns
spec:
  ...
  hooks:
  ...
  workflows:
  - name: pre-hook
    priority: 4
    sequence:
    - hook: mysql-pod-exec/flush-tables-with-read-lock
  - name: post-hook
    priority: 4
    sequence:
    - hook: mysql-deployment-check/replicasReady
  - name: backup
    sequence:
    - group: mysql-resources
    - workflow: pre-hook
    - group: mysql-volumes
  - name: restore
    sequence:
    - group: mysql-volumes
    - group: mysql-resources
    - workflow: post-hook
```

Ensure that the names can be any valid string or similar to other workflow names like `backup` and `restore`. In this scenario, pre-hook is used in the backup workflow and post-hook is used in the `restore` workflow.
If a child recipe defines the same workflow, then it dynamically retrieves and runs.

Creating a child recipe

After a recipe linked to a another recipe through these labels `dp.isf.ibm.com/parent-recipe` and `dp.isf.ibm.com/parent-recipe-namespace`, then parent-child relationship gets created.

For example:

```
apiVersion: spp-data-protection.isf.ibm.com/v1alpha1
kind: Recipe
metadata:
  labels:
    dp.isf.ibm.com/parent-recipe: parent-recipe
    dp.isf.ibm.com/parent-recipe-namespace: ibm-spectrum-fusion-ns
  name: child-1
  namespace: ibm-spectrum-fusion-ns
spec:
  appType: mysql-child-1
  groups:
  ...
  hooks:
  ...
  workflows:
  - name: pre-hook
    priority: 5000
    sequence:
    - group: cm-child-1
    - hook: mysql-pod-exec-child-1/create-temp-file-pre
  - name: post-hook
    priority: 5000
    sequence:
    - hook: mysql-pod-exec-child-1/create-temp-file-post
    - group: cm-child-1
```

Adding new workflows with priorities

A child recipe can define workflows that are present in the parent recipe.

Adding a new workflow

For example, add a new workflow named `demo-workflow` with priority 1 in the parent recipe.

```
workflows:
  - name: demo-workflow
    priority: 1
    sequence:
      - hook: demo-hook/echo-test
```

Now, you can define the same workflow in the child recipe. The execution behavior depends on the assigned priority:

- High priority - Runs before the parent workflow

- Low priority - Runs after the parent workflow
- Same priority - Run order is arbitary (either the parent or child might run first).

Child workflow with higher priority
: For example, the child recipe defines a demo-workflow with priority 5, then it runs before the parent workflow.

```
workflows:
  - name: demo-workflow
    priority: 5
    sequence:
      - hook: demo-child-hook/echo-child-test
```

Adding new workflows without a sequence

A parent recipe can define workflows that have no sequence. This allows child recipes to use these workflows and define their own sequences.

Defining a workflow in the parent recipe
: For example, add a new workflow named `demo-workflow` in the parent recipe with no sequence and priority 0:

```
workflows:
  - name: demo-workflow
    priority: 0
    sequence: []
```

Defining a workflow in the child recipe
: Now, the child recipe can define the same workflow and specify its own sequence:

```
workflows:
- name: demo-workflow
priority: 5
sequence:
- hook: demo-child-hook/echo-child-test1
- hook: demo-child-hook/echo-child-test2
```

Important:

- If the parent has only one child and parent workflow sequence is empty, then priority does not matter, and the child workflow executes as defined.
- If the parent has multiple children defining the same workflow, execution occurs according to the assigned priorities.

Pre-recipe inventory workflow (Optional)

This is an optional workflow for the parent recipe, run before the backup workflow.

- It allows an application to define and run its own custom exec hook in the parent recipe before any child recipe is merged into parent.
- Upon successful execution, IBM Fusion continues processing and takes the child recipe inventory for dynamic recipe execution.
- The main objective is to construct child recipes that reflect the current state of the application in the cluster.

For example:

```
apiVersion: spp-data-protection.isf.ibm.com/v1alpha1
kind: Recipe
metadata:
  name: parent-recipe
  namespace: ibm-spectrum-fusion-ns
spec:
  appType: mysql-ns
  groups:
  ...
  hooks:
  - name: my-child-recipes
    type: exec
    ...
    - name: "manage"
...
  workflows:
  - name: prerecipeinventory
    sequence:
    - hook: my-child-recipes/manage
  ...
```

Parallel backup hook execution

Running sequences in parallel within dynamic recipe workflows can significantly speed up the overall process. It is useful for workflows that involve tasks that depend heavily on I/O operations. If any one sequence fails, then the entire job aborts, but it waits for any currently running sequences in other parallel blocks to finish first.
Note: Undo operations are still performed sequentially, but in the reverse order of workflow execution.
Parallel hook execution can be requested in a IBM Fusion recipe by assigning the same `priority` to the workflow step in each recipe.

Key considerations

- Parallel execution is available for both backup and restore workflows.
- Parallel execution only can be used with dynamic recipes (when there is a parent recipe and at least one child recipe).
- If there are multiple recipes, consider running them simultaneously to increase efficiency.
  - Workflows with same priority run in parallel only if all sequences are hooks.
  - Workflows with same priority run sequentially if they have at least one `group` sequence within the workflow.
  - Workflows with different priorities run sequentially.
  - Workflows with mixed priorities: For example, there is a shared workflow that is defined in three recipes, two have the same priority and one is a different priority. In this particular case, the workflows with identical priorities run in parallel and the recipe with the different priority runs

sequentially, before or after depending on the relative priorities.

- Number of parallel workers are controlled by flag **hook-parallel-workers** in **guardian-configmap** (ibm-backup-restore namespace). The default value is 1 and can be set to higher values with maximum of 35. Minimum value 0 have special meaning - internally translates to number of usable CPUs.

Parallel restore hook execution

IBM Fusion's parallel restore mechanism follows the same approach as its backup process. During restoration, the system dynamically merges saved parent-child recipes at runtime to build the final restore workflow. If a child recipe is not eligible for restoration to an alternate namespace, the system skips it during the merge. The status field in the Fusion Restore Custom Resource (CR) records details of any skipped recipes.

To perform a parallel restore with modified recovery steps (distinct from the saved parent recipes), create a custom **Restore** CR. Specify the new recipe **name** and **namespace** in the **spec** field.

Key considerations

Parallel restore adheres to all the key considerations outlined for backup. Additionally, it possesses specific capabilities that further contribute to successful application recovery.

- The system excludes child recipes that are labeled with **dp.isf.ibm.com/alternate-namespace-restore=false** when you target alternate namespaces.
- To introduce slight variations in restore steps, you can specify a new recipe **name** and **namespace** in the custom **Restore** CR. This existing capability is valuable especially during parallel restores.

Sample custom **Restore** CR

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Restore
metadata:
  name: <custom-restore-name>
  namespace: <custom-restore-namespace>
spec:
  recipe:
    name: <new-parent-recipe-name>
    namespace: <new-parent-recipe-namespace>
```

Parallel backup and restore can significantly optimize data protection times when processing Fusion recipes, which have multiple activities that can be executed in parallel.

Assigning a Recipe

You can assign a recipe to your backup and restore operations.

1. Assign a recipe with a backup policy assignment CR for backup and restore operations as follows:

   Assign a recipe for backup operations

   You can assign a recipe to an application or applications by specify the recipe in the PolicyAssignment CR to assign it to one or more applications. You can specify a single recipe in a PolicyAssignment CR.
   To assign a recipe to an application, you can fix the PolicyAssignment CR by using the following example:

   ```
   oc -n ibm-spectrum-fusion-ns patch policyassignment POLICY-ASSIGNMENT-NAME --type merge -p '{"spec":{"recipe":
   {"name":"RECIPE_NAME", "namespace":"RECIPE_NAMESPACE", "apiVersion":"spp-data-protection.isf.ibm.com/v1alpha1"}}}'
   ```

   - **POLICY_ASSIGNMENT-NAME** is the name as specified in the PolicyAssignment CR.
   - **RECIPE_NAME** is the name of the recipe as specified in the Recipe CR.
   - **RECIPE_NAMESPACE** is the namespace where the Recipe CR is located.

   Assign a recipe for restore operations when restoring to the original namespace

   During a restore operation, a recipe must be started with the following precedence:
   - If there is a Recipe CR with the same name and in the same namespace as the Recipe CR that was used during the backup operation, it must take precedence. It can be helpful in situations when you want to specify a different workflow or different hooks during the restore.
   - If there is no Recipe CR with the same name and in the same namespace, the recipe that was used during the backup operation must be run. This information is stored as part of the backup metadata.

   Assign a recipe for restore operations when restoring to a different namespace

   If you are restoring to an alternative namespace, any namespaces that are specified in a hook cannot be run correctly as they refer to the original namespace. If you want the hooks to run in the alternative namespace, change the Recipe CR as mentioned in Assign a recipe for restore operations when restoring to the original namespace. So that any hook namespaces are referring to the alternative namespace.

2. Assign a recipe CR with an application CR as follows:

   Reserved variables in application CR

   The Backup & Restore service initializes the following reserved variables. Do not redefine these variables in your application CR, except for **ISF_ALLOWED_UNRESOLVED_VARS**.
   - **ISF_BACKUP_UID**: the UUID of the backup job.
   - **ISF_NAMESPACE_MAPPINGS**: namespace mappings in a comma-separated string format, accessible in exec hook commands.

     ```
     "<source_namepace-1>:<target_namespace-1>,<source_namepace-2>:<target_namespace-2>,…"
     ```

   - **ISF_NOT_EQUAL_NAMESPACE_MAPPINGS**: an empty string when restoring to the same namespace, or the value of **ISF_NAMESPACE_MAPPINGS** when restoring to different namespaces.
   - **SERVICE_NAMESPACE**: a predefined value, "**common-service**", used by cp4d.
   - **${APP.${SERVICE_NAMESPACE}.nameEmptyIfNotMapped}**: checks if **SERVICE_NAMESPACE** exists, used by cp4d.

- `ISF_ALLOWED_UNRESOLVED_VARS`: This is Application variable used to skip verification for not found variables. Possible values are: `"true"` or `"false"`, default is `"false"`.

`ISF_ALLOWED_UNRESOLVED_VARS` is an application variable that skips verification for unresolved variables when set to `"true"`. Default is `"false"`. Set it to `"true"` if your application recipe uses shell variables and array syntax instead of substituting variables from the recipe.

```
spec:
  variables:
    - name: ISF_ALLOWED_UNRESOLVED_VARS
      value: 'true'
```

Working with application CRs for single namespace

For applications that only has a single namespace, there are no changes that are required in the application CR to support a recipe. The recipe is associated with the application, in this case with the PoliyAssignment CR, as defined in the Assign a recipe for backup operations.

Working with application CRs for applications that span multiple namespaces

If you have an application that spans multiple namespaces and want to use a recipe for the application, you need to specify all of the namespaces included in the application that is in the application CR.

- **Using variable substitutions**:
  - Variables can be used to permit a single recipe to be used in multiple, similar applications. Variables are defined in the application CR, and any number of variables can be defined.

    ```
    - variables:
       - name: <variable name>
           value: <variable value>
    ```

  - The variables that can be referenced within the Recipe CR.

    ```
    ${<variable name>}
    ```

    Valid characters for variables are: a-zA-Z0-9_-

  - Each group exposes to the following variables automatically.
    - Effective namespaces

      These can be used within recipe by these patterns.

      ```
      ${GROUP.<groupname>.namespaces} - effective namespaces of group
      ${GROUP.<groupname>.namespace}  - effective namespace of group - single namespace enforced, (runtime) validation error otherwise
      ```

  - Variables can solely be used for the following fields:
    - Namespace field of hooks.
    - Within commands of exec hooks.

  Tip: When creating a Recipe CR, place the Recipe CR in the `ibm-spectrum-fusion-ns` (or another common namespace). During a restore operation, you might need to change the recipe and don’t want the recipe CR to be in the application namespace, as this namespace might not be available during restore.

  For example,

  ```
  apiVersion: application.isf.ibm.com/v1aplha1
  kind: Application
  metadata:
    name: wp-app1
    namespace: ibm-spectrum-fusion-ns
  spec:
    appType: wordpress
    includedNamespaces:
      - wordpress
    variables:
    - name: WORDPRESS_NAMESPACE
      value: wordpress
  ---

  apiVersion: spp-data-protection.isf.ibm.com/v1alpha1
  kind: Recipe
  metadata:
    name: wp-recipe
    namespace: wordpress
  spec:
    appType: wordpress
    groups:
      - name: mysql_data
        type: volume
        labelSelector: app=wordpress
      - name: frontend
        type: resource
        labelSelector: tier in (frontend),app=wordpress
      - name: backend
        type: resource
        labelSelector: tier notin (frontend)
    hooks:
    - name: demoexechook
      type: exec
      namespace: ${WORDPRESS_NAMESPACE}
      nameSelector: wordpress-mysql.*
      ops:
      - name: pre
        command: >
          ["/bin/bash", "-c", "echo 'This is pretest' > /tmp/cfg_map_pre.txt"]
        container: mysql
      - name: post
        command: >
          ["/bin/bash", "-c", "echo 'This is posttest' > /tmp/cfg_map_post.txt"]
        container: mysql
  ```

```
  - name: demoscalehook
    type: scale
    namespace: ${WORDPRESS_NAMESPACE}
    selectResource: statefulset
    nameSelector: wordpress$
  workflows:
  - name: backup
    sequence:
    - hook: demoscalehook/down
    - group: frontend
    - group: backend
    - hook: demoexechook/pre
    - group: mysql_data
    - hook: demoexechook/post
    - hook: demoscalehook/up
  - name: restore
    sequence:
    - group: mysql_data
    - group: backend
    - group: frontend
```

Resource Transformation

Use the instructions to create a transform CR.

Before you begin

Ensure that the Backup & Restore service is installed. For steps to install Backup & Restore, [Install Backup & Restore](#).

About this task

When restoring data to a different namespace or cluster, some resource definitions might need to be updated. For example, hardcoded storage classes or domain names that do not match the new environment.

While IBM Fusion Backup & Restore supports you to restore applications to alternative namespaces or storage classes. However, sometimes you need to change a resource before restoring it. This process is called a transformation.

Transformations are in-memory modifications that are applied to resources during the restore process, ensuring that they are correctly configured for the target environment before being written to disk or applied to the cluster.

Procedure

1. Define a transform CR that contains all the necessary transformations. The transform CR must be referenced in the restore CR, which ensures that the transformations get applied.
   - Create transform CR
   - Create restore CR with a transform CR
   - Apply the restore CR
2. A transform CR can consist of multiple transform rules. Each transform must have the following three sections:

   Name
   : The name of each transform rule must be unique.

   Patch type
   : There are four types of patches that can be defined:
     - `Json patch`
     - `Merge patch`
     - `Strategic patch`
     - `String patch`

   Subject
   : The subject specifies the resources on which the patch gets to apply.
     - `groupResource` - Fully qualified resource name, such as
       - Core API group (v1) - pods, services, configmaps, secrets, persistentvolumeclaims, and so on.
       - Named API groups (apps/v1) - deployments.apps, statefulsets.apps, and so on.
       - Batch (batch/v1): jobs.batch, cronjobs.batch
       - networking.k8s.io/v1 - ingress.networking.k8s.io, networkpolicies.networking.k8s.io
     - `resourceNameRegex` - matching resource name
     - `labelSelector` - matching resource label
     - `namespaces` - the namespace where the resource is available

3. The four kinds of patches that can be applied in transform CR.

   JSON patch
   : A JSON patch supports operations like test, add, remove, and replace on specific paths. There are three fields need to define in the JSON patch:
     - **op** - operation name
       - `test` - to test if specific value is present on the given path
       - `add` - to add the value on the given path
       - `remove` - to remove the value on the given path
       - `replace` - to replace the value on the given path
       - `copy` - to copy the value from path to the path

- `move` - to move the value from path to the path
- **path** - on which the operation applies
- **value** - value on which these operations acted upon
- **from** - from which the value is copied

Note: For PVC, `copy` and `move` operation are supported for labels and annotations

For example, transform CR to transform multiple storage classes with JSON patch:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Transform
metadata:
  name: transform-copy-move
  namespace: ibm-spectrum-fusion-ns
spec:
  transforms:
  - name: configmap-move-labels-to-annotations
    json:
    - op: move
      from: /metadata/labels/a
      path: /metadata/annotations/a
    subject:
      groupResource: configmaps
      labelSelector: c=d
  - name: configmap-move-annotations-to-labels
    json:
    - op: move
      from: /metadata/annotations/m
      path: /metadata/labels/m
    subject:
      groupResource: configmaps
      labelSelector: c=d
  - name: pvc-move-labels-to-annotations
    json:
    - op: move
      from: /metadata/labels/a
      path: /metadata/annotations/a
    subject:
      groupResource: persistentvolumeclaims
      labelSelector: c=d
  - name: pvc-move-annotations-to-labels
    json:
    - op: move
      from: /metadata/annotations/m
      path: /metadata/labels/m
    subject:
      groupResource: persistentvolumeclaims
      labelSelector: c=d
  - name: pvc-copy-labels-to-annotations
    json:
    - op: copy
      from: /metadata/labels/i
      path: /metadata/annotations/i
    subject:
      groupResource: persistentvolumeclaims
      labelSelector: i=j
  - name: pvc-copy-annotations-to-labels
    json:
    - op: copy
      from: /metadata/annotations/p
      path: /metadata/labels/p
    subject:
      groupResource: persistentvolumeclaims
      labelSelector: i=j
  - name: sts-copy-image
    json:
    - op: copy
      from: /spec/template/spec/containers/0/image
      path: /spec/template/spec/initContainers/0/image
    subject:
      groupResource: statefulsets.apps
      labelSelector: a=b
  - name: sts-copy-memory-request-to-limit
    json:
    - op: copy
      from: /spec/template/spec/containers/0/resources/requests/cpu
      path: /spec/template/spec/containers/0/resources/limits/cpu
    subject:
      groupResource: statefulsets.apps
      labelSelector: a=b
```

Note: `${SOURCE_STORAGE_CLASS_FS}`, `${TARGET_STORAGE_CLASS_FS}`, `${SOURCE_STORAGE_CLASS_RBD}`, `${TARGET_STORAGE_CLASS_RBD}`, these are the placeholders here which can be pass dynamically using `transformMapping` in restore CR.

Merge patch

Merges provided fields into the target resource using a simple JSON structure. A specific field needs to be defined.

patchData - provide a JSON structure

For example, transform CR to add labels to `statefulsets` using merge patch.

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Transform
metadata:
  name: transform-using-merge-patch
  namespace: ibm-spectrum-fusion-ns
 - name: statefulsets-with-merge-patch-labels
```

```
    mergePatches:
    - patchData: |
        {
          "metadata": {
            "labels": {
              "modify-by": "${MODIFY_BY}"
            }
          }
        }
    subject:
      groupResource: statefulsets.apps
      labelSelector: 'a=b'
```

Note: `{MODIFY_BY}` is a placeholder here that can be pass dynamically using `transformMapping` in restore CR.

Strategic patch
: Merges fields intelligently based on resource schema, preserving arrays like containers by name. A specific field needs to be defined.
patchData - provide a JSON structure

For example, transform CR to update image version and add a `env` variable using strategic patch.

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Transform
metadata:
  name: transform-using-strategic-patch
  namespace: ibm-spectrum-fusion-ns
  - name: deployment-with-strategic-patch
    strategicPatches:
    - patchData: |
        {
          "spec":
            {
              "template":
                {
                  "spec": {
                    "containers": [
                      {
                        "name": "nginx-main",
                        "image": "nginx:1.25.6",
                        "imagePullPolicy": "Always",
                        "env": [
                          {
                            "name": "test",
                            "value": "${MODIFY_BY}"
                          },
                        ]
                      }
                    ]
                  }
                }
            }
        }
    subject:
      groupResource: deployments.apps
      resourceNameRegex: nginx-multi-container
```

String patch
: Replaces specific strings in a target resource using key-value pairs. Two specific fields needs to be defined.
- **key** - fully qualified path where string need to be replaced
- **replaceStrings**
  - `Old` - old string, which needs to be replaced
  - `New` - new string, which needs to be replaced on the place of old string.

For example, transform CR to transform data in the configmap using string patch.

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Transform
metadata:
  name: transform-using-string-patch
  namespace: ibm-spectrum-fusion-ns
spec:
  transforms:
  - name: configmap-with-string-patch
    stringPatch:
    - key: data/instance.json
      replaceStrings:
      - new: '\"persistence.cachingpv.storageClass\": \"${TARGET_STORAGE_CLASS_FS}\"'
        old: '\"persistence.cachingpv.storageClass\": \"transform\"'
    subject:
      groupResource: configmaps
      resourceNameRegex: dv-instance-config
```

4. After the transform CR is define, then create a restore CR with its reference.
   For example:

```
apiVersion: data-protection.isf.ibm.com/v1alpha1
kind: Restore
metadata:
  name: restore-with-transform
  namespace: ibm-spectrum-fusion-ns
spec:
  backup: <backup-name>
  targetCluster: <target-cluster-name>
  transform: transform-storage-classes
  transformMapping:
```

```
INGRESS_HOST:  transformed.com
SOURCE_STORAGE_CLASS_FS: ocs-storagecluster-cephfs
TARGET_STORAGE_CLASS_FS: ocs-storagecluster-cephfs-1
SOURCE_STORAGE_CLASS_RBD: ocs-storagecluster-ceph-rbd
TARGET_STORAGE_CLASS_RBD: ocs-storagecluster-ceph-rbd-1
MODIFY_BY: fusion-transform
```

Define the two new fields **transform** and **transformMapping** in the restore CR.

- **transform** - name of the transform CR.
- **transformMapping** - If required, pass the variable that can be substituted in the transform CR.

5. Apply the restore CR.

Note:

- Transform CR should be present on the target cluster where restoring the application.
- For **persistentvolumeclaims**, only JSON patch is supported, and all the 4 patches are supported for resources.
- The test operation in JSON patch is used in conjunction with other operations (**add, replace, remove**). If the test operation **fails**, the entire patch is aborted and none of the subsequent operations are run in that transform rule.
- Strategic Merge patch operates based on the name field within arrays (such as containers):
  - If the name field is missing, then the merge is skipped.
  - If a matching name field is present, then the existing entry is merged.
  - If a matching name field is not present, then a new entry is added to the array.
  - If a label key contains a slash (/), replace it with a (~) for it to be valid.
- For more examples of using Transform CR, see Sample Transform CR.

Transformation with custom recipe:

Note: With default recipe, this setting is not applicable.

To use the transformation with custom recipe, you need to add the following field in the group of the recipe

```
"disableTransform: false"
```

For example:

```
apiVersion: spp-data-protection.isf.ibm.com/v1alpha1
kind: Recipe
metadata:
  name: db2-prod-backup-recipe
  namespace: ibm-spectrum-fusion-ns
spec:
  appType: db2
  groups:
    - includedNamespaces:
        - db2-prod
      name: db2-volumes
      type: volume
    - includedNamespaces:
        - db2-prod
      includedResourceTypes:
        - configmaps
        - secrets
        - services
        - serviceaccounts
      name: db2-resources
      type: resource
    - includedNamespaces:
        - db2-prod
      includedResourceTypes:
        - statefulsets.apps
      name: db2-statefulsets
      type: resource
...
```

To enable the transformation, include **disableTransform: false** for this group **db2-statefulsets**.

```
- includedNamespaces:
        - db2-prod
      includedResourceTypes:
        - statefulsets.apps
      name: db2-statefulsets
      type: resource
      disableTransform: false
```

Backup and restore of IBM Software Hub to the same or different cluster

IBM Software Hub is integrated with IBM Fusion HCI, which enables you to create online backups and to restore them to the same cluster or to a different cluster.

To create IBM Software Hub backups with IBM Fusion or IBM Fusion HCI, see Creating and scheduling online backups of IBM Software Hub with IBM Fusion.

To restore IBM Software Hub online backup and restore to the same cluster with IBM Fusion or IBM Fusion HCI, see IBM Software Hub online backup and restore to the same cluster with IBM Fusion.

To restore IBM Software Hub backups to a different cluster with IBM Fusion or IBM Fusion HCI, see IBM Software Hub online backup and restore to a different cluster with IBM Fusion.

Remember: IBM Cloud Paks instance backups cannot be created when continuous backup is enabled on the MongoDB service.

Note: You can restore backup data to a different StorageClass provisioner on an alternate cluster, for example, restore data that was originally backed-up from a CephFS provisioned StorageClass to a Scale provisioned StorageClass.

The Restore CR includes an optional **spec** field, **targetStorageClass** where you can specify a specific storage class for the restore. It is only available through manually created Restore CRs from the command line.

- If the `targetStorageClass` is not provided, the same storage class as the backup is attempted for the restore.
- If the `targetStorageClass` field is not provided and the same storage class as the backup is not available, the default storage class of the system is used.

The `targetStorageClass` is not supported for IBM Software Hub.

IBM Fusion repository for recipes

The IBM Fusion public GitHub repository provides helpful utilities for IBM Fusion and allows members of the broader IBM Fusion community to share and contribute tools and knowledge.

This repository hosts the following categories of artifacts that can be used with IBM Fusion HCI:

- Multiple application backup and restore recipes using IBM Fusion. It includes recipes for applications such as IBM Db2, PostgreSQL, Postgres EnterpriseDB, MongoDB, Redis, and others. Check the Recipes repository for the complete inventory of applications.
- Maximo Core and Manage application recipes. For the IBM Fusion GitHub repository, see Maximo Fusion recipes.
- Helpful utilities and scripts to explore backup and restore
  For example, you can run getRecipeWorkflow.sh scripts from the command line to view recipe workflow logs and run script getResources.sh to get complete list of resources backed up or restored.

If you like to develop recipes for other popular applications, read the following blog series on creating backup and restore recipes:

1. How IBM Fusion Simplifies Backup and Recovery of Complex OpenShift Applications - Part 1 "So Why is this Complex?"
2. How IBM Fusion Simplifies Backup and Recovery of Complex OpenShift Applications - Part 2 "Orchestrations and Recipes"
3. How IBM Fusion Simplifies Backup and Recovery of Complex OpenShift Applications - Part 3 “A Deeper Dive into Recipes"

Also, for more recipes, see Orchestrate a backup or restore.

Performance and scaling

- **Backup & restore resource self-monitoring**
  The imposed load on the Backup & restore service has a direct correlation to the number of concurrent Backup & restore jobs in progress. While the service can catch up from periodic spikes in activity, a sustained elevated level of activity may cause any one Backup & restore microservice to fall behind in job processing, which may result in job failures.
- **Backup & restore hub performance and scaling**
  Use this topic to understand the number of spokes you can scale up to and the number of spoke clusters that can be connected to a single hub. It outlines the capacities and the scalability considerations to ensure efficient backup and restore operations.
- **Backup & restore configuration parameters**
  Configuration parameters allow customization of IBM Fusion Backup & restore settings.

Backup & restore resource self-monitoring

The imposed load on the Backup & restore service has a direct correlation to the number of concurrent Backup & restore jobs in progress. While the service can catch up from periodic spikes in activity, a sustained elevated level of activity may cause any one Backup & restore microservice to fall behind in job processing, which may result in job failures.

The **Backup and Restore hub microservices**—such as `backup-service`, `job-manager`, `application-service`, `backup-policy` and `backup-location`—now include built-in resource monitoring for Memory and CPU consumption. This monitoring capability helps identify potential resource shortages by providing a Warning alert through IBM Fusion events. This alert notifies the user of a sustained level of activity the current microservice cannot sustain and recommends scaling out the microservice with another replica to ensure successful Backup & restore job operations.

Resource Usage Monitoring and Thresholds

When Memory or CPU usage exceeds a defined threshold (default is 85%) for a specific duration (default is 5 minutes), a IBM Fusion event with a warning severity is triggered to notify about the increased resource usage. For instance, if the Memory or CPU usage for the `backup-service` pod crosses the threshold, the following event gets generated:

```
Pod backup-service-xxx in namespace ibm-backup-restore exceeded 85% of memory/CPU limit for 5 minutes.
Consider scaling the deployment with an additional replica.
```

These alerts are marked as fixed after the Memory or CPU consumption falls behind the threshold limit.

To scale out with another replica, use the following command to get the existing replica count for the deployment from the ‘AVAILABLE’ column:

```
oc get deployment -n <backup-restore-namespace> <deployment-name>
```

Then perform the following command with an increase of 1 to the existing replica count:

```
oc scale deployment -n <backup-restore-namespace> <deployment-name> --replicas=<new-replica-count>
```

Configurable parameters

You can customize the threshold levels and monitoring duration by adding or modifying parameters in the `guardian-configmap`. Here are the parameters available for configuration:

memoryThresholdPercentage

Sets the memory usage threshold as a percentage. Default is “85” for 85%.

cpuThresholdPercentage
Sets the CPU usage threshold as a percentage. Default is “85” for 85%.

monitorDurationInSeconds
Defines the duration (in seconds) that usage must exceed the threshold to trigger an alert. Default is 300 seconds (5 minutes).

Example configuration:

```
memoryThresholdPercentage: '85'
cpuThresholdPercentage: '85'
monitorDurationInSeconds: '300'
```

Backup & restore hub performance and scaling

Use this topic to understand the number of spokes you can scale up to and the number of spoke clusters that can be connected to a single hub. It outlines the capacities and the scalability considerations to ensure efficient backup and restore operations.

The Backup & restore hub service is efficiently designed to handle large-scale backup and restore concurrent jobs across clusters. With proven scalability and performance. The system is tested successfully to handle up to 1000 concurrent jobs. It serves as a reference point and not a limitation; the hub can scale further with appropriate resource allocation.

Sizing blueprints for varying workloads

The following table provides the recommended configurations for each required service when handling different numbers of concurrent jobs. As the number of concurrent jobs increases, you can scale out (increase the replicas) or scale up (increase the resource limits) for the required services as outlined:

Note:

- Services not listed in the blueprint table do not require scaling. These services can handle the load without modification because they do not have scaling requirements for up to 1000 concurrent jobs.
- Some services, such as **`mongodb-v2`** or **`operator-control-manager`** pods in the **`ibm-backup-restore`** namespace, cannot be scaled out by increasing replicas. For these services, increase the CPU and memory resources to ensure optimal performance as the number of concurrent jobs increases.
- The **`guardian-bridge`** service automatically scales to handle the additional load and provides backward compatibility when connected spokes are at a version earlier than IBM Fusion 2.13.

Table 1. Sizing blueprints

Clusters	Concurrent Jobs	Mongodb-v2				Kafka broker pool	Backup service	Backup policy service	Application service	Job manager service	Window end time (hr)
		Replicas	CPU	Memory	Ephemeral	Replicas	Replicas	Replicas	Replicas	Replicas	
10	100	3	1	1 GiB	0.5 GiB	3	1	1	1	1	4
20	200	3	2	1 GiB	0.5 GiB	3	1	1	1	1	4
30	300	3	3	1.5 GiB	1 GiB	6	1	1	1	1	5

40	400	3	3.5	1.5 GiB	1 GiB	6	1	1	1	1	5
50	500	3	4	2 GiB	1.5 GiB	9	1	1	1	1	6
60	600	3	4.5	2 GiB	1.5 GiB	9	1	1	1	1	6
70	700	3	5	2 GiB	1.5 GiB	12	1	1	1	1	7
80	800	3	6	2.5 GiB	1.75 GiB	12	2	1	1	2	8
90	900	3	7	2.5 GiB	1.75 GiB	15	2	2	2	2	9
100	1000	3	8	3 GiB	2 GiB	15	3	2	2	2	10

Scaling Services

Use the following commands to scale out the desired replicas or scale up resource limits for the required services:

mongodb-v2 - scale up resource limits

The mongodb-v2 `StatefulSet` cannot be scaled out. Increase CPU, memory, and ephemeral storage limits as the workload grows:

```
oc set resources sts mongodb-v2 --limits=cpu=<desired-cpu>,memory=<desired-memory>,ephemeral-storage=<desired-ephemeral> -
-containers=mongodb -n <backup-restore-namespace>
```

For example:

```
oc set resources sts mongodb-v2 --limits=cpu=8,memory=3Gi,ephemeral-storage=2Gi --containers=mongodb -n ibm-backup-restore
```

Kafka broker-pool - scale out replicas

The Kafka broker-pool is managed as a `KafkaNodePool` resource. Scale its replicas using:

```
oc patch kafkanodepool broker-pool -n <backup-restore-namespace> --type='json' -p='[{"op": "replace", "path":
"/spec/replicas", "value": <desired-replicas>}]'
```

For example:

```
oc patch kafkanodepool broker-pool -n ibm-backup-restore --type='json' -p='[{"op": "replace", "path": "/spec/replicas",
"value": 15}]'
```

`backup-service` - scale out replicas

```
oc scale deployment backup-service --replicas=<desired-replicas> -n <backup-restore-namespace>
```

For example with 100 clusters/1000 jobs (3 replicas):

```
oc scale deployment backup-service --replicas=3 -n ibm-backup-restore
```

`backuppolicy-deployment` - scale out replicas

```
oc scale deployment backuppolicy-deployment --replicas=<desired-replicas> -n <backup-restore-namespace>
```

For example with 90-100 clusters (2 replicas):

```
oc scale deployment backuppolicy-deployment --replicas=2 -n ibm-backup-restore
```

`applicationsvc` - scale out replicas

```
oc scale deployment applicationsvc --replicas=<desired-replicas> -n <backup-restore-namespace>
```

For example with 80-100 clusters (2 replicas):

```
oc scale deployment applicationsvc --replicas=2 -n ibm-backup-restore
```

`job-manager` - scale out replicas

```
oc scale deployment job-manager --replicas=<desired-replicas> -n <backup-restore-namespace>
```

For example with 80-100 clusters (2 replicas):

```
oc scale deployment job-manager --replicas=2 -n ibm-backup-restore
```

`windowEndTime` - update through `BackupPolicy` CR

The `windowEndTime` is a field under `spec.schedule` in the `BackupPolicy` CR. The value is in HH:MM format.
Note: `windowEndTime` must be set to a minimum of 4 hours later than the scheduled (cron) start time. This can also be updated from the IBM Fusion user interface by navigating to Backup & Restore > Policies and by editing the required policy.
Update the `windowEndTime` field (HH:MM format) under `spec.schedule`. Ensure the value is at least 4 hours after the policy cron schedule time:

```
oc patch fbp <policy-name> -n <ibm-spectrum-fusion-ns> --type='json' -
p='[{"op":"replace","path":"/spec/schedule/windowEndTime","value":"<HH:MM>"}]'
```

For example with 100 clusters/1000 jobs (window ending at 10:00):

```
oc patch fbp <policy-name> -n <ibm-spectrum-fusion-ns> --type='json' -
p='[{"op":"replace","path":"/spec/schedule/windowEndTime","value":"10:00"}]'
```

`cancelJobAfter` - update in job-manager deployment

The `cancelJobAfter` value is specified in milliseconds and is configured as an environment variable in the job-manager deployment.
Patch the `env` variable value (index 3 corresponds to `cancelJobAfter`):

```
oc patch deployment job-manager -n <backup-restore-namespace> --type='json' -
p='[{"op":"replace","path":"/spec/template/spec/containers/0/env/3/value","value":"<milliseconds>"}]'
```

For example with 80 clusters/800 jobs (10800000 ms = 3 hours):

```
oc patch deployment job-manager -n ibm-backup-restore --type='json' -
p='[{"op":"replace","path":"/spec/template/spec/containers/0/env/3/value","value":"10800000"}]
```

`MaxConcurrentReconciles` - update in `dp-operator` CSV

`MaxConcurrentReconciles` is configured as an environment variable in the `guardian-dp-operator` cluster service version.
Patch the value (index 3 corresponds to `MaxConcurrentReconciles` based on the current CSV):

```
oc patch csv <guardian-dp-operator-csv-name> -n <backup-restore-namespace> --type='json' -p='[{"op": "replace", "path":
"/spec/install/spec/deployments/0/spec/template/spec/containers/1/env/3/value", "value": "<desired-value>"}]'
```

For example with 80-100 clusters (20 reconciles):

```
oc patch csv guardian-dp-operator.v2.13.0-20260402145657 -n ibm-backup-restore --type='json' -p='[{"op": "replace",
"path": "/spec/install/spec/deployments/0/spec/template/spec/containers/1/env/3/value", "value": "20"}]'
```

Backup & restore configuration parameters

Configuration parameters allow customization of IBM Fusion Backup & restore settings.

Change defaults for IBM Fusion Backup & restore agent

deleteBackupWait
: Timeout for restic command to delete backup in S3 storage. Set in minutes. The default value is 20 minutes, and the allowed range is 10 to 120.

pvcSnapshotMaxParallel
: Number of threads available to take concurrent snapshots. The default value is 20.

backupDatamoverTimeout
: Maximum amount of time in minutes for the datamover to complete backup. The default value is 1200 seconds, and the allowed range is 10 to 14400. After you modify backupDatamoverTimeout, update cancelJobAfter.

restoreDatamoverTimeout

Maximum amount of time in minutes for datamover to complete restore. The default value is 1200 seconds, and the allowed range is 10 to 14400. After you modify restoreDatamoverTimeout, update cancelJobAfter.

snapshotRestoreJobTimeLimit
: This parameter is not used.

pvcSnapshotRestoreTimeout
: Timeout for creating PVC from snapshot in minutes. The default value is 15 minutes.

kafka-thread-size
: Controls the number of concurrent backup or restore jobs processed by a transaction manager pod. The default value is 10.

snapshotTimeout
: Timeout for snapshot to resolve to the ready state in minutes. The default value is 20, and the allowed range is 10 to 120.

datamoverJobpodEphemeralStorageLimit
: Datamover pod ephemeral storage limit. The default value is 2000Mi.

datamoverJobPodDataMinGB
: Minimum PVC capacity for each datamover pod before a new datamover pod is started. It is set in GB, and the default value is 10 GB.

datamoverJobpodMemoryLimit
: It is the Datamover pod memory limit, and the default value is 15000Mi.

datamoverJobpodCPULimit
: It is the Datamover pod CPU limit, and the default value is 2.

cancelJobAfter
: If you modify backupDatamoverTimeout or restoreDatamoverTimeout, update the job-manager deployment configuration parameter cancelJobAfter. It is the maximum amount of time in milliseconds that the job-manager waits before it cancels the long-running job. The default value is 3600000 (1 hour).

MaxNumJobPods
: Implement configurable MaxNumJobPods for datamovers.
This field helps to control the number of PersistentVolumeClaims that is attached to datamover pods during backup and restore to the **`BackupStorageLocation`**. It is set on a per install basis. If you have spoke clusters installed in your IBM Fusion installation, then set on each spoke cluster individually. It is not a global field that applies to all clusters in the installation.

It helps to distribute the storage load across multiple nodes of the cluster when available. Some StorageClasses impose a maximum number of PVCs that can be attached to an individual node of the cluster. This field helps to manage this StorageClass limitation. To find out whether your StorageClass has this limitation, check whether the CSINode of your storage provider has the spec.drivers[].allocatable.count field set. The VPC Block on IBM Cloud is one such storage provider with this limitation, typically 10 per node. If you increase the number of pods and the application has more than 30 PVCs, it decreases the number of PVCs attached to each node. If it goes below this, you can use the default, which is more than sufficient.

This field can increase or decrease the maximum number of datamover pods that are assigned in each backup or restore. More pods use more resources such as CPU and memory, and can help improve performance for backups and restores with larger numbers of PVCs. Increasing this value may help if the number of PVCs to be backed up or restored at the same time is more than 30.

This field does not guarantee the creation of more datamover pods. A number of heuristics are used at runtime to help determine the assignment of PVCs to datamovers, including total PVC capacity, number of PVCs, amount of data transferred during previous backups, total number of PVCs handled, storage providers involved, among others. This field changes the maximum allowed, and it does not guarantee the specified number of datamovers.

For more information about how to change the value, see Change defaults for IBM Fusion Backup & restore agent.

DeleteBackupRequest CR cleanup
: The **`DeleteBackupRequest`** CRs in the Completed state gets automatically deleted after a default retention of 14 days. You can set the **`dbrCRRetention`** configuration parameter in ConfigMap guardian-configmap to change the default. By default, the cleanup thread runs every 2 hours and cleans up a maximum of 100 **`DeleteBackupRequest`** CRs in one run. These two values can be overridden by specifying **`dbrCleanupCheckInterval`** and **`dbrCleanupBatchSize`** configuration parameters in the **`guardian-configmap`** ConfigMap.

dbrCRRetention
: Number of days after which **`DeleteBackupRequest`** CRs in the Completed state get deleted. If not specified, then 14 days is the default.

dbrCleanupCheckInterval
: Interval, in hours, at which the **`DeleteBackupRequest`** cleanup thread runs. If not specified, 2 hours is the default.

dbrCleanupBatchSize
: Number of **`DeleteBackupRequest`** CRs to cleanup in one run of the cleanup thread. If not specified, 100 is the default.

Change defaults for IBM Fusion Backup & restore agent

Do the following steps to change the value:

1. In the OpenShift® Container Platform console, click Operators > Installed Operators.
2. Change the project to the IBM Spectrum Fusion Backup and Restore namespace. For example **`ibm-backup-restore`**.
3. Click to open IBM Fusion Backup & Restore Agent.
4. Click the Data Protection Agent tab and click the dpagent install. Alternatively, if you want to use the OC command:

   ```
   oc edit -n ibm-backup-restore dataprotectionagent
   ```

5. Go to the YAML tab.
6. Edit spec.transactionManager.datamoverJobPodCountLimit. The value must be numeric and in quotes. For example, '3', '5', '10'

Backup and restore large number of files

When you back up and restore a large number of files located on CephFS, you must perform the following additional steps for the operations to succeed. The optimal values depend on your individual environment, but the following values are representative for a backup and restore of a million files.

- Prevent the transaction manager from failing long running backup jobs. In the **ibm-backup-restore** project, edit the config map named **guardian-configmap**. Look for **backupDatamoverTimeout**. This value is in minutes, and the default is 20 minutes. For example, increase this value to 8 hours (480).
- Prevent the job manager from canceling long running jobs. In the **ibm-backup-restore** project, edit the **job-manager** deployment. Under **env**, look for **cancelJobAfter**. This value is in milliseconds, and the default is 1 hour. For example, increase this value to 20 hours (72000000).
- Prevent the transaction manager from failing long running restore jobs. In the **ibm-backup-restore** project, edit the config map named **guardian-configmap**. Look for **restoreDatamoverTimeout**. This value is in minutes, and the default is 20 minutes. For example, increase this value to 20 hours (1200).
- In the same config map, increase the amount of ephemeral storage the data mover is allowed to use by increasing datamoverJobpodEphemeralStorageLimit to 4000Mi or more.
- OpenShift Data Foundation parameters.
  - Increase the resources available to OpenShift Data Foundation. Increase the limits and requests for the two MDS pods, a and b, for example to 2 CPU and 32 Gi memory. For more information about the changes, see Changing resources for the OpenShift Data Foundation components.
  - Prevent SELinux relabelling
    At restore time, OpenShift will attempt to relabel each of the files. If it takes too long, the restored pod will fail with CreateContainerError. This article explains the situation and some of the possible workarounds to prevent the relabeling:https://access.redhat.com/solutions/6221251.

Additional parameters are available when you backup and restore large number of files that are located on CephFS. The optimal values depend on your individual environment, but the values in this example represent backup and restore of a million files. For such large number of files, you must be on OpenShift Container Platform 4.14 or later, and Data Foundation 4.12 or later.

Set up the following Backup & Restore parameters:

- In the same config map, increase the amount of ephemeral storage the data mover is allowed to use. Increase **datamoverJobpodEphemeralStorageLimit** to 4000Mi or more.

Set up the following Red Hat® OpenShift Data Foundation parameters:

- Increase the resources available to Red Hat OpenShift Data Foundation. Increase the limits and requests for the two MDS pods, for example, to 2 cpu and 32 Gi memory. For the procedure to change, see Changing resources for the OpenShift Data Foundation components.
- Set up to Prevent SELinux relabeling:
  At restore time, OpenShift attempts to relabel each of the files. If it takes too long, the restored pod fail with **CreateContainerError**. This article explains the situation and some of the possible workarounds to prevent the relabeling: https://access.redhat.com/solutions/6221251.

Use the following steps to understand whether your backups fail due to a large number of files:

1. Run the following command to check whether the MDS pods are restarting:

   ```
   oc get pod -n openshift-storage |grep mds
   ```

2. If they are restart, check the termination reason:
   a. Describe the pod.
   b. Check whether the termination is **OOM Kill**.
3. Run the following command to check the memory usage by the MDS pods and monitor for memory usage:

   ```
   oc adm top pod -n openshift-storage
   ```

4. If the memory usage keeps spiking until the pod restarts, then see Changing resources for the OpenShift Data Foundation components.

Backup and restore large number of resources

When you back up and restore a large number of cluster resources, the velero pod resource limits must be increased for the operations to succeed. The optimal values depend on your individual environment, but the following values are representative for backup and restore of 1500 resources.

```
oc patch dpa velero -n <backup-restore-namespace> --type merge -p '{"spec": {"configuration": {"velero": {"podConfig": 
{"resourceAllocations": {"limits": {"ephemeral-storage": "1Gi"}}}}}}}'
```

```
oc patch dpa velero -n <backup-restore-namespace> --type merge -p '{"spec": {"configuration": {"velero": {"podConfig": 
{"resourceAllocations": {"limits": {"memory": "2Gi"}}}}}}}'
```

Configure DataMover type for Backup and Restore

The DataMover type dictates both the entity responsible for backing up data on PersistentVolumeClaims and the method of storage. **kopia** is the only allowed value.

- The default value is **kopia**.
- The **legacy** DataMover for versions equal to or lower than 2.8

The value can be set globally and per PolicyAssignment. The two different DataMover types are not cross-compatible. Backups with one DataMover type use that type during restore. Existing backups continue to expire normally per their Policy.

Change the DataMover type used across all new backups
From the OpenShift console:

1. Log in to OpenShift Console.
2. Go to Workloads > ConfigMaps.
3. Select the IBM Fusion install namespace.
4. Open **isf-data-protection-config** and update **data.Datamover** field.

Example command:

```
oc patch configmap -n ibm-spectrum-fusion-ns isf-data-protection-config -p '{"data":{"DataMover":"kopia"}}'
```

Here, replace namespace **ibm-spectrum-fusion-ns** with your Fusion namespace. Also, this example shows **kopia** DataMover. For **legacy** DataMover, replace **kopia** to **legacy**.

Choose DataMover during policy assignment

To facilitate a trial of the DataMover options, select it by annotating PolicyAssignment objects. Each PolicyAssignment object associates an existing Policy and BackupStorageLocation with an application to backup. By setting this value according to the following instructions, the global setting gets ignored. Consequently, all new backups associated with the PolicyAssignment use the specified DataMover type in the annotation.

1. Search by label to find the existing PolicyAssignment for your applications.
   - To search by application - `dp.isf.ibm.com/application-name=<\application name>`
   - To search by policy - `dp.isf.ibm.com/backuppolicy-name=<\policy name>`
   - To search by backpropagation - `dp.isf.ibm.com/backupstoragelocation-name=<\backupstoragelocation name>`

     The general naming format is as follows:

     ```
     <application name>-<\policy name>-<cluster url>
     ```

     Example:

     ```
     aws-20240716-220437-awsdaily-apps.bnr-hcp-munch.apps.blazehub01.mydomain.com
     ```

2. Change the value of DataMover.

   From the OpenShift console:

   a. Go to Operators > Installed Operators.
   b. Select the IBM Fusion install namespace.
   c. Open IBM Fusion.
   d. Go to the Policy Assignment tab and change the DataMover value.

   Using commands:

   a. Get the policy assignment object:

   ```
   oc get policyassignment -n ibm-spectrum-fusion-ns
   ```

   b. Add the annotation to the PolicyAssignment object.

   ```
   dp.isf.ibm.com/dataMover: <datamover type>
   ```

   Example:

   ```
   dp.isf.ibm.com/dataMover: legacy
   ```

Choose nodes for DataMovers for OADP DataMover

From 2.9 version onwards, the Backup & restore is transitioning from the in-house DataMover to OADP. DataMover controller using restic as the data transfer software to OADP, which uses its own implementation of DataMover and controller with kopia.

Path to locate the custom resource

On Backup & restore namespace, do the following steps:

1. Go to Install Operators > OADP Operator.
2. On the Operator details page, select DataProtectionApplication tab and velero instance.
3. Select YAML tab to update configuration parameters.

spec.datamoverConfiguration

The field **`spec.datamoverConfiguration`** is used to configure the new OADP DataMovers. For backwards compatibility, you can continue to use the fields under **`spec.transactionManager`** to configure the DataMovers from the previous releases.

spec.datamoverConfiguration.nodeAgentConfig.env

This array allows users to specify environment variables inserted into the node-agent pods. The example value of HOME allows setting the folder used by kopia for local files.

It controls the on-container folder where cache files related to the storage repository are stored. The default value is `/home/velero` and it uses local ephemeral storage.

```
spec:
  datamoverConfiguration:
    nodeAgentConfig:
      env:
      - name: HOME
      value: /home/velero
```

spec.datamoverConfiguration.nodeAgentConfig.labels

This field adds additional labels to the node-agent daemonset and pods. It is an optional field to organize pods and does not affect backup and restore behavior.This field is a YAML object.

```
 spec:
  datamoverConfiguration:
    nodeAgentConfig:
      labels:
        cloudpakbackup: datamover
```

spec.datamoverConfiguration.nodeAgentConfig.tolerations

This is an array field that specifies the conditions for nodes that permit the execution of Pods in resource-constrained environments or multi-architecture clusters. For more information about this field, see Kubernetes documentation.

```
 spec:
  datamoverConfiguration:
    nodeAgentConfig:
      tolerations:
      - key: "kubernetes.io/arch"
```

```
        operator: "Equal"
        effect: "amd64"
```

spec.datamoverConfiguration.nodeAgentConfig.resourceAllocations
: This field defines the resource allocation for the node-agent controller, including both scheduling requirements and limits.
If backups or restores are causing evictions in the node-agent pods, increase the value of this field. Go through the Events in the Backup and Restore install namespace to check resource violations:

- The status field of the Daemonset node-agent
- The OpenShift Dashboard for resource monitoring in the Backup and Restore namespace
  See OpenShift documentation.
- Grafana in the Backup and Restore namespace if installed
  See OpenShift documentation
- Other applications that use Prometheus metrics.

Example:

```
spec:
 datamoverConfiguration:
   nodeAgentConfig:
       resourceAllocations:
         limits:
           cpu: "2"
           ephemeral-storage: "50Mi"
           memory: "2048Mi"
         requests:
           cpu: "200m"
           ephemeral-storage: "25Mi"
           memory: "256Mi"
```

spec.configuration.nodeAgent.podConfig.nodeSelector
: This field decides the nodes where the DaemonSet node agent must run. This feature is a crucial security feature to restrict the HostPath mount of the node-agent that exposes the following values:

- All PersistentVolumeClaims
- Projected API volumes such as configmaps and secrets through volume mount
- Kubernetes API tokens that are in use on the node

Note: As this field causes Restore failures, remove it before restore operations.
If needed, this field can isolate the Daemonset node-agent Pods to Nodes without any security concerns.

The pods are assigned to Nodes that match the labels attached to the Nodes. If more than one label is used, the labels are treated in a logical AND. Only Nodes that match all of the labels runs the Pod.

For example, if you set the nodeSelector to "kubernetes.io/hostname: bnr-hcp-munch-6df6702b-2bv7z", then the Daemonset node-agent Pods run only the node with this label.If there is only one node with this label, Daemonset node-agent is restricted to deploying Pods to only the labeled node.

```
spec:
   configuration:
      nodeAgent:
         podConfig:
            nodeSelector:
               kubernetes.io/hostname: bnr-hcp-munch-6df6702b-2bv7z
```

spec.datamoverConfiguration.datamoverPodConfig
: This setting manages the PVCs, resource allocations, and the location within the cluster where the datamover pods operate. The LoadConcurrency provides the number and LoadAffinity provides the location.

spec.datamoverConfiguration.datamoverPodConfig.loadConcurrency
: This field controls the maximum number of datamovers assigned to nodes in the cluster.
If there is anything that appears to be in conflict, see Velero documentation.

This setting controls the number of datamovers deployed per node-agent controller pod, and it consists of two parts:

- `spec.datamoverConfiguration.datamoverPodConfig.loadConcurrency.globalConfig`
  This sets the maximum number of datamovers deployable from a single node-agent controller pod for a generic node in the cluster. The subsequent field overrides globalConfig and manages this setting independently. It is of type integer with default value of five and minimum value of 1. If loadConcurrency is added to the config, this value is REQUIRED.
- `spec.datamoverConfiguration.datamoverPodConfig.loadConcurrency.perNodeConfig` is optional.
  This allows setting a maximum number of datamovers assigned to a node. Example:

  ```
  loadConcurrency:
    globalConfig: 2
    perNodeConfig:
    - nodeSelector:
        matchLabels:
          kubernetes.io/hostname: node1
      number: 3
    - nodeSelector:
        matchLabels:
          beta.kubernetes.io/instance-type: Standard_B4ms
      number: 5
  ```

  The perNodeConfig uses the default nodeSelector elements from Kubernetes to select nodes by using labels. When multiple labels are specified in a matchLabels, the selected nodes are found using an AND operation, meaning all the labels must be found. These also take a more complex matchExpressions field as an alternative to nodeSelector. For more information about assigning pods to nodes, see [Kubernetes documentation](). In this example, the node with a label **kubernetes.io/hostname=node1** can run up to 3 DataMovers at once. And a node with label **beta.kubernetes.io/instance-type=Standard_B4ms** can run up to 5 DataMovers. With the globalConfig set to 2, all other nodes with matching storage to the PVC may run up to 2 DataMovers at the same time.

spec.datamoverConfiguration.datamoverConfiguration.datamoverPodConfig.LoadAffinity

This field controls the nodes in the cluster where the DataMovers must run. For more information, see [Velero reference documentation]().

```
loadAffinity:
- nodeSelector:
    matchLabels:
      beta.kubernetes.io/instance-type: Standard_B4ms
    matchExpressions:
    - key: kubernetes.io/hostname
      values:
      - node-1
      - node-2
      - node-3
      operator: In
    - key: xxx/critial-workload
      operator: DoesNotExist
```

The value is a list of **nodeSelectors**. The selected nodes are controlled through a nodeSelector field that matches the labels. Each member of the list is evaluated independently. The nodes that match the element are combined to create the final list where the DataMovers can run. If the **loadConcurrency.globalConfig** is set to 0, then only the selected nodes in **loadAffinity** and **perNodeConfig** can run the backup and restore jobs.

Ensure that you meet the following criteria.

- The intersection of nodes matching loadConcurrency.perNodeConfig and loadConcurrency.loadAffinity is where the jobs run. Ensure the size of the intersection is at least one node.
- If the storage used during the backup or restore process is only available on a subset of nodes and **loadConcurrency.globalConfig** is set to 0, you must select at least one node where the storage is accessible. Otherwise, the job fails.
- If the **loadConcurrency.globalConfig** is set to 0 and the selected nodes do not have enough resources in CPU and memory to schedule a DataMover pod, then the backup or restore job fails.

spec.datamoverConfiguration.datamoverPodConfig.podResources

Controls the amount of CPU and memory that is made available by the DataMovers. It has the following sub-fields and default values are set for the missing fields. Example:

```
podResources:
  cpuRequest: 2
  memoryRequest: 4Gi
  ephemeralStorageRequest: 5Gi
  cpuLimit: 4
  memoryLimit: 16Gi
  ephemeralStorageLimit: 5Gi
```

For more information about the default Velero resource profile for large clusters, see [CPU and memory requirements in Red Hat documentation]().

Resource limits must be sufficient to back up the most resource-intensive volumes in the cluster.

spec.datamoverConfiguration.datamoverPodConfig.backupPVC

This section defines the format for the PersistentVolumeClaim used in the backup process. See [Velero documentation]().
During backup, each execution of a volumegroup in a Recipe (default Recipe is all PersistentVolumeClaims), a VolumeSnapshot gets created from a PersistentVolumeClaim representing the volume state at the time of the VolumeSnapshot timestamp. Some storage systems, such as CephFS from Data Foundation and IBM Storage Scale, implement shallow-copy backup volume exposure without file copy to address performance concerns compared to regular volume restores.

By default, the agent configures CephFS, IBM Storage Scale, and NFS PersistentVolumeClaims to be created in ReadOnlyMany mode during backups.

Setting values in **backupPVC** you can configure PersistentVolumeClaims from any StorageClass to use their "shallow-copy" implementation. This way, IBM Fusion does not need the specifics, and you can leverage alternative StorageClasses to access different storage features.

Example:

```
backupPVC:
  storage-class-1:
    storageClass: backupPVC-storage-class
    readOnly: true
  storage-class-2:
    storageClass: backupPVC-storage-class
  storage-class-3:
    readOnly: true
```

In this example, a data transfer of PersistentVolumeClaims of StorageClass "storage-class-1" makes use of an alternative StorageClass "backupPVC-storage-class" to create the volume with **accessModes: ReadOnlyMany**. The storage-class-2 is same as storage-class-1 except that the **accessModes** remains unchanged.

In storage-class-3, no alternate StorageClass can be used, and the volume gets created with **accessModes: ReadOnlyMany**. All DataMovers mount the backup volume as ReadOnly, and this affects the behavior of storage on volume creation. The original application **PersistentVolumeClaim** is not modified. Only the **PersistentVolumeClaim** used for the backup is modified.

If the StorageClass belongs to storage systems CephFS or IBM Storage Scale, the setting overwrites the agent's internal behavior with the one configured.

Not all storage systems support **PersistentVolumeClaims** with **accessModes: ReadOnlyMany**. See your appropriate storage documentation. The OpenShift does not provide a way to check what **accessModes** are supported prior to **PersistentVolumeClaim** creation.

To use PVC storage as a local cache for kopia datamovers:

1. Update ConfigMap node-agent-config.

```
kind: ConfigMap
apiVersion: v1
metadata:
data:
  node-agent-config: '{"loadConcurrency": {"globalConfig": 6}, "podResources": {"cpuRequest": "500m", "cpuLimit": 
"4", "memoryRequest": "500Mi", "memoryLimit": "4Gi", "ephemeralStorageRequest": "5Gi", "ephemeralStorageLimit": 
"5Gi"}, "backupPVC": {"ibm-spectrum-fusion-mgmt-sc": {"storageClass": "ibm-spectrum-fusion-mgmt-sc", "readOnly": 
true, "spcNoRelabeling": true}, "ocs-storagecluster-cephfs": {"storageClass": "ocs-storagecluster-cephfs", 
"readOnly": true, "spcNoRelabeling": true}}}'
```

2. Add the following field highlighted with ** ***:

```
kind: ConfigMap
apiVersion: v1
metadata:
data:
  node-agent-config: '{**"storage":{"storageClassName":"<storage class to use for PVC>","size":"<size of 
PVC>"}**,"loadConcurrency":{"globalConfig":6},"podResources":
{"cpuRequest":"500m","cpuLimit":"4","memoryRequest":"500Mi","memoryLimit":"4Gi","ephemeralStorageRequest":"5Gi","ephe
meralStorageLimit":"5Gi"},"backupPVC":{"ibm-spectrum-fusion-mgmt-sc":{"storageClass":"ibm-spectrum-fusion-mgmt-
sc","readOnly":true,"spcNoRelabeling":true},"ocs-storagecluster-cephfs":{"storageClass":"ocs-storagecluster-
cephfs","readOnly":true,"spcNoRelabeling":true}}}'
```

   Where the `storageClassName` is the `StorageClass` to allocate the storage from. Example: `ibm-spectrum-fusion-mgmt-sc`. The size is the Kubernetes resource size field. Examples: "500Mi" "5Gi" "10Gi" "20Gi" "50Gi".
3. Save the ConfigMap.
4. Restart the pods of `daemonset node-agent`.
5. Replace the namespace of `ibm-backup-restore` with the Fusion Backup & Restore install namespace.

```
oc delete pod -n ibm-backup-restore -l name=node-agent
```

spec.datamoverConfiguration.datamoverPodConfig.maintenanceConfig

Kopia requires periodic maintenance jobs on the backup repository. See Velero documentation.
For each namespace, a Kopia repository is created and maintained. If a cloud object storage repository is used, automatic object expiration must not be used as it triggers repository corruption. The maintenance tasks enhance efficiency and reduce resource usage in future backup and restore jobs, typically lasting less than a minute. Their resource usage is similar to backup expiration tasks. These maintenance jobs run by default every 4 hours. The maintenance frequency can be adjusted by edited the `OADP DataProtectionAgents` under the field `spec.configuration.velero.args.default-repo-maintain-frequency`. These are the "Quick Maintenance Tasks" described by Kopia. See https://kopia.io/docs/advanced/maintenance/.

The "Full Maintenance Tasks" are run during backup expiration. The only available configuration item through the `DataProtectionAgent` object is the pod resources used during maintenance jobs.

spec.datamoverConfiguration.datamoverPodConfig.maintenanceConfig.resourceAllocations

This field is set using the Kubernetes `resourceAllocations` object format. https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/
The following example is also the default values. Missing values such as `requests.cpu` are set to unlimited.

```
datamoverConfiguration:
    datamoverPodConfig:
      podResources:
        cpuLimit: '4'
        cpuRequest: '2'
        memoryLimit: 16Gi
        memoryRequest: 4Gi
        ephemeralStorageRequest: 5Gi
        ephemeralStorageLimit: 5Gi
```

The default is 5Gi.

spec.datamoverConfiguration.maintenanceConfig.jobSettings.ttl

This adds a Time To Live to Velero maintenance jobs. Velero defaults to keeping a record of 3 previously completed maintenance jobs. IBM Fusion changes this default value to 1. When you add this field, the Velero maintenance jobs are removed within seconds after completion.

```
  datamoverConfiguration
    maintenanceConfig:
      jobSettings:
        ttl: 240
      podResources:
        cpuLimit: '4'
        cpuRequest: '2'
        memoryLimit: '4Gi'
        memoryRequest: '500Mi'
        ephemeralStorageLimit: '5Gi'
        ephemeralStorageRequest: '5Gi'
```

If you do not set this field and you have `spec.datamoverConfiguration.maintenanceConfig.storage`, then it is automatically set to 300. For more information, see https://kubernetes.io/docs/concepts/workloads/controllers/ttlafterfinished/.

spec.datamoverConfiguration.maintenanceConfig.storage

This section enables you to use a PersistentVolumeClaim to act as a cache during Velero maintenance jobs. Most usage of ephemeral-storage during maintenance jobs originates from the kopia datamover cache. The addition of this field with a sufficiently sized PersistentVolumeClaim to handle the required cached data eliminates the aforementioned ephemeral-storage usage.

```
    maintenanceConfig:
      jobSettings:
        ttl: 240
      storage:
```

```
        storageClassName: "ocs-external-storagecluster-ceph-rbd"
        size: 25Gi
      podResources:
        cpuLimit: '4'
        cpuRequest: '2'
        memoryLimit: '4Gi'
        memoryRequest: '500Mi'
        ephemeralStorageLimit: '5Gi'
        ephemeralStorageRequest: '5Gi'
```

If storage is configured, then you need the following fields:

- **storageClassName**
  The name of the **StorageClass** to be used with the **PersistentVolumeClaim**. The **StorageClass** must support **accessMode: ReadWriteOnce**.
- **size**
  The size of the volume to be created. This field uses the existing Kubernetes resources request format such as "5Gi" and "250Mi".

  The size of the created PVC is to be used by each data mover pod.

During maintenance jobs, a PersistentVolumeClaim gets created. It is deleted when the job is removed based on the **spec.datamoverConfiguration.maintenanceConfig.jobSettings.ttl** field. If **spec.datamoverConfiguration.maintenanceConfig.jobSettings.ttl** is not specified and the storage configuration exists, then the ttl gets set to 300 (five minutes).

Advanced: Per Namespace Configuration

The maintenance config values mentioned previously are applied globally. These fields can also be applied manually by editing the appropriate **maintenance-config** ConfigMap. It allows overrides by namespace in case of resource usage differences.
The format is the same as **DataProtectionAgent spec.datamoverConfiguration.maintenanceConfiguration** in JSON instead of YAML.

The configuration is applied using a JSON format where the key is the repository. If a value is set for a repository, the value overrides the global values. For instance, if the global values are as described in the following example:

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: maintenance-config
  namespace: ibm-backup-restore
data:
  global: '{"jobSettings":{"ttl":300},"podResources":
{"cpuRequest":"500m","cpuLimit":"4","memoryRequest":"500Mi","memoryLimit":"4Gi","ephemeralStorageLimit":"5Gi","ephemeralSt
orageRequest":"5Gi"}, "storage":{"storageClassName":"ocs-external-storagecluster-ceph-rbd","size":"20Gi"}}'
```

Additional keys are in the **<namespacename>-<bsl name>-kopia** format:

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: maintenance-config
  namespace: ibm-backup-restore
data:
  global: '{"jobSettings":{"ttl":300},"podResources":
{"cpuRequest":"500m","cpuLimit":"4","memoryRequest":"500Mi","memoryLimit":"4Gi","ephemeralStorageLimit":"5Gi","ephemeralSt
orageRequest":"5Gi"}}'
  zen-s3-kopia: '{"jobSettings":{"ttl":300},"podResources":
{"cpuRequest":"500m","cpuLimit":"4","memoryRequest":"500Mi","memoryLimit":"4Gi"}, "storage":{"storageClassName":"ocs-
external-storagecluster-ceph-rbd","size":"25Gi"}}'
```

It ensures that the maintenance jobs related to data from the zen namespace connected to BSL S3 use a 25Gi PersistentVolumeClaim cache and do not set resource requests or limits on ephemeral storage.

node-agent-config

To configure the data mover pods to use PVCs as cache in place of ephemeral storage, add the following field into the node-agent-config config map:

```
storage":{"storageClassName":"ocs-external-storagecluster-ceph-rbd","size":"25Gi"}
```

To configure the data mover pods to use PVCs as cache in place of ephemeral storage.
Example value of the node-agent-config:

```
{"loadConcurrency": {"globalConfig": 5}, "podResources": {"cpuRequest": "2", "cpuLimit": "4", "memoryRequest": "4Gi",
"memoryLimit": "16Gi", "ephemeralStorageRequest": "5Gi", "ephemeralStorageLimit": "5Gi"}, "storage":
{"storageClassName":"ocs-external-storagecluster-ceph-rbd","size":"25Gi"}, "backupPVC": {"ocs-storagecluster-cephfs":
{"storageClass": "ocs-storagecluster-cephfs", "readOnly": true, "spcNoRelabeling": true}}}
```

Latest permissible start time for the backup process

You can set the latest permissible start time for a scheduled backup. This setting is useful whenever multiple backups use the same policy or schedule and the agent gets overloaded frequently with jobs as they all start at the same time. Without the windowEndTime setting, jobs have, by default, up to one hour for the agent to start processing before they are considered hung and subsequently cancelled. You can manually spawn additional agent replicas to handle the spike in job activity and process the backlog of jobs sooner, but these replicas remain idle for the remainder of the day. With the windowEndTime option, the job can remain queued up for processing by the agent for a custom period of time before getting cancelled. This allows for a single agent replica to be fully utilized for a longer period of time throughout the day without the need to manually create multiple policies or schedules.

Set the windowEndTime field in the **BackupPolicy** CR in the 24 hour "HH:MM" format.

Example:

```
spec:
  backupStorageLocation:  kn-aws
  provider: isf-backup-restore
  retention:
    number: 5
    unit: days
  schedule:
    cron: `45 14 * * * '
    timezone: America/Los_Angeles
    windowEndTime: `17:45'
```

In this example, the schedule is set to run daily at 14:45 (2.45 PM). By specifying a windowEndTime of 17:45, you can ensure that the backup has until 5:45 PM to start. If the backup does not start by 5.45 PM due to the load on the agent, it gets cancelled. If the agent does not pick it up and the windowEndTime setting does not exist, then backup gets cancelled after one hour.
Note: For all new `BackupPolicy` CRs that have a specific start time (not just a repeat interval), if the windowEndTime is not explicitly specified, a default end time gets set to 4 hours after the scheduled start time. Existing `BackupPolicy` CRs maintain their original behavior without windowEndTime, resulting in a one-hour limit before they are considered hung and subsequently cancelled. You can edit the existing Policy CRs to add the windowEndTime option.

Failed backup retention

By default, failed backups are automatically deleted after 7 days. You can change this retention period by updating the `failedBackupRetention` configuration parameter in the `guardian-configmap` configmap. To do so, add the following entry to the configmap: `failedBackupRetention: '5'`. This sets the retention period to 5 days.

FIPS-140-2

The Federal Information Processing Standard Publication 140-2 (FIPS-140-2) is a standard that defines a set of security requirements for the use of cryptographic modules. Law mandates this standard for the US government agencies and contractors and is also referenced in other international and industry specific standards.

IBM Fusion Backup & Restore service is FIPS compliant. It uses FIPS validated cryptographic modules provided by Red Hat Enterprise Linux OS/CoreOS (RHCOS).

Backup & Restore service installation and upgrade issues

Use this troubleshooting information to resolve install and upgrade problems that are related to Backup & Restore service.

One or more Backup & Restore pods are in CrashLoopBackOff state due to startup/liveness/readiness probe failures

Problem statement
: During installation or upgrade of IBM Fusion Backup & Restore service, a few pods might be in `CrashLoopBackOff` state due to startup/liveness/readiness probe failures. This causes the installation or upgrade to not progress further.

Cause
: One or more Backup & Restore service pods enter a `CrashLoopBackOff` state due to startup/liveness/readiness probe failures. This can happen when the cluster load causes the pod CPU to be throttled leading to longer startup times.

Resolution
: Increase the `periodSeconds` and `failureThreshold` values for the startup/liveness/readiness probes in the Backup & Restore service deployment. Updating these values allows additional time for the pods to start successfully and enables the service to complete the upgrade.

Backup & Restore service installation may hang at 75%

If the provided custom install namespace exceeds 24 characters in length, then the Fusion Backup & Restore service installation may hang at 75% completion. To resolve this issue, provide a custom namespace that is no more than 24 characters in length.

Application information not repopulated after service reinstallation

Problem statement
: The application information gets removed during Backup & Restore service uninstallation. It does not get repopulated in MongoDB after the service reinstallation. Restore jobs validation fail with the following error message.

```
postJobRequest: Application Info not returned from application service for applicationId
```

Resolution
: The workaround is to restart the `isf-application-operator-controller-manager` pod in the IBM Fusion namespace.

Backup & Restore server installation displays an Invalid Input error

Problem statement
: The Backup & Restore service deployment fails with the following error in the IBM Fusion user interface:

```
Install Error: Invalid Input: Both the api Server and bootstrapToken fields need to be populated
```

Resolution
: As a resolution, try with a private browser window.

Backup & Restore hub service in a custom namespace

Problem statement
: In general, the Backup & Restore service is installed or upgraded to the default **ibm-backup-restore** namespace. To avoid issues during the installation or upgrade of the service with custom namespace, do the resolution steps.

Resolution

1. Go to Operators > Installed Operators > Fusion Service Definition.
2. Open the **ibm-backup-restore-service** CR and go to the YAML tab.
3. In spec.onboarding.parameters, search for parameter with **name** as namespace and change the defaultValue to the custom-namespace where the service must be installed.
   Example:

```
parameters:
      - dataType: string
        name: namespace
        defaultValue: <custom-namespace>
        userInterface: false
        required: true
        descriptionCode: BMYSRV00003
        displayNameCode: BMYSRV00004
```

4. Store the following YAML in server-fsi.yaml, and replace **<custom-namespace>** with the namespace where the service must be installed.

```
apiVersion: service.isf.ibm.com/v1
kind: FusionServiceInstance
metadata:
  name: ibm-backup-restore-service-instance
  namespace: ibm-spectrum-fusion-ns
spec:
  parameters:
    - name: doInstall
      provided: false
      value: 'true'
    - name: namespace
      provided: false
      value: <custom-namespace>
    - name: storageClass
      provided: true
      value: lvms-lmvg
  serviceDefinition: ibm-backup-restore-service
  triggerUpdate: false
  enabled: true
  doInstall: true
```

5. Run the following command to apply the changes:

```
oc apply -f server-fsi.yaml
```

6. For the upgrade procedure, upgrade the IBM Fusion operator, repeat step 1 of the resolution, and then upgrade the service.

Pods in Crashloopbackoff state after upgrade

Problem statement
: The Backup & Restore service health changes to unknown and two pods go into Crashlookbackoff state.

Resolution
: In the resource settings of **guardian-dp-operator** pod that is in **ibm-backup-restore** namespace, set the value of IBM Fusion operator memory limits to 1000 mi.

Example:

```
resources:
          limits:
            cpu: 1000m
            memory: 1000Mi
          requests:
            cpu: 500m
            memory: 250Mi
```

Backup & Restore service goes into unknown state

Problem statement
: This Backup & Restore service might go into unknown state when you upgrade IBM Fusion from 2.6.x to 2.13.0.

Resolution
: It automatically shows healthy on the IBM Fusion user interface after you upgrade the Backup & Restore service.

MongoDB pod crashes with `CrashLoopBackOff` status

Problem statement
The MongoDB pod crashes due to an OOM error.

Resolution
To resolve the error, increase the memory limit from 256Mi to 512Mi. Do the following steps to change the memory limit:

1. Log in to the Red Hat® OpenShift® web console as an administrator.
2. Go to Workloads > StatefulSet.
3. Select the project **`ibm-backup-restore`**.
4. Select the MongoDB pod, and go to the YAML tab.
5. In the YAML, change the memory limit for MongoDB container from 256Mi to 512Mi.
6. Click Save.

Backup & Restore service installation and upgrade issue with the IBM Storage Protect Plus catalog source

Problem statement
The backups can fail with a **`FailedValidation`** error after you install or upgrade the IBM Fusion and Backup & Restore service to 2.8.0.

```
validationErrors': ['an existing backup storage location wasn't specified at backup creation time and the default guardian-min
```

```
Error: BackupStorageLocation.velero.io "guardian-minio" not found'
```

It occurs when the IBM Storage Protect Plus catalog exists in the OpenShift Container Platform clusters.

Resolution
The workaround is to upgrade the OADP, installed in the **`ibm-backup-restore`** namespace, to version 1.3.

Troubleshooting Backup & Restore service issues

List of known Backup & Restore issues in IBM Fusion.

Small IBM Scale PersistentVolumeClaims (PVCs) can cause failures

Small IBM Scale PersistentVolumeClaims (PVCs) are more prone to VolumeSnapshot failures. The smaller the PVC, the higher the likelihood. However, failures can occur regardless of size. According to IBM Scale, 10Gi is the threshold above which failures are quite unlikely, and below which they may begin to occur on a semi-regular basis.

Common issues

This section outlines common issues and their resolutions across multiple subcategories, such as backup, restore, service protection.

Rack shutdown in a multi-rack setup

Problem statement
A rack shutdown in a multi-rack HA setup can cause the Backup & Restore service to go to an unknown state.

Resolution
In the case of a rack shutdown, if a graceful shutdown is not working or the node is in a non-recoverable state, you can manually add an **`out-of-service`** taint on the node. For more information about how it works, see Kubernetes Blogs.
Remove the taint on the affected node after the node is recovered.

Backup or restore fails because of PVCs

Problem statement
Backup & Restore fails with the following error message:

```
error to create persistentvolumeclaims "<claim name>" is forbidden: exceeded quota"
```

Cause
During backup or restore operations, the Backup & Restore service temporarily allocates additional storage through PersistentVolumeClaims to stage application data. The requested storage size can even match the application size backed up or restored.

Resolution

- **Scenario 1: Failure due to ResourceQuota**:
Adjust or remove the storage.capacity field from the **`ResourceQuota`** object to allow the Backup & Restore service to allocate staging PersistentVolumeClaims as needed. The quota must equal the total size of the largest application's PersistentVolumeClaims, plus an additional 10Gi for local in-place policy storage. A higher quota may be required when you perform parallel operations that span all applications backed up or restored simultaneously.

- **Scenario 2: Failure due to ClusterResourceQuota**:
During backup or restore operations, Backup & Restore service temporarily consumes additional storage using PersistentVolumeClaims to stage data. To ensure successful backup and restore, either exclude the Backup & Restore namespace from the quota or increase it sufficiently. The quota must be set to the total size of the largest application's PersistentVolumeClaims, plus an additional 10Gi for local in-place policy storage. A higher quota may be necessary when you perform parallel operations that span all applications backed up or restored simultaneously.

For more information about resource quotas, see Red Hat Documentation.

- **Backup issues**
  List of backup issues in the Backup & Restore service of IBM Fusion.
- **Restore issues**
  List of restore issues in Backup & Restore service of IBM Fusion.
- **IBM Software Hub backup and restore issues**
  This section lists the troubleshooting tips and tricks during backup and restore of IBM Software Hub.
- **Hub and spoke connection issues**
  Procedure to debug issue in the hub and spoke connections. Backup & Restore service uses connection CR to setup hub and spoke connection.
- **Backup & restore service from OpenShift Container Platform console**
  List of issues in the Backup & Restore managed and monitored from the OpenShift® Container Platform console.

Backup issues

List of backup issues in the Backup & Restore service of IBM Fusion.

Backup size for block volumes is not correct

When the policy backup storage location gets changed, the backup size for application backups with Ceph RBD block volumes is incorrectly displayed during the first backup. As the correct information is displayed after the first backup following the change in the backup storage location, you can ignore this issue.

Multiple VM backups failing with snapshot deadline exceeded error

Resolution

1. Run the following command to enable full auditing:

```
sudo auditctl -w /etc/shadow -p
```

2. Rerun the backup and run the following command to identify the file that is causing ownership or permission issues.

```
sudo ausearch -m avc -ts recent
```

3. If the **ausearch** command reports issues, then run the following commands to generate a local policy to allow access.

```
sudo ausearch -c 'qemu-ga' --raw | audit2allow -M my-qemuga
sudo semodule -X 300 -i my-qemuga.pp
```

Failed to create snapshot content

Problem statement
: Failed to create snapshot content with the following error:
`Cannot find CSI PersistentVolumeSource for directory-based static volume`

Resolution
: To resolve the error, see Create a VolumeSnapshot.

Assign a backup policy operation fails

Problem statement
: If you have a PolicyAssignment for an application on the hub and you create a PolicyAssignment for the same application on the spoke, then your attempt to assign a backup policy for the application fails. In both assignments, the application, backup policy, and short-form cluster name are the same. The current format of the PolicyAssignment CR name is `appName-backupPolicyName-shortFormClusterName`. The issue happens when the first string of the cluster names is identical. In this scenario, the creation gets rejected because the PolicyAssignment name exists in OpenShift® Container Platform.
For example:

Hub assignment creates `app1-bp1-apps`:

- Application - `app1`
- BackupPolicy - `bp1`
- AppCluster - `apps.cluster1`

Spoke assignment creates `app1-bp1-apps` (The OpenShift Container Platform rejects it)

- Application - `app1`
- BackupPolicy - `bp1`
- AppCluster - `apps.cluster2`

Resolution
: To create the PolicyAssignment for the spoke application, delete the PolicyAssignment CR for the hub application assignment and attempt spoke application assignment again.

Backups do not work as defined in the backup policies

Problem statement
: Sometimes, backups do not work as defined in the backup policies, especially when you set hourly policies. For example, if you set a policy for two hours and it does not run every two hours, then gaps exist in the backup history. The possible reason might be that during pod crash and restart, scheduled jobs were not accounting for the time zone, causing gaps in run intervals.

Diagnosis
: The following are the observed symptoms:

- Policies with custom every X hour at minute YY schedules: the first scheduled run of this policy will run at minute YY after X hours + time zone offset from UTC instead of at minute YY after X hours.
- Monthly and yearly policies run more frequently.

Resolution
: You can start backups manually until the next scheduled time.

Backup jobs are stuck in a running state for a long time and are not canceled

Resolution
: Do the following steps to resolve the issue:

1. Ensure that all jobs are finished and the queue is empty before you do any disruptive actions like node restarts.
2. If jobs are running for a long period and do not progress, follow the steps to delete the backup or restore CR directly.
    a. Log in to IBM Fusion.
    b. Go to Backup & Restore > Jobs > Queue and get the name of the job that is stuck.
    c. Run the following command to delete backup job.

    ```
    oc delete fbackup <job_name>
    ```

    d. Run the following command to delete restore job.

    ```
    oc delete frestore <job_name>
    ```

Policy creation

Problem statement
: Sometimes, when you create a backup policy, the following errors can occur:

```
Error: Policy daily-snapshot could not created.
```

Resolution
: Restart the **`isf-data-protection-operator-controller-manager-* pod`** in IBM Fusion namespace. It triggers the recreation of the in-place-snapshot BackupStorageLocation CR.

Policy assignment from Backup & Restore service page of the OpenShift Container Platform console

Problem statement
: In the Backup & Restore service page of the OpenShift Container Platform console, the backup policy assignment to an application fails with a gateway timeout error.

Resolution
: Use your IBM Fusion user interface.

Backup of multiple VMs attempt is failed

Problem statement
: This issue occurs when some VMs are in a migrating state. The OpenShift Container Platform does not support snapshot of the VMs in migrating state.

Resolution
: Follow the steps to resolve this issue:

1. Check whether the virtual machine is in a migrating state:
2. Run the following command to check migrating VM.

    ```
    oc get virtualmachineinstancemigrations -A
    ```

    Example output:

    ```
    NAMESPACE           NAME                                          PHASE         VMI
    fb-bm1-fs-1-5g-10   rhel8-lesser-wildcat-migration-8fhbo          Failed        rhel8-lesser-wildcat
    vm-centipede-bm2    centos-stream9-chilly-hawk-migration-57jyk    Failed        centos-stream9-chilly-hawk
    vm-centos9-bm1-1    centos-stream9-instant-toad-migration-bfyz6   Failed        centos-stream9-instant-toad
    vm-centos9-bm1-1    centos-stream9-instant-toad-migration-d9547   Failed        centos-stream9-instant-toad
    vm-windows10-bm2-1  kubevirt-workload-update-4dm57                Failed        win10-zealous-unicorn
    vm-windows10-bm2-1  kubevirt-workload-update-f2s5w                Failed        win10-zealous-unicorn
    vm-windows10-bm2-1  kubevirt-workload-update-gt6nj                Failed        win10-zealous-unicorn
    vm-windows10-bm2-1  kubevirt-workload-update-rjwmn                Failed        win10-zealous-unicorn
    vm-windows10-bm2-1  kubevirt-workload-update-vfxfl                TargetReady   win10-zealous-unicorn
    vm-windows10-bm2-1  kubevirt-workload-update-z2thw                Failed        win10-zealous-unicorn
    vm-windows11-bm2-1  kubevirt-workload-update-9gr6v                Failed        win11-graceful-coyote
    vm-windows11-bm2-1  kubevirt-workload-update-clbck                Failed        win11-graceful-coyote
    vm-windows11-bm2-1  kubevirt-workload-update-j6pmx                Failed        win11-graceful-coyote
    vm-windows11-bm2-1  kubevirt-workload-update-sfbbx                Pending       win11-graceful-coyote
    vm-windows11-bm2-1  kubevirt-workload-update-th5dd                Failed        win11-graceful-coyote
    vm-windows11-bm2-1  kubevirt-workload-update-zl679                Failed        win11-graceful-coyote
    vm-windows11-bm2-2  kubevirt-workload-update-7dp6g                Failed        win11-conservative-moth
    vm-windows11-bm2-2  kubevirt-workload-update-9nb9m                TargetReady   win11-conservative-moth
    vm-windows11-bm2-2  kubevirt-workload-update-cdrf5                Failed        win11-conservative-moth
    vm-windows11-bm2-2  kubevirt-workload-update-dm8fz                Failed        win11-conservative-moth
    ```

```
vm-windows11-bm2-2   kubevirt-workload-update-kwr6c              Failed        win11-conservative-moth
vm-windows11-bm2-2   kubevirt-workload-update-zt8wx              Failed        win11-conservative-moth
```

3. Exclude the migrating virtual machine from the backup. Reattempt it after the migration is complete.

Backup applications table does not show the new backup times for the backed-up applications

Problem statement
: The backup applications table does not show the new backup times for the backed-up applications.

Resolution
: Go to the Applications and Jobs view to see the last successful backup job for a given application. For applications on the hub, the Applications table has the correct last backup time.

Backups are failing for the virtual machines

Problem statement
: The backups and snapshots are failing for the virtual machines that is mounted with second disk.

Resolution

1. Run the following command to get disks details for the virtual machine.

   ```
   oc get virtualmachine -A -o json | jq '.items[] | [{name:.metadata.name, namespace:.metadata.namespace, volumes:.spec.template.spec.volumes}] | select(.[].volumes[].dataVolume | length > 1) | {name :.[].name, namespace:.[].namespace}'
   ```

   Example output:

   ```
   {
   "name": "rhel9-absent-basilisk",
   "namespace": "vmtesting"
   }
   ```

2. If you find the virtual machines are mounted with second disk, then follow the steps mentioned in the Red Hat solution to resolve the issue.

Known issues and limitations

- When the Backup & Restore Hub auto-upgrade setting is disabled, the Backup & Restore service may still be upgraded to IBM Fusion 2.13.0.
- Note: This issue only occurs with the first job after installation.
  After a fresh installation, the first job might not process as expected. To resolve this issue, cancel the job and wait for the cancellation to complete. Then, retry the job to process it successfully.
- The OpenShift Container Platform cluster can have problems and become unusable. After you recover the cluster, rejoin the connections. For the steps to clean the connection and setup the connection between two clusters again, see Connection setup after OpenShift Container Platform cluster recovery. OpenShift Container Platform cluster can have problems and become unusable.
- The S3 bucket must not have an expiration policy or an archive rule. For more information about this known issue, see S3 buckets must not enable expiration policies.
- The Azure Endpoint URL must not contain the name of the bucket.
- When you backup an application present in namespace with high security context constraints privileges. The restored namespace will not have the same security context constraints privileges, resulting in restored pods in crashloopbackoff status.
  Workaround:
  - Restart the application pod.
- When you configure node affinity for the Backup & Restore, consider the following limitations:
  - There is no way to add `nodeAffinity` in `DataProtectionApplication` CR. The Velero pod may get scheduled on the GPU nodes, even if other nodes are eligible for scheduling.
  - There is one `Node-agent` pod per node, which continues to run on GPU nodes as well.
    For more information about Configuring the OpenShift API for Data Protection with Multicloud Object Gateway, see Red Hat documentation.
  - There is no way to specify `nodeAffinity` for AMQ and OAPD operator pods. These operator pods may get scheduled on the GPU nodes, even if other nodes are eligible for scheduling.
  - There is no way to specify anti `NodeAffinity` for OLM jobs pods, so these pods may get scheduled on the GPU nodes, even if other nodes are eligible for scheduling. However, these are very short lived pods.
  - The `datamover` pod for restore work may get scheduled on a GPU node. Currently, Kopia honors the setting for backup job pod, but not for restore job pod.
  - In summary, the following pods do not have the logic to avoid GPU nodes:
    - Velero
    - Datamover (for restore)
    - AMQ Operator
    - OAPD operator
    - Node Agent
    - OLM jobs

Restore issues

List of restore issues in Backup & Restore service of IBM Fusion.

Restore Fails with `Timeout on Preparing Data Download` error

Problem statement

The restore operation is inconsistent and sometimes fails with the error timeout on preparing data download in the data mover pod due to reason: failed to add finalizer. This issue is observed in certain environments, and the cause is not immediately apparent.

Symptoms

- Restore operation fails with a timeout error.
- Error message indicates a problem with preparing data download in the data mover pod.
- The specific error message is failed to add finalizer.

Cause

The cause of this issue is not definitively known but is related to the functioning of the IBM Storage Scale Container Storage Interface (CSI) pods. The pods may be experiencing issues that prevent the successful completion of the restore operation.

Resolution

To resolve this issue, follow these steps:

1. Run the following command to delete the IBM Storage Scale Container Storage Interface (CSI) pods. For versions of IBM Storage Scale Container Native Storage Access (CNSA) prior to 6.0.1, use **`ibm-spectrum-scale-csi`** as the namespace instead of **`ibm-spectrum-scale`**.

   ```
   kubectl delete pod -l product=ibm-spectrum-scale-csi -n ibm-spectrum-scale
   ```

   Example output:

   ```
   pod "ibm-spectrum-scale-csi-attacher-86c4864c94-tlcqn" deleted
   pod "ibm-spectrum-scale-csi-attacher-86c4864c94-wn6sr" deleted
   pod "ibm-spectrum-scale-csi-bqtwj" deleted
   pod "ibm-spectrum-scale-csi-dqsgr" deleted
   pod "ibm-spectrum-scale-csi-mskc6" deleted
   pod "ibm-spectrum-scale-csi-provisioner-67b878b57-6lftl" deleted
   pod "ibm-spectrum-scale-csi-resizer-84cf579d49-bx4jr" deleted
   pod "ibm-spectrum-scale-csi-rrggd" deleted
   pod "ibm-spectrum-scale-csi-snapshotter-8b69bc4dd-xkt4s" deleted
   pod "ibm-spectrum-scale-csi-srh2z" deleted
   ```

2. Run the following command to verify that the IBM Storage Scale Container Storage Interface (CSI) pods restarted successfully.

   ```
   oc get pods -l product=ibm-spectrum-scale-csi -n ibm-spectrum-scale
   ```

   Example output:

   ```
   NAME                                                READY   STATUS    RESTARTS   AGE
   ibm-spectrum-scale-csi-attacher-86c4864c94-55z9p    1/1     Running   0          7s
   ibm-spectrum-scale-csi-attacher-86c4864c94-t9sww    1/1     Running   0          7s
   ibm-spectrum-scale-csi-frmtq                        3/3     Running   0          6s
   ibm-spectrum-scale-csi-g4mqr                        3/3     Running   0          6s
   ibm-spectrum-scale-csi-hxb4q                        3/3     Running   0          6s
   ibm-spectrum-scale-csi-provisioner-67b878b57-ltj4g  1/1     Running   0          7s
   ibm-spectrum-scale-csi-q5jgp                        3/3     Running   0          6s
   ibm-spectrum-scale-csi-resizer-84cf579d49-wvgv2     1/1     Running   0          7s
   ibm-spectrum-scale-csi-rgv5d                        3/3     Running   0          7s
   ibm-spectrum-scale-csi-snapshotter-8b69bc4dd-7cpq5  1/1     Running   0          7s
   ```

Restore fails because of ephemeral local storage exceeding limits error in Velero pod

Resolution

When you back up and restore a large number of cluster resources, Velero pod resource limits must be increased for the operations to succeed. The optimal values depend on your individual environment, but the following values are representative for backup and restore of 1500 resources. To change the local storage the Velero Pod is allowed to use from the default 500Mi (Mebibytes) to 1Gi (Gibibytes):

```
oc patch dpa velero -n <backup-restore-namespace> --type merge -p '{"spec": {"configuration": {"velero": {"podConfig": {"resourceAllocations": {"limits": {"ephemeral-storage": "1Gi"}}}}}}}'
```

To change the memory limit of the Velero Pod from default 2Gi (Gibibytes) to 4Gi (Gibibytes):

```
oc patch dpa velero -n <backup-restore-namespace> --type merge -p '{"spec": {"configuration": {"velero": {"podConfig": {"resourceAllocations": {"limits": {"memory": "4Gi"}}}}}}}'
```

Restore of applications from hub or spoke 2.9.0 to spoke at version 2.8.1 fails

Problem statement

Restore of applications from hub or spoke 2.9.0 to spoke at version 2.8.x fails with error in Transaction Manager service:

```
Argument of type 'NoneType' is not iterable.
```

Resolution

This is a known limitation so do not attempt restore 2.9.0 (kopia) based backups to a lower version 2.8.x (restic).

IBM Cloud Limitations with OADP DataMover

Problem statement

Backups of raw volumes fail for IBM Cloud with OADP 1.4.0 or lower.

Cause

The OADP 1.3 or higher expose the volume of the underlying host during backup or restore. The folders on the host are exposed in the Pods that are associated with the Daemonset node-agent. The /var/lib/kubelet/{plugins,pods} folder is exposed by default. The folders required to work on IBM Cloud are /var/data/kubelet/{plugins,pods}. As a result, the backup and restore of volumeMode: block volumes fail with the following example error:

```
Failed transferring data
```

```
[BMYBR0009](https://ibm.com/docs/SSFETU_2.9/errorcodes/BMYBR0009.html) There was an error when processing the job in the Trans
```

The ID of the individual DataUpload varies for jobs.

Resolution

1. Set the DataMover type to `legacy` by either the global method or the Per PolicyAssignment method. For the procedure to update the type, see [Configure DataMover type for Backup and Restore]().
   Though this workaround allows continued use of of DataMover `kopia` type, it has the following drawbacks:

   It disables further changes to DataMover and Velero configurations, such as the ability to change resource allocations (CPU, Memory, and Ephemeral-storage) and nodeSelectors (datamover node placement) for Datamover type `kopia`.

   Legacy is not affected by this workaround.

   To avoid job failures, do not make these changes while a backup or restore job is in progress.

2. In the OpenShift Console, go to Workloads > Deployments.
3. Select `openshift-adp-controller-manager` and scale the number of Pods to 0.
4. Go to Workloads > Daemonsets and select node-agent.
5. Select the YAML tab.
6. Under the volumes section, add the additional volume `host-data` as shown in the following example.
   Note: It exposes an additional folder on the host other than the folders mention in the Cause of this issue.

```
    volumes:
      - name: host-pods
        hostPath:
          path: /var/lib/kubelet/pods
          type: ''
      - name: host-plugins
        hostPath:
          path: /var/lib/kubelet/plugins
          type: ''
      - name: host-data
        hostPath:
          path: /var/data/kubelet
          type: ''
      - name: scratch
        emptyDir: {}
      - name: certs
        emptyDir: {}
```

7. Under `volumeMounts`, add the `host-data` volume as shown in the following example.

```
          volumeMounts:
            - name: host-pods
              mountPath: /host_pods
              mountPropagation: HostToContainer
            - name: host-plugins
              mountPath: /var/lib/kubelet/plugins
              mountPropagation: HostToContainer
            - name: host-data
              mountPath: /var/data/kubelet
              mountPropagation: HostToContainer
            - name: scratch
              mountPath: /scratch
            - name: certs
              mountPath: /etc/ssl/certs
```

8. Save the changes and wait for a couple minutes for the Pods to restart.
   The backups and restores of PersistentVolumeClaims with volumeMode: block succeeds on Red Hat® OpenShift® of IBM Cloud.

Restore fails on IBM Power Systems

Restore failures are observed in a IBM Power Systems environment. If you have cluster(s) running on IBM Power Systems, do not upgrade the Backup & Restore service to 2.8.1 instead contact [Contact IBM Support]().

exec format error

Problem statement

Sometimes, you may observe the following error message:

```
"exec <executable name>": exec format error
```

For example:

```
The pod log is empty except for this message: exec /filebrowser
```

The example error can be due to the wrong architecture of the container. For example, an amd64 container on s390x nodes or an s90x container on amd64 nodes.

Resolution

As a resolution, check whether the container that you want to restore and the local node architecture match.

Restore of namespaces that contains admission webhooks fails

Problem statement
: Restore of namespaces that contains admission webhooks fails.

Example error in IBM Fusion restore job:

```
"Failed restore <some  resource>" "BMYBR0003
      RestorePvcsFailed There was an error when  processing the job in the Transaction Manager
      service"
```

Example error in Velero pod:

```
level=error msg="Namespace
      domino-platform, resource restore error: error restoring
      certificaterequests.cert-manager.io/domino-platform/hephaestus-buildkit-client-85k2v:
      Internal error occurred: failed calling webhook  "webhook.cert-manager.io": failed to call
      webhook: Post  "https://cert-manager-webhook.domino-platform.svc:443/mutate?timeout=10s\\":
      service "cert-manager-webhook" not found"
```

Resolution

1. Identify the admission webhooks that is applicable to the namespace being restored:

```
oc get mutatingwebhookconfigurations
oc describe mutatingwebhookconfigurations
```

2. Change the failure Policy parameter from **Fail** to **Ignore** to temporarily disable webhook validation prior to restore:

```
failurePolicy: Ignore
```

Restore before upgrade fails with a BMYBR0003 error

Problem statement
: When you try to restore backups before upgrade, it fails with a BMYBR0003 error.

Diagnosis
: After you upgrade, your jobs may fail:

- Backup jobs with the status:

```
"Failed transferring data" "BMYBR0003 There was an error when processing the job in the Transaction Manager service"
```

- Restore jobs with the status:

```
"Failed restore <some resource>" "BMYBR0003 There was an error when processing the job in the Transaction Manager service
```

Confirm the issue in the logs of the manager container of the Data Mover pod.

A sample error message:

```
2023-07-26T03:39:47Z    ERROR   Failed with error.      {"controller": "guardiancopyrestore", "controllerGroup": "guardian.isf
github.ibm.com/ProjectAbell/guardian-dm-operator/controllers/util.Retry
        /workspace/controllers/util/utils.go:39
github.ibm.com/ProjectAbell/guardian-dm-operator/controllers/kafka.(*kafkaWriterConnection).PublishMessage
        /workspace/controllers/kafka/kafka_native_connection.go:71
github.ibm.com/ProjectAbell/guardian-dm-operator/controllers.(*GuardianCopyRestoreReconciler).updateOverallCrStatus
        /workspace/controllers/status.go:191
github.ibm.com/ProjectAbell/guardian-dm-operator/controllers.(*GuardianCopyRestoreReconciler).doRestore
        /workspace/controllers/guardiancopyrestore_controller.go:187
github.ibm.com/ProjectAbell/guardian-dm-operator/controllers.(*GuardianCopyRestoreReconciler).Reconcile
        /workspace/controllers/guardiancopyrestore_controller.go:92
sigs.k8s.io/controller-runtime/pkg/internal/controller.(*Controller).Reconcile
        /go/pkg/mod/sigs.k8s.io/controller-runtime@v0.14.6/pkg/internal/controller/controller.go:122
sigs.k8s.io/controller-runtime/pkg/internal/controller.(*Controller).reconcileHandler
        /go/pkg/mod/sigs.k8s.io/controller-runtime@v0.14.6/pkg/internal/controller/controller.go:323
sigs.k8s.io/controller-runtime/pkg/internal/controller.(*Controller).processNextWorkItem
        /go/pkg/mod/sigs.k8s.io/controller-runtime@v0.14.6/pkg/internal/controller/controller.go:274
sigs.k8s.io/controller-runtime/pkg/internal/controller.(*Controller).Start.func2.2
        /go/pkg/mod/sigs.k8s.io/controller-runtime@v0.14.6/pkg/internal/controller/controller.go:235
```

Resolution:
: Search for the **guardian-dm-controller-manager** and kill it. A new pod starts in a minute. After the pod reaches a healthy state, retry backup and restores.

"Failed restore snapshot" error occurs with applications using IBM Storage Scale storage PVCs

- A "Failed restore snapshot" error occurs with applications using IBM Storage Scale storage PVCs.

  Cause
  : The "disk quota exceeded" error occurs whenever you restore from an object storage location having applications that use IBM Storage Scale PVC with a size less than 10 GB.

  Resolution
  : Increase the IBM Storage Scale PVC size to a minimum of 10 GB and do a backup and restore operation.

Restore to a cluster that does not have an identical storage class as the source cluster

You cannot restore to a cluster that does not have an identical storage class as the source cluster. However, the transaction manager still attempts to create PVCs with the non-existent storage class on the spoke cluster and eventually fails with **Failed restore snapshot** status.

Applications page does not show the details of the application

Problem statement

The new backed up applications page does not show the details of the application when you upgrade IBM Fusion to the latest version while leaving the Backup & Restore service in the older version.

Resolution

As a resolution, it is recommended to upgrade the Backup & Restore service to the latest version after an IBM Fusion upgrade.

S3 buckets must not enable expiration policies

Backup & Restore currently uses restic to move the Velero backups to repositories on S3 buckets. When a restic repository gets initialized, it creates a configuration file and several subdirectories for its snapshots. As restic does not update it after initialization, the modification timestamp of this configuration file is never updated.

If you configure an expiration policy on the container, the restic configuration objects get deleted at the end of the expiration period. If you have an archive rule set, the configuration objects get archived after the time defined in the rule. In either case, the configuration is no longer accessible and subsequent backup and restore operations fail.

Note: The S3 buckets must not enable expiration policies. Also, the bucket must not have an archive rule set.

Virtual machine restore failure

Problem statement

By default, the VirtualMachineClone objects are not restored due to the following unexpected behavior:

- If you create a VirtualMachineClone object and delete the original virtual machine, then the restore fails because the object gets rejected.
- If you create a VirtualMachineClone object and then delete the clone virtual machine, then the restore fails because the Virtualization ignores the `status.phase` "Succeeded" and clones the virtual machine again.
  As a result, the clone gets re-created every time you delete it.
- If you create a VirtualMachineClone and then do a backup with the original and clone virtual machine, the restore fails because it ignores the `status.phase` "Succeeded" and tries to clone again to the virtual machine that exists.
  The Openshift Virtualization creates a snapshot of the original VirtualMachine, which adds an unwanted VirtualMachineSnasphot and a set of associated VolumeSnapshots after restore whose name starts with "tmp-". The clone operation does not complete and remains stuck in "RestoreInProgress" state because the requested VirtualMachine exists in the VirtualMachineClone.

Resolution

As a resolution, force the restore of the VirtualMachineClone objects by explicitly including it in the Recipe.
Change the OADP `DataProtectionApplication` object "velero" and add in the `spec.configuration.velero.args.restore-resource-priorities` field as follows:

```
    velero:
      args:
        restore-resource-priorities: 
"securitycontextconstraints,customresourcedefinitions,namespaces,managedcluster.cluster.open-cluster-
management.io,managedcluster.clusterview.open-cluster-management.io,klusterletaddonconfig.agent.open-cluster-
management.io,managedclusteraddon.addon.open-cluster-
management.io,storageclasses,volumesnapshotclass.snapshot.storage.k8s.io,volumesnapshotcontents.snapshot.storage.k8s.io,vo
lumesnapshots.snapshot.storage.k8s.io,datauploads.velero.io,persistentvolumes,persistentvolumeclaims,serviceaccounts,secre
ts,configmaps,limitranges,pods,replicasets.apps,clusterclasses.cluster.x-
k8s.io,endpoints,services,-,clusterbootstraps.run.tanzu.vmware.com,clusters.cluster.x-
k8s.io,clusterresourcesets.addons.cluster.x-k8s.io,virtualmachines.kubevirt.io,virtualmachineclones.clone.kubevirt.io"
```

Service recovery involving a large number of CRs

Problem statement

When service recovery processes a large number of CRs, service discovery may fail after you add the backup storage location.

Resolution

1. Check whether `backup-location` pod is evicted due to the ephemeral storage limit exceeded.
2. If no `backup-location` pod is evicted, check the logs of backup-location pod for any of the following messages:
   - `Processing service protection discovery was successful`
   - `Processing service protection discovery failed`
   - `Error processing service protection discovery`

   If you do not see any of these messages, then the service discovery may be still in running state.
3. If the `backup-location` pod restarts because of ephemeral storage limit, update the resources of backup location deployment temporarily. Increase the ephemeral storage limit from 512 Mi to 1 Gi, and the memory limit from 512 Mi to 2 Gi. These limits must be changed back after the service discovery completes populating the service backups that are available for restore.

   ```
   oc set resources deployment backup-location-deployment --limits=ephemeral-storage=1Gi,memory=2Gi -n <backup-restore-namespace>
   ```
4. After the service discovery completes, search the `backup-location` pod logs for the following message, which includes the number of resources found in the latest, recoverable service backup:

   ```
   Restic backup validation processed
   ```

   Example resources:
   - 363 Application CRs
   - 40 Fusion BackupStorageLocation CRs

- 385 Secrets for BackupStorageLocation CRs
- 25 BackupPolicy CRs
- 565 PolicyAssignment CRs
- 12126 Backup CRs

Note: There may be more than one such message if service backups are not suitable for recovery, such as containing no Backup CRs.
For more information about Backup & Restore Hub performance and scaling, see [Backup & restore hub performance and scaling](#).

5. Update the resources of the transaction manager to temporarily increase the memory limit from 500 Mi to 2 Gi.
   For example:

   ```
   oc set resources deployment transaction-manager --limits=memory=2Gi -n <backup-restore-namespace>
   ```

6. Update the resources of backup services to temporarily increase the number of replicas. For example, 8 for the previous step example, as well as increasing the memory limit from 1 Gi to 2 Gi.

   ```
   oc set resources deployment backup-service --limits=memory=2Gi -n <backup-restore-namespace>
   ```

   Sample output:

   ```
   The following command isn't working; scale the number of replicas in OpenShift UI.
   oc scale deployment backup-service --replicas=8 -n <backup-restore-namespace>
   ```

7. Update the resources of mongodb based on the number of CRs to be processed. In this example, temporarily increase the CPU limit to 4 and the memory limit to 4 Gi.

   ```
   oc set resources sts mongodb --limits=cpu=4,memory=4Gi -n <backup-restore-namespace>
   ```

8. Update the resources of velero resources. Modify the `DataProtectionApplication` CR named Velero:

   ```
       velero:
         customPlugins:
           - image: 'cp.icr.io/cp/fbr/guardian-kubevirt-velero-
   plugin@sha256:bce6932338d8e2d2ce0dcca2c95bdfa8ab6e337a758e039ee91773f3b26ceb06'
             name: isf-kubevirt
         defaultPlugins:
           - openshift
           - aws
         noDefaultBackupLocation: false
         podConfig:
           labels:
             app.kubernetes.io/part-of: ibm-backup-restore
           resourceAllocations:
             limits:
               cpu: '2'
               ephemeral-storage: 500Mi
               memory: 2Gi
             requests:
               cpu: 200m
               ephemeral-storage: 275Mi
               memory: 256Mi
   ```

   In this example, update the ephemeral-storage limit from 500Mi to 1Gi.
9. Wait for all pods that were modified to restart and be in running state.
   Note: After the service recovers successfuly, it may take time for the services to processes the restored CRs.
10. Wait for all of the backup storage locations to be connected.
11. Check the backup service pods to determine if any CRs are still in progress.
12. After processing completes, revert the resources to their original values or adjust them based on the expected number of concurrent jobs.

More than one BackupRepository found for workload namespace

Problem statement
: When you restore a namespace with multiple PVCs, the restore job might fail with the following error message:

```
error to initialize data path: error to ensure backup repository <repo>: failed to wait BackupRepository, errored early: more
```

Cause
: It is due to a race condition during the creation of the Backup repository.

Resolution
: To workaround the problem, clean the restored namespaces and try again.

IBM Software Hub backup and restore issues

This section lists the troubleshooting tips and tricks during backup and restore of IBM Software Hub.

Original namespace as destination for IBM Software Hub application

Problem statement
: OpenShift® restore from IBM Storage Protect Plus does not force you to select the original namespace as destination for IBM Software Hub application.

Impact
: From IBM Storage Protect Plus user interface, IBM Software Hub restore fails whenever you do not select the original namespace as the target destination to restore to.

Resolution

Whenever you restore IBM Software Hub from IBM Storage Protect Plus user interface, ensure that you select the original namespace as the restore destination. These steps are applicable while you backup and test restore of IBM Software Hub applications. For more information about the procedure, see https://www.ibm.doc/docs/en/SSQNUZ_4.5_test/cpd/admin/online_bar_fusion_spp.html.

Tethered namespace issues

Problem statement
: For IBM Software Hub Application type, you can tether secondary or other namespaces to the primary IBM Software Hub platform namespace. However, after a successful restore, the IBM Fusion application associated with the IBM Software Hub namespace does not restore the tethered namespace.

Resolution

1. Uninstall `cpdbr-oadp` and reinstall it.
2. After you add the tethered namespaces, restart the `isf-application-controller-manager-xxxx` pod that is within the `ibm-spectrum-fusion-ns` to force a reconcile..
3. In about five minutes, run the following command to verify whether the `fapp` is updated to include the tethered namespace in `spec.includedNamespaces`

   ```
   oc get fapp -n ibm-spectrum-fusion-ns cpd-instance -o yaml
   ```

IBM Software Hub backup of zen namespace fails partially in IBM Storage Protect Plus user interface

Problem statement
: For an on-demand backup of a single application from a policy that has multiple applications assigned, a snapshot backup is taken only for that selected application but the copy backup is initiated for all of the applications assigned to the policy. This behavior causes the copy backup of the applications without a snapshot backup to fail. This issue does not impact the application selected for the ad-hoc backup.

Backup fails for the platform with error in EDB Postgres cluster

Cause
: Labels and annotations in the EDB Postgres cluster resources were not updated after a switchover of the EDB Postgres cluster's primary instance and replica.

Diagnosis
: For example, in IBM Fusion, the backup fails at the `Hook: br-service hooks/pre-backup` stage in the backup sequence.
In the `cpdbr-oadp.log` file, you see the following error:

```
time=<timestamp> level=info msg=cmd stderr: Error: cannot take a cold backup of the primary instance
or a target primary instance if the k8s.enterprisedb.io/snapshotAllowColdBackupOnPrimary annotation is not set to enabled
```

Resolution
: The following workaround will automatically run before you create a backup. This workaround is useful if you set up automatic backups.
Note: This resolution works only for online backup and restore.

1. Download the edb-patch-aux-ckpt-cm.yaml file.
2. Run the following command:

   ```
   oc apply -n ${PROJECT_CPD_INST_OPERATORS} -f edb-patch-aux-ckpt-cm.yaml
   ```

3. Retry the backup.

Hub and spoke connection issues

Procedure to debug issue in the hub and spoke connections. Backup & Restore service uses connection CR to setup hub and spoke connection.

You might encounter an error when you attempt setup connections between clusters.

Bootstrap token in init secret is not correct or expired: Unauthorized

Problem statement
: Connection setup fails with the following message in the connection CR:

```
apiVersion: application.isf.ibm.com/v1
kind: Connection
metadata:
  name: <connection-name>
  namespace: <connection-namespace>
spec:
  remoteCluster:
    apiEndpoint: <cluster api endpoint>
    connectionOperatorNamespace: <connection-namespace>
    heartBeatInterval: 10m
    initSecretName: <init-secret-name>
status:
  conditions:
    - lastTransitionTime: '2023-06-15T02:31:01Z'
      message: 'Bootstrap token in init secret is not correct or expired: Unauthorized'
      reason: CreateBootstrapSecret
```

```
      status: 'False'
      type: BootstrapSecretAvaliable
  connectionFromRemoteClusterHealth:
    message: ''
    messageCode: ''
    messageType: ''
  connectionState: Failed
  connectionToRemoteClusterHealth:
    message: ''
    messageCode: ''
    messageType: ''
```

Cause
: The bootstrap token in the `init` secret is not correct or expired.

Resolution

1. Get the bootstrap token again.

   ```
   oc create token isf-application-operator-cluster-bootstrap -n <connection-namespace>
   ```

2. Replace the token in `init` secret:

   ```
   oc edit secret <init-secret-name> -n <connection-namespace>
   ```

CA certificate of peer cluster is not correct

Problem statement
: The `CA certificate of peer cluster is not correct` error occurs in connection CR.

Cause
: The CAcert in the configmap `kube-root-ca.crt` in namespace `kube-public` of the remote cluster is not correct.

Resolution
: In the remote cluster, place the right CAcert in the configmap `kube-root-ca.crt` and namespace `kube-public`. Connection pkg also provides a customized configmap.
If it is not possible to place the right CAcert in configmap `kube-root-ca.crt` and namespace `kube-public`, then place the right CAcert in `custom-ca.crt` and Fusion namespace:

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: custom-ca.crt
  namespace: <connection-namespace>
data:
  ca.crt: <right CAcert>
```

Backup & restore service from OpenShift Container Platform console

List of issues in the Backup & Restore managed and monitored from the OpenShift® Container Platform console.

fusion-console plugin issue

Problem statement
: The OpenShift Container Platform console pod gets into panic state initially with a connection error.

Cause
: This error occurs because the inter-namespace communication gets blocked between OpenShift Container Platform console pod (`openshift-console` namespace) and `fusion-console` pod (running in IBM Fusion operator install namespace).

Resolution

1. Check whether any typo exists in the *.apps definition or there is a bad name resolution.
2. If the error persists even after you fix the issue in step 1, restart the Fusion proxy pods.
3. Verify that no `NetworkPolicy/NetworkPolicies` or networking plugin configurations exist that can prevent or interfere with inter-namespace communication.

Known issue

The Fusion Native Console plugin follows the n, n-1 & n-2 support matrix of OpenShift Container Platform, where "n" is an even OpenShift Container Platform version. For example, Fusion Console 2.10 on OpenShift Container Platform 4.18 release. In case of versions that are not supported, run the following command to disable the plugin:

```
oc patch -n ibm-spectrum-fusion-ns ISFConsolePlugin isf-console --type=merge -p '{"spec":{"enabled": 'false'}}'
```

Here, the `ibm-spectrum-fusion-ns` is the namespace where the IBM Fusion operator is installed.

Data Cataloging

Companies need the ability to use unstructured data to meet their business priorities.

IBM Data Cataloging is a container native modern metadata management software that provides data insight for exabyte-scale heterogeneous file, object, backup, and archive storage on premises and in the cloud. The software easily connects to these data sources to rapidly ingest, consolidate, and index metadata for billions of files and objects.

IBM Data Cataloging provides a rich metadata layer that enables storage administrators, data stewards, and data scientists to efficiently manage, classify, and gain insights from massive amounts of data. It improves storage economics, helps mitigate risk, and accelerates large-scale analytics to create competitive advantage and speed critical research.

Many companies face significant challenges to manage their data. Some difficult challenges that companies face include:

- Pinpointing and activating relevant data for large-scale analytics.
- Lacking the fine-grained visibility needed to map data to business priorities.
- Removing redundant, trivial, and obsolete data.
- Identifying and classifying sensitive data.

Benefits of IBM Data Cataloging

IBM Data Cataloging can help you manage your unstructured data by reducing the data storage costs, uncovering hidden data value, and reducing the risk of massive data stores. See Table 1.

Table 1. Benefits of IBM Data Cataloging

Optimize - Improve storage usage	Analyze - Uncover hidden data value	Govern - Mitigate risk and improve data quality	Data Management
Decreases storage capital expenditure (CapEx) by facilitating data movement to colder, cheaper storage.	Accelerates data identification for large-scale analytics.	Perform data inspection and classification.	Automate tags for custom insight.
Increases storage efficiency by eliminating trivial or redundant data.	Operationalize tasks to reduce the burden of data preparation.	Helps ensure that data is compliant with governance policies by labeling sensitive data.	Create reports for analysis.
Reduces storage operating expenditure (OpEx) by improving storage administrator productivity.	Orchestrates the ML/DL and Platform Symphony® MapReduce process.	Helps reduce risk that is hidden in heterogeneous data sources.	GUI search for real-time results Search content for fast discovery.

- **IBM Data Cataloging architecture**
  IBM Data Cataloging is an extensible platform that provides exabyte scale data ingest, data visualization, data activation, and business-oriented data mapping.
- **Installing IBM Data Cataloging**
  IBM Data Cataloging service is a modern metadata management software that provides data insight for exabyte-scale heterogeneous file, object, backup, and archive storage on premises and in the cloud. It can help you manage your unstructured data by reducing the data storage costs, uncovering hidden data value, and reducing the risk of massive data stores.
- **Uninstalling IBM Data Cataloging**
  Steps to uninstall IBM Data Cataloging by using command line interface.
- **Configuring IBM Data Cataloging**
  This section provides information on how to deploy and configure IBM Data Cataloging capabilities.
- **Administering IBM Data Cataloging**
  This section provides information on how to administering the IBM Data Cataloging capabilities.
- **Monitoring IBM Data Cataloging**
  This section provides information on how to monitor the IBM Data Cataloging capabilities.
- **Collecting logs and metrics**
  Steps to collect logs and metrics for the IBM Data Cataloging.
- **Data Cataloging Harvester CLI**
  IBM Data Cataloging Harvester is a new capability that is designed to import external data to the IBM Data Cataloging service catalog database. It imports the data even if it is not coming from a Db2 database that uses the same schema.
- **FKEY migration script**
  The FKEY migration script is a fix to avoid potential rare collisions in the FKEY file identifier when ingesting billions of records.
- **Configurable Db2 log trimmer**
  Db2 log trimmer tool provides a mechanism to trim the informative logs that are generated by scanning a data source. It uses the Harvester CLI to ingest data into the IBM Data Cataloging.
- **IBM Data Cataloging service installation and upgrade issues**
  Use this troubleshooting information to resolve install and upgrade problems that are related to the IBM Data Cataloging service.
- **IBM Fusion IBM Data Cataloging known issues**
  List of all issues in the IBM Data Cataloging service along with its resolution.

IBM Data Cataloging architecture

IBM Data Cataloging is an extensible platform that provides exabyte scale data ingest, data visualization, data activation, and business-oriented data mapping.

Note: All references to "Spectrum Discover" in the images refer to "Data Cataloging".

Exabyte-scale data ingest

- Scan billions of files and objects in a day
- Real-time event notifications
- Automatic indexing

Data Visualization

- Fast queries of billions of records
- Multi-faceted search
- Drilldown Dashboard

Data Activation

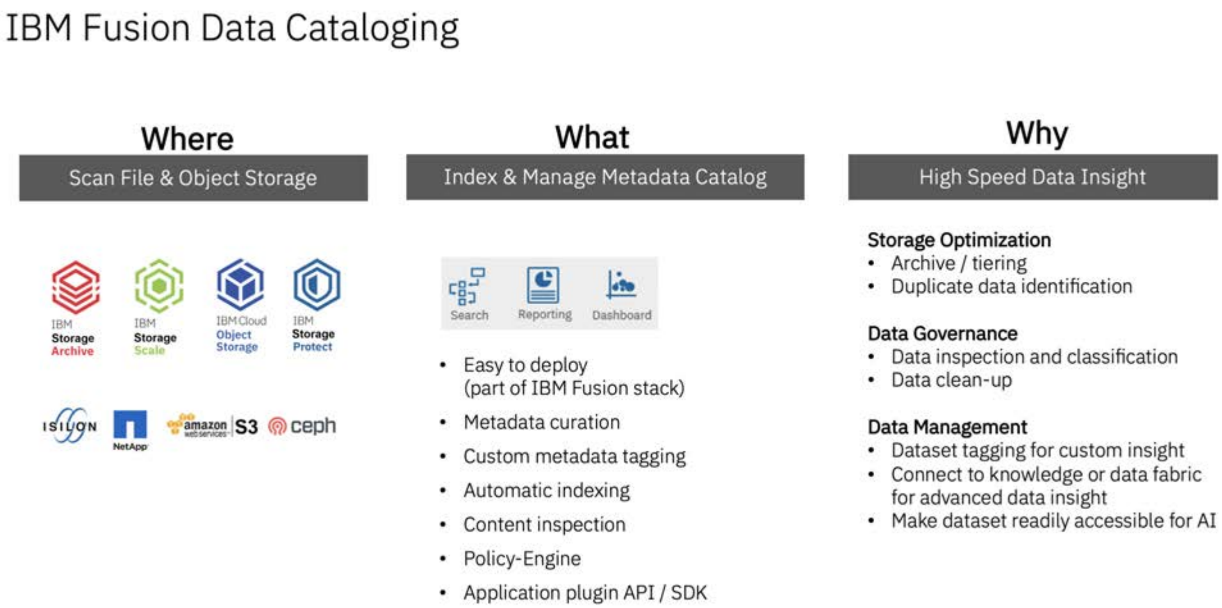
- Application software development kit (SDK)
- Extensible architecture
- Solution blueprints

Business-oriented data mapping

- System-level data tagging
- Contextual data tagging
- Policy-driven workflows

The following figure illustrates a high-level view of the IBM Data Cataloging architecture.

Figure 1. IBM Data Cataloging architecture



Data management

IBM Data Cataloging connects to the data sources shown in the architecture image (*IBM Data Cataloging architecture*), and automatically harvests and indexes the system metadata where the system metadata refers to certain information. This might include the following information.

- It might include the names of the files and objects.
- It might include the bucket or path where the data resides.
- It might include the size.
- It might include the time the data sources were last modified.

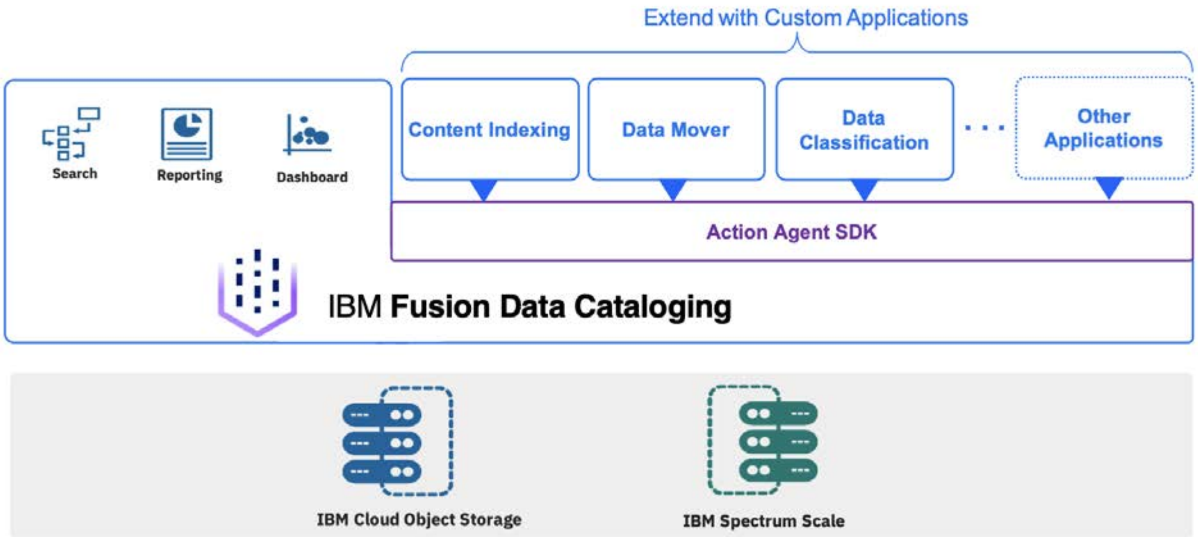
After the data is ingested, analytics are automatically applied to classify and group the data according to the different system metadata attributes. The data can be inspected automatically in IBM Data Cataloging by using built-in content search capabilities to identify sensitive and personally identifiable information and perform data classification. The content inspection capabilities can also be used by researchers and data scientists to extract content from their data sets. This easy-to-use extraction ability assists with data discovery.

The records that are maintained by IBM Data Cataloging can also be further enriched with custom metadata tags that map the data to business constructs and further increase the value of the data.

You can use IBM Data Cataloging catalog to gain insight about your data and to find your data easily.

The IBM Data Cataloging architecture also supports a community-supported catalog of open source applications that enhance and customize the capabilities of IBM Data Cataloging with third-party extensions. Users can find and install available applications and can develop and share new applications that use an SDK that contains sample code and a fully published API. For more information, see Creating your own applications to use in the Data Cataloging application catalog. For more information, see the topic *Creating your own applications to use in the IBM Data Cataloging application catalog* in the Administration section.

Figure 2. Application SDK architecture



- **Role-based access control**
  IBM Data Cataloging provides access to resources based on roles. You can restrict access to information based on roles.
- **Data source connections**
  A data source connection specifies the parameters for cataloging of metadata from a source system to IBM Data Cataloging.
- **Cataloging metadata**
  System metadata is created and updated by the host system, and not the application software. IBM Data Cataloging allows the addition of tags that can capture non-system metadata-specific attributes, which are stored in the IBM Data Cataloging catalog.
- **Enriching metadata**
  IBM Data Cataloging can enrich the metadata from supported platforms with additional information by using policies, content inspection, custom tags, and custom applications.
- **Graphical user interface**
  The IBM Data Cataloging graphical user interface is a portal that is used for running data searches, report generation, policy and tag management, and user Access Management. Based on a user's role, they might have access to one or more of these areas.
- **Reports for IBM Data Cataloging**
  Reports for IBM Data Cataloging are grouped or non-grouped. Grouped reports have information for count and sum in columns and non-grouped reports have information in rows.

Role-based access control

IBM Data Cataloging provides access to resources based on roles. You can restrict access to information based on roles.

The role that is assigned to a user or group determines the privileges for that user or group. Users and groups can be associated with collections, which use policies that determine the metadata that is available to view.

User and group access can be authenticated by IBM Data Cataloging, an LDAP server, or the IBM Cloud® Object Storage System. The administrator can manage the user access functions.

Roles

Roles determine how users and groups access records or the IBM Data Cataloging environment.
Remember: If a user or group is assigned to multiple roles, the least restrictive role is applicable.
For example, if you are assigned the role of Data User, and you are also assigned the role of a Data Admin, you have the privileges of a Data Admin.

Admin
: An admin can create users, groups, collections, manage LDAP, and IBM Cloud Object Storage connections for user access management.

Data Admin
: Users with the Data Admin role can access all metadata that is collected by IBM Data Cataloging and is not restricted by collections.

Collection Admin
: The Collection Admin role is as a bridge between the Data Admin role and the Data User role.

  - Users with the Collection Admin role can list any type of tag and create or modify `Characteristic` tags. Users with the Collection Admin role cannot create, modify, or delete `Open` and `Restricted` tags. These permissions are the same permissions as the Data User role.
    Note: The built-in `Collection` tag is a special tag that can be set only by users with the Data Admin role. All other tags can be set by any user with the Data User or Data Admin or Collection Admin role.
  - Users with the Collection Admin role can
    - Create, update, and delete the policies for the collections they administer.
    - View, update, and delete policies of data users for the collections they administer. They cannot delete a policy if it has a collection that they do not administer.
    - Add users to collections that they administer. These data users can access a particular collection, which means that they can access the records that are marked with that collection value.

Collection User
: Users with this role can access metadata that is collected by IBM Data Cataloging, but metadata access can be restricted by the collections that are assigned to users in this role.

- Users assigned with the Collection User role can:
  - Run scans of collections the user is assigned.
  - View policies of the collections the user is assigned.
  - List any type of tag.
- Users assigned with the Collection User role cannot:
  - Create, update, and delete any policies.
  - Create, modify, and delete any tags.

Data User
: Users with the Data User role can access metadata that is collected by IBM Data Cataloging. Metadata access might be restricted by policies in the collections that are assigned to users in this role. A user with the Data User role can also define tags and policies based on the collections to which the role is assigned.

Service User
: The Service User role is assigned to accounts for IBM® service and support personnel.

Data source connections

A data source connection specifies the parameters for cataloging of metadata from a source system to IBM Data Cataloging.

Without the proper connection information, ingesting metadata from a connected system fails. You can use the Data Source Connections page to view connection information for the data sources that are connected to your environment.

For more information, see Configure data source connections.

Cataloging metadata

System metadata is created and updated by the host system, and not the application software. IBM Data Cataloging allows the addition of tags that can capture non-system metadata-specific attributes, which are stored in the IBM Data Cataloging catalog.

Scans are jobs that are scheduled or on demand, and occur at a data source level. For example, a file system or object vault. A set of metadata records is generated with each record that captures the state of an individual file or object within the data source at the time of the scan.

IBM Data Cataloging supports scanning the following data sources:

- IBM Storage Scale
- IBM Elastic Storage® Server
- IBM Cloud® Object Storage
- IBM Storage Protect
- IBM Spectrum® Archive
- Red Hat® Ceph® Storage
- NetApp Storage Solutions
- Dell EMC Isilon Scale Out Network Attached Storage
- Amazon Simple Storage Service (Amazon S3)
- NFSv3 and NFSv4
- SMB

Live event notifications are triggered by user actions on the source data. Examples are reading, writing, moving, deleting data, changing permissions, or ownership. The events generate a metadata record in real time that is stored in IBM Data Cataloging. IBM Data Cataloging supports live event notifications for the following data sources:

- IBM Cloud Object Storage
- IBM Elastic Storage Server
- IBM Storage Scale
- Red Hat Ceph Storage

With IBM Storage Scale you can enable live events to start a watch folder on the specified file system. The IBM Storage Scale watch folder works with IBM Data Cataloging to capture file system event notifications and deliver them to IBM Data Cataloging by using Kafka.
Important: If the connection from IBM Storage Scale to IBM Data Cataloging is interrupted, the watch suspends. Additionally, events are no longer captured in IBM Data Cataloging, which requires a file system rescan to capture the lost updates.

Enriching metadata

IBM Data Cataloging can enrich the metadata from supported platforms with additional information by using policies, content inspection, custom tags, and custom applications.

Policies

Policies are used to add additional information about the source data that is indexed in IBM Data Cataloging. A policy determines the set of files to add tag values to, or to send to the built-in content inspection capabilities of IBM Data Cataloging, or to a custom application through filtering criteria. The policies give you the ability to run actions one time or on a set schedule. Policies work in batches and can be paused, resumed, stopped, or restarted. You can control the load on the IBM Data Cataloging system or source storage system for content inspection policies and policies that start custom applications.

Applications

A deep inspect application extracts information from source data records and returns it to IBM Data Cataloging to be indexed. For example, by using a custom application, you might create a DEEP-INSPECT policy to extract key characteristics from files of a certain type. The characteristics are applied to the metadata records for the files in IBM Data Cataloging as custom tags and made searchable. You can search for data by name, size, and content.

You can use custom applications to extend the capabilities that are performed by IBM Data Cataloging.

- **Policy engine**
  Policies offer a method whereby you can schedule one-time or repetitive actions on a filtered set of records.
- **Applications**
  IBM Data Cataloging policies might contain applications in the actions parameters.

Policy engine

Policies offer a method whereby you can schedule one-time or repetitive actions on a filtered set of records.

The policy management API service is a RESTful web service that is designed to create, list, update, and delete policies. You can use a policy to initiate action on a select set of indexed documents or data. You can do a task immediately or on a set schedule.

Several types of policies that are supported by IBM Data Cataloging enrich the metadata records. You can create policies with information to determine which set of documents to run, the action to take, and when to run policies periodically.

A policy includes

Policy ID
: Name of the policy.

Filter
: Selects a set of documents to work.

Action
: ID, parameters, and schedule.

The following list is a description of the policies.

AUTOTAG
: A policy that tags a set of records based on filter criteria with a pre-defined set of tags.

CONTENT SEARCH
: A policy that uses the built-in content inspection capabilities of IBM Data Cataloging to extract content from source data and index it automatically into the IBM Data Cataloging catalog.

DEEP-INSPECT
: A policy that passes lists of files based on filter criteria to the analytics application that opens the source data file and extracts metadata information from it. The policy passes the data back to IBM Data Cataloging in the form of tags so you can do a search, and do the following activities:

  - Set up a filter to do a search query that finds the candidates to apply the policy.
    For example, you can set an action for filtered candidates AUTOTAG: **`tag1: value`**, **`tag2: value`**
  - Set a schedule to apply the policy by specifying the following methods:
    - Immediately
    - Periodically

Applications

IBM Data Cataloging policies might contain applications in the actions parameters.

Use an application when you want to do a specific action on data or metadata on IBM Data Cataloging.

You can define an application when you create a new DEEP-INSPECT policy. You can add parameters for an application during the process of creating a DEEP-INSPECT policy.

When you open the window for applications, you can see a view of a table with the following information:

Application
: The name of the application.

Parameters
: The parameters that were assigned to the application when the policy was created.

Action ID
: Actions that are supported by the application for enriching metadata. For example, CONTENTSEARCH, DEEP-INSPECT.

View or Delete
: Use the delete trashcan icon to remove the application from the database.

Graphical user interface

The IBM Data Cataloging graphical user interface is a portal that is used for running data searches, report generation, policy and tag management, and user Access Management. Based on a user's role, they might have access to one or more of these areas.

The IBM Data Cataloging environment provides access to users and groups. The role that is assigned to a user or group determines the functions that are available. Users and groups can also be associated with collections, which use policies that determine the metadata that is available to view.

User and group access can be authenticated by IBM Data Cataloging, an LDAP server, or the IBM Cloud® Object Storage System. The administrator can manage the user access functions.

Roles

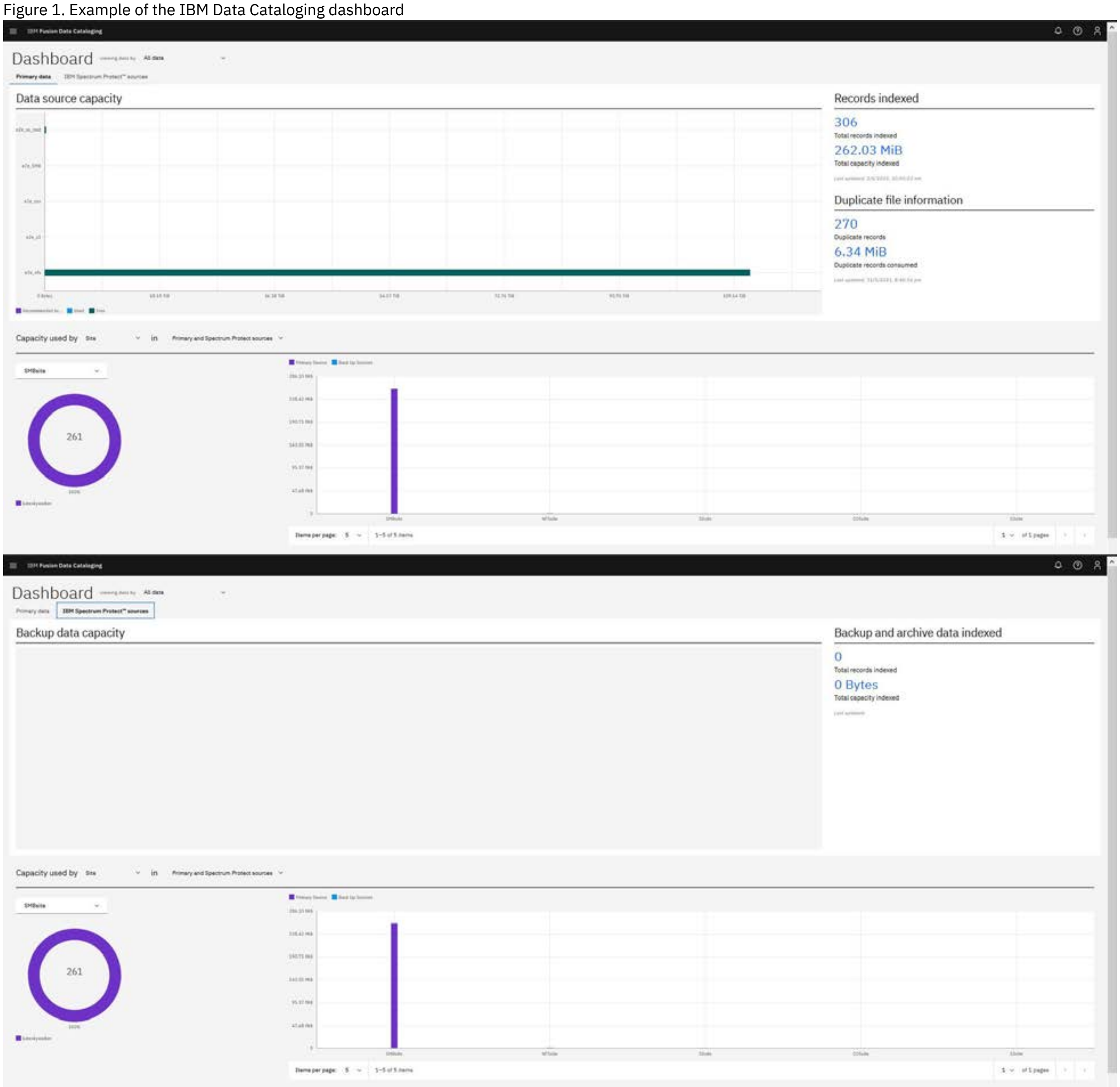
Roles determine how users and groups can access records on the IBM Data Cataloging environment.

If a user or group is assigned to multiple roles, the least restrictive role is used. For example, if a user is assigned a role of Data User, and is included in a data administrator role, the user has the privileges of a data administrator.

Dashboard

An example of the IBM Data Cataloging dashboard is shown.

Figure 1. Example of the IBM Data Cataloging dashboard



Data administrators and users can view the following:

- Metrics for the overall capacity used by every data source
- Total number of files
- Amount of capacity that is used by records with specific tags and facets, for example, owner, cluster, and size range
- Distribution of those records across data sources

Users can click any of the dashboard widgets to initiate a search and further explore and drill down into the data. Administrators and user can also perform the following:

- Monitor storage usage and data recommendations
- View total indexed data and capacity

- View duplicate file or object candidates. For example:
  - Number
  - Capacity used
- Preview capacity use by data facet. For example:
  - Classification
  - Owner
  - File type
- View data capacity by group or collection. For example:
  - Customer defined
  - Lab or project

Understanding size and capacity differences

IBM Data Cataloging collects size and capacity information:

- Size refers to the size of a file or object in bytes.
- Capacity refers to the amount of space the file or object consumes on the source storage in bytes.

For objects, size and capacity values always match. For files, size and capacity values can be different because of file system block overhead or sparsely populated files.

Note: Storage protection overhead (such as RAID values or erasure coding) and replication overhead are not captured in the capacity values.

Reports for IBM Data Cataloging

Reports for IBM Data Cataloging are grouped or non-grouped. Grouped reports have information for count and sum in columns and non-grouped reports have information in rows.

Data Curation Reports are a way for administrators, also known as data curators, to view the state of their storage environment in different ways. They can range from high-level grouped information to individual record level information.

For example, you can sort a report by owner, project, and department, or you can generate a list of records that meet a specific criterion. Additionally, you can create a report that lists the records in a project, not been reviewed for over a year. The owner of the data can evaluate whether to archive or delete the report.

For more information, see Reports*Reports* in *Data Cataloging: Administration Guide*.

For more information, see REST API for IBM Data Cataloging. For more information, see *REST API* in the *IBM Data Cataloging: REST API Guide*.

Installing IBM Data Cataloging

IBM Data Cataloging service is a modern metadata management software that provides data insight for exabyte-scale heterogeneous file, object, backup, and archive storage on premises and in the cloud. It can help you manage your unstructured data by reducing the data storage costs, uncovering hidden data value, and reducing the risk of massive data stores.

Before you begin

- Meet the system requirements to install the IBM Data Cataloging service.
- The following details are a base line for finding the resources that are needed for IBM Fusion IBM Data Cataloging service deployment. Based on the following tables, the resources can be estimated based on the number of approximate files that are required. The following are the resource values that are calculated per compute node: You must have at least two worker nodes, each with the same amount of resources available.
- IBM Fusion IBM Data Cataloging service must have dedicated compute resources. Ensure that you have enough to cover the resources limits to perform as expected:

Table 1. Profile requirements

CPU	RAM	Disk space	Network	Storage	Workload
77	162 GB	120 GB	10 GB	1 TB	1 B

- The standard deployment for IBM Data Cataloging service project requests and limits. Ensure you set the resource quota values for IBM Data Cataloging as follows:

```
kind: ResourceQuota
apiVersion: v1
metadata:
  name: dcs-resource-quota
  namespace: ibm-data-cataloging
spec:
  hard:
    limits.cpu: '72'
    limits.memory: 115Gi
    pods: '120'
    requests.cpu: '18'
    requests.memory: 26Gi
```

- Important: For the IBM Data Cataloging service to run successfully on all platforms, ensure that the storage classes have the following attributes:
  - ReadWriteMany (RWX) permissions
  - volumeBindingMode set to Immediate
  - AllowVolumeExpansion set to true
- If you have not configured the IBM operator catalog, then configure it. For the procedure to add IBM operator catalog, see Adding the IBM operator catalog.
- Go through troubleshooting information related to the installation of IBM Data Cataloging. See IBM Data Cataloging service installation and upgrade issues.

Procedure

1. Go to Services page in IBM Fusion or IBM Fusion HCI user interface.
2. In the Available section, click Data cataloging tile.
3. In the Data cataloging window, go through the details of the service and click Install.
4. In the Install service message box, select a Storage class.
   Important:
   - If you want to use Global Data Platform as the storage provider, then it is recommended to select the default storage class **`ibm-spectrum-fusion`**. Otherwise, if you want to use Fusion Data Foundation, then select the **`ocs-storagecluster-cephfs`** storage class. You can also use a custom storage class that matches the requirements.
   - For deployments operating in FIPS mode, you must specify or select a **`storage class`** that uses **`encrypted storage`**.
     - If the default storage classes (**`ibm-spectrum-fusion`** or **`ocs-storagecluster-cephfs`**) are **`FIPS-compliant`** and use encrypted storage, you can continue to use them.
     - If not, you must configure and select a custom encrypted storage class that meets FIPS requirements.
     - For storage based on node SSDs, ensure the SSDs are encrypted at deployment time using encryption methods that meets FIPS requirements.
     - For storage based on Red Hat® OpenShift® Data Foundation, configure encrypted Red Hat OpenShift Data Foundation storage and use a storage class from that encrypted pool.
5. Click Install. In case of failures, go through the downloaded logs to understand the cause of the failure and fix the issue. For more information about service issues in IBM Fusion, see IBM Data Cataloging service installation and upgrade issues.
6. Validate the installation.
   - IBM Fusion user interface:
     After you enable the IBM Data Cataloging service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification gets displayed. The notification disappears automatically after some time.

     Table 2. Health states IBM Data Cataloging service

State	Description
`Installing`	Service installation is in progress
`Upgrading`	Service upgrade is in progress
`Healthy`	Service is healthy
`Degraded`	Service is not healthy

   - Verify the IBM Data Cataloging installation from the OpenShift Container Platform web console:
     - Run the following command to verify the installed operators.

       ```
       oc -n ibm-data-cataloging get csv
       ```

       ```
       Expected output
       NAME                                           DISPLAY                          VERSION              REPLAC
       amqstreams.v2.3.0-1                            Red Hat Integration - AMQ Streams  2.3.0-1            amqstr
       db2u-operator.v3.1.0                           IBM Db2                          3.1.0
       ibm-spectrum-discover-operator.v210.0.0        IBM Storage Discover             210.0.0-1678324770
       ```

     - Run the following command to verify IBM Data Cataloging status.

       ```
       oc -n ibm-data-cataloging get isd
       ```

       ```
       Expected output
       NAME                             READY   SUMMARY                        ERROR   AGE   HEALTHSTATUS
       data-cataloging-service-instance True    Awaiting next reconciliation           9d    Healthy
       ```

     - Run the following command to verify that all Persistent Volume Claims are bounded.

       ```
       oc -n ibm-data-cataloging get pvc
       ```

       ```
       Expected output
       NAME                         STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLA
       activelogs-c-isd-db2u-0      Bound    pvc-7e98eead-9a38-48b6-a880-16d730120796   150Gi      RWO            ocs-storag
       activelogs-c-isd-db2u-1      Bound    pvc-7af71786-1529-4b3c-bbd1-b5d1ee733d33   150Gi      RWO            ocs-storag
       c-isd-meta                   Bound    pvc-1ba20699-ddb8-4600-bf9f-4d637b14da49   50Gi       RWX            ocs-storag
       data-c-isd-db2u-0            Bound    pvc-488540db-c951-4f48-b65c-814bf634ca33   300Gi      RWO            ocs-storag
       data-c-isd-db2u-1            Bound    pvc-56a3105a-347e-4b55-b11a-d0a1d351d622   300Gi      RWO            ocs-storag
       data-isd-sasl-kafka-0        Bound    pvc-3924c9cc-173d-4244-90fa-a234325dc04b   100Gi      RWO            ocs-storag
       data-isd-sasl-zookeeper-0    Bound    pvc-3c66397b-ac03-4ed8-841a-e33d83355a23   64Gi       RWO            ocs-storag
       data-isd-ssl-kafka-0         Bound    pvc-ce0ebeaf-d2b6-4e40-8c31-e31d45a208e5   100Gi      RWO            ocs-storag
       data-isd-ssl-zookeeper-0     Bound    pvc-7fdc9ac5-9158-48ac-bad3-3c24590f1f2b   64Gi       RWO            ocs-storag
       isd-backup                   Bound    pvc-2e077c8e-e992-4e1a-8b98-d24bfdbb5343   250Gi      RWX            ocs-storag
       isd-data                     Bound    pvc-b98e124f-1f98-43e5-9bf6-c8758c681192   250Gi      RWX            ocs-storag
       tempts-c-isd-db2u-0          Bound    pvc-cd3fdd20-343b-49ca-ad1a-690f8eec3e60   50Gi       RWO            ocs-storag
       tempts-c-isd-db2u-1          Bound    pvc-a2adb668-a362-4f61-8ed3-ae103f9b2cd3   50Gi       RWO            ocs-storag
       ```

     - Run the following command to verify Db2u cluster detailed progress.

       ```
       oc -n ibm-data-cataloging get formations.db2u.databases.ibm.com isd -o go-template='{{range .status.components}}
       {{printf "%s,%s,%s\n" .kind .name .status.state}}{{end}}' | column -s, -t
       ```

       ```
       Expected output
       account                account-ibm-data-cataloging-isd  OK
       PersistentVolumeClaim  c-isd-meta                       OK
       secret                 c-isd-sshkeys-db2uhausr          OK
       secret                 c-isd-sshkeys-db2uadm            OK
       secret                 c-isd-sshkeys-db2instusr         OK
       secret                 c-isd-ldappassword               OK
       secret                 c-isd-ldapblueadminpassword      OK
       secret                 c-isd-instancepassword           OK
       secret                 c-isd-db2u-lic                   OK
       secret                 c-isd-certs-wv-rest              OK
       ```

```
secret                c-isd-certs-db2u-api            OK
configmap             c-isd-db2uconfig                OK
configmap             c-isd-db2regconfig              OK
configmap             c-isd-db2dbmconfig              OK
configmap             c-isd-db2dbconfig               OK
service               c-isd-ldap                      OK
Deployment            c-isd-ldap                      OK
service               c-isd-tools                     OK
Deployment            c-isd-tools                     OK
configmap             c-isd-db2u-api                  OK
service               c-isd-db2u-engn-svc             OK
service               c-isd-db2u-head-engn-svc        OK
Job                   c-isd-instdb                    OK
service               c-isd-etcd                      OK
StatefulSet           c-isd-etcd                      OK
service               c-isd-db2u-rest-svc             OK
networkpolicy         c-isd-rest-ext                  OK
Deployment            c-isd-rest                      OK
service               c-isd-db2u-graph-svc            OK
networkpolicy         c-isd-graph-ext                 OK
Deployment            c-isd-graph                     OK
service               c-isd-db2u-internal             OK
service               c-isd-db2u                      OK
StatefulSet           c-isd-db2u                      OK
networkpolicy         c-isd                           OK
networkpolicy         c-isd-ext                       OK
Job                   c-isd-restore-morph             OK
```

- Run the following command to verify the Db2u cluster CR status.

```
oc -n ibm-data-cataloging get db2ucluster
```

```
Expected output
NAME   STATE   MAINTENANCESTATE   AGE
isd    Ready   None               9d
```

- Run the following command to verify that the ports 50000 and 50001 are listing.

```
oc exec -n ibm-data-cataloging -it $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --selector
name=dashmpp-head-0 | awk '{print $1}') -- su - db2inst1 -c 'netstat -ntlp'
```

```
Expected output
Defaulted container "db2u" out of: db2u, init-labels (init), init-kernel (init)
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 10.254.4.11:60000       0.0.0.0:*               LISTEN      23707/db2sysc 0
tcp        0      0 10.254.4.11:60001       0.0.0.0:*               LISTEN      23712/db2sysc 1
tcp        0      0 0.0.0.0:9443            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:50022           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:50000           0.0.0.0:*               LISTEN      23707/db2sysc 0
tcp        0      0 0.0.0.0:50001           0.0.0.0:*               LISTEN      23707/db2sysc 0
tcp6       0      0 :::50052                :::*                    LISTEN      -
tcp6       0      0 :::50022                :::*                    LISTEN      -
```

- Run the following command to verify that workload is PUREDATA_OLAP.

```
oc exec -n ibm-data-cataloging -c db2u -it $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --
selector name=dashmpp-head-0 | awk '{print $1}') -- sudo su - db2inst1 -c "db2set -all | grep DB2_WORKLOAD= |
awk '{print \$2}'"
```

```
Expected output
DB2_WORKLOAD=PUREDATA_OLAP
```

- Run the following command to verify the database encryption configurations.

```
oc -n ibm-data-cataloging exec -c db2u $(oc -n ibm-data-cataloging get po --no-headers --show-labels=true --
selector name=dashmpp-head-0 | awk '{print $1}') -- su - db2inst1 -c "db2 get db cfg for bludb | grep 'Encrypted
database'"
```

```
 Encrypted database                                      = YES
```

- Run the following command to verify the Kafka CRs readiness.

```
oc -n ibm-data-cataloging get kafka
```

```
Expected output
NAME        DESIRED KAFKA REPLICAS   DESIRED ZK REPLICAS   READY   WARNINGS
 isd-sasl   1                        1                     True
 isd-ssl    1                        1                     True
```

- Run the following command to verify that the IBM® Spectrum Discover pods are running.

```
oc -n ibm-data-cataloging get pod -l component=discover
```

```
Expected output
 NAME                                           READY   STATUS      RESTARTS     AGE
isd-api-7564475d98-cs6nx                       1/1     Running     0            9d
isd-auth-5fcb8bd95f-jvzwp                      1/1     Running     0            9d
isd-backup-restore-689fc64f7-pgx22             1/1     Running     0            9d
isd-connmgr-7d4cdb4cc7-xnkzm                   1/1     Running     1 (9d ago)   9d
isd-consumer-ceph-le-5f9b5d4676-4fsgg          1/1     Running     0            9d
isd-consumer-ceph-le-5f9b5d4676-9k9cs          1/1     Running     0            9d
isd-consumer-ceph-le-5f9b5d4676-bphqp          1/1     Running     0            9d
isd-consumer-ceph-le-5f9b5d4676-cc2pz          1/1     Running     0            9d
```

```
isd-consumer-ceph-le-5f9b5d4676-chlvk        1/1     Running     0            9d
isd-consumer-ceph-le-5f9b5d4676-fchp5        1/1     Running     0            9d
isd-consumer-ceph-le-5f9b5d4676-jlbrw        1/1     Running     0            9d
isd-consumer-ceph-le-5f9b5d4676-l78cc        1/1     Running     0            9d
isd-consumer-ceph-le-5f9b5d4676-lpcsn        1/1     Running     0            9d
isd-consumer-ceph-le-5f9b5d4676-n2wqt        1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-54qm9         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-6hsq9         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-9p4p5         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-dhlgc         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-hr76h         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-mkvx8         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-tvscz         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-vrrd4         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-w59c2         1/1     Running     0            9d
isd-consumer-cos-le-846f5fd97f-xqncl         1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-4klcc       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-8r6w7       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-fqs5m       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-fwb8p       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-h78g5       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-lvwl5       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-nhznb       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-q2l8b       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-z6tfj       1/1     Running     0            9d
isd-consumer-cos-scan-79fb979585-zv2zw       1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-5mrrg      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-8rphj      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-b5dbv      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-b5lv2      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-d9gbm      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-f5rxf      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-m8nw6      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-ntfbb      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-ql55x      1/1     Running     0            9d
isd-consumer-file-scan-5c7565bf7b-qvggm      1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-6s75x    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-845bx    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-8kmnf    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-gntbt    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-mlnfb    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-p9lpt    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-rckm8    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-vkmpd    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-vmh8x    1/1     Running     0            9d
isd-consumer-protect-scan-67d5ffb65-zm4gf    1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-6dw9h       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-9nhsz       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-dnwpw       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-dz2mf       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-h2ffj       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-hdrxg       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-nf779       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-pf6n4       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-q8gqf       1/1     Running     0            9d
isd-consumer-scale-le-5d56449994-xzmvh       1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-49vkb     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-54pb8     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-5sr89     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-98lm9     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-d4zlj     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-fc82j     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-mnsn2     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-r2kc6     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-tglsz     1/1     Running     0            9d
isd-consumer-scale-scan-7bdc7957c6-zbhgv     1/1     Running     0            9d
isd-contentsearchagent-5d7bfbd887-r487x      1/1     Running     1 (9d ago)   9d
isd-db-schema-xm5cn                          0/1     Completed   0            9d
isd-db2whrest-6c6bbfcbc6-9v6v2               1/1     Running     1 (9d ago)   9d
isd-generate-license-6g94v                   0/1     Completed   0            9d
isd-importtags-64498dbdd8-5kqpb              1/1     Running     1 (9d ago)   9d
isd-keystone-7dc84b69f5-sw6nn                1/1     Running     0            9d
isd-license-check-27982200-c9gkm             0/1     Completed   0            2d21h
isd-license-check-27983640-ldxv2             0/1     Completed   0            45h
isd-license-check-27985080-rhzjl             0/1     Completed   0            21h
isd-policyengine-598f565867-zg5sn            1/1     Running     1 (9d ago)   9d
isd-producer-ceph-le-76b9885854-557b7        1/1     Running     0            9d
isd-producer-cos-le-85d446c9d8-zvzmg         1/1     Running     0            9d
isd-producer-cos-scan-d584f646d-2t669        1/1     Running     0            9d
isd-producer-file-scan-5586b48fd-zkt7q       1/1     Running     0            9d
isd-producer-protect-scan-7b66bb7c64-wz6qq   1/1     Running     0            9d
isd-producer-scale-le-8d7bc64b4-nvqbl        1/1     Running     0            9d
isd-producer-scale-scan-5df647956-qfl6p      1/1     Running     0            9d
isd-proxy-5f4d7dd8ff-44fwt                   1/1     Running     0            9d
isd-reports-c6kmq                            0/1     Completed   0            9d
isd-scaleafmdatamover-85cc44b566-644vx       1/1     Running     1 (9d ago)   9d
isd-scaleilmdatamover-6b86dd7b58-j28sn       1/1     Running     1 (9d ago)   9d
isd-sdmonitor-6fffc999b7-s2srd               1/1     Running     0            9d
isd-tikaserver-79548bdb4-ssft5               1/1     Running     0            9d
isd-ui-backend-f6c4b65d7-dwzrs               1/1     Running     0            9d
isd-ui-frontend-9659fc847-vcbmv              1/1     Running     0            9d
```

- Run the following command to verify the console route availability.

```
oc -n ibm-data-cataloging get route console
```

```
Expected output
NAME      HOST/PORT                                        PATH   SERVICES   PORT    TERMINATION   WILDCARD
console   console-ibm-data-cataloging.apps.ocp2.vmlocal           isd-svc    <all>   reencrypt     None
```

Uninstalling IBM Data Cataloging

Steps to uninstall IBM Data Cataloging by using command line interface.

Procedure

1. Run the following command and export IBM Fusion namespace as an environmental variable.

   ```
   export FUSION_NS="<Fusion namespace>"
   ```

   Important: Replace `<Fusion namespace>` with your namespace.
2. Run the following command to prevent catalog source creation.

   ```
   oc -n ${FUSION_NS} patch fusionservicedefinition data-cataloging-service-definition --type='json' -p='[{"op": "replace",
   "path": "/spec/onboarding/serviceOperatorSubscription/triggerCatSrcCreate", "value":false}]'
   ```

3. Run the following command to scale down prerequisite operator.

   ```
   oc -n ${FUSION_NS} scale --replicas=0 deployment/isf-prereq-operator-controller-manager
   ```

4. Run the following commands to delete all the IBM Data Cataloging workload, networking and storage resources.

   ```
   oc project default
   export DCS_CR=$(oc get SpectrumDiscover -n ibm-data-cataloging -o jsonpath="{.items[*].metadata.name}")
   oc delete SpectrumDiscover ${DCS_CR} -n ibm-data-cataloging --timeout=2m --wait=true
   oc delete project ibm-data-cataloging --timeout=1m --wait=true
   ```

5. Run the following command to delete the IBM Fusion IBM Data Cataloging service instance object.

   ```
   oc -n ${FUSION_NS} delete fusionserviceinstance data-cataloging-service-instance
   ```

6. Run the following command to delete the `SecurityContextConstraints` objects.
   Important: The SCC objects are based on their namespace, and the namespace must be included in the name of the SCC object. The default namespace is `ibm-data-cataloging`, but you must change it if you use a different one.

   ```
   oc delete securitycontextconstraints/isd-scc-ibm-data-cataloging
   oc delete securitycontextconstraints/ibm-data-cataloging-c-isd-scc
   ```

7. Run the following command to delete the IBM Data Cataloging `ConsoleLink` object.

   ```
   oc delete consolelink/data-cataloging
   ```

8. Run the following command to delete IBM Data Cataloging `CustomResourceDefinitions` from the cluster.

   ```
   oc delete customresourcedefinition/spectrumdiscovers.spectrum-discover.ibm.com
   oc delete customresourcedefinition/spectrumdiscoverapplications.spectrum-discover.ibm.com
   ```

9. Run the following command to scale up prerequisite operator.

   ```
   oc -n ${FUSION_NS} scale --replicas=1 deployment/isf-prereq-operator-controller-manager
   ```

Configuring IBM Data Cataloging

This section provides information on how to deploy and configure IBM Data Cataloging capabilities.

- **Setting up IBM Data Cataloging custom user and password**
  Use this information to create a custom user and password for the IBM Data Cataloging service.
- **Configure data source connections**
  Data source connections describe the source data systems for which IBM Data Cataloging indexes metadata.
- **Editing and using the TimeSinceAccess and Size Range buckets**
  Users can group or aggregate data into three user-defined bucket ranges. The three user-defined bucket ranges are TimeSinceAccess, Size Range and FileGroup.
- **IBM Spectrum Storage software requirements**
  Use IBM Data Cataloging to index metadata from other applications and to orchestrate the data management.
- **IBM Data Cataloging installation with alternative VLAN**
  Install and configure IBM Data Cataloging that uses additional VLAN connected through the IBM Fusion upstream links.
- **Multiple connection managers**
  Multiple connection managers are a new capability that is designed to enhance scanning performance and enable parallel ingestion. It proves especially valuable in scenarios where data sources are geographically dispersed and need to be scanned as remote sources.
- **Enabling feature for skipping the snapshot directories**
  When the skipping feature is enabled, you can skip the metadata ingestion to the snapshot directories.
- **Managing IBM Data Cataloging rules when integrated with CAS**
  IBM Data Cataloging provides the ability to integrate with Content-Aware Storage to configure file ingestion filtering. This feature enables you to filter files from a data source based on specific criteria, such as file type or content, and ingest them into the corresponding domain.

Setting up IBM Data Cataloging custom user and password

Use this information to create a custom user and password for the IBM Data Cataloging service.

About this task

It is required to perform this procedure before installation to prevent errors caused by the user being already initialized with the default values.

Procedure

1. Run the following command to create the **`ibm-data-cataloging`** namespace.

   ```
   oc create namespace ibm-data-cataloging
   ```

2. Run the following command to create the keystone secret to hold the custom user and password.

   ```
   oc -n ibm-data-cataloging create secret generic keystone --from-literal=user=<CUSTOM_USER> --from-literal=password=
   <CUSTOM_PASSWORD>
   ```

3. Deploy the IBM Data Cataloging service from theIBM Fusion user interface.

Configure data source connections

Data source connections describe the source data systems for which IBM Data Cataloging indexes metadata.
Creating data source connections in IBM Data Cataloging identifies source storage systems that are to be indexed by IBM Data Cataloging.

For some data source types, a network connection is (optionally) created to allow for automated scanning and indexing of the source system metadata. IBM Data Cataloging will not index data from unknown sources, so creating a data source connection is the first step towards cataloging any source storage system.

Remember: Depending on when the scan is stopped, stopping a running scan might result in an inconsistent database state for the connection.
You can add data connections to the source storage systems from the IBM Data Cataloging graphical user interface and REST API. For more information on configuring data source connections offline, see Configuring data source connections offline. *Configuring data source connections offline* in the *IBM Data Cataloging: Concepts, Planning, and Deployment Guide*.

IBM Data Cataloging discards any data that comes in from an unknown connection. Therefore, connections must be established before data ingestion. To see the list of defined connections, use the Connections tab under the Data source management window of the GUI.
Remember: If you use a MAC, you might have to adjust the scroll bar settings in System Settings to see all available connection types. For example, activate the Show scroll bars: Always option.
Typically, a data source is equivalent to a single file system or object vault or bucket. A data source connection is an alias for the combination of a cluster name and a data source within the cluster. This allows multiple file systems or buckets or vaults with the same name to be indexed by IBM Data Cataloging when they are in separate clusters.

Remember:

- IBM Data Cataloging does not support file or file path names that use characters that are not part of the UTF-8 character set.
- After you do a manual scan, run a Metadata summarization database refresh:
  1. In Data source management > Metadata summarization database, go to Data connections > Discover information.
  2. Click refresh and wait for the process to complete successfully. The expected time of completion is based on the amount of records to process. For example, 30 minutes.

- **IBM Storage Scale data source connections**
  You can create an IBM Storage Scale data source connection, scan a data source, and manually initiate a scan.
- **IBM Storage Archive data source connections**
  You can define tags and policies in IBM Data Cataloging based on values that are derived from IBM Spectrum® Archive metadata to help in searching and categorizing files.
- **IBM Cloud Object Storage data source connection**
  You can create the IBM Cloud® Object Storage (COS) connection and initiate a scan.
- **IBM Data Cataloging and S3 object storage data source connections**
  Use this information to understand how IBM Data Cataloging works with S3-compliant object store.
- **Scanning an Elastic Storage Server data source connection**
  Use the IBM Data Cataloging GUI to scan an Elastic Storage Server (ESS) data connection.
- **Creating a Network File System data source connection**
  You can use the IBM Data Cataloging graphical user interface to create a Network File System (NFS) data connection.
- **Creating an SMB data source connection**
  Creating an SMB data connection by using the IBM Data Cataloging graphical user interface.
- **Creating an IBM Storage Protect data source connection**
  Creating the IBM Storage Protect data connection by using the IBM Data Cataloging graphical user interface.
- **Configuring data source connections offline**
  Multiple source data connections can be added to the IBM Data Cataloging system through offline mode.

IBM Storage Scale data source connections

You can create an IBM Storage Scale data source connection, scan a data source, and manually initiate a scan.

Tip: If the data source connection is used by the ScaleILM application to tier data by using IBM Spectrum® Archive, the `host` setting of the IBM Storage Scale connection must specify one of the IBM Storage Archive nodes. For more information, see Tiering data by using ScaleILM application.For more information, see the topic *Tiering data by using ScaleILM application* in the *Data Cataloging: Administration Guide*.
IBM Data Cataloging supports the following method of scanning IBM Storage Scale data sources:

Automated scanning
A data source connection is defined on IBM Data Cataloging, including details of how to connect to the IBM Storage Scale system. IBM Data Cataloging connects to the IBM Storage Scale system by using these details, scans the file system and sends the details back to IBM Data Cataloging. For more information, see Prerequisites for automated scanning.For more information, see the topic *Prerequisites for automated scanning* in the *Data Cataloging: Administration Guide*.

- **IBM Storage Scale scanning considerations**
  The following sections comprise considerations that you need to understand to scan IBM Storage Scale effectively.
- **Prerequisites for automated scanning**
  You can use IBM Storage Scale automated scanning features.
- **Creating an IBM Storage Scale data source connection**
  You can use the IBM Data Cataloging graphical user interface to create data connections from the source storage systems.
- **Automated scanning of an IBM Storage Scale data source**
  As an administrator, you can initiate an IBM Storage Scale scan from IBM Data Cataloging to collect system metadata from IBM Storage Scale file system.
- **Manual scanning of an IBM Storage Scale data source**
  How to configure IBM Data Cataloging to connect to IBM Spectrum Scale.

IBM Storage Scale scanning considerations

The following sections comprise considerations that you need to understand to scan IBM Storage Scale effectively.

- **Security considerations**
  Use this information to securely scan a system connection.
- **Performance considerations**
  Use this information to scan a system connection without degrading performance.

Security considerations

Use this information to securely scan a system connection.

Scanning an IBM Storage Scale instance involves the use of the **mmapplypolicy** command on the IBM Storage Scale system, which requires superuser permissions. When you are creating the data source connection for the target IBM Storage Scale system in the IBM Data Cataloging interface, you are prompted for a *userid* and *password* to enable automated scans. You are not required to provide these credentials if scans are run only manually on the target IBM Storage Scale system by an administrator. However, if you want to run automation and/or schedule scans, then the authentication credentials are required. By default, IBM Data Cataloging uses password authentication to the Scale cluster to run commands remotely. However, you can supply your own RSA private key by selecting the shared key authentication option when you are configuring the connection if you want to avail passwordless authentication.

Rather than providing root login credentials, an administrator must create a special user ID with limited permissions on the IBM Storage Scale system. The administrator must also enable a password-less **sudo** for the user ID, to the binaries needed for scanning. This prevents someone from gaining root access to the target IBM Storage Scale system if the IBM Data Cataloging system is somehow compromised.

Changing passwordless SSH keys

You can rotate RSA authentication key pairs for passwordless SSH on a frequency and remove old security keys from the authorized_hosts file on the IBM Storage Scale node that IBM Data Cataloging connects to. To update the authentication keys, follow these steps:

1. Ensure that the id_rsa.pub contents for the new authentication key pair are in the ~/.ssh/authorized_hosts file for the user that is specified in the IBM Data Cataloging connection document for the IBM Storage Scale target file system.
2. Edit the connection and paste the contents of the new private key file (id_rsa) in the input form.

After you edit the connection with the new private key file, IBM Data Cataloging uses it to connect to the IBM Storage Scale target system.

Performance considerations

Use this information to scan a system connection without degrading performance.

Running a scan policy on an IBM Storage Scale system can be resource intensive and cause noticeable performance degradation on the IBM Storage Scale system. Often, system administrators choose to designate certain nodes or node classes for running the scans. The IBM Data Cataloging interface has an input field when creating IBM Storage Scale connections for the administrator to specify which nodes or node class(es) they would like to run the scan on. The value `all` will run the scan across all nodes in the cluster. Any other list (comma separated) will be treated as a list of nodes or node classes on which to run the scan. Scan times vary by the size of the filesystem, how many nodes are used in the scan, how many CPUs are used per node, and whether or not the IBM Storage Scale cluster metadata is in flash memory.

Prerequisites for automated scanning

You can use IBM Storage Scale automated scanning features.

IBM Data Cataloging supports two levels of automated scanning of IBM Spectrum Scale systems. Both levels require that IBM Data Cataloging must establish a password-less Secure Shell (SSH) connection to the IBM Storage Scale clustered system that is being scanned.

The difference between the two levels lies in the manner in which the output is handled. The factors that are considered inspects whether:

- The output of the IBM Storage Scale policy that is run to do the scan is stored in a file (which then must be automatically copied back to IBM Data Cataloging and ingested locally); Or
- If the output is, instead, pushed to the ingest Kafka queue of IBM Data Cataloging system directly from the IBM Storage Scale policy output.

The Kafka queue of IBM Data Cataloging system is more space-efficient and time-efficient but has certain dependencies on the IBM Storage Scale clustered system that must be met so that it can function. The IBM Data Cataloging automated scan code determines whether the dependencies are met on the IBM Storage Scale clustered system.

If the dependencies are met on the IBM Storage Scale clustered system, it attempts to scan the system by using the optimized path. If the dependencies are not met on the IBM Storage Scale clustered system, it defaults to the file copy path.

The following sections list the prerequisites for creating a connection and performing automated scanning. These prerequisites consider security and performance factors. For more information, see IBM Storage Scale scanning considerations .

- **Identify the IBM Storage Scale cluster and node list**
  Identify a node to connect with the GPFS cluster.
- **Creating or identifying a user ID and password for scanning**
  Identify a user to perform scanning or create a new user ID.
- **Validate scan user permissions and configuration**
  Ensure that the user has the required permissions and configurations for scanning.
- **Check the dependencies for optimized scanning**
  Identify the dependencies that must be met on the IBM Storage Scale system to optimize automated scan ingest.

Identify the IBM Storage Scale cluster and node list

Identify a node to connect with the GPFS cluster.

Identify a node in the target IBM Storage Scale cluster to use for the IBM Data Cataloging connection to the GPFS cluster.

You must identify the node list or node class that participates in the scanning activity.

Creating or identifying a user ID and password for scanning

Identify a user to perform scanning or create a new user ID.

About this task

You can identify an existing user to perform scanning or follow these steps on the IBM Storage Scale system to create a special user ID for scanning.

Procedure

1. Log in to the IBM Storage Scale management node as **root**.
   Alternatively, you can **sudo** to root from another user ID.
2. Use the following `adduser` steps to ensure that you are able to ssh into the cluster:
   a. `adduser` <user> `-m`
   b. `passwd` <user>
3. Run: **visudo**.
   a. Add the following line in the users section:
      <user> `ALL=NOPASSWD: /usr/lpp/mmfs/bin/mmapplypolicy, /usr/lpp/mmfs/bin/mmrepquota`

   b. Write and quit: `:wq`
4. Create an IBM Data Cataloging working directory and ensure that <user> has write permissions.
   For example: **mkdir -p /gpfs/fs1/sd_scan -m 770; chown <user> /gpfs/fs1/sd_scan**
5. To execute a Data Mover policy for tiering data on the IBM Storage Scale cluster by using the ScaleILM application on the IBM Storage Scale connection , do the following:
   a. Run: **visudo**
   b. Add or update the following line in the users section:
      <user> `ALL=NOPASSWD: /usr/lpp/mmfs/bin/mmapplypolicy, /usr/lpp/mmfs/bin/mmrepquota, /usr/lpp/mmfs/bin/mmlspool, /opt/ibm/ltfsee/bin/eeadm,/usr/bin/ls`

   c. Write and quit: `:wq`
   For more information, see Tiering data by using ScaleILM application. For more information, see the topic *Tiering data by using ScaleILM application* in the *Data Cataloging: Administration Guide*.
6. To execute a Data Mover policy for copying data to the IBM Storage Scale cluster by using the ScaleAFM application on the IBM Storage Scale connection, do the following:
   a. Run: **visudo**.
   b. Add or update this line in the users section:
      ```
      <user>ALL=NOPASSWD:
      /usr/lpp/mmfs/bin/mmapplypolicy, /usr/lpp/mmfs/bin/mmrepquota, /usr/lpp/mmfs/bin/mmlspool,
      /usr/lpp/mmfs/bin/mmafmctl, /usr/lpp/mmfs/bin/mmafmcosctl, /usr/lpp/mmfs/bin/mmaddcallback,
      /usr/lpp/mmfs/bin/mmdelcallback
      ```

   c. Write and quit: `:wq`

For more information, see [Copying data using ScaleAFM application](). For more information, see the topic *Copying data using ScaleAFM application* in the *Data Cataloging: Administration Guide*.

Validate scan user permissions and configuration

Ensure that the user has the required permissions and configurations for scanning.

You need to ensure that it is possible to use Secure Shell (SSH) to log in to the IBM Storage Scale system with the scanning user ID and password.

Run the **sudo /usr/lpp/mmfs/bin/mmapplypolicy**.

You also need to validate the listed factors for the scanning-related working directory:

- It exists.
- It is globally accessible by the scan worker nodes.
- The scan user has write permissions to the directory (ownership of the directory is preferred but not mandatory).

Place the **`id_rsa`** and **`id_rsa.pub`** files in **`/opt/ibm/metaocean/data/connections/scale/`** directory on the IBM Data Cataloging instance if a specific RSA key pair for password-less SSH is wanted.

You need to validate if the listed things are installed on the IBM Storage Scale node. This node is identified while you are identifying a node in the target IBM Storage Scale cluster. The node in the target is identified for use in the IBM Data Cataloging connection to the GPFS cluster:

- A python2 level of at least Python 2.7.5
- A sufficient level of librarian
- An appropriate level of confluent-Kafka

Note: This validation is recommended for optimized scan ingestion. However, it is optional and can be skipped.

Check the dependencies for optimized scanning

Identify the dependencies that must be met on the IBM Storage Scale system to optimize automated scan ingest.

The dependencies that must be satisfied on the IBM Storage Scale system to optimize automated scan ingest from IBM Data Cataloging are:

- A `librdkafka` library 0.11.4 or later
- A Python 3.0 or later with accompanying Python package installer (pip3)
- A `confluent-kafka` version that is greater than or equal to the installed `librdkafka` version.

If these dependencies are met, the scan output is pushed to the ingest Kafka queue of IBM Data Cataloging system directly from the IBM Storage Scale policy output.

An administrator can determine whether `librdkafka` is installed on the IBM Storage Scale node by running the **`find /usr -name "*librdkafka*"`** or **`ls /lib64/librdkafka*`** commands. The `librdkafka` package is included with newer levels of IBM Storage Scale on x86 and ppc64le platforms. However, it can also be built from the source code on older levels of IBM Storage Scale or ppc64 platforms. If the IBM Storage Scale system runs on Red Hat® Enterprise Linux® (RHEL) and is connected to a Red Hat Satellite, you can install it by running the Yellowdog Updater Modified (YUM) command **`yum install librdkafka`** as **`root`**. You can find source packages of `librdkafka` here: https://github.com/edenhill/librdkafka

The user ID specified in the data source connection must be able to locate the following two binaries by using the OS shell path:

1. A Python 3 binary as either **`python`** or **`python3`**
2. A Python package installer as **`pip3`**
   Note: Symbolic links or aliases may be used to locate the Python executables.

After you install a sufficient version of Python, you can install `confluent-kafka` by using pip. To get pip, you must install the `python-setuptools` package, which provides a binary called `easy_install`. For more information, see https://pypi.org/project/setuptools/#files

After `easy_install` is available, you can install pip by running **`easy_install-2.7`** pip as **`root`**. After you install pip, you can install `confluent-kafka` by running **`pip install confluent-kafka`** as **`root`**.

Creating an IBM Storage Scale data source connection

You can use the IBM Data Cataloging graphical user interface to create data connections from the source storage systems.

Procedure

1. Log in to the IBM Data Cataloging web interface with a user ID that has the Data Admin role that is associated with it.
   The data admin access role is required for creating connections. For more information, see [Managing user access](). For more information, see *Managing User Access* in the *Data Cataloging: Administration Guide*.
2. Click menu and go to Data connections > Connections to display the different types of data source connection names, connection type, clusters, data source, site, state, scan status, next scan, and Add Connection button.
3. Click Add connection to display a new window that shows Add data source connection.
   You can enter in the connection name and connection type. The connection types are:
   - IBM Storage Scale

- IBM Cloud® Object Storage
- Network File System (NFS)

4. Complete the following steps:
   a. In the field for Connection name, define a Connection name.
   b. Choose type of connection from Connection type drop-down list.
5. Set the connection type to IBM Storage Scale. The page displays the connection name, user, password, working directory, and scan directory information that you can enter. You can also schedule a data scan, select a collection, or enable live events.
   If you click Enable live events, you can enable the IBM Storage Scale watch folder on the specified file system.
6. Complete updating values for all the fields to add the IBM Storage Scale connection type, and click Submit Connection.

   For IBM Storage Scale connections, you can enter the following information:

   Connection name
   : The name of the connection, an identifier for the user. For example, filesystem1.
     Note: It must be a unique name within IBM Data Cataloging.

   User
   : A user ID that has permissions to connect to the data source system and initiate a scan.

   Password
   : The password for the user ID specified in user.

   Authentication Type
   : The password authentication can be done by using the password that is provided to authenticate with the IBM Storage Scale cluster. The shared RSA key authentication performs a password less authentication by using a private key that is provided by the system administrator and whose public key exists in the authorized keys for the specified user on the Scale host.

   Note: IBM Data Cataloging 2.0.3.1 removes the support for self-generated RSA key pair for IBM Data Cataloging. Any existing connections that use that method is updated to password based authentication and the self-generated key pair is removed during the upgrade to 2.0.3.1 or later. If the password for the scan user that is stored in IBM Data Cataloging is no longer valid, that can result in scan failures after the update. To rectify this, you must edit the connection and provide a valid password for the scan user or a valid RSA private key for authentication.

   Working Directory
   : A scratch directory on the source data system where IBM Data Cataloging can put its temporary files.
     Note: When you edit an existing connection and change the User from a root user to a non-root user, you must also change the Working Directory. This change is necessary because the non-root User cannot access the files that are previously created by the root user in the existing Working Directory.

   Scan Directory
   : The root directory of the scan. All files and directories under this directory are scanned. Typically, this directory is the base directory of the file system. For example, /gpfs/fs1.

   Connection Type
   : The type of source storage system this connection represents.

   Site
   : An optional physical location tag that an administrator can provide to see the physical distribution of their data.

   Cluster
   : The IBM Storage Scale or GPFS cluster name. To obtain, run the following command from the IBM Storage Scale file system: /usr/lpp/mmfs/bin/mmlscluster.

   Host
   : The hostname or IP address of an IBM Storage Scale node from which a scan can be initiated, for example a quorum-manager node.

   File system
   : The short name (omit /dev/) of the file system to be scanned. For example, fs1.
     Note: It is important to exactly match the file system name (data source) that IBM Storage Scale populates in the scan file. Run the following command on the IBM Storage Scale system: `/usr/lpp/mmfs/bin/mmlsmount all`

   Node list
   : The comma-delimited list of nodes or node classes that participates in the scan of an IBM Storage Scale file system. For example, `scale01,scale02`.

     ```
     Node  Daemon node name  IP address    Admin node name  Designation
     --------------------------------------------------------------------
        1   msys111-10g       172.16.8.111  msys111-dmz      quorum-manager-perfmon
        2   msys112-10g       172.16.8.112  msys112-dmz      quorum-manager-perfmon
        3   msys113-10g       172.16.8.113  msys113-dmz      quorum-manager-perfmon
     ```

     Note: When you create data source connections for IBM Storage Scale file systems, it is important to exactly match the cluster name and the file system name (data source) that IBM Storage Scale populates in the scan file.
     Run the following commands on the IBM Storage Scale system.

     a. Run the following command to display information about the GPFS cluster:

     ```
     $ /usr/lpp/mmfs/bin/mmlscluster

     GPFS cluster information
     ========================
      GPFS cluster name:         modevvm19.tuc.example.com,
      GPFS cluster id:           7146749509622277333
      GPFS UID domain:           modevvm19.tuc.example.com
      Remote shell command:      /usr/bin/ssh
      Remote file copy command:  /usr/bin/scp
      Repository type:           CCR
     Node  Daemon node name           IP address    Admin node name            Designation
     ---------------------------------------------------------------------------------------
     1     modevvm19.tuc.example.com  203.0.113.24  modevvm19.tuc.example.com  quorum-manager
     ```

     b. Run the following command to display information about file systems that are mounted: on the nodes:

     ```
     $ /usr/lpp/mmfs/bin/mmlsmount all
     File system gpfs0 is mounted on 1 nodes
     File system Data_Science_8M is mounted on 7 nodes.
     File system icp4D_data_fs_master1 is mounted on 8 nodes.
     File system icp4D_data_fs_master2 is mounted on 8 nodes.
     File system icp4D_data_fs_master3 is mounted on 8 nodes.
     ```

Automated scanning of an IBM Storage Scale data source

As an administrator, you can initiate an IBM Storage Scale scan from IBM Data Cataloging to collect system metadata from IBM Storage Scale file system.

Before you begin

You can include or exclude the files during initial IBM Storage Scale scan process by configuring the following environment variable:

INCLUDE_SCALE_SNAPSHOTS
: When the *INCLUDE_SCALE_SNAPSHOTS* variable value is set to 'false' (default value), the IBM Storage Scale scan excludes all the files that are inside the .snapshots directories, otherwise, if the variable value is set to 'true', the scan includes all the files, including the .snapshots directories.

About this task

To set the *INCLUDE_SCALE_SNAPSHOTS* variable by using configmap, see Enabling skip snapshot directories feature on Red Hat® OpenShift®*Enabling skip snapshot directories feature on Red Hat® OpenShift* in the *IBM Storage Scale: Administration Guide*.

When a scan is initiated from the IBM Data Cataloging graphical user interface, the data moves asynchronously back to the IBM Data Cataloging.
Remember: Before you initiate a scan, see IBM Storage Scale scanning considerations.
Automated scanning and data ingestion relies on an established and active network connection between the IBM Data Cataloging instance and the source IBM Storage Scale management node. If the connection cannot be established, the state of the data source connection displays `'unavailable'` and the option for automated scanning does not appear in the IBM Data Cataloging GUI for that connection.

Note: You cannot run scans unless you add override warnings in the configuration file.

Procedure

1. Log in to IBM Data Cataloging web interface.
2. Click menu and go to Data connections > Connections.
3. Select the data source connection name that you want to scan. Ensure that the connection is online for your system ready to scan. (There is an indicator in the State column.)
4. Select Scan now to start the scan, and a small message appears to confirm that the connection name you specify is being scanned.
   You can view the status of the scan on the table in the Scan Status column for the target connection. After the Scan Status has a check mark next to it, the scan is complete.
   Remember: You can also specify a time to begin the scan. Any time zones specified default to Coordinated Universal Time (UTC) time. So, if you specify your scan for 12 noon, it is 12 noon in UTC.

- **Automated scanning of an IBM Storage Scale fileset**
  As an administrator, you can initiate the IBM Storage Scale scan from IBM Data Cataloging to collect system metadata from the IBM Storage Scale file set or file sets.

Automated scanning of an IBM Storage Scale fileset

As an administrator, you can initiate the IBM Storage Scale scan from IBM Data Cataloging to collect system metadata from the IBM Storage Scale file set or file sets.

Before you begin

This feature adds a requirement for non-root user IDs that are used for scanning IBM Storage Scale data source systems. This feature uses the **mmlsfileset** command to retrieve the list of available file sets from the target system when you have root-level permissions. So, if you use a non-root user ID it must have sudo access to **mmlsfileset** for this function to work.

There is already a requirement for a non-root scan user to have sudo access to **mmapplypolicy**, so this requirement adds **mmlsfileset** as an extra required command.
Note: You cannot query the available file sets on a target IBM Storage Scale connection or initiate a file set level scan unless you fulfill this requirement.

About this task

Scan the IBM Storage Scale file set or file sets to insert or update the records for the files that are found by IBM Data Cataloging in that file set or file sets. The scan is scoped to the specified file set, which ensures a faster total scan than scanning the entire file system. Multiple file sets can be specified in a single scan operation, but the scanning of each file set is done successively.

As the scan progresses, the status message is updated to indicate the following information:

- The status message indicates which file set is being scanned.
- The status message indicates when data operations (such as transferring files or indexing data) occur.

This status message can be seen in the GUI on the data source connections table or it can be queried by using the REST API.
This feature works irrespective of whether the data is returned to IBM Data Cataloging by using a direct Kafka connection or by using the file copy method. After a file set level scan completes, a scan generation is recorded or committed.

Additionally, an internal reclamation policy is generated to remove any deleted files that did not appear in the updated scan. The scope of this reclamation policy is limited to the file set that is scanned and does not affect other file sets or the actual file system. This limitation helps you achieve consistency with the source IBM Storage Scale system at file set level granularity.

Procedure

1. Go to the IBM Data Cataloging GUI.
2. Click menu and go to Data connections > Connections table.
   Select the data source connection name and click Scan now, which opens the Select scan type dialog box.
   You can select whether to scan the entire file system or to scan a list of file sets.
   Important: Connection types other than IBM Storage Scale and SMB/CIFS do not open this dialog box. Additionally, Scan now continues to function as it has, which means that there is an immediate initiation of a full connection scan.
3. Select either Scan all to scan all file sets or Select Filesets to scan a specific file set.
   Selecting Scan all initiates a full scan of the file system. If you choose to scan all file sets, click Scan to run the scan.

   Selecting Select Filesets initiates a specific file scan. Click Next to open the Select Individual Filesets dialog box. Use this dialog box to select the specific file sets that you want to scan. Search the table by using the table search header:
   a. You can select file sets by clicking the row of the table that represents that file set. Clicking the row highlights that row and the count under View X selected filesets increases by 1.
   b. You can also select file set by filtering the search criteria. The table can be filtered to show only the selected file sets by clicking View X selected filesets, for ease of review. For example, you can enter `fs` to display all file set with those characters in that order. Click the file set in the table row that you want to select to run the scan on that file set.

   To go back to viewing all available file sets, click View X selected filesets again. The button changes to View all filesets when you view only the selected file sets.

4. After you select all wanted file sets, you can initiate the scan by clicking Scan. Clicking Scan takes you to the Connections table.
   A notification indicates when the scan starts (or that the scan fails if there is a problem). You can view the status of the scan on the table in the Scan Status column for the target connection.
   Remember: After the Scan Status has a check mark next to it, the scan is complete for all selected file sets.

Manual scanning of an IBM Storage Scale data source

How to configure IBM Data Cataloging to connect to IBM Spectrum Scale. After completing these steps, data can be ingested from an IBM Spectrum Scale data source to IBM Data Cataloging for metadata indexing.

Before you begin

Create the data source connection to IBM Storage Scale. For more information, see Configure data source connections.
You can include or exclude the files during initial IBM Storage Scale scan process by configuring the following environment variable:

INCLUDE_SCALE_SNAPSHOTS
    When the *INCLUDE_SCALE_SNAPSHOTS* variable value is set to 'false' (default value), the IBM Storage Scale scan excludes all the files that are inside the .snapshots directories, otherwise, if the variable value is set to 'true', the scan includes all the files, including the .snapshots directories.

To set the *INCLUDE_SCALE_SNAPSHOTS* variable by using configmap, see Enabling skip snapshot directories feature on Red Hat® OpenShift®*Enabling skip snapshot directories feature on Red Hat® OpenShift* in the *IBM Storage Scale: Administration Guide*.

The minimum connection parameters required for manual scanning are:

- Connection Name
- Connection Type
- Cluster
- Filesystem

Restriction: IBM Data Cataloging uses a unit separator (ASCII code 0x1F) as the field delimiter for ingestion into the database. This means that data which contains this character in path/file/object names results in improper parsing of the input data and the records are rejected by IBM Data Cataloging.

Procedure

1. Perform a file system scan to collect system metadata from IBM Spectrum Scale to be ingested into IBM Data Cataloging. For more information, see Performing file system scan to collect metadata from IBM Storage Scale.
2. Copy the output of the file system scan to the IBM Data Cataloging master node. For more information, see Copying the output of the IBM Storage Scale file system scan to the IBM Data Cataloging master node.
3. Ingest data from the file system scan in IBM Data Cataloging. For more information, see Ingesting metadata from IBM Storage Scale file system scan in IBM Data Cataloging.
4. Ingest quota information from the file system. For more information, see Ingesting quota information from the file system.

Performing file system scan to collect metadata from IBM Storage Scale

You can use the file system scanning tool, IBM Storage Scale Scanner, to collect system metadata from IBM Storage Scale to be ingested into IBM Data Cataloging.

About this task

The IBM Storage Scale Scanner tool uses the IBM Storage Scale information lifecycle management (ILM) policy engine to obtain the system metadata about the files stored on the file system. The system metadata is written to a file, which is then transferred to the IBM Data Cataloging master node. The file is then ingested within the node and analytics is carried out to provide search, duplicate file detection, archive data detection, and capacity show-back report generation. The following system metadata is collected from the file system scan:

Key name	Description

Key name	Description
Site	The site where the file or object resides.
Platform	The source storage platform that contains the file or object.
Size	The size of the file.
Owner	The owner of the file.
Path	The subdirectory where the data resides.
Name	The name of the data.
Permissions	The permissions for the file (mode).
ctime	The change time of the file metadata (inode).
mtime	The time when the data was last modified.
atime	The time when the data was last accessed.
Filesystem	The name of the IBM Storage Scale file system that is storing the data.
Cluster	The name of the IBM Storage Scale cluster.
inode	The IBM Storage Scale inode that is storing the data.
Group	The Linux® group associated with the file.
Fileset	The file set that stores the file.
Pool	The storage pool where the file resides.
Migstatus	If applicable, indicates whether the data is migrated to tape or object.
migloc	If applicable, indicates the location of the data if migrated to tape or object.
ScanGen	Scan generation - useful to track rescans.

The IBM Storage Scale Scanner tool also collects quota information by calling **mmrepquota**.

The tool comprises the following files:

- scale_scanner.py: The tool that starts the IBM Storage Scale ILM policy.
- scale_scanner.conf: The configuration file used to customize the behavior of the scale_scanner.py tool.
- createScanPolicy: The script that is called internally by the tool.

Procedure

Install the IBM Storage Scale Scanner tool by unpacking the utility from the IBM Data Cataloging node to the required location on the IBM Storage Scale cluster node.

1. Log in to the IBM Data Cataloging node through Secure Shell (SSH) with the `moadmin` username and password:

   ```
   ssh modadmin@data.cataloging.ibm.com
   ```

2. Change to the directory that contains the IBM Storage Scale scanning utility

   ```
   /opt/ibm/metaocean/spectrum-scale
   ```

3. Copy the createScanPolicy, _init_.py, scale_scanner.conf, and scale_scanner.py files to a node in the IBM Storage Scale cluster:

   ```
   scp * root@spectrumscale.ibm.com:/my_scanner_directory
   ```

   ```
   createScanPolicy 100% 3320 3.2KB/s 00:00
   init.py 100% 427 0.4KB/s 00:00
   scale_scanner.conf 100% 1595 1.6KB/s 00:00
   scale_scanner.py 100% 13KB 13.2KB/s 00:00
   ```

4. On the IBM Storage Scale node where you install the scanning utility, edit the configuration file (scale_scanner.conf) as follows:
   a. Use the IBM Data Cataloging UI to create a connection to the SS system on which you start a manual scan for. Set the **`filesystem`** and **`scandir`** fields, and optionally set the **`outputdir`** and **`site`** fields in the [spectrumscale] stanza of the file.

   ```
   [spectrumscale]
   # Spectrum Scale Filesystem which hosts the scan directory
   # example:  /dev/gpfs0
   filesystem=/dev/gpfs0
   # The directory path on Spectrum Scale Filesystem to perform scan on
   # example: /gpfs0
   # specifies a global directory to be used for temporary storage during
   # mmapplypolicy command processing. The specified directory must be
   #mounted with read/write access within a shared file system
   mountpoint=mount point of the gpfs filesystem
   # It is unclear what the mount_point should be, but setting the mount point
   # to the mount point of the scale file system on the IBM Spectrum Scale node works.
   scandir=/gpfs0
   # The directory to store output data from the scan in (default is
   # scandir)
   outputdir=
   # The site tag to specify a physical location or organization identifier.
   # If you use this field, remove the comment (#)
   #site=
   ```

   b. Set the **`scale_connection`**, **`master_node_ip`**, and **`username`** fields in the **`[datacataloging]`** stanza of the file.
   Note: **`scale_connection`** refers to the name of the IBM Storage Scale file system that is scanned and ingested into IBM Data Cataloging. The **`scale_connection`** value must match the value that is defined in the **`Data Source`** column of the Data Connections page in the IBM Data Cataloging GUI. The username must be a valid name of the IBM Data Cataloging user who has the **`dataadmin`** role. The username field takes the format of <domain_name>/<username>. To determine a domain and username with the **`dataadmin`** role, go to the Access Users page in the IBM Data Cataloging GUI and click the view for the defined users.

   For the local domain, it is not necessary to specify the domain as part of the username field as it is the default domain. For example, to define username for user1 in the local domain that is assigned the **`dataadmin`** role, in the configuration file, enter the following value: **`username=user1`**

.

```
[datacataloging]
# Name of the Spectrum Scale connection to scan files from
# Check using the Spectrum Discover connection manager APIs
scale_connection=fs3
# Data Cataloging Master Node IP
master_node_ip=203.0.113.23
# Data Cataloging user name, having 'dataadmin' role
# Use format <domain_name>/<username>
# e.g. username=Scale/scaleuser1
username=user1
```

Note: The scanner output file generates approximately 1 K of metadata for every file in the system. If there are 12 M files, the size is expected to be approximately 12 GB. By default, the output file is written to the same directory that is being scanned. The log file output location can be customized by setting the `outputdir` field.

5. Run the scan by using the following command:

```
./scale_scanner.py
```

Note: While you run the **./scale_scanner.py** command, you can start another scan. If you start another scan, ensure that you run the scan with another connection that is online and is not being scanned currently. When the scanner is running, the scanner hides the scan now automatically.
Note: As you run the scale_scanner.py script, you are prompted for the password for the IBM Data Cataloging user that is configured in the scale_scanner.conf file with the username under the `datacataloging` section. You must provide the correct password for the configured user. As described in the configuration file, this user needs to be a valid user configured in the IBM Data Cataloging Authentication service (Access management). Also, this user must be assigned to the `dataadmin` role.
For example:

```
$ ./scale_scanner.py
Enter password for SD user 'user1':
Scale Scan Policy is created at: ./scanScale.policy
```

Note:
- After you see a line similar to "`0 'skipped' files and/or errors`", press enter to return to the command prompt.
- The scan takes approximately 2 minutes 30 seconds for every 10 M files on the following configuration:

```
x86 -based Spectrum Scale Cluster
•4 M4 NSD client nodes
•2 M4 NSD server nodes
•DCS3700 350 2TB NL SAS drives & 20 200GB SSD
•QDR InfiniBand cluster network
```

Copying the output of the IBM Storage Scale file system scan to the IBM Data Cataloging master node

After you have scanned your IBM Spectrum Scale file system and have the list.metaOcean output file, copy it to the IBM Data Cataloging master node.

Procedure

As an IBM Data Cataloging administrator, use the **scp** command to copy the list.metaOcean file from the scan output directory to the /opt/ibm/metaocean/data/producer directory on the master node.
Note: If there are multiple file systems in the same cluster that are being scanned, you can rename the list.metaOcean file to avoid name conflicts and to not overwrite an existing list.metaOcean file that is in use. For example:

```
$ mv list.metaOcean list.metaOcean.myfilesystem
```

```
$ scp list.metaOcean.myfilesystem moadmin@MasterNodeIP:/opt/ibm/metaocean/data/producer
```

Ingesting metadata from IBM Storage Scale file system scan in IBM Data Cataloging

Records are inserted into IBM Data Cataloging for indexing when they are pushed to a Kafka connector topic corresponding to the type of data being ingested. In the case of IBM Storage Scale, the Kafka connector topic type is `scale-scan-connector-topic`.

About this task

A Kafka client producer is required to put the IBM Storage Scale file system scan file records onto the Kafka connector topic. The following steps show how to use the **ingest** alias command to push the records in the list.metaOcean file (or another named file) onto the Kafka connector topic.

Procedure

1. Run the following command to ingest the data:

```
$ ingest /opt/ibm/metaocean/data/producer/list.metaOcean
```

2. Replace the list.metaOcean path with the path of the file that you want to ingest.

Ingesting quota information from the file system

The file system scanning tool, IBM Storage Scale Scanner, has the ability to harvest and send quota information to IBM Data Cataloging.

Procedure

To perform quota ingestion, run the following command on the IBM Storage Scale cluster node:

```
./scale_scanner.py --quota-only
```

For example:

```
$ sudo ./scale_scanner.py --quota-only
Enter password for SD user 'user1':
```

IBM Storage Archive data source connections

You can define tags and policies in IBM Data Cataloging based on values that are derived from IBM Spectrum® Archive metadata to help in searching and categorizing files.

IBM Data Cataloging integrates with IBM Storage Archive to display search results that include the following archive state of files:

Migration status `migstatus`
: Search results display details for the following migration status:

  `migrtd`
  : Indicates that the file is migrated to tape.

  `resdnt`
  : Indicates that the file is resident in the file system.

  `premig`
  : Indicates that the file is pre-migrated to tape.

Migration "location" `migloc`
: Search results display information on the tape cartridge in the following format: "1 *tape cartridge volser@tape storage pool id@tape library serial number*". Any additional copies must be separated by colons.

Actual size of file in the associated IBM Storage Scale file system `Size Consumed Bytes`
: IBM Data Cataloging displays a zero if the file is moved to tape.

Remember: The migration state information is collected and summarized in the IBM Data Cataloging State facet. You can access this facet by using IBM Data Cataloging visual search. For more information, see Searching.For more information, see the topic *Searching* in the *Data Cataloging: Administration Guide*.
The IBM Data Cataloging interface displays the search results including the metadata information.

Important: The location information that is displayed in IBM Data Cataloging is provided by IBM Storage Scale. It corresponds to the `dmapi.IBMTPS` attribute for the file. Run the **mmlsattr -L** command for more details.

IBM Cloud Object Storage data source connection

You can create the IBM Cloud® Object Storage (COS) connection and initiate a scan.

IBM® COS uses a connector residing on the storage system to push events to a Kafka topic residing in the IBM Data Cataloging cluster. When configured, the IBM Data Cataloging consumes the events and indexes them into the IBM Data Cataloging database.
Restriction: IBM Data Cataloging uses a unit separator (ASCII code 0x1F) as the field delimiter for ingestion into the database. This means that data which contains this character in path/file/object names results in improper parsing of the input data and the records are rejected by IBM Data Cataloging.

- **Prerequisites**
  The IBM Cloud Object Storage Scanner and Replay prerequisites are listed:
- **Creating an IBM Cloud Object Storage data source connection**
  You can create an IBM Cloud Object Storage (COS) data source connection and from the storage system.
- **Scanning an IBM Cloud Object Storage data connection**
  You can initiate an IBM connection scan to collect system metadata from an IBM Cloud Object Storage system.
- **Best practices for scanning IBM Cloud Object Storage systems**
  Use best practices for scanning IBM Cloud Object Storage (IBM COS) systems.
- **Enabling bucket notifications for Ceph Object Storage**
  Use this information to enable bucket notifications for Ceph® Object Storage.
- **Enabling bucket notifications for Ceph Object Storage on Red Hat OpenShift**
  Use this information to enable bucket notifications for Ceph Object Storage on Red Hat® OpenShift®.
- **Replaying IBM Cloud Object Storage notifications**
  Use the IBM Cloud Object Storage (COS) Replay feature to resend notifications that failed because of an outage or loss of data.
- **Configure IBM Cloud Object Storage notifications for IBM Data Cataloging**
  Ingesting IBM Cloud Object Storage event records into IBM Data Cataloging requires the user to enable the Notification service on the IBM Cloud Object Storage system. Thereafter, the user must connect the IBM Cloud Object Storage system to the IBM Cloud Object Storage connector Kafka topic on the IBM Data Cataloging cluster. The name of this connector topic is `cos-le-connector-topic`.
- **Enabling IBM Cloud Object Storage notification services**
  The IBM Cloud Object Storage notification service can be enabled with the information that follows:
- **Testing the IBM Cloud Object Storage notification service**
  To test the IBM Cloud Object Storage notification service, the tester can populate the IBM Cloud Object Storage vault with test data.

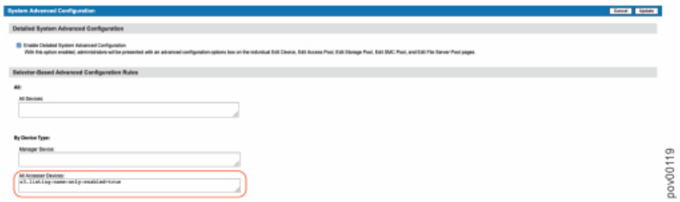
Prerequisites

The IBM Cloud® Object Storage Scanner and Replay prerequisites are listed:

IBM Cloud Object Storage Scanner prerequisite
: For the Scanner, you must enable the Get Bucket Extension for all accesser devices.
: To enable the Get Bucket Extension, you must set the `s3.listing-name-only-enabled` equal to true in the Manager System Advanced Configuration.

  See Figure 1.

  Figure 1. Example of the system advanced configuration

  Remember: You do not need to restart the Accesser, but you might need to wait for 5 minutes before the setting takes effect if you do not restart it.

IBM Cloud Object Storage Replay prerequisite
: For the Replay, **`access_logs`** are uploaded to the management vaults within 1 hour after rotation. Rotation can be triggered earlier by setting the Rotation Period to the minimum value of 15 minutes in Manager under Maintenance/Logs/Device Log Configuration. Refer to the IBM Cloud Object Storage System documentation to make sure that this is configured, and the relevant access logs are present before you run the Replay.

Creating an IBM Cloud Object Storage data source connection

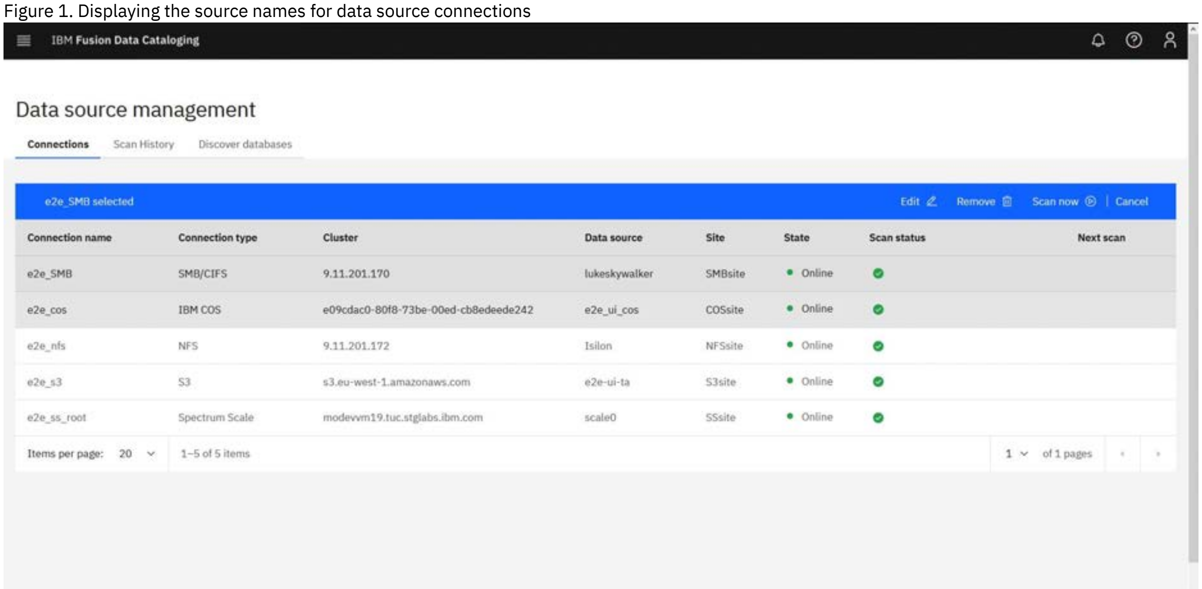
You can create an IBM Cloud® Object Storage (COS) data source connection and from the storage system.

Procedure

1. Log in to the IBM Data Cataloging web interface with a user ID that has the Data Admin role that is associated with it.
   The data admin access role is required for creating connections. For more information, see Role-based access control.

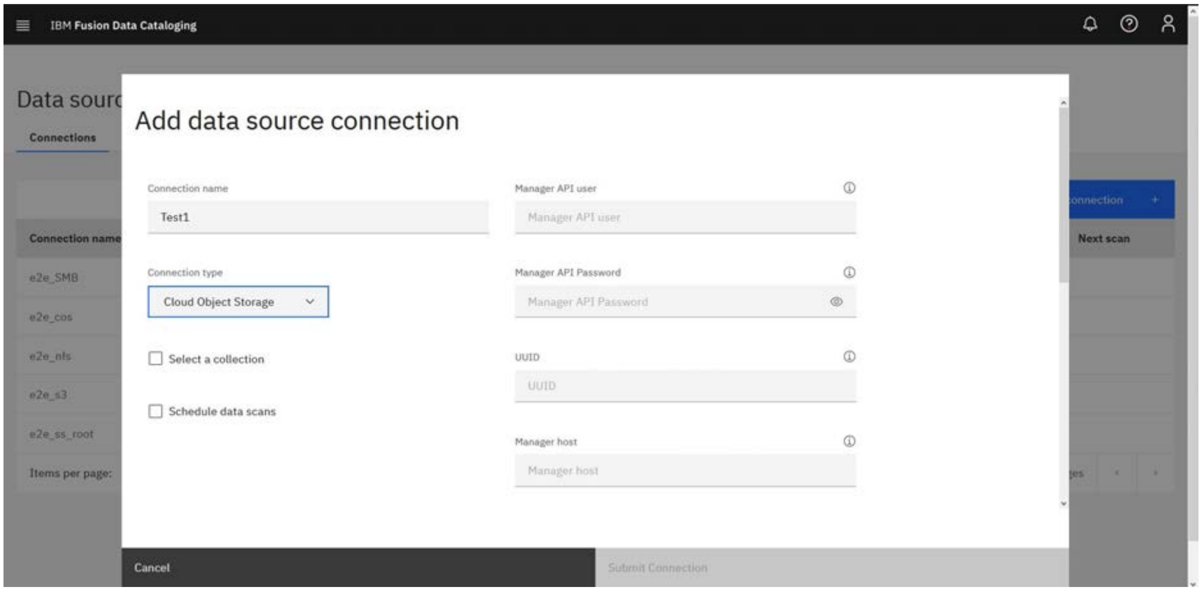
2. Click  menu and click Data connections.
   Clicking Connections displays the different types of data source connection names, connection type, clusters, data source, site, state, scan status, next scan , and Add Connection.

   Figure 1. Displaying the source names for data source connections

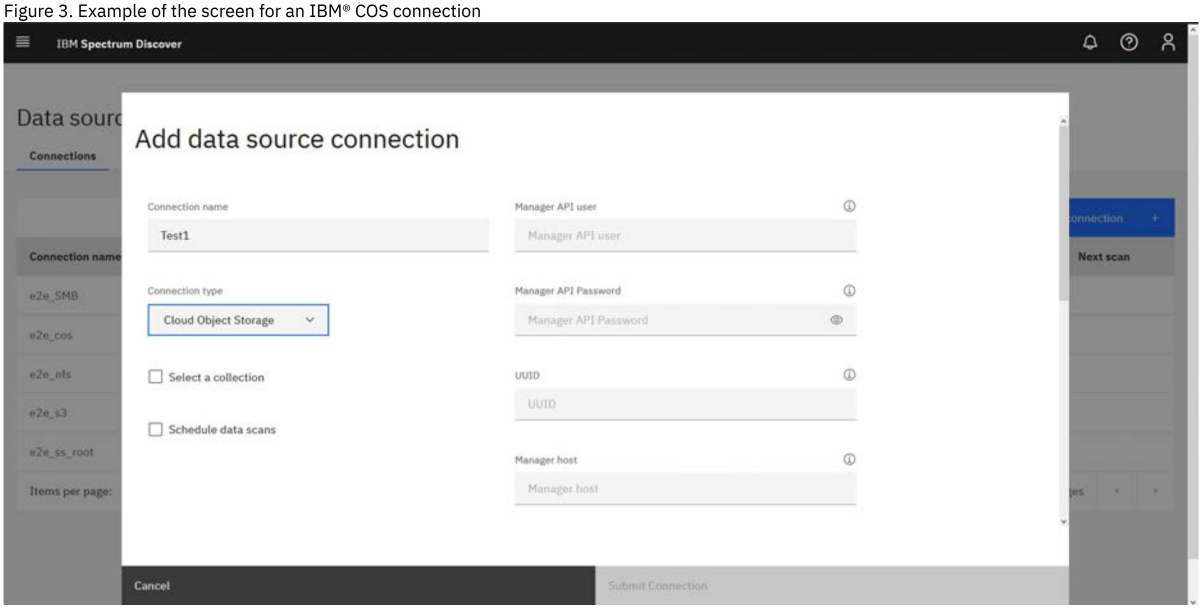
3. Click Add connection to display a new window that shows Add data source connection.

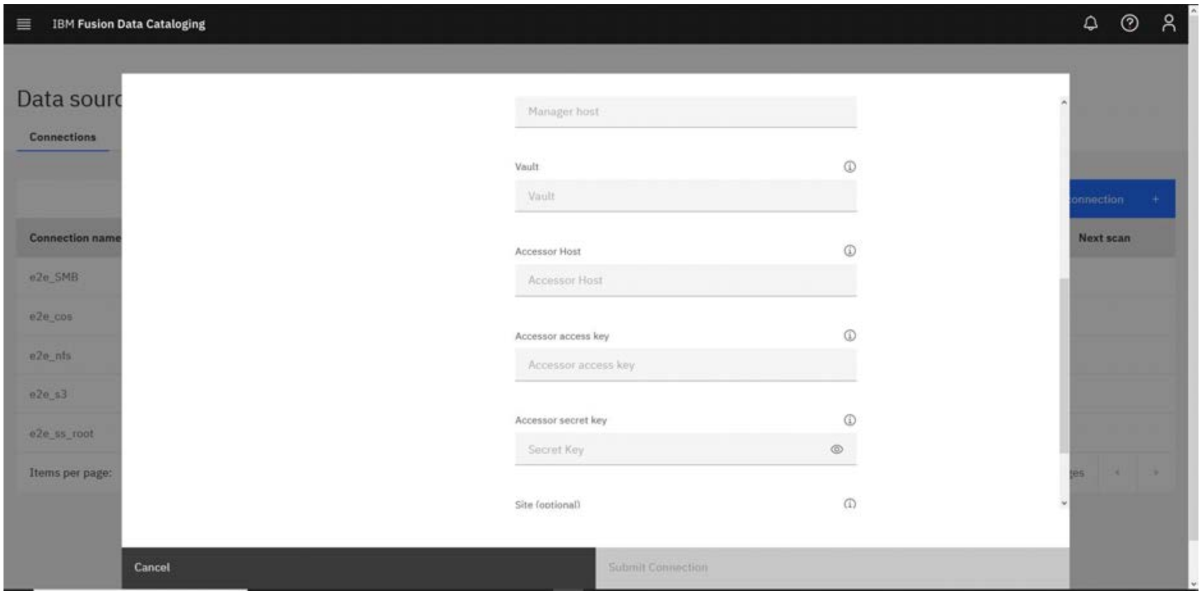
   Figure 2. Example of window that shows Data Connections Add data source Connection



4. Do the following steps:
    a. In the field for Connection name, define a Connection name.
    b. Choose the type of data source connection from the Connection type list.
5. Select the connection type Cloud Object Storage.
   Figure 3

Figure 3. Example of the screen for an IBM® COS connection





6. In the screen for Cloud Object Storage, complete fields, and click Submit Connection.
   For Cloud Storage Object Connections Manager

   Manager API user
   : A user ID that has permissions to connect to the data source system.

   Manager API Password
   : The password for the user ID specified above.

   UUID
   : The unique ID of the DSNet cluster. To obtain the UUID, log in to the COS Manager GUI and click Help > About this system on the upper-right corner of the window.

   Host
   : The IP or hostname of the manager node within the DSNet.

   Vault
   : The specific data vault represented by this connection.

   Site
   : An optional physical location tag that an administrator can provide to see the physical distribution of their data.

   Accesser
   : The IP address or hostname of the Accesser® node on DSNet.

   Accesser access key
   : The Accesser access key that has permission to access data in the data vault that is to be scanned. If the accessor access key value is blank, the value is retrieved (for the manager API user) from the manager API.

   Accesser secret key
   : The Accesser secret key that has permission to access data in the vault that is to be scanned. If the secret access key value is blank, the value is retrieved (for the manager API user) from the manager API.

Scanning an IBM Cloud Object Storage data connection

You can initiate an IBM® connection scan to collect system metadata from an IBM Cloud® Object Storage system.

About this task

When you initiate a scan from the IBM Data Cataloging graphical user interface (GUI), the metadata is transferred asynchronously back to the IBM Data Cataloging instance.

Note:
IBM Data Cataloging does not support scanning of vaults in a dsNet that has any of the following things:

- Proxy vault
- Mirrored vault
- Vault setup for migration

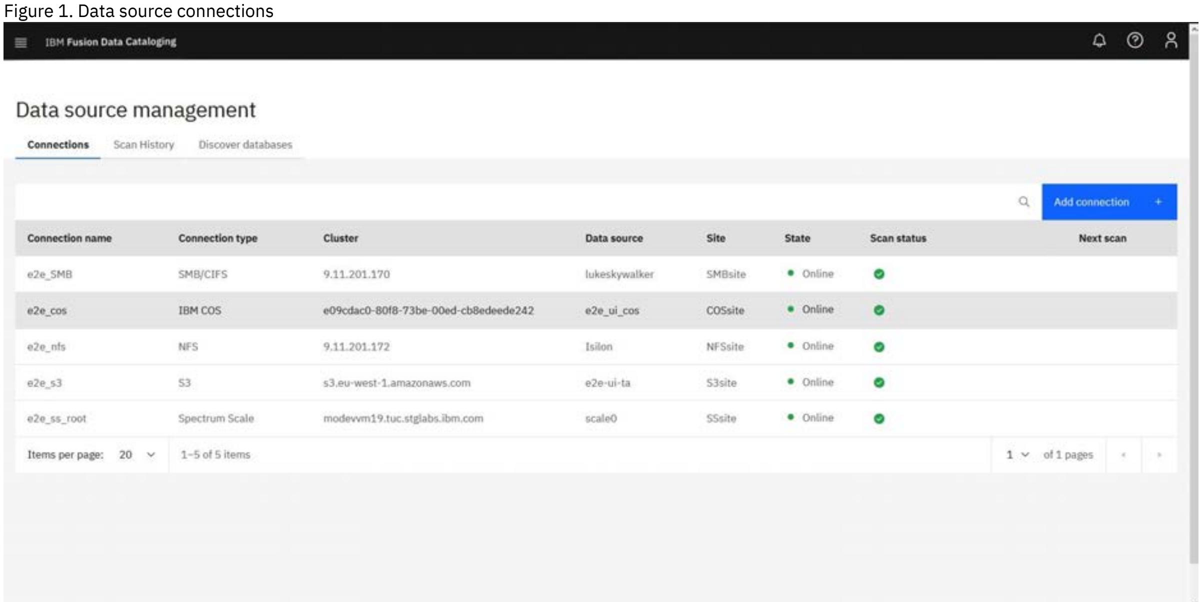
Automated scanning and data ingestion relies on an established and active network connection between the IBM Data Cataloging instance and the IBM Cloud Object Storage storage source. If the connection cannot be established, the state of the data source connection shows as unavailable, and the option for automated scanning does not appear in the IBM Data Cataloging GUI for that connection.

Note: If a scan does not complete successfully, check the log file for errors and warnings. If the error or warning message indicates a need to check the configurations file or the settings file, then you must modify the file as required. For example, in some cases you must update the override warnings in the settings file by adding: `"override_warnings": true` at the root level.
The settings or the configuration file is available in the following location: /opt/ibm/metaocean/data/connections/cos/scan/scanner-settings.json

Procedure

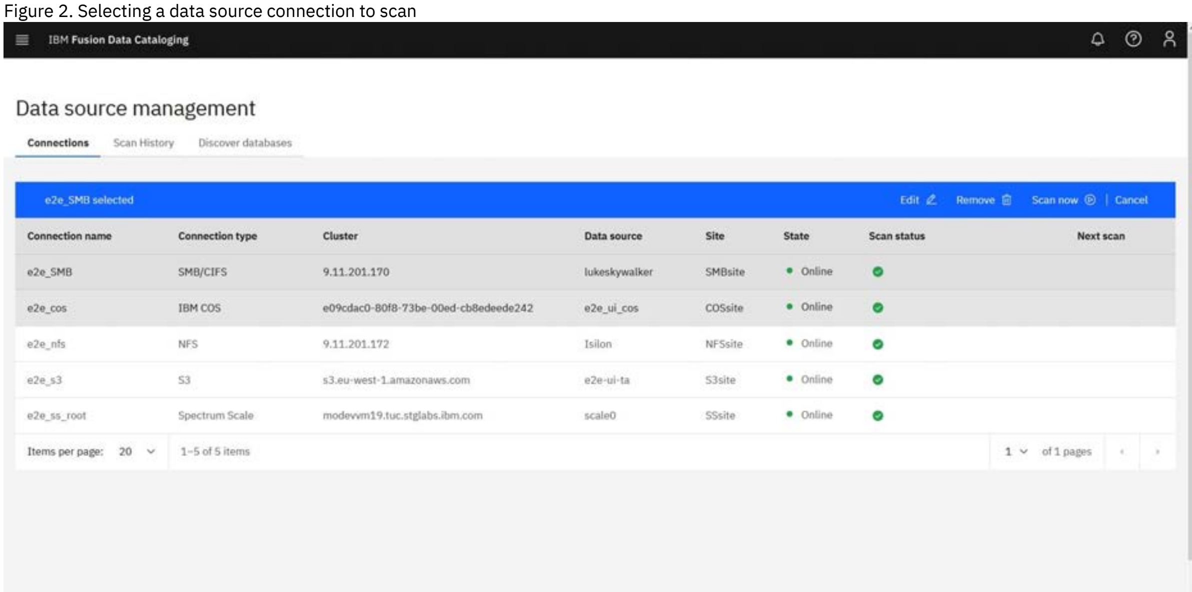
1. Log in to the IBM Data Cataloging graphical user interface (GUI).
2. Click menu and go to Data connections > Connections.
   The following example shows the Data connections menu page:

   Figure 1. Data source connections

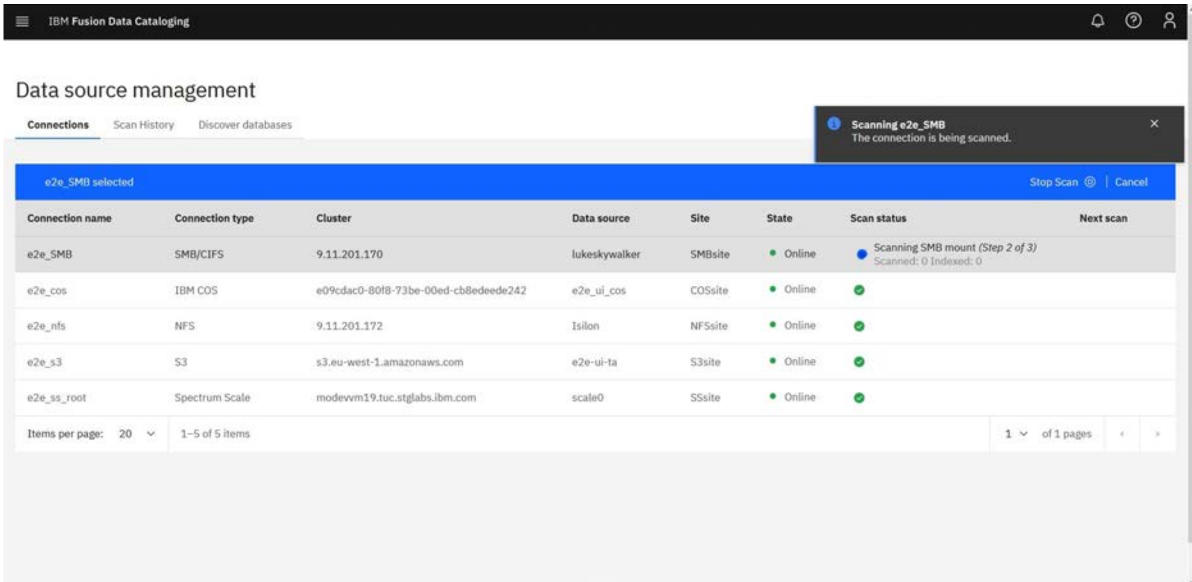
3. Select the data source connection name that you want to scan. Ensure that the `State` is listed as `Online` to make your system scan ready.
   The following example shows how to connect to the IBM Data Cataloging library.

   Figure 2. Selecting a data source connection to scan

4. Select Scan now to change the status to Scanning.
   The following example shows an active scan.

   Figure 3. Active scans



5. When the scan finishes, the state field returns to a status of Online.

Best practices for scanning IBM Cloud Object Storage systems

Use best practices for scanning IBM Cloud® Object Storage (IBM® COS) systems.

It is recommended to check the log files in the following directories after each scan:

/opt/ibm/metaocean/data/connections/cos/<connection_name>/debug/<scan_timestamp>/scanner.debug indicates whether the scan was successful or not.

/opt/ibm/metaocean/data/connections/cos/<connection_name>/error/<scan_timestamp>/scanner.error contains a list of all the messages that are not delivered to IBM Data Cataloging.

This file contains a list of all the messages that are not delivered to IBM Data Cataloging.

/opt/ibm/metaocean/data/connections/cos/<connection_name>/data/<scan_timestamp>/ contains a subfolder with the scanned data source name. There is a stats folder inside this folder that contains information about the number of objects in the data source or the number of objects or scanned files.

You can also compare the total size of the bucket that is reported in IBM Data Cataloging with the total size of the IBM COS at its source (if it is available).

Enabling bucket notifications for Ceph Object Storage

Use this information to enable bucket notifications for Ceph® Object Storage.

Before you begin

Ensure you have the following features set up:

- IBM Data Cataloging 2.0.2.1
- Red Hat® Ceph Storage 4.0 (available starting with version Beta 8)
- A Ceph Object Gateway node that is set up with an HTTPS endpoint

Restriction:

- Ceph Object bucket names must be unique across all data sources. You cannot use the same bucket name to reach a Ceph data source. For example, if there is the IBM Cloud® Object Storage or Amazon S3 bucket with the name `"my_bucket"`, you cannot reach a Ceph data source with the bucket name `"my_bucket"`.
- Notifications from versioned buckets are not supported.
- Only one IBM Data Cataloging node can be configured for push notifications from Ceph Object Storage cluster at a time.

About this task

Use the following steps to enable bucket notifications for Ceph Object Storage.

Procedure

1. Create a data source connection to the Ceph Object Storage cluster.
   A Ceph Object Storage source is established as an Amazon S3 data source connection.
   Remember: Each bucket must have its own data source connection entry in IBM Data Cataloging.
2. To enable Ceph Object Storage bucket notifications:
   a. Copy the ca.crt file from IBM Data Cataloging node to a directory on the Ceph Object Gateway nodes.

    b. Locate the file in the /etc/kafka directory on the IBM Data Cataloging node.
    c. Give this file a unique name on the Ceph node after it is copied over.
    Remember: Ensure that the file has the same name and in the same location on each Ceph Object Gateway node.
    You can choose to use /etc/ssl/certs as the copy target directory on the Ceph Object Gateway node.

3. Create a topic entity by using Ceph bucket notification REST API. The topic contains the push endpoint on IBM Data Cataloging where the notifications are sent to.
   Remember: To enable notifications to be sent to IBM Data Cataloging you must provide push endpoint parameters when you create the topic entity.
   These parameters include the IBM Data Cataloging Kafka topic and credentials that are required to securely produce messages to the topic. For more information about the REST API, see Create a Topic*Create a Topic* in the Ceph documentation.
   The following parameters must be in the POST request:

   ```
   POST
   Action=CreateTopic
   &Name=ceph-le-connector-topic
   &push-endpoint=<endpoint>
   &Attributes.entry.5.key=use-ssl&Attributes.entry.5.value=true
   &Attributes.entry.6.key=ca-location&Attributes.entry.6.value=<file path>
   ```

   In this example:

   <endpoint>
   Indicates the URI of the IBM Data Cataloging Kafka broker in this format: kafka://cos:<password>@<discover_fqdn>:9092
   <password>
   Indicates the password that can be obtained by an administrator on the IBM Data Cataloging node from the following location: /etc/kafka/sasl_password
   <discover_fqdn>
   Indicates the fully qualified domain name of the IBM Data Cataloging node.
   <file path>
   Indicates the location and file name of the Kafka certificate authority (CA) file on the Ceph Object Gateway Node.

   The following example shows topic creation by using the s3curl utility:

   ```
   $ ./s3curl.pl --id=rhceph -- -k -X POST https://<ceph object gateway address>:8080/ -d
   "Action=CreateTopic&Name=ceph-le-connector-topic&push-endpoint=kafka://cos:
   <password>@<discover_fqdn>:9092&Attributes.entry.5.key=use-ssl&Attributes.entry.5.value=true&
   Attributes.entry.6.key=ca-location&Attributes.entry.6.value=/etc/ssl/certs/ca.crt"
   ```

   The --id parameter identifies the credentials to use in the `s3curl` configuration file.

4. Create a notification entity by using the Ceph bucket REST API. This associates events on a specific bucket to a topic. For more information, see CREATE NOTIFICATION *CREATE NOTIFICATION* in the Ceph documentation.
   The following example shows how to establish a bucket notification by using the s3curl utility:

   ```
   $ ./s3curl.pl --id=rhceph --put=notif.xml -- -k https://<ceph object gateway address>:8080/<bucket>?notification
   ```

   ```
   Contents of notif.xml:
   <NotificationConfiguration xmlns="http://s3.amazonaws.com/doc/2010-03-31/">
       <TopicConfiguration>
           <Id>id1</Id>
           <Topic>arn:aws:sns:default::ceph-le-connector-topic</Topic>
       </TopicConfiguration>
   </NotificationConfiguration>
   ```

   You can now capture events on objects within the configured buckets.

Enabling bucket notifications for Ceph Object Storage on Red Hat OpenShift

Use this information to enable bucket notifications for Ceph® Object Storage on Red Hat® OpenShift®.

Before you begin

Ensure you have the following features set up:

- IBM Data Cataloging 2.0.4
- Red Hat Ceph Storage 4.1z2 or later
- A Ceph Object Gateway node that is set up with an HTTPS endpoint

Restriction:

- Ceph Object bucket names must be unique across all data sources. You cannot use the same bucket name to reach a Ceph data source. For example, if an IBM Cloud® Object Storage or Amazon S3 bucket exists with the name `"my_bucket"`, you cannot reach a Ceph data source with the bucket name `"my_bucket"`.
- Notifications from versioned buckets are not supported.
- Only one IBM Data Cataloging node can be configured for push notifications from Ceph Object Storage cluster at a time.

About this task

Use the following steps to enable bucket notifications for Ceph Object Storage.

Procedure

1. Create a data source connection to the Ceph Object Storage cluster.
   A Ceph Object Storage source is established as an Amazon S3 data source connection.

Remember: Each bucket must have its own data source connection entry in IBM Data Cataloging.

2. To enable Ceph Object Storage bucket notifications:
    a. Issue the following command to transfer the ca.crt file from IBM Data Cataloging node to the Ceph Object Gateway nodes.

    ```
    oc get secret kafka -n ibm-data-cataloging -o jsonpath='{.data.sasl_ca\.crt}' | base64 -d > ca.crt
    ```

    b. Give this file a unique name on the Ceph node after it is copied over.
    Remember: Ensure that the file has the same name and in the same location on each Ceph Object Gateway node.
    You can choose to use /etc/ssl/certs as the copy target directory on the Ceph Object Gateway node.
3. Create a topic entity by using Ceph bucket notification REST API. The topic contains the push endpoint on IBM Data Cataloging where the notifications are sent to.
Remember: To enable notifications to be sent to IBM Data Cataloging, you must provide endpoint parameters when you create the topic entity.
These parameters include the IBM Data Cataloging Kafka topic and credentials that are important to securely produce messages to the topic. For more information about the REST API, see https://docs.ceph.com/docs/master/radosgw/notifications/#create-a-topic.
The following parameters must be in the POST request:

```
POST
Action=CreateTopic
&Name=ceph-le-connector-topic
&push-endpoint=<endpoint>
&Attributes.entry.5.key=use-ssl&Attributes.entry.5.value=true
&Attributes.entry.6.key=ca-location&Attributes.entry.6.value=<file path>
```

The parameters that are shown in the example are explained in the following section.

<endpoint>
    Indicates the URI of the IBM Data Cataloging Kafka broker in this format: kafka://cos:<password>@<discover_fqdn>:443
<password>
    Indicates the password that can be obtained by an administrator on the Red Hat OpenShift node. Issue the following command to obtain the password:

    ```
    oc get secret kafka-sasl-password -n ibm-data-cataloging -o jsonpath='{.data.password}'
    ```

<discover_fqdn>
    Indicates the fully qualified domain name of the IBM Data Cataloging node.
<file path>
    Indicates the location and file name of the Kafka certificate authority (CA) file on the Ceph Object Gateway Node.

The following example shows topic creation by using the s3curl utility:

```
$ ./s3curl.pl --id=rhceph -- -k -X POST https://<ceph object gateway address>:8080/ -d
"Action=CreateTopic&Name=ceph-le-connector-topic&push-endpoint=kafka://cos:
<password>@<discover_fqdn>:9092&Attributes.entry.5.key=use-ssl&Attributes.entry.5.value=true&
Attributes.entry.6.key=ca-location&Attributes.entry.6.value=/etc/ssl/certs/ca.crt"
```

The --id parameter identifies the credentials to use in the **s3curl** configuration file.

4. Create a notification entity by using the Ceph bucket REST API. This associates events on a specific bucket to a topic. For more information, see: https://docs.ceph.com/docs/master/radosgw/s3/bucketops/#create-notification
The following example shows how to establish a bucket notification by using the s3curl utility:

```
$ ./s3curl.pl --id=rhceph --put=notif.xml -- -k https://<ceph object gateway address>:8080/<bucket>?notification


Contents of notif.xml:
<NotificationConfiguration xmlns="http://s3.amazonaws.com/doc/2010-03-31/">
    <TopicConfiguration>
        <Id>id1</Id>
        <Topic>arn:aws:sns:default::ceph-le-connector-topic</Topic>
    </TopicConfiguration>
</NotificationConfiguration>
```

You can now capture events on objects within the configured buckets.

Replaying IBM Cloud Object Storage notifications

Use the IBM Cloud® Object Storage (COS) Replay feature to resend notifications that failed because of an outage or loss of data.

The IBM® COS Replay reads object metadata from vaults and submits the metadata to IBM Data Cataloging by using Kafka notifications.

- **Overview of architecture**
This topic describes a high-level overview of IBM Cloud Object Storage Scanner architecture.
- **Configuration file**
The configuration file is used by the Notifier and Replay.
- **Replay performance**
The number of requests that are issued by IBM Cloud Object Storage Replay is throttled to ensure that overall dsNet performance remains at the agreed level.
- **Replay tasks and vault settings**
A few scenarios exist that prevent the Replay from operating correctly.
- **Including and excluding vaults**
You can set the vaults that you scan with various settings in the configuration file.
- **Stats files**
The IBM Cloud Object Storage Scanner tracks each LIST process status to a stats file.
- **Replay**
When a severe outage occurs and causes the loss of notifications sent by the system to IBM Data Cataloging, the IBM Cloud Object Storage Scanner Replay feature can be used to recover lost notifications.

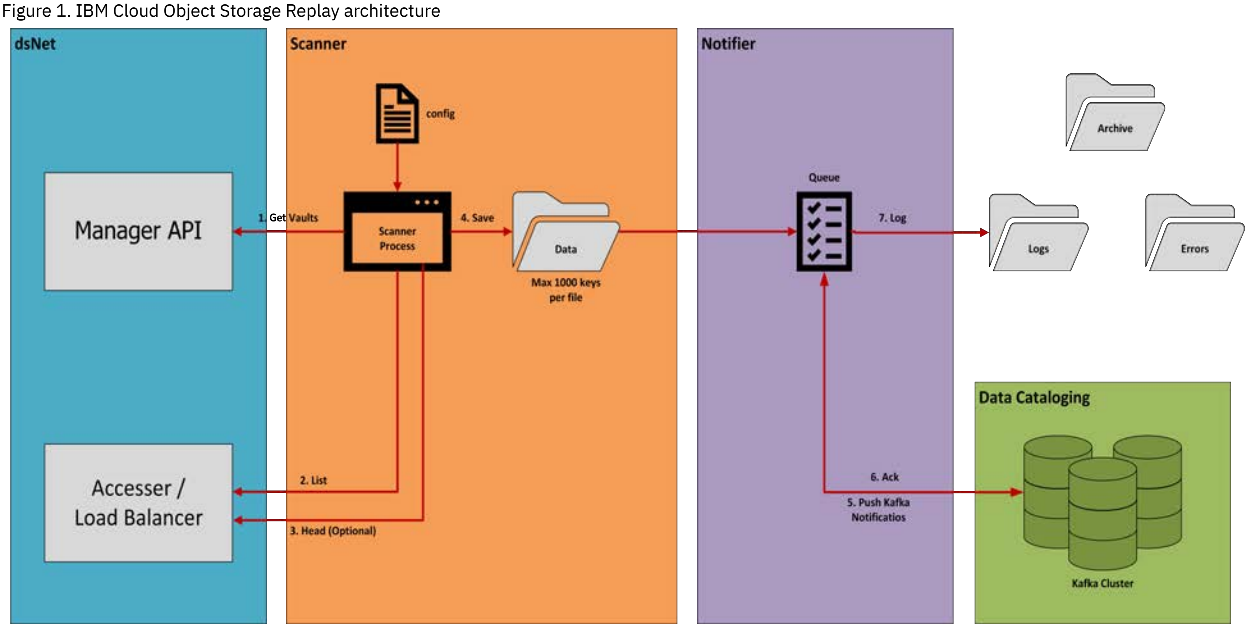
- **Initialization for Replay**
  During the startup, Replay reads the configuration file and issues requests to the Manager of the dsNet device similar to the Scanner.
- **Error conditions**
  Sometimes Replay does not have enough information to replay the original notification. If this occurs, you must fix the problems manually.
- **Output**
  Messages are batched by 1,000 or to the Scanner list objects size configuration setting, if specified.
- **Renaming a vault for Replay**
  When you rename a vault, it is possible that Replay can abort.
- **Starting the Replay**
  The guidelines and rules for using Replay are documented in this topic.
- **Debug mode for Replay**
  Run Replay in debug mode to troubleshoot problems.
- **Notifier**
  The Notifier is the component that reads the JSON notifications that are written by Scanner or Replay and sends notifications to the Kafka cluster.
- **Limitations**
  Limitations apply when the Notifier uses a Kafka configuration retrieved from the Manager API.
- **Starting the Notifier**
  Running the Notifier has rules and limitations.
- **Notifier operation**
  The Notifier enumerates and processes all .log files in the Scanner data directory.
- **Stopping the Notifier**
  You might need to stop and restart the Notifier.
- **Restarting the Notifier**
  When you stop the Notifier following a shutdown with the kill.notifier file, you must rename the file manually or delete the file before you do a restart.
- **Progress report**
  The Progress Report provides an instant snapshot of status for the Scanner and Notifier.
- **Logging**
  You can view the list of directories that are generated by scanner, notifier, and replay.
- **IBM Cloud Object Storage Scanner output data**
  The Scanner generates a directory beneath the output data directory for each vault or vault prefix as defined in the configuration file.
- **Appendix**
  The appendix shows an example of a log file and examples of Scanner debug data.

Overview of architecture

This topic describes a high-level overview of IBM Cloud® Object Storage Scanner architecture.

The following figure shows a high-level overview of IBM Cloud Object Storage Replay architecture.

Figure 1. IBM Cloud Object Storage Replay architecture



The /opt/ibm/metaocean/data/connections/cos/replay/output/data folder puts the replay output and the notifier reads Kafka messages from the directory. Putting replay output onto disk means that a cold restart is possible.

The IBM Cloud Object Storage Replay consists of two major components:

Replay
: Downloads system's logs and re-creates notifications that are sent during a defined time period.

Notifier
: Submits the extracted information to IBM Data Cataloging.

Configuration file

The configuration file is used by the Notifier and Replay.

The configuration file includes:

- Information regarding the net
- Runtime parameters for the Notifier and Replay
- A list of vaults to scan

The configuration file is named scanner-settings.json and must sit in the /opt/ibm/metaocean/data/connections/cos/replay directory.

The rules for IBM Cloud® Object Storage Replay settings are:

- All access logs are scanned.
- All objects that are created or updated since Coordinated Universal Time (UTC) 00:00:01 from 11 April 2018 to Coordinated Universal Time (UTC) 10:01:53 on 21 September 2018 are scanned in batches of 1000.
- Custom metadata is retrieved for each object or version.
- Ten vaults are processed in parallel.
- Each vault has a single process LIST that issues requests and 15 processes that issue HEAD requests.

The following example shows every setting. Most settings have default values and can be omitted, but these screens show a typical example by using default values.

Example of the Cloud Object Storage Replay settings

```
{
     "system": {
     "name": "Test dsnet",
     "uuid": "00000000-0000-0000-0000-000000000000",
     "manager_ip": "172.1.1.1",
     "accesser_ip": "172.1.1.2",
     "accesser_supports_https": false,
     "manager_username": "admin",
     "manager_password": "password",
     "is_ibm_cos": true
},
   "timestamps": {
     "min_utc": "2018-01-01T00:00:00Z",
     "max_utc": "2018-09-21T10:01:53Z"
},
   "policy_engine" : {
     "spectrum_discover_host": "modevvm32.tuc.stglabs.ibm.com"
     "user": "sdadmin",
     "password": "password"
},
   "scanner": {
     "max_requests_per_second": 5000,
     "max_parallel_list": 10,
     "parallel_head_per_list": 5,
     "list_objects_size": 100
},
"notifier":{
     "kafka_format": 1,
     "kafka_endpoint": "192.168.1.1:9092",
     "kafka_topic": "cos-le-connector-topic",
     "kafka_username": "cos",
     "kafka_password": "password",
     "kafka_pem": "-----BEGIN CERTIFICATE------...\n-----END CERTIFICATE-----\n"
},
   "logging": {
     "debug_log_max_bytes": 10000000,
     "debug_log_backup_count": 10000,
     "notification_log_max_bytes": 10000000,
     "notification_log_backup_count": 10000,
     "notification_log_all": true
},
   "include_all_vaults": false,
   "has_custom_metadata": true,
   "override_warnings": true,
   "exclude-vaults": ["Manager"],
   "vaults": [
     {
       "vault_name": "Vault-1"
     },
     {
       "vault_name": "Vault-2",
       "has_custom_metadata": false
     },
     {
       "vault_name": "Vault-3",
       "has_custom_metadata": false,
       "prefix": "customers/live"
     }
  ]
}
```

Typical Cloud Object Storage configuration settings

```
{
```

```
    "dsnet": {
    "name": "Test dsnet",
    "uuid": "00000000-0000-0000-0000-000000000000",
    "manager_ip": "172.1.1.1",
    "accesser_ip": "172.1.1.2",
    "accesser_supports_https": false,
    "manager_username": "admin",
    "manager_password": "password",
    "is_ibm_cos": true
},
  "timestamps": {
    "min_utc": "2018-01-01T00:00:00Z",
    "max_utc": "2018-09-21T10:01:53Z"
},
  “policy_engine” : {
    “spectrum_discover_host”: ”modevvm32.tuc.stglabs.ibm.com”
    “user”: “sdadmin”,
    “password”: “password”
},
  "scanner": {
    "max_requests_per_second": 5000,
    "max_parallel_list": 10,
    "parallel_head_per_list": 5,
    "list_objects_size": 100
},
  "notifier":{
    "kafka_format": 1,
    "kafka_endpoint": "192.168.1.1:9092",
    "kafka_topic": "cos-le-connector-topic",
    "kafka_username": "cos",
    "kafka_password": "password",
    "kafka_pem": "-----BEGIN CERTIFICATE-----...\n-----END CERTIFICATE-----\n"
},
  "logging": {
    "debug_log_max_bytes": 10000000,
    "debug_log_backup_count": 10000,
    "notification_log_max_bytes": 10000000,
    "notification_log_backup_count": 10000,
    "notification_log_all": true
},
  "include_all_vaults": false,
  "has_custom_metadata": true,
  "override_warnings": true,
  "exclude-vaults": ["Manager"],
  "vaults": [
    {
      "vault_name": "Vault-1"
    },
    {
      "vault_name": "Vault-2",
      "has_custom_metadata": false
    },
    {
      "vault_name": "Vault-3",
      "has_custom_metadata": false,
      "prefix": "customers/live"
    }
 ]
}


{
    "dsnet": {
    "manager_ip": "192.168.2.106",
    "accesser_ip": "192.168.2.111"
},
    "timestamps": {
    "min_utc": "2018-04-11T00:00:01.000Z",
    "max_utc": "2018-09-21T10:01:53Z"
},
    "scanner":{
       "max_requests_per_second": 5000
},
    "include_all_vaults": true
}

{
    "system": {
    "manager_ip": "192.168.2.106",
    "accesser_ip": "192.168.2.111"
},
   “policy_engine” : {
    “spectrum_discover_host”: ”modevvm32.tuc.stglabs.ibm.com”
},
    "timestamps": {
    "min_utc": "2018-04-11T00:00:01.000Z",
    "max_utc": "2018-09-21T10:01:53Z"
},
    "scanner":{
       "max_requests_per_second": 5000
},
    "include_all_vaults": true
}
```

IBM Cloud Object Storage Scanner is highly configurable. Each element in the file is described in Table 1.

Remember: IBM Data Cataloging does not support file or file path names that use characters that are not part of the UTF-8 character set.

Table 1. Explanation of the configuration file

Element	Description	Optional	Default value	Restart scanner if changed	Restart notifier if changed
System section					
`name`	Free-text name of the dsNet. Appears in the `'system_name'` element in all Kafka messages.	√	Retrieved from Manager API if configured. If not, the name does not appear in Kafka messages.	√	✗
`uuid`	UUID of the dsNet. Appears in the `'system_uuid'` element in all Kafka messages.	√	Retrieved from Manager API.	√	✗
`manager_ip`	Single IP address or host name of the manager device.	✗	Not applicable	√	✗
`accesser_ip`	Single IP address or host name of an accesser device or load balancer that routes to the accessers.	✗	Not applicable	√	✗
`accesser_supports_https`	Boolean value that indicates whether http or https can be used when you send requests to the accesser or load balancer.	√	true	√	✗
`manager_username`	Username for accessing the manager API. For testing only. Not to be used in production.	√	Supplied by user at prompt	√	✗
`manager_password`	Password for accessing the Manager API. For testing only. Not to be used in production.	√	Supplied by user at prompt	√	✗
`is_ibm_cos`	Boolean value that indicates whether the system is an IBM Cloud Object Storage or another s3 compliant system. If true, the IBM® Get Bucket Extension is used to retrieve object keys from the vaults. Note: Setting the value to false is not currently supported by the Scanner and Notifier.	√	True	√	✗
`accesser_access_key`	Access key ID for S3 calls to the accesses or load balancer. For testing only. Not to be used in production.	√	Supplied by user at prompt if you cannot retrieve it from Manager API for the user account that is specified in dsNet/manager_ username.	√	✗
`accesser_secret_key`	Secret key for S3 calls to the accesser or load balancer. For testing only. Not to be used in production.	√	Supplied by user at prompt if you cannot retrieve from Manager API.	√	✗
Time stamps section					
`min_utc`	Only objects or version in the vaults that have a **`LastModified`** datetime on or after this timestamp is submitted to IBM Data Cataloging. Needs to be less than the **`max_utc`** value. Note: Changing **`min_utc`** and restarting scanner applies only to objects not yet scanned. Objects scanned before restart might have a **`LastModifiedDate`** value that is earlier than the **`min_utc`** value.	✗		√ See note.	✗
`max_utc`	Only objects or version in the vaults that have a **`LastModified`** datetime on or before this time stamp is submitted to IBM Data Cataloging. Needs to be more than min_utc and less than current time. Note: Changing **`max_utc`** to a more recent time and restarting does not mean that new objects written since the old **`max_utc`** is scanned. The scanner continues from the last object's key that is scanned in lexicographic order. This means that new objects with names smaller than the last object scanned are not scanned.	√		√ See note.	✗
Policy engine section		(Only required for IBM Data Cataloging 2.0.0.3 and later)			
`spectrum_discover_host`	Host name or IP address of the policy engine service from which the Kafka certificate is retrieved.	✗	none	√	√
`user`	Username for authorization on policy engine.	✗	none	√	√
`password`	Password for authorization on policy engine.	✗	none	√	√
Replay section					
`access_log_directory`	The `access_log_directory` is where the dsNet access log files are stored after download. Access logs must be in the root input folder. Files in subdirectories are not processed.	√	[IBM Cloud Object Storage Replay]/ access_logs	Restart Replay if changed	Restart Replay if changed
`download`	If download is set to false, access logs are not downloaded and are assumed to already be present in access_log_directory.	√	true	Restart Replay if changed	Restart Replay if changed
Notifier section		√			
`kafka_format`	Format of the Kafka message.	√	1	✗	√
`kafka_endpoint`	IP address and port of the Kafka endpoint.	√	Retrieved from Manager API	✗	√
`kafka_topic`	Name of the Kafka topic.	√	Retrieved from Manager API	✗	√

Element	Description	Optional	Default value	Restart scanner if changed	Restart notifier if changed
System section					
`kafka_username`	The username for authentication with Kafka. Note: For testing only. Not to be used in production.	√	Supplied by user at prompt if you cannot retrieve from Manager API.	✗	√
`kafka_password`	The password for authentication with Kafka. Note: For testing only. Not to be used in production.	√	Supplied by user at prompt if cannot be retrieved from Manager API.	✗	√
`kafka_pem`	The certificate PEM for authentication with Kafka. Must include '\n' characters to ensure correct formatting. Note: For testing only. Not to be used in production.	√	Supplied by user at prompt if it cannot be retrieved from the system	✗	√
Logging section					
`debug_log_max_bytes`	The scanner.debug and notifier.debug roll over when this size is reached.	√	1,000,000	√	√
`debug_log_backup_count`	The number of scanner.debug and notifier.debug files to retain.	√	10	√	√
`notification_log_max_b`	The notification.log rolls over when this size is reached.	√	1,000,000	√	√
`notification_log_backup_count`	The number of notification.log files to retain.	√	10	√	√
`notification_log_all`	Boolean value that controls the level of Notifier logging. When true: an entry is written to notification.log for message you send to the Kafka cluster. When false: only failed sends are written to notification.log.	√	False	✗	√
Root-level items					
`include_all_vaults`	Boolean value that determines whether all vaults in the dsNet are scanned. If false, the details of the vaults to be scanned must be specified in the 'vaults' element. Boolean value that determines whether custom metadata and content type are retrieved for each object by using individual HEAD requests.	√	False	√	✗
`has_custom_metadata`	This value is only relevant when a versioned vault is scanned. For IBM Cloud Object Storage systems, non-versioned vaults always require a HEAD request for every object. Can be overridden for each vault in the 'vaults' element.	√	True	√	✗
`override_warnings`	Boolean value that allows the scanner to run and ignore any warnings that are generated on start-up. For example, a warning is raised on start-up if versioning is suspended on a vault.	√	False	√	✗
`exclude_vaults`	Comma-separated list of vault names to be excluded from scanning, such as: `"exclude-vaults": ["COSVault", "COSVault-V"]`	√	[] Empty list	√	✗
`vaults`	List of vaults to be scanned. If `include_all_vaults` is true, the vaults list can be left empty. This list can be used to define more detailed scanning parameters for individual vaults. Any settings that are defined here take precedence over the settings that are described. Each element in the list contains: The `vault_name` is the name of the vault. The `has_custom_metadata` is an optional Boolean that overrides the `has_custom_metadata` that is described. The `prefix` is an optional string that is used to filter the objects or versions that are retrieved from the vault.	√	Dependent on settings include_all_vaults and exclude_vaults	√	✗

Replay performance

The number of requests that are issued by IBM Cloud® Object Storage Replay is throttled to ensure that overall dsNet performance remains at the agreed level.

You can control throttling by the number of settings in a configuration file. All settings are optional. The following screen shows an example of the default values.

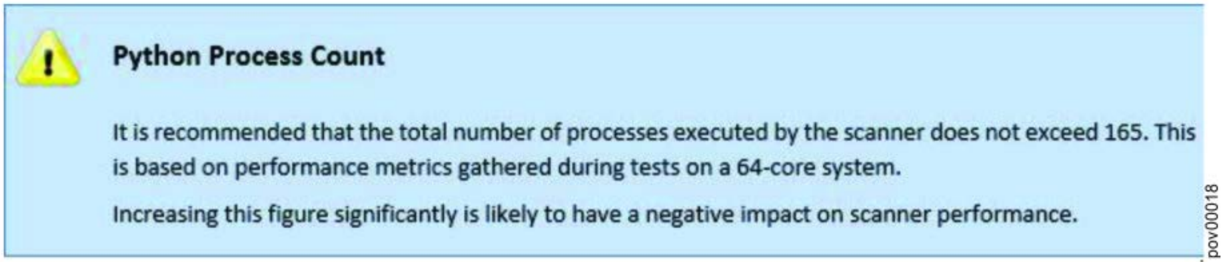
```
"replay": {
     "max_requests_per_second": 1000,
     "max_parallel_list": 10,
     "parallel_head_per_list": 15,
     "list_objects_size": 1000
}
```

Process count

The following list shows an example of how 161 processes are divided. Figure 1 shows a caution message of how the number of processes should not exceed 161.

- One main process
- 10 List worker processes
- 150 HEAD worker processes

Figure 1. Python process count



Maximum Replay performance

Replay and Notifier maximize performance on a 64 core Intel(R) Xeon(R) CPU E5-2690 v3 @ 2.60GHz server is 2300 objects that are scanned and notified per second with a dsNet with 6 accessers and 12 slicestors under customer load at 50 percent capacity.

```
"replay": {
     "max_requests_per_second": 2300,
     "max_parallel_list": 10,
     "parallel_head_per_list": 15,
     "list_objects_size": 1000
 }
```

The recommendation is to start the replay at a rate of 1000 objects scanned per second. Measure the latency degradation of customer traffic and increase the scanning rate until the maximum acceptable degradation is reached.

One thousand objects per second on the net, which is a 5 - 27 percent increase of write operations, the latency (larger increase for smaller size files) and around 10 percent for read operations latency were measured.

At 2000 objects a second, a 10 - 50 percent increase of write operations latency and in the range 18 - 28 percent, and 10 percent for read operations latency were measured.

Replay tasks and vault settings

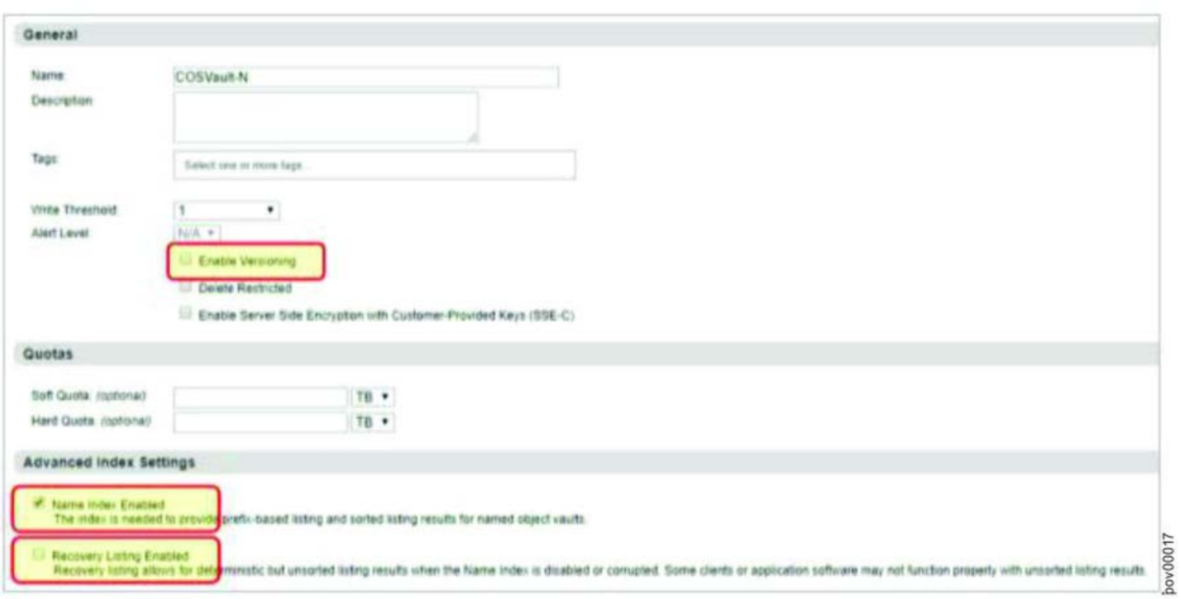
A few scenarios exist that prevent the Replay from operating correctly.

Certain combinations of the following IBM Cloud® Object Storage vault settings prevent the Replay from running a full scan:

- Vault versioning
- Name index
- Recover listing

Figure 1 shows settings for the first three items on the vault configuration page in the DsNet Manager user interface.

Figure 1. Settings for three items on the vault configuration page in the net Manager user interface



The scenarios that are invalid are reported at startup.
Remember: You must correct the scenarios before you can run the Replay.

If a scan does not complete successfully, make sure that you check the log file for errors and warnings. In some cases, you must modify the settings file as detailed in the errors and warning messages. The settings file is located at: /opt/ibm/metaocean/data/connections/cos/scan/scanner-settings.json

Table 1 shows the behavior for the Replay for different combinations of the four variables.

Table 1. Behaviors for Replay for four variables

ID	Name index	Recovery listing	Versioned	Cloud Object Storage Replay behavior
0	✗	✗	✗	☒ Stop start-up and report error in config file: Error: Objects cannot be listed because Name Index and Recovery Index are both disabled. You might enable Recovery Listing on the vault or add this vault to the "`exclude_vaults`" list in the configuration file. For example: `"exclude-vaults": ["vault-name"]`
1	✗	✗	√	☒ Stop start-up and report error in config file: Error: Objects cannot be listed because Name Index and Recovery Index are both unavailable. You might enable Recovery Listing on the vault or add this vault to the "`exclude_vaults`" list in the configuration file. For example: `"exclude-vaults": ["vault-name"]`
2	✗	√	✗	☑ Object Listing is run.
3	✗	√	√	☑ Object Listing runs. Only the most recent version of each object is listed. A warning is logged: Warning: Versions cannot be listed as Name Index is unavailable. An object scan is run and only the most recent version of each object is listed. You must add **`override_warnings: true`** in the configuration file to ignore this warning. Switching Name Index on does not enable scanning of a full version history. Objects created while Name Index is off is not present when it is enabled.
4	√	✗	✗	☑ Object Listing is run.
5	√	✗	√	☑ Object Listing is run.
6	√	√	✗	☑ Object Listing is run.
7	√	√	√	! Stop start-up and report warning: This is a versioned vault but version scanning is not possible as Recovery Listing is enabled. You might either disable Recovery Listing on the vault to allow version scanning, or rerun the Replay with the argument `override-warnings: true` to allow object scanning.

Important: You might receive system errors about records not being scanned to the database if you scan a IBM Cloud Object Storage vault with Name Index disabled and Recovery Listing enabled. You cannot specify a prefix listing on a vault that has Recovery Listing enabled (you cannot specify a blank prefix either).

Including and excluding vaults

You can set the vaults that you scan with various settings in the configuration file.

Use the following settings in the configuration file to scan the vaults:

- **`include_all_vaults`** (Boolean)
- **`exclude_vaults`** (List)
- **`vaults`** (Dictionary)

When **`include_all_vaults`** is true, all vaults in the system are scanned except for any vaults specified in the **`exclude_vaults`** list.

You might consider **`exclude_vaults`** a list of vaults to ignore and **`vaults`** is a list that specifies details of individual vaults to be scanned.

If **`include_all_vaults`** is true and the vaults list is populated, the list of vaults that are scanned is the superset of all vaults that are returned by the Manager that are merged with the vaults list from the config file.

An error is raised and the Scanner aborts on start-up if the same vault appears in both **`vaults`** and **`exclude_vaults`**.

Mirror, Proxy, Data Migration

IBM Cloud® Object Storage Scanner does not support scanning of the following:

- Mirrored vaults
- Proxy vaults
- Vaults that are set up for migration

Any vaults of these types are ignored by the scanner and a warning logged in the debug log.

Examples for including and excluding vaults

To summarize the rules for including and excluding vaults, following are some examples:

Example 1

- The system contains 1000 vaults.
- Five of the 1000 vaults are management vaults (named mgmt-1 to mgmt-5).

- The scan includes all vaults except the management vaults.

```
"include_all_vaults": true,
"exclude-vaults": ["mgmt-1", "mgmt-2", "mgmt-3", "mgmt-4", "mgmt-5"]
```

Example 2

- The system contains 1000 vaults.
- 5 of the 1000 vaults are management vaults (named mgmt-1 to mgmt-5).
- The scan includes all vaults except the management vaults.
- The scan includes a filter for scanning a vault that is named vault-x.
- The scan includes only a scan of the objects whose key starts with **production/finance**.

```
"include_all_vaults": true,
"exclude-vaults": ["mgmt-1", "mgmt-2", "mgmt-3", "mgmt-4", "mgmt-5"],
"vaults": [
     {"vault_name":"vault-x", "prefix":"production/finance"}
  ]
```

Example 3

- The system contains 1000 vaults.
- 5 of the 1000 vaults are management vaults (named mgmt-1 to mgmt-5).
- The scan includes all vaults except the management vaults.
- The scan includes a filter for scanning a vault that is named vault-x.
- The scan includes only a scan of the objects whose key starts with **production/finance** or **production/marketing**.

```
"include_all_vaults": true,
"exclude-vaults": ["mgmt-1", "mgmt-2", "mgmt-3", "mgmt-4", "mgmt-5"],
"vaults": [
     {"vault_name":"vault-x", "prefix":"production/finance"},
     {"vault_name":"vault-x", "prefix":"production/marketing"}
   ]
```

Example 4

- The system contains 1000 vaults.
- Run a test on three vaults named vault-a, vault-b, and versioned-vault-c.
- Run a scan on versioned-vault-c and issue LIST requests. Do not issue HEAD requests because the objects do not have custom amz headers.

```
"include_all_vaults": false,
"vaults": [
     {"vault_name":"vault-a"
     {"vault_name":"vault-b"
     {"vault_name":"vault-c",  "has_custom_metadata":false}
   ]
```

Stats files

The IBM Cloud® Object Storage Scanner tracks each LIST process status to a stats file.

During a scan, the Scanner runs multiple processes. Each `LIST` processes and tracks the progress, saves the **`next_key`**, and optionally the **`next_version`** to a stats file named task.stats that is stored with the log files in the /opt/ibm/metaocean/data/connections/cos/replay/output/data directory.

```
{
     "estimated_object_count": 1000,
     "list_objects_size": 100,
      "next_key": "",
      "next_version": "",
      "prefix": "",
      "scan_type": "Object Scan",
      "status": "Complete",
      "total_bytes_output": 1126809,
      "total_bytes_scanned": 1126809,
      "total_objects_output": 47,
      "total_objects_scanned": 47,
      "vault_name": "dsmgmt-sp1",
      "vault_uuid": "868daa21-9e56-4c41-b6fd-845a4c85cea9"
}
```

From the Scanner, you can start, stop, recover files from a crash, and restart at the point where the scan was interrupted.

When you start the scanner:

1. Processing of the Scanner continues from **`next_key`** and **`next_version`**.
2. Queue of the Notifier is optimized by reloading from the files in the data folder instead of requerying the dsNet.
3. Batches that were processed partially are reprocessed. Duplicate Kafka notifications might occur, but are handled safely by the IBM Data Cataloging system.

Replay

When a severe outage occurs and causes the loss of notifications sent by the system to IBM Data Cataloging, the IBM Cloud® Object Storage Scanner Replay feature can be used to recover lost notifications.

Replay parses the access logs of a system and reconstitutes the notifications. Also, the Notifier can resend the notifications.

Initialization for Replay

During the startup, Replay reads the configuration file and issues requests to the Manager of the dsNet device similar to the Scanner.

Data from the configuration file is validated to ensure that appropriate permissions are granted in the dsNet. This allows access to management vaults and regular vaults. Startup errors or warnings are logged and printed to the console.

After initialization, Replay extracts accesser log files from the management vaults of dsNet and enables Replay to process and write notifications to the output directory.

Error conditions

Sometimes Replay does not have enough information to replay the original notification. If this occurs, you must fix the problems manually.

For example, if vault versioning was suspended when you made the request and you receive an s3 DeleteObject for an object or delete marker, the following error is logged:

```
error_code=True, error_description="Delete operation with [no version_id|null
version_id|version_id] for vault with versioning = [suspended/enabled]"
```

The error message displays because Replay cannot distinguish when a notification with s3:CreateDeleteMarker or s3:CreateDeleteMarker:NullVersionDeleted is sent.

If vault versioning is disabled, and an s3 PutObject request is received for an object that is deleted, the following error is logged:

```
error_code=404, error_message="Not Found"
```

The error message displays because Replay cannot determine the tag of the object that was deleted.

Output

Messages are batched by 1,000 or to the Scanner list objects size configuration setting, if specified.

The messages are written to the output folder with the same Notification format used by the Scanner.

```
{
     "system_name": "Test",
     "object_etag": "\"de37d2cee49596916f62a233dfc790a4\"",
     "request_time": "2018-09-24T18:49:29.383Z",
     "format": 1,
     "bucket_uuid": "ac89915b-d4ec-7ff1-00be-9c32b2aca580",
     "system_uuid": "f7d033c2-9066-499a-a883-829860d4d865",
     "object_length": "12319",
     "object_name": "test_version",
     "bucket_name": "vault3",
     "content_type": "binary/octet-stream",
     "request_id": "17451c3d-e81e-40ed-939a-4534780daaa8",
     "operation": "s3:PutObject"
}
```

If an error occurs, the error messages are written to the /opt/ibm/metaocean/data/connections/cos/replay/output/data/access_log_error/ directory. Take note of the extra `error_code` and `error_description` elements.

```
{
     "system_name": "Test",
     "object_version": "null",
     "request_time": "2018-09-24T17:07:59.471Z",
     "format": 1,
     "bucket_uuid": "ac89915b-d4ec-7ff1-00be-9c32b2aca580",
     "system_uuid": "f7d033c2-9066-499a-a883-829860d4d865",
     "object_length": "12319",
     "object_name": "object5.2",
     "bucket_name": "vault3",
     "request_id": "ebc472a1-f955-4605-895b-840867b12e01",
     "operation": "s3:PutObject",
     "error_description": "Not Found",
     "error_code": 404
}
```

Renaming a vault for Replay

When you rename a vault, it is possible that Replay can abort.

Replay aborts when you:

- Delete the vault.
- Rename the vault.
- Discover that the read permission is revoked for the credentials that are supplied by the operator or manager API.

All other scans of a vault continue scanning until complete.

You can find the details of the errors that include stack trace in the replay.debug file in the /opt/ibm/metaocean/data/connections/cos/replay/debug/replay/[timestamp] directory.

Starting the Replay

The guidelines and rules for using Replay are documented in this topic.

To start Replay, run the following command:

```
cos-replay
```

The following rules apply for Replay:

- Configure Replay according to the guidelines in Table 1.
- Replay component requires `min_utc` and `max_utc` time stamps defined in the Configuration file.
- Only notifications sent between `min_utc` and `max_utc` are parsed and replayed.
- Replay automatically shuts down when all accesser logs are downloaded and processed. The message Complete Replay Process appears in the console.

This is an example of how to start Replay:

```
Starting COS Replay - Version 0.1 Log file and config file are in directory /Users/weebrew/
Documents/Development/ibmworkspace/cosscanner/output/debug/scanner/20180925-125528-232131
Starting Accessor Log Extraction Downloading files...
('Downloaded', 10, 'of', 36)
('Downloaded', 20, 'of', 36)
('Downloaded', 30, 'of', 36)
Download complete.
Total files: 36
Complete Accessor Log Extraction
Starting Replay process
Complete Replay process
```

Debug mode for Replay

Run Replay in debug mode to troubleshoot problems.

To start debug mode, run the following command:

```
cos-replay --log=DEBUG
```

Running debug mode creates large log files and creates a significant drop in performance. Do not run debug mode for long periods especially when you are in production mode.

Notifier

The Notifier is the component that reads the JSON notifications that are written by Scanner or Replay and sends notifications to the Kafka cluster.

When notifications are acknowledged by Kafka, the Notifier moves the file to the archive folder.

On start-up, Notifier calls the Manager API and retrieves details of any Notification Service Configurations (NSC) configured in the dsNet for IBM Data Cataloging. If more than one is found, the first one is used.

Retrieval of NSCs is overridden by defining the details of the Kafka configuration in the config file.

```
 "notifier":{
   "kafka_format": 1,
   "kafka_endpoint": "192.168.1.34:9092",
   "kafka_topic": "cos-le-connector-topic"
}
```

Limitations

Limitations apply when the Notifier uses a Kafka configuration retrieved from the Manager API.

- If more than one NSC exists, the first one is used for all vaults.
- If more than one host name is defined in the NSC, the first one is used for all vaults.

Starting the Notifier

Running the Notifier has rules and limitations.

To start the Notifier, run the following command:

```
cos-notify
```

After you start the Notifier, you are prompted for security credentials for the manager API and Kafka cluster.

```
Starting COS Notifier - Version 0.1
Enter the Manager API username: admin
Enter the Manager API password:
Enter the Kafka username: cos
Enter the Kafka password:
Enter the Kafka pem:
Creating Kafka producer...
Done
Notifier is running
Log file and config file are in directory C:\dev\cos-scanner\output\debug\notifier\20180912-121641-283000
Checking for files in   \data
- 11 files found Checking for files in output\data
- 256 files found
```

Rules and limitations

The following rules and limitations apply to the Notifier:

- You cannot start the Notifier in the background because the Notifier requires user input at the terminal window.
- You can stop the Notifier and force the Notifier to run in the background.
- The passwords and pem do not display when you type and paste the passwords in the console.
- The certificate pem is approximately 1600 characters. If you use an SSH connection, the certificate pem might be truncated to 1000 characters.
- If the number is truncated to 1000 characters, include the certificate pem in the config file.

Notifier operation

The Notifier enumerates and processes all .log files in the Scanner data directory.

After all files are processed, the Notifier repeats the process so that new .log files that are generated by the Scanner are processed. The Notifier sleeps repeatedly in 1-second intervals if no new files are found in the /opt/ibm/metaocean/data/connections/cos/replay/output/data directory.

The Notifier does not automatically shut down. The Notifier continues to monitor the Scanner data directory for new .log files. Monitor the progress of the Notifier by using the status report. When the operator or administrator determines that all scanned objects are submitted successfully to the IBM Data Cataloging, shut down the Notifier by using the kill switch.

Stopping the Notifier

You might need to stop and restart the Notifier.

Before you stop and restart the Notifier:

1. Create a file named kill.notifier in the /opt/ibm/metaocean/data/connections/cos/replay/output/command directory.
2. Ensure that the processing of any batches is complete before you stop the Notifier.

Stopping the Notifier displays the following output:

The shutdown is complete when the "**Shutdown is complete**" message displays.

```
Starting COS Notifier - Version 0.1
Enter the Manager API username: admin
Enter the Manager API password:
Enter the Kafka username: cos Enter the Kafka password:
Enter the Kafka pem: Creating Kafka producer...
Done
Notifier is running
Log file and config file are in directory C:\dev\cos-scanner\output\
   debug\notifier\20180912-121641-283000
Checking for files in output\data
- 11 files found Checking for files in output\data
- 256 files found Detected the kill trigger file. Shutting down...
Shutdown is complete
```

Restarting the Notifier

When you stop the Notifier following a shutdown with the kill.notifier file, you must rename the file manually or delete the file before you do a restart.

If you do not rename or delete the kill.notifier file, the system finds the file and displays the following message:

```
C:\dev\cos-scanner>python main_notifier.py Starting COS Notifier - Version 0.1
The file 'kill.notifier' is preventing the notifier from running.
```

```
You should delete or rename the file and re-start the notifier.
File location: 'C:\dev\cos-scanner\command\kill.notifier
```

The Scanner and Notifier are separate solutions that share the config file. The files operate independently, so you can start and stop either file independently any time.

Progress report

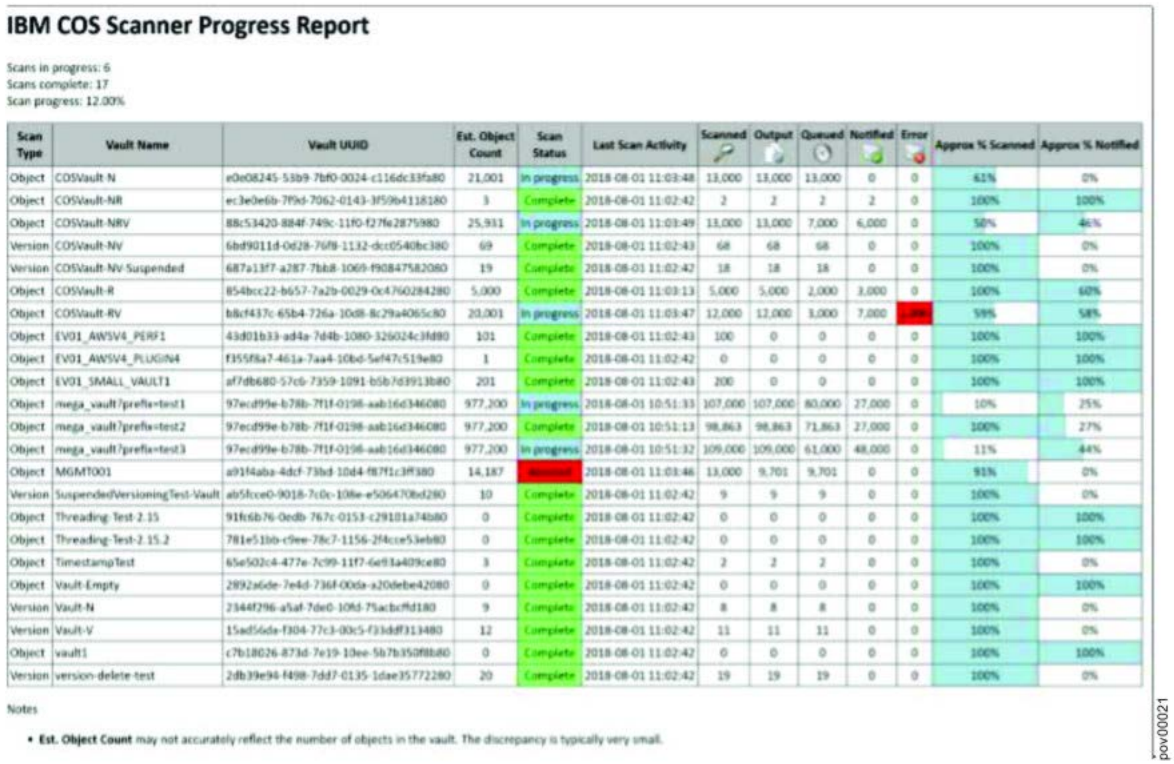
The Progress Report provides an instant snapshot of status for the Scanner and Notifier.

To create a progress report, run the following command:

```
cos-report
```

The progress report displays in plain text format to the console in a static HTML file that is named: /opt/ibm/metaocean/data/connections/cos/replay/output/cos-scanner-report.html
If a progress report exists, the new progress report overwrites the existing progress report. See Figure 1.

Figure 1. IBM Cloud Object Storage Scanner progress report



IBM COS Scanner Progress Report

Scans in progress: 6
Scans complete: 17
Scan progress: 12.00%

Scan Type	Vault Name	Vault UUID	Est. Object Count	Scan Status	Last Scan Activity	Scanned	Output	Queued	Notified	Error	Approx % Scanned	Approx % Notified
Object	COSVault-N	e0e08245-53b9-7bf0-0024-c116dc33fa80	21,001	In progress	2018-08-01 11:03:48	13,000	13,000	13,000	0	0	61%	0%
Object	COSVault-NR	ec3e0e6b-7f9d-7062-0143-3f59b4118180	3	Complete	2018-08-01 11:02:42	2	2	2	2	0	100%	100%
Object	COSVault-NRV	88c53420-884f-749c-11f0-f27fe2875980	25,931	In progress	2018-08-01 11:03:49	13,000	13,000	7,000	6,000	0	50%	46%
Version	COSVault-NV	6bd9011d-0d28-76f8-1132-dcc0540bc380	69	Complete	2018-08-01 11:02:43	68	68	68	0	0	100%	0%
Version	COSVault-NV-Suspended	687a13f7-a287-7bb8-1069-f90847582080	19	Complete	2018-08-01 11:02:42	18	18	18	0	0	100%	0%
Object	COSVault-R	854bcc22-b657-7a2b-0029-0c4760284280	5,000	Complete	2018-08-01 11:03:13	5,000	5,000	2,000	3,000	0	100%	60%
Object	COSVault-RV	b8cf437c-65b4-726a-10d8-8c29a4065c80	20,001	In progress	2018-08-01 11:03:47	12,000	12,000	3,000	7,000	[illegible]	59%	58%
Object	EV01_AWSV4_PERF1	43d01b33-ad4a-7d4b-1080-326024c3fd80	101	Complete	2018-08-01 11:02:43	100	0	0	0	0	100%	100%
Object	EV01_AWSV4_PLUGIN4	f355f8a7-461a-7aa4-10bd-5ef47c519e80	1	Complete	2018-08-01 11:02:42	0	0	0	0	0	100%	100%
Object	EV01_SMALL_VAULT1	af7db680-57c6-7359-1091-b5b7d3913b80	201	Complete	2018-08-01 11:02:43	200	0	0	0	0	100%	100%
Object	mega_vault?prefix=test1	97ecd99e-b78b-7f1f-0198-aab16d346080	977,200	In progress	2018-08-01 10:51:33	107,000	107,000	80,000	27,000	0	10%	25%
Object	mega_vault?prefix=test2	97ecd99e-b78b-7f1f-0198-aab16d346080	977,200	Complete	2018-08-01 10:51:13	98,863	98,863	71,863	27,000	0	100%	27%
Object	mega_vault?prefix=test3	97ecd99e-b78b-7f1f-0198-aab16d346080	977,200	In progress	2018-08-01 10:51:32	109,000	109,000	61,000	48,000	0	11%	44%
Object	MGMT001	a91f4aba-4dcf-73bd-10d4-f87f1c3ff380	14,187	[illegible]	2018-08-01 11:03:46	13,000	9,701	9,701	0	0	91%	0%
Version	SuspendedVersioningTest-Vault	ab5fcce0-9018-7c0c-108e-e506470bd280	10	Complete	2018-08-01 11:02:42	9	9	9	0	0	100%	0%
Object	Threading-Test-2.15	91fc6b76-0edb-767c-0153-c29101a74b80	0	Complete	2018-08-01 11:02:42	0	0	0	0	0	100%	100%
Object	Threading-Test-2.15.2	781e51bb-c9ee-78c7-1156-2f4cce53eb80	0	Complete	2018-08-01 11:02:42	0	0	0	0	0	100%	100%
Object	TimestampTest	65e502c4-477e-7c99-11f7-6e93a409ce80	3	Complete	2018-08-01 11:02:42	2	2	2	0	0	100%	0%
Object	Vault-Empty	2892a6de-7e4d-736f-00da-a20debe42080	0	Complete	2018-08-01 11:02:42	0	0	0	0	0	100%	100%
Version	Vault-N	2344f296-a5af-7de0-10fd-75acbcffd180	9	Complete	2018-08-01 11:02:42	8	8	8	0	0	100%	0%
Version	Vault-V	15ad56da-f304-77c3-00c5-f33ddf313480	12	Complete	2018-08-01 11:02:42	11	11	11	0	0	100%	0%
Object	vault1	c7b18026-873d-7e19-10ee-5b7b350f8b80	0	Complete	2018-08-01 11:02:42	0	0	0	0	0	100%	100%
Version	version-delete-test	2db39e94-f498-7dd7-0135-1dae35772280	20	Complete	2018-08-01 11:02:42	19	19	19	0	0	100%	0%

Notes

- **Est. Object Count** may not accurately reflect the number of objects in the vault. The discrepancy is typically very small.

pov00021

See Table 1 for a description of information in the progress report.

Table 1. Description for IBM Cloud Object Storage Scanner progress report

Column name	Description
Scan Type	Either Object or Version. Non-Versioned vaults show Object. Versioned vaults show Version. However, there are some exceptions. If the Name Index for a versioned vault is unavailable but Recovery Listing is enabled, an object scan might be run. The user is alerted that an object scan can be done, but this object scan requires changes to the configuration file.
Vault Name	The name of the vault. Any prefix that is defined in the configuration file is also shown. Example: `mega_vault?prefix=test`
Vault UUID	The UUID of the vault.
Estimated Object Count	The estimated number of objects in the vault, as reported by the Manager API. This value is refreshed from the Manager API each time the scanner is started, regardless of the status of each scan. Given that the number of objects in each vault might be constantly changing, the number of objects that are reported in this column becomes out of date during long running scans. Note: This issue affects only the status report but does not affect the data integrity of the Scanner.

Column name	Description
Scan Status	Shows the status of the scanner. Not started The task is queued but not started. In progress The task is running. Complete The task finished. Aborted The task encountered an unrecoverable error and aborted. Shut down the Scanner and the debug file, and inspect the file to investigate the problems. After you resolve the problems, restart the Scanner. The debug file is in the /opt/ibm/metaocean/data/connections/cos/replay/output/data/<vault-name>/<prefix> directory. For each vault, see the /opt/ibm/metaocean/data/connections/cos/replay/output/data/<vault-name>/<prefix> directories.
Last scan activity	The last time data was retrieved from the vault.
Scanned	Number of objects or versions that are scanned. For a versioned vault, this value shows a figure that is higher than the Estimated Object Count.
Output	Number of objects or versions that scanned AND whose `LastModified` time stamp is inside the time window that is defined in the configuration file. The figure in the column is Queued + Notified + Error.
Queued	Number of objects or versions that are Output and are waiting to be sent to the Kafka cluster.
Notified	Number of objects or versions that are submitted successfully to the Kafka cluster.
Error	Number of objects or versions that failed to send to the Kafka cluster. Details of all errors are logged to notifier.debug.
Approximate percentage scanned	Scanned as a percentage of Est. Object Count. The cell background shows a progress bar.
Approximate percentage scanned	Notified as a percentage of Output. The cell background shows a progress bar.

Table 2. What is reported beneath the report title

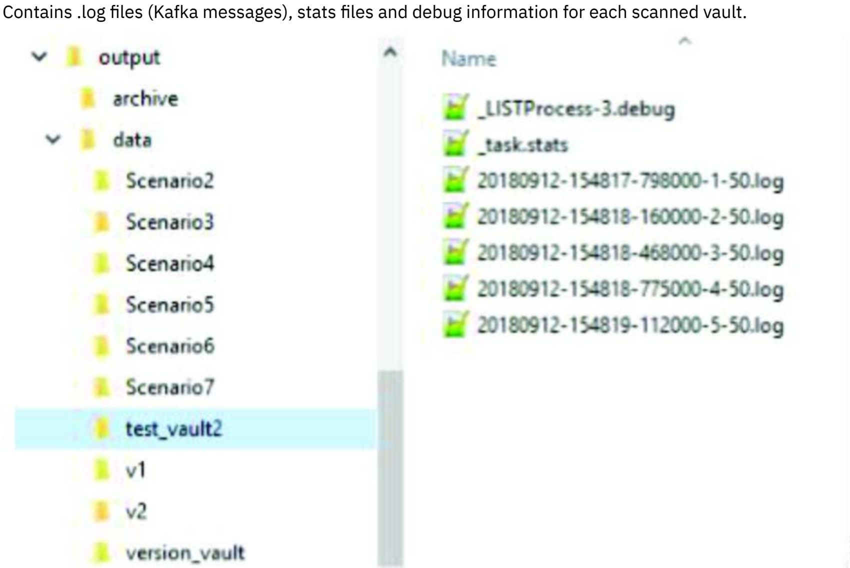
Measure	Description
Scans in progress	Number of scans with the status `"In Progress"`. Applies to Scanner only.
Scans complete	Number of scans with the status `"Complete"`. Applies to Scanner only.
Scan progress	Sum (number of objects scanned) as a percentage of sum (estimated object count).

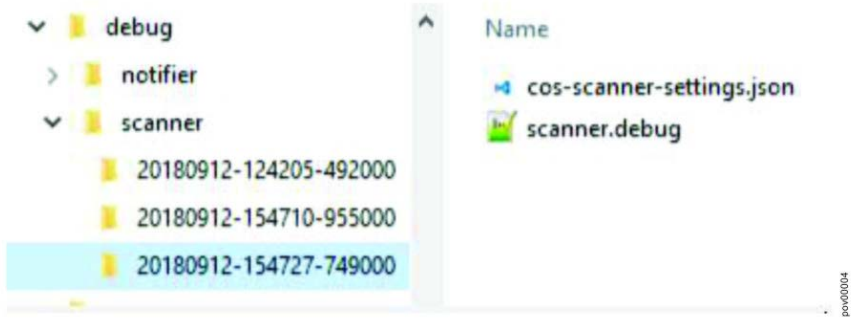
Logging

You can view the list of directories that are generated by scanner, notifier, and replay.

Table 1 lists the directories that are generated on start-up by the Scanner, Notifier, and Replay.

Table 1. List of directories generated by scanner, notifier, and replay

Directory	Description
For IBM Data Cataloging: `/opt/ibm/metaocean/data/connections/cos/replay/output/data`	Contains .log files (Kafka messages), stats files and debug information for each scanned vault. 

<table>
<tr><th>Directory</th><th>Description</th></tr>
<tr><td>For IBM Data Cataloging:
<code>/opt/ibm/metaocean/data/connections/cos/replay/output/debug/[scanner|replay]/</code></td><td>Contains Scanner and Replay debug or troubleshooting information. A new subdirectory is created each time the Scanner and Replay starts. Each subdirectory contains a copy of the configuration file and scanner.debug (replay.debug).

</td></tr>
<tr><td>For IBM Data Cataloging:
<code>/opt/ibm/metaocean/data/connections/cos/replay/output/debug/notifier</code></td><td>Contains Notifier debug or troubleshooting information. A new subdirectory is created each time the Notifier starts. It contains a copy of the configuration file and notifier.debug.
Same directory-naming convention as shown for the scanner.

The notifier.debug rolls over when it reaches a predefined size as defined in the configuration file. See Configuration file.</td></tr>
<tr><td>For IBM Data Cataloging:
<code>/opt/ibm/metaocean/data/connections/cos/replay/output/archive/</code></td><td>Contains all .log files that are successfully processed by the Notifier, and their contents are successfully submitted to the Kafka cluster.
Remember: Files in this directory are truncated – they contain only the object key and they can also contain the version.</td></tr>
<tr><td>For IBM Data Cataloging:
<code>/opt/ibm/metaocean/data/connections/cos/replay/output/error/</code></td><td>Contains any .log files that failed to submit to the Kafka cluster.</td></tr>
<tr><td>For IBM Data Cataloging:
<code>/opt/ibm/metaocean/data/connections/cos/replay/output/notification_log/</code></td><td>The notification.log file contains details of any errors (including stack trace) that occur when it attempts to send notifications to the Kafka cluster.
If <code>logging/notification_log_all</code> is true in the config file, all successful sends are also logged.

The notification.log file rolls over when it reaches a predefined size as defined in the configuration file. See Configuration file.</td></tr>
</table>

IBM Cloud Object Storage Scanner output data

The Scanner generates a directory beneath the output data directory for each vault or vault prefix as defined in the configuration file.

The /opt/ibm/metaocean/data/connections/cos/replay/output/data directory is the Scanner output data directory.

The following screen shows an example of a configuration file and also shows that all vaults are scanned, but mega_vault has four separate prefixes that are defined winch means the four scans of the vault occurred.

```
"include_all_vaults": true,
  "vaults": [
  {"vault_name": "mega_vault", "prefix": "main/production/finance"},
  {"vault_name": "mega_vault", "prefix": "main/production/sales"},
  {"vault_name": "mega_vault", "prefix": "main/production/marketing"},
  {"vault_name": "mega_vault", "prefix": "main/production/hr"}
]
```

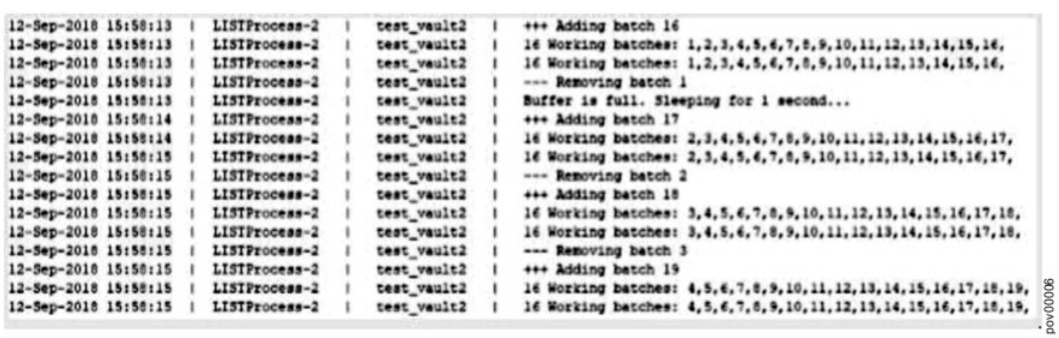
Figure 1 shows the directory structure.

Figure 1. Directory structure from the configuration file

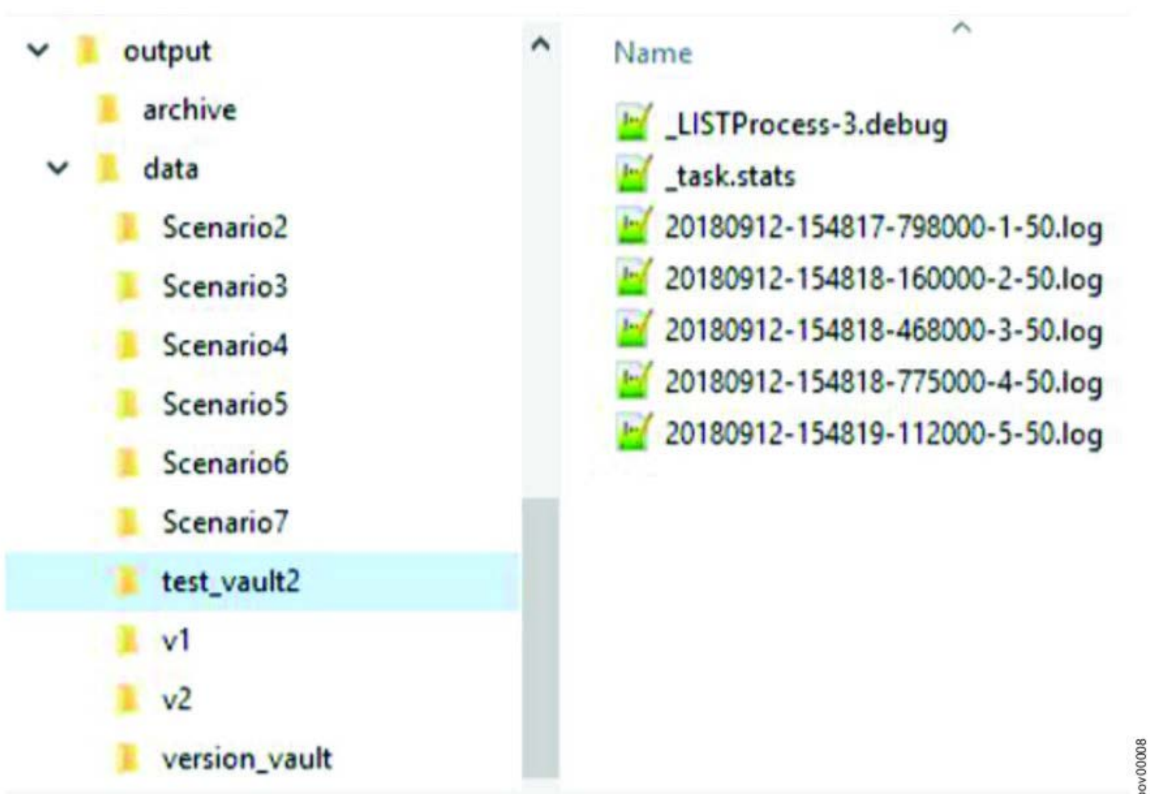
The status and progress of each scan must be maintained so a separate directory structure is created for each scan. Table 1 shows the leaf directories that contain the file names and description.

Table 1. Leaf directory file names

<table>
<tr><th>File name</th><th>Description</th></tr>
<tr><td>_LISTProcessN.debug</td><td>The N in the file name is different for each process (0 - 9 if there are 10 processes).
Contains detailed debug information and details of any errors that are encountered when you scan the vault. Figure 2 shows an example of running in debug mode.

Figure 2. Example of running in debug mode

12-Sep-2018 15:58:13 | LISTProcess-2 | test_vault2 | +++ Adding batch 16
12-Sep-2018 15:58:13 | LISTProcess-2 | test_vault2 | 16 Working batches: 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,
12-Sep-2018 15:58:13 | LISTProcess-2 | test_vault2 | 16 Working batches: 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,
12-Sep-2018 15:58:13 | LISTProcess-2 | test_vault2 | --- Removing batch 1
12-Sep-2018 15:58:13 | LISTProcess-2 | test_vault2 | Buffer is full. Sleeping for 1 second...
12-Sep-2018 15:58:14 | LISTProcess-2 | test_vault2 | +++ Adding batch 17
12-Sep-2018 15:58:14 | LISTProcess-2 | test_vault2 | 16 Working batches: 2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | 16 Working batches: 2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | --- Removing batch 2
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | +++ Adding batch 18
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | 16 Working batches: 3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | 16 Working batches: 3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | --- Removing batch 3
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | +++ Adding batch 19
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | 16 Working batches: 4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,
12-Sep-2018 15:58:15 | LISTProcess-2 | test_vault2 | 16 Working batches: 4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,
pov00006</td></tr>
</table>

<table>
<tr><th>File name</th><th>Description</th></tr>
<tr><td>task.stats</td><td>Scanner starts in JSON format for a single vault. Updated following successful processing of each batch of objects.

</td></tr>
<tr><td>*.log</td><td>The Scanner creates multiple .log files for each vault. Each .log file contains up to 1000 Kafka messages, ready to be submitted to the Kafka cluster by the Notifier.
The naming convention for the log files is
<code><date>-<time>-<milliseconds>-<batch number>-<number of messages in file>.log</code>

</td></tr>
</table>

Appendix

The appendix shows an example of a log file and examples of Scanner debug data.

Log file

Figure 1 shows an example with extra line breaks.

Figure 1. Example of a log file

{"system_name": "Test", "object_version": "38d811e6-dba1-4830-859d-6275f2016bc3", "object_etag":
"\"73b8c5dcca9cc6aab928396a2d98a340\"", "request_time": "2018-08-24T18:39:16Z", "format": 1, "bucket_uuid": "cbc03649-
d218-727d-10ec-df8c22873280", "system_uuid": "f7d033c2-9066-499a-a883-829860d4d865", "meta_headers": [], "object_length":
12, "object_name": "my-object-black", "bucket_name": "version_vault", "content_type": "text/plain", "request_id":
"7ed39768-1184-4d5f-8c0c-7912515bf8fe", "operation": "s3:PutObject"}

{"system_name": "Test", "object_version": "8a68208b-9b5f-4107-9eae-fa08200c7913", "object_etag":
"\"73b8c5dcca9cc6aab928396a2d98a340\"", "request_time": "2018-08-24T18:39:15Z", "format": 1, "bucket_uuid": "cbc03649-
d218-727d-10ec-df8c22873280", "system_uuid": "f7d033c2-9066-499a-a883-829860d4d865", "meta_headers": [], "object_length":
12, "object_name": "my-object-black", "bucket_name": "version_vault", "content_type": "text/plain", "request_id":
"6ed2a774-f384-4cba-96fd-81dfbb681482", "operation": "s3:PutObject"}

{"system_name": "Test", "object_version": "322d9ed2-ca86-4efd-b95a-a2467ded9202", "object_etag":
"\"73b8c5dcca9cc6aab928396a2d98a340\"", "request_time": "2018-08-24T18:39:15Z", "format": 1, "bucket_uuid": "cbc03649-
d218-727d-10ec-df8c22873280", "system_uuid": "f7d033c2-9066-499a-a883-829860d4d865", "meta_headers": [], "object_length":
12, "object_name": "my-object-black", "bucket_name": "version_vault", "content_type": "text/plain", "request_id":
"ced1de9e-e62f-4966-b803-7aff3ddab245", "operation": "s3:PutObject"}

{"system_name": "Test", "object_version": "0e2b0369-a506-4ce3-8336-ea42eae16489", "object_etag":
"\"73b8c5dcca9cc6aab928396a2d98a340\"", "request_time": "2018-08-24T18:39:15Z", "format": 1, "bucket_uuid": "cbc03649-
d218-727d-10ec-df8c22873280", "system_uuid": "f7d033c2-9066-499a-a883-829860d4d865", "meta_headers": [], "object_length":
12, "object_name": "my-object-black", "bucket_name": "version_vault", "content_type": "text/plain", "request_id":
"ee5b8ded-86af-4e9e-9054-8902f7a7a5c0", "operation": "s3:PutObject"}

{"system_name": "Test", "object_version": "41518cb8-632f-45c4-989b-44b4d2b19b2e", "object_etag":
"\"73b8c5dcca9cc6aab928396a2d98a340\"", "request_time": "2018-08-24T18:39:15Z", "format": 1, "bucket_uuid": "cbc03649-
d218-727d-10ec-df8c22873280", "system_uuid": "f7d033c2-9066-499a-a883-829860d4d865", "meta_headers": [], "object_length":
12, "object_name": "my-object-black", "bucket_name": "version_vault", "content_type": "text/plain", "request_id":
"5adaa4d9-7aa6-4c46-bcd1-1260c074a972", "operation": "s3:PutObject"}

{"system_name": "Test", "object_version": "c75c6faf-e7a9-41ce-a576-6bfc9d374dfc", "object_etag":
"\"73b8c5dcca9cc6aab928396a2d98a340\"", "request_time": "2018-08-24T18:39:14Z", "format": 1, "bucket_uuid": "cbc03649-
d218-727d-10ec-df8c22873280", "system_uuid": "f7d033c2-9066-499a-a883-829860d4d865", "meta_headers": [], "object_length":
12, "object_name": "my-object-black", "bucket_name": "version_vault", "content_type": "text/plain", "request_id":
"35aa2dee-8700-4cfb-a62e-dc7f7ec7b8e1", "operation": "s3:PutObject"}

pov00009

Scanner debug data

Figure 2, Figure 3, Figure 4, and Figure 5 show that throttling settings are logged, and multiple HEAD processes are started for each LIST process.

Figure 2. Scanner debug

```
12-Sep-2018 15:57:25 |
12-Sep-2018 15:57:25 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:25 | Python package setup
12-Sep-2018 15:57:25 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:25 | """Setup"""
12-Sep-2018 15:57:25 | from setuptools import setup, find_packages
12-Sep-2018 15:57:25 | setup(
12-Sep-2018 15:57:25 |     name='ibm_cos_scanner',
12-Sep-2018 15:57:25 |     version='2.0.0',
12-Sep-2018 15:57:25 |     packages=find_packages(),
12-Sep-2018 15:57:25 |     include_package_data=True,
12-Sep-2018 15:57:25 |     zip_safe=True,
12-Sep-2018 15:57:25 |     url='www.ibm.com',
12-Sep-2018 15:57:25 |     license='See LICENSE folder',
12-Sep-2018 15:57:25 |     author=IBM',
12-Sep-2018 15:57:25 |     description='IBM COS Scanner / Spectrum Discover Notifier'
12-Sep-2018 15:57:25 | )
12-Sep-2018 15:57:25 |
12-Sep-2018 15:57:25 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:25 | Initialising IBM COS Scanner. Reading config
12-Sep-2018 15:57:25 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:25 | Retrieving System Advanced Configuration
12-Sep-2018 15:57:25 | Calling https://172.19.17.38/manager/api/json/1.0/viewSystemConfiguration.adm
12-Sep-2018 15:57:26 |    |- OK
12-Sep-2018 15:57:26 |    |- dsNet Name: est
12-Sep-2018 15:57:26 |    |- dsNet UUID: f7d033c2-9066-499a-a883-829860d4d865
12-Sep-2018 15:57:26 |
12-Sep-2018 15:57:26 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:26 | Retrieving user's access keys
12-Sep-2018 15:57:26 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:26 | Calling https://172.19.17.38/manager/api/json/1.0/listMyAccessKeys.adm
12-Sep-2018 15:57:26 |    |- OK
12-Sep-2018 15:57:26 |    |- Accesser credentials successfully retrieved from Manager API
12-Sep-2018 15:57:26 |
12-Sep-2018 15:57:26 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:26 | Retrieving vault information from dsNet
12-Sep-2018 15:57:26 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:26 | Calling https://172.19.17.38/manager/api/json/1.0/viewSystemConfiguration.adm
12-Sep-2018 15:57:27 |    |- OK
12-Sep-2018 15:57:27 |
12-Sep-2018 15:57:27 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:27 | Retrieving vault size information from dsNet
12-Sep-2018 15:57:27 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:27 | Calling https://172.19.17.38/manager/api/json/1.0/listVaults.adm
12-Sep-2018 15:57:27 |    |- OK
12-Sep-2018 15:57:27 |
12-Sep-2018 15:57:27 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:27 | Retrieving dsNet device information
12-Sep-2018 15:57:27 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:27 | Calling https://172.19.17.38/manager/api/json/1.0/listDevices.adm
12-Sep-2018 15:57:28 |    |- OK
12-Sep-2018 15:57:28 | 3 devices found
12-Sep-2018 15:57:28 |    |- device_id:1    manager     172.19.17.38   0e547a7f-849a-79e8-102c-2026af755443vm-eqx20420l-
mgr-03
12-Sep-2018 15:57:28 |    |- device_id:2    accesser    172.19.17.39   39a14ec8-090c-79d8-10f9-255249197f43vm-eqx20420l-
```

Figure 3. Scanner debug (continued)

```
12-Sep-2018 15:57:25  |
12-Sep-2018 15:57:25  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:25  |  Python package setup
12-Sep-2018 15:57:25  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:25  |  """Setup"""
12-Sep-2018 15:57:25  |  from setuptools import setup, find_packages
12-Sep-2018 15:57:25  |  setup(
12-Sep-2018 15:57:25  |      name='ibm_cos_scanner',
12-Sep-2018 15:57:25  |      version='2.0.0',
12-Sep-2018 15:57:25  |      packages=find_packages(),
12-Sep-2018 15:57:25  |      include_package_data=True,
12-Sep-2018 15:57:25  |      zip_safe=True,
12-Sep-2018 15:57:25  |      url='www.ibm.com',
12-Sep-2018 15:57:25  |      license='See LICENSE folder',
12-Sep-2018 15:57:25  |      author=IBM',
12-Sep-2018 15:57:25  |      description='IBM COS Scanner / Spectrum Discover Notifier'
12-Sep-2018 15:57:25  |  )
12-Sep-2018 15:57:25  |
12-Sep-2018 15:57:25  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:25  |  Initialising IBM COS Scanner. Reading config
12-Sep-2018 15:57:25  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:25  |  Retrieving System Advanced Configuration
12-Sep-2018 15:57:25  |  Calling https://172.19.17.38/manager/api/json/1.0/viewSystemConfiguration.adm
12-Sep-2018 15:57:26  |     |- OK
12-Sep-2018 15:57:26  |     |- dsNet Name: est
12-Sep-2018 15:57:26  |     |- dsNet UUID: f7d033c2-9066-499a-a883-829860d4d865
12-Sep-2018 15:57:26  |
12-Sep-2018 15:57:26  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:26  |  Retrieving user's access keys
12-Sep-2018 15:57:26  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:26  |  Calling https://172.19.17.38/manager/api/json/1.0/listMyAccessKeys.adm
12-Sep-2018 15:57:26  |     |- OK
12-Sep-2018 15:57:26  |     |- Accesser credentials successfully retrieved from Manager API
12-Sep-2018 15:57:26  |
12-Sep-2018 15:57:26  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:26  |  Retrieving vault information from dsNet
12-Sep-2018 15:57:26  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:26  |  Calling https://172.19.17.38/manager/api/json/1.0/viewSystemConfiguration.adm
12-Sep-2018 15:57:27  |     |- OK
12-Sep-2018 15:57:27  |
12-Sep-2018 15:57:27  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:27  |  Retrieving vault size information from dsNet
12-Sep-2018 15:57:27  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:27  |  Calling https://172.19.17.38/manager/api/json/1.0/listVaults.adm
12-Sep-2018 15:57:27  |     |- OK
12-Sep-2018 15:57:27  |
12-Sep-2018 15:57:27  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:27  |  Retrieving dsNet device information
12-Sep-2018 15:57:27  |  ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:27  |  Calling https://172.19.17.38/manager/api/json/1.0/listDevices.adm
12-Sep-2018 15:57:28  |     |- OK
12-Sep-2018 15:57:28  |  3 devices found
12-Sep-2018 15:57:28  |     |- device_id:1    manager     172.19.17.38   0e547a7f-849a-79e8-102c-2026af755443vm-eqx20420l-
mgr-03
12-Sep-2018 15:57:28  |     |- device_id:2    accesser    172.19.17.39   39a14ec8-090c-79d8-10f9-255249197f43vm-eqx20420l-
```

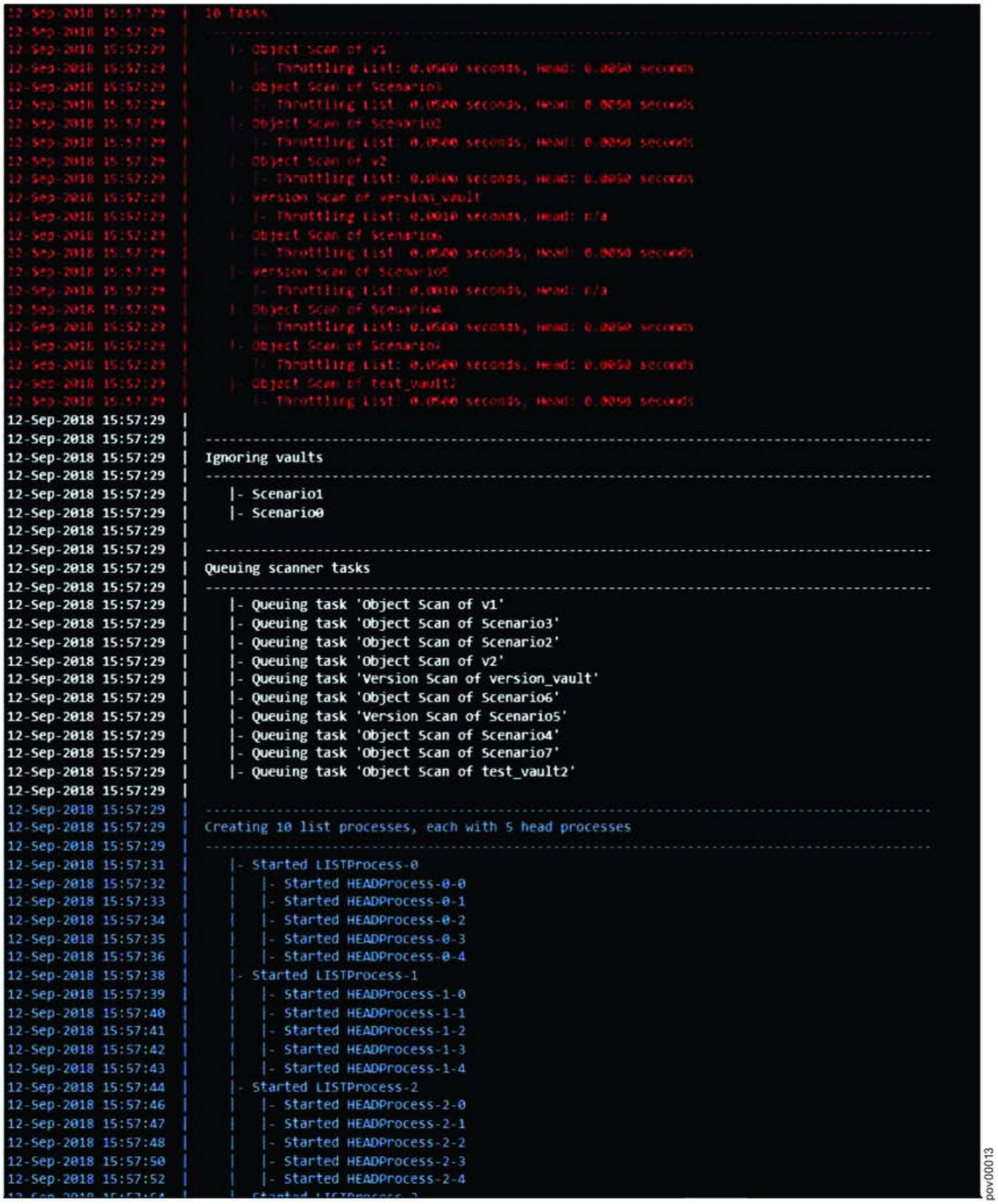
pov00011

Figure 4. Scanner debug (continued)

```
12-Sep-2018 15:57:25 |
12-Sep-2018 15:57:25 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:25 | Python package setup
12-Sep-2018 15:57:25 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:25 | """Setup"""
12-Sep-2018 15:57:25 | from setuptools import setup, find_packages
12-Sep-2018 15:57:25 | setup(
12-Sep-2018 15:57:25 |     name='ibm_cos_scanner',
12-Sep-2018 15:57:25 |     version='2.0.0',
12-Sep-2018 15:57:25 |     packages=find_packages(),
12-Sep-2018 15:57:25 |     include_package_data=True,
12-Sep-2018 15:57:25 |     zip_safe=True,
12-Sep-2018 15:57:25 |     url='www.ibm.com',
12-Sep-2018 15:57:25 |     license='See LICENSE folder',
12-Sep-2018 15:57:25 |     author=IBM',
12-Sep-2018 15:57:25 |     description='IBM COS Scanner / Spectrum Discover Notifier'
12-Sep-2018 15:57:25 | )
12-Sep-2018 15:57:25 |
12-Sep-2018 15:57:25 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:25 | Initialising IBM COS Scanner. Reading config
12-Sep-2018 15:57:25 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:25 | Retrieving System Advanced Configuration
12-Sep-2018 15:57:25 | Calling https://172.19.17.38/manager/api/json/1.0/viewSystemConfiguration.adm
12-Sep-2018 15:57:26 |    |- OK
12-Sep-2018 15:57:26 |    |- dsNet Name: est
12-Sep-2018 15:57:26 |    |- dsNet UUID: f7d033c2-9066-499a-a883-829860d4d865
12-Sep-2018 15:57:26 |
12-Sep-2018 15:57:26 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:26 | Retrieving user's access keys
12-Sep-2018 15:57:26 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:26 | Calling https://172.19.17.38/manager/api/json/1.0/listMyAccessKeys.adm
12-Sep-2018 15:57:26 |    |- OK
12-Sep-2018 15:57:26 |    |- Accesser credentials successfully retrieved from Manager API
12-Sep-2018 15:57:26 |
12-Sep-2018 15:57:26 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:26 | Retrieving vault information from dsNet
12-Sep-2018 15:57:26 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:26 | Calling https://172.19.17.38/manager/api/json/1.0/viewSystemConfiguration.adm
12-Sep-2018 15:57:27 |    |- OK
12-Sep-2018 15:57:27 |
12-Sep-2018 15:57:27 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:27 | Retrieving vault size information from dsNet
12-Sep-2018 15:57:27 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:27 | Calling https://172.19.17.38/manager/api/json/1.0/listVaults.adm
12-Sep-2018 15:57:27 |    |- OK
12-Sep-2018 15:57:27 |
12-Sep-2018 15:57:27 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:27 | Retrieving dsNet device information
12-Sep-2018 15:57:27 | --------------------------------------------------------------------------------
12-Sep-2018 15:57:27 | Calling https://172.19.17.38/manager/api/json/1.0/listDevices.adm
12-Sep-2018 15:57:28 |    |- OK
12-Sep-2018 15:57:28 | 3 devices found
12-Sep-2018 15:57:28 |    |- device_id:1    manager     172.19.17.38   0e547a7f-849a-79e8-102c-2026af755443vm-eqx20420l-
mgr-03
12-Sep-2018 15:57:28 |    |- device_id:2    accesser    172.19.17.39   39a14ec8-090c-79d8-10f9-255249197f43vm-eqx20420l-
```

pov00012

Figure 5. Scanner debug (continued)



```
12-Sep-2018 15:57:29 | 10 tasks
12-Sep-2018 15:57:29 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:29 |    |- Object Scan of v1
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0500 seconds, Head: 0.0050 seconds
12-Sep-2018 15:57:29 |    |- Object Scan of Scenario3
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0500 seconds, Head: 0.0050 seconds
12-Sep-2018 15:57:29 |    |- Object Scan of Scenario2
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0500 seconds, Head: 0.0050 seconds
12-Sep-2018 15:57:29 |    |- Object Scan of v2
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0500 seconds, Head: 0.0050 seconds
12-Sep-2018 15:57:29 |    |- Version Scan of version_vault
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0010 seconds, Head: n/a
12-Sep-2018 15:57:29 |    |- Object Scan of Scenario6
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0500 seconds, Head: 0.0050 seconds
12-Sep-2018 15:57:29 |    |- Version Scan of Scenario5
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0010 seconds, Head: n/a
12-Sep-2018 15:57:29 |    |- Object Scan of Scenario4
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0500 seconds, Head: 0.0050 seconds
12-Sep-2018 15:57:29 |    |- Object Scan of Scenario7
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0500 seconds, Head: 0.0050 seconds
12-Sep-2018 15:57:29 |    |- Object Scan of test_vault2
12-Sep-2018 15:57:29 |        |- Throttling List: 0.0500 seconds, Head: 0.0050 seconds
12-Sep-2018 15:57:29 |
12-Sep-2018 15:57:29 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:29 | Ignoring vaults
12-Sep-2018 15:57:29 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:29 |    |- Scenario1
12-Sep-2018 15:57:29 |    |- Scenario0
12-Sep-2018 15:57:29 |
12-Sep-2018 15:57:29 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:29 | Queuing scanner tasks
12-Sep-2018 15:57:29 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:29 |    |- Queuing task 'Object Scan of v1'
12-Sep-2018 15:57:29 |    |- Queuing task 'Object Scan of Scenario3'
12-Sep-2018 15:57:29 |    |- Queuing task 'Object Scan of Scenario2'
12-Sep-2018 15:57:29 |    |- Queuing task 'Object Scan of v2'
12-Sep-2018 15:57:29 |    |- Queuing task 'Version Scan of version_vault'
12-Sep-2018 15:57:29 |    |- Queuing task 'Object Scan of Scenario6'
12-Sep-2018 15:57:29 |    |- Queuing task 'Version Scan of Scenario5'
12-Sep-2018 15:57:29 |    |- Queuing task 'Object Scan of Scenario4'
12-Sep-2018 15:57:29 |    |- Queuing task 'Object Scan of Scenario7'
12-Sep-2018 15:57:29 |    |- Queuing task 'Object Scan of test_vault2'
12-Sep-2018 15:57:29 |
12-Sep-2018 15:57:29 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:29 | Creating 10 list processes, each with 5 head processes
12-Sep-2018 15:57:29 | ------------------------------------------------------------------------------------------
12-Sep-2018 15:57:31 |    |- Started LISTProcess-0
12-Sep-2018 15:57:32 |    |    |- Started HEADProcess-0-0
12-Sep-2018 15:57:33 |    |    |- Started HEADProcess-0-1
12-Sep-2018 15:57:34 |    |    |- Started HEADProcess-0-2
12-Sep-2018 15:57:35 |    |    |- Started HEADProcess-0-3
12-Sep-2018 15:57:36 |    |    |- Started HEADProcess-0-4
12-Sep-2018 15:57:38 |    |- Started LISTProcess-1
12-Sep-2018 15:57:39 |    |    |- Started HEADProcess-1-0
12-Sep-2018 15:57:40 |    |    |- Started HEADProcess-1-1
12-Sep-2018 15:57:41 |    |    |- Started HEADProcess-1-2
12-Sep-2018 15:57:42 |    |    |- Started HEADProcess-1-3
12-Sep-2018 15:57:43 |    |    |- Started HEADProcess-1-4
12-Sep-2018 15:57:44 |    |- Started LISTProcess-2
12-Sep-2018 15:57:46 |    |    |- Started HEADProcess-2-0
12-Sep-2018 15:57:47 |    |    |- Started HEADProcess-2-1
12-Sep-2018 15:57:48 |    |    |- Started HEADProcess-2-2
12-Sep-2018 15:57:50 |    |    |- Started HEADProcess-2-3
12-Sep-2018 15:57:52 |    |    |- Started HEADProcess-2-4
```

pov00013

Configure IBM Cloud Object Storage notifications for IBM Data Cataloging

Ingesting IBM Cloud® Object Storage event records into IBM Data Cataloging requires the user to enable the Notification service on the IBM Cloud Object Storage system. Thereafter, the user must connect the IBM Cloud Object Storage system to the IBM Cloud Object Storage connector Kafka topic on the IBM Data Cataloging cluster. The name of this connector topic is `cos-le-connector-topic`.

A combination of SASL and TLS is used to authenticate and encrypt the connection between the IBM Cloud Object Storage source system and the Kafka brokers which reside in the IBM Data Cataloging cluster. The certificate and credentials required to establish this connection might be obtained directly from the IBM Data Cataloging cluster by the IBM Data Cataloging storage administrator.

For information on how to enable and configure the IBM Cloud Object Storage Notification service with the IBM Data Cataloging provided credentials, see IBM Cloud Object Storage Administration System documentation.

The following information is required to establish a secure connection between IBM Cloud Object Storage and IBM Data Cataloging:

Hosts

One or more of the IBM Data Cataloging Kafka brokers is in the format: *host1:port,host2:port*. The Kafka producers on the IBM Cloud Object Storage system will retrieve the full list of IBM Data Cataloging Kafka brokers from the first host that is alive and responding. The broker's host and port (the list configured might contain more than one broker) for SASL SSL can be obtained by the IBM Data Cataloging storage administrator from the following location on the IBM Data Cataloging master node: /etc/kafka/server.properties.

Authentication credentials
: The username is `cos` and the password can be obtained by the IBM Data Cataloging storage administrator from the following location on the IBM Data Cataloging master node: /etc/kafka/sasl_password.

Certificate PEM for TLS encryption
: This the CA certificate that is used to sign the Kafka server and client certificates for the IBM Data Cataloging cluster. It might be obtained by the IBM Data Cataloging storage administrator from the following location on the IBM Data Cataloging master node: /etc/kafka/ca.crt.
This file is in the PEM format and the entirety of its contents must be pasted into the Certificate PEM field of the COS Notifications configuration panel.

Enabling IBM Cloud Object Storage notification services

The IBM Cloud® Object Storage notification service can be enabled with the information that follows:

Procedure

1. Log in to the IBM Cloud Object Storage Manager Admin console (https://manager_host/manager/login.adm) with a username of **admin** and a password of **password**.
   If you defined your own password, use your pre-defined password. If you do not have a pre-defined password use the default password.
2. Select the Administration tab.
3. Scroll to the end of the page and select Configure the Notification Service.

   Figure 1. Configurations

   Before you add a notification service to the IBM Cloud Object Storage platform, you must obtain some information from the IBM Data Cataloging server.

   To authenticate the IBM Cloud Object Storage notification service, you can capture the Kafka user name and password from the files on the IBM Data Cataloging platform.

   If the Transport Layer Security (TLS) is enabled in the IBM Cloud Object Storage notification service, you can also copy the certificate authority (CA) in PEM format from the IBM Data Cataloging platform. After you collect the information, you can add the information to the notification service configuration.

- **Authenticating, encrypting, and enabling**
  Configure Kafka authentication, TLS encryption, and notification service settings by extracting certificates and credentials from the IBM Spectrum Discover server configuration files.
- **Authenticating, encrypting, and enabling services on Red Hat OpenShift**
  Follow the procedure shown to authenticate, encrypt, and enable services on Red Hat® OpenShift®:

Authenticating, encrypting, and enabling

Configure Kafka authentication, TLS encryption, and notification service settings by extracting certificates and credentials from the IBM Spectrum Discover server configuration files.

- Log in to the IBM Data Cataloging server and extract the information from the following example, which contains an example of Kafka user name and password.

```
moadmin@server kafka]$ cd /etc/kafka

[moadmin@server kafka]$ cat kafka_server_jaas.conf
KafkaServer {
org.apache.kafka.common.security.plain.PlainLoginModule required
user_cos="meezDMxFNZJMSxdyWQKSjVbs";
};
```

```
User= cos
Password = meezDMxFNZJMSxdyWQKSjVbs
```

Encryption

The following information shows an example of a certificate of authority for the PEM file.

1. Log in to the IBM Data Cataloging server as moadmin.
2. In the /etc/kafka/ca.crt file, copy the block of text that starts with **`BEGIN CERTIFICATE`** and that ends with **`END CERTIFICATE`**. The following example displays what the copied block of text might look like:


```
-----BEGIN CERTIFICATE-----
MIIExTCCA62gAwIBAgIJAKMX/n6ULb6YMA0GCSqGSIb3DQEBCwUAMIGYMQswCQYD
VQQGEwJHQjEOMAwGA1UECAwFSEFOVFMxEDAOBgNVBAcMB0h1cnNsZXkxDDAKBgNV
BAoMA0lCTTEZMBcGA1UECwwQc3BlY3RydW1kaXNjb3ZlcjEZMBcGA1UEAwwQc3Bl
Y3RydW1kaXNjb3ZlcjEjMCEGCSqGSIb3DQEJARYUbWxhd3JlbmNlQHVrLmlibS5j
b20wHhcNMTkwMTAyMTY1MDU5WhcNMzgxMjI4MTY1MDU5WjCBmDELMAkGA1UEBhMC
R0IxDjAMBgNVBAgMBUhBTlRTMRAwDgYDVQQHDAdIdXJzbGV5MQwwCgYDVQQKDANJ
Qk0xGTAXBgNVBAsMEHNwZWN0cnVtZGlzY292ZXIxGTAXBgNVBAMMEHNwZWN0cnVt
ZGlzY292ZXIxIzAhBgkqhkiG9w0BCQEWFG1sYXdyZW5jZUB1ay5pYm0uY29tMIIB
IjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAwg7z4gDeWlkeJjPvj3wobDBB
JrHJngooDbPLicRSf/yjl1NgwbWbjIjIeL9R8My+24hRUGfym9IWCM8qMWyEHG+w
+Rr/6jdQyD89j+m1c2ly3nDhXYsTQZR03UylC/TimF6fc07CfuQ1E2ljHf/JXVK4
ESVilhZR23/tWIfbITZmLvdftJSx0Kgu0Ow4BIr9kpQ3bXwt/eoDvAhdKztDouWN
lYCGmdzFOi6E3asspxHhcsGW3bcMu5mqzT6BEnSzrxr8kRbRDL6Q0Pqv33XVxP6z
OHIvv1uFg9Vq6XHIZLBhWNDqPgYoAbT0Q43vUxk7mJ3uJQY6bgbfuEa+PxygQwID
AQABo4IBDjCCAQowHQYDVR0OBBYEFEKxmmHeSfxgHuFL1dd82WMyf190MIHNBgNV
HSMEgcUwgcKAFEKxmmHeSfxgHuFL1dd82WMyf190oYGepIGbMIGYMQswCQYDVQQG
EwJHQjEOMAwGA1UECAwFSEFOVFMxEDAOBgNVBAcMB0h1cnNsZXkxDDAKBgNVBAoM
A0lCTTEZMBcGA1UECwwQc3BlY3RydW1kaXNjb3ZlcjEZMBcGA1UEAwwQc3BlY3Ry
dW1kaXNjb3ZlcjEjMCEGCSqGSIb3DQEJARYUbWxhd3JlbmNlQHVrLmlibS5jb22C
CQCjF/5+lC2+mDAMBgNVHRMEBTADAQH/MAsGA1UdDwQEAwIBBjANBgkqhkiG9w0B
AQsFAAOCAQEANINRvyeuJh69iRK5dPJssmcISXcZv4X33ukAyRt4zLNFToSkTfj2
ZAtQCNgQNl9Ln7Twuit+e6wifxAkA+UD7wrxMzb32+Mpw/XNzo5DnhInfvkAfC62
SHqWIaqTLXDeGbE8O7ieFsI7kAgEQCf23z/vESB2+m1XBI1UcuxMioYwX4YTb14/
GLDJkqhXMLWV+h/7NU7KbERSBia24N5zlR6Ed/rx83uD2AwBnBqt24sD6Q8Gbm+e
HLMv0JrH1vty1vGsfkZnSHb+E6V/5+GsnpIaDyIpsCvM1LqS/wMzBg9hlT5sii8l
mmqMTK6yqcqS7CfWFv/DjQr/i9ECyJ8fAQ==
-----END CERTIFICATE-----
```


Notification service configuration setup

1. Check Enable Configuration.

```
NAME: <NAME>
Topic: cos-le-connector-topic
Hosts: <SD hostname> :9093
Type: IBM Spectrum Discover
```

Enabling authentication

- Check Enable authentication.

```
Username: cos
Password: <PASSWORD>
```

Enabling encryption

1. Check Enable TLS for Apache Kafka network connections.
2. Add the certificate PEM file from the IBM Data Cataloging platform. See Figure 1.

Figure 1. Add a storage vault to the configuration

Authenticating, encrypting, and enabling services on Red Hat OpenShift

Follow the procedure shown to authenticate, encrypt, and enable services on Red Hat® OpenShift®:

- Log in to IBM Data Cataloging server and issue the following command to extract the password information:

```
oc get secret kafka-sasl-password -n ibm-data-cataloging -o jsonpath='{.data.password}'
```

Encryption

The following information shows an example of a certificate of authority for the PEM file:

1. Log in to the IBM Data Cataloging server as moadmin.
2. Issue the following command to extract the information:

```
oc get secret kafka -n ibm-data-cataloging -o jsonpath='{.data.sasl_ca\.crt}' | base64 -d
```

3. Copy the block of text that starts with **BEGIN CERTIFICATE** and ends with **END CERTIFICATE**. The following example displays what the copied block of text might look like:


```
-----BEGIN CERTIFICATE-----
MIIExTCCA62gAwIBAgIJAKMX/n6ULb6YMA0GCSqGSIb3DQEBCwUAMIGYMQswCQYD
VQQGEwJHQjEOMAwGA1UECAwFSEFOVFMxEDAOBgNVBAcMB0h1cnNsZXkxDDAKBgNV
BAoMA0lCTTEZMBcGA1UECwwQc3BlY3RydW1kaXNjb3ZlcjEZMBcGA1UEAwwQc3Bl
Y3RydW1kaXNjb3ZlcjEjMCEGCSqGSIb3DQEJARYUbWxhd3JlbmNlQHVrLmlibS5j
b20wHhcNMTkwMTAyMTY1MDU5WhcNMzgxMjI4MTY1MDU5WjCBmDELMAkGA1UEBhMC
R0IxDjAMBgNVBAgMBUhBTlRTMRAwDgYDVQQHDAdIdXJzbGV5MQwwCgYDVQQKDANJ
Qk0xGTAXBgNVBAsMEHNwZWN0cnVtZGlzY292ZXIxGTAXBgNVBAMMEHNwZWN0cnVt
ZGlzY292ZXIxIzAhBgkqhkiG9w0BCQEWFG1sYXdyZW5jZUB1ay5pYm0uY29tMIIB
IjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAwg7z4gDeWlkeJjPvj3wobDBB
JrHJngooDbPLicRSf/yjl1NgwbWbjIjIeL9R8My+24hRUGfym9IWCM8qMWyEHG+w
+Rr/6jdQyD89j+m1c2ly3nDhXYsTQZR03UylC/TimF6fc07CfuQ1E2ljHf/JXVK4
ESVilhZR23/tWIfbITZmLvdftJSx0Kgu0Ow4BIr9kpQ3bXwt/eoDvAhdKztDouWN
lYCGmdzFOi6E3asspxHhcsGW3bcMu5mqzT6BEnSzrxr8kRbRDL6Q0Pqv33XVxP6z
OHIvv1uFg9Vq6XHIZLBhWNDqPgYoAbT0Q43vUxk7mJ3uJQY6bgbfuEa+PxygQwID
AQABo4IBDjCCAQowHQYDVR0OBBYEFEKxmmHeSfxgHuFL1dd82WMyf190MIHNBgNV
HSMEgcUwgcKAFEKxmmHeSfxgHuFL1dd82WMyf190oYGepIGbMIGYMQswCQYDVQQG
EwJHQjEOMAwGA1UECAwFSEFOVFMxEDAOBgNVBAcMB0h1cnNsZXkxDDAKBgNVBAoM
A0lCTTEZMBcGA1UECwwQc3BlY3RydW1kaXNjb3ZlcjEZMBcGA1UEAwwQc3BlY3Ry
dW1kaXNjb3ZlcjEjMCEGCSqGSIb3DQEJARYUbWxhd3JlbmNlQHVrLmlibS5jb22C
CQCjF/5+lC2+mDAMBgNVHRMEBTADAQH/MAsGA1UdDwQEAwIBBjANBgkqhkiG9w0B
AQsFAAOCAQEANINRvyeuJh69iRK5dPJssmcISXcZv4X33ukAyRt4zLNFToSkTfj2
ZAtQCNgQNl9Ln7Twuit+e6wifxAkA+UD7wrxMzb32+Mpw/XNzo5DnhInfvkAfC62
SHqWIaqTLXDeGbE8O7ieFsI7kAgEQCf23z/vESB2+m1XBI1UcuxMioYwX4YTb14/
GLDJkqhXMLWV+h/7NU7KbERSBia24N5zlR6Ed/rx83uD2AwBnBqt24sD6Q8Gbm+e
HLMv0JrH1vty1vGsfkZnSHb+E6V/5+GsnpIaDyIpsCvM1LqS/wMzBg9hlT5sii8l
mmqMTK6yqcqS7CfWFv/DjQr/i9ECyJ8fAQ==
-----END CERTIFICATE-----
```


Notification service configuration setup

- Check Enable Configuration.

```
NAME: <NAME>
Topic: cos-le-connector-topic
Hosts: <SD ipaddress> :443
Type: IBM Spectrum Discover
```

Enabling authentication

- Check Enable authentication.

```
Username: cos
Password: <PASSWORD>
```

Enabling encryption

1. Check Enable TLS for Apache Kafka network connections.
2. Add the certificate PEM file from the IBM Data Cataloging platform. See Figure 1.

Figure 1. Add a storage vault to the configuration

Testing the IBM Cloud Object Storage notification service

To test the IBM Cloud® Object Storage notification service, the tester can populate the IBM Cloud Object Storage vault with test data.

About this task

You can use a number of methods to write files to an IBM Cloud Object Storage vault, but you can use cURL directly on IBM Data Cataloging platform. cURL is a computer software project that provides a library and command-line tool for transferring data that uses various protocols.

Procedure

1. Create a test file, for example, object_1.txt.
   The test file can be any file that contains data.
2. Write a file to the IBM Cloud Object Storage vault by using cURL.
   Requirements

   IBM® COS Vault Name (vault1) [anonymous access enabled]
   IBM COS Accesser IP address

Example

- curl -X PUT -i -T object_1.txt http://9.11.200.208/vault1/object_1.txt
- HTTP/1.1 100 ContinueHTTP/1.1 200 OK
- Date: Fri, 04 Jan 2019 13:21:14 Greenwich mean time
- X-Clv-Request-Id: a9ad657a-a919-4b13-9b72-961ae8c57e3c
- Server: 3.14.0.23
- X-Clv-S3-2.5
- x-amz-request-id: a9ad657a-a919-4b13-9b72-961ae8c57e3c
- ETag: "7c517c7108f7180377e7b37db2e39261"
- Content-Length: 0

- **Monitoring the IBM Cloud Object Storage accesser logs**
  To determine whether a file is successfully written to the IBM Cloud Object Storage vault and a notification is successfully sent to the IBM Data Cataloging server, the accesser logs can be monitored on the IBM Cloud Object Storage Accesser server.
- **Monitoring the IBM Data Cataloging producer IBM Cloud Object Storage logs**
  When the IBM Data Cataloging server receives a notification from the IBM Cloud Object Storage platform, the IBM Data Cataloging producer IBM Cloud Object Storage records a transaction.
- **Monitoring the IBM Data Cataloging dashboard for IBM Cloud Object Storage ingestion**
  You can monitor the IBM Data Cataloging dashboard for IBM Cloud Object Storage ingestion.

Monitoring the IBM Cloud Object Storage accesser logs

To determine whether a file is successfully written to the IBM Cloud® Object Storage vault and a notification is successfully sent to the IBM Data Cataloging server, the accesser logs can be monitored on the IBM Cloud Object Storage Accesser server.

In the following example, an object that is written to vault1 results in the sending of one notification to the IBM Data Cataloging server. The user must have access privileges to log on to the IBM Cloud Object Storage Accesser host to check the log files.

Confirm that an object is stored in the IBM Cloud Object Storage vault.

```
root@ibm_accesser:/var/log/dsnet-core# tail -f http.log
9.11.201.78 - ""  -  [04/Jan/2019:13:21:14 +0000] "PUT /vault1/object_1.txt HTTP/1.1" 200 0 "-" "curl/7.29.0" 22
```

Confirm that a notification is sent to the IBM Data Cataloging server.

```
root@ibm_accesser:/var/log/dsnet-core# tail -f notification.log
{"time":"2019-01-04T13:21:14.668Z","request_id":"a9ad657a-a919-4b13-9b72-961ae8c57e3c","retried":true,"success":true,
"request_time":"2019-01-04T13:21:14.567Z","kafka_config_uuid":"d842c7a0-9c36-412e-8908-8ad5120a261e","topic":"cos-le-connector-
topic"
```

Monitoring the IBM Data Cataloging producer IBM Cloud Object Storage logs

When the IBM Data Cataloging server receives a notification from the IBM Cloud® Object Storage platform, the IBM Data Cataloging producer IBM Cloud Object Storage records a transaction.

A successful notification is recorded as an offset value of one, when a notification is received from IBM Cloud Object Storage platform.

```
[moadmin@data_cataloging]$ oc logs -f -n producercos  kindled-alligator-producer-cos-producer-9f6966b4-8jsg7
break time. waiting for work...
2019-01-04 13:21:19.187 > offset_commit_cb: success, offsets:[{part: 0, offset: 1, err: none}]
```

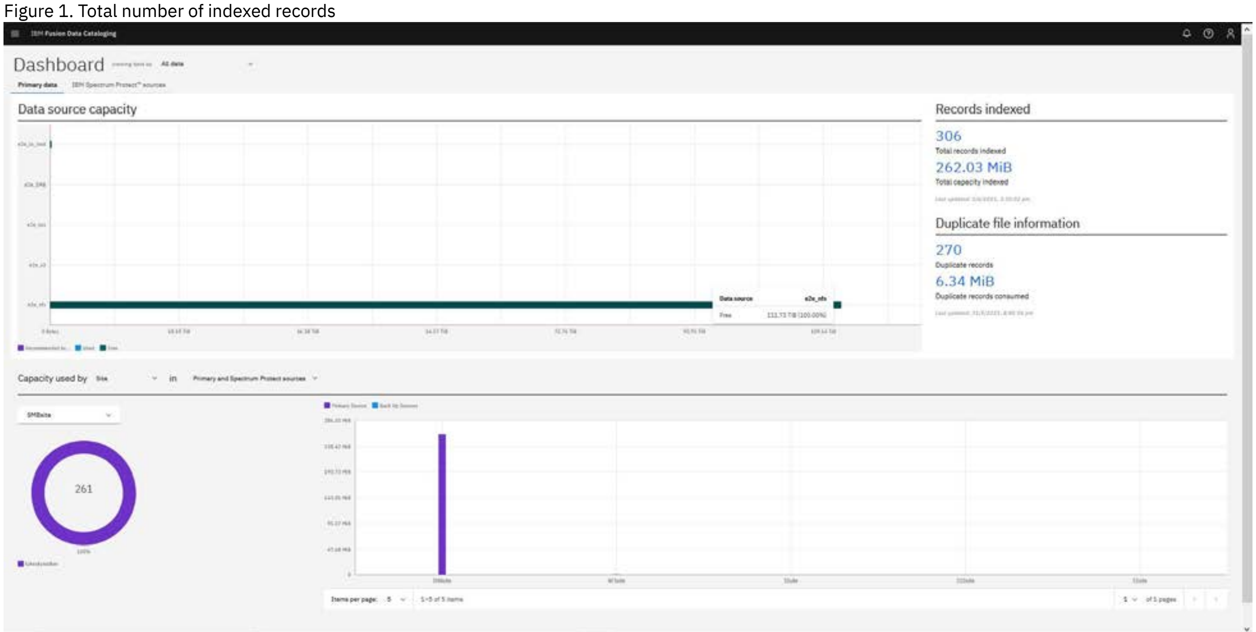
Monitoring the IBM Data Cataloging dashboard for IBM Cloud Object Storage ingestion

You can monitor the IBM Data Cataloging dashboard for IBM Cloud® Object Storage ingestion.

After IBM Cloud Object Storage notifications are ingested from the IBM Cloud Object Storage platform, the IBM Data Cataloging dashboard displays the total number of indexed records.

Note: The IBM Data Cataloging dashboard can take approximately 30 minutes to display the total number of indexed records.
See Figure 1.

Figure 1. Total number of indexed records



IBM Data Cataloging and S3 object storage data source connections

Use this information to understand how IBM Data Cataloging works with S3-compliant object store.

- **Creating an S3 object storage data connection**
  Use this information to create a connection to an S3-compliant object store.
- **Scanning an S3 object storage data connection**
  Use this information to scan a connection for an S3-compliant object store.
- **Best practices for scanning an S3 object storage system**
  Use best practices for scanning S3-compliant object storage systems.

Creating an S3 object storage data connection

Use this information to create a connection to an S3-compliant object store.

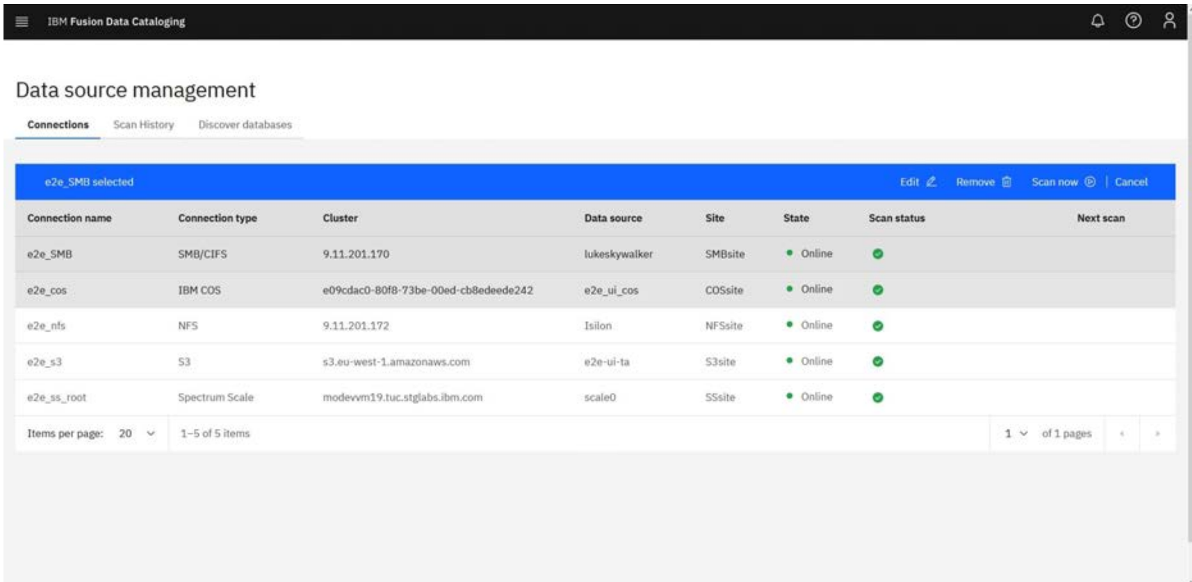
About this task

To create the S3-compliant connection:
Note: The storage host supports SSL by default. If `http` is required, then explicitly add `http://` prior to the hostname or IP address.

Procedure

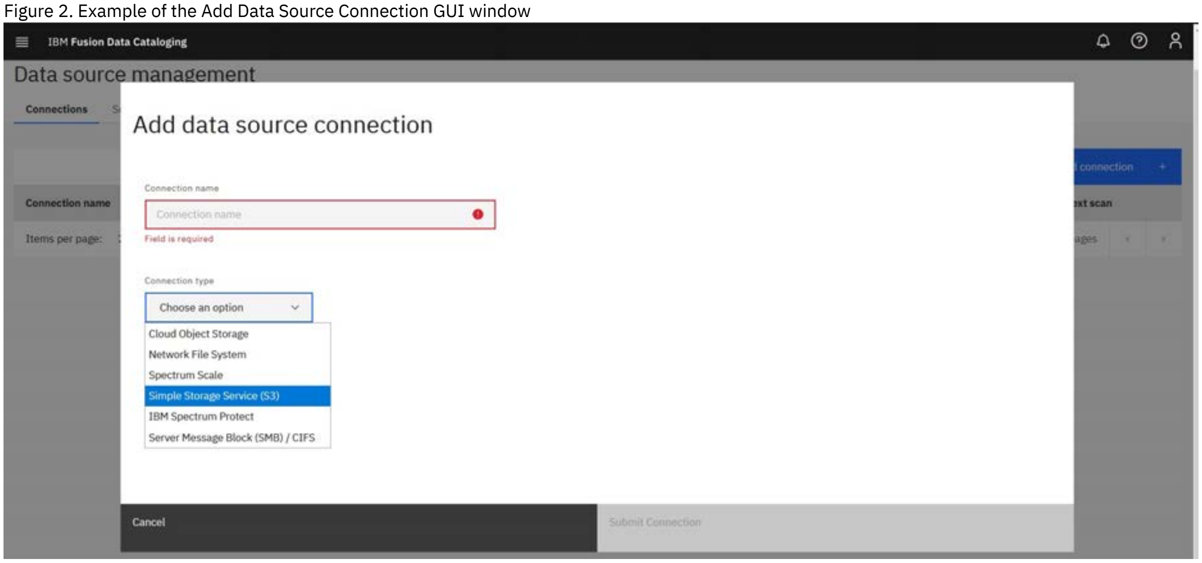
1. Log in to the IBM Data Cataloging graphical user interface (GUI) with a user ID that is associated with data administration role.
   The data administration access role is required for creating connections. For more information about role-based access control, go to Role-based access control.
2. Click and go to Data connections > Connections.
   Clicking Connections displays the different types of data source connection names, platforms, clusters, data source, size, and Add Connection.

   Figure 1. Displaying the source names for Data Source Connections



3. Click Add Connection to display a new window that shows Add Data Source Connection.

Figure 2. Example of the Add Data Source Connection GUI window

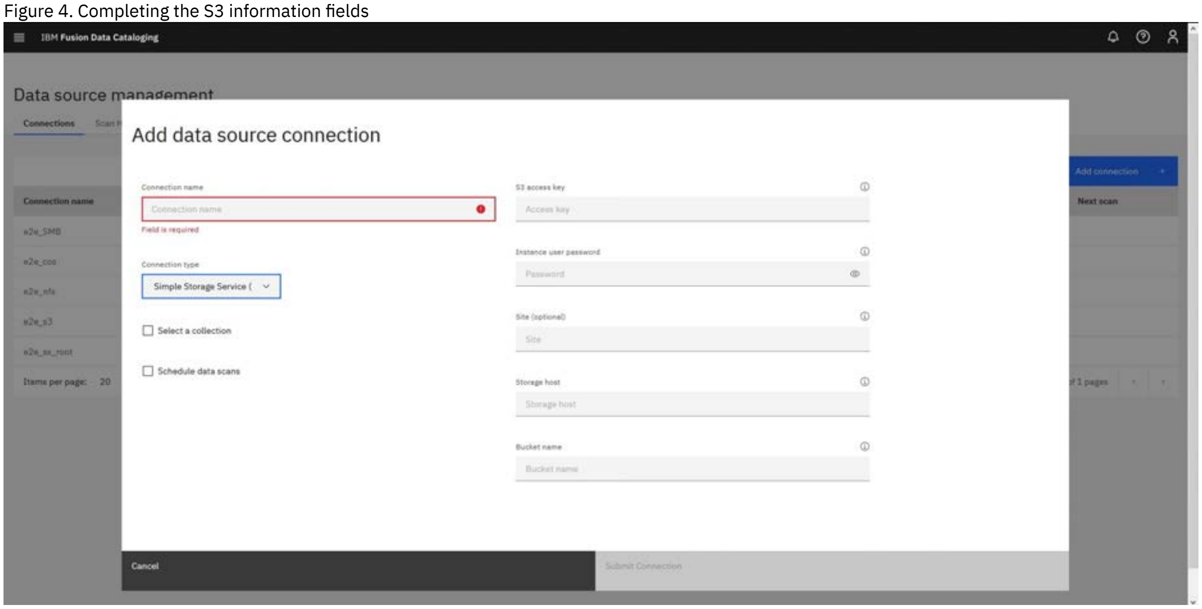


4. Complete the following steps:
   a. In the field for Connection name, define a Connection name.
   b. Choose data source connection type from Connection type list.
5. Select the connection type Simple Storage Service (S3).

Figure 3. Selecting Simple Storage Service (S3)

6. In the screen for S3, complete the fields and then click Submit Connection for the S3 connections manager.

Figure 4. Completing the S3 information fields



S3 access key
: Indicates the access key ID for the S3 object store.

S3 secret key
: Indicates the secret key ID for the S3 object store.

Site
: Indicates the physical location of the records.

Storage host
: Indicates the IP address or host of the storage system.

Bucket name
: Indicates the name of the bucket that you are going to scan.

Scanning an S3 object storage data connection

Use this information to scan a connection for an S3-compliant object store.

About this task

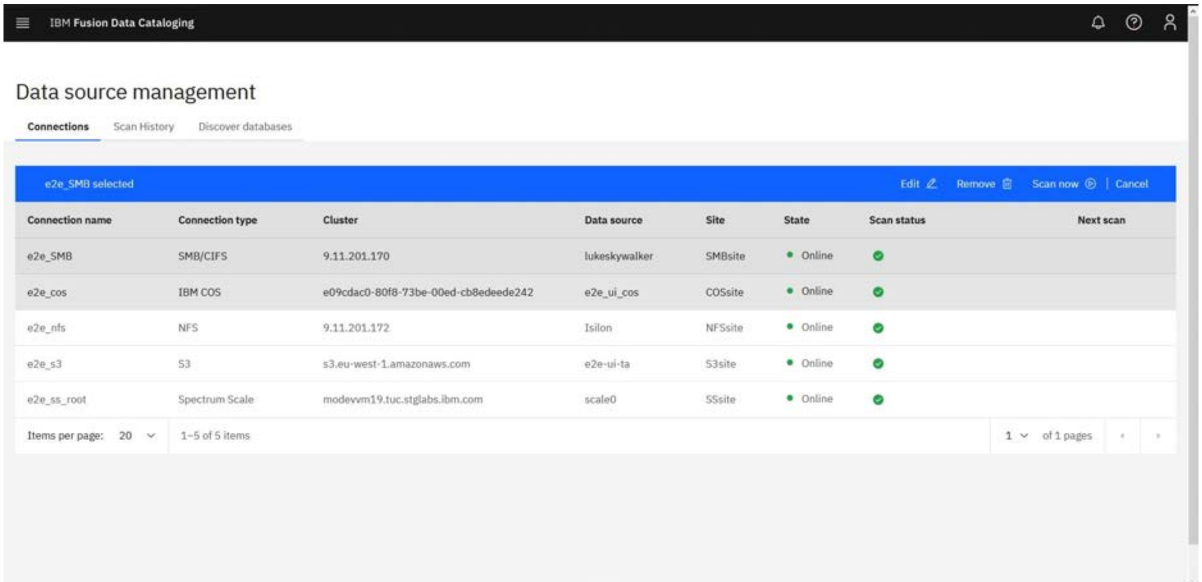
When you initiate a scan from the IBM Data Cataloging graphical user interface (GUI), the metadata is transferred asynchronously back to the IBM Data Cataloging instance.

Automated scanning and data ingestion relies on an established and active network connection between the IBM Data Cataloging instance and the S3 storage source. If the connection cannot be established, the state of the data source connection shows as unavailable, and the option for automated scanning does not appear in the IBM Data Cataloging GUI for that connection.

Procedure

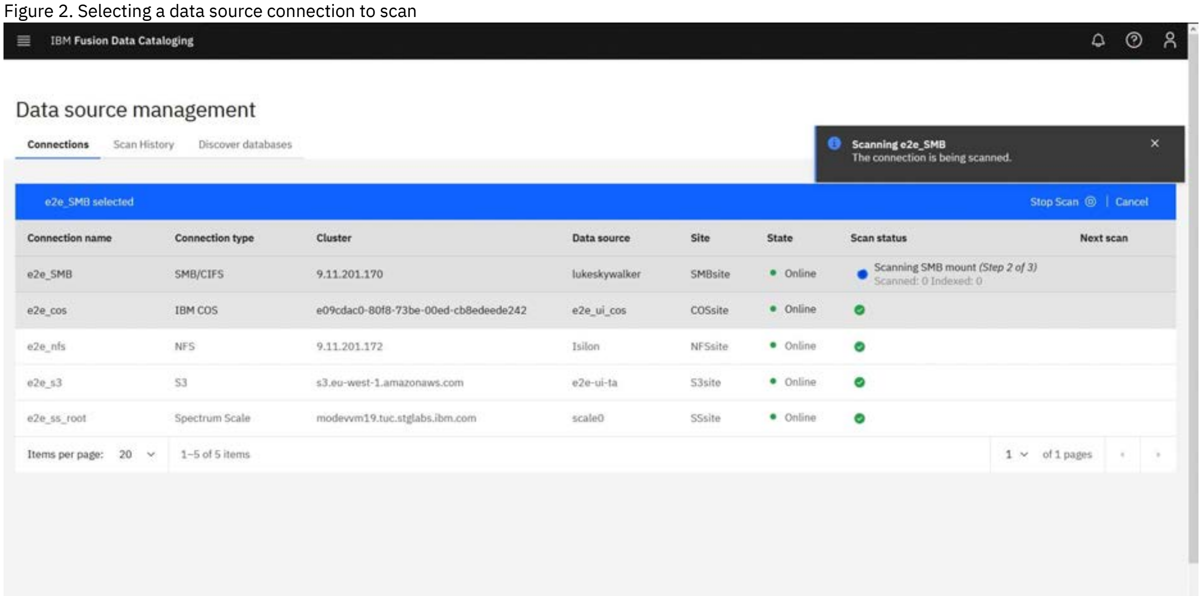
1. Log in to IBM Data Cataloging web interface.
2. Click go to Data connections > Connections.
   The following figure shows the data connections lists:

   Figure 1. Data source connections



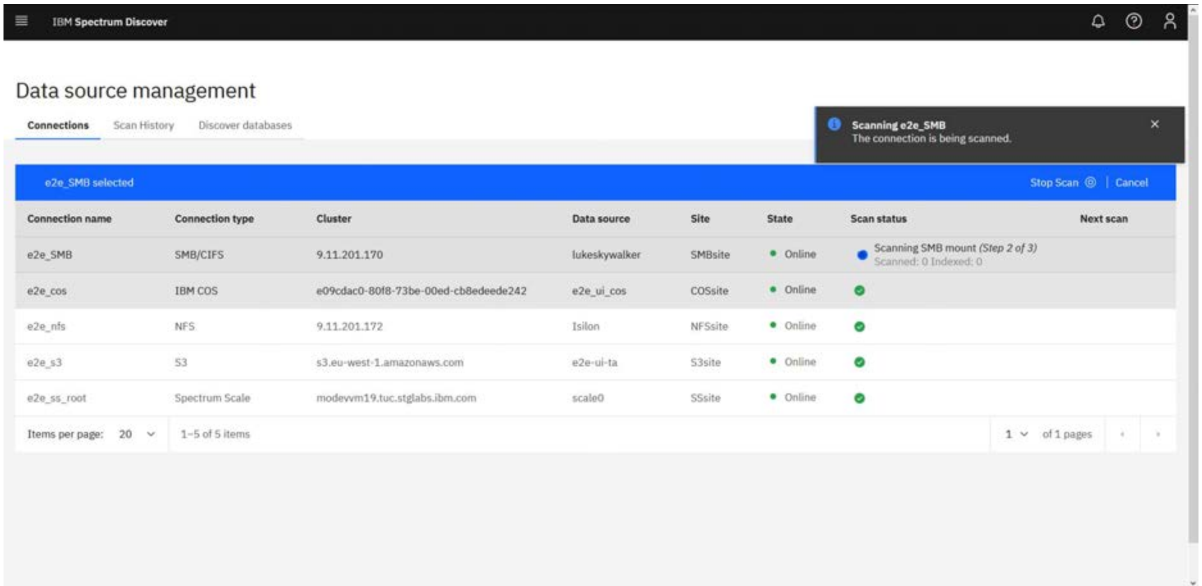
3. Select the data source connection name that you want to scan. Ensure that the **`State`** is listed as **`Online`** to make your system scan ready.
   The following figure shows how to select a data source connection to scan.

   Figure 2. Selecting a data source connection to scan



4. Select Scan now to change the status to Scanning.
   The following figure shows an active scan.

   Figure 3. Active scans



5. When the scan finishes, the state field returns to a status of Online.

Best practices for scanning an S3 object storage system

Use best practices for scanning S3-compliant object storage systems.

It is recommended to check the log files in the following directories after each scan:

/opt/ibm/metaocean/data/connections/s3/<connection_name>/debug/<scan_timestamp>/scanner.debug indicates whether the scan was successful or not.

/opt/ibm/metaocean/data/connections/s3/<connection_name>/error/<scan_timestamp>/scanner.error contains a list of all the messages that are not delivered to IBM Data Cataloging.

/opt/ibm/metaocean/data/connections/s3/<connection_name>/data/<scan_timestamp>/ contains a subfolder with the scanned data source name. There is a stats folder inside this folder that contains information about the number of objects in the data source or the number of objects or scanned files.

You can also compare the total size of the bucket that is reported in IBM Data Cataloging with the total size of the S3 object store at source (if it is available).

Scanning an Elastic Storage Server data source connection

Use the IBM Data Cataloging GUI to scan an Elastic Storage Server (ESS) data connection.

Procedure

1. Log in to IBM Data Cataloging web interface with data admin privileges.
2. Open an internet browser to the IP address or host name of your IBM Data Cataloging cluster.
   The default credentials to put into the IBM Spectrum Discover dialog box are:

   ```
   Default user: sdadmin
   Default Password: Passw0rd
   ```

3. Click menu and go to Data connections > Add data source connection.
4. Enter the EMS node as the host system name:

   Connection Name
      Indicates the connection name, such as `ESS_test`.
   Connection Type
      Indicates the connection type, which is `Spectrum Scale`.
   User
      Indicates the user that initiates the scan, such as `root`.
   Password
      Indicates the password for the user that initiates the scan.
   Working Directory
      Indicates the working directory for the scan, such as `/gpfd/icp4D_data_fs_master1/sd_scan`.
   Scan Directory
      Indicates the scan directory, such as `/gpfd/icp4D_data_fs_master1`.
   Site (optional)
      Indicates the scan site. This field is optional.
   Cluster
      Indicates the clustered system name that is being scanned, such as `heliumess.tuc.stglabs.ibm.com`.

Host
: Indicates the host system name that is being scanned, such as `9.11.103.45`.

Filesystem
: Indicates the file system name that is being scanned.

Node List
: Indicates the node list of nodes that are being scanned. Enter `gss_ppc64` as the node class to perform the scan.

5. After you complete all fields, select Submit Connection.
6. Select your data source from the table (click the row) and then select Scan Now.
7. You can monitor the status of the scans from Data Connections Overview.
8. You can also search for documents from that data source by using the search navigation icon in the left side of the GUI.
   In the search icon you must:
   a. Enter a query.
   b. Click Search.
   c. Indicate the files that are found in the search results.

Creating a Network File System data source connection

You can use the IBM Data Cataloging graphical user interface to create a Network File System (NFS) data connection.

About this task

Note: For NFS scanning that uses Data ONTAP 8.1.2 or later, export the file system with the following configuration:

Protocol
: NFSv3 and NFSv4

Security type
: UNIX

Client permissions
: A minimum of read-only access is necessary.

Anonymous users
: Root access must be granted.

Setuid and Setgid executable routines are not necessary.
Creating data source connections in IBM Data Cataloging identifies source storage systems that are to be indexed by IBM Data Cataloging.

For some data source types, a network connection can be created to allow for automated scanning and indexing of the source system metadata. IBM Data Cataloging will not index data from unknown sources, so creating a data source connection is the first step towards cataloging any source storage system.

Procedure

1. Log in to the IBM Data Cataloging web interface with a user ID that has the data admin role that is associated with it.
   Note: The data admin access role is necessary for creating connections.
2. Click menu and go to Data connections to display the connection's source name, platform, cluster, data source, site, next run time, and state of existing data source connections.
3. Click Add connection to display the Add data source connection window.
4. Enter a name in the Connection name field.
5. Select the connection type Network File System from the Connection type list.
6. Complete the fields for the NFS parameters, and click Submit Connection.

Parameters for NFS connections

Connection name
: The name of the connection, an identifier for the user, for example `filesystem1`.
  Note: It must be a unique name within IBM Data Cataloging.

Connection type
: The type of source storage system this connection represents.

Enable NFSv4 ACL scans
: Enables the scan of NFSv4 ACL permission entries.

Data source
: The full name of the data source.

Export path
: The data path from which data is to be exported.

Host
: The host name of the node from which a scan can be initiated.

Site
: The location in which the data source facility is located.

Creating an SMB data source connection

Creating an SMB data connection by using the IBM Data Cataloging graphical user interface.

About this task

Create data source connections in IBM Data Cataloging to identify and index source storage systems.

For some data source types, a network connection is (optionally) created to allow for automated scanning and indexing of the source system metadata. IBM Data Cataloging does not index data from unknown sources, so creating a data source connection is the first step towards cataloging any source storage system.

Procedure

1. Log in to the IBM Data Cataloging web interface with a user ID that is associated with the data admin role.
   Note: The Data admin access role is required for creating connections.
2. Click menu and go to Data connections > Connections to display the source connection name, platform, cluster, data source, site, next run time, and state of existing data source connections.
3. Click Add connection to display the Add data source connection window.
4. Enter a name in the Connection name field.
5. Select the connection type Server Message Block (SMB) / CIFS from the Connection type list.
6. Complete the fields for the SMB parameters, and click Submit Connection.

Parameters for SMB connections

Connection Name
: Indicates the name of the connection, an identifier for the user, for example Share1.
  Note: It must be a unique name within IBM Data Cataloging.

Share
: Indicates the path name of the SMB/CIFS file share, for example //server/share.

User
: Indicates the user ID that has access permissions to the SMB share, for example DOMAIN1\user1

Password
: Indicates the authentication password for the user ID provided.

Site (Optional)
: Indicates the location in which the data source facility is located.

Creating an IBM Storage Protect data source connection

Creating the IBM Storage Protect data connection by using the IBM Data Cataloging graphical user interface.

About this task

Creating data source connections in IBM Data Cataloging identifies source storage systems that are to be indexed by IBM Data Cataloging.

For some data source types, a network connection is (optionally) created to allow for automated scanning and indexing of the source system metadata. IBM Data Cataloging will not index data from unknown sources, so creating a data source connection is the first step towards cataloging any source storage system.

Procedure

1. Log in to the IBM Data Cataloging web interface with a user ID that has the data admin role that is associated with it.
   Note: The data admin access role is required for creating connections.
2. Click menu and go to Data connectios > Connections to display the source name, platform, cluster, data source, site, next run time, and state of existing data source connections.
3. Click Add connection to display the Add data source connection window.
4. Enter a name in the Connection name field.
5. Select the connection type IBM Spectrum Protect from the Connection type list.
6. Complete the fields for the IBM Storage Protect connection type, and click Submit Connection.

Parameters for IBM Storage Protect connections

Connection Name
: The name of the connection, an identifier for the user, for example filesystem1.
  Note: It must be a unique name within IBM Data Cataloging.

Spectrum Protect Server IP
: The IP address or host name of the IBM Storage Protect server.

ODBC Port
: The ODBC (open database connector) port for the Protect server (default is 51500).

Instance User
: The name of the IBM Storage Protect database instance user. The default is `tsminst1`.

Instance user password
: The password for the database instance user.

Site (Optional)
: The location in which the data source facility is located.

Configuring data source connections offline

Multiple source data connections can be added to the IBM Data Cataloging system through offline mode.

Before you begin

- You must have the **SD_USER,SD_PASSWORD & OVA** environment variables set in the IBM Data Cataloging system. To verify the details, enter `gettoken`.
- Enter the connection values inside the JSON file and add it to IBM Data Cataloging host by using **load_connection <nameofjsonfile>**.
  A sample JSON file with connection values as shown.

```
{
 "name": "SpectrumScale2Million",
 "platform": "Spectrum Scale",
 "cluster": "ctolib.cluster1",
 "datasource": "scale0",
 "site":"Offline Scale"
}
```

Procedure

1. Log in to the IBM Data Cataloging host system with following credentials.

```
Host: <IBM Data Cataloging_hostname>
login: moadmin
Password: Passw0rd
```

2. SCP the list.metaocean file to the IBM Data Cataloging VM by using the following command before you run the ingestion script.

```
sudo scp moadmin@modevdump:/mnt/datadump/scans/<file_name>/home/moadmin/
```

3. After the list.metaocean file is copied to the IBM Data Cataloging host, issue the following command to ingest the file.

```
$ ingest <file_name>
```

4. To monitor the ingestion process by following the producer log, issue the following commands.

```
$ kp | grep spectrum-discover-producer-<connection_name>-scan
```

Note: The preceding command returns the producer pod log.

```
$ oc logs -f -n ibm-data-cataloging spectrum-discover-producer-<connection_name>-scan-<randomchars>
```

Editing and using the TimeSinceAccess and Size Range buckets

Users can group or aggregate data into three user-defined bucket ranges. The three user-defined bucket ranges are TimeSinceAccess, Size Range and FileGroup.

The TimeSinceAccess bucket groups files and objects based on the time they were last accessed. The SizeRange bucket groups files and objects based on their size. The File Group bucket groups files based on their file type or extension. All three buckets can be customized to better align with the user's requirements. Both the TimeSinceAccess and the SizeRange buckets have up to five custom ranges with user-defined labels.

Note: The **FileGroup** bucket cannot be edited through the user interface and must be modified using REST APIs. For more information, see/db2whrest/v1/buckets/<bucket>: PUT. For more information, see the topic */db2whrest/v1/buckets/<bucket>: PUT* in the *IBM Data Cataloging: REST API Guide*
To access the SizeRange bucket groups, select Metadata > Tags > edit icon for the SizeRange tag. For example, SizeRange can be broken up into 'T-shirt size' ranges where the ranges and labels are:

Table 1. Examples of size ranges and sizes of buckets with user-defined labels

Size range	Size
0 - 4 K	XS
4 K - 1 M	S
1 M - 1 G	M
1 G - 1T B	L
1 TB+	XL

After you change or update a bucket definition, IBM Data Cataloging summarizes the current set of files and objects into their respective bucket ranges. The changes are updated periodically every half an hour; thus, it may take a half an hour or more before the changes are reflected in the Spectrum Discover GUI.
Note: Ensure that the maximum value for each bucket is greater than the value assigned to the previous bucket.
See the Figure 1

Figure 1. Example of how to define the settings for a SizeRange bucket

To open the menu for the TimeSinceAccess buckets select Metadata > Tags > edit icon for the TimeSinceAccess tag. See the Figure 1 for an example.

Figure 2 shows an example of how to modify and define the settings of a bucket that is older than one year.

Figure 2. Example of how to modify and define the settings of a bucket that is older than one year old

pov00055

IBM Spectrum Storage software requirements

Use IBM Data Cataloging to index metadata from other applications and to orchestrate the data management.

IBM Cloud Object Storage

IBM Data Cataloging indexes metadata from IBM Cloud® Object Storage by receiving notifications that contain metadata from IBM Cloud Object Storage. IBM Data Cataloging also supports scanning IBM Cloud Object Storage to harvest metadata.

The following table shows the minimum required IBM Cloud Object Storage software version to enable metadata harvesting with IBM Data Cataloging:

Table 1. IBM Cloud Object Storage software requirements

Component	Version
IBM Cloud Object Storage	3.14.0 and higher

IBM Storage Scale

IBM Data Cataloging indexes metadata from IBM Storage Scale by scanning IBM Storage Scale file systems. The IBM Storage Scale watch folders technology preview also enables IBM Storage Scale to send events that contain metadata to IBM Data Cataloging.

The following table lists the minimum required IBM Storage Scale software versions:

Table 2. IBM Spectrum Scale software requirements

Component	Feature	Version
IBM Storage Scale	Scanning	4.2.3.x and higher
IBM Storage Scale	Live events	(Advanced and Data Management Editions, only) 5.0.4.1 and higher
IBM Storage Scale	Data management that uses ScaleAFM Application	5.1 and higher

The following requirements are needed to enable live events:

- You must use IBM Storage Scale 5.0.3.x. Due to an IBM Storage Scale performance issue that might result in the unexpected suspension of the IBM Storage Scale watch, IBM Data Cataloging recommends the use of IBM Storage Scale 5.0.4.x.
- Watch folder must be enabled for the Scale cluster.
- A minimum of three nodes on the IBM Storage Scale cluster are required to act as Kafka brokers.
- The IBM Storage Scale nodes that act as brokers must meet a minimum local space requirement of 20 GB each to successfully enable the watch with a secondary sink.

IBM Storage Protect

IBM Data Cataloging indexes metadata from IBM Storage Protect by scanning IBM Storage Protect file systems.

The following table lists the minimum required IBM Storage Protect software versions to enable metadata harvesting with IBM Data Cataloging:

Table 3. IBM Spectrum Protect software requirements

Component	Version
IBM Storage Protect	7.x and higher

IBM Storage Archive

IBM Data Cataloging supports the advanced tiering function with the ScaleILM application.

The following table lists the minimum IBM Spectrum® Archive software version that is required to control the data placement by IBM Data Cataloging.

Table 4. IBM Storage Archive software requirements

Component	Version
IBM Storage Archive Enterprise Edition (EE)	1.3.0.6 and higher

IBM Data Cataloging installation with alternative VLAN

Install and configure IBM Data Cataloging that uses additional VLAN connected through the IBM Fusion upstream links.

For security reasons, the target VLAN network is not available for the IBM Fusion cluster through the default IBM Fusion public network.

There are several approaches available to access alternate VLANs that do not have routable access through the default IBM Fusion network.

The reminder of this document covers the option of OpenShift® `NetworkAttachmentDefinitions` to provide IBM Data Cataloging access to alternative VLAN access. It explains the setup and implementation of `NetworkAttachmentDefinitions` and the required modifications to the IBM Data Cataloging definitions.

IBM Fusion VLAN Configuration

1. Add VLAN. For procedure, see Adding VLANs.
2. Add VLAN to link.
3. Check switches to ensure that VLAN is available on bond250 and ports.
4. Check the compute node to ensure that VLAN is added.

IBM Data Cataloging Configuration

1. Log in to the Red Hat® OpenShift Container Platform web console.
2. Go to Networking > NetworkAttachmentDefinitions.
3. Click Project drop-down and select ibm-data-cataloging namespace from the list.
4. Click the NAD that you want and check the Container and VM NAD details.
   For example:

   For container access

   Note: Adding extra VLANs to the IBM Fusion switch link creates an additional bridge interface for each additional VLAN on the worker nodes `bond0` interface automatically.

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
annotations:
k8s.v1.cni.cncf.io/resourceName: bridge.network.kubevirt.io/br4001
name: br4001-c
namespace: ibm-data-cataloging
spec:
config: >-
{"cniVersion": "0.3.1", "name": "br4001", "type": "macvlan", "master": "br4001","mode": "bridge", "ipam": { "type":
"static" , "routes": [ {"dst":"10.140.20.0/24","gw":"10.100.100.1" } ] }}
```

Item	Value	Description
Name	br4001-c	`Network Attachment Definition`
`dst`	10.140.20.0	Destination subnet
`gw`	10.100.100.1	Router
Name	br4001	Bridge 4001 (auto generated)

   For VM access

   Note: After attached to any VM, additional configuration on the VM interface must be necessary, including IP and routing information.

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
 name: br4001
 namespace: ibm-datacataloging
spec:
```

```
nodeSelector:
  vmtest: "true"
config: >-
  {"name":"br4001","type":"cnv-bridge","cniVersion":"0.3.1","bridge":"br4001","macspoofchk":true,"ipam":{}}
```

Item	Value	Description
Name	br4001	`Network Attachment Definition`
Bridge	br4001	Bridge interface (auto generated)

Container updates

Important: The container updates might change based on your connection type such as NFS, S3, SMB, and Scale. The following example shows the Scale connection type.

The following IBM Data Cataloging containers require updates for Scale scanning through secondary VLAN as follows:

- `isd-connmgr-main`
- `isd-consumer-scale-le`
- `isd-consumer-scale-scan`

Assign a unique IP address for each container pod set.

For example:

Container	IP address	Resource
`isd-connmgr-main`	10.100.100.11	`Statefulset`
`isd-consumer-scale-le`	10.100.100.12	`Deployment`
`isd-consumer-scale-scan`	10.100.100.13	`Deployment`

```
apiVersion: apps/v1
metadata:
  name: isd-connmgr-main
....
spec:
  template:
    annotations:
        k8s.v1.cni.cncf.io/networks: |-
          [
            {
              "name": "br4001-c",
              "interface": "br4001",
              "namespace": "ibm-data-cataloging",
              "ips": ["10.100.100.11/24"]
            }
          ]
```

Activate changes

1. Do a graceful shutdown of the IBM Data Cataloging service. For the procedure, see Graceful shutdown.
2. Edit `Statefulset isd-conn-mgr-main` in the components. You can append the new annotation to the current annotations.
3. Edit `Deployments` for the consumer type. In this case it depends on the connection type.
   For example for scale it must be the `isd-consumer-scale-scan`.
4. Return the IBM Data Cataloging service to the running state. For the procedure, see Returning IBM Data Cataloging to a running state.

Example Scaling Commands:

- Scale down IBM Data Cataloging:

  ```
  oc get deployments |grep consumer
  oc get deployments |grep consumer|awk '{ print $1}'|xargs -L 1 echo oc scale --\ replicas=0 deployment |bash

  oc get deployments |grep producer
  oc get deployments |grep producer|awk '{ print $1}'|xargs -L 1 echo oc scale --\ replicas=0 deployment |bash

  oc get deployments |grep isd-producer-scale-le
  oc get deployments |grep isd-producer-scale-le|awk '{ print $1}'|xargs -L 1 echo oc \ scale --replicas=0 deployment
  |bash

  oc get statefulset |grep connmgr
  oc scale --replicas=0 statefulset.apps/isd-connmgr-main

  oc get deployments |grep db2whrest
  oc scale --replicas=0 deployment db2whrest
  ```

- Scale up IBM Data Cataloging:

  ```
  $oc scale --replicas=1 deployment <deployment_name isd-db2whrest
  $oc get deployments |grep db2whrest
  isd-db2whrest                   1/1     1            1           6d20h

  oc get deployments |grep consumer
  oc get deployments |grep consumer|awk '{ print $1}'|xargs -L 1 echo oc scale --\ replicas=10 deployment |bash

  oc get deployments |grep producer
  oc get deployments |grep producer|awk '{ print $1}'|xargs -L 1 echo oc scale --\ replicas=10 deployment |bash

  oc get deployments |grep connmgr
  oc scale --replicas=1 statefulset.apps/isd-connmgr-main
  ```

VLAN Testing

Use the following commands to verify that the VLAN is present in the containers.

Note: Addresses are hex byte reverse.

```
sh-4.4# cat /proc/net/route
Iface  Destination      Gateway   Flags  RefCnt  Use  Metric  Mask   MTU     Window  IRTT

br4001 0064640A         00000000  0001     0       0     0    00FFFFFF  0       0       0
br4001 0064640A         0164640A  0003     0       0     0    00FFFFFF  0       0       0
sh-4.4#
```

This is what expecting to see for the `10.100.100.0` network through `10.100.100.1` router.

A utility container can be used with more commands like **apline** or **centos** to confirm that the network endpoints are reachable.

Using **/proc** to determine assigned IP address.

```
sh-4.4# cat /proc/net/fib_trie | grep "|--" | egrep -v "0.0.0.0"
                    |-- 10.100.100.0
                    |-- 10.100.100.11
...
            |-- 127.0.0.0
           |-- 127.0.0.1
                    |-- 10.100.00.0
                    |-- 10.100.100.11
...
            |-- 127.0.0.0
           |-- 127.0.0.1
        |-- 127.255.255.255
```

Testing pods:
From the virtual machine or target cluster, ping the container or pod IP through VLAN.

After completion of all steps, create IBM Data Cataloging data connection. For the procedure, see Creating an IBM Storage Scale data source connection.

Multiple connection managers

Multiple connection managers are a new capability that is designed to enhance scanning performance and enable parallel ingestion. It proves especially valuable in scenarios where data sources are geographically dispersed and need to be scanned as remote sources.

With this new capability, users can take advantage of two primary deployment scenarios:

- The first scenario involves deploying multiple connection managers within the same cluster, allowing for efficient coordination and distribution of scanning tasks. It enables optimized resource utilization and faster processing of data.
- The second scenario involves adding external nodes to the system and deploying one or more connection managers on these nodes. This distributed setup further enhances scanning performance by using extra computing resources and enabling parallel execution of scanning operations across multiple nodes.
  Note:
  - Multiple connection managers improve scanning performance, but it is necessary to understand that the increase in scanning speed does not increase in the performance of indexing records into the database.
    The indexing process might have its own limitations and dependencies that might impact overall performance.

  Example deployment scenario:
  - Main cluster located in Mexico with 2 connection managers deployments.
  - One remote worker located in France with 3 connection manager deployed to scan France data sources.
  - One remote worker located in Canada with 2 connection manager deployed to scan Canada data sources.
    Example:

    ```
    kind: SpectrumDiscover
    apiVersion: spectrum-discover.ibm.com/v1alpha1
    metadata:
      name: spectrumdiscover-sample
      namespace: discover
    spec:
      license:
        accept: true
      doInstall: true
      rwx_storage_class: ibmc-file-gold-gid
      connmgr:
        site: mexico
        replicas: 2
        extraLocations:
          - site: france
            locationType: remote
            replicas: 2
          - site: canada
            locationType: remote
            replicas: 3
            affinity:
      tolerations:
      - effect: PreferNoSchedule
        key: isd
        operator: Exists
    ```

    In case it's required to modify main location needs to be specified it onsite property as follows:

    ```
    connmgr:
        site: france
    ```

  Note: In case site does not exist in a statefulset as example.com or empty, for instance, internal scheduler assigns it to any type local connection manager available.

Enabling feature for skipping the snapshot directories

When the skipping feature is enabled, you can skip the metadata ingestion to the snapshot directories.

Before you begin

- You must set the following environment variables in the IBM Data Cataloging:
  - *NFS_SKIP_SNAPSHOT_DIRS*
  - *INCLUDE_SCALE_SNAPSHOTS*
- Set the environment variable by using the configmap '**connmgr**'.

About this task

To enable the feature for skipping *NFS_SKIP_SNAPSHOT_DIRS* the snapshot directories, you need to set the following environment variable in IBM Data Cataloging.

NFS_SKIP_SNAPSHOT_DIRS
: This environment variable is used specifically for an NFS connection. When the variable is set to 'True', IBM Data Cataloging skips the scanning of snapshot directories. When the variable is set to 'False' or not set at all, it scans the snapshot directories.

INCLUDE_SCALE_SNAPSHOTS
: When the *INCLUDE_SCALE_SNAPSHOTS* variable value is set to 'false' (default value), the IBM Data Cataloging scan excludes all the files that are inside the .snapshots directories, otherwise, if the variable value is set to 'true', the scan includes all the files, including the .snapshots directories.

- **Enabling skip snapshot directories feature on Red Hat OpenShift**
  The following procedure helps you set the configuration for skipping snapshot directories by using *NFS_SKIP_SNAPSHOT_DIRS* and *INCLUDE_SCALE_SNAPSHOTS* variable.

Enabling skip snapshot directories feature on Red Hat® OpenShift®

The following procedure helps you set the configuration for skipping snapshot directories by using *NFS_SKIP_SNAPSHOT_DIRS* and *INCLUDE_SCALE_SNAPSHOTS* variable.

Before you begin

Add the following variables as needed to the configmap and set the value to 'True'.

- *NFS_SKIP_SNAPSHOT_DIRS*
- *INCLUDE_SCALE_SNAPSHOTS*

Procedure

1. Issue the following command to change the current project:

```
oc project ${PROJECT_NAME}
```

2. Issue the following command to edit the configmap:

```
oc edit configmap connmgr
```

After you run the preceding command, the output looks similar to the following as shown here. You can modify the following file and add or update the variables and its value in the **data** section.

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: v1
data:
  CONNMGR_ENDPOINT: /connmgr/v1/
  CONNMGR_PROTOCOL: http
  NFS_SKIP_SNAPSHOT_DIRS: "True"
kind: ConfigMap
metadata:
  creationTimestamp: "2022-08-26T19:55:39Z"
  name: connmgr
  namespace: spectrum-discover
  ownerReferences:
  - apiVersion: spectrum-discover.ibm.com/v1alpha1
    kind: SpectrumDiscover
    name: spectrumdiscover
    uid: 001fb331-c2ef-4b28-8827-5c7d6a702904
  resourceVersion: "13761646"
  uid: bf621847-4c00-44c9-a7f2-4d3430ede67b
```

3. Save the preceding updated object and issue the following command to verify that data is updated.

```
oc get configmap connmgr -o yaml
```

After you run the preceding command, the output looks similar to the following as shown here:

```
apiVersion: v1
data:
  CONNMGR_ENDPOINT: /connmgr/v1/
  CONNMGR_PROTOCOL: http
  NFS_SKIP_SNAPSHOT_DIRS: "True"
kind: ConfigMap
metadata:
  creationTimestamp: "2022-08-23T11:33:40Z"
  name: connmgr
  namespace: spectrum-discover
  ownerReferences:
  - apiVersion: spectrum-discover.ibm.com/v1alpha1
    kind: SpectrumDiscover
    name: spectrumdiscover
    uid: 64aa30c1-d621-47f5-9ee8-763ee9386bbd
  resourceVersion: "293269249"
  uid: e54ebffa-be11-4d05-82e4-03bc94edbce6
```

4. Issue the following command to verify that the update is made in the connection manager pod:

```
oc -n ibm-data-cataloging delete sts/isd-connmgr-main
```

Managing IBM Data Cataloging rules when integrated with CAS

IBM Data Cataloging provides the ability to integrate with Content-Aware Storage to configure file ingestion filtering. This feature enables you to filter files from a data source based on specific criteria, such as file type or content, and ingest them into the corresponding domain.

1. Ensure that you have IBM Fusion 2.11.0 version.
2. Install the IBM Data Cataloging to 2.3.0 version. For more information, see Installing IBM Data Cataloging.
   Important: If you face any issues during the installation of the IBM Data Cataloging 2.3.0, then see Data Cataloging upgrade issues to resolve the issue.
3. Connection to Scale Remote Filesystem from IBM Data Cataloging with live events enabled. For more information, see IBM Storage Scale data source connections.
4. Configure file-ingestion filtering using IBM Data Cataloging on the Content-Aware Storage. For more information, see Integrating IBM Data Cataloging for file filtering.
5. When IBM Fusion Data Cataloging is enabled to perform file filtering on Content-Aware Storage, a default rule is automatically created. This rule filters files based on the file types such as `pdf`, `doc`, `docx`, `txt`, `html`, `markdown`, `md`, `pptx`, `jpeg`, `png`, `bmp`, `xslx`, `asciidoc`, `adoc`, `xhtml`, `csv`, `webp`, `wav`, `ogg`, `opus`, `mp3`.
   For example:

```
apiVersion: datacataloging.ibm.com/v1alpha1
metadata:
  name: rule-allowed-filetypes
  namespace: ibm-data-cataloging
spec:
  metadata:
    - field: filetype
      operator: in
      value:
        - pdf
        - doc
        - docx
        - txt
        - html
        - markdown
        - md
        - pptx
        - jpeg
        - png
        - bmp
        - xslx
        - asciidoc
        - adoc
        - xhtml
        - csv
        - webp
        - wav
        - ogg
        - opus
        - mp3
```

6. Use the following command to extend the list of supported files, you can update the default rule by patching the current Rule object.
   For example, to include the file types such as `pdf`, `txt`, `html`, `json`, `md`, `docx`, `pptx`, `bmp`, `jpeg`, `png`, `tiff`, and `sh`.

```
DCS_NS=ibm-data-cataloging
oc patch rule allowed-filetypes -n $DCS_NS --type merge -p '{"spec": {"metadata": [{"field": "filetype","operator":
"in","value": ["doc", "docx", "pdf", "txt", "gif", "html", "json", "md", "pptx", "bmp", "jpeg", "png", "tiff", "sh" ]}]}}'
```

7. In addition to metadata rule filtering, you can perform a complementary filter based on content using regex-based content search capability. For more information, see Identifying the required regex expressions.
   For example, to filter files based on the presence of an email pattern, you can add a content requirement to the default rule:

```
DCS_NS=ibm-data-cataloging
oc -n $DCS_NS patch rule allowed-filetypes --type=merge -p '{"spec":{"content":>
[{"field":"email","operator":"regex","value":"\b[\w\.=-]+@[\w\.-]+\.[\w]{2,3}\b"}]}}'
```

Administering IBM Data Cataloging

This section provides information on how to administering the IBM Data Cataloging capabilities.

- **AI-enabled capabilities**
  IBM Fusion Data Cataloging is expanding its AI-driven data management capabilities with two significant innovations. The new IBM Data Cataloging Model Context Protocol (MCP) server that enables multiple large language models (LLMs) to connect to and operate across a unified set of cataloging tools.
- **Managing user access**
  The Data cataloging environment provides access to users and groups. The role that is assigned to a user or group determines the functions that are available. Users and groups can also be associated with collections that use policies that determine the metadata that is available to view.
- **Managing metadata policies**
  Policies might be used to automatically tag a set of documents on a periodic basis. In addition, policies might be used to send sets of documents to be deep-inspected by a registered application.
- **Using content search policies**
  You can define regular expressions to search for and create policies that use the regular expressions.
- **Tiering data by using ScaleILM application**
  Use the IBM Data Cataloging ScaleILM application to move data to different tiers (pools) that are configured on the IBM Storage Scale connection.
- **Copying data by using ScaleAFM application**
  The ScaleAFM application supports copy function between IBM Storage Scale connection and IBM Cloud® Object Storage connection.
- **Exporting metadata to IBM Watson Knowledge Catalog**
  IBM Data Cataloging Watson Knowledge Catalog (WKC) connector supports export of the metadata to either an IBM Cloud instance or to an On-Premise Instance of the Watson Knowledge Catalog .
- **Importing externally curated tags for COS/S3 using import tags application**
  The import tags application is used to import a set of externally curated tags for Cloud Object Storage and S3 services.
- **Performing retention analytics on IBM Storage Protect archive data**
  Use IBM Data Cataloging to perform retention analytics on archive data managed by IBM Storage Protect.
- **Managing tags**
  A tag is a custom metadata field used to enhance system metadata with organization-specific information. For instance, an organization may categorize its storage by project, cost center, or region but the attributes not provided by the storage system itself. Tags enable data classification, search, and reporting based on these organizational dimensions. Use tags to store this additional information and to manage or locate data using metadata that reflects your organizational context.
- **Discover data**
  By discovering your data, you can apply policies that assign tags to your data. You can apply tags to the results of a single search, or you can use policies to automatically apply tags on a periodic basis.
- **Managing applications**
  An application is a program that interfaces with IBM Data Cataloging and can access the source storage. There are many use cases for application, including data content inspection for enriching metadata, data movement or migration, data scrubbing or sanitation, and more. Data is identified by IBM Data Cataloging by policy filter and passed to the application as pointers through a messaging queue. Then, the application performs whatever work is appropriate on the source data and returns a completion status back to IBM Data Cataloging, which might or might not include enriched metadata for the records. If it does include enriched metadata, IBM Data Cataloging catalogs that metadata and makes it immediately searchable.
- **Reports**
  Reports can be generated upon applying tags to a set of data.
- **Creating a IBM Data Cataloging application for metadata-based policies**
  Provides information about how to create IBM Data Cataloging application for metadata-based policies.
- **Graceful shutdown**
  Provides detailed instructions to put IBM Data Cataloging in idle state on an OpenShift® environment.
- **REST API for IBM Data Cataloging**
  REST API reference documentation provides endpoint specifications, request and response formats, and authentication details for programmatic access to IBM Spectrum Discover services.
- **Data protection**
  Data protection features in IBM Spectrum Discover support backup, disaster recovery, and restore operations to protect catalog metadata and system configurations.
- **Using a third-party data movement application to manage data**
  Introduction to data movement with IBM Data Cataloging.

AI-enabled capabilities

IBM Fusion Data Cataloging is expanding its AI-driven data management capabilities with two significant innovations. The new IBM Data Cataloging Model Context Protocol (MCP) server that enables multiple large language models (LLMs) to connect to and operate across a unified set of cataloging tools.

Complementing this, IBM Fusion Data Cataloging is introducing a technology preview of Content-based AI-enabled autotagging, which allows users to selectively analyze data and automatically generate tags based on document content.

Together, these capabilities enhance governance, automation, and intelligent discovery across enterprise data ecosystems.

- **IBM Data Cataloging MCP server**
  IBM Data Cataloging Service offers a new capability of a Model Context Protocol (MCP) server. It is designed to enable seamless interoperability between multiple large language models (LLMs) and a unified suite of cataloging and data management tools. Through a centralized control layer, the MCP server allows LLMs to securely connect, interact, and run operations by using tools that are registered within the MCP environment.
- **Content-based AI-enabled autotagging**
  IBM Data Cataloging provides AI-powered content-based tagging to automatically classify unstructured data by using the actual content of files or objects, not only their metadata. This capability leverages an inference model that is accelerated by external GPUs to propose and apply business-relevant tags across large datasets with minimal manual effort.

IBM Data Cataloging MCP server

IBM Data Cataloging Service offers a new capability of a Model Context Protocol (MCP) server. It is designed to enable seamless interoperability between multiple large language models (LLMs) and a unified suite of cataloging and data management tools. Through a centralized control layer, the MCP server allows LLMs to securely connect, interact, and run operations by using tools that are registered within the MCP environment.

Integrated with IBM Data Cataloging, the MCP server enhances how AI-driven systems access, analyze, and organize metadata from diverse data sources. It provides a consistent interface for tool discovery, execution, and lifecycle management, reducing the need for direct integrations between individual LLMs and the underlying cataloging infrastructure.

Key capabilities

Natural language queries on Data Cataloging Services (DCS)
: It enables the user to explore and query cataloged information by using natural language, without requiring knowledge of query syntax. This capability improves data accessibility and empowers business users to retrieve insights more intuitively.

LLM-based tags and labels catalog suggestions
: It assists the user in creating, managing, and discovering tags more efficiently. The LLM can use the MCP tools to suggest relevant tags based on dataset content and highlight existing tags across the catalog. This capability helps maintain metadata consistency, reduces manual effort, and enhances the overall accuracy of data classification.

Policy-driven auto-tagging and management
: It allows the user to view existing catalog policies and create or run auto-tagging policies that are powered by AI. These policies automatically classify and label datasets based on predefined rules and model insights. This helps enforce governance standards, improves compliance, and ensures that metadata remains accurate and up to date over time.

Before you begin

Before you enable and use the MCP server within the IBM Data Cataloging environment, ensure that your system meets the following requirements. These prerequisites are necessary to authenticate with your cluster, enable the MCP components, and verify the available service endpoints.

Ensure that the following tools are installed and accessible from your local environment:

- Red Hat OpenShift® CLI (`oc`)
- `curl`

After verifying the prerequisites, log in to your OpenShift cluster by using CLI.

Setting up the MCP server

To enable MCP server capabilities, ensure that the Data Catalog has the required features that are enabled for the MCP server.

Note: The `DCS_NS` variable specifies the namespace where the DCS instance is deployed. By default, it is `ibm-data-cataloging`, and adjusts according to the namespace where it is installed.
Run the following command:

```
DCS_NS=ibm-data-cataloging
```

```
oc -n ${DCS_NS} patch SpectrumDiscover $(oc -n ${DCS_NS} get spectrumdiscover -o jsonpath='{.items[*].metadata.name}') \
  --type=merge -p '{
    "spec": {
      "enabled_features": {
        "natural-language-queries": {
          "enabled": true
        }
      }
    }
  }'
```

After the patch is successfully applied, the `SpectrumDiscover` loads the MCP server with new capabilities.

Verifying that MCP server is running
: After enabling the features, verify that the MCP server pods are up and running.

  Run the following command:

  ```
  oc -n ${DCS_NS} get pod -l 'app=isd,role=dcs-ai-mcp-server'
  ```

  Example output:

  ```
  isd-dcs-ai-mcp-server 1/1 Running 0 3h
  ```

  If you see pods in the Running state, the MCP server is successfully up and operational.

Connecting to MCP server

After the MCP server is deployed and prerequisites are met, you can retrieve the service endpoints to enable LLMs or interact with its tools. The MCP server exposes both SSE and HTTP endpoints for integration.

Retrieve MCP endpoints on Linux/macOS

```
oc get routes -n ${DCS_NS} -o json \
  | jq -r '.items[]
  | select(.metadata.name | test("mcp"))
  | "Route: \(.metadata.name)\nHost: \(.spec.host)\nSSE Endpoint: https://\(.spec.host)/mcp\nHTTP Endpoint: https://\
(.spec.host)/mcp/http\n"'
```

Example output:

```
Route: dcs-mcp-route
Host: dcs-mcp-route-ibm-data-cataloging.apps.com
SSE Endpoint: https://dcs-mcp-route-ibm-data-cataloging.ibm.com/mcp
HTTP Endpoint: https://dcs-mcp-route-ibm-data-cataloging.com/mcp/http
```

The command outputs -

- `Route`: The OpenShift route for the MCP server.
- `Host`: The host URL assigned by OpenShift.
- `SSE Endpoint`: The URL to connect through server-sent events.
- `HTTP Endpoint`: The URL to connect through standard HTTP requests.
  This connection provides the necessary endpoints to integrate LLMs or other clients with the MCP server.

Initialization

Example of how to initialize the MCP server:

```
curl -i -X POST \
  <MCP http endpoint> \
  -H "Content-Type: application/json" \
  -H "Accept: application/json, text/event-stream" \
  -d '{
    "jsonrpc": "2.0",
    "id": 1,
    "method": "initialize",
    "params": {}
  }' --insecure
```

Replace all placeholder values before sending the request.

Example Output:

```
HTTP/1.1 200 OK
access-control-allow-headers: Content-Type
access-control-allow-methods: GET, POST, OPTIONS
access-control-allow-origin: *
cache-control: no-cache, no-transform
content-type: text/event-stream
mcp-session-id: xxxxxxxxxxxxxxxxxxxx
date: Tue, 16 Dec 2025 19:17:50 GMT
transfer-encoding: chunked
set-cookie: xxxxxxxxxxxxxxxxxxxxxxxxx; path=/; HttpOnly; Secure; SameSite=None
```

Save the `mcp-session-id`, it is necessary to send requests.

List Tools

```
curl -X POST \
  <MCP http endpoint> \
  -H "Content-Type: application/json" \
  -H "Accept: application/json, text/event-stream" \
  -H "mcp-session-id: <mcp-session-id>" \
  -d '{
    "jsonrpc": "2.0",
    "id": 1,
    "method": "tools/list",
    "params": {}
  }' \
  --insecure
```

Replace all placeholder values before sending the request.

MCP server - Tools

The following section describes the new tools that are introduced in the MCP server, which is designed to improve interaction with data sources, manage credentials, policies, and tagging. Each tool serves a specific purpose and can be used independently or together to optimize data management and automation workflows.

`dcs_set_credentials`
The Set Credentials tool allows the user to configure the credentials that are required to connect the MCP server to the Data Cataloging Service.

Tool name	Arguments	Request

`dcs_set_credentials`	DCS URL, Username, Password	`curl -X POST`
		`<mcp http endpoint>`
		`-H "Content-Type: application/json"`
		`-H "Accept: application/json, text/event-stream"`
		`-H "mcp-session-id: <mcp-session-id>"`
		`-d '{ "jsonrpc": "2.0", "id": 1, "method": "tools/call", "params": { "name": "dcs_set_credentials", "arguments": { "cluster_url": "<DCS URL>", "username": "user", "password": "password" } } }'`
		`--insecure`

	DCS URL, Authentication token	`curl -X POST`
		`<mcp http endpoint>`
		`-H "Content-Type: application/json"`
		`-H "Accept: application/json, text/event-stream"`
		`-H "mcp-session-id: <mcp-session-id>"`
		`-d '{ "jsonrpc": "2.0", "id": 1, "method": "tools/call", "params": { "name": "dcs_set_credentials", "arguments": { "cluster_url": "<DCS URL>", "token": "<token>" } } }'`
		`--insecure`

Usage: It is used when starting a new session or when you need to update your connection credentials.

Replace all placeholder values before sending the request.

Obtaining a username and password:

You can retrieve the user and password by using the following commands:

macOS

```
#User
oc get secret keystone -n ${DCS_NS} -o jsonpath="{.data.user}" | base64 -D; echo
```

```
#Password
oc get secret keystone -n ${DCS_NS} -o jsonpath="{.data.password}" | base64 -D; echo
```

Linux

```
#User
oc get secret keystone -n ${DCS_NS} -o jsonpath="{.data.user}" | base64 --decode; echo
```

```
#Password
oc get secret keystone -n ${DCS_NS} -o jsonpath="{.data.password}" | base64 --decode; echo
```

Obtaining DCS URL:
Retrieve the DCS URL by using the following command:

```
DCS_CONSOLE=$(oc -n ${DCS_NS} get route console -o jsonpath='{"https://"}{.spec.host}')
echo $DCS_CONSOLE
```

Obtaining token:
Retrieve the token by using the following command:

```
curl -k -I -u <username>:<password> ${DCS_CONSOLE}/auth/v1/token | grep x-auth-token
```

Response:

```
x-auth-token: <token>
```

`dcs_file_search`

The File Search tool enables you to search across cataloged data sources by using natural language queries. It automatically converts user input into structured queries that retrieve relevant files, datasets, or metadata entries.

With this capability, you can find information quickly without needing to understand query syntax or underlying database languages. By bridging natural language with structured search, File Search improves accessibility for both technical and non-technical users, and accelerates data discovery.

Tool name	Argument	Request
`dcs_file_search`	Query	`curl -X POST <mcp http endpoint> -H "Content-Type: application/json" -H "Accept: application/json, text/event-stream" -H "mcp-session-id: <mcp-session-id>" -d '{ "jsonrpc": "2.0", "id": 1, "method": "tools/call", "params": { "name": "dcs_file_search", "arguments": { "query": "SELECT * FROM METAOCEAN LIMIT 1" } } }' --insecure`

Replace all placeholder values before sending the request. Replace the query as needed.

Example prompt:

```
Show all CSV files containing customer transactions from 2023
```

The tool retrieves matching entries from the registered data sources.

`dcs_get_registered_tags`

The Get Registered Tags tool retrieves a complete list of all tags that are registered in the cataloging system. It helps users and AI agents review existing classifications before creating or suggesting new ones, ensuring that tag definitions remain consistent and non-duplicative.

This tool is useful for maintaining a clean and standardized tagging structure across multiple datasets and departments.

Tool name	Request
`dcs_get_registered_tags`	`curl -X POST <mcp http endpoint> -H "Content-Type: application/json" -H "Accept: application/json, text/event-stream" -H "mcp-session-id: <mcp-session-id>" -d '{ "jsonrpc": "2.0", "id": 1, "method": "tools/call", "params": { "name": "dcs_get_registered_tags", "arguments": { } } }' --insecure`

Replace all placeholder values before sending the request.

Example use case:

Before adding a new Financial tag, you can check whether a similar tag exists in the catalog.

`dcs_get_recommend_tags`

The Recommend Tags tool uses LLM-powered analysis to suggest relevant tags for cataloged files and datasets. By examining metadata and contextual information, the tool recommends tags that help classify data more accurately and consistently.

This feature significantly reduces the time that is spent on manual tagging and enhances catalog quality by ensuring that all datasets are properly labeled.

Tool name	Argument	Request

Tool name	Argument	Request
dcs_get_reco mmend_tags	Query	curl -X POST <mcp http endpoint> -H "Content-Type: application/json" -H "Accept: application/json, text/event-stream" -H "mcp-session-id: <mcp-session-id>" -d '{ "jsonrpc": "2.0", "id": 1, "method": "tools/call", "params": { "name": "dcs_get_recommend_tags", "arguments": { "query": "SELECT * FROM METAOCEAN LIMIT 10" } } }' --insecure

Replace all placeholder values before sending the request. Replace the query as needed.

Example prompt:

```
Recommend me some tags to classify my CSV files
```

The tool retrieves tags recommendation based on the metadata.

`dcs_create_tag`

The Create Tag tool enables you to define and create new tags.

Tool name	Arguments	Request
dcs_create_t ag	tagName, type	curl -X POST <mcp http endpoint> -H "Content-Type: application/json" -H "Accept: application/json, text/event-stream" -H "mcp-session-id: <mcp-session-id>" -d '{ "jsonrpc": "2.0", "id": 1, "method": "tools/call", "params": { "name": "dcs_create_tag", "arguments": { "tagName": "TEST", "type": "VARCHAR(256)" } } }' --insecure

Replace all placeholder values before sending the request.

Example prompt:

```
Create the tags that you recommend me.
```

The tool creates tags based on the recommendations.

`dcs_get_policies`

The Get Policies tool retrieves and lists all existing cataloging policies within the system. Policies define automated rules or governance conditions that are applied to data, metadata, or tags.

This tool provides visibility into the active policies that govern cataloged assets. It also helps users understand which rules are being enforced and how they affect data classification or compliance workflows.

Tool name	Request
dcs_get_polici es	curl -X POST <mcp http endpoint> -H "Content-Type: application/json" -H "Accept: application/json, text/event-stream" -H "mcp-session-id: <mcp-session-id>" -d '{ "jsonrpc": "2.0", "id": 1, "method": "tools/call", "params": { "name": "dcs_get_policies", "arguments": { } } }' --insecure

Replace all placeholder values before sending the request.

Example use case:

A data steward can use this tool to view all current auto-tagging and compliance policies before creating new ones.

`dcs_create_policy`

The Create AutoTag Policy tool enables you to define and deploy new auto-tagging policies that automatically assign tags to cataloged assets based on predefined rules and AI insights. It streamlines governance by automating repetitive classification tasks and ensuring that policies remain consistent across data sources.

This tool currently supports auto-tag policy types, allowing AI models to dynamically apply tags based on metadata patterns or user-defined contextual rules.

Tool name	Arguments	Request
dcs_create_p olicy	polFilter, policyName, schedule, tags	curl -X POST <mcp http endpoint> -H "Content-Type: application/json" -H "Accept: application/json, text/event-stream" -H "mcp-session-id: <mcp-session-id>" -d '{ "jsonrpc": "2.0", "id": 1, "method": "tools/call", "params": { "name": "dcs_create_policy", "arguments": { "polFilter": "filetype='pdf'", "policyName": "TEST", "schedule": "NOW", "tags": {'tagtest':'valtest'} } }' --insecure

Replace all placeholder values before sending the request.

Example prompt:

```
Explore the catalog for security-related flaws and recommend me 5 different tags with proposed values that can be used for
identify and classify the data on the catalog, make sure they are not duplicate.
```

Content-based AI-enabled autotagging

IBM Data Cataloging provides AI-powered content-based tagging to automatically classify unstructured data by using the actual content of files or objects, not only their metadata. This capability leverages an inference model that is accelerated by external GPUs to propose and apply business-relevant tags across large datasets with minimal manual effort.

Overview

Traditional metadata-based tagging relies on file attributes such as name, size, or type. AI-powered content-based tagging goes further: the system reads file content, combines it with a defined business context and then proposes the most relevant tag names and values by consuming a reasoning model.

The feature operates in two stages:

Stage 1: Tag proposal (dry run)
: The system analyzes a sample of files in a pre-configured path, generates tag name suggestions that are aligned with the defined business context, and stores them in the business context ConfigMap. No tags are created or applied in IBM Data Cataloging service at this point. Review the proposal and activate the ones that are approved.

Stage 2: Tag application
: On subsequent runs, the system detects the tags that are activated and proceeds to apply them: It creates the tags in IBM Data Cataloging if they do not exist already, retrieves untagged files or objects from the catalog, uses the reasoning model to determine the appropriate value for each active tag, and enriches the catalog.

The two stages are controlled entirely through the business context ConfigMap and run automatically on a 30-minute schedule. This approach enables consistent tags assignments with no manual categorization, standarizing data management across large data sources and improves discovery and governance capabilities.

Before you begin

Before enabling this feature, ensure that the following prerequisites are met:

- IBM Fusion 2.13 is deployed with access to the IBM Storage Scale filesystem through a Global Data Platform remote mount.
- IBM Data Cataloging is installed at 2.5.3 version and a IBM Data Cataloging service connection to an IBM Storage Scale filesystem is configured and online.
- An AI inference engine is accessible from the cluster.
- The `ai-suggestions` feature toggle is available in your `SpectrumDiscover` custom resource.

Activating content-based autotagging

To activate content-based autotagging, follow these steps:

1. Create the business context ConfigMap.
   Create a ConfigMap named `ai-tag-extraction-context` in the IBM Data Cataloging namespace. This ConfigMap defines the data source, the path to analyze, and the business context that guides the reasoning model.

   ```
   apiVersion: v1
   kind: ConfigMap
   metadata:
     name: ai-tag-extraction-context
     namespace: ibm-data-cataloging
   data:
     connection_names: "<dcs-connection-name>"
     filesystem: "<storage-scale-filesystem-cr-name>"
     path: "/gpfs/<filesystem>/path/to/your/dataset"
     allowed_filetypes: "pdf,txt,docx"
     context: "<<Describe your business use case here. The more specific the context,
       the more relevant the proposed tags. For example: This dataset contains
       HR profiles including job role, employment status, and salary information.
       The goal is to classify these files for talent management and compliance.>>"
   ```

Field	Description
`connection_names`	The name of the IBM Data Cataloging service connection as registered in the catalog.
`filesystem`	Full name of the `filesystems.scale.spectrum.ibm.com` custom resource in the `ibm-spectrum-scale` namespace. Run `oc get filesystems.scale.spectrum.ibm.com -n ibm-spectrum-scale` to find the correct name.
`path`	Full IBM Storage Scale path to the directory containing the files to tag.
`allowed_filetypes`	Comma-separated list of file extensions to process.
`context`	Business context description that guides the reasoning model tag proposals.

2. Enable the feature in the `SpectrumDiscover` custom resource.
   Edit the `SpectrumDiscover` custom resource to enable both the reasoning model proxy and the AI suggestions feature:

   ```
   enabledFeatures:
     aiSuggestions:
       enabled: true
       inferenceEngineHost: "http://<reasoning-model-host>:<port>"
       model: "granite4"
   ```

Field	Description
`inferenceEngineHost`	URL of the reasoning or compatible inference engine accessible from the cluster.
`model`	Model alias to use. The default is `granite4`.

   After saving the custom resource, the operator deploys the following resources automatically:

   - LLM proxy deployment and service (`isd-llm-proxy`)
   - Remote content search deployment and service (`isd-rcs-ai-tags`)
   - AI tag extraction CronJob (`isd-ai-tag-extraction`, runs every 30 minutes)

Using content-based autotagging

First run: tag proposal
: On the first CronJob execution, the system performs a dry run:

1. Reads the business context ConfigMap.
2. Analyzes a sample of files in the configured path by using the reasoning model.
3. Proposes tag names and their possible values.
4. Writes the proposals to a new key **suggested_tags** in the same ConfigMap.
5. Exits without creating or applying any tags in the IBM Data Cataloging service.

After the first run, inspect the ConfigMap to review the proposals:

```
oc get cm ai-tag-extraction-context -n ibm-data-cataloging -o jsonpath='{.data.suggested_tags}' | python3 -m json.tool
```

Example output:

```
{
  "EMPLOYEE_STATUS": { "active": false },
  "JOB_ROLE":        { "active": false },
  "LOCATION":        { "active": false },
  "SALARY_INDICATOR": { "active": false },
  "WORK_ARRANGEMENT": { "active": false }
}
```

On subsequent dry runs without any active tags, the system forwards the existing proposals to the reasoning model. Hence, it does not re-propose the same names. New complementary tags might be added to the list.

Activating tags

Edit the **suggested_tags** value in the ConfigMap to set **"active": true** on the tags you want to apply:

```
oc edit cm ai-tag-extraction-context -n ibm-data-cataloging
```

Change the desired entries:

```
{
  "EMPLOYEE_STATUS": { "active": true },
  "JOB_ROLE":        { "active": true },
  "LOCATION":        { "active": false },
  "SALARY_INDICATOR": { "active": true },
  "WORK_ARRANGEMENT": { "active": false }
}
```

On the next CronJob run (within 30 minutes, or triggered manually), the system:

1. Detects the tags marked **active: true**.
2. Creates those tags in IBM Data Cataloging service if they do not exist.
3. Scans the configured connection to refresh the catalog.
4. Queries for untagged files matching the allowed filetypes and path.
5. Calls the reasoning model to extract tag values.
6. Applies the values to the catalog.

Triggering a manual run

To run the job immediately without waiting for the next scheduled execution:

```
oc create job --from=cronjob/isd-ai-tag-extraction manual-run-$(date +%s) -n ibm-data-cataloging
```

Monitor the run:

```
oc logs -n ibm-data-cataloging -l job-name=manual-run-<timestamp> --follow
```

Monitoring

Verify deployed resources

```
# Business context ConfigMap
oc get cm ai-tag-extraction-context -n ibm-data-cataloging

# Remote content search service (must exist before the CronJob can reach the LLM)
oc get svc isd-rcs-ai-tags -n ibm-data-cataloging

# CronJob schedule and last execution
oc get cronjob isd-ai-tag-extraction -n ibm-data-cataloging

# Recent job runs
oc get jobs -n ibm-data-cataloging -l role=ai-tag-extraction

# RemoteContentSearch CR status
oc describe remotecontentsearch ai-tags -n ibm-data-cataloging
```

Check feature conditions

```
oc get spectrumdiscover -n ibm-data-cataloging \
  -o jsonpath='{.items[0].status.conditions}' | python3 -m json.tool
```

Verify by using **AISuggestions** with **status: "True"** to confirm that the feature is running correctly.

Disabling content-based autotagging

Set **enabled: false** in the **SpectrumDiscover** custom resource:

```
spec:
  enabledFeatures:
```

```
  aiSuggestions:
    enabled: false
```

Troubleshooting

Symptom	Likely cause	Resolution
The **isd-rcs-ai-tags** service does not exist after enabling the feature.	The **filesystem** field in the ConfigMap does not match the exact CR name in **ibm-spectrum-scale** namespace.	1. Run the `oc get filesystems.scale.spectrum.ibm.com -n ibm-spectrum-scale` command. 2. Update the ConfigMap with the exact name.
PVC **isd-rcs-ai-tags** remains in **Pending** state.	A previous PV with the same name is in **Released** state.	Run the `oc delete pv isd-rcs-ai-tags` command and trigger a reconcile by annotating the **ai-tags RemoteContentSearch** CR: `oc annotate remotecontentsearch ai-tags -n ibm-data-cataloging reconcile-trigger="$(date +%s)" --overwrite`
CronJob exits with **no valid connections**.	The **connection_names** value does not match an online IBM Storage Scale connection.	Verify that the connection is online and the name matches exactly.
All files fail with **AI response contained no valid tag values**.	The inference engine is unreachable or returned an unexpected response.	Check LLM proxy logs: `oc logs -n ibm-data-cataloging -l role=llm-proxy`

Use cases

Human Resources document classification
: HR departments store large volumes of sensitive documents including contracts, performance reviews, and compliance records. Content-based tagging automatically classifies these files with tags such as **employee_status**, **job_role**, **salary_indicator**, or **compliance_category**, enabling governance and discovery without manual effort.

Workforce analytics
: Data engineers building analytics pipelines can tag files with dimensions such as **location**, **department**, or **contract_type** to create filtered subsets for downstream processing without modifying the source data.

Managing user access

The Data cataloging environment provides access to users and groups. The role that is assigned to a user or group determines the functions that are available. Users and groups can also be associated with collections that use policies that determine the metadata that is available to view.

User and group access can be authenticated by Data Cataloging, a Lightweight Directory Access Protocol (LDAP) server, the IBM Cloud® Object Storage, or using the Red Hat®OpenShift® credentials. If you use Data Cataloging, LDAP, or IBM COS as authentication methods, then it is required to follow the same authentication process by entering the associated username and password to access the Data Cataloging user interface.

If you use Red HatOpenShift as authentication method, then log in by default through Single-Sign On, unless this option is explicitly disabled. The administrator role can manage the user access functions. For more information, see Initial login.

Note: A local user that is created on the Data cataloging system must use a user name and password to log in. Users from an external LDAP or IBM Cloud® Object Storage domain must include the domain name as a prefix to the user name with a forward slash (/), such as **"<domain>/<user>"**. The domain name is the name that is given to the external authentication domain in Data cataloging.

Roles

Roles determine how users and groups can access records or the Data cataloging environment.

If a user or group is assigned to multiple roles, the least restrictive role is used. For example, if a user is assigned to the Data User role but is also included in a group that is assigned to the Data Admin role, that user has the privileges of the Data Admin role.

The following roles are available:

Admin
: This role can create users, groups, and collections. This role can also manage connections to Lightweight Directory Access Protocol (LDAP) and IBM Cloud Object Storage domains. This role can use the Application Management APIs to install, upgrade, or delete Data cataloging applications that use the Data cataloging API service.

Data Admin
: Users with this role can access all metadata that is collected by Data cataloging and is not restricted by policies or collections. This role can also define tags and policies, including policies that assign a collection value to a set of records.
Note:
The built-in **Collection** tag is a special tag. This tag can be set only by users with the Data Admin role. All other tags can be set by any user with the Data User or Data Admin or Collection Admin role.

  Users with this role can also edit local users and local groups and assign roles and collections to users and groups.

Collection Admin
: The Collection Admin role is a bridge between the Data Admin role and the Data User role. Users with the Collection Admin role can:

- Create, update, and delete the policies for the collections that they administer.
- View, update, and delete policies of data users for the collections they administer. They cannot delete a policy if it has a collection that they do not administer.
- Add users to collections that they administer. These data users can access to a particular collection, which means that they can access to the records marked with that collection value.

- List any type of tag and create or modify `Characteristics` tags. They cannot create, modify, or delete `Open` and `Restricted` tags. These permissions are the same as the ones associated with the Data User role.

Collection User
: Users with this role can access metadata that is collected by IBM Data Cataloging, but metadata access can be restricted by the collections that are assigned to users in this role.

- Users assigned with the Collection User role can:
  - Run scans of collections the user is assigned.
  - View policies of the collections the user is assigned.
  - List any type of tag.
- Users assigned with the Collection User role cannot:
  - Create, update, and delete any policies.
  - Create, modify, and delete any tags.

Data User
: Users with this role can access metadata that is collected by Data cataloging, but metadata access can be restricted by the collections that are assigned to users in this role. This role can also define tags and policies, based on the collections to which the role is assigned.

Service User
: This role is assigned to accounts for IBM® service and support personnel.

- **Initial login**
  To log in IBM Data Cataloging graphical user interface, use the following information.
- **Resetting the sdadmin password**
  If the `sdadmin` password changes and you forget the password, you can access the keystone container and run the **reset_sdadmin_details.sh** script to reset the password to the original password.
- **Password policies**
  Data cataloging 2.0.2.1 introduces password policies for the local users who are configured in the default authentication domain.
- **Changing password**
  To change the password, you need to provide the existing and new password.
- **Managing user accounts**
  The administrator can add and manage local user accounts, which are authenticated by IBM Data Cataloging. The administrator can also assign local or LDAP and IBM Cloud Object Storage-managed users to roles and collections.
- **Managing groups**
  The administrator can create and manage local groups that are authenticated by IBM Data Cataloging. The administrator can also assign local or Lightweight Directory Access Protocol (LDAP) and IBM Cloud Object Storage system-managed groups to roles and collections.
- **Managing collections**
  Collections are logical groups of metadata that share a common member access list. For example, a collection can restrict metadata within a research project to the members of the project only. Members outside of the project cannot see the metadata.
- **Managing LDAP and IBM Cloud Object Storage System connections**
  The administrator can create and manage connections to LDAP or IBM Cloud Object Storage System servers that provide authentication for IBM Data Cataloging users.

Initial login

To log in IBM Data Cataloging graphical user interface, use the following information.

IBM Data Cataloging service provides SSO capability. To know more about SSO, see https://www.ibm.com/topics/single-sign-on .

Use your OpenShift® Container Platform credentials to log in IBM Data Cataloging user interface by using SSO.

When Single Sign-On (SSO) is disabled as the default authentication method, the following default login credentials can be used for the IBM Data Cataloging graphical user interface:

Username
: `sdadmin`

Password
: `Passw0rd`

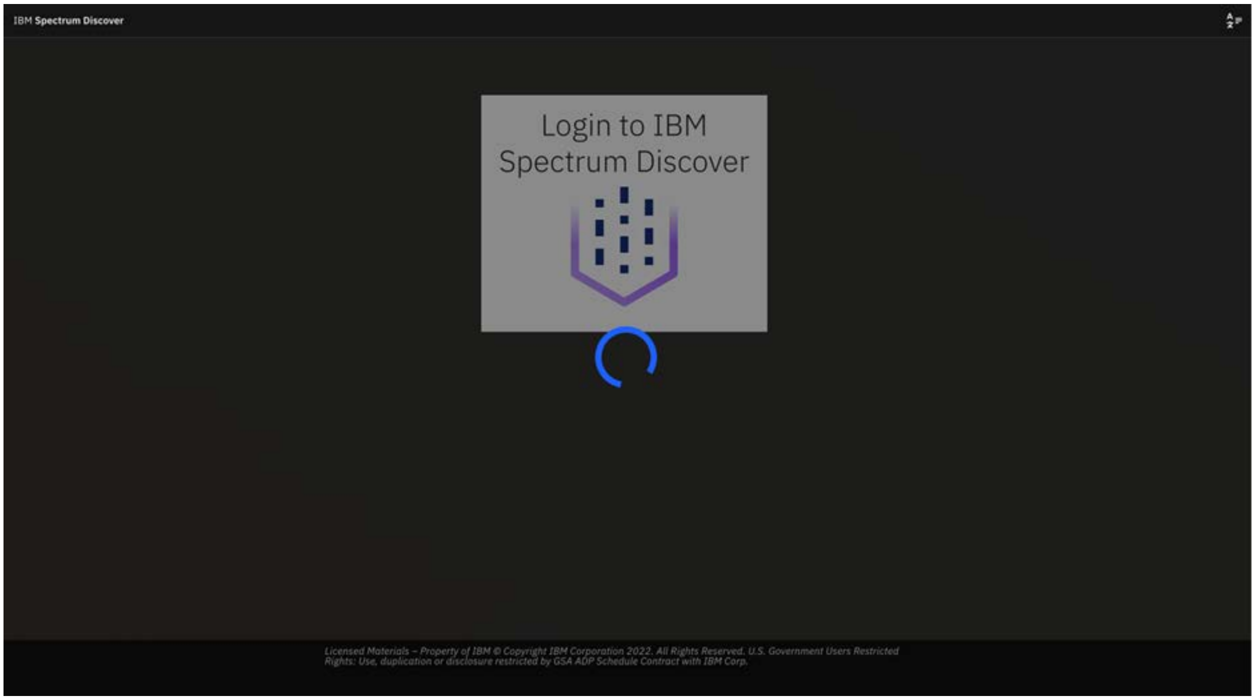
Important:

- When the SSO is disabled, the administrator must change the password during the initial login.
- If you have been using SSO login for a long time, it is not recommended to switch to traditional USERNAME/PASSWORD login because you can face authentication issues.

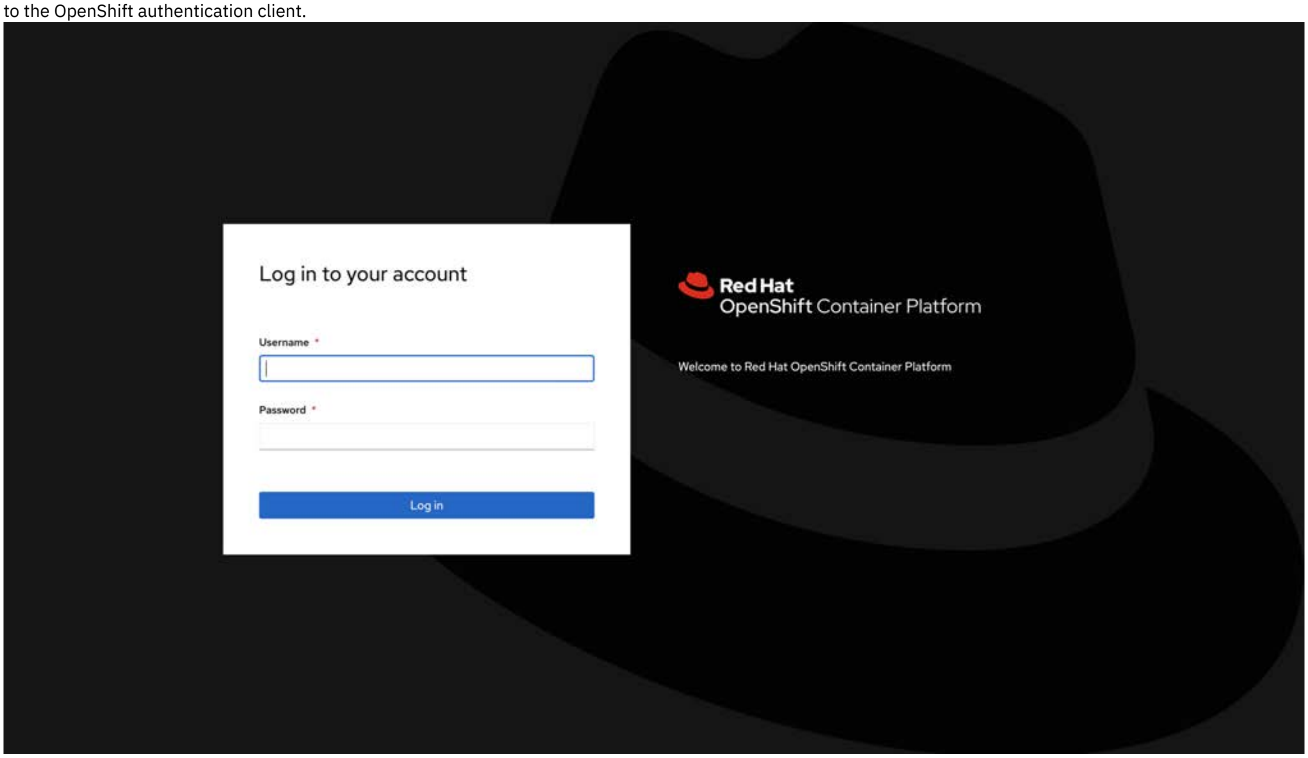
Log in to IBM Data Cataloging Dashboard with SSO enabled

Follow the steps to log in to the IBM Data Cataloging Dashboard:

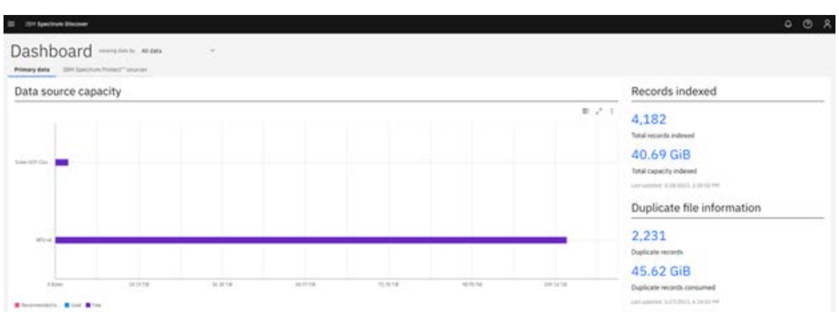
1. Start the IBM Data Cataloging graphical user interface, and you get the logging screen validation as follows.



2. Wait for the validation for OpenShift Platform session. If you have an active session, you are redirected to Data Cataloging Dashboard, otherwise you are redirected to the OpenShift authentication client.



3. Enter your OpenShift credentials and click Log in button.
   You are redirected to IBM Data Cataloging main dashboard.

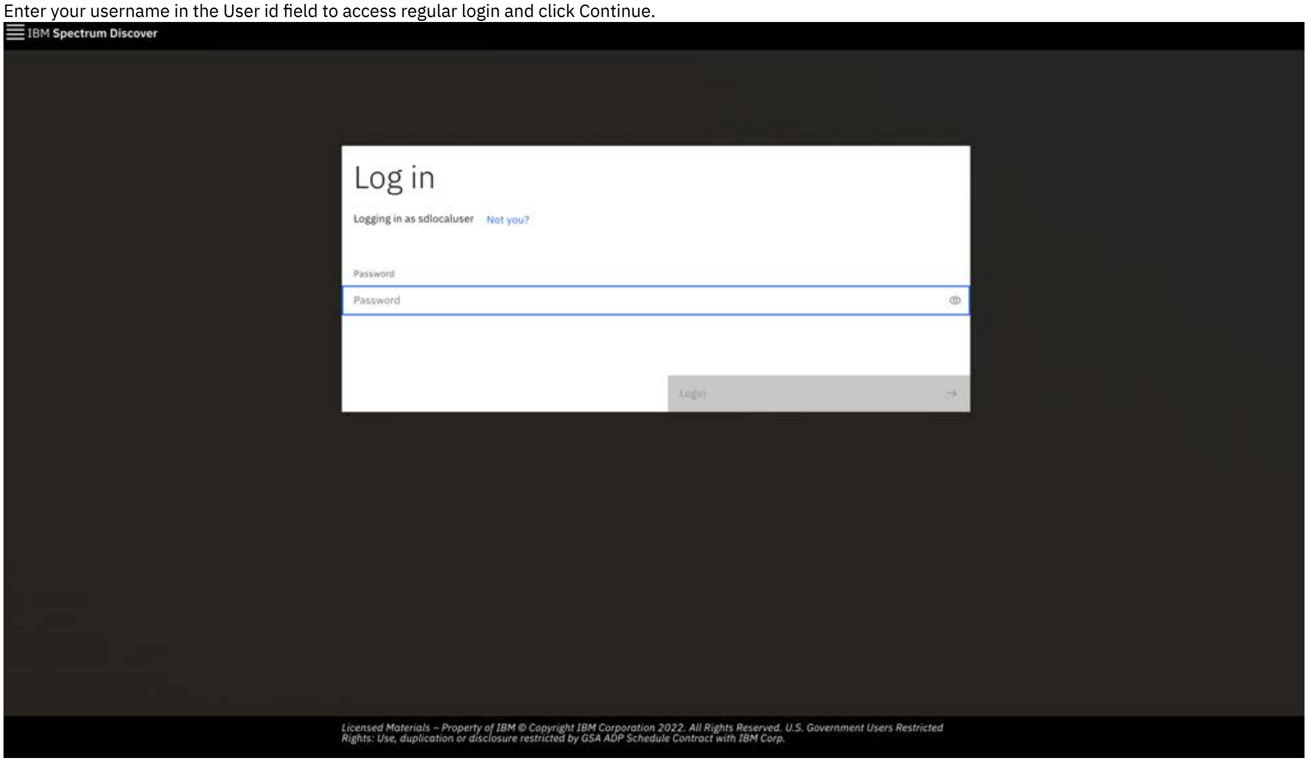


Note: Consider that IBM Data Cataloging for security renews the session every hour, and there are cases in which it is necessary to re-enter the OpenShift Container Platform credentials.

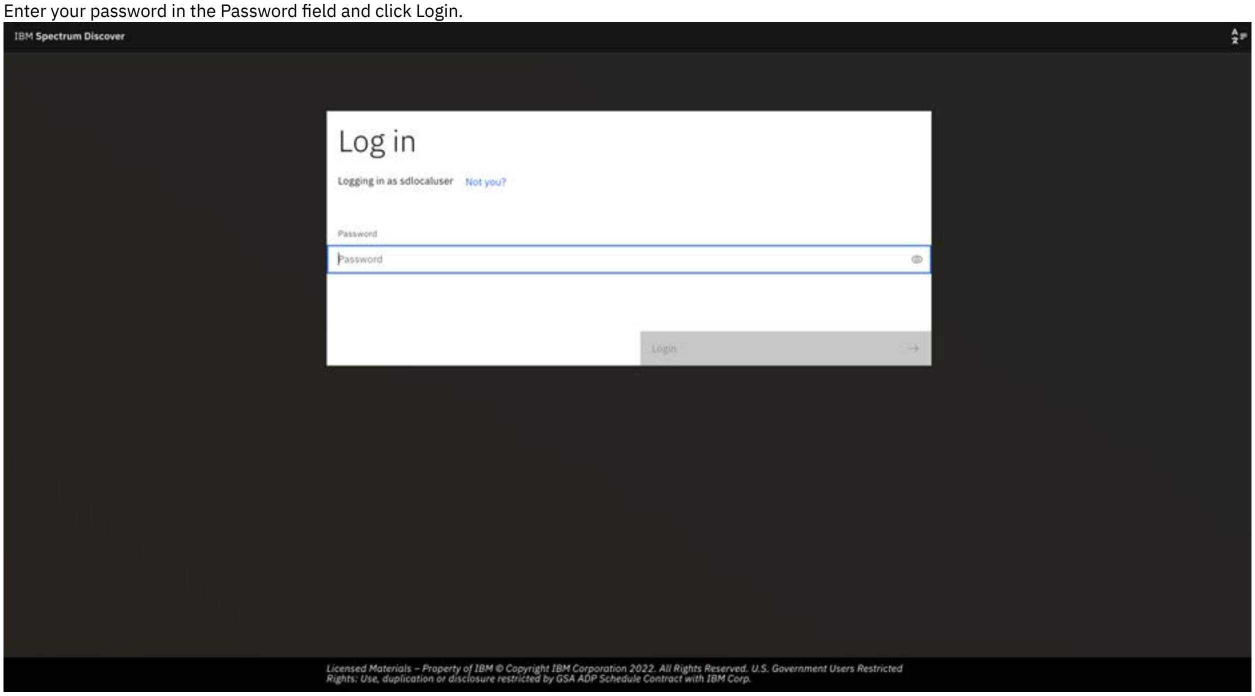
Log in to IBM Data Cataloging Dashboard with SSO disabled

If you see a previous user logged in, you need to click on the Not you? button to see the following screen.

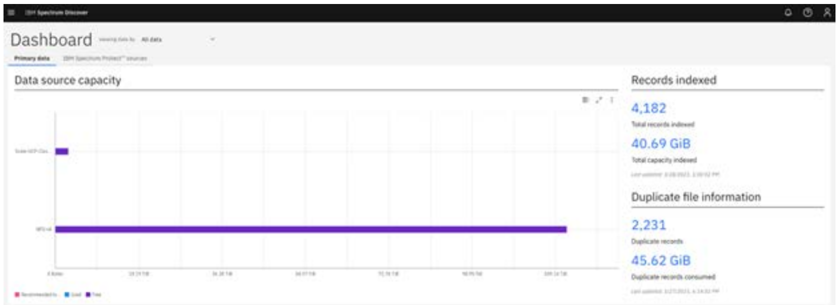
1. Enter your username in the User id field to access regular login and click Continue.



2. Enter your password in the Password field and click Login.



3. After the credentials are validated, IBM Data Cataloging main dashboard is displayed as follows.



Note:
Regardless of which user appears in the User id field, clicking the Login with OpenShift Platform button will redirect you to the OpenShift authentication client and you can access using your OpenShift credentials.

Disabling SSO login

The single sign-on (SSO) enable by default when you install the IBM Data Cataloging service. Follow the steps to disable the SSO:

1. Login to OpenShift Container Platform web console.
2. Go to Operators > Installed Operators > and select the IBM Data Cataloging.
3. Under the `data-cataloging-service` instance, go to Details tab.
4. Scroll down to the Enabled Features section and clear the Single Sign-On checkbox.
5. Go to Workloads > Deployments and look for the `isd-ui-backend` deployment.
6. Go to YAML tab and change the `OCPAUTH_DEFAULT` envvar value from `True` to `False`.
7. Click Save and wait until a pod is created.
8. After the pod is created, clean cookies, cache, and reload the IBM Data Cataloging GUI.

When SSO is disabled as the default authentication method, the following default login credentials can be used for the IBM Data Cataloging graphical user interface:

Username
: `sdadmin`

Password
: `Passw0rd`

Resetting the sdadmin password

If the `sdadmin` password changes and you forget the password, you can access the keystone container and run the **reset_sdadmin_details.sh** script to reset the password to the original password.

Procedure

1. Issue the following command to get the keystone pod name:

   ```
   c -n ibm-data-cataloging rsh deployment/isd-keystone
   ```

   The system returns the pod details similar to the following:

   ```
   Name                                         READY   STATUS    RESTARTS   AGE
   spectrum-discover-keystone-599cf9fb77-bpqc8  1/2     Running    0         3d21h
   ```

2. Go to the directory `cd /` and run the reset_sdadmin_details.sh script to reset the details or the original password.

   ```
   : ussuri
   : 127.0.0.1
   : 127.0.0.1
   : 3
   : RegionOne
   : keystone
   : default
   : spectrum-discover
   : sdadmin
   : Passw0rd
   : admin
   : http://127.0.0.1:5000/v3
   : http://127.0.0.1:5000/v3
   : http://127.0.0.1:5000/v3
   keystone-manage bootstrap --bootstrap-project-name spectrum-discover --bootstrap-username sdadmin --bootstrap-password 
   Passw0rd --bootstrap-role-name admin --bootstrap-service-name keystone --bootstrap-region-id RegionOne --bootstrap-admin-
   url http://127.0.0.1:5000/v3 --bootstrap-public-url http://127.0.0.1:5000/v3 --bootstrap-internal-url 
   http://127.0.0.1:5000/v3
   ```

Password policies

Data cataloging 2.0.2.1 introduces password policies for the local users who are configured in the default authentication domain.

The password policies that are introduced for all local user accounts, enhance their security.

Note:
Data cataloging does not enforce password policies for the user accounts that are imported to the Data cataloging authentication scheme. These policies include all user accounts imported from the external domains like LDAP or IBM® Cloud Object Store that are configured with Data cataloging. Any password policies that are configured for these external authentication providers (LDAP/IBM Cloud Object Store), would apply to the corresponding users from these authentication domains.

Password policies

Data cataloging local users must follow the password policies that are defined in the 2.0.2.1 release.

Password strength requirements

- Passwords must have a minimum length of seven characters.
- Passwords must contain at least one letter.
- Passwords must contain at least one digit.

Unique password history requirements

- Users must create a unique password each time the password is changed. The new password cannot be any of the last five passwords previously used.

Password expiration requirements

- The User password expires after 90 days from the time it is changed.

Password change requirements
Data cataloging users with Admin roles (like the "sdadmin" user) can create a new user or reset the password of an existing user. However, this password expires when the user logs in for the first time and must be changed immediately.

Account lockout requirements
A user account is locked out for 1 hour after five successive failed login attempts.

Password upgrade for existing users
Data cataloging deployments, upgraded from versions earlier than 2.0.2.1, include the new password policies that are applied to local user accounts. Existing user accounts are also impacted in the following ways:

- Existing users can continue to use their current passwords to log in to the system.
- Passwords for existing user accounts expire only in the following situations:
  - Passwords expire when users change their password. In this scenario, the new password will expire after 90 days.
  - Passwords expire when the administrative user resets the user password. In this scenario, the updated password expires immediately after the first login and the user must create a new unique password.
- When the user password is changed, the following password policies are enforced:
  - **Password strength requirements**
  - **Unique Password history requirements** - This policy restricts users from reusing any of the last five passwords.
- On completing the product upgrade, the **Account lockout requirements** policy is immediately enforced for all local users that includes all existing users.

Note: To apply all the password policies to the local user accounts after they upgrade to 2.0.2.1 or later releases, follow the listed recommendations:

- The Admin user resets passwords for all the existing local user accounts and communicates the new password to the respective users.
- All the local users use the UI Password change REST API to change their passwords. For more information, see Changing password and /auth/v1/users/<user_ID>/password: POST */auth/vi/users/<user_ID>/password: Post* in *IBM Data Cataloging: REST API Guide*.

Changing password

To change the password, you need to provide the existing and new password.

About this task

To change your password, follow these steps.

Procedure

1. Click the user icon.
2. Under Account Settings, click Change Password.
3. Enter the existing password, new password, and new password confirmation.
4. Click Save.
   Note: When the password is changed successfully, you are logged out of IBM Data Cataloging. You need to relogin with the new password.

Managing user accounts

The administrator can add and manage local user accounts, which are authenticated by IBM Data Cataloging. The administrator can also assign local or LDAP and IBM Cloud® Object Storage-managed users to roles and collections.

Use the Users tab on the Access page to view information about user accounts that are authenticated by the local domain or either an LDAP or IBM Cloud Object Storage server. You can also use the tab to create, edit, or delete local users. You cannot create or delete either LDAP or IBM Cloud Object Storage user accounts, but you can assign these users to roles and collections.

Adding a local user account
: To add a local user account that is authenticated by IBM Data Cataloging, click Create new user and the Add local User window opens. For more information, see Creating user accounts.

Editing a user account
: You can edit account information for a local user. You cannot edit the details of either Lightweight Directory Access Protocol (LDAP) or IBM Cloud Object Storage user accounts, but you can assign these users to roles and collections.
: To edit a local user account, select the user that you want to edit and click Edit. Use the Edit User window to edit the local user account.

: To edit an LDAP or IBM Cloud Object Storage user account roles, select the user that you want to edit and click Edit. Use the Edit User window to assign these users to roles and collections.

Deleting a local user account
: To delete a local user account, select the user that you want to delete and click Delete.

User information

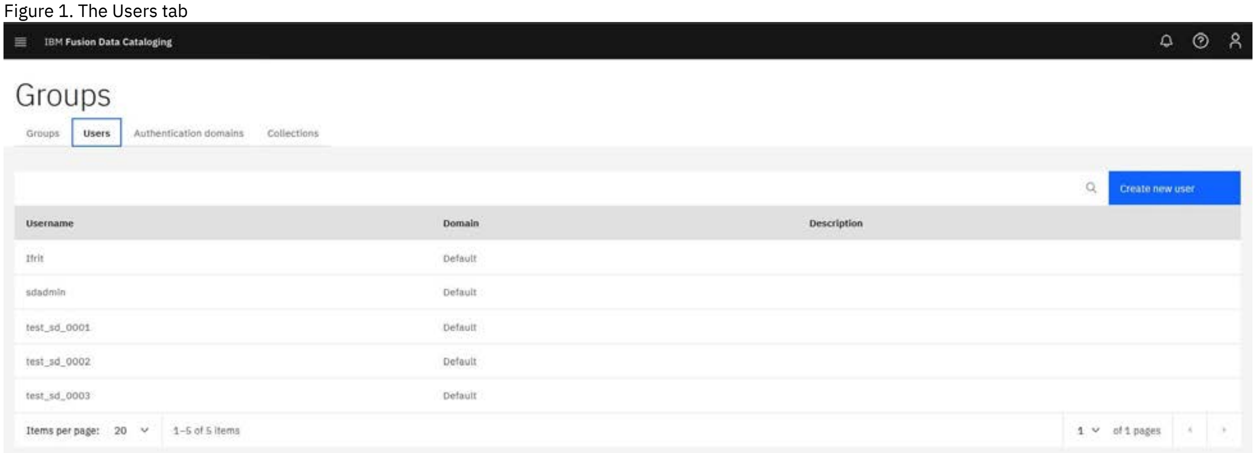
The Users tab lists the users that are available from the local domain and from either LDAP or IBM Cloud Object Storage connections. The tab includes the following user account information:

User Name
: Indicates the username for the account.

Domain
: Indicates the domain that provides authentication for the user. For authentication by IBM Data Cataloging, the domain name is Default.

Description
: Indicates the description of the user.

Figure 1. The Users tab



- **Creating user accounts**
  The administrator can add local user accounts, which are authenticated by IBM Data Cataloging, and assign roles to users.

Creating user accounts

The administrator can add local user accounts, which are authenticated by IBM Data Cataloging, and assign roles to users.

About this task

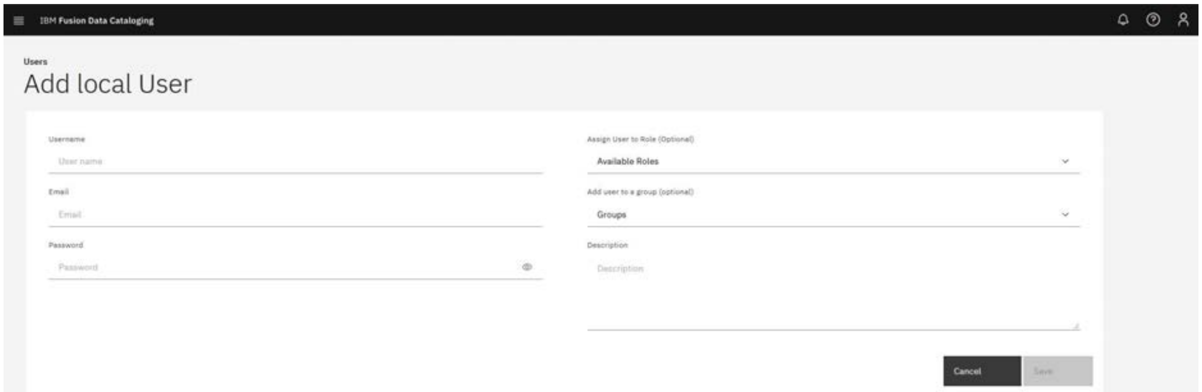
Go to the Users tab on the Access management page to add a local account.

- You can also assign roles and passwords to users.
- You can also add a user to a group.

Procedure

1. Log in to the IBM Data Cataloging web interface.
2. Click  menu and go to Access management > Users. Click Create new users to open Add local User window.

   Figure 1. The Add local User window



3. Enter a User Name and Email address for the user.
4. Enter a Password for the user.
5. Choose the available role from the Assign User to Role list to assign one or more roles to the user.
   Note:
   - This step is optional. For more information about roles, see Roles.
   - Users that are assigned the Data User or Collection Admin role must also be associated with at least one collection.
6. Use the Assign User to Group list to assign the user to one or more user groups.
   Note: This step is optional. You can also use the Groups tab to assign users to groups.
7. Provide description about user in the Description field.
   Note: This step is optional.
8. Click Save.

Managing groups

The administrator can create and manage local groups that are authenticated by IBM Data Cataloging. The administrator can also assign local or Lightweight Directory Access Protocol (LDAP) and IBM Cloud® Object Storage system-managed groups to roles and collections.

Use the Groups tab on the Access management page to view information about groups accounts that are authenticated by either a local domain, an LDAP server, or the IBM Cloud Object Storage server. You can also use the tab to add, edit, or delete local groups. You cannot edit or delete LDAP or IBM Cloud Object Storage groups, but you can assign these groups to roles and collections.

Adding a local group
   To add a local group, click Create new group to open the Add Local Group window. For more information, see Creating groups.

Editing a group
   To edit a group, select the group that you want to edit and click Edit. Use the Edit Group window to edit the local group.

Deleting a local group
   To delete a local group, select the group that you want to delete and click Delete.

Group information

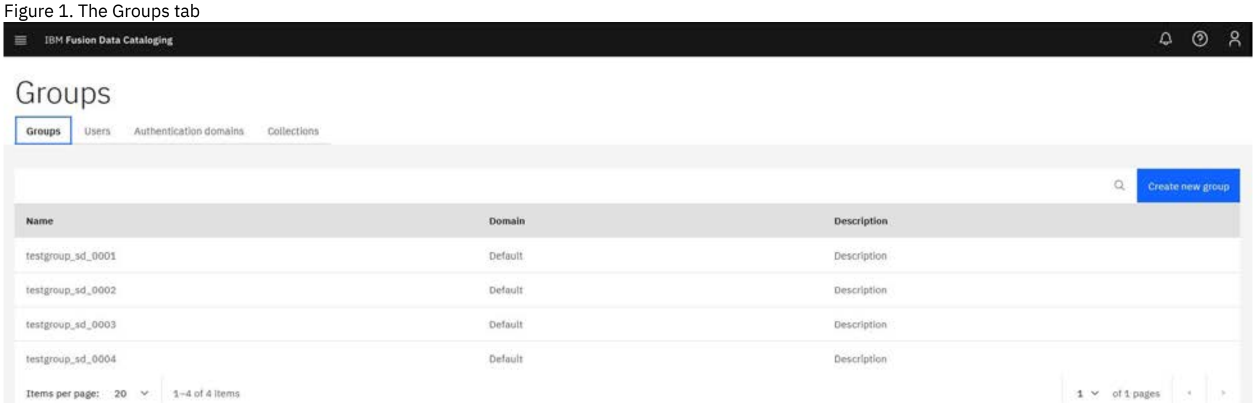
The Groups tab includes the following information:

Group Name
   Indicates the name for the group.

Domain
   Indicates the domain that provides authentication for the group. For authentication by IBM Data Cataloging, the domain name is Default.

Description
   Indicates the description of the group.

Figure 1. The Groups tab



- **Creating groups**
  The administrator can add local groups that are authenticated by IBM Data Cataloging and assign users and roles to the groups.

Creating groups

The administrator can add local groups that are authenticated by IBM Data Cataloging and assign users and roles to the groups.

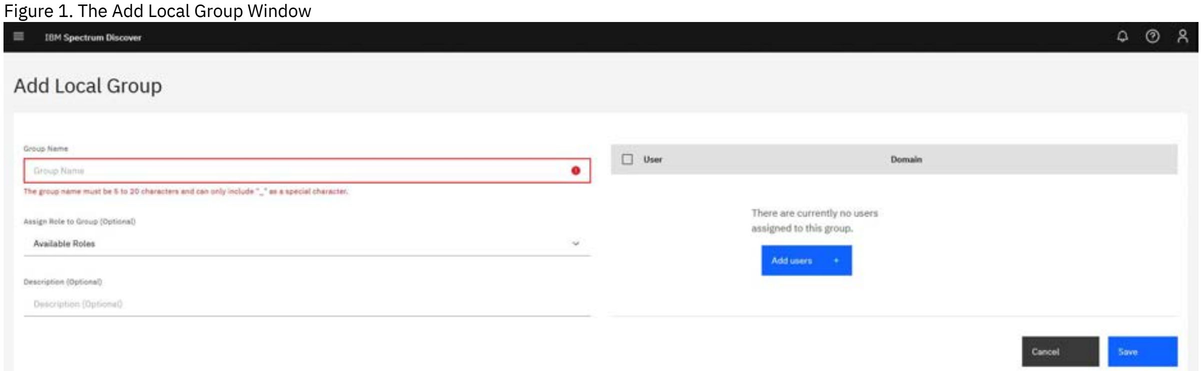
About this task

Go to the Groups tab on the Access management page to add local groups. You can assign users and roles to the group and add the group to a collection.

Procedure

1. From the Groups tab of the Access management page, click Create new group to open the Add Local Group window.

   Figure 1. The Add Local Group Window

2. Enter a Group Name.
3. Use the Assign Role to Group list to assign one or more roles to the group.
   Note:
   - This field is optional.
   - For more information, see Roles.
   - Groups that are assigned the Data User role or Collection Admin role must be associated with at least one collection.
4. Click Add Users to open the Add Users window and add one or more local users to the collection.
   a. Enter a username that you want to add to the group and press Enter. The window lists each name that you enter. Click a name to remove it from the list. Click Add to add the users to the group.
   The Users list displays the following details for users that are added to the group.

   Username
      The username or email address of the member.
   Domain
      The domain that provides authentication for the member.

5. Enter the group detail in the Description field for the group.
   Note: This step is optional.
6. Click Save.

Managing collections

Collections are logical groups of metadata that share a common member access list. For example, a collection can restrict metadata within a research project to the members of the project only. Members outside of the project cannot see the metadata.

The administrator can:

- Create collections.
- Assign users and groups to collections.
- Create a policy to associate specific metadata that is collected by IBM Data Cataloging with the collection.
- Assign a collection to a connection to associate specific metadata from that connection data source that is collected by IBM Data Cataloging with the collection.

Users with the Data Admin role can view all metadata that is collected by IBM Data Cataloging and are not restricted by collections. Users with the Data Admin role can create policies that assign a collection value to a set of records, thus grouping a set of records under a collection.

Users with the Collection Admin role can add the Data User role to user for collections that they administer. Adding this role gives data users access to a particular collection, which allows the users to access the records that are marked with that collection value.

Collection User
   Users with this role can access metadata that is collected by IBM Data Cataloging, but metadata access can be restricted by the collections that are assigned to users in this role.

   - Users assigned with the Collection User role can:
     - Run scans of collections the user is assigned.
     - View policies of the collections the user is assigned.
     - List any type of tag.
   - Users assigned with the Collection User role cannot:

- Create, update, and delete any policies.
- Create, modify, and delete any tags.

Use the Collections page to manage collections.

Creating a collection
: To create a collection, click Create Collection. For more information, see Creating collections.

Editing a collection
: To edit a collection, select the collection that you want to edit and click Edit Collection. Use the Edit Collection window to edit the collection.

Deleting a collection
: To delete a collection, select the collection that you want to edit and click Delete Collection.

Collections information

The Collections page includes the following information:

Collection Name
: Indicates the name of the collection.

Description
: Indicates the description of the collection.

Figure 1. The Collections tab

- **Creating collections**
  The administrator can create collections or assign users and groups to collections. A Collection Admin administrator can assign users and groups only to collections that they administer. A Data Admin administrator can use the AUTOTAG policy to associate metadata records with a collection.

Creating collections

The administrator can create collections or assign users and groups to collections. A Collection Admin administrator can assign users and groups only to collections that they administer. A Data Admin administrator can use the AUTOTAG policy to associate metadata records with a collection.

About this task

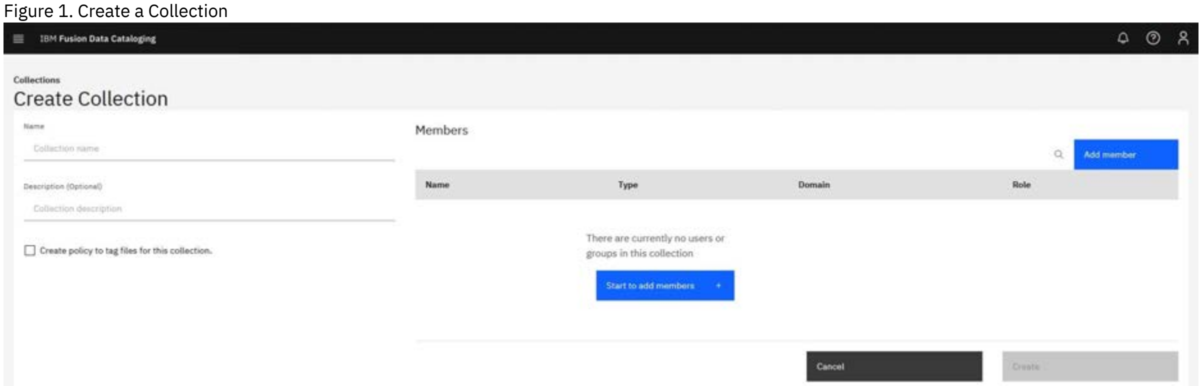
Collections are logical groups of records. Access to these record groups is restricted to specific users or groups. The administrator can associate policies with an appropriate collection value so that searches can be restricted to only the scope that a user or group has permissions to see.

Use the Collections tab on the Access management page to create collections.

Procedure

1. From the Collections tab of the Access management page, click Create collection to open the Create Collection window.

   Figure 1. Create a Collection

2. Enter a collection Name and the details in the Description field.
   Note: The Description field is optional.
3. Click Start to Add Members to open the Add Members window and add one or more users or groups to the collection.
   a. Enter a username, group name, or email address of a member to include in the collection and press Enter. The window lists each name or address that you enter. Click a name or address to remove it from the list. Click Add member to add the members to the collection.
   b. Select the role for the member on the collection. The default role is Data User.

The Members area lists the following details for the members of the collection.

Name
: The username, group name, or email address of the member.

Type
: The account type: user or group.

Domain
: The domain that provides authentication for the member.

Role
: The role on the collection that is assigned to the member.

4. To create a policy for the collection, select Create policy to tag files for this collection. For more information, see Managing metadata policies.
5. Click Create.

Managing LDAP and IBM Cloud Object Storage System connections

The administrator can create and manage connections to LDAP or IBM Cloud® Object Storage System servers that provide authentication for IBM Data Cataloging users.

Use the Authentication Domains tab on the Access page to create, test, manage, or delete LDAP connections.

You can create a connection that includes all users and groups that are authenticated by an LDAP server or only users or groups within a specified LDAP member range.
Note: You cannot specify a member range for users and groups that are managed by the IBM Cloud Object Storage System.

Creating a connection
: To create a connection to an authentication domain, click Add Domain Connection.
For steps to create a connection to an LDAP server, see Creating an LDAP connection.

For steps to create a connection to an IBM Cloud Object Storage system server, see Creating an IBM Cloud Object Storage connection.

Editing a connection
: To edit a connection, click Edit.

Deleting a connection
: To delete a connection, click Delete.

Note: You cannot map the LDAP roles from your connection and must use an admin user to make the role assignment because the roles between IBM Data Cataloging and your LDAP server might be different.

- **Creating an LDAP connection**
  The administrator can create a connection to a Lightweight Directory Access Protocol (LDAP) server that provides authentication for IBM Data Cataloging users.
- **Creating an IBM Cloud Object Storage connection**
  The administrator can create a connection to an IBM Cloud Object Storage server that provides authentication for IBM Spectrum Discover users and groups from the corresponding domain.

Creating an LDAP connection

The administrator can create a connection to a Lightweight Directory Access Protocol (LDAP) server that provides authentication for IBM Data Cataloging users.

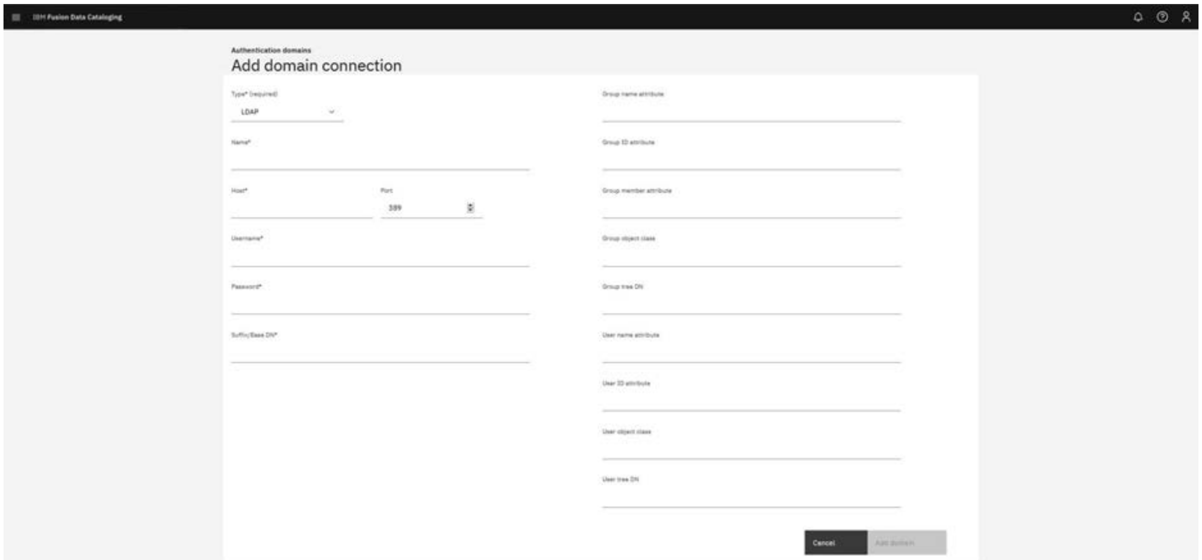
About this task

Use the Authentication Domains tab on the Access page to create an LDAP or secure LDAP (LDAPS) connection. You can create a connection that includes all users and groups that are authenticated by an LDAP server or only users or groups within a specified LDAP member range.
Note: You cannot map the LDAP roles from your connection and must use an admin user to make the role assignment because the roles between IBM Data Cataloging and your LDAP server might be different.

Procedure

1. From the Authentication Domains tab of the Access page, click Add Domain Connection to open the Add Domain Connection window.
2. From the Type list, select LDAP or LDAPS.

   Figure 1. Create an LDAP connection



3. Enter the following information for the LDAP directory:

   Name
   : Indicates a name that IBM Data Cataloging associates with the connection to the directory that provides authentication.

   Port
   : Indicates the LDAP server port that provides the connection.

   Username
   : Indicates the distinguished name (DN) for the user that is used to access directory name entries. Use the following format:

   ```
   cn=relative_distinguished_name dc=domain_component
   ```

   For example,

   ```
   cn=Randy Marsh,dc=example,dc=com
   ```

   Password
   : Indicates the password for the user name.

   Suffix/Base DN
   : Indicates the DN that is the base of entry searches in the directory. For example:
     - `dc=test`
     - `dc=org`

   Group Name Attribute
   : Indicates the LDAP attribute that is mapped to the group name.

   Group ID Attribute
   : Indicates the LDAP attribute that is mapped to the group ID.

   Group Member Attribute
   : Indicates the LDAP attribute that is mapped to show group membership.

   Group Object Class
   : Indicates the LDAP object class for groups.

   Group Tree DN
   : Indicates the DN that is the base for group searches.

   Username Attribute
   : Indicates the LDAP attribute that is mapped to the user name.

   User ID Attribute
   : Indicates the LDAP attribute that is mapped to the user ID.

   User Object Class
   : Indicates the LDAP object class for users.

   User Tree DN
   : Indicates the DN that is the base for user searches.

4. Click Connect.

- **Configuring self-signed certificates to secure LDAPS connection**
  Establish secure LDAPS connections with self-signed certificates to protect the confidential data.

Configuring self-signed certificates to secure LDAPS connection

Establish secure LDAPS connections with self-signed certificates to protect the confidential data.

About this task

IBM Data Cataloging supports LDAPS domain connections with LDAP servers that are deployed with trusted (CA signed) certificates. To use self-signed certificates, you need to add the self-signed certificate to the IBM Data Cataloging keystone pod trusted certificates list.

Procedure

1. Ensure that the cacerts directory exists in the following path, which is accessible to the keystone pod:

   ```
   /opt/ibm/metaocean/data/keystone/cacerts/
   ```

   If the directory is not available, create it by running the following command:

   ```
   oc -n ibm-data-cataloging exec deployment/isd-keystone -- mkdir -p /opt/ibm/metaocean/data/keystone/cacerts
   ```

   Then, retrieve the keystone pod name and copy the self-signed certificate to the directory:

   ```
   KEYSTONE_POD=$(oc -n ibm-data-cataloging get pod -o jsonpath='{.items[0].metadata.name}' -l role=keystone)
   oc -n ibm-data-cataloging cp cert.crt $KEYSTONE_POD:/opt/ibm/metaocean/data/keystone/cacerts/cert.crt
   ```

2. Update the certificates within the keystone pod by running the following command:

   ```
   oc -n ibm-data-cataloging exec deployment/isd-keystone /update-cacerts.sh
   ```

3. Restart the keystone pod to apply the changes by executing:

   ```
   oc -n ibm-data-cataloging delete pod -l role=keystone
   ```

Creating an IBM Cloud Object Storage connection

The administrator can create a connection to an IBM Cloud® Object Storage server that provides authentication for IBM Spectrum® Discover users and groups from the corresponding domain.

About this task

Use the Authentication Domains tab on the Access page to create a connection to an IBM® Cloud Object Storage System server. You must provide credentials for the IBM Cloud Object Storage security administrator.

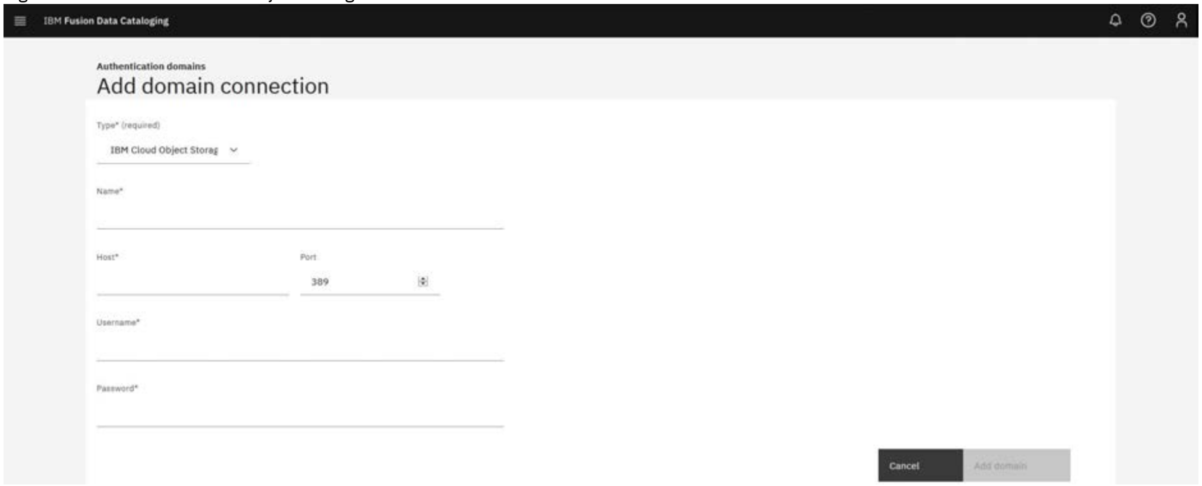
All users and groups that are managed by the IBM Cloud Object Storage are available for IBM Data Cataloging. You cannot specify a member range for these connections.

Note: You cannot run scans unless you add override warnings in the configuration file.

Procedure

1. From the Authentication Domains tab of the Access page, click Add Domain Connection to open the Add Domain Connection window.
2. From the Type list, select IBM Cloud Object Storage.

   Figure 1. Create an IBM Cloud Object Storage Connection

3. Enter the following information for the IBM Cloud Object Storage connection:

   Name
   : Indicates the IBM Cloud Object Storage domain name.

   Host
   : Indicates the IBM Cloud Object Storage hostname.

   Port
   : Indicates the IBM Cloud Object Storage port number.

   User name
   : Indicates the IBM Cloud Object Storage security administrator name.

   Password
   : Indicates the IBM Cloud Object Storage security administrator password.

4. Click Connect.

Managing metadata policies

Policies might be used to automatically tag a set of documents on a periodic basis. In addition, policies might be used to send sets of documents to be deep-inspected by a registered application.

Roles and permissions

Data User
: Users with this role can create, modify, and view their policies. Policies can be applied only to the collections the user has access to.
Users with the Data User role cannot use a `COLLECTION` tag when they create or modify policies.

Collection Admin
: Users with this role can create, modify, and view their policies. Policies can be applied only to the collections that they administer. Users with the Collection Admin role cannot use a `COLLECTION` tag when they create or modify policies.

Collection User
: Users with this role can access metadata that is collected by IBM Data Cataloging, but metadata access can be restricted by the collections that are assigned to users in this role.

- Users assigned with the Collection User role can:
  - Run scans of collections the user is assigned.
  - View policies of the collections the user is assigned.
  - List any type of tag.
- Users assigned with the Collection User role cannot:
  - Create, update, and delete any policies.
  - Create, modify, and delete any tags.

Data Administrator
: Users with this role can create, modify, view, and delete policies.

Security Administrators
: Users with this role cannot create, modify, view, or delete policies.

Service User
: Users with this role cannot create, modify, view, or delete policies.

- **Adding policies**
  You can add policies to help with data management.
- **Running policies**
  You can configure policies to run at specified or scheduled times, at policy creation time, or when the system is manually started, paused, restarted, or stopped.
- **Viewing policies**
  You can view your policy information.
- **Modifying policies**
  You can modify your policies to work with and use your data more effectively.
- **Deleting policies**
  You can delete policy information.

Adding policies

You can add policies to help with data management.

About this task

To add a policy, you must define the policy and its parameters, establish a schedule to run the policy, and save the policy.
You can add custom metadata values to all or a subset of the records based on filter criteria. For example, you can add a project name to records based on their location within the file system or owner ID.

You can add a policy filter, which is similar to the **where** clause in an SQL query. The filter must be constructed by using standard SQL syntax:

- To enact a policy on all files not accessed in one year, the filter might be written as:

  ```
  atime < (NOW() - 365 DAYS)
  ```

- To enact a policy on all files owned by `Smithers`, the filter might be written as:

  ```
  owner='Smithers'
  ```

- To enact a policy on all PDF files in cluster `cl1` and data source `fs1`, the filter might be written as:

  ```
  cluster='cl1' and datasource='fs1' and filetype='pdf'
  ```

Procedure

1. Go to Metadata > Tag Management.
2. Under Policies click Add Policy.
3. Enter a name for the policy in the Name box (for example, `MyPolicy`).
4. Select the required policy type from the Policy Type menu.
   The policy types are:

- `AUTOTAG`
- `CONTENT SEARCH`
- `DEEP-INSPECT`

5. Click Next step.
6. Complete the policy information:
   a. Select the list of Collections the policy applies to. If no collections are selected, the policy applies to all collections available to the user when run.
   b. You must use a filter for the policy. The filter defines which set of records the policy acts on. Enter your filter into the Filter box (for example, `size > 100`).
   c. Complete the policy-specific parameters based on which policy type you select in step 3:
      i. You can set parameters for **`AUTOTAG`** policy types. For more information, see Adding auto-tagging policy parameters.
      ii. You can set parameters for **`DEEP-INSPECT`** policy types. For more information, see Adding deep-inspection policy parameters.
      iii. You can set parameters for **`CONTENT SEARCH`** policy types. For more information, see Adding content search policy parameters.
7. Click Next step
8. Now that your policy parameters are defined, you must schedule the frequency.
9. Click the slider control to set the status to one of the following values:

   Active
   An *Active* policy is run whenever its scheduling event is reached.
   Inactive
   An *Inactive* policy is not run when its scheduling event is reached, including the **Now** event.

10. Select a Schedule to apply the policy. Indicate whether you want to schedule the policy `Now`, `Daily`, `Weekly`, or `Monthly`.
    Note: Policy schedule times are entered in Coordinated Universal Time (UTC).

    Now
    Indicates that the policy is applied immediately unless the policy's status is **`Inactive`**.
    Daily
    Indicates a specific time of day to apply the policy. Enter the time of day by clicking the hour and minute from the widget that is shown. The policy is applied daily at the specified time.
    Weekly
    Indicates a specific day and time in which to apply the weekly policy:
    a. Enter the time of day to apply the policy by clicking the hour and minute from the widget that is shown.
    b. Select the day of the week from the list of days.
    The policy is applied once a week on the specified day and time.
    Monthly
    Indicates a specific day and time in which to apply the monthly policy:
    a. Enter the time of day to apply the policy by clicking the hour and minute from the widget that is shown.
    b. Select the date by clicking the month and day from the widget that is shown.
    The policy is applied once a month on the specified day and time.

11. Click Next step.
12. Review the data and click Save to save the new policy.
    The new policy displays in the list of policies under the Policies tab.

- **Adding auto-tagging policy parameters**
  You can add specific parameters for auto-tagging policies after you define the policy type and set up the mandatory filter or optional collection information.
- **Adding deep-inspection policy parameters**
  You can add DEEP-INSPECT policies.
- **Adding content search policy parameters**
  You can add search content parameters for your policies.

Adding auto-tagging policy parameters

You can add specific parameters for auto-tagging policies after you define the policy type and set up the mandatory filter or optional collection information.

Procedure

1. Associate one or more tags with this policy by using one of the following methods:
   a. Click +Add Tag.
   b. Select a tag name from the Field menu.
   The Fields can also be specified by going to Metadata > Tags.
   c. Select which Tag to apply (for example, `TEMPERATURE`) and then enter the appropriate Values.
   Note: If you do not know the valid values for a tag, go to Search in the main menu and select the tag from the Start a visual exploration list. Click the Go "circle arrow" icon. The valid values of the tag are displayed.
   d. To delete a Field, click the Delete "minus" icon next to a field.
   e. You can add more tag values by clicking the +Add Tag control.
   Each new Field defaults to the next item in the Field menu.
   Or you can automatically extract the tag from the directory path by clicking the Extract tag from box. For example, you might have files in a directory structure by department and want to extract the department name into a tag. Automatically extract the tag:
      i. Select the Extract tag from path checkbox.
      ii. Select a tag from the Field menu.
      iii. Specify the Depth in the path to be used as the value of the tag.
   To create a new tag, see Creating tags.
2. Click Next Step.

Adding deep-inspection policy parameters

You can add DEEP-INSPECT policies.

About this task

You can enrich metadata through an external deep inspection application. For more information, see Managing applications. For example, you can extract patient names from medical records and index the names. Indexing the names helps when you search for files that pertain to patients by name. Deep inspection policies can send lists of files to an application, which can examine the contents of files and return the values that it finds paired with defined tag keys.

Important: If the deep-inspect application returns a **`tag:value`** pair that is not requested by the deep-inspect policy, the tag:value pair is still updated by the deep-inspect policy as long as the tag resides in IBM Data Cataloging.

Procedure

Do the following to add parameters for deep-inspection policies:

1. Add the Application name (for example, `example-application)`.
   Note: To add an application, see /policyengine/v1/applications: POST*/policyengine/v1/applications: POST* in the *IBM Data Cataloging: REST API Guide*.
2. Click +Add Tag.
3. Select which Parameter to apply (for example, `extract-tags`), and then select the appropriate Values (for example, `TEMPERATURE`).
4. You can add more parameters by clicking the +Add parameter control. This is optional.
5. You can delete a parameter by clicking the Delete "minus" icon next to the parameter's Value. This is optional.
6. Click Next Step.

Adding content search policy parameters

You can add search content parameters for your policies.

About this task

You can enrich metadata through the built-in content search application. For more information, see Using content search policies.

You can add specific parameters for content search policies by using the following steps:

Procedure

1. Select `contentsearchagent` for the Application.
2. Click +Add Row to open the Parameter dialog.
3. Enter a tag name in the Parameter box and select one or more Search Expressions from the dropdown list.
4. For the Value, select either True/False or Value matching expression.
5. Repeat steps 1 - 3 to set other tags that use the same policy.
6. Click Next Step.

Running policies

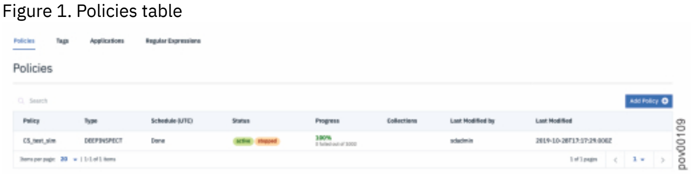
You can configure policies to run at specified or scheduled times, at policy creation time, or when the system is manually started, paused, restarted, or stopped.

Procedure

To configure a policy to run:

1. Go to Metadata
2. Click a policy to select it. The following screen displays:

   Figure 1. Policies table

   From the screen, you can perform the following actions on the selected policy:

   Pause
   : Click the "vertical bars" icon to pause a policy that is running. When a policy is paused, it enters a `Paused` state. A policy cannot be paused until the current batch that is being processed finishes.

   Start
   : Click the "right-arrow" icon to resume a paused policy or to start a stopped policy from its beginning. When a policy is started, it enters a `Running` state.

   Stop
   : Click the "square" icon to stop a policy that is in progress. When a policy is stopped, it enters a `Stopped` state.

   When a policy completes, it enters a `Stopped` state. The progress column indicates the success or failure status of the policy. If there are failures, you can examine the per policy execution log files to obtain more details of the failures.

For more information, see Viewing policies.

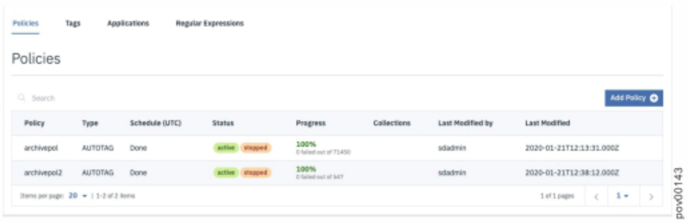
Viewing policies

You can view your policy information.

About this task

You can see a list of the active and inactive policies and their status.

Figure 1. Policies table



Procedure

1. Go to Metadata > Tag Management
2. Under Policies, view a table of the specified policies with the following aspects:

   Policy
   : Displays the name of a policy.

   Type
   : Displays the policy type. There are three types:
     - **AUTOTAG**: You can apply custom metadata values to some, or all of the records based on filter criteria.
     - **CONTENT SEARCH**: You can enrich metadata by using the built-in CONTENTSEARCH application.
     - **DEEP-INSPECT**: You can enrich metadata through content inspection of source data.

   Schedule
   : Displays the frequency at which a policy is applied. Policy schedule times are displayed in Coordinated Universal Time (UTC).

     Done
     : Indicates that the policy is applied.

     Daily: 00:00
     : Indicates that the policy is applied one time a week on the displayed day and time.

     Weekly:[day], 00:00
     : Indicates that the policy is applied one time a week on the displayed day and time.

     Monthly:[date], 00:00
     : Indicates that the policy is applied one time a month on the displayed date and time.

   Status
   : Displays the status and current state of the policy.
     - A policy's **Inactive** status shows the **None** state.
     - A policy's **Active** status can have a state of **Initialized**, **Running**, **Paused**, or **Stopped**.

   Progress
   : Displays the percentage of completion of a policy.
     If there is an Error "yellow triangle" icon, you can hover the cursor over it to see more information.

     If there are some records that met the filter criteria but fail to be tagged, the number of failed records and the total number of records are displayed below the percentage of completion.

     If there are failures, the per policy execution log files can be examined to obtain more details of the failures. For more information, see Viewing policy history details and logs.

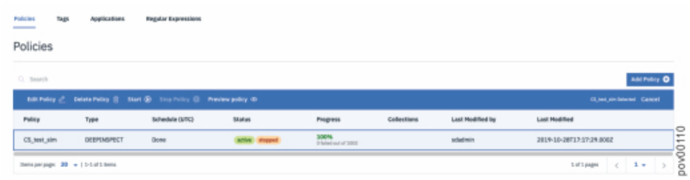
   Collections
   : Displays the name of the collection that the policy applies to or the number of collections the policy applies to if there is more than one collection.

   Last modified by
   : Displays the name of the user who last modified the collection.

   Last modified
   : Displays the date and time that the collection was last modified.

   You can add a policy by clicking Add Policy +. For more information, see Adding auto-tagging policy parameters or Adding deep-inspection policy parameters on page 19.
3. Click a policy to select it. The following screen displays:

   Figure 2. Policies table



From the screen, you can perform the following actions:

Start, Pause, Stop, or delete
: You can start, pause or stop a policy. For more information, see [Running policies](#).

Edit/Delete
: Use the Edit "pencil" icon to modify a policy. Use the Delete "trashcan" icon to delete a policy. You cannot delete a policy that is running (The "trashcan" icon is made unavailable).

View
: Preview the details of the selected policy.

- **Viewing policy history details and logs**
  You can view the history details, execution statistics and debug log data for all older or current metadata policies.

Viewing policy history details and logs

You can view the history details, execution statistics and debug log data for all older or current metadata policies.

About this task

Follow the procedure to view policy history details and logs.

Procedure

1. Go to Metadata > Tag Management.
2. Under Policy History, view the following information for the policy that ran recently or was run in the past.

   Policy name
   : The name of the policy that was run.

   Type
   : The policy type that was run.

   Started
   : The time when the policy execution started.

   Ended
   : The time when the policy execution was completed. This column remains blank if the policy is running.

   Status
   : The current or final status of the policy execution.

   Total records
   : The number of indexed records that matched the policy filter criteria for this policy execution.

   Successful
   : The number of records that were processed successfully by the policy execution.

   Skipped
   : The number of records that matched the filter criteria, but were skipped by the policy execution.

   Failed
   : The number of records that the application failed to process.

   Note: Click the table header for each column to sort the tables. You can sort tables in both ascending or descending order.
3. Click to select a row in the policy history table.
4. Click View log to view the run log contents for the selected policy execution.
   Note: The run log also displays the details of all failed records. You can download and debug the logs if necessary.

Modifying policies

You can modify your policies to work with and use your data more effectively.

About this task

You can modify a policy from the table on the Policies page. You cannot change the Name or Type of a policy.

Procedure

1. Go to Metadata > Tag management
2. Under Policies, select a policy you want to modify and click the Edit "pencil" icon on the header.
   The Edit policy page appears. The policy parameters displayed depends on the policy type. The policy configuration area changes depending on the Extract tag from path setting.

Note: You cannot modify the name and the policy type.
3. Modify the collections that the policy applies to (if required).
4. Modify or enter a filter in the Filter box.
   The filter must be constructed by using standard SQL syntax. For more information, see example filters on page 17.
5. Modify the parameters specific to the policy type (if required).
   - For more information about **`AUTOTAG`** policy types, see Adding auto-tagging policy parameters.
   - For more information about **`DEEP-INSPECT`** policy types, see Adding deep-inspection policy parameters.
   - For more information about **`CONTENT SEARCH`** policy types, see Adding content search policy parameters, Adding deep-inspection policy parameters, or Managing user accounts.
6. Click the slider control to set the status to one of the following values:

   Active
      An *Active* policy is run whenever its scheduling event is reached.
   Inactive
      An *Inactive* policy is not run when its scheduling event is reached, including the **Now** event.

7. Specify a Schedule to apply the policy. Policy schedule times are entered in Coordinated Universal Time (UTC).

   Now
      The policy is applied immediately, unless the policy's status is **Inactive**
   Daily
      Indicates a specific time of day to apply the policy. Enter the time of day to apply the policy by clicking the hour and minute from the widget that displays. The policy is applied daily at the specified time.
   Weekly
      Indicates a specific day and time in which to apply the weekly policy:
      a. Enter the time of day to apply the policy by clicking the hour and minute from the widget that displays.
      b. Select the day of the week from the list of days.
      The policy is applied one time a week on the specified day and time.
   Monthly
      Indicates a specific day and time in which to apply the monthly policy:
      a. Enter the time of day to apply the policy by clicking the hour and minute from the widget that displays.
      b. Select the date by clicking the month and day from the widget that displays.
      The policy is applied one time a month on the specified day and time.

8. Under Review check all the changes you have made.
9. Click Save to save the changes.
   The modified policy is displayed in the list of policies on the Policies tab.

Deleting policies

You can delete policy information.

About this task

A policy can be deleted from the table on the Policies page. You cannot delete a policy that is running. A user with the role Data User can delete only their own policies.

Procedure

1. Go to Metadata > Policies.
2. Click the Delete "trashcan" icon in the Edit/Delete column of the policy you want to delete.
   If the "trashcan" icon is unavailable, then the policy is not available for deletion.
3. Click Delete in the confirmation window.
   The policy is removed from the table in the Policies tab.

Using content search policies

You can define regular expressions to search for and create policies that use the regular expressions.

About this task

You can enrich metadata through content inspection of source data by using the built-in CONTENTSEARCH application. To use this function, you can define regular expressions to search for and create policies that use these regular expressions.

When the policy runs, the documents are retrieved from the source system by the CONTENTSEARCH application, converted to text format if necessary, and searched by using the defined regular expressions. The results of the search are returned to Data cataloging and the metadata of the files that are updated. To create a CONTENT SEARCH policy, see Adding policies.

When you create a CONTENT SEARCH policy, you can select:

- Any tag type (including **`Open`**, **`Restricted`**, and **`Characteristics`** tags) for the **`CONTENTSEARCH`** application.
- Any regular expression.
- Either **`"True/False"`** or **`"Value matching expression"`**. **`"True/False"`** sets the tag value to either **`True`** or **`False`** if a match is found or not found. **`"Value matching expression"`** sets the tag value to the extracted content match.

Remember:

- If you select a `Restricted` tag with a defined set of values and choose to extract the value from a document, the value that is extracted must match one of the `RESTRICTED` values in the tag.
- If the content extracted exceeds the minimum tag value length, the extracted content is truncated.

- **Identifying the required regex expressions**
  The following information can help you identify required regex expressions.
- **Creating a content search policy**
  You can create a content search policy to filter a search query that finds candidate documents to which you can apply the policy.
- **Viewing content search application logs**
  The following information describes how to view the content search application logs.
- **Hints and tips for using content search**
  Following are some best practices for using the content search feature.

Identifying the required regex expressions

The following information can help you identify required regex expressions.

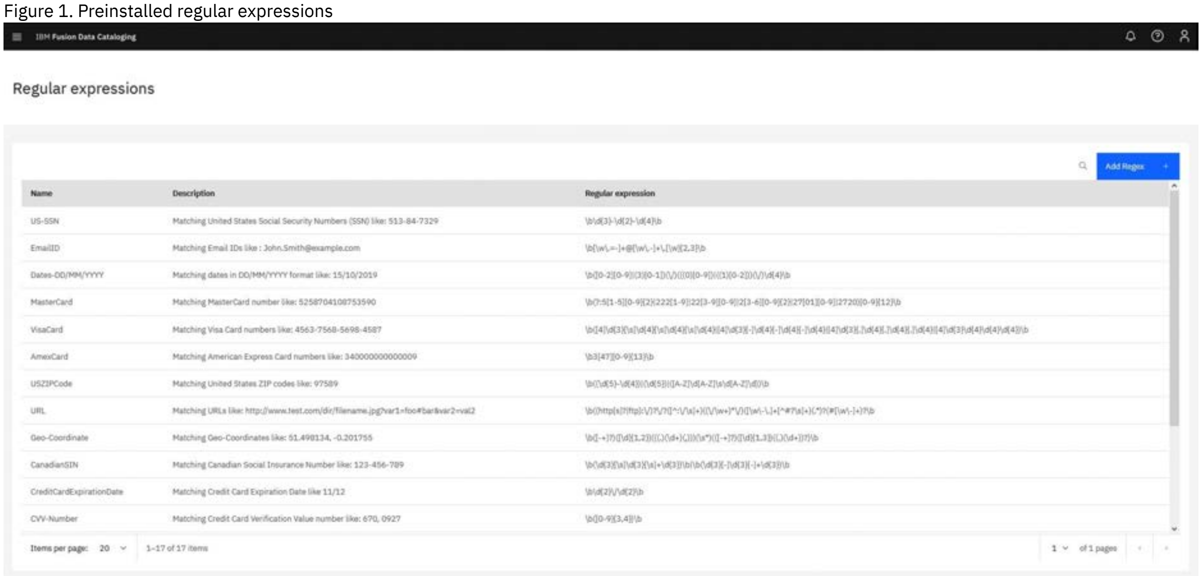
About this task

Validate that the regular expressions that are required for the policy are present. You can create or modify them if necessary.

Procedure

1. On the metadata page, select the Regular Expression tab.
   The list of available expressions is displayed.

   Figure 1. Preinstalled regular expressions

2. Search through the list of regular expressions to find any that match the content to be searched.
   As shown in Figure 1 that includes a selection of regular expressions.
   For example, an expression with an embedded value that might be extracted.

   ```
   ^([\w\.=-]+@[\w\.-]+\.[\w]{2,3})$
   ```

   This regular expression matches an email address, and the value that is returned for the tag is the matched email address. This sort of regex is appropriate to use in a value type match.
   Another example is an expression with no embedded value, for a straight match.

   ```
   ^Patient Name:.*$
   ```

   This expression detects the presence of the literal string **"Patient Name:"** and subsequent string in a fixed format file, but it does not extract the value. This is appropriate for use in a Boolean find search.
3. If there are no suitable regular expressions, select the Create icon.
4. If a regular expression exists but requires modifications select the expression and click the "Modify" icon. If this regular expression is in use by any other policy, this modification affects those policies.
5. Enter a suitable name, description, and the regular expression pattern.
6. Select Save.

Creating a content search policy

You can create a content search policy to filter a search query that finds candidate documents to which you can apply the policy.

About this task

For information about creating a content search policy, see Managing metadata policies and Adding policies.

Viewing content search application logs

The following information describes how to view the content search application logs.

About this task

If you want to view the content search application logs, perform the following actions.

Procedure

1. Run the following command for enter in the project:

   ```
   oc project ibm-data-cataloging
   ```

2. Any failures of download or inspection are logged in the application log file.
   The following command provides the application log file:

   ```
   oc logs deployment/isd-contentsearchagent
   ```

Hints and tips for using content search

Following are some best practices for using the content search feature.

Testing on a subset of documents

Running a content search policy on a set of documents has several steps that includes retrieving the document, formatting it as text, if necessary, and searching the document. Depending on the number, formatting, and size of the documents, searching the document can be a time-consuming process.

Therefore, it is best to test the policy and corresponding expressions on a small subset of documents to determine whether the policy and the regular expressions you select are correct. One way to run the test is to use a policy filter that selects only a small set of documents. After you confirm that the policy and search criteria is operating as expected, you can run it against the required set of documents.

The test on a subset of documents can also help you estimate how long the policy might take to run on the complete set of documents.

Avoiding retagging

When you rerun a policy against a set of documents that is previously tagged, the documents are retagged. If the values returned are different than the previous search, they are updated. This difference might occur if the policy or the set of expressions is modified, or if the set of documents is modified.

To avoid retagging the documents, add a criteria to the filter to not select documents that are already tagged.

Modifying regular expressions

If you modify a regular expression, it affects all policies that use that expression. Rerunning these policies might cause the documents to be tagged differently. To avoid changing the behavior of existing policies, create a new regular expression and use it in the specific policies where it is required.

Converting files with Apache Tika

Data cataloging uses Apache Tika to convert files to text before it searches the content. This conversion has an impact on the overall content search performance.

Therefore, files that are text format must be configured in the `contentsearch` agent to prevent them from being processed unnecessarily by Apache Tika. The default configuration treats JavaScript Object Notation (JSON) and Variant Call Format (VCF) file types as text. To add more text file types to the configuration, edit the file:

```
/opt/ibm/metaocean/data/agents/contentsearch/conf/contentsearch.conf
```

And add more types to the line:

```
text_filetypes=vcf,json
```

Apache Tika runs in a Kubernetes pod within Data cataloging. You can increase Apache Tika pod instances to improve performance. For example, run this command to scale the number of Tika instances to three:

```
oc -n ibm-data-cataloging scale --replicas=3 deploy/isd-tikaserver
```

Apache Tika is resource-intensive, so make sure that the number of Apache Tika instances does not exceed the host resources.

Supported connection types

Content search on Data cataloging supports the following connection types:

- Spectrum Scale
- COS
- NFS
- S3
- SMB

For more information, see the topic *IBM Spectrum® Scale data source connection* in *IBM Data Cataloging: Concepts, Planning, and Deployment Guide*.For more information, see IBM Storage Scale data source connections.

Tiering data by using ScaleILM application

Use the IBM Data Cataloging ScaleILM application to move data to different tiers (pools) that are configured on the IBM Storage Scale connection.

Before you begin

- Configure IBM Storage Scale with one or more internal pools. For more information, see: Internal storage poolsFor more information, see *Internal Storage pools* in the *IBM Storage Scale: Administration Guide*.
- If you want to, you can configure IBM Spectrum® Archive and its pools.
- Create the IBM Storage Scale connection in IBM Data Cataloging. If the external pool that is managed by IBM Storage Archive is used as the destination tier, then the "host" setting of IBM Storage Scale connection must specify one of the IBM Storage Archive nodes.
- IBM Data Cataloging ScaleILM application uses the same user, that is configured in IBM Data Cataloging to scan the IBM Storage Scale and run the Data Movement policies. For this data movement, the IBM Storage Scale connection is used as the 'source_connection'. For more information, see Creating or identifying a user ID and password for scanning. For more information, see the topic *Creating or identifying a user ID and password for scanning* in the *IBM Data Cataloging: Concepts, Planning, and Deployment Guide*.

About this task

The IBM Data Cataloging ScaleILM application provides the advanced data tiering function through the IBM Storage Scale connection. It uses the system metadata and custom metadata of documents that are collected by IBM Data Cataloging for this advanced data tiering function. This capability eliminates the need to scan file systems during IBM® License Manager (ILM) execution. It also provides cognitive tiering capability by using the results of IBM Data Cataloging's AUTOTAG, CONTENT SEARCH, and DEEP-INSPECT policies.

IBM Storage Scale provides the built-in Information Lifecycle Management (ILM) capability that optimizes the cost-effectiveness of data by moving the physical location of data between the storage pools with different cost or performance characteristics. IBM Storage Scale's ILM supports the data movement between its internal storage pools and between both the internal pool and the external storage repository. The movement from external storage repository is managed by external applications, such as IBM Storage Archive Enterprise Edition (EE) and IBM Storage Protect for Space Management.

Ensure that you know the IBM Storage Archive Enterprise Edition (EE) software versions that are supported. For more information, see IBM Spectrum Storage software requirements. For more information, see the topic *IBM Spectrum Storage software requirements* in the *IBM Data Cataloging: Concepts, Planning, and Deployment Guide*.

While tiering data in external pools, the current location of the data is indicated in its file state. The data locations and the corresponding file states that indicate these locations, are listed in the following table:

Table 1. Data locations and File States

Data locations	File states
The data location is only in the internal pool.	The file state is resident.
The data location is both internal and external pool.	The file state is premigrated.
The data location is only in the external pool.	The file state is migrated.

Note: The functions of the ScaleILM application depend on the capability of underlying storage hardware and its management software. For more information, see the following topics:

- *Information Lifecycle Management for IBM Storage Scale* in the *IBM Storage Scale: Administration Guide*
- *IBM Storage Archive Enterprise Edition (EE) IBM Documentation*

Procedure

1. Log in to IBM Data Cataloging GUI.
2. Go to Admin > Management Policies page.
3. Click Add Policy to create a policy.
4. Click the slider control and set the status to one of the following values:

   Active
   : An active policy is run whenever its scheduling event is reached.

   Inactive
   : An inactive policy is not run even when its scheduling event is reached, including the Now event.

5. Enter a policy name.
6. Enter a Policy filter. The policy filter includes the criteria for selecting the files for tiering, such as `filetype="pdf"`.
   Note: If the destination tier is the external pool that is managed by IBM Storage Archive, define the policy filter criteria as "state in 'resdnt'". For example, `filetype ='txt'` and `state in 'resdnt'`.
   If the filter criteria do not include `state in 'resdnt'` while tiering data to an external pool, the ScaleILM application skips the files that are not in resdnt state.
7. Click Next Step to select the type of policy.
8. Select `TIER` as the Policy type.

9. Select `ScaleILM` as the Agent.
10. Select the `source_connection` from the drop-down list. The source connection is the source from where the data is being moved, such as the name of the defined IBM Storage Scale connection.
11. Enter the `destination_tier` where you want to move your data, such as:
    - Internal Pools
      Gold, silver, bronze, flash system
    - External Pools
      - **archive:pool1@library1**
      - **archive:pool2@library2**
      - **archive:pool1@library1, pool2@library3**

    Note: When you specify an IBM Storage Scale internal pool as the "destination_tier", you need to ensure that you specify a valid internal pool name. That internal pool name must be configured in the IBM Storage Scale source connection for the corresponding data source (file system). These internal pools can be listed by using the following IBM Storage Scale command:

    ```
    mmlspool <device> all
    ```

    When you specify an external pool that is managed by IBM Storage Archive as the "destination_tier", you must specify a valid archive pool. This archive pool must be defined in the IBM Storage Archive that is configured on the IBM Storage Scale cluster node.

    Additionally, you must specify the name of the pools that are defined in the IBM Storage Archive configuration, with the prefix "archive:". The pool names must be specified in the same format as defined in the -p option of IBM Storage Archive Enterprise Edition (EE) CLI (eeadm). The syntax must be as follows:

    ```
    archive:<poolName>@<libraryName>
    ```

    or

    ```
    archive:<poolName1>@<libraryName1>,<poolName2>@<libraryName2>, ...
    ```

    The policy execution fails when you do not follow the instructions.
12. Select Next Step to enter a schedule. The schedule indicates when you want to start the tiering.
13. Select Next Step to review the policy.
14. Select Submit to create the policy.
15. When the policy is created, view the files on the IBM Data Cataloging Search catalog page to ensure that they are moved to the new tier. The following metadata are updated:

    Tier
    : Displays the name of the internal pool in IBM Storage Scale where the data is stored.
      Note: Even if the file is in the migrated state (migrtd), the tier field shows the name of the original internal pool.

    State
    : Displays the current state as one of the following values:
      - resdnt
      - premig
      - migrtd

    Migloc
    : Displays the location information of the external pool when the file is in premigrated (premig) or migrated (migrtd) state. If the file is in a resident (resdnt) state, this field shows `NA`.

    SizeConsumed
    : Displays the actual size of the file in bytes, that is associated with the IBM Storage Scale file system field SizeConsumed Bytes.

- **Viewing ScaleILM application logs**
  The following section describes how to view the ScaleILM Data Mover application logs.

Viewing ScaleILM application logs

The following section describes how to view the ScaleILM Data Mover application logs.

About this task

Follow the procedure to view the ScaleILM Data Mover application logs.

Procedure

1. Run the following command:

   ```
   oc -n ibm-data-cataloging logs deployment/isd-scaleilmdatamover
   ```

2. Any failures that occur while the files are processed for tiering by ScaleILM Data Mover application, are logged in the application log file. A sample error log from the ScaleILM Data Mover application logs is shown in the following section. These error logs occur when the file is not on the specified `'source_connection'` while the application is processing the Tiering policy.

   ```
   2020-04-28 14:44:36,439 - __main__ - ERROR - File not found:
   /ibm/scale0/unicode_test/migrateDir.popFSDir.19824/unicodedir/test010.\n.test
   2020-04-28 14:44:36,440 - __main__ - ERROR - File not found:
   /ibm/scale0/unicode_test/migrateDir.popFSDir.19824/unicodedir/test137..test
   2020-04-28 14:44:36,440 - __main__ - ERROR - File not found:
   /ibm/scale0/unicode_test/migrateDir.popFSDir.19824/unicodedir/test147..test
   ```

```
2020-04-28 14:44:36,440 - __main__ - ERROR - File not found:
/ibm/scale0/unicode_test/migrateDir.popFSDir.19824/unicodedir/test157..test
```

A sample error log from the ScaleILM Data Mover application logs for an invalid `'destination_tier'`, is shown in the following section:

```
2020-04-28 14:35:39,566 - __main__ - ERROR - Invalid destination_tier: 'gold:abcd'.
```

A sample error log from the ScaleILM Data Mover application logs when the `'destination_tier'` does not exist on the IBM Storage Scale connection, is shown in the following section:

```
2020-05-12 14:32:38,436 - __main__ - ERROR - Provided internal pool goldflower does not exist on system: 9.11.212.29
```

3. Run the following command to view policy engine pod logs:

```
oc -n ibm-data-cataloging logs deployment/isd-policyengine
```

A sample policy engine pod log is shown in the following section. This log shows information that is received from the ScaleILM Data Mover application. It describes errors that are encountered while a Tiering policy is processed and when a file is not found on the specified `'source_connection'`:

```
2020-04-28 14:44:37,455 - policy.policyapiservice - ERROR - [policy_id: scale-tier-unicode-modevvm19-pol]: Agent reported
'status: failed' for fkey: 'modevvm19.tuc.stglabs.ibm.comscale0121547', path:
'/ibm/scale0/unicode_test/migrateDir.popFSDir.19824/unicodedir/test237..test', reason: 'File not found:
/ibm/scale0/unicode_test/migrateDir.popFSDir.19824/unicodedir/test237..test', errno: 'ENOENT'
2020-04-28 14:44:37,455 - policy.policyapiservice - ERROR - [policy_id: scale-tier-unicode-modevvm19-pol]: Agent reported
'status: failed' for fkey: 'modevvm19.tuc.stglabs.ibm.comscale0121557', path:
'/ibm/scale0/unicode_test/migrateDir.popFSDir.19824/unicodedir/test247..test', reason: 'File not found:
/ibm/scale0/unicode_test/migrateDir.popFSDir.19824/unicodedir/test247..test', errno: 'ENOENT'
```

A sample error log from the Policy engine pod logs for an invalid `'destination_tier'`, is shown in the following section:

```
2020-04-28 14:35:40,064 - policy.policyapiservice - ERROR - [policy_id: scale-tier-modevvm19-invalid-destn-pol]: Agent
reported 'status: failed' for fkey:  'modevvm19.tuc.stglabs.ibm.comscale0300770', path: '/ibm/scale0/covid-19-
dataset/inference/NWPU-Chest-X-Ray/2b16346f-8fe4-45ee-b3c7-eaa46fd26041.png', reason: 'Invalid destination_tier:
'gold:abcd'.', errno: 'EINVAL'
```

A sample error log from the Policy engine pod logs when the `'destination_tier'` does not exist on the IBM Storage Scale connection, is shown in the following section:

```
2020-05-12 07:40:34,819 - policy.policyapiservice - ERROR - [policy_id: scale-tier-pol]: Agent reported 'status: failed'
for fkey: 'nikkogpfs26728', path: '/ibm/gpfs2/scale0/customer_e2e/financial_report_file6.pdf', reason: 'Provided internal
pool goldflower does not exist on system: 9.11.212.29', errno: 'EINVAL'
2020-05-12 07:40:34,819 - policy.policyapiservice - ERROR - [policy_id: scale-tier-pol]: Agent reported 'status: failed'
for fkey: 'nikkogpfs26748', path: '/ibm/gpfs2/scale0/customer_e2e/HR_report_US_file6.pdf', reason: 'Provided internal pool
goldflower does not exist on system: 9.11.212.29', errno: 'EINVAL'
```

Copying data by using ScaleAFM application

The ScaleAFM application supports copy function between IBM Storage Scale connection and IBM Cloud® Object Storage connection.

Before you begin

- You must know the IBM Storage Scale versions that are supported. For more information, see IBM Spectrum® Storage software requirements. For more information, see *IBM Spectrum Storage software requirements* in the *IBM Data Cataloging: Concepts, Planning, and Deployment Guide*.
- You need to configure the Active File Management (AFM) functions on IBM Storage Scale filesets under a filesystem. With this configuration, AFM now works as a cache for a bucket or vault on a IBM Cloud Object Storage, or other Cloud Object Storage data source that supports Amazon S3 protocol as a target.

  For more information, see the topic *Introduction to AFM to cloud object storage* in the IBM Storage Scale *Concepts, Planning, and Installation Guide*.
- You need to create the IBM Storage Scale connection in IBM Data Cataloging by configuring the filesystem in the IBM Storage Scale cluster, which contains the AFM fileset, with the Cloud Object Store target vault that is already configured.
- IBM Data Cataloging ScaleAFM application uses the same user that is configured in IBM Data Cataloging to scan the IBM Storage Scale connection and run the data movement policies. For this data movement, the IBM Storage Scale connection is used as the destination_connection . For more information, see Creating or identifying a user ID and password for scanning. For more information, see *Creating or identifying a user ID and password for scanning* in the *IBM Data Cataloging: Concepts, Planning, and Deployment Guide*.

About this task

The IBM Data Cataloging ScaleAFM application provides the advanced copy function between the IBM Storage Scale connection and the IBM Cloud Object Storage connection (IBM Cloud Object Store or an S3 connection).

The application uses the system metadata and custom metadata of the documents that are collected by IBM Data Cataloging. The IBM Data Cataloging collects the metadata for advanced data management function to copy the select set of data from a vault on the IBM Cloud Object Storage connection to an AFM fileset configured on a IBM Storage Scale connection.

The application also provides the cognitive data management capability by analyzing the results of IBM Data Cataloging's AUTOTAG, Content Search, and Deep Inspect policies.

The AFM to Cloud Object Storage is an IBM Storage Scale feature that enables placement of files or objects in an IBM Storage Scale cluster to a Cloud Object Storage. Cloud object services such as Amazon S3 and IBM Cloud Object Storage offers industry-leading scalability, data availability, security, and performance.

The AFM to Cloud Object Storage allows associating an IBM Storage Scale fileset with a Cloud Object Storage. You can use a Cloud Object Storage to store large amount of data for use cases: mobile applications, backup and restore, enterprise applications, and big data analytics. The data is stored locally on IBM Storage Scale filesets and on a Cloud Object Storage. For more information, see the topic *Introduction to AFM to cloud object storage* in the IBM Storage Scale *Concepts, Planning, and Installation Guide*.

Note:
The functions of ScaleAFM application depend on the capability of the underlying storage hardware and its management software, specifically the IBM Storage Scale's AFM functions for Cloud Object Store.

For more information, see the topic *Active File Management for IBM Spectrum Scale* in the *IBM Spectrum Scale: Administration Guide*.

Procedure

1. Log in to IBM Data Cataloging GUI.
2. Go to Administration > Data Management page.
3. Click Add Policy to create a policy.
4. Enter a policy name.
5. Select `COPY` from the Type list.
6. Click Proceed to Configure.
7. Select `ScaleAFM` from the Application list.
8. Enter a Policy filter in the Filter field.
   The policy filter includes the criteria for selecting the files for copying from source Cloud Object Store datasource to a destination IBM Storage Scale datasource. For example, the policy filter can be like **`filetype="pdf"`**.
9. Select the connection type from the Source Connection list.
   It comprises connections that are configured with IBM Data Cataloging of type "Cloud Object Storage" or "Simple Storage Service (S3)". The source connection is the source from where the data is being copied, such as the name of the defined IBM Cloud Object Storage connection.
10. Select the connection type from the Destination Connection list.
    It comprises connections that are configured with IBM Data Cataloging of type 'Spectrum Scale'. The destination connection is the target where the data is being copied, such as the name of the defined IBM Spectrum Scale connection.
    Note: The IBM Storage Scale file system which is configured as the datasource for the destination IBM Storage Scale connection needs to be already configured with a fileset with IBM Storage Scale AFM functions enabled and configured with a Cloud Object Storage vault as the target. This vault must match with the one configured as `Source Connection` in step 9.
11. Enter the fileset value in Spectrum scale_afm_fileset field.
    It is configured as the AFM fileset under the destination IBM Storage Scale connection, as explained in step 10.
12. Enter the directory path in the Target base dir field.
    The Target Base Dir defines the absolute path of a base directory on the destination IBM Storage Scale system to which files from the source system are to be downloaded. It must be a directory path that is expressed as absolute path under the link path of the destination IBM Storage Scale fileset. The field name must begin with a '/'. If not specified, the link path of the IBM Storage Scale AFM fileset that is configured with IBM Storage Scale connection, is used as the **`target_base_dir`**.
13. Click Proceed to Schedule to configure a schedule to the policy.
14. Click the slider control to select one of the following policy status:

    Active
    : An active policy is run whenever its scheduling event is reached.

    Inactive
    : An inactive policy is not run even when its scheduling event is reached, including the Now event.

15. Select a policy schedule under Frequency.
    The schedule indicates when you want to schedule the policy for an execution.
16. Click Proceed to review to review the policy.
17. Click Submit to create the policy.
    - When the policy is completed, you can update the IBM Data Cataloging catalog for the destination IBM Storage Scale connection by manually starting a scan of the connection.
    - If the destination IBM Storage Scale connection is already created enabling live events, then the IBM Data Cataloging catalog automatically gets updated in a while with the entries relevant to the newly copied files from the Cloud Object Storage datasource by using the ScaleAFM data management policy, recently executed.

- **Viewing ScaleAFM application logs**
  Procedure to view ScaleAFM application logs.

Viewing ScaleAFM application logs

Procedure to view ScaleAFM application logs.

Procedure

1. Issue the following command to view the ScaleAFM application logs:

   ```
   oc -n ibm-data-cataloging logs deployment/isd-scaleafmdatamover
   ```

2. Any failures that occur while the files are processed for copying by ScaleAFM Data Mover application, are logged in the application log file. A sample error log from the ScaleAFM Data Mover application when a file (object) does not exist in the vault. The vault is configured with the source Cloud Object Storage data source, while the application is processing the Copy policy.

   ```
   2020-09-30 15:07:47,032 - __main__ - ERROR - Failed queuing the file 'vault1/1ws3711.txt' during AFM data transfer
   ```

   This error log occurs when the IBM Storage Scale AFM Fileset name entered in the policy definition, does not exist on the IBM Storage Scale filesystem that is configured as datasource in the destination IBM Storage Scale connection:

   ```
   2020-09-30 15:02:12,549 - __main__ - ERROR - An exception occurred while processing message: Fileset 'DAAA_ICOS123' not found in file system 'fs1'
   ```

3. Run the following command to view policy engine pod logs:

```
oc -n ibm-data-cataloging logs deployment/isd-policyengine
```

A sample policy engine pod log error from the ScaleAFM Data Mover application when a Copying policy is processed and when a file is not found on the specified 'source_connection':

```
2020-09-30 15:07:47,080 - policy.policyapiservice - ERROR - [policy_id: DAAA_Pol1]: Agent reported 'status: failed' for 
fkey: 'vault11ws3711.txt'
, path: 'vault1/1ws3711.txt', reason: 'Failed queuing the file 'vault1/1ws3711.txt' during AFM data transfer', errno: 
'ENOENT'
```

A sample error log from the policy engine pod logs, when the IBM Storage Scale AFM Fileset name entered in the policy definition, does not exist on the IBM Storage Scale filesystem. The IBM Storage Scale filesystem is configured as datasource in the destination IBM Storage Scale connection:

```
2020-09-30 15:02:13,100 - policy.policyapiservice - ERROR - [policy_id: DAAA_InvFileset]: Agent reported 'status: failed' 
for fkey: 'vault1analyt
ics_results1.txt', path: 'vault1/analytics_results1.txt', reason: 'An exception occurred while processing message: Fileset 
'DAAA_ICOS123' not fou
nd in file system 'fs1'', errno: 'EIO'
```

Exporting metadata to IBM Watson® Knowledge Catalog

IBM Data Cataloging Watson Knowledge Catalog (WKC) connector supports export of the metadata to either an IBM Cloud® instance or to an On-Premise Instance of the Watson Knowledge Catalog .

The default deployment of the WKC connector contains the parameters that are associated with the IBM Cloud instance of Watson Knowledge Catalog . On start, the IBM Data Cataloging WKC connector waits for WKC credential values. In the absence of appropriate credentials, the Export button is not visible on the IBM Data Cataloging user interface. For more information, see Configuring Watson Knowledge Catalog connector.

- **Configuring Watson Knowledge Catalog connector**
  A utility script is provided to simplify and automate the Watson Knowledge Catalog connector configuration. Alternatively, you can configure the Watson Knowledge Catalog connector manually.
- **Exporting metadata**
  Export data with relevant metadata tags from IBM Data Cataloging to Watson Knowledge Catalog .
- **Mapping similar source connections in Watson Knowledge Catalog**
  IBM Data Cataloging supports mapping source connections with Watson Knowledge Catalog connections (WKC) through WKC connector App.
- **Troubleshooting export issues**
  This topic describes some issues faced while exporting data by using the Watson Knowledge Catalog.

Configuring Watson Knowledge Catalog connector

A utility script is provided to simplify and automate the Watson Knowledge Catalog connector configuration. Alternatively, you can configure the Watson Knowledge Catalog connector manually.

Before you begin

Collect details of the IBM Cloud® Watson Catalog instance. The details that are required can vary depending on whether the IBM Watson® Catalog is on Cloud or On-premise.
To add IBM Cloud Watson Catalog details to the IBM Data Cataloging, obtain the following details:

1. Obtain the API key for accessing the Watson Knowledge Catalog application. To access WKC application on the IBM® Cloud, Log in to IBM Cloud and select Create an IBM Cloud API key.
2. Export or copy the key.
3. Copy the base URI of your WKC instance. For IBM Cloud, the WKC instance URI is based on the geographical location.

To add the IBM on-premises Watson Knowledge Catalog details to IBM Data Cataloging, obtain the following details:

- Obtain the values for the following Watson Knowledge Catalog (WKC) parameters from the systems administrator of the On-Premises instance:
  - WKC_USER
  - WKC_PASSWORD
  - WKC_BASE_URI
  - WKC_AUTH_URI

To successfully export document-related metadata from IBM Data Cataloging to the Watson Knowledge Catalog , WKC application must have access to the original documents. This step is achieved by using either a linked or a non-linked data source. Exporting metadata from a linked data source indicates that both Watson Knowledge Catalog and IBM Data Cataloging have access to the original data source location, eliminating the need to copy the original files.

Exporting metadata from a non-linked data source is used when Watson Knowledge Catalog does not have access to original IBM Data Cataloging data source location. In this case, IBM Data Cataloging copies the files that are associated with the export metadata to a location where the Watson Knowledge Catalog service can access these documents. A Watson Knowledge Catalog Cloud Object Storage (COS) connection must be available to export from non-linked data sources.

You must know the details of linked or non-linked data sources before you start configuring the WKC Connector. For more information, see Mapping similar source connections in Watson Knowledge Catalog.

- **Configuring Watson Knowledge Catalog using the utility script**
  Utility script is used to configure the Watson Knowledge Catalog (WKC) and the script automates the WKC connector app configuration process.

- **Configuring Watson Knowledge Catalog manually**
  Procedure to configure Watson Knowledge Catalog (WKC) connector app manually.

Configuring Watson Knowledge Catalog using the utility script

Utility script is used to configure the Watson Knowledge Catalog (WKC) and the script automates the WKC connector app configuration process.

Procedure

Log in to the IBM Data Cataloging instance and run the following command:

```
/opt/ibm/metaocean/utility-scripts/configureWKC.sh
```

The script prompts for credentials and data source connections and configures the WKC application.

The script can be rerun to change the configuration. This is applicable for the following scenarios:

- Exporting from IBM Data Cataloging to a new or different Watson Knowledge Catalog connection.
- A new connection is added to IBM Data Cataloging.

Configuring Watson Knowledge Catalog manually

Procedure to configure Watson Knowledge Catalog (WKC) connector app manually.

Before you begin

Before you perform any changes to the Watson Knowledge Catalog (WKC) connector application configuration, ensure that no WKC export policies are in running state. If you do any changes in the WKC connector application configuration, while policies are running it can cause the WKC connector application to fail to start and resulting in a `CrashLoopBackoff` state. If you face this issue, stop any WKC export policies that are running and the WKC connector application starts normally.

Procedure

1. Log in to the IBM Data Cataloging instance and issue the following command to configure the WKC connector app:

   ```
   oc -n ibm-data-cataloging edit deploy/isd-wkcconnector
   ```

2. For IBM® Watson Knowledge Catalog cloud deployment, define the following parameters:
   - name:WKC_API_KEY
   - value:<API-KEY-VALUE>

   For cloud deployment, ensure that `WKC_USER` and `WKC_PASSWORD` value remain undefined in the environment. To delete WKC connector secret, issue the following command:

   ```
   oc delete secret wkcconnector --ignore-not-found
   ```

   You can set WKC_BASE_URI value for WKC cloud, depending on the on-premises geographical location. The parameter value is the default when WKC_BASE_URI is not defined.

   Example

   ```
   name: WKC_BASE_URI
   value:https://api.dataplatform.cloud.ibm.com/v2/
   ```

3. For IBM Watson Knowledge Catalog on deployment, add the parameter details in the following sample format:

   ```
   name:WKC_BASE_URI
   value:https://<wkc_hostname>/v2/
   name:WKC_AUTH_URI
   value:https:// <wkc_hostname>/icp4d-api/v1/authorize
   ```

   a. To define the WKC connector secret manually, issue the following command:

      ```
      oc create secret generic wkcconnector --from-literal="user=$WKC_USER" --from-literal="password=$WKC_PASSWORD"
      ```

4. Configure any linked data sources if needed by setting the value of the `WKC_CONNECTION_MAP` field.
   For more information see, Mapping similar source connections in Watson Knowledge Catalog.Mapping similar source connections in Watson Knowledge Catalog on page 40.
   Note:
   - If you switch to a different IBM Watson Knowledge Catalog , you can edit the relevant WKC environment variables in the deployment editor to add the values that are associated with the new account. The application automatically restarts and recognizes the new account that is linked.
   - If you add new catalogs to the WKC instance, IBM Data Cataloging retains the old registration information and continues to point to the old catalog IDs. To resolve this issue, follow the procedure:
     - Go to metadata > Applications.
     - To restart the WKC application instance, issue the following commands:

       ```
       oc -n ibm-data-cataloging delete isd-wkcconnector
       ```

Exporting metadata

Export data with relevant metadata tags from IBM Data Cataloging to Watson Knowledge Catalog .

About this task

The Watson Knowledge Catalog is a data catalog system that is not always able to scan relevant data sources and capture the relevant metadata from those files. IBM Data Cataloging helps to bridge this critical gap by helping to export data to Watson Knowledge Catalog with all relevant metadata tags.

Procedure

1. On the IBM Data Cataloging Dashboard, search for the data to be exported by using a specific filter criteria.
2. Click Export Data. The Export Data to Watson Knowledge Catalog window appears.
3. Under Destination Catalog, select the catalog in Watson Knowledge Catalog where you want to export the data.
4. Select the tags that you want to export from the list in Metadata Tags to Export.
5. Click Submit.
6. After completion of the process, the exported data is displayed in the Watson Knowledge Catalog with the tags that are imported from IBM Data Cataloging.
   Now you have an alternative option to create a data management policy to export data to the Watson Knowledge Catalog by using the DATASET_EXPORT policy type.
   Note: Tags in IBM Data Cataloging represent a name (for example, SizeRange) and a value (for example, small, large, or medium). In Watson Knowledge Catalog , the tags represent a value. The exported data maps both of these attributes and it creates a single label.
   For example, `SizeRange:Small`.

Mapping similar source connections in Watson Knowledge Catalog

IBM Data Cataloging supports mapping source connections with Watson Knowledge Catalog connections (WKC) through WKC connector App.

When metadata is being exported to WKC, IBM Data Cataloging might use a source connection that WKC can also connect to. In such a scenario, you can configure the connection mapping within the WKC Connector App.

The connections that can be linked to are:

- Amazon S3
- IBM Cloud® Object Storage
- IBM Storage Scale

Note: Connections with IBM Storage Scale are established through an S3 connection as WKC does not support IBM Storage Scale directly.
The details for configuring the connection maps with each of these source connections are described.

S3

For an IBM Data Cataloging S3 connection, the WKC connection must contain the following details:

Bucket
: If the bucket name is configured, then you do not need to provide any further configuration details. The WKC Connector App can infer the details from the global namespace of Amazon S3 buckets.
  If the bucket name is not provided, then configure the *WKC_Connection_Map* environment variable by using the following format: `<datasource>;<cluster>:<wkc connection name>`. A sample variable value is shown. All documents, that the WKC App receives in its work message corresponding to the data source and cluster pair that is defined in the variable, maps to the WKC connection of that name.

```
WKC_CONNECTION_MAP=testbucket1.sd.ibm.com;s3.eu-west-1.amazonaws.com:s3_con_no_bucket
```

Note: It is mandatory to provide the bucket name while you are configuring details in IBM Data Cataloging but it is optional for WKC.

IBM Cloud Object Storage Infrastructure

For an IBM Data Cataloging IBM Cloud Object Storage connection the WKC connection must contain the following details:

Login URL
: The login URL must be that of the accessor defined in IBM Data Cataloging. The data source for IBM Cloud Object Storage is the vault. However, since it is not possible to provide a vault here, you must provide the mapping within the environment variable *WKC_CONNECTION_MAP* in the following format: `<datasource>;<cluster>:<wkc connection name>`. A sample variable value is shown.

```
WKC_CONNECTION_MAP=vault1;e09cdac0-80f8-73be-00ed-cb8edeede242:local_cos_con
```

Multiple IBM Cloud Object Storage connections to the same system can map to the same WKC connection (as it is at a higher level and can see all vaults).

IBM Storage Scale

Connection mapping with IBM Storage Scale must be done through an S3 connection as WKC cannot connect directly with it.

To establish an S3 connection, configure the following details in the WKC connector app:

Endpoint URL
: The S3 Endpoint URL on the IBM Storage Scale mode. For example, `http://modevvm19.tuc.stglabs.ibm.com:9000`.

Access Key
: Type the S3 access key.

Secret Key
: Type the S3 secret key.

Bucket
: Do not enter values in the bucket field.

Define the mapping within the environment variable *WKC_CONNECTION_MAP* in the following format: `<datasource>;<cluster>:<wkc connection name>`. A sample variable value is shown:

```
WKC_CONNECTION_MAP=scale0;modevvm19.tuc.stglabs.ibm.com:s3_scale
```

Mapping multiple connections

You can map multiple connections within the same environment variable by using commas to separate the values. A sample is shown:

```
WKC_CONNECTION_MAP=vault1;e09cdac0-80f8-73be-00ed-
cb8edeede242:local_cos_con,scale0;modevvm19.tuc.stglabs.ibm.com:s3_scale,testbucket1.sd.ibm.com;s3.eu-west-
1.amazonaws.com:s3_con_no_bucket
```

Run the following command to edit the WKC Connector app deployment and add the mapping:

oc -n spectrum-discover edit deploy/spectrum-discover-wkcconnector

The WKC connections are configured in the following format:

```
name: WKC_CONNECTION_MAP
value: vault1;e09cdac0-80f8-73be-00ed-
cb8edeede242:local_cos_con,scale0;modevvm19.tuc.stglabs.ibm.com:s3_scale,testbucket1.sd.ibm.com;s3.eu-west-
1.amazonaws.com:s3_con_no_bucket
```

Note: The **edit** command configures the WKC connection mapping and automatically restarts the WKC pod.

Troubleshooting export issues

This topic describes some issues faced while exporting data by using the Watson Knowledge Catalog.

- **Authentication failure with Watson Knowledge Catalog - both on-Premise and IBM Cloud**
  Resolve authentication failure with the Watson Knowledge Catalog (WKC).
- **Incorrect WKC URL configuration**
  Resolve the errors that occur due to an incorrect WKC URL configuration.
- **Invalid connection type in catalog**
  Resolve the error that occurs due to an invalid connection type in catalog.
- **No linked connection type**
  Resolve the errors that occur when links cannot be identified for the connection type.
- **WKC connector pod in CrashLoopBackoff state**
  Check the following settings if the Watson Connector pod is in the CrashLoopBackoff or Error state.
- **S3 connection issues**
  Resolve the issues faced owing to S3 connection.

Authentication failure with Watson Knowledge Catalog - both on-Premise and IBM Cloud

Resolve authentication failure with the Watson Knowledge Catalog (WKC).

The following error is displayed on the Watson Knowledge Catalog connector logs, when the connector fails to authenticate with the Watson Knowledge Catalog instance.

```
2020-07-16 12:46:04,870 - WKCConnector - ERROR - User authentication failed with response code 401
2020-07-16 12:46:04,871 - WKCConnector - INFO - No valid token available, cannot connect to WKC. Please set API Key or User 
Credentials
```

When the WKC authentication fails, check to ensure that the correct values are defined for the following environment variables for the on-premises connection:

```
WKC_USER, WKC_PASSWORD
```

If you are using an IBM Cloud® connection type, you must check the value of the environment variable `WKC_API_KEY`.

Run the following command to check and confirm the configuration for the specified environment variables in the pod helm chart: **oc edit deployment spectrum-discover-wkcconnector**

Incorrect WKC URL configuration

Resolve the errors that occur due to an incorrect WKC URL configuration.

The following errors occur:

```
HTTPSConnectionPool(host='machine.ibm.com', port=443): Max retries exceeded with url: /v2/catalogs (Caused by
NewConnectionError('<urllib3.connection.HTTPSConnection object at 0x7f63e7973c18>: Failed to establish a new connection: [Errno
-2] Name or service not known',))
2020-07-09 22:59:29,412 - WKCConnector - ERROR - Cannot communicate with WKC to retrieve catalog information. Check config
URIs.
```

A possible reason for the error is that invalid values are defined for the environment variable **WKC_BASE_URI**. You can get similar errors if you define incorrect values for the **WKC_AUTH_URI** environment variable.

Run the following command to check and confirm the configuration of the said environment variables in the pod helm chart: **oc edit deployment spectrum-discover-wkcconnector**.

The following error occurs, when incorrect API key is provided for WKC cloud connections:

```
2020-09-28 13:29:30,189 - WKCConnector - ERROR - Supplied API Key failed with response code 400
2020-09-28 13:29:30,190 - WKCConnector - ERROR - Unable to connect to WKC, resetting agent registration
```

The following error occurs, when incorrect user/password is provided for WKC on-premises connections:

```
2020-09-28 13:34:24,530 - WKCConnector - INFO - API Key not set, checking for user/password
2020-09-28 13:34:24,767 - WKCConnector - ERROR - User authentication failed with response code 401
2020-09-28 13:34:24,768 - WKCConnector - ERROR - Unable to connect to WKC, resetting agent registration
```

Invalid connection type in catalog

Resolve the error that occurs due to an invalid connection type in catalog.

When a valid connection type cannot be detected, the following error occurs:

```
2020-06-30 22:37:09,628 - WKCConnector - ERROR - Could not find a supported connection in the supplied catalog
```

This error might occur in the scenario when the connection type within the Watson Knowledge catalog (WKC) does not match the connection type on Spectrum Discover. For example, if you try to export metadata from an IBM Data Cataloging Cloud Object Storage (COS) source to a WKC catalog, which does not have an IBM Cloud® Object Storage connection. The error can also occur if you have an IBM Cloud Object Storage bucket on WKC mapped to an on-Premises IBM Cloud Object Storage bucket that is configured on IBM Data Cataloging.

No linked connection type

Resolve the errors that occur when links cannot be identified for the connection type.

The following errors can occur resulting in an export failure:

```
2020-07-01 13:58:34,129 - WKCConnector - ERROR - Cannot export /Changing ACEs/Not for Luke and Amy/Full Gamora/Partial
Luke/Partial Amy/No Clara/I mean not clara/TinyDoc.txt as document is not on a linked connection, and was unable to be
downloaded from the SD connection for copy
2020-07-01 13:58:34,129 - WKCConnector - ERROR - Failed to export /Changing ACEs/Not for Luke and Amy/Full Gamora/Partial
Luke/Partial Amy/No Clara/I mean not clara/TinyDoc.txt
```

An exported document must either be on a linked connection, or must be available for copying to backup IBM Cloud® Object Storage in the WKC catalog, if that exists.

WKC connector pod in CrashLoopBackoff state

Check the following settings if the Watson Connector pod is in the CrashLoopBackoff or Error state.

Check that there are no Watson Knowledge Catalog export policies that are in running state when you perform any changes to Watson Knowledge Catalog connector application configuration.

Check the settings for the following environment variables if you are using IBM Cloud® Watson Knowledge Catalog:

```
WKC_API_KEY
```

Check the correct value for the following parameter if you are using IBM Cloud Watson Knowledge Catalog:

```
WKC_BASE_URI
```

Check the settings for the following environment variables if you are using an on-premises Watson Knowledge Catalog :

```
WKC_BASE_URI
WKC_AUTH_URI
WKC_CONNECTION_MAP
```

After successfully exporting records, if you change an environment variable to an invalid value and then export records again, you can end up with unprocessed messages in the Kafka buffer. You must clear these messages before you can change the invalid environment variable to the correct value. Run the following commands to clear the Kafka queue and then restart the Watson Knowledge Catalog connector application.

Clear Queue

/opt/kafka/bin/kafka-topics.sh --delete --zookeeper localhost:2181 --topic WKCConnector_work

Recycle Pod

oc -n ibm-data-cataloging delete pod -l role=wkcconnector

If the Watson Knowledge Catalog connector pod remains down after you have performed the above steps, do the following:

1. If the Watson Knowledge Catalog connector (WKC connector) pod in IBM Data Cataloging cannot be started, delete the existing pod using the following command:

   ```
   oc -n ibm-data-cataloging delete pod -l role=wkcconnector
   ```

2. Retrieve the logs from the WKC connector pod using the following command:

   ```
   oc -n ibm-data-cataloging logs deployment/isd-wkcconnector
   ```

3. Check for errors in the logs by running the following command:

   ```
   oc -n ibm-data-cataloging logs deployment/isd-wkcconnector | grep error
   ```

4. If the log terminates with an exception, check the policy engine log for errors by using the following command:

   ```
   pollog | grep error
   ```

5. Check if the following error message appears at the time of the exception:

   ```
   <time_stamp> - policy.policyapiservice - ERROR - b'Error: Cannot update application, 'WKCConnector, it is in use by the follow
   ```

6. The WKC connector pod might not start until the error shown in step 5 is resolved.
   Go to the Tag Management Policies section in the user interface and stop all running policies.

7. After the policies are stopped, then restart the WKC connector pod by deleting it:

   ```
   oc -n ibm-data-cataloging delete pod -l role=wkcconnector
   ```

S3 connection issues

Resolve the issues faced owing to S3 connection.

To enable export of IBM Storage Scale file metadata without copying the actual file to Watson Knowledge Catalog, the latter must be connected to IBM Storage Scale through an S3 interface.

If you are experiencing issues with IBM Storage Scale file exports, check whether the connection is present in the connector connection map and that the details are made available in Watson Knowledge Catalog. You must check to see that correct values are defined for the IP, port, access, and secret key. To preview the documents in the catalog, you must check that the S3 interface is available on the IBM Storage Scale system.

Importing externally curated tags for COS/S3 using import tags application

The import tags application is used to import a set of externally curated tags for Cloud Object Storage and S3 services.

Before you begin

The S3/COS (Cloud Object Storage) data source that is associated with the objects in the external tags file must be scanned before you run an import tags policy.

Tag names must be defined in IBM Data Cataloging before the Import Tags policy is run. You must refrain from using Restricted tag type and use Open or Characteristics tag types.

Note:

- You can create limited number of Open type tags and the tags must correspond to values in the header row of the comma-separated values (CSV) file. Tags that are not defined before you trigger the policy are not imported.
- The CSV file must be in the bucket that is defined in the data source.
- Only a single tag file is supported per policy.

Requirements of the external CSV file are listed as shown.

- The tags file must be in CSV format.
- The first row in the file must be a header row or line.
- The first column must be the full object path or name. For example, if the bucket is auto_data, and the object name is car1/image1.png, then the first column entry is auto_data/car1/image1.png.
- The value in the header row for the first column is not restricted by IBM Data Cataloging.
- The subsequent columns in the CSV file represent the tag values that can be imported into IBM Data Cataloging for the associated object records.
- The second through Nth entries in the header row must correspond to valid tags in IBM Data Cataloging that are defined before you run the import tags policy.
- Each entry in the CSV file must represent a unique file in the data source.

```
Example contents of a CSV file:
objectname,bus,tree,stop_sign,red_light,yellow_light,green_light,pedestrian
auto_data/car1/image1.png,1,3,0,1,1,0,1
auto_data/car2/image1.png,1,6,0,0,0,0,12
auto_data/car2/image2.png,1,3,0,2,1,0,1
auto_data/car3/image1.png,1,3,0,2,1,0,2
```

The following tags can be defined in IBM Data Cataloging from the records available in the CSV file:

- bus
- tree

- stop_sign
- red_light
- yellow_light
- green_light
- pedestrian

Note: Only a single tag import policy can be run at a time.

About this task

The IBM Data Cataloging import tags application allows a user with DATA ADMIN role to apply a pre-curated set of labels (tags) that are available in an external CSV file to S3/COS object records stored in IBM Data Cataloging.

For example, an external analytics job might generate tag information for a set of S3/COS objects, and save this information into a CSV file. The CSV file comprises an entry for each object that contains an object name and an associated list of labels or tags.

The import tags application can merge these tags into the associated object records in IBM Data Cataloging, extending the records with new information.

Procedure

1. Configure a COS/S3 data source connection for IBM Data Cataloging, with the vault or bucket that gets scanned later.
   The resulting system metadata that is indexed in IBM Data Cataloging is then enriched with the imported tag data.

   Example
   COS/S3 datasource connection name is **`camera_vault`** and the vault or bucket name is **`camera_lidar_semantic`**. Connection is created and scan is run.

   For more information about configuring and scanning COS/S3 data sources, see Configure data source connections. For more information about configuring and scanning COS/S3 data sources, see *Configure data source connections* in the *IBM Data Cataloging: Concepts, Planning, and Deployment Guide*
2. Place external tag file on to the data source so that IBM Data Cataloging can access it.
   A single CSV file comprises a list of objects and tags. The objects and tags are applied to the indexed records for a COS/S3 data source. These objects and tags must be uploaded to the configured source storage bucket by using any IBM® COS/S3 compatible data management utility.

   Example
   IBM Data Cataloging data source with connection name **`camera_vault`** is configured to scan vault and bucket **`camera_lidar_semantic`**.
3. Define tag names in IBM Data Cataloging.
   a. Use the headers from the CSV file, starting with the second column.
   b. Create corresponding tags in IBM Data Cataloging for any of the columns where you want to import data.
   For example, for a column you want to import with header value ColumnA, create a tag ColumnA. The columns for which you do not want to import data, must not store tags that are defined with the header value.

   Example
   CSV file comprises columns with header values front, rear, center, car, bicycle, pedestrian, and truck that you want to import. Characteristics tags are created with names: front, rear, center, car, bicycle, pedestrian, and truck.

   For more information about creating tags, see Creating tags.
   For more information about creating multiple tags through REST API see, /policyengine/v1/tags/:POST.For more information about creating multiple tags through REST API, see */policyengine/v1/tags/:POST* in the *IBM Data Cataloging: REST API Guide*.
4. Initiate the policy by the using REST API interface. For more information, see Initiating Policy Using REST API.

- **Initiating Policy Using REST API**
  Initiating a policy using the REST API triggers the import tag policy engine to process metadata tags for the configured data sources.

Initiating Policy Using REST API

Initiating a policy using the REST API triggers the import tag policy engine to process metadata tags for the configured data sources.

- **Registering import tags application**
  The process of registering an import tags application.
- **Defining the import tag policy type**
  Procedure to define the import tags policy in a JSON payload.
- **Viewing the policy status**
  You can view policy status either from the IBM Data Cataloging GUI or by running read (GET) request.
- **Viewing the import tags application log**
  You can view the import tag application logs through a simple procedure.

Registering import tags application

The process of registering an import tags application.

About this task

The import tags application is built-in and pre-registered in IBM Data Cataloging. To view the registered application, you can use the IBM Data Cataloging GUI.

For more information, see Managing applications. Managing applications on page 63.

Defining the import tag policy type

Procedure to define the import tags policy in a JSON payload.

About this task

You can create the import tags policies by using the policy management REST API. A sample policy that is defined in a JSON payload is shown:

```
{
        "pol _id": "importtags_pol",
        "action_id": "IMPORT_TAGS",
        "action_params": {
                "agent":"Import Tags",
                "source_connection":"camera_vaul t",
                "tag_file_path":"camera_lidar _semantic/A2D2_labels.csv",
                "tag_file_type":"csv"
        },
        "schedule": "NOW",
        "pol_state": "active",
        "pol_filter": "datasource=' camera_lidar_semanti c' "
}
```

The following table lists the action parameters for defining the import tags policy type:

Table 1. List of action parameter for an import tags policy type

Action Parameter	Description
souce_connection	The Cloud Object Storage (COS) or S3 connection name that is defined in IBM Data Cataloging for the records that the imported tags can be applied to.
tag_file_path	The absolute path (bucket or object name) of the CSV file that contains the list of objects and associated tags to import.

IMPORT_TAGS is the new action_id that is defined for the import tags policies.

Procedure

1. Send the following query request to list all the supported action IDs on IBM Data Cataloging node:

   ```
   curl -H "Authorization:Bearer<token>"-k https://<data_cataloging_host>:443/policyengine/ v1/action_ids
   ```

2. Send the following query request to the IBM Data Cataloging node to obtain the current Import Tags application schema:

   ```
   curl-H" Authorization:Bearer <token>"-k https://<data_cataloging_host>:443/policyengine/v1/applications/ImportTags/schema?
   action_id=I MPORT_TAGS
   ```

3. To define the policy, and schedule or run it immediately, use the /policyengine/v1/policies -d '<data>': POST endpoint to process the request.
   For more information, see /policyengine/v1/policies -d '<data>': POST
   For more information, see */policyengine/v1/policies -d '<data>': POST* in the *IBM Data Cataloging: REST API Guide*.

Example

Send the following request to create an Import Tags policy for data source connection camera_vault that runs immediately by using the external tag file A2D2_labels.csv:

```
curl - k - H ' Authorization: Bearer <token>'
https: //<data_cataloging_host>/policyengine/v1/policies-d'
        {
                "pol_id": "importtags",
                "action_id": "IMPORT_TAGS",
                "action_params": {
                        "agent": "Import Tags",
                        "source_connection": "camera_vault",
                        "tag_file_path": "camera_lidar_semantic/A2D2_labels.csv",
                        "tag_file_type": "csv"
                },
                "schedule": "NOW",
                "pol_state": "active",
                "pol _filter": "datasource='camera_lidar _semantic' "
        }
'-XPOST -H" Content-Type:application/json"
```

The following response is returned:

```
Policy 'importtags_ut'added
```

Viewing the policy status

You can view policy status either from the IBM Data Cataloging GUI or by running read (GET) request.

About this task

Follow the procedure shown to view the policy status:

Procedure

1. Log in to IBM Data Cataloging GUI.
2. Go to Administration > Data Management tab.
   The policy status can also be checked with the following REST API read request:

   ```
   /policyengine/v1/policies: GET and /policyengine/v1/policies/<policy_name>: GET
   ```

   For more information, see /policyengine/v1/policies: GET and /policyengine/v1/policies/<policy_name>: GET.For more information, see */policyengine/v1/policies: GET and /policyengine/v1/policies/<policy_name>: GET* in the *IBM Data Cataloging: REST API Guide*.
   Note: As import tags policies rely on an external list of objects/file and not on a policy filter, the policy preview service does not apply to this policy type. The number of records updated in IBM Data Cataloging will be no more than the number of entries in the external tag file.
3. Go to Policies tab.
   Policies are listed with their status as active or inactive.

Example

A sample REST API request for checking import tag policy and the corresponding response is shown:

```
curl -k -H' Authorizati on:Bearer <token>'
https://<data_cataloging_host>/policyengine/v1/policies/importtags_ut

Response:
{
        "pol_id": "importtags_ut",
        "action_id": "IMPORT_TAGS",
        "action_params":"{\"agent\":\"Import Tags\", \"source_connection\": \"camera_vault\", \"tag_file_path\":
\"camera_lidar_semantic/A2D2_labels.csv\",\"tag_file_type\":\"csv\"}",
        "schedule":"NOW",
        "pol_filter":"datasource='camera_lidar_semantic' ",
        "pol_state":"active",
        "pol_status":"running",
        "explicit": "true",
        "execution_info":"{\"start_time\":\"2020-09-17  21: 04: 35\",\"total _count\":350,\"submitted_count\":
400,\"failed_count\": 8,\"completed_count\":300,\"ski pped_count\":0,\"i mport _tags_count\": 0,\"run_id\":\"a3b0f
7647647443e8c5237aebe2c5408\"}",
        "policy_owner": "sdadmin",
        "last_updated_by": "sdadmin",
        "last_updated":"2020-09-17T21:04:35.000Z",
        "collections":[]
}
```

During processing, the import tags application does some basic checking on the CSV file. If a row has fewer columns than the header row, the row is skipped. If a header value contains an invalid or missing tag name, the values for that column are not imported. These records are added to the failed count in the policy.

Viewing the import tags application log

You can view the import tag application logs through a simple procedure.

About this task

Follow the procedure to view the import tag application logs:

Procedure

Issue the following command to view the import tag application logs:

```
oc -n ibm-data-cataloging logs deployment/isd-importtags
```

A sample error log from the import tag application is shown in the following section. This error occurs when there exist rows in the CSV file that contain fewer columns than the header row:

```
2020-09-17 21:04:41,538-Import Tags-ERROR-Skipped  entry. Ragged row: expected 44 columns, found 36
```

A sample error log from the import tag application is shown in the following section. This error occurs when there exist empty rows in the CSV file (newline):

```
2020-09-17 21:04:41,540-Import Tags-ERROR-Skipped entry. Empty row.
```

A sample error log from the import tag application is shown in the following section. This error occurs when there exist rows with an invalid file/object name in the first column:

```
2020-09-17 21:04:56,239-Import Tags-ERROR-Skipped entry. Object/file name not found in csv column 0 (camera_lidar_semantic).
```

Performing retention analytics on IBM Storage Protect archive data

Use IBM Data Cataloging to perform retention analytics on archive data managed by IBM Storage Protect.

Before you begin

You must have an admin role to do this task.

About this task

IBM Data Cataloging feature is leveraged to perform search, analytics, and reporting on the retention period for archive data managed by the IBM Storage Protect.

The existing IBM Storage Protect data connection and scan function in the IBM Data Cataloging is enhanced to obtain the additional retention information from the IBM Storage Protect servers for the archive data.

Grouping expiration values to tag values
: You can group the expiration values and run SQL queries to track the retention time period for the archive data managed by IBM Storage Protect. The custom tags are applied by `AUTOTAG` policies that look for all the data that is about to expire by a certain time period. The tagged policies can be scheduled to run periodically. After you apply `AUTOTAG` policies, the IBM Data Cataloging GUI and REST APIs can be used to search, discover, and perform analytics on the data that is going to expire during the various time periods.

Perform the following steps to apply retention analytics on the IBM Storage Protect archive data.

Procedure

1. Enable retention analytics feature. For more information, see, Enabling the retention analytics feature.
2. Add IBM Storage Protect data source and initiating a scan. For more information, see Adding IBM Storage Protect data source and initiating scan.
3. Search the expiry time. For more information, see Searching expiry time.

- **Enabling the retention analytics feature**
  Enable the retention analytics feature in IBM Data Cataloging.
- **Adding IBM Storage Protect data source and initiating scan**
  Add IBM Storage Protect data source before you initiate a scan on the archive data.
- **Searching expiry time**
  Search the expiration time for the IBM Storage Protect archive data that is going to expire in certain time.

Enabling the retention analytics feature

Enable the retention analytics feature in IBM Data Cataloging.

About this task

To enable the retention analytics for IBM Storage Protect archive data feature in IBM Data Cataloging, you must first set an environment variable in IBM Data Cataloging.

Following are the list of environment variable that you can set in the IBM Data Cataloging. The expiration data is not scanned if the expiration variables are not set.

PROTECT_ARCHIVE_WITH_EXPIRE_ONLY
: When the variable is set to 'True', the IBM Storage Protect scan feature scans the archive data only and includes the expiration data that is needed for the expiration date analysis. When the variable is set to 'False' or not set at all, the IBM Storage Protect performs the default action of scanning all the backup and archive records without including the expiration data that is needed for the expiration date analysis.
  Important: When the variable value is set, the IBM Data Cataloging server performs IBM Data Cataloging scans on the archive data only that includes the expiration data. This action takes precedence over any other environment variables that are defined for the retention analytics feature.

PROTECT_ARCHIVE_EXPIRATION
: When the variable is set to 'True', IBM Storage Protect scans the archive data and checks that expiration data is also scanned. When the variable is set to 'False' or not set at all, the expiration data is not included, unless the value of *PROTECT_ARCHIVE_WITH_EXPIRE_ONLY* is set to true.

PROTECT_ARCHIVE_DISABLED
: When the variable value is set to 'True', IBM Storage Protect scan feature does not scan the archive data unless the value of *PROTECT_ARCHIVE_WITH_EXPIRE_ONLY* is set to true. When set to 'False' or not set at all, IBM Storage Protect scan feature scans the archive data.

PROTECT_BACKUP_DISABLED
: When the variable is set to 'True', IBM Storage Protect scan feature does not scan the backup data. When set to 'False' or not set at all, IBM Storage Protect scan feature scans the backup data unless the value of *PROTECT_ARCHIVE_WITH_EXPIRE_ONLY* is set to true.

Important: The *PROTECT_ARCHIVE_EXPIRATION* value must be set to "True", if a complete scan of the IBM Storage Protect connection with the archive retention feature is to be completed.
Perform the following steps for enabling retention analytics feature in the IBM Data Cataloging environment:

Procedure

1. Issue the following command to log in to the IBM Data Cataloging by using ssh:

   ```
   ssh <adminuser>@<data_cataloging_host>
   ```

   Note: The **`<adminuser>`** is a IBM Data Cataloging server user who is assigned with an administrator role and the **`<data_cataloging_host>`** is either the FDQN or IP of the IBM Data Cataloging server.
2. Issue the following command to edit the configmap:

   ```
   oc edit configmap connmgr
   ```

   After you run the preceding command, the output looks similar to the following as shown here. You can modify the following file and add or update the variables and its value in the **`data`** section.

   ```
   apiVersion: v1
   data:
     CONNMGR_ENDPOINT: /connmgr/v1/
   ```

```
  CONNMGR_PROTOCOL: http
  PROTECT_ARCHIVE_WITH_EXPIRE_ONLY: "True"
kind: ConfigMap
metadata:
  creationTimestamp: "2022-08-26T19:55:39Z"
  name: connmgr
  namespace: spectrum-discover
  ownerReferences:
  - apiVersion: spectrum-discover.ibm.com/v1alpha1
    kind: SpectrumDiscover
    name: spectrumdiscover
    uid: 001fb331-c2ef-4b28-8827-5c7d6a702904
  resourceVersion: "13761646"
  uid: bf621847-4c00-44c9-a7f2-4d3430ede67b
```

3. Save the preceding updated file and issue the following command to verify that the data is updated.

```
oc get configmap connmgr -o yaml
```

4. Issue the following commands to verify that the update is made in the connection manager pod:

```
oc -n ibm-data-cataloging delete sts/isd-connmgr
```

Adding IBM Storage Protect data source and initiating scan

Add IBM Storage Protect data source before you initiate a scan on the archive data.

About this task

After the retention analytics feature is enabled in the IBM Data Cataloging, you can add one, or more IBM Storage Protect data sources and initiate a scan. For more information, see Creating an IBM Storage Protect data source connection.

Important: After the scan is complete, the database needs to be refreshed before you add the tags and the policies.
Do the following to tag expiry time:

Procedure

1. Log in to the IBM Data Cataloging GUI.
2. Click and go to Metadata > Tag Management
3. On the Tags tab, click the Add Tag.
4. In the New organizational tags window, enter the tag name in the Name field.
   For example, enter `ExpirationDate`, `ExpirationDays`, or `Expiration` as a tag name.
5. Select `Restricted` from the tag Type list.

   Restricted
      A Restricted tag can be anything that describes groups of records, but is restricted to a set of pre-defined values, such as data classification or billing department number.

6. Enter the expiration values into the Values field and press Enter key to save each expiration value.
   For example, to group all the data that is going to expire by the end of a year, create a tag that is called 'Expiration Year'. Similarly, to group all the data that is going to expire after few days from now, create a tag that is called 'ExpirationDays'.
   Each saved expiration values (date or number of days) are grouped and displayed underneath the Values field. These expiration values are grouped and associated with a custom tag that is created. For example, tag values can be 01-January-2021 for a date-specific expiration or, 1 day, 1 month, and 1 year for days-specific expiration.
7. Click Submit.
   The tag name, types, and values are displayed on the Tags tab after you submit the record.
   Important: You must add all the expected values before you submit the record.
8. Go to the Policies tab and click Add policy.
9. On the Define page, enter the policy name in the Name field.
10. Select `AUTOTAG` from Policy Type list.
11. On the Policy page, enter the values in Filter field.

    Sample filter input for 30-days from today as expiration period
       `mtime > NOW() AND mtime < (NOW() + 30 DAYS) AND platform like 'Spectrum Protect' and state like 'Archive'`
    Sample filter input for all data that is going to expire by the end of year 2021
       `State in 'Archive' AND mtime between '2021-01-01' and '2021-12-31'`
       Important: The date in the query must be in the format of 'YYYY-MM-DD'.

12. Choose the custom tag created to track the expiration time period from Tag list.
    The tag list contains all the custom tags created with their saved tag name, type, and value details created on Tags tab. You can add more than one tag here to tag a policy.
13. Choose the expiration value (date or number of day's) from the Values list.
    The expiration values that you choose from the Values list are tagged with custom tag that you choose from the Tag list.
    For example, if you choose `ExpirationDays` as custom tag, you can choose the expiration value `30DayExpiration` as the number of days for data to expire to tag with the custom tag. Alternatively, if you want to group records by all the data that is going to expire by the end of a year, select the 'ExpirationYear' value.

14. Schedule the policy to run periodically by using the available schedule options.

    Schedule options
       Now
       Daily

Weekly
Monthly

15. Verify the policy details on Review page before you submit the save the policy.

    Sample policy detail on review page as shown

```
Name:
30DayExpirationPolicy
Action id:
AUTOTAG
Pol filter:
mtime > NOW() AND mtime < (NOW() + 30 DAYS) AND platform like '%Spectrum Protect%' and state like '%Archive%'
Pol state:
active
Collections:
```

Searching expiry time

Search the expiration time for the IBM Storage Protect archive data that is going to expire in certain time.

About this task

After the IBM Storage Protect archive record expiration dates are tagged, do the following to search and apply further analytics on the records.

Important: Each policy that is defined must be executed before you perform search.

Procedure

1. Log in to the IBM Data Cataloging GUI.
2. Click  and go to Search > Query builder > SQL query page.
3. Choose groupings based on the tag that is created from Group by list.
   For example, ExpirationDate.
4. Enter the SQL query in the query box.
   For example, `State in 'Archive' AND mtime between '2021-01-01' and '2021-12-31'`
   Important: The mtime date in the query must be in the format of 'YYYY-MM-DD'.
5. Set the limit for the number of records returned from the Limit results list.
6. Click View results.
   The Query Results page displays the grouped results with expiration time, total files filtered, and the total size.
   Note: Where, Total size represents the original size of the files that are stored in the backend. Total size that is returned in the report does not account for space savings from compression and data deduplication. Therefore, can over-state the amount of storage that is going to expire on the IBM Storage Protect server. Currently, the data expiration support within the IBM Data Cataloging does not consider the amount of space that is saved in the size of the file for container pools that supports compression or data deduplication.
7. To generate a report, select the checkbox and click Generate report. On the Generate report dialog, Enter the report name in the Name field, and click Submit.
8. To view the report, go to Search > Reports.
   After you submit the report, the new report gets generated and listed on Reports page in a table with following metadata details:
   - Report name
   - Last run
   - Duration
   - Status
   - Output size
9. Select a report from the table lists and click Report summary.
   A View Data Report window is displayed with all the details of a report.

   Sample report details as shown

```
View Data Report

Report
SProtectDemoReport
Last Run
Wed Jun 30 2021 19:09:52 GMT+0530 (India Standard Time)
Duration
0 seconds
Status
Complete
Size
247 Bytes
Name
"SProtectDemoReport"
Group By
["TAG7","datasource","filespace","nodename"]
Filters
[]
Sort By
""
Query
"(daystoexpire = '180' AND datasource = 'WIN-RUT6M1TE74U' AND filespace = '\\\\win-rut6m1te74u\\c$' AND nodename =
'WIN-RUT6M1TE74U') OR (daystoexpire = '720' AND datasource = 'WIN-RUT6M1TE74U' AND filespace = '\\\\win-
```

```
rut6m1te74u\\c$' AND nodename = 'WIN-RUT6M1TE74U')"
See on table
```

Managing tags

A tag is a custom metadata field used to enhance system metadata with organization-specific information. For instance, an organization may categorize its storage by project, cost center, or region but the attributes not provided by the storage system itself. Tags enable data classification, search, and reporting based on these organizational dimensions. Use tags to store this additional information and to manage or locate data using metadata that reflects your organizational context.

Permissions

Security Administrators
: Cannot create, update, delete or list any type of tag.

Data Administrators
: Can create, modify, delete, and list both `Label` and `Tag` tags.

Data Users
: Can list all tags.
: Cannot create, modify or delete `Tag` tags.
: Can create and modify `Label` tags.

Types of tags

Label
: Lightweight identifiers used for simple tagging. Useful for short descriptors like `project`, `region`, or `team`. Value size can be up to `256`, `1024`, or `4096` characters.

Tag
: Structured tags with a predefined list of accepted values. Ideal for controlled vocabularies such as `status`, `compliance level`, or `department`. Value size is always limited to `4096` characters.

- **Creating tags**
  Use the following information when you create tags.
- **Viewing and searching tags**
  View and search tags from the Tags tab on the Metadata page to locate specific tags by name, type, or value, and sort or filter results as needed.
- **Editing tags**
  Edit tag values from the Tags tab on the Metadata page to update the allowed values for existing organizational tags in IBM Spectrum Discover.
- **Deleting tags**
  Delete tags from the Tags tab on the Metadata page to remove metadata classifications that are no longer needed from the IBM Spectrum Discover catalog.

Creating tags

Use the following information when you create tags.

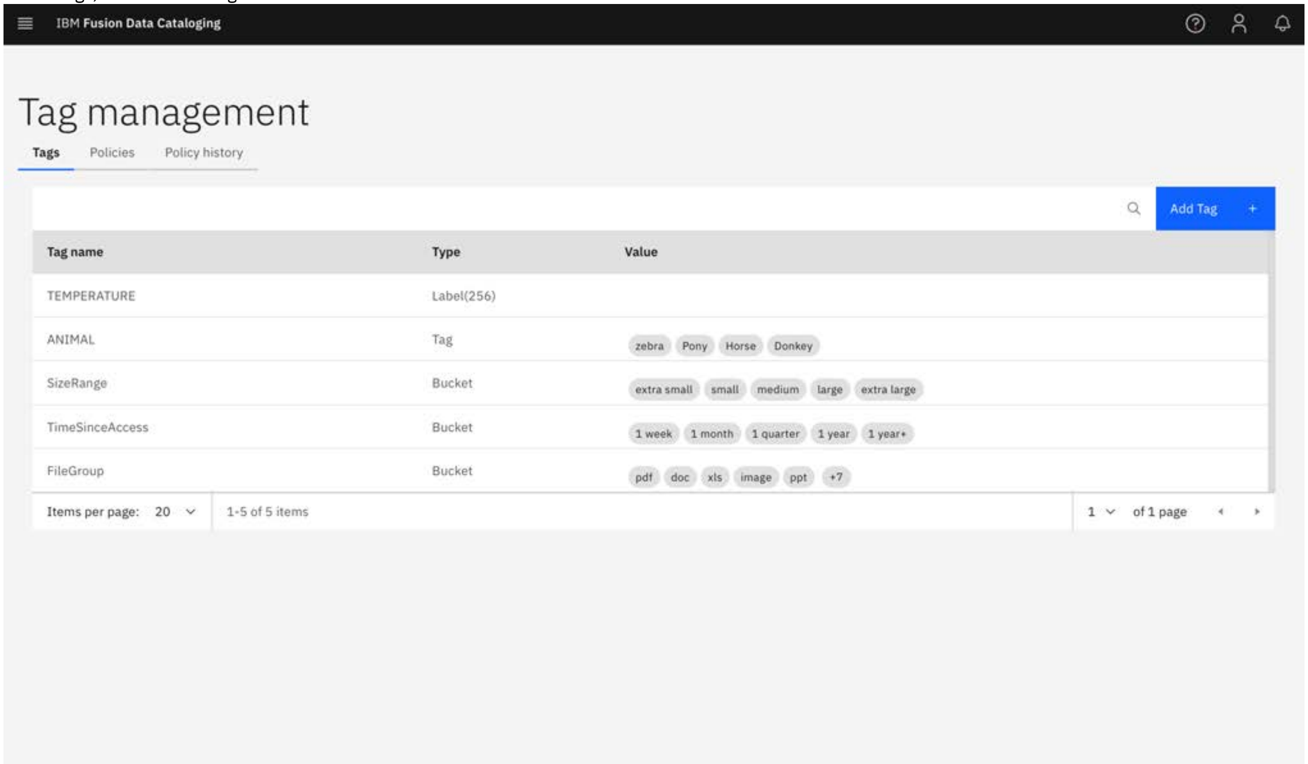
About this task

Use the Tags tab to create new tags. The table lists the tag name in the Field Name column, the Type, and the tag values in the Tags column. Use the icons to Edit or Delete a tag.

Procedure

1. Go to Metadata > Tag Management

2. Under Tags, click the Add Tag button.

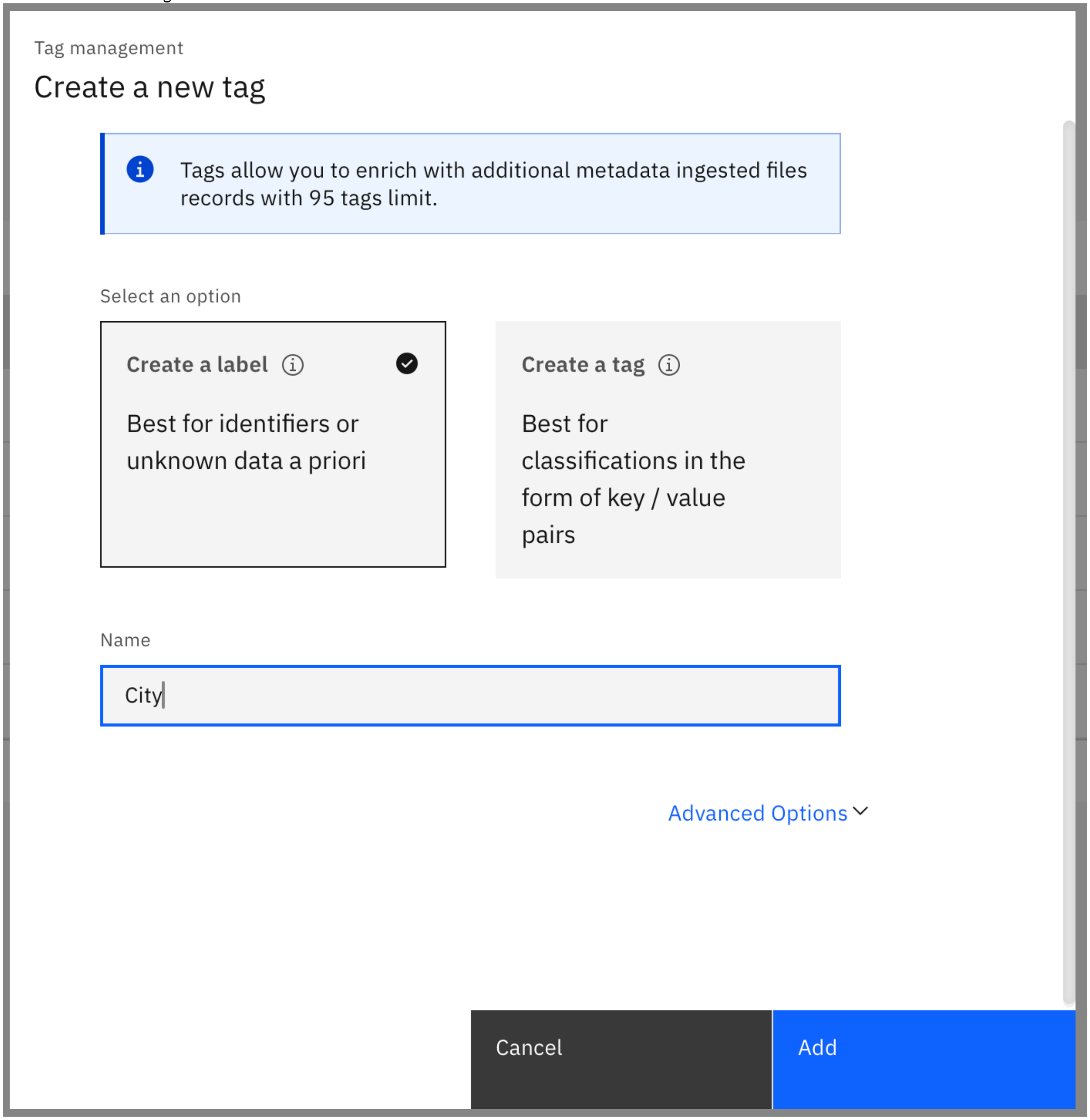


3. In the Create a new tag page, select the type of tag you want to create.

Label
: A lightweight tag used to describe records with simple identifiers such as project name, department, or device ID. Labels are not restricted to predefined values unless explicitly configured.

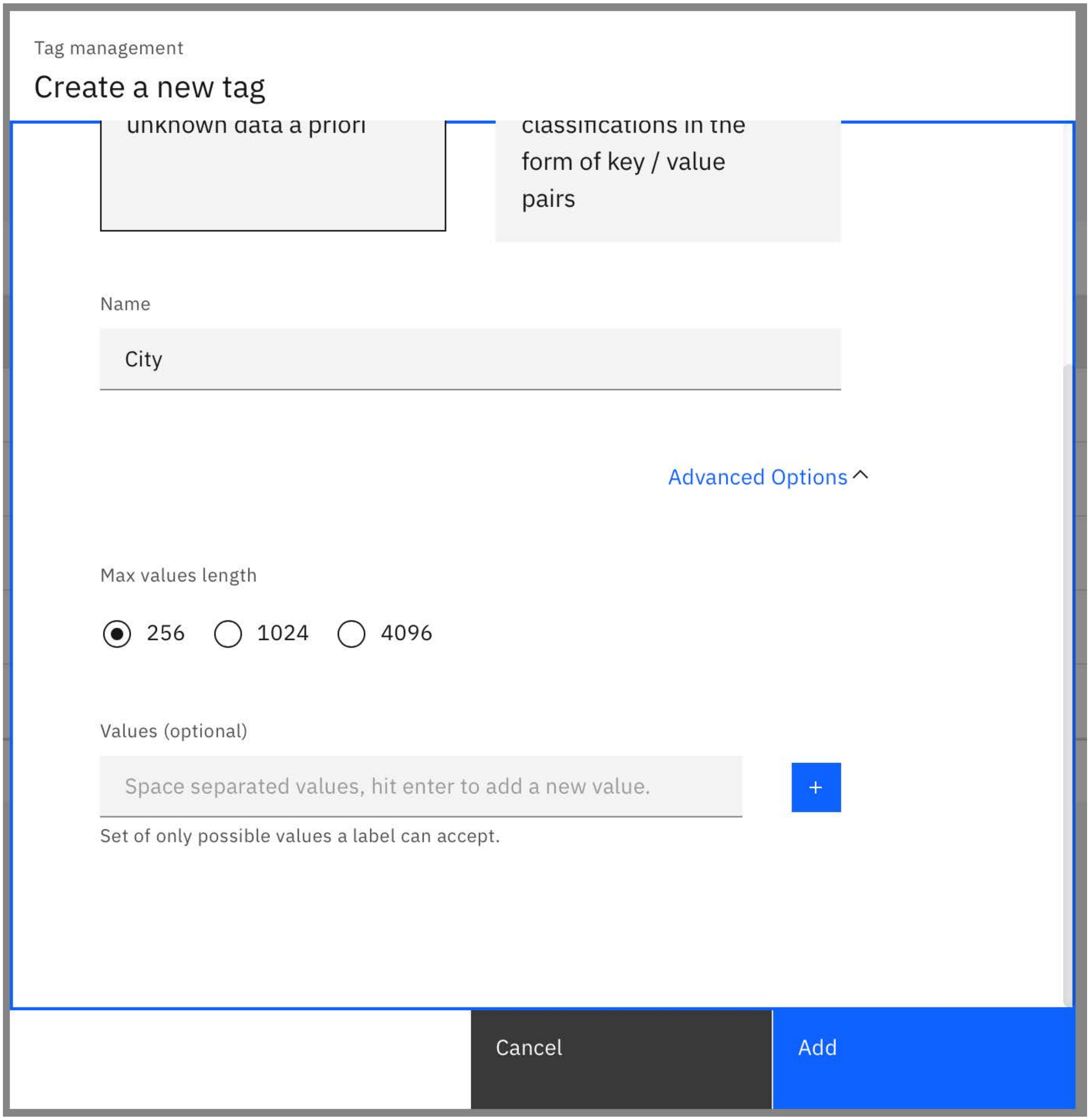
Tag
: A structured tag that enforces a predefined set of accepted values. Useful for controlled vocabularies like classification levels or billing codes.

4. Enter the name of the tag in the Name field.



5. Optional: If the tag type is Label, then click Advanced Options you can select a maximum value length of `256`, `1024`, or `4096` characters.
6. Define the following values for the tag or label:
   - These values are only accepted inputs when the tag is applied to metadata.
   - Enter one or more values into the **Values** box.
   - Press the Enter key after each value. Saved values appears in the input box.
     Note: The total combined length of all values must not exceed `1024` characters.

7. Click Add.



The tags, their types, and values are displayed in the table on the Tags tab.

Viewing and searching tags

View and search tags from the Tags tab on the Metadata page to locate specific tags by name, type, or value, and sort or filter results as needed.

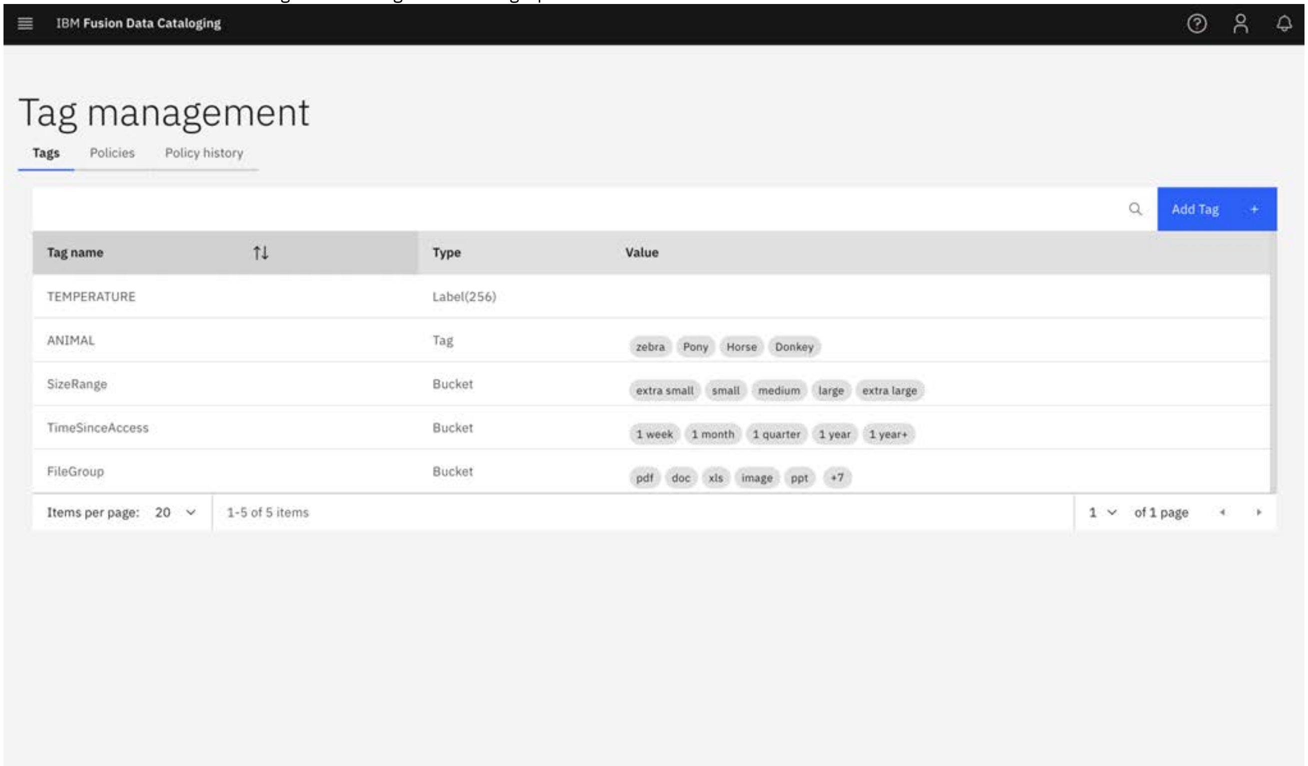
About this task

You can see a list of all tags or search for a specific ones from the Tags tab on the Metadata page.

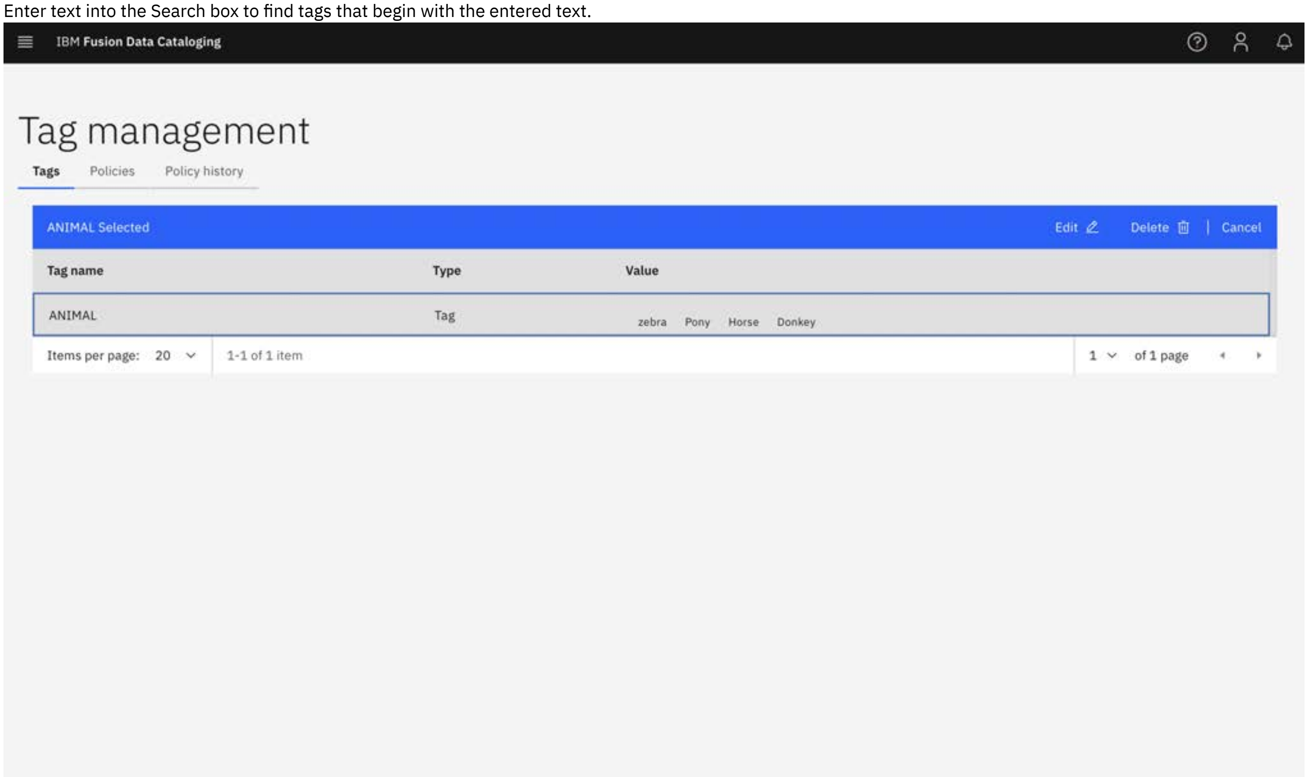
Procedure

1. Go to Metadata > Tag Management.
2. Under Tags tab, a list of tag **Names**, **Types**, and **Values** are displayed.

3. Click the column headers to sort tags in ascending or descending alphabetical order.



4. Enter text into the Search box to find tags that begin with the entered text.



As you type, a filtered list of tags containing the text string is automatically displayed.

5. Select a tag and click Edit or Delete from the header row to perform the desired action.

Editing tags

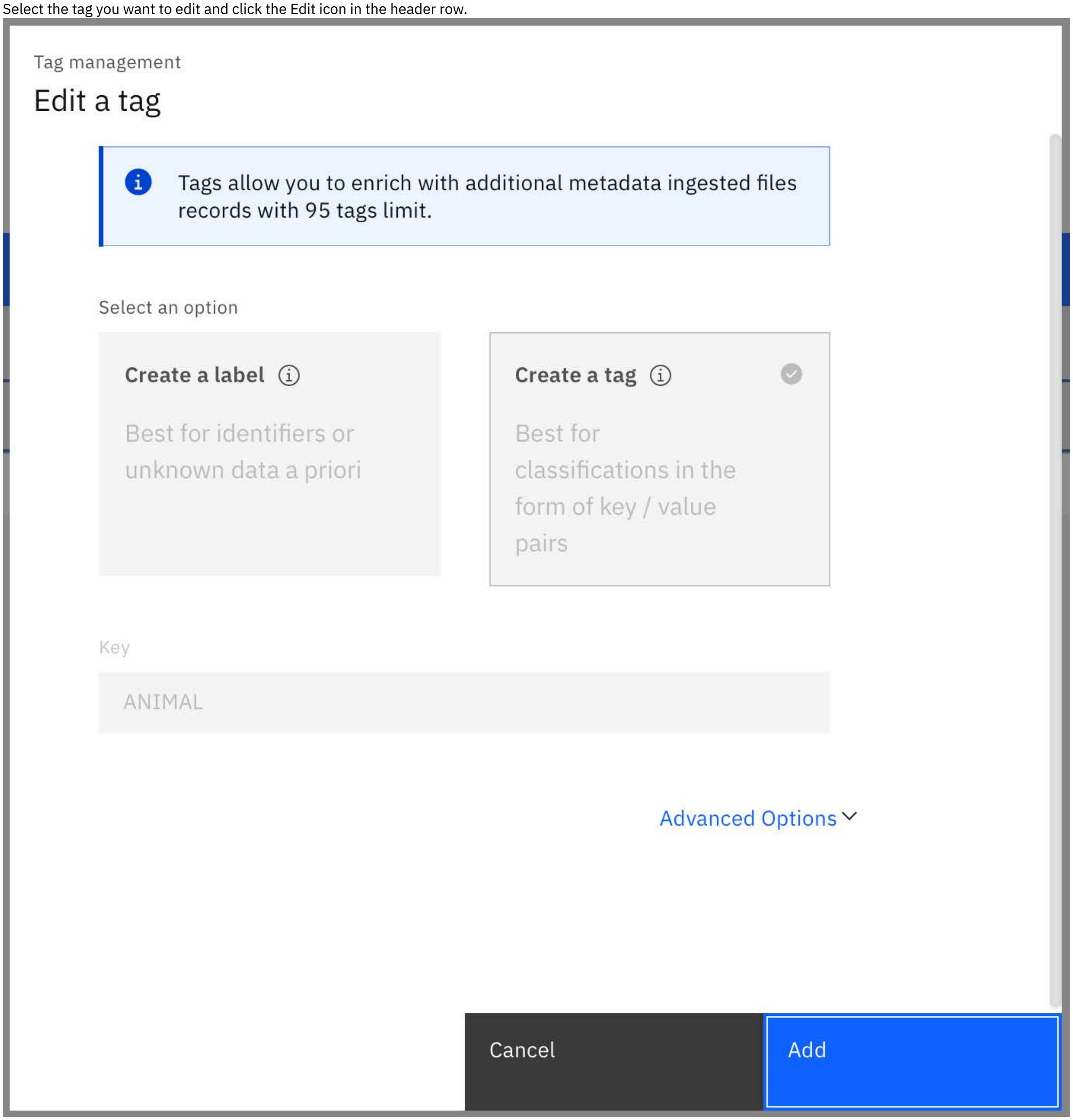
Edit tag values from the Tags tab on the Metadata page to update the allowed values for existing organizational tags in IBM Spectrum Discover.

About this task

You can edit tag values from the **Tags** tab on the **Metadata** page.

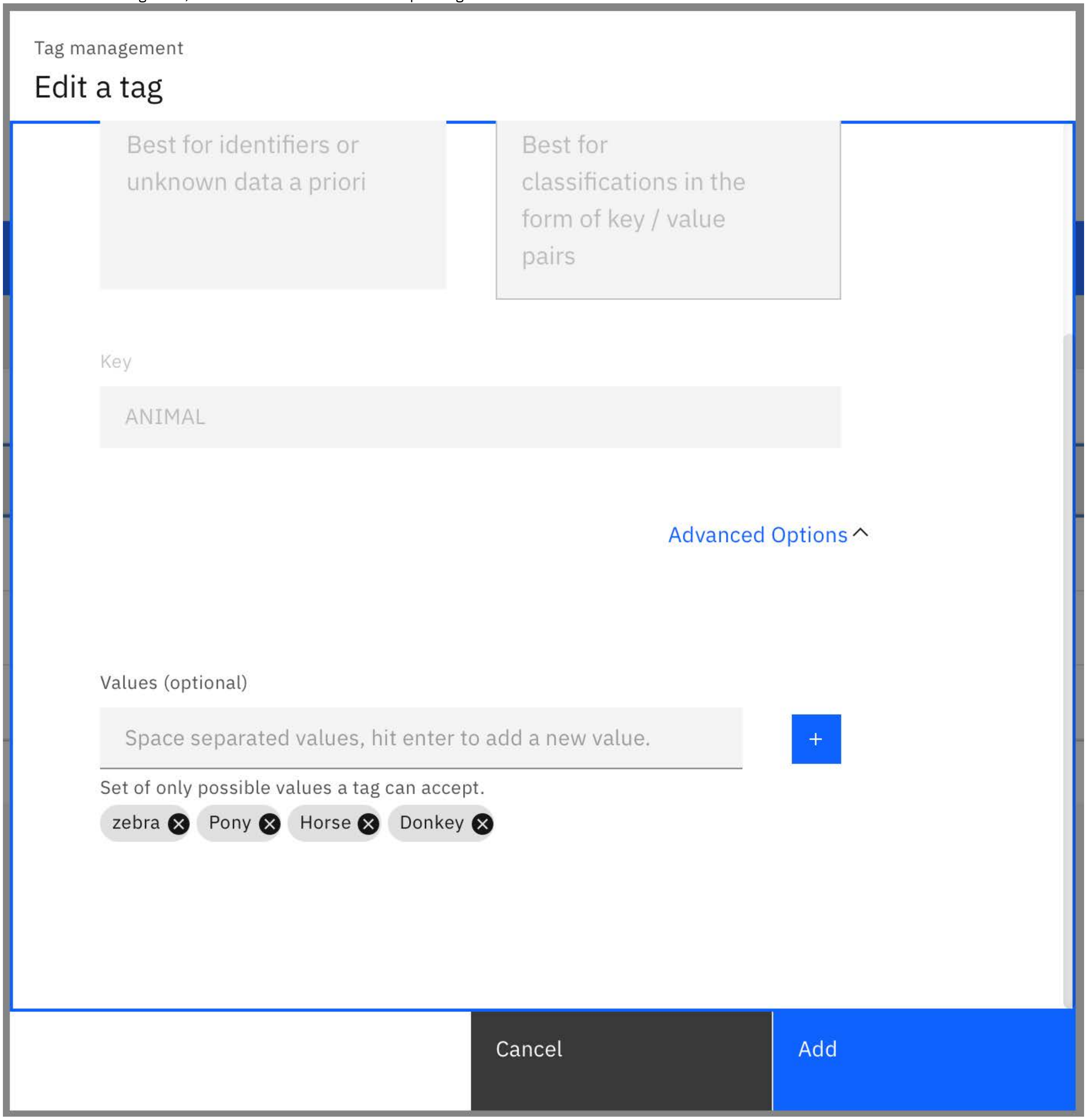
Procedure

1. Go to Metadata > Tag Management.
2. Select the tag you want to edit and click the Edit icon in the header row.



3. In the Edit a tag window that appears, enter one or more new values in the Values field and click Enter.
   Saved values are displayed in the input box.

4. To remove an existing value, click the x icon next to the corresponding value.



Note: The Name and Type fields of the tag cannot be modified.

5. Click Submit.
   The modified values are listed in the table under the Tags tab.

Deleting tags

Delete tags from the Tags tab on the Metadata page to remove metadata classifications that are no longer needed from the IBM Spectrum Discover catalog.

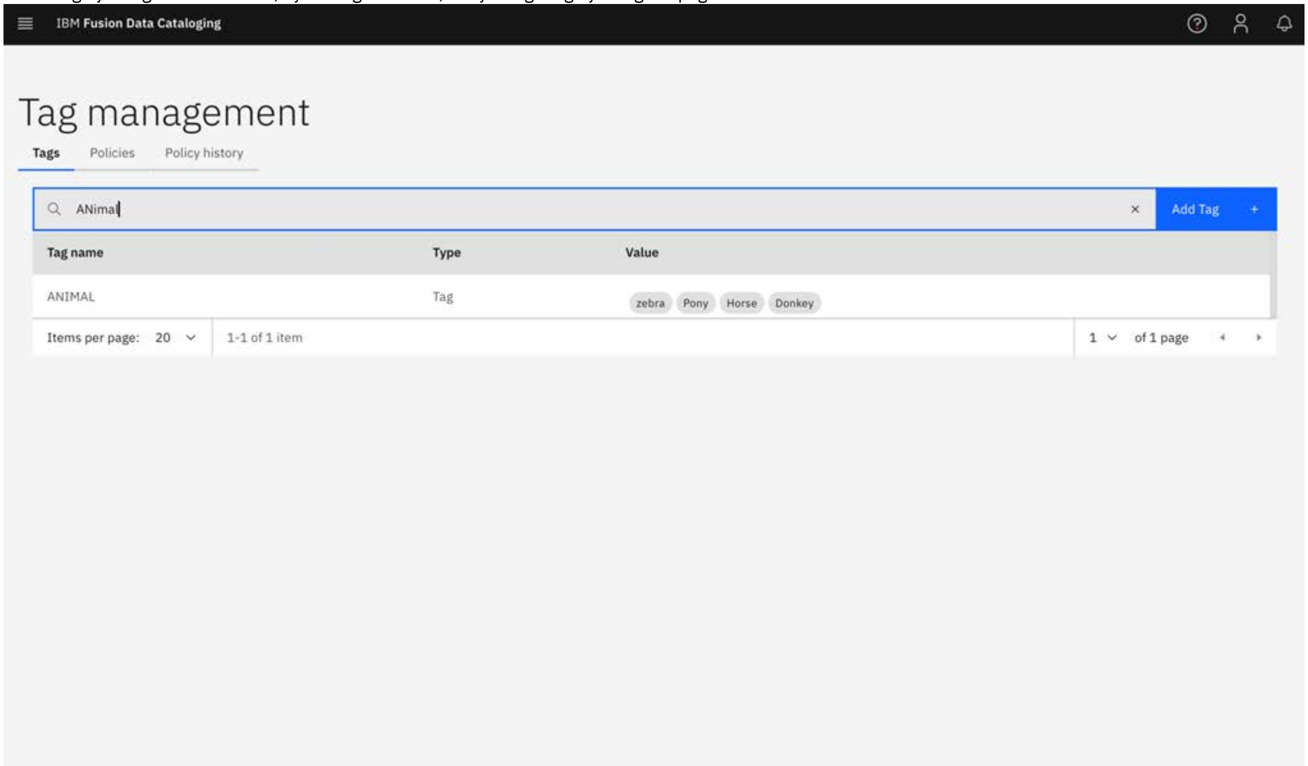
About this task

You can delete tags from the Tags tab on the Metadata page.

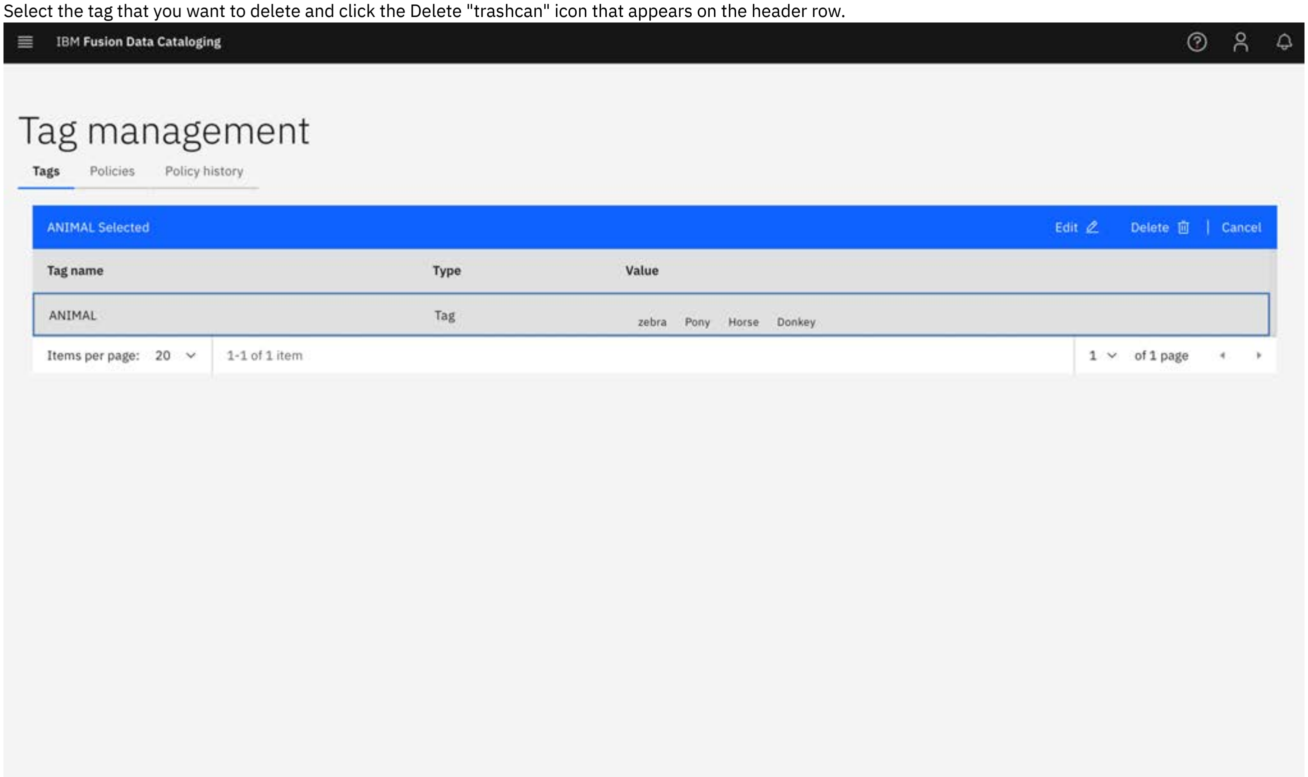
Procedure

1. Go to Metadata > Tag Management.

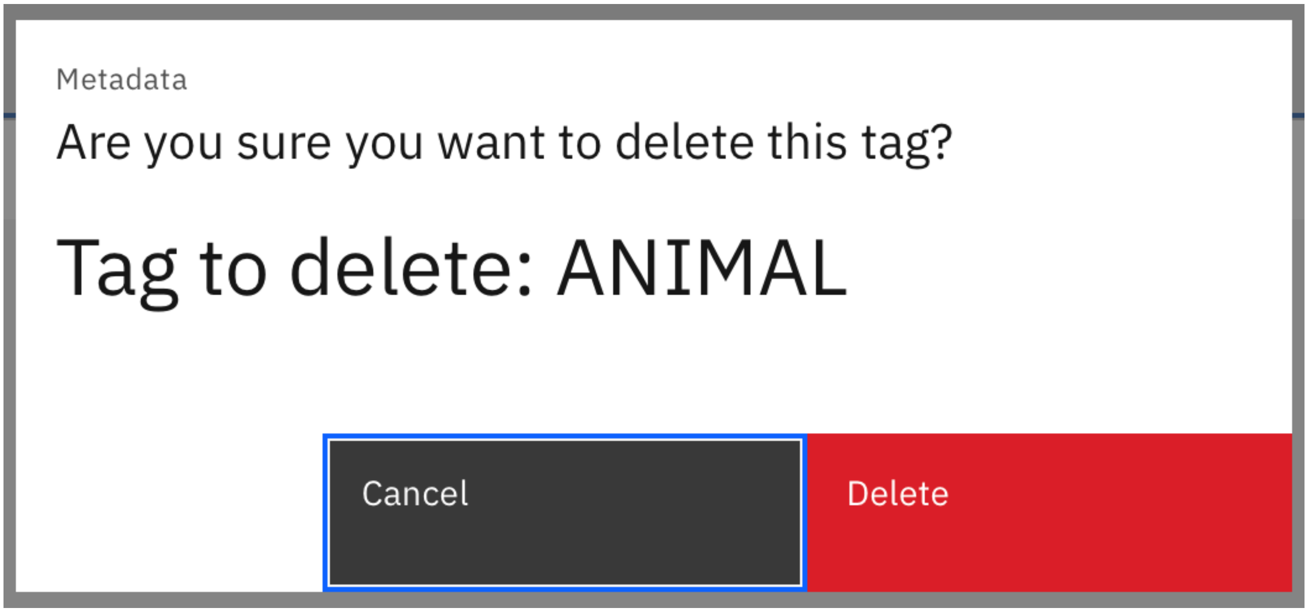
2. Find a tag by using the Search box, by sorting a column, or by navigating by using the page arrows at the bottom of the table.



3. Select the tag that you want to delete and click the Delete "trashcan" icon that appears on the header row.



4. Click Delete in the confirmation box.



The tag is removed from the table in the Tags tab.

Discover data

By discovering your data, you can apply policies that assign tags to your data. You can apply tags to the results of a single search, or you can use policies to automatically apply tags on a periodic basis.

There are three ways to discover data:

- Content-based keyword and tagging. The search is based on regular expression patterns that are defined within IBM Data Cataloging. For more information, see Creating a content search policy.
- Create a policy by using tags with known values. A policy is automatically run against all data that meets criteria that are specified in a filter. For more information about creating and using policies, see Managing metadata policies.
- Search your data by using a query in standard SQL grammar or do a visual exploration of tags by point-and-click. For more information, see Searching system and custom metadata fields.

- **Searching**
  Build visual or SQL queries in the Query Builder to search for data in IBM Spectrum Discover, then filter, sort, and group the results to drill down to the relevant records.
- **Grouping data by file type**
  Create policies to group data by file type.
- **Searching system and custom metadata fields**
  Search system and custom metadata fields using SQL queries in the Query Builder to filter, sort, and retrieve file records from the IBM Spectrum Discover data catalog.

Searching

Build visual or SQL queries in the Query Builder to search for data in IBM Spectrum Discover, then filter, sort, and group the results to drill down to the relevant records.

About this task

You can build a visual query or a custom SQL query to begin the search. In a visual query, you can add tags from the search page and view the values in the form of charts, tables, or directories for a more enhanced experience. The following procedure helps you build queries to search for relevant results.

Procedure

1. Click on the upper left of the IBM Data Cataloging interface and then click Search.
   The Query Builder page appears.
2. To build a visual query, click Add corresponding to the Tag that you want to include in your search.
3. On the Select Values page, select the values in the table that is displayed beneath the chart to build your query.
4. Click Update Query to preview your query on the Query Builder page.
   Note:
   - The Potential Record Count under Query Summary reflects the expected count of records in the search result that is displayed for the selected tags.
   - Click Reset Query to delete your current selection and build the query again.
   - Click on the Query Preview box to copy the icon for future use.
5. Click View Results to display the search results.

6. To build a custom query, click the SQL Query tab on the Query Builder page.
7. Type your query in the text area provided.
   Note:
   - Click the + icon on the right to select from the query field suggestions.
   - Click to view all recent search queries that were previously used.
8. Click View Results to display the search results.
   On the Query Result page, the search results are displayed under the Individual tab.
9. Select the tags that are provided under the Grouped By list on the left to drill down the results further.
   The Query results are now displayed under the Condensed tab.
   Note:
   - Click to edit the displayed columns from the list that appears.
   - Select a search result row and then click Generate Report to generate a report for the set of records.
   - Select a search result row and then click View Individual Records to view the records with no grouping selected.
   - Click Back to query builder to return to the query builder screen from any search query pages.

Grouping data by file type

Create policies to group data by file type.

About this task

You can create AUTOTAG policies and run them on a periodic basis to group data by specific file types.

Procedure

1. Create an Open tag to group data. For example, FileTypeGroup.
   a. Go to Metadata > Tag Management
   b. Under Tags click Add Tags.
   c. Select `Open` in the Type menu.
   d. Click Submit.
2. Create an AUTOTAG policy with a relevant filter criteria. For example, `Filetype = 'pdf'`.
   a. Go to Metadata > Tag Management.
   b. Under Policies click Add Policy.
   c. Enter a name for the policy in the Name box. For example, `FileTypeGroupTagging`.
   d. Select `Autotag` from the Policy Type menu.
   e. Click Next step.
   f. Type `filetype = 'pdf'` as the filter criteria.
      Note: You can modify the filter criteria based on the file type you want to group. For example, if you want to group all JPG images, type `filetype = 'jpg'`.
   g. Click Add Tag.
   h. Select the file type group tag that you previously defined. In this example, the tag name is `FileTypeGroup`.
   i. Match the tag value with the filetype in the filter criteria. For example, if the filter criteria is `filetype = 'pdf'` use `pdf` as the tag value and then click Next step.
   j. Click the slider control to set the policy status to `active`.
   k. Under Schedule select the frequency for running the policy and then click Next step. For example, Now, Daily, Weekly, Monthly
   l. Review the policy and click Save to run the AUTOTAG policy.
   Note: You can manually refresh the IBM Data Cataloging database to confirm whether the policy is run properly. Refreshing the database is optional and not an essential prerequisite for the process. To manually refresh the database, follow the procedure shown:
   a. Go to Data Connections > Discover Databases.
   b. Click Run table refresh under the Metadata Summarization database
3. Create a policy to group data with a different file type tag value or modify the existing policy.
4. Define a schedule to rerun policies on a periodic basis.

Searching system and custom metadata fields

Search system and custom metadata fields using SQL queries in the Query Builder to filter, sort, and retrieve file records from the IBM Spectrum Discover data catalog.

1. On the Query Builder page, click SQL query.
   a. Enter your query directly and click View results. Use the standard grammar that is used in an SQL query.
   b. Click + icon to select a suggested field from the list that appears, complete your query, and click View results.
   c. Click and select from the search queries that were previously used.
   d. Modify the query if necessary, and click View results.
   The query language is SQL. The underlying code takes care of certain semantics, for example,
   - Keyword
   - Columns to select
   - Name of the databases
   - Where clause
   - Limits
   - Offsets
   - Order by clauses
   The search clause that is input by the user is only the body of the query that would appear after the where clause and before the limit/offset/order qualifiers.

- **System metadata fields to search on**
  The list in this section provides definitions of items on which you can search system metadata fields.
- **Access control list metadata to search on**
  Access Control List (ACL) metadata is collected from SMB/CIFS data source search results.
- **Search on custom metadata fields**
  You can do a search on custom metadata fields.
- **Examples of search filters**
  This section provides a list of examples for search filters.
- **Search results table**
  The search results table displays information about the records that met the search criteria.
- **Refine search results**
  After a set of search results is returned, you can further refine the data by using the following options:
- **Sort search results**
  You can sort search results by column.
- **Tag search results manually**
  After a filtered set of records is displayed on the search pane, you can select all or some of those documents and apply organizational tags to them.

System metadata fields to search on

The list in this section provides definitions of items on which you can search system metadata fields.

The list below shows search filters that you can use for a search.

Datasource
: The name of the datasource where the record originated. The datasource refers to the label of the source storage system that was defined in the IBM Data Cataloging connection management panel.

Platform
: The type of storage system from which this record originated.

Site
: The physical site for the data as input by the user at scan time.

Cluster
: The name of the IBM Storage Scale cluster to which the record belongs. This term applies only to IBM Storage Scale.

NodeName
: For IBM Storage Protect, this indicates the node or client system to which the backup or archive record belongs.

Fileset
: The file set to which the record belongs for IBM Storage Scale. This term applies only to IBM Storage Scale.

MgmtClass
: For IBM Storage Protect, this indicates the management class to which the backup or archive record belongs.

Owner
: The system metadata owner of the record (file only).

Group
: The system metadata group owner of the record (file only).

UID
: The numeric ID of file owner (file only).

GID
: The numeric ID of file group (file only).

Path
: The file path or object storage bucket of the file that is represented by this record.

Filename
: The name of the file or object represented by the record.

Filespace
: For IBM Storage Protect, this indicates the file space to which the backup or archive record belongs.

Filetype
: The type of the file or `object.MtimeLast` modified time for the file (file only).

State
: The state of the file or object. Possible values for IBM Storage Scale are:

  premig
  : Premigrated

  migrtd
  : Migrated

  resdnt
  : Resident

  Possible values for IBM Storage Protect are:

  ACTIVE
  : Active backup copy

  INACTIVE
  : Inactive backup copy

  ARCHIVE
  : Archive copy

Mtime
: Last modified time for the file (file only).

Atime
: Last accessed time for the file (file only).

Time

Creation time of the file (file only).

Size
: Size of the file or object.

Inode
: The inode of the file (file only).

Permissions
: The permissions of the file (file only).

sizeConsumed
: The size of the consumed capacity (file only).

recordversion
: The version enabled for the cloud storage system buckets or vaults. IBM Data Cataloging stores the S3 versioning identifier in the recordversion tag of the corresponding object. Cloud versioning allows the user to store multiple copies, or versions, of the same object.

Access control list metadata to search on

Access Control List (ACL) metadata is collected from SMB/CIFS data source search results.

For SMB/CIFS data sources, IBM Data Cataloging collects Access Control List (ACL) metadata in addition to the standard system metadata. This information is stored in two tables, which are the Access Control Owner and Group (ACOG) and the Access Control Entries (ACES).

Important: Since ACL data is not displayed by default and is unavailable using the "visual query", a special SQL query is needed to view ACL data.
Follow the steps to view ACL data for an SQL query:

1. Log in to IBM Data Cataloging GUI.
2. From the main menu, click Search > Query builder
   The Query builder page is displayed.
3. Click SQL query tab.
4. In the Select * from Spectrum Discover where field, use the following query and click View results to view all ACL data.

   `"aces.entrytype like '%ACL'"`

Note:

- To limit the ACL results to NFS or SMB, use the following query:

  `"aces.entrytype = 'NFS4ACL'", "aces.entrytype = 'DACL'", or "aces.entrytype = 'SACL'"`

- For any other specific aces or acog query, use the following query:

  `"aces.accesstype = 'AUDIT'" or "acog.groupname ='UNIX_GROUPS\wheel'"`

For information on how ACL data is used in policies, see How to use ACL data in policies.

Type the criteria in the search bar to search the ACL metadata. Possible fields to search on are:

acog.ownername
: Indicates the owner of the file.

acog.ownerid
: Indicates the security identifier for the owner of the file.

Note: To search for files based on owner name or security identifier, see the following example:

`acog.ownername='DOMAIN\user'`

`acog.ownerid='S-1-1-1-1000'`

acog.groupname
: Indicates the group of owner of the file.

acog.groupid
: Indicates the security identifier for the group of the owner of the file.

aces.username
: Indicates the user or group name for which this ACE applies.

aces.userid
: Indicates the security identifier for the user or group name for which this ACE applies.

aces.entrytype
: Indicates that the entry type can either be DACL or SACL and NFS4ACL for NFSv4.

aces.accesstype
: Indicates that the access type can be either one of the following options:

  - ALLOWED
  - DENIED
  - AUDIT

Note: To search files with an access control entry that allows everyone to access, see the following example:

`aces.username='\Everyone' and aces.entrytype='DACL' and aces.access_type='ALLOWED'`

To search for files that are on a particular data source and display a Deny Access Control Entry, see the following example:

```
datasource in ('smb1') and aces.entrytype='DACL' and aces.accesstype='DENIED'
```

aces.permissions
: Indicates the possible permission levels that include:

- R - Read
- W - Write
- X - Execute
- D - Delete
- P - Write access controls
- O - Owner

The valid permission combinations are:

- READ - R + X
- CHANGE - R + W + X + D
- FULL - R + W + X+ D + P + O

For NFSv4 the permissions vary as shown:

- r - read-data
- w - write-data
- a - append-data
- x - execute
- d - delete
- t - read the attributes of the file
- T - write the attribute of the file
- n - read the named attributes of the file
- N - write the named attributes of the file
- c - read the file ACL
- C - write the file ACL
- o - change ownership of the file

Multiple NFSv4 permissions can be used in combinations like "rxtncy" for every entry.

aces.flags
: Indicates the flags displaying the type of access control entry (ACE).

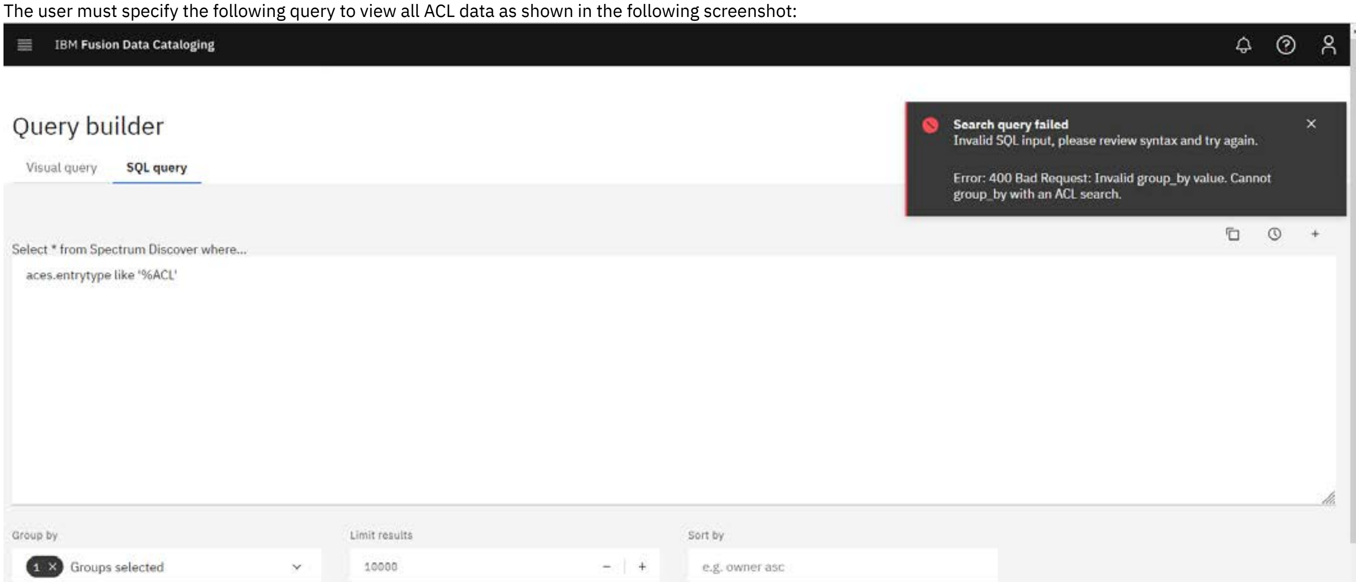
Note: A file can have more than one access control entry (ACE) associated with it. Search results that contain ACL metadata, repeat the file metadata for each ACE. Therefore, reports are the preferred method for using IBM Data Cataloging search results with ACL metadata.

- **How to use ACL data in policies**
  This topic describes how to use ACL data in your policies.

How to use ACL data in policies

This topic describes how to use ACL data in your policies.

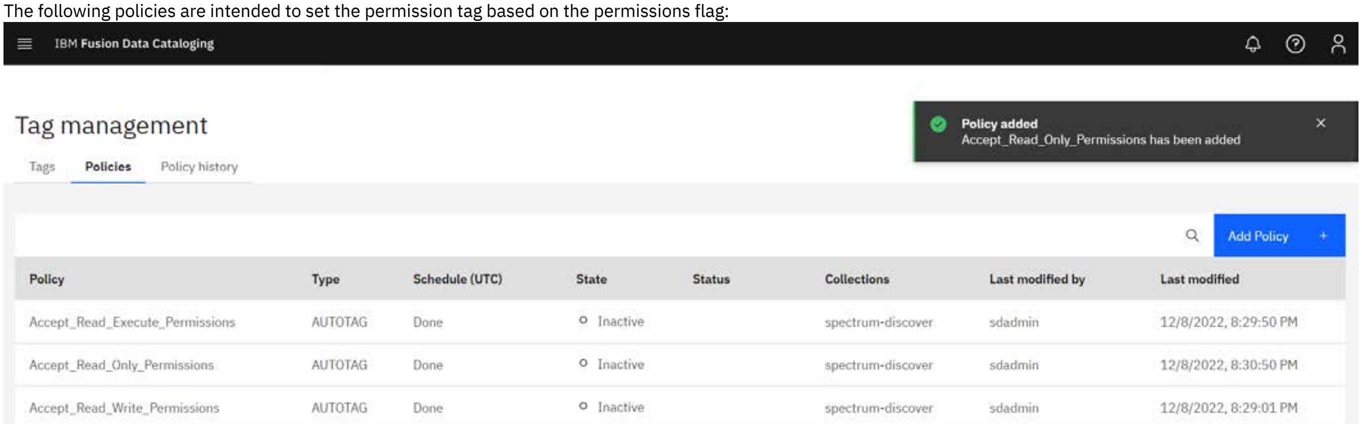
The user must specify the following query to view all ACL data as shown in the following screenshot:



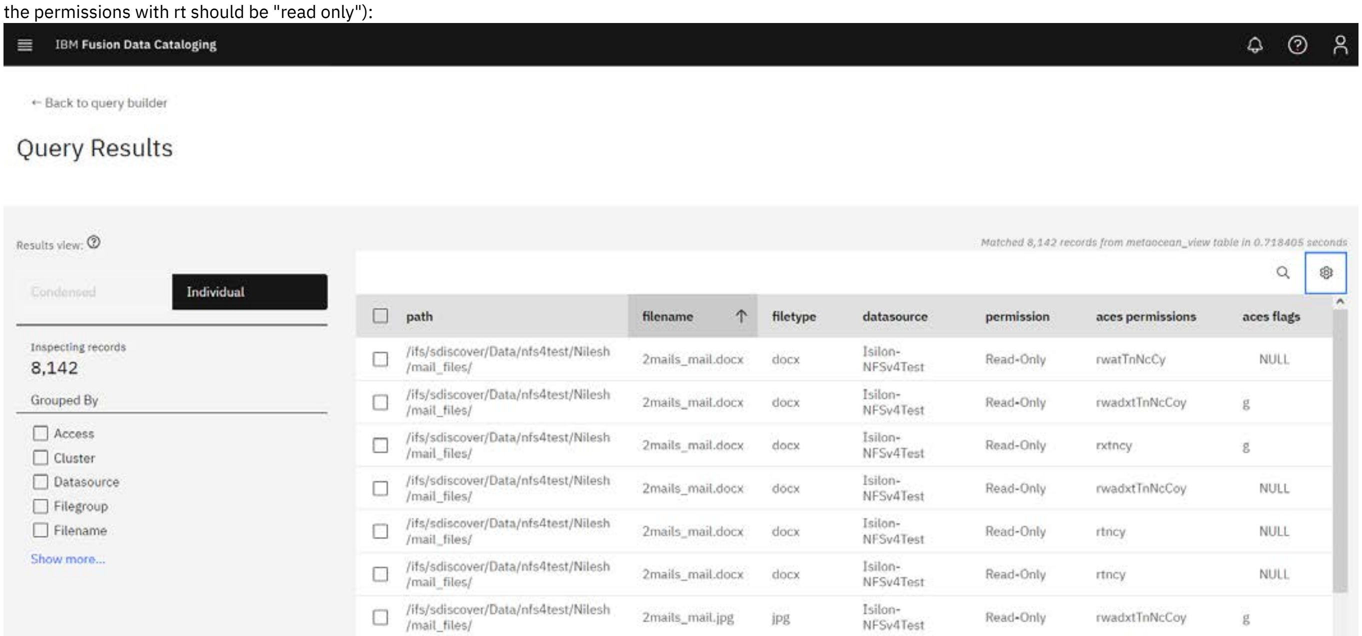
Note: ACL grouping is not supported and in order to perform such groupings, an external DB client like the DB2 warehouse client should be used against the Spectrum Discover database.
Important: When using acog or aces data to make tagging policies, it is important to understand that a single selection of acog or aces data may result in tags not being meaningful.

For example, the following tag is defined to show that a row has a permission type of either read-write, read-execute, or read only:

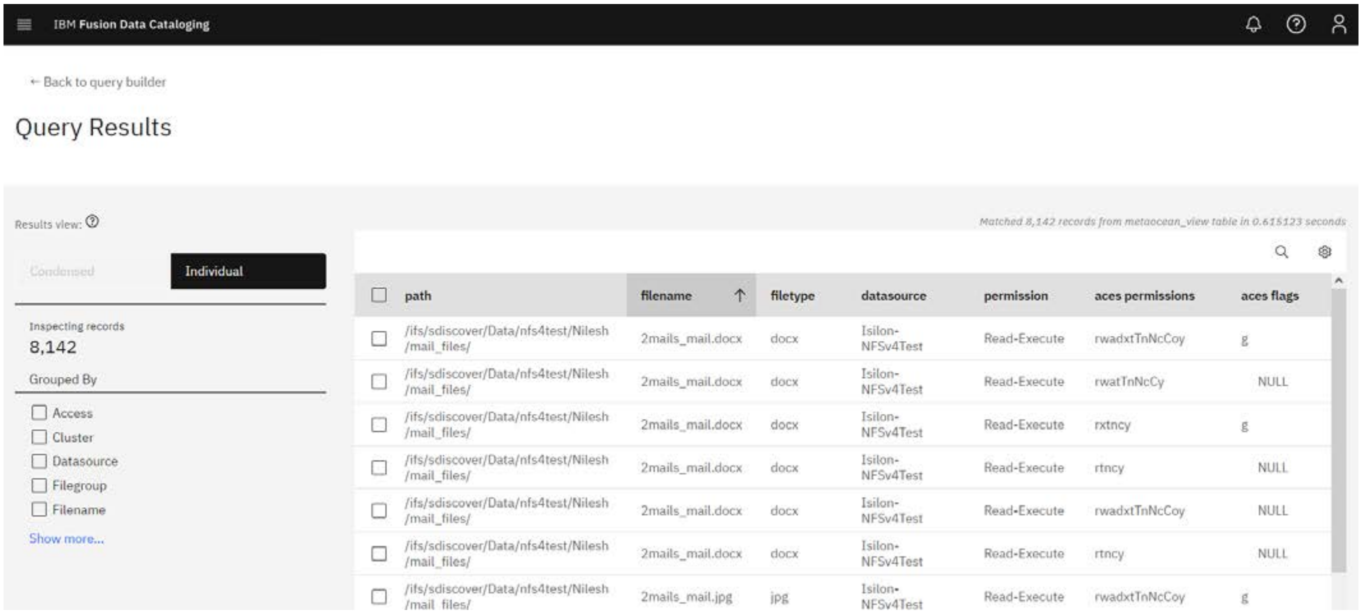
The following policies are intended to set the permission tag based on the permissions flag:



After running the Read Only policy, the Permission tag is populated, but as seen from the following row for the same file, the tag does not match the real permissions (only the permissions with rt should be "read only"):



After running the Read Execute policy, the Permission tag is populated but the permission tag is changed from Read-Only to Read-Execute which makes it meaningless and effectively useless:



To solve this problem, you can create tags and set the value as "True" if the condition exists.
Note: You cannot add a "false" value because if you were to run a policy that found a condition that doesn't exist, you would run into the same meaningless results.
When tags have a "True" value, you can now set the policies to have a "True" value. By doing this, you can determine what permissions are allowed for each file.
Note: You can customize your own tags and use aces/acog information that is meaningful to you.

Search on custom metadata fields

You can do a search on custom metadata fields.

Comparators
To do a search, you can also use the following comparators:

=
: Is equal to.

<>
: Is not equal to.

<
: Is less than.

>
: Is greater than.

<=
: Is less than or equal to.

>=
: Is greater than or equal to.

is
: When you search for `null` values:

  is null
  : Indicates a null (or no) value.

  is not null
  : Indicates a valid value that is not null.

Conjunctions
You can also use conjunctions.

AND
: Tie together multiple filter criteria.

OR
: Meet at least one of multiple filter criteria.

Helpers
You can also use helpers.

NOW()
: Get the current TIMESTAMP.

DAYS/MONTHS/YEARS
: Compares TIMESTAMP/DATE values.

Wildcards
You can also use a wildcard.

%
: You can use a wildcard like % with the keyword LIKE to form a wildcard search.

Note: Both single quotation marks (') and backslashes (\) need to be escaped with a leading backslash. For example, if the search statement is: `mytag in ('first', 'can't', 'with\slash')`, the following convention needs to be followed to escape it:
`mytag in ('first', 'can\'t', 'with\\slash').`

Examples of search filters

This section provides a list of examples for search filters.

Note: You must wrap string values in single quotation marks but you cannot wrap numeric values in single quotes.

Owner=**`'bob'`**
    # All files owned by **`'bob'`**.
Fileset=**`'bobs project'`**
    # All files in the file set **`bobs_project`**.
Filetype = **`'pdf'`** AND size > 500000
    # All PDF files that are larger than 500000 bytes.
Filetype is **`('txt', 'pdf', 'doc')`**
    # All files of type TXT, PDF, or DOC.
Atime < (NOW() - 180 DAYS)
    # All files not accessed in the last 180 days.
Filesystem = **`'big_fs '`** AND owner <> **`'root'`**
    # All files in the **`big_fs`** filesystem that are not owned by root.
collection = **`‘proj_xylem’`**
    Search for all records that are tagged with the user-defined tag **`‘Project’`** set to **`‘proj_xylem’`**.
collection <> ’’
    Search for all records that have a collection that is assigned.
filename LIKE **`‘the_quick_brown_%’`**
    Returns all records for which the file name begins with **`“the_quick_brown_”.`**
department= **`‘department_xylem’`**
    Search for all records that are tagged with the user-defined tag **`‘Department’`** set to **`‘proj_xylem’`**.
`custom_tag` is null
    # All files for which **`custom_tag`** is not set to any value.

Search results table

The search results table displays information about the records that met the search criteria.

By default, certain columns are shown, and others are hidden. Click next to the search bar to customize the column view.

Refine search results

After a set of search results is returned, you can further refine the data by using the following options:

- Under Grouped By, click the relevant criteria to group the search results.
- Select a record row and click View individual record on the header to view the record details.
- Click Back to Query builder and select the size of search results you want to view from Limit results.
- Click and select the Used capacity column to view a comparative study of the file size that is reported and the amount of space that is being used in the storage.

You can use a combination of any or all of the filtering criteria. The search results are immediately refined based on the criteria you select.

Sort search results

You can sort search results by column.

When you click the column header, you can sort the results in ascending order. When you click the column a second time, you can sort the column in descending order. The time it takes to sort depends on the size of the result set.
Note: Sorting by a second column loses the order of the data in the first column. A combination sort view is not supported.

Tag search results manually

After a filtered set of records is displayed on the search pane, you can select all or some of those documents and apply organizational tags to them.

For example, if a drill-down search result identifies all of the records for a particular project, you can assign a specific tag to it. Click Add Tags and specify that an organizational tag called 'Project' is set to the name of the project that is represented by the filtered set. The tag application runs as a background task and you get a notification when the processing is complete.

You can apply more than one tag at the same time.

Managing applications

An application is a program that interfaces with IBM Data Cataloging and can access the source storage. There are many use cases for application, including data content inspection for enriching metadata, data movement or migration, data scrubbing or sanitation, and more. Data is identified by IBM Data Cataloging by policy filter and passed to the application as pointers through a messaging queue. Then, the application performs whatever work is appropriate on the source data and returns a completion status back to IBM Data Cataloging, which might or might not include enriched metadata for the records. If it does include enriched metadata, IBM Data Cataloging catalogs that metadata and makes it immediately searchable.

Permissions

Data Administrator
: Create (register), update, delete (unregister), and view the applications.

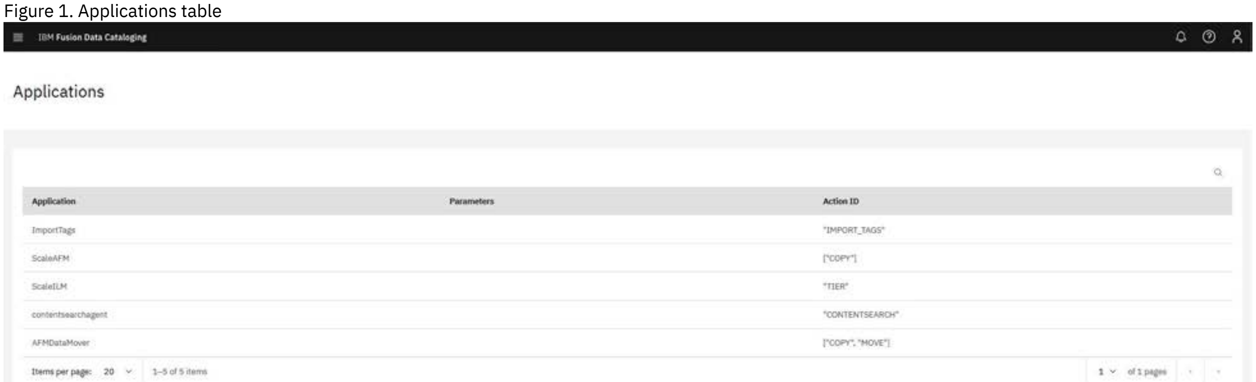
Data User
: View the applications created by a Data Administrator.

Security Administrator
: Cannot create, modify, view, or delete any application.

Management

Applications might be viewed and deleted by navigating to Metadata > Applications. You can define an application when you are creating a new **DEEP-INSPECT** policy. In addition, you can add Parameters for an application during the process of creating a **DEEP-INSPECT** policy. For more information, see Adding deep-inspection policy parameters .

Figure 1. Applications table



The Applications table displays the following information:

Application
: The name of the application.

Parameters
: The parameters that were assigned to the application when the policy was created.

Action ID
: **AUTOTAG** or **deepinspect** - the policy type that the application is assigned to.

For more information, see Application management by using APIs.

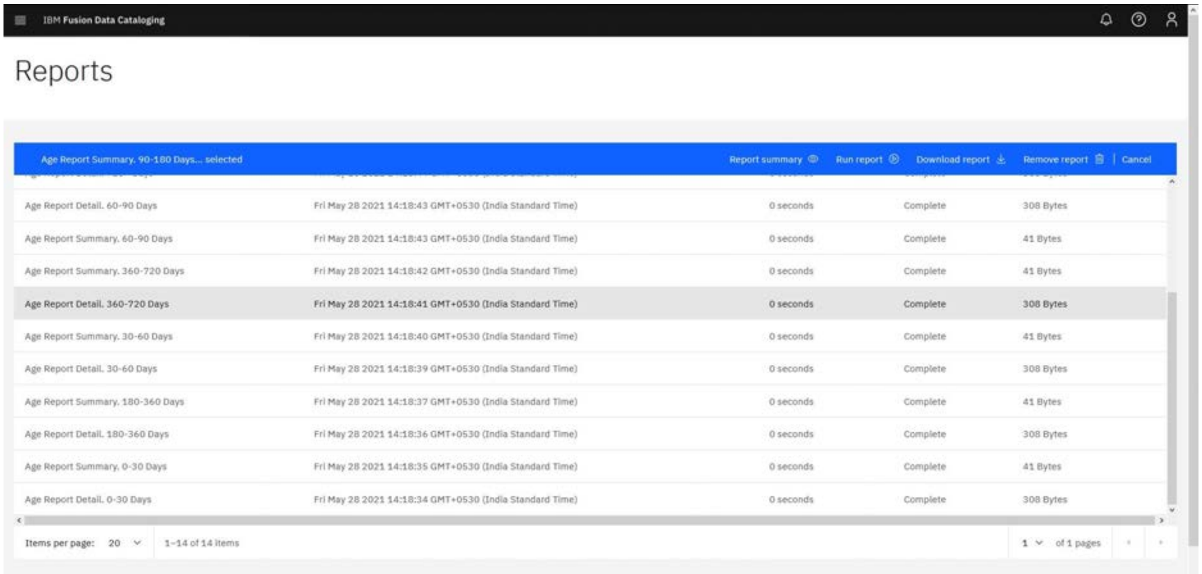
Reports

Reports can be generated upon applying tags to a set of data.

Procedure

1. Reports can be generated by using the following methods:
   - Discover data by performing a Search in IBM Data Cataloging. The search results provide an option to Generate Reports. For more information, see Searching for details.
   - Use the Graphical User Interface (GUI)Graphical User Interface (GUI) to automatically run the reports during deployment.
2. Click  menu and go to Search > Reports .

   Figure 1. Reports table

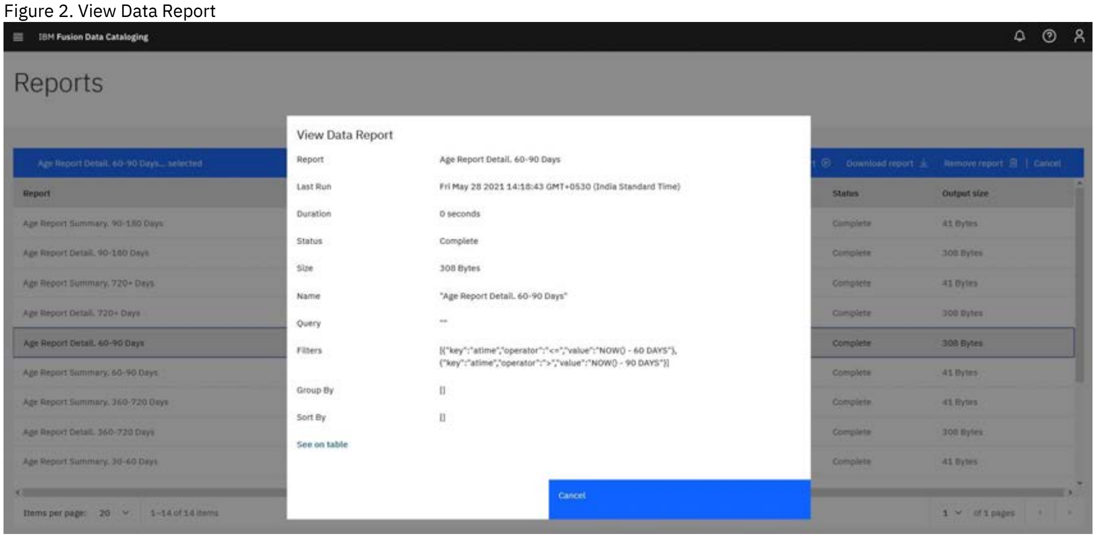


3. The following actions can be completed in a table:

Report Summary

a. To view a report, select a report and click Report summary. The report's statistics are displayed in a View Data Report dialog.

Figure 2. View Data Report



b. Click See on table to view all the records of a report. The Query Results window displays the results of the search.

Download report

Click Download report to open a report with a text editor, or to save the report to local storage.

Run report

To run a report, select a report and click Run report.

Remove report

To remove a report, select a report and click Remove report.

Creating a IBM Data Cataloging application for metadata-based policies

Provides information about how to create IBM Data Cataloging application for metadata-based policies.

By default, a IBM Data Cataloging application creates a privileged container prepared to process a policy that requires both metadata and content inspection, which oftentimes mounts a data source to fetch the required files. To modify this behavior and create non-root applications for metadata-only policies, we can add the non-root annotation to the resource.

For example:

```
apiVersion: spectrum-discover.ibm.com/v1alpha1
kind: SpectrumDiscoverApplication
metadata:
  annotations:
    nonroot: 'true'
  name: non-root-app
  namespace: ibm-data-cataloging
spec:
  application_name: nonroot
```

```
log_level: DEBUG
replicas: 1
repo_name: <PRIVATE_REGISTRY>/metadata-example-application
tag: v1.0.0
```

Graceful shutdown

Provides detailed instructions to put IBM Data Cataloging in idle state on an OpenShift® environment.

About this task

The procedure might be helpful in situations where IBM Data Cataloging is behaving unexpectedly or there is a maintenance window of not more than one hour.

Procedure

1. Stop or wait for all running scans and policies to stop. Do not skip this step until everything looks stopped.
2. Do `oc` login to your OpenShift Container Platform cluster.
3. Do `oc project <project>` to select namespace where IBM Data Cataloging service is installed.
4. Stop all pods and set replicas to zero for the following deployments. and respect the following order:
   Note: Follow the suggested sequence.

   a. Consumers

      i. Run the following command to get the list of the deployments and filter by consumer.

      ```
      $oc get deployments |grep consumer
      ```

      Example output:

      ```
      isd-consumer-ceph-le                10/10   10          10          6d20h
      isd-consumer-cos-le                 10/10   10          10          6d20h
      isd-consumer-cos-scan               10/10   10          10          6d20h
      isd-consumer-file-scan              10/10   10          10          6d20h
      isd-consumer-protect-scan           10/10   10          10          6d20h
      isd-consumer-scale-le               10/10   10          10          6d20h
      isd-consumer-scale-scan             10/10   10          10          6d20h
      ```

      ii. Run the following to command to scale down replicas to 0 for every consumer deployment.

      ```
      $oc scale --replicas=0 deployment <deployment_name>
      ```

      Example output:

      ```
      $oc scale --replicas=0 deployment isd-consumer-file-scan
      deployment.apps/ isd-consumer-file-scan scaled
      ```

      iii. Ensure you scale down replicas to 0 for every consumer deployment.
      Example output:

      ```
      $oc get deployments|grep consumer-cos-scan
      isd-consumer-file-scan                0/0     0            0            6d20h
      ```

   b. Producers

      i. Run the following command to get the list of the deployments and filter by producer.

      ```
      $oc get deployments |grep producer
      ```

      Example output:

      ```
      isd-producer-ceph-le                1/1     1           1           6d20h
      isd-producer-cos-le                 1/1     1           1           6d20h
      isd-producer-cos-scan               1/1     1           1           6d20h
      isd-producer-file-scan              1/1     1           1           6d20h
      isd-producer-protect-scan           1/1     1           1           6d20h
      isd-producer-scale-le               1/1     1           1           6d20h
      isd-producer-scale-scan             1/1     1           1           6d20h
      ```

      ii. Run the following to command to scale down replicas to 0 for every producer deployment.
      Important: Ensure that you scale down replicas to 0 for every producer deployment.

      ```
      $oc scale --replicas=0 deployment <deployment_name>
      ```

      Example output:

      ```
      $oc scale --replicas=0 deployment isd-producer-file-scan
      deployment.apps/ isd-producer-file-scan scaled
      ```

      iii. Ensure that you have scale down replicas to 0 for every producer deployment.
      Example output:

      ```
      $oc get deployments|grep producer-cos-scan
      isd-producer-cos-scan                0/0     0            0            6d20h
      ```

   c. Connection manager

      i. Run the following command to get the list of the statefulsets and filter by connection manager.

      ```
      $oc get statefulsets |grep connmgr
      ```

Example output:

```
$oc get statefulsets |grep connmgr
connmgr                        0/0      0              0            6d20h
```

ii. Run the following to command to scale down replicas to 0 for every connection manager statefulset.
Important: Ensure that you have scale down replicas to 0 for every connection manager statefulset.

```
$oc scale --replicas=0 statefulset
```

Example output:

```
$oc scale --replicas=0 statefulset connmgr
statefulset.apps/ connmgr scaled
```

iii. Ensure that you have scale down replicas to 0 for every connection manager statefulset.
Example output:

```
$oc get statefulsets|grep connmgr
connmgr                       0/0      0              0            6d20h
```

d. Check whether db2whrest stopped inserting data properly

Ensure that no database activity is present before stopping db2whrest. It is necessary to check that specific csv files created are empty, which means there are no pending jobs being processed.

i. Run the following command to go to db2whrest.

```
oc rsh deployment/isd-db2whrest
```

Now, list all CSV files related to merging that are empty.

For example:

```
$for file in $(find /gpfs/gpfs0/db2wh/home/bluadmin -name *LOAD*.csv); do file $file; done |grep data
/gpfs/gpfs0/db2wh/home/bluadmin/ACOG_LOAD4.csv: data
/gpfs/gpfs0/db2wh/home/bluadmin/ACOG_LOAD1.csv: data
$
```

ii. Wait for a few seconds and try again until you get an empty list.

e. DB2WHRest

i. Run the following command to get the list of the deployments and filter by db2whrest.

```
$oc get deployments |grep db2whrest
```

Example output:

```
$oc get deployments |grep db2whrest
isd-db2whrest                       1/1      1              1            6d20h
```

ii. Run the following command to scale down replicas to 0 for every db2whrest deployment.

```
$oc scale --replicas=0 deployment <deployment_name>
```

Example output:

```
$oc scale --replicas=0 deployment db2whrest
deployment.apps/ db2whrest scaled
```

iii. Ensure that all scale down replicas are 0 for every db2whrest deployment.
Example output:

```
$oc get deployments|grep db2whrest
isd-db2whrest                      0/0      0              0            6d20h
```

5. Stop Db2 instance. For procedure, see Stopping and starting a Db2 instance.
6. After checking that everything is down, IBM Data Cataloging is successfully in an idle state. It is recommended to be in this state for a short period, not more than one hour. Warning messages on other IBM Data Cataloging pods are expected.

- **Flushing Kafka topics**
  You must have kafka to avoid issues related to restore the specified flushed out topic.
- **Returning IBM Data Cataloging to a running state**
  Provides detailed instructions to put IBM Data Cataloging in running state on an OpenShift environment.

Flushing Kafka topics

You must have kafka to avoid issues related to restore the specified flushed out topic.

Procedure

1. Run the following command to get Kafka topics.

```
oc get kafkatopic |grep file-scan-topic
```

Example output:

```
File-scan-topic                  isd      10       1        True
```

2. Run the following command to back up Kafka topic.

```
oc get kafkatopic file-scan-topic -o yaml
```

Example output:


```
apiVersion: kafka.strimzi.io/v1beta2
kind: KafkaTopic
metadata:
  resourceVersion: '15457887'
  name: file-scan-topic
  namespace: ibm-data-cataloging
  ownerReferences:
    - apiVersion: spectrum-discover.ibm.com/v1alpha1
      kind: SpectrumDiscover
      name: discover-instance
      uid: 52fd9b6c-0823-4f88-b43d-525dd6992b1b
  labels:
    strimzi.io/cluster: isd-ssl
spec:
  partitions: 10
  replicas: 1
```


3. Run the following command to delete the Kafka topic to flush the data.

```
oc delete kafkatopic file-scan-topic
```

4. Run the following command to apply the backed-up YAML file on the cluster to recreate the topic until it deletes the Kafka topic.

```
oc apply -f <topic.yaml>
```

Example output:


```
apiVersion: kafka.strimzi.io/v1beta2
kind: KafkaTopic
metadata:
  resourceVersion: '15457887'
  name: file-scan-topic
  namespace: ibm-data-cataloging
  ownerReferences:
    - apiVersion: spectrum-discover.ibm.com/v1alpha1
      kind: SpectrumDiscover
      name: discover-instance
      uid: 52fd9b6c-0823-4f88-b43d-525dd6992b1b
  labels:
    strimzi.io/cluster: isd-ssl
spec:
  partitions: 10
  replicas: 1
```


Returning IBM Data Cataloging to a running state

Provides detailed instructions to put IBM Data Cataloging in running state on an OpenShift® environment.

About this task

Follow the steps in a specific sequence to put IBM Data Cataloging in to a running state from idle.

Procedure

1. Start Db2 instance. For procedure, see Stopping and starting a Db2 instance.
2. It is needed to come back to a running state and ensure that you make replicas back to the previous state in the following order.
   Note: It is important to scale up smoothly for those deployments that have two or more replicas and wait for them to be at the ready stage to scale up the next two replicas.

   a. DB2WHRest

      i. Run the following command to get the list of the deployments and filter by db2whrest.

      ```
      $oc get deployments |grep db2whrest
      ```

      Example output:

      ```
      $oc get deployments |grep db2whrest
      isd-db2whrest                    0/0     0            0          6d20h
      ```

      ii. Run the following to command to scale up replicas to 1 for every db2whrest deployment.

      ```
      $oc scale --replicas=1 deployment <deployment_name>
      ```

      Example output:

      ```
      $oc scale --replicas=1 deployment db2whrest
      deployment.apps/ db2whrest scaled
      ```

      iii. Ensure that you have scale up replicas to 1 for every db2whrest deployment.
      Example output:

```
$oc get deployments |grep db2whrest
isd-db2whrest                    1/1     1            1           6d20h
```

b. Connection manager

i. Run the following command to get the list of the deployments and filter by connection manager.

```
$oc get deployments |grep connmgr
```

Example output:

```
$oc get deployments |grep connmgr
isd-connmgr                    0/0     0            0           6d20h
```

ii. Run the following to command to scale down replicas to 1 for every connection manager deployment.

```
$oc scale --replicas=1 deployment <deployment_name>
```

Example output:

```
$oc scale --replicas=1 deployment connmgr
deployment.apps/ connmgr scaled
```

iii. Ensure that you have scale up replicas to 1 for every connection manager deployment.
Example output:

```
$oc get deployments |grep connmgr
isd-connmgr                    1/1     1            1           6d20h
```

c. Producers

i. Run the following command to get the list of the deployments and filter by producer.

```
$oc get deployments |grep producer
```

Example output:

```
isd-producer-ceph-le                    0/0     0            0           6d20h
isd-producer-cos-le                     0/0     0            0           6d20h
isd-producer-cos-scan                   0/0     0            0           6d20h
isd-producer-file-scan                  0/0     0            0           6d20h
isd-producer-protect-scan               0/0     0            0           6d20h
isd-producer-scale-le                   0/0     0            0           6d20h
isd-producer-scale-scan                 0/0     0            0           6d20h
```

ii. Run the following to command to scale up replicas to 1 for every producer deployment.

```
$oc scale --replicas=1 deployment <deployment_name>
```

Example output:

```
$oc scale --replicas=1 deployment isd-producer-file-scan
deployment.apps/ isd-producer-file-scan scaled
```

iii. Tip: It is recommended to scale up replicas for two producers and continue with the next two only when they are ready, up-to-date, and available. The object of this is to give OpenShift time to handle these requests properly.
Ensure that you have scale up replicas to 1 for every producer deployment.
Example output:

```
$oc get deployments|grep producer-cos-scan
isd-producer-cos-scan                   1/1     1            1           6d20h
```

d. Consumers

i. Run the following command to get the list of the deployments and filter by consumer.

```
$oc get deployments |grep consumer
```

Example output:

```
isd-consumer-ceph-le                    0/0     0            0           2d
isd-consumer-cos-le                     0/0     0            0           2d
isd-consumer-cos-scan                   0/0     0            0           2d
isd-consumer-file-scan                  0/0     0            0           2d
isd-consumer-protect-scan               0/0     0            0           2d
isd-consumer-scale-le                   0/0     0            0           2d
isd-consumer-scale-scan                 0/0     0            0           2d
```

ii. Run the following to command to scale up replicas to 2 for every consumer deployment.

```
$oc scale –replicas=2 deployment isd-consumer-file-scan
```

Example output:

```
$oc scale –replicas=2 deployment isd-consumer-file-scan
deployment.apps/isd-consumer-file-scan scaled
```

iii. Tip: It is recommended to scale up replicas for two producers and continue with the next two only when they are ready, up-to-date, and available. The object of this is to give OpenShift time to handle these requests properly.
Ensure that you have scaled up replicas to 10 for every consumer deployment, but you must scale up two at a time.
Example output:

```
$oc get deployments|grep isd-consumer-file-scan
isd-consumer-file-scan          2/2     2        2       2d
```

iv. Some pods might face issues when scaling up the replicas back to 10 in a row. If that happens, then you need to wait for a few minutes, but if it is back to the Running or Ready (1/1) state for a long time, then you need to check for failing pods and delete them. OpenShift will create them again after you delete them.

1. Run the following to get the list of pods that are not running.

```
$oc get pods | grep -v Running
```

Example output:

```
isd-consumer-file-scan-6d6bbf8487-2asd3                        0/1     Pending   0               2h
isd-consumer-file-scan-6d6bbf8487-ljkws                          0/1     Pending   0               2h
isd-consumer-file-scan-6d6bbf8487-m96hc                         0/1     Pending   0               2h
isd-consumer-file-scan-6d6bbf8487-w55lr                          0/1     Pending   0               2h
isd-consumer-file-scan-6d6bbf8487-z8hqd                          0/1     Pending   0               2h
```

2. Run the following command to delete pods that are not running.

```
$oc delete pod isd-consumer-file-scan-6d6bbf8487-ljkws
```

Example output:

```
$oc delete pod isd-consumer-file-scan-6d6bbf8487-ljkws
pod "isd-consumer-file-scan-6d6bbf8487-ljkws" deleted

$ oc get pods isd-consumer-file-scan-6d6bbf8487-ljkws
NAME                                     READY   STATUS        RESTARTS    AGE
isd-consumer-file-scan-6d6bbf8487-ljkws  1/1     Terminating   0           2h
```

3. OpenShift will create them again after you delete them.
For example:

```
$oc get pods|grep consumer-file-scan
isd-consumer-file-scan-6d6bbf8487-41x8r                            0/1      Running      0
64s

#Check until having 1/1 Running state

$oc get pods|grep consumer-file-scan
isd-consumer-file-scan-6d6bbf8487-41x8r                            1/1      Running      0
2m39s
```

REST API for IBM Data Cataloging

REST API reference documentation provides endpoint specifications, request and response formats, and authentication details for programmatic access to IBM Spectrum Discover services.

- **IBM Data Cataloging REST APIs**
  The Data Cataloging REST APIs are REST-style APIs that provide interoperability between a client and server over a network. These APIs allow authenticated users to perform management tasks.
- **Endpoints for working with a Db2 Warehouse**
  The Db2® Warehouse API provides endpoints for working with a Db2 Warehouse.
- **Endpoints for working with policy management**
  With the policy management API service, you can create, list, update, and delete the policies.
- **Endpoints for working with connection management**
  With the connection management API service you can create, update, delete, and get information about data source connection entries in the connection table.
- **Application management by using APIs**
  The application management APIs provide options to get the details of the available applications, install or upgrade an application, and delete an application. You must have the Admin role to access the application management endpoints.
- **RBAC management by using APIs**
  The IBM Data Cataloging resource-based access control (RBAC) is a REST API service that enables role-based access to the IBM Data Cataloging services. This service uses OpenStack Keystone as a backend for providing Identity and Access Management (IAM) across multiple domains that are attached to the IBM Data Cataloging.

IBM Data Cataloging REST APIs

The Data Cataloging REST APIs are REST-style APIs that provide interoperability between a client and server over a network. These APIs allow authenticated users to perform management tasks.

The following list shows the significant features of REST-style APIs:

- REST-style APIs are resource-based.
- REST-style APIs are stateless.
- REST-style APIs are client or server.
- REST-style APIs are cacheable.
- REST-style APIs are a layered system.

A representational state transfer (REST) system is a resource-based service system in which requests are made to the resource's universal resources identifier (URI). These requests start a response from the resource in the JSON or CSV format.

The operations that you can perform on the resources or a resource element are directed by the HTTP methods such as GET, POST, PUT, and DELETE and in some cases by the parameters of the HTTPS request. The following list provides the meanings of the basic HTTP methods that are used in the requests:

GET
: Reads a specific resource or a collection of resources and provides the details as the response.

PUT
: Updates a specific resource or a collection of resources.

DELETE
: Removes or deletes a specific resource.

POST
: Creates a resource.

- **API endpoints**
  The Data Cataloging REST APIs include several API services for monitoring data and performing other management tasks.
- **Asynchronous endpoints**
  Some REST API endpoints run asynchronously.
- **Status codes**
  Each API request that is sent to the server returns a response that includes an HTTP status code and any requested information.
- **Authentication process**
  The REST API services require token-based authentication rather than authentication with a user ID and password.

API endpoints

The Data Cataloging REST APIs include several API services for monitoring data and performing other management tasks.

IBM Data Cataloging provides the following REST APIs:

- An API for working with a DB2® warehouse
- A policy management API and an autotags API
- A connection management API
- An application management API
- A resource-based access control (RBAC) API

The endpoints of each API have a characteristic basic syntax. In the following code blocks, `data_cataloging_host` is the host name or IP address of the API server:

- Endpoints for the Db2® Warehouse API:

  ```
  https://<data_cataloging_host>/db2whrest/v1/<endpoint_ID>
  ```

  For example:

  ```
  curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ sql_query_async?<sql>
  ```

- Endpoints for the policy management API:

  ```
  https://<data_cataloging_host>/policyengine/v1/policies/<endpoint_ID>
  ```

  For example:

  ```
  curl -k -H 'Authorization: Bearer <token>' "https://<data_cataloging_host>/policyengine/v1/ policies/<policy_ID>
  ```

- Endpoints for the resource-based access control (RBAC) API:

  ```
  https://<data_cataloging_host>/auth/v1/<endpoint_ID>
  ```

  For example:

  ```
  curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users
  ```

Asynchronous endpoints

Some REST API endpoints run asynchronously.

This behavior is helpful if the action that is performed has a long waiting period or has a negative impact on system performance. Some endpoints are explicitly asynchronous, such as the `/db2whrest/v1/sql_query_async?<sql>: GET` endpoint. Other endpoints are initially synchronous but are automatically converted by the system to asynchronous operations if they take more than 10 seconds to complete.

When you start an endpoint that results in an asynchronous operation, you must poll the asynchronous operation until it is complete. Follow these steps:

1. Find the *location URI* of the asynchronous operation. In response to the request that starts the endpoint, the system returns a synchronous response that indicates that asynchronous operation is accepted for processing but is not yet complete. This response includes a status code of `202 ACCEPTED` and a response header that contains the location of the asynchronous operation. In the following example, the status code appears at line 2 and the location URI appears in line 5:

   ```
   * HTTP 1.0, assume close after body
   < HTTP/1.0 202 ACCEPTED
   < Content-Type: text/html; charset=utf-8
   < Content-Length: 28
   < Location: https://labsrv04/db2whrest/v1/task_status/9c0db090-bd33-41c5-969f-a1603ddf49ab
   < Server: Werkzeug/0.14.1 Python/3.4.5
   < Date: Thu, 15 Feb 2018 20:37:10 GMT
   ```

2. Run the `/db2whrest/v1/task_status/<task_id>: GET` endpoint on the location URI to get the status of the asynchronous operation. The following example uses the location URI from the example in Step 1:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://labsrv04/db2whrest/v1/task_status/9c0db090-bd33-41c5-969f-a1603ddf49ab
   -X GET
   ```

3. If the status code that is returned from the task_status endpoint in Step 2 is `303`, then the asynchronous operation is not completed. Repeat Step 2.

4. If the status code of the task_status endpoint in Step 2 is `200`, then the asynchronous operation completes and the response header contains the location URI of the results. Run the **`/db2whrest/v1/task_status/<task_id>: GET`** endpoint on this location URI to get the results.

For more information see the topic /db2whrest/v1/task_status/<task_id>: GET and <task_id>/peek.

Status codes

Each API request that is sent to the server returns a response that includes an HTTP status code and any requested information.

The following are some of the common HTTP status codes:

200 OK
: The endpoint operation was successful.

201 Created
: The endpoint operation was successful and resulted in the creation of a resource.

202 Accepted
: The request is accepted for processing, but the processing is not yet completed. Asynchronous endpoints return this status code in the response to the original request.

204 No content
: The endpoint operation was successful, but no content is returned in the response.

303 [interim response status]
: The endpoint operation is in progress. Asynchronous endpoints return this status code in response to a request for status.

The following are some common HTTP status error codes:

400 Bad Request
403 Forbidden
404 Not Found
500 Internal Server Error

Along with the HTTP status codes, the Db2® Warehouse API service also returns an error reason in the **`reason`** field of a JSON dictionary. In the following example, a request to add a record to a database is rejected with the status error code 400 "BAD REQUEST". The reason code contains more information, including the following error message:

```
curl -k -v -H 'Authorization: Bearer <token>'
https://data_cataloging_host/db2whrest/v1/sql_query -X POST -d"select *
from labsrv04 where ownerzzz='root'"
<HTTP/1.1 400 BAD REQUEST
{"reason": "('42S22', '[42S22] [IBM][CLI Driver][DB2/LINUXX8664] SQL0206N  \"OWNERZZZ\" is not
valid in the context where it is used.  SQLSTATE=42703\\n')", "status": "error"}
```

Authentication process

The REST API services require token-based authentication rather than authentication with a user ID and password.

Users who need to access the IBM Data Cataloging APIs need to get an authentication token first by using their username and password. Then, use that token to get authenticated to the IBM Data Cataloging system to perform various operations by using APIs.

Authentication is achieved in IBM Data Cataloging through the following steps:

1. The administrator registers an enterprise domain, which can be either Lightweight Directory Access Protocol (LDAP) or Cloud Storage Object, with the authentication service. Registering the enterprise domain with the authentication service includes the following steps:
   a. The administrator gets an authentication token by using the credentials.
      When the administrator (who is in the LDAP domain) logs in to IBM Data Cataloging, the credentials are passed from IBM Data Cataloging to LDAP for authentication. Then, the administrator gets an authentication token by using the credentials.
      Note: Users from an external LDAP or Cloud Storage Object domain, need to include domain name in the user name as **`"<domain>/<user>"`** to get an authentication token by using the REST APIs.
   b. Register the domain by using the obtained authentication token.
      IBM Data Cataloging integrates with both LDAP users and Cloud Storage Object users. Administrators can add the Cloud Storage Object domain to IBM Data Cataloging and the users are imported into IBM Data Cataloging where the administrator can add the users to the appropriate collections.

      In this case, the Cloud Storage Object users can either use their user name and password or the Cloud Storage Object native Amazon Simple Storage Service (Amazon S3) access key and secret key pair to get authenticated with the IBM Data Cataloging authentication service to get an authentication token.

      Like the users from LDAP domain, the Cloud Storage Object domain users can use this authentication token to access the IBM Data Cataloging services and their scope is restricted to the projects to which they can access.
2. The administrator adds collections to the authentication service and adds users to these collections by assigning them appropriate roles. The records that the users can see or apply policies to are restricted according to the collections to which they have access.
3. The user requests for an authentication token by using their user name and password.
   The IBM Data Cataloging RESTful service like DB2WH REST, policy engine, tags, applications, and the various authentication service endpoints require a bearer auth-token to be passed to it in the authorization header.

   The authentication token needs to be obtained by using the authentication token service endpoint, which is the endpoint for a user to log in with user name and password credentials, by using HTTP basic authentication. This token then can be used for authorization across various service endpoints. After a user receives the authentication token, it is valid for 1 hour. Using this token, a user can use various IBM Data Cataloging services.

4. Users can access the IBM Data Cataloging services by using the authentication token.

For more information, see /auth/v1/token: GET.

Endpoints for working with a Db2 Warehouse

The Db2® Warehouse API provides endpoints for working with a Db2 Warehouse.

- **/db2whrest/v1/search: POST**
  Searches a database for data.
- **/db2whrest/v1/report: POST**
  Creates a curation report definition and runs it immediately.
- **/db2whrest/v1/report: GET**
  Gets information on all the curation reports that are available in the system.
- **/db2whrest/v1/report/<report_id>: GET**
  Gets information about a single curation report.
- **/db2whrest/v1/report/<report_id>/download: GET**
  Downloads the output file of the specified curation report.
- **/db2whrest/v1/bucket: GET**
  Gets information on all buckets that are supported.
- **/db2whrest/v1/buckets/<bucket>: GET**
  Gets information on a specific bucket that is supported.
- **/db2whrest/v1/report/<report_id>: PUT**
  Runs the specified curation report. The old output file is replaced by the new one.
- **/db2whrest/v1/buckets/<bucket>: PUT**
  Modifies the bucket details.
- **/db2whrest/v1/report/<report_id>: DELETE**
  Deletes the specified curation report definition and its output file.
- **/db2whrest/v1/sql_query?<sql>: GET**
  Retrieves data from a database with an SQL query in a GET command. You can use any standard SQL query. You must include the SQL query as a parameter of the URL.
- **/db2whrest/v1/sql_query: POST**
  Gets data from a database with an SQL query in a POST command. You can use any standard SQL query. You must include the SQL query in the message body of the POST command.
- **/db2whrest/v1/sql_query_async?<sql>: GET**
  Gets data from a database asynchronously with an SQL query in a GET command.
- **/db2whrest/v1/sql_query_async: POST**
  Gets data from a database asynchronously with an SQL query in a POST command. You can use any standard SQL query. You must include the SQL query in the message body of the POST command.
- **/db2whrest/v1/task_status/<task_id>: GET and <task_id>/peek**
  Gets the status of the specified asynchronous task.
- **/db2whrest/v1/summary_tables -X GET**
  Gets information on all the summary tables that are available in the system.
- **/db2whrest/v1/summary_tables/<table> -X GET**
  Gets information for a particular summary table available in the system.
- **/db2whrest/v1/summary_tables/<table> -X PUT**
  Change the summary table configuration for a particular summary table available in the system.
- **/db2whrest/v1/summary_tables/<table>/<action> -X PUT**
  Triggers a start or scheduled start action for a particular summary table.
- **/db2whrest/v1/bulk_add_tags/docs: POST**
  Searches a database for data.

/db2whrest/v1/search: POST

Searches a database for data.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√[1]	X	X

[1]The search is restricted to documents that are tagged with collections to which the user ID has a datauser role assigned.

The **search** endpoint has the following format:

```
/db2whrest/v1/search -H 'Authorization: Bearer <token>' -X POST -d@<data.json>
-H "Content-Type: application/json"
```

The endpoint accepts requests in JSON format and returns a response in a self-describing data set. It also returns the query time and number of elements in the result set. If an endpoint operation takes more than 10 seconds to complete it is converted to an asynchronous operation. For more information on asynchronous endpoint operations, see Asynchronous endpoints.

You can modify a **search** endpoint with the following fields. An example follows this list:

query
: Specifies a search query string.

filters
: Specifies an array of dictionaries that filter the results of the query. Each dictionary must contain the following three fields:

  key

The name of the field or column to be returned.

operator

One of the following operators: **=, >, <, <>, <=, >=, in, like**.

value

The value of the field or column to filter on.

group_by

Specifies a list of fields to be used to summarize search results. If the group_by field is not specified, the search returns record-level information.

Note: The combination **"group_by": ["filename"]** causes the query to be applied to the duplicate file table. All other group_by combinations cause the query to be applied to the summary tables.

sort_by

Specifies an array of dictionary objects. Each dictionary object must specify a field or column name to be sorted on and a sort direction. Valid sort directions are asc and desc.

limit

Specifies the maximum number of rows that can be returned by the response.

The following example illustrates how to specify these fields:

```
{
    "query": "platform='Spectrum Scale'",
    "filters": [
      {
        "key": "project",
        "operator": "is",
        "value": "null"
      }
    ],
    "group_by": ["Filesystem","Owner","Site"],
    "sort_by": [{"Filesystem": "asc"},{"Owner": "asc"}],
    "limit": 100000
}
```

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/search -X POST -d@search.json -H 
"Content-Type: application/json"
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

1. The following example shows how to define search parameters and format the data that is returned:
    a. Step 1: Define the search parameters in a file named search.json:

    ```
    {
        "query": "platform='Spectrum Scale'",
        "filters": [
          {
            "key": "project",
            "operator": "is",
            "value": "null"
          }
        ],
        "group_by": ["Filesystem","Owner","Site"],
        "sort_by": [{"Filesystem": "asc"},{"Owner": "asc"}],
        "limit": 100000
    }
    ```

    b. Step 2: Submit the request:

    ```
    curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ search -X POST -d@search.json
    -H "Content-Type: application/json
    ```

    The following response is returned:


```
{
"facet_tree": {
"OWNER": "[{\"owner\":\"nobody\",\"count\":8},{\"owner\":\"benjamin\",\"count\":2},
{\"owner\":\"_NULL_\",\"count\":989},{\"owner\":\"borgato\",\"count\":2},
{\"owner\":\"boston\", \"count\":4554},{\"owner\":\"root\",\"count\":366104932},
{\"owner\":\"beale\",\"count\":203545},
{\"owner\":\"baldwin\",\"count\":2},{\"owner\":\"behr\",\"count\":785144},
{\"owner\":\"babcock\", \"count\":9082943},{\"owner\":\"broadwood\",\"count\":375}]",
"PLATFORM": "[{\"platform\":\"Spectrum Scale\",\"count\":376182496}]",
"FILESYSTEM": "[{\"filesystem\":\"fs11-1m-me1\",\"count\":185493076},
{\"filesystem\":\"filesys1\",\"count\":10077057},{\"filesystem\":\"fs10-1m-me1\",
\"count\":180612363}]",
"CLASSIFICATION": "[{\"classification\":null,\"count\":376182496}]",
"DEPARTMENT": "[{\"department\":null,\"count\":376182496}]",
"CLUSTER": "[{\"cluster\":\"host.ibm.com\",\"count\":366105439},{\"cluster\":
```

```
\"filesys1.university.edu\",\"count\":10077057}]",
"TIER": "[{\"tier\":\"system\",\"count\":376182496}]",
"ARCHIVE": "[{\"archive\":null,\"count\":376182496}]",
"SITE": "[{\"site\":\"host\",\"count\":366105439},{\"site\":\"\",\"count\":10077057}]",
"PROJECT": "[{\"project\":null,\"count\":376182496}]"},
"query_time_secs": 1.174846,
"rows": "[{\"filesystem\":\"fs10-1m-me1\",\"owner\":\"_NULL_\",\"site\":\"host\",
\"count\":468,
\"sum\":52933001066},{\"filesystem\":\"fs10-1m-me1\",\"owner\":\"nobody\",\"site\":
\"host\", \"count\":4,\"sum\":20976},{\"filesystem\":\"fs10-1m-me1\",\"owner\":\"root\",
\"site\":\"host\",\"count\":180611891,\"sum\":2002705351263},{\"filesystem\":
\"fs11-1m-me1\",\"owner\":\"_NULL_\", \"site\":\"host\",\"count\":521,\"sum\":58473947666},
{\"filesystem\":\"fs11-1m-me1\",\"owner\":\"nobody\",\"site\":\"host\",\"count\":4,\"sum\":20976},{\"filesystem\":\"fs11-
1m-me1\",
\"owner\":\"root\",\"site\":\"host\",\"count\":185492551,\"sum\":2174830065250},
{\"filesystem\":\"filesys1\",\"owner\":\"baldwin\",\"site\":\"\",\"count\":2,
\"sum\":16388},{\"filesystem\":\"filesys1\",\"owner\":\"behr\",\"site\":\"\",
\"count\":785144,\"sum\":5000771899189}, {\"filesystem\":\"filesys1\",
\"owner\":\"boston\",\"site\":\"\",\"count\":4554,\"sum\":57670240959030},
{\"filesystem\":\"filesys1\",\"owner\":\"beale\",\"site\":\"\",\"count\":203545,
\"sum\":69505800364825},{\"filesystem\":\"filesys1\",\"owner\":\"root\",\"site\":
\"\",\"count\":490,\"sum\":265209686947},{\"filesystem\":\"filesys1\",\"owner\":
\"broadwood\",\"site\":\"\",\"count\":375,\"sum\":48214210142151},{\"filesystem\":
\"filesys1\",\"owner\":\"benjamin\",\"site\":\"\",\"count\":2,\"sum\":3140759161},
{\"filesystem\":\"filesys1\",\"owner\":\"babcock\",\"site\":\"\",\"count\":9082943,
\"sum\":48534270142857},{\"filesystem\":\"filesys1\",\"owner\":\"borgato\",
\"site\":\"\",\"count\":2,\"sum\":4415704787}]","count": 15,"doc_count": 376182496
}
```


In the following code block, the information from the rows field of the response is reflowed so that it is easier to read. The sum column is omitted. You can see that the rows are grouped by file system, owner, and site and are sorted by file system and owner:

```
"rows": "[
{\"filesystem\":\"fs10-1m-me1\", \"owner\":\"_NULL_\", \"site\":\"host\", \"count\":468,
{\"filesystem\":\"fs10-1m-me1\", \"owner\":\"nobody\", \"site\":\"host\", \"count\":4,
{\"filesystem\":\"fs10-1m-me1\", \"owner\":\"root\",   \"site\":\"host\", \"count\":1806,
{\"filesystem\":\"fs11-1m-me1\", \"owner\":\"_NULL_\", \"site\":\"host\", \"count\":521,
{\"filesystem\":\"fs11-1m-me1\", \"owner\":\"nobody\", \"site\":\"host\", \"count\":4,
{\"filesystem\":\"fs11-1m-me1\", \"owner\":\"root\",   \"site\":\"host\", \"count\":1854,
{\"filesystem\":\"filesys1\",    \"owner\":\"baldwin\",\"site\":\"\",     \"count\":2,
{\"filesystem\":\"filesys1\",    \"owner\":\"behr\",   \"site\":\"\",     \"count\":785144,
{\"filesystem\":\"filesys1\",    \"owner\":\"boston\", \"site\":\"\",     \"count\":4554,
{\"filesystem\":\"filesys1\",    \"owner\":\"beale\",  \"site\":\"\",     \"count\":203545,
{\"filesystem\":\"filesys1\",    \"owner\":\"root\",   \"site\":\"\",     \"count\":490,
{\"filesystem\":\"filesys1\",    \"owner\":\"broadwood\",\"site\":\"\",   \"count\":375,
{\"filesystem\":\"filesys1\",    \"owner\":\"benjamin\", \"site\":\",     \"count\":2,
{\"filesystem\":\"filesys1\",    \"owner\":\"babcock\",\"site\":\"\",     \"count\":9082943,
{\"filesystem\":\"filesys1\",    \"owner\":\"borgato\",\"site\":\"\",     \"count\":2,
```

2. The following example shows how to search for duplicate files with a size greater than 1 MiB:
   a. Step 1: Define the search parameters in a file named search.json:

   ```
   {
       "query": "",
       "filters": [
         {
           "key": "size",
           "operator": ">",
           "value": 1048576
         }
       ],
       "group_by": ["size", "filename"],
       "sort_by": [{"size": "asc"}, {"filename": "asc"}],
       "limit": 100
   }
   ```

   b. Step 2: Submit the request:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ search -X POST -d@search.json
   -H "Content-Type: application/json"
   ```

The following response is returned. Some of the rows of the response are omitted:


```
{
"facet_tree":{"OWNER": "[{\"owner\":\"behr\",\"count\":160096},{\"owner\":\"babcock\",
\"count\":14609},{\"owner\":\"beale\",\"count\":612},{\"owner\":\"root\",\"count\":34}]",
"DEPARTMENT":"[{\"department\":null,\"count\":175351}]","FILESYSTEM": "[{\"filesystem\":
\"filesys1\",\"count\":175351}]",    "PROJECT": "[{\"project\":\"TCGA_kirc\",\"count\":86},
{\"project\":\"TCGA_ucs\",\"count\":14},{\"project\":\"TCGA_stad\",\"count\":21},
{\"project\":\"TCGA_lusc\",\"count\":20},{\"project\":\"acc\",\"count\":2},{\"project\":
\"TCGA_meso\",\"count\":54},{\"project\":\"TCGA_skcm\",\"count\":107},{\"project\":
\"TCGA_ov\",\"count\":81},{\"project\":\"kich\",\"count\":10},{\"project\":\"TCGA_lgg\",
\"count\":137},{\"project\":\"TCGA_prad\",\"count\":662},{\"project\":\"TCGA_laml\",
\"count\":6},{\"project\":\"TCGA_thym\",\"count\":54},{\"project\":\"TCGA_thca\",\"count\":
255},{\"project\":\"kirp\",\"count\":256},{\"project\":\"hnsc\",\"count\":22},{\"project\":
\"TCGA_pcpg\",\"count\":2},{\"project\":\"Eichler\",\"count\":877},{\"project\":
\"TCGA_ucec\",\"count\":238},{\"project\":\"TCGA_luad\",\"count\":1417},{\"project\":
\"Level1\",\"count\":6},{\"project\":\"brca\",\"count\":14},{\"project\":null,\"count\":
168855},{\"project\":\"TCGA_paad\",\"count\":196},{\"project\":\"TCGA_read\",\"count\":244}
,{\"project\":\"cesc\",\"count\":613},{\"project\":\"coad\",\"count\":176},{\"project\":
\"dlbc\",\"count\":20},{\"project\":\"blca\",\"count\":471},{\"project\":\"TCGA_sarc\",
\"count\":90},{\"project\":\"TCGA_lihc\",\"count\":206},{\"project\":\"gbm\",\"count\":120},
{\"project\":\"esca\",\"count\":19}]","CLASSIFICATION": "[{\"classification\":null,
\"count\":175351}]","ARCHIVE": "[{\"archive\":null,\"count\":175351}]"},"query_time_secs":
1.377808,"rows": "[{\"size\":1048608,\"filename\":\"NA-C-ms22-cm0.mcdat\",\"count\":4106,
```

```
\"sum\":4305584448},{\"size\":1048608,\"filename\":\"asm_g100-C-ms22-cm0.mcdat\",
\"count\":2,\"sum\":2097216},{\"size\":1048608,\"filename\":\"asm_g1083-C-ms22-cm0.mcdat\",
\"count\":2,\"sum\":2097216},{\"size\":1048608,\"filename\":\"asm_g111-C-ms22-cm0.mcdat\",
\"count\":2,\"sum\":2097216},{\"size\":1048608,\"filename\":\"asm_g1122-C-ms22-cm0.mcdat\",
\"count\":2,\"sum\":2097216},{\"size\":1048608,\"filename\":\"asm_g134-C-ms22-cm0.mcdat\",
\"count\":2,\"sum\":2097216},{\"size\":1048608,\"filename\":\"asm_g145-C-ms22-cm0.mcdat\",
\"count\":2,\"sum\":2097216},{\"size\":1048608,\"filename\":\"asm_g147-C-ms22-cm0.mcdat\",
\"count\":3,\"sum\":3145824},{\"size\":1048608,\"filename\":\"asm_g149-C-ms22-cm0.mcdat\",
\"count\":4,\"sum\":4194432},{\"size\":1048608,\"filename\":\"asm_g151-C-ms22-cm0.mcdat\",
\"count\":2,\"sum\":2097216},{\"size\":1048608,\"filename\":\"asm_g153-C-ms22-cm0.mcdat\",
\"count\":2,
...
8583},{\"size\":1053910,\"filename\":\"asm_g2981.asm\",\"count\":2,\"sum\":2107820},
{\"size\":1053914,\"filename\":\"seqDB.v006.dat\",\"count\":2,\"sum\":2107828},
{\"size\":1054721,\"filename\":\"asm_g3384.asm\",\"count\":2,\"sum\":2109442}]","count":
100,"doc_count":4318
}
```

In the following code block, the information from the rows field of the response is reflowed so that it is easier to read. You can see that the rows are grouped by size and file name and that they are sorted by size and file name in ascending order:

```
"rows":
"[
{\"size\":1048608,\"filename\":\"NA-C-ms22-cm0.mcdat\",   \"count\":4106,\"sum\":4305584448},
{\"size\":1048608,\"filename\":\"asm_g100-C-ms22-cm0.mcdat\",   \"count\":2,\"sum\":2097216},
{\"size\":1048608,\"filename\":\"asm_g1083-C-ms22-cm0.mcdat\",  \"count\":2,\"sum\":2097216},
{\"size\":1048608,\"filename\":\"asm_g111-C-ms22-cm0.mcdat\",   \"count\":2,\"sum\":2097216},
{\"size\":1048608,\"filename\":\"asm_g1122-C-ms22-cm0.mcdat\",  \"count\":2,\"sum\":2097216},
{\"size\":1048608,\"filename\":\"asm_g134-C-ms22-cm0.mcdat\",   \"count\":2,\"sum\":2097216},
{\"size\":1048608,\"filename\":\"asm_g145-C-ms22-cm0.mcdat\",   \"count\":2,\"sum\":2097216},
{\"size\":1048608,\"filename\":\"asm_g147-C-ms22-cm0.mcdat\",   \"count\":3,\"sum\":3145824},
{\"size\":1048608,\"filename\":\"asm_g149-C-ms22-cm0.mcdat\",   \"count\":4,\"sum\":4194432},
{\"size\":1048608,\"filename\":\"asm_g151-C-ms22-cm0.mcdat\",   \"count\":2,\"sum\":2097216},
{\"size\":1048608,\"filename\":\"asm_g153-C-ms22-cm0.mcdat\",   \"count\":2,
...
{\"size\":1053910,\"filename\":\"asm_g2981.asm\",               \"count\":2,\"sum\":2107820},
{\"size\":1053914,\"filename\":\"seqDB.v006.dat\",              \"count\":2,\"sum\":2107828},
{\"size\":1054721,\"filename\":\"asm_g3384.asm\",               \"count\":2,\"sum\":2109442}
]",
```

3. The following example shows how to perform a nongrouped search (record level results) for files owned by **benjamin**:
   a. Step 1: Define the search parameters in a file named search.json:

      ```
      {
          "query": "",
          "filters": [
            {
              "key": "owner",
              "operator": "=",
              "value": "benjamin"
            }
          ],
          "group_by": [],
          "sort_by": [],
          "limit": 100
      }
      ```

   b. Step 2: Submit the following request:

      ```
      curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ search -X POST -d@search.json
      -H "Content-Type: application/json"
      ```

   The following response is returned:

```
* Connection #0 to host localhost left intact
{
"query_time_secs": 1422.651552,"rows": "[{\"filesystem\":\"filesys1\",\"revision\":\"MO1\",
\"site\":\"\",\"platform\":\"Spectrum Scale\",\"cluster\":\"filesys1.university.edu\",
\"inode\":257173,\"owner\":\"benjamin\",\"group\":\"iacs\",\"permissions\":\"-r--r--r--\",
\"fileset\":\"hg19\",\"uid\":null,\"gid\":null,\"path\":\"\\/filesys1\\/hg19\\/\",
\"filename\":\"Homo_sapiens_assembly19.fasta.fai\",\"filetype\":\"fasta\",\"migstatus\":
\"resdnt\",\"migloc\":\"NA\",\"mtime\":\"2014-08-08T19:37:13.000Z\",\"atime\":
\"2017-08-27T22:21:11.000Z\",\"ctime\":\"2016-02-22T19:54:30.000Z\",\"inserttime\":
\"2018-08-02T15:47:25.000Z\",\"tier\":\"system\",\"size\":2780,\"qpart\":1,\"fkey\":
\"filesys1.university.edufilesys1257173\",\"project\":null,\"department\":null,
\"archive\":null,\"classification\":null,\"tag5\":null,\"tag6\":null,\"tag7\":null,
\"tag8\":null,\"tag9\":null,\"tag10\":null,\"tag11\":null,\"tag12\":null,\"tag13\":null,
\"tag14\":null,\"tag15\":null,\"tag16\":null},{\"filesystem\":\"filesys1\",\"revision\":
\"MO1\",\"site\":\"\",\"platform\":\"Spectrum Scale\",\"cluster\":
\"filesys1.university.edu\",\"inode\":322176,\"owner\":\"benjamin\",\"group\":\"iacs\",
\"permissions\":\"-r--r--r--\",\"fileset\":\"hg19\",\"uid\":null,\"gid\":null,\"path\":
\"\\/filesys1\\/hg19\\/\",\"filename\":\"Homo_sapiens_assembly19.fasta\",\"filetype\":
\"fasta\",\"migstatus\":\"resdnt\",\"migloc\":\"NA\",\"mtime\":
\"2014-08-08T19:37:13.000Z\",\"atime\":\"2017-08-27T22:21:15.000Z\",\"ctime\":
\"2016-02-22T19:44:09.000Z\",\"inserttime\":\"2018-08-02T15:47:25.000Z\",\"tier\":
\"system\",\"size\":3140756381,\"qpart\":4,
\"fkey\":\"filesys1.university.edufilesys1322176\",\"project\":null,\"department\":null,
\"archive\":null,\"classification\":null,\"tag5\":null,\"tag6\":null,\"tag7\":null,\"tag8\":null,
\"tag9\":null,\"tag10\":null,\"tag11\":null,\"tag12\":null,\"tag13\":null,\"tag14\":null,
\"tag15\":null,\"tag16\":null}]","doc_count": 2,"count": 2,"facet_tree": {"OWNER":
 "[{\"owner\":\"benjamin\",\"count\":2.0}]","FILESYSTEM": "[{\"filesystem\":\"filesys1\",
\"count\":2.0}]","ARCHIVE": "[{\"archive\":null,\"count\":2.0}]","CLUSTER":
"[{\"cluster\":\"filesys1.university.edu\",\"count\":2.0}]","SITE": "[{\"site\":\"\",
\"count\":2.0}]","CLASSIFICATION": "[{\"classification\":null,\"count\":2.0}]",
```

```
"DEPARTMENT": "[{\"department\":null,\"count\":2.0}]","PLATFORM": "[{\"platform\":
\"Spectrum Scale\",\"count\":2.0}]","TIER": "[{\"tier\":\"system\",\"count\":2.0}]",
"PROJECT": "[{\"project\":null,\"count\":2.0}]"}
}
```

In the following code block, the information from the rows field of the response is reflowed for better viewing. Many columns are omitted. You can see that the response returns two rows in which the owner field is benjamin:

```
"rows": "[
{\"filesystem\":\"filesys1\",\"revision\":\"MO1\",\"site\":\"\", ...\"owner\":\"benjamin\", ...
{\"filesystem\":\"filesys1\",\"revision\":\"MO1\",\"site\":\"\", ...\"owner\":\"benjamin\", ...}
]",
```

/db2whrest/v1/report: POST

Creates a curation report definition and runs it immediately.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/report -X POST -d@report.json -H
"Content-Type: application/json"
```

This endpoint has the same parameters as the `/db2whrest/v1/search` endpoint and also has a name parameter:

query
: Specifies a search query string.

filters
: Specifies an array of dictionaries that filter the results of the query. Each dictionary must contain the following three fields:

  key
  : The name of the field or column to be returned.

  operator
  : One of the following operators: `=, >, <, <>, <=, >=, is, like`.

  value
  : The value of the field or column to filter on.

group_by
: Specifies a list of fields to be used to summarize search results. Grouped queries return output columns for count and sum, whereas nongrouped queries return all columns for each row. If the `group_by` field is not specified, the search returns record-level information.

sort_by
: Specifies an array of dictionary objects. Each dictionary object must specify a field or column name to be sorted on and a sort direction. Valid sort directions are asc and desc.

limit
: Specifies the maximum number of rows that can be returned by the response.

name
: Specifies a name for the report output file. If this parameter is not specified, the endpoint uses the randomly generated UUID as the file name base.

The following example illustrates how to specify these fields:

```
{
    "name": "Unassigned Project Report",
    "query": "platform='Spectrum Scale'",
    "filters": [
      {
        "key": "project",
        "operator": "is",
        "value": "null"
      }
    ],
    "group_by": ["Filesystem","Owner","Site"],
    "sort_by": [{"Filesystem": "asc"},{"Owner": "asc"}],
    "limit": 100000
}
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Status codes

201
: The operation is successful.

All other status code values
: The operation failed.

- 201: The operation is successful.
- All other status code values: The operation failed.

Examples

1. The following example shows how to create a report:
    a. Step 1: Define the search parameters in a file named report.json:

    ```
    {
        "name": "Unassigned Project Report",
        "query": "platform='Spectrum Scale'",
        "filters": [
          {
            "key": "project",
            "operator": "is",
            "value": "null"
          }
        ],
        "group_by": ["Filesystem","Owner","Site"],
        "sort_by": [{"Filesystem": "asc"},{"Owner": "asc"}],
        "limit": 100000
    }
    ```

    b. Step 2: Submit the following request:

    ```
    curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/report -X POST -d@report.json
    -H "Content-Type: application/json"
    ```

2. The response includes the ID of the new report and the status of the operation ("report created"):

    ```
    {"report": "b5ff3126-353d-4d7d-857a-750cc20b8bab", "status": "report created"}
    ```

Report scripts

IBM Data Cataloging contains a set of predefined JSON and scripts to generate specific reports.

1. JSON Examples
    - The following example lists the files or objects that are accessed in last 0 - 30 days. This report is created in the UI.

      age_report_0-30_days_since_access_detail.json

      ```
      {
          "name": "Age Report Detail. 0-30 Days",
          "query": "",
          "filters": [
            {
              "key": "atime",
              "operator": ">",
              "value": "NOW() - 30 DAYS"
            }
          ],
          "group_by": [],
          "sort_by": []
      }
      ```

    - This example summarizes the files or objects that are accessed in last 0 - 30 days and are grouped by data source. This report is created in the UI.

      age_report_0-30_days_since_access_summary.json

      ```
      {
          "name": "Age Report Summary. 0-30 Days",
          "query": "",
          "filters": [
            {
              "key": "atime",
              "operator": ">",
              "value": "NOW() - 30 DAYS"
            }
          ],
          "group_by": ["datasource"],
          "sort_by": [{"datasource": "asc"}]
      }
      ```

    - This example lists the files or objects that are accessed in last 30 - 60 days. This report is created in the UI.

      age_report_30-60_days_since_access_detail.json

      ```
      {
          "name": "Age Report Detail. 30-60 Days",
          "query": "",
          "filters": [
            {
              "key": "atime",
              "operator": "<=",
              "value": "NOW() - 30 DAYS"
            },
      ```

```
        {
          "key": "atime",
          "operator": ">",
          "value": "NOW() - 60 DAYS"
        }
      ],
      "group_by": [],
      "sort_by": []
}
```

- This example summarizes the files or objects that are accessed in last 30 - 60 days and are grouped by data source. This report is created in the UI.

  age_report_30-60_days_since_access_summary.json

```
{
    "name": "Age Report Summary. 30-60 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 30 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
        "value": "NOW() - 60 DAYS"
      }
    ],
    "group_by": ["datasource"],
    "sort_by": [{"datasource": "asc"}]
}
```

- This example lists the files or objects that are accessed in last 60 - 90 days. This report is created in the UI.

  age_report_60-90_days_since_access_detail.json

```
{
    "name": "Age Report Detail. 60-90 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 60 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
        "value": "NOW() - 90 DAYS"
      }
    ],
    "group_by": [],
    "sort_by": []
}
```

- The following example summarizes the files or objects that are accessed in last 60 - 90 days and are grouped by data source. This report is created in the UI.

  age_report_60-90_days_since_access_summary.json

```
{
    "name": "Age Report Summary. 60-90 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 60 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
        "value": "NOW() - 90 DAYS"
      }
    ],
    "group_by": ["datasource"],
    "sort_by": [{"datasource": "asc"}]
}
```

- The following example lists the files or objects that are accessed in last 90 - 180 days. This report is created in the UI.

  age_report_90-180_days_since_access_detail.json

```
{
    "name": "Age Report Detail. 90-180 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 90 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
```

```
        "value": "NOW() - 180 DAYS"
      }
    ],
    "group_by": [],
    "sort_by": []
}
```

- The following example summarizes the files or objects that are accessed in last 90 - 180 days and are grouped by data source. This report is created in the UI.

  age_report_90-180_days_since_access_summary.json

```
{
    "name": "Age Report Summary. 90-180 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 90 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
        "value": "NOW() - 180 DAYS"
      }
    ],
    "group_by": ["datasource"],
    "sort_by": [{"datasource": "asc"}]
}
```

- The following example lists the files or objects that are accessed in last 180 - 360 days. This report is created in the UI.

  age_report_180-360_days_since_access_detail.json

```
{
    "name": "Age Report Detail. 180-360 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 180 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
        "value": "NOW() - 360 DAYS"
      }
    ],
    "group_by": [],
    "sort_by": []
}
```

- The following example summarizes the files or objects that are accessed in last 180 - 360 days and are grouped by data source. This report is created in the UI.

  age_report_180-360_days_since_access_summary.json

```
{
    "name": "Age Report Summary. 180-360 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 180 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
        "value": "NOW() - 360 DAYS"
      }
    ],
    "group_by": ["datasource"],
    "sort_by": [{"datasource": "asc"}]
}
```

- The following example lists the files or objects that are accessed in last 360 - 720 days. This report is created in the UI.

  age_report_360-720_days_since_access_detail.json

```
{
    "name": "Age Report Detail. 360-720 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 360 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
        "value": "NOW() - 720 DAYS"
```

```
        }
    ],
    "group_by": [],
    "sort_by": []
}
```

- The following example summarizes the files or objects that are accessed in last 360 - 720 days and are grouped by data source. This report is created in the UI.

  age_report_360-720_days_since_access_summary.json

```
{
    "name": "Age Report Summary. 360-720 Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 360 DAYS"
      },
      {
        "key": "atime",
        "operator": ">",
        "value": "NOW() - 720 DAYS"
      }
    ],
    "group_by": ["datasource"],
    "sort_by": [{"datasource": "asc"}]
}
```

- The following example lists the files or objects accessed that are not accessed in the last 720 days. This report is created in the UI.

  age_report_720+_days_since_access_detail.json

```
{
    "name": "Age Report Detail. 720+ Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 720 DAYS"
      }
    ],
    "group_by": [],
    "sort_by": []
}
```

- The following example summarizes the files or objects that are accessed in last 720 days and are grouped by data source. This report is created in the UI.

  age_report_720+_days_since_access_summary.json

```
{
    "name": "Age Report Summary. 720+ Days",
    "query": "",
    "filters": [
      {
        "key": "atime",
        "operator": "<=",
        "value": "NOW() - 720 DAYS"
      }
    ],
    "group_by": ["datasource"],
    "sort_by": [{"datasource": "asc"}]
}
```

2. SQL scripts
   - This script provides the count of potentially duplicate files across the heterogeneous environment. This report is not created in the UI.

     duplicate_files_by_count.sql

```
select filename, size, count(fkey) from metaocean group by filename,
    size order by count(fkey) desc limit 20;
```

   - This script provides the size of potentially duplicate files across the heterogeneous environment. This report is not created in the UI.

     duplicate_files_by_total_size.sql

```
select filename,entrysize,entrycount,totalsize from (select filename,
    size as entrysize, count(fkey) as entrycount, count(fkey)*size as TotalSize
    from metaocean group by filename,size) where entrycount>1 order by totalsize desc;
```

   - This script provides the size of potentially duplicate files across the heterogeneous environment. This report is not created in the UI.

     size_snap.sql

```
select datasource,count(*),sum(size)/1024/1024/1024,max(mtime),max(atime)
    from metaocean group by datasource with ur
```

   - This script provides a view of the capacity that is consumed per collection. This report is not created in the UI.

     space_per_collection.sql

```
select metaocean.collection,count(*),sum(size)/1024/1024/1024,max(mtime),datasource,
    tier from metaocean group by metaocean.collection,datasource,tier order by max(mtime)
```

```
    desc with ur
```

- This script provides a view of the capacity that is consumed per file type. This report is not created in the UI.

  space_per_filetype.sql

  ```
  select filetype,datasource,tier,count(*),sum(size)/1024/1024/1024 from metaocean
      group by filetype,datasource,tier order by filetype,datasource,tier desc with ur
  ```

- This script provides a view of the capacity that is consumed per user. This report is not created in the UI.

  space_per_user.sql

  ```
  select owner,tier,count(*),sum(size)/1024/1024/1024,max(mtime),max(atime) from metaocean
      group by owner,tier order by owner,tier with ur
  ```

Reports can also be generated by using one of two CLI utilities that are provided with IBM Data Cataloging. Both exist on the IBM Data Cataloging nodes:

- /opt/ibm/metaocean/reports/generate_report.py
  To run the utilities, log in to an IBM Data Cataloging node and run generate_report.py with the following usage:

  ```
  python generate_report.py [-h] [-o filename] -u username [infile]
  ```

  - Run a report with an SQL input file:

    ```
    python generate_report.py -u sdadmin -o report.csv sql/space_per_user.sql
    Enter password for SD user 'sdadmin':
    ```

  - Run a report with a JSON input file:

    ```
    python generate_report.py -u sdadmin  sql/age_report_0-30_days_since_access_detail.json
    Enter password for SD user 'sdadmin':
    ```

  This requires a IBM Data Cataloging user name and input file. You must have the Data Admin role to generate reports. The tool prompts for a password

  Input file examples are stored in the /opt/ibm/metaocean/reports/sql directory. There is a mix of JSON and SQL files in the directory. All JSON files create a report in the IBM Data Cataloging UI. The SQL files create a CSV file.

- /opt/ibm/metaocean/reports/generate_path_age_report.py
  Important: This tool does not create reports in the IBM Data Cataloging UI.
  Here is the usage of the tool:

  ```
  generate_path_age_report.py [-h] -u username -r report -p pathlevel -a archive

  optional arguments:
    -h, --help            show this help message and exit
    -u username, --user username
                          User name with authority to create reports
    -r report, --report report
                          The report type to be generated (ARDS, ARDSOW, AROW, ARPL,
                          CPPL, FTMB, CPFT, FTMB50)
    -p pathlevel, --pathlevel pathlevel
                          The level of path used in some reports
    -a archive, --archive archive
                          The archive threshold in months
  ```

  The `generate_path_age_report.py` tool needs a minimum of three parameters:
  - `pathlevel`
    The level of path used in some reports, for example:

    ```
    1 = /x/, 2=/x/y/
    ```

    If your report does not use this parameter a value of 1 must be used.
  - `archive`
    The archive threshold is the number of months since the last time a file is considered relevant for archiving for reporting purposes. If your report does not use this parameter a value of 1 must be used.

  - `report`
    The report type is one of the following codes:

    ```
    ARDS   = Summary of archivable capacity grouped by datasource
    ARDSOW = Summary of archivable capacity grouped by datasource
    AROW   = Summary of archivable capacity grouped by owner
    ARPL   = Summary of archivable capacity grouped by specified path level
    CPPL   = Summary of capacity grouped by specified path level
    FTMB   = Summary of filetype usage by month, previous 12 months
    CPFT   = Summary of capacity grouped by filetype
    FTMB50 = Summary of filetype usage by month, previous 12 months' top 50 filetypes
    ```

Here is an example of running the `generate_path_age_report.py` tool:

```
python generate_path_age_report.py -u sdadmin -r ARDSOW -p 2 -a 12
Starting to create report type 'ARDSOW' for user: sdadmin
Enter password for SD user 'sdadmin':
Setting up path level summary table
Generating path level summary table
Generating path level summary table complete.
Generating report
Generating report - building temporary table.
Generating report - querying temporary table.
Generating report - writing output file.
Report generated successfully. Results are in 'rpt_ar_ds_ow.csv'.
```

/db2whrest/v1/report: GET

Gets information on all the curation reports that are available in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ report -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

1. The following example shows how to get information on all the curation reports that are available in the system.
   a. Submit the request:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/ v1/report -X GET -H "Accept: application/json"
   ```

   b. The following response is returned. In this example the system contained a total of only two reports, and the response displays the available information about both reports. The response data is in JSON format. The data has been manually reflowed to make it more readable.

   ```
   [
     {
       "name": "Age Report Detail. 0-30 Days",
       "query": "{\"filters\": [{\"value\": \"NOW() - 30 DAYS\", \"key\": \"atime\", \"operator\":
       \">\"}], \"sort_by\": [], \"name\": \"Age Report Detail. 0-30 Days\", \"query\": \"\",
       \"group_by\": []}",
       "size": 208,
       "report": "34d03123-4611-4f9f-9730-ffe89c6c53cb",
       "duration": 0,
       "filename": "Age Report Detail. 0-30 Days-2018-11-01_18:58:04.csv",
       "lastrun": "2018-11-01T18:58:04.000Z",
       "status": "complete",
       "schedule": null
     },
     {
       "name": "Age Report Summary. 0-30 Days",
       "query": "{\"filters\": [{\"value\": \"NOW() - 30 DAYS\", \"key\": \"atime\", \"operator\":
       \">\"}], \"sort_by\": [{\"datasource\": \"asc\"}], \"name\": \"Age Report Summary. 0-30 Days\",
       \"query\": \"\", \"group_by\": [\"datasource\"]}",
       "size": 21,
       "report": "b5ff3126-353d-4d7d-857a-750cc20b8bab",
       "duration": 0,
       "filename": "Age Report Summary. 0-30 Days-2018-11-01_18:58:14.csv",
       "lastrun": "2018-11-01T18:58:14.000Z",
       "status": "complete",
       "schedule": null
     }
   ]
   ```

   Note: To download the output of a report, use the download endpoint. For more information, see /db2whrest/v1/report/<report_id>/download: GET.

/db2whrest/v1/report/<report_id>: GET

Gets information about a single curation report.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/report/ <report_id> -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

1. The following example shows how to get information about a single curation report.
   a. Submit a request to display information about report ID "b5ff3126-353d-4d7d-857a-750cc20b8bab":

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ report/b5ff3126-353d-4d7d-
   857a-750cc20b8bab -X GET -H "Accept: application/json"
   ```

   b. The following response is returned. The response data is in JSON format:

   ```
   {"name": "Age Report Summary. 0-30 Days", "query":"{\"filters\":[{\"value\":\"NOW() -
   30 DAYS\",\"key\":\"atime\",\"operator\":\">\"}],\"sort_by\":[{\"datasource\":\"asc\"}],
   \"name\":\"Age Report Summary. 0-30 Days\",\"query\":\"\",\"group_by\":[\"datasource\"]}",
   "size":21,"report":"b5ff3126-353d-4d7d-857a-750cc20b8bab","duration":0,"filename":"Age
   Report Summary. 0-30 Days-2018-11-01_18:58:14.csv","lastrun":"2018-11-01T18:58:14.000Z",
   "status":"complete","schedule":null}
   ```

   In the following code block, the response data has been manually reflowed to make it easier to read:

   ```
   {
   "name":"Age Report Summary. 0-30 Days","query":
   "{\"filters\":[{\"value\":\"NOW() - 30 DAYS\",\"key\":\"atime\",\"operator\":\">\"}],
   \"sort_by\":[{\"datasource\":\"asc\"}],\"name\":\"Age Report Summary. 0-30 Days\",
   \"query\":\"\",\"group_by\":[\"datasource\"]}",
   "size":21,
   "report":"b5ff3126-353d-4d7d-857a-750cc20b8bab",
   "duration":0,
   "filename":"Age Report Summary. 0-30 Days-2018-11-01_18:58:14.csv",
   "lastrun":"2018-11-01T18:58:14.000Z",
   "status":"complete",
   "schedule":null
   }
   ```

   Note: To download the output of a report, use the download endpoint. For more information, see /db2whrest/v1/report/<report_id>/download: GET.

/db2whrest/v1/report/<report_id>/download: GET

Downloads the output file of the specified curation report.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/report/ <report_id>/download -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- CSV

Status codes

- 200: The operation is successful.
- Any other status code value: The operation failed.

Examples

1. The following example shows how to download the output file of a specific report.
   a. Submit a request to display the output from report ID "6baae44c-2b85-4954-ba44-d3e637d4b48d":

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ report/6baae44c-2b85-4954-
   ba44-d3e637d4b48d/download
   ```

This report searches for files greater than 1 MiB in size and grouped by datasource.

b. The following response is returned. The output is in CSV format:

```
datasource,count,sum
filesystem1,346.0,685733316581
filesystem2,370213.0,249855575516039
```

/db2whrest/v1/bucket: GET

Gets information on all buckets that are supported.

Buckets influence the manner in which IBM Data Cataloging groups data for search faceting and in the user interface widgets. The predefined bucket types that are supported are:

- SizeRange
- TimeSinceAccess
- FileGroup

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/buckets -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

1. The following example shows how to get information on all the buckets that are supported by the bucket service.
   a. Submit the request:

```
curl -k -H 'Authorization: Bearer <token> https://<data_cataloging_host>/db2whrest/v1/buckets -X GET -H "Accept: application/json"
```

   b. The following response is returned.


```
[
  {
    "data": "{\"type\": \"int\", \"source_field\": \"size\", \"ranges\": {\"extra small\": [0, 4096], \"small\": [4096, 1048576], \"extra large\": [1099511627776, \"INFINITY\"], \"large\": [1073741824, 1099511627776], \"medium\": [1048576, 1073741824]}, \"name\": \"SizeRange\"}",
    "name": "SizeRange"
  },
  {
    "data": "{\"ranges\": {\"1 year+\": [365, \"INFINITY\"], \"1 quarter\": [30, 90], \"1 year\": [90, 365], \"1 month\": [7, 30], \"1 week\": [0, 7]}, \"source_field\": \"atime\", \"type\": \"date\", \"name\": \"TimeSinceAccess\"}",
    "name": "TimeSinceAccess"
  },
  {
    "name": "FileGroup",
    "data": "{\"name\": \"FileGroup\", \"type\": \"list\", \"ranges\": {\"pdf\": [\"pdf\"], \"doc\": [\"doc\", \"docx\", \"odt\"], \"xls\": [\"xls\", \"xlsm\", \"xlsx\", \"ods\"], \"image\": [\"jpg\", \"jpeg\", \"png\", \"gif\", \"tif\"], \"ppt\": [\"pptx\", \"ppt\", \"pps\", \"odp\"], \"compressed\": [\"zip\", \"tar\", \"gz\", \"z\", \"7z\", \"xz\"], \"text\": [\"txt\", \"rtf\"], \"audio\": [\"mp3\", \"wav\"], \"video\": [\"mp4\", \"mov\", \"mpg\", \"mkv\"], \"medical_image\": [\"img\", \"hdr\", \"nii\", \"mnc\", \"dcm\"], \"genomics\": [\"bam\", \"sam\", \"vcf\"], \"semi_structured\": [\"csv\", \"tsv\", \"avro\", \"json\", \"xml\", \"parquet\"]}, \"source_field\": \"filetype\"}"
  }
]
```


/db2whrest/v1/buckets/<bucket>: GET

Gets information on a specific bucket that is supported.

Buckets influence the manner in which IBM Data Cataloging groups data for search faceting and in the user interface widgets. The predefined bucket types that are supported are:

- SizeRange
- TimeSinceAccess
- FileGroup

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/buckets/<bucket> -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

1. The following example shows how to get information on all the buckets that are supported by the bucket service.
   a. Submit the request:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/buckets/SizeRange -X GET -H
   "Accept: application/json"
   ```

   b. The following response is returned.

   ```
   [
     {
       "data": "{\"type\": \"int\", \"source_field\": \"size\", \"ranges\": {\"extra small\": [0, 4096], \"small\":
   [4096, 1048576], \"extra large\": [1099511627776, \"INFINITY\"], \"large\": [1073741824, 1099511627776], \"medium\":
   [1048576, 1073741824]}, \"name\": \"SizeRange\"}",
       "name": "SizeRange"
     }
   ]
   ```

/db2whrest/v1/report/<report_id>: PUT

Runs the specified curation report. The old output file is replaced by the new one.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/report/ <report_id> -X PUT
```

Supported request types and response formats

Supported request types:

- PUT

Supported response formats:

- JSON

/db2whrest/v1/buckets/<bucket>: PUT

Modifies the bucket details.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

To modify the details of a bucket, you must create a JSON object that is modeled after the original (or existing) bucket definition as shown in the sample:

```
{
  "name": "SizeRange",
  "source_field": "size",
  "type": "int"
  "ranges": {
    "extra small": [ 0, 4096 ],
    "small": [ 4096, 1048576 ],
    "medium": [ 1048576, 1073741824 ],
    "large": [ 1073741824, 1099511627776 ],
    "extra large": [ 1099511627776, "INFINITY" ]
  }
}

{
  "name": "TimeSinceAccess",
  "source_field": "atime",
  "type": "date",
  "ranges": {
    "1 week\": [ 0, 7 ],
    "1 month": [ 7, 30 ],
    "1 quarter": [ 30, 90 ],
    "1 year": [ 90, 365 ],
    "1 year+": [ 365, "INFINITY" ]
    }
}

{
  "name": "FileGroup",
  "type": "list",
  "ranges": {
    "pdf": [ "pdf" ],
    "doc": [ "doc", "docx", "odt" ],
    "xls": [ "xls", "xlsm", "xlsx", "ods" ],
    "image": [ "jpg", "jpeg", "png", "gif", "tif" ],
    "ppt": [ "pptx", "ppt", "pps", "odp" ],
    "compressed": [ "zip", "tar", "gz", "z", "7z", "xz" ],
    "text": [ "txt", "rtf" ],
    "audio": [ "mp3", "wav" ],
    "video": [ "mp4", "mov", "mpg", "mkv" ],
    "medical_image": [ "img", "hdr", "nii", "mnc", "dcm" ],
    "genomics": [ "bam", "sam", "vcf" ],
    "semi_structured": [ "csv", "tsv", "avro", "json", "xml", "parquet" ]
  },
  "source_field": "filetype"
}
```

The JSON object must be based on the default values of the bucket. You can modify the range labels or values. The bucket name, source_field, and bucket types cannot be modified.

The three bucket types that are supported include:

int
: For the "int" types, each range consists of a label and a minimum and maximum value. The only 'int' type bucket that is supported currently is *SizeRange*. The units for 'SizeRange' are in bytes. A sample int type appears as shown:

```
"small": [4096, 8192]
```

date
: For the "date" types, each range consists of a label and a minimum and maximum value that represents the number of days. The only 'date' type bucket that is supported in the current version is *TimeSinceAccess*. A sample date range appears as shown:

```
"1_to_2_years": [365, 730]
```

list
: For the "list" types, each range consists of a label and a list of at least one string value. The only 'list' type bucket that is supported in the current version is *FileGroup*. The list of values must be a list of file extensions to be grouped. A sample list range appears as shown:

```
"images": ["jpg", "jpeg", "gif", "png", "bmp"]
```

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/buckets/<bucket> -X PUT -d@bucket.json -
H "Content-Type: application/json
```

Supported request types and response formats

Supported request types:

- PUT

Supported response formats:

- JSON

Examples

The following example shows how to modify information for a single bucket.

1. Submit the request:

```
curl -k -H 'Authorization: Bearer <token>' https://<master_node>/db2whrest/v1/buckets/SizeRange -X PUT -d@bucket.json -H
"Content-Type: application/json
```

2. The following response is returned:

```
[
  {
    "status": "success"
  }
]
```

/db2whrest/v1/report/<report_id>: DELETE

Deletes the specified curation report definition and its output file.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/report/ <report_id> -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

/db2whrest/v1/sql_query?<sql>: GET

Retrieves data from a database with an SQL query in a GET command. You can use any standard SQL query. You must include the SQL query as a parameter of the URL. To send the query in an HTTP message body, see /db2whrest/v1/sql_query: POST.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√[1]	X	X
[1]The search is restricted to documents that are tagged with collections to which the user has a datauser role assigned.				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/sql_query?<sql_statements> -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- CSV
- JSON

Examples

1. The following example shows how to get data from a database with an SQL query in a GET command. The SQL query is included as a parameter of the URL. This example specifies that the data must be returned in JSON format:
   a. Submit the request:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/sql_query?
   select%20*%20from%20meteor%20limit%202 -X GET -H "Accept: application/json"
   ```

   b. The following response is returned:

```
[{"datasource":"ctolib","operation":"INDEX","ingesttype":"SCAN","revision":"MO1","site":
"","platform":"Spectrum Scale","cluster":"ctolib.cluster1","inode":4042,"owner":"user1",
"group":"group1","permissions":"-rw-r--r--","fileset":"brother","uid":null,"gid":null,
"path":"\/ctolib\/brother\/dir1\/","filename":"file1.txt","filetype":"txt",
"recordversion":"","migstatus":"resdnt","migloc":"NA","mtime":"2015-08-20T21:47:09.000Z",
"atime":"2016-06-08T17:20:41.480Z","ctime":"2016-03-12T15:12:49.565Z","inserttime":
"2018-10-24T04:19:37.000Z","updatedtime":"2018-10-24T04:19:37.000Z","requesttime":
"2018-10-24T04:19:37.000Z","scangen":1,"tier":"system","size":28185,"qpart":0,"fkey":
"ctolib.cluster1ctolib4042","collection":null,"temperature":null,"duplicate":null,
"some-tag":null,"tag2":null,"tag3":null,"tag4":null,"tag5":null,"tag6":null,"tag7":null,
"tag8":null,"tag9":null,"tag10":null,"tag11":null,"tag12":null,"tag13":null,"tag14":null,
"tag15":null,"tag16":null},
{"datasource":"ctolib","operation":"INDEX","ingesttype":"SCAN","revision":"MO1","site":
"","platform":"Spectrum Scale","cluster":"ctolib.cluster1","inode":285102,"owner":"user1",
"group": "group1","permissions":"-rw-r--r--","fileset":"brother","uid":null,"gid":null,
"path":"\/ctolib\/brother\/dir1\/","filename":"file2.txt","filetype":"txt",
"recordversion":"","migstatus":"resdnt","migloc":"NA","mtime":"2015-08-21T00:55:33.000Z",
"atime":"2016-06-24T00:51:00.797Z","ctime":"2016-03-10T22:55:06.160Z","inserttime":
"2018-10-24T04:19:50.000Z","updatedtime":"2018-10-24T04:19:50.000Z","requesttime":
"2018-10-24T04:19:50.000Z","scangen":1,"tier":"system","size":435947,"qpart":0,"fkey":
"ctolib.cluster1ctolib28510 2","collection":null,"temperature":null,"duplicate":null,
"some-tag":null,"tag2":null,"tag3":null,"tag4":null,"tag5":null,"tag6":null,"tag7":null,
"tag8":null,"tag9":null,"tag10":null,"tag11":null,"tag12":null,"tag13":null,"tag14":
null,"tag15":null,"tag16":null}]
```

2. The following example makes the same query as the preceding example but does not specify an output format:
   a. Submit the request:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/sql_query?
   select%20*%20from%20meteor%20limit%202 -X GET"
   ```

   b. The following response is returned. The data is in CSV format:

   ```
   0,ctolib,INDEX,SCAN,MO1,,Spectrum Scale,ctolib.cluster1,250692,user1,group1,-rw-r--r--,
   brother,,,/ctolib/brother/dir1/,file1.txt,txt,,resdnt,NA,2015-08-20 11:26:29.000000,
   2016-06-2323:26:00.113925,2016-03-10 19:33:36.639688,2018-10-24 04:19:50.000000,
   2018-10-2404:19:50.000000,2018-10-24 04:19:50.000000,1,system,266314,0,
   ctolib.cluster1ctolib250692,
   ,,,,,,,,,,,,,,,,,,
   1,ctolib,INDEX,SCAN,MO1,,Spectrum Scale,ctolib.cluster1,250662,user1,group1,-rw-r--r--,
   brother,,,/ctolib/brother/dir1,file2..txt,txt,,resdnt,NA,2015-08-20 11:26:25.000000
   ,2016-06-23 23:25:59.990565,2016-03-10 19:33:36.070314,2018-10-24 04:19:50.000000,
   2018-10-24 04:19:50.000000,2018-10-24 04:19:50.000000,1,system,305365,0,
   ctolib.cluster1ctolib250662,
   ,,,,,,,,,,,,,,,,,,
   ```

/db2whrest/v1/sql_query: POST

Gets data from a database with an SQL query in a POST command. You can use any standard SQL query. You must include the SQL query in the message body of the POST command.
To send the query as a parameter of the URL, see /db2whrest/v1/sql_query?<sql>: GET. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√[1]	X	X
[1]The search is restricted to documents that are tagged with collections to which the user ID has a datauser role assigned.				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token> https://<data_cataloging_host>/db2whrest/v1/ sql_query -X POST -d@<sql.dat>
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- CSV
- JSON

Examples

1. The following example shows how to get the data from a database with an SQL query in a POST command. The SQL query is sent in the POST message body. The output format is plain text.
   a. Write the record to be added to the database into a file named sql.dat.
   b. Submit the request:

   ```
   curl -k -H 'Authorization: Bearer <token> https://<data_cataloging_host> /db2whrest/v1/sql_query -X POST -d"select *
   from datasrv04 where owner='root'"
   ```

   c. The following response is returned. The output is in plain text format:

```
0,filesys1,INDEX,SCAN,MO1,,Spectrum Scale,filesys1.university.edu,3067343,root,root,
-rw-r--r--,root,9,10,/filesys1/cellranger-2.0.0/refdata-cellranger-ercc92-1.2.0/star/,
sjdbList.out.tab,tab,resdnt,NA,2016-11-14 19:40:10,2017-06-02 21:30:26,
2017-06-02 21:30:26,2018-07-24 16:32:47,system,0,1,filesys1.university.edufilesys13067343,
,,,,,,,,,,,,,,,
1,filesys1,INDEX,SCAN,MO1,,Spectrum Scale,filesys1.university.edu,3067333,root,root,
-rw-r--r--,root,9,10,/filesys1/cellranger-2.0.0/refdata-cellranger-ercc92-1.2.0/star/,
chrLength.txt,txt,resdnt,NA,2016-11-14 19:40:10,2017-06-02 23:28:21,2017-06-02 21:30:26,
2018-07-24 16:32:47,system,412,1,filesys1.university.edufilesys13067333,
,,,,,,,,,,,,,,,
...
```

In the following code block, the information from the response is curtailed so that it is easier to read. Only the first 14 columns are shown here. You can see that both rows seem to have root as the owner:

```
0,filesys1,INDEX,SCAN,MO1,,Spectrum Scale,filesys1.university.edu,3067343,root,root,
-rw-r--r--,root,9,10,/filesys1/. . . .
1,filesys1,INDEX,SCAN,MO1,,Spectrum Scale,filesys1.university.edu,3067333,root,root,
-rw-r--r--,root,9,10,/filesys1/. . . .
...
```

2. The following example shows how to get the data from a database with an SQL query in a POST command. The SQL query is sent in the POST message body. The output format is JSON.
   a. Write the SQL record in a file named sql.dat.
   b. Submit the request.

```
curl -k -H 'Authorization: Bearer <token> https://<data_cataloging_host>/db2whrest/v1/ sql_query -X POST -d@"sql.dat"
-H "accept: application/json"
```

   c. The following response is returned. The output is in JSON format:


```
[{"filesystem":"filesys1","operation":"INDEX","source":"SCAN","revision":"MO1","site":"",
"platform":"Spectrum Scale","cluster":"filesys1.university.edu","inode":3067112,"owner":
"root","group":"root","permissions":"-r--r--r--","fileset":"root","uid":9,"gid":10,"path":
"\/filesys1\/blast_db\/","filename":"other_genomic.75.tar.gz","filetype":"gz","migstatus":
"resdnt","migloc":"NA","mtime":"2014-12-31T06:00:00.000Z","atime":
"2016-09-26T16:17:14.000Z","ctime":"2016-09-22T17:46:21.000Z","inserttime":
"2018-07-24T16:32:59.000Z","tier":"system","size":929541048,"qpart":0,"fkey":
"filesys1.university.edufilesys13067112","project":null,"department":null,"backup":null,
"tag4":null,"tag5":null,"tag6":null,"tag7":null,"tag8":null,"tag9":null,"tag10":null,
"tag11":null,"tag12":null,"tag13":null,"tag14":null,"tag15":null,"tag16":null},
{"filesystem":"filesys1","operation":"INDEX","source":"SCAN","revision":"MO1","site":"",
"platform":"Spectrum Scale","cluster":"filesys1.university.edu","inode":3067092,"owner":
"root","group":"root","permissions":"-r--r--r--","fileset":"root","uid":9,"gid":10,
"path":"\/filesys1\/blast_db\/","filename":"other_genomic.65.tar.gz","filetype":"gz",
"migstatus":"resdnt","migloc":"NA","mtime":"2014-12-31T06:00:00.000Z","atime":
"2016-09-26T16:17:07.000Z","ctime":"2016-09-22T17:24:09.000Z","inserttime":
"2018-07-24T16:32:59.000Z","tier":"system","size":933683784,"qpart":0,"fkey":
"filesys1.university.edureflib3067092","project":null,"department":null,"backup":null,
"tag4":null,"tag5":null,"tag6":null,"tag7":null,"tag8":null,"tag9":null,"tag10":null,
"tag11":null,"tag12":null,"tag13":null,"tag14":null,"tag15":null,"tag16":null},
...
```


/db2whrest/v1/sql_query_async?<sql>: GET

Gets data from a database asynchronously with an SQL query in a GET command.

You can use any standard SQL query. You must include the SQL query as a parameter of the URL. To send the query in an HTTP message body, see /db2whrest/v1/sql_query_async: POST. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√[1]	X	X
[1]The search is restricted to documents that are tagged with collections to which the user ID has a datauser role assigned.				

The asynchronous endpoints are useful alternatives in situations where the operation takes a long time to complete or has a negative effect on the overall performance of the system. For more information, see the topic Asynchronous endpoints.

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/sql_query_async?<sql_statements> -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- CSV
- JSON

Examples

1. The following example shows how to get data from a database asynchronously with an SQL query in a GET command. The SQL query is included as a parameter of the URL.
   a. Submit the request. The endpoint responds by displaying the field `{"status": "work scheduled"}` to indicate that the operation is running asynchronously.

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/sql_query_async?
   select%20*%20from%20datasrv04%20where%20owner%3D%27root%27 -X GET {"status": "work scheduled"}
   ```

   The synchronous response to the request contains information about the asynchronously running task. The following is an example. For more information, see the topic Asynchronous endpoints.

   ```
   * HTTP 1.0, assume close after body
   < HTTP/1.0 202 ACCEPTED
   < Content-Type: text/html; charset=utf-8
   < Content-Length: 28
   < Location: https://data_cataloging_host/db2whrest/v1/task_status/25b2df40-8f0c-4e32-a2f9-999ff3b18ada
   < Server: Werkzeug/0.14.1 Python/3.4.5
   < Date: Thu, 15 Feb 2018 20:37:10 GMT
   ```

   b. Later, poll the URL that is provided in the `Location` field of the synchronous response header in Step (a).

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ task_status/25b2df40-8f0c-
   4e32-a2f9-999ff3b18ada
   ```

   c. When the operation is complete, the endpoint displays the rows of data that are returned in response to the SQL query in the request. In the following example only the first two rows of output are shown:

   ```
   0,reflib,INDEX,SCAN,MO1,,Spectrum Scale,reflib.university.edu,3067343,root,root,
   -rw-r--r--,root,9,10,/reflib/cellranger-2.0.0/refdata-cellranger-ercc92-1.2.0/star/,
   sjdbList.out.tab,tab,resdnt,NA,2016-11-14 19:40:10,2017-06-02 21:30:26,
   2017-06-02 21:30:26,2018-07-24 16:32:47,system,0,1,reflib.university.edureflib3067343,
   ,,,,,,,,,,,,,,,
   .......
   1,reflib,INDEX,SCAN,MO1,,Spectrum Scale,reflib.university.edu,3067333,root,root,
   -rw-r--r--,root,9,10,/reflib/cellranger-2.0.0/refdata-cellranger-ercc92-1.2.0/star/,
   chrLength.txt,txt,resdnt,NA,2016-11-14 19:40:10,2017-06-02 23:28:21,
   2017-06-02 21:30:26,2018-07-24 16:32:47,system,412,1,reflib.university.edureflib3067333,
   ,,,,,,,,,,,,,,,
   ...
   ```

/db2whrest/v1/sql_query_async: POST

Gets data from a database asynchronously with an SQL query in a POST command. You can use any standard SQL query. You must include the SQL query in the message body of the POST command.
To send the query as a parameter of the URL, see /db2whrest/v1/sql_query_async?<sql>: GET.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√[1]	X	X

[1]The search is restricted to documents that are tagged with collections to which the user ID has a datauser role assigned.

The asynchronous endpoints are useful alternatives in situations where the operation takes a long time to complete or has a negative effect on the overall performance of the system. For more information, see the topic Asynchronous endpoints.

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token> https://<data_cataloging_host>/db2whrest/v1/sql_query_async -X POST -d@<sql.dat>
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- CSV
- JSON

Examples

1. The following example shows how to get data from a database asynchronously with an SQL query in a POST command. The SQL query is sent in the POST message body. The output format is JSON.
   a. Write the record to be added to the database into a file named sql.dat.
   b. Submit the request:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/sql_query_async -X POST -
   d"select * from metaocean where owner='root'" -H "accept: application/json"
   ```

   The synchronous response to the request contains information about the asynchronously running task. The following is an example. For more information, see the topic /db2whrest/v1/sql_query_async?<sql>: GET.

```
* HTTP 1.0, assume close after body
< HTTP/1.0 202 ACCEPTED
< Content-Type: text/html; charset=utf-8
< Content-Length: 28
< Location: https://<data_cataloging_host>/db2whrest/v1/task_status/3e40cfeb-efc9-4091-8bc8-
b194ff49d728
< Server: Werkzeug/0.14.1 Python/3.4.5
< Date: Thu, 15 Feb 2018 20:37:10 GMT
```

c. Later, poll the URL that is provided in the `Location` field of the synchronous response header in Step (b):

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/task_status/ 25b2df40-8f0c-
4e32-a2f9-999ff3b18ada
```

d. When the operation is complete, the endpoint displays the rows of data that are returned in response to the SQL query in the request. In the following example only the first two rows of output are shown. The output format is JSON:


```
[{"filesystem":"reflib","operation":"INDEX","source":"SCAN","revision":"MO1","site":"",
"platform":"Spectrum Scale","cluster":"reflib.university.edu","inode":3067112,"owner":
"root","group":"root","permissions":"-r--r--r--","fileset":"root","uid":9,"gid":10,
"path":"\/reflib\/blast_db\/","filename":"other_genomic.75.tar.gz","filetype":"gz",
"migstatus":"resdnt","migloc":"NA","mtime":"2014-12-31T06:00:00.000Z","atime":
"2016-09-26T16:17:14.000Z","ctime":"2016-09-22T17:46:21.000Z","inserttime":
"2018-07-24T16:32:59.000Z","tier":"system","size":929541048,"qpart":0,"fkey":
"reflib.university.edureflib3067112","project":null,"department":null,"backup":
null,"tag4":null,"tag5":null,"tag6":null,"tag7":null,"tag8":null,"tag9":null,"tag10":
null,"tag11":null,"tag12":null,"tag13":null,"tag14":null,"tag15":null,"tag16":null},

{"filesystem":"reflib","operation":"INDEX","source":"SCAN","revision":"MO1","site":"",
"platform":"Spectrum Scale","cluster":"reflib.university.edu","inode":3067092,"owner":
"root","group":"root","permissions":"-r--r--r--","fileset":"root","uid":9,"gid":10,
"path":"\/reflib\/blast_db\/","filename":"other_genomic.65.tar.gz","filetype":"gz",
"migstatus":"resdnt","migloc":"NA","mtime":"2014-12-31T06:00:00.000Z","atime":
"2016-09-26T16:17:07.000Z","ctime":"2016-09-22T17:24:09.000Z","inserttime":
"2018-07-24T16:32:59.000Z","tier":"system","size":933683784,"qpart":0,"fkey":
"reflib.university.edureflib3067092","project":null,"department":null,"backup":
null,"tag4":null,"tag5":null,"tag6":null,"tag7":null,"tag8":null,"tag9":null,"tag10":
null,"tag11":null,"tag12":null,"tag13":null,"tag14":null,"tag15":null,"tag16":null},
...
]
```


/db2whrest/v1/task_status/<task_id>: GET and <task_id>/peek

Gets the status of the specified asynchronous task.

The following table shows which roles can access these two REST API endpoints:

Table 1. Access by role

Endpoint	Data admin	Data user	Collection Admin	Admin	Service user
/db2whrest/v1/task_status/<task_id>: GET	√	√	√	X	X
/db2whrest/v1/task_status/<task_id>/peek: GET	√	√	√	X	X

The peek variation of this endpoint returns the status of the asynchronous operation. The nonpeek version returns both the status of the operation and a response header that includes further information. For more information about using these endpoints, see the following examples, and also see the topic Asynchronous endpoints.

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/task_status/ <task_id> -X GET
```

or

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/task_status/ <task_id>/peek -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- CSV
- JSON

Examples

1. In this example, the user discovers that the asynchronous operation completes and finds the location URI of the results of the operation. For more information on these steps, see the topic Asynchronous endpoints. The user runs the `/db2whrest/v1/task_status/<task_id>: GET` endpoint on the location URI to get the results of the operation:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ task_status/9c0db090-bd33-41c5-
   969f-a1603ddf49ab
   ```

   The results are returned in the response. Only the first two rows of the results are shown in this example:

```
0,reflib,INDEX,SCAN,MO1,,Spectrum Scale,reflib.university.edu,3067343,root,root,
-rw-r--r--,root,9,10,/reflib/cellranger-2.0.0/refdata-cellranger-ercc92-1.2.0/star/,
sjdbList.out.tab,tab,resdnt,NA,2016-11-14 19:40:10,2017-06-02 21:30:26,2017-06-02
21:30:26,2018-07-24 16:32:47,system,0,1,reflib.university.edureflib3067343,
,,,,,,,,,,,,,,,
1,reflib,INDEX,SCAN,MO1,,Spectrum Scale,reflib.university.edu,3067333,root,root,
-rw-r--r--,root,9,10,/reflib/cellranger-2.0.0/refdata-cellranger-ercc92-1.2.0/star/,
chrLength.txt,txt,resdnt,NA,2016-11-14 19:40:10,2017-06-02 23:28:21,2017-06-02 21:30:26,
2018-07-24 16:32:47,system,412,1,reflib.university.edureflib3067333,
,,,,,,,,,,,,,,,
...
```

2. In this example, the user learns that the asynchronous operation is in progress and finds the location URI of the asynchronous operation. For more information on these tasks, see the topic Asynchronous endpoints. The user can do the following steps:
   a. Monitor the status of the asynchronous operation by running the `/db2whrest/v1/task_status/<task_id>/peek: GET` endpoint on the location URI of the asynchronous operation several times:

      ```
      curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/ task_status/9c0db090-bd33-
      41c5-969f-a1603ddf8888/peek
      ```

      In the first few tries the response to the peek endpoint indicates that the asynchronous operation is still running:

      ```
      {"status":work scheduled}
      ```

      On the final try the response to the peek endpoint indicates that the asynchronous operation completes:

      ```
      {"status":work scheduled}
      ```

   b. Find the location URI of the asynchronous operation. This location URI is already available in Step 2(a), where the peek endpoint is run on it.
   c. Get the location URI of the results of the operation. To accomplish this task, run the `/db2whrest/v1/task_status/<task_id>: GET` endpoint on the location URI of the asynchronous operation, which you obtained in Step 2(b). The response contains the location URI of the results of the operation.
   d. Get the results of the asynchronous operation. To accomplish this task, run the `/db2whrest/v1/task_status/<task_id>: GET` endpoint on the location URI of the results of the operation, which you obtained in Step 2(c):

      ```
      curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/task_status/9c0db090-bd33-
      41c5-969f-a1603ddf8888
      ```

      The results of the asynchronous operation are returned in the response. Only the first two rows of the results are shown in this example:

      ```
      0,reflib,INDEX,SCAN,MO1,,Spectrum Scale,reflib.university.edu,3067343,root,root,
      -rw-r--r--,root,9,10,/reflib/cellranger-2.0.0/refdata-cellranger-ercc92-1.2.0/star/,
      sjdbList.out.tab,tab,resdnt,NA,2016-11-14 19:40:10,2017-06-02 21:30:26,2017-06-02
      21:30:26,2018-07-24 16:32:47,system,0,1,reflib.university.edureflib3067343,
      ,,,,,,,,,,,,,,,
      1,reflib,INDEX,SCAN,MO1,,Spectrum Scale,reflib.university.edu,3067333,root,root,
      -rw-r--r--,root,9,10,/reflib/cellranger-2.0.0/refdata-cellranger-ercc92-1.2.0/star/,
      chrLength.txt,txt,resdnt,NA,2016-11-14 19:40:10,2017-06-02 23:28:21,2017-06-02 21:30:26,
      2018-07-24 16:32:47,system,412,1,reflib.university.edureflib3067333,
      ,,,,,,,,,,,,,,,
      ...
      ```

   For more information, see the topic Asynchronous endpoints.

/db2whrest/v1/summary_tables -X GET

Gets information on all the summary tables that are available in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	CollectionAdmin	Data user	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

1. The following example shows how to get information on all the summary tables that are available in the system.
   a. Submit the request:

      ```
      curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables -X GET
      ```

b. The following response is returned. The response data is in JSON format. The data has been manually reflowed to make it more readable.

```
{
  "duplicates": {
    "enabled": false,
    "last_duration": null,
    "last_updated_time": "2019-07-10T18:39:58.348Z",
    "schedule": {
      "hour": 0,
      "minute": "14"
    },
    "update_running": false,
    "update_start_time": null
  },
  "mrcapacity": {
    "enabled": true,
    "last_duration": "0:01:10",
    "last_updated_time": "2019-07-10T21:45:02.462Z",
    "schedule": {
      "minute": "*/5"
    },
    "update_running": true,
    “update_start_time": "2019-07-10_21:50:01”
  }
}
```

/db2whrest/v1/summary_tables/<table> -X GET

Gets information for a particular summary table available in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	CollectionAdmin	Data user	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables/<table> -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

1. The following example shows how to get information on a particular summary table that is available in the system.
   a. Submit the request:

```
curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables/mrcapacity
```

   b. The following response is returned. The response data is in JSON format. The data has been manually reflowed to make it more readable.

```
{
  "enabled": true,
  "last_duration": "0:01:10",
  "last_updated_time": "2019-07-10T21:45:02.462Z",
  "schedule": {
    "minute": "*/5"
  },
  "update_running": true,
  "update_start_time":  “2019-07-10T22:00:00.342Z”
}
```

/db2whrest/v1/summary_tables/<table> -X PUT

Change the summary table configuration for a particular summary table available in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	CollectionAdmin	Data user	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables/<table> -X PUT -d@data.json -H "Content-Type: application/json"
```

Supported request types and response formats

Supported request types:

- PUT

Supported response formats:

- JSON

Examples

1. The following example shows how to disable auto scheduling for a particular summary table that is available in the system.
    a. Submit the request:

    ```
    curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables/mrcapacity -X PUT -d'{"enabled": false}' -H "Content-Type: application/json"
    ```

    b. The following response is returned.

    ```
    {"status": "success"}
    ```

2. The following example shows how to enable auto scheduling for a particular summary table that is available in the system.
    a. Submit the request:

    ```
    curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables/mrcapacity -X PUT -d'{"enabled": true, "schedule": {"minute": "*/5"}}' -H "Content-Type: application/json"
    ```

    b. The following response is returned.

    ```
    {"status": "success"}
    ```

3. The following example shows how to change the configuration schedule for a particular summary table that is available in the system.
    a. Submit the request:

    ```
    curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables/mrcapacity -X PUT -d'{"enabled": true, "schedule": {"minute": "*/5"}}' -H "Content-Type: application/json"
    ```

    b. The following response is returned.

    ```
    {"status": "success"}
    ```

/db2whrest/v1/summary_tables/<table>/<action> -X PUT

Triggers a start or scheduled start action for a particular summary table.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	CollectionAdmin	Data user	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables/<table>/<action> -X PUT
```

Supported request types and response formats

Supported request types:

- PUT

Supported response formats:

- JSON

Examples

1. The following example shows how to start a table refresh on demand for a particular summary table that is available in the system.
    a. Submit the request:

    ```
    curl -u <user>:<pass> https://<master_node>/db2whrest/v1/summary_tables/mrcapacity/start -X PUT
    ```

b. The following response is returned.

```
{"status": "success"}
```

/db2whrest/v1/bulk_add_tags/docs: POST

Searches a database for data.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
✔	✔[1]	✔[1]	X	X
[1]The search is restricted to documents that are tagged with collections to which the user ID has a datauser role assigned.				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/bulk_add_tags/docs -X POST -
d@bulk_of_docs.json -H "Content-type: application/json"
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

1. The following example shows how to define search parameters and format the data that is returned:
   a. Step 1: Define the search parameters in a file named bulk_of_docs.json:

```
{
  "docs": {
      "fkey1": {
          "tags": {
            "tag1": "This is the value for tag1",
            "tag2": "This is the value for tag2",
            "tag3": "This is the value for tag3",
          }
      },
      "fkey2": {
          "tags": {
            "tag4": "This is the value for tag4",
            "tag5": "This is the value for tag5",
            "tag6": "This is the value for tag6",
          }
      },
  }
}
```

   b. Step 2: Submit the request:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/db2whrest/v1/bulk_add_tags/docs -X POST -
d@bulk_of_docs.json -H "Content-Type: application/json"
```

   The following response is returned:

```
(200 OK)
{"status": "success"}
```

Endpoints for working with policy management

With the policy management API service, you can create, list, update, and delete the policies.

A policy API consists of the following elements:

pol_id
: The name of the policy.

action_id
: The action to perform.

action_params
: The action contains the following elements:

  tags

The type of actions. Currently, the only type of action is the AUTOTAG action. For information about using the AUTOTAG action, see the topic Adding fields with the AUTOTAG action.

pol_filter
: A search filter to select a set of documents to work on.

schedule
: The schedule for processing the operation:

NOW
: The action is done immediately.

dayOfWeek, hour, month, timezone, minute, dayOfMonth
: The action is done at the specified time.
: Note: The API supports the creation of a schedule by using a timezone parameter. Internally, this information is converted to Coordinated Universal Time (UTC). It is recommended that the timezone parameter is not used or that the default of Coordinated Universal Time (UTC) is used in the API, because the policy schedule time is not displayed in the GUI. The GUI assumes that all times are in Coordinated Universal Time (UTC).

The policy engine runs the specified policy based on the schedule and the action ID that are specified in the API. The policy operations are applied to the documents that are specified by the filter.

- **/policyengine/v1/policies: GET and /policyengine/v1/policies/<policy_name>: GET**
  Lists the attributes of one or more policies.
- **/policyengine/v1/policies/<policy_name>/preview: GET**
  Previews a policy.
- **/policyengine/v1/policies/<policy_name>/status:GET**
  Gets policy status.
- **/policyengine/v1/policyhistory: GET**
  Retrieves policy history information for all the policies.
- **/policyengine/v1/policyhistory/<pol_id>/<log_id>: GET**
  Retrieves the policy run log for a policy execution.
- **/policyengine/v1/policies/<policy_name>/<action>: POST**
  Applies action to policy.
- **/policyengine/v1/policies -d '<data>': POST**
  Creates a policy.
- **/policyengine/v1/policies/<policy_name> -d '<data>': PUT**
  Updates a policy.
- **/policyengine/v1/policies/<policy_name>: DELETE**
  Deletes a policy.
- **/policyengine/v1/policyhistory: DELETE**
  Deletes the policy execution history and logs.
- **/policyengine/v1/tags: GET**
  Retrieves tag definitions.
- **/policyengine/v1/tags/: POST**
  Updates tags definitions.
- **/policyengine/v1/tags/: PUT**
  Adds one or more new tags to the list of existing definitions.
- **/policyengine/v1/tags/: DELETE**
  Deletes an existing tag definition.
- **/policyengine/v1/regex: POST**
  Registers the regular expression in the IBM Data Cataloging system.
- **/policyengine/v1/regex: GET**
  Retrieves list of all the regular expressions registered in the IBM Data Cataloging system.
- **/policyengine/v1/regex/<regex_name>: GET**
  Retrieves a single regular expression detail from the IBM Data Cataloging system.
- **/policyengine/v1/regex: PUT**
  Updates the regular expressions in the IBM Data Cataloging system.
- **/policyengine/v1/regex: DELETE**
  Deletes the regular expression with value from the IBM Data Cataloging system.
- **Adding fields with the AUTOTAG action**
  With the AUTOTAG action you can create policies that automatically add fields either to a specified set of documents or to all the documents in the system.

/policyengine/v1/policies: GET and /policyengine/v1/policies/<policy_name>: GET

Lists the attributes of one or more policies.

These two endpoints list the attributes either of a specified policy or of all the policies in the system. The following table shows which roles can access these two REST API endpoints:

Table 1. Access by role

Endpoints	Data admin	Data user	Collection Admin	Collection user	Admin	Service user
/policyengine/v1/policies: GET	√	√	√[1]	√	X	X
/policyengine/v1/policies/<policy_name>: GET	√	√	√[1]	√	X	X
[1] Collection Admin user can list, update, and delete policies applied to the collections to which they have the Collection Admin role assigned.						

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/ policies -X GET -H ’Accept: application/json’
```

or

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/ policies/<policy_name> -X GET -H
'Accept: application/json'
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

1. The following example returns information about all the policies that are configured in the system:
   a. Issue the following request on one line:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host> /policyengine/v1/policies -X GET -H
   'Accept: application/json'
   ```

   b. The following response is returned:

   ```
   [
      {
         "pol_id": "pol1",
         "action_id": "AUTOTAG",
         "action_params": {
            "tags": {
            "tag2": "val2",
            "tag3": "val3"
            }
         },
         "schedule": {
         "dayOfWeek": 4,
         "hour": 5,
         "minute": 15
         }
         "pol_state": "active"
         "pol_filter": "filetype='jpg'",
      },
   {
      "pol_id": "pol2",
         "action_id": "AUTOTAG",
         "action_params": {
            "tags": {
            "tag4": "val4",
            "tag5": "val5"
            }
         },
         "schedule": "NOW",
         "pol_state": "active"
         "pol_filter": "filetype='jpg'",
      }
   ]
   ```

2. The following example returns information about the specified policy:
   a. Issue the following request on one line:

   ```
   curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host> /policyengine/v1/policies/pol1 -X GET -H
   'Accept: application/json'
   ```

   b. The following response is returned:

   ```
   [
      {
         "pol_id": "pol1",
         "action_id": "AUTOTAG",
         "action_params": {
            "tags": {
            "tag2": "val2",
            "tag3": "val3"
            }
         },
         "schedule": {
         "dayOfWeek": 4,
         "hour": 5,
         "minute": 15
         }
         "pol_filter": "filetype='jpg'",
         "pol_state": "active"
      }
   ]
   ```

/policyengine/v1/policies/<policy_name>/preview: GET

Previews a policy.

This endpoint helps to display policy preview information for a specific policy. The following information is available for preview:

Estimation information
: Displays the total count and size of documents. The size of documents on the disk is calculated based on the list of documents the policy applies to and those that are returned.

Execution information
: Displays whether the policy has been previously executed or is currently running. The information returned also includes the number and size of documents completed, the start time and, the duration of the policy in seconds.

Policy schedule
: Displays the policy schedule, if any.

Policy run time
: Displays the future run date and time that has been scheduled for the policy.

The following table displays the roles that can access the REST API endpoint.

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√	√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host> /policyengine/v1/policies/policy1/preview -X GET -H 'Accept: application/json'
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns the policy preview information about a specific policy that is configured in the system.

Issue the following request in one line:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/ policies/<policy_name>/preview -X GET -H 'Accept: application/json'
```

The following response is returned:


```
{
  "schedule": {
    "minute": "18",
    "hour": "12",
    "dayOfMonth": "*",
    "month": "*",
    "dayOfWeek": "*",
    "timezone": "UTC"
  },
  "next_run_time": "2020-03-19 12:18PM",
  "estimation_info" : {
    "document_count": 152467,
    "total_size": 24935050549,
    "total_size_on_disk": 25610976130
   },
  "execution_info": {
    "start_time": "",
    "duration": 3000,
    "tier_count": 10345,
   "tier_size": 1000000,
   "failed_count": 0
  }
}
```


/policyengine/v1/policies/<policy_name>/status:GET

Gets policy status.

The status endpoint retrieves the policy status information for a specific policy.

The response includes the following information:

- total_count
- completed_count

- failed_count
- submitted_count
- start_time
- end_time

The response returned depends on three possible policy scenarios. These include Never Run, Running, and Finished. If the policy is never executed (Never Run), then the response only returns the policy status.

If the policy is running, the counts mentioned in the preceding list contain the current values and an end time is not returned.

If the policy is finished then the counts contain the final values and an end time is returned.

The following table displays the roles that can access the REST API endpoint.

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√	[1]√	[1]√	√	X	X
[1] Users with Collection Admin role can access this endpoint only if they are owners of the policy or can access one or more collections that are included in the policy. Users with Data User role can access the endpoint only if they are owners of the policy.					

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/ policies/<policy_name>/status -X GET
-H 'Accept: application/json'
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns policy status information about a specific policy.

Issue the following request in one line:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host> /policyengine/v1/policies/policy1/status -X GET -H
'Accept: application/json'
```

The following response is returned:

```
{
    "status": "running",
    "total_count": 15000,
    "completed_count": 14000,
    "failed_count": 0,
    "submitted_count": 15000,
    "start_time": "2020-03-27_11:12:10"
}
```

/policyengine/v1/policyhistory: GET

Retrieves policy history information for all the policies.

The /policyengine/v1/policyhistory retrieves the policy execution history details for all the policies that you have permission to view within the system. The entries contain detailed information on the specified policy execution. It also provides the location of the run log file.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√	√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/policyhistory -X GET -H 'Accept:
application/json'
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns the policy history information for all policies for which you have view permissions.

1. Issue the following request in one line.

```
curl -k -H 'Authorization: Bearer ${TOKEN}' https://<data_cataloging_host>/policyengine/v1/policyhistory -X GET
```

2. The following response is returned.

```
[
  {
    "id": 893909178,
    "policyid": "patent_pol",
    "state": "complete",
    "status": null,
    "starttime": "2020-09-22T20:34:32.000Z",
    "endtime": "2020-09-22T20:35:19.000Z",
    "logfile": "/policy_logs/patent_pol-2020-09-22 20:34:32/run.log",
    "startedby": null,
    "scheduled": null,
    "totalcount": 7,
    "failedcount": 0,
    "skippedcount": 0,
    "successcount": 7,
    "totalsize": null,
    "policyfilter": "path = '/mnt/datadump/patent_dataset/'",
    "type": "CONTENTSEARCH",
    "additional_info": "{\"explicit\": \"true\"}"
  },
  {
    "id": 3465755378,
    "policyid": "dicom_hdr_extract",
    "state": "complete with fail",
    "status": null,
    "starttime": "2020-09-22T21:30:59.000Z",
    "endtime": "2020-09-22T21:31:09.000Z",
    "logfile": "/policy_logs/dicom_hdr_extract-2020-09-22 21:30:59/run.log",
    "startedby": null,
    "scheduled": null,
    "totalcount": 26,
    "failedcount": 9,
    "skippedcount": 0,
    "successcount": 17,
    "totalsize": null,
    "policyfilter": "project='watson_health' and filetype='dcm'",
    "type": "CONTENTSEARCH",
    "additional_info": "{\"explicit\": \"true\"}"
  }
]
```

Response data

The response data can be explained as shown.

id
: The response returns the unique run id for the specified policy execution.

policy_id
: The response returns the name of the policy that is being run.

state
: The response returns the current or final state of the policy execution.

status
: The response returns the additional information about the run state (for example, a failure message).

starttime
: The response returns the date or timestamp when the policy execution was started.

endtime
: The response returns the date or timestamp when the policy execution completes if the policy is not still running.

logfile
: The response returns the log file identifier for this policy execution.

startedby
: This information field is reserved for future use.

scheduled
: This information field is reserved for future use.

totalcount
: The response returns the total number of records that meet the policy filter criteria.

failedcount
: The response returns the number of records that are marked as failed during the policy execution.

skippedcount
: The response returns the number of records that are marked as skipped during the policy execution.

successcount
: The response returns the number of records that are marked as successfully processed after the policy execution is completed.

totalsize
: This information field is reserved for future use.

policyfilter

The response returns the filter criteria that defined the set of records to be processed in the policy execution.

type
: The response returns the type of policy that is being run.

additional_info/explicit
: The response returns the Boolean value that displays *True* if the policy execution was for a user-defined policy or *False* if it was run for a system-generated policy.

/policyengine/v1/policyhistory/<pol_id>/<log_id>: GET

Retrieves the policy run log for a policy execution.

This endpoint helps you retrieve and view the policy run log information for all policy executions that you have access to. It provides you with a direct access to the policy execution log data without having to search for it on the file system.

The following table displays the roles that can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√	√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/policyhistory/<pol_id>/<log_id> -X
GET -H ’Accept: application/json’
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns the policy log execution details for the policies that you have access to:

1. Issue the following request in one line.

```
curl -k -H 'Authorization: Bearer ${TOKEN}'
https://<data_cataloging_host>/policyengine/v1/policyhistory/patent_pol/893909178 -X GET -H ’Accept: application/json’
```

2. The following response is returned.


```
[
  {
  "log": "[2020-09-22 20:34:32] - Execution beginning for policy (patent_pol)\n[2020-09-22 20:34:33] - Policy status
change to (running)\n[2020-09-22 20:34:33] - Policy stats update: {'pol_id': 'patent_pol', 'execution_info':
'{\"start_time\": \"2020-09-22 20:34:32\", \"total_count\": 0, \"submitted_count\": 0, \"failed_count\": 0,
\"completed_count\": 0, \"skipped_count\": 0, \"ctrl_submitted_count\": 0, \"ctrl_completed_count\": 0,
\"contentsearch_count\": 0, \"run_id\": null}'}\n[2020-09-22 20:34:33] - Policy stats update: {'pol_id': 'patent_pol',
'execution_info': '{\"start_time\": \"2020-09-22 20:34:32\", \"total_count\": 7, \"submitted_count\": 0, \"failed_count\":
0, \"completed_count\": 0, \"skipped_count\": 0, \"ctrl_submitted_count\": 0, \"ctrl_completed_count\": 0,
\"contentsearch_count\": 0, \"run_id\": null}'}\n[2020-09-22 20:34:35] - Applying action 'CONTENTSEARCH' to 7
documents\n[2020-09-22 20:34:38] - Policy stats update: {'pol_id': 'patent_pol', 'execution_info': '{\"start_time\":
\"2020-09-22 20:34:32\", \"total_count\": 7, \"submitted_count\": 7, \"failed_count\": 0, \"completed_count\": 0,
\"skipped_count\": 0, \"ctrl_submitted_count\": 0, \"ctrl_completed_count\": 0, \"contentsearch_count\": 0, \"run_id\":
\"ab0d0623dd844fd7b21276fce892d6ed\"}'}\n[2020-09-22 20:34:38] - Policy stats update: {'pol_id': 'patent_pol',
'execution_info': '{\"start_time\": \"2020-09-22 20:34:32\", \"total_count\": 7, \"submitted_count\": 7, \"failed_count\":
0, \"completed_count\": 0, \"skipped_count\": 0, \"ctrl_submitted_count\": 0, \"ctrl_completed_count\": 0,
\"contentsearch_count\": 0, \"run_id\": \"ab0d0623dd844fd7b21276fce892d6ed\"}'}\n[2020-09-22 20:34:45] - Policy stats
update: {'pol_id': 'patent_pol', 'execution_info': '{\"start_time\": \"2020-09-22 20:34:32\", \"total_count\": 7,
\"submitted_count\": 7, \"failed_count\": 0, \"completed_count\": 0, \"skipped_count\": 0, \"ctrl_submitted_count\": 0,
\"ctrl_completed_count\": 0, \"contentsearch_count\": 0, \"run_id\": \"ab0d0623dd844fd7b21276fce892d6ed\"}'}\n[2020-09-22
20:34:56] - Policy stats update: {'pol_id': 'patent_pol', 'execution_info': '{\"start_time\": \"2020-09-22 20:34:32\",
\"total_count\": 7, \"submitted_count\": 7, \"failed_count\": 0, \"completed_count\": 0, \"skipped_count\": 0,
\"ctrl_submitted_count\": 0, \"ctrl_completed_count\": 0, \"contentsearch_count\": 0, \"run_id\":
\"ab0d0623dd844fd7b21276fce892d6ed\"}'}\n[2020-09-22 20:35:06] - Policy stats update: {'pol_id': 'patent_pol',
'execution_info': '{\"start_time\": \"2020-09-22 20:34:32\", \"total_count\": 7, \"submitted_count\": 7, \"failed_count\":
0, \"completed_count\": 0, \"skipped_count\": 0, \"ctrl_submitted_count\": 0, \"ctrl_completed_count\": 0,
\"contentsearch_count\": 0, \"run_id\": \"ab0d0623dd844fd7b21276fce892d6ed\"}'}\n[2020-09-22 20:35:16] - Policy stats
update: {'pol_id': 'patent_pol', 'execution_info': '{\"start_time\": \"2020-09-22 20:34:32\", \"total_count\": 7,
\"submitted_count\": 7, \"failed_count\": 0, \"completed_count\": 0, \"skipped_count\": 0, \"ctrl_submitted_count\": 0,
\"ctrl_completed_count\": 0, \"contentsearch_count\": 0, \"run_id\": \"ab0d0623dd844fd7b21276fce892d6ed\"}'}\n[2020-09-22
20:35:19] - Policy stats update: {'pol_id': 'patent_pol', 'execution_info': '{\"start_time\": \"2020-09-22 20:34:32\",
\"total_count\": 7, \"submitted_count\": 7, \"failed_count\": 0, \"completed_count\": 7, \"skipped_count\": 0,
\"ctrl_submitted_count\": 0, \"ctrl_completed_count\": 0, \"contentsearch_count\": 0, \"run_id\":
\"ab0d0623dd844fd7b21276fce892d6ed\"}'}\n[2020-09-22 20:35:20] - Policy stats update: {'pol_id': 'patent_pol',
'execution_info': '{\"start_time\": \"2020-09-22 20:34:32\", \"total_count\": 7, \"submitted_count\": 7, \"failed_count\":
0, \"completed_count\": 7, \"skipped_count\": 0, \"ctrl_submitted_count\": 0, \"ctrl_completed_count\": 0,
\"contentsearch_count\": 0, \"run_id\": \"ab0d0623dd844fd7b21276fce892d6ed\", \"end_time\": \"2020-09-22
20:35:19\"}'}\n[2020-09-22 20:35:20] - Policy patent_pol run ending\n[2020-09-22 20:35:20] - Policy patent_pol run
completed\n[2020-09-22 20:35:20] - Policy status change to (complete)\n"
}
```

```
]
```

Response data

The response data can be explained as shown.

log
: The response returns the policy run log output details that is displayed as a string.

/policyengine/v1/policies/<policy_name>/<action>: POST

Applies action to policy.

This API endpoint helps you start, kill, pause, and resume a policy. You need to add the required action start, pause, resume, kill to the following path: /policyengine/v1/policies/{pol_id}/{action}.

The table shows the roles that can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√	[1] √	[1] √	X	X	X
[1] Users with Collection Admin role can access this endpoint only if they are owners of the policy or can access one or more collections that are included in the policy. Users with Data User role can access this endpoint only if they are owners of the policy.					

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/policies/policy_name/action -X POST -
H 'Content-type: application/json'
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

The following example shows how to start a policy by applying the **start** action.

Issue the following request on one line:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/pol_2/start -X POST -H 'Content-type:
application/json'
```

The following response is returned:

```
Accepted
```

/policyengine/v1/policies -d '<data>': POST

Creates a policy.

The **/policyengine/v1/policies: POST** endpoint creates a policy with the characteristics that are specified in the request data. For a description of the request parameters, see Endpoints for working with policy management. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Collection Admin	Collection user	Data user	Admin	Service user
√	√	X	√ [1, 2]	X	X
[1] The command allows access only if the schedule attribute is set to NOW. [2] The command applies the policy only in collections where the user has a datauser role.					

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/policies -d '<data>' -X POST -H
"Content-Type: application/json"
```

Supported request types, input fields, and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

1. The following example creates an AUTOTAG policy named `pol2` to start immediately:
    a. Issue the following request on one line:

    ```
    curl -k -H 'Authorization: Bearer <token>'
    https://<data_cataloging_host>/policyengine/v1/policies
    -d '{
       "pol_id":"pol2",
       "pol_filter": "user='research1'",
       "action_id": "AUTOTAG",
       "action_params": {"tags": {"tag4":"val4", "tag5":"val5"}},
       "schedule":"NOW"
    }'
     -X POST -H "Content-Type: application/json"
    ```

    b. The following response is returned:

    ```
    Policy 'pol2' added
    ```

2. The following example creates a AUTOTAG policy named `pol2` to start at a scheduled time:
   Note: For information about using the AUTOTAG action, see the topic Adding fields with the AUTOTAG action.
    a. Issue the following request on one line:

    ```
    curl -k -H 'Authorization: Bearer <token>'
    https://<data_cataloging_host>/policyengine/v1/policies
    -d '
    {
      "pol_id":"pol2",
      "action_id":"AUTOTAG",
      "action_params":{
          "tags": { "test_tag1": "test1" }
        },
      "pol_filter":"filename LIKE 'f0%'",
      "schedule": {
          "dayOfWeek": "*",
          "hour": "13",
          "month": "*",
          "timezone": "UTC",
          "minute": "31",
          "dayOfMonth": "*"
        }
    }'
     -X POST -H "Content-Type: application/json"
    ```

    Note: The API supports the creation of a schedule using a **timezone** parameter. Internally this is converted to Coordinated Universal Time (UTC). It is recommended that the **timezone** parameter is not used or that the default of Coordinated Universal Time (UTC) is used in the API, because the policy schedule time is not displayed in the GUI. The GUI assumes all times are in UTC.
    b. The following response is returned:

    ```
    Policy 'pol2' added
    ```

   To verify that the policy is created, issue a GET request to list the information about the specified policy or about all the policies in the system. For more information, see /policyengine/v1/policies: GET and /policyengine/v1/policies/<policy_name>: GET.

/policyengine/v1/policies/<policy_name> -d '<data>': PUT

Updates a policy.

The **/policyengine/v1/policies/*<policy_id>* -d '*<data>*': PUT** endpoint updates an existing policy with the attribute values that are specified in the request data. Attributes that are not updated keep the same values that they had before the update. The following attributes can be updated: `action_parameters`, `schedule`, `pol_filter`, and `pol_state`. The `action_id` attribute cannot be updated. However, you can delete an existing policy and then create a new one with the same name and attributes as the deleted policy but with a different `action_id`. For more information about the attributes in the request data, see Endpoints for working with policy management.

Note: You cannot update a policy while it is running.
The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√	√[1, 2]	X	X	X	X

[1]The command allows access only if the schedule attribute is set to NOW.
[2]The command applies the policy only in collections where the user has a datauser role.

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/policies/<policy_name> -d '<data>' -X
PUT -H "Content-Type: application/json"
```

Supported request types and response formats

Supported request types:

- PUT

Supported response formats:

- JSON

Examples

1. The following example updates a policy named pol2:
   a. Issue the following request on one line:

```
curl -k -H 'Authorization: Bearer \<token>'
https://<data_cataloging_host>
/policyengine/v1/policies/<policy_name>
-d '{
   "pol_filter": "user='research1'",
   "pol_state": "active"
   "action_params": {"tags": {"tag4":"val4", "tag5":"val5"}},
   "schedule":"NOW"
}'
 -X POST -H "Content-Type: application/json"
```

   b. The following response is returned:

```
Policy 'pol2' updated
```

   To verify that the policy is created, issue a GET request to list the information about the specified policy or about all the policies in the system. For more information, see /policyengine/v1/policies: GET and /policyengine/v1/policies/<policy_name>: GET.

/policyengine/v1/policies/<policy_name>: DELETE

Deletes a policy.

The `/policyengine/v1/policies/<policy_name>: DELETE` endpoint deletes the specified policy. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	X	X	X

Important: You cannot delete the `"Collection"` and `"Temperature"` tags.

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/policies/pol2 -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

1. The following example deletes a policy named pol2:
   a. Issue the following request on one line:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/policies/<policy_name> -X
POST
```

   b. The following response is returned:

```
Deleted policy pol2
```

   To verify that the policy is deleted, issue a GET request to list the information about the specified policy or about all the policies in the system. For more information, see /policyengine/v1/policies: GET and /policyengine/v1/policies/<policy_name>: GET.

/policyengine/v1/policyhistory: DELETE

Deletes the policy execution history and logs.

This endpoint helps you delete the policy execution history and logs of all policies that you can access.

The following table displays the roles that can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√	√	√	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/policyengine/v1/policyhistory -X DELETE -H ’Accept: application/json’ [-d ‘{“days_older_than”: <int>}’]
```

Note: You can specify an override to the default retention of 30 days in the JSON input data field, `{"days_older_than": <int>}`. When specified, it ensures that only the policy history records and log files that are older than the input value provided, are removed.

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

The following example deletes the policy history entries and run logs, for any policy executions that you have permission to view and that are older than 30 days.

1. Issue the following request in one line.

   ```
   curl -k -H 'Authorization: Bearer ${TOKEN}' https://<data_cataloging_host>/policyengine/v1/policyhistory -X DELETE -H ’Accept: application/json’
   ```

2. The following response is returned.

   ```
   [
     {"status": "Success. Deleted 12 policy history records."}
   ]
   ```

Response data

The response data can be explained as shown.

status
: The response returns an indication whether the request succeeded or failed. For a successful request, this information field also indicates the number of policy execution history entries and the corresponding log files that were purged.

/policyengine/v1/tags: GET

Retrieves tag definitions.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√	√	√	√	X	X

Synopsis of the request URL

```
$ curl -H "Authorization: Bearer <token>" -k https://<data_cataloging_host>:443/policyengine/v1/tags
```

Supported request types and response formats

Supported request types:

- GET

Note: When a request type is not specified, curl defaults to a GET operation.
Supported response formats:

- JSON

Examples

The following response is returned:

```
[
  {
    "tag": "project",
    "type": "Open",
    "value": "[]"
  },
  {
    "tag": "chargeBack",
    "type": "Restricted",
    "value": "[\"dept1\",\"dept2\",\"dept3\"]"
  },
  {
    "tag": "classification",
    "type": "Restricted",
    "value": "[\"public\",\"confidential\",\"internal\",\"restricted\",\"other\"]"
  },
  {
    "tag": "demotag",
    "type": "Open",
    "value": "[]"
  },
  {
    "tag": "instrument",
    "type": "Restricted",
    "value": "[\"spectrometer\",\"microscopeA\",\"microscopeB\"]"
  }
]
```

/policyengine/v1/tags/: POST

Updates tags definitions.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√[1]	√[2]	√	X	X	X

[1] Has read/write access to `'Open'`, `'Restricted'`, and `'Characteristics'` types of tags.

[2] Has read-only access to `'Open'` and `'Restricted'` types of tags, and read/write access to `'Characteristics'` tags.

To create an `'Open'` tag, run this command:

```
$ curl -k -H "Authorization: Bearer <token>" https://<data_cataloging_host>:443/policyengine/v1/tags -d '{"tag": "project", 
"type": "Open"}' -X POST -H "Content-Type: application/json"
```

Synopsis of the request URL

```
$ curl -k -H "Authorization: Bearer <token>"
https://<data_cataloging_host>:443/policyengine/v1/tags -d '{"tag": "department",
"type": "Restricted", "value": ["Finance", "HR", "IT"]' -X POST -H "Content-Type:
application/json"
```

```
$ curl -k -H "Authorization: Bearer <token>"
https://<data_cataloging_host>:443/policyengine/v1/tags -d '{"tag": "annotations",
"type": "Characteristics"}' -X POST -H "Content-Type: application/json"
```

Supported request types and response formats

Supported request types:

- POST
- Specify one of the following tag `"type"` options:

  ```
  "Open" | "Restricted" | "Characteristics"
  ```

Note:

- `"Open"` allows the unrestricted setting of tag values with a maximum string length of 256 characters.
- `"Restricted"` allows the definition of specified tag values with a maximum string length of 256 characters.
- `"Characteristics"` allows the definition of unspecified tag values with a maximum string length of 4096 characters.

Supported response formats:

- JSON

/policyengine/v1/tags/: PUT

Adds one or more new tags to the list of existing definitions.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection admin	Collection user	Admin	Service user
√[1]	√[2]	√	X	X	X

[1]READ/WRITE access to 'Open', 'Restricted', and 'Characteristics' types of tags.

[2]READ Only access to 'Open' and 'Restricted' types of tags, READ/WRITE access to 'Characteristics' tags.

Synopsis of the request URL

```
$ curl -H "Authorization: Bearer <token>" -k
https://<data_cataloging_host>:443/policyengine/v1/tags/<tag_name> -d '{"value":
["dept1", "dept2", "dept3", "dept4"]}' -X PUT -H "Content-Type: application/json"
```

Supported request types and response formats

Supported request types:

- PUT

Supported response formats:

- JSON

Note: Tag `payload` update requests are applied to the existing tag. Update tag requests overwrite the existing attributes of a tag. However, you cannot change the tag type.

/policyengine/v1/tags/: DELETE

Deletes an existing tag definition.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Collection user	Admin	Service user
√[1]	X[2]	X	X	X	X

[1] For read/write access to `'Open'`, `'Restricted'`, and `'Characteristics'` types of tags.

[2] For read-only access to `'Open'` and `'Restricted'` types of tags, and read/write access to `'Characteristics'` tags.

Important: You cannot delete the `"Collection"` and `"Temperature"` tags.

Synopsis of the request URL

```
$ curl -H "Authorization: Bearer <token>" -k https://<data_cataloging_host>:443/policyengine/v1/tags/<tag_name> -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

1. If a tag specified from the set of currently defined tags is "in-use" (its nondefault value is set in at least one document) the delete tag operation returns an error:

```
< HTTP/1.0 403 FORBIDDEN
< Content-Type: text/html; charset=utf-8
< Content-Length: 102
< Server: Werkzeug/0.14.1 Python/2.7.5
< Date: Wed, 09 May 2018 17:20:05 GMT
<
Cannot delete tag <tag_name> because it is in use in '38' records. Force deletion with
'force' option.
```

2. To force deletion of an in-use tag, use the `'force'` option as follows:

```
$ curl -H "Authorization: Bearer <token>" -k
https://<data_cataloging_host>:443/policyengine/v1/tags/<tag_name> -X DELETE -d
'{"force":"true"}' -H "Content-Type: application/json"
```

Important: The force option results in the deletion of the tag and its values are set by default to `NULL` or empty string.

/policyengine/v1/regex: POST

Registers the regular expression in the IBM Data Cataloging system.

The table shows the roles that can perform create operation on the endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>'
https://$SDHOST/policyengine/v1/regex -d @regexp.json -H "Content-type:application/json" -X POST
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

The following example shows how to register a regular expression.

Issue the following request on one line:

```
curl -k -H "Authorization: Bearer ${TOKEN}" https://$SDHOST/policyengine/v1/regex -d @regexp.json -H "Content-
type:application/json" -X POST
```

Example: Request JSON file

```
{
  "regex_id": "US-SSN-blank-delimited",
  "pattern": "\\b\\d{3}\\s\\d{2}\\s\\d{4}\\b",
  "description": "US SSN delimited by white space, not dashes."
}
```

The following response is returned:

```
201(Created)
```

/policyengine/v1/regex: GET

Retrieves list of all the regular expressions registered in the IBM Data Cataloging system.

The following table shows roles that can perform read operation on the endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://$SDHOST/policyengine/v1/regex -X GET -H 'Accept: application/json'
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns all the regular expressions registered for which you have read permission.

1. Issue the following request in one line.

   ```
   curl -k -H "Authorization: Bearer ${TOKEN}" https://$SDHOST/policyengine/v1/regex -X GET
   ```

2. The following response is returned.

   ```
   200(OK)
   ```

   ```
   [
   ```

```
  {
    "regex_id": "EmailID",
    "pattern": "\\b[\\w\\.=-]+@[\\w\\.-]+\\.[\\w]{2,3}\\b",
    "description": "Matching Email IDs like : John.Smith@example.com"
  },
  {
    "regex_id": "US-SSN",
    "pattern": "\\b\\d{3}-\\d{2}-\\d{4}\\b",
    "description": "Matching United States Social Security Numbers (SSN) like: 513-84-7329"
  },
  {
    "regex_id": "IPV4-Address",
    "pattern": "\\b\\d{1,3}[.]\\d{1,3}[.]\\d{1,3}[.]\\d{1,3}\\b",
    "description": "Matching IPV4 address like: 192.168.1.1"
  },
  {
    "regex_id": "Dates-MM/DD/YYYY",
    "pattern": "\\b(((0)[0-9])|((1)[0-2]))(\\/)([0-2][0-9]|(3)[0-1])(\\/)\\d{4}\\b",
    "description": "Matching dates in MM/DD/YYYY format like: 05/21/2019"
  },
  {
    "regex_id": "Dates-DD/MM/YYYY",
    "pattern": "\\b([0-2][0-9]|(3)[0-1])(\\/)(((0)[0-9])|((1)[0-2]))(\\/)\\d{4}\\b",
    "description": "Matching dates in DD/MM/YYYY format like: 15/10/2019"
  },
  {
    "regex_id": "MasterCard",
    "pattern": "\\b(?:5[1-5][0-9]{2}|222[1-9]|22[3-9][0-9]|2[3-6][0-9]{2}|27[01]
            [0-9]|2720)[0-9]{12}\\b",
    "description": "Matching MasterCard number like: 5258704108753590"
  },
  {
    "regex_id": "VisaCard",
    "pattern": "\\b([4]\\d{3}[\\s]\\d{4}[\\s]\\d{4}[\\s]\\d{4}|[4]\\d{3}[-]\\d{4}[-]\\d{4}[-]
            \\d{4}|[4]\\d{3}[.]\\d{4}[.]\\d{4}[.]\\d{4}|[4]\\d{3}\\d{4}\\d{4}\\d{4})\\b",
    "description": "Matching Visa Card numbers like: 4563-7568-5698-4587"
  },
  {
    "regex_id": "AmexCard",
    "pattern": "\\b3[47][0-9]{13}\\b",
    "description": "Matching American Express Card numbers like: 340000000000009"
  },
  {
    "regex_id": "USZIPCode",
    "pattern": "\\b((\\d{5}-\\d{4})|(\\d{5})|([A-Z]\\d[A-Z]\\s\\d[A-Z]\\d))\\b",
    "description": "Matching United States ZIP codes like: 97589"
  },
  {
    "regex_id": "URL",
    "pattern": "\\b((http[s]?|ftp):\\/)?\\/?([^:\\/\\s]+)((\\/\\w+)*\\/)
            ([\\w\\-\\.]+[^#?\\s]+)(.*)?(#[\\w\\-]+)?\\b",
    "description": "Matching URLs like: http://www.test.com/dir
                /filename.jpg?var1=foo#bar&var2=val2"
  },
  {
    "regex_id": "Geo-Coordinate",
    "pattern": "\\b([-+]?)([\\d]{1,2})(((.)(\\d+)(,)))(\\s*)(([-+]?)
            ([\\d]{1,3})((.)(\\d+))?)\\b",
    "description": "Matching Geo-Coordinates like: 51.498134, -0.201755"
  },
  {
    "regex_id": "CanadianSIN",
    "pattern": "\\b(\\d{3}[\\s]\\d{3}[\\s]+\\d{3})\\b|\\b(\\d{3}[-]\\d{3}[-]+\\d{3})\\b",
    "description": "Matching Canadian Social Insurance Number like: 123-456-789"
  },
  {
    "regex_id": "CreditCardExpirationDate",
    "pattern": "\\b\\d{2}\\/\\d{2}\\b",
    "description": "Matching Credit Card Expiration Date like 11/12"
  },
  {
    "regex_id": "CVV-Number",
    "pattern": "\\b([0-9]{3,4})\\b",
    "description": "Matching Credit Card Verification Value number like: 670, 0927"
  },
  {
    "regex_id": "Currency",
    "pattern": "\\b(\\d+(\\.\\d{2})?)\\b",
    "description": "Matching currency like: 123, 25.50"
  }
]
```


/policyengine/v1/regex/<regex_name>: GET

Retrieves a single regular expression detail from the IBM Data Cataloging system.

The following table shows the roles that can perform read operation on the endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	X	X

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer ${TOKEN}" https://$SDHOST/policyengine/v1/regex/$REGEXNAME -H "Content-type:application/json"
-X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns the single regular expression detail registered for which you have read permission.

1. Issue the following request in one line.

   ```
   curl -k -H "Authorization: Bearer ${TOKEN}" https://$SDHOST/policyengine/v1/regex/EmailID -H "Content-
   type:application/json" -X GET
   ```

2. The following response is returned.

   ```
   200(OK)
   ```

   ```
   {
           "regex_id": "EmailID",
           "pattern": "\\b[\\w\\.=-]+@[\\w\\.-]+\\.[\\w]{2,3}\\b",
           "description": "Matching Email IDs like : John.Smith@example.com"
   }
   ```

/policyengine/v1/regex: PUT

Updates the regular expressions in the IBM Data Cataloging system.

The table shows the roles that can perform update operation on the endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>'
https://$SDHOST/policyengine/v1/regex -d @regexp.json -H "Content-type:application/json" -X PUT
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

The following example shows how to update a regular expression.

Issue the following request on one line:

```
curl -k -H "Authorization: Bearer ${TOKEN}" https://$SDHOST/policyengine/v1/regex -d @regexp.json -H "Content-
type:application/json" -X PUT
```

Example: Request JSON payload

```
{
  "regex_id": "US-SSN-blank-delimited",
  "pattern": "\\b\\d{3}\\s\\d{2}\\s\\d{4}\\b",
  "description": "US SSN delimited by white space, not dashes."
}
```

The following response is returned:

```
200(OK)
Accepted
```

/policyengine/v1/regex: DELETE

Deletes the regular expression with value from the IBM Data Cataloging system.

The table shows the roles that can perform delete operation on the endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -s -o response.txt -H "Authorization: Bearer ${TOKEN}" https://${SDHOST}/policyengine/v1/regex/$REGEX_NAME -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

The following example shows how to delete a regular expression record.

Issue the following delete request:

```
curl -k -s -o response.txt -H "Authorization: Bearer ${TOKEN}" https://${SDHOST}/policyengine/v1/regex/US-SSN-blank-delimited -
X DELETE
```

The following response is returned:

```
204(No Content)
```

Note: When you perform DELETE operation, the resource gets deleted successfully but there is no response message in the output.

Adding fields with the AUTOTAG action

With the AUTOTAG action you can create policies that automatically add fields either to a specified set of documents or to all the documents in the system.

Two methods are available for adding new fields. One method is to add fields that contain initial values. The other method is to add fields that contain values that are extracted from existing fields of the same document. The following topics describe these two methods.

For more information about creating a policy, see the topic /policyengine/v1/policies -d '<data>': POST.

- **Adding fields that contain initial values**
  With the AUTOTAG action you can create a policy that automatically adds fields with initial values to a specified set of documents.
- **Adding fields that are extracted by rule from existing fields**
  Create a new field whose value is extracted from an existing field.

Adding fields that contain initial values

With the AUTOTAG action you can create a policy that automatically adds fields with initial values to a specified set of documents.

This method is useful when you want to add fields to a set of documents and you know beforehand the values that you want the fields to contain. For example, you might want to add a "billingaddress" field that contains the initial value "100 Corporation Street, Metropolis, USA" to every document that has a "username" field that contains the value "mjsmith". Or you might want to add an "ownerid" field that contains the initial value "admin02" to every document that lies in a certain directory path.

For more information about creating a policy, see the topic /policyengine/v1/policies -d '<data>': POST.

The following example shows how to create a policy that adds a project number and a department code to each document that contains a particular user name:

```
$ curl -k  -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/policyengine/v1/policies/autotagpol1
-d '{
"pol_filter":"user='research1'",
"action_id": "AUTOTAG",
"action_params": {
    "tags": {"ProjectNumber":"1404", "DepartmentCode":"H8AC"}
    }
```

```
"schedule":"NOW"
}'
-X POST -H "Content-Type: application/json"
```

The preceding example adds the fields "ProjectNumber":"1404" and "DepartmentCode":"H8AC" to every document in which the "user" field is set to 'research1'.

Adding fields that are extracted by rule from existing fields

Create a new field whose value is extracted from an existing field.

With the **AUTOTAG** action you can create a policy that adds a field to a document, where the new field contains a value that is extracted by rule from an existing field in the same document. Two types of rule are available. With the first type of rule, you can extract a value from an existing field by parsing tokens in the field value that are separated by a specified delimiter. With the second type of rule, you can extract a value from an existing field by searching the field value with a regular expression.
Note: The new field can contain a maximum of 256 characters. If the extracted value contains more than 256 characters, it is trimmed to 256 characters before it is assigned to the new field.
In the following two examples, the documents contain a **"filepathActiveUsers"** field in which the parts of the path are separated by forward slashes (/) and in which the second part of the path contains the project name. For example:

```
"filepathActiveUsers":"'/Lab1/ProjectHA48/users/active'"
```

You want to add a field **"projectname"** that contains the project name, and you want to add the field to all documents that contain the field **"user":"'research1'"**.

1. The first example shows how to extract the project name by parsing the path into tokens that are separated by a forward slash. Follow these steps:
   a. Create a file autotag_rule_pol_split.dat that contains the parameters for the new policy. The following code block shows the contents of the file:

   ```
   {
       "action_id": "AUTOTAG",
       "action_params":
       {
           "rule" : {
               "name": "setFromExistingField",
               "action": "split",
               "existingField": "filepathActiveUsers",
               "delimiter": "/",
               "fieldNo": 2,
               "newField": "projectname"
           }
       },
       "schedule": "NOW",
       "pol_filter": "user='research1'
   }
   ```

   b. Create the policy and apply it:

   ```
   curl -k -H 'Authorization: Bearer <token>'
   https://<data_cataloging_host>/policyengine/v1/policies/autotagpol2
   -d @autotag_rule_pol_split.dat
   -X POST
   -H "Content-Type: application/json"
   ```

   The policy extracts the second token, the project name, from the path and assigns the value to the new field **"projectname"**.

2. The second example shows how to extract the project name from the path with a regular expression. Follow these steps:
   a. Create a file autotag_rule_pol_regex.dat that contains the parameters for the new policy. The following code block shows the contents of the file:

   ```
   {
       "action_id": "AUTOTAG",
       "action_params": {
           "rule" : {
               "name": "setFromExistingField",
               "action": "regex",
               "existingField": "filepathActiveUsers",
               "regexPattern": "[^/]+",
               "matchNo": 3,
               "newField": "projectname"
           }
       },
       "schedule": "NOW",
       "pol_filter": "user='research1'"
   }
   ```

   b. Create the policy and apply it:

   ```
   curl -k -H 'Authorization: Bearer <token>'
   https://<data_cataloging_host>/policyengine/v1/policies/autotagpol2
   -d @autotag_rule_pol_regex.dat
   -X POST
   -H "Content-Type: application/json"
   ```

   The policy extracts the third match to the regex pattern, the project name, from the path and assigns the value to the new field **"projectname"**.

Endpoints for working with connection management

With the connection management API service you can create, update, delete, and get information about data source connection entries in the connection table.

A data source connection contains the following attributes:

name
: Indicates the name of the connection.

type
: Indicates the type of the connection.

cluster
: Indicates the cluster in which the data source is located.

datasource
: Indicates the file system or vault in which the data is stored.

site
: Indicates the location in which the cluster is maintained.

Note: To determine cluster and data source values for a connection in an IBM Spectrum Scale cluster, follow these steps:

1. Open a console on a node in the IBM Spectrum Scale cluster.
2. Run the following command to get information about the cluster:

   ```
   /usr/lpp/mmfs/bin/mmlscluster
   ```

   The command output displays information about the cluster similar to the following example:

   ```
   GPFS cluster information
   ========================
     GPFS cluster name:         modevvm19.metro.labs.cpr.com
     GPFS cluster id:           7146749509622277333
     GPFS UID domain:           modevvm19.metro.labs.cpr.com
     Remote shell command:      /usr/bin/ssh
     Remote file copy command:  /usr/bin/scp
     Repository type:           CCR

    Node  Daemon node name              IP address   Admin node name               Designation
   -------------------------------------------------------------------------------------------
      1   modevvm19.metro.labs.cpr.com  9.11.201.74  modevvm19.metro.labs.cpr.com quorum-manager
   ```

3. Make a note of the IBM Spectrum Scale cluster name, which in the preceding example is `modevvm19.metro.labs.cpr.com`. Use this value as the `cluster` attribute for the connection.
4. Run the following command to get information about all the mounted file systems in the cluster:

   ```
   /usr/lpp/mmfs/bin/mmlsmount all
   ```

   The command output displays information about mounted file systems as in the following example:

   ```
   File system gpfs0 is mounted on 1 nodes.
   ```

5. Make a note of the name of the file system that you want to create a connection for. For example, in the preceding example only one file system is mounted and its name is `gpfs0`. Use this value as the `datasource` attribute for the connection.

- **/connmgr/v1/connections: GET and /connmgr/v1/connections/<connection_name>: GET**
  Lists the attributes of one or more connections.
- **/connmgr/v1/scan/<connection>/partitions: GET**
  Gets the list of fileset (IBM Storage Scale) or Shares (SMB/CIFS) on the target system.
- **/connmgr/v1/connections -d '<data>': POST**
  Creates a connection.
- **/connmanager/v1/scan/<connection>/partial: POST**
  Initiates a partial scan of the datasource connection.
- **/connmgr/v1/connections/<connection_name> -d '<data>': PUT**
  Updates a connection.
- **/connmgr/v1/connections/<connection_name>: DELETE**
  Deletes a connection.
- **/connmgr/v1/scan/<connection_name>: POST**
  Starts the scan of a connection.
- **/connmgr/v1/scan/<connection_name>: GET**
  Gets the current scan status of a single connection.
- **/connmgr/v1/scan: GET**
  Gets the scan status of all running scans.
- **/connmgr/v1/scan/<connection_name>: PUT**
  Stops a scan.

/connmgr/v1/connections: GET and /connmgr/v1/connections/<connection_name>: GET

Lists the attributes of one or more connections.

These two endpoints list the attributes either of a specified connection or of all the connections in the connection table. The following table shows which roles can access these two REST API endpoints:

Table 1. Access by role

Endpoints	Data admin	Data user	Collection Admin	Admin	Service user
/connmgr/v1/connections: GET	√	X	√	X	X
/connmgr/v1/connections/<connection_name>: GET	√	X	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/connmgr/v1/ connections -X GET
```

or

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/connmgr/v1/ connections/<connection_name> -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON
- text/plain

Examples

1. The following example returns information about all the connections in the connection table:
   a. Run the following request on one line:

   ```
   curl -k -H 'Authorization: Bearer <token>'
   https://<data_cataloging_host>/connmgr/v1/connections -X GET -H 'Accept: application/json'
   ```

   The following response is returned. The response lists the information for all the connections in the connections table. For example, this table contains two connections, `con01` and `con02`:

   ```
   [
      {
         "name": "con01",
         "type": "general",
         "cluster": "modevvm19.metro.labs.cpr.com",
         "datasource": "gpfs0",
         "site":"datasite03"
      }
      {
         "name": "con02",
         "type": "general",
         "cluster": "modevvm19.metro.labs.cpr.com",
         "datasource": "gpfs1",
         "site":"datasite03"
      }
   ]
   ```

2. The following example returns information about connection `con02`:
   a. Run the following request on one line:

   ```
   curl -k -H 'Authorization: Bearer <token>'
   https://<data_cataloging_host>/connmgr/v1/connections/con02 -X GET -H 'Accept: application/json'
   ```

   The following response is returned:

   ```
   [
      {
         "name": "con02",
         "type": "general",
         "cluster": "modevvm19.metro.labs.cpr.com",
         "datasource": "gpfs1",
         "site":"datasite03"
      }
   ]
   ```

/connmgr/v1/scan/<connection>/partitions: GET

Gets the list of fileset (IBM Storage Scale) or Shares (SMB/CIFS) on the target system.

This API endpoint fetches the list of filesets (IBM Storage Scale) or shares (SMB/CIFS) on the target system. Only an IBM Data Cataloging user with the data admin role can get the list of partitions.

The following table displays the roles that can access the REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	[1]√	X	X
[1] The Collection Admin user, can list the partitions for connections assigned to the collections which are assigned to them.				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/connmgr/v1/scan/<connection_name>/partitions -X GET
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

1. The following example returns partition (fileset) information from an IBM Storage Scale connection:
   - Run the following request on one line:

     ```
     curl -k -H 'Authorization: Bearer <token>'
     https://< data_cataloging_host>/connmgr/v1/scan/ss1 -X GET -H 'Accept: application/json'
     ```

   ```
   ["root", "fileset1", "fileset2"]
   ```

2. The following example returns partition (share) information from an SMB/CIFS connection:
   Run the following request on one line:

   ```
   curl -k -H 'Authorization: Bearer <token>'
   https://<data_cataloging_host>/connmgr/v1/scan/smb1 -X GET -H 'Accept: application/json'
   ```

   ```
   ["root", "dir1"]
   ```

/connmgr/v1/connections -d '<data>': POST

Creates a connection.

The /connmgr/v1/connections -d *'<data>'*: POST endpoint creates a new connection entry in the connections table. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/connmgr/v1/
connections/ -d '<data>' -X POST -H "Content-Type: application/json"
```

Supported request types, input fields, and response formats

Supported request types:

- POST

Supported input fields:
JSON input is expected. Use a **Content type: application/json** header in the HTTP protocol request.

name
: The unique name of this connection.

cluster
: The cluster ID of the IBM Storage Scale cluster or IBM Cloud® Object Storage system UUID.

platform
: ["Spectrum Scale" | "IBM® COS" | "NFS" | "S3" | "IBM Storage Protect"] | "SMB/CIFS"

datasource
: Name of individual filesystem or vault. (For example, `scratch`)

site
: Physical location of data from this connection. This field is optional.

host
: The hostname or IP address of the interface node for scanning. This field is not mandatory for creating a connection, but is required for automated scan support.

mount_point
: The network path to be scanned. This field is only used for NFS and SMB/CIFS connections. For any other connection types, it is an optional field. For NFS, the 'mount_point' is the export path, for example, /export1/dir1. For SMB/CIFS, the 'mount_point' is the network share path, for example, \\SMBSERVER\share1\.

user
: The username to connect to the source data system with. This field is not mandatory for creating a connection but is required for automated scan support.

password
: The password on the source data system belonging to the user ID specified earlier. This field is not mandatory for creating a connection, but is required for automated scan support for connections other than NFS.

additional_info
: Connection-type specific information. This field is not mandatory for creating a connection, but is required for automated scan support. This information varies by platform as follows:

  IBM Cloud Object Storage (IBM COS)

  accesser_address
  : The hostname or IP address of the IBM COS accessor system.

accesser_access_key
: the access key for the corresponding accessor address specified by the `accesser_address`.

accesser_secret_key
: The accessor's secret key, corresponding to the IBM COS `accesser_address`.

manager_username
: User name of the manager for the IBM COS system.

manager_password
: Password of the IBM COS manager user name specified earlier.

SMB/CIFS

auth_type
: Denotes SMB authentication type parameter.

IBM Storage Scale

working_dir
: Directory on the IBM Storage Scale system to be used for the IBM Data Cataloging files used during scans of the IBM Storage Scale file system. This directory must exist before the connection is established.

nodes
: Specify a list of one or more nodes in the IBM Storage Scale cluster or the keyword `all` to enable the IBM Data Cataloging scans to execute across all nodes. The node names must match the names used by the IBM Storage Scale cluster configuration. For example, the nodes shown by the **mmlscluster** command executed on a node.

scan_dir
: The directory path within the IBM Storage Scale file system that is to be scanned. For example, `/scale/zoo`.

NFS

NFSv4
: Denotes NFS protocol.

S3

access_key
: Access key for the S3 data connection.

secret_key
: The secret key corresponding to this S3 data connection.

IBM Storage Protect

port
: The Open Database Connector (ODBC) port for the IBM Storage Protect server. This field is set to `51500` by default.

Supported response formats:

- JSON

Examples

1. The following example creates a connection that is named con01:
   a. Issue the following request on one line:

   ```
   curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/connmgr/v1/
   connections -d '{"name":"con01","platform":"Spectrum Scale", "cluster":
   "modevvm19.metro.labs.cpr.com", "datasource": "gpfs1", "site":"datasite03"}' -X POST
   -H "Content-Type: application/json"
   ```

   b. The following response is returned:

   ```
   Connection 'con01' added
   ```

   To verify that the connection is created, issue a GET request to list the information about the specified connection or about all the connections in the connection table. For more information, see /connmgr/v1/connections: GET and /connmgr/v1/connections/<connection_name>: GET.
2. The following examples demonstrates to create a connection using SMB parameter `auth_type` and NFSv4 protocol:
   - POST request using SMB parameter `auth_type`

   ```
   curl -s -k -H "Authorization: Bearer <token> -H "Content-Type: application/json" -X POST
   http://<data_cataloging_host>/connmgr/v1/
   connections -d'{"name":"perffs","platform":"Spectrum Scale","cluster":"gkscale4app-nd-
   1.fyre.ibm.com","datasource":"perffs","site":"svl","host":"host_ip","user":"root","password":"P@ssw0rd","additional_i
   nfo":"{"working_dir":"/gpfs/perffs","nodes":"all","scan_dir":"/gpfs/perffs","auth_type":"password"}"}'
   ```

   - POST request using NFS protocol `NFSv4`

   ```
   curl -s -k -H "Authorization: Bearer <token> -H "Content-Type: application/json" -X POST
   <data_cataloging_host>/connmgr/v1/
   connections -
   d'{"name":"nfs4test","platform":"NFS","cluster":"cluster1","datasource":"nfs4test","site":"Tucson","host":"
   <host_ip>","mount_point":"/ifs/sdiscover/Data/nfs4test","protocol":"nfs"}'
   ```

/connmanager/v1/scan/<connection>/partial: POST

Initiates a partial scan of the datasource connection.

This API endpoint initiates a partial scan of the datasource connection by specifying a list of one or more partitions to be scanned.

The table shows the roles that can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	[1]√	X	X
[1]Collection Admin users can initiate a partial scan of connections for collections to which they have been assigned.				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/connmanager/v1/scan/connection/partial -X POST -H 'Content-type: application/json'
```

Supported request types and response formats

Supported request types:

- POST

Supported input fields:

- JSON input in the following form:

  ```
  {"partitions": ["partition1", "partition2", "etc."]}
  ```

- JSON input in the following form for NFS partial scanning:

  ```
  {"partitions": [{"dir": "/exports/example/a", "recursive": true}, {"dir": "/exports/example/b", "recursive": false}]}
  ```

  where each subdirectory to scan is an item of the array with the following structure:

  ```
  {
      "dir": "/exports/example/a", // Directory path that starts with the mount point of the NFS connection
      "recursive": false // true if directories inside of the provided path should be scanned recursively
  }
  ```

Supported response formats:

- JSON
  Returns the status of whether or not the scan was started successfully. Error messages are displayed if JSON is malformed or if a specified partition is not found on the filesystem.

Examples

Issue the following request in one line:

```
curl -k -H 'Content-Type: application/json' -X POST
https://<data_cataloging_host>/connmgr/v1/scan/smb1/partial -H "Accept:application/json" -H "Content-type: application/json"
-X POST -d'{"partitions": ["dir1"]}
```

Issue the following NFS partial scan request:

```
curl -k -H 'Content-Type: application/json' -X POST
https://<data_cataloging_host>/connmgr/v1/scan/nfs-1/partial -H "Accept:application/json" -H "Content-type: application/json"
-X POST -d'{"partitions":[{"dir":"/exports/data","recursive":true}]}'
```

The following response is returned:

```
{"status": "Success"}
```

/connmgr/v1/connections/<connection_name> -d '<data>': PUT

Updates a connection.

The /connmgr/v1/connections/*<connection_name>* -d '*<data>*': PUT endpoint updates an existing connection with the attribute values that are specified in the request data. Attributes that are not updated keep the same values that they had before the update. The following attributes can be updated: `name`, `type`, `cluster`, `datasource`, and `site`. For more information about the attributes in the request data, see Endpoints for working with connection management. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/connmgr/v1/ connections/<connection_name> -d '<data>'
-X PUT -H "Content-Type: application/json"
```

Supported request types, input fields, and response formats

Supported request types:

- PUT

Supported input fields:
JSON input is expected. Use a **Content type: application/json** header in the HTTP protocol request.

name
: The unique name of this connection.

cluster
: The cluster ID of the IBM Storage Scale cluster or IBM Cloud® Object Storage system UUID.

platform
: ["Spectrum Scale" | "IBM® COS" | "NFS" | "S3" | "IBM Storage Protect"].

datasource
: Name of individual file system or vault. For example, `scratch`.

site
: Physical location of data from this connection. This field is optional.

host
: The hostname or IP address of the interface node for scanning. This field is not mandatory for creating a connection, but is required for automated scan support.

user
: The username to connect to the source data system with. This field is not mandatory for creating a connection, but is required for automated scan support.

password
: The password on the source data system belonging to the user ID specified above. This field is not mandatory for creating a connection, but is required for automated scan support.

additional_info
: Connection-type specific information. This field is not mandatory for creating a connection, but is required for automated scan support.

Supported response formats:

- JSON

Examples

1. The following example updates connection `con01` to `con02`:
   a. Issue the following request on one line:

   ```
   curl -k -H 'Authorization: Bearer <token>'https://<data_cataloging_host>/connmgr/v1/
   connections/con01 -d'{"name":"con02","platform":"Spectrum Scale","cluster":
   "modevvm19.metro.labs.cpr.com","datasource":"gpfs1", "site":"datasite03"}' -X PUT
   -H "Content-Type:application/json"
   ```

   b. The following response is returned:

   ```
   Connection 'con02' updated
   ```

   To verify that the policy is updated, issue a GET request to list the information about the specified connection or about all the connections in the system. For more information, see /connmgr/v1/connections: GET and /connmgr/v1/connections/<connection_name>: GET.

/connmgr/v1/connections/<connection_name>: DELETE

Deletes a connection.

The /connmgr/v1/connections/*<connection_name>*: DELETE endpoint deletes the specified connection from the connection table. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/connmgr/v1/ connections/<connection_name> -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

1. The following example deletes a connection named con02:
    a. Issue the following request on one line:

    ```
    curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/connmgr/v1/connections/con02 -X DEL
    ```

    b. The following response is returned:

    ```
    Connection 'con02' deleted
    ```

    To verify that the connection is deleted, issue a GET request to list the information about the specified connection or about all the connections in the connection table. For more information, see /connmgr/v1/connections: GET and /connmgr/v1/connections/<connection_name>: GET.

/connmgr/v1/scan/<connection_name>: POST

Starts the scan of a connection.

The /connmgr/v1/scan/*<connection_name>*: POST endpoint starts the scan of a connection. Only an IBM Data Cataloging user with data admin role or collection admin role can start a scan. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	√[1]	X	X
[1] Collection Admin user can start, stop, and get the status of the scans that are applied to the collections to which they have the Collection Admin role assigned.				

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <token>" -X POST https://<data_cataloging_host>/connmgr/v1/scan/<connection_name>
```

Supported request types, input fields, and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

Follow these steps to start a scan:

1. Obtain an auth token by using the credentials of the data admin user as shown:

    ```
    curl -k https://<data_cataloging_host>/auth/v1/token -u "<user_name>:<password>
    ```

    For a valid user, the auth token is returned in the `X-Auth-Token` response header.

2. Start the scan of a connection with name `sdconnection` using the following request:

    ```
    curl -k -H "Authorization: Bearer <token>" -X POST https://<data_cataloging_host>/connmgr/v1/scan/sdconnection
    ```

Response

The following table displays the common response statuses:

Table 2. Common response statuses

Status code	Message	Description
400	Must specify a connection name for the scan endpoint.	Connection name in the URL is not specified.
	Scan already currently in-progress for *<connection_name>*.	Scan already in progress.
404	Could not locate connection document for *<connection_name>*. Aborting scan.	Connection with the given name does not exist.
403	User does not have permission to view connection.	User does not have access to the collection to which the connection belongs.
200	-	Scan successfully started.

Note: A user with a collection admin role can only start a scan for a connection if an associated collection is set, and if it is a collection the collection admin user administers.
A successful scan displays the following response:

```
"Status": "Success"
```

The following displays an error response example:

```
"status": "Connection name in URL not specified."
```

/connmgr/v1/scan/<connection_name>: GET

Gets the current scan status of a single connection.

The /connmgr/v1/scan/*<connection_name>*: GET endpoint gets the current scan status of a single connection. Only an IBM Data Cataloging user with data admin role or collection admin role can get the status of a scan. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	√[1]	X	X
[1] Collection Admin user can start, stop, and get the status of the scans that are applied to the collections to which they have the Collection Admin role assigned.				

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <token>" -X GET https://<data_cataloging_host>/connmgr/v1/scan/<connection_name>
```

Supported request types, input fields, and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

Follow these steps to get the scan status of a connection:

1. Obtain an auth token by using the credentials of the data admin user as shown:

   ```
   curl -k https://<data_cataloging_host>/auth/v1/token -u "<user_name>:<password>
   ```

   For a valid user, the auth token is returned in the **X-Auth-Token** response header.

2. Get the scan status of a connection with name **sdconnection** using the following request:

   ```
   curl -k -H "Authorization: Bearer <token>" -X GET https://<data_cataloging_host>/connmgr/v1/scan/sdconnection
   ```

Response

The following table displays the common response statuses:

Table 2. Common response statuses

Status code	Message	Description
200	Scan status JSON.	Scan status successfully retrieved.

Note: A user with a collection admin role can only get the scan status for a connection if an associated collection is set, and if it is a collection the collection admin user administers. Therefore, in the response, a collection admin user is only presented with the status of connections that belong to the connections the collection admin user administers.

A successful scan displays the following response:

```
[
{
"Name":"sdconnection",
"status":"Running",
"message":"Crawling NFS mount",
"Phase":2,
"Total_phases":3
},
{
"Name":"ss",
"status":"Running",
"message":"Preparing to scan connection",
"Phase":1,
"total_phases":4
}
]
```

/connmgr/v1/scan: GET

Gets the scan status of all running scans.

The /connmgr/v1/scan endpoint gets the current scan status of all connections. Only an IBM Data Cataloging user with data admin role or collection admin role can get the status of all running scans. The following table shows which roles can access /connmgr/v1/scan endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user

Data admin	Data user	Collection Admin	Admin	Service user
√	X	√[1]	X	X
[1] Collection Admin user can start, stop, and get the status of the scans that are applied to the collections to which they have the Collection Admin role assigned.				

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <token>" -X GET https://<data_cataloging_host>/connmgr/v1/scan
```

Supported request types, input fields, and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

Follow these steps to get the scan status of all the running scans:

1. Obtain an auth token by using the credentials of the Data Admin user as shown:

   ```
   curl -k https://<data_cataloging_host>/auth/v1/token -u "<user_name>:<password>
   ```

   For a valid user, the authentication token is returned in the **X-Auth-Token** response header.
2. Get the scan status of all the running scans by using the following request:

   ```
   curl -k -H "Authorization: Bearer <token>" -X GET https://<data_cataloging_host>/connmgr/v1/scan
   ```

Response

The following table displays the common response statuses:

Table 2. Common response statuses

Status code	Message	Description
404	No scan is in progress for *<connection_name>*.	No scan is in progress for the specified connection.
403	User does not have permission to get scan status for connection *<connection_name>*.	User cannot access the collection to which the connection belongs.
200	Scan status JSON.	Scan status successfully retrieved.

Note: A user with a collection admin role can get only the current scan status for a connection if an associated collection is set, and if it is a collection the Collection Admin user administers.

A successful scan displays the following response:

```
{
"Name":"sdconnection",
"status":"Running",
"message":"Crawling NFS mount",
"Phase":2,
"total_phases":3
}
```

The following displays an error response example:

```
{
        "status": "Connection name in URL not specified."
}
```

/connmgr/v1/scan/<connection_name>: PUT

Stops a scan.

The /connmgr/v1/scan/*<connection_name>*: PUT endpoint stops a running scan. Only an IBM Data Cataloging user with data admin role or collection admin role can stop a scan. The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	√[1]	X	X
[1] Collection Admin user can start, stop, and get the status of the scans that are applied to the collections to which they have the Collection Admin role assigned.				

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <token>" -X PUT https://<data_cataloging_host>/connmgr/v1/scan/<connection_name>/stop
```

Supported request types, input fields, and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

Follow these steps to stop a scan:

1. Obtain an auth token by using the credentials of the data admin user as shown:

   ```
   curl -k https://<data_cataloging_host>/auth/v1/token -u "<user_name>:<password>
   ```

   For a valid user, the auth token is returned in the `X-Auth-Token` response header.

2. Stop the scan using the following request:

   ```
   curl -k -H "Authorization: Bearer <token>" -X PUT https://<data_cataloging_host>/connmgr/v1/scan/sdconnection/stop
   ```

Response

The following table displays the common response statuses:

Table 2. Common response statuses

Status code	Message	Description
400	Must specify a connection name for the scan endpoint.	Connection name in the URL is not specified.
	Must specify an action for the scan endpoint. Available action endpoints are: `/pause`, `/resume`, and `/stop`.	The stop action is not specified in the URL.
	No scan is currently in progress for *<connection_name>*.	No scans currently in progress for the given connection.
	Unsupported action `<action specified in url>`.	Unsupported action specified.
404	Could not locate connection document for *<connection_name>*.	Connection with the given name does not exist.
403	User does not have permission to control connection.	User does not have access to the collection to which the connection belongs.
200	-	Scan successfully stopped.

Note: A user with a collection admin role can only stop a scan for a connection if an associated collection is set, and if it is a collection the collection admin user administers.

A successful scan displays the following response:

```
"Status": "Stop of sdconnection scan submitted."
```

The following displays an error response example:

```
"Status": "No scan is currently in progress for sdconnection."
```

Application management by using APIs

The application management APIs provide options to get the details of the available applications, install or upgrade an application, and delete an application. You must have the Admin role to access the application management endpoints.

You can change the self-signed certificate that IBM Data Cataloging uses. For more information, see IBM Data Cataloging REST APIs.For more information, see the topic *IBM Spectrum® Discover REST APIs* in the *Data Cataloging: Administration Guide*.

- **/policyengine/v1/applications: POST**
  The /policyengine/v1/applications registers an action agent. Only an IBM Data Cataloging user with *data admin* role can register the application with IBM Data Cataloging.
- **/policyengine/v1/applications/<application name>: GET**
  Gets the details of a specific application that is configured in the system.
- **/policyengine/v1/tlscert: GET**
  Gets a CA-certified TLS certificate.
- **/policyengine/v1/action_ids: GET**
  Retrieves and displays a list of actions IDs of all the registered applications.
- **/policyengine/v1/applications/<deployment_name>/schema?action_id=<action_id>: GET**
  Gets the application schema for a specified application.
- **/policyengine/v1/application_names?action_id=<action_id>: GET**
  Gets a list of applications that support a specified action ID.
- **/policyengine/v1/applications: DELETE**
  Deletes an application.

/policyengine/v1/applications: POST

The /policyengine/v1/applications registers an action agent. Only an IBM Data Cataloging user with *data admin* role can register the application with IBM Data Cataloging.

The following table shows which roles can access /policyengine/v1/applications endpoint:

Table 1. Access by role

Data admin	Collection Admin	Data user	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <token>" -H "Content-Type: application/json" -X POST
https://<data_cataloging_host>/policyengine/v1/applications
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

Registering an application involves the following steps.

1. Obtain an auth token by using credentials of the data admin user as shown in the following example:

   ```
   curl -k https://<data_cataloging_host>/auth/v1/token -u "<user_name>:<password>
   ```

   For a valid user, the auth token is returned in the **"X-Auth-Token"** response header.

2. Create a .json file with application details:

   ```
   cat agentregmsg.json

   {
   "action_agent": "extractapplication",
   "action_id": "DEEPINSPECT",
   "action_params": ["extract_tags"]
   }
   ```

   Note: Only `DEEPINSPECT` is supported as a valid `action_id` and must always be in upper case. The `action_agent` parameter must be a maximum of 64 characters long and can contain alpha-numeric characters and '.', '_' or '-' characters. However, the action parameters that are converted to string can be a maximum of 256 characters long.
3. Submit the following request:

   ```
   curl -k https://<data_cataloging_host>/policyengine/v1/applications -H "Content-Type: application/json" -X POST -d
   @agentregmsg.json -H "Authorization: Bearer <token>"
   ```

Response:

```
Content-Type: application/json
    {
      "broker_ip": 9.11.200.114,
      "broker_port": 9093,
      "work_q": "extractapplication_work",
      "completion_q": "extractapplication_compl"
    }
```

Multiple application instances

If you want to run multiple copies of an application, then only the first application registration attempt receives a success response. Any subsequent applications receive a `409 Conflict` response. In such cases, use the /policyengine/v1/agents/<agent name> endpoint to find the required Kafka information to proceed.

Example GET response:

```
{
"broker_ip": "localhost",
"work_q": "extractapplication_work",
"auth": "extractapplication_user:extractapplication_password",
"params": "["tags"]",
"agent": "extractagent",
"broker_port": "9093",
"completion_q": "extractapplication_compl",
"action_id": "\"DEEPINSPECT\""
}
```

- **Example: Create a DeepInspect policy using the application**
  Use the POST /policyengine/v1/policies/extractpol endpoint to create a policy that makes use of the registered application.

Example: Create a DeepInspect policy using the application

Use the POST /policyengine/v1/policies/extractpol endpoint to create a policy that makes use of the registered application.

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <token>" -H "Content-Type: application/json" -X POST
https://<data_cataloging_host>:443/policyengine/v1/policies/extractpol
```

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

Registering an application involves the following steps.

1. Define the policy details in a JSON file as shown in the following example:

```
cat testpol
{
        "action_params": {
                "extract_tags": ["vin", "sensor"],
                "agent": "extractapplication"
        },
        "pol_id": "extractpol",
        "schedule": "NOW",
        "pol_filter": "size>23",
        "action_id": "DEEPINSPECT"
}
```

2. Submit the following request:

```
curl -k-H "Authorization: Bearer <token>"https://<data_cataloging_host>/policyengine/v1/policies/extractpol -X POST -d
@testpol -H "Content-Type: application/json"
```

As the policy is created, as the policy schedule is 'NOW'. The job request messages are immediately pushed to the Kafka work topic corresponding to the registered application as shown in the following example:

The following example provides a JSON message request.

```
{
    "mo_ver": "1.0",
    "action_id": "deepinspect",
    "action_params": {
        "agent": "extractapplication",
          "tags": {"extract_tags": ["vin", "sensor"]}
    },
    "agent": " extractapplication",
    "policy_id": " extractpol",
    "docs": [
        {"path": "/fs1/path1/file1.txt", "fkey": "spectrumscale.cluster.example"},
      {"path": "/fs1/path1/file2.txt", "fkey": "spectrumscale.cluster.example"},
          ......
          {"origpath": "/fs1/path1/file3.txt", "fkey": "spectrumscale.cluster.example"}
     ]
}
```

The following table lists the job request message format.

Table 1. Job request message format

Field	Value Type	Description
mo_ver	Float	The message version
policy_id	String	The name of the policy ID that requested the job.
action_id	String	The name of the action ID
agent	String	The name of the application
action_params	Object	JSON object of custom application parameters
docs	Array	The array of JSON objects, each containing information about documents (files or objects) to be inspected.

The following example provides a JSON message response:

```
{
    "mo_ver": "1.0",
    "policy_id": " extractpol",
    "docs": [
        {"status": "success", "tags": {"vin": "vin-value", "sensor": "sensor-value"}, "path": "/fs1/path1/file1.txt", "fkey":
"spectrumscale.cluster.example "},
        {"status": "success", "tags": {"vin": "vin-value", "sensor": "sensor-value"}, "path": "/fs1/path1/file1.txt", "fkey":
"spectrumscale.cluster.example "},
        {"status": "failed", "tags": {}, "path": "/fs1/path1/file1.txt", "fkey": "spectrumscale.cluster.example "}

     ]
}
```

The following table lists the job response message format.

Table 2. Job response message format

Field	Value Type	Description
mo_ver	Float	The message version
policy_id	String	The name of the policy ID that requested the job.
docs	Array	The array of JSON objects, each containing information about documents (files/objects) to be inspected.

/policyengine/v1/applications/<application name>: GET

Gets the details of a specific application that is configured in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	CollectionAdmin	Data user	Admin	Service user
√	X	√	X	X

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <token>" -H "Content-Type: application/json"
https://<data_cataloging_host>/policyengine/v1/application/<application _name>
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of a specific application.

Request:

```
curl -k -H "Authorization: Bearer <token>" -H "Content-Type: application/json"
"https://<data_cataloging_host>/policyengine/v1/application/extractapplication"
```

Response:


```
{
"broker_ip": "localhost",
"work_q": "
extractagent_work",
"auth": "extractapplication_user:extractapplication_password",
"params": "["tags"]",
"agent": "extractagent",
"broker_port": "9093",
"completion_q": "extractapplication_compl",
"action_id": "deepinspect"
}
```


/policyengine/v1/tlscert: GET

Gets a CA-certified TLS certificate.

The IBM Data Cataloging uses TLS protocol for encrypting the communication in-flight between the IBM Data Cataloging nodes (the Kafka topics) and the applications. It uses TLS client certificates to securely authenticate the applications. The certificates that are provided by the IBM Data Cataloging admin upon registration are used by applications to authenticate to the Kafka brokers in IBM Data Cataloging.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	CollectionAdmin	Data user	Admin	Service user
√	X	√	X	X

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <token>" https://<data_cataloging_host>/policyengine/v1/ tlscert
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get a CA-certified TLS certificate.

Request:

```
curl -k -H "Authorization: Bearer <token>" https://<data_cataloging_host>/policyengine/v1/ tlscert
```

Response:


```
-----BEGIN CERTIFICATE-----
Samplej1CAmoCCQDoQCvwi/bLdzANBgkqhkiG9w0BQsFADCBmDELMAkGA1UEBhMCR0IxDjAMBgNV
BAgMBUhBTlRTMRAwDgYDVQQHDAdIdXJzbGV5MQwwCgYDVQQKDANJQk0xGTAXBgNVBAsMEHNwZWN0
cnVtZGlzY292ZXIxGTAXBgNVBAMMEHNwZWN0cnVtZGlzY292ZXIxIzAhBgkqhkiG9w0BCQEWFG1s
YXdyZW5jZUB1ay5pYm0uY29tMB4XDTE4MTAxNTIyNTEzOVoXDTE5MTAxNTIyNTEzOVowbTELMAkG
A1UEBhMCR0IxDjAMBgNVBAgTBUhhbnRzMRAwDgYDVQQHEwdIdXJzbGV5MQwwCgYDVQQKEwNJQk0x
CzAJBgNVBAsTAklEMSEw4wYDVQQDExh0aG9yLnR1Yy5zdGdsYWJzLmlibS5jb20wggEiMA0GCSqG
SIb3DQEBAQUAA4IBDwAwggEKAoIBAQCJFq8t8rk2fZf86TQWTE6R0VhFxmn9daqWyTQGz56zqGnX
UxyF1wnIZQp7EGS3rBgHemV48X7xgxbshNVf1cr0jANvgElB66JOaESJUz1m/s3B728qpYHO4Wom
2ii17hF3VdQhAvd72hd2kpJ3XVtp95yktLJ40Cr6x/4Kgsm8iKIiYYN3LbqqtfNB5CyKV9qNzTGb
ERPeSujS4HjUbaXQ+xeex6PhT6MDHsuSCnT6p6W+WoAw1C+W8eKpDIRiQFBPxzFM1buvJutQ+H8u
5ipXKgi6ki5qYeKr5pdZKI/tHgAhZnuQOAbnGjumqRr1Ddx/0y4Ay5ZPcyRYeZJaUtJXAgMBAAEw
DQYJKoZIhvcNAQELBQADggEBAIo8Su+9I3wnbypy+c9Gee9p0zIv5ksJTC9KmUziVF4dr8J8t5Mx
cA42yB2Q8hL5OyGgIyYm3Kq+tbaq1jUgqdcOzKKxMIKnS5ut6uifoL3oEq4Gx6Qsjo+RLduEAqZC
LrCVM5Nm8i9mY5g4NIK/o3G2Mud5MKeHc/XnMw4ZNzqfsEIkiG1DAgLHVJVokdo4gtoB7pN9bdtA
Qdm7nH/N5vJL7HNTtupcyHQqTIot+WCJ1YqiKqZVnMqqjFm4zMAbzrNzMqIFdSNkJn9IU+6LJn7v
CrOu6Whgkh+37Plc2mnViWRmD2+gsHcPOUhhsJMltB+RghqEjStF4TW4v29E36E=
-----END CERTIFICATE-----
Bag Attributes
    friendlyName: kafka_client
    localKeyID: 54 69 6D 65 20 31 35 33 39 36 34 33 39 31 31 37 39 31
    1.3.18.0.2.28.24: IBM_SDK_JAVA_8_PKCS12
Key Attributes: <No Attributes>
-----BEGIN PRIVATE KEY-----
SampleIBADANBgkqhkiG9w0BAQEFAASCBKYwggSiAgEAAoIBAQCJFq8t8rk2fZf8
6TQWTE6R0VhFxmn9daqWyTQGz56zqGnXUxyF1wnIZQp7EGS3rBgHemV48X7xgxbs
hNVf1cr0jANvgElB66JOaESJUz1m/s3B728qpYHO4Wom2ii17hF3VdQhAvd72hd2
kpJ3XVtp95yktLJ40Cr6x/4Kgsm8iKIiYYN3LbqqtfNB5CyKV9qNzTGbERPeSujS
4HjUbaXQ+xeex6PhT6MDHsuSCnT6p6W+WoAw1C+W8eKpDIRiQFBPxzFM1buvJutQ
+H8u5ipXKgi6ki5qYeKr5pdZKI/tHgAhZnuQOAbnGjumqRr1Ddx/0y4Ay5ZPcyRY
eZJaUtJXAgMBAAECggEAGXLSlk8GvuzVuCd6eeE0fLFGF73bkyeywUccl18XcPaE
n2pYUyg2MrM/HNVQPSB/XXQFPkxmamMRcFrAWYgQoxES2jIQZ2JcwP1uK6f+MA6B
F81g76PF0yfZUtkQ0aUeLO8L8Azi8GTuEh7nfa0g+Sz0MUcyEmdicyeJK15YIN1e
3Z0bl6kment4xvcLjPa2iFa1UJZ8gMNT6aS+NBaexjNlDW+NlftplRH6z0iqUObd
MeNvi7kiqOpre/QiPUzJSmmPsDSwCCtob9oAHa9JrfLH19VnBQGqee4T/KPQEiE8
q5M5IV187I2SHj0DEOl7YEwfdaQN2ZQ+gC7yZiuo8QKBgQC8xNvOqpiz4OVuauFK
2FLITKuvCSIEjaeVQAEDlTh6/XIjOl6yQOhkQ5Shq2XUu+U5A7R/HR8EFPjRdS6o
jVcftO16DNzUNVkXE/cSfwdudKKfXunC4O/KLMH7aTp/R8UvuLDYx2rgL7SKnBWL
XL8LK1YguJpjFb1n9Ey4VIGf6wKBgQC56dUi6wkifNfQqr36e6QehV2n9LXYEudS
H1ZOzmdl5UqEWIeEjaeZ+iwxB1qQzysrxuG76GNvDWBKxC6y7/3iDMwe0yJErMe2
y3PPl5mPdbBSc35cFVH0OnMObx0O9rX461eP6wuOdC2JZtgMClhoKl9YaGWl6dTg
ct/TI7IoRQKBgAu50smgsUGjO71DKGGE1Ieas6N41sMfpYRQ6H83er2xFrD3hEBg
Zc8sn4P3aKjxTwjeoQywW2fegaJJPC++L+ZJT+Mf/2uzMKiNVfCxMV38IsnxlGXF
oT+Wzin23Gpho+8V8u5UXOA2NmnRzkjM5xUjiQMZVIXJKO3XMBjXLwInAoGAcRxM
OYdoMlxkbadEGJ6a9prLpYEkFcdCh9tbBm1byUncsXKvWL1aiDsGIfpIZqw0ptjE
xC+aN5mayUJc6y9Kdvq0QdjRE5lPlO5pkpa4rei7kx6DdYDfMMlS8Sg3NMBzbHO6
/WvSxyP2BI/WyFSO/ykwgzhZB72hb8FUXkBsh0UCgYAikQFEAJmgCDPWO1a4Ts3C
pUEH8ZlPDj77aL/otxdr+UiEOBzmWknlb0Nq+KtEHbq3bk0wQ9aC7lnUT+HgJJ2z
th/mxilgYeWF2CH5DvtZRNIDR0ihuMQe4dDcFyA/cGxrGwYUXX+XTnLT8krGZ1Nv
O6I9o1BVn8DVCy1LaI7e0Q==
-----END PRIVATE KEY-----
-----BEGIN CERTIFICATE-----
SampleCCA62gAwIBAgIJAIBi92yKX7vIMA0GCSqGSIb3DQEBCwUAMIGYMQswCQYD
VQQGEwJHQjEOMAwGA1UECAwFSEFOVFMxEDAOBgNVBAcMB0h1cnNsZXkxDDAKBgNV
BAoMA0lCTTEZMBcGA1UECwwQc3BlY3RydW1kaXNjb3ZlcjEZMBcGA1UEAwwQc3Bl
Y3RydW1kaXNjb3ZlcjEjMCEGCSqGSIb3DQEJARYUbWxhd3JlbmNlQHVrLmlibS5j
b20wHhcNMTgxMDE1MjI0OTM4WhcNMzgxMDEwMjI0OTM4WjCBmDELMAkGA1UEBhMC
R0IxDjAMBgNVBAgMBUhBTlRTMRAwDgYDVQQHDAdIdXJzbGV5MQwwCgYDVQQKDANJ
Qk0xGTAXBgNVBAsMEHNwZWN0cnVtZGlzY292ZXIxGTAXBgNVBAMMEHNwZWN0cnVt
ZGlzY292ZXIxIzAhBgkqhkiG9w0BCQEWFG1sYXdyZW5jZUB1ay5pYm0uY29tMIIB
IjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAxPPx8WO7TxpKbmMmOWd5MwPf
4CenKrgpYmeU+gQbgFIINEPMa9jG2lUetxk36rN4GxUlsvJpYaFIDDyG8or2KrZo
Q7yBagYmVbN6dhrKwH8YlBS0sJnockM8w8kGYc6ENT9dNH9/jaaBZXX9dIlgdjQE
oSOuu1CJgu+vGK2cwFqp/LSmtJDKdgM8/aSa8YTyEUg7HNz+fv2m7DkGyjvTkRem
sXzZCXIzFbvN4boJ7QnZgVDjvunYrTCMJiOautIFaYv00mzhPxXf+dzbkffcR4Hu
qZbPGyAbN+ydwt+8QSJfuDaEQAhHkSKVnCxiwMhaAawOE8/DyPRzzZnHk6AV8wID
AQABo4IBDjCCAQowHQYDVR0OBBYEFJz0+e8N+mU+diD3emP9E5/MAwdxMIHNBgNV
HSMEgcUwgcKAFJz0+e8N+mU+diD3emP9E5/MAwdxoYGepIGbMIGYMQswCQYDVQQG
EwJHQjEOMAwGA1UECAwFSEFOVFMxEDAOBgNVBAcMB0h1cnNsZXkxDDAKBgNVBAoM
A0lCTTEZMBcGA1UECwwQc3BlY3RydW1kaXNjb3ZlcjEZMBcGA1UEAwwQc3BlY3Ry
dW1kaXNjb3ZlcjEjMCEGCSqGSIb3DQEJARYUbWxhd3JlbmNlQHVrLmlibS5jb22C
CQCAYvdsil+7yDAMBgNVHRMEBTADAQH/MAsGA1UdDwQEAwIBBjANBgkqhkiG9w0B
AQsFAAOCAQEAkP3buR6yS5CQ634jisvv+vSTCbYauV8ZNaNoudBBl1QiiWR0SPVn
aR+oPtASZ7Ab3uICdo1/Bf9Wo1LzdTEpnfMmYQOG7C2UNj8TDvFIhCUutkMmO5MB
NDIH7Q/XUqSXqrBq+KkrfEl5MqqPLU8yelEF1L0U7MHLBP3h2fz528hnLBL5rjI+
NbM7yJFIA4nsnVoqUZbWZOGiAqZ0uPCFLC6hyD/DzT2rfHgfiS3Y9sLUdRMhHd8o
j6w8+G8J3B5lhqbAueKcNXS6ZCBIPJtiovZtsaT4AnslpB2GRJoWJQvxJzr1OLCm
```

```
KfRlkCJTnbPKfI34HxOE/k/wpOJSample==
-----END CERTIFICATE-----
```

The response contains the client certificate and the client key, followed by the root certificate. Based on the response output, the organization of the response is as follows:

```
-----BEGIN CERTIFICATE-----
client certificate
-----END CERTIFICATE-----
Bag Attributes
...
Key Attributes:
...
-----BEGIN PRIVATE KEY-----
client key
-----END PRIVATE KEY-----
-----BEGIN CERTIFICATE-----
root certificate
-----END CERTIFICATE-----
```

These certificates and key are required for the secure communication connection to the Kafka brokers in IBM Data Cataloging.

Note: Before you connect to the Kafka brokers in IBM Data Cataloging, the application must complete the additional settings for the secure connection.

/policyengine/v1/action_ids: GET

Retrieves and displays a list of actions IDs of all the registered applications.

This API endpoint provides a unique list of action IDs for the registered applications.

The following table displays the roles that can access the REST API endpoint.

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<spectrum_discover_host>/policyengine/v1/
action_ids -X GET -H ’Accept: application/json’
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns a list of action IDs of registered applications that are configured in the system.

Issue the following request in one line:

```
curl -k -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>
/policyengine/v1/action_ids -X GET -H ’Accept: application/json’
```

The following response is returned:

```
[
    "CONTENTSEARCH",
    "COPY",
    "AUTOTAG",
    "TIER"
]
```

/policyengine/v1/applications/<deployment_name>/schema?action_id=<action_id>: GET

Gets the application schema for a specified application.

This endpoint returns the schema from the application registration message. The schema is used by the UI to dynamically generate policy creation views for the registered application. If the action ID is not specified as a query parameter, the endpoint returns the whole schema. For the specified action ID, a subset of the registration schema is returned.

The following table displays the roles that can access the REST API endpoint.

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<spectrum_discover_host>/policyengine/v1/
applications/<deployment_name>/schema?action_id=<action_id>  -X GET -H 'Accept: application/json'
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns the application schema for a specified application and a specified action ID that is registered on the system.

Issue the following request in one line:

```
curl -k -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/policyengine/v1/applications/ScaleILM/schema?action_id=MOVE -X GET -H 'Accept: application/json'
```

The following response is returned:


```
{
  "properties": {
    "destination_tier": {
      "description": "Tier name which can be a Spectrum Scale internal pool name or an external pool name like 'tape-archive'
to tier the files to a LTFS Tape library configured on the Spectrum Scale cluster. Internal pool names are mentioned as-is,
while the external pools are mentioned as 'external:tape-archive:pool1@lib1' or 'external:cos' (future use). External pool name
'tape-archive' is an IBM Spectrum Discover defined pool name for the LTFSEE Tape pool configured on the Spectrum Scale
system.",
      "type": "string"
    },
    "source_connection": {
      "description": "Defines platform name of the source data connection.",
      "type": "string"
    }
  },
  "required": [
    "source_connection",
    "destination_tier"
  ],
  "type": "object"
}
```


/policyengine/v1/application_names?action_id=<action_id>: GET

Gets a list of applications that support a specified action ID.

This API endpoint provides a unique list of application names with the specified action ID.

If an action ID is not provided as a query parameter then the endpoint returns all the application names on the system.

The following table displays the roles that can access the REST API endpoint.

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	X	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<spectrum_discover_host>/policyengine/v1/
application_names?action_id=<action_id>  -X GET -H 'Accept: application/json'
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example returns a list of application names with the given action id that are configured in the system.

Issue the following request in one line:

```
curl -k -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>
/policyengine/v1/action_ids -X GET -H 'Accept: application/json'
```

The following response is returned:

```
[
    "WKCConnector",
    "contentsearchagent",
    "ScaleILM"
]
```

/policyengine/v1/applications: DELETE

Deletes an application.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	CollectionAdmin	Data user	Admin	Service user
√	X	X	X	X

Synopsis of the request URL

```
curl -k -H "Authorization: Bearer <_token_>" -X DELETE
https://<data_cataloging_host>/policyengine/v1/applications/<application_name>
```

Supported request types and response formats

Supported request types:

- DELETE

Examples

Deleting an application involves the following steps.

1. Obtain an auth token by using credentials of the data admin user as shown in the following example:

   ```
   curl -k https://<data_cataloging_host>/auth/v1/token -u "<user_name>:<password>" -v
   ```

   For a valid user, the auth token is returned in the **"X-Auth-Token"** response header.

2. Submit the following request:

   ```
   curl -k https://<data_cataloging_host>/policyengine/v1/applications/<application_name> " -X DELETE -H "Authorization:
   Bearer <_token_> "
   ```

   Response:

   ```
   204 No Content
   ```

RBAC management by using APIs

The IBM Data Cataloging resource-based access control (RBAC) is a REST API service that enables role-based access to the IBM Data Cataloging services. This service uses OpenStack Keystone as a backend for providing Identity and Access Management (IAM) across multiple domains that are attached to the IBM Data Cataloging.

The authentication service uses a default user with user name *sdadmin* and password *Passw0rd* in the domain named *default*. This user has the administrative role and can be used to do the following actions:

- This user can create other users and user groups.
- This user can register new domains with the authentication service.
- This user can create projects.
- This user can assign roles to the users at project or domain level.

The following are the predefined user roles with the corresponding access levels:

admin
: Default user role created by the system. Users with this role can create other users, projects, domains, and assign roles. Users with this role cannot see metadata records.

dataadmin
: The users with this role can see all metadata records across projects.

collectionadmin

The users with this role can access metadata that is collected. But, metadata access is restricted to the records that are associated with the collections to which the user has the Collection Admin or Data User role assigned.

Important: The Collection Admin role is available as a technology preview in the 2.0.1.1 release. For limitations on the usage of the Collection Admin role, see the *IBM Spectrum® Discover Release Notes*.

datauser
: This role is ideal for a researcher or data scientist. Users with this role can see records that are associated with projects to which they belong.

serviceuser
: This user role intended for service personnel. Users with this role have read-only access to the system logs.

IBM Data Cataloging integrates with the enterprise LDAP connected to IBM Spectrum Scale. Using the authentication service APIs, admin users can add an LDAP domain definition to IBM Data Cataloging. The users and groups from the registered LDAP domain are automatically imported into IBM Data Cataloging and the administrators can add these users and groups to different projects while they assign them the Data User role.

Admin users can also assign the Data Admin role to some of the users and user groups to give access to the entire IBM Data Cataloging index for searches and policies.

The authentication API service endpoint has the following basic structure: `https://<host address>/auth/v1/<endpoint>`

For example, the following endpoint gets authentication token for the users: `curl -k -u <user>:<pass> https://<host address>/auth/v1/token`

- **Managing tokens**
  The users need to get an authentication token from the authentication server by using their user name and password. Then, they need to use that token to access the services offered by the IBM Data Cataloging system.
- **Managing users**
  All the create, read, update, and delete operations are allowed on the *default* domain. But the read-only (GET) access is allowed for the users and groups that are imported from the registered Lightweight Directory Access Protocol (LDAP) or Cloud Storage Object domains.
- **Managing user roles**
  The authorization or access privileges of users and user groups are defined with the help of user roles.
- **Managing user groups**
  You can create user groups for better manageability of users. Defining user groups also provides flexibility of assigning specific roles to a group of users rather than assigning the roles individually.
- **Managing collections**
  You can create and manage collections by using REST APIs.
- **Managing domains**
  You can create and manage user domains by using REST APIs.

Managing tokens

The users need to get an authentication token from the authentication server by using their user name and password. Then, they need to use that token to access the services offered by the IBM Data Cataloging system.

For more information on the authentication process, see Authentication process.

The following endpoint is available to manage tokens:

- **/auth/v1/token: GET**
  Gets authentication token for the user.

/auth/v1/token: GET

Gets authentication token for the user.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	√	√

Synopsis of the request URL

Users from an external LDAP or COS domain must include domain name in the user name as *<domain>/<user>* to get an authentication token.

```
curl -k -u ldap/user1:pass https://<data_cataloging_host>/auth/v1/token -v
```

```
curl -k -u cos/user1:pass https://<data_cataloging_host>/auth/v1/token -v
```

```
curl -k -u cos/<access_key_id>:<secret_key> https://<data_cataloging_host>/auth/v1/token -v
```

Without specifying domain name as part of user name:

```
curl -k -u user1:pass https://<data_cataloging_host>/auth/v1/token -v
```

Note: When using curl, add the -v parameter to see the response headers that contain the token.

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Default response data is in CSV format. For results in JSON format, use the "Accept: application/json" header.

Examples

The following example shows a request and the corresponding response.

Request that contains domain as part of the user name:

```
curl -k -u ldap/user1:pass https://<data_cataloging_host>/auth/v1/token -v
```

Request that does not contain domain as part of the user name:

```
curl -k -u user1:pass https://<data_cataloging_host>/auth/v1/token -v
```

Response:

```
< HTTP/1.0 200 OK
< X-Auth-Token: gAAAAABbUdgLOLk67Zk_XQO0bbJt_qe4suz449m7XLM-D6e0jUzDxBW574L38y5xF5y3wc-
Xq0Bp43uQH13QN-wTsioPoOfOlbyrN5avBag2iHUkCOgXyA46TvY5LdGu46LEs-gO-
qidQgUCg5im4QF3Zvw5enCTvQd2N1bAg186CVOc8Qk5o1ldJGd51nL3ELts0LsVg9F4oPhv9HixwZflhlCsuPg1Gw
< Content-Type: text/html; charset=utf-8
< Content-Length: 0
< Server: Werkzeug/0.14.1 Python/2.7.5
< Date: Fri, 20 Jul 2018 12:39:39 GMT
```

You need to use the token that is obtained in the previous step to access the services offered by IBM Data Cataloging.

The following example shows how to use the token that is obtained to get the list of from the policy engine:

```
curl -v -k -H ‘Authorization: Bearer gAAAAABbUdgLOLk67Zk_XQO0bbJt_qe4suz449m7XLM-
D6e0jUzDxBW574L38y5xF5y3wc-Xq0Bp43uQH13QN-wTsioPoOfOlbyrN5avBag2iHUkCOgXyA46TvY5LdGu46LEs-
gO-qidQgUCg5im4QF3Zvw5enCTvQd2N1bAg186CVOc8Qk5o1ldJGd51nL3ELts0LsVg9F4oPhv9HixwZflhlCsuPg1Gw’
https://<data_cataloging_host>/policyengine/v1/policies
```

Managing users

All the create, read, update, and delete operations are allowed on the *default* domain. But the read-only (GET) access is allowed for the users and groups that are imported from the registered Lightweight Directory Access Protocol (LDAP) or Cloud Storage Object domains.

- **/auth/v1/users: GET**
  Gets the details of the users that are created in the system.
- **/auth/v1/users: POST**
  Creates a user.
- **/auth/v1/users/<user_ID>/password: POST**
  Changes the user password.
- **/auth/v1/users/<user ID>: GET**
  Gets the details of a specific user.
- **/auth/v1/users/<user_ID>: PATCH**
  Updates an existing user.
- **/auth/v1/users/<user_ID>: DELETE**
  Deletes a specific user.
- **/auth/v1/users/<user_ID>/groups: GET**
  Gets the details of the groups to which a specific user belongs.
- **/auth/v1/users/<user_ID>/collections: GET**
  Gets the list of collections that are assigned to a user.
- **/auth/v1/users/<user_ID>/roles: GET**
  Gets the list of roles that are assigned to a user.
- **/auth/v1/users/<user_ID>/role_assignment: GET**
  Returns a list of the roles and associated collections assigned to the user where applicable. This does not include any role assignments inherited from groups that the user belongs to.
- **/auth/v1/users/users_summary: GET**
  Gets the list of users and their domain details.

/auth/v1/users: GET

Gets the details of the users that are created in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√	√	√[1]
[1]The specified user ID must be the same as the requesting user ID. Otherwise the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the uses who are created in the system.

Request:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users
```

Response:

```
(200 OK):
{
    "users": [
        {
            "domain_id": "default",
            "enabled": true,
            "id": "2e905adec130453f888a425dfc038f9c",
            "name": "sdadmin",
            "options": {},
            "password_expires_at": null
        },
        {
            "domain_id": "default",
            "enabled": true,
            "id": "4ee5b47f439640d29b6fac7253a64290",
            "name": "kris",
            "options": {},
            "password_expires_at": null
        },
        {
            "domain_id": "default",
            "enabled": true,
            "id": "a8dbddcd66b746b9958c34fd5a28855c",
            "name": "dataadmin-user",
            "options": {},
            "password_expires_at": null
        },
        {
            "domain_id": "default",
            "enabled": true,
            "id": "f731680239d7457aa2bad27b276d01f4",
            "name": "kris-tsv-txt",
            "options": {},
            "password_expires_at": null
        },
        {
            "domain_id": "bbd9f2993849402490cefc015013b6e9",
            "id": "b7be4be4f78f6c371583cf4b3617e47900477f6d09f732c2493a22b79d99a2a3",
            "name": "dataadmin",
            "options": {},
            "password_expires_at": null
        },
        {
            "domain_id": "bbd9f2993849402490cefc015013b6e9",
            "id": "0337d1845a17fa29ff6ba83c2fb62870273030dfb4d10de8dfab2e9fbe68aeef",
            "name": "user1",
            "options": {},
            "password_expires_at": null
        },
        {
            "domain_id": "bbd9f2993849402490cefc015013b6e9",
            "id": "d523a89af4e3c7042b564d241dcb058cd892030b63d00091128ba1d1d98a5d73",
            "name": "user2",
            "options": {},
            "password_expires_at": null
        }
    ]
}
```

/auth/v1/users: POST

Creates a user.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X POST
https://<data_cataloging_host>/auth/v1/users -d '<json details of user>'
```

The details of a user can be the following:

- Name
- Password
- Email
- Description
- Project
- Default project

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

The following example shows how to create a user.

Request:

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X POST
https://<data_cataloging_host>/auth/v1/users -d '{"name":"testuser","password":"testpass",
"email":"testuser@ibm.com"}'
```

Response:

```
201 Created
```

/auth/v1/users/<user_ID>/password: POST

Changes the user password.

The table shows the roles that can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√	√	√	√

Synopsis of the request URL

```
curl -k -H 'Content-Type: application/json' -X POST
https://<data_cataloging_host>/auth/v1/users/user_id/password -d'<json details of user>'
```

Required Parameters in the JSON payload:

- password - new password
- original_password - old/current password

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

The following example shows how to change a user password.

Request:

```
curl -k -H 'Content-Type: application/json' -X POST
https://<data_cataloging_host>/auth/v1/users/a26f98ab046449bebce70c4832b22ac2/password -X POST-H"Content-Type:application/json"
-d '{"password":"NewPassw0rd","original_password":"Passw0rd"}'
```

Response:

```
204 No Content
```

/auth/v1/users/<user ID>: GET

Gets the details of a specific user.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√	√	√[1]

[1]The specified user ID must be the same as the requesting user ID. Otherwise the command fails with error 403, "Not authorized". An exception is if the user ID is "`me`", the result is the details of the current user.

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/<user_ID>
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of a specific user.

Request:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/4ee5b47f439640d29b6fac7253a64290
```

Response:

```
200 OK:
{
    "domain_id": "default",
    "email": "testuser@ibm.com",
    "enabled": true,
    "id": "5b3cd6af1c38479aa3a8cb220230c651",
    "name": "testuser",
    "options": {},
    "password_expires_at": null
}
```

The following special case returns the current user's details.

Request:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/me
```

/auth/v1/users/<user_ID>: PATCH

Updates an existing user.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/users/<user_ID> -X PATCH -d '<json details to update>
```

Supported request types and response formats

You can update the following attributes of a user:

- Name
- Password
- Email
- Description
- Project
- Default project

Supported request types:

- PATCH

Supported response formats:

- JSON

Examples

The following example shows how to update an existing user.

Request:

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/users/4ee5b47f439640d29b6fac7253a64290
-X PATCH -d '{"name": "testuser-updated", "password": "newpassword"}'
```

Response:

```
200 OK
```

/auth/v1/users/<user_ID>: DELETE

Deletes a specific user.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/<user_ID> -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

The following example shows how to delete a user.

Request:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/4ee5b47f439640d29b6fac7253a64290
```

Response:

```
204 No Content
```

/auth/v1/users/<user_ID>/groups: GET

Gets the details of the groups to which a specific user belongs.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√	√	√[1]
[1]The specified user ID must be the same as the requesting user ID. Otherwise the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/<user_ID>/groups
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the groups to which a specific user belongs.

Request:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/
4ee5b47f439640d29b6fac7253a64290/groups
```

Response:

```
{
    "groups": [
        {
            "name": "test-group1",
            "description": "Test Group 1",
            "domain": "default"
        },
        {
            "name": "test-group2",
            "description": "Test Group 2",
            "domain": "default"
        }
    ]
}
```

/auth/v1/users/<user_ID>/collections: GET

Gets the list of collections that are assigned to a user.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√	√	√[1]
[1]The specified user ID must be the same as the requesting user ID. Otherwise the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/<user_ID>/collections
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the collections to which the user is assigned.

Request:

```
curl -k -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/users/4ee5b47f439640d29b6fac7253a64290/collections
```

Response:

```
(200 OK):
{
    "collections": [
        {
            "description": "",
```

```
            "domain_id": "default",
            "enabled": true,
            "id": "1d657422853c4c33bde953837c709655",
            "is_domain": false,
            "name": "JPGProj",
            "parent_id": "default",
            "tags": []
        },
        {
            "description": "",
            "domain_id": "default",
            "enabled": true,
            "id": "28bafe9031af4aa7a637b00aa436fb01",
            "is_domain": false,
            "name": "XMLProj",
            "parent_id": "default",
            "tags": []
        },
        {
            "description": "Bootstrap collection for initializing the cloud.",
            "domain_id": "default",
            "enabled": true,
            "id": "c64cafe7f69349d0ba79a8fe931be8d2",
            "is_domain": false,
            "name": "spectrum-discover",
            "parent_id": "default",
            "tags": []
        }
    ]
}
```

/auth/v1/users/<user_ID>/roles: GET

Gets the list of roles that are assigned to a user.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√	√	√[1]
[1]The specified user ID must be the same as the requesting user ID. Otherwise the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k -H ‘Authorization: Bearer <token>’ https://<data_cataloging_host>/auth/v1/users/<user_ID>/roles
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the list of roles that are assigned to a user.

Request:

```
curl -k -H ‘Authorization: Bearer <token>’ https://<data_cataloging_host>/auth/v1/users/4ee5b47f439640d29b6fac7253a64290/roles
```

Response:

```
(200 OK):
{
    "roles": [
        {
            "domain_id": null,
            "id": "4a5415cb9cc5460aafe12a6f6206448e",
            "name": "datauser"
        }
    ]
}
```

/auth/v1/users/<user_ID>/role_assignment: GET

Returns a list of the roles and associated collections assigned to the user where applicable. This does not include any role assignments inherited from groups that the user belongs to.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	√	√	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/<user_id>/role_assignments
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the roles that is assigned to a specific user:

Request:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/
4ee5b47f439640d29b6fac7253a64290/role_assignments
```

Response:

```
Returns 200 OK:
{
    "roles":[
        {
            "collections":[
                {
                    "description":"Bootstrap project for initializing the cloud.",
                    "domain_id":"default",
                    "enabled":true,
                    "id":"09dd9136eaa6416784d6c5bdf3fa2f78",
                    "is_domain":false,
                    "name":"data-cataloging",
                    "parent_id":"default",
                    "tags":[]
                }
            ],
            "domain_id":"domain_id",
            "id":"5f365cf289d14c93a986b8bea976f92b",
            "name":"collectionadmin"
        }
    ]
}
```

/auth/v1/users/users_summary: GET

Gets the list of users and their domain details.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√	√	√[1]

[1] The specified user ID must be the same as the requesting user ID. Otherwise the command fails with error 403, "Not authorized". Data users can access this API to retrieve only those user details they are entitled to view.

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/users_summary
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the user details.

Request:

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/users/users_summary
```

Response:

```
(200 OK):
 {
    "users": [
        {
               "domain_id": "default",
               "domain_name": "Default",
               "enabled": true,
               "id": "7d040d976f924d60b5f39f0648950d1b",
               "name": "sdadmin",
               "options": {},
               "password_expires_at": null
        },
        {
               "domain_id": "a69001e854174dc1a4e0c98f712744bb",
               "domain_name": "lldap",
               "id": "1e8155daee6f021008a09aca09e1473d823ca7128d16ba25b3a10a7c1aa3433e",
               "name": "user2_NoGroup",
               "options": {},
               "password_expires_at": null
        },
        {
               "domain_id": "a69001e854174dc1a4e0c98f712744bb",
               "domain_name": "lldap",
               "id": "38f5434fe9d12fcb3278deb29fb104441b87d040f5774bd545a2d1d0c7adaa7f",
               "name": "user5_SystemAdmin",
               "options": {},
               "password_expires_at": null
        }
    ]
}
```

Managing user roles

The authorization or access privileges of users and user groups are defined with the help of user roles.

The following are the predefined user roles with the corresponding access levels:

admin
: Users with this role can create other users, projects, domains, and assign roles. Users with this role cannot see metadata records. This role is a default user role that is created by the system.

dataadmin
: Users with this role can see all metadata records across projects.

collectionadmin
: Users with this role can access metadata that is collected. However, metadata access is restricted to the records that are associated with the collections to which the user has the Collection Admin or Data User role assigned.
: Important: The Collection Admin role is available as a technology preview in the 2.0.1.1 release. For limitations on the usage of the Collection Admin role, see the *IBM Spectrum® Discover Release Notes*.

datauser
: Users with this role can see records that are associated with projects to which they belong. This role is ideal for a researcher or data scientist.

serviceuser
: Users with this role have read-only access to the system logs. This user role intended for service personnel.

- **/auth/v1/roles/<role_ID>: PUT**
  Assigns a role to users or user groups.
- **/auth/v1/roles/<role_ID >: GET**
  Gets the details of a specific user role.
- **/auth/v1/roles: GET**
  Gets the details of user roles.
- **/auth/v1/roles/<role_ID>: DELETE**
  Removes the role that is assigned to a user, group, collection, or domain.

/auth/v1/roles/<role_ID>: PUT

Assigns a role to users or user groups.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	√	√	X

The following roles can be assigned to a user:

admin
: Default user role created by the system. Users with this role can create other users, projects, domains, and assign roles. Users with this role cannot see metadata records.

dataadmin
: The users with this role can see all metadata records across projects.

datauser
: This role is ideal for a researcher or data scientist. Users with this role can see records that are associated with projects to which they belong.

serviceuser
: This user role intended for service personnel. Users with this role have read only access to the system logs.

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' -X PUT https://<data_cataloging_host>/auth/v1/roles/<role_ID> -d '<details of where role is to be assigned>'
```

You can specify the following details in the **roles** endpoint to assign roles to the corresponding component. An example follows this list.

user_id
: The user ID to which the role must be assigned.

group_id
: The group to which the role must be assigned.

project_id
: The project to which the role must be assigned.

domain_id
: The domain to which the role must be assigned.

The following examples illustrate how to specify these fields to assign a role to a domain or collection:

```
{
    "user_id": <string> [optional],
    "group_id": <string> [optional],
    "collection_id": <string> [optional],
    "domain_id": <string> [optional]
}
example [assign role to user in default domain]
{
    "user_id": "5b3cd6af1c38479aa3a8cb220230c651"
}
example [assign role to user in specific domain]
{
    "user_id": "5b3cd6af1c38479aa3a8cb220230c651",
    "domain_id": "4bfa9a9f71154dbeb554261a1c8d9a53"
}
example [assign role to user in specific collection]
{
    "user_id": "5b3cd6af1c38479aa3a8cb220230c651",
    "collection_id": "3b8c501b173c4ebcb99f322a4f7cf60e"
}
```

Supported request types and response formats

Supported request types:

- PUT

Supported response formats:

- JSON

Examples

Example 1: Assigning a role to a specific user.

Request:

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X PUT
https://<data_cataloging_host>/auth/v1/roles/4a5415cb9cc5460aafe12a6f6206448e
-d '{"user_id": "5b3cd6af1c38479aa3a8cb220230c651"}'
```

Response:

```
200 OK
```

Example 2: Assigning a role to a group.

Request:

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X PUT
https://<data_cataloging_host>/auth/v1/roles/4a5415cb9cc5460aafe12a6f6206448e
-d '{"group_id":  "6c4de7af1c38479aa3a8cb220230d762"}'
```

Response:

```
200 OK
```

Example 3: Assigning role for a user in a collection.

Request:

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X PUT
https://<data_cataloging_host>/auth/v1/roles/4a5415cb9cc5460aafe12a6f6206448e
-d '{"user_id": "5b3cd6af1c38479aa3a8cb220230c651", "domain_id": "af1c38479a5b3cd6c651a3a8cb220230"}'
```

Response:

```
200 OK
```

/auth/v1/roles/<role_ID >: GET

Gets the details of a specific user role.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√[1]	√	√[1]
[1]The specified role must be one that is assigned to the requesting user ID. Otherwise the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/roles/<role_ID>
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of a specific user role.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/roles/4a5415cb9cc5460aafe12a6f6206448e
```

Response:

```
200 (OK):
{
    "domain_id": null,
    "id": "4a5415cb9cc5460aafe12a6f6206448e",
    "name": "datauser"
}
```

/auth/v1/roles: GET

Gets the details of user roles.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√[1]	√	√[1]
[1]The specified role must be one that is assigned to the requesting user ID. Otherwise the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/roles
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of user roles.

Request:

```
curl -k  -H ‘Authorization: Bearer <token>’ https://<data_cataloging_host>/auth/v1/roles
```

Response:

```
{
  "roles": [
    {
      "domain_id": null,
      "id": "0ce17850c81a4e5da06ecd93f8ea0393",
      "name": "admin"
    },
    {
      "domain_id": null,
      "id": "383d4b371f244a2a8529a35b665b5c01",
      "name": "datauser"
    },
    {
      "domain_id": null,
      "id": "7d9f6d213b6d4317a395bbf248fdd9b6",
      "name": "collectionadmin"
    },
    {
      "domain_id": null,
      "id": "103bc9b8f1864c19af5663ec357f8d51",
      "name": "dataadmin"
    },
    {
      "domain_id": null,
      "id": "8388af928ce546bea7a0cd804a427b7c",
      "name": "serviceuser"
    }
  ]
}
```

/auth/v1/roles/<role_ID>: DELETE

Removes the role that is assigned to a user, group, collection, or domain.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	√	√	X

Synopsis of the request URL

```
curl -k  -H ‘Authorization: Bearer <token>’ -X DELETE https://<data_cataloging_host>/auth/v1/
roles/<role_ID> -d ‘<the component for which the role must be revoked >’
```

You can specify one of the following items for which you need to revoke the role:

- user_id
- group_id
- collection_id
- domain_id

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

Example 1: Revoking role from a user.

Request:

```
curl -k  -H 'Content-Type: application/json' -H ‘Authorization: Bearer <token>’ -X DELETE
https://<data_cataloging_host>/auth/v1/roles/4a5415cb9cc5460aafe12a6f6206448e
```

```
-d '{"user_id": "5b3cd6af1c38479aa3a8cb220230c651"}'
```

Response:

```
204 No Content
```

Example 2: Revoking role from a group.

Request:

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X DELETE
https://<data_cataloging_host>/auth/v1/roles/4a5415cb9cc5460aafe12a6f6206448e
-d '{"group_id": "6c4de7af1c38479aa3a8cb220230d762"}'
```

Response:

```
204 No Content
```

Example 3: Revoking role for a user of a collection.

Request:

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X DELET
https://<data_cataloging_host>/auth/v1/roles/4a5415cb9cc5460aafe12a6f6206448e
-d '{"user_id": "5b3cd6af1c38479aa3a8cb220230c651", "collection_id": "c38479aa35b3cd6af1a8cb220230c651"}'
```

Response:

```
204 No Content
```

Example 4: Revoking role for a group of a domain.

Request:

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X DELETE
https://<data_cataloging_host>/auth/v1/roles/4a5415cb9cc5460aafe12a6f6206448e
-d '{"user_id": "5b3cd6af1c38479aa3a8cb220230c651", "domain_id": "af1c38479a5b3cd6c651a3a8cb220230"}'
```

Response:

```
204 No Content
```

Managing user groups

You can create user groups for better manageability of users. Defining user groups also provides flexibility of assigning specific roles to a group of users rather than assigning the roles individually.

- **/auth/v1/groups: GET**
  Gets the list of groups that are created in the system.
- **/auth/v1/groups: POST**
  Creates a group in the system.
- **/auth/v1/groups/<group ID>: GET**
  Gets the details of a specific group.
- **/auth/v1/groups/<group_ID>: PATCH**
  Updates the attributes of an existing group.
- **/auth/v1/groups/<group_ID>: DELETE**
  Deletes a specific user group.
- **/auth/v1/groups/<group_ID>/collections: GET**
  Gets the details of the collections that are assigned in a group.
- **/auth/v1/groups/<group_ID>/roles: GET**
  Gets the details of the user roles that are assigned to a group.
- **/auth/v1/groups/<group_id>/role_assignments: GET**
  Returns a list of the roles and associated collections assigned to the group where applicable.
- **/auth/v1/groups/<group_ID>/users: GET**
  Gets the details of the users who are part of a specific group.
- **/auth/v1/groups/<group_ID>/user/<user_ID>: DELETE**
  Removes a specific user from a group.
- **/auth/v1/groups/<group_ID>/user/<user_ID>: PUT**
  Assigns a user to a group.
- **/auth/v1/groups/groups_summary: GET**
  Gets the list of groups and their domain details.

/auth/v1/groups: GET

Gets the list of groups that are created in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√	√	√[1]

Data admin	Data user	Collection Admin	Admin	Service user
[1]The specified group must be one that is assigned to the requesting user ID. Otherwise the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the uses that are created in the system.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups
```

Response:

```
(200 OK):
{
    "groups": [
        {
"id": "a51f84bfeb034b12a9a362109d78bec3",
            "name": "test-group1",
            "description": "Test Group 1",
            "domain": "default"
        },
        {
"id": "b62g95bfeb034b12a9a362109d89cfd4",
            "name": "test-group2",
            "description": "Test Group 2",
            "domain": "default"
        }
    ]
}
```

/auth/v1/groups: POST

Creates a group in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	√	√	X

Synopsis of the request URL

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X POST
https://<data_cataloging_host>/auth/v1/groups -d '<json details of group>'
```

You can specify the following details of a group:

- "name": Name of the group.
- "description": A meaningful description for the group.
- "domain_id": The domain to which the group belongs.

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

The following example shows how to create a group.

Request:

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/groups -d '{"name": "test-group"}'
```

Response:

```
201 Created
```

/auth/v1/groups/<group ID>: GET

Gets the details of a specific group.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√[1]	√[1]	√[1]	√	√[1]
[1]The specified group must be one that is assigned to the requesting user ID. Otherwise the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/<group_ID>
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of a specific group.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/a51f84bfeb034b12a9a362109d78bec3
```

Response:

```
(200 OK):
{
    "id": "a51f84bfeb034b12a9a362109d78bec3",
    "name": "test-group1",
    "description": "Test Group 1",
    "domain": "default"
}
```

/auth/v1/groups/<group_ID>: PATCH

Updates the attributes of an existing group.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/groups/<group_ID> -X PATCH -d '<json details to update>'
```

Supported request types and response formats

You can update the following attributes of a group.

- Name
- Description
- Domain ID

You can update one or more attributes in a single call.
Supported request types:

- PATCH

Supported response formats:

- JSON

Examples

The following example shows how to update the attributes of a group:

Request:

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/groups/a51f84bfeb034b12a9a362109d78bec3 -X PATCH -d '{"name": "AdminGroup",
"description":"Verifies installations", "domain_id": "Default"}'
```

Response:

```
200 OK
```

/auth/v1/groups/<group_ID>: DELETE

Deletes a specific user group.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/<group_id> -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

The following example shows how to delete a group.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/a51f84bfeb034b12a9a362109d78bec3 -X
DELETE
```

Response:

```
204 No Content
```

/auth/v1/groups/<group_ID>/collections: GET

Gets the details of the collections that are assigned in a group.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/<group_ID>/collections
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the collections assigned within a user group.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' 
https://<data_cataloging_host>/auth/v1/groups/a51f84bfeb034b12a9a362109d78bec3/collections
```

Response:

```
((200 OK):
[
    {
        "description": "This is a test collection",
        "domain_id": "default",
        "enabled": true,
        "id": "51d5bf6c7c4e4fd3b35c53ca91d95705",
        "is_domain": false,
        "name": "test-collection",
        "parent_id": "default",
        "tags": [
            "tag1",
            "tag2"
        ]
    }
]
```

/auth/v1/groups/<group_ID>/roles: GET

Gets the details of the user roles that are assigned to a group.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/<group_ID>/roles
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the user roles that are assigned to a user group.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/
a51f84bfeb034b12a9a362109d78bec3/roles
```

Response:

```
(200 OK):
[
    {
        "domain_id": "default",
               "id": "1bdcad2ec7f2414793e71b8b505515b1",
               "name": "datauser"
    }
]
```

/auth/v1/groups/<group_id>/role_assignments: GET

Returns a list of the roles and associated collections assigned to the group where applicable.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	√	√	X

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/<group_id>/role_assignments
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the roles that is assigned to a specific group:

Request:

```
curl -k -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/groups/a51f84bfeb034b12a9a362109d78bec3/role_assignments
```

Response:

```
Returns 200 OK:
{
    "roles":[
        {
            "collections":[
                {
                    "description":"Bootstrap project for initializing the cloud.",
                    "domain_id":"default",
                    "enabled":true,
                    "id":"09dd9136eaa6416784d6c5bdf3fa2f78",
                    "is_domain":false,
                    "name":"spectrum-discover",
                    "parent_id":"default",
                    "tags":[
                    ]
                },
                {
                    "description":"kutle",
                    "domain_id":"default",
                    "enabled":true,
                    "id":"40b0ea6fca7a4bafa38581cb508aa373",
                    "is_domain":false,
                    "name":"some-collection",
                    "parent_id":"default",
                    "tags":[
                    ]
                }
            ],
            "domain_id":null,
            "id":"5f365cf289d14c93a986b8bea976f92b",
            "name":"collectionadmin"
        }
    ]
}
```

/auth/v1/groups/<group_ID>/users: GET

Gets the details of the users who are part of a specific group.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/<group_ID>/users
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of the users that are part of a user group.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/
a51f84bfeb034b12a9a362109d78bec3/users
```

Response:

```
(200 OK):
[
    {
        "domain_id": "default",
               "email": "testuser@ibm.com",
               "enabled": true,
               "id": "dfdb2cadca0d4e9da26e6c40e5cadbd5",
               "name": "datauseruser",
               "options": {},
               "password_expires_at": null
    },
    {
        "domain_id": "default",
               "email": "testuser@ibm.com",
               "enabled": true,
               "id": "106999c5eb7f4148932be6d24b6b772c",
               "name": "test-datauser-1",
               "options": {},
               "password_expires_at": null
    }
]
```

/auth/v1/groups/<group_ID>/user/<user_ID>: DELETE

Removes a specific user from a group.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/<group_id>/user/<user_ID> -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

The following example shows how to remove a user from a group.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/groups/
a51f84bfeb034b12a9a362109d78bec3/user/4ee5b47f439640d29b6fac7253a64290 -X DELETE
```

Response:

```
204 No Content
```

/auth/v1/groups/<group_ID>/user/<user_ID>: PUT

Assigns a user to a group.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k -H 'Content-Type: application/json' -H ‘Authorization: Bearer <token>
https://<data_cataloging_host>/auth/v1/groups/<group_ID>/user/<user_ID> -X PUT
```

Examples

The following example shows how to assign a user to a group:

Request:

```
curl -k -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/groups/a51f84bfeb034b12a9a362109d78bec3/user/4ee5b47f439640d29b6fac7253a64290  -X PUT
-H "Content-Type: application/json"
```

Response:

`200` if the command is successful.

/auth/v1/groups/groups_summary: GET

Gets the list of groups and their domain details.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√	√	√[1]
[1] The specified user ID must be the same as the requesting user ID. Otherwise the command fails with error 403, "Not authorized". Data users can access this API to retrieve details only for the groups to which they belong.				

Synopsis of the request URL

```
curl -k -H ‘Authorization: Bearer <token>’ https://<data_cataloging_host>/auth/v1/groups/groups_summary
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the domain names to which the group is associated.

Request:

```
curl -k -H ‘Authorization: Bearer <token>’ https://<data_cataloging_host>/auth/v1/groups/groups_summary
```

Response:

```
(200 OK):
{
        "groups": [
                {
                        "description": "",
                        "domain_id": "default",
                        "domain_name": "Default",
                        "id": "86f7a42ea54d4659bb60fc6d5c620f2d",
                        "name": "bugfix5"
                },
                {
                        "domain_id": "a69001e854174dc1a4e0c98f712744bb",
                        "domain_name": "lldap",
                        "id": "d3b1ae0d1b5efb0c064797bf8239165cce32aaf3c057151279198a780bc8cf0b",
                        "name": "Sales"
                },
                {
                        "domain_id": "a69001e854174dc1a4e0c98f712744bb",
                        "domain_name": "lldap",
                        "id": "18326a0d724b4ffaa26a228fc88b6852198a1d10a3a51bc732632afe4d112dc9",
```

```
                        "name": "HumanResources"
                }
        ]
}
```

Managing collections

You can create and manage collections by using REST APIs.

- /auth/v1/collections: GET
  Gets the details of the collections that are available in the system.
- /auth/v1/collections: POST
  Creates a collection.
- /auth/v1/collections/<collection_ID>: GET
  Gets the details of a specific collection.
- /auth/v1/collections/<collection_ID>: PATCH
  Updates the attributes of an existing collection.
- /auth/v1/collections/<collection_ID>: DELETE
  Deletes a specific collection.
- /auth/v1/collections/<collection_ID>/groups: GET
  Gets a list of the user groups that can access a collection.
- /auth/v1/collections/<collection_ID>/users: GET
  Gets a list of the users who can access a collection.
- /auth/v1/collections/<collection_id>/role_assignments: GET
  Returns a list of the users and groups and their roles that are assigned to the collection.

/auth/v1/collections: GET

Gets the details of the collections that are available in the system.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√[1]	√	X
[1] The list that is returned comprises the set of collections that are assigned to the requesting User ID. If no collections are assigned, then an empty response is returned.				

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/collections
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of collections that are available in the system.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/collections
```

Response:

```
(200 OK):
{
    "collections": [
        {
            "description": "Bootstrap project for initializing the cloud.",
            "domain_id": "default",
            "enabled": true,
            "id": "dfd993527cb34b0bbcd700fab121264e",
            "is_domain": false,
            "name": "spectrum-discover",
            "parent_id": "default",
            "tags": []
        },
        {
            "description": "This is a test collection",
            "domain_id": "default",
            "enabled": true,
```

```
            "id": "51d5bf6c7c4e4fd3b35c53ca91d95705",
            "is_domain": false,
            "name": "test-collection",
            "parent_id": "default",
            "tags": [
                "tag1",
                "tag2"
            ]
        }
    ]
}
```


/auth/v1/collections: POST

Creates a collection.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X POST
https://<data_cataloging_host>/auth/v1/collections -d '{details of the collection}'
```

The details that can be specified are:

- Name
- Description
- Tags
  Important: IBM Data Cataloging does not use collection tags. These collection tags are not related to the IBM Data Cataloging metadata tags.

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

The following example shows how to create a collection.

Request:

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/collections -d '{"name": "test-collection", "description": "Test Collection 1","tags":
["tag1", "tag2"]}'
```

Response:

```
201 Created
```

/auth/v1/collections/<collection_ID>: GET

Gets the details of a specific collection.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	√[1]	√[1]	√	X
[1] The specified collection must be the one that is assigned to the requesting user ID. If no collection is assigned, then the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/collections/<collection_ID>
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the details of a specific collection.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/collections/51d5bf6c7c4e4fd3b35c53ca91d95705
```

Response:

```
(200 OK):
{
    "description": "This is a test collection",
    "domain_id": "default",
    "enabled": true,
    "id": "51d5bf6c7c4e4fd3b35c53ca91d95705",
    "is_domain": false,
    "name": "test-collection",
    "parent_id": "default",
    "tags": [
        "tag1",
        "tag2"
    ]
}
```

/auth/v1/collections/<collection_ID>: PATCH

Updates the attributes of an existing collection.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	√	√	X

Synopsis of the request URL

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/collections/<collection_ID> -X PATCH
-d '<json_details_to_update>'
```

Supported request types and response formats

You can update the following attributes of a collection.

- Name
- Description
- Tags

You can update one or more attributes in a single call.
Supported request types:

- PATCH

Supported response formats:

- JSON

Examples

The following example shows how to update the attributes of a collection:

Request:

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/collections/a51f84bfeb034b12a9a362109d78bec3
-X PATCH -d '{"name": "General", "description":"ThirdQuarter", "tags": {"ProjectNumber":"1404", "DepartmentCode":"H8AC"}'
```

Response:

```
200 OK
```

/auth/v1/collections/<collection_ID>: DELETE

Deletes a specific collection.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Content-Type: application/json' -H ‘Authorization: Bearer <token>’ -X DELETE
https://<data_cataloging_host>/auth/v1/collections/<collection_ID>
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

The following example shows how to delete a specific collection.

Request:

```
curl -k  -H ‘Authorization: Bearer <token>’ https://<data_cataloging_host>/auth/v1/collections/dfd993527cb34b0bbcd700fab121264e
-X DELETE
```

Response:

```
204 No Content
```

/auth/v1/collections/<collection_ID>/groups: GET

Gets a list of the user groups that can access a collection.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	[1]√	√	X
[1] The specified collection must be one that is assigned to the requesting user ID. If no collection is assigned, then the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k  -H ‘Authorization: Bearer <token>’ https://<data_cataloging_host>/auth/v1/collections/<collection_ID>/groups
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get details of user groups that belong to a specific collection.

Request:

```
curl -k  -H ‘Authorization: Bearer <token>’
https://<data_cataloging_host>/auth/v1/collections/a51f84bfeb034b12a9a362109d78bec3/groups
```

Response:

```
(200 OK):
[
    {
        "description": "",
        "domain_id": "default",
        "id": "d1c1f64d62df4585ad8c76b345e0a651",
        "name": "test-group-1"
```

```
    }
]
```

/auth/v1/collections/<collection_ID>/users: GET

Gets a list of the users who can access a collection.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	[1]√	√	X
[1] The specified collection must be one that is assigned to the requesting user ID. If no collection is assigned, then the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/collections/<collection_ID>/users
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get details of users who belong to a specific collection.

Request:

```
curl -k  -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/collections/a51f84bfeb034b12a9a362109d78bec3/users
```

Response:

```
(200 OK):
[
    {
        "domain_id": "default",
        "email": "testuser@ibm.com",
        "enabled": true,
        "id": "106999c5eb7f4148932be6d24b6b772c",
        "name": "test-datauser-1",
        "options": {},
        "password_expires_at": null
    },
    {
        "domain_id": "default",
        "enabled": true,
        "id": "35470c08a699441394a0822da0161a16",
        "name": "sdadmin",
        "options": {},
        "password_expires_at": null
    },
    {
        "domain_id": "default",
        "email": "testuser@ibm.com",
        "enabled": true,
        "id": "dfdb2cadca0d4e9da26e6c40e5cadbd5",
        "name": "datauseruser",
        "options": {},
        "password_expires_at": null
    }
]
```

/auth/v1/collections/<collection_id>/role_assignments: GET

Returns a list of the users and groups and their roles that are assigned to the collection.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
√	X	[1]√	√	X
The specified collection must be one that is assigned to the requesting user ID. If no collection is assigned, then the command fails with error 403, "Not authorized".				

Synopsis of the request URL

```
curl -k -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/collections/<collection_id>/role_assignments
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get the list of groups to which the collection belongs:

Request:

```
curl -k -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/collections/a51f84bfeb034b12a9a362109d78bec3/role_assignments
```

Response:


```
{
    "groups": [
       {
            "description": "",
            "domain_id": "default",
            "id": "fccc7d34ee1d43128f2fda479f9f63e9",
            "name": "test_data_user_darko_group",
            "roles": [
                {
                    "domain_id": null,
                    "id": "73f471083b444117ba469d867c98c9b0",
                    "name": "datauser"
                }
            ]
        }
    ],
    "users": [
        {
            "description": "Test collection admin 3",
            "domain_id": "default",
            "email": "test_cadmin3@x.cmo",
            "enabled": true,
            "id": "154e1a1bb1fc40569924a0bd12ef7c0c",
            "name": "test_cadmin3",
            "options": {},
            "password_expires_at": null,
            "roles": [
                {
                    "domain_id": null,
                    "id": "55567000f3594155a1158edb6b5a9158",
                    "name": "collectionadmin"
                }
            ]
        }
}
```


Managing domains

You can create and manage user domains by using REST APIs.

- **/auth/v1/domains/<domain_ID>: GET**
  Gets the details of a specific domain.
- **/auth/v1/domains: POST**
  Creates a domain.
- **/auth/v1/domains/<domain_ID>: PATCH**
  Updates the attributes of an existing domain.
- **/auth/v1/domains/<domain_ID>: DELETE**
  Deletes a specific domain.

/auth/v1/domains/<domain_ID>: GET

Gets the details of a specific domain.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/domains/<domain ID>
```

Supported request types and response formats

Supported request types:

- GET

Supported response formats:

- JSON

Examples

The following example shows how to get details of a specific domain.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/domains/7a2251243611437cae707fb1aa5acf2c
```

Response:

```
(200 OK):
{
    "description": "Open LDAP Domain",
    "enabled": true,
    "id": "7a2251243611437cae707fb1aa5acf2c",
    "name": "ldap",
    "tags": []
}
```

/auth/v1/domains: POST

Creates a domain.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>' -X POST
https://<data_cataloging_host>/auth/v1/domains -d '<json domain details>'
```

In case of LDAP domain, you can specify the following attributes:

- "id"
- "type"
- "host"
- "port"
- "binddn"
- "bindpassword"
- "basedn"

Supported request types and response formats

Supported request types:

- POST

Supported response formats:

- JSON

Examples

The following example shows how to create the domain and pull in a list of LDAP users.

```
Type LDAP
Name LDAPForum
URL ( LDAP IP or host.domainame ) ldap.forumsys.com
Port 389
```

```
user cn=read-only-admin,dc=example,dc=com
password password
Suffix / Base DN dc=example,dc=com
Group Name Attribute cn
Group Object Class
Group Tree DN ou=mathematicians,dc=example,dc=com
Username Attribute uid
User Object Class person
User Tree DN dc=example,dc=com
```

/auth/v1/domains/<domain_ID>: PATCH

Updates the attributes of an existing domain.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/domains/<domain_ID> -X PATCH -d '<json_details_to_update>'
```

Supported request types and response formats

In an LDAP domain, you can update the following attributes:

- "id"
- "type"
- "host"
- "port"
- "binddn"
- "bindpassword"
- "basedn"

You can update one or more attributes in a single call.
Supported request types:

- PATCH

Supported response formats:

- JSON

Examples

The following example shows how to update the attributes of a domain:

Request:

```
curl -k -H 'Content-Type: application/json' -H 'Authorization: Bearer <token>'
https://<data_cataloging_host>/auth/v1/domains/a51f84bfeb034b12a9a362109d78bec3 -X PATCH
-d '{"id":"7a2251243611437cae707fb1aa5acf2c", "type":"Default"}'
```

Response:

```
200 OK
```

/auth/v1/domains/<domain_ID>: DELETE

Deletes a specific domain.

The following table shows which roles can access this REST API endpoint:

Table 1. Access by role

Data admin	Data user	Collection Admin	Admin	Service user
X	X	X	√	X

Synopsis of the request URL

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/domains/<domain_id> -X DELETE
```

Supported request types and response formats

Supported request types:

- DELETE

Supported response formats:

- JSON

Examples

The following example shows how to delete a specific domain.

Request:

```
curl -k  -H 'Authorization: Bearer <token>' https://<data_cataloging_host>/auth/v1/domains/
7a2251243611437cae707fb1aa5acf2c -X DELETE
```

Response:

```
204 No Content
```

Data protection

Data protection features in IBM Spectrum Discover support backup, disaster recovery, and restore operations to protect catalog metadata and system configurations.

- **Backup and restore**
  IBM Spectrum® Discover includes a set of procedures for safely backing up and restoring your database and file system for both Virtual appliance and the Red Hat® OpenShift® deployment.
- **Disaster recovery procedures**
  Disaster recovery procedures describe how to recover an IBM Spectrum Discover system from the loss of a single node or an entire deployment, including deployments on Red Hat OpenShift.

Backup and restore

IBM Spectrum® Discover includes a set of procedures for safely backing up and restoring your database and file system for both Virtual appliance and the Red Hat® OpenShift® deployment.

- **Backup and restore scripts**
  Backup and restore script is written in Python to run Db2 commands inside the Db2u pod in the OpenShift environment.

Backup and restore scripts

Backup and restore script is written in Python to run Db2 commands inside the Db2u pod in the OpenShift® environment.

Before you begin

1. Install the Python 3 and `oc` CLI and logged in to the environment.
2. Export the project name variable with the name of the environment. The default variable is `ibm-data-cataloging`.
3. For restore, check whether the `dcs-Backup` folder contains the tar.gz file that is generated from the previous backup.

About this task

The backup and restore script works in various steps, which require a stable IBM Data Cataloging environment. It must not be in use because it needs to scale IBM Data Cataloging pods to 0. The process consists of the following steps:

- Scale pods to 0
- Shutdown Db2
- Run backup or restore with Db2 move.
- Get the tar archive of backup files.
- Clean files from the Db2 pod
- Start Db2
- Scale pods to 1

Procedure

Follow the steps to run the script:

a. Get the script from the utility scripts repo resources repo then run it with the following command.

```
python3 backup_restore.py
```

b. The script contains a basic menu with the following options:
   i. Backup
   ii. Restore
   iii. Shutdown Db2

iv. Start Db2
v. Exit

The first option does not need any steps. It must start with the steps that are described in the overview. At the end, the backup tar.gz file must be in a directory that is named `dcs-Backup`, where the script runs.

The second option lists `dcs-Backup`tar.gz files. You can select one from those and run it with the steps that are described in the overview. After the restore is done, you need to go to the IBM Data Cataloging user interface and refresh the summary database to see the imported records.

After one of the first two options is finished, the system starts the scale pods. Some of them can fail in crashloop or error, but after the depending pods are running, they must recover after running state pods.

The third and fourth options are only used if the backup fails, leaving the environment in `maintenance mode`. To restore the state of IBM Data Cataloging, you need to scale pods manually.

Disaster recovery procedures

Disaster recovery procedures describe how to recover an IBM Spectrum Discover system from the loss of a single node or an entire deployment, including deployments on Red Hat OpenShift.

Use this process to recover from a disaster that involves an IBM Data Cataloging system and discusses the following scenarios:

- Recovery from the entire loss of a single node IBM Data Cataloging deployment.
- Recovery from the entire loss of an IBM Data Cataloging deployment on Red Hat® OpenShift®.

- **Preparations for disaster recovery**
  Before you need to perform disaster recovery, there are several tasks that must be accomplished to ensure the ability to recover.
- **Running disaster recovery for Red Hat OpenShift**
  The disaster recovery procedure for Red Hat OpenShift is similar to its backup and restore procedures.

Preparations for disaster recovery

Before you need to perform disaster recovery, there are several tasks that must be accomplished to ensure the ability to recover.

Procedure

1. Take a backup of the IBM Data Cataloging system as described in Backup and restore.
2. Record the installation configuration, including:
   - Network settings
   - Storage settings
   - CPU and memory
   - IBM Data Cataloging version
3. Ensure that the physical and virtual infrastructure is available to replace the system that might fail.
4. You cannot recover to a different version of IBM Data Cataloging. If a change of version is required, you need to recover to the current version before you install the upgrade.
   If you perform a recovery on a system that is upgraded, it is recovered directly to the upgraded IBM Data Cataloging version. For example, if IBM Data Cataloging 2.0.0.3 is deployed and then upgraded to IBM Data Cataloging 2.0.1 and then recovered from a failure, the recovery goes directly to the 2.0.1.

Running disaster recovery for Red Hat OpenShift

The disaster recovery procedure for Red Hat® OpenShift® is similar to its backup and restore procedures.

On successfully repairing or rebuilding your Red Hat OpenShift cluster, you can restore your system from an appropriate backup.

For more information, see Restoring deployed on.

.

Using a third-party data movement application to manage data

Introduction to data movement with IBM Data Cataloging.

Before you begin

- The third-party application must be registered with IBM Data Cataloging.
- IBM Data Cataloging must scan both the source and destination data sources before the data movement.
- The application must be configured to access the same source and destination data sources.

You can see the third-party application documentation for details.

About this task

The capabilities of IBM Data Cataloging can be extended by applications; for example, data mover applications can be used to move data between data sources, based on the IBM Data Cataloging insights. Applications register with IBM Data Cataloging providing details of the operations they support and the parameters they need to fulfill the operation.

The following table lists the supported data movement operations:

Table 1. Data movement operations

Operation	Source Type	Destination Type
MOVE	NFS S3 COS SMB/CIFS	NFS S3 COS SMB/CIFS
COPY	NFS S3 COS SMB/CIFS	NFS S3 COS SMB/CIFS
TIER	NFS S3 COS SMB/CIFS	NFS S3 COS SMB/CIFS

Note:
For now, it is not possible for an application to register new operations beyond the records listed in the Data movement operations table.

You can create data movement policies that identify files and objects that are candidates for data operations, by using registered data movement Applications. The IBM Data Cataloging policy engine generates job request messages in JSON format. It contains a batch of files or objects with the source and target information, and any additional options such as preserving source system timestamps.

The job request messages are put onto a Kafka egress topic. The data movement application reads the messages from the topic and performs the necessary data movement operation, and provides response messages in JSON format on a Kafka ingress topic.

IBM Data Cataloging can interact with third-party data movement applications to move, copy, or tier data between data sources. Create data management policies on IBM Data Cataloging, and specify the set of documents to process by using a policy filter. The policy filter can be based on the system metadata or the custom metadata of documents that are collected by IBM Data Cataloging.

The third-party application registers with IBM Data Cataloging providing the operations that it supports (move, copy or tier). For each operation, it gives the list of parameter values that it needs to perform the operation.

When you create the data management policy in IBM Data Cataloging the user defines the filter, the parameter values, and when the policy must run. When the policy runs, IBM Data Cataloging sends the list of files to the data movement application and any additional parameters. The application processes the files and returns a status summary to IBM Data Cataloging. The summary is displayed to the user.

Note:

- During data migration, the migrated files need to preserve the IBM Data Cataloging tags. You can follow a manual procedure to preserve the tags. For more information, see Preserving tags during data movement.
- A third-party data movement application can register a schema, defining the parameters it requires for data movement. You need to set these parameters in the application user interface (UI) while creating the policy. Invalid parameters cause an error when you submit a policy.

  However, in few cases, due to the complexity of the schema, the error may not point to the precise location of the problem or may indicate that a valid parameter is invalid. Therefore, when an error is raised when you submit a policy, and it is not definite where the problem is, it is recommended that you check all the policy parameters.

Procedure

1. Log in to the IBM Data Cataloging GUI.
2. Go to Admin > Management Policies.
3. To create a policy, click Add Policy.
4. Click the slider control and set the status to one of the following values:

   Active
   : An active policy runs whenever its scheduling event is reached.

   Inactive
   : An inactive policy does not run even when its scheduling event is reached (including a NOW event).

5. Enter a policy name.
6. Enter a policy filter.
   The policy filter includes the criteria for selecting the files for moving or copying. For example, filetype="pdf" selects all files of type PDF.
   Note:
   - IBM Data Cataloging tracks the last known migration status for each file in the ‘STATE’ facet. This can be leveraged when defining data movement policies to either target or avoid files in a particular state. The following values represent the file status:

migrtd (migrated)
: File contents are only present on the target system but a stub file exists on the source.

resdnt (resident)
: File contents are only present on the source system.

- The following are the scenarios where it is useful to filter the file status by 'STATE' facet are:
  - In a TIER policy where files that are already migrated shouldn't be migrated again.
  - In a COPY policy where files that are migrated shouldn't be copied as this would result in a recall of the data.
  - In a TIER policy where some files are being recalled in preparation for a workload. Any files that are already resident don't need to be recalled.
- To target files that are resident, simply add "state = 'resdnt'" to the filter criteria. To target files that are NOT resident, simply add "state <> 'resdnt'" to the filter criteria.

7. To select the policy type, click Next Step.
8. Select MOVE, COPY, or TIER as the policy type.
9. Select the agent name as the Agent.
10. Enter the remaining parameters. The parameters that are displayed depend on the application, and these parameters might include:

    Source connection type
    : Indicates the type of connection that the files currently reside on.

    Source connection
    : Indicates the name of the connection that the files currently reside on.

    Destination connection type
    : Indicates the type of connection that the files are being moved or copied to.

    Destination connection
    : Indicates the name of the connection name that the files are being moved or copied.

    Force migrate
    : Indicates whether to force demigration or recall of the file at source location when it is migrated to other location before you perform the operation.

    Overwrite
    : Indicates what value to give when a file exists at the destination.

    Preserve attributes, timestamp, or permissions
    : Indicates the parameters to control whether the files metadata is preserved.

11. To enter a schedule, select Next Step.
    The schedule indicates when you want to start the move of the copy.
12. To review the policy, select Next Step.
13. To create the policy, select Submit. The policy runs at the scheduled time.
14. When the policy runs or completes an execution status summary, view it by clicking Policy Preview.

- **Data movement with IBM Data Cataloging and Moonwalk**
  Policy-based data movement and management with integration of IBM Data Cataloging and Moonwalk applications.
- **Preserving tags during data movement**
  Preserve tags during data movement by generating a report of files and their tags before migration, then reapplying the tags using AUTOTAG policies after the destination connection is rescanned.

Data movement with IBM Data Cataloging and Moonwalk

Policy-based data movement and management with integration of IBM Data Cataloging and Moonwalk applications.

Moonwalk data mover application

Moonwalk is a heterogeneous Data Management System. It automates and manages the movement of data from primary storage locations to lower-cost file systems, object stores, tape, or cloud storage services. Files can be moved from source storage locations to target storage locations. These files are demigrated transparently when accessed by a user or an application. Moonwalk also provides capability to copy and move files along with a range of Disaster Recovery options.

Moonwalk provides a data movement application that integrates with IBM Data Cataloging, enabling to move, copy, or migrate/tier data from the IBM Data Cataloging user interface.

The moonwalk application is certified against the IBM Data Cataloging 2.0.4.

For more information about Moonwalk application and integrating the application with IBM Data Cataloging, see Moonwalk documentation.

Preserving tags during data movement

Preserve tags during data movement by generating a report of files and their tags before migration, then reapplying the tags using AUTOTAG policies after the destination connection is rescanned.

Before you begin

Generate a report of the files to be moved or copied before you start the data movement process. This report includes the relevant tags for each source file. A sample report is shown in the following table:

Table 1. Report generated for moving files

Path	Filename	Filetype	Datasource	tagA	tagB
/	Chrysanthemum.jpg	jpg	dir1	image	value1
/	Lighthouse.jpg	jpg	dir1	image	value1
/	newtextfile.txt	txt	dir1	text	value2

Path	Filename	Filetype	Datasource	tagA	tagB
/	Hydrangeas.jpg	jpg	dir1	jpg	value1
/	Thumbs.db	db	dir1	unknown	value1
/	Koala.jpg	jpg	dir1	jpg	value1
/	Desert.jpg	jpg	dir1	jpg	value1

About this task

If IBM Data Cataloging tags must be preserved during data migration, then you can follow a manual procedure to ensure that the tags are preserved in the moved or the copied files.

Procedure

1. Run the data movement process.
2. Rescan the destination connection after the data movement completes successfully.
3. Refer to the information provided in the report, shown in Table 1 that was generated before you started the data movement process, to identify the files for which you need to preserve the tags.
   Note: You can define a filter to identify and select the correct set of files that must be tagged. For example, if we want to move files shown in Table 1 to a new datasource (ddsource), you can define filters that include the actions to be taken for the tag values. For a filter defined as shown:

   ```
   datasource in ('ddsource') and filetype in ('jpg')
   ```

   You can create an AUTOTAG policy that sets the value of **tagA** to **image**.
   This process assumes either of the two things.
   - No other files belonging to these filetypes exist in the destination datasource.
   - Those files can also be tagged with the **tagA** values.
4. Reapply the tags (for example, **tagA**) on the moved or copied files by using the auto tagging capability of IBM Data Cataloging.
   Note:
   You must perform a search using the above filters to check that it retrieves the correct documents that are being selected. You can also sort, based on the report headings in MS Excel, to identify a suitable filter.

   It is not easy to identify a suitable filter to reapply the **tagB** values shown in Table 1 which indicates the difficulty in manually reapplying the tags.
   This solution is feasible only where you can define a small number of filters that identify the groups of files to which you need to apply the tags. If filters cannot be defined then you must run several AUTOTAG policies, which might be an impractical and tedious process.

Monitoring IBM Data Cataloging

This section provides information on how to monitor the IBM Data Cataloging capabilities.

- **Monitoring data sources**
  You can use the Home page to monitor the data sources that are connected to your IBM Spectrum® Discover environment. Use the Data Source Connections page view details about data source connections.
- **Monitoring the IBM Data Cataloging service environment**
  You can monitor the health and status of the IBM Data Cataloging service environment and obtain audit log information.
- **Health check monitoring**
  Provides series of checkpoints to check IBM Data Cataloging health status.

Monitoring data sources

You can use the Home page to monitor the data sources that are connected to your IBM Spectrum® Discover environment. Use the Data Source Connections page view details about data source connections.

- **Viewing data source status**
  Use the Home page to monitor your environment for storage system capacity, used capacity, records indexed, and duplicate files. You can also view data usage for a specific area of your organization.
- **Viewing data source connections**
  Use the Data Source Connections page to view connection information for the data sources that are connected to your IBM Data Cataloging environment.
- **Recommended to move**
  In the IBM Data Cataloging dashboard, you can categorize data as Recommended to move.
- **Deleting or editing a connection**
  Use the following information to delete or edit a connection.

Viewing data source status

Use the Home page to monitor your environment for storage system capacity, used capacity, records indexed, and duplicate files. You can also view data usage for a specific area of your organization.

The data in the home page is updated periodically. The last update is indicated by a time stamp.

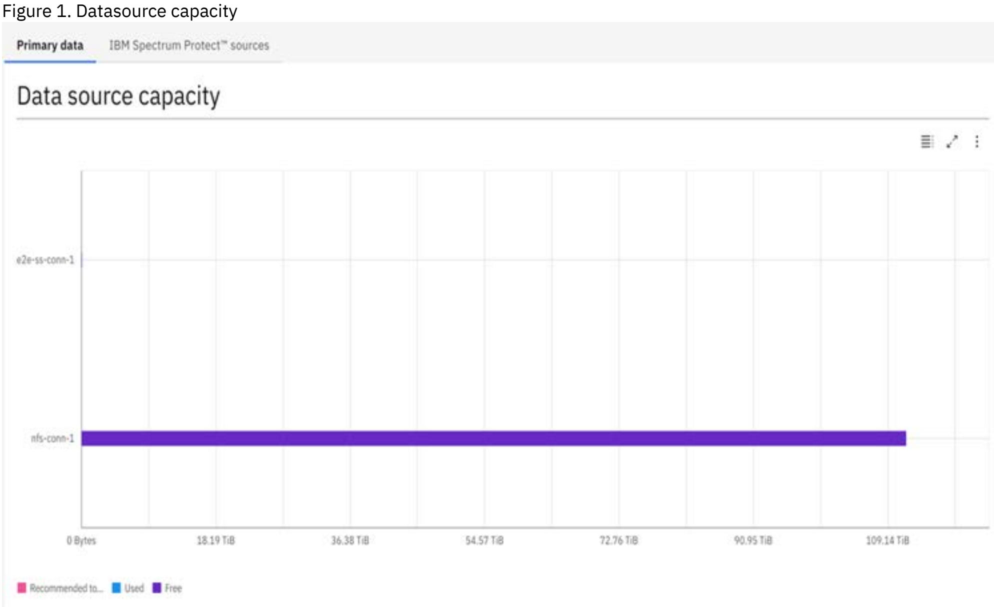
Viewing storage system capacity

Use the Data source capacity area to view capacity usage compared to the allocated capacity for all data sources that are registered with IBM Spectrum® Discover. The data sources can be a mixture of file systems and object vaults. A graph provides a convenient view of the current capacity of data sources and whether any are close to running out of space. This view also indicates the number of files to move or archive, based on user-defined policies.

Hover over a data source in the graph to view details about the data source. Click a data source to open the Search page and perform a search of the selected data source.

Note: Data sources that do not have data residing in them are not displayed in the graph.

Figure 1. Datasource capacity



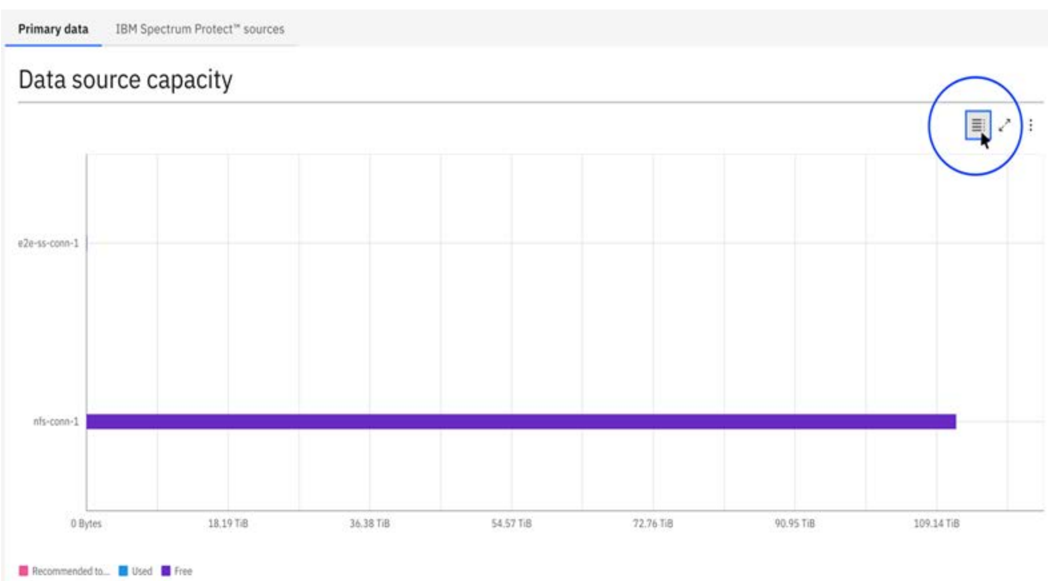
Understanding size and capacity differences

IBM Spectrum Discover collects size and capacity information. Generally:

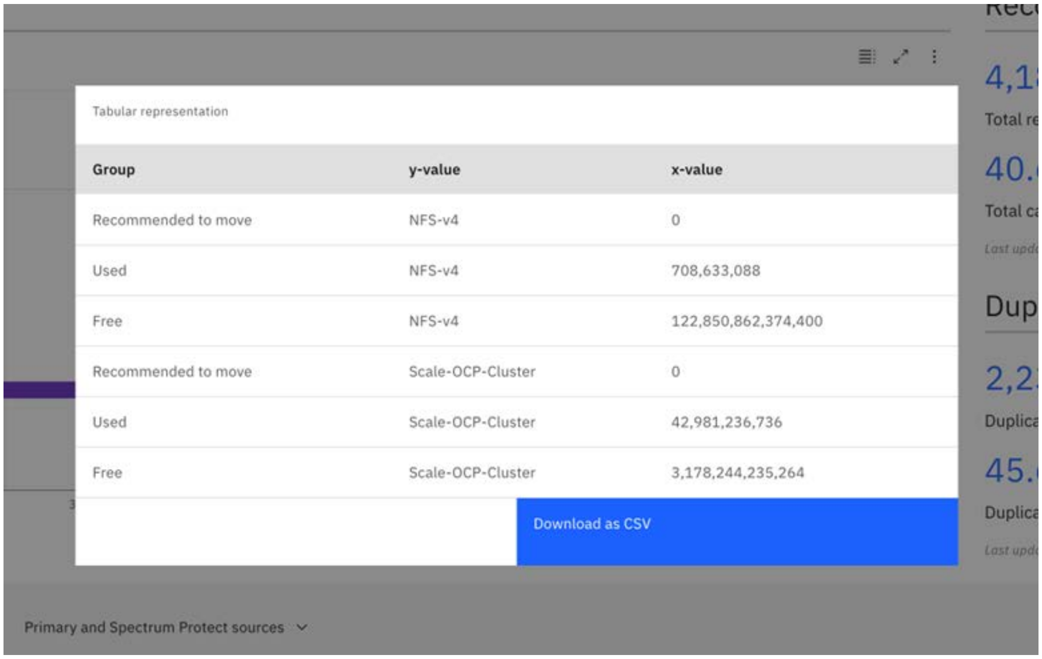
- Size refers to the size of a file or object in bytes.
- Capacity refers to the amount of space the file or object consumes on the source storage in bytes.

For objects, size and capacity values always match. For files, size and capacity values can be different because of file system block overhead or sparsely populated files.

Note: Storage protection overhead (such as RAID values or erasure coding) and replication overhead are not captured in the capacity values.
The Data source capacity area contains a view to visualize the data in a tabular representation. To see the view, click on the table button on the chart toolbar.



The tabular representation provides a complementary view to the analysis of the information on the chart. On the Group column, you can see the recommended to move, used, and free values that indicate the number of files to move or archive based on user-defined policies. On the y-value column, you can see the names of the data sources, and on the x-value column, you can see the size and capacity information in bytes.



Group	y-value	x-value
Recommended to move	NFS-v4	0
Used	NFS-v4	708,633,088
Free	NFS-v4	122,850,862,374,400
Recommended to move	Scale-OCP-Cluster	0
Used	Scale-OCP-Cluster	42,981,236,736
Free	Scale-OCP-Cluster	3,178,244,235,264

Viewing used capacity

Use the Capacity Used by area to view graphs with an aggregated display of capacity usage for selected metadata attributes. You can view capacity for both primary and backup sources. The graphs provide details about capacity usage by aggregating across different attributes that are available from standard system metadata.

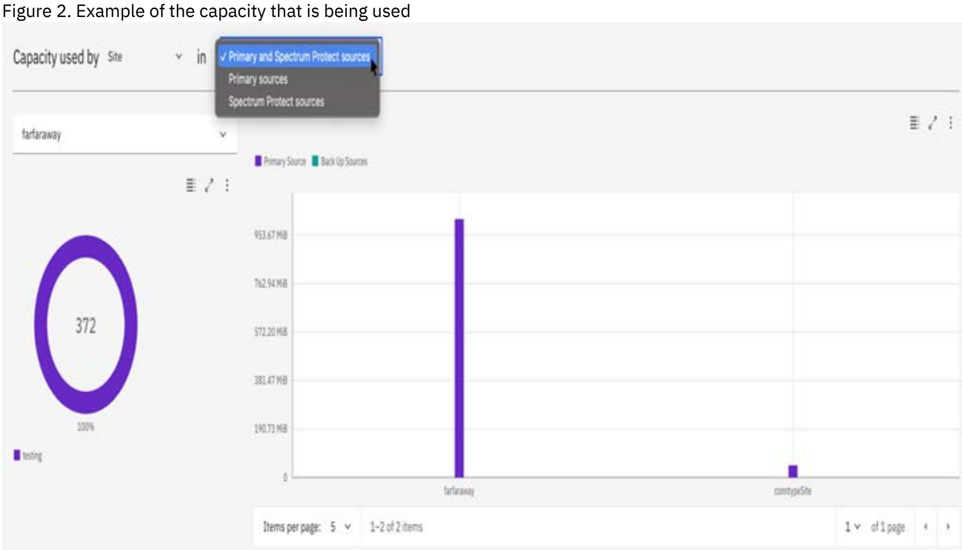
Use the Capacity Used by list to select an attribute and display the capacity consumers of that attribute in the graphs.

The Used graph displays the highest consumers of capacity for the selected attribute, in order of consumption.

The data source graph displays the percentage of overall usage per data source for the selected attribute. You can select a specific capacity consumer to display in the graph.

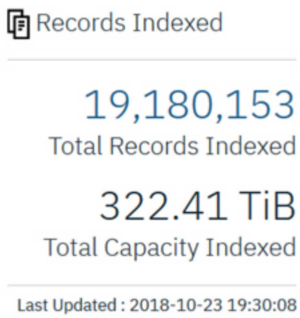
Hover over a value in a graph to view details. Click a value in a graph to open the Search page and search the selected item.

Figure 2. Example of the capacity that is being used



Viewing records indexed

Use the Records Indexed area to view both the total number of records and the capacity of the records that are indexed by IBM Spectrum Discover. This view provides a summary view of total storage usage.

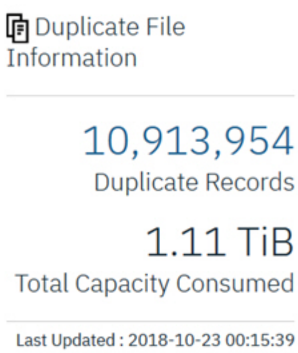


Click the Total Records Indexed value to open the Search page and perform a search of the indexed records.

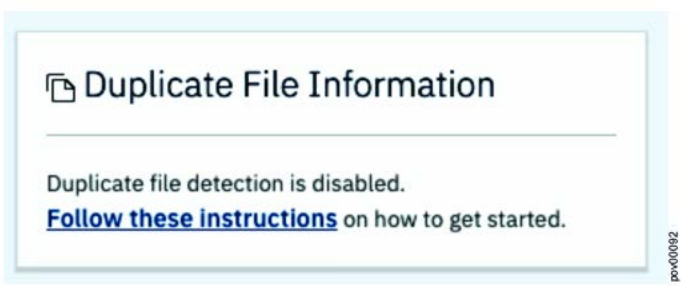
Viewing duplicate file information

Use the Duplicate File Information area to view information about possible duplicate files within the storage environment. Possible duplicate files are files with the same name and size but different paths or object names. The number of duplicates and the capacity that is consumed by these files is displayed. You can also use a report that provides detailed and sorted information for the potential duplicates.

Click the Duplicate Records value to open the Search page and perform a search of duplicate records.

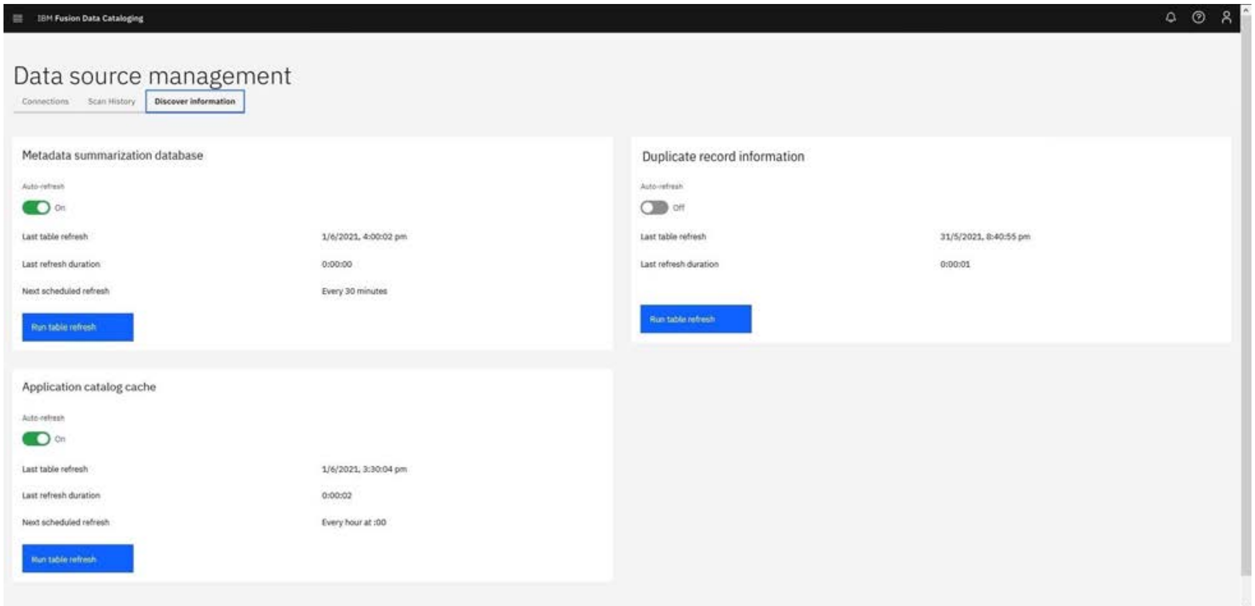


Identifying potential duplicates can be resource-intensive on IBM Data Cataloging. By default, the background task that refreshes potential duplicate information is disabled. However, you can update potential duplicate information either on demand or on a specific schedule. If you disable duplicate background task, the dashboard shows the following message:



To view and manage how often data in the home page is updated, navigate to Data connections > Discover database under Data source management window.

Figure 3. Run table refresh button in the Discover database window



From here, you can enable or disable the automatic updating of summary information. You can update information on the home page on demand by clicking the Run table refresh.

Viewing data source connections

Use the Data Source Connections page to view connection information for the data sources that are connected to your IBM Data Cataloging environment.

The following connections details are available:

Source Name
: A name that uniquely identifies the connection to the data source. A data source can have multiple connections.

Platform
: The platform of the data source - IBM Storage Scale system or IBM Cloud® Object Storage system.

Cluster
: The cluster address of the data source.

Data source name
: The full name of the data source.

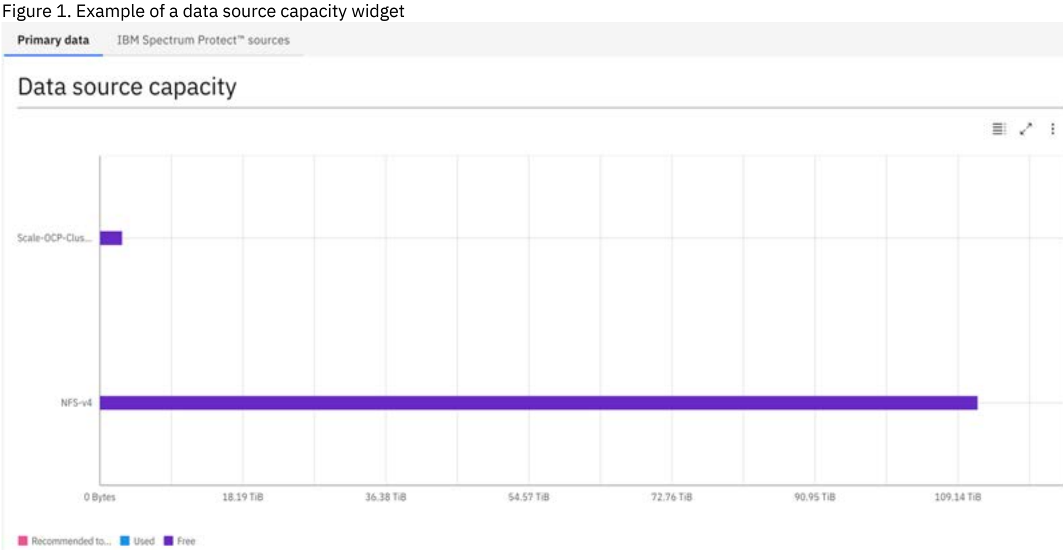
Site
: The physical location of the data source.

Recommended to move

In the IBM Data Cataloging dashboard, you can categorize data as Recommended to move.

Figure 1 shows an example of a data source capacity widget.

Figure 1. Example of a data source capacity widget



Use the **data source capacity** area to view capacity usage compared to the allocated capacity for all data sources that are registered with IBM Data Cataloging. The data sources can be a mixture of file systems and object vaults. A graph provides a convenient view of the current capacity of data sources and whether any are close to running out of space. This view also indicates the number of files to move or archive, based on user-defined policies.

Hover over a data source in the graph to view details about the data source. Click a data source to open the Search page and perform a search of the selected data source.

The data source capacity widget displays any files or objects that have the TEMPERATURE tag set to a value of ARCHIVE as Recommended to move. You can create an AUTOTAG policy to look for files and objects, which meet your archive criteria and set the TEMPERATURE tag to a value of ARCHIVE.

Any files that match the criteria of the AUTOTAG policy filter are tagged as ARCHIVE. The filter might be age-based or more complex. For example, the filter might match only certain file types, or files over some size threshold.

Figure 2 shows an example of a screen that shows the TEMPERATURE tag.

Figure 2. Example of a screen that shows the TEMPERATURE tag

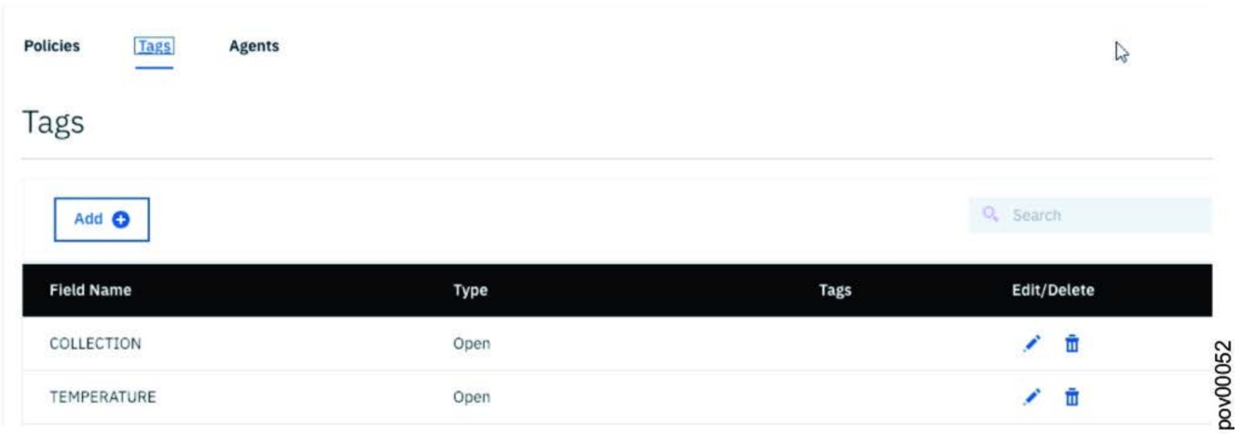
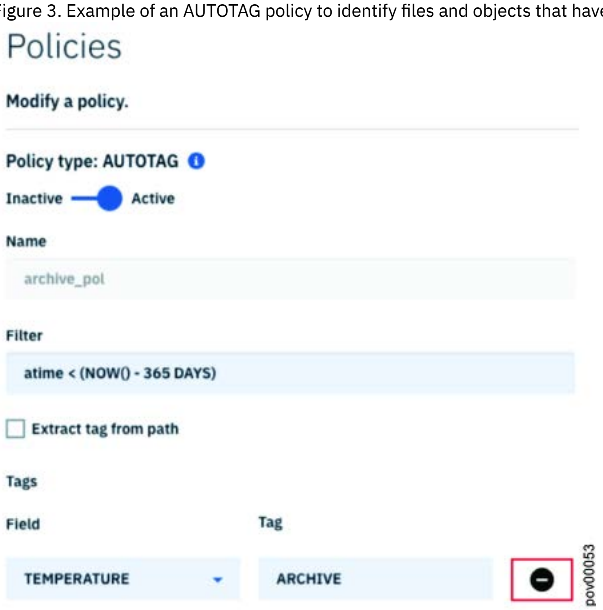


Figure 3 shows an example of an AUTOTAG policy to identify files and objects that have not been accessed for more than a year.

Figure 3. Example of an AUTOTAG policy to identify files and objects that have not been accessed in more than one year



Deleting or editing a connection

Use the following information to delete or edit a connection.

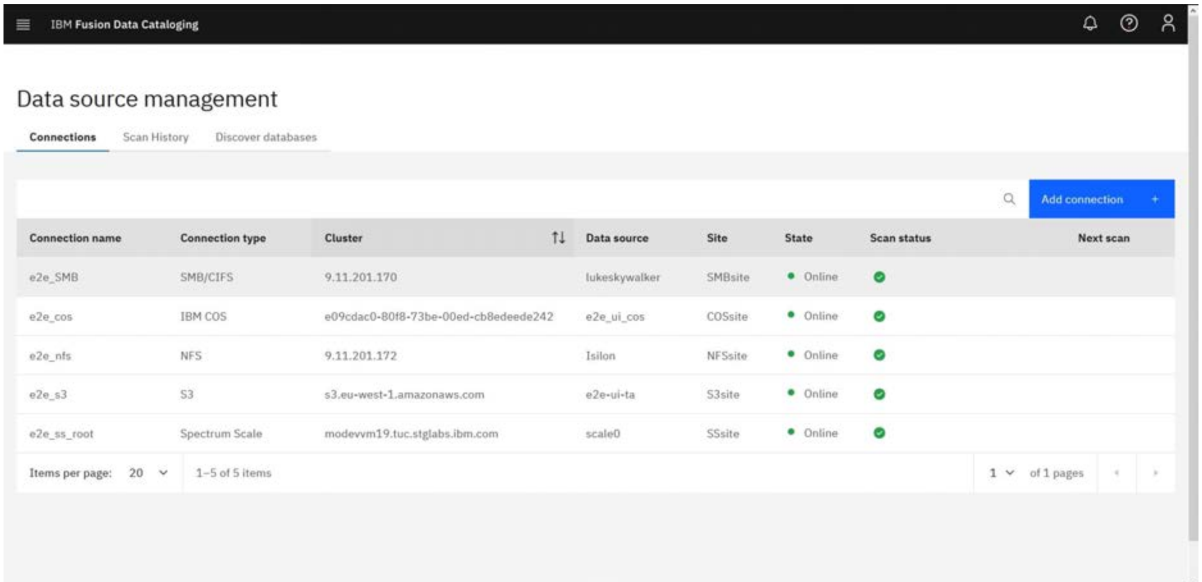
About this task

You can delete or edit a connection by using the graphical user interface.

Procedure

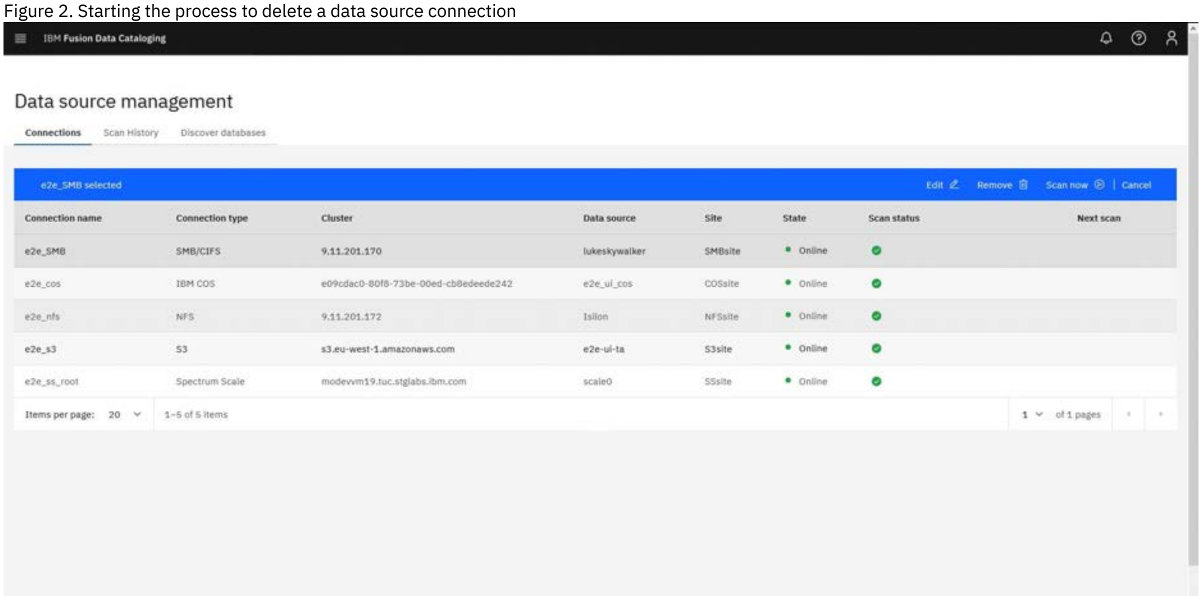
1. Click Data connections to display a listing of existing connections as shown in Figure 1

   Figure 1. Example of a listing of existing connections



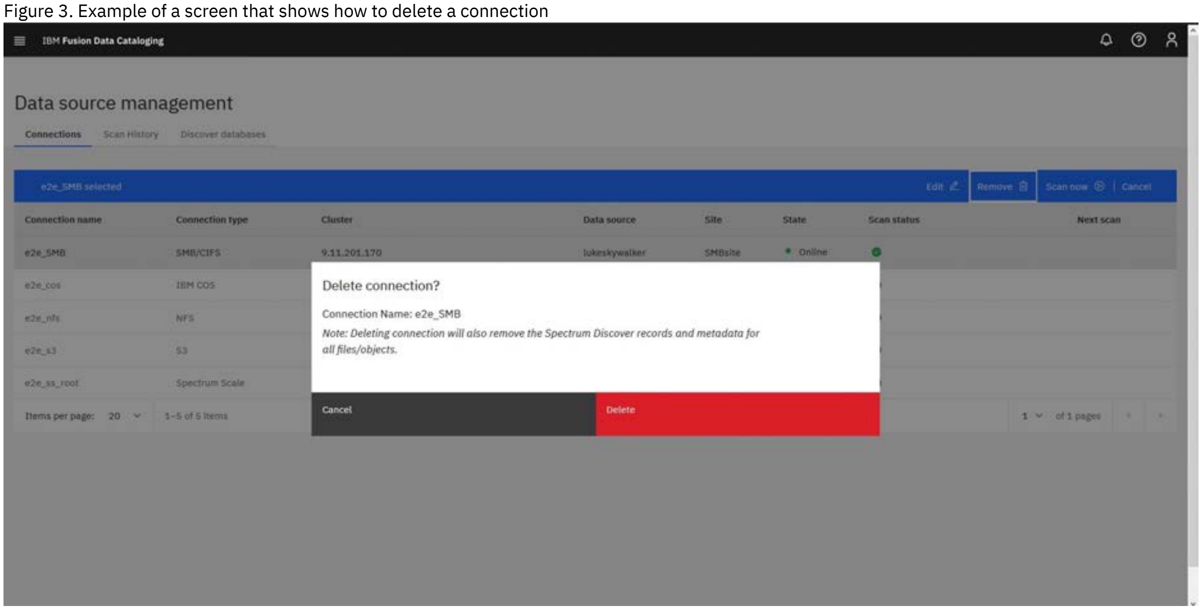
2. Click Remove to start the process to remove the data source connection as shown in Figure 2.

Figure 2. Starting the process to delete a data source connection



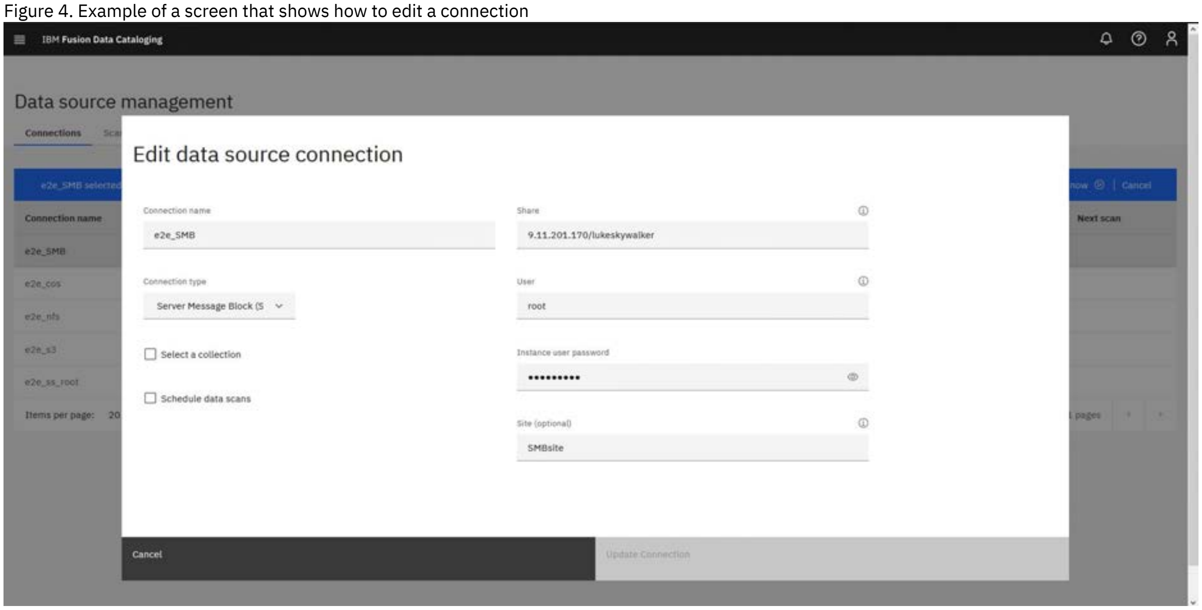
3. Clicking Remove displays a screen as shown in Figure 3. If you are sure you want to delete the connection, click Delete.

Figure 3. Example of a screen that shows how to delete a connection



4. To edit a connection, click Edit.
    a. Edit the appropriate fields in the window for Edit Data Source Connection.
    b. Click Update Connection.

Figure 4. Example of a screen that shows how to edit a connection



Monitoring the IBM Data Cataloging service environment

You can monitor the health and status of the IBM Data Cataloging service environment and obtain audit log information.

- **Audit log**
  Use the audit log entries to monitor activity of REST API calls within the IBM Spectrum® Discover environment, including the API endpoint that was used.
- **Configurable log trimmer handler**
  Log trimmer configurable feature provides the capability to interact with IBM Db2 components to remove unnecessary logs with a certain age, whose accumulation might impact storage and filesystem performance.
- **SD-Monitor**
  SD monitor obtains metrics about the Kafka messages that are fetched from the Kafka topics.

Audit log

Use the audit log entries to monitor activity of REST API calls within the IBM Spectrum® Discover environment, including the API endpoint that was used.

To view audit log entries, extract the output from the compressed file that is generated by the FFDC script. You can use a text editor to read the FFDC output. Audit log entries are in JSON format and are identified in the FFDC output by the string AUDIT in the type field.

For more information about API endpoints in the IBM Spectrum Discover environment, see *REST API* in *IBM Data Cataloging: REST API Guide*REST API for IBM Data Cataloging.

The audit log includes the following fields:

service
: The service that processed the request. The service and node name are included. The following details are optional: `namespace`, `serviceInstance`, and `containerId`.

requestId
: The request ID that is returned back to the client, or a correlation tag that is used for internal tracking.

timestampStart
: The time that the request was received.

request
: The API endpoint that made the request.

serverAddress
: The IP address of the server or node that processed the request.

userAgent
: The identification string of the user agent that made the request.

type
: The log entry type: `AUDIT`

responseSize
: Size of the response, in bytes, sent back to the client.

hostname
: The IP address from which the request originates.

protocol
: The protocol of the request.

requestLatency
: The latency of the request in milliseconds.

responseStatus
: The return code that is provided to the client.

auth
: The user name and the authentication scheme, bearer (for LDAP) or basic (for local authentication).

Configurable log trimmer handler

Log trimmer configurable feature provides the capability to interact with IBM Db2 components to remove unnecessary logs with a certain age, whose accumulation might impact storage and filesystem performance.

By default, the job to erase log files is configured to run at midnight daily and erase logs older than 12 hours.

To configure age for log clean up, a config map is defined. Use the following example to configure it:

```
export NAMESPACE=ibm-data-cataloging
export TIME_DELETION=720
oc patch configmap db2-log-trimmer -n $NAMESPACE --type merge -p "{\"data\":{\"time-limit-min\":\"${TIME_DELETION}\"}}"
```

TIME_DELETION vary in minutes.

SD-Monitor

SD monitor obtains metrics about the Kafka messages that are fetched from the Kafka topics.

About this task

The SD monitor provides information about the Kafka messages that are fetched from the Kafka topics.

In the IBM Data Cataloging service, each Kafka topic represents a type of connection, such as NFS, IBM Storage Scale, COS, and S3, which has multiple consumers fetch the messages that are sent from the producers when they connect to a data source and scan data from it.

You can retrieve the following metrics from the SD monitor:

- Current committed offset for the set of topics and partitions.
- Log end offset (highest offset in the partition).
- Lag, the difference between the current committed offset and log end offset values.

Procedure

Follow the steps to get SD monitor information is as follows:

a. Run the following command to get the logs of the SD monitor pod.

```
(oc logs $( oc get pods -n ibm-data-cataloging | grep sdmonitor | awk '{print $1}' ) -n ibm-data-cataloging) | grep Fetch
```

b. Locate the array that contains the metrics about the connection type that you want to debug.
An example array for NFS connections.

```
'file-scan-topic': {0: [30768, 30768, 0], 1: [31050, 31050, 0], 2: [30897, 30897, 0], 3: [30788, 30788, 0], 4 : [30608,
30608, 0], 5: [30525, 30525, 0], 6: [30908, 30908, 0], 7: [30610, 30610, 0], 8: [30726, 30726, 0], 9 : [30959, 30959, 0]}
```

Note: Each sub-array `0: [30768, 30768, 0]` indicates the metrics that are related to one of the consumers. In this sub-array, the first value represents the current committed offset. The second value is the log end offset, and the third value represents the lag.
It is important to consider the following points when you read the log information:

- The SD monitor provides a condensed metric of the Kafka messages that are received by each one of the consumers.
  For example, if you run two different NFS scans, the expectation is to get the metrics of all the Kafka messages sent by the two scans.
- The SD monitor runs every 30 seconds, which means that you can see a small lag between the progress of the scans and the metrics they are reported in the SD monitor log.

Health check monitoring

Provides series of checkpoints to check IBM Data Cataloging health status.

It provides IBM Data Cataloging health status and different commands that might be used for continuous monitoring of the health check.

When scanning is running, different checkpoints can be made to ensure that they are scanned properly.

Monitor connection manager and db2whrest pods frequently to avoid overhead on database
: One of the issues that might disrupt IBM Data Cataloging function is overrunning the database, causing malfunctions in different components such as the user interface or connection manager.
The top memory and CPU consumption for connection manager and db2whrest pods is around 85% of their limits.
It is recommended to run the following commands to ensure that those limits are met for regular usage:

1. Check the current usage of connmgr pod and db2whrest.

```
$oc adm top pod isd-connmgr-795544f666-l25mc
$oc adm top pod isd-db2whrest-7c574c7bf-x7l9t
```

   Example output:

```
$oc adm top pod isd-connmgr-795544f666-l25mc
NAME                                     CPU(cores)   MEMORY(bytes)
isd-connmgr-scheduler-795544f666-l25mc   0m           34Mi

$oc adm top pod isd-db2whrest-7c574c7bf-x7l9t
NAME                             CPU(cores)   MEMORY(bytes)
isd-db2whrest-7c574c7bf-x7l9t    21m          512Mi
```

2. Compare current usage with the limits set and make sure that it is not greater than 85% for a long period of time.

```
$oc describe pod isd-connmgr-795544f666-l25mc|grep -A2 Limits
    Limits:
      cpu:     1
      memory:  4Gi
$oc describe pod isd-db2whrest-7c574c7bf-x7l9t|grep -A2 Limits
    Limits:
      cpu:     2
      memory:  8Gi
```

   Note: IBM Data Cataloging recommends having 10 parallel scans at most to avoid malfunction or reduce significantly performance on the product.

Check request latency of a connection filtering db2whrest log
: This checkpoint is important for checking the time for a request query to be processed, the following example uses one connection to see the time for the request to respond. Values < 2000 (milliseconds) are expected.

1. This checkpoint is important for checking the time for a request query to be processed, the following example uses one connection to see the time for the request to respond. Values < 2000 (miliseconds) are expected.
2. Run the following command to check the pod that is associated with db2whrest.

```
$oc get pods | grep db2whrest
```

   Example output:

```
$oc get pods | grep db2whrest
isd-db2whrest-7x574c7bf-c7l9t           1/1     Running         0       43h
```

3. Get the logs and filter them.
   For example:

```
$oc logs isd-db2whrest-7x574c7bf-c7l9t|grep -E '${connectionName}.*requestLatency'
{"type": "AUDIT", "hostname": "172.17.47.184", "serverAddress": "172.21.166.189", "userAgent": "python-requests/2.31.0",
```

4. It is also acceptable if the requested latency value is less than 2000. This is an important checkpoint that can give you clues about database behavior when you submit a new scan.

Monitor Kafka lag as complementary signal of a healthy product
: When a scan is being processed, one signal of a healthy environment is the lag, which is the difference between indexed records and scanned records.
A healthy state would be when the lag is less than 30%. A state of unhealthiness occurs when the number of scanned records continues to increase without progress in indexing, resulting in a significant lag and a sustained difference more than 30% for an extended period.
To resolve that situation, it is recommended to reduce the number of parallel scans, giving more time for the ingestion rate to increase properly and reducing the lag.

Check average insert rate of a scan and batch size
: The following checkpoints give clues to the health of the insert rate into the database and the batch size of Kafka.

1. Insert rate values might vary depending on different factors. However, the expected rate is around 200 messages per second. It means 200 messages per second get inserted into the database per consumer.
2. Regarding the batch size of Kafka, it is the amount of data that is accumulated and sent as a batch by a Kafka producer. This value is expected to be around 50,000 overall. If the value decreases significantly for a long period, it might be a sign of misbehavior.
3. Run the following command to get the consumer pod details.

```
$oc get pods|grep consumer-file-scan
```

   Note: In this case, the consumer is the file, because it is an NFS scan and should be 10 pods.
   Example output:

```
isd-consumer-file-scan-6d6bbf8487-45c6c                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-6zfv9                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-77g7v                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-97cfv                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-bwmr5                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-c5cb9                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-ljkws                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-m96hc                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-w55lr                          1/1     Running     0               7d13h
isd-consumer-file-scan-6d6bbf8487-z8hqd                          1/1     Running     0               7d13h
```

4. Get the logs and filter them.
   For example:

```
$oc logs isd-consumer-file-scan-6d6bbf8487-45c6c|grep -A5 "Avg Insert"
… omitted …
DB Avg Insert Rate (msg/sec): { current: 210, min: 1, max: 245 }
2023-06-22 13:54:40.611 > offset_commit_cb: success, offsets:[{part: 1, offset: 79452963, err: none}]
2023-06-22 13:56:25.588 > Consumed= 50014 Dup_id= 13 Skipped= 0 |Last_Batch: Att= 50000 Succ= 50000 Fail= 0 |Total:
```

```
Att= 50001 Succ= 50001 Fail= 0 | Batch_Time(ms): E= 160 B= 36155 L= 68659 KC= 0 TH= 0 Tot= 104974
 DB Avg Insert Rate (msg/sec): { current: 192, min: 1, max: 245 }
2023-06-22 13:56:25.605 > offset_commit_cb: success, offsets:[{part: 1, offset: 79502976, err: none}]
2023-06-22 13:59:44.298 > Consumed= 100024 Dup_id= 23 Skipped= 0 |Last_Batch: Att= 50000 Succ= 50000 Fail= 0 |Total:
Att= 100001 Succ= 100001 Fail= 0 | Batch_Time(ms): E= 139 B= 11888 L= 186663 KC= 1 TH= 0 Tot= 198691
 DB Avg Insert Rate (msg/sec): { current: 196, min: 1, max: 245 }
2023-06-22 13:59:44.314 > offset_commit_cb: success, offsets:[{part: 1, offset: 79552986, err: none}]
2023-06-22 14:03:04.724 > Consumed= 150031 Dup_id= 30 Skipped= 0 |Last_Batch: Att= 50000 Succ= 50000 Fail= 0 |Total:
Att= 150001 Succ= 150001 Fail= 0 | Batch_Time(ms): E= 155 B= 1594 L= 197868 KC= 1 TH= 0 Tot= 200408
 DB Avg Insert Rate (msg/sec): { current: 206, min: 1, max: 245 }
```

Check Qpart assignation when a new connection is submitted for scan

This checkpoint shows that when a scan is assigned to the different partitions, the health status must be associated with the correct assignment of the 10 Qparts (0–9), and scangen must be constant.

1. Run the following command to get the producer pod details.

   ```
   $oc get pods|grep producer-file-scan
   ```

   Note: In this case, the producer is the file, because it is an NFS scan only one producer is acceptable.
   Example output:

   ```
   $oc get pods|grep producer-file-scan
   isd-producer-file-scan-79c45bbc77-jm5v6                        1/1     Running     4 (7d14h ago)   7d14h
   ```

2. Get the logs from one and filter the connection name and **COMMITSCAN**.
   For example:

   ```
   $oc logs isd-producer-file-scan-79c45bbc77-jm5v6|grep -E ${connectionName}.*COMMITSCAN'
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:0]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:1]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:2]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:3]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:4]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:5]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:6]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:7]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:8]
   Built commit scan message [connection:${connectionName},eventname:COMMITSCAN,scangen:1,qpart:9]
   ```

   In case 1, it is acceptable for the connection to be assigned to 10 different partitions and scangen to be constant.

Check connections progressing and scan in progress state when a new scan started

These requests can be made to the connection manager API endpoint to check the status of a connection. It would be necessary to create a token to be ready to consume the API properly. For more information about how to get the token, see /auth/v1/token: GET.
At this stage, the token is saved in the **${TOKEN}** variable, **hostRoute** and **connectionName** are identified, and **jq** is installed for JSON formatting purposes.

For example:

```
$ curl -q -H "Authorization: Bearer $TOKEN" https://<<hostRoute>/connmgr/v1/connections/<<connectionName>>|jq
{
  "name": "exports",
  "platform": "NFS",
  "cluster": "nfs-1.nfs.svc.cluster.local",
  "datasource": "exports",
  "current_gen": 3,
  "password": null,
  "site": "",
  "online": 1,
  "scan_topic": "file-scan-connector-topic",
  "le_topic": "file-le-connector-topic",
  "le_enabled": 1,
  "host": "nfs-1.nfs.svc.cluster.local",
  "mount_point": "/exports",
  "protocol": "nfs",
  "user": null,
  "additional_info": "{\"working_dir\": \"/nfs-scanner/\", \"local_mount\": \"/nfs-
scanner/da68f4b426134c5d9fb498a15219857f\"}",
  "scan_in_progress": 1,
  "total_records": 10162,
  "scan_compl_rec": 0,
  "schedule": null
}
```

As shown, there is a scan in progress for that specific connection, and total records must be augmenting over time. It would be necessary to give some seconds to get data reflected on the endpoint.
These are some of the checkpoints that can be used for checking the state of the connections. This was one example of the many endpoints that can be consumed by the IBM Data Cataloging API. For more information, see REST API for IBM Data Cataloging.

Check database records increasing while scan is progressing

If querying the API or scanning progress on the user interface does not show any progress, it might give the impression that the system is not responding properly. However, that might not be accurate. It is necessary to establish a connection to the database and make some queries to ensure that the system is actually making insertions into the proper table.
Warning: The database is stressed when doing several transactions. Be careful when you query the database directly. Also, avoid doing complex queries when one or several scans are occurring; this might cause unnecessary overhead and impact the unusual behavior of the database.

1. Run the following command to get db2 head pod and access it.

   ```
   $ headPodName=$(oc -n ${projectName} get po --selector name=dashmpp-head-0|grep isd|awk '{print $1}')
   $ oc -n ${projectName} rsh ${headPodName}
   ```

2. Change user to db2inst1 and connect to bludb database.

```
sh-4.4$ su db2inst1 -
[db2inst1@c-isd-db2u-0 - Db2U ]$ db2 connect to bludb

   Database Connection Information

 Database server        = DB2/LINUXX8664 11.5.7.0
 SQL authorization ID   = DB2INST1
 Local database alias   = BLUDB

#Do couple counting queries agains metaocean table and make sure it's progressing leaving couple of seconds from one 
query to another

[db2inst1@c-isd-db2u-0 - Db2U ]$ select count(fkey) from bluadmin.metaocean

1
---------------------------------
                           10364.

  1 record(s) selected.

[db2inst1@c-isd-db2u-0 - Db2U ]$ select count(fkey) from bluadmin.metaocean

1
---------------------------------
                           10952.

  1 record(s) selected.
```

Collecting logs and metrics

Steps to collect logs and metrics for the IBM Data Cataloging.

Procedure

1. Run the following `oc` login command.

   ```
   oc login --token=<OCP_TOKEN> --server=<OCP_SERVER>
   ```

   For more information about CLI login, see CLI login.

2. Export `OCP_TOKEN`, `PROJECT_NAME`, and `TIMESTAMP` environment variables.

   ```
   export OCP_TOKEN=$(oc whoami --show-token) # or the actual token if not logged in
   export PROJECT_NAME=ibm-data-cataloging
   export TIMESTAMP=$(date +"%Y%m%d%H%M%S")
   ```

3. Run the following command to create a secret with a token to use in the script that collects resources.

   ```
   oc -n $PROJECT_NAME create secret generic dcs-must-gather --from-literal=token=$OCP_TOKEN
   ```

4. Create dcs-must-gather-pod.yaml file.

   ```
   cat >> dcs-must-gather-pod.yaml << EOF

   apiVersion: v1
   kind: Pod
   metadata:
     name: dcs-must-gather
     labels:
       app: isd
       component: discover
       role: must-gather
   spec:
     containers:
     - name: must-gather
       env:
         - name: WORKDIR
           value: /tmp
         - name: OCP_TOKEN
           valueFrom:
             secretKeyRef:
               name: dcs-must-gather
               key: token
       image: cp.icr.io/cp/ibm-spectrum-discover/must-gather:2.1.3
       imagePullPolicy: Always
       resources:
         requests:
           cpu: 50m
           memory: 128Mi
           ephemeral-storage: 2Gi
         limits:
           cpu: 2
           memory: 8Gi
           ephemeral-storage: 5Gi
       command:
         - '/bin/bash'
         - '-c'
         - '/opt/must-gather/must-gather.sh; while true; do sleep 60; done'

   EOF
   ```

5. Run the following command to create a must gather pod.

```
oc -n $PROJECT_NAME create -f dcs-must-gather-pod.yaml
```

6. Monitor the logs until it reports that the **tar** file has been created.

```
oc -n $PROJECT_NAME logs -f dcs-must-gather
```

Example output:

```
2023-09-12 04:12:12 - INFO - Archiving completed: dcs-must-gather.tar.xz
```

7. Copy the **tar** file.

```
oc -n $PROJECT_NAME cp dcs-must-gather:/tmp/dcs-must-gather.tar.xz dcs-must-gather-$TIMESTAMP.tar.xz
```

8. Remove the must gather resources.

```
oc -n $PROJECT_NAME delete pod dcs-must-gather
oc -n $PROJECT_NAME delete secret dcs-must-gather
```

Data Cataloging Harvester CLI

IBM Data Cataloging Harvester is a new capability that is designed to import external data to the IBM Data Cataloging service catalog database. It imports the data even if it is not coming from a Db2 database that uses the same schema.

Before you begin

- Ensure you have the following installations:
  - Python +3.11 or later version
  - **curl tar gzip sed pip3** packages
  - Red Hat® OpenShift® CLI.

With the IBM Data Cataloging Harvester, it is possible to import the external data to the catalog database even if it is not coming from a Db2 table using the exact same schema. IBM Fusion 2.8.0 supports importing data associated to a single data source also known as a connection. After using the CLI against a valid SQLite file, then it should create a connection and import all metadata stored in the import file records or additional tags depending on the command that is executed.

Supported import formats

The Harverster CLI helps to read records in the specified format and sends them to the import service component that runs on the cluster to use by Db2 after it is validated. Not only the input file is in the format that the Harvester supports, but also the schema rules that succeed during analyzing tasks.

The IBM Data Cataloging harverster needs the following schemas. The IBM Data Cataloging harvester requires at least two tables to work with:

Connections table
: It is used to store information about the data source.

```
CREATE TABLE CONNECTIONS (
    name TEXT,
    platform TEXT,
    cluster TEXT,
    datasource TEXT,
    site TEXT,
    host TEXT,
    mount_point TEXT,
    application_code TEXT,
    local_mount TEXT,
    total_records INTEGER,
    current_gen TEXT,
    password TEXT,
    online TEXT,
    scan_topic TEXT,
    le_topic TEXT,
    le_enabled INTEGER
)
```

Metaocean table
: It contains all the base metadata present in the data source.

```
CREATE TABLE METAOCEAN (
    INODE INTEGER,
    OWNER TEXT,
    [GROUP] TEXT,
    PERMISSIONS TEXT,
    UID INTEGER,
    GID INTEGER,
    PATH TEXT,
    FILENAME TEXT,
    MTIME TEXT,
    ATIME TEXT,
    CTIME TEXT,
    SIZE INTEGER,
    TYPE TEXT,
    STATE TEXT
)
```

The SQLite database might have more optional tables if you want to import tag values for additional metadata. The additional tables cannot be processed unless the tags mode in the Harvester CLI defines tables by their names.

Example of additional tags table:
Note: Inode and filename are only the required columns, and rest are the names of already created tags on Data Cataloging.

```
CREATE TABLE META (
    inode TEXT,
    filename TEXT,
    fkey TEXT,
    ExampleCustomTag1 TEXT,
    ExampleCustomTag2 TEXT,
    ExampleCustomTag3 TEXT,
)
```

Setting up the external host

1. Log in to the Red HatOpenShift CLI. For procedure, see Red Hat OpenShift CLI.
2. If the Harvester CLI runs on hosts outside of the OpenShift cluster, then it is possible to face SSL problems caused by self-signed certificates. To resolve SSL problems, you can use the following commands to download the router certificate in the host that running the Harvester and then use environment variables to use them.
    a. Run the following command to download the OpenShift router TLS certificate.

    ```
    oc -n openshift-ingress get secret router-certs-default -o json | jq -r '.data."tls.crt"' | base64 -d >
    $(pwd)/router_tls.crt
    ```

    b. Run the following command to export the environment variables to define the path of the TLS certificate.

    ```
    export IMPORT_SERVICE_CERT=$(pwd)/router_tls.crt
    export DCS_CERT=$(pwd)/router_tls.crt
    ```

3. Download the Harvester CLI.

    ```
    export DCS_NAMESPACE=ibm-data-cataloging
    export DCS_IMPORT_SVC_HOST=$(oc -n $DCS_NAMESPACE get route isd-import-service -o jsonpath="{.spec.host}")
    export DCS_CONSOLE_HOST=$(oc -n $DCS_NAMESPACE get route console -o jsonpath="{.spec.host}")
    export DCS_USERNAME=$(oc -n $DCS_NAMESPACE get secret keystone -o jsonpath="{.data.user}" | base64 -d)
    export DCS_PASSWORD=$(oc -n $DCS_NAMESPACE get secret keystone -o jsonpath="{.data.password}" | base64 -d)

    curl "https://${DCS_IMPORT_SVC_HOST}/v1/download_harvester" --output harvester.tar.gz --insecure
    tar xf harvester.tar.gz
    rm -vf harvester.tar.gz
    ```

4. Run the following set up script to start by using the Harvester CLI.

    ```
    chmod +x ./harvester/setup.sh
    ./harvester/setup.sh
    ```

Running the Harvester CLI

1. Ensure that you export the records with all the base metadata.

    ```
    SQLITE_FILE_PATH=/tmp/example.sqlite
    CONFIG_FILE=$(pwd)/harvester/config.ini
    python3 harvester metaocean $SQLITE_FILE_PATH -c $CONFIG_FILE
    ```

2. Use `-p nfs` option if you want to import NFS-based metadata.

    ```
    SQLITE_FILE_PATH=/tmp/example.sqlite
    CONFIG_FILE=$(pwd)/harvester/config.ini
    python3 harvester metaocean $SQLITE_FILE_PATH -c $CONFIG_FILE -p nfs
    ```

3. Run the following command to import values after they are created by using either the user interface or API.

    ```
    SQLITE_FILE_PATH=/tmp/example.sqlite
    CONFIG_FILE=$(pwd)/harvester/config.ini
    python3 harvester tags $SQLITE_FILE_PATH -c $CONFIG_FILE
    ```

4. Use `-p nfs` option if you want to import NFS-based metadata.

    ```
    SQLITE_FILE_PATH=/tmp/example.sqlite
    CONFIG_FILE=$(pwd)/harvester/config.ini
    python3 harvester metaocean $SQLITE_FILE_PATH -c $CONFIG_FILE -p nfs
    ```

Configuring the Harvester

The `config.ini` file inside the harvester directory provides critical support in the data import behavior. The following are the available variables to configure:

BlockSizeBytes
: It indicates the block size configured in the datasource that is being imported. Normally, the value is set to 4096 bytes, which helps to calculate the real capacity allocated on the filesystem based on the blocks required to store the file.

AvailableMOVW
: It must be set to 10 as it is the only supported value for optimal data ingestion. It helps the harvester to calculate the pieces of a set of records that can be divided for parallel insertion.

MaxProcessingWorkers

It is an ignored variable, and use a one worker process always. If a higher level of parallelism is required, then run another harvester to process an additional SQLite file.

IngestRequestRetries
: After the harvester finishes processing the metadata from the SQLite file, it sends it using an HTTP request to the import service for it to be indexed in the database. But it is possible to fail to do so for different reasons, such as a temporary network unavailability or a considerable API overload blocking incoming requests.
By default, the harvester tries to send the metadata 120 times, depending on the highest value and the time limit. The suggested range is 120 to 240.

StatusRequestRetries
: After the ingest request is sent, the harvester follows up with additional requests to get the status of the ingest operation to confirm whether it finished successfully. It might take a long time, so retrying is required. The suggested range is 240 to 360.

TagsWorkdir
: The name of the directory that is presented inside the harvester directory is where temporary files are stored during tag harvesting. By default, `tmp` is the name of this directory, but it can be updated in case it is necessary to have a more descriptive directory name.

MOBatchSize and TagsBatchSize
: Number of rows to read from the SQLite and load into memory at a time, a higher value is associated to a higher memory usage. It is not recommended to use a value higher than 1,00,000 because it does not represent a performance benefit. The suggested values are 10000, 50000 and 100000.

MOBatchesPerSegment and TagsBatchesPerSegment
: Number of batches to process before sending them using an ingestion request, a higher value is associated with bigger files. Using a value higher than 100 does not translate to a direct performance benefit. The suggested values are 10, 50 and 100.

Data ingestion system requirements

Ensure that you must have the following system requirements:

- 4 CPU cores and 16GB of RAM.
- At least 1.5x the size of the SQLite file to import must be available as free space on the filesystem. For example, if the SQLite file is 1GB, then make sure at least 1.5GB of storage is available.
- Python 3.11 or higher.

Installation
: Run the following commands to download and install IBMFDC.
Optionally, you can set up a virtual environment to benefit from an isolated installation.

Note: Before proceeding, ensure you have an active and authenticated session in the OpenShift console. This is necessary for the `oc` command to retrieve route information correctly. For more information, see Red Hat Documentation.

```
DCS_NS="ibm-data-cataloging"
DCS_IMPORT_SVC_HOST=$(oc -n ${DCS_NS} get route isd-import-service -o jsonpath='{.spec.host}')
```

```
curl -O -J "https://${DCS_IMPORT_SVC_HOST}/v1/ibmfdc" --insecure
WHEEL_FILE=$(ls ibmfdc-*.whl)
pip3 install ${WHEEL_FILE}
rm -f ${WHEEL_FILE}
```

Preparing the CLI configuration file
: By default, the ibmfdc command looks for a file named config.ini in the current working directory. If you want to specify a different configuration file, use the `oc` command-line argument.
Configuration file format:

```
ConsoleRouteURL = <DCS_CONSOLE_URL>
ImportCSVRouteURL = <DCS_IMPORT_SVC_URL>
AdminUsername = <DCS_USERNAME>
AdminPassword = <DCS_PASSWORD>
```

For example:

```
ConsoleRouteURL = https://console-ibm-data-cataloging.apps.<domain>
ImportCSVRouteURL = https://isd-import-service-ibm-data-cataloging.apps.<domain>
AdminUsername = sdadmin
AdminPassword = Passw0rd
```

Ingesting remote metadata using the IBMFDC Harvester
: The IBMFDC Harvester supports ingesting both base and additional metadata simultaneously. However, all necessary tags must be created beforehand, otherwise the meta table in the SQLite file will be ignored. This is because none of its columns matches any existing tag names.
Importing base and additional metadata:

```
ibmfdc harvester -m meta <path_to_sqlite_file>
```

Importing only base metadata (ignores tags):

```
ibmfdc harvester <path_to_sqlite_file>
```

FKEY migration script

The FKEY migration script is a fix to avoid potential rare collisions in the FKEY file identifier when ingesting billions of records.

About this task

The FKEY migration script fix consists of a script that needs to be run in the Db2 terminal of IBM Data Cataloging.

Procedure

1. Run the following command to do SSH to the `c-Db2u-0 pod`.

```
oc -n ibm-data-cataloging rsh c-isd-db2u-0
      su - db2inst1
```

2. Create a file named FKEY_Updater.sh with the FKEY migration script.
   The script iterates over a list of data sources, fixes the FKEY in all the files, and ingests them in each data source of the previous IBM Data Cataloging service versions.

```
      # ========================================================
      # How to invoke
      # Send the list of datasources to fix as follows:
      # ./FKEY_Updater.sh datasource1 datasource2 datasource3
      # ========================================================

      start_date=`date +%s.%N`
      datasourcearray=( "$@" )

      echo "Connecting to database..."
      db2 connect to bludb
      echo "Connected to BLUDB."
      printf '\n'
      echo "List of datasources to update:"
      printf ' - %s\n' "${datasourcearray[@]}"
      printf '\n'

      for datasource in "${datasourcearray[@]}"
      do
        echo "Starting update procedures for datasource '${datasource}'..."
        printf '\n'
        echo "Updating table ACESMAPLOADBASE..."
        db2 "update bluadmin.acesmaploadbase amlb set amlb.fkey=(mo.cluster || '_' || mo.datasource || '_' || mo.inode) 
from bluadmin.metaocean mo where amlb.fkey=mo.fkey and mo.datasource='${datasource}';"
        echo "ACESMAPLOADBASE table updated successfully."
        printf '\n'
        echo "Updating table ACOGMAPLOADBASE..."
        db2 "update bluadmin.acogmaploadbase acmlb set acmlb.fkey=(mo.cluster || '_' || mo.datasource || '_' || mo.inode) 
from bluadmin.metaocean mo where acmlb.fkey=mo.fkey and mo.datasource='${datasource}';"
        echo "ACOGMAPLOADBASE table updated successfully."
        printf '\n'
        echo "Updating table ACESMAP (This action could take several minutes)..."
        db2 "update bluadmin.acesmap am set am.fkey=(mo.cluster || '_' || mo.datasource || '_' || mo.inode) from 
bluadmin.metaocean mo where am.fkey=mo.fkey and mo.datasource='${datasource}';"
        echo "ACESMAP table updated successfully."
        printf '\n'
        echo "Updating table ACOGMAP (This action could take several minutes)..."
        db2 "update bluadmin.acogmap acm set acm.fkey=(mo.cluster || '_' || mo.datasource || '_' || mo.inode) from 
bluadmin.metaocean mo where acm.fkey=mo.fkey and mo.datasource='${datasource}';"
        echo "ACOGMAP table updated successfully."
        printf '\n'
        echo "Updating table METAOCEAN (This action could take several minutes)..."
        db2 "update bluadmin.metaocean mo set mo.fkey=(mo.cluster || '_' || mo.datasource || '_' || mo.inode) where not 
REGEXP_LIKE(mo.fkey, mo.cluster || '_' || mo.datasource || '_' || mo.inode) and mo.datasource='${datasource}';"
        echo "METAOCEAN table updated successfully."
        printf '\n'
        echo "Updates on datasource '${datasource}' done successfully."
        printf '\n'
      done
      printf '\n'
      end_date=`date +%s.%N`
      runtime=$(echo "$end_date - $start_date" | bc -l)

      echo "Execution time was $runtime seconds."
```

3. Run the script.

```
./FKEY_Updater.sh datasource1 datasource2 datasource3
```

   The final execution time of the script depends on the number of files that are ingested in the database.

Configurable Db2 log trimmer

Db2 log trimmer tool provides a mechanism to trim the informative logs that are generated by scanning a data source. It uses the Harvester CLI to ingest data into the IBM Data Cataloging.

Configurable Db2 log trimmer tool helps to better handle the filesystem usage by deleting informative files placed in the Db2 PVC.

From IBM Fusion 2.8.x release, the capabilities of the Db2 log trimmer are extended by a `ConfigMap` that gives an option of setting the currency for running the trimmer (in minutes). It selects and deletes the old files and `.BAD` extension files.
Important:

- The Db2 `.BAD` files are log files that are generated by failed transactions in Db2 when trying to ingest data.
- If there is failure at the Db2 level that requires the guidance of a Db2 expert, those `.BAD` files can be indispensable to perform the root cause analysis and it is recommended NOT TO DELETE those files unless needed.

The fields in the `ConfigMap` are the following:

- **time-limit-min** - default value is "720" (files with more than 720 minutes of last modification)
- **bad-file-erase** - default value is "0" (not to delete the `.BAD` files)
- **frequency-min** - default value is "720" (run the log trimmer each 720 minutes)

You can edit the values of the `ConfigMap` by using the following command.

```
oc edit configmap db2-log-trimmer -n ibm-data-cataloging
```

IBM Data Cataloging service installation and upgrade issues

Use this troubleshooting information to resolve install and upgrade problems that are related to the IBM Data Cataloging service.

Installation stuck because of resource quota

Problem statement
: The IBM Data Cataloging installation is stuck because of resource quota set on IBM Fusion and IBM Data Cataloging namespaces. tuntil both the resource quota were removed

Resolution
: Set the resource quota values for IBM Data Cataloging as follows:

```
kind: ResourceQuota
apiVersion: v1
metadata:
  name: dcs-resource-quota
  namespace: ibm-data-cataloging
spec:
  hard:
    limits.cpu: '100'
    limits.memory: 180Gi
    pods: '120'
    requests.cpu: '20'
    requests.memory: 40Gi
```

Data cataloging upgrade stuck for more than 1 hour

Problem statement
: IBM Data Cataloging upgrade is stuck for more than 1 hour and the progress shows 15%.

Symptoms
: The IBM Data Cataloging upgrade runs for more than 1 hour and gets stuck at 15%. On the OpenShift® Console, the installed operator shows just the IBM Data Cataloging operator and not `IBM DB2` and `Streams for Apache Kafka` operators under the `ibm-data-cataloging` namespace.
Also, the Subscription for the IBM Data Cataloging operator shows `InstallPlanPending, Reason "RequiresApproval`.

Resolution

1. In the OpenShift console, go to Operators > Installed Operators tab.
2. Select IBM Storage Discover operator.
3. Go to Subscriptions tab, and select InstallPlan.
4. Approve the InstallPlan.
   There is an alternative method using OpenShift CLI as follows:
   a. Run the following command to get the InstallPlan that is not approved.

   ```
   oc get ip -n ibm-data-cataloging -o=jsonpath='{.items[?(@.spec.approved==false)].metadata.name}'
   ```

   b. Run the following command to get the InstallPlan name.

   ```
   oc patch installplan $(oc get ip -n ibm-data-cataloging -o=jsonpath='{.items[?
   (@.spec.approved==false)].metadata.name}') -n ibm-data-cataloging --type merge --patch '{"spec":
   {"approved":true}}'
   ```

5. Wait for the upgrade process to complete.

Data Cataloging import service pod in CrashLoopBackOff state

Diagnosis

1. Check for CrashLoopBackOff on the import service pod.

   ```
   oc -n ibm-data-cataloging get pod -l role=import-service
   ```

2. Confirm that the logs show `permission denied` error:

   ```
   oc -n ibm-data-cataloging logs -l role=import-service
   ```

Resolution

1. Debug import service pod.

   ```
   oc -n ibm-data-cataloging debug deployment/isd-import-service --as-user=0
   ```

2. Update the directory permissions.

```
chmod 775 /uploads
mkdir -p /uploads/failed_requests
chmod 775 /uploads/failed_requests
exit
```

IBM Data Cataloging database schema job is not in a completed state during installation or upgrade

Note: This procedure is applicable to recover DB2 that goes into an unavailable mode after the service installation and related to a degraded state of the IBM Data Cataloging service.

Symptoms

The `isd-db2whrest` or `isd-db-schema` pods report a not ready or error state.
Run the following command to view the common logs:

```
oc -n ibm-data-cataloging logs -l 'role in (db2whrest, db-schema)' --tail=200
```

Go through the logs to check whether the following error exists:

```
Waiting on c-isd-db2u-engn-svc port 50001...

db2whconn - ERROR - [FAILED]: [IBM][CLI Driver] SQL1224N The database manager is not able to accept new requests, has 
terminated all requests in progress, or has terminated the specified request because of an error or a forced interrupt. 
SQLSTATE=55032

Connection refused
```

Resolution

1. Restart Db2:

```
oc -n ibm-data-cataloging rsh c-isd-db2u-0
sudo wvcli system disable -m "Disable HA before Db2 maintenance"
su - ${DB2INSTANCE}
db2stop
db2start
db2 activate db BLUDB
exit
sudo wvcli system enable -m "Enable HA after Db2 maintenance"
```

2. Confirm that Db2 HA-monitoring is active:

```
sudo wvcli system status
exit
```

3. Check whether the problem occurred during installation or upgrade or a post installation problem.
4. If this occurs during an upgrade or installation, recreate the `isd-db-schema` job and monitor the pod until it gets to a completed state.

```
SCHEMA_OLD="isd-db-schema-old.json"
SCHEMA_NEW="isd-db-schema-new.json"
oc -n ibm-data-cataloging get job isd-db-schema -o json > $SCHEMA_OLD
jq 'del(.spec.template.metadata.labels."controller-uid") | del(.spec.selector) | del (.status)' $SCHEMA_OLD > 
$SCHEMA_NEW
oc -n ibm-data-cataloging delete job isd-db-schema
oc -n ibm-data-cataloging apply -f $SCHEMA_NEW

oc -n ibm-data-cataloging get pod | grep isd-db-schema
```

5. If this is a post installation problem, restart db2whrest.

```
oc -n ibm-data-cataloging delete pod -l role=db2whrest
```

IBM Data Cataloging installation is stuck for more than 1 hour

Note: This procedure must be used only during installation problems, not for upgrades or any subsequent issues.

Symptoms

The IBM Data Cataloging installation running for more than an hour, and it stuck somewhere between 35% and 80% (both inclusive).

Resolution

1. Run the following command to scale down the operator.

```
oc -n ibm-data-cataloging scale --replicas=0 deployment/spectrum-discover-operator
```

2. Run the following command to scale down workloads.

```
oc -n ibm-data-cataloging scale --replicas=0 deployment,statefulset -l component=discover
```

3. Run the following command to remove the DB schema job if present.

```
oc -n ibm-data-cataloging delete job isd-db-schema --ignore-not-found
```

4. Run the following commands to delete the Db2 instance and the password secret.

```
oc -n ibm-data-cataloging delete db2u isd
oc -n ibm-data-cataloging delete secret c-isd-ldapblueadminpassword --ignore-not-found
```

5. Wait until the Db2 pods and persistent volume claims get removed.

```
oc -n ibm-data-cataloging get pod,pvc -o name | grep c-isd
```

6. Run the following command to scale up the operator.

```
oc -n ibm-data-cataloging scale --replicas=1 deployment/spectrum-discover-operator
```

IBM Data Cataloging service is not installed successfully on IBM Fusion HCI with GPU nodes

Problem statement

The data catalog service is in installing state for hours.

Resolution

To resolve the problem, do the following steps:

1. Patch FSD with new affinity to not schedule `isd` workload on those nodes:

```
oc -n <Fusion_namespace> patch fusionservicedefinitions.service.isf.ibm.com data-cataloging-service-definition  --
patch "$(cat fsd_dcs_patch.yaml)"
```

The fsd_dcs_patch.yaml file:

```
cat >> fsd_dcs_patch.yaml << EOF

apiVersion: service.isf.ibm.com/v1
kind: FusionServiceDefinition
metadata:
  name: data-cataloging-service-definition
  namespace: <Fusion_namespace>
spec:
  onboarding:
    parameters:
      - dataType: string
        defaultValue: ibm-data-cataloging
        descriptionCode: BMYSRV00003
        displayNameCode: BMYSRV00004
        name: namespace
        required: true
        userInterface: false
      - dataType: storageClass
        defaultValue: ''
        descriptionCode: BMYDC0300
        displayNameCode: BMYDC0301
        name: rwx_storage_class
        required: true
        userInterface: true
      - dataType: bool
        defaultValue: 'true'
        descriptionCode: descriptionCode
        displayNameCode: displayNameCode
        name: doInstall
        required: true
        userInterface: false
      - dataType: json
        defaultValue: '{"accept": true}'
        descriptionCode: descriptionCode
        displayNameCode: displayNameCode
        name: license
        required: true
        userInterface: false
      - dataType: json
        defaultValue: '{"nodeAffinity":{"requiredDuringSchedulingIgnoredDuringExecution":{"nodeSelectorTerms":
[{"matchExpressions":[{"key":"nvidia.com/gpu","operator":"NotIn","values":["Exists"]}]}]}}}'
        descriptionCode: descriptionCode
        displayNameCode: displayNameCode
        name: affinity
        required: true
        userInterface: false

EOF
```

If the output shows this error message `Error from server (UnsupportedediaType): the body of the request was in an unknown format - accepted media types include: application/json-patch+json, application/merge-patch+j son, application/apply-patch+yaml`, then you need to follow the steps to resolve this issue:

a. Go to OpenShift Container Platform web console.
b. Go to Operators > Installed Operators tab, under `<Fusion_namespace>`, select the IBM Fusion operator.
c. Select the IBM Fusion service instance tab, and select the `data-cataloging-service-instance`.
d. Select the YAML tab, and edit the YAML file for the `data-cataloging-service-instance`. Under `spec.onboarding.parameters`, ensure you add the following lines.

```
- dataType: json
           defaultValue: '{"nodeAffinity":{"requiredDuringSchedulingIgnoredDuringExecution":
{"nodeSelectorTerms":[{"matchExpressions":[{"key":"isf.ibm.com/nodeType","operator":"NotIn","values":
["gpu"]}]}]}}}'
           descriptionCode: descriptionCode
           displayNameCode: displayNameCode
           name: affinity
           required: true
           userInterface: false
```

2. Display the patch FSD:

```
oc -n <Fusion_namespace> get fusionservicedefinitions.service.isf.ibm.com data-cataloging-service-definition -o yaml
```

3. Install from the user interface.

IBM Data Cataloging database pods stuck during initialization phase

Symptoms

Multiple Db2 pods use the host port 5002, which can cause pods to stay in the `init` phase.

Resolution

1. Uninstall the IBM Data Cataloging service. For procedure, see Uninstalling IBM Data Cataloging.
2. Create a file with the IBM Data Cataloging `FusionServiceDefinition` patch.

```
cat >> fsd_dcs_patch.yaml << EOF
apiVersion: service.isf.ibm.com/v1
kind: FusionServiceDefinition
metadata:
  name: data-cataloging-service-definition
  namespace: ibm-spectrum-fusion-ns
spec:
  onboarding:
    parameters:
      - dataType: string
        defaultValue: ibm-data-cataloging
        descriptionCode: BMYSRV00003
        displayNameCode: BMYSRV00004
        name: namespace
        required: true
        userInterface: false
      - dataType: storageClass
        defaultValue: ''
        descriptionCode: BMYDC0300
        displayNameCode: BMYDC0301
        name: rwx_storage_class
        required: true
        userInterface: true
      - dataType: bool
        defaultValue: 'true'
        descriptionCode: descriptionCode
        displayNameCode: displayNameCode
        name: doInstall
        required: true
        userInterface: false
      - dataType: json
        defaultValue: '{"accept": true}'
        descriptionCode: descriptionCode
        displayNameCode: displayNameCode
        name: license
        required: true
        userInterface: false
      - dataType: json
        defaultValue: '{"size":1,"mln":2,"storage":{"activelogs":{"requests":"300Gi"},"data":
{"requests":"600Gi"},"meta":{"requests":"100Gi"},"activelogs":{"tempts":"100Gi"}}}'
        descriptionCode: descriptionCode
        displayNameCode: displayNameCode
        name: dbwh
        required: true
        userInterface: false
EOF
```

3. Apply the patch to downsize the Db2 cluster.

```
oc -n ibm-spectrum-fusion-ns patch fusionservicedefinitions.service.isf.ibm.com data-cataloging-service-definition --
type=merge --patch-file fsd_dcs_patch.yaml
```

4. Install IBM Data Cataloging service from the IBM Fusion user interface. For procedure, see Installing IBM Data Cataloging.

IBM Fusion IBM Data Cataloging known issues

List of all issues in the IBM Data Cataloging service along with its resolution.

The following known issues exist in IBM Data Cataloging service, with workarounds included wherever possible. If you come across an issue that cannot be solved by using these instructions, contact IBM support .

Resource quota limitation

With resource quota set on IBM Fusion and IBM Data Cataloging namespaces, the IBM Data Cataloging installation gets stuck until both the resource quota are removed.

Live events stopped working after IBM Data Cataloging

Problem statement

Live events functionality stopped updating record events after the IBM Data Cataloging is upgraded.

Resolution

Follow the steps to resolve this issue:

- Delete the watcher on IBM Storage Scale without deleting the connection on the IBM Data Cataloging service as follows:
  1. Run the following command to list the watchers.

     ```
     /usr/lpp/mmfs/bin/mmwatch all list
     ```

     Example output:

     ```
     Filesystem testing has 1 watcher(s):
     Cluster ID             WatchID/PID     Type       Start Time                 Path                         Description
     ------------------------------------------------------------------------------------------------------------------------
     13748412159870826098   CLW1733515374   FSYS       Fri Dec 06 12-02-55 2024   /ibm/testing
     ```

  2. Run the following command to delete the watchers.

     ```
     sudo /usr/lpp/mmfs/bin/mmwatch testing disable --watch-id CLW1733515374
     ```

     Example output:

     ```
     [I] Successfully checked or deleted Clustered Watch policy skip partitions for watch: CLW1733515374
     [I] Successfully removed Clustered Watch policy rules for watch: CLW1733515374
     [I] Successfully removed Clustered Watch configuration file from the CCR for watch: CLW1733515374
     [I] Successfully removed Clustered Watch configuration directory for watch: CLW1733515374
     [I] Successfully checked or removed Clustered Watch global catchall and config fileset skip partitions for watch: CL
     [I] Successfully disabled Clustered Watch: CLW1733515374
     ```

  3. Run the command again to list the watchers to verify the watcher deletion.

     ```
     /usr/lpp/mmfs/bin/mmwatch all list
     ```

     Example output:

     ```
     Cluster ID       WatchID/PID   Type    Start Time        Path           Description
      ---------------------------------------------------------------------------------------------------------------------
     Filesystem testing has no watchers.
     ```

  4. Run the following command to export cluster's URL and IBM Storage Scale working directory as environmental variables.

     ```
     export CLUSTER_URL = "yourcluster.ibm.com"
     export WD_PATH="/example/path/wd"
     ```

  5. Run the following command to manually add the watcher on the IBM Storage Scale.

     ```
     sudo /usr/lpp/mmfs/bin/mmwatch testing enable --event-handler kafkasink --sink-brokers "isd-kafka-tlsext-
     bootstrap-ibm-data-cataloging.apps.qa-bm-$CLUSTER_URL:443" --sink-topic scale-le-connector-topic --sink-auth-
     config $WD_PATH/kafka/auth.file --events
     IN_ATTRIB,IN_CLOSE_WRITE,IN_MODIFY,IN_CREATE,IN_DELETE,IN_MOVED_FROM,IN_MOVED_TO
     ```

     Example output:

     ```
     [I] Beginning enablement of Clustered Watch with newly created watch ID: CLW1733516766
     [I] Verified the watch type is FSYS for filesystem testing
     [I] Successfully added Clustered Watch configuration file into CCR for watch: CLW1733516766
     [I] Successfully added Clustered Watch configuration in the Spectrum Scale file system: testing for watch: CLW173351
     [I] Successfully added Clustered Watch policy rules for watch: CLW1733516766
     [I] Successfully checked or created Clustered Watch global catchall and config fileset skip partitions for watch: CL
     [I] Successfully checked or created Clustered Watch policy skip partitions for watch: CLW1733516766
     [I] Successfully enabled Clustered Watch: CLW1733516766
     ```

  6. Run the command again to list the watchers to verify the watcher creation.

     ```
     /usr/lpp/mmfs/bin/mmwatch all list
     ```

     Example output:

     ```
     Filesystem testing has 1 watcher(s):
      Cluster ID       WatchID/PID   Type    Start Time        Path           Description
      ---------------------------------------------------------------------------------------------------------------------
      13748412159870826098  CLW1733516766  FSYS    Fri Dec 06 12-26-07 2024 /ibm/testing
     ```

  7. On IBM Data Cataloging, re-scan the IBM Storage Scale connection.
  8. Any modifications on the scanned IBM Storage Scale folder should be monitored through live events and automatically updated on IBM Data Cataloging.

IBM Data Cataloging service in Metro-DR setup shows in Degraded state

Diagnosis

1. IBM Data Cataloging service is in degraded state
2. Run the following command to check whether the Pod **isd-db2whrest** is not ready:

   ```
   oc -n ibm-data-cataloging get pod -l role=db2whrest
   ```

3. Run the following command to check whether Db2 retries the network check and fails because of the timeout:

   ```
   oc -n ibm-data-cataloging logs -l type=engine --tail=100
   ```

   Example output:

   ```
   + timeout 1 tracepath -l 29 c-isd-db2u-1.c-isd-db2u-internal
   + [[ 17 -lt 120 ]]
   ```

```
+ (( n++ ))
+ echo 'Command failed. Attempt 18/120:'
Command failed. Attempt 18/120:
```

Resolution
: Increase the time before timeout, usually change from 1 second to 3-5 seconds.

1. Modify the timeout from 1 to 3 in `isd-db2u-0`:

   ```
   oc -n ibm-data-cataloging exec c-isd-db2u-0 -- sudo sed -i 's/timeout 1 tracepath/timeout 3 tracepath/g'
   /db2u/scripts/include/common_functions.sh
   ```

2. Wait until the current attempt exceeds the predefined 120 retries. After it restarts, it picks the updated value:

   ```
   oc -n ibm-data-cataloging logs -l type=engine --tail=50
   ```

3. Monitor `db2whrest` pod readiness:

   ```
   oc -n ibm-data-cataloging get pod -l role=db2whrest -w
   ```

COS connection reporting scan aborted due to inactivity

If a COS connection scan fails with the error "Scan aborted because of a long period of inactivity", it can be resolved by editing the settings file connections/cos/scan/scanner-settings.json within the data PV and choosing a higher value for notifier_timeout than the default value of 120 seconds. The change will be picked on the next scan. No pod restart is required.

Database connection issue after reboot

If an unexpected cluster update or node reboot causes database connection issues. For resolution, see the steps mentioned in the *IBM Data Cataloging database schema job is not in a completed state during installation or upgrade* section. .

Image pull error due to authentication failure

Problem statement
: The OpenShift® Container Platform login token expires occasionally, and as this is the container image registry password, this breaks the service account access to the registry.

Resolution
: If a pod is failing to pull an image from the registry with an authentication error, then re-create the image-registry-pull-secret and relink the service accounts to the new secret:

```
oc delete secret image-registry-pull-secret
HOST=$(oc get route default-route -n openshift-image-registry --template='{{ .spec.host }}')
oc create secret docker-registry image-registry-pull-secret \
    --docker-server="${HOST}" \
    --docker-username=kubeadmin \
    --docker-password="$(oc whoami -t)"
for account in spectrum-discover-operator strimzi-cluster-operator spectrum-discover-ssl-
zookepper spectrum-discover-sasl-zookeeper; do oc secrets link $account image-registry-pull-
secret --for=pull; done
```

Visual query builder search terms overrides SQL search when going into individual mode

If a search is started in the query builder, then changed to SQL mode, the initial group search is as expected but if expanded to individual records it uses the query builder terms as the base. A workaround is to clear the visual query before changing to SQL query.

LDAPS configuration failing if dollar sign is in password

Currently, the dollar sign is not supported on passwords for ldaps configuration. A workaround is to create a password without the dollar sign in it.

Content search policy missing files

If there are issues with the incorrect expected data count while running a policy, you must verify that the connection is active, and rescan to get the latest data ingested to IBM Data Cataloging. After successful upgrade of IBM Data Cataloging, a rescan of existing connections is recommended.

REST API returns token with unprintable characters

It is a noted issue that a carriage return (\r) is included at the end of HTTP response headers due to an issue with curl. This has been known to occasionally break scripts that use an auth token from the IBM Data Cataloging appliance as shown here:

```
$ curl -k -H "Authorization: Bearer ${TOKEN}" https://$SDHOST/policyengine/v1/tags
curl: (92) HTTP/2 stream 0 was not closed cleanly: PROTOCOL_ERROR (err 1)
```

As such, it is recommended to filter out the \r character. If you have a line like the following in bash:

```
`TOKEN=$(curl -i -k https://$SDHOST/auth/v1/token -u "$SDUSER:$SDPSWD" | grep -i x-auth-token |
awk '{print $2}')`
```

Simply add a | tr -d '\r' at the end to avoid running into this issue:

```
`TOKEN=$(curl -i -k https://$SDHOST/auth/v1/token -u "$SDUSER:$SDPSWD" | grep -i x-auth-token |
awk '{print $2}' | tr -d '\r')`
```

Adding S3 connection gives false negative

Problem statement
: When a connection of type S3 is added through IBM Data Cataloging user interface, it gives an undefined error message.

Resolution
: Refreshing the browser removes the error message, and the connections table shows that the S3 connection was successful.

Scale datamover AFM and ILM capabilities not working properly due to SDK misleading function when deploying an application

Problem statement
: When deploying IBM Data Cataloging service deployments pods `scaleafmdatamover` and `scaleilmdatamover` might show an error on logs when application is deployed.
For example:

```
2023-07-20 02:51:54,311 - ibm_spectrum_discover_application_sdk.ApplicationLib - INFO - Invoking conn manager at 
http://172.30.255.202:80/connmgr/v1/internal/connections
Traceback (most recent call last):
File "/application/ScaleAFMDataMover.py", line 1023, in
APPLICATION = ScaleAFMApplicationBase(REGISTRATION_INFO)
File "/application/ScaleAFMDataMover.py", line 112, in init
self.conn_details = self.get_connection_details()
File "/usr/local/lib/python3.9/site-packages/ibm_spectrum_discover_application_sdk/ApplicationLib.py", line 492, in 
get_connection_details
raise Exception(err)
UnboundLocalError: local variable 'err' referenced before assignment
2023-07-20 02:51:54,367 INFO exited: scaleafm-datamover (exit status 1; not expected)
2023-07-20 02:51:55,368 INFO gave up: scaleafm-datamover entered FATAL state, too many start retries too quickly
```

Cause
: SDK bug when deploying applications on DCS, causing deployed applications not to behave properly and pods in the incorrect state to show errors.

Resolution
: After identified this behavior, then follow the steps to resolve this issue:

1. Verify the connmgr API is running and accessible through HTTP (curl to the connmgr service would be enough).
2. Check the application pod and remove it to be redeployed.

Policies are not finished, resulting in a hanging state

Problem statement
: The policies are not finished, which results in a hanging state.

Cause
: Inconsistent behavior on policies results in a no finish status. The issue is still under investigation.

Resolution
: Identify the policy engine pod and eliminate it; OpenShift Container Platform will create another pod automatically, and policies will be executed and finished properly after the pod creation.
Example:

```
oc -n ibm-data-cataloging delete pod -l role=policyengine
```

IBM Data Cataloging service goes degraded state after IBM Fusion HCI rack restart

Problem statement
: The IBM Data Cataloging service goes degraded after some of the nodes are restarted or after IBM Fusion HCI rack restart. Several pods go pending with the following errors `Unable to attach or mount volumes: unmounted volumes=[spectrum-discover-db2wh], unattached volumes=[], failed to process volumes=[]: timed out waiting for the condition` and `MountVolume.SetUp failed for volume "xxx" : rpc error: code = Internal desc = staging path yyy for volume zzz is not a mountpoint`.

Resolution
: Run the following steps to resolve this issue:

1. Run the following commands to make the compute nodes unschedule and drain them one after one.

   ```
   oc adm cordon worker4.fusion-test-zlinux.cp.fyre.ibm.com
   oc adm drain worker4.fusion-test-zlinux.cp.fyre.ibm.com --ignore-daemonsets --force --delete-emptydir-data
   ```

2. After the node is drained, ensure it schedules again and then proceed to another node with the same process.
   It removes stale directory entries from nodes that are detected as mount points.
3. The issue automatically resolves and IBM Data Cataloging service must be in a healthy state after all the nodes are backed up.

Db2 license does not display correctly on the upgrade set up

Problem statement
: The Db2 license displays incorrectly on the IBM Data Cataloging service upgrade set up.

Resolution
: Run the following steps to resolve this issue:

1. Run the following command to get the subsequent IBM Data Cataloging scoped content.

   ```
   oc project ibm-data-cataloging
   ```

2. For the new license to take effect, delete the Db2 engine pods for the `Db2uCluster` or `Db2uInstance`:

   ```
   oc delete $(oc get po -l type=engine,formation_id=isd -oname)
   ```

3. After the new Db2 pod is ready, verify the updated Db2 license:

   ```
   oc exec -it  c-isd-db2u-0 -- su - db2inst1 -c "db2licm -l"
   ```

   For more about Db2 community edition license certificate key, see [Upgrading your Db2 Community Edition license certificate key]

ConstraintsNotSatisfiable for db2u-operator

Problem statement
: The Db2 version used by IBM Data Cataloging service 2.1.6 got removed from the latest version of the IBM Operator Catalog.

Resolution
: Follow the steps to resolve the issue:

1. Add or update the IBM Catalog Source object `ibm-operator-catalog` in the `openshift-marketplace` namespace during install or upgrade.

   ```
   apiVersion: operators.coreos.com/v1alpha1
   kind: CatalogSource
   metadata:
     name: ibm-operator-catalog-old
     namespace: openshift-marketplace
   spec:
     displayName: IBM Operator Catalog
     publisher: IBM
     sourceType: grpc
     image: icr.io/cpopen/ibm-operator-catalog:@sha256:c2538264cb1882b1c98fea5ef162f198ce38ed8c940e82e3b9db458a9a46cb15
     updateStrategy:
       registryPoll:
         interval: 45m
   ```

2. Update the `ibm-operator-catalog` source with the latest tag after the installation or upgrade completed.

   ```
   apiVersion: operators.coreos.com/v1alpha1
   kind: CatalogSource
   metadata:
     name: ibm-operator-catalog-old
     namespace: openshift-marketplace
   spec:
     displayName: IBM Operator Catalog
     publisher: IBM
     sourceType: grpc
     image: icr.io/cpopen/ibm-operator-catalog:latest
     updateStrategy:
       registryPoll:
         interval: 45m
   ```

ConstraintsNotSatisfiable for db2u-operator

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   ```
   apiVersion: operators.coreos.com/v1alpha1
   kind: CatalogSource
   metadata:
     name: ibm-operator-catalog-old
     namespace: openshift-marketplace
   spec:
     displayName: IBM Operator Catalog
     publisher: IBM
     sourceType: grpc
     image: icr.io/cpopen/ibm-operator-catalog@sha256:5d606e4eb2b875e0b975f892e80343105ea5fb0d67f96e1400d77a715f6df72a
    updateStrategy:
       registryPoll:
         interval: 45m
   ```

2. Update the `ibm-operator-catalog` source with the latest tag after the installation or upgrade completed.

   ```
   apiVersion: operators.coreos.com/v1alpha1
   kind: CatalogSource
   metadata:
     name: ibm-operator-catalog-old
     namespace: openshift-marketplace
   spec:
     displayName: IBM Operator Catalog
     publisher: IBM
     sourceType: grpc
     image: icr.io/cpopen/ibm-operator-catalog:latest
     updateStrategy:
   ```

```
  registryPoll:
    interval: 45m
```

Content-Aware Storage (CAS)

The IBM Fusion Content-Aware Storage service is engineered to enhance the value of customers' AI applications, such as retrieval augmented generation (RAG) for faster time to insights, reduced cost, improved performance, enhanced security, and streamlined operations.

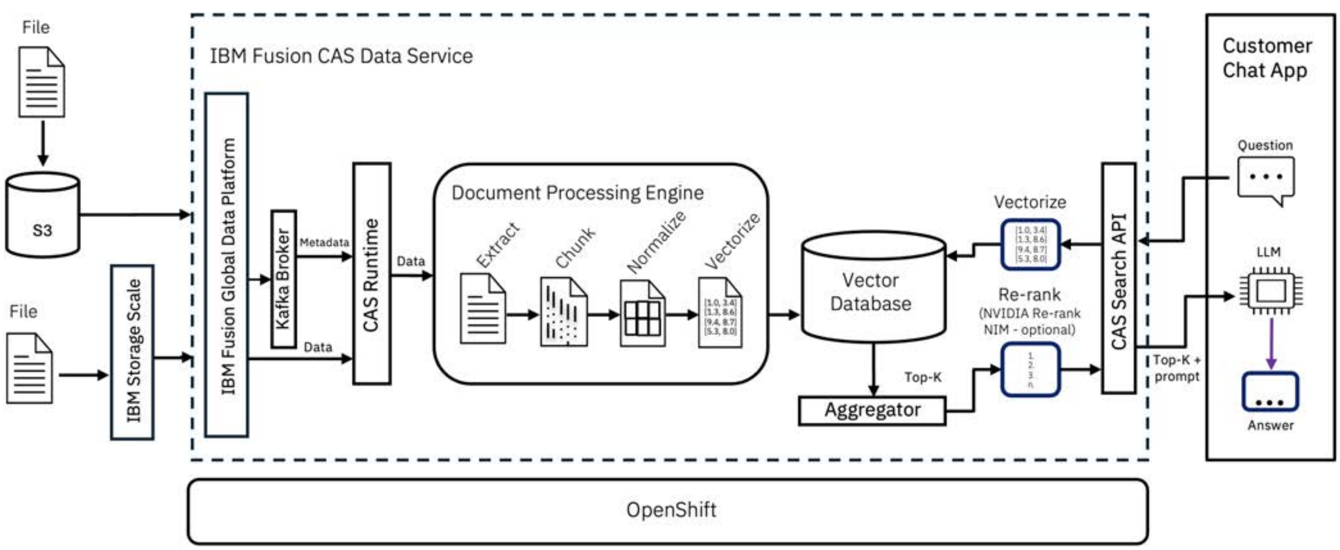
CAS provides turn-key processing of unstructured data for use in multi-modal RAG applications using IBM Docling Multimodal services and NVIDIA NeMo Retriever Extraction microservices. It supports automated extraction of text, tables, and charts from Enterprise documents and makes the extracted information available for use in RAG applications. NVIDIA GPUs are used to accelerate the data extraction processing. For more information on CAS-related frequently asked questions, see Content-Aware Storage FAQs.

CAS uses IBM Storage Scale to cache existing Enterprise data residing on S3 sources to optimize processing with the IBM Docling Multimodal services and NVIDIA NeMo Retriever Extraction microservices without having to manage multiple copies of data. CAS also supports automated processing of data residing in the IBM Storage Scale filesystem. Incremental change processing is leveraged to detect when data is modified in the IBM Storage Scale filesystem and automatically process the changed data for use in RAG applications.

- **Architecture**
  The objective of the Content-Aware Storage (CAS) architecture is to facilitate smooth interactions between LLMs and extensive quantities of unstructured data, amplifying insight derivation and suggestion functionalities.
- **Installing Content-Aware Storage (CAS)**
  Initiate the installation of CAS from the user interface of IBM Fusion. It installs Kafka, CAS operator, and CNPG.
- **Configuring Content-Aware Storage (CAS)**
  This section provides instructions for configuring CAS.
- **Client application with Content-Aware Storage (CAS)**
  You are required to provide a client application that uses the CAS search API to leverage the data ingested by CAS as part of your RAG application. This topic outlines the workflow for querying the CAS database using the exposed API.
- **Monitoring**
  This page provides a comprehensive view of monitoring capabilities, structured into two key sections: monitoring using dashboard in Content-Aware Storage (CAS) user interface and metric collection through Prometheus.
- **Uninstalling Content-Aware Storage (CAS) service**
  When you do not need CAS service, uninstall it from IBM Fusion.
- **Troubleshooting issues and known limitations in Content-Aware Storage (CAS)**
  The common troubleshooting steps and limitations in CAS service.
- **Related information**
  IBM publications, IBM MediaCenter videos, and external videos that contain information that is related to Content-Aware Storage (CAS).

Architecture

The objective of the Content-Aware Storage (CAS) architecture is to facilitate smooth interactions between LLMs and extensive quantities of unstructured data, amplifying insight derivation and suggestion functionalities.



The IBM Fusion CAS service uses IBM Fusion Global Data Platform service to remote mount a IBM Storage Scale cluster. This cluster can be Storage Scale software-based, or Storage Scale Server 3500 or 6000-based. CAS automatically processes files that are stored on the remote IBM Storage Scale cluster into parsed content, and metadata is normalized and stored along with vector embeddings in its managed vector database. Document processing is available with IBM's Docling Multimodal services or NVIDIA NeMo Retriever Extraction microservices.

IBM Storage Scale watch folders are leveraged to detect changed data in the remote IBM Storage Scale filesystem, which triggers ingest processing by CAS of just the changed content.

Files stored in external S3 data sources are ingested through Active File Management (AFM) running in Global Data Platform. No external IBM Storage Scale filesystem is required. CAS performs a scan of the AFM-S3 fileset automatically to determine the data that needs to be ingested.

The CAS service provides a search API that aggregates semantic and keyword search to return the most accurate results. NVIDIA's reranking is an optional search component that can be added to provide more accuracy.

Customer applications such as a chatbot can send natural language searches to the CAS search API. Upon receiving the request, CAS creates a vector representation of the natural language search and performs a query against the vector database that uses the search method that is specified in the API (semantic, keyword, or hybrid). In the NVIDIA Multimodal pipeline, the top 100 results are sent to a CAS aggregator component that uses additional keyword and optionally vendor reranking to produce the top number of requested results. The optimized results are passed to a reranking function, which performs an additional optimization of the results before returning them to the search API.

In a customer provided RAG client, the CAS search API can be used to retrieve relevant search results, which can then be combined with the original prompt or question and sent to a large language model (LLM) to generate a response for the end user. IBM provides a sample RAG client in the form of a chatbot that customers can use to explore CAS capabilities or as a foundation for developing their own chatbot or RAG-based solution.

Note: CAS leverages natural language processing (NLP) and AI technologies to extract semantic meaning from only unstructured text data.
Important: To know about the resource usage like, vCPU, memory, storage, and GPU, see Planning and prerequisites.

- **Understanding user personas**
  The personas of IBM Fusion Content-Aware Storage (CAS) service include Scale Admin, Data Engineer, and Developer. The Developer connects RAG application for working with the ingested content.

Understanding user personas

The personas of IBM Fusion Content-Aware Storage (CAS) service include Scale Admin, Data Engineer, and Developer. The Developer connects RAG application for working with the ingested content.

Scale Admins

Scale Admins are individuals responsible for managing storage software, such as IBM Storage Scale clusters and storage infrastructure. The Scale Admin deploys CAS with IBM Fusion and AI pipeline onto OpenShift® Container Platform cluster. The Scale Admin can see events for CAS in the same interface as IBM Fusion for a consistent serviceability experience.

The key responsibilities of a Scale Admin are as follows:

- Deploy, configure, and manage IBM Storage Scale clusters to ensure optimal performance and reliability.
- Configure and fine-tune IBM Storage Scale to efficiently handle high-throughput, low-latency AI data processing, ensuring integration with GPUs and AI frameworks.
- Work with Data Engineer to oversee data ingestion, preprocessing, and movement across storage tiers.
- Monitor system health of the Scale environment, implement caching strategies, and optimize data placement to prevent bottlenecks in AI model training and deployment.
- Leverage automation tools and scripts to streamline data lifecycle management, scaling storage resources dynamically to meet AI workload demands.

Data Engineer

Data engineers leverage data lakehouses as a flexible and scalable solution to store, process, and analyze diverse data types. Data Engineer defines a fileset on its storage content and connects it to an AI pipeline. In addition, the Data Engineer monitors the execution of the AI pipeline to understand the amount and timeliness of processing changes in content. As a Data Engineer, I can configure an Nvidia NIM pipeline to my Scale Fileset through CRs.

The key responsibilities of a Data Engineer are as follows:

- Storage of raw data in its original format. Including structured, semi-structured and unstructured data.
- Design and implement data ingestion pipelines that extract data from various sources, ensuring that all relevant data is available via a single interface for analysis and reporting.
- Maintain/curate metadata catalogs that capture information about data sources, data transformations, and data dependencies.
- Transform, augment, cleanse, and derive new datasets from existing data in accordance with their consumer's needs.
- Define access controls, data security measures, and data privacy policies to ensure compliance with policy and regulatory requirements.
- Enable data scientists, analysts, and other stakeholders to explore and analyze data.

Application Developer

The Application Developer queries and chats using the client enterprise application.

Installing Content-Aware Storage (CAS)

Initiate the installation of CAS from the user interface of IBM Fusion. It installs Kafka, CAS operator, and CNPG.

Before you begin

- Proceed with the installation only after all the prerequisites are met. For more information about prerequisites, see Planning and prerequisites.
- Ensure that you have storage class and is marked as default. For example, `ibm-spectrum-fusion`. For more information about how to mark a storage class as default, see Prerequisites topic.
- Ensure all Scale projects and pods are running.

About this task

Use this procedure to install the CAS service from the Services page of the IBM Fusion user interface. You can also install the CAS service using CLI. For more information about CLI usage, see Configuring Content-Aware Storage (CAS) through CLI.

Follow the procedure if:

- Your data resides on an external IBM Storage Scale filesystem.
- You prefer to use an external IBM Storage Scale filesystem as cache filesystem for your S3 data source.

Otherwise, follow the procedure in Installing CAS without a remotely mounted IBM Storage Scale filesystem.

Procedure

1. In the IBM Fusion user interface, go to Services page.
2. In the Available section, click the CAS tile.
3. In the CAS page, go through the features and capabilities of the service and click Install.
   If the pre-check raises a warning, then resolve it at OpenShift level.
   The installation starts and a notification appears on the Services page. You can see the progress of the installation in the Services > **Installed** section. After the installation is completed successfully, you can see the status as Healthy. A success notification gets displayed. It disappears automatically after some time.

   From the ellipsis menu, you can download logs and view documentation.
4. In the Install service window, select storage class and click Install.
5. Validate the installation.
   - Verify the CAS installation from the IBM Fusion user interface:
     After you enable the CAS service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.
   - Verify the CAS installation from the OpenShift® Container Platform web console:
     - In the OpenShift console, go to Home > Projects.
     - In the Projects page, search and check for the availability of a namespace with string `ibm-cas`.
     - Go to Operators > Installed Operators and check for the availability of CAS.
     - Open Content Aware Storage and check whether a CAS service definition is available.
     - After you enable the CAS service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

What to do next

- To verify the percentage completion of prerequisites post installation, do the following steps:
  1. Open IBM Fusion user interface.
  2. Go to Content-Aware Storage > Overview.
  3. In the Jump right in section, click Getting started with CAS tile.
  4. A side panel opens with the status of the following prerequisites:
     - Integrate NVIDIA NIM
     - Set up of Kafka Broker
     - Connect caching file system
- After you successfully install CAS, set up a Scale user to enable watch creation, configure domains and data sources. Map the domain to the data source. For more information about these procedures, see Configuring Content-Aware Storage (CAS).

- **Planning and prerequisites**
  Go through the Content-Aware Storage (CAS) installation prerequisites before you install the service.
- **Installing CAS without a remotely mounted IBM Storage Scale filesystem**
  Install CAS on bare metal without a remotely mounted IBM Storage Scale filesystem using local Data Foundation storage and S3 buckets as the data source.
- **Installing NVIDIA NeMo Retriever Library**
  The Content-Aware Storage (CAS) Document Processor Engine can either use IBM Docling Multimodal or NVIDIA NeMo Retriever Extraction for document extraction.
- **Configuring NVIDIA Inference Microservices (NIMs)**
  After you install CAS, the **`cas-config`** configmap must be created. It provides the NeMo Retriever Library endpoint information.
- **Configuring a Multi-Instance GPU (MIG) with CAS**
  Multi-Instance GPU (MIG) partitions a single NVIDIA GPU into multiple isolated instances. Each instance operates as an independent GPU, enabling resource sharing across multiple deployments.
- **Configuring IBM Storage Scale user to enable watch creation**
  Configure an IBM Storage Scale user with all the needed privileges to enable watch creation.

Planning and prerequisites

Go through the Content-Aware Storage (CAS) installation prerequisites before you install the service.

General requirements

Ensure that you meet the following prerequisites for the CAS solution:

- Ensure that you have IBM Fusion 2.13.0 version.
  Important:

- CAS service installation can be performed only if IBM Fusion is installed by using the default `ibm-spectrum-fusion-ns` namespace. If the IBM Fusion installation is performed by using a custom namespace, the CAS installation cannot be completed.
- IBM Fusion Access for Storage Area Network (SAN), Global Data Platform, and CAS are mutually exclusive services. If IBM Fusion Access for SAN is planned or installed, you cannot deploy the Global Data Platform or CAS services. Review your architectural requirements and deployment goals before installing these services.

- Resource requirements
- * IBM Storage Scale requirements. See IBM Storage Scale requirements.
- * Configure Scale remote filesystems. See Configure Scale remote file systems.
- * Ensure that you have the latest IBM Storage Scale 5.2.3.x version.
- * After at least one IBM Storage Scale remote filesystem is configured, in the OpenShift® console, go to Storage > Storage Class and verify whether the storage class created from the IBM Storage Scale remote filesystem is available for CAS. For example, if you enter `ibm-spectrum-fusion`, verify whether it is available.
- Configure NIMs. See Configuring NVIDIA Inference Microservices (NIMs).

* Skip these requirements if you are installing CAS without a remotely mounted IBM Storage Scale filesystem.

Configuring flags in CAS ConfigMap

To use the NVIDIA re-ranker service, you can add the `NVMM_NEMO_RANKER` and `NVMM_NEMO_RANKER_SERVICE` flags to the cas-config ConfigMap.

For more information, see Configuring NVIDIA Inference Microservices (NIMs).

Allowlist for proxy

Add the following URLs to the allowlist for proxy installation:

- `apigee.googleapis.com`: Using Apigee Hybrid runtime, you can learn about proxies, shared flows, and other key components. It also provides information about configuration and system health.
- `apigeeconnect.googleapis.com`: Needed for `apigee-mart-server` and `apigee-connect` communication when `VPC-SC` is enabled.
- `binaryauthorization.googleapis.com`: Optional, only for Anthos if the binary authorization is enabled.
- `gcr.io`: Google Container Registry where the container images are hosted.
- `raw.githubusercontent.com`: To install the operator manifest.

Resource requirements

Work with IBM representative to ensure you have all the hardware components that are necessary for the CAS solution. Allocate three OpenShift compute nodes.

- The following resources are required for CAS to run on the three compute nodes:
  - 3 x NVIDIA L40S or H100 GPUs
  - An additional L40S or H100 GPU is needed for the optional re-ranker service.
  - 166 vCPU (83 physical cores)
  - 1024GB memory
  - 1TB IBM Storage Scale
  - Most common IBM Storage Scale ESS configuration is 48 NVMe drives with the capacity of 30TB in the ESS 6000 with 4 CX-7 network adapters (1.5 TB memory).
  - When using the Docling Multimodal Processing Engine, see System requirements.
- For CAS service system requirements, see System requirements for each of the components in addition to the resource requirements.
  Note: Content-Aware Storage (CAS) depends upon IBM Fusion management software and the Global Data Platform service.

Enabling GPUs in OpenShift

To enable GPUs in the OpenShift cluster, following components must be installed to support GPU workloads:

- Red Hat’s Node Feature Discovery Operator - See the Red Hat documentation.
- NVIDIA’s GPU operator version - See the NVIDIA documentation.

To deploy GPU operators in a disconnected or airgapped environment, see the NVIDIA documentation.

NVIDIA Multimodal Document Processing Engine requirements

- The NVIDIA GPU operator, multi-modal RAG blueprint, and NVIDIA NIM operator are pulled from external NVIDIA NGC registry. As part of this process, you must provide NVIDIA NGC license keys to pull the NVIDIA components.
  NVIDIA NIM is a set of optimized cloud-native microservices that are designed to simplify the deployment of generative AI models anywhere, across cloud, data center, and GPU-accelerated workstations. Initially, you must deploy the NVIDIA NeMo Retriever. For more information about the procedure to deploy, see NVIDIA documentation.

- Optional: NVIDIA text reranking:
  The NVIDIA text reranker is an optional component that requires an additional supported GPU. This component takes the initial set of search results and uses additional scoring from NVIDIA to return the top ranked results. For more information about deploying NeMo Retriever text reranking NIM, see NVIDIA documentation.

- Contact IBM for the following details:
  - For optimization and performance of L40S or H100 GPU utilization of the NVIDIA NeMo Retriever Library multimodal PDF Ingest blueprint.
  - To configure NeMo Retriever Library NIM components with Multi-Instance GPU (MIG) for optimal use of supported GPUs or with time slicing. For more information, see Configuring a Multi-Instance GPU (MIG) with CAS.
  - Additional tuning parameters required to optimize pipeline performance.

Compatibility matrix for NVIDIA's NeMo Retriever Library Multimodal blueprint with CAS

Content-Aware Storage (version)	NVIDIA NeMo Retriever Library Multimodal blueprint

Content-Aware Storage (version)	NVIDIA NeMo Retriever Library Multimodal blueprint
v1.0.6, v1.0.7, v1.1.0	v25.9.0
v1.1.1, v1.1.2, v1.1.3, v1.1.4	v26.1.2
v1.1.5 and newer	v26.3.0

IBM Storage Scale requirements

If your datasource is on S3:

- Active File Management (AFM) is required to be configured on IBM Storage Scale remote file system. For more information, see Configuring Active File Management.
- IBM Storage Scale cluster must be at version 5.2.3.1 or above.

Configure Scale remote file systems

To configure remote file systems, see Connecting to remote IBM Storage Scale file systems. For software requirement levels, see Software requirement levels.
Important:

- When you setup the Global Data Platform user to create the connection to the remote file system, the user must be part of either the `CsiAdmin` or `ContainerOperator` groups.
  However, these groups do not have enough privileges to allow watcher creation. As a prerequisite to enable successful watcher creation, add these users to the `StorageAdmin` group. Create a user with all the privileges needed as a prerequisite. For more information about this configuration, see Configuring IBM Storage Scale user to enable watch creation.

- After the remote file system installation is complete, mark your storage class, for example, `ibm-spectrum-fusion`, as default, by running the following command:

  ```
  oc patch storageclass ibm-spectrum-fusion -p '{"metadata": {"annotations":{"storageclass.kubernetes.io/is-default-class":"true"}}}'
  ```

Offline installation

Content-Aware Storage supports usage of the Docling Multimodal document processing engine in an offline environment. For more information about the procedure to deploy offline, see Mirroring Content-Aware Storage (CAS) images.

Installing CAS without a remotely mounted IBM Storage Scale filesystem

Install CAS on bare metal without a remotely mounted IBM Storage Scale filesystem using local Data Foundation storage and S3 buckets as the data source.

Before you begin

- Proceed with the installation only after all the prerequisites are met. For more information about prerequisites, see Planning and prerequisites.
- For IBM Fusion services and platform support matrix, see Support matrix for IBM Fusion versions.

Supported environment

- IBM Fusion Bare Metal
- Data Foundation is installed or intended to be used as the local storage provider.
- Data resides on AWS or IBM S3 buckets only.

About this task

To install CAS without a remotely mounted IBM Storage Scale filesystem, follow the instructions in README.md.

Limitations

- The automation script creates a local IBM Storage Scale filesystem named `cache-fs`. This filesystem is reserved exclusively as a cache for S3 data and must not be used by other applications.
- The `cache-fs` filesystem is not expandable.
- If your CAS data source is connected but I/O ingestion is slow, check whether:
  1. Data Foundation shows a warning alert `SystemMemoryExceedsReservation`.
  2. Any Object Storage Daemon (OSD) pod has `restarts`:

     ```
     oc get pods -n openshift-storage -l app=rook-ceph-osd
     ```

     For pods that have `restarts`, check if termination reason is `OOMKilled`:

     ```
     oc describe pod rook-ceph-osd-xxx -n openshift-storage
     ```

     Use the following procedure to increase the OSD pod memory from the default 8 GB:

     https://access.redhat.com/solutions/6959127

- The system supports the shutdown or failure of only one quorum node at any specific time.

Installing NVIDIA NeMo Retriever Library

The Content-Aware Storage (CAS) Document Processor Engine can either use IBM Docling Multimodal or NVIDIA NeMo Retriever Extraction for document extraction.

Note: NVIDIA Ingest (nv-ingest) has been renamed to NeMo Retriever Library.
For more information about installing the GPU operator, see Enabling GPUs in OpenShift.

For more information about installing NVIDIA NeMo Retriever Library, see the NVIDIA documentation.

Verify the PodPidsLimit

To ensure compatibility and stability, you must update your cluster and deployment settings when using NeMo Retriever Library v26.3.0 with CAS v1.1.5 or newer.

1. To retrieve the current configuration, run the following command:

```
oc get KubeletConfig
```

2. To increase the pod PID limit, modify the **kubeletConfig** by adding the following **podPidsLimit** to the **spec** section:

```
spec:
  kubeletConfig:
    podPidsLimit: 12228
```

Note: After updating the configuration, all worker nodes in the cluster get restarted which can cause a temporary disruption.
For more information on changing the **podPidsLimit** when using the ROSA environment, see the Red Hat documentation.

Install the NIM Operator

NeMo Retriever Library requires that the NIM Operator is deployed and running. It is required to install the NIM Operator first before running the NeMo Retriever Library Helm chart. The operator can be installed from the OpenShift® marketplace. Search for The NVIDIA NIM Operator for Kubernetes, and install using the default options.

Select NVIDIA Inference Microservices (NIMs)

Before installing NIMs from the Helm chart, select the NIMs that are needed to extract your documents. For example, if your documents include audio files, installation of the audio NIM is required. Each NIM requires one GPU resource, which can either be a full GPU, or a GPU slice using a Multi-Instance GPU (MIG). The audio, reranking, and image captioning NIMs require a full GPU to run properly. By default, the NeMo Retriever Library Helm chart deploys five NIMs as indicated in the following table. For more information about MIG, see Configuring a Multi-Instance GPU (MIG) with CAS.

NIM	Usage	Parameter in Helm chart	Default setting
`llama-nemotron-embed-1b-v2`	Text embedding	`nimOperator.embedqa.enabled`	`true`
`nemotron-page-elements-v3`	Object detection within documents	`nimOperator.page_elements.enabled`	`true`
`nemotron-graphic-elements-v1`	Chart extraction	`nimOperator.graphic_elements.enabled`	`true`
`nemotron-table-structure-v1`	Table extraction	`nimOperator.table_structure.enabled`	`true`
`nemotron-ocr-v1`	Optical character recognition (OCR)	`nimOperator.ocr.enabled`	`true`
`nemotron-nano-12b-v2-v1`	Vision language model (VLM) image captioning	`.nemotron_nano_12b_v2_vl.enabled`	`false`
`audio`	Audio extraction*	`nimOperator.audio.enabled`	`false`
`llama-nemotron-rerank-1b-v2`	Reranker for query accuracy	`nimOperator.rerankqa.enabled`	`false`

* Audio extraction is in the early access mode from NVIDIA with version 26.1.2. For more information, see Use NeMo Retriever Extraction with Riva for Audio Processing.

Install NeMo Retriever Library

To install NeMo Retriever Library, follow these steps:

1. Create a new project called **nv-ingest**:

```
oc new-project nv-ingest
```

2. In your command line session, set the following bash variables:

```
NGC_API_KEY=<api key>
NAMESPACE=nv-ingest
```

Tip: To generate an NVIDIA NGC API key, see Generate Your NGC Keys.

3. Add **helm** repositories:

```
# Nvidia nemo-microservices NGC repository
helm repo add nemo-microservices https://helm.ngc.nvidia.com/nvidia/nemo-microservices --username='$oauthtoken' --
password=<NGC_API_KEY>

# Nvidia NIM NGC repository
helm repo add nvidia-nim https://helm.ngc.nvidia.com/nim/nvidia --username='$oauthtoken' --password=<NGC_API_KEY>

# NVIDIA NIM Base repository
helm repo add nim https://helm.ngc.nvidia.com/nim --username='$oauthtoken' --password=<NGC_API_KEY>

# Nvidia NIM baidu NGC repository
helm repo add baidu-nim https://helm.ngc.nvidia.com/nim/baidu --username='$oauthtoken' --password=<NGC_API_KEY>
```

4. Deploy NeMo Retriever Library using the **helm** chart. To select NIMs, see Select NVIDIA Inference Microservices (NIMs).
Example 1: Deploy NeMo Retriever Library with the five default NIM services deployed

```
helm upgrade \
   --install \
   nv-ingest \
   https://helm.ngc.nvidia.com/nvidia/nemo-microservices/charts/nv-ingest-26.3.0.tgz \
   -n ${NAMESPACE} \
   --username '$oauthtoken' \
   --password "${NGC_API_KEY}" \
   --set ngcImagePullSecret.create=true \
   --set ngcImagePullSecret.password="${NGC_API_KEY}" \
   --set ngcApiSecret.create=true \
   --set ngcApiSecret.password="${NGC_API_KEY}" \
   --set image.repository="nvcr.io/nvidia/nemo-microservices/nv-ingest" \
   --set image.tag="26.3.0" \
   --set milvusDeployed=false
```

Example 2: Deploy NeMo Retriever Library with the default NIM services (same as Example 1) along with the audio Riva NIM

```
helm upgrade \
   --install \
   nv-ingest \
   https://helm.ngc.nvidia.com/nvidia/nemo-microservices/charts/nv-ingest-26.3.0.tgz \
   -n ${NAMESPACE} \
   --username '$oauthtoken' \
   --password "${NGC_API_KEY}" \
   --set ngcImagePullSecret.create=true \
   --set ngcImagePullSecret.password="${NGC_API_KEY}" \
   --set ngcApiSecret.create=true \
   --set ngcApiSecret.password="${NGC_API_KEY}" \
   --set nimOperator.audio.enabled=true \
   --set image.repository="nvcr.io/nvidia/nemo-microservices/nv-ingest" \
   --set image.tag="26.3.0" \
   --set milvusDeployed=false
```

Example 3: Deploy NeMo Retriever Library with the default NIM services (same as example 1) along with the audio Riva NIM and the reranking NIM

```
helm upgrade \
   --install \
   nv-ingest \
   https://helm.ngc.nvidia.com/nvidia/nemo-microservices/charts/nv-ingest-26.3.0.tgz \
   -n ${NAMESPACE} \
   --username '$oauthtoken' \
   --password "${NGC_API_KEY}" \
   --set ngcImagePullSecret.create=true \
   --set ngcImagePullSecret.password="${NGC_API_KEY}" \
   --set ngcApiSecret.create=true \
   --set ngcApiSecret.password="${NGC_API_KEY}" \
   --set nimOperator.audio.enabled=true \
   --set nimOperator.rerankqa.enabled=true \
   --set image.repository="nvcr.io/nvidia/nemo-microservices/nv-ingest" \
   --set image.tag="26.3.0" \
   --set milvusDeployed=false
```

Example 4: Minimum configuration

For a minimum configuration installation that extracts only text from files (text or PDF), the text embedding NIM is required. However, the other NIM services can be disabled.

```
helm upgrade \
   --install \
   nv-ingest \
   https://helm.ngc.nvidia.com/nvidia/nemo-microservices/charts/nv-ingest-26.3.0.tgz \
   -n ${NAMESPACE} \
   --username '$oauthtoken' \
   --password "${NGC_API_KEY}" \
   --set ngcImagePullSecret.create=true \
   --set ngcImagePullSecret.password="${NGC_API_KEY}" \
   --set ngcApiSecret.create=true \
   --set ngcApiSecret.password="${NGC_API_KEY}" \
   --set nimOperator.graphic_elements.enabled=false \
   --set nimOperator.page_elements.enabled=false \
   --set nimOperator.table_structure.enabled=false \
   --set nimOperator.ocr.enabled=false \
   --set image.repository="nvcr.io/nvidia/nemo-microservices/nv-ingest" \
   --set image.tag="26.3.0" \
   --set milvusDeployed=false
```

5. Check the PVC sizes created for the **NIMCache**:

```
oc get pvc -n nv-ingest
```

Note: You are recommended to increase the audio NIM to 250Gi, reranker NIM to 100Gi, and the **embedqa** NIM to 100Gi to avoid issues downloading the required models.
Expand the following PVCs:

   a. **embedqa** NIM PVC

   ```
   oc patch pvc llama-nemotron-embed-1b-v2-pvc  -n nv-ingest --type merge -p '{"spec":{"resources":{"requests":
   {"storage":"100Gi"}}}}'
   ```

   b. Audio NIM PVC (only applicable when the audio NIM is enabled)

   ```
   oc patch pvc audio-pvc  -n nv-ingest --type merge -p '{"spec":{"resources":{"requests":{"storage":"250Gi"}}}}'
   ```

   c. Reranker NIM PVC (only applicable when the reranking NIM is enabled):

```
oc patch pvc llama-nemotron-rerank-1b-v2-pvc  -n nv-ingest --type merge -p '{"spec":{"resources":{"requests":
{"storage":"100Gi"}}}}'
```

6. Verify that the pods are running:

```
oc get deployments -n nv-ingest
```

Example output:

```
NAME                                    READY   UP-TO-DATE   AVAILABLE   AGE
audio                                   1/1     1            1           5d23h
llama-nemotron-embed-1b-v2              1/1     1            1           5d23h
llama-nemotron-rerank-1b-v2             1/1     1            1           5d23h
nemotron-graphic-elements-v1            1/1     1            1           5d23h
nemotron-nano-12b-v2-vl                 1/1     1            1           5d23h
nemotron-ocr-v1                         1/1     1            1           5d23h
nemotron-page-elements-v3               1/1     1            1           5d23h
nemotron-table-structure-v1             1/1     1            1           5d23h
nv-ingest                               1/1     1            1           5d18h
nv-ingest-opentelemetry-collector       1/1     1            1           5d18h
nv-ingest-zipkin                        1/1     1            1           5d18h
```

Configuring NVIDIA Inference Microservices (NIMs)

After you install CAS, the `cas-config` configmap must be created. It provides the NeMo Retriever Library endpoint information.

To determine the NeMo Retriever Library endpoint information, you must know the namespace in which NeMo Retriever Library is installed. The endpoints are constructed by using the pattern `http(s)://..svc.cluster.local`. When you use the default installation namespace of NeMo Retriever Library and because it does not support `https` by default, it results in the following URLs:

- Embed service URL: `http://llama-32-nv-embedqa-1b-v2.nv-ingest.svc.cluster.local`
- Ingest service URL: `http://nv-ingest.nv-ingest.svc.cluster.local`

Note: These URLs must not end with the `/` character.
To complete this activity, use the **oc** command line tool and run the following command:

```
oc apply -f - <<EOF
apiVersion: v1
kind: ConfigMap
metadata:
  name: cas-config
  namespace: ibm-cas
data:
  NVMM_EMBED_SERVICE: <Embed Service URL as determined above>
  NVMM_NIM_SERVICE:  <Ingest Service URL as determined above>
EOF
```

NVIDIA re-ranker

Optionally, you can set `NVMM_NEMO_RANKER` in the `CasInstall` CR to enable the NVIDIA re-ranker service. This service analyzes semantic relevance and reorders search results to improve precision in enterprise search and streamline AI-driven workflows. For more information, see the NVIDIA documentation.

- Re-ranker service URL: `http://llama-32-nv-rerankqa-1b-v2.nv-ingest.svc.cluster.local`

Note: The service URL must not end with the `/` character.
To enable the NVIDIA re-ranker service, set the `NVMM_NEMO_RANKER` flag to `YES` and the `NVMM_NEMO_RANKER_SERVICE` flag to the re-ranker service URL in the `cas-config` ConfigMap:

```
spec:
oc patch configmap cas-config \
  -n ibm-cas \
  --type merge \
  -p '{"data":{"NVMM_NEMO_RANKER": "YES", "NVMM_NEMO_RANKER_SERVICE": "<Re-ranker service URL as determined above>"}}'
```

Configuring a Multi-Instance GPU (MIG) with CAS

Multi-Instance GPU (MIG) partitions a single NVIDIA GPU into multiple isolated instances. Each instance operates as an independent GPU, enabling resource sharing across multiple deployments.

Before you begin

MIG support requirements
: Only specific NVIDIA GPUs support MIG. For more information on the supported models and their configurations, see the NVIDIA documentation.
  Note: The NeMo Retriever Library NIMs support only a subset of the GPUs that support MIG.

Before you configure MIG, plan how to partition each GPU by using the supported profiles. Each profile defines the amount of memory and compute engines that the slice can use.

For example, an A100 GPU with 80 GB of memory can be partitioned in the following way:

- One 40 GB slice with 3 compute engines

- Two 20 GB slices with 2 compute engines each

Each A100 GPU contains seven compute engines. Ensure that the total number of compute engines across all slices on a GPU do not exceed seven.
Note: For optimal ingestion performance, assign a 7g.80gb MIG profile to the **embedqa** NIM service to increase GPU compute capacity.

Example: Partitioning two A100 - 80 GB GPUs to support NeMo Retriever Library

The following example shows how to configure MIG on a Red Hat OpenShift Container Platform cluster with two A100 - 80 GB GPUs.
The A100 supports 40 GB, 20 GB, and 10 GB slices. The reranker NIM requires at least 22 GB of memory, so you must assign it a 40 GB slice. The **embedqa** NIM also receives a 40 GB slice due to its footprint. Assign the remaining NIMs to 20 GB slices.

Use the following partitioning approach:

- GPUs 0 and 1:
  - One 40 GB slice (3g.40gb)
  - Two 20 GB slices (2g.20gb)

About this task

This task configures MIG on NVIDIA GPUs that support the MIG feature. The steps involve enabling MIG, defining GPU partitions, applying the MIG configuration to your nodes, and assigning deployment to specific GPU slices.

To configure MIG on NVIDIA GPUs, perform the following steps:

Procedure

1. Apply the following configmap in the `nvidia-gpu-operator` namespace:

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nv-ingest-mig
data:
  config.yaml: |
    version: v1
    mig-configs:
      all-disabled:
        - devices: all
          mig-enabled: false
      nv-ingest-mig-config:
        - devices: [0, 1]
          mig-enabled: true
          mig-devices:
            "2g.20gb": 2
            "3g.40gb": 1
```

2. Temporary stop any pods that use the GPUs. The MIG manager cannot reconfigure GPUs that are in use.
3. Patch the `gpu-cluster-policy` and set the MIG strategy value to "mixed".

```
kubectl patch clusterpolicies.nvidia.com/gpu-cluster-policy \
    --type='json' \
    -p='[{"op":"replace", "path":"/spec/mig/strategy", "value":"mixed"}]'
```

4. Label one or more nodes with the GPUs that you want to configure your MIG settings, referencing the MIG configuration defined in the configmap. Then set the correct value to the `NODE` variable:

```
NODE=ip-10-0-9-79.us-west-2.compute.internal
MIG_CONFIGURATION=nv-ingest-mig-config
oc label node $NODE nvidia.com/mig.config=$MIG_CONFIGURATION --overwrite
```

5. Monitor the pods in the `nvidia-gpu-operator` namespace. The configuration must recycle several pods, including the `gpu-feature-discovery` pods.
6. To verify the MIG configuration, run the following command:

```
nvidia-smi
```

You can see MIG slices with memory usage metrics after NeMo Retriever Library is activated. Initially, the memory usage appears as 1 MiB until the pods start to use the slices.

```
+-----------------------------------------------------------------------------------+
| MIG devices:                                                                      |
+------------------+----------------------------------+-----------+-----------------------+
| GPU  GI  CI  MIG |                     Memory-Usage |        Vol|        Shared         |
|      ID  ID  Dev |                       BAR1-Usage | SM     Unc| CE  ENC  DEC  OFA  JPG |
|                  |                                  |        ECC|                       |
|==================+==================================+===========+=======================|
|  0    2   0   0  |           3590MiB / 40192MiB     | 42      0 |  3   0    2    0    0 |
|                  |               2MiB / 32767MiB    |           |                       |
+------------------+----------------------------------+-----------+-----------------------+
|  0    3   0   1  |           2959MiB / 19968MiB     | 28      0 |  2   0    1    0    0 |
|                  |               2MiB / 16383MiB    |           |                       |
+------------------+----------------------------------+-----------+-----------------------+
|  0    4   0   2  |           2971MiB / 19968MiB     | 28      0 |  2   0    1    0    0 |
|                  |               2MiB / 16383MiB    |           |                       |
+------------------+----------------------------------+-----------+-----------------------+
|  1    2   0   0  |          25386MiB / 40192MiB     | 42      0 |  3   0    2    0    0 |
|                  |               2MiB / 32767MiB    |           |                       |
+------------------+----------------------------------+-----------+-----------------------+
|  1    3   0   1  |           2955MiB / 19968MiB     | 28      0 |  2   0    1    0    0 |
|                  |               2MiB / 16383MiB    |           |                       |
```

```
+------------------+--------------------------------+-----------+-----------------------+
|  1    4    0    2 |           2281MiB / 19968MiB   | 28       0 |  2    0    1    0    0 |
|                  |                2MiB / 16383MiB  |           |                       |
+------------------+--------------------------------+-----------+-----------------------+
```

Note: If the MIG configuration fails to apply, retrigger the MIG manager by disabling and re-enabling it in the cluster policy configuration:

```
migManager:
    config:
      default: all-disabled
      name: nv-ingest-mig
    enabled: true -> false -> true
```

7. Before you scale up your microservices that require GPUs, such as NeMo Retriever Library NIMs, you must add the following snippets to their deployments:
   - For larger NIMs, such as the NVIDIA re-ranker:

     ```
     containers:
             - resources:
                 limits:
                   nvidia.com/mig-3g.40gb: '1'
     ```

   - For smaller NIMs:

     ```
     containers:
             - resources:
                 limits:
                   nvidia.com/mig-2g.20gb: '1'
     ```

8. Scale up the deployments. The pods are then allocated to the correct GPU slice.

Configuring IBM Storage Scale user to enable watch creation

Configure an IBM Storage Scale user with all the needed privileges to enable watch creation.

IBM Storage Scale has a spectrum of user roles, each with varying degree of privileges. For example, Security Administrator, Administrator, System Administrator, Storage Administrator, CSI Administrator, Container Operator. For more information about the user roles, see Scale Administration Guide.

As part of doing the IBM Storage Scale remote mount, the Global Data Platform asks the user to ensure that they are a part of both CSI Administrator and Container Operator. The StorageAdmin is needed to enable CAS to manage filesystem configurations such as clustered watch. For more information about the configuration steps, see Connecting to remote IBM Storage Scale file systems.

It is not sufficient for the CAS user to be a part of the Storage Administrator group, in addition to the `CsiAdmin`, `ContainerOperator` groups. As a IBM Storage Scale admin, run the following example command from command line:

```
/usr/lpp/mmfs/gui/cli/chuser csi-storage-gui-user -g CsiAdmin,ContainerOperator,StorageAdmin
```

Example output:

```
EFSSG0020I The user cast-user has been successfully changed.
EFSSG1000I The command completed successfully.
```

It involves the alignment of the CSI user name to match the one previously created during the remote cluster setup by using the Global Data Platform.

Configuring IBM Storage Scale for change detection

Determine your IBM Storage Scale version first:

```
mmdiag --version
```

For IBM Storage Scale 6.0.0.x and later, no manual configuration is required.

If you are running IBM Storage Scale version 5.2.3.x, you need to manually enable S3 change detection. As an IBM Storage Scale administrator, run the following commands:

```
mmchconfig syncReadWFConfig=yes --force -i
mmchconfig afmPtrashOpt=3 -i
```

Note: Run these commands only once for the entire IBM Storage Scale cluster.

Configuring Content-Aware Storage (CAS)

This section provides instructions for configuring CAS.

Do the following step to configure CAS from the IBM Fusion user interface:

- Creating a data source or Creating a domain and connecting with data source

Tip: Starting from CAS v1.1.1, CAS installation automatically configures the Kafka broker connection on IBM Storage Scale.
You can also configure CAS service using CLI. For more information about CLI usage, see Configuring Content-Aware Storage (CAS) through CLI.
Note: Before you begin the configuration, ensure you have met all the prerequisites and installed CAS successfully. For the procedure to install, see Installing Content-Aware Storage (CAS).
Quick start to configuration:
Note: If the Getting started with CAS section is not available, it indicates that all the prerequisites are not met.

1. In the IBM Fusion user interface, go to Content-Aware Storage > Overview.
2. In the Jump right in section, you can click the following tiles to quick start configuration to streamline data preparation, chunking and delivery to embedding or inference systems:
    - Connect a data source
    - Create a domain
    - View Search APIs
3. To view the percentage completion of the configuration, do the following steps:
    a. In the Jump right in section, click Getting started with CAS tile.
    b. A side panel opens with the status of the following Data ingestion:
        - Connect a data source
        - Create a domain
        - View Search APIs

- **Integrating CAS and IBM Data Cataloging**
  Rule-based integration between IBM Data Cataloging and Content-Aware Storage (CAS) ensures selective, policy-driven data ingestion into CAS vector stores.
- **Creating a data source**
  A data source is a repository or a system that stores and provides data for use in applications or systems. They are storage locations that provide data to use in AI workflows.
- **Creating a domain and connecting with data source**
  Domains collect, transform, and transfer data from multiple data sources to a destination, enabling efficient data flow for analysis and AI models.
- **Configuring Document Processor Engine**
  Steps to configure Document Processor Engines: IBM Docling Multimodal and NVIDIA Multimodal.
- **Configuring Content-Aware Storage (CAS) through CLI**
  You can use the CRs to configure CAS.
- **Using your own Hugging Face access token with CAS for NVIDIA Multimodal**
  NVIDIA NeMo Retriever Library uses models that are downloaded from Hugging Face to do token-based splitting on the content extracted from documents. NVIDIA recommends to use the meta-llama/Llama-3.2-1B tokenizer, however this model is gated and requires a Hugging Face token in order access it. A Hugging Face access token is a personal API key that is used to securely authenticate programmatic access to Hugging Face services.
- **Configuring CAS Query Access Control and Security**
  Content-Aware Storage (CAS) supports domain-level authorization, Identity Provider integration, and optional file-level security to control query access.

Integrating CAS and IBM Data Cataloging

Rule-based integration between IBM Data Cataloging and Content-Aware Storage (CAS) ensures selective, policy-driven data ingestion into CAS vector stores.

By integrating IBM Data Cataloging with CAS, organizations can apply advanced pre-filtering through configurable, rule-based policies that control which files are ingested into CAS.

Administrators can define criteria such as file type, size, location, and content patterns to ensure that only relevant data is stored in CAS vector stores. Custom tags and attributes further enhance this capability by providing fine-grained control over the ingestion process. For more information, see Managing IBM Data Cataloging rules when integrated with CAS.

- **Integrating IBM Data Cataloging for file filtering**
  You can enable integration of IBM Data Cataloging with Content-Aware Storage (CAS) to configure file ingestion filtering. Any file from a data source that meets the filtering criteria can be ingested into the corresponding domain. Filtering is enforced across all domains within the same IBM Fusion cluster.
- **Using IBM Data Cataloging attributes with CAS**
  Content-Aware Storage (CAS) integrates with IBM Data Cataloging service (DCS) to support automated file categorization through flexible, attribute-based tagging.
- **IBM Data Cataloging and Content-Aware Storage insights**
  IBM Data Cataloging and Content-Aware Storage (CAS) introduce data insights dashboards that provide immediate visibility into data once sources are connected and scanned. This experience is enabled when both IBM Data Cataloging and CAS are integrated. The provided insights surface key indicators of data composition on each established data source.

Integrating IBM Data Cataloging for file filtering

You can enable integration of IBM Data Cataloging with Content-Aware Storage (CAS) to configure file ingestion filtering. Any file from a data source that meets the filtering criteria can be ingested into the corresponding domain. Filtering is enforced across all domains within the same IBM Fusion cluster.

Before you begin

- Ensure that you have IBM Fusion 2.12.0 version.
- Install IBM Data Cataloging 2.3.0. For more information, see Installing IBM Data Cataloging.
- Connect to Scale Remote Filesystem from IBM Data Cataloging by using these instructions: Creating an IBM Storage Scale data source connection. For more information, see IBM Storage Scale data source connections.
- Understand the default filtering criteria. For more information, see Managing IBM Data Cataloging rules when integrated with CAS.

Enable or disable filtering through CLI

Enable filtering
: To enable filtering by using IBM Data Cataloging, run the following command:

```
oc patch casinstall ibm-cas-service-instance -n ibm-cas --type=merge -p '{"spec":{"enabled_features":{"dcs":
{"enable":true,"instance_name":"data-cataloging-service-instance","namespace":"ibm-data-cataloging"}}}}'
```

To confirm filtering is enabled, run the following command:

```
DCS_CR=$(oc get SpectrumDiscover -n ibm-data-cataloging -o jsonpath="{.items[*].metadata.name}")
oc get SpectrumDiscover $DCS_CR -n ibm-data-cataloging -o jsonpath='{.status.conditions[*]}{"\n"}'
```

Tip: You must see the type `ContentAwareStorage` listed as `SuccessfullyEnabled`:

```
{"lastTransitionTime":"2025-09-
10T08:33:42Z","message":"","reason":"SuccessfullyEnabled","status":"True","type":"ContentAwareStorage"}
```

Disable filtering

To disable filtering, run the following command:

```
oc patch casinstall ibm-cas-service-instance -n ibm-cas --type=merge -p '{"spec":{"enabled_features":{"dcs":
{"enable":false,"instance_name":"data-cataloging-service-instance","namespace":"ibm-data-cataloging"}}}}'
```

To confirm filtering is disabled, run the following command:

```
DCS_CR=$(oc get SpectrumDiscover -n ibm-data-cataloging -o jsonpath="{.items[*].metadata.name}")
oc get SpectrumDiscover $DCS_CR -n ibm-data-cataloging -o jsonpath='{.status.conditions[*]}{"\n"}'
```

Tip: You must no longer see the type `ContentAwareStorage` listed in the output.

One-click installation and enablement integration experience

CAS provides a one-click installation of IBM Data Cataloging directly from the Content-Aware Storage > Settings page. This integration streamlines the setup process by enabling administrators to install and configure IBM Data Cataloging without leaving the CAS user interface, improving the integration between both data services.

Overview

When IBM Data Cataloging is not installed, the Advanced filtering and scanning section in the Content-Aware Storage > Settings page displays an Install Data Cataloging button. Clicking this button launches the same installation flow that is used in the Services page. As a result, no additional configuration is required outside of CAS.

After installation completes, the same section enables controls to manage IBM Data Cataloging filtering:

- Modify rulesets: Opens the OpenShift® Container Platform custom resource editor to configure filtering rulesets. For more information, see Managing IBM Data Cataloging rules when integrated with CAS.
- Enable / Disable toggle: Enables or disables IBM Data Cataloging-based advanced filtering without uninstalling the service.

Prerequisites

Before starting the installation, ensure that you meet the following prerequisites:

- CAS v1.1.5 is deployed and the Content-Aware Storage > Settings page is accessible.
- The cluster meets the resource requirements for IBM Data Cataloging.

Installing IBM Data Cataloging

1. In the IBM Fusion console, go to Content Aware Storage > Settings.
2. Locate the Advanced filtering and scanning section. If IBM Data Cataloging is not installed, the section displays an Install Data Cataloging button.
3. Click Install Data Cataloging. An installation progress indicator that shows the current percentage replaces the button.
   Installation follows the same setup process as the Services page. If the installation fails, an error message is displayed in the same section with details to help you troubleshoot or check Fusion Events.
4. Wait for the installation to complete. The progress indicator disappears when IBM Data Cataloging is fully installed.

Managing IBM Data Cataloging after installation

After IBM Data Cataloging is installed, the Advanced filtering and scanning section updates to show the management controls.

Control	Description
Modify rulesets	Opens the OpenShift Container Platform custom resource for IBM Data Cataloging, where you can configure the filtering rulesets that determine which content CAS processes.
Enabled / Disabled toggle	Enables or disables IBM Data Cataloging based advanced filtering. Disabling does not uninstall the service.

Modifying rulesets

Click Modify rulesets to open the OpenShift Container Platform custom resource editor. Update the ruleset configuration as needed and save the resource. The changes take effect immediately. For more information, see Managing IBM Data Cataloging rules when integrated with CAS.

Enabling and disabling filtering

Use the toggle in the Advanced filtering and scanning section to enable or disable IBM Data Cataloging filtering at any time. When filtering is disabled, CAS continues to operate without IBM Data Cataloging-based content analysis until filtering is re-enabled.

Using IBM Data Cataloging attributes with CAS

Content-Aware Storage (CAS) integrates with IBM Data Cataloging service (DCS) to support automated file categorization through flexible, attribute-based tagging.

Attributes can be applied in DCS either before or after files are ingested into CAS, enabling administrators to enrich data with meaningful metadata such as document type, department, project, or other custom attributes. After attributes are applied in DCS, you can use the CAS Query Service API to perform filtered queries that return only the most relevant content chunks.

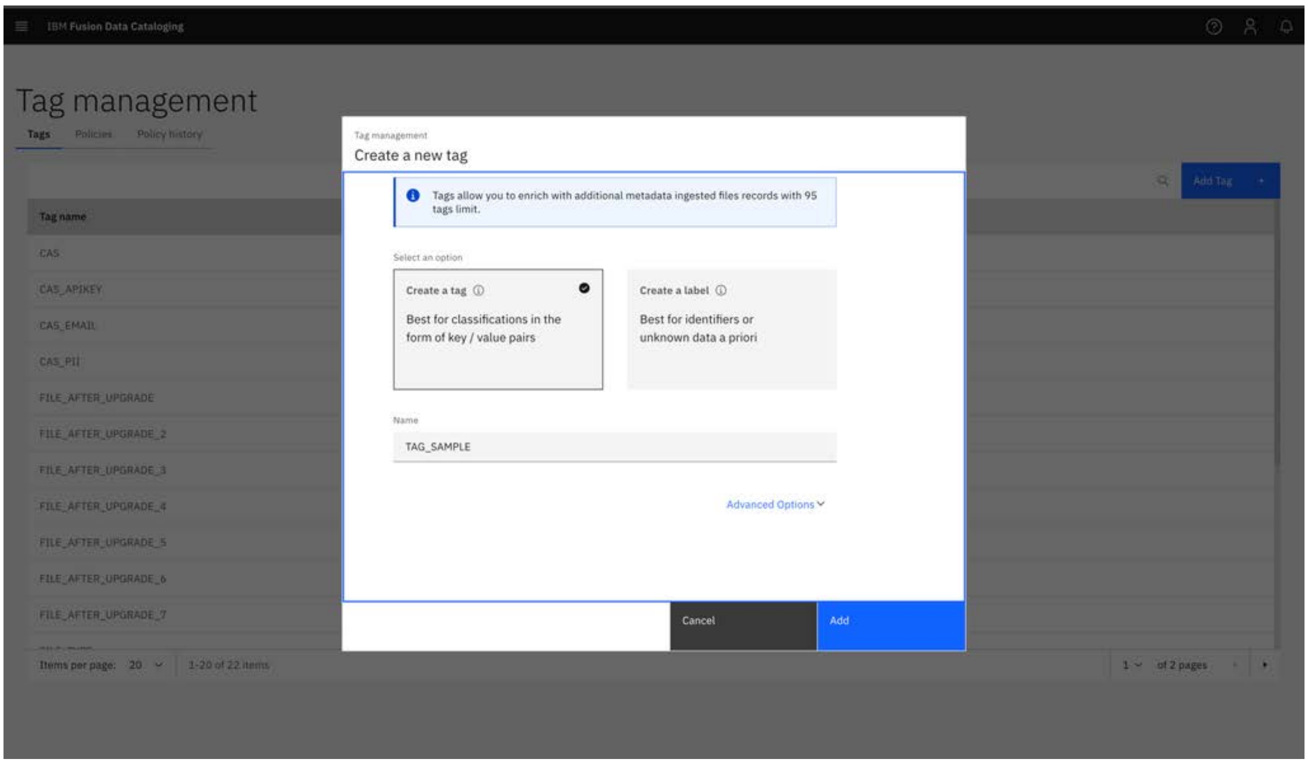
Creating a tag

In DCS, additional metadata is applied by creating tags. A tag is a custom metadata field that enhances existing metadata with organization-specific information and supports data classification, search, and reporting.

To create a tag, follow these steps:

1. In the IBM Fusion Data Cataloging UI, open the menu and go to Metadata > Tag management.
2. Open the Tags section, and click Add Tag.

The Create a new tag window opens.



3. In the Name field, specify a tag name, and click Add.

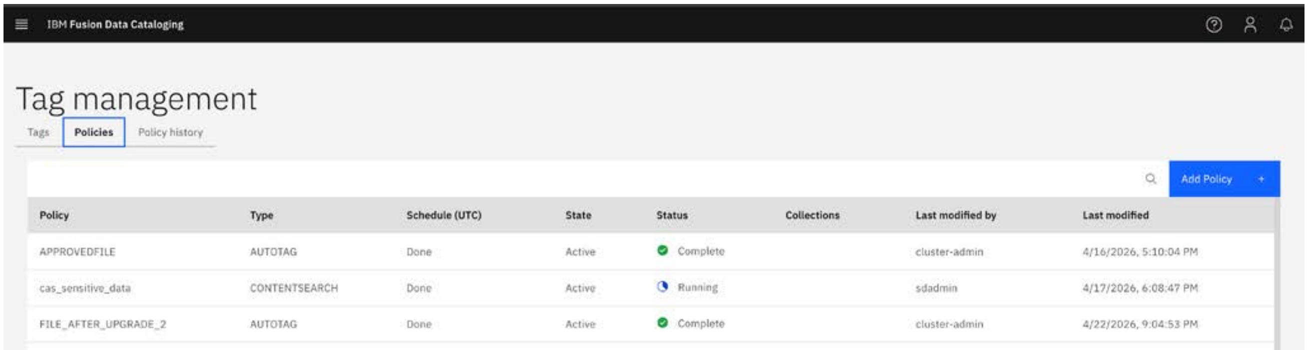
For additional details about tags and how to create them, see Managing tags.

Creating a policy

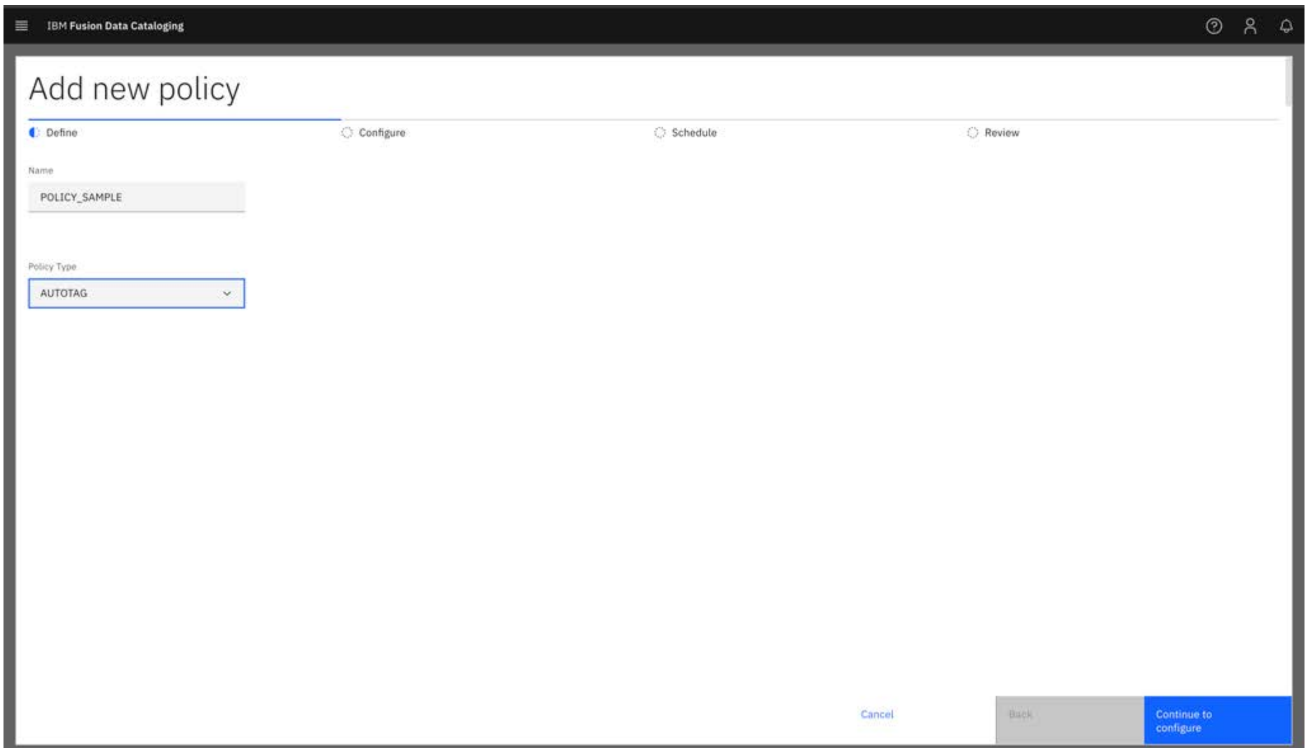
After a tag is created, a DCS policy must be defined to associate documents with that tag. The policy uses a user-defined query to classify documents, and when the policy is executed, the tag is automatically populated with values for all matching documents.

To create a policy, follow these steps:

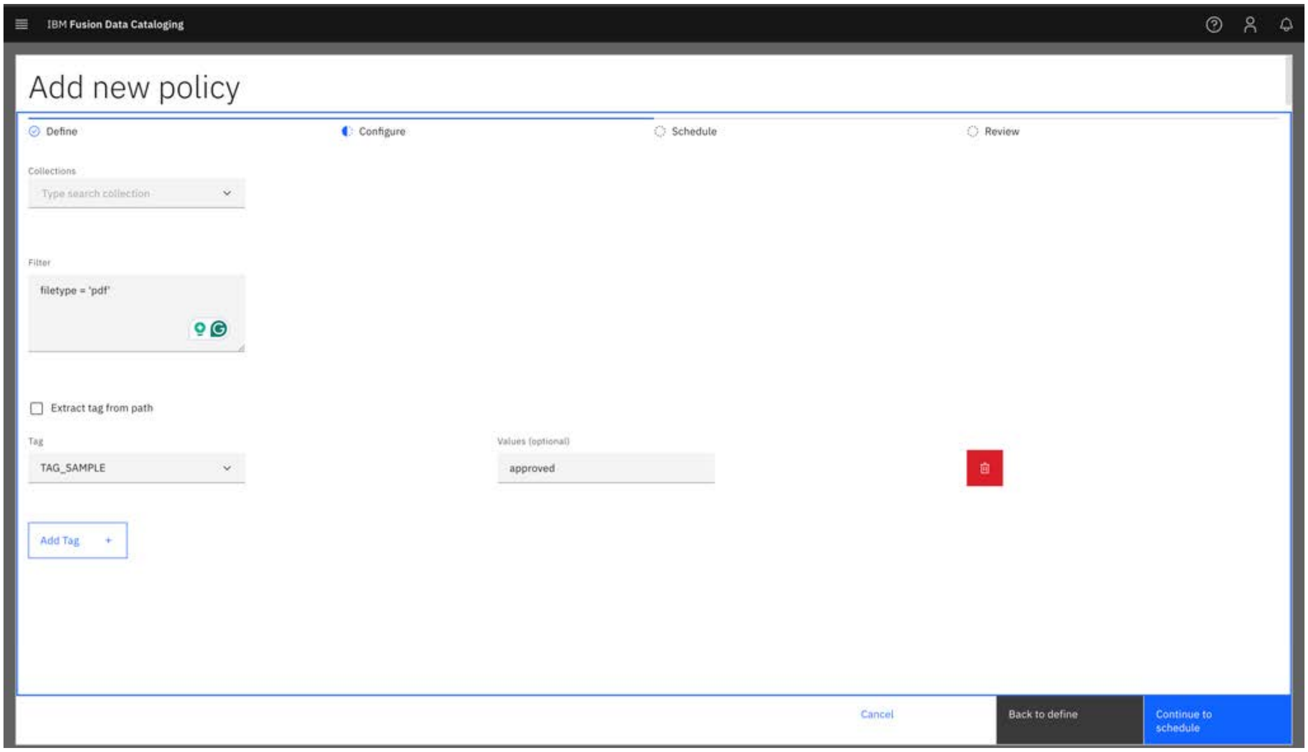
1. In the IBM Fusion Data Cataloging UI, open the menu and go to Metadata > Tag management.
2. Open the Policies section, and click Add Policy.



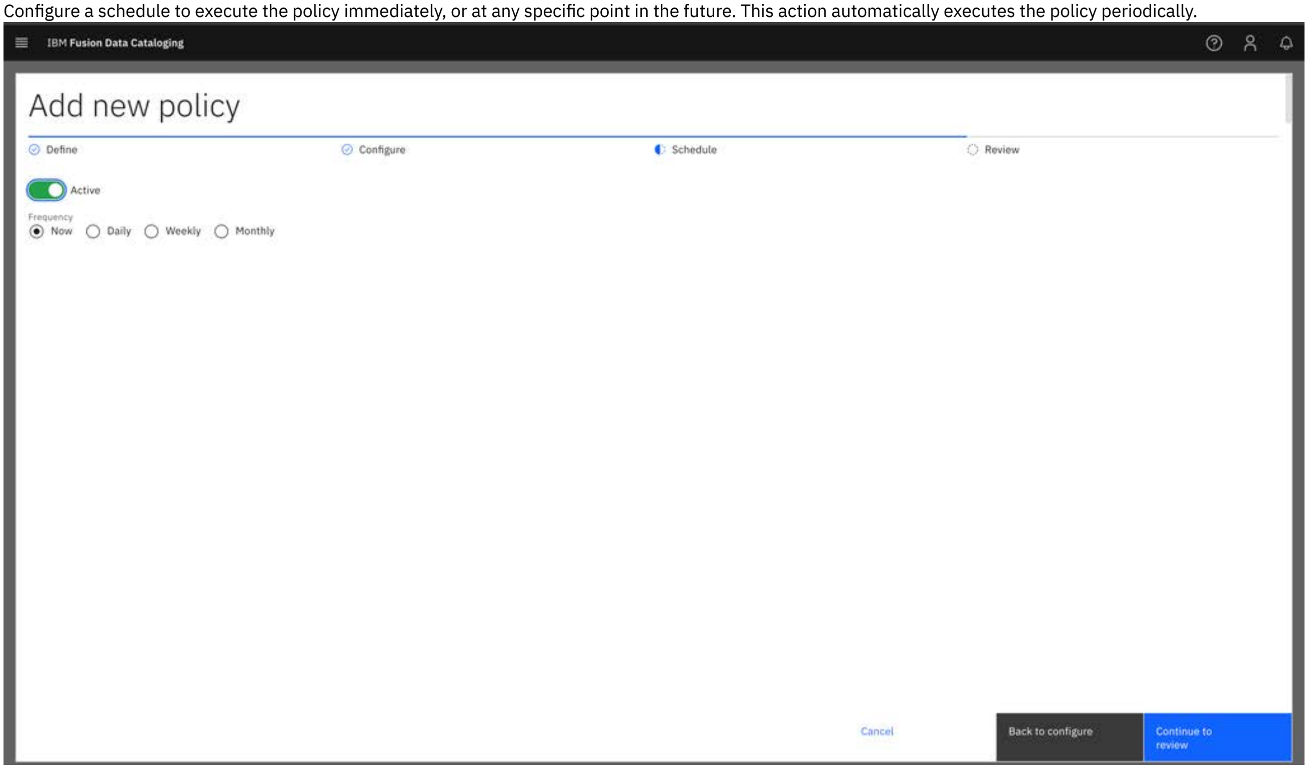
The Add new policy window opens.



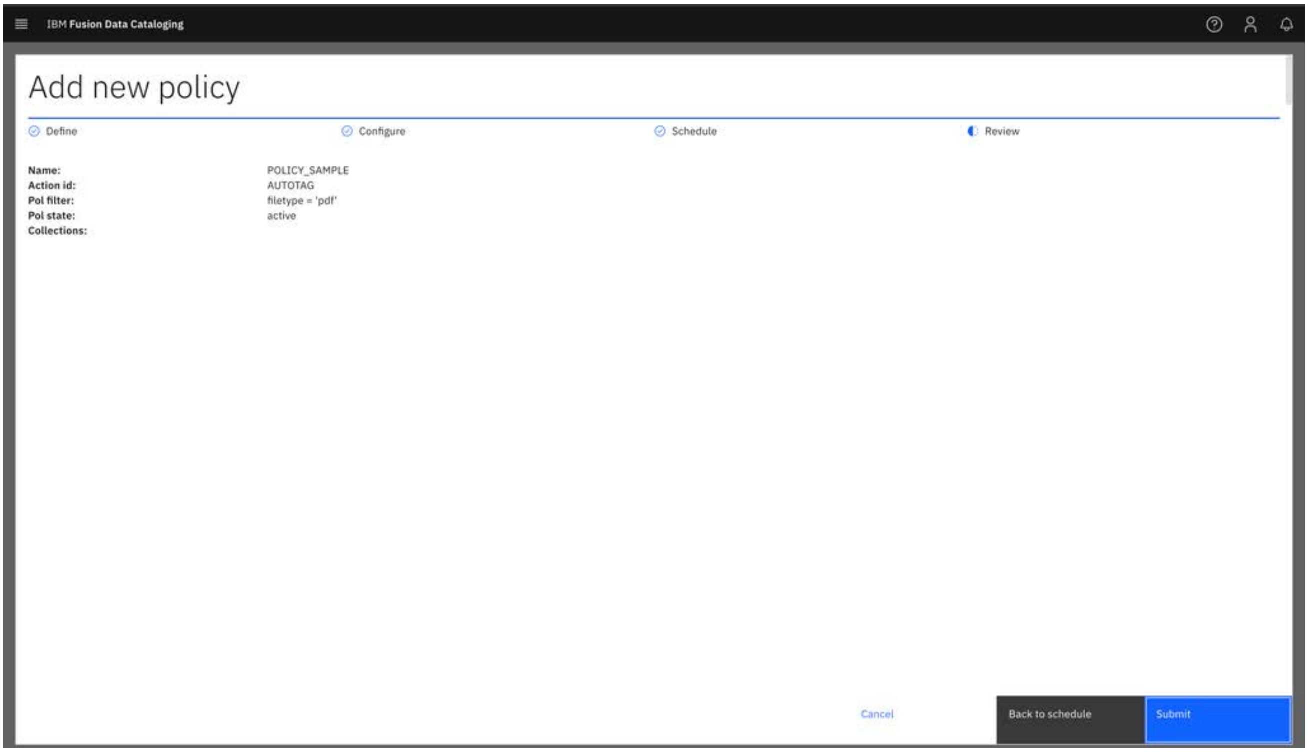
3. In the Name field, specify a name for the new policy.
4. Select the Policy Type. The default value is AUTOTAG policy.
5. Click Continue to configure.
6. Configure the policy. This action includes specifying a filter, which is used by the mechanism to tag files by tagging only those files that match the specified filter. In addition, specify the tag to be used and the value to assume.



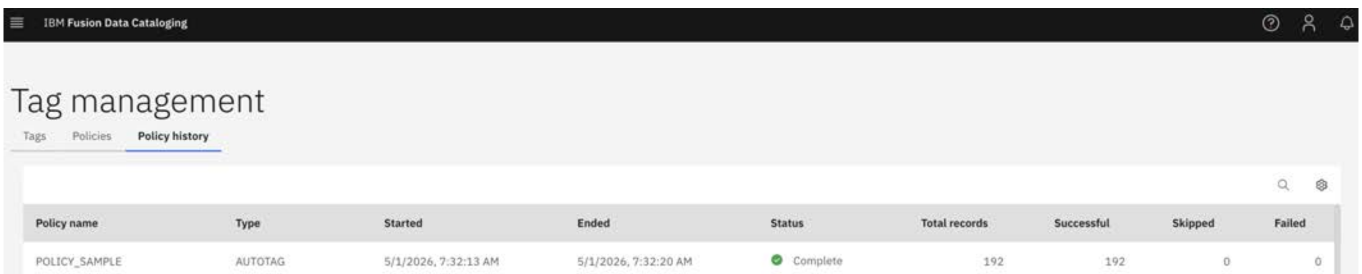
7. Click Continue to schedule.
8. Configure a schedule to execute the policy immediately, or at any specific point in the future. This action automatically executes the policy periodically.



9. Click Continue to review.
10. Review the information before creating the policy. Then, click Submit.



You can monitor the policy's progress by reviewing the Policy history records.



For additional instructions on creating policies, see Adding policies.

IBM Data Cataloging and Content-Aware Storage insights

IBM Data Cataloging and Content-Aware Storage (CAS) introduce data insights dashboards that provide immediate visibility into data once sources are connected and scanned. This experience is enabled when both IBM Data Cataloging and CAS are integrated. The provided insights surface key indicators of data composition on each established data source.

The UI dashboard surfaces foundational indicators of the associated data sources from the storage. No additional configuration is required to start using the dashboard.

Prerequisites

- Content-Aware Storage v1.1.5
- IBM Data Cataloging v2.5.3
- IBM Storage Scale v6.0.0.2 or later
- Ensure that both services are integrated. For more information, see Integrating IBM Data Cataloging for file filtering.

Overview

When a data source is added on CAS, the dashboards display insights that are scoped to that source.

You can go to Fusion UI > Content-Aware Storage > Data Sources > Details View.

The experience is organized into the following sections:

- Data source summary: High-level file and storage metrics.
- Data composition: Breakdown of file types and file size distribution.

Data source summary

At the top of the Data source view, four summary cards display key metrics for the selected data source.

Metric	Description
Total files	Total number of files that are indexed by IBM Data Cataloging across all data sources and prefixes for the data source.
Total storage	Aggregate storage capacity consumed, reported in GiB.
Ready to process	Number and percentage of files that match the CAS ingestion criteria defined in the active ruleset. For more information, see Managing IBM Data Cataloging rules when integrated with CAS. When IBM Data Cataloging and CAS are integrated, this value reflects the result of the rules that are specified in the custom resource.
Sensitive data	Count of files flagged as containing sensitive content. See the Sensitive data section for details on how this count is determined.

Data composition

The Data composition section provides a structural breakdown of the cataloged dataset across two dimensions.

File types and formats

Files are grouped by category (Documents, Images, Videos, Audio, Other). Each category shows the total file count and storage size, and distinguishes between formats that CAS can process and those it cannot. Expanding a category reveals the individual extensions within each group.

The following file formats are supported for CAS ingestion:

pdf, **doc**, **docx**, **txt**, **html**, **markdown**, **md**, **pptx**, **jpeg**, **png**, **bmp**, **xlsx**, **asciidoc**, **adoc**, **xhtml**, **csv**, **webp**, **wav**, **ogg**, **opus**, **mp3**

For more information, see Managing IBM Data Cataloging rules when integrated with CAS. All other extensions appear under the Not Supported label within their category.

File size distribution

A horizontal bar chart shows the number of files that are grouped by size range. The buckets used are: < 1 MB, 1-10 MB, 10-100 MB, 100 MB-1 GB, and > 1 GB.

This metric is derived from a single aggregated API query against the IBM Data Cataloging catalog.

Creating a data source

A data source is a repository or a system that stores and provides data for use in applications or systems. They are storage locations that provide data to use in AI workflows.

Before you begin

- For prerequisites specific to S3 type data source, see S3 type data source.
- For prerequisites and limitations specific to NFS data sources, see Network File System (NFS) type data source.

About this task

- The data source serves as the foundation for feeding information into the data pipeline.
  You have the options of:
  - Object Storage such as Amazon S3
  - IBM Storage Scale
  - Network File System (NFS)
- Content-Aware Storage (CAS) does not directly interface with S3 storage providers. It accesses S3 storage through the IBM Storage Scale S3 filesystem and with Active File Management (AFM) feature enablement provides a cached copy of the S3 content.
- Up to 25 CAS data sources are supported. Each data source can be one of the following:
  - An external S3 bucket using a IBM Storage Scale AFM fileset to ingest the S3 content.
  - A fileset residing on the IBM Storage Scale filesystem attached to CAS (without AFM).
- CAS supports multiple IBM Storage Scale filesystems. However, all remote filesystems must belong to a single remote IBM Storage Scale cluster.
- The following constraints exist for change notifications for filesets that reside in the IBM Storage Scale file system is remote mounted by IBM Fusion Global Data Platform service (without AFM):
  - Up to 10 million files per fileset
  - Up to 100 million files total monitored
- Only independent fileset is supported.

Procedure

1. From the menu, go to Content-aware storage > Data source.
2. In the Data sources page, click Connect data source.
3. Enter the name of the Data source.
4. Select the storage type and click Next.
   The available storage types are IBM Cloud, Storage Scale, AWS, S3 Compliant, and NFS.
5. Enter the following details in the Connection details page.
   The Connection details page vary based on your storage type selection.
   - IBM Cloud, AWS, or S3 Compliant
     - Enter the Endpoint.
       It refers to the URL through which you can access the bucket and its contents. For more information about Endpoint rules, see AWS documentation.
     - Enter the name of the S3 Bucket.
       For more information about the bucket naming guidelines, see AWS documentation.
     - For AWS, enter the Region. It is the Amazon Web Services region where the bucket is located.

- Enter the Access key and Secret access key
  These are security credentials needed to access the contents of the bucket.

- In the Certificate settings section, enter the Secret name for certificate.
  This is an optional parameter. SSL secured object storage locations require certificates for authentication. Create an OpenShift TLS secret in the **ibm-spectrum-fusion-ns** namespace or your IBM Fusion namespace. Provide the secret name for the credentials. For more information about creating a secret, see Creating a secret.

- Storage Scale
  Enter the Path. It is the junction path.

- NFS
  Provide the following information:
  - NFS server: Enter the NFS server hostname or IP address.
    Note: Although this field is defined as a list, only a single NFS server is supported in the CAS 1.1.5 release.
  - Export path: Enter the NFS export path to monitor. For example: /home/user/data

6. In the Caching filesystem section, select the file system where the cache of this data source must be stored.
   If there is only one remote file system detected, it gets automatically selected and this field is not available for selection.
7. Click Connect to submit the information so that CAS can be enabled for the data source.

What to do next

For IBM Storage Scale type data source, determine the group owner that has read and execute permission to the junction path to allow CAS to read its files and view the directory.
Log in to your Scale cluster to check the permission. For example, if your junction path is /gpfs/gpfs3/my-data/, run the following command to change the path:

```
cd /gpfs/gpfs3/
```

```
ls -la
```

Example output:

```
drwxr-x---  2 root cas    4096 May 22 19:50 my-data
[root@tc11scale1 gpfs3]# getent group cas | cut -d: -f3
310
```

If the group owner is not root, then run the following command to add the annotation to the previously created datasource. In this example, the group owner is **cas**.

```
oc annotate DataSource datasource-name group-id='310' --overwrite
```

Here, the **datasource-name** and **310** are example values that can change depending on the datasource name and the GID set in Scale.

- **Prerequisites for S3 and NFS type data sources**
  Prerequisites for creating S3 and NFS type data sources in CAS, including Active File Management requirements, version constraints, and AWS permissions.
- **Deploying an S3 data source with custom SSL certificate**
  Deploy an S3 data source with a custom SSL certificate on IBM Storage Scale 6.0.0.x by manually creating the AFM fileset before connecting the data source.

Prerequisites for S3 and NFS type data sources

Prerequisites for creating S3 and NFS type data sources in CAS, including Active File Management requirements, version constraints, and AWS permissions.

S3 type data source

Prerequisites
Ensure that you meet the following prerequisites if you plan to use an S3 type data source:

- If you choose a remote filesystem as the cache filesystem for your S3 type data source,
  - Enable Active File Management (AFM) on the remote IBM Storage Scale cluster. For more information, see the Scale documentation.
  - The remote IBM Storage Scale cluster must be at least at version 5.2.3 and the IBM Fusion cluster must be at least at IBM Fusion level 2.10.
- Starting with IBM Storage Scale 6.0.1, the Amazon Web Services account used to provide the access key and secret key for bucket authentication must be granted both **AmazonSNSFullAccess** and **AmazonSQSFullAccess** permissions. For more information, see the Scale AFM documentation.

Network File System (NFS) type data source

Prerequisites
Ensure that you meet the following prerequisites if you plan to use an NFS type data source:

- If you choose a remote filesystem as the cache filesystem for your NFS type data source,
  - Enable Active File Management (AFM) on the remote IBM Storage Scale cluster. For more information, see the Scale documentation.
  - The remote IBM Storage Scale cluster must be at least at version 5.2.3.
- The IBM Fusion cluster must be at least at IBM Fusion level 2.13 and Fusion Data Foundation must be at 4.21.x.

Limitations and considerations
The following limitations apply to NFS data sources:

- In this CAS release 1.1.5, the integration with IBM Data Cataloging is not supported.
- Only a single NFS server per NFS export path is supported.
- Only new NFS filesets that are created through CAS are supported. NFS data source does not support existing NFS filesets that are previously created in IBM Storage Scale.
- Kerberos-enabled NFS exports are not supported.

- File-level access control lists (ACLs) are not supported. Only domain-level ACLs are applicable.
- The NFS server and export path are fixed at data source creation time and cannot be updated later due to AFM limitations.

Deploying an S3 data source with custom SSL certificate

Deploy an S3 data source with a custom SSL certificate on IBM Storage Scale 6.0.0.x by manually creating the AFM fileset before connecting the data source.

Important: The following procedure applies only to IBM Storage Scale version 6.0.0.x.
For IBM Storage Scale 6.0.1.x and later, custom Transport Layer Security (TLS) certificates are configured automatically when you create a data source. Specify the certificate secret name in the Certificate settings section. Certificate rotation occurs automatically when the secret is updated. Follow the standard procedure described in Creating a data source.

Procedure (IBM Storage Scale 6.0.0.x only)

1. For Filesystem with IBM Storage Scale version 6.0.0.x, create a data source for an S3 bucket, which has a custom Secure Sockets Layer (SSL) certificate by adding bucket name, end-point, secret key, and access key details as mentioned in Creating a data source.
   The data source status is updated to `Authentication Failure` as the SSL cannot be processed.
2. Log in to the IBM Storage Scale 6.0.0.x cluster.
3. Get the AFM cache fileset name by listing the configuration details on the filesystem with the following command, and identify the one for the S3 bucket you just tried to add:

   ```
   mmafmcoskeys all get --report
   ```

4. Retrieve the `fileset-name` from the export map, which is in the following format:

   ```
   <bucket-name>:<fileset-name>-exportmap=<keys>
   ```

5. Copy the TLS certificate on a path, for example at /var/cas/tls-cert/sample-cert.pem.
6. Run the following command to re-create the fileset and make it `Active`:

   ```
   mmafmcosconfig <fs> <fileset-name> --endpoint 'https://<fileset-name>-exportmap:443' --object-fs --bucket <bucket-name> --
   tmpdir '.cachevolumetmp' --gid <gid> --mode 'iw' --ssl-cert-verify --ssl-cert-path <cert-with-full-path> --async-prefetch-
   interval 60 --makeactive
   ```

   The fileset must be deployed with the following output:

   ```
   Partition <fileset-name> is created and policy broadcasted to all nodes.
   ```

7. Check the status of the fileset:

   ```
   /usr/lpp/mmfs/bin/mmafmctl <fs> getstate -j <fileset-name>
   ```

   Remember: The fileset must be in the `Active` state and the contents of S3 bucket must be readable in the fileset.
8. Delete the previously failed data source and redeploy it with the same bucket information.

Results

The data source gets `Connected` as it can be able to use the manually created fileset.

Creating a domain and connecting with data source

Domains collect, transform, and transfer data from multiple data sources to a destination, enabling efficient data flow for analysis and AI models.

About this task

When you configure a domain, you automatically configure the `DocumentProcessorCR`. The processor streamlines data flow for analysis, reporting, and powering AI models effectively.

Procedure

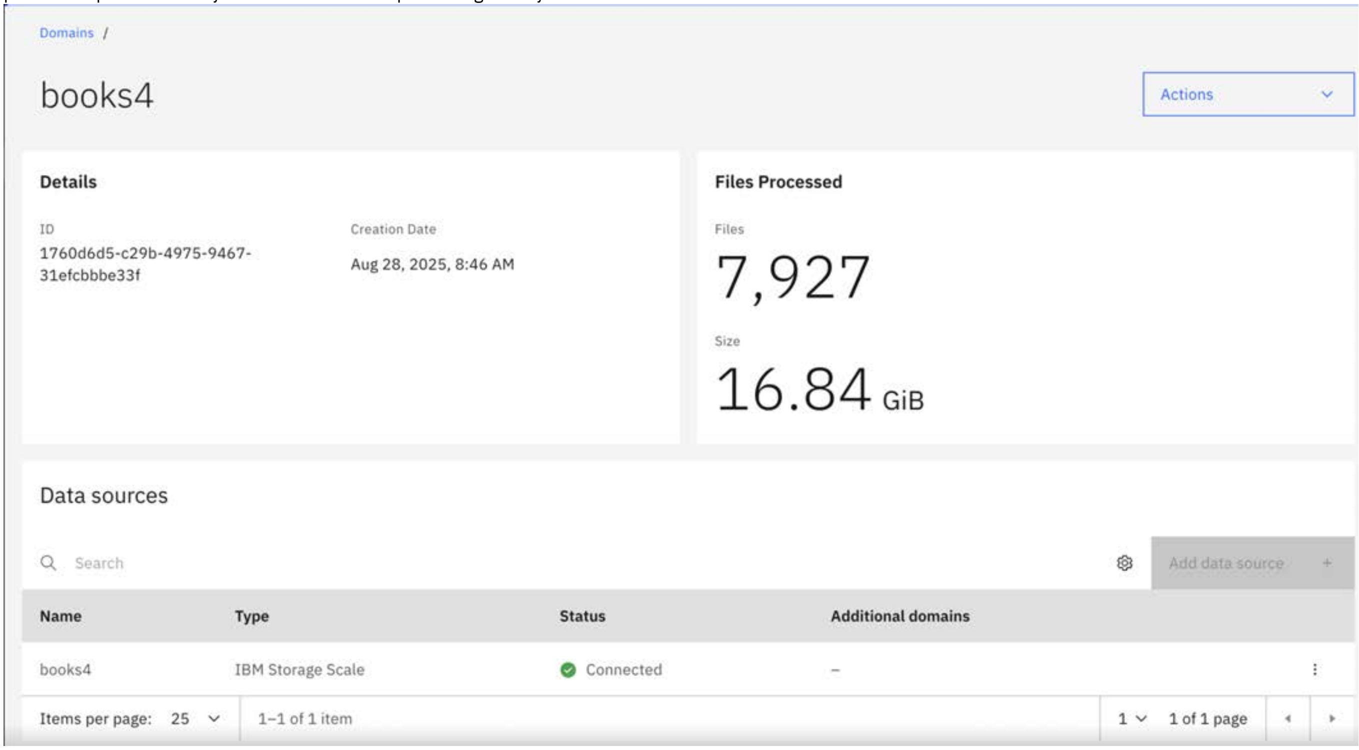
1. Go to Content-aware Storage > Domains.
2. Click Create domain.
3. In the Create domain dialog, enter the name of the domain and select the document processor engine.
   Note: NVIDIA multimodal requires NVIDIA NIM to be configured first.
4. Click Next.
5. Select the content types to be processed by the domain (only applicable to NVIDIA multimodal). Then click Next.
6. Select an option to either connect a new data source or use an existing one.
   a. If you choose to use an existing data source, click Finish. The domain gets created.
   b. If you choose to connect a new data source, click Next and proceed with the next steps to enter the required data source details. For more information about the data source fields, see Creating a data source. Upon completing, both the data source and domain get created.

   If the status of the data source gets stuck in connecting state, ensure that you enable watch creation. For more information about the configuration, see Configuring Scale user to enable watch creation.

   Note: You can delete a domain from the Domain details page only if no data source exists for it.

What to do next

You can view file processing metrics and data source status for each domain. Metrics include the total number of files and the file size within the selected domain that provides improved visibility into data volume and processing activity.



Tip: To customize the metrics polling interval, set the `METRICS_POLLING_INTERVAL_MS` parameter in the `cas-config` ConfigMap located in the CAS installation namespace:

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: cas-config
  namespace: ibm-cas
data:
  ENABLE_FILE_LEVEL_SECURITY: 'true'
  HF_ACCESS_TOKEN_SECRET_NAME: hf-access-token
  METRICS_POLLING_INTERVAL_MS: '60000' ---> Metrics Polling interval in millisecond
  NVMM_EMBED_SERVICE: 'http://nv-ingest-embedqa.nv-ingest.svc.cluster.local'
  NVMM_NIM_SERVICE: 'http://nv-ingest.nv-ingest.svc.cluster.local'
```

Configuring Document Processor Engine

Steps to configure Document Processor Engines: IBM Docling Multimodal and NVIDIA Multimodal.

- **IBM Docling Multimodal processing engine**
  IBM Docling Multimodal is an open source, document processing engine that is designed to convert unstructured documents into structured, machine-readable formats for use in AI applications. It enables you to extract, parse, embed, and store document vectors by using Docling with Granite embedding models.
- **NVIDIA Multimodal processing engine settings**
  This section enables you to configure the NVIDIA Multimodal pipeline.

IBM Docling Multimodal processing engine

IBM Docling Multimodal is an open source, document processing engine that is designed to convert unstructured documents into structured, machine-readable formats for use in AI applications. It enables you to extract, parse, embed, and store document vectors by using Docling with Granite embedding models.

The supported document types for text-based extraction include: `.pdf`, `.txt`, `.md`, `.asciidoc`, `.csv`, `.html`, `.xhtml`, `.docx`, `.pptx`, `.xlsx`. As of now, other file types without text are not supported (for example, image and audio files).

Enabling Docling Multimodal document processor engine

To enable the Docling Multimodal document processor engine through command line, access the `CasInstall` Custom Resource (CR) and add the following configuration under the spec section:

```
documentProcessingEngine:
  docling_multimodal:
    enabled: true
```

Alternatively, the `CasInstall` CR instance can be patched to enable `docling_multimodal`, as shown in the following example:

```
oc patch casinstalls.cas.isf.ibm.com ibm-cas-service-instance -n ibm-cas --type=merge -p '{"spec":{"documentProcessingEngine":{"docling_multimodal":{"enabled":true}}}}'
```

Disabling Docling Multimodal document processor engine

Disabling the Docling Multimodal document processor engine removes the Docling services and no longer allows ingestion or query to any domains, which are configured to use Docling Multimodal.

To disable the Docling Multimodal document processor engine through command line, access the `CasInstall` CR and add the following configuration under the spec section:

```
documentProcessingEngine:
  docling_multimodal:
    enabled: false
```

Alternatively, the `CasInstall` CR instance can be patched to disable `docling_multimodal`, as shown in the following example:

```
oc patch casinstalls.cas.isf.ibm.com ibm-cas-service-instance -n ibm-cas --type=merge -p '{"spec":{"documentProcessingEngine":{"docling_multimodal":{"enabled":false}}}}'
```

- **Configuring Docling Multimodal GPUs**
  This section describes how to configure Docling Multimodal GPUs.

Configuring Docling Multimodal GPUs

This section describes how to configure Docling Multimodal GPUs.

The default GPU resource that is used by the Docling services is `nvidia.com/gpu`. If the GPUs are configured to use Multi-Instance GPU (MIG), then a corresponding MIG profile must be assigned to each service for it to start properly. The IBM Docling Multimodal resources can be overridden in the `CasInstall` Custom Resource (CR). Each deployment that requires a GPU has an override flag for use with alternative resource configurations.

To alter the Docling Multimodal GPU targets from the command line, access the `CasInstall` CR and add the following flags under the `spec` section:

```
spec:
  flags:
    - VLLM_VISION_GPU=<gpu target>
    - VLLM_EMBEDDING_GPU=<gpu target>
    - DOCLING_GPU=<gpu target>
```

For example, to specify an alternative Multi-Instance GPU (MIG) indicator for all, add the following flags under the `spec` section:

```
spec:
  flags:
    - VLLM_VISION_GPU=nvidia.com/mig-7g.80gb
    - VLLM_EMBEDDING_GPU=nvidia.com/mig-7g.80gb
    - DOCLING_GPU=nvidia.com/mig-7g.80gb
```

Result: The GPU resource indicator is then automatically detected and a new deployment is rolled out.

NVIDIA Multimodal processing engine settings

This section enables you to configure the NVIDIA Multimodal pipeline.

The `DocumentProcessor` custom resource (CR) supports the following configuration options for controlling the NVIDIA Multimodal file processing engine. Each option affects content chunking, processing performance, and search quality. When you select a processing approach, consider the tradeoffs among these factors.

NVIDIA Multimodal `DocumentProcessor` CR configuration options and considerations

Option	Valid option values	Description	Considerations
`spec` > `tasks` > `name` > `nvidiaExtract` > `extractText`	`True/False` (Default: `True`)	Extracts the actual text from a document, including the text in tables, charts, and infographics.	Extract text is the most performant for ingestion.
`spec` > `tasks` > `name` > `nvidiaExtract` > `extractTables`	`True/False` (Default: `True`)	Extracts text by summarizing each table.	A second chunk is created if `extractText` is also selected.
`spec` > `tasks` > `name` > `nvidiaExtract` > `extractCharts`	`True/False` (Default: `True`)	Extracts text by summarizing each chart.	A second chunk is created if `extractText` is also selected.
`spec` > `tasks` > `name` > `nvidiaExtract` > `extract` > `infographics`	`True/False` (Default: `True`)	Extracts text by summarizing each infographic.	A second chunk is created if `extractText` is also selected.
`spec` > `tasks` > `name` > `nvidiaExtract` > `extractImages`	`True/False` (Default: `True`)	Extracts caption by summarizing each image.	Impacts the time of ingestion.

Option	Valid option values	Description	Considerations
`spec` > `tasks` > `name` > `nvidiaExtract` > `tableOutputFormat`	Enumeration: `markdown`, `pseudo_markdown`, `simple` (Default: `pseudo_markdown`)	Extracted tables are returned in a particular format to be returned as the text that is vectorized.	N/A
`spec` > `tasks` > `name` > `nvidiaExtract` > `extractTextDepth`	Enumeration: `page`, `document` (Default: `page`)	Extracts text at levels of granularity.	Impacts the number of chunks that are stored and the size of text returned.
`spec` > `tasks` > `name` > `nvidiaSplit` > `chunkSize`	Numeric (Default: `1024`)	Size of each chunk in number of bytes.	For more information on all splitting options, see Split Documents.
`spec` > `tasks` > `name` > `nvidiaSplit` > `chunkOverlap`	Numeric (Default: `150`)	Number of bytes to overlap between consecutive chunks.	
`spec` > `tasks` > `name` > `nvidiaSplit` > `tokenizer`	String (Default: `meta-llama/Llama-3.2-1B`)	Identifies the embedding model to split the content with.	
`spec` > `tasks` > `name` > `nvidiaSplit` > `splitSourceTypes`	Array of Strings (Default: `empty`)	Identifies the file types to apply the prior `nvidiaSplit` configuration.	

During the Creating a domain and connecting with data source procedure, the CAS user interface creates a `Domain` custom resource (CR) and an associated NVIDIA Multimodal `DocumentProcessor` CR. The `DocumentProcessor` CR configures the processing engine to use the default settings, which include extracting text only and processing content on a per-page basis. To apply more configuration options, you must create or modify the NVIDIA Multimodal `DocumentProcessor` CR by using the OpenShift console or the CLI.

Configuring NVIDIA Multimodal pipeline

After you create a `Domain` CR in the CAS user interface, a corresponding `DocumentProcessor` CR is generated for that domain. Update the `spec` section of the `DocumentProcessor` CR with the required configuration options.
Note: Changes to processing options affect only newly ingested or updated documents. Previously processed documents are reingested with the new settings only when the original document is modified.

1. Find the `DocumentProcessor` CR that is associated with the `Domain`.
   Tip: The `DocumentProcessor` CR has the same name as the `Domain`.
   Example:
   - `oc get domains.cas.isf.ibm.com -n ibm-cas`
   - `oc get documentprocessors.cas.isf.ibm.com -n ibm-cas`
2. View the current configuration of the `DocumentProcessor` CR.
   Example:
   ```
   oc get documentprocessors.cas.isf.ibm.com mydomain -n ibm-cas -o yaml
   ```
3. Ensure that the type is `nvidia_multimodal`. Example: `type: nvidia_multimodal`
4. Modify the `DocumentProcessor` CR according to the options added in the table.
5. Save the `DocumentProcessor` CR.

Results
: The CAS operator automatically reconciles the `Deployment` associated with the `DocumentProcessor` CR and the corresponding processing engine pods are restarted with new settings.

Configuring Content-Aware Storage (CAS) through CLI

You can use the CRs to configure CAS.

Ensure all the prerequisites are met. See Planning and prerequisites. To install CAS from the user interface of IBM Fusion, see Installing Content-Aware Storage (CAS).

Create a Scale datasource

```
oc apply -f - <<EOF
apiVersion: cas.isf.ibm.com/v1beta1
kind: DataSource
metadata:
  name: <datasource name>
spec:
 scale:
  fileSystemName: <filesystem name>
  path: <fileset path>
EOF
```

Create a S3 datasource

For prerequisites, create a key/value secret from the OpenShift® console in CAS namespace. For other prerequisites, see Prerequisites for S3 and NFS type data sources.

The following example is based on Amazon AWS.

```
apiVersion: cas.isf.ibm.com/v1beta1
kind: DataSource
metadata:
  name: <datasource name>
  namespace: ibm-cas
spec:
 s3:
  bucket: <bucket name>
  endpoint: <endpoint>
  credentials:
    accessKeyRef:
      key: ""
    secretKeyRef:
      key: ""
    secretName: <Secret name>
  fileSystemName: <filesystem name>
  provider: aws
```

If a Group ID is defined for the underlying fileset, run the following command to add the annotation to the previously created datasource previously:

```
oc annotate DataSource <datasource name> group-id='310' --overwrite
```

Where, `<datasource name>` and `310` are example values that can change depending on the datasource name and the GID set in Scale.

Create a NFS data source

For prerequisites, see Prerequisites for S3 and NFS type data sources.

```
apiVersion: cas.isf.ibm.com/v1beta1
kind: DataSource
metadata:
  name: <datasource name>
  namespace: ibm-cas
spec:
  nfs:
    servers:
      - <server>
    path: "<path>"
    fileSystemName: <filesystem name>
```

Create a domain and add datasource to it

```
kubectl apply -f - <<EOF
apiVersion: cas.isf.ibm.com/v1beta1
kind: Domain
metadata:
  name: <domain name>
spec:
  dataSources:
  - <datasource name>
  collections:
  - name: "<domain name>"
EOF
```

Create a document processor

Document processor **`type`** field refers to the document processing engine that is used to ingest documents. CAS can be configured to use either NVIDIA Multimodal Services or Docling Multimodal Services.

The keys for each type are as follows:

- NVIDIA Multimodal Processing Engine - **`nvidia_multimodal`**

```
oc apply -f - <<EOF
apiVersion: cas.isf.ibm.com/v1beta1
kind: DocumentProcessor
metadata:
  name: <domain name>
spec:
  type: nvidia_multimodal
  domains:
  - <domain name>
EOF
```

- Docling Multimodal Processing Engine - **`docling_multimodal`**

```
oc apply -f - <<EOF
apiVersion: cas.isf.ibm.com/v1beta1
kind: DocumentProcessor
metadata:
  name: <domain name>
spec:
  type: docling_multimodal
  domains:
  - <domain name>
EOF
```

Advanced configuration for the NVIDIA Multimodal document processor

To extract content from tables, charts, and images, from the documents in the dataset, enable additional settings in the document processor custom resource (CR).
Note: Enabling these options increases document processing time.
Supported file types for NVIDIA Multimodal document processor: `.pdf`, `.txt`, `.html`, `.json`, `.md`, `.docx`, `.pptx`, `.bmp`, `.jpeg`, `.png`, `.tiff`, `.sh`
Supported options for `nvidia_multimodal` extract task:

- `extractCharts`: Extracts chart graphs from documents.
- `extractTables`: Extracts tabular data from documents.
- `extractImages`: Extracts content from image files and images that are contained within documents.

The following example shows how to configure these options in the `nvidia_multimodal` extract task.

```
oc apply -f - <<EOF
apiVersion: cas.isf.ibm.com/v1beta1
kind: DocumentProcessor
metadata:
  name: <domain name>
spec:
  type: nvidia_multimodal
  tasks:
    - name: Extract
      nvidiaExtract:
        extractCharts: true
        extractImages: true
        extractTables: true
        extractText: true
        extractTextDepth: page
    - name: Split
      nvidiaSplit:
        chunkOverlap: 150
        chunkSize: 1024
        tokenizer: meta-llama/Llama-3.2-1B
    - name: Embed
      nvidiaEmbed: {}
  domains:
  - <domain name>
EOF
```

If you want to configure file ingestion filtering by using Data Cataloging, add the additional supported file types for NVIDIA Multimodal document processor to Data Cataloging's filtering rule:

```
oc patch rule allowed-filetypes -n ibm-data-cataloging --type merge -p '{"spec": {"metadata": [{"field": 
"filetype","operator": "in","value": ["doc", "docx", "pdf", "txt", "gif", "html", "json", "md", "pptx", "bmp", "jpeg", 
"png", "tiff", "sh" ]}]}}'
rule.datacataloging.ibm.com/allowed-filetypes patched
```

Note: You must update the filtering rule every time you enable Data Cataloging filtering, if you later disable it.

Create a CAS Resource Access Control (CRAC) CR

```
apiVersion: cas.isf.ibm.com/v1beta1
kind: CasResourceAccessControl
metadata:
  name: <CR name: "<Domain name>-access">
  namespace: <CAS installation namespace>
spec:
  #Determines whether file level permission checking can be done for a given domain
  fileACL: <true/false>
  resourceRef:
    name: <Domain name>
    type: Domain
  subjects:
    groups:
      - name: <Name of the IDP/OpenShift group who needs access of given domain>
        role: READER
      # Special group for granting access to all authenticated users
      - name: cas-all-authenticated
        role: READER
    users:
      - name: <Name of the IDP/OpenShift user who needs access of given domain>
        role: READER
```

Note: The `cas-all-authenticated` special group grants domain access to any user who has a valid OpenShift or identity provider (IDP) authentication token. When this group is configured, the system bypasses individual user and group access control (ACL) checks for the domain. This capability is useful for environments where simplified access control is required.
The CAS Resource Access Control (CRAC) CR status contains validation details for resources, users, and groups.

- If resource is not valid, the condition status indicates the exact error.
- If users in the specific user list are not valid, the condition status indicates the name of invalid users.
- If groups in the specific group list are not valid, the condition status indicates the name of invalid groups.
- If the resource, users, and groups are valid, the condition status reflects the validation status message.
  The following example shows how the status message appears in the CRAC CR:

```
status:
  conditions:
    - lastTransitionTime: '2025-08-29T02:15:42Z'
      message: 'Resource: gt20 is validated'
      observedGeneration: 1
```

```
      reason: AsExpected
      status: 'True'
      type: ValidatedResource
    - lastTransitionTime: '2025-08-29T02:15:42Z'
      message: Users are validated
      observedGeneration: 1
      reason: AsExpected
      status: 'True'
      type: ValidatedUsers
    - lastTransitionTime: '2025-08-29T02:15:42Z'
      message: Groups are validated
      observedGeneration: 1
      reason: AsExpected
      status: 'True'
      type: ValidatedGroups
```

To configure CAS to restrict specific users to specific CAS domains through user interface, see [Configuring CAS Resource Access Control (CRAC)](#).

To remove search access for a domain through CLI, delete the relevant users and groups from the CRAC CR and save the changes. Removed users and groups then lose authorization to access the query search API for the domain.

Configuring network policies

CAS automatically deploys and manages Kubernetes network policies that enforce a zero-trust communication model across its internal services.

To disable the network policies:

```
oc patch casinstall <install name> -n <cas namespace> \
  --type merge -p '{"spec":{"networkPolicy":{"enabled":false}}}'
```

To enable the network policies:

```
oc patch casinstall <install name> -n <cas namespace> \
  --type merge -p '{"spec":{"networkPolicy":{"enabled":true}}}'
```

Note: The changes are applied automatically, and a restart is not required. You can disable network policies if you want to apply custom policies.

Using your own Hugging Face access token with CAS for NVIDIA Multimodal

NVIDIA NeMo Retriever Library uses models that are downloaded from Hugging Face to do token-based splitting on the content extracted from documents. NVIDIA recommends to use the meta-llama/Llama-3.2-1B tokenizer, however this model is gated and requires a Hugging Face token in order access it. A Hugging Face access token is a personal API key that is used to securely authenticate programmatic access to Hugging Face services.

Before you begin

You must have a valid token to access Meta's Llama 3.2 language models and evals gated repository.

About this task

In Content-Aware Storage (CAS), a Hugging Face access token is used to securely interact with Hugging Face's APIs and services that support intelligent content processing.

For more information about Hugging Face access tokens, see [Hugging Face User Access Tokens](#).

To provide your Hugging Face access token for the meta-llama/Llama-3.2-1B model, perform the following steps:

Procedure

1. Encode the generated Hugging Face access token in Base64 format by using the following command. The resulting value is refereed to as the HUGGING_FACE_ACCESS_TOKEN in the next step.

   ```
   echo "hf_xxxxxxxxxxx" | base64
   ```

2. Create a secret in the CAS installation namespace:

   ```
   kind: Secret
   apiVersion: v1
   metadata:
     name: hf-access-token
     namespace: ibm-cas
   data:
     hf_access_token: HUGGING_FACE_ACCESS_TOKEN
   type: Opaque
   ```

3. Use the encoded access token and update the `cas-install` CR 'spec' section:

   ```
   spec:
     flags:
       - HF_ACCESS_TOKEN_SECRET_NAME=hf-access-token
   ```

Configuring CAS Query Access Control and Security

Content-Aware Storage (CAS) supports domain-level authorization, Identity Provider integration, and optional file-level security to control query access.

You can configure Content-Aware Storage (CAS) query access control and security to suit your requirements and environment by selecting one of the following methods:

- Authorize CAS query users to CAS domains:
  - By default, all users of the Red Hat OpenShift cluster where CAS is deployed can issue CAS queries, depending on the domains they are authorized to access. In this context, they are referred to as CAS query users.
  - Configure CAS to authorize users to specific CAS domains. If an unauthorized CAS query user of a domain runs CAS queries, the query API fails with status code 403 (Invalid Access). For more information, see Configuring CAS Resource Access Control (CRAC).
- Manage CAS users with your own Identity Provider (IDP):
  - Configure CAS to connect to your own IDP. Without this configuration, CAS serves only the Red Hat OpenShift users. For more information, see Configuring CAS to connect to your Identity Provider (IDP).
- Limit certain users to certain files:
  - To enforce stricter access control, you can configure CAS to serve only the content from files for which users have explicit file-level read permissions. Without this configuration, CAS returns results from all ingested data within a domain to any user who has domain-level access, regardless of file-level read permissions. For more information, see Configuring file-level access control in CAS.

Security type	Red Hat OpenShift users	External IDP users
Domain-level access control (mandatory)	Supported	Supported
File-level access control (optional)	Unsupported	Supported; an external IDP is required to federate user information from the Directory server used by the file system.

  - CAS query user setup: To designate users of Red Hat OpenShift or external IDP or Directory server as CAS query users, you must either configure the users or their groups through CAS Resource Access Control (CRAC). For more information, see Configuring CAS Resource Access Control (CRAC).

- **Configuring CAS Resource Access Control (CRAC)**
  Configure users and groups that can access specific Content-Aware Storage (CAS) domains by using the CRAC custom resource.
- **Configuring CAS to connect to your Identity Provider (IDP)**
  An Identity Provider is a system or service that creates, maintains, and manages identity information for users and provides authentication services to applications or other systems. It acts as a trusted third party that authenticates users and shares their identity information with other services.
- **Configuring file-level access control in CAS**
  File-level security in Content-Aware Storage (CAS) ensures that users can access only the data they are authorized to read from the file system.

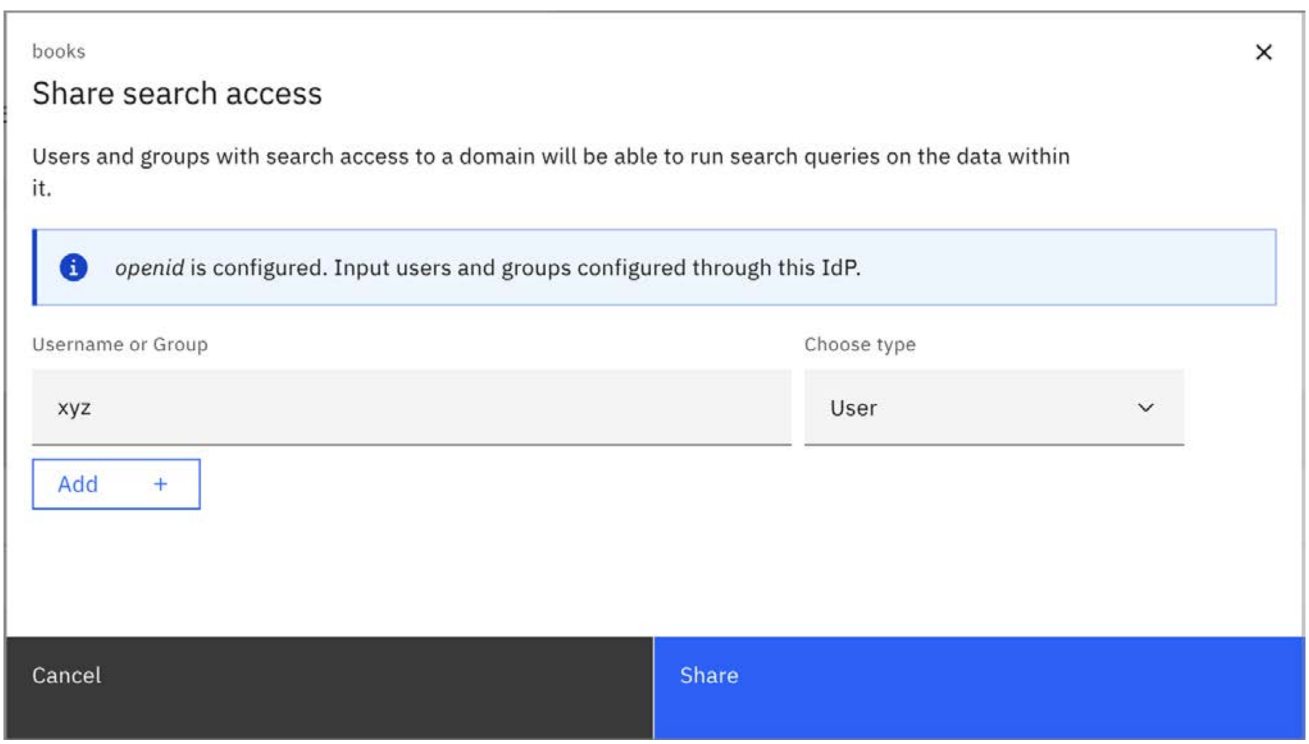
Configuring CAS Resource Access Control (CRAC)

Configure users and groups that can access specific Content-Aware Storage (CAS) domains by using the CRAC custom resource.

You can define users and groups that can have access to a domain by using CRAC API Custom Resource (CR).

To configure CAS to restrict specific users to specific CAS domains, follow these steps:

1. Open the IBM Fusion UI.
2. Go to Content-aware Storage > Domains.
3. Click the domain name that you want to open.
4. Click Actions and select Share search access.
   The Share search access dialog opens.



5. In the Username or Group field, add the username or group name to which you want to grant the domain access.
6. From the Choose type dropdown, select the type as User or Group.
   Tip: To add all users or groups at the same time, click Add.
7. Click Share.
   Note: When IDP CR is configured on your system, you can provide the external IDP token to the query search API for semantic search execution.

Granting global access to all authenticated users

To simplify access control, you can grant domain access to all authenticated users at once by using the `cas-all-authenticated` special group. This approach removes the need to add individual users or groups.
Important: The `cas-all-authenticated` group grants access to any user with a valid OpenShift® or identity provider (IDP) authentication token. Specific user and group assignments provide more granular security control and are recommended when restricted access is required.

Removing search access

You can remove users and groups that have access to specific domains.

To remove users or groups from specific CAS domains, follow these steps:

1. Open the IBM Fusion UI.
2. Go to Content-aware Storage > Domains.
3. Click the domain name that you want to open.
4. Click Actions and select Remove search access.
   The Remove search access dialog opens.
5. Select the users or groups whose search access is to be removed.
6. Click Remove.
   Removed users and groups lose authorization to access the query search API for the domain.

Configuring CAS to connect to your Identity Provider (IDP)

An Identity Provider is a system or service that creates, maintains, and manages identity information for users and provides authentication services to applications or other systems. It acts as a trusted third party that authenticates users and shares their identity information with other services.

About this task

An IDP can be integrated with CAS to enhance security, access control, and compliance. It helps CAS by providing:

- **Secure Access**: It ensures that only authorized users access sensitive content.
- **Single Sign-On (SSO)**: It enables you to log in once to access multiple systems.
- **Compliance**: It helps meet regulatory requirements with audit trails and access controls.
- **User Management**: It simplifies onboarding, offboarding, and role-based access.

Configure CAS with client ID and client secret

- Client ID: It is the ID created in the IDP to allow CAS API access.
- Client secret: It is the password that is associated with the client ID for secure access to IDP APIs.

To enable user and group validation through the Security Identity Manager (SIM) client, CAS uses the client ID and client secret to authenticate with the IDP. To obtain the client ID and client secret, perform the following steps:

1. Create and enable API access in your IDP.
2. Configure CAS with the credentials that are provided by the IDP, as the client ID and client secret.
   CAS then uses this client ID and client secret to query the user and group APIs from the IDP.

To connect your IDP to the CAS deployment, perform the following steps:

Procedure

1. Configure an IDP secret for CAS by creating a secret that is named `idp-sec` in CAS installation namespace with `client_id` and `client_secret` in Base64 format:

```
kind: Secret
apiVersion: v1
metadata:
  name: idp-sec
  namespace: ibm-cas
data:
  client_id: <client ID in base64 format>
  client_secret: <client secret in base64 format>
type: Opaque
```

2. Create a CAS `IdentityProvider` CR to specify the custom configuration details of your IDP. Specify the following fields based on your IDP configurations:
   - email: It is the list of claims whose value must be used as the email address.
   - groups: It is the list of claims whose value must be used as groups.
   - name: It is the list of claims whose value must be used as name.
   - sub: It is the list of claims whose value must be used as subject.
   - issuer: It is the issuer URL of the external IDP. For example: https://keycloak-keycloak-automation.apps.com/auth/realms/master
   - groupUrl: It is the IDP URL for retrieving all groups. For example: https://keycloak-keycloak-automation.apps.com/auth/admin/realms/master/groups

- userUrl: It is the IDP URL for retrieving all users. For example: https://keycloak-keycloak-automation.apps.com/auth/admin/realms/master/users

```
apiVersion: cas.isf.ibm.com/v1beta1
kind: IdentityProvider
metadata:
  name: identityprovider-sample
  namespace: ibm-cas
spec:
  identityProvider:
    claims:
      email:
        - <email>
      groups:
        - <groups>
      name:
        - <preferred_username>
      sub:
        - <subject>
    issuer: <Issuer URL of IDP>
    mappingMethod: claim
    name: openid
    scimClient:
      clientSecretName: <OpenShift secret storing client_id and client_secret parameters of IDP client>
      groupUrl: <IDP URL to retrieve all groups>
      userUrl: <IDP URL to retrieve all users>
    type: OpenidProvider
```

3. To enable file-level security, add the following claims of the external IDP in the `claims` section:

```
spec:
  identityProvider:
    claims:
      groupID:
        - gidNumber
      supplementalGroups:
        - supplementalGroups
      userID:
        - uidNumber
```

Configuring file-level access control in CAS

File-level security in Content-Aware Storage (CAS) ensures that users can access only the data they are authorized to read from the file system.

Overview

File-level security uses an external identity provider (IDP) integrated with a directory server to retrieve your numeric user ID, numeric group ID, and supplemental group IDs. At the time of query, it performs a real-time access check to verify whether the user has the read permission for the parent file of each matching vector.

When file-level security is enabled, users and groups that are authorized through CAS Resource Access Control (CRAC) can still run CAS queries. However, query results are limited to files for which the user has 'read' access. File-level security is configured at the CAS deployment level and applies to all document processors and domains, both existing and newly created.

This feature requires CAS to be integrated with an external IDP that federates users from the same LDAP directory that is used to authenticate file system users.
Note: Without file-level security configuration, CAS returns results from all ingested data within a domain to any user who has domain-level access, regardless of file-level read permissions.

Before you begin

File-level access control in CAS requires the following setup:

- A directory server, such as OpenLDAP must store information about a user's numeric user ID and numeric group ID, and supplemental group IDs.
- An external Identity Provider (IDP) that uses OpenID standard, such as Keycloak. The IDP must use the directory server for user federation to retrieve and provide the following user attributes to the `/userinfo` endpoint:
  - `userID`
  - `groupID`
  - `supplementalGroups`
    Note: The `supplementalGroups` must contain the numeric IDs for the user's groups, such as `userID` and `groupID`.
    Note: The attribute names (`userID`, `groupID`, and `supplementalGroups`) are used as examples. You can use custom attribute names in the directory server. During the claim-mapping configuration (in the CRAC CR), these custom attribute names are mapped to the standard names recognized by CAS: `userID`, `groupID`, and `supplementalGroups`.
- File system clients
  - File system clients remote mount the storage file system of the CAS data source. For more information, see Mounting a remote GPFS file system in IBM Storage Scale documentation.
  - The file system client machines are then configured to use the same directory server for user authentication. For more information, see Setting up authentication servers to configure protocol user access in IBM Storage Scale documentation.
- CAS
  - Configure CAS to connect to the IDP that is configured to federate users from the same directory server used by the file system client machines. For more information, see Configuring CAS to connect to your Identity Provider (IDP).

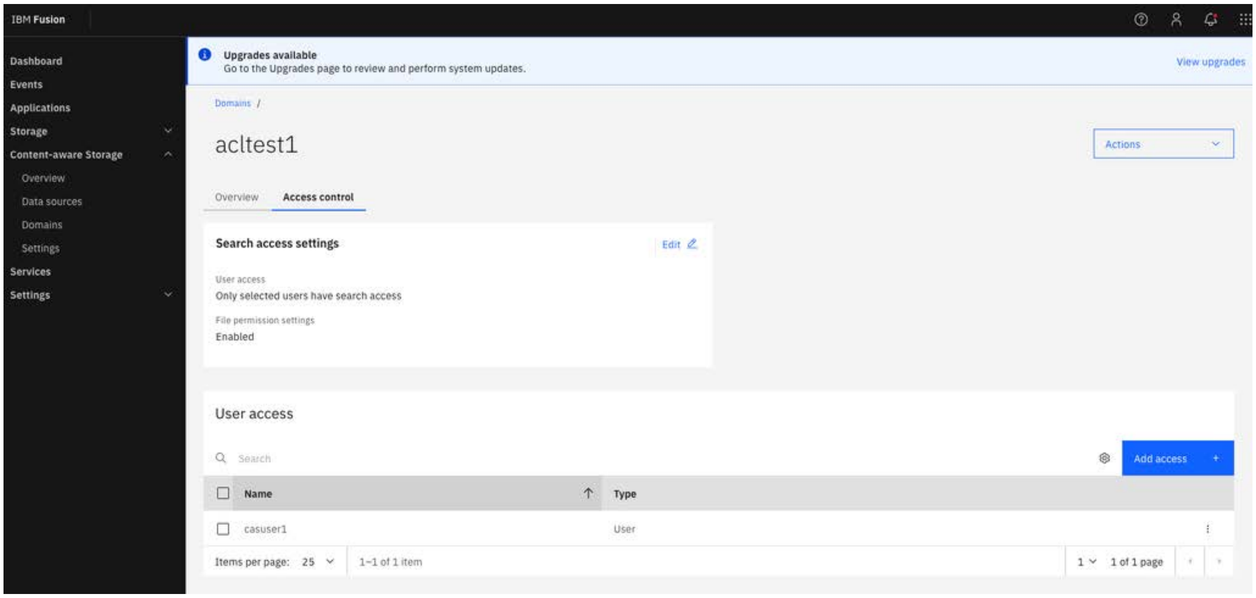
Modifying file-level access control for a domain

Using command line

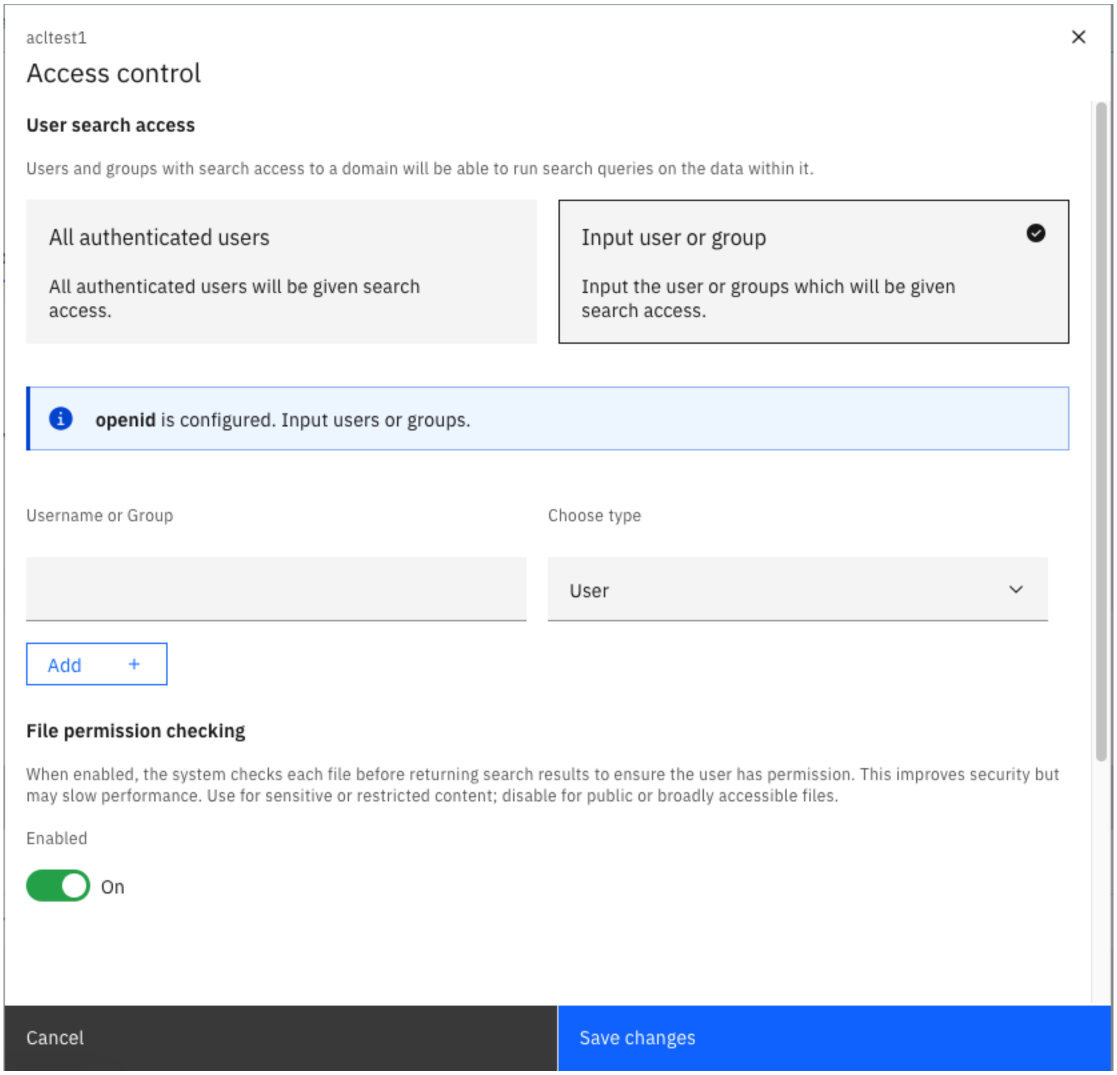
To enable or disable file-level permissions for a specific domain, add `fileACL: true/false` under the `spec` section in the `CasResourceAccessControl` instance for the domain. For more information, see Create a CAS Resource Access Control (CRAC) CR.

Using user interface

1. Open the IBM Fusion UI and go to Content-aware Storage > Domains.
2. Select the Domain to modify.
3. Select the Access control tab.



4. In the Search access settings section, select Edit.
5. Use the toggle to enable or disable File permission checking.



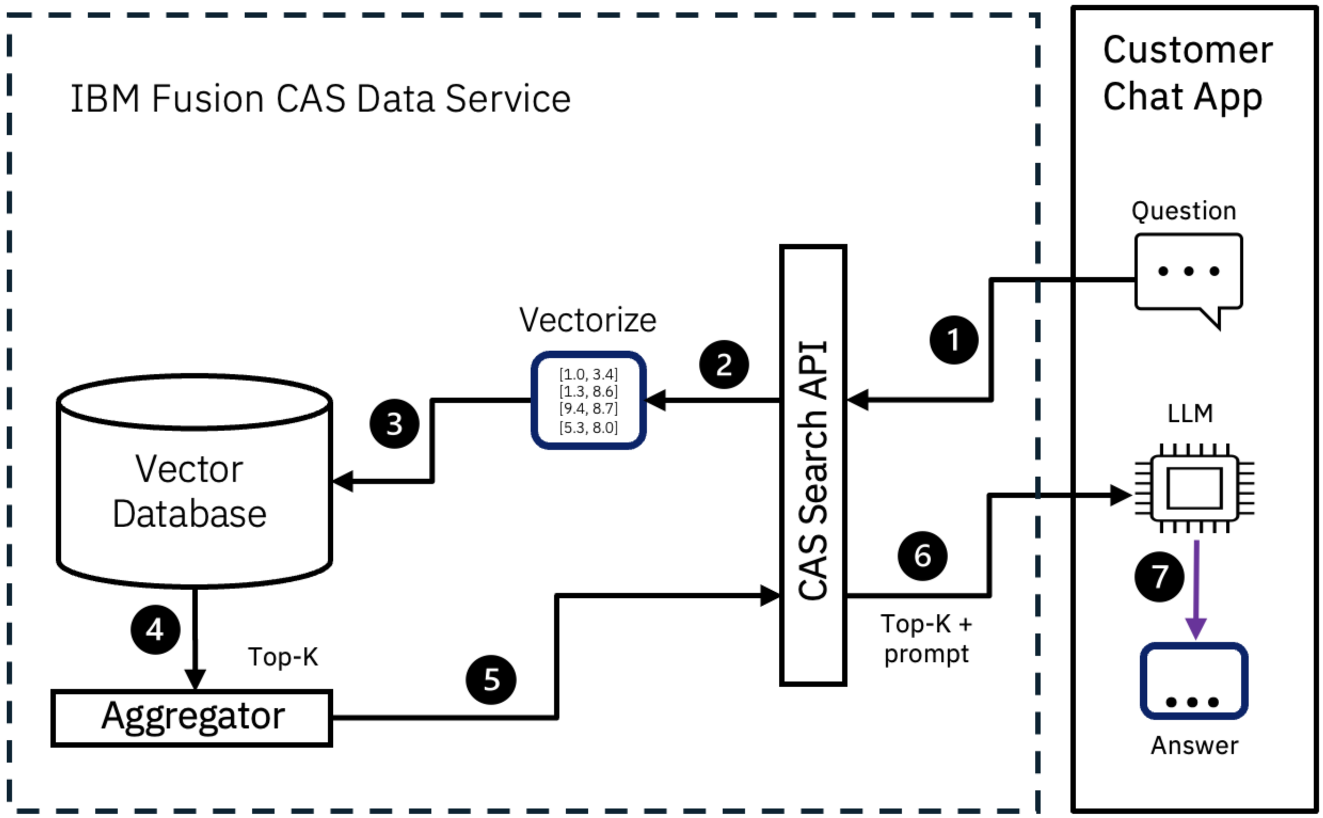
Client application with Content-Aware Storage (CAS)

You are required to provide a client application that uses the CAS search API to leverage the data ingested by CAS as part of your RAG application. This topic outlines the workflow for querying the CAS database using the exposed API.

Before you begin, ensure the following prerequisites are met for the client application:

- NVIDIA llama-3.1 NIM and GPU support.
- Network connectivity to issue REST APIs to CAS.

The following diagram illustrates the workflow where a user client application runs query REST calls on the CAS database and displays the results in the front end.



The entire workflow is divided into two parts: One part retrieves results from CAS, while the other obtains results from a Large Language Model (LLM).

The Vector database within the CAS namespace is designed to store data ingested from various files. The namespace also includes the query service pod, which features a FAST API (REST API). To retrieve information from the Vector database, there exists an installation of a document processing engine outside the namespace. Document processing is available with IBM Docling Multimodal services or NVIDIA NeMo Retriever Extraction microservices. NVIDIA re-ranking service is an optional search component that can be added to provide more accuracy. Your task involves developing an application (client application), that communicates with this REST API by using the OpenShift® cluster authentication. For more information about the FAST API methods, syntax, description, or to try the APIs, see Content-Aware Storage (CAS) APIs.

In addition, you must have a large language model of your choice that is accessible from your application. If you want to use the IBM Watson granite model LLM, then to download and use the different IBM Granite models, see IBM Granite.

When the user of the client application raises a question from its front end, a query REST Call is sent to the query service pod through the FAST API REST call. The query service in turn goes with the query to the document processing engine to get the embedding. This service generates embeddings for the data, enabling the query service to understand and process the request of the user effectively. After the results are fetched, a SQL query is sent with the embedding to the database for top "K" number of Vector results. When the user runs query from the front end, they can specify the value of this "K". For example, `Select, column from table order by, limit by vector, with embedding`. The top "K" results are returned to the client application which in turn sends it to the LLM. While embeddings make the retrieval process efficient, LLMs add a layer of contextual understanding, transforming raw data into meaningful information. Finally, the end results are published on the front end of the client application.

- **Content-Aware Storage (CAS) APIs**
  List of all REST API references for CAS service.
- **Integrating Model Context Protocol (MCP) with CAS**
  Content-Aware Storage (CAS) supports Model Context Protocol (MCP), enabling AI chat applications to access and search vector stores through natural language interactions.
- **Integrating CAS with watsonx Orchestrate by using MCP**
  Large language models move from experimentation into production. Enterprises build AI agents that require access to real-time business data. watsonx Orchestrate provides conversational AI for workflow automation. Integrating watsonx Orchestrate with enterprise systems such as Content-Aware Storage (CAS) often requires custom integration code for each data source.

Content-Aware Storage (CAS) APIs

List of all REST API references for CAS service.

`/cas/api/v1`

For more information about Swagger docs, see **`https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/docs#`**.

Authentication
: For authentication of REST APIs, ensure that you have a valid OpenShift® Container Platform bearer token in the Authorization header:
Retrieve the token:

1. In the OpenShift console, click your profile name and select Copy login command.

2. Log in with your credentials. Then click the Display Token link.
3. Copy the token value available under the Log in with this token section.
   Note: For authentication of REST APIs when an external Identity Provider (IDP) is configured, ensure that you have a valid IDP bearer token in the Authorization header.

Swagger documentation

1. In your browser, go to Swagger: `https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/docs#`.
   The Swagger document specifies the list of resources that are available in the REST API and the operations that can be called on those resources. The Swagger document also specifies the list of parameters for an operation, including the name and type of the parameters and information about acceptable values for those parameters.

2. In the FastAPI page, click Authorize.
3. In the Available authorizations window, enter a value for HTTPBearer.
4. Click Authorize.

Try APIs from Swagger
In your browser, go to Swagger: `https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/docs#`.
In Swagger, expand the API and click Try it out.

- `/vector_stores`
  - Description: It returns a document processor "vector stores" query that you have access to.
  - Method: GET
  - Query parameters:

Parameter:Type	Description
"after":string	Optional. A cursor for use in pagination. This parameter is not yet implemented.
"before":string	Optional. A cursor for use in pagination. This parameter is not yet implemented.
"limit":int	Optional. A limit on the number of objects to be returned. The default limit is 10.
"order":string	Optional. Sorts order by the vector store ID - `asc` for ascending order and `desc` for descending order. Default value is `desc`.

  - Usage: Check Swagger.
  - Response: Check Swagger.

- `/vector_stores/{vector_store_id}/search`
  - Description: Matches the exact words and uses Natural Language Processing (NLP) and machine learning to find results that are semantically relevant. If you do not have access to a specific domain, the API raises a 403 (Invalid access) error. If file-level security is enabled, semantic search results are restricted to only those files, which the CAS query user is authorized to read in the IBM Storage Scale file system.
  - Method: POST
  - Path parameter:

Parameter:Type	Description
"vector_store_id":string	Required. Name of a particular document processor that ingested documents from a particular domain.

  - Request body parameters:

Parameter:Type	Description
"query":string	Required. The natural language question that is used by the embedding model to return a list of relevant snippets of text from the ingested documents.
"filters":dict	Optional. A filter to apply based on file attributes. For more information, see Filter parameters.
"max_num_results":int	Optional. The maximum number of results returned. The default value is "10".
"ranking_options":dict	Optional. Ranking options for search default is null: "ranker":str - Optional. The default value is "auto". "score_threshold":float - Optional. The default value is "0". This parameter is not yet implemented.
"rewrite_query":bool	Optional. Whether to rewrite the natural language query for vector search. This parameter is not yet implemented.
"enable_source":boolean	Optional. If true, any source information about the document is returned. Default value is "False".
"enable_content_metadata":boolean	Optional. If true, any metadata about the relevant document is returned. Default value is "False".

  - Usage: Check Swagger.
  - Response: Check Swagger.

- `/vector_stores/{vector_store_id}/files/{file_id}/content`
  - Description: It returns all the content (text chunks) for a specific `vector-store` and file by their IDs.
  - Method: GET
  - Query parameters:

Parameter	Description
enable_source	Optional. When enabled, it returns top-level source file `metadata` attributes, or `null`, if not enabled.
enable_content_metadata	Optional. When enabled, it returns content `metadata` attributes, or `null`, if not enabled.

  - Usage: Check Swagger.
  - Response: Check Swagger.

Filter parameters
Two types of filters can be used:

- Comparison filter: A filter used to compare a specified attribute key to a defined value using a defined comparison operation.

Parameter	Type	Description
key	`str`	Required. The key to compare against the value.

Parameter	Type	Description
type	`str`	Required. Comparison operators: `eq`, `ne`, `gt`, `gte`, `lt`, `lte`, `in`, `nin` The set of valid operators for a comparison filter depends on the key being compared.
value	`str`, `int`, `float`, `bool`, `ISO8601 str`	Required. The value to compare against the key, which can be an array for `in` or `nin` operators.

- Compound filter: A filter used to combine an array of filters using `and` or `or`.

Parameter	Type	Description
type	`str`	Required. A type of logical operation: `and`, `or`.
filters	`array`	Required. An array of filters to combine. Items can be Comparison filter or Compound filter.

Supported keys:

Key	Type	Description
`file_id`	`str`	The file identifier.
`file_size`	`int`, `float`	The size of the file in bytes.
`file_path`	`str`	The full path of the file including the file name.
`file_name`	`str`	The full name of the file after the last path delimiter.
`file_type`	`str`	The type of source file as determined by the ingester (for example: PDF).
`file_created`	`ISO8601 str`	The creation date of the file in ISO8601 format.
`file_modified`	`ISO8601 str`	The last modified date of the file in ISO8601 format.
`attributes`	`str`, `int`, `bool`	The custom attributes set on the file through IBM Data Cataloging service tags. For more information, see Managing tags.

Supported comparison operators for key types:

Key type	Operators
`str`	`eq`, `ne`, `in`, `nin`, `contains`
`int`, `float`	`eq`, `ne`, `gt`, `gte`, `lt`, `lte`, `in`, `nin`
`ISO8601 str`	`eq`, `ne`, `gt`, `gte`, `lt`, `lte`, `in`, `nin`
`bool`	`eq`, `ne`

Note: When using the `in` or `nin` operators, the `value` parameter of the comparison must be an array of the appropriate type.

Examples

- Comparison filter:

```
"filters": {"key":"file_name","type":"eq","value":"sg248497.pdf"}
```

- Compound filter:

```
"filters": {
    "type": "and",
    "filters": [ {"key":"file_path","type":"eq","value":"/gpfs/fs1/gt20"},
                 {"key":"file_size","type":"lt","value":28000000},
                 {"key":"file_size","type":"gt","value":24000000}
               ]
}
```

- Compound filters support two nesting levels:

```
"filters": {
    "type":"or",
    "filters": [ {"key":"file_size","type":"gt","value":2000000},
                 {"type":"and",
                  "filters": [ {"key":"date_created","type":"gt","value":"2021-05-01"},
                               {"key":"date_created","type":"lt","value":"2021-05-31"}
                             ]
                 }
               ]
}
```

/api/v1/querysearch

Note: The `/api/v1/querysearch` APIs are deprecated with IBM Fusion 2.12.0 (CAS 1.1.0) release.

Authentication

For authentication of REST APIs, ensure that you have a valid OpenShift Container Platform bearer token in the Authorization header:

Retrieve the token:

1. In the OpenShift console, click your profile name and select Copy login command.
2. Log in with your credentials. Then click the Display Token link.
3. Copy the token value available under the Log in with this token section.
   Note: For authentication of REST APIs when an external Identity Provider (IDP) is configured, ensure that you have a valid IDP bearer token in the Authorization header.

Swagger documentation

1. In your browser, go to Swagger: `https://<fusionconsoleurl>/api/v1/querysearch/docs`.
   The Swagger document specifies the list of resources that are available in the REST API and the operations that can be called on those resources. The Swagger document also specifies the list of parameters for an operation, including the name and type of the parameters and information about acceptable values for those parameters.
2. In the FastAPI page, click Authorize.
3. In the Available authorizations window, enter a value for HTTPBearer.
4. Click Authorize.

Try APIs from Swagger

In your browser, go to Swagger: `https://<fusionconsoleurl>/api/v1/querysearch/docs`.
In Swagger, expand the API and click Try it out.

- **`/tables`**
  - Description: It returns a document processor "tables" query that you have access to.
  - Method: GET
  - Usage: Check Swagger.
  - Response: Check Swagger.
- **`/semantic_search`**
  - Description: Matches the exact words and uses Natural Language Processing (NLP) and machine learning to find results that are semantically relevant. If you do not have access of a specific domain, the API raises a 403 (Invalid access) error. If file-level security is enabled, semantic search results are restricted to only those files, which the CAS query user is authorized to read in the IBM Storage Scale file system.
  - Method: POST
  - Parameters:

Parameter: Type	Description
"query": string	The natural language question that is used by the embedding model to return a list of relevant snippets of text from the ingested documents.
"table": string	Name of a particular document processor that ingested documents from a specific domain.
"limit": int	The maximum number of results returned, default is 5.
"enable_source": boolean	If true, any source information about the document is returned.
"enable_content_metadata": boolean	If true, any metadata about the relevant document is returned.

  - Usage: Check Swagger.
  - Response: Check Swagger.

End-user CLI chatbot sample

The CLI-powered chatbot enables a secured, authenticated access to Content-Aware Storage (CAS) through a conversational CLI. It is powered by LLMs such as IBM watsonx Granite, Open AI, NVIDIA NIM, and Ollama. It uses CAS's semantic APIs to deliver efficient, AI-enabled insights, accelerating RAG workflows with improved performance, security, and efficiency.

For more information, follow the reference example in GitHub.

Access Control List (ACL) users control the query search process. The system retrieves user identities from the Identity Provider (such as Keycloak) and validates them against CAS ACL users based on roles and permissions. If validation succeeds, the system issues a token, which the user uses to run the CAS query API.

```
                                CAS ENTERPRISE CHATBOT
                              v1.0.0 - Enhanced Edition

+------------------------------------ ABOUT CAS ------------------------------------+
| Content-Aware Storage powers AI applications like RAG with faster insights, lower costs, better performance, stronger security, and simplified operations. |
+-----------------------------------------------------------------------------------+

[17:51:23] INFO     Configuration loaded from config.yaml
           INFO     CAS Chatbot initialized

Starting CAS Chatbot...
  Authenticating with OpenShift...
Authentication successful
[17:51:29] INFO     Successfully authenticated with OpenShift
  Checking CAS service status...
Query Search API: OK
   Welcome to IBM Fusion Content Aware Storage
Health Check: OK
[17:51:30] INFO     CAS service status check passed
```

```
[fusion-assist-sre] 👤 You: help
                        Available Commands
┏━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━━━━━━━━┓
┃ Command       ┃ Description            ┃ Example                  ┃
┡━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━━━━━━━━┩
│ help          │ Show this help menu    │ help                     │
│ bye/exit/quit │ Exit the chatbot       │ bye                      │
│ history       │ Show query history     │ history                  │
│ clear         │ Clear query history    │ clear                    │
│ tables        │ List available tables  │ tables                   │
│ table <name>  │ Switch to a specific table │ table customers      │
│ stats         │ Show session statistics │ stats                   │
│ export        │ Export chat history    │ export                   │
│ <query>       │ Ask a question         │ What are the sales trends? │
└───────────────┴────────────────────────┴──────────────────────────┘
[fusion-assist-sre] 👤 You: table gt20
Switched to table: gt20
[gt20] 👤 You: list down research papers.
⠸ 🔍 Searching...
[17:55:09] INFO     Query executed successfully for table: gt20
Trying LLM provider: nvidia
Based on the provided data, I found the following research papers relevant to the topic:

1. IBM Redbook: IBM System Storage SAN Volume Controller and Storwize V7000 Best Practices and Performance Guidelines, SG24-7521
2. IBM Redbook: Implementing IBM Spectrum Virtualize for Public Cloud Version 8.3, REDP-5466
3. IBM Redbook: Implementing the IBM Storwize V7000 with IBM Spectrum Virtualize V8.2.1, SG24-7938
4. IBM Redbook: Implementing the IBM System Storage SAN Volume Controller with IBM Spectrum Virtualize V8.1, SG24-7933
5. IBM Redpaper: IBM Storage for Red Hat OpenShift Container Platform V3.11 Blueprint Version 1 Release 1, REDP-5564
6. IBM Redpaper: IBM Storage for Red Hat OpenShift Blueprint Version 1 Release 4, REDP-5565
7. IBM Redpaper: Red Hat OpenShift and IBM Cloud Paks on IBM Power Systems: Volume 1, SG24-8459
8. IBM Knowledge Center: IBM Real-time Compression and performance guidelines summary, REDP-5466
9. IBM Knowledge Center: IBM Disk Magic, REDP-5466
10. IBM Knowledge Center: IBM Storage for Red Hat OpenShift Container Platform V3.11 Blueprint Version 1 Release 1, REDP-5564

Please note that these papers are based on the provided data and might not be the most relevant or up-to-date research papers.
[gt20] 👤 You: █
```

Integrating Model Context Protocol (MCP) with CAS

Content-Aware Storage (CAS) supports Model Context Protocol (MCP), enabling AI chat applications to access and search vector stores through natural language interactions.

Model Context Protocol (MCP)

Model Context Protocol (MCP) is a standardized JSON-RPC 2.0 protocol that allows AI assistants and chat applications to interact with external services. MCP enables AI clients to discover available capabilities and use them through conversational interfaces.

Benefits of MCP server

- AI assistants can discover and use CAS vector stores automatically.
- It supports natural language queries for vector search operations.
- It provides both standard and streaming response options.

Available MCP tools

CAS uses two primary tools through the MCP interface for working with vector stores.

list_vector_stores

- Description: Lists all vector stores available in the CAS instance. This tool retrieves the complete inventory of vector stores that can be accessed for search operations.
- Parameters: The parameters (`server_url` and `auth_token`) are needed as input for the CAS server you want to search in.

Parameter	Type	Description
`server_url`	string	Required. Base URL of the CAS server (for example, `https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>`).
`auth_token`	string	Required. Authentication token for API access.

- Usage example: An AI assistant calls this tool to discover which vector stores are available before performing searches.

search_vector_stores

- Description: Performs semantic search within a specified vector store. This tool uses vector embeddings to find content similar to the provided query text that returns ranked results based on semantic similarity.
- Parameters:

Parameter	Type	Description
`server_url`	string	Required. Base URL of the CAS server (for example, `https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>`).
`auth_token`	string	Required. Authentication token for API access.
`vector_store_id`	string	Required. ID of the vector store to search.
`query`	string	Required. Search query text.
`limit`	integer	Optional. Maximum number of results to return. The default value is '10'.

- Usage example: To find documents about 'machine learning best practices' in a vector store named 'technical_docs', the AI assistant calls this tool with the query text and store ID. The tool returns semantically similar documents that are ranked by relevance score.

get_vector_store_file_content

- Description: Retrieves the complete content of a specific file from a vector store. Use this tool when you need the complete document after finding it through search results. The `file_id` can be obtained from `search_vector_stores` results.
- Parameters:

Parameter	Type	Description
`server_url`	string	Required. Base URL of the CAS server (for example, `https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>`).
`auth_token`	string	Required. Authentication token for API access.
`vector_store_id`	string	Required. ID of the vector store to search.
`file_id`	string	Required. The ID of the file to fetch content for.

- Usage example: After searching for `storage configuration` and finding relevant results, use this tool with the `file_id` from those results to fetch the complete document content for detailed analysis.
- Example: Standard MCP endpoint

```
curl -k -N -X POST "https://<console-ibm-spectrum-fusion-ns.apps.openshift>/cas/api/v1/mcp" \
  -H "Content-Type: application/json" \
  -H "Accept: application/json" \
  -d '{
    "jsonrpc": "2.0",
    "id": 6,
    "method": "tools/call",
    "params": {
      "name": "get_vector_store_file_content",
      "arguments": {
        "server_url": "https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>",
        "auth_token": "your_auth_token",
        "vector_store_id": "your_store_id",
        "file_id": "5243905"
      }
    }
  }'
```

- Example: MCP streamable endpoint

```
curl -k -N -X POST \ https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/mcp-streamable/ \
  -H "Content-Type: application/json" \
  -H "Accept: application/json, text/event-stream" \
  -d '{
    "jsonrpc": "2.0",
    "id": 6,
    "method": "tools/call",
    "params": {
      "name": "get_vector_store_file_content",
      "arguments": {
        "server_url": "https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>",
        "auth_token": "your_auth_token",
        "vector_store_id": "your_store_id",
        "file_id": "5243905"
      }
    }
  }'
```

Authentication

For more information on authentication, see Content-Aware Storage (CAS) APIs.

All MCP tool calls require authentication. Provide the CAS authentication token in the `auth_token` parameter when you call any tool. The token is validated by using the same authentication system as the REST APIs.

MCP endpoints

/cas/api/v1/mcp

- Description: Standard MCP endpoint that provides JSON-RPC 2.0 protocol support for synchronous operations.
- Method: POST
- Content-Type: `application/json`

Initialize example:

```
curl -X POST \
  https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/mcp \
  -H "Content-Type: application/json" \
  -d '{
    "jsonrpc": "2.0",
    "id": 1,
    "method": "initialize",
    "params": {}
  }'
```

List available tools:

```
curl -X POST \
  https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/mcp \
  -H "Content-Type: application/json" \
  -d '{
    "jsonrpc": "2.0",
    "id": 2,
    "method": "tools/list",
    "params": {}
  }'
```

Call MCP tools:

```
curl -X POST \
  https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/mcp \
  -H "Content-Type: application/json" \
  -d '{
    "jsonrpc": "2.0",
    "id": 3,
    "method": "tools/call",
    "params": {
      "name": "list_vector_stores",
      "arguments": {
        "server_url": "https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>",
        "auth_token": "your_auth_token"
      }
    }
  }'
```

/cas/api/v1/mcp-streamable/

- Description: Streaming MCP endpoint that provides real-time response delivery by using Server-Sent Events (SSE). Use this endpoint when continuous or progressive responses are needed.
- Method: POST
- Content-Type: `application/json`
- Accept: `text/event-stream`

Initialize example:

```
curl -N -X POST \
  https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/mcp-streamable/ \
  -H "Content-Type: application/json" \
  -H "Accept: text/event-stream" \
  -d '{
    "jsonrpc": "2.0",
    "id": 1,
    "method": "initialize",
    "params": {
      "protocolVersion": "2024-11-05",
      "capabilities": {},
      "clientInfo": {
        "name": "test-client",
        "version": "1.0.0"
      }
    }
  }'
```

Call streaming tools:

```
curl -k -N -X POST \
https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/mcp-streamable/ \
-H "Content-Type: application/json" \
-H "Accept: application/json, text/event-stream" \
-d '{
  "jsonrpc": "2.0",
  "id": 3,
  "method": "tools/call",
  "params": {
    "name": "list_vector_stores",
    "arguments": {
      "server_url": "https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>",
      "auth_token": "your-token-here"
    }
  }
}'
```

```
curl -N -X POST \
  https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>/cas/api/v1/mcp-streamable/ \
  -H "Content-Type: application/json" \
  -H "Accept: text/event-stream" \
  -d '{
    "jsonrpc": "2.0",
    "id": 4,
    "method": "tools/call",
```

```
    "params": {
      "name": "search_vector_stores",
      "arguments": {
        "server_url": "https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>",
        "auth_token": "your_auth_token",
        "vector_store_id": "your_store_id",
        "query": "your search query",
        "limit": 10
      }
    }
  }'
```

Getting started using MCP endpoints in your application

To use MCP with CAS, follow these steps:

1. Obtain the CAS server URL and authentication token.
2. Configure the AI chat client to connect to the MCP endpoint.
   The AI assistant discovers the available tools automatically.
3. Begin with using natural language queries that require information from vector stores.

Error handling

MCP follows JSON-RPC 2.0 error conventions. If a request fails, the response includes an error object with a code and message.

Common error codes:

- -32700: Parse error (invalid JSON)
- -32600: Invalid request
- -32601: Method not found
- -32602: Invalid parameters
- -32603: Internal error

Integrating CAS with watsonx Orchestrate by using MCP

Large language models move from experimentation into production. Enterprises build AI agents that require access to real-time business data. watsonx Orchestrate provides conversational AI for workflow automation. Integrating watsonx Orchestrate with enterprise systems such as Content-Aware Storage (CAS) often requires custom integration code for each data source.

Model Context Protocol (MCP) removes this requirement. MCP provides a standardized, conversational interface that allows AI agents to discover and invoke enterprise capabilities without custom integration code. For more information about integrating MCP with CAS, see Integrating Model Context Protocol (MCP) with CAS.

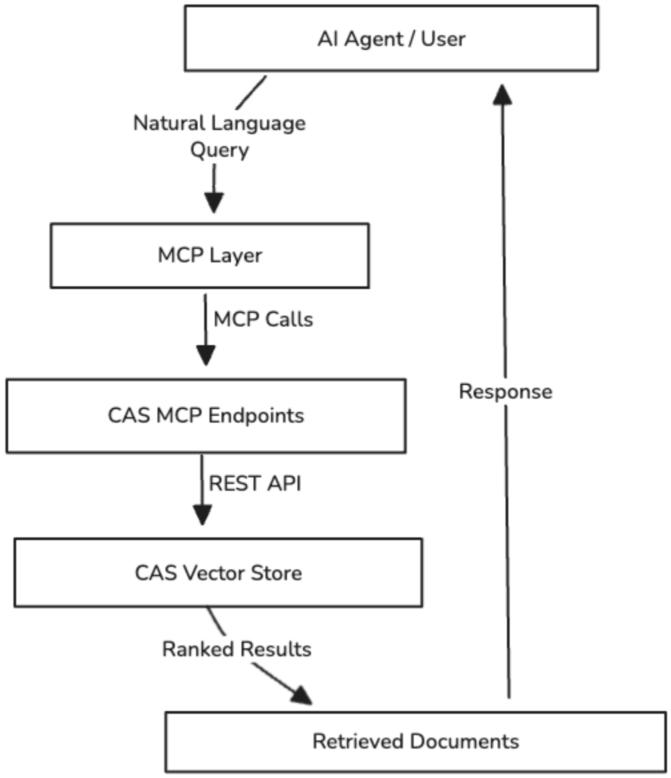
The following section explains how MCP enables watsonx Orchestrate to access CAS, describes the technical configuration, and demonstrates the integration through natural language queries.

Integration challenges

Traditional integrations between AI agents and enterprise systems introduce several challenges:

- Each vector database integration requires a custom API client code.
- AI agents cannot discover new capabilities without redeployment.
- Authentication, error handling, and connection management increase development overhead.
- Adding multiple data sources multiplies integration complexity.

Integration with watsonx Orchestrate



Validation of the architecture uses watsonx Orchestrate. The workflow follows three phases:

1. Connect CAS to watsonx Orchestrate by using MCP.
2. Create an AI agent that uses CAS tools.
3. Validate queries through natural language interaction.

Before you begin

Ensure that the following components are available:

- watsonx Orchestrate instance: An active watsonx Orchestrate deployment with CLI access. For more information about setting up this deployment, see Installing watsonx Orchestrate service.
- Model configured in watsonx Orchestrate: A model enabled to run AI agents that query CAS. For more information about adding a model, see Enabling a specific model.
- IBM Fusion instance with CAS deployed: CAS running on OpenShift®. For more information about installing CAS, see Installing Content-Aware Storage (CAS).
- CAS authentication token: Bearer token for CAS API access.
- CAS URL: The IBM Fusion console URL of the cluster. For example: `https://<console-ibm-spectrum-fusion-ns.apps.openshifturl>`
- Network connectivity: Network access from watsonx Orchestrate instance to the CAS server route.

Configuration

1. Set up the watsonx Orchestrate CLI environment.
   The watsonx Orchestrate CLI manages MCP servers, toolkits, and agents.
   a. Download and install the watsonx Orchestrate CLI for your operating system. For more information, see https://developer.watson-orchestrate.ibm.com/getting_started/installing.
   b. Verify the installation:
   ```
   orchestrate --version
   ```
   c. Set up a Python virtual environment (recommended):
   ```
   python3 -m venv orchestrate-env
   source orchestrate-env/bin/activate  # On Windows: orchestrate-env\Scripts\activate
   ```
2. Configure the watsonx Orchestrate environment.
   a. In the watsonx Orchestrate web console, retrieve the watsonx Orchestrate instance URL from Profile Settings. Add and activate the environment:
   ```
   orchestrate env add -n wxo-instance -u <your orchestrate url>
   ```
   ```
   orchestrate env activate wxo-instance
   ```
   b. When prompted, enter the watsonx Orchestrate credentials. Credentials can be obtained from the watsonx Orchestrate documentation.
3. Configure the CAS MCP server URL.
   a. To construct the URL, append `/cas/api/v1/mcp-streamable/` to the base CAS URL described in the Before you begin section. For example: `https://console-ibm-spectrum-fusion-ns.apps.<openshift-cluster-domain>/cas/api/v1/mcp-streamable/`
4. Register the CAS MCP server as a toolkit in watsonx Orchestrate:

```
orchestrate toolkits add \
  --kind mcp \
  --name cas \
  --description "MCP server for CAS vector stores" \
  --url <your-cas-mcp-url> \
  --transport streamable_http \
  --tools "search_vector_stores,list_vector_stores,get_vector_store_file_content"
```

This command registers the CAS MCP server and exposes the following tools to watsonx Orchestrate agents:

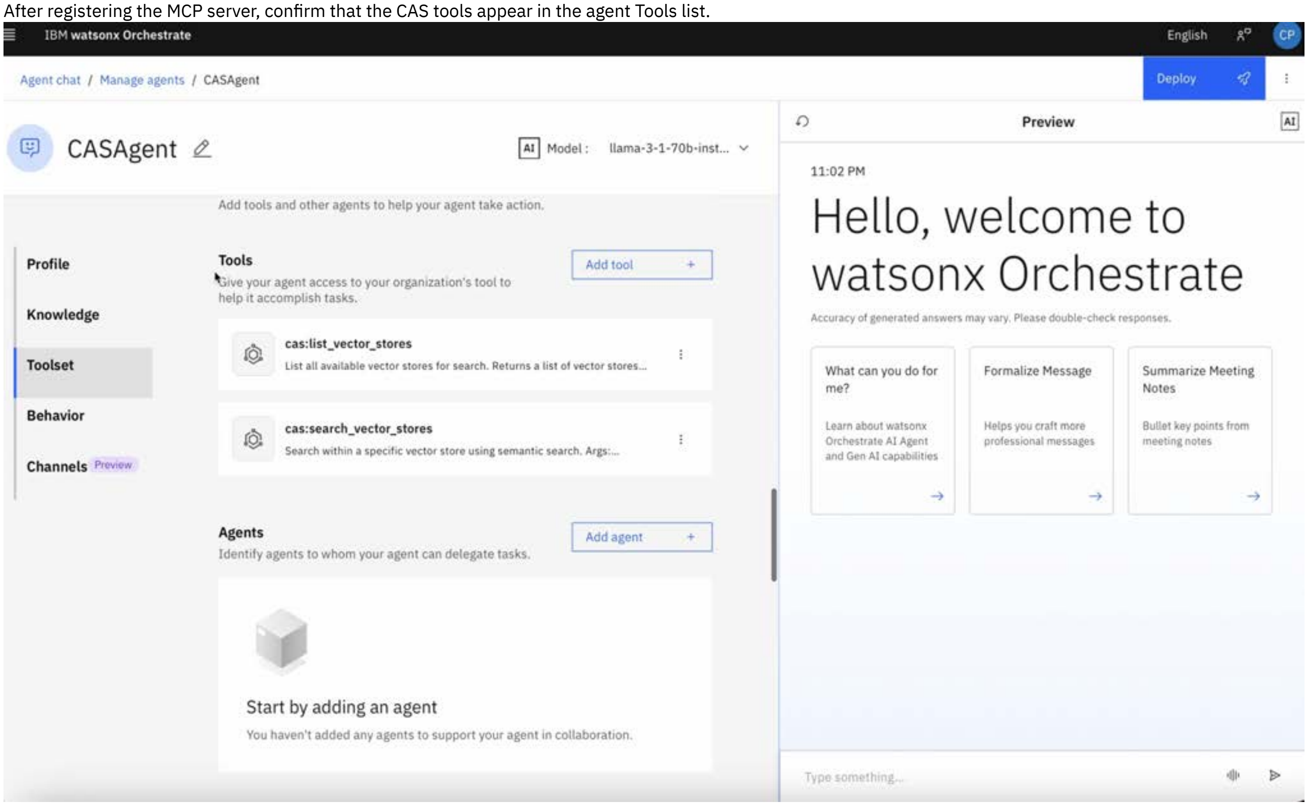
- `search_vector_stores`
- `list_vector_stores`
- `get_vector_store_file_content`

5. Create and configure the AI agent.
   a. Open the watsonx Orchestrate web console and click Create new agent.
   b. Enter the required details and click Create.
   c. In the Tools section of agent configuration, select Add Tool.
   d. Select tools from the cas toolkit.
   e. Click Add tool and add the following tools:
      - `list_vector_stores`
      - `search_vector_stores`
      - `get_vector_store_file_content`
   f. Click Save.
   g. Click Deploy.

Validation

Verify tool registration

After registering the MCP server, confirm that the CAS tools appear in the agent Tools list.



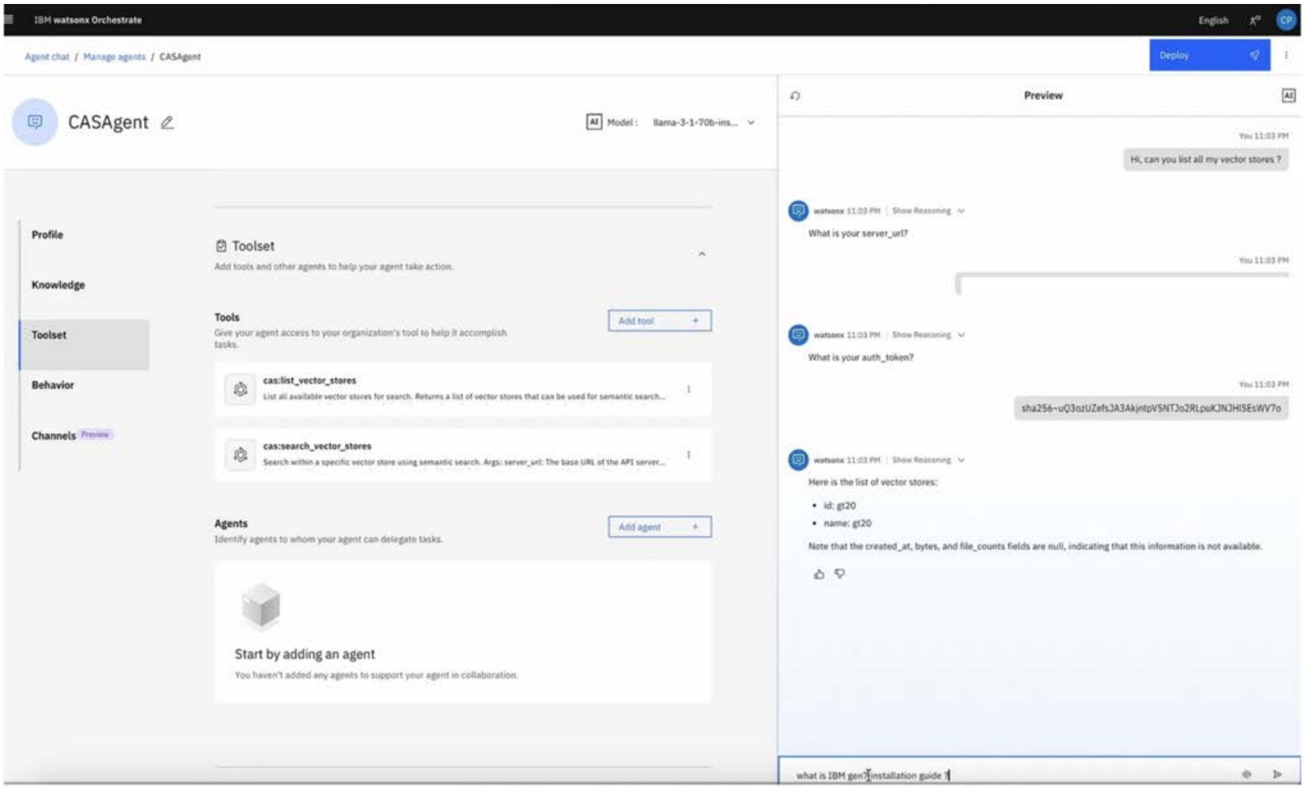
Verify tool execution

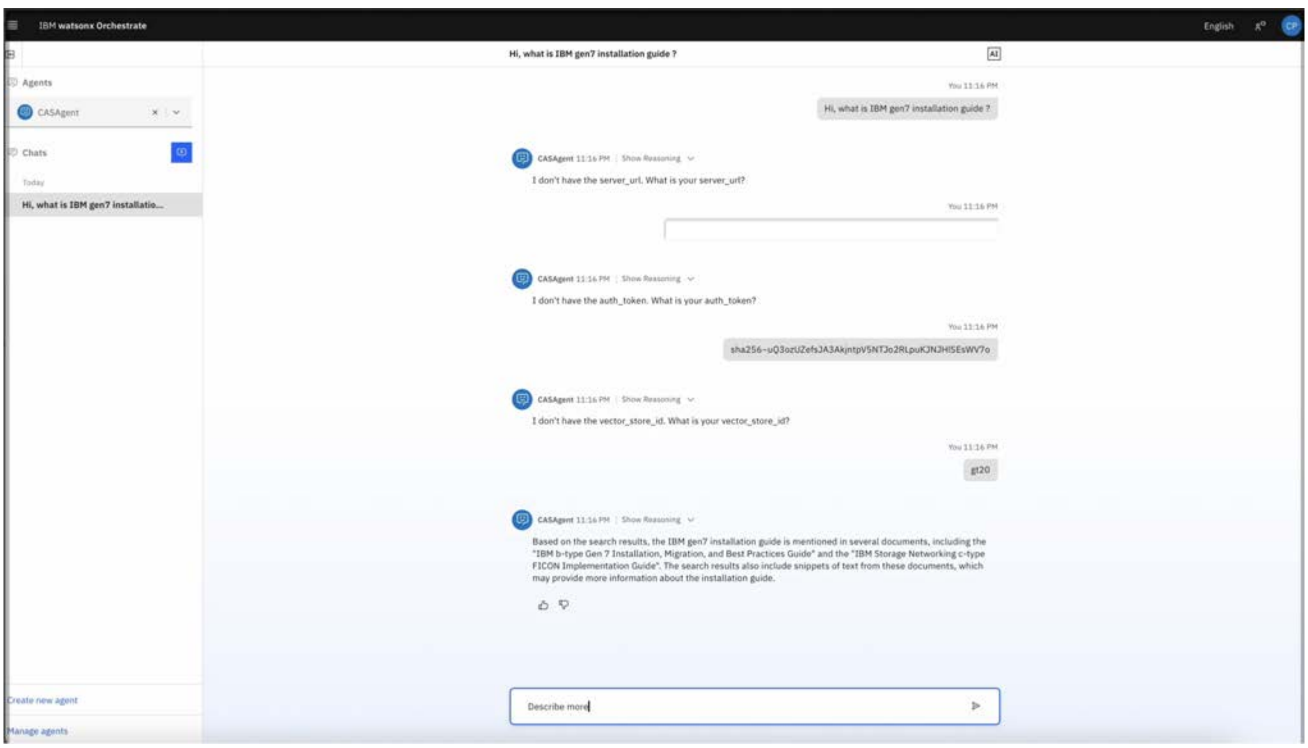
Test the deployed agent by using natural language prompts.

Example prompt:

```
What vector stores are available in CAS?
```

Note: When prompted, provide the CAS route URL and bearer token in the chat interface.





Results

The integration connects CAS with watsonx Orchestrate through MCP. AI agents access enterprise vector stores directly by using natural language.

With this configuration, you can:

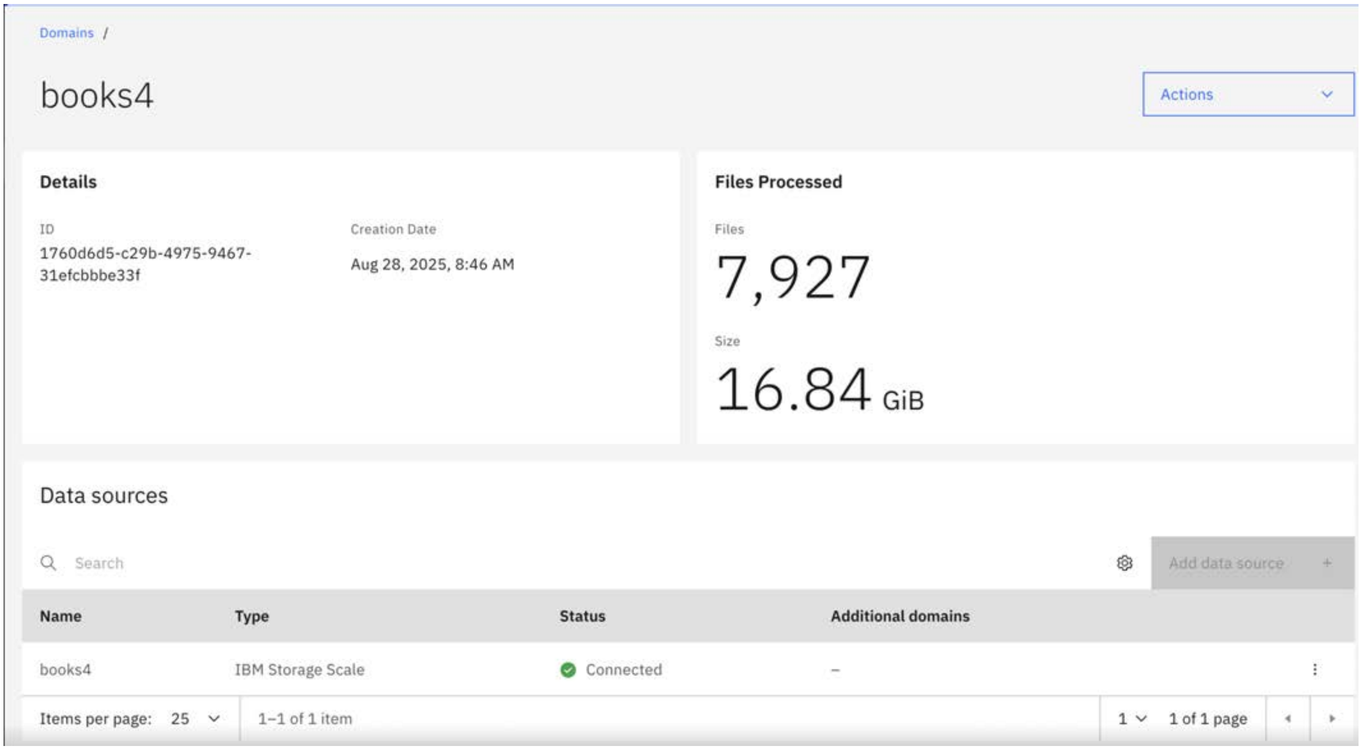
- Run agentic workflows that retrieve and act on enterprise knowledge.
- Integrate retrieval-augmented generation (RAG) pipelines with CAS vector stores.
- Eliminate custom integration code and reduce operational complexity.

Monitoring

This page provides a comprehensive view of monitoring capabilities, structured into two key sections: monitoring using dashboard in Content-Aware Storage (CAS) user interface and metric collection through Prometheus.

Monitor from CAS dashboard

Go to Content-Aware Storage > Overview page of the IBM Fusion user interface to view the CAS dashboard. It provides an inventory of domains and data sources and newly created domain entries. You can use this information to get insights of system activity.



CAS metrics

CAS provides a set of Prometheus metrics to give you insights on processing time of a batch of documents with respect to the document sizes.

Metric	Description
`cas_documents_batch_duration_seconds [label: document_processor_name, event_type]`	Overall time taken by CAS to successfully process a batch of new, updated, or deleted documents.
`cas_ingest_service_batch_duration_seconds [label: document_processor_name]`	Time taken by document processor to successfully process a batch of new or updated documents. This is a subset of the overtime time measured by `cas_documents_batch_duration_seconds`.
`cas_documents_size_bytes_total [label: document_processor_name, event_type]`	Total size of all documents in a batch of new, updated, or deleted documents. This metric is published regardless of whether the batch processing is successfully or not. It gives you insights on whether there is any unexpected spike of document size.
`cas_document_size_bytes [label: document_processor_name, event_type]`	Size of each document. This metric is published regardless of whether the batch processing is successfully or not. It gives you insights on whether there is any unexpected large file.
`cas_documents_count [label: file_type, domain, documentProcessor]`	Total number of files in the domain that the specified document processor handles. This metric is published only after the document processor processes at least one file.
`cas_documents_size_bytes [label: file_type, domain, documentProcessor]`	Total size in bytes of all files present in the domain that the specified document processor handles. This metric is published only after the document processor processes at least one file.

The following table provides label definitions:

Label	Value
`document_processor_name`	`domain name`
`event_type`	`update: new or updated documents`
	`delete: deleted documents`

Uninstalling Content-Aware Storage (CAS) service

When you do not need CAS service, uninstall it from IBM Fusion.

About this task

After you run the script, the following high-level steps occur in sequence:

- Uninstalls Apache Kafka service.
- Uninstalls the CloudNativePG (CNPG) service.
- Deletes the CAS namespace.

Procedure

1. Log in to your cluster.
2. In your browser, go to this URL - https://github.com/IBM/storage-fusion/blob/master/CAS/cas-cleanup/customer_cas_cleanup.sh.
3. Download the customer_cas_cleanup.sh file and run the script.

- **Cleaning up Content-Aware Storage (CAS) data**
  You can use CRs to clean the data source.

Cleaning up Content-Aware Storage (CAS) data

You can use CRs to clean the data source.

Clean up the data source

Clean up the data source through CLI
  To delete the data source, run the following command:

```
oc delete datasource <datasource-name>
```

  Note: After you delete the data source, the respective PV, PVC, Kafka Topic, and Clustered Watch Folder also get deleted.

Clean up the data source through user interface
  To clean up the data source through UI, follow these steps:

1. Open the IBM Fusion UI.
2. Go to Content-aware Storage > Data sources.
3. Select the data source that you want to delete.
4. In the ellipsis overflow menu, click Remove.

Cleanup steps for an S3 data source
  To clean up an S3 data source, follow these steps:

1. From the data source YAML, get the associated `filesystem-name` and `fileset-name`.
2. Log in to the IBM Storage Scale cluster.
3. Unlink the fileset with the following command:

   ```
   mmunlinkfileset <filesystem-name> <fileset-name>
   ```

4. Delete the fileset:

   ```
   mmdelfileset <filesystem-name> <fileset-name>
   ```

5. Delete the bucket configuration information:

   ```
   mmafmcoskeys <bucket-name>:<fileset-name>-exportmap delete
   ```

6. Delete the data source.

Clean up the domain

To delete the Domain, run the following command:

```
oc delete Domain <domain-name>
```

Clean up the Document Processor

To delete the Document Processor, run the following command:

```
oc delete DocumentProcessor <document-processor-name>
```

Note: After you delete the Document Processor, the respective deployment, PVC, Service, and Database Table also get deleted.

Troubleshooting issues and known limitations in Content-Aware Storage (CAS)

The common troubleshooting steps and limitations in CAS service.

CAS operator reconcile stalls: Missing EventManager NetworkPolicy

Problem statement
  The CAS operator logs repeat the following pattern approximately every 60 seconds throughout the entire install and any subsequent reconcile:

```
Event Manager service url: https://<clusterIP>:10666/api/v1/eventmanager/alerts
Error POSTing info request (...): Post "https://<clusterIP>:10666/api/v1/eventmanager/alerts":
  context deadline exceeded (Client.Timeout exceeded while awaiting headers)
```

Each timeout adds approximately 60 seconds of dead time to every reconcile cycle, visibly slowing installation and upgrade progress.

Important: This fix is non-recurrent. Apply it once before CAS installation. It persists across installations and upgrades unless **isf-serviceability-operator** reconciles the policy away.

Cause

The **eventmanager-network-policy** in **ibm-spectrum-fusion-ns** has no ingress rule permitting traffic from the **ibm-cas** namespace to port **10666**. Every outbound POST from the CAS operator is silently dropped, and the HTTP client blocks until the OS TCP timeout expires.

When to apply

Apply this fix before CAS installation begins. Every reconcile cycle stalls without it.

Re-apply if the symptom returns after an **isf-serviceability-operator** reconcile overwrites the policy.

Resolution

Save the following manifest to a file named allow-cas-operator-to-eventmanager.yaml and apply it:

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-cas-operator-to-eventmanager
  namespace: ibm-spectrum-fusion-ns
  labels:
    app.kubernetes.io/managed-by: cas-operator
spec:
  podSelector:
    matchLabels:
      app: isfemdep
  policyTypes:
  - Ingress
  ingress:
  - from:
    - namespaceSelector:
        matchLabels:
          kubernetes.io/metadata.name: ibm-cas
      podSelector:
        matchLabels:
          app.kubernetes.io/component: operator
          app.kubernetes.io/name: cas.isf.ibm.com
    ports:
    - port: 10666
      protocol: TCP
```

```
oc apply -f allow-cas-operator-to-eventmanager.yaml
```

Result

After applying the policy, the CAS operator logs no longer show `context deadline exceeded` for Event Manager POST calls.

Docling, vllm, and cast-runtime pods crash with permission errors: SCC fsGroup override

Problem statement

After a CAS upgrade or on a fresh install, one or more of the following symptoms occur:

- The **docling** pod shows **1/2 CrashLoopBackOff**. Logs show:

  ```
  FileNotFoundError: .../ch_PP-OCRv4_det_mobile.onnx does not exist
  ```

  A stale **huggingface-model-data** PVC contains _**infer**-suffix model files. CAS 1.1.5 expects _**mobile**-suffix files.

- The **vllm-embedding** or **vllm-vision** pod shows **1/2 CrashLoopBackOff**. Logs show:

  ```
  PermissionError: [Errno 13] Permission denied: '/models/modules'
  ```

- The **cast-runtime** pod logs show:

  ```
  Error publishing messages [Errno 13] Permission denied: '/shared/<jobID>'
  ```

  Inspecting the volume mount inside the pod confirms root ownership:

  ```
  $ stat /shared
  Access: (0755/drwxr-xr-x)  Uid: (0/root)  Gid: (0/root)
  ```

Important: This fix is non-recurrent. Apply it once before the affected pods first start. It persists until the SCC or RBAC binding is removed.

Cause

The **ibm-cas-setuid-setgid-scc** is configured with **fsGroup: RunAsAny** at priority 10. When this SCC wins pod admission, the kubelet skips **fsGroup** injection and does not chown volume mounts. PVC mounts remain owned by **root:root**, and pods running as a non-root UID cannot write to them.

When to apply

Apply all steps before the **DocumentProcessor**, **vllm**, and **docling** pods first start. The SCC must exist at pod admission time.

If the affected pods have already started under the wrong SCC, complete all steps and then bounce the pods (step 4).

Resolution

1. Create an override SCC at priority 11, which takes precedence over the existing SCC at priority 10:

```
oc apply -f - <<'EOF'
apiVersion: security.openshift.io/v1
kind: SecurityContextConstraints
metadata:
  name: ibm-cas-cast-runtime-mustrunasgroup
allowPrivilegeEscalation: false
allowPrivilegedContainer: false
allowedCapabilities: [SETUID, SETGID]
fsGroup:
  type: MustRunAs
groups: []
users: []
runAsUser:
  type: MustRunAsNonRoot
seLinuxContext:
  type: MustRunAs
supplementalGroups:
  type: RunAsAny
volumes: [configMap, emptyDir, persistentVolumeClaim, secret]
EOF
```

2. Create a ClusterRole that grants use of the new SCC:

```
oc apply -f - <<'EOF'
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: ibm-cas-cast-runtime-mustrunasgroup-use
rules:
- apiGroups: [security.openshift.io]
  resourceNames: [ibm-cas-cast-runtime-mustrunasgroup]
  resources: [securitycontextconstraints]
  verbs: [use]
EOF
```

3. Bind the `ClusterRole` to the relevant service accounts:

```
oc apply -f - <<'EOF'
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: ibm-cas-cast-runtime-mustrunasgroup-use
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: ibm-cas-cast-runtime-mustrunasgroup-use
subjects:
- kind: ServiceAccount
  name: ibm-isf-cas-operator-system
  namespace: ibm-cas
- kind: ServiceAccount
  name: default
  namespace: ibm-cas
EOF
```

4. Bounce the affected pods so that they are re-admitted under the new SCC:

```
# Replace <domain> with your DocumentProcessor domain name (for example, deucalion-domain)
oc delete pod -n ibm-cas \
  -l app.kubernetes.io/component=cast-runtime,app.kubernetes.io/part-of=<domain>
oc delete pod -n ibm-cas -l app.kubernetes.io/component=vllm
oc delete pod -n ibm-cas -l app.kubernetes.io/name=docling
```

Result

After the pods restart, confirm that they are admitted under the new SCC:

```
oc get pod <pod-name> -n ibm-cas \
  -o jsonpath='{.metadata.annotations.openshift\.io/scc}{"\n"}'
# Expected: ibm-cas-cast-runtime-mustrunasgroup
```

Upgrade stalls: Kafka CR is patched to an unsupported version

Problem statement

After enabling the cache filesystem or triggering any upgrade, the CAS operator logs repeat indefinitely:

```
Kafka not yet ready.
```

Inspecting the Kafka custom resource (CR) shows:

```
oc get kafka kafka -n ibm-cas -o jsonpath='{.status.conditions}'
[{
  "message": "Unsupported Kafka.spec.kafka.version: 3.9.0. Supported versions are: [4.0.0, 4.1.0]",
  "reason": "UnsupportedKafkaVersionException",
  "status": "True",
  "type": "NotReady"
}]
```

Important: This issue is recurrent. Re-apply the fix each time the CAS operator patches the Kafka CR to an unsupported version, that is, on every install re-entry or upgrade that triggers `UpdateKafkaVersion`.

Cause

The CAS operator hardcodes Kafka version 3.9.0 when patching the Kafka CR during upgrades and install re-entry. AMQ Streams 3.1.0-14 supports only Kafka versions 4.0.0 and 4.1.0. As a result, the Kafka CR transitions to a `NotReady` state immediately after the update, and the `IsHealthyKafkaCluster` check never returns `true`.

When to apply
: Apply this fix before triggering install re-entry, that is, before applying the `CasInstall` status patch that re-enters the cache filesystem install path. Strimzi requires approximately 30 seconds to reconcile the Kafka CR to `Ready` after this patch.

  If the operator has already entered the loop, apply the patch and wait for the next reconcile cycle to proceed.

Resolution
: Patch the Kafka CR to set a supported version:

```
oc patch kafka kafka -n ibm-cas --type=merge \
  -p '{"spec":{"kafka":{"version":"4.0.0"}}}'
```

Result
: Within approximately 30 seconds, the Kafka CR shows `type=Ready`, `status=True`.

  Verify by running the following command:

```
oc get kafka kafka -n ibm-cas \
  -o jsonpath='{.status.conditions[?(@.type=="Ready")].status}{"\n"}'
# Expected: True
```

  The operator log stops repeating `Kafka not yet ready` after the next reconcile cycle.

CAS installation fails with CAS Cache Filesystem Failed error

Problem statement
: When installing CAS with Local Data Caching for Content Aware Storage (CAS) enabled, the installation stops and eventually fails with a message similar to:

```
CAS Cache Filesystem Failed
```

  The installation does not progress past the cache filesystem validation stage.

Cause
: When Local Data Caching for Content Aware Storage (CAS) is enabled, CAS waits for the IBM Storage Scale cache filesystem (`cache-fs`) to reach a healthy state before continuing the installation.

  By default, CAS waits up to 20 minutes for the cache filesystem to become healthy. If the filesystem does not reach a healthy state within this period, the installation is canceled and marked as failed.

  This situation can occur when:

  - Filesystem creation is taking longer than expected.
  - The `cache-fs` filesystem requires additional time to become healthy.

Resolution
: If additional time is required for the cache filesystem to become healthy, configure the CAS installation to ignore the filesystem health timeout.

  1. Verify the CAS installation resource:

     ```
     oc get casinstall -n ibm-cas
     ```

  2. Patch the CAS installation resource to add the following annotation:

     ```
     oc patch casinstall ibm-cas-service-instance \
       -n ibm-cas \
       --type merge \
       -p '{"metadata":{"annotations":{"ignore-filesystem-health-timeout":"true"}}}'
     ```

  3. Verify that the annotation is applied:

     ```
     oc get casinstall ibm-cas-service-instance \
       -n ibm-cas \
       -o yaml
     ```

Result
: After you apply the annotation, CAS overrides the default 20-minute cache filesystem health timeout. The installation continues to wait until the IBM Storage Scale cache filesystem becomes healthy, and then proceeds.

  Note: This annotation only affects installations that use Local Data Caching for Content Aware Storage (CAS).
  Important: Use this workaround only when the cache filesystem is expected to become healthy eventually.

EventManager NetworkPolicy blocks CAS operator alert POSTs

Problem statement
: The CAS operator (`ibm-isf-cas-operator-controller-manager`) calls `PostInfo()` and `PostAlert()` from the `isf-common` event client library during every `setInstallationStageAndStatus` reconcile step. These calls POST to https://<clusterIP>:10666/api/v1/eventmanager/alerts. As a result, no CAS events are reported in the IBM Fusion UI events section.

Cause
: The `eventmanager-network-policy` in `ibm-spectrum-fusion-ns` has no ingress rule permitting traffic from `ibm-cas`. Hence, every outbound POST is silently dropped. The HTTP client also lacks a short timeout, so each POST blocks for the full OS or kernel TCP timeout before returning `context deadline exceeded`.

Resolution

Apply the following custom NetworkPolicy in **ibm-spectrum-fusion-ns** to grant the CAS operator pod ingress to eventmanager on port **10666**.

1. Create a YAML file with the following content:

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-cas-operator-to-eventmanager
  namespace: ibm-spectrum-fusion-ns
  labels:
    app.kubernetes.io/managed-by: cas-operator
spec:
  podSelector:
    matchLabels:
      app: isfemdep
  policyTypes:
  - Ingress
  ingress:
  - from:
    - namespaceSelector:
        matchLabels:
          kubernetes.io/metadata.name: ibm-cas
      podSelector:
        matchLabels:
          app.kubernetes.io/component: operator
          app.kubernetes.io/name: cas.isf.ibm.com
    ports:
    - port: 10666
      protocol: TCP
```

2. Apply with the following command:

```
oc apply -f allow-cas-operator-to-eventmanager.yaml
```

Important: This workaround is not durable. The **isf-serviceability-operator** (label **app.kubernetes.io/managed-by: isf-serviceability-operator**) manages **eventmanager-network-policy**, which might reconcile over or conflict with the standalone policy. The custom policy is also not applied automatically on fresh installations.

PVC remains in Pending state

Problem statement

The data source remains in the Connecting state and the associated PersistentVolumeClaim (PVC) stays in the **Pending** state.

The PVC events might contain an error similar to the following:

```
rpc error: code = Internal desc = failed to update comment for fileset [...]
rpc error: code = Internal desc = remote call failed with response
400 Invalid value in 'filesetName'
```

Cause

This issue can occur when the NFS export path that is specified during data source creation is no longer valid or accessible.

Resolution

1. Verify that the configured NFS export path still exists on the NFS server.
2. If the export path is changed:
   a. Delete the data source that references the outdated NFS export path.
   b. Obtain the current NFS export path from the NFS server.
   c. Create a new data source by using the updated NFS export path.

After the data source is re-created with a valid export path, the PVC must progress from the **Pending** state and the data source must eventually transition to the Connected state.

PVC remains in Pending state because the volume cannot be provisioned

Problem statement

The data source remains in the Connecting state and the associated PersistentVolumeClaim (PVC) stays in the **Pending** state.

Review the PVC events:

1. Open the data source PVC.
2. Select the Events tab.
3. Look for messages similar to the following:

```
Waiting for a volume to be created either by the external provisioner 'spectrumscale.csi.ibm.com' or manually by the
system administrator. If volume creation is delayed, please verify that the provisioner is running and correctly
registered.
```

Cause

The prior message indicates that IBM Storage Scale has not yet provisioned the volume that is requested by the PVC.

Possible causes include:

- The IBM Storage Scale cluster is running low on available capacity.
- The IBM Storage Scale cluster is temporarily unable to create new filesets or volumes.
- Volume provisioning is taking longer than expected due to cluster activity or resource constraints.

Resolution

1. Verify that the IBM Storage Scale cluster has sufficient free capacity.
2. If the cluster is full or nearly full, free up storage space and wait for provisioning to resume.
3. Monitor the PVC status and events for updates.
4. Wait for the PVC to transition from `Pending` to `Bound`.

Note: By default, an NFS data source can take up to one hour to transition to the Connected state. If the PVC is still progressing and no provisioning errors are reported, allow sufficient time for the provisioning process to complete before taking corrective action.
After the PVC reaches the `Bound` state, the data source must continue the connection process and eventually transition to the Connected state.

IBM Fusion or Global Data Platform upgrade gets stuck due to ibm-cas/kafka-broker-nodepool eviction

Problem statement
Global Data Platform upgrade gets stuck in between as ibm-cas/kafka-broker-nodepool-0 fails to evict.

Cause #1
The nodepools have anti-affinity configured correctly, but the eviction logic is restrictive. After one pod is evicted, the system does not allow any additional pods to be taken down, which prevents further rescheduling.

Cause #2
IBM Storage Scale includes a Pod Disruption Budget (PDB) for its node pods with a similar restrictiveness to the Kafka pods. Sometimes, a conflict can occur between the Kafka PDB and the IBM Storage Scale PDB. When this happens, the `scale-core` pod might remain in the `Terminating` state, and block the rolling update of a Machine Config Pool (MCP).

Resolution

1. Update Kafka custom resource (CR)
   a. In the OpenShift® console, go to Installed Operators > Streams for Apache Kafka (ibm-cas ns) > All Instances > Kafka > YAML.
   b. Locate the `Kafka` section and add:

   ```
   template:
     podDisruptionBudget:
       maxUnavailable: 2
   ```

   c. Save and apply the updated `Kafka` CR.
2. If one or more relevant scale-core pods are stuck in `Terminating` state for an extended period, follow these steps:
   a. Log in to the OpenShift cluster and check the pod status in the `openshift-machine-config-operator` namespace:

   ```
   oc get pods -n openshift-machine-config-operator
   ```

   b. Identify the `machine-config-controller` pod (for example: `machine-config-controller-669b794b8f-6p4bl`).
   c. View the logs to determine where MCP is blocked:

   ```
   oc logs machine-config-controller-669b794b8f-6p4bl -n openshift-machine-config-operator
   ```

   d. If the logs indicate that a pod is blocking MCP, manually evict that pod from the node:

   ```
   oc adm drain compute-1-ru4.racke.mydomain.com --force=true --ignore-daemonsets=true --grace-period=-1 --delete-emptydir-data
   ```

   e. Delete the `scale-core` pod:

   ```
   kubectl delete pod <scale-core-pod-name> --grace-period=0 --force
   ```

Strimzi migration from ZooKeeper to KRaft gets stuck at early CAS upgrade stage

Problem statement
The Strimzi cluster operator pod gets stuck in a no-op reconciliation loop. The following `cas-operator` log message reconciles multiple times:

```
2025-10-30T00:49:10Z  INFO  KRaft migration in progress... requeueing    {"controller": "casinstall", "controllerGroup": "cas.isf.ibm.com", "controllerKind": "CasInstall", "CasInstall": {"name":"ibm-cas-service-instance","namespace":"ibm-cas"}, "namespace": "ibm-cas", "name": "ibm-cas-service-instance", "reconcileID": "07107592-ca70-4d44-9a96-b66fa29f20af"}
```

If the metadata state of the Kafka Custom Resource (CR) is in `KRaftMigration` for a long time, it means that it is stuck:

```
% oc get Kafka
NAME  DESIRED KAFKA REPLICAS  DESIRED ZK REPLICAS  READY  METADATA STATE  WARNINGS
kafka  3            3            True  KRaftMigration
```

`amq-streams-cluster-operator` logs:

```
2025-10-30 01:03:33 INFO StrimziPodSetController:349 - Reconciliation #632(watch) StrimziPodSet(ibm-cas/kafka-broker-nodepool): StrimziPodSet will be reconciled
2025-10-30 01:03:33 INFO StrimziPodSetController:385 - Reconciliation #632(watch) StrimziPodSet(ibm-cas/kafka-broker-nodepool): reconciled
2025-10-30 01:03:33 INFO StrimziPodSetController:349 - Reconciliation #633(watch) StrimziPodSet(ibm-cas/kafka-broker-nodepool): StrimziPodSet will be reconciled
2025-10-30 01:03:33 INFO StrimziPodSetController:385 - Reconciliation #633(watch) StrimziPodSet(ibm-cas/kafka-broker-nodepool): reconciled
2025-10-30 01:03:33 INFO StrimziPodSetController:349 - Reconciliation #634(watch) StrimziPodSet(ibm-cas/kafka-zookeeper): StrimziPodSet will be reconciled
2025-10-30 01:03:33 INFO StrimziPodSetController:385 - Reconciliation #634(watch) StrimziPodSet(ibm-cas/kafka-zookeeper): reconciled
2025-10-30 01:03:33 INFO StrimziPodSetController:349 - Reconciliation #635(watch) StrimziPodSet(ibm-cas/kafka-controller-nodepool): StrimziPodSet will be reconciled
```

```
2025-10-30 01:03:33 INFO StrimziPodSetController:385 - Reconciliation #635(watch) StrimziPodSet(ibm-cas/kafka-controller-nodepool): reconciled
2025-10-30 01:03:35 INFO ClusterOperator:140 - Triggering periodic reconciliation for namespace ibm-cas
2025-10-30 01:03:35 INFO AbstractOperator:266 - Reconciliation #636(timer) Kafka(ibm-cas/kafka): Kafka kafka will be checked for creation or modification
2025-10-30 01:03:38 INFO AbstractOperator:546 - Reconciliation #636(timer) Kafka(ibm-cas/kafka): reconciled
2025-10-30 01:05:35 INFO ClusterOperator:140 - Triggering periodic reconciliation for namespace ibm-cas
2025-10-30 01:05:35 INFO AbstractOperator:266 - Reconciliation #637(timer) Kafka(ibm-cas/kafka): Kafka kafka will be checked for creation or modification
2025-10-30 01:05:38 INFO AbstractOperator:546 - Reconciliation #637(timer) Kafka(ibm-cas/kafka): reconciled
```

Resolution
: Restart the **amq-streams-cluster-operator** pod. This resumes the migration process.

Errors in CAS runtime pod after Scale upgrade

Problem statement
: Data ingestion fails with errors in CAS runtime pod.

Cause #1
: The filesystem might be unmounted. This error can be observed in the **cas-runtime** logs as a **NodeNotReadyError**, or a pod is stuck at **Init:0/1** or in a **CrashLoopBackOff** state with a **FailedMount** event.

Resolution

1. Ensure that the filesystem is mounted correctly on all required nodes where Kafka pods and other affected pods are running.
2. To check which nodes the filesystem is mounted on, go to the Scale cluster and run the following command:

   ```
   mmlsmount all -L
   ```

   This lists all the nodes that filesystem is mounted on.
3. On the OpenShift Container Platform cluster, in the **ibm-spectrum-scale** namespace, identify the corresponding pod.
4. In the **ibm-spectrum-scale** namespace, find a healthy pod that is one where the filesystem is correctly mounted.
5. Run the following command to mount the filesystem on all the necessary pods:

   ```
   mmmount all -a
   ```

   It also resolves any filesystem mount-related errors.
6. Restart all the Kafka pods, **cas-runtime** pod, and **nv-ingest** pods.

Cause #2
: The **nv-ingest** job fails.

Resolution

1. Check whether the NV-ingest Redis pod is restarted. It can cause **nv-ingest** job to fail.
2. Restart the pods corresponding to **nv-ingest**.

Cause #3
: The Kafka pods fail.

Resolution

1. This issue can be related to the filesystem mount problem described in Cause #1. Follow the resolution steps in Cause #1 if the error is related to filesystem mount.
2. Restart the Kafka pods.

Inaccurate search results using the Docling Multimodal Processing Engine

Problem statement
: Intermittent issues are observed where search results from a domain using Docling Multimodal Processing Engine return random results.

Cause
: The vector index becomes corrupted and results in inaccurate search results.

Resolution
: To fix the issue, drop and recreate the vector index on the affected domain by following these steps:

1. Log in to the OpenShift cluster by running the following command through command line:

   ```
   oc login <cluster-url>
   ```
2. Switch to the **ibm-cas** project:

   ```
   oc project ibm-cas
   ```
3. Save the database password for later use:

   ```
   oc get secret cluster-parade-app -oyaml -n ibm-cas | yq '.data.password' | base64 -d
   ```
4. Access the parade database pod:

   ```
   oc exec -it cluster-parade-1 -- psql -U app -d app -h localhost -W
   ```

   Enter the database password retrieved in Step 3.

5. Run the following command to view table details:

```
\d nameOfDomain
```

Locate the index labeled `diskann (vector)`.

Example output:

```
app=> \d red1
                                              Table "public.red1"
      Column      |       Type        | Collation | Nullable |              Default
------------------+-------------------+-----------+----------+----------------------------------
 pk               | bigint            |           | not null | nextval('red1_pk_seq'::regclass)
 fileid           | character varying |           |          |
 text             | character varying |           |          |
 vector           | vector(768)       |           |          |
 source           | jsonb             |           |          |
 content_metadata | jsonb             |           |          |
Indexes:
    "red1_pkey" PRIMARY KEY, btree (pk)
    "red1_file_idx" btree (fileid)
    "red1_vector_idx" diskann (vector)
```

6. Drop the `diskann (vector)` index:

```
DROP INDEX nameOfDiskAnnIndex
```

Example:

```
app=> DROP INDEX red1_vector_idx;
DROP INDEX
```

7. Verify the table details and confirm that the vector index no longer appears:

```
\d nameOfTable
```

8. Recreate the vector index with the following command:

```
CREATE INDEX nameOfDomain_vector_idx on public.nameOfDomain USING diskann (vector vector_cosine_ops);
```

Example:

```
app=> CREATE INDEX red1_vector_idx on public.red1 USING diskann (vector vector_cosine_ops);
NOTICE:  Starting index build with num_neighbors=-1, search_list_size=100, max_alpha=1.2,
storage_layout=SbqCompression.
WARNING:  Indexed 96 tuples
CREATE INDEX
```

9. Verify the table details and confirm that the new vector index appears in the list:

```
\d nameOfTable
```

Example:

```
app=> \d red1
                                              Table "public.red1"
      Column      |       Type        | Collation | Nullable |              Default
------------------+-------------------+-----------+----------+----------------------------------
 pk               | bigint            |           | not null | nextval('red1_pk_seq'::regclass)
 fileid           | character varying |           |          |
 text             | character varying |           |          |
 vector           | vector(768)       |           |          |
 source           | jsonb             |           |          |
 content_metadata | jsonb             |           |          |
Indexes:
    "red1_pkey" PRIMARY KEY, btree (pk)
    "red1_file_idx" btree (fileid)
    "red1_vector_idx" diskann (vector)
```

Errors in CAS operator after upgrading to CAS v1.0.6

Problem statement

If IBM Data Cataloging is installed, errors can be observed in the CAS operator:

```
`2025-10-02T15:17:35Z    ERROR   Failed to get DCS Instance`
```

Cause

If IBM Data Cataloging is installed by using Fusion GUI but it is not enabled for filtering, CAS operator repeatedly shows an error when it looks up the IBM Data Cataloging instance. This issue does not happen if you install IBM Data Cataloging through CLI.

Resolution

After you upgrade CAS to version 1.0.6, if IBM Data Cataloging is installed but you do not plan to enable filtering for CAS, follow these steps:

1. Scale down the CAS operator:

```
oc scale deployment ibm-isf-cas-operator-controller-manager --replicas=0 -n ibm-cas
```

2. Patch the `CasInstall` custom resource (CR):

```
oc patch casinstall ibm-cas-service-instance -n ibm-cas --type=merge -p '{"spec":{"enabled_features":{"dcs":
{"enable":false,"instance_name":"data-cataloging-service-instance","namespace":"ibm-data-cataloging"}}}}'
```

3. Scale up the CAS operator:

```
oc scale deployment ibm-isf-cas-operator-controller-manager --replicas=1 -n ibm-cas
```

Errors in existing document processor after upgrading to CAS v1.0.6

Problem statement
: After upgrading CAS from v1.0.5 to v1.0.6, errors can be observed in ingestion in the document processor logs. If you are running one or more Docling Multimodal pipelines, the following error can be observed in the document processor (`cast-runtime`) pod logs:

```
`2025-10-02T07:05:26Z - xxx - ERROR - xxx - Repo id must be in the form 'repo_name' or 'namespace/repo_name':
'/models/granite-embedding-278m-multilingual'. Use `repo_type` argument if needed.`
```

Cause
: With CAS 1.0.6 release, the deployment `spec` for the Docling Multimodal document processors (`cast-runtime`) is changed to mount a PVC with the required LLM models. You need to force the operator to deploy a new deployment for each document processor that is running `docling_multimodal` type pipeline.

Resolution
: After you upgrade CAS to version 1.0.6, redeploy all the document processors that are running the `docling_multimodal` type pipeline:

1. Scale down the CAS operator.

   ```
   oc scale deployment ibm-isf-cas-operator-controller-manager --replicas=0 -n ibm-cas
   ```

2. Identify `DocumentProcessor` CRs that are running the `docling_multimodal` type pipeline.

   ```
   oc get DocumentProcessor -n ibm-cas
   ```

   Alternatively, you can use `jq` to filter by type.

   ```
   oc get documentprocessor -ojson| jq '.items[] | select(.spec.type == "docling_multimodal") | .metadata.name'
   ```

3. Delete all the deployments associated with those CRs.

   ```
   oc delete deployment <DocumentProcessor CR name>  -n ibm-cas
   ```

4. Scale up the CAS operator.

   ```
   oc scale deployment ibm-isf-cas-operator-controller-manager --replicas=1 -n ibm-cas
   ```

Data Cataloging `SpectrumDiscover` CR fails to display `SuccessfullyEnabled` status

Problem statement
: When you update `CasInstall` Custom Resource (CR) to enable filtering by Data Cataloging, `SpectrumDiscover` CR fails to display `SuccessfullyEnabled` status for `ContentAwareStorage` feature.

Cause
: Run the following command and check if `isd-metadata-extender` pod is having any issue to start:

```
oc get pod -n ibm-data-cataloging
```

Resolution
: To identify the issue that is preventing the pod to start, run the following command:

```
oc describe pod <isd-metadata-extender pod name>
```

No data ingested after domain creation with Data Cataloging filtering enabled

Problem statement
: Data Cataloging is successfully enabled for filtering. When you create a domain by using a data source through UI (or a domain and document processor through CLI), the files that meet the filtering criteria are not ingested into the vector database as expected.

The `isd-metadata-extender` pod logs under the `ibm-data-cataloging` namespace shows the following error:

```
{"level":"error","ts":"2025-09-27T06:56:58Z","caller":"watch/document_processor.go:88","msg":"Error running scale partial
scan","error":"unexpected status code 404 while running partial scan on connection
fs1","stacktrace":"github.ibm.com/ProjectAbell/dcs-metadata-
extender/pkg/watch.TriggerPartialScans\n\t/tmp/app/pkg/watch/document_processor.go:88"}
```

Cause
: A connection to the Scale filesystem is not created.

Resolution

1. Go to the Data Cataloging UI and create a connection for the Scale filesystem.
2. Delete the domain and document processor CR and redeploy them.

No data ingested for any domain

Problem statement
: No indication of data ingestion in logs of CAS runtime pods for any of the corresponding `DocumentProcessor`.

Cause
: One possible area to check is the health of `cas-core` pod. Check whether it is up and running with the following command:

```
oc get pod -n ibm-cas
```

Resolution

Restart the `cas-core` pod or determine the issue by running the following command:

```
oc describe pod <cas-core pod name>
```

Error in `query_service` configmap and deployment

Problem statement
: When you upgrade CAS from v1.0.3 to v1.0.4, a ConfigMap and some deployment configs must be added to prevent errors in Access Control List (ACL) processing.

Cause
: Query service API fails to work when an external Identity Provider (IDP) token is provided.

Resolution

1. Create the following ConfigMap:

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: query-search-config
  namespace: ibm-cas
data:
  use_self_cert: 'false'
```

2. Update the query search deployment:
   a. Set the `readOnlyRootFilesystem` attribute to 'true'.
   b. In the query service deployment's `env` section, add the following entry:

```
- name: ENABLE_FILE_LEVEL_SECURITY
  value: false
```

   c. In the `volumes` section, add the following section:

```
- name: query-search-config
  configMap:
    name: query-search-config
    defaultMode: 420
- name: ibm-cas-cabundle
  configMap:
    name: openshift-service-ca.crt
    items:
    - key: service-ca.crt
      path: service-ca.crt
```

   d. In the `volumeMounts` section, add the following section:

```
- name: "query-search-config"
  readOnly: true
  mountPath: "/etc/config/cm/"
- name: ibm-cas-cabundle
  readOnly: true
  mountPath: /etc/config/cabundle
```

The nv-ingest pod is in `imagepullback` error

Problem statement
: The `nv-ingest` pod is in `imagepullback` error state due to too many requests for `docker.io`.

Cause and resolution

1. **Cause**:
   Too many requests for `docker.io`.

   Resolution: Authenticate to `docker.io`.

2. **Cause**:
   The `nv-ingest` pod is not able to pull the image correctly.

   Resolution: Check NVIDIA guide to fix the issue.

CAS ingestion fails

Problem statement
: The CAS ingestion fails with the following error in the logs of the `cast-runtime` pods:

```
2025-03-26 22:46:36,725 - ERROR - Error during fetching, retrying...
Error: HTTPSConnectionPool(host='nv-ingest.nv-ingest.svc.cluster.local', port=7670):
Max retries exceeded with url: /v1/fetch_job/ (Caused by SSLError(SSLError(1, '[SSL] record layer failure (_ssl.c:1006)')))
```

Cause
: It implies that the `https` NVIDIA NIM service is not available.

Resolution

1. Go to `cast-runtime` deployment.
2. In the value parameter, change `https` to `http`:

```
- name: NVMM_NIM_SERVICE
value: https://nv-ingest.nv-ingest.svc.cluster.local
```

to

```
- name: NVMM_NIM_SERVICE
value: http://nv-ingest.nv-ingest.svc.cluster.local
```

Error in semantic_search

Problem statement
: In the query search pod, you can get the following errors during a **`semantic_search`**:

```
ERROR: querysearch/semantic_search failed Connection error.
Traceback (most recent call last):
File "/opt/app-root/lib64/python3.11/site-packages/httpx/_transports/default.py", line 101, in map_httpcore_exceptions
yield
File "/opt/app-root/lib64/python3.11/site-packages/httpx/_transports/default.py", line 250, in handle_request
resp = self._pool.handle_request(req)
File "/opt/app-root/lib64/python3.11/site-packages/httpcore/_backends/sync.py", line 154, in start_tls
with map_exceptions(exc_map):
File "/usr/lib64/python3.11/contextlib.py", line 158, in __exit__
self.gen.throw(typ, value, traceback)
File "/opt/app-root/lib64/python3.11/site-packages/httpcore/_exceptions.py", line 14, in map_exceptions
raise to_exc(exc) from exc
```

Cause
: It implies that the TLS version of the NVIDIA service is not available.

Resolution

1. Scale down the query-search deployment.
2. In the query-search deployment, change the value of variable:

   ```
   - name: NVMM_EMBED_SERVICE
   value: 'https://llama-nemotron-embed-1b-v2.nv-ingest.svc.cluster.local'
   ```

   to

   ```
   - name: NVMM_EMBED_SERVICE
   value: 'http://llama-nemotron-embed-1b-v2.nv-ingest.svc.cluster.local'
   ```

3. If the NVMM Nemo Ranking service is enabled in CAS Install CR, change the below value in the query-search deployment. Change the value of variable:

   ```
   - name: NVMM_NEMO_RANKER_SERVICE
   value: 'https://llama-nemotron-rerank-1b-v2.nv-ingest.svc.cluster.local'
   ```

   to

   ```
   - name: NVMM_NEMO_RANKER_SERVICE
   value: 'http://llama-nemotron-rerank-1b-v2.nv-ingest.svc.cluster.local'
   ```

4. Scale the deployment back up to its original number.

Datasource stuck in "Connecting" state error

Problem statement
: The create Datasource may get stuck in a "connecting" state due to different reasons. To do a proper diagnostic, check the CAS operator logs to find the error message.

Cause and resolution
: The possible causes of this error are as follows:

1. **Cause:**
   Scale CSI user does not have enough privileges for Watcher creation.

   ```
   ERROR RETURNED BY sConn.CreateWatch4Fileset =========>
           =====> [EFSSG0012C Permission denied: Your role(s): [csiadmin, containeroperator], required
           role(s): [admin, storageadmin, securityadmin]] Unable to enable clustered watch for source
           castFS:root
   ```

   Resolution:

   Add the Scale CSI user to the Storage Administrator group in Scale as specified [Configuring Scale user to enable watch creation](#).

2. **Cause:**
   Kafka authentication ConfigMap is missing.

   ```
   2025-03-28T18:21:10Z ERROR Reconciler error {"controller": "datasource", "controllerGroup": "cas.isf.ibm.com", "controlle
   ```

   Resolution:

   Create the ConfigMap for Kafka authentication.

3. **Cause:**
   Access is denied to static PV that is created by the Datasource. If a functional issue is detected regarding static PV accessibility, it could be related to an access denied error. To validate, access the mount path by using the pod generated by the DocumentProcessor. The pod has the same name as the DocumentProcessor.

```
(app-root) sh-5.1$ cd /gpfs/gpfs3/fileset_sample
bash: cd: /app-root: Permission denied
```

Resolution:

Add a known GID to the Fileset in Scale and add an annotation to the Datasource in CAS as described in Step 8.

Known limitations

- When you open the data source Details page from the IBM Fusion UI, an error message about failing to fetch data source insights might appear. If IBM Data Cataloging is not installed on the cluster, you can ignore this message. The error is relevant only when IBM Data Cataloging is installed.
- When you deploy the `CasInstall` version 1.1.0 with `docling_multimodal enabled: true`, the `vllm-vision` pod results in `CrashLoopBackoff` status. The pod and its status can be ignored as the failing pod causes no functional impact.
- Deploying an S3 data source with a custom SSL certificate is not supported for Filesystem with IBM Storage Scale version 5.2.x.
- When you upgrade CAS from v1.0.5 to v1.0.6, two PVCs are orphaned and must be manually deleted to reclaim storage. To delete these PVCs, run the following commands:

  ```
  oc delete pvc vllm-embedding-pvc
  oc delete pvc vllm-vision-pvc
  ```

- When you create a domain in the user interface, the `docling_multimodal` pipeline is automatically enabled. To disable it, update the `DocumentProcessor` custom resource (CR) and set the `dataType` of the new domain to `nvidia_multimodal`.
- Support for CAS on Azure Red Hat OpenShift (ARO) has additional licensing requirements from Red Hat due to the need to change the maximum number of threads for the pods. For more information, see the Red Hat knowledge base.
- Each domain must use a unique datasource location, that is, unique Scale path or unique S3 location. CAS currently does not support sharing same datasource location among multiple domains.

Related information

IBM publications, IBM MediaCenter videos, and external videos that contain information that is related to Content-Aware Storage (CAS).

IBM publications

- Content-Aware IBM Storage Scale
- Accelerating AI with IBM Storage Scale and NVIDIA

IBM MediaCenter videos

- IBM Content Aware Storage
- Content Aware Storage (CAS) with IBM Fusion
- AI Data Assistance with IBM Content Aware Storage (CAS)

External videos

- What is Content-Aware Storage? Unlocking AI's Potential with RAG
- Content-Aware Storage: Powering AI Agents & Assistants with RAG

Software Hub

The Software Hub is a comprehensive platform that simplifies the deployment and management of IBM software solutions, including IBM Software Hub, watsonx.ai and watsonx Orchestrate, on IBM Fusion HCI. It provides a streamlined and automated experience, reducing the complexity and manual effort associated with traditional deployment processes.

The Software Hub is designed to simplify the deployment and management of IBM software solutions on IBM Fusion HCI. The platform automates many of the manual tasks associated with traditional deployment processes, reducing the time and effort required for initial setup and ongoing management.
Important:

- To install services such as Software Hub, watsonx.ai, and watsonx Orchestrate, IBM Software Hub requires containers to run with root privileges. For more information about installing the services, see Installing Software Hub services.
- If you have installed any Software Hub or IBM watsonx services previously and not through the IBM Fusion HCI user interface, and the installation of remaining services is stalled, contact IBM support .

Key features and benefits

- Simplifies the deployment of IBM software solutions, including watsonx.ai and watsonx Orchestrate, on IBM Fusion HCI.
- Automates pre-requisites and configuration for Software Hub, eliminating the need for manual documentation review.
- Provides basic resource pre-checks (CPU, memory, storage) to ensure successful installation.
- Auto-selects storage classes, such as `ibm-spectrum-fusion-cp` or recommended default, for optimal performance
- Supports the deployment of watsonx.ai, watsonx Orchestrate, and watsonx.data.
- Utilizes Software Hub 5.4.0, enabling customers to deploy the latest version of Software Hub.
- Initializes the cp-br service in Software Hub, enabling recipe-based backups without additional manual configuration.

- **Prerequistes**
  Ensure that you meet all the prerequisites for a successful installation of the Software Hub service.
- **Installing Software Hub services**
  Initiate the installation of Software Hub service from the user interface of IBM Fusion.
- **Troubleshooting issues and known limitations in Software Hub**
  The common troubleshooting steps and limitations in the Software Hub service.

Prerequistes

Ensure that you meet all the prerequisites for a successful installation of the Software Hub service.

1. One of the following storage services must be available:
   - Fusion Data Foundation
   - Global Data Platform
2. Verify that the required storage classes exist:
   - For Fusion Data Foundation: `ocs-storagecluster-ceph-rbd` and `ocs-storagecluster-cephfs`.
   - For Global Data Platform: `ibm-storage-fusion-cp-sc`.
3. MCG must be set up for installing watsonx Orchestrate. If you are using Fusion Data Foundation, MCG is installed automatically.
4. If you are using Global Data Platform storage class, install MCG separately before installing IBM watsonx OC.
5. If possible, install watsonx.ai and watsonx Orchestrate separately, rather than concurrently. Wait for one service to complete installation successfully before installing the other. For more information, see Preparing to install IBM watsonx.ai and Preparing to install watsonx Orchestrate.
6. Before installing the Software Hub, watsonx.ai, and watsonx Orchestrate services, create the required image pull secret by following the instructions steps:
   a. Log in to the cluster using the `oc login` command.
   b. Create a `dockerconfig.json` file using your IBM Entitlement Key:

   ```
   cat <<EOF > dockerconfig.json
   {
     "auths": {
       "cp.icr.io": {
         "auth": "${IMAGE_PULL_CREDENTIALS}"
       },
       "icr.io":{
         "auth": "${IMAGE_PULL_CREDENTIALS}"
       }
     }
   }
   EOF
   ```

   c. Create the `ibm-entitlement-key` secret in both the operator and instance namespaces.

   ```
   oc create secret docker-registry ibm-entitlement-key \
   --from-file ".dockerconfigjson=dockerconfig.json" \
   --namespace=cpd-operator
   ```

   ```
   oc create secret docker-registry ibm-entitlement-key \
   --from-file ".dockerconfigjson=dockerconfig.json" \
   --namespace=cpd-instance
   ```

7. IBM watsonx services stack on IBM Fusion HCI is not supported with AMD GPU cards. It is currently supported with only NVIDIA GPU cards.
8. For OpenShift® Container Platform version 4.21, the options to install Software Hub, watsonx.ai, and watsonx Orchestrate services are not visible on the IBM Fusion user interface by default. To make these options visible, users must access the respective FSDs and update the `ocpVersionMaxReq` field to `4.21`. This manual update enables the display of these services on the IBM Fusion user interface for OpenShift Container Platform version 4.21.
9. Ensure that you must install Software Hub service to install the Watsonx services.

Installing Software Hub services

Initiate the installation of Software Hub service from the user interface of IBM Fusion.

Before you begin

Ensure that you meet all the prerequisites for a successful installation of the Software Hub service. For more information, see Prerequistes.

About this task

Use this procedure to install the IBM Software Hub service from the Services page of the IBM Fusion user interface. You can also install the IBM Software Hub service using CLI. For more information about CLI usage, see Installing the IBM Software Hub command-line interface.
Important: The installation procedure for the Software Hub, watsonx.ai, and watsonx Orchestrate services are identical. This topic provides the steps to install the Software Hub service.
Note:

- It is recommended to install the Software Hub before you install the watsonx.ai or watsonx Orchestrate, as it provides the foundational components required for both services.
- Installing watsonx.ai and watsonx Orchestrate automatically installs the Software Hub.
- Red Hat OpenShift AI version 2.22 is installed automatically as part of the watsonx.ai and watsonx Orchestrate services. For more information, see Installing Red Hat OpenShift AI.

Procedure

1. Before installing Software Hub, ensure that IBM **`Cert Manager`** is not installed and that the **`ibm-cert-manager`** namespace does not exist. If either is present, uninstall IBM **`Cert Manager`** and delete the namespace.
   Note: As part of a fresh installation, the latest version of Red Hat® certificate Manager is installed with Software Hub.
2. In the IBM Fusion user interface, go to Services page.
3. In the Available section, click the Software Hub tile.
4. In the Software Hub page, go through the features and capabilities of the service and click Install.
   If the pre-check raises a warning, then resolve it at OpenShift level.
   The installation starts and a notification appears on the Services page. You can see the progress of the installation in the Services > **Installed** section. After the installation is completed successfully, you can see the status as Healthy. A success notification gets displayed. It disappears automatically after some time.

   From the ellipsis menu, you can download logs and view documentation.

5. Validate the installation.
   - Verify the IBM Software Hub installation from the IBM Fusion user interface:
     After you enable the IBM Software Hub service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

   - Verify the IBM Software Hub installation from the OpenShift® Container Platform web console:
     - In the OpenShift console, go to Home > Projects.
     - In the Projects page, search and check for the availability of a namespace with string `ibm-software-hub-operator`.
     - Go to Operators > Installed Operators and check for the availability of IBM Software Hub.
     - After you enable the IBM Software Hub service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

What to do next

1. If a failure occurs, check for pods in the **`cpd-instance`** and **`cpd-operator`** namespaces.
2. For post installation instructions, check the following details:
   - For watsonx.ai, see Post-installation setup for IBM watsonx.ai.
   - For watsonx Orchestrate, see Post-installation setup for watsonx Orchestrate.
3. You can access the watsonx dashboard by using the information in **`softwarehubmanager`** CR in the **`ibm-software-hub-operator`** namespace.

```
apiVersion: softwarehub.fsh.ibm.com/v1
kind: SoftwarehubManager
metadata:
  labels:
    app.kubernetes.io/managed-by: kustomize
    app.kubernetes.io/name: fsh-ai-operator
  name: ibm-softwarehub-service-instance
  namespace: ibm-software-hub-operator
spec: {}
status:
  cpdServiceEndpoint: 'https://xxxx.ibm.com'
  cpdServiceEndpointSecret: ibm-iam-bindinfo-platform-auth-idp-credentials
  health: Healthy
  healthStatuses:
    - message:
        message: Operators & Operands are healthy
        messageCode: BMYSH0106
      serviceName: SoftwareHub
      status: Healthy
  installStatus:
    messages:
      - message: Installed SoftwareHub installed and healthy
        messageCode: BMYSH0102
    progressPercentage: 100
    retryOnFailure: true
    status: Completed
  installedVersion: 5.3.1
  upgradeStatus: {}
```

After successfully installing Software Hub, you can access the IBM watsonx dashboard by clicking the URL present in field **`cpdServiceEndpoint`** and use the credentials that are present in secret **`ibm-iam-bindinfo-platform-auth-idp-credentials`**.
Note: The secret is present in the **`cpd-instance`** namespace.

- **Installing watsonx.ai service**
  Initiate the installation of watsonx.ai service from the user interface of IBM Fusion.
- **Installing watsonx Orchestrate service**
  Initiate the installation of watsonx Orchestrate service from the user interface of IBM Fusion.

Installing watsonx.ai service

Initiate the installation of watsonx.ai service from the user interface of IBM Fusion.

Before you begin

Ensure that you meet all the prerequisites for a successful installation of the watsonx.ai service. For more information, see Prerequistes.

About this task

Use this procedure to install the watsonx.ai service from the Services page of the IBM Fusion user interface.

Procedure

1. In the IBM Fusion user interface, go to Services page.
2. In the Available section, click the Watsonx.ai tile.
3. In the Watsonx.ai page, go through the features and capabilities of the service and click Install.
   If the pre-check raises a warning, then resolve it at OpenShift level.
   The installation starts and a notification appears on the Services page. You can see the progress of the installation in the Services > **Installed** section. After the installation is completed successfully, you can see the status as Healthy. A success notification gets displayed. It disappears automatically after some time.

   From the ellipsis menu, you can download logs and view documentation.

4. Validate the installation.
   - Verify the watsonx.ai installation from the IBM Fusion user interface:
     After you enable the watsonx.ai service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

   - Verify the watsonx.ai installation from the OpenShift® Container Platform web console:
     - In the OpenShift console, go to Home > CustomResourceDefinition.
     - In the CustomResourceDefinition page, search and check for the `WatsonxAIManager` CR.
     - After you enable the watsonx.ai service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

What to do next

1. If a failure occurs, check for pods in the `cpd-instance` and `cpd-operator` namespaces.
2. For post installation instructions, check the following details:
   - For watsonx.ai, see Post-installation setup for IBM watsonx.ai.

Installing watsonx Orchestrate service

Initiate the installation of watsonx Orchestrate service from the user interface of IBM Fusion.

Before you begin

- Ensure that you meet all the prerequisites for a successful installation of the watsonx Orchestrate service. For more information, see Prerequistes.
- For watsonx Orchestrate, a new flag, `EnableGPUPrechecks`, is introduced. By default, this flag is set to `false`, indicating that you are utilizing an external Large Language Model (LLM). In this case, prechecks for the GPU operator and Node Feature Discovery (NFD) operator are not performed. To use an internal LLM, users can set the `EnableGPUPrechecks` flag to `true`, which triggers additional prechecks for the NFD and GPU operators.

About this task

Use this procedure to install the watsonx Orchestrate service from the Services page of the IBM Fusion user interface.

Procedure

1. In the IBM Fusion user interface, go to Services page.
2. In the Available section, click the Watsonx.Orchestrate tile.
3. In the Watsonx.Orchestrate page, go through the features and capabilities of the service and click Install.
   If the pre-check raises a warning, then resolve it at OpenShift level.
   The installation starts and a notification appears on the Services page. You can see the progress of the installation in the Services > **Installed** section. After the installation is completed successfully, you can see the status as Healthy. A success notification gets displayed. It disappears automatically after some time.

   From the ellipsis menu, you can download logs and view documentation.

4. Validate the installation.
   - Verify the watsonx Orchestrate installation from the IBM Fusion user interface:
     After you enable the watsonx Orchestrate service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

   - Verify the watsonx Orchestrate installation from the OpenShift® Container Platform web console:
     - In the OpenShift console, go to Home > CustomResourceDefinition.
     - In the CustomResourceDefinition page, search and check for the `WatsonxOrchestrateManager` CR.
     - After you enable the watsonx Orchestrate service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

What to do next

1. If a failure occurs, check for pods in the `cpd-instance` and `cpd-operator` namespaces.
2. For post installation instructions, check the following details:
   - For watsonx Orchestrate, see Post-installation setup for watsonx Orchestrate.

Troubleshooting issues and known limitations in Software Hub

The common troubleshooting steps and limitations in the Software Hub service.

Software Hub, IBM watsonx, and watsonx Orchestrate services installation fails due to Podman runtime errors

Problem statement

During the installation of Software Hub, the deployment can fail due to container runtime errors related to Podman. These errors prevent required installation steps from completing.

In the operator logs, the failure appears as follows:

```
Ansible execution failed: exit status 2

User specified container image: icr.io/cpopen/cpd/olm-utils-v4:5.3.1, but error detected:
cannot clone: Operation not permitted
Error: cannot re-exec process

Attempting pull container image icr.io/cpopen/cpd/olm-utils-v4:5.3.1

Pulling image output:
cannot clone: Operation not permitted
Error: cannot re-exec process

Command exception: exit status 125
RunPluginCommand: Execution error: exit status 1
```

Impact

- The Software Hub installation fails during the Ansible execution phase.
- The `olm-utils-v4` container image cannot be pulled or executed.

Root cause

This issue occurs due to insufficient container runtime permissions. A recent change introduced enforces stricter security constraints that prevent Podman from performing required operations.

As a result, containers fail to execute unless elevated privileges are explicitly allowed.

Resolution

To resolve this issue, update the Software Hub ClusterServiceVersion (CSV) to allow the required elevated privileges for Podman.

- Option 1: Update the security context in the CSV
  Add the following security context settings:

  ```
  securityContext:
    privileged: true
    allowPrivilegeEscalation: true
  ```

  And:

  ```
  securityContext:
    runAsNonRoot: false
    runAsUser: 0
  ```

- Option 2: Patch the CSV by using the CLI
  Run the following command to apply the required security context settings:

  ```
  oc patch csv ibm-fsh-operator.v1.0.3 \
  -n ibm-software-hub-operator \
  --type='json' \
  -p='[
    {
      "op": "add",
      "path": "/spec/install/spec/deployments/0/spec/template/spec/securityContext/runAsUser",
      "value": 0
    },
    {
      "op": "add",
      "path": "/spec/install/spec/deployments/0/spec/template/spec/securityContext/runAsNonRoot",
      "value": false
    },
    {
      "op": "add",
      "path": "/spec/install/spec/deployments/0/spec/template/spec/containers/0/securityContext/privileged",
      "value": true
    },
    {
      "op": "add",
      "path":
  "/spec/install/spec/deployments/0/spec/template/spec/containers/0/securityContext/allowPrivilegeEscalation",
      "value": true
    }
  ]'
  ```

  Existing configuration:

```
securityContext:
   capabilities:
       drop:
         - ALL
   privileged: false
   allowPrivilegeEscalation: false
```

New configuration:

```
securityContext:
   privileged: true
   allowPrivilegeEscalation: true
```

Known limitations

- When installing an Inference Foundation Model, such as `llama3-1-8b`, the installation starts but the `watsonxaiifm` operator reconciliation stops at 60%. Do not upgrade to NVIDIA GPU Operator version 25.10. For more information on troubleshooting this issue, see [Watsonx.ai model installation stuck at 60% reconciliation](). Note: This limitation is not applicable to IBM Software Hub v5.3.x.
- If you upgrade from an older version of OpenShift® Container Platform to version 4.20, IBM Software Hub 5.2.2 is not supported. If you plan to install any Blue Stack services, perform a fresh installation of IBM Fusion 2.12 only on OCP versions up to 4.19.
- In the IBM Fusion user interface, IBM watsonx service tiles showing version 1.0.0, while the installed version is 5.2.2.
- When upgrading Software Hub from version 5.2.2 to 5.3.0, the IBM Fusion user interface might display a `Precheck Failed` status for Watsonx services. This indication is false and does not necessarily signify an actual failure. To confirm the upgrade status, check the status of the `WatsonxAIManager CR` `ibm-watsonxai-service-instance` or `WatsonxOrchestrateManager` `ibm-watsonxorchestrate-service-instance`, specifically the `environmentReadiness` section. If all entries in `environmentReadiness`, such as `SoftwareHubHealth`, `NFDCRHealth`, and `GPUClusterPolicyHealth`, report a healthy or ready state with `INFO` messages, the environment is valid. In this case, you can safely upgrade the watsonx services, disregarding the `Precheck Failed` status in the IBM Fusion user interface.
  For example:

  ```
  environmentReadiness:
    - category: SoftwareHubHealth
      message: Software Hub has been upgraded to latest version.
      messageCode: BMYSH0111
      messageType: INFO
      noRedirection: false
      prechecks:
        - ibm-watsonxorchestrate-service-instance
      skipEnvironmentalCheck: false
    - category: NFDCRHealth
      message: NFD CR is in Ready state.
      messageCode: BMYCO0305
      messageType: INFO
      noRedirection: false
      prechecks:
        - ibm-watsonxorchestrate-service-instance
      skipEnvironmentalCheck: false
    - category: GPUClusterPolicyHealth
      message: GPU ClusterPolicy gpu-cluster-policy is in Ready state.
      messageCode: BMYCO0355
      messageType: INFO
      noRedirection: false
      prechecks:
        - ibm-watsonxorchestrate-service-instance
  ```

Lakehouse-Table [Technology preview]

The Lakehouse-Table service is a Kubernetes-native solution for building and managing lakehouse infrastructure on IBM Fusion. It provides a simplified experience for administrators to provision and manage Data Lakehouse capabilities.

Important:
Lakehouse-Table service is a Technology Preview feature and is subject to Technology Preview support limitations. Technology Preview features are not supported with Red Hat production service level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using them in production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process.

For more information, see [Technology Preview Features Support Scope]().

The Lakehouse-Table service enables administrators to create a foundational lakehouse environment, including object storage integration, Iceberg catalog management, and integration with the IBM watsonx.data. This service is the entry point for building a complete data lakehouse experience on IBM Fusion.

Key features and benefits

- It provides a simplified experience for building and managing lakehouse infrastructure.
- It is tightly integrated with IBM watsonx.data, enabling seamless data management and analytics.
- It automates infrastructure provisioning by using operators, reducing manual configuration and minimizing errors.
- It provides a Kubernetes-native experience, enabling administrators to manage lakehouse infrastructure using familiar tools and workflows.

- **Installing Lakehouse-Table service**
  Initiate the installation of Lakehouse-Table service from the user interface of IBM Fusion.
- **Configuring Lakehouse-Table service**
  Use the instructions to configure the Lakehouse-Table service by creating an `OpenTableCatalog` resource and verifying its creation.
- **Troubleshooting issues and known limitations in Lakehouse-Table**
  The common troubleshooting steps and limitations in the Lakehouse-Table service.

Installing Lakehouse-Table service

Initiate the installation of Lakehouse-Table service from the user interface of IBM Fusion.

Before you begin

To ensure a successful installation of the Software Hub service, verify that the following prerequisites are met:

1. IBM Fusion is installed and configured.
2. The necessary storage classes are available:
    - Block storage (RBD)
    - Filesystem storage (CephFS)
    - Object storage (RGW)
3. Ensure that you apply the IBM Fusion service definition.

```
apiVersion: service.isf.ibm.com/v1
kind: FusionServiceDefinition
metadata:
  name: ibm-lakehouse-service
  namespace: ibm-spectrum-fusion-ns
spec:
  disableFIPSSupport: false
  hasRelatedDefinition: false
  createResourceConfigMap: false
  supportedPlatforms:
    - BareMetalHCI
    - BareMetal
    - BareMetalSDS
    - VSphere
  subService: false
  serviceability:
    logcollector:
      namespaces:
        - ibm-software-hub-operator
  autoDeploy: false
  serviceInformation:
    lastUpdated: 1733952411788
    vendor: IBM
    name: BMYWD0001
    dialogDescription: BMYWD0201
    version: 5.4.0
    icon:
'data:image/svg+xml;base64,PD94bWwgdmVyc2lvbj0iMS4wIiBlbmNvZGluZz0iVVRGLTgiPz4KPHN2ZyBpZD0iQnVpbGRfaWNvbl9oZXJlIiBkYXRhLW5
hbWU9IkJ1aWxkIGljb24gaGVyZSIgeG1sbnM9Imh0dHA6Ly93d3cudzMub3JnLzIwMDAvc3ZnIiB4bWxuczp4bGluaz0iaHR0cDovL3d3dy53My5vcmcvMTk5O
S94bGluayIgdmlld0JveD0iMCAwIDMyIDMyIj4KICA8ZGVmcz4KICAgIDxzdHlsZT4KICAgICAgLmNscy0xIHsKICAgICAgICBtYXNrOiB1cmwoI21hc2spOwo
gICAgICB9CgogICAgICAuY2xzLTIgewogICAgICAgIGZpbGw6ICMwMDFkNmM7CiAgICAgIH0KCiAgICAgIC5jbHMtMyB7CiAgICAgICAgZmlsbDogdXJsKCNsa
W5lYXItZ3JhZGllbnQtMik7CiAgICAgIH0KCiAgICAgIC5jbHMtNCB7CiAgICAgICAgZmlsbDogdXJsKCNsaW5lYXItZ3JhZGllbnQpOwogICAgICB9CgogICA
gICAuY2xzLTUgewogICAgICAgIGZpbGw6ICNmZmY7CiAgICAgIH0KCiAgICAgIC5jbHMtNiB7CiAgICAgICAgZmlsbDogdXJsKCNCbHVlX0lCTV9JY29uX0dyY
WRpZW50KTsKICAgICAgfQogICAgPC9zdHlsZT4KICAgIDxsaW5lYXJHcmFkaWVudCBpZD0ibGluZWFyLWdyYWRpZW50IiB4MT0iLTM0ODEuMTUiIHkxPSItODA
xLjA4IiB4Mj0iLTM0ODEuMTUiIHkyPSItODA2LjA4IiBncmFkaWVudFRyYW5zZm9ybT0idHJhbnNsYXRlKDM0ODcuNjUgODMyLjU4KSIgZ3JhZGllbnRVbml0c
z0idXNlclNwYWNlT25Vc2UiPgogICAgICA8c3RvcCBvZmZzZXQ9IjAiIHN0b3AtY29sb3I9IiMwMDAiIHN0b3Atb3BhY2l0eT0iMCIvPgogICAgICA8c3RvcCB
vZmZzZXQ9Ii42OCIgc3RvcC1jb2xvcj0iIzAwMCIvPgogICAgPC9saW5lYXJHcmFkaWVudD4KICAgIDxsaW5lYXJHcmFkaWVudCBpZD0ibGluZWFyLWdyYWRpZ
W50LTIiIHgxPSItNDE0Ni42NSIgeTE9Ijc2LjQyIiB4Mj0iLTQxNDYuNjUiIHkyPSI3MS40MiIgZ3JhZGllbnRUcmFuc2Zvcm09InRyYW5zbGF0ZSg0MTc1LjY
1IC01NS40MikiIHhsaW5rOmhyZWY9IiNsaW5lYXItZ3JhZGllbnQiLz4KICAgIDxtYXNrIGlkPSJtYXNrIiB4PSIwIiB5PSIwIiB3aWR0aD0iMzIiIGhlaWdod
D0iMzIiIG1hc2tVbml0cz0idXNlclNwYWNlT25Vc2UiPgogICAgICA8Zz4KICAgICAgICA8cGF0aCBjbGFzcz0iY2xzLTUiIGQ9Ik0yNy44NywySDQuMTNjLTE
uMTgsMC0yLjEzLjk1LTIuMTMsMi4xM3YyMy43NGMwLDEuMTguOTUsMi4xMywyLjEzLDIuMTNoMy44N3YtMmgtMy44N2MtLjA3LDAtLjEzLS4wNi0uMTMtLjEzV
jEwLjVoMjR2OS41aDJWNC4xM2MwLTEuMTgtLjk1LTIuMTMtMi4xMy0yLjEzWk0yOCw4LjVINHYtNC4zN2MwLS4wNy4wNi0uMTMuMTMtLjEzaDIzLjc0Yy4wNyw
wLC4xMy4wNi4xMy4xM3Y0LjM3WiIvPgogICAgICAgIDxyZWN0IGNsYXNzPSJjbHMtNCIgeD0iNS41IiB5PSIyNi41IiB3aWR0aD0iMiIgaGVpZ2h0PSI1IiB0c
mFuc2Zvcm09InRyYW5zbGF0ZSgzNS41IDIyLjUpIHJvdGF0ZSg5MCkiLz4KICAgICAgICA8cmVjdCBjbGFzcz0iY2xzLTMiIHg9IjI4IiB5PSIxNiIgd2lkdGg
9IjIiIGhlaWdodD0iNSIgdHJhbnNmb3JtPSJ0cmFuc2xhdGUoNTggMzcpIHJvdGF0ZSgxODApIi8+CiAgICAgIDwvZz4KICAgIDwvbWFzaz4KICAgIDxsaW5lY
XJHcmFkaWVudCBpZD0iQmx1ZV9JQk1fSWNvbl9HcmFkaWVudCIgZGF0YS1uYW1lPSJCbHVlIElCTSBJY29uIEdyYWRpZW50IiB4MT0iMCIgeTE9IjE2IiB4Mj0
iMzIiIHkyPSIxNiIgZ3JhZGllbnRVbml0cz0idXNlclNwYWNlT25Vc2UiPgogICAgICA8c3RvcCBvZmZzZXQ9Ii4xIiBzdG9wLWNvbG9yPSIjMDBiOGIyIi8+C
iAgICAgIDxzdG9wIG9mZnNldD0iLjkiIHN0b3AtY29sb3I9IiMwMDVhZmYiLz4KICAgIDwvbGluZWFyR3JhZGllbnQ+CiAgPC9kZWZzPgogIDxnIGNsYXNzPSJ
jbHMtMSI+CiAgICA8cGF0aCBjbGFzcz0iY2xzLTYiIGQ9Ik0wLDMyVjBoMzJWMzJIMFoiLz4KICA8L2c+CiAgPHBhdGggY2xhc3M9ImNscy0yIiBkPSJNMjgsM
jJjLTEuMzktLjA4LTIuNi0xLTMuMDUtMi4zMi0uMTQtLjQtLjUyLS42OC0uOTUtLjY4cy0uODEuMjctLjk1LjY4Yy0uNDUsMS4zMi0xLjY2LDIuMjQtMy4wNSw
yLjMyLTEuMzktLjA4LTIuNi0xLTMuMDUtMi4zMi0uMTQtLjQtLjUyLS42OC0uOTUtLjY4LS40My0uMDEtLjgyLjI3LS45NS42OC0uNDUsMS4zMi0xLjY2LDIuM
jQtMy4wNSwyLjMyaC0xYy0uNTUsMC0xLC40NS0xLDFoMGMwLC41NS40NSwxLDEsMWgxYzEuNTcuMDEsMy4wNi0uNzIsNC0xLjk5LDEuNjcsMi4yMSw0LjgyLDI
uNjUsNy4wMy45Ny4zNy0uMjguNjktLjYuOTctLjk3Ljk0LDEuMjcsMi40MywyLDQsMS45OWgxYy41NSwwLDEtLjQ1LDEtMWgwYzAtLjU1LS40NS0xLTEtMWgtM
VoiLz4KICA8cGF0aCBjbGFzcz0iY2xzLTIiIGQ9Ik0yOCwyOGMtMS4zOS0uMDgtMi42LTEtMy4wNS0yLjMyLS4xNC0uNC0uNTItLjY4LS45NS0uNjhzLS44MS4
yNy0uOTUuNjhjLS40NSwxLjMyLTEuNjYsMi4yNC0zLjA1LDIuMzItMS4zOS0uMDgtMi42LTEtMy4wNS0yLjMyLS4xNC0uNC0uNTItLjY4LS45NS0uNjgtLjQzL
S4wMS0uODIuMjctLjk1LjY4LS40NSwxLjMyLTEuNjYsMi4yNC0zLjA1LDIuMzJoLTFjLS41NSwwLTEsLjQ1LTEsMWgwYzAsLjU1LjQ1LDEsMSwxaDFjMS41Ny4
wMSwzLjA2LS43Miw0LTEuOTksMS42NywyLjIxLDQuODIsMi42NSw3LjAzLjk3LjM3LS4yOC42OS0uNi45Ny0uOTcuOTQsMS4yNywyLjQzLDIsNCwxLjk5aDFjL
jU1LDAsMS0uNDUsMS0xaDBjMC0uNTUtLjQ1LTEtMS0xaC0xWiIvPgo8L3N2Zz4='
    description: BMYWD0002
    features:
      - heading: BMYWD0202
        text: BMYWD0203
      - heading: BMYWD0204
        text: BMYWD0205
      - heading: BMYWD0206
        text: BMYWD0207
      - heading: BMYWD0208
        text: BMYWD0209
    category: BMYAI0202
  relatedDefinitionSpec: {}
  skipServiceDiscovery: false
  onboarding:
    isSingleton: true
    resourceKind: LakehouseManager
    syncServiceCRSpec: false
```

```
    groupName: lakehouse-table.fsh.ibm.com
    parameters:
      - dataType: string
        defaultValue: ibm-software-hub-operator
        descriptionCode: BMYSRV00003
        displayNameCode: BMYSRV00004
        name: namespace
        required: true
        userInterface: false
    multiVersionService: false
    serviceOperatorSubscription:
      triggerCatSrcCreate: false
      isClusterWide: false
      channel: v1.0
      skipDeployOperator: false
      readImageFromFSI: false
      catalogSourceDetails:
        displayName: Fusion Sofewarehub Catalog
        imageName: ibm-fsh-operator-catalog
        imageTag: v1.0
        publisher: IBM
        registryPath: icr.io/cpopen
        skipCatSrcCreation: false
      operatorName: ibm-fsh-operator
      globalCatalogSource: false
      createCatalogSource: true
      catalogSourceName: ibm-fsh-operator-catalog
    version: v1
    resourcePluralName: lakehousemanagers
  preferredStorageClassList:
    - ocs-storagecluster-ceph-rbd
    - ocs-storagecluster-cephfs
  serviceInstallPrerequisite:
    config: {}
    kubeobject:
      - StorageClass
    ocpVersionMaxReq: '4.21'
    ocpVersionMinReq: '4.14'
```

4. The following table outlines the compatibility matrix for the components involved, including the minimum supported versions.

Table 1. Compatibility Matrix for Components

Component	Minimum supported version
IBM Cloud Paks for Data	5.4
watsonx.data	2.4
Data Lakehouse-Table	1.0.0
Fusion Data Foundation	4.21

Note: The versions listed are the minimum supported versions. Using versions later than those listed is supported unless otherwise noted.

About this task

Use this procedure to install the Lakehouse-Table service from the Services page of the IBM Fusion user interface.

Procedure

1. Before installing Lakehouse-Table, ensure that IBM `Cert Manager` is not installed and that the `ibm-cert-manager` namespace does not exist. If either is present, uninstall IBM `Cert Manager` and delete the namespace.
   Note: As part of a fresh installation, the latest version of Red Hat® certificate Manager is installed with Software Hub.
2. In the IBM Fusion user interface, go to Services page.
3. In the Available section, click the Lakehouse-Table tile.
   It opens a Lakehouse-Table service page.
4. In the Lakehouse-Table page, go through the features and capabilities of the service and click Install.
   It opens a Lakehouse-Table install service page.
   The installation starts and a notification appears on the Services page. You can see the progress of the installation in the Services > Installed section. After the installation is completed successfully, you can see the status as Healthy. A success notification gets displayed. It disappears automatically after some time.

   From the ellipsis menu, you can download logs and view documentation.

5. Validate the installation.
   - Verify the Lakehouse-Table installation from the IBM Fusion user interface:
     After you enable the Lakehouse-Table service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

   - Verify the Lakehouse-Table installation from the OpenShift® Container Platform web console:
     - In the OpenShift console, go to Home > CustomResourceDefinition.
     - In the CustomResourceDefinition page, search and check for the `LakehouseManager` CR.
       Lakehouse-Table service also installs Software Hub in a separate namespace with the `SoftwareHubManager` custom resource named `lakehouse-table-softwarehub`.

     - After you enable the Lakehouse-Table service, you can view the service version and health status. From the ellipsis menu, you can download logs and view documentation. After you successfully collect the logs, a success notification is displayed. The notification disappears automatically after some time.

Configuring Lakehouse-Table service

Use the instructions to configure the Lakehouse-Table service by creating an `OpenTableCatalog` resource and verifying its creation.

Before you begin

Ensure that you met all the prerequisites before you configure the Lakehouse-Table.

1. Ensure that the Security Token Service (STS) must be enabled on your Ceph RGW cluster for IAM role-based access.
   a. Run the following command to enable the Ceph Toolbox pod:

   ```
   oc patch storagecluster ocs-storagecluster -n openshift-storage \
     --type json \
     --patch '[{ "op": "replace", "path": "/spec/enableCephTools", "value": true }]'
   ```

   b. Wait for the Toolbox application to initialize.

   ```
   oc get pods -n openshift-storage | grep rook-ceph-tools
   ```

   Example output:

   ```
   rook-ceph-tools-xxxxx   1/1     Running   0          30s
   ```

   c. Run the following command to access the Toolbox pod:

   ```
   TOOLBOX_POD=$(oc get pods -n openshift-storage -l app=rook-ceph-tools -o jsonpath='{.items[0].metadata.name}')
   oc -n openshift-storage rsh $TOOLBOX_POD
   ```

   d. Run the following command to find the RGW client name:

   ```
   ceph auth list | grep "client.rgw"
   ```

   Inside the toolbox pod, it lists all Ceph auth clients to find your RGW client.
   Example output:

   ```
   client.rgw.ocs-storagecluster-cephobjectstore.a
   ```

   The client name is formatted as follows: `client.rgw.<Ceph object store name>.<instance ID>`.
   e. To enable STS for the RGW client, update the `CephObjectStore` configuration.
      - Run the following commands to replace `<cephobjectstore>` with the name of your `CephObjectStore` (for example, `ocs-storagecluster-cephobjectstore`) and `<instance-id>` with the instance identifier (for example, `a`).

        ```
        ceph config set client.rgw.<cephobjectstore>.<instance-id> rgw_s3_auth_use_sts true
        ceph config set client.rgw.<cephobjectstore>.<instance-id> rgw_sts_key "abcdefghijklmnop"
        ```

        For example:

        ```
        ceph config set client.rgw.ocs-storagecluster-cephobjectstore.a rgw_s3_auth_use_sts true
        ceph config set client.rgw.ocs-storagecluster-cephobjectstore.a rgw_sts_key "abcdefghijklmnop"
        ```

        Important: The `rgw_sts_key` must be a secure, random, alphanumeric string that is 16 characters in length. The example that is provided uses a placeholder value. Ensure that you generate a secure key for your configuration.

   f. Run the following command to verify STS Configuration:

   ```
   ceph config get client.rgw.<cephobjectstore>.<instance-id> rgw_s3_auth_use_sts
   ceph config get client.rgw.<cephobjectstore>.<instance-id> rgw_sts_key
   ```

   g. To apply the changes, exit the Toolbox pod and restart the RGW pods by running the following commands:

   ```
   exit  # Exit toolbox pod
   oc delete pods -n openshift-storage -l app=rook-ceph-rgw
   ```

   h. Wait for the RGW pods to restart:

   ```
   oc get pods -n openshift-storage -l app=rook-ceph-rgw
   ```

2. Configure the global configmap as follows:
   Important:
   - Before creating catalogs, configure the global configmap for the operator to provide default values. This way, you can create minimal catalog custom resources without duplicating the configuration.
   - The global configmap is automatically populated by IBM Fusion. If necessary, you can update the details as required.

   a. Run the following command to edit the global configmap:

   ```
   oc edit configmap ibm-isf-lakehouse-tables-operator-global-config -n lakehouse-tables-operator-system
   ```

   b. Configure the required settings.
   For example:

   ```
   apiVersion: v1
   kind: ConfigMap
   metadata:
     name: ibm-isf-lakehouse-tables-operator-global-config
     namespace: lakehouse-tables-operator-system
   data:
     # RGW Configuration
     RGW_ADMIN_USER_SECRET_NAME: "rook-ceph-object-user-ocs-storagecluster-cephobjectstore-lakehouse-admin"
     RGW_NAMESPACE: "openshift-storage"
     RGW_TLS_DISABLED: "false"
     RGW_TLS_CERT_CONFIGMAP_REF: "openshift-service-ca.crt"
     RGW_TLS_KEY: "service-ca.crt"

     # MDS Configuration
   ```

```
  MDS_URI: "https://ibm-lh-lakehouse-mds-rest-svc.lakehouse-table-cpd-
instance.svc.cluster.local:8180/mds/metadata/v1"
  MDS_USERNAME: "cpadmin"
  MDS_PASSWORD_SECRET_NAME: "ibm-iam-bindinfo-platform-auth-idp-credentials"
  MDS_NAMESPACE: "lakehouse-table-cpd-instance"
  MDS_AUTH_INSTANCE_ID: "1767781376189241"  # Get from ibm-lh-lakehouse-config-cm
  AMS_URI: "https://internal-nginx-svc.lakehouse-table-cpd-instance.svc.cluster.local:12443"

  # MDS Token Configuration
  MDS_TOKEN_TYPE: "Bearer"
  MDS_TOKEN_ENDPOINT: "https://internal-nginx-svc.lakehouse-table-cpd-instance.svc.cluster.local:12443/icp4d-
api/v1/authorize"

  # MDS TLS Configuration
  MDS_TLS_DISABLED: "false"
  MDS_TLS_CERT_SECRET_REF: "ibm-lh-tls-secret"
  MDS_TLS_KEY: "cabundle.crt"
```

Procedure

To create an `OpenTableCatalog` resource, follow these steps:

a. You can create an `OpenTableCatalog` resource by using either a minimal catalog configuration or an explicit catalog configuration.

Minimal catalog configuration

i. Create a file named `my-catalog-minimal.yaml` with the following content:

```
apiVersion: lakehouse-tables.isf.ibm.com/v1alpha1
kind: OpenTableCatalog
metadata:
  name: my-iceberg-catalog
  namespace: lakehouse-catalog-admin
spec:
  catalog:
    tableFormat: "iceberg"
```

ii. Run the following command to apply the resource:

```
oc apply -f my-catalog-minimal.yaml
```

Explicit catalog configuration (optional)

If you prefer to specify all values explicitly or override the global configuration, create a file that is named `my-catalog.yaml` with the following content:

```
apiVersion: lakehouse-tables.isf.ibm.com/v1alpha1
kind: OpenTableCatalog
metadata:
  name: my-iceberg-catalog
  namespace: lakehouse-table-cpd-instance
spec:
  objectStore:
    adminUser:
      secretName: rook-ceph-object-user-ocs-rgw-rgw-user
      secretNamespace: openshift-storage
    tls:
      disabled: true
  bucketName: my-data-bucket
  iamResources:
    account:
      accountId: "RGW12345678901234567"
      accountName: "my-data-account"
      rootUserName: "my-account-root"
    userName: "my-data-user"
    roleName: "my-data-role"
    policyName: "my-data-policy"
  catalog:
    catalogName: "my_catalog"
    server:
      lakehouse:
        mdsUri: "https://ibm-lh-lakehouse-mds-rest-svc.lakehouse-table-cpd-
instance.svc.cluster.local:8180/mds/metadata/v1"
        userName: "cpadmin"
        passwordSecretRef:
          secretName: "ibm-iam-bindinfo-platform-auth-idp-credentials"
          secretNamespace: "lakehouse-table-cpd-instance"
        token:
          type: "Bearer"
          endpoint: "https://cpd-lakehouse-table-cpd-instance.<your-domain>/icp4d-api/v1/authorize"
      tls:
        disabled: true
    tableFormat: "iceberg"
```

b. Run the following command to apply the resource:

```
oc apply -f my-catalog.yaml
```

c. Run the following commands to verify whether the `OpenTableCatalog` resource is created successfully:

````
```bash
oc -n lakehouse-tables-operator-system get otc my-iceberg-catalog -o jsonpath='{.status}' | jq -r .
{
"accountId": "RGW59281713583741298",
"info": {
"amsUri": "<https://internal-nginx-svc.lakehouse-table-cpd-instance.svc.cluster.local:12443",>
"authSecretName": "my-iceberg-catalog-bearer-fetch-credentials", "authSecretNamespace": "lakehouse-table-cpd-instance",
````
````

~~~
"catalogName": "my_iceberg_catalog", "catalogStorageBackend": "my-data-bucket", "mdsUri": "<https://ibm-lh-lakehouse-mds-rest-svc.lakehouse-table-cpd-instance.svc.cluster.local:8180/mds/metadata/v1",> "objectStoreEnpoint": "<https://rook-ceph-rgw-ocs-storagecluster-cephobjectstore.openshift-storage.svc:443",> "userName": "my-iceberg-catalog-zen-admin"
},
"lastSyncTime": "2026-06-05T17:40:59Z",
"phase": "Ready"
}
```
~~~

The operator creates the following resources:

- S3 bucket: **my-data-bucket**
- IAM account with root user
- IAM user, role, and policy
- Iceberg catalog registered in watsonx.data
- Secret access with credentials: **my-iceberg-catalog-fetch-bearer-token**.

d. To use the **OpenTableCatalogUser**, you must first obtain an OAuth token. Follow the steps to retrieve the token:

i. Run the following command to obtain the authentication secret name:

```
AUTH_SECRET="$(oc get opentablecatalog "$CAT_NAME" -n "$CAT_NS" -o jsonpath='{.status.info.authSecretName}')"
```

ii. Run the following command to display all authentication details:

```
oc get secret "$AUTH_SECRET" -n "$CAT_NS" -o json | jq -r '.data | to_entries[] | "KEY=\(.key)\nVALUE=\
(.value|@base64d)\n---"'
```

From the output, identify the **username** and **password** values.

iii. Use the **curl** command to generate the OAuth token. Replace **<username>** and **<PASSWORD>** with the values obtained in the previous step:

```
curl -k -X POST '<https://cpd-lakehouse-table-cpd-instance.apps.datalake.qe.ceph.tuc.ibm.com/icp4d-api/v1/authorize'>
\
-H 'Content-Type: application/json' \
-d '{"password":"<PASSWORD>","username":"<username>"}'
```

iv. Run the following command to get the **OpenTableCatalogUser** name:

```
oc describe otc <catalog_name> -n <otc_namespace>
```

e. Create additional **OpenTableCatalogUser**, add the CRD to the configuration.

Minimal CRD (catalog use only)

```
apiVersion: lakehouse-tables.isf.ibm.com/v1alpha1
kind: OpenTableCatalogUser
metadata:
  name: analyst-user
  namespace: cpd-instance
spec:
  catalogUser:
    userName: analyst
  resource:
    catalogRef: opentablecatalog-sample
    permission: use
```

Minimal CRD with L3 data policies

```
apiVersion: lakehouse-tables.isf.ibm.com/v1alpha1
kind: OpenTableCatalogUser
metadata:
  name: invoice-user
  namespace: cpd-instance
spec:
  catalogUser:
    userName: invoice-user
  resource:
    catalogRef: opentablecatalog-sample
    permission: use
    dataPolicies:
      - scope: table
        target: sales.invoices
        permissions: ["read"]
```

f. Optional: Run the following command to delete the resource:

```
oc delete otc my-iceberg-catalog -n lakehouse-tables-operator-system
```

The data is retained even if you delete the resource.

Troubleshooting issues and known limitations in Lakehouse-Table

The common troubleshooting steps and limitations in the Lakehouse-Table service.

Catalog creation fails

Problem statement

The **OpenTableCatalog** resource shows an error in its status, indicating a failure in catalog creation.

Cause

The cause of the failure can be attributed to several factors, including:

- Incorrect RGW admin credentials.
- Network connectivity issues to RGW and watsonx.data.
- Invalid TLS certificates if TLS is enabled.
- Detailed error messages available in the operator logs.

Resolution
To resolve the issue, follow these steps:

1. Verify that the RGW admin credentials are correct.
2. Check the network connectivity to RGW and watsonx.data.
3. Ensure that the TLS certificates are valid if TLS is enabled.
4. Check the operator logs for detailed error messages to diagnose the issue.

Cannot access catalog from Spark or Trino

Problem statement
Authentication errors occur when querying the catalog from Spark or Trino.

Cause
The cause of the authentication errors can be due to:

- Missing or incorrect bearer token secret.
- Expired bearer token.
- Vended credentials not enabled in the catalog configuration.
- Inaccessibility of the S3 endpoint from the query engine pods.

Resolution
To resolve the issue, follow these steps:

1. Verify that the bearer token secret exists using the command `oc get secret <catalog-name>-fetch-bearer-token`.
2. Check the token expiration and refresh it if necessary.
3. Ensure that vended credentials are enabled in the catalog configuration.
4. Verify that the S3 endpoint is accessible from the query engine pods.

Out-of-Memory (OOM) issues

Problem statement
Exceeding the specified limits in the catalog table configuration can result in OOM issues on IO engines, MDS, and other components.

Symptoms

- OOM errors occur on IO engines, MDS, or other components.
- System performance degrades or becomes unresponsive.

Cause:
The following limits, when exceeded, can cause OOM issues:

- Total artifacts exceeding 900 in a single catalog.
- Total number of schemas exceeding 250.
- Total number of tables exceeding 1250.
- Number of rows exceeding 80,000 (with 5 columns) or 250,000 (with 5 columns) in a table.
- Number of `OpenTableCatalogUser` exceeding 250.
- Number of IO engines (Trino) exceeding 5 with 4G RAM.

Resolution
Follow the steps to resolve the issue:

1. Ensure that the total number of artifacts, schemas, tables, and rows are within the specified limits.
2. Distribute data across multiple catalogs, schemas, or tables to avoid exceeding the specified limits.
3. Limit the number of IO engines (Trino) to 5 or fewer with 4G RAM, or increase the RAM allocation as needed.
4. Regularly monitor system performance and adjust configuration as needed to prevent OOM issues.

Known limitations

- When installing Lakehouse-Table service, the `ibm-cpd-ae-operator` pod may encounter an `ImagePullBackOff` error. As a resolution, run the following command to patch the `ibm-cpd-ae-operator` deployment:

```
oc patch deployment ibm-cpd-ae-operator -n cpd-operator \
  --type='json' \
  -p='[
    {
      "op": "replace",
      "path": "/spec/template/spec/containers/0/image",
      "value": "cp.stg.icr.io/cp/ibm-cpd-analyticsengine-
operator@sha256:7f9ceed629755a77a4443059fee31b677a3d9f8dcad82336892ee77fd252b83e"
    }
  ]'
```

Disaster recovery

Disaster recovery offerings of IBM Fusion HCI.

Disaster recovery (DR) is the ability to recover and continue business-critical applications from natural or human created disasters. It is a component of the overall business continuance strategy of any major organization as designed to preserve the continuity of business operations during major adverse events.

IBM Fusion HCI with Fusion Data Foundation storage offers a resilient disaster recovery solution that protects against regional failures, maintaining predictable data loss and business continuity. For more information, see Disaster recovery configuration for Fusion Data Foundation.

IBM Fusion HCI with Global Data Platform storage offers a robust disaster recovery solution that helps minimize data loss and downtime during critical events. It allows two systems to form a stretched storage layer, enabling seamless failover and failback between clusters. For more information, see Disaster recovery configuration for Global Data Platform.

- **Disaster recovery configuration for Fusion Data Foundation**
  Disaster recovery offerings of IBM Fusion HCI with Fusion Data Foundation storage.

Disaster recovery configuration for Global Data Platform

Disaster recovery offerings of IBM Fusion HCI with Global Data Platform storage.

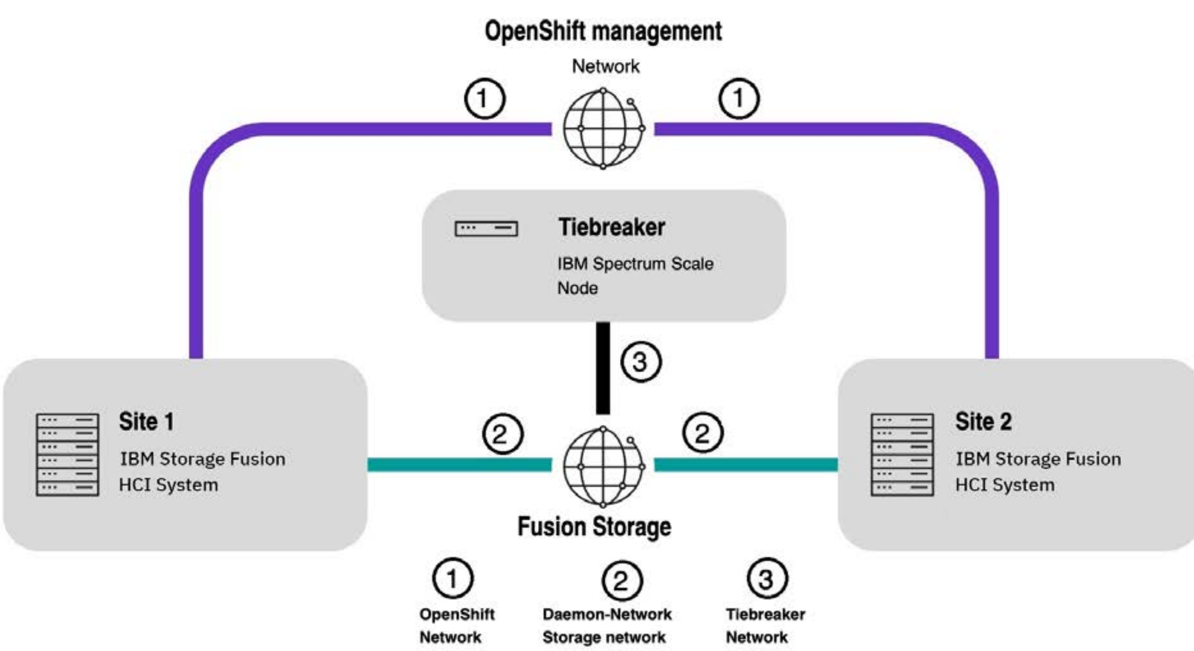
IBM Fusion HCI with Global Data Platform storage supports the metropolitan disaster recovery (Metro-DR). It allows two IBM Fusion HCIs to form a stretched storage layer, enabling seamless failover and failback between clusters. For more information, see Metro-DR configuration for Global Data Platform.
Note: In a disaster recovery solution, two clusters must be connected to failover and failback applications. If the cluster recovers after the expiry of the client certificate, clean and set up the connection to rejoin the recovered cluster. For the procedure to rejoin, see Reconnecting OpenShift Container Platform cluster.

Metro-DR configuration for Global Data Platform

Metropolitan disaster recovery (Metro-DR provides two-way synchronous data replication between IBM Fusion HCI clusters installed at two sites. When a site disaster occurs, applications can be failed over to the second site. Because replication is synchronous, the Metro-DR solution is only available for metropolitan distance data centers with 40-millisecond latency or less.

You can deploy IBM Fusion HCI in data center site 1 and data center site 2, resulting in Red Hat® OpenShift® Container Platform clusters that run at each site. Two clusters are configured such that they share a synchronously replicated storage layer, allowing for data mirroring with zero Recovery Point Objective (RPO) between the two clusters. Both the Red Hat OpenShift clusters can host active workloads at the same time rather than keeping one cluster to be a cold standby. The IBM Fusion HCI storage network is stretched between both Red Hat OpenShift clusters so that data written to one cluster is automatically mirrored to the second cluster. A special tie breaker node is hosted at a third site and is used to determine which cluster is in charge of the data when the network between the two clusters is severed. Configuring a Metro-DR topology requires several network connections to be made between the two clusters and the tie breaker.

Note: In Metro-DR, the terminology scale admin network refers to the OpenShift Container Platform pod network, and scale daemon network refers to the IBM Fusion HCI storage network.
The following picture depicts the connectivity between Site 1 and Site 2.



The Metro-DR setup includes two sets of networks, namely internal storage network and Red Hat OpenShift network. In Metro-DR, the internal storage network of one site must connect to the other site and tiebreaker, that is, externalize your storage network. Similarly, the Red Hat OpenShift network of one site needs to connect to the other site.

OpenShift management network
: The Red Hat OpenShift management network provides communication between the two OpenShift clusters, and is used by IBM Fusion HCI to coordinate replication activities, such as determining which applications are mirrored or triggering application failover. Every node in both clusters has an IP address in this network. The Red Hat OpenShift network across the sites must not be connected over network address translation (NAT). You can decide to have the Red Hat OpenShift network across the sites in the same or different subnet.

IBM Fusion storage network
: The IBM Fusion HCI storage network is used for storage communication between nodes. In a Metro-DR configuration, the storage network must be connected between the two clusters to allow data replication. The storage network must also be connected to a tie breaker and the tie-breaker needs connectivity only on the storage network. This allows the tie breaker to choose which cluster is in charge of the data when the two clusters cannot talk to each other. The IBM Fusion HCI Storage network uses the 100G interface of each node and requires a high-bandwidth connection across the sites. The IBM Fusion HCI Storage network between the sites must not be connected through Network Address Translation (NAT). The IBM Fusion HCI storage network for each site must be in a different subnet than the other site and the tiebreaker.

Upgrade and upsize considerations in Metro-DR

As a prerequisite, know these upgrade and upside considerations before you use Metro-DR.

- When any of the following activities are in progress on a site, then do not attempt any other activity from the list on either this site or the other site:
  - Upgrade
  - Scale out
  - Scale up
    For example, when scale up is in progress, do not trigger upgrade and vice versa.
  - Node maintenance / replacement or repair
  - Disk maintenance /replacement or repair
  - Unhealthy state of open shift or storage cluster
- When one of the site is at a lower version level than the other, do not any attempt any operations on either of the sites in the Metro-DR cluster. For example, site 1 is at IBM Fusion HCI 2.8.2 (lower level) and site 2 is at IBM Fusion HCI 2.9.0.
- In case of the following situations, do an upgrade or scale up to obtain parity between the two sites:
  Note: Ensure you bring them both to the same level at the earliest as the tolerant limit is only couple of days.
  - Ensure that the IBM Fusion version is at the same level for both site 1 and site 2.
  - Ensure that the OpenShift® Container Platform version is at the same level for both site 1 an site 2.
  - Ensure that the submariner version is at the same level for both site 1 and site 2.
  - Add nodes when mismatch in the number of nodes between sites exist.
  - Add disks when mismatch in the number of disks between sites exist.

If these mismatches exist, the tolerant limit is 2 days and you need to bring them both at the same level at the earliest.

Configuring Metro-DR set up

Configure a Metro-DR relationship between two racks by connecting Site 1 and Site 2 through the Topology page and completing the setup wizard.

Before you begin

Before you set up Metro-DR, ensure that all the mentioned prerequisites are met:

- For general Metro-DR prerequisites, see General Metro-DR prerequisites.
- For tiebreaker prerequisites, see Preparing the tiebreaker.
- In disaster recovery, two clusters must be connected to failover and failback applications. If the cluster recovers after the expiry of the client cert, the connection must be cleaned and setup to rejoin the recovered cluster. For the procedure to rejoin, see Reconnecting OpenShift Container Platform cluster.
  During the installation planning, you can enable disaster recovery for your rack in either of these methods:
  - You have already set up your first rack as Site 1. During the installation of Site 2, connect it to Site 1 in a Metro-DR relationship.
  - You already got a standalone rack set up but not designated it as Site 1. During the installation of Site 2, utilize the connection snippet of Site 1 to establish a Metro-DR relationship between the two racks.

About this task

For more information about Metro-DR, see Metro-DR configuration for Global Data Platform.

Procedure

1. Go to the Topology page of the primary site to see a pictorial representation of the connection between primary site and secondary site.
   Example failure statuses for Metro-DR are connection state failed, admin network connectivity failure, Daemon network connectivity failure, stretch cluster addition failure, Minio setup failure, RamenDr failure.
2. Click Connect in the pictorial representation or click Connect tiebreaker.
3. Enter the following tiebreaker details in the Connect tiebreaker window:

   IP address
   : Enter the IP address of the tiebreaker.

   User ID
   : The user id to log in to the tiebreaker

Password
: The password for the tiebreaker user.

4. Click Save
   The `Tiebreaker connecting` notification message is displayed. After the tiebreaker connects successfully, the Connect tiebreaker changes to Connect cluster. You cannot add more than two clusters.
   Note: The sites (clusters) and tiebreaker statuses are as follows:

   Healthy
   : The status of the cluster or tiebreaker is healthy.

   Degraded
   : Whenever a mismatch exists between the two sites.

   Failed
   : When the connectivity is lost between the sites or with tiebreaker.

   Running
   : All clusters and tiebreaker are running.

   If you want to edit details of the tiebreaker, do the following steps:
   a. Go to the Topology page.
   b. In the pictorial representation, click the tiebreaker.
   c. Click Edit.
   d. In the Edit tiebreaker credentials window, update the details of the tiebreaker and click Save.

Enabling and disabling applications for Metro-DR

You must first register your applications for disaster recovery.

Before you begin

The prerequisite for this step is that the Metro-DR setup must be complete.

Procedure

1. Go to the Applications page.
   Alternatively, you can enroll applications from the Replicated applications page:
2. Click the ellipsis overflow menu of your application record and click Manage disaster recovery.
3. In the Manage disaster recovery window, select Metro tile and click Save.
   Disaster recovery is enabled for the selected application. After you register your applications, you can see a table with a list of applications that are protected by this disaster recovery in the Disaster recovery page.

4. To validate, go to the Disaster recovery > Replicated applications page.
5. Check whether the enrolled application is available in the Replicated applications page.
   If you no longer need disaster recovery for an application, then do the following steps to disable from the Replicated applications page or the Applications page:
   a. Scale down the applications before you remove the enrollment.
      Otherwise, it leads to inconsistent Ramen CR and Replication error.
   b. Search for the application and click the ellipsis overflow menu of of the record and click Manage disaster recovery.
   c. In the Manage disaster recovery window, select No disaster recovery tile.
   d. Click Save.
   e. Scale up the applications.

Failover applications from site 1 to site 2

Application failover is the capability of automatically switching over to the secondary site when the primary site becomes unavailable.

Before you begin

Consider the following points before you failover applications:

- If backup policies are applied to the applications on site 1, they continue to run until those polices are removed from the failover application.
- The applications must be enabled for disaster recovery.
- Scale down of an application before initiating planned failover.
- Before unplanned failover of applications to surviving site, do Metro-DR Data Fencing. For the procedure, see Metro-DR data fencing.

About this task

The planned failover must be done from the site where you want the applications to be failed over.

Note: If you plan to failover virtual machines, also see Disaster recovery for virtual machines.

Procedure

1. In the Replicated applications page, click Actions > Failover.
   The Failover remote applications window gets displayed. It also indicates the number of applications available for failover. This Failover does a failover for all remote applications.
2. In the Failover remote applications window, select the applications and click Failover.
   It creates the Persistent Volumes (PVs) that belong to the applications in the local site. In addition, it creates namespace and application CR on the cluster from where you initiated the failover. After relocation is complete to remote cluster, a success message is displayed on the screen.
3. In the Replicated applications page, check the status of the Primary cluster.
   Initially, the status of the Primary cluster for the application is Failing over. After successful completion of the failover, the primary cluster of the given application must be changed to secondary site or partner cluster. After completion, it displays

   icon next to the name of the Primary cluster, and a Failover complete notification is displayed. The IBM Fusion HCI prepares the cluster for deployment of the relocated applications.
4. Scale up the applications.

Failback applications from the mirrored site to the original site

In a Metro-DR setup, application failback involves the process of restoring applications and virtual machines back to the original site.

About this task

You failback after a failover operation completes successfully and the primary site is restored from disaster. It helps ensure minimal disruption to the business operations and allows for the gradual migration of workloads back to the original site.

Note: If you plan to failover virtual machines, also see Disaster recovery for virtual machines.

Procedure

1. Log in to the original site of the application.
2. Go to the Replicated applications page.
3. Select one or more applications marked with  symbol and click Failover.
4. Click Actions > Failover.
5. In the Failover remote applications window, select the applications and click Failover.
   A success notification is displayed on the screen.

Metro-DR data fencing

Before failover of applications to surviving site, do the following Metro-DR Data Fencing.

About this task

Note: Fencing is needed only in case of unplanned failover.

Procedure

1. Designate two new additional Scale quormn nodes.
2. Add label `scale.spectrum.ibm.com/designation=quorum` for any two worker nodes.

   ```
   oc label node <node> scale.spectrum.ibm.com/designation=quorum
   ```

   For example:

   ```
   oc label node compute-1-ru6.rackae2.mydomain.com   scale.spectrum.ibm.com/designation=quorum
   node/compute-1-ru6.rackae2.mydomain.com labeled
   ```

3. Verify the node status.

   ```
   oc get nodes -l scale.spectrum.ibm.com/designation=quorum | grep compute
   ```

   Sample output:

   ```
   NAME                                 STATUS   ROLES    AGE   VERSION
   compute-1-ru6.rackae2.mydomain.com   Ready    worker   14d   v1.23.17+16bcd69
   compute-1-ru7.rackae2.mydomain.com   Ready    worker   14d   v1.23.17+16bcd69
   ```

4. Uninstall submariner from site-2 to disrupt admin network connectivity across sites.
   Submariner goes down due to unavailability of the other site.
5. Edit **mni** CR and add `uninstall` to **installSubmariner**. When other site comes up, it gets uninstalled. This step avoids network connection to establish between the sites accidentally.

   ```
   oc edit mni metrodr-network
   ```

   Run the following command to check the edited YAML.

   ```
   oc get mni metrodr-network -oyaml -n ibm-spectrum-fusion-ns
   ```

Example YAML:


```
apiVersion: network.isf.ibm.com/v1
kind: MetroDRNetworkInstall
metadata:
  creationTimestamp: "2023-07-10T06:11:53Z"
  finalizers:
  - metrodrnetworkinstall.network.isf.ibm.com/finalizer
  generation: 6
  name: metrodr-network
  namespace: ibm-spectrum-fusion-ns
  ownerReferences:
  - apiVersion: metrodr.isf.ibm.com/v1
    blockOwnerDeletion: true
    controller: true
    kind: MetroDR
    name: metrodrsite
    uid: b5501ab2-4865-4866-aa78-1ef8ca5199f1
  resourceVersion: "39680914"
  uid: 7a2b39e6-9d21-4f8c-9bc2-4e6ce4139415
spec:
  installSubmariner: uninstall
  vlanAdditionRequeDuration: 2m
```


6. Modify network definition in the network CR on the surviving site.
   a. Take the backup of **networks.operator.openshift.io/cluster**.

      ```
      oc get networks.operator.openshift.io/cluster -oyaml >/tmp/networks.operator.openshift.io_cluster.yaml
      ```

   b. Edit and remove **daemon-network** entries.

      ```
      oc edit networks.operator.openshift.io/cluster
      ```

   c. Delete the following route entries pointing to the other site:

      ```
      name: daemon-network
      namespace: ibm-spectrum-scale
      rawCNIConfig: '{ "cniVersion": "0.3.1", "name": "daemon-network", "type": "bridge",
      "bridge": "br3201", "mtu": 9000, "ipam": { "type": "static","routes": [{ "dst":
      "192.168.192.0/18", "gw": "192.168.128.1"}, { "dst": "192.168.192.200/32", "gw":
      "192.168.128.1"}]}}'
      type: Raw
      ```

   d. Restart the scale pods for the changes to take effect.
      Note: Check that the Scale PODs are restarted after this step and make a note.
   e. Do not break route to Tiebreaker.
      Run the following commands to validate whether the daemon connectivity to Tie-breaker is available.

      ```
      oc project ibm-spectrum-scale
      oc get po
      ```

      Example output:

      ```
      NAME                                  READY   STATUS    RESTARTS       AGE
      compute-1-ru5                         2/2     Running   2              14d
      compute-1-ru6                         2/2     Running   9 (18h ago)    14d
      compute-1-ru7                         2/2     Running   9 (18h ago)    14d
      control-1-ru2                         2/2     Running   2              14d
      control-1-ru3                         2/2     Running   2              14d
      control-1-ru4                         2/2     Running   2              14d
      ```

      ```
      oc rsh compute-1-ru5 //any Running node)
      mmgetstate -a
      ```

      Example output:

      ```
      Node number  Node name                                           GPFS state
      13      control-1-ru2.daemon.ibm-spectrum-scale.stg.rackae1  unknown
                 2  control-1-ru3.daemon.ibm-spectrum-scale.stg.rackae1  unknown
                 3  control-1-ru4.daemon.ibm-spectrum-scale.stg.rackae1  unknown
                 4  compute-1-ru5.daemon.ibm-spectrum-scale.stg.rackae1  unknown
                 5  compute-1-ru6.daemon.ibm-spectrum-scale.stg.rackae1  unknown
                 6  compute-1-ru7.daemon.ibm-spectrum-scale.stg.rackae1  unknown
                 7  compute-1-ru5.daemon.ibm-spectrum-scale.stg.rackae2  active
                 8  compute-1-ru6.daemon.ibm-spectrum-scale.stg.rackae2  active
                 9  compute-1-ru7.daemon.ibm-spectrum-scale.stg.rackae2  active
                10  control-1-ru2.daemon.ibm-spectrum-scale.stg.rackae2  active
                11  control-1-ru3.daemon.ibm-spectrum-scale.stg.rackae2  active
                12  control-1-ru4.daemon.ibm-spectrum-scale.stg.rackae2  active
                13  gpfs-tiebreaker                                      active
      ```

      ```
      ping gpfs-tiebreaker
      ```

      Example output:

      ```
      PING gpfs-tiebreaker (192.168.192.200) 56(84) bytes of data.
      64 bytes from gpfs-tiebreaker (192.168.192.200): icmp_seq=1 ttl=64 time=0.239 ms
      64 bytes from gpfs-tiebreaker (192.168.192.200): icmp_seq=2 ttl=64 time=0.218 ms
      ```

   f. Click Exit.
7. Check for the presence of static routes on each scale CORE POD. If they are present, then manually remove entry on each scale CORE PODs to immediately break daemon connection to the failed site.

a. Check route entries on each scale pod

```
oc project ibm-spectrum-scale
oc get po
```

Example output:

```
NAME                                 READY   STATUS    RESTARTS       AGE
compute-1-ru5                        2/2     Running   2              14d
compute-1-ru6                        2/2     Running   9 (18h ago)    14d
compute-1-ru7                        2/2     Running   9 (18h ago)    14d
control-1-ru2                        2/2     Running   2              14d
control-1-ru3                        2/2     Running   2              14d
control-1-ru4                        2/2     Running   2              14d
```

```
oc rsh compute-1-ru5 //any Running node
```

b. Check existing routes.

```
route -n
```

```
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         10.135.0.1      0.0.0.0         UG    0      0        0 eth0
10.135.0.0      0.0.0.0         255.255.254.0   U     0      0        0 eth0
192.168.128.0   192.168.192.1   255.255.192.0   UG    0      0        0 net1
192.168.192.0   0.0.0.0         255.255.192.0   U     0      0        0 net1
192.168.192.200 192.168.192.1   255.255.255.255 UGH   0      0        0 net1
```

c. Delete route with daemon network between two sites.

```
route del -net 192.168.128.0 gw 192.168.192.1 netmask 255.255.192.0
```

d. Ensure that the route is deleted.

```
route -n
```

Example output:

```
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         10.135.0.1      0.0.0.0         UG    0      0        0 eth0
10.135.0.0      0.0.0.0         255.255.254.0   U     0      0        0 eth0
192.168.192.0   0.0.0.0         255.255.192.0   U     0      0        0 net1
192.168.192.200 192.168.192.1   255.255.255.255 UGH   0      0        0 net1
```

e. Click Exit.
f. Repeat steps from 7.a to 7.d for all the compute nodes.

8. Capture a Scale must-gather for any future reference.
9. Deploy the application on the surviving site.
10. Check whether the application deployment works correctly.
11. Failover applications and wait for its completion.
12. Recover the failed site.
13. Delete failed over VMs on the failed site.
14. Remove the added label `scale.spectrum.ibm.com/designation=quorum` for the two compute nodes.

```
oc get nodes -l scale.spectrum.ibm.com/designation=quorum | grep compute
```

Sample output:

```
NAME                               STATUS ROLES AGE VERSION
compute-1-ru6.rackae2.mydomain.com Ready worker 14d v1.23.17+16bcd69
compute-1-ru7.rackae2.mydomain.com Ready worker 14d v1.23.17+16bcd69

Run the command to remove labels added
oc label node compute-1-ru6.rackae2.mydomain.com scale.spectrum.ibm.com/designation-
oc label node compute-1-ru7.rackae2.mydomain.com scale.spectrum.ibm.com/designation-
```

15. Verify whether the labels are removed from the compute nodes.

```
oc get nodes -l scale.spectrum.ibm.com/designation=quorum | grep compute
```

16. Scale down deployments and delete VMs of all failed-over applications or VMs on the recovered site:
    a. Check for application deployments under each namespace:

```
oc get deployments --namespace=<namespace>
```

Example command:

```
oc get deployments --namespace=wordpressapp1-ns
```

```
NAME                    READY   UP-TO-DATE   AVAILABLE   AGE
Wordpressapp1         0/0     0            0           21h
wordpressapp1-mysql   0/0     0            0           21h
```

b. Scale down application deployments to 0.

```
oc scale deployments/wordpressapp1 --namespace=wordpressapp1-ns --replicas 0
oc scale deployments/wordpressapp1-mysql --namespace=wordpressapp1-ns --replicas 0
```

c. Run the following command to delete the VMs.

```
oc delete vm -n myvmnamespace
```

17. Reapply network configurations on the surviving site.
    a. Re-install submariner.
        i. Get the status of Submariner.

        ```
        oc get mni metrodr-network -oyaml -n ibm-spectrum-fusion-ns
        ```

        Sample YAML:


        ```
        apiVersion: network.isf.ibm.com/v1
            ..
            reason: LastReconcileCycleSucceded
            status: "True"
            type: Available
          networkInstallStatus:
            messageCode:
              message: Submariner uninstalled successfully.
            progressPercentage: 100
        ```


        ii. Edit Submariner CR.

        ```
        oc edit mni metrodr-network
        ```

        ```
        Spec:
          installSubmariner: install
        ```

        iii. Check installation status:

        ```
        oc get mni metrodr-network -oyaml -n ibm-spectrum-fusion-ns
        ```

        Output:

        ```
        message: Submariner installation completed successfully
        ```

    b. Add the following entries as follows:
        i. Run the following `oc`command.

        ```
        oc edit networks.operator.openshift.io/cluster
        ```

        ii. Copy the `daemon-network` entries (removed with step 6.a) from the backup file.
        For example:

        ```
        name: daemon-network
        namespace: ibm-spectrum-scale
        rawCNIConfig: '{ "cniVersion": "0.3.1", "name": "daemon-network", "type": "bridge",
        "bridge": "br3201", "mtu": 9000, "ipam": { "type": "static","routes": [{ "dst":
        "192.168.192.0/18", "gw": "192.168.128.1"}, { "dst": "192.168.192.200/32", "gw":
        "192.168.128.1"}]}}'
        type: Raw
        ```

        iii. Run the following command to check whether `daemon-network` entries are added correctly.

        ```
        oc get networks.operator.openshift.io/cluster -oyaml
        ```

        iv. Restart the scale core pods to update the configuration.
    c. Check if the routes are present on scale code pods, else add route entries for each scale pod.

    ```
    oc project ibm-spectrum-scale
    oc get po
    ```

    Sample output:

    ```
    NAME                                  READY   STATUS    RESTARTS      AGE
    compute-1-ru5                         2/2     Running   2             14d
    compute-1-ru6                         2/2     Running   9 (18h ago)   14d
    compute-1-ru7                         2/2     Running   9 (18h ago)   14d
    control-1-ru2                         2/2     Running   2             14d
    control-1-ru3                         2/2     Running   2             14d
    control-1-ru4                         2/2     Running   2             14d
    ```

    ```
    oc rsh compute-1-ru5 //any Running node
    ```

    ```
    route add -net 192.168.128.0 gw 192.168.192.1 netmask 255.255.192.0
    route -n
    ```

    Sample output:

    ```
    Kernel IP routing table
    Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
    0.0.0.0         10.135.0.1      0.0.0.0         UG    0      0        0 eth0
    10.135.0.0      0.0.0.0         255.255.254.0   U     0      0        0 eth0
    192.168.128.0   192.168.192.1   255.255.192.0   UG    0      0        0 net1
    192.168.192.0   0.0.0.0         255.255.192.0   U     0      0        0 net1
    192.168.192.200 192.168.192.1   255.255.255.255 UGH   0      0        0 net1
    ```

    d. Run the following command to check whether the routes are present.

```
route -n
```

If routes are not present, then add the routes and verify using the following commands.

```
route add -net 192.168.128.0 gw 192.168.192.1 netmask 255.255.192.0
route -n
```

e. Check the scale pod status. The status must be active.

```
oc project ibm-spectrum-scale
oc get po
```

Sample output:

```
NAME                                  READY   STATUS    RESTARTS       AGE
compute-1-ru5                         2/2     Running   2              14d
compute-1-ru6                         2/2     Running   9 (18h ago)    14d
compute-1-ru7                         2/2     Running   9 (18h ago)    14d
control-1-ru2                         2/2     Running   2              14d
control-1-ru3                         2/2     Running   2              14d
control-1-ru4                         2/2     Running   2              14d
```

```
oc rsh compute-1-ru5 //any Running node
mmgetstate -a
```

Sample output:

```
Node number  Node name                                                   GPFS state
-------------------------------------------------------------------------------
          1  control-1-ru2.daemon.ibm-spectrum-scale.stg.rackae1  active
          2  control-1-ru3.daemon.ibm-spectrum-scale.stg.rackae1  active
          3  control-1-ru4.daemon.ibm-spectrum-scale.stg.rackae1  active
          4  compute-1-ru5.daemon.ibm-spectrum-scale.stg.rackae1  active
          5  compute-1-ru6.daemon.ibm-spectrum-scale.stg.rackae1  active
          6  compute-1-ru7.daemon.ibm-spectrum-scale.stg.rackae1  active
          7  compute-1-ru5.daemon.ibm-spectrum-scale.stg.rackae2  active
          8  compute-1-ru6.daemon.ibm-spectrum-scale.stg.rackae2  active
          9  compute-1-ru7.daemon.ibm-spectrum-scale.stg.rackae2  active
         10  control-1-ru2.daemon.ibm-spectrum-scale.stg.rackae2  active
         11  control-1-ru3.daemon.ibm-spectrum-scale.stg.rackae2  active
         12  control-1-ru4.daemon.ibm-spectrum-scale.stg.rackae2  active
         13  gpfs-tiebreaker                                      active
```

Metro-DR issues

Use these troubleshooting information to resolve issues when you work with Metro-DR.

Metro-DR failback blockage due to ISO `DataVolume`

Problem statement
: During a Metro-DR failback operation, the process is blocked, and the virtual machine fails to recover. The failback operation remains in a pending state, preventing the successful completion of the disaster recovery process.

Cause
: The failback blockage is caused by the handling of the ISO **`DataVolume`** in the virtualization or Containerized Data Importer (CDI) layer. When an ISO **`DataVolume`** is attached to a VM, additional temporary Persistent Volume Claims (PVCs) are created during failback and remain in a **`Pending`** state. This prevents the failover or failback workflow from progressing.

Resolution
: When a failback operation is triggered with ISO PVC protection enabled, the workflow might be blocked due to the presence of an ISO **`DataVolume`**. To resolve this issue, complete the following steps:

1. Delete the additional temporary PVCs created during the failback operation.
2. After deleting the temporary PVCs, the failback workflow can proceed successfully.
3. To avoid potential issues with ISO PVC protection during failback operations, follow these best practices:
   a. Detach the installation ISO after OS installation.
   b. Configure the VM to reference the ISO PVC directly using **`persistentVolumeClaim`**.

For more information about Virtual Machine Disaster Recovery, see virtual machine disaster recovery documentation.

Metro-DR compute node issue

In a Metro-DR setup, if a compute node is a part of the scale cluster at any site, the Global Data Platform service status may appear as Degraded. To validate the scale health for that site, in the IBM Fusion user interface, go to Storage > Local Storage and check the storage cluster status.

Usable capacity displayed as 0 in the user interface

In Metro-DR setup, the IBM Fusion user interface may display the usable capacity as 0. It is a user interface issue and does not affect the actual capacity or functionality of the setup.

Failback issues

Problem statement
: During unplanned failover, some applications can be in a "`replication error`" state after recovery of the failed site with fencing.

Resolution

- If the application failover is not successful, then check for the **`volumeattachments`** of the application by using the following command.

  ```
  oc get volumeattachment -n <namespace name> | grep < pvc name of given application>
  ```

  If the **`volumeattachments`** exist, then try to delete by using the following command.

  ```
  oc delete volumeattachment -n < namespace name>
  ```

IBM Fusion operator status is pending with errors

Problem statement
: The IBM Fusion operator status is pending with the following errors in 2.7.2 version:

- `RequirementsNotMe`
- `asyncreplications.dataprotection.isf.ibm.com`

Resolution
: Apply the following CRD YAML:

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  annotations:
    controller-gen.kubebuilder.io/version: v0.8.0
  creationTimestamp: null
  name: asyncreplications.dataprotection.isf.ibm.com
spec:
  group: dataprotection.isf.ibm.com
  names:
    kind: AsyncReplication
    listKind: AsyncReplicationList
    plural: asyncreplications
    singular: asyncreplication
  scope: Namespaced
  versions:
  - name: v1
    schema:
      openAPIV3Schema:
        description: AsyncReplication is the Schema for the asyncreplications API
        properties:
          apiVersion:
            description: 'APIVersion defines the versioned schema of this representation
              of an object. Servers should convert recognized schemas to the latest
              internal value, and may reject unrecognized values. More info: 
https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources'
            type: string
          kind:
            description: 'Kind is a string value representing the REST resource this
              object represents. Servers may infer this from the endpoint the client
              submits requests to. Cannot be updated. In CamelCase. More info: 
https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds'
            type: string
          metadata:
            type: object
          spec:
            description: AsyncReplicationSpec defines the desired state of AsyncReplication
            properties:
              consistencyGroup:
                description: Foo is an example field of AsyncReplication. Edit asyncreplication_types.go
                  to remove/update
                type: string
              recoveryPointObjective:
                type: string
              remoteSite:
                type: string
              targetRole:
                type: string
            type: object
          status:
            description: AsyncReplicationStatus defines the observed state of AsyncReplication
            properties:
              currentRole:
                description: 'INSERT ADDITIONAL STATUS FIELD - define observed state
                  of cluster Important: Run "make" to regenerate code after modifying
                  this file'
                type: string
              phase:
                type: string
            type: object
        type: object
    served: true
    storage: true
    subresources:
      status: {}
status:
  acceptedNames:
    kind: ""
```

```
    plural: ""
  conditions: []
  storedVersions: []
```

Global Data Platform upgrade fails in site 1

Problem statement

- Whenever you do the following upgrades on site 1 of the Metro-DR, the "`Global Data platform service not active on all the nodes`" error message gets displayed:
  - IBM Fusion 2.7.2 to 2.8.0
  - Global Data Platform 5.19.0 to 5.2.0
- When you upgrade IBM Fusion 2.7.2 to 2.8.0, the Global Data Platform upgrade of 5.1.9.0 to 5.2.0.0 of Metro-DR Site 1 gets stuck with error "`ECE service not active on all the nodes`".

Resolution

Run the following command to patch Scale recovery group with 'Active' condition:

```
oc patch recoverygroup rg201 -n ibm-spectrum-scale --type=merge --subresource status --type='json' -p '[{ "op": "add",
"path": "/status/conditions/-","value": { "type": "Active", "status": "True", "message": "The recovery group is active",
"reason": "Active", "lastTransitionTime":"'"`date +%FT%T`Z"'"}}]'
```

Less descriptor disks

Problem statement

The site monitoring checks whether enough descriptor disks are assigned for each site. For multiple RGs per site, only one RG sees the descriptor disks. The other RG servers send `disk_missing` that is interpreted as `less descriptor disks`.

Resolution

As a resolution, add the **`site_fs_desc_fail`** event to the "ignore list" to suppress the entire event. Run the command **`mmchconfig mmhealth-events-ignore="site_fs_desc_fail" --force`** once on the cluster and then run **`mmsysmoncontrol restart`** on the cluster manager node.

Latest tiebreaker version

After the upgrade of tiebreaker to the latest version, Metro-DR CR does not show the latest version. You can ignore this as it is a known issue.

Application removals fail in the local tab of failed site after recovery or reconnect

You can observe this issue only in case of fail over of fail back of applications on the surviving site. You can ignore this issue as the CR status is correct, and further relocation or fail back of these applications happen properly.

Add and Remove DR does not work as expected when a remote site is down

When multiple DR protected applications exist in the local site and the other site goes down, the add and remove operations of disaster recovery takes longer than usual.

Connection setup after OpenShift Container Platform cluster recovery

Problem statement

The OpenShift® Container Platform cluster can have problems and become unusable.

Resolution

After you recover the cluster, rejoin the connections. For the steps to clean the connection and setup the connection between two clusters again, see Connection setup after OpenShift Container Platform cluster recovery.

IBM Fusion HCI installation hangs because of disaster recovery

Resolution

If minio pod does not come up for a long time and the Metro-DR installation gets stuck even after the minio pod goes to the running state, then restart Metro-DR deployment in **`ibm-spectrum-fusion-ns`** (or your Fusion namespace) to trigger reconciliation.

RamenDR operator crashes after the restoration of a failed site

Resolution

If RamenDR operator crashes after the restoration of a failed site, then apply the following workaround:

1. Retrieve **`s3CompatibleEndpoint`** from **`s3StoreProfiles`** of the local site.
2. Retrieve the secret of local site that is specified in **`s3SecretRef`** of **`s3StoreProfiles`**.
3. Run the following command from **`isf-minio`** pod in **`ibm-spectrum-fusion-ns`** namespace to connect to the local Minio server.

```
mc alias set myminio <s3CompatibleEndpoint retrieved in step 1> <AWS_ACCESS_KEY_ID parameter of secret retrieved in
step 1> < AWS_SECRET_ACCESS_KEY parameter of secret retrieved in step 1>
```

4. Run the following command to find out all zero-bytes files in Minio store:

```
mc ls myminio --recursive --insecure | grep -i VolumeReplicationGroup | grep 0B
```

5. Run the following command to delete the zero-bytes file that is retrieved in step 3:

```
mc rm <path of files retrieved in step 3>
```

6. Restart **ramen-dr-cluster-operator** deployment in the **ibm-spectrum-fusion-ns** namespace.

If the Volumereplicationgroup's ReplicationState of the application fail over does not change to secondary after the restoration of a failed site, then do the following workaround steps:

1. Go to Red Hat® OpenShift Container Platform web management console.
2. Go to CustomResourceDefinitions > volumereplicationgroups.ramendr.openshift.io > instances.
3. Edit YAMLs of selected **Volumereplicationgroups** and change **spec.replicationState = secondary**.
4. Save YAMLs.

Failovers of concurrent applications take a long time after network recovery

Problem statement
: During Disaster Recovery scenario, when either of the Metro-DR site is powered Off or network is disconnected.

Resolution
: As a resolution, run the following steps before you initiate failover of applications on the surviving site:

1. In Red Hat OpenShift Container Platform console, go to Workloads > ConfigMap.
2. Open the **ramen-dr-cluster-operator-config** configmap in the **ibm-spectrum-fusion-ns** namespace.
3. Add MaxConcurrentReconcile parameter in this configmap with value as `50`.
4. Save the changes and restart **ramen-dr-cluster-operator** deployment in the **ibm-spectrum-fusion-ns** namespace.

Connection between IBM Storage Scale pods

Problem statement
: Unable to ping between IBM Storage Scale pods after you reestablish the network connection between site 1 and site 2.
Workaround:

1. Change in the CR **Spec** of the **installSubmariner** field to **uninstall**.

```
oc edit mni
```

```
Spec:
    installSubmariner: uninstall
```

2. Delete the namespace, if it is not already deleted.
3. Change in the CR **Spec** of the **installSubmariner** field to **install**.

```
oc edit mni
```

```
Spec:
    installSubmariner: install
```

4. If the installation is not successful, then try to install the submariner by using the following manual steps:
   a. Label the control nodes as gateways on both sites of Metro-DR.

```
oc label node control-0.isf-rackc.rtp.raleigh.ibm.com submariner.io/gateway=true
oc label node control-1.isf-rackc.rtp.raleigh.ibm.com submariner.io/gateway=true
oc label node control-2.isf-rackc.rtp.raleigh.ibm.com submariner.io/gateway=true
```

   b. Log in to MetroDR-operator pod.

```
[root@hci-dev1 ~]# oc get po | grep metr
isf-metrodr-operator-controller-manager-bddffd98-dhvgt            2/2     Running   0                13h
[root@hci-dev1 ~]# oc rsh isf-metrodr-operator-controller-manager-bddffd98-dhvgt
```

   c. Deploy the broker for submariner.

```
subctl deploy-broker  --operator-debug  --kubeconfig=/tmp/site-1-kubeconfig
```

   d. Join to broker.

```
subctl join --kubeconfig=/tmp/site-2-kubeconfig --operator-debug --pod-debug  --clusterid=site-2  broker-
info.subm
subctl join --kubeconfig=/tmp/site-1-kubeconfig  --operator-debug --pod-debug  --clusterid=site-1 broker-
info.subm
```

Disaster recovery connections in installing state for hours

Diagnosis
: You might encounter authentication or authorization issues while you connect to the remote cluster. To confirm that the issue is related to authorization, check whether you see the following messages in the Metro-DR pod logs:

- `Incorrect token, need to regenerate kubeconfig file`
- `UnAuthorized`

Cause

- If Metro-DR instance displays any of the following error messages, then do the following workaround:

```
oc get mdr -oyaml
```

`"Submariner is not installed"`

- `oc get mni -oyaml`

```
"Failed to uninstall Submariner"
OR
"Submariner installation failed"
```

- Submariner connectivity might get affected when any of the nodes go to a bad state.
  If problems exist in pod connectivity between racks even after all nodes are back in working condition, then do the following workaround.

Resolution

1. Update site 2 password on site 1 and regenerate the kubeconfig file:
   a. Get the site 2 token from site 2 service account:

```
[root@hci-dev1 ~]# oc get sa fusion-admin-controller-manager -oyaml
secrets:
- name: fusion-admin-controller-manager-dockercfg-vmp2m
- name: fusion-admin-controller-manager-token-wj96d


[root@hci-dev1 ~]# oc get secret fusion-admin-controller-manager-token-wj96d -oyaml
data:
  token: site2Token
```

   b. Update the site 2 token in secret `isf-metrodr-config-secret` on site 1:

```
[root@hci-dev1 ~]# oc edit secret isf-metrodr-config-secret
Data:
   Site2KubePassword: site2Token
```

   c. Log in to the Metro-DR DR pod on site 1 and delete the incorrect site 2 kubeconfig file:

```
root@hci-dev1 ~]# oc get po | grep metro
isf-metrodr-operator-controller-manager-7fd8c4fbc-wwkc8           2/2     Running              0              13h

[root@hci-dev1 ~]# oc exec -ti isf-metrodr-operator-controller-manager-7fd8c4fbc-wwkc8 -c manager bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] --
[COMMAND] instead.
bash-4.4$ ls /tmp/
ks-script-c9hd6mir  ks-script-jrnjxyf4  site-1-kubeconfig  site-2-kubeconfig
bash-4.4$ rm -f /tmp/site-2-kubeconfig
```

2. Update site 1 password on site 2 and regenerate the kubeconfig file:
   a. Get the site 1 token from site 1 service account:

```
[root@hci-dev1 ~]# oc get sa fusion-admin-controller-manager -oyaml
secrets:
- name: fusion-admin-controller-manager-dockercfg-vmp2m
- name: fusion-admin-controller-manager-token-wj96d


[root@hci-dev1 ~]# oc get secret fusion-admin-controller-manager-token-wj96d -oyaml
data:
  token: site1Token
```

   b. Update the site 1 secret token in `isf-metrodr-config-secret` on site 2:

```
[root@hci-dev1 ~]# oc edit secret isf-metrodr-config-secret
Data:
   Site1KubePassword: site1Token
```

   c. Log in to the Metro-DR pod on site 2 and delete the incorrect site 1 kubeconfig file:

```
root@hci-dev1 ~]# oc get po | grep metro
isf-metrodr-operator-controller-manager-7fd8c4fbc-wwkc8           2/2     Running              0              13h

[root@hci-dev1 ~]# oc exec -ti isf-metrodr-operator-controller-manager-7fd8c4fbc-wwkc8 -c manager bash
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exec [POD] --
[COMMAND] instead.
bash-4.4$ ls /tmp/
ks-script-c9hd6mir  ks-script-jrnjxyf4  site-1-kubeconfig  site-2-kubeconfig
bash-4.4$ rm -f /tmp/site-1-kubeconfig
```

Issues with more than 3 quorum nodes on a cluster

Problem statement

The following error occurs only when the OpenShift Container Platform contains more than 10 nodes for the initial installation in a Metro-DR:

```
mmchnode: The number of quorum nodes exceeds the maximum (8) allowed.
```

Cause

The Scale works fine when both sites are healthy. As the quorum nodes are not balanced on both sites, when the site that has five quorum nodes goes down, the whole scale cluster goes down.

Resolution

1. Run the following command to get the quorum nodes:

```
oc get nodes -l scale.spectrum.ibm.com/designation=quorum
```

2. Run the following command to remove the label from two non-control nodes:

```
oc label nodes <node_name> scale.spectrum.ibm.com/designation-
```

Metro-DR goes into a critical state after scale up or scale down operation

Problem statement

The Metro-DR pod goes into critical state critical state after scale up or scale down operation.

Resolution

Follow the steps on the sites to resolve this issue:

1. Log in to the OpenShift Container Platform web console.
2. Go to Administration > CustomResourceDefinitions.
3. Click Filesystem CR.
4. Go to YAML tab and copy the specification details.
5. Similarly, check the RecoveryGroup CR and note the available RVGs dedicated for the sites.
   For example, rg1 and rg201 for the primary site and rg2 and rg202 for the secondary site.
6. Check the number of NVME disks attached to the storage nodes on both the sites. You can also execute `lsblk` command on the storage nodes to get the NVME disk count.
7. Calculate the Vdisk Size:
   - If there are two disks, then it is 100%
   - If there are four disks, then it is 500%
   - If there are six disks, then it is 33%
   - If there are eight disks, then it is 25%
8. Add the respective entries for the recovery group.
   For example, the entry order must be rg1, rg2, rg201 and rg202.
9. Check whether the Vdisk array count for the respective RVG should be half of the disk count. ex. Suppose
   For example:
   - If there are two disks, then the Vdisk array should appear only once
   - If there are four disks, then the Vdisk array should appear twice.
10. Ensure to adjust the Vdisk set array for the site RVG and create a set.
11. Add another site RVG entries as it is in the `Filesystem` CR spec.
12. Save the `Filesystem` CR.

Storage configuration failed on the Metro-DR site

Problem statement

Storage configuration on the secondary site is stuck because of the following error in the `isf-storage-operator-controller-manager` pod.

```
ERROR    controllers.scale       Unable to set the values        {"scale": {"name":"storagemanager","namespace":"ibm-spectrum-f
```

Resolution

Follow the steps to create secrets for the service account to resolve this issue:

1. Run the following command to list the service accounts in the `ibm-spectrum-scale-operator` namespace to verify whether the `ibm-spectrum-scale-operator` service account is present.

   ```
   oc get sa -n ibm-spectrum-scale-operator
   ```

2. Run the following command to list the secrets in the `ibm-spectrum-scale-operator` namespace to verify whether the `ibm-spectrum-scale-operator-token` secret is present.

   ```
   oc get secrets -n ibm-spectrum-scale-operator
   ```

3. If the secret is not present, then create a file `tokensecret.yaml` with the following content.

   ```
   kind: Secret
   apiVersion: v1
   metadata:
     name: ibm-spectrum-scale-operator-token-metro
     namespace: ibm-spectrum-scale-operator
     annotations:
       kubernetes.io/service-account.name: ibm-spectrum-scale-operator
   data:
     namespace: aWJtLXNwZWN0cnVtLXNjYWxlLW9wZXJhdG9y
   type: kubernetes.io/service-account-token
   ```

4. Run the following command to apply the `tokensecret.yaml` file in the `ibm-spectrum-scale-operator` namespace.

   ```
   oc apply -f tokensecret.yaml -n ibm-spectrum-scale-operator
   ```

5. Run the following command to list the secrets again in the `ibm-spectrum-scale-operator` namespace and ensure that the `ibm-spectrum-scale-operator-token` secret is added.

   ```
   oc get secrets -n ibm-spectrum-scale-operator
   ```

Ramen DR pod goes into a `CrashLoopBackOff` state

Problem statement

Ramen DR pod is in a `CrashLoopBackOff` goes into a `CrashLoopBackOff` state with the following errors.

```
ERROR controller/controller.go:200 Could not wait for Cache to sync {"controller": "drclusterconfig", "controllerGroup": "rame
ERROR setup cmd/main.go:285 problem running manager {"error": "failed to wait for drclusterconfig caches to sync: timed out wa
```

Cause

This issue might occur due to the `DRClusterConfig` CRD is missing from the cluster.

Diagnosis

To confirm that the issue is related to CRD missing, check whether you see the following message in the Metro-DR pod logs:

- `missing drclusterconfigs.ramendr.openshift.io`

Resolution

Contact IBM support to resolve the issue.

Application DR configuration for Metro-DR

For Metro-DR configuration, from the disaster recovery page of IBM Fusion HCI user interface, you can manage disaster recovery for your storage and Kubernetes storage resources. You can use the **VolumeReplicationGroup** (VRG) to manage disaster recovery for all other Kubernetes resources. Enable **kubeObjectProtection** flag to protect all Kubernetes resources in the namespace where VRG is associated. If it is in disabled mode, then only PV metadata is protected.

See the instructions in the sub-sections on how you can setup Application DR, enable DR for an application, failover an application, and disable Application DR.

Application DR

Steps to setup an Application DR, enroll an application, failover applications, and disable applications.

Step 1-setup Application DR

You can setup an Application DR anytime after you set up the Metro-DR and before you enable DR for an application.

1. Set up Metro-DR. For the procedure to setup, see Configuring Metro-DR set up. After Site 1, Site 2, and Tiebreaker are installed, proceed with the next steps.
2. Do the following steps on both Site 1 and Site 2 of Metro-DR:
   a. If OADP (version that is used by Backup & Restore) operator does not exist, then create a namespace and install it.
      Note: The installed OADP must align with the version that is used by Fusion Backup & Restore so that the following issues do not exist:
      - OADP inconsistencies
      - CRD conflicts between Fusion Backup & Restore and Metro-DR
      i. Go to OperatorHub > Operator Installation.
      ii. Find OADP, the Red Hat® distribution of Velero.
      iii. In the Install Operator window, select the default namespace **openshift-adp**. You can also install it in another namespace.
      iv. Click Install.
   b. Create **DataProtectionApplication** instance to start the **velero** operator.

```
apiVersion: oadp.openshift.io/v1alpha1
kind: DataProtectionApplication
metadata:
  labels:
    app.kubernetes.io/component: velero
  name: velero
  namespace: openshift-adp
spec:
  backupImages: false
  configuration:
    restic:
      enable: false
    velero:
      defaultPlugins:
        - openshift
        - aws
      noDefaultBackupLocation: true
      podConfig:
        resourceAllocations:
          limits:
            cpu: '1'
            ephemeral_storage: 25Mi
            memory: 1Gi
          requests:
            cpu: 100m
            ephemeral_storage: 25Mi
            memory: 256Mi
  podDnsConfig: {}
```

3. On the site1, do the following steps:
   a. Obtain the minio secret content.

```
site1_keyid=$(oc extract -n ibm-spectrum-fusion-ns secret/isf-metrodr-minio-site1 --keys=AWS_ACCESS_KEY_ID --
to=-);site1_key=$(oc extract -n ibm-spectrum-fusion-ns secret/isf-metrodr-minio-site1 --keys=AWS_SECRET_ACCESS_KEY --
to=-);echo "[default]" > cloud-credentials-site1.yaml;echo "aws_access_key_id = ${site1_keyid}" >> cloud-credentials-
site1.yaml;echo "aws_secret_access_key = ${site1_key}" >> cloud-credentials-site1.yaml
```

   It creates a cloud-credentials-site1.yaml.
   b. Create secret **cloud-credentials-site1** from the file generated in the previous step.

```
oc create secret generic cloud-credentials-site1 --namespace openshift-adp --from-file cloud=./cloud-credentials-
site1.yaml
```

   c. Obtain the minio secret for site2.

```
site2_keyid=$(oc extract -n ibm-spectrum-fusion-ns secret/isf-metrodr-minio-site2 --keys=AWS_ACCESS_KEY_ID --to=-);site2_key=$(oc extract -n ibm-spectrum-fusion-ns secret/isf-metrodr-minio-site2 --keys=AWS_SECRET_ACCESS_KEY --to=-);echo "[default]" > cloud-credentials-site2.yaml;echo "aws_access_key_id = ${site2_keyid}" >> cloud-credentials-site2.yaml;echo "aws_secret_access_key = ${site2_key}" >> cloud-credentials-site2.yaml
```

d. Create `cloud-credentials-site2` from the file generated in the previous step.

```
oc create secret generic cloud-credentials-site2 --namespace openshift-adp --from-file cloud=./cloud-credentials-site2.yaml
```

e. Check whether the cloud credential secrets are created for both the sites. Example output:

```
[root@roadiem9 cpd-cli-linux-EE-13.1.2-89]# oc get secret -n openshift-adp |grep -E "NAME|cloud"
NAME                                            TYPE                                  DATA   AGE
cloud-credentials-site1                         Opaque                                1      57s
cloud-credentials-site2                         Opaque                                1      31s
```

4. Repeat the previous step and substeps on site2.
5. On site1, create secret `cloud-crentials-site1` from the file that is generated in step 3.
6. Create secrets in the Velero operator namespace to access S3 storage on both the sites of Metro-DR. The S3 credentials in this secret are in the following Amazon Web Services format:

```
[default]
aws_access_key_id=<AWS_ACCESS_KEY_ID>
aws_secret_access_key=<AWS_SECRET_ACCESS_KEY>
```

7. Access details in the `ibm-spectrum-fusion-ns` (or your Fusion namespace) Velero secret:
   - For Velero secret to access Site 1 S3 storage, see `isf-metrodr-minio-site1` secret.
   - For Velero secret to access Site 2 S3 storage, see `isf-metrodr-minio-site2` secret.

   Example `cloud-credentials-site1` secret content to access the S3 storage of Metro-DR sites:
   - Metro-DR Site 1.

```
Key: cloud
Value:
[default]
aws_access_key_id=minio
aws_secret_access_key=04ie0oO7x46g0i5
```

   - Metro-DR Site 2:

```
Key: cloud
Value:
[default]
aws_access_key_id=minio
aws_secret_access_key=84ie0oO9x46g0i8
```

8. Enable `kubeObjectProtection` in `ramen-dr-cluster-operator-config` map of `ibm-spectrum-fusion-ns` namespace:

a. To enable `kubeObjectProtection`, set disabled to `false` and specify the Velero installation namespace.

```
kubeObjectProtection:
      disabled: false
      veleroNamespaceName: openshift-adp
```

b. In `S3StoreProfiles` section, for every `S3StoreProfile`, specify the Velero secret details for the corresponding site, including key name and secret name:
Example for Site 1:

```
veleroNamespaceSecretKeyRef:
        key: cloud
        name: cloud-credentials-site1
```

Example for Site 2:

```
veleroNamespaceSecretKeyRef:
        key: cloud
        name: cloud-credentials-site2
```

c. From site 1, get the ingress certificate details by using the following command:

```
oc get cm default-ingress-cert -n openshift-config-managed -o jsonpath="{.data.ca-bundle\.crt}"
```

Note: Encode `ingresscertificate` to base64 format.

d. Repeat the previous step on site 2.

e. On site1, edit the ConfigMap `ramen-dr-cluster-operator-config` in namespace `ibm-spectrum-fusion-ns` to add `caCertificates` under `s3StoreProfiles`. Ensure that you add the appropriate `caCertificates`for the S3 profile.

f. Repeat the previous step on site 2.

g. Configure `ramen-dr-cluster-operator-config` ConfigMap by using the following example:

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: ramen-dr-cluster-operator-config
  namespace: ibm-spectrum-fusion-ns
data:
  ramen_manager_config.yaml: |
    apiVersion: ramendr.openshift.io/v1alpha1
    drClusterOperator: {}
```

```
    health:
      healthProbeBindAddress: :8081
    kind: RamenConfig
    kubeObjectProtection:
      disabled: false
      veleroNamespaceName: openshift-adp
    leaderElection:
      leaderElect: true
      leaseDuration: 0s
      renewDeadline: 0s
      resourceLock: ""
      resourceName: dr-cluster.ramendr.openshift.io
      resourceNamespace: ""
      retryPeriod: 0s
    metrics:
      bindAddress: 127.0.0.1:9289
    ramenControllerType: dr-cluster
    s3StoreProfiles:
    - s3Bucket: isf-minio-site1
      s3CompatibleEndpoint: https://isf-minio-ibm-spectrum-fusion-ns.apps.rackag2.mydomain.com
      s3ProfileName: site1
      s3Region: site1
      s3SecretRef:
        name: isf-minio-site1
        namespace: ibm-spectrum-fusion-ns
      veleroNamespaceSecretKeyRef:
        key: cloud
        name: cloud-credentials-site1
      caCertificates:
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSURiRENDQWxTZ0F3SUJBZ0lIWjc3OHV4V3QrREFOQmdrcWhraUc5dzBCQVFzRkFEQW1NU1F3SWdZR
FZRUUQKREJ0cGJtZHlaWE56TFc5d1pYSmhkRzl5UURFMk9ESXpNVGs1TlRBd0hoY05Nak13TkRJME1EY3dOVFE1V2hjTgpNalV3TkRJek1EY3dOVFV3V2
pBbU1TUXdJZ1lEVlFRRERCc3FMbUZ3Y0hNdWNtRmphMkZuTWk1dGVXUnZiV0ZwCmJpNWpiMjB3Z2dFaU1BMEdDU3FHU0liM0RRRUJBUVVBQTRJQkR3QXd
nZ0VLQW9JQkFRQ3BiaWI5dURTTTVjTk4KdzdKVGRSWGppd2JNdUlWRGhXbmhIMU9oejA4SEY4ck9McEplVEpiYXZJKzBPdjdnaHcwb2p3MWVoaW9adktr
aApVdWg2R1dyWUtiTk5DSm0xVkhrSWt6ZDhXYlpDcklkdTFXcCtnN21zaHpkcTBYZEtNOXVkblB3aG9VNURtV2RzCmNuRjU0bnl0bkhoOXJVM0ZNU1hHU
GhyU3J0cnVEQ01EUkQzQVJ6UmlOL09VNVlyZHFtYk1teHM3RUJtZmRJY1QKWGlFQkoxb09TTC85enkzNnZiSFFkckptc2hPUEN6VG1LWkJ6MklVL3NvR2
NraVBrT2tXL1gvRERUemZNTEJNWApab3NxWWRpNWxjOTR0ek9wUjNMOGU1bVhrR2k4cnNHamJpcURmTHpMZFFtN1pmT1BqYWdsQjE1M0JvdnRMM1hjCmh
KQzJiWGJsQWdNQkFBR2pnWjR3Z1pzd0RnWURWUjBQQVFIL0JBUURBZ1dnTUJNR0ExVWRKUVFNTUFvR0NDc0cKQVFVRkJ3TUJNQXdHQTFVZEV3RUIvd1FD
TUFBd0hRWURWUjBPQkJZRUZJVy9UdjdtY1BBS1JVWlVJNlRDMmVMdwpaZWhTTUI4R0ExVWRJd1FZTUJhQUZDNmpGaXJiZzdVeXdUeEc2UTFDWHMyMWVwV
jJNQ1lHQTFVZEVRUWZNQjJDCkd5b3VZWEJ3Y3k1eVlXTnJZV2N5TG0xNVpHOXRZV2x1TG1OdmJUQU5CZ2txaGtpRzl3MEJBUXNGQUFPQ0FRRUEKdENDUG
ZkUWs2Ui9XeGZ3VC9SWGdlOFczWlFwNUJjL3A1eEJ0MG1QODcwbVBFc0ZmRGhBVVpiYStZeFJNWWp0SApOblBhemlKQkdmUVpFamRoblVGb0lVVjNYd0V
ENUQrSzZtVCtxVWgyNUNNazBuc1YveEZ0eUZuTlYyWElFcSthCmtjclowZTcvVVRpcVJsR3lYelgwSUpFYnlsYWxrL1lHb1pqMG10SDZ6aEVNVjNaczV1
UVlJYmNJSmh0N1Z0UHkKR2lJQmdxaHBhS2NPU2VJWUpaUzlyVDE0NEdIL2lvc2xQTTNpUFJDQkxmUVNWSnJqMGRndVpKN0czQS9Vc3BEbAozOFp5QnNFZ
XZ1bXgxTDY5M0xoOElsN2NzcnBTVVBFZG8zc3ZZcXdtWVBlQzdVZTN3RGEyaCtjQkEyMW5CYW5zCllLT1ErYTV5eXFUR0MyKzBoVnpiaHc9PQotLS0tLU
VORCBDRVJUSUZJQ0FURS0tLS0tCi0tLS0tQkVHSU4gQ0VSVElGSUNBVEUtLS0tLQpNSUlERERDQ0FmU2dBd0lCQWdJQkFUQU5CZ2txaGtpRzl3MEJBUXN
GQURBbU1TUXdJZ1lEVlFRRERCdHBibWR5ClpYTnpMVzl3WlhKaGRHOXlRREUyT0RJek1UazVOVEF3SGhjTk1qTXdOREkwTURjd05UUTVXaGNOTWpVd05E
SXoKTURjd05UVXdXakFtTVNRd0lnWURWUVFEREJ0cGJtZHlaWE56TFc5d1pYSmhkRzl5UURFMk9ESXpNVGs1TlRBdwpnZ0VpTUEwR0NTcUdTSWIzRFFFQ
kFRVUFBNElCRHdBd2dnRUtBb0lCQVFEc1VhME9IOG96TFBtS3VxTVhSV1h4CnNVWFgyN0tuY2pqdnMzNFc5b2V0aGp2ZWh5U0NLMHJvYkpDQmt2enpHel
pRaXRTT2NmUUNnV1BUQ2NIVWl5YVMKNWNKUUt1bGFSbjBYUTFrVld4MVQxbFo2NXZHcmliVnhYblhTUTVhVzR3T0w1bVJNT1hyVnkyZ0ZLK2NHMzZ2Rgp
rQ2VBMmZXcXZONi8ycnZ3aEFpU2VYTHpOc3dEWHk1Q3puOGZiajl3K2Q0YWE2VFhUSG96VWZlQ21aaCtpK3MyCjlpQ0MzSzhjSGtSdTN4K0xwZldWK2Mz
NGd3c0JCbGtJYU1xWkpNY094WGhNemJ3dHJtRGJOeHpLc3ZUeXJ2bVoKcnhiWHVEblpyMExWZ3hvSS9qNjhvUEdreWpRdVQxd0VEcEdtZ1JHczhVdFRNa
VdrcG1QY2VPTmVrTzVDbmltbgpBZ01CQUFHalJUQkRNQTRHQTFVZER3RUIvd1FFQXdJQ3BEQVNCZ05WSFJNQkFmOEVDREFHQVFIL0FnRUFNQjBHCkExVW
REZ1FXQkJRdW94WXEyNE8xTXNFOFJ1a05RbDdOdFhxVmRqQU5CZ2txaGtpRzl3MEJBUXNGQUFPQ0FRRUEKMjRJN3JvYmJTZmtXdXZGVjZQNGNEeXdoQ0x
WKzZjWHQzdnVLWnFCWmxjSlB4ODFpNnl3YktyTjZidXNQRWkxQwpDWkIzdVpJcTBrZUhVRGFoQndwM1BYVWlhcG9LMHRRNC9CcWpBd21DejZXVHc2VmVR
RDlaZWErcFpINWlaNHZDClBIV1VkSHdId0ZUOUZWRkVxNWxRa3RUYVdqclRLdTE1YnpHT3RpNTE3RFkyaHNrMDJFcVJXdk1DUzZTeWhNb3IKbDN5MlNiM
TFnVWMyZ1czaU1TSFo4ZzkzcHVIZGZRUXVEMElqbXFSMWp4ZHc1K05tcFlka3N1M1ArbGE4aW1lawpxa1U3eUp4UzhnRzZ1NmFrdXk0YU9YakhwVUFtST
VOTG0xdW9wNXdiUTlGV2RVZkNxQmd0cmxRblhhaVNXQ3lBCmJXTkxjR0V1a3RnZzUrcGVVVjJzMHc9PQotLS0tLUVORCBDRVJUSUZJQ0FURS0tLS0t
    - s3Bucket: isf-minio-site2
      s3CompatibleEndpoint: https://isf-minio-ibm-spectrum-fusion-ns.apps.rackag3.mydomain.com
      s3ProfileName: site2
      s3Region: site2
      s3SecretRef:
        name: isf-minio-site1
        namespace: ibm-spectrum-fusion-ns
      veleroNamespaceSecretKeyRef:
        key: cloud
        name: cloud-credentials-site2
      caCertificates:
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSURiVENDQWxXZ0F3SUJBZ0lJVTc3MFdhck5pdVl3RFFZSktvWklodmNOQVFFTEJRQXdKakVrTUNJR
0ExVUUKQXd3YmFXNW5jbVZ6Y3kxdmNHVnlZWFJ2Y2tBeE5qZ3lORGc0TXpFeE1CNFhEVEl6TURReU5qQTFOVEUxTVZvWApEVEkxTURReU5UQTFOVEUxTW
xvd0pqRWtNQ0lHQTFVRUF3d2JLaTVoY0hCekxuSmhZMnRoWnpNdWJYbGtiMjFoCmFXNHVZMjl0TUlJQklqQU5CZ2txaGtpRzl3MEJBUUVGQUFPQ0FROEF
NSUlCQ2dLQ0FRRUFxN1JYaXA2WW04UlkKYVJEYVhZTGdZaWpTVjNaOEx6S3UvWDVnakR4NUtqd2tBUW5IeWZJazBpOWVrSnRMR2RabWgzbEFCZi9jRS9Y
NAoyUzlZLzFiRUhPN3J2cHpLVHhZajVtQ1BSRXRLT3BBY1B1bEY3SjJqcnBYODRMYzJJdmNtZHB1blR0TVhUVHBtCkxFS0JVdWMyVUZkOHdrcUpKV3pRR
VlhVFpueDZxVVVYOEM2SkRYVzlNM1B6THJIOHBza2xxS0ZvWk9yaEtzNnYKYjhOeUJZMjd1eTNaZWFhS21KU2l2K1R4M2RURGR1amxwNTdoUnd3Y004Wn
NRdy93MjB2cTQ2S2JPcWZ2YmpmZApIZ0pEbHV3WjhBQzZGWld2OCtwRVk4YTlaVDVtUE1MTnZ5OUZnS2NRMzczdjZEd2VTSGZudWxsZjJVYldBTkJLCjJ
LOTNTTUwyc3dJREFRQUJvNEdlTUlHYk1BNEdBMVVkRHdFQi93UUVBd0lGb0RBVEJnTlZIU1VFRERBS0JnZ3IKQmdFRkJRY0RBVEFNQmdOVkhSTUJBZjhF
QWpBQU1CMEdBMVVkRGdRV0JCUTVCelVrVkdic3BVUEU2cjhrSkZrZApxV01lTGpBZkJnTlZIU01FR0RBV2dCUW5nOHErVlFMSnRzUWtEbTk1S0x1RTJYY
TRrVEFtQmdOVkhSRUVIekFkCmdoc3FMbUZ3Y0hNdWNtRmphMkZuTXk1dGVXUnZiV0ZwYmk1amIyMHdEUVlKS29aSWh2Y05BUVVMQlFBRGdnRUIKQUN3Nm
FVTkRNY0VtaVBMUWZWcTEzY0laZHZVaWE3Y252Q3hLNWR1WDdReVZKOU5jZ21QMUo0WDJ6Vmh0enJMawptNnAybjhCcHF0WXZGSDBSRkxJUHpCVTd6eVN
0N0szUDJ5clhHM3laV1N6aXVIYURPdUQ5YXBmTVY3QnlrOVdYCk8xQnVxeU0ydzA3NDBPajU3bndybTNKd3pBK0dsVStVc0RTaSs0RlRuT0lBTHF6R05F
WG5IMy80eHN5WVgwejUKNDNpNXdpbUJPYVE0ZWE4K1hlalNOeGJBVzRTNUF3VGlTQnVHaWNSN1RrR0NaNGo3WDZ2eEF0dWZ3Q2pnNElxZAo3RGNWcW11W
EdxREhZTFBtVVhKMWVBaXYrRmRCTkQ3dE8xVWhQdXdMdEZxWlAyTERiNHpZSkF5Tnd4NHZPYjJGCnBNZ3dzVmNFZXdkbklmaGduM1hLZGZNPQotLS0tLU
VORCBDRVJUSUZJQ0FURS0tLS0tCi0tLS0tQkVHSU4gQ0VSVElGSUNBVEUtLS0tLQpNSUlERERDQ0FmU2dBd0lCQWdJQkFUQU5CZ2txaGtpRzl3MEJBUXN
GQURBbU1TUXdJZ1lEVlFRRERCdHBibWR5ClpYTnpMVzl3WlhKaGRHOXlRREUyT0RJME9EZ3pNVEV3SGhjTk1qTXdOREkyTURVMU1UVXdXaGNOTWpVd05E
STEKTURVMU1UVXhXakFtTVNRd0lnWURWUVFEREJ0cGJtZHlaWE56TFc5d1pYSmhkRzl5UURFMk9ESTBPRGd6TVRFdwpnZ0VpTUEwR0NTcUdTSWIzRFFFQ
kFRVUFBNElCRHdBd2dnRUtBb0lCQVFEWFp3SFVoTTJ0Z2pZMG5LOEJGZkFtCllMZkFTOHAwaEdKZ3B3SmtCZFllTUUyTlg2Q1RSeFU0RmZWMVRaZm8vQl
d3K3dpcHpzY1hzWE9OMUEzWDczVUIKcnhUOXM4OFJTZDJtWlI3VllPVThqbVNPZXR2N1NOeUlqYWtic3hJcmJJanRJOHVpZlpQNGlJT2VlMmFUaWJSYQo
rWEY4MURuLzdCNldwdmxrYUpxa0ZXekQ3Q2pQeFdUd1NqRjJRZllyOVVYYldYZWI2SnMwaHNwbGRGNDhzQ2pqClQrRXE4VnNxRmdjNFJaQWxjRFVMZEE3
VUs3OE1YTnI2NkU4Z2NqdlhtaS9RdjJvZ3dHZGczeHZHTlNDUGw5YW0KQzFkSGJ4Uk9vSkZjNkx6THpiZ3BqeVJHbnRJblNTbCtSRmsvTk13cUUrci8rQ
khpWXFpOTlReWxUR2lZY1JQbgpBZ01CQUFHalJUQkRNQTRHQTFVZER3RUIvd1FFQXdJQ3BEQVNCZ05WSFJNQkFmOEVDREFHQVFIL0FnRUFNQjBHCkExVW
REZ1FXQkJRbmc4cStWUUxKdHNRa0RtOTVLTHVFMlhhNGtUQU5CZ2txaGtpRzl3MEJBUXNGQUFPQ0FRRUEKelRoOHhsS3FneWlPU2V4NS9ZblFob0ZycHN
5c0x0UGJmMkE2QUVCSy9acGVORlc1T3VmZm5QU25EUEtqdGtORwp5UTlUOFJuS2FrVTRKK0NldGNlUTdKM0l4R3ZzdjNqVGZ3ZXlVMjEra1R2TTJhaGZX
Vmt3NTZEZlRUZmU4dm5sCnQrcVpqRnBmV2p4NnU1Nnp5S0t1NXcwTm12WllXSUdDOFFUWlJNOE9VRmhudjJFeCtnT2NRb2dPcEtNdXdnVWkKemEzWVRxU
C93aVJ2Q2l6ZURXUmsxUmdnR0xVWkcvRGUxZUYxcVp0U0xaZHlFelQ3N3hxMEdCbVpXQjh6NktISAp1eW5VS0ZBa0tZSDJObTAydmlmdVN1MEU2TVN4a2
lsVm5qM1ZJQUgrVnJTNU15OHh6MkJzcGdYVnVSSmtnL1lICnFmR1Ywb3VYenZCczV2SHJUZFdNdlE9PQotLS0tLUVORCBDRVJUSUZJQ0FURS0tLS0t
    volSync:
      disabled: true
```

```
    webhook:
      port: 9443
```

h. Go to Workloads > Pods in the `ibm-spectrum-fusion-ns` namespace on site1 and site2 (or IBM Fusion namespace) and then delete the pod with the prefix `ramen-dr-cluster-operator-` to reflect the Ramen configuration changes.

Step 2-enroll an application

1. Enable DR from IBM Fusion user interface and update `VolumeReplicationGroup` by adding `kubeObjectProtection`.
   For example:

   ```
   apiVersion: ramendr.openshift.io/v1alpha1
   kind: VolumeReplicationGroup
   metadata:
     name: shioramen
     namespace: shioramen
   spec:
     kubeObjectProtection: {}
     pvcSelector: {}
     replicationState: primary
     s3Profiles:
       - site2
       - site1
     sync: {}
     volSync:
       disabled: true
   ```

   Important: Ensure that all the required parameters are defined in spec section during the VRG creation. If you edit VRG parameters later on, it can lead to an inconsistent behavior. If changes to the VRG are required after creation, then you must recreate the VRG.
2. If the Application is already enrolled by using the UI, then use the following command to patch the existing VRG:

   ```
   kubectl patch -n <namespace> vrg/<namespace> --type json -p'[{"op":"add", "path":"/spec/kubeObjectProtection", "value":
   {}}]'
   ```

3. The VRG status must show `ClusterDataProtected` as true. If you see `AnnotationFailed` reason, then go to the PVC and remove the following annotation from it:

   ```
   volumereplicationgroups.ramendr.openshift.io/vr-archived: archiveV1-0
   ```

Step 3-failover applications

1. Scale down the applications deployment on site1 after the `KubeObjects` are protected.
2. Update the `VolumeReplicationGroup (VRG)` by setting the `replicationState` to `secondary` to initiate the failover on the site1.
3. Delete the PVC on Site1.
4. Create a `VRG` with `replicationState` set to `primary` and `kubeObjectProtection: {}`, and verify whether the applications deployment starts successfully.

Step 4-disable Application DR

For each application that is DR enabled, disable Metro-DR for that application to clean it up, including deletion of its `VolumeReplicationGroup`.

1. Disable DR for applications from the IBM Fusion HCI. For the procedure to disable, see Disable disaster recovery point in Backing up your applications topic.
2. Update `Ramen` ConfigMap and restart the `Ramen` controller:

   ```
   kubectl -nibm-spectrum-fusion-ns patch cm/ramen-dr-cluster-operator-config --type json -
   p[{\"op\":add\,\"path\":/data/ramen_manager_config.yaml\,\"value\":\""$(kubectl -nibm-spectrum-fusion-ns get cm/ramen-dr-
   cluster-operator-config -ojsonpath='{.data.ramen_manager_config\.yaml}'|sed -n '/^kubeObjectProtection:/{:1;n;/^
   /b1};p;$!s/$/\\n/'|tr -d '\n')"\"}]
   sleep 120
   kubectl -nibm-spectrum-fusion-ns rollout restart deploy/ramen-dr-cluster-operator;kubectl -nibm-spectrum-fusion-ns rollout
   status -w deploy/ramen-dr-cluster-operator
   ```

   Note: Update Ramen ConfigMap, wait for two minutes, and then restart the Ramen controller.

Kubernetes Resource Protection

Kubernetes applications rely on Kubernetes API resources to function within a cluster. For example, deployments can specify the program components of an application, ConfigMap modifies how a user wants to run an application, and custom resources control the overall operation of the application through application operators. Protecting Kubernetes resources against data loss and disasters in an application-independent manner simplifies Kubernetes application development for applications, requiring those backups and restore services.

Selective protection and recovery of Kubernetes resources for disaster recovery

To minimize recovery time, an application can pre-deploy some of its Kubernetes resources on the recovery cluster. Such an application wants a method to avoid duplicate recovery of its Kubernetes resources to ensure the benefits of pre-deployment. Some Kubernetes resources are created dynamically by Kubernetes itself and an application does not need to preserve the history of those resources so that the protection is not needed. Kubernetes events are a prime example of such resources. In the case of events, applications might need a method to prevent protection and recovery. Hence, the RamenDR Recipe custom resource for disaster recovery provides a flexible mechanism to support these examples and has generalized the mechanism for other use cases.

The general technique of selective protection and recovery mechanism is to filter Kubernetes resources by its kind and label. Resources can be selectively protected and recovered by its kind. It uses an include and exclude mechanism that is provided within the VRG. In addition to include and exclude, the standard Kubernetes label selector mechanism is used to protect specific resources.

Kubernetes resource protection and recovery order

Kubernetes applications must be designed to run on a specified system with a desired state, and the desired state is continuously attempted, achieved, and maintained over time. It is up to the application to deal with asynchronous behavior that is required by this set and attempt, achieve, maintain architecture. However, this level of asynchrony cannot be maintained in highly complex applications with stringent user expectations. First, an asynchronous scope breaks the ability of programmers to predict all sequences of events. Second, asynchronous scope can be a source of failed dependencies. It results in back-off retry loops that can violate the application's Recovery Time Objective (RTO). Restoring resources in a prescribed order can avoid both of these problems. So the Ramen VRG provides a mechanism to support capturing and restoring resources in a prescribed order.

An example of Kubernetes resource protection specification

```
apiVersion: ramendr.openshift.io/v1alpha1
kind: Recipe
metadata:
  name: recipe-sample
  namespace: my-app-ns
spec:
  appType: demo-app  # required, but not currently used
- name: volumes
  type: volume
  labelSelector: app=my-app
- name: config
  backupRef: config
  type: resource
  includedResourceTypes:
  - configmap
  - secret
- name: deployments
  backupRef: deployments
  type: resource
  includedResourceTypes:
  - deployment
- name: instance-resources
  backupRef: instance-resources
  type: resource
  excludedResourceTypes:
  - configmap
  - secret
  - deployment
hooks:
- name: service-hooks
  namespace: my-app-ns
  labelSelector: shouldRunHook=true
  ops:
  - name: pre-backup
    container: main
    command: ["/scripts/pre_backup.sh"]  # must exist in 'main' container
    timeout: 1800
  - name: post-restore
    container: main
    command: ["/scripts/post_restore.sh"]  # must exist in 'main' container
    timeout: 3600
workflows:
- name: capture  # referenced in VRG
  sequence:
  - group: config
  - group: deployments
  - hook: service-hooks/pre-backup
  - group: instance-resources
- name: recover  # referenced in VRG
  sequence:
  - group: config
  - group: deployments
  - group: instance-resources
  - hook: service-hooks/post-restore
```

VRG sample that uses this Recipe

```
apiVersion: ramendr.openshift.io/v1alpha1
kind: VolumeReplicationGroup
metadata:
  name: vrg-sample
  namespace: my-app-ns  # same Namespace as Recipe
spec:
  kubeObjectProtection:
    recipeRef:
      name: recipe-sample
      captureWorkflowName: capture
      recoverWorkflowName: recover
      volumeGroupName: volumes
```

Explanation of the capture and recovery specifications in VRG

The scope of a VRG disaster protection is a single Kubernetes namespace. The VRG protects persistent volumes that are associated with the namespace and optionally protects Kubernetes resources in the namespace. This documentation contains the product previews for protecting Kubernetes resources so does not explain the persistent volume disaster protection.

VRG allows Kubernetes resources to be captured (backed up) and restored as part of disaster recovery. This is achieved through the RamenDR recipe. If a RamenDR recipe specification is not included in VRG, Kubernetes resources are not protected as part of VRG disaster protection.

The RamenDR has two workflows, namely capture and recover. The capture order workflow provides instructions on how to capture a namespace Kubernetes resource. The recover workflow provides instructions on how to recover a namespaces Kubernetes resource after a disaster. This indicates that the order of capture and recovery workflow can be different and need not include the same resources. Each capture and recover workflow contains a list of resource instructions. Each item in the list is acted upon, even if it duplicates the work that is done by other items in the list. So care must be taken to avoid duplication in backup or recover lists so that best RPO and RTO are achieved.

The list items must meet the following requirements:
Note: If the requirements are not met, then the operations of the lists are undefined.

1. The name of each item in the captureOrder list must be unique.
2. The backupName of each item in the recoverOrder list much matches a name in the recoverOrder list.
3. A labelSelector in a list item only applies to that item in the list.
4. If a list item contains multiple labelSelectors, then any resource that matches either label selector is operated upon.
5. IncludeClusterResources in a list item apply only to that item in the list.
6. Each list item can contain either an includedResources section or an excludedResources section, but not both.

Disaster recovery configuration for Fusion Data Foundation

Disaster recovery offerings of IBM Fusion HCI with Fusion Data Foundation storage.

Disaster recovery provides the ability to fail over an application from one Red Hat® OpenShift® Container Platform cluster to another. It also allows relocating the same application back to the original primary cluster.

IBM Fusion HCI with Fusion Data Foundation storage supports the regional disaster recovery (Regional-DR) capability that protects applications and virtual machines (VMs) against regional failures and disaster scenarios like data center failures, maintaining predictable data loss and business continuity. For more information, see Regional-DR configuration for Fusion Data Foundation.

Region failure in Regional-DR is expressed by using the terms, **Recovery Point Objective (RPO)** and **Recovery Time Objective (RTO)**.

- **RPO** is a measure of how frequently you take backups or snapshots of persistent data. In practice, the RPO indicates the amount of data that will be lost or need to be reentered after an outage.
- **RTO** is the amount of downtime a business can tolerate. The RTO answers the question, “How long can it take for our system to recover after we are notified of a business disruption?”

- **Regional-DR configuration for Fusion Data Foundation**
  Regional disaster recovery (Regional-DR) ensures business continuity during the unavailability of an IBM Fusion HCI site in a geographical region, accepting some loss of data in a predictable amount.
- **Monitoring disaster recovery health**
  This section provides all the necessary configuration and commands for setting up the disaster recovery dashboard that help to monitor the health of your disaster recovery solution.
- **Troubleshooting disaster recovery**
  This troubleshooting section provides guidance or workarounds on how to fix some of the disaster recovery configuration issues.
- **Disabling disaster recovery for a disaster recovery-enabled application**
  Learn how to disable disaster recovery for disaster recovery-enabled managed and discovered applications.
- **Uninstalling disaster recovery from hub and managed clusters**
  Learn how to uninstall disaster recovery (DR) partially or completely from hub and managed clusters.

Regional-DR configuration for Fusion Data Foundation

Regional disaster recovery (Regional-DR) ensures business continuity during the unavailability of an IBM Fusion HCI site in a geographical region, accepting some loss of data in a predictable amount.

IBM Fusion HCI with Fusion Data Foundation storage supports the Regional-DR capability that is built on asynchronous data replication. While Regional-DR uses asynchronous replication with some risk of data loss, it protects business-critical applications on the IBM Fusion HCI from region failures.

To configure Regional-DR, use Red Hat® Advanced Cluster Management (RHACM) to set disaster recovery policies on base clusters and hosted control plane (HCP) clusters deployed across two IBM Fusion HCI sites. To manage them, a hub cluster is required for the RHACM operator. For more information, see Components required for Regional-DR.

- **Components required for Regional-DR**
  Regional disaster recovery (Regional-DR) relies on RHACM and Fusion Data Foundation components to support workloads and data mobility between clusters deployed across two IBM Fusion HCI sites.
- **Configuring Regional-DR**
  Configure regional disaster recovery (Regional-DR) capability in the IBM Fusion HCI with Fusion Data Foundation to replicate and recover applications and virtual machines (VMs) across clusters.
- **Validating Regional-DR with applications**
  Regional disaster recovery (Regional-DR) supports disaster recovery of applications deployed on managed clusters regardless of whether the Red Hat Advanced Cluster Management (RHACM) manages the applications or not. It also supports disaster recovery of virtual machines (VMs).
- **Viewing Recovery Point Objective values for disaster recovery enabled applications**
  Recovery Point Objective (RPO) value is the most recent sync time of persistent data from the cluster where the application is currently active to its peer. This sync time helps determine duration of data lost during failover.
- **Enabling granular disaster recovery for individual or groups of virtual machines in a namespace [Technology preview]**
  You can enable granular disaster recovery (DR) for individual virtual machines (VMs) or VM groups within the same namespace, allowing independent failover and relocation actions.

Components required for Regional-DR

Regional disaster recovery (Regional-DR) relies on RHACM and Fusion Data Foundation components to support workloads and data mobility between clusters deployed across two IBM Fusion HCI sites.

Clusters

The Regional-DR brings the ability to failover applications and virtual machines (VMs) from one cluster to another cluster and then failback the same applications and VMs to the original primary cluster.

The following figure illustrates an example of the Regional-DR configuration for clusters deployed across two IBM Fusion HCI sites:

Figure 1. Illustration of a Regional-DR setup for base clusters

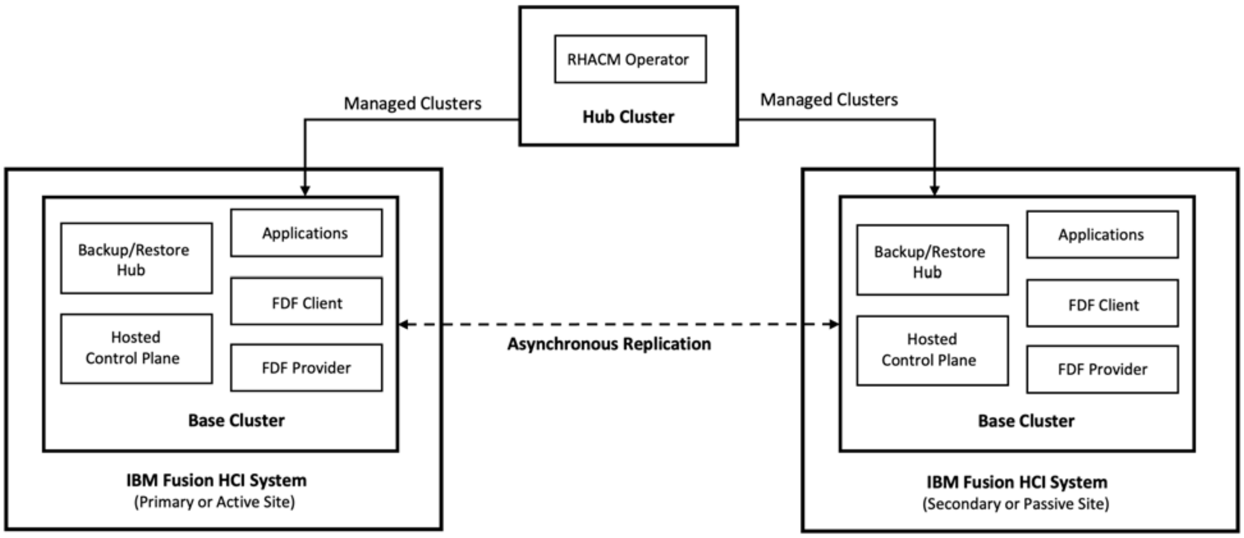
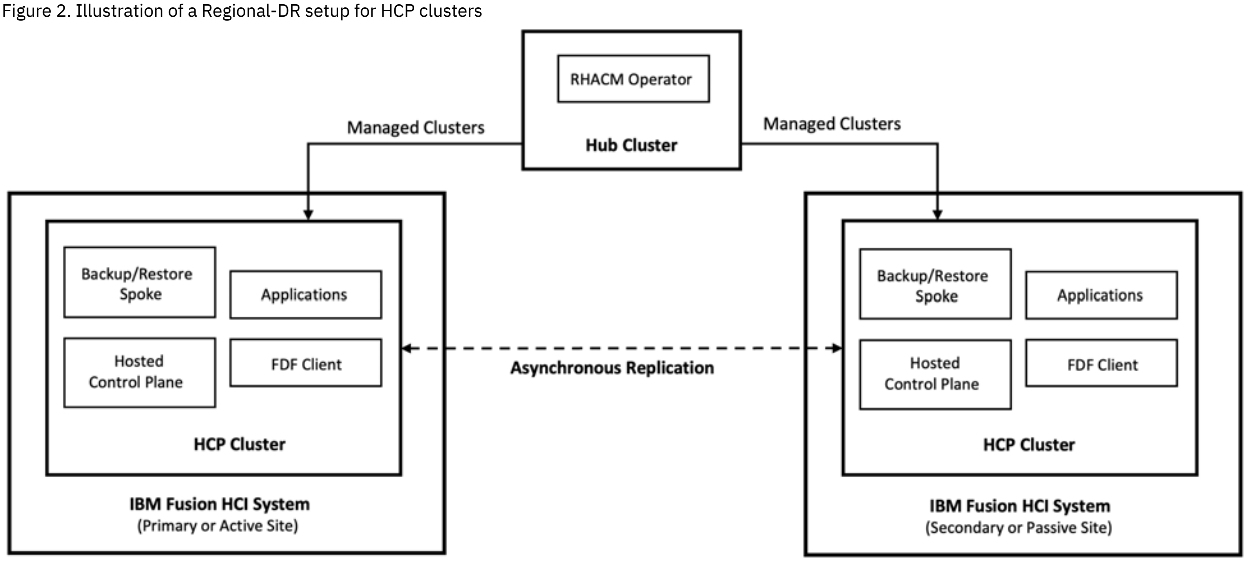


Figure 2. Illustration of a Regional-DR setup for HCP clusters



As shown in the figures Figure 1 and Figure 2, you can have the following clusters in the Regional-DR setup:

Table 1. List of clusters for Regional-DR setup

Cluster	Description
Base cluster	The cluster where the Fusion Data Foundation is installed in a provider mode and serves storage to different hosted control plane (HCP) clusters. Each IBM Fusion HCI site must have a minimum of one base cluster.
HCP cluster	The virtualized or bare metal hosted control plane (HCP) cluster where the Fusion Data Foundation client is installed. You can have multiple virtualized and bare metal HCP clusters in each site. The version of the HCP clusters must be 4.18 or higher.
Hub cluster	An external cluster where the RHACM operator is installed and created outside the IBM Fusion HCI sites. This hub cluster manages base clusters and hosted control plane (HCP) clusters of both IBM Fusion HCI sites.

Red Hat® Advanced Cluster Management for Kubernetes (RHACM) operator

The RHACM operator manages multiple clusters and application lifecycle. Hence, the RHACM hub cluster serves as a control plane in the Regional-DR setup.

RHACM operator is split into two parts:

- **RHACM hub:** Run on the hub cluster or multicluster control plane.
- **Managed clusters:** Imported base and HCP clusters to the hub cluster. The base or HCP cluster from the primary site is called a primary-managed cluster, while the one from the secondary site is called a secondary-managed cluster.

For more information about the RHACM, see Red Hat Advanced Cluster Management for Kubernetes documentation.

OpenShift Multicluster Orchestrator operator

The OpenShift® Multicluster Orchestrator operator orchestrates configuration and peering of the Fusion Data Foundation clusters for the Regional-DR configuration. It needs to be installed on the hub cluster.

Configuring Regional-DR

Configure regional disaster recovery (Regional-DR) capability in the IBM Fusion HCI with Fusion Data Foundation to replicate and recover applications and virtual machines (VMs) across clusters.

Before you begin

- If an external Red Hat® Advanced Cluster Management (RHACM) hub cluster is already available, you can use it for the Regional-DR configuration. If you don't have it, install the RHACM operator to create a hub cluster outside the IBM Fusion HCI sites.

  For instructions, see the Install guide in the RHACM documentation set.

  If hosted control plane (HCP) clusters deployed on IBM Fusion HCI sites are upgraded from a previous version and the RHACM operator is already installed on an HCP cluster, migrate the RHACM operator from the HCP cluster to a new cluster. For assistance with migration, contact IBM support.

- It is recommended that the VMs should use the `ocs-storagecluster-ceph-rbd-virtualization` storage class.
- Ensure that the storage class with the same name exists on the secondary-managed cluster.
- Ensure that the version of the HCP clusters is 4.18 or higher.

Procedure

1. Deploy primary and secondary IBM Fusion HCI sites and configure them to have the network reachability between them.
   Each site should have one IBM Fusion HCI. If only one IBM Fusion HCI site is present, install an IBM Fusion HCI in another site by following the instructions mentioned in Installing IBM Fusion HCI.
   Important: The IBM Fusion HCI version number should be the same across both sites.
2. Open the following ports to ensure that the pods are reachable from other clusters:
   Note: In provider mode, host networking is used for MONs and OSD.
   - Ceph mon v1: 6789
   - Ceph mon v2: 3300
   - Ceph OSD: 6800–7300
   - Ceph Manager: 9283
3. Optional: Set up an SSL connection between clusters deployed on IBM Fusion HCI sites.
   For instructions, see Establishing an SSL connection between clusters.
   Note: If a signed and valid set of certificates has been used for the deployment of base or HCP clusters in both IBM Fusion HCI sites, skip this step.
4. Install the Fusion Data Foundation in newly created clusters.
   - **For base clusters:**
     Install the Fusion Data Foundation in provider mode by following the instructions mentioned in Installing Fusion Data Foundation.

   - **For HCP clusters:**
     Install the Fusion Data Foundation client by following the instructions mentioned in Installing Fusion Data Foundation on a Hosted Control Plane cluster.

   Important: The Fusion Data Foundation version number must be the same on base and HCP clusters deployed across IBM Fusion HCI sites.
5. Import the clusters of primary and secondary IBM Fusion HCI sites to the RHACM hub cluster.
   - **For base clusters:**
     Import base clusters to the RHACM hub by following the instructions mentioned in Discovering multicluster engine operator hosted clusters in Red Hat Advanced Cluster Management of the RHACM documentation.

   - **For HCP clusters:**
     Discover HCP clusters from the multicluster engine (MCE) operator by following the instructions mentioned in Discovering hosted clusters of the RHACM documentation.

     Discovered HCP clusters can be automatically imported into the RHACM hub if the `spec.importAsManagedCluster` field is set to `true`. You need to set this field in the individual discovered cluster's custom resource (CR). For instructions, see Configuring settings for automatic import of the RHACM documentation.

6. Connect the manged clusters and service networks by using the Submariner add-on of the RHACM.
   Verify that the clusters have non-overlapping Classless Inter-Domain Routings (CIDRs) and cluster private networks. For overlapping CIDRs, enable Globalnet during the Submariner add-on installation to connect clusters.

   Run the following command on each managed cluster to verify that the clusters have overlapping CIDRs or non-overlapping CIDRs:

   ```
   oc get networks.config.openshift.io cluster -o json | jq .spec
   ```

   The following example is for clusters with overlapping CIDRs, so the Globalnet needs to be enabled to connect clusters.

   Example output for primary-managed cluster:

```
{
  "clusterNetwork": [
    {
      "cidr": "10.5.0.0/16",
      "hostPrefix": 23
    }
  ],
  "externalIP": {
    "policy": {}
  },
  "networkType": "OVNKubernetes",
  "serviceNetwork": [
    "10.15.0.0/16"
  ]
}
```

Example output for secondary-managed cluster:

```
{
  "clusterNetwork": [
    {
      "cidr": "10.5.0.0/16",
      "hostPrefix": 23
    }
  ],
  "externalIP": {
    "policy": {}
  },
  "networkType": "OVNKubernetes",
  "serviceNetwork": [
    "10.15.0.0/16"
  ]
}
```

Additionally, if Submariner and Fusion Data Foundation are already installed on the managed clusters, use the OpenShift Data Foundation command-line interface (CLI) tool to get additional information about the managed clusters. This information determines the need for enabling Globalnet during Submariner installation based on the cluster service and private networks. Download the OpenShift Data Foundation CLI tool from the customer portal.

Run the following command on one of the two managed clusters, where *PeerManagedClusterName(ClusterID)* is the name of the peer Fusion Data Foundation cluster:

```
odf get dr-prereq <PeerManagedClusterName(ClusterID)>
```

If Submariner is installed without Globalnet on managed clusters with overlapping services, the following output appears:

```
Info: Submariner is installed.
Info: Globalnet is required.
Info: Globalnet is not enabled.
```

Note: This requires Submariner to be uninstalled and then reinstalled with Globalnet enabled.
If Submariner is installed with Globalnet on managed clusters with overlapping services, the following output appears:

```
Info: Submariner is installed.
Info: Globalnet is required.
Info: Globalnet is enabled.
```

If Submariner is installed without Globalnet on managed clusters with non-overlapping services, the following output appears:

```
Info: Submariner is installed.
Info: Globalnet is not required.
Info: Globalnet is not enabled.
```

If Submariner is installed with Globalnet on managed clusters with non-overlapping services, the following output appears:

```
Info: Submariner is installed.
Info: Globalnet is not required.
Info: Globalnet is enabled.
```

For more information on Submariner add-ons and Globalnet configuration, see Submariner multicluster networking and service discovery of the RHACM documentation.

7. Install the OpenShift® Multicluster Orchestrator operator on the hub cluster.
   For instructions, see Installing Multicluster Orchestrator operator.
8. To connect clusters with providerAPIServerServiceType: ClusterIP by using the Submariner add-on do one of the following methods:
   If MetalLB is configured and the IP range is provided, do method 1. If MetalLB is not configured and Globalnet was enabled when setting up Submariner, do method 2.
   Note: If a proxy is configured on the clusters, ensure that the `No_Proxy` list is configured based on the method that you use.
   - If you use method 1 or method 2, add the `api-server-exported-address` to the `No_Proxy` list.
   - If you do not use either method, add the service network of the peer managed cluster to the `No_Proxy` list.
   - **Method 1**:
     If MetalLB is configured and the `ocs-provider-server-load-balancer` service has assigned IP addresses, add the following annotation (key-value pair) to the `StorageCluster` resource on both managed clusters:
     - **Key**: `ocs.openshift.io/api-server-exported-address`
     - **Value**: Endpoint of the `<LoadBalancer ip>:50051`.

     If the `ocs-provider-server-load-balancer` service is in the **Pending** state and has no assigned IP addresses, assign IP addresses to the `LoadBalancer` service. If no IP addresses are available, use method 2.

     To fetch the IP address of the `ocs-provider-server-load-balancer` service, run the following command:

     ```
     oc get svc -n openshift-storage -o wide | grep provide
     ```

     Example output:

```
ocs-provider-server-load-balancer   LoadBalancer   172.30.213.146   172.200.88.109   50051:30506/TCP   6d   app=ocsProvid
```

After the annotation is added, the `odf-info` configMap will detect this endpoint, which the MCO will use to create the `StorageClusterPeer`.

For example:

```
ocs.openshift.io/api-server-exported-address: '172.200.88.109:50051'
```

- **Method 2**:
  If you do not want to use MetalLB and no IP addresses are available, create a service export and use Submariner for cross-cluster communication between the two clusters. Do the following steps:

  - Create a `ServiceExport` for the `ocs-provider-server` service.

    ```
    apiVersion: multicluster.x-k8s.io/v1alpha1
    kind: ServiceExport
    metadata:
      name: ocs-provider-server
      namespace: openshift-storage
    ```

  - Add the following annotation (key-value pair) to the `StorageCluster` resource on both managed clusters:
    - **Key**: `ocs.openshift.io/api-server-exported-address`
    - **Value**: `<managerClusterName>.ocs-provider-server.openshift-storage.svc.clusterset.local:50051`

    `managedClusterName` refers to the name used during cluster import into the hub cluster. For example, if `rackm03` was the name provided, then:

    ```
    ocs.openshift.io/api-server-exported-address:rackm03.ocs-provider-server.openshift-storage.svc.clusterset.local:50051
    ```

    Command example to add the annotation:

    ```
    oc -n openshift-storage annotate storagecluster ocs-storagecluster 'ocs.openshift.io/api-server-exported-address="173.100.89.108:50051"'
    ```

    Where,
    - `openshift-storage`: The namespace where Fusion Data Foundation is installed.
    - `ocs-storagecluster`: The storage cluster name.
    - `key`: ocs.openshift.io/api-server-exported-address
    - `value`: "173.100.89.108:50051"

    Note: Either RHACM, MCE, or Submariner brings the `ServiceExport` CRD. So, you can create the `ServiceExport` after Submariner is installed for the clusters that do not have MCE installed.
    After the annotation is added, the `odf-info` configMap will detect this endpoint, which the MCO will use to create the `StorageClusterPeer`.

9. Check whether a `KlusterletAddonConfig` resource already exists for both managed clusters on the RHACM hub cluster.
   Run the following command on the RHACM hub cluster to verify that `KlusterletAddonConfig` resources exist for both managed clusters:

   ```
   oc get klusterletaddonconfig -A
   ```

   If the `KlusterletAddonConfig` resources for both managed clusters are listed in the output, proceed to the next step. Otherwise, create the missing resource by applying the following YAML configuration:

   For example:

   ```
   apiVersion: agent.open-cluster-management.io/v1
   kind: KlusterletAddonConfig
   metadata:
     name: <rack name>
     namespace: <rack name>
   spec:
     clusterName: <rack name>
     clusterNamespace: <rack name>
     clusterLabels:
       name: <rack name>
       cloud: auto-detect
       vendor: auto-detect
     applicationManager:
       enabled: true
     policyController:
       enabled: true
     searchCollector:
       enabled: true
     certPolicyController:
       enabled: true
   ```

   Remember: In the YAML configuration, provide the same cluster name that you get in the output of the following command:

   ```
   oc get managedcluster
   ```

10. Create a disaster recovery policy (DRPolicy) for the managed clusters.
    For instructions, see Creating a disaster recovery policy on hub cluster.
11. Verify whether the Regional-DR is successfully configured.
    For instructions, see Verifying Regional-DR configuration.

What to do next

Validate Regional-DR with applications by following the instructions mentioned in Validating Regional-DR with applications.

- **Installing Multicluster Orchestrator operator**
  Install the Fusion Data Foundation Multicluster Orchestrator operator for orchestrating configuration and peering of the Fusion Data Foundation clusters for the Regional-DR configuration.

- **Establishing an SSL connection between clusters**
  Establish an SSL connection between clusters deployed on IBM Fusion HCI sites so that metadata can be stored on the Multicloud Object Gateway (MCG) bucket of the base clusters by using a secure transport protocol.
- **Creating a disaster recovery policy on hub cluster**
  The disaster recovery policy (DRPolicy) defines an asynchronous replication schedule with a specified replication interval for the managed clusters. Use this policy to replicate and recover applications across those managed clusters.
- **Disaster recovery protection for cloned and restored RBD volumes**
  The CephCSI driver does necessary operations on the cloned or restored RADOS Block Device (RBD) volumes to prepare them for disaster recovery protection when you enable the disaster recovery support for restored and cloned PersistentVolumeClaims.
- **Verifying Regional-DR configuration**
  Verify that all required components for regional disaster recovery (Regional-DR) are properly configured and operational.

Installing Multicluster Orchestrator operator

Install the Fusion Data Foundation Multicluster Orchestrator operator for orchestrating configuration and peering of the Fusion Data Foundation clusters for the Regional-DR configuration.

Before you begin

- Ensure that you created the following `CatalogSource` manifest on the hub cluster:

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: isf-data-foundation-catalog
  namespace: openshift-marketplace
spec:
  displayName: Data Foundation Catalog
  grpcPodConfig:
    affinity:
      nodeAffinity:
        preferredDuringSchedulingIgnoredDuringExecution:
          - preference:
              matchExpressions:
                - key: gpu.isf.ibm.com
                  operator: DoesNotExist
            weight: 50
    securityContextConfig: restricted
  icon:
    base64data: ""
    mediatype: ""
  image: icr.io/cpopen/isf-data-foundation-catalog:v4.21
  publisher: IBM
  sourceType: grpc
  updateStrategy:
    registryPoll:
      interval: 60m
```

  Note: Ensure that the version number you specify in the `image` field of the `CatalogSource` manifest matches the Fusion Data Foundation version installed on the managed cluster.
- Ensure that you are logged in to the Red Hat® OpenShift® Container Platform as an administrator.
- Ensure that `ibm-spectrum-fusion-ns` is selected in the Project field.

Procedure

1. Go to Ecosystem > Software Catalog tab.
2. Search for the `Multicluster Orchestrator` operator in the All Items field and select the IBM-provided Multicluster Orchestrator tile.
3. On the Fusion Data Foundation Multicluster Orchestrator window, keep all default settings and click Install.
   Fusion Data Foundation Multicluster Orchestrator operator is installed.
   Note:
   The Fusion Data Foundation Multicluster Orchestrator operator also installs the Openshift DR Hub operator on the hub cluster.
4. Verify that the resources of the Fusion Data Foundation Multicluster Orchestrator operator are installed in the openshift-operators project and available to all namespaces.
5. Verify that Pods of operators are in the `Running` status.
   Note: Check whether the Openshift DR Hub operator is also in the `Running` status.
   Run the following command:

```
oc get pods -n openshift-operators
```

   Example output:

```
NAME                                        READY   STATUS      RESTARTS    AGE
odf-multicluster-console-6845b795b9-blxrn   1/1     Running     0           4d20h
odfmo-controller-manager-f9d9dfb59-jbrsd    1/1     Running     0           4d20h
ramen-hub-operator-6fb887f885-fss4w         2/2     Running     0           4d20h
```

Establishing an SSL connection between clusters

Establish an SSL connection between clusters deployed on IBM Fusion HCI sites so that metadata can be stored on the Multicloud Object Gateway (MCG) bucket of the base clusters by using a secure transport protocol.

About this task

Here, clusters refer to the base or hosted control plane (HCP) clusters imported into the hub cluster. For more information, see List of clusters for Regional-DR setup.

If a signed and valid set of certificate has been used for the deployment of base or HCP clusters in IBM Fusion HCI sites, skip this procedure.

Procedure

1. Extract the ingress certificate of the primary-managed cluster and save the output to the primary.crt file.
   Run the following command:

   ```
   oc get cm default-ingress-cert -n openshift-config-managed -o jsonpath="{['data']['ca-bundle\.crt']}" > primary.crt
   ```

2. Extract the ingress certificate of the secondary-managed cluster and save the output to the secondary.crt file.
   Run the following command:

   ```
   oc get cm default-ingress-cert -n openshift-config-managed -o jsonpath="{['data']['ca-bundle\.crt']}" > secondary.crt
   ```

3. Copy the certificate contents from both managed clusters into a new config map file and save it in the namespace of clusters.
   For base clusters, save it in the namespace where the base cluster object is created, such as `openshift-config`. For HCP clusters, save it in the namespace where the `HostedCluster` object is created.

   For example, the config map file name is user-ca-bundle.yaml.

   Note:
   - Ensure that the certificate contents are correctly indented after you copy and paste it from the primary.crt and secondary.crt files.
   - Each cluster might have more or fewer than three certificates, as shown in the following example.

   An example of a config map file is as follows:

   ```
   apiVersion: v1
   data:
     ca-bundle.crt: |
       -----BEGIN CERTIFICATE-----
       <copy contents of cert1 from primary.crt here>
       -----END CERTIFICATE-----

       -----BEGIN CERTIFICATE-----
       <copy contents of cert2 from primary.crt here>
       -----END CERTIFICATE-----

       -----BEGIN CERTIFICATE-----
       <copy contents of cert3 primary.crt here>
       -----END CERTIFICATE----

       -----BEGIN CERTIFICATE-----
       <copy contents of cert1 from secondary.crt here>
       -----END CERTIFICATE-----

       -----BEGIN CERTIFICATE-----
       <copy contents of cert2 from secondary.crt here>
       -----END CERTIFICATE-----

       -----BEGIN CERTIFICATE-----
       <copy contents of cert3 from secondary.crt here>
       -----END CERTIFICATE-----
   kind: ConfigMap
   metadata:
     name: user-ca-bundle
     namespace: openshift-config
   ```

4. Add the name of the newly created config map to the `spec.configuration.proxy.trustedCA` in the `HostedCluster` CR.
   Note: This step applies only to HCP clusters.
   For example:

   ```
   configuration:
       proxy:
         trustedCA:
           name: user-ca-bundle
   ```

5. Create the config map on primary-managed cluster, secondary-managed cluster, and hub cluster.
   Run the following command:

   ```
   oc create -f user-ca-bundle.yaml
   ```

   Example output:

   ```
   configmap/user-ca-bundle created
   ```

6. Patch the default proxy resource on primary-managed cluster, secondary-managed cluster, and hub cluster.
   Run the following command:

   ```
   oc patch proxy cluster --type=merge  --patch='{"spec":{"trustedCA":{"name":"user-ca-bundle"}}}'
   ```

   Example output:

   ```
   proxy.config.openshift.io/cluster patched
   ```

Creating a disaster recovery policy on hub cluster

The disaster recovery policy (DRPolicy) defines an asynchronous replication schedule with a specified replication interval for the managed clusters. Use this policy to replicate and recover applications across those managed clusters.

Before you begin

Ensure that primary and secondary-managed clusters are imported to the hub cluster as instructed in step 5 of *Configuring Regional-DR*.

Procedure

1. Log in to the OpenShift® Container Platform web console and select All clusters.
2. From the Data Services menu, select the Disaster recovery tab.
3. On the Overview tab of the Disaster recovery window, click Create a disaster recovery policy or go to the Policies tab and click Create DRPolicy.
4. In the Policy name field, enter a policy name that is not already in use.
5. Select two clusters from the list of managed clusters to associate with this new policy.
   The Replication policy field is automatically set with the value **`Asynchronous`**.
6. Set the time in the Replication interval field.
   Important: If you are creating a disaster recovery policy for VM workloads, enable flattening to ensure reliable and supported regional disaster recovery behavior. To enable flattening, follow step 7.
7. Optional: Expand Advanced settings and select the Enable disaster recovery support for restored and cloned PersistentVolumeClaims (For Data Foundation only) checkbox.
   For more information on disaster recovery protection for cloned and restored RADOS Block Device (RBD) volumes, see Disaster recovery protection for cloned and restored RBD volumes.
   Important: Use this option only with discovered applications and if cloned and restored RADOS Block Device (RBD) volumes are available.
8. Click Create.

Disaster recovery protection for cloned and restored RBD volumes

The CephCSI driver does necessary operations on the cloned or restored RADOS Block Device (RBD) volumes to prepare them for disaster recovery protection when you enable the disaster recovery support for restored and cloned PersistentVolumeClaims.

Cloned and restored RBD volumes are copy-on-write clones of the corresponding data sources. Such RBD volumes are still linked to the parent volume to optimize space consumption and Ceph cluster performance. To mirror RBD images that have such volumes, the cloned and restored RBD volumes need to be separated from the parent RBD image.

Selecting the Enable disaster recovery support for restored and cloned PersistentVolumeClaims (For Data Foundation only) checkbox while creating a disaster recovery policy instructs CephCSI driver to do necessary operations on the cloned or restored RBD volumes to prepare them for disaster recovery protection. The necessary operation has the following consequences:

- Space consumption may increase up to the total capacity of all cloned and restored RBD volumes that are protected. Ensure that enough space available in the cluster so that the cluster does not full.
- Resource usage of the Ceph cluster increases.
- The time taken for the operation to complete depends on the number of the RBD volumes, size of each volume, Ceph cluster resource availability, and client IO going on the RBD volume. Therefore, the amount of time that is required for completion of the operation cannot be determined.

Verifying Regional-DR configuration

Verify that all required components for regional disaster recovery (Regional-DR) are properly configured and operational.

Procedure

1. Verify that the DRPolicy is created successfully.
   Run this command on the Hub cluster for each of the DRPolicy resources created, where *<drpolicy_name>* is replaced with your unique name.

   ```
   oc get drpolicy <drpolicy_name> -o jsonpath='{.status.conditions[].reason}{"\n"}'
   ```

   Example output: `Succeeded`

   When a DRPolicy is created, along with it, two DRCluster resources are also created. It might take up to 10 minutes for all three resources to be validated and for the status to show as `Succeeded`.

   Note: Editing of **`SchedulingInterval`**, **`ReplicationClassSelector`**, **`VolumeSnapshotClassSelector`** and **`DRClusters`** field values are not supported in the DRPolicy.
2. Verify the object bucket access from the Hub cluster to both the **Primary-managed cluster** and the **Secondary-managed cluster**.
   a. Get the names of the DRClusters on the Hub cluster.

      ```
      oc get drclusters
      ```

      Example output:

      ```
      NAME        AGE
      ocp4bos1    4m42s
      ```

```
ocp4bos2   4m42s
```

b. Check S3 access to each bucket created on each managed cluster.
Use the DRCluster validation command, where *<drcluster_name>* is replaced with your unique name.
Note: Editing of **Region** and **S3ProfileName** field values are non supported in DRClusters.

```
oc get drcluster <drcluster_name> -o jsonpath='{.status.conditions[2].reason}{"\n"}'
```

Example output: `Succeeded`

Note: Ensure to run command for both DRClusters on the Hub cluster.

3. Verify that the IBM Fusion Data Foundation DR Cluster operator installation was successful on the **Primary-managed cluster** and the **Secondary-managed cluster**.

```
oc get csv,pod -n openshift-dr-system
```

Example output:

```
NAME                                                                                      DISPLAY
clusterserviceversion.operators.coreos.com/metallb-operator.v4.18.0-202505081435          MetalLB Operator
clusterserviceversion.operators.coreos.com/odr-cluster-operator.v4.18.4-rhodf             Openshift DR Cluster Operator
clusterserviceversion.operators.coreos.com/openshift-gitops-operator.v1.16.0-0.1746014725.p   Red Hat OpenShift GitOps

NAME                                            READY   STATUS    RESTARTS   AGE
pod/ramen-dr-cluster-operator-5458dbcb5c-jtzgm   2/2     Running   0          15s
```

You can also verify that IBM Fusion Data Foundation DR Cluster Operator is installed successfully on the **OperatorHub** of each managed cluster.

Note: On the initial run, VolSync operator is installed automatically. VolSync is used to set up volume replication between two clusters to protect CephFs-based PVCs. The replication feature is enabled by default.

4. Verify that the status of the IBM Fusion Data Foundation mirroring **daemon** health on the **Primary managed cluster** and the **Secondary managed cluster**.

```
oc get cephblockpoolradosnamespaces -n openshift-storage -o jsonpath='{range .items[*]}{.status.mirroringStatus.summary}
{"\n"}{end}'
```

Example output:

```
{"daemon_health":"OK","health":"OK","image_health":"OK","states":{}}
```

CAUTION:
It could take up to 10 minutes for the daemon_health and health to go from Warning to OK. If the status does not become OK eventually, then use the RHACM console to verify that the Submariner connection between managed clusters is still in a healthy state. Do not proceed until all values are OK.

Validating Regional-DR with applications

Regional disaster recovery (Regional-DR) supports disaster recovery of applications deployed on managed clusters regardless of whether the Red Hat® Advanced Cluster Management (RHACM) manages the applications or not. It also supports disaster recovery of virtual machines (VMs).

After successfully configuring Regional-DR on the managed clusters, you can validate the setup by replicating and recovering applications across those clusters.

Applications are categorized as follows:

- **Managed applications:** Workloads deployed and managed through RHACM, such as ApplicationSet-based application. For more information on ApplicationSet-based application, see ApplicationSet in *Red Hat Advanced Cluster Management for Kubernetes product documentation*.
- **Discovered applications:** Workloads deployed directly on one of the managed clusters without using RHACM. Although these applications are appear in the RHACM console, their lifecycle (create, edit, delete) is not managed by RHACM.

To validate with VMs, if the VM is configured with a secondary VLAN in the IBM Fusion HCI site, ensure that the same secondary VLAN is configured and a corresponding Network Attachment Definition (NAD) is created on the other site. For the secondary VLAN configuration, see Provisioning additional networking.

Note: Regional-DR now supports leveraging RBD and CephFS volumes using non-default replica-2 storage classes that are managed by Fusion Data Foundation. Support has also been extended to compression-enabled storage classes.
Validate Regional-DR with the following applications:

ApplicationSet-based application

To validate Regional-DR with a Subscription-based application, do as follows:

1. Create a sample ApplicationSet-based application by following the instructions mentioned in Creating an ApplicationSet-based application.
2. Apply the DRPolicy to the sample application by following the instructions mentioned in Applying DRpolicy to an ApplicationSet-based application.
3. Do the failover operation of the sample application between the managed clusters by following the instructions mentioned in Subscription-based application failover between managed clusters.
4. Do the relocate operation of the sample application between the managed clusters by following the instructions mentioned in Relocating an ApplicationSet-based application between managed clusters.

After validation, you can delete the sample application by following the instructions mentioned in Deleting a sample application.

Discovered application

To validate Regional-DR with a discovered application, do as follows:

1. Ensure that all prerequisites mentioned in Prerequisites for disaster recovery protection of discovered applications are met.
2. Create a sample discovered application by following the instructions mentioned in Creating a sample discovered application.
3. Enroll the DRPolicy to the sample application by following the instructions mentioned in Enrolling a sample discovered application for disaster recovery protection.

4. Do the failover operation of the sample application between the managed clusters by following the instructions mentioned in Failover disaster recovery protected discovered application.
5. Do the relocate operation of the sample application between the managed clusters by following the instructions mentioned in Relocate disaster recovery protected discovered application.

While the Backup & Restore service and Regional-DR can be configured for the same application, the Backup & Restore service does not currently recognize the disaster recovery state of an application. Even when a hub and spoke configuration is used between two sites, a failed-over application is treated by the Backup & Restore service as a new application.

As a result:

- The Backup & Restore service continues to attempt scheduled backups for the original application on site 1, which might require manual disabling.
- The failed-over application on site 2 does not inherit Backup & Restore service protection from site 1. The application remains unprotected until a new `PolicyAssignment` custom resource (CR) is created. Backups created for the application on site 2 are independent of the backups created for the same application on site 1.

To configure Backup & Restore service in a Regional-DR configuration, complete the following steps:

1. Enroll the application for both Regional-DR and the Backup & Restore service.
2. Fail over the application from site 1 to site 2.
3. Delete the `PolicyAssignment` CR for the application on site 1 to prevent scheduled backups.
4. Relocate the application from site 2 back to site 1.
5. Create a new `PolicyAssignment` CR for the application on site 1 to resume backups.

After validation, you can delete the sample application by following the instructions mentioned in Disable disaster recovery for protected applications.

Multi-volume consistency for disaster recovery

By default, Fusion Data Foundation disaster recovery (DR) provides crash-consistent multi-volume consistency groups for Regional-DR to be used by applications that are deployed across multiple volumes.

Note: If you have upgraded from a previous Fusion Data Foundation version to 4.21, to enable consistency groups you must disable Regional-DR after the upgrade, and then re-enable Regional-DR.

- **Virtual machine garbage collection automation**
  Virtual machine garbage collection automation provides an automated mechanism to clean up virtual machine resources on the source cluster during disaster recovery failover or relocation workflows, reducing manual effort and supporting lower recovery time objectives.

Creating an ApplicationSet-based application

Create an ApplicationSet-based application on the Hub cluster using the Argo CD Pull model to test Regional-DR failover and relocate across managed clusters.

Before you begin

- Ensure that the Red Hat OpenShift GitOps operator is installed on all three clusters: **Hub cluster**, **Primary managed cluster** and **Secondary managed cluster**. For instructions, see Installing Red Hat OpenShift GitOps Operator in web console.
- On the Hub cluster, ensure that both Primary and Secondary managed clusters are registered to GitOps. For registration instructions, see Application management and GitOps > GitOps > Registering managed clusters to Red Hat OpenShift GitOps operator in Red Hat Advanced Cluster Management for Kubernetes product documentation. Then check if the Placement used by `GitOpsCluster` resource to register both managed clusters, has the tolerations to deal with cluster unavailability. You can verify if the following tolerations are added to the Placement using the command.

  `oc get placement <placement-name> -n openshift-gitops -o yaml.`

  ```
  tolerations:
  - key: cluster.open-cluster-management.io/unreachable
    operator: Exists
  - key: cluster.open-cluster-management.io/unavailable
    operator: Exists
  ```

  In case the tolerations are not added, see Application management and GitOps > GitOps in Red Hat Advanced Cluster Management for Kubernetes product documentation.
- Ensure that you have created the `ClusterRoleBinding` yaml on both the **Primary** and **Secondary managed** clusters. For instruction, see Application management and GitOps > GitOps > Deploying Argo CD with Push and Pull model > Prerequisites in Red Hat Advanced Cluster Management for Kubernetes product documentation.

Procedure

1. On the Hub cluster, navigate to All Clusters > Applications > Create application.
2. Select type as Argo CD ApplicationSet - Pull model.
3. In General step, enter your Application set name.
4. Select Argo server `openshift-gitops` and Requeue time as `180` seconds.
5. Click Next.
6. In the Repository location for resources section, select Repository type `Git`.
7. Enter the Git repository URL for the sample application, the GitHub Branch, and Path where the resources `busybox` Pod and PVC will be created.
   a. Use the sample application repository as https://github.com/red-hat-storage/ocm-ramen-samples
   b. Select Revision as `main`.
   c. Choose one the following Path:
      - `workloads/deployment/odr-regional-rbd` to use RBD Regional-DR.
      - `workloads/deployment/odr-regional-cephfs` to use CephFS Regional-DR.

8. Enter Remote namespace value. (example, busybox-sample) and click Next.
9. Select Sync policy settings and click Next.
   You can choose one or more options.
10. Add a label *<name>* with its value set to the managed cluster name.
11. Click Next.
12. Review the setting details and click Submit.

Applying DRpolicy to an ApplicationSet-based application

Apply a disaster recovery policy to an ApplicationSet-based application from the Hub cluster console to enable volume replication and protect the application against cluster failure.

Before you begin

Ensure that both managed clusters referenced in the DRPolicy are reachable. If not, the application will not be DR protected untill both clusters are online.

Procedure

1. On the Hub cluster, navigate to All Clusters > Applications.
2. Click the Actions menu at the end of application to view the list of available actions.
3. Click Manage disaster recovery.
4. Click Enroll application.
5. Select Policy name and click Next.
6. Select an Application resource and then use PVC label selector to select **`PCV label`** for the selected resource.
   Note: You can select more than one PVC label for the selected application resources.
7. After adding all the application resources, click Next.
8. In the Enroll managed application modal, review the policy configuration details and click Assign. The newly assigned Data policy details are displayed on the Manage disaster recovery modal.
9. Verify that you can view the assigned policy details on the Applications page.
   a. On the Applications page, navigate to the DR Status column and view the status.
      The status can be any of the following:
      - Protecting: Shown during initial DR setup before data synchronization begins
      - Healthy: All volumes and Kubernetes resources are syncing normally
      - Warning: Volumes or Kubernetes resources are syncing slower than usual
      - Critical: Volumes or Kubernetes resources are not syncing
      - Protection Error: An error occurred during application protection setup

      You can click the status to view the last synced time for application volumes and the DR policy assigned.

      Note: It may take a few minutes for the DR Status to move from Protecting to Healthy after disaster recovery is enabled. The DR status will show Protecting during initial setup, then automatically transition to health-based statuses (Healthy/Warning/Critical) once data synchronization begins.
   b. Failover and relocate status can be viewed in the DR Status column when initiated. These statuses can be clicked to view more details about the action.
      DR Status shows either Protecting, Healthy, Warning, or Critical when failover or relocate is not taking place. During failover or relocate operations, the status will show the operation-specific status (FailingOver, Relocating, and so on).
10. In the user interface, verify the volume consistency groups:
    a. In the Overview tab, at the end of the protected application row from the action menu (⋮), select Manage disaster recovery.
    b. In the modal, verify Volume group replication is *Enabled*.
    c. In the same modal, select View volume groups.
    d. Verify the volume consistency groups have the correct volumes and a *Synced* status.
11. Optional: For Rados block device (RBD), verify **`volumereplication`**, **`volumereplicationgroup`**, and **`volumegroupreplication`** on the primary cluster.
    For **`volumereplication`**:

    ```
    oc get volumereplication -A
    ```

    Example output:

    ```
    NAMESPACE        NAME                                        AGE    VOLUMEREPLICATIONCLASS                  SOURCEKIND
    SOURCENAME                                                      DESIREDSTATE   CURRENTSTATE
    busybox-sample   vr-bc4a166e-13bc-45cd-99f0-8bd516f7ffcc     112s   rbd-volumereplicationclass-1625360775
    VolumeGroupReplication    vgr-07a0050bee1f25720f5cae8d5daa6230-busybox-placement-drpc   primary        Primary
    ```

    For **`volumereplicationgroup`**:

    ```
    oc get volumereplicationgroup -A
    ```

    Example output:

    ```
    NAMESPACE        NAME                     DESIREDSTATE   CURRENTSTATE
    busybox-sample   busybox-placement-drpc   primary        Primary
    ```

    For **`volumegroupreplication`**:

    ```
    oc get volumegroupreplication -A
    ```

    Example output:

    ```
    NAMESPACE        NAME                                                          VOLUMEGROUPREPLICATIONCLASS
    VOLUMEGROUPREPLICATIONCONTENT                     DESIREDSTATE   CURRENTSTATE   AGE
    busybox-sample   vgr-07a0050bee1f25720f5cae8d5daa6230-busybox-placement-drpc   rbd-volumegroupreplicationclass-1625360775-
    1602718344   vgrcontent-bc4a166e-13bc-45cd-99f0-8bd516f7ffcc   primary        Primary        114s
    ```

12. Optional: For CephFS, verify `ReplicationGroupSource` and `ReplicationSource` on the Primary cluster.
    For **ReplicationGroupSource**:

    ```
    oc get ReplicationGroupSource -A
    ```

    Example output:

    ```
    NAMESPACE       NAME                                              LAST SYNC              DURATION          NEXT SYNC
    SOURCE                                                                                                        LAST SYNC
    START
    busybox-cephfs  busybox-cephfs-bbd241b65b6589782fddaa82636e358d   2026-03-18T11:20:14Z   1m16.462467415s   2026-03-
    18T11:25:00Z   {"matchLabels":{"ramendr.openshift.io/consistency-group":"busybox-cephfs-
    bbd241b65b6589782fddaa82636e358d"}}
    ```

    For **ReplicationSource**:

    ```
    oc get ReplicationSource -A
    ```

    Example output:

    ```
    NAMESPACE       NAME          SOURCE          LAST SYNC              DURATION        NEXT SYNC
    busybox-cephfs  busybox-pvc   vs-busybox-pvc  2026-03-18T11:20:14Z   16.450725976s
    ```

13. Optional: For CephFS, verify `ReplicationGroupDestination` and `ReplicationDestination` on the Secondary cluster.
    For **ReplicationGroupDestination**:

    ```
    oc get ReplicationGroupDestination -A
    ```

    Example output:

    ```
    NAMESPACE       NAME                                              LAST SYNC              DURATION        LAST SYNC START
    busybox-cephfs  busybox-cephfs-b1f2b5bac27b3d8805437424895d212c   2026-03-18T11:20:15Z   47.700328282s   2026-03-
    18T11:20:15Z
    ```

    For **ReplicationDestination**:

    ```
    oc get ReplicationDestination -A
    ```

    Example output:

    ```
    NAMESPACE       NAME          LAST SYNC              DURATION        NEXT SYNC
    busybox-cephfs  busybox-pvc   2026-03-18T11:20:15Z   47.401909636s
    ```

ApplicationSet-based application failover between managed clusters

Fail over an ApplicationSet-based application from a primary managed cluster to a secondary managed cluster to maintain application availability during a disaster or cluster failure.

Before you begin

- When primary cluster is in a state other than Ready, check the actual status of the cluster as it might take some time to update.
  1. Go to RHACM console > Infrastructure > Clusters > Cluster list tab.
  2. Check the status of both the managed clusters individually before performing a failover operation.

  However, failover operation can still be run when the cluster you are failing over to is in a *Ready* state.
- Run the following command on the Hub Cluster to check if `lastGroupSyncTime` is within an acceptable data loss window, when compared to current time.

  ```
  oc get drpc -o yaml -A | grep lastGroupSyncTime
  ```

  Example output:

  ```
  [...]
  lastGroupSyncTime: "2023-07-10T12:40:10Z"
  ```

About this task

Failover is a process that transitions an application from a primary cluster to a secondary cluster in the event of a primary cluster failure. While failover provides the ability for the application to run on the secondary cluster with minimal interruption, making an uninformed failover decision can have adverse consequences, such as complete data loss in the event of unnoticed replication failure from primary to secondary cluster. If a significant amount of time has gone by since the last successful replication, it's best to wait until the failed primary is recovered.

`LastGroupSyncTime` is a critical metric that reflects the time since the last successful replication occurred for all PVCs associated with an application. In essence, it measures the synchronization health between the primary and secondary clusters. So, prior to initiating a failover from one cluster to another, check for this metric and only initiate the failover if the `LastGroupSyncTime` is within a reasonable time in the past.

Note: During the course of failover the Ceph-RBD mirror deployment on the failover cluster is scaled down to ensure a clean failover for volumes that are backed by Ceph-RBD as the storage provisioner.

Procedure

1. Failover the protected application using either of the following paths:
   - **Option 1**: From the Applications page

     On the Hub cluster, navigate to Applications. Click Actions (⋮) and select Failover application.

- **Option 2**: From the Protected Applications page

  On the Hub cluster, navigate to Data Services > Disaster recovery > Protected Applications > Actions (⋮) > Failover.

2. When the Failover application modal is shown, verify the details that are presented are correct and check the status of the Failover readiness. If the status is *Ready* with a green tick, it indicates that the target cluster is ready for failover to start.
   Important:
   If there are data inconsistencies caused by synchronization delays, a warning message appears stating **Inconsistent data on target cluster**. This alerts to the possibility of data loss if the failover is initiated. The message is no longer displayed when data synchronization is complete.

3. Click Initiate.
   All the system workloads and their available resources are now transferred to the target cluster.
4. In the DR status column, you can view the progress of the failover. The modal shows the progress of the failover as Preparing > Failover > Restoring > Clean up.
5. Verify that the activity status shows as *FailedOver* for the application.
   a. Go to the Applications > Overview tab.
   b. In the Data policy column, click the policy link for the application you applied the policy to.
   c. On the Data Policy popover page, verify that you can see one or more policy names and the ongoing activities associated with the policy in use with the application.

If volume synchronization does not occur after a failover and ApplicationSet-based applications continue running on both clusters, apply the following workaround:

From the hub cluster, delete the `manifestwork` resource that continues to run the ApplicationSet-based applications on the cluster from which the applications were failed over.

For example:

```
oc delete manifestwork -n rackm14 app-busybox-cephfs-1-rackm14-48b8c
```

Relocating an ApplicationSet-based application between managed clusters

Relocate an application to its preferred location when all managed clusters are available.

Before you begin

- When primary cluster is in a state other than Ready, check the actual status of the cluster as it might take some time to update. Relocate can only be performed when both primary and preferred clusters are up and running.
  1. Go to RHACM console > Infrastructure > Clusters > Cluster list tab.
  2. Check the status of both the managed clusters individually before performing a relocate operation.
- Perform relocate when `lastGroupSyncTime` is within the replication interval (for example, 5 minutes) when compared to current time. This is recommended to minimize the Recovery Time Objective (RTO) for any single application.
  Run this command on the Hub Cluster:

  ```
  oc get drpc -o yaml -A | grep lastGroupSyncTime
  ```

  Example output:

  ```
  [...]
  lastGroupSyncTime: "2023-07-10T12:40:10Z"
  ```

  Compare the output time (UTC) to current time to validate that all `lastGroupSyncTime` values are within their application replication interval. If not, wait to Relocate until this is true for all `lastGroupSyncTime` values.

Procedure

1. Relocate the protected application using either of the following paths:

   Option 1: From the Applications page
   a. On the Hub cluster, navigate to All Clusters > Applications.
   b. In the Overview tab, at the end of the application row from the action menu **(⋮)**, select Manage diaster recovery.
   c. Click Relocate.

   Option 2: From the Protected Applications page
   - On the Hub cluster, navigate to Data Services > Disaster recovery > Protected Applications > Actions (⋮) > Relocate.

2. When the Relocate application modal is shown, select Policy and Target cluster to which the associated application will relocate to in case of a disaster.
   Important:
   If there are data inconsistencies caused by synchronization delays, a warning message appears stating **Inconsistent data on target cluster**. This alerts to the possibility of data loss if the failover is initiated. The message is no longer displayed when data synchronization is complete.

3. Click Initiate.
   All the system workloads and their available resources are now transferred to the target cluster.
4. Close the modal window and track the status using the Data policy column on the Applications page.
   The status moves from Preparing > Syncing > Restoring > Clean up.
5. Verify that the activity status shows as *Relocated* for the application.
   a. Go to Applications > Overview.
   b. In the Data policy column, click the policy link for the application you applied the policy to.
   c. On the Data Policy popover page, verify that you can see one or more policy names and the relocation status associated with the policy in use with the application.

Deleting a sample application

The sample application **busybox** can be deleted by using the Red Hat Advanced Cluster Management for Kubernetes (RHACM) console.

Before you begin

Important: When deleting a DR protected application, access to both clusters that belong to the DRPolicy is required. This is to ensure that all protected API resources and resources in the respective S3 stores are cleaned up as part of removing the DR protection. If access to one of the clusters is not healthy, deleting the `DRPlacementControl` resource for the application, on the hub, would remain in the `Deleting` state.
Do not begin deleting a sample application until the failover and relocate testing is completed and the application is ready to be removed from RHACM and the managed clusters.

About this task

Procedure

1. On the **Hub cluster**, navigate to Applications.
2. Search for the sample application to be deleted (for example, `busybox`).
3. Click the Action menu next to the application that you want to delete.
4. Click Delete application.
   When the Delete application is selected a new screen appears asking if the application-related resources should also be deleted.
5. Select Remove application related resources to delete the Subscription and PlacementRule.
6. Click Delete.
   This deletes the `busybox` application on the Primary-managed cluster (or the cluster that the application was running on).
7. In addition to the resources deleted by using the RHACM console, delete the `DRPlacementControl` if it is not auto-deleted after deleting the `busybox` application.
   a. Log in to the Red Hat OpenShift web console for the Hub cluster and navigate to Installed Operators for the project `busybox-sample`.
      For ApplicationSet applications, select the project as `openshift-gitops`.
   b. Click OpenShift DR Hub Operator > DRPlacementControl.
   c. Click the Action menu next to the `busybox` application DRPlacementControl that you want to delete.
   d. Click Delete DRPlacementControl.
   e. Click Delete.
   Note: This process can be used to delete any application with a `DRPlacementControl` resource.

Subscription-based application failover between managed clusters

Fail over a subscription-based application from a primary managed cluster to a secondary managed cluster to maintain application availability during a disaster or cluster failure.

Before you begin

- When primary cluster is in a state other than Ready, check the actual status of the cluster as it might take some time to update.
  1. Go to RHACM console > Infrastructure > Clusters > Cluster list tab.
  2. Check the status of both the managed clusters individually before performing a failover operation.
  However, failover operation can still be run when the cluster you are failing over to is in a *Ready* state.
- Run the following command on the Hub Cluster to check if `lastGroupSyncTime` is within an acceptable data loss window, when compared to current time.

  ```
  oc get drpc -o yaml -A | grep lastGroupSyncTime
  ```

  Example output:

  ```
  [...]
  lastGroupSyncTime: "2023-07-10T12:40:10Z"
  ```

About this task

Failover is a process that transitions an application from a primary cluster to a secondary cluster in the event of a primary cluster failure. While failover provides the ability for the application to run on the secondary cluster with minimal interruption, making an uninformed failover decision can have adverse consequences, such as complete data loss in the event of unnoticed replication failure from primary to secondary cluster. If a significant amount of time has gone by since the last successful replication, it's best to wait until the failed primary is recovered.

`LastGroupSyncTime` is a critical metric that reflects the time since the last successful replication occurred for all PVCs associated with an application. In essence, it measures the synchronization health between the primary and secondary clusters. So, prior to initiating a failover from one cluster to another, check for this metric and only initiate the failover if the `LastGroupSyncTime` is within a reasonable time in the past.

Note: During the course of failover the Ceph-RBD mirror deployment on the failover cluster is scaled down to ensure a clean failover for volumes that are backed by Ceph-RBD as the storage provisioner.

Procedure

1. Failover the protected application using either of the following paths:

- **Option 1**: From the Applications page

  On the Hub cluster, navigate to Applications. Click Actions (⋮) and select Failover application.

- **Option 2**: From the Protected Applications page

  On the Hub cluster, navigate to Data Services > Disater recovery > Protected Applications > Actions (⋮) > Failover.

2. After the Failover application popup is shown, select Policy and Target cluster to which the associated application will failover in a disaster.
3. Click the Select subscription group dropdown to verify the default selection or modify this setting.
   By default, the subscription group that replicates for the application resources is selected.
4. Check the status of the Failover readiness.
   - If the status is *Ready* with a green tick, it indicates that the target cluster is ready for failover to start. Proceed to step 5.
   - If the status is *Unknown* or *Not ready*, then wait until the status changes to *Ready*.

   Important:
   If there are data inconsistencies caused by synchronization delays, a warning message appears stating **Inconsistent data on target cluster**. This alerts to the possibility of data loss if the failover is initiated. The message is no longer displayed when data synchronization is complete.
5. Click Initiate.
   All the system workloads and their available resources are now transferred to the target cluster.
6. In the DR status column, you can view the progress of the failover. The modal shows the progress of the failover as Preparing > Failover > Restoring > Clean up.
7. Verify that the activity status shows as *FailedOver* for the application.
   a. Go to the Applications > Overview tab.
   b. In the Data policy column, click the policy link for the application you applied the policy to.
   c. On the Data Policy popover page, click the View more details link.
   d. Verify that you can see one or more policy names and the ongoing activities (Last sync time and Activity status) associated with the policy in use with the application.

Relocating a Subscription-based application between managed clusters

Relocate an application to its preferred location when all managed clusters are available.

Before you begin

- When the primary cluster is in a state other than Ready, check the actual status of the cluster as it might take some time to update. Relocate can be performed only when both primary and preferred clusters are up and running.
  1. Go to RHACM console > Infrastructure > Clusters > Cluster list tab.
  2. Check the status of both the managed clusters individually before performing a relocate operation.
- Perform relocate when `lastGroupSyncTime` is within the replication interval (for example, 5 minutes) when compared to current time. This is recommended to minimize the Recovery Time Objective (RTO) for any single application.
  Run this command on the Hub Cluster:

  ```
  oc get drpc -o yaml -A | grep lastGroupSyncTime
  ```

  Example output:

  ```
  [...]
  lastGroupSyncTime: "2023-07-10T12:40:10Z"
  ```

  Compare the output time (UTC) to current time to validate that all `lastGroupSyncTime` values are within their application replication interval. If not, wait to Relocate until this is true for all `lastGroupSyncTime` values.

Procedure

1. Relocate the protected application using either of the following paths:

   Option 1: From the Applications page
   a. On the Hub cluster, navigate to All Clusters > Applications.
   b. In the Overview tab, at the end of the application row from the action menu **(⋮)**, select Manage diaster recovery.
   c. Click Relocate.

   Option 2: From the Protected Applications page
   - On the Hub cluster, navigate to Data Services > Disater recovery > Protected Applications > Actions (⋮) > Relocate.
2. When the Relocate application modal is shown, select Policy and Target cluster to which the associated application will relocate to in case of a disaster.
3. By default, the subscription group that will deploy the application resources is selected. Click the Select subscription group dropdown to verify the default selection or modify this setting.
4. Check the status of the Relocation readiness.
   - If the status is *Ready* with a green tick, it indicates that the target cluster is ready for failover to start. Proceed to step 5.
   - If the status is *Unknown* or *Not ready*, then wait until the status changes to *Ready*.

   Important:
   If there are data inconsistencies caused by synchronization delays, a warning message appears stating **Inconsistent data on target cluster**. This alerts to the possibility of data loss if the failover is initiated. The message is no longer displayed when data synchronization is complete.
5. Click Initiate.
   All the system workloads and their available resources are now transferred to the target cluster.
6. In the DR status column, you can view the progress of the failover. The modal shows the progress of the failover as Preparing > Syncing > Restoring > Clean up.
7. Verify that the activity status shows as *Relocated* for the application.
   a. Go to the Applications > Overview tab.

b. In the Data policy column, click the policy link for the application you applied the policy to.
c. On the Data Policy popover page, click the View more details link.
d. Verify that you can see one or more policy names and the ongoing activities (Last sync time and Activity status) associated with the policy in use with the application.

Deleting a sample application

The sample application `busybox` can be deleted by using the Red Hat Advanced Cluster Management for Kubernetes (RHACM) console.

Before you begin

Important: When deleting a DR protected application, access to both clusters that belong to the DRPolicy is required. This is to ensure that all protected API resources and resources in the respective S3 stores are cleaned up as part of removing the DR protection. If access to one of the clusters is not healthy, deleting the `DRPlacementControl` resource for the application, on the hub, would remain in the `Deleting` state.
Do not begin deleting a sample application until the failover and relocate testing is completed and the application is ready to be removed from RHACM and the managed clusters.

About this task

Procedure

1. On the **Hub cluster**, navigate to Applications.
2. Search for the sample application to be deleted (for example, `busybox`).
3. Click the Action menu next to the application that you want to delete.
4. Click Delete application.
   When the Delete application is selected a new screen appears asking if the application-related resources should also be deleted.
5. Select Remove application related resources to delete the Subscription and PlacementRule.
6. Click Delete.
   This deletes the `busybox` application on the Primary-managed cluster (or the cluster that the application was running on).
7. In addition to the resources deleted by using the RHACM console, delete the `DRPlacementControl` if it is not auto-deleted after deleting the `busybox` application.
   a. Log in to the Red Hat OpenShift web console for the Hub cluster and navigate to Installed Operators for the project `busybox-sample`.
      For ApplicationSet applications, select the project as `openshift-gitops`.
   b. Click OpenShift DR Hub Operator > DRPlacementControl.
   c. Click the Action menu next to the `busybox` application DRPlacementControl that you want to delete.
   d. Click Delete DRPlacementControl.
   e. Click Delete.
   Note: This process can be used to delete any application with a `DRPlacementControl` resource.

Prerequisites for disaster recovery protection of discovered applications

This section provides instructions to guide you through the prerequisites for protecting discovered applications. This includes tasks such as assigning a data policy and initiating DR actions such as failover and relocate.

1. Ensure that all the DR configurations have been installed on the Primary managed cluster and the Secondary managed cluster.
2. Install the OADP 1.4 operator.
   Note: Any version before OADP 1.4 will not work for protecting discovered applications.
   a. On the **Primary** and **Secondary managed cluster**, navigate to OperatorHub and use the keyword filter to search for `OADP`.
   b. Click the OADP tile.
   c. Keep all default settings and click Install. Ensure that the operator resources are installed in the `openshift-adp` project.
      Note: If OADP 1.4 is installed after DR configuration has been completed then the `ramen-dr-cluster-operator` pods on the **Primary managed cluster** and the **Secondary managed cluster** in namespace `openshift-dr-system` must be restarted (deleted and recreated).
3. [Optional] Add CACertificates to `ramen-hub-operator-config` **ConfigMap**.
   Configure network (SSL) access between the primary and secondary clusters so that metadata can be stored on the alternate cluster in a Multicloud Gateway (MCG) object bucket using a secure transport protocol and in the Hub cluster for verifying access to the object buckets.
   Note: If all of your OpenShift clusters are deployed using a signed and valid set of certificates for your environment then this section can be skipped.
   If you are using self-signed certificates, then you have already created a **ConfigMap** named `user-ca-bundle` in the `openshift-config` namespace and added this ConfigMap to the default Proxy cluster resource.
   a. Find the encoded value for the CACertificates.

```
oc get configmap user-ca-bundle -n openshift-config -o jsonpath="{['data']['ca-bundle\.crt']}" |base64 -w 0
```

   b. Add this base64 encoded value to the configmap `ramen-hub-operator-config` on the Hub cluster. Example below shows where to add CACertificates.

```
oc edit configmap ramen-hub-operator-config -n openshift-operators
```

```
[...]
    ramenOpsNamespace: openshift-dr-ops
    s3StoreProfiles:
    - s3Bucket: odrbucket-36bceb61c09c
      s3CompatibleEndpoint: https://s3-openshift-storage.apps.hyper3.vmw.ibmfusion.eu
      s3ProfileName: s3profile-hyper3-ocs-storagecluster
      s3Region: noobaa
      s3SecretRef:
```

```
        name: 60f2ea6069e168346d5ad0e0b5faa59bb74946f
      caCertificates: {input base64 encoded value here}
    - s3Bucket: odrbucket-36bceb61c09c
      s3CompatibleEndpoint: https://s3-openshift-storage.apps.hyper4.vmw.ibmfusion.eu
      s3ProfileName: s3profile-hyper4-ocs-storagecluster
      s3Region: noobaa
      s3SecretRef:
        name: cc237eba032ad5c422fb939684eb633822d7900
      caCertificates: {input base64 encoded value here}
```

4. Verify that there are DR secrets created in the OADP operator default namespace `openshift-adp` on the **Primary managed cluster** and the **Secondary managed cluster**. The DR secrets that were created when the first DRPolicy was created, will be similar to the secrets below. The DR secret name is preceded with the letter `v`.

   ```
   oc get secrets -n openshift-adp
   ```

   ```
   NAME                                          TYPE     DATA   AGE
   v60f2ea6069e168346d5ad0e0b5faa59bb74946f      Opaque   1      3d20h
   vcc237eba032ad5c422fb939684eb633822d7900      Opaque   1      3d20h
   [...]
   ```

   Note: There will be one DR created secret for each managed cluster in the `openshift-adp` namespace.
5. Verify if the Data Protection Application (DPA) is already installed on each managed cluster in the OADP namespace `openshift-adp`. If not already created then follow the next step to create this resource.
   a. Create the DPA by copying the following YAML definition content to `dpa.yaml`.

   ```
   apiVersion: oadp.openshift.io/v1alpha1
   kind: DataProtectionApplication
   metadata:
     labels:
       app.kubernetes.io/component: velero
     name: velero
     namespace: openshift-adp
   spec:
     backupImages: false
     configuration:
       nodeAgent:
         enable: false
         uploaderType: restic
       velero:
         defaultPlugins:
           - openshift
           - aws
           - kubevirt
         noDefaultBackupLocation: true
   ```

   b. Create the DPA resource.

   ```
   oc create -f dpa.yaml -n openshift-adp
   ```

   ```
   dataprotectionapplication.oadp.openshift.io/velero created
   ```

   c. Verify that the OADP resources are created and are in `Running` state.

   ```
   oc get pods,dpa -n openshift-adp
   ```

   ```
   NAME                                                  READY   STATUS    RESTARTS   AGE
   pod/openshift-adp-controller-manager-7b64b74fcd-msjbs 1/1     Running   0          5m30s
   pod/velero-694b5b8f5c-b4kwg                           1/1     Running   0          3m31s


   NAME                                                AGE
   dataprotectionapplication.oadp.openshift.io/velero  3m31s
   ```

Creating a sample discovered application

To test failover from the **Primary managed cluster** to the **Secondary managed cluster** and relocate for discovered applications, you need a sample application that is installed without using the RHACM create application capability.

Before you begin

Note: Regional-DR now supports leveraging CephRBD volumes using non-default replica-2 storage classes that are managed by Fusion Data Foundation. Support has also been extended to compression-enabled storage classes.

Procedure

1. Log in to the **Primary managed cluster** and clone the sample application repository.

   ```
   git clone https://github.com/red-hat-storage/ocm-ramen-samples.git
   ```

2. Verify that you are on the main branch.

   ```
   cd ~/ocm-ramen-samples
   git branch
   * main
   ```

   The correct directory should be used when creating the sample application based on your scenario, metro or regional. To find your directory:

```
ls workloads/deployment | egrep -v 'cephfs|k8s|base'
```

```
odr-metro-rbd
odr-regional-rbd
odr-regional-cephfs
```

3. Create a project named `busybox-discovered` on both the **Primary and Secondary managed clusters**.

```
oc new-project busybox-discovered
```

4. Create the `busybox` application on the **Primary managed cluster**.
   This sample application example is for Regional-DR using a block (Ceph RBD) volume. CephFS volumes can be used as well.

```
oc apply -k workloads/deployment/odr-regional-rbd -n busybox-discovered
```

```
persistentvolumeclaim/busybox-pvc created
deployment.apps/busybox created
```

   Note:
   Fusion Data Foundation disaster recovery solution now extends protection to discovered applications that span across multiple namespaces.

5. Verify that busybox is running in the correct project on the **Primary managed cluster**.

```
oc get pods,pvc,deployment -n busybox-discovered
```

```
NAME                              READY   STATUS    RESTARTS   AGE
pod/busybox-796fccbb95-qmxjf      1/1     Running   0          18s


NAME                                    STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS                          VOLUMEATTRIBUTESCLASS   AGE
persistentvolumeclaim/busybox-pvc   Bound    pvc-b20e4129-902d-47c7-b962-040ad64130c4   1Gi        RWO            ocs-
storagecluster-ceph-rbd   <unset>                 18s


NAME                      READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/busybox   1/1     1            1           18
```

Enrolling a sample discovered application for disaster recovery protection

This section guides you on how to apply an existing DR Policy to a discovered application from the Protected applications tab.

Before you begin

- Ensure that Disaster Recovery has been configured and that at least one DR Policy has been created.

Procedure

1. On RHACM console, navigate to Disaster recovery > Protected applications tab.
2. Click Enroll application to start configuring existing applications for DR protection.
3. Select ACM discovered applications.
4. In the Namespace page, choose the DR cluster which is the name of the **Primary managed cluster** where `busybox` is installed.
5. Select namespace where the application is installed. For example, `busybox-discovered`.
   Note: If you have workload spread across multiple namespaces then you can select all of those namespaces to DR protect.
6. Choose a unique Name, for example `busybox-rbd`, for the discovered application and click Next.
7. In the Configuration page, select either Resource label or Recipe.
8. Resource label is used to protect your resources where you can set which resources will be included in the kubernetes-object backup and what volume's persistent data will be replicated. If you selected Resource label, provide label expressions and PVC label selector. Choose the label `appname=busybox` for both the kubernetes-objects and for the PVC(s).
9. If you selected Recipe, then from the Recipe list select the name of the recipe.
   Important: The recipe resource must be created in the application namespace on both managed clusters before enrolling an application for disaster recovery.
10. If you are using a recipe and want to add the recipe's parameters, under Recipe parameters, add the Key and Value (optional), then click Add parameter.
11. Click Next.
12. In the Replication page, select an existing DR Policy and the kubernetes-objects backup interval.
    Note: It is recommended to choose the same duration for the PVC data replication and kubernetes-object backup interval (for example, 5 minutes).
13. Click Next.
14. Review the configuration and click Save.
    To check the status, run the following command:

```
oc get drpc <project-name>-n openshift-dr-ops -oyaml
```

    Note: It may take up to 3–4 minutes for both application volumes (PVCs) and Kubernetes objects to reach a healthy state.
    Use the Back button to go back to the screen to correct any issues.

15. Verify that you can view the assigned policy details on the Applications page.
    a. On the Applications page, navigate to the DR Status column and view the status.
       The status can be any of the following:

       - Protecting: Shown during initial DR setup before data synchronization begins
       - Healthy: All volumes and Kubernetes resources are syncing normally
       - Warning: Volumes or Kubernetes resources are syncing slower than usual
       - Critical: Volumes or Kubernetes resources are not syncing
       - Protection Error: An error occurred during application protection setup

You can click the status to view the last synced time for application volumes and the DR policy assigned.

Note: It may take a few minutes for the DR Status to move from Protecting to Healthy after disaster recovery is enabled. The DR status will show Protecting during initial setup, then automatically transition to health-based statuses (Healthy/Warning/Critical) once data synchronization begins.

b. Failover and relocate status can be viewed in the DR Status column when initiated. These statuses can be clicked to view more details about the action. DR Status shows either Protecting, Healthy, Warning, or Critical when failover or relocate is not taking place. During failover or relocate operations, the status will show the operation-specific status (FailingOver, Relocating, and so on).

16. Verify that the volume consistency groups have been enabled from the Protected applications tab:
    a. Click the arrow next to the DRPC to expand the details.
    b. Click the number under the Volume Consistency groups column to open the modal.
    c. In the Volume consistency groups modal, verify the volume consistency group(s) has the correct volumes and a *Synced* status.
17. View the status of DRPC and VolumeReplicationGroup (VRG).
    a. To see the status of the DRPC, run the following command on the Hub cluster:

```
oc get drpc {drpc_name} -o wide -n openshift-dr-ops
```

The discovered applications store resources such as DRPlacementControl (DRPC) and Placement on the Hub cluster in a new namespace called **openshift-dr-ops**. The DRPC name can be identified by the unique Name configured in prior steps (for example, **busybox-rbd**).

    b. To see the status of the VRG for discovered applications, run the following command on the managed cluster where the busybox application was manually installed.

```
oc get vrg {vrg_name} -n openshift-dr-ops
```

The VRG resource is stored in the namespace **openshift-dr-ops** after a DR Policy is assigned to the discovered application. The VRG name can be identified by the unique Name configured in prior steps (for example, **busybox-rbd**).

18. Optional: For Rados block device (RBD), verify **volumereplication** and **volumegroupreplication** on the primary cluster.
For **volumereplication**:

```
oc get volumereplication -A
```

Example output:

```
NAMESPACE            NAME                                            AGE    VOLUMEREPLICATIONCLASS                    SOURCEKIND
SOURCENAME                                      DESIREDSTATE   CURRENTSTATE
busybox-discovered   vr-f9fc057c-5e95-47cb-b515-b4ee19d4ca8e   109s   rbd-volumereplicationclass-1625360775
VolumeGroupReplication   vgr-e8b7cca8211a41455790dadc2707a08c-busybox-rbd   primary        Primary
```

For **volumegroupreplication**:

```
oc get volumegroupreplication -A
```

Example output:

```
NAMESPACE            NAME                                                      VOLUMEGROUPREPLICATIONCLASS
VOLUMEGROUPREPLICATIONCONTENT                          DESIREDSTATE   CURRENTSTATE   AGE
busybox-discovered   vgr-e8b7cca8211a41455790dadc2707a08c-busybox-rbd   rbd-volumegroupreplicationclass-1625360775-
1602718344   vgrcontent-f9fc057c-5e95-47cb-b515-b4ee19d4ca8e   primary        Primary        113s
```

19. Optional: For CephFS, verify **ReplicationGroupSource** and **ReplicationSource** on the Primary cluster.
For **ReplicationGroupSource**:

```
oc get ReplicationGroupSource -A
```

Example output:

```
NAMESPACE        NAME                                                  LAST SYNC              DURATION          NEXT SYNC
SOURCE                                                                                                           LAST SYNC
START
busybox-cephfs   busybox-cephfs-bbd241b65b6589782fddaa82636e358d   2026-03-18T11:20:14Z   1m16.462467415s   2026-03-
18T11:25:00Z   {"matchLabels":{"ramendr.openshift.io/consistency-group":"busybox-cephfs-
bbd241b65b6589782fddaa82636e358d"}}
```

For **ReplicationSource**:

```
oc get ReplicationSource -A
```

Example output:

```
NAMESPACE        NAME          SOURCE           LAST SYNC              DURATION        NEXT SYNC
busybox-cephfs   busybox-pvc   vs-busybox-pvc   2026-03-18T11:20:14Z   16.450725976s
```

20. Optional: For CephFS, verify **ReplicationGroupDestination** and **ReplicationDestination** on the Secondary cluster.
For **ReplicationGroupDestination**:

```
oc get ReplicationGroupDestination -A
```

Example output:

```
NAMESPACE        NAME                                                  LAST SYNC              DURATION        LAST SYNC START
busybox-cephfs   busybox-cephfs-b1f2b5bac27b3d8805437424895d212c   2026-03-18T11:20:15Z   47.700328282s   2026-03-
18T11:20:15Z
```

For **ReplicationDestination**:

```
oc get ReplicationDestination -A
```

Example output:

```
NAMESPACE        NAME          LAST SYNC              DURATION         NEXT SYNC
busybox-cephfs   busybox-pvc   2026-03-18T11:20:15Z   47.401909636s
```

Failover disaster recovery protected discovered application

Fail over a disaster recovery-protected discovered application to the secondary managed cluster and complete the required cleanup on the primary managed cluster.

Before you begin

- Ensure that the application namespace is created in both managed clusters (for example, `busybox-discovered`).
- Never initiate a failover operation of an application when one or both resource types are in a Warning or Critical status.

About this task

This procedure guides you on how to fail over a discovered application that is disaster recovery protected.

Procedure

1. In the RHACM console, navigate to Disaster Recovery > Protected applications tab.
2. At the end of the application row, click on the Actions menu and choose to initiate Failover.
3. In the Failover application modal window, review the status of the application and the target cluster.
   Important:
   If there are data inconsistencies caused by synchronization delays, a warning message appears stating **Inconsistent data on target cluster**. This alerts to the possibility of data loss if the failover is initiated. The message is no longer displayed when data synchronization is complete.
4. Click Initiate.
   Important:
   If you initiate a failover while either the primary or secondary site is down, the disaster recovery status (DR Status) displays as critical and a warning message appears: Volumes and Kubernetes resources are not syncing. This is expected behavior. After you verify that the persistent volume claims (PVCs), deployments, applications, and virtual machine (VM) instances are running correctly, you can continue to use the applications.
5. In the DR status column, you can view the progress of the failover. The modal shows the progress of the failover as Preparing > Failover > Restoring > Clean up.
6. Verify that the busybox application is running on the **Secondary managed cluster**.

   ```
   oc get pods,pvc -n busybox-discovered
   ```

   ```
   NAME                          READY   STATUS    RESTARTS   AGE
   pod/busybox-796fccbb95-qmxjf   1/1     Running   0          2m46s


   NAME                                STATUS   VOLUME                                     CAPACITY   ACCESS MODES
   STORAGECLASS                     VOLUMEATTRIBUTESCLASS   AGE
   persistentvolumeclaim/busybox-pvc   Bound    pvc-b20e4129-902d-47c7-b962-040ad64130c4   1Gi        RWO              ocs-
   storagecluster-ceph-rbd   <unset>                 2m57s


   NAME                                                               AGE     VOLUMEREPLICATIONCLASS                   PVCNAME
   DESIREDSTATE   CURRENTSTATE
   volumereplication.replication.storage.openshift.io/busybox-pvc   2m45s   rbd-volumereplicationclass-1625360775   busybox-
   pvc    primary         Primary
   ```

7. Check the progression status of Failover until the result is `WaitOnUserToCleanup`. The DRPC name can be identified by the unique Name configured in prior steps (for example, `busybox-rbd`).

   ```
   oc get drpc {drpc_name} -n openshift-dr-ops -o jsonpath='{.status.progression}{"\n"}'
   ```

   ```
   WaitOnUserToCleanUp
   ```

   Important:
   If the result doesn't appear as `WaitOnUserToCleanup` and instead moves to `Completed`, you might encounter the error "Secondary cluster transition failed as PVC is potentially in use by a pod." This indicates that application cleanup was not completed on the primary cluster. In this case, complete the cleanup on the primary cluster to release the PVC. After cleanup, replication resumes automatically. If replication does not resume, perform a relocate operation to move the application to the secondary cluster and continue operations with the last synchronized data.
8. Remove the `busybox` application from the Primary managed cluster to complete the *Failover* process.
   a. Go to the Protected applications tab. You will see a message to remove the application.
   b. Go to the cloned repository for `busybox` and run the following commands on the Primary managed cluster where you failed over from. Use the same directory that was used to create the application (for example, `odr-regional-rbd`).

      ```
      cd ~/ocm-ramen-samples/
      git branch
      ```

      ```
      * main
      ```

      ```
      oc delete -k workloads/deployment/odr-regional-rbd -n busybox-discovered
      ```

      ```
      persistentvolumeclaim "busybox-pvc" deleted
      deployment.apps "busybox" deleted
      ```
9. After deleting the application, navigate to the Protected applications tab and verify that the busybox resources are both in *Healthy* status.

Relocate disaster recovery protected discovered application

Relocate a disaster recovery protected discovered application from a secondary managed cluster to a primary managed cluster by using the Red Hat Advanced Cluster Management (RHACM) console.

Before you begin

- Ensure that the application namespace is created in both managed clusters (for example, `busybox-discovered`).
- Never initiate a relocate operation of an application when one or both resource types are in a Warning or Critical status.

About this task

This procedure guides you on how to failover a discovered application which is disaster recovery protected.

Procedure

1. In the RHACM console, navigate to Disaster Recovery > Protected applications tab.
2. At the end of the application row, click on the Actions menu and choose to initiate Relocate.
3. In the Relocate application modal window, review the status of the application and the target cluster.
   Important:
   If there are data inconsistencies caused by synchronization delays, a warning message appears stating **Inconsistent data on target cluster**. This alerts to the possibility of data loss if the failover is initiated. The message is no longer displayed when data synchronization is complete.
4. Click Initiate.
5. In the DR status column, you can view the progress of the relocation. The progress shows Preparing > Clean up, at which time you need to perform the following clean up activities.
6. The DRPC name can be identified by the unique Name configured in prior steps (for example, `busybox-rbd`).

   ```
   oc get drpc {drpc_name} -n openshift-dr-ops -o jsonpath='{.status.progression}{"\n"}'
   ```

   ```
   WaitOnUserToCleanUp
   ```

7. Remove the busybox application from the **Secondary managed cluster** before *Relocate* to the **Primary managed cluster** is completed
   a. Go to the cloned repository for `busybox` and run the following commands on the Secondary managed cluster where you relocated from. Use the same directory that was used to create the application (for example, `odr-regional-rbd`).

      ```
      cd ~/ocm-ramen-samples/
      git branch
      ```

      ```
      * main
      ```

      ```
      oc delete -k workloads/deployment/odr-regional-rbd -n busybox-discovered
      ```

      ```
      persistentvolumeclaim "busybox-pvc" deleted
      deployment.apps "busybox" deleted
      ```

8. After deleting the application, navigate to the Protected applications tab and verify that the busybox resources are both in *Healthy* status.
9. Verify that the busybox application is running on the **Primary managed cluster**.

   ```
   oc get pods,pvc -n busybox-discovered
   ```

   ```
   NAME                           READY   STATUS    RESTARTS   AGE
   pod/busybox-796fccbb95-qmxjf   1/1     Running   0          2m46s
   ```

   ```
   NAME                                    STATUS   VOLUME                                     CAPACITY   ACCESS MODES
   STORAGECLASS                       VOLUMEATTRIBUTESCLASS   AGE
   persistentvolumeclaim/busybox-pvc   Bound    pvc-b20e4129-902d-47c7-b962-040ad64130c4   1Gi        RWO            ocs-
   storagecluster-ceph-rbd   <unset>                 2m57s
   ```

   ```
   NAME                                                              AGE     VOLUMEREPLICATIONCLASS                 PVCNAME
   DESIREDSTATE   CURRENTSTATE
   volumereplication.replication.storage.openshift.io/busybox-pvc   2m45s   rbd-volumereplicationclass-1625360775   busybox-
   pvc    primary        Primary
   ```

10. After the clean up activities are performed, the DR Status column will move from Syncing > Restoring.

Disable disaster recovery for protected applications

This section guides you to disable disaster recovery resources when you want to delete the protected applications or when the application no longer needs to be protected.

Procedure

1. Login to the Hub cluster.
2. List the `DRPlacementControl` (DRPC) resources. Each DRPC resource was created when the application was assigned a DR policy.

   ```
   oc get drpc -n openshift-dr-ops
   ```

3. Find the **DRPC** that has a name that includes the unique identifier that you chose when assigning a DR policy (for example, `busybox-rbd`) and delete the DRPC.

```
oc delete {drpc_name} -n openshift-dr-ops
```

4. List the **Placement** resources. Each Placement resource was created when the application was assigned a DR policy.

```
oc get placements -n openshift-dr-ops
```

5. Find the **Placement** that has a name that includes the unique identifier that you chose when assigning a DR policy (for example, `busybox-rbd-placement-1`) and delete the Placement.

```
oc delete placements {placement_name} -n openshift-dr-ops
```

Virtual machine garbage collection automation

Virtual machine garbage collection automation provides an automated mechanism to clean up virtual machine resources on the source cluster during disaster recovery failover or relocation workflows, reducing manual effort and supporting lower recovery time objectives.

Overview

Virtual machine (VM) garbage collection automation provides an automated mechanism to clean up VM resources on the source cluster during Disaster Recovery (DR) failover or relocation workflows. This reduces manual effort and helps users perform DR operations more quickly, supporting lower Recovery Time Objectives (RTO).

Prerequisites

- Virtual machine protection must be enabled through the Fusion Data Foundation Virtualization console.

How it works

During DR failover or relocation, VM resources that are protected and orchestrated by the Fusion Data Foundation Virtualization console are automatically deleted from the source cluster.

This deletion uses Kubernetes' owner-based garbage collection principles, ensuring that associated child resources, such as disks and pods, are also removed cleanly as part of normal Kubernetes garbage collection.

Known risks and limitations

- Automation applies only to VM resources whose protection is managed by the Fusion Data Foundation Virtualization console.
- Other discovered application workload types, including GitOps-managed and other non-VM resources, remain out of scope and require manual cleanup.
- Automation is limited to resources directly owned by the VM lifecycle.

Viewing Recovery Point Objective values for disaster recovery enabled applications

Recovery Point Objective (RPO) value is the most recent sync time of persistent data from the cluster where the application is currently active to its peer. This sync time helps determine duration of data lost during failover.

About this task

You can view the Recovery Point Objective (RPO) value of all the protected volumes for their workload on the Hub cluster.
Note: This RPO value is applicable only for Regional-DR during failover. Relocation ensures there is no data loss during the operation, as all peer clusters are available.

Procedure

1. On the Hub cluster, navigate to Applications > Overview tab.
2. In the Data policy column, click the policy link for the application you applied the policy to.
   A Data Policies modal page appears with the number of disaster recovery policies applied to each application along with failover and relocation status.
3. On the Data Policies modal page, click the View more details link.
   A detailed **Data Policies** modal page is displayed that shows the policy names and the ongoing activities (Last sync, Activity status) associated with the policy that is applied to the application.

   The **Last sync time** reported in the modal page, represents the most recent sync time of all volumes that are DR protected for the application.

Enabling granular disaster recovery for individual or groups of virtual machines in a namespace [Technology preview]

You can enable granular disaster recovery (DR) for individual virtual machines (VMs) or VM groups within the same namespace, allowing independent failover and relocation actions.

Important:

Granular disaster recovery for individual or groups of virtual machines in a namespace is a Technology Preview feature and is subject to Technology Preview support limitations. Technology Preview features are not supported with Red Hat production service level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using them in production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process.

For more information, see Technology Preview Features Support Scope.

Key highlights

- **Granular VM Control**:

  Each discovered and ACM-managed VM can have its own DR policy, enabling independent DR operations without impacting other VMs in the namespace.

- **Namespace Partitioning**:

  Supports multiple DRPCs within a namespace, avoiding the limitations of namespace-level DR protection.

- **Scenarios Supported**:

  - Independent DR for discovered and ACM-managed VMs.
  - Namespace-level DR remains for non-VM applications.
- **Upgrade Considerations**:

  - Existing DRPCs are treated as namespace-level.
  - Users must disable DR before switching to VM-specific protection.
- **Manage disaster recovery of virtual machines from Virtual Machines page**
  Note: Existing UI for namespace-level DR remains unchanged.

Important:
During failover of a CNV discovered application, if the primary managed cluster is down and other CNV applications are running in different namespaces on either of the managed clusters, there have been cases where the VM fails to start on the failover cluster. This issue is due to MAC address conflicts.

To prevent this issue, it is recommended to configure disjoint MAC address ranges for each cluster from the Day 1. For example:

```
oc patch configmap kubemacpool-mac-range-config -n openshift-cnv --type merge -p '{"data":
{"RANGE_START":"02:00:00:00:00:00","RANGE_END":"02:00:FF:ff:FF:FF"}}'
```

```
oc patch configmap kubemacpool-mac-range-config -n openshift-cnv --type merge -p '{"data":
{"RANGE_START":"02:01:00:00:00:00","RANGE_END":"02:01:FF:FF:FF:FF"}}'
```

Then restart the `kubemacpool-mac-controller-manager`.

Command example:

```
oc get pods -n openshift-cnv|grep kubemacpool-mac-controller-manager
```

Example output:

```
kubemacpool-mac-controller-manager-544ff96b4f-dgb7x    2/2     Running   3               30h
```

Monitoring disaster recovery health

This section provides all the necessary configuration and commands for setting up the disaster recovery dashboard that help to monitor the health of your disaster recovery solution.

- **Enable monitoring for disaster recovery**
  Enable basic monitoring for your disaster recovery setup by adding a cluster-monitoring label to the `openshift-operators` namespace on the Hub cluster.
- **Enabling disaster recovery dashboard on Hub cluster**
  Enable the disaster recovery dashboard on the Hub cluster by creating a custom metrics allowlist configmap so that RHACM observability collects Fusion Data Foundation metrics from managed clusters.
- **Viewing health status of disaster recovery replication relationships**
  View the health status of disaster recovery replication relationships from the Hub cluster dashboard to monitor operator health, cluster status, and DR policy details for protected applications.
- **Viewing health status of ApplicationSet-based and Subscription-based applications**
  Learn how to view the health status of ApplicationSet-based and Subscription-based applications to identify replication delays and troubleshoot issues.
- **Disaster recovery metrics**
  Ramen disaster recovery metrics, scraped by Prometheus from the hub cluster, provide per-application data on synchronization timestamps, duration, bytes transferred, and workload protection status.
- **Disaster recovery alerts**
  Reference list of disaster recovery alerts for Fusion Data Foundation.

Enable monitoring for disaster recovery

Enable basic monitoring for your disaster recovery setup by adding a cluster-monitoring label to the `openshift-operators` namespace on the Hub cluster.

About this task

Use this procedure to enable basic monitoring for your disaster recovery setup.

Procedure

1. On the Hub cluster, open a terminal window.
2. Add the following label to **openshift-operator** namespace.

```
oc label namespace openshift-operators openshift.io/cluster-monitoring='true'
```

Enabling disaster recovery dashboard on Hub cluster

Enable the disaster recovery dashboard on the Hub cluster by creating a custom metrics allowlist configmap so that RHACM observability collects Fusion Data Foundation metrics from managed clusters.

Before you begin

- Ensure that you have already installed the following:
    - Red Hat® OpenShift® Container Platform 4.18 or higher and have administrator privileges.
    - ODF Multicluster Orchestrator with the console plugin enabled.
    - Red Hat Advanced Cluster Management for Kubernetes (RHACM) 2.12 or higher from Operator Hub. For instructions on how to install, see Installation and upgrade > Install in Red Hat Advanced Cluster Management for Kubernetes product documentation.
- Ensure you have enabled observability on RHACM. See Operations > Observability > Observability service > Enabling the Observability service > Enabling Observability from the command line interface in Red Hat Advanced Cluster Management for Kubernetes product documentation.

Procedure

1. On the Hub cluster, open a terminal window and perform the next steps.
2. Create the configmap file named **observability-metrics-custom-allowlist.yaml**.
   You can use the following YAML to list the disaster recovery metrics on Hub cluster. For details, see Operations > Observability > Observability service > Observability advanced confihuration > Adding custom metrics in Red Hat Advanced Cluster Management for Kubernetes product documentation. To know more about ramen metrics, see Disaster recovery metrics.

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: observability-metrics-custom-allowlist
  namespace: open-cluster-management-observability
data:
  metrics_list.yaml: |
    names:
      - ceph_rbd_mirror_snapshot_sync_bytes
      - ceph_rbd_mirror_snapshot_snapshots
    matches:
      - __name__="csv_succeeded",exported_namespace="openshift-dr-system",name=~"odr-cluster-operator.*"
      - __name__="csv_succeeded",exported_namespace="openshift-operators",name=~"volsync.*"
```

3. In the **open-cluster-management-observability** namespace, run the following command:

```
oc apply -n open-cluster-management-observability -f observability-metrics-custom-allowlist.yaml
```

4. After **observability-metrics-custom-allowlist** yaml is created, RHACM will start collecting the listed Fusion Data Foundation metrics from all the managed clusters.
   To exclude a specific managed cluster from collecting the observability data, add the following cluster label to the **clusters: observability: disabled**.

Viewing health status of disaster recovery replication relationships

View the health status of disaster recovery replication relationships from the Hub cluster dashboard to monitor operator health, cluster status, and DR policy details for protected applications.

Before you begin

Ensure that you have enabled the disaster recovery dashboard for monitoring. For instructions, see chapter Enabling disaster recovery dashboard on Hub cluster.

Procedure

1. On the Hub cluster, ensure that All Clusters option is selected.
2. Refresh the console to make the DR monitoring dashboard tab accessible.
3. Go to Data Services and click Data policies.
4. On the Overview tab, you can view the health status of the operators, clusters, and applications. Green tick indicates that the operators are running and available.
5. Click Disaster recovery tab to view a list of DR policy details and connected applications.

Viewing health status of ApplicationSet-based and Subscription-based applications

Learn how to view the health status of ApplicationSet-based and Subscription-based applications to identify replication delays and troubleshoot issues.

Procedure

1. On the Hub cluster, go to All Clusters > Applications.
2. In the DR Status column of the application, view the status:
    - Healthy: The last group sync time is less than 2X that of the sync interval.
    - Warning: The last group sync time is greater than 2X and less than 3X of the sync interval.
    - Critical: The last group sync time is greater than 3X of the sync interval.

What to do next

You can also view the application status for virtual machine (VM) based workloads:

- On the Hub cluster, go to All Clusters > Infrastructure > Virtual Machines > Status column.

Disaster recovery metrics

Ramen disaster recovery metrics, scraped by Prometheus from the hub cluster, provide per-application data on synchronization timestamps, duration, bytes transferred, and workload protection status.

These are the ramen metrics that are scrapped by prometheus.

- ramen_last_sync_timestamp_seconds
- ramen_policy_schedule_interval_seconds
- ramen_last_sync_duration_seconds
- ramen_last_sync_data_bytes
- ramen_workload_protection_status

Run these metrics from the Hub cluster where Red Hat Advanced Cluster Management for Kubernetes (RHACM operator) is installed.

Last synchronization timestamp in seconds

This is the time in seconds which gives the time of the most recent successful synchronization of all PVCs per application.

Metric name
: **`ramen_last_sync_timestamp_seconds`**

Metrics type
: Gauge

Labels

- **`ObjType`**: Type of the object, here its DPPC
- **`ObjName`**: Name of the object, here it is DRPC-Name
- **`ObjNamespace`**: DRPC namespace
- **`Policyname`**: Name of the DRPolicy
- **`SchedulingInterval`**: scheduling interval value from DRPolicy

Metric value
: Value is set as Unix seconds which is obtained from **`lastGroupSyncTime`** from DRPC status.

Policy schedule interval in seconds

This gives the scheduling interval in seconds from DRPolicy.

Metric name
: **`ramen_policy_schedule_interval_seconds`**

Metrics type
: Gauge

Labels

- **`Policyname`**: Name of the DRPolicy

Metric value
: Set to scheduling interval in seconds which is taken from DRPolicy.

Last synchronization duration in seconds

This represents the longest time taken to sync from the most recent successful synchronization of all PVCs per application.

Metric name
: **`ramen_last_sync_duration_seconds`**

Metrics type
: Gauge

Labels

- **`obj_type`**: Type of the object, here its DPPC

- `obj_name`: Name of the object, here it is DRPC-Name
- `obj_namespace`: DRPC namespace
- `scheduling_interval`: Scheduling interval value from DRPolicy

Metric value
: The value is taken from **`lastGroupSyncDuration`** from DRPC status.

Total bytes transferred from most recent synchronization

This value represents the total bytes transferred from the most recent successful synchronization of all PVCs per application.

Metric name
: **`ramen_last_sync_data_bytes`**

Metrics type
: Gauge

Labels

- `obj_type`: Type of the object, here its DPPC
- `obj_name`: Name of the object, here it is DRPC-Name
- `obj_namespace`: DRPC namespace

Metric value
: The value is taken from **`lastGroupSyncBytes`** from DRPC status.

Workload protection status

This value provides the application protection status per application that is DR protected.

Metric name
: **`ramen_workload_protection_status`**

Metrics type
: Gauge

Labels

- `obj_type`: Type of the object, here its DPPC
- `obj_name`: Name of the object, here it is DRPC-Name
- `obj_namespace`: DRPC namespace
- `scheduling_interval`: Scheduling interval value from DRPolicy

Metric value
: The value is either a "1" or a "0", where "1" indicates application DR protection is healthy and a "0" indicates application protection degraded and potentially unprotected.

Disaster recovery alerts

Reference list of disaster recovery alerts for Fusion Data Foundation.

This section provides a list of all supported alerts associated with IBM Fusion Data Foundation within disaster recovery environment.

Recording rules

- Record: **`ramen_sync_duration_seconds`**

  Expression

  ```
  sum by (obj_name, obj_namespace, obj_type, job, policyname)(time() - (ramen_last_sync_timestamp_seconds > 0))
  ```

  Purpose
  : The time interval between the volume group's last sync time and the time now in seconds.

- Record: **`ramen_rpo_difference`**

  Expression

  ```
  ramen_sync_duration_seconds{job="ramen-hub-operator-metrics-service"} / on(policyname, job) group_left() (ramen_policy_sc
  ```

  Purpose
  : The difference between the expected sync delay and the actual sync delay taken by the volume replication group.

- Record: **`count_persistentvolumeclaim_total`**

  Expression

  ```
  count(kube_persistentvolumeclaim_info)
  ```

  Purpose
  : Sum of all PVC from the managed cluster.

Alerts

- Alert: **`VolumeSynchronizationDelay`**

Impact
: Critical

Purpose
: Actual sync delay taken by the volume replication group is thrice the expected sync delay.

YAML

```
alert: VolumeSynchronizationDelay
expr: ramen_rpo_difference >= 3
for: 5s
labels:
  severity: critical
annotations:
  description: "The syncing of volumes is exceeding three times the scheduled snapshot interval, or the volumes have been recently protected. (DRPC: {{ $labels.obj_name }}, Namespace: {{ $labels.obj_namespace }})"
  alert_type: "DisasterRecovery"
```

- Alert: `VolumeSynchronizationDelay`

  Impact
  : Warning

  Purpose
  : Actual sync delay taken by the volume replication group is twice the expected sync delay.

  YAML

  ```
  alert: VolumeSynchronizationDelay
  expr: ramen_rpo_difference > 2 and ramen_rpo_difference < 3
  for: 5s
  labels:
    severity: warning
  annotations:
    description: "The syncing of volumes is exceeding two times the scheduled snapshot interval, or the volumes have been recently protected. (DRPC: {{ $labels.obj_name }}, Namespace: {{ $labels.obj_namespace }})"
    alert_type: "DisasterRecovery"
  ```

- Alert: `WorkloadUnprotected`

  Impact
  : Warning

  Purpose
  : Application protection status is degraded for more than 10 minutes.

  YAML

  ```
  alert: WorkloadUnprotected
  expr: ramen_workload_protection_status == 0
  for: 10m
  labels:
    severity: warning
  annotations:
    description: "Workload is not protected for disaster recovery (DRPC: {{ $labels.obj_name }}, Namespace: {{ $labels.obj_namespace }})."
    alert_type: "DisasterRecovery"
  ```

Troubleshooting disaster recovery

This troubleshooting section provides guidance or workarounds on how to fix some of the disaster recovery configuration issues.

- **Troubleshooting Regional-DR**
  Administrators can use this troubleshooting information to understand how to troubleshoot and fix their Regional-DR solution.

Troubleshooting Regional-DR

Administrators can use this troubleshooting information to understand how to troubleshoot and fix their Regional-DR solution.

- **rbd-mirror daemon health is in warning state**
  If the `rbd-mirror` daemon health reports a `warning` state after a network disconnection, the daemon may have a startup issue caused by unreliable connectivity to the secondary cluster.
- **volsync-rsync-src pod is in error state as it is unable to resolve the destination hostname**
  If the `volsync-rsync-src` pod enters an error state with an SSH hostname resolution failure, the `submariner-lighthouse-agent` may need to be restarted on both nodes.
- **Cleanup and data sync for ApplicationSet workloads remain stuck after older primary managed cluster is recovered post failover**
  Troubleshoot DR policies that protect all applications in the same namespace.
- **VM recipe known issues**
  Lists known VM recipe issues.

rbd-mirror daemon health is in warning state

If the **rbd-mirror** daemon health reports a **warning** state after a network disconnection, the daemon may have a startup issue caused by unreliable connectivity to the secondary cluster.

Problem

There appears to be numerous cases where WARNING gets reported if mirror service **::get_mirror_service_status** calls **Ceph** monitor to get service status for **rbd-mirror**.

Following a network disconnection, **rbd-mirror** daemon health is in the **warning** state while the connectivity between both the managed clusters is fine.

Resolution

Run the following command in the toolbox and look for **leader:false**

```
rbd mirror pool status --verbose ocs-storagecluster-cephblockpool | grep 'leader:'
```

If you see the following in the output:

Output	Workaround
leader: false	It indicates that there is a daemon startup issue and the most likely root cause could be due to problems reliably connecting to the secondary cluster. Workaround: Move the **rbd-mirror** pod to a different node by simply deleting the pod and verify that it has been rescheduled on another node.
leader: true or no output	Contact IBM Support

BZ reference: [2118627]

volsync-rsync-src pod is in error state as it is unable to resolve the destination hostname

If the **volsync-rsync-src** pod enters an error state with an SSH hostname resolution failure, the **submariner-lighthouse-agent** may need to be restarted on both nodes.

Problem

VolSync source pod is unable to resolve the hostname of the VolSync destination pod. The log of the VolSync Pod consistently shows an error message over an extended period of time similar to the following log snippet.

```
oc logs -n busybox-workloads-3-2 volsync-rsync-src-dd-io-pvc-1-p25rz
```

Example output

```
VolSync rsync container version: ACM-0.6.0-ce9a280
Syncing data to volsync-rsync-dst-dd-io-pvc-1.busybox-workloads-3-2.svc.clusterset.local:22 ...
ssh: Could not resolve hostname volsync-rsync-dst-dd-io-pvc-1.busybox-workloads-3-2.svc.clusterset.local: Name or service not known
```

Resolution

Restart **submariner-lighthouse-agent** on both nodes.

```
oc delete pod -l app=submariner-lighthouse-agent -n submariner-operator
```

Cleanup and data sync for ApplicationSet workloads remain stuck after older primary managed cluster is recovered post failover

Troubleshoot DR policies that protect all applications in the same namespace.

Problem

ApplicationSet based workload deployments to managed clusters are not garbage collected in cases when the hub cluster fails. It is recovered to a standby hub cluster, while the workload has been failed over to a surviving managed cluster. The cluster that the workload was failed over from, rejoins the new recovered standby hub.

ApplicationSets that are DR protected, with a regional DRPolicy, hence starts firing the **VolumeSynchronizationDelay** alert. Further such DR protected workloads cannot be failed over to the peer cluster or relocated to the peer cluster as data is out of sync between the two clusters.

Resolution

The workaround requires that

openshift-gitops
operators can own the workload resources that are orphaned on the managed cluster that rejoined the hub post a failover of the workload was performed from the new recovered hub. To achieve this the following steps can be taken:

1. Determine the Placement that is in use by the ArgoCD ApplicationSet resource on the hub cluster in the **openshift-gitops** namespace.
2. Inspect the placement label value for the ApplicationSet in this field: **spec.generators.clusterDecisionResource.labelSelector.matchLabels**. This would be the name of the Placement resource *<placement-name>*
3. Ensure that there exists a **PlacemenDecision** for the ApplicationSet referenced **Placement**.

```
oc get placementdecision -n openshift-gitops --selector cluster.open-cluster-management.io/placement=<placement-name>
```

This results in a single `PlacementDecision` that places the workload in the currently desired failover cluster.

4. Create a new `PlacementDecision` for the ApplicationSet pointing to the cluster where it should be cleaned up.

   For example:

```
apiVersion: cluster.open-cluster-management.io/v1beta1
kind: PlacementDecision
metadata:
  labels:
    cluster.open-cluster-management.io/decision-group-index: "1" # Typically one higher than the same value in the 
esisting PlacementDecision determined at step (2)
    cluster.open-cluster-management.io/decision-group-name: ""
    cluster.open-cluster-management.io/placement: cephfs-appset-busybox10-placement
  name: <placemen-name>-decision-<n> # <n> should be one higher than the existing PlacementDecision as determined in 
step (2)
  namespace: openshift-gitops
```

5. Update the newly created `PlacementDecision` with a status **subresource**.

```
decision-status.yaml:
status:
  decisions:
  - clusterName: <managedcluster-name-to-clean-up> # This would be the cluster from where the workload was failed 
over, NOT the current workload cluster
    reason: FailoverCleanup
```

```
oc patch placementdecision -n openshift-gitops <placemen-name>-decision-<n> --patch-file=decision-status.yaml --
subresource=status --type=merge
```

6. Watch and ensure that the Application resource for the ApplicationSet has been placed on the desired cluster.

```
oc get application -n openshift-gitops  <applicationset-name>-<managedcluster-name-to-clean-up>
```

   In the output, check if the SYNC STATUS shows as `Synced` and the HEALTH STATUS shows as `Healthy`.

7. Delete the PlacementDecision that was created in step (3), such that ArgoCD can garbage collect the workload resources on the <managedcluster-name-to-clean-up>

```
oc delete placementdecision -n openshift-gitops <placemen-name>-decision-<n>
```

ApplicationSets that are DR protected, with a regional DRPolicy, stops firing the `VolumeSynchronizationDelay` alert.

BZ reference: [2268594]

VM recipe known issues

Lists known VM recipe issues.

`ramendr.openshift.io/k8s-resource-selector` label is not automatically applied when a PVC is added to an already protected workload NS

Problem
: When an additional PV or PVC is attached to an already protected VM on Day 2, the newly created PV/PVC is not automatically protected.

Resolution
: To ensure the new PV/PVC is protected, manually add the following label on the new PV/PVC:

```
("ramendr.openshift.io/k8s-resource-selector": "<value same as that of base PVC of associated VM>")
```

This label links the new PV/PVC to the existing protected group of resources.

DRPolicy without flattening applied to a cloned volume leads to errors

Problem
: Regional-DR currently does not support volumes created from CSI snapshots or clones in protected applications. These volumes rely on a parent volume that isn't replicated.

Resolution
: To avoid these issues, use a DRPolicy that flattens PVs before replication if cloning is required.

For more information, see this knowledgebase article.

Disabling disaster recovery for a disaster recovery-enabled application

Learn how to disable disaster recovery for disaster recovery-enabled managed and discovered applications.

- **Disabling disaster recovery for managed applications**
  Use this procedure to disable disaster recovery for disaster recovery-managed applications.

- **Disabling disaster recovery for discovered applications**
  Use this procedure to disable disaster recovery for disaster recovery-discovered applications.

Disabling disaster recovery for managed applications

Use this procedure to disable disaster recovery for disaster recovery-managed applications.

Procedure

1. In the RHACM console, go to All Clusters > Data Services > Protected applications tab.
2. At the end of the application row, click Acions menu and select Remove disaster recovery.
3. Click Remove.

Results

Your application loses disaster recovery protection, preventing volume synchronization (replication) between clusters.

The application is no longer in the Protected applications tab when the disaster recovery is removed.

Disabling disaster recovery for discovered applications

Use this procedure to disable disaster recovery for disaster recovery-discovered applications.

Procedure

1. In the RHACM console, go to All Clusters > Data Services > Protected applications tab.
2. At the end of the application row, click Acions menu and select Remove disaster recovery.
3. Click Remove.

Results

Your application loses disaster recovery protection, preventing volume synchronization (replication) between clusters.

The application is no longer in the Protected applications tab when the disaster recovery is removed.

Uninstalling disaster recovery from hub and managed clusters

Learn how to uninstall disaster recovery (DR) partially or completely from hub and managed clusters.

You can uninstall Regional-DR in the following ways:

Partial uninstallation

Partial uninstallation involves removing all DR resources and configurations from two or more peer-managed clusters that are defined in a DRPolicy.

Complete uninstallation

Complete uninstallation is a full uninstallation of the DR setup and includes the following steps:

- Remove all DRPolicies and mirrorpeers from all peer clusters.
- Remove the DR operators and associated resources from the hub cluster.

- **Prerequisites for disaster recovery uninstallation**
  Before uninstalling disaster recovery (DR), first remove disaster recovery protection from all protected applications, both managed and discovered.
- **Uninstalling disaster recovery for peer-managed clusters**
  Learn how to uninstall disaster recovery (DR) resources for the peer-managed clusters.
- **Uninstalling disaster recovery operators**
  Learn how to uninstall disaster recovery (DR) completely from DR operators.

Prerequisites for disaster recovery uninstallation

Before uninstalling disaster recovery (DR), first remove disaster recovery protection from all protected applications, both managed and discovered.

Before you begin

Ensure that both managed clusters are online and healthy.

Procedure

1. Find all DRPolicies associated with the peer clusters you want to remove DR from using one of the following options:
   - **Using the RHACM console**:
     - On the hub cluster, go to All Clusters > Data Services > Disaster Recovery > Policies.
     - Identify any DRPolicies associated with the peer clusters you want to remove DR from.
   - **Using the CLI**:
     - Get the names of the peer clusters from the RHACM console. On the hub cluster, go to All Clusters > Infrastructure > Clusters.
     - Run the following commands on the hub cluster, replacing the cluster names with the peer clusters found in the previous step:

       ```
       CLUSTER_A="{cluster1_name}"
       ```

       ```
       CLUSTER_B="{cluster2_name}"
       ```

       ```
       oc get drpolicies -o json | jq -r --arg c1 "$CLUSTER_A" --arg c2 "$CLUSTER_B" '.items[]
       |select((.spec.drClusters | sort) == ([$c1, $c2] | sort)) .metadata.name'
       ```

     - Identify any DRPolicies associated with the peer clusters you want to remove DR from.
2. Find any existing DRPlacementControl (DRPC) resources that are associated with your peer-managed clusters.
   Run the following command on the hub cluster for each DRPolicy identified in step 1:

   ```
   DRPOLICY_NAME="{drpolicy_name}"
   ```

   Replace `drpolicy_name` with the DRPolicy.

   ```
   oc get drplacementcontrols -A -o json | jq -r --arg drpc "$DRPOLICY_NAME" \ '.items[] | select(.spec.drPolicyRef.name ==
   $drpc) | "\(.metadata.namespace)/\(.metadata.name)"'
   ```

3. If any DRPCs are found, remove DR protection from the associated applications.
   a. Remove DR protection from managed applications.
      i. On the hub cluster, go to All Clusters > Applications.
      ii. In the Overview tab, at the end of the protected application row from the action menu (⋮), select Manage disaster recovery.
      iii. Click Remove disaster recovery.
      iv. Click Confirm remove.
   b. Remove DR protection from discovered applications.
      i. In the RHACM console, go to All Clusters > Data Services > Protected applications tab.
      ii. At the end of the application row, click the action menu (⋮) and select Remove disaster recovery.
      iii. Click Remove in the next prompt.
4. Run the following command on the hub cluster to find the associated mirrorpeer:

   ```
   export DR_POLICY={drpolicy_name}
   ```

   Replace `drpolicy_name` with one of the DRPolicy names found in step 1.

   ```
   DR_CLUSTERS=$(oc get drpolicy $DR_POLICY -o json | jq -r '.spec.drClusters | sort | join(",")')
   ```

   ```
   oc get mirrorpeers -A -o json | jq -r --arg dr_clusters "$DR_CLUSTERS" '.items[] | {name: .metadata.name,clusters:
   (.spec.items | map(.clusterName) | sort | join(",")) } |select(.clusters == $dr_clusters) | .name'
   ```

Uninstalling disaster recovery for peer-managed clusters

Learn how to uninstall disaster recovery (DR) resources for the peer-managed clusters.

Before you begin

Ensure that the cluster names, DRPolicies, and mirrorpeer are identified as instructed in Prerequisites for disaster recovery uninstallation.

Procedure

1. From the hub cluster, delete DRPolices associated with the peer-managed clusters that are slated for DR removal.
   The DRPolicies that are found in step 1 of Prerequisites for disaster recovery uninstallation needs to be deleted.

   Command example:

   ```
   oc delete drpolicy {drpolicy_name1 drpolicy_name2 ...}
   ```

   Replace `drpolicy_name1` with the DRPolicy names that are found in step 2 of Prerequisites for disaster recovery uninstallation needs to be deleted.

2. From the hub cluster, delete the mirrorpeer that is associated with the peer-managed clusters that are slated for DR removal.
   One mirrorpeer per DRPolicy that are found in step 4 of Prerequisites for disaster recovery uninstallation needs to be deleted.

   Command example:

   ```
   oc delete mirrorpeer {mirrorpeer_name}
   ```

   Important: If you are removing DR for specific peer clusters, do not proceed to uninstall DR operators or other shared resources. This concludes the partial DR uninstall process.

Uninstalling disaster recovery operators

Learn how to uninstall disaster recovery (DR) completely from DR operators.

Before you begin

If your goal is to completely remove DR from your OpenShift® environment, ensure that you followed all the steps that are mentioned in Prerequisites for disaster recovery uninstallation and Uninstalling disaster recovery for peer-managed clusters to uninstall DR from every peer-managed cluster.

Procedure

1. On the hub cluster, delete the `subscriptions.operators` and `clusterserviceversion` (csv) for the Red Hat® OpenShift Data Foundation Multicluster Orchestrator and OpenShift DR Hub Operator.
   Command example:

```
oc delete subscriptions.operators.coreos.com -l operators.coreos.com/odr-hub-operator.openshift-operators -n openshift-
operators
oc delete subscriptions.operators.coreos.com -l operators.coreos.com/odf-multicluster-orchestrator.openshift-operators -n
openshift-operators
oc delete csv -l operators.coreos.com/odr-hub-operator.openshift-operators -n openshift-operators
oc delete csv -l operators.coreos.com/odf-multicluster-orchestrator.openshift-operators -n openshift-operators
```

2. Alternatively, you can delete the Red Hat OpenShift Data Foundation Multicluster Orchestrator and OpenShift DR Hub Operator from the RHACM console.
   a. Go to Operators > Installed Operators.
   b. Locate the DR Hub Operator.
   c. Click the action menu **(⋮)** and select Uninstall Operator.
   d. Repeat the same steps for the Red Hat OpenShift Data Foundation Multicluster Orchestrator.
3. From the hub cluster, delete the `managedclusterview` for each peer cluster.
   Command example:

```
oc delete managedclusterview -l multicluster.odf.openshift.io/created-by=odf-multicluster-managedcluster-controller -A
```

4. From the hub cluster, delete the dynamic console plug-in for the Red Hat OpenShift Data Foundation Multicluster Orchestrator operator.
   Command example:

```
oc delete consoleplugins.console.openshift.io odf-multicluster-console
```

5. From the hub cluster, delete the configmap `odf-client-info`.
   Command example:

```
oc delete configmap odf-client-info -n openshift-operators
```

6. From the hub cluster, delete the `odf-multicluster-console` service.
   Command example:

```
oc delete service odf-multicluster-console -n openshift-operators
```

Results

All DR-related operator components and configurations are fully removed from the hub cluster and all managed clusters.

High availability

High availability ensures that system services and data remain accessible with minimal downtime by distributing compute, control, and storage components across independent failure domains.

In the IBM Fusion HCI solution, high availability refers to deployment configurations that maintain continuous operation and data integrity even in the event of major infrastructure failures, such as the loss of a rack, availability zone, or network domain.

The architecture eliminates single points of failure by distributing control-plane nodes, storage replicas, and compute resources across discrete failure domains or availability zones.

The IBM Fusion HCI offers the following multi rack topology to support the high availability for the OpenShift® Container Platform cluster:

2AZ and arbiter node topology
: A deployment configuration that spans two availability zones with an additional arbiter node located in a third zone. The arbiter node maintains cluster quorum and allows continued operation in case of a zone or rack failure. This option offers a cost-effective balance between high availability and resource utilization. For more information, see 2AZ and arbiter node configuration with Fusion Data Foundation.

- **2AZ and arbiter node configuration with Fusion Data Foundation**
  Deploy Fusion Data Foundation in a two-availability-zone and arbiter node (2AZ + arbiter node) configuration to maintain data availability across availability zones.

2AZ and arbiter node configuration with Fusion Data Foundation

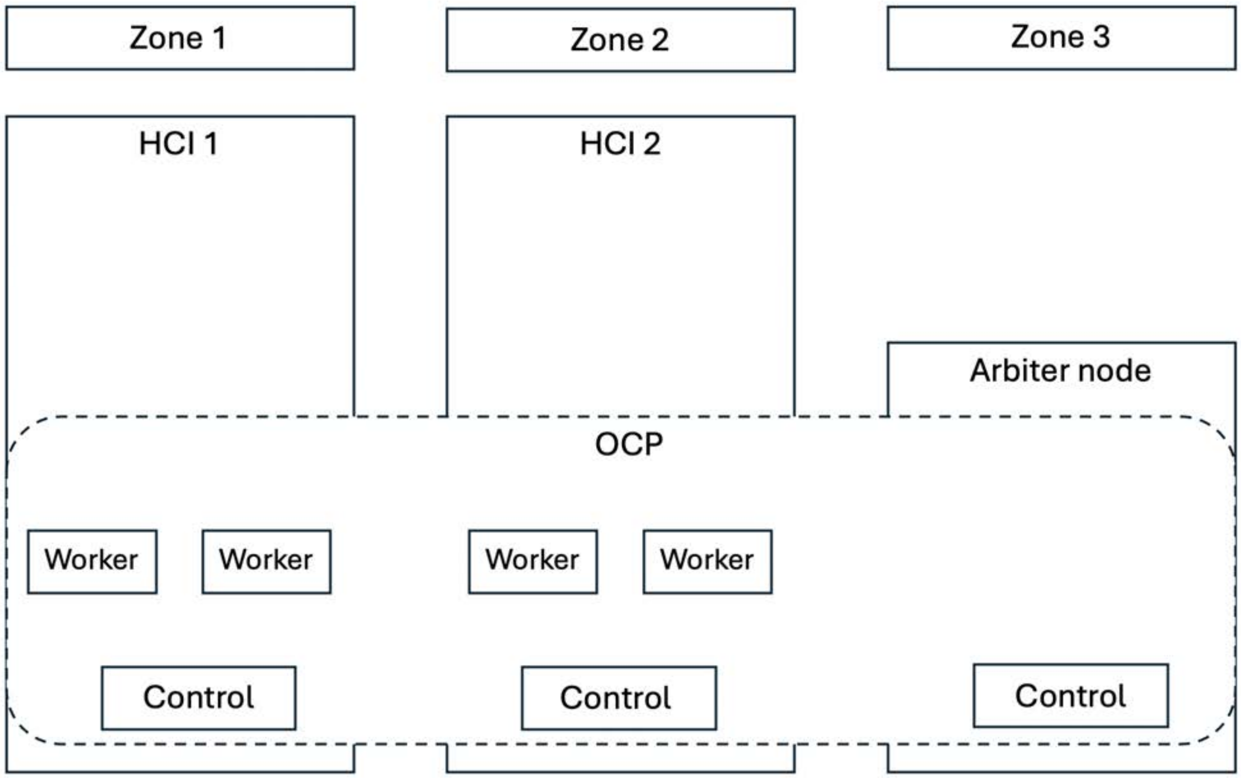
Deploy Fusion Data Foundation in a two-availability-zone and arbiter node (2AZ + arbiter node) configuration to maintain data availability across availability zones.

IBM Fusion HCI supports a 2AZ + arbiter node configuration, in which two racks are deployed across separate availability zones within a region, while an arbiter node (or control node) is deployed in a third zone. This configuration ensures that one zone or rack can remain isolated if a failure occurs in another zone. For more information on the available deployment configurations of IBM Fusion HCI, see Deployment configurations.

In this configuration, you can run a single OpenShift® Container Platform cluster that includes all nodes from both racks and the arbiter node to achieve high availability. The OpenShift Container Platform control plane is formed using the control node of each rack and the arbiter node, which acts as the tie-breaker to maintain cluster quorum.

An example configuration of 2AZ + arbiter node configuration is as follows:

Figure 1. Sample 2AZ + arbiter node configuration



Deploy the Fusion Data Foundation in the cluster, which is stretched across the two racks to provide the storage infrastructure with disaster recovery capabilities. In the event of a disaster, such as one of the two zones being partially or fully unavailable, the application workloads in the cluster continue running, ensuring business continuity.

For Fusion Data Foundation deployment instructions in the 2AZ + arbiter node configuration, see Installing Fusion Data Foundation in 2AZ and arbiter node configuration.

- **Installing Fusion Data Foundation in 2AZ and arbiter node configuration**
  Install and configure Fusion Data Foundation in a two-availability-zone and arbiter node (2AZ + arbiter node) configuration to provide storage resiliency and disaster recovery capabilities.
- **Installing zone-aware sample application**
  Install a zone-aware sample application to validate of resiliency and high availability in the two-availability-zone and arbiter node (2AZ + arbiter node) configuration.
- **Recovering applications**
  Understand the recovery methods for applications and storage in a two-availability-zone and arbiter node (2AZ + arbiter node) configuration to ensure resiliency and high availability.

Installing Fusion Data Foundation in 2AZ and arbiter node configuration

Install and configure Fusion Data Foundation in a two-availability-zone and arbiter node (2AZ + arbiter node) configuration to provide storage resiliency and disaster recovery capabilities.

Before you begin

- Complete the network planning and meet all prerequisites mentioned in Network for two-availability-zone and arbiter node (2AZ + arbiter node) configuration.
- Install IBM Fusion HCI by following the instructions mentioned in Installing IBM Fusion HCI.
- Use Fusion Data Foundation version 4.19 or higher.
- Use OpenShift® Container Platform version 4.19 or higher.
- Hosted Control Plane is not supported for the 2AZ + arbiter node deployment.
- Flexible scaling is not supported for the 2AZ + arbiter node deployment.
- To install and configure Fusion Data Foundation offline, see Mirroring Fusion Data Foundation images.

Procedure

1. In the IBM Fusion HCI web console, go to the Services tab and install Fusion Data Foundation.
   Note: If the Install button is not available, upgrade OpenShift Container Platform to version 4.21 or higher.
   The installation of Fusion Data Foundation completes successfully.

2. Configure the Fusion Data Foundation storage by following the instructions mentioned in [Configuring Fusion Data Foundation in provider mode](#).
   Note: After configuring Fusion Data Foundation storage, ensure that you create **NetworkFenceClass** CRs based on the number of storage clusters connected to the OpenShift Container Platform cluster. The **NetworkFenceClass** CRs are necessary for fencing and tainting nodes during non-graceful node shutdowns to prevent volume corruption. For instructions, see [Creating a NetworkFenceClass CR](#).
   The configuration of Fusion Data Foundation storage completes successfully.
3. Verify that the arbiter configuration details are available in ocs-storagecluster:
   a. Log in to the OpenShift Container Platform web console and select the required project.
   b. Go to Storage > Data Foundation > Storage Systems.
   c. On the StorageSystems page, click the ellipses (⋮) menu next to ocs-storagecluster, and select Edit Storage System.
   d. Click YAML tab and check whether the following arbiter configuration details are present:

```
   arbiter:
     enable: true
   nodeTopologies:
     arbiterLocation: arbiter

----

   replica: 4
```

Results

Fusion Data Foundation storage is successfully deployed in the 2AZ + arbiter node configuration.

What to do next

Validate whether the 2AZ + arbiter node topology is successfully configured by checking resiliency and high availability by following the instructions as follows:

- Install a zone-aware sample application by following the instructions mentioned in [Installing zone-aware sample application](#).
- Recover the sample application by following the instructions mentioned in [Recovering applications](#).

Installing zone-aware sample application

Install a zone-aware sample application to validate of resiliency and high availability in the two-availability-zone and arbiter node (2AZ + arbiter node) configuration.

About this task

Install a zone-aware sample application in the stretched OpenShift® Container Platform cluster to validate of resiliency and high availability in the 2AZ + arbiter node configuration.

Important: With latency between the data zones, one can expect to see performance degradation compared to an OpenShift Container Platform cluster with low latency between nodes and zones (for example, all nodes in the same location). How much the performance gets degraded, depends on the latency between the zones and on the application behavior that uses the storage (such as heavy write traffic). Ensure that you test the critical applications with stretch cluster configuration to ensure sufficient application performance for the required service levels.
A ReadWriteMany (rwx) Persistent Volume Claim (PVC) is created by using the **ocs-storagecluster-cephfs** storage class. Multiple pods use the newly created rwx PVC at the same time. The application that is used in the example here is called File Uploader.

This topic demonstrates how an application is spread across topology zones so that it is still available when there is a site outage.
Note: This demonstration is possible since this application shares the same rwx volume for storing files. It works for persistent data access as well because Fusion Data Foundation is configured as a stretched cluster with zone awareness and high availability.

Procedure

1. Create a new project.

   ```
   oc new-project my-shared-storage
   ```

   Note: Enure that there is an image registry available before creating the PHP application `file-uploader`.
2. Deploy the example PHP application called **file-uploader**.

   ```
   oc new-app openshift/php:7.4-ubi8~https://github.com/red-hat-storage/ocm-ramen-samples --name=file-uploader
   ```

   Example Output:

   ```
   Found image 4f2dcc0 (9 days old) in image stream "openshift/php" under tag "7.4-ubi8" for "openshift/php:7.4-
   ubi8"

   Apache 2.4 with PHP 7.4
   -----------------------
   PHP 7.4 available as container is a base platform for building and running various PHP 7.4 applications and frameworks.
   PHP is an HTML-embedded scripting language. PHP attempts to make it easy for developers to write dynamically generated web
   pages. PHP also offers built-in database integration for several commercial and non-commercial database management
   systems, so writing a database-enabled webpage with PHP is fairly simple. The most common
   use of PHP coding is probably as a replacement for CGI scripts.

   Tags: builder, php, php74, php-74

   * A source build using source code from https://github.com/christianh814/openshift-php-upload-demo will be cr
   eated
   * The resulting image will be pushed to image stream tag "file-uploader:latest"
   ```

```
* Use 'oc start-build' to trigger a new build

--> Creating resources ...
    imagestream.image.openshift.io "file-uploader" created
    buildconfig.build.openshift.io "file-uploader" created
    deployment.apps "file-uploader" created
    service "file-uploader" created
--> Success
    Build scheduled, use 'oc logs -f buildconfig/file-uploader' to track its progress.

    Application is not exposed. You can expose services to the outside world by executing one or more of the commands 
below:
     'oc expose service/file-uploader'

    Run 'oc status' to view your app.
```

3. View the build log and wait until the application is deployed.

```
oc logs -f bc/file-uploader -n my-shared-storage
```

Example Output:

```
Cloning "https://github.com/christianh814/openshift-php-upload-demo" ...

    [...]
    Generating dockerfile with builder image image-registry.openshift-image-regis
try.svc:5000/openshift/php@sha256:d97466f33999951739a76bce922ab17088885db610c
0e05b593844b41d5494ea
STEP 1: FROM image-registry.openshift-image-registry.svc:5000/openshift/php@s
ha256:d97466f33999951739a76bce922ab17088885db610c0e05b593844b41d5494ea
STEP 2: LABEL "io.openshift.build.commit.author"="Christian Hernandez <christ
ian.hernandez@yahoo.com>"       "io.openshift.build.commit.date"="Sun Oct 1 1
7:15:09 2017 -0700"       "io.openshift.build.commit.id"="288eda3dff43b02f7f7
b6b6b6f93396ffdf34cb2"       "io.openshift.build.commit.ref"="master"       "
io.openshift.build.commit.message"="trying to modularize"       "io.openshift
.build.source-location"="https://github.com/christianh814/openshift-php-uploa
d-demo"       "io.openshift.build.image"="image-registry.openshift-image-regi
stry.svc:5000/openshift/php@sha256:d97466f33999951739a76bce922ab17088885db610
c0e05b593844b41d5494ea"
STEP 3: ENV OPENSHIFT_BUILD_NAME="file-uploader-1"     OPENSHIFT_BUILD_NAMESP
ACE="my-shared-storage"     OPENSHIFT_BUILD_SOURCE="https://github.com/christ
ianh814/openshift-php-upload-demo"     OPENSHIFT_BUILD_COMMIT="288eda3dff43b0
2f7f7b6b6b6f93396ffdf34cb2"
STEP 4: USER root
STEP 5: COPY upload/src /tmp/src
STEP 6: RUN chown -R 1001:0 /tmp/src
STEP 7: USER 1001
STEP 8: RUN /usr/libexec/s2i/assemble
---> Installing application source...
=> sourcing 20-copy-config.sh ...
---> 17:24:39     Processing additional arbitrary httpd configuration provide
d by s2i ...
=> sourcing 00-documentroot.conf ...
=> sourcing 50-mpm-tuning.conf ...
=> sourcing 40-ssl-certs.sh ...
STEP 9: CMD /usr/libexec/s2i/run
STEP 10: COMMIT temp.builder.openshift.io/my-shared-storage/file-uploader-1:3
b83e447
Getting image source signatures

[...]
```

The command prompt returns out of the tail mode once you see `Push successful`.
Note: The new-app command deploys the application directly from the Git repository and does not use the Red Hat OpenShift template, hence Red Hat OpenShift route resource is not created by default. You need to create the route manually.

What to do next

- Scaling the application after installation

- **Scaling the application after installation**
  Scale the application to four replicas and attach a ReadWriteMany PVC to make the application zone-aware and available across both data zones.
- **Modifying deployment to be zone-aware**
  Modify an application deployment to use topology spread constraints so that pods are distributed across availability zones and remain available during a single zone failure.

Scaling the application after installation

Scale the application to four replicas and attach a ReadWriteMany PVC to make the application zone-aware and available across both data zones.

Procedure

1. Scale the application to four replicas and expose its services to make the application zone-aware and available.

```
oc expose svc/file-uploader -n my-shared-storage
```

```
oc scale --replicas=4 deploy/file-uploader -n my-shared-storage
```

```
oc get pods -o wide -n my-shared-storage
```

You should have four file-uploader pods in a few minutes. Repeat the above command until there are 4 file-uploader pods in the **Running** status.

2. Create a PVC and attach it into an application.

```
oc set volume deploy/file-uploader --add --name=my-shared-storage \
-t pvc --claim-mode=ReadWriteMany --claim-size=10Gi \
--claim-name=my-shared-storage --claim-class=ocs-storagecluster-cephfs \
--mount-path=/opt/app-root/src/uploaded \
-n my-shared-storage
```

This command:

- Creates a PVC.
- Updates the application deployment to include a volume definition.
- Updates the application deployment to attach a volume mount into the specified mount-path.
- Creates a new deployment with the four application pods.

3. Check the result of adding the volume.

```
oc get pvc -n my-shared-storage
```

Example Output:

```
NAME                STATUS   VOLUME                                     CAPACITY   ACCESS MODES   STORAGECLASS
AGE
my-shared-storage   Bound    pvc-5402cc8a-e874-4d7e-af76-1eb05bd2e7c7   10Gi       RWX            ocs-storagecluster-
cephfs   52s
```

Notice the **ACCESS MODE** is set to rwx.

All the four **file-uploader** pods are using the same rwx volume. Without this access mode, Red Hat OpenShift does not attempt to attach multiple pods to the same Persistent Volume (PV) reliably. If you attempt to scale up the deployments that are using ReadWriteAfter (RWO) PV, the pods may get colocated on the same node.

Modifying deployment to be zone-aware

Modify an application deployment to use topology spread constraints so that pods are distributed across availability zones and remain available during a single zone failure.

About this task

Currently, the **file-uploader** deployment is not zone-aware and can schedule all the pods in the same zone. In this case, when there is a site outage then the application is no longer available. For more information, see Configure > Nodes > Controlling pod placement onto nodes (scheduling) > Controlling pod placement by using pod topology spread constraints in [Red Hat OpenShift Container Platform](#) product documentation.

Procedure

1. Add the pod placement rule in the application deployment configuration to make the application zone-aware.
   a. Run the following command, and review the output:

   ```
   oc get deployment file-uploader -o yaml -n my-shared-storage | less
   ```

   Example Output:

   ```
   [...]
   spec:
     progressDeadlineSeconds: 600
     replicas: 4
     revisionHistoryLimit: 10
     selector:
       matchLabels:
         deployment: file-uploader
     strategy:
       rollingUpdate:
         maxSurge: 25%
         maxUnavailable: 25%
       type: RollingUpdate
     template:
       metadata:
         annotations:
           openshift.io/generated-by: OpenShiftNewApp
         creationTimestamp: null
         labels:
           deployment: file-uploader
         spec:  # <-- Start inserted lines after herecontainers:  # <-- End inserted lines before here
           - image: image-registry.openshift-image-registry.svc:5000/my-shared-storage/file-
   uploader@sha256:a458ea62f990e431ad7d5f84c89e2fa27bdebdd5e29c5418c70c56eb81f0a26b
             imagePullPolicy: IfNotPresent
             name: file-uploader
   [...]
   ```

   b. Edit the deployment to use the topology zone labels.

```
oc edit deployment file-uploader -n my-shared-storage
```

Add add the following new lines between the **Start** and **End** (shown in the output in the previous step):

```
[...]
    spec:
      topologySpreadConstraints:
        - labelSelector:
            matchLabels:
              deployment: file-uploader
          maxSkew: 1
          topologyKey: topology.kubernetes.io/zone
          whenUnsatisfiable: DoNotSchedule
        - labelSelector:
            matchLabels:
              deployment: file-uploader
          maxSkew: 1
          topologyKey: kubernetes.io/hostname
          whenUnsatisfiable: ScheduleAnyway
      nodeSelector:
        node-role.kubernetes.io/worker: ""
      containers:
[...]
```

Example output:

```
deployment.apps/file-uploader edited
```

2. Scale down the deployment to **zero** pods and then back to **four** pods. This is needed because the deployment changed in terms of pod placement.
   Scaling down to **zero** pods

   ```
   oc scale deployment file-uploader --replicas=0 -n my-shared-storage
   ```

   Example output:

   ```
   deployment.apps/file-uploader scaled
   ```

   Scaling up to **four** pods

   ```
   oc scale deployment file-uploader --replicas=4 -n my-shared-storage
   ```

   Example output:

   ```
   deployment.apps/file-uploader scaled
   ```

3. Verify that the four pods are spread across the four nodes in datacenter1 and datacenter2 zones.

   ```
   oc get pods -o wide -n my-shared-storage | egrep '^file-uploader'| grep -v build | awk '{print $7}' | sort | uniq -c
   ```

   Example output:

   ```
   1 perf1-mz8bt-worker-d2hdm
      1 perf1-mz8bt-worker-k68rv
      1 perf1-mz8bt-worker-ntkp8
      1 perf1-mz8bt-worker-qpwsr
   ```

   Search for the zone labels used.

   ```
   oc get nodes -L topology.kubernetes.io/zone | grep datacenter | grep -v master
   ```

   Example output:

   ```
   perf1-mz8bt-worker-d2hdm   Ready    worker   35d   v1.20.0+5fbfd19   datacenter1
   perf1-mz8bt-worker-k68rv   Ready    worker   35d   v1.20.0+5fbfd19   datacenter1
   perf1-mz8bt-worker-ntkp8   Ready    worker   35d   v1.20.0+5fbfd19   datacenter2
   perf1-mz8bt-worker-qpwsr   Ready    worker   35d   v1.20.0+5fbfd19   datacenter2
   ```

4. Use the file-uploader web application by using your browser to upload new files.
   a. Find the route that is created.

      ```
      oc get route file-uploader -n my-shared-storage -o jsonpath --template="http://{.spec.host}{'\n'}"
      ```

      Example Output:

      ```
      http://file-uploader-my-shared-storage.apps.cluster-ocs4-abdf.ocs4-abdf.sandbox744.opentlc.com
      ```

   b. Point your browser to the web application by using the route in the previous step.
      The web application lists all the uploaded files and offers the ability to upload new ones and download the existing data.

   c. Select an arbitrary file from your local computer and upload it to the application.
      i. Click Choose file to select an arbitrary file.
      ii. Click Upload.
   d. Click List uploaded files to see the list of all currently uploaded files.

   Note: The OpenShift® Container Platform image registry, ingress routing, and monitoring services are not zone-aware.

Recovering applications

Understand the recovery methods for applications and storage in a two-availability-zone and arbiter node (2AZ + arbiter node) configuration to ensure resiliency and high availability.

How the application is designed determines how soon it becomes available again on the active zone.

There are different methods of recovery for applications and their storage depending on the site outage. The recovery time depends on the application architecture.

- **Understanding zone failure**
  Zone failure occurs when all nodes in a data zone lose communication with the second zone; recovery succeeds only when communication between zones is fully disconnected.
- **Recovering zone-aware HA applications with rwx storage**
  Zone-aware HA applications using ReadWriteMany CephFS volumes experience pod termination in the failed zone and may require a node restart if pods enter CrashLoopBackOff during recovery.
- **Recovering HA applications with rwx storage**
  If all application replicas using ReadWriteMany storage are in the failed zone, the application takes 6–8 minutes to recover on the active zone after the pod eviction timeout expires.
- **Recovering applications with RWO storage**
  Applications using ReadWriteAfter storage have pods that become stuck in `Terminating` status after a zone failure and require manual intervention such as powering off the node to recover.
- **Recovering StatefulSet pods**
  StatefulSet pods behave like pods with ReadWriteAfter volumes during a zone failure and must be force-deleted with the node powered off to reschedule on the active zone.
- **Fencing and tainting nodes to prevent volume corruption during non-graceful node shutdowns**
  Fence and taint nodes to protect storage volumes from corruption during unexpected or non-graceful node shutdowns.

Understanding zone failure

Zone failure occurs when all nodes in a data zone lose communication with the second zone; recovery succeeds only when communication between zones is fully disconnected.

In this section, zone failure is considered as a failure where all OpenShift® Container Platform, master and worker nodes in a zone are no longer communicating with the resources in the second data zone (for example, powered down nodes). If communication between the data zones is still partially working (intermittently up or down), the cluster, storage, and network admins should disconnect the communication path between the data zones for recovery to succeed.
Important: Before installing the sample application, power off the OpenShift Container Platform nodes (at least the nodes with Fusion Data Foundation devices) to test the failure of a data zone to validate that your file-uploader application is available, and you can upload new files.
Note: To prevent volume corruption during non-graceful node shutdowns, follow the steps in Fencing and tainting nodes to prevent volume corruption during non-graceful node shutdowns.

Recovering zone-aware HA applications with rwx storage

Zone-aware HA applications using ReadWriteMany CephFS volumes experience pod termination in the failed zone and may require a node restart if pods enter CrashLoopBackOff during recovery.

Applications that are deployed with `topologyKey: topology.kubernetes.io/zone`, have one or more replicas that are scheduled in each data zone, and are using shared storage, that is, ReadWriteMany (rwx) CephFS volume, terminate themselves in the failed zone after few minutes and new pods are rolled in and stuck in pending state until the zones are recovered.

An example of this type of application is detailed in the Installing zone-aware sample application section.
Important: During zone recovery if application pods go into CrashLoopBackOff (CLBO) state with permission denied error while mounting the CephFS volume, then restart the nodes where the pods are scheduled. Wait for some time and then check whether the pods are running again.
Note: To prevent volume corruption during non-graceful node shutdowns, follow the steps in Fencing and tainting nodes to prevent volume corruption during non-graceful node shutdowns.

Recovering HA applications with rwx storage

If all application replicas using ReadWriteMany storage are in the failed zone, the application takes 6–8 minutes to recover on the active zone after the pod eviction timeout expires.

Applications that are using `topologyKey: kubernetes.io/hostname` or no topology configuration, have no protection against all the application replicas being in the same zone.
Note: This can happen even with *podAntiAffinity* and `topologyKey: kubernetes.io/hostname` in the **Pod** spec because this anti-affinity rule is host-based and not zone-based.
If this happens and all replicas are located in the zone that fails, the application that uses ReadWriteMany (rwx) storage takes 6-8 minutes to recover on the active zone. This pause is for the OpenShift® Container Platform nodes in the failed zone to become `NotReady` (60 seconds) and then for the default pod eviction timeout to expire (300 seconds).

Note: To prevent volume corruption during non-graceful node shutdowns, follow the steps in Fencing and tainting nodes to prevent volume corruption during non-graceful node shutdowns.

Recovering applications with RWO storage

Applications using ReadWriteAfter storage have pods that become stuck in `Terminating` status after a zone failure and require manual intervention such as powering off the node to recover.

Applications that use ReadWriteAfter (RWO) storage have a known behavior that is described in this Kubernetes issue. Because of this issue, if there is a data zone failure, any application pods in that zone mounting RWO volumes (for example, **cephrbd** based volumes) are stuck with **Terminating** status after 6-8 minutes and is not re-created on the active zone without manual intervention.

Check the OpenShift® Container Platform nodes with a status of **NotReady**. There may be an issue that prevents the nodes from communicating with the Red Hat OpenShift control plane. However, the nodes may still be performing I/O operations against Persistent Volumes (PVs).

If two pods are concurrently writing to the same RWO volume, there is a risk of data corruption. Ensure that processes on the **NotReady** node are either terminated or blocked until they are terminated.

Example solutions:

- Use an out of band management system to power off a node, with confirmation, to ensure process termination.
- Withdraw a network route that is used by nodes at a failed site to communicate with storage.
  Note: Before restoring service to the failed zone or nodes, confirm that all the pods with PVs have terminated successfully.

To get the **Terminating** pods to re-create on the active zone, you can either force delete the pod or delete the finalizer on the associated PV. After one of these two actions is completed, the application pod should re-create on the active zone, and successfully mount its RWO storage.

Force deleting the pod
: Force deletions do not wait for confirmation from the kubelet that the pod has been terminated.

```
oc delete pod <PODNAME> --grace-period=0 --force --namespace <NAMESPACE>
```

<PODNAME>
: Is the name of the pod

<NAMESPACE>
: Is the project namespace

Deleting the finalizer on the associated PV
: Find the associated PV for the Persistent Volume Claim (PVC) that is mounted by the Terminating pod and delete the finalizer by using the **oc patch** command.

```
oc patch -n openshift-storage pv/<PV_NAME> -p '{"metadata":{"finalizers":[]}}' --type=merge
```

<PV_NAME>
: Is the name of the PV

An easy way to find the associated PV is to describe the Terminating pod. If you see a multi-attach warning, it should have the PV names in the warning (for example, **pvc-0595a8d2-683f-443b-aee0-6e547f5f5a7c**).

```
oc describe pod <PODNAME> --namespace <NAMESPACE>
```

<PODNAME>
: Is the name of the pod

<NAMESPACE>
: Is the project namespace

Example output:

```
[...]
Events:
  Type     Reason                  Age   From                     Message
  ----     ------                  ----  ----                     -------
  Normal   Scheduled               4m5s  default-scheduler        Successfully assigned openshift-storage/noobaa-db-pg-0 
to perf1-mz8bt-worker-d2hdm
  Warning  FailedAttachVolume      4m5s  attachdetach-controller  Multi-Attach error for volume "pvc-0595a8d2-683f-443b-
aee0-6e547f5f5a7c" Volume is already exclusively attached to one node and can't be attached to another
```

Warning: Fusion Data Foundation does not support taints relating to non-graceful node shutdown for automated pod eviction and volume detachment operations. For information, see Non-graceful node shutdown handling. It is mandatory to ensure that the node is shutdown, or network route to the node is withdrawn, prior to force deleting pods.

Recovering StatefulSet pods

StatefulSet pods behave like pods with ReadWriteAfter volumes during a zone failure and must be force-deleted with the node powered off to reschedule on the active zone.

Pods that are part of a StatefulSet have a similar issue as pods mounting ReadWriteAfter (RWO) volumes. More information is referenced in the Kubernetes resource StatefulSet considerations.

To get the pods part of a StatefulSet to re-create on the active zone after 6-8 minutes you need to force delete the pod with the same requirements (that is, OpenShift® Container Platform node powered off or communication disconnected) as pods with RWO volumes.

Fencing and tainting nodes to prevent volume corruption during non-graceful node shutdowns

Fence and taint nodes to protect storage volumes from corruption during unexpected or non-graceful node shutdowns.

Before you begin

Ensure that the `NetworkFenceClass` CRs are created in the cluster based on the number of storage clusters connected to the OpenShift® Container Platform cluster. For instructions, see Creating a NetworkFenceClass CR.

Procedure

1. Create a `NetworkFence` CR to fence the IP addresses of storage nodes that are down.
    a. Get the node name that needs to be fenced.
    Command example:

    ```
    oc get node
    ```

    b. Get the `csiaddonsnode` object corresponding to the node found in step 1.a.
    Command example:

    ```
    oc get csiaddonsnode -n openshift-storage |grep -i <nodename> |grep -i daemonset
    ```

    c. Get the IP address from the `csiaddonsnode` object found in step 1.b.
    Command example:

    ```
    oc get csiaddonsnode -n openshift-storage \
    compute-2-ru5.f091004.fusion.tadn.ibm.com-openshift-storage-daemonset-openshift-storage.rbd.csi.ceph.com-nodeplugin-csi-addons \
    -o jsonpath='{.status.networkFenceClientStatus}'
    [{"ClientDetails":[{"cidrs":["10.48.120.82/32"],"id":"04a26bdf-1e5b-4d09-b052-871c5e32f960"}],"networkFenceClassName":"odf-networkfenceclass"},{"ClientDetails":[{"cidrs":["10.48.120.82/32"],"id":"04a26bdf-1e5b-4d09-b052-871c5e32f960"}],"networkFenceClassName":"ocs-storagecluster-ceph-rbd-networkfenceclass"}]
    ```

    Replace `<csi-addons-name>` with the `csiaddonsnode` object found in step 1.b.

    d. Create the `NetworkFence` CR for the CIDR used in step 1.c with the OpenShift node name for easy identification.
    For example:

    ```
    apiVersion: csiaddons.openshift.io/v1alpha1
    kind: NetworkFence
    metadata:
      name: <openshift-node-name>
    spec:
      cidrs:
      - 10.244.0.1/32
      driver: openshift-storage.rbd.csi.ceph.com
      fenceState: Fenced
      parameters:
        clusterID: openshift-storage
      secret:
        name: rook-csi-rbd-provisioner
        namespace: openshift-storage
    ```

    - `parameters.clusterID`: Specifies the cluster id. Fetch it from the `parameters.clusterID` field in the `ocs-storagecluster-ceph-rbd` StorageClass.
    - `secret.name`: Specifies the name of the secret that is required for network fencing. Fetch it from the `parameters.csi.storage.k8s.io/provisioner-secret-name` field in the `ocs-storagecluster-ceph-rbd` StorageClass.
    - `secret.namespace`: Specifies the namespace in which the secret is located. Fetch it from the `parameters.csi.storage.k8s.io/provisioner-secret-namespace` field in the `ocs-storagecluster-ceph-rbd` StorageClass.

    e. Verify that network fencing is done successfully.
    Command example:

    ```
    oc get networkfence
    ```

    Example output:

    ```
    NAME                     DRIVER                                CIDRS                FENCESTATE    AGE    RESULT
    <openshift-node-name>    openshift-storage.rbd.csi.ceph.com    ["10.244.0.1/32"]    Fenced        42h    Succeeded
    ```

2. Add taint for the nodes that are down.
Command example:

    ```
    oc patch node <node-name> --type='merge' -p '{"spec":{"taints":[{"key":"node.kubernetes.io/out-of-service","value":"nodeshutdown","effect":"NoExecute"}]}}'
    ```

3. Monitor application relocation to surviving nodes.
4. Repeat step 1, 2, and 3 for all nodes that are down.
5. After the nodes are back online, do as follows:
Note: Repeat the following steps for all nodes.
    a. Change the state of `NetworkFence` to `Unfenced` in the CR.
    Command example:

    ```
    oc patch networkfence <network-fence-object-name> -p '{"spec":{"fenceState":"Unfenced"}}' --type=merge
    ```

    b. Verify that the network unfencing is done successfully.
    Command example:

    ```
    oc get networkfence
    ```

    Example output:

```
NAME                    DRIVER                            CIDRS               FENCESTATE   AGE   RESULT
<openshift-node-name>   openshift-storage.rbd.csi.ceph.com   ["10.244.0.1/32"]   unfenced     42h   Succeeded
```

c. Untaint the node.
   Command example:

```
oc adm taint nodes <node-name> node.kubernetes.io/out-of-service=nodeshutdown:NoExecute-
```

Multi-cluster IBM Fusion HCI using Hosted Control Plane

Create a Hosted Control Plane with Red Hat® OpenShift® Virtualization clusters or Bare Metal clusters (both internal and external) on IBM Fusion HCI.

For more information about internal and external, see Choose your configuration.

Important:

- Bare Metal and virtualized Hosted Control Plane clusters are supported in connected environments including proxy.
- Bare Metal and virtualized Hosted Control Plane are supported in the offline environment.

For these clusters, the control plane components are hosted on a IBM Fusion HCI management cluster, and the compute nodes are provisioned as virtual machines or Bare Metal servers. A hosted control plane refers to a solution that offers a decoupled management plane where you can install and run the core control plane components on a IBM Fusion HCI. Instead of running control plane components on a dedicated physical or virtual machine, you can use the hosted control plane feature to run these components as lightweight pods on the IBM Fusion HCI cluster. You can host the control plane components, API server, etcd, and control manager, separately from the compute nodes.

The IBM Fusion HCI can now host multiple IBM Fusion clusters. You can create clusters that match your specific workload needs, and clusters can be created and deleted on demand. IBM Fusion HCI supports Hosted Control Plane clusters using OpenShift virtualization and Bare Metal nodes (both internal and external).

Benefits of hosted control plane:

- Cost efficiency
  The reduction in the number of nodes makes clusters more cost-effective:
  - You do not need control nodes per cluster.
  - Reduced number of Bare Metal nodes needed for clusters.
  - Red Hat OpenShift Virtualization VMs can be used as compute nodes. The control pods on a hosting cluster allows for the two variants: Red Hat OpenShift Virtualization and Bare Metal.
- Ability to provide Cluster on demand
  You can delete clusters when they are no longer needed. The resources are freed up for new provisioning.

  For more information and benefits of the Hosted Control Plane, see Red Hat documentation.

OpenShift Virtualization Hosted Control Plane clusters

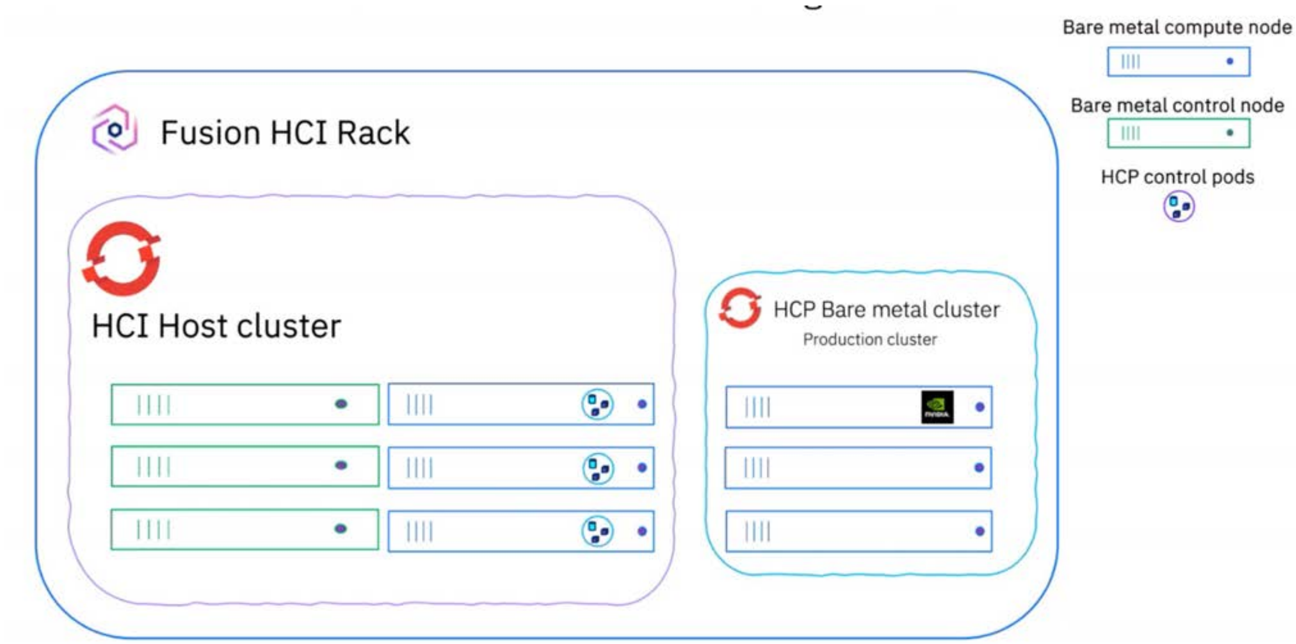
IBM Fusion HCI supports hosted control plane in the configurations of virtual clusters and Bare Metal clusters. The virtual clusters are virtualized VMs as nodes within the IBM Fusion HCI.

For more information about the deployment, see Deployment of virtualized clusters with Fusion Data Foundation.

Bare Metal Hosted Control Plane clusters

For bare metal clusters, an IBM Fusion HCI admin can provision on demand OpenShift clusters on Bare Metal IBM Fusion HCI, and support is available for both internal and external. Though the control plane is hosted in a similar manner as virtualized clusters, the compute nodes are deployed on physical servers.

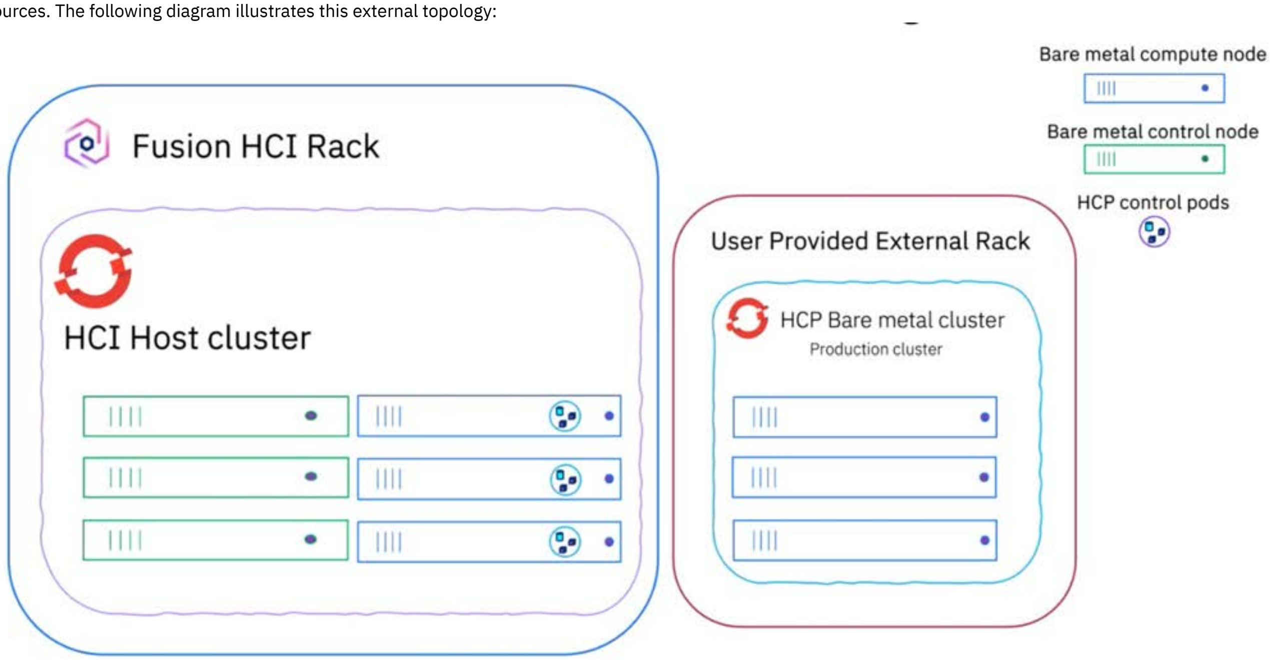
Hosted Control Plane Bare Metal internal servers in a rack
: Using servers inside the IBM Fusion HCI rack, you can set up workloads to run on your Bare Metal Hosted Control Plane cluster using a GPU node. For example, watsonx.ai is a workload.
  The IBM Fusion HCI rack houses all the servers required for the system, with some serving as OpenShift hosts. A Hosted Control Plane Bare Metal cluster is created using a subset of these servers, which takes on the workload of a GPU node. By converting three control servers into control pods on this hosted cluster, you save three control servers. The following diagram illustrates this managed topology:



Hosted Control Plane Bare Metal external servers

External servers outside of the IBM Fusion HCI cluster can be used to create Bare Metal Hosted Control Plane clusters with Fusion data services. Here, bring your own Bare Metal hosts and turn them into nodes. The control planes of this Hosted Control Plane cluster are hosted in the IBM Fusion HCI management cluster. You can create more Bare Metal compute nodes out of your stock of x86 servers.

On the IBM Fusion HCI rack, you can set up your own host cluster. The external rack you bring can be utilized for OpenShift clusters, where a Hosted Control Plane Bare Metal cluster is created. Meanwhile, the control pods operate on the IBM Fusion HCI cluster, enabling you to make the most of your existing hardware resources. The following diagram illustrates this external topology:



For more information about the deployment, see Deployment of Bare Metal clusters with Fusion Data Foundation.

Benefits of IBM Fusion in Hosted Control Plane

- Provision OpenShift Container Platform clusters with services like Fusion Data Foundation, Backup & Restore, IBM Data Cataloging, and disaster recovery. You can add labels during the time of hosted cluster creation so that you need not log in to your cluster to install the services. For an overview on each of the Fusion services, see Fusion services.
  Note: You can also add service label later based on your need. Only Data Foundation Backup & Restore can be done through labeling.
- Support is available for Provider and consumer mode in Fusion Data Foundation. In the provider mode, you run a storage cluster locally that can serve multiple mode clusters. The provider mode is similar to an external Fusion Data Foundation configuration. It acts as the provider and base storage on the host cluster. You can provision new clusters within a internal rack and use storage from the central provider cluster. Here, the provider and consumer are in a hosted control plane relationship.
- Data services are automatically cleaned up upon the deletion of Hosted Control Plane clusters.
- Single dashboard for managing Backup & Restore and storage.
- Manage logs and events of multiple clusters.
- Provide dynamic storage for applications in Hosted Control Plane clusters.

- Disaster recovery is available for applications on Hosted Control Plane clusters.
- Provide Backup & Restore for applications in Hosted Control Plane clusters.
- With IBM Fusion HCI, you can order GPU nodes and install them in your rack so that you can run a dedicated cluster for your AI workloads.
- Support for deployment of IBM Cloud Pak for Integration and IBM Software Hub on Hosted Control Plane.

- **Interoperability matrix**
  Use this section to get the detailed support matrix for the Hosted Control Plane.
- **Hosted Control Plane sizing on IBM Fusion**
  Plan and size your Hosted Control Plane on IBM Fusion to ensure optimal performance, scalability, and resilience.
- **Prerequisites for Hosted Control Plane clusters**
  The prerequisites before you create virtualized or Bare Metal Hosted Control Plane clusters.
- **Configuring a Hosted Control Plane cluster in a proxy environment**
  Use these instructions to gather the required configuration and define the necessary Kubernetes resources to configure a `HostedCluster` in a proxy environment.
- **Deployment of virtualized clusters with Fusion Data Foundation**
  Create virtualized (Hosted Control Plane) clusters with the Fusion Data Foundation as storage type.
- **Deploying Bare Metal Hosted Control Plane cluster in disconnected environment**
  A disconnected environment deployment operates without internet access. You can deploy Hosted Control Plane (HCP) in a disconnected environment on Bare Metal infrastructure.
- **Deploying virtualized Hosted Control Plane cluster in disconnected environment**
  A disconnected environment deployment operates without internet access. You can deploy Hosted Control Plane (HCP) in a disconnected environment on the virtualized infrastructure.
- **Deployment of Bare Metal clusters with Fusion Data Foundation**
  Create Bare Metal hosted clusters with Fusion Data Foundation as the storage type. It supports both Hosted Control Plane Bare Metal internal servers in a rack and Hosted Control Plane Bare Metal external servers.
- **Installation of data services on a Hosted Control Plane cluster**
  Provision Hosted Control Plane clusters with services like Fusion Data Foundation, Global Data Platform remote mount, and Backup & Restore.
- **Backup & Restore of applications in Hosted Control Plane clusters**
  Using the Backup & Restore service, you can backup and restore your applications on Hosted Control Plane clusters.
- **Backup and restore of the Hosted Control Plane cluster**
  You can backup and restore your Hosted Control Plane cluster.
- **Updating the storage quota of a Hosted Control Plane cluster**
  Update the allotted storage quota of the Fusion Data Foundation client on a Hosted Control Plane cluster to meet the storage requirement of the application workloads.
- **Cleaning up of a Hosted Control Plane cluster**
  If you no longer need a hosted cluster, use this procedure to remove it.
- **Troubleshooting issues and known limitations in Hosted Control Plane cluster**
  Use these troubleshooting information to know the problem and workaround for the installation of spoke cluster and

Interoperability matrix

Use this section to get the detailed support matrix for the Hosted Control Plane.

Run the following command to retrieve the list of supported OpenShift® Container Platform versions for an Hosted Control Plane cluster with the installed MCE version.

```
oc get configmap supported-versions -n hypershift -o yaml
```

Hub OpenShift Container Platform version		MCE version				
		2.6	**2.7**	**2.8**	**2.9**	**2.10**
	Supported Hosted Control Plane OpenShift Container Platform versions	4.16 ***	4.17	4.16 to 4.18	4.17 to 4.19	4.18 to 4.20
		Is supported by MCE	**Is supported by MCE**	**Is supported by MCE**	**Is supported by MCE**	**Is supported by MCE**
4.16		Yes	Yes	Yes	Yes	Yes
4.17		Yes	Yes	Yes	Yes	Yes
4.18		No	No	Yes	Yes	Yes
4.19		No	No	No	Yes	Yes
4.20		No	No	No	Yes	Yes

***- only virtualized Hosted Control Plane is supported, and not Bare Metal Hosted Control Plane.

Hosted Control Plane sizing on IBM Fusion

Plan and size your Hosted Control Plane on IBM Fusion to ensure optimal performance, scalability, and resilience.

When planning the sizing of a Hosted Control Plane on IBM Fusion, several factors must be carefully evaluated to ensure optimal performance, scalability, and resilience. This topic provides guidance on the key considerations and formulas to help you estimate the required infrastructure for Hosted Control Plane deployments.

Key considerations

Control and compute node specification
: The resource allocation for control nodes and compute nodes (vCPUs, memory, and storage) directly impacts the stability and responsiveness of the hosted clusters. However, control nodes vCPUs and memory are not considered as available capacity for Hosted Control Plane in this sizing guide. Storage from control nodes is considered.

Number of Hosted Control Plane clusters
: The total count of Hosted Control Plane clusters to be deployed determines the aggregate resource requirement. Each additional cluster adds to CPU, memory, and storage usage as well as LVM storage.

vCPU allocation
: Proper sizing of vCPUs is essential for workload execution, cluster operations, and ensuring smooth control plane performance. Over or under provisioning can affect latency and throughput.

Memory requirements
: Adequate memory allocation is critical for API responsiveness, workload scheduling, and supporting large-scale workloads across clusters.

Storage capacity and drive count
: Persistent, ephemeral, and LVM storage must be considered. This ensures sufficient capacity for workloads, logs, and cluster metadata.

OpenShift® reserve
: A portion of system resources must be reserved for OpenShift to maintain cluster operations, monitoring, and system services.

High availability
: To prevent downtime, HA considerations should include replication of control plane components, redundant nodes, and reserved resources to handle failures seamlessly.

Pod capacity per node
: Each OpenShift node supports a maximum number of pods, which defines its pod density. Proper planning of the required number of Hosted Control Plane clusters must account for this limitation to ensure that workloads are evenly distributed and performance is maintained.

Queries Per Second (QPS) limits
: Hosted control plane sizing is about control-plane load and workloads that cause heavy API activity, etcd activity, or both.

  - For the first 1000 QPS, 11.9 vCPU and 13.6 memory are used.
  - As the load increases by 1000 QPS, the hosted control plane resource utilization increases by 9 vCPUs and 2.5 GB memory.

IBM Fusion resource consumption

Fusion resource usage includes:

Fusion base
: Resources required for IBM Fusion services. IBM Fusion services run on both control and compute nodes. However, for sizing purposes, we consider only compute node resource consumption.

Fusion Data Foundation
: Resources consumed based on the number of nodes and the number of drives per node participating in storage.
  Note: Fusion Data Foundation is considering only of compute nodes that participate in Fusion Data Foundation cluster.

Backup & Restore
: Additional resources are required to support backup and restore operations.

For guidelines and formulas to estimate infrastructure requirements, incorporating system overhead and high availability considerations. See Planning Hosted Control Plane deployments on IBM Fusion.

- **Planning Hosted Control Plane deployments on IBM Fusion**
  When planning Hosted Control Plane deployments on IBM Fusion, accurate sizing is essential to ensure performance, scalability, and resilience. Since Hosted Control Plane clusters rely on shared infrastructure, understanding how vCPU, memory, storage, and OpenShift platform reserves are consumed becomes critical for effective capacity planning.

Planning Hosted Control Plane deployments on IBM Fusion

When planning Hosted Control Plane deployments on IBM Fusion, accurate sizing is essential to ensure performance, scalability, and resilience. Since Hosted Control Plane clusters rely on shared infrastructure, understanding how vCPU, memory, storage, and OpenShift® platform reserves are consumed becomes critical for effective capacity planning.

Examples of Hosted Control Plane sizing:

Guidelines and formulas

The following guidelines and formulas provide a structured approach to estimate infrastructure requirements, incorporating system overhead and high availability considerations. These estimates serve as a baseline for right-sizing clusters.

Understanding Hosted Control Plane sizing requirements
: To estimate infrastructure requirements for Hosted Control Plane deployments, consider the following factors:

  - Number of Hosted Control Plane clusters
  - HCP size (same or different sizes with distinct vCPU, memory, and storage requirements)
  - Control node and compute node configuration
    For example:
    - Control node: 32 core, 512 GiB , 4 disks x 3.2 TiB, and 500 pod limit
    - Compute node: 64 core, 1024 GiB, 4 disks x 3.2 TiB, 1 LVM disk x 1-2 TiB, and 500 pod limit

Calculating Hosted Control Plane resource requirements
: Use the following formulas to calculate Hosted Control Plane resource requirements:
  Note:

- When sizing a KubeVirt based Hosted Control Plane cluster, consider the following resource consumption guidelines.
- For Bare Metal Hosted Control Plane clusters, Hosted Control Plane data plane resources are provided directly by the Bare Metal nodes and do not require additional sizing considerations.

- `vCPU requirement` - total required vCPUs = (vCPUs per node × node count) / overcommit_ratio
  Parameters:
  - `vCPUs per node` - vCPUs required per Hosted Control Plane cluster node.
  - `node count` - number of nodes in the Hosted Control Plane cluster.
  - `overcommit ratio` - factor to consider when you overcommit vCPU resources.
- `Memory requirement` - total required memory (GiB) = memory per node × node count
  Parameters:
  - `memory per node` - memory required per Hosted Control Plane cluster node.
  - `node count` - number of nodes in the HCP cluster.
- `Storage requirement` - total required storage (GiB) = storage per node × node count
  Parameters:
  - `storage per node` - storage required per Hosted Control Plane cluster node.
  - `node count` - number of nodes in the Hosted Control Plane cluster.

Examples for Hosted Control Plane specification and infrastructure resources

Hosted Control Plane specification overview

Hosted Control Plane specification	Description
KubeVirt	Each Hosted Control Plane node requires 4 vCPUs, 8 GiB of memory, and 32 GiB of storage (root disk) + minimum control plane resources.
Bare Metal	Minimum control plane resources.

Infrastructure resource overview

Infrastructure resources	Description
Control nodes	32 cores and 512 GiB of memory
Compute nodes	64 cores and 1024 GiB of memory
Storage per node	4 disks x 3.2 TiB

Examples for KubeVirt Hosted Control Plane cluster

For 10 KubeVirt Hosted Control Plane clusters with 3 nodes:

Hosted Control Plane control plane component (runs on the host cluster):

Component	Formula	Calculation
vCPU	Number of Hosted Control Plane clusters x vCPU per Hosted Control Plane control plane	10 × 11 vCPUs = 110 vCPUs
Memory	Number of Hosted Control Plane clusters x memory per Hosted Control Plane control plane	10 × 19 GiB = 190 GiB
Pods	Number of Hosted Control Plane clusters x pods per Hosted Control Plane control plane	10 × 78 = 780 pods

Hosted Control Plane data plane component (worker VMs capacity for OpenShift Container Platform ingress or console):

Component	Formula	Calculation
vCPU	Number of Hosted Control Plane clusters x number of nodes per Hosted Control Plane cluster x vCPU per Hosted Control Plane data plane	10 × 3 x 4 vCPUs = 120 vCPUs
Memory	Number of Hosted Control Plane clusters x number of nodes per Hosted Control Plane cluster x memory per Hosted Control Plane data plane	10 × 3 x 8 GiB = 240 GiB
Storage	Number of Hosted Control Plane clusters x number of nodes per Hosted Control Plane cluster x storage per Hosted Control Plane data plane	10 × 3 × 32 GiB = 960 GiB

Hosted Control Plane LVM storage

Component	Formula	Calculation
Storage	Number of Hosted Control Plane clusters x LVM storage per Hosted Control Plane cluster	10 × 24 GiB = 240 GiB

Net resources needed for 10 virtual Hosted Control Plane clusters (prior to overhead consideration)

Component	Formula	Calculation
Total CPUs	vCPUs per Hosted Control Plane control plane + vCPUs per Hosted Control Plane data plane	110 + 120 = 230 vCPUs
Total memory	Memory per Hosted Control Plane control plane + memory per Hosted Control Plane data plane	190 + 240 = 430 GiB
Total Storage	Storage (Fusion Data Foundation) per Hosted Control Plane control plane + storage (Fusion Data Foundation) per Hosted Control Plane data plane	960 GiB (Fusion Data Foundation storage)
	Storage (LVM) per Hosted Control Plane control plane + storage (LVM) per Hosted Control Plane data plane	240 GiB (LVM storage)

Component (IBM Fusion HCI)	Values
Overcommit ratio (vCPU)	2:1
High availability reserve (both vCPU and memory)	10%
Hosted Control Plane Hub QPS (pods based limit)	Minimum 78 pods per Hosted Control Plane - control plane
Hosted Control Plane Hub QPS (load based limit)	Additional for first, 1000 QPS, 11.9 vCPU and 13.6 GiB memory As the load increases by 1000 QPS, the hosted control plane resource utilization increases by 9 vCPUs and 2.5 GB memory.
IBM Fusion base (on managed or hub cluster)	Control: 2 vCPU, 7 GiB memory Compute: 2 vCPU, 7 GiB memory Storage: 0.5GiB

Component (IBM Fusion HCI)	Values
Backup & Restore (on managed or hub cluster)	Compute: 4.46 vCPU, 17.78 GiB and 3.35 GiB Storage
Fusion Data Foundation (on managed or hub cluster)	Compute: 13 + (number of drives x 2) vCPU Compute: 26 + (number of drives x 5) GiB Storage: 50.5 GiB
OpenShift reserve	Start with 0.06 CPU for a 1-core system For each extra core, add 0.012 CPU

Net resources needed for 10 virtual Hosted Control Plane clusters (post overhead consideration)

Table 1.

Metric	Formula	Calculation	Result
vCPU (demand)	Total Hosted Control Plane vCPU + IBM Fusion base + Backup & Restore Hub + FDF hub vCPU + OpenShift reserve vCPU	230 + 2 + 4.46 + (13 + 2x4) + 11.9	269.36 vCPU
Physical cores to provision (after 2:1 overcommit & 10% HA)	(vCPU demand) ÷ (2 × (1 - HA%))	269.36 ÷ (2 × 0.9)	149.64 physical cores
Memory (demand)	Total Hosted Control Plane memory + IBM Fusion base + Backup & Restore Hub + FDF hub memory + OpenShift reserve memory	430 + 7 + 17.78 + 46 + 13.6	514.38 GiB
Memory to provision (after 10% HA)	(Memory demand) ÷ (1 - HA%)	(Memory demand) ÷ 0.9	571.53 GiB
Fusion Data Foundation storage	Total Hosted Control Plane Fusion Data Foundation storage + IBM Fusion base Fusion Data Foundation + Fusion Data Foundation hub storage + Backup & Restore hub storage	960 + 0.5 + 50.5 + 3.35	1014.35 GiB
LVM or other storage	Total Hosted Control Plane LVM storage	240	240 GiB
Storage total	Total storage (Fusion Data Foundation and LVM storage combined)	Fusion Data Foundation (1014.35 GiB) + LVM (240 GiB)	1254.35 GiB
Hosted Control Plane control pods	Total Hosted Control Plane pods	78 pods × 10 clusters	780 pods

Approximate resources required:

Component	Requirement (after HA & overcommit)	Per node capacity	Nodes needed
Physical cores	149.64 cores	Compute: 64 cores	2 compute nodes
Memory	571.53 GiB	Compute: 1024 GiB	1 compute node
Fusion Data Foundation storage (considering 3-way replica for Fusion Data Foundation)	Fusion Data Foundation 1014.35 GiB Fusion Data Foundation (3×) = 3043.05 GiB	1 node = 12.8 TiB (13,107 GiB)	0 compute node (3 control node disks are sufficient)
LVM storage	240 GiB	1TiB	3 compute nodes (for high availability)
HCP control pods	780 pods	500 pods	2 compute nodes
Control plane (HA minimum)	-	Control node = 32 cores, 512 GiB	3 control nodes

Examples for Bare Metal Hosted Control Plane cluster

Hosted Control Plane control plane component

Component	Formula	Calculation
vCPU	Number of Hosted Control Plane clusters x vCPU per Hosted Control Plane control plane	3 × 11 vCPUs = 33 vCPUs
Memory	Number of Hosted Control Plane clusters x memory per Hosted Control Plane control plane	3 × 19 GiB = 57 GiB
Pods	Number of Hosted Control Plane clusters x pods per Hosted Control Plane control plane	3 × 78 = 234 pods

Hosted Control Plane LVM storage

Component	Formula	Calculation
Storage	Number of Hosted Control Plane clusters x LVM storage per Hosted Control Plane cluster	3 × 24 GiB = 72 GiB

Net resources needed for 3 Bare Metal Hosted Control Plane clusters (prior to overhead consideration)

Component	Formula	Calculation
Total CPUs	vCPUs per Hosted Control Plane control plane	33 vCPUs
Total memory	Memory per Hosted Control Plane control plane	57 GiB
Total storage	Storage (LVM) per Hosted Control Plane control plane	240 GiB (LVM storage)

Component (IBM Fusion HCI)	Values
Overcommit ratio (vCPU)	1:2
High availability reserve (both vCPU and memory)	10%
Hosted Control Plane Hub QPS (pods based limit)	Minimum 78 pods per Hosted Control Plane - control plane

Component (IBM Fusion HCI)	Values
Hosted Control Plane Hub QPS (load based limit)	For first, 1000 QPS, 11.9 vCPU and 13.6 GiB memory As the load increases by 1000 QPS, the hosted control plane resource utilization increases by 9 vCPUs and 2.5 GB memory.
IBM Fusion base	Control: 2 vCPU, 7 GiB memory Compute: 2 vCPU, 7 GiB memory Storage: 0.5GiB
Backup & Restore (IBM Fusion HCI)	Compute: 4.46 vCPU, 17.78 GiB and 3.35 GiB Storage
Fusion Data Foundation	Compute: 13 + (number of drives x 2) vCPU Compute: 26 + (number of drives x 5) GiB Storage: 50.5 GiB
OpenShift reserve	Start with 0.06 CPU for a 1-core system For each extra core, add 0.012 CPU

Net resources needed for 3 Bare Metal Hosted Control Plane clusters (post overhead consideration)

Table 2.

Metric	Formula	Calculation	Result
vCPU (demand)	Total Hosted Control Plane vCPU + IBM Fusion base + Backup & Restore Hub + FDF hub vCPU + OpenShift reserve vCPU	33 + 2 + 4.46 + (13 + 2x4) + 11.9	72.36 vCPU
Physical vCPU cores to provision (after 1:2 overcommit & 10% HA)	(vCPU demand) ÷ (2 × (1 - HA%))	(vCPU demand) ÷ (2 × 0.9)	40.2 physical cores
Memory (demand)	Total Hosted Control Plane memory + IBM Fusion base + Backup & Restore Hub + FDF hub memory + OpenShift reserve memory	57 + 7 + 17.78 + 46 + 13.6	141.38 GiB
Memory to provision (after 10% HA)	(Memory demand) ÷ (1 - HA%)	(Memory demand) ÷ 0.9	157.08 GiB
Fusion Data Foundation storage	Total Hosted Control Plane Fusion Data Foundation storage + IBM Fusion base Fusion Data Foundation + Fusion Data Foundation hub storage + Backup & Restore hub storage	NA	NA
LVM or other storage	Total Hosted Control Plane LVM storage	72	72 GiB
Storage total	Total storage (Fusion Data Foundation and LVM storage combined)	72 GiB LVM	240 GiB
Hosted Control Plane control pods	Total Hosted Control Plane pods	78 pods × 3 clusters	234 pods

Approximate resources required:

Component	Requirement	Per node capacity	Nodes needed
vCPU (physical cores)	40.2 cores	Compute: 64 cores	1 compute node
Memory	157.08 GiB	Compute: 1024 GiB	1 compute node
Total storage (LVM)	72 GiB	Per node = 12.8 TiB ≈ 13,107.20 GiB	3 compute nodes (for high availability)
HCP control pods	234 pods (78 × 3 clusters)	500 pods	1 compute node
Control plane (HA minimum)	-	Control node = 32 cores, 512 GiB	3 control nodes

Prerequisites for Hosted Control Plane clusters

The prerequisites before you create virtualized or Bare Metal Hosted Control Plane clusters.

Before you begin

- The single or multi-rack installation of IBM Fusion HCI with a minimum of three compute nodes.
- The version of OpenShift® Container Platform must be 4.16 or higher.
- Install and configure Fusion Data Foundation provider mode storage type on the hub cluster.
- If you plan to install Backup & Restore service, ensure that 12 GiB memory is available post the installation.
- Verify that you have all the necessary LVM drives before you initiate LVM configuration on the hub cluster. For more information about LVM drives, see Dedicated drives for LVM filesystems.
- Ensure that the following ports are open on the hub cluster to deploy Hosted Control Plane clusters and support management and data traffic:

Table 1. Ports required for Fusion Data Foundation server and client

Type	Protocol	Ports	Description
Custom TCP	TCP	6789	Ceph Monitor
		3300	Ceph Monitor
		6800-7300	Ceph OSD, MGR, MDS
		9283	Ceph MGR Prometheus Exporter
		31659	API Server

Procedure

1. From the Red Hat® Operator Catalog, install MetalLB 4.16 or higher.
   Add the MetalLB Operator to your cluster so that when a service of type `LoadBalancer` is added to the cluster, MetalLB can add a fault-tolerant external IP address for the service. For the procedure to install and validate MetalLB, see https://docs.openshift.com/. Go to your specific version of OpenShift Container

Platform and check MetalLB details.
Note: When you set up the load balancer, other applications can also get the advertised addresses. You must have enough addresses for any workloads on this cluster and created OpenShift Container Platform clusters.

2. Create MetalLB CR based on the following example:

```
apiVersion: metallb.io/v1beta1
kind: MetalLB
metadata:
  name: metallb
  namespace: metallb-system
```

3. Create an IPaddress pool.
   Note:
   - Reserve a set of unused IPs on the same CIDR as the Bare Metal network of the IBM Fusion HCI cluster for MetalLB. The MetalLB serves these IPs to any load-balancer service that is installed on the cluster and not just the Hosted Control Plane.
   - When planning IP addresses, consider the number of Hosted Control Plane to be created and allocate additional IPs for load balancer services. Reserve three IP addresses for Fusion Data Foundation storage load balancer services. If other workloads utilizes load balancers, factor those requirements into your IP address planning to ensure sufficient addresses are available.

   Example:

```
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: metallb
  namespace: metallb-system
spec:
  addresses:
  - 9.9.0.51-9.9.0.70
```

   Note: When you set up the load balancer, other applications can also get the advertised addresses. You must have enough addresses for any workloads on this cluster and created OpenShift Container Platform clusters.
4. Create `L2advertisement` based on the following example:

```
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: l2advertisement
  namespace: metallb-system
spec:
  ipAddressPools:
   - metallb
```

5. Run the following command to patch ingress:
   Note: This requirement is only for virtualized Hosted Control Plane clusters.

```
oc patch ingresscontroller -n openshift-ingress-operator default --type=json -p '[{ "op": "add", "path": 
"/spec/routeAdmission", "value": {wildcardPolicy: "WildcardsAllowed"}}]'
```

6. Install Multi Cluster Engine for Kubernetes operator.
7. Create a Multi Cluster Engine CR instance based on the following example:

```
apiVersion: multicluster.openshift.io/v1
kind: MultiClusterEngine
metadata:
  name: multiclusterengine
spec:
  targetNamespace: multicluster-engine
```

   Change the parameter values according to your environment.
8. Wait until the Multi Cluster Hub instance is created, available, and in Running state.
   The Multi Cluster Engine instance must be available.
   Run the following command to check whether the Multi Cluster Engine instance is available.

```
$ oc get multiclusterengine
```

   Example output:

```
$ oc get multiclusterengine
NAME STATUS AGE
multiclusterengine Available 170m
```

9. Download hosted control plane CLI from OpenShift Container Platform console:
   a. Go to the Command Line Tools page.
   b. From the Hosted Control Plane - Hosted Control Plane Command Line Interface (CLI) section, download the CLI tar and extract from archive based on your platform.
      Note: In MCE, you can use the user interface instead of CLI. However, the CLI gives more options than the user interface.
10. On the hub cluster, configure LVM by using the local drives for the etcd pods of hosted clusters. To ensure optimal etcd performance, LVM must be set up with three dedicated LVM drives across three compute or compute-storage nodes.
    Note: Use the `lvms-hcp-etcd` StorageClass.
    For more information about dedicated drives for LVM filesystems, see [Dedicated drives for LVM filesystems].

11. Increase the maximum number of pods per node from the default value of 250 to 500. For the procedure to update the value, see [Managing the maximum number of pods per node].
12. For Hosted Control Plane deployments, ensure the following on the hub cluster:
    - A valid pull secret must exist in the namespace where the Hosted Control Plane cluster is defined (the namespace containing the `HostedCluster` resource).
    - The `HostedCluster` CR must include a reference to this pull secret through the `pullSecret` field.
      For example:
      Note: Credentials in pull-secret must include username, password and auth.

```
pullSecret:
    name: <hosted_cluster_name>-pull-secret
```

What to do next

1. Create the hosted control plane clusters. For the procedure to create, see Creating a virtualized Hosted Control Plane cluster.
2. Deploy virtualized or bare metal clusters. For the procedure to install the hosted cluster, see Prerequisites for Hosted Control Plane clusters or Installing Backup & Restore on the hosted cluster.

- **Dedicated drives for LVM filesystems**
  You need dedicated drives for Logical Volume Manager (LVM) filesystems to host control planes. The NVMe drives must be allocated to IBM Fusion HCI with Fusion Data Foundation storage.

Dedicated drives for LVM filesystems

You need dedicated drives for Logical Volume Manager (LVM) filesystems to host control planes. The NVMe drives must be allocated to IBM Fusion HCI with Fusion Data Foundation storage.

Key considerations for dedicated drives for LVM filesystems:

- **Ordering guidance:**
  - You can order these LVM drives of 1.6 TB and add them to a new rack configuration or MES to a pre-ordered rack. By default, when it comes in a new rack, the LVM drive is placed in bay 9.
  - If your earlier IBM Fusion HCI order did not include LVM drives, you can submit a MES order to add them.
  - The MES kit delivers NVMe drives that must be placed in bay 9 of any three compute-only or compute-storage nodes, with a maximum of one drive per node.
- **Rack placement guidance for stand-alone and multi-rack:**
  - If you select the LVM drives option, it is not possible to add more than 9 NVMe drives to storage nodes in the rack. Racks that have 10 NVMe drives in each of the storage nodes are not supported.
  - **In high-availability (HA) multi-rack system**, drives must be distributed across one compute or compute-storage node per rack, spanning three racks, to ensure resiliency.
  - The LVM drives for hosted control plane must not be placed on control nodes or GPU nodes.
  - When you place them in any three compute nodes, the IBM Fusion HCI automatically detects the presence of the control plane drives and configures them for LVM.

Configuring a Hosted Control Plane cluster in a proxy environment

Use these instructions to gather the required configuration and define the necessary Kubernetes resources to configure a `HostedCluster` in a proxy environment.

Before you begin

- Hosted Control Plane is deployed in a restricted environment with limited internet access.
- Proxy server is configured to provide access to external resources.

About this task

In environments where direct internet access is restricted, clusters often rely on a proxy to access required resources. This topic provides instructions on how to configure a Hosted Control Plane cluster in a proxy environment.

Procedure

1. Ensure to add the proxy configuration to the `spec` section of the `HostedCluster` CR while creating `HostedCluster` CR in hub cluster.

   ```
   spec:
     configuration:
       proxy:            # Proxy Configuration
         httpProxy: <http-proxy>
         httpsProxy: <https-proxy>
         noProxy: <list of domains to not route through proxy>
         trustedCA:
           name: <trusted-ca-secret>
   ```

   The following proxy configuration settings are required:
   - `httpProxy`: The HTTP proxy URL.
   - `httpsProxy`: The HTTPS proxy URL.
   - `noProxy`: A list of domains that should not be routed through the proxy.
   - `trustedCA`: The name of the trusted CA secret.
     Note: The filed `trustedCA` should be empty if no custom certificate is used in proxy server.
2. To retrieve the existing proxy configuration values, run the following command on the Hub cluster:

   ```
   oc get proxy cluster -o yaml
   ```

   Example output:

   ```
   apiVersion: config.openshift.io/v1
   kind: Proxy
   metadata:
   ```

```
  name: cluster
  resourceVersion: '532'
spec:
  httpProxy: '<http://squid:squid@172.x.y.z:3128'>
  httpsProxy: '<http://squid:squid@172.x.y.z:3128'>
  noProxy: '.mydomain.com,9999,metal3-state.openshift-machine-api,5050,6385,8089,6180,80,8084,169.253.0.0/16'
  trustedCA:
    name: ''
status:
  httpProxy: '<http://squid:squid@172.x.y.z:3128'>
  httpsProxy: '<http://squid:squid@172.x.y.z:3128'>
  noProxy: 
'.cluster.local,.mydomain.com,.svc,10.128.0.0/14,127.0.0.1,169.253.0.0/16,172.a.b.c/24,172.x.y.z/16,5050,6180,6385,80,8084
,8089,9999,api-int.isfdelace.mydomain.com,localhost,metal3-state.openshift-machine-api'
```

3. Ensure that the following additional settings are added to the `noProxy` field in the `HostedCluster` CR while creating the CR:
    - `localhost`: `127.0.0.1`
    - Cluster subdomain (For example, `isfdelace.mydomain.com`)
    - Cluster base domain (For example, `.mydomain.com`)
    - Service URL extension: `.svc`
    - Service network and cluster network of the base Cluster
    - Service network and cluster network of the HCP cluster
    - First IP of the service network from the HCP
4. To retrieve the cluster network and service network values, run the following command on the hub cluster:

    ```
    oc get network cluster -o yaml
    ```

5. When adding hosts to a Bare Metal hosted control plane cluster, you must download the `discovery.iso` from the MCE user interface. When selecting Add Hosts > With Discovery ISO, click Show proxy settings and add the following values:
    - HTTP proxy URL
    - HTTPS proxy URL
    - No proxy domains

    These field values must match the values used in the earlier steps.

Deployment of virtualized clusters with Fusion Data Foundation

Create virtualized (Hosted Control Plane) clusters with the Fusion Data Foundation as storage type.

- **Planning and prerequisites for your virtualized Hosted Control Plane clusters**
  The prerequisites before you create OpenShift Virtualization Hosted Control Plane clusters.
- **Creating a virtualized Hosted Control Plane cluster**
  Create an OpenShift virtualization Hosted Control Plane cluster.

Planning and prerequisites for your virtualized Hosted Control Plane clusters

The prerequisites before you create OpenShift® Virtualization Hosted Control Plane clusters.

- From the Red Hat® Operator Catalog, deploy Red Hat OpenShift Virtualization. Use the latest version available in your Red Hat operator catalog as the version is driven by underlying the OpenShift Container Platform version. For more information about the supported versions, see Interoperability matrix.
  It is a software component of OpenShift Container Platform to run and manage virtual machine workloads alongside container workloads. For the procedure to install OpenShift Virtualization, see OpenShift virtualization on IBM Fusion HCI. To validate the installation, see Red Hat Documentation.
- Create hyperconverged CR based on the following example:

  ```
  apiVersion: hco.kubevirt.io/v1beta1
  kind: HyperConverged
  metadata:
    name: kubevirt-hyperconverged
    namespace: openshift-cnv
  spec: {}
  ```

- Wait for five to ten minutes for the CR status to be available and the OpenShift Container Platform console to refresh and show up a new Virtualization menu.

Creating a virtualized Hosted Control Plane cluster

Create an OpenShift® virtualization Hosted Control Plane cluster.

Before you begin

- On a six node IBM Fusion HCI, you must create only a maximum of five Hosted Control Plane clusters. On a larger configuration rack having ten or more nodes, you must create a maximum of ten Hosted Control Plane clusters. Go through the sizing guidelines. See Hosted control plane sizing guidance.
- The IBM Fusion HCI with the Multi-Cluster Engine version 2.7 or higher supports hosted clusters, which run on OpenShift version 4.16. For more information about the supported versions, see Interoperability matrix. The upgrade of OpenShift Container Platform is supported on Hosted Control Plane clusters. For creating a hosted cluster with KubeVirt platform by using CLI, see Red Hat Documentation.
- Ensure you meet the following prerequisites:
    - Prerequisites for Hosted Control Plane clusters

- Planning and prerequisites for your virtualized Hosted Control Plane clusters

About this task

If the virtual cluster is not created or has issues, see Troubleshooting hosted control planes.

Procedure

1. Log in to OpenShift Container Platform console, and go to Multi Cluster Engine (MCE) view.
2. Go to Infrastructure > Clusters.
3. In the Cluster list tab, click Create cluster.
4. Select Red Hat OpenShift Virtualization as your infrastructure provider.
5. In the Control plane type, select Hosted. A Create cluster wizard page gets displayed.
6. In the Create cluster wizard, enter your cluster details:
    a. In the Cluster details wizard page, enter your cluster details.

        Release image
            Select your OpenShift version. For more information about the supported versions, see Interoperability matrix.
            Note: No multi arch images must be used while provisioning Hosted Control Plane clusters.

        Additional labels
            Enter `isf.ibm.com/fusion-base=`, `isf.ibm.com/fusion-fdf=`, and `isf.ibm.com/fusion-backup=` labels to install IBM Fusion, Fusion Data Foundation, and Backup & Restore service on the newly created cluster.

        Pull secret
            A pull secret is required for the creation of a Hosted Control Plane cluster. Include the `cp.icr.io` for the Fusion services.

    b. Click Next to go to the Node pools page of the wizard.
    c. In the Node pools page, enter the location and size of the nodes.
    d. Click Next to go to the Review and create page of the wizard.
    e. In the Review and create page, go through the details and click Create.

    Alternatively, use the following command line example to create a Hosted Control Plane for OpenShift Virtualization by using the KubeVirt platform type.
    Note: Ensure you check the following points before you proceed:
    - Ensure you export the variables used in the example with their right values.
    - Also, check whether you have the latest Hosted Control Plane Command Line interface (CLI). If you do not have the latest version, go to Red Hat® OpenShift Container Platform console and download the latest version from Command line tools page.

```
# Set environment variables
export CLUSTER_NAME=example
export PULL_SECRET="$HOME/pull-secret"
export MEM="6Gi"
export CPU="2"
export WORKER_COUNT="2"

hcp create cluster kubevirt \
 --name $CLUSTER_NAME \
 --node-pool-replicas $WORKER_COUNT \
 --pull-secret $PULL_SECRET \
 --memory $MEM \
 --cores $CPU \
 --etcd-storage-class lvms-hcp-etcd \
 --infra-availability-policy HighlyAvailable \
 --root-volume-storage-class ocs-storagecluster-ceph-rbd-virtualization \
 --root-volume-volume-mode Block \
--root-volume-access-modes ReadWriteMany
```

    Run the following command to get a list of available parameters:

```
hcp create cluster kubevirt --help
```

7. To validate the clusters, do the following steps:
    a. Log in to the OpenShift Container Platform console of the hub, and go to multi cluster view.
    b. Go to Infrastructure > Clusters.
    c. From the Clusters list, click a hosted cluster name link to view the details.

    Note: Hosted Control Plane cluster creation can take more than 30 minutes.

What to do next

1. Install the IBM Fusion base on the Hosted Control Plane cluster. For instructions, see Installing the IBM Fusion base.
2. Install the Fusion Data Foundation storage client on the Hosted Control Plane cluster. For instructions, see Installing Fusion Data Foundation on a Hosted Control Plane cluster.

Deploying Bare Metal Hosted Control Plane cluster in disconnected environment

A disconnected environment deployment operates without internet access. You can deploy Hosted Control Plane (HCP) in a disconnected environment on Bare Metal infrastructure.

For the procedure to deploy Bare Metal Hosted Control Plane cluster in disconnected network, see Red Hat documentation.

Important: While you follow the Red Hat documentation, take note of the following two steps that must be done differently:

1. Log in from the service node as KNI user to run the commands.
2. The pull secret for Hosted Control Plane in its namespace must reference the host only, without including the repository path.
3. Skip *Deploying a registry for hosted control planes in a disconnected environment* procedure. See Red Hat documentation.
4. If it is a IBM Fusion HCI deployment with a service node, there is no need to set up a new web server. You can use the existing web server on the service node instead of setting up a new one as follows:
   a. Create a new directory at `/var/www/html/hcp/images`.
   b. Copy the extracted files into this new directory.
   c. The web server on the service node runs on port 8118.

   However, if the deployment is IBM Fusion SDS, a web server needs to be set up manually.
5. Due to a failure in the extraction of ISO from the private image, the ISO is sourced from `Quay.io` and manually copied to the service node.
6. When you create the hosted cluster CR, remove OIDC service.

```
- service: OIDC
  servicePublishingStrategy:
    type: Route
```

7. Skip *Creating bare metal hosts for the hosted cluster* procedure. See Red Hat documentation.
   Important: To add hosts using discovery ISO, follow the steps 4 and 5 mentioned in the Planning and prerequisites for your Bare Metal Hosted Control Plane.
8. When creating an infrastructure environment to configure host inventory, select the following network types: Static, Bond, and Bridge.
9. After the Bare Metal cluster deployment is complete, follow the steps to enable changes to be reflected in the Bare Metal Hosted Control Plane user interface:
   a. To configure ingress, allow access to the Bare Metal Hosted Control Plane user interface and ensure that your Bare Metal cluster is accessible. For more information about setting up ingress, see Configuring ingress for Bare Metal Hosted cluster.
   b. Before proceeding with the ingress setup, ensure that the **Red Hat Catalog Source** is deployed in the Bare Metal Hosted Control Plane cluster. This catalog source should include the **MetalLB operator**.

Examples of CRs and CMs created:

Configuring `AgentServiceConfig` resources

1. Create a configmap named `custom-registries` in the `multicluster-engine` namespace.

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: custom-registries
  namespace: multicluster-engine
data:
  <registry_name>..<port>: |
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----
  <registry_name>..<port>: |
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----
  <registry_name>..<port>: |
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----
  registries.conf: |
    unqualified-search-registries = ["registry.access.redhat.com", "docker.io"]
    short-name-mode = ""
    [[registry]]
       prefix = ""
       location = "quay.io/openshift-release-dev/ocp-v4.0-art-dev"
       mirror-by-digest-only = true
       [[registry.mirror]]
         location = "$TARGET_PATH/openshift/release"

    [[registry]]
       prefix = ""
       location = "quay.io/openshift-release-dev/ocp-release"
       mirror-by-digest-only = true
       [[registry.mirror]]
         location = "$TARGET_PATH/openshift/release-images"

    [[registry]]
       prefix = ""
       location = "registry.redhat.io/multicluster-engine"
       mirror-by-digest-only = true
       [[registry.mirror]]
         location = "$TARGET_PATH/multicluster-engine"

    [[registry]]
       prefix = ""
       location = "registry.redhat.io/openshift4"
       mirror-by-digest-only = true
       [[registry.mirror]]
         location = "$TARGET_PATH/openshift4"

    [[registry]]
       prefix = ""
       location = "registry.redhat.io/rhel9"
       mirror-by-digest-only = true
       [[registry.mirror]]
         location = "$TARGET_PATH/rhel9"

    [[registry]]
       prefix = ""
       location = "registry.redhat.io/rhel8"
       mirror-by-digest-only = true
       [[registry.mirror]]
```

```
        location = "$TARGET_PATH/rhel8"

    [[registry]]
       prefix = ""
       location = "registry.redhat.io/redhat"
       mirror-by-digest-only = true
       [[registry.mirror]]
         location = "$TARGET_PATH/redhat"
```

Note:

- Replace `<registry_name>..<port>` with the actual registry name and port.
- Provide the CA root certificate and intermediate certificate of the registry.
- The `unqualified-search-registries` is mandatory.
- Mirrors list can be found in the workspace folder created when mirroring OpenShift® Container Platform and multi-cluster engine or `metalLB` images.

2. Create an `AgentServiceConfig` resource named `agent`.

```
apiVersion: agent-install.openshift.io/v1beta1
kind: AgentServiceConfig
metadata:
  name: agent
spec:
  databaseStorage:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 200Gi
    storageClassName: ocs-storagecluster-ceph-rbd
  filesystemStorage:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 200Gi
    storageClassName: ocs-storagecluster-ceph-rbd
  imageStorage:
    accessModes:
      - ReadWriteOnce
    resources:
      requests:
        storage: 100Gi
  mirrorRegistryRef:
    name: custom-registries
  osImages:
    - cpuArchitecture: x86_64
      openshiftVersion: 4.18.20
      rootFSUrl: 'http://172.20.x.y:8118/hcp/images/rhcos-418.94.202501221327-0-live-rootfs.x86_64.img'
      url: 'http://172.20.x.y:8118/hcp/images/rhcos-418.94.202501221327-0-live.x86_64.iso'
      version: 418.94.202501221327-0
```

Note:

- Specify the configmap name created earlier in the `multicluster-engine` namespace under `mirrorRegistryRef`.
- Add one entry for each OpenShift Container Platform version you wish to deploy hosted cluster with under `osImages`.
- The `rootFSUrl` and `url` are links to files extracted earlier and hosted on a web server.
- The `version` is the core OS version of the downloaded ISO.

Configuring TLS certificates for a disconnected installation of Hosted Control Plane

Create a configmap named `registry-config` in the `openshift-config` namespace.

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: registry-config
  namespace: openshift-config
data:
   <registry_name>..<port>: |
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----
  <registry_name>..<port>: |
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----
  <registry_name>..<port>: |
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----
```

Note:

- Replace `<registry_name>..<port>` with the actual registry name and port.
- Provide the CA root certificate and intermediate certificate of the registry.

Creating a `hostedcluster` on Bare Metal

1. Create two namespaces: `fusion-bm` and `fusion-bm-fusion-bm`.

```
apiVersion: v1
kind: Namespace
metadata:
  name: fusion-bm-fusion-bm
spec: {}
```

```
apiVersion: v1
kind: Namespace
metadata:
  name: fusion-bm
spec: {}
```

2. Create a configmap named `user-ca-bundle` in the `fusion-bm` namespace.

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: user-ca-bundle
  namespace: fusion-bm
data:
  ca-bundle.crt: |
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----
```

Note: Provide the registry CA root certificate.

3. Create a secret named `fusion-bm-pull-secret` and `sshkey-cluster-fusion-bm` in the `fusion-bm` namespace.

```
apiVersion: v1
kind: Secret
metadata:
  name: fusion-bm-pull-secret
  namespace: fusion-bm
data:
  .dockerconfigjson: xxxxxxxxx
type: kubernetes.io/dockerconfigjson
---
apiVersion: v1
kind: Secret
metadata:
  name: sshkey-cluster-fusion-bm
  namespace: fusion-bm
data:
  id_rsa.pub: xxxxxxxxx
type: Opaque
```

4. Create a role named `capi-provider-role` in the `fusion-bm-fusion-bm` namespace.

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: capi-provider-role
  namespace: fusion-bm-fusion-bm
rules:
  - verbs:
      - '*'
    apiGroups:
      - agent-install.openshift.io
    resources:
      - agents
```

5. Create a `hostedcluster` CR named `fusion-bm` in the `fusion-bm` namespace.

```
apiVersion: hypershift.openshift.io/v1beta1
kind: HostedCluster
metadata:
  name: fusion-bm
  namespace: fusion-bm
spec:
  fips: false
  release:
    image: $TARGET_PATH/openshift/release-images:4.18.20-x86_64
  dns:
    baseDomain: xyz.com
  controllerAvailabilityPolicy: HighlyAvailable
  etcd:
    managed:
      storage:
        persistentVolume:
          size: 8Gi
          storageClassName: ocs-storagecluster-ceph-rbd
        type: PersistentVolume
    managementType: Managed
  infrastructureAvailabilityPolicy: SingleReplica
  platform:
    agent:
      agentNamespace: fusion-bm-fusion-bm
    type: Agent
  additionalTrustBundle:
    name: user-ca-bundle
  networking:
    clusterNetwork:
      - cidr: 10.132.0.0/14
    networkType: OVNKubernetes
    serviceNetwork:
      - cidr: 172.31.0.0/16
  pullSecret:
    name: fusion-bm-pull-secret
  configuration:
    operatorhub:
      disableAllDefaultSources: true
  capabilities: {}
```

```
  sshKey:
    name: sshkey-cluster-fusion-bm
  autoscaling: {}
  imageContentSources:
    - mirrors:
        - $TARGET_PATH/openshift/release
      source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
    - mirrors:
        - $TARGET_PATH/redhat
      source: registry.redhat.io/redhat
    ...
  olmCatalogPlacement: management
  services:
    - service: APIServer
      servicePublishingStrategy:
        type: LoadBalancer
    - service: OAuthServer
      servicePublishingStrategy:
        type: Route
    ...
```

Note:
- Specify the configmap name created earlier in the `fusion-bm` namespace under `additionalTrustBundle`.
- Specify the secret name created earlier in the `fusion-bm` namespace under `pullSecret` and `sshKey`.
- Specify the image content sources and mirrors.

6. Wait for the `hostedcluster` CR to get created and the API to become available.
7. Create a `NodePool` CR named `fusion-bm-np` in the `fusion-bm` namespace.

```
apiVersion: hypershift.openshift.io/v1beta1
kind: NodePool
metadata:
  name: fusion-bm-np
  namespace: fusion-bm
spec:
  arch: amd64
  clusterName: fusion-bm
  management:
    autoRepair: false
    replace:
      rollingUpdate:
        maxSurge: 1
        maxUnavailable: 0
      strategy: RollingUpdate
    upgradeType: InPlace
  nodeDrainTimeout: 0s
  platform:
    agent:
      agentLabelSelector:
        matchLabels:
          inventory.agent-install.openshift.io/hcp: fusion-bm
    type: Agent
  release:
    image: '$TARGET_PATH/openshift/release-images:4.18.20-x86_64'
  replicas: 3
```

Note:
- Specify the cluster name and release image.
- Specify the number of replicas.

Create a host inventory

1. Create a secret named `pullsecret-fusion-bm` in the `fusion-bm-fusion-bm` namespace.

```
apiVersion: v1
kind: Secret
metadata:
  name: pullsecret-fusion-bm
  namespace: fusion-bm-fusion-bm
data:
  .dockerconfigjson: xxxxxxxx
type: kubernetes.io/dockerconfigjson
```

2. Create an `InfraEnv` CR named `fusion-bm` in the `fusion-bm-fusion-bm` namespace.

```
apiVersion: agent-install.openshift.io/v1beta1
kind: InfraEnv
metadata:
  name: fusion-bm
  namespace: fusion-bm-fusion-bm
  labels:
    agentclusterinstalls.extensions.hive.openshift.io/location: us
    networkType: static
spec:
  agentLabels:
    agentclusterinstalls.extensions.hive.openshift.io/location: us
  imageType: minimal-iso
  mirrorRegistryRef:
    name: custom-registries
    namespace: multicluster-engine
  ipxeScriptType: DiscoveryImageAlways
  cpuArchitecture: x86_64
  sshAuthorizedKey: xxxxxxxx
  pullSecretRef:
```

```
    name: pullsecret-fusion-bm
  nmStateConfigLabelSelector:
    matchLabels:
      infraenvs.agent-install.openshift.io: fusion-bm
```

Note:

- Specify the mirror registry reference and namespace.
- Specify the image type and CPU architecture.
- Specify the SSH authorized key and pull secret reference.

Deploying virtualized Hosted Control Plane cluster in disconnected environment

A disconnected environment deployment operates without internet access. You can deploy Hosted Control Plane (HCP) in a disconnected environment on the virtualized infrastructure.

For the procedure to deploy virtualized Hosted Control Plane cluster in disconnected network, see Red Hat documentation.

Important: While you follow the Red Hat documentation, take note of the following two steps that must be done differently:

1. The pull secret for the Hosted Control Plane in its namespace must only reference the host. Do not include the repository path.
2. When you create the virtualized Hosted Control Plane cluster CR, remove OIDC service.

```
- service: OIDC
  servicePublishingStrategy:
    type: Route
```

Examples of CRs and CMs created while deploying virtualized Hosted Control Plane:

1. Create two namespaces: `clusters` and `clusters-hosted-cluster-name`.
   For example, `clusters-disconnected1example`

```
apiVersion: v1
kind: Namespace
metadata:
  creationTimestamp: null
  name: clusters-disconnected1
spec: {}
status: {}


apiVersion: v1
kind: Namespace
metadata:
  creationTimestamp: null
  name: clusters
spec: {}
status: {}
```

2. Copy the registry CA cert and create a configmap in the `clusters` namespace.

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: user-ca-bundle
  namespace: clusters
data:
  ca-bundle.crt: |
    // Registry1 CA
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----

    // Registry2 CA
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----

    // Registry3 CA
    -----BEGIN CERTIFICATE-----
    -----END CERTIFICATE-----
```

3. Create a pull secret in the `clusters` namespace.

```
kind: Secret
apiVersion: v1
metadata:
  name: disconnected-secret
  namespace: clusters
data:
  .dockerconfigjson: xxxxxxxx
type: kubernetes.io/dockerconfigjson
```

4. Create a `hostedcluster` CR in the `clusters` namespace.

```
apiVersion: hypershift.openshift.io/v1beta1
kind: HostedCluster
metadata:
  name: disconnected1
  namespace: clusters
spec:
  fips: false
  release:
```

```
    image: $TARGET_PATH/openshift/release-images:4.18.20-x86_64
  dns:
    baseDomain: xyz.com
  controllerAvailabilityPolicy: HighlyAvailable
  etcd:
    managed:
      storage:
        persistentVolume:
          size: 8Gi
          storageClassName: ocs-storagecluster-ceph-rbd
        type: PersistentVolume
    managementType: Managed
  infrastructureAvailabilityPolicy: SingleReplica
  platform:
    kubevirt:
      baseDomainPassthrough: true
    type: KubeVirt
  additionalTrustBundle:
    name: user-ca-bundle
  networking:
    clusterNetwork:
      - cidr: 10.132.0.0/14
    networkType: OVNKubernetes
    serviceNetwork:
      - cidr: 172.31.0.0/16
  pullSecret:
    name: disconnected-secret
  configuration:
    operatorhub:
      disableAllDefaultSources: true
  capabilities: {}
  sshKey:
    name: ''
  autoscaling: {}
  imageContentSources:
    - mirrors:
        - $TARGET_PATH/openshift/release
      source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
    - mirrors:
        - $TARGET_PATH/redhat
      source: registry.redhat.io/redhat
    - mirrors:
        - $TARGET_PATH/rhel9
      source: registry.redhat.io/rhel9
    - mirrors:
        - $TARGET_PATH/rhel8
      source: registry.redhat.io/rhel8
    - mirrors:
        - $TARGET_PATH/openshift/release-images
      source: quay.io/openshift-release-dev/ocp-release
    - mirrors:
        - $TARGET_PATH/openshift4
      source: registry.redhat.io/openshift4
    - mirrors:
        - $TARGET_PATH/multicluster-engine
      source: registry.redhat.io/multicluster-engine
  olmCatalogPlacement: management
  services:
    - service: APIServer
      servicePublishingStrategy:
        type: LoadBalancer
    - service: Ignition
      servicePublishingStrategy:
        type: Route
    - service: Konnectivity
      servicePublishingStrategy:
        type: Route
    - service: OAuthServer
      servicePublishingStrategy:
        type: Route
```

Note: While you follow the Red Hat documentation, take note of the following two steps that must be done differently:

a. The pull secret for the Hosted Control Plane in its namespace must only reference the host. Do not include the repository path.
b. When you create the virtualized Hosted Control Plane cluster CR, remove OIDC service.

```
- service: OIDC
  servicePublishingStrategy:
    type: Route
```

5. Create a `NodePool` CR in the `clusters` namespace.

```
apiVersion: hypershift.openshift.io/v1beta1
kind: NodePool
metadata:
  name: disconnected1
  namespace: clusters
spec:
  arch: amd64
  clusterName: disconnected1
  management:
    autoRepair: false
    upgradeType: Replace
  nodeDrainTimeout: 0s
  nodeVolumeDetachTimeout: 0s
  platform:
```

```
    kubevirt:
      attachDefaultNetwork: true
      compute:
        cores: 2
        memory: 6Gi
      networkInterfaceMultiqueue: Enable
      rootVolume:
        persistent:
          size: 32Gi
        type: Persistent
    type: KubeVirt
  release:
    image: $TARGET_PATH/openshift/release-images:4.18.20-x86_64
  replicas: 2
```

Note:

- Specify the cluster name and release image.
- Specify the number of replicas along with configuration such as cores, memory and so on.

Deployment of Bare Metal clusters with Fusion Data Foundation

Create Bare Metal hosted clusters with Fusion Data Foundation as the storage type. It supports both Hosted Control Plane Bare Metal internal servers in a rack and Hosted Control Plane Bare Metal external servers.

The following are the high-level steps to create a Bare Metal Hosted Control Plane cluster:
Note: Ensure you meet all the prerequisites before you deploy:

- Prerequisites for Hosted Control Plane clusters
- Planning and prerequisites for your Bare Metal Hosted Control Plane

1. Plan for the Bare Metal Hosted Control Plane
2. Create the Infrastructure Environment.
3. Import the hosts by using Multi Cluster Engine (MCE):
   a. Create the NMStateConfig for each host.
   b. Download the `discovery.iso`.
   c. Boot the host with the `discovery.iso`.
   d. Accept host into MCE inventory.
4. Create the Hosted Control Plane cluster.
5. Configuring ingress for newly created Bare Metal hosted cluster.

- **Planning and prerequisites for your Bare Metal Hosted Control Plane**
  As a prerequisite, gather networking information of the hosting cluster. Define and apply DHCP and DNS information, and configure host inventory settings.
- **Configuring ingress for Bare Metal Hosted cluster**
  After you deploy the Bare Metal clusters with Fusion Data Foundation, you must configure the ingress for you to view the changes in the user interface.
- **Booting up the Bare Metal Hosted Control Plane nodes through CLI**
  Use the steps to boot the Dell and Lenovo nodes from the service node for Bare Metal Hosted Control Plane.
- **Expanding your Hosted Control Plane Bare Metal cluster**
  Add a new host to the exiting Bare Metal Hosted Control Plane cluster.

Planning and prerequisites for your Bare Metal Hosted Control Plane

As a prerequisite, gather networking information of the hosting cluster. Define and apply DHCP and DNS information, and configure host inventory settings.

Plan for the Bare Metal Hosted Control Plane:

1. Choose your configuration.
2. Gather networking information about IBM Fusion HCI hosting cluster.
3. Define the new networking information.
4. Apply the networking information to the DNS and DHCP in your environment

Choose your configuration

There are two configurations of Bare Metal Hosted Control Plane cluster.

Internal
: In the internal Bare Metal Hosted Control Plane, the servers that are used in the cluster are within the IBM Fusion HCI. When the servers are internal to the IBM Fusion HCI, most of the network information required is collected and present in the Custom Resources of the IBM Fusion HCI.

External
: In external Bare Metal Hosted Control Plane, the servers used in the cluster are external to the IBM Fusion HCI. For external servers, you must gather the network information from your lab network administrator.
Connecting your external servers to IBM Fusion HCI:

External servers must be connected to the IBM Fusion HCI through a switch to allow connectivity. These servers can then be imported into Multi Cluster Engine (MCE) as part of a Host Inventory. For instructions to import and its usage in a Hosted Control Plane, see Deploying Bare Metal clusters with Fusion Data Foundation.

Networking prerequisites

Step 1: Gather network information

The networking information consists of the entries that the network administrator needs to add to the network. You need the following information about the host cluster:

Keep the following data ready to use when you create a cluster and `nmstate` config for each node.

Cluster wide information:

Note: The cluster name, sub domain, and domain are all for one hosted cluster. Create them for each of your hosted clusters.

Details	Description	Example values
Base Domain	Cluster base domain you want to use for Hosted Control Plane cluster to be created	`mydomain.com`
Cluster name	Cluster name you want to use for HCP cluster to be created.	`fusion-bm`
Sub Domain	This is always <cluster name>.<base domain>	`fusion-bm.mydomain.com`

Individual nodes in the cluster:

Entry	Description	Example values
BMC address	This is value of field named ipv6ULA or ipv4 in the kickstart CM. You can use ipv6ULA or ipv4.	Example for ipv6ULA - fd9c:316d:179e:c0de:a94:efff:fefe:2ec1 Example value for ipv4 - 170.254.2.9
IMM USER/Password	To get this value, search for `secretName` field in the `kickstart config`. Inside that secret, search for `defaultUserName` and `defaultUserPasswrd` fields for IMM user and Password respectively.	
Bare metal primary interface IP	Unused IP that you plan yo use for Bare Metal host in Hosted Control Plane cluster	172.17.x.y
Macaddress	It is available in the `kickstart config` `networkInterfaces`, the bond0 macAddress. You need both Bare Metal Primary and secondary interface Mac addresses. You can find the macaddress in the slot section that includes the first macaddress. This guidance is for internal HCI managed nodes. Consult your network administrator for external bare metal nodes	
Gateway IP	Run the following command to get this from one of HCI nodes: `ip r │grep default│grep br-ex│awk '{print $3}'` If the Hosted Control Plane nodes use the same gateway server as the base HCI cluster. Otherwise, get it from the network administrator. It is only needed when you use the static IP assignment for nodes in Hosted Control Plane cluster. For DHCP managed nodes, this field is not needed.	172.17.x.1
MTU	This can be 9000 or 1500 based on base HCI rack Bare metal MTU used during installation. Find this value from field `mtuCount` in secret `userconfig-secret` in namespace `ibm-spectrum-fusion-ns`.	9000
Bare metal Primary interface's first port's Mac address	This is the value of primary macaddress in kickstart config map for a given node where the interfaceType is baremetal.	08:c0:eb:d4:16:4e
Bare metal Primary interface's second port's Mac address	This is the value of secondary macaddress in kickstart config map for given node where interfaceType is baremetal.	08:c0:eb:d4:16:4f
Bare metal Primary interface's Mac address	This is value of field `networkInterfaces` -> macAddress in kickstart config map for a given node where the `interfaceType` is `baremetal`.	08:c0:eb:d4:16:4e
Bare metal Primary interface's (bond0) first port/interface name	This is value of field networkInterfaces -> interfaceLeg1 in kickstart config map for a given node, where interfaceType is `baremetal`.	ens3f0np0
Bare metal Primary interface's (bond0) second port/interface name	This is value of field networkInterfaces -> interfaceLeg2 in kickstart config map for given node, where interfaceType is `baremetal`.	

Example servers external to IBM Fusion HCI:

Server	BMC address of external Bare Metal node	HCP Cluster name	BMC address	Switch bond0 info	mac Address	Type
sr650immru22 tc11-m04-ru22.mydomain.com	fd8c:215d:178e:c0de:3a68:ddff:fe57:2e95	extbarehcp01m04	1.23.45.178	ens1f0np0 b8:3f:d2:3c:6d:60 ens1f1np1 b8:3f:d2:3c:6d:61	b8:3f:d2:3c:6d:60	External

Example servers internal to IBM Fusion HCI:

Server	macaddress Example	HCP Cluster name	BMC address	BMC address	Type/ Label
compute-1-ru8	1070FDB8DF72	fusion-bm	1.23.45.163	fd8c:215d:178e:c0de:a94:efff:fefd:e7e1	Compute

DHCP update for the cluster:

Note: This section is needed only when you plan to use DHCP to manage IP and hostname assignment for nodes in Hosted Control Plane. If you use static IP assignment, then skip this section.

You need DHCP only if you want to manage your server IPs using the DHCP for hosted clusters.

To ensure that the individual servers have an IP address that can be contacted by the Hosted Control Plane cluster, make an entry for each host that links its macaddress to an IP address that can be contacted by the IBM Fusion HCI cluster. The macaddress can be found on the back of each physical server.

Example:

Server	macaddress	IP address
compute-1-ru8.fusion-bm.mydomain.com	10:70:FD:B8:DF:72	172.17.x.y

DNS update for Cluster:

Note: Add DNS entry of every host user who wants to create Bare Metal Hosted Control Plane cluster.
The DNS update table entries for each Hosted Control Plane cluster that links a lab provided Server IP address to the `*.apps.NameofCluster.FQDN`. This IP address is used to create a load balancer on the Hosted Control Plane cluster to allow ingress to that cluster.

Load balancer IP	Entry for DNS	FQDN Host	Ingress Type	HCP Clustername	Domain name
1.23.45.910	*.apps.barehcp01m04.mydomain.com	compute-1-ru8.fusion-bm.mydomain.com	loadbalancer	fusion-bm	mydomain.com

Server name	Entry for DNS	IP address
compute-1-ru8	compute-1-ru8.fusion-bm.mydomain.com	172.17.x.y

Step 2: Apply the networking information to the DNS and DHCP in your environment
After you gather hosting cluster information, apply the following in your environment (lab network environment):

- DHCP update
- DNS update for host

Multi Cluster Engine (MCE)

Step 1: Configure Host Inventory Settings
After the initial setup, configure the host inventory settings. For more information about the host inventory settings, see Red Hat documentation.

Step 2: Create project in host IBM Fusion OpenShift Container Platform cluster for host inventory for Hosted Control Plane cluster to be created
From the OpenShift Container Platform console of IBM Fusion, create a project designated for the host inventory along with other necessary resources to support the Hosted Control Plane cluster.
You can use any name. In this example, cluster name is used for simple identification.

Step 3: Create your infrastructure environment
The next step is to create an infrastructure environment in MCE or using the Hosted Control Plane CLI. This infrastructure environment contains the list of Bare Metal hosts that you can select to create a Bare Metal Hosted Control Plane.
Note: Create one infrastructure environment per hosted control plane cluster. Each Hosted Control Plane can only add hosts from a single infrastructure.

1. In the MCE user interface, go to Infrastructure > Host Inventory.
2. Select Create infrastructure environment.
3. On the Create infrastructure environment page, enter the following details:

Name
Name of the infrastructure. No specific name is required.

Network type
Select Static IP, Bridges, and Bonds.

Location
Location of hosts

Labels
Optional value.

Pull Secret
The pull secret must include the following:
- cloud.openshift.com
- cp.icr.io
- quay.io
- registry.connect.redhat.com
- registry.redhat.io

SSH public key
Generate a private public ssh key pair from bastion host and specify public key in ssh field. It is optional but recommended.

Step 4: Import the hosts by using MCE
With the infrastructure environment created, the next steps are to boot individual servers with an ISO to import them into the host inventory.

Step 4.1: Create the NMStateConfig for each host
The **`NMStateConfig`** allows the server to be recognized by the IBM Fusion HCI cluster. For additional information, see Red Hat documentation.
The NMStateConfig varies based on whether static IP assignment or DHCP IP assignment is used. For a fresh IBM Fusion HCI 2.10 installation or an upgrade from 2.9 to 2.10 with the network migration script applied, use the new NMStateConfig. If IBM Fusion HCI is upgraded from version 2.9 to 2.10 without applying the network migration script, retain the old NMStateConfig configuration.

Static IP
New **`NMStateConfig`** for static IP assignment:

```
apiVersion: agent-install.openshift.io/v1beta1
kind: NMStateConfig
metadata:
  labels:
    infraenvs.agent-install.openshift.io: <same as infra name>
  name: <any unique name for host within infra for example fusion-bm-ru8>
  namespace: <same namespace as used for infra>
spec:
  config:
    dns-resolver:
      config:
        server:
          - <DNS server IP>
    interfaces:
      - ipv6:
          enabled: false
        link-aggregation:
```

```
          mode: 802.3ad
          options:
            lacp_rate: '1'
            miimon: '140'
            xmit_hash_policy: '1'
          ports:
            - <Bare metal Primary interface's first port/interface name>
            - <Bare metal Primary interface's second port/interface name>
        mac-address: <Bare metal Primary interface's first port's Mac address>
        mtu: <MTU value>
        name: bond0
        state: up
        type: bond
      - ipv4:
          address:
            - ip: <Bare metal primary interface IP>
              prefix-length: 24
          dhcp: false
          enabled: true
        ipv6:
          enabled: false
        name: <bond name followed by VLAN ID of type OpenShift Customer VLAN present on the cluster for example 
bond0.VLAN_ID>
        state: up
        type: vlan
        vlan:
          base-iface: bond0
          id: <VLAN ID of type OpenShift Customer VLAN present on the cluster>
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: <Gateway IP>
          next-hop-interface: <bond name followed by VLAN ID of type OpenShift Customer VLAN present on the cluster 
for example bond0.VLAN_ID>
  interfaces:
    - macAddress: <Bare metal Primary interface's first port's Mac address>
      name: Bare metal Primary interface's first port/interface name>
    - macAddress: <Bare metal Primary interface's second port's Mac address>
      name: Bare metal Primary interface's second port/interface name>
```

Old **NMStateConfig** for static IP Assignment:

```
apiVersion: agent-install.openshift.io/v1beta1
kind: NMStateConfig
metadata:
  labels:
    infraenvs.agent-install.openshift.io: <same as infra name>
  name: <any unique name for host within infra for example fusion-bm-ru8>
  namespace: <same namespace as used for infra>
spec:
  config:
    dns-resolver:
      config:
        server:
          - <DNS server IP>
    interfaces:
      - ipv4:
          address:
            - ip: <Bare metal primary interface IP>
              prefix-length: 24
          dhcp: false
          enabled: true
        ipv6:
          enabled: false
        link-aggregation:
          mode: 802.3ad
          options:
            lacp_rate: '1'
            miimon: '140'
            xmit_hash_policy: '1'
          ports:
            - <Bare metal Primary interface's first port/interface name>
            - <Bare metal Primary interface's second port/interface name>
        mac-address: <Bare metal Primary interface's first port's Mac address>
        mtu: <MTU value>
        name: bond0
        state: up
        type: bond
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: <Gateway IP>
          next-hop-interface: bond0
  interfaces:
    - macAddress: <Bare metal Primary interface's first port's Mac address>
      name: <Bare metal Primary interface's first port/interface name>
    - macAddress: <Bare metal Primary interface's second port's Mac address>
      name: <Bare metal Primary interface's second port/interface name>
```

DHCP

New **NMStateConfig** for DHCP IP assignment:

```
apiVersion: agent-install.openshift.io/v1beta1
kind: NMStateConfig
metadata:
```

```
  labels:
    infraenvs.agent-install.openshift.io: <same as infra name>
  name: <any unique name for host within infra for example fusion-bm-ru8>
  namespace: <same namespace as used for infra>
spec:
  config:
    interfaces:
     - ipv6:
        enabled: false
       link-aggregation:
        mode: 802.3ad
        options:
          lacp_rate: "1"
          miimon: "140"
          xmit_hash_policy: "1"
        ports:
        - <Bare metal Primary interface's first port/interface name>
        - <Bare metal Primary interface's second port/interface name>
      mac-address: <Bare metal Primary interface's first port's Mac address>
      mtu: <MTU value>
      name: bond0
      state: up
      type: bond
     - ipv4:
          dhcp: true
          enabled: true
        ipv6:
          enabled: false
        name: <bond name followed by VLAN ID of type OpenShift Customer VLAN present on the cluster. For example 
bond0.VLAN_ID>
        state: up
        type: vlan
        vlan:
          base-iface: bond0
          id: <VLAN ID of type OpenShift Customer VLAN present on the cluster>
  interfaces:
    - macAddress: <Bare metal Primary interface's first port's Mac address>
      name: <Bare metal Primary interface's first port/interface name>
    - macAddress: <Bare metal Primary interface's second port's Mac address>
      name: <Bare metal Primary interface's second port/interface name>
```

Old NMStateConfig:

```
apiVersion: agent-install.openshift.io/v1beta1
kind: NMStateConfig
metadata:
  labels:
    infraenvs.agent-install.openshift.io: <same as infra name>
  name: <any unique name for host within infra for example fusion-bm-ru8>
  namespace: <same namespace as used for infra>
spec:
  config:
    interfaces:
    - ipv4:
        dhcp: true
        enabled: true
      ipv6:
        enabled: false
      link-aggregation:
        mode: 802.3ad
        options:
          lacp_rate: "1"
          miimon: "140"
          xmit_hash_policy: "1"
        ports:
        - <Bare metal Primary interface's first port/interface name>
        - <Bare metal Primary interface's second port/interface name>
      mac-address: <Bare metal Primary interface's first port's Mac address>
      mtu: <MTU value>
      name: bond0
      state: up
      type: bond
  interfaces:
    - macAddress: <Bare metal Primary interface's first port's Mac address>
      name: <Bare metal Primary interface's first port/interface name>
    - macAddress: <Bare metal Primary interface's second port's Mac address>
      name: <Bare metal Primary interface's second port/interface name>
```

The `NMStateConfig` is dependent on the network environment and must be created in consultation with your network administrators.

Enter the following fields in the NMStateConfig.yaml:
Note: These fields within the `NMStateConfig` are unique per host.

Field	Description
name	Unique name for each node.
namespace	Namespace is the infraenv namespace.
labels	`infraenvs.agent-install.openshift.io: infraNamespace`. The `infraNamespace` must match the name of the `infraenvironment` into which the host is imported.
ports	The correct ports can be found in the `kickstart config` CR. The `kickstart config` for the particular node to be imported to MCE. Within the `kickstart config`, there is a section “NetworkInterfaces”. Within this section, the interfaceName with “baremetal” has the name of the two interfaces interfaceLeg1 and interfaceLeg2. The values under interfaceLeg1 and interfaceLeg2 are the ports. This guidance is for internal. Consult your network administrator for external.
ipaddress	It is the ipv4 address for that node or server.

Field	Description
interfaces > port: macaddress: and port: macaddress:	You can find these values in the `kickstart config` CR. Each port is an entry from the ports. The `macaddress` is for that port. The `macaddress` can be `kickstart config` for the node by looking for the `networkType port: baremetal`. Map that `macaddress` to the `networkCards: -slot-#:` There will be one matching `macaddress` and one new one, use those in the `NMStateConfig.yaml`. Important: change any letters to lower case. Example: `ens1f0np1: macaddress:` macaddress can be found in the `kickstart config` for the node under `networkCards: slot-1: macaddress:` This guidance is for internal. Consult your network administrator for external.

NMState information is sourced from the switch. For unmanaged servers, you must manually provide the required values.

After the NMStateConfig.yaml is created for a server, it must be added to IBM Fusion HCI cluster using the following commands:

1. Go to the `infraenvnamespace` namespace.

   ```
   oc project infraenvnamespace
   ```

2. Apply the YAML.

   ```
   oc apply -f nmstate.yaml
   ```

Step 4.2: Download the discovery.iso

Use the **wget** command to download the `discovery.iso` to the service node or another jumpbox.
Note: Keep this ISO secure as you can use this to add a server to the cluster with sensitive information.
To get the retrieval command do the following:

1. Log into the hub IBM Fusion HCI cluster.
2. In the MCE section, go to Infrastructure > Host inventory.
3. In the infrastructure environment, select the infrastructure you intend to import the host into and select that infrastructure environment.
4. Select Add Hosts > With Discovery ISO. There exists either a URL or a **wget** command. Use the command or URL to transfer the ISO to your jumpbox.

Step 4.3: Boot the host with the discovery.iso

This task requires you to log into the IMM, mount the ISO image, and reboot the server.
Note: For internal, go through the steps in this section. For external, go through the Red Hat documentation.
Mount the ISO on to the server through the IMM:

1. Get the IPv6 or IPv4 from kickstart config for that server.
2. Log in to the IMM.
3. To get this value, search for `secretName` field in the `kickstart config`. Inside that secret, search for `defaultUserName` and `defaultUserPasswrd` fields for IMM user and Password respectively.
4. Select Remote Console tab.
5. In the Remote Console tab, select Launch remote console and Media.
6. In the Media page, go to the Mount media from Client Browser section.
7. Select the discovery.iso image and mount it.
8. Select one virtual media to boot on the next restart.
9. Select the mounted discovery.iso and restart immediately.
10. Close windows and monitor that the remote console is showing the reboot.

Step 5: Accept the host into the MCE Inventory

After the server gets restarted with discovery.iso, accept the host into the host inventory.

1. In the MCE user interface, go to Infrastructure > Host inventory.
2. From the table, select the infrastructure environment where the host got imported.
   Note: In the hosts table, you can see the approve host after the server gets booted up completely. It can take 10-15 minutes. Verify the server before you accept. If the server name is not recognizable, then look at the details to confirm. You can change the host name but ensure it is unique.
3. Accept the host into the inventory.

(Optional) Step 6: Labeling hosts

After you accept the hosts, add labels to them through the Agent CR. It can be useful to identify a specific cluster or nodepool.
To add the label, do the following steps:

1. Log into the OpenShift console.
2. Go to Administration > CustomResourceDefinitions.
3. Search for Agent of type `agent-install-openshift.io`.
4. In the instances, find the agent to modify the label.
5. Edit the label of the agent to add a new label.
   Run the following command to label the agent:

   ```
   oc -n <hosted-cluster-namespace> label agent <agent-name> inventory.agent-install.openshift.io/<label-key>=<label-value>
   ```

   For example, If there exists an agent `0176c90f-1111-78cf-1dc6-8d987fe794b2` under namespace `fusion-bm`, then run the following command to apply the label:

   ```
   oc -n fusion-bm label agent 0176c90f-1111-78cf-1dc6-8d987fe794b2 inventory.agent-install.openshift.io/hcp=hcp1
   ```

Deploying Bare Metal clusters with Fusion Data Foundation

You can create a Hosted Control Plane cluster by using Bare Metal hosts either through the OpenShift® Console or the Hosted Control Plane CLI.

Before you begin

- Ensure you meet the following prerequisites:
  - Prerequisites for Hosted Control Plane clusters
  - Planning and prerequisites for your Bare Metal Hosted Control Plane

Procedure

1. Go to Multi Cluster Engine (MCE) user interface.
2. Go to Infrastructure > Clusters > Create cluster.
3. Select Host Inventory and then Hosted.
4. Enter the credentials.
   The credentials should include the following:
   - cloud.openshift.com
   - cp.icr.io
   - quay.io
   - registry.connect.redhat.com
   - registry.redhat.io
5. Enter Cluster name.
   Important: It is crucial that the cluster name is the same name used in the network planning. Do not use a name that does not have an alias entry. The cluster is a part of the base domain. For example, `mydomain.com`. Controller and Infrastructure availability are environment dependent.
   Though the recommendation is to have 3 nodes for a resilient cluster, you can have a single node Hosted Control Plane cluster.
6. Enter Namespace. It is the namespace of the infrastructure environment of the hosts.
7. Enter Labels. These are the labels found on hosts in this particular infrastructure environment. To ensure certain hosts are in a particular cluster, use the labels. Otherwise, a random host from that infrastructure environment gets chosen.
8. Enter the Networking type.
   The value is environment dependent, but you can choose LoadBalancer type for resilience. The SSH public key must be the same that you used during host import.
9. Open the YAML and add the following in the `spec` section: HCP cluster to use the specially created Local Volume Storage for HCP

```
spec:
  etcd:
    managed:
      storage:
        persistentVolume:
          size: 8Gi
          storageClassName: lvms-hcp-etcd
        type: PersistentVolume
    managementType: Managed
```

   It causes the Hosted Control Plane cluster to use the Local Volume Storage created for it.

10. Click Create.
    The cluster creation starts and it can take up to half an hour. After the nodepool hosts are in the ready state, add the load balancer for the console to be available.

What to do next

After you deploy the Bare Metal clusters with Fusion Data Foundation, configure the ingress to view the changes in the user interface. For more information about the steps to configure ingress, see Configuring ingress for Bare Metal Hosted cluster.
For more information about how to deploy and manage virtual machines on Bare Metal clusters, see Managing virtual machines on Bare Metal Hosted Control Plane cluster.

Configuring ingress for Bare Metal Hosted cluster

After you deploy the Bare Metal clusters with Fusion Data Foundation, you must configure the ingress for you to view the changes in the user interface.

Before you begin

You must have deployed the Bare Metal cluster successfully. For more information about the deployment procedure, see Deploying Bare Metal clusters with Fusion Data Foundation.

Procedure

1. If you choose LoadBalancer networking type, then add a load balancer to the newly created Hosted Control Plane cluster.
   You need the load balancer to gain external access to the Hosted Control Plane cluster. Using a load balancer allows the nodes to be resilient as opposed to the nodeport approach. The load balancer gets added to the newly created Hosted Control Plane cluster, and not the IBM Fusion HCI hub cluster. To access the new Hosted Control Plane cluster, download the kubeconfig. Steps to download the kubeconfig:
   a. Log in to the IBM Fusion HCI hub OpenShift console.
   b. Go to the ACM user interface and select Infrastructre > Clusters.
   c. In the clusters list, select the newly created Hosted Control Plane cluster.
   d. In the Cluster nodepools section, click Download kubeconfig. It downloads the kubeconfig for the cluster.
      After the kubeconfig is available, use the OC commands to create YAMLs on the new Hosted Control Plane cluster.
2. Create metallb operator.
   a. Create a new YAML file.Example YAML:

```
apiVersion: v1
kind: Namespace
metadata:
  name: metallb
  labels:
    openshift.io/cluster-monitoring: "true"
  annotations:
    workload.openshift.io/allowed: management
---
apiVersion: operators.coreos.com/v1
kind: OperatorGroup
metadata:
  name: metallb-operator-operatorgroup
  namespace: metallb
---
apiVersion: operators.coreos.com/v1alpha1
kind: Subscription
metadata:
  name: metallb-operator
  namespace: metallb
spec:
  channel: "stable"
  name: metallb-operator
  source: redhat-operators
  sourceNamespace: openshift-marketplace
```

b. Apply the YAML:

```
oc --kubeconfig=kubeconfig.yaml apply -f metallb-operator-config.yaml
```

c. Wait for all the pods to be up and running. Run the following command to confirm:

```
oc --kubeconfig=kubeconfig.yaml get pods -n metallb
```

3. Create an instance of metallb
   a. Create a new file metallb-instance-config.yaml.
   Example:

```
apiVersion: metallb.io/v1beta1
kind:  MetalLB
metadata:
   name: metallb
   namespace:  metallb
```

b. Apply the file:

```
oc --kubeconfig=kubeconfig.yaml apply -f metallb-instance-config.yaml
```

c. Wait for the pods to be up and running

```
oc --kubeconfig=kubeconfig.yaml get pods -n metallb
```

4. Create an `IPAddressPool` and `L2Advertisement`.
   a. Create an `IPAddressPool named ipaddresspool-l2advertisement-config.yaml`.
   Example:

```
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  name: ippool
  namespace: metallb
spec:
  protocol: layer2
  autoAssign: false
  addresses:
    - 1.23.45.678-1.23.45.678
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  name: l2-advertisement
  namespace: metallb
spec:
  ipAddressPools:
    - ippool
```

The name of the `IPAddressPool` must match the one in the `L2Advertisement`.

The addresses must be the IPAddress in the DNS that is the `*.apps.NAMEofCluster` in the DNS network table set up by the network administrator.

b. Apply the ipaddresspool-l2advertisement-config.yaml.

```
oc --kubeconfig=kubeconfig.yaml apply -f ipaddresspool-l2advertisement-config.yaml
```

5. Create a service for the `loadbalancer`.
   a. Create metallb-loadbalancer-service.yaml.
   Note: The `address-pool` must match the name of the `addresspool` in previous step.
   Example:

```
kind: Service
apiVersion: v1
metadata:
```

```
  annotations:
    metallb.universe.tf/address-pool: ippool
  name: metallb-ingress
  namespace: openshift-ingress
spec:
  ports:
    - name: http
      protocol: TCP
      port: 80
      targetPort: 80
    - name: https
      protocol: TCP
      port: 443
      targetPort: 443
  selector:
    ingresscontroller.operator.openshift.io/deployment-ingresscontroller: default
  type: LoadBalancer
```

b. Apply the load balancer service YAML.

```
oc --kubeconfig=kubeconfig.yaml apply -f metallb-loadbalancer-service.yaml
```

c. Monitor the Cluster Operator console for issues:

```
oc --kubeconfig=kubeconfig.yaml get co
```

Booting up the Bare Metal Hosted Control Plane nodes through CLI

Use the steps to boot the Dell and Lenovo nodes from the service node for Bare Metal Hosted Control Plane.

Before you begin

Ensure that you met the following prerequisites:

1. Log in to the service node as **kni** user.
2. Collect the BMC addresses, username, and passwords for all target nodes. For procedure, see Planning and prerequisites for your Bare Metal Hosted Control Plane.
3. Download and host the discovery ISO as follows:
   a. Get the discovery ISO download URL from the MCE user interface.
   b. Download the ISO on the bastion node, then transfer it to the service node under the directory.

   ```
   mkdir /var/www/html/discovery-isos
   # move the ISO to:
   mv discovery.iso /var/www/html/discovery-isos
   ```

   Note: By default, the service node runs an **http** web server on port **8118**.

   c. Access URL for the ISO using only service node provisioning IP. Run the following command to get the provisioning IP.

   ```
   ip -4 -o addr show dev provisioning | awk '{print $4}' | cut -d/ -f1
   ```

   Example output:

   ```
   169.253.1.67
   ```

   Example URL to access the discovery ISO:

   ```
   http://169.253.1.67:8118/discovery-isos/discovery.iso
   ```

   d. Use node-specific tools for configuration:
   - For Dell nodes, use **racadm**
   - For Lenovo nodes, use **onecli** and **ipmitool**.
     Note: These tools are available on the service node.

Procedure

1. Note: The **racadm** command is preinstalled and included in the systems path. You can run it directly from the command line.
   Use the following steps to boot the Dell nodes:
   a. Run the following command to enable the iDRAC web server interface (HTTP or HTTPS).

   ```
   racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn set iDRAC.Webserver.Enable 1
   ```

   Example output:

   ```
   [Key=iDRAC.Embedded.1#WebServer.1]
   Object value modified successfully
   ```

   b. Run the following command to set the HTTP port for the iDRAC web interface to port 80.

   ```
   racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn set iDRAC.Webserver.HttpPort 80
   ```

   Example output:

   ```
   [Key=iDRAC.Embedded.1#WebServer.1]
   Object value modified successfully
   ```

   c. Run the following command to set the HTTPs port for the iDRAC web interface to port 443.

   ```
   racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn set iDRAC.Webserver.HttpPort 443
   ```

Example output:

```
[Key=iDRAC.Embedded.1#WebServer.1]
Object value modified successfully
```

d. Run the following command to power off the node.

```
racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn serveraction powerdown
```

Example output:

```
Server is already powered OFF.
```

e. Run the following command to set the boot order of the node to virtual CD or DVD.

```
racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn set
iDRAC.ServerBoot.FirstBootDevice VCD-DVD
```

Example output:

```
[Key=iDRAC.Embedded.1#ServerBoot.1]
Object value modified successfully
```

f. Run the following command to unmount the ISO image if it is already mounted on the node.

```
racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn remoteimage -d
```

Example output:

```
Disable Remote File Started. Please check status using -s
option to know Remote File Share is ENABLED or DISABLED.
```

g. Run the following command to check the status of ISO image unmount.

```
racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn remoteimage -s
```

Example output:

```
Remote File Share is Disabled
UserName
Password
ShareName
```

h. Run the following command to mount the downloaded discovery ISO image and host it on the service node web server.

```
racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn remoteimage -c -l
http://169.253.1.67:8118/discovery-isos/discovery.iso
```

Example output:

```
Remote Image is now Configured
```

i. Run the following command to check the status of the ISO image that is mounted on node.

```
racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn remoteimage -s
```

Example output:

```
Remote File Share is Enabled
UserName
Password
ShareName http://169.253.1.67:8118/discovery-isos/discovery.iso
```

j. Run the following command to power on the node to boot it with the mounted ISO image.

```
racadm -r <BMC_NODE_IP> -u <BMC_NODE_USERNAME> -p <BMC_NODE_PASSWORD> --nocertwarn serveraction powerup
```

Example output:

```
Server power operation initiated successfully
```

k. Repeat the same steps for all other nodes that you want to add to the hosted cluster.

2. Use the following steps to boot the Lenovo nodes:
   a. Run the following command to power off the node.

```
ipmitool -I lanplus -H <BMC_NODE_IP> -U <BMC_USERNAME> -P <BMC_PASSWORD> power off
```

   b. Run the following command to set the boot order of the node to CD or DVD.

```
export LC_ALL=en_US;export LC_TYPE=en_US;sudo -E /home/kni/isfconfig/onecli/onecli config set BootOrder.BootOrder
"CD/DVD Rom=Network=USB Storage=Hard Disk" -N --bmc <BMC_USERNAME>:<BMC_PASSWORD>@[<BMC_NODE_IP>]
```

Example output:

```
Lenovo XClarity Essentials OneCLI lxce_onecli02f-3.1.0

(C) Lenovo 2013-2020 All Rights Reserved

OneCLI License Agreement and OneCLI Legal Information can be found at the following location:

    "/root/onecli/Lic"

Invoking SET command ...

Connected to BMC at IP address <BMC_NODE_IP> by IPMI
```

```
BootOrder.BootOrder=CD/DVD Rom=Network=USB Storage=Hard Disk

Succeed.
```

c. Run the following command to set the node boot mode to UEFI.

```
export LC_ALL=en_US;export LC_TYPE=en_US;sudo -E /home/kni/isfconfig/onecli/onecli config set
BootModes.SystemBootMode "UEFI Mode" -N --bmc <BMC_USERNAME>:<BMC_PASSWORD>@[<BMC_NODE_IP>]
```

Example output:

```
Lenovo XClarity Essentials OneCLI lxce_onecli02f-3.1.0

(C) Lenovo 2013-2020 All Rights Reserved

OneCLI License Agreement and OneCLI Legal Information can be found at the following location:

    "/root/onecli/Lic"

Invoking SET command ...

Connected to BMC at IP address <BMC_NODE_IP> by IPMI

BootModes.SystemBootMode=UEFI Mode

Configure successfully, please reboot system.

Succeed.
```

d. Run the following command to mount the ISO image.

```
echo | sudo -E /home/kni/isfconfig/onecli/rdmount -d /var/www/html/discovery-isos/discovery.iso -s <BMC_NODE_IP> -l
<BMC_USERNAME> -p <BMC_NODE_PASSWORD>
```

Example output:

```
Extracting...

Executing...

drivePath: /var/www/html/discovery-isos/discovery.iso

spName: <BMC_NODE_IP>

loginName: USERID

password: ******

No existing remote disk used on the target BMC.

/var/www/html/discovery-isos/discovery.iso is an ISO file.

Starting to determine BMC type

XCC found: XCC

XCC found so will try HTTP with password authentication.

Version: 4.0.0.3 AVMP

1.0.0.10 VMIO

Connecting to: <BMC_NODE_IP> at port 3900

SSL Start handshake

SSL Handshake Completed

Sending login with password: USERID

Remote device list updated.

Remote device list updated.

Mapping /var/www/html/discovery-isos/discovery.iso to Remote Device[0] as Read Only. Success.

Succeed mapping. Wait for notify on status.

Remote device list updated.

Remote device list updated.

Mount 0: /var/www/html/discovery-isos/discovery.iso as type 105

Succeed in mounting.

return code: 0
```

e. Run the following command to power up the node.

```
ipmitool -I lanplus -H <BMC_NODE_IP> -U <BMC_NODE_USERID> -P <BMC_PASSWORD> power on
```

f. Run the following command to verify the power state of the node.

```
ipmitool -I lanplus -H <BMC_NODE_IP> -U <BMC_NODE_USERID> -P <BMC_PASSWORD> power state
```

g. Repeat the same steps for all other nodes that you want to add to the hosted cluster.

Expanding your Hosted Control Plane Bare Metal cluster

Add a new host to the exiting Bare Metal Hosted Control Plane cluster.

About this task

After a new Bare Metal host is available in the infrastructure > Nodes > Discovered nodes tab, you can optionally choose to add node to the Hosted Control Plane Bare Metal cluster.

Procedure

1. Create **`nmstate`** config for new node following **`nmstate`** YAML template.
   For more information about **`nmstate`**, see Create the NMStateConfig for each host.
2. Add this new host in the host inventory and boot from the discovery ISO. In MCE, approve the host in the host inventory.
   For more information about these steps, see Download the discovery.iso and Boot the host with the discovery.iso.
   Wait for sometime for the host to be available in MCE. After you approve, it changes to available state. For more information about the steps to accept, see Accept the host into the MCE Inventory.
3. Go to Infrastructure > Clusters > <Your Bare Metal HCP Cluster Name>.
4. Expand Cluster nodepools section and click the ellipsis menu of the node pool.
5. Increase the value of Number of hosts based on the number of new hosts that you plan to add. For example, if you have two nodes in the cluster and want to expand it by one more, then the value of Number of hosts is `3`.
6. Click Update.
   It initiates the addition of a new node to an existing cluster and completes within a few minutes.

Installation of data services on a Hosted Control Plane cluster

Provision Hosted Control Plane clusters with services like Fusion Data Foundation, Global Data Platform remote mount, and Backup & Restore.

- **Installing the IBM Fusion base**
  Procedure to install the IBM Fusion base on a Hosted Control Plane cluster.
- **Installing Fusion Data Foundation on a Hosted Control Plane cluster**
  Install the Fusion Data Foundation client as the storage type for the hosted cluster.
- **Installing Global Data Platform remote mount on the hosted cluster**
  From the IBM Fusion HCI hub cluster, install Global Data Platform remote mount on the hosted cluster so that applications can provision storage from a remote-mounted filesystem. It you can use a label-based approach for configuration of IBM Storage Scale CNSA cluster, similar to existing services.
- **Installing Backup & Restore on the hosted cluster**
  From the IBM Fusion HCI hub cluster, install Backup & Restore on the hosted cluster with Fusion Data Foundation as the storage type.

Installing the IBM Fusion base

Procedure to install the IBM Fusion base on a Hosted Control Plane cluster.

About this task

Note: If you plan to use IBM Maximo® in Hosted Control Plane with IBM Fusion, log in to the Hosted Control Plane cluster and deploy Fusion operator.
For the procedure to install IBM Fusion on Bare Metal, see Installing IBM Fusion HCI.

- Install Backup & Restore Spoke. See Backup & Restore spoke

Procedure

1. Log in to the OpenShift® Container Platform web console of the IBM Fusion HCI hub, and select All clusters.
2. Go to Infrastructure > Clusters.
   The Clusters page is displayed, and the Cluster list tab is open by default.
3. Go through the list of the associated clusters and their details.
4. You can click the name value link of the cluster to view more information.
   In the cluster details page, the Overview tab is open by default.
5. In the Overview tab page, edit the Labels to include `isf.ibm.com/fusion-base`, and click Save.
   Alternatively, you can edit the label by using the **`oc`** commands:
   a. Log in to the hub cluster.

   ```
   oc login to the hub cluster
   ```

   b. Add label `isf.ibm.com/fusion-base` to the hosted control plan clusters on hub to auto install IBM Fusion consumer cluster:

   ```
   oc label managedcluster <consumer_cluster_name> isf.ibm.com/fusion-base=''
   ```

Installing Fusion Data Foundation on a Hosted Control Plane cluster

Install the Fusion Data Foundation client as the storage type for the hosted cluster.

Before you begin

- Ensure that the **isf.ibm.com/fusion-base** label is present on the hosted cluster. If the label is not present, install IBM Fusion on the hosted cluster. For more information about the steps to install the IBM Fusion base, see Installing the IBM Fusion base.
- Ensure that prerequisites for Hosted Control Plane clusters are met.

Remember: Fusion Data Foundation is supported on hosted clusters running OpenShift® Container Platform versions N, N-1, N-2, N-3, and N-4, where N is the OpenShift Container Platform version on the hub cluster.

About this task

During the installation of the Fusion Data Foundation client, you can set a storage quota to limit its storage usage.

If the storage quota not assigned during installation, then the storage quota defaults to unlimited for Fusion Data Foundation versions 4.17 and higher.

After the storage quota is set, you cannot reduce it later. Therefore, it is recommended to allocate required storage quota for Fusion Data Foundation clients during the installation of the hosted clusters to ensure the better storage management. For more information about how to set the quota value, see step 4.

Procedure

1. Log in to the OpenShift Container Platform web console and select All clusters.
2. From the Infrastructure menu, select the Clusters tab.
   The Cluster list page appears.
3. In the Cluster list tab, click the ellipsis menu of the required cluster and select Edit labels.
   The Edit labels window appears.
4. To allocate a specific storage quota for the Fusion Data Foundation client, do one of the following methods before adding the `isf.ibm.com/fusion-fdf` label:
   Important: If you skip this step and proceed to step 5, the storage quota is set to unlimited by default for this Fusion Data Foundation client (if version is 4.17 or higher). Also, the `isf.ibm.com/fusion-fdf-quota` label appears on the Labels field.
   - **By modifying the Labels field of the hosted cluster:**
     - Add the `isf.ibm.com/fusion-fdf-quota=<value>` label specifying the required storage quota.

       Remember: The storage quota value must match the pattern `<integer-value><Gi/Ti>`, such as 10Gi, 10Ti, or so on. Otherwise, an error message appears on the screen.
       For example:

       ```
       isf.ibm.com/fusion-fdf-quota=300Gi
       ```

       Alternatively, if you can add the `isf.ibm.com/fusion-fdf-quota=<value>` label from the hub cluster by using the following **oc** command:

       ```
       oc label managedcluster <consumer_cluster_name> isf.ibm.com/fusion-fdf-quota='300Gi'
       ```

     - Click Save.
   - **By modifying the configmap of the hub cluster:**
     - Go to the hub cluster.
     - Select the IBM Fusion namespace from the Project list.

       For example, **ibm-spectrum-fusion-ns** namespace.

     - Go to Workloads > ConfigMaps.
     - Select the **fusion-addon-config** configmap and go to the YAML tab.
     - Add the default-fdf-storage-quota: <value> parameter and click Save.

       Remember: The storage quota value must match the pattern `<integer-value><Gi/Ti>`, such as 10Gi, 10Ti, or so on. Otherwise, an error message appears on the screen.
       For example:

       ```
       default-fdf-storage-quota: 200Gi
       ```

       Important: This storage quota value is set as a default storage quota for all the hosted clusters associated with this hub cluster whenever you add the `isf.ibm.com/fusion-fdf` label. To override it, add the `isf.ibm.com/fusion-fdf-quota=<value>` label with the required value before adding the `isf.ibm.com/fusion-fdf` label.
     - Go back to the All clusters > Infrastructure > Clusters.

       The Cluster list page appears.

     - Click the ellipsis menu of the required cluster and select Edit labels.

       The Edit labels window appears.

5. In the Edit labels window, add the `isf.ibm.com/fusion-fdf` label and click Save.
   Alternatively, if you can add the `isf.ibm.com/fusion-fdf` label from the hub cluster by using the following **oc** command:

   ```
   oc label managedcluster <consumer_cluster_name> isf.ibm.com/fusion-fdf=''
   ```

   The Fusion Data Foundation service starts to install automatically, and it might take 5–10 minutes to complete the installation.
   Note: A notification appears after the successful installation.
6. Verify whether the Fusion Data Foundation client installation is complete and healthy as follows.

a. Go to the IBM Fusion user interface of the hosted cluster and open the Services tab.
   Ensure that the Fusion Data Foundation service status is healthy.
b. On the Storage tab, check that the storage status shows as normal and the connection status shows as connected for the local storage.
c. Go to the hub cluster in the OpenShift Container Platform web console and select the Storage Consumers tab under the Storage menu.
   The Storage Consumers tab displays installed storage consumers with the allotted storage quota.

   To increase the storage quota, see Updating the storage quota of a Hosted Control Plane cluster.

If you face any issues during the installation of the Fusion Data Foundation client, see Troubleshooting issues and known limitations in Hosted Control Plane cluster.

What to do next

To backup applications on the hosted cluster, see Backup & Restore of applications in Hosted Control Plane clusters.

Installing Global Data Platform remote mount on the hosted cluster

From the IBM Fusion HCI hub cluster, install Global Data Platform remote mount on the hosted cluster so that applications can provision storage from a remote-mounted filesystem. It you can use a label-based approach for configuration of IBM Storage Scale CNSA cluster, similar to existing services.

Before you begin

- Install IBM Fusion consumer. The label **`isf.ibm.com/fusion-base`** is required before you add the Global Data Platform remote mount label.
- Meet all the prerequisites for the hosted cluster. See Prerequisites for Hosted Control Plane clusters.

About this task

Note: Global Data Platform remote mount on Hosted Control Plane clusters is supported only on Bare Metal Hosted Control Plane clusters.
You can have a Data Foundation client alongside the CNSA remote service. The Backup & Restore service can be installed with remote-mounted filesystems and it can protect workloads using storage from remote-mounted filesystems.

Procedure

1. Log in to the OpenShift® Container Platform web console of the IBM Fusion consumer cluster.
2. Go to Operators > Installed operators.
3. Check whether the IBM Fusion consumer mode cluster is installed.
   The Status changes from Installing to Succeeded.
4. On the hub cluster, go back to the cluster details > Overview tab page and edit the Labels to include `isf.ibm.com/fusion-gdp-rm`, and click Save.
   Alternatively, you can edit the label from the hub cluster by using the following **`oc`** command:
   Add label `isf.ibm.com/fusion-gdp-rm` to the hosted control plan clusters on hub to auto install IBM Fusion consumer cluster:

   ```
   oc label managedcluster <consumer_cluster_name> isf.ibm.com/fusion-gdp-rm=''
   ```

5. Go to the IBM Fusion consumer cluster user interface and open the Services page.
   The Global Data Platform remote mount starts to install automatically. If it does not, then refresh and wait for five to ten minutes.
6. Wait for the Global Data Platform remote mount installation to complete.
   A notification is displayed confirming the success of the installation.
7. To backup applications on your hosted plan cluster, see Backup & Restore of applications in Hosted Control Plane clusters.

Installing Backup & Restore on the hosted cluster

From the IBM Fusion HCI hub cluster, install Backup & Restore on the hosted cluster with Fusion Data Foundation as the storage type.

Before you begin

- Install IBM Fusion consumer and Fusion Data Foundation storage type.
- The label **`isf.ibm.com/fusion-base`** is required before you add **`fusion-backup`** label.
- If a specific storageclass is to be used during Backup & Restore installation, then do the following:
  1. Go to the OpenShift® Console on the Hub cluster.
     Note: Changes made in **`fusion-addon-config`** ConfigMap is applicable to all Hosted Control Planes deployed on the Hub cluster.
  2. In the IBM Fusion namespace, locate the **`fusion-addon-config`** configmap.
  3. Add an entry:

     ```
     backup-storage-class-name: "NameOfStorageClass"
     ```

     To provide the custom storage class, update the followings fields:

     ```
     data:
       clusters: |
         <cluster-name>: |
           "fusion-install-namespace": "ibm-spectrum-fusion-ns"
           "backup-storage-class-name": "<custom-storageclass-name>"
     ```

     Example YAML:

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: fusion-addon-config
  namespace: ibm-spectrum-fusion-ns
data:
  sds-catalog-source-image: <sds-image>
  storage-quota-min-supported-version: "v4.17"
  clusters: |
    cluster1: |
      "fusion-install-namespace": "ibm-spectrum-fusion-ns"
      "backup-storage-class-name": "ocs-storagecluster-ceph-rbd"
      "backup-restore-spoke-ns": "ibm-backup-restore"
    cluster2: |
      "fusion-install-namespace": "custom-ns"
      "backup-storage-class-name": "ocs-storagecluster-ceph-rbd"
      "backup-restore-spoke-ns": "ibm-backup-restore"
```

Note: If the storage class name is not filled, the Backup & Restore installation makes use of the default storage class on the hosted cluster, prioritizing a storage class with a volume snapshot class. The default values are same as that of the hub cluster values.

Procedure

1. Log in to the OpenShift Container Platform web console of the IBM Fusion HCI hub, and select All clusters.
2. Go to Infrastructure > Clusters.
   The Clusters page is displayed and the Cluster list tab is open by default.
3. Go through the list of the associated clusters and their details.
4. You can click the name value link of the cluster to view more information.
   In the cluster details page, the Overview tab is open by default.
5. In the Overview tab page, edit the Labels to include `isf.ibm.com/fusion-backup`, and click Save.
   Use the default storage class that got created as a part of the Data Foundation for Backup & Restore.
   Use the Backup & Restore to protect your applications but not the entire cluster.

Backup & Restore of applications in Hosted Control Plane clusters

Using the Backup & Restore service, you can backup and restore your applications on Hosted Control Plane clusters.

Backup and restore your applications on Hosted Control Plane clusters. Here, IBM Fusion HCI hub cluster is where you install the Backup & Restore hub service and Backup & Restore spoke is installed on the hosted cluster. For the steps to backup and restore, see Backup & Restore overview.

Backup and restore of the Hosted Control Plane cluster

You can backup and restore your Hosted Control Plane cluster.

Red Hat® provides instructions for backing up and restoring the **etcd** database for Hosted Control Plane. For more information about this process, see Red Hat Documentation on Kubevirt platform.

Updating the storage quota of a Hosted Control Plane cluster

Update the allotted storage quota of the Fusion Data Foundation client on a Hosted Control Plane cluster to meet the storage requirement of the application workloads.

Before you begin

Points to remember before starting this procedure:

- You can only increase the storage quota from a defined value to a higher limited value, such as from 90Gi to 100Gi, 100Gi to 200Gi, 9Gi to 1Ti, and so on.
- You can't increase the storage quota from a defined value to unlimited.
- You can't decrease the storage quota.
- You can't modify the storage quota from unlimited to any specific value, as this action is considered a decrease.
- When you set the storage quota, it sets limits on the total storage consumption across the cluster. Therefore, while updating the storage quota, ensure that:
  - The value meets the storage requirements for all application workloads in the cluster.
  - The value meets the storage requirements of all PersistentVolumeClaims (PVCs), regardless of their state within the cluster.

About this task

Use this procedure if you have configured the storage quota for Fusion Data Foundation clients during their installation on hosted clusters and now need to increase it. For more information on storage quota configuration, see Installing Fusion Data Foundation on a Hosted Control Plane cluster.

Whenever the storage consumption of a Fusion Data Foundation client exceeds 80% of its quota limit, you might face issues in creating new applications on the cluster.

You can verify the usage by checking the Storage quota utilization field on the Storage Consumers tab in the OpenShift® Container Platform web console.

If the client's storage consumption exceeds 80% of the quota, a warning icon appears next to the value of the Storage quota utilization field along with the StorageQuotaUtilizationThresholdReached alert. In such cases, increase the client's storage quota to clear the alert.

Note: To see the details of the StorageQuotaUtilizationThresholdReached alert, go to Notifications > Other Alerts.

Procedure

1. Log in to the OpenShift Container Platform web console and select All clusters.
2. Go to Infrastructure > Clusters.
   The Cluster list tab appears on the screen.
3. Click the ellipsis menu of the required cluster and select Edit labels.
   The Edit labels window appears on the screen.
4. Update the `isf.ibm.com/fusion-fdf-quota=<value>` with the required value.
   Remember: The storage quota value must match the pattern `<integer-value><Gi/Ti>`, such as 10Gi, 10Ti, or so on. Otherwise, an error message appears on the screen.
   For example:

   Old storage quota value:

   ```
   isf.ibm.com/fusion-fdf-quota=100Gi
   ```

   New storage quota value:

   ```
   isf.ibm.com/fusion-fdf-quota=1Ti
   ```

5. Click Save.
6. Verify that the Storage Consumers tab displays the updated storage quota for the client as follows.
   a. Go to the hub cluster in the OpenShift Container Platform web console.
   b. Select the Storage Consumers tab under the Storage menu.
   c. Check whether the Storage quota field displays the updated storage quota and the value of the Storage quota utilization field is below 80%.

Results

The Storage Consumers tab displays the updated storage quota for the relevant storage client.

Cleaning up of a Hosted Control Plane cluster

If you no longer need a hosted cluster, use this procedure to remove it.

Before you begin

- Get the cluster ID:
  1. Go to the Red Hat® OpenShift® console of the hosted cluster.
  2. Go to Home > Overview and copy the Cluster ID from the Details section.
- Take a backup of the applications that you want to protect.
- Before you remove the Fusion Data Foundation service, ensure all workloads are stopped: delete the associated PVCs and clean up the storage namespace linked to it.

About this task

If the hosted cluster does not clean up after you remove the base or services label, then give it some time for the changes to take effect. If it still does not clean up, then see Troubleshooting issues and known limitations in Hosted Control Plane cluster.

Procedure

1. On the hub cluster, go back to the hosted cluster details > Overview tab page and edit the Labels to remove `isf.ibm.com/fusion-backup`, and click Save.
2. Remove all the workloads that use Fusion Data Foundation storage classes.
   Important: Fusion Data Foundation cannot be uninstalled if PVCs or VolumeSnapshots exist on the hosted cluster.
3. Edit the Labels again to remove `isf.ibm.com/fusion-fdf`, and click Save.
   Remember: Removing the `isf.ibm.com/fusion-fdf` label also removes the `isf.ibm.com/fusion-fdf-quota=<value>` label, if available.
4. Do the following steps to verify the removal of the Data Foundation storage:
   a. Go to the user interface of the IBM Fusion hosted cluster.
   b. Confirm that the Data Foundation menu option is no longer available.
   c. Go to the Services tab and check whether the Data Foundation service tile is available in the Get started section.
   d. Go to the Red Hat OpenShift console of the hosted cluster and go to Storage > StorageClasses and verify whether all Data Foundation storage classes are removed.
   e. Go to the Storage Clients tab and verify whether the storage client is removed.
   Note: Removing the storage client may take up to 5 minutes.
5. To clean up the Global Data Platform remote mount, complete the following steps:
   a. Delete all applications and PVCs that use storage classes provided by Global Data Platform remote mount.
   Important: Global Data Platform remote mount cannot be uninstalled if applications and PVCs exist on the hosted cluster.
   b. Remove the **`isf.ibm.com/fusion-gdp-rm`** label from the Hosted Control Plane cluster that you want to delete.
6. Delete the hosted cluster.
   a. Log in to the OpenShift Container Platform web console of the IBM Fusion HCI hub, and select All clusters.
   b. Go to Infrastructure > Clusters.
   The Clusters page is displayed and the Cluster list tab is open by default.
   c. Click the name value link of the cluster to view more information.
   In the cluster details page, the Overview tab is open by default.

Alternatively, you can also select the Destroy clusters from the ellipses overflow menu of the cluster.

   d. In the Actions menu, select Destroy clusters to remove the cluster.
   e. In the Permanently destroy clusters window, go through details.
   f. Enter the name of the cluster in the Confirm by typing "<cluster name>" below.
   g. Click Destroy.
      The cluster is removed and its components are deleted. The related namespace is also automatically deleted.
7. To verify the deletion of the hosted cluster, do the following steps:
   a. Log in to the OpenShift Container Platform web console of the IBM Fusion HCI hub, and select All clusters.
   b. Go to Infrastructure > Clusters.
      The Clusters page is displayed and the Cluster list tab is open by default.
   c. Go through the list of the associated clusters and confirm that the Cluster ID of the deleted hosted cluster does not exist.
8. Remove the Backup & Restore connection.
   For the procedure to remove connections, see Disabling the connection.

Troubleshooting issues and known limitations in Hosted Control Plane cluster

Use these troubleshooting information to know the problem and workaround for the installation of spoke cluster and

Unable to delete internal and external bare metal Hosted Control Plane clusters using HCP CLI of Red Hat

Problem statement
: When you use the HCP CLI, the hosted cluster does not get deleted. The cluster remains in a "destroying" state for over an hour.

Resolution
: Check the pods logs of the hypershift namespace for error messages

Hosted Control Plane in Red Hat documentation

For Hosted Control Plane troubleshooting documentation by Red Hat®, see Red Hat Documentation.

PVCs on Hosted Control Plane clusters are stuck in terminating state

If PVCs on the Hosted Control Plane clusters are stuck in terminating state, then do the following steps for all applications that have PVCs associated to Fusion Data Foundation storage.

1. Scale down deployment of the PVC application to 0.
2. Delete application.
3. Check the PVC status to confirm whether it is deleted.
   Note: Remove all PVCs from the applications before you remove the Fusion Data Foundation label.
4. Check whether the Fusion Data Foundation client on the Hosted Control Plane cluster is deleted.

Issues in hosted cluster clean up

Issues in Fusion base cleanup
: If the hosted cluster does not clean up after you remove the base label (`isf.ibm.com/fusion-base`), then give it some time for the changes to take effect. If it still does not clean up, follow these steps to understand the cause of the issue:

  1. Log in to the hosted cluster.
  2. Go to pods in the `open-cluster-management-agent-addon` namespace.
  3. Get the terminal access to the `fusion-cleanup-agent` pod.
  4. Run the following **cleanup** script:

     ```
     bash /scripts/sds/sds-cleanup.sh
     ```

  5. If the script stuck at any resource deletion, remove the finalizer from that resource.

Issues in Fusion Data Foundation service clean up
: **Before you begin:**

  - Before you proceed with Fusion Data Foundation deletion, stop the backup service and clean up the associated PVCs.
  - If Fusion Data Foundation sizes are large and workloads are running continuously, such as Cloud Pak or Watsonx, then do the following steps:
    1. Scale the pods down to zero.
    2. Stop the application.
    3. Delete the associated PVCs.

  If the Fusion Data Foundation service does not clean up after you remove its label (`isf.ibm.com/fusion-fdf`), then give it some time for the changes to take effect. If it still does not clean up, follow these steps to understand the cause of the issue:

  1. Log in to the hosted cluster.
  2. Go to pods in the `open-cluster-management-agent-addon` namespace.
  3. Get the terminal access to the `fusion-odf-cleanup-agent` pod.
  4. Run the following **cleanup** script:

     ```
     bash /scripts/odf/delete-fusion-odf.sh
     ```

5. If the script stuck at any resource deletion, remove the finalizer from that resource.

Issues in Backup & Restore service clean up

If the Backup & Restore service does not clean up after you remove its label (`isf.ibm.com/fusion-backup`), then give it some time for the changes to take effect. If it still does not clean up, follow these steps to understand the cause of the issue:

1. Log in to the hosted cluster.
2. Go to pods in the `open-cluster-management-agent-addon` namespace.
3. Get the terminal access to the `fusion-bnr-cleanup-agent` pod.
4. Run the following **cleanup** script:

   ```
   bash /scripts/backup-restore/uninstall-backup-restore.sh
   ```

5. If the script stuck at any resource deletion, remove the finalizer from that resource.

Pull image issue occurs during the installation of IBM Fusion in a hosted cluster

Problem statement

If a pull image issue occurs, ensure that the pull-secret for `cp.icr.io` is available to the Hosted Control Plane cluster.

Resolution

To check and update the pull-secret, do the following steps:

1. On the IBM Fusion HCI hub cluster, go to Administration > CustomResourceDefinition and search for `HostedCluster`.
2. Look for the instance that corresponds with the Hosted Control Plane cluster and open it in YAML view.
3. Search for `pullSecret` and note down the name of the secret.
4. In the clusters namespace, search for that secret and open it in edit mode.
5. Check to see whether the correct value of `cp.icr.io` available. If not, modify the secret with the right value.
   It can take a while to propagate. You may even have to approve the changes from the Compute > Nodes page of the Hosted Control Plane cluster.

It is recommended to install a Hosted Control Plane cluster with the following credentials:

- `cloud.openshift.com`
- `cp.icr.io`
- `quay.io`
- `registry.connect.redhat.com`
- `registry.redhat.io`

Issues in installation of IBM Fusion in a hosted cluster

Resolution

1. Check the status of `manifestwork` `fusion-install` in `clusternamespace` of the hub cluster:

   ```
   oc login to the hub
   oc get manifestwork fusion-install -n <spoke_cluster_name> -o yaml
   ```

2. If no error exists in the status of the `manifestwork`, check the status of IBM Fusion installation in the spoke cluster:

   ```
   oc login to the spoke
   oc get csv -n ibm-spectrum-fusion-ns
   ```

Issues in the installation of Fusion Data Foundation in the hosted cluster

Resolution

As a resolution, check the status of `manifestwork` `odfclient-install` in `clusternamespace` in the hub cluster:

```
oc login to the hub
oc get manifestwork odfclient-install -n <spoke_cluster_name> -o yaml
```

Backup issues in Hosted Control Plane with Fusion Data Foundation

Problem statement

Concurrent backups fail during a Velero snapshot with the following error message:

```
The  operation has timed out because Velero has failed to report status.'
      However, the backup phase is updated as 'DataTransferfailed.
```

The timeout for Velero to pickup request is set to 30 minutes in the transaction manager.

Resolution

To resolve the issue, increase it to 60 minutes.

Hosted Control Plane cluster does not get created because of unavailability of IP addresses

Resolution

Check whether the installed load balancer has sufficient IP addresses for the Hosted Control Plane cluster. Check the `IPAddressPool` object for the IP range. Run the following command to check whether an IP is available:

```
oc get svc -A | grep LoadBalancer
```

Known issues and limitations

- During the installation or upgrade of the Hosted Control Plane, you might encounter an issue where the `control-plane-operator` pods restart continuously and increase in number in the `clusters-<hcp_name>` namespaces on the host cluster.
- The use of the AMD GPU operator is currently not supported with Hosted Control Plane. Internal nodes with AMD GPUs can be installed into the hub cluster. Stand alone clusters with AMD GPUs can be managed by the multi-cluster engine. In case of external servers, you can run AMD GPUs only on the hub cluster and not on the Hosted Control Plane.
  Note: This is a limitation of AMD/RH, and not that of IBM Fusion.
- Random "image pull" failure can occur on the Hosted Control Plane due to secrets.
- Sometimes, the hosted cluster status goes in offline mode after deletion. Contact IBM Support to resolve the issue.
- The **hcp destroy** command can cause the hosted cluster to get stuck indefinitely during cleanup. Contact IBM Support to resolve the issue.
- The YAML tab view on the OpenShift® Container Platform console is not working as expected during the installation of the multi cluster engine operator.
- The hosted clusters expect the `storageprofile` of the used storage class, but it is not available during cluster creation.
- The following issues occur whenever you remove the disks and place them back in the rack:
  - Disks are not reflected in the node.
  - LVM cluster and Data Foundation go into a degraded state.

  As a resolution, restart the affected node.

Red Hat OpenShift Virtualization

IBM Fusion enables the coexistence of containers and VMs, with OpenShift Virtualization allowing VMs to be managed as Kubernetes resources. Beyond VM hosting, Fusion and OpenShift Virtualization offer virtual machine mobility, backup and restore, and additional capabilities to ensure high availability and disaster recovery.

- **Overview**
  The Red Hat OpenShift provides a platform for running virtual machines and container workloads side by side. It relieves you of maintaining separate infrastructure, platforms, tools, skills, and teams for virtual machines and containers. It also enables virtual machines to be managed with the same modern practices that are used for containers.
- **Red Hat OpenShift Virtualization Sizing on IBM Fusion**
  Sizing in Red Hat OpenShift Virtualization is to arrive at the optimal configuration of the infrastructure resources to efficiently accommodate the number and types of virtual machines (VMs) you plan to deploy.
- **Setting storage for virtual machines**
  Virtual machines use block storage and file storage.
- **Deploying Red Hat OpenShift Virtualization Operator**
  Procedure to deploy the Red Hat OpenShift Virtualization Operator.
- **Managing virtual machines on Bare Metal Hosted Control Plane cluster**
  In a Bare Metal Hosted Control Plane cluster, you create, delete, manage, clone, and monitor virtual machines.
- **Migrating VMs to OpenShift Virtualization**
  To move from traditional VM-based infrastructure to a Kubernetes-native environment, migrate virtual machines (VMs) to OpenShift Virtualization (OSV), which integrates VMs and containers within the same platform.
- **Backup and restore of virtual machines**
  IBM Fusion protects virtual machines from data loss and corruption by backing up the virtual machines regularly. The virtual machines can then be restored from a previous backup.
- **Creating a virtual machine by using Clone PVC**
  A clone is an exact duplicate of an existing storage volume that can be used like any standard volume. You create a clone to capture a point-in-time snapshot of the data.
- **Disaster recovery for virtual machines**
  Disaster Recovery of virtual machines involves creating regular backups, maintaining up-to-date replicas, and implementing robust failover mechanisms. Implementing failover mechanisms ensures high availability and minimizes potential data loss, reducing downtime, and ensuring long-term resilience against disasters.

Overview

The Red Hat® OpenShift® provides a platform for running virtual machines and container workloads side by side. It relieves you of maintaining separate infrastructure, platforms, tools, skills, and teams for virtual machines and containers. It also enables virtual machines to be managed with the same modern practices that are used for containers.

IBM Fusion provides high-performance infrastructure, storage, encryption at rest, and resiliency to run virtual machines on Red Hat OpenShift.

The RWX (ReadWriteMany) mode is crucial because it enables Live Migration for virtual machines in Red Hat OpenShift. With RWX block storage support, virtual machines can seamlessly move between nodes within the OpenShift cluster during Live Migration while maintaining the performance benefits of the block storage.

IBM Fusion HCI provides the infrastructure to run Red Hat OpenShift on Bare Metal- x86 compute workers, a high-speed network, and storage. The appliance is designed with redundancy in mind, allowing for servers, switches, and drives to fail without impacting virtual machines that run in the cluster. IBM Fusion HCI uses an IPI installation of Red Hat OpenShift, which enables the cluster to run without an external load balancer.

Virtual machine resiliency can be provided in the following ways:

- High availability is achieved by using three-way replication to mirror data across availability zones. It allows workloads to continue to run even when storage infrastructure fails.
- RWX storage enables Live Migration, which allows virtual machines to be moved between nodes in the cluster.
- Backup capabilities protect virtual machines from data loss or corruption, enabling them to be restored to an earlier point in time. You can restore the virtual machines to an alternative cluster to recover from a complete cluster failure or practice recovery.

Red Hat OpenShift Virtualization Sizing on IBM Fusion

Sizing in Red Hat® OpenShift® Virtualization is to arrive at the optimal configuration of the infrastructure resources to efficiently accommodate the number and types of virtual machines (VMs) you plan to deploy.

With proper sizing, your infrastructure can efficiently manage the expected workload, allowing you to allocate the right resources to optimize both performance and cost.

Sizing considerations

Some of the factors that influence the sizing of virtual machines on IBM Fusion are as follows:

CPU
: The number of vCPUs depend on the computational demands of your VMs. VMs that run resource-intensive applications or multiple workloads require more vCPUs to perform efficiently.

Memory
: Memory allocation is crucial for VMs to run smoothly and must be sized according to the specific workload requirements of each VM. Applications that perform data-intensive operations or handle large in-memory datasets require more memory to maintain optimal performance.

Storage
: The storage depends on the volume of data that each VM is expected to handle. It includes the storage for operating systems, application data, and any additional storage for backups or high-availability configurations.

Networking
: While not directly affecting the number of VMs, network bandwidth and latency are crucial for the performance of VMs, especially in environments with high data transfer rates or low-latency requirements.

Workload Characteristics
: The type of applications and their resource usage patterns significantly impact the sizing considerations. For example, CPU-bound, memory-bound, I/O-bound.

Redundancy and High Availability
: Configurations for redundancy and high availability (HA) may require additional resources, such as extra storage or memory, to help ensure that VMs can failover without service interruption.

Extra capacity considerations

When you plan your sizing, consider the capacities that are needed for CoreOS, OpenShift Container Platform, and IBM Fusion HCI to ensure that your VMs have access to the necessary resources.

Go through the following resource requirements for the various components:

OpenShift Container Platform
: OpenShift Container Platform Resource Requirements

Data Foundation
: https://ibm.com/docs/en/fusion-software/2.10.0?topic=foundation-planning-fusion-data-deployment

Before your allocate resources to VMs, subtract these overhead values from your total available resources.

- **Sizing calculator**
  Use the following formulas to provide an estimated infrastructure for VM deployments. Considering the overhead and high availability, these estimates offer a close approximation for effective resource planning.

Sizing calculator

Use the following formulas to provide an estimated infrastructure for VM deployments. Considering the overhead and high availability, these estimates offer a close approximation for effective resource planning.

Expected input

Number of VMs
: Total number of virtual machines planned.

VM Types
: Are the VMs of the same type or different types with distinct vCPU, Memory, and Storage requirements?

Formulas

- **CPU Capacity:** The following formula calculates the total vCPU capacity that is available for VMs. Subtract the overhead for control plane and other necessary services:
  Total Available (vCPUs) = (vCPUs_per_node × node_count − control_plane_requirements − overhead) × (1−ha_reserve_percent) × (overcommit_ratio)

Parameter	Description
vCPUs_per_node	The number of vCPUs per compute node. For example, with 64 cores and 2 threads per core, you would have 128 vCPUs per node.
node_count	The number of compute nodes.
control_plane_requirements	CPU is reserved for control plane functions.

Parameter	Description
overhead	Overhead for management services. For example, Data Foundation in IBM Fusion.
ha_reserve_percent	Percentage of capacity reserved for high availability.
overcommit_ratio	Factor to consider when you overcommit CPU resources. For example, 4:1.

- **Memory Capacity:** Similarly, the memory capacity calculation accounts for the overhead and reserves for high availability:

  Total Available Memory (GiB) = (total_node_memory × node_count − control_plane_requirements − overhead) × (1 − ha_reserve_percent)

Parameter	Description
total_node_memory	Total physical memory per compute node.
control_plane_requirements	Memory reserved for control plane operations.
overhead	Overhead for management services. For example, Data Foundation in IBM Fusion.
node_count	Number of compute nodes.
ha_reserve_percent	Percentage of memory reserved for high availability.

- **Storage Capacity:** Storage capacity is calculated after accounting for the overhead needed for management and other services:
  Total Available Storage (TiB) = total_storage_capacity − storage_overhead

Parameter	Description
total_storage_capacity	Total storage capacity available.
storage_overhead	Overhead for management services and storage redundancy.

Example

Table with example VM specification:

VM specification	Description
50 VMs of Type 1	Each VM requires 2 vCPUs, 8 GiB of memory, and 100 GiB of storage.
20 VMs of Type 2	Each VM requires 4 vCPUs, 16 GiB of memory, and 200 GiB of storage.

Table with example infrastructure Resources (Before Overhead and HA)

Infrastructure Resources (Before Overhead and HA)	Description
Compute Nodes	13 nodes, each with 64 cores and 2 TB (2048 GiB) of memory.
Total Raw Storage	16 nodes (13 compute + 3 control), each with 10 disks of 7.68 TiB each.

Table with example overhead and high availability reserves

Overhead and high availability reserves	
CPU: Overhead	32x4 = 128 (assuming each core provides 4 vCPUs) reserved for management services.
CPU: High Availability (HA) Reserve	20% of the remaining vCPUs.
CPU: Control Plane Requirements	4x4= 16 vCPUs (assuming each core provide 4 vCPUs) reserved for control plane operations.
Memory: Overhead	72 GiB reserved for management services.
Memory: High Availability (HA) Reserve	20% of the remaining memory.
Memory: Control Plane Requirements	24 GiB reserved for control plane operations.
Storage: Total Raw Storage	16×10×7.68 TiB = **1228.8 TiB**
Storage: Usable Storage After 3-Way Replication	(1228.8)/3 TiB = **409.6 TiB**
Storage: Storage Overhead	Assume 10% overhead for management services and system needs: **40.96 TiB**
Storage: Total Available Storage After Overhead	Total Available Storage= (Total Usable Storage) - (Storage Overhead)= 409.6 TiB - 40.96 TiB= **368.64 TiB**

Table showing CPU calculation

CPU	Value
Total Physical CPU Cores	Total Physical CPU Cores= 64×13= 832 cores
Total vCPUs	256x13 = 3328 vCPUs (assuming each core provides 4 vCPUs, per node would be 64x4 = 256)
Control Plane Requirements	4x4= 16 vCPUs (assuming each core provide 4 vCPUs) are reserved for control plane operations.
Overhead	32x4 = 128 (assuming each core provides 4 vCPUs) are reserved for management services.
High Availability (HA) Reserve	20% of the remaining vCPUs.
Total Available vCPUs After Control Plane, Overhead, and HA Reserve	Total Available (vCPUs) = (vCPUs_per_node × node_count − control_plane_requirements − overhead) × (1−ha_reserve_percent) × (overcommit_ratio)
Substituting the values	Total Available vCPUs = (3328 − 16 − 128) × 0.8 × 4 = 10188.8 vCPUs (Overcommit ratio is assumed to be 4:1) Rounding it down to 10189 vCPUs available for VM deployments.
Comparison with Available CPU	180 vCPUs required < 10189 vCPUs available. Rounding it down to 10189 vCPUs available for VM deployments.

Result: The infrastructure has sufficient CPU resources to support the required VMs even after considering the control plane requirements.

Table showing memory calculation

Memory	Values
Total Physical Memory	Total Physical Memory=2048×13=26624 GiB
Control Plane Requirements	24 GiB reserved for control plane operations.
Overhead	72 GiB reserved for management services.
High Availability (HA) Reserve	20% of the remaining memory.
Total Available Memory After Control Plane, Overhead, and HA Reserve	*Total Available Memory (GiB) = (total_node_memory × node_count − control_plane_requirements − overhead) × (1 − ha_reserve_percent)*
Substituting the values	Total Available Memory (GiB)=(2048×13 − 24 − 72) × 0.8= 26528×0.8= 21222.4 GiB Rounding it down to **21222 GiB** available for VM deployments.
Comparison with Available Memory	720 GiB required < 21222 GiB available

Result: The infrastructure has sufficient memory resources to support the required virtual machines even after considering the control plane requirements, overhead, and HA.

Table showing storage calculation

Storage	Values
Total Raw Storage	Total Raw Storage=16×10×7.68 TiB= 1228.8 TiB
Total Usable Storage After 3-Way Replication	Total Usable Storage=(1228.8)/3 TiB= 409.6 TiB
Storage Overhead	Storage Overhead= 40.96 TiB
Total Available Storage After Overhead:	Total Available Storage=409.6 TiB − 40.96 TiB= 368.64 TiB
Total Storage required for Type 1 VMs:	Storage for Type 1 VMs= 50×100 GiB= 5000 GiB= 5 TiB
Total Storage required for Type 2 VMs:	Storage for Type 2 VMs= 20×200 GiB= 4000 GiB= 4 TiB
Total Storage required for all VMs:	Total Storage required= 5 TiB+4 TiB= 9 TiB
Comparison with Available Storage:	9 TiB required < 368.64 TiB available

Conclusion

- CPU: 10189 vCPUs available; 180 vCPUs required (Sufficient)
- Memory: 21222 GiB available; 720 GiB required (Sufficient)
- Storage: 368.64 TiB available; 9 TiB required (Sufficient)

The infrastructure can support the deployment of 50 Type 1 VMs and 20 Type 2 VMs, with sufficient CPU, memory, and storage resources. These resources are available after accounting for overhead and high availability reserves.

Setting storage for virtual machines

Virtual machines use block storage and file storage.

Block-mode storage through the **`ocs-storagecluster-ceph-rbd-virtualization`** storage class is recommended as it provides optimized performance for block IO.

However, for Windows VMs, it is always recommended to use **`ocs-storagecluster-ceph-rbd-virtualization`** storage class. If you use the **`ocs-storagecluster-ceph-rbd`** storage class, then Windows VMs may report bad CRC or signature errors due to I/O patterns. The performance of the cluster performance can degrade due to excessive re-transmissions caused by CRC errors that leads to network saturation.

VMs must request RWX storage so that they can be moved between nodes non-disruptively using Live Migration.

- File storage is supported for virtual machines, but it lacks the performance advantage of block storage.
- RWO volumes are supported for virtual machines but are not recommended, as they block Live Migration. VMs using RWO volumes cannot move between nodes without causing disruption.

For more information about live migration, see Live migration for virtual machines.

Fusion Data Foundation can be configured for encryption, with keys managed within the OpenShift cluster or by an external Key Management Service (KMS). For more information about encryption, see Preparing to connect to an external KMS server in Fusion Data Foundation.

Deploying Red Hat OpenShift Virtualization Operator

Procedure to deploy the Red Hat® OpenShift® Virtualization Operator.

About this task

OpenShift Virtualization provides a platform to bring virtual machines (VMs) into containerized workflows by running a virtual machine within a container where you develop, manage, and deploy virtual machines side-by-side with containers in one platform. For more information about how to create and manage VMs, see Red Hat OpenShift Virtualization and What you can do with OpenShift Virtualization.

A list of example VM features that the Red Hat OpenShift Virtualization supports:

- Create and manage Linux and Windows virtual machines
- Connect to virtual machines through a variety of consoles and CLI tools
- Import and clone existing virtual machines
- Manage network interface controllers and storage disks attached to virtual machines

Procedure

1. Go to the Red Hat OpenShift Container Platform Operator Hub.
2. Search for OpenShift Virtualization.
3. Install the OpenShift Virtualization Operator.
4. Configure the installation parameters to either custom or recommended namespace.
5. Wait for the installation to complete.
6. Go to OpenShift Virtualization in Installed Operators page and click Create Hyperconverged to configure the Hyperconverged resource.
   Note: When you create the **`HyperConverged`** CR, ensure you configure the appropriate eviction strategy for your virtual machines in the **`HyperConverged`** CR. For more information, refer to the Eviction strategy documentation.
7. When the Operator deployment is complete go to Installed Operators and look for OpenShift Virtualization within the **`openshift-cnv`** or your namespace.
8. Check whether the status is successful.

A new Virtualization menu appears. It indicates that your cluster is now ready to deploy virtual machines within a project /or namespace of your choice.

What to do next

- You can create virtual machines with any default size or custom sizes.
  For more information about how to deploy and manage virtual machines on Bare Metal clusters, see Managing virtual machines on Bare Metal Hosted Control Plane cluster.
- Use `ocs-storagecluster-ceph-rbd-virtualization` storage class for the OS disk of VMs. For more information about setting storage, see Setting storage for virtual machines.
- Optionally, create a VLAN of type Virtualization VLAN and add the created VLAN to Link. If the primary pod network in OpenShift Virtualization becomes congested, you can improve performance by isolating that traffic onto a dedicated secondary network. For more information about dedicated secondary network, see Creating dedicated secondary Network in Virtualization.
- **Configuring GPU access with NVIDIA vGPU**
  NVIDIA vGPU technology enables a single physical GPU to be divided into multiple virtual instances, allowing multiple virtual machines to share GPU resources. This improves GPU utilization, reduces infrastructure costs, and maintains performance isolation for AI, data analytics, and graphics workloads.

Configuring GPU access with NVIDIA vGPU

NVIDIA vGPU technology enables a single physical GPU to be divided into multiple virtual instances, allowing multiple virtual machines to share GPU resources. This improves GPU utilization, reduces infrastructure costs, and maintains performance isolation for AI, data analytics, and graphics workloads.

Important: Ensure that your GPU hardware is supported by NVIDIA vGPU technology. To get the complete list of supported GPU cards, see NVIDIA documentation.
For the procedure to configure NVIDIA vGPU for virtual machines in OpenShift® Virtualization, see NVIDIA documentation.
Important: While you follow the NVIDIA documentation, take note of the following two steps that must be done differently:

1. To enable sandbox workloads for vGPU, you must configure additional parameters in the `ClusterPolicy` for the GPU operator.

   ```
   sandboxWorkloads.enabled=true
   sandboxDevicePlugin.enabled=true
   sandboxWorkloads.defaultWorkload=vm-vgpu
   vfioManager.enabled=false
   ```

2. After completing the step creating a `ClusterPolicy` for the GPU operator in the NVIDIA documentation, you must label the GPU node with the specific vGPU device profile.

   ```
   oc label node <gpu-node-name> \
     nvidia.com/vgpu.config=<vgpu-device-name> \
     --overwrite
   ```

   Replace `<gpu-node-name>` with the name of your GPU-enabled node, and `<vgpu-device-name>` with the vGPU profile that you want to enable. For example, on NVIDIA RTX Pro 6000 Blackwell GPUs, a valid vGPU profile is `DC-48Q`. You can find the appropriate `vgpu-device-name` in the `default-vgpu-devices-config ConfigMap`, or in a custom `ConfigMap` if created.

   To identify the correct profile, search for the `device_id` that matches your GPU, using the last four characters of the device ID as a reference.

Managing virtual machines on Bare Metal Hosted Control Plane cluster

In a Bare Metal Hosted Control Plane cluster, you create, delete, manage, clone, and monitor virtual machines.

1. Provision a Bare Metal Hosted Control Plane cluster. For the procedure, see Deploying Bare Metal clusters with Fusion Data Foundation.
2. Install Fusion Data Foundation service. See Installing Fusion Data Foundation on a Hosted Control Plane cluster.
3. Install OpenShift Virtualization operator.
   For the procedure to install OpenShift Virtualization operator, see Deploying Red Hat OpenShift Virtualization Operator.
4. After the cluster is available, deploy virtual machines on it. You can deploy VMs using default or custom sizes based on resource needs. For more information about the steps to create virtual machines from OpenShift console, see Red Hat documentation.
5. After the VMs are in running state, manage their lifecycle, such as clone, monitor, and decommission tasks.
   For more information about these management tasks, see Creating a virtual machine by using Clone PVC and Deleting virtual machines Red Hat Documentation.

Migrating VMs to OpenShift Virtualization

To move from traditional VM-based infrastructure to a Kubernetes-native environment, migrate virtual machines (VMs) to OpenShift® Virtualization (OSV), which integrates VMs and containers within the same platform.

- **Migrating VMs from VMware vSphere to OpenShift**
  OpenShift Virtualization supports the migration of VMs from VMware vSphere to OpenShift.

Migrating VMs from VMware vSphere to OpenShift

OpenShift® Virtualization supports the migration of VMs from VMware vSphere to OpenShift.

Before you begin

- Ensure that the internal OpenShift registry setup is complete. Use an OpenShift internal registry image from a single namespace to provision workloads across multiple namespaces. After the internal OpenShift registry setup is complete, follow this procedure to use an OpenShift internal registry image that is stored in a single namespace to provision workloads across desired namespaces.
- Deploy the Migration Toolkit for Virtualization (MTV) operator. For more information about deploying MTV operator, see Installing the Migration Toolkit for Virtualization.

About this task

OpenShift Virtualization supports the migration of VMs from VMware vSphere to OpenShift. To accelerate this process, it uses Virtual Disk Development Kit (VDDK) images. VDDK enables faster, block-level access to VM disks, reducing transfer times by skipping unused data and improving overall performance. It is useful for large VMs or bulk migrations. For more information about how to build and configure the VDDK image, see OpenShift.

Note: In this procedure, Origin Namespace refers to where the internal image is stored and Destination Namespace refers to where the image are pulled and used by pods.

Procedure

1. Create a Service Account in the destination namespace.
   It is the namespace where your pods run and provides access to the image.

   ```
   oc create serviceaccount registry-puller -n <destination-namespace>
   ```

2. Create an Image Pull Secret in the destination namespace.
   Go through the internal OpenShift registry and provide your credentials.

   ```
   oc create secret docker-registry registry-pull-secret \
     --docker-server=<internal-openshift-image-registry> \
     --docker-username=<your_username> \
     --docker-password=<your_password> \
     --docker-email=<your_email> \
     -n <destination-namespace>
   ```

3. Link the Secret to the Service Account in the destination namespace.

   ```
   oc secrets link registry-puller registry-pull-secret --for=pull -n <destination-namespace>
   ```

4. Grant Permission to the Service Account to Pull from the origin namespace.
   It allows the service account in the destination namespace to pull images from the origin namespace.

   ```
   oc adm policy add-role-to-user system:image-puller \
     system:serviceaccount:<destination-namespace>:registry-puller \
     -n <origin-namespace>
   ```

High-availability of virtual machines

High availability of virtual machines (VMs) ensure critical applications and services are available even in the event of unplanned scenarios.

IBM Fusion supports the following features for the high availability of VMs.

Live Migration
: Use live migration to move VMs from one host to another without downtime. This is useful for load balancing and maintenance. See Live migration for virtual machines.

Failover Clustering
: Configure VMs in a failover cluster, which allows VMs to automatically restart on a different host if the original host fails. It is very useful in the event of any unexpected hardware failures or other unplanned incidents. See Virtual machine failover.

Live migration for virtual machines

High availability of virtual machines can be achieved by live migration as the virtual machine continues to be operative and accessible. Live migration is the process of moving a virtual machine that is running from one physical host to another seamlessly without interrupting the workload.

Before you begin

In the `openshift-cnv` namespace, edit the `HyperConverged` CR and add the necessary live migration parameters:

```
oc edit hyperconverged kubevirt-hyperconverged -n openshift-cnv
```

Example:

```
apiVersion: hco.kubevirt.io/v1beta1
kind: HyperConverged
metadata:
  name: kubevirt-hyperconverged
  namespace: openshift-cnv
spec:
  liveMigrationConfig:
```

```
  bandwidthPerMigration: 64Mi
  completionTimeoutPerGiB: 800
  parallelMigrationsPerCluster: 5
  parallelOutboundMigrationsPerNode: 2
  progressTimeout: 150
```

About this task

Manually initiate a live migration of a virtual machine instance to another node using either the web console or the CLI. For more information about live migration, see Red Hat documentation.

If you want to offload live migration traffic from the default pod network to a dedicated secondary network using KubeVirt and Multus, see Live migration over a dedicated secondary network.

Live migration from the user interface

Procedure

1. As an administrator user, log in to the OpenShift® Container Platform console.
2. Go to Virtualization > VirtualMachines.
3. Search for the virtual machine that you want to migrate.
4. Click the ellipses overflow menu of the virtual machine and select Migrate or you can go to the virtual machine Overview screen and select Migrate Virtual Machine from the Actions menu.
   The status of the virtual machine changes from Running to Migrating.
5. After the migration completes, in the virtual machine details page, check the hosting node name to confirm the migration.

Live migration from the command line

Procedure

1. Create a migrate.yaml configuration file for the virtual machine instance to migrate.
   migrate.yaml:

   ```
   apiVersion: kubevirt.io/v1
   kind: VirtualMachineInstanceMigration
   metadata:
     name: {{ vm_name }}-migrate
     namespace: {{ namespace }}
   spec:
     vmiName: {{ vm_name }}
   ```

   Example:

   ```
   apiVersion: kubevirt.io/v1
   kind: VirtualMachineInstanceMigration
   metadata:
     name: migration-job
     namespace: openshift-cnv
   spec:
     vmiName: icsvm-storage-gui-kocmddajec
   ```
2. Apply the YAML file to initiate the migration.

   ```
   oc apply migrate.yml
   ```

Live migration over a dedicated secondary network

You can offload live migration traffic from the default pod network to a dedicated secondary network using KubeVirt and Multus.

Before you begin

- IBM Fusion with OpenShift® Virtualization installed.
- Multus and Whereabouts (for IPAM) enabled. The default is in OpenShift.
- KubeVirt CLI (`virtctl`) installed.

About this task

By default, all traffic including live migrations flows through the pod network. Though it works well for small workloads, it becomes a problem in production clusters where the migration traffic is memory-intensive or the pod network can become a bottle neck. Hence, as a solution you can route live migration traffic through a Secondary Network using NAD (Network Attached Definition).

Procedure

1. Create a VLAN.
   For the procedure to create a VLAN, see Adding VLANs.
2. Define the Migration Network using Network Attachment Definition (NAD).

Create a NetworkAttachmentDefinition (NAD) that represents the dedicated migration network in OpenShift Virtualization. For example, **`virtualized-nad`**.
   a. Go to **Networking** in the OpenShift Web Console.
   b. Click NetworkAttachmentDefinitions (NAD).
      Ensure you are in the **`openshift-cnv`** namespace or the namespace used for KubeVirt components.
   c. Click Create NAD.
   d. Go to the YAML tab of your NetworkAttachmentDefinitions (NAD) and replace the **`range`**, **`name`**, and **`bridge`** values as per your environment.
   e. Click Create to save and apply the NAD.
3. Configure Live Migration to use the secondary network.
   After the NetworkAttachmentDefinition (NAD) is created, update the OpenShift Virtualization HyperConverged resource to instruct KubeVirt to use this secondary network for live migrations.
   a. In the OpenShift web console, go to Operators > Installed Operators.
   b. Select OpenShift Virtualization from the list.
   c. Click the HyperConverged tab and select the HCI resource. In general, it is **`kubevirt-hyperconverged`**.
   d. Click Edit either through YAML or Form edit.
   e. Locate or add **`liveMigrationConfig`** section and specify **`network`** parameter to match the name of your NAD. For example, **`virtualized-nad`**
4. Attach the Secondary Network (NAD) to your Virtual Machine.
   a. Go to Virtualization > VirtualMachines.
   b. Choose the VM (or create a new one) that you want to enable for live migration over the secondary network.
   c. Click the VM name to open its details page.
   d. Go to the Network Interfaces tab and click Add Network Interface.
   e. Enter the following details:
      - Provide a Name. For example, `virtualized-nad`.
      - Enter the Model. The default is **`virtio`**.
      - In Network, select the previously created NAD in the **`openshift-cnv`** namespace. In this example, **`virtualized-nad`**.
      - Select Bridge or masquerade as per your setup. For live migration, it is Bridge.
   f. Click Save and restart the VM if required.
5. Validate the live Migration over the secondary network.
   a. Migrate the VM Manually:
      - Go to Virtualization > VirtualMachines.
      - Select the running VM.
      - Click Actions > Migrate to trigger a live migration.
   b. Monitor the migration status using the UI or CLI and look for **`Completed`** or **`Succeeded`** status:

      ```
      oc get vmim -n <namespace>
      ```

   c. Monitor the IP Address:
      i. Check for **`TargetNodeAddress`**.

         ```
         oc get vmi <vm name> -o jsonpath='{.status.migrationState.targetNodeAddress}' -n <namespace>
         ```

      ii. Check Virt-Handler Logs:

         ```
         oc logs -n openshift-cnv <virt-handler-pod-name> -c virt-handler
         ```

         Search for the following log lines:

         ```
         proxy started listening ... outbound: <migNet-socket>
         ```

      iii. Validate that the traffic used the secondary interface.
   d. Verify the Network Interface on VM:

      ```
      ip a
      ```

      Ensure the secondary interface is attached and has an IP from the NAD. For example, **`10.200.5.x`** if using **`whereabouts`**.

Virtual machine failover

In IBM Fusion, you can achieve automatic failover during unplanned node failures by using MachineHealthCheck and Self Node Remediation operators. During Stage 2 of the IBM Fusion installation, the **`ComputeInit`** Custom Resource (CR) automates the full lifecycle of Workload Failover in OpenShift® environments. From installation to cleanup, it interprets declarative intent and ensures that all dependent resources are created, updated, and tracked. It eliminates error-prone manual steps. Use this procedure to enable or disable the feature as needed. It is enabled by default during installation.

About this task

A node can be shutdown in a planned graceful way or unexpectedly because of reasons such as power outage or other external factors. A node shutdown could lead to workload failure if the node is not drained before the shutdown. A node shutdown can be either graceful or non-graceful.

Without VM failover automation, a worker node failure leaves the VMs on that node in a state of uncertainty, waiting for recovery. This downtime can cause the application to become unavailable until the node is back online. Using this workload failover feature, the VMs running on the failed node are automatically migrated to another available healthy node. It ensures high availability and minimizes downtime without user intervention. It does not cover live migration for maintenance or manual node migrations. For more information about live migration in IBM FusionIBM Fusion HCI, see Live migration for virtual machines.

Note: By default, the Workload Failover feature is enabled in the Compute Operator. You can use the following procedure to disable it and re-enable it later if needed.

Procedure

1. Go to the **`ComputeInit`** custom resource:
2. Open the **`ComputeInit`** CR.
   - In the OpenShift Console, go to **Administration > CustomResourceDefinitions**.

   a. In OpenShift console, go to Administration > CustomResourceDefinition.
   b. Search and open `ComputeInit` CR.
   c. Select your CR instance and click YAML to edit.
3. Locate the `workloadFailover` section:
   Example `workloadFailover` section:

   ```
   workloadFailover:
       disableWorkloadFailover: false
       remediationStrategy: OutOfServiceTaint
       softwareRebootEnabled: false
       unhealthyConditionDuration: "15s"
       maxUnhealthy: "40%"
   ```

4. To disable, change the `workloadFailover` value of `disableWorkloadFailover` to `true` and save the CR.
   If you want to enable the feature, change the value from `true` to `false` in the CR.
5. Re-enable the feature later, if needed:
   a. Edit the `disableWorkloadFailover` field and set it to `false`:
      `disableWorkloadFailover: false`
   b. Save the CR.

Backup and restore of virtual machines

IBM Fusion protects virtual machines from data loss and corruption by backing up the virtual machines regularly. The virtual machines can then be restored from a previous backup.

Backup virtual machines

Backup policies define the frequency of backups, their storage locations, and the duration for which they must be retained. A best practice is to store backups in an object storage outside of the OpenShift® cluster, as this allows virtual machines to be recovered even when the original cluster is lost or compromised.

Virtual machine backups are incremental, meaning that only changed data is part of each backup. This approach reduces the amount of network traffic that is needed to transfer backups to object storage, and reduces the object storage requirements.

The IBM Fusion offers the following virtual machine backups through the Backup & Restore service.

In-place backup of virtual machine and Red Hat® OpenShift resources
: Virtual machines can be restored in place, or in a different Red Hat OpenShift cluster.

Option to copy backup to object storage
: Backups can be stored in object storage that is on a site or in the public cloud.

Integrated backup policy management
: Virtual machines are protected by using backup policies, which define the frequency of backups, their storage locations, and the duration for which they must be retained.

Incremental backup of virtual machine
: Only changes are included in each backup, reducing the amount of network traffic and object storage utilization.

To protect a group of virtual machines that represents an application, assign a backup policy to the namespace that the virtual machines belong to. It automatically backs up all the virtual machines in the namespace that uses the backup policy.
Note: A Red Hat OpenShift best practice is to create a namespace for a collection of virtual machines that represent an individual application.
To assign backup policies to an application that contains the virtual machines, go to the Applications page in the IBM Fusion user interface. Select the namespace that the virtual machines belong to and then assign a backup policy. Alternatively, you can manage backup policies declaratively by using the Custom Resources. It enables you to manage a large fleet of applications by using a GitOps approach, where a `PolicyAssignment` CR is deployed to the cluster alongside the application.

Restore virtual machines

To recover an application that is made up of virtual machines, select a backup to restore. Use the timestamp of each backup to select the appropriate point in time for restoration. The application can be restored in place, or you can restore the application to a different Red Hat OpenShift cluster.

IBM Fusion provides the following recovery options through its Backup & Restore service:

- You can recover a virtual machine to the same or alternative cluster.
- You can recover all virtual machines in a namespace to an alternative namespace and then delete the virtual machines that are no longer needed.
- You can recover all virtual machines in a namespace to an alternative namespace and then copy files from the target virtual machine by using the command line.
- You can manage the assignment of a backup policy to a high volume of virtual machines by directly working with the Fusion data protection custom resources, specifically the resources that are used to create backup policies and backup policy assignments.
- You can restore subsets of virtual machines that belong to an application by using the following two methods:
  - You can restore the entire application into an alternative namespace and then delete the virtual machines that are no longer needed.
  - You can restore the entire application into a namespace to an alternative namespace and then copy files from the target virtual machine by using the command line.

To validate, restart the virtual machine after recovery.

Achieving data consistency for virtual machine workloads

Achieving data consistency for virtual machines by using file system consistency, crash consistency, or application consistency approaches. For more information, see Achieving data consistency for virtual machine workloads.

Creating a virtual machine by using Clone PVC

A clone is an exact duplicate of an existing storage volume that can be used like any standard volume. You create a clone to capture a point-in-time snapshot of the data.

Procedure

1. Go to Virtualization > Virtual Machines.
2. Click Create virtual machine and select the option from template.
3. Select the virtual machine template used earlier to deploy on primary cluster and click disk source.
   If the option does not exist, then click Customise virtual machine.
4. Click Clone PVC and select the PVC project (virtual machine namespace).
5. Select the PVC name.
6. Click Create virtual machine.

Disaster recovery for virtual machines

Disaster Recovery of virtual machines involves creating regular backups, maintaining up-to-date replicas, and implementing robust failover mechanisms. Implementing failover mechanisms ensures high availability and minimizes potential data loss, reducing downtime, and ensuring long-term resilience against disasters.

About this task

The topic provides the procedure on how to enable protection for virtual machines through disaster recovery by using IBM Fusion HCIIBM Fusion.

Procedure

1. Set up Metro-DR configuration between two clusters.
   For the procedure to set up, see Configuring Metro-DR set up.
2. Enable disaster recovery protection for the virtual machine.
   For the procedure to enable or enroll virtual machines for disaster protection, see Enabling and disabling applications for Metro-DR. After the application is enrolled for disaster recovery, go the virtual machine Overview tab page and check whether the Metro status is synchronized in the Disaster recovery section. Go to the Replicated applications page to see the virtual machine with details of the primary and partner cluster.
   Note: Before enabling disaster recovery protection for a virtual machine, remove any installation ISO volumes. These volumes are typically only required during the initial operating system installation and are not necessary for workload recovery. If an ISO volume must remain attached, reference the ISO persistent volume claim (PVC) directly using `persistentVolumeClaim` instead of creating an upload-based data volume.
3. From the IBM Fusion HCI user interface of the partner cluster, failover virtual machines.
   Important: As a prerequisite for virtual machine failover, shut down the virtual machine on the primary cluster.
   a. In the Red Hat® OpenShift® console of the partner cluster, go to Virtualization > VirtualMachines.
   b. Search for the virtual machine and from the ellipsis overflow menu, select Stop.

   Before you delete a virtual machine, make sure to copy the YAML file so you can redeploy the virtual machine using that at a failover site. If you stopped the virtual machine before copying the YAML, restart it after the deployment.

   For the procedure to failover, see Failover applications from site 1 to site 2.
4. After the failover is complete, remove the original virtual machine from the primary cluster.
   a. Go to the Red Hat OpenShift console of the original virtual machine on the primary cluster.
   b. Go to Virtualization > Virtual Machines.
   c. Search for the and from the ellipsis overflow menu, select Delete.
   d. In the Delete VirtualMachine window, click Delete.
   e. Select the resources of the virtual machine and click Delete.
5. Deploy the virtual machine on the current primary cluster.
   a. In the virtualMachines page of the Open shift console, click create virtual machine and select the option with YAML.
   b. Copy paste the YAML file copied from the original cluster.
      The status of the newly deployed virtual machine is in Running state.
   c. To validate, do the following steps:
      i. Go to Storage PersistenVolumeClaims and check whether the virtual machine is operational. Also, check whether the used capacity of the persistentvolumeclaim is correct.
      ii. Alternatively, check the Used (GiB) capacity column value of the virtual machine in the Applications page of IBM Fusion HCI user interface.

      Note: If a virtual machine is created using Red Hat template catalogs, then you can also create Virtual Machine using Clone PVC option. For the procedure to clone, see Creating a virtual machine by using Clone PVC.
6. Initiate the planned failback.
   a. Stop the virtual machine in the current primary cluster.
   b. Go to the the IBM Fusion HCI user interface of the partner cluster and failover the virtual machine.
      For the procedure to The primary cluster reverts to the original primary where the virtual machine was set up originally.
   c. Remove the virtual machine from the current primary cluster.
7. Deploy the virtual machine on the partner cluster or the original primary cluster.
   a. In the virtualMachines page of the Open shift console, click create virtual machine and select the option with YAML.
   b. Copy paste the YAML file copied from the original cluster.
      The status of the newly deployed virtual machine is in Running state.
   c. To validate, do the following steps:
      i. Go to Storage PersistenVolumeClaims and check whether the virtual machine is operational. Also, check whether the used capacity of the persistentvolumeclaim is correct.
      ii. Alternatively, check the Used (GiB) capacity column value of the virtual machine in the Applications page of IBM Fusion HCI user interface.

      Note: If a virtual machine is created using Red Hat template catalogs, then you can also create Virtual Machine using Clone PVC option. For the procedure to clone, see Creating a virtual machine by using Clone PVC.

Advanced workloads

IBM Fusion is built to provide enterprise ready deployments for IBM Cloud Paks, IBM watsonx, IBM Maximo®, and Db2 Warehouse.

- **IBM Cloud Paks support for IBM Fusion**
  Mapping of IBM Cloud Paks with IBM Fusion.
- **IBM Maximo® Applications support for IBM Fusion**
  Mapping of IBM Maximo® Applications with IBM Fusion.
- **IBM watsonx**
  IBM Fusion HCI enables optimized Data and AI solutions. It is built to provide enterprise-ready deployments of IBM watsonx.
- **AI workloads on IBM Fusion**
  IBM Fusion HCI is built to handle complex container AI and VM workloads as it leverages its advanced GPU nodes. IBM Fusion HCI combined with IBM watsonx offers a consistent high performance across a wide range of container AI and VM workloads. Every release of IBM Fusion HCI is validated with IBM watsonx to ensure that the infrastructure is compatible.

IBM Cloud Paks support for IBM Fusion

Mapping of IBM Cloud Paks with IBM Fusion.

IBM Cloud Paks	Version	Backup and restore support	Storage support	Hosted Control Plane
IBM Software Hub (formerly Cloud Pak for Data) Important: For virtualization, install the OpenShift® virtualization operator provided by Red Hat®. Do not install the KubeVirt HyperConverged cluster operator from the community operator catalog source. IBM watsonx support in IBM Software Hub: • watsonx Assistant • watsonx.ai • watsonx Code Assistant • watsonx Code Assistant for Red Hat Ansible Lightspeed • watsonx Code Assistant for Z • watsonx Code Assistant for Z Code Explanation • watsonx.data • watsonx.governance • watsonx Orchestrate	5.3.x or later	• Online backup and recovery to the same or alternate cluster using IBM Fusion • Backup and restore to the same cluster using IBM Software Hub native tools	• IBM Storage Scale • Fusion Data Foundation • IBM Software Hub storage considerations	Supported See IBM Software Hub cloud deployment environments
IBM Cloud Pak for Business Automation	24.0 or later	• Online backup and restore to same or alternate cluster using IBM Fusion • Backup and restore using IBM Cloud Pak for Business Automation native tools	• IBM Storage Scale • Fusion Data Foundation • IBM Cloud Pak for Business Automation storage considerations	Not supported
IBM Cloud Pak for Integration	16.1.0 or later	• Backup and restore using IBM Cloud Pak for Integration native tools	• IBM Storage Scale • Fusion Data Foundation • IBM Cloud Pak for Integration storage considerations	Supported See IBM Cloud Pak for Integration documentation
IBM Cloud Pak for AIOps	4.10.x or later	• Online backup and recovery to the same or alternate cluster using IBM Fusion • Backup and restore using IBM Cloud Pak for AIOps native tools	• IBM Storage Scale • Fusion Data Foundation • Storage considerations for IBM Cloud Pak for AIOps AI Manager	Supported See IBM Cloud Pak for AIOps documentation
IBM Cloud Pak for Security	1.10 or later	Backup and restore using IBM Cloud Pak for Security native tools	• IBM Storage Scale • Fusion Data Foundation • IBM Cloud Pak for Security storage considerations	Not supported

IBM Cloud Paks	Version	Backup and restore support	Storage support	Hosted Control Plane
IBM Cloud Pak for Network Automation	2.6.x or later	Backup and restore using IBM Cloud Pak for Network Automation native tools	• IBM Storage Scale • Fusion Data Foundation • IBM Cloud Pak for Network Automation storage considerations	Not supported

For the storage class to use with IBM Cloud Paks, IBM Storage Scale version (All), see Storage class to use with Cloud Paks.

To use IBM Cloud Paks in IBM Fusion, the storage class must include **`shared: true`**.

- **Validation tools for IBM Cloud Paks**
  Run the storage readiness and performance suites on your IBM Fusion environments. The suites run on OpenShift Container Platform to evaluate the setup and performance of your storage setup.

Validation tools for IBM Cloud Paks

Run the storage readiness and performance suites on your IBM Fusion environments. The suites run on OpenShift® Container Platform to evaluate the setup and performance of your storage setup.

Storage validation tool

Run the IBM Software Hub storage validation tool on your Red Hat® OpenShift cluster to verify your storage setup for use with IBM Software Hub. Set up your environment and before you install your IBM Cloud Paks, run the validation tool. For more information about the tool, see storage validation tool.

Performance validation tool

Run the storage performance tool to ensure that your storage performs properly and you meet the recommended performance guidelines. Collect Storage performance metrics on your Red Hat OpenShift cluster. For more information about the tool, see k8s-storage-perf GitHub repository.

As a prerequisite to run storage performance suite, run the following command to install selinux:

```
pip install selinux
```

The tool validates whether your performance (for example, network, environment) runs at a speed that enables IBM Cloud Paks to run successfully.
An example of the output:

```
Summary
Cluster Name                 PVC               Storage Type  Environment  Test Name   Thread Count  write MiB/s        Requiremen
storage-performance-cluster  pvc-sysbench-rwo  Scale            HCI      rndwr_4k_8      8            46.4         Recommended to r
storage-performance-cluster  pvc-sysbench-rwx  Scale            HCI      seqwr_1g_2      2            3766.0       Recommended to r
storage-performance-cluster  pvc-sysbench-rwo  Scale            HCI      seqwr_1g_2      2            3901.2       Recommended to r
storage-performance-cluster  pvc-sysbench-rwx  Scale            HCI      rndwr_4k_8      8            46.1         Recommended to r
```

Here, the PVCs are running against the tests. The thread count specifies the number of simultaneous requests. The write Mb/s indicates the output.

Example of the partial output (from the "all" full metrics run) of a minimum IBM Fusion HCI configuration, that is, for a smallest environment. Depending on your configuration, environment, and network speed, the output would vary.

Summary							
Cluster Name	PVC	Storage Type	Environment	Test Name	Thread Coun	write MiB/s	Requirement
storage-perforr	pvc-sysbencl	Scale	HCI	rndwr_4k_8	8	46.4	Recommended to meet the requirement of 11 MiB/s or higher
storage-perforr	pvc-sysbencl	Scale	HCI	seqwr_1g_2	2	3766	Recommended to meet the requirement of 128 MiB/s or higher
storage-perforr	pvc-sysbencl	Scale	HCI	seqwr_1g_2	2	3901.2	Recommended to meet the requirement of 128 MiB/s or higher
storage-perforr	pvc-sysbencl	Scale	HCI	rndwr_4k_8	8	46.1	Recommended to meet the requirement of 11 MiB/s or higher

Detailed Measurements														
Cluster Name	PVC	Storage Type	Environment	Test Name	Thread Coun	write MiB/s	Writes/s	read MiB/s	Reads/s	Total Time	Latency Min	Latency Avg	Latency Max	Latency 95th
storage-perforr	pvc-sysbencl	Scale	HCI	rndwr_4k_8	8	46.4	11893.6	0	0	10	0.3	0.7	95.8	1
storage-perforr	pvc-sysbencl	Scale	HCI	seqrd_1g_2	2	0	0	7380.8	7.2	10.2	217.2	279.6	583.2	555.4
storage-perforr	pvc-sysbencl	Scale	HCI	rndrd_4k_8	8	0	0	3142.7	804539.1	10	0	0	2.1	0
storage-perforr	pvc-sysbencl	Scale	HCI	seqwr_1g_2	2	3766	3.7	0	0	10.3	374.7	540	821.4	670.2
storage-perforr	pvc-sysbencl	Scale	HCI	seqwr_1g_2	2	3901.2	3.8	0	0	10.4	354.6	521.3	737	695.1
storage-perforr	pvc-sysbencl	Scale	HCI	rndwr_4k_8	8	46.1	11803.4	0	0	10	0.2	0.7	81.4	1.1
storage-perforr	pvc-sysbencl	Scale	HCI	seqrd_1g_2	2	0	0	7404.1	7.2	10.2	207.3	280.6	585.3	548.3
storage-perforr	pvc-sysbencl	Scale	HCI	rndrd_4k_8	8	0	0	3278.2	839227.1	10	0	0	37.7	0

Here, the cluster name is **`storage-performance-cluster`** of storage type `ocs`. The *Summary* section of the table also includes the requirement recommendations. In this example, for the **`rndwr_4k_8 test`**, the write MiB/s is 46.4 and the recommended is 11 MiB/s or higher. In all the rows, the output is either 2X or more than 2X of the recommendation. It helps you validate whether your environment meets the recommended performance requirements for IBM Cloud Paks.

You can use this feature regardless of IBM Cloud Paks.

IBM Software Hub sizing in the Sales Configurator

The starting point is a minimum IBM Software Hub configuration. If you select services from the list for inclusion, it shows the minimum required configuration. For example, it displays the number of compute nodes, resource pools, and memory to deploy. It also lists the dependencies. For example, if you want to use Watson OpenScale, then you need Watson Machine Learning, Common Core Services, and Analytics Engine Powered by Apache Spark.
Note: Ensure that the IBM Fusion/storage sizing are also considered in your IBM Software Hub sizing.
To view the sales configurator, go to https://app.ibmsalesconfigurator.com/#/zen/configure.

IBM Maximo® Applications support for IBM Fusion

Mapping of IBM Maximo® Applications with IBM Fusion.

IBM Maximo® Applications	Version	Backup & Restore support	Storage support	Hosted Control Plane
Maximo Core	• 8.11.6 and later • 9.0.0 and later	Online backup and restore for Maximo Core using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat® OpenShift® Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.
Maximo Manage	• 8.7.4 and later • 9.0.0 and later	Online backup and restore for Maximo Manage using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat OpenShift Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.
Maximo Optimizer	9.0.0 and later	Online backup and restore for Maximo Optimizer using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat OpenShift Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.
Maximo Health	Deployed as part of Manage	Online backup and restore for Maximo Health using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat OpenShift Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.
Maximo IOT	9.0 and later	Online backup and restore for Maximo IOT using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat OpenShift Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.
Maximo Monitor	9.0 and later	Online backup and restore for Maximo Monitor using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat OpenShift Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.
Maximo Predict	9.0 and later	Online backup and restore for Maximo Predict using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat OpenShift Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.

IBM Maximo® Applications	Version	Backup & Restore support	Storage support	Hosted Control Plane
Maximo Collaborate (formerly Assist)	9.0 and later	Online backup and restore for Maximo Collaborate (formerly Assist) using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat OpenShift Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.
Maximo Visual Inspection Note: To check supported GPU models, see Maximo Visual Inspection documentation.	9.0 and later	Online backup and restore for Visual Inspection using IBM Fusion	• IBM Storage Scale • Fusion Data Foundation • Red Hat OpenShift Data Foundation	• Supported on Bare Metal Hosted Control Plane • Supported on Virtualized Hosted Control Plane For more information about Hosted Control Plane with IBM Fusion, see IBM Maximo® on Hosted Control Plane.

For the general Maximo BR procedure, see https://github.com/IBM/storage-fusion/tree/master/backup-restore/recipes/Maximo/2.9.x.

- **IBM Maximo® Application Suite**
  IBM Maximo® Application Suite is a collection of AI-driven asset management applications that gains enhanced backup, restore, and storage capabilities when deployed with IBM Fusion.
- **IBM Maximo® on Hosted Control Plane**
  Use the IBM Maximo® IBM Operator catalog and monthly updates for the latest information about the Fusion operator.

IBM Maximo® Application Suite

IBM Maximo® Application Suite is a collection of AI-driven asset management applications that gains enhanced backup, restore, and storage capabilities when deployed with IBM Fusion.

IBM Maximo® Application Suite provides powerful AI-driven asset and analytics solutions for your enterprise needs.

IBM Maximo® Application Suite is a powerful collection of applications, industry solutions, add-ons, and tools.

When you deploy it from IBM Fusion, it enhances backup, restore, and storage capabilities across IBM Maximo® Application Suite.

Install

The IBM Maximo® CLI is the preferred way to install IBM Maximo® as it provides an automated deployment for this type of workload. For more information, see IBM Maximo® KC and IBM Maximo® Application Suite CLI.
Tip: To customize IBM Maximo® Application Suite instances with extra features that might be unavailable in IBM Maximo® Application Suite CLI, you can use ansible-devops for advanced configuration. For more information, seeMAS Devops Ansible Collection.
For installation sizing calculator guidance, see Sizing guidance.

Hosted Control Plane support

To install IBM Maximo® Application Suite on a Hosted Control Plane, see IBM Maximo® on Hosted Control Plane.

Limitations

If you plan to use any IBM Maximo® application dependent on Kafka such as IOT and Monitor, do not use the Strimzi open source operator in conjunction with IBM Fusion Backup & Restore. Instead, use the Streams for Apache Kafka by Red Hat® (formerly AMQ-Streams). Ensure that you match the version available in the IBM Fusion Backup & Restore namespace with the Streams operator that you are planning to use for IBM Maximo® Application Suite. For more information about the limitations, see IBM Maximo® limitations.

Backup & Restore is not supported for IBM Maximo® IoT, IBM Maximo® Monitor, and IBM Maximo® Predict, but they can be installed on any cluster that has OpenShift® Virtualization installed.

IBM Maximo® on Hosted Control Plane

Use the IBM Maximo® IBM Operator catalog and monthly updates for the latest information about the Fusion operator.

For the list of IBM Maximo® deployments, see IBM Maximo® Applications support for IBM Fusion.

If you plan to use IBM Maximo® in Hosted Control Plane with IBM Fusion you must deploy Fusion operator and Fusion Services by logging into the Hosted Control Plane cluster and use the following procedure:

- Install IBM Fusion on Hosted Control Plane cluster. See Installing IBM Fusion on On-premises Bare Metal.
- Install Fusion Data Foundation. See Installing Fusion Data Foundation on the hosted cluster.
- Install Backup & Restore Spoke. See Backup & Restore .

Install

The CLI is the preferred way to install as it provides an automated deployment for this type of workload. For more information, see Maximo KC and MAS CLI.
Tip: To customize IBM Maximo® Application Suite instances with extra features that might be unavailable in IBM Maximo® Application Suite CLI, you can use ansible-devops for advanced configuration. For more information, see Maximo Application Suite Devops Ansible Collection.
For installation sizing calculator guidance, see Sizing guidance.

IBM watsonx

IBM Fusion HCI enables optimized Data and AI solutions. It is built to provide enterprise-ready deployments of IBM watsonx.

Up and running quickly with IBM Fusion HCI
: IBM Fusion HCI provides a Bare Metal appliance to get up and running quickly with IBM watsonx. The hyper-converged solution provides Bare Metal compute nodes, GPU nodes, high-speed switches, and everything that you need to host your Data and AI platform. The automated installer deploys Red Hat® OpenShift®. High-performance container native storage is preconfigured to run IBM Software Hub that further simplifies the deployment.

Enterprise-grade resiliency for watsonx
: IBM Fusion HCI is built with resiliency in mind. The appliance features a fully redundant networking, and its container native storage protects against node or drive failures by using erasure coding. The IBM watsonx configuration is protected and it uses built in backup and recovery capabilities to recover your IBM watsonx deployment to another cluster in an event of a disaster.

Accelerating watsonx.data queries with IBM Fusion HCI
: IBM Fusion HCI accelerates watsonx.data queries by providing a high speed-distributed cache based on IBM technology that is used for High-Performance Computing. When you run queries, intelligent caching serves metadata and data, cutting down on the queries that go to the object storage.

- **Deploying watsonx.data capabilities**
  The deployment of watsonx.data On-premises is made easy by the combination of IBM Fusion HCI and IBM Storage Ceph, which provide all of the infrastructure that you need for your stand-alone data lakehouse.
- **Deploying watsonx.ai capabilities**
  You can access watsonx.ai services on the IBM Fusion HCI.
- **Deploying IBM® watsonx.governance capabilities**
  The deployment of IBM® watsonx.governance On-premises is made easy by the combination of IBM Fusion HCI.

Deploying watsonx.data capabilities

The deployment of watsonx.data On-premises is made easy by the combination of IBM Fusion HCI and IBM Storage Ceph, which provide all of the infrastructure that you need for your stand-alone data lakehouse.

Important: Do not install KubeVirt HyperConverged Cluster Operator. It can cause problems when you install IBM Software Hub software.
IBM Fusion HCI is a hosting platform for IBM watsonx.data and it provides the following capabilities:

- Data sharing and self-service access
- Data caching to accelerate watsonx.data performance.
- Local S3 object store.
- IBM Storage Ceph provides an external S3 object store for watsonx.data. This S3 object store can be the main S3 object store for watsonx.data, or alongside S3 object store with other on-premise or public cloud object stores. The IBM Storage Ceph object storage interface (Ceph Object Gateway) is compatible with a large subset of the Amazon S3 RESTful API. For more information about IBM Storage Ceph, see https://www.ibm.com/docs/en/storage-ceph/6.
- Ability to include GPU servers in the IBM Watsonx solution

These high-level instructions show how to prepare IBM Fusion HCI so that you can install watsonx.data and make use of buckets with storage acceleration.

There are three main steps in this process:

For detailed information about planning, implementation, monitoring, and backup and restore, see Accelerating IBM watsonx.data with IBM Fusion HCI Redpaper.

1. Upgrade OpenShift® Container Platform 4.14 or higher.
2. Configure the Multicloud Object Gateway that provides access to accelerated buckets.
   The Multicloud Object Gateway (MCG) provides an object endpoint that watsonx.data and other workloads can connect to access multiple buckets, including storage acceleration buckets. The Multicloud Object Gateway is provided by the Red Hat® OpenShift Data Foundation operator.

   Install the Red Hat OpenShift Data Foundation operator into Red Hat OpenShift Data Foundation.

   a. Search for Red Hat OpenShift Data Foundation in OperatorHub.
   b. Install the Red Hat OpenShift Data Foundation operator.
   c. Follow the instructions to Deploy standalone Multicloud Object Gateway. When you are prompted for a StorageClass in *step 3 of section 3.3*, use the **`ibm-storage-fusion-cp-sc`** StorageClass that is configured by default in IBM Fusion HCI.
3. Configure Advanced File Management (AFM nodes) in the HCI appliance to enable storage acceleration.
   IBM Fusion HCI uses special Advanced File Management (AFM) nodes to connect to external object storage and provide data caching that is used for storage acceleration. Install these AFM nodes in the IBM Fusion HCI. You must have already installed them as a part of setting up the physical appliance.

   To use, add the AFM nodes to the OpenShift cluster. Follow the Upsizing nodes instructions to add the AFM nodes to the OpenShift cluster and configure them for storage.

   Attach a storage accelerated bucket to a watsonx.data catalog:
   a. Create an AFM fileset and attach it to a bucket.
   b. Expose the accelerated bucket through the Multicloud Gateway.
   c. Connect the accelerated bucket to a catalog.

i. Log in to watsonx.data console."
ii. From the navigation menu, select Infrastructure Manager.
iii. To define and connect a bucket, click Add component and select Add bucket.
iv. In the **Add bucket** window, provide the details to connect to existing externally managed object storage.

Deploying watsonx.ai capabilities

You can access watsonx.ai services on the IBM Fusion HCI.

To leverage watsonx.ai on premises, purchase IBM Fusion HCI and install watsonx.ai application as part of IBM Software Hub. You can also install watsonx.ai on IBM Fusion HCI by using the IBM container registry through an operator.

For more information about watsonx.ai, see https://www.ibm.com/products/watsonx-ai.

The watsonx.ai is supported on a single IBM Fusion HCI with up to four G03 GPU nodes: Each node can have up to 8x H100 NVL 94GB GPU PCIe cards or 8x Nvidia L40S 48GB GPU cards (32 total GPU cards per rack).

Optionally, you can add tolerances for watsonx.ai workloads to run on the GPU nodes.

Important: Install KubeVirt HyperConverged Cluster Operator only from the Red Hat® Operators catalog. The version provided by the Community Operators catalog cause problems when you install the IBM Software Hub software.
IBM Fusion HCI is a hosting platform for watsonx.ai and it provides the following capabilities:

- Leverage the watsonx.ai user interface and REST API interfaces.
- Schedule regular online backups of a watsonx.ai instance to recover to an earlier point in time in the event of data loss or corruption.
- Monitor the model serving and tuning pipeline KPIs by using built in OpenShift® monitoring capabilities.
- Leverage the watsonx.ai health and performance monitoring through the IBM Software Hub common services.
- Extend the Backup & Restore service recipes of IBM Fusion HCI for the watsonx.ai component as well.

Note: In this release, watsonx.ai support is not available for high-availability IBM Fusion HCI.

Deploying IBM® watsonx.governance capabilities

The deployment of IBM® watsonx.governance On-premises is made easy by the combination of IBM Fusion HCI.

The IBM® watsonx.governance includes tools for governing models and the usage resources for evaluating and explaining models. When you run model evaluations and explanations with IBM® watsonx.governance, you consume resources. Based on your requirements, set the amount of usage resources to consume.

An IBM Cloud Object Storage service instance gets automatically provisioned to provide storage for the assets you create or add to workspaces. Information in IBM Cloud Object Storage is encrypted and resilient. Each workspace has its dedicated bucket.

You can use IBM® watsonx.governance tools to govern models in the following ways:

- **Monitor and evaluate models:** You can monitor model output and explain model predictions. Your IBM® watsonx.governance model monitoring and evaluation tools are in projects and deployment spaces.
- **Track and document AI use cases:** You can view model lifecycle status, general model and deployment details, training information and metrics, and deployment metrics. Your IBM® watsonx.governance model tracking tools in AI use cases.

Important: Do not install KubeVirt HyperConverged Cluster Operator. It can cause problems when you install IBM Software Hub software.
For more information about planning, prerequisites, implementation, and usage, see IBM Software Hub documentation.

AI workloads on IBM Fusion

IBM Fusion HCI is built to handle complex container AI and VM workloads as it leverages its advanced GPU nodes. IBM Fusion HCI combined with IBM watsonx offers a consistent high performance across a wide range of container AI and VM workloads. Every release of IBM Fusion HCI is validated with IBM watsonx to ensure that the infrastructure is compatible.

You can select the most suitable GPU node based on your specific AI workload requirements. For details on supported GPU types and their hardware configurations, see GPU nodes.

Node labeling and GPU virtualization techniques can effectively enhance performance and ensure efficient resource management for both complex AI container workloads and virtual machine (VM) workloads.

Node labeling for GPU reservation

IBM Fusion HCI uses labels to reserve GPU nodes for AI workloads.

For example, the IBM Fusion Backup & Restore uses the `gpu.isf.ibm.com` label to control the placement of its pods. To prevent Backup & Restore workloads on GPU nodes, run the following command from the OpenShift® cluster to label the GPU nodes.

```
oc label node/<node name> gpu.isf.ibm.com=""
```

The IBM Fusion service pods avoid nodes with such a label. You can use this approach for other workloads.
For example, the IBM Fusion Backup & Restore utilizes the `gpu.isf.ibm.com` label to manage the placement of its pods effectively. To restrict IBM Fusion Backup & Restore workloads from running on GPU nodes, run the following command to apply this label to these nodes in the OpenShift cluster. After the nodes are labeled, the

Backup & Restore pods automatically avoid GPU nodes with this label. This approach can also be adapted for other workloads to ensure resource optimization and placement control.

GPU virtualization options

IBM Fusion provides the following GPU virtualization options:

- PCI Passthrough for VMs to access GPU
- VGPU for VMs

The Peripheral Component Interconnect (PCI) passthrough feature enables virtual machines to directly interact and manage hardware devices. When you configure this feature, it allows these devices to function as if they are physically connected to the guest operating system, thereby ensuring smooth and uninterrupted operation. This setup is particularly useful for applications requiring direct hardware access, such as graphics-intensive tasks or specialized hardware control. For more information about this feature, see Red Hat documentation.

In case of vGPU for VMs, some graphics processing unit (GPU) cards support the creation of virtual GPUs (vGPUs). OpenShift Virtualization can automatically create vGPUs and other devices whenever an administrator provides configuration details in the `HyperConverged` custom resource (CR). This automation is especially useful for large clusters. For more information about this feature, see Red Hat documentation and NVIDIA documentation.

Administering

You can manage and administer hardware, monitor and collect logs, understand events, raise call tickets, work with APIs, manage users, and understand the UI.

- **Managing the infrastructure and storage of IBM Fusion HCI**
  You can manage your appliance infrastructure by up sizing compute nodes and storage disks. In addition, you can administer and configure aspects of storage integration with the IBM Fusion HCI offering.
- **User management**
  Use the role-based user restrictions and user management capabilities to administer IBM Fusion HCI.
- **Knowing your IBM Fusion user interface**
  Learn the various key elements of the IBM Fusion user interface for ease of use and navigation.
- **Upgrading IBM Fusion**
  Upgrade to the latest version so you can have access to the new functionalities and improvements that enhance your overall experience with IBM Fusion HCI.

Managing the infrastructure and storage of IBM Fusion HCI

You can manage your appliance infrastructure by up sizing compute nodes and storage disks. In addition, you can administer and configure aspects of storage integration with the IBM Fusion HCI offering.

- **Administering nodes and racks**
  Administering nodes and service node in the system includes day-to-day operations such as power-on, power-off, placing node in maintenance, upscale, and up size storage disks. You can add new racks to your cluster or edit its details.
- **Shutting down and restarting IBM Fusion HCI rack**
  Procedures to gracefully shut down and restart IBM Fusion HCI. The process vary depending on whether the storage is Fusion Data Foundation or Global Data Platform.
- **Networking**
  Configure and administer network settings for IBM Fusion.
- **Customer Replacement Units (CRU)**
  List of Customer Replacement units for your reference.
- **Observability**
  Observability in IBM Fusion enables centralized, multi-cluster monitoring that uses open source technologies like Prometheus, Thanos, and Grafana. This solution collects metrics from managed clusters, provides global querying and long-term storage, and delivers unified dashboards for visualization and alerting.
- **Administering and configuring storage in IBM Fusion HCI**
  This section discusses the administration and configuration aspects of the IBM Storage Scale storage integration to the IBM Fusion HCI offering and for the successful functioning of the IBM Fusion appliance with the storage integration.

Administering nodes and racks

Administering nodes and service node in the system includes day-to-day operations such as power-on, power-off, placing node in maintenance, upscale, and up size storage disks. You can add new racks to your cluster or edit its details.

For any user interface related issues, see IBM Fusion HCI user interface issues.

- **Administering the service node**
  Administering the service node in the system includes day-to-day operations such as monitoring the health, performing power operations, collecting logs, and doing firmware updates whenever it is available.
- **Adding a Day 2 service node**
  This section provides the steps for adding a Day 2 service node to an existing IBM Fusion HCI rack.
- **Administering the node**
  Administering the nodes in the system includes day-to-day operations such as monitoring the health, putting a node into maintenance mode, and performing power operations whenever it is available.
- **Restarting Baseboard Management Controller (BMC)**
  The Baseboard Management Controller (BMC) restart is a process used to perform a restart of BMC. This restart is typically performed when the BMC is not functioning correctly or when there are any active node health events. The BMC restart might remove any false health events raised.

- **Scaling the cluster**
  As workloads in the cluster increase, you must add more CPU, memory, GPU, or storage to accommodate them. CPU and memory can be increased by adding additional compute nodes to the cluster. For workloads that require GPUs, you can add specialized GPU nodes to the cluster. Storage can be scaled by adding additional disks to existing storage nodes, or adding additional storage nodes to the cluster. If you want to scale your cluster beyond a single rack, then you can add up to two expansion racks, enabling your cluster to host more compute, storage, and GPU nodes.
- **Node firmware**
  IBM Fusion HCI continuously monitors the firmware installed on each node. If a node is found to be running a firmware version lower than the Fusion-certified level, the system automatically initiates a firmware upgrade. If the upgraded firmware requires a reboot to take effect, the node is marked as Upgrade available.
- **Upgrading the NVIDIA GPU driver with the NVIDIA GPU operator**
  After installing IBM Fusion HCI and adding a GPU node to the OpenShift® cluster, the Node Feature Discovery (NFD) component and the NVIDIA GPU operator are automatically installed. The NFD component labels the hardware capabilities of the GPU node, and the NVIDIA GPU operator automates the deployment and management of NVIDIA drivers and related components required for GPU accelerated workloads.
- **Customizing Node Feature Discovery (NFD) CR**
  Use the instructions to customize the Node Feature Discovery (NFD) configuration to meet your specific requirements.
- **Converting compute node to AFM node**
  Convert a Lenovo compute node to a AFM node through backend CR.
- **Prerequisite for removing worker node from the IBM Fusion HCI cluster**
  The following section provides the prerequisite to remove worker node from the IBM Fusion HCI cluster.
- **Configuring GPU upsize on Dell systems**
  Configure GPU upsize for Dell systems in IBM Fusion HCI.
- **Node issues**
  This topic provides a comprehensive list of common troubleshooting steps and known issues that may arise while working with compute nodes.

Administering the service node

Administering the service node in the system includes day-to-day operations such as monitoring the health, performing power operations, collecting logs, and doing firmware updates whenever it is available.

View service node details

1. From the IBM Fusion HCI user interface, click Infrastructure > Overview page.
2. Click the service node that is placed at RU23, close to the optional KVM service console.
   It opens a Service node slide out pane.
3. Click view full details to see Service node page that includes the front, rear, and internal graphical views, recent events, and other details of the service node and its internal components.

Download service node logs

1. In the service node details page, click Actions > Download logs to download the logs of the service node.
2. In the confirmation window, click Collect.

Power operations on the service node

Power operations on a node server, such as powering on, powering off, or restarting, are crucial for maintenance, troubleshooting, system recovery.

Power on and power off operations on a service node:

1. In the service node details page, click Actions > Power off or Power on to power off or power on the service node respectively.
2. In the confirmation window, select Power off node to shut down and Power on node to start up.

Restart operations on a service node

1. From the Actions drop-down menu, click Restart to restart the service node.
2. In the confirmation window, click Restart node.

Adding a Day 2 service node

This section provides the steps for adding a Day 2 service node to an existing IBM Fusion HCI rack.

Before you begin

Ensure that you meet these prerequisites before adding a Day 2 service node:

1. Create DNS entry:
   a. Create a DNS record for the service node by using the format `servicenode-1.<domain.com>`.
   b. Ensure that both forward (A/AAAA) and reverse (PTR) lookups are properly configured.
2. DHCP configuration (if applicable):
   If the existing cluster uses DHCP-based IP assignment, create a DHCP reservation for the service node by using its MAC address. For more information about DHCP setup, see Setting up DHCP.
3. Cluster information:
   Keep the following cluster details readily available:

- OpenShift® Bare metal VLAN ID
- Rack generation type (Gen1 or Gen2)

4. Switch IP addresses:
   Retrieve the IP addresses of the following switches:
   - High-Speed Switch 1 (RU20)
   - High-Speed Switch 2 (RU21)
   - Management Switch 1 (RU18)
   - Management Switch 2 (RU19)

   To obtain the IP addresses, follow these steps:
   a. Log in to the OpenShift console user interface and go to Workloads > ConfigMaps.
   b. Select the project `ibm-spectrum-fusion-ns`.
   c. Search for `kickstart` and open the `kickstart-<serial>` ConfigMap.
   d. Scroll down to locate the IPv4 addresses of the switches. If IPv4 is not available, retrieve the IPv6 addresses.
5. Switch credentials (ISFUSER):
   Obtain the password for user `ISFUSER` on all switches (RU18, RU19, RU20, RU21).

   To retrieve the passwords:
   a. Log in to the OpenShift console user interface and go to Workloads > Secrets.
   b. Select the project `ibm-spectrum-fusion-ns`.
   c. Search by using:
      - `hspeed1` (High-Speed Switch 1), `hspeed2` (High-Speed Switch 2)
      - `mgmt1` (Management Switch 1)
      - `mgmt2` (Management Switch 2)

   d. Open the secret named in the format `<switch-name>-<serial>-<secret>`.
6. Static IP configuration (if applicable):
   If the cluster uses static IP addressing, keep the following details available for the service node:
   - Bare metal network IP (for example: `10.1.1.100/24`)
   - Gateway IP
   - DNS server IP
   - NTP server IP
7. Upsize the service node only when the cluster, switches and nodes are healthy and no maintenance activities are in progress.

Procedure

To add a Day 2 service node to an existing IBM Fusion HCI rack, follow these steps:

1. Install the service node in the existing IBM Fusion HCI rack.
2. Place the service node at `RU33` for a Gen1 rack, or place it at `RU23`.
   Tip: A rack is considered as a Gen1 rack if its control nodes are models 9155-C01 or 9155-C05.
3. Cable the service node at your site.
   Note: The connection and wiring remain the same for Gen1, Gen2, and later versions of existing racks.
   a. Connect the high-speed network interfaces:
      i. Connect Service Node 25GbE Port 1 to High-Speed Switch 1 (RU20), Port 1 using a QSFP-to-SFP adapter, and a 25GbE cable.
      ii. Connect Service Node 25GbE Port 2 to High-Speed Switch 2 (RU21), Port 1 using a QSFP-to-SFP adapter, and a 25GbE cable.
   b. Connect the service node to the management switches:
      i. Connect the Service Node IMM/BMC port to Management Switch 1 (RU18), Port 33.
      ii. Connect Service Node 4-port 1GbE NIC – Port 1 to Management Switch 2 (RU19), Port 33.
      iii. Connect Service Node 4-port 1GbE NIC – Port 2 to Management Switch 1 (RU18), Port 35.
      iv. Connect Service Node 4-port 1GbE NIC – Port 3 to Management Switch 2 (RU19), Port 35.
   c. Connect the out-of-band interface of the service node:
      i. Connect Service Node 4-port 1GbE NIC - Port 4 to the customer data center network to carry out-of-band traffic.
4. Power on the service node and connect with KVM. Then log in to the RHEL OS using default user `kni` and password `passw0rd`.
5. Change to the following directory:

   ```
   cd /home/kni/isfconfig
   ```

6. Run the following command to change the permission for the script:

   ```
   chmod +x servicenode_script_1.sh
   ```

7. Locate the script `servicenode_script_1.sh` in the current directory and run the script 1:

   ```
   ./servicenode_script_1.sh
   ```

   After script 1 completes successfully, IBM Fusion HCI automatically discovers the node, performs the required scale-up operation, and displays the node in the IBM Fusion HCI user interface.
   To set up vault secrets and certification, execute the `servicenode_script2_2.13.sh` script as mentioned in the following steps.

   Important:
   - If the IBM Fusion HCI rack is using an IPv6 stack on the provisioning network, run the `./servicenode_script_2.sh` script. After running the script, the service node appears in the IBM Fusion HCI user interface.
   - If the IBM Fusion HCI rack is using an IPv4 stack on the provisioning network, skip running the `./servicenode_script_2.sh` script as mentioned in the following steps.
8. Set the IBM Fusion HCI namespace environment variable to `ibm-spectrum-fusion-ns`.
   For example:

   ```
   export FUSION_NAMESPACE="ibm-spectrum-fusion-ns"
   ```

9. Ensure that the following files are present in the `/home/kni/isfconfig` directory:
   - `kickstart.json`
   - `appliance-info.json`
10. Run the following `oc login` command to log in to the OpenShift Container Platform cluster:

```
KUBECONFIG=/tmp/remote-ocp-config oc login --token=sha256~TOKEN --server=https://api.rackName.fusion.tadn.ibm.com:6443 --insecure-skip-tls-verify=true
```

11. Run the following command to verify whether you are logged in to the cluster or not.

    ```
    KUBECONFIG=/tmp/remote-ocp-config oc get secret -n ${FUSION_NAMESPACE}
    ```

    If you are unable to see the secret list in `ibm-spectrum-fusion` namespace, then check the `oc` login command again and make sure that you get the correct remote access to the cluster.
12. Run the following command to change the permission for the script:

    ```
    chmod +x servicenode_script_2_2.13.sh
    ```

13. Run the following command to keep a backup of the vault files:

    ```
    sudo cp -r /var/vault /var/vault_backup/
    ```

14. Check if the service node entry is present in the `KickStart` configmap by running the following command:

    ```
    KUBECONFIG=/tmp/remote-ocp-config KICKSTART_CM=$(kubectl get configmap appliance-info -n ibm-spectrum-fusion-ns -o json | jq -r '.data | to_entries[0].value | fromjson | .kickstartCM') && kubectl get configmap "$KICKSTART_CM" -n ibm-spectrum-fusion-ns -o json | jq -r '.data."kickstart.json"' | jq -r '.computeNodeIntegratedManagementModules[] | select(.type=="servicenode")'
    ```

    If the command returns a result for `"servicenode-*"`, proceed to the next step. Otherwise, wait until the entry is added to the `KickStart` configmap. If the service node entry is not added to the `KickStart` configmap after 10 minutes, contact IBM Support.

15. Important: To run the `servicenode_script_2_2.13.sh` script, remote login to the OpenShift Container Platform cluster is required. Ensure that you have the correct access credentials before executing the script. If login issues occur, resolve them before proceeding with script execution.
    Run the `servicenode_script_2_2.13.sh` script that is located in the `/home/kni/isfconfig/`directory by using the following command:

    ```
    KUBECONFIG=/tmp/remote-ocp-config /home/kni/isfconfig/servicenode_script_2_2.13.sh
    ```

    Ensure that the script execution is successful.
16. Ensure that you must back up the `vault-secret` and `vault-login-secret` from the OpenShift Container Platform for future reference.
    Important: Download and securely store the `vault-secret` and `vault-login-secret` YAML from OpenShift Container Platform in `ibm-spectrum-fusion-ns` namespace after a successful `servicenode_script_2_2.13.sh` execution.
17. After completing the previous steps, run the following command to delete the temporary configuration file.

    ```
    rm /tmp/remote-ocp-config ; rm /tmp/extracted_*
    ```

18. If you backed up the vault secret and vault login secret, then you can delete the `/var/vault_backup` directory from the newly added service node.

    ```
    rm -rf /var/vault_backup/
    ```

19. To enable automated credential rotation, manually update the base rack `PlatformConfig` file after successfully adding the service node. For more information, see Enabling password rotation for IBM Fusion HCI system.

If the service node does not appear in the IBM Fusion HCI user interface, contact IBM Support.

Administering the node

Administering the nodes in the system includes day-to-day operations such as monitoring the health, putting a node into maintenance mode, and performing power operations whenever it is available.

For any node drain issues during maintenance, see Issues related to IBM Fusion HCI node drains.

Node inventory tabs: Configured, Discovered and Hosted inventory

From the IBM Fusion HCI menu, click Infrastructure > Nodes. The Node page includes Configured nodes and Discovered nodes tabs. By default, the Node page opens the Configured nodes tab.

Go to Discovered nodes tab to add discovered nodes in a rack that were not added to the OpenShift® cluster earlier.

Node details

The following table provides the node details in the nodes list:

	Description
Name	The name of the node. • Naming convention of nodes added by installation or upsize: `compute-<rackid>-<rack unit number>.<domainname>` For a single rack, rackid is always 1.

	Description
Hardware status	The node Status column also displays the health of the node. • indicates that the status is healthy • indicates warning status • indicates critical status The node Status column that is shown on the node inventory page is the node hardware monitoring state on the IBM Fusion user interface. A node is monitored by connecting to its remote management module. Hence, the node state is a reflection of its connectivity to the monitoring module and its ability to fetch the monitoring data. Important: • This state does not reflect node operating system state and their readiness state in the OpenShift cluster. As a result, the node status between IBM Fusion and OpenShift user interface might not be the same. : • When the IBM Fusion user interface shows Running and the OpenShift user interface shows Not Ready, then follow the steps that are mentioned in the Compute node issues to resolve the OpenShift node status issue. To see the OpenShift state, go to Dashboard > OpenShift section.
Type	The options are Compute only, Compute storage, AFM, or GPU.
Firmware	The firmware status of the node.
S/N	The hardware serial number.
Rack	The name of the rack.
Rack unit	The rack unit position of the node in the rack.
CPU cores	The amount of CPU in cores.
Memory (GB)	The amount of memory in GB.

Use the settings icon to customize the columns. You can customize the number of rows that must be displayed in a page, jump to a specific page, or go to the previous or next page.
Use the Search text box to filter and find a specific node.

Node power operations

Node actions from the ellipsis overflow menu of a node record:

- Enable maintenance or Disable maintenance When the node is moved to maintenance successfully, the ellipsis overflow menu option shows Disable maintenance option and the Enable maintenance option is available when the node is not already in maintenance.
- **Power operations:**

  Power on and power off operations on a node:
  Note: Do all power operations only from the IBM Fusion HCI user interface.
  1. Move the node to maintenance.
  2. In the ellipsis menu of the node, click Power off node to power off the node.
  3. In the confirmation window, click Power off.
  4. In the ellipsis menu of the node, click Power on node to power on the node.
  5. In the confirmation window, click Power on.
  6. After you complete the maintenance operations, from the ellipsis overflow menu of the node, click Power on node to power on the node.

  Restart and shutdown operations on a node
  1. Move the node to maintenance.
  2. In the ellipsis menu of the node, click Restart node or Shutdown node. Alternatively, you can click the Manage resources in the node details > Inventory tab page and then do power operations.
  3. In the confirmation window for restart or shutdown actions, click Restart node or Shutdown node accordingly.
     Note: If you shutdown a node, then the node goes offline. While offline, all data collection for this node stops and no changes or upgrades can be made.

  If you want to understand Scale behavior during node reboots, see Scale behavior during node restarts.

Enable and disable node maintenance

When you put a node into maintenance mode, it marks the node to Scheduling disabled and also drains workload from it. You can place a node to maintenance from the user interface or through any operation that needs a server reboot.

To run power operations, place a node in a maintenance mode.

The following procedure guides you with the steps to enable or disable maintenance mode on a node from the user interface.

Procedure

1. In the Nodes page, click the ellipsis menu of the node you want to move to maintenance and click Enable maintenance mode.
2. In the Enable maintenance mode confirmation window, click Enable.
3. Wait for node to go to maintenance mode.
4. After all the required maintenance operations are completed, move the node out of maintenance. From the ellipsis overflow menu of a node that is in maintenance, select Disable maintainance mode option.

Maintenance node operation timeout is configurable now. The default value is 15 minutes and configurable time can go up to 40 minutes. Edit the `MaintenanceWarningTimeout` value to configure the maintenance operation.

Important:

- You cannot move more than one node to maintenance at a time. Also, if the GPFS cluster health is degraded, node maintenance will not succeed. If you ignore, then the IBM Fusion HCI user interface shows a failed message:

  ```
  "Detected problem in previous maintenance operation"
  ```

  Fix the root cause of the issue in events or CR status:

  ```
  oc describe cmt <instance node name> -n <fusion namespace>
  ```

- If you face an issue retrieving the compute nodes and network components after maintenance mode operation, log out and log in from the user interface.
- The maintenance mode on a node can take four minutes to 30 minutes to succeed, depending on the workload on the node and the Scale PodDisruptionBudget. If it takes more than 30 minutes, then the operation gets timed out eventually. For more information about this issue, see Issues related to IBM Fusion HCI node drains.
- When you put a node into maintenance mode from IBM Fusion HCI user interface, it marks a node to Scheduling disabled and also drains workload from it.

Restarting Baseboard Management Controller (BMC)

The Baseboard Management Controller (BMC) restart is a process used to perform a restart of BMC. This restart is typically performed when the BMC is not functioning correctly or when there are any active node health events. The BMC restart might remove any false health events raised.

Before you begin

The BMC restart causes the BMC to restart, which temporarily disrupts system management functions. Therefore, critical functions like monitoring and firmware upgrade for a node are paused until the BMC is back online.

About this task

Use the following instruction to restart the BMC.

Procedure

1. Log in to the IBM Fusion HCI user interface.
2. Go to Infrastructure > Nodes.
3. In the Nodes page, click the ellipsis menu of the node you want to reset and click Restart BMC.
4. In the Restart Baseboard Management Controller (BMC) confirmation window, click Restart.
5. Alternatively, you can perform the BMC restart from the node page as follows:
   a. In the Nodes page, click the node that you want to restart.
   b. From the Actions drop-down menu, click Restart BMC to restart the node.
   Note: After resolving a monitoring degraded error, it may take few minutes for the node status to update to `Normal` in the user interface.

Scaling the cluster

As workloads in the cluster increase, you must add more CPU, memory, GPU, or storage to accommodate them. CPU and memory can be increased by adding additional compute nodes to the cluster. For workloads that require GPUs, you can add specialized GPU nodes to the cluster. Storage can be scaled by adding additional disks to existing storage nodes, or adding additional storage nodes to the cluster. If you want to scale your cluster beyond a single rack, then you can add up to two expansion racks, enabling your cluster to host more compute, storage, and GPU nodes.

- **Adding nodes to the cluster**
  After installing new node hardware in an existing rack, or adding an additional rack of hardware, you complete a procedure to configure the nodes so that they are monitored and added to the OpenShift® cluster.
- **Adding storage**
  You can add additional storage nodes or add disks to existing nodes.

Adding nodes to the cluster

After installing new node hardware in an existing rack, or adding an additional rack of hardware, you complete a procedure to configure the nodes so that they are monitored and added to the OpenShift® cluster.

- **Configuring nodes for management**
  How to configure the newly installed nodes so that they are monitored and added to the OpenShift cluster. For nodes that contribute to storage, add them to the storage cluster.
- **Adding expansion racks**
  You can expand your IBM Fusion HCI with an additional rack. You only need as few as one node, along with the necessary switches, to add the rack successfully.

Configuring nodes for management

How to configure the newly installed nodes so that they are monitored and added to the OpenShift cluster. For nodes that contribute to storage, add them to the storage cluster.

Before you begin

Meet the following prerequisites for all node types in the specified sequence:

1. Ensure switches are in healthy state and connections between node and switches are correct.
2. The purchased quantity of nodes are delivered to your location with the following information:
   - For Compute-storage nodes, the received new node must have a specific number of NVMe drives. That is, the new node must have the same number of disks as the ones that are already there.
   - All new nodes must have a MAC address from IBM.

     Note: Update your Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) Configuration to include reservations for all nodes in the IBM Fusion HCI. For more information about the prerequisite, see Setting up DHCP for IBM Fusion HCI and Setting up DNS. In case of Static IP, you must update the DNS for the new nodes. For more information about the prerequisites, see Setting up Static IP and Setting up DNS. These configurations are based on the MAC addresses shared with you by IBM.
3. Ensure that the IBM Support Representative inserted the nodes in the appropriate rack slots and connected the network and power cables as per the wiring prescription of IBM Fusion HCI rack.
4. The IBM Fusion appliance is up and running.
5. Ensure that all the prechecks are successful and node is in Ready to add state in the Provision Status column.
6. Red Hat® OpenShift® is up and running and the existing nodes are healthy.
7. When you add a node to Red Hat OpenShift, the operation might get stuck for a long time in the node network configuration policy creation stage. To avoid this issue, do the steps that are mentioned in the troubleshooting section before you add the node. For the workaround steps, see Node issues.
8. Naming convention of nodes that are added: `compute-<rackid>-<rack unit number>.<domainname>`

   For a single rack, the rackid is always 1. For multi-rack, rackid is the rack number where the node is added. For an example guidance for the base system, see host name conventions.

Upsizing when backup or restore is in progress

When a backup or restore job is in progress, you must not upsize the nodes. For scale-out or scale-up operation, turn off the IBM Fusion backup or restore jobs, if any. Do not back up or restore when an upsize action is in progress.

Upsize nodes to brand-new racks

1. Upsize node must be powered on only on a running IBM Fusion HCI (stage 2 complete). Otherwise, keep the upsized node powered off (remove the power cables from the node).
2. Complete the installation.
3. Plug-in the power and switch on the node.

About this task

- The following node types are supported in this release:
  - Compute-only and compute-storage server models: 9155-C10 (32 core node), 9155-C14 (64 core node), 9155-D10 (32 core node), 9155-D14 (64 core node)
  - GPU node models: 9155-G01, 9155-G03, 9155-G04, and 9155-DG1, DG2, DG3, and DG4
  - Active File Management (AFM) server model 9155-F01
  - Service node model: 9155-B01

  For hardware configuration of compute-storage or compute-only nodes, see Hardware overview.
- In IBM Fusion HCI rack, when nodes are added they must be added into the lowest available rack position to maintain mechanical stability.
- Allowed node configuration combinations for high-availability multi-rack:
  - Select same number of nodes from all three racks. A minimum of three nodes are required for the configuration.
  - For scale out, add an equal number of storage nodes from each rack.
  - For scale up, add same number of disks across each rack.

  Note: Do not deviate from these specification as it can result in an imbalanced configuration of a Data Foundation cluster.
- For the two-availability-zone and arbiter node (2AZ + arbiter node) configuration, during capacity scaling, add disks in multiples of four and distribute them evenly across both zone.

Important: Upgrading Gen2 storage to Gen1 racks is supported for Red Hat OpenShift Data Foundation configurations, but not supported for Global Data Platform configurations.

Procedure

1. After a node is physically added to a rack, networking is done, and powered on, log in to IBM Fusion HCI user interface.
2. Go to Infrastructure > Nodes.
3. Check whether the node you added to the physical hardware is available in the Discovered nodes tab.

   If there are any errors in the configuration, then `Error connecting to node` message gets displayed in the State column. All of the error conditions have built-in retries. Whenever the event persists for a longer duration, support tickets get raised to check the appliance. If the condition is transient, it recovers automatically. Otherwise, IBM Support checks the appliance.

   Check whether the node is in Ready to add state in the Provision Status column. If the node is in Not ready to add state, then prechecks are not successful. Select the node that is in error state and click View details. It opens a Pre-check list side pane where you can see failure prechecks details.
4. To add a node to the OpenShift cluster, select the nodes and click Add to cluster.

   The user interface displays the progress of the node addition. To view the nodes progress for addition, select that specific node.
5. In the Add node window, click Add to confirm.
6. Go to the Configured nodes tab to see whether the newly added node is in the list with the State as Adding to OpenShift.

   Note: If the node is compute-only and the storage is Global Data Platform, proceed with adding it to the storage cluster. If the storage is Fusion Data Foundation, add the node to the OpenShift Container Platform cluster, then skip all subsequent steps in this procedure.
7. When all the nodes are ready for storage configuration, add them to the storage cluster: This final step allows workloads on the nodes to access storage, so complete this step before stateful workloads are scheduled on the nodes.

Option	Description

Option	Description
Global Data Platform	To configure node for Global Data Platform, do the following steps: a. Go to the Global Data Platform page. The following notification is displayed: `Nodes are ready for storage` The Storage status of the nodes in the Nodes list is Ready for storage and the Type is Storage. b. Click Configure nodes link in the notification. c. In the Configure <node type> nodes window, click Install. The Storage status of the nodes in the Nodes list changes to Installing Storage software. After the configuration is complete, a success notification is displayed and the Storage status of the node changes to Ready. d. Repeat steps for the remaining available node types one by one.
Fusion Data Foundation	a. Go to Local storage page. b. Click Add nodes. If nodes are available for addition, it is enabled. c. In the Add storage nodes page, go through the details and Add. After the selected nodes are added successfully, the status of the node changes to Ready.

8. To verify storage nodes, do the following steps:
    a. Go to the node details page.
    b. In the Inventory tab, go to Disks tab.
    c. Check the new size. Also, check whether the disk information reflects the newly added drive.
    d. Check the total disk count and Raw capacity after you add the disk.
        i. Go to Infrastructure > Dashboard.
        ii. In the Disks tile, check total disk count and Raw capacity.

Adding expansion racks

You can expand your IBM Fusion HCI with an additional rack. You only need as few as one node, along with the necessary switches, to add the rack successfully.

Before you begin

The IBM Service Support Representative must have completed the setup stage installation for the expansion rack.
Important: To add a new rack and expand your current IBM Fusion HCI infrastructure, contact IBM support.

Procedure

1. From the IBM Fusion HCI menu, click Infrastructure > Overview.
2. Click Add rack.
   The Add rack wizard page is displayed.
3. In the Getting started wizard page, go through the details and click Next to initiate the discovery process.
4. In the Discovery wizard page, wait for the rack discovery to complete.
   In this page, IBM Fusion HCI enables expansion of rack, connects to the provisioned node, and discovers rack data.
   Errors might occur during the discovery step because of the following reasons:
    - Unable to establish connection to the provisioned node on the new rack. To resolve this issue, revisit your DNS configuration and retry.

   There are Retry and Cancel options on this page. If the discovery fails, it can be retried by clicking retry and if you wish to cancel, click Cancel.
5. Choose a zone from the Availability zone drop-down list.
6. Click Next.
7. In the Validation wizard page, check whether all the nodes are validated properly.
   The IBM Fusion HCI validates the configuration of each node as a preparatory step so as to join the cluster. You need as few as one node, along with the necessary switches to add the rack successfully.
   This page continuously shows the number of nodes that are successfully validated. If the validation does not complete within the stipulated time, the validation operation fails and message corresponding to that gets displayed.

   If issues exist, then check your connections or fix problematic nodes and retry. You cannot proceed with errors, so click Save & Quit. Click Collect logs to download and view logs.
   In this page, if rack addition fails with the following error message, then contact IBM Support:

   ```
   Rack import timedout
   ```

   The Retry on this page refreshes the node validation status.
8. If the session is lost during the add rack operation, do the following steps:
    a. Log in to IBM Fusion HCI user interface.
    b. Go to Infrastructure > Nodes.
       A `Scale out operation not completed` notification is displayed on the page.
    c. Click Resume process link in the `Scale out operation not completed` notification.
       Use Resume process to continue the add rack operation from where you left off.

       For any errors in rack addition due to `timedout` issues, go to the Infrastructure > Nodes > Discovered nodes tab page to know more details.
9. Click Next.
   The optional Node selection wizard page is displayed.

10. Optionally, in the Node selection wizard page, you can open the Edit node selection and edit node assignments.
    By default, IBM Fusion HCI automatically configures all nodes in the rack. The Edit node selection shows a table where you can deselect nodes. If you do not deselect, then the nodes get added automatically to the OpenShift® cluster. Your deselection depends on the amount of nodes required for your storage building block. For example, you have twenty nodes and six nodes are required for storage building block. The table displays only the remaining fourteen nodes. Also, a summary section is displayed with the following details:
    - Nodes
    - CPUs
    - Memory (GB)
    - Usable capacity (TiB)

    The values of the summary section changes dynamically based on your selection and deselection.
11. Click Next.
    The Finish wizard page is displayed.
12. In Finish wizard page, click Finish to close the Add rack wizard.
    The IBM Fusion HCI adds these nodes to the OpenShift cluster. After the add rack operation is complete, the user interface shows auxiliary rack related nodes in the Discovered node tab. To add more racks, click Actions > Add rack again and follow the steps mentioned in this procedure.
13. Select nodes and click Add to cluster.
    The IBM Fusion HCI starts to add nodes to the OpenShift cluster.

- **Viewing rack details**
  You can view the details of all the racks in your high availability cluster.

Viewing rack details

You can view the details of all the racks in your high availability cluster.

Procedure

1. From the IBM Fusion HCI menu, click Infrastructure > Nodes.
2. In the Configured nodes tab, go to Actions and click Rack details.
   The Rack details window is displayed with the following information:
   - Link name
   - Baremetal VLAN ID
   - Baremetal VLAN
   - Storage VLAN ID
   - Port type
   - Native VLAN ID
   - Transceiver
3. Expand the individual racks to view network details added during the setup installation stage.

Adding storage

You can add additional storage nodes or add disks to existing nodes.

- **Adding additional storage nodes**
  As the disk slots fill up on existing storage nodes in the cluster, you can add additional storage nodes to increase the amount of available storage. All storage nodes must have an equal number of disks, so new storage nodes must match the disk configuration of the storage nodes that are already in the rack.
- **Adding new disks to existing Global Data Platform nodes**
  If storage gets added at the hardware level, then the new storage information is available on the IBM Fusion HCI user interface. You can add storage to the cluster by adding additional disks to the storage nodes in the cluster. Each storage node must have the same number of disks, so make sure to add an equal number of disks to each node in the cluster.
- **Adding new disks to existing Fusion Data Foundation nodes**
  After you add additional disks to Fusion Data Foundation storage nodes, it auto detects the new disks and auto expands the capacity to Fusion Data Foundation storage cluster. Each storage node must have the same number of disks, so make sure to add an equal number of disks to each node in the cluster.

Adding additional storage nodes

As the disk slots fill up on existing storage nodes in the cluster, you can add additional storage nodes to increase the amount of available storage. All storage nodes must have an equal number of disks, so new storage nodes must match the disk configuration of the storage nodes that are already in the rack.

About this task

After the storage node hardware is added to the rack, configure them for management. For the actual procedure, see Configuring nodes for management.

Adding new disks to existing Global Data Platform nodes

If storage gets added at the hardware level, then the new storage information is available on the IBM Fusion HCI user interface. You can add storage to the cluster by adding additional disks to the storage nodes in the cluster. Each storage node must have the same number of disks, so make sure to add an equal number of disks to each node in the cluster.

Before you begin

Ensure that the following conditions are met:

- No node is in maintenance mode.
- Ensure that operations leading to node reboot are not performed.
- Do not initiate an upgrade of OpenShift®, IBM Fusion HCI, or node firmware until the new storage addition is complete.
- Number of disks is the same for all servers in the storage cluster.
  If you want disk upsize, then add same number of disks to all storage nodes available in the cluster. For example, a cluster has a base minimum of six storage nodes, then at least 6*2=12 disks with two disks each must be added. For nodes, if the cluster has six storage nodes with two disks each, you can add one more node with two disks. Now, your storage cluster has seven storage nodes with two disks each.

- Storage nodes must have an even number of disks.
- During installation or upgrade, the pod may get stuck in **`container creating`** state and the crash loop might fail with attach volume. For such error scenarios, do the following workaround steps:
  1. Unschedule the node from which you were trying to attach volume.
  2. Delete the pod that got stuck in **`container creating`** state.
  3. Run upsize again.
- For hardware configuration of compute-storage or compute-only nodes, see Hardware overview.

About this task

Note: If you are on a Metro-DR setup, then ensure that both sites have the same storage capacity. If you upsize storage disks on one site, ensure that you upsize the storage capacity on the second site at the earliest. If you do operations on the local and remote cluster during this mismatch, IBM is not responsible for any data loss or inconsistencies. If you face any disk failures, ensure that you rectify them and bring both site 1 and site 2 at the same level at the earliest.
If you face any issues during disk upsize, see Global Data Platform service issues.

Procedure

1. In the IBM Fusion menu, go to Global Data Platform depending on your storage type.
2. Click Actions > Add disk.
   A window opens with details of the newly added storage at the hardware level.
   If you are not able to see additional disks even after 1 hour, contact IBM Support.

3. Go through the storage details and click Add.
   Note: It is a best practice to add a bunch of storage nodes at a time and not one-by-one.
   The following message is displayed:

   ```
   Request accepted successfully
   ```

   All new disks on all nodes get added automatically and disk selection is not required.

4. To verify, go to the node details page and check the Disks tab. For more information about the procedure, see Node or service node details.

Adding new disks to existing Fusion Data Foundation nodes

After you add additional disks to Fusion Data Foundation storage nodes, it auto detects the new disks and auto expands the capacity to Fusion Data Foundation storage cluster. Each storage node must have the same number of disks, so make sure to add an equal number of disks to each node in the cluster.

Before you begin

Ensure that the following conditions are met:

- No node is in maintenance mode.
- Ensure that operations leading to node reboot are not performed.
- Do not initiate an upgrade of OpenShift®, IBM Fusion HCI, or node firmware until the new storage addition is complete.
- Number of disks is the same for all servers in the storage cluster.
  If you want disk upsize, then add same number of disks to all storage nodes available in the cluster. For example, a cluster has a base minimum of six storage nodes, then at least 6*2=12 disks with two disks each must be added. For nodes, if the cluster has six storage nodes with two disks each, you can add one more node with two disks. Now, your storage cluster has seven storage nodes with two disks each.

- Storage nodes must have an even number of disks.
- During installation or upgrade, the pod may get stuck in **`container creating`** state and the crash loop might fail with attach volume. For such error scenarios, do the following workaround steps:
  1. Unschedule the node from which you were trying to attach volume.
  2. Delete the pod that got stuck in **`container creating`** state.
  3. Run upsize again.
- For hardware configuration of compute-storage or compute-only nodes, see Hardware overview.

About this task

If you face any issues during disk upsize, see Global Data Platform service issues.

Procedure

1. In the IBM Fusion menu, go to Data Foundation.
   Wait for the new capacity to be shown in the Data Foundation page.
2. To verify, go to the node details page and check the Disks tab. For more information about the procedure, see Node or service node details.

Node firmware

IBM Fusion HCI continuously monitors the firmware installed on each node. If a node is found to be running a firmware version lower than the Fusion-certified level, the system automatically initiates a firmware upgrade. If the upgraded firmware requires a reboot to take effect, the node is marked as Upgrade available.

Before you begin

Ensure that you mirror Red Hat® operator images (`node-maintenance-operator` and `self-node-remediation`) to your enterprise registry when working in an air-gapped environment. For more information, see Mirroring Red Hat operator images to enterprise registry.

About this task

Note:
This automated firmware management applies to the following node types:

- Nodes in the base OpenShift® Container Platform cluster
- Nodes in Hosted Control Plane clusters
- Service nodes
- Nodes not part of any cluster

For service nodes and unclustered nodes, IBM Fusion HCI automatically handles the reboot process.
You can verify the status of the firmware in the Infrastructure > Nodes page. In the Firmware level column, you can find the firmware status that is associated with the release. In the following table, you can see some of the possible firmware statuses:

Status	Description
Up to date	It indicates that the node firmware matches the Fusion-certified level.
Upgrade available	A new firmware update is available for the node, user needs to trigger upgrade from the Upgrades page on the IBM Fusion HCI user interface.
Upgrade pending	The node is scheduled for reboot.
Upgrade failed	Firmware upgrade failed; contact IBM support for assistance. For cause, resolution, and troubleshooting for Update failed, see Troubleshooting update failures section.
Upgrade stalled	Firmware upgrade failed during the maintenance mode or reboot stage. For cause, resolution, and troubleshooting for Update stalled, see Remediating a maintenance drain failure.

Do not start a Hosted Control Plane node firmware upgrade in the following conditions:

- If a node is in maintenance mode.
- If a node is pending in maintenance mode operation.
- If a node health is critical.

Procedure

1. Go to Infrastructure > Nodes in the IBM Fusion HCI user interface.
2. Identify nodes marked as Upgrade available.
3. Do the following steps to reboot the nodes:
   a. Go to Settings > Upgrades in the IBM Fusion HCI user interface.
   b. Click the Node firmware under the Rolling upgrades section.
      A side panel opens with the nodes details.

   c. Select the cluster that you want to reboot and click Upgrade.
      The upgrade progress in detail can be monitored from the side panel.

   IBM Fusion HCI automatically checks the environment before node reboot. If it is safe to proceed, the selected nodes get restarted in a rolling fashion to minimize impact to the OpenShift cluster.
   Note:
   - Do not trigger the Add Node to Cluster action for nodes that are listed under the Discovered tab if their firmware upgrade is still in progress.
   - IBM Fusion HCI performs pre-reboot checks to ensure it is safe to proceed. If conditions are met, nodes are rebooted in a rolling fashion to minimize impact on cluster operations.
   - If a node is in an upgrade-failed state, first resolve the underlying issues. Then, delete the `cfw` Custom Resource (CR) to restart the upgrade process. After you delete the CR, the system automatically retries the operation.
   - During the node firmware upgrade, the node health status can change from Normal to Monitoring Degraded or Critical because system components are involved. This status usually resolves automatically after the upgrade completes. If the status remains Monitoring Degraded or Critical after the upgrade, contact IBM Support.

   Troubleshooting update failures:
   Firmware Update failed status can occur because of the following reasons:

   Silent firmware flashing failed OR new firmware not reflected after reboot
      This may be due to hardware errors. As a resolution, follow the steps to resolve the issue:
      a. If Call Home is configured, a support case event **BMYCO0009** is automatically opened; no further action is required.
      b. If Call Home is not configured, then contact IBM support for assistance.

Node failed to move to maintenance mode
: When a user triggers a node or cluster reboot, the upgrade may fail if the node cannot drain pods before reboot. As a resolution, drain the pods manually and then resume the upgrade. For more information about draining the pods, see Remediating a maintenance drain failure.

Upgrade gets stalled due to node being stuck in **`NotReady`** state
: This may be due to multiple errors such as issues in node restart, certificate errors, and OpenShift Container Platform node status shows **`NotReady`**. As a resolution, follow the steps to resolve the issue:
    a. For node restart issues, see Issues in node restart.
    b. For certificate errors, see Node in NotReady state due to certificate errors in OpenShift pods.
    c. For node status shows **`NotReady`** in the OpenShift Container Platform, see OpenShift Container Platform node status shows NotReady but IBM Fusion HCI user interface node status shows as Running.

Upgrading the NVIDIA GPU driver with the NVIDIA GPU operator

After installing IBM Fusion HCI and adding a GPU node to the OpenShift® cluster, the Node Feature Discovery (NFD) component and the NVIDIA GPU operator are automatically installed. The NFD component labels the hardware capabilities of the GPU node, and the NVIDIA GPU operator automates the deployment and management of NVIDIA drivers and related components required for GPU accelerated workloads.

During Stage 2 deployment, if the system detects an NVIDIA GPU node, the Fusion operator automatically sets the **`manageNvidiaGPUOperator`** field to **`true`** in the **`ComputeInit`** custom resource and installs the latest supported version of the NVIDIA GPU operator.

The **`manageNvidiaGPUOperator`** setting controls the automatic installation and upgrade of the NVIDIA GPU Operator.

Value	Description
True (default)	Automatically installs and upgrades the NVIDIA GPU operator when GPU nodes are detected
False	NVIDIA GPU operator installation and upgrades are managed manually.

Manual NVIDIA GPU operator management
: To manually manage the NVIDIA GPU operator lifecycle, update the **`manageNvidiaGPUOperator`** field in the **`ComputeInit`** custom resource before you perform NVIDIA GPU node upsize operations or use a NVIDIA GPU operator version other than the latest supported stable version that is installed by IBM Fusion HCI.

1. Run the following command to update the **`manageNvidiaGPUOperator`** field to false:

```
oc patch computeinit computeinit \
  -n ibm-spectrum-fusion-ns \
  --type=merge \
  -p '{"spec":{"manageNvidiaGPUOperator":false}}'
```

2. Run the following command to verify that the update is applied successfully:

```
oc get computeinit computeinit \
  -n ibm-spectrum-fusion-ns \
  -o jsonpath='{.spec.manageNvidiaGPUOperator}'
```

When the **`manageNvidiaGPUOperator`** attribute is set to **`false`**, IBM Fusion HCI does not automatically install, upgrade, or reinstall the NVIDIA GPU operator. You must manually manage the NVIDIA GPU operator lifecycle, including maintaining a version other than the latest supported stable version that is installed by IBM Fusion HCI.

To restore automatic NVIDIA GPU operator lifecycle management, update the field to true.

Upgrading the NVIDIA GPU operator
: In IBM Fusion HCI 2.13 and later releases, the NVIDIA GPU operator is automatically upgraded to the latest supported stable version.

Automatic upgrade behavior

- The latest supported NVIDIA GPU operator version is installed by default.
- Upgrading to IBM Fusion HCI 2.13 or later automatically upgrades the NVIDIA GPU operator.

Manual upgrade procedure
: Manual upgrade procedures are required only when the **`manageGPUOperator`** parameter is set to **`false`**.

NVIDIA DCGM exporter dashboard
: The NVIDIA DCGM exporter dashboard displays GPU related metrics and monitoring graphs. The configuration of the NVIDIA DCGM exporter dashboard is automated.

Viewing GPU metrics
: To monitor GPU performance, you can access GPU metrics through the OpenShift Container Platform web console. Ensure that you must have the OpenShift Container Platform web console access and NVIDIA DCGM exporter dashboard available in the console.

Administrator perspective:
: In the OpenShift Container Platform web console from the side menu, switch to the Administrator perspective, then navigate to Observe > Dashboards and select NVIDIA DCGM Exporter Dashboard from the Dashboard list.

Developer perspective:
: If the dashboard is added to the developer perspective, in the OpenShift Container Platform web console from the side menu, switch to the Developer perspective, navigate to Observe > Dashboards and select NVIDIA DCGM Exporter Dashboard from the Dashboard list.

For more information, see NVIDIA documentation.

Customizing Node Feature Discovery (NFD) CR

Use the instructions to customize the Node Feature Discovery (NFD) configuration to meet your specific requirements.

The NFD operator and node feature discovery CR are automatically created by IBM Fusion if one or more GPU nodes exist in the cluster. The NFD CR provides a way to configure the NFD daemon set to orchestrate all resources needed to run the NFD daemon set.

The NFD CR configuration is defined in a YAML file that specifies the parameters for the NFD daemon set. The configuration includes settings for the operand, topology updater, and worker configuration.

```
apiVersion: nfd.openshift.io/v1
kind: NodeFeatureDiscovery
metadata:
  name: nfd-instance
  namespace: openshift-nfd
spec:
  instance: ''
  operand:
    imagePullPolicy: Always
    image: 'registry.redhat.io/openshift4/ose-node-feature-discovery-rhel9:v4.18'
  topologyUpdater: false
  workerConfig:
    configData: |
      core:
      #  labelWhiteList:
      #  noPublish: false
        sleepInterval: 60s
      #  sources: [all]
      #  klog:
      #    addDirHeader: false
      #    alsologtostderr: false
      #    logBacktraceAt:
      #    logtostderr: true
      #    skipHeaders: false
      #    stderrthreshold: 2
      #    v: 0
      #    vmodule:
      ##   NOTE: the following options are not dynamically run-time 
      ##          configurable and require a nfd-worker restart to take effect
      ##          after being changed
      #    logDir:
      #    logFile:
      #    logFileMaxSize: 1800
      #    skipLogHeaders: false
      sources:
      #  cpu:
      #    cpuid:
      ##     NOTE: whitelist has priority over blacklist
      #      attributeBlacklist:
      #        - "BMI1"
      #        - "BMI2"
      #        - "CLMUL"
      #        - "CMOV"
      #        - "CX16"
      #        - "ERMS"
      #        - "F16C"
      #        - "HTT"
      #        - "LZCNT"
      #        - "MMX"
      #        - "MMXEXT"
      #        - "NX"
      #        - "POPCNT"
      #        - "RDRAND"
      #        - "RDSEED"
      #        - "RDTSCP"
      #        - "SGX"
      #        - "SSE"
      #        - "SSE2"
      #        - "SSE3"
      #        - "SSE4.1"
      #        - "SSE4.2"
      #        - "SSSE3"
      #      attributeWhitelist:
      #  kernel:
      #    kconfigFile: "/path/to/kconfig"
      #    configOpts:
      #      - "NO_HZ"
      #      - "X86"
      #      - "DMI"
        pci:
          deviceClassWhitelist:
            - "0200"
            - "03"
            - "12"
          deviceLabelFields:
      #      - "class"
            - "vendor"
      #      - "device"
      #      - "subsystem_vendor"
      #      - "subsystem_device"
      #  usb:
      #    deviceClassWhitelist:
      #      - "0e"
      #      - "ef"
      #      - "fe"
      #      - "ff"
      #    deviceLabelFields:
```

```
#       - "class"
#       - "vendor"
#       - "device"
#   custom:
#     - name: "my.kernel.feature"
#       matchOn:
#         - loadedKMod: ["example_kmod1", "example_kmod2"]
#     - name: "my.pci.feature"
#       matchOn:
#         - pciId:
#             class: ["0200"]
#             vendor: ["15b3"]
#             device: ["1014", "1017"]
#         - pciId :
#             vendor: ["8086"]
#             device: ["1000", "1100"]
#     - name: "my.usb.feature"
#       matchOn:
#         - usbId:
#           class: ["ff"]
#           vendor: ["03e7"]
#           device: ["2485"]
#         - usbId:
#           class: ["fe"]
#           vendor: ["1a6e"]
#           device: ["089a"]
#     - name: "my.combined.feature"
#       matchOn:
#         - pciId:
#             vendor: ["15b3"]
#             device: ["1014", "1017"]
#           loadedKMod : ["vendor_kmod1", "vendor_kmod2"]
```

If you want to customize the existing NFD CR based on your requirements, see Configuring the Node Feature Discovery Operator in *Red Hat OpenShift Container Platform* documentation.

Converting compute node to AFM node

Convert a Lenovo compute node to a AFM node through backend CR.

Before you begin

- Compute only node must be a part of the storage cluster. To upsize the node to the storage cluster, see Configuring nodes for management.
- The Global Data Platform service must be healthy before you begin the node conversion process.

Procedure

1. Run the following command to provide a name of the node in the **spec** section > **nodeToBeDesignatedAsAfm** field of the **storagemanager** custom resource.

   ```
   oc patch Scale storagemanager -n ibm-spectrum-fusion-ns --type='json' -p='[{"op": "add", "path": "/spec/nodeToBeDesignatedAsAfm", "value": "<compute-only_node_name>"}]'
   ```

2. Run the following command to verify whether the **storagemanager** custom resource field is updated with the provided node name:

   ```
   oc get Scale storagemanager -n ibm-spectrum-fusion-ns -oyaml | grep nodeToBeDesignatedAsAfm
   ```

   The progress can be tracked through the status field **userDesignatedAfmNodeStatus** on the **storagemanager** custom resource.

   ```
   oc get Scale storagemanager -n ibm-spectrum-fusion-ns -oyaml | grep userDesignatedAfmNodeStatus
   ```

   Example output:

   ```
   userDesignatedAfmNodeStatus: IN-PROGRESS
   ```

   After you successfully convert the node from Compute to AFM, the status of the **userDesignatedAfmNodeStatus** in the **storagemanager** custom resource displays as **COMPLETED**.

   ```
   oc get Scale storagemanager -n ibm-spectrum-fusion-ns -oyaml | grep userDesignatedAfmNodeStatus
   ```

   Example output:

   ```
   userDesignatedAfmNodeStatus: COMPLETED
   ```

   After you complete the node conversion process, the node name gets added to the **designatedAfmNodes** field in the status section.

3. Run the following command to obtain a list of all the converted nodes:

   ```
   oc get Scale storagemanager -n ibm-spectrum-fusion-ns -o=jsonpath='{.status.designatedAfmNodes[*]}'
   ```

4. Run the following command to verify the role of the node:

   ```
   oc get node <node_name> -oyaml | grep 'scale.spectrum.ibm.com/role'
   ```

   Example command:

   ```
   oc get node compute-1-ru10.rackae1.mydomain.com -oyaml | grep 'scale.spectrum.ibm.com/role'
   ```

   Example output:

```
scale.spectrum.ibm.com/role: afm
```

5. Run the following command to verify whether the required taints are applied on the node.

```
oc get node <node_name> -o=jsonpath='{.spec.taints}'
```

Example:

```
oc get node compute-1-ru10.rackae1.mydomain.com -o=jsonpath='{.spec.taints}'
```

6. Go to the Global Data Platform page on the IBM Fusion HCI user interface and ensure that the node type displays AFM correctly.

Prerequisite for removing worker node from the IBM Fusion HCI cluster

The following section provides the prerequisite to remove worker node from the IBM Fusion HCI cluster.

Procedure

1. Start a debug session on the node by running the following command:

```
oc debug node/<NODE_NAME>
```

2. When the debug pod starts, switch to the host environment:

```
chroot /host
```

3. Set the fallback option:

```
mokutil --set-fallback-noreboot true
```

Example

```
❯ oc debug node/compute-1-ru5.isf-rackae4.mydomain.ibm.com
Starting pod/compute-1-ru5isf-rackae4mydomainibmcom-debug-c4jxg ...
To use host binaries, run `chroot /host`
Pod IP: 172.25.100.28
If you don't see a command prompt, try pressing enter.

sh-5.1# chroot /host

sh-5.1# mokutil --set-fallback-noreboot true
```

Configuring GPU upsize on Dell systems

Configure GPU upsize for Dell systems in IBM Fusion HCI.

Before you begin

Note: If a Dell GPU upsize operation is in progress, do not start an upsize operation on any other node. IBM Fusion HCI supports only one Dell GPU node upsize operation at a time, and parallel upsize operations are not supported.
Contact IBM Support to determine the following information:

- The switches that are connected to the GPU.
- The ports that are assigned for GPU connectivity.

Procedure

1. Identify the available ports on the switch.
   a. In the OpenShift® console, go to Administration > CustomResourceDefinition > Switch.
   b. Select the high-speed switch for the corresponding rack.
   c. Verify port configuration.
      For example:

```
swp1 - configured: yes
swp2 - configured: yes
swp3 - configured: yes
swp4 - configured: yes
swp5 - configured: yes
swp6 - configured: no
```

   d. Identify the last configured port and the next two consecutive available ports.
      For example:
      - Last configured port: `swp5`
      - Next two consecutive available ports: `swp6` and `swp7`
2. Update the Link CR with the available ports.
   a. In the OpenShift console, go to Administration > CustomResourceDefinition > Links.
   b. Select the Link instance for the rack.
   c. Update `portList` by adding the available ports in the `spec` section:

```
spec:
  availableLinkTypes:
  - availableTransceivers:
    - Single QSFP
    - 2x Splitter QSFP
    - 4x Splitter QSFP
    linkType: customer_links
  switches:
  - portList:
    - '32'
    - '31'
    - '6'
    - '7'
    switchList:
    - hspeed1-f791047
    - hspeed2-f791047
```

3. If ports are already allocated, run the following command to allow deletion of existing `bond0_links`:

```
oc patch link <link_instance_name> \
  -n ibm-spectrum-fusion-ns \
  --type=merge \
  --subresource=status \
  -p '{"status":{"allowBond0LinkDeletion":true}}'
```

   Tip: Ports 6 and 7 might already be associated with `bond0_links`. This step enables their removal.
4. Remove the existing bond links from the Link CR `spec`.
   Important: Copy configuration details from `b_6` before removing it, as this information is required in the next step.
   a. In the Link CR `spec`, remove the following entries:
      - `b_6`
      - `b_7`
   b. Wait a few minutes and verify that the entries are removed from both `spec` and `status`.

   Note: If you encounter the following webhook error, complete the steps to resolve it before you continue:

```
admission webhook "vlinkuser.network.isf.ibm.com" denied the request:
Another operation is in progress on Link resource.
```

   a. Go to Workloads > ConfigMaps > isf-network-coordination.
   b. Check the YAML `data` section:

```
data:
  entryTypes: 'bond0_links'
  operationInProgress: 'True'
  operationType: 'add'
  resource: 'link'
```

   c. Update `operationInProgress` to `'False'`.
5. Add a new GPU link in the Link CR `spec` with the following parameters:
   - `name`: `gpu_bond<starting_port>`
   - `ports`: Two consecutive ports (for example, `6,7`)
   - `uuid`: Same as the name
   - `vlanNames` / `vlanIds`: Copy from removed `b_6`
   - `linkType`: `gpu_dell_2x2_links`

   Example configuration:

```
spec:
  aggregation: true
  switches:
    - hspeed2-f791047
    - hspeed1-f791047
  transceiver: 2x Splitter QSFP
  vlanNames: 'vlan1512,Default Native,testvlan2004'
  name: gpu_bond6
  ports: '6,7'
  linkType: gpu_dell_2x2_links
  vlanIds: '1512,1,2004'
  nativeVlan: Default Native
  uuid: gpu_bond6
  vlanType: trunk
```

   After a few minutes, the status updates. Expected status example:

```
status:
  gpu_bond6:
    aggregation: true
    switches:
      - hspeed2-f791047
      - hspeed1-f791047
    transceiver: 2x Splitter QSFP
    vlanNames: 'vlan1512,Default Native,testvlan2004'
    name: gpu_bond6
    linkCreated: true
    ports: '6,7'
    linkType: gpu_dell_2x2_links
    vlanIds: '1512,1,2004'
    linkState: true
    nativeVlan: Default Native
    uuid: gpu_bond6
    vlanType: trunk
```

   Ensure that `linkCreated: true` and `linkState: true` are present.

6. Remove ports 6 and 7 from the switch `portList` in the Link CR:

```
switches:
  - portList:
    - '32'
    - '31'
```

Wait until the ports are unconfigured on the switch. Verify that `uplinkPortConfigStatus` shows the following state:

```
uplinkPortConfigStatus:
  '31': ready
  '32': ready
```

7. Update the compute network handshake.
   a. In the OpenShift console, go to Workloads > ConfigMaps > computenetworkhandshake.
   b. Select the instance for the node. For example: `computenetworkhandshake-1-ru6`.
   c. Run the following command to patch the instance:

   ```
   oc patch computenetworkhandshake computenetworkhandshake-1-ru6 \
     -n ibm-spectrum-fusion-ns \
     --subresource=status \
     --type=merge \
     -p '{"status":{"message":"SwitchConfigurationSucceeded","switchConfigured":true}}'
   ```

8. Run the following command to set `allowBond0LinkDeletion` back to `false`:

```
oc patch link <link_instance_name> \
  -n ibm-spectrum-fusion-ns \
  --type=merge \
  --subresource=status \
  -p '{"status":{"allowBond0LinkDeletion":false}}'
```

Node issues

This topic provides a comprehensive list of common troubleshooting steps and known issues that may arise while working with compute nodes.

- **Lenovo node issues**
  This topic provides a comprehensive list of common troubleshooting steps and known issues that may arise while working with the nodes.
- **Dell node issues**
  This topic provides a comprehensive list of common troubleshooting steps and known issues that may arise while working with the Dell nodes.
- **Cleaning up localhost node from the IBM Fusion HCI cluster**
  Follow the instructions to clean up the localhost node from the IBM Fusion HCI cluster.

Lenovo node issues

This topic provides a comprehensive list of common troubleshooting steps and known issues that may arise while working with the nodes.

Known issues
: The following known issues exist while you work with the compute nodes. The workarounds are included wherever possible. If you come across an issue that cannot be solved by using these instructions, contact IBM Support.

  - Incorrect port speed display for up Ports on configured and discovered nodes
  - Management switch high availability is lost

Troubleshooting issues
: The following troubleshooting issues exist while you work with the compute nodes with workarounds included wherever possible. If you come across an issue that cannot be solved by using these instructions, contact IBM support.

  - Unable to add node to OpenShift Container Platform
  - Gen2 node upsize failed on Gen 1 rack
  - Blue screen appears in G03 and G04 GPU nodes
  - mmaddnode command fails during node upsize on storage cluster
  - NVMe drives missing on XCC but seen on CoreOS
  - Issues in node restart
  - GPU node capacity value shows 0
  - Compute nodes in a hung state
  - Pod migration gets stuck in the Terminating or ContainerCreating state
  - Compute node can abruptly become unreachable
  - Add rack workflow from base rack fails in node validation
  - OpenShift Container Platform node status shows NotReady but IBM Fusion HCI user interface node status shows as Running
  - Node goes missing from the IBM Fusion HCI rack graphical view
  - Node in NotReady state due to certificate errors in OpenShift pods

Unable to add node to OpenShift Container Platform

Problem statement
: The node cannot be added to the OpenShift® Container Platform cluster.

Cause
: The issue might be caused by a hung power operation (powerop) for the node. This can be identified by checking the logs on the system.

Symptoms

- The power operation is not progressing.
- The live powerop CR logs for the node are blank or do not show any activity.
- The powerop CR is not getting updated.

Resolution

To resolve this issue, follow these steps:

1. Check the live logs on the system for the powerop operation.
2. If the logs are blank or not updating, delete the powerop CR by running the following commands in the `ibm-spectrum-fusion-ns` namespace:

   ```
   oc get cpr
   ```

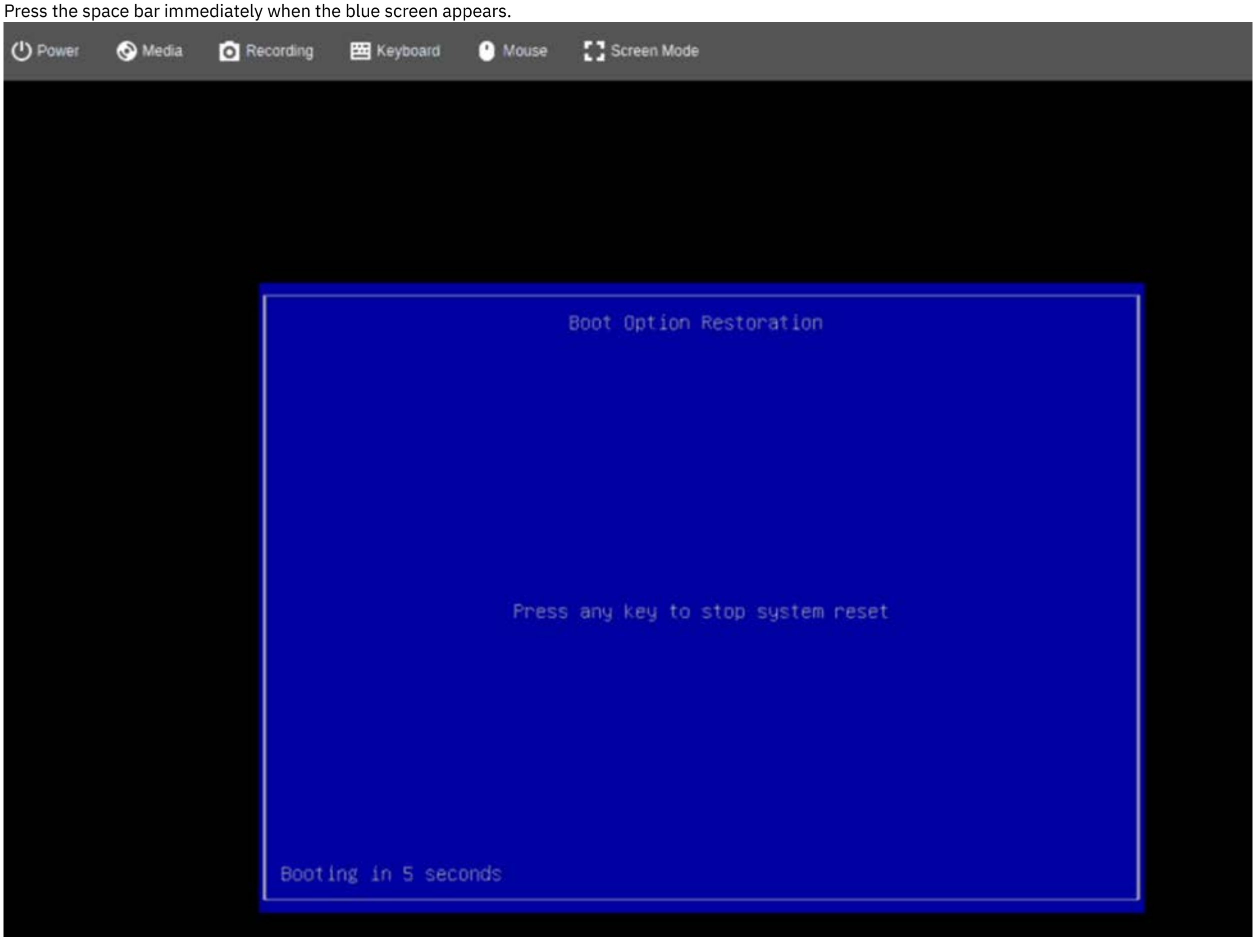
   ```
   oc delete cpr <node-powerop-cr-name>
   ```

   Important: Wait for the command to delete the powerop CR. Avoid forceful deletion as it may cause further issues.
3. After deleting the powerop CR, verify if the node can be added to the OpenShift Container Platform cluster.
4. If the issue persists, contact IBM support .

Gen2 node upsize failed on Gen 1 rack

Resolution

1. Log in to the BMC console of the node.
2. Press the space bar immediately when the blue screen appears.



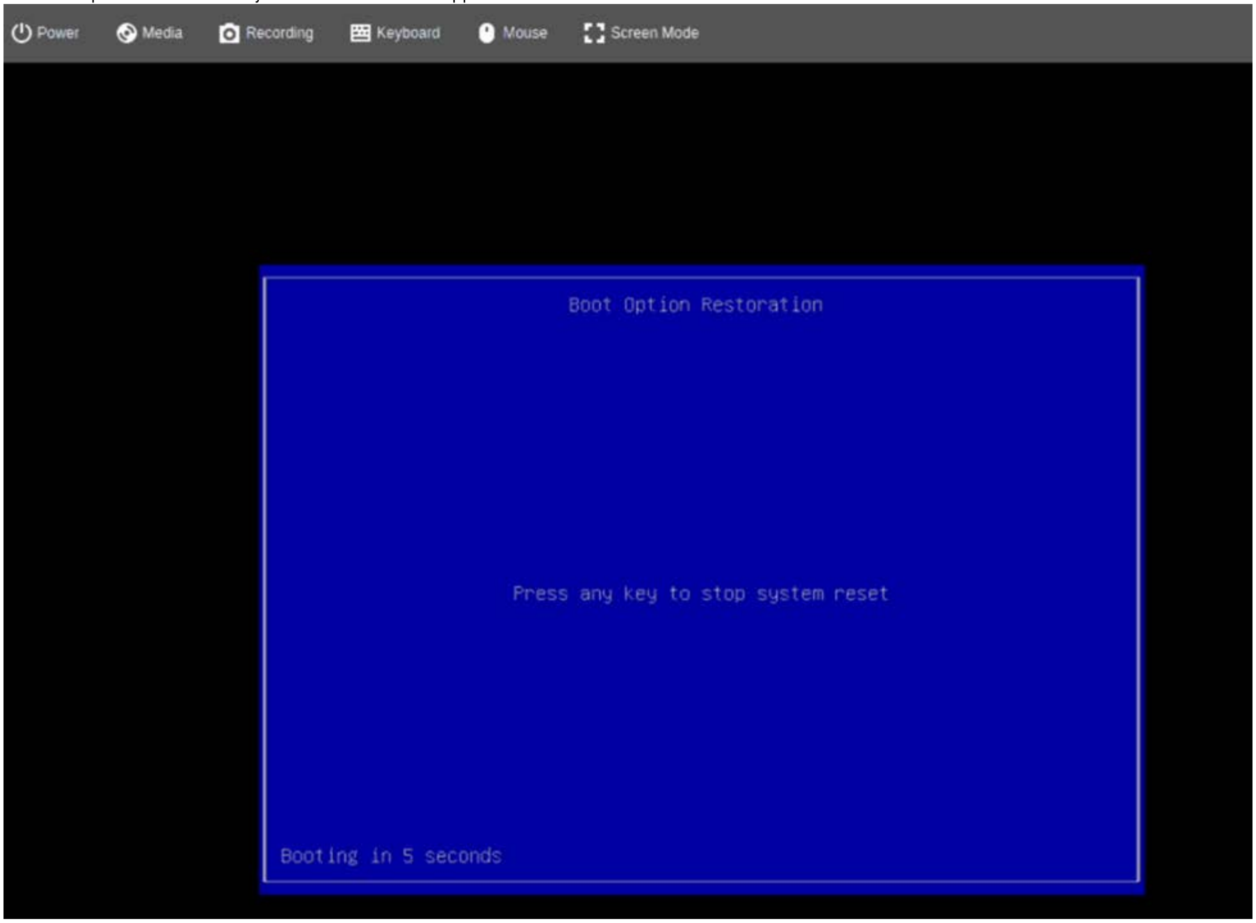
3. Use the arrow keys to select Always continue boot.
4. Press Enter to confirm the selection.
   The node reboots and resolves the issue.

Blue screen appears in G03 and G04 GPU nodes

Resolution

1. Log in to the BMC console of the node.

2. Press the space bar immediately when the blue screen appears.



3. Use the arrow keys to select Always continue boot.
4. Press Enter to confirm the selection.
   The node reboots and resolves the issue.

mmaddnode command fails during node upsize on storage cluster

Resolution
: After the initial failure of the **mmaddnode** command during node addition, run the following command to remove the files that do not get cleaned up properly:

```
oc exec -it compute-1-ru13 -c config -- /bin/sh
sh-5.1# rm /var/mmfs/gen/mmsdrfs
```

NVMe drives missing on XCC but seen on CoreOS

Resolution

1. Run **resetsp** to restart the BMC.
2. If the step 1 does not resolve the issue, run **OS reboot** command to reboot the node.
3. If both step 1 and step 2 do not resolve the issue, then contact IBM support to replace the faulty drive or faulty NVMe backplane.

Issues in node restart

Problem statement
: Sometimes, after you restart a node, it may not change its state to ready and the `ovs-configuration.service` may fail.

Resolution
: As a resolution, restart the node. For more information about Scale behavior during node restarts, see Scale behavior during node restarts.

GPU operator error

Certain Linux kernels that have KASLR enabled might experience a known issue with `High Memory Mode` initialization. This issue can cause CUDA initialization to fail. As a workaround, disable the `High Memory Mode`. For more information, see Disable High Memory Mode.

Management switch high availability is lost

When the management switch high availability is lost due to the management switch at RU18 being down, the cluster continues to function, but autodiscovery of nodes is not supported.

Incorrect port speed display for up Ports on configured and discovered nodes

Sometimes, the port speed for all up ports on configured and discovered nodes may be displayed as 0 Gbps instead of the actual speed.

GPU node capacity value shows 0

Problem statement
: It is observed that sometimes users might see GPU node capacity as 0 and also that the NFD discovery instance has the `nfd-worker-xxx` in a constant crashloopback.

Resolution
: This issue was observed in OpenShift Container Platform release 4.10.21. Upgrade the OpenShift Container Platform to 4.10.61, reinstall the NFD operator with the NFD Discovery CR, and install the NVIDIA operator and create the cluster policy that detects the GPUs.

Compute nodes in a hung state

Resolution
: Steps to restart when the compute nodes get into a hung state:
: As a prerequisite, you must have administrator access to Red Hat® OpenShift and OC (Red Hat OpenShift CLI) command access to the cluster.

1. Log in to OpenShift UI.
2. Go to Compute > Bare Metal Hosts.
3. Ensure to select `openshift-machine-api` project on the Bare Metal Hosts page.
4. From the ellipsis menu, click Power Off for the node.
5. Power it on again and check whether the issue is resolved.

Pod migration gets stuck in the Terminating or ContainerCreating state

Problem statement
: A controller node goes down and the migration of the pod gets stuck in Terminating or ContainerCreating state.

Resolution
: As a workaround, delete the container creating pod so that it can get scheduled in another available controller node.

Compute node can abruptly become unreachable

Problem statement
: A compute node can abruptly become unreachable either due to a power outage or network disruption.

Cause
: The pods that run on that specific compute node might have not got evacuated cleanly and they remain stuck in `Terminating` or `Unknown` state.

Resolution
: To ensure a complete evacuation and migration of the pod that is stuck on the failed node, force delete the pod manually by using the following command:

```
oc delete pod <pod name> --grace-period=0 --force --namespace <namespace>
```

Add rack workflow from base rack fails in node validation

Problem statement
: Add rack workflow from base rack fails in node validation for the nodes of auxiliary rack in case of high availability cluster.

Cause
: The issue might occur because monitoring failed for the nodes of the auxiliary rack.

Resolution

1. Before performing the add rack operation, ensure that all the nodes on the auxiliary rack are healthy and that there are no warnings, critical events or any component failures on any of the nodes.
2. Restart the BMC on the failing nodes to clear the events on the compute node.

OpenShift Container Platform node status shows NotReady but IBM Fusion HCI user interface node status shows as Running

Problem statement
: The Running state of the node on the IBM Fusion HCI user interface indicates that the Baremetal node is healthy and running. Whenever you see IBM Fusion HCI node status is Running and OpenShift Container Platform node status is Not ready, then follow the steps in the resolution.

Resolution

1. Log in to OpenShift user interface.
2. Go to Compute > Bare Metal.
3. From the ellipsis menu, click Power Off for the node.
4. Power it on again and check if the issue is resolved.

Node goes missing from the IBM Fusion HCI rack graphical view

Problem statement

OpenShift Container Platform node status shows **`Ready`** even though the IBM Fusion HCI user interface node status shows the node state as **`Error connecting to node`**
For more information about node status, see Monitoring hardware from IBM Fusion HCI user interface.

This issue might occur due to the power or network issues.

Resolution

The node can be viewed graphically in the user interface of the IBM Fusion HCI rack after it gets into a healthy state. For more information, see Monitoring hardware from IBM Fusion HCI user interface.

1. Get the name of the compute-operator pod name:

   ```
   oc get po -n ibm-spectrum-fusion-ns | grep isf-compute-operator
   ```

2. Copy the OneCli folder from the operator pod:

   ```
   oc rsync -c manager isf-compute-operator-controller-manager-<pod name>:/fw/onecli
   ```

   Note: For the subsequent commands, create a separate session.
3. Start a debug pod on the target node (BMC has error connecting to the node) that is in Ready state.

   ```
   oc debug node/<nodename>
   ```

4. Get the debug pod name:

   ```
   oc get po -A -o wide | grep <nodename> | grep debug
   ```

5. Copy the local copy of the OneCli folder to debug pod.

   ```
   oc rsync onecli pod/<debug pod>:/tmp
   ```

6. From the other session, run the following command to restart the node:

   ```
   cd /tmp/onecli
   ./onecli misc rebootbmc
   ```

7. Go to the Infrastructure > Nodes and wait for the following error message to go away:

   ```
   Error connecting to node
   ```

   It takes around three minutes for the error to disappear from the page.

Node in `NotReady` state due to certificate errors in OpenShift pods

Problem statement

Issues can occur when a node is in the **`NotReady`** state. One common reason is that the OpenShift network related pods go into a **`crashloopbackoff`** state due to an expired certificate file.

Diagnose

Follow the steps to diagnose the issue:

1. Check the output using the following command.

   ```
   oc get co
   ```

2. If you see any error message related to network, then check if any of the pods in namespace **`openshift-ovn-kubernetes`** are in crashing state.
3. If a pod is crashing state, then check the logs for it using following command.

   ```
   oc logs <pod-name> -c ovnkube-controller -n openshift-ovn-kubernetes
   ```

   Check for error related to wrong certificate.

   ```
   I1210 14:50:40.305586   78501 certificate_store.go:130] Loading cert/key pair from "/etc/ovn/ovnkube-node-certs/ovnkube-c
   I1210 14:50:40.305640   78501 certificate_store.go:130] Loading cert/key pair from "/etc/ovn/ovnkube-node-certs/ovnkube-c
   F1210 14:50:40.305690   78501 ovnkube.go:136] failed to start the node certificate manager: failed to initialize the cert
   ```

4. In a similar way, check if any pod contains issue in namespace **`openshift-multus`**.

Resolution

Follow the steps to resolve the issue:

1. Use the following steps to resolve the crashing pod issue in **`openshift-ovn-kubernetes`** namespace:
   a. Run the following command to change the project.

      ```
      oc project openshift-ovn-kubernetes
      ```

   b. Run the following command to log in to the failing pod.

      ```
      oc rsh <pod-name>
      ```

   c. Run **`rm -rf /etc/ovn/ovnkube-node-certs/ovnkube-client-current.pem`**
   d. Run the following command to restart the crashing pod.

      ```
      oc delete <pod-name>
      ```

2. Use the following steps to resolve the crashing pod issue in **`openshift-multus`** namespace:

a. Run the following command to check the node of the failing pod.

```
oc get pods -n openshift-multus -owide
```

b. Run the following command to log in to the respective node .

```
oc debug node/<node-name>
```

c. Run the following command to delete the **symlink**.

```
rm -rf etc/cni/multus/certs/multus-client-current.pem
```

After fixing the pods in both the namespaces, check if the error message is gone in the output of `oc get co`. The node should come into **Ready** state in sometime.

- **Add node failure**
  Node addition failures and their resolutions.
- **Issues related to IBM Fusion HCI node drains**
  Use these common troubleshooting tips and tricks when you work with IBM Fusion HCI.

Add node failure

Node addition failures and their resolutions.

Add node failure due to inspection error

Cause

This issue occurred because of ipmitool version in the rack. Run the following command to know the version number of the IPMI:

```
portcontrol -ipmi on
```

Example response:

```
The current security settings require incoming IPMI over LAN to use cipher suite ID 17.
If you are using the IPMItool utility (prior to version 1.8.19), you must specify the option <-C 17>.
To effectively enable this service, you must ensure that 'ipmi' is selected in the '-ai' option of the 'users' command.
ok
```

```
ipmitool -V
```

Example response:

```
ipmitool version 1.8.18
```

Diagnosis

1. Log in to the compute operator pod by using the manager container.

   ```
   oc rsh -c manager <pod_name>
   ```

2. Go to fw/onecli directory.
3. Run the following command to create a output directory:

   ```
   ./OneCli  config set IMM.Security_Mode Compatibility --bmc
   USERID:'Wo6k5tSKT0'@[fd8c:215d:178e:c0de:ea80:88ff:fe07:b66b] --never-check-trust
   ```

   Example output:

   ```
   Create output directory at /fw/onecli/logs/OneCli-20241007-062329-988988/ failed with error is 13

   [WARNING]: No permission to create output directory at /fw/onecli/logs/OneCli-20241007-062329-988988/
   All output file will not be saved. You can specify another directory by "--output" parameter or with higher permission.

   Start to connect BMC at fd8c:215d:178e:c0de:ea80:88ff:fe07:b66b to apply config set
   Connected to BMC at IP address  by REDFISH
   Invoking SET command ...
   IMM.Security_Mode=Compatibility
   /fw/onecli/logs/OneCli-988988-20241007-062329/common_result.xml: cannot open file

   No permission to create output directory at /fw/onecli/logs/OneCli-20241007-062329-988988/
   All output file will not be saved.
   Succeed.
   ```

Resolution

To resolve the issue, see IPMI check Error.

Add node failure to OpenShift Container Platform cluster

Node addition to the Red Hat® OpenShift® Container Platform cluster fails.

Severity

Warning or Error

Explanation

Node addition to the Red Hat OpenShift Container Platform cluster reports a failure. As a part of the node addition process, IBM Fusion HCI validates and runs a few steps. If any of these validations or steps fail, a node addition failure gets reported.

Some of the reported failures can auto resolve after a few minutes because the IBM Fusion HCI retries the same validation and steps in every reconciliation cycle. You can view the actual failure reason in the status of the **computeprovisionworker** CR.

In the following example, the **message** field indicates a node DNS validation failure. Determine the failure message from the **computeprovisionworker** CR status, and follow the steps in the *Recommended actions* section.

```
oc -n ibm-spectrum-fusion-ns get cpw provisionworker-compute-1-ru5 -o yaml
apiVersion: install.isf.ibm.com/v1
kind: ComputeProvisionWorker
metadata:
 creationTimestamp: "2024-05-01T21:39:53Z"
 generation: 1
 name: provisionworker-compute-1-ru5
 namespace: ibm-spectrum-fusion-ns
 resourceVersion: "8733225"
 uid: 3aa59cb6-c032-4b31-98ec-26284ed4588a
spec:
 location: RU5
 rackSerial: 8Y2DXXX
status:
 hostname: compute-1-ru5.mycluster.mydomain.com
 installStatus: Installing
 ipAddress: XX.YY.ZZ.15
 location: RU5
 macAddress: 08:xx:eb:ff:yy:zz
 machineSetScaled: true
 message: DNS validation for node failed.
 messageCode: IOCPW0011
 name: provisionworker-compute-1-ru5
 nodeType: storage
 ocpRole: compute-1-ru5
 oneTimeNetworkBoot: Completed
 rackName: mycluster
 startTimeStamp: "2024-05-01T21:39:54Z"
```

Recommended actions

DNS validation for node failed

The error indicates a node DNS validation failure, which can be intermittent because of an unsuccessful DNS call. It eventually succeeds in the next cycle. However, if the issue persists for more than 10 minutes, check the DHCP and DNS configurations to confirm that the correct FQDN to IP exists for the DNS lookup and reverse lookup. After you fix the issue, the node addition in IBM Fusion HCI proceeds automatically to the next step.

Unable to add a node and failed to set boot order

Usually, this failure gets resolved in the subsequent reconcile cycles. If it persists more than 10 minutes, then a Baseboard Management Controller (BMC) restart can resolve the issue. Sometimes, the IMM does not respond to **ipmi** or **redfish** commands. Contact IBM support to restart the BMC.
After your restart the BMC, the error gets resolved in the subsequent reconciliation cycles.

Failed to read userconfig secret object

The error indicates that an intermittent API error exists in reading the OpenShift Container Platform secret. It can occur due to a delay in the API response or some other network interruption. This error self-resolves automatically in the successive reconciliation cycles.

Failed to read kickstart configmap data

The error indicates that an intermittent API error exists in reading the OpenShift Container Platform configmap. It can occur due to a delay in the API response or some other network interruption. It self-resolves automatically in the successive reconciliation cycles.

Unable to add a node and the Red Hat OpenShift node did not transition to Ready state

The CSRs pending for approval is the most common reason for this error. The IBM Fusion HCI operator approves these CSRs as a part of the node addition process. If it does not occur as a rare case scenario, manually approve them to resolve the issue. Run the following commands to check and approve the pending CSRs:

1. Run the following command to look for pending CSRs:

```
oc get csr | grep -i pending
```

2. Run the following command to approve pending CSRs:

```
oc get csr -o go-template='{{range .items}}{{if not .status}}{{.metadata.name}}{{"\n"}}{{end}}{{end}}' | xargs 
oc adm certificate approve
```

Failed to scale machineset during new node addition

As a part of the Red Hat OpenShift cluster expansion process, a Bare Metal host object gets created for every node that is added to the cluster. This Bare Metal host object go through the following states: **registering** > **inspecting** > **available** > **provisioning** > **provisioned**.
After the Bare Metal host becomes **available**, the replica of the Bare Metal machineset CR in the OpenShift Container Platform cluster increases by one to start the machine creation process of that host. If that step fails, then you can see a reported error in the **computeprovisionworker** status. The Fusion operator tries to increase it in the next reconcile cycle again and that eventually succeeds. If it does not succeed in a rare case scenario, do the following steps to manually increase the replica count of the machineset CR.

1. Run the following command to get the machineset name:

```
oc get machinesets -n openshift-machine-api | tail -1 | awk '{print $1}'
```

2. Run the following command to get the machineset current replica count:

```
oc get machinesets -n openshift-machine-api | grep worker | awk '{print $2}'
```

3. Run the following command to increase the replica by one:

```
oc scale --replicas=<old replica count from step2> + 1 machineset <machineset name from step 1> -n openshift-
machine-api
```

In the subsequent reconciliation cycles, the node addition process moves to the next steps and gets reflected in the IBM Fusion HCI user interface.

Node might contain an invalid hostname (localhost)
: The error indicates that the added host did not get the expected fully qualified hostname assigned from DHCP. A common reason for this error is that the DHCP server does not get configured correctly. Check the following content in the DHCP server:

1. Hostname option is set for this host entry
2. No mistakes exist in the mac address that is used for this host entry
3. DHCP service is running and it is reachable from the host

After you fix the issue on the DHCP side, restart the node by using the Bare Metal host object from the Red Hat OpenShift console. To restart the node, do the following steps from the OpenShift Container Platform web console:

1. Go to Home > Search.
2. Search for CRs with `ComputePowerOp`. It lists CRs and nodes in the cluster.
3. Go to the CR of the node to restart and then go to the YAML tab.
4. Add the following line to `Spec` to power off the node.

   ```
   powerOP: GracefulShutdown
   ```

   if you want to power on the node, change the line to the following line:

   ```
   powerOP: On
   ```

Unable to add one or more nodes as they contain an invalid hostname (localhost)
: This error indicates that the host you want to add did not get the expected valid fully qualified hostname assigned from DHCP. A common reason for this error is that the DHCP server is not configured correctly. In a DHCP server, check whether the following conditions exist:

- Hostname option is set for this host entry
- No mistake in the mac address used for this host entry
- DHCP service is running and DHCP is reachable from the host

After you fix the DHCP issues, restart the node to proceed with the node addition. To restart node, do the following steps from the Red Hat OpenShift Container Platform web console:

1. Go to Home > Search.
2. Search for CRs with `ComputePowerOp`.
3. Go to the CR of the node to restart and then go to the YAML tab.
4. Add the following line to `Spec` to power off the node.

   ```
   powerOP: GracefulShutdown
   ```

   if you want to power on the node, change the line to the following line:

   ```
   powerOP: On
   ```

Node hardware failure
: This error indicates that add node fails due to any problem with node hardware.
Check compute monitoring CR of the node and take an appropriate action to resolve the issue.

For example:

```
NodeType: storage
      nodeMonStatus:
        eventCode: BMYCO0003
        message: One or more component on the node 169.253.1.6 is in critical state.
        messageArg:
          - 169.253.1.6
        messageCode: BMYMO1004E
        state: Critical
        activeEvents:
        - Power Supply 2 has Failed.
        - Sensor PSU 2 Failure has transitioned to critical from a less severe state.
        - 'Non-redundant:Sufficient Resources from Redundancy Degraded or Fully Redundant for Power Resource has 
asserted.'
```

Issues related to IBM Fusion HCI node drains

Use these common troubleshooting tips and tricks when you work with IBM Fusion HCI.

Issues related to draining IBM Fusion HCI node

Pod Distribution Budget (PDB) limits the number of Pods of a replicated application that are down simultaneously from voluntary disruptions. Scale uses PDB to ensure that storage does not get corrupted and minimum number nodes are available. You may experience delays or failure in node drain or node restart during any of the following operations:
Note: You must never reboot a node before it is drained successfully. Always perform node power operations from the IBM Fusion HCI user interface as it ensures successful node drain.

- Configuration updates
  - Red Hat® OpenShift® Machine Config Operator (MCO)
  - IBM Fusion component specification changes
- Upgrade
  - OpenShift Container Platform upgrades

    - IBM Fusion software upgrades
    - Firmware upgrades
- User Initiated
    - Maintenance operation
    - kdump enabling/disabling
    - Operations that result in node restart
    - Node drain

Cause
: The PDB configuration is set by Scale to control the node drain so that the Scale cluster remains healthy always, and you can drain only an allowed quantity of nodes at any time. For more details, see [Scale behavior during node restarts](#).

Resolution
: When node drain hangs due to any of the suggested operations, then do the following steps:

1. Run the following command to identify the pod that causes the issue.

```
oc describe cmt <target node name> -n ibm-spectrum-fusion-ns
```

   The command lists pods pending for eviction.
2. Go through events of the node having the issue:

   Scenario where drains are prevented by an application
   : If drains are prevented by an application, consult the owner of the application to proceed with the drain, and manually drain the node. For more information about the issue and to troubleshoot, see Identifying applications preventing cluster maintenance in *IBM Storage Scale Container Native* documentation.

   Scenario where drains are prevented by VM
   : Check Identifying applications preventing cluster maintenance in *IBM Storage Scale Container Native* documentation to determine the node that is waiting for reboot and check whether live migration is set up properly to allow VM to migrate to a different node. For more information about setting up live migration, see Virtual machine live migration in *Red Hat OpenShift Container Platform* documentation.

   Scenarios where issue is due to node maintenance
   - The maintenance mode on a node can take from four minutes to thirty minutes to succeed. If the maintenance mode operation is taking more than 30 minutes, then Fusion continues to retry draining the pods with a warning event BMYCO0012, and gets timed out eventually. If you want to stop the retries, then delete the `Computemaintenance` CR instance.
     Run the following command to delete `Computemaintenance` CR instance.

```
oc delete cmt <instnace-nodename> -n ibm-spectrum-fusion-ns
```

   - If it takes a long time and you want to continue the operation, check the pod that is pending by using the **oc get pods** command. If the pod name belongs to IBM Storage Scale, see Identifying applications preventing cluster maintenance in *IBM Storage Scale Container Native* documentation. If issue persists, collect Scale system health and Scale logs, and contact IBM support.

   If you want to understand Scale behavior during node reboots, see *Scale behavior during node restarts* section.

Scale behavior during node restarts

Restart of a node can happen due to various reasons such as an MCO roll out, user initiated, or as part of firmware or software upgrades. Scale has a maximum tolerance on the number of nodes that are in unavailable or in not ready state. It is based on the erasure code that is configured for the storage cluster. To stay within this tolerance, Scale uses the POD Disruption Budget (PDB) feature of OpenShift.

Scale CNSA implements PDB with a "maxUnavailable=0", which means that it does not allow a Scale pod to go down without its knowledge. Even with a "maxUnavailable=0" PDB, the design does allow for cluster updates. In this configuration, MCO is prevented from taking down the node and draining the CNSA core pod. The Scale CNSA pod detects the operation, drains the applications, and then exits itself, thus freeing the node up for the operation to continue.

If any application refuses to unmount, or if the scale core pod itself determines the cluster, then it goes into a bad or hung state. If it goes down, then things pause until the condition gets resolved.

For more information about identifying applications that prevent cluster maintenance, see Identifying applications preventing cluster maintenance in *IBM Storage Scale Container Native* documentation.

Remediating a maintenance drain failure

To remediate a maintenance drain failure, follow these steps:

1. In the OpenShift web console, check the status of the stalled upgrade by using one of the following methods:
   - On the individual node's page, click View Pods.
   - Go to Settings > Upgrades, locate the Node firmware card under Rolling Upgrades, select the node with the status Upgrade Stalled, and click View Pods.
     Note: The View Pods option in the OpenShift console applies to OpenShift Container Platform nodes for IBM Fusion HCI version 2.12.0. For Hosted Control Plane nodes, eviction must be performed manually.
2. After accessing the View Pods page, you can see a list of all pods, including their current state and namespace.
3. Check the pods listed under pendingPods of the `Compute Maintenance` custom resource (CR) that are scheduled for eviction as part of this CR. To access the `Compute Maintenance` CR, follow these steps:
   - Command line:

```
oc get cmt -n ibm-spectrum-fusion-ns // to fetch the instances
oc get cmt <cmt-name> -oyaml -n ibm-spectrum-fusion-ns
```

   - OpenShift console:
     - Go to Administration > CustomResourceDefinitions, and search for ComputeMaintenance.
     - Click ComputeMaintenance.
     - Select Instances to view the CR.

4. Troubleshoot stalled pods:
    a. Copy the name of the stalled pod.
    b. Search for the pod on the View Pods page.
    c. Check the pod state and investigate why it is not draining. For detailed troubleshooting steps, see Issues related to draining IBM Fusion HCI node.
5. If the issue persists, contact IBM Support.

Dell node issues

This topic provides a comprehensive list of common troubleshooting steps and known issues that may arise while working with the Dell nodes.

Unable to add node to OpenShift Container Platform

Problem statement
    The node cannot be added to the OpenShift® Container Platform cluster.

Cause
    The issue might be caused by a hung power operation (powerop) for the node. This can be identified by checking the logs on the system.

Symptoms

- The power operation is not progressing.
- The live powerop CR logs for the node are blank or do not show any activity.
- The powerop CR is not getting updated.

Resolution
    To resolve this issue, follow these steps:

1. Check the live logs on the system for the powerop operation.
2. If the logs are blank or not updating, delete the powerop CR by running the following commands in the **ibm-spectrum-fusion-ns** namespace:

    ```
    oc get cpr
    ```

    ```
    oc delete cpr <node-powerop-cr-name>
    ```

    Important: Wait for the command to delete the powerop CR. Avoid forceful deletion as it may cause further issues.
3. After deleting the powerop CR, verify if the node can be added to the OpenShift Container Platform cluster.
4. If the issue persists, contact IBM support .

Unable to add GPU node to compute cluster

Problem statement
    Unable to add a GPU node to the compute cluster due to a failure in importing the system configuration.

Symptoms
    When attempting to add a Dell GPU DG4 node to the compute cluster, the following issues occur:

- Dell iDRAC issue (LC068) causes the GPU node configuration to fail in IBM Fusion compute configuration CR.
- Failure to set the network boot order while adding the node to the OpenShift Container Platform due to a Redfish call error. The error message indicates: **errorCode: SYS011, message: Pending configuration values are already committed**, unable to perform another set operation.

Cause
    The cause of the issue is attributed to a Dell iDRAC issue (LC068) and a Redfish call failure to set PXE settings.

Resolution
    To resolve this issue, contact IBM support.

Dell nodes job queue creation fails

Problem statement
    Dell nodes job queue creation fails due to internal memory overload.

Cause
    Due to a bug in the **LCLogAggregation** function in current versions of **iDRAC**, an issue has arisen with the management of event logs within the internal **Lifecycle Controller (LCC)** partition. This malfunction causes iDRAC to encounter an internal error, which prevents the creation of configuration jobs. As a result, node configuration changes cannot be applied because the internal memory lacks sufficient space to accommodate new tasks. An following is an example error that is seen from the terminal when you try to create the job:

```
racadm>>jobqueue create BIOS.Setup.1-1 -s TIME_NOW -r pwrcycle
ERROR: JCP012: The operation failed due to an internal iDRAC error.
Retry the operation. If the issue persists, reset iDRAC and retry the operation.
racadm>>
```

Resolution

1. Get the **ini** file to send to Dell support team:

    ```
    racadm -r <iDRAC ip> -u <iDRAC user> -p <iDRAC password> debug unblock rootshellash <date(yyyy-mm-dd)> <date next month(yyyy-mm-dd)> -f <service tag>.ini
    ```

    For example:

```
racadm -r 169.253.1.3 -u root -p calvin debug unblock rootshellash 2025-08-20 2025-09-20 -f JW47NW3.ini
```

2. With the new `ini` file from Dell support team enable "debug grant":

   ```
   racadm -r <iDRAC IP> -u <iDRAC user> -p <iDRAC password> debug grant -f <service tag>.ini
   ```

   For example:

   ```
   racadm -r <iDRAC IP> -u root -p calvin debug grant -f Case_#214177489_-_4X47NW3.ini
   ```

3. Ssh to iDRAC:

   ```
   ssh <iDRAC IP>
   ```

4. Open debug mode:

   ```
   debug invoke rootshellash
   ```

5. Change to root:

   ```
   /tmp $ su -
   ```

6. Extract the event.log:

   ```
   /home/root# scp /flash/data1/LCL/event.log root@1<iDRAC IP>:/tmp
   ```

7. Remove event-log:

   ```
   /home/root# rm /flash/data1/LCL/event.log
   ```

8. Disable LCLogAggregation:

   ```
   set iDRAC.Logging.LCLogAggregation 0
   ```

9. Reset iDRAC:

   ```
   racreset
   ```

10. Wait for two to five minutes and check if the **jobqueue** command is working.

XLR660 iDRAC dedicated network port failover limitation

Problem statement
: The XLR660 includes a dedicated iDRAC (Integrated Dell Remote Access Controller) network port to provide out-of-band management capabilities and this dedicated iDRAC port does not support network failover. If the iDRAC port loses connectivity, the system does not automatically reroute management traffic through other available network interfaces.

Service impact
: In such failover scenarios, the IBM Fusion user interface may display a `Error Connecting to Node` message on the nodes page. It impacts the hardware monitoring and firmware monitoring capabilities, which retrieve the system information, adapter information, storage information, firmware status and power operation of IBM Fusion get fail.

Resolution
: If you see this issue, contact IBM support.

Cleaning up localhost node from the IBM Fusion HCI cluster

Follow the instructions to clean up the localhost node from the IBM Fusion HCI cluster.

About this task

The localhost node must not be added to the OpenShift® cluster, as it creates an issue at a later stage. A couple of OpenShift objects are created when the localhost node is added to the IBM Fusion HCI cluster.
Important: The example commands and outputs in this topic are for guidance only. Do not copy and run the example commands and YAMLs as they are presented. Instead, ensure you replace the placeholders with your relevant values.

Procedure

Follow the steps to clean up the localhost node so that the node addition can retry after you resolve the DHCP or DNS misconfiguration issues.

a. Run the following command to edit the `ComputeProvisionWorker CR` of the compute node that indicated an error message `This node might contain an invalid hostname (localhost).`
   For example:

   ```
   oc edit cpw provisionworker-compute-1-ru5
   ```

   Edit the respective CR to update the location information to an empty string to ensure that the `ComputeProvisionWorker` controller is not involved when you clean different objects of the compute node.

   For example:

   ```
   oc edit cpw provisionworker-compute-1-ru5
   ```

```
...
spec:
  location: ""
```

b. Delete the respective machine object of the compute node.
   You need to identify the machine object first and then mark it with an annotation. Then, scale down the machineset object to delete the machine object.

   i. Run the following command to get the machine object name from the BMH object.

   ```
   oc -n openshift-machine-api get bmh,machine
   ```

   Example output:

   ```
   NAME                                  STATE                    CONSUMER                              ONLINE   ERROR   AGE
   baremetalhost.metal3.io/compute-1-ru5   provisioned              isf-rackae6-42ps4-worker-0-r6fmw   true             22h
   baremetalhost.metal3.io/compute-1-ru6   provisioned              isf-rackae6-42ps4-worker-0-t922n   true             22h
   baremetalhost.metal3.io/compute-1-ru7   provisioned              isf-rackae6-42ps4-worker-0-g842m   true             22h
   baremetalhost.metal3.io/control-1-ru2   externally provisioned   isf-rackae6-42ps4-master-0         true             44h
   baremetalhost.metal3.io/control-1-ru3   externally provisioned   isf-rackae6-42ps4-master-1         true             44h
   baremetalhost.metal3.io/control-1-ru4   externally provisioned   isf-rackae6-42ps4-master-2         true             44h

   NAME                                                        PHASE     TYPE   REGION   ZONE   AGE
   machine.machine.openshift.io/isf-rackae6-42ps4-master-0         Running                          44h
   machine.machine.openshift.io/isf-rackae6-42ps4-master-1         Running                          44h
   machine.machine.openshift.io/isf-rackae6-42ps4-master-2         Running                          44h
   machine.machine.openshift.io/isf-rackae6-42ps4-worker-0-g842m   Running                          22h
   machine.machine.openshift.io/isf-rackae6-42ps4-worker-0-r6fmw   Running                          22h
   machine.machine.openshift.io/isf-rackae6-42ps4-worker-0-t922n   Running                          22h
   ```

   In this example, the BMH object of **`compute-1-ru5`** maps to the **`isf-rackae6-42ps4-worker-0-r6fmw`** of the machine object.

   ii. Mark the machine object to delete by a special annotation **`machine.openshift.io/cluster-api-delete-machine: delete-me`**. The special marking helps to override the machine deletion policy rule.
   For example:

   ```
   oc -n openshift-machine-api edit machine isf-rackae6-42ps4-worker-0-r6fmw

   apiVersion: machine.openshift.io/v1beta1
   kind: Machine
   metadata:
     annotations:
       machine.openshift.io/cluster-api-delete-machine: delete-me
   ...
   ```

   iii. Now you need to scale down the machineset object so that the deletion of the machine object gets initiated.
   Note: After the machineset scale down is performed, the machine objects corresponding to **`compute-1-ru5`** are cleaned up, and the status of the BMH object corresponding to **`compute-1-ru5`** changes to deprovisioning.

   ```
   oc -n openshift-machine-api get machineset
   ```

   Example output:

   ```
   NAME                         DESIRED   CURRENT   READY   AVAILABLE   AGE
   isf-rackae6-cltp4-worker-0   3         3         3       3           8h
   ```

   ```
   oc -n openshift-machine-api get machineset -oyaml | grep replicas
   ```

   Example output:

   ```
   replicas: 3
   ```

   Example command:

   ```
   oc -n openshift-machine-api scale --replicas=<old replica value - 1> machineset <machine set name>
   ```

   ```
   machineset.machine.openshift.io/isf-rackae6-42ps4-worker-0 scaled
   ```

   iv. After the machineset scale down is performed successfully, the status of the BMH object corresponding to **`compute-1-ru5`** changes to ready.
   Important: This activity might take few minutes, and wait for it to be reflected against the BMH object before proceed with the further steps.

c. Delete the compute nodes in the node object from the OpenShift Container Platform cluster if anything present.

   ```
   oc get nodes
   ```

   Example output:

   ```
   NAME                                               STATUS   ROLES           AGE   VERSION
   compute-1-ru6.yourdomain.com   Ready    worker          21h   v1.23.17+16bcd69
   compute-1-ru7.yourdomain.com   Ready    worker          21h   v1.23.17+16bcd69
   control-1-ru2.yourdomain.ibm.com   Ready    master,worker   43h   v1.23.17+16bcd69
   control-1-ru3.yourdomain.com   Ready    master,worker   43h   v1.23.17+16bcd69
   control-1-ru4.yourdomain.com   Ready    master,worker   43h   v1.23.17+16bcd69
   localhost                                          Ready    worker          20h   v1.23.17+16bcd69
   ```

   ```
   oc delete node localhost
   ```

d. Delete the BMH object of the compute node.

   ```
   # oc -n openshift-machine-api get bmh
   ```

   Example output:

   ```
   NAME            STATE                    CONSUMER                              ONLINE   ERROR   AGE
   compute-1-ru5   available                                                      false            24h
   compute-1-ru6   provisioned              isf-rackae6-42ps4-worker-0-t922n   true             24h
   ```

```
compute-1-ru7   provisioned              isf-rackae6-42ps4-worker-0-g842m   true              24h
control-1-ru2   externally provisioned   isf-rackae6-42ps4-master-0         true              46h
control-1-ru3   externally provisioned   isf-rackae6-42ps4-master-1         true              46h
control-1-ru4   externally provisioned   isf-rackae6-42ps4-master-2         true              46h
```

Example command

```
oc -n openshift-machine-api delete bmh compute-1-ru5
baremetalhost.metal3.io "compute-1-ru5" deleted
```

e. Delete all pending `CertificateSigningRequests` corresponding to the localhost node.

```
for i in `oc get csr --no-headers | grep -i system:node:localhost | grep -i pending | awk '{ print $1 }'`;do oc delete csr
$i; done
```

f. Fix the DNS or DHCP issue to get the correct hostname for the corresponding compute node instead of the local host.
g. Delete the `ComputeProvisionWorker` object of the compute node.

```
oc -n ibm-spectrum-fusion-ns get cpw
```

Example output:

```
NAME                            AGE
provisionworker-compute-1-ru5   24h
provisionworker-compute-1-ru6   24h
provisionworker-compute-1-ru7   24h
```

Example command:

```
oc -n ibm-spectrum-fusion-ns delete cpw provisionworker-compute-1-ru5
computeprovisionworker.install.isf.ibm.com "provisionworker-compute-1-ru5" deleted
```

h. If the issue occurs during installation at the time of OpenShift configuration, the node conversion resumes automatically. If the issue occurs during the node upsize, then retry the node addition.

Shutting down and restarting IBM Fusion HCI rack

Procedures to gracefully shut down and restart IBM Fusion HCI. The process vary depending on whether the storage is Fusion Data Foundation or Global Data Platform.

- **Shutting down and restarting IBM Fusion rack with Data Foundation storage**
  Procedure to gracefully restart the IBM Fusion rack with Fusion Data Foundation storage.
- **Shutting down and restarting IBM Fusion HCI rack with Global Data Platform**
  Procedure to gracefully restart the IBM Fusion rack with Global Data Platform storage.

Shutting down and restarting IBM Fusion HCI rack with Data Foundation storage

Procedure to gracefully restart the IBM Fusion rack with Fusion Data Foundation storage.

Before you begin

Install OpenShift® Command Line Interface (CLI):

1. Log in to OpenShift Container Platform web console.
2. Click ? in the title bar, and click Command Line Tools.
   The Command Line Tools page is displayed.
3. In the Command Line Tools, click Download oc for <your platform>.
4. Save the file.
5. Unpack the downloaded archive file.
6. Move the **oc** binary to a directory on your path.
7. Run the file to install the OpenShift CLI.

Procedure

1. Capture system health check report before bringing down the rack. It helps to check for any preexisting issues post power-on.
   a. Ensure no machine config or update is in progress or no node is not ready
   b. To verify, run the following commands:

      ```
      oc get co
      ```

      ```
      oc get clusterversion
      ```

   c. Run the following commands to list the pods, cluster operators, and nodes.

      ```
      oc get po -A | grep -v Running | grep -v Completed
      oc get nodes
      ```

   d. Check that the Fusion Data Foundation cluster is in a healthy state.
      i. In the Red Hat® OpenShift web console, click Storage > Data Foundation.
      ii. In the Status card of the Overview tab, clickStorage system.
      iii. In the notification window, click the Storage system link.

    iv. In the Status card of theBlock and File tab, verify that the storage cluster is in a healthy state.
    v. In the Details card, verify the cluster information.

   Note: The health check must be saved to a different system.
2. Check whether there exists any active Backup & Restore jobs. If Backup & Restore or application synch is in progress, then wait for them to complete. Wait for in progress workload operations to complete. Before you proceed with the shutdown of the storage cluster, ensure that no data is in progress for any job or application.
3. Stop the applications that consume storage from the Fusion Data Foundation to stop the I/O.
   The shutdown sequence is important to avoid any data corruption and pod failures. The shutdown of the application components must to be done before you shut down the nodes. Scale down the application deployments so that the application pods do not get started on other nodes in case node selectors are not set. Also, this must stop storage I/O from the applications.
4. If you have enabled IBM Data Cataloging, then place the service in an idle state on the Red Hat OpenShift environment. For more information about the shut down procedure in IBM Data Cataloging, see Graceful shutdown.
5. Shut down the Red Hat OpenShift Container Platform cluster.
   a. If the cluster-wide proxy is enabled, be sure to export the NO_PROXY, HTTP_PROXY, and HTTPS_PROXY environment variables, on bastion node from where you intend to run `oc` commands. To check whether the proxy is enabled run below command:

      ```
      oc get proxy cluster -o yaml
      ```

   b. Take etcd backup.

      ```
      oc debug node/<node_name> (any one control node)
      ```

      ```
      sh-4.15# /usr/local/bin/cluster-backup.sh /home/core/assets/backup
      ```

   c. Copy the etcd backup to external system. These files are generated during the previous step b.

      ```
      snapshot_.db and static_kuberesources_.tar.gz
      ```

      You can use the `oc rsync` command to copy the files to an external system. You need two terminals for this operation.
      i. Open terminal one.
      ii. Run the following commands for etcd backup:

         ```
         oc debug node/<node_name>
         sh-4.15# /usr/local/bin/cluster-backup.sh /home/core/assets/backup
         ```

         In `oc debug node/<node_name>` command, use any one control node.
      iii. Run the following command and record the new pod name:
         It is the source pod, and the backup files reside inside the pod.

         ```
         oc debug
         ```

         Do not close the terminal 1.
      iv. Open terminal two and run the following command to copy the file to the local folder:

         ```
         oc -n <namespace_of_debug_pod> rsync <source_podname_in_above_step>:/home/core/assets/backup/snapshot_.db
         <local_folder_path>
         ```

         If required, add the namespace of the debug node pod location.

      v. Repeat the step ii to copy another backup file to the external system.
      vi. Close the terminal windows after all the files are copied.
         Note: If `oc rsync` command fails to work, use `scp` command from inside the source pod:

         ```
         scp /home/core/assets/backup/snapshot_.db root@:/destinationatexternalsystem
         ```

      For more information about the procedure, see Copy local files to or from a remote directory in *Red Hat OpenShift Container Platform* documentation.
   d. Ensure that you take off the workloads before you shut down the nodes.
   e. Run the following commands to shut down the nodes:
      Ensure that the control node hosting the IBM Fusion operators is powered off at the end. Shutting down this node prematurely results in loss of access to both the IBM Fusion and OpenShift Container Platform user interfaces.

      For Fusion Data Foundation, you must first shut down the application nodes and then shutdown the Fusion Data Foundation nodes.

      ```
      for i in `oc get nodes  |grep  compute|awk '{print $1}'`; do oc debug nodes/$i -- chroot /host shutdown -h 1 ; done
      ```

      Finally, shutdown the OpenShift control plane nodes.

      ```
      for i in `oc get nodes  |grep -v compute|grep -v NAME|awk '{print $1}'`; do oc debug nodes/$i -- chroot /host
      shutdown -h 1 ; done
      ```

      After 3 to 5 minutes, the Red Hat OpenShift Container Platform becomes inaccessible.
      This step brings down all the software on the rack. The rack is ready to be powered off.

6. After all the nodes are powered off, physically press the power off button of the nodes.
   Note:
   - This physical power off indicates to the Baseboard Management Controller (BMC) that you intend to keep the node powered down and prevents automatic restart.
   - The switches do not have the option to shutdown, and they can only be rebooted. When you power off the entire rack (unplugged), the switches shut down automatically. Similarly, when the power is restored to the rack, the switches comes up automatically.
7. Power on the rack.
   a. Power on the rack.
   b. Go to the physical node and click the power button to power on all the nodes.
      Power on all control nodes. After all control nodes are up, power on compute nodes.
      For Fusion Data Foundation, power on the nodes in this sequence:
      i. Start the Control Plane nodes.
      ii. Start the Fusion Data Foundation nodes.

iii. Verify that these pods are in a running state:

```
oc get pods -n openshift-storage
```

It is possible that all Fusion Data Foundation pods may not be Running by now, but it is expected to have `rook-ceph-{mds-*,mon-*,mgr-*,osd-*}` pods to be running to serve storage. If the Ceph cluster is not in a good shape, it results in a FailedMounts with infra and application pods.

iv. Start the application nodes.

c. After all the nodes are up and cluster operators are up (except image registry), run the following commands to ensure that the OpenShift cluster is up along with the IBM Fusion operators.

```
oc get po -A | grep -v Running | grep -v Completed
oc get co
oc get nodes
```

d. For Fusion Data Foundation, scale up the application deployments.
Give it a few minutes and check the cluster or storage dashboard.

e. Run the following commands to ensure that the storage pods are up:

Fusion Data Foundation
Run the following command to validate all pods are up in the `openshift-{storage,monitoring,logging,image-registry}` and the application namespaces:

```
oc get pods -n <NAMESPACE>
```

8. Bring back IBM Data Cataloging to a running state.
For the procedure, see Returning IBM Data Cataloging to a running state.

Shutting down and restarting IBM Fusion HCI rack with Global Data Platform

Procedure to gracefully restart the IBM Fusion rack with Global Data Platform storage.

Before you begin

Install OpenShift® Command Line Interface (CLI):

1. Log in to OpenShift Container Platform web console.
2. Click ? in the title bar, and click Command Line Tools.
The Command Line Tools page is displayed.
3. In the Command Line Tools, click Download oc for <your platform>.
4. Save the file.
5. Unpack the downloaded archive file.
6. Move the **oc** binary to a directory on your path.
7. Run the file to install the OpenShift CLI.

Procedure

1. Capture health check report before bringing down the rack. It helps to check for any preexisting issues post power-on.
Note: The health report must be saved to a different system.

Global Data Platform
For Global Data Platform storage, run the following OC commands to get the system health status before shutdown:

a. Run the following commands to list the pods, cluster operators, and nodes.

```
oc get po -A | grep -v Running | grep -v Completed
oc get co
oc get nodes
```

b. Change to `ibm-spectrum-scale` namespace:

```
oc project ibm-spectrum-scale
```

c. Log in to a running pod. For example, compute-1-ru5 pod:

```
oc rsh compute-1-ru5
```

d. Run the following command to get the state of the GPFS daemon on one or more nodes.

```
mmgetstate -a
```

e. Run the following command to display the current configuration information for a GPFS cluster.

```
mmlscluster
```

2. Check whether there exists any active Backup & Restore jobs. If Backup & Restore or application synch is in progress, then wait for them to complete. Wait for in progress workload operations to complete. Before you proceed with the shutdown of the storage cluster, ensure that no data is in progress for any job or application.
3. Run the following steps based on whether your rack is a stand alone or is in a disaster recovery setup (Metro-DR):

Stand-alone

a. Run the following command to shut down:

```
mmshutdown -a
```

b. Run the following command to verify whether all nodes are down:

```
mmgetstate -a
```

c. Exit from the pod

```
exit
```

Metro-DR
If you plan to shut down a site, ensure that you failover your applications to the other site.

a. Shutdown scale pods on affected site by using the **mmshutdown** directly in the pod Terminal.
b. Run `exit` to exit from the pod

4. Run the following storage commands to shut down the storage cluster.

Global Data Platform cluster
a. Switch the project to `ibm-spectrum-scale-operator`.

```
oc project ibm-spectrum-scale-operator
```

b. Set the replicas in the deployment configuration:

```
oc scale --replicas=0 deployment ibm-spectrum-scale-controller-manager
```

c. Switch the project to `ibm-spectrum-scale`.

```
oc project ibm-spectrum-scale
```

d. Log in to compute-1-ru<x>:

```
oc rsh compute-1-ru<x>
```

5. Place the IBM Data Cataloging service in an idle state on the Red Hat® OpenShift environment. For more information about the procedure, see Graceful shutdown.
6. Shut down the Red Hat OpenShift Container Platform cluster.
   a. If the cluster-wide proxy is enabled, be sure to export the NO_PROXY, HTTP_PROXY, and HTTPS_PROXY environment variables. To check if the proxy is enabled, run `oc get proxy cluster -o yaml`.
   b. Take etcd backup.

```
oc debug node/<node_name> (any one control node)
```

```
sh-4.15# /usr/local/bin/cluster-backup.sh /home/core/assets/backup
```

   c. Copy the etcd backup to external system.

```
snapshot_.db and static_kuberesources_.tar.gz
```

   You can use the `oc rsync` command to copy the files to an external system. You need two terminals for this operation.
      i. Open terminal one.
      ii. Run the following commands for etcd backup:

```
oc debug node/<node_name>
sh-4.15# /usr/local/bin/cluster-backup.sh /home/core/assets/backup
```

      In `oc debug node/<node_name>` command, use any one control node.
      iii. Run the following command and record the new pod name:
      It is the source pod, and the backup files reside inside the pod.

```
oc debug
```

      Do not close the terminal 1.
      iv. Open terminal two and run the following command to copy the file to the local folder:

```
oc -n <namespace_of_debug_pod> rsync <source_podname_in_above_step>:/home/core/assets/backup/snapshot_.db
<local_folder_path>
```

      If required, add the namespace of the debug node pod location.

      v. Repeat the step ii to copy another backup file to the external system.
      vi. Close the terminal windows after all the files are copied.
   For more information about the procedure, see Copy local files to or from a remote directory in *Red Hat OpenShift Container Platform* documentation.
   d. Ensure that you take off the workloads before you shut down the nodes.
   e. Run the following commands to shut down the nodes:
   Ensure that the control node hosting the IBM Fusion operators is powered off last. Shutting down this node prematurely results in loss of access to both the IBM Fusion and OpenShift Container Platform user interfaces.
   Finally, shutdown the OpenShift control plane nodes.

```
for node in $(oc get nodes -o jsonpath='{.items[*].metadata.name}');
do oc debug node/${node} -- chroot /host shutdown -h 1;
done
```

   After 3 to 5 minutes, the Red Hat OpenShift Container Platform becomes inaccessible.
   This step brings down all the software on the rack. The rack is ready to be powered off.

7. Physically press the power off button of the nodes and the rack.
   Note:

- This physical power off indicates to the Baseboard Management Controller (BMC) that you intend to keep the node powered down and prevents automatic restart.
- The switches do not have the option to shutdown, and they can only be rebooted. When you power off the entire rack (unplugged), the switches shut down automatically. Similarly, when the power is restored to the rack, the switches comes up automatically.

8. Power on the rack.
   a. Power on the rack.
   b. Go to the physical node and click the power button to power on all the nodes.
      Power on all control nodes. After all control nodes are up, power on compute nodes.
   c. After all the nodes are up and cluster operators are up (except image registry), run the following commands to ensure that the OpenShift cluster is up along with the IBM Fusion operators.

      ```
      oc get po -A | grep -v Running | grep -v Completed
      oc get co
      oc get nodes
      ```

   d. For Global Data Platform, bring back the Scale.

      ```
      oc project ibm-spectrum-scale-operator
         oc scale --replicas=1 deployment ibm-spectrum-scale-controller-manager
      ```

      Give it a few minutes and check the cluster or storage dashboard.

   e. Run the following commands to ensure that the storage pods are up:

      Global Data Platform
      i. Switch namespace to **ibm-spectrum-scale**:

         ```
         oc project ibm-spectrum-scale
         ```

      ii. Verify whether all pods are in running state in the **ibm-spectrum-scale** project:

         ```
         oc get pods
         ```

      iii. To run commands on a node, run the following **rsh** command:

         ```
         oc rsh compute-t-ru<x>
         ```

      iv. Run the following command to get the state of the GPFS daemon on one or more nodes.

         ```
         mmgetstate -a
         ```

      v. Switch project to **ibm-spectrum-scale-csi**:

         ```
         oc project ibm-spectrum-scale-csi
         ```

      vi. Verify whether all pods are in running state in the **ibm-spectrum-scale-csi** project. This may take sometime.

         ```
         oc get pods
         ```

9. Bring back IBM Data Cataloging to a running state.
   For the procedure, see Returning IBM Data Cataloging to a running state.

Networking

Configure and administer network settings for IBM Fusion.

Configuring network settings

Configure your network settings so that you can connect to the system, modify the existing network configurations.

- **Configuring network connections**
  To configure and manage network connections of your IBM Fusion appliance, add VLANs and them link to the system.
- **Upgrading switch firmware**
  Upgrade switch firmware.
- **Creating dedicated secondary Network in Virtualization**
  If the primary pod network in OpenShift® Virtualization becomes congested, you can improve performance by isolating that traffic onto a dedicated secondary network.
- **Provisioning additional networking**
  Provision additional networking for direct network access to a container or to a virtual machine.
- **Networking issues**
  A list of all troubleshooting and known issues that exist in the networking of IBM Fusion HCI.

Configuring network connections

To configure and manage network connections of your IBM Fusion appliance, add VLANs and them link to the system.

About this task

- **VLANs**
  A virtual local area network (VLAN) is a network concept and is central in directing traffic to your resources. It is an isolated broadcast domain that is created within

a switch and is used to isolate broadcast traffic on the public and private networks.
- **Links**
  In the Links section, you add links and configure their settings based on the VLANs that are defined in the VLANs section.

VLANs

A virtual local area network (VLAN) is a network concept and is central in directing traffic to your resources. It is an isolated broadcast domain that is created within a switch and is used to isolate broadcast traffic on the public and private networks.

Overview

VLANs are managed automatically with no need to interact directly with them, and they are assigned or removed according to the requirements. Each VLAN created within a switch is isolated from other VLANs. The network traffic can pass from one VLAN to another by adding a routing device. The routing functions must be provided at the data center core network. VLANs provide one method of packet identification, and they allow multiple workloads to coexist on the same system.

Work with your VLANs to add connections to the system as part of your network configuration.

- **Adding VLANs**
  As part of your network configuration, you can add one or more VLANs.
- **Editing VLANs**
  As part of your network configuration, you can edit a VLAN.
- **Deleting VLANs**
  As part of your network configuration, you can delete a VLAN.

Adding VLANs

As part of your network configuration, you can add one or more VLANs.

Before you begin

Note:

- You can create VLAN only of type **`Custom VLAN`**, **`Native VLAN`**, and **`Virtualization VLAN`**.
- **`Virtualization VLAN`** addition is supported on all new IBM Fusion HCI installations with version 2.10 or later.
- If you need to create a **`Virtualization VLAN`** on a IBM Fusion HCI cluster that is already upgraded to version 2.10, then you must first run a migration script. To run the migration script, reach out to IBM support.

Procedure

1. Click Infrastructure > Network.
   The Network page gets displayed.
2. Click VLANs tab.
   The VLANs tab page gets displayed. It lists all VLANs in a table format with Name, Type, VLAN ID, Allow multiple VLAN link, and VLAN status columns. Use the settings icon to customize the column headers.
3. Click Create VLAN.
   The Add VLAN slide out pane gets displayed.
4. Provide the VLAN details.
   - Name
     Enter a name of the VLAN.
   - VLAN ID
     Enter VLAN ID to uniquely identify it.
   - Type
     Select a type of VLAN from the drop-down list. For example, `Native VLAN`.
   - Allow multiple link
     Select or clear the Allow multiple link check-box. If you select Allow multiple link, the VLAN can be used for more than one link.

   Note: To add a VLAN to the uplinks of all racks in a multitrack cluster, ensure that Allow multiple link is always selected.
5. Click Add to apply your changes.
   You can see the new record in the VLANs table of the Network page. For more information, see Switch details.
   Note: In case of failures, check whether any high-speed switches are down.

Editing VLANs

As part of your network configuration, you can edit a VLAN.

Procedure

1. From the menu, go to the Infrastructure > Network page.
   The Network page gets displayed.
2. Click VLANs tab to open the VLANs tab page.
3. Select a VLAN record that you want to edit.
   To search for a specific record, see Switch details.
4. At the end of the record, click the ellipsis menu to view more actions and then click Edit.
   The Edit VLAN slide out panel gets displayed.
5. Edit the name of the VLAN or change the Allow multiple link check box selection.
   Note: The VLAN ID and VLAN Type values are read-only and cannot be edited.
6. Click Save to apply your changes.
   You can see the edited record in the VLANs table of the Network page. For more information, see Switch details.

Deleting VLANs

As part of your network configuration, you can delete a VLAN.

Procedure

1. From the menu, go to the Infrastructure > Network page.
   The Network page gets displayed.
2. Click VLANs tab to open the VLANs tab page.
3. Select a VLAN record that you want to delete.
   To search for a specific record, see Switch details.
4. Click the ellipsis icon and select Delete.
   A confirmation message gets displayed. Any changes to network configuration may result in connectivity issues.
5. Click Delete to proceed with the delete operation.

Links

In the Links section, you add links and configure their settings based on the VLANs that are defined in the VLANs section.

Work with the links to add network connections to your IBM Fusion appliance as part of your network configuration.
Note: If you are using a multi-rack setup, you must manage your Alternative network configuration and the links.

- **Adding links**
  As part of your network configuration, you can add links based on the VLANs that you defined in the VLANs section.
- **Editing links**
  As part of your network configuration, you can configure the link settings based on the VLANs that you defined in the VLANs section.
- **Deleting Links**
  As part of your network configuration, you can delete links.
- **Adding additional ports for uplink**
  You can enable additional switch ports for network uplinks by configuring the ports for connectivity.

Adding links

As part of your network configuration, you can add links based on the VLANs that you defined in the VLANs section.

About this task

If you are using a multi-rack setup, you must manage your Alternative network configuration and the links. The Add link page is not available for you.
Important: Ensure that you add links in a sequential order, and do not add multiple links at a time.

Procedure

1. Click Infrastructure > Network.
   The Network page is displayed.
2. Click Links tab to open it.
3. Click Create link.
   The Add link slide out pane gets displayed.
4. Provide the link details.

   - Link name
     Specify a name in Link name to uniquely identify a link.

   - Port
     Select one of the available options from the Port list. The port selection is made on both switches. You can select more than one port.

   - Port type
     Select `Access` or `Trunk` from the Port Type list. The `Access` port provides access to a single VLAN only.
     Important: Always choose `Trunk` as port type. Select `Access` only when the uplink switch is not VLAN aware.

- Transceiver
  Select one of the available options from the Transceiver list. The transceiver defines the speed and wire type to be used for the link.
- Aggregation
  Select one of the available options from the Aggregation list.
- VLANs
  Select one or more VLANs from the VLANs list. You can assign multiple VLANs from this multi-select label list. The Default Native list is populated with the selected VLANs. The Internal Storage VLAN option is available to externalize local storage.
- Native VLAN
  Select native VLAN whenever the port type is `Trunk`.

5. Click Save to apply your changes and to add the link.
   You can see the new record in the Links table of the Network page. For more information, see Switch details.

Editing links

As part of your network configuration, you can configure the link settings based on the VLANs that you defined in the VLANs section.

Procedure

1. Click Infrastructure > Network.
   The Network page is displayed.
2. Click Links tab to display the Links section.
3. Select a link record that you want to edit.
4. At the end of the record, click the ellipsis menu to view more actions and then click Edit.
   The Edit links pane is displayed.
5. Edit the link details.
   You can edit all the values of a link.
6. Click Save to apply your changes and to add the link.
   You can see the edited record in the Links table of the Network page. For more information, see Switch details.

Deleting Links

As part of your network configuration, you can delete links.

Procedure

1. From the menu, go to the Infrastructure > Network page.
   The Network page gets displayed.
2. Click Links tab to open the Links tab page.
3. Select a link record that you want to delete.
4. Click the ellipsis icon and select Delete.
   A confirmation message gets displayed. Any changes to network configuration may result in connectivity issues.
5. Click Delete to proceed with the delete operation.

Adding additional ports for uplink

You can enable additional switch ports for network uplinks by configuring the ports for connectivity.

Before you begin

Ensure that the following conditions are met before you begin:

- No existing connections are present on the ports.
- The ports are not configured as split ports.
- The ports do not appear in any port group or list other than **`customerPorts`**.

About this task

By default, you can connect your network to a rack by using only two fixed switch ports, 31 and 32. These ports are statically reserved for customer-facing connectivity, which limits the number of physical uplinks available for customers to connect to their network infrastructure.

This fixed port assignment restricts scalability and flexibility in rack networking. The two-port model restricts the customers who require additional uplinks for redundancy, higher bandwidth, or custom network topologies. To address this limitation, dynamic customer port support allows additional switch ports to be configured for customer network connectivity.

With dynamic port support, you can:

- Use any available customer uplink ports.

- Improve redundancy and fault tolerance.
- Distribute traffic more efficiently across multiple uplinks.
- Support complex or custom network topologies.
- Continue using existing configurations on ports 31 and 32 without modification.

Procedure

1. Verify switch port availability:
   a. In the IBM Fusion HCI user interface, go to Administration > CustomResourceDefinition > Switch.
   b. Select the high-speed switch for the corresponding rack.
   c. Verify that ports do not appear in any list other than `customerPorts` and are not part of any other port group or list in the `spec` section.

   Example switch names:

   - `hspeed1-f42051`
   - `hspeed2-f42051`
2. Add ports to the `Available` section:
   a. Go to Administration > CustomResourceDefinition > Links.
   b. Select the Link instance for the corresponding rack.
   c. Add ports to the `portList`.
   Example:

```
availableLinkTypes:
  - availableTransceivers:
      - Single QSFP
      - 2x Splitter QSFP
      - 4x Splitter QSFP
    linkType: customer_links
    switches:
      - portList:
          - '32'
          - '31'
          - '28'
          - '27'
        switchList:
          - hspeed1-f421051
          - hspeed2-f421051
```

   d. Save the changes.
3. Create a link from the new ports:
   a. Wait a few minutes for the added ports to appear in the IBM Fusion HCI UI.
   b. Go to Infrastructure > Network > Links.
   c. Click Create link.
   The Add link slide out pane opens.

   d. Enter the required information:
      - Link Name
      - Rack
      - Port
      - Port type
      - Transceiver
      - Aggregation
      - VLANs
      - Native VLAN
   e. Click Add.
4. Verify that the controller configures the ports:
   a. Wait while the controller applies the configuration to the switches.
   b. Confirm that the `Link` CR status changes to `Configured`.
   c. If configuration fails, review the error details.

Results

- The selected ports are available as customer uplinks.
- You can configure up to four uplink ports per rack, and the controller automatically configures the switch ports.

Upgrading switch firmware

Upgrade switch firmware.

Before you begin

- Upgrade the IBM Fusion HCI management software.
- Post the upgrade of IBM Fusion HCI management software, check whether the firmware upgrade exists by using any of the following methods:
  - Open the Red Hat® OpenShift® Container Platform web console. In the Overview page, go to Activity > Recent events section and check whether you see the following message:

    ```
    IBM Fusion firmware upgrade available for switches.
    ```

  - Open the user interface. It displays notifications in the following locations:
    - In the bell icon, you can see the following messages for the switch :

```
Switch firmware upgrades are available
```

- In the Infrastructure > Network, you can see the following message:

```
Firmware upgrade to version <version_number> is available
```

About this task

- The following are the tasks that the administrator must be aware of during the firmware upgrade.
  - The administrator can upgrade the high-speed switches and spine switches with the model SN3700C/SN3700V, and management switches with the model SN2201.
  - The firmware upgrade triggers on high-speed switches, followed by management and spine switches.
  - The firmware upgrade happens on the primary and secondary switches.
  - The secondary switches are upgraded first, followed by the primary switches during the upgrade process.
  - If the secondary switch firmware upgrade fails, then the related primary switch is not marked for the upgrade.
  - If a firmware update fails, a warning is raised for that switch. That warning can be seen in the Recent Events or Events sections.
- The switch firmware upgrade supports high-speed switches and spine switches with the model SN3700C/SN3700V, and management switches with model SN2201 in IBM Fusion HCI 2.11.

Note:

- Do the switch firmware upgrade in the specified sequence.
- Upcoming software upgrades are scheduled for the following network switches:
  - The high speed switches with model SN3700C or SN3700V should be upgraded to 5.15.1.
  - The management switches with model SN2201 should be upgraded to 5.15.1.
  - The management switches with model AS4610 should be upgraded to 4.3.4.

Procedure

1. Go to Infrastructure > Network.
   The Network page gets displayed. The Switches tab is open by default.
2. Click Upgrade all.
   The Upgrade firmware window gets displayed. It shows how many switches to upgrade along with the estimated time.
   Note: After the switch firmware upgrade is triggered, do not click anything in this page until the upgrade process is complete.
   The Upgrade firmware window gets displayed. The IBM Fusion HCI runs prechecks on the Spanning Tree Protocol (STP) , registry access, Fusion operator status, and DNS resolution.
3. If there are blocker issues, contact IBM Support.
   a. To know more the fixes for blockers or warnings, contact IBM Support.
      The IBM Fusion HCI user interface takes you to the Network page wherein a Upgrade pre-check report window displays all the Blocking issues and Issues (optional). If the upgrade fails, check the **updateStatus** in the **switchupgrade** CR.
   b. Contact IBM to know more about how to fix an issue.
   c. Fix all the blocking issues.
4. After the pre-checks complete without any blocker issues, click Upgrade now.
   The following message appears to confirm the initiation of upgrade:

   ```
   Switch Firmware upgrade in-progress
   ```

   For example:

   ```
   Switch Firmware upgrade in-progress
   1 out of 2 switches upgraded
   ```

   To know more details about the upgrade that is in progress, click View details in the notification tab of the Network page. A slide out pane displays more information about the upgrade that is in progress, including errors and warnings. Guidance links are available against the corresponding warnings and errors.

Creating dedicated secondary Network in Virtualization

If the primary pod network in OpenShift® Virtualization becomes congested, you can improve performance by isolating that traffic onto a dedicated secondary network.

Procedure

1. Open IBM Fusion user interface.
2. Create VLAN and wait until the VLAN status is Normal.
   For more information about how to add VLANs, see Adding VLANs.
3. Go to Links page and edit an existing link that supports the VLAN that you just created.
   In the Edit link pane, select `Virtualization` option from the VLANs drop-down list.
4. Create Network Attached Definition (NAD).
   a. In the OpenShift console, go to Networking > NetworkAttachmentDefinition.
   b. In the Create NetworkAttachmentDefinition, select `Linux Bridge` in the Network Type.
   c. Update Bridge name.
      It is part of Config Map in **isf-spectrum-fusion-ns** namespace under **isf-virtualization-bridge**.
      Example ConfigMap with bridgename:

      ```
      kind: ConfigMap
      apiVersion: v1
      metadata:
        name: isf-virtualization-bridge
        namespace: ibm-spectrum-fusion-ns
        uid: a107f372-acaa-445c-bd19-0aab9a97aaa4
      ```

```
  resourceVersion: '11882214'
  creationTimestamp: '2025-04-02T07:29:58Z'
data:
  '805': br805
```

5. Create Virtual Machine and bridge bind it with the NAD you created in the previous step.
   a. In the OpenShift console, go to Virtualization > VirtualMachines.
   b. In create virtual machines, select the Red Hat Enterprise Linux 9 VM template.
   c. In the Customize and create VirtualMachine, click Add network interface.
   d. In the Add network interface window, select the VLAN you created for the Network and click Save.
6. Open the newly created virtual machine and go to the Console tab.
7. Log in with your credentials.
8. Run the following commands to list the IP addresses:

   ```
   ip addr
   ```

   Get the interface name from the virtual machine console.
9. Create the file and update it will details:

   ```
   cat <<EOF >/etc/sysconfig/network-scripts/ifcfg-<Interface-name>
   BOOTPROTO=static
   DEVICE=<Interface-name>
   # HWADDR=$HW_ADDR
   ONBOOT=yes
   TYPE=Ethernet
   USERCTL=no
   IPADDR=<IP_ADDR>
   GATEWAY=<GATEWAY>
   NETMASK=<NETWMASK>
   DNS1=<DNS_IP>
   EOF
   ```

   You can get the values of Gateway, Netmask, DNS, VLAn from your network team. Validate the file:

   ```
   cat /etc/sysconfig/network-scripts/ifcfg-eth1
   ```

10. Run the following commands to assign the IP and make secondary as primary:

    ```
    sudo ip link set <interface-name> down
    sudo ip link set <interface-name> up
    sudo systemctl restart NetworkManager
    ```

    Wait for the IP to get assigned.
    If the multiple network interfaces exist, then disable the other one so that the secondary network interfaces are enabled and ping google.com to verify.

11. Validate the Network Manager status:
    a. List all connections:

       ```
       nmcli connection show
       ```

    b. Disable eh0 (primary connection) and enable eth1 (secondary connection):

       ```
       nmcli connection down <UUID>
       ```

    c. Show network status:

       ```
       nmcli general status
       ```

    d. Show active connections:

       ```
       nmcli connection show --active
       ```

       The secondary network is now enabled and primary default eth0 is disabled.
    e. Ping google.com to check traffic on the secondary network.

Provisioning additional networking

Provision additional networking for direct network access to a container or to a virtual machine.

Before you begin

As a prerequisite, deploy the Red Hat® OpenShift® Virtualization Operator within IBM Fusion HCI. For the procedure to deploy, see Deploying Red Hat OpenShift Virtualization Operator.

Procedure

1. Define the network range and the VLAN for which the external network traffic will traverse.
   a. Log in to IBM Fusion HCI user interface.
   b. Go to Networking page and go to VLAN tab.
   c. Click Add VLAN to define a new VLAN.
   d. Select the VLAN Type as `Virtualization VLAN`.
   e. Add the newly Defined VLAN to the Uplink ports on the high speed switches.
      In the Edit LINK slide out panel, select value for VLANs and Save.
2. Log in to OpenShift Container Platform web console.

3. Go to Administration > CustomResourceDefinitions.
4. Search for **nodenetworkconfigurationpolicies.nmstate.io** and click to open it.
5. In the Instances tab, view the NNCP policy created with the previously created VLAN ID. For example, if the created VLAN ID is "2222", the NNCP policy name is "br2222-bond0-policy".
6. Copy the bridge name. In this example, it is "br2222".
7. Select your project or namespace and go to Networking > NetworkAttachmentDefinitions.
8. In the NetworkAttachmentDefinitions, click Create Network Attachment Definition to create a new Network Attachment Definition specific to the desired namespace.
   You can reuse the same network attachment definitions across multiple namespaces.
9. Select Linux bridge from the Network Type drop-down list.
   Note: When creating the NAD, include only the bridge details. Do not add VLAN information, as this disrupts VM communication.
10. Add **br2222** in the Bridge name and click Create.
11. Check whether the Bridge Name is same as the one created by the node network policy. In this example, it is "br2222".
12. You can now attach a second network adapter to your virtual machine at the time of network configuration, and assign it to the defined network attachment definition. It allows access to the external network.

What to do next

Ensure you delete the secondary VLAN. For steps, see Secondary VLAN detaching.

- **Secondary VLAN detaching**
  Use these instructions to detach the secondary VLAN.

Secondary VLAN detaching

Use these instructions to detach the secondary VLAN.

If there are any NADs created for the secondary VLAN and are used as additional NIC for VMs, then the additional interface must be detached.

1. Log in to the OpenShift® Container Platform web console.
2. Go to Network > NetworkAttachmentDefinition.
3. Select the NAD and go to the YAML tab, and change the namespace to the appropriate namespace.
4. Click the ellipsis icon and select Delete NetworkAttachmentDefinition.
5. From the IBM Fusion HCI user interface, go to Infrastructure > Network.
6. Click Links tab.
   The default Link detail appears .
7. Edit the default link and remove the secondary VLAN. For steps, see Editing links.
8. Go to the VLANs tab and delete the VLAN that needs to be deleted. For steps, see Deleting VLANs.

Networking issues

A list of all troubleshooting and known issues that exist in the networking of IBM Fusion HCI.

Configuring the F5 load balancer

Problem statement
: You can use an external load balancer in place of the default load balancer to access the Red Hat® OpenShift® cluster after the installation of IBM Fusion HCI.

Resolution
: You can use node placement and toleration to place **router-default** pods to the infrastructure nodes on your environment, then you can configure external load balancer to route traffic to the infrastructure nodes.

  1. Run the following command to label worker nodes as infrastructure nodes. Use the correct compute nodes for your environment.

```
oc label nodes compute-1-ru5.rackd.mydomain.com node-role.kubernetes.io/infra=''
oc label nodes compute-1-ru6.rackd.mydomain.com node-role.kubernetes.io/infra=''
```

  2. Create a **default.yaml** file to place **router-default** pods to the infrastructure nodes.

```
apiVersion: operator.openshift.io/v1
kind: IngressController
metadata:
  name: default
  namespace: openshift-ingress-operator
spec:
  nodePlacement:
    nodeSelector:
      matchLabels:
        node-role.kubernetes.io/infra: ""
    tolerations:
    - effect: NoSchedule
      operator: Exists
  replicas: 2
status: {}
```

  3. Run the following command to apply the change.

```
oc apply -f default.yaml
```

4. Run the following command to verify that the `router-default` pods are running on the infrastructure nodes.

```
oc -n openshift-ingress get pod -o wide
```

Example output:

```
NAME                               READY   STATUS    RESTARTS   AGE   IP              NODE                                NOM
router-default-6c8f978c-9bd9j      1/1     Running   0          23m   172.20.240.29   compute-1-ru6.rackd.mydomain.com    <no
router-default-6c8f978c-zglxz      1/1     Running   0          23m   172.20.240.28   compute-1-ru5.rackd.mydomain.com    <no
```

5. Configure your external load balancer. For more information, see Configuring an external load balancer.

Missing network events that are raised during port down or up on a management switch

Problem statement

If a **cumulusLinkDown** event is observed on the IBM Fusion HCI user interface for any high-speed or management switches, then follow the steps that are mentioned in the following resolution section.

Resolution

Do the following workaround steps:

1. Go to Infrastructure > Network > Switches .
2. In the ellipsis menu of the switch, click Run Command.
3. Select net show interface all from the list and click Run.
4. Check whether the state of the port is **UP**, **DN**, or **ADMDN**.
   a. If the state of the port is **UP**, then the port is recovered and is up and running. In this scenario, the event is not a new entry and can be ignored.
   b. If the state of the port is **DN** or **ADMDN**, then contact IBM support to check the switch.

Management switch unavailability

Problem statement

A management switch is not available.

Resolution

Do the following steps to log in to the management switche and resolve the issue.

1. Run the following `oc` command to get the corresponding switch IPv6 or IPv4 address.

```
 oc get switches <switchName> -o yaml | grep switchIpadress
```

For example:

If the switch name is mgmt1-rackc, then run the following command:

```
 oc get switches mgmt1-rackc -o yaml | grep switchIpadress
```

2. Run the following `oc` command to get the switch login credentials from the switch secret.

```
oc get secrets <switchName>-secret -oyaml | grep defaultUserName
oc get secrets <switchName>-secret -oyaml | grep defaultUserPasswrd
```

For example:

```
 oc get secrets mgmt1-rackc-secret -oyaml | grep defaultUserName
oc get secrets mgmt1-rackc-secret -oyaml | grep defaultUserPasswrd
```

3. Run the following command to decode both the username and password, obtained from the previous step.

```
echo <username/password> | base64 -d
```

4. Use the username and password that is decoded from the previous step to login to the switch from one of the compute nodes. From the Red Hat OpenShift user interface, go to one of the compute node terminals.

```
ssh `<username>@<switchIpadress>`
```

5. SSH to log in to the switch and go to the root user.

```
sudo su
```

6. Check the values of `ClientAliveInterval` and `ClientAliveCountMax` in `sshd_config`.

```
root@mgmt1-rackc:mgmt:~$ cat  /etc/ssh/sshd_config | egrep "ClientAliveInterval|ClientAliveCountMax"
          #ClientAliveInterval 0
          #ClientAliveCountMax 3
```

7. Update /etc/ssh/sshd_config file with `ClientAliveInterval` as 600 and `ClientAliveCountMax` as 0, and save the file.

```
vi /etc/ssh/sshd_config
          "Remove "#" on both lines and change the values as given below and Save the file(Press Esc to enter Command
mode, and then type :wq to write and quit the file)"
          ClientAliveInterval 600
          ClientAliveCountMax 0
```

8. Check that the values of `ClientAliveInterval` and `ClientAliveCountMax` are correctly set.

```
root@mgmt1-rackc:mgmt:~$ cat  /etc/ssh/sshd_config | egrep "ClientAliveInterval|ClientAliveCountMax"
          ClientAliveInterval 600
          ClientAliveCountMax 0
          root@mgmt1-rackc:mgmt:~$
```

9. Restart the **sshd** service.

```
root@mgmt1-rackc:mgmt:~$ sudo systemctl restart sshd
```

10. Run the following command to clear the SSH sessions that belong to ISFUSER (most of the SSH session belongs to ISFUSER).

```
for i in `ps -aef | grep ssh | grep ISFUSER | awk {'print $2'}`; do echo $i; kill -9 $i; done
```

Switches moved to critical state after you power off TOR

Problem statement
: Switches are in critical state because the high availability cluster connectivity breaks between the racks.

Cause
: The switches from one rack do not share loopback Anycast IP to the spine and other switches.

Resolution
: Important: This issue and its workaround steps are applicable only to a high availability cluster.

1. Log in to all high-speed switches
2. Run the following `oc` command to get the corresponding switch IPv6 or IPv4 address.

```
oc get switches <switchName> -o yaml | grep switchIpadress
```

For example, if the switch name is hspeed1-rackc, then run the following command:

```
oc get switches hspeed1-rackc -o yaml | grep switchIpadress
```

3. Run the following `oc` command to get the switch login credentials from the switch secret.

```
oc get secrets <switchName>-secret -oyaml | grep defaultUserName
oc get secrets <switchName>-secret -oyaml | grep defaultUserPasswrd
```

For example, if the switch name is hspeed1-rackc, then run the following command:

```
oc get secrets hspeed1-rackc-secret -oyaml | grep defaultUserName
oc get secrets hspeed1-rackc-secret -oyaml | grep defaultUserPasswrd
```

4. Run the following command to decode both the username and password, obtained from the previous step.

```
echo <username/password> | base64 -d
```

5. Use the decoded username and password from the previous step to login to the switch from one of the compute nodes. From the Red Hat OpenShift user interface, go to one of the compute node terminals.

```
ssh `<username>@<switchIpadress>`
```

6. Run the following command to restart `frr` service on the switch.

```
sudo systemctl restart frr
```

Adding a VLAN fails on a expansion rack setup

Problem statement
: Adding a VLAN fails on a expansion rack setup with the combination of gen1 and gen2 racks.

Resolution
: If you face any issues during adding a VLAN on a expansion rack setup, then contact IBM support.

Network adapters validation fails

Problem statement
: The network adapter validation fails during replacing a network adapter.

Resolution
: As a resolution, ensure you update the kickstart file whenever you replaces a network adapter. Also, update the kickstart with the MAC address of the replaced network adapter.

Admin network in a critical state

Problem statement
: The admin network is in a critical state, showing an error on the IBM Fusion HCI user interface because pods are unable to communicate between sites.

Cause
: The submariner gateway pods are unable to establish a connection after Site1 is recovered, preventing the submariner from connecting successfully.

Diagnosis
: Follow the steps to diagnose through user interface:

1. Log in to the site2 OpenShift Container Platform user interface.
2. Go to Administration > CustomResourceDefinitions.
3. Search for MetroDR and select MetroDR from the list.
4. Go to Instances tab and select metrodrsite.
5. Go to YAML tab and check **submarinerMonitoringCommandOutput** under **metroDRSiteStatus**.
6. Check whether you see the following error for the node.

```
control-1-ru4.rackae1.mydomain   active 0 connections out of 1 are established
```

7. Note down the node name.

Follow the steps to diagnose through CLI:

1. Log in to the site2 OpenShift Container Platform user interface.
2. Run the following command.

   ```
   oc get mdr -o yaml
   ```

3. Check **submarinerMonitoringCommandOutput** under **metroDRSiteStatus** in the output.
4. Check whether you see the following error for the node.

   ```
   control-1-ru4.rackae1.mydomain   active 0 connections out of 1 are established
   ```

5. Note down the node name.

Resolution

Follow the steps to resolve the issue through user interface:

1. Log in to the OpenShift Container Platform user interface of the site2.
2. Go to Workloads > Pods
3. Select submariner-operator from the Project drop-down.
4. Click Manage columns icon which is present beside the search bar.
5. Select Node under the Additional columns.
   The pods details appears.
6. Restart or delete the submariner-gateway pod which is on the node name that you have noted from the diagnose steps.
7. The error resolves automatically after you applying the resolution steps. If the issue persists, contact IBM support.

Follow the steps to resolve the issue through CLI:

1. Log in to the OpenShift Container Platform user interface of the site 2.
2. Run the following command to get the list of submariner gateway pods.

   ```
   oc get pods -o wide -n submariner-operator | grep submariner-gateway
   ```

   Example output:

   ```
   submariner-gateway-jc6pt                  1/1     Running   1              16h   172.20.102.26   control-1-ru3.racka
   submariner-gateway-kpqpc                  1/1     Running   1              16h   172.20.102.25   control-1-ru2.racka
   submariner-gateway-mp7jt                  1/1     Running   2 (16h ago)    16h   172.20.102.27   control-1-ru4.racka
   ```

3. Run the following command to delete the gateway pod which is on the node name that you noted from the diagnose steps.

   ```
   oc delete pod <podname> -n submariner-operator
   ```

   Example output:

   ```
   oc delete pod submariner-gateway-mp7jt -n submariner-operator
   pod "submariner-gateway-mp7jt" deleted
   ```

4. The error resolves automatically after you applying the resolution steps. If the issue persists, contact IBM support.

Customer Replacement Units (CRU)

List of Customer Replacement units for your reference.

The list of Machine Type Models (MTMs) for IBM Fusion HCI and their CRU part numbers.

Dell hardware

The list of Machine Type Models (MTMs) for IBM Fusion HCI and their CRU part numbers.

Table 1. IBM Fusion HCI machine type models (MTMs)

Name	Machine type	Model
IBM Fusion HCI Compute/Storage Server	9155	D10
IBM Fusion HCI Compute-storage Server	9155	D14
IBM Fusion HCI 1GbE Switch	9155	S04
IBM Fusion HCI 200GbE High-Speed Switch	9155	S05
IBM Fusion HCI GPU Server	9155	DG1
IBM Fusion HCI GPU Server	9155	DG2
IBM Fusion HCI GPU Server	9155	DG3
IBM Fusion HCI GPU Server	9155	DG4
IBM Fusion HCI Service node	9155	DB1
IBM Fusion HCI Rack	9155	R42
IBM Fusion HCI KVM	9155	TF5

IBM Fusion HCI compute server and compute-storage server (9155-D10 and 9155-D14)

PART NUMBER	PART NAME
03MU287	7.68TB SSD U.2 FIPS Gen4 PCIe NVMe

PART NUMBER	PART NAME
03MU318	3.84TB SSD U.2 FIPS Gen4 PCIe NVMe
03MU300	1.6TB SSD U.2 FIPS Gen4 PCIe NVMe
03MU290	Power Supply 1100W Hot Plug Redundant Delta (D10)
03MU317	Power Supply 1400W Hot Plug Redundant Delta (D14)
03MU299	960GB SSD M.2 Gen4 PCIe

IBM Fusion HCI service node (9155-DB1)

PART NUMBER	PART NAME
03MU290	Power Supply 1100W Hot Plug Redundant Delta
03MU299	960GB SSD M.2 Gen4 PCIe

Lenovo hardware

The list of Machine Type Models (MTMs) for IBM Fusion HCI and their CRU part numbers.

Table 2. IBM Fusion HCI machine type models (MTMs)

Name	Machine type	Model
IBM Fusion HCI Compute/Storage Server	9155	C01
IBM Fusion HCI Compute-storage Server	9155	C05
IBM Fusion HCI Compute-only Server	9155	C00
IBM Fusion HCI Compute Server	9155	C04
IBM Fusion HCI Compute-storage Server	9155	C10
IBM Fusion HCI Compute-storage Server	9155	C14
IBM Fusion HCI 100GbE High-Speed Switch	9155	S01
IBM Fusion HCI 1GbE Switch	9155	S02
IBM Fusion HCI 1GbE Switch	9155	S04
IBM Fusion HCI 200GbE High-Speed Switch	9155	S05
IBM Fusion HCI GPU Server Gen 02	9155	G02
IBM Fusion HCI GPU Server Gen 03	9155	G03
IBM Fusion HCI Service node	9155	B01
IBM Fusion HCI Rack	9155	R42
IBM Fusion HCI KVM	9155	TF5

IBM Fusion HCI compute server (9155-C00 and 9155-C04) and IBM Fusion HCI compute or storage server (9155-C01 and 9155-C05)

PART NUMBER	PART NAME
39M5509	Power Cord, 2.8M
00WF660	2.5" HDD Filler
00WF663*	
01KP619	1U Front VGA Filler
02JK752	7.68TB NVMe PCIe4.0
02JJ452	1100W PT 1+1 Power Supply
02JJ988	1100AB-19 A 1100W PT 1+1 Power Supply
02YF631	1100W 230Vac/115Vac AC HS Power Supply
02YF634	1100W 230Vac/115Vac AC HS Power Supply
03HC902	1100W 230Vac/115Vac AC Hot Swap Power Supply
03KK301	1100W 230Vac/115Vac AC Hot Swap Power Supply
03KK302	1100W 230Vac/115Vac AC Hot Swap Power Supply
03KK326	1100W 230Vac/115Vac AC Hot Swap Power Supply

* Only IBM Fusion HCI compute server (9155-C00 and 9155-C04) part numbers.

IBM Fusion HCI compute server (9155-C10) and IBM Fusion HCI compute or storage server (9155-C14)

PART NUMBER	PART NAME
00AM463	Miscellaneous Parts Kit
00MP352	Alcohol Wipe
00WF660	ThinkSystem 1x1 2.5" HDD Filler
00WF663	ThinkSystem 2x2 2.5" HDD Filler
02JG989	ThinkSystem Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
02YF232	Power Supply Acbell 1100W 230Vac/115Vac AC Hot Swap
02YF233	Power Supply GreatWall 1100W 230Vac/115Vac AC Hot Swap
03KH389	Tool, Torx T20 and Torx T30
03LD899	Power Supply Acbel 1100W 230Vac/115Vac AC Hot Swap
03KH160	3.84TB NVMe SSD
03KH161	7.68TB NVMe SSD

IBM Fusion GPU server (9155-G02)

PART NUMBER	PART NAME
00WF660	2.5" HDD Filler
00WF662	2.5" HDD Filler
02JG989	Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
03GX687	2.5" 3.2TB SAS Hot Swap SSD

PART NUMBER	PART NAME
02JJ991	1800W 230Vac AC HS Power Supply
02YF613	1800W 230Vac AC HS Power Supply
02YF637	1800W 230Vac AC HS Power Supply
03KK305	1800W 230Vac AC HS Power Supply
03KK306	1800W 230Vac AC HS Power Supply

IBM Fusion GPU server (9155-G03)

PART NUMBER	PART NAME
02JG989	Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter
02YF226	2600W 230Vac AC HS Power Supply
02YF227	2600W 230Vac AC HS Power Supply

IBM Fusion HCI 100GbE high-speed switch (9155-S01)

PART NUMBER	PART NAME
39M5509	Power Cord, 2.8M
01PG798	1100W Power Supply
49Y7930	40GbE QSFP 1M cable
01FT722	100GbE AOC 3M cable
01FT739	100GbE QSFP 2M cable
01FT718	100GbE cbl 0.5M cable
01PG799	Fan module

IBM Fusion HCI 1GbE switch (9155-S02)

PART NUMBER	PART NAME
39M5509	Power Cord, 2.8M
02YF121	150W Power Supply
01AF039	1.5M CAT5e (blue) cable
01AF054	1.5M CAT5e (yellow) cable
02CL334	1.02M CAT5e (blue) cable
01KL755	1.0M CAT5e (yellow) cable
01AF044	0.6M CAT5e (blue) cable
01AF208	0.6M CAT5e (yellow) cable
01AF210	0.6M CAT5e (green) cable
01KL948	10GbE DAQ 1M cable

- **Removing and replacing Customer Replacement Units (CRU)**
  Removing and replacing CRU includes the guidelines and instructions that need to take care when you remove the CRU.

Removing and replacing Customer Replacement Units (CRU)

Removing and replacing CRU includes the guidelines and instructions that need to take care when you remove the CRU.

- **Removing and replacing the power supply unit**
  Remove the existing power supply unit of the server and replace it with a new one.
- **Replacing an NVMe disk**
  Follow the instructions to replace an NVMe disk.
- **Disconnecting and connecting power cords**
  Steps to disconnect and connect existing power cords to the server.

Removing and replacing the power supply unit

Remove the existing power supply unit of the server and replace it with a new one.

Before you begin

Prerequisites for removing the power supply unit

- The server is shipped with two power supply by default. In this case, the power supply is hot pluggable and redundant.
- Ensure that you power off the server and disconnect all the power cords and network cables before you remove the power supply unit.
- Ensure that you put the node into maintenance mode. For more about node maintenance, see Administering the node.
- Ensure that you install an additional hot-swap power supply to support redundancy.
- 

Ensure that you wear an ESD wrist strap when you remove the power supply unit.
- Ensure that you handle the component carefully and include more than one qualified person to perform the task if the component weight is more than 18 kg (40 lb).

Prerequisites for replacing the power supply unit

- The server is shipped with only one power supply by default. In this case, the power supply is non-hot swap.
- Ensure that you power off the server and disconnect all the power cords and network cables before you replace the power supply unit.
- Ensure that you put the node into maintenance mode. For more about node maintenance, see Administering the node.
- Ensure that you install an additional hot-swap power supply to support redundancy.
-  Ensure that you wear an ESD wrist strap when you replace the power supply unit.
- Ensure that you handle the component carefully and include more than one qualified person to perform the task if the component weight is more than 18 kg (40 lb).

About this task

Remove and replace the power supply unit in the specified order. For more information about the part numbers, see

Procedure

1. Remove the power supply unit as follows:
   a. Disconnect the power cord from the hot-swap power supply and the electrical outlet.
   b. Slide the release tab toward the handle and carefully pull the handle backwards at the same time to remove the hot-swap power supply out of the chassis.
2. Replace the power supply unit as follows:
   a. Take the new part out of the package and place it on a static-protective surface.
   b. Remove the power supply filler if it is installed.
   c. Place the new hot-swap power supply into the bay and push until it gets into position.
   d. Connect the power supply unit to a properly grounded electrical outlet.

What to do next

1. Install a power supply filler or a new power supply to cover the slot.
2. If you are instructed to return an old hot-swap power supply, follow all packaging instructions and use any packaging materials provided.
3. If the server is turned off, turn on the server. Ensure that both the power input LED and the power output LED on the power supply are lit, indicating that the power supply is operating properly.

Replacing an NVMe disk

Follow the instructions to replace an NVMe disk.

About this task

The procedure steps are applicable only for the Global Data Platform storage. For Data Foundation storage, see Disk replacement and OpenShift dedicated OSD failure recovery.

Procedure

1. Open the IBM Fusion HCI user interface.
2. Click the App Switcher icon in title bar and click Storage outbound arrow from the list.
   As soon as you click the Storage outbound arrow, the IBM Storage Scale user interface is displayed.
3. Click Sign In to login to the IBM Storage Scale user interface.
4. Click the Events icon and check whether there is any error event showing the bad disk.
5. Click Run Fix Procedure and follow the instructions to replace the bad disk.
   As soon as you click the Run Fix Procedure, the Fix Procedure: Replace Disks page is displayed.
6. Click Next.
7. Replace the disk physically and select the The disk has been replaced check box.
8. Click Finish.
   The message is displayed in the Fix Procedure: Replace Disks page:

   ```
   Successfully replaced the disks
   ```

9. Click Close.

What to do next

1. Ensure that the error event of a bad disk disappears from the Events page.
2. Ensure that the new disk shows up with the different serial number in the Physical Disks page.
3. Check the file system mount and physical disks in a core scale pod.
   - Run the OC command to check Scale core pod.

     ```
     oc get pod
     NAME READY STATUS RESTARTS AGE
     compute-0 2/2 Running 0 17d
     compute-1 2/2 Running 0 17d
     ```

```
compute-2 2/2 Running 0 17d
control-0 2/2 Running 0 17d
control-1 2/2 Running 0 17d
control-2 2/2 Running 0 17d
```

- Run the OC command to see all the containers in this pod.

```
oc describe pod/compute-2 -n ibm-spectrum-scale
```

- Run the following command to check the file system mount.

```
mmlsmount all
```

- Run the following commands to check the current physical disks health.

```
mmvdisk pdisk list --rg all --not-ok
mmvdisk: All pdisks are ok.
mmlsrecoverygroup rg1 -L --pdisk
declustered current allowable
recovery group arrays vdisks pdisks format version format version
----------------- ----------- ------ ------ -------------- --------------
rg1 1 25 12 5.1.5.0 5.1.5.0
declustered needs replace scrub background activity
array service vdisks pdisks spares threshold BER trim free space duration task progress priority
----------- ------- ------ ------ ------ --------- ------- ---- ---------- -------- -------------------------
DA1 no 25 12 1,3 1 enable no 326 GiB 14 days rebalance 2% low
n. active, declustered state,
pdisk total paths array free space remarks
----------------- ----------- ----------- ---------- -------
n001p001 1, 1 DA1 212 GiB ok
n001p002 1, 1 DA1 208 GiB ok
n002p001 1, 1 DA1 300 GiB ok
n002p002 1, 1 DA1 252 GiB ok
n003p001 1, 1 DA1 236 GiB ok
n003p002 1, 1 DA1 234 GiB ok
n004p001 1, 1 DA1 4552 GiB ok
n004p002 1, 1 DA1 512 GiB ok
n005p001 1, 1 DA1 232 GiB ok
n005p002 1, 1 DA1 266 GiB ok
n006p001 1, 1 DA1 250 GiB ok
n006p002 1, 1 DA1 222 GiB ok
```

Disconnecting and connecting power cords

Steps to disconnect and connect existing power cords to the server.

Before you begin

Prerequisites for removing the power supply unit

- The server is shipped with two power supply by default. In this case, the power supply is hot pluggable and redundant.
- Ensure that you power off the server and disconnect all the power cords and network cables before you remove the power supply unit.
- Ensure that you put the node into maintenance mode. For more about node maintenance, see Administering the node.
- Ensure that you install an additional hot-swap power supply to support redundancy.
- 

Ensure that you wear an ESD wrist strap when you remove the power supply unit.

- Ensure that you handle the component carefully and include more than one qualified person to perform the task if the component weight is more than 18 kg (40 lb).

Prerequisites for replacing the power supply unit

- The server is shipped with only one power supply by default. In this case, the power supply is non-hot swap.
- Ensure that you power off the server and disconnect all the power cords and network cables before you replace the power supply unit.
- Ensure that you put the node into maintenance mode. For more about node maintenance, see Administering the node.
- Ensure that you install an additional hot-swap power supply to support redundancy.
- Ensure that you wear an ESD wrist strap when you replace the power supply unit.
- Ensure that you handle the component carefully and include more than one qualified person to perform the task if the component weight is more than 18 kg (40 lb).

Procedure

1. Important:
   - Ensure that all power cables are properly connected when connecting power to the product. Connect all power cords to a properly wired and grounded electrical outlet for the racks with AC power. Ensure that the outlet supplies proper voltage and phase rotation according to the PDU or system rating plate.
   - If IBM supplied the power cords, then connect power to this unit only with the IBM provided power cord.
   - The product might be equipped with multiple power cords.

   Disconnect and connect the power cords in the specified order.
   a. Ensure that the mating connector of the power supply is free of any dirt or obstacles.
   b. For AC power, remove the power cords from the outlets.
   c. Unplug the power cord from the power supply.
2. Connect the power cord as follows:
   a. Connect the power cord to the power supply.
   b. For AC power, attach the power cords to the outlets.

Observability

Observability in IBM Fusion enables centralized, multi-cluster monitoring that uses open source technologies like Prometheus, Thanos, and Grafana. This solution collects metrics from managed clusters, provides global querying and long-term storage, and delivers unified dashboards for visualization and alerting.

- **AI-enabled observability with Red Hat OpenShift Lightspeed**
  Simplify observability across clusters through smart agents.
- **Monitoring IBM Fusion HCI**
  IBM Fusion HCI provides multiple ways to monitor the health of the hardware and software.
- **IBM Fusion metrics**
  This section describes the metrics that are used in IBM Fusion.
- **Enabling observability for IBM Fusion clusters**
  Use these instructions to enable multi-cluster observability for IBM Fusion clusters.
- **IBM Fusion Call Home Telemetry**
  IBM Fusion Call Home Telemetry enhances supportability by providing IBM with valuable insights into system configuration and performance. Telemetry data is collected from various components and securely transmitted to IBM, helping support teams proactively identify issues and optimize system behavior based on real-world usage.

AI-enabled observability with Red Hat OpenShift Lightspeed

Simplify observability across clusters through smart agents.

Red Hat OpenShift® Lightspeed provides an AI-based chat window that understands OpenShift and the current cluster state. When integrated through the Model Context Protocol (MCP), the chat window retrieves live data from the cluster when configured. As a result, responses reflect the current state rather than static documentation. You interact exclusively through the chat window: you submit a query, Lightspeed uses MCP to retrieve the required data, and then explains and summarizes the results. This approach provides a single location for monitoring health, troubleshooting issues, and reviewing summarized information without switching between the OpenShift console and other tools.

One place to ask questions
: Use the Lightspeed chat window, either in the OpenShift console or from an IDE through MCP, to ask about cluster health, workloads, and issues. The chat window uses MCP to access live cluster state and, when available, metrics and alerts, providing a consolidated conversational view.

On-demand debugging and analysis
: Ask questions such as *What's wrong with the cluster?*, *Why is this failing?*, or *Show me what's running in this namespace*. The LLM retrieves current data through MCP and combines it with the knowledge of OpenShift to provide focused, context-aware responses without running anything in the background.

Summarized insights
: Request high-level overviews in plain language, such as *Summarize what's unhealthy* or *What's going on in this project?* The chat window used MCP to retrieve the relevant resources and return concise summaries, enabling quick triage and deeper investigation only when required.

Integration with existing tools
: MCP complements existing monitoring tools rather than replacing them by adding a chat window on top. Continue to use Grafana for dashboards and Prometheus for metrics, while using Lightspeed for quick queries, summaries, and guided troubleshooting based on the same data.

Monitoring IBM Fusion HCI

IBM Fusion HCI provides multiple ways to monitor the health of the hardware and software.

- **Monitoring hardware from IBM Fusion HCI user interface**
  The IBM Fusion HCI user interface provides an enhanced hardware monitoring experience. With hardware monitoring, you can quickly determine a hardware component in the IBM Fusion HCI that needs your attention.
- **Monitoring and logging by using any monitoring tool**
  IBM Fusion provides multiple ways to monitor the health of the hardware and software.

Monitoring hardware from IBM Fusion HCI user interface

The IBM Fusion HCI user interface provides an enhanced hardware monitoring experience. With hardware monitoring, you can quickly determine a hardware component in the IBM Fusion HCI that needs your attention.

Hardware monitoring provides overall and individual health and inventory information of servers and switches, and their components such as CPUs, DIMMs, Storage Drives, Adapters, Fans, PSUs.

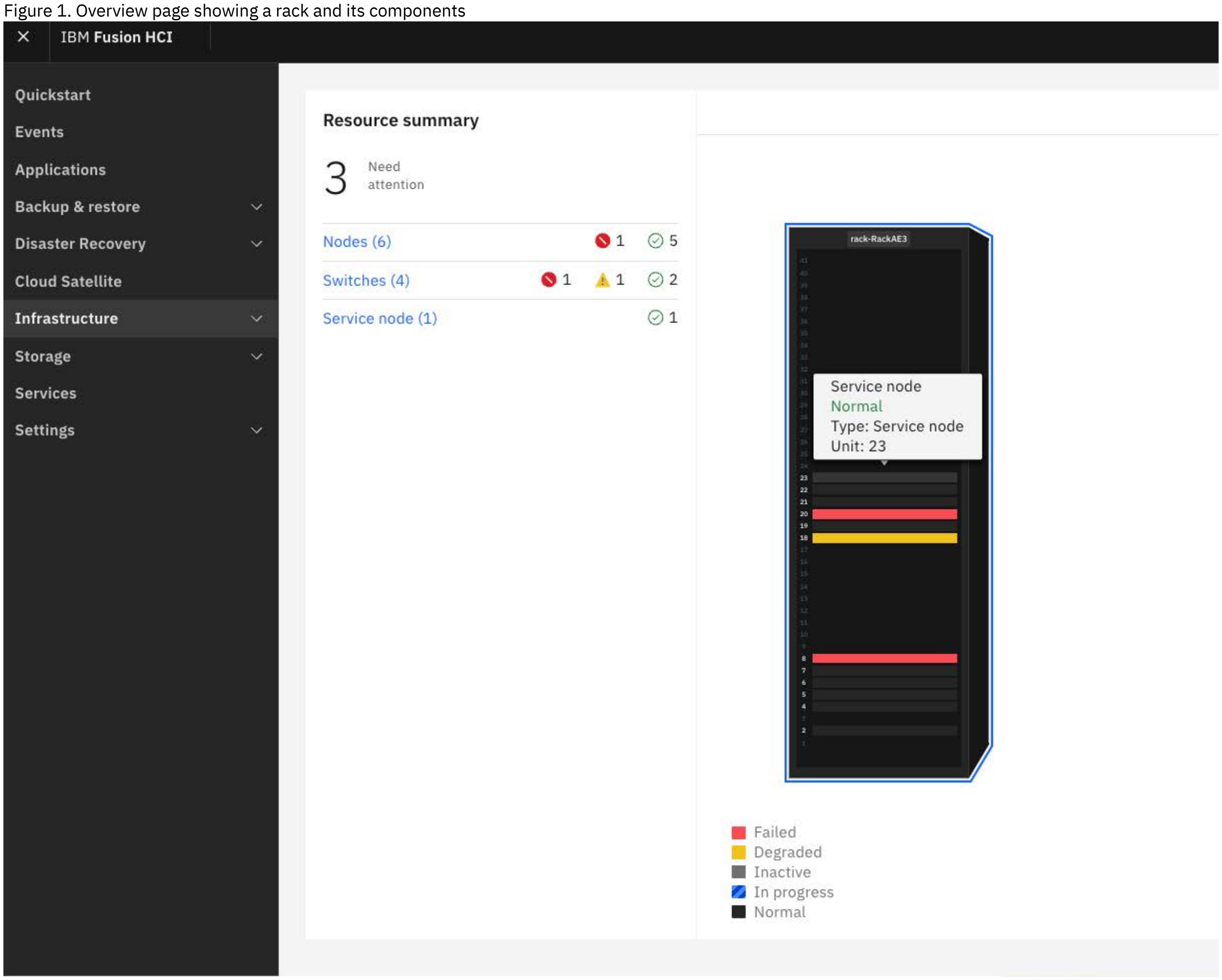
Go to Infrastructure > Overview in the IBM Fusion HCI user interface to view the graphical view of the hardware appliance. Under the Resource summary section, you can find the total number of nodes, switches, and service node along with their health statuses. Hover over a graphical view of the hardware appliance to identify units in the rack. Also, you can zoom in and out to easily focus on specific sections of the rack for detailed inspection and use Add rack to add extra racks and useful when there are more than three racks to look at.

- For node, it shows the name of the hardware, the health status, the type of node, inventory, and the rack unit.
- For switch, it shows the name of hardware, the health status, the type of switch, inventory, and the rack unit.
- For service node, it shows the name hardware, the health status, the type of node, inventory, and the rack unit.

In a graphical view of the IBM Fusion HCI rack, the color of the hardware component represents the health status. Additionally, you can view the different colors and their prominence under the graphical view of the hardware appliance. The nodes and switches are shown as per their appropriate locations in the actual physical rack. For more information on rack unit numbers and location, see Hardware overview.

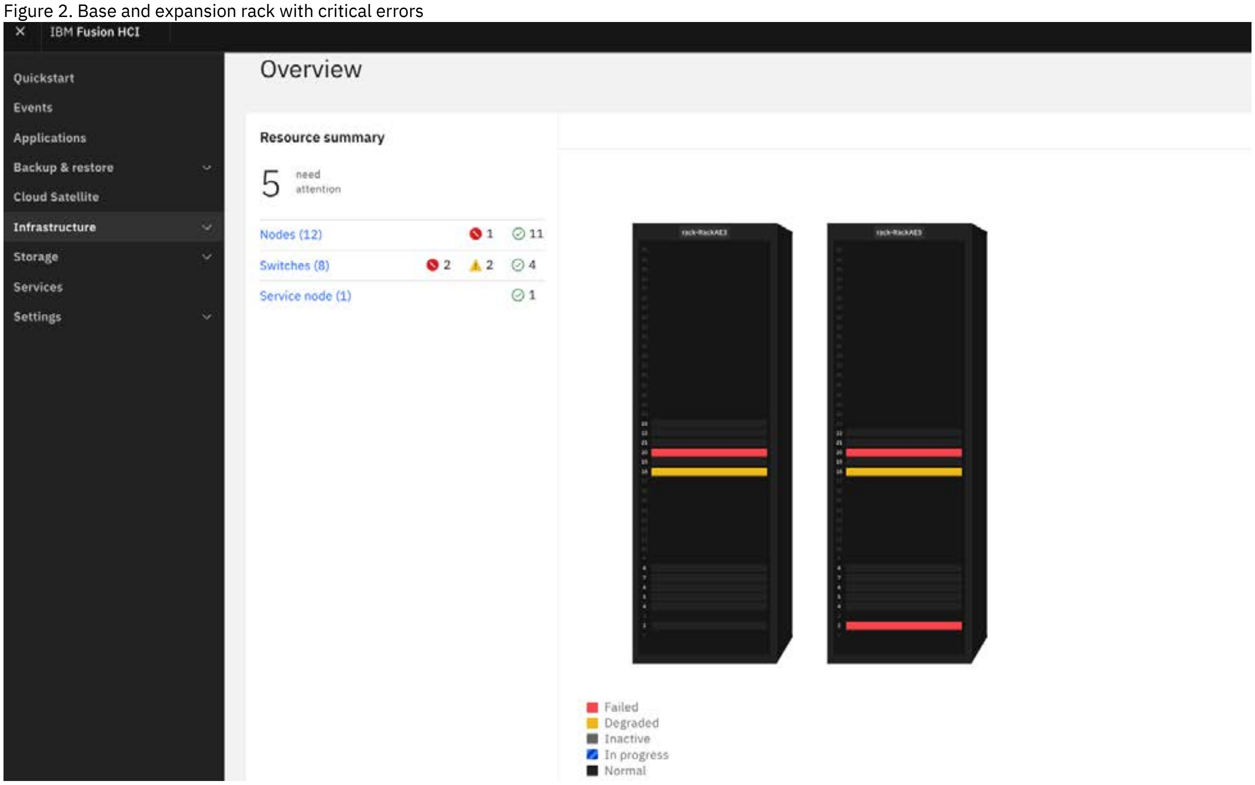
The following picture shows a sample Overview dashboard page with a single rack. In this example, the rack shows degraded state for management switch in Rack Unit 18 and critical state for a storage node at Rack Unit 8 and high speed switch at Rack Unit 20.

Figure 1. Overview page showing a rack and its components



The following figure shows a base and expansion rack setup. In this example, the power supply is lost for management switch in base rack at Rack Unit 18 and storage node in expansion rack at Rack Unit 2:

Figure 2. Base and expansion rack with critical errors



You can identify the node that has a hardware issue based on the legend displayed in the IBM Fusion HCI user interface:

- The color green indicates that the status of the hardware is in normal and healthy state.
- The color red indicates critical errors in the hardware and needs attention.
- The color gray indicates that the hardware is in inactive and power off state.
- The color blue with diagonal stripes indicates that an action is in progress on the hardware, such as power up, power down, or a firmware upgrade.
- The color yellow denotes warning.

For more information about and node and switch monitoring, see Node monitoring and Switch monitoring.

- **Node monitoring**
  Node monitoring provides continuous tracking details of the health, performance, and availability of nodes within a hardware appliance.
- **Switch monitoring**
  Switch monitoring provides continuous tracking details of the health, performance, and availability of switches within a hardware appliance. With switch monitoring, you can track switch status and quickly determine used and unused ports, fans, loosen or unseated connection details of the switch in the IBM Fusion HCI. This topic provides you instructions and guidelines to monitor switches from the Overview dashboard page.

Node monitoring

Node monitoring provides continuous tracking details of the health, performance, and availability of nodes within a hardware appliance.

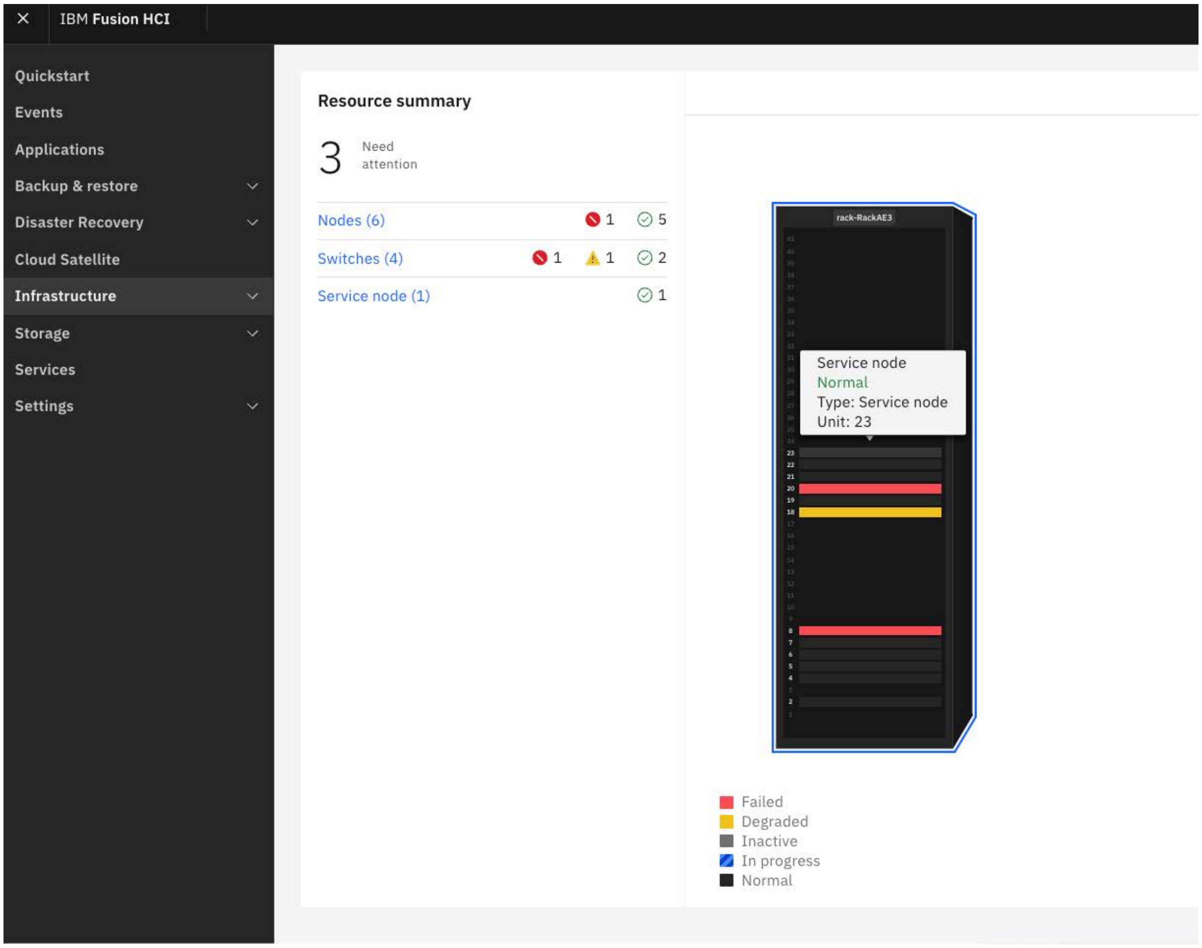
It helps administrators quickly identify the following issues:

- Active and inactive drives
- Invalid configurations
- Loose or unseated hardware connections
  It provides real-time insights into node status, allowing for proactive maintenance and efficient troubleshooting.

Use these instructions and guidelines to monitor nodes from the Overview dashboard page.

1. Go to Infrastructure > Overview in the IBM Fusion HCI user interface to view the graphical view of the hardware appliance.
   Under the Resource summary section, you can find the following details:
   - The total number of nodes
   - The current health status of each node
   - A graphical representation of the hardware appliance

   Figure 1. Example Overview page showing error for a node



In the graphical view of the hardware appliance, the color indicates the health status of a node.

2. Hover over a graphical view of the hardware to identify node details.
   For node or service node, it shows the name of the hardware, the health status, the type of node, and the rack unit.
3. Go through the color indicators and decode their statuses.
4. Fix errors, failures, and warnings based on the guidance.
5. After all the errors and failures are fixed, go to the Overview dashboard page and check the health status of problematic nodes. For more information about nodes, see Node or service node details.

Understanding color indicators and decoding status

Node or service node in Green color
: It indicates that the node or service node is in a healthy or normal state, and no action is required.

Node or service node in Red or Yellow color
: It indicates that the node is failed or in a degraded state. Do the following steps to resolve the issue:

- Click the component that is in red or yellow color.
  A slide out window is displayed with the Hardware status, type, firmware, S/N, rack, and rack unit details. For more information about nodes, see Node or service node details.
- Go through the details in the slide out window. If you want more information about the node, click View full details.
  It opens Nodes page that includes the front, rear, and inside view of the node, recent events, and additional details of the node and its internal components.
- Select Front in the Components section to see front view of the node and hover over a graphical view to check internal components such as used and unused drive slots along with their connection status.
  To debug further, click the internal component that is in red color. It opens a new slide out pane for drives with more details such as name, slot, type, status, total capacity (GB), and serial number.
- Select Back in the Components section to see back view of the node and hover over a graphical view to check internal components such as used and unused adapter slots along with their invalid configuration and loosen or unseated connection details.
  To debug further, click the internal component that is in red or yellow color. It opens a new slide out pane for adapters with more details such as slot, port, type, speed, status, network address, adapter, bond, and connected to.
- Select Internal in the Components section to see internal view of the node and hover over a graphical view to check internal components such as used and unused DIMM slots, CPUs, and fans status.
  To debug further, click the internal component that is in red or yellow color. It opens a new slide out pane with more details. For example, for a DIMM, you can see details such as slot, state, capacity, memory type, and serial number.
- Click View table to check all internal components and their status in the table format.

It opens Components table page that includes Storage Drives, OS Drives, Ports, CPUs, DIMMs, Fans, and PSUs in seven tab pages. For more information about node details, see Node or service node details.

- Go through the Recent events section to understand the error.
  Note: The recent events pane includes details of the last five events that occurred for the node.
- Click View all to go to the events page and view all recent events on the hardware component.
  The BMYxxx code and the error message inform you about the error and possible corrective actions.

- Go through the Details section to get more details of the node such as type, model, S/N, IP address, rack, rack unit, firmware, architecture, CPU cores, frequency, memory, energy consumption, and temperature.
  Important: Click Details under Firmware in the Details section to get more details about the firmware versions.
- Click Actions drop down to download the logs of the node and to do power operations, restarting the node, and node maintenance activities. For more information, see Administering nodes and racks.
- To diagnose and take corrective action, try the following options:
  - Check whether the hardware is in the correct position. To view the rack positions of the nodes, see Hardware overview.
  - Check the BMYxxx errors. For more information about individual BMYxxx codes and their corrective action, see Compute events and error codes.
    If you are not able to fix the issue, contact IBM support.

Node or service node in Gray color
: It indicates that the node is in a disabled or powered off state.
  To power on or power off the hardware, see *Node power operations* section in Administering nodes and racks.

Node or service node in color blue with diagonal stripes
: It indicates that an action is in progress on the hardware, such as power on and power down.
  For more information about the actions on a node hardware, see the following links:

  - Power operations. See Administering nodes and racks.
  - Configuration expansion. See Scaling the cluster. Here, you can see the procedure to add nodes to the cluster or add storage to a node. To add a rack, see Adding expansion racks.

- **Node or service node details**
  The Nodes page in the IBM Fusion HCI user interface provides information about nodes and their health status.

Node or service node details

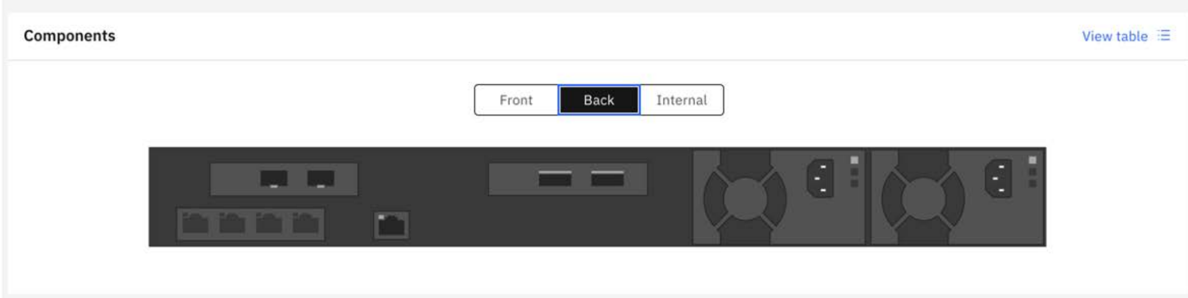
The Nodes page in the IBM Fusion HCI user interface provides information about nodes and their health status.

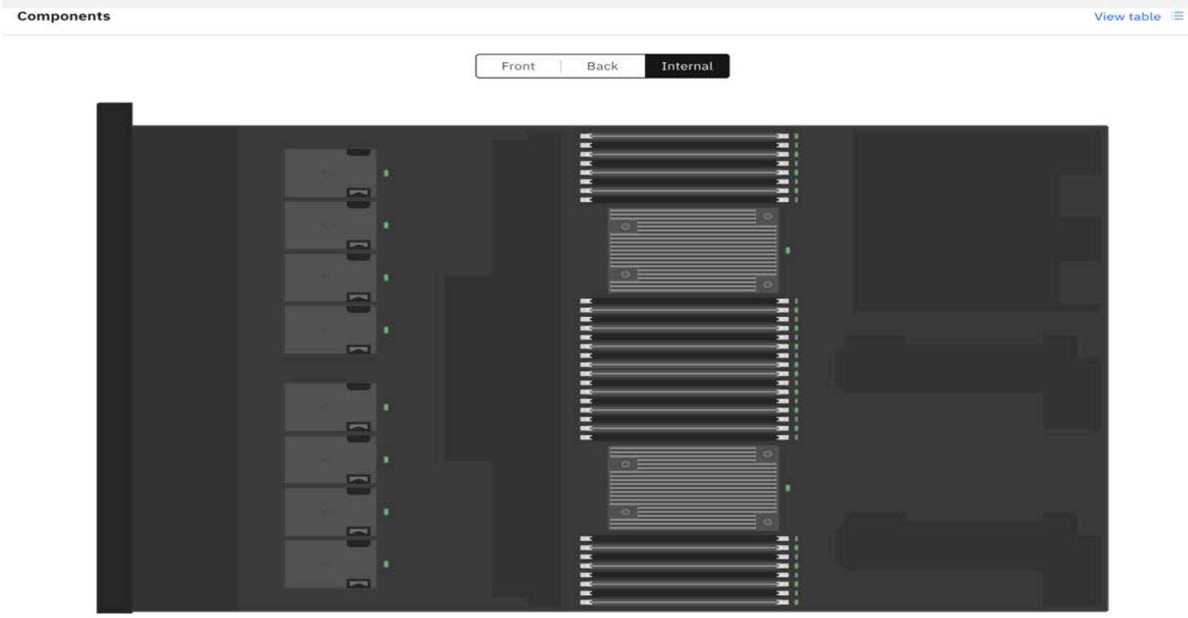
From the IBM Fusion HCI menu, click Infrastructure > Nodes. The Nodes page provides the details of the nodes in a table format. The column headers that are displayed by default in the table are Name, Hardware status, Type, Firmware level, Serial number, Rack name, Rack Unit number, CPU cores, and Memory (GB) details. Use the Search field to filter and find records. The settings icon helps you to customize the column headings of these tables. Decide the number of records per page by using the Items per page drop-down list of the table. You can traverse to the previous or next page views. Select a number from the drop-down list of the page numbers to jump to a specific page view.

Click the node name link that you want to manage. It opens Nodes page that includes the front, rear, and inside graphical view, recent events, and other details of the node and its internal components. Select Front, Back, and Internal in the Components section to see front, back, and inside views of the node and hover over a graphical view to check internal components and their status. To debug further, click the internal component that is in red color. It opens a new slide out pane and provides more details about the drives and adapters.

The following picture shows an example of a front and rear views of the Lenovo compute storage node.

Figure 1. Compute storage node





Click View table to check all internal components and its status in the table format. It opens Components table page that includes Storage Drives, OS Drives, Ports, CPUs, DIMMs, Fans, and PSUs in seven tabs.

Component	Details
Storage drives	It displays a list of storage drives. The column headers that are displayed by default in the table are Name, Slot, Status, Type, and Total Capacity (GB).
OS drives	It displays a list of OS drives. The column headers that are displayed by default in the table are Name, Slot, Status, Type, and Total Capacity (GB).
Ports	It displays a list of ports. The column headers that are displayed by default in the table are Slot, Status, Speed (Gbps), Network address, Adapter, Bond, and Connected to details.
CPUs	It displays a list of CPUs. The column headers that are displayed by default in the table are Name, Health, State, Cores, Threads, and Clock speed (MHz).
DIMMs	It displays a list of DIMMs. The column headers that are displayed by default in the table are Slot, Status, Serial number, Type, and Capacity.
Fans	It displays a list of Fans. The column headers that are displayed by default in the table are Name, Health, and Speed (RPM).
PSUs	It displays a list of PSUs. The column headers that are displayed by default in the table are Name, Status, Capacity (watts), Manufacturer, Model, Part number, and Firmware.

Click Actions drop down to download the logs of the node and to do power operations, restarting the node, and node maintenance activities. For more information, see Administering nodes and racks.

Click View all in the Recent events panel to go to the events page and view all recent events on the hardware component.

Note: The recent events pane includes details of the last five events that occurred for the node.

If you want more information about the node, go through Details section to get more details of the node as follows:

Node details	Description
Type	The type of the node (the options are Compute only, Compute storage, AFM, or GPU).
Model	The model number of the node.
S/N	The hardware serial number.
IP address	This is IP address of the BMC of the node and not the Host/OS/payload.

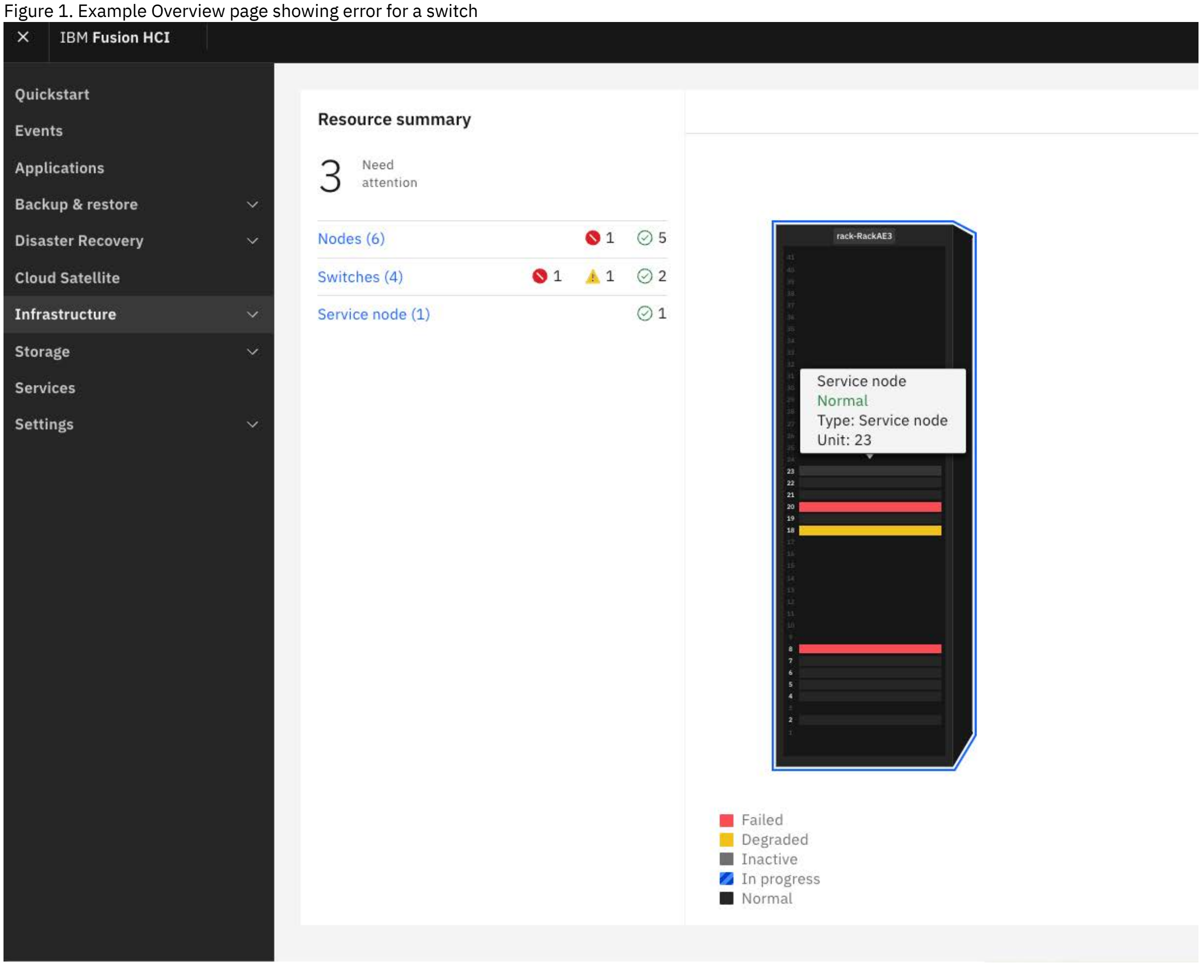
Node details	Description
Rack	The name of the rack.
Rack unit	The rack position of the node in the rack.
Host name	The host name of the rack.
Firmware	The firmware version of the node. Click View all to find the detailed firmware versions of the hardware components.
Architecture	The architecture details of the node.
CPU cores	The number of CPU in cores.
Frequency	Each CPU frequency of the node.
Memory	The amount of memory in GB.
Energy consumption	The energy that is used by the CPU and Server in watts.
Temperatures	Ambient CPU 1, CPU 2, and Exhaust temperature in Centigrade.

Switch monitoring

Switch monitoring provides continuous tracking details of the health, performance, and availability of switches within a hardware appliance. With switch monitoring, you can track switch status and quickly determine used and unused ports, fans, loosen or unseated connection details of the switch in the IBM Fusion HCI. This topic provides you instructions and guidelines to monitor switches from the Overview dashboard page.

1. Go to Infrastructure > Overview in the IBM Fusion HCI user interface to view the graphical view of the hardware appliance.
   Under the Resource summary section, you can find the total number of switches along with their health statuses.

   Figure 1. Example Overview page showing error for a switch

   In the graphical view of the hardware appliance, the color indicates the health status of a node.
2. Hover over a graphical view of the hardware to identify switch details.
   For switch, it shows the name of the hardware, the health status, the type of switch, and the rack unit.
3. Go through the color indicators and decode their statuses.
4. Fix errors, failures, and warnings based on the guidance.
5. After all the errors and failures are fixed, go to the Overview dashboard page and check the health status of problematic nodes. For more information about nodes, see Switch details.

Understanding color indicators and decoding status

Important: The color gray does not exist for the switches as failed, degraded, or normal are the available states.

Switch in Green color
: It indicates that the switch is in a healthy or normal state, and no action is required.

Switch in Red or Yellow color
: It indicates that the switch is failed or in a degraded state. Do the following steps to resolve the issue:

  - Click the component that is in red or yellow color.
    The slide out pane is displayed with the Hardware status, type, firmware, s/n, rack, and rack unit.
  - Go through the details in the slide out pane. If you want more information about the switch, click View full details. Alternatively, go to the Network page > Switches tab and click the switch name link.
    It opens Network page that includes the graphical view, recent events, and other details of the switch and its internal components.
  - Select Front in the Components section to see front view of the switch and hover over a graphical view to check internal components status such as used and unused port slots along with their working status.
  - Select Back in the Components section to see back view of the switch and hover over a graphical view to check internal components used and unused fan slots along with their working status.
    To debug further, click the internal components such as fans and PSUs that are in red or yellow color. It opens a new slide out pane for fans and PSUs with more details such as slot, speed, and state.
  - Click View table to check all internal components and its status in the table format.
    It opens Components table page that includes Ports, Fans, and Power supply in three tab pages. For network details, see [Switch details](#).
  - Go through the Recent events section to understand the error.
  - Click View all to go to the events page and view all recent events on the hardware component.
    The BMYxxx code and the error message informs you about the error.
  - Go through the Details section to get more details of the switch such as type, model, IBM S/N, manufacture S/N, rack unit, rack, and firmware.
  - To diagnose and take corrective action, try the following options:
    - Check whether the hardware is in the correct position. To view the rack positions of the switch, see Hardware overview.
    - Check the BMYxxx errors. For more information about individual BMYxxx network codes and their corrective action, see Network events error codes.
      If you are not able to fix the issue, contact IBM support.

Switch in color blue with diagonal stripes
: It indicates that a firmware upgrade is in progress. See Upgrading switch firmware.

- **Switch details**
  The Network page in the IBM Fusion user interface provides information about switches, VLANs, and links in the system.

Switch details

The Network page in the IBM Fusion user interface provides information about switches, VLANs, and links in the system.

From the IBM Fusion HCI menu, click Infrastructure > Network. The Network page displays Switches, Links, and VLAN in three tab pages. Click each tab to view the respective details in a table format. Use the Search field to filter and find records. The settings icon helps you to customize the column headings of these tables. Decide the number of records per page by using the Items per page drop-down list of the table. You can traverse to the previous or next page views. Select a number from the drop-down list of the page numbers to jump to a specific page view.

Switches tab page
: The Switches is the default tab page that gets displayed when you click Infrastructure > Network. It displays a list of high speed, internal management, and spine switches (for high-availability cluster and expansion cluster) in a table format. The column headers that are displayed by default in the table are Name, Hardware status, Type, Model, Firmware, IBM S/N, Rack, and Rack unit. Click  (Run command icon) to run a command to work with the switch. For more information about run command, see Running commands for switches. You cannot delete a switch from this tab page.

  Switch details
  : Click the switch name link that you want to manage. It opens the Network page that includes the graphical view (front and back), recent events, and other details of the switch and its internal components and statuses. Hover over a graphical view of the switch to check internal components and status. To debug further, click the internal component that is in a critical or degraded state. It opens a new slide out pane for ports with more details such as port, speed, and status.

    The following images shows examples of management and high-speed switches of the Lenovo hardware.

    Figure 1. Management switch

    Figure 2. High-speed switch

Click the View table to check all internal components and its status in the table format. It opens Components table page that includes Ports, Fans, and Power supply in three tabs.

Component	Details
Ports	It displays a list of ports. The column headers that are displayed by default in the table are Port, Status, and Speed (Gbps).
Fans	It displays the status of the fans of the switch.
Power supply	It displays the status of the power supply of the switch.

Click View all in the Recent events panel to go to the events page and view all recent events on the hardware component.
Note: The recent events pane includes details of the last five events that occurred for the switch.
If you want more information about the switch, go through Details section to get more details of the switch.

	Description
Type	The type of the switch (the options are Spine, Management, and High-speed).
Model	The model number of the switch.
IBM S/N	The hardware serial number.
Manufacture S/N	The IPv6 address of the switch.
Rack unit	The rack unit position of the switch in the rack.
Rack	The name of the rack.
Firmware	The firmware version of the switch.

Click Run command to run a command to work with the switch. For more information about run command, see Running commands for switches.

Links tab page

Click Links tab to view a list of all available links in a table format. The column headers that are displayed by default are Name, Type, Port number, Transceiver, Aggregation, VLANs, and Native VLAN. To add a new link, click Create link. For more information about adding new link, see Adding links. Click edit icon to edit the link or delete icon to delete it.

VLANs tab page

Click VLAN tab to view a list of all available VLANs in a table format. The column headers that are displayed by default in the table are Name, Type, Status, ID, and Multiple links. To add a new VLAN, click Create VLAN. For more information about creating VLAN, see Adding VLANs. Click edit icon to edit the VLAN or delete icon to delete it. If a VLAN is associated to a link, then you cannot delete it.

- **Running commands for switches**
  As part of your network configuration, you can run a command to work with the switches. The Network page in the IBM Fusion user interface provides information about switches in the system.

Running commands for switches

As part of your network configuration, you can run a command to work with the switches. The Network page in the IBM Fusion user interface provides information about switches in the system.

Procedure

1. Click Infrastructure > Network.
   The Network page is displayed.
2. Click Switches to display the Switches section.
3. Determine the switch record that you want to work with.
4. Click Run command.
   The Run command pane is displayed.
5. Click Command drop-down list to view the available command options.
6. Select a command option from the list.
7. Click Run to apply the command to the switch.
   A window with the command information is displayed. If the command information is extensive, click Show more or Show less to expand or collapse the selected option to view more information about the command. You can also use the Copy to clipboard option to copy the command information for reference.
8. Click Done to complete the changes.
   You can see the updated record in the Switches section of the Network page. For more information, see Switch details.
9. Optional: Click Reset if you want to undo the current changes.

Monitoring and logging by using any monitoring tool

IBM Fusion provides multiple ways to monitor the health of the hardware and software.

- Events are fed into the Red Hat® OpenShift® event manager
  The following example IBM Fusion events for your reference:
  - Hardware events for IBM Fusion HCI, including switches and nodes
  - Events that are related to the data services of IBM Fusion, such as backup and restore

  As IBM Fusion feeds events into Red Hat OpenShift event manager, they show up in the integration that you use to monitor OpenShift events.

- Metrics are fed into Prometheus
  IBM Fusion feeds metrics into a Red Hat OpenShift Prometheus instance.

  The advantage is that they are available in any monitoring tool you are using with Prometheus, such as Grafana. IBM Fusion provides a set of default Grafana dashboards and an access link to the interface.
- IBM Fusion provides logs for its services
- IBM Fusion provides a user interface that shows the status of the system
- IBM Fusion provides CRs that reflect the status of the system

Events and metrics provide observability that can be fed into other monitoring systems, such as Grafana and ELK.

Important:

- IBM Fusion does not install Grafana and ELK as part of its installation.
- If you already have monitoring tools that you use in your environment, then you can feed IBM Fusion observability into these existing tools. For example, if you use Grafana to monitor your OpenShift environment, then you can import metrics and dashboards of IBM Fusion into its instance.

Monitoring

The metrics are collected from storage, network, and compute components and sent to Prometheus, which is then propagated to Grafana for visualization.

Follow the steps to access IBM Fusion information in Grafana. After Grafana is installed and Prometheus data source is configured, you can import dashboards of IBM Fusion with the following steps:
Important:

- You must have a working Grafana instance.
- The OpenShift Container Platform version must be at 4.14 or later.
- The following steps 1 - 5 explain how to configure Prometheus data sources. If it is already configured, then you can directly go to step 6.

Important: Ensure that you change the namespace or project before following the steps. Run the following command to change the namespace.

```
oc project <your-grafana-namespace>
```

1. Create a ServiceAccount associated with your Grafana, and replace the namespace.

   ```
   oc create serviceaccount grafana -n <your-grafana-namespace>
   ```

2. Create a Role Binding to a ServiceAccount that is associated with your data source in Grafana, and replace the namespace. So, that can use the Prometheus cluster.

   ```
   oc create clusterrolebinding grafana-cluster-monitoring-view \
     --clusterrole=cluster-monitoring-view \
     --serviceaccount=<your-grafana-namespace>:grafana
   ```

3. Create a secret of type `service-account-token` by using the following sample YAML.

   ```
   apiVersion: v1
   kind: Secret
   type: kubernetes.io/service-account-token
   metadata:
     name: grafana-secret
     annotations:
       kubernetes.io/service-account.name: grafana
   ```

4. Run the following command to get a token.

   ```
   oc get secret grafana-secret -ojsonpath='{.data.token}' | base64 -d
   ```

5. Configure Grafana with a Prometheus data source and authentication.
   a. Log in to Grafana user interface.
   b. From the menu, go to Configurations and select Data Sources.
   c. Click Add data sources and select Prometheus.
   d. Update the following values:

      URL
         Run the following command to get the address of the Cluster Prometheus Route.

         ```
         oc get route prometheus-k8s -n openshift-monitoring
         ```

         Enable Skip TLS Verify parameter under Auth section.

      Custom HTTP Headers
         ```
         HTTP Method: GET
         Header: Authorization
         Value: bearer <token-from-step-4>
         ```

6. Download the `JSON` file for each of the dashboard:
   - For network, compute, and storage (Global Data Platform) `JSON` files, see https://www.ibm.com/support/pages/node/7027780.
7. Import the files in the Grafana dashboard. For the procedure to import files in Grafana dashboard, see Importing files.
   Note: Select Upload JSON file option and upload `JSON` downloaded in step 6.
8. Verify whether that Grafana dashboard is imported correctly.
9. Click Dashboards menu and choose the newly imported dashboard.

Viewing dashboard

In Grafana, you can view different dashboards for storage, network, and compute wherein the following metrics are captured for each component and represented graphically:

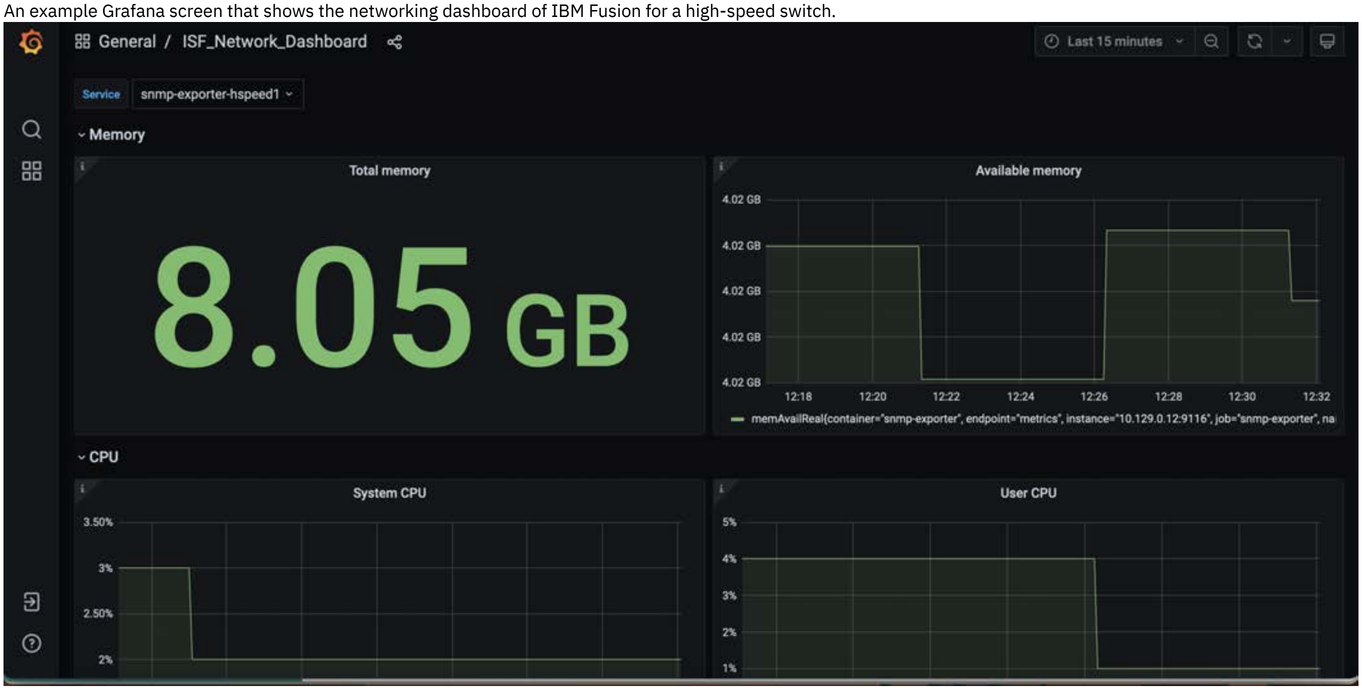
The following metrics are captured for each of the components:

- **Network**
  - High-speed switches
  - Management switches
  - Spine switches
- **Compute**
  - Energy status
  - Thermal status

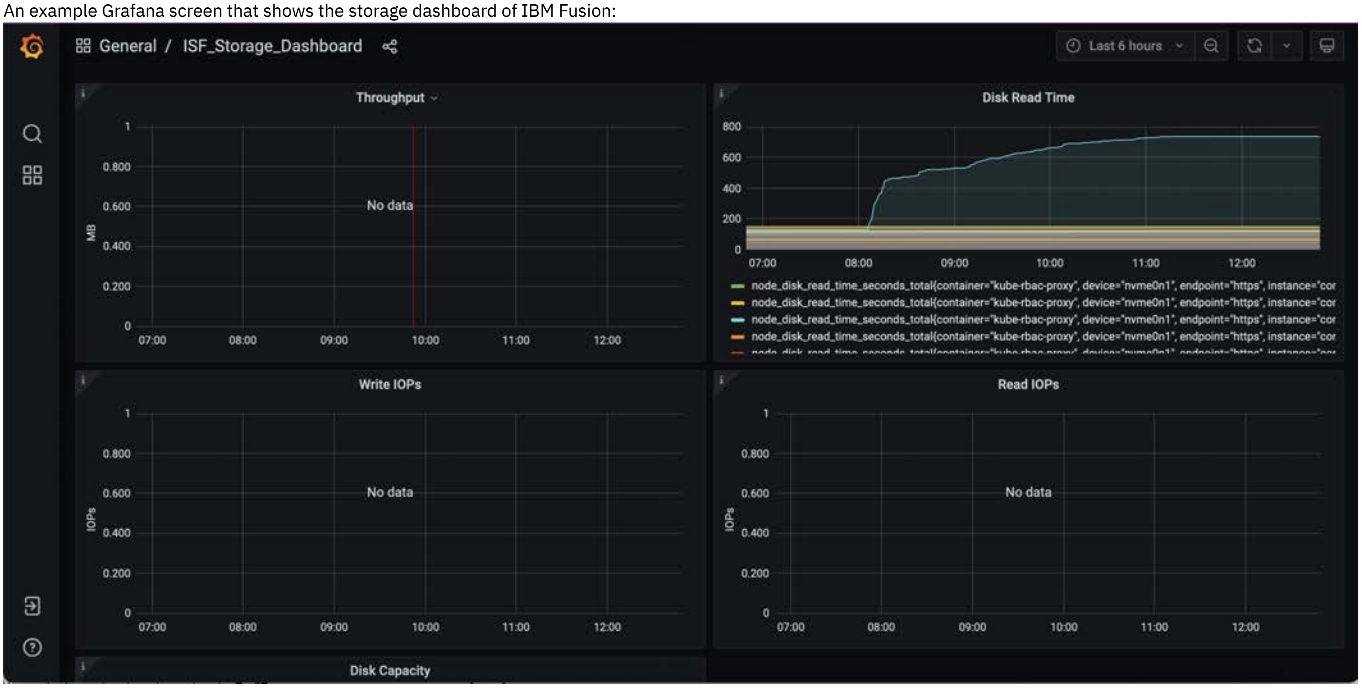
Note: The following metrics are deprecated after IBM Fusion 2.11.0:

- read_operations
- write_operations
- disk_capacity
- fs_read_throughput
- fs_write_throughput
- bytes_read
- bytes_written
- bytes_sent
- bytes_received

An example Grafana screen that shows the networking dashboard of IBM Fusion for a high-speed switch.



An example Grafana screen that shows the storage dashboard of IBM Fusion:



An example Grafana screen that shows the compute dashboard of IBM Fusion:

Note: For Dell hardware, the **`Energy Stats`** panels do not display any data.

For more information about Grafana, see https://grafana.com/docs/. For more information about Prometheus, see https://prometheus.io/docs/introduction/overview/.

- **Paid account access to Docker Hub for Grafana installation and deployment**
  The default installation configuration for Grafana uses the free account quota to pull images from Docker Hub. If you see that the Grafana POD is in an error state due to **`image pull`** error, then it means that you must use a paid account to pull Grafana images from Docker Hub to complete the deployment. You will need to redeploy Grafana by using an API key provided by Docker Hub. If redeployment is not done, then you must wait for few hours till the free quota is reset by Docker Hub and then images can be pulled again.
- **Creating an audit index pattern and visualize audit data in Kibana**
  From the Kibana console, create a basic index pattern.

Paid account access to Docker Hub for Grafana installation and deployment

The default installation configuration for Grafana uses the free account quota to pull images from Docker Hub. If you see that the Grafana POD is in an error state due to **`image pull`** error, then it means that you must use a paid account to pull Grafana images from Docker Hub to complete the deployment. You will need to redeploy Grafana by using an API key provided by Docker Hub. If redeployment is not done, then you must wait for few hours till the free quota is reset by Docker Hub and then images can be pulled again.

Complete these steps to for Grafana installation and redeployment.

1. Log in to Docker. For more information, see Log in to Docker.
2. Create a Pull Secret. For more information, see Create a Secret based on existing Docker credentials.
3. Update the pull secret in **`ServiceAccounts`**. Do these steps:
   a. Log in to the OpenShift® Container Platform web management console.
   b. Click User Management > ServiceAccounts. The ServiceAccounts page is displayed.
   c. Search for grafana-serviceaccount by Name. Click to open it in the ServiceAccount details page, then click the YAML tab.
   d. Update the secret name under imagePullSecrets. For example, specify the pull secret name as follows.

   ```
   name: ibm-spectrum-fusion-registrykey
   ```
4. Go to Workloads > Pods.
5. Select ibm-spectrum-fusion-ns namespace from the Project list.
6. Restart the Grafana deployment Pod. You can see that Pods are up and running. The status indicates as **`Running`** against each Pod.

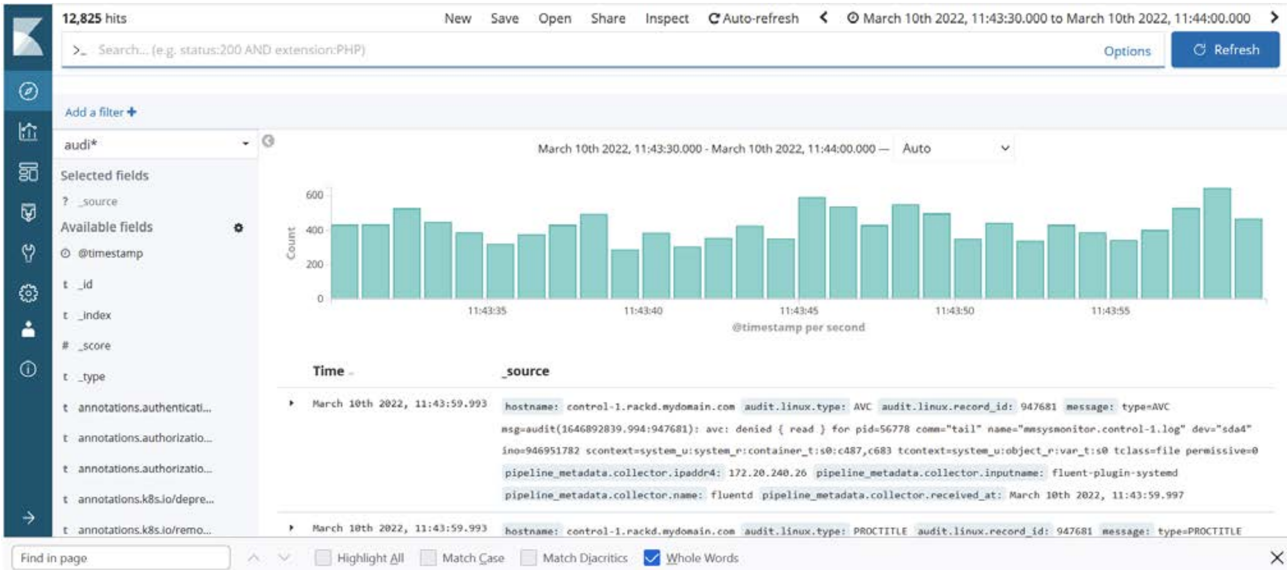
Creating an audit index pattern and visualize audit data in Kibana

From the Kibana console, create a basic index pattern.

By using the Kibana console, you can create a basic index pattern. Do these steps.

1. From the IBM Fusion user interface, click the App launcher (the icon with nine dots).
2. Click the outbound arrow to open the Kibana console in another window.
3. Log in to the Kibana console with the user credentials.
   Note: The Kibana console shares a single sign-on authentication with the IBM Fusion user interface.
4. Click Management > Index Patterns. The Create index pattern page is displayed if you are adding the first index pattern. For more information, see Red Hat OpenShift Container Platform documentation.

5. In Step 1 of 2: Define index pattern, enter the Index pattern. You must create index patterns when logged into Kibana the first time for the **app**, **infra**, and **audit** indices. You can enter wildcard characters (*). For example, enter `audi*`.
6. Click Next step.
7. In the Step 2 of 2: Configure settings, use the **@timestamp** time field to view their container logs.
8. Click Create index pattern to add the index pattern.
   For example, if you entered `audi*` in the index pattern. The Fields tab page lists every field in the `audi*` index and the field's associated core type as recorded by Elasticsearch.
9. Click Discover menu to visualize the audit data.



IBM Fusion metrics

This section describes the metrics that are used in IBM Fusion.

Platform

Metric	Description
`isf_cluster_details`	It displays the details about the cluster (version, platform, and environment).
`isf_service`	It displays the health status of the installed services ('1' for healthy, '0' for unhealthy).
`isf_environment_risk`	It displays the count of active environment risks on the cluster.
`isf_service_upgrade_risk_count`	It displays the count of active upgrade pre-check risks per service on the cluster.

Hardware

Metric	Description
`isf_switches_info`	It displays the switch details (Model, Type, Vendor, IBM MTM).
`isf_switches_port`	It displays the overall health of the port group ('1' for healthy, '0' for unhealthy).
`isf_nodes_info`	It displays the nodes hardware details (Node Type, IBM-MTM, Vendor).
`isf_node_upgrade_risk_count`	It displays the count of active upgrade pre-check risks per node on the cluster.
`isf_switch_upgrade_risk_count`	It displays the count of active upgrade pre-check risks per switch on the cluster.

Fusion events

Metric	Description
`fusion_events_total`	It displays the count of total unique fusion events created.
`fusion_duplicate_events_total`	It displays the count of total fusion events that are identified as duplicate.
`fusion_hardware_events_total`	It displays the count of total fusion events that are associated with a hardware asset.

Storage

Metric	Description
`isf_local_storage_total_gib`	It displays the local storage type and its usable capacity metric in GiB.
`isf_local_storage_used_percentage`	It displays the percentage of local storage used.

Switches

CPU metrics

Metric	Description
`ssCpuUser`	It displays the percentage of CPU time that is spent for running user-level processes.
`ssCpuSystem`	It displays the percentage of CPU time that is spent for running system or kernel-level processes.
`ssCpuIdle`	It displays the percentage of time when the CPU is idle.

Memory metrics

Metric	Description
`memTotalReal`	It displays the total physical memory that is installed on the switch.
`memAvailReal`	It displays the amount of physical memory currently free or available.

Interface traffic metrics

Metric	Description
`ifInOctets`	It displays the total bytes received on an interface.
`ifOutOctets`	It displays the total bytes transmitted out of an interface.

Sensor and temperature metrics

Metric	Description
`psu1TempOrAmbientTemp`	It displays the PSU1 temperature or ambient temperature based on switch model.
`psu2Temp`	It displays the temperature of the PSU2.
`cpuPackageSensor`	It displays the temperature of the CPU package.
`cpuCoreSensor0`	It displays the temperature of CPU core 0.
`cpuCoreSensor1`	It displays the temperature of CPU core 1.
`portAmbientSensor`	It displays the ambient temperature near front-panel ports.
`mainBoardAmbientSensor`	It displays the ambient temperature on the main system board.
`asicTempSensor`	It displays the temperature of the switch's main forwarding Application-Specific Integrated Circuit (ASIC).

Interface status metrics

Metric	Description
`ifOperStatus`	It displays the operational state of the interface (1 = up, 0 = down).
`ifAdminStatus`	It displays the admin-configured state of the interface (0 = up, 1 = down).

Password rotation

Global password rotation status

Metric	Description
`password_rotation_current_status`	It indicates the overall password rotation status for an asset and component. The value is `0` for success and `1` for failure. The metric is labeled by `asset_name` and `component_name`, allowing status tracking per asset and per system component.

Serviceability password rotation metrics

Metric	Description
`serviceability_pwd_rotation_total`	It displays the total number of password rotation attempts for serviceability credentials. The `result` label indicates whether the rotation attempt is successful or failed.
`serviceability_pwd_rotation_current_error_code`	It displays the current password rotation error state for a serviceability asset (`0` for pass, `1` for fail).
`serviceability_pwd_rotation_last_success_timestamp_seconds`	It displays the Unix timestamp of the last successful password rotation for a serviceability asset.
`serviceability_pwd_rotation_period_seconds`	It displays the configured password rotation period in seconds for a serviceability credential type.

Compute password rotation metrics

Metric	Description
`compute_pwd_rotation_total`	It displays the total number of password rotation attempts for compute assets. The `result` label indicates whether the rotation attempt was a `success` or `failure`.
`compute_pwd_rotation_current_error_code`	It displays the current error state for compute password rotation. The value indicates status (`0` = healthy or no failure, `1` = password reset failed, `2` = Kubernetes secret update failed).
`compute_pwd_rotation_last_success_timestamp_seconds`	It displays the Unix timestamp of the last successful password rotation for a compute asset.

Install password rotation metrics

Metric	Description
`install_pwd_rotation_total`	It displays the total number of password rotation attempts for installation assets. The `result` label indicates whether the rotation attempt was a `success` or `failure`.
`install_pwd_rotation_current_error_code`	It displays the current error state for installation password rotation (`0` for pass, `1` for fail).
`install_pwd_rotation_last_success_timestamp_seconds`	It displays the Unix timestamp of the last successful password rotation for an installation asset.

Enabling observability for IBM Fusion clusters

Use these instructions to enable multi-cluster observability for IBM Fusion clusters.

Before you begin

Ensure that you met the following prerequisites:

- Network connectivity between clusters and the hub or query endpoint.
- S3-compatible object storage for long-term metrics retention.

About this task

Achieve a comprehensive multi-cluster observability solution for IBM Fusion by using Red Hat® Advanced Cluster Management or an existing Thanos and Grafana infrastructure. Multi-cluster observability uses open source technologies to create a centralized monitoring solution:

- `Thanos` - Provides global querying across multiple Prometheus instances and long-term metrics storage in S3-compatible object stores.
- `Grafana` - Delivers visualization dashboards for metrics analysis and alerting.
- `Prometheus` - Collects metrics from each managed cluster and sends data to the hub cluster.

It delivers **centralized fleet-wide monitoring** through unified dashboards, enabling proactive issue identification and management across the following IBM Fusion components:

- Fleet overview
- Unhealthy data services
- Active environmental risks
- Unhealthy statuses of nodes and switches
- Event metrics
- Upgrade risks for IBM Fusion services, nodes, and switches
- Storage cluster health
- Switch monitoring:
  - Resource utilization (CPU and memory)
  - Network interface health
  - Thermal monitoring (PSU, CPU core, port, board, ASIC)

Procedure

Resources for custom IBM Fusion observability dashboards or configuration YAMLs are available through the GitHub repository.

What to do next

The centralized dashboard represents a fundamental shift from reactive to proactive infrastructure management, ensuring your IBM Fusion fleet maintains optimal performance while minimizing the risk of undetected issues escalating into service outages.

IBM Fusion Call Home Telemetry

IBM Fusion Call Home Telemetry enhances supportability by providing IBM with valuable insights into system configuration and performance. Telemetry data is collected from various components and securely transmitted to IBM, helping support teams proactively identify issues and optimize system behavior based on real-world usage.

Telemetry automatically collects and transmits the data from IBM Fusion components, such as software applications or hardware devices. Also, it is used to gather information about system configuration, usage patterns, and performance metrics. For steps to enable call home telemetry, see Enabling Call Home for IBM Fusion HCI.
Note: The system uploads Call Home telemetry data every 24 to 26 hours.
The following details are collected and uploaded as part of call home telemetry support:

IBM Fusion cluster details
: Collects statistical insights into IBM Fusion resources, such as the number of pods, deployments, and other key components, along with version and health status details for both OpenShift® and IBM Fusion. Also, it collects node storage consumption details for OpenShift.

IBM Fusion license data
: Collects IBM Fusion licensing data. For more information on licensing metrics, see Understanding the IBM Fusion license report.

IBM Fusion events data
: Collects the summary of IBM Fusion events by severity and event ID.

IBM Fusion Services
: Collects the details such as name, version, health, and upgrade details of different fusion services installed in the cluster. If Backup & Restore service is installed, then it shows the success % of backups and restores.

Hosted clusters
: Collects basic count and health status of hosted clusters that are deployed.

Virtual machines
: Collects basic count and health status of virtual machines that are created on cluster.

Basic rack details
: Collects configuration details including serial number, model, storage type, multi-rack setup, proxy enablement, Disaster Recovery (DR) enablement, IBM serial number, and other IBM Fusion HCI rack-specific parameters.

Node and switch health monitoring
: Collects hardware and health details for nodes and switches, including name, vendor, model, firmware, and type. Also, it captures component-level stats such as memory, processors, fans, GPU nodes, cards, power supplies (for nodes), and relevant metrics for switches.

Administering and configuring storage in IBM Fusion HCI

This section discusses the administration and configuration aspects of the IBM Storage Scale storage integration to the IBM Fusion HCI offering and for the successful functioning of the IBM Fusion appliance with the storage integration.

About storage administration and configuration

The IBM Fusion appliance supports customer workloads on OpenShift® Container Platform.

IBM Fusion supports only file-based storage classes that are based on IBM Storage Scale CSI driver.

The IBM Spectrum® Storage Scale Erasure Code Edition (ECE) layer does the hardware storage aggregation of the NVMe disks that are attached to the individual compute nodes of the appliance. The CSI layer provides virtualized storage over the physical storage aggregated by the ECE for consumption by the workloads that run on OpenShift Container Platform.

Note: IBM Fusion HCI brings out of the box IBM Storage Scale Erasure Code Edition (ECE) as the cloud native storage option.

- **Monitoring storage options from IBM Storage Scale Erasure Code Edition (ECE) console**
  From the IBM Storage Scale Erasure Code Edition (ECE) console, you can manage file systems, node addition, disk replacement, AFM management. In addition, you can monitor the health check, and serviceability. From the IBM Fusion user interface, you can manage storage node addition.

Monitoring storage options from IBM Storage Scale Erasure Code Edition (ECE) console

From the IBM Spectrum® Storage Scale Erasure Code Edition (ECE) console, you can manage file systems, node addition, disk replacement, AFM management. In addition, you can monitor the health check, and serviceability. From the IBM Fusion user interface, you can manage storage node addition.

IBM Fusion includes a storage platform layer based on IBM Storage Scale. IBM Fusion brings together the unstructured data management capabilities of IBM Storage Scale. IBM Fusion combines IBM Storage Scale functions with backup and restore software. Operations can protect data directly to cloud or to a remote IBM Storage Scale system and then even archive to tape.

Access the IBM Storage Scale Erasure Code Edition (ECE) console from the IBM Fusion user interface to work with the storage options.

1. From the Dashboard menu in the IBM Fusion user interface, click Storage Software to open the IBM Storage Scale Erasure Code Edition (ECE) console.
2. Log in to the IBM Storage Scale Erasure Code Edition (ECE) console for storage management and to leverage the features for advanced configurations.
   Note: The IBM Storage Scale Erasure Code Edition (ECE) console shares a single sign-on authentication with the IBM Fusion user interface.

See the following menu options on the IBM Storage Scale Erasure Code Edition (ECE) console that you can use to work with the storage options of IBM Fusion.

Home
: It provides an overall summary of the IBM Storage Scale system configuration, status of its components, and services that are hosted on it.

Monitoring
: It provides a dashboard to display the predefined performance charts and the customized chart. You can use more of the monitoring options that are available in this menu for working with the storage options.

Nodes
: It monitors the performance and health status of the nodes that are configured in the cluster. You can also create and manage user-defined node classes from the Nodes page. As part of the IBM Fusion tasks, click Files > File Systems to manage nodes or disks for storage.

  - **`Add nodes or disks`**: To add nodes or disks from the IBM Fusion user interface, see Configuring nodes for management and Adding additional storage nodes. Separate procedures are involved for adding AFM, GPU, and storage nodes.
  - **`Disk replacement`**: Replace the disks from the IBM Storage Scale Erasure Code Edition (ECE) console. For more information, see Disk Replacement and Replacing disks in a GPFS file system.

Cluster
: It monitors the network and network adapter performance and health statuses.

Storage
: It helps to monitor the performance, health status, and configuration aspects of all available pools in the IBM Storage Scale cluster.

Services
: It monitors and manages various services that are hosted on the IBM Fusion system.

Support
: You can download diagnostic data to troubleshoot the reported IBM Fusion issues. You can also configure the call home feature that automatically notifies the IBM® Support for the reported issues.

For more information, see the IBM Storage Scale documentation.

User management

Use the role-based user restrictions and user management capabilities to administer IBM Fusion HCI.

IBM Fusion HCI user interface is configured with OpenShift® Container Platform to have a single sign-on. For the first time login to IBM Fusion HCI user interface and OpenShift Container Platform web management console, use the **`kubeadmin`**, which is the default user for both. To authenticate the default user login, use the password that got generated during the installation of IBM Fusion HCI.
Note: For security reasons, create a user with **`cluster-admin`** role and delete the default **`kubeadmin`** user.
For more information about roles, see Default cluster roles in *Red Hat OpnShift Container Platform* documentation.

User roles

Role-based access control (RBAC) objects determine whether a user is allowed to do an action within a project. By using role-based access control, you can set the resources and permissions available to a user. Role can be assigned to a user or group with role bindings. Role binding has the mapping of a role to a user or user group. You can bind your users to the following two default OpenShift cluster level roles:

Role	Permissions
`cluster-admin`	A super-user that can do any action in any project. When bound to a user with a local binding, they have full control over quota and every action on every resource in the project.
`view`	A user who cannot do any modifications, but can see most of the objects in a project. They cannot view or modify roles or bindings.

You can create more users and user groups. You can also update or delete existing users by using the `cluster-admin` user roles.

Add user to a role

Step 1: Get user and their role details:

To get all users and their cluster wide roles, run the following command:

```
oc get clusterrolebindings -o json | jq -r '.items[] | {role: .roleRef.name, users: .subjects[]?.name} | select(.users !=
null)'
```

Alternatively, you can run the following commands:

- Find all Users with a specific role:

```
oc get clusterrolebindings -o json | jq -r '.items[] | select(.roleRef.name == "<role>")
      | {role: .roleRef.name, users: .subjects[]?.name}'
```

For example, you want the list of all users in cluster-admin role:

```
oc get clusterrolebindings -o json | jq -r '.items[] | select(.roleRef.name == "cluster-admin")
      | {role: .roleRef.name, users: .subjects[]?.name}'
```

Example output:

```
{
  "role": "cluster-admin",
   "user": "ft-admin"
}
{
  "role": "cluster-admin",
   "user": "openshift-apiserver-operator"
}
{
  "role": "cluster-admin",
   "user": "openshift-config-operator"
}{
  "role": "cluster-admin",
   "user": "service-ca-operator"
}
```

- Find a specific user role in a cluster:

```
oc get clusterrolebindings -o json | jq -r '.items[] | select(.subjects[]?.name ==
      "<username>") | {role: .roleRef.name, user: "<username>"}'
```

For example, you want to find the role of ft-admin in a cluster:

```
oc get clusterrolebindings -o json | jq -r '.items[] | select(.subjects[]?.name ==
      "ft-admin") | {role: .roleRef.name, user: "ft-admin"}'
```

Example response:

```
{
  "role": "cluster-admin",
   "user": "ft-admin"
}
```

Step 2: Add user to a role:

Add user to cluster-admin role:

```
oc adm policy add-cluster-role-to-user cluster-admin <username>
```

For example, add sf-admin user to cluster-admin role:

```
oc adm policy add-cluster-role-to-user cluster-admin sf-admin
```

Add user to view role:

```
oc adm policy add-cluster-role-to-user view <username>
```

For example, add sf-usr user to view role:

```
oc adm policy add-cluster-role-to-user view sf-usr
```

For more information about OpenShift Container Platform RBAC, see Using RBAC to define and apply permissions in *Red Hat OpnShift Container Platform* documentation.

For more information about authentication and authorization, see Understanding authentication in *Red Hat OpnShift Container Platform* documentation.

RBAC permissions for IBM Fusion HCI user interface

The following table displays the RBAC permissions for IBM Fusion HCI user interface.

Table 1. RBAC for IBM Fusion HCI users

User interface page or menu option	Cluster-admin	View user
Events	• You can mark events as fixed	• You cannot mark events as fixed
Applications	• You can assign policies to applications • You can restore application backup • You can edit application details • You can enable or disable applications for disaster recovery	• You cannot assign policies to applications • You cannot restore application backup • You cannot edit application details • You cannot enable applications for disaster recovery
Backup policies	• You can add, edit, or delete policies • You can add, edit, or delete backup locations	• You cannot add, edit, or delete policies • You cannot add, edit or delete backup locations For the procedure to manage the backup and restore of your application namespace as a view user through CRs, see Self-service Backup & Restore
Backup policies		
Infrastrucure > Compute page	• Can upsize nodes or add storage disks. • Can manage node resources like moving a node to maintenance, power on a node, and so on.	• Cannot upsize nodes or add storage disks. • Cannot manage node resources.
Infrastructure > Network page	• Can run commands on switches • Can add VLANs and Links.	• Cannot run commands on switches. • Cannot add VLANs and Links.
Settings > Call Home page.	• Can enable IBM Call Home or edit Call Home details.	• Cannot enable Call Home or edit Call Home details.
Settings > Encryption page.	• Can edit encryption settings. • Can delete encryption settings.	• Cannot edit encryption settings. • Cannot delete encryption settings.
From the title bar, click the help icon and select Collect support logs.	• Can generate logs and log sets. • Can enable Call Home.	• Cannot generate logs or log sets. • Can download generated logs.
App Switcher icon in title bar > Storage outbound arrow	• Can replace disk (pdisk replacement). • Can download snap. • Can manage events (mark as resolved, fix, hide tip, notification, and others).	• Cannot configure, modify, or manage the system or its resources.
Disaster recovery	• Can set up site 1, site 2, and tiebreaker in Metro Sync DR. • Can upgrade local, remote and tiebreaker clusters • Can failover applications from one site to another	• Cannot set up site 1, site 2, and tiebreaker in Metro Sync DR. • Cannot upgrade local, remote and tiebreaker clusters • Cannot failover applications from one site to another
Services	• Can enable and disable IBM Fusion services • Can upgrade IBM Fusion services	• Cannot enable and disable IBM Fusion services • Cannot upgrade IBM Fusion services
Applications icon in title bar >OpenShift outbound arrow	For more information about the permissions of the role, see Using RBAC to define and apply permissions in *Red Hat OpnShift Container Platform* documentation. Note: Menu option is available to navigate to OpenShift console with same login credentials.	

Configure identity providers

Step 1: Configure identify providers

- **LDAP**
  Configure your organizations LDAP with OpenShift to access IBM Fusion HCI user interface. For the more information and procedure, see Configuring an LDAP identity provider in *Red Hat OpnShift Container Platform* documentation.

- **httpasswd**
  Configure httpassword identity provider to create users that can access OpenShift and IBM Fusion HCI user interface. To configure the user with httpasswd identity provider, see Configuring an htpasswd identity provider in *Red Hat OpnShift Container Platform* documentation.

Step 2: Bind your user to a role or to a group.
The user or group can have `cluster-admin` or `view` roles.

Step 3: Log in to IBM Fusion HCI user interface
Log in to IBM Fusion HCI user interface by using the newly created or added user.

Knowing your IBM Fusion user interface

Learn the various key elements of the IBM Fusion user interface for ease of use and navigation.

To open the user interface from any of the supported browsers:

1. Log in to OpenShift® Container Platform using kubeadmin.
2. Click the Applications icon in the title bar and click IBM Fusion.
3. In the IBM Fusion user interface, click Let's get started in the Welcome to IBM Fusion page.

The Quick start page is the landing page of IBM Fusion user interface.
From this Quick start page, you can provide storage for container workloads, backup and restore the applications and control plane data, manage IBM Fusion HCI hardware, upgrade components, investigate events and open support tickets.

The following options are available in the landing page of IBM Fusion user interface:

- You can do the following tasks from the navigation menu:
  - Click Events to view IBM Fusion events. They are Kubernetes native events that include filter-specific labels. For more information about the events page, see Analyzing events in IBM Fusion HCI.
  - Click Applications to backup and restore application and its resources. For more information about how to backup and restore applications, see Backup & Restore of your applications and virtual machines.
  - Click Backup policies to define parameters that are applied to backup jobs. For steps to create new policies or work with existing policies, see Backup policies.
  - Click Infrastructure to manage the compute and network for the IBM Fusion HCI appliance. For more information about compute and network, see Administering the node and Networking.
  - Click Settings to do the following:
    - IBM Fusion
      You can set up encrypted data at rest by connecting to a Security Key Lifecycle Manager. For more information about how to set up encryption, see Configuring encryption for Global Data Platform storage.
    - IBM Fusion HCI
      Enable Call Home and upgrade IBM Fusion HCI components. For more information about Call Home and upgrade, see Enabling Call Home for IBM Fusion HCI and Implementing a comprehensive upgrade procedure respectively.
- From the applications icon in the title bar, you can quickly navigate to the following consoles by using their corresponding outbound arrows:
  - Click OpenShift console to go to the OpenShift web management console. From the title bar of OpenShift console, click Red Hat applications > IBM Fusion to navigate to IBM Fusion.
  - IBM Fusion HCI specific outbound links
    Apart from OpenShift console link, IBM Fusion HCI has the following additional outbound links:
    - Click Prometheus to go to Prometheus console. It is a monitoring solution for storing metrics. You must enable Grafana to use this option. For procedure to enable, see Monitoring and logging by using any monitoring tool.
    - Click Storage to go to the IBM Storage Scale ECE user interface.
    - Click IBM Cloud to go to IBM Cloud.
- The following details and options are available in the title bar:
  - Displays the cluster ID of IBM Fusion.
  - Click the help icon on the title bar of the page to do the following tasks:
    - Click Open support case to create a support ticket with IBM.
    - Click Collect support logs to download support logs.
    - Click IBM Documentation to view IBM Documentation for IBM Fusion.
    - Click About to know the version of IBM Fusion.
  - The bell icon is visible on the IBM Fusion user interface only when any event or alert occurs and is available for the user. You can also click View all events to go the Events page.
  - Click the user icon to do the following tasks:
    - View the currently logged in user. Click user to view and manage the profile details of the logged in user.
    - Click Logout to log out of the IBM Fusion.
      Note: To close all open sessions in IBM Fusion, log out of both IBM Fusion UI console and OpenShift Container Platform UI console.
- **Working with the dashboard**
  Dashboard provides an overview of the infrastructure, managed clusters, hardware status, services health, storage, backup jobs, and environmental risks of the IBM Fusion HCI.
- **Browser requirements**
  List of supported web browsers for IBM Fusion HCI user interface.
- **Icons used in the user interface**
  A reference to the icons used in the user interface and their naming convention.
- **IBM Fusion HCI user interface issues**
  A list of all known issues in the IBM Fusion user interface.

Working with the dashboard

Dashboard provides an overview of the infrastructure, managed clusters, hardware status, services health, storage, backup jobs, and environmental risks of the IBM Fusion HCI.

Account
: The Account menu includes the following information:

- IBM Fusion license and audit metrics report. Go to Account > License usage and click the download icon to get the detailed license and audit metrics report. To understand the metrics, see Understanding the IBM Fusion license report.

Details
: The Details section includes the following hardware information:

- Cluster ID details.
- Infrastructure details.

- OpenShift® version details. Click Upgrade pre-check next to OpenShift version to open a slide out pane Upgrade pre-check to find the details of issues during the OpenShift upgrade. To debug further, click the outbound arrows to find exact problem details and possible prevention steps.
  Important: The upgrade pre-check identifies issues to be resolved before an upgrade is attempted, not the results of an upgrade operation.
- Total number of nodes added to the cluster.

Services
: The Services section includes the following information:

  - It provides a detailed list of services that are installed on the IBM Fusion HCI and their health status.
  - Click View services to go to Services page to view the list of available and installed services on the IBM Fusion HCI along with their health status.

Hardware status
: The Hardware status section includes the following information:

  - It provides the number of hardware components in the IBM Fusion HCI that needs your attention.
  - Click View hardware to go to Overview page to view the graphical view of the hardware appliance where you can find the total number of nodes, switches, and service node along with their health statuses.

Environmental risks
: The Environmental risks section includes the following information:

  - It provides the number of environmental risks present in the IBM Fusion HCI that needs your attention.
  - Click View risks to open a slide out pane Environmental risks to check a detailed list of risks that might impact your workloads and upgrades. To debug further, click the outbound arrows to find exact problem details and possible prevention steps.

Managed clusters
: The Managed clusters section provides the number of hosted cluster created. Click Learn how to create a hosted cluster to view the steps to create a hosted cluster.
  Important:

  - Ensure that a local storage service (either Data Foundation or Global Data Platform) is installed.
  - The local storage must be configured.

Backup jobs over time
: The Backup jobs over time section includes the total number of backups available.
  Important:

  - Ensure that the Backup & Restore service is installed to view backup jobs details. To get steps to install Backup & Restore service, click View Backup & Restore.
  - Ensure that a local storage service (either Data Foundation or Global Data Platform) is installed.
  - The local storage must be configured.

Local storage
: The Local storage section provides a circular pie chart that is used versus the available capacity of the storage. The storage capacity is represented in GiB and the Used capacity is represented graphically and in percentage. Click View storage to go to Local storage page to view the usable capacity, services health, and storage nodes details.
  Important:

  - Ensure that a local storage service (either Data Foundation or Global Data Platform) is installed.
  - The local storage must be configured.

Replication
: The Replication section includes the following information.

  - Status of the both site 1 and site 2, such as connected or not.
  - Type of the disaster recovery.
  - Total numbers of applications added to the backup information.
    Click View topology to find the tiebreaker details whether the tiebreaker is connected between the site 1 and site 2.

    Important:
    - A replication card is displayed unless the Data Foundation local storage service is installed (replication is only supported for Global Data Platform storage).
    - Also the content of the card available only after Global Data Platform local storage is configured.

- **Understanding the IBM Fusion license report**
  Use these instructions to get a detailed explanation of each metric and how to interpret them to ensure compliance with IBM Fusion licensing terms.

Understanding the IBM Fusion license report

Use these instructions to get a detailed explanation of each metric and how to interpret them to ensure compliance with IBM Fusion licensing terms.

IBM Fusion licenses require entitlements for the following key components:

- Cores managed
- Threads per core (SMT)
- Backup subscribed storage
- Services deployed
- Storage capacity provisioned or accessed

Cores managed

- Cores on all applicable nodes in the cluster must be entitled by using IBM Fusion licenses with a VPC metric or a managed server metric. For IBM Fusion HCI, for calculating the metric, the `PROCESSOR_CORE` metric considers all nodes by default, whereas in IBM Fusion, only worker nodes are considered.

- The reported value of **`PROCESSOR_CORE`** is obtained from a native OpenShift® API for each node and aggregated by the IBM Fusion license service.
- The quantity of cores that are included in IBM Fusion licenses with a compute metric must meet or exceed the **`PROCESSOR_CORE`** reported value.

IBM clients who purchase certain IBM Software products, such as IBM Cloud Pak, Watson, or Maximo Application Suite, may be entitled to use IBM Fusion as a supporting program within their license agreement.

Key entitlement details:

- Restrictions may apply to services and storage capacity, but not to the number of cores, unless otherwise stated.
- Clients using IBM Fusion as a supporting program must replace their entitlement with paid licenses to remove the restrictions.

Deployment requirements:

- IBM Fusion licenses with a managed server metric can only be deployed on x86 bare metal nodes.
- IBM Fusion licenses with a VPC metric can be deployed on any node type, including physical, virtual, on-premises, or public cloud.

Threads per core (SMT)

The value reflects the maximum value of threads per core among all the nodes in the environment. The metric is reported in the licensing report and named as **`NODE_SMT_MAX`**.

Backup subscribed storage

This value helps understand the total storage capacity, which is covered with backup and assigned a backup policy. The metric is reported in the licensing report and named as **`BACKUP_STORAGE_GB`**, and the value is reported in GiB.
Note: If two clusters are connected and configured as a hub and spoke, for spoke applications that are backed up on the hub, the metric value does not reflect the total storage capacity.

Services deployed

- Clients can install services from IBM Fusion using operators that are included in the product.
- Each service flag should be validated against the Fusion licenses purchased or applied to the observed cluster to ensure compliance.
- Restrictions of use in other products, such as IBM Cloud Paks, are applicable if IBM Fusion is deployed using terms from those products.

When a client installs services from IBM Fusion, they must ensure that the services are licensed for use. Each installed service has a flag with a value of 1. To validate, check the flag against the purchased or applied IBM Fusion licenses for the cluster.
Key considerations:

- Validate each service flag against the applicable license terms.
- If using IBM Cloud Paks, follow the restrictions of use for those products.
- Non-compliance may occur if services are used without proper licensing.

For example, if the BACKUP_RESTORE_SERVICE metric is 1 and IBM Cloud Pak license terms are used, the deployment may not be compliant.

Storage capacity provisioned or accessed

- IBM Fusion licenses usually include an allowance for provisioning or accessing storage in the cluster such as Fusion Data Foundation, IBM Storage Scale, IBM Storage Ceph Pro.
  Other types of storage can be accessed using Fusion services such as object storage, or a third-party storage array with Fusion Access.
- Each type of permitted provisioning or access method is reported in a metric with the prefix STORAGE_.
  For example, **`STORAGE_FDF`** can be provisioned within the cluster (locally) or provisioned outside the cluster (externally).
- Clients should validate that the storage capacities and types that are deployed are covered by IBM Fusion licenses and any product-specific licenses.
  For example, client may already own IBM Storage Scale and then add to it using their Fusion entitlements when setting up their Fusion cluster. Combining licenses for the same type of IBM storage from different sources is usually permitted.
- If the cluster is configured with the GDP remote mount metric, the **`STORAGE_GDP_REMOTE_MOUNT`** metric reports the total usable capacity in GiB.

Some Fusion licenses include an entitlement for primary storage and allow core deployment. If this entitlement is not enough, you can buy additional primary storage expansion parts to increase the capacity.
Primary storage:
Primary storage refers to the usable capacity available to a cluster.

IBM Software Entitlements:
If you are using IBM Software entitlements with limited storage capacity, you can increase your storage entitlements by replacing the included licenses with paid Fusion licenses.

- Replace the licenses with paid Fusion licenses that meet the processor core requirements.
- Buy additional capacity entitlements to cover any remaining storage needs.

Browser requirements

List of supported web browsers for IBM Fusion HCI user interface.

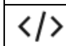
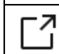
In general, IBM Fusion HCI user interface works with most of the modern web browsers. The following table lists the browsers that are tested against IBM Fusion HCI user interface.Keep your browser version up to date with at least the last two versions so that the user interface works without issues.

Browsers	Supported
Mozilla Firefox	Supported
Google Chrome	Supported
Safari	Supported
Microsoft ® Edge	Supported

Icons used in the user interface

A reference to the icons used in the user interface and their naming convention.

Table 1. Symbol references

Icon	Naming convention used
	Green tick mark to indicate healthy, completed,
	Warning
	Critical, Not connected, Failed transferring data
	Canceled
	Search filter
	Information
	Launch YAML
	Restore
	Ellipsis overflow menu
	Settings
	External link
	Outbound arrow
	Run command for switch

IBM Fusion HCI user interface issues

A list of all known issues in the IBM Fusion user interface.

Known issues in the node management pages

- If you use the IBM Fusion user interface to move a node to maintenance mode, then the user interface might show the status of the node as Ready.
- Only one node can go down at a time for maintenance and power down operation. If you place a second node in maintenance, then the user interface does not show any error message instead shows the node with status as Ready (maintenance mode in progress).
- During backend upgrades, the BMC reboots automatically for changes to take effect. As a result, monitoring may temporarily fail to connect. The health status incorrectly shows an error during this reboot period. The update status incorrectly defaults to Up-to-date when the upgrade is still in progress. However, the status recovers to Updating after the BMC connection is successful.

General issues

- To close or terminate all open sessions in IBM Fusion, log out of both IBM Fusion UI console and OpenShift® Container Platform UI console.
- Sometimes during upgrades and the nodes are down, a red toast notification appear on the user interface when you configure the remote filesystem even though file system is recovered and shows connected. In such case, you need to be explicitly close the notification.

API reference

List of all REST API references for IBM Fusion HCI.

For authentication of IBM Fusion REST APIs, ensure you have a valid Red Hat® OpenShift® Container Platform bearer token in Authorization header:

For example,

`Authorization : Bearer sha256~4rpK-5GEwvgGKnXbBtMJ9dcaRDhIqOxt5BihZ1NVt20`

IBM Fusion console route is the host for all these APIs. To access the console route, do the following steps:

1. Log in to Red Hat OpenShift Container Platform.

2. Go to Network > Routes page.
3. In the Routes page, search for Console. Ensure that you are in **`ibm-spectrum-fusion-ns`** project.
   Note: All the APIs mentioned in swagger are not applicable to IBM Fusion HCI.

If you want to use the API, then use the following command to get the route:

```
oc get route console -n <namespace where fusion is installed>
```

Example:

```
oc get route console -n ibm-spectrum-fusion-ns
```

Example result:

```
NAME      HOST/PORT                                                          PATH    SERVICES   PORT     TERMINATION   WI
console   console-ibm-spectrum-fusion-ns.apps.domain.com                isf-svc      <all>        reencrypt          None
```

The list of APIs to understand the endpoints, methods (GET, POST, PUT, DELETE), parameters, and authentication requirements:

- IBM Fusion appliance management API
- Active file management (AFM) APIs
- Serviceability APIs
- Services APIs
  For Backup & Restore service APIs, see Backup and restore as code.

  For IBM Data Cataloging service APIs, see IBM Data Cataloging REST APIs.

IBM Fusion appliance management API

The IBM Fusion HCI API references.

- Compute APIs
- Networking APIs
- Upgrade APIs
- Active File Management APIs

Compute APIs

List of compute monitoring CRs

```
GET /api/v1/compute-monitorings
```

Get a specific **`ComputeMonitoring`** CR

```
GET /api/v1/compute-monitorings/<node-name>
```

List of **`ComputeConfiguration`** CRs

```
GET /api/v1/compute-configurations
```

Enable maintenance mode for compute node

```
POST /api/v1/compute/update/maitenancemode/<node-name>
```

Payload:

```
{"maintenanceMode":"true"}
```

Power operation for compute node

```
POST /api/v1/compute/update/poweroperation/<node-name>
```

Payload:

```
{"maintenanceMode":"true"}
```

Note: The node must be in maintenance mode before calling this API. To enable the maintenance mode, call the *Enable maintenance mode for compute node* API.

Add new compute node

```
POST /api/v1/computeconfig
```

Payload:

```
[{"linkLocal":"<linklocaladdress","ru":"<RUnumber>","ibmSerialNumber":"<ibmserialnumber>","rackSerialNumber":"
<rackname>"}]
```

To check the added node, check the **`successNodes`** field in the response. An empty list means that the compute node failed to add to the cluster.

Add node to storage cluster (scale out)

```
PUT /api/v1/compute/scaleout
```

Payload:

```
{"scaleOutNodes":"compute-0,compute-2,compute-3"}
```

To confirm success of scale out, call the **`GET compute node`** API.

Check compute node storage

```
GET /api/v1/addstorage/info
```

Add storage disks (scale up)

```
PUT /api/v1/compute/scaleup
```

This API is applicable only if additional storage disks are available.

```
GET /api/v1/addstorage/info before this.
```

For successful scale, check storage information. To check the information, call the *Check compute node storage* API before and after this PUT API call.

Networking APIs

Note: If you using Alternative network configuration, then the links related APIs are not applicable.

List all network switches

```
GET /api/v1/network/switches
```

List all network VLANs

```
GET /api/v1/network/vlans
```

List all network links

```
GET /api/v1/network/links
```

List all network switch commands

```
GET /api/v1/network/listcommands
```

Run network switch command

```
GET /api/v1/networkcommands/<switch-name>/<command>
```

Create new VLAN

```
POST /api/v1/network/vlans/rack-vlans/add/
```

Payload:

```
{"vlanId":<vlan-id>,"vlanName":<vlan-name> ,"vlanType":<vlan-type>,"multipleLinks": <yes/no>}
```

Update VLAN

```
POST /api/v1/network/vlans/rack-vlans/update/<vlan-id>
```

Payload:

```
{"vlanId":<vlan-id>,"vlanName":<vlan-name> ,"vlanType":<vlan-type>,"multipleLinks": <yes/no>}
```

Create new link

```
POST /api/v1/network/links/rack-links/add/
```

Payload:

```
{"name":<link-name> ,"ports":<port>,"vlanType": <type>,"transceiver":<transceiver-value>
,aggregation":true,"nativeVlan":<native-vlan-name>,"vlanNames":<vlan-names> }
```

Edit the link

```
POST /api/v1/network/links/rack-links/update/<uuid>
```

Payload:

```
{"name":<link-name> ,"ports":<port>,"vlanType": <type>,"transceiver":<transceiver-value>
,aggregation":true,"nativeVlan":<native-vlan-name>,"vlanNames":<vlan-names> }
```

Upgrade

IBM Fusion upgrade status

```
GET /api/v1/upgrade
```

Serviceability APIs

The serviceability API references for Call Home, Events, Logs, and Remote support.

- Call Home APIs
- Event APIs
- Log APIs
- Remote support APIs

IBM Call Home APIs

Get IBM Call Home data

`GET /api/v1/callhome`

GET response:

```
{
    "context": "",
    "clusterName": "isf-rackb",
    "domainName": "rtp.raleigh.ibm.com",
    "organization": "ABS_Corp_Dummy",
    "customerid": "2434-4343",
    "email": "test_user@in.ibm.com",
    "customername": "Test User",
    "servicetype": "fullService",
    "agreesprivacy": true,
    "customeraddress": "",
    "contact": "(000)-123456-78",
    "machineaddress": "13th Street. 47 W 13th St, New York, NY 10011, USA.",
    "machinezip": "123456",
    "machinecity": "Newyork",
    "machinestate": "Newyork",
    "machinecountry": "US",
    "skipCallHome": false,
    "isBYOS": false
}
```

Update IBM Call Home data

`PUT /api/v1/callhome/update`

The payload is the same as the response to GET Call Home Data. All the fields are mandatory, even when we plan to update a few of them. Sending an incomplete payload may result in an error.

If Call Home is not configured, then the GET does not return data.

```
{
        "agreesPrivacy": true,
        "servicetype": "fullService",
        "organization": "ABS_Corp_Dummy",
        "customerid": "2434-4343",
        "customername": "Test User",
        "contact": "(000)-123456-78",
        "email": "test_user@in.ibm.com",
        "machineaddress": "13th Street. 47 W 13th St, New York, NY 10011, USA. 20 Cooper Square",
        "machinecountry": "US",
        "machinecity": "Newyork",
        "machinestate": "Newyork",
        "machinezip": "123456"
}
```

Turn off Call Home

`PUT /api/v1/callhome/turnoff`

Payload:

```
data: {
             removeconfig: "true"
            }
```

Upload Call Home Telemetry

Post API to upload telemetry data to IBM Call Home.

`POST /api/v1/callhomeclient/telemetryupload`

Event APIs

Get Events

`GET /api/v1/eventmanager/events?${queryString}`

Get event categories

`GET /api/v1/eventmanager/eventcats`

Get event stats

`GET /api/v1/eventmanager/eventstats?${query}`

For example, query `'Fixed=false'` gets unfixed events

Mark an event as fixed

`PUT /api/v1/eventmanager/events/${eventId}`

Payload:

```
{ annotations: { isf_fixed: 'true' } }
```

Note: This API is applicable only for unfixed and open Call Home tickets.

Log APIs

Get all log collector sets

```
GET /api/v1/logcollector
```

Create log collector set

```
POST /api/v1/logcollector
```

Payload:

```
{"requests":{"storage":{"type":"list-entry","source-list":"isf-collection-sets","list-entry":"storage-sds"}}}
```

Payload for OpenShift Data Foundation (ODF):

```
{
 "requests": {
   "odf": {
     "type": "list-entry",
     "source-list": "isf-collection-sets",
     "list-entry": "odf-sds"
   }
 }
}
```

Response:

```
{"jobid":"odf-20221103062933"}
```

The `jobid` returned by POST call is used for getting the status.

For example:

```
GET /api/v1/logcollector/odf-20221103062933
```

Log set download

```
GET /api/v1/logcollector/<job-id>/results
```

This GET command downloads log sets.

Delete log set

```
Delete /api/v1/logcollector/${jobid}
```

Remote support APIs

You can call these API's from the cluster Fusion ui route.

Get remote support status

```
GET /api/v1/service-node/remote-support
```

Sample request:

```
curl --request GET --url https://<fusion-url>/api/v1/service-node/remote-support
```

Sample GET response:

```
{
    "status":"running",
    "startTime":"2024-06-10T11:31:30Z",d
    "stopTime":"2024-06-10T11:33:30Z",
    "isConfigured": true,
    "isProxy": false
}
```

Start remote support connection

```
POST /api/v1/service-node/remote-support/start
```

Sample request:
Note: Ensure you update the fields with the actual values.

```
curl --request POST --url https://servicenode-host:3000/api/v1/service-node/remote-support/start --header 'X-Username: 
sn_username'  --header 'X-Password: sn_password'  --cacert /etc/pki/tls/certs/server.crt  --data 
'{

        "activeHours": 24

}'
```

Sample POST response:

```
{
  "status": "success",
  "message": "",
  "messageCode": "",
  "messageArgs": null,
  "errorMessage": ""
}
```

Stop remote support connection

```
POST /api/v1/service-node/remote-support/stop
```

Sample request:

```
curl --request POST  --url https://<fusion-url>/api/v1/service-node/remote-support/stop
```

Sample POST response:

```
{
  "status": "success",
  "message": "",
  "messageCode": "",
  "messageArgs": null,
  "errorMessage": ""
}
```

Test remote support connection

```
GET /api/v1/service-node/remote-support/connection-status
```

Sample request:

```
curl --request GET  --url https://<fusion-url>/api/v1/service-node/remote-support/connection-status
```

Sample GET response:

```
{
  "status": "success"
}
```

Configure remote support connection

```
POST /api/v1/service-node/remote-support/configuration
```

Sample request:
Note: Ensure you update the fields with the actual values.

```
curl --request POST --url https://servicenode-host:3000/api/v1/service-node/remote-support/configuration --header 'X-
Username: sn_username'  --header 'X-Password: sn_password'  --cacert /etc/pki/tls/certs/server.crt --data
'{

        "customerNumber": "customerName",

        "company": "company"

}'
```

Sample POST response:

```
 {
  "status": "success",
  "message": "",
  "messageCode": "",
  "messageArgs": null,
  "errorMessage": ""
}
```

Disable remote support connection

```
Delete /api/v1/service-node/remote-support/configuration
```

Sample request:

```
curl --request DELETE --url https://<fusion-url>/api/v1/service-node/remote-support/configuration
```

Sample response:

```
{
"status": "success"
}
```

Get proxy details

GET API to get proxy details for remote support.
Sample request:

```
curl --request GET --url 'https://<fusion-url>/api/v1/service-node/proxy?proxyType=remoteSupport'
```

Sample response:

```
 {
"host": "samplehost",
"port": "8080",
"username": "proxy_username",
"password": ""
}
```

Set up proxy for remote support

POST API to set up proxy for remote support.

```
curl --request POST    --url https://<fusion-url>/api/v1/service-node/proxy    --data '{ "proxyType": "remoteSupport",
"host": "samplehost", "port": "8080", "username": "proxy_username",  "password": "proxy_password" }
```

```
{
 "status": "success"
}
```

Delete proxy
: Delete API to clean up proxy for remote support.

  Sample request:

```
curl --request DELETE  --url https://localhost:3001/api/v1/service-node/proxy --data '{ "proxyType": "remoteSupport" }'
```

```
{
 "status": "success"
}
```

Note: Sometimes, though the APIs return 200, the operation for PUT and POST may be unsuccessful. In such cases, use other corresponding GET APIs to confirm the operation.

Active file management (AFM) APIs

The Active file management (AFM) API references.

Retrieve the authorization token
: For the CURL command, see Active file management (AFM) fileset creation

Retrieve the filesystem name (*filesystem_name*) that is mounted on the OpenShift® Container Platform cluster:

```
curl -k -H "Authorization:Token <token retreived from the previous command>" -X GET --header 'content-
type:application/json' https://<spectrum_scale_gui_route>/scalemgmt/v2/filesystems
{
  "filesystems" : [ {
    "name" : "ibmspectrum-fs"
  } ],
  "status" : {
    "code" : 200,
    "message" : "The request finished successfully."
  }
```

Example:

```
curl -k -H "Authorization:Token 07b0a89c-5a62-4b8d-b709-b5b51b2a394c" -X GET --header 'content-type:application/json' 
https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf.ibm.com/scalemgmt/v2/filesystems
{
"filesystems" : [ {
"name" : "ibmspectrum-fs"
} ],
"status" : {
"code" : 200,
"message" : "The request finished successfully."
}
```

Create AFM fileset

```
curl -k -H "Authorization:Token <"Token Value> -X POST --header 'content-type:application/json' --header 
'accept:application/json' -d '{

    "filesetName": "<"Name of fileset">",
    "path": "<"filesystem mount point">/<"fileset name">",
    "afmTarget": "<"NFS export path">",
    "afmMode": "independent-writer"
    }' <"Scale route URL">/scalemgmt/v2/filesystems//filesets
    {
    "jobs" : [ {
    "jobId" : 2000000011591,
    "status" : "RUNNING",
    "submitted" : "2023-01-12 12:33:46,219",
    "completed" : "N/A",
    "runtime" : 3,
    "request" : {
    "data" : {
    "afmMode" : "independent-writer",
    "afmTarget" : "<"NFS export path">",
    "filesetName" : "<"fileset Name">",
    "path" : "<"filesystem mount point">/<"fileset name">"
    },
    "type" : "POST",
    "url" : "/scalemgmt/v2/filesystems//filesets"
    },
    "result" : { },
    "pids" : [ ]
    } ],
    "status" : {
    "code" : 202,
    "message" : "The request was accepted for processing."
    }
    }
```

Example:

```
curl -k -H "Authorization:Token 07b0a89c-5a62-4b8d-b709-b5b51b2a394c" -X POST --header 'content-type:application/json' --
header 'accept:application/json' -d '{

    "filesetName": "afmScale",
    "path": "/mnt/ibmspectrum-fs/afmScale",
    "afmTarget": "nfs://9.30.44.36/mnt/fs1/fset1",
    "afmMode": "independent-writer"
    }' https://scalemgmt/v2/filesystems/ibmspectrum-fs/filesets
    {
    "jobs" : [ {
    "jobId" : 2000000011591,
    "status" : "RUNNING",
    "submitted" : "2023-01-12 12:33:46,219",
    "completed" : "N/A",
    "runtime" : 3,
    "request" : {
    "data" : {
    "afmMode" : "independent-writer",
    "afmTarget" : "nfs://9.30.44.36/mnt/fs1/fset1",
    "filesetName" : "afmScale",
    "path" : "/mnt/ibmspectrum-fs/afmScale"
    },
    "type" : "POST",
    "url" : "/scalemgmt/v2/filesystems/ibmspectrum-fs/filesets"
    },
    "result" : { },
    "pids" : [ ]
    } ],
    "status" : {
    "code" : 202,
    "message" : "The request was accepted for processing."
    }
    }
```

Get cache status

```
curl -k -H "Authorization:Token “<"Token value">” -X GET --header 'content-type:application/json' <"Scale Route
URL">/scalemgmt/v2/filesystems/<"filesystem Name">/afm/state

For eg:
```

Example:

```
curl -k -H "Authorization:Token 07b0a89c-5a62-4b8d-b709-b5b51b2a394c" -X GET --header 'content-type:application/json'
https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racki.rtp.raleigh.ibm.com/scalemgmt/v2/filesystems/ibmspectrum-
fs/afm/state
{
"filesetAfmStateList" : [ {
"cacheState" : "Active",
"filesetName" : "afmfsetnfs",
"filesystemName" : "ibmspectrum-fs"
}, {
"cacheState" : "Active",
"filesetName" : "afmpallavi",
"filesystemName" : "ibmspectrum-fs"
}, {
"cacheState" : "Active",
"filesetName" : "afms",
"filesystemName" : "ibmspectrum-fs"
}, {
"cacheState" : "Active",
"filesetName" : "cosfset1",
"filesystemName" : "ibmspectrum-fs"
}, {
"cacheState" : "Active",
"filesetName" : "afmScale",
"filesystemName" : "ibmspectrum-fs"
}, {
"cacheState" : "Active",
"filesetName" : "afmScalePal",
"filesystemName" : "ibmspectrum-fs"
}, {
"cacheState" : "Active",
"filesetName" : "afmScaleS",
"filesystemName" : "ibmspectrum-fs"
} ],
"status" : {
"code" : 200,
"message" : "The request finished successfully."
}
}
```

Get job status

```
curl -k -H "Authorization:Token <“<"Token Value">” -X GET --header 'content-type:application/json' <"Scale Route
URL">/scalemgmt/v2/jobs/
```

Example:

```
curl -k -H "Authorization:Token 07b0a89c-5a62-4b8d-b709-b5b51b2a394c" -X GET --header 'content-type:application/json'
https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racki.rtp.raleigh.ibm.com/scalemgmt/v2/jobs/2000000011591
{
```

```
"jobs" : [ {
"jobId" : 2000000011591,
"status" : "COMPLETED",
"submitted" : "2023-01-12 12:33:46,219",
"completed" : "2023-01-12 12:33:55,845",
"runtime" : 9626,
"request" : {
"data" : {
"afmMode" : "independent-writer",
"afmTarget" : "nfs://9.30.44.36/mnt/fs1/fset1",
"filesetName" : "afmScale",
"path" : "/mnt/ibmspectrum-fs/afmScale"
},
"type" : "POST",
"url" : "/scalemgmt/v2/filesystems/ibmspectrum-fs/filesets"
},
"result" : {
"progress" : [ ],
"commands" : [ "mmcrfileset 'ibmspectrum-fs' 'afmScale' --inode-space new --inode-limit '1M:50K' --allow-permission-change
'chmodAndSetAcl' -p 'afmTarget=nfs://9.30.44.36/mnt/fs1/fset1' -p 'afmMode=independent-writer' ", "mmlinkfileset
'ibmspectrum-fs' 'afmScale' -J '/mnt/ibmspectrum-fs/afmScale' " ],
"stdout" : [ "EFSSG0070I File set afmScaleS created successfully.", "EFSSG0078I File set afmScale successfully linked.\n"
],
"stderr" : [ ],
"exitCode" : 0
},
"pids" : [ ]
} ],
"status" : {
"code" : 200,
"message" : "The request finished successfully."
}
}
```


Upgrading IBM Fusion

Upgrade to the latest version so you can have access to the new functionalities and improvements that enhance your overall experience with IBM Fusion HCI.

To achieve the upgrade, you must follow these steps in sequence:

1. Upgrade IBM Fusion HCI software to the most recent version.
2. Upgrade all services in the specified sequence. For more information about the service upgrades, see Upgrading IBM Fusion HCI services.
3. Upgrade Red Hat® OpenShift® Container Platform.

Each step builds on the previous one, so ensure they are performed in the correct order for a successful upgrade. Before you begin each of the upgrades, thoroughly review the prerequisites, support matrix, limitations, and known issues.
The IBM Fusion software is upgraded from the OpenShift console.

You can upgrade the services from the Services page or the Settings > Upgrades page of the IBM Fusion user interface.

When you upgrade from the Services page, you can initiate upgrade of individual services based on the availability of a new version. In the Upgrade page, you can initiate multiple non-disruptive and rolling upgrades of service in parallel. For both non disruptive and rolling upgrades, you can view the status of the prechecks and monitor the progress of the upgrade. In the Services page, click View upgrade details of a ongoing service upgrade and in the Upgrade page, you can click the service tile of an ongoing upgrade to monitor the pre-check results and the progress of the upgrade in a slide out pane.

In addition, in the Upgrade page, you can do rolling disruptive upgrade where the upgrade mandates a node restart. To minimize the impact on the workloads, these restarts happen one at a time.

- **Implementing a comprehensive upgrade procedure**
  As an administrator user, upgrade components of IBM Fusion HCI from 2.11.x or 2.12.x to 2.13.0. The IBM Fusion HCI upgrade involves both software and firmware as applicable.
- **Prerequisites**
  Prerequisites for IBM Fusion HCI upgrade procedure. Review this topic before you begin an upgrade.
- **Upgrading a disaster recovery setup**
  Use this procedure to upgrade IBM Fusion HCI with Metro-DR setup.
- **Upgrading tiebreaker from IBM Storage Scale 5.2.3.x to 5.2.3.7**
  When you want to upgrade from IBM Fusion 2.12.x Metro-DR setup to IBM Fusion 2.13.0 Metro-DR setup, then upgrade your tiebreaker from IBM Storage Scale 5.2.3.x to 5.2.3.7.
- **Upgrading IBM Fusion HCI management software**
  This topic outlines the steps to upgrade IBM Fusion HCI management software from 2.12.x release to 2.13.0.
- **Upgrading IBM Fusion HCI services**
  You can choose to upgrade the installed services from the IBM Fusion HCI user interface.
- **Upgrading Hosted Control Plane Fusion and its services**
  When IBM Fusion and services on the Hosted Control Plane cluster are installed using labels, they are automatically upgraded to the latest supported version available on the IBM Fusion hub.
- **Upgrading OpenShift Container Platform**
  Upgrade OpenShift in IBM Fusion HCI. For offline upgrades of OpenShift Container Platform, follow the Red Hat steps. However, follow the guidelines and versions that are specified in this section.
- **Firmware upgrade**
  The firmware on all nodes, whether they are part of the base OpenShift Container Platform cluster or Hosted Control Plane hosted nodes, is automatically updated each time the IBM Fusion operator is upgraded. Firmware updates are applied automatically in the background. Firmware updates for switches are orchestrated by IBM Fusion, but require a manual step to trigger the update.
- **Upgrading Red Hat OpenShift Virtualization**
  Upgrade of Red Hat OpenShift Virtualization on IBM Fusion HCI is done based on Red Hat guidelines.

- **Upgrading the service node**
  This section provides the upgrade procedure of the service node.
- **Troubleshooting upgrade issues**
  Troubleshooting IBM Fusion HCI upgrade issues.

Implementing a comprehensive upgrade procedure

As an administrator user, upgrade components of IBM Fusion HCI from 2.11.x or 2.12.x to 2.13.0. The IBM Fusion HCI upgrade involves both software and firmware as applicable.

Before you begin

- Go through the prerequisites before you upgrade. For more information about the prerequisites, see Prerequisites.
- Go through the following considerations that impact the IBM Fusion HCI upgrade procedure:
  - Follow the sequence mentioned in the procedure of this topic. Wait for each step to complete before you begin the next step in sequence.
  - Service upgrades are mandatory after the IBM Fusion HCI software upgrades.
- Go through the issues related to upgrade. For more information about troubleshooting issues, see Troubleshooting upgrade issues.

About this task

The procedure ensures that IBM Fusion software, data services, and OpenShift® Container Platformare brought up to their most recent version.

Note: Follow the sequence mentioned in this procedure.

Procedure

1. Upgrade IBM Fusion HCI management software.
   See Upgrading IBM Fusion HCI management software. For procedure to upgrade IBM Fusion HCI with disaster recover setup, see Upgrading a disaster recovery setup.
2. Upgrade the storage service. For more information about upgrade, see Upgrading IBM Fusion HCI services.
3. Upgrade the Backup & Restore service. For more information about the steps to upgrade, see Upgrading IBM Fusion HCI services.
4. Upgrade the IBM Data Cataloging service. For more information about the steps to upgrade, see Upgrading IBM Fusion HCI services.
5. Optionally, if you want to upgrade OpenShift Container Platform, see Upgrading OpenShift Container Platform. For more information about Red Hat OpenShift Container Platform, see Red Hat OpenShift Container Platform documentation.
   For the supported version of Red Hat OpenShift Container Platform, see Support matrix for IBM Fusion HCI versions.

What to do next

- Before you proceed to work with IBM Fusion HCI, go through the known issues and troubleshooting mentioned in Troubleshooting upgrade issues.

Prerequisites

Prerequisites for IBM Fusion HCI upgrade procedure. Review this topic before you begin an upgrade.

General

- You must be on IBM Fusion HCI version 2.12.x to move to 2.13.0.
- For supported OpenShift® Container Platform versions, see Support matrix.
  Note: For the procedure to upgrade OpenShift Container Platform, see OpenShift Container Platform upgrade . For storage support related information, see Support matrix.
- If you installed IBM Fusion HCI in the previous version by using offline or online installation mode, then do not change the mode during the upgrade to the new version. To change the installation mode, reinstall IBM Fusion HCI.
- Logs that you collected by using the IBM Fusion Collect logs user interface page gets deleted after the upgrade process completes. Download the needed logs before you begin the upgrade. In addition, check system health from the log collections.
- The IBM Fusion HCI operator upgrade can get stuck during upgrade due to a known Red Hat OLM issue. The issue is specific to Red Hat OpenShift Container Platform versions 4.14, 4.15, 4.16, 4.17, and 4.18. Resolve the problem before you proceed with the upgrade. For the resolution steps, see Hot fix page.
- For upgrading Hosted Control Plane (HCP) clusters in offline environment, IBM Fusion catalog source must be created in IBM Fusion HCI. For more information, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.

Setup enterprise registry

If you installed the earlier version of IBM Fusion HCI by using your enterprise registry, then follow the steps to mirror images in your enterprise registry.

- Mirror IBM Fusion HCI images, IBM Storage Scale images, Fusion Data Foundation, Backup & Restore, and IBM Data Cataloging images. For steps to mirror, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.
- Update the global pull secret with the mirror registry credentials to which the current version images are mirrored. If you are mirroring to the same enterprise registry that you used for the previous version, skip this step.
- Modify the image content source policy `isf-operator-index` to add the new mirror that points to the new registry for each source defined in the image content source policy. If you are mirroring to the same enterprise registry that you used for the previous version, skip this step.
  See the sample for image content source policy:

Note: After the IBM Fusion HCI is upgraded, you can see all the new IBM Fusion services that are introduced in 2.13.0. If you want to install the new services, add image content source policy and the related image. For more information, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.

```
apiVersion: operator.openshift.io/v1alpha1
kind: ImageContentSourcePolicy
metadata:
  name: isf-operator-index
spec:
  repositoryDigestMirrors:
  # for scale
  - mirrors:
    - "<Old ISF enterprise registry>/<Old ISF target-path>"
    - "<Old ISF enterprise registry>/<New ISF target-path>"
    source: cp.icr.io/cp/spectrum/scale
  - mirrors:
    - "<Old ISF enterprise registry>/<Old ISF target-path>"
    - "<Old ISF enterprise registry>/<New ISF target-path>"
    source: icr.io/cpopen
  #for IBM Spectrum Fusion operator
  - mirrors:
    - "<old ISF enterprise registry>/<Old ISF target-path>"
    - "<Old ISF enterprise registry>/<New ISF target-path>"
    source: cp.icr.io/cp/isf

  # for ose-kube-rbac-proxy
  - mirrors:
    - "<New ISF enterprise registry>/<New ISF target-path>/openshift4/ose-kube-rbac-proxy"
    source: registry.redhat.io/openshift4/ose-kube-rbac-proxy
```

- For offline, check whether the `ibm-opencloud` and `ibm-operator-catalog` catalog source are in an error state. If they are in an error state and not in use, then delete them.
  - To validate if `catalogsource` are in use or not, run the following command to list all the subscriptions in IBM Fusion HCI cluster.

    ```
    oc get sub -A
    ```

    If the catalog source name under source fields does not show `ibm-opencloud` and `ibm-operator-catalog` in the output, then it is not in use.
  - Run `oc get catsrc -A` command to ensure that the following catalog sources are present and in ready state:
    - `isf-catalog` or `fusion-catalog`
    - `redhat-operators`
  - Run the following command to check whether `olm pod catalog-operator` is running in `openshift-operator-lifecycle-management` project:

    ```
    oc -n openshift-operator-lifecycle-manager get po
    ```

Metro-DR

- **CSI configmap**
  Run the following command to tune CSI configmap:

  ```
  oc -n ibm-spectrum-scale-csi create configmap ibm-spectrum-scale-csi-config --from-
  literal=VAR_DRIVER_DISCOVER_CG_FILESET=DISABLED
  ```

Network

- If you are using the proxy in an online or offline environment, then make sure to add all the `ipv6LLA` of the rack nodes to the OpenShift Proxy. You can find the `ipv6LLA` values in the kickstart-config map in the IBM Fusion namespace.
  The `ipv6LLA` values are added in the `noProxy` key inside the `spec` section.

  ```
  oc edit proxy cluster
  ```

  Example:

  ```
  apiVersion: [config.openshift.io/v1](http://config.openshift.io/v1)
  kind: Proxy
  metadata:
    name: cluster
  spec:
    httpProxy: http://<IP>:<PORT>/
    httpsProxy: http://<IP>:<PORT>/
    noProxy: .[base.test.net](http://base.test.net/),9999,metal3-state.openshift-machine-
  api,5050,6385,8089,6180,80,8084,fe80::72e2:84ff:fe23:78df,fd8c:215d:178e:c0de:72e2:84ff:fe23:78df,fe80::72e2:84ff:fe23:77f
  3,fd8c:215d:178e:c0de:72e2:84ff:fe23:77f3,fe80::72e2:84ff:fe23:78c3,fd8c:215d:178e:c0de:72e2:84ff:fe23:78c3
  ```

Node

- Do not start node upsize or scale disk scale out until upgrade is complete.
- Ensure that no operation is done on OpenShift Container Platform cluster that causes machine configuration rollout, for example:
  - Node maintenance
  - Node reboot
  - Old firmware upgrade
  - Image content source policy update
  - Pull secret updates

Upgrading a disaster recovery setup

Use this procedure to upgrade IBM Fusion HCI with Metro-DR setup.

About this task

In Metro-DR upgrade, you must follow the upgrade of sites in the specified order.
In IBM Fusion HCI release, multi rack is only supported with Data Foundation storage and not with Global Data Platform.

Important: Contact IBM support before you begin the upgrade of Metro-DR deployment.

Procedure

1. Upgrade site 1.
   For the procedure to upgrade, see the single rack upgrade procedure in Implementing a comprehensive upgrade procedure.
2. Upgrade site 2.
   For the procedure to upgrade, see the single rack upgrade procedure in Implementing a comprehensive upgrade procedure.
3. For Metro-DR, upgrade Tiebreaker. For the tiebreaker upgrade procedure, see Upgrading tiebreaker from IBM Storage Scale 5.2.3.x to 5.2.3.7.

Upgrading tiebreaker from IBM Storage Scale 5.2.3.x to 5.2.3.7

When you want to upgrade from IBM Fusion 2.12.x Metro-DR setup to IBM Fusion 2.13.0 Metro-DR setup, then upgrade your tiebreaker from IBM Storage Scale 5.2.3.x to 5.2.3.7.

Before you begin

- You must have the entitlement access to download the scale code package.
- You must have a minimum of 2 GB available to download the package.

Procedure

1. SSH into the tiebreaker virtual machine.
2. Download the IBM Storage Scale Data Management 5.2.3.7 from IBM Entitled System Support.
   a. Log in to IBM Entitled System Support - https://www.ibm.com/servers/eserver/ess/landing/index.html.
   b. Go to My entitled software > Software Downloads.
   c. In the Product, search for 5771-PP7 (IBM Fusion HCI) and click Add Product.
      The Storage_Scale_Data_Management-5.2.3.7-x86_64-Linux.tar.gz is available for IBM Storage Scale Data Management.
   d. Select the Storage_Scale_Data_Management-5.2.3.7-x86_64-Linux.tar.gz file to download.
3. Run the following command to extract the IBM Storage Scale package:

```
chmod +x Storage_Scale_Data_Management-5.2.3.7-x86_64-Linux.tar.gz
./Storage_Scale_Data_Management-5.2.3.7-x86_64-Linux.tar.gz
```

4. Go to directory /usr/lpp/mmfs/5.2.3.7/ansible-toolkit.
5. Run setup command to install ansible.

```
./spectrumscale setup -s <Tiebreaker IP>
```

6. Run the following command to add the tiebreaker node:

```
./spectrumscale node add <Tiebreaker IP> -a
```

7. Run the following command for upgrade precheck:

```
./spectrumscale upgrade precheck --skip ssh
```

8. Run the following command to upgrade tiebreaker:

```
./spectrumscale upgrade run --skip ssh
```

9. Reboot the tiebreaker.

Upgrading IBM Fusion HCI management software

This topic outlines the steps to upgrade IBM Fusion HCI management software from 2.12.x release to 2.13.0.

Before you begin

- Go through the prerequisites before you upgrade. For more information about the prerequisites, see Prerequisites.
- If you upgrade IBM Fusion HCI that is part of a Metro-DR configuration, then see Upgrading a disaster recovery setup.
- If you are using a private image registry, mirror the IBM Fusion HCI images to the registry before you upgrade. For the procedure and images to download, see Mirroring IBM Fusion HCI images.
- If you are using a private image registry, mirror the Red Hat® images to the registry before you upgrade. For the procedure and images to download, see Mirroring Red Hat operator images to enterprise registry.
- Before you upgrade Fusion operator, you must perform preventive steps so that the operator upgrade does not get stuck. For more information about the steps, see Operator upgrade can get stuck for IBM Fusion HCI.

- Check the compatibility of your underlying storage before you proceed with the upgrade. For more information about the version support, see Support matrix, Global Data Platform storage support for multi rack, and Fusion Data Foundation storage compatibility across rack configurations.

About this task

- The IBM Fusion HCI does automatic prechecks during upgrade to help you detect and fix errors.
  Important: From IBM Fusion HCI version 2.13.0, you can directly upgrade from 2.11.x to 2.13.0. For more information, see Enhanced upgrade experience.

Procedure

1. Log in to the OpenShift® Container Platform management console as a cluster administrator.
2. From the navigation menu, click Operators > Installed Operators and ensure that you select **ibm-spectrum-fusion-ns** project.
3. From the Installed Operators list, select IBM Fusion that is on your existing version. The Details tab opens by default.
4. Go to Subscription tab and check whether the Update approval is Manual or Automatic. If it is Automatic, change the Update approval to Manual.
5. Find CatalogSource section in the same tab, then click the fusion-catalog link.
6. In the YAML tab, update the value of the **image** under the **spec** section to reference the target image.
   a. For online upgrade:

```
spec:
  displayName: IBM Spectrum Fusion Catalog
  image: 'icr.io/cpopen/isf-operator-catalog:2.13.0-linux.amd64'
  publisher: IBM
  sourceType: grpc
```

   b. Apply the catalog source from the mirrored location. See Mirroring IBM Fusion HCI images.
   Apply the catalog source from the mirrored location
      i. In the OpenShift Container Platform management console, change the namespace to **ibm-spectrum-fusion-ns** and click the installed IBM Fusion HCI operator.
      ii. Go to the Subscription tab and click Edit subscription.
      iii. Update the source to the newly generated IBM Fusion catalog name.

```
spec:
  channel: v2.0
  installPlanApproval: Manual
  name: isf-operator
  source: ibm-spectrum-fusion-catalog
  sourceNamsepace: openshift-marketplace
status:
```

7. Save the YAML.
8. In the OpenShift Container Platform management console, change the namespace to the **ibm-spectrum-fusion-ns** and then click the installed IBM Fusion HCI operator.
9. Approve the install plan manually:
   a. Go to the Subscription tab.
   b. Click Upgrade available in the Upgrade status section. Alternatively, go to Operators > Installed Operators and click Upgrade available in the Status of the IBM Fusion record.
   c. In the InstallPlan details, click Preview InstallPlan. It lists all the install components before approving.
   d. Click Approve to begin upgrade.
10. Wait for the upgrade to complete.
11. Verify whether the IBM Fusion HCI management software is in succeeded state and the version is displayed correctly. Also, in the Subscription tab, ensure that the upgrade status displays Up to date.

What to do next

Important: After you upgrade IBM Fusion HCI, the Backup & Restore service goes to a Unknown state. Upgrade Backup & Restore service for the state to change to Healthy.

- **Enhanced upgrade experience**
  This topic describes the process for upgrading IBM Fusion HCI from version 2.11.x directly to 2.13.0, bypassing the intermediate 2.12.x releases, on supported configurations.

Enhanced upgrade experience

This topic describes the process for upgrading IBM Fusion HCI from version 2.11.x directly to 2.13.0, bypassing the intermediate 2.12.x releases, on supported configurations.

To upgrade IBM Fusion HCI from 2.11.x to 2.13.0 directly, ensure the following prerequisites are met:

- IBM Fusion HCI version must be on 2.11.x.
- The OpenShift® version is either 4.18.x or 4.19.x (for Multi-Rack 2AZ topologies only).
- Ensure that the firmware for both the switch and node is updated to IBM Fusion HCI version 2.11.0.
- Ensure that the Fusion Data Foundation service is installed and configured..
  Important: Global Data Platform topologies are not supported for direct upgrade from 2.11.x to 2.13.0
- If using Content-Aware Storage 1.0.x, upgrade to at least Content-Aware Storage 1.0.7 before initiating the IBM Fusion HCI upgrade.

Table 1. Supported Enhanced Upgrade Topologies

Support matrix for enhanced upgrade experience from 2.11.x to 2.13.0					
Topology	**Fusion Data Foundation**	**Global Data Platform**	**Backup & Restore**	**IBM Data Cataloging**	**Content-Aware Storage**

Support matrix for enhanced upgrade experience from 2.11.x to 2.13.0					
Single rack and expansion rack	Supported	Not supported	Supported	Supported	Supported
Multi-Rack 2AZ (OpenShift Container Platform)	Supported	Not supported	Supported	Supported	Supported

Important: Any topology not listed here must first be upgraded to IBM Fusion HCI 2.12.x to upgrade to version 2.13.0.

Upgrading IBM Fusion HCI services

You can choose to upgrade the installed services from the IBM Fusion HCI user interface.

Before you begin

If you choose to upgrade IBM Fusion HCI services using the Enterprise Registry, ensure that you mirror the images to your Enterprise Registry before you begin the upgrade. For more information, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.

About this task

Important:

- For an offline installation, ensure that you have completed mirroring and updated `ImageContentSourcePolicy/ImageDigestMirrorSet` for deployed services. Otherwise, you see catalog source pods failing with an `imagepullbackoffs` error and no upgrade button shows for services.
- Before upgrading OADP or AMQ version for an offline install, ensure that the catalogsource pods are in running state.

You can view the availability of an upgrade in the following user interface pages:

- In the IBM Fusion user interface:
  - Click the bell icon and view notifications in System notifications.
  - Notification banner on the Services page.
  - Settings > Upgrade page.

- **Migrating Global Data Platform remote mount**
  Migrate the Global Data Platform remote mount service when installing or upgrading Fusion Data Foundation 4.21 or later.
- **Upgrading Fusion Data Foundation**
  Steps to upgrade the Fusion Data Foundation service.
- **Upgrading Global Data Platform**
  Upgrade Global Data Platform service.
- **Upgrading Backup & Restore service**
  After a IBM Fusion operator upgrade, if a Backup & Restore upgrade is available, IBM Fusion automatically initiates prechecks and then proceeds with the upgrade.
- **Upgrading IBM Data Cataloging service**
  Steps to upgrade IBM Data Cataloging service.
- **Upgrading Content-Aware Storage (CAS)**
  Steps to upgrade Content-Aware Storage (CAS) service.
- **Upgrading Software Hub service**
  Steps to upgrade the Software Hub service.

Migrating Global Data Platform remote mount

Migrate the Global Data Platform remote mount service when installing or upgrading Fusion Data Foundation 4.21 or later.

Before you begin

Ensure that the following conditions are met before starting the migration:

- No backup or restore jobs are running.
- The IBM Storage Scale cluster is healthy:
  - Run the `mmhealth` command.
  - Verify that all components report a healthy status.
- No MachineConfigPool (MCP) rollout is in progress.
- Global Data Platform version is 6.0.0.2. If the current version is 6.0.0.1, upgrade it to 6.0.0.2 before proceeding. For instructions, see Upgrading Global Data Platform.

About this task

This procedure does not apply when IBM Spectrum® Storage Scale Erasure Code Edition (ECE) is used as local storage. It applies only when Fusion Data Foundation is used as local storage.

Starting with Red Hat® OpenShift® Container Platform 4.21 and Fusion Data Foundation 4.21, the Global Data Platform remote mount service is integrated into Fusion Data Foundation as External Scale.

In earlier releases, Global Data Platform was bundled separately with IBM Fusion. From Fusion Data Foundation 4.21 onward:

- No separate upgrade option is available for version for 6.0.0.2 and later. Global Data Platform remote mount upgrades and migration of IBM Storage Scale remote mount (CNSA) from version 6.0.0.2 to 6.0.1.0 or later are performed automatically in the background during Fusion Data Foundation installation or upgrade.
- Each Fusion Data Foundation release includes an updated IBM Storage Scale build, which is applied automatically.

- Future updates to IBM Storage Scale are delivered through Fusion Data Foundation upgrades.

The migration workflow depends on your deployment scenario.

Procedure

Scenario 1: Only Global Data Platform remote mount is installed

If Global Data Platform remote mount 6.0.0.2 is installed and Fusion Data Foundation is not installed:

- Install Fusion Data Foundation 4.21 or later. For instructions, see Installing Fusion Data Foundation.

During installation:

- The system automatically triggers the Global Data Platform remote mount migration.
- Global Data Platform remote mount is upgraded to version 6.0.1.0 or later as part of the process.

Scenario 2: Both Fusion Data Foundation and Global Data Platform remote mount are installed

If Fusion Data Foundation 4.20 or earlier and Global Data Platform 6.0.0.2 are already installed:

1. Disable the Fusion Data Foundation automatic upgrade to prevent unintended upgrade behavior.
2. Manually trigger the Fusion Data Foundation upgrade from version 4.20 to 4.21 or later. For instructions, see Upgrading Fusion Data Foundation.

During the upgrade:

- Global Data Platform remote mount migration runs and completes automatically.
- No separate action is required for migration.

Results

After the Fusion Data Foundation installation or upgrade and the Global Data Platform migration are complete:

- The Global Data Platform remote mount service is upgraded to 6.0.1.0 or later.
- The Global Data Platform remote mount service is in healthy state.
- All IBM Storage Scale pods run in a single namespace, `ibm-spectrum-scale`.

Upgrading Fusion Data Foundation

Steps to upgrade the Fusion Data Foundation service.

Before you begin

- Important: If you upgrade to Fusion Data Foundation 4.21 or later and the Global Data Platform remote mount service installed and configured, the service is automatically migrated to version 6.0.1.0. If automatic updates are enabled, disable automatic upgrades before proceeding. Review the supported migration scenarios before you proceed with the upgrade.
- For the Fusion Data Foundation service upgrade, follow this upgrade sequence: 4.16.x > 4.17.x> 4.18.x> 4.19.x> 4.20.x.

The version of Data Foundation service is dependent on the version of OpenShift® Container Platform.

About this task

For more information about the storage version support, see IBM Fusion support matrix.

When you order and upsize Logical Volume Manager (LVM) drives in the upgraded IBM Fusion HCI 2.13 version, IBM Fusion HCI automatically detects the presence of control plane drives and configures them for LVM. However, if LVM is already enabled in your existing installation of IBM Fusion HCI and you add additional 1.6 TB drives again, the newly added drives remain unutilized.

Procedure

1. Log in to IBM Fusion HCI user interface.
2. Click Services menu to go to the Services page.
   Note: Alternatively, you can go to the Settings > Upgrades page and view the upgrade status and in progress upgrades. The Data Foundation goes into rolling upgrade. If you can run service upgrades in parallel in a non-disruptive manner, click Upgrade (<number of upgrades available>). If there are any errors found in the automatic prechecks, then click the service tile for more details about the error.
3. In the Installed section, click Upgrade from the ellipsis menu of Data Foundation.
   Only if an upgrade is available for Data Foundation, the Upgrade option is displayed.
   The Upgrade pre-check window gets displayed.
4. In the Upgrade pre-check window, monitor the progress of the pre-checks.
   The IBM Fusion HCI runs prechecks to ensure that your system is ready for an upgrade.
   If all the checks pass or there are some conditions in warning state, you can proceed with the upgrade. However, it is highly recommended to fix the warnings to prevent any potential impacts on the system during the upgrade process. Immediately address any blocker errors encountered during prechecks, as they require fixing before proceeding with the upgrade. After the completion of all pre-checks, a summary of total number of issues found gets displayed. If there are no blocker issues, go to step 6.
5. If there are blocker issues, do the following steps:

   a. For more information about blockers or warnings, click View upgrade details link in the Services page. The IBM Fusion HCI user interface takes you to the Services page wherein a Upgrade pre-check report window displays all the Blocking issues and Issues (optional). Alternatively, click the service tile in error state in the Upgrade page to view the Upgrade pre-check report with Blocking issues and Issues (optional) issues.
   b. Click the corresponding BMYxxxxxx code to know more about how to fix the issue.
      If you find error resources in the Resources section of the individual issues, you can click the resource link to know more details about it.
   c. Fix all the blocking issues based on the guidance.
   d. Return to the Services page of the IBM Fusion HCI user interface and click Upgrade.
6. After the pre-checks complete without any blocker issues, click Upgrade now.
   To know more details about the upgrade that is progress, click View details in the notification tab of the Services page. A slide out pane displays the following details:
   - Progress percentage
   - Upgrade pre-check. As this is complete, a check mark appears and the status shows complete.
   - Upgrade in progress.
7. Wait for a minute or more for the upgrade to complete.
   After the completion of upgrade, a success notification gets displayed.
   To view the status of the upgrade, go to Settings > Upgrade or the Services page.

   Important:
   - If auto-upgrade is not enabled during installation, you may encounter a situation where some components of the Data Foundation operators do not upgrade as expected. To ensure all components are upgraded correctly, you must verify that all **`InstallPlans`** in the **`openshift-storage`** namespace are approved. For steps, see [Data Foundation operator upgrade is not initiated].
   - The upgrade consumes more time depending on the number of compute nodes in the rack.
8. When you upgrade Data Foundation from version 4.16.z to 4.17.z or 4.18.z or 4.19.z or 4.20.z, you must also upgrade the Local Storage Operator (LSO) to ensure that OpenShift Container Platform, Data Foundation, and LSO are on compatible versions.
   If LSO is not upgraded automatically at the same OpenShift Container Platform level post OpenShift Container Platform upgrade, follow these steps to change the Update Channel and set the Update approval:
   a. Change the Update Channel.
      To upgrade LSO to a new minor (y-stream) version, update the channel as follows:
      i. Go to Operators > Installed Operators > Local Storage > Subscription tab > Update Channel.
      ii. Set the update channel to match your current OpenShift Container Platform version (For example, 4.19 or 4.20).
   b. Set Update Approval:
      If Update Approval is set to Automatic, the upgrade proceeds automatically.
      If set to Manual, you must manually approve and apply the upgrade.

Upgrading Global Data Platform

Upgrade Global Data Platform service.

Before you begin

- Optionally, before you upgrade the Global Data Platform, enable kdump to save the core that gets created in the event of any IBM Storage Scale crash. You need the core dump for debugging and analysis. For the steps to enable kdump, see [Enabling kdump] in *Red Hat OpenShift Container Platform* documentation.
- Run the following command to check the pidslimits.

  ```
  for node in `oc get nodes -o name`; do oc debug $node -- chroot /host grep podPidsLimit /etc/kubernetes/kubelet.conf; done
  ```

  Example output:

  ```
  for node in `oc get nodes -o name`; do  oc debug $node -- chroot /host grep podPidsLimit  /etc/kubernetes/kubelet.conf; done
  Starting pod/compute-1-ru5rackm02mydomaincom-debug ...
  To use host binaries, run `chroot /host`
    "podPidsLimit": 12228,

  Removing debug pod ...
  Starting pod/compute-1-ru6rackm02mydomaincom-debug ...
  To use host binaries, run `chroot /host`
    "podPidsLimit": 12228,

  Removing debug pod ...
  Starting pod/compute-1-ru7rackm02mydomaincom-debug ...
  To use host binaries, run `chroot /host`
    "podPidsLimit": 12228,

  Removing debug pod ...
  ```

  Note: Ensure that the pidslimit is not lesser than 12228 for each node in any condition.
- If you plan to do a offline installation, mirror the Global Data Platform 5.2.3.0 images to your enterprise registry. For the procedure and images to download, see [Mirroring IBM Storage Scale images].
- Ensure no Backup & Restore jobs are running.

About this task

You can perform service upgrades from the Services page or the Settings > Upgrades page of the IBM Fusion user interface.
For more information about the storage version support, see [Support matrix].

Procedure

1. Log in to IBM Fusion HCI user interface.

2. Click Services menu to go to the Services page.
   Note: Alternatively, you can go to the Settings > Upgrades page and view the upgrade status and in progress upgrades. The Global Data Platform goes into rolling upgrade. If you can run service upgrades in parallel in a non-disruptive manner, click Upgrade (<number of upgrades available>). If there are any errors found in the automatic prechecks, then click the service tile for more details about the error.
3. In the Installed section, click Upgrade from the ellipsis menu of Global data platform.
   Only if an upgrade is available for Global data platform, the Upgrade option is displayed.
   The Upgrade pre-check window gets displayed.
4. In the Upgrade pre-check window, monitor the progress of the pre-checks.
   The IBM Fusion HCI runs prechecks on conditions that impact the upgrade of the service. For more information about the prechecks and its coverage, see the user interface.
   If all the checks pass or there are some conditions in warning state, you can proceed with the upgrade. However, it is highly recommended to fix the warnings to prevent any potential impacts on the system during the upgrade process. Immediately address any blocker errors encountered during prechecks, as they require fixing before proceeding with the upgrade. After the completion of all pre-checks, a summary of total number of issues found gets displayed. If there are no blocker issues, go to step 6.

5. If there are blocker issues, do the following steps:
   a. For more information about blockers or warnings, click View upgrade details link in the Services page. The IBM Fusion HCI user interface takes you to the Services page wherein a Upgrade pre-check report window displays all the Blocking issues and Issues (optional). Alternatively, click the service tile in error state in the Settings > Upgrades page to view the Upgrade pre-check report with Blocking issues and Issues (optional) issues.
      You can also view the parallel upgrades from Settings > Upgrades page.
   b. Click the corresponding BMYxxxxxx code to know more about how to fix the issue.
      If you find error resources in the Resources section of the individual issues, you can click the resource link to know more details about it.
   c. Fix all the blocking issues based on the guidance.
   d. Return to the Services page or the Upgrade page of the IBM Fusion HCI user interface and click Upgrade.
6. After the pre-checks complete without any blocker issues, click Upgrade now.
7. When the upgrade is in progress, in the side panel, you can click view details link for advanced monitoring.
   If you find error resources in the View details section of the individual sections, you can click the resource link to know more details about it.
8. Wait for the upgrade to complete.
   After the completion of the upgrade, a success notification gets displayed.
   If you want to disable the kdump after a successful upgrade of Global data platform, see and follow the instructions for Disabling kdump.
   Note: The upgrade consumes more time depending on the number of compute nodes in the rack.
   To view the status of the upgrade, go to Settings > Upgrade or the Services page.

   If you see “Service upgrade has failed” message, see View details on the right pane for more details. If you are not able to fix the issue, contact IBM support.

Upgrading Backup & Restore service

After a IBM Fusion operator upgrade, if a Backup & Restore upgrade is available, IBM Fusion automatically initiates prechecks and then proceeds with the upgrade.

Important: When you upgrade to version 2.13.0, the Backup & Restore database is upgraded and the data is migrated to the new version of MongoDB. Ensure that the Backup & Restore storage class contains sufficient available capacity. The upgrade process requires an extra 60 GiB of storage temporarily to migrate the Backup & Restore database to the new version. After the migration is complete, the old Persistent Volume Claims (PVCs) are deleted, releasing the extra storage.
If you plan to do an offline installation in IBM Fusion 2.13.0, mirror the Backup & Restore 2.13.0 images to your enterprise registry for the automatic upgrade to initiate. For the procedure and images to download, see Mirroring Backup & Restore images.

To view the status of the upgrade, go to Settings > Upgrade or the Services page.

Upgrading IBM Data Cataloging service

Steps to upgrade IBM Data Cataloging service.

About this task

Sometimes, the IBM Data Cataloging service upgrade may not be available after you upgrade IBM Fusion. To know more about such issues and its resolutions, see Troubleshooting Data Cataloging service upgrade issues.

Procedure

1. Log in to IBM Fusion HCI user interface.
2. Click Services menu to go to the Services page.
   Note: Alternatively, you can go to the Settings > Upgrades page and view the upgrade status and in progress upgrades. If you can run service upgrades in parallel in a non-disruptive manner, click Upgrade (<number of upgrades available>). If there are any errors found in the automatic prechecks, then click the service tile for more details about the error.
3. In the Installed section, click Upgrade from the ellipsis menu of IBM Data Cataloging.
   Only if an upgrade is available for IBM Data Cataloging, the Upgrade option is displayed.
   The Upgrade pre-check window gets displayed.
4. In the Upgrade pre-check window, monitor the progress of the pre-checks.
   The IBM Fusion HCI runs some common prechecks.
   If all the checks pass or there are some conditions in warning state, you can proceed with the upgrade. However, it is highly recommended to fix the warnings to prevent any potential impacts on the system during the upgrade process. Immediately address any blocker errors encountered during prechecks, as they require fixing before proceeding with the upgrade. After the completion of all pre-checks, a summary of total number of issues found gets displayed. If there are no blocker issues, go to step 6.

5. If there are blocker issues, do the following steps:

    a. For more information about blockers or warnings, click View details link.
    b. The IBM Fusion HCI user interface takes you to the Services page wherein a Upgrade pre-check report window displays all the Blocking issues and Issues (optional). Alternatively, click the service tile in error state in the Settings > Upgrades page to view the Upgrade pre-check report with Blocking issues and Issues (optional) issues.
    You can also view the parallel upgrades from Settings > Upgrades page.
    c. Click the corresponding BMYxxxxxx code to know more about how to fix the issue.
    If you find error resources in the Resources section of the individual issues, you can click the resource link to know more details about it.
    d. Fix all the blocking issues based on the guidance.
    e. Return to the Services page of the IBM Fusion HCI user interface and click Upgrade.
6. After the pre-checks complete without any blocker issues, click Upgrade now.
7. Wait for a minute or more for the upgrade to complete.
   After the completion of upgrade, a success notification gets displayed.
   Note: The upgrade consumes more time depending on the number of compute nodes in the rack.
   To view the status of the upgrade, go to Settings > Upgrade or the Services page.

   If you see "Service upgrade has failed" message, see View details on the right pane for more details. If you are not able to fix the issue, contact IBM support.

Upgrading Content-Aware Storage (CAS)

Steps to upgrade Content-Aware Storage (CAS) service.

Before you begin

Open the IBM Fusion user interface, go to the Services tab and check the current version of CAS in the Installed section.
Note: You cannot skip version when upgrading CAS to the latest version. For example, if CAS v1.0.4 is installed as your current version, upgrading to v1.1.0 requires an upgrade path of 1.0.4 > 1.0.5 > 1.0.6 > 1.0.7 > 1.1.0.

Procedure

1. Open the IBM Fusion user interface.
2. Go to the Services tab. In the Installed section, click Upgrade service in the ellipsis overflow menu of Content-Aware Storage.
   Alternatively, you can go to the Settings > Upgrades page, and view the upgrade status and in progress upgrades. If you can run service upgrades in parallel in a non-disruptive manner, click Upgrade (<number of upgrades available>).
3. In the Upgrade service window, click Upgrade.
   The upgrade starts and you can see the upgrade status. After the completion of upgrade, a success notification is displayed.
   Note: Wait for a minute or more to see the upgrade status in the Services tab of the IBM Fusion user interface. You can also find the latest version in the Services tab when the upgrade completes.

Upgrade considerations for CAS v1.0.6 and earlier to v1.1.0
If you are upgrading from CAS v1.0.6 or earlier, start by upgrading to CAS v1.0.7 through the IBM Fusion user interface. After IBM Fusion initiates the upgrade, you must manually approve the process in OpenShift® before it can continue to CAS v1.1.0 by following these steps:

1. Manual approval that is required after CAS v1.0.7
   After Fusion completes the upgrade to CAS v1.0.7, the next upgrade step (from v1.0.7 to v1.1.0) does not start automatically.

   Before you continue, verify that the CAS subscription in OpenShift is configured as follows:

   - **Update approval: Manual**

   OpenShift path:

   Operators > Installed Operators > Project: ibm-cas > Content Aware Storage > Subscription

2. Verify whether CAS v1.0.7 is installed correctly
   Before you approve the upgrade to version 1.1.0, verify that the environment is ready. To confirm Kafka readiness for the next upgrade step, ensure that Kafka is running in KRaft mode.

   ```
   oc get kafka -n ibm-cas -o yaml | yq '.items[0].status.kafkaMetadataState'
   ```

   Expected output:

   ```
   KRaft
   ```

   When the value is `KRaft`, you can approve the next upgrade safely.

3. Approve upgrade to CAS v1.1.0
    a. After confirming readiness, go to Operators > Installed Operators and select the project as ibm-cas.
    b. Open Content Aware Storage, go to the Subscription tab, and click 1 requires approval.
    c. Select Preview install plan.
    d. Click Approve.
   The installation process of CAS v1.1.0 is triggered.

   While the upgrade is waiting for manual approval in OpenShift, IBM Fusion displays the service status as Upgrading. This behavior is expected. IBM Fusion remains in this state until you approve the upgrade to version 1.1.0 in the OpenShift UI. After approval and installation are complete, IBM Fusion updates the version and marks the service as Healthy.

Upgrading Software Hub service

Steps to upgrade the Software Hub service.

Before you begin

Ensure that you met the following prerequisites for a successful upgrade of the Software Hub service.

- Before upgrading the Software Hub service, create the required image pull secret by following the instructions steps:
  1. Log in to the cluster using the `oc login` command.
  2. Create a `dockerconfig.json` file using your IBM Entitlement Key:

```
cat <<EOF > dockerconfig.json
{
  "auths": {
    "cp.icr.io": {
      "auth": "${IMAGE_PULL_CREDENTIALS}"
    },
    "icr.io":{
      "auth": "${IMAGE_PULL_CREDENTIALS}"
    }
  }
}
EOF
```

  3. Create the `ibm-entitlement-key` secret in both the operator and instance namespaces.

```
oc create secret docker-registry ibm-entitlement-key \
--from-file ".dockerconfigjson=dockerconfig.json" \
--namespace=cpd-oeprator
```

```
oc create secret docker-registry ibm-entitlement-key \
--from-file ".dockerconfigjson=dockerconfig.json" \
--namespace=cpd-instance
```

  Note:
  - During upgrade, automated the backup and uninstalling of IBM cert manager operator.
  - The Red Hat® cert manager latest version is installed as part of upgrade the Software Hub service.

Procedure

1. Log in to IBM Fusion HCI user interface.
2. Click Services menu to go to the Services page.
   Note: Alternatively, you can go to the Settings > Upgrades page and view the upgrade status and in progress upgrades. The Software Hub goes into rolling upgrade. If you can run service upgrades in parallel in a non-disruptive manner, click Upgrade (<number of upgrades available>). If there are any errors found in the automatic prechecks, then click the service tile for more details about the error.
3. In the Installed section, click Upgrade from the ellipsis menu of Software Hub.
   Only if an upgrade is available for Software Hub, the Upgrade option is displayed.
   The Upgrade pre-check window gets displayed.
4. In the Upgrade pre-check window, monitor the progress of the pre-checks.
   If an empty page for the upgrade precheck is displayed. To check the status of these prechecks, run the following command:

```
oc get softwarehubmanager ibm-softwarehub-service-instance \
-n ibm-software-hub-operator \
-o jsonpath='{.status.environmentReadiness}'
```

   In the `Environment Readiness` category, check the message type for each precheck. If the message type is `INFO`, the precheck is passed. If any precheck has a message type of `ERROR`, check the Events page for details on the failure.
5. After the prechecks complete without issues, click Upgrade.
6. During the upgrade, you can click the View details link in the side panel for advanced monitoring. If error resources are listed, click the resource link for more information.
7. Wait for the upgrade to complete. A success notification is displayed upon completion.
8. To view the status of the upgrade, go to Settings > Upgrade or the Services page.
   If the service upgrade failed message is displayed, check the Events page for details. If you cannot resolve the issue, contact IBM Support.

- **Upgrading watsonx.ai service**
  Steps to upgrade the watsonx.ai service.
- **Upgrading watsonx Orchestrate service**
  Steps to upgrade the watsonx Orchestrate service.

Upgrading watsonx.ai service

Steps to upgrade the watsonx.ai service.

Before you begin

Ensure that you met the following prerequisites for a successful upgrade of the watsonx.ai service.

1. Before upgrading the Software Hub service, create the required image pull secret by following the instructions steps:
   a. Log in to the cluster using the `oc login` command.
   b. Create a `dockerconfig.json` file using your IBM Entitlement Key:

```
cat <<EOF > dockerconfig.json
{
  "auths": {
    "cp.icr.io": {
      "auth": "${IMAGE_PULL_CREDENTIALS}"
    },
    "icr.io":{
      "auth": "${IMAGE_PULL_CREDENTIALS}"
    }
  }
}
EOF
```

c. Create the `ibm-entitlement-key` secret in both the operator and instance namespaces.

```
oc create secret docker-registry ibm-entitlement-key \
--from-file ".dockerconfigjson=dockerconfig.json" \
--namespace=cpd-oeprator
```

```
oc create secret docker-registry ibm-entitlement-key \
--from-file ".dockerconfigjson=dockerconfig.json" \
--namespace=cpd-instance
```

2. Ensure that you upgrade the Software Hub service to latest version before upgrading the watsonx.ai service.

Procedure

1. Log in to IBM Fusion HCI user interface.
2. Click Services menu to go to the Services page.
   Note: Alternatively, you can go to the Settings > Upgrades page and view the upgrade status and in progress upgrades. The watsonx.ai goes into rolling upgrade. If you can run service upgrades in parallel in a non-disruptive manner, click Upgrade (<number of upgrades available>). If there are any errors found in the automatic prechecks, then click the service tile for more details about the error.
3. In the Installed section, click Upgrade from the ellipsis menu of Software Hub.
   Only if an upgrade is available for watsonx.ai, the Upgrade option is displayed.
   The Upgrade pre-check window gets displayed.
4. In the Upgrade pre-check window, monitor the progress of the pre-checks.
   If an empty page for the upgrade precheck is displayed. To check the status of these prechecks, run the following command:

   ```
   oc get watsonxaimanager ibm-watsonxai-service-instance \
   -n ibm-software-hub-operator \
   -o jsonpath='{.status.environmentReadiness}'
   ```

   In the `Environment Readiness` category, check the message type for each precheck. If the message type is `INFO`, the precheck is passed. If any precheck has a message type of `ERROR`, check the Events page for details on the failure.

5. After the prechecks complete without issues, click Upgrade.
6. During the upgrade, you can click the View details link in the side panel for advanced monitoring. If error resources are listed, click the resource link for more information.
7. Wait for the upgrade to complete. A success notification is displayed upon completion.
8. To view the status of the upgrade, go to Settings > Upgrade or the Services page.
   If the service upgrade failed message is displayed, check the Events page for details. If you cannot resolve the issue, contact IBM Support.

Upgrading watsonx Orchestrate service

Steps to upgrade the watsonx Orchestrate service.

Before you begin

Ensure that you met the following prerequisites for a successful upgrade of the watsonx Orchestrate service.

1. Before upgrading the watsonx Orchestrate service, create the required image pull secret by following the instructions steps:
   a. Log in to the cluster using the `oc login` command.
   b. Create a `dockerconfig.json` file using your IBM Entitlement Key:

```
cat <<EOF > dockerconfig.json
{
  "auths": {
    "cp.icr.io": {
      "auth": "${IMAGE_PULL_CREDENTIALS}"
    },
    "icr.io":{
      "auth": "${IMAGE_PULL_CREDENTIALS}"
    }
  }
}
EOF
```

c. Create the `ibm-entitlement-key` secret in both the operator and instance namespaces.

```
oc create secret docker-registry ibm-entitlement-key \
--from-file ".dockerconfigjson=dockerconfig.json" \
--namespace=cpd-oeprator
```

```
oc create secret docker-registry ibm-entitlement-key \
--from-file ".dockerconfigjson=dockerconfig.json" \
--namespace=cpd-instance
```

2. Ensure that you upgrade the Software Hub service to latest version before upgrading the watsonx Orchestrate service.

Procedure

1. Log in to IBM Fusion HCI user interface.
2. Click Services menu to go to the Services page.
   Note: Alternatively, you can go to the Settings > Upgrades page and view the upgrade status and in progress upgrades. The watsonx Orchestrate goes into rolling upgrade. If you can run service upgrades in parallel in a non-disruptive manner, click Upgrade (<number of upgrades available>). If there are any errors found in the automatic prechecks, then click the service tile for more details about the error.
3. In the Installed section, click Upgrade from the ellipsis menu of Software Hub.
   Only if an upgrade is available for watsonx Orchestrate, the Upgrade option is displayed.
   The Upgrade pre-check window gets displayed.
4. In the Upgrade pre-check window, monitor the progress of the pre-checks.
   If an empty page for the upgrade precheck is displayed. To check the status of these prechecks, run the following command:

   ```
   oc get watsonxorchestratemanager ibm-watsonxorchestrate-service-instance \
   -n ibm-software-hub-operator \
   -o jsonpath='{.status.environmentReadiness}'
   ```

   In the **Environment Readiness** category, check the message type for each precheck. If the message type is **INFO**, the precheck is passed. If any precheck has a message type of **ERROR**, check the Events page for details on the failure.

5. After the prechecks complete without issues, click Upgrade.
6. During the upgrade, you can click the View details link in the side panel for advanced monitoring. If error resources are listed, click the resource link for more information.
7. Wait for the upgrade to complete. A success notification is displayed upon completion.
8. To view the status of the upgrade, go to Settings > Upgrade or the Services page.
   If the service upgrade failed message is displayed, check the Events page for details. If you cannot resolve the issue, contact IBM Support.

Upgrading Hosted Control Plane Fusion and its services

When IBM Fusion and services on the Hosted Control Plane cluster are installed using labels, they are automatically upgraded to the latest supported version available on the IBM Fusion hub.

It ensures that IBM Fusion and services versions (Fusion Data Foundation, Global Data Platform remote mount, and Backup & Restore) on the hub and the Hosted Control Plane cluster remains synchronized.
Note: When you upgrade HCP clusters on IBM Fusion HCI from version 2.12.x to 2.13.0, you must manually approve the **installplan** generated for the upgrade on respective HCP cluster.

- **Upgrading OpenShift Container Platform on Hosted Control Plane**
  Upgrade the OpenShift Container Platform version for Hosted Control Planes (HCP) clusters by following the Red Hat documentation and updating the NodePool custom resource to point to the mirrored release image for offline environments.

Upgrading OpenShift Container Platform on Hosted Control Plane

Upgrade the OpenShift Container Platform version for Hosted Control Planes (HCP) clusters by following the Red Hat documentation and updating the NodePool custom resource to point to the mirrored release image for offline environments.

To upgrade the OpenShift® Container Platform version for Hosted Control Plane clusters, see the Updating a control plane in a hosted cluster in *Red Hat OpenShift Container Platform* documentation.
Note: For offline Hosted Control Plane clusters:

- If you are upgrading OpenShift Container Platform in an offline Hosted Control Plane cluster, first mirror the images for the target OpenShift Container Platform version. For more information, see Mirroring OpenShift Container Platform release images to enterprise registry.
- In addition to the steps provided in the Red Hat documentation, make sure to update the **release-image** field in the **NodePool** custom resource (CR) to point to the mirrored OpenShift Container Platform release image.

Upgrading OpenShift Container Platform

Upgrade OpenShift in IBM Fusion HCI. For offline upgrades of OpenShift Container Platform, follow the Red Hat steps. However, follow the guidelines and versions that are specified in this section.

Before you begin

Important: If you are upgrading to OpenShift Container Platform 4.21 for the Fusion Data Foundation service and you selected Enable automatic updates during installation, disable automatic upgrades if the Global Data Platform remote mount service is installed or configured. Before you proceed with the upgrade, review the supported migration scenarios.

Important: Check the compatibility of your underlying storage before you proceed with the OpenShift Container Platform upgrade. For more information about the version support, see Support matrix.

- Common prerequisites:
  - For OpenShift Container Platform upgrades, the following upgrade sequences are recommended by IBM and are validated with IBM Fusion Lab:
    - 4.18.22 > 4.19.9 > 4.20.z
    - 4.17.28 > 4.18.18 > 4.18.25
    - 4.17.z > 4.18.25

    Note: If you choose to use a different OpenShift upgrade path than those paths listed prior, you must plan the upgrade in coordination with Red Hat or by using the OpenShift Upgrade Planner tool: https://access.redhat.com/labs/ocpupgradegraph/update_path/

    Important: The OpenShift Container Platform 4.14.x and 4.15.x versions are not supported from IBM Fusion HCI 2.12.0. Upgrade to any supported OpenShift Container Platform version before you upgrade IBM Fusion HCI.
  - After every OpenShift Container Platform upgrade, check whether you have an installation of OpenShift virtualization operator. If it is available, then upgrade it. For more information, see Updating OpenShift Virtualization in *Red Hat OpenShift Container Platform* documentation.
  - If you are using an offline method, see the following *Updating clusters topics* based on your version of OpenShift Container Platform:
    - Red Hat OpenShift Container Platform 4.16 documentation
    - Red Hat OpenShift Container Platform 4.17 documentation
    - Red Hat OpenShift Container Platform 4.18 documentation
    - Red Hat OpenShift Container Platform 4.19 documentation
    - Red Hat OpenShift Container Platform 4.20 documentation
    - Red Hat OpenShift Container Platform 4.21 documentation
  - Verify whether all services and OpenShift Container Platform are healthy.
  - Both high speed switches must be available, functional, and healthy.

    Note: Review the supported and allowed upgrade path on OpenShift Container Platform upgrade graph tool. Do this to avoid the upgrade to a version that does not have any further upgrade paths due to unavailability of stable channels. For more information about upgrade version details, see Red Hat OpenShift Container Platform Update Graph.
- When the submariner auto upgrade fails or you observe any submariner issue during OpenShift Container Platform upgrade, contact IBM Support.
- Ensure that the appropriate eviction strategy is configured for the virtual machine before starting the OpenShift Container Platform upgrade. For more information, see Eviction strategies in *Red Hat OpenShift Container Platform* documentation.

About this task

In the following procedure, version 4.18 is used as an example. Replace it with your version of OpenShift Container Platform.

Procedure

1. Run the following command to change the channel to stable 4.18 or higher.

   ```
   oc patch clusterversion version --type merge -p '{"spec": {"channel": "stable-4.18"}}'
   ```

2. Upgrade OpenShift Container Platform to 4.18.18.

   Important: When you upgrade OpenShift Container Platform from version 4.20 to 4.21, run the following command to acknowledge that the Sigstore signatures are mirrored and your cluster is ready update:

   ```
   oc -n openshift-config patch configmap admin-acks --patch '{"data":{"ack-4.20-sigstore-in-4.21":"true"}}' --type=merge
   ```

3. Wait for upgrade to complete.
4. Verify whether all services and OpenShift Container Platform are healthy.

   IBM Fusion HCI user interface:
   - Go to the Services page and check the health of all services.
   - Go to Infrastructure > Overview and check the health of all nodes and switches.

   OpenShift Container Platform console:
   Go to Operators > Installed operator in IBM Fusion namespace and check that the ISF operator shows in Succeed state.

   OC commands
   - Run the following command to check the ready state of the OpenShift Container Platform nodes:

     ```
     oc get nodes
     ```

   - Run the following command to check that all operators are available.

     ```
     oc get co
     ```

Firmware upgrade

The firmware on all nodes, whether they are part of the base OpenShift® Container Platform cluster or Hosted Control Plane hosted nodes, is automatically updated each time the IBM Fusion operator is upgraded. Firmware updates are applied automatically in the background. Firmware updates for switches are orchestrated by IBM Fusion, but require a manual step to trigger the update.

For more information about automatic node firmware upgrade, see Upgrading node firmware.

Note:

- Power operations must not be performed on any node in the cluster during a firmware upgrade.
- IBM Fusion HCI manages firmware upgrades for both switches and nodes in a rolling fashion, ensuring that there is a minimal impact to the Red Hat® OpenShift cluster during the upgrades. Firmware update for switches happens one at a time.

Upgrading Red Hat OpenShift Virtualization

Upgrade of Red Hat® OpenShift® Virtualization on IBM Fusion HCI is done based on Red Hat guidelines.

For the procedure to upgrade Red Hat OpenShift Virtualization, see Updating OpenShift Virtualization in *Red Hat OpenShift Container Platform* documentation.

Important: If you created virtual machines on the OpenShift Container Platform cluster by using the OpenShift virtualization, then the PVCs of the live migration VMs use Read Write Many (RWX) access modes. Otherwise, these VMs block drain during roll-outs, and you must manually force-drain the nodes and delete the VMs.

Upgrading the service node

This section provides the upgrade procedure of the service node.

Before you begin

Before you start the service node upgrade, ensure that the following prerequisites are met:

Version and system requirements

- The service node must be on the previous **`major.minor`** versions. For example, the service node must be on versions 2.11 or 2.12 before you upgrade to version 2.13.
- The service node must be running Red Hat® Enterprise Linux 9.6 or later.
- The service node must have a minimum of 80 GiB of free storage space.
- The service node firmware must be at the newest version. The compute operator manages the firmware update.

Credentials and access

- The correct Kubernetes-native Infrastructure (KNI) user credentials must be stored in the OpenShift® Container Platform secret. For example, the secret name can be **`<rack>-servicenode-kni-secret`**.
- The service node agent uses port **`3001`** on the service node during the upgrade.

Air-gapped environment requirements

- The upgrade image must be mirrored to the private registry. For more information, see Mirroring IBM Fusion HCI images.
- The IDMS or ICSP configuration must be correct.

If all prerequisites are met and the KNI user secret is updated correctly, you can trigger the service node upgrade from the IBM Fusion UI.

Upgrade process

The service node upgrade process occurs in the following steps:

1. OpenShift Container Platform cluster operations
   a. The user triggers the upgrade from the IBM Fusion UI.
   b. The registry CA certificates are synchronized to the service node.
   c. The upgrade image is resolved. In air-gapped environments, this phase includes IDMS or ICSP handling.
   d. The upgrade image is downloaded to the service node, and the upgrade agent is extracted and started.
2. Service node operations
   a. The upgrade agent upgrades the service node.
   b. The upgrade agent installs additional software bundles such as AWX and Vault.
3. Populate and configure Vault. The Vault is configured and populated with the latest data.
   Remember: Download and securely store the Vault secrets after a successful service node upgrade as follows:
   a. Select **`ibm-spectrum-fusion-ns`** namespace in projects drop-down, and navigate to workload > secrets > vault secret.
   b. In the YAML section, click the Download to download the secret.
   c. Repeat steps 3.a and 3.b for the **`vault-login-secret`** secret.
   d. Verify that you have downloaded the YAML files for both **vault-secret** and **vault-login-secret**.

Failure scenarios and troubleshooting

Pre-upgrade validation failures

- RHEL version check failed: The service node must be running RHEL 9.6.
- Insufficient storage: Free up the disk space so that at least 80 GiB is available.
- Firmware not at newest version: The compute operator automatically upgrades the service node firmware.

Connectivity failures

- Invalid credentials: Update the KNI user credentials in the OpenShift Container Platform secret.

Image download failures

- Image pull failed: Verify the pull secret, the IDMS or ICSP configuration for air-gapped environments, and the image specified in the **`cr-version-cm`** ConfigMap. In air-gapped environments, ensure that the image is mirrored correctly.
- Certificate issues: Ensure that **`additionalTrustedCA`** is configured for air-gapped environments.

Upgrade agent failures

- The agent not responding: Collect the service node logs.
- Stage-specific failures: Check the upgrade status for specific stage errors.
- Maximum retry reached: After a failure, a **CRITICAL** event is raised. Fix the root cause and then retrigger the upgrade.

Vault-related failures during service node upgrade
: If the service node upgrade fails, perform the following checks:

  - Confirm that the service node is accessible and there are no network connectivity issues.
  - Confirm that Vault is installed on the service node by running the `vault version` command.
  - Verify whether the service node was restarted or powered off during the upgrade process.
  - If the service node was restarted, check the Vault service status. Vault might start in a sealed or masked state after a node reboot or service interruption.
  - If Vault is sealed, stopped, or masked, resolve the condition and retry the upgrade. For the recovery steps, see Troubleshooting Vault.

Monitoring

You can monitor the service node upgrade from the IBM Fusion UI.

Events

- `INFO: BMYUP1136` - Service node upgrade triggered
- `INFO: BMYUP1172` - Service node upgrade completed
- `CRITICAL: BMYUP1141` - Service node upgrade failed

Logs
: If a failure occurs, collect the service node log package for additional debugging.

Re-trigger upgrade

1. Ensure that the `update-operator` pod is up and running.
2. Re-verify that the version and system requirements are met by running the following command:

```
oc get healthmanager systemhealth -n ibm-spectrum-fusion-ns -o json | jq -e '
[
  .status.environmentReadiness[]
  | select(
      .category=="RHELVersionCompatability" or
      .category=="ServiceNodeStorageAvailability" or
      .category=="ServiceNodeFirmwareStatus"
    )
  | select(.messageType=="INFO")
]
| length == 3
' > /dev/null && echo "PASSED" || echo "FAILED"
```

3. Proceed according to the command output:
   - `PASSED`: Safe to retrigger the service node upgrade.
   - `FAILED`: Do not retrigger the upgrade. First investigate and resolve the failing prechecks.
4. Retrigger a failed upgrade only after you fix the observed failure:

```
oc patch updatemanager version --type=merge -p '{"spec":{"serviceNodeUpgrade":{"triggerUpdate":true}}}'
```

   - This action resolves previous failure events.
   - The agent resumes from the last successful stage, but you must retrigger the upgrade.

Troubleshooting upgrade issues

Troubleshooting IBM Fusion HCI upgrade issues.

Node upgrade pending

When you use a rack as a service node and upgrade to version 2.11, you may see all nodes in the "Upgrade Pending" state on the Nodes page. It indicates that a firmware upgrade is in progress. Check the `computeinit` Custom Resource (CR) to identify the node that is currently undergoing the upgrade.

Restore failures post upgrade

Problem statement
: After upgrade, you might encounter some restore failures. Check whether the job logs of the failed restore jobs contain the following error:

```
Invalid value: true: Privileged containers are not allowed, spec.containers[0].securityContext.capabilities.add: Invalid
value: "SYS_ADMIN": capability may not be added, provider "nonroot":
```

Resolution
: Contact IBM Support to resolve this known issue.

Community operator catalog is shown as missing

Resolution
: If the Community operator catalog is shown as missing, then create it before you attempt upgrade.

Machine config roll out error

Resolution
If an operation causes **machine config roll out** and gets stuck for a long time, then check whether the node to be updated is pingable and has an IP after restart. If there exist any DHCP or network issues that prevent the node from getting a hostname, then fix them and restart the node.

On-demand backup failures post upgrade

Problem statement
Post upgrade, on-demand backup failures might happen for existing applications.

Resolution
Do the following manual steps after you upgrade to avoid this problem:

1. Run the following command to display the phase status of the backup policies associated with all your applications:

   oc get fpa -A

   Example output:

```
oc get fpa -A

NAMESPACE               NAME                                    PROVIDER     APPLICATION           BACKUPPOLICY      DATAC
ibm-spectrum-fusion-ns  deptest2-azure-hourly-30                isf-ibmspp   deptest2              azure-hourly-30
ibm-spectrum-fusion-ns  new-generic-1-azure-hourly-45           isf-ibmspp   new-generic-1         azure-hourly-45
ibm-spectrum-fusion-ns  new-mongo-project-1-azure-hourly-15     isf-ibmspp   new-mongo-project-1   azure-hourly-15
ibm-spectrum-fusion-ns  new-mongo-project-azure-hourly-30       isf-ibmspp   new-mongo-project     azure-hourly-30
```

2. Verify whether your **policyassignment** CR corresponds to any application in **InitializeError** phase. In this example, the **new-mongo-project-1** application is in **InitializeError** phase.
3. Log in to IBM Fusion HCI user interface.
4. Go to Applications > Backups tab.
5. Unassign the backup policy that is assigned to the application in **InitializeError** phase and wait for its unassignment. In this example, unassign **azure-hourly-15** policy from **new-mongo-project-1** application.
6. Reassign the backup policy.

ImagePull failure

Resolution
If an **ImagePull** failure occurs due to intermittent network or registry issue during an upgrade, then restart the pod and retry. If the issue persists, contact IBM support.

DeadlineExceeded error

Problem statement
IBM Cloud Paks foundational services operator ClusterServiceVersion (CSV) status shows Failed and its InstallPlan status shows Failed after the subscription gets created.

Resolution
If you notice that the operator installation or upgrade fails with **DeadlineExceeded error**, see Operator installation or upgrade fails with DeadlineExceeded error.

Operator `OOMKilled` error in IBM Fusion namespace

Problem statement
The pods go into crash loop state with the OOMKilled error after OpenShift® Container Platform upgrade.

Resolution
Follow steps 1 through 4 below to address the **OOMKilled** error related to the **isf-update-operator**:

1. Go to IBM Fusion **clusterserviceversion** object (Operators > Installed Operators > IBM Fusion operator > YAML tab).
2. Search for the deployment name of the **isf-update-operator** (**isf-update-operator-controller-manager**) from the list of deployments in the **clusterserviceversion** object under **spec.install.spec.deployments**.
3. In the specified deployment object, search for the container name manager under the **spec.template.spec.containers** and increase the memory limit in the **resources.limits.memory**.
4. After changing the limits in the IBM Fusion **clusterserviceversion**, the update operator pod restarts with the new limits.
5. If the OOMKilled issue still persists, then follow the steps 1 - 4 again.

watsonx.ai upgrade progress stalled at 60%

Problem statement
The watsonx.ai service upgrade is stalled at 60%. This issue occurs when the WML CR is fixed at 22.5%.

Resolution
Follow the steps to resolve the issue:
Note: In the following commands, **$instance_ns** represents the namespace where the watsonx.ai instance is installed.

1. Run the following command to verify the status of the WML custom resource:

   oc describe wmlbases wml-cr -n $instance_ns | grep Progress

Example output:

```
Progress:                  22.5%
Progress Message:          Waiting for dependency CCS to upgrade
Wml Status:                InProgress
```

If the WML CR is stuck at 22.5% with the above message, continue to the next step.

2. Run the following command to verify that the CCS CR is completed successfully:

   ```
   oc describe ccs ccs-cr -n $instance_ns | grep Progress
   ```

   Example output:

   ```
   Progress:          100%
   Progress Message:  Last reconciliation succeeded
   ```

   If the CCS CR progress is 100% and the WML CR is still stuck at 22.5%, then proceed with the following resolution steps.

3. To resolve the WML upgrade issues, follow the workaround instruction mentioned in the Watson Machine Learning upgrade to version 5.3 fails with imagepull error.
4. Run the following command to verify whether the `WatsonxAIIFM` CR is stuck at 9%:

   ```
   oc describe watsonxaiifm watsonxaiifm-cr -n $instance_ns | grep Progress
   ```

   Example output:

   ```
   Progress:         9%
   Progress Message: Waiting for CCS cr ccs-cr to be in 'Completed' state, before continuing with watsonxaiifm install or up
   Watsonxaiifm Status: InProgress
   ```

   If the CR is in this state, proceed to the next step.

5. To resolve the `WatsonxAIIFM` upgrade issue, follow the workaround instruction mentioned in the Watson Machine Learning upgrade to version 5.3 fails with imagepull error.

watsonx Orchestrate upgrade progress stalled at 83%

Problem statement
: The watsonx Orchestrate service upgrade is stalled at 83%. This issue occurs when watsonx Orchestrate `Kafka` resource does not enter the verifying state during upgrade..

Resolution
: To resolve the issue, follow the workaround instruction mentioned in the During upgrade watsonx Orchestrate Kafka is not in ready state.

Troubleshooting and serviceability

The steps to troubleshoot an issue vary depending on the problem. To help make relevant information available to you as quickly as possible, the log files and other tools for troubleshooting problems are consolidated.

- **Contacting IBM Support Center**
  The IBM® Support Center is available for various types of IBM hardware and software problems that IBM Fusion customers might encounter.
- **How to provide feedback**
  Customers can submit feedback on the IBM Documentation platform and on product documentation using the "Was this topic helpful?" rating on every page.
- **Serviceability in IBM Fusion HCI**
  The serviceability of IBM Fusion includes Call Home client, event manager, and log collector. They gather event logs to track down problematic events, help troubleshooting issues, and update relevant information in the support ticket automatically.
- **Events and error codes message references in IBM Fusion HCI**
  This reference information provides a consolidated view of all the events or error messages that can be displayed when the IBM Fusion system might result in an event or an error.
- **Troubleshooting issues in IBM Fusion HCI**
  This topic covers the most common issues and their workarounds, which you might encounter while you work with the IBM Fusion HCI appliance.

Contacting IBM Support Center

The IBM® Support Center is available for various types of IBM hardware and software problems that IBM Fusion customers might encounter.

Before you begin

Points to note when you contact the IBM Support Center:

1. For IBM Fusion HCI, ensure that you have your system’s serial number and its associated account number.
2. Based on your issue, collect appropriate log files and diagnostic data.
3. IBM Support provides a time period within which IBM representative returns your call. Be sure that the person that you identified as your contact can be reached in the phone number.
4. An online case gets created to track the problem you are reporting. Note down the number for future reference and communication. If you need to make subsequent calls to discuss the problem, use the number to identify the problem.

About this task

If you enabled Call Home, then it connects your system to service representatives who can monitor issues and respond to problems efficiently and quickly to keep your system up and running. Enabling Call Home significantly improves your IBM Support experience. For more information about Call Home and Service level in Call Home, see Enabling Call Home for IBM Fusion HCI.

If you havean IBM Hardware and Software Maintenance service contract, then use this procedure to open a case on IBM Support site.

The following table lists alternative ways of contacting IBM Support Center:

Your location	Method of contacting the IBM Support Center
In the United States	Call **1-800-IBM-SERV** for support.
Outside the United States	Contact your local IBM Support Center or see the Directory of worldwide contacts (www.ibm.com/planetwide).

If you do not have an IBM Software Maintenance service contract, then contact your IBM sales representative to find out how to proceed. For non-IBM failures, follow the problem-reporting procedures provided with that product.

Procedure

1. Log in to IBM support page by using your IBMid and click Continue.
2. In Let's troubleshoot, click Open a case tile.
3. In the Open a case page, enter the following details:
   a. Select a value for the Type of support that you need.
   b. Enter the Case title to describe your case. The maximum length of the case title is 255 characters.
   c. Enter `IBM` in Product manufacturer.
   d. Enter `Fusion` or `Fusion HCI Physical Appliance` for Product.
   e. For `Fusion HCI Physical Appliance`, enter the Machine serial number.
   f. Enter Product version.
   g. Select Service Type.
   h. Check Address where product is located.
   i. Select Severity.
   j. For `Fusion HCI Physical Appliance`, enter Machine Type Model.
      Note: For `Fusion HCI Physical Appliance` the machine models are currently: 9155-C10, 9155-C14, 9155-G03, 9155-S01, 9155-S02, 9155-R42, 9155-TF5.
   k. Enter System down (yes or no).
   l. Enter Failing Component (Software or Hardware)
   m. Enter Account name and Case description.
   n. Enter Case contact phone number.
   o. You can Add team members based on the need.

   Note: If a diagnostic file is required for this case, you can upload it after you submit this case.
4. Go through the terms and conditions and click Submit case.

How to provide feedback

Customers can submit feedback on the IBM Documentation platform and on product documentation using the "Was this topic helpful?" rating on every page.

IBM customers can provide feedback on two levels: feedback on the the platform (IBM Documentation site) and feedback on your product's documentation.

How to provide feedback on IBM Documentation platform

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We welcome any feedback that you have. You will not receive a direct response to your feedback, but your comments help us improve your IBM Documentation experience.

How to provide feedback on your product's documentation

IBM Documentation provides a single standardized process to leave feedback on your product using "Was this topic helpful?", located at the top of every page. When you rate the content with a thumbs up or thumbs down, a dialog appears where you can leave optional comments. IBM Documentation automatically captures both the product and the URL of the page you are leaving your feedback on.
We welcome any feedback that you have, including comments on the clarity, accuracy, or completeness of the information. You will not receive a direct response to your feedback, but your comments help us improve our product documentation.
If you require support and a response to technical questions, visit IBM's technical support site to open a defect.

Serviceability in IBM Fusion HCI

The serviceability of IBM Fusion includes Call Home client, event manager, and log collector. They gather event logs to track down problematic events, help troubleshooting issues, and update relevant information in the support ticket automatically.

Call Home

The Call Home feature alerts you and IBM Support about hardware failures and potentially serious configuration or environmental events. It constantly monitors the health and availability of your storage so that you can address issues before they impact your business.

You can enable this feature to receive requests from any IBM Fusion source. For more information about benefits and how to enable, see Benefits of enabling Call Home and Enabling Call Home for IBM Fusion HCI.

For more information about how to open a support ticket, see **Support ticket**.

Event manager

The event manager contains the logic for servicing the basic event handling. Event Manager does basic checks for duplicate events and forwards them to Call Home Client. The expectation is that Call Home Client determines whether an event must be ignored or must open a Salesforce ticket.

You can work with the event manager from both IBM Fusion user interface and OpenShift® user interface:

For more information about how to filter events, see Analyzing events in IBM Fusion HCI.

Log collector

The log collector collects and returns appliance logs dynamically at run time. You can configure and manage collection sets for the log collector by using both user interface and APIs. A list of all the appliance logs and collection sets that can be returned are maintained in related config maps.

For more information about logs in IBM Fusion HCI, see Logs in IBM Fusion HCI.

- **Analyzing events in IBM Fusion HCI**
  The Events page provides you with the necessary features to view event records in a table view, filter them based on severity and category, and view the status of support tickets.
- **Logs in IBM Fusion HCI**
  The IBM Fusion HCI has in-built log capability to collect the logs. You can configure and manage the log package by using both the user interface and APIs.
- **Benefits of enabling Call Home**
  Configure and enable the call home feature as it improves the user experience when IBM Fusion is used.
- **Remote support**
  Remote support is a service that enables IBM Support representative to remotely access your rack and accomplish essential tasks, such as installation, maintenance, and troubleshooting.
- **Password rotation**
  The password rotation feature automatically updates hardware (nodes) credentials at regular intervals to enhance system security and meet enterprise compliance requirements. It minimizes the risk of credential compromise by eliminating the need for manual password updates.
- **Custom certificate expiration alert**
  This section provides information about custom certificate expiration monitoring and alerting for critical cluster components.
- **Configuring License Service Reporter for IBM Fusion**
  Configure License Service Reporter to obtain a unified view of IBM Fusion licensing metrics. License Service Reporter supports this feature and can be extended to IBM Fusion. In addition to the IBM Fusion user interface, it provides an additional view of licensing metrics, offering deeper insights. It can also be configured across multiple clusters, enabling centralized monitoring and reporting of license usage in distributed environments.
- **Troubleshooting serviceability issues**
  List of all troubleshooting and known issues in Events, Log collection, and Call Home.

Analyzing events in IBM Fusion HCI

The Events page provides you with the necessary features to view event records in a table view, filter them based on severity and category, and view the status of support tickets.

Note: The IBM Storage Scale controller instance name and the `isf_node` label (as seen in CNI) can exceed 63 characters, which are the Source column value in the Events dashboard. Due to the Red Hat® OpenShift® limitation for labels, an error occurs when it is processed and displayed. When this error takes place for events processing, IBM Fusion truncates the name to 60 characters to display on the dashboard and in Events or Call Home data. For more information, see Host name validation errors.

Viewing events from the IBM Fusion HCI user interface

Events are listed as table rows with Severity, Category, Time, Description, Support ticket, and Resource columns. By default, all critical, warning, and informational events are displayed in the list. You can sort the columns Severity, Category, and Support ticket.

1. Log in to IBM Fusion HCI user interface.
2. Click Events menu.
3. In the Events page, enter keywords in Search text to search for records. Also, filter records based on Severity and Category.
   The events are listed with the following headers by default:
   - Severity - Type of event. It can be Critical, Warning, or Informational.
   - Category - Category of the event. For example, Backup and restore, Compute, Network, Other, Storage, and System.
   - Time - Date and time of the occurrence of the event
   - Description - Description of the event.
   - Support ticket - Indicates the ticket number associated with the event.
   - Fixed - Displays the events that are already fixed or resolved. Events are either fixed by operators or manually.

   You can use the downloads icon to download all events that have registered in the last 48 hours.

   You can use the settings icon to select or clear the column headings from the table display. If you want to reset to default column headings, then click Reset to default.

   You can also decide the number rows to display in a view from the Items per page drop-down list. Use the arrow keys to jump between the list of pages or use the drop-down list next to the arrow keys to go to a specific page.

The events exist in the system for the following time duration:

- Information events lasts for 3 to 4 hours
- Warning events lasts for 7 days
- Critical events lasts for 14 days

If any existing `Warning` or `Critical` event is known to be resolved, it can be marked as fixed from the IBM Fusion events page.

For a list of IBM Fusion HCI events and error messages, see Events and error codes message references in IBM Fusion HCI.

Viewing events from the Red Hat OpenShift Container Platform console

1. Log in to OpenShift Container Platform console.
2. Go to Overview > Events and view events.
3. In the Events page, you can filter based on resources and type. In You can also type a name or message to filter.

Important: Also, you can enable notifications of IBM Fusion events using OpenShift Container Platform `Alertmanager`. For steps, see IBM Fusion event alerts.

Command line or command prompt

You can run oc commands to view events:

```
oc get event --field-selector reason=ISFEventManager
```

For specific component, use the appropriate label and annotation label.

- **IBM Fusion event alerts**
  Use these instructions to enable notifications of IBM Fusion events using OpenShift Container Platform `Alertmanager`.

IBM Fusion event alerts

Use these instructions to enable notifications of IBM Fusion events using OpenShift® Container Platform `Alertmanager`.

About this task

IBM Fusion supports notifications for `CRITICAL` and `WARNING` events, which are visible in the IBM Fusion user interface. By configuring supported receivers in the OpenShift Container Platform, notifications can be routed through preferred channels such as Slack, email, and webhooks. This setup simplifies incident management by delivering alerts directly to configured notification channels. Additionally, multiple IBM Fusion clusters can be configured with similar settings to centralize event notifications in a single location.

Procedure

1. Log in to the OpenShift Container Platform web console.
2. Go to Administration > Cluster Settings.
3. In the Configuration tab, select the Alertmanager.
4. In the Alertmanager, click Create Receiver.
   The Create Receiver page opens.
5. Enter the Receiver name and select the channel that you prefer from the Receiver type drop-down list.
   For example, if you select the Slack as a preferred channel. Ensure that you must update the following details.
   a. Enter the Slack API URL.
   b. Enter the Channel name to send the notifications.
   c. Use Show advanced configuration drop-down to customize the Icon, Username, Title, and Text.
      Important: You can use the following labels and annotations to customize your message:

      ```
      Severity: {{ .CommonLabels.severity }}
      Hardware Address: {{ .CommonLabels.hwAddr }}
      Event Message: {{ .CommonLabels.alertMessage }}
      Alert Summary: {{ .CommonAnnotations.summary }}
      Alert Description: {{ .CommonAnnotations.description }}
      ```

      - Cluster - Add the cluster name manually.
      - Severity - `Critical` or `Warning`.
      - Address - Hardware address generating the event (cannot be present for software).
      - Event message - Description of the event.
      - Alert summary - Static string of `IBM Fusion Event Alert`.
      - Alert description - Static string `Alert sent from IBM Fusion Event Manager`.
   d. Enter the label details in the Routing labels.
      Important:
      - Ensure that you enter `source = ISFEventManager` label in the Routing labels.
      - Do not mark Send resolved alerts to this receiver? to true.
      Optionally, you can also add `severity = CRITICAL` or `severity = WARNING` labels.
6. Click Create.

What to do next

After the receiver is configured, you start receiving notifications for IBM Fusion events based on the routing label specified during setup.

By default, when the routing label source is set to `ISFEventManager`, the receiver is notified of all events with severity levels `WARNING` and `CRITICAL`. This setup ensures that important alerts are delivered promptly without needing extra filtering.

To streamline monitoring across multiple racks, you can apply the same receiver configuration to each rack. This enables centralized alerting, allowing all events to be routed to a common notification channel for unified visibility and management.

Note: The cluster name is not automatically detected, so it is recommended to manually specify the cluster name in the receiver configuration to ensure accurate identification and traceability of events.

Logs in IBM Fusion HCI

The IBM Fusion HCI has in-built log capability to collect the logs. You can configure and manage the log package by using both the user interface and APIs.

You can create your log package by using one or more component logs. IBM may also request you for logs from IBM Fusion HCI when troubleshooting a support case. You may be asked to collect a particular set of logs package and upload the log package to an IBM site. The IBM Fusion HCI gathers the current logs and includes them in a compressed archive file. For the procedure to collect log package, see Collecting log packages for IBM Fusion HCI. Decompress the collected log zip file and use the sub-folders and files as diagnostic data to resolve issues or errors in IBM Fusion HCI. For more guidance on the download log package, see Log package.

If you have enabled Call Home, then the system automatically collects appropriate log packages and raises a IBM Support ticket.

- **Collecting log packages for IBM Fusion HCI**
  You can create your log package by using one or more component logs.
- **Log package**
  A quick overview of the IBM Fusion HCI log packages.
- **Automatic cleanup of log packages**
  This section describes the automatic cleanup feature for diagnostic log packages that are collected from the IBM Fusion environment.

Collecting log packages for IBM Fusion HCI

You can create your log package by using one or more component logs.

Procedure

1. From the title bar, click the help icon and select Support logs.
2. In the Support logs page, click Create.
   The Create a log package page gets displayed.
3. In the Create a log package page, select one or more components to create the log package. The log package contains the folders, namespaces, and relevant files. For more guidance on the log package, see Log package.
   You can collect logs for individual nodes and switches. Click Add and select the nodes or switches that you want to collect the logs. Alternatively, you can download logs for the nodes and switches from the Nodes and Network menu on the IBM Fusion HCI user interface as follows:
   a. Go to Infrastructure > Nodes.
   b. In the ellipsis menu of the node, click Collect logs to get the logs of the node.

   System logs
      System logs contains the information about the operation, status, and potential issues of the system health and administration in the IBM Fusion HCI.

      System health check
         Use this option to collect logs that provide an overall assessment of the health of IBM Fusion HCI. This log package includes the following information:
         - The health of the components that make up the IBM Fusion HCI.
         - The health of IBM Fusion services such as the Global Data Platform and Backup & Restore.
         - The health of the IBM Fusion HCI operator
         - The Red Hat® OpenShift® configuration of Bare Metal hosts, machine sets, machines, and nodes.

      Administration
         Use this option to collect logs related to Red Hat OpenShift, administration, and audit logging.

   Hardware logs
      Hardware logs contains the information about the operation, status, and potential issues of the hardware components such as nodes, service node, and switches in the IBM Fusion HCI.

      Nodes
         Using this option to collect logs for the nodes in the IBM Fusion HCI appliance. Troubleshoot issues related to node health, firmware upgrades, storage configuration, Metro-DR, and Regional DR configurations. For example, issues like offline nodes and issues with firmware upgrade.

      Network switches
         Use this option to collect logs for the switches in the IBM Fusion HCI appliance. Troubleshoot issues related to the IBM Fusion HCI network. The logs that are collected in this package include information on the switches, Vlans, and links that make up the appliance network. For example, issues like network outages, degraded performance, connecting the IBM Fusion HCI appliance to the data center network, and problems related to Metro-DR recovery.
         Attention: Network switches log collection package does not include the spine switches logs.

   Software and services
      Software and services logs contains the information about the operation, status, and potential issues of the services such as Global Data Platform, IBM Data Cataloging, Backup & Restore, and Data Foundation in the IBM Fusion HCI.

      Global Data Platform
         Important:
         - Enable scale traces before you collect the Global Data Platform logs. For more information about how to enable traces, see Generating GPFS trace reports.

         Use this option to collect logs related to the physical storage configuration of IBM Fusion HCI and Global Data Platform. These logs are used to troubleshoot storage issues. For example, issues like loss of storage access, problems with CSI provisioning, problems related to disaster recovery (Metro-DR and Regional DR), problems with the overall health of the Global Data Platform.

Backup & Restore
: Use this option to collect logs related to the Backup & Restore service. These logs are used to troubleshoot issues that are related to setup, backup, and restore. These logs contain mainly the **`ibm-backup-restore`** namespace resources.
For Backup & Restore service, you can also select logs for the specific clusters as follows:

a. In the Summary slide out pane, click edit icon next to Backup & Restore.
The Select spoke clusters page gets displayed with all the available clusters.

b. Select the spoke clusters that you want to include in the log package.
Important:
- You can collect logs for spokes from Hub by selecting the spoke clusters from the Hub IBM Fusion HCI user interface.
- The logs can also be collected from the individual Spoke clusters.

Backup & Restore agent
: Use this option to collect logs related to the Backup & Restore service on the spoke cluster. These logs are used to troubleshoot issues that are related to setup, backup, and restore. These logs contain mainly the **`ibm-backup-restore`** namespace resources.

IBM Data Cataloging
: Use this option to collect logs that are related to the IBM Data Cataloging service. These logs contain mainly the **`ibm-data-cataloging`** namespace resources.

Data Foundation
: Use this option to collect logs that are related logs related to Kubernetes resources of openshift-local-storage, openshift-storage-client, openshift-lvm-storage, and openshift-storage.

Important: If a service is not enabled, the user interface does not display the service component in the log collection.

4. Click Create.
The created log package gets added to the log list with status as Creating. After the collection is complete, the status changes to Success.
The log package creation might take several minutes and the amount of time might vary based on the type of logs collected. You can view the progress of the log package creation in the table that is shown on the Support logs page.

The Support logs page lists all created log packages with the following details:

Name
: The name of the log package.
Note: The name of the log package consists of two parts: component name and timestamp. For example, in a nodes package, the file name must be `cn-20220324124442`. The name of the log package change for every component.
- **`Global Data Platform: gdp`**
- **`System heath check: sh`**
- **`Administration: ad`**
- **`Network: nw`**
- **`Data foundation: odf`**

Contents
: It contains details of which component logs comprise the log package.

Status
: The values of the status field are Collecting, Success, and Partial.

Note: The Partial status means that some of the requested logs are collected successfully, but there are error messages that are generated in the process.

Created
: The date and time of the creation of the log package.

Expires
: The expiry date and time of the log set.

Note: The collected log packages are auto-deleted after 24 hours.

5. After a log package is successfully added to the log list, you can upload it to IBM directly by using the Call Home feature, download it to your local, or delete it.

Upload the collected log package to IBM Support
: If Call Home is enabled, you can use the IBM Fusion HCI user interface to automatically upload logs to a ticket. For steps to enable call home, see [Enabling Call Home for IBM Fusion HCI](#).

a. Click the ellipsis menu of the log record and click Upload.
The Upload log package window gets displayed. The collected logs get uploaded so that the IBM Support team can use for diagnosis.
b. In the Upload log package window, choose whether you want to create a new support case or attach to an existing case.
c. If it is an existing case, then enter the Ticket number.
Note: A log collection package can be uploaded only once to any ticket through this user interface page.
d. Click Upload log.

Download collected log package
: Click the ellipsis menu of the log record and click Download. The Downloading log package window gets displayed indicating that the download is in progress.

Delete a log package
: a. Click the ellipsis menu of the log record and click Delete.
b. In the Confirm delete log package window, click Delete.

If you did not enable Call Home, then download the log package and upload it to an IBM site. In the table of log packages, click the ellipsis menu for the log package that you want to upload and select the Download action. It triggers the download of the log package file in your web browser. The user interface provides three methods of uploading logs to IBM.

FTP transfer
: This option is the fastest approach to upload logs to IBM. It uses FTP to upload logs to the IBM Enhanced Customer Data Repository (ECUREP) site.

Browser upload
: Using the browser upload option, upload the log package from your web browser. This option is limited to files that are smaller than 200 MiB.

Blue Diamond
: HIPPA customers designated as Blue Diamond must upload logs to the IBM Blue Diamond site.

What to do next

1. After a log package is successfully completed, you can check the folders, namespaces, and relevant files of the log package to read and troubleshoot. See Log package.

Log package

A quick overview of the IBM Fusion HCI log packages.

The following log packages are available in the Logs page:

1. Nodes
2. Network switches
3. Global Data Platform
4. System health check
5. Administration
6. Backup and restore
7. Data cataloging
8. Data Foundation

Every log package contains a HealthManagerCR folder expect for the Backup & Restore service.

By default, every log package contains a Must Gather folder except for the on-boarded services such as IBM Data Cataloging, Global Data Platform and Backup & Restore. The Data Foundation service must gather logs collected in the `data-foundation-ext` log collection set. For more information about Must Gather, see Must Gather section.
Note:

- The Must Gather folder contains common data that is collected for all the log packages.
- The `isf-serviceability-operator-manager` service account uses cluster admin role as it is required to collect `oc adm must-gather` logs.

Nodes

Logs related to compute nodes, service node, firmware upgrades, and node health. The nodes log package mainly contains the folders for the following namespace:

ibm-spectrum-fusion-ns
: It contains the logs and object definition files for all major resources in this namespace. For example, routes, pods logs, deployments, daemon sets, persistent volumes, and config maps.
Note: It includes the Kubernetes events that are associated with the resource. If any resource does not exist, then skip it.

openshift-machine-api
: It contains the logs and object definition files for all major resources in this namespace. For example, routes, pods logs, deployments, daemon sets, persistent volumes, and config maps.

openshift-machine-config-operator
: It contains the logs and object definition files for all major resources in this namespace. For example, routes, pods logs, deployments, daemon sets, persistent volumes, and config maps.

IMM-logs
: Important: The IMM logs package is collected only for the Lenovo nodes.
The bmc-logs logs folder contains the IMM_NAME.ffdc and IMM_NAME.inventory files for every IMM.
Note: The log contents vary based on the Node vendor `imm` logs for Lenovo and Support Assist package for Dell.

Dell Support Assist logs
: Important: The Dell Support Assist Logs package is collected only for the Dell nodes.
The Dell support assist logs contains the debug logs, hardware logs, and storage logs for the Dell nodes.

OCP must gather
: The folder contains OpenShift® Container Platform must-gather logs for analysis.

Service node logs
: The service node logs contain diagnostic information from the service node to help troubleshooting and system diagnostics. These include Baseboard Management Controller (BMC) logs and retrieval of operating system (OS) logs from the service node.

OS files and directories collected by default

- /home/kni/isfLogs/
- /opt/fusion/remote-support/challengeManagerClient.log.*
- /var/opt/ibm/trc/target/*.log
- /var/opt/ibm/trc/target/profiles/lightsoutprofile.properties
- /home/kni/logs
- /etc/ibmtrct.conf
- /var/log/fusion

Diagnostic command outputs

The command-output or directory contains outputs from system diagnostic commands:

- System information (**uname -a**)
- Operating system details (**cat /etc/os-release**)
- Hostname and system information (**hostnamectl**)
- Disk space usage (**df -h**)
- Memory usage (**free -h**)
- System uptime and load average (**uptime**)

Vault must gather logs
It retrieves vault status details and diagnostic logs to assist with troubleshooting internal vault issues.

Error logging

- **error.log** - Lists any files or directories that failed to be collected due to permission issues, file not found, or other issues. This log is created only if errors occur.
  Note:
  - If a file or directory path is not found, it is logged in **error.log** but does not mark the log collection as partial. The log collection is marked as partial only if a file or directory exists but cannot be retrieved due to permission issues or other access-related issues.
  - If a directory path exists but contains no content, it does not appear in the collected log package.
- **stderr.log** - Contains error output from diagnostic commands. This log is created only if commands fail.

Important:

- Service node logs must be collected by using the Support Log page. Do not use the Service Node page under Infrastructure > Overview, as it does not collect OS logs.
- Service node logs are collected only if a service node is present in the rack.
- All files and directories must have read permission for the **kni** user on the service node.
- Additional custom paths can be specified through the **isf-serviceability-operator-collection-sets** ConfigMap.

Nodes log packages also contain the custom resources such as **baremetalhosts**, **machinesets**, **machines**, **spectrumfusions**, **computeprovisionworkers**, **computemonitorings**, **rackinfoes**, **computeinits**, **computeconfigurations**, **computemaintenances**, **computepowerops**, **computediscoveries**, **computefirmwares**, and **updatemanagers**.

Network switches

Logs related to appliance network errors and connectivity. The network switches log package mainly contains the folders for the following namespaces:

ibm-spectrum-fusion-ns
It contains the logs and object definition files for all major resources in this namespace. For example, routes, pods logs, deployments, daemon sets, persistent volumes, and config maps.
Note: It includes the Kubernetes events that are associated with the resource. If any resource does not exist, then skip it.

openshift-monitoring
It contains the logs and object definition files for all major resources in this namespace. For example, routes, pods logs, deployments, daemon sets, persistent volumes, and config maps.

Vlans
The Vlans folder contains a vlans.json file that includes a Vlan list.

Links
The Links directory contains a links.json file that includes a link list.

Switch
The Switch directory contains a switches.json file that includes only the definition of the switches.

Switch logs
The switch log packages include directories that contain the actual switch logs. These packages are organized by switch type, and each package includes the following directories:

- Juniper switches
  - juniper-switch-logs
  - clncl
  - gencl
- Cumulus switches
  - cumulus-switch-logs
  - clncl
  - gencl

Network log packages also contain the custom resources such as **networkinits**, **switchfirmwares**, **network-attachment-definitions**, and **updatemanagers**.

Global Data Platform

Logs related to storage availability. The Global Data Platform log package mainly contains the folders for the following namespaces:
Important: By default, it collects the logs from the IBM Fusion namespace and the namespace where the service is deployed.

- **ibm-spectrum-scale-namespace**
- **ibm-spectrum-scale-csi-namespace**
- **ibm-spectrum-scale-operator-namespace**
- **openshift-monitoring**

Global Data Platform log packages contain both the custom and cluster resources. The custom resources such as **spectrumfusions**, **scalemanagers**, **scaleclusters**, and **nodes**. The cluster resources such as **storageclasses.**

Follow the steps to collect `Velero` namespace logs:

1. On the Global Data Platform service, manually add the `Velero` namespace under the namespaces in the serviceability section of the `FusionServiceDefinition` instance.
2. It ensures that the `Velero` namespace logs are collected as part of log collection for the Global Data Platform service.
3. The `Velero` namespace is specified in `ramen-dr-cluster-operator-config` configmap in the `ibm-spectrum-fusion-ns` namespace.

It also includes the GPFS snap logs and must-gather that is related to IBM Storage Scale.
Important: To obtain GPFS logs, gather them from IBM Fusion HCI user interface and not from OpenShift Container Platform.

System health check

Logs related to overall assessment of the health of IBM Fusion HCI. The system health check log package mainly contains the folders for the following namespaces:

Node metrics
: It contains the object definition files for the NodeMetricsList type.

`ibm-spectrum-fusion-pod-metric`
: It contains the PodMetrics files for every pod present in this namespace.

Switch logs
: The switch log packages include directories that contain the actual switch logs. These packages are organized by switch type, and each package includes the following directories:

- Juniper switches
  - juniper-switch-logs
  - clncl
  - gencl
- Cumulus switches
  - cumulus-switch-logs
  - clncl
  - gencl

It also includes the Baremetalhosts, Machinesets, Machines, and Node network configuration policy folders with the relevant log files.

Administration

Logs related to Red Hat® OpenShift, administration, and audit logging. The administration log package mainly contains the folder for the following namespace:

Audit logs
: Audit logs contain the Control-0-rack, Kube, OAuth, and OpenShift folders with the relevant log files.

Backup and restore

Logs related to the backup and restore of your workload applications. The backup and restore log package mainly contains the folders for the following namespaces:
Important: By default, it collects the logs from the IBM Fusion namespace and the namespace where the service is deployed.

`ibm-backup-restore`
: It contains the logs and object definition files for all major resources in this namespace. For example, routes, pods logs, deployments, daemon sets, persistent volumes, and config maps.
Note: It includes the Kubernetes events that are associated with the resource. If any resource does not exist, then it can be skipped.

If Cloud Pak for Data (CPD) is installed and custom resource (CR) is created, the backup and restore log packages also include backup and restore–specific CPD logs, which contain log data from all CPD-related namespaces.
Note: The CPD log collection process might fail to collect logs for Velero backup and restore operations. The failure occurs when the CPD script is unable to validate or recognize self-signed TLS certificates configured for the backup storage location. To prevent this, ensure that a properly configured and trusted TLS certificate is associated with the backup storage location.

Data cataloging

Logs related to the IBM Data Cataloging service. The IBM Data Cataloging log package mainly contains the folders for the following namespace:
Important: By default, it collects the logs from the IBM Fusion namespace and the namespace where the service is deployed.

`ibm-data-cataloging`
: It contains the logs and object definition files for all major resources in this namespace. For example, routes, pods logs, deployments, daemon sets, persistent volumes, and config maps.
Note: It includes the Kubernetes events that are associated with the resource. If any resource does not exist, then skip it.

Data Foundation

Logs related to storage availability. The Red Hat OpenShift Data Foundation log package mainly contains the folders for the following namespaces:
Important: By default, it collects the logs from the IBM Fusion namespace and the namespace where the service is deployed.
Red Hat OpenShift Data Foundation log packages contains both the custom and cluster resources. The custom resources such as **spectrumfusions**, **odfmanagers**, **odfclusters**, **fusionservicedefinitions**, **fusionserviceinstances**, **storageclusters**, **cephclusters**, **localvolumediscoveries**, **localvolumediscoveryresults**, **localvolumesets**, **catalogsources**, **volumesnapshotclasses**, and **clustercsidrivers**.

The cluster resources such as **persistentvolumes**, **clusterrolebindings**, **storageclasses**, and **storageclassclaims**.

Follow the steps to collect `Velero` namespace logs:

1. On the Data Foundation service, manually add the `Velero` namespace under the namespaces in the serviceability section of the `FusionServiceDefinition` instance.
2. It ensures that the `Velero` namespace logs are collected as part of log collection for the Data Foundation service.
3. The `Velero` namespace is specified in `ramen-dr-cluster-operator-config` configmap in the `ibm-spectrum-fusion-ns` namespace.

It includes the `Openshift-storage` and `Openshift-local-storage` folder with the relevant log files.

Note: Though Data Foundation log status shows partial due to the unavailability of one or more namespaces, you can still upload and download partial logs.

Must gather

The Must Gather folder contains the common data that is collected for all the log packages. It mainly contains the following files, which might be helpful in debugging to get more information about the environment. It contains information about nodes, nodes config, kickstart.json, rackinfo config map, and persistent volumes related files.

Automatic cleanup of log packages

This section describes the automatic cleanup feature for diagnostic log packages that are collected from the IBM Fusion environment.

After a log package collection completes, the system automatically removes or sanitizes sensitive information before the package is available to download. This behavior protects sensitive data and supports compliance with enterprise security and data-protection requirements.

Key benefits

- Protects sensitive information by removing or masking confidential data.
- Supports compliance with data-protection and privacy requirements.
- Enables safe sharing of log packages with IBM Support.
- Operates automatically without requiring additional user actions.
- Preserves diagnostic information that is needed for analysis and troubleshooting.

Protected information

The log cleanup process removes or obfuscates sensitive data in collected log packages.

Removed data

- Kubernetes secrets, including `secret` resources and `secret.yaml` files
- Credentials such as passwords, tokens, and API keys
- Authentication files, including `kubeconfig`, pull secrets, and service account tokens
- Private keys, including SSH keys and TLS private keys (`.key`, `.pem`)
- Certificate files, including SSL and TLS certificates (`.crt`)

Masked or obfuscated data

- IP addresses are replaced with consistent placeholder values
- MAC addresses are replaced with consistent placeholder values
- Internal domain names are obfuscated
- Credentials (API keys) and tokens are obfuscated
- PEM certificate content found in files

Unchanged content

The cleanup process removes or obfuscates only sensitive information. All logs, resources, and configuration data that is required for debugging remain intact.

User experience

By default, automatic log cleanup is enabled and requires no manual intervention.

1. Collect logs by using the IBM Fusion UI.
2. The system automatically sanitizes sensitive data in the log package.
3. Download the clean and secure log package.

Configuration

When to change the configuration (optional)

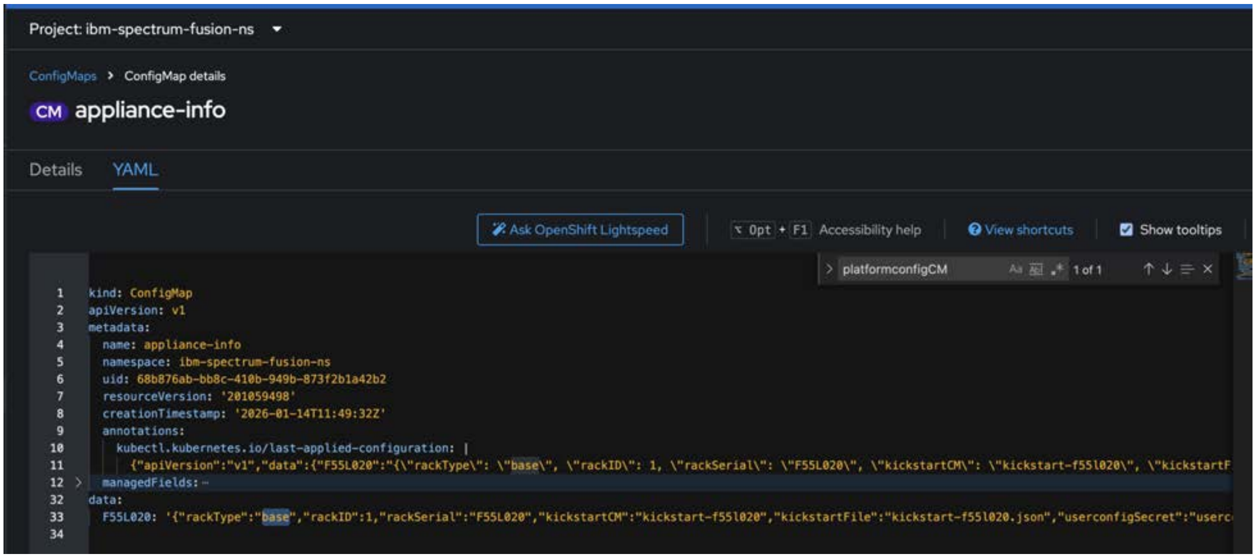
The default configuration is secure and recommended for all environments. Disable automatic log cleanup only in the following scenarios:

- Logs remain strictly within the organization for internal debugging
- IBM Support explicitly requests for unsanitized log packages

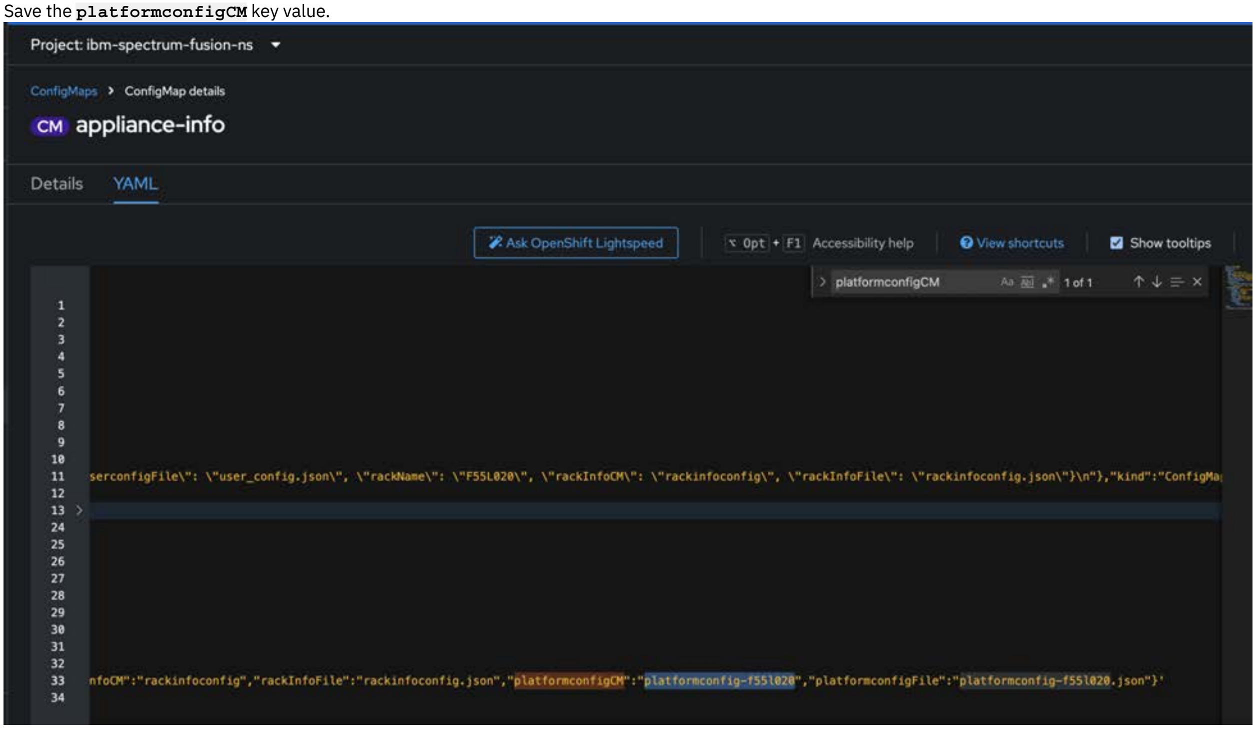
Disabling automatic log package cleanup

To disable automatic log package sanitization by using the OpenShift console:

1. Log in to the OpenShift® web console.
2. Switch to Project `<FusionNamespace>` (typically `ibm-spectrum-fusion-ns`).
3. Go to Workloads > ConfigMaps, locate the `appliance-info` ConfigMap, and open it in YAML view.
4. Search for `platformconfigCM` and locate it in the `rackType: base` entry.

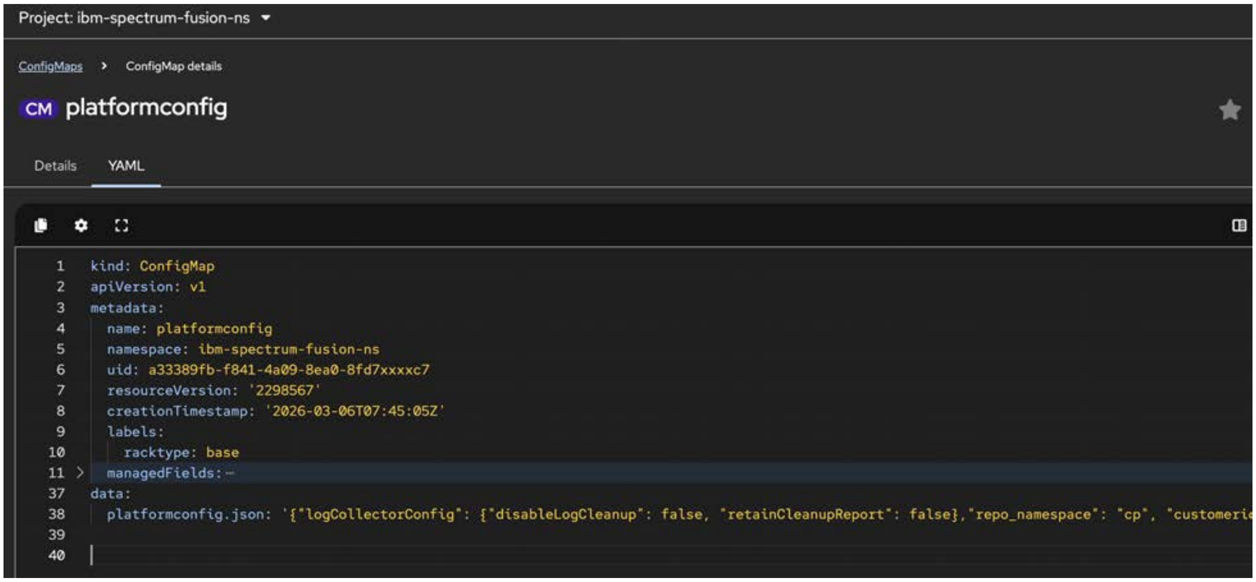


5. Save the **`platformconfigCM`** key value.



6. Return to Workloads > ConfigMaps, search for **`platformconfig`** ConfigMap along with its value from the previous step, and open it in YAML view.
7. Locate the **`logCollectorConfig`** entry in the ConfigMap and update it with the following configuration. If the **`logCollectorConfig`** entry does not exist, add the following entry to the JSON content of the key **`<platform-config-name>.json`** (retrieved in step 4):

```
"logCollectorConfig": {"disableLogCleanup": true},
```



8. Save the changes.
   The automatic log package cleanup is now disabled.

Benefits of enabling Call Home

Configure and enable the call home feature as it improves the user experience when IBM Fusion is used.

The call home feature when enabled provides the following benefits:

1. Improves customer service response time
   - *Scheduled data uploads:*
     If the Call Home feature is enabled, then the IBM® support representatives can start with the analysis of the issue that is reported without any delay. The IBM support team can use the cluster data from the Call Home scheduled uploads to do an immediate analysis of the issue. For more information, see Logs in IBM Fusion HCI.

     However, if the Call Home feature is not enabled, then the IBM support team first collects debugging data from the IBM Fusion user interface. The whole process takes time to request, collect, and deliver the data, which might cause a delay in resolving the issue on the site.
   - *Selected failure events trigger automatic Call Home uploads:*
     If the Call Home feature is enabled, then the event-based uploads feature automatically collects the relevant component-specific information and uploads it to the IBM ECuRep server. The uploaded data enables the IBM support team to immediately find the root cause of the issue that is reported on the customer site.
2. Proactively detects issues
   - Detects violations against best practices and common misconfiguration.
   - Finds out which specific customer site is affected by the reported issue by using tools like High Impact Programming Error Authorized Program Analysis Report (HIPER APARs).
   - Constantly extends the list of automatically detected issues, best practice violations, and common misconfiguration over time.

   Note: IBM Fusion user interface displays the list of events.
3. Improves customer support experience in areas of consumability and ease of usage
   - *Uploads gpfs snaps directly from the IBM Fusion system:*
     If the Call Home feature is enabled, then a customer can easily find out the customer number. The requested debugging data is transferred from the customer site cluster or node to the IBM support team by raising a service ticket. For more information, see Enabling Call Home for IBM Fusion HCI.
4. Creates service tickets automatically for reported failures
   If the call home feature is enabled, then the system automatically creates Salesforce service tickets for any hardware failures that are reported on customer sites.
   Note: A user can configure the Call Home settings on the IBM Fusion user interface.

- **Data privacy with Call Home**
  A licensee can configure the IBM Fusion to automatically send specific cluster information to the IBM support team by using the Call Home feature.
- **Enabling Call Home for IBM Fusion HCI**
  You can enable Call Home and remote access from the Support settings page.
- **Configuring Call Home with proxy**
  You can configure Call Home to use a proxy server for outbound connectivity to IBM Support.
- **IBM Fusion Call Home Telemetry**
  IBM Fusion Call Home Telemetry enhances supportability by providing IBM with valuable insights into system configuration and performance. Telemetry data is collected from various components and securely transmitted to IBM, helping support teams proactively identify issues and optimize system behavior based on real-world usage.
- **Custom Certificate Authority (CA) support for Call Home**
  Support for custom Certificate Authority (CA) for IBM Fusion Call Home.

Data privacy with Call Home

A licensee can configure the IBM Fusion to automatically send specific cluster information to the IBM® support team by using the Call Home feature.

When a licensee enables the Call Home feature, the licensee provides the support contact information, such as names, phone numbers, or email addresses, which are needed during the Call Home feature configuration and activation process.

By default, when a licensee decides to activate and use the Call Home feature, the licensee agrees to allow IBM and its subsidiaries to store and use the licensee's support contact information. This information is processed and used with IBM business relationship and might be shared with third-parties under the direction of IBM. For example, IBM support center representatives or assignees of IBM and its subsidiaries.

The support contact information can be used for processing business orders, business promotions, problem determination, or market research.

When the call home feature is configured and activated, the IBM Fusion collects all monitoring information that is related to system utilization, performance, capacity planning, and service maintenance. The service information includes system failure logs, part numbers, machine serial number, software version, maintenance levels, installed patches, and configuration values.

When the Call Home feature is enabled, the licensee allows IBM to use and share the data, which is gathered from system monitoring functions, within IBM and with IBM business partners and third-parties, such as IBM support centre sub-contractors or assignees. The shared data is used only under the direction of IBM for the following defined purposes:

- Determining a problem.
- Assisting licensee with performance and capacity planning.
- Assisting IBM to enhance IBM products and services.
- Notifying licensee about the licensee's system status and available solutions.

Note: The licensee information excludes the collection and transmission of licensee's financial, statistical, and personal data, and licensee's business plans.
When the Call Home feature is enabled, a licensee agrees to the fact that the licensee's support contact information can be transferred to countries that might not be a member of the European Union. A licensee can disable the Call Home feature at any time.

Enabling Call Home for IBM Fusion HCI

You can enable Call Home and remote access from the Support settings page.

Before you begin

- Ensure that the Firewall sites are configured for Call Home.
- If your environment requires a proxy, confirm the proxy details and credentials before you proceed.

Note: Telemetry is no longer configured as a separate option. When you enable Call Home, telemetry collection and transmission are enabled automatically. For more information, see IBM Fusion Call Home Telemetry.

About this task

Support status banner
    A yellow banner might appear at the header of the Support settings page with messages similar to:

- `Support features disabled. Call Home or other support features are not enabled. Review support settings to restore full support capabilities.`
  This banner is an expected behavior in the following case:
  - Call Home and/or telemetry is disabled
- `Support disconnected. One or more support features are not functioning. Review your support settings to restore full support capabilities.`
  This banner is an expected behavior in the following cases:
  - Call Home or telemetry is enabled but in a failed or degraded state
  - Required contact information is missing or incomplete

After Call Home is successfully enabled and connectivity is restored, the banner clears automatically.

Procedure

1. Open Support settings.
   In the IBM Fusion user interface, go to Support > Support settings.

   Note: If contact information is not configured, the page displays a warning: `Contact details must be provided before enabling any support settings`. The Enable button for Call Home is disabled until this information is added.
2. Add contact information.
   a. In the Contact information section, click Add.
   b. Enter contact information:
   In the Contact information dialog, provide the primary support contact details:
      i. Enter Company name.
      ii. Enter Customer number.
      iii. Enter Primary contact name.
      iv. Enter Primary email address.
      v. Enter Primary phone number:
         1. Select the country code.
         2. Enter the phone number.
   c. Provide the data center location:
   In the Data center section, enter the physical location details:
      i. Enter Address line 1.

ii. Enter Address line 2 (optional).
iii. Enter City.
iv. Enter State/Province/Region.
v. Enter Zip/Postal code.
vi. Select the appropriate Country from the drop-down list.

d. Click Add to save the information.
After you save the information, the warning message is cleared and the Call Home and remote access options become available.

3. Enable Call Home.
   a. In the Call Home section, click Enable.
   b. Select a connection type:
      - Direct connection
      - Proxy connection, if your network requires a proxy
   c. If you use a proxy, enter the proxy configuration details:
      - Host and port
      - Authentication details or certificates, if required
   d. Click Test connection to validate connectivity.
   - If the test succeeds, a success confirmation is displayed and Call Home is enabled.
   - If the test fails, an error message explains the issue, for example, a proxy authentication failure or connectivity error. Correct the configuration and test again.
4. Optional: Enable remote access.
   a. In the Remote access section, click Enable.
   b. Confirm the action when prompted.

   Remote access is enabled and ready for use when IBM Support assistance is required.

   Note: Enabling remote access does not start a support session automatically. Sessions are initiated only with customer approval.

What to do next

After successful configuration:

- The Call Home section shows an enabled status.
- Any connectivity or configuration issues are surfaced directly on the Support settings page.

These steps ensure that Call Home is correctly enabled and ready to support proactive monitoring and faster issue resolution.

Configuring Call Home with proxy

You can configure Call Home to use a proxy server for outbound connectivity to IBM Support.

About this task

Call Home supports both HTTP and HTTPS proxies, including HTTP, HTTPS (TLS), and HTTPS with mutual TLS (mTLS). Select the connection type based on your environment's ability to allow outbound internet access.

Call Home supports direct connectivity and different proxy configurations to align with common enterprise network and security models.

Important: The Proxy host field must include the protocol: `http://` or `https://`. A hostname without the protocol prefix is not accepted. For example: `http://proxy.example.com` or `https://proxy.example.com`.

Procedure

1. In the IBM Fusion user interface, go to Settings > Support.
2. Review and complete the contact information.
3. In the Call Home section, click Edit.
4. Select a connection type.
   Choose the appropriate connection type based on your network configuration:

   Direct connection
      Use a direct connection when the system can reach IBM Support endpoints directly without routing through a proxy. No additional fields are required for this option.

   Open proxy
      Use an open proxy when outbound traffic must pass through a proxy that does not require authentication. This option supports both HTTP and HTTPS proxies. Provide a Certificate Authority (CA) certificate only if the proxy uses a non-public CA.

   Basic authentication
      Use basic authentication when the proxy requires a username and password for access. This option supports HTTP and HTTPS proxies. If HTTPS is used and a private CA issues the proxy certificate, a CA certificate must be provided.

   Certificate authentication
      Use Certificate authentication when the proxy uses certificate-based security. This option supports TLS and mutual TLS (mTLS), allowing configuration of the CA certificate for server validation and, when required, a client certificate and private key. It is typically used in environments with stricter security or compliance requirements. If required by the proxy, username and password can also be provided.

5. Configure the proxy details as required.
   The following fields apply whenever a proxy is configured, regardless of authentication type:

Field	Required	Description
Proxy host	Yes (for proxy connections)	Fully qualified proxy URL. It must start with `http://` or `https://`.

Field	Required	Description
Port	Yes (for proxy connections)	Network port on which the proxy listens (for example, 8080 or 3128).
Username	Optional	Proxy username, if required by the proxy configuration.
Password	Optional	Proxy password, if required by the proxy configuration.
CA certificate	Conditional	Certificate Authority (CA) is used to validate the proxy server certificate. Required when the proxy uses HTTPS (TLS or mTLS) with a private or internal CA. Supported formats: `.crt`, `.cer`, `.cert`, `.ca`
Client certificate	Conditional	A client certificate is presented to the proxy during mutual TLS (mTLS). Required only when the proxy enforces client certificate authentication. Supported formats: `.crt`, `.cer`, `.cert`, `.ca`
Client key	Conditional	Private key corresponding to the client certificate. Required only for mTLS authentication. Supported formats: `.pem`, `.key`, `.crt`, `.cer`, `.cert`, `.ca`

6. Save and test the configuration.
   a. Click Save to apply the configuration.
   b. Click Test connection to validate the configuration.
   - If the test succeeds, a success confirmation is displayed and Call Home is configured with the proxy settings.
   - If the test fails, an error message explains the issue. Verify the proxy configuration and test again.

IBM Fusion Call Home Telemetry

IBM Fusion Call Home Telemetry enhances supportability by providing IBM with valuable insights into system configuration and performance. Telemetry data is collected from various components and securely transmitted to IBM, helping support teams proactively identify issues and optimize system behavior based on real-world usage.

Telemetry automatically collects and transmits the data from IBM Fusion components, such as software applications or hardware devices. Also, it is used to gather information about system configuration, usage patterns, and performance metrics. For steps to enable call home telemetry, see Enabling Call Home for IBM Fusion HCI.
Note: The system uploads Call Home telemetry data every 24 to 26 hours.
The following details are collected and uploaded as part of call home telemetry support:

IBM Fusion cluster details
: Collects statistical insights into IBM Fusion resources, such as the number of pods, deployments, and other key components, along with version and health status details for both OpenShift® and IBM Fusion. Also, it collects node storage consumption details for OpenShift.

IBM Fusion license data
: Collects IBM Fusion licensing data. For more information on licensing metrics, see Understanding the IBM Fusion license report.

IBM Fusion events data
: Collects the summary of IBM Fusion events by severity and event ID.

IBM Fusion Services
: Collects the details such as name, version, health, and upgrade details of different fusion services installed in the cluster. If Backup & Restore service is installed, then it shows the success % of backups and restores.

Hosted clusters
: Collects basic count and health status of hosted clusters that are deployed.

Virtual machines
: Collects basic count and health status of virtual machines that are created on cluster.

Basic rack details
: Collects configuration details including serial number, model, storage type, multi-rack setup, proxy enablement, Disaster Recovery (DR) enablement, IBM serial number, and other IBM Fusion HCI rack-specific parameters.

Node and switch health monitoring
: Collects hardware and health details for nodes and switches, including name, vendor, model, firmware, and type. Also, it captures component-level stats such as memory, processors, fans, GPU nodes, cards, power supplies (for nodes), and relevant metrics for switches.

Custom Certificate Authority (CA) support for Call Home

Support for custom Certificate Authority (CA) for IBM Fusion Call Home.

Overview

IBM Fusion Call Home supports custom Certificate Authorities (CAs) to enable seamless operation in enterprise environments that use TLS-intercepting proxies. This support allows Call Home to establish secure connections to IBM support services even when your network uses non-standard certificate authorities.

Importance of custom CA support for Call Home

In many enterprise environments, standard TLS certificates do not meet security policy or network architecture requirements. Custom CA support addresses these challenges by enabling Call Home to trust certificates signed by your organization's certificate authorities.

Use cases and scenarios

Custom CA support is required in the following scenarios:

TLS-intercepting proxies

Many organizations use security proxies that intercept outbound HTTPS connections to inspect traffic for threats. These proxies decrypt the original connection, inspect the content, and then re-encrypt it using an enterprise certificate before forwarding to the destination.
Example flow:

```
Call Home  →  Corporate proxy  →  IBM support
              (intercepts and     (esupport.ibm.com)
               re-signs with
               enterprise CA)
```

Without custom CA support, Call Home rejects the re-signed certificates as untrusted, which prevents communication with IBM support services.

Private certificate authorities
: Organizations that use private Certificate Authorities sign certificates for internal services and infrastructure. When Call Home communicates through infrastructure secured by these private CAs, it must trust the certificate chain to establish secure connections.

Custom CA integration behavior

Call Home integrates with OpenShift® cluster-wide CA configuration through an automated process:

- Automatic discovery: Call Home automatically detects and uses custom CAs configured at the OpenShift cluster level.
- Dynamic updates: When cluster administrators update CA configurations, Call Home automatically receives and applies these changes.
- Zero configuration: Call Home does not require manual CA configuration.
- Operator-managed: The ISF serviceability operator manages CA injection and updates.

Remote support

Remote support is a service that enables IBM Support representative to remotely access your rack and accomplish essential tasks, such as installation, maintenance, and troubleshooting.

When you contact IBM Support teams with an issue or question, the support team can initiates a remote connection for problem determination and resolution provided you enable the remote support feature. For the procedure to configure remote support, see Enabling remote support.

Remote support uses the latest security technology to protect the data exchanged during a remote session. All transferred information, including screen views, file transfer data, and identities, are encrypted. Encryption and decryption are from end-to-end, so data cannot be intercepted during transit and can be viewed only from the console of the remote support team. For more information about security considerations, see Security considerations.

Benefits of the remote support

Use remote support to reduce disruptions, optimize workflows, and save time and resources. It leads to an overall enhancement in efficiency and cost savings.

Key benefits of remote support are as follows:

Faster problem solving
: You can contact technical experts in your support region to help resolve problems and do not have to wait for data, such as logs, memory dumps, and traces. As a result, problems can be solved faster.

Connection with a worldwide network of experts
: IBM Technical support team can call on other worldwide subject experts available through chat, phone, email, or remote assistance to assist with problem determination.

Closer monitoring and enhanced collaboration
: You can monitor the activities and join conference calls with IBM Support teams to discuss the next steps.

Save time and money
: Many of your problems can be solved without sending an IBM Support representative to your site.

- **Security considerations**
  Security considerations information to help you to create for a remote support connection.
- **Enabling remote support through user interface**
  Enable remote support connection so that IBM Support representatives can remotely access your service node and storage system.
- **Enabling remote support manually**
  Enable remote support connection manually when the IBM Fusion HCI user interface is not available.
- **Forwarding additional ports for remote support**
  By default, port 22 is used for SSH based command line access to the service node using remote support. You can configure additional ports to let IBM Service Support Representative access other services like installation stage 2 user interface.

Security considerations

Security considerations information to help you to create for a remote support connection.

Remote support uses the latest security technology to ensure that the data exchanged between support teams and clients is secure. The identities are verified and protected with industry standard authentication technology, and remote support connection sessions are kept secure and private with the use of randomly generated session keys and advanced encryption.

The following are the some key security aspects:

Authorization and access Control

Remote support maintains multiple-level internal authorizations for any privileged access to the IBM Fusion HCI components. Only approved IBM service personnel can gain access to the tools that provide the security codes for IBM Fusion service node command-line access. Remote support sessions can only be initiated by a customer.
After a session is ended, then the support can no longer connect to the service node. Client needs to set timeout in hours while starting a remote support session. Starting a remote support session is considered as a consent from the client for support engineer to connect to the system remotely. The timeout to automatically stop remote support session prevents customers from accidentally keeping connections open.

Strong password protection
: Remote support sessions are protected by strong password authentication. Support engineers are authenticated using a challenge and response password exchange. Multi factor authentication ensures that auditing can be achieved at a greater level. Additionally challenge response authentication can be used to grant different level of access to different support engineers.

Advanced Encryption
: Remote support implements outbound connections protected by 128-bit MARS or TLS encryption to prevent intruder access to the information exchanged during all remote support sessions. Chat, screen viewing, screen sharing and file transfer data are encrypted end to end, and packets are never decrypted in transit by the communication servers.

Uncompromising firewalls and proxy
: Remote support works seamlessly with most firewalls and the connections are possible without any firewall reconfiguration. It requires access to outbound ports at both ends of a connection, so no need to open holes in firewalls. It also supports configuring a proxy for communication.

For more information about how to collect audit logs that includes record of all activities related a remote support session, see Audit logs.

- **Audit logs**
  Audit logs track and record all activities related to remote access and actions taken during remote support session.

Audit logs

Audit logs track and record all activities related to remote access and actions taken during remote support session.

Audit logs in a remote support connection provides a detailed record of activities, mainly related to security, system operations, and system actions. It tracks all the activities performed by the IBM support representative and customer. The audit logs mainly consists the service node, OpenShift® Container Platform, and node audit details.

Service node audit

The service node audit contains utilities such as **`Rsyslog`**, **`Auditd(Audit Daemon)`**, and **`Logrotate`** are used which come by default in RHEL.

`Rsyslog`
: **`Rsyslog`** is used as a logging utility to generate log files for the commands directly executed only by the IBM support representative in the shell.
For example:

```
$ ls /var/log/fusion/
```

Example output:

```
audit.log  audit.log.1  audit.log-20241013
```

The latest logs are stored in the **`audit.log`** file. Only users with **`sudo`** access to service node can read these files.

`Auditd(Audit Daemon)`
: The **`Auditd`** is configured to track all system calls that users execute. It provides detailed audits and tracks the ones that may not be run directly from the command line. This detailed audit captures even the implicit commands that SSR might run from a script. To simplify this process, a wrapper script (**`fusion-audit`**) is provided. The utility simply asks for start and end date time as a parameter lists the output, and saves the detailed output to a file.
For example:

```
[kni@tc11gen2001svcnode ~]$ fusion-audit
***********************************************************************
Welcome to IBM Fusion - Audit Daemon
This utility facilitates the generation of an audit report actions
performed by IBM support during the specified time filter.
***********************************************************************

Your current date in locale-specific format is as follows:
10/15/2024

Enter start date (in above format) and time, eg: DATE<space>HH:MM:SS
10/11/2024

Enter End date and time in similar format. Leave it blank for current time.

Current time is taken for end time filter

COMMAND: sudo ausearch -sc execve -ts 10/11/2024 -te now -ga ibmsupport -i

***********************************************************************
Output Summary from 10/11/2024 to now
***********************************************************************
type=PROCTITLE msg=audit(10/11/2024 18:09:45.548:10668) : proctitle=bash
type=PROCTITLE msg=audit(10/11/2024 18:09:45.634:10684) : proctitle=basename /usr/bin/bash
type=PROCTITLE msg=audit(10/11/2024 18:10:40.713:10694) : proctitle=oc get pods
type=PROCTITLE msg=audit(10/11/2024 18:10:43.984:10697) : proctitle=ls --color=auto
type=PROCTITLE msg=audit(10/11/2024 18:10:48.146:10702) : proctitle=ssh
***********************************************************************
```

```
Detailed Ausearch output saved as AuditReport_2024-10-15_05-45-04.log
[kni@tc11gen2001svcnode ~]$
```

Logrotate

Multiple files available in this folder as a log rotation rule is applied for this file. By default, weekly rotation is configured. If you want to change it, then see the Linux man page documentation.

For example:

```
$ cat /etc/logrotate.d/fusion
/var/log/fusion/audit.log {
    weekly
    rotate 7
    compress
    delaycompress
    missingok
    notifempty
    create 0640 root adm
    postrotate
        /bin/systemctl reload rsyslog > /dev/null
    endscript
}
```

OpenShift Container Platform audit

OpenShift Container Platform provides various logging profiles based on the needs. For more information on profile and the level of content that gets logged, see Configuring the audit log policy in *Red Hat OpenShift Container Platform* documentation.

For more information on how these logs can be queried based on filters, see Viewing audit logs in *Red Hat OpenShift Container Platform* documentation.

Node audit

Node audit logs of compute hardware contains the record of actions performed by the user or services on the hardware. It helps you with ensuring accountability, traceability, and regulatory compliance for data access, modification, and security within the IBM Fusion HCI.

Follow the steps to download the node audit logs:

1. Go to Events in the IBM Fusion HCI.
2. Enter the text `CEUSER` in the search description.
   It lists all the alerts raised for the actions performed by the IBM support representative .
3. Click Download.
4. For any further auditing of actions performed on the node, collect system health check logs and contact IBM Support. For steps download logs, see Collecting log packages for IBM Fusion HCI

Enabling remote support through user interface

Enable remote support connection so that IBM Support representatives can remotely access your service node and storage system.

About this task

From the Support settings page of the user interface, enter the contact information, optional proxy server details, and enable the remote support option. After you enable the remote support, you can start or stop remote sessions.

Procedure

1. From the IBM Fusion HCI user interface, go to Settings > Support.
   The Support settings page gets displayed.
2. In the Contact information section, enter the following information.

   Company
   : Enter the name of your organization.

   Customer number
   : Enter the IBM ID assigned to your organization.

   Primary name
   : Enter the primary name of your contact.

   Primary phone
   : Enter the phone number of your primary contact.

   Primary email
   : Enter the official email address of your primary contact.

3. Click Save.
   A success message for successfully saving the contact information is displayed.
   Note: The Configure button in the Proxy settings section and the Remote access disabled toggle button in the Remote support section are enabled only after the contact information is saved successfully.
4. Optional: In the Proxy settings section, click Configure and enter the following optional Proxy settings information.

Host address
: Enter the host address.

Port
: Enter the port details.

Username (optional)
: Enter the username of the server.

Password (optional)
: Enter the password of the server.

The username and password details are optional.

5. Click Save
6. In the Remote support section, click the toggle to enable the remote support connection.
   The Start session is visible in the user interface after you click the Remote support toggle button.
7. If you want to initiate a remote session, then click Start session.
   The Start remote support session window gets displayed.
8. Enter the Session time and click Start session.
   The IBM support team can log in to the service node after starting the remote support session. Share necessary credentials with the IBM Support, such as OpenShift® token, login details for switch and nodes.
9. The Stop session button gets displayed instead of the Start session. The session automatically closes after the scheduled time. Alternatively, you can choose to end the session anytime with the Stop session button. If you enabled the session during installation, you can click Stop session button in the Remote support bar that is located near to the title bar.

Enabling remote support manually

Enable remote support connection manually when the IBM Fusion HCI user interface is not available.

About this task

The following instructions can be used to configure and start remote support session in case the IBM Fusion HCI user interface is not available. The following steps are to be run only when necessary as it requires manual steps which could be error prone.

To forward additional ports for remote support, see Forwarding additional ports for remote support.

Important: As this is a manual procedure, be sure to explicitly stop the session after you are done; otherwise, it continues to remain active.

Procedure

1. Log in to the service node with **kni** user details.
2. Optional: Configure proxy.
   a. Update the **/etc/ibmtrct.conf** file.
   b. Ensure you update the **ProxyURL** field.
      Example of **ProxyURL** filed with username and password:
      ```
      ProxyURL= dummyuser:dummypassword@9.23.x.x:xx
      ```
      Example of **ProxyURL** filed without username and password:
      ```
      ProxyURL= 9.23.x.x:xx
      ```
3. Configure remote support.
   a. Edit the **/var/opt/ibm/trc/target/profiles/lightsoutprofile.properties** file.
   b. Ensure you update the following fields:
      ```
      Custom.CustNumber=<IBM_assigned_customer_number>
      Custom.CustName=<Company_name >
      ProfileEnabled flag set to "true"
      ```
4. Run the following command to restart the service.
   ```
   sudo service ibmtrct restart
   ```
5. Validate the logs.
   The logs are available in the **/var/opt/ibm/trc/target** directory. The log files are collected on a daily basis and you find the log file for each day in the **trc_base_Day.log** format.
   For example:
   ```
   [kni@servicenode-1 target]$ ls trc_base_*.log
   ```
   Example output:
   ```
   trc_base_Fri.log  trc_base_Mon.log  trc_base_Sat.log  trc_base_Sun.log  trc_base_Tue.log  trc_base_Wed.log
   ```
6. Set flag to false to stop the remote support session.
   .
   ```
   Custom.CustNumber=<IBM_assigned_customer_number>
   Custom.CustName=<Company_name >
   ProfileEnabled flag set to "false"
   ```
7. Restart the service for the changes to take effect.

```
sudo service ibmtrct restart
```

Forwarding additional ports for remote support

By default, port 22 is used for SSH based command line access to the service node using remote support. You can configure additional ports to let IBM Service Support Representative access other services like installation stage 2 user interface.

About this task

All remote support communication is conducted over HTTPS, with no need to open any additional ports.

Procedure

1. Log in to the service node with `kni` user details.
2. Edit the `/var/opt/ibm/trc/target/profiles/lightsoutprofile.properties` file.
3. Update the `TunnelList` field.
   For example:

   ```
   Old value: TunnelList = command_line\=tcp@localhost\:22
   New value: TunnelList = command_line\=tcp@localhost\:22,install_ui\=tcp@localhost\:3000
   ```

4. Save and close the properties file.
   Note: If there is an ongoing remote support session, then stop and the start session for the changes to take effect.

Password rotation

The password rotation feature automatically updates hardware (nodes) credentials at regular intervals to enhance system security and meet enterprise compliance requirements. It minimizes the risk of credential compromise by eliminating the need for manual password updates.

The system manages password rotation internally and applies it to supported hardware components without affecting normal operations. The system securely stores updated credentials and allows only authorized components to access them.

The system logs all password rotation activities to support auditing and traceability. It ensures that administrators can monitor credential changes and verify adherence to security policies.

- **Enabling password rotation for IBM Fusion HCI system**
  Steps to configure and manage password rotation for hardware credentials in the IBM Fusion HCI system.
- **External secret rotation**
  This section describes how to integrate an external vault or any other external secret management solution.

Enabling password rotation for IBM Fusion HCI system

Steps to configure and manage password rotation for hardware credentials in the IBM Fusion HCI system.

Before you begin

- Ensure that you have the required administrative privileges to modify cluster configuration.
- If you plan to use an external Vault, ensure that the Vault is installed, configured, and accessible from the cluster.

About this task

Password rotation feature is introduced with multiple configurable strategies to enhance security and meet enterprise compliance requirements.

You can configure password rotation by updating the `credentialRotation` section in the `platformconfig` ConfigMap. The system supports internal management, external secret management integration, or disabling the feature entirely.

Procedure

1. Log in to the cluster by using the OpenShift® CLI or user interface.
2. Go to the `ibm-spectrum-fusion-ns` namespace.
3. Edit the `platformconfig` ConfigMap for the base rack.
4. In the `credentialRotation` section, choose one of the following strategies:
   - Internal (default):
     - Internal strategy is supported only on service node rack.
     - Passwords are stored on the service node rack.
     - Credentials are automatically rotated every 90 days.
     - The system fully manages the rotation.

     Example configuration:

     ```
     "credentialsPolicy": {
       "strategy": {
     ```

```
    "type": "internal",
    "kmsProvider": {
      "name": "k8sSecret",
    },
    "rotationPeriod": 90
  },
  "sshKeyRotationPeriod": 90
}
```

- External:
  - Enables integration with an external secret management system. As of now, HashiCorp vault is supported. To configure your own secret management system, you must adhere to the requirements stated in External secret rotation.
  - The external vault manages the password rotation based on its policies.

  Example configuration:

```
"credentialsPolicy": {
  "strategy": {
    "type": "external"
  },
  "sshKeyRotationPeriod": 90
}
```

- None:
  - Disables password rotation completely.

  Example configuration:

```
"credentialsPolicy": {
  "strategy": {
    "type": "none"
  }
}
```

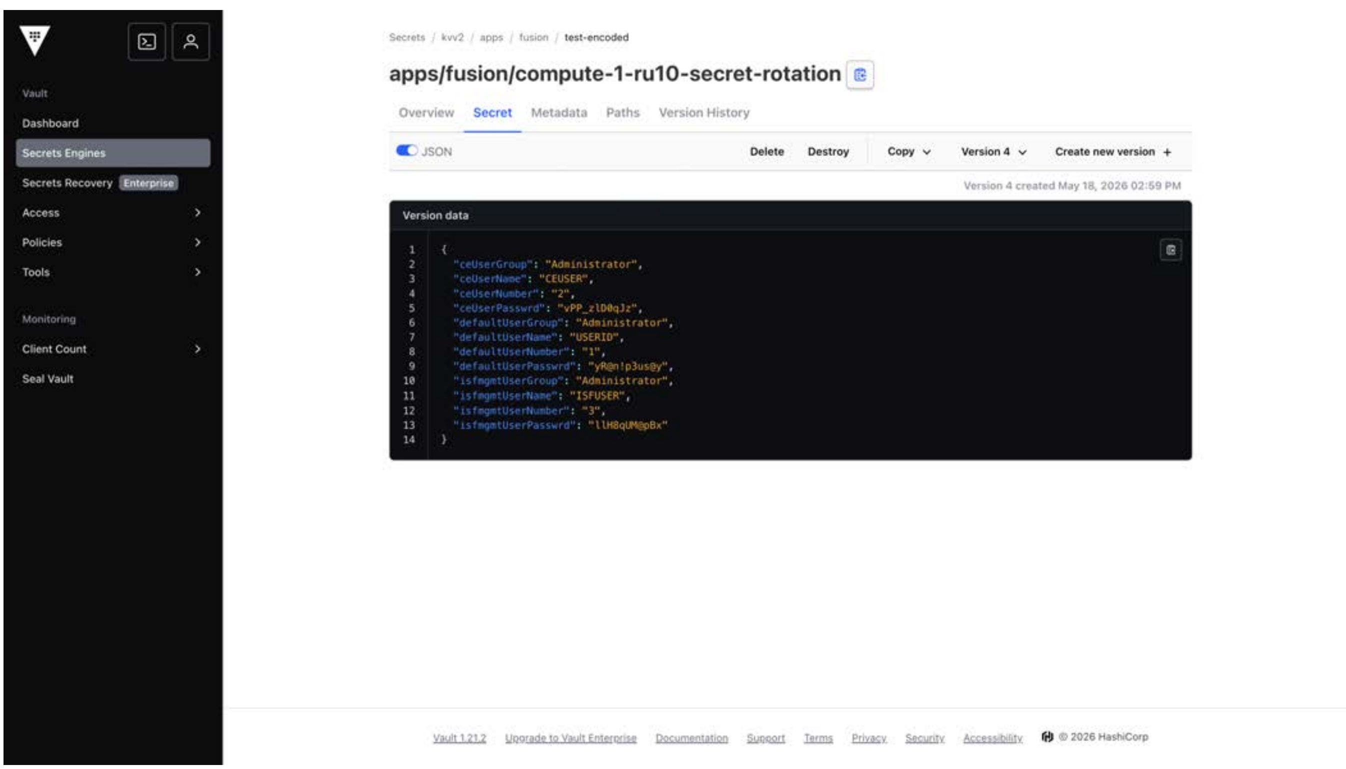
The password rotation configuration is applied to the system based on the selected strategy.

SSH key rotation

- SSH key rotation is supported for the KNI user.
- It is internally managed and operates independently of password rotation.
- If the `none` strategy is selected, SSH key rotation is also disabled.

External Vault configuration

To use an external Vault, configure and integrate the HashiCorp Vault with the system.



Important: When mapping secrets from the external Vault, ensure that the Vault secret contains the same key-value pairs as the original secret, and that the values are stored in plain text. The password must strictly comply with the hardware complexity requirements.

Example configuration:

```
apiVersion: secrets.hashicorp.com/v1beta1
kind: VaultStaticSecret
metadata:
  name: <vss-name>
```

```
  namespace: <fusion-ns>
spec:
  type: kv-v2
  mount: kv
  path: <secret-path>
  destination:
    name: <rotation-secret-name>
    create: false
  refreshAfter: 30s
  vaultAuthRef: <static-auth-name>
```

Important: The `destination.name` field must follow this format: `<original-secret-name>-rotation`

Troubleshooting password rotation

You can monitor and troubleshoot password rotation issues in IBM Fusion HCI by using events, metrics, and the `SpectrumFusion` custom resource (CR). The status of password rotation is reported for each node in the CRs. For node-specific details, see the `ComputeMonitoring` CR.

Monitoring password rotation

- Track events and metrics to verify that rotation operations are proceeding as expected.
- Check the `SpectrumFusion` CR for the final status of password rotation.
- For compute nodes, consult the `ComputeMonitoring` CR for rotation status specific to each node.
- For service node SSH key, consult the `SpectrumFusion` CR for rotation.

Common issues and resolution

- Node is down:
  - Symptom: Rotation status shows failure for a node.
  - Resolution: Ensure that the affected node is up and operational. Resolve any hardware or network issues before attempting rotation again.
- Password does not meet hardware complexity requirements:
  - Symptom: Rotation fails for a specific component or node.
  - Resolution: Update the password configuration to comply with the hardware complexity rules. Ensure that the new password matches the hardware-defined rotation policy.
- Internal Vault update issues:
  - Symptom: IBM Fusion HCI console shows events indicating a Vault secret update failure.
  - Resolution: Verify that the Vault service is running on the service node. Start the Vault service if required, and then reattempt password rotation. For more information about Vault issues and their resolutions, see Troubleshooting Vault.
- External Vault integration issues:
  - Symptom: Rotation fails when you use an external vault.
  - Resolution: Verify that the Vault connection is valid, the secret exists, and key-value pairs match the required format. Ensure that the `VaultStaticSecret` resource is correctly configured.

External secret rotation

This section describes how to integrate an external vault or any other external secret management solution.

Overview

You can integrate an external vault or any other external secret management solution to manage periodic credential rotation. You can also implement a custom process to rotate secrets at regular intervals.
Important: Externally rotated secrets or custom-managed secrets must follow the required format.

- Use the prescribed secret naming convention
- Preserve the exact key-value structure of the original secret

Failure to comply with these requirements might result in rotation or integration failures.

Requirements and constraints

- The rotation secret name must use the original secret name with the `-rotation` suffix. For example: `<original-secret-name>-rotation`
- All values must be Base64 encoded.
- The secret must use the same keys as the original secret.
- Do not add fields and do not remove fields.
- The namespace must match the IBM Fusion namespace, for example, `ibm-spectrum-fusion-ns`.
- The password must strictly comply with the hardware complexity requirements.

Secret template

Use the following template when you update a rotation secret:

```
apiVersion: v1
kind: Secret
metadata:
  name: <rotation-secret-name>                # Format: <original-secret-name>-rotation
  namespace: <namespace>
type: Opaque
```

```
data:
  ceUserGroup: <base64-ce-user-group>
  ceUserName: <base64-ce-user-name>
  ceUserNumber: <base64-ce-user-number>
  ceUserPasswrd: <base64-ce-user-password>

  defaultUserGroup: <base64-default-user-group>
  defaultUserName: <base64-default-user-name>
  defaultUserNumber: <base64-default-user-number>
  defaultUserPasswrd: <base64-default-user-password>

  isfmgmtUserGroup: <base64-isfmgmt-user-group>
  isfmgmtUserName: <base64-isfmgmt-user-name>
  isfmgmtUserNumber: <base64-isfmgmt-user-number>
  isfmgmtUserPasswrd: <base64-isfmgmt-user-password>
```

Password hardware complexity requirements

Length: Exactly 15 characters

Required character types (minimum 1 each)

- Uppercase letters (`A-Z`)
- Lowercase letters (`a-z`)
- Numbers (`0-9`)
- Special characters from: `! ( ) @ #`

Critical rules

- The first character must be a letter (cannot be a number or special character).
- Only these special characters are allowed: `! ( ) @ #`
- The following characters are prohibited:
  - `<`
  - `>`
  - `\`
  - Space
  - `/`
  - `&`

Note: Password complexity requirements can vary depending on the hardware model and firmware version.

Custom certificate expiration alert

This section provides information about custom certificate expiration monitoring and alerting for critical cluster components.

About this task

The certificate expiration alert feature proactively monitors the validity of custom certificates used by critical cluster services such as the API server, cluster proxy, and ingress, and generates alerts before certificates expire to help prevent service disruptions.

When you enable this feature, the system periodically checks certificate validity and raises warning and critical alerts based on configurable thresholds. It ensures that you can take timely action to renew certificates and maintain uninterrupted cluster operations.

The monitoring process is lightweight and does not affect system performance or availability. The system logs all alert events to support auditing, observability, and operational tracking.

Components monitored

The system generates alerts for certificate expiration of the following components:

- API server
- Cluster proxy
- Ingress

Configuration of custom certificate expiration alerts

Certificate expiration monitoring is configured in the `platformconfig` ConfigMap for the base rack by using the `certificateExpiryCheck` section. You can enable or disable this feature and define alert thresholds for warning and critical events.

Example configuration:

```
"certificateExpiryCheck": {
 "enable": true,
 "warningEventPeriod": 120,
 "criticalEventPeriod": 90
}
```

- `warningEventPeriod`: Number of days before certificate expiration when the system generates a warning alert. In the prior example, the system raises a warning 120 days before expiration.
- `criticalEventPeriod`: Number of days before certificate expiration when the system generates a critical alert. In the prior example, the system raises a critical alert 90 days before expiration.

Procedure

Updating the configuration

1. Edit the **platformconfig** ConfigMap in the base rack.
2. Modify the **certificateExpiryCheck** section as required.
   - Set **enable** to **true** or **false** to enable or disable monitoring.
   - Adjust **warningEventPeriod** and **criticalEventPeriod** thresholds based on your operational requirements.
3. Save and apply the changes.

Results

The updated configuration takes effect after you apply it. The system begins monitoring certificate expiration based on the configured thresholds.

Alerts or event IDs

The system generates the following event IDs for certificate expiration monitoring:

Event ID	Severity	Description	Parameters
BMYSVC2004	Warning	Custom certificate for component %s expiring in %d days.	• %s: Component name (API server, cluster proxy, or ingress) • %d: Number of days until expiration
BMYSVC3003	Critical	Custom certificate for component %s expiring in %d days.	

Configuring License Service Reporter for IBM Fusion

Configure License Service Reporter to obtain a unified view of IBM Fusion licensing metrics. License Service Reporter supports this feature and can be extended to IBM Fusion. In addition to the IBM Fusion user interface, it provides an additional view of licensing metrics, offering deeper insights. It can also be configured across multiple clusters, enabling centralized monitoring and reporting of license usage in distributed environments.

Before you begin

- Ensure that the newest version of License Service Reporter is installed, with support for Usage Metering Operator as a data source.
- Review the prerequisites for IBM License Service Reporter in the Installing and configuring License Service Reporter section.

Install and configure License Service Reporter

Ensure that the IBM License Service Reporter is installed on the cluster. If it is not installed, follow the instructions in the Installing License Service Reporter from public container registry section.

Configure a connection to the License Service Reporter

Follow the steps in the Configuring a connection to License Service Reporter by using the OpenShift console section to connect the Usage Metering Operator in the Fusion namespace to License Service Reporter.

Use the following high-level steps and documentation links to simplify the setup process for IBM Fusion.

Create the following resources in the namespace where IBM Fusion is installed:

Create the ConfigMap

Edit the following YAML to match your environment. Use the URL and certificate that is provided by License Service Reporter.

```
kind: ConfigMap
apiVersion: v1
metadata:
 name: <name>
data:
  url: '<License Service Reporter URL>'
  tls.crt: |
    <License Service Reporter certificate>
```

For more information on obtaining the License Service Reporter's URL and certificate, see Creating the ConfigMap.

Create the secret

To create the secret that is used to configure the connection to the License Service Reporter, update the following YAML to be in line with your environment:

```
kind: Secret
apiVersion: v1
metadata:
  name: <name>
data:
  token: <License Service Reporter token>
type: Opaque
```

For more information on retrieving the License Service Reporter token, see Creating the secret.

Create the configuration Custom Resource (CR)

1. Log in to the OpenShift® console of the cluster where IBM Fusion is deployed.
2. Open the IBM Fusion namespace.
3. Go to Operators > Installed Operators.
4. Select IBM Usage Metering Operator and open the IBM Usage Metering tab.
5. Click ibm-usage-metering-fusion and open the YAML tab.
6. Add the following content to the **spec** section and save:

```
spec:
   sender:
     reporter:
       configMapName: <Name of the ConfigMap>
       secretName: <Name of the secret>
```

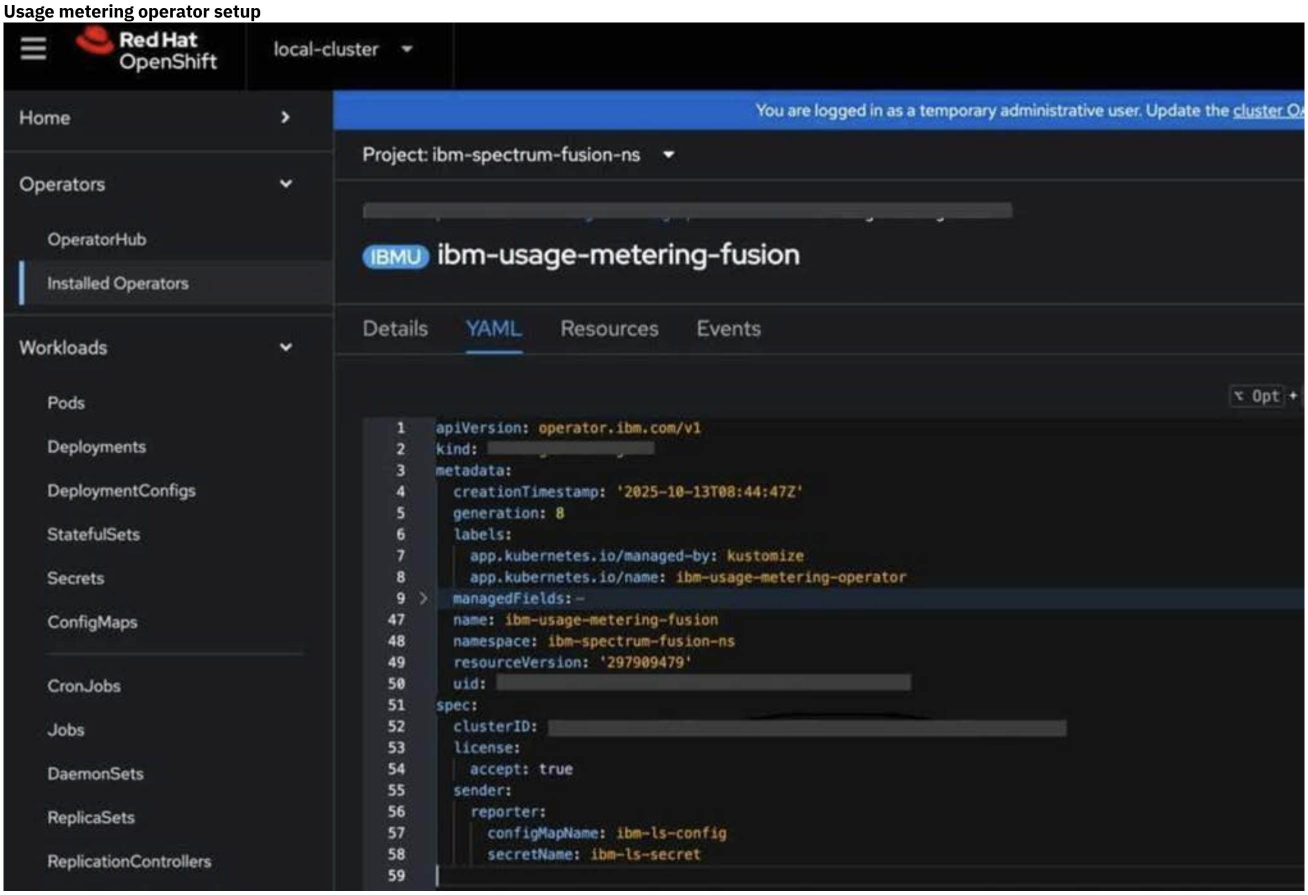
Note: The `configMapName` and `secretName` are the names of the ConfigMap and secret that are created in the previous steps.

Results

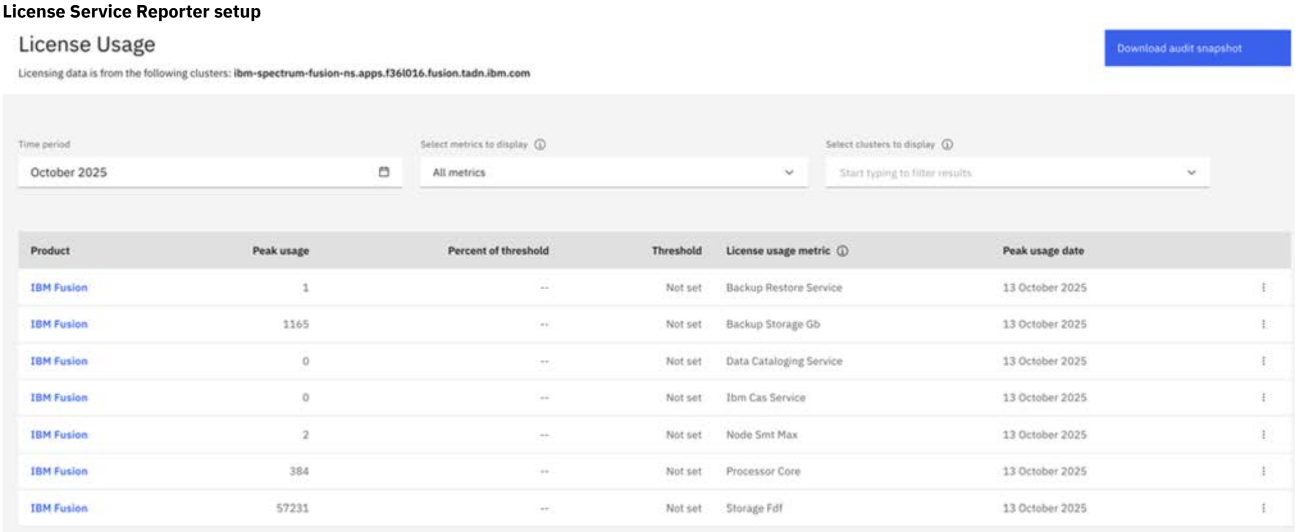
After completing the configuration, a connection is set and the Usage Metering Operator begins sending data to License Service Reporter. It enables a unified view of IBM Fusion licensing metrics.

Example of the unified view of IBM Fusion Licensing metrics in License Service Reporter.

- **Usage metering operator setup**



- **License Service Reporter setup**



What to do next

To log in to License Service Reporter console, follow these steps:

- For more information on getting the console route, see Retrieving License Service Reporter console route.
- For more information on how to retrieve the credentials to login, see Retrieving credentials for basic authentication.

Troubleshooting serviceability issues

List of all troubleshooting and known issues in Events, Log collection, and Call Home.

IBM Fusion licensing audit snapshot is unavailable

Problem statement
: The IBM Fusion licensing audit snapshot is unavailable .

Resolution
: Follow the steps to resolve the issue:

1. Run the following steps to check the IBM Fusion license service configuration:
   a. Log in to the OpenShift® Container Platform web console.
   b. Go to Workloads > CronJobs.
   c. Select the Licensig-CronJob and go to Jobs tab.
   d. Ensure that the cronjob is set up and running successful.
   e. If the cronjob is in a failed state, then create a new job by selecting Start Job under Actions drop-down.
   f. If the issue still persists after creating a new job, then contact IBM support.
2. Run the following steps to check for UMS state:
   a. Log in to the OpenShift Container Platform web console.
   b. Switch to Project **ums-fusion**.
   c. Go to Installed Operators.
   d. Check whether IBM Usage Metering Operator status is Succeeded.
   e. Go to Workloads > Pods and make sure that one pod of **ibm-usage-metering-operator** (with prefix **ibm-usage-metering-operator**) is in a running state.
   f. Switch to Project **<fusion-namespace>** and make sure that one pod of **ibm-usage-metering-instance** is in a running state.

Log collector does not work in IBM Fusion without primary storage

Run the following commands to start logcollector without persistent storage.

```
oc patch isflcdep logcollector --type=merge -p '{"spec":{"storageclass":"","replicas":1}}'
oc delete deployment logcollector
```

Note: New pods are created within a few minutes, after which log collection becomes available.

Unable to delete proxy config from IBM Fusion user interface

Problem statement
: Unable to delete proxy configurations available on the IBM Fusion user interface.

Resolution
: Follow the steps to resolve the issue:

1. Important: Ensure that you disable the remote support connection before you update or delete the proxy configuration.
   Disable remote support connection from the IBM Fusion HCI user interface as follows:
   a. From the IBM Fusion HCI user interface, go to Settings > Support.
      The Support settings page gets displayed.
   b. In the Remote support section, click the toggle button to disable the remote support connection.
      Note: Disable the remote support connection only if it is enabled.
2. Log in to the service node as a **kni** user.
3. Remove the proxy credential from pass tool on the service node.

   ```
   pass rm -f remoteSupport
   ```

4. Clean up the proxy details related to remote support from **/home/kni/isf-ui/conf/proxydata.json** on the service node.

   ```
   jq 'del(.remoteSupport)' /home/kni/isf-ui/conf/proxydata.json > temp.json && mv temp.json /home/kni/isf-ui/conf/proxydata.json
   ```

5. Refresh the IBM Fusion HCI or install user interface.

Events that go through the trap server do not get created in IBM Fusion

Problem statement
: Sometimes, events that go through the trap server do not get created in IBM Fusion.

Resolution
: If you see the following details in the trapserver logs, follow the steps as a workaround:

```
Listening for traps on 0.0.0.0:31620
 [fd8c:215d:178e:c0de:a94:efff:fef3:35cd]:41493 byte array parsed in is not a sequence
 [fd8c:215d:178e:c0de:a94:efff:fef3:3561]:35748 failed to parse sequence length0
 2021/10/13 14:20:48.124 [D] Recovered in resetServer, r=runtime error: slice bounds out of range [:-2592903709665718705]
```

```
Listening for traps on 0.0.0.0:31620
 [fd8c:215d:178e:c0de:a94:efff:fef3:3555]:58314 failed to parse sequence length0
 [fd8c:215d:178e:c0de:a94:efff:fef3:35cd]:34801 byte array parsed in is not a sequence
 [fd8c:215d:178e:c0de:a94:efff:fef3:35cd]:52070 byte array parsed in is not a sequence
 [fd8c:215d:178e:c0de:a94:efff:fef3:3555]:47601 parse error
 [fd8c:215d:178e:c0de:a94:efff:fef3:3585]:56764 length parse error @ idx 2
 [fd8c:215d:178e:c0de:a94:efff:fef3:3399]:44497 failed to parse sequence length0
 [fd8c:215d:178e:c0de:a94:efff:fef3:35cd]:57884 failed to parse sequence length0
 [fd8c:215d:178e:c0de:a94:efff:fef3:3585]:59094 length parse error @ idx 2
 [fd8c:215d:178e:c0de:a94:efff:fef3:3555]:40214 parse error
 [fd8c:215d:178e:c0de:a94:efff:fef3:3399]:37409 byte array parsed in is not a sequence
```

- Restart the trapserver pod in the `ibm-spectrum-fusion-ns`.
- Run the following command to delete all the `ComputeMonitoring` CRs that are present in the `ibm-spectrum-fusion-ns` namespace.

  ```
  oc delete cmo --all -n ibm-spectrum-fusion-ns
  ```

  Wait for the custom resource instances to get recreated.
- Restart BMC of all the compute nodes by executing `resetsp` command from the BMC command line.

Log status shows complete for downloaded log file with 0-bytes size

Problem statement

The Log status shows complete for downloaded log file with of 0-bytes size. The status should be failed if logs are not collected, but it shows completed with 0-bytes size.

Cause

1. Pods get evicted because of a lack of storage space.
2. The ongoing jobs are taking time to get storage space, but if it takes more time, then they are marked as stale and automatically get cleaned up.

Resolution

There are two methods to resolve this issue:

Method 1

1. Delete the unnecessary logs through the IBM Fusion HCI user interface to get storage space. For more information, see Delete a log package.
2. Delete the on going jobs that are taking a long time and retry the job again.

Important: Method 2 is applicable only when the log collector is using PV.

Method 2

- You can increase the log collector PVC size by following the steps:
  1. Log in to the Red Hat® OpenShift Container Platform web console.
  2. Go to Storage > PersistentVoulmeClaims.
     The PersistentVolumeClaims page gets displayed.
  3. Select the log collecter PVC that you want to modify.
  4. Click the ellipsis icon and select Expand PVC.
     The Expand PersistentVolumeClaims page gets displayed.
  5. Set the PVC value that you want and click Expand.

Log collector deployment fails with `CrashLoopBackOff` due to `Init` container failure

Problem statement

When using Global Data Platform storage in an IBM Fusion HCI cluster, the `Init` container of the log collector pod may fail to set permissions on the logs directory. As a result `Init` container fails and the log collector pod enters a `CrashLoopBackOff` state.

Resolution

Follow the steps to resolve the issue:

1. Delete the log collector deployment and wait for it to be automatically recreated by the serviceability operator.
   Note: It may take 5 minutes for the deployment to get recreated.
2. Important: If step 1 does not fix the problem and the pod is still in the `CrashLoopBackOff` state, try this step:
   If the issue persists, verify the presence of the `init-set-ownership` `init` container and patch the deployment as follows:
   a. Run the following command to check for the `init` container.

      ```
      oc get pod <log-collector-pod> -n ibm-spectrum-fusion-ns -o jsonpath='{.spec.initContainers[*].name}'
      ```

   b. If present, apply the patch using the command for your operating system:

      For Linux or macOS:

      Use the following command:

      ```
      oc patch deployment logcollector -n ibm-spectrum-fusion-ns -p '{"spec":{"template":{"spec":
      {"initContainers":[{"name":"init-set-ownership","securityContext":{"capabilities":{"add":["CHOWN"]}}}]}}}}'
      ```

      For windows:

      Use the following command:

      ```
      oc patch deployment logcollector -n ibm-spectrum-fusion-ns -p "{\"spec\":{\"template\":{\"spec\":
      {\"initContainers\":[{\"name\":\"init-set-ownership\",\"securityContext\":{\"capabilities\":{\"add\":
      [\"CHOWN\"]}}}]}}}}"
      ```

   This patch allows the `init` container to change ownership of the logs directory, ensuring the log collector starts successfully.
3. If the issue still persists, then IBM support.

Known issues

- In the Logs page of the IBM Fusion user interface, if you select System Health Check option during log collection, then it takes longer than usual time to complete. It is observed that the log collection process might take 20 to 25 minutes. In some cases, it can be due to many directories and multiple IMM log file collection process.
- For IBM Storage Scale warning events, the fixed status might be incorrect.
- Sometimes, in the Events page, the Source column of the events list might be incorrect.
- Events are not created for IBM Storage Scale events with entity_name fields that do not conform to the URL path components (^(([A-Za-z0-9][-A-Za-z0-9_.]*)?[A-Za-z0-9])?$).
- Call Home ticket creation can have a failed state when the system is not entitled or the Call Home server does not respond with the ticket number. Click Verify connection to check the test connection.
- To prevent a deadlock condition, the event manager must be restarted every 24 hours. Run the following command to restart the event manager:

  ```
  oc rollout restart deployment eventmanager
  ```

- Sometimes, the automatic upload of logs might not happen and the Call Home would fail. In such cases, manually upload the logs.
- If you power off a control node wherein the trap server pod is running, then the migration of trap server pods to a different node fails. As a result, the trap server pod may get stuck in the terminating state, and some SNMP trap events may fail to capture and display.
- In rare scenarios, the Events page comes up as empty because of a backend error due to a load with a 504 Gateway error. Generally, this page comes up after sometime automatically as the system recovers, so please try again after sometime.
- One possible reason is log collector pod runs out of space and no space is available on it. This can happen when too many logs are collected in short period. The solution is to delete already collected logs from Fusion UI which are no longer required.

  Workaround
  - From the title bar, click the help icon and select Support logs.
  - Identify and delete logs through ellipsis menu in the Support logs page.
- If Data Foundation log collection request or requests for multiple log packages collected simultaneously fails or stuck for more than 6 hours, then update the pod memory to 12000MiB instead of 6000MiB in the log collector deployment at `spec.containers[0].resources.limits.memory` and then attempt the log collection again one by one.
- Sometimes, an intermittent partial log collection error occurs for node log packages in case of multi-rack set up.

  Workaround
  - Retry to collect node logs again. For steps, see Collecting log packages for IBM Fusion HCI.
  - If the issue persists, then contact IBM team to get the commands (FFDC) to collect the individual node logs.

Events and error codes message references in IBM Fusion HCI

This reference information provides a consolidated view of all the events or error messages that can be displayed when the IBM Fusion system might result in an event or an error.

About event or error messages

When you encounter an event or error message in a log or on other parts of the IBM Fusion user interface, look up the event or error message by its prefix to find more information. Use the events and error messages to diagnose and solve problems when you troubleshoot the IBM Fusion system.

Message prefix format

A message number format denotes the type of message and with which component it is associated with. The default prefix starts as `BMY` in the code, which is followed by one of these tags:

- The compute node-related events tag is associated with `CO` in the message code.
- The installation-related events tag is associated with `IN` in the message code.
- The backup and restore-related events tag is associated with `BR` in the message code.
- The network-related events tag is associated with `NW` in the message code.
- The upgrade-related events tag is associated with `UP` in the message code.
- The call home related events tag is associated with `CH` in the message code.
- The disaster recovery-related events tag is associated with `DR` in the message code.
- The hardware-related events tag is associated with `XC` in the message code.
- The 4 digits after the tag denote the number of the event or error message. For example, `0000`.
- The Cumulus Network Switch events tag is associated to `BMYCUxxxx` in the message code.
- The list of hardware trap events that can open tickets tag is associated to `BMYXC` and `BMYDL`in the message code.

An example of a message prefix format is `BMYCO0001`.

Message severity tag

A severity tag is a one-character alphabetic code, followed by a message. You can determine the message severity by examining the text or by contacting the IBM® Support. To collect logs, see Collecting log packages for IBM Fusion HCI.

Table 1. Message severity tags

Severity tag	Message type	Meaning
C	CRITICAL	Indicates a critical condition that must be corrected immediately. The system discovered an internal inconsistency of some kind. The system functioning might be halted or the system might attempt to continue despite the inconsistency. Report these errors to IBM® Support.
I	INFORMATIONAL	Indicates normal operation. This message by itself indicates that nothing is wrong; no action is required.
W	WARNING	Indicates a problem, but the system functioning continues. The problem can be a transient inconsistency. It can be that the action skipped some operations on some objects, or is reporting an irregularity that could be of interest.

- **Hardware events and error codes**
  IBM Fusion HCI supports Lenovo and Dell servers. It monitors these servers for hardware events such as CPU, DIMM, drive, and fan issues, as well as BMC-related events like misconfigurations and firmware errors. Event IDs start with BMYDL for Dell and BMYXC for Lenovo.
- **Install events and error codes**
  List of all the events that you might encounter during installation.
- **Upgrade events and error codes**
  List of all the events that you might encounter during upgrade.
- **Compute events and error codes**
  List of all the events that you might encounter in compute nodes.
- **Network events error codes**
  List of all the events that you might encounter in network node.
- **Metro-DR events and error codes**
  List of all the events and error codes that you might encounter in Metro-DR setup.
- **Global Data Platform events and error codes**
  List of all the events and error codes that you might encounter while you work with storage.
- **Data foundation events and error codes**
  List of all the events and error codes that you might encounter while you work with Data Foundation storage.
- **High availability cluster events and error codes**
  List of all the events that you might encounter during High availability cluster setup.
- **IBM Data Cataloging events and error codes**
  List of all the events that you might encounter while you work with IBM Data Cataloging for the IBM Fusion HCI .
- **Backup & Restore events and error codes**
  List of all the events that you might encounter during Backup & Restore for the IBM Fusion HCI.
- **Connection services events and error codes**
  List of all the events that you might encounter during connection service for the IBM Fusion HCI .
- **Content-Aware Storage events and error codes**
  List of all the events that you might encounter while you work with CAS for the .
- **Serviceability events and error codes**
  List of all the events that you might encounter that related to serviceability.
- **Hosted Control Plane events and error codes**
  List of all the events that you might encounter that related to Hosted Control Plane.
- **Software Hub events and error codes**
  List of all the events and error codes that you might encounter while you work with storage.

Hardware events and error codes

IBM Fusion HCI supports Lenovo and Dell servers. It monitors these servers for hardware events such as CPU, DIMM, drive, and fan issues, as well as BMC-related events like misconfigurations and firmware errors. Event IDs start with BMYDL for Dell and BMYXC for Lenovo.

Refer to the respective vendor documentation for detailed descriptions and recommended actions.

- **Events and error codes for Dell hardware**
  Hardware events or alerts raised for Dell hardware failures.
- **Events and error codes for Lenovo hardware**
  Hardware events or alerts raised for Lenovo hardware failures.

Events and error codes for Dell hardware

Hardware events or alerts raised for Dell hardware failures.

Hardware trap events

IBM Fusion HCI converts Dell hardware trap events into IBM Fusion events for streamlined observability. Each event includes a `Reference ID (RefID)` matching the original Dell event identifier. Search for the `RefID` within the following Dell documentation for detailed information on the specific error or alert: Dell error and event reference guide.

Call Home events

- BMYDL2153
  Fan arg1 RPM is less than the lower critical threshold.
- BMYDL2161
  The system board arg1 temperature is less than the lower critical threshold.
- BMYDL2185
  Power supply arg1 failed.
- BMYDL2225
  The system board battery is absent.
- BMYDL2265
  Multi-bit memory errors are detected on the memory device at location(s) arg1. Immediately replace the DIMM.
- BMYDL2297
  The drive is removed from the drive bay.

- BMYDL2417
A bus time-out was detected on a component at bus arg1 device arg2 function arg3.
- BMYDL2497
CPU Exception Type 0x00: Divide by Zero (Software).
- BMYDL3249
The air Filter in the bezel has a connectivity issue.
- BMYDL5604
CPU arg1 has a thermal trip (over-temperature) event.

BMYDL2153

Fan arg1 RPM is less than the lower critical threshold.

Severity

Critical

Problem description

It indicates the fan is not performing optimally. The fan may be installed improperly or may be failing. Fan speeds may have increased to protect the system.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL2161

The system board arg1 temperature is less than the lower critical threshold.

Severity

Critical

Problem description

It indicates the Ambient air temperature is too cool.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL2185

Power supply arg1 failed.

Severity

Critical

Problem description

It indicates the power supply is failed.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL2225

The system board battery is absent.

Severity

Critical

Problem description

It indicates the system settings may be preserved if input power is not removed from the power supplies.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL2265

Multi-bit memory errors are detected on the memory device at location(s) arg1. Immediately replace the DIMM.

Severity

Critical

Problem description

It indicates the memory has an uncorrectable error. System performance may be degraded and the operating system and/or applications may fail as a result of this.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL2297

The drive is removed from the drive bay.

Severity

Critical

Problem description

It indicates the controller detected that the drive is removed.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL2417

A bus time-out was detected on a component at bus arg1 device arg2 function arg3.

Severity

Critical

Problem description

It indicates the system performance may be degraded. The device has failed to respond to a transaction (source/ target abort or completion timeout).

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL2497

CPU Exception Type 0x00: Divide by Zero (Software).

Severity

Critical

Problem description

It indicates the fan is not performing optimally. The fan may be installed improperly or may be failing. Fan speeds may have increased to protect the system.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL3249

The air Filter in the bezel has a connectivity issue.

Severity

Critical

Problem description

It indicates the air Filter in the bezel has a connectivity issue.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.

2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYDL5604

CPU arg1 has a thermal trip (over-temperature) event.

Severity

Critical

Problem description

It indicates that the processor temperature increased beyond the operational range. Thermal protection shut down the processor. Factors external to the processor may have induced this exception.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

Events and error codes for Lenovo hardware

Hardware events or alerts raised for Lenovo hardware failures.

- **Call Home events and error codes**
  List of Call Home events that you might encounter in the Lenovo hardware.
- **BMYXC0021**
  OS timeout value exceeded.
- **BMYXC0022**
  Refer to error conditions for a specific condition.
- **BMYXC0023**
  Power off.
- **BMYXC0024**
  Power on.
- **BMYXC0025**
  System boot failure.
- **BMYXC0026**
  OS loader timeout.
- **BMYXC0027**
  Predictive Failure Analysis (PFA) information.
- **BMYXC0030**
  Event remote login.
- **BMYXC0035**
  System log 75% full.
- **BMYXC0037**
  Network change notification.
- **BMYXC0038**
  Host NIC link state down or up.
- **BMYXC0039**
  Drive Hotplug.
- **BMYXC0060**
  The component mentioned in the error message is in degraded state.
- **BMYXC0164**
  Power error.
- **BMYXC0165**
  Fan error.
- **BMYXC0255**
  IP address is active.
- **BMYXC9999**
  Trap server events raised for the hardware failures.

Call Home events and error codes

List of Call Home events that you might encounter in the Lenovo hardware.

- **BMYXC0000**
  Temperature threshold exceeded.

- **BMYXC0001**
  Voltage threshold exceeded.
- **BMYXC0004**
  Power failure.
- **BMYXC0005**
  Hard disk drive failure.
- **BMYXC0009**
  Redundant Power Supply failure.
- **BMYXC0010**
  Redundant Power Supply failure.
- **BMYXC0011**
  Single Fan failure.
- **BMYXC0012**
  Temperature threshold exceeded.
- **BMYXC0013**
  Voltage threshold exceeded.
- **BMYXC0036**
  Incompatible hardware configuration.
- **BMYXC0040**
  CPU Error.
- **BMYXC0041**
  Memory Error.
- **BMYXC0042**
  CPU error.
- **BMYXC0043**
  Memory error.
- **BMYXC0050**
  The component mentioned in the error message is in critical or unhealthy state.

BMYXC0000

Temperature threshold exceeded.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0001

Voltage threshold exceeded.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0004

Power failure.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0005

Hard disk drive failure.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0009

Redundant Power Supply failure.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0010

Redundant Power Supply failure.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0011

Single Fan failure.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0012

Temperature threshold exceeded.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0013

Voltage threshold exceeded.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0036

Incompatible hardware configuration.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0040

CPU Error.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0041

Memory Error.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0042

CPU error.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0043

Memory error.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the System health check logs and contact IBM support.

BMYXC0050

The component mentioned in the error message is in critical or unhealthy state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects compute and system health check collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes and System health check logs and contact IBM support.

BMYXC0021

OS timeout value exceeded.

Severity

Informational

User actions

No action is required.

BMYXC0022

Refer to error conditions for a specific condition.

Error conditions

The error message can be for the following conditions:

1. The operating System status changed to offline.
2. The server is restarted with a chassis control command.
3. The server is powered with a chassis control command.
4. User LXPM has mounted the file tdm_image.img from local.
5. The port 1 of the PCIe device Intel I350 1GbE RJ45 4-port OCP Ethernet Adapter onboard is linked down.
6. Disk related errors. For example, disk removal, addition, rebuild, hot spare and so on.
7. User password modified by user from at IP address.

Severity

Informational, warning, or critical
Note: The critical severity occurs whenever there are disk related errors.

User actions

Do the following steps to diagnose the error:

1. Error conditions one, two, three, four, and seven are informational messages, and no action is required.
2. For error condition five, ensure that you check the network cable and power cable connectivity to the respective ports.
3. For error condition six, collect the System health check and Nodes logs and contact IBM support.

BMYXC0023

Power off.

Severity

Informational

User actions

No action is required.

BMYXC0024

Power on.

Severity

Informational

User actions

No action is required.

BMYXC0025

System boot failure.

Severity

Informational

User actions

No action is required.

BMYXC0026

OS loader timeout.

Severity

Informational

User actions

No action is required.

BMYXC0027

Predictive Failure Analysis (PFA) information.

Severity

Informational

User actions

No action is required.

BMYXC0030

Event remote login.

Severity

Informational

User actions

No action is required.

BMYXC0035

System log 75% full.

Severity

Informational

User actions

No action is required.

BMYXC0037

Network change notification.

Severity

Informational

User actions

No action is required.

BMYXC0038

Host NIC link state down or up.

Severity

Informational

User actions

No action is required.

BMYXC0039

Drive Hotplug.

Severity

Informational

User actions

No action is required.

BMYXC0060

The component mentioned in the error message is in degraded state.

Severity

Warning

User actions

Collect the System health check and Nodes logs and contact IBM support.

BMYXC0164

Power error.

Severity

Warning

User actions

Collect the System health check logs and contact IBM support.

BMYXC0165

Fan error.

Severity

Warning

User actions

Collect the System health check logs and contact IBM support.

BMYXC0255

IP address is active.

Severity

Informational

Error conditions

The error message can be for the following conditions:

- IPv6 or IPv4 Link Local address is active.
- Management controller IP address and configuration assigned statically.

User actions

No action is required.

BMYXC9999

Trap server events raised for the hardware failures.

Severity

Critical or warning or informational

User actions

Do the following steps to diagnose the error:

1. Note down the `IBM-MTM: 9155-xxx` model number and `EVENT ID: xxxx` details of the BMYXC9999.
2. Check the following table to get details of the IBM MTMs with their respective Lenovo server model and publications details.

Table 1. IBM and Lenovo server model Mapping

Description	IBM MTM	Lenovo server model	Publication
IBM Fusion HCI Compute Server	9155-C00	SR645	SR645 publication
IBM Fusion HCI Compute/Storage Server	9155-C01	SR645	SR645 publication
IBM Fusion HCI Compute Server	9155-C04	SR645	SR645 publication
IBM Fusion HCI Compute/Storage Server	9155-C05	SR645	SR645 publication
IBM Fusion HCI AFM Server	9155-F01	SR630	SR630 publication
IBM Fusion HCI GPU Server	9155-G01	SR665	SR665 publication
IBM Fusion HCI GPU Server	9155-G02	SR650v3	SR650v3 publication
IBM Fusion HCI GPU Server	9155-G03	SR675v3	SR675v3 publication
IBM Fusion HCI Compute/Storage Server	9155-C10	SR630v3	SR630v3 publication
IBM Fusion HCI Compute/Storage Server	9155-C14	SR630v3	SR630v3 publication

3. To get workaround steps or more information about the event, click and open the Lenovo server model publication based on your IBM MTM model number and search with the event id.

Install events and error codes

List of all the events that you might encounter during installation.

For Call Home events, see Install Call Home events and error codes.

For Informational, Warning and Critical events, see the following:

- **Install Call Home events and error codes**
  List of Call Home events that you might encounter in the install.
- **BMYIN1101**
  Node addition to OpenShift® Container Platform is in progress.
- **BMYIN1104**
  Bare metal host is not yet provisioned.

- **BMYIN1105**
  OpenShift node is not yet ready.
- **BMYIN1106**
  Node network configuration policy is not yet configured.
- **BMYIN1107**
  Node is now added to OpenShift cluster.
- **BMYIN1109**
  One time network boot and power off was successful.
- **BMYIN1117**
  Waiting for new node to come up.
- **BMYIN1501**
  Application has been created successfully.
- **BMYIN1502**
  Installing service Global Data Platform.
- **BMYIN1503**
  Installed service Global Data Platform.
- **BMYIN1504**
  Installing service Backup & Restore.
- **BMYIN1505**
  Installed service Backup & Restore.
- **BMYIN1508**
  Service is installed.
- **BMYIN1509**
  Service installation is complete.
- **BMYIN1510**
  Service instance is discovered.
- **BMYIN1512**
  Catalog source is created.
- **BMYIN1513**
  Catalog source creation is completed.
- **BMYIN2102**
  Location for the node is not specified.
- **BMYIN2103**
  Failed to read kickstart config map data.
- **BMYIN2110**
  Failed to read userconfig secret object.
- **BMYIN2111**
  DNS validation failed for the node.
- **BMYIN2112**
  Invalid name is used in ComputeProvisionWorker CR.
- **BMYIN2116**
  Failed to scale machineset during new node addition.
- **BMYIN2117**
  Rack serial for the node is not specified.
- **BMYIN2118**
  Unable to read OpenShift Container Platform location for the node from kickstart configmap.
- **BMYIN2119**
  Both location and rack serial for the node are not specified.
- **BMYIN2501**
  Error installing service Global Data Platform.
- **BMYIN2502**
  Error installing service Backup & Restore.
- **BMYIN3115**
  OpenShift node did not transition to the `Ready` state.
- **BMYIN3116**
  Service is Unhealthy.
- **BMYIN3117**
  Service Catalog source creation is failed.
- **BMYIN3118**
  A node might contain an invalid hostname (local host).
- **BMYIN3119**
  Service installation failed.
- **BMYIN3129**
  Failed to taint the node.
- **BMYIN3130**
  Failed to update the cluster scheduler.
- **BMYIN4005**
  `vault_secret_not_found`
- **BMYIN4006**
  `vault_sealed`
- **BMYIN4007**
  `vault_authentication_failed`
- **BMYIN4008**
  `vault_update_success`
- **BMYIN4009**
  `vault_delete_files_success`
- **BMYIN4010**
  `vault_fetch_info_success`
- **BMYIN4011**
  `vault_healthy`

- BMYIN4012
  `vault_get_secret_success`
- BMYIN4013
  `vault_seal_status_check_success`
- BMYIN4014
  `vault_userpass_login_success`
- BMYIN4015
  `vault_unseal_success`
- BMYIN4016
  `vault_get_client_failed`
- BMYIN4017
  `vault_create_secret_failed`
- BMYIN4018
  `vault_unseal_status_check_failed`
- BMYIN4019
  `vault_userpass_login_failed`
- BMYIN4020
  `vault_delete_vault_files_failed`
- BMYIN4021
  `vault_fetch_info_failed`
- BMYIN4022
  `vault_in_configure_success`
- BMYIN4023
  `vault_in_configure_failed`
- BMYIN4024
  `vault_info_on_ocp_success`
- BMYIN4025
  `vault_info_on_ocp_failure`
- BMYIN4026
  `vault_populate_success`
- BMYIN4027
  `vault_populate_failure`
- BMYIN4028
  `vault_fetch_secret_fail`
- BMYIN4029
  `ServiceNodeKeyRotationSuccess`
- BMYIN4030
  `ServiceNodeKeyRotationFailed`
- BMYIN4101
  `UnsupportedPlatform`
- BMYIN4102
  `RelatedServiceInstalled`
- BMYIN4103
  `RelatedStorageServiceInstalled`
- BMYIN4104
  `StorageClassPrereqNotFound`
- BMYIN4105
  `OCPVersionPrereqNotFound`
- BMYIN4106
  `CRDMissingPrereqNotFound`
- BMYIN4107
  `DataJSONKeyNotFoundInCM`
- BMYIN4108
  `SkipServiceInSpectrumFusionCRTrue`
- BMYIN4110
  `PrereqServiceNotInstalled`
- BMYPC5001
  Cluster operator is available but is in progress state.
- BMYPC5002
  A `MachineConfigPool` is updating.
- BMYPC5004
  A pod PodDisruptionBudget (PDB) is set to zero.
- BMYPC5005
  Virtual machine is using PVC with non-RWX access mode.
- BMYPC5006
  A pod in the `openshift-operator-lifecycle-manager` namespace is unhealthy.
- BMYPC5010
  IBM Fusion operator installation is in progress.
- BMYPC5013
  Unable to check DNS resolution for the registry.
- BMYPC5027
  The OpenShift Container Platform is currently operating at the latest version that is supported by IBM Fusion.
- BMYPC5028
  A virtual machine is configured with PCI pass-through.
- BMYPC5029
  The OpenShift Container Platform is currently operating at the latest version that is supported by Fusion Data Foundation.
- BMYPC6001
  Cluster operator is not healthy.
- BMYPC6002
  A `MachineConfigPool` is degraded.

- **BMYPC6003**
  A node is not ready or unscheduleable.
- **BMYPC6006**
  A pod in the `openshift-operator-lifecycle-manager` namespace is not ready.
- **BMYPC6007**
  A `Catalogsource` is not ready.
- **BMYPC6008**
  Inaccessible registry.
- **BMYPC6010**
  IBM Fusion operator is in a `Failed` state or not found.
- **BMYPC6013**
  The DNS resolution for image registry failed.
- **BMYPC6014**
  Global Data Platform service upgrade is in progress.
- **BMYPC6015**
  It is a switch upgrade precheck error. STP is enabled on the high-speed switches.
- **BMYPC6023**
  `Pidslimit` on the nodes is less than 12228.
- **BMYPC6040**
  Backup & Restore agent instance not found.
- **BMYPC6042**
  The Subscription is missing the `CatalogSource`.
- **BMYPC6044**
  Backup & Restore server instance is unhealthy.
- **BMYPC6045**
  Backup & Restore agent instance is unhealthy.
- **BMYPC6076**
  The DNS resolution for the node failed.
- **Configuration error**
  Message: Encountered a configuration error. Red Hat® OpenShift Container Platform installation did not start successfully.
- **Timed out creating bootstrap VM**
  Message: Bootstrap VM creation on provisioner node (also known as service node) has not started.
- **Bootstrap VM creation did not complete**
  Message: Bootstrap VM creation on the provisioner node (also known as service node) did not complete within the expected time.
- **Failed control nodes inspection**
  Message: Inspection failed for one or more control nodes.
- **Control nodes creation failed**
  Message: Failed to create the Red Hat OpenShift Container Platform cluster control nodes.
- **Cluster bootstrapping failed**
  Cluster bootstrapping fails when one or more control nodes do not boot during installation of Red Hat OpenShift Container Platform.
- **Catalog source pod is healthy**
  Message: IBM Fusion catalog source pod is not found or is not healthy.
- **Operator subscription isf-subscription not found**
  Message: Subscription for IBM Fusion operator not found.
- **Bootstrap VM is not pingable**
  A temporary bootstrap VM is created on provisioner node to create Red Hat OpenShift Container Platform control plane.
- **Configuration error**
  The certificate does not contain a wildcard domain for cluster.
- **Failed to install CSV for isf-operator**
  When you install IBM Fusion, the installation fails with "`Failed to install CSV for ifs-operator`" error.
- **Failed to install IBM Fusion operator**
  Failed to install the IBM Fusion operator. Check the log file /home/kni/logs/installoperator_playbook.log for more details.
- **The catlogsource pod for Red Hat operators is not running**
  The `catlogsource` pod for `redhat-operators` is not running in `openshift-marketplace` namespace. Check the log file %s for more details.
- **Failed to deploy nmstate-operator**
  Failed to deploy `nmstate-operator`. Check the log file /home/kni/logs/installoperator_playbook.log for more details.
- **Failed to install CSV for kubernetes-nmstate-operator**
  While installing IBM Fusion HCI, the installation fails with the error message `Failed to install CSV for kubernetes-nmstate-operator`. Check log file /home/kni/logs/installoperator_playbook.log for more details.
- **Delete files from provisioned node**
  The provisioner node, also know as service node, remains intact after IBM Fusion HCI initialization. For security reasons, all the configuration files are deleted from the service node.

Install Call Home events and error codes

List of Call Home events that you might encounter in the install.

Install Call Home events

If you have enabled Call Home feature, then the system automatically collects relevant log package and raises a Call Home ticket.

If you have not enabled Call Home feature, collect the relevant log package and contact IBM support.

- **BMYIN3108**
  Unable to add a new node. Failed to set boot order.
- **BMYIN3113**
  Unable to add a new node. Bare Metal host did not transition to Provisioned state.

- **BMYIN3114**
  Unable to add a new node. OpenShift® node not found after Bare Metal host provisioning.
- **BMYIN3115**
  Unable to add a new node. OpenShift node did not transition to Ready state.

BMYIN3108

Unable to add a new node. Failed to set boot order.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect system logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN3113

Unable to add a new node. Bare Metal host did not transition to Provisioned state.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect system logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN3114

Unable to add a new node. OpenShift® node not found after Bare Metal host provisioning.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect system logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN3115

Unable to add a new node. OpenShift® node did not transition to Ready state.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect system logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN1101

Node addition to OpenShift® Container Platform is in progress.

Severity

Informational

User actions

No action is required.

BMYIN1104

Bare metal host is not yet provisioned.

Severity

Informational

User actions

No action is required.

BMYIN1105

OpenShift® node is not yet ready.

Severity

Informational

User actions

No action is required.

BMYIN1106

Node network configuration policy is not yet configured.

Severity

Informational

User actions

No action is required.

BMYIN1107

Node is now added to OpenShift® cluster.

Severity

Informational

User actions

No action is required.

BMYIN1109

One time network boot and power off was successful.

Severity

Informational

User actions

No action is required.

BMYIN1117

Waiting for new node to come up.

Severity

Informational

User actions

No action is required.

BMYIN1501

Application has been created successfully.

Severity

Informational

User actions

No action is required.

BMYIN1502

Installing service Global Data Platform.

Severity

Informational

User actions

No action is required.

BMYIN1503

Installed service Global Data Platform.

Severity

Informational

User actions

No action is required.

BMYIN1504

Installing service Backup & Restore.

Severity

Informational

User actions

No action is required.

BMYIN1505

Installed service Backup & Restore.

Severity

Informational

User actions

No action is required.

BMYIN1508

Service is installed.

Severity

Informational

User actions

No action is required.

BMYIN1509

Service installation is complete.

Severity

Informational

User actions

No action is required.

BMYIN1510

Service instance is discovered.

Severity

Informational

User actions

No action is required.

BMYIN1512

Catalog source is created.

Severity

Informational

User actions

No action is required.

BMYIN1513

Catalog source creation is completed.

Severity

Informational

User actions

No action is required.

BMYIN2102

Location for the node is not specified.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect compute logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN2103

Failed to read kickstart config map data.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect compute logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN2110

Failed to read userconfig secret object.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect compute logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN2111

DNS validation failed for the node.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check whether the correct DNS record is added to the DNS server.
   After you fix this issue, the install process resumes automatically.
2. If the problem persists, contact IBM Support.

BMYIN2112

Invalid name is used in ComputeProvisionWorker CR.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect compute logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN2116

Failed to scale machineset during new node addition.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect compute logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN2117

Rack serial for the node is not specified.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect compute logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN2118

Unable to read OpenShift® Container Platform location for the node from kickstart configmap.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect compute logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN2119

Both location and rack serial for the node are not specified.

Severity

Critical

User actions

If this occurs during installation, collect logs for install operator and contact IBM support. For steps to collect install operator logs, see Collecting log files of final installation.

If this occurs post installation, then go to Logs UI page and collect compute logs. For more information about logs page, see Collecting log packages for IBM Fusion HCI.

BMYIN2501

Error installing service Global Data Platform.

Severity

Warning

User actions

Collect Storage logs and contact IBM support.

BMYIN2502

Error installing service Backup & Restore.

Severity

Warning

User actions

Collect Backup and Restore logs and contact IBM support.

BMYIN3115

OpenShift® node did not transition to the `Ready` state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command to verify the pending certificates.

```
oc get csr --no-headers | grep -i pending
```

2. Run the following command to approve the pending certificates.

```
for i in `oc get csr --no-headers | grep -i pending |  awk '{ print $1 }'`; do oc adm certificate approve $i; done
```

3. Restart the node.

```
oc debug node/<NODE_NAME>
chroot /host
reboot
```

4. If the problem persists, then collect Administration logs and contact IBM support.

BMYIN3116

Service is Unhealthy.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Go to OpenShift® Container Platform web console and check the FusionServiceInstance for the specific service and check the conditions section to find the error.
2. If the issue persists, run the following oc command, and check the status section and debug the error.

```
oc get fusionserviceinstance <instance-name-of-service> -o yaml
```

3. If this occurs during installation or post service installation, collect logs for the corresponding service that shows as unhealthy and contact IBM support. Follow the steps to collect logs from IBM Fusion.
    - Go to Services page in IBM Fusion user interface.
    - Click the ellipses menu for the corresponding service.
    - Select the Collect logs button. It downloads all the necessary logs.

BMYIN3117

Service Catalog source creation is failed.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command and try to debug the results.

```
oc describe catalogsource <catlogsource-name> -n openshift-marketplace
```

2. If the result says Username/Password missing, manually update the secret for the required registry.
3. If the issue persists, contact the IBM support.

BMYIN3118

A node might contain an invalid hostname (local host).

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Correct the DHCP configuration of the affected node.
2. Contact IBM support for local host node clean up.

BMYIN3119

Service installation failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Go to OpenShift® Container Platform web console and check the FusionServiceInstance for the specific service and check the conditions section to find the error.
2. If the issue persists, run the following oc command, and check the status section and debug the error.

   ```
   oc get fusionserviceinstance <instance-name-of-service> -o yaml
   ```

3. If this occurs during installation or post service installation, collect logs for the corresponding service that shows as unhealthy and contact IBM support. Follow the steps to collect logs from IBM Fusion.
   - Go to Services page in IBM Fusion user interface.
   - Click the ellipses menu for the corresponding service.
   - Select the Collect logs button. It downloads all the necessary logs.

BMYIN3129

Failed to taint the node.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Update taint based on node type when the underlying storage is Global Data Platform only.
2. Run the following commands to update the taint for the AFM node.

   ```
   oc adm taint nodes <NODE_NAME> isf.ibm.com/noworkload:NoSchedule
   oc adm taint nodes <NODE_NAME> afm.isf.ibm.com/noworkload:NoSchedule
   ```

3. Run the following commands to update the taint for the other nodes.

   ```
   oc adm taint nodes <NODE_NAME> isf.ibm.com/noworkload:NoSchedule
   ```

BMYIN3130

Failed to update the cluster scheduler.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command to update the cluster scheduler.

   ```
   oc edit schedulers.config.openshift.io cluster
   ```

2. set `mastersSchedulable` field to true and save.

```
mastersSchedulable: true
```

BMYIN4005

`vault_secret_not_found`

Severity

Warning

Problem description

It indicates that Vault secret not found.

User actions

Contact IBM support.

BMYIN4006

`vault_sealed`

Severity

Critical

Problem description

It indicates that Vault is sealed and needs to be unsealed.

User actions

Contact IBM support.

BMYIN4007

`vault_authentication_failed`

Severity

Critical

Problem description

It indicates that Vault authentication failed.

User actions

Contact IBM support.

BMYIN4008

`vault_update_success`

Severity

Informational

Problem description

It indicates that successfully created or updated secret in Vault.

User actions

No action is required.

BMYIN4009

`vault_delete_files_success`

Severity

Informational

Problem description

It indicates that successfully deleted vault related files.

User actions

No action is required.

BMYIN4010

`vault_fetch_info_success`

Severity

Informational

Problem description

It indicates that successfully fetched vault information from the Vault.

User actions

No action is required.

BMYIN4011

`vault_healthy`

Severity

Informational

Problem description

It indicates that Vault is healthy.

User actions

No action is required.

BMYIN4012

`vault_get_secret_success`

Severity

Informational

Problem description

It indicates that successfully retrieved secret from Vault.

User actions

No action is required.

BMYIN4013

`vault_seal_status_check_success`

Severity

Informational

Problem description

It indicates that successfully checked Vault seal status.

User actions

No action is required.

BMYIN4014

`vault_userpass_login_success`

Severity

Informational

Problem description

It indicates that successfully logged in to Vault using `userpass` authentication.

User actions

No action is required.

BMYIN4015

`vault_unseal_success`

Severity

Informational

Problem description

It indicates that successfully unsealed Vault.

User actions

No action is required.

BMYIN4016

`vault_get_client_failed`

Severity

Critical

Problem description

It indicates that failed to obtain Vault client.

User actions

Contact IBM support.

BMYIN4017

`vault_create_secret_failed`

Severity

Critical

Problem description

It indicates that failed to create or update secret in Vault.

User actions

Contact IBM support.

BMYIN4018

`vault_unseal_status_check_failed`

Severity

Critical

Problem description

It indicates that failed to unseal vault.

User actions

Contact IBM support.

BMYIN4019

`vault_userpass_login_failed`

Severity

Critical

Problem description

It indicates that failed to login to Vault using **`userpass`** authentication.

User actions

Contact IBM support.

BMYIN4020

`vault_delete_vault_files_failed`

Severity

Critical

Problem description

It indicates that failed to delete vault-related files.

User actions

Contact IBM support.

BMYIN4021

`vault_fetch_info_failed`

Severity

Critical

Problem description

It indicates that failed to fetch vault information from the Vault.

User actions

Contact IBM support.

BMYIN4022

`vault_in_configure_success`

Severity

Informational

Problem description

It indicates that successfully installed and configured Vault.

User actions

No action is required.

BMYIN4023

`vault_in_configure_failed`

Severity

Critical

Problem description

It indicates that failed to install and configure Vault.

User actions

Contact IBM support.

BMYIN4024

`vault_info_on_ocp_success`

Severity

Informational

Problem description

It indicates that created Vault secret on OCP.

User actions

No action is required.

BMYIN4025

`vault_info_on_ocp_failure`

Severity

Critical

Problem description

It indicates that failed to create Vault secret on OCP.

User actions

Contact IBM support.

BMYIN4026

`vault_populate_success`

Severity

Informational

Problem description

It indicates that successfully populated Vault with data.

User actions

No action is required.

BMYIN4027

`vault_populate_failure`

Severity

Critical

Problem description

It indicates that failed to populate Vault with data.

User actions

Contact IBM support.

BMYIN4028

`vault_fetch_secret_fail`

Severity

Critical

Problem description

It indicates that failed to fetch secrets' information from the vault.

User actions

Contact IBM support.

BMYIN4029

`ServiceNodeKeyRotationSuccess`

Severity

Informational

Problem description

It indicates that the service node SSH key rotation completed successfully.

User actions

No action is required.

BMYIN4030

`ServiceNodeKeyRotationFailed`

Severity

Critical

Problem description

It indicates that the service node SSH key rotation failed after maximum retry attempts.

User actions

1. Collect the service node logs for further debugging.
2. If there are issues with service node connectivity, contact IBM support.

BMYIN4101

`UnsupportedPlatform`

Severity

Informational

Problem description

It indicates that the service is not supported on this platform.

User actions

No action is required.

BMYIN4102

`RelatedServiceInstalled`

Severity

Informational

Problem description

It indicates that the related service is already installed.

User actions

No action is required.

BMYIN4103

`RelatedStorageServiceInstalled`

Severity

Informational

Problem description

It indicates that the related storage service is already installed on this setup.

User actions

No action is required.

BMYIN4104

`StorageClassPrereqNotFound`

Severity

Informational

Problem description

It indicates that the `StorageClass` is not present to install this service.

User actions

Check the availability of the storage class specified in the FSD/FSI.

BMYIN4105

`OCPVersionPrereqNotFound`

Severity

Informational

Problem description

It indicates that the OpenShift® Container Platform version is not in line with the service to be installed.

User actions

Upgrade the current OpenShift Container Platform version if services are supported on higher version than the current version.

BMYIN4106

`CRDMissingPrereqNotFound`

Severity

Informational

Problem description

It indicates that Custom Resource Definition (CRD) is not present on the setup to install the service.

User actions

Check with IBM Support for CRD.

BMYIN4107

`DataJSONKeyNotFoundInCM`

Severity

Informational

Problem description

It indicates that the ConfigMap information is invalid to install the service.

User actions

Validate whether the ConfigMap is present on the setup and its values are valid, which are specified in the FSD.

BMYIN4108

`SkipServiceInSpectrumFusionCRTrue`

Severity

Informational

Problem description

It indicates that the service installation is skipped in the **`Spectrum Fusion`** CR.

User actions

If you want to install the service, you can remove the **`skipservice`** list values from the **`Spectrum Fusion`** CR.

BMYIN4110

`PrereqServiceNotInstalled`

Severity

Informational

Problem description

It indicates that a dependent service is not installed on this setup.

User actions

Install the dependent service first. After the dependent service is installed successfully, the **`prereq`** service can be deployed.

BMYPC5001

Cluster operator is available but is in progress state.

Severity

Warning

Problem description

It indicates that the specified OpenShift®cluster operator is available but not all replicas are available. Do not start any upgrade until all cluster operators are available and are not in progress or degraded state.

User actions

Do the following steps to resolve the issue:

1. Check whether a machine configuration rollout is in progress, as it might be the reason for an unhealthy cluster operator. In such a case, wait for the machine configuration rollout to complete and the cluster operator to turn healthy.
2. After you resolve the health and status of the cluster operator, the upgrade precheck detects the fix and resumes the upgrade operation automatically.

BMYPC5002

A **`MachineConfigPool`** is updating.

Severity

Error or warning

Problem description

It indicates that a **`MachineConfigPool`** update is in progress. It impacts the upgrade as the next machine configuration rollout in the queue fails because of its dependency on the successful rollout of the ongoing change.

For example, an ongoing rollout is for a registry auth change for all the nodes. This change must get rolled out to all nodes before you start the upgrade so that no pod failures due to image pullback errors cause upgrade failures.

User actions

Do the following steps to resolve the issue:

1. The issue might occur for a short duration whenever the Machine config rollout, OpenShift® Container Platform, and Global Data Platform operations are in progress. Wait for these operations to complete, then check that the **`MachineConfigPool`** is healthy and available.
2. If a pool remains in a degraded state for a long time, then use the following command to find the root cause.

   ```
   oc describe machineconfigpool <poolname>
   ```

3. Check for the problems indicated in the description, and debug further to identify the root cause. The following are some possible reasons:
   - If a node is stuck in a **`NotReady`** state, restart the node to fix the issue.
   - Though the node completes the roll out, it might not be in a schedulable state. Compare the required and current configurations. If the configurations are the same, manually remove taint of the unscheduleable state.

After the **`MachineConfigPool`** is available and up to date, the upgrade precheck detects the fix and resumes upgrade automatically.

BMYPC5004

A pod PodDisruptionBudget (PDB) is set to zero.

Severity

Warning

Problem description

It indicates that the PodDisruptionBudget (PDB) of a pod is zero. During the upgrade, each node restarts in a rolling fashion. Any pod with a PDB set to zero blocks the node restarts because of the incomplete node drain operation. It might cause the upgrade to get stuck until you delete the pod manually. The upgrade operation restarts after the pod gets deleted and the node drains completely.

User actions

Do the following steps to resolve the issue:

1. To avoid this issue, delete the pod and set the PDB to zero.
2. If you proceed with the upgrade without deleting the pod, the upgrade gets stuck during the node restart where the pod is running. Delete the pod whenever the IBM Fusion user interface shows the upgrade as blocked.
3. Select a method based on how disruptive the pod deletion is for its associated workloads. If you can tolerate a pod in unscheduled state during the upgrade, then pod deletion before the upgrade causes it to run unblocked.
4. If you require active replicas of the pod throughout the upgrade, then you must manually manage the upgrade.
5. Every time the upgrade gets blocked because a nod drain is in progress, delete the pod that causes the issue and allow it to reschedule on another node.

The upgrade restarts after the pod deletion and node drain operations are complete.

BMYPC5005

Virtual machine is using PVC with non-RWX access mode.

Severity

Warning

Problem description

It indicates that a virtual machine is using PVCs with RWO access mode. It blocks the upgrade by preventing nodes from restarting that is necessary for machine configuration rollout and firmware upgrade.

User actions

Do the following steps to resolve the issue:

1. Manually stop the virtual machine until the upgrade is complete.
2. Manually stop the virtual machine before you start the upgrade or wait for the node on the virtual machine to drain completely.
3. If you choose to wait, then the upgrade can be paused while draining the node, and the upgrade should not proceed until you stop the virtual machine.
4. It is always recommended to migrate all virtual machines to use RWX PVC by cloning the existing RWO PVC. It prevents issues with upgrades and also enables virtual machines to take advantage of live migration.

Continue with the upgrade operation after you migrate all virtual machines to use RWX PVC by cloning the existing RWO PVC.

BMYPC5006

A pod in the **`openshift-operator-lifecycle-manager`** namespace is unhealthy.

Severity

Error

Problem description

It indicates that pod in the **`openshift-operator-lifecycle-manager`** namespace is not ready. Pods in the **`openshift-operator-lifecycle-manager`** namespace are responsible for managing catalogs, packages, and operator lifecycle in the OpenShift®. If pods are not running or have errors, then it can cause operator upgrade failure in the IBM Fusion HCI.

User actions

Do the following steps to resolve the issue:

1. Check for the events of the failed pods and logs to find the root cause of failure and take an appropriate action to fix the issues. Restarting the pod might also solve the problem.
2. After all the pods in the **`openshift-operator-lifecycle-manager`** namespace are ready, the upgrade precheck detects the fix and resumes upgrade automatically.

BMYPC5010

IBM Fusion operator installation is in progress.

Severity

Error

Problem description

The IBM Fusion operator is in `installready` or `installing` or `pending` state, which indicates that the operator install is in progress. It can block the ability of IBM Fusion operator or other IBM Fusion components ability to install, upgrade, or function.

User actions

Do the following steps to resolve the issue:

1. Check for IBM Fusion operator CSV events and conditions to find the root cause for this state. Some of the common reasons for the `Pending` state of the IBM Fusion operator are as follows:
    - Check whether IBM Fusion operator is installing or upgrading.
    - Some changes might corrected in any deployment spec in CSV for IBM Fusion operator. It recreates pods and reconciles to succeeded state in some time.
    - One of the IBM Fusion operator pod is not running. It might be due to an OOM killed error where the pod is requesting more memory than its set limit. To resolve this, update the memory limit for that deployment in CSV.
    - A pod might be in a `crashloopbackoff` due to another reason. Check the pod log and events to find the root cause and take an appropriate action.
    - It is possible that a `Custom Resource Definition` is missing due to an accidental deletion. In such case, contact IBM support to get the CRD YAML and create it in the cluster.

    You can continue with the upgrade operation after the IBM Fusion operator is successful.

BMYPC5013

Unable to check DNS resolution for the registry.

Severity

Warning

Problem description

Unable to determine whether the DNS resolution for the image registry is successful or not. It might block the upgrade due to image pull failures.

User actions

Do the following steps to resolve the issue:

1. Check whether the image registry is used to host images for either IBM Fusion or any of its data service services or firmwares.
2. If the image registry hosts any of images, then manually verify that each cluster node can resolve registry hostname to its IP since this registry will be used to pull images during upgrade and upgrade fails if registry hostname can not be resolved.
3. If any node can not resolve hostname then fix that issue.

BMYPC5027

The OpenShift® Container Platform is currently operating at the latest version that is supported by IBM Fusion.

Severity

Warning

User actions

Do not upgrade OpenShift Container Platform to the next minor version as it is not supported by the IBM Fusion.

BMYPC5028

A virtual machine is configured with PCI pass-through.

Severity

Warning

Problem description

It indicates that a virtual machine is configured with PCI pass-through and it blocks OpenShift® node draining and upgrades of Global Data Platform service, OpenShift and compute nodes.

User actions

Stop the virtual machine mentioned in the precheck so that it does not block draining of nodes.

If the issue still persists, then go to virtual machines YAML definition, and under the **spec** field, add **evictionStrategy: LiveMigrateIfPossible**. Save the YAML file and restart the virtual machine so that the draining of the nodes cannot be blocked once the virtual machine is up.

BMYPC5029

The OpenShift® Container Platform is currently operating at the latest version that is supported by Fusion Data Foundation.

Severity

Warning

User actions

Do not upgrade OpenShift Container Platform to the next minor version as it is not supported by the Fusion Data Foundation service.

BMYPC6001

Cluster operator is not healthy.

Severity

Error

Problem description

It indicates that the specified OpenShift® cluster operator is not available or degraded. It can impact the upgrade because different cluster operators are responsible for the various functions in the OpenShift cluster. If any function is not working due to an unavailable cluster operator, it might result in an upgrade failure.

User actions

Do the following steps to resolve the issue:

1. Check whether a machine configuration rollout is in progress as it can be a reason for the health status of the cluster operator. In this case, wait for machine configuration rollout to complete and cluster operator to turn healthy.
2. Check whether a node is in a **NotReady** state as it be a reason for the health status of the cluster operator. Debug the reason for **NotReady** state and check that it is a part of the machine configuration rollout progress. If it is a part of the machine configuration rollout, then wait for the roll out to complete.
3. If a node remains in a **NotReady** state for more than 30 minutes, then check the network for the node by pinging the IPv4 address or hostname. In case the node does not respond, then restart the node.
4. If node is in a **Ready** state but scheduling is disabled, then it might be due to a machine configuration rollout, node in a maintenance mode, or ongoing firmware upgrade. Wait for the operation to complete for the node to become scheduleable.
   After the cluster operator is available and not degraded, the upgrade precheck detects the fix and resumes upgrade automatically.

If you want to override the prechecks that are blocking the upgrade, do the following steps:

It converts the **Error** prechecks to **Warning** prechecks and triggers the upgrade.
Note: Before using the **precheck-acks** configmap to override the pre-checks, ensure that the unhealthy resource does not affect the upgrade.

1. Run the following command and export IBM Fusion namespace as an environmental variable.

   ```
   export FUSION_NS="namespace-where-fusion-is-installed"
   ```

2. Run the following command to create a **precheck-acks** configmap in the namespace where the IBM Fusion is installed.

   ```
   oc create configmap precheck-acks -n $FUSION_NS
   ```

3. In the **precheck-acks** config map, add the key **ClusterOperatorHealth** and set the value to **true**.

   ```
   oc patch configmap precheck-acks -n $FUSION_NS --type merge -p '{"data": {"ClusterOperatorHealth": "true"}}'
   ```

BMYPC6002

A **MachineConfigPool** is degraded.

Severity

Error

Problem description

It indicates that a `MachineConfigPool` is in a degraded state in the OpenShift® cluster. It can impact the upgrade because the different IBM Fusion component upgrades also start machine configuration roll outs and they might not succeed if `MachineConfigPool` is in a degraded state. A new machine configuration must be rolled out when all `MachineConfigPool` are healthy.

User actions

Do the following steps to resolve the issue:

1. The issue might occur for a short duration whenever the Machine config rollout, OpenShift Container Platform, and Global Data Platform operations are in progress. Wait for these operations to complete, then check that the `MachineConfigPool` is healthy and available.
2. If a `MachineConfigPool` remains in a degraded state for a long time, then use following command to find the root cause.

   ```
   oc describe machineconfigpool <poolname>
   ```

3. Check for the problems indicated in the description, and debug further to identify the root cause. The following are some possible reasons:
   - A node is stuck in a `NotReady` state, then restart the node to fix the issue.
   - Though the node completes the roll out, it might not be in a schedulable state. Compare the required and current configurations. If the configurations are the same, manually remove taint of the unscheduleable.
4. After the `MachineConfigPool` is available and up to date, the upgrade precheck detects the fix and resumes upgrade automatically.

BMYPC6003

A node is not ready or unscheduleable.

Severity

Error

Problem description

It indicates that a node is not ready or unschedulable. Ensure that all nodes are available before you proceed with the upgrade.

User actions

Do the following steps to resolve the issue:

1. Check whether the node is placed in a maintenance mode for a planned work. In this case, bring the node out of the maintenance mode after the planned work is complete.
2. If a machine configuration roll out is in progress, then wait for the roll out to complete and the node to return to a ready state.
3. If the node is in a `Notready` state for more than 30 minutes, then it might be an error with the node in the joining cluster. Do the following steps to resolve issues that are related to the joining cluster:
   a. Run the following command to check for the pending CSRs.

      ```
      oc get csr|grep -i pending
      ```

   b. If you find any pending CSRs, then run the following command to approve them.

      ```
      oc get csr -o go-template='{{range .items}}{{if not .status}}{{.metadata.name}}{{"\n"}}{{end}}{{end}}' | xargs oc adm
      certificate approve
      ```

   c. If you do not find any pending CSRs, then restart the node.
   d. Connect to the node by using the core OS private SSH key and run the following command.

      ```
      ip a
      ```

   e. Recover the node during the following scenarios:
      - IPv4 address assigned to the bond0 interface
      - `br-ex` interface not created

   You can continue with the upgrade operation after the node is in a `Ready` state.

BMYPC6006

A pod in the `openshift-operator-lifecycle-manager` namespace is not ready.

Severity

Error

Problem description

It indicates that pod in the `openshift-operator-lifecycle-manager` namespace is not ready. Pods in the `openshift-operator-lifecycle-manager` namespace are responsible for managing catalogs, packages, and operator lifecycle in the OpenShift®. If pods are not running or have errors, then it can cause operator upgrade failure in the IBM Fusion HCI.

User actions

Do the following steps to resolve the issue:

1. Check for the events of the failed pods and logs to find the root cause of failure and take an appropriate action to fix the issue. Pod restart can also resolve the problem.
2. After all the pods in the `openshift-operator-lifecycle-manager` namespace are ready, the upgrade precheck detects the fix and resumes upgrade automatically.

If you want to override the prechecks that are blocking the upgrade, do the following steps:

It converts the `Error` prechecks to `Warning` prechecks and triggers the upgrade.
Note: Before using the `precheck-acks` configmap to override the pre-checks, ensure that the unhealthy resource does not affect the upgrade.

1. Run the following command and export IBM Fusion namespace as an environmental variable.

   ```
   export FUSION_NS="namespace-where-fusion-is-installed"
   ```

2. Run the following command to create a `precheck-acks` configmap in the namespace where the IBM Fusion is installed.

   ```
   oc create configmap precheck-acks -n $FUSION_NS
   ```

3. In the `precheck-acks` config map, add the key `OpenShiftLifeCycleManagementPodStatus` and set the value to `true`.

   ```
   oc patch configmap precheck-acks -n $FUSION_NS --type merge -p '{"data": {"OpenShiftLifeCycleManagementPodStatus": 
   "true"}}'
   ```

BMYPC6007

A `Catalogsource` is not ready.

Severity

Error

Problem description

It indicates that a `Catalogsource` in the OpenShift® is either not ready or not healthy. It is required for any operator lifecycle management operation, such as installation or upgrade to function. All catalog sources in the OpenShift cluster must be healthy and ready. Any unhealthy catalog source impacts the OLM functions in the cluster.

User actions

Do the following steps to resolve the issue:

- Some of the possible reasons with diagnostic steps for a catalog source to not work are as follows:
  1. Invalid image name in the catalog source:
     - Check the pod for the given catalog source in its dedicated namespace or `openshift-marketplace` namespace.
     - Run the following command to describe the pod.

       ```
       oc describe pod <pod name> -n <namespace>
       ```

     - If you find an error with the message `invalid image reference` under events section, then fix the image field in the catalog source YAML. Wait for the catalog source to turn healthy.
  2. Missing images in the registry for disconnected setup:
     - Check the pod for the given catalog source in its dedicated namespace or `openshift-marketplace` namespace.
     - Run the following command to describe the pod.

       ```
       oc describe pod <pod name> -n <namespace>
       ```

     - If you find an error with the message `unknown manifest` under events section, mirror images to enterprise registry and wait for catalog source to turn healthy.
  3. Missing `ImageContentSourcePolicy` for disconnected set up:
     - Check the pod for the given catalog source in its dedicated namespace or `openshift-marketplace` namespace.
     - Run the following command to describe the pod.

       ```
       oc describe pod <pod name> -n <namespace>
       ```

     - If you find an error with the messages `unknown manifest` and `Mirrors failed` under events section, then review the `imagecontentsourcepolicy` applied for any incorrect paths or ports or spelling mistakes. Fix them and wait for catalog source to turn healthy.
  4. Missing credentials for the registry in the global pull-secret of the `openshift-config` namespace. Check the credentials in the global pull-secret in the `openshift-config` namespace:
     - Check the pod for the given catalog source in its dedicated namespace or `openshift-marketplace` namespace.
     - Run the following command to describe the pod.

       ```
       oc describe pod <pod name> -n <namespace>
       ```

- If you find an error with the messages `Unknown desc = unable to retrieve auth token: invalid` and `username/password: unknown: Authentication is required` under events section, then review the `pull-secret` in `openshift-config` namespace for correct credentials for registry where the image is referenced. Fix auth for registry in secret and wait for catalog source to turn healthy.

5. Connection error to the registry. Ensure that the firewall and proxy do not block registry and port access.
    - Check the pod for the given catalog source in its dedicated namespace or `openshift-marketplace` namespace.
    - Run the following command to describe the pod.

    ```
    oc describe pod <pod name> -n <namespace>
    ```

    - If you find an error with the messages `pinging container registry registry.redhat.io: Get` and `"https://<registryhost>": dial tcp <registryip>: connect: connection refused` under events section, then review the firewall and proxy settings to make sure that registry access is allowed to the cluster and wait for catalog source to turn healthy.

After the `CatalogSource` is in a ready state, the upgrade precheck detects the fix and resumes the upgrade operation automatically.

If you want to override the prechecks that are blocking the upgrade, do the following steps:

It converts the `Error` prechecks to `Warning` prechecks and triggers the upgrade.
Note: Before using the `precheck-acks` configmap to override the pre-checks, ensure that the unhealthy resource does not affect the upgrade.

1. Run the following command and export IBM Fusion namespace as an environmental variable.

    ```
    export FUSION_NS="namespace-where-fusion-is-installed"
    ```

2. Run the following command to create a `precheck-acks` configmap in the namespace where the IBM Fusion is installed.

    ```
    oc create configmap precheck-acks -n $FUSION_NS
    ```

3. In the `precheck-acks` config map, add the key `CatalogSourceStatus` and set the value to `true`.

    ```
    oc patch configmap precheck-acks -n $FUSION_NS --type merge -p '{"data": {"CatalogSourceStatus": "true"}}'
    ```

BMYPC6008

Inaccessible registry.

Severity

Warning

Problem description

It indicates that the specified image registry is not accessible from one or more cluster nodes. It blocks the upgrade because image pulls might fail. It also causes other issues outside of upgrades as newly scheduled or rescheduled pods are unable to access their images.

For the registries that have non-empty auth specified in the pull-secret in the `openshift-config` namespace, check for the connectivity to make sure that the images are pulled successfully for IBM Fusion components and the OpenShift® Container Platform. If you fail to address the registry access might result in pods can get into an `ImagePullbackOff` error.

User actions

Do the following steps to resolve the issue:

1. Check the network connectivity between the OpenShift cluster and the image registry.
2. A firewall can prevent the access to the registry.
3. After the connectivity issue is resolved, the upgrade precheck detects the fix and resumes upgrade automatically.

BMYPC6010

IBM Fusion operator is in a `Failed` state or not found.

Severity

Error

Problem description

It indicates that the IBM Fusion operator is in a `Failed` state or not found. It can block the ability of the IBM Fusion operator or other IBM Fusion components ability to install, upgrade, or do other function.

User actions

Do the following steps to resolve the issue:

1. Check for IBM Fusion operator CSV events and conditions to find the root cause for this state. Some of the common reasons for the **Failed** state of the IBM Fusion operator are as follows:
   - One of the IBM Fusion operator pod is not running. It might be due to an OOM killed error where the pod is requesting more memory than its set limit. To resolve this, update the memory limit for that deployment in CSV.
   - A pod might be in a `crashloopbackoff` due to another reason. Check the pod log and events to find the root cause and take an appropriate action.
   - It is possible that a `Custom Resource Definition` is missing due to an accidental deletion. In this case, contact IBM support to get the CRD YAML and create it in the cluster.

You can continue with the upgrade operation after the IBM Fusion operator is successful.

If you want to override the prechecks that are blocking the upgrade, do the following steps:

It converts the `Error` prechecks to `Warning` prechecks and triggers the upgrade.
Note: Before using the `precheck-acks` configmap to override the pre-checks, ensure that the unhealthy resource does not affect the upgrade.

1. Run the following command and export IBM Fusion namespace as an environmental variable.

   ```
   export FUSION_NS="namespace-where-fusion-is-installed"
   ```

2. Run the following command to create a `precheck-acks` configmap in the namespace where the IBM Fusion is installed.

   ```
   oc create configmap precheck-acks -n $FUSION_NS
   ```

3. In the `precheck-acks` config map, add the key `FusionOperatorStatus` and set the value to `true`.

   ```
   oc patch configmap precheck-acks -n $FUSION_NS --type merge -p '{"data": {"FusionOperatorStatus": "true"}}'
   ```

BMYPC6013

The DNS resolution for image registry failed.

Severity

Error

Problem description

It indicates that the hostname resolution specified for the image registry failed for one or more cluster nodes. It blocks the upgrade because of image pull failure. It can also cause other issues outside of the upgrade because the newly scheduled or rescheduled pods cannot access the images.

The registries can have a non-empty auth specified in the pull-secret of the `openshift-config` namespace. Check the connectivity for successful image pulls of IBM Fusion components and the OpenShift® Container Platform. Failure to address the registry access can result in an `ImagePullbackOff` error of the pod.

User actions

Do the following steps to resolve the issue:

1. Check the DNS settings for the image registry.
2. A firewall can prevent the access to the registry
3. After the DNS resolution issue is resolved, the upgrade precheck detects the fix and resumes upgrade automatically.

If you want to override the prechecks that are blocking the upgrade, do the following steps:

It converts the `Error` prechecks to `Warning` prechecks and triggers the upgrade.
Note: Before using the `precheck-acks` configmap to override the pre-checks, ensure that the unhealthy resource does not affect the upgrade.

1. Run the following command and export IBM Fusion namespace as an environmental variable.

   ```
   export FUSION_NS="namespace-where-fusion-is-installed"
   ```

2. Run the following command to create a `precheck-acks` configmap in the namespace where the IBM Fusion is installed.

   ```
   oc create configmap precheck-acks -n $FUSION_NS
   ```

3. In the `precheck-acks` configmap, add the key `DNSResolution` and set the value to `true`.

   ```
   oc patch configmap precheck-acks -n $FUSION_NS --type merge -p '{"data": {"DNSResolution": "true"}}'
   ```

BMYPC6014

Global Data Platform service upgrade is in progress.

Severity

Error

Problem description

It indicates that the Global Data Platform upgrade is in progress. The upgrades for compute nodes firmware, switch firmware and OpenShift® should not be triggered when the Global Data Platform service upgrade is in progress as the nodes get rebooted in a rolling pattern during the Global Data Platform upgrade.

User actions

Wait until the Global Data Platform service upgrade to be complete before triggering the upgrades for compute nodes, switches, or OpenShift.

BMYPC6015

It is a switch upgrade precheck error. STP is enabled on the high-speed switches.

Severity

Error

Problem description

It indicates that the STP is enabled on the high-speed switches. For the switch upgrade, the STP needs to be disabled.

User actions

Collect Network switches logs and contact IBM support.

BMYPC6023

`Pidslimit` on the nodes is less than 12228.

Severity

Error

Problem description

The node `Pidslimits` is set to a value less than 12228, which is the minimum required value for the Global Data Platform to run on IBM Fusion HCI. It results in a non-functional storage cluster that impacts the applications that require storage.

User actions

Do the following steps to resolve the issue:

1. Follow steps 1 to 4 to apply `kubeletconfig`. It ensures the `pidslimits` are as expected on nodes.
2. After the `Pidslimit` is set correctly, the upgrade precheck detects the fix and resumes upgrade automatically.

BMYPC6040

Backup & Restore agent instance not found.

Severity

Error

Problem description

It indicates that there is no Backup & Restore agent instance on the cluster. This can occur if the `DataProtectionAgent` CR is deleted from the cluster.

User actions

Contact IBM support to resolve this issue.

BMYPC6042

The Subscription is missing the `CatalogSource`.

Severity

Error

Problem description

It indicates that the listed subscription is missing in the `CatalogSource`. This can occur if the `CatalogSource` is missing from the subscription CR, or if the `CatalogSource` listed in the subscription CR is missing on the cluster.

User actions

Do the following steps to resolve the issue:

1. Check the `CatalogSource` required by the subscription exists on the cluster and create the `CatalogSource` if it does not exist
2. If you want to override the prechecks that are blocking the upgrade, do the following steps:
   Note: Before using the `precheck-acks configmap` to override the prechecks, ensure that the unhealthy resource does not affect the upgrade.
   a. Run the following command and export IBM Fusion namespace as an environmental variable.

      ```
      export FUSION_NS="namespace-where-fusion-is-installed
      ```

   b. Run the following command to create a `precheck-acks configmap` in the namespace where the IBM Fusion is installed.

      ```
      oc create configmap precheck-acks -n $FUSION_NS
      ```

   c. In the `precheck-acks configmap`, add the key `BNRSubscriptionCatalogs` and set the value to true.

      ```
      oc patch configmap precheck-acks -n $FUSION_NS --type merge -p '{"data": {"BNRSubscriptionCatalogs": "true"}}'
      ```

BMYPC6044

Backup & Restore server instance is unhealthy.

Severity

Error

Problem description

It indicates that the Backup & Restore server instance is unhealthy. This can occur if one or more services listed the `DataProtectionServer` CR contain issues.

Recommended actions

Contact IBM support.

BMYPC6045

Backup & Restore agent instance is unhealthy.

Severity

Error

Problem description

It indicates that the Backup & Restore agent instance is unhealthy. This can occur if one or more services listed the `DataProtectionServer` CR contain issues.

Recommended actions

Contact IBM support.

BMYPC6076

The DNS resolution for the node failed.

Severity

Error

Problem description

It indicates that the hostname resolution failed for the specified node.

User actions

Follow the steps described in the Setting up DNS section to verify the DNS configuration of the affected node.

After you correct the DNS configuration, the node addition process proceeds successfully.

Configuration error

Message: Encountered a configuration error. Red Hat® OpenShift® Container Platform installation did not start successfully.

Diagnosis

1. As a kni user on a provisioner node (also known as service node), review the log file **/home/kni/logs/install_<timestamp>.log**.
2. Scroll to end of file and check whether it reports any failure. Following is an example of a successful run, however, if you find value for **failed** not equal to **0**, then check the log for a step that failed.

```
PLAY RECAP *********************************************************************
localhost                  : ok=68   changed=3    unreachable=0    failed=0    skipped=45   rescued=0    ignored=0
```

Next actions

1. Take appropriate action to resolve the reported problem in the log.
2. In the installation user interface, click **Retry** to restart the installation.

Timed out creating bootstrap VM

Message: Bootstrap VM creation on provisioner node (also known as service node) has not started.

Diagnosis

1. As a kni user, on the provisioner node, review the log file **/home/kni/clusterconfigs/.openshift_install.log**.
2. Rectify the reported problem.

Next actions

In the installation user interface, click **Retry** to restart the installation.

If problem persists, collect all files in **/home/kni/logs** and **/home/kni/clusterconfigs**, and contact IBM support.

Bootstrap VM creation did not complete

Message: Bootstrap VM creation on the provisioner node (also known as service node) did not complete within the expected time.

Diagnosis

1. As a kni user, on the provisioner node, review the log file **/home/kni/clusterconfigs/.openshift_install.log**.
2. Rectify the reported problem.

Next actions

In the installation user interface, click Retry to restart the installation.

If problem persists, do the following steps:

1. Collect all files from **/home/kni/logs**, **/home/kni/clusterconfigs**, and **/home/kni/install/results.json**.
2. Contact IBM support.

Failed control nodes inspection

Message: Inspection failed for one or more control nodes.

Diagnosis

Note: Run all commands as a `kni` user from provisioner node.

1. Check the `.openshift_install.log` for errors.
2. If the logs do not provide the necessary information, then continue the steps in this procedure for further diagnostics.
3. Ping node by using the IP and hostname that are reserved for the node in DHCP.
4. If ping does not work, the do the following steps to access the IMM user interface of the Baremetal node:
   Find IPv6 or IPv4 address of node and debug by using the following command:

```
oc get bmh -A -o wide |grep control
openshift-machine-api   control-1-ru2   OK       externally provisioned   isf-rackae4-2bztv-master-0
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:2f95]:623   unknown              true              3h28m
openshift-machine-api   control-1-ru3   OK       externally provisioned   isf-rackae4-2bztv-master-1
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefd:cecd]:623   unknown              true              3h28m
openshift-machine-api   control-1-ru4   OK       externally provisioned   isf-rackae4-2bztv-master-2
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:3031]:623   unknown              true              3h28m
```

5. Note down the IPv6 or IPv4 address of the node you are checking.
6. Go to `/home/kni/isfconfig` folder and open `kickstart-.json` file to find the password of the IMM user `USERID` of node.
7. Open the file and look for the string `OCPRole` with value equal to hostname (for example, control-1-ru3) of the node.
   Sample section for IPv6:

```
  "ipv6ULA": "XXXXXXXXXXXXXXXXXXXXXX",
       "ipv6LLA": "XXXXXXXXXXXXX",
       "serialNum": "J1025PXX",
       "mtm": "7D2XCTO1WW",
       "ibmSerialNumber": "rackae402",
       "ibmMTM": "9155-C01",
       "type": "storage",
       "OCPRole": "control-1-ru3",
       "location": "RU2",
       "name": "IMM_RU2",
       "bootDevice": "/dev/sda",
       "users": [
          {
             "user": "CEUSER",
             "password": "XXXXX",
             "group": "Administrator",
             "number": 2
          },
          {
             "user": "ISFUSER",
             "password": "XXXXXX",
             "group": "Administrator",
             "number": 3
          },
          {
             "user": "USERID",
             "password": "XXXXXX",
             "group": "Administrator",
             "number": 1
          }
```

   Sample section for IPv4:

```
{
       "OCPRole": "compute-1-ru7",
       "bootDevice": "/dev/disk/by-path/pci-0000:01:00.0-nvme-1",
       "ibmMTM": "9155-D10",
       "ibmSerialNumber": "delld07",
       "ipv4": "169.253.1.7",
       "ipv6LLA": "",
       "ipv6ULA": "",
       "location": "RU7",
       "manufacturer": "Dell",
       "mtm": "2668",
       "name": "IMM_RU7",
       "networkCards": [
        {
         "slot-1": [
          "30:3E:A7:10:80:2A",
          "30:3E:A7:10:80:2B",
          "30:3E:A7:10:80:2C",
          "30:3E:A7:10:80:2D"
         ],
         "slot-2": [
          "5C:25:73:C9:94:3E",
          "5C:25:73:C9:94:3F"
         ]
        }
       ],
       "networkInterfaces": [
        {
         "interfaceLeg1": "ens2f0np0",
         "interfaceLeg2": "ens2f1np1",
         "interfaceName": "bond0",
         "interfaceType": "baremetal",
         "macAddress": "5c:25:73:c9:94:3e",
         "secondaryMacAddress": "5C:25:73:C9:94:3F"
```

```
        },
        {
         "interfaceLeg1": "eno12409",
         "interfaceLeg2": "eno12419",
         "interfaceName": "bond1",
         "interfaceType": "provisioning",
         "macAddress": "30:3e:a7:10:80:2b",
         "secondaryMacAddress": "30:3E:A7:10:80:2C"
        }
       ],
       "nodeIndex": "6",
       "nvme": [
        {
         "capacity": "3576.983 GB",
         "location": "PCIe SSD in Slot 0 in Bay 1",
         "manufacture": "Micron Technology Inc",
         "productName": "Dell DC NVMe ISE 7450 RI U.2 3.84TB",
         "serialNumber": "2415487E9851"
        },
        {
         "capacity": "3576.983 GB",
         "location": "PCIe SSD in Slot 1 in Bay 1",
         "manufacture": "Micron Technology Inc",
         "productName": "Dell DC NVMe ISE 7450 RI U.2 3.84TB",
         "serialNumber": "2415487EDAAC"
        }
       ],
       "pxeSlotLabel": "NIC.Integrated.1-2-1",
       "secretName": "compute-1-ru7-secret",
       "serialNum": "9Y47NW3",
       "type": "storage",
       "unusedInterfaces": [
        "ens4f0np0",
        "ens4f1np1",
        "ens4f2np2",
        "ens4f3np3"
       ],
       "uuid": "uuid"
      }
     ],
```


8. Run the following command to create a tunnel to the IMM of the node:

```
ssh -N -f -L :1443:[IPv6 or IPv4 address of IMM]:443 -L :3900:[IPv6 or IPv4 address of IMM]:3900 root@localhost
```

9. To access the IMM user interface using the provisioner node IP address, open your browser and access the following IP.

```
https://<provisioner ip>:1443
```

   - user - Enter `USERID`.
   - password - Use the value obtained in the previous step.

10. From the IMM user interface page, open the remote console and check whether the node is up and shows a valid hostname prompt.

   - If the console shows localhost, then review your DHCP/DNS settings to ensure whether correct reservation is made for the node.
   - If the node shows a Red Hat Linux login prompt instead of Core OS, then it infers that the node Base management controller (BMC) is not responsive.

   Do the following steps to resolve the issue:

   a. Identify IMM of the missing node. Here is mapping of nodes to IMM:

   ```
   control-1-ru2 ==> imm_ru2
   control-1-ru3 ==> imm_ru3
   control-1-ru4 ==> imm_ru4
   ```

   b. For node(s) that did not show up in the previous steps, run the following sampe IMM command to connect to their IMMs: For example, if node `control-1-ru3` did not show up, then from RU7/provisioner (compute-1-ru7), run the `imm_ru3` command as a `kni` user.
   c. Wait for the successful connection to IMM. c. In the system prompt, run the `resetsp` on IMM prompt.
   d. Wait for 10 minutes.

Next actions

In the installation user interface, click the Retry to restart the installation.

Control nodes creation failed

Message: Failed to create the Red Hat® OpenShift® Container Platform cluster control nodes.

Diagnosis

Note: Run all commands as a `kni` user from provisioner node.

1. Check the `.openshift_install.log` for errors.
2. If the logs do not provide the necessary information, then continue next steps in this procedure for further diagnostics.
3. As a kni user, run `oc get nodes` on the provisioner (service node).
4. Check whether the following three ready nodes are not listed in the output:

```
NAME                                    STATUS   ROLES                         AGE   VERSION
control-1-ru2.rackm03.rtp.raleigh.ibm.com   Ready    control-plane,master,worker   55m   v1.27.3+4aaeaec
control-1-ru3.rackm03.rtp.raleigh.ibm.com   Ready    control-plane,master,worker   56m   v1.27.3+4aaeaec
control-1-ru4.rackm03.rtp.raleigh.ibm.com   Ready    control-plane,master,worker   56m   v1.27.3+4aaeaec
```

5. Identify the node that is not booted. If you do not see these three nodes, then note down the missing node(s) from the output.
6. Identify the IMM of the missing node. Here is mapping of nodes to IMM:

```
control-1-ru2 ==> imm_ru2
control-1-ru3 ==> imm_ru3
control-1-ru4 ==> imm_ru4
```

7. For node(s) that did not show up in the step 4 output, then run the corresponding IMM command to connect to its IMM. For example, if node `control-1-ru3` did not show up, then from RU7/provisioner (compute-1-ru7), run the `imm_ru3` command as a `kni` user.
8. Run `resetsp` command on the IMM prompt.
9. Wait for 10 minutes and retry the installation.
10. If issue persists, ping the control nodes by using the IP and hostname that are reserved for the node in DHCP.
11. If ping is not working, then do the following steps to access the IMM user interface of the Baremetal node:
    a. Run the following command to find the IPv6 or IPv4 address of the node to debug:

    ```
    oc get bmh -A -o wide |grep control
    ```

    Example output:

    ```
    openshift-machine-api   control-1-ru2   OK       externally provisioned   isf-rackae4-2bztv-master-0
    ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:2f95]:623   unknown             true              3h28m
    openshift-machine-api   control-1-ru3   OK       externally provisioned   isf-rackae4-2bztv-master-1
    ipmi://[fd8c:215d:178e:c0de:a94:efff:fefd:cecd]:623   unknown             true              3h28m
    openshift-machine-api   control-1-ru4   OK       externally provisioned   isf-rackae4-2bztv-master-2
    ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:3031]:623   unknown             true              3h28m
    ```

    b. Note down the IPv6 or IPv4 address of the node you are checking.
    c. Go to `/home/kni/isfconfig/` and open the `kickstart-.json`
    d. Find the password for the IMM user (USERID) of the node and search for the string OCPRole` with value equal to hostname (for example control-1-ru3) of node.
    Sample section for IPv6:

    ```
    ```
      "ipv6ULA": "XXXXXXXXXXXXXXXXXXXXXX",
           "ipv6LLA": "XXXXXXXXXXXXXX",
           "serialNum": "J1025PXX",
           "mtm": "7D2XCTO1WW",
           "ibmSerialNumber": "rackae402",
           "ibmMTM": "9155-C01",
           "type": "storage",
           "OCPRole": "control-1-ru3",
           "location": "RU2",
           "name": "IMM_RU2",
           "bootDevice": "/dev/sda",
           "users": [
              {
                 "user": "CEUSER",
                 "password": "XXXXX",
                 "group": "Administrator",
                 "number": 2
              },
              {
                 "user": "ISFUSER",
                 "password": "XXXXXX",
                 "group": "Administrator",
                 "number": 3
              },
              {
                 "user": "USERID",
                 "password": "XXXXXX",
                 "group": "Administrator",
                 "number": 1
              }

      ```
    ```

    Sample section for IPv4:

    ```
    {
           "OCPRole": "compute-1-ru7",
           "bootDevice": "/dev/disk/by-path/pci-0000:01:00.0-nvme-1",
           "ibmMTM": "9155-D10",
           "ibmSerialNumber": "delld07",
           "ipv4": "169.253.1.7",
           "ipv6LLA": "",
           "ipv6ULA": "",
           "location": "RU7",
           "manufacturer": "Dell",
           "mtm": "2668",
           "name": "IMM_RU7",
           "networkCards": [
            {
             "slot-1": [
              "30:3E:A7:10:80:2A",
              "30:3E:A7:10:80:2B",
              "30:3E:A7:10:80:2C",
              "30:3E:A7:10:80:2D"
             ],
    ```

```
        "slot-2": [
         "5C:25:73:C9:94:3E",
         "5C:25:73:C9:94:3F"
        ]
       }
      ],
      "networkInterfaces": [
       {
        "interfaceLeg1": "ens2f0np0",
        "interfaceLeg2": "ens2f1np1",
        "interfaceName": "bond0",
        "interfaceType": "baremetal",
        "macAddress": "5c:25:73:c9:94:3e",
        "secondaryMacAddress": "5C:25:73:C9:94:3F"
       },
       {
        "interfaceLeg1": "eno12409",
        "interfaceLeg2": "eno12419",
        "interfaceName": "bond1",
        "interfaceType": "provisioning",
        "macAddress": "30:3e:a7:10:80:2b",
        "secondaryMacAddress": "30:3E:A7:10:80:2C"
       }
      ],
      "nodeIndex": "6",
      "nvme": [
       {
        "capacity": "3576.983 GB",
        "location": "PCIe SSD in Slot 0 in Bay 1",
        "manufacture": "Micron Technology Inc",
        "productName": "Dell DC NVMe ISE 7450 RI U.2 3.84TB",
        "serialNumber": "2415487E9851"
       },
       {
        "capacity": "3576.983 GB",
        "location": "PCIe SSD in Slot 1 in Bay 1",
        "manufacture": "Micron Technology Inc",
        "productName": "Dell DC NVMe ISE 7450 RI U.2 3.84TB",
        "serialNumber": "2415487EDAAC"
       }
      ],
      "pxeSlotLabel": "NIC.Integrated.1-2-1",
      "secretName": "compute-1-ru7-secret",
      "serialNum": "9Y47NW3",
      "type": "storage",
      "unusedInterfaces": [
       "ens4f0np0",
       "ens4f1np1",
       "ens4f2np2",
       "ens4f3np3"
      ],
      "uuid": "uuid"
     }
    ],
```


e. Run the following command to create a tunnel to the IMM of the node:

~~~~
```
ssh -N -f -L :1443:[IPv6 or IPv4 address of IMM]:443 -L :3900:[IPv6 or IPv4 address of IMM]:3900 root@localhost
```
~~~~

f. To access the IMM user interface using the provisioner node IP address, open your browser and access the following IP.

~~~~
 ```
 https://<provisioner ip>:1443
 ```
 - user - Enter `USERID`.
 - password - Use the value obtained in the previous step.
~~~~

g. From the IMM user interface page, open the remote console and check whether the node is up and shows a valid hostname prompt.
   - If the console shows localhost, then review your DHCP/DNS settings to ensure whether correct reservation is made for the node.
   - If the node shows a Red Hat linux login prompt instead of Core OS, then it infers that the node Base management controller (BMC) is not responsive.

   Do the following steps to resolve the issue:
   i. Identify IMM of the missing node. Here is mapping of nodes to IMM:

   ~~~~
   ```
   control-1-ru2 ==> imm_ru2
   control-1-ru3 ==> imm_ru3
   control-1-ru4 ==> imm_ru4
   ```
   ~~~~

   ii. For node(s) that did not show up in the previous steps, run the following sample IMM command to connect to their IMMs: For example, if node `control-1-ru3` did not show up, then from RU7/provisioner (compute-1-ru7), run the following command as a `kni` user: `imm_ru3`
   iii. Wait for the successful connection to IMM.
   iv. In the system prompt, run the `resetsp` on IMM prompt.
   v. Wait for 10 minutes.

Next actions

In the installation user interface, click the Retry to restart the installation.

Cluster bootstrapping failed

Cluster bootstrapping fails when one or more control nodes do not boot during installation of Red Hat® OpenShift® Container Platform.

Message: Bootstrapping failed for Red Hat OpenShift Container Platform cluster.

Diagnosis

Note: Run all commands as a `kni` user from provisioner node.

1. Check the `.openshift_install.log` for errors.
2. If the logs do not provide the necessary information, then continue the steps in this procedure for further diagnostics.
3. As a kni user, run `oc get nodes` on the provisioner (service node).
4. Check whether the following three ready nodes are not listed in output:

    ```
    NAME                                      STATUS   ROLES                        AGE   VERSION
    control-1-ru2.rackm03.rtp.raleigh.ibm.com Ready    control-plane,master,worker  55m   v1.27.3+4aaeaec
    control-1-ru3.rackm03.rtp.raleigh.ibm.com Ready    control-plane,master,worker  56m   v1.27.3+4aaeaec
    control-1-ru4.rackm03.rtp.raleigh.ibm.com Ready    control-plane,master,worker  56m   v1.27.3+4aaeaec
    ```

5. Identify the node that is not booted. If you do not see the listed three nodes, then note down the missing node(s) from the output.
6. Identify the IMM of the missing node. Here is mapping of nodes to IMM:

    ```
    control-1-ru2 ==> imm_ru2
    control-1-ru3 ==> imm_ru3
    control-1-ru4 ==> imm_ru4
    ```

7. For node(s) that did not show up in output in step 4, run the corresponding IMM command to connect to its IMM. For example, if node `control-1-ru3` did not show up, then from the RU7/provisioner (compute-1-ru7), run the `imm_ru3` command as `kni` user. Wait for the successful connection to IMM.
8. In the system prompt, run the `resetsp` on IMM prompt.
9. Wait for 10 minutes and retry installation.
10. If none of the nodes show up, ping the control nodes by using the IP and hostname that are reserved for the node in DHCP.
11. If ping does not work, then do the following steps to access the IMM user interface of the Baremetal node:
    a. Find IPv6 or IPv4 address of node to debug using the following command:

    ```
    oc get bmh -A -o wide |grep control
    ```

    Example output:

    ```
    openshift-machine-api   control-1-ru2   OK       externally provisioned   isf-rackae4-2bztv-master-0
    ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:2f95]:623   unknown            true             3h28m
    openshift-machine-api   control-1-ru3   OK       externally provisioned   isf-rackae4-2bztv-master-1
    ipmi://[fd8c:215d:178e:c0de:a94:efff:fefd:cecd]:623   unknown            true             3h28m
    openshift-machine-api   control-1-ru4   OK       externally provisioned   isf-rackae4-2bztv-master-2
    ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:3031]:623   unknown            true             3h28m
    ```

    b. Note down the IPv6 or IPv4 address of the node you are checking.
    c. Go to `/home/kni/isfconfig` folder and open `kickstart-.json` file to find the password of the IMM user `USERID` of node
    d. Open the file and look for the string `OCPRole` with value equal to hostname (for example, control-1-ru3) of the node.
    Sample section for IPv6:

    ```
    "ipv6ULA": "XXXXXXXXXXXXXXXXXXXXXX", "ipv6LLA": "XXXXXXXXXXXXX", "serialNum": "J1025PXX", "mtm": "7D2XCTO1WW",
    "ibmSerialNumber": "rackae402", "ibmMTM": "9155-C01", "type": "storage", "OCPRole": "control-1-ru3", "location":
    "RU2", "name": "IMM_RU2", "bootDevice": "/dev/sda", "users": [ { "user": "CEUSER", "password": "XXXXX", "group":
    "Administrator", "number": 2 }, { "user": "ISFUSER", "password": "XXXXXX", "group": "Administrator", "number": 3 }, {
    "user": "USERID", "password": "XXXXXX", "group": "Administrator", "number": 1 }
    ```

    Sample section for IPv4:

    ```
    "ipv4: 165.253.1.7", "serialNum": "J1025PXX", "mtm": "7D2XCTO1WW", "ibmSerialNumber": "rackae402", "ibmMTM": "9155-
    C01", "type": "storage", "OCPRole": "control-1-ru3", "location": "RU2", "name": "IMM_RU2", "bootDevice": "/dev/sda",
    "users": [ { "user": "CEUSER", "password": "XXXXX", "group": "Administrator", "number": 2 }, { "user": "ISFUSER",
    "password": "XXXXXX", "group": "Administrator", "number": 3 }, { "user": "USERID", "password": "XXXXXX", "group":
    "Administrator", "number": 1 }
    ```

    e. Run the following command to create a tunnel to the IMM of the node:

    ```
    ssh -N -f -L :1443:[IPv6 or IPv4 address of IMM]:443 -L :3900:[IPv6 or IPv4 address of IMM]:3900 root@localhost
    ```

    f. Access the IMM user interface using the provisioner node IP address, open your browser and access the following IP.

    ```
    https://<provisioner ip>:1443
    ```

    - User - Enter `USERID`.
    - Password - Use the value obtained in the previous step.
12. From the IMM user interface page, open the remote console and check whether the node is up and shows a valid hostname prompt.
    - If the console shows localhost, then review your DHCP/DNS settings to ensure whether correct reservation is made for the node.
    - If the node shows a Red Hat linux login prompt instead of Core OS, then it infers that the node Base management controller (BMC) is not responsive.

    Do the following steps to resolve the issue:
    a. Identify IMM of the missing node. Here is mapping of nodes to IMM:

    ```
    control-1-ru2 ==> imm_ru2
    control-1-ru3 ==> imm_ru3
    control-1-ru4 ==> imm_ru4
    ```

b. For node(s) that did not show up in the previous steps, run commands to connect to their IMMs. For example, if node `control-1-ru3` did not show up, then from RU7/provisioner (compute-1-ru7), run `imm_ru3` command as a kni user:
c. Wait for the successful connection to IMM.
d. In the system prompt, run the `resetsp` on IMM prompt.
e. Wait 10 minutes.

Next actions

In the installation user interface, click the Retry to restart the installation.

Catalog source pod is healthy

Message: IBM Fusion catalog source pod is not found or is not healthy.

Diagnosis

Note: Run all commands as a kni user from the provisioner node.

1. Run the following command to verify whether the catalogsource exists or not.

   ```
   oc -n openshift-marketplace get catsrc
   ```

   Sample output:

   ```
   NAME  DISPLAY                     TYPE              PUBLISHER        AGE
   certified-operators            Certified             Operators       grpc Red Hat      11d
   community-operators            Community             Operators       grpc Red Hat      11d
   fusion-catalog                 IBM Storage Fusion    Catalog         grpc IBM          11d
   isf-data-foundation-catalog    Data Foundation       Catalog         grpc IBM          9d
   redhat-marketplace             Red Hat               Marketplace     grpc Red Hat      11d
   redhat-operators               Red Hat               Operators       grpc Red Hat      11d
   ```

   From the output, check for catalog source with name `fusion-catalog` exists.

2. If catalog source exists, then run the following command to check for the next `catalogsource` pod:

   ```
   oc -n openshift-marketplace get po
   ```

3. Check for a pod that starts with `fusion-catalog-` and is in `Running` state with `1/1` containers in `READY` field.

   For example

   ```
   oc -n openshift-marketplace get po
   ```

   Sample output:

   ```
   NAME                     READY          STATUS          RESTARTS        AGE
   certified-operators-ztp8f                     1/1           Running          0            14h
   community-operators-78vrd                     1/1           Running          0            14h
   fusion-catalog-kbw6h                          1/1           Running          0            2d12h
   isf-data-foundation-catalog-tv6pc             1/1           Running          0            19h
   marketplace-operator-f84d7df4f-f2zrg          1/1           Running          0            2d11h
   redhat-marketplace-8dgwp                      1/1           Running          0            2d11h
   redhat-operators-qj6pk                        1/1           Running          0            14h
   ```

4. If the pod is in `ImagePullBackOff` or `ErrImagePull` state, then describe pod to check the exact reason.

   - If describe shows timeout in pulling image, then check network bandwidth.
   - Offline installation:
     - If the listed error is related to an image manifest not known, then check whether the image is present in the enterprise registry.
     - If describe shows authentication error, then check whether the secret `pull-secret` in namespace `openshift-configs` has the correct credentials for the enterprise registry.

Next actions

Take corrective measures based on the diagnostics and rerun the installation.

Operator subscription isf-subscription not found

Message: Subscription for IBM Fusion operator not found.

Diagnosis

Note: Run all commands as a kni user from provisioner node.

1. Run the following command and in the output check whether the subscription with name isf-subscription exists.

   ```
   oc get sub -n ibm-spectrum-fusion-ns
   ```

Sample output:

```
NAME                PACKAGE             SOURCE                CHANNEL
grafana-operator    grafana-operator    community-operators   v4
isf-subscription    isf-operator        isf-catalog           v2.0
```

2. If subscription does not exist, then go to `/home/kni/logs` and check the log file that starts with `installoperator` for any errors:

Next actions

Take corrective measures based on the diagnostics and rerun installation. If issue is not resolved, contact IBM support with all logs from `/home/kni/` and archive that is downloaded from installation user interface.

Bootstrap VM is not pingable

A temporary bootstrap VM is created on provisioner node to create Red Hat OpenShift Container Platform control plane.

Message: Either bootstrap VM is not created or did not receive hostname and IP from DHCP.

Diagnosis

Note: Run all commands from the provisioner node as a kni user.

1. Run the following command on the provisioner node as a `kni` user to check whether the bootstarp VM is created:

   ```
   sudo virsh list
   ```

2. Empty output of previous command indicates that bootstrap VM is not created. To debug further, check the log file `/home/kni/clusterconfigs/.openshift_install.log` for errors.

   ```
   sudo virsh list
   setlocale: No such file or directory
   Id   Name   State
   --------------------
   ```

3. If output from first command is not empty. Example when bootstrap VM is created.

   ```
   sudo virsh list

   setlocale: No such file or directory
   Id   Name                           State
   ---------------------------------------------
   1    isf-rackae4-2bztv-bootstrap   running
   ```

4. If VM is listed in step 3, then run the following command to connect to it and check whether it has a valid hostname assigned.

   ```
   ping bootstrap.<subdomain>
   ```

   Here, subdomain is your `<cluster name>.<basedomain name>`.
   - If bootstrap is not pinging, then verify your DHCP and DNS setting for bootstrap VM.
   - An IP reservation in DHCP server must be made for bootstrap VM with correct MAC address as received in TDA sheet.
   - Lookup and reverse lookup records in DNS server for bootstrap VM.
   - Rectify the identified problems.

Next actions

In the installation user interface, click **Retry** to restart the installation.

Configuration error

The certificate does not contain a wildcard domain for cluster.

Diagnosis

1. Provided ingress certificate does not have a wildcard sub-domain of the OpenShift® cluster.
2. Provided certificate must adhere to guidelines from OpenShift outlined here - Understanding the default ingress certificate in *Red Hat OpenShift Container Platform* documentation.

Next actions

1. Obtain a new certificate with valid wildcard OpenShift subdomain in it.
2. In the installation user interface, click Back and provide a new certificate with a private key. The installation restarts with the new certificate.

Failed to install CSV for isf-operator

When you install IBM Fusion, the installation fails with "`Failed to install CSV for ifs-operator`" error.

For more information, check the log file **`/home/kni/logs/installoperator_playbook.log`**.

Diagnosis

Run all the commands as a **`kni`** user from the provisioner node.

1. Open the log file **`/home/kni/logs/installoperator_playbook.log`** and search for the string **`"Wait for CSV isf-operator to get installed"`** for errors to find the root cause.
2. If the previous logs do not contain details of the root cause, then continue with next steps for further diagnostics.
3. Run **`oc get csv -n ibm-spectrum-fusion-ns`** on the provisioner (also known as RU7 or compute-1-ru7 node). Example output:

```
NAME                             DISPLAY              VERSION          REPLACES              PHASE
isf-operator.v2.8.0-13104160    IBM Storage Fusion   2.8.0            isf-operator.v2.7.1   Failed
```

4. Check which pods are not in **`Running`** or **`Completed`** state by using the following command:

```
oc get pods -n ibm-spectrum-fusion-ns | grep -ivE 'Running|Completed
```

Sample output with one pod in **`ImagePullBackOff`** error:

```
NAME                                                              READY   STATUS             RESTARTS   AGE
isf-application-operator-controller-manager-7578464c4b-tld5w     0/2     ImagePullBackOff   0          116m
logcollector-5cf49dbc55-jzczf                                     0/1     Pending            0          116m
logcollector-5cf49dbc55-pjgl5                                     0/1     Pending            0          116m
```

5. Run the following describe command to view the details of pod:

```
oc descibe pod <POD NAME> -n ibm-spectrum-fusion-ns
```

6. Scroll the output and examine the **`Events`** section of the output.
   Example output:

```
Events:
  Type     Reason          Age    From                 Message
  ----     ------          ----   ----                 -------
  Normal   Scheduled       98m    default-scheduler    Error: ImagePullBackOff
```

7. If error from the describe command indicates authorisation or authentication error, then check whether the credentials you provided during the stage2 installation are correct.
   - For online installation, it is the entitlement key that you provided. In this case, the user is always **`cp`**.
   - For online installation, also check whether the specified entitlement key includes an IBM Fusion HCI entitlement.
   - For installation from enterprise registry, check the provided credentials for your enterprise registry (for both multiple and single repositories).
   - Try to pull the image manually on one OpenShift® Container Platform node by connecting to the node from provisioner by using **`oc debug node/<NODE-NAM>`**.
8. If the error from the previous describe command indicates manifest unknown error, then do the following checks:
   - For installation from enterprise registry, check whether your enterprise registry is configured properly and reachable from provisooner (RU7).
   - For installation from enterprise registry, make sure that the IBM Fusion HCI images are mirrored correctly.
   - Ensure that the user mirror images with right digest to right path exist in the enterprise registry.

Next actions

Take corrective actions and click Retry in the installation user interface.

Failed to install IBM Fusion operator

Failed to install the IBM Fusion operator. Check the log file /home/kni/logs/installoperator_playbook.log for more details.

Diagnosis

Run all commands as a kni user from the provisioner node.

1. Run the following command to check whether the catalogsource exists:

```
oc get catsrc -n openshift-marketplace
```

2. Verify whether the catalog source with name **`isf-catalog`** exists.
   Sample output:

```
NAME                          DISPLAY                   TYPE   PUBLISHER   AGE
certified-operators           Certified Operators       grpc   Red Hat     3d9h
community-operators           Community Operators       grpc   Red Hat     3d9h
ibm-operator-catalog          IBM Operator Catalog      grpc   IBM         3d2h
isf-data-foundation-catalog   Data Foundation Catalog   grpc   IBM         3d5h
isf-catalog.                                            grpc               8h
redhat-marketplace            Red Hat Marketplace       grpc   Red Hat     3d9h
redhat-operators              Red Hat Operators         grpc   Red Hat     3d9h
```

3. Run the following command to check whether the `catalogsource` pod exists:

```
oc get pods -n openshift-marketplace
```

4. Check whether a pod that starts with `isf-catalog-` exists with the `Status` as `Running` and `Ready` as `1/1`.
   Example:

```
NAME                                   READY   STATUS              RESTARTS   AGE
isf-catalog-wh8zn                      1/1     Running             0          8h
marketplace-operator-dd4dff8f-jxwm4    1/1     Running             0          32h
redhat-marketplace-d9dl8               1/1     Running             0          32h
redhat-operators-gzk4q                 1/1     Running             0          32h
```

5. If you find that the pod is not in running status, then run the following command to view the details:

```
oc descibe pod <POD NAME> -n openshift-marketplace
```

6. Scroll to the bottom and examine the `Events` section of the output.

```
Events:
  Type     Reason          Age   From                Message
  ----     ------          ----  ----                -------
  Normal   Scheduled       98m   default-scheduler   Error: ImagePullBackOff
```

7. If error from the describe command indicates authorisation or authentication error, then check whether the credentials you provided during the stage2 installation are correct.
   - For online installation, it is the entitlement key that you provided. In this case, the user is always `cp`.
   - For online installation, also check whether the specified entitlement key includes an IBM Fusion HCI entitlement.
   - For installation from enterprise registry, check the provided credentials for your enterprise registry (for both multiple and single repositories).
   - Try to pull the image manually on one OpenShift® Container Platform node by connecting to the node from provisioner by using `oc debug node/<NODE-NAM>`.
8. If the error from the previous describe command indicates manifest unknown error, then do the following checks:
   - For installation from enterprise registry, check whether your enterprise registry is configured properly and reachable from provisoner (RU7).
   - For installation from enterprise registry, make sure that the IBM Fusion HCI images are mirrored correctly.
   - Ensure that the user mirror images with right digest to right path exist in the enterprise registry.
9. If `isf-catalog` pod is running successfully, then run the following command to check for IBM Fusion HCI namespace:

```
oc get namespace | grep ibm-spectrum-fusion-ns
```

   Sample output:

```
ibm-spectrum-fusion-ns                             Active   3d9h
```

   If the command `oc get namespace | grep ibm-spectrum-fusion-ns` gives a blank output, then it means that the IBM Fusion HCI namespace is not created. Check the log file /home/kni/logs/installoperator_playbook.log for the following message:

```
message":"etcdserver: request timed out","code":500}\n'", "reason": "Internal
      Server Error", "status": 500}
```

   The failure indicates that this API issue is caused by the network bandwidth or any other latency. If the installation retries fail, contact IBM Support. Even if you do not find any API related issue in your logs, then contact IBM Support.
10. If the namespace is created, then run the following command to check for IBM Fusion `operator group`:

```
oc get og -n ibm-spectrum-fusion-ns
```

   Sample output:

```
NAME      AGE
isf-og    3d9h
```

   If the previous `oc get og -n ibm-spectrum-fusion-ns` command gives a blank output, then it means that the operator group is not created. Check the log file /home/kni/logs/installoperator_playbook.log for the following message:

```
message":"etcdserver: request timed out","code":500}\n'", "reason": "Internal
      Server Error", "status": 500}
```

   The failure indicates that this API issue is caused by the network bandwidth or any other latency. If the installation retries fail, contact IBM Support. Even if you do not find any API related issue in your logs, then contact IBM Support.
11. If the operator group is created, then run the following command to check for IBM Fusion `subscription`:

```
oc get sub  -n ibm-spectrum-fusion-ns
```

   Sample output:

```
NAME                PACKAGE        SOURCE             CHANNEL
isf-subscription    isf-operator   iso-catalog-new    v2.0
```

12. If the previous `oc get og -n ibm-spectrum-fusion-ns` command gives a blank output, then it means that the operator group is not created. Check the log file /home/kni/logs/installoperator_playbook.log for the following message:

```
message":"etcdserver: request timed out","code":500}\n'", "reason": "Internal
      Server Error", "status": 500}
```

   The failure indicates that this API issue is caused by the network bandwidth or any other latency. If the installation retries fail, contact IBM Support. Even if you do not find any API related issue in your logs, then contact IBM Support.
13. If subscription is created, then run the following command to check for IBM Fusion `install plan`:

```
oc get ip -n ibm-spectrum-fusion-ns
```

   Sample output:

```
NAME            CSV                            APPROVAL   APPROVED
install-bcqb4   isf-operator.v2.8.0-13321732   Manual     true
```

If the command `oc get ip -n ibm-spectrum-fusion-ns` gives a blank output, then it means that the install plan is not created.

14. If `install plan` is not found, then run the following commands to check jobs:

    `oc get job -n openshift-marketplace`

    Example output:

    ```
    NAME                                                               COMPLETIONS   DURATION   AGE
    0308d219fbaf8eec9372a4314df5af975f06f2ff3cd55c8fc33489e88deb346   1/1           9s         3d2h
    08dd142dfe6be404a597f0c82c8bad5787997272aa8867842e1018e06b3c32f   1/1           8s         3d4h
    0f59dc23d14241f8977ca45073457ddfaca24a68c7d0a6c8e14edf4d9bca337   1/1           7s         3d6h
    19a4f4d1b807398820a95a3b292d4f595c442ae7861c49d5b27d09624c0e9b1   1/1           10s        3d
    235cc047aad86ecb3c6b1d2d45ebf4bcbd54d89c352e386745edaa2b5e89b33   1/1           9s         3d2h
    31c5fff269b3664c17b94430b42e652ab175ec2ebee85e6c7df01a7da6ff600   1/1           7s         3d3h
    324ae09e681c9662339ae2dd9a38b2ad70c2430272c3229da527fd0f2b77fb8   1/1           12s        3d9h
    56d513408bd5a601c1701a5ca18fa71ff4e8dc049c5f205ecfcde1897ee03bd   1/1           7s         3d6h
    63d794fe8f947104f804d2db67793404dd4c95756e405bf30f5bd4ac59409b1   1/1           9s         3d
    69cd2f81af46d28954bf54d01ba81582a3ee4059da62aba89fc4762f7c1d199   1/1           10s        3d6h
    6f568dbc4626d4adcf92316d6d9e0a7250ae118b5cad7d97cac9525bf810ee3   1/1           9s         3d3h
    7a16b39766293d24b1ea674bf2221c218603f148aa5b01def75c3c792dd6ee5   1/1           27s        3d6h
    82067d1666f17dd7acd36ec050bfa065850240fcaaef5aba3ad11a91d6aa63f   1/1           9s         3d
    963e64e4cb5615267ab3821d47a8acb6f04586013df3472949178b49f049c64   1/1           8s         3d6h
    96eb43df54bdc1c822600b707e1078090ed2cfdd3587360a7545525097f52a6   1/1           9s         3d6h
    c7906c42e8405161bde11a3e044dbc23c87bf31e17393e061c917b6074ff28a   1/1           12s        3d2h
    c7f12c9dc97b5a771181f4fb58ecf7cd812f8d8a5756f810d40e5295f345f73   1/1           38s        3d
    d0d80a9e2553e077d37d4a407fe3a58b4b44596df02efad7d5d962e0505680b   1/1           22s        3d
    d68f3512c112308d77831d6c2765ee4c8e8d1960661a33dad59f0419e9fd732   1/1           9s         3d3h
    f31ed4a99e495429a1d47697faef595eb7a0f59b40d43617f8f3e2018bd1842   1/1           7s         3d6h
    ```

15. If any job does not have COMPLETION as 1/1, then run the following command to check for errors:

    `oc get jobname -n openshift-marketplace`

Next actions

Take corrective actions and click Retry in the installation user interface.

The catlogsource pod for Red Hat operators is not running

The `catlogsource` pod for `redhat-operators` is not running in `openshift-marketplace` namespace. Check the log file %s for more details.

Diagnosis

Run all commands as a kni user from the provisioner node.

1. Run the following command to verify whether the `catalogsource` exists or not.

   `oc -n openshift-marketplace get catsrc`

2. Verify whether the catalog source with name `redhat-operators` exists.
   Sample output:

   ```
   NAME                    DISPLAY                       TYPE   PUBLISHER   AGE
   certified-operators     Certified Operators           grpc   Red Hat     3d22h
   community-operators     Community Operators           grpc   Red Hat     3d22h
   ibm-operator-catalog    IBM Operator Catalog          grpc   IBM         3d12h
   isf-catalog             IBM Spectrum Fusion Catalog   grpc   IBM         3d21h
   opencloud-operators     IBMCS Operators               grpc   IBM         3d12h
   redhat-marketplace      Red Hat Marketplace           grpc   Red Hat     3d22h
   redhat-operators        Red Hat Operators             grpc   Red Hat     3d22h
   ```

3. If the catalog source exists, then run the following command to check for `catalogsource` pod:

   `oc -n openshift-marketplace get po`

4. Check whether a pod that starts with `redhat-operators-` exists with the `Status` as `Running` and `Ready` as `1/1`.
   Example:

   ```
   NAME                                    READY   STATUS    RESTARTS   AGE
   certified-operators-2j2jx               1/1     Running   0          3d11h
   community-operators-9wxcp               1/1     Running   0          14h
   ibm-operator-catalog-d768c              1/1     Running   0          3d10h
   isf-catalog-d972s                       1/1     Running   0          3d21h
   marketplace-operator-84558d7577-9nmdj   1/1     Running   0          3d21h
   opencloud-operators-bghhj               1/1     Running   0          3d12h
   redhat-marketplace-czb2z                1/1     Running   0          3d21h
   redhat-operators-49nf8                  1/1     Running   0          142m
   ```

5. If the pod is in `ImagePullBackOff` or `ErrImagePull` state, then describe pod to check the exact reason:
   - If describe shows timeout in pulling image, then check the network bandwidth.
   - If error listed is the image manifest is not known and it is an offline installation, then check whether the image is present in enterprise registry.
   - If describe shows authentication error and it is an offline installation, then check the secret `pull-secret` in namespace `openshift-config` for correct enterprise registry credentials.

Next actions

Take corrective actions and rerun installation.

Failed to deploy nmstate-operator

Failed to deploy **nmstate-operator**. Check the log file /home/kni/logs/installoperator_playbook.log for more details.

Diagnosis

Run all commands as a kni user from the provisioner node.

1. Run the following command to check whether the **catalogsource** exists or not:

   ```
   oc get catsrc -n openshift-marketplace
   ```

2. Verify whether the catalog source with name **redhat-operators** exists.

   ```
   NAME                          DISPLAY                   TYPE   PUBLISHER   AGE
   certified-operators           Certified Operators       grpc   Red Hat     3d9h
   community-operators           Community Operators       grpc   Red Hat     3d9h
   ibm-operator-catalog          IBM Operator Catalog      grpc   IBM         3d2h
   isf-data-foundation-catalog   Data Foundation Catalog   grpc   IBM         3d5h
   isf-catalog.                                            grpc               8h
   redhat-marketplace            Red Hat Marketplace       grpc   Red Hat     3d9h
   redhat-operators              Red Hat Operators         grpc   Red Hat     3d9h
   ```

3. If catalog source exists, then run the following command to check for **catalogsource** pod:

   ```
   c get pods -n openshift-marketplace
   ```

4. Verify a pod starting with **redhat-operators-** is present and is in **Running** state with **1/1** containers in **READY** field. For example:

   ```
   NAME                                   READY   STATUS              RESTARTS   AGE
   isf-catalog-wh8zn                      1/1     Running             0          8h
   marketplace-operator-dd4dff8f-jxwm4    1/1     Running             0          32h
   redhat-marketplace-d9dl8               1/1     Running             0          32h
   redhat-operators-gzk4q                 1/1     Running             0          32h
   ```

5. If the pod is not running, then run the following command to see the details of pod:

   ```
   oc descibe pod <POD NAME> -n openshift-marketplace
   ```

6. Scroll and examine the **Events** section of the output.

   ```
   Events:
     Type     Reason          Age    From                 Message
     ----     ------          ----   ----                 -------
     Normal   Scheduled       98m    default-scheduler    Error: ImagePullBackOff
   ```

7. If error from the describe command in authorisation or authentication error, then check the credentials you provided in stage2 installation GUI page:
   - For online installation, check your pull-secret that you provided.
   - For installation from enterprise registry, check the provided credentials for your enterprise registry (for both multiple and single repositories).
   - Try to pull the image manually on one OpenShift® Container Platform node by connecting to the node from provisioner by using **oc debug node/<NODE-NAM>**.
8. If error from above describe command (step#3) indicated manifest unknown error then make sure below checks:
   - For installation from enterprise registry, check whether your enterprise registry is configured properly and reachable from provisooner (RU7).
   - For installation from enterprise registry, make sure that the **kubernetes-nmstate-operator** images are mirrored correctly.
   - Ensure that the user mirror images with right digest to right path exist in the enterprise registry.
9. If the **redhat-operators** pod is running successfully, then run the following command to check the namespace **openshift-nmstate**:

   ```
   oc get namespace | grep openshift-nmstate
   ```

   Sample output:

   ```
   openshift-nmstate                                   Active   4d1h
   ```

   If the previous **oc get og -n openshift-nmstate** command gives a blank output, then it means that the operator group is not created. Check the log file /home/kni/logs/installoperator_playbook.log for the following message:

   ```
   message":"etcdserver: request timed out","code":500}\n'", "reason": "Internal
         Server Error", "status": 500}
   ```

   The failure indicates that this API issue is caused by the network bandwidth or any other latency. If the installation retries fail, contact IBM Support. Even if you do not find any API related issue in your logs, then contact IBM Support.
10. If namespace is created, then check for the operator group with name **openshift-nmstate**:

    ```
    oc get og -n openshift-nmstate
    ```

    Sample output:

    ```
    NAME                AGE
    openshift-nmstate   4d2h
    ```

If the previous `oc get og -n openshift-nmstate` command gives a blank output, then it means that the operator group is not created. Check the log file /home/kni/logs/installoperator_playbook.log for the following message:

```
message":"etcdserver: request timed out","code":500}\n'", "reason": "Internal
      Server Error", "status": 500}
```

The failure indicates that this API issue is caused by the network bandwidth or any other latency. If the installation retries fail, contact IBM Support. Even if you do not find any API related issue in your logs, then contact IBM Support.

11. If the operator group is created, then run the following command to check for `kubernetes-nmstate-operator` `subscription`:

    ```
    oc get sub -n openshift-nmstate
    ```

    Sample output:

    ```
    NAME                          PACKAGE                       SOURCE             CHANNEL
    kubernetes-nmstate-operator   kubernetes-nmstate-operator   redhat-operators   stable
    ```

    If the previous `oc get og -n openshift-nmstate` command gives a blank output, then it means that the operator group is not created. Check the log file /home/kni/logs/installoperator_playbook.log for the following message:

    ```
    message":"etcdserver: request timed out","code":500}\n'", "reason": "Internal
          Server Error", "status": 500}
    ```

    The failure indicates that this API issue is caused by the network bandwidth or any other latency. If the installation retries fail, contact IBM Support. Even if you do not find any API related issue in your logs, then contact IBM Support.
12. If the subscription is created, then check for `kubernetes-nmstate-operator` installplan:

    ```
    oc get ip  -n openshift-nmstate
    ```

    Sample output:

    ```
    NAME            CSV                                                 APPROVAL    APPROVED
    install-kz5sb   kubernetes-nmstate-operator.4.14.0-202311021650   Automatic   true
    ```

13. If the `oc get ip -n openshift-nmstate` command gives a blank output, then it means that the install plan is not created.If `install plan` is not found then please check jobs by execute command `oc get job -n openshift-marketplace`.

    ```
    NAME                                                               COMPLETIONS   DURATION   AGE
    0308d219fbaf8eec9372a4314df5af975f06f2ff3cd55c8fc33489e88deb346   1/1           9s         3d2h
    08dd142dfe6be404a597f0c82c8bad5787997272aa8867842e1018e06b3c32f   1/1           8s         3d4h
    0f59dc23d14241f8977ca45073457ddfaca24a68c7d0a6c8e14edf4d9bca337   1/1           7s         3d6h
    19a4f4d1b807398820a95a3b292d4f595c442ae7861c49d5b27d09624c0e9b1   1/1           10s        3d
    235cc047aad86ecb3c6b1d2d45ebf4bcbd54d89c352e386745edaa2b5e89b33   1/1           9s         3d2h
    31c5fff269b3664c17b94430b42e652ab175ec2ebee85e6c7df01a7da6ff600   1/1           7s         3d3h
    324ae09e681c9662339ae2dd9a38b2ad70c2430272c3229da527fd0f2b77fb8   1/1           12s        3d9h
    56d513408bd5a601c1701a5ca18fa71ff4e8dc049c5f205ecfcde1897ee03bd   1/1           7s         3d6h
    63d794fe8f947104f804d2db67793404dd4c95756e405bf30f5bd4ac59409b1   1/1           9s         3d
    69cd2f81af46d28954bf54d01ba81582a3ee4059da62aba89fc4762f7c1d199   1/1           10s        3d6h
    6f568dbc4626d4adcf92316d6d9e0a7250ae118b5cad7d97cac9525bf810ee3   1/1           9s         3d3h
    7a16b39766293d24b1ea674bf2221c218603f148aa5b01def75c3c792dd6ee5   1/1           27s        3d6h
    82067d1666f17dd7acd36ec050bfa065850240fcaaef5aba3ad11a91d6aa63f   1/1           9s         3d
    963e64e4cb5615267ab3821d47a8acb6f04586013df3472949178b49f049c64   1/1           8s         3d6h
    96eb43df54bdc1c822600b707e1078090ed2cfdd3587360a7545525097f52a6   1/1           9s         3d6h
    c7906c42e8405161bde11a3e044dbc23c87bf31e17393e061c917b6074ff28a   1/1           12s        3d2h
    c7f12c9dc97b5a771181f4fb58ecf7cd812f8d8a5756f810d40e5295f345f73   1/1           38s        3d
    d0d80a9e2553e077d37d4a407fe3a58b4b44596df02efad7d5d962e0505680b   1/1           22s        3d
    d68f3512c112308d77831d6c2765ee4c8e8d1960661a33dad59f0419e9fd732   1/1           9s         3d3h
    f31ed4a99e495429a1d47697faef595eb7a0f59b40d43617f8f3e2018bd1842   1/1           7s         3d6h
    ```

14. If any job does not have COMPLETION as '1/1', then run the `oc get jobname -n openshift-marketplace` comment and check for errors.

Next actions

Take corrective actions and rerun installation.

Failed to install CSV for kubernetes-nmstate-operator

While installing IBM Fusion HCI, the installation fails with the error message `Failed to install CSV for kubernetes-nmstate-operator`. Check log file /home/kni/logs/installoperator_playbook.log for more details.

Diagnosis

Run all commands as a kni user from the provisioner node.

1. See log file `/home/kni/logs/installoperator_playbook.log` and search for the following string:

   ```
   Wait for CSV kubernetes-nmstate-operator to get installed on second attempt
   ```

2. If the logs do not give enough information, then continue further diagnostics.
3. As a kni user, run `oc get csv -n openshift-nmstate` on the provisioned (also known as RU7 or compute-1-ru7 node). Check for `Failed' status.

   ```
   NAME                                              DISPLAY                      VERSION                  REPLACES   PHASE
   kubernetes-nmstate-operator.4.14.0-202311021650   Kubernetes NMState Operator  4.14.0-202311021650               Failed
   ```

4. Check which pods are not `Running` or in `Completed` state:

oc get pods -n openshift-nmstate

See the sample output for pods that are not in 'Running' STATUS:

```
NAME                                     READY   STATUS    RESTARTS        AGE
nmstate-cert-manager-64474fd576-v75sq    1/1     Running   0               2d2h
nmstate-console-plugin-5bb9bbf97c-ssjf2  1/1     Running   0               2d2h
nmstate-handler-2x16x                    1/1     Running   4 (4d1h ago)    4d1h
nmstate-handler-8xktl                    1/1     Running   0               4d2h
nmstate-handler-97pzc                    1/1     Running   3 (4d1h ago)    4d1h
nmstate-handler-ggbpt                    1/1     Running   0               4d2h
nmstate-handler-p9wsn                    1/1     Running   0               4d2h
nmstate-handler-pjjdr                    1/1     Running   3 (4d1h ago)    4d1h
nmstate-operator-c89955d9-snmh4          1/1     Running   0               2d2h
nmstate-webhook-69477dbc4f-6v719         1/1     Running   0               2d2h
nmstate-webhook-69477dbc4f-mqwmg         1/1     Running   0               2d2h
```

5. See the details of pod which are not in 'Running' STATUS:

   oc descibe pod <POD NAME> -n openshift-nmstate

6. Scroll and examine the **Events** section of the output:

```
Events:
  Type      Reason          Age    From                 Message
  ----      ------          ----   ----                 -------
  Normal    Scheduled       98m    default-scheduler    Error: ImagePullBackOff
```

7. If error from the describe command indicates authorisation or authentication error, then check whether the credentials you provided during the stage2 installation are correct.
   - For online installation, check pull-secret that you provided.
   - For installation from enterprise registry, check the provided credentials for your enterprise registry (for both multiple and single repositories).
   - Try to pull the image manually on one OpenShift® Container Platform node by connecting to the node from provisioner by using **oc debug node/<NODE-NAM>**.
8. If the error from the previous describe command indicates manifest unknown error, then do the following checks:
   - For installation from enterprise registry, check whether your enterprise registry is configured properly and reachable from provisooner (RU7).
   - For installation from enterprise registry, make sure that the **redhat kubernetes-nmstate-operator** images are mirrored correctly.
   - Ensure that the user mirror images with right digest to right path exist in the enterprise registry.

Next actions

Take corrective actions and rerun installation.

Delete files from provisioned node

The provisioner node, also know as service node, remains intact after IBM Fusion HCI initialization. For security reasons, all the configuration files are deleted from the service node.

Diagnostics

Note: Run all commands as kni user from the provisioner node.

1. Check /home/kni/logs/servicenode_cleanup_playbook.log for more details.
2. Scroll to end of file and check if it reports a failure.

The following example shows a successful run:

```
PLAY RECAP ***********************************************************************
localhost                  : ok=68    changed=3     unreachable=0     failed=0     skipped=45    rescued=0     ignored=0
```

If you find a failed value that is not equal to 0, then review the log to identify the step that failed.

Next actions

1. Log in to service node using kni user.
2. Check for /home/kni/install/cleanup_files_manually.json
3. Delete all the service node files manually, which are listed in the cleanup_files_manually.json.

Upgrade events and error codes

List of all the events that you might encounter during upgrade.

- **BMYUP1103**
  Upgrade of Red Hat® OpenShift® cluster is in progress.
- **BMYUP1104**
  Red Hat OpenShift cluster upgrade is complete.
- **BMYUP1105**
  IBM Fusion firmware upgrade available for nodes.

- **BMYUP1106**
  IBM Fusion firmware upgrade available for switches.
- **BMYUP1107**
  Firmware update succeeded for the compute node.
- **BMYUP1108**
  Firmware update is in progress on the compute node.
- **BMYUP1109**
  Firmware update succeeded for the switch.
- **BMYUP1110**
  Firmware Update of the switch is in progress.
- **BMYUP1111**
  Started upgrade of the Global Data Platform service.
- **BMYUP1112**
  Completed upgrade of the Global Data Platform service.
- **BMYUP1113**
  Started upgrade of the Data Protection service.
- **BMYUP1114**
  Completed upgrade of the Data Protection service.
- **BMYUP1117**
  IBM Fusion service upgrade available for the on-boarded service.
- **BMYUP1118**
  Service upgrade is available.
- **BMYUP1119**
  Service upgrade in progress.
- **BMYUP1120**
  Service upgrade is completed.
- **BMYUP1121**
  Operator Upgrade in progress.
- **BMYUP1123**
  Operator Upgrade completed.
- **BMYUP1136**
  Service node upgrade triggered.
- **BMYUP1141**
  Service node upgrade failed.
- **BMYUP1172**
  Service node upgrade completed.
- **BMYUP2106**
  Firmware update Job failed for the compute node.
- **BMYUP2107**
  Firmware Update of the switch has failed.
- **BMYUP2108**
  Failed to login to the switch with default user credentials after firmware upgrade.
- **BMYUP2109**
  Failed to change the password of ISFUSER on the switch.
- **BMYUP2110**
  Failed to login to the switch with ISFUSER credentials.
- **BMYUP2111**
  Failed to login to switch during firmware upgrade.
- **BMYUP3101**
  Failed to upgrade the Global Data Platform service.
- **BMYUP3102**
  Failed to upgrade Backup & Restore service.
- **BMYUP3104**
  Service upgrade is failing.
- **BMYUP3105**
  Operator Upgrade Failed.
- **BMYUP3106**
  Operator upgrade failed for the service.
- **BMYUP3153**
  Failed to ensure port is allowed.
- **BMYUP3154**
  Failed to ensure upgrade directory is created.
- **BMYUP3155**
  Failed to extract upgrade payload.
- **BMYUP3156**
  Failed to untar upgrade payload.
- **BMYUP3157**
  Failed to restart Install API service.
- **BMYUP3158**
  Failed to ensure port is free.
- **BMYUP3159**
  Failed to check upgrade health.
- **BMYUP3160**
  Failed to monitor upgrade agent.
- **BMYUP3161**
  Regenerate certificate failed.
- **BMYUP3162**
  Sync certificate from Service Node to OCP failed.
- **BMYPC4076**
  DNS resolution is successful for the node.

- **BMYPC4080**
  PXE port status with address is up.
- **BMYPC4082**
  Node adapters are available.
- **BMYPC4083**
  Bare Metal network bond contains sufficient port availability.
- **BMYPC4085**
  Node firmware is currently in progress with current state as %s.
- **BMYPC5055**
  One AMQ subscription on the cluster is not at a version that is compatible with the Backup & Restore upgrade.
- **BMYPC5056**
  Multiple AMQ subscriptions on the cluster are not at a version compatible with the Backup & Restore upgrade.
- **BMYPC5057**
  One other OADP subscription on the cluster is not at a version compatible with the Backup & Restore upgrade.
- **BMYPC5058**
  Multiple OADP subscriptions on the cluster are not at a version compatible with the Backup & Restore upgrade. Proceeding with the upgrade may cause issues with backup functionality.
- **BMYPC6020**
  OpenShift Container Platform upgrade is in progress.
- **BMYPC6021**
  Compute nodes firmware upgrade is in progress.
- **BMYPC6022**
  Compute node reboot is in progress.
- **BMYPC6053**
  Switch firmware upgrade is in progress.
- **BMYPC6054**
  Found a failed OLM job in namespace.
- **BMYPC6055**
  Found multiple failed OLM jobs in the namespace.
- **BMYPC6072**
  Kickstart entry is incorrect.
- **BMYPC6073**
  Node health is not normal.
- **BMYPC6074**
  Kickstart entry is correct.
- **BMYPC6075**
  Node health is normal.
- **BMYPC6076**
  The DNS resolution for the node failed.
- **BMYPC6080**
  PXE port status with address is not up.
- **BMYPC6082**
  Node adapters are missing.
- **BMYPC6083**
  All ports %s of Bare Metal network bond are down.
- **BMYPC6085**
  Node firmware is currently not in progress or already up-to-date.

BMYUP1103

Upgrade of Red Hat® OpenShift® cluster is in progress.

Severity

Informational

User actions

No action is required.

BMYUP1104

Red Hat® OpenShift® cluster upgrade is complete.

Severity

Informational

User actions

No action is required.

BMYUP1105

IBM Fusion firmware upgrade available for nodes.

Severity

Informational

User actions

No action is required.

BMYUP1106

IBM Fusion firmware upgrade available for switches.

Severity

Informational

User actions

No action is required.

BMYUP1107

Firmware update succeeded for the compute node.

Severity

Informational

User actions

No action is required.

BMYUP1108

Firmware update is in progress on the compute node.

Severity

Informational

User actions

No action is required.

BMYUP1109

Firmware update succeeded for the switch.

Severity

Informational

User actions

No action is required.

BMYUP1110

Firmware Update of the switch is in progress.

Severity

Informational

User actions

No action is required.

BMYUP1111

Started upgrade of the Global Data Platform service.

Severity

Informational

User actions

No action is required.

BMYUP1112

Completed upgrade of the Global Data Platform service.

Severity

Informational

User actions

No action is required.

BMYUP1113

Started upgrade of the Data Protection service.

Severity

Informational

User actions

No action is required.

BMYUP1114

Completed upgrade of the Data Protection service.

Severity

Informational

User actions

No action is required.

BMYUP1117

IBM Fusion service upgrade available for the on-boarded service.

Severity

Informational

User actions

No action is required.

BMYUP1118

Service upgrade is available.

Severity

Informational

User actions

No action is required.

BMYUP1119

Service upgrade in progress.

Severity

Informational

User actions

No action is required.

BMYUP1120

Service upgrade is completed.

Severity

Informational

User actions

No action is required.

BMYUP1121

Operator Upgrade in progress.

Severity

Informational

User actions

No action is required.

BMYUP1123

Operator Upgrade completed.

Severity

Informational

User actions

No action is required.

BMYUP1136

Service node upgrade triggered.

Severity

Informational

User actions

No action is required.

BMYUP1141

Service node upgrade failed.

Severity

Critical

User actions

See ../upgrade/upgrading_service_node.html#upgrading_service_node__recovery-procedure. If the issue persists, contact IBM support.

BMYUP1172

Service node upgrade completed.

Severity

Informational

User actions

No action is required.

BMYUP2106

Firmware update Job failed for the compute node.

Error conditions

The error message can be for the following conditions:

- Failed to put the node in maintenance mode for firmware update.
- Failed to move the node out of maintenance after firmware update attempt.
- Power operation of graceful restart of the node failed after firmware update attempt.
- Firmware update failed on the node, not reflecting the latest version after the node restart and IMM reset.
- Firmware update operation timed out on the compute node.

- Scale cluster did not turn healthy post firmware update on the node.
  Note: If the storage is configured with Data Foundation, then also the message shows that `Scale cluster did not turn healthy post firmware update on the node` in the IBM Fusion user interface because of a typographical error in the event message.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Check whether any other node is in maintenance mode. Ensure that all the nodes must be up and in a ready state to perform firmware update operations.
2. An administrator needs to move the node out of maintenance mode by using the OpenShift® or IBM Fusion user interface.
3. The administrator should power-off and power-on the node, and after it gets completed successfully, move the node out of maintenance mode manually. This issue might get resolved on its own. This issue happens mostly when power operation takes longer than expected.
4. Run the OC Command to check whether the scale cluster is healthy.

   ```
   oc -n ibm-spectrum-fusion-ns get scales storagemanager -oyaml | grep storageClusterStatus
   ```

5. Run the OC Command to check whether the Data Foundation cluster is healthy.

   ```
   oc get odfcluster odfcluster -o yaml -n ibm-spectrum-fusion-ns | grep odfcluster.cephClusterHealth
   ```

6. Retrigger firmware update if compute CR shows upgradeRequired=true. If the firmware update fails again, contact IBM Support.
7. The administrator should collect the Nodes logs and then retrigger the firmware update on the node after ensuring all the prerequisites are met. If the retrigger fails again, contact IBM Support.
8. The firmware update timed out. The administrator should collect the Nodes logs and retrigger the firmware update after some time. If the firmware update continues to fail, contact the IBM Support.
9. Collect the Nodes logs. Perform a power-off and power-on of the target node. See Administering the node.
10. After successfully powering on the node, wait 15 minutes to see whether the scale cluster state changed to healthy. If the scale cluster remains in a DEGRADED state, contact the IBM Support.

BMYUP2107

Firmware Update of the switch has failed.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYUP2108

Failed to login to the switch with default user credentials after firmware upgrade.

Severity

Warning

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command to get the switch IP address.

   ```
   oc get switch mgmt1-rackag5 -o yaml | grep switchIpadress
   ```

   Example output:

   ```
   switchIpadress: fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
   ```

2. Check connectivity to the switch from one of the compute node using the following steps:
   a. Run the following command to get the node details.

      ```
      oc get node
      ```

      Example output:

      ```
      NAME                                STATUS                     ROLES     AGE     VERSION
      compute-1-ru27.rackag5.mydomain.com   Ready                      worker    5d14h   v1.25.14+bcb9a60
      compute-1-ru6.rackag5.mydomain.com    Ready,SchedulingDisabled   worker    9d      v1.25.14+bcb9a60
      ```

```
compute-1-ru7.rackag5.mydomain.com    Ready,SchedulingDisabled   worker                9d      v1.25.14+bcb9a60
control-1-ru2.rackag5.mydomain.com    Ready                      control-plane,master   9d      v1.25.14+bcb9a60
control-1-ru3.rackag5.mydomain.com    Ready                      control-plane,master   9d      v1.25.14+bcb9a60
control-1-ru4.rackag5.mydomain.com    Ready                      control-plane,master   9d      v1.25.14+bcb9a60
```

b. Run the following command to log in to node.

```
oc debug node/compute-1-ru27.rackag5.mydomain.com
```

Example output:

```
Temporary namespace openshift-debug-svmds is created for debugging node...
Starting pod/compute-1-ru27rackag5mydomaincom-debug ...
To use host binaries, run `chroot /host`
Pod IP: 172.88.55.32
If you don't see a command prompt, try pressing enter.
sh-4.4# ping fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
PING fd8c:215d:178e:c0de:e69d:73ff:fe69:b400(fd8c:215d:178e:c0de:e69d:73ff:fe69:b400) 56 data bytes
64 bytes from fd8c:215d:178e:c0de:e69d:73ff:fe69:b400: icmp_seq=1 ttl=64 time=5.48 ms
64 bytes from fd8c:215d:178e:c0de:e69d:73ff:fe69:b400: icmp_seq=2 ttl=64 time=0.464 ms
^C
--- fd8c:215d:178e:c0de:e69d:73ff:fe69:b400 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.464/2.971/5.478/2.507 ms
sh-4.4#
```

3. If ping to switch IP is not working, then collect Network switches logs and contact IBM support.
4. If ping to switch IP works, then run the following commands to get the default username and password.

```
oc get secret mgmt1-rackag5-secret -o yaml | grep defaultUserName
oc get secret mgmt1-rackag5-secret -o yaml | grep defaultUserPasswrd
```

5. Decode the username and password obtained from the previous step.

```
echo <username/password> | base64 --decode
```

6. Try to SSH from one of the compute node using the default credentials.

```
ssh cumulus@fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
```

Example output:

```
The authenticity of host 'fd8c:215d:178e:c0de:e69d:73ff:fe69:b400 (fd8c:215d:178e:c0de:e69d:73ff:fe69:b400)' can't be establis
ECDSA key fingerprint is SHA256:n9XPEXgyP8A5fgKBCFB4NVwnwwtBYfNnoS5mIu27L7c.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'fd8c:215d:178e:c0de:e69d:73ff:fe69:b400' (ECDSA) to the list of known hosts.
cumulus@fd8c:215d:178e:c0de:e69d:73ff:fe69:b400's password:
Linux mgmt1-rackag5 4.19.0-cl-1-iproc #1 SMP PREEMPT Cumulus 4.19.149-1+cl4.3u1 (2021-01-28) armv7l

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basis.
Last login: Mon Jan 29 04:11:14 2024 from fd8c:215d:178e:c0de:d8a0:53d6:6eb6:6a55


*****************************************************************
Please send these support file(s) to support@cumulusnetworks.com:
 /var/support/cl_support_mgmt1-rackag5_20230721_092859.txz
 /var/support/cl_support_mgmt1-rackag5_20230821_041846.txz
 /var/support/cl_support_mgmt1-rackag5_20230822_151434.txz
 /var/support/cl_support_mgmt1-rackag5_20230910_040832.txz
 /var/support/cl_support_mgmt1-rackag5_20230911_061727.txz
 /var/support/cl_support_mgmt1-rackag5_20231026_072515.txz
*****************************************************************


cumulus@mgmt1-rackag5:mgmt:~$
```

7. If the credentials are not working, then collect Network switches logs and contact IBM support.

BMYUP2109

Failed to change the password of ISFUSER on the switch.

Severity

Warning

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command to get the switch IP address.

```
oc get switch mgmt1-rackag5 -o yaml | grep switchIpadress
```

Example output:

```
switchIpadress: fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
```

2. Check connectivity to the switch from one of the compute node using the following steps:
   a. Run the following command to get the node details.

   ```
   oc get node
   ```

   Example output:

   ```
   NAME                                 STATUS                     ROLES                  AGE     VERSION
   compute-1-ru27.rackag5.mydomain.com  Ready                      worker                 5d14h   v1.25.14+bcb9a60
   compute-1-ru6.rackag5.mydomain.com   Ready,SchedulingDisabled   worker                 9d      v1.25.14+bcb9a60
   compute-1-ru7.rackag5.mydomain.com   Ready,SchedulingDisabled   worker                 9d      v1.25.14+bcb9a60
   control-1-ru2.rackag5.mydomain.com   Ready                      control-plane,master   9d      v1.25.14+bcb9a60
   control-1-ru3.rackag5.mydomain.com   Ready                      control-plane,master   9d      v1.25.14+bcb9a60
   control-1-ru4.rackag5.mydomain.com   Ready                      control-plane,master   9d      v1.25.14+bcb9a60
   ```

   b. Run the following command to log in to node.

   ```
   oc debug node/compute-1-ru27.rackag5.mydomain.com
   ```

   Example output:

   ```
   Temporary namespace openshift-debug-svmds is created for debugging node...
   Starting pod/compute-1-ru27rackag5mydomaincom-debug ...
   To use host binaries, run `chroot /host`
   Pod IP: 172.88.55.32
   If you don't see a command prompt, try pressing enter.
   sh-4.4# ping fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
   PING fd8c:215d:178e:c0de:e69d:73ff:fe69:b400(fd8c:215d:178e:c0de:e69d:73ff:fe69:b400) 56 data bytes
   64 bytes from fd8c:215d:178e:c0de:e69d:73ff:fe69:b400: icmp_seq=1 ttl=64 time=5.48 ms
   64 bytes from fd8c:215d:178e:c0de:e69d:73ff:fe69:b400: icmp_seq=2 ttl=64 time=0.464 ms
   ^C
   --- fd8c:215d:178e:c0de:e69d:73ff:fe69:b400 ping statistics ---
   2 packets transmitted, 2 received, 0% packet loss, time 1002ms
   rtt min/avg/max/mdev = 0.464/2.971/5.478/2.507 ms
   sh-4.4#
   ```

3. If ping to switch IP is not working, then collect Network switches logs and contact IBM support.
4. If ping to switch IP works, then run the following commands to get the default username and password.

```
oc get secret mgmt1-rackag5-secret -o yaml | grep defaultUserName
oc get secret mgmt1-rackag5-secret -o yaml | grep defaultUserPasswrd
```

5. Decode the username and password obtained from the previous step.

```
echo <username/password> | base64 --decode
```

6. Try to SSH from one of the compute node using the default credentials.

```
ssh cumulus@fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
```

Example output:

```
The authenticity of host 'fd8c:215d:178e:c0de:e69d:73ff:fe69:b400 (fd8c:215d:178e:c0de:e69d:73ff:fe69:b400)' can't be establis
ECDSA key fingerprint is SHA256:n9XPEXgyP8A5fgKBCFB4NVwnwwtBYfNnoS5mIu27L7c.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'fd8c:215d:178e:c0de:e69d:73ff:fe69:b400' (ECDSA) to the list of known hosts.
cumulus@fd8c:215d:178e:c0de:e69d:73ff:fe69:b400's password:
Linux mgmt1-rackag5 4.19.0-cl-1-iproc #1 SMP PREEMPT Cumulus 4.19.149-1+cl4.3u1 (2021-01-28) armv7l

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basis.
Last login: Mon Jan 29 04:11:14 2024 from fd8c:215d:178e:c0de:d8a0:53d6:6eb6:6a55


*****************************************************************
Please send these support file(s) to support@cumulusnetworks.com:
 /var/support/cl_support_mgmt1-rackag5_20230721_092859.txz
 /var/support/cl_support_mgmt1-rackag5_20230821_041846.txz
 /var/support/cl_support_mgmt1-rackag5_20230822_151434.txz
 /var/support/cl_support_mgmt1-rackag5_20230910_040832.txz
 /var/support/cl_support_mgmt1-rackag5_20230911_061727.txz
 /var/support/cl_support_mgmt1-rackag5_20231026_072515.txz
*****************************************************************


cumulus@mgmt1-rackag5:mgmt:~$
```

7. If the credentials are not working, then collect Network switches logs and contact IBM support.

BMYUP2110

Failed to login to the switch with ISFUSER credentials.

Severity

Warning

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command to get the switch IP address.

   ```
   oc get switch mgmt1-rackag5 -o yaml | grep switchIpadress
   ```

   Example output:

   ```
   switchIpadress: fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
   ```

2. Check connectivity to the switch from one of the compute node using the following steps:
   a. Run the following command to get the node details.

      ```
      oc get node
      ```

      Example output:

      ```
      NAME                                   STATUS                       ROLES                 AGE     VERSION
      compute-1-ru27.rackag5.mydomain.com    Ready                        worker                5d14h   v1.25.14+bcb9a60
      compute-1-ru6.rackag5.mydomain.com     Ready,SchedulingDisabled     worker                9d      v1.25.14+bcb9a60
      compute-1-ru7.rackag5.mydomain.com     Ready,SchedulingDisabled     worker                9d      v1.25.14+bcb9a60
      control-1-ru2.rackag5.mydomain.com     Ready                        control-plane,master  9d      v1.25.14+bcb9a60
      control-1-ru3.rackag5.mydomain.com     Ready                        control-plane,master  9d      v1.25.14+bcb9a60
      control-1-ru4.rackag5.mydomain.com     Ready                        control-plane,master  9d      v1.25.14+bcb9a60
      ```

   b. Run the following command to log in to node.

      ```
      oc debug node/compute-1-ru27.rackag5.mydomain.com
      ```

      Example output:

      ```
      Temporary namespace openshift-debug-svmds is created for debugging node...
      Starting pod/compute-1-ru27rackag5mydomaincom-debug ...
      To use host binaries, run `chroot /host`
      Pod IP: 172.88.55.32
      If you don't see a command prompt, try pressing enter.
      sh-4.4# ping fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
      PING fd8c:215d:178e:c0de:e69d:73ff:fe69:b400(fd8c:215d:178e:c0de:e69d:73ff:fe69:b400) 56 data bytes
      64 bytes from fd8c:215d:178e:c0de:e69d:73ff:fe69:b400: icmp_seq=1 ttl=64 time=5.48 ms
      64 bytes from fd8c:215d:178e:c0de:e69d:73ff:fe69:b400: icmp_seq=2 ttl=64 time=0.464 ms
      ^C
      --- fd8c:215d:178e:c0de:e69d:73ff:fe69:b400 ping statistics ---
      2 packets transmitted, 2 received, 0% packet loss, time 1002ms
      rtt min/avg/max/mdev = 0.464/2.971/5.478/2.507 ms
      sh-4.4#
      ```

3. If ping to switch IP is not working, then collect Network switches logs and contact IBM support.
4. If ping to switch IP works, then run the following commands to get the default username and password.

   ```
   oc get secret mgmt1-rackag5-secret -o yaml | grep defaultUserName
   oc get secret mgmt1-rackag5-secret -o yaml | grep defaultUserPasswrd
   ```

5. Decode the username and password obtained from the previous step.

   ```
   echo <username/password> | base64 --decode
   ```

6. Try to SSH from one of the compute node using the default credentials.

   ```
   ssh ISFUSER@fd8c:215d:178e:c0de:e69d:73ff:fe69:b400
   ```

   Example output:

   ```
   The authenticity of host 'fd8c:215d:178e:c0de:e69d:73ff:fe69:b400 (fd8c:215d:178e:c0de:e69d:73ff:fe69:b400)' can't be establis
   ECDSA key fingerprint is SHA256:n9XPEXgyP8A5fgKBCFB4NVwnwwtBYfNnoS5mIu27L7c.
   Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
   Warning: Permanently added 'fd8c:215d:178e:c0de:e69d:73ff:fe69:b400' (ECDSA) to the list of known hosts.
   ISFUSER@fd8c:215d:178e:c0de:e69d:73ff:fe69:b400's password:
   Linux mgmt1-rackag5 4.19.0-cl-1-iproc #1 SMP PREEMPT Cumulus 4.19.149-1+cl4.3u1 (2021-01-28) armv7l

   Welcome to NVIDIA Cumulus (R) Linux (R)

   For support and online technical documentation, visit
   http://www.cumulusnetworks.com/support

   The registered trademark Linux (R) is used pursuant to a sublicense from LMI,
   the exclusive licensee of Linus Torvalds, owner of the mark on a world-wide
   basis.
   Last login: Mon Jan 29 04:11:14 2024 from fd8c:215d:178e:c0de:d8a0:53d6:6eb6:6a55


   *****************************************************************
   Please send these support file(s) to support@cumulusnetworks.com:
    /var/support/cl_support_mgmt1-rackag5_20230721_092859.txz
    /var/support/cl_support_mgmt1-rackag5_20230821_041846.txz
   ```

```
 /var/support/cl_support_mgmt1-rackag5_20230822_151434.txz
 /var/support/cl_support_mgmt1-rackag5_20230910_040832.txz
 /var/support/cl_support_mgmt1-rackag5_20230911_061727.txz
 /var/support/cl_support_mgmt1-rackag5_20231026_072515.txz
******************************************************************


ISFUSER@mgmt1-rackag5:mgmt:~$
```

7. If the credentials are not working, then collect Network switches logs and contact IBM support.

BMYUP2111

Failed to login to switch during firmware upgrade.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYUP3101

Failed to upgrade the Global Data Platform service.

Severity

Critical

User actions

Collect Global Data Platform logs and contact IBM support.

BMYUP3102

Failed to upgrade Backup & Restore service.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Ensure that you performed the before you begin steps of Implementing a comprehensive upgrade procedure.
2. If the issue persists, contact IBM support.

BMYUP3104

Service upgrade is failing.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Go to OpenShift® Container Platform web console and check the FusionServiceInstance for the specific service and check the conditions section to find the error.
2. If the issue persists, run the following oc command, and check the status section and debug the error.

```
oc get fusionserviceinstance <instance-name-of-service> -o yaml
```

3. If this occurs during installation or post service installation, collect logs for the corresponding service that shows as unhealthy and contact IBM support. Follow the steps to collect logs from IBM Fusion.
   - Go to Services page in IBM Fusion user interface.
   - Click the ellipses menu for the corresponding service.
   - Select the Collect logs button. It downloads all the necessary logs.

BMYUP3105

Operator Upgrade Failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Go to OpenShift® Container Platform web console and check the FusionServiceInstance for the specific service and check the conditions section to find the error.
2. If the issue persists, run the following oc command to check the status section and try to debug the error.

```
oc get fusionserviceinstance <instance-name-of-service> -o yaml
```

3. If the issue still persists, do the following:
   - If this occurs during Data Cataloging upgrade, Collect Data Cataloging logs and contact IBM support.
   - If this occurs during Backup and Restore upgrade, Collect Backup and Restore logs and contact IBM support.

BMYUP3106

Operator upgrade failed for the service.

Severity

Error

Problem description

It indicates that the operator upgrade for the service is failed. You can find reason for the upgrade failure in the displayed message on the user interface.

User actions

Contact IBM support to resolve this issue.

BMYUP3153

Failed to ensure port is allowed.

Severity

Critical

User actions

No action is required.

BMYUP3154

Failed to ensure upgrade directory is created.

Severity

Critical

User actions

No action is required.

BMYUP3155

Failed to extract upgrade payload.

Severity

Critical

User actions

No action is required.

BMYUP3156

Failed to untar upgrade payload.

Severity

Critical

User actions

No action is required.

BMYUP3157

Failed to restart Install API service.

Severity

Critical

User actions

No action is required.

BMYUP3158

Failed to ensure port is free.

Severity

Critical

User actions

No action is required.

BMYUP3159

Failed to check upgrade health.

Severity

Critical

User actions

No action is required.

BMYUP3160

Failed to monitor upgrade agent.

Severity

Critical

User actions

No action is required.

BMYUP3161

Regenerate certificate failed.

Severity

Critical

User actions

No action is required.

BMYUP3162

Sync certificate from Service Node to OCP failed.

Severity

Critical

User actions

No action is required.

BMYPC4076

DNS resolution is successful for the node.

Severity

Informational

User actions

No action is required.

BMYPC4080

PXE port status with address is up.

Severity

Informational

User actions

No action is required.

BMYPC4082

Node adapters are available.

Severity

Informational

User actions

No action is required.

BMYPC4083

Bare Metal network bond contains sufficient port availability.

Severity

Informational

User actions

No action is required.

BMYPC4085

Node firmware is currently in progress with current state as %s.

Severity

Error

Problem description

It indicates that the node firmware is currently in progress with current state as %s.

User actions

Follow the steps to resolve the issue:

1. Check the reason for failure and take appropriate action to resolve the issue.
2. If the issue persists, contact IBM support to resolve this issue.

BMYPC5055

One AMQ subscription on the cluster is not at a version that is compatible with the Backup & Restore upgrade.

Severity

Warning

Problem description

It indicates that there is an AMQ subscription found outside the Backup & Restore namespace is not at a compatible version. For Backup & Restore version 2.10, the AMQ version must be at 2.9.x. If there is an AMQ operator that is installed outside the Backup & Restore namespace is not at 2.9.x, then it can cause conflicts and failures as Backup & Restore try to upgrade its own AMQ to 2.9.0.

User actions

Upgrade the other AMQ to 2.9.x version.

BMYPC5056

Multiple AMQ subscriptions on the cluster are not at a version compatible with the Backup & Restore upgrade.

Severity

Warning

Problem description

It indicates that there are multiple AMQ subscriptions found outside of the Backup & Restore namespace that are not at a compatible version. For Backup & Restore version 2.10, the AMQ version must be at 2.9.x. If there is an AMQ operator installed outside of the Backup & Restore namespace that is not at 2.9.x, then it can cause conflicts and failures as Backup & Restore try to upgrade its own AMQ to 2.9.0.

User actions

Upgrade the other AMQ installation to the version 2.9.x.

BMYPC5057

One other OADP subscription on the cluster is not at a version compatible with the Backup & Restore upgrade.

Severity

Warning

Problem description

It indicates that there is an OADP subscription found outside of the Backup & Restore namespace that is not at a compatible version. For Backup & Restore version 2.10, the OADP version must be at 1.4.x. If there is an OADP operator installed outside of the Backup & Restore namespace that is not at 1.4.x, then it can cause conflicts and failures as Backup & Restore try to upgrade its own OADP to 1.4.0.

User actions

Upgrade the other OADP installation to the version 1.4.x.

BMYPC5058

Multiple OADP subscriptions on the cluster are not at a version compatible with the Backup & Restore upgrade. Proceeding with the upgrade may cause issues with backup functionality.

Severity

Warning

Problem description

It indicates that there are multiple OADP subscriptions found outside of the Backup & Restore namespace that are not at a compatible version. For Backup & Restore version 2.10, the OADP version must be at 1.4.x. If there is an OADP operator installed outside of the Backup & Restore namespace that is not at 1.4.x, then it can cause conflicts and failures as Backup & Restore try to upgrade its own OADP to 1.4.0.

User actions

Upgrade the other OADP installation to the version 1.4.x.

BMYPC6020

OpenShift® Container Platform upgrade is in progress.

Severity

Error

Problem description

It indicates that the OpenShift Container Platform upgrade is in progress. The upgrades for compute nodes firmware, switch firmware and Global Data Platform service should not be triggered when the OpenShift Container Platform upgrade is in progress since nodes are rebooted in a rolling pattern.

User actions

Wait until the OpenShift Container Platform service upgrade to be complete.

BMYPC6021

Compute nodes firmware upgrade is in progress.

Severity

Error

Problem description

It indicates that the upgrade for the compute node firmwares are in progress. In such case, the upgrades for OpenShift® cluster, switches, and Global Data Platform service must not be triggered because the firmware upgrade requires rolling restart of OpenShift nodes.

User actions

Wait until the compute nodes silent upgrades to finish.

BMYPC6022

Compute node reboot is in progress.

Severity

Error

Problem description

It indicates that the compute node reboot is in progress. In such case, the upgrades for OpenShift® cluster, switches, and Global Data Platform service should not be triggered.

User actions

Wait until the compute node reboot to finish.

BMYPC6053

Switch firmware upgrade is in progress.

Severity

Error

Problem description

It indicates that the switch firmware upgrade is in progress. In such a case, the upgrades for OpenShift® cluster, Global Data Platform, and compute firmware must not be triggered.

User actions

Wait until the switch upgrade to finish.

BMYPC6054

Found a failed OLM job in namespace.

Severity

Error

Problem description

It indicates that there is a failed OLM job is identified in the namespace and it might cause issues when you upgrade the service.

User actions

Investigate and resolve the cause of job failure and then delete the configmap of each job linked in the event message. Each failed job contains a corresponding configmap in the same namespace with the same name, and removing the configmap also removes its related job.

BMYPC6055

Found multiple failed OLM jobs in the namespace.

Severity

Error

Problem description

It indicates that there are multiple failed OLM jobs identified in the namespace and it might cause issues when you upgrade the service.

User actions

Investigate and resolve the cause of job failures and then delete the configmaps of each job linked in the event message. Each failed job contains a corresponding configmap in the same namespace with the same name, and removing these configmaps also removes their related jobs.

BMYPC6072

Kickstart entry is incorrect.

Severity

Error

Problem description

It indicates that there is an incorrect entry in the kickstart and reported keys are missing.

User actions

Follow the steps to resolve the issue:

1. Check and update the missing keys in the kickstart file.
2. If the issue persists, contact IBM support to resolve this issue.

BMYPC6073

Node health is not normal.

Severity

Error

Problem description

It indicates that the node health is not normal because of the reported issues.

User actions

Follow the steps to resolve the issue:

1. Restart service processor on the BMC or IDRAC.
2. If the issue persists, contact IBM support to resolve this issue.

BMYPC6074

Kickstart entry is correct.

Severity

Informational

User actions

No action is required.

BMYPC6075

Node health is normal.

Severity

Informational

User actions

No action is required.

BMYPC6076

The DNS resolution for the node failed.

Severity

Error

Problem description

It indicates that the hostname resolution failed for the specified node.

User actions

Follow the steps described in the Setting up DNS section to verify the DNS configuration of the affected node.

After you correct the DNS configuration, the node addition process proceeds successfully.

BMYPC6080

PXE port status with address is not up.

Severity

Error

Problem description

It indicates that the PXE status with address is not up.

User actions

Follow the steps to resolve the issue:

1. Check the Ports tab in the IBM Fusion UI under the node monitoring page to verify the status of the first port of the 1G adapter. Also, confirm the status of the connected switch port.
2. If the issue persists, contact IBM support to resolve this issue.

BMYPC6082

Node adapters are missing.

Severity

Error

Problem description

It indicates that the node adapters are missing.

User actions

Follow the steps to resolve the issue:

1. Check the BMC or IDRAC and reseat the adapter or any loose connection.
2. If the issue persists, contact IBM support to resolve this issue.

BMYPC6083

All ports %s of Bare Metal network bond are down.

Severity

Error

Problem description

It indicates that the all ports of the Bare Metal network bond are down.

User actions

Follow the steps to resolve the issue:

1. Check the cable connectivity, switch port configuration.
2. If the issue persists, contact IBM support to resolve this issue.

BMYPC6085

Node firmware is currently not in progress or already up-to-date.

Severity

Informational

User actions

No action is required.

Compute events and error codes

List of all the events that you might encounter in compute nodes.

For Call Home events, see Compute Call Home events and error codes.

For the events that you might encounter during a compute node firmware upgrade, see Upgrade events and error codes.

For Informational, Warning, and Critical events, see the following:

- **Compute Call Home events and error codes**
  List of Call Home events that you might encounter in the compute.
- **BMYCO0001**
  Unable to login, unable to ping, invalid credentials.
- **BMYCO0004**
  Failed to get hardware monitoring data for the node.
- **BMYCO0005**
  The system board hardware is replaced on the existing node located at RU.
- **BMYCO0007**
  Failed to set and update Processors.
- **BMYCO0008**
  Default password successfully reset to a complex password upon first login.
- **BMYCO0010**
  Firmware upgrade success.
- **BMYCO0011**
  Unable to put the node into maintenance mode as scale cluster status is not healthy.
- **BMYCO0014**
  IBM Fusion software error recovery.
- **BMYCO0015**
  Hardware validation error.
- **BMYCO0016**
  Informational event for updating user about upsize success.
- **BMYCO0017**
  Discovered a new node at RU, link-locations and rack serials or failed to get the location of a new node from rackinfo configmap.
- **BMYCO0019**
  Power operation (on, off) does not succeed within the stipulated time of 75 seconds.
- **BMYCO0020**
  Successfully performed power operation [Gracefulshutdown/GracefulRestart] on the node.
- **BMYCO0021**
  Successfully performed power operation [On] on the node.
- **BMYCO0022**
  This event indicates a validation or configuration failure during node provisioning or management.
- **BMYCO0048**
  Network adapters with addresses not found on the node.
- **BMYCO0049**
  Error in getting the port status from switch CR.
- **BMYCO0055**
  Node is a storage node and contains zero NVMe drives.
- **BMYCO0062**
  Password rotation failed for node.
- **BMYCO0066**
  Password rotation completed successfully for node.
- **BMYCO0067**
  Failed to access the secret for node.
- **BMYCO0068**
  Failed to update the master secret for node.
- **BMYCO0069**
  Password complexity validation failed for the node.
- **BMYCO0210**
  Power off warning about successful power operation execution on the node.
- **BMYCO0150**
  Workload Failover Enabled.
- **BMYCO0151**
  Workload Failover Disabled.
- **BMYCO0161**
  Self Node Remediation Operator Installed
- **BMYCO0162**
  Self Node Remediation Operator Installation Failed
- **BMYCO0171**
  SelfNodeRemediationConfig Updated
- **BMYCO0173**
  SelfNodeRemediationConfig Update Skipped
- **BMYCO0180**
  SNR Template Creation Started
- **BMYCO0181**
  SNR Template Created
- **BMYCO0182**
  SNR Template Creation Failed
- **BMYCO0183**
  SNR Template Already Exists
- **BMYCO0190**
  MHC Creation Started
- **BMYCO0191**
  MHC Created
- **BMYCO0192**
  MHC Already Exists
- **BMYCO0194**
  MHC Creation Failed
- **BMYCO0195**
  MHC Deletion Started

- **BMYC00196**
  MHC Deleted
- **BMYC00197**
  MHC Deletion Failed
- **BMYC00211**
  Power on information about the successful power operation execution on the node.
- **BMYC00300**
  NFD installation is started and is in progress.
- **BMYC00302**
  NFD installation is completed successfully.
- **BMYC00303**
  NFD CR status verification failed.
- **BMYC00304**
  NFD CR is not in a required state.
- **BMYC00305**
  NFD CR is in `Ready` state.
- **BMYC00306**
  NFD operator installation failed from operator hub.
- **BMYC00307**
  NFD operator pod is not in running state.
- **BMYC00308**
  NFD operator pod is in running state.
- **BMYC00309**
  NFD CR creation failed.
- **BMYC00310**
  NFD CR is created successfully.
- **BMYC00311**
  All pods in `openshift-nfd` namespace are not running.
- **BMYC00312**
  All pods in `openshift-nfd` namespace are running.
- **BMYC00313**
  Label `pci-10de` is not present on all NVIDIA GPU nodes. Here the label referred is `feature.node.kubernetes.io/pci-10de` and this is not present on all GPU nodes having NVIDIA GPU cards.
- **BMYC00314**
  Label `pci-10de` is present on all NVIDIA GPU nodes. Here the label referred is `feature.node.kubernetes.io/pci-10de` and this is present on all GPU nodes having NVIDIA GPU cards.
- **BMYC00350**
  NVIDIA GPU operator installation is started and is in progress.
- **BMYC00351**
  NVIDIA GPU operator installation failed.
- **BMYC00352**
  NVIDIA GPU operator installation is completed successfully.
- **BMYC00353**
  GPU cluster policy `gpu-cluster-policy` is created successfully.
- **BMYC00354**
  GPU cluster policy `gpu-cluster-policy` creation failed.
- **BMYC00355**
  GPU cluster policy `gpu-cluster-policy` is in `Ready` state.
- **BMYC00356**
  GPU cluster policy `gpu-cluster-policy` is in `NotReady` state.
- **BMYC00357**
  GPU cluster policy `gpu-cluster-policy` is in `Ignored` state.
- **BMYC00358**
  GPU cluster policy `gpu-cluster-policy` is in `Disabled` state.
- **BMYC00359**
  GPU cluster policy `gpu-cluster-policy` is in `Unknown` state.
- **BMYC00360**
  NVIDIA driver `DaemonSet` check failed.
- **BMYC00361**
  NVIDIA driver `DaemonSet` pods check failed.
- **BMYC00362**
  GPU cluster policy `gpu-cluster-policy` status not yet initialized (pending). Cluster policy status is not initialized yet. Waiting for the operator to update the status.
- **BMYC00363**
  All pods in NVIDIA GPU operator namespace not running.
- **BMYC00364**
  The NVIDIA GPU Operator upgrade is in progress, following an update to the operator subscription to enable automatic approval on the stable channel.
- **BMYC00365**
  NVIDIA GPU operator upgrade failed.
- **BMYC00366**
  Failed to read NVIDIA DCGM exporter dashboard JSON configuration.
- **BMYC00368**
  Failed to apply required labels to configmap `nvidia-dcgm-exporter-dashboard` for NVIDIA DCGM exporter dashboard.
- **BMYC00367**
  Failed to create configmap `nvidia-dcgm-exporter-dashboard` for NVIDIA DCGM exporter dashboard.
- **BMYC00400**
  NMO installation started and is in progress in cluster in the specified namespace.
- **BMYC00401**
  NMO installation completed successfully in the cluster in the specified namespace.

- **BMYCO0402**
  NMO installation failed in the cluster in the specified namespace.
- **BMYCO0403**
  NMO operator pod is not in running state in the cluster in the specified namespace.
- **BMYCO0404**
  NMO operator pod is in running state in the cluster in the specified namespace.
- **BMYFW0001E**
  Failed to put a node into maintenance mode for the firmware update.
- **BMYFW0002E**
  Failed to move a node out of maintenance mode after the firmware upgrade.
- **BMYFW0012E**
  Firmware update operation timed out on the compute node.
- **BMYFW0013E**
  Storage cluster does not turn healthy after a node firmware upgrade.
- **BMYFW0016E**
  Firmware upgrade stalled.
- **BMYFW0032E**
  Failed to move a node into maintenance mode as the storage cluster status is unhealthy.
- **BMYFW0036E**
  Unable to move a node to a maintenance mode due to an unsafe IBM Storage Scale core pod eviction.
- **BMYFW0037E**
  Failed to move a node into a maintenance mode because an MCO rollout is in progress on the cluster.
- **BMYFW0041E**
  Unable to move a node into maintenance mode. Failed to evict the workload pods from a node.
- **BMYPC5030**
  The node is degraded due to an active event.
- **BMYPC6030**
  Node hardware is not reachable.
- **BMYPC6060**
  The node is in a critical state due to an active event.
- **BMYPC6061**
  Bond1 IP address is missing from the node.

Compute Call Home events and error codes

List of Call Home events that you might encounter in the compute.

Compute Call Home events

If you have enabled Call Home feature, then the system automatically collects compute log package and raises a Call Home ticket.

If you have not enabled Call Home feature, collect the Nodes logs and contact IBM support.

- **BMYCO0003**
  This error code is related to scale out operation.
- **BMYCO0006**
  Any critical set failures.
- **BMYCO0009**
  Firmware upgrade failures
- **BMYCO0013**
  IBM Fusion software error.
- **BMYCO0023**
  Failed to upsize the node.

BMYCO0003

This error code is related to scale out operation.

Severity

Critical

Error conditions

The error message can be for the following conditions:

- Failed to connect to the BMC on the Lenovo servers.
- Failed to connect to iDRAC in case of the Dell servers.
- If one or more components are in critical state on the Lenovo servers.

Note: All of the error conditions have built-in retries. Whenever the event persists for a longer duration, support tickets get raised to check the appliance. If the condition is transient, it recovers automatically. Otherwise, IBM Support checks the appliance.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects compute collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes logs and contact IBM support.

BMYCO0006

Any critical set failures.

Severity

Critical

Error conditions

The error message can be for the following conditions:

- Unable to change first time password
- Configuring IPv6 or IPv4 address failed with output
- IMM's new IPv6 or IPv4 address configured is not reachable - Onecli config set failed with output (Boot order and System Boot Mode)
- Onecli failed to clear SysInfoProdIdentifier for SR645 with output
- Onecli failed to set Processors.PreferredIOBus=Preferred
- Onecli failed to set Processors.PreferredIOBusNumber
- Configuring ports failed with output
- Configuring alertentries failed with output
- Configuring power restoration policy failed
- Configuring snmp failed
- Configuring snmpAlerts failed with output
- Configuring password complexity failed with output
- Adding user %s failed with output
- Clearing existing raid config failed for controller
- Unable to config drv mkgood for node
- Unable to config drive mkgood for node
- Error clearing mvstor for node
- Configuring disks in raid mode failed with output
- Configuring disks in raid mode failed with output
- Enabling IPMI over VLAN failed with output
- Onecli config set failed with output (Settling Network Link Type to Ethernet)
- Failed to set Mellanox Network Adapter NetworkLinkType to Ethernet

Note: All of the error conditions have built-in retries. However, there shall be support tickets to check the appliance if the event persists for a longer duration. If the condition is transient, it recovers automatically. Otherwise, IBM Support checks the appliance.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects compute collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes logs and contact IBM support.

BMYCO0009

Firmware upgrade failures

Severity

Critical

Error conditions

The error message can be for the following conditions:

- The system components on the server fail to upgrade, resulting in a silent flashing failure.
- After rebooting, a firmware mismatch occurs because the new firmware version is not applied.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects compute collection set logs and raises a Call Home ticket.

2. If you have not enabled Call Home feature, collect the Nodes logs and contact IBM support.

BMYCO0013

IBM Fusion software error.

Severity

Critical

Error conditions

The error message can be for the following conditions:

- Unable to find if rackSerial number in appliance-info config map
- Unable to read configmap.
- Unable to update kickstart configmap with new node entry
- Storage node upsize on multi-rack is not supported
- Failed to read the kickstart configmap, it could be missing or empty.
- Failed to parse the kickstart, malformed kickstart configmap
- Malformed kickstart configmap
- Internal error: Failed to create the appliance-info configmap.
- Failed to read the appliance-info configmap, it has no data.
- Failed to read kickstart data, either rack serial " + rackSerial + " is missing in the appliance-info or reading/parsing failed
- Failed to perform power operation on the node
- Failed to update scale CR post disk upsize, error
- Failed to detect NVME after performing resetsp
- Failed to read appliance-info configmap data
- Failed to get Rack details from appliance-info configmap
- Failed read kickstart configmap
- Failed to parse kickstart configmap
- Failed to get management switch information from kickstart configmap
- Failed to retrieve information from management switch
- Failed to get the location of a new node from rackinfo configmap
- Failed to get rack-id from appliance-info configmap
- Failed to get management switch credentials from secret
- IBM serial number is not set for the node located at rack unit.
- Failed to connect to the BMC of the node.
- API server on service node is not reachable.

Note: This error code is related to the error conditions that are encountered in the controller due to IMM connectivity or other Kubernetes get operations.
All of the error conditions have built-in retries. However, there shall be support tickets to check the appliance if the event persists for a longer duration. If the condition is transient, it recovers automatically. Otherwise, IBM Support checks the appliance.

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects compute collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes logs and contact IBM support.

BMYCO0023

Failed to upsize the node.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects compute collection set logs and raises a Call Home ticket.
2. If you have not enabled Call Home feature, collect the Nodes logs and contact IBM support.

BMYCO0001

Unable to login, unable to ping, invalid credentials.

Error conditions

The error message can be for the following conditions:

IMM

- Failed to connect to IMM for changing the default user's password.
- Unable to login to IMM of new node.
- Unable to connect or log in to the IMM of new node.
- Unable to create remote OneCli connection with IMM.
- Unable to create remote OneCli connection with IMM using LLA.
- Connected the cable to an empty port of the management switch 1 that is not configured.

Node
Failed to change the default user's password after node configuration.

Other messages

- Onecli executable not found.
- Unable to fetch link local.
- Failed detect system type.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Do a visual inspection of the physical network connectivity.
2. Check that the cables are connected and LEDs show a connected state for the network interface of the node.
3. Check whether any device is connected to an empty port of the management switch 1 that is not configured.
4. If the issue persists, collect the Nodes and Network switches logs and contact IBM support.

BMYCO0004

Failed to get hardware monitoring data for the node.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. This is a temporary error and it gets resolved on its own.
2. If the issue persists for a longer period, collect the compute-monitoring CR status and Nodes logs, and contact IBM support.

BMYCO0005

The system board hardware is replaced on the existing node located at RU.

Severity

Informational

User actions

No action is required.

BMYCO0007

Failed to set and update Processors.

Error conditions

The error message can be for the following conditions:

- Failed to update Processors. PreferredIOBus to be Preferred.
- Failed to set Processors. PreferredIOBusNumber for the node.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Reboot the node manually.
2. If the issue persists after rebooting the node, then collect the Nodes logs and contact IBM support.

BMYCO0008

Default password successfully reset to a complex password upon first login.

Severity

Informational

User actions

No action is required.

BMYCO0010

Firmware upgrade success.

Severity

Informational

User actions

No action is required.

BMYCO0011

Unable to put the node into maintenance mode as scale cluster status is not healthy.

Severity

Critical

Error conditions

The error message can be for the following conditions:

- Failed to move node out of maintenance
- Failed to move the node into maintenance mode because the node cordoning not succeeded
- Failed to move the node into maintenance mode because the node exists in maintenance mode.
- Failed to move node into maintenance mode because the cluster status might not be retrieved.
- Failed to move the node into maintenance mode as because the IBM Fusion Data Foundation cluster status is not healthy.
- Failed to move the node into maintenance mode as because the IBM Storage Scale cluster status is not healthy.
- Unable to move node to maintenance mode. Some other process may be preventing the node from draining, such as an open file handle, an open pod login, or a workload with a Pod Disruption Budget set to zero.
- Failed to move node into maintenance because the MCO rollout is in progress on the cluster.
- Multiple nodes cannot be put in maintenance mode simultaneously, if a node exists in maintenance mode.
- Maintenance mode success.
  Note: This is an informational message and no action is required for this message.

User actions

Do the following steps to diagnose the error:

1. Check whether any node exists in maintenance mode.
2. Move the node out of maintenance and retry.
3. Run the `oc` command to check whether the scale cluster is healthy.

```
oc -n ibm-spectrum-fusion-ns get scales storagemanager -oyaml | grep storageClusterStatus
```

4. If the Scale cluster is not healthy, check and bring it to a healthy state.
5. Run the `oc` command to check whether the Data Foundation cluster is healthy.

```
oc -n ibm-spectrum-fusion-ns get odfcluster odfcluster -o yaml | grep status
```

6. If the Data Foundation cluster is not healthy, check and bring it to a healthy state.

BMYCO0014

IBM Fusion software error recovery.

Error conditions

The error message can be for the following conditions:

- Unable to read port status from any switch to validate the cabling.
- Provided RU number is invalid because it is not sequentially incremental New node should be stacked up in bottom up direction. Last used RU is <RU number>.

Severity

Informational

User actions

No action is required.

BMYCO0015

Hardware validation error.

Error conditions

The error message can be for the following conditions:

Hardware validation

- The new node model is not recognized.
- The provided RU number is invalid because it is not sequentially incremental. Stack up the new node in a bottom-up direction.
- Provide a valid RU.
- Unable to read network CRS, internal runtime error".
- Unable to create remote OneCli connection with IMM using LLA.
- Management switch port is not up for the switch. If the port is not up even after 20 minutes, check cable connectivity.
- Unable to read port status from any switch to validate the cabling.
- High speed switch port is not up for the switch hence retrying connectivity. If the port is not up even after 20 minutes, check cable connectivity.
- Unable to read the port status of switch.
- Unable to read port status from any switch to validate the cabling.
- Unable to read/parse user_config secret.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYCO0016

Informational event for updating user about upsize success.

Severity

Informational

User actions

No action is required.

BMYC00017

Discovered a new node at RU, link-locations and rack serials or failed to get the location of a new node from rackinfo configmap.

Severity

Informational

User actions

This is an informational event when a new node is discovered on a rack. No action is required.

BMYC00019

Power operation (on, off) does not succeed within the stipulated time of 75 seconds.

Severity

Critical

User actions

Check the power state of the node on the IBM Fusion HCI nodes page and perform the operation manually. If the issue persists, collect the Nodes logs and contact IBM support.

BMYC00020

Successfully performed power operation [Gracefulshutdown/GracefulRestart] on the node.

Severity

Warning

User actions

No action is required.

BMYC00021

Successfully performed power operation [On] on the node.

Severity

Informational

User actions

No action is required.

BMYC00022

This event indicates a validation or configuration failure during node provisioning or management.

Severity

Informational

Error conditions

1. Failed to validate UUID.
2. Failed to validate `Pxelabel`.
3. Failed to set privilege of user as did not get IMM credentials from the secret.
4. Failed to set privilege of user `<username>` with output `<command-output>`.
5. Node type from Kickstart (`<data from kickstart>`) and BMC (`<data from bmc>`) for node that is located at `<ru>` are not matching.
6. Updating `nvme` information in kickstart failed for node `<node name>`.
7. Network adapters with addresses `<list of adapter addresses>` not found on node `<node name>`.
8. Error in getting the port status from `switch` custom resource.

User actions

No action is required.

BMYCO0048

Network adapters with addresses not found on the node.

Severity

Warning

User actions

Contact IBM support.

BMYCO0049

Error in getting the port status from switch CR.

Severity

Warning

User actions

Contact IBM support.

BMYCO0055

Node is a storage node and contains zero NVMe drives.

Severity

Warning

User actions

Contact IBM support.

BMYCO0062

Password rotation failed for node.

Severity

Critical

User actions

Contact IBM support.

BMYCO0066

Password rotation completed successfully for node.

Severity

Information

User actions

No action is required.

BMYCO0067

Failed to access the secret for node.

Severity

Critical

User actions

Contact IBM support.

BMYCO0068

Failed to update the master secret for node.

Severity

Critical

User actions

Contact IBM support.

BMYCO0069

Password complexity validation failed for the node.

Severity

Critical

User actions

Contact IBM support.

BMYCO0210

Power off warning about successful power operation execution on the node.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Check the power connection.
2. Check whether the node is down.
3. To proceed, check and re-power.

BMYCO0150

Workload Failover Enabled.

Description

It indicates that the workload failover is successfully enabled for this compute node, and the system is now monitoring the nodes for failures.

Severity

Information

User actions

No action.

BMYCO0151

Workload Failover Disabled.

Description

It indicates that the workload failover is successfully disabled.

Severity

Information

User actions

No action.

BMYCO0161

Self Node Remediation Operator Installed

Description

It indicates that the SNR Operator is successfully installed in the cluster, and the failover remediation component is ready.

Severity

Information

User actions

No action.

BMYCO0162

Self Node Remediation Operator Installation Failed

Description

It indicates that the installation of the Self Node Remediation Operator did not complete successfully.

Severity

Critical

User actions

1. Check OLM logs and ensure all required permissions and catalog sources are available.
2. Check whether the CatalogSource is available and healthy.
   OLM fetches operator metadata from a CatalogSource.

   ```
   oc get catalogsource -n openshift-marketplace
   ```

3. Check the Subscription.

   ```
   oc get subscription self-node-remediation -n openshift-workload-availability -o yaml
   ```

BMYCO0171

SelfNodeRemediationConfig Updated

Description

SelfNodeRemediationConfig successfully updated and synchronized with the desired specification.

Severity

Information

User actions

No action

BMYCO0173

SelfNodeRemediationConfig Update Skipped

Description

SelfNodeRemediationConfig is already up to date. No changes necessary.

Severity

Information

User actions

No action

BMYCO0180

SNR Template Creation Started

Description

The controller has started creating the SelfNodeRemediationTemplate object.

Severity

Information

User actions

No action

BMYC00181

SNR Template Created

Description

SelfNodeRemediationTemplate created successfully; remediation templates are ready.

Severity

Information

User actions

No action

BMYC00182

SNR Template Creation Failed

Description

The controller failed to create the SelfNodeRemediationTemplate.

Severity

Critical

User actions

Check whether the namespace and API group self-node-remediation.medik8s.io are accessible.

Validate controller permissions:

The controller must have the following permissions:

```
- apiGroups: ["self-node-remediation.medik8s.io"]
  resources: ["selfnoderemediationtemplates"]
  verbs: ["get", "list", "create", "update", "patch"]
```

BMYC00183

SNR Template Already Exists

Description

The required SNRTemplate already exists with the desired configuration. Existing template will be used.

Severity

Information

User actions

No action

BMYC00190

MHC Creation Started

Description

The system is initiating the creation of MachineHealthCheck object.

Severity

Information

User actions

No action

BMYCO0191

MHC Created

Description

MachineHealthCheck resource created successfully, enabling compute node health monitoring.

Severity

Information

User actions

No action

BMYCO0192

MHC Already Exists

Description

MachineHealthCheck already exists with the expected configuration.

Severity

Information

User actions

No action

BMYCO0194

MHC Creation Failed

Description

Controller failed to create the MachineHealthCheck resource.

Severity

Critical

User actions

Check if the namespace and API group **`machine.openshift.io`** are accessible. Validate controller permissions.

Ensure that the controller ServiceAccount has the following RBAC rule in the Role or ClusterRole:

```
- apiGroups: ["machine.openshift.io"]
  resources: ["machinehealthchecks"]
  verbs: ["get", "list", "watch", "delete"]
```

BMYC00195

MHC Deletion Started

Description

Initiated deletion of the MachineHealthCheck resource as workload failover has been disabled.

Severity

Information

User actions

No action

BMYC00196

MHC Deleted

Description

Successfully removed MachineHealthCheck. It confirms failover deactivation.

Severity

Information

User actions

No action

BMYC00197

MHC Deletion Failed

Description

Failed to delete the MachineHealthCheck resource.

Severity

Critical

User actions

Check whether the object still exists, verify API connectivity, and ensure the controller has delete permissions.

Check controller permissions:

Ensure the controller ServiceAccount has the following RBAC rule in the Role or ClusterRole:

```
- apiGroups: ["machine.openshift.io"]
  resources: ["machinehealthchecks"]
  verbs: ["get", "list", "watch", "delete"]
```

BMYC00211

Power on information about the successful power operation execution on the node.

Severity

Informational

User actions

No action is required.

BMYCO0300

NFD installation is started and is in progress.

Severity

Informational

User actions

No action is required.

BMYCO0302

NFD installation is completed successfully.

Severity

Informational

User actions

No action is required.

BMYCO0303

NFD CR status verification failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command to check whether node feature discovery CR is present on the cluster:

   ```
   oc get nodefeaturediscovery nfd-instance -o yaml -n openshift-nfd
   ```

2. If the issue persists, then contact IBM support.

BMYCO0304

NFD CR is not in a required state.

Severity

Warning/Critical

User actions

Do the following steps to diagnose the error:

1. If NFD CR is in degraded state.
   a. Run the following command to check if pods are running or not in the namespace `openshift-nfd`.

      ```
      oc get pods -n openshift-nfd
      ```

   b. Run the following command to check events for the pods that are not running:

      ```
      oc describe pod <pod-name> -n openshift-nfd
      ```

2. If NFD CR is in progressing state.
    a. Run the following command to check if pods are running or not in the namespace `openshift-nfd`.

    ```
    oc get pods -n openshift-nfd
    ```

    b. Run the following command to check events for the pods that are not running:

    ```
    oc describe pod <pod-name> -n openshift-nfd
    ```

3. If NFD CR is in unknown state.
    - Run the following command to check the status of NFD CR.

    ```
    oc get nodefeaturediscovery nfd-instance -o json -n openshift-nfd | jq '.status'
    ```

4. If the issue persists, then contact IBM support.

BMYCO0305

NFD CR is in `Ready` state.

Severity

Informational

User actions

No action is required.

BMYCO0306

NFD operator installation failed from operator hub.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following commands to check if subscription in namespace `openshift-nfd` is in `LatestKnown` state:

    ```
    oc get subscription.operators.coreos.com nfd -n openshift-nfd
     oc describe subscription.operators.coreos.com <subscription name> -n openshift-nfd
    ```

2. Run the following command to check if install plan phase is complete:

    ```
    oc get installplan -n openshift-nfd -o jsonpath='{range .items[*]}{.metadata.name}{"\t"}{.status.phase}{"\n"}{end}'
    ```

3. If the issue persists, then contact IBM support.

BMYCO0307

NFD operator pod is not in running state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command to check the events of the `nfd` operator pod:

    ```
    oc describe pod -n openshift-nfd
    ```

2. If the issue persists, then contact IBM support.

BMYCO0308

NFD operator pod is in running state.

Severity

Informational

User actions

No action is required.

BMYCO0309

NFD CR creation failed.

Severity

Informational

User actions

No action is required.

BMYCO0310

NFD CR is created successfully.

Severity

Informational

User actions

No action is required.

BMYCO0311

All pods in **openshift-nfd** namespace are not running.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command:

   `oc describe pod -n openshift-nfd`

2. Run the following command to check each of the pods which are not in running state.

   `oc describe pod <pod-name> -n openshift-nfd`

3. If the issue persists, then contact IBM support.

BMYCO0312

All pods in **openshift-nfd** namespace are running.

Severity

Informational

User actions

No action is required.

BMYC00313

Label **pci-10de** is not present on all NVIDIA GPU nodes. Here the label referred is **feature.node.kubernetes.io/pci-10de** and this is not present on all GPU nodes having NVIDIA GPU cards.

Severity

Critical

User actions

Get the list of all GPU nodes having NVIDIA GPU cards as follows:

1. Run the following command to check the `ComputeMonitoring` CRs for GPU nodes in namespace `ibm-spectrum-fusion-ns`.

   ```
   oc get computemonitorings -n ibm-spectrum-fusion-ns
   ```

2. For each monitoring CR of GPU node, run the following command:

   ```
   oc get computemonitorings <monitoringcr_name>  -n ibm-spectrum-fusion-ns  -o json | jq '.status.nodes[0].processorInfos'
   ```

   Check if manufacturer is NVIDIA for GPU cards such as GPU1, GPU2.

   ```
   processorInfos:
           GPU 1:
             processor:
               Memory: 80GB
               Model: A100
               Name: ThinkSystem NVIDIA A100 80GB PCIe Gen4 Passive GPU
               SlotNo.: slot-2
               DeviceType: GPU
               Manufacturer: NVIDIA
               Generation: Gen4
           GPU 2:
             processor:
               Memory: 80GB
               Model: A100
               Name: ThinkSystem NVIDIA A100 80GB PCIe Gen4 Passive GPU
               SlotNo.: slot-5
               DeviceType: GPU
               Manufacturer: NVIDIA
               Generation: Gen4
           GPU 3:
             processor:
               Memory: 80GB
               Model: A100
               Name: ThinkSystem NVIDIA A100 80GB PCIe Gen4 Passive GPU
               SlotNo.: slot-7
               DeviceType: GPU
               Manufacturer: NVIDIA
               Generation: Gen4
   ```

3. Check if the GPU nodes having NVIDIA GPU cards have this label `"feature.node.kubernetes.io/pci-10de.present" : "true"`.
4. If label does not exist, then contact IBM support.

BMYC00314

Label **pci-10de** is present on all NVIDIA GPU nodes. Here the label referred is **feature.node.kubernetes.io/pci-10de** and this is present on all GPU nodes having NVIDIA GPU cards.

Severity

Informational

User actions

No action is required.

BMYC00350

NVIDIA GPU operator installation is started and is in progress.

Severity

Informational

User actions

No action is required.

BMYC00351

NVIDIA GPU operator installation failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check whether the subscription in the namespace `nvidia-gpu-operator` is in a `AtLatestKnown` state using the following commands:

```
oc get subscription.operators.coreos.com -n nvidia-gpu-operator
oc describe subscription.operators.coreos.com <subscription name> -n nvidia-gpu-operator
```

2. Run the following command to check whether the install plan phase is complete:

```
oc get installplan -n nvidia-gpu-operator -o jsonpath='{range .items[*]}{.metadata.name}{"\t"}{.status.phase}{"\n"}{end}'
```

BMYC00352

NVIDIA GPU operator installation is completed successfully.

Severity

Informational

User actions

No action is required.

BMYC00353

GPU cluster policy `gpu-cluster-policy` is created successfully.

Severity

Informational

User actions

No action is required.

BMYC00354

GPU cluster policy `gpu-cluster-policy` creation failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of the `nvidia-driver-daemonset` pods in the `nvidia-gpu-operator` namespace.
2. Look for pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. Run the following command to get detailed event logs and error messages and contact IBM support.

```
oc describe pod -n nvidia-gpu-operator
```

BMYCO0355

GPU cluster policy `gpu-cluster-policy` is in `Ready` state.

Severity

Informational

User actions

No action is required.

BMYCO0356

GPU cluster policy `gpu-cluster-policy` is in `NotReady` state.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Check the status of the `nvidia-driver-daemonset` pods in the `nvidia-gpu-operator` namespace.
2. Look for pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. Use `kubectl describe pod -n nvidia-gpu-operator` to get detailed event logs and error messages.
4. If the `daemonset` pods are running, then execute one of them and run `nvidia-smi` to confirm successful driver installation and GPU detection.

```
oc exec -it <nvidia-driver-daemonset-pod> -n nvidia-gpu-operator -- nvidia-smi
```

5. If the issue persists, then contact IBM support.

BMYCO0357

GPU cluster policy `gpu-cluster-policy` is in `Ignored` state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of the `nvidia-driver-daemonset` pods in the `nvidia-gpu-operator` namespace.
2. Look for pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. Use `kubectl describe pod -n nvidia-gpu-operator` to get detailed event logs and error messages.
4. If the `daemonset` pods are running, then execute one of them and run `nvidia-smi` to confirm successful driver installation and GPU detection.

```
oc exec -it <nvidia-driver-daemonset-pod> -n nvidia-gpu-operator -- nvidia-smi
```

5. If the issue persists, then contact IBM support.

BMYCO0358

GPU cluster policy `gpu-cluster-policy` is in `Disabled` state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of the **nvidia-driver-daemonset** pods in the **nvidia-gpu-operator** namespace.
2. Look for pods in states like **ImagePullBackOff**, **CrashLoopBackOff**, or **Error**.
3. Use **kubectl describe pod -n nvidia-gpu-operator** to get detailed event logs and error messages.
4. If the **daemonset** pods are running, then execute one of them and run **nvidia-smi** to confirm successful driver installation and GPU detection.

   ```
   oc exec -it <nvidia-driver-daemonset-pod> -n nvidia-gpu-operator -- nvidia-smi
   ```

5. If the issue persists, then contact IBM support.

BMYCO0359

GPU cluster policy **gpu-cluster-policy** is in **Unknown** state.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Check the status of the **nvidia-driver-daemonset** pods in the **nvidia-gpu-operator** namespace.
2. Look for pods in states like **ImagePullBackOff**, **CrashLoopBackOff**, or **Error**.
3. Use **kubectl describe pod -n nvidia-gpu-operator** to get detailed event logs and error messages.
4. If the **daemonset** pods are running, then execute one of them and run **nvidia-smi** to confirm successful driver installation and GPU detection.

   ```
   oc exec -it <nvidia-driver-daemonset-pod> -n nvidia-gpu-operator -- nvidia-smi
   ```

5. If the issue persists, then contact IBM support.

BMYCO0360

NVIDIA driver **DaemonSet** check failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command to get the NVIDIA driver **DaemonSet** deployed by the GPU operator.

   ```
   oc get daemonset -n nvidia-gpu-operator -l openshift.driver-toolkit
   ```

   Example output:

   ```
   oc get daemonset -n  nvidia-gpu-operator  -l openshift.driver-toolkit
   NAME                                             DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELECTOR
   nvidia-driver-daemonset-416.94.202502260030-0   1         1         1       1            1           feature.node.kubernetes.i
   ```

2. Verify that the values are equal:

   ```
   DESIRED = CURRENT
   CURRENT = READY
   ```

3. If the issue persists, then contact IBM support.

BMYCO0361

NVIDIA driver **DaemonSet** pods check failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command to get the expected number of scheduled pods from the NVIDIA driver **DaemonSet**:

```
oc get daemonset -n nvidia-gpu-operator -l openshift.driver-toolkit -o
jsonpath='{.items[0].status.desiredNumberScheduled}'
```

2. Run the following command to get the list of pods created by the NVIDIA driver **DaemonSet**:

```
oc get pods -n nvidia-gpu-operator | grep nvidia-driver-daemonset
```

   Example output:

```
oc get pods -n nvidia-gpu-operator | grep nvidia-driver-daemonset

nvidia-driver-daemonset-416.94.202502260030-0-6x6dd   2/2     Running     0               4d16h
```

3. Compare the number of running pods (from the output in step 2) with the **desiredNumberScheduled** value (from step 1).
4. If both numbers match, it confirms that all expected NVIDIA driver pods are up and running as intended.
5. If the issue persists, then contact IBM support.

BMYC00362

GPU cluster policy **gpu-cluster-policy** status not yet initialized (pending). Cluster policy status is not initialized yet. Waiting for the operator to update the status.

Severity

Informational

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Run the following command to check if the cluster policy status is updated:

```
oc get clusterpolicy -n nvidia-gpu-operator
```

2. After the status shows as ready, describe the cluster policy to confirm conditions:

```
oc describe clusterpolicy gpu-cluster-policy -n nvidia-gpu-operator
```

3. If it is still in pending state, wait for the operator to complete initialization and ensure the **gpu-operator** deployment pod is running.

```
oc get pods -n nvidia-gpu-operator
```

BMYC00363

All pods in NVIDIA GPU operator namespace not running.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Run the following command to verify whether all pods in the NVIDIA GPU operator namespace are running:

```
oc get pods -n nvidia-gpu-operator
```

2. Run the following command to get the events of pods which are not in running state.

```
oc describe pod nvidia-operator-validator-k5dvn  -n nvidia-gpu-operator | grep -A5 -i "Events:"
```

3. If the issue persists, then contact IBM support.

BMYC00364

The NVIDIA GPU Operator upgrade is in progress, following an update to the operator subscription to enable automatic approval on the stable channel.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Run the following command to verify the NVIDIA GPU operator subscription configuration:

   ```
   oc get subscription gpu-operator-certified -n nvidia-gpu-operator -o yaml
   ```

   Ensure that the subscription uses automatic approval and the stable channel.

2. Run the following command to verify the current installed and target CSV versions:

   ```
   oc get subscription gpu-operator-certified -n nvidia-gpu-operator -o jsonpath='{.status.installedCSV}{"\n"}
   {.status.currentCSV}{"\n"}'
   ```

   If the installed CSV and current CSV are different, then an upgrade is still in progress.
3. Run the following command to check pod health in the NVIDIA GPU operator namespace:

   ```
   oc get pods -n nvidia-gpu-operator
   ```

4. If pods are not progressing to `Running` or `Completed` state, inspect operator logs:

   ```
   oc logs -n nvidia-gpu-operator deploy/gpu-operator-controller-manager
   ```

5. If the issue persists, then contact IBM support.

BMYC00365

NVIDIA GPU operator upgrade failed.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Run the following command to verify the NVIDIA GPU operator subscription configuration exists:

   ```
   oc get subscription gpu-operator-certified -n nvidia-gpu-operator
   ```

   If the subscription is missing, then the NVIDIA GPU operator is not installed successfully.

2. Run the following command to inspect the subscription configuration:

   ```
   oc get subscription gpu-operator-certified -n nvidia-gpu-operator -o yaml
   ```

   Verify that `spec.installPlanApproval` is set to `Automatic`, `spec.channel` is set to `stable`, and `spec.startingCSV` is empty.

3. Run the following command to inspect related operator lifecycle manager resources:

   ```
   oc get installplan,csv -n nvidia-gpu-operator
   ```

4. If there are failed install plans or unhealthy CSV resources, inspect their details using the following commands:

   ```
   oc describe installplan -n nvidia-gpu-operator
   oc describe csv -n nvidia-gpu-operator
   ```

5. If the issue persists, then contact IBM support.

BMYC00366

Failed to read NVIDIA DCGM exporter dashboard JSON configuration.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Verify that the operator image or deployment contains the DCGM exporter dashboard asset expected by the controller:

```
oc logs -n ibm-spectrum-fusion-ns deploy/isf-compute-operator-controller-manager -c manager | grep -i "DCGM Exporter Dashboard JSON"
```

If the log shows a file read or embedded asset error, the dashboard JSON resource may be missing or corrupted in the operator build.

2. Restart the operator pod after confirming the deployment image is correct:

```
oc rollout restart deploy/isf-compute-operator-controller-manager -n ibm-spectrum-fusion-ns
```

3. Check the operator logs again for repeated dashboard read failures:

```
oc logs -n ibm-spectrum-fusion-ns deploy/isf-compute-operator-controller-manager -c manager
```

4. If the issue persists, then contact IBM support.

BMYCO0368

Failed to apply required labels to configmap **nvidia-dcgm-exporter-dashboard** for NVIDIA DCGM exporter dashboard.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command to inspect the configmap labels:

```
oc get configmap nvidia-dcgm-exporter-dashboard -n openshift-config-managed --show-labels
```

Verify that the following labels are present and set to **true**:
   - **console.openshift.io/dashboard=true**
   - **console.openshift.io/odc-dashboard=true**
2. If the labels are missing, inspect the full configmap definition:

```
oc get configmap nvidia-dcgm-exporter-dashboard -n openshift-config-managed -o yaml
```

3. Verify that the operator has permission to patch configmaps in the target namespace:

```
oc auth can-i patch configmaps -n openshift-config-managed --as system:serviceaccount:ibm-spectrum-fusion-ns:isf-compute-operator-controller-manager
```

4. Check operator logs for the patch error:

```
oc logs -n ibm-spectrum-fusion-ns deploy/isf-compute-operator-controller-manager -c manager
```

5. If the issue persists, then contact IBM support.

BMYCO0367

Failed to create configmap **nvidia-dcgm-exporter-dashboard** for NVIDIA DCGM exporter dashboard.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following command to verify whether the target configmap already exists:

```
oc get configmap nvidia-dcgm-exporter-dashboard -n openshift-config-managed
```

2. If the configmap already exists, inspect it for conflicts or invalid data:

```
oc get configmap nvidia-dcgm-exporter-dashboard -n openshift-config-managed -o yaml
```

3. Verify that the operator has permission to create configmaps in the target namespace:

```
oc auth can-i create configmaps -n openshift-config-managed --as system:serviceaccount:ibm-spectrum-fusion-ns:isf-compute-operator-controller-manager
```

4. Check the operator logs for the exact create failure returned by the Kubernetes API:

```
oc logs -n ibm-spectrum-fusion-ns deploy/isf-compute-operator-controller-manager -c manager
```

5. If the issue persists, then contact IBM support.

BMYC00400

NMO installation started and is in progress in cluster in the specified namespace.

Severity

Information

User actions

No action is required.

BMYC00401

NMO installation completed successfully in the cluster in the specified namespace.

Severity

Information

User actions

No action required.

BMYC00402

NMO installation failed in the cluster in the specified namespace.

Severity

Critical

User actions

Contact IBM support.

BMYC00403

NMO operator pod is not in running state in the cluster in the specified namespace.

Severity

Critical

User actions

Contact IBM support.

BMYC00404

NMO operator pod is in running state in the cluster in the specified namespace.

Severity

Information

User actions

No action is required.

BMYFW0001E

Failed to put a node into maintenance mode for the firmware update.

It indicates that a node could not be moved into maintenance mode for the firmware upgrade. Check the diagnostic messages that are available on the IBM Fusion user interface and take an appropriate action before you retry the firmware upgrade operation on other nodes.

Severity

Error

Diagnosis and resolution

Do the following steps to diagnose and resolve the issue:

1. Check whether the node is placed in a maintenance mode for a planned work. In this case, bring the node out of maintenance mode after the planned work is complete.
2. If a machine configuration roll out is in progress, then wait for the roll out to complete and the node to return to a ready state.
3. If the node is in a `Notready` state for more than 30 minutes, then it might be an error with the node in the joining cluster. Do the following steps to resolve the issues that are related to the joining cluster:
    a. Run the following command to check for the pending CSRs.

    ```
    oc get csr|grep -i pending
    ```

    b. If you find any pending CSRs, then run the following command to approve them:

    ```
    oc get csr -o go-template='{{range .items}}{{if not .status}}{{.metadata.name}}{{"\n"}}{{end}}{{end}}' | xargs oc adm
    certificate approve
    ```

    c. If you do not find any pending CSRs, then restart the node.
    d. Connect to the node by using the core OS private SSH key and run the following command.

    ```
    ip a
    ```

    e. Recover the node during the following scenarios:
        - IPv4 address assigned to the bond0 interface
        - `br-ex` interface not created
4. The storage cluster health can be degraded. Check the current storage cluster type.
    - If the storage cluster type is IBM Storage Scale or IBM Storage Scale ECE, then see IBM Storage Scale documentation to resolve the issue.
    - If the storage cluster type is Data Foundation, then see Fusion Data Foundation documentation to resolve the issue.

BMYFW0002E

Failed to move a node out of maintenance mode after the firmware upgrade.

It indicates that the node cannot be moved out of maintenance mode. It can happen whenever the `ComputeMaintenance` CR instance deletion does not succeed.

Severity

Error

Diagnosis and Resolution

Check whether the instance of a `ComputeMaintenance` CR is present in the `ibm-spectrum-fusion-ns` namespace and delete it manually. After the `ComputeMaintenance` CR is deleted successfully, the node comes out of the maintenance mode automatically.

BMYFW0012E

Firmware update operation timed out on the compute node.

It indicates that a node firmware upgrade workflow is taking more than expected time to complete the upgrade. Some of the possible reasons for this issue are as follows:

- Moving a node into maintenance might be taking more time than expected due to pending pods for eviction.
- Firmware flash operation might be taking more time to complete than expected.

Severity

Error

Diagnosis and Resolution

Do the following steps to diagnose and resolve the issue:

1. Cancel the firmware upgrade operation. For steps to cancel, see Upgrading node firmware.
2. Wait for firmware upgrade message on the IBM Fusion user interface.
3. On the IBM Fusion user interface, check the timeout reason in the status and take appropriate action.

BMYFW0013E

Storage cluster does not turn healthy after a node firmware upgrade.

It indicates that the storage cluster (Data Foundation or IBM Storage Scale ECE) state is unhealthy after the node rejoined the OpenShift® Container Platform cluster because of a firmware upgrade.

Severity

Error

Diagnosis and resolution

Do the following steps to diagnose and resolve the issue:

1. Wait for the storage cluster to turn healthy.
2. Check the current storage cluster type.
   - If the storage cluster type is IBM Storage Scale or IBM Storage Scale ECE, then see IBM Storage Scale documentation to resolve the issue.
   - If the storage cluster type is Data Foundation, then see Fusion Data Foundation documentation to resolve the issue.

BMYFW0016E

Firmware upgrade stalled.

If a user triggers a node or cluster reboot, the upgrade can fail if the node is unable to drain pods before the reboot occurs.

Severity

Error

User action

As a resolution, drain the pods manually and then resume the upgrade. For more information about draining the pods, see Remediating a maintenance drain failure.

BMYFW0032E

Failed to move a node into maintenance mode as the storage cluster status is unhealthy.

It indicates that the storage cluster status is unhealthy and IBM Fusion cannot move a node to maintenance mode. Proceed with node maintenance or firmware upgrade only when the storage cluster is healthy.

Severity

Error

Diagnosis and Resolution

Do the following steps to diagnose and resolve the issue:

1. Check the current storage cluster type.
   - If the storage cluster type is IBM Storage Scale or IBM Storage Scale ECE, then see IBM Storage Scale documentation to resolve the issue.
   - If the storage cluster type is Data Foundation, then see Fusion Data Foundation documentation to resolve the issue.
2. Retry the firmware upgrade after the storage cluster turns to a healthy state.

BMYFW0036E

Unable to move a node to a maintenance mode due to an unsafe IBM Storage Scale core pod eviction.

It indicates that the node maintenance did not succeed because the eviction of a IBM Storage Scale core pod on the node did not occur. Firmware upgrade workflow continues to drain the pods from the node until workflow timeout. You cannot cancel it because of the following reasons:

- Workflow is in an in-progress state
- Firmware upgrade cannot be initiated on any other node

The pod eviction might succeed because the workflow continuously retries to drain the pod from the node.

Severity

Error

Diagnosis and resolution

Check the reason for the pod eviction. To know more information about the error and debug steps, see IBM Storage Scale documentation. After successful eviction of a scale core pod, the firmware upgrade operation resumes automatically.

BMYFW0037E

Failed to move a node into a maintenance mode because an MCO rollout is in progress on the cluster.

It indicates that a **`MachineConfigPool`** update is in progress. It impacts the upgrade as the next machine configuration rollout in the queue fails because of its dependency on the successful rollout of the ongoing change.

For example, an ongoing rollout is for a registry auth change that is applicable for all the nodes. This change must get rolled out to all nodes before you start the upgrade so that no pod failures due to image pullback errors cause upgrade failures.

Severity

Error

Diagnosis and resolution

Do the following steps to diagnose and resolve the issue:

1. The issue might occur for a short duration whenever the following operations are in progress:
    - Machine config rollout
    - OpenShift® Container Platform upgrade
    - Global Data Platform upgrade

    Wait for these operations to complete, then check whether the **`MachineConfigPool`** is healthy and available.
2. If a pool remains in a degraded state for a long time, then use the following command to find the root cause.

    ```
    oc describe machineconfigpool <poolname>
    ```
3. Check for the problems indicated in the description, and debug further to identify the root cause. The following are some of the possible reasons:
    - A node is stuck in a **`NotReady`** state, restart the node to fix the issue.
    - Though the node completes the roll out, it might not be in a schedulable state. Compare the required and current configurations. If the configurations are the same, manually remove taint of the unscheduleable.
4. After the **`MachineConfigPool`** is available and up to date, the firmware upgrade workflow detects the fix and resumes upgrade automatically.

BMYFW0041E

Unable to move a node into maintenance mode. Failed to evict the workload pods from a node.

It indicates that the node maintenance did not succeeds because there are pods pending for eviction.

Severity

Error

Diagnosis and resolution

Do the following steps to diagnose and resolve the issue:

1. Check the list of pods pending for eviction.
2. If the application pods are safe to drain, manually drain them.
3. After the manual draining of the pods, if the firmware upgrade status is still in progress, then the upgrade proceeds automatically.
4. If the firmware upgrade fails, reattempt the failed firmware upgrade.

    Note: You can modify the pod drain threshold timeout value by updating the field **`MaintenanceWarningTimeout`** in the **`configmap`** and **`compute-operator-config`** under the **`ibm-spectrum-fusion-ns`** namespace. The possible value can be between 5 to 40 and the default time units are in minutes. The modified value is applicable for the fresh firmware upgrade only and not applicable for the upgrade operation that is in progress.

BMYPC5030

The node is degraded due to an active event.

It indicates that the node health is in a warning state.
The reason can be due to the warning state of any of the following hardware components or the loss of redundancy associated with the component.

- Cooling devices
- Power modules
- Local storage
- Processors
- Memory
- System

It is recommended not to start node firmware upgrade until the node status turns healthy.

Severity

Warning

User actions

Collect the System health check logs and contact IBM support.

BMYPC6030

Node hardware is not reachable.

It indicates that the node is not reachable because of the following reasons:

- Fault in the network
- Loss of power supply
- Hardware failure

Do not to start node firmware upgrade until the node is reachable.

Severity

Error

User actions

Collect the System health check logs and contact IBM support.

BMYPC6060

The node is in a critical state due to an active event.

Critical state of the node indicates that node is in the verge of failure, immediate attention is required. The reason might be the failure of any of the following hardware components:

- Cooling devices
- Power modules
- Local storage
- Processors
- Memory
- System

It is recommended not to start the node firmware upgrade until node status turns healthy.

Severity

Critical

User actions

Collect the System health check logs and contact IBM support.

BMYPC6061

Bond1 IP address is missing from the node.

This condition indicates that the Bond1 IP address is missing on a node in the OpenShift® cluster. The IP address must be configured and present to ensure proper network functionality and to avoid undefined or unexpected behavior.

Severity

Error

User actions

Complete the following steps to resolve the issue:

1. Ensure that the node is in the Ready state. Then log in to the node by running the following command:

   `oc debug node/<node-name>`

2. After logging in to the node, switch to the host filesystem within the node:

   `chroot /host`

3. Run the following commands sequentially:

   `nmcli conn down bond1`

   `nmcli conn up bond1`

4. To verify whether the Bond1 IP address is restored, run:

   `ip a s bond1`

   The Bond1 IP address must now be visible on the node, and the pre-check must clear within a few minutes. If the IP address is still not assigned, contact IBM support for further assistance.

Network events error codes

List of all the events that you might encounter in network node.

For Call Home events, see Network Call Home events and error codes.

For the events that you might encounter during a switch firmware upgrade, see Upgrade events and error codes.

For Informational, Warning, and Critical events, see the following:

- **Network Call Home events and error codes**
  List of Call Home events that you might encounter in the network.
- **BMYNW0001**
  Switch interface up.
- **BMYNW0003**
  Switch interface down.
- **BMYNW0005**
  <switch-name:switch IBM Serial Number>: Fan is not OK
- **BMYNW0007**
  <switch-name:switch IBM Serial Number>: Fan is OK
- **BMYNW0009**
  <switch-name:switch IBM Serial Number>: PSU Fan is not OK.
- **BMYNW0011**
  <switch-name:switch IBM Serial Number>: PSU Fan is OK.
- **BMYNW0013**
  <switch-name:switch IBM Serial Number>: PSU temperature is not OK.
- **BMYNW0015**
  <switch-name:switch IBM Serial Number>: PSU temperature is OK.
- **BMYNW0017**
  <switch-name:switch IBM Serial Number>: Ambient temperature is not OK.
- **BMYNW0019**
  <switch-name:switch IBM Serial Number>: Ambient temperature is OK.
- **BMYNW0021**
  <switch-name:switch IBM Serial Number>: CPU Package temperature not OK.
- **BMYNW0023**
  <switch-name:switch IBM Serial Number>: CPU Package temperature OK.
- **BMYNW0025**
  <switch-name:switch IBM Serial Number>: CPU Core temperature not OK.
- **BMYNW0027**
  <switch-name:switch IBM Serial Number>: CPU Core temperature OK.
- **BMYNW0029**
  <switch-name:switch IBM Serial Number>: Port ambient temperature is not OK.
- **BMYNW0031**
  <switch-name:switch IBM Serial Number>: Port Ambient temperature is OK.
- **BMYNW0033**
  <switch-name:switch IBM Serial Number>: Main board ambient temperature is not OK
- **BMYNW0035**
  <switch-name:switch IBM Serial Number>: Main Board Ambient temperature is OK.
- **BMYNW0037**
  <switch-name:switch IBM Serial Number>: ASIC temperature is not OK.

- **BMYNW0039**
  <switch-name:switch IBM Serial Number>: ASIC temperature is OK.
- **BMYNW0041**
  <switch-name:switch IBM Serial Number>: Greater than 90% L3 entries.
- **BMYNW0043**
  <switch-name:switch IBM Serial Number>: L3 entries count normal.
- **BMYNW0045**
  <switch-name:switch IBM Serial Number>: Greater than 90% L2 entries.
- **BMYNW0047**
  <switch-name:switch IBM Serial Number>: L2 entries count normal.
- **BMYNW0049**
  <switch-name:switch IBM Serial Number>: Lesser than 5% memory available.
- **BMYNW0051**
  <switch-name:switch IBM Serial Number>: Memory availability normal.
- **BMYNW0053**
  <switch-name:switch IBM Serial Number>: CumulusLinkUP / JuniperLinkUP - port: <port number>.
- **BMYNW0055**
  <switch-name:switch IBM Serial Number>: CumulusLinkDOWN / JuniperLinkDOWN - port: <port number>.
- **BMYNW0057**
  <switch-name:switch IBM Serial Number> : 15 min Load Average too high (= value).
- **BMYNW0111**
  Failed to login to the switch.
- **BMYNW0112**
  Switch IP address is invalid.
- **BMYNW0113**
  Secret name is missing in the switch CR.
- **BMYNW0114**
  Switch secret is missing.
- **BMYNW0115**
  Failed to add SNMP trap configuration in switch.
- **BMYNW0116**
  Failed to create SNMP exporter resources.
- **BMYNW0117**
  Spine switch is not reachable.
- **BMYNW0118**
  Rack is isolated as both the High Speed Switches of the rack are not reachable.
- **BMYNW0131**
  SNMP exporter image name or version information is missing.
- **BMYNW0132**
  SNMP service is not starting on the switch. The SNMP daemon failed to start or is in a failed state.
- **BMYNW0151**
  SNMP service is in stopped state. This is an informational message indicating SNMP monitoring is not active.
- **BMYNW0201**
  VLAN configuration failed on the switch. The operator cannot create or delete VLANs on the switch.
- **BMYNW0231**
  There are extra VLANs on the switch.
- **BMYNW0301**
  Link configuration failed on the switch. The operator cannot configure network links (bonds, uplinks) on the switch.
- **BMYNW0331**
  Links CR status and Links on the switch are not the same. There is a mismatch between the desired state (CR) and actual state (switch).
- **BMYNW0332**
  Specific link status on CR and link on switch are not the same. Individual link configuration does not match between CR and switch.
- **BMYNW0351**
  There are extra links on the switch.
- **BMYNW0401**
  Rack import failed. The network initialization process might not complete successfully.
- **BMYNW0411**
  Failed to create switch CR.
- **BMYNW0412**
  Failed to create VLAN CR.
- **BMYNW0413**
  Failed to create link CR.
- **BMYNW0414**
  Failed to delete old link custom resource during network reconfiguration.
- **BMYNW0431**
  PolicyUpdateError.
- **BMYNW0451**
  Kickstart config map is missing.
- **BMYNW0452**
  Failed to read kickstart config map data.
- **BMYNW0453**
  The rackinfoconfig config map is missing.
- **BMYNW0454**
  Failed to read rackinfoconfig config map data.
- **BMYNW0455**
  The user config-secret secret is missing.
- **BMYNW0457**
  Failed to read appliance-info config map data.
- **BMYPC6016**
  It is a switch upgrade precheck error. One of the high-speed switches is not healthy.

- **BMYPC6017**
  It is a switch upgrade precheck error. No redundancy for uplink port.
- **BMYPC6034**
  It is a switch upgrade precheck error. One of the switches contains misconfiguration.
- **BMYPC6035**
  Switch is healthy.
- **BMYPC6036**
  Ports related to the uplink are up and running for the mentioned switch.
- **BMYPC6037**
  SNMP exporter pod for the mentioned switch is up and running.
- **BMYPC6038**
  It is a switch upgrade precheck error. Unable to monitor the SNMP exporter pod.
- **BMYPC6039**
  All the configurations on the switches are in a consistent state.
- **BMYPC6047**
  Storage VLAN is tagged on the required ports on the high-speed switches.
- **BMYPC6048**
  It is a switch upgrade precheck error.
- **BMYPC6049**
  Custom VLAN is tagged on the required ports on the high-speed switches.
- **BMYPC6050**
  It is a switch upgrade precheck error.
- **BMYPC6051**
  OpenShift® VLAN is tagged on the required ports on the high-speed switches.
- **BMYPC6052**
  It is a switch upgrade precheck error.
- **Configuration error**
  Message: Encountered a configuration error. Red Hat® OpenShift Container Platform installation did not start successfully.
- **Timed out creating bootstrap VM**
  Message: Bootstrap VM creation on provisioner node (also known as service node) has not started.
- **Bootstrap VM creation did not complete**
  Message: Bootstrap VM creation on the provisioner node (also known as service node) did not complete within the expected time.
- **Failed control nodes inspection**
  Message: Inspection failed for one or more control nodes.
- **Control nodes creation failed**
  Message: Failed to create the Red Hat OpenShift Container Platform cluster control nodes.
- **Cluster bootstrapping failed**
  Cluster bootstrapping fails when one or more control nodes do not boot during installation of Red Hat OpenShift Container Platform.
- **Catalog source pod is healthy**
  Message: IBM Fusion catalog source pod is not found or is not healthy.
- **Operator subscription isf-subscription not found**
  Message: Subscription for IBM Fusion operator not found.
- **Bootstrap VM is not pingable**
  A temporary bootstrap VM is created on provisioner node to create Red Hat OpenShift Container Platform control plane.
- **Configuration error**
  The certificate does not contain a wildcard domain for cluster.

Network Call Home events and error codes

List of Call Home events that you might encounter in the network.

Network Call Home events

If you have enabled Call Home feature, then the system automatically collects network log package and raises a Call Home ticket.

If you have not enabled Call Home feature, collect the Network logs and contact IBM support.

- **BMYNW0101**
  Connectivity check failed for switch.

BMYNW0101

Connectivity check failed for switch.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. If you have enabled Call Home feature, then the system automatically collects network collection set logs and raises a Call Home ticket.

2. If you have not enabled Call Home feature, collect the Network switches logs and contact IBM support.

BMYNW0001

Switch interface up.

Severity

Informational

User actions

No action is required.

BMYNW0003

Switch interface down.

Severity

Warning

User actions

This can be a hardware problem. Do the following steps to diagnose the problem:

1. Check whether ports 31 and 32 are down.
2. Do a physical examination of their cables and switches.
3. If the problem is not resolved, collect the Network switches logs and contact IBM support.

BMYNW0005

<switch-name:switch IBM Serial Number>: Fan is not OK

Error conditions

The error message can be for the following conditions:

- <switch-name:switch IBM Serial Number>: Fan1 Tray 1 Not OK
- <switch-name:switch IBM Serial Number>: Fan2 Tray 1 Not OK
- <switch-name:switch IBM Serial Number>: Fan3 Tray 2 Not OK
- <switch-name:switch IBM Serial Number>: Fan4 Tray 2 Not OK
- <switch-name:switch IBM Serial Number>: Fan5 Tray 3 Not OK
- <switch-name:switch IBM Serial Number>: Fan6 Tray 3 Not OK
- <switch-name:switch IBM Serial Number>: Fan7 Tray 4 Not OK
- <switch-name:switch IBM Serial Number>: Fan8 Tray 4 Not OK
- <switch-name:switch IBM Serial Number>: PSU1 Fan1 Not OK
- <switch-name:switch IBM Serial Number>: PSU2 Fan1 Not OK

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Inspect power connection from PSU to fan tray (1~4), for disconnections or loose connections.
2. If the connection is fine, collect the Network switches logs and contact IBM support.

BMYNW0007

<switch-name:switch IBM Serial Number>: Fan is OK

Error conditions

The error message can be for the following conditions:

- <switch-name:switch IBM Serial Number>: Fan1 Tray 1 OK
- <switch-name:switch IBM Serial Number>: Fan2 Tray 1 OK
- <switch-name:switch IBM Serial Number>:Fan3 Tray 2 OK
- <switch-name:switch IBM Serial Number>: Fan4 Tray 2 OK
- <switch-name:switch IBM Serial Number>: Fan5 Tray 3 OK
- <switch-name:switch IBM Serial Number>: Fan6 Tray 3 OK
- <switch-name:switch IBM Serial Number>: Fan7 Tray 4 OK
- <switch-name:switch IBM Serial Number>: Fan8 Tray 4 OK
- <switch-name:switch IBM Serial Number>: PSU1 Fan1 OK
- <switch-name:switch IBM Serial Number>: PSU2 Fan1 OK

Severity

Informational

User actions

No action is required.

BMYNW0009

<switch-name:switch IBM Serial Number>: PSU Fan is not OK.

Error conditions

The error message can be for the following conditions:

- <switch-name:switch IBM Serial Number>: PSU1 Fan1 Not OK
- <switch-name:switch IBM Serial Number>: PSU2 Fan1 Not OK

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check the LED state.
2. Check loose PWR connection and the cable connecting power PDU.
3. If the issue is not resolved,contact IBM support.

BMYNW0011

<switch-name:switch IBM Serial Number>: PSU Fan is OK.

Error conditions

The error message can be for the following conditions:

- <switch-name:switch IBM Serial Number>: PSU1 Fan1 OK
- <switch-name:switch IBM Serial Number>: PSU2 Fan1 OK

Severity

Informational

User actions

No action is required.

BMYNW0013

<switch-name:switch IBM Serial Number>: PSU temperature is not OK.

Error conditions

The error message can be for the following conditions:

- <switch-name:switch IBM Serial Number>: PSU1 Temperature Not OK
- <switch-name:switch IBM Serial Number>: PSU2 Temperature Not OK

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check the temperature in and around the appliance.
2. Check whether the rack has airflow gap. If airflow problem, then the node also gets heated up.
3. Check if the fan is working.
4. If the problem persists, collect Network switches logs and contact IBM support.

BMYNW0015

<switch-name:switch IBM Serial Number>: PSU temperature is OK.

Error conditions

The error message can be for the following conditions:

- <switch-name:switch IBM Serial Number>: PSU1 Temperature OK
- <switch-name:switch IBM Serial Number>: PSU2 Temperature OK

Severity

Informational

User actions

No action is required.

BMYNW0017

<switch-name:switch IBM Serial Number>: Ambient temperature is not OK.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check the temperature in and around the appliance.
2. Check whether the rack has airflow gap. If airflow problem, then the node also gets heated up.
3. Check if the fan is working.
4. If the problem persists, collect Network switches logs and contact IBM support.

BMYNW0019

<switch-name:switch IBM Serial Number>: Ambient temperature is OK.

Severity

Informational

User actions

No action is required.

BMYNW0021

<switch-name:switch IBM Serial Number>: CPU Package temperature not OK.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check the temperature in and around the appliance.
2. Check whether the rack has airflow gap. If airflow problem, then the node also gets heated up.
3. Check if the fan is working.
4. If the problem persists, collect Network switches logs and contact IBM support.

BMYNW0023

<switch-name:switch IBM Serial Number>: CPU Package temperature OK.

Severity

Informational

User actions

No action is required.

BMYNW0025

<switch-name:switch IBM Serial Number>: CPU Core temperature not OK.

Error conditions

The error message can be for the following conditions:

- hspeed2-racki:racki21: CPU Core 0 temperature not OK
- hspeed2-racki:racki21: CPU Core 1 temperature not OK

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check the temperature in and around the appliance.
2. Check whether the rack has airflow gap. If airflow problem, then the node also gets heated up.
3. Check if the fan is working.
4. If the problem persists, collect Network switches logs and contact IBM support.

BMYNW0027

<switch-name:switch IBM Serial Number>: CPU Core temperature OK.

Error conditions

The error message can be for the following conditions:

- hspeed2-racki:racki21: CPU Core 0 temperature OK
- hspeed2-racki:racki21: CPU Core 1 temperature OK

Severity

Informational

User actions

No action is required.

BMYNW0029

<switch-name:switch IBM Serial Number>: Port ambient temperature is not OK.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check the temperature in and around the appliance.
2. Check whether the fan is working.
3. If the problem persists, collect Network switches logs and contact IBM support.

BMYNW0031

<switch-name:switch IBM Serial Number>: Port Ambient temperature is OK.

Severity

Informational

User actions

No action is required.

BMYNW0033

<switch-name:switch IBM Serial Number>: Main board ambient temperature is not OK

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check the temperature in and around the appliance.
2. Check whether the rack has airflow gap. If airflow problem, then the node also gets heated up.
3. Check whether the fan is working.
4. If the problem persists, collect Network switches logs and contact IBM support.

BMYNW0035

<switch-name:switch IBM Serial Number>: Main Board Ambient temperature is OK.

Severity

Informational

User actions

No action is required.

BMYNW0037

<switch-name:switch IBM Serial Number>: ASIC temperature is not OK.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check the temperature in and around the appliance.
2. Check whether the rack has airflow gap. If airflow problem, then the node also gets heated up.
3. Check if the fan is working.
4. If the problem persists, collect Network switches logs and contact IBM support.

BMYNW0039

<switch-name:switch IBM Serial Number>: ASIC temperature is OK.

Severity

Informational

User actions

No action is required.

BMYNW0041

<switch-name:switch IBM Serial Number>: Greater than 90% L3 entries.

Severity

Critical

User actions

If the problem persists, collect Network switches logs and contact IBM support.

BMYNW0043

<switch-name:switch IBM Serial Number>: L3 entries count normal.

Severity

Informational

User actions

No action is required.

BMYNW0045

<switch-name:switch IBM Serial Number>: Greater than 90% L2 entries.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support .

BMYNW0047

<switch-name:switch IBM Serial Number>: L2 entries count normal.

Severity

Informational

User actions

No action is required.

BMYNW0049

<switch-name:switch IBM Serial Number>: Lesser than 5% memory available.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0051

<switch-name:switch IBM Serial Number>: Memory availability normal.

Severity

Informational

User actions

No action is required.

BMYNW0053

<switch-name:switch IBM Serial Number>: CumulusLinkUP / JuniperLinkUP - port: <port number>.

Severity

Informational

User actions

No action is required.

BMYNW0055

<switch-name:switch IBM Serial Number>: CumulusLinkDOWN / JuniperLinkDOWN - port: <port number>.

Severity

Critical

User actions

If this occurs during maintenance as one time error, ignore. If it does not recover by itself, then run commands to see the status. For the commands, see Running commands for switches. If the error still persists, collect Network switches logs and contact IBM support.

BMYNW0057

<switch-name:switch IBM Serial Number> : 15 min Load Average too high (= value).

Severity

Warning

User actions

If event `BMYNW0057 - <switch-name> <: IBM Serial No>: <no value>` gets generated, then it can be either due to `1 min Load Average too high` or `5 min Load Average too high` event generated from the switch. Collect the Network switches logs and contact IBM support.

For all other values associated to this error code, collect Network switches logs and contact IBM support.

BMYNW0111

Failed to login to the switch.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0112

Switch IP address is invalid.

Severity

Critical

User actions

Contact IBM support for the IP issue.

BMYNW0113

Secret name is missing in the switch CR.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

- Check the kickstart config.
- Collect Network switches logs and installation stage 2 logs.
- Contact IBM support.

BMYNW0114

Switch secret is missing.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

- Check the kickstart config.
- Collect Network switches logs and installation stage 2 logs.
- Contact IBM support.

BMYNW0115

Failed to add SNMP trap configuration in switch.

Severity

Critical

User actions

Collect Network switches logs. Contact IBM support.

BMYNW0116

Failed to create SNMP exporter resources.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0117

Spine switch is not reachable.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0118

Rack is isolated as both the High Speed Switches of the rack are not reachable.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Check the physical connectivity.
2. If the connectivity is fine, collect the Network switches logs and contact IBM support.

BMYNW0131

SNMP exporter image name or version information is missing.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0132

SNMP service is not starting on the switch. The SNMP daemon failed to start or is in a failed state.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0151

SNMP service is in stopped state. This is an informational message indicating SNMP monitoring is not active.

Severity

Informational

User actions

Collect Network switches logs and contact IBM support.

BMYNW0201

VLAN configuration failed on the switch. The operator cannot create or delete VLANs on the switch.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0231

There are extra VLANs on the switch.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0301

Link configuration failed on the switch. The operator cannot configure network links (bonds, uplinks) on the switch.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0331

Links CR status and Links on the switch are not the same. There is a mismatch between the desired state (CR) and actual state (switch).

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Revert the changes done the switch manually.
2. If they are no changes, then collect the Network switches logs and contact IBM support.

BMYNW0332

Specific link status on CR and link on switch are not the same. Individual link configuration does not match between CR and switch.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Revert the changes done the switch manually.
2. If they are no changes, then collect the Network switches logs and contact IBM support.

BMYNW0351

There are extra links on the switch.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0401

Rack import failed. The network initialization process might not complete successfully.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0411

Failed to create switch CR.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0412

Failed to create VLAN CR.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0413

Failed to create link CR.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYNW0414

Failed to delete old link custom resource during network reconfiguration.

Severity

Warning/Informational

User actions

Collect Network switches logs and contact IBM support.

BMYNW0431

PolicyUpdateError.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0451

Kickstart config map is missing.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0452

Failed to read kickstart config map data.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0453

The rackinfoconfig config map is missing.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0454

Failed to read rackinfoconfig config map data.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYNW0455

The user config-secret secret is missing.

Severity

Warning

User actions

Collect the Network switches logs and contact IBM support.

BMYNW0457

Failed to read appliance-info config map data.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYPC6016

It is a switch upgrade precheck error. One of the high-speed switches is not healthy.

Severity

Error

Problem description

It indicates that one of the high-speed switches is not healthy. The issue causes because the IBM Fusion network operator is unable to log in to the respective switch. Reason for unsuccessful login might be as follows:

1. Switch down
2. Switch IP removed on the switch
3. Switch credentials changed.

User actions

Collect Network switches logs and contact IBM support.

BMYPC6017

It is a switch upgrade precheck error. No redundancy for uplink port.

Severity

Error

Problem description

It indicates that no redundancy for uplink port. The uplink port of one of the high-speed switches is either down or not configured. An upgrade in such scenarios could cause network connectivity issues.

User actions

Collect Network switches logs and contact IBM support.

BMYPC6034

It is a switch upgrade precheck error. One of the switches contains misconfiguration.

Severity

Error

Problem description

It indicates that one of the switches contains misconfiguration. The missing configuration might be a VLAN not tagged on the required ports on the switch. It causes if a VLAN is added during manual configuration of the switches.

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command to see the VLANs that are not tagged on the required ports on the switch.

   ```
   oc get switch -o yaml | grep -A 2 "is not tagged"
   ```

2. If the issue persists, collect the Network switches logs and contact IBM support.

BMYPC6035

Switch is healthy.

Severity

Informational

User actions

No action is required.

BMYPC6036

Ports related to the uplink are up and running for the mentioned switch.

Severity

Informational

User actions

No action is required.

BMYPC6037

SNMP exporter pod for the mentioned switch is up and running.

Severity

Informational

User actions

No action is required.

BMYPC6038

It is a switch upgrade precheck error. Unable to monitor the SNMP exporter pod.

Severity

Error

Problem description

It indicates that unable to monitor the SNMP exporter pod. It occurs if the SNMP export pod for the related switch is not created.

User actions

Collect Network switches logs and contact IBM support.

BMYPC6039

All the configurations on the switches are in a consistent state.

Severity

Informational

User actions

No action is required.

BMYPC6047

Storage VLAN is tagged on the required ports on the high-speed switches.

Severity

Informational

User actions

No action is required.

BMYPC6048

It is a switch upgrade precheck error.

Severity

Error

Problem description

It indicates that the storage VLAN is not tagged on the required ports of the switch on one of the high-speed switches. It blocks the triggering of the switch upgrade.

User actions

Collect the Network switches logs and contact IBM support.

BMYPC6049

Custom VLAN is tagged on the required ports on the high-speed switches.

Severity

Informational

User actions

No action is required.

BMYPC6050

It is a switch upgrade precheck error.

Severity

Error

Problem description

It indicates that one of the custom VLAN is not tagged on the required ports of the switch on one of the high-speed switches. It blocks the triggering of the switch upgrade.

User actions

Collect the Network switches logs and contact IBM support.

BMYPC6051

OpenShift® VLAN is tagged on the required ports on the high-speed switches.

Severity

Informational

User actions

No action is required.

BMYPC6052

It is a switch upgrade precheck error.

Severity

Error

Problem description

It indicates that OpenShift® VLAN is not tagged on the required ports of the switch on one of the high-speed switches. It blocks the triggering of the switch upgrade.

User actions

Collect the Network switches logs and contact IBM support.

Configuration error

Message: Encountered a configuration error. Red Hat® OpenShift® Container Platform installation did not start successfully.

Diagnosis

1. As a kni user on a provisioner node (also known as service node), review the log file **`/home/kni/logs/install_<timestamp>.log`**.
2. Scroll to end of file and check whether it reports any failure. Following is an example of a successful run, however, if you find value for **`failed`** not equal to `0`, then check the log for a step that failed.

```
PLAY RECAP *********************************************************************
localhost                  : ok=68   changed=3    unreachable=0    failed=0    skipped=45   rescued=0    ignored=0
```

Next actions

1. Take appropriate action to resolve the reported problem in the log.
2. In the installation user interface, click **Retry** to restart the installation.

Timed out creating bootstrap VM

Message: Bootstrap VM creation on provisioner node (also known as service node) has not started.

Diagnosis

1. As a kni user, on the provisioner node, review the log file **/home/kni/clusterconfigs/.openshift_install.log**.
2. Rectify the reported problem.

Next actions

In the installation user interface, click **Retry** to restart the installation.

If problem persists, collect all files in **/home/kni/logs** and **/home/kni/clusterconfigs**, and contact IBM support.

Bootstrap VM creation did not complete

Message: Bootstrap VM creation on the provisioner node (also known as service node) did not complete within the expected time.

Diagnosis

1. As a kni user, on the provisioner node, review the log file **/home/kni/clusterconfigs/.openshift_install.log**.
2. Rectify the reported problem.

Next actions

In the installation user interface, click Retry to restart the installation.

If problem persists, do the following steps:

1. Collect all files from **/home/kni/logs**, **/home/kni/clusterconfigs**, and **/home/kni/install/results.json**.
2. Contact IBM support.

Failed control nodes inspection

Message: Inspection failed for one or more control nodes.

Diagnosis

Note: Run all commands as a `kni` user from provisioner node.

1. Check the `.openshift_install.log` for errors.
2. If the logs do not provide the necessary information, then continue the steps in this procedure for further diagnostics.
3. Ping node by using the IP and hostname that are reserved for the node in DHCP.
4. If ping does not work, the do the following steps to access the IMM user interface of the Baremetal node:
   Find IPv6 or IPv4 address of node and debug by using the following command:

```
oc get bmh -A -o wide |grep control
openshift-machine-api   control-1-ru2   OK       externally provisioned   isf-rackae4-2bztv-master-0
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:2f95]:623   unknown             true              3h28m
openshift-machine-api   control-1-ru3   OK       externally provisioned   isf-rackae4-2bztv-master-1
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefd:cecd]:623   unknown             true              3h28m
openshift-machine-api   control-1-ru4   OK       externally provisioned   isf-rackae4-2bztv-master-2
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:3031]:623   unknown             true              3h28m
```

5. Note down the IPv6 or IPv4 address of the node you are checking.
6. Go to `/home/kni/isfconfig` folder and open `kickstart-.json` file to find the password of the IMM user `USERID` of node.
7. Open the file and look for the string `OCPRole` with value equal to hostname (for example, control-1-ru3) of the node.
   Sample section for IPv6:

```
  "ipv6ULA": "XXXXXXXXXXXXXXXXXXXXXX",
       "ipv6LLA": "XXXXXXXXXXXXX",
       "serialNum": "J1025PXX",
       "mtm": "7D2XCTO1WW",
       "ibmSerialNumber": "rackae402",
       "ibmMTM": "9155-C01",
       "type": "storage",
       "OCPRole": "control-1-ru3",
       "location": "RU2",
       "name": "IMM_RU2",
       "bootDevice": "/dev/sda",
       "users": [
```

```
        {
           "user": "CEUSER",
           "password": "XXXXX",
           "group": "Administrator",
           "number": 2
        },
        {
           "user": "ISFUSER",
           "password": "XXXXXX",
           "group": "Administrator",
           "number": 3
        },
        {
           "user": "USERID",
           "password": "XXXXXX",
           "group": "Administrator",
           "number": 1
        }
```

Sample section for IPv4:

```
{
       "OCPRole": "compute-1-ru7",
       "bootDevice": "/dev/disk/by-path/pci-0000:01:00.0-nvme-1",
       "ibmMTM": "9155-D10",
       "ibmSerialNumber": "delld07",
       "ipv4": "169.253.1.7",
       "ipv6LLA": "",
       "ipv6ULA": "",
       "location": "RU7",
       "manufacturer": "Dell",
       "mtm": "2668",
       "name": "IMM_RU7",
       "networkCards": [
        {
         "slot-1": [
          "30:3E:A7:10:80:2A",
          "30:3E:A7:10:80:2B",
          "30:3E:A7:10:80:2C",
          "30:3E:A7:10:80:2D"
         ],
         "slot-2": [
          "5C:25:73:C9:94:3E",
          "5C:25:73:C9:94:3F"
         ]
        }
       ],
       "networkInterfaces": [
        {
         "interfaceLeg1": "ens2f0np0",
         "interfaceLeg2": "ens2f1np1",
         "interfaceName": "bond0",
         "interfaceType": "baremetal",
         "macAddress": "5c:25:73:c9:94:3e",
         "secondaryMacAddress": "5C:25:73:C9:94:3F"
        },
        {
         "interfaceLeg1": "eno12409",
         "interfaceLeg2": "eno12419",
         "interfaceName": "bond1",
         "interfaceType": "provisioning",
         "macAddress": "30:3e:a7:10:80:2b",
         "secondaryMacAddress": "30:3E:A7:10:80:2C"
        }
       ],
       "nodeIndex": "6",
       "nvme": [
        {
         "capacity": "3576.983 GB",
         "location": "PCIe SSD in Slot 0 in Bay 1",
         "manufacture": "Micron Technology Inc",
         "productName": "Dell DC NVMe ISE 7450 RI U.2 3.84TB",
         "serialNumber": "2415487E9851"
        },
        {
         "capacity": "3576.983 GB",
         "location": "PCIe SSD in Slot 1 in Bay 1",
         "manufacture": "Micron Technology Inc",
         "productName": "Dell DC NVMe ISE 7450 RI U.2 3.84TB",
         "serialNumber": "2415487EDAAC"
        }
       ],
       "pxeSlotLabel": "NIC.Integrated.1-2-1",
       "secretName": "compute-1-ru7-secret",
       "serialNum": "9Y47NW3",
       "type": "storage",
       "unusedInterfaces": [
        "ens4f0np0",
        "ens4f1np1",
        "ens4f2np2",
        "ens4f3np3"
       ],
       "uuid": "uuid"
      }
     ],
```

8. Run the following command to create a tunnel to the IMM of the node:

```
ssh -N -f -L :1443:[IPv6 or IPv4 address of IMM]:443 -L :3900:[IPv6 or IPv4 address of IMM]:3900 root@localhost
```

9. To access the IMM user interface using the provisioner node IP address, open your browser and access the following IP.

```
https://<provisioner ip>:1443
```

   - user - Enter **USERID**.
   - password - Use the value obtained in the previous step.

10. From the IMM user interface page, open the remote console and check whether the node is up and shows a valid hostname prompt.
   - If the console shows localhost, then review your DHCP/DNS settings to ensure whether correct reservation is made for the node.
   - If the node shows a Red Hat Linux login prompt instead of Core OS, then it infers that the node Base management controller (BMC) is not responsive.

   Do the following steps to resolve the issue:
   a. Identify IMM of the missing node. Here is mapping of nodes to IMM:

```
control-1-ru2 ==> imm_ru2
control-1-ru3 ==> imm_ru3
control-1-ru4 ==> imm_ru4
```

   b. For node(s) that did not show up in the previous steps, run the following sampe IMM command to connect to their IMMs: For example, if node `control-1-ru3` did not show up, then from RU7/provisioner (compute-1-ru7), run the `imm_ru3` command as a `kni` user.
   c. Wait for the successful connection to IMM. c. In the system prompt, run the `resetsp` on IMM prompt.
   d. Wait for 10 minutes.

Next actions

In the installation user interface, click the Retry to restart the installation.

Control nodes creation failed

Message: Failed to create the Red Hat® OpenShift® Container Platform cluster control nodes.

Diagnosis

Note: Run all commands as a `kni` user from provisioner node.

1. Check the `.openshift_install.log` for errors.
2. If the logs do not provide the necessary information, then continue next steps in this procedure for further diagnostics.
3. As a kni user, run `oc get nodes` on the provisioner (service node).
4. Check whether the following three ready nodes are not listed in the output:

```
NAME                                      STATUS   ROLES                        AGE   VERSION
control-1-ru2.rackm03.rtp.raleigh.ibm.com  Ready    control-plane,master,worker  55m   v1.27.3+4aaeaec
control-1-ru3.rackm03.rtp.raleigh.ibm.com  Ready    control-plane,master,worker  56m   v1.27.3+4aaeaec
control-1-ru4.rackm03.rtp.raleigh.ibm.com  Ready    control-plane,master,worker  56m   v1.27.3+4aaeaec
```

5. Identify the node that is not booted. If you do not see these three nodes, then note down the missing node(s) from the output.
6. Identify the IMM of the missing node. Here is mapping of nodes to IMM:

```
control-1-ru2 ==> imm_ru2
control-1-ru3 ==> imm_ru3
control-1-ru4 ==> imm_ru4
```

7. For node(s) that did not show up in the step 4 output, then run the corresponding IMM command to connect to its IMM. For example, if node `control-1-ru3` did not show up, then from RU7/provisioner (compute-1-ru7), run the `imm_ru3` command as a `kni` user.
8. Run `resetsp` command on the IMM prompt.
9. Wait for 10 minutes and retry the installation.
10. If issue persists, ping the control nodes by using the IP and hostname that are reserved for the node in DHCP.
11. If ping is not working, then do the following steps to access the IMM user interface of the Baremetal node:
   a. Run the following command to find the IPv6 or IPv4 address of the node to debug:

```
oc get bmh -A -o wide |grep control
```

   Example output:

```
openshift-machine-api   control-1-ru2   OK       externally provisioned   isf-rackae4-2bztv-master-0
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:2f95]:623   unknown            true             3h28m
openshift-machine-api   control-1-ru3   OK       externally provisioned   isf-rackae4-2bztv-master-1
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefd:cecd]:623   unknown            true             3h28m
openshift-machine-api   control-1-ru4   OK       externally provisioned   isf-rackae4-2bztv-master-2
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:3031]:623   unknown            true             3h28m
```

   b. Note down the IPv6 or IPv4 address of the node you are checking.
   c. Go to `/home/kni/isfconfig/` and open the `kickstart-.json`
   d. Find the password for the IMM user (USERID) of the node and search for the string OCPRole` with value equal to hostname (for example control-1-ru3) of node.
   Sample section for IPv6:

```
```
  "ipv6ULA": "XXXXXXXXXXXXXXXXXXXXXX",
```
```

```
        "ipv6LLA": "XXXXXXXXXXXXX",
        "serialNum": "J1025PXX",
        "mtm": "7D2XCTO1WW",
        "ibmSerialNumber": "rackae402",
        "ibmMTM": "9155-C01",
        "type": "storage",
        "OCPRole": "control-1-ru3",
        "location": "RU2",
        "name": "IMM_RU2",
        "bootDevice": "/dev/sda",
        "users": [
           {
              "user": "CEUSER",
              "password": "XXXXX",
              "group": "Administrator",
              "number": 2
           },
           {
              "user": "ISFUSER",
              "password": "XXXXXX",
              "group": "Administrator",
              "number": 3
           },
           {
              "user": "USERID",
              "password": "XXXXXX",
              "group": "Administrator",
              "number": 1
           }
```

Sample section for IPv4:

```
{
        "OCPRole": "compute-1-ru7",
        "bootDevice": "/dev/disk/by-path/pci-0000:01:00.0-nvme-1",
        "ibmMTM": "9155-D10",
        "ibmSerialNumber": "delld07",
        "ipv4": "169.253.1.7",
        "ipv6LLA": "",
        "ipv6ULA": "",
        "location": "RU7",
        "manufacturer": "Dell",
        "mtm": "2668",
        "name": "IMM_RU7",
        "networkCards": [
         {
          "slot-1": [
           "30:3E:A7:10:80:2A",
           "30:3E:A7:10:80:2B",
           "30:3E:A7:10:80:2C",
           "30:3E:A7:10:80:2D"
          ],
          "slot-2": [
           "5C:25:73:C9:94:3E",
           "5C:25:73:C9:94:3F"
          ]
         }
        ],
        "networkInterfaces": [
         {
          "interfaceLeg1": "ens2f0np0",
          "interfaceLeg2": "ens2f1np1",
          "interfaceName": "bond0",
          "interfaceType": "baremetal",
          "macAddress": "5c:25:73:c9:94:3e",
          "secondaryMacAddress": "5C:25:73:C9:94:3F"
         },
         {
          "interfaceLeg1": "eno12409",
          "interfaceLeg2": "eno12419",
          "interfaceName": "bond1",
          "interfaceType": "provisioning",
          "macAddress": "30:3e:a7:10:80:2b",
          "secondaryMacAddress": "30:3E:A7:10:80:2C"
         }
        ],
        "nodeIndex": "6",
        "nvme": [
         {
          "capacity": "3576.983 GB",
          "location": "PCIe SSD in Slot 0 in Bay 1",
          "manufacture": "Micron Technology Inc",
          "productName": "Dell DC NVMe ISE 7450 RI U.2 3.84TB",
          "serialNumber": "2415487E9851"
         },
         {
          "capacity": "3576.983 GB",
          "location": "PCIe SSD in Slot 1 in Bay 1",
          "manufacture": "Micron Technology Inc",
          "productName": "Dell DC NVMe ISE 7450 RI U.2 3.84TB",
          "serialNumber": "2415487EDAAC"
         }
        ],
        "pxeSlotLabel": "NIC.Integrated.1-2-1",
```

```
        "secretName": "compute-1-ru7-secret",
        "serialNum": "9Y47NW3",
        "type": "storage",
        "unusedInterfaces": [
         "ens4f0np0",
         "ens4f1np1",
         "ens4f2np2",
         "ens4f3np3"
        ],
        "uuid": "uuid"
       }
     ],
```

e. Run the following command to create a tunnel to the IMM of the node:

```
```
ssh -N -f -L :1443:[IPv6 or IPv4 address of IMM]:443 -L :3900:[IPv6 or IPv4 address of IMM]:3900 root@localhost
```
```

f. To access the IMM user interface using the provisioner node IP address, open your browser and access the following IP.

```
 ```
 https://<provisioner ip>:1443
 ```
 - user - Enter `USERID`.
 - password - Use the value obtained in the previous step.
```

g. From the IMM user interface page, open the remote console and check whether the node is up and shows a valid hostname prompt.
- If the console shows localhost, then review your DHCP/DNS settings to ensure whether correct reservation is made for the node.
- If the node shows a Red Hat linux login prompt instead of Core OS, then it infers that the node Base management controller (BMC) is not responsive.

Do the following steps to resolve the issue:

i. Identify IMM of the missing node. Here is mapping of nodes to IMM:

```
```
control-1-ru2 ==> imm_ru2
control-1-ru3 ==> imm_ru3
control-1-ru4 ==> imm_ru4
```
```

ii. For node(s) that did not show up in the previous steps, run the following sample IMM command to connect to their IMMs: For example, if node `control-1-ru3` did not show up, then from RU7/provisioner (compute-1-ru7), run the following command as a `kni` user: `imm_ru3`

iii. Wait for the successful connection to IMM.

iv. In the system prompt, run the `resetsp` on IMM prompt.

v. Wait for 10 minutes.

Next actions

In the installation user interface, click the Retry to restart the installation.

Cluster bootstrapping failed

Cluster bootstrapping fails when one or more control nodes do not boot during installation of Red Hat® OpenShift® Container Platform.

Message: Bootstrapping failed for Red Hat OpenShift Container Platform cluster.

Diagnosis

Note: Run all commands as a `kni` user from provisioner node.

1. Check the `.openshift_install.log` for errors.
2. If the logs do not provide the necessary information, then continue the steps in this procedure for further diagnostics.
3. As a kni user, run `oc get nodes` on the provisioner (service node).
4. Check whether the following three ready nodes are not listed in output:

```
NAME                                        STATUS   ROLES                         AGE   VERSION
control-1-ru2.rackm03.rtp.raleigh.ibm.com   Ready    control-plane,master,worker   55m   v1.27.3+4aaeaec
control-1-ru3.rackm03.rtp.raleigh.ibm.com   Ready    control-plane,master,worker   56m   v1.27.3+4aaeaec
control-1-ru4.rackm03.rtp.raleigh.ibm.com   Ready    control-plane,master,worker   56m   v1.27.3+4aaeaec
```

5. Identify the node that is not booted. If you do not see the listed three nodes, then note down the missing node(s) from the output.
6. Identify the IMM of the missing node. Here is mapping of nodes to IMM:

```
control-1-ru2 ==> imm_ru2
control-1-ru3 ==> imm_ru3
control-1-ru4 ==> imm_ru4
```

7. For node(s) that did not show up in output in step 4, run the corresponding IMM command to connect to its IMM. For example, if node `control-1-ru3` did not show up, then from the RU7/provisioner (compute-1-ru7), run the `imm_ru3` command as `kni` user. Wait for the successful connection to IMM.
8. In the system prompt, run the `resetsp` on IMM prompt.
9. Wait for 10 minutes and retry installation.
10. If none of the nodes show up, ping the control nodes by using the IP and hostname that are reserved for the node in DHCP.
11. If ping does not work, then do the following steps to access the IMM user interface of the Baremetal node:

a. Find IPv6 or IPv4 address of node to debug using the following command:

```
oc get bmh -A -o wide |grep control
```

Example output:

```
openshift-machine-api   control-1-ru2   OK       externally provisioned   isf-rackae4-2bztv-master-0
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:2f95]:623   unknown            true             3h28m
openshift-machine-api   control-1-ru3   OK       externally provisioned   isf-rackae4-2bztv-master-1
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefd:cecd]:623   unknown            true             3h28m
openshift-machine-api   control-1-ru4   OK       externally provisioned   isf-rackae4-2bztv-master-2
ipmi://[fd8c:215d:178e:c0de:a94:efff:fefe:3031]:623   unknown            true             3h28m
```

b. Note down the IPv6 or IPv4 address of the node you are checking.
c. Go to `/home/kni/isfconfig` folder and open `kickstart-.json` file to find the password of the IMM user `USERID` of node
d. Open the file and look for the string `OCPRole` with value equal to hostname (for example, control-1-ru3) of the node.
Sample section for IPv6:

```
"ipv6ULA": "XXXXXXXXXXXXXXXXXXXXXX", "ipv6LLA": "XXXXXXXXXXXXX", "serialNum": "J1025PXX", "mtm": "7D2XCTO1WW",
"ibmSerialNumber": "rackae402", "ibmMTM": "9155-C01", "type": "storage", "OCPRole": "control-1-ru3", "location":
"RU2", "name": "IMM_RU2", "bootDevice": "/dev/sda", "users": [ { "user": "CEUSER", "password": "XXXXX", "group":
"Administrator", "number": 2 }, { "user": "ISFUSER", "password": "XXXXXX", "group": "Administrator", "number": 3 }, {
"user": "USERID", "password": "XXXXXX", "group": "Administrator", "number": 1 }
```

Sample section for IPv4:

```
"ipv4: 165.253.1.7", "serialNum": "J1025PXX", "mtm": "7D2XCTO1WW", "ibmSerialNumber": "rackae402", "ibmMTM": "9155-
C01", "type": "storage", "OCPRole": "control-1-ru3", "location": "RU2", "name": "IMM_RU2", "bootDevice": "/dev/sda",
"users": [ { "user": "CEUSER", "password": "XXXXX", "group": "Administrator", "number": 2 }, { "user": "ISFUSER",
"password": "XXXXXX", "group": "Administrator", "number": 3 }, { "user": "USERID", "password": "XXXXXX", "group":
"Administrator", "number": 1 }
```

e. Run the following command to create a tunnel to the IMM of the node:

```
ssh -N -f -L :1443:[IPv6 or IPv4 address of IMM]:443 -L :3900:[IPv6 or IPv4 address of IMM]:3900 root@localhost
```

f. Access the IMM user interface using the provisioner node IP address, open your browser and access the following IP.

```
https://<provisioner ip>:1443
```

- User - Enter `USERID`.
- Password - Use the value obtained in the previous step.

12. From the IMM user interface page, open the remote console and check whether the node is up and shows a valid hostname prompt.
    - If the console shows localhost, then review your DHCP/DNS settings to ensure whether correct reservation is made for the node.
    - If the node shows a Red Hat linux login prompt instead of Core OS, then it infers that the node Base management controller (BMC) is not responsive.

    Do the following steps to resolve the issue:

    a. Identify IMM of the missing node. Here is mapping of nodes to IMM:

    ```
    control-1-ru2 ==> imm_ru2
    control-1-ru3 ==> imm_ru3
    control-1-ru4 ==> imm_ru4
    ```

    b. For node(s) that did not show up in the previous steps, run commands to connect to their IMMs. For example, if node `control-1-ru3` did not show up, then from RU7/provisioner (compute-1-ru7), run `imm_ru3` command as a kni user:
    c. Wait for the successful connection to IMM.
    d. In the system prompt, run the `resetsp` on IMM prompt.
    e. Wait 10 minutes.

Next actions

In the installation user interface, click the Retry to restart the installation.

Catalog source pod is healthy

Message: IBM Fusion catalog source pod is not found or is not healthy.

Diagnosis

Note: Run all commands as a kni user from the provisioner node.

1. Run the following command to verify whether the catalogsource exists or not.

   ```
   oc -n openshift-marketplace get catsrc
   ```

   Sample output:

   ```
   NAME  DISPLAY                   TYPE             PUBLISHER        AGE
   certified-operators            Certified            Operators        grpc Red Hat      11d
   community-operators            Community            Operators        grpc Red Hat      11d
   fusion-catalog                 IBM Storage Fusion   Catalog          grpc IBM          11d
   isf-data-foundation-catalog    Data Foundation      Catalog          grpc IBM          9d
   redhat-marketplace             Red Hat              Marketplace      grpc Red Hat      11d
   redhat-operators               Red Hat              Operators        grpc Red Hat      11d
   ```

   From the output, check for catalog source with name `fusion-catalog` exists.

2. If catalog source exists, then run the following command to check for the next `catalogsource` pod:

   ```
   oc -n openshift-marketplace get po
   ```

3. Check for a pod that starts with `fusion-catalog-` and is in `Running` state with `1/1` containers in `READY` field.

   For example

   ```
   oc -n openshift-marketplace get po
   ```

   Sample output:

   ```
   NAME                    READY          STATUS         RESTARTS       AGE
   certified-operators-ztp8f                    1/1           Running          0            14h
   community-operators-78vrd                    1/1           Running          0            14h
   fusion-catalog-kbw6h                         1/1           Running          0            2d12h
   isf-data-foundation-catalog-tv6pc            1/1           Running          0            19h
   marketplace-operator-f84d7df4f-f2zrg         1/1           Running          0            2d11h
   redhat-marketplace-8dgwp                     1/1           Running          0            2d11h
   redhat-operators-qj6pk                       1/1           Running          0            14h
   ```

4. If the pod is in `ImagePullBackOff` or `ErrImagePull` state, then describe pod to check the exact reason.

   - If describe shows timeout in pulling image, then check network bandwidth.
   - Offline installation:
     - If the listed error is related to an image manifest not known, then check whether the image is present in the enterprise registry.
     - If describe shows authentication error, then check whether the secret `pull-secret` in namespace `openshift-configs` has the correct credentials for the enterprise registry.

Next actions

Take corrective measures based on the diagnostics and rerun the installation.

Operator subscription isf-subscription not found

Message: Subscription for IBM Fusion operator not found.

Diagnosis

Note: Run all commands as a kni user from provisioner node.

1. Run the following command and in the output check whether the subscription with name isf-subscription exists.

   ```
   oc get sub -n ibm-spectrum-fusion-ns
   ```

   Sample output:

   ```
   NAME               PACKAGE            SOURCE                CHANNEL
   grafana-operator   grafana-operator   community-operators   v4
   isf-subscription   isf-operator       isf-catalog           v2.0
   ```

2. If subscription does not exist, then go to `/home/kni/logs` and check the log file that starts with `installoperator` for any errors:

Next actions

Take corrective measures based on the diagnostics and rerun installation. If issue is not resolved, contact IBM support with all logs from `/home/kni/` and archive that is downloaded from installation user interface.

Bootstrap VM is not pingable

A temporary bootstrap VM is created on provisioner node to create Red Hat OpenShift Container Platform control plane.

Message: Either bootstrap VM is not created or did not receive hostname and IP from DHCP.

Diagnosis

Note: Run all commands from the provisioner node as a kni user.

1. Run the following command on the provisioner node as a `kni` user to check whether the bootstarp VM is created:

   ```
   sudo virsh list
   ```

2. Empty output of previous command indicates that bootstrap VM is not created. To debug further, check the log file `/home/kni/clusterconfigs/.openshift_install.log` for errors.

   ```
   sudo virsh list
   setlocale: No such file or directory
   Id   Name   State
   --------------------
   ```

3. If output from first command is not empty. Example when bootstrap VM is created.

```
sudo virsh list
```

```
setlocale: No such file or directory
Id   Name                                     State
--------------------------------------------------
1    isf-rackae4-2bztv-bootstrap   running
```

4. If VM is listed in step 3, then run the following command to connect to it and check whether it has a valid hostname assigned.

   ```
   ping bootstrap.<subdomain>
   ```

   Here, subdomain is your `<cluster name>.<basedomain name>`.
   - If bootstrap is not pinging, then verify your DHCP and DNS setting for bootstrap VM.
   - An IP reservation in DHCP server must be made for bootstrap VM with correct MAC address as received in TDA sheet.
   - Lookup and reverse lookup records in DNS server for bootstrap VM.
   - Rectify the identified problems.

Next actions

In the installation user interface, click **Retry** to restart the installation.

Configuration error

The certificate does not contain a wildcard domain for cluster.

Diagnosis

1. Provided ingress certificate does not have a wildcard sub-domain of the OpenShift® cluster.
2. Provided certificate must adhere to guidelines from OpenShift outlined here - Understanding the default ingress certificate in *Red Hat OpenShift Container Platform* documentation.

Next actions

1. Obtain a new certificate with valid wildcard OpenShift subdomain in it.
2. In the installation user interface, click Back and provide a new certificate with a private key. The installation restarts with the new certificate.

Metro-DR events and error codes

List of all the events and error codes that you might encounter in Metro-DR setup.

- **Metro-DR event codes**
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Metro-DR event codes

List of all the event codes that you might encounter in Metro-DR setup.

- **BMYDR102**
  Site role is invalid.
- **BMYDR201**
  Network configuration failed.
- **BMYDR202**
  Admin network connectivity is down.
- **BMYDR203**
  Daemon network connectivity is down.
- **BMYDR204**
  Tie breaker connectivity is down.
- **BMYDR205**
  Network latency exceeds the threshold of 40 ms.
- **BMYDR401**
  Failed to retrieve S3 object store Configuration.
- **BMYDR402**
  Failed to update S3 object store Configuration.
- **BMYDR403**
  Failed to configure disaster recovery.
- **BMYDR404**
  Failed to update RamenDR config map with S3 object store profiles.
- **BMYDR405**
  Failed to create VolumeReplicationGroup.

- **BMYDR406**
  Failed to protect application.
- **BMYDR407**
  Failed to relocate application.
- **BMYDR408**
  Failed to create namespace.
- **BMYDR409**
  Failed to update replication State of VolumeReplicationGroup.
- **BMYDR410**
  Failed to update replication label of VolumeReplicationGroup.
- **BMYDR411**
  Failed to sync between Metro-DR sites for relocation/failover status changes.
- **BMYDR412**
  Failed to retrieve remote applications list.
- **BMYDR413**
  Failed to create minio user.
- **BMYDR0414**
  Failed to delete old link custom resource during network reconfiguration.
- **BMYDR415**
  Failed to enable read only policy for minio user.
- **BMYDR416**
  Failed to enable read write policy for minio user.
- **BMYDR417**
  Failed to scale down remote deployment.
- **BMYDR418**
  Failed to update ReplicationState of RemoteVolumeReplicationGroup.
- **BMYDR419**
  Application is successfully relocated.
- **BMYDR420**
  Application is successfully protected.
- **BMYDR421**
  Failed to configure minio user.
- **BMYDR422**
  Failed to scale down local deployment.
- **BMYDR423**
  Failed to delete local persistent volume claims.
- **BMYDR424**
  Failed to delete remote persistent volume claims.
- **BMYDR425**
  Failed to delete VolumeReplicationGroup.
- **BMYDR426**
  Failed to update relocation status of application.
- **BMYDR427**
  Failed to relocate application since application namespace already exists.
- **BMYDR428**
  Failed to relocate application since application CR already exists.
- **BMYDR481**
  Failed to achieve `RecoveryPointObjective` of an application.
- **BMYDR482**
  Application protection degraded.
- **BMYDR483**
  Application planned failover failed.
- **BMYDR484**
  Application forced failover failed.
- **BMYDR485**
  Started restarting replication.
- **BMYDR486**
  Application unenrollment is successful.
- **BMYDR487**
  Metro - DR upgrade failed.
- **BMYDR489**
  PeerVRGStatusUpdationFailed.

BMYDR102

Site role is invalid.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYDR201

Network configuration failed.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYDR202

Admin network connectivity is down.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYDR203

Daemon network connectivity is down.

Severity

Critical

User actions

Collect Network switches logs and contact IBM support.

BMYDR204

Tie breaker connectivity is down.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network connectivity of Tie breaker from primary and secondary site, and resolve if you find any issues.
2. If the issue persists, collect Network switches logs and contact IBM support.

BMYDR205

Network latency exceeds the threshold of 40 ms.

Severity

Warning

User actions

Collect Network switches logs and contact IBM support.

BMYDR401

Failed to retrieve S3 object store Configuration.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR402

Failed to update S3 object store Configuration.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR403

Failed to configure disaster recovery.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR404

Failed to update RamenDR config map with S3 object store profiles.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYDR405

Failed to create VolumeReplicationGroup.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYDR406

Failed to protect application.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR407

Failed to relocate application.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR408

Failed to create namespace.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYDR409

Failed to update replication State of VolumeReplicationGroup.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYDR410

Failed to update replication label of VolumeReplicationGroup.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYDR411

Failed to sync between Metro-DR sites for relocation/failover status changes.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR412

Failed to retrieve remote applications list.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR413

Failed to create minio user.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0414

Failed to delete old link custom resource during network reconfiguration.

Severity

Warning/Informational

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR415

Failed to enable read only policy for minio user.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR416

Failed to enable read write policy for minio user.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR417

Failed to scale down remote deployment.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR418

Failed to update ReplicationState of RemoteVolumeReplicationGroup.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR419

Application is successfully relocated.

Severity

Informational

User actions

No action is required.

BMYDR420

Application is successfully protected.

Severity

Informational

User actions

No action is required.

BMYDR421

Failed to configure minio user.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR422

Failed to scale down local deployment.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR423

Failed to delete local persistent volume claims.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR424

Failed to delete remote persistent volume claims.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR425

Failed to delete VolumeReplicationGroup.

Severity

Critical

User actions

Do the following actions to resolve the issue:

1. Ensure that PVC is not referenced by any pods in the API server.
2. Ensure that PVCs should not contain any `VolumeAttachments`. To check the `VolumeAttachments`, follow the steps:
    a. If the application failover is not successful, then check for the `VolumeAttachments` of the application by using the following command.

    ```
    oc get volumeattachment -n <namespace name> | grep < PV referenced by the PVC of the given application>
    ```

    b. If the `VolumeAttachments` exist, then delete it by using the following command.

    ```
    oc delete volumeattachment -n < namespace name>
    ```

3. If the problem persists, then collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR426

Failed to update relocation status of application.

Severity

Critical

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR427

Failed to relocate application since application namespace already exists.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the compute logs of target cluster in case of failover.
2. If no failover is identified, then collect the Nodes logs and contact IBM support.

BMYDR428

Failed to relocate application since application CR already exists.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the compute logs of target cluster in case of failover.
2. If no failover is identified, then collect the Nodes logs and contact IBM support.

BMYDR481

Failed to achieve `RecoveryPointObjective` of an application.

Severity

Warning

User actions

Collect Nodes and Storage logs and contact IBM support.

BMYDR482

Application protection degraded.

Severity

Warning

User actions

Collect Nodes and Storage logs and contact IBM support.

BMYDR483

Application planned failover failed.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check whether the planned failover triggered from secondary site of an application might leading to this error.
2. Ensure that planned failover needs to be triggered from primary site of an application.

BMYDR484

Application forced failover failed.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check whether the forced failover triggered from secondary site of an application might leading to this error.
2. Ensure that forced failover needs to be triggered from primary site of an application.

BMYDR485

Started restarting replication.

Severity

Informational

User actions

No action is required.

BMYDR486

Application unenrollment is successful.

Severity

Informational

User actions

No action is required.

BMYDR487

Metro - DR upgrade failed.

Severity

Critical

User actions

Collect Nodes and Storage logs and contact IBM support.

BMYDR489

PeerVRGStatusUpdationFailed.

Severity

Critical

User actions

Do the following steps to resolve the issue:

1. Wait for the reconciliation process to complete so that the status of the application in VRG might be updated.
2. Verify whether VRG is created and application status is updated.
3. If the probe persists, then collect Storage logs and contact IBM support.

Metro-DR error codes

List of all the error codes that you might encounter in Metro-DR setup.

- **BMYSS1001I**
  Global Data Platform service installation completed successfully.
- **BMYSS1002I**
  Global Data Platform service installation is in progress.
- **BMYSS1003I**
  Global Data Platform service installation failed.
- **BMYSS1004I**
  Global Data Platform service site is healthy.
- **BMYSS1005E**
  Heartbeats are missing from the site, which could indicate network, resource, or configuration issues.

- **BMYSS1006E**
Many nodes face missing heartbeat events at the site, which could indicate network, resource, or configuration issues.
- **BMYSS1007E**
GPFS is reported down or unavailable at the site.
- **BMYSS1008E**
Many nodes are facing GPFS down events at the site, which could indicate network, resource, or configuration issues.
- **BMYSS1009E**
Replication issues exist at the site.
- **BMYSS1010E**
Quorum nodes cannot communicate with each other that is causing GPFS daemon process to lose quorum.
- **BMYSS1011E**
Site nodes are unable to contact the quorum nodes at another site.
- **BMYSS1012E**
File system is down or unavailable on all nodes at the site.
- **BMYSS1013E**
Many nodes face file system events at the site, which indicate network, resource, or configuration issues.
- **BMYNW1001I**
Metro sync DR network installation not started.
- **BMYNW1003E**
Submariner installation failed.
- **BMYNW1004I**
Submariner installation completed successfully.
- **BMYNW1005I**
Added storage VLAN to link.
- **BMYNW1006E**
Failed to add storage VLAN to uplink.
- **BMYNW1007I**
Added static routes for daemon network.
- **BMYNW1008E**
Failed to add static routes for daemon network.
- **BMYNW1009I**
Successfully restarted scale pods after adding static routes.
- **BMYNW1010E**
Failed to restart scale pods after adding static routes.
- **BMYNW1011I**
Successfully verified daemon network connectivity.
- **BMYNW1012E**
Daemon network is not reachable across sites.
- **BMYNW1013I**
Metro sync DR network installation successful.
- **BMYNW1014E**
Metro sync DR network installation failed.
- **BMYNW1015I**
Admin network connectivity is up.
- **BMYNW1016E**
Admin network connectivity is down.
- **BMYNW1017I**
Daemon network connectivity is up.
- **BMYNW1018E**
Daemon network connectivity is down.
- **BMYNW1019I**
Tie breaker connectivity is up.
- **BMYNW1020E**
Tie breaker connectivity is down.
- **BMYNW1021I**
Successfully uninstalled Submariner.
- **BMYNW1022E**
Failed to uninstall Submariner.
- **BMYDR0002E**
RamenDR installation failed.
- **BMYDR0003E**
Failed to retrieve S3 object store configuration.
- **BMYDR0004E**
Failed to update S3 object store configuration.
- **BMYDR0005E**
Failed to access S3 object store secrets.
- **BMYDR0006E**
Failed to create S3 object store profile.
- **BMYDR0007E**
Failed to update RamenDR config map with S3 object store profiles.
- **BMYDR0008E**
Failed to create VolumeReplicationGroup.
- **BMYDR0009E**
Failed to protect application.
- **BMYDR0010E**
Failed to relocate application.
- **BMYDR0011E**
Failed to create namespace.
- **BMYDR0012E**
Failed to update replication State of VolumeReplicationGroup.

- **BMYDR0013E**
  Failed to update relocation label of VolumeReplicationGroup.
- **BMYDR0014E**
  Failed to sync between Metro-DR sites.
- **BMYDR0015E**
  Failed to retrieve remote applications list.
- **BMYDR0016E**
  Failed to create minio user.
- **BMYDR0017E**
  Failed to create secret for minio user.
- **BMYDR0018E**
  Failed to enable Read Only policy for minio user.
- **BMYDR0019E**
  Failed to enable ReadWrite policy for minio user.
- **BMYDR0020E**
  Failed to scale down remote deployment.
- **BMYDR0021E**
  Failed to update ReplicationState of RemoteVolumeReplicationGroup.
- **BMYDR0025E**
  Failed to scale down local deployment.
- **BMYDR0026E**
  Failed to delete local persistent volume claims.
- **BMYDR0027E**
  Failed to delete remote persistent volume claims.
- **BMYDR0028E**
  Failed to delete VolumeReplicationGroup.
- **BMYDR0029E**
  Failed to update relocation status of application.
- **BMYDR0030E**
  Failed to relocate application since application namespace already exists.
- **BMYDR0031E**
  Failed to relocate application since application CR already exists.
- **BMYDR0103E**
  Minio installation failed.
- **BMYDR0104E**
  Failed to provision persistent volume for Minio object store.
- **BMYDR0105E**
  Failed to create Minio service.
- **BMYDR0106E**
  Failed to create Minio route.
- **BMYDR0107E**
  Failed to create Minio access secret.
- **BMYDR0108E**
  Failed to create Minio deployment.
- **BMYDR0109E**
  Failed to create S3 bucket on Minio object store.
- **BMYDR0120E**
  Failed to read Minio object store credentials.
- **BMYDR0121E**
  Failed to create Minio object store client.
- **BMYDR0130E**
  Failed to list S3 buckets on Minio object store.
- **BMYDR7001E**
  Error in configuring switches.
- **BMYDR7002E**
  Unable to run commands on the switches. Ping is disabled on the sites.
- **BMYDR7003E**
  Daemon network connectivity issue.
- **BMYDR7005E**
  Unable to verify MTU between the sites.
- **BMYDR7007E**
  Unable to clear high-speed switches configuration.

BMYSS1001I

Global Data Platform service installation completed successfully.

Severity

Informational

User actions

No action is required.

BMYSS1002I

Global Data Platform service installation is in progress.

Severity

Informational

User actions

No action is required.

BMYSS1003I

Global Data Platform service installation failed.

Severity

Informational

User actions

No action is required.

BMYSS1004I

Global Data Platform service site is healthy.

Severity

Informational

User actions

No action is required.

BMYSS1005E

Heartbeats are missing from the site, which could indicate network, resource, or configuration issues.

Severity

Error

User actions

All nodes are missing heartbeats and you cannot reach them through TCP communication. Any of the following statements can be the fix for the problem:

- Check network connectivity between the sites.
- Run the following command to check the status of IBM Storage Scale pods on the other side.

  ```
  oc get pods -n ibm-spectrum-scale-operator
  ```

  If pods are down, check power loss on the entire site.
- Check OpenShift® Container Platform cluster to see if worker nodes are down due to hardware failure or power down:

  ```
  oc get nodes
  ```

  Check operator logs to see why pods do not start. For guidance on how to look at the operator logs, see Debugging the IBM Spectrum Scale operator.
- If pods do not start still, check the OpenShift Container Platform dashboard to see if nodes have run out of resources. If pods are in tuning state, then you do not have to check its resources.
- Check whether there are physical network problems. For example, switch broken, submariner issues. In case of physical network problems, then contact your networking team.

If all above steps do not resolve the issue, then collect storage logs and contact IBM support.
For reference, the equivalent IBM Storage Scale error code is **site_missing_heartbeats**.

BMYSS1006E

Many nodes face missing heartbeat events at the site, which could indicate network, resource, or configuration issues.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYSS1007E

GPFS is reported down or unavailable at the site.

Severity

Error

User actions

IBM Storage Scale daemon process is reported as down or unavailable at the site. For example, from site 1, you notice that all nodes in site 2 are down. The node may be up and OS running but IBM Storage Scale is not working.

1. Log in to IBM Storage Scale pods (all pods one after the other) and run the following command to check the health of GPFS services at the site:

   `mmhealth node show`

   The output gives the status of the node. Either the IBM Storage Scale daemon process is down or is in a failed state.
2. If it is a IBM Storage Scale daemon process down event, then run **mmstartup**.
3. If this does not fix the problem, collect Global Data Platform logs and contact IBM support.

For reference, the equivalent IBM Storage Scale error code is **site_gpfs_down**.

BMYSS1008E

Many nodes are facing GPFS down events at the site, which could indicate network, resource, or configuration issues.

Severity

Error

User actions

Some IBM Storage Scale daemon process nodes are down in the site. For example, from site 1, you notice that some nodes in site 2 are down. The node may be up and OS running but IBM Storage Scale is not working.

1. Log in to the IBM Storage Scale pods that have a problem and run the following command to check the health of GPFS services at the site:

   `mmhealth node show`

   The output gives the status of the node. Either the IBM Storage Scale daemon process is down or is in a failed state.
2. If it is a IBM Storage Scale daemon process down event, then run **mmstartup**.
3. If this does not fix the problem, collect Global Data Platform logs and contact IBM support.

For reference, the equivalent IBM Storage Scale error code is **site_gpfs_warn**.

BMYSS1009E

Replication issues exist at the site.

Severity

Error

User actions

Data might not have got replicated properly, so check the status of the disk. Run **mmlsdisk** command for specific filesystem to see if all disks are up (status). If disks are down, then do the following steps:

1. Run the following command to start all disks that are down:

   ```
   mmchdisk  <filesystem name> start -a
   ```

   Check the recovery group status.

   ```
   mmhealth node show
   ```

   Health status gives input on what the problem is. For example, physical disks can be down. If too many physical disks go down, then the recovery group also goes down.

2. Go to the nodes that show degraded and check events. From the events, see reasons and notices for the cause of the degradation.
3. Run command **mmhealth** with specific events based on the previous results to get more information about the degradation. For more information about the command, see mmhealth command.
4. Run the following command to restore the application:

   ```
   mmrestripefs <filessystem name> -r
   ```

For reference, the equivalent IBM Storage Scale error code is `site_degraded_replication`.

BMYSS1010E

Quorum nodes cannot communicate with each other that is causing GPFS daemon process to lose quorum.

Severity

Error

User actions

Sometimes, quorum nodes cannot be reached in the site, so check the connectivity to quorum nodes. Some scale pods can also be designated as quorum nodes.

1. In OpenShift® Container Platform, check whether the pods are running for the quorum nodes:

   ```
   oc get nodes
   ```

2. If they are running, check the health and network connectivity between nodes:

   ```
   mmheath node show
   ```

   For more information about **mmhealth** command, see mmhealth command.

3. Try to ping or run **mmnetverfiy** to check the network.

   ```
   "site_quorum_down":{
   "message":"Quorum down is reported by site {id}",
       "site_quorum_error":{
           "message":"Site {id} is experiencing quorum issues with site {0}.",
   ```

4. If the issue still persists, collect storage logs and contact IBM support.

For reference, the equivalent IBM Storage Scale error code is `site_quorum_down`.

BMYSS1011E

Site nodes are unable to contact the quorum nodes at another site.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYSS1012E

File system is down or unavailable on all nodes at the site.

Severity

Error

User actions

All nodes in the site have a problem with the filesystem. The site is up but filesystem is not available or the other side has no access to data however replication is fine on both sides. Check the health of the file system at the site and ensure that it is properly mounted on all nodes. Some GPFS nodes are down in the site.

1. Log in to pod (one after the other all pods) and run the following command:

   `mmhealth node show`

   The output gives the health status of the node. Either the GPFS is down or is in a failed state.

2. If the status does not indicate anything, run `mmlsmount` to understand if the filesystem is mounted
3. If the filesystem is not mounted, then run the `mmmount` command to mount it.
4. If `mmmount` does not fix the issue, then contact IBM support.

For reference, the equivalent IBM Storage Scale error code is `site_fs_down`.

BMYSS1013E

Many nodes face file system events at the site, which indicate network, resource, or configuration issues.

Severity

Error

User actions

Many nodes face file system events at the site, which indicate network, resource, or configuration issues. Some nodes in the site have a problem with the filesystem. The site is up but filesystem is not available or the other side has no access to data however replication is fine on both sides. Check the health of the file system at the site and ensure that it is properly mounted on all nodes.

1. Log in to pods one by one that have problems and run the following command:

   `mmhealth node show`

   The output gives the status of the node. Either the GPFS is down or is in a failed state.

2. If the status does not indicate anything, run `mmlsmount` to understand if the filesystem is mounted
3. If the filesystem is not mounted, then run the `mmmount` command to mount it.
4. If `mmmount` does not fix the issue, then contact IBM support.

For reference, the equivalent IBM Storage Scale error code is `site_fs_warn`.

BMYNW1001I

Metro sync DR network installation not started.

Severity

Informational

User actions

No action is required.

BMYNW1003E

Submariner installation failed.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYNW1004I

Submariner installation completed successfully.

Severity

Informational

User actions

No action is required.

BMYNW1005I

Added storage VLAN to link.

Severity

Informational

User actions

No action is required.

BMYNW1006E

Failed to add storage VLAN to uplink.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYNW1007I

Added static routes for daemon network.

Severity

Informational

User actions

No action is required.

BMYNW1008E

Failed to add static routes for daemon network.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYNW1009I

Successfully restarted scale pods after adding static routes.

Severity

Informational

User actions

No action is required.

BMYNW1010E

Failed to restart scale pods after adding static routes.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check status of scale core pods and scale installation.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYNW1011I

Successfully verified daemon network connectivity.

Severity

Informational

User actions

No action is required.

BMYNW1012E

Daemon network is not reachable across sites.

Severity

Informational

User actions

No action is required.

BMYNW1013I

Metro sync DR network installation successful.

Severity

Informational

User actions

No action is required.

BMYNW1014E

Metro sync DR network installation failed.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYNW1015I

Admin network connectivity is up.

Severity

Informational

User actions

No action is required.

BMYNW1016E

Admin network connectivity is down.

Severity

Critical

User actions

Collect Nodes logs and contact IBM support.

BMYNW1017I

Daemon network connectivity is up.

Severity

Informational

User actions

No action is required.

BMYNW1018E

Daemon network connectivity is down.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Ping gateways and check whether they are reachable across sites.
2. Customer must verify that their network connectivity is fine.
3. If everything is fine, then check with your network admin to fix the issue.
4. If the issue persists, collect Global Data Platform logs and contact IBM support.

BMYNW1019I

Tie breaker connectivity is up.

Severity

Informational

User actions

No action is required.

BMYNW1020E

Tie breaker connectivity is down.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Ping gateways and check if they are reachable across sites.
2. Customer should verify that their TB connectivity is fine.

BMYNW1021I

Successfully uninstalled Submariner.

Severity

Informational

User actions

No action is required.

BMYNW1022E

Failed to uninstall Submariner.

Severity

Warning

User actions

Collect Nodes logs and contact IBM support.

BMYDR0002E

RamenDR installation failed.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Run the oc command to collect Metro Sync DR CR status.

   ```
   oc get mdr metrodrsite -o yaml
   ```

2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0003E

Failed to retrieve S3 object store configuration.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro Sync DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0004E

Failed to update S3 object store configuration.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro Sync DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0005E

Failed to access S3 object store secrets.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro Sync DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0006E

Failed to create S3 object store profile.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0007E

Failed to update RamenDR config map with S3 object store profiles.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0008E

Failed to create VolumeReplicationGroup.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0009E

Failed to protect application.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0010E

Failed to relocate application.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0011E

Failed to create namespace.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0012E

Failed to update replication State of VolumeReplicationGroup.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0013E

Failed to update relocation label of VolumeReplicationGroup.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0014E

Failed to sync between Metro-DR sites.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro-DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0015E

Failed to retrieve remote applications list.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro Sync DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0016E

Failed to create minio user.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0017E

Failed to create secret for minio user.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0018E

Failed to enable Read Only policy for minio user.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0019E

Failed to enable ReadWrite policy for minio user.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0020E

Failed to scale down remote deployment.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0021E

Failed to update ReplicationState of RemoteVolumeReplicationGroup.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check the network reachability between the two Metro Sync DR racks, and resolve if you find any issues.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0025E

Failed to scale down local deployment.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0026E

Failed to delete local persistent volume claims.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0027E

Failed to delete remote persistent volume claims.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0028E

Failed to delete VolumeReplicationGroup.

Severity

Error

User actions

Do the following actions to resolve the issue:

1. Ensure that PVC is not referenced by any pods in the API server.
2. Ensure that PVCs should not contain any `VolumeAttachments`. To check the `VolumeAttachments`, follow the steps:
   a. If the application failover is not successful, then check for the `VolumeAttachments` of the application by using the following command.

   ```
   oc get volumeattachment -n <namespace name> | grep < PV referenced by the PVC of the given application>
   ```

   b. If the `VolumeAttachments` exist, then delete it by using the following command.

   ```
   oc delete volumeattachment -n < namespace name>
   ```

3. If the problem persists, then collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0029E

Failed to update relocation status of application.

Severity

Error

User actions

Collect the Nodes logs of the both site 1 and site 2 and contact IBM support.

BMYDR0030E

Failed to relocate application since application namespace already exists.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the compute logs of target cluster in case of failover.
2. If no failover is identified, then collect the Nodes logs and contact IBM support.

BMYDR0031E

Failed to relocate application since application CR already exists.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check the compute logs of target cluster in case of failover.
2. If no failover is identified, then collect the Nodes logs and contact IBM support.

BMYDR0103E

Minio installation failed.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0104E

Failed to provision persistent volume for Minio object store.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Check whether any scale or storage events raised and resolve it.
2. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0105E

Failed to create Minio service.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0106E

Failed to create Minio route.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0107E

Failed to create Minio access secret.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0108E

Failed to create Minio deployment.

Severity

Error

User actions

Do the following steps to resolve the error:

1. Run the oc command to identify the minio pod using prefix "ifs-minio".

   ```
   oc get po | grep isf-minio
   ```

2. Run the oc command to check whether any issues related to Image Pull. If yes, resolve it try again.

   ```
   oc describe po isf-minio-xxxx
   ```

3. If the issue persists, collect Nodes logs and contact IBM support.

BMYDR0109E

Failed to create S3 bucket on Minio object store.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0120E

Failed to read Minio object store credentials.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0121E

Failed to create Minio object store client.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR0130E

Failed to list S3 buckets on Minio object store.

Severity

Error

User actions

Collect Nodes logs and contact IBM support.

BMYDR7001E

Error in configuring switches.

Severity

Error

User actions

Collect Network switches logs and contact IBM support.

BMYDR7002E

Unable to run commands on the switches. Ping is disabled on the sites.

Severity

Error

User actions

Contact your network admin to allow ICMP on the network.

BMYDR7003E

Daemon network connectivity issue.

Severity

Error

User actions

Do the following steps to resolve the issue:

1. Contact your network admin to check whether any VLAN mismatch between the IBM Fusion cluster and switches.
2. Contact your network admin to check whether the routing is connected properly between the sites.

BMYDR7005E

Unable to verify MTU between the sites.

Severity

Error

User actions

Do the following steps to resolve the issue:

1. Check whether you disabled the jumbo frame while configuring storage on the site 2. If jumbo frame is disabled, then the MTU is 1500 or else the MTU is 9000 for IBM Fusion racks.
2. Contact your network admin to check the MTU configured on the IBM Fusion racks and the MTU set on the data center.

BMYDR7007E

Unable to clear high-speed switches configuration.

Severity

Error

User actions

Collect Network switches logs and contact IBM support.

Global Data Platform events and error codes

List of all the events and error codes that you might encounter while you work with storage.

For physical disk events, see https://www.ibm.com/docs/en/ess/6.0.0?topic=events-physical-disk.

For NVMe events, see https://www.ibm.com/docs/en/storage-scale/6.0.1?topic=events-nvme.

- **Global Data Platform event codes**
  List of all the event codes that you might encounter while you work with storage.
- **Global Data Platform error codes**
  List of all the error codes that you might encounter while you work with storage.

Global Data Platform event codes

List of all the event codes that you might encounter while you work with storage.

- **BMYSS0017**
  IBM Storage Scale remote cluster is disconnected.
- **BMYSS0018**
  IBM Storage Scale remote cluster is connected.
- **BMYSS0019**
  IBM Storage Scale remote filesystem is disconnected.
- **BMYSS0020**
  IBM Storage Scale remote filesystem is connected.
- **BMYSS0021**
  IBM Storage Scale remote filesystem is added.
- **BMYSS0022**
  IBM Storage Scale remote filesystem is removed.
- **BMYSS1361**
  IBM Storage Scale cluster installation status.
- **BMYSS1362**
  IBM Storage Scale upgrade started.
- **BMYSS1363**
  IBM Storage Scale Scaleout is in progress.
- **BMYSS1364**
  IBM Storage Scale scale up is in progress.
- **BMYSS1365**
  IBM Storage Scale Scaleout is started.
- **BMYSS1366**
  IBM Storage Scale scale up started.
- **BMYSS1367**
  IBM Storage Scale cluster status.
- **BMYSS1368**
  IBM Storage Scale cluster installation is in progress.
- **BMYSS1369**
  IBM Storage Scale cluster installation is completed successfully.
- **BMYSS1370**
  IBM Storage Scale upgrade failed.
- **BMYSS1371**
  IBM Storage Scale upgrade is in progress.
- **BMYSS1372**
  IBM Storage Scale upgrade completed successfully.
- **BMYSS1373**
  IBM Storage Scale scale up completed successfully.
- **BMYSS1374**
  IBM Storage Scale scaleout completed successfully.
- **BMYSS1375**
  Metro sync DR scale installation started.
- **BMYSS1376**
  Metro sync DR scale installation is in progress.
- **BMYSS1377**
  Metro sync DR scale installation is completed successfully.
- **BMYSS1378**
  IBM Storage Scale status is healthy.
- **BMYSS1379**
  IBM Storage Scale status is degraded.
- **BMYSS1380**
  IBM Storage Scale status is critical.
- **BMYSS1381**
  IBM Storage Scale status is failed.
- **BMYSS1382**
  All ECE or CSI pods are not running properly.

- **BMYSS1383**
  Rack does not have sufficient number of storage nodes for scale install.
- **BMYSS1384**
  Scaleout process failed for the nodes. New nodes are of mixed type. The Scaleout is supported only if the provided nodes are of the same type.
- **BMYSS1386**
  IBM Storage Scale disk count mismatch.
- **BMYSS1387**
  IBM Storage Scale service is not active on storage or non-storage pods .
- **BMYSS1389**
  Filesystem is not mounted on all the storage nodes.
- **BMYSS1390**
  IBM Storage Scale upgrade prerequisite check failed.
- **BMYSS1391**
  IBM Storage Scale Compute to AFM node conversion completed.
- **BMYSS1392**
  IBM Storage Scale Compute to AFM node conversion is in-progress.
- **BMYSS1393**
  IBM Storage Scale Compute to AFM node conversion failed.
- **BMYSS1394**
  IBM Storage Scale upgrade is in progress and pods are failed to evict.
- **BMYSS1395**
  The CSI Pods are not in running state in the `ibm-spectrum-scale-csi` namespace.
- **BMYSS1396**
  Conversion of the Compute to AFM node is not required because the requested node is AFM node.
- **BMYSS1397**
  Conversion of the Compute to AFM node failed due to an incorrect node name.
- **BMYSS1399**
  The scale operator pod is not running in the `ibm-spectrum-scale-operator` namespace.
- **BMYSS1400**
  IBM Storage Scale upgrade is in progress and pods are failed to remove.
- **BMYSS1401**
  Disk capacity does not match on all the nodes.
- **BMYSS1402**
  All disk capacity on a node does not match.
- **BMYSS1403**
  Rack does not have sufficient number of storage nodes in ready state for the scale install.
- **BMYSS1404**
  Labelled nodes for RG are less than required minimum supported nodes.
- **BMYSS1405**
  Recovery group is not created successfully.
- **BMYSS1406**
  Filesystem is not created successfully.
- **BMYPC5018**
  Global Data Platform status is degraded.
- **BMYPC6018**
  Global Data Platform status is in a critical or down state.
- **BMYPC6024**
  A scale up operation on the Red Hat® OpenShift® cluster is in progress.
- **BMYPC6025**
  A scale out operation on the Red Hat OpenShift cluster is in progress.
- **BMYPC6026**
  The current version of the Global Data Platform is already up to date.
- **BMYPC6031**
  A scale up operation on the Red Hat OpenShift cluster is triggered.
- **BMYPC6032**
  A scale out operation on the Red Hat OpenShift cluster is triggered.
- **BMYPC6033**
  The OpenShift Container Platform version is not compatible with the Global Data Platform version.

BMYSS0017

IBM Storage Scale remote cluster is disconnected.

Severity

Warning

User actions

Collect Global Data Platform logs and contact IBM support.

BMYSS0018

IBM Storage Scale remote cluster is connected.

Severity

Informational

User actions

No action is required.

BMYSS0019

IBM Storage Scale remote filesystem is disconnected.

Severity

Warning

User actions

Collect Global Data Platform logs and contact IBM support.

BMYSS0020

IBM Storage Scale remote filesystem is connected.

Severity

Informational

User actions

No action is required.

BMYSS0021

IBM Storage Scale remote filesystem is added.

Severity

Informational

User actions

No action is required.

BMYSS0022

IBM Storage Scale remote filesystem is removed.

Severity

Informational

User actions

No action is required.

BMYSS1361

IBM Storage Scale cluster installation status.

Severity

Informational

User actions

No action is required.

BMYSS1362

IBM Storage Scale upgrade started.

Severity

Informational

User actions

No action is required.

BMYSS1363

IBM Storage Scale Scaleout is in progress.

Severity

Informational

User actions

No action is required.

BMYSS1364

IBM Storage Scale scale up is in progress.

Severity

Informational

User actions

No action is required.

BMYSS1365

IBM Storage Scale Scaleout is started.

Severity

Informational

User actions

No action is required.

BMYSS1366

IBM Storage Scale scale up started.

Severity

Informational

User actions

No action is required.

BMYSS1367

IBM Storage Scale cluster status.

Severity

Informational

User actions

No action is required.

BMYSS1368

IBM Storage Scale cluster installation is in progress.

Severity

Informational

User actions

No action is required.

BMYSS1369

IBM Storage Scale cluster installation is completed successfully.

Severity

Informational

User actions

No action is required.

BMYSS1370

IBM Storage Scale upgrade failed.

Severity

Critical

User actions

Contact IBM support for Global data platform upgrade assistance.

BMYSS1371

IBM Storage Scale upgrade is in progress.

Severity

Informational

User actions

No action is required.

BMYSS1372

IBM Storage Scale upgrade completed successfully.

Severity

Informational

User actions

No action is required.

BMYSS1373

IBM Storage Scale scale up completed successfully.

Severity

Informational

User actions

No action is required.

BMYSS1374

IBM Storage Scale scaleout completed successfully.

Severity

Informational

User actions

No action is required.

BMYSS1375

Metro sync DR scale installation started.

Severity

Informational

User actions

No action is required.

BMYSS1376

Metro sync DR scale installation is in progress.

Severity

Informational

User actions

No action is required.

BMYSS1377

Metro sync DR scale installation is completed successfully.

Severity

Informational

User actions

No action is required.

BMYSS1378

IBM Storage Scale status is healthy.

Severity

Informational

User actions

No action is required.

BMYSS1379

IBM Storage Scale status is degraded.

Severity

Informational

User actions

1. Run the following command to collect must-gather logs:

```
oc adm must-gather --image=icr.io/cpopen/ibm-spectrum-scale-must-gather:v5.2.3.0
```

2. Contact IBM support.

BMYSS1380

IBM Storage Scale status is critical.

Severity

Informational

User actions

No action is required.

BMYSS1381

IBM Storage Scale status is failed.

Severity

Informational

User actions

No action is required.

BMYSS1382

All ECE or CSI pods are not running properly.

Severity

Warning

User actions

Do the following steps to diagnose the problem:

1. Check whether the ECE or CSI pods are in ready status.
2. Check and resolve this issue before you upgrade scale operation.

BMYSS1383

Rack does not have sufficient number of storage nodes for scale install.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check whether the storage nodes are sufficient for install.
2. Add storage nodes if they are not sufficient for a scale installation.

BMYSS1384

Scaleout process failed for the nodes. New nodes are of mixed type. The Scaleout is supported only if the provided nodes are of the same type.

Severity

Critical

User actions

Provide nodes of the same type for the Scaleout to be successful.

BMYSS1386

IBM Storage Scale disk count mismatch.

Severity

Critical

User actions

NVMe disk count is not same on all the nodes. Ensure that NVME disk count is same on all the nodes.

BMYSS1387

IBM Storage Scale service is not active on storage or non-storage pods .

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Run the following command to change the project to `ibm-spectrum-scale`.

   `oc project ibm-spectrum-scale`

2. Run the following command to get the pods in the `ibm-spectrum-scale` namespace.

   `oc get pods -n ibm-spectrum-scale`

3. Run the following command on the scale core pod that is in a running state for remote terminal connection.

   `oc rsh <podname>`

4. Run the following command to check the scale service state on all scale core pods.

   `mmgetstate -a`

BMYSS1389

Filesystem is not mounted on all the storage nodes.

Severity

Warning

User actions

Do the following steps to diagnose the problem:

1. Run the following command to change the project to `ibm-spectrum-scale`.

   `oc project ibm-spectrum-scale`

2. Run the following command to get the pods in the `ibm-spectrum-scale` namespace.

   `oc get pods -n ibm-spectrum-scale`

3. Run the following command on the scale core pod that is in a running state for remote terminal connection.

   `oc rsh <podname>`

4. Run the following command to check whether filesystem is mounted on all nodes.

   `mmlsmount all`

BMYSS1390

IBM Storage Scale upgrade prerequisite check failed.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Run the following command to check the MCP rollout status.

```
oc get mcp
```

2. If the MCP rollout status is in progress, then wait for it to finish.

BMYSS1391

IBM Storage Scale Compute to AFM node conversion completed.

Severity

Informational

User actions

No action is required.

BMYSS1392

IBM Storage Scale Compute to AFM node conversion is in-progress.

Severity

Informational

User actions

No action is required.

BMYSS1393

IBM Storage Scale Compute to AFM node conversion failed.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Check whether the Global Data Platform is in a healthy state.
2. Ensure that the requested node for conversion is a compute node.
3. Ensure that a valid node name is provided for Compute to AFM node conversion.

BMYSS1394

IBM Storage Scale upgrade is in progress and pods are failed to evict.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Run the following command to get all the pods scheduled on the node.

   ```
   oc get pods --field-selector spec.nodeName=<node name> --all-namespaces
   ```

2. Delete the pods that are failed to evict.

BMYSS1395

The CSI Pods are not in running state in the **ibm-spectrum-scale-csi** namespace.

Severity

Warning

User actions

Do the following steps to diagnose the problem:

1. Run the following command to get all the pods in the `ibm-spectrum-scale-csi` namespace.

   ```
   oc get pods -n ibm-spectrum-scale-csi
   ```

2. Check whether all the pods are in a running state and containers of each pod are in a ready state.

BMYSS1396

Conversion of the Compute to AFM node is not required because the requested node is AFM node.

Severity

Informational

User actions

No action is required.

BMYSS1397

Conversion of the Compute to AFM node failed due to an incorrect node name.

Severity

Critical

User actions

Ensure that you provide the correct node name to convert the node from Compute to AFM.

BMYSS1399

The scale operator pod is not running in the **ibm-spectrum-scale-operator** namespace.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Run the following command to get all the pods in the `ibm-spectrum-scale-operator` namespace.

   ```
   oc get pods -n ibm-spectrum-scale-operator
   ```

2. Check whether all the pods are in a running state and containers of each pod are in a ready state.

BMYSS1400

IBM Storage Scale upgrade is in progress and pods are failed to remove.

Severity

Critical

User actions

Do the following steps to diagnose the problem:

1. Run the following command to get all the pods scheduled on the node.

   ```
   oc get pods --field-selector spec.nodeName=<node name> --all-namespaces
   ```

2. Delete the pods that are failed to remove.

BMYSS1401

Disk capacity does not match on all the nodes.

Severity

Critical

User actions

Do the following steps to resolve the problem:

1. Run the following command to get the names of all nodes in the cluster.

   ```
   oc get nodes
   ```

2. Run the following command for each node to check the disk capacity.

   ```
   oc debug node/<node_name> -- lsblk
   ```

   Example output:

   ```
   Temporary namespace openshift-debug-nrcbd is created for debugging node...
   Starting pod/compute-1-ru5rackm03rtpraleighibmcom-debug-8b2gq ...
   To use host binaries, run `chroot /host`
   NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
   sda           8:0    0 894.2G  0 disk
   |-sda1        8:1    0     1M  0 part
   |-sda2        8:2    0   127M  0 part
   |-sda3        8:3    0   384M  0 part /host/boot
   `-sda4        8:4    0 893.7G  0 part /host/sysroot
   nvme0n1     259:1    0     7T  0 disk
   `-nvme0n1p1 259:2    0     7T  0 part
   nvme1n1     259:4    0     7T  0 disk
   `-nvme1n1p1 259:5    0     7T  0 part
   ```

3. Confirm that each node in the cluster contains the same disk configuration and capacity.
4. If you find any discrepancies in disk sizes, then ensure that should be addressed and maintain uniform capacity on all nodes.

BMYSS1402

All disk capacity on a node does not match.

Severity

Critical

User actions

Do the following steps to resolve the problem:

1. Run the following command to get the names of all nodes in the cluster.

   ```
   oc get nodes
   ```

2. Run the following command for each node to check the disk capacity.

   ```
   oc debug node/<node_name> -- lsblk
   ```

   Example output:

   ```
   Temporary namespace openshift-debug-nrcbd is created for debugging node...
   Starting pod/compute-1-ru5rackm03rtpraleighibmcom-debug-8b2gq ...
   To use host binaries, run `chroot /host`
   NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
   sda           8:0    0 894.2G  0 disk
   |-sda1        8:1    0     1M  0 part
   |-sda2        8:2    0   127M  0 part
   |-sda3        8:3    0   384M  0 part /host/boot
   `-sda4        8:4    0 893.7G  0 part /host/sysroot
   ```

```
nvme0n1      259:1    0      7T  0 disk
`-nvme0n1p1 259:2    0      7T  0 part
nvme1n1      259:4    0      7T  0 disk
`-nvme1n1p1 259:5    0      7T  0 part
```

3. Confirm that each node in the cluster contains the same disk configuration and capacity.
4. If you find any discrepancies in disk sizes, then ensure that should be addressed and maintain uniform capacity on all nodes.

BMYSS1403

Rack does not have sufficient number of storage nodes in ready state for the scale install.

Severity

Critical

User actions

Do the following steps to resolve the problem:

1. Run the following command to get the names of all nodes in the cluster.

   `oc get nodes`

   Example output:

```
NAME                                       STATUS   ROLES                  AGE   VERSION
compute-1-ru5.rackm03.rtp.raleigh.ibm.com  Ready    worker                 75d   v1.28.13+2ca1a23
compute-1-ru6.rackm03.rtp.raleigh.ibm.com  Ready    worker                 75d   v1.27.13+fd36fb9
compute-1-ru7.rackm03.rtp.raleigh.ibm.com  Ready    worker                 75d   v1.27.13+fd36fb9
control-1-ru2.rackm03.rtp.raleigh.ibm.com  Ready    control-plane,master   75d   v1.28.13+2ca1a23
control-1-ru3.rackm03.rtp.raleigh.ibm.com  Ready    control-plane,master   75d   v1.28.13+2ca1a23
control-1-ru4.rackm03.rtp.raleigh.ibm.com  Ready    control-plane,master   75d   v1.28.13+2ca1a23
```

2. Ensure the minimum number of storage nodes is in the ready state.

BMYSS1404

Labelled nodes for RG are less than required minimum supported nodes.

Severity

Critical

User actions

Do the following steps to resolve the problem:

1. Run the following command to list all recovery groups in the cluster.

   `oc get rg -n ibm-spectrum-scale`

   Example output:

```
NAME   AGE
rg1    18d
```

2. Run the following command to list all nodes with a specific recovery group label.

   `oc get nodes -l scale.spectrum.ibm.com/rg=<recovery group name>`

   Example output:

```
NAME                                       STATUS   ROLES                  AGE   VERSION
compute-1-ru5.gen2005.rtp.raleigh.ibm.com  Ready    worker                 18d   v1.29.8+632b078
compute-1-ru6.gen2005.rtp.raleigh.ibm.com  Ready    worker                 18d   v1.29.8+632b078
compute-1-ru7.gen2005.rtp.raleigh.ibm.com  Ready    worker                 18d   v1.29.8+632b078
control-1-ru2.gen2005.rtp.raleigh.ibm.com  Ready    control-plane,master   18d   v1.29.8+632b078
control-1-ru3.gen2005.rtp.raleigh.ibm.com  Ready    control-plane,master   18d   v1.29.8+632b078
control-1-ru4.gen2005.rtp.raleigh.ibm.com  Ready    control-plane,master   18d   v1.29.8+632b078
```

3. Verify the number of nodes for each recovery group and ensure that the minimum required nodes have the recovery group label.

BMYSS1405

Recovery group is not created successfully.

Severity

Critical

User actions

Do the following steps to resolve the problem:

1. Run the following command to list all recovery groups in the cluster.

   ```
   oc get rg -n ibm-spectrum-scale
   ```

   Example output:

   ```
   NAME   AGE
   rg1    18d
   ```

2. Run the following command to describe the details of a specific recovery group.

   ```
   oc describe rg <rg name> -n ibm-spectrum-scale
   ```

3. Check the status of the recovery group to find the reason for unsuccessful recovery group creation.

BMYSS1406

Filesystem is not created successfully.

Severity

Critical

User actions

Do the following steps to resolve the problem:

1. Run the following command to list filesystem in the cluster.

   ```
   oc get fs -n ibm-spectrum-scale
   ```

   Example output:

   ```
   NAME              ESTABLISHED   AGE
   ibmspectrum-fs    True          18d
   ```

2. Run the following command to describe the details of a specific filesystem.

   ```
   oc describe fs <fs name> -n ibm-spectrum-scale
   ```

3. Check the status of the filesystem to find the reason for unsuccessful filesystem creation.

BMYPC5018

Global Data Platform status is degraded.

Severity

Warning

Problem description

It indicates that the Global Data Platform status is degraded. The Global Data Platform status must be healthy for a successful upgrade.

User actions

Check for the issues that cause the degraded state of the Global Data Platform and take an appropriate action to resolve them. For more information about the reasons for the degraded state, see Installing Global Data Platform.

BMYPC6018

Global Data Platform status is in a critical or down state.

Severity

Warning

Problem description

It indicates that the Global Data Platform status is in a critical or down state. The Global Data Platform status must be healthy for a successful upgrade.

User actions

Check for the issues that cause the critical or down state of the Global Data Platform take an appropriate action to resolve them. For more information about the reasons for the degraded state, see Installing Global Data Platform.

BMYPC6024

A scale up operation on the Red Hat® OpenShift® cluster is in progress.

Severity

Critical

Problem description

It indicates that the scale up operation on the Red Hat OpenShift cluster is in progress. To ensure a successful upgrade, wait for the scale up operation to complete successfully.

User actions

Wait for all nodes to be added to the cluster before you proceed with the upgrade.

BMYPC6025

A scale out operation on the Red Hat® OpenShift® cluster is in progress.

Severity

Critical

Problem description

It indicates that the scale out operation on the Red Hat OpenShift cluster is in progress. To ensure a successful upgrade, wait for the scale out operation to complete successfully.

User actions

Wait for all nodes to be added to the cluster before you proceed with the upgrade.

BMYPC6026

The current version of the Global Data Platform is already up to date.

Severity

Critical

Problem description

It indicates that the version of the Global Data Platform is already up to date. The current scale version is identical to the available IBM Storage Scale version.

User actions

An upgrade is not required because the Global Data Platform version is already up to date.

BMYPC6031

A scale up operation on the Red Hat® OpenShift® cluster is triggered.

Severity

Critical

Problem description

It indicates that the scale up operation on the Red Hat OpenShift cluster is triggered. To ensure a successful upgrade, wait for the scale up operation to complete successfully.

User actions

Wait for all nodes to be added to the cluster before you proceed with the upgrade.

BMYPC6032

A scale out operation on the Red Hat® OpenShift® cluster is triggered.

Severity

Critical

Problem description

It indicates that the scale out operation on the Red Hat OpenShift cluster is triggered. To ensure a successful upgrade, wait for the scale out operation to complete successfully.

User actions

Wait for all nodes to be added to the cluster before you proceed with the upgrade.

BMYPC6033

The OpenShift® Container Platform version is not compatible with the Global Data Platform version.

Severity

Critical

Problem description

It indicates that the OpenShift Container Platform version is not compatible with the Global Data Platform version. The IBM Storage Scale 5.2.3 is incompatible with the current OpenShift Container Platform version.

User actions

Upgrade the OpenShift Container Platform version to 4.16 or later for a successful upgrade.

Global Data Platform error codes

List of all the error codes that you might encounter while you work with storage.

- **BMYSS0101E**
  Block size is not set correctly.
- **BMYSS0102E**
  Error in getting and applying the 100G workaround.
- **BMYSS01015E**
  The scale operator pod is not in a running state.
- **BMYSS01017E**
  The CSI pods are not up and in a running state.

- **BMYSS01019E**
  Pods fail to evict the node that is in the drain state.
- **BMYSS01024W**
  Issues in fetching details of the IBM Storage Scale cluster upgrade approval.
- **BMYSS01025W**
  Issues in fetching details of the IBM Storage Scale filesystem upgrade approval.
- **BMYSS01026E**
  The Global Data Platform service is not active on all the scale pods.
- **BMYSS01027E**
  The IBM Storage Scale filesystem is not mounted on all the scale pods.
- **BMYSS01028E**
  Storage service image is not matched in the `cr-version-cm` configmap.
- **BMYSS01034E**
  Failed to scale down the IBM Storage Scale operator.
- **BMYSS01035I**
  Node upgrade is in progress.
- **BMYSS01037W**
  Storage cluster status is degraded since some of the IBM Storage Scale or IBM Storage Scale CSI pods are down.
- **BMYSS01038W**
  The scale pods are not up and running.
- **BMYSS01040E**
  Pods fail to evict on the node.
- **BMYSS01041E**
  Failed to delete IBM Storage Scale security context constraints.
- **BMYSS01042E**
  Failed to delete IBM Storage Scale role binding configuration.
- **BMYSS01043E**
  Failed to delete IBM Storage Scale webhook configuration.
- **BMYSS01045I**
  All scale core pods in `ibm-spectrum-scale ns` is up and running.
- **BMYSS01046I**
  Nodes is successfully upgraded during the Global Data Platform upgrade.
- **BMYSS01047I**
  All nodes are successfully upgraded during the Global Data Platform upgrade.

BMYSS0101E

Block size is not set correctly.

Severity

Error

User actions

Do the following steps to diagnose the problem:

1. Check whether the block size in scale CR is properly set.
2. If the block size is not set correctly, then specify the correct block size.

BMYSS0102E

Error in getting and applying the 100G workaround.

Severity

Error

User actions

Do the following steps to diagnose the problem:

1. Run the oc command to check for the node that has a problem.

   ```
   oc get nncp
   ```

2. If the issue persists, then collect the Global Data Platform and contact IBM support.

BMYSS01015E

The scale operator pod is not in a running state.

Severity

Error

Problem description

It indicates that the scale operator pod is not in a running state. The scale operator pod must be in a running state for a successful upgrade.

User actions

Do the following steps to resolve the issue:

1. Run the following command to get the list of pods in the `ibm-spectrum-scale-operator` namespace.

   ```
   oc get pods -n ibm-spectrum-scale-operator
   ```

2. If the pod is in a `ImagePullBackOff` state, then resolve the issue.
3. If the pod is in the `CrashLoopBackOff` state, then collect the Global Data Platform logs and contact IBM support.

BMYSS01017E

The CSI pods are not up and in a running state.

Severity

Error

Problem description

It indicates that the CSI pods are not up and in a running state. All CSI pods must be in running state for a successful upgrade.

User actions

Do the following steps to resolve the issue:

1. Run the following command to get the list of pods in the `ibm-spectrum-scale-csi` namespace.

   ```
   oc get pods -n ibm-spectrum-scale-csi
   ```

2. Wait until all CSI pods are in the running state.
3. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01019E

Pods fail to evict the node that is in the drain state.

Severity

Error

Problem description

It indicates that the pods fail to evict the node that is draining. The node restart initiates only after the node is fully drained. All nodes must be rebooted for a successful upgrade.

User actions

Do the following steps to resolve the issue:

1. Check the upgrade details in the IBM Fusion user interface to retrieve the name of the failed pods to evict on the node.
2. Terminate the failing pods to evict.
   The upgrade operation resumes automatically after the issue is resolved.

BMYSS01024W

Issues in fetching details of the IBM Storage Scale cluster upgrade approval.

Severity

Warning

Problem description

It indicates the issues in fetching details of the IBM Storage Scale cluster upgrade approval. Upgrade approval resources are needed to enable the new functions of the installed release and lock in the new level. You must resolve this issue even though it does not impact the upgrade.

User actions

Do the following steps to resolve the issue:

1. Run the following command to get the cluster `UpgradeApprovals` instances of `TYPE=cluster` in the `ibm-spectrum-scale` namespace.

   ```
   oc get UpgradeApprovals -n ibm-spectrum-scale
   ```

2. Wait for some time for the status to be updated.
   The upgrade operation resumes automatically after the issue is resolved.

3. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01025W

Issues in fetching details of the IBM Storage Scale filesystem upgrade approval.

Severity

Warning

Problem description

It indicates the issues in fetching details of the IBM Storage Scale cluster upgrade approval. Upgrade approval resources are needed to enable the new function of the installed release and lock in the new level. You must resolve this issue even though it does not impact the upgrade.

User actions

Do the following steps to resolve the issue:

1. Run the following command to get the cluster `UpgradeApprovals` instances of `TYPE=cluster` in the `ibm-spectrum-scale` namespace.

   ```
   oc get UpgradeApprovals -n ibm-spectrum-scale
   ```

2. Wait for some time for the status to be updated.
   The upgrade operation resumes automatically after the issue is resolved.

3. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01026E

The Global Data Platform service is not active on all the scale pods.

Severity

Error

Problem description

It indicates that the Global Data Platform service is not active on all the scale pods. The Global Data Platform service must be active on all pods with the latest scale configuration for the upgrade to be successful.

User actions

Do the following steps to resolve the issue:

1. Run the following command to get the list of pods in the `ibm-spectrum-scale` namespace.

   ```
   oc get pods -n ibm-spectrum-scale
   ```

2. Run the following command on the Global Data Platform core pod that is running for remote terminal connection.

   ```
   oc rsh <podname>
   ```

3. Run the following command to check the Global Data Platform service state on all the nodes.

   `mmgetstate -a`

4. Wait for the Global Data Platform service to be active on all the nodes.
   The upgrade operation resumes automatically after the Global Data Platform service is active on all the pods.

5. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01027E

The IBM Storage Scale filesystem is not mounted on all the scale pods.

Severity

Error

Problem description

It indicates that the IBM Storage Scale filesystem is not mounted on all the scale pods. The IBM Storage Scale filesystem is mounted on all the scale pods for the upgrade to be successful.

User actions

Do the following steps to resolve the issue:

1. Run the following command to get the list of pods in the **ibm-spectrum-scale** namespace.

   `oc get pods -n ibm-spectrum-scale`

2. Run the following command on the IBM Storage Scale core pod that is running for remote terminal connection.

   `oc rsh <podname>`

3. Run the following command to check the IBM Storage Scale filesystem is mounted on all nodes.

   `mmlsmount all`

4. Wait for the IBM Storage Scale filesystem is mounted on all nodes.
   The upgrade operation resumes automatically after the IBM Storage Scale filesystem is mounted.

5. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01028E

Storage service image is not matched in the **cr-version-cm** configmap.

Severity

Error

Problem description

It indicates that the storage service image is not matched in the **cr-version-cm** configmap. The **isf-storage-services** image in **cr-version-cm** configmap and **isf-storage-service-dep** pod image must match for the upgrade to be successful.

User actions

Do the following steps to resolve the issue:

1. Run the following command to check the **isf-storage-services** image in the **cr-version-cm** configmap.

   `oc get cm cr-version-cm -n ibm-spectrum-fusion-ns -oyaml`

2. Run the following command to get the list of pods in the **ibm-spectrum-fusion-ns** namespace.

   `oc get pods -n ibm-spectrum-fusion-ns`

3. Run the following command to get the image of the **isf-storage-service-dep** pod.

   `oc get pod <pod name> -n ibm-spectrum-fusion-ns -oyaml | grep 'image:'`

4. Check whether the **isf-storage-service** image is same on configmap and deployment.
5. Wait for the image to be updated. If images are different, then replace **isf-storage-services** deployment image with **isf-storage-services** image in the **cr-version-cm** configmap.
   The upgrade operation resumes automatically after the image mismatch is resolved.

BMYSS01034E

Failed to scale down the IBM Storage Scale operator.

Severity

Error

Problem description

It indicates that the IBM Storage Scale operator is failed to scale down. The IBM Storage Scale is not scaled down for a successful upgrade.

User actions

Do the following steps to resolve the issue:

1. Check the IBM Storage Scale operator pod logs for errors in scaling down of the operator and take an appropriate action.
   The upgrade operation resumes automatically after the issue is resolved.
2. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01035I

Node upgrade is in progress.

Severity

Informational

User actions

No action is required.

BMYSS01037W

Storage cluster status is degraded since some of the IBM Storage Scale or IBM Storage Scale CSI pods are down.

Severity

Warning

Problem description

It indicates that the storage cluster status is degraded since some of the IBM Storage Scale or IBM Storage Scale CSI pods are down. The storage cluster status needs to be healthy for a successful upgrade.

User actions

Do the following steps to resolve the issue:

1. After all the nodes are updated, wait for storage cluster status to turn healthy.
2. If storage cluster status is unhealthy even after waiting, then check the logs of the `isf-storage-operator` pod in the `isf-spectrum-fusion-ns namespace` for any errors and take an appropriate action to resolve them.
3. Check the logs of the `ibm-spectrum-scale-controller-manager` pod in the `ibm-spectrum-scale-operator` namespace for any errors and take an appropriate action to resolve them.
   The upgrade operation resumes automatically after the issue is resolved.
4. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01038W

The scale pods are not up and running.

Severity

Warning

Problem description

It indicates that the scale pods are not up and running. The scale core pods must be up and running for the scale upgrade to be successful.

User actions

Do the following steps to resolve the issue:

1. Check the logs of the `ibm-spectrum-scale-controller-manager` pod in the `ibm-spectrum-scale-operator` for any errors and take an appropriate action to resolve them.
2. Check the logs of the `isf-storage-operator` pod in the `isf-spectrum-fusion-ns` namespace for any errors and take an appropriate action to resolve them.
   The upgrade operation resumes automatically after the issue is resolved.
3. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01040E

Pods fail to evict on the node.

Severity

Error

Problem description

It indicates that the pods fail to evict on the node. The node restart initiates only after the node is fully drained. All nodes must be rebooted for the upgrade to finish.

User actions

Do the following steps to resolve the issue:

1. Check the upgrade details in the IBM Fusion user interface to retrieve the name of the failed pods to evict on the node.
2. Terminate the failing pods to evict.
   The upgrade operation resumes automatically after the issue is resolved.

BMYSS01041E

Failed to delete IBM Storage Scale security context constraints.

Severity

Error

Problem description

It indicates that the IBM Storage Scale security context constraints are failed to delete. Delete the existing security context constraints **ibm-spectrum-scale-privileged** so that it creates a new security context constraint when a new manifest is applied.

User actions

Do the following steps to resolve the issue:

1. Check the logs of the `isf-storage-operator` pod within the `isf-spectrum-fusion-ns` namespace to identify the issues and take an appropriate action to resolve them.
2. Check the logs of the `ibm-spectrum-scale-controller-manager` pod within the `ibm-spectrum-scale-operator` namespace to identify the issues and take an appropriate action to resolve them.
3. Check the logs of the `isf-storage-operator` pod in the `isf-spectrum-fusion-ns` namespace for any errors and take an appropriate action to resolve them.
   The upgrade operation resumes automatically after the issue is resolved.
4. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01042E

Failed to delete IBM Storage Scale role binding configuration.

Severity

Error

Problem description

It indicates that the IBM Storage Scale role binding configuration is failed to delete. Delete the existing role binding configuration **`ibm-spectrum-scale-privileged`** so that it creates a new role binding configuration when a new manifest is applied.

User actions

Do the following steps to resolve the issue:

1. Check the logs of the `isf-storage-operator` pod within the `isf-spectrum-fusion-ns` namespace to identify the issues and take an appropriate action to resolve them.
2. Check the logs of the `ibm-spectrum-scale-controller-manager` pod within the `ibm-spectrum-scale-operator` namespace to identify the issues and take an appropriate action to resolve them.
3. Check the logs of the `isf-storage-operator` pod in the `isf-spectrum-fusion-ns` namespace for any errors and take an appropriate action to resolve them. The upgrade operation resumes automatically after the issue is resolved.
4. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01043E

Failed to delete IBM Storage Scale webhook configuration.

Severity

Error

Problem description

It indicates that the IBM Storage Scale webhook configuration is failed to delete. Delete the existing webhook configuration **`ibm-spectrum-scale-mutating-webhook-configuration`** or **`ibm-spectrum-scale-validating-webhook-configuration`** so that it creates a webhook configuration when a new manifest is applied.

User actions

Do the following steps to resolve the issue:

1. Check the logs of the `isf-storage-operator` pod within the `isf-spectrum-fusion-ns` namespace to identify the issues and take an appropriate action to resolve them.
2. Check the logs of the `ibm-spectrum-scale-controller-manager` pod within the `ibm-spectrum-scale-operator` namespace to identify the issues and take an appropriate action to resolve them.
3. Check the logs of the `isf-storage-operator` pod in the `isf-spectrum-fusion-ns` namespace for any errors and take an appropriate action to resolve them. The upgrade operation resumes automatically after the issue is resolved.
4. If the issue persists for a long time, then collect the Global Data Platform logs and contact IBM support.

BMYSS01045I

All scale core pods in **`ibm-spectrum-scale ns`** is up and running.

Severity

Informational

User actions

No action is required.

BMYSS01046I

Nodes is successfully upgraded during the Global Data Platform upgrade.

Severity

Informational

User actions

No action is required.

BMYSS01047I

All nodes are successfully upgraded during the Global Data Platform upgrade.

Severity

Informational

User actions

No action is required.

Data foundation events and error codes

List of all the events and error codes that you might encounter while you work with Data Foundation storage.

- **BMYSS0100**
  Data Foundation installation is in progress.
- **BMYSS0101**
  Data Foundation installation succeeded.
- **BMYSS0102**
  Data Foundation installation failed.
- **BMYSS0103**
  Data Foundation status is degraded.
- **BMYSS0104**
  Data Foundation status is healthy.
- **BMYSS0105**
  Data Foundation upgrade is in progress.
- **BMYSS0106**
  Data Foundation upgrade succeeded.
- **BMYSS0107**
  Data Foundation upgrade failed.
- **BMYSS0108**
  Data Foundation storage cluster installation is in progress.
- **BMYSS0109**
  Data Foundation storage cluster installation succeeded.
- **BMYSS0110**
  Data Foundation storage cluster status is degraded.
- **BMYSS0111**
  Data Foundation storage cluster status is healthy.
- **BMYSS0112**
  Data Foundation is scaling up automatically.
- **BMYSS0121**
  Failed to validate Data Foundation encryption configuration.
- **BMYSS0122**
  Data Foundation capacity is adding.
- **BMYSS0125**
  Data Foundation capacity is added.
- **BMYSS0126**
  Nodes are added to Data Foundation storage cluster successfully.
- **BMYSS0127**
  Data Foundation is auto-scaled up successfully.
- **BMYSS0128**
  Storage Class `ibm-spectrum-fusion-mgmt-sc` is not provisioned in accordance with Fusion Data Foundation guidelines.
- **BMYSS0131**
  The LVM disk is present either on a control or GPU node.
- **BMYSS0132**
  Nodes contain the LVM disks in the specified disk slot, but the total number of disks are less than three.
- **BMYSS0133**
  Existing LVM drives cannot be used for LVM cluster configuration due to not meet the prerequisites.
- **BMYSS0134**
  Selected worker nodes with LVM drives for LVM cluster configuration with respective disk slots.
- **BMYSS0135**
  A node from each zone must contribute an LVM disk for LVM cluster configuration. The current node selection does not satisfy the criteria.
- **BMYSS0136**
  The LVM cluster is successfully configured.

- **BMYSS0137**
  The LVM cluster is in a degraded or failed state.
- **BMYSS0138**
  The LVM configmap in the specified namespace contains fewer than three nodes in the `data.computeNodes` field.
- **BMYSS0139**
  The LVM configmap in the specified namespace contains an empty `data.drives` field without disk path.
- **BMYSS0140**
  No LVM drives are available for configuring the LVM cluster.

BMYSS0100

Data Foundation installation is in progress.

Severity

Informational

User actions

No action is required.

BMYSS0101

Data Foundation installation succeeded.

Severity

Informational

User actions

No action is required.

BMYSS0102

Data Foundation installation failed.

Error conditions

The error message can be for the following conditions:

- Unable to install Data Foundation operators.
- Unable to install local storage operator.

Severity

Warning

User actions

Collect Data Foundation logs and contact IBM support.

BMYSS0103

Data Foundation status is degraded.

Error conditions

The error message can be for the following conditions:

- Some Data Foundation operators are not running.
- The CSV phase state of Data Foundation is not succeeded.

Severity

Warning

User actions

Collect Data Foundation logs and contact IBM support.

BMYSS0104

Data Foundation status is healthy.

Severity

Informational

User actions

No action is required.

BMYSS0105

Data Foundation upgrade is in progress.

Severity

Informational

User actions

No action is required.

BMYSS0106

Data Foundation upgrade succeeded.

Severity

Informational

User actions

No action is required.

BMYSS0107

Data Foundation upgrade failed.

Error conditions

The error message can be for the following condition:

- Unable to upgrade Data Foundation operators.

Severity

Warning

User actions

Collect Data Foundation logs and contact IBM support.

BMYSS0108

Data Foundation storage cluster installation is in progress.

Severity

Informational

User actions

No action is required.

BMYSS0109

Data Foundation storage cluster installation succeeded.

Severity

Informational

User actions

No action is required.

BMYSS0110

Data Foundation storage cluster status is degraded.

Error conditions

The error message can be for the following condition:

- The phase of the Data Foundation storage cluster is not ready

Severity

Warning

User actions

Collect Data Foundation logs and contact IBM support.

BMYSS0111

Data Foundation storage cluster status is healthy.

Severity

Informational

User actions

No action is required.

BMYSS0112

Data Foundation is scaling up automatically.

Severity

Informational

User actions

No action is required.

BMYSS0121

Failed to validate Data Foundation encryption configuration.

Error conditions

The error message can be for the following conditions:

- Encryption configuration provided is not correct.
- Error occurs when validating encryption configuration.

Severity

Warning

User actions

Do the following steps to diagnose the problem:

1. Check the event description for detailed reason and ensure that the encryption configuration provided is valid.
2. If the issue persists, collect Data Foundation logs and contact IBM support.

BMYSS0122

Data Foundation capacity is adding.

Severity

Informational

User actions

No action is required.

BMYSS0125

Data Foundation capacity is added.

Severity

Informational

User actions

No action is required.

BMYSS0126

Nodes are added to Data Foundation storage cluster successfully.

Severity

Informational

User actions

No action is required.

BMYSS0127

Data Foundation is auto-scaled up successfully.

Severity

Informational

User actions

No action is required.

BMYSS0128

Storage Class **ibm-spectrum-fusion-mgmt-sc** is not provisioned in accordance with Fusion Data Foundation guidelines.

Severity

Critical

User actions

To resolve this critical event, manually delete the storage claim **ibm-spectrum-fusion-mgmt-sc**.

1. Check for PVCs that are using the storage class **ibm-spectrum-fusion-mgmt-sc**.

```
oc get pvc --all-namespaces -o json | jq '.items[] | select(.spec.storageClassName == "ibm-spectrum-fusion-mgmt-sc") |
{namespace: .metadata.namespace, name: .metadata.name, storageClass: .spec.storageClassName}'
```

2. Migrate PVCs (if any) from **ibm-spectrum-fusion-mgmt-sc** storage class to **ocs-storagecluster-cephfs** storage class.
   If there are any application PVCs using this storage class, plan to migrate them to the **ocs-storagecluster-cephfs** storage class. Go through the Red Hat documentation for guidance on migrating PVCs.
   Note: If any service (Backup & Restore,IBM Data Cataloging, or CAS) is installed on the cluster and is using the **ibm-spectrum-fusion-mgmt-sc** storage class, contact IBM support.
3. Delete the storage claim.
   After all PVCs are successfully migrated and none are using the **ibm-spectrum-fusion-mgmt-sc** storage class, run the following command to manually delete the storage claim:

```
oc delete storageclaim ibm-spectrum-fusion-mgmt-sc
```

4. Verify deletion of the storage class.
   After you delete the storage claim, run the following command to ensure that the storage class does not exist:

```
oc get storageclass | grep ibm-spectrum-fusion-mgmt-sc
```

BMYSS0131

The LVM disk is present either on a control or GPU node.

Severity

Critical

Problem description

It indicates that a 1.6 TB drive is placed either on a control or GPU node. For that reason, the drive cannot be used for LVM cluster configuration.

User actions

Ensure that you placed a drive on a worker node for a successful LVM configuration.

BMYSS0132

Nodes contain the LVM disks in the specified disk slot, but the total number of disks are less than three.

Severity

Warning

Problem description

It indicates that the set of nodes that are listed contain the LVM disks in the disk slot that is mentioned in the event, but the total number of disks are less than three.

User actions

Ensure that a minimum of three worker nodes must have LVM disks in the same disk slots to configure the LVM cluster.

BMYSS0133

Existing LVM drives cannot be used for LVM cluster configuration due to not meet the prerequisites.

Severity

Critical

Problem description

It indicates that the existing LVM drives cannot be used for LVM cluster configuration because they do not meet the prerequisites.

User actions

Ensure that you must meet the following conditions for a successful LVM cluster configuration:

1. The LVM disks (1.6 TB) must be available on worker nodes (excluding GPU nodes).
2. Disks must be installed in the same slot across all nodes.
3. Global Data Platform service must not be installed.
4. The Data Foundation service must be installed with `BackingStorageType` set to provider mode.

BMYSS0134

Selected worker nodes with LVM drives for LVM cluster configuration with respective disk slots.

Severity

Informational

User actions

No action is required.

BMYSS0135

A node from each zone must contribute an LVM disk for LVM cluster configuration. The current node selection does not satisfy the criteria.

Severity

Critical

Problem description

It indicates that a node from each zone must contribute an LVM disk for LVM cluster configuration. The current selection does not satisfy the criteria.

User actions

Ensure that each zone has a worker node with an LVM drive in the same disk slot.

BMYSS0136

The LVM cluster is successfully configured.

Severity

Informational

User actions

No action is required.

BMYSS0137

The LVM cluster is in a degraded or failed state.

Severity

Critical

User actions

Contact IBM support to resolve the issue.

BMYSS0138

The LVM configmap in the specified namespace contains fewer than three nodes in the `data.computeNodes` field.

Severity

Critical

Problem description

It indicates that the `lvm-config` configmap in the `ibm-spectrum-fusion-ns` namespace is custom created and contains fewer than threenodes in the `data.computeNodes` field.

User actions

Ensure that the minimum three worker nodes are listed under `data.computeNodes` field.

BMYSS0139

The LVM configmap in the specified namespace contains an empty `data.drives` field without disk path.

Severity

Critical

Problem description

It indicates that the `lvm-config` configmap in the `ibm-spectrum-fusion-ns` namespace is custom created without a disk path.

User actions

Ensure that a minimum one drive path is specified.

BMYSS0140

No LVM drives are available for configuring the LVM cluster.

Severity

Critical

Problem description

It indicates that the `lvm-config` configmap in the `ibm-spectrum-fusion-ns` namespace is custom created without LVM drivers.

User actions

Ensure that minimum one LVM drive is present for configuring the LVM cluster.

High availability cluster events and error codes

List of all the events that you might encounter during High availability cluster setup.

For Informational, Warning, and Critical events, see the following:

- **BMYCO0004**
  Failed to get hardware monitoring data for the node.
- **BMYCO0031**
  Auxiliary details validation failed.
- **BMYCO0032**
  Add Rack succeeded.
- **BMYCO0033**
  Add Rack discovery failed.

BMYCO0004

Failed to get hardware monitoring data for the node.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. This is a temporary error and it gets resolved on its own.
2. If the issue persists for a longer period, collect the compute-monitoring CR status and Nodes logs, and contact IBM support.

BMYCO0031

Auxiliary details validation failed.

Error conditions

The number of NVMe disks on the auxiliary rack is less than that of the base rack storage nodes.

The number of storage nodes are insufficient to build storage cluster.

Severity

Warning

User actions

Ensure you check the sufficient number of storage nodes on the auxiliary rack to meet the raid parity of the base rack storage cluster.

BMYCO0032

Add Rack succeeded.

Severity

Informational

User actions

No action is required.

BMYCO0033

Add Rack discovery failed.

Error conditions

The error message can be for the following conditions:

1. Provisioner is not accessible.
2. Provisioner is invalid.
3. Invalid OCP Role in the auxiliary kickstart.
4. Insufficient number of storage nodes on the auxiliary rack.
5. Insufficient number of NVMe disks per node on the auxiliary rack.
6. SCP failed for stage-1 generated JSON files.
7. Malformed stage-1 generated files.
8. Duplicate entries in the appliance-info configmap.
9. Failed to detect the provisioner node.
10. Insufficient nodes; current status of the nodes that were successfully added to compute cluster.
11. Internal error caused by K8S resource creation.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. For error condition one and two, check the DNS configuration and retry. If the issue persists, then collect the Nodes and Network switches logs and contact IBM support.
2. For error condition three, check the stage 1 installation generated kickstart JSON and ensure that the stage 1 installation is done successfully with expansion rack as the selected option.
3. For error condition four, the number of nodes available on the auxiliary rack and ensure that all the nodes are connected.
4. For error condition five, check that all the storage node should have disks for storage.
5. For error condition six, copying stage 1 generated files from the auxiliary rack provisioner node failed. Check access and connectivity to the provisioner host.
6. For error condition seven, the files that are generated in Stage 1 installation on the provisioner node of the auxiliary rack are malformed. Retry stage 1 installation and choose an expansion rack for the additional rack to succeed.
7. For error condition eight, the auxiliary details are already available on the base rack, and this issue is caused by incorrect inputs during stage1 install of the auxiliary rack installation. Retry stage 1 installation with valid inputs.
8. For error condition nine, check the network and DNS configuration and retry.
9. For error condition 10, all the discovered auxiliary nodes cannot be configured within the stipulated timeout. This issue resolves itself. If the issue persists for more than 30 minutes, then check the network connectivity to the failed auxiliary nodes.
10. For error condition 11, this issue resolves itself. If the issue persists for a long time, then you can collect Nodes and Network switches logs and contact IBM support.

IBM Data Cataloging events and error codes

List of all the events that you might encounter while you work with IBM Data Cataloging for the IBM Fusion HCI .

For Informational, Warning, and Critical events, see the following:

- **BMYDC0001**
  IBM Data Cataloging service is healthy.
- **BMYDC0002**
  IBM Data Cataloging service health is degraded.
- **BMYDC0003**
  IBM Data Cataloging service workload resources are ready.
- **BMYDC0004**
  IBM Data Cataloging service workload resources are not ready.
- **BMYDC0005**
  IBM Data Cataloging service API is available.
- **BMYDC0006**
  IBM Data Cataloging service API is not completely available.
- **BMYDC0100**
  IBM Data Cataloging service installation in progress.
- **BMYDC0101**
  IBM Data Cataloging service installation succeeded.
- **BMYDC0102**
  IBM Data Cataloging service installation failing.

- **BMYDC0103**
  IBM Data Cataloging service is already installed.
- **BMYDC0104**
  IBM Data Cataloging service upgrade in progress.
- **BMYDC0105**
  IBM Data Cataloging service upgrade succeeded.
- **BMYDC0106**
  IBM Data Cataloging service upgrade failing.
- **BMYDC0107**
  IBM Data Cataloging service is not ready for upgrade.
- **BMYDC0108**
  New version is available for IBM Data Cataloging service upgrade.
- **BMYDC0109**
  No new versions available for IBM Data Cataloging service upgrade.
- **BMYDC0110**
  IBM Data Cataloging service prerequisites are ready.
- **BMYDC0113**
  IBM Data Cataloging service Kafka instances are ready.
- **BMYDC0114**
  IBM Data Cataloging service Kafka instances are not ready.
- **BMYDC0115**
  IBM Data Cataloging service Db2 is ready.
- **BMYDC0116**
  IBM Data Cataloging service Db2 is not ready.
- **BMYDC0117**
  IBM Data Cataloging service license was not accepted.
- **BMYDC0118**
  Cluster API that is required by the IBM Data Cataloging service is not available.

BMYDC0001

IBM Data Cataloging service is healthy.

Severity

Informational

User actions

No action is required.

BMYDC0002

IBM Data Cataloging service health is degraded.

Severity

Warning

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command and check whether if there is any issue with the specific components listed.

```
oc -n ibm-data-cataloging get deployment -l 'component=discover,role in (api,connmgr,db2whrest,policyengine,isd-proxy,ui)'
```

```
Expected output
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
isd-api           1/1     1            1           6d11h
isd-connmgr       1/1     1            1           6d11h
isd-db2whrest     1/1     1            1           6d11h
isd-policyengine  1/1     1            1           6d11h
isd-proxy         1/1     1            1           6d11h
isd-ui-backend    1/1     1            1           6d11h
isd-ui-frontend   1/1     1            1           6d11h
```

2. If you find any issue with the specific components listed, collect the Data Cataloging logs and contact IBM support.

BMYDC0003

IBM Data Cataloging service workload resources are ready.

Severity

Informational

User actions

No action is required.

BMYDC0004

IBM Data Cataloging service workload resources are not ready.

Severity

Warning

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command and check whether if there is any issue with the specific components listed.

```
oc -n ibm-data-cataloging get deployment -l 'component=discover'
```

```
Expected output
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
isd-api                      1/1     1            1           6d11h
isd-auth                     1/1     1            1           6d11h
isd-backup-restore           1/1     1            1           6d11h
isd-connmgr                  1/1     1            1           6d11h
isd-consumer-ceph-le         10/10   10           10          6d11h
isd-consumer-cos-le          10/10   10           10          6d11h
isd-consumer-cos-scan        10/10   10           10          6d11h
isd-consumer-file-scan       10/10   10           10          6d11h
isd-consumer-protect-scan    10/10   10           10          6d11h
isd-consumer-scale-le        10/10   10           10          6d11h
isd-consumer-scale-scan      10/10   10           10          6d11h
isd-contentsearchagent       1/1     1            1           6d11h
isd-db2whrest                1/1     1            1           6d11h
isd-importtags               1/1     1            1           6d11h
isd-keystone                 1/1     1            1           6d11h
isd-policyengine             1/1     1            1           6d11h
isd-producer-ceph-le         1/1     1            1           6d11h
isd-producer-cos-le          1/1     1            1           6d11h
isd-producer-cos-scan        1/1     1            1           6d11h
isd-producer-file-scan       1/1     1            1           6d11h
isd-producer-protect-scan    1/1     1            1           6d11h
isd-producer-scale-le        1/1     1            1           6d11h
isd-producer-scale-scan      1/1     1            1           6d11h
isd-proxy                    1/1     1            1           6d11h
isd-scaleafmdatamover        1/1     1            1           6d11h
isd-scaleilmdatamover        1/1     1            1           6d11h
isd-sdmonitor                1/1     1            1           6d11h
isd-tikaserver               1/1     1            1           6d11h
isd-ui-backend               1/1     1            1           6d11h
isd-ui-frontend              1/1     1            1           6d11h
isd-wkcconnector             1/1     1            1           6d11h
```

2. If you find any issue with the specific components listed, collect the Data Cataloging logs and contact IBM support.

BMYDC0005

IBM Data Cataloging service API is available.

Severity

Informational

User actions

No action is required.

BMYDC0006

IBM Data Cataloging service API is not completely available.

Severity

Warning

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command and check whether if there is any issue with the specific components listed.

```
oc -n ibm-data-cataloging get deployment -l 'component=discover,role in (api,connmgr,db2whrest,policyengine,isd-proxy,ui)'
```

```
Expected output
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
isd-api             1/1     1            1           6d11h
isd-connmgr         1/1     1            1           6d11h
isd-db2whrest       1/1     1            1           6d11h
isd-policyengine    1/1     1            1           6d11h
isd-proxy           1/1     1            1           6d11h
isd-ui-backend      1/1     1            1           6d11h
isd-ui-frontend     1/1     1            1           6d11h
```

2. If you find any issue with the specific components listed, collect the Data Cataloging logs and contact IBM support.

BMYDC0100

IBM Data Cataloging service installation in progress.

Severity

Informational

User actions

No action is required.

BMYDC0101

IBM Data Cataloging service installation succeeded.

Severity

Informational

User actions

No action is required.

BMYDC0102

IBM Data Cataloging service installation failing.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Run the following commands and collect the `ibm_data_cataloging_fsd.yaml, ibm_data_cataloging_fsi.yaml, ibm_data_cataloging_cr.yaml and ibm_data_cataloging_operator_logs.yaml` files information.

```
oc -n ibm-spectrum-fusion-ns get fusionservicedefinition data-cataloging-service-definition -o yaml > 
ibm_data_cataloging_fsd.yaml
oc -n ibm-spectrum-fusion-ns get fusionserviceinstance data-cataloging-service-instance -o yaml > 
ibm_data_cataloging_fsi.yaml
oc -n ibm-data-cataloging get isd -o yaml > ibm_data_cataloging_cr.yaml
oc -n ibm-data-cataloging logs deployment/spectrum-discover-operator > ibm_data_cataloging_operator_logs.yaml
```

2. Contact IBM support and provide the `ibm_data_cataloging_fsd.yaml, ibm_data_cataloging_fsi.yaml, ibm_data_cataloging_cr.yaml and ibm_data_cataloging_operator_logs.yaml` files information.

BMYDC0103

IBM Data Cataloging service is already installed.

Severity

Informational

User actions

No action is required.

BMYDC0104

IBM Data Cataloging service upgrade in progress.

Severity

Informational

User actions

No action is required.

BMYDC0105

IBM Data Cataloging service upgrade succeeded.

Severity

Informational

User actions

No action is required.

BMYDC0106

IBM Data Cataloging service upgrade failing.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. Run the following commands and collect the `ibm_data_cataloging_fsd.yaml, ibm_data_cataloging_fsi.yaml, ibm_data_cataloging_cr.yaml and ibm_data_cataloging_operator_logs.yaml` files information.

```
oc -n ibm-spectrum-fusion-ns get fusionservicedefinition data-cataloging-service-definition -o yaml > 
ibm_data_cataloging_fsd.yaml
oc -n ibm-spectrum-fusion-ns get fusionserviceinstance data-cataloging-service-instance -o yaml > 
ibm_data_cataloging_fsi.yaml
oc -n ibm-data-cataloging get isd -o yaml > ibm_data_cataloging_cr.yaml
oc -n ibm-data-cataloging logs deployment/spectrum-discover-operator > ibm_data_cataloging_operator_logs.yaml
```

2. Contact IBM support and provide the `ibm_data_cataloging_fsd.yaml, ibm_data_cataloging_fsi.yaml, ibm_data_cataloging_cr.yaml and ibm_data_cataloging_operator_logs.yaml` files information.

BMYDC0107

IBM Data Cataloging service is not ready for upgrade.

Severity

Informational

User actions

No action is required.

BMYDC0108

New version is available for IBM Data Cataloging service upgrade.

Severity

Informational

User actions

No action is required.

BMYDC0109

No new versions available for IBM Data Cataloging service upgrade.

Severity

Informational

User actions

No action is required.

BMYDC0110

IBM Data Cataloging service prerequisites are ready.

Severity

Informational

User actions

No action is required.

BMYDC0113

IBM Data Cataloging service Kafka instances are ready.

Severity

Informational

User actions

No action is required.

BMYDC0114

IBM Data Cataloging service Kafka instances are not ready.

Severity

Warning

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command and check whether if there is any issue with the specific components listed.

   `oc -n ibm-data-cataloging get kafka isd-ssl isd-sasl`

   ```
   Expected output
   NAME       DESIRED KAFKA REPLICAS   DESIRED ZK REPLICAS   READY   WARNINGS
   isd-ssl    1                        1                     True
   isd-sasl   1                        1                     True
   ```

2. If you find any issue with the specific components listed, collect the Data Cataloging logs and contact IBM support.

BMYDC0115

IBM Data Cataloging service Db2 is ready.

Severity

Informational

User actions

No action is required.

BMYDC0116

IBM Data Cataloging service Db2 is not ready.

Severity

Warning

User actions

Do the following steps to diagnose and report the problem:

1. Run the following command and check whether if there is any issue with the specific components listed.

   `oc -n ibm-data-cataloging get db2u`

   ```
   Expected output
   NAME   STATE   MAINTENANCESTATE   AGE
   isd    Ready   None               6d15h
   ```

2. If you find any issue with the specific components listed, collect the Data Cataloging logs and contact IBM support.

BMYDC0117

IBM Data Cataloging service license was not accepted.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. This is not expected, by default IBM Data Cataloging license is accepted.
2. Run the following command to check the IBM Data Cataloging.

```
ISD_INSTANCE=$(oc -n ibm-data-cataloging get isd -o name | head -n1)
oc -n ibm-data-cataloging patch $ISD_INSTANCE --type='merge' -p '{"spec":{"license":{"accept":true}}}'
```

```
Expected output
spectrumdiscover.spectrum-discover.ibm.com/data-cataloging-service-instance patched
```

BMYDC0118

Cluster API that is required by the IBM Data Cataloging service is not available.

Severity

Critical

User actions

Do the following steps to diagnose and report the problem:

1. To avoid installations outside of most OpenShift® clusters, this event is only triggered when a cluster API that should be available by default in OpenShift is not present. The action to take is to attempt the install on any supported OpenShift version.
2. Attempt installation in an OpenShift cluster running any of the supported versions (4.14, 4.15, 4.16, 4.17, and 4.18).

Backup & Restore events and error codes

List of all the events that you might encounter during Backup & Restore for the IBM Fusion HCI.

For Informational, Warning, and Critical events, see the following:

- **BMYBR0009**
  There was an error when processing the job in the service.
- **BMYDP1101**
  Created backup storage location.
- **BMYDP1102**
  Updated backup storage location.
- **BMYDP1105**
  Backup storage location is available.
- **BMYDP1121**
  Backup policy created.
- **BMYDP1122**
  Backup policy updated.
- **BMYDP1141**
  Backup policy applied.
- **BMYDP1142**
  Backup policy assigned.
- **BMYDP1143**
  Backup policy unassigned.
- **BMYDP1146**
  Policy assignment is created initial backup for an application.
- **BMYDP1161**
  Created backup for application from backup policy.
- **BMYDP1163**
  Completed backup for application from backup policy.
- **BMYDP1181**
  Starting restore from backup.
- **BMYDP1182**
  Completed restore from backup.
- **BMYDP1364**
  Failed for application from backup policy.
- **BMYDP2103**
  Deleted backup storage location.
- **BMYDP2123**
  Backup policy deleted.
- **BMYDP2162**
  Deleted backup for application from the backup policy.
- **BMYDP3104**
  Failed to register backup storage location.
- **BMYDP3106**
  Backup storage location is unavailable.
- **BMYDP3124**
  Failed to create the backup policy.
- **BMYDP3144**
  Creation of `PolicyAssignment` failed, referenced custom resource could not be found or is in a error state.

- **BMYDP3145**
  Application validation failed.
- **BMYDP3165**
  Canceled for application from backup policy.
- **BMYDP3166**
  Failed to start a job for backup for application from backup policy.
- **BMYDP3183**
  Failed restore from backup.
- **BMYDP3184**
  Canceled restore from backup.
- **BMYDP3185**
  Failed to start a job for restore from backup.

BMYBR0009

There was an error when processing the job in the service.

Severity

Critical

User actions

Error condition: Backup fails with the following FailedValidation error

The backup fails with the following FailedValidation error and the namespace contains encrypted Ceph RBD PVC in Block volume mode:

```
BMYBR0009
There was an error when processing the job in the Transaction Manager service. The underlying error was:
Found unsupported PVC: encrypted Ceph RBD PVC in Block volume mode cannot be safely restored <namespace/pvc>. Only Filesystem
```

Delete encrypted Ceph RBD PVC in Block volume mode from the namespace. Use Filesystem volume mode for encrypted Ceph RBD PVCs.
Note: LUKS (Linux Unified Key Setup) encrypted Ceph RBD PVC with Block volume mode are not supported. LUKS (Linux Unified Key Setup) encrypted Ceph RBD PVC with Filesystem volume mode are supported.

In general for these processing errors, collect Backup and Restore logs and contact IBM support.

BMYDP1101

Created backup storage location.

Severity

Informational

User actions

No action is required.

BMYDP1102

Updated backup storage location.

Severity

Informational

User actions

No action is required.

BMYDP1105

Backup storage location is available.

Severity

Informational

User actions

No action is required.

BMYDP1121

Backup policy created.

Severity

Informational

User actions

No action is required.

BMYDP1122

Backup policy updated.

Severity

Informational

User actions

No action is required.

BMYDP1141

Backup policy applied.

Severity

Informational

User actions

No action is required.

BMYDP1142

Backup policy assigned.

Severity

Informational

User actions

No action is required.

BMYDP1143

Backup policy unassigned.

Severity

Informational

User actions

No action is required.

BMYDP1146

Policy assignment is created initial backup for an application.

Severity

Informational

User actions

No action is required.

BMYDP1161

Created backup for application from backup policy.

Severity

Informational

User actions

No action is required.

BMYDP1163

Completed backup for application from backup policy.

Severity

Informational

User actions

No action is required.

BMYDP1181

Starting restore from backup.

Severity

Informational

User actions

No action is required.

BMYDP1182

Completed restore from backup.

Severity

Informational

User actions

No action is required.

BMYDP1364

Failed for application from backup policy.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Verify that the backup storage location is available and has sufficient capacity.
2. Correct the issue that caused the backup to fail, and then re-attempt the backup.
3. If the issue persists, then collect Backup and Restore logs and contact IBM support.

BMYDP2103

Deleted backup storage location.

Severity

Informational

User actions

No action is required.

BMYDP2123

Backup policy deleted.

Severity

Informational

User actions

No action is required.

BMYDP2162

Deleted backup for application from the backup policy.

Severity

Informational

User actions

No action is required.

BMYDP3104

Failed to register backup storage location.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Verify the Information in the backup storage location CR is valid.
2. Verify both connectivity and credentials for the backup storage location.
3. If you do not find any issue related to backup storage location, then collect Backup and Restore logs and contact IBM support.

BMYDP3106

Backup storage location is unavailable.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Verify whether the backup storage location is operational and that the backup storage location has sufficient capacity.
2. If you do not find any issue related to backup storage location, then collect Backup and Restore logs and contact IBM support.

BMYDP3124

Failed to create the backup policy.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Verify the backup storage location is correctly specified and available.
2. If you do not find any issue related to backup storage location, then collect Backup and Restore logs and contact IBM support.

BMYDP3144

Creation of `PolicyAssignment` failed, referenced custom resource could not be found or is in a error state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Verify that the application and policy are correctly specified.
2. Correct the custom resource and re-attempt the policy assignment.
3. If the issue persists, then collect Backup and Restore logs and contact IBM support.

BMYDP3145

Application validation failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Verify that the application and policy are correctly specified.
2. Ensure the application is valid, then retry the policy assignment.
3. If the issue persists, then collect Backup and Restore logs and contact IBM support.

BMYDP3165

Canceled for application from backup policy.

Severity

Critical

User actions

Collect Backup and Restore logs and contact IBM support.

BMYDP3166

Failed to start a job for backup for application from backup policy.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Verify that the backup storage location is available and has sufficient capacity.
2. Check and correct the issue that caused the backup to fail, and then re-attempt the backup.
3. If the issue persists, then collect Backup and Restore logs and contact IBM support.

BMYDP3183

Failed restore from backup.

Severity

Critical

User actions

Collect Backup and Restore logs and contact IBM support.

BMYDP3184

Canceled restore from backup.

Severity

Critical

User actions

Collect Backup and Restore logs and contact IBM support.

BMYDP3185

Failed to start a job for restore from backup.

Severity

Critical

User actions

Collect Backup and Restore logs and contact IBM support.

Connection services events and error codes

List of all the events that you might encounter during connection service for the IBM Fusion HCI .

Connection service is a common service for the Backup & Restore.

For Informational, Warning and Critical events, see the following:

- **BMYCS0001**
  Failed to send heartbeat to remote cluster `<remote-apiendpoint>: <Error>`.
- **BMYCS0002**
  No heartbeat received from the remote cluster `<remote-apiendpoint>`.
- **BMYCS0003**
  Connection to remote cluster is healthy.
- **BMYCS0004**
  Connection from remote cluster is healthy.
- **BMYCS0005**
  Init secret has the same API endpoint with the local cluster.
- **BMYCS0006**
  Bootstrap token in init secret is not correct or expired: `<Error>`.
- **BMYCS0007**
  Failed to get CA certificate from remote cluster: `<Error>`.
- **BMYCS0008**
  CA certificate of peer cluster is not correct: `<Error>`.
- **BMYCS0009**
  Failed to get infrastructure provider of the remote cluster: `<Error>`.
- **BMYCS0010**
  Failed to get cluster name of remote cluster: `<Error>`.
- **BMYCS0011**
  Bootstrap secret created.
- **BMYCS0012**
  Failed to wait for the client config to be ready: `<Error>`.
- **BMYCS0013**
  Kubeconfig secret created.
- **BMYCS100**
  Create a bootstrap secret in progress.
- **BMYCS101**
  Create bootstrap secret completed.
- **BMYCS102**
  Create a kubeconfig secret in progress.
- **BMYCS103**
  Create kubeconfig secret completed.
- **BMYCS104**
  Connection to remote API endpoint `<remote-apiendpoint>` is disabled.
- **BMYCS105**
  Connection to remote API endpoint `<remote-apiendpoint>` is healthy.
- **BMYCS106**
  Connection from remote API endpoint `<remote-apiendpoint>` is healthy.
- **BMYCS300**
  Init secret has the same API endpoint as the local cluster.
- **BMYCS301**
  The Bootstrap token in the init secret is not correct or expired.
- **BMYCS302**
  Failed to get infrastructure provider of the remote cluster.
- **BMYCS303**
  Failed to get CA certificate from the remote cluster.
- **BMYCS304**
  The CA certificate of the remote cluster is not correct.
- **BMYCS305**
  Failed to get the cluster name of the remote cluster.
- **BMYCS306**
  Failed to wait for the client configuration of the remote cluster to be ready.

- **[BMYCS307]**
  Connection to remote API endpoint is unhealthy.
- **[BMYCS308]**
  Connection from remote API endpoint is unhealthy.

BMYCS0001

Failed to send heartbeat to remote cluster **`<remote-apiendpoint>:`**
`<Error>`.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

BMYCS0002

No heartbeat received from the remote cluster **`<remote-apiendpoint>`**.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

BMYCS0003

Connection to remote cluster is healthy.

Severity

Informational

User actions

No action is required.

BMYCS0004

Connection from remote cluster is healthy.

Severity

Informational

User actions

No action is required.

BMYCS0005

Init secret has the same API endpoint with the local cluster.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Copy the code snippet from the remote cluster instead of local cluster to set up connection.
2. If the issue persists, then collect Storage logs and contact IBM support.

BMYCS0006

Bootstrap token in init secret is not correct or expired: `<Error>`.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Re-generate the bootstrap token and replace it in the init secret.
2. If the issue persists, then collect Storage logs and contact IBM support.

BMYCS0007

Failed to get CA certificate from remote cluster: `<Error>`.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check whether the `kube-root-ca.crt` configmap in the `kube-public` namespace contains CA certs in the remote cluster.
2. If the issue persists, then collect Storage logs and contact IBM support.

BMYCS0008

CA certificate of peer cluster is not correct: `<Error>`.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check whether the `kube-root-ca.crt` confimap in the `kube-public` namespace contains the correct CA certificate.
2. If it does not contain the correct CA certificate, then put the correct CA certificate in the configmap.
3. If the issue persists, then collect Storage logs and contact IBM support.

BMYCS0009

Failed to get infrastructure provider of the remote cluster: `<Error>`.

Severity

Informational

User actions

No action is required.

BMYCS0010

Failed to get cluster name of remote cluster: `<Error>`.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

BMYCS0011

Bootstrap secret created.

Severity

Informational

User actions

No action is required.

BMYCS0012

Failed to wait for the client config to be ready: `<Error>`.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

BMYCS0013

Kubeconfig secret created.

Severity

Informational

User actions

No action is required.

BMYCS100

Create a bootstrap secret in progress.

Severity

Informational

User actions

No action is required.

BMYCS101

Create bootstrap secret completed.

Severity

Informational

User actions

No action is required.

BMYCS102

Create a kubeconfig secret in progress.

Severity

Informational

User actions

No action is required.

BMYCS103

Create kubeconfig secret completed.

Severity

Informational

User actions

No action is required.

BMYCS104

Connection to remote API endpoint `<remote-apiendpoint>` is disabled.

Severity

Informational

User actions

No action is required.

BMYCS105

Connection to remote API endpoint `<remote-apiendpoint>` is healthy.

Severity

Informational

User actions

No action is required.

BMYCS106

Connection from remote API endpoint `<remote-apiendpoint>` is healthy.

Severity

Informational

User actions

No action is required.

BMYCS300

Init secret has the same API endpoint as the local cluster.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Edit the init secret with the right remote cluster API endpoint.
2. If the issue persists, then collect Storage logs and contact IBM support.

BMYCS301

The Bootstrap token in the init secret is not correct or expired.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Re-generate the bootstrap token and replace it in the init secret.
2. If the issue persists, then collect Storage logs and contact IBM support.

BMYCS302

Failed to get infrastructure provider of the remote cluster.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

BMYCS303

Failed to get CA certificate from the remote cluster.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check if the configmap `kube-root-ca.crt` in namespace kube-public has CA certs in the remote cluster.
2. If the issue persists, then collect Storage logs and contact IBM support.

BMYCS304

The CA certificate of the remote cluster is not correct.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Check if the configmap `kube-root-ca.crt` in namespace kube-public has the right CA. If not, put the right CA in this configmap.
2. If the issue persists, then collect Storage logs and contact IBM support.

BMYCS305

Failed to get the cluster name of the remote cluster.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

BMYCS306

Failed to wait for the client configuration of the remote cluster to be ready.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

BMYCS307

Connection to remote API endpoint is unhealthy.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

BMYCS308

Connection from remote API endpoint is unhealthy.

Severity

Critical

User actions

Collect Storage logs and contact IBM support.

Content-Aware Storage events and error codes

List of all the events that you might encounter while you work with CAS for the .

For Informational, Warning, and Critical events, see the following:

- **BMYCA0201**
  `KAFKAInstallFail`
- **BMYCA0204**
  `QueryServiceInstallFail`
- **BMYCA0216**
  `DProcCRAbsent`
- **BMYCA0217**
  `PipelineInstanceRetrivalFailed`
- **BMYCA0225**
  `KAFKAApplyYamlFailed`
- **BMYCA0227**
  `KAFKAClusterDeploymentFailed`
- **BMYCA0228**
  `KAFKAClusterNotHealthy`
- **BMYCA0229**
  `KAFKAInstanceRetrivalFailed`
- **BMYCA0230**
  `KAFKAClusterStatusRetrievalFailed`
- **BMYCA0231**
  `KAFKAInstanceNotReady`
- **BMYCA0232**
  `ParadeDBClusterRetrivalFailed`
- **BMYCA0233**
  `ParadeDBClusterDeploymentFailed`
- **BMYCA0234**
  `ParadeDBClusterNotHealthy`
- **BMYCA0235**
  `ParadeDBPoolerInstanceRetrivalFailed`
- **BMYCA0243**
  `ParadeDBConnectionFail`
- **BMYCA0244**
  `QueryServicePodFailed`
- **BMYCA0245**
  `QueryServiceNotHealthy`
- **BMYCA0258**
  `GetUsersFail`
- **BMYCA0259**
  `ValidateUsersFail`
- **BMYCA0260**
  `GetGroupsFail`
- **BMYCA0261**
  `ValidateGroupsFail`
- **BMYCA0263**
  `DBConnectionFail`
- **BMYCA0264**
  `DBPopulationFail`
- **BMYCA0266**
  `DeleteEntriesOfUsersAndGroupFromACLTableFail`
- **BMYCA0267**
  `DeleteEntriesOfDomainsFromACLTableFail`
- **BMYCA0268**
  `ClusterWatchDisableFail`

BMYCA0201

`KAFKAInstallFail`

Severity

Critical

Problem description

It indicates that Kafka has failed to deploy.

User actions

Follow the steps to resolve the issue:

1. Review the `cas-operator` logs to identify the cause of the Kafka installation failure.
2. Collect CAS logs and check whether you find any errors that are related to CAS operator.
3. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0204

QueryServiceInstallFail

Severity

Critical

Problem description

It indicates that the query service failed to install correctly.

User actions

Follow the steps to resolve the issue:

1. Check whether `query-service` pods are running in the `ibm-cas` namespace.
2. Check whether image is being correctly pulled. If not, check the `icr.io` pull-secret.
3. Check logs in `ibm-cas-operator-controller` pod for query service errors.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0216

DProcCRAbsent

Severity

Critical

Problem description

The `DocumentProcessor` CR is missing.

User actions

Review the `cas-operator` logs to identify the cause of the CR creation failure.

BMYCA0217

PipelineInstanceRetrivalFailed

Severity

Critical

Problem description

The pipeline instance retrieval has failed.

User actions

Review the `cas-operator` logs to identify the cause of the pipeline instance retrieval failure.

BMYCA0225

`KAFKAApplyYamlFailed`

Severity

Critical

Problem description

It indicates that the Kafka installation has failed.

User actions

Follow the steps to resolve the issue:

1. Check whether the Kafka operator is installed successfully.
2. Check whether any errors or waring messages in CAS installation.
3. Collect CAS logs and check whether you find any errors that are related to CAS operator.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0227

`KAFKAClusterDeploymentFailed`

Severity

Critical

Problem description

It indicates that the Kafka installation has failed.

User actions

Follow the steps to resolve the issue:

1. Check whether the Kafka operator is installed successfully.
2. Check whether any errors or waring messages in CAS installation.
3. Collect CAS logs and check whether you find any errors that are related to CAS operator.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0228

`KAFKAClusterNotHealthy`

Severity

Low

Problem description

It indicates that the Kafka cluster is not in a healthy state.

User actions

Follow the steps to resolve the issue:

1. Wait until the Kafka instance is present to let the log info fix by itself.
2. Check whether the Kafka operator is installed successfully.
3. Check whether any errors or warning messages in CAS installation.
4. Collect CAS logs and check whether you find any errors that are related to CAS operator.
5. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0229

`KAFKAInstanceRetrivalFailed`

Severity

Low

Problem description

It indicates that the Kafka instance is not present or is being installed.

User actions

Follow the steps to resolve the issue:

1. Wait until the Kafka instance is present to let the log info fix by itself.
2. Check whether the streams for Apache Kafka operator are installed successfully.
3. Collect CAS logs and check whether you find any errors that are related to CAS operator.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0230

`KAFKAClusterStatusRetrievalFailed`

Severity

Low

Problem description

It indicates that the Kafka cluster status has failed to retrieve.

User actions

Follow the steps to resolve the issue:

1. Wait until the Kafka instance is present to let the log info fix by itself.
2. Check whether the streams for Apache Kafka operator are installed successfully.
3. Collect CAS logs and check whether you find any errors that are related to CAS operator.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0231

`KAFKAInstanceNotReady`

Severity

Low

Problem description

It indicates that the Kafka instance is not in the ready state.

User actions

Follow the steps to resolve the issue:

1. Wait until the Kafka instance is present to let the log info fix by itself.
2. Check whether the streams for Apache Kafka operator are installed successfully.
3. Collect CAS logs and check whether you find any errors that are related to CAS operator.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0232

`ParadeDBClusterRetrivalFailed`

Severity

Critical

Problem description

It indicates that the **ParadeDB** cluster has failed to retrieve.

User actions

Follow the steps to resolve the issue:

1. Check whether the CNPG operator is installed successfully.
2. Check whether the **RemoteFilesystem** status is connected.
3. Check whether all PVCs in the **ibm-cas** namespace in a bound state.
4. Collect CAS logs and check whether you find any errors that are related to CAS operator.
5. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0233

ParadeDBClusterDeploymentFailed

Severity

Critical

Problem description

It indicates that the Parade DB Cluster has failed to deploy.

User actions

Follow the steps to resolve the issue:

1. Check whether the CloudNativePG operator is installed successfully.
2. Check whether the Parade DB pods are running in the **ibm-cas** namespace.
3. Collect CAS logs and check whether you find any errors that are related to CAS operator and CloudNativePG operator.
4. Check the logs whether there is any other related error code.
5. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0234

ParadeDBClusterNotHealthy

Severity

Critical

Problem description

It indicates that the **ParadeDB** cluster is not in a healthy state.

User actions

Follow the steps to resolve the issue:

1. Check whether the CNPG operator is installed successfully.
2. Check whether the **RemoteFilesystem** status is connected.
3. Check whether all PVCs in the **ibm-cas** namespace in a bound state.
4. Collect CAS logs and check whether you find any errors that are related to CAS operator.
5. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0235

ParadeDBPoolerInstanceRetrivalFailed

Severity

Critical

Problem description

It indicates that the `ParadeDB` pooler instance has failed to retrieve.

User actions

Follow the steps to resolve the issue:

1. Check whether the CNPG operator is installed successfully.
2. Check whether the `RemoteFilesystem` status is connected.
3. Check whether all PVCs in the `ibm-cas` namespace in a bound state.
4. Collect CAS logs and check whether you find any errors that are related to CAS operator.
5. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0243

`ParadeDBConnectionFail`

Severity

Critical

Problem description

It indicates that the `ParadeDB` connection has failed.

User actions

Follow the steps to resolve the issue:

1. Check whether the CNPG operator is installed successfully.
2. Check whether the `RemoteFilesystem` status is connected.
3. Check whether all PVCs in the `ibm-cas` namespace in a bound state.
4. Collect CAS logs and check whether you find any errors that are related to CAS operator.
5. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0244

`QueryServicePodFailed`

Severity

Critical

Problem description

It indicates that the query service pod fails to maintain a running state.

User actions

Follow the steps to resolve the issue:

1. Check whether `query-service` pods are running in the `ibm-cas` namespace.
2. Check whether image is being correctly pulled. If not, check the `icr.io` pull-secret.
3. Check logs in `ibm-cas-operator-controller` pod in the `ibm-cas` namespace for query service errors.
4. Check logs in `query-service` pod in the `ibm-cas` namespace for query service errors.
5. Check for errors in the `query-service` deployment in the `ibm-cas` namespace.
6. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0245

`QueryServiceNotHealthy`

Severity

Critical

Problem description

It indicates that the query service is missing one or more expected resources.

User actions

Follow the steps to resolve the issue:

1. Check whether `query-service` pods are running in the `ibm-cas` namespace.
2. Check whether image is being correctly pulled. If not, check the `icr.io` pull-secret.
3. Check logs in `ibm-cas-operator-controller` pod in the `ibm-cas` namespace for query service errors.
4. Check logs in `query-service` pod in the `ibm-cas` namespace for query service errors.
5. Check for errors in the `query-service` deployment in the `ibm-cas` namespace.
6. Check for the presence of the `ibm-cas` route in the `ibm-cas` namespace.
7. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0258

`GetUsersFail`

Severity

Critical

Problem description

It indicates that the CAS service failed to retrieve users from the identity provider (Red Hat OpenShift or external identity provider depending on configuration).

User actions

Follow the steps to resolve the issue:

1. Collect CAS logs and check CAS operator pod logs to identify the cause of the failure while retrieving users from the identity provider.
2. If external identity provider is configured, validate identity provider configuration including `issuerURL`, and `client_id` and `client_secret` parameters defined in the secret.
3. Ensure that the identity provider is up and accessible to the cluster.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0259

`ValidateUsersFail`

Severity

Critical

Problem description

It indicates that one or more users who are specified in the `CasResourceAccessControl` (CRAC) custom resource (CR) are not present in Red Hat OpenShift or the configured external identity providers depending on identity provider configuration.

User actions

Follow the steps to resolve the issue:

1. Check the `status.conditions` field of the CRAC CR for the `ValidatedUsers` condition, which lists the users that failed validation.
2. Ensure that the users are correctly defined in the CRAC CR.
3. Verify that the specified users exist either in the Red Hat OpenShift cluster or the external identity provider.
4. If the issue persists, collect relevant logs and contact IBM support for assistance.

BMYCA0260

`GetGroupsFail`

Severity

Critical

Problem description

It indicates that the CAS service failed to retrieve groups from the identity provider (Red Hat OpenShift or external identity provider depending on configuration).

User actions

Follow the steps to resolve the issue:

1. Collect CAS logs and check CAS operator pod logs to identify the cause of the failure while retrieving groups from the identity provider.
2. If the external identity provider is configured, validate identity provider configuration including `issuerURL`, and `client_id` and `client_secret` parameters defined in the secret.
3. Ensure that the identity provider is up and accessible to the cluster.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0261

`ValidateGroupsFail`

Severity

Critical

Problem description

It indicates that one or more groups that are specified in the `CasResourceAccessControl` (CRAC) custom resource (CR) are not present in Red Hat OpenShift or the configured external identity providers depending on identity provider configuration.

User actions

Follow the steps to resolve the issue:

1. Check the `status.conditions` field of the CRAC CR for the `ValidatedGroups` condition, which lists the groups that failed validation.
2. Ensure that the groups are correctly defined in the CRAC CR.
3. Verify that the specified groups exist either in the Red Hat OpenShift cluster or the external identity provider.
4. If the issue persists, collect relevant logs and contact IBM support for assistance.

BMYCA0263

`DBConnectionFail`

Severity

Critical

Problem description

It indicates that the database (DB) connection has failed.

User actions

Follow the steps to resolve the issue:

1. Check whether the CNPG operator is installed successfully.
2. Check whether the `RemoteFilesystem` status is connected.
3. Check whether all PVCs in the `ibm-cas` namespace in a bound state.
4. Collect CNPG operator logs.
5. Collect CAS logs and check whether you find any errors that are related to CAS operator.
6. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0264

`DBPopulationFail`

Severity

Critical

Problem description

It indicates that the database (DB) population has failed.

User actions

Follow the steps to resolve the issue:

1. Check whether the CNPG operator is installed successfully.
2. Check whether the `RemoteFilesystem` status is connected.
3. Check whether all PVCs in the `ibm-cas` namespace in a bound state.
4. Collect CNPG operator logs.
5. Collect CAS logs and check whether you find any errors that are related to CAS operator.
6. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0266

DeleteEntriesOfUsersAndGroupFromACLTableFail

Severity

Critical

Problem description

It indicates that the deletion of user and group entries from the Access Control List (ACL) database table failed.

User actions

Follow the steps to resolve the issue:

1. Review the `cas-operator` logs to identify the cause of the failure when deleting user and group entries from the Access Control List (ACL) table.
2. Collect CAS logs and check whether you find any errors that are related to CAS operator.
3. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0267

DeleteEntriesOfDomainsFromACLTableFail

Severity

Critical

Problem description

It indicates that the deletion of domain entries from the Access Control List (ACL) database table failed.

User actions

Follow the steps to resolve the issue:

1. Review the `cas-operator` logs to identify the cause of the failure when deleting domain entries from the Access Control List (ACL) table.
2. Collect CAS logs and check whether you find any errors that are related to CAS operator.
3. Take appropriate action based on the error or contact IBM support to resolve the issue.

BMYCA0268

ClusterWatchDisableFail

Severity

Critical

Problem description

It indicates that the Watch Folder Deletion of associated data source has failed.

User actions

Follow the steps to resolve the issue:

1. Check whether the CAS operator is installed successfully.
2. Check whether the Remote Scale Cluster is connected.
3. Collect CAS logs and check whether you find any errors that are related to CAS operator.
4. Take appropriate action based on the error or contact IBM support to resolve the issue.

Serviceability events and error codes

List of all the events that you might encounter that related to serviceability.

For Informational, Warning, and Critical events, see the following:

- **Serviceability Call Home events and error codes**
  List of Call Home events that you might encounter that related to serviceability.
- **BMYLC0001**
  Remaining storage space is less than 30 percent.
- **BMYLC0002**
  Remaining storage space is less than 10 percent.
- **BMYSVC1001**
  Credential rotation started.
- **BMYSVC1002**
  Credential rotation completed.
- **BMYSVC2001**
  Rotation secret missing for: %s
- **BMYSVC2002**
  Failed to update secret: %s
- **BMYSVC2003**
  Password generation failed for: %s
- **BMYSVC2005**
  Vault health check failed for password rotation.
- **BMYSVC2006**
  Vault health check failed for SSH key rotation.
- **BMYSVC3001**
  Failed to read `SpectrumFusion` custom resource.
- **BMYSVC3002**
  Failed to update `RotationStatus` ConfigMap.

Serviceability Call Home events and error codes

List of Call Home events that you might encounter that related to serviceability.

The event that is raised if the issue exists in Call Home functions.
Note: The event does not generate Call Home events, though it is critical as the backend itself is not reachable.

- **BMYCH1001**
  Failure in establishing a TCP connection to the IBM eSupport endpoint.
- **BMYCH2001**
  Upload success.
- **BMYCH2002**
  Upload failed.
- **BMYCH2003**
  Upload timeout.

BMYCH1001

Failure in establishing a TCP connection to the IBM eSupport endpoint.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check whether the network configuration is correct or not.
2. Ensure that the endpoint is accessible and properly configured.
3. Check whether any potential firewall or security settings that might be blocking the connection.

4. If the issue persists for long time, then contact IBM support.

BMYCH2001

Upload success.

Severity

Informational

User actions

No user action is required.

BMYCH2002

Upload failed.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Check whether the network configuration is correct or not.
2. Ensure that the endpoint is accessible and properly configured.
3. Check whether any potential firewall or security settings that might be blocking the connection.
4. If the issue persists for long time, then contact IBM support.

BMYCH2003

Upload timeout.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Check whether the network configuration is correct or not.
2. Ensure that the endpoint is accessible and properly configured.
3. Check whether any potential firewall or security settings that might be blocking the connection.
4. If the issue persists for long time, then contact IBM support.

BMYLC0001

Remaining storage space is less than 30 percent.

Severity

Warning

User actions

Do the following steps to resolve the error:

1. Delete the unnecessary logs through the IBM Fusion HCI and IBM Fusion user interface to get storage space. For more information, see Delete a log package.
2. If the issue persists, then contact IBM support.

BMYLC0002

Remaining storage space is less than 10 percent.

Severity

Critical

User actions

Do the following steps to resolve the error:

1. Delete the unnecessary logs through the IBM Fusion HCI and IBM Fusion user interface to get storage space. For more information, see Delete a log package.
2. If the issue persists, then contact IBM support.

BMYSVC1001

Credential rotation started.

Severity

Informational

User actions

No action is required.

BMYSVC1002

Credential rotation completed.

Severity

Informational

User actions

No action is required.

BMYSVC2001

Rotation secret missing for: %s

Severity

Warning

User actions

Contact IBM support.

BMYSVC2002

Failed to update secret: %s

Severity

Warning

User actions

Contact IBM support.

BMYSVC2003

Password generation failed for: %s

Severity

Warning

User actions

Contact IBM support.

BMYSVC2005

Vault health check failed for password rotation.

Severity

Warning

User actions

Contact IBM support.

BMYSVC2006

Vault health check failed for SSH key rotation.

Severity

Warning

User actions

Contact IBM support.

BMYSVC3001

Failed to read `SpectrumFusion` custom resource.

Severity

Critical

User actions

Contact IBM support.

BMYSVC3002

Failed to update `RotationStatus` ConfigMap.

Severity

Critical

User actions

Contact IBM support.

Hosted Control Plane events and error codes

List of all the events that you might encounter that related to Hosted Control Plane.

For Informational, Warning, and Critical events, see the following:

- **BMYFA1001**
  Labels are added to the cluster.
- **BMYFA1002**
  Label removed from the cluster.
- **BMYFA1003**
  Label removed from the cluster.
- **BMYFA1004**
  The `fusion-addon-config` configmap exists.
- **BMYFA1005**
  Successfully removed storage from the cluster.
- **BMYFA1301**
  The ACM operator installation is preventing the addition of labels.
- **BMYFA1302**
  Unable to uninstall the service.
- **BMYFA1303**
  Failed to remove storage from the cluster.
- **BMYFA1304**
  Unable to remove the storage from the clusters.
- **BMYFA1305**
  The service client is waiting for the service installation to complete on the cluster.
- **BMYFA1306**
  Failed to delete the `NetworkPolicy` for clusters.
- **BMYFA1307**
  The fusion-base label is missing for clusters.
- **BMYFA1601**
  The `fusion-addon-config` configmap does not exist in the namespace.
- **BMYFA1604**
  The add-on is missing resources.
- **BMYFA1605**
  Failed to create install agent on cluster.
- **BMYFA1606**
  Failed to create install agent on the cluster where Global Data Platform is the storage provider.
- **BMYFA1607**
  Failed to install Backup & Restore agent.
- **BMYFA1608**
  Failed to delete the Backup & Restore connection for the clusters.
- **BMYFA1610**
  Failed to retrieve the storage provider.
- **BMYHC9001**
  Nodes are not ready.
- **BMYHC9002**
  Updating config for `nodepool`
- **BMYHC9003**
  This is single node cluster, updating config might not work
- **BMYHC7001**
  All nodes are healthy.
- **BMYHC7002**
  Nodepools in HCP cluster are healthy

BMYFA1001

Labels are added to the cluster.

Severity

Informational

User actions

No action is required.

BMYFA1002

Label removed from the cluster.

Severity

Informational

User actions

No action is required.

BMYFA1003

Label removed from the cluster.

Severity

Informational

User actions

No action is required.

BMYFA1004

The `fusion-addon-config` configmap exists.

Severity

Informational

User actions

No action is required.

BMYFA1005

Successfully removed storage from the cluster.

Severity

Informational

User actions

No action is required.

BMYFA1301

The ACM operator installation is preventing the addition of labels.

Severity

Warning

User actions

Install advanced cluster management operator on the IBM Fusion HCI or provided cluster.

BMYFA1302

Unable to uninstall the service.

Severity

Warning

User actions

Do the following steps to resolve the issue:

1. To uninstall the IBM Fusion service, uninstall the backup and restore agent and Fusion Data Foundation service from Hosted Control Plane cluster by removing their respective labels.
2. To uninstall the Fusion Data Foundation, ensure that you have to uninstall backup and restore agent from the Hosted Control Plane cluster by removing their respective labels.

BMYFA1303

Failed to remove storage from the cluster.

Severity

Warning

User actions

Do the following steps to resolve the issue:

1. Check whether any `openshift-storage` namespace is stuck in a terminating state.
2. In the status, check for the resource which the namespace is waiting for.
3. Remove the finalizer for the resource from the `openshift-storage` namespace status list.
4. Check whether namespace is deleted successfully.

BMYFA1304

Unable to remove the storage from the clusters.

Severity

Warning

User actions

Do the following steps to resolve the issue:

1. Check whether any `openshift-storage` namespace is stuck in a terminating state.
2. In the status, check for the resource which the namespace is waiting for.
3. Remove the finalizer for the resource from the `openshift-storage` namespace status list.
4. Check whether namespace is deleted successfully.

BMYFA1305

The service client is waiting for the service installation to complete on the cluster.

Severity

Warning

User actions

Wait for the installation of the service to be completed.

BMYFA1306

Failed to delete the `NetworkPolicy` for clusters.

Severity

Warning

User actions

Run the following command to delete the `NetworkPolicy` from the `openshift-storage-egress` clusters namespace.

```
oc delete networkpolicy openshift-storage-egress -n <clusters-<cluster_name>>
```

BMYFA1307

The fusion-base label is missing for clusters.

Severity

Warning

User actions

Add `isf.ibm.com/fusion-base` label to the cluster

BMYFA1601

The `fusion-addon-config` configmap does not exist in the namespace.

Severity

Critical

User actions

Add `isf.ibm.com/fusion-base` label to the cluster.

BMYFA1604

The add-on is missing resources.

Severity

Critical

User actions

Collect the Storage logs and contact IBM support.

BMYFA1605

Failed to create install agent on cluster.

Severity

Critical

User actions

Collect the Storage logs and contact support team.

BMYFA1606

Failed to create install agent on the cluster where Global Data Platform is the storage provider.

Severity

Critical

User actions

Remove the label from the cluster because the Fusion Data Foundation can not be installed if the primary storage is Global Data Platform.

BMYFA1607

Failed to install Backup & Restore agent.

Severity

Critical

User actions

Do the following steps to resolve the issue:

1. Create the storage class for the Backup & Restore agent.
2. If you use Fusion Data Foundation storage class, then ensure that the Fusion Data Foundation client is installed.

BMYFA1608

Failed to delete the Backup & Restore connection for the clusters.

Severity

Critical

User actions

Delete the connection from IBM Fusion user interface manually.

BMYFA1610

Failed to retrieve the storage provider.

Severity

Critical

User actions

Collect the Storage logs and contact IBM support.

BMYHC9001

Nodes are not ready.

Severity

Critical

User actions

It indicates that a Node in Hosted Control Plane cluster is not ready. To trigger firmware upgrade, all nodes of the cluster must be healthy state.

BMYHC9002

Updating config for **`nodepool`**

Severity

Critical

Description

It indicates that the `nodepool` config update is in progress for the Hosted Control Plane cluster.

User actions

Ensure that the `nodepool` config update is complete before you start the firmware upgrade for the cluster.

If you want to override the prechecks that are blocking the upgrade, do the following steps:
Note: It converts the Error prechecks to Warning prechecks and triggers the upgrade. Before using the `precheck-acks configmap` to override the pre-checks, ensure that these not ready nodes do not affect the upgrade.

1. Run the following command and export IBM Fusion namespace as an environmental variable.

   ```
   export FUSION_NS="namespace-where-fusion-is-installed"
   ```

2. Run the following command to create a `precheck-acks` configmap in the namespace where the IBM Fusion is installed.

   ```
   oc create configmap precheck-acks -n $FUSION_NS
   ```

3. In the `precheck-acks` config map, add the key `HCPNodePoolUpdateConfigInProgress` and set the value of Hosted Control Plane cluster name.

   ```
   oc edit configmap precheck-acks -n $FUSION_NS
   ```

   ```
   apiVersion: v1
   kind: ConfigMap
   metadata:
     name: precheck-acks
     namespace: ibm-spectrum-fusion-ns
   data:
     HCPNodeNotReady: <HCP-cluster1>,<HCP-cluster2>
   ```

   ```
   apiVersion: v1
   kind: ConfigMap
   metadata:
     name: precheck-acks
     namespace: ibm-spectrum-fusion-ns
   data:
     HCPNodePoolUpdateConfigInProgress: <HCP-cluster1>,<HCP-cluster2>
   ```

BMYHC9003

This is single node cluster, updating config might not work

Severity

Warning

Description

It indicates that this is a single node cluster, so node upgrade or reboot may make the cluster unreachable.

User actions

No action

BMYHC7001

All nodes are healthy.

Severity

Information

User actions

No action

BMYHC7002

Nodepools in HCP cluster are healthy

Severity

Information

User actions

No action

Software Hub events and error codes

List of all the events and error codes that you might encounter while you work with storage.

For Informational, Warning, and Critical events, see the following:

- **BMYOA0101**
  OpenShift® AI installation is in progress.
- **BMYOA0102**
  OpenShift AI installation is completed.
- **BMYOA0103**
  OpenShift AI installation failed.
- **BMYOA0104**
  OpenShift AI is in a healthy state.
- **BMYOA0105**
  OpenShift AI is in a degraded state.
- **BMYOA0106**
  OpenShift AI operator pods are running.
- **BMYOA0107**
  OpenShift AI DSCI CR is created.
- **BMYOA0108**
  OpenShift AI DSCI CR is in a ready state.
- **BMYOA0109**
  OpenShift AI DSC CR is created.
- **BMYOA0110**
  OpenShift AI DSC CR is in a ready state.
- **BMYOA0111**
  OpenShift AI application pods are running.
- **BMYOA0112**
  OpenShift AI inference config is patched.
- **BMYOA0113**
  OpenShift AI inference config is verified.
- **BMYOA0114**
  OpenShift AI DSCI CR is in progressing state.
- **BMYOA0115**
  OpenShift AI DSC CR is in progressing state.
- **BMYOA0116**
  OpenShift AI installation completed from the OperatorHub.
- **BMYSH0101**
  Software Hub installation is in progress.
- **BMYSH0102**
  Software Hub installation is completed.
- **BMYSH0103**
  Software Hub installation failed.
- **BMYSH0104**
  CPD CR is not in a healthy state.
- **BMYSH0105**
  CPD CR is in a healthy state.
- **BMYSH0106**
  Software Hub CR is in a healthy state.
- **BMYSH0107**
  Software Hub CR is not in a healthy state.
- **BMYSH0108**
  Software Hub installation is in progress before installing watsonx.ai services.
- **BMYWA0101**
  watsonx.ai installation is started.
- **BMYWA0102**
  watsonx.ai installation is completed.
- **BMYWA0103**
  watsonx.ai installation is in progress.
- **BMYWA0104**
  watsonx.ai installation failed.

- **BMYWA0105**
  watsonx.ai CR not in a healthy state.
- **BMYWA0106**
  watsonx.ai CR in a healthy state.
- **BMYWO0101**
  watsonx Orchestrate installation is started.
- **BMYWO0102**
  watsonx Orchestrate installation is completed.
- **BMYWO0103**
  watsonx Orchestrate installation is in progress.
- **BMYWO0104**
  watsonx Orchestrate installation failed.
- **BMYWO0105**
  watsonx Orchestrate CR not in a healthy state.
- **BMYWO0106**
  watsonx Orchestrate CR is in a healthy state.

BMYOA0101

OpenShift® AI installation is in progress.

Severity

Informational

User actions

No action is required.

BMYOA0102

OpenShift® AI installation is completed.

Severity

Informational

User actions

No action is required.

BMYOA0103

OpenShift® AI installation failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Run the following commands to check whether all pods running or completed with the `READY` state.

   ```
   oc get pods -n redhat-ods-operator -o wide
   oc get pods -n redhat-ods-applications -o wide
   ```

2. Run the following command to check whether the pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.

   ```
   oc describe pod/ <pod> -n <ns>
   ```

3. Run the following command to check CR health (DSCI/DSC) and verify that `Ready condition` is set to `True`.

   ```
   oc get dsci,dsc -A
   ```

   If the issue persists, identify the failing component (for example, KServe, ModelMesh, or Pipelines) and check its deployment status and events in the `redhat-ods-applications` namespace.

4. Run the following command to verify that exactly one `OperatorGroup` is configured to target the correct namespace.

```
oc get og -n redhat-ods-operator
```

5. Run the following command to check whether the `InstallPlan` is completed.

   ```
   oc get csv,installplan -n redhat-ods-operator
   ```

   If the `InstallPlan` status is `Pending` or `Failed` for version 2.19.1, either approve or correct the source. If the CSV status is `Failed`, delete the failed CSV file to retry the process.
6. Run the following command to verify whether the subscription is installed.

   ```
   oc describe subscriptions.operators.coreos.com/rhods-operator -n redhat-ods-operator
   ```

7. Run the following commands to verify that the catalog source status is `Ready`.

   ```
   oc get catalogsource -n openshift-marketplace
   oc describe catalogsource redhat-operators -n openshift-marketplace
   ```

   If the source is not ready, restart the `Marketplace` pods.

8. If the issue persists, then contact IBM support.

BMYOA0104

OpenShift® AI is in a healthy state.

Severity

Informational

User actions

No action is required.

BMYOA0105

OpenShift® AI is in a degraded state.

Severity

Warning

User actions

Do the following steps to diagnose the error:

1. Run the following command to check whether `redhat-ods-operator` and `redhat-ods-applications` namespaces are active.

   ```
   oc get ns redhat-ods-operator redhat-ods-applications
   ```

2. Ensure that an `OperatorGroup` is available in the `redhat-ods-operator` namespace.

   ```
   oc get og -n redhat-ods-operator
   ```

3. Run the following command to verify that the `InstallPlan` status is `Complete` and the `CSV PHASE` status is `Succeeded`.

   ```
   oc get  ip,csv -n redhat-ods-operator
   ```

4. Run the following commands to check pods status in the `redhat-ods-operator` and `redhat-ods-applications` namespaces.

   ```
   oc get pods -n redhat-ods-operator
   oc get pods -n redhat-ods-applications
   ```

   All Pods must be running or completed state.
5. If the issue persists, then contact IBM support.

BMYOA0106

OpenShift® AI operator pods are running.

Severity

Informational

User actions

No action is required.

BMYOA0107

OpenShift® AI DSCI CR is created.

Severity

Informational

User actions

No action is required.

BMYOA0108

OpenShift® AI DSCI CR is in a ready state.

Severity

Informational

User actions

No action is required.

BMYOA0109

OpenShift® AI DSC CR is created.

Severity

Informational

User actions

No action is required.

BMYOA0110

OpenShift® AI DSC CR is in a ready state.

Severity

Informational

User actions

No action is required.

BMYOA0111

OpenShift® AI application pods are running.

Severity

Informational

User actions

No action is required.

BMYOA0112

OpenShift® AI inference config is patched.

Severity

Informational

User actions

No action is required.

BMYOA0113

OpenShift® AI inference config is verified.

Severity

Informational

User actions

No action is required.

BMYOA0114

OpenShift® AI DSCI CR is in progressing state.

Severity

Informational

User actions

No action is required.

BMYOA0115

OpenShift® AI DSC CR is in progressing state.

Severity

Informational

User actions

No action is required.

BMYOA0116

OpenShift® AI installation completed from the OperatorHub.

Severity

Informational

User actions

No action is required.

BMYSH0101

Software Hub installation is in progress.

Severity

Informational

User actions

No action is required.

BMYSH0102

Software Hub installation is completed.

Severity

Informational

User actions

No action is required.

BMYSH0103

Software Hub installation failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of pods in the `cpd-instance` and `cpd-operator` namespace.
2. Check whether the pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. If the issue persists, then contact IBM support.

BMYSH0104

CPD CR is not in a healthy state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of pods in the `cpd-instance` and `cpd-operator` namespace.
2. Check whether the pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. If the issue persists, then contact IBM support.

BMYSH0105

CPD CR is in a healthy state.

Severity

Informational

User actions

No action is required.

BMYSH0106

Software Hub CR is in a healthy state.

Severity

Informational

User actions

No action is required.

BMYSH0107

Software Hub CR is not in a healthy state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of pods in the `cpd-instance` and `cpd-operator` namespace.
2. Check whether the pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. If the issue persists, then contact IBM support.

BMYSH0108

Software Hub installation is in progress before installing watsonx.ai services.

Severity

Informational

User actions

No action is required.

BMYWA0101

watsonx.ai installation is started.

Severity

Informational

User actions

No action is required.

BMYWA0102

watsonx.ai installation is completed.

Severity

Informational

User actions

No action is required.

BMYWA0103

watsonx.ai installation is in progress.

Severity

Informational

User actions

No action is required.

BMYWA0104

watsonx.ai installation failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of pods in the `cpd-instance` and `cpd-operator` namespace.
2. Check whether the pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. If the issue persists, then contact IBM support.

BMYWA0105

watsonx.ai CR not in a healthy state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of pods in the `cpd-instance` and `cpd-operator` namespace.
2. Check whether the pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. If the issue persists, then contact IBM support.

BMYWA0106

watsonx.ai CR in a healthy state.

Severity

Informational

User actions

No action is required.

BMYWO0101

watsonx Orchestrate installation is started.

Severity

Informational

User actions

No action is required.

BMYWO0102

watsonx Orchestrate installation is completed.

Severity

Informational

User actions

No action is required.

BMYWO0103

watsonx Orchestrate installation is in progress.

Severity

Informational

User actions

No action is required.

BMYWO0104

watsonx Orchestrate installation failed.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of pods in the `cpd-instance` and `cpd-operator` namespace.
2. Check whether the pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. If the issue persists, then contact IBM support.

BMYWO0105

watsonx Orchestrate CR not in a healthy state.

Severity

Critical

User actions

Do the following steps to diagnose the error:

1. Check the status of pods in the `cpd-instance` and `cpd-operator` namespace.
2. Check whether the pods in states like `ImagePullBackOff`, `CrashLoopBackOff`, or `Error`.
3. If the issue persists, then contact IBM support.

BMYWO0106

watsonx Orchestrate CR is in a healthy state.

Severity

Informational

User actions

No action is required.

Troubleshooting issues in IBM Fusion HCI

This topic covers the most common issues and their workarounds, which you might encounter while you work with the IBM Fusion HCI appliance.

Component / Feature	Link
Installation and upgrade	Installation issues
Fusion services installation and upgrade	• Fusion Data Foundation service- Fusion Data Foundation service install and upgrade issues • Global Data Platform service- Global Data Platform service install and upgrade issues • Backup & Restore service- Backup & Restore service installation and upgrade issues • IBM Data Cataloging service- IBM Data Cataloging service installation and upgrade issues
Offline mirroring	Troubleshooting and known issues in offline mirroring
Backup and restore service	Troubleshooting Backup & Restore service issues
Global Data Platform storage	Global Data Platform service issues
Fusion Data Foundation storage	Fusion Data Foundation service error scenarios
Node and IBM Fusion HCI node drains	Node issues and Issues related to IBM Fusion HCI node drains
Networking	Networking issues
Serviceability	Troubleshooting serviceability issues
Connection setup after OpenShift® Container Platform cluster recovery	Connection setup after OpenShift Container Platform cluster recovery
Multi-rack HA, 3-zone HA, expansion rack issues	Multi-rack issues
Virtualization	Virtualization issues

- **Connection setup after OpenShift Container Platform cluster recovery**
  The OpenShift Container Platform cluster can have problems and become unusable. After you recover the cluster, rejoin the connections.
- **Multi-rack issues**
  This topic covers issues related to multi-rack HA, 3-zone HA, and expansion racks.
- **Using Vault with IBM Fusion HCI**
  This section describes about the diagnostic utility for IBM Fusion HCI.
- **Virtualization issues**
  This topic covers issues related to virtualization and their resolution.

Connection setup after OpenShift Container Platform cluster recovery

The OpenShift® Container Platform cluster can have problems and become unusable. After you recover the cluster, rejoin the connections.

Disaster recovery

- Two clusters must be connected to failover and failback applications
- Reconnect to another cluster

Backup & Restore

- One hub cluster and more than one spoke clusters can be connected easily to manage application backups in multi clusters.
- Reconnect to the hub or spoke

If the cluster recovers before the expiration time, then no action is needed as the cluster rejoins the connection automatically. If the cluster recovers after the client cert expires, then clean the connection and setup again to rejoin the recovered cluster. You can get the client cert effective time from the connection CR status, for example:

```
oc get connection <connection_name> -n ibm-spectrum-fusion-ns
```

```
  - lastTransitionTime: '2024-01-03T10:23:20Z'
    message: >-
      client certificate rotated starting from 2024-01-03 10:18:31 +0000 UTC
      to 2024-01-19 13:12:00 +0000 UTC
    reason: ClientCertificateUpdated
    status: 'True'
    type: ClusterCertificateRotated
```

Clean the connection and setup the connection between cluster-a and cluster-b again.

1. Clean the connection:
   Delete the connection CR in both cluster-a and cluster-b clusters:

   - In cluster-a

     ```
     oc delete connection <connection_cr_with_cluster-b_endpoint_in_spec> -n ibm-spectrum-fusion-ns
     ```

   - In cluster-b

     ```
     oc delete connection <connection_cr_with_cluster-a_endpoint_in_spec> -n ibm-spectrum-fusion-ns
     ```

2. Set up the connection between clusters again.
   a. Get the bootstrap token in cluster-a:

      ```
      kubectl create token isf-application-operator-cluster-bootstrap -n <namespace of isf-application-operator>
      ```

   b. Create connection CR and `init` secret in cluster-b with the bootstrap token and API endpoint of cluster-a.
      For example, create `init` secret in cluster-b:

      ```
      apiVersion: v1
      kind: Secret
      metadata:
        name: init-<cluster-a>
        namespace: ibm-spectrum-fusion-ns
      stringData:
        apiserver: <cluster-a api endpoint>
        bootstrapToken: <token generated in step 2.1>
      ```

3. Create connection CR with this `init` secret in `spec` in cluster-b:

   ```
   apiVersion: application.isf.ibm.com/v1
   kind: Connection
   metadata:
     name: <connection-name>
     namespace: ibm-spectrum-fusion-ns
   spec:
     remoteCluster
       apiEndpoint: <cluster-a api endpoint>
       initSecretName: init-<cluster-a>
   ```

Global Data Platform service issues

This section lists the troubleshooting tips and tricks when you use IBM Fusion storage.

Warning: Do NOT delete IBM Storage Scale pods. Deletion of Scale pods in many circumstances has implications on availability and data integrity.

Filesystem status from Remote file systems is stuck in Connecting state

Cause
: The secure boot is not supported by Scale and CNSA.

Resolution

1. Check whether the nodes are running with secure boot:

   ```
   mokutil --sb-state
   ```

   ```
   SecureBoot enabled
   ```

2. As a root password, you can disable the secure boot with the following command:

   ```
   mokutil --disable-validation
   ```

   ```
   password length: 8~16
   input password:
   input password again:
   ```

Increased CPU usage on Scale pods after upgrading to Cloud Pak for Integration (CP4I)

Problem statement
: Scale pod CPU usage has increased due to changes in MQ streams application. It is because of IBM MQ's liveness and readiness probes.

Resolution
: The file handle reservation can be turned off by setting the environment variable `AMQ_NO_RESERVE_FD=1`. It fixes the high CPU usage.

Scale operator failed to add quorum label

Problem statement
: During Storage configuration, the recovery group status can be "`Waiting on some daemons in the recovery group to be restarted`".

Resolution

1. Run the following command to check the count of the quorum node:

```
oc get nodes -l scale.spectrum.ibm.com/designation=quorum
```

2. Check the number of quorum nodes. Whenever the node count is less than 10, there must 3 quorum nodes. If not, apply the following quorum label for the missing nodes:

```
scale.spectrum.ibm.com/designation=quorum
```

Remote filesystem connection goes to a Disconnected state

Problem statement
: If even one scale core pod is not ready due to node restart or other issues, then a problem occurs on the filesystem mount of that node and the filesystem CR reports errors. As a result, the filesystem goes to a Disconnected and the same is displayed on the IBM Fusion user interface.

Resolution
: Wait for the scale core pod to go to Ready state, and the filesystem automatically changes to Connected state.

PVCs stuck in pending state

Resolution
: If PVCs are stuck in a pending state for a long time, restart Scale GUI pods.

Nodes with pods scheduled for storage results in a `crashloobackoff` or container creating errors

Problem statement
: When install, upgrade, or upsize operations are in progress and the storage configuration is not yet done, the nodes with pods scheduled for storage results in a `crashloobackoff` or container creating errors.

Resolution
: To resolve this issue, disable schedule on these nodes before you begin these operations, and schedule the pods on nodes that are configured for storage.

File system is not mounted on some nodes

Problem statement
: After IBM Storage Scale upgrade, the file system is not mounted on some nodes.

Resolution
: Do the following workaround steps:

1. Delete the problematic pod where the file system is not mounted. Delete one pod at a time.
2. Refresh the pod overview page until the pod appears. Run `watch oc get pods`.
3. When the problematic pod comes up, set the replicas to 0 in the `Spec` section in the `ibm-spectrum-scale-operator` namespace. You must stop the operator deployment immediately. Ensure that no operator pod is running in `ibm-spectrum-scale-operator` namespace. Command for setting the operator replica to 0:

```
oc patch deploy ibm-spectrum-scale-controller-manager \
 --type='json' -n ibm-spectrum-scale-operator \
 -p='[{"op": "replace", "path": "/spec/replicas", "value": 0}]'
```

```
oc get deployment -n ibm-spectrum-scale-operator
```

Sample output:

```
I0420 16:56:23.579340    4184 request.go:645] Throttling request took 1.192496025s, request: GET:https://api.isf-rackk.rtp.raleigh.ibm.com:6443/apis/nmstate.io/v1beta1?timeout=32s
NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
ibm-spectrum-scale-controller-manager   0/0     0            0           28h
```

4. Wait till the problematic pod is in `init` container state and run the following command:

```
oc exec podname -- mmsdrrestore -p <any working core ip/name[Specify pod IP where FS is mounted]>
```

5. After the previous step is successful, set the replicas to 1 again in the `ibm-spectrum-scale-operator` deployment. Ensure that the operator pod is running in `ibm-spectrum-scale-operator` namespace.

   Run the following command for setting the operator replica to 0:

```
oc patch deploy ibm-spectrum-scale-controller-manager \
 --type='json' -n ibm-spectrum-scale-operator \
 -p='[{"op": "replace", "path": "/spec/replicas", "value": 1}]'
```

```
oc get deployment -n ibm-spectrum-scale-operator
```

Sample output:

```
I0420 16:56:23.579340    4184 request.go:645] Throttling request took 1.192496025s, request: GET:https://api.isf-
rackk.rtp.raleigh.ibm.com:6443/apis/nmstate.io/v1beta1?timeout=32s
NAME                                        READY   UP-TO-DATE   AVAILABLE   AGE
ibm-spectrum-scale-controller-manager       1/1     0            0           28h
```

6. Wait for some time for the problematic pod to come in running state and file system gets mounted on this problematic node.

IBM Storage Scale user interface does not load properly or IBM Storage Scale CSI pods are in `CrashLoopBack` state

Problem statement
: Sometimes, IBM Storage Scale user interface might not load properly or IBM Storage Scale CSI pods are in **`CrashLoopBack`** state.

Resolution
: As a workaround, do the following steps:

1. Restart the GUI pods and check whether it is working properly.
2. If GUI pod restart does not open the IBM Storage Scale console, then restart the **`ibm-spectrum-scale-controller-manager`** pod and then restart GUI pod.

Not able to log in to IBM Storage Scale user interface

Cause
: The secret that is mounted in GUI pods /etc/gui-oauth/oauth.properties file and secret present in the **`ibm-spectrum-scale-gui-oauthclient`** are different.

```
gui-1
sh-4.4$ cd /etc/gui-oauth/
sh-4.4$ cat oauth.properties
secret=tjCK6kE0pmCe2d8b7azw
oauthServer=https://oauth-openshift.apps.isf-racka.rtp.raleigh.ibm.com
redirectURI=https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racka.rtp.raleigh.ibm.com/auth/redirect
clientID=ibm-spectrum-scale-gui-oauthclient
k8sAPIServer=https://172.30.0.1:443
sh-4.4$
gui_0
secret=tjCK6kE0pmCe2d8b7azw
oauthServer=https://oauth-openshift.apps.isf-racka.rtp.raleigh.ibm.com
redirectURI=https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racka.rtp.raleigh.ibm.com/auth/redirect
clientID=ibm-spectrum-scale-gui-oauthclient
k8sAPIServer=https://172.30.0.1:443
Oauth client config (API Explorer -> OAuthClient -> ibm-spectrum-scale-gui-oauthclient)
kind: OAuthClient
apiVersion: oauth.openshift.io/v1
metadata:
  name: ibm-spectrum-scale-gui-oauthclient
  uid: a9eb9d85-33b9-4cd7-99c8-ae0213032380
  resourceVersion: '314932'
  creationTimestamp: '2022-03-29T17:09:38Z'
  labels:
    app.kubernetes.io/instance: ibm-spectrum-scale
    app.kubernetes.io/name: gui
  ownerReferences:
apiVersion: scale.spectrum.ibm.com/v1beta1
      kind: Gui
      name: ibm-spectrum-scale-gui
      uid: da23c696-5c71-405b-b9ac-c7d0ccb4dd43
      controller: true
      blockOwnerDeletion: true
  managedFields:
manager: ibm-spectrum-scale/ibm-spectrum-scale-gui
      operation: Apply
      apiVersion: oauth.openshift.io/v1
      time: '2022-03-29T17:09:38Z'
      fieldsType: FieldsV1
      fieldsV1:
        'f:grantMethod': {}
        'f:metadata':
          'f:labels':
            'f:app.kubernetes.io/instance': {}
            'f:app.kubernetes.io/name': {}
          'f:ownerReferences':
            'k:{"uid":"da23c696-5c71-405b-b9ac-c7d0ccb4dd43"}':
              .: {}
              'f:apiVersion': {}
              'f:blockOwnerDeletion': {}
              'f:controller': {}
              'f:kind': {}
              'f:name': {}
              'f:uid': {}
        'f:redirectURIs':
          'v:"https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racka.rtp.raleigh.ibm.com/auth/redirect"': {}
        'f:secret': {}
*secret: bk2_q1FqwhVhoNqOl7ob*
redirectURIs:
>-
    https://ibm-spectrum-scale-gui-ibm-spectrum-scale.apps.isf-racka.rtp.raleigh.ibm.com/auth/redirect
```

```
grantMethod: auto
```

Resolution
: As a resolution, do the following steps:

1. As a resolution, restart the GUI pods.
2. If the previous step (step 1) does not resolve the issue, then restart IBM Storage Scale Operator pod and then restart the GUI pods.

Download log files of all deployments

If you cannot download log files of all deployments, individually download the log files.

Issues in node drain and node restart

Resolution
: To resolve such node related issues, see Issues related to IBM Fusion HCI node drains.

IBM Storage Scale cluster failed to recover after powering off and on the compute nodes

Diagnosis
: Identify if the nodes joining the cluster are in an out-of-tolerance situation.

1. mmhealth reported UNKNOWN RG status on some nodes.
2. mmrepquota commands have been generated repeatedly, likely by the GUI, which has caused many long waits.
3. Due to the large number of long waits, the cluster manager node was not responding to any **ts cmd** other than **tsctl nqStatus**.
   As a resolution, follow the steps defined in the Recover storage cluster from an unplanned multi-node restart or failure.

MCO rollout is not progressing for more than 5 hours during the scale upgrade

Cause
: Whenever the nodes are over committed and no pod limit is available, you might see delays in the policy rollout, ICSP, pull secret, and MCO rollout changes.

Resolution
: As a resolution, do the following step:

- IBM Fusion HCI nodes allow the maximum number of pods on a node as defined by OpenShift® Container Platform default and ensure to keep that count within the limit. Otherwise, you might see this issue.

IBM Storage Scale cluster status is in a critical state after power on/off the rack

Cause
: IBM Storage Scale cluster status is in critical state due to `unmounted_fs_check` is present on the `filesystem` CR.

Resolution
: As a resolution, wait until this `filesystem` CR status gets cleared; after that, the IBM Storage Scale cluster status automatically returns to a healthy state. If the issue persists for a long time, then contact IBM support.

Disk upsize issues

- Manual steps to test the `/dev` mount:
  After you add a new NVMe disk to the server, the new disk cannot be discovered in the IBM Storage Scale pod. The disks do not show up with /dev/nvme* device names in the POD, IBM Storage Scale need the device name /dev/nvme* to access the disks. Without the /dev/nvme* device name, the new disks cannot be added into recovery group.

  Resolution
  : Do the following manual steps to mount disks to /dev in the pod.

  1. Log in to each IBM Storage Scale storage pods.
     ```
     oc exec -it <pod name> -c <container> /bin/sh
     ```
  2. Stop the GPFS daemons on one or more nodes:
     ```
     mmshutdown
     ```
  3. Mount `/dev` from the host running the following command:
     ```
     mount -t devtmpfs none /dev
     ```
  4. Validate that the content of /dev is from host and you can see sub directories like block, bus, char.
     ```
     mmstartup
     ```

- **Recover storage cluster from an unplanned multi-node restart or failure**
  The service of IBM Storage Scale file system might hang if the storage nodes are restarted.
- **IBM Storage Scale file system not showing newly added file system size**
  The IBM Storage Scale file system is not showing the newly added file system size after adding six nodes as upsize.

Recover storage cluster from an unplanned multi-node restart or failure

The service of IBM Storage Scale file system might hang if the storage nodes are restarted.

The storage nodes might get restarted unexpectedly and might cause the disk to miss in the storage cluster. If the missing disks exceed the fault tolerance RAID code used that is used in the storage cluster, the file system might stop to provide service. It cannot finish the log recovery process for the lost nodes. It might also block the restarting nodes to join the cluster again.

Follow these steps to recover IBM Spectrum® Storage Scale Erasure Code Edition (ECE) from the out of fault tolerance.
Note: Out of fault tolerance in the IBM Storage Scale Erasure Code Edition (ECE) environment can be caused due to different reasons.

1. Identify the problem.
   a. Run the **mmfsadm dump waiters** command on the running node of the application. It displays the long waiters of I/O threads.
   b. Run the **mmgetstate** command. It displays the restart nodes in arbitrating state and others in active state.
   c. Run the **tslsrecgroup rg_1 --server --v2** command several times. It displays that Log Groups jumps between nodes.
   d. Check the mmfs.log file on all storage nodes. The log message shows up in loop on different nodes. See the message as follows:

   ```
   [E] Beginning to resign log group LG002 in recovery group rg1 ....
   ```

   It means that the user log group continues to resign.
2. Start the recovery process.
   a. Run the **mmlscluster** command to find all the quorum nodes.
   b. Run the **mmshutdown** command on all quorum nodes. This action causes all nodes to unmount from the file system.
   c. Run the **mmstartup** command on all quorum nodes.
   d. Run the **mmlsrecoverygroup rg1 -L --pdisk** to check whether the recovery group is active again and that no missing `pdisk` exists.
   e. Mount the file system again on all the client nodes and restart the application.

IBM Storage Scale file system not showing newly added file system size

The IBM Storage Scale file system is not showing the newly added file system size after adding six nodes as upsize.

About this task

A system with a single recovery group (setup with 4+2p erasure code) needs to do node upsizing with six nodes all together.

Procedure

1. Follow the steps to upsize nodes:
   a. Follow the standard node upsize process. For more information, see Configuring nodes for management.
   b. IBM Fusion HCI user interface shows the Running state for the newly added nodes.
   c. In the IBM Fusion HCI user interface, use the app switcher to go to the IBM Storage Scale user interface.
   d. Click Recovery groups on the IBM Storage Scale RAID tile.
   The entry for the recovery group server nodes column shows up empty. If true, perform the following steps. Otherwise, contact the IBM support.
   e. Edit the labels of newly added nodes, change the label `scale.spectrum.ibm.com/rg: rg1` to `scale.spectrum.ibm.com/rg: rg2`.
   f. Repeat the following process for each newly added node.
      i. Open the OpenShift® Container Platform user interface.
      ii. Go to Compute > Nodes > Details > Actions > Edit labels.
      iii. Remove the label `scale.spectrum.ibm.com/rg: rg1` and add the label `scale.spectrum.ibm.com/rg: rg2`.
      iv. Click Save.
2. Follow the steps to validate the upsized nodes:
   a. In the IBM Fusion HCI user interface, use the app switcher to go to the IBM Storage Scale user interface.
   b. Click Recovery groups on the IBM Storage Scale RAID tile.
   c. Click the entry of the recovery group server nodes column for rg2.
   d. Check all the newly added nodes are listed and in healthy state.
   e. Create a sample PVC to check that storage can be provisioned.

Multi-rack issues

This topic covers issues related to multi-rack HA, 3-zone HA, and expansion racks.

Rack shutdown issues

Problem statement
: A rack shutdown in a multi-rack HA setup can cause the Backup & Restore service to go to an unknown state.

Resolution
: In the case of a rack shutdown, if a graceful shutdown is not working or the node is in a non-recoverable state, you can manually add an `out-of-service` taint on the node. For more information about how it works, see Kubernetes Blogs.
Remove the taint on the affected node after the node is recovered.

Using Vault with IBM Fusion HCI

This section describes about the diagnostic utility for IBM Fusion HCI.

IBM Fusion HCI uses Vault to securely store and manage sensitive credentials, secrets, and cryptographic keys. Vault provides centralized secrets management with strong access controls, audit logging, and encryption.

Key features

The Vault provides the following key features:

- Centralized secret storage: All sensitive credentials are stored in one secure location.
- Audit logging: Complete audit trail of all access and operations.
- Access control: Fine-grained policies for secret access.
- TLS encryption: HTTPS communication with self-signed certificates.

Initialization parameters

- Unseal keys: The system generates five unseal keys.
- Threshold: Three unseal keys are required to unseal the Vault.

Note:

- After stage 2 installation completes, Vault files are deleted from the /var/vault directory if the user selects the option to delete sensitive data. The Vault files (`vaultLogin` and `vault.config`) are included in the `keysAndPassword` archive.
- During a service node upgrade or service node addition (Day 2 activity), no files are available for download. After the operation completes successfully, you must download the `vault-secret` and `vault-login-secret` YAML files from OpenShift® as a backup.

Security features

Vault provides the following security features.

Seal and unseal mechanism

Sealed state: The Vault starts in a sealed state, where it encrypts all data and prevents access.

Unsealing: The unseal process requires three of the five generated unseal keys to reconstruct and decrypt the master key.

Encryption

Data encryption: Vault protects data by using the following encryption mechanisms.

- Encrypts all data at rest by using AES-256-GCM
- Encrypts the master key by using Shamir's Secret Sharing with unseal keys
- Secures all network communication with TLS by using self-signed certificates

File encryption: Vault stores sensitive files in the /var/vault/ directory in Base64-encoded format.

Troubleshooting Vault

Prerequisites

- Access to the OpenShift cluster.
- Vault-related secrets available, either from /var/vault (if present) or the `vault-secrets` resource in OpenShift.
- Access to the service node to execute Vault-related commands.
- The Vault is installed and checked using the `vault version` command.

Vault version

To check the installed Vault version, run the following command:

```
vault --version
```

Vault status

1. Before you begin troubleshooting, run the following command on the service node to check the Vault service status:

```
systemctl status vault
```

2. Review the output to determine the current state of Vault, and then run the appropriate troubleshooting commands on the service node.

The following sections describe common Vault issues and their resolutions:

Vault is sealed

Symptoms

- The Vault API returns a *Vault is sealed* error.
- The `vault status` command shows `Sealed: true`.

Resolution

- Unsealing Vault using IBM Fusion-managed `vault-secret`:
  You can unseal Vault by using the `vault-secret` stored in the `ibm-spectrum-fusion-ns` namespace. This secret contains the required unseal keys and credentials.

Procedure

1. Verify that the Vault is sealed:

   ```
   vault status
   ```

   When Vault is sealed, the output includes entries similar to the following:

   ```
   Sealed: true
   Seal Type: shamir
   Initialized: true
   Total Shares: 5
   Threshold: 3
   Unseal Progress: 0/3
   ```

2. Extract unseal keys from the `vault-secret`.
   The `vault-secret` in the `ibm-spectrum-fusion-ns` namespace stores all unseal keys. Extract the required keys and store them in environment variables:

   ```
   # Extract unseal keys from the secret
   oc get secret vault-secret -n ibm-spectrum-fusion-ns -o jsonpath='{.data.unsealKey1}' | base64 -d
   oc get secret vault-secret -n ibm-spectrum-fusion-ns -o jsonpath='{.data.unsealKey2}' | base64 -d
   oc get secret vault-secret -n ibm-spectrum-fusion-ns -o jsonpath='{.data.unsealKey3}' | base64 -d
   oc get secret vault-secret -n ibm-spectrum-fusion-ns -o jsonpath='{.data.unsealKey4}' | base64 -d
   oc get secret vault-secret -n ibm-spectrum-fusion-ns -o jsonpath='{.data.unsealKey5}' | base64 -d

   # Store the extracted keys in environmental variables for easy use
   UNSEAL_KEY_1=$(oc get secret vault-secret -n ibm-spectrum-fusion-ns -o jsonpath='{.data.unsealKey1}' |
   base64 -d)
   UNSEAL_KEY_2=$(oc get secret vault-secret -n ibm-spectrum-fusion-ns -o jsonpath='{.data.unsealKey2}' |
   base64 -d)
   UNSEAL_KEY_3=$(oc get secret vault-secret -n ibm-spectrum-fusion-ns -o jsonpath='{.data.unsealKey3}' |
   base64 -d)
   ```

3. Unseal Vault.
   The Vault requires three of the five unseal keys. Apply the keys sequentially:

   ```
   # Set Vault address
   export VAULT_ADDR='https://127.0.0.1:8200'
   export VAULT_SKIP_VERIFY=true

   # Unseal with first key
   vault operator unseal $UNSEAL_KEY_1

   # Expected output:
   # Key                Value
   # ---                -----
   # Seal Type          shamir
   # Initialized        true
   # Sealed             true
   # Total Shares       5
   # Threshold          3
   # Unseal Progress    1/3    <-- Progress indicator
   # Unseal Nonce       <nonce-value>
   # Version            1.21.1
   # Storage Type       raft
   # HA Enabled         true

   # Unseal with second key
   vault operator unseal $UNSEAL_KEY_2

   # Expected output:
   # Unseal Progress    2/3    <-- Progress indicator

   # Unseal with third key (final)
   vault operator unseal $UNSEAL_KEY_3

   # Expected output when successfully unsealed:
   # Sealed             false   <-- Vault is now unsealed
   # Unseal Progress    0/3
   ```

4. Verify that the Vault is unsealed:

   ```
   # Check vault status
   vault status

   # Expected output:
   ## Sealed: false
   ## Initialized: true
   ## HA Enabled: true
   ```

   Alternative: Unseal in a single command
      You can also unseal Vault by running a single command that applies three unseal keys sequentially:

      ```
      export VAULT_ADDR='https://127.0.0.1:8200'
      export VAULT_SKIP_VERIFY=true

      vault operator unseal $(oc get secret vault-secret -n ibm-spectrum-fusion-ns -o
      jsonpath='{.data.unsealKey1}' | base64 -d) && \
      vault operator unseal $(oc get secret vault-secret -n ibm-spectrum-fusion-ns -o
      jsonpath='{.data.unsealKey2}' | base64 -d) && \
      vault operator unseal $(oc get secret vault-secret -n ibm-spectrum-fusion-ns -o
      jsonpath='{.data.unsealKey3}' | base64 -d)
      ```

Troubleshooting the unseal process

This section describes common issues that can occur during the Vault unseal process and the corresponding resolutions.

`vault-secret` not found in the `ibm-spectrum-fusion-ns` namespace

Symptom

- The `vault-secret` does not appear in the `ibm-spectrum-fusion-ns` namespace.

Resolution

1. Review the `vault.config` file (Base64-encoded) located in the /var/vault directory.
2. Run the following command to decode the file and get the unseal keys:

```
cat /var/vault/vault.config | base64 -d
```

Invalid unseal key

Symptom

- The unseal command fails with an *invalid unseal key* error.

Resolution

Verify that the unseal key is correctly decoded and that the secret stores the keys in Base64-encoded format:

```
# Verify that the key is correctly decoded
echo $UNSEAL_KEY_1 | wc -c  # Must be 44 characters (Base64-encoded key)

# Check if keys are Base64-encoded in the secret
oc get secret vault-secret -n ibm-spectrum-fusion-ns -o yaml | grep unsealKey
```

Connection refused

Symptoms

- Vault commands fail with *connection refused* errors.
- You cannot connect to the Vault API.

Resolution

Verify that the Vault service runs and listens on the expected port:

```
# Check if the Vault service is running
systemctl status vault

# Check if the Vault is listening on port 8200
ss -tlnp | grep 8200

# Verify the Vault address
echo $VAULT_ADDR
```

- Unsealing Vault using /var/vault artifacts on the service node:
  Note: During a service node upgrade or when the service node is added as part of Day 2 operations, the required Vault artifacts are available in the /var/vault directory in the encoded format. These files contain sensitive information and are automatically deleted after the operation completes.
  1. Decode the unseal keys:

  ```
  base64 -d /var/vault/unsealKey1
  base64 -d /var/vault/unsealKey2
  base64 -d /var/vault/unsealKey3
  ```

  2. Check seal status:

  ```
  vault status
  ```

  3. Run the following command three times with different keys to unseal Vault:

  ```
  vault operator unseal <decoded-unseal-key-1>
  vault operator unseal <decoded-unseal-key-2>
  vault operator unseal <decoded-unseal-key-3>
  ```

  4. Verify that the Vault is unsealed:

  ```
  vault status
  ```

Vault service not running

Symptoms

- You cannot connect to the Vault.
- Connection attempts return *connection refused* errors.

Resolution

```
# Check the Vault service status
systemctl status vault

# Start the Vault service
systemctl start vault

# Check the Vault logs
journalctl -u vault -n 100 --no-pager

# Verify that Vault listens on the expected port
ss -tlnp | grep 8200
```

Vault service is masked

Symptoms

- You cannot connect to the Vault.

Resolution

1. Check whether the vault status is masked on the service node:

```
[kni@servicenode-1 ~]$ systemctl status vault
○ vault.service
     Loaded: masked (Reason: Unit vault.service is masked.)
     Active: inactive (dead)
[kni@servicenode-1 ~]$ vault status
WARNING! VAULT_ADDR and -address unset. Defaulting to https://127.0.0.1:8200.
Error checking seal status: Get "https://127.0.0.1:8200/v1/sys/seal-status": dial tcp 127.0.0.1:8200: connect: 
connection refused
```

2. Run the following commands to unmask the vault:

```
# Unmask the service
sudo systemctl unmask vault

# Reload systemd units
sudo systemctl daemon-reload

# Enable Vault at boot
sudo systemctl enable vault

# Start Vault
sudo systemctl start vault
```

3. Verify the vault status. The Vault must be in a running state.

```
[kni@servicenode-1 ~]$ vault status
WARNING! VAULT_ADDR and -address unset. Defaulting to https://127.0.0.1:8200.
Key                     Value
---                     -----
Seal Type               shamir
Initialized             true
Sealed                  true
Total Shares            5
Threshold               3
Unseal Progress         0/3
Unseal Nonce            n/a
Version                 1.21.1
Build Date              2025-11-18T13:04:32Z
Storage Type            raft
Removed From Cluster    false
HA Enabled              true
[kni@servicenode-1 ~]$ systemctl status vault
● vault.service - "HashiCorp Vault - A tool for managing secrets"
     Loaded: loaded (/usr/lib/systemd/system/vault.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-05-25 13:33:40 UTC; 14s ago
       Docs: https://developer.hashicorp.com/vault/docs
   Main PID: 215850 (vault)
      Tasks: 12 (limit: 818971)
     Memory: 25.6M
        CPU: 138ms
     CGroup: /system.slice/vault.service
             └─215850 /usr/bin/vault server -config=/etc/vault.d/vault.hcl

[kni@servicenode-1 ~]$ ss -lntp | grep 8200
LISTEN 0      4096          0.0.0.0:8200       0.0.0.0:*

[kni@servicenode-1 ~]$ curl -k https://127.0.0.1:8200/v1/sys/health
{"initialized":true,"sealed":true,"standby":true,"performance_standby":false,"replication_performance_mode":"unknown"
,"replication_dr_mode":"unknown","server_time_utc":1779716110,"version":"1.21.1","enterprise":false,"echo_duration_ms
":0,"clock_skew_ms":0,"replication_primary_canary_age_ms":0,"removed_from_cluster":false}
```

Service node is down

Symptoms

- Unable to access the service node.
- As the service node is down, unable to access the vault.

Resolution

Wait until the service node is fully operational. The Vault service restarts automatically and remains in a sealed state. After the service node is up and the Vault service is active, unseal Vault by following the standard unseal procedure.

Token expired

Symptoms

- Requests fail with *permission denied* errors after previously succeeding.
- Commands return *token expired* errors.

Resolution

Check token time-to-live (TTL) and renew the token if it supports renewal:

```
# Check token TTL
vault token lookup

# Renew the token if it is renewable
vault token renew
```

Support

If you encounter Vault-related issues, collect the following logs from the service node and attach it to the IBM Support ticket:

1. Collect the service node logs. For more information, see Service node logs.
2. Collect the /var/vault_logs/vault_audit.log file from the service node.
3. Attach both logs that are collected in step 1 and step 2 to the IBM Support ticket.

Virtualization issues

This topic covers issues related to virtualization and their resolution.

GPU node power cycle issues

Problem statement
After a planned or unplanned power cycle of a GPU node, any virtual machines (VMs) that use a vGPU enter a `CrashLoopBackOff` state.

Resolution

1. Restart the `vgpu-device-config` pod in the `nvidia-gpu-operator` namespace.
2. Verify the available vGPU profiles by running the following command:

```
oc get node <gpu_node_name> -o json | jq '.status.allocatable | with_entries(select(.key |
startswith("nvidia.com/"))) | with_entries(select(.value != "0"))'
```

Example output:

```
{
  "nvidia.com/NVIDIA_RTX_Pro_6000_Blackwell_DC-48Q": "4"
}
```

3. Start the VM.

Frequently asked questions

Commonly asked question and answers on deployment, storage, backup, infrastructure, and serviceability of IBM Fusion HCI.

- **General**
  Common question and answers related to IBM Fusion HCI.
- **Workloads**
  Common question and answers related to IBM Fusion HCI.
- **Install**
  Answers to questions that are related to installation of IBM Fusion HCI.
- **Multi cluster IBM Fusion HCI using Hosted Control Plane**
  This section answers questions that are related to the Hosted Control Plane in the IBM Fusion HCI.
- **Red Hat Advanced Cluster Management (RHACM)**
  This section answers questions related to RHACM in IBM Fusion HCI.
- **Expansion rack**
  This section answers questions that are related to expansion rack in IBM Fusion HCI.
- **Rack reset**
  Answers to questions that are related to rack reset in IBM Fusion HCI.
- **Upgrade**
  This section answers questions that are related to upgrade in IBM Fusion HCI.
- **Global Data Platform**
  This section answers questions that are related to Global Data Platform storage in IBM Fusion HCI.
- **Global Data Platform Metro-DR**
  This section answers questions that are related to Metro-DR in IBM Fusion HCI.
- **Data Foundation Storage**
  This section answers questions that are related to storage in IBM Fusion.
- **Network**
  This section answers questions related to networking in IBM Fusion HCI.
- **Compute**
  This section answers questions related to compute/storage node management and upsizing.
- **IBM Data Cataloging**
  This section answers questions that are related to IBM Data Cataloging in IBM Fusion HCI.
- **Content-Aware Storage (CAS)**
  This section answers questions that are related to Content-Aware Storage in IBM Fusion HCI.
- **Remote support**
  This section answers questions that are related to remote support in IBM Fusion HCI.
- **Backup and Restore**
  This section answers questions related to backup and restore feature of IBM Fusion.
- **Support and serviceability**
  This section answers questions related to serviceability.
- **Security**
  This section answers questions related to security in IBM Fusion HCI.
- **Role-based access control (RBAC)**
  RBAC related questions and answers for IBM Fusion HCI.

General

Common question and answers related to IBM Fusion HCI.

1. What is the official name of the hardware offering?
   IBM Fusion HCI is the name of the hardware. Three additional software parts get installed to complete a solution:

   - Red Hat® OpenShift® Container Platform
   - IBM Fusion HCI Management Software
   - IBM Fusion HCI Storage Software

2. Does IBM Fusion HCI support both Red Hat OpenShift and VMware?
   Red Hat OpenShift Virtualization supports both native VMWare images and converted OpenShift VM images. IBM Fusion HCI is a perfect platform to run both Virtual Machines and Containers. Acquire guidance on what virtual workloads work best on OpenShift with IBM Fusion HCI. For more information about Red Hat OpenShift Virtualization, see About OpenShift Virtualization in *Red Hat OpenShift Container Platform* documentation.

3. Can IBM Fusion HCI with OpenShift Virtualization be used to migrate clients off VMware?
   Yes, IBM Fusion HCI with OpenShift Virtualization can be used to migrate you off VMware. For more information about migration, see https://access.redhat.com/products/migration-toolkit-virtualization.

4. Differences between the IBM Fusion HCI and IBM Fusion versions?
   IBM Fusion can be deployed two different ways:
   - IBM Fusion is an operator that is installed on Red Hat OpenShift Container Platform.
   - IBM Fusion HCI is an entire rack of devices (servers, switches, and so on) which will have Red Hat OpenShift Container Platform and IBM Fusion software installed on it.

   User experience varies because of those different deployment models:
   - IBM Fusion HCI has hardware-centric experience and features (monitoring, troubleshooting, adding drives/nodes), whereas forIBM Fusion, those actions take place in Red Hat OpenShift Container Platform.
   - The storage management experience between IBM Fusion HCI and IBM Fusion will vary. IBM Fusion HCI lets you define multiple storage classes but you can only have one file system.

5. **What is the difference between IBM Fusion HCI and Cloud Pak System? Is IBM Fusion HCI a replacement for Cloud Pak System?**
   IBM Fusion HCI provides a cost-effective computing platform, comprised of storage rich servers and network hardware, for OpenShift container-based workloads. It is easy to deploy, simple to maintain, and easy to scale-out capacity as workload demand grows. IBM Fusion HCI is purpose-built for modern container-based workloads and makes it easy for Infrastructure and Operations (I&O) teams to achieve cloud operating models in their private datacenters. As IBM Fusion HCI is purpose-built for modern container-based workloads, it eliminates unnecessary baggage and operational complexity of systems that were designed for last-generation VM based workloads. The Cloud Pak System is designed to run VM based applications on VMware.

6. Where can I learn more about IBM Fusion HCI and IBM Fusion?
   You can see IBM Documentation at IBM Fusion documentation.

7. Can we use Fusion Advanced with Red Hat OpenShift Virtualization Engine VE and OpenShift Kubernetes Engine (OKE)?
   Yes

8. What is supported to run IBM Fusion HCI in a customer supplied rack?
   There is a recommended option of purchasing IBM Fusion HCI without a rack. You can bring your own rack but must ensure that the rack meets the necessary specifications and requirements for the IBM Fusion HCI. If you are interested in using the own racks, contact your IBM Sales representatives for this offering.

9. What is the reasoning behind having 6 PDUs for all config? Why do you have to configure 6 PDUs in econfig?
   IBM Fusion HCI no longer ships 6 PDUs instead only the number of PDUs that are needed are shipped. The number of PDUs that need to be energized depends on the specific rack configuration.

   For more information, see Hardware overview.

10. IBM Fusion HCI has which OS (Red Hat core OS or RHEL)?
    IBM Fusion HCI uses CoreOS as the base platform to deploy Red Hat OpenShift Clusters.

11. IBM Fusion HCI comes with preinstalled Red Hat OpenShift or do we have to deploy it?
    It has a automated installer to deploy Red HatOpenShift at customer site.

12. What would happen if a customer deletes the IBM Fusion HCI or IBM Fusion operators from Red Hat OpenShift Container Platform console > Installed operator page?
    It is not recommended to delete the operator. If you delete, then it leaves the appliance in an inconsistent mode. Contact IBM support for further assistance.

13. Is KEDA (Kubernetes-based Event Driven Autoscaling) supported with IBM Fusion HCI?
    Yes, you should be able to use KEDA on IBM Fusion HCI. In general, see the following document on using KEDA with Red Hat OpenShift, and IBM Fusion HCI is based on an Red HatOpenShift on Bare Metal cluster - https://developer.ibm.com/patterns/scaling-an-event-driven-architecture-with-kafka-on-openshift/

14. How to configure or order IBM Fusion services (IBM Data Cataloging, Global Data Platform, Backup & Restore, Fusion Data Foundation)?
    Yes, you can install IBM Fusion services from the IBM Fusion HCI user interface. For the procedure to install, see Installing services.

15. Do the AFM nodes for IBM Fusion HCI run as containers or bare metal AFM servers?
    The AFM nodes are compute nodes in the Red Hat OpenShift cluster that host the containers running the code that implements AFM. For more information about how IBM Storage Scale handles AFM, see Sharing data.

16. Does IBM Fusion HCI support the usage of NFS protocol RWM protection in 3-site configuration?
    IBM Storage Scale AFM DR does not support 3-site configuration. You can configure IBM Storage Scale stretch cluster with three copies if you have low latency networks available between three sites. Alternatively, you can configure two sites by using IBM Storage Scale Stretch cluster configuration over low latency network and configure AFM-DR over NFS RWX for the 3rd site.

17. If the GPU enhanced nodes are not sufficient, then is there another option apart from acquiring another IBM Fusion HCI appliance? (Like adding GPU worker nodes outside the IBM Fusion HCI rack?)
    IBM Fusion HCI can be ordered with up to four GPU nodes. Each of these GPU nodes can hold from 1 to 8 GPU cards. These cards can be either NVIDIA L40S or NVIDIA H100 NVL GPU cards.

18. IBM Visual Insights will run on Maximo Red Hat OpenShift Container Platform instead of IBM Power but it requires GPU. Will IBM Fusion HCI include GPUs?
    IBM Fusion HCI can be ordered with 6 NVIDIA A100 GPUs. If required, you can order another 2 GPU units.

19. What to do if we see a red icon for some of the widgets in the IBM Fusion HCI infrastructure overview user interface page?
    The red icon indicates a problem with that component and it needs attention. Refer to the widget page or to the events page for further details. For more information about infrastructure overview and events, see Monitoring hardware from IBM Fusion HCI user interface and Analyzing events in IBM Fusion HCI. If the problem persists, then contact IBM support for assistance.

20. On top of the embedded Prometheus monitoring, can the user go further with monitoring integration of their systems. Can they use Alert manager to extract data and send it to an external monitoring tools? Which other options are available to centralize the IBM Fusion HCI appliance on a standard customer monitoring platform?
    For every metric that is seen in Prometheus, you can configure the OpenShift alert manager to create alerts and send them to destinations supported by alert manager. As IBM Fusion feeds events into OpenShift event manager, they show up in any integration that you use to monitor OpenShift events. For more information, see Monitoring and logging by using any monitoring tool.

21. What configurations are available for IBM Fusion HCI? Can the hardware be ordered in different configurations? Can the hardware be upgraded in the field?
    Yes, the hardware can be upgraded in the field. For more information about upgrade, see Upgrading node firmware and Upgrading switch firmware.

    The base and minimal configuration consists of a 42U rack, dual TORs, dual management switches, and 6 dual-socket servers each populated with two 7.68 TB NVMe drives or 3.84TB drives.

    64 core servers with 1024GB of RAM are also available for the base.

22. **What are the minimum and maximum number of nodes?**
    This procedure installs and configures all the default nodes (Three controllers and three compute nodes). If you ordered additional nodes, then they get installed as well. The maximum number of nodes in a rack is 16 and the minimum is 6 nodes.

23. **Can two IBM Fusion HCI racks be clustered in an active-active configuration?**
    We are calling this configuration 'Metro sync DR'. To achieve synchronous replication between the racks, the racks must reside within a metropolitan area (for low latency) and must be connected by a high-bandwidth, high-quality network link. For the exact specification, see Upgrade and upsize considerations in Metro-DR.

24. Can three IBM Fusion HCI racks be clustered in a multi-zone configuration? What are the RPO characteristics?
    We are calling this configuration as 'High-availability cluster', also known as AFM DR in the IBM Storage Scale community. The Recovery Point Objective (RPO) depends on several factors: the latency and bandwidth of the link connecting racks across zones as well as the size and quantity of the transactions that need to be replicated.

25. **Is IBM Spectrum® Storage Scale Erasure Code Edition (ECE) deployed as a container?**
    Yes. IBM Storage Scale Erasure Code Edition (ECE) is deployed container native and managed through an operator.

26. Can IBM Fusion HCI support traditional VMs and containers on the same system?
    Yes, IBM Fusion HCI can support running containers and VMs is side-by-side on the same system with Red Hat OpenShift Virtualization. OpenShift Virtualization comes with Red Hat OpenShift Container Platform.
    Note: The OpenShift Virtualization is a technology based on KubeVirt.

27. Can user run content from Red Hat Marketplace on IBM Fusion HCI?
    Yes, user can run content from Red Hat Marketplace on IBM Fusion HCI.

28. What Cloud Paks will run on IBM Fusion HCI?
    For existing Cloud Pak support, see IBM Cloud Paks support for IBM Fusion.

29. Can IBM Fusion HCI support Windows?
    Yes, IBM Fusion HCI can support both Windows and Linux virtual machines with Red Hat OpenShift Virtualization.

30. Is support available for "protocol"?
    Protocol is the ability to access data in IBM Storage Scale via a mechanism other than native Scale. The IBM Storage Scale supports NFS and SMB.

31. If a IBM Fusion HCI deployed spans 3 racks, then can the control nodes be deployment across the 3 OpenShift Container Platform clusters?
    When the 3 racks are initially setup, all control nodes are in the first rack (basically, you set up and configure the first rack,and then add the other two). After the initial installation is complete, control nodes can be dispersed over 3 racks.
    In expansion, the control nodes remain in one rack. The reason because the cluster is already created and you cannot change the control nodes. In expansion rack, the control nodes remain in one rack because the cluster is already created and you cannot change the control nodes.

32. Can IBM Fusion HCI be used to expand an existing cluster?
    If you use an existing HCI rack, then the new rack can be extended with an existing OpenShift Container Platform cluster. It is called the expansion rack.

33. Do we support single OpenShift Container Platform cluster across multiple IBM Fusion HCI racks?
    Although there are restrictions, you can cluster 3 racks together in one big OpenShift Container Platform cluster.

34. What is the minimum size of the Client environment? (3/4 x86 bare servers dedicated to containers)?
    The minimum size of a IBM Fusion HCI is currently 6 nodes in a rack. The rack can scale up to 16 nodes, and 3 racks can be interconnected.

35. What is the max power draw and cooling figures for IBM Fusion HCI especially with the newly announced hardware?
    Use the StorM tool to calculate the power consumption of a particular configuration.

36. Is mixing of 9155-C10 / 9155-C14 compute nodes supported (for example: 3 x 32 core servers + 3 x 64 cores servers)? Also, in StorM is there a possibility to choose only one GPU server?
    Yes, mixing of 9155-C10 / 9155-C14 servers is allowed. The only server requirement is a minimum of 6x 9155-C10 or 9155-C14 servers with storage to form the base storage cluster. All servers, including GPU servers, can be purchased in increments of one after the base minimum server requirement (mentioned in this topic) is met.

37. Can I add an expansion rack to an existing IBM Fusion HCI rack?
    Yes, you can add a new rack to an existing rack using multi-rack capability.

38. Can I deploy more than one rack in single IBM Fusion HCI cluster?
    Yes, you can deploy two or three racks as a single cluster IBM Fusion HCI using a multi-rack set up.

Workloads

Common question and answers related to IBM Fusion HCI.

1. Can we customize IBM Fusion HCI to deploy OpenShift® Virtualization for general Linux VM workloads?
   You can directly use OpenShift Virtualization to deploy virtual machines on the IBM Fusion HCI.

2. In IBM Fusion HCI, how much storage capacity is available for customer workloads after system has been fully installed?
   Usable Storage Capacity in TB Using 4+2P Coding.
   The 8+2P and and 4+3P encodings are also supported.

   The units of the drive size are 7.68 TB:
   - Depends on how many flash modules you deploy and how many storage nodes you deploy
   - Base configuration - 6 nodes with 2 drives each

   For more information storage capacity, see Installing IBM Fusion HCI.

3. Is MyInvenio Processing Mining (within CP4BA) a tested service for IBM Fusion HCI?
   For Cloud Paks support with IBM Fusion HCI and IBM Fusion, see IBM Cloud Paks support for IBM Fusion.

Install

Answers to questions that are related to installation of IBM Fusion HCI.

1. Is it possible to configure custom ingress certificate?
   Yes, it can be done at installation time by using the IBM Fusion HCI installer or manually as Day 2 following Red Hat OpenShift Container Platform documentation.

2. Is it possible to setup cluster-wide proxy?
   Yes, you can setup cluster-wide proxy.

3. Is offline installation supported for air-gapped data centers?
   Yes, offline installation is supported for air-gapped data centers.
   It is common in production to have a cluster that does not have internet access (air gap, offline, disconnected). In such cases, you can still install the IBM Fusion HCI solution in your air gap environment. The experience is the same as an online installation where you use the IBM and Red Hat entitled registries. However, you must do additional steps to mirror images to your enterprise registry. For more information about mirroring images, see End-to-end mirroring of IBM Fusion HCI and its services images to the enterprise registry.

4. Is Quay registry supported for air gap installation?
   Yes, from 2.8 release onwards, support is available for Quay registry.

5. Is self-signed certificate registry support available for air gap installation?
   Yes, from 2.8 release onwards, support is available for registries with self-signed certificate.

6. What are the stages of the IBM Fusion HCI installation and approximately how long does each stage take?

Installation stage	Approximate time duration
Stage 1: Network configuration	12 minutes
Stage 2: IBM Fusion HCI software installation	90 minutes

   Total: 1 hour 42 minutes

7. Is it possible to change the network configuration of the customer after the initiation of an installation?
   For example, if the customer wants to make changes to their DNS configuration (including changing the cluster name and/or domain name), or make changes to their DHCP configuration?
   It is possible to make network-related changes and restart the installation. To make network changes and restart installation from stage 1. For more information, see Installing IBM Fusion HCI.

8. Can the customer customize the following during IBM Fusion HCI installation - (a) Subnet (b) Red Hat has one of the default pod networks?
   Yes, user interface support exists in stage 2 to customize the pages for OpenShift® Container Platform network customization.

9. Is there a limit on the number of Red Hat OpenShift Container Platform clusters that can be deployed on a single IBM Fusion HCI?
   A single IBM Fusion HCI appliance is limited to one Red Hat® OpenShift Container Platform cluster.

10. Is Deployment Guide available for IBM Fusion HCI?
    The deployment guide is available in the following customer-facing IBM Documentation location: Planning and prerequisites.

11. Who is responsible for running the rack initialization script?
    IBM installs the IBM Fusion HCI. Customers must provide information about the network where IBM Fusion HCI is installed. The client will also need to bring entitlement for Red Hat OpenShift Container Platform.

12. Where can IBM Fusion HCI be deployed?
    IBM Fusion HCI can be deployed in datacenters and in edge locations that can accommodate 42 u racks that meet all the Planning and prerequisites of the IBM Fusion HCI.

Multi cluster IBM Fusion HCI using Hosted Control Plane

This section answers questions that are related to the Hosted Control Plane in the IBM Fusion HCI.

1. Can I use Multi Cluster Engine (MCE) for creating hosted control planes?
   Yes
2. Is Red Hat® Advanced Cluster Management for Kubernetes (ACM) required for creating hosted control planes?
   ACM is not necessary for creating hosted control planes, as MCE alone is sufficient.
3. Do I need OPP license to create hosted control planes?
   No, hosted control planes can be created with OpenShift® Container Platform license by using MCE operator.
4. How do I size my cluster for hosted control planes?
   For sizing information, see Red Hat OpenShift Container Platform documentation.
5. What must be the values of cluster and service network for hosted control plane cluster?
   For more information about the values, see Red Hat OpenShift Container Platform documentation.
6. Can I create VMs in Virtualized hosted control plane cluster?
   No, Red Hat does not support nested virtualization.
7. Can I create VMs in Bare Metal Hosted Control Plane cluster?
   Yes
8. Can I create Bare Metal hosted cluster with GPUs nodes?
   Yes
9. Can I create Hosted Control Plane clusters on IBM Fusion HCI with Dell hardware?
   Yes
10. I have hosted control plane clusters running on IBM Fusion HCI with ACM. Can I configure IBM Fusion HCI in a Regional-DR relation?
    Yes. You must migrate ACM to an external OpenShift Container Platform cluster leaving MCE and hosted clusters in IBM Fusion HCI.
11. Can I provide storage through remote mount scale to Hosted Control Plane?
    Yes, only in Bare Metal hosted clusters with minimum three compute nodes.
12. Can I restrict how much storage a hosted cluster can consume from Fusion Data Foundation provider?
    Yes, using storage quota option of Fusion Data Foundation starting version 4.17.
13. How do I pin a specific IP for a Hosted Control Plane cluster?
    Define MetalLB ipaddresspool. For the procedure, see MetalLB documentation.
14. How many clusters can I create?
    With the base rack, 5 Hosted Control Plane clusters are supported. Extended racks are 3 additional clusters per node.
15. Is Global Data Platform supported as a storage for Hosted Control Plane clusters?
    No, this is still a Technical Preview.
16. What are the network requirements?
    All Hosted Control Plane clusters must have a private or external IP assigned to them to communicate with the control pods on the hub cluster. A load balancer, such as Metallb, must be installed for this purpose.
17. Can hosted clusters use the Fusion Data Foundation storage?
    Yes, by installing the Fusion Data Foundation client on each hosted cluster, the cluster receives the Data Foundation storage from the Fusion Data Foundation provider on IBM Fusion HCI.

Red Hat Advanced Cluster Management (RHACM)

This section answers questions related to RHACM in IBM Fusion HCI.

1. Is RHACM license available with IBM Fusion HCI?
   No, RHACM license is not included in IBM Fusion HCI. Contact Red Hat for RHACM license queries.
2. What is the deployment model for application onboarding load balancing etc? How to ensure that an application is HA across at least 2 "sites" and not all installing on a single rack within a site.
   Each IBM Fusion HCI today is in its own cluster. You can use Red Hat Advanced Cluster Management (RHACM) to manage multiple clusters and deploy applications to and across clusters. RHACM also has a policy engine that can lay down policies on OpenShift® and enforce how users configure the cluster and deploy applications. For more information on RHACM, see https://www.redhat.com/en/technologies/management/advanced-cluster-management.
3. Are there any major limitations for customization after this stack is deployed?
   For disk drives, all nodes that are a part of the storage cluster must have the same number of drives (this is just the way 4+2p ECE works). Otherwise, this is a fully functional OpenShift on bare metal cluster, and its yours to customize and install apps.

Expansion rack

This section answers questions that are related to expansion rack in IBM Fusion HCI.

1. On a base rack with a 4+2p scale cluster, can we add only 3 storage nodes from the expansion rack with 6 storage nodes to OpenShift® Container Platform and reserve the rest of the nodes for future consumption?
   No. To maintain the scale cluster quorum, on a 4+2p scale cluster, all 6 storage nodes must be added to the cluster.
2. Is rack Expansion support on Data Foundation cluster.
   Yes, it supports on Data Foundation cluster. The auxiliary rack must have minimum three storage nodes, and all the NVMe disks size must be of same size across all the nodes.
3. If storage nodes on the initial rack have 4 NVMe drives per node. Can I add nodes with 2 NVMe drives per node to the expansion racks of the cluster?
   Yes, for Data Foundation. The storage nodes in the initial and expansion racks can have different numbers of NVMe drives.
   For example, the initial rack can have 6 NVMe drives in the storage nodes, and the storage nodes in the expansion rack can have 4 NVMe drives.
4. Can I expand the rack only to increase the compute capacity by using IBM Fusion HCI expansion multi-rack?
   No, it is not possible to add compute capacity without adding the minimum of six storage nodes. To increase the compute capacity, add them along with minimum required storage nodes.

Rack reset

Answers to questions that are related to rack reset in IBM Fusion HCI.

Troubleshooting questions

1. What if rack reset fails?
   Follow these steps:
   a. Review error logs in UI.
   b. Fix the underlying issue.
   c. Re-launch rack reset.
   d. Workflow resumes from the last checkpoint.
   e. If the issue persists, contact IBM Support.
2. Can I restart a failed rack reset?
   Yes, simply re-launch from the UI:
   - Workflow automatically resumes from the last checkpoint.
   - No need to start from the beginning.
   - Skips already completed phases.

Post-reset questions

1. Where are the output files located?
   Output files are in /home/kni/isfconfig:
   - kickstart.json - Node configurations
   - rackInfo.json - Rack metadata
2. How do I verify rack reset was successful?
   Confirm that the Workflow status shows `Completed`.

Upgrade

This section answers questions that are related to upgrade in IBM Fusion HCI.

1. Is firmware upgrade available in IBM Fusion HCI?
   Yes, firmware upgrade available in IBM Fusion HCI. For firmware upgrade, see Firmware upgrade.
2. Is OpenShift Container Platform upgrade automated?
   No, OpenShift Container Platform upgrade is not automated.
3. **Are IBM Fusion HCI upgrades disruptive?**
   No, upgrades are not disruptive. Rack firmware can be updated non-disruptively. The IBM Fusion HCI management software makes it easy to evacuate compute nodes and switches so that they can be updated without disrupting workloads.
4. Can I upgrade the version of IBM Fusion HCI services independently?
   No, components of IBM Fusion HCI must not be upgraded. You should upgrade on the IBM Fusion software, which would then orchestrate and upgrade other components as needed.
5. If I run the service node upgrade script and it exited with an error, what is the issue?
   The script output prints the exact error that indicates what action needs to be taken. The output also specifies a log file name (in the `tmp/log` directory) where all the script output logs are present. It can help further troubleshoot the issue.
6. How can I get the help options for the service node upgrade script?
   Run the following command:

   ```
   ./upgrade_servicenode_to_2_12.sh --help
   ```

Global Data Platform

This section answers questions that are related to Global Data Platform storage in IBM Fusion HCI.

1. Does IBM Fusion HCI (with IBM Storage Scale) support block and object storage?
   Support is available for object storage and RWO & RWX storage. IBM Fusion HCI does not support block. Most container workloads use RWO and do not actually need block. Many people conflate block and RWO.
2. In IBM Fusion HCI, the IBM Storage Scale ECE filesystem is considered as container-native or container-ready?
   IBM Storage Scale Fusion deploys containerized ECE filesystem, which works together with IBM Storage Scale Container Storage Interface (CSI) plug-in to provide persistent storage service for applications. From technical perspective, applications can use ECE filesystem almost the same way as ODF/NFS to define PVC and get storage service from ECE with CSI. You need to use different storage class and other properties for configuration.
3. FlashSystems can provide persistent storage for containers on block storage. Why is block storage not mentioned within the context of IBM Fusion HCI?
   IBM Fusion HCI is hyperconverged infrastructure that includes compute, storage, and network. It currently does not support accessing external block storage.
   For the IBM Fusion variant (SDS), OpenShift® can use any storage supported via CSI drivers, including IBM FlashSystems and other 3rd party systems. Also, note that when a container uses "block Storage", it is internally formatted with a filesystem.
4. Is it supported to setup Advanced File Management (AFM) DR manually between two IBM Fusion HCI without integration with Fusion Data Foundation and the operators?
   No. The proposed disaster recovery solutions are as follows:
   - Metro DR
   - AFM cache to external storage array and do DR on external storage.
5. Does IBM Fusion HCI support OpenShift?
   Yes, the IBM Fusion HCI supports OpenShift.
6. Is Fusion Data Foundation available as a service?
   Yes, Fusion Data Foundation is available as a service in IBM Fusion HCI.
7. Can and how does IBM Fusion HCI connect to external storage?
   Yes. The IBM Fusion HCI supports AFM cache capabilities, which means that you can configure IBM Fusion HCI to connect to popular NFS filers and S3 compliant object stores. Data residing in these external storage can be cached locally on the IBM Fusion HCI for performance.
8. Can IBM Fusion HCI connect to external object storage?
   Yes, IBM Fusion HCI can connect to external S3 object stores. This requires the AFM compute nodes 9155-F01.
9. Does IBM Fusion HCI support S3 compliant object storage?
   Yes
10. Can IBM Fusion HCI connect to external Scale/ESS storage?
    For this requirement, the IBM Fusion HCI must have AFM nodes. The AFM nodes can be either 9155-F01 servers or 32-core compute-only servers (9155-C00 or 9155-C10) that have been configured to provide AFM features.
11. Can IBM Fusion HCI connect to external NFS storage?
    Yes. Leverage the AFM cache capabilities provided by IBM Storage Scale ECE. This requires the AFM compute nodes 9155-F01.
12. Is it possible to integrate with third-party storage?
    Yes, IBM Fusion HCI utilizes IBM Storage Scale AFM capabilities to access data from any vendor's NAS.
13. Is the IBM Fusion HCI storage presented as NFS? Can it be accessed via NFS?
    No.
14. Does IBM Fusion HCI deliver a curated Kubernetes and container native storage solution?
    IBM Fusion HCI provides OpenShift cluster, and best practices-based compute, storage and network infrastructure stack. It is a highly available enterprise-ready OpenShift platform.
15. Does IBM Fusion HCI support the access of object storage from applications that run on OpenShift?
    To access object storage from applications that run on OpenShift, use standard S3 interfaces. IBM Fusion HCI itself provides a S3-compatible object store.
16. Does IBM Fusion HCI support encrypted S3?
    Fusion does not provide S3 protocol support such as providing S3 protocol access to data for container native applications or applications outside IBM Fusion HCI.
17. Does the third-party file system manage the data and write to IBM Fusion HCI as if it is a storage resource?
    IBM Fusion HCIdoes not provide storage to 3rd party file systems. IBM Fusion HCI itself includes a file system (via IBM Storage Scale). It can also consume storage from external third-party file systems using the AFM technology of IBM Storage Scale.
18. Is IBM Fusion Data Foundation support included with Cloud Pak for Data (CP4D)?
    Depending on the CP4D version, you might have limitations on what OpenShift Container Platform and Fusion Data Foundation version you can use.
    From an Red Hat® OpenShift Data Foundation perspective, the version is linked to the OpenShift Container Platform version you have deployed IBM Software Hub on. For example, OpenShift Container Platform 4.16.x, you can run Red Hat OpenShift Data Foundation 4.16 only (except when doing an upgrade of OpenShift Container Platform where you can have a mismatch between the 2 versions). Red Hat OpenShift Data Foundation scalability will depend on the amount of CPU and RAM you have on each node where Red Hat OpenShift Data Foundation is deployed but Red Hat lifted limits on the number of Red Hat OpenShift Dedicated per node (although 9 is still the recommended number to limit the failure domain size). The last limit published is 2000 nodes for Red Hat OpenShift Data Foundation. Check here https://access.redhat.com/articles/5001441

    Based on this it gives you 2000x9x4TB = 24PB theoretical limit. One thing to consider for Red Hat OpenShift Data Foundation vs IBM Fusion HCI is the data services you need as IBM Fusion HCI offers more features than Red Hat OpenShift Data Foundation.
19. What is the version of IBM Storage Scale ECE that will be deployed in IBM Fusion HCI?
    For the supported version of IBM Storage Scale ECE, see https://supportcontent.ibm.com/support/pages/node/6824743.
20. With IBM Fusion HCI and encryption, what policy is defined or is there a scope to define the policy? Is the encryption/policy user defined or is it fixed?
    It is for the whole filesystem.

21. Does it allow for different key per worker nodes? Can workload on the node "A" decrypt data from worker "B" or it is per HCI cluster?
    No. It is per IBM Fusion HCI cluster (two racks are using same key for Metro sync DR scenarios)
22. Is IBM Fusion HCI license required for persistence storage on Infrastructure nodes?
    You do not need to license Fusion on infra nodes when you only host the persistent storage offered by Fusion. The deployment of Fusion services other than storage on the infra nodes (for example, catalog or backup) are called application-level services. It means the infra nodes require OCP and Fusion licensing like a worker node. When infra nodes need to be licensed, they use the same model as normal worker nodes; you can use Bare Metal licensing for x86 Bare Metal nodes, or else you can use VPC licensing.
23. Does IBM Fusion HCI support Ceph now?
    Yes, IBM Fusion HCI supports IBM Ceph Storage.
24. Does Fusion Data Foundation support encryption?
    No, encryption is not supported for Fusion Data Foundation storage type in IBM Fusion HCI.
25. Does applications using remote Global Data Platform storage applicable for Backup & Restore?
    No
26. Does applications using remote Global Data Platform storage applicable for disaster recovery?
    No
27. Where to check node drain or MCP rollout logs?
    Run the following command to get the node drain or MCP rollout logs.

    ```
    oc logs -n openshift-machine-config-operator machine-config-controller-xxxxx -c
          machine-config-controller
    ```

    for details

Global Data Platform Metro-DR

This section answers questions that are related to Metro-DR in IBM Fusion HCI.

1. What is the admin network?
   The default OpenShift® container network is referred to as the admin network.
2. What is a daemon or storage network?
   The network that is used by scale on the 100GbE ports is called the storage/daemon network. All scale pods require an extra IP address on this interface.
3. What is a submariner?
   Submariner is an open source software that is used to bring connectivity across the admin network of both sites.
4. What are the site 1 and site 2 network requirements?
   Following are the site 1 and site 2 network requirements:
   a. The default pod IPs on the scale pods must be routable across both the sites.
   b. The storage IPs configured on the scale pods both sites must be routable.
   c. The tie breaker must be routable to both the sites and daemon network.
5. What is a Tie Breaker?
   In a disaster recovery setup, tiebreaker is a mechanism that is used to resolve conflicts between replicated data or services. It determines which site must be considered the primary site during disasters like hardware failure, or other issues.
6. How does Metro-DR ensures data consistency?
   Metro-DR ensures integrity and data consistency through synchronous data replication, that is, data is replicated on both sites always .
7. How does Metro-DR perform failover and failback operations?
   Metro-DR performs failover and failback operations automatically, ensuring minimum disturbance to active services or data. In a disaster situation, failover occurs to the secondary site. After the primary site is healthy, failback is triggered to return services to the primary site.

Data Foundation Storage

This section answers questions that are related to storage in IBM Fusion.

1. Does the Data Foundation support Object Storage?
   Yes, the Data Foundation support Object Storage.
2. What is the supported Data Foundation version?
   The Data Foundation version is the same as the OpenShift® Container Platform version, for example, on OpenShift Container Platform 4.16, install Data Foundation 4.16.
   For a complete list of supported versions, see Support matrix for IBM Fusion HCI versions.
3. What is the difference between Fusion Data Foundation and Red Hat® OpenShift Data Foundation?
   Fusion Data Foundation and Red Hat OpenShift Data Foundation provide similar functionality for provisioning container-native storage in an OpenShift cluster. IBM allows Red Hat to resell support for ODF through a subscription model, with some minor differences in capability. For example, Provider mode is not available in ODF. Red Hat offers ODF in two editions: ODF Essentials and ODF Advanced, with ODF Advanced including Disaster Recovery (DR) capabilities. Currently, neither IBM Fusion nor IBM Cloud Paks include a subscription to Red Hat ODF. The FDF image is sourced from the IBM image repository, while the ODF image is sourced from the Red Hat image repository.
4. What levels of encryption are supported?

Both cluster-wide encryption and storageclass encryption are supported.

5. How to handle and support key management?
   Support is available for external Key Management System, either HashiCorp Vault or Thales CipherTrust Manager.
6. Does Fusion Data Foundation support encryption of data in flight?
   Yes, it utilizes Transport Layer Security (TLS).

Network

This section answers questions related to networking in IBM Fusion HCI.

1. Each node has both a dual-port 25GbE network interface card and a dual-port 100GbE network interface card. Are there two cards of each?
   There is one card of each.
2. Is any of this information in stage 1 network setup screen auto-populated? Must the Service Support Representative (SSR) get this information from the TDA or other planning worksheets?
   Some of the fields are auto-populated, and most of the information should come from the TDA worksheets.
3. Are the connectors to the customer switches DAC or AOC or QSFP?
   Both DAC and AOC cables are supported, and QSA adapter is also supported for using SFP.
4. Are there options to physically isolate data network traffic between applications running inside the POD by using dedicated physical network ports?
   It is not possible to create physically separate data traffic like that.
5. Does the switch firmware upgrade process supports failover/failback?
   When upgrading switch firmware, only one switch can be upgraded at a time. While this is happening, the other switch in the pair carries the traffic so that applications remain connected.
6. Does IBM Fusion HCI include all power and network cables (intra-rack)?
   Yes, all network cables needed within the rack are included. Only the cables needed for the servers ordered are included. However, you need to supply the network cables that connect the IBM Fusion HCI high-speed switches to your networks switches. For multi-rack appliance, you must purchase cables. For more information about cables and transceivers, see Network cable and transceiver options.
7. Which Network topology does IBM Fusion HCI support?
   Currently, the configuration is OpenShift® Container Platform CNI networkType Governesses. IPVLAN is used for the multi-NIC. The two NICs on baremetal are bonded and used for CNI POD network. The NICS are SRIOV compatible, but it is not configured, tuned, or enabled. For more information about network topology, see Planning and prerequisites.
8. What does connecting these switches upstream look like? Is there a prerequisite document?
   Network planning is here: configurable 4 x 100GB uplinks to your datacenter switch (see Network topologies and Network cable requirements) Planning and prerequisites.
9. How many switches are there in IBM Fusion HCI?
   There are 2 management switches and 2 high-speed switches. These switches are paired via MLAG to form a single logical switches.
10. What is the switch model for high-speed switch?
    The switch is NVIDIA switch. The switch model is SN3700V.
11. What is the switch model for management switch?
    The switch is NVIDIA switch. The switch model is SN2201.
12. Which ports are used for storage network?
    The 100 Gb ports are used for storage network.
13. Which ports are used for application network?
    The 25 Gb ports in Lenovo are used for application network.
14. Should the CIDR be unique across sites in Metro sync DR setup?
    Yes, it must be unique. For more information on CIDR, see General Metro-DR prerequisites.

Compute

This section answers questions related to compute/storage node management and upsizing.

1. If I add four compute nodes, is each node added one at a time or would all four pull in at the same time?
   Nodes can be purchased in any number of units, and you can add them in groups from the user interface.
   Node addition is a 3-step procedure:
   a. Compute configuration that readies the hardware and configures XCC.
   b. Add node to OpenShift® Container Platform.
   c. Add node to IBM Storage Scale.
   All three steps can be done with multiple nodes at a time. However, in Step a, it is recommended to fill the nodes from lower RU to higher RU to maintain the center of gravity of the rack.
2. How long does each node take to add?
   It takes approximately 30 minutes to add one node into IBM Fusion HCI ecosystem. Contact IBM support to help with node addition. For more information about node upsize, see Adding nodes to the cluster.
   Compute configuration steps take 20 minutes to configure. If in case the node needs firmware updates, it will take another 90 minutes to update the firmware and complete the configuration.

3. Where does the Service Support Representative (SSR) configure the XCC from?
   The SSR does not configure the XCC. Based on the Link-Local Address (LLA), the IBM Fusion HCI management software connects to the XCC using the default credentials. It also performs the remaining configurations.
4. After the SSR does the rack-and-stack all at once, can I add the nodes later? Does the SSR verify the correctness of the installation or connectivity?
   SSR assists you with *Step 1: Configure node* wherein new nodes are added to cluster for configuration. The configuration completion includes verification. After the completion, you can do the remaining two phases of adding to OpenShift Container Platform and IBM Storage Scale.
5. Would it be possible to add on drives to the nodes after they join the IBM Fusion HCI cluster? For example, there is a need for more storage over time.
   Yes, if you started with less than the full disk count, you can add more drives to the storage nodes. Drives need to be added across all the storage nodes. You can also add more storage nodes but they also have to match the number of existing disks.
6. Do the 64-core compute nodes provide double the number of cores and quadruple the amount of RAM compared to the original compute nodes that were released a year ago?
   IBM Fusion HCI supports 64-core servers (9155-C04, 9155-C05, and 9155-C14) that come with a base memory of 1024 GB RAM, upgradeable to 2048 GB RAM. It is an alternative to 32-core servers (9155-C00, 9155-C01, and 9155-C10), so you can get the most out of demanding applications that are ideal for businesses with high data processing needs.
7. Are there any interop concerns if compute nodes are not on the same firmware release?
   It is always recommended to have all the compute nodes at the same firmware level.
8. Is it possible to remove the expansion rack from the cluster or IBM Fusion HCI after adding an expansion rack to a base rack?
   There is no provision to remove the expansion after the nodes are added to the OpenShift Cluster. If such a need arises, then contact IBM support.

IBM Data Cataloging

This section answers questions that are related to IBM Data Cataloging in IBM Fusion HCI.

1. What is IBM Data Cataloging?
   IBM Data Cataloging is an IBM Fusion service that provides the capability to analyze the metadata of your data sources. It provides rapid automated data discovery and robust metadata capture, curation an enrichment.
2. How can I perform that data analysis and metadata capture?
   In first place, you need to create connections to the wanted data source. After that, you need to select the connection and run a scan over it. You see different statuses when running your scan (as scanning or indexing, for example). After it is terminated, you can see your data that is indexed in the IBM Data Cataloging database.
3. What other actions can you do with your data?
   You can do more complex management with your data by adding tag management or running specific policies.
4. What kind of data sources are supported?
   IBM Data Cataloging supports connection to several kinds of data sources, such as IBM Storage Scale, NFSv4, S3, Cloud Object Storage, IBM Storage Protect, and Server Message Block.
5. Can I use IBM Storage Scale as storage provider for IBM Data Cataloging?
   Yes, IBM Storage Scale can be used as storage provider to install IBM Data Cataloging. You need to have a Scale cluster to be set as remote storage provider, and use the Global Data Platform service available in IBM Fusion to connect your Fusion cluster with the remote Scale.
6. Can I use Red Hat® OpenShift® Data Foundation as a storage provider for IBM Data Cataloging?
   Yes, Red Hat OpenShift Data Foundation can be used as a storage provider for IBM Data Cataloging. You just need to install the IBM Data Cataloging service available in IBM Fusion. After it is installed, you can install IBM Data Cataloging. When installing IBM Data Cataloging as a storage class, be sure that you select the `ocs-storagecluster-cephfs` as storage class.
7. Can I use any storage class to install IBM Data Cataloging?
   IBM Data Cataloging is designed to be storage-agnostic, which means that you can use any storage provider. Only ensure that the storage class selected meets with the requirements that are specified in the prerequisites section.

Content-Aware Storage (CAS)

This section answers questions that are related to Content-Aware Storage in IBM Fusion HCI.

1. What is Content-Aware Storage (CAS) in IBM Fusion Systems?
   CAS is a storage service that helps unlock the value of unstructured data through AI technologies. It enables the access to this value in new data-driven solutions, without the complexities of implementing and managing large AI solutions.
2. What is the primary objective of the CAS architecture?
   The CAS architecture is designed to enable seamless interaction between large language models (LLMs) and vast volumes of unstructured data, enhancing insight generation and suggestion capabilities.
3. How does CAS process unstructured data?
   CAS automatically processes files that are stored on a remote IBM Storage Scale cluster. It extracts parsed content and normalizes metadata, which is then stored along with vector embeddings in a managed vector database.
4. What type of storage platforms does CAS support?
   CAS supports IBM Storage Scale software-based clusters, and Storage Scale Server 3500 and 6000 platforms. It uses remote mounting by using the IBM Fusion Global Data Platform service.
5. How does CAS detect and ingest modified data?

CAS uses IBM Storage Scale watch folders to detect changes (additions, deletions, or updates) in the file system. Only the modified content is ingested, optimizing performance and resource usage.

6. Can CAS ingest data from external S3 sources?
   Yes, CAS supports ingest processing of files that are stored in external S3 sources by using Active File Management (AFM). It automatically scans AFM-S3 filesets to identify data for ingestion.
7. What type of document processing services does CAS support?
   CAS integrates with IBM's Docling Multimodal services and NVIDIA NeMo Retriever Extraction microservices for document parsing and content extraction.
8. How does CAS perform search operations?
   CAS provides a search API that supports semantic, keyword, and hybrid search methods. It aggregates results and optionally uses NVIDIA's reranking for enhanced accuracy.
9. How do customer applications interact with CAS?
   Applications such as chatbots can send natural language queries to the CAS search API. CAS converts the query into a vector representation and performs a search against its vector database, returning optimized results.
10. What is the role of NVIDIA reranking service in CAS search?
    In the NVIDIA multimodal pipeline, CAS uses an aggregator component to refine the top 100 search results. These results are further optimized by using keyword-based and optional reranking before they are returned to the API.
11. Does CAS support Retrieval-Augmented Generation (RAG) workflows?
    Yes, CAS search results can be combined with user prompts and sent to an LLM to generate responses, enabling RAG-based client applications.
12. What types of data does CAS support for semantic extraction?
    CAS focuses on unstructured text data. It uses natural language processing (NLP) and AI technologies to extract semantic meaning from documents.
13. Which IBM Fusion version is required to install CAS?
    CAS requires the newest IBM Fusion version. Installation must be performed by using the default `ibm-spectrum-fusion-ns` namespace. Custom namespaces are not supported for CAS installation.
14. What are the hardware requirements for CAS?
    To run CAS on three OpenShift® compute nodes, the following resources are required:
    - 3 × NVIDIA GPUs
    - One other GPU for optional reranker service
    - 166 vCPUs (83 physical cores)
    - 1024 GB memory
    - 1 TB IBM Storage Scale
15. What is the recommended IBM Storage Scale configuration?
    The most common configuration includes:
    - 48 NVMe drives
    - 30 TB capacity (Enterprise Storage Server 6000)
    - 4 CX-7 network adapters
    - 1.5 TB memory
16. What software versions are required for IBM Storage Scale?
    Ensure that IBM Storage Scale version 5.2.3.1 or later is installed. Remote file systems must be configured and available for CAS.
17. How do you verify the availability of a storage class for CAS?
    In the OpenShift console, go to Storage > Storage Class and confirm that the storage class (for example, `ibm-spectrum-fusion`) created from the IBM Storage Scale remote file system is available and marked as default.
18. What are the prerequisites for enabling GPU workloads in OpenShift?
    Install the following components:
    - Red Hat Node Feature Discovery Operator
    - NVIDIA GPU Operator (compatible version)
19. What is NVIDIA NIM and why is it required?
    NVIDIA NIM is a set of optimized microservices for deploying generative AI models. CAS uses NVIDIA NeMo Retriever for document processing. Provide NVIDIA NGC license keys to pull components from the NVIDIA NGC registry.
20. Is the NVIDIA text reranker mandatory?
    No, the NVIDIA text reranker is optional and requires an additional supported GPU. It enhances search result accuracy by using advanced scoring.
21. What type of proxy URLs must be allowlisted for CAS installation?
    Add the following URLs to your proxy allowlist:
    - `apigee.googleapis.com`
    - `apigeeconnect.googleapis.com`
    - `binaryauthorization.googleapis.com` (optional)
    - `gcr.io`
    - `raw.githubusercontent.com`
22. How do you configure CAS to use the NVIDIA reranker service?
    Add the following flags to the `cas-config` ConfigMap:
    - `NVMM_NEMO_RANKER`
    - `NVMM_NEMO_RANKER_SERVICE`
23. What user permissions are required for remote file system setup?

The Global Data Platform user must be part of the **StorageAdmin** group to enable watch creation. Membership in **CsiAdmin** or **ContainerOperator** alone is insufficient.

24. How do you initiate the installation of CAS?
    You can start the installation from the IBM Fusion user interface by navigating to the Services page and selecting the Content-Aware Storage tile.

25. What components are installed with CAS?
    The CAS installation includes Kafka, the CAS operator, and CNPG (Cloud Native PostgreSQL).

26. What prerequisites must be met before installing CAS?
    Ensure the following items:
    - All prerequisites that are listed in the Planning and prerequisites section are fulfilled.
    - A default storage class (for example, **ibm-spectrum-fusion**) is available and marked as default.
    - All IBM Storage Scale projects and pods are running.

27. What can you do if the installation pre-check raises a warning?
    Resolve the warning at the OpenShift level before proceeding. The pre-check ensures that the environment is ready for CAS deployment.

28. How can you monitor the CAS installation progress?
    After initiating installation, a notification appears on the Services page. You can track progress under Services > Installed. After completed, the CAS service status shows as Healthy.

29. How do you verify that CAS is installed successfully?
    You can verify installation from:
    - **IBM Fusion UI**: Check service version and health status. Use the ellipsis menu to download logs and view documentation.
    - **OpenShift Console**: Go to Home > Projects and search for a namespace that contains **ibm-cas**. Then check Operators > Installed Operators for CAS availability and service version.

30. How do you configure CAS after installation?
    Set up a Scale user to enable watch creation, configure domains and data sources, and map domains to data sources. For more information, see Configuring Content-Aware Storage (CAS).

31. Can you download installation logs?
    Yes. From the ellipsis menu in the CAS service window, you can download logs and view related documentation. A success notification confirms log collection.

32. What happens if the CAS service status is not healthy after installation?
    Check the logs for errors, verify that all prerequisites are met, and ensure that all required services (Kafka, CNPG, CAS operator) are running. Revalidate the environment and retry installation if needed.

Remote support

This section answers questions that are related to remote support in IBM Fusion HCI.

1. When would be a service node available for remote connection?
   The IBM support engineer able to connect to the service node only when the remote support connection is enabled by IBM Fusion administrator.

2. Is IBM support engineer authorized to view and connect to a customer environment?
   Yes, access control is managed by the w3 Blue groups. So, the IBM support engineers who are part of the IBM Fusion remote support Blue group can view the available systems and connect to it.

3. Is remote support connection process secured?
   Yes, after the remote support connection is enabled, it creates a secure channel between the service node and remote support connection servers. IBM support engineers can connect to servers using remote support controller application. In such a way, all the actions performed by the IBM Support engineer will be routed through the secured channel.

4. Is there any second layer of authentication for remote support connection?
   Yes, second layer authentication exists. After SSR is connected to the customer network using remote connection, they need to log in to the service node via CLI and complete the challenge response authentication. The service node generates a challenge and that needs to obtain corresponding response from the another server which is managed using Blue groups. The challenge response validation is a combination of IBM support engineers ID and serial number of the service node.

5. Can i terminate ongoing remote support session?
   Yes, as soon as the remote support connection is turned off from the customers end, then all active connections will be terminated.

6. What customer data is used by the remote support session?
   The remote support session is used to support customer storage devices reporting active support issues. It collects personal information such as name, customer organization name and email address which is then be stored in the application database and log files. This information is used to allow IBM support personnel to identify the devices require support and the contact information of the user Is responsible for the system should remote attempts at resolution should fail.
   The contact information is stored in the database and log files and is only accessible to IBM support personnel with a legitimate business need.

7. Which ports are used for remote support session connectivity
   All remote support session communication happens through a secured **https** port 443.

8. How are switch and node accessed by IBM support representative?
   After logging in to the service node, customer need to provide the credentials of the switch or node that the IBM support representative wants to log in

9. How to log in to the OpenShift® Container Platform?
   Similar to nodes and switches, IBM support engineer needs to get the OpenShift Container Platform log in credentials from the customer.

10. what level of access does a remote support engineer have on a service node?
    On service node, support engineer gets non **sudo** Linux user. The level of access for the OpenShift is in hands of the customer.
11. what happens if a service engineer credentials get compromised?
    Only clients who have enabled a remote support session gets impacted. At maximum intruder will be able to login to the service node. Further accesses requires customer to explicitly share more credentials.
12. How can I request audit logs for the previous IBM support remote sessions?
    Yet to get inputs.

Backup and Restore

This section answers questions related to backup and restore feature of IBM Fusion.

1. Can I install the Backup & restore as a day 2 operation?
   Yes, from the Services user interface page of IBM Fusion.
2. How to verify whether Velero was installed and what gets backed up behind the scenes?
   Velero runs as a pod in the **ibm-backup-restore ns** namespace. If you are in the OpenShift® Container Platform console, look at the pods in the **ibm-backup-restore ns** namespace or run the following **oc** command:

   ```
   oc get pods -n ibm-backup-restore ns
   ```
3. What are the supported backup storage locations?
   You can use IBM Fusion to create both local backups as well as copy data to an external, even offsite, S3 compliant object store.
4. Does IBM Fusion backs up application data consistently across all the Persistent Volumes (PV)? Which applications are supported in an application consistent backup mode within IBM Fusion?
   Yes, it backs up application data consistently across all the PVs. There are two ways to achieve consistency. One is to quiesce the application. That can be done for virtually any application, but it does imply the willingness to accept some downtime. Another way is crash consistent backup. Not all applications may be able to restore themselves using a crash consistent backup. It depends on the application and how it is designed. For more information about data consistency support in IBM Fusion, see Achieving data consistency for containers.
5. Will IBM Fusion data protection support ODF remote mount?
   Yes, ODF supports a remote mount. For more information, see Deploying OpenShift Data Foundation in external mode in *Red Hat OpenShift Data Foundation* documentation.

Support and serviceability

This section answers questions related to serviceability.

1. How can I display events from the command line?
   The following OC command lists a summary of each IBM Fusion HCI event currently in the event queue:

   ```
   oc get events --field-selector reason=ISFEventManager
   ```

   The following OC command lists the contents of each IBM Fusion HCI event currently in the event queue:

   ```
   oc get events --field-selector reason=ISFEventManager -o yaml
   ```
2. How long do events stay in the event queue?
   - INFO events stay in the queue 3 hours just like regular OCP events
   - WARNING events stay in the event queue 7 days
   - CRITICAL events stay in the event queue 14 days
3. How can I delete events?
   The events are deleted as per default rules specified in question 2 of this section.
4. Which events will open Call Home tickets?
   The list of events that open Call Home tickets is in the configmap allow-tickets. This configmap also contains the list of logs that will be collected automatically when the ticket is opened.

   ```
   lenovoxcc:
                allowed: true
                defaultlogs:
                    loglel01: isf-collection-sets:compute
   ```

   This section indicates that when a ticket is opened for the lenovoxcc component, it will automatically generate the compute collection set in the isf-collection-sets configmap.
   For more information about Call Home tickets, see *Providing service for server hardware* chapter of the Service guide.
5. How can I mark a Call Home ticket as closed?
   There is no automatic update of the ISF events from the Call Home server. After a ticket gets closed, you must go to the ISF Events UI, select the event that opened the ticket, and then in the events context menu, choose "Mark as fixed".
6. Is the IBM Call Home feature mandatory?
   Call Home is an optional feature. Customers can disable the feature if they have concerns that prevent them from giving IBM Fusion HCI access to the internet.
7. What collection sets are defined?

There are 2 main config maps that define what logs will be collected. isf-logs is a list of individual requests

```
logcollector-dep:
      type: k8s-resource
      group: apps
      version: v1
      kind: deployments
      namespace: ibm-spectrum-fusion-ns
      name: logcollector
      description: Log Collector log
```

This entry in `isf-logs` indicates that when the `logcollector-dep` request is made, Log Collector will gather all information about the `logcollector` deployment in the `ibm-spectrum-fusion-ns` namespace (including its pods and pod logs). The second config map is `isf-collection-sets`, which can group requests from `isf-logs`, other entries in `isf-collection-sets`, and individual requests.

```
compute:
      type: collection-set
      collections:
        must-gather:
          type: list-entry
          source-list: isf-collection-sets
          list-entry: must-gather
        ibm-spectrum-fusion-namespace:
          type: list-entry
          source-list: isf-logs
          list-entry: ibm-spectrum-fusion-namespace
```

This entry in `isf-collection-sets` indicates that when the compute collection set is requested, Log Collector will gather the must-gather collection set (defined in `isf-collection-sets`) and `ibm-spectrum-fusion-namespace` (defined in `isf-logs`).

8. What files are included in each collection set shown on the Log Collector UI?
   The specific contents will change as more functions are added to the product, but for now a good summary is in the "Collection Contents" section of this blog post: https://community.ibm.com/community/user/storage/blogs/byron-williams/2021/10/28/viewing-and-examining-logs-in-ibm-spectrum-fusion?CommunityKey=e596ba82-cd57-4fae-8042-163e59279ff3
9. How long should it take for logs to be collected from the UI?
   All collection sets must finish within 10 minutes, except for the "System health check", which takes 2 to 3 hours. Sometimes, it can take up to a maximum of 6 hours to run.
10. How long will log collections remain on the server?
   Logs will be automatically deleted after 24 hours. Logs can be intentionally deleted from the UI page.
11. If my log collection status is partial, does it indicate a log collection failure?
   Partial state of log collection does not mean that the process failed during log collection. It indicates the failure of at least one item in the collection, but no impact to other logs in the collection.
12. What filter options are available for the IBM Fusion HCI Events page?
   See the "Event UI" section of this blog post: https://community.ibm.com/community/user/storage/blogs/byron-williams/2021/10/26/viewing-and-interpreting-events-in-ibm-spectrum-fu?CommunityKey=e596ba82-cd57-4fae-8042-163e59279ff3&tab=recentcommunityblogsdashboard
13. What services are sold with IBM Fusion HCI?
   Today, support will be provided via IBM Expert Care contracts. Add on services will be developed later.
14. What is IBM Storage Expert Care?
   Storage Expert Care is a simplified method of selecting services and support for storage systems at the time of purchase. Storage Expert Care is designed to simplify and standardize the support approach on IBM Fusion HCI, FlashSystem 5200, 7200 & 9200 with pricing calculated as a percentage of the net system (HW & SW). Clients may choose the support level and duration, to align with their business needs.
15. Must clients purchase Storage Expert Care?
   A client has the option of 1) buying the hardware with the Warranty alone, or 2) choosing from Storage Expert Care Basic or Premium on IBM Fusion HCI. The standard Warranty is: 1 Year of Warranty services, 9x5xNBD, Parts-only, IOL Limited and 90-day SWMA. IBM does not offer separate HW or SW maintenance for these products except for Storage Expert Care.
16. Which Storage Expert Care tiers are available?
   Basic and Premium for IBM Fusion HCI was announced on 8/10/21.Support terms are available for1, 2, 3, 4, and 5 years.
17. What are the components of the Premium tier?
   Premium tier includes IBM Software Maintenance (SWMA), Technical Account Manager (TAM), 30 Min. Enhanced Response (Sev1 and Sev2), and Predictive Support.
18. Is remote support available for upgrade?
   IBM offers twice a year updates and they can be applied remotely. However, the client must enable remote access to IBM Fusion HCI. Support is not yet available for IBMers to remotely access client systems on-demand and without client interaction.
19. Where to find more information about Remote support?
   For more information about remote support, see Chapter 16 of our internal service guide- http://worklodepid1.fyre.ibm.com:8081/spectrum_fusion/9155_spectrum_fusion_24_service_book.pdf.
20. Will clients be able to purchase Storage Expert Care only if they selected Warranty at initial purchase?
   Storage Expert Care will be available for separate purchase in TSS Systems (CONGA) in 1Q22. IBM highly recommends selling Storage Expert Care together with the system at time of purchase to ensure the client's systems are protected beyond the standard Warranty. Standard warranty provides 90-day SWMA and 9x5xNBD, Parts-only, IOL Limited.
21. What services are sold with IBM Fusion HCI?
   Today, support will be provided via IBM Expert Care contracts. Add on services will be developed later.
22. When will Renewals become available in TSS Systems (ISAT/CONGA)?
   Renewals will be available in TSS Systems (CONGA) in 1Q22. Prices will be based on the original services price and discount.

23. Will TSS be able to quote additional years than what is available in the configurator (years 1-5)?
    Yes, clients can request special bid pricing for years 6 and 7 of Storage Expert Care.
24. Which date will IBM consider as a start date for Expert Care after the order?
    Storage Expert Care starts at the same time as Warranty start. IBM has published rules for Warranty start for Customer Setup machines (CSU) and IBM Installed (IBI) machines.
25. What is Predictive Support?
    IBM provides predictive alerts for performance, space capacity, and other system problems. If there is a significant need for immediate action to be taken to avoid or prevent an incident, an action plan will be discussed with Client.
26. How will customers get support for IBM Fusion HCI? Will they need to manage multiple support contacts or will there be a single point of contact to receive support?
    There will be a single point of contact for support. The customer will open support tickets against IBM Fusion HCI to receive support for both hardware and software. In some instances, the customer may be required to open support tickets with Red Hat to obtain support for OpenShift® Container Platform. IBM and Red Hat will collaborate to make the support experience as unified as possible.
27. Is Lab service team involved to help the client? If so, is it charged service or included in Expert Care premium?
    After IBM Fusion HCI is installed and ready for customer usage, customers can then begin creation and deployment of containers. Customers will also be able to monitor the system containers/workloads, system resource utilization and alerts. IBM Fusion HCI will support Premium Expert Care and will include the installation services.
28. Is Lab Services required to implement IBM Fusion HCI in the Customer location?
    IBM Fusion HCI does not require lab services. IBM Fusion HCI is physically installed into the customer datacenter by an IBM SSR. The SSR will inspect the hardware and connect power and network cables. The SSR will additionally perform necessary configuration steps to integrate the appliance in the customer's network.

Security

This section answers questions related to security in IBM Fusion HCI.

- Does IBM Fusion HCI support Multi-factor authentication(MFA)?
  Yes, IBM Fusion HCI supports MFA with IBM Security Verify. For more information, see https://www.redbooks.ibm.com/Redbooks.nsf/RedpieceAbstracts/redp5662.html?Open.
- Does IBM Fusion HCI integrate with Identity providers?
  IBM Fusion HCI leverages the authentication & authorization mechanisms provided by Red Hat® OpenShift® Container Platform. For more information, see Red Hat OpenShift Container Platform documentation.
- How IBM Fusion HCI is a secure platform to deploy customer applications?
  IBM Fusion HCI is a secure platform to deploy your applications. For more information, see Security in IBM Fusion.
- Can security patching be done against Red Hat repositories or must wait IBM validated patches?
  Red Hat OpenShift Container Platform patching will come directly from Red Hat image repositories. The patching or updates are applied in the usual OpenShift way. IBM Fusion HCI does not introduce or add any impediments on top of OpenShift. OpenShift admins can easily work with IIBM Fusion HCI.

Role-based access control (RBAC)

RBAC related questions and answers for IBM Fusion HCI.

1. Users of which identity providers can be authenticated with IBM Fusion HCI?
   Any identity provider that can be configured with OpenShift® can be used to authenticate to IIBM Fusion HCI. Labs has tested only with `htpasswd` and LDAP.
2. Which OpenShift roles are authorized to access IBM Fusion HCI user interface?
   Any user with clusteradmin or view roles can access IBM Fusion HCI.
3. What happens when a user with an invalid role tries to login the user interface?
   The user will not get authenticated to IBM Fusion HCI.
4. Does the IBM Fusion HCI GUI have multi-factor authentication (MFA) for sign in?
   IBM Fusion HCI uses SSO through Red Hat® OpenShift Data Foundation authentication. Red Hat OpenShift Data Foundation can also be configured to enforce MFA. IBM Fusion HCI supports all of the OAuth authenticators that OpenShift supports.
5. How does SSO work between IBM Fusion HCI and other integrated applications?
   - Red Hat OpenShift:
     Yes, SSO is configured for IBM Fusion HCI and Red HatOpenShift. It works for Kubeadmin. For identity providers, there is a RFE raised against OpenShift team for enabling SSO login for identity providers.
   - IBM Storage Scale:
     Yes, SSO is configured for IBM Fusion HCI and IBM Storage Scale.
6. How does authorization and authentication work in OpenShift?
   For more information about authorization and authentication, see https://docs.redhat.com/en/documentation/openshift_container_platform/4.21/html/authentication_and_authorization/understanding-authentication.
7. Who are the primary personas/roles to whom Fusion is targeted?
   There are two primary personas:
   a. The team that is setting up and managing the OpenShift infrastructure.
   b. The application team that works on developing and deploying applications to OpenShift.

Accessibility features for IBM Fusion

Accessibility features assist users who have a disability, such as restricted mobility or limited vision, to use information technology content or products successfully.

Accessibility features

IBM Fusion includes the following major accessibility features:

- Keyboard-only operation
- Operations that use a screen reader
- Interfaces that are commonly used by screen readers
- Keys that are discernible by touch but do not activate just by touching them
- Industry-standard devices for ports and connectors
- The attachment of alternative input and output devices

IBM® Documentation, and its related publications, are accessibility-enabled. The IBM Fusion online product documentation in IBM Documentation (IBM Docs) is enabled for accessibility. The accessibility features are described in IBM Documentation at Accessibility.

IBM Fusion uses the recent W3C Standard, WAI-ARIA 1.0 (www.w3.org/TR/wai-aria/), to ensure compliance with US Section 508 (www.access-board.gov/guidelines-and-standards/communications-and-it/about-the-section-508-standards/section-508-standards) and Web Content Accessibility Guidelines (WCAG) 2.0 (www.w3.org/TR/WCAG20/). To take advantage of accessibility features, use the recent release of your screen reader and the recent web browser that is supported by IBM Fusion.

Keyboard navigation

This product uses standard Microsoft Windows navigation keys and the following keyboard shortcuts.

Table 1. Keyboard shortcuts in IBM Fusion

Action	Shortcut for Internet Explorer	Shortcut for Firefox
Move to and around the navigation menu	Tab and up/down/left/right	Tab and up/down/left/right
Exit the navigation menu or close the dialog	Esc	Esc
Move to the Contents View frame	Enter	Enter

Interface information

The IBM Fusion user interfaces do not have content that flashes 2 - 55 times per second.

The IBM Fusion web user interfaces rely on cascading stylesheets to render content properly and to provide a usable experience. The application provides an equivalent way for low-vision users to use a user's system display settings, including high-contrast mode. You can control font size by using the device or web browser settings.

The IBM Fusion web user interface includes WAI-ARIA navigational landmarks that you can use to quickly go to functional areas in the application.

Exception and work-around

The following pages do not provide keyboard-only operation and a screen reader:

- Deploy Distribute UI
- Monitoring UI of deployment manage inlet
- Pattern builder UI

Related accessibility information

In addition to standard IBM help desk and support websites, IBM established a TTY telephone service for use by deaf or hard of hearing customers to access sales and support services:

TTY service 800-IBM-3383 (800-426-3383) (within North America)

IBM and accessibility

For more information about the commitment that IBM has to accessibility, see the IBM Human Ability and Accessibility Center.

IBM Fusion considerations for GDPR readiness

IBM Fusion considerations for GDPR readiness cover data classification, lifecycle management, and user account controls for protecting personal data.

Notice

This document is intended to help you in your preparations for GDPR readiness. It provides information about features of IBM Fusion that you can configure, and aspects of the product's use, that you must consider to help your organization with GDPR readiness. This information is not an exhaustive list, due to the many ways that clients can choose and configure features, and the large variety of ways that the product can be used in itself and with third-party applications and systems.

Clients are responsible for ensuring their own compliance with various laws and regulations, including the European Union General Data Protection Regulation. Clients are solely responsible for obtaining advice of competent legal counsel as to the identification and interpretation of any relevant laws and regulations that might affect the clients' business and any actions the clients might need to take to comply with such laws and regulations.

The products, services, and other capabilities that are described here are not suitable for all client situations and might have restricted availability. IBM does not provide legal, accounting, or auditing advice or represent or warrant that its services or products ensure that clients are in compliance with any law or regulation.

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GDPR

General Data Protection Regulation (GDPR) is adopted by the European Union ("EU") and applies from 25 May 2018.

Why is GDPR important?
GDPR establishes a stronger data protection regulatory framework for processing of personal data of individuals. GDPR brings:

- New and enhanced rights for individuals
- Widened definition of personal data
- New obligations for processors
- Potential for significant financial penalties for non-compliance
- Compulsory data breach notification

Read more about GDPR

- EU GDPR Information Portal
- ibm.com/GDPR website

Product configuration – considerations for GDPR readiness

The following sections provide considerations for configuring IBM Fusion to help your organization with GDPR readiness.

Data life cycle

User accounts
The IBM Fusion system administrator or security administrator creates a user by providing a user ID, email address, full name, and password to grant the user access to the system. This personal data is stored in the database on the client's hardware and can be fully managed by the system administrator or security administrator and edited by the user.

Information on managing users is documented in IBM Documentation (IBM Docs) for IBM Fusion.

System logs
Personal data, including IP addresses, session IDs, user IDs, web-page URLs, and cookie names, might exist within operating system and application logs. The data within these logs is captured automatically as part of the offering and is beyond the control of the client. The logs are retained on disk when sufficient space is available. As more space is needed, older log files are removed. The log files might not be modified or deleted by the client.

The purpose of the system log files is for use during troubleshooting situations. As needed, the log files might be collected and downloaded from the offering for transfer to IBM Support. The log files are not included in the system backups and are therefore constrained to the node unless involved while logs are collected for troubleshooting activity.

Information on system logs is documented in IBM Documentation for IBM Fusion.

Personal data used for online contact with IBM
IBM Fusion clients can submit online comments/feedback/requests to contact IBM about IBM Fusion subjects in various ways, primarily:

- Public comments area on pages in the IBM Fusion community.
- Public comments area on pages of IBM Fusion documentation in IBM Documentation.
- Public comments in the IBM Fusion space of dWAnswers
- Feedback forms in the IBM Fusion community

Typically, only the client name and email address are used, to enable personal replies for the subject of the contact, and the use of personal data conforms to the IBM Online Privacy Statement.

Data collection

For more information, see Data life cycle.

Data storage

Personal data is contained within backups of the offerings. Such personal data includes the personal data that is associated with user accounts that are stored within the database. The IBM Documentation provides information pertaining to creating the backups within the IBM Fusion offering.

The backup feature enables the client to transfer the backup archives to an external location. However, management of any external backup archives is beyond the scope of the offering. The client must implement a set of established 'best practices' for managing and securing such backup files. Information on managing backups is documented in IBM Documentation for IBM Fusion.

Data access

General security measures (for example, disk encryption, physical and remote access) either directly implemented by the offering or suggested actions for the client's preparedness to deploy the offering are documented in IBM Documentation.

For user account data, read or write access can be given to specific users.

Data processing

General security measures are directly implemented by the offering.

Data deletion

Personal data associated with user accounts (as described in Data life cycle) can be fully managed by the system administrator or security administrator, including deletion. Users are not authorized to delete the personal data associated with the accounts. Information on managing users is documented in IBM Documentation for IBM Fusion.

Personal data, including IP addresses, session IDs, and user IDs, might exist within operating system and application logs. The log files might not be modified or deleted by the client. The logs are retained on disk when sufficient space is available. As more space is needed, older log files are removed. The log files might not be modified or deleted by the client. Information on system logs is documented in IBM Documentation for IBM Fusion.

Data monitoring

IBM Fusion does not monitor operating system or application logs, which are collected by the system and remain on the node as space permits. When needed for troubleshooting, logs might be downloaded from the console. Typically, such files remain local to the offering and cannot be managed or altered by users or administrators. Administrators might be able to review some log files (for troubleshooting purposes and without context of any personal data that is contained within) from the offering console. For more complex troubleshooting situations, such logs might be collected and downloaded from the offering for transmission to IBM Support.

Responding to data subject rights

IBM Fusion meets the following data subject rights: right to access, modify, forgotten, and portability.

Glossary

Glossary terms and definitions cover key storage, networking, Kubernetes, and AI concepts used across storage and data management products.

This glossary provides terms and definitions for the #REPLACE_ME_PRODUCTS# software and products.

The following cross-references are used in this glossary:

- *See* refers you from a nonpreferred term to the preferred term or from an abbreviation to the spelled-out form.
- *See also* refers you to a related or contrasting term.

A B C D E F G H I J K L M N O P Q R S T U V W X Z

A

activation key
: See license key.

advisory lock
: A type of lock that a process holds on a region of a file that signals any other process to not use or lock the region or an overlapping region. Other processes are not forced to comply.

AFM-S3 fileset
: A logical grouping of files in an Active File Management (AFM) enabled IBM Storage Scale system that connects to an external S3 bucket for data ingestion.

allocatable extent limit
: A maximum total capacity for the system. The allocatable extent limit is calculated from pool extent sizes.

allowlist
: A list of items, such as usernames, email addresses, or IP addresses, that are granted access to a certain system or function. When an allowlist is used for access control, all entities are denied access, except for those that are included in the allowlist.

application key
: See license key.

array
: An ordered collection, or group, of physical devices (disk drive modules) that are used to define logical volumes or devices. An array is a group of drives designated to be managed with a Redundant Array of Independent Disks (RAID).

asynchronous replication
: A type of replication in which control is given back to the application as soon as the write operation is made to the source volume. Some time later, the write operation is made to the target volume. See also synchronous replication.

audit log

An unalterable record of all commands or user interactions that are issued to the system.

authenticated user
: A user who has logged in to the system with a valid account (user ID and password).

authentication
: The mechanism by which a system determines what permissions a particular authenticated user has to access specific resources or actions. See also authorization.

authorization
: The mechanism by which a system determines what permissions a particular authenticated user has to access specific resources or actions. See also authentication.

authorization code
: An alphanumeric code generated for administrative functions, such as password resets or two-factor authentication bypass.

available capacity
: The amount of usable capacity that is not yet used in a system, pool, array, or MDisk.

B

bearer token
: A security token used in the HTTP Authorization header to authenticate API requests. In Content-Aware Storage, it can be obtained from OpenShift or an external identity provider (IDP).

block storage
: A unit of data storage on a device.

C

cache
: Storage or memory that is used to improve access times to instructions, data, or both. For example, data that resides in cache memory is normally a copy of data that resides elsewhere in slower, less expensive storage, such as on a disk or on another network node.

cache eviction
: A process by which data associated with a file is removed from the cache system. The data is removed either by using a Least Recently Used (LRU) algorithm when configured General Parallel File System (GPFS) hard or soft quota limits are exceeded or by issuing a command. When referenced again in the cache system, the data that is associated with the file is retrieved from the home system.

caching file system
: A storage mechanism that temporarily stores frequently accessed data to improve performance and reduce latency in data retrieval.

caching I/O group
: The I/O group in the system that performs the cache function for a volume.

call home
: A communication link established between a product and a service provider. The product can use this link to place a call to a service provider when it requires service. With access to the machine, service personnel can perform service tasks, such as viewing error and problem logs or initiating trace and dump retrievals.

capacity
: The amount of data that can be contained on a storage medium.

capacity recycling
: The amount of provisioned capacity that can be recovered without causing stress or performance degradation. This capacity identifies the amount of resources that can be reclaimed and provisioned to other objects in an environment.

capacity threshold
: The percent of total usable physical capacity that used capacity must exceed before a notification is sent. See also total usable physical capacity.

CasInstall custom resource
: A Kubernetes custom resource used to configure the installation parameters of the Content-Aware Storage (CAS) service, including enabling optional features like NVIDIA reranking.

CAS namespace
: A dedicated Kubernetes namespace where CAS components (such as pods, services, and configurations) are deployed. Deleting this namespace removes all CAS-related resources.

CAS operator
: A Kubernetes operator that automates the deployment, configuration, and lifecycle management of the CAS service within the OpenShift® environment.

CAS query user
: A user that is authorized to perform search queries against CAS domains. This includes users from Red Hat OpenShift or external identity providers, depending on the access control configuration.

certificate
: A digital document that binds a public key to the identity of the certificate owner, thereby enabling the certificate owner to be authenticated. A certificate is issued by a certificate authority and is digitally signed by that authority.

change volume
: A volume that is used in Global Mirror that holds earlier consistent revisions of data when changes are made.

child pool
: A user-defined capacity that is formed from capacity that is defined either in another pool or a system. See also parent pool.

CIFS
: See Common Internet File System.

CIM
: See Common Information Model.

CIM object manager (CIMOM)
: The common conceptual framework for data management that receives, validates, and authenticates the CIM requests from the client application. It then directs the requests to the appropriate component or service provider.

CIMOM
: See CIM object manager.

CKD
: See count key data.

CLI
: See command-line interface.

client
: A software program or computer that requests services from a server. See also host, server.

cloud account

An agreement with a cloud service provider to use storage or other services at that service provider. Access to the cloud account is granted by presenting valid credentials.

cluster
: A group of computers and other resources that operate together as a single system.

command-line interface (CLI)
: A computer interface in which the input and output are text based.

Common Information Model (CIM)
: An implementation-neutral, object-oriented schema for describing network management or systems management information. The Distributed Management Task Force (DMTF) develops and maintains CIM specifications.

Common Internet File System (CIFS)
: A protocol that manages shared, remote file access for applications to files, printers, serial ports, and so on over a TCP/IP network.

compression
: A function that removes repetitive characters, spaces, strings of characters, or binary data from the data being processed and replaces characters with control characters. Compression reduces the amount of storage space that is required for data.

compute node
: An independent machine that contains one or more microprocessors, memory, storage, and network controllers and runs its own operating system and applications.

concurrent copy
: A function of the DFSMSdss component that is used to back up any collection of data at a point in time with minimum down time for the database or application that uses the collection of data.

ConfigMap
: An API object that holds key-value pairs of configuration data that can be consumed in pods or used to store configuration data. Pods can consume ConfigMaps as environment variables, command-line arguments, or as configuration files in a volume.

copyback
: A process that moves data back to its expected or preferred location to maintain an array in a more efficient configuration after a failed drive is replaced.

count-key-data record
: See data record.

count key data (CKD)

1. An architecture for a direct access storage device (DASD) device or logical device that specifies the access mechanisms for the logical data units on the device through a specific set of supported channel commands. Extensions to the CKD command set form the basis of Extended CKD.
2. A data recording format that uses self-defining record formats in which each record on a volume is represented by up to three fields: a count field identifying the record and specifying its format, an optional key field that can be used to identify the data area contents, and an optional data field that typically contains the user data. See also data record, storage architecture type.

CRU
: See customer-replaceable unit.

customer-replaceable unit (CRU)
: An assembly or part that can be replaced in its entirety by a user when any one of its components fails.

custom resource
: A Kubernetes extension that is used to define and manage custom configurations for applications and services.

cylinder
: A unit of storage on a count-key-data (CKD) device with a fixed number of tracks.

D

data consistency
: A characteristic of the data at the target site where dependent write order is maintained to guarantee the recoverability of applications.

data lakehouse
: A unified data storage and processing architecture that combines the flexibility of a data lake with the structured querying and performance optimizations of a data warehouse, enabling scalable and efficient data analysis for AI and analytics applications.

data record
: A basic unit of data recording format. See also count key data, fixed-block architecture.

data reduction
: A set of techniques that can be used to reduce the amount of usable capacity that is required to store data. Examples of data reduction include data deduplication and compression. See also data reduction savings, stored capacity.

data reduction savings
: The total amount of usable capacity that is saved in a system, pool, or volume through the application of an algorithm such as compression or deduplication on the written data. This saved capacity is the difference between the written capacity and the used capacity. See also data reduction.

data source
: A storage location that hosts documents for use in AI workflows. Also see Creating a data source.

destage
: To move data from cache to a nonvolatile storage medium.

directory server
: A server that can add, delete, change, or search directory information on behalf of a client.

distributed RAID
: An alternative RAID scheme where the number of drives that are used to store the array can be greater than the equivalent, typical RAID scheme. The same data stripes are distributed across a greater number of drives, which increases the opportunity for parallel I/O and hence improves overall array performance. See also rebuild area.

DNS
: See Domain Name System.

domain
: A configuration that collects, transforms, and transfers data from multiple sources to a destination, enabling efficient data flow for analysis and AI models. Also see Creating a domain and connecting with data source.

Domain Name System (DNS)
: The distributed database system that maps domain names to IP addresses.

drive
: A data storage device. A drive can be either a magnetic disk drive or a solid-state drive (SSD).

drive class

A combination of drive technology and speed, which uniquely defines a class of drives that have approximately the same performance characteristics.

drive technology
: A category of a drive that pertains to the method and reliability of the data storage techniques being used on the drive. Possible values include enterprise (ENT) drive, nearline (NL) drive, or solid-state drive (SSD).

E

ECKD
: See extended count key data.

effective capacity
: The amount of provisioned capacity that can be created in a system or pool without running out of usable capacity given the current data reduction savings being achieved. This capacity equals the usable capacity divided by the data reduction savings percentage.

enclosure
: The metal structure in which various electronic components are mounted.

encryption deadlock
: The inability to access encryption keys to decrypt data. See also encryption recovery key.

encryption key label
: The list of encryption key labels used by the storage system to identify keys that will be used on the key server.

encryption key manager
: See encryption key server.

encryption key server
: An internal or external system that runs a key manager that receives and then serves existing encryption keys or certificates to a storage system.

encryption recovery key
: An encryption key that allows a method to recover from an encryption deadlock situation where the normal encryption key servers are not available. See also encryption deadlock.

enterprise
: Pertaining to a type of data storage device that has higher error recovery limits, vibration tolerance, and end-to-end error detection than standard desktop hard drives.

extended count key data (ECKD)
: An extension of the count-key-data (CKD) architecture. It includes additional commands that can be used to improve performance.

extent type
: See storage architecture type.

F

failback
: The restoration of an appliance to its initial configuration after detection and repair of a failed network or component.

failover
: An automatic operation that switches to a redundant or standby system or node in the event of a software, hardware, or network interruption.

FAST API
: A high-performance Python web framework used to build RESTful APIs. In Content-Aware Storage, it powers the query service pod that handles search requests from client applications.

FBA
: See fixed-block architecture.

FC
: See Fibre Channel.

FC-AL
: See Fibre Channel Arbitrated Loop.

FCIP
: See Fibre Channel over IP.

FCP
: See Fibre Channel Protocol.

feature activation code
: See license key.

Fibre Channel (FC)
: A technology for transmitting data between computer devices. It is especially suited for attaching computer servers to shared storage devices and for interconnecting storage controllers and drives. See also zoning.

Fibre Channel Arbitrated Loop (FC-AL)
: An implementation of the Fibre Channel standards that uses a ring topology for the communication fabric; refer to American National Standards Institute (ANSI) INCITS 272-1996, (R2001). In this topology, two or more Fibre Channel end points are interconnected through a looped interface.

Fibre Channel connection (FICON)
: A Fibre Channel communication protocol designed for IBM mainframe computers and peripherals.

Fibre Channel extender
: A device used to extend a Fibre Channel link over a greater distance than is supported by the standard, usually a number of miles or kilometers. Devices must be deployed in pairs at each end of a link.

Fibre Channel over IP (FCIP)
: A network storage technology that combines the features of the Fibre Channel Protocol and the Internet Protocol (IP) to connect distributed SANs over large distances.

Fibre Channel Protocol (FCP)
: The serial SCSI command protocol used on Fibre Channel networks. See also open system.

FICON
: See Fibre Channel connection.

field-replaceable unit (FRU)
: An assembly that is replaced in its entirety when any one of its components fails.

file module
: A component that provides file systems to network users. A file module must be provided with storage for the file systems.

fileset
: A logical subdivision of a file system used to organize and manage data independently within a larger storage environment.

file system (FS)
: A collection of files and certain attributes associated with those files.

file system storage
: Data storage that is organized into files and directories.

fixed-block architecture (FBA)
: An architecture for a virtual device that specifies the format of and access mechanisms for the virtual data units on the device. The virtual data unit is a block. All blocks on the device are the same size (fixed size). The system can access them independently. See also data record, storage architecture type.

fixed block
: See fixed-block architecture.

FlashCopy
: Pertaining to a point-in-time copy where a virtual copy of a volume is created. The target volume maintains the contents of the volume at the point in time when the copy was established. Any subsequent write operations to the source volume are not reflected on the target volume.

flash drive
: A data storage device, which is typically removable and rewritable, that uses solid-state memory to store persistent data. See also flash module.

flash module
: A modular hardware unit containing flash memory, one or more flash controllers, and associated electronics. See also flash drive.

flush-through mode
: See write-through mode.

form factor
: The industry-standard physical dimensions of a storage system drive enclosure. Possible values include “3.5 inch”, “2.5 inch”, and “1.8 inch.”

frame
: The hardware support structure, covers, and all electrical parts mounted therein that are packaged as one entity for shipping.

FRU
: See field-replaceable unit.

FS
: See file system.

full restore operation
: A copy operation where a local volume is created by reading an entire a volume snapshot from cloud storage.

full snapshot
: A type of volume snapshot that contains all the volume data. When a full snapshot is created, an entire copy of the volume data is transmitted to the cloud.

G

General Parallel File System (GPFS)
: A high-performance shared-disk file system that can provide data access from nodes in a clustered system environment.

Global Mirror
: A method of an asynchronous replication that maintains data consistency across multiple volumes within or across multiple systems. Global Mirror is generally used where distances between the source site and target site cause increased latency beyond what the application can accept.

GPFS
: See General Parallel File System.

GPU resource indicator
: A Kubernetes resource identifier (for example, nvidia.com/gpu, nvidia.com/mig-7g.80gb) used to specify the type and configuration of GPU resources allocated to a container or service.

Granite Embedding model
: One of a collection of embedding models that generate high-quality text embeddings and a reranker model to improve the relevance and quality of search results or recommendations.

group id (GID)
: In the UNIX or Linux operating system, an integer that uniquely identifies each group of users to the operating system.

H

Hardware Management Console (HMC)
: A system that controls managed systems, including the management of logical partitions and use of Capacity Upgrade on Demand. Using service applications, the HMC communicates with managed systems to detect and consolidate information, which can then be sent for analysis.

HMC
: See Hardware Management Console.

host
: A physical or virtual computer system that hosts computer applications, with the host and the applications using storage. See also client, host, server.

host cluster
: A configured set of physical or virtual hosts that share one or more storage volumes in order to increase scalability or availability of computer applications.

host interface card
: See interface card.

host object
: A logical representation of a host within a storage system that is used to represent the host for configuration tasks.

hot-spare
: Pertaining to redundant hardware (such as an adapter, a disk, a drive, or a server) that is installed and available in the event of a hardware failure.

hybrid search
: A search approach that combines semantic and keyword search methods to improve accuracy and relevance.

HyperSwap
: Pertaining to a function that provides continuous, transparent availability against storage errors and site failures, and is based on synchronous replication.

I

identity provider (IDP)
: A service that creates, maintains, and manages identity information and is the source of identities that are communicated to service providers in federated relationships.

IDP token

A secure token issued by an external identity provider (IDP) used to authenticate users when accessing CAS services, including semantic search APIs.

I/O
: See input/output.

I/O enclosure
: A hardware unit in a storage system where data is transferred into and out of the system.

incremental restore operation
: A copy operation where a local volume is modified to match a volume snapshot by reading from cloud storage only the parts of the volume snapshot that differ from the local volume.

incremental snapshot
: A type of volume snapshot where the changes to a local volume relative to the volume's previous snapshot are stored on cloud storage.

input/output (I/O)
: Pertaining to a device, process, channel, or communication path involved in data input, data output, or both.

interface card
: An optional part of a node canister that provides the system with additional host and storage connectivity options.

interface node
: A node that connects a system to an Internet Protocol (IP) network for file-serving capabilities by using service protocols.

Internet Small Computer System Interface (iSCSI)
: An IP-based standard for linking data storage devices over a network and transferring data by carrying SCSI commands over IP networks. See also Small Computer System Interface.

iSCSI
: See Internet Small Computer System Interface.

J

junction path
: A mount point in IBM Storage Scale that connects a fileset to the file system hierarchy, allowing access to stored data.

K

Kafka
: An open-source distributed event streaming platform used for building real-time data pipelines and streaming applications. In Content-Aware Storage, Kafka is used to manage messaging and data flow between components.

Kafka broker
: A server within a Kafka cluster that stores and distributes messages to consumers. It handles incoming data streams and ensures reliable delivery.

Kafka topic
: A category or feed name used in Apache Kafka to which records are sent and from which consumers read.

key server
: A server that negotiates the values that determine the characteristics of a dynamic virtual private network (VPN) connection that is established between two endpoints.

keyword search
: A traditional search method that retrieves results based on exact word or phrase matches.

L

large language model (LLM)
: A language model with many parameters, trained on a large quantity of text.

licensed capacity
: The amount of capacity on a storage system that a user is entitled to configure.

license key
: An alphanumeric code that activates a licensed function on a product.

license key file
: A file that contains one or more licensed keys.

M

machine signature
: A string of characters that identifies a system. A machine signature might be required to obtain a license key.

management node
: A node that is used for configuring, administering, and monitoring a system.

maximum replication delay
: The number of seconds that Metro Mirror or Global Mirror replication can delay a write operation to a volume.

Metro Global Mirror
: A cascaded solution where Metro Mirror synchronously copies data to the target site. This Metro Mirror target is the source volume for Global Mirror that asynchronously copies data to a third site. This solution has the potential to provide a disaster recovery with no data loss at Global Mirror distances when the intermediate site does not participate in the disaster that occurs at the production site.

Metro Mirror
: A method of synchronous replication that maintains data consistency across multiple volumes within the system. Metro Mirror is generally used when the write latency caused by the distance between the source site and target site is acceptable to application performance.

N

natural language processing (NLP)
: A field of artificial intelligence and linguistics that studies the problems inherent in the processing and manipulation of natural language, with an aim to increase the ability of computers to understand human languages.

nearline
: Pertaining to a type of storage in which data is available in a short amount of time, but not instantly.

nearline SAS drive
: A drive that combines the high capacity data storage technology of a Serial Advanced Technology Attachment (SATA) drive with the benefits of a serial-attached SCSI (SAS) interface for improved connectivity.

node
: A single processing unit within a system. For redundancy, multiple nodes are typically deployed to make up a system.

NVIDIA NIM pipeline
: A pipeline configuration that uses NVIDIA's NeMo Inference Microservices (NIM) to process and extract insights from data, typically integrated with IBM Storage Scale for AI workloads.

O

open system
: A system that complies with industry-defined interoperability standards. An open system can be connected to other systems complying with the same standards. See also Small Computer System Interface, Fibre Channel Protocol.

order confirmation code
: See authorization code.

overhead capacity
: An amount of usable capacity that is occupied by metadata in a system or pool and other data that is used for system operations.

overprovisioned ratio
: The ratio of provisioned capacity to usable capacity in a system or pool.

overprovisioning
: The result of creating more provisioned capacity in a storage system or pool than there is usable capacity. Overprovisioning occurs when thin provisioning or data reduction techniques ensure that the used capacity of the provisioned volumes is less than their provisioned capacity.

P

parent pool
: A storage pool that receives its capacity from MDisks and has, or will have, some of its capacity allocated to child pools. See also child pool.

performance group
: A collection of volumes that is assigned the same performance characteristics. See also performance policy.

performance policy
: A policy that specifies performance characteristics, for example quality of service (QoS). See also performance group.

PFC
: See priority flow control.

pod
: A group of containers that are running on a Kubernetes cluster. A pod is a runnable unit of work, which can be a either a stand-alone application or a microservice.

pool
: See storage pool.

pool pair
: Two storage pools that are required to balance workload. Each storage pool is controlled by a separate node.

port
: The physical entity within a host, system, or storage system that performs the data communication (transmitting and receiving) over the Fibre Channel.

priority flow control (PFC)
: A link-level flow control mechanism, IEEE standard 802.1Qbb. PFC operates on individual priorities. Instead of pausing all traffic on a link, PFC is used to selectively pause traffic according to its class.

projected capacity
: The estimated volume capacity that is available for volume creation, given the current average performance of any data compression, excluding thin-provisioning savings. See also thin-provisioning savings.

protocol
: A set of rules controlling the communication and transfer of data between two or more devices or systems in a communication network.

provisioned capacity
: The total capacity of all volumes and volume copies in a system or pool.

persistent volume (PV)
: Networked storage in a cluster that is provisioned by an administrator.

persistent volume claim (PVC)
: A request for cluster storage. It allows a pod to use a persistent volume (PV) by claiming a portion of it.

Q

query service pod
: A Kubernetes pod within the CAS namespace that exposes the search API through FAST API, enabling client applications to query the vector database.

R

rack
: A free-standing structure that can hold multiple servers, storage systems, chassis, switches, and other devices.

RAID
: See Redundant Array of Independent Disks.

RAID 0
: A data striping technique, which is commonly called RAID Level 0 or RAID 0 because of its similarity to common, RAID, data-mapping techniques. It includes no data protection, however, so, strictly speaking, the appellation RAID is a misnomer. RAID 0 is also known as data striping.

RAID 1
: A form of storage array in which two or more identical copies of data are maintained on separate media.

RAID 10
: A collection of two or more physical drives that present to the host an image of one or more drives. In the event of a physical device failure, the data can be read or regenerated from the other drives in the RAID due to data redundancy.

RAID 5

A form of parity RAID in which the disks operate independently, the data stripe size is no smaller than the exported block size, and parity check data is distributed across the array's disks.

RAID 6
: A form of RAID that can continue to process read and write requests to all of an array's virtual disks in the presence of two concurrent disk failures.

RAID level
: The level of protection provided by the specific techniques of striping, mirroring, or parity used by a Redundant Array of Independent Disks (RAID).

RAID type
: See RAID level.

raw capacity
: The reported capacity of the drives in the system before formatting or RAID (Redundant Array of Independent Disks) is applied.

rebuild area
: Reserved capacity that is distributed across all drives in a redundant array of drives. If a drive in the array fails, the lost array data is systematically restored into the reserved capacity, returning redundancy to the array. The duration of the restoration process is minimized because all drive members simultaneously participate in restoring the data. See also distributed RAID.

reclaimable capacity
: The amount of provisioned capacity that can be recovered without causing stress or performance degradation. This capacity identifies the amount of resources that can be reclaimed and provisioned to other objects in an environment.

reclaimed capacity
: See reclaimable capacity.

recovery key
: See encryption recovery key.

Redundant Array of Independent Disks (RAID)
: A collection of two or more physical disk drives that present to the host an image of one or more logical disk drives. In the event of a physical device failure, the data can be read or regenerated from the other disk drives in the array due to data redundancy.

repo
: See repository.

repository (repo)
: A persistent storage area for data and other application resources.

reranker
: A specialized AI model that is designed to improve the relevance and accuracy of search and retrieval results, especially in retrieval-augmented generation (RAG) pipelines.

reranking
: A post-processing step in search workflows that reorders retrieved results based on additional criteria or AI models to improve relevance.

reserved capacity
: The amount of used capacity that is made up of capacity reserved for system use. See also total usable physical capacity.

retrieval augmented generation (RAG)
: A technique in which a large language model is augmented with knowledge from external sources to generate text. In the retrieval step, relevant documents from an external source are identified from the user’s query. In the generation step, portions of those documents are included in the LLM prompt to generate a response grounded in the retrieved documents.

S

SCIM client
: A configuration block in the `IdentityProvider` custom resource that enables CAS to retrieve user and group information from the IDP using the SCIM (System for Cross-domain Identity Management) protocol.

SCSI
: See Small Computer System Interface.

SCSI-FCP
: See SCSI Fibre Channel Protocol.

SCSI device
: A product, such as a drive or adapter, connected to a host through an I/O interface using the Small Computer System Interface (SCSI) protocol. A SCSI device is either an initiator,target, or both. See also Small Computer System Interface.

SCSI Fibre Channel Protocol (SCSI-FCP)
: A standard that defines the protocol used to transfer Small Computer System Interface (SCSI) commands over the transport physical layer of the Fibre-Channel interface. This standard is published by ANSI as X3.269-1996.

SCSI initiator
: The system component that initiates communications with attached targets.

SCSI target
: A device that acts as a subordinate to a SCSI initiator and consists of a set of one or more logical units (LUs), each with an assigned logical unit number (LUN). The LUs on the SCSI target are typically I/O devices.

search API
: An application programming interface that enables external applications to perform semantic, keyword, or hybrid searches on ingested content.

security context constraint (SCC)
: A Kubernetes policy object used to define permissions and capabilities for pods, such as allowing containers to run with elevated privileges like SETUID and SETGID.

semantic search
: A search technique that uses the meaning of words and phrases rather than exact keyword matches to retrieve relevant results.

server
: A computer program or a device that provides functions for other programs or devices, called clients. See also client, host.

Server Message Block (SMB)
: A protocol that manages requests and responses in a client/server environment so that clients on a network can share files, directories, and devices.

Small Computer System Interface (SCSI)
: An ANSI-standard electronic interface that allows personal computers to communicate with peripheral hardware, such as disk drives, tape drives, CD-ROM drives, printers, and scanners faster and more flexibly than previous interfaces. See also open system, Internet Small Computer System Interface, SCSI device.

SMB
: See Server Message Block.

solid-state drive (SSD)

1. A storage device that contains nonvolatile flash memory. A solid-state drive (SSD) has no moving mechanical components.

space
: See capacity.

space efficient
: See thin provisioning.

spare drive
: A drive reserved in an array for rebuilding a failed drive in a RAID. Should a drive fail in a RAID, a spare drive from within that device adapter (DA) pair will be selected to rebuild it.

SSD
: See solid-state drive.

standard-provisioned volume
: A volume that completely uses storage at creation.

standard provisioning
: The ability to completely use a volume's capacity for that specific volume.

storage architecture type (storage type)
: The type of storage architecture, either count key data (CKD) or fixed block (FB), for which an array, pool, or volume is provisioned. See also count key data, fixed-block architecture.

storage class
: A storage class is a Kubernetes resource that defines different types of storage (for example, performance and replication) available in a cluster.

storage enclosure
: A specialized chassis that is designed to hold and power drives while providing a mechanism to allow them to communicate to one or more separate computers.

storage node
: A component of a storage system that provides internal storage or a connection to one or more external storage systems.

storage pod
: A subcomponent of a network-attached storage (NAS) system that consists of two or more storage nodes and one or more supported storage systems.

storage pool (pool)
: A collection of storage that identifies an underlying set of resources. These resources provide the capacity and management requirements for a volume or set of volumes.

storage system
: A system that provides persistent storage within a network. A storage system can include facilities for host attachment, user role authentication, a command-line interface (CLI), a graphical user interface (GUI), and storage devices that most often include Redundant Array of Independent Disks (RAID) controllers. It might also include agents for enabling third-party management software to monitor or manage the storage devices.

storage type
: See storage architecture type.

stored capacity
: The amount of capacity that is used to store data that is written by a host after data reduction See also data reduction, total usable physical capacity.

support assistance
: A function that is used to provide support personnel access to the system to complete troubleshooting and maintenance tasks.

synchronous replication
: A type of replication in which the application write operation is made to both the source volume and target volume before control is given back to the application. See also asynchronous replication.

syslog
: A standard for transmitting and storing log messages from many sources to a centralized location to enhance system management.

T

thin-provisioned volume
: A volume that allocates storage when data is written to it.

thin-provisioning savings
: The total amount of usable capacity that is saved in a system, pool, or volume by consuming usable capacity only when needed as a result of write operations. The capacity that is saved is the difference between the provisioned capacity minus the written capacity. See also volume capacity, projected capacity, written capacity.

thin provisioning
: The ability to defer capacity allocation on a storage resource until data is actually written to it.

time slicing
: A GPU resource management technique that allows multiple workloads to share a GPU by allocating time slots for execution, improving utilization.

TLS secret
: A Kubernetes secret that stores TLS certificates and keys used to secure communication with external services, such as S3 buckets.

tokenizer
: A text segmentation program that scans text and determines if and when a series of characters can be recognized as a token.

total capacity savings
: The total amount of usable capacity that is saved in a system, pool, or volume through thin provisioning and data reduction techniques. This saved capacity is the difference between the used usable capacity and the provisioned capacity.

total usable physical capacity
: The amount of physical configured storage space that is available for stored capacity or reserved capacity. This capacity can consist of both internal storage through arrays and external storage through MDisks. See also reserved capacity, capacity threshold, stored capacity.

transparent cloud tiering
: The functions that use cloud storage as an extension of on-premises storage.

trial license
: A temporary entitlement to use a licensed function.

TSE for FlashCopy
: A thin-provisioning method in which storage space is allocated from a TSE repository on an as needed basis. See also TSE repository.

TSE repository
: The amount of capacity in a storage pool reserved for volumes that use a thin-provisioning method of TSE for FlashCopy. See also TSE for FlashCopy.

U

unmapped volume capacity
: The amount of volume capacity that is not mapped to a host. See also volume capacity.

update

1. Software maintenance such as a manufacturing refresh, refresh pack, or fix pack that changes the modification level of a product.
2. To modify a file or data set with current information.
3. To apply fixes to a system.

upgrade

1. Any hardware or software change to a later release, or any hardware addition or software addition.
2. To install a new version or release of a product to replace an earlier version or release of the same product.

usable capacity
: The amount of capacity that is provided for storing data on a system, pool, array, or MDisk after formatting and RAID techniques are applied.

used capacity
: The amount of usable capacity that is taken up by data or overhead capacity in a system, pool, array, or MDisk after data reduction techniques have been applied.

user role
: An identifier that is assigned to a user that defines the set of permissions that are granted to that user.

V

vector database
: A specialized database within the CAS namespace that stores vector embeddings generated from ingested documents, enabling semantic search and retrieval.

vector embedding
: A numerical representation of text data that is used to capture semantic meaning, enabling similarity-based search and retrieval in vector databases.

virtual capacity
: See provisioned capacity.

virtualized capacity
: The amount of capacity that is contributed to a storage pool by a given provisioning group.

virtual machine (VM)
: An emulation of a particular computer system. Virtual machines operate based on the computer architecture and functions of a real or hypothetical computer. Their implementations might involve specialized hardware, software, or a combination of both.

VM
: See virtual machine.

volume
: A fixed amount of physical or virtual storage on a data storage medium.

volume access set
: The set of I/O groups that allows host access to a volume. This set can optionally include the caching I/O group.

volume capacity
: The total capacity for all volumes in a system or storage pool. Volume capacity is defined by the client when a volume is created and surfaced to the host. See also thin-provisioning savings, unmapped volume capacity.

volume snapshot
: A collection of objects on a cloud storage account that represents the data of a volume at a particular time.

W

watch folder
: A designated directory in a file system that is monitored for changes, triggering automated processing workflows when new or modified files are detected.

worldwide ID (WWID)
: A name identifier that is unique worldwide and that is represented by a 64-bit value that includes the IEEE-assigned organizationally unique identifier (OUI).

worldwide name (WWN)
: A 64-bit, unsigned name identifier that is unique.

worldwide node name (WWNN)
: A unique 64-bit identifier for a host containing a Fibre Channel port. See also worldwide port name.

worldwide port name (WWPN)
: A unique 64-bit identifier associated with a Fibre Channel adapter port. The WWPN is assigned in an implementation-independent and protocol-independent manner. See also worldwide node name.

write-through mode
: A process in which data is written to a storage device at the same time as the data is cached.

written capacity
: The amount of usable capacity that would have been used to store written data in a system or pool if data reduction was not applied. See also thin-provisioning savings.

written capacity limit
: The largest amount of capacity that can be written to a drive, array, or MDisk. The limit can be reached even when usable capacity is still available.

WWID
: See worldwide ID.

WWN
: See worldwide name.

WWNN
: See worldwide node name.

WWPN
: See worldwide port name.

X

X.509 certificate
: A certificate that contains information that is defined by the X.509 standard.

Z

z Global Mirror

A method of an asynchronous replication function that maintains data consistency across multiple volumes that are attached to a z/OS system. Time-based data consistency is maintained through the Data Facility Storage Management Subsystem (DFSMS) system data mover (SDM) component.

zoning
: The grouping of multiple ports to form a virtual, private, storage network. Ports that are members of a zone can communicate with each other, but are isolated from ports in other zones. See also Fibre Channel.

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EU Data Act (Regulation 2023/2854)

EU Data Act (Regulation 2023/2854) compliance information describes the types of product data generated by storage products, including vital product data, configuration data, and logging data, along with their retention, access, and control characteristics.

Overview

This Data Information Page provides information regarding data generated by this product. To the extent that access to and use of this data is governed by Regulation (EU) 2023/2854 ("Data Act") the EU Data Act Terms available at this link apply.

Vital Product Data (VPD) / Machine Readable Product Data (MRPD)

Description
    Human readable information on product data including Serial Number, Machine Type and Model, Date of manufacture, and similar identifying information.

Data characteristics

- Format: Text files, CSV files, HTTP/HTTPS, JSON or XML files, and other human readable file types
- Volume: Varies by product, but VPD/MRPD size does not typically change; as of 2025, this data could be up to 4 GB
- Generation: Real-time when the product is powered on; most VPD/MRPD data remains static

Storage and retention

- Storage products can send product data to IBM, sometimes through a service or management console
- Products and consoles can store product data sent to IBM temporarily, deleting data based on space management requirements and/or time
- IBM servers can store product data for at least 180 days

User access and control

- Product data can be obtained by customers through GUIs, CLIs, SNMP, REST interfaces, host commands, and other mechanisms
- Users cannot erase VPD/MRPD data
- Access available through:
  - Service console GUI, CLI and/or REST interface
  - Product-specific interfaces (GUIs, CLIs, REST interface, SNMP, host commands)

Configuration Data

Description
    Human readable information describing the configuration of the product, identifying parts such as FRUs (Field Replaceable Units) and Feature Codes.

Data characteristics

- Format: Text files, CSV files, HTTP/HTTPS, JSON or XML files, and other human readable file types
- Volume: Varies by product and is expected to change with future code product updates and as new features are added; as of 2025, this data could be up to 4 GB
- Generation: Real-time when the product is powered on; most configuration/inventory data is stable until the customer upgrades the product

Storage and retention

- Storage products can send product data to IBM, sometimes through a service or management console
- Products and consoles can store product data sent to IBM temporarily, deleting data based on space management requirements and/or time
- IBM servers can store product data for at least 180 days

User access and control

- Product data can be obtained by customers through GUIs, CLIs, SNMP, REST interfaces, host commands, and other mechanisms
- Access available through:
  - Service console GUI, CLI and/or REST interface
  - Product-specific interfaces (GUIs, CLIs, REST interface, SNMP, host commands)

Logging Data

Description
    Files containing data generated by system use and available to customers, including:

- High level error logs tracking error information
- Command logs showing information such as read or write commands sent to the storage device
- Access logs tracking users logging into the system and running admin commands

Data characteristics

- Format: Text files, CSV files, HTTP/HTTPS, JSON or XML files, and other human readable file types
- Volume: Varies by product and is expected to change with future code product updates and as new features are added; as of 2025, this data could be up to 4GB
- Generation: Real-time when the product is powered on; logging runs continuously and is finite in length, with older data removed as newer data is generated

Storage and retention

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- Products and consoles can store product data sent to IBM temporarily, deleting data based on space management requirements and/or time
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